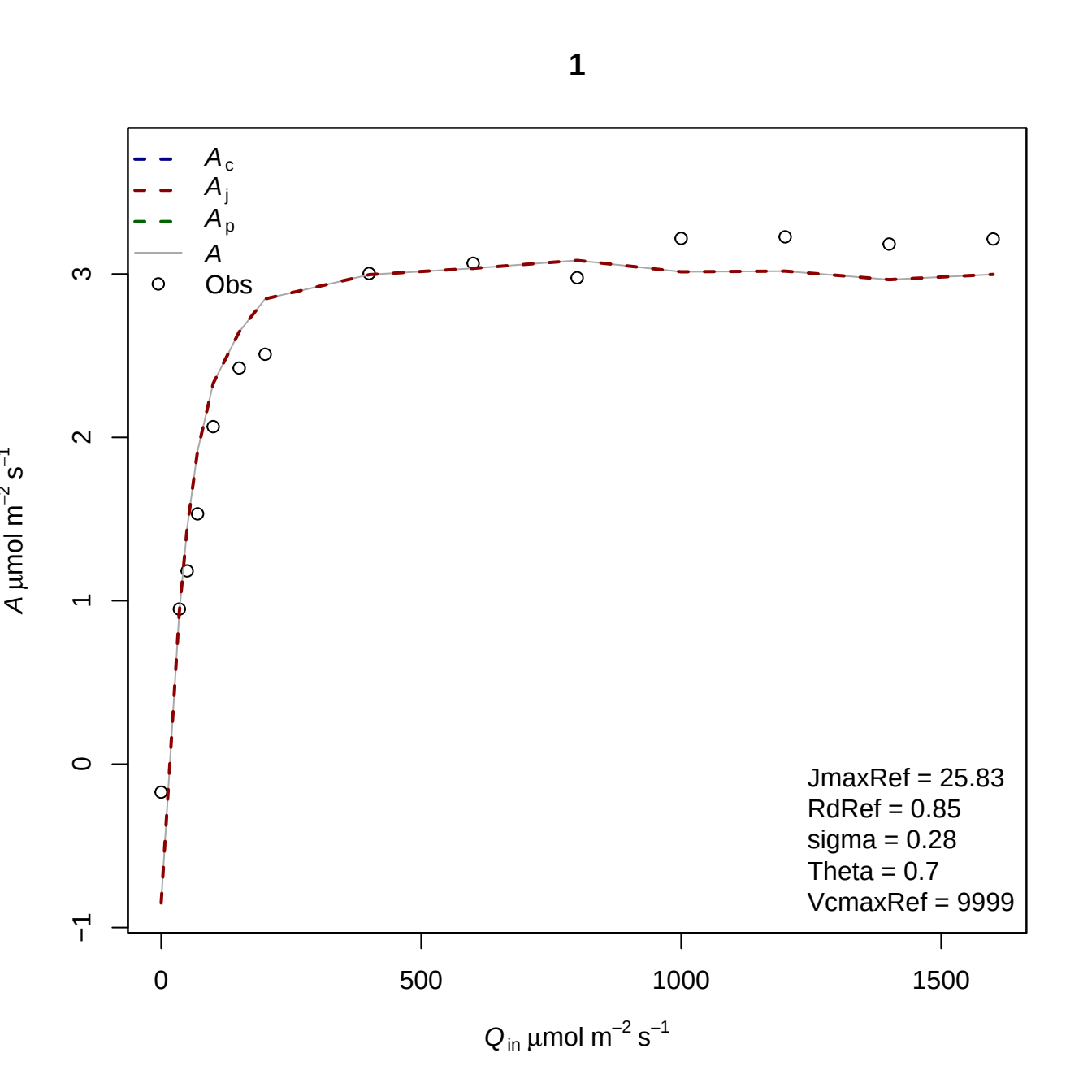
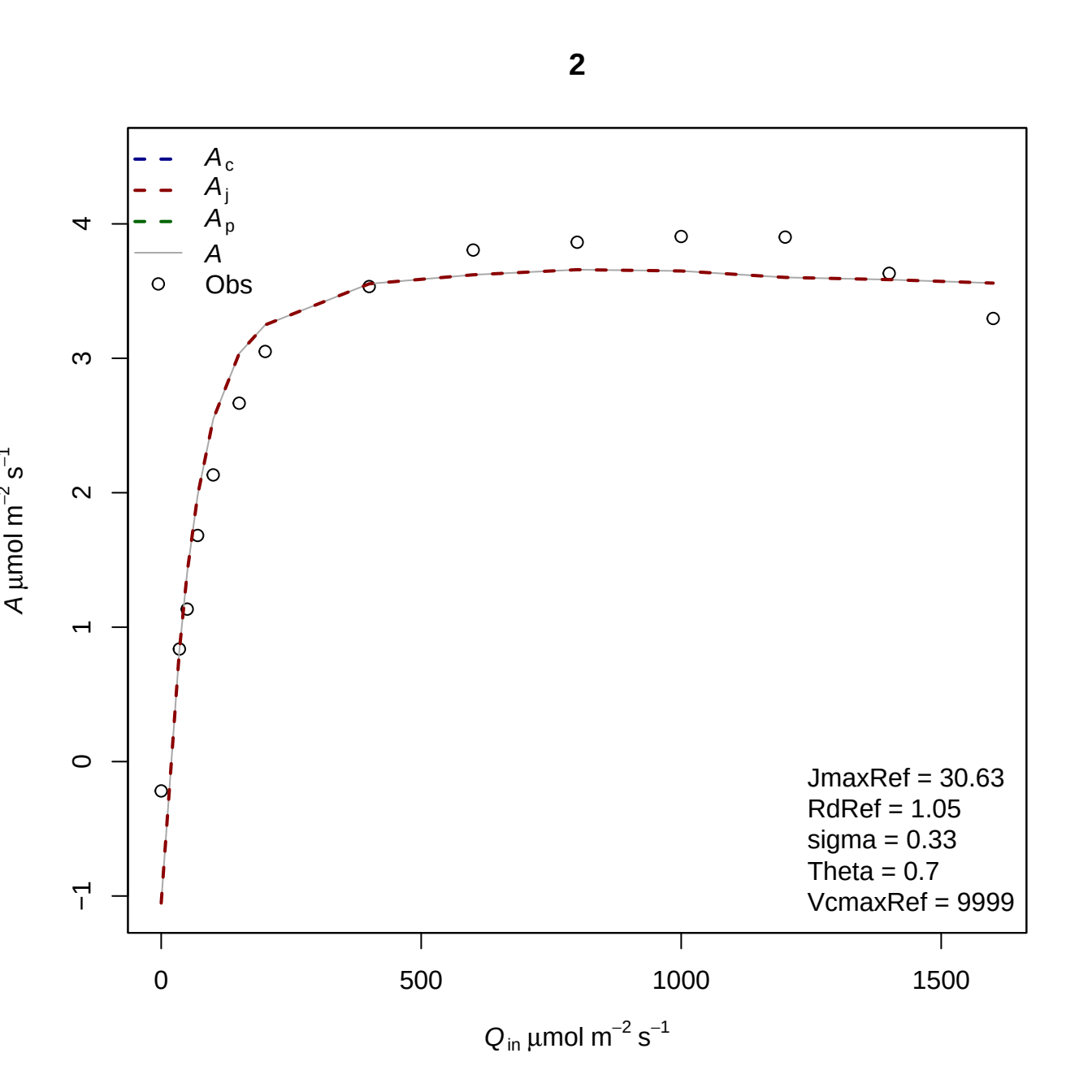
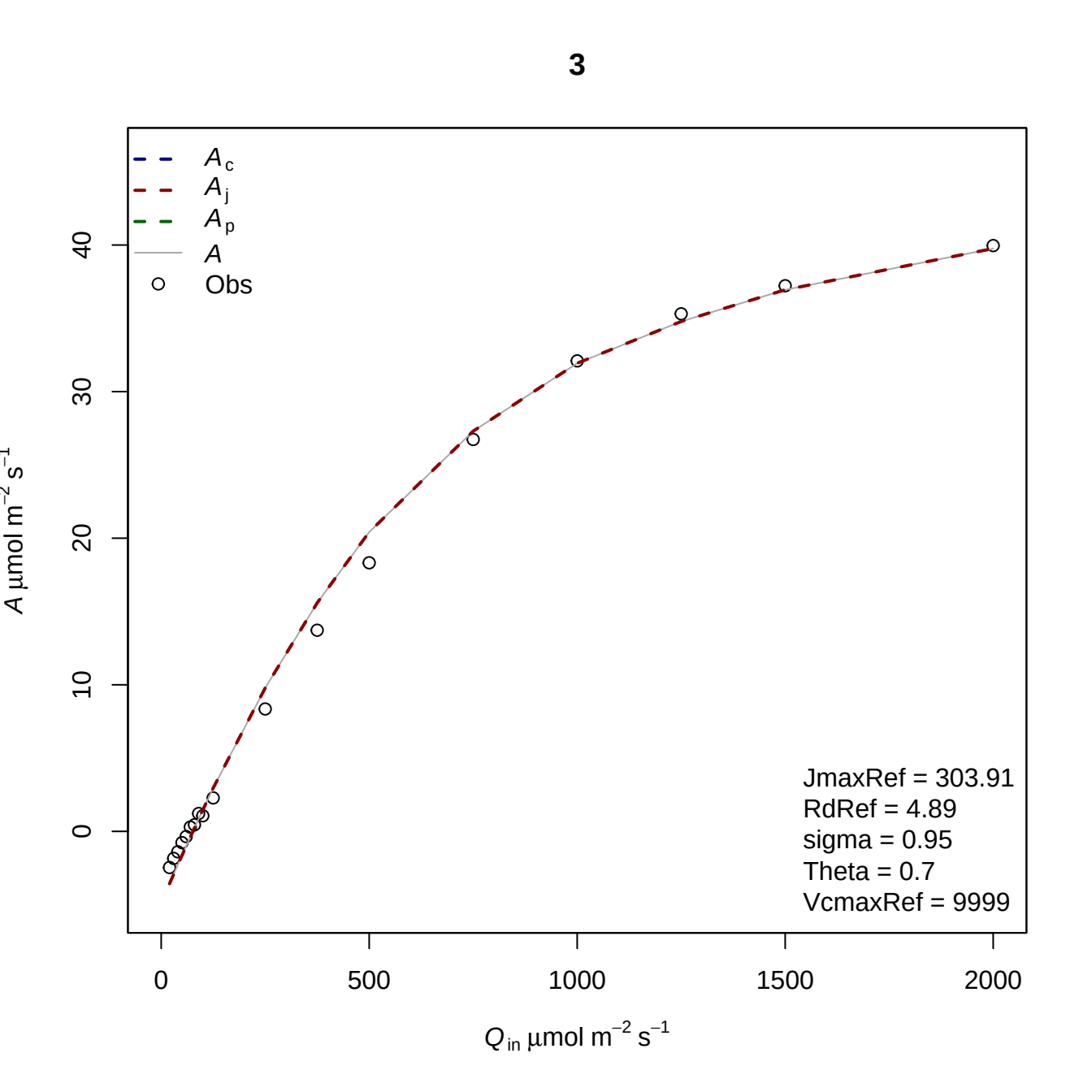
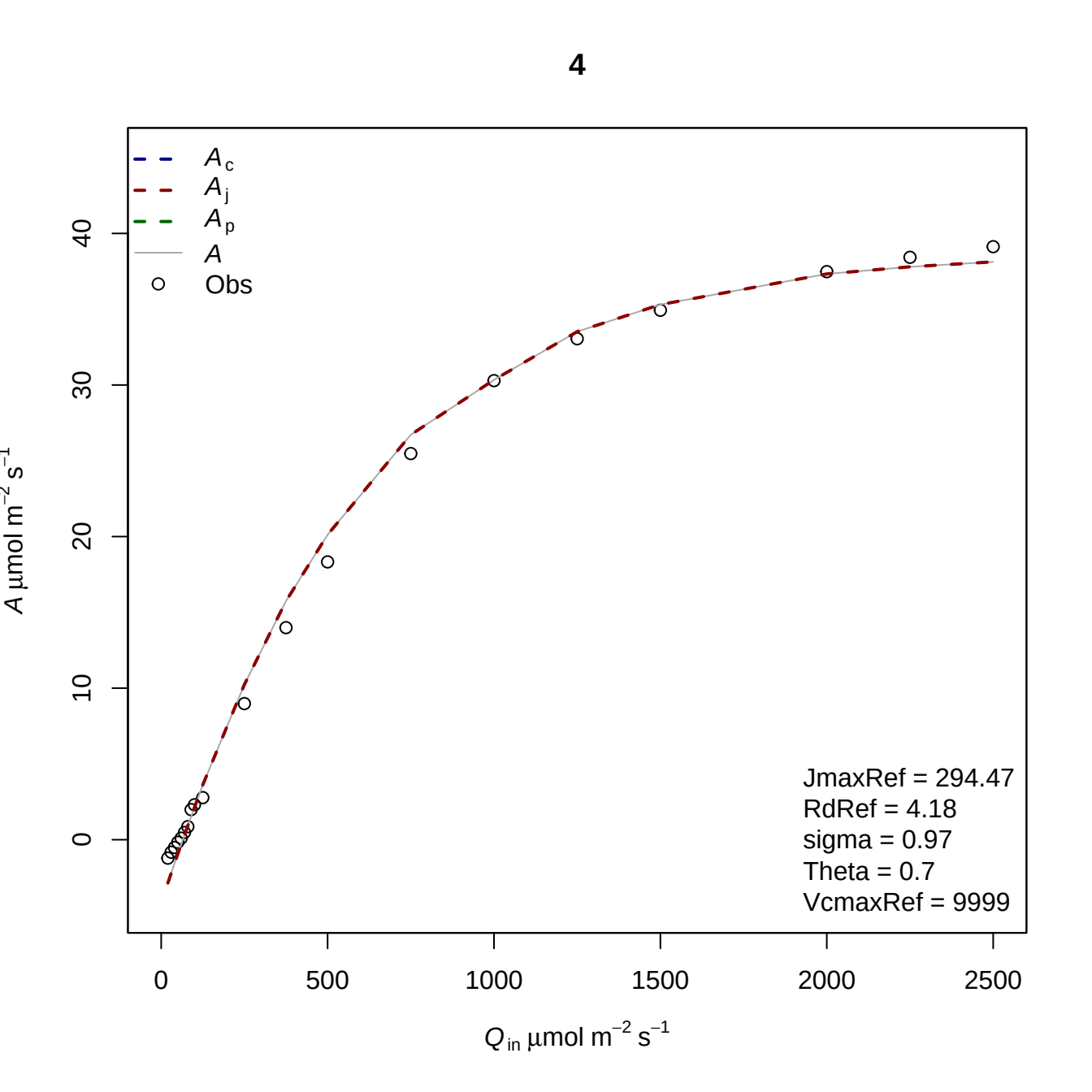
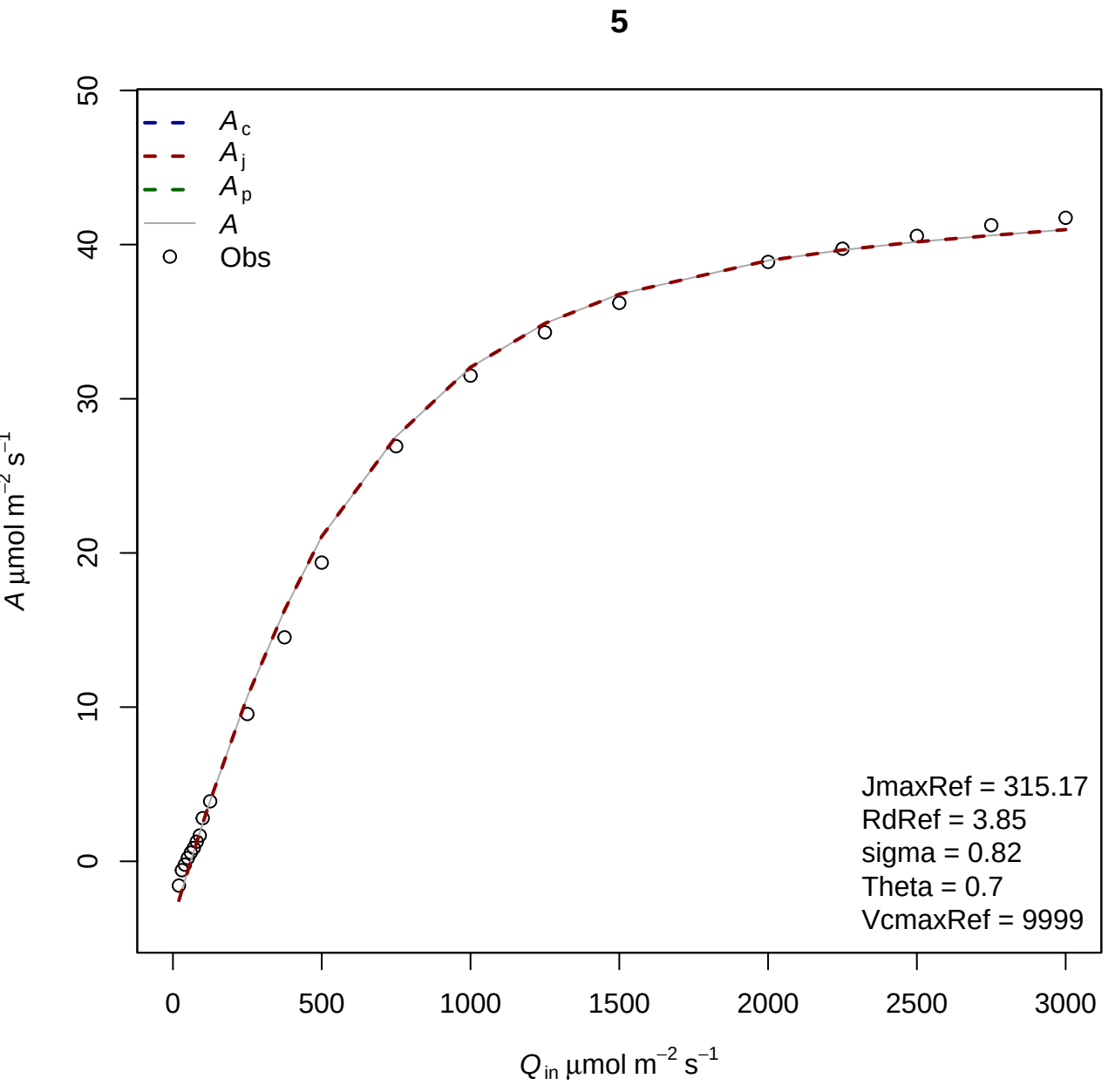
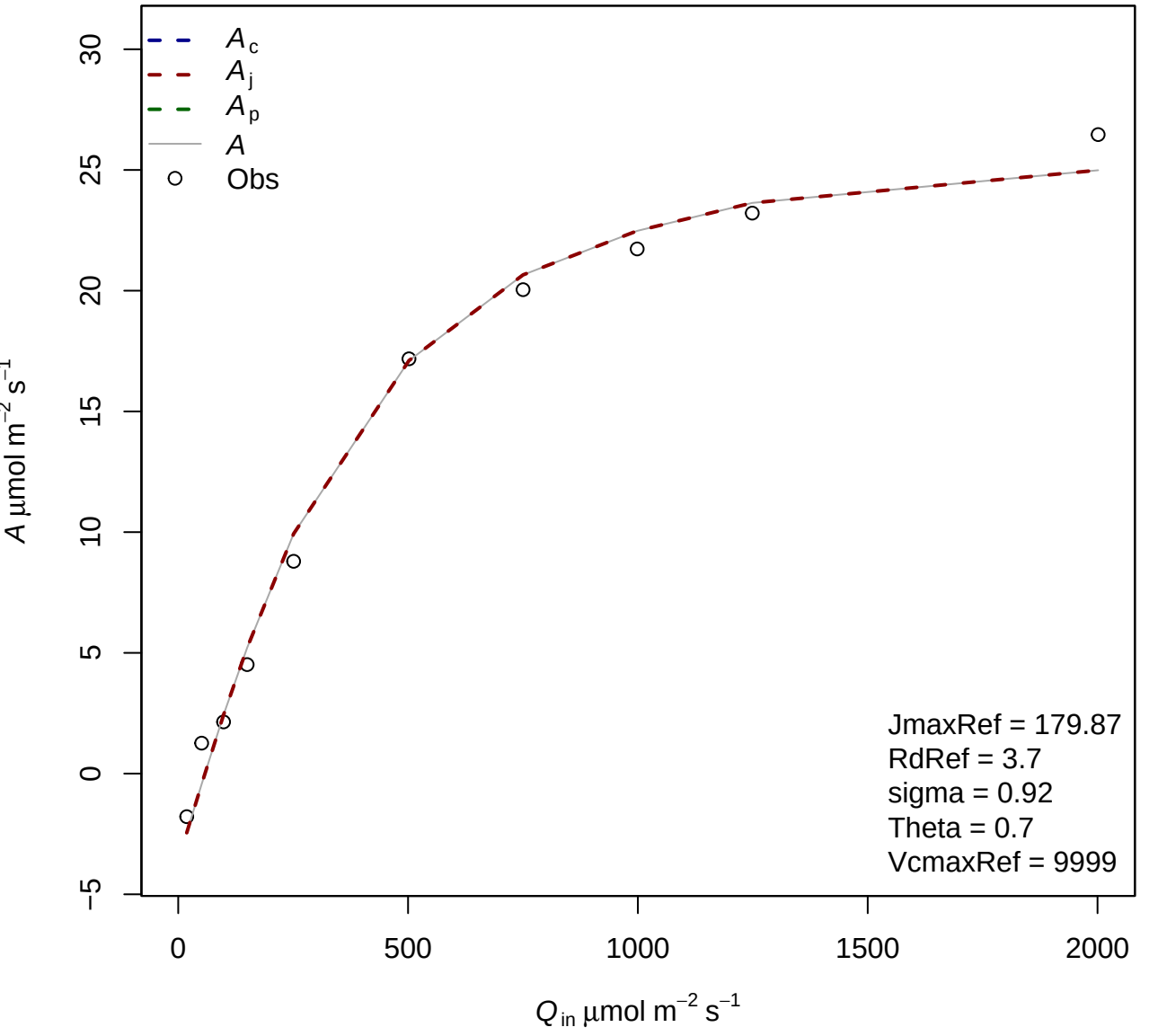

1
A_c
A_j
A_p
A
Obs
JmaxRef = 25.83
RdRef = 0.85
sigma = 0.28
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

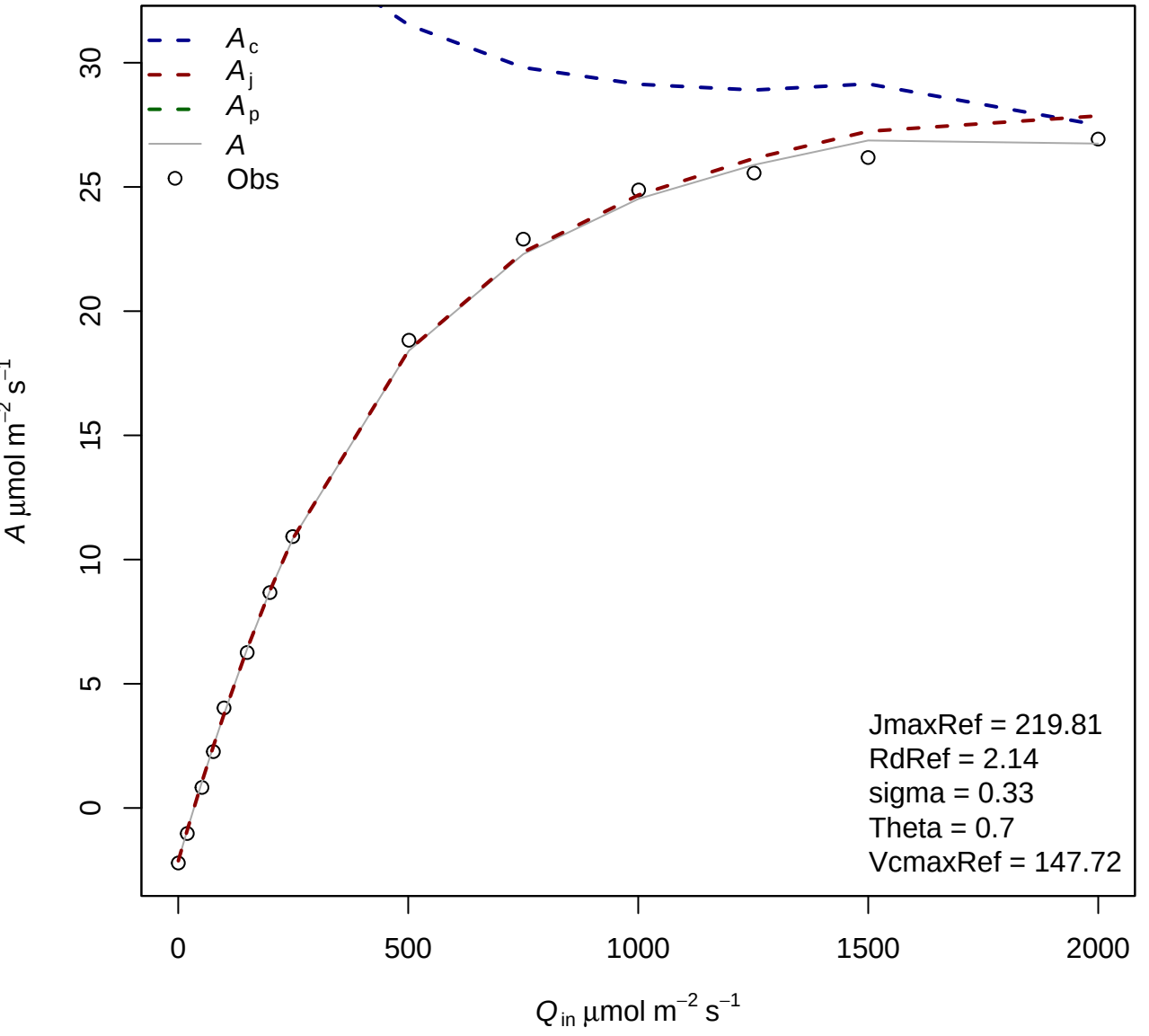

2
A_c
A_j
A_p
A
Obs
JmaxRef = 30.63
RdRef = 1.05
sigma = 0.33
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

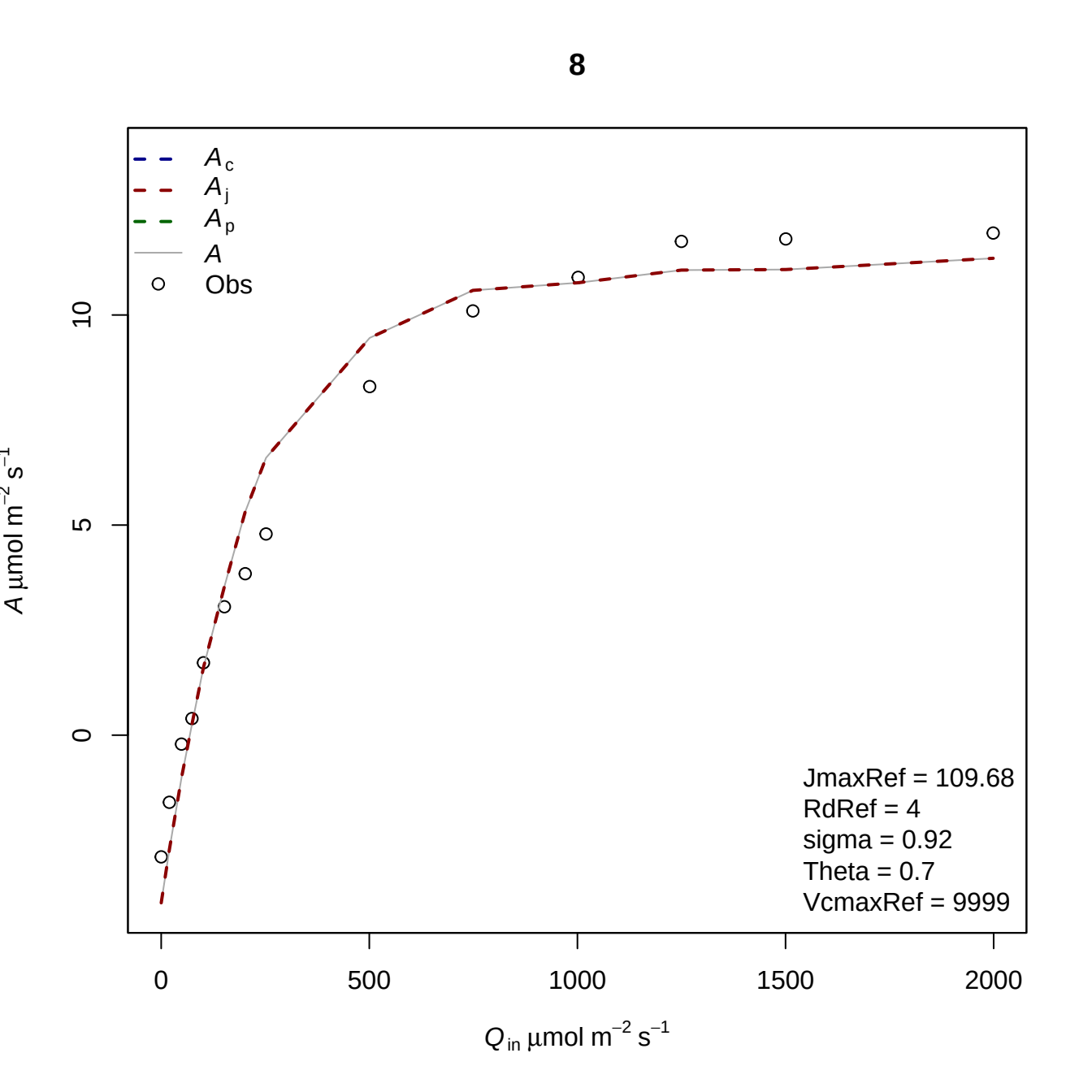

3
A_c
A_j
A_p
A
Obs
JmaxRef = 303.91
RdRef = 4.89
sigma = 0.95
Theta = 0.7
VcmaxRef = 9999
0
10
20
30
40
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

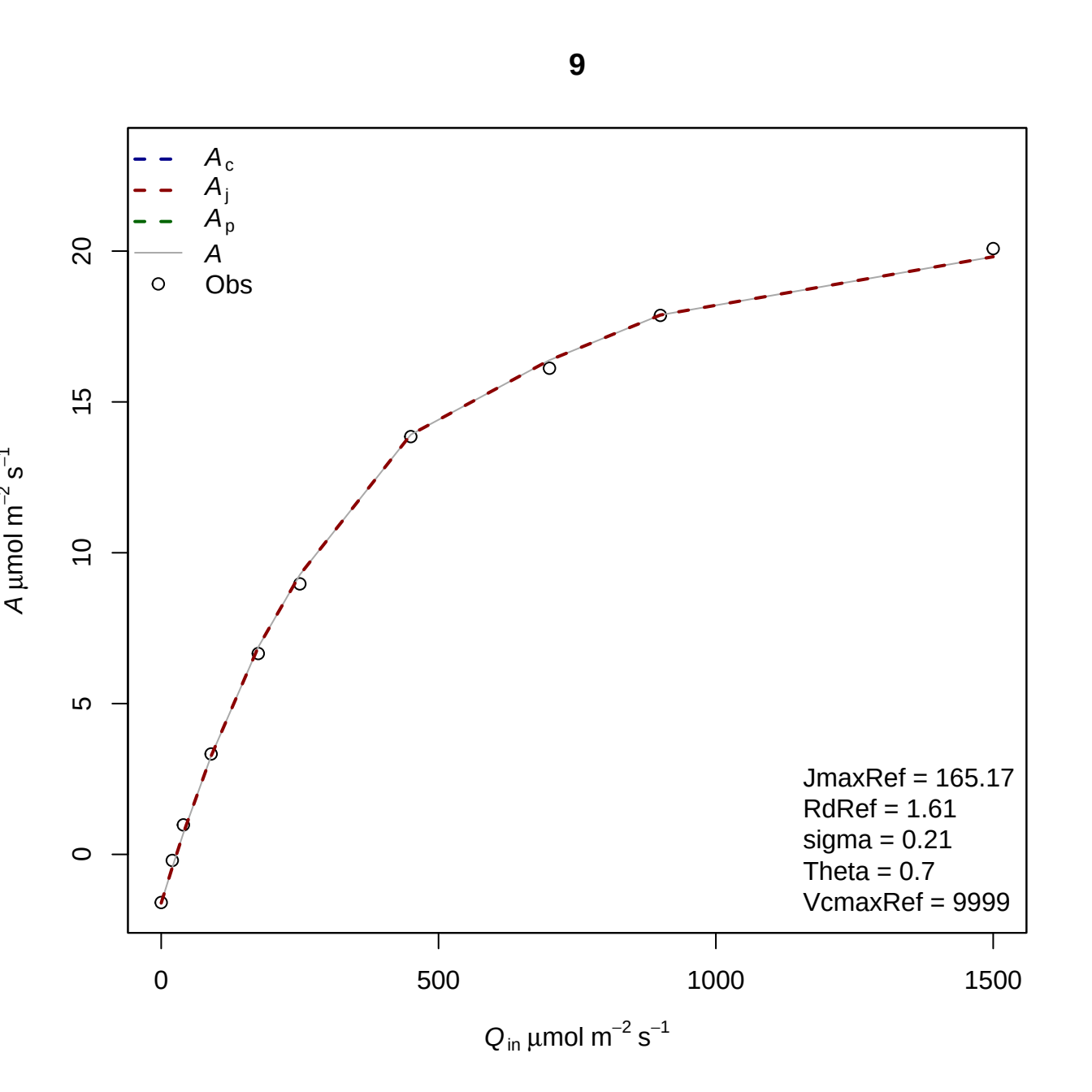

4
A_c
A_j
A_p
A
Obs
JmaxRef = 294.47
RdRef = 4.18
sigma = 0.97
Theta = 0.7
VcmaxRef = 9999
0
10
20
30
40
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
2500
Q_{in} µmol m^{-2} s^{-1}

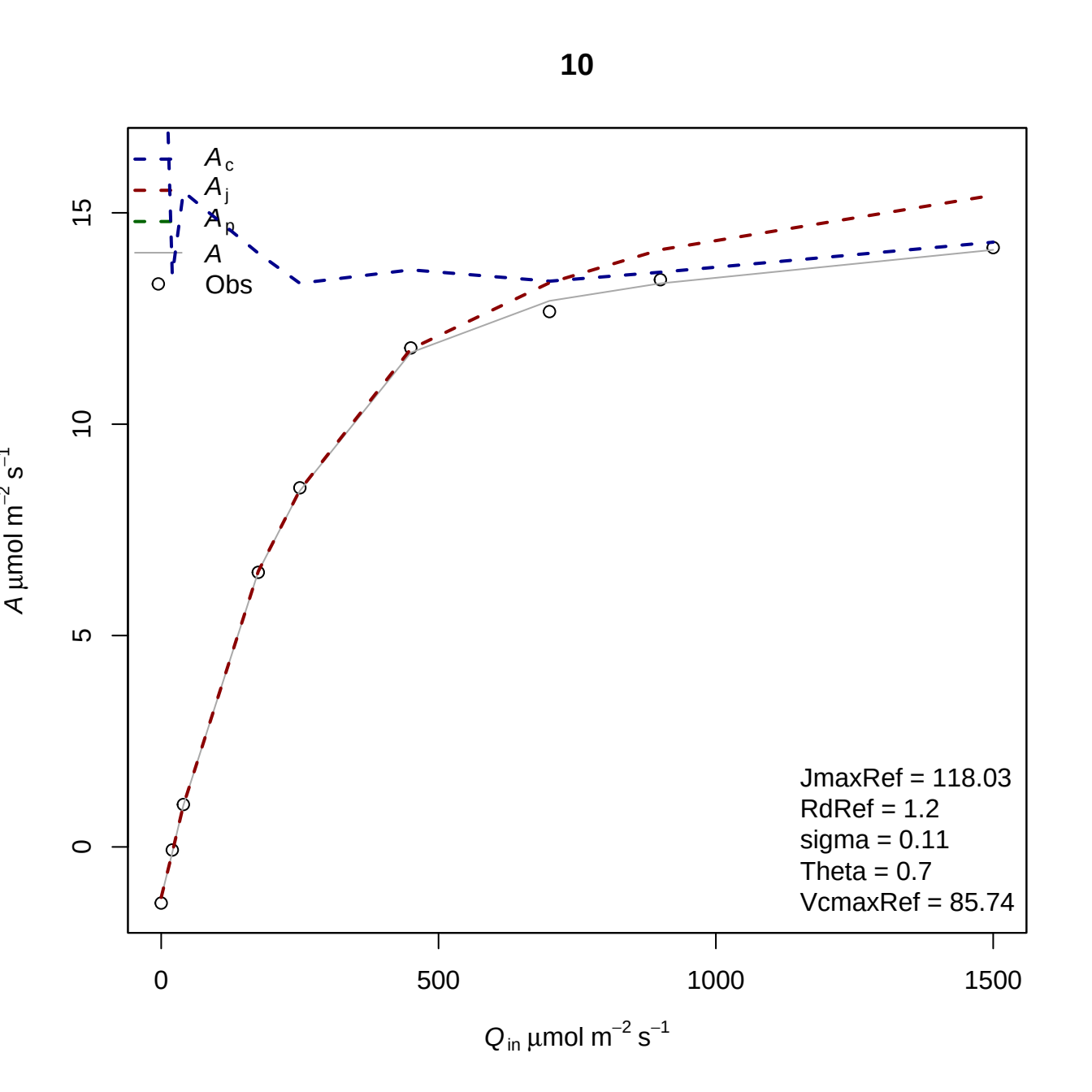

5
A_c
A_j
A_p
A
Obs
JmaxRef = 315.17
RdRef = 3.85
sigma = 0.82
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
40
50
0
500
1000
1500
2000
2500
3000

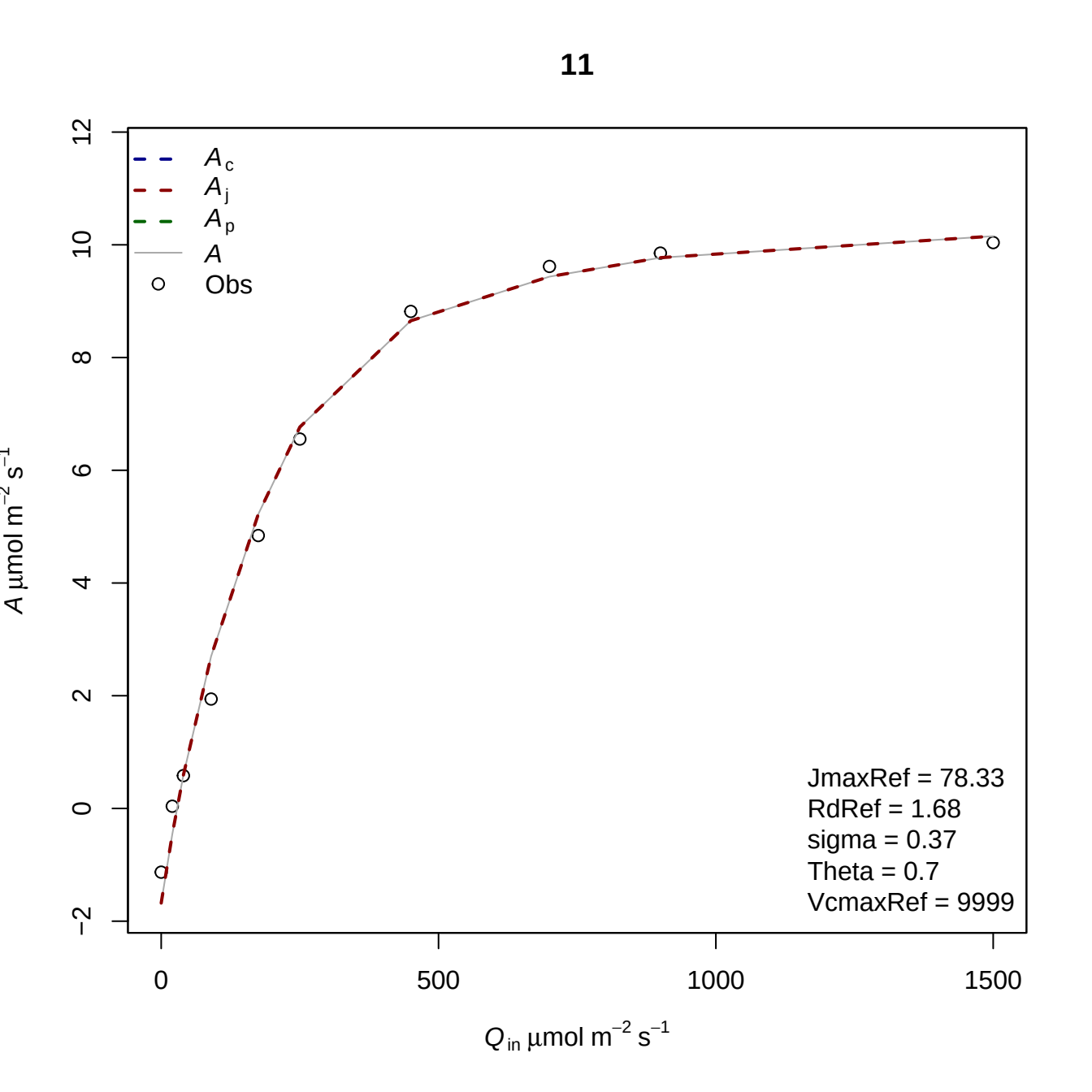

A_c
A_j
A_p
A
Obs
JmaxRef = 179.87
RdRef = 3.7
sigma = 0.92
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
25
30
0
500
1000
1500
2000

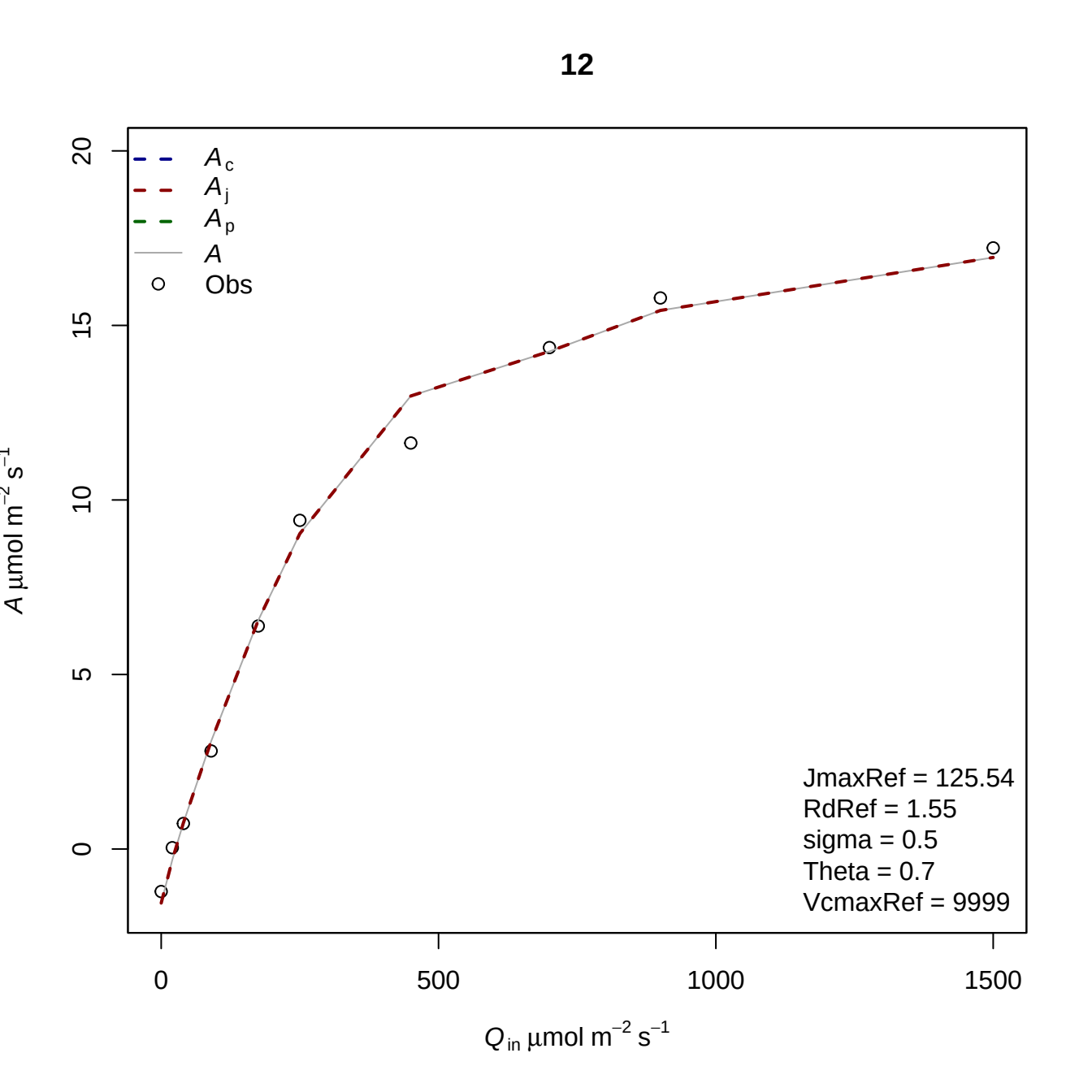

A_c
A_j
A_p
A
Obs
JmaxRef = 219.81
RdRef = 2.14
sigma = 0.33
Theta = 0.7
VcmaxRef = 147.72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

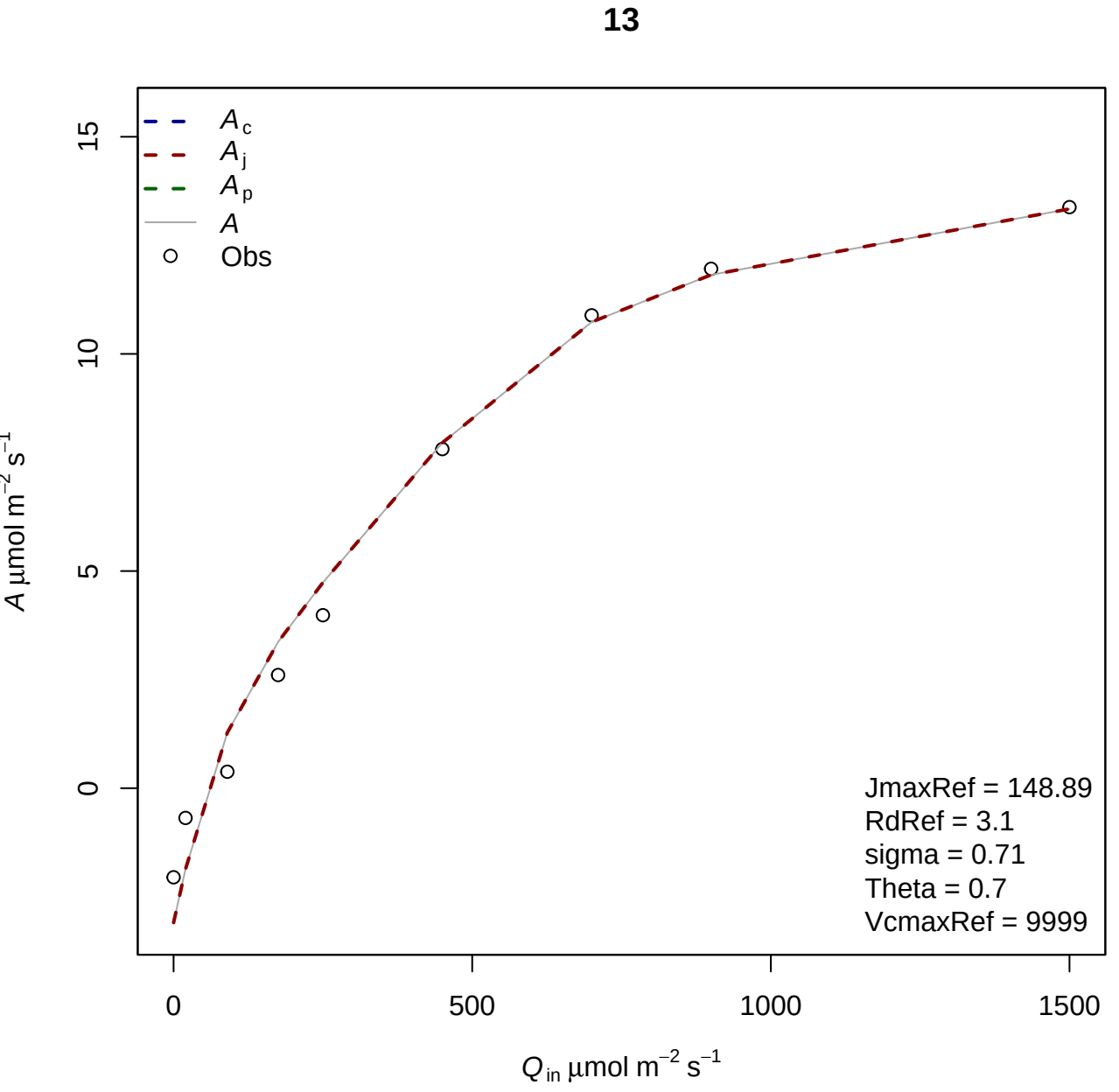

8
A_c
A_j
A_p
A
Obs
JmaxRef = 109.68
RdRef = 4
sigma = 0.92
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

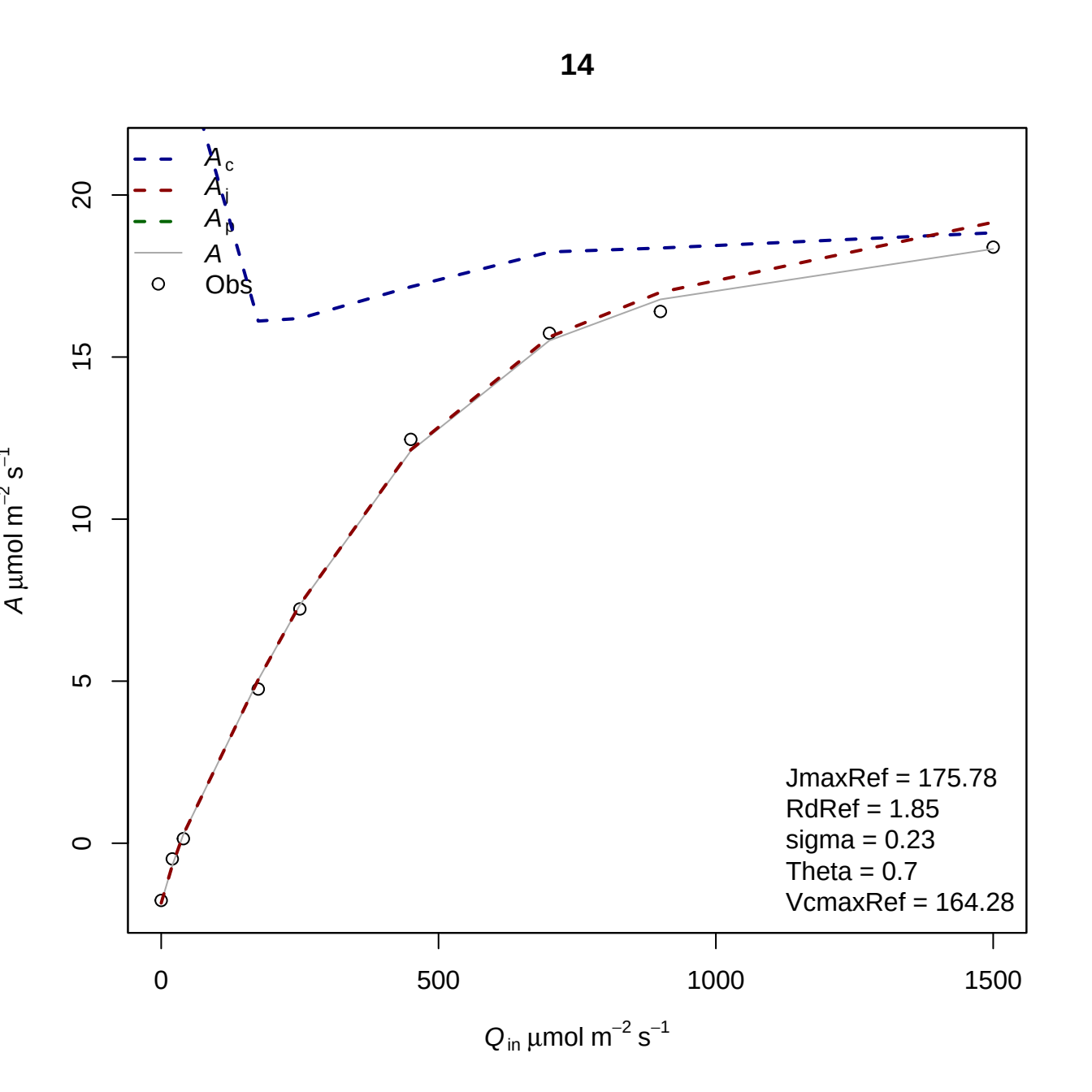

9
A_c
A_j
A_p
A
Obs
JmaxRef = 165.17
RdRef = 1.61
sigma = 0.21
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

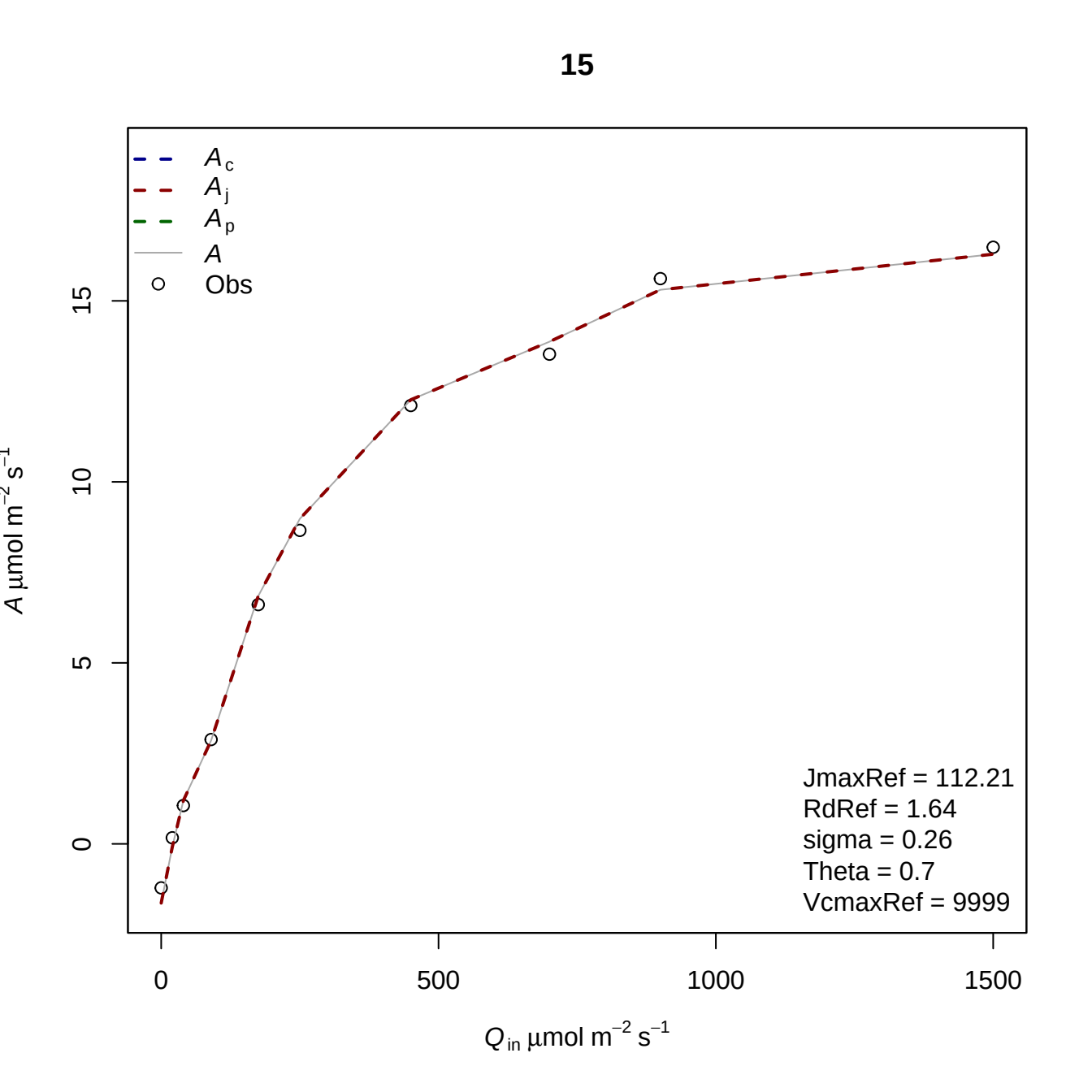

10
A_c
A_j
A_p
A
Obs
JmaxRef = 118.03
RdRef = 1.2
sigma = 0.11
Theta = 0.7
VcmaxRef = 85.74
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

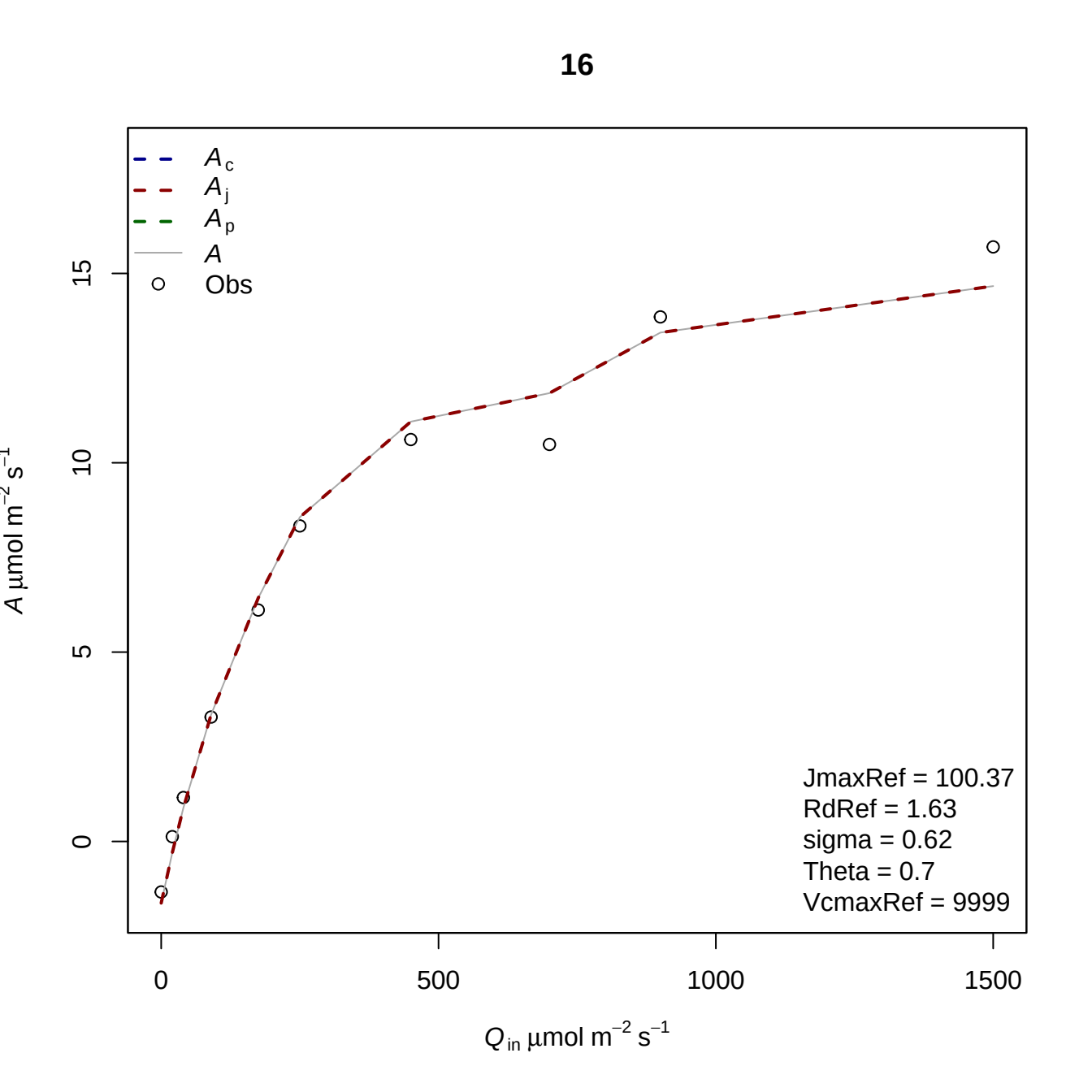

11
A_c
A_j
A_p
A
Obs
JmaxRef = 78.33
RdRef = 1.68
sigma = 0.37
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

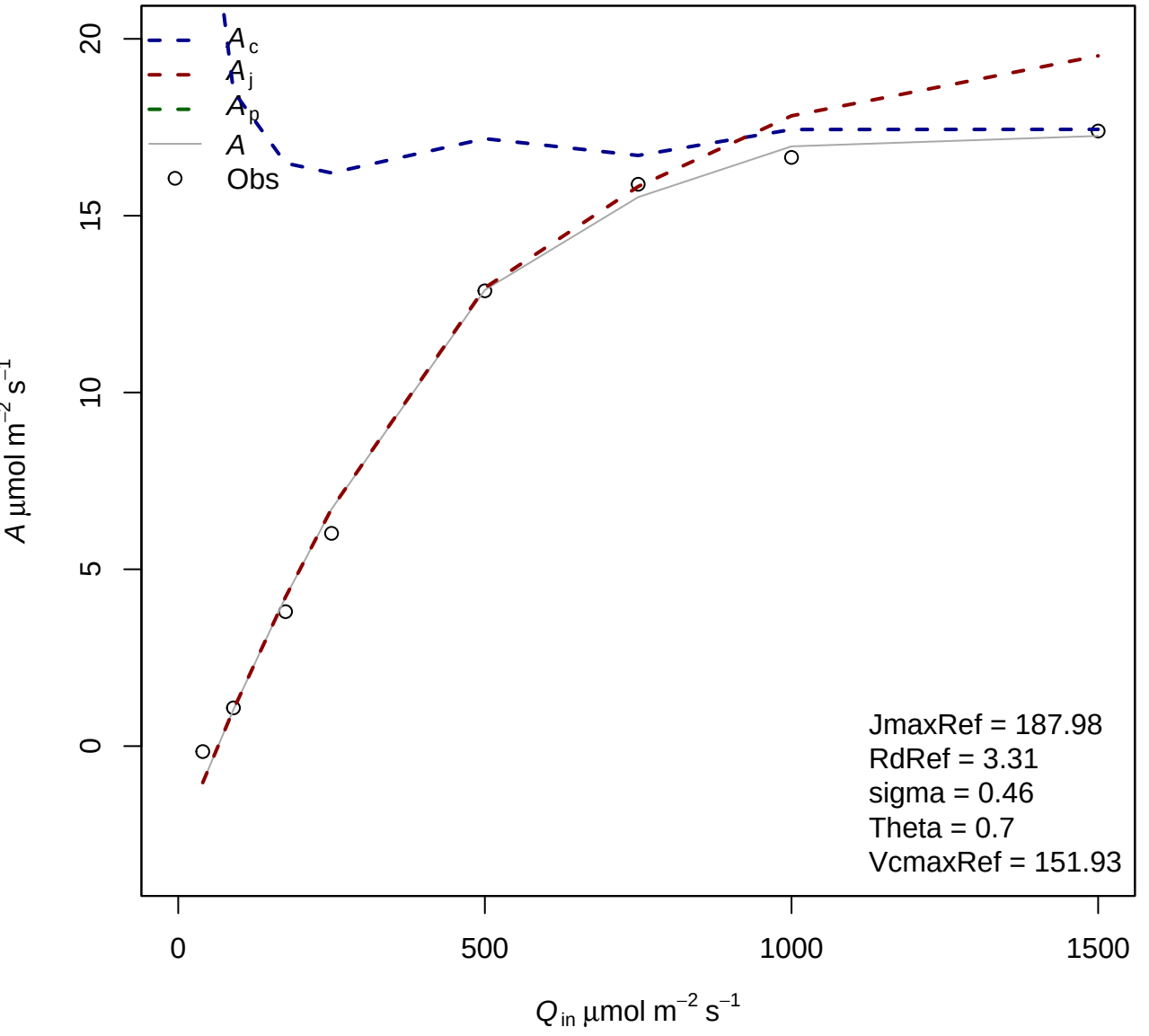

12
A_c
A_j
A_p
A
Obs
JmaxRef = 125.54
RdRef = 1.55
sigma = 0.5
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

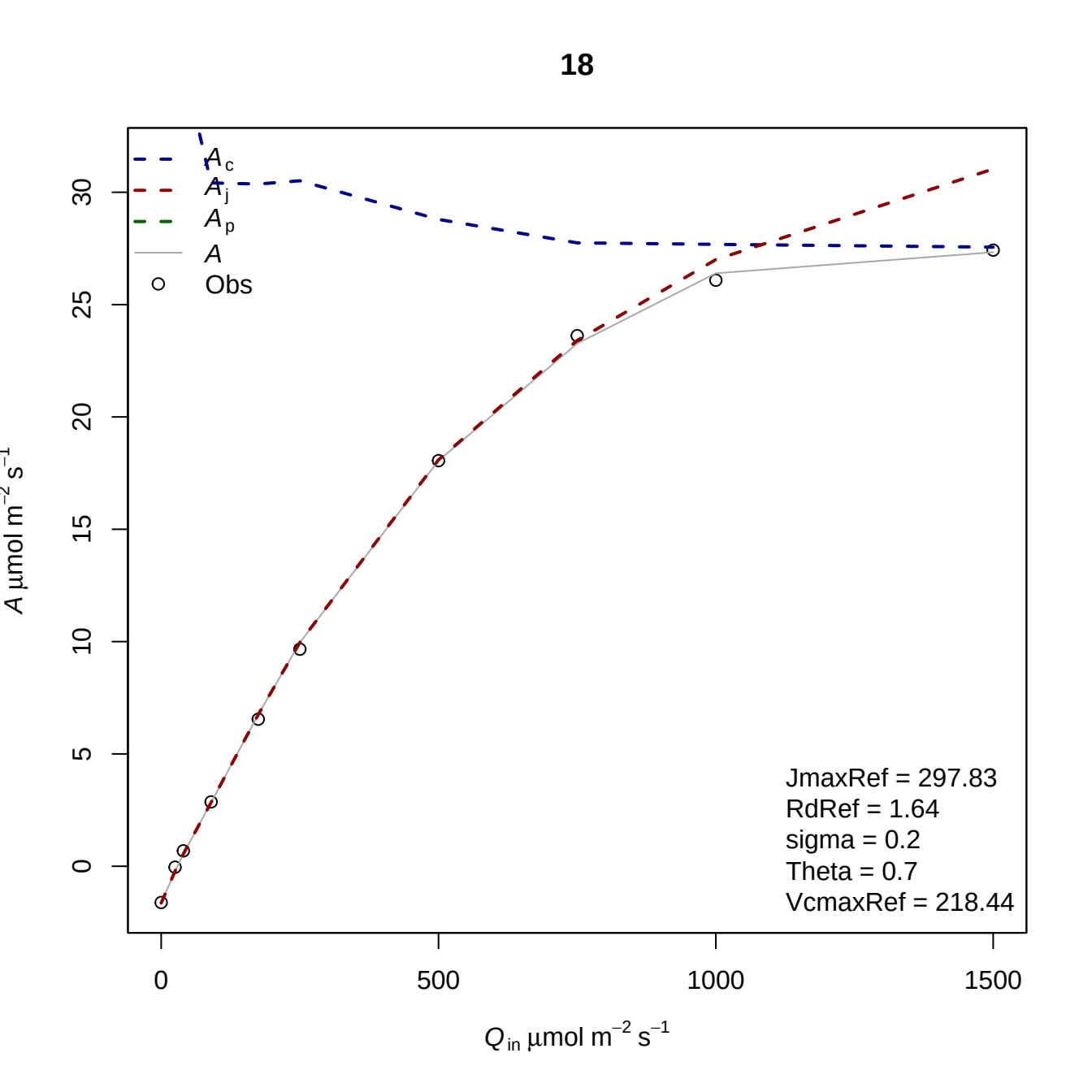

13
A_c
A_j
A_p
A
Obs
15
10
5
0
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}
JmaxRef = 148.89
RdRef = 3.1
sigma = 0.71
Theta = 0.7
VcmaxRef = 9999

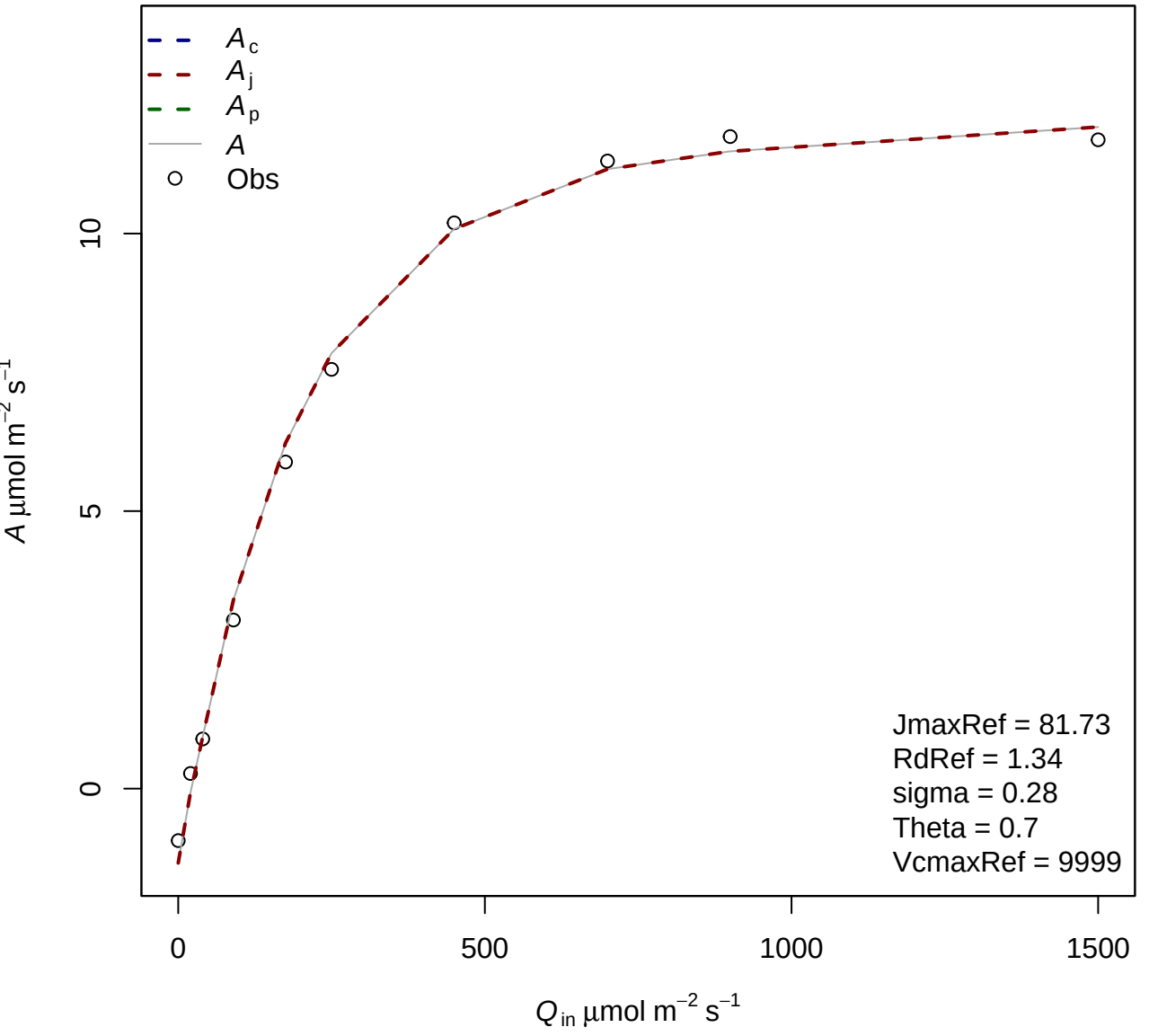

14
A_c
A_j
A_p
A
Obs
JmaxRef = 175.78
RdRef = 1.85
sigma = 0.23
Theta = 0.7
VcmaxRef = 164.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

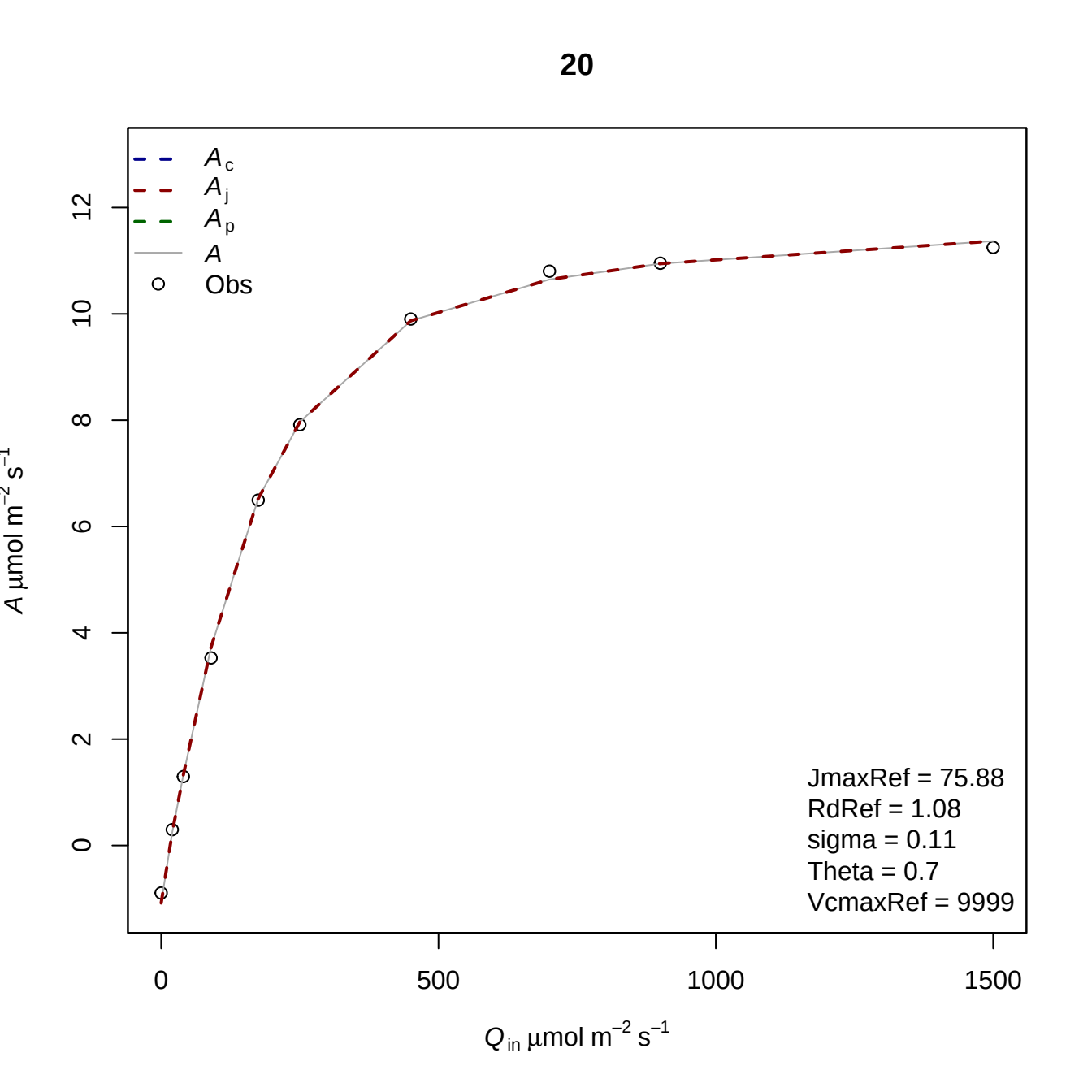

15
A_c
A_j
A_p
A
Obs
JmaxRef = 112.21
RdRef = 1.64
sigma = 0.26
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

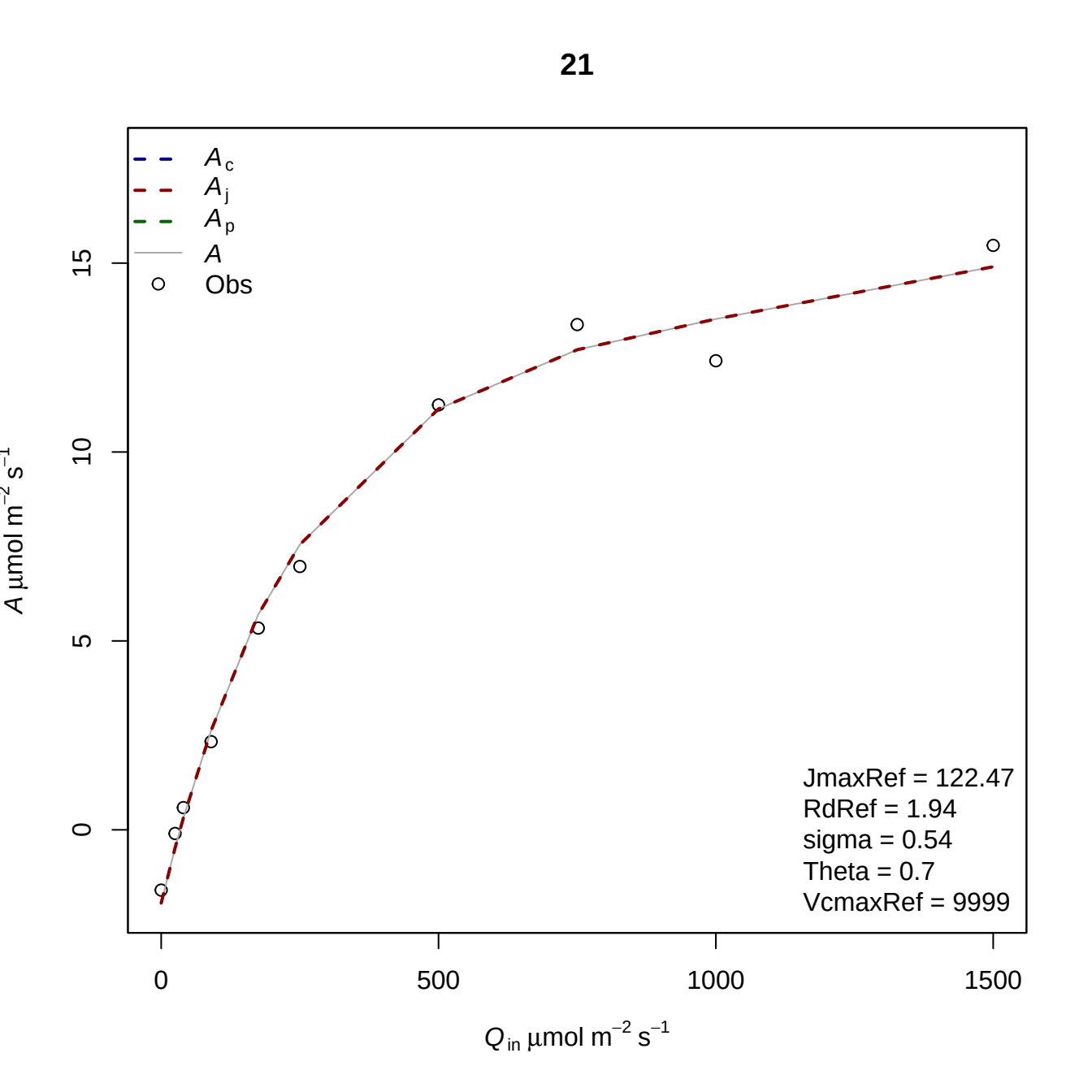

16
A_c
A_j
A_p
A
Obs
JmaxRef = 100.37
RdRef = 1.63
sigma = 0.62
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

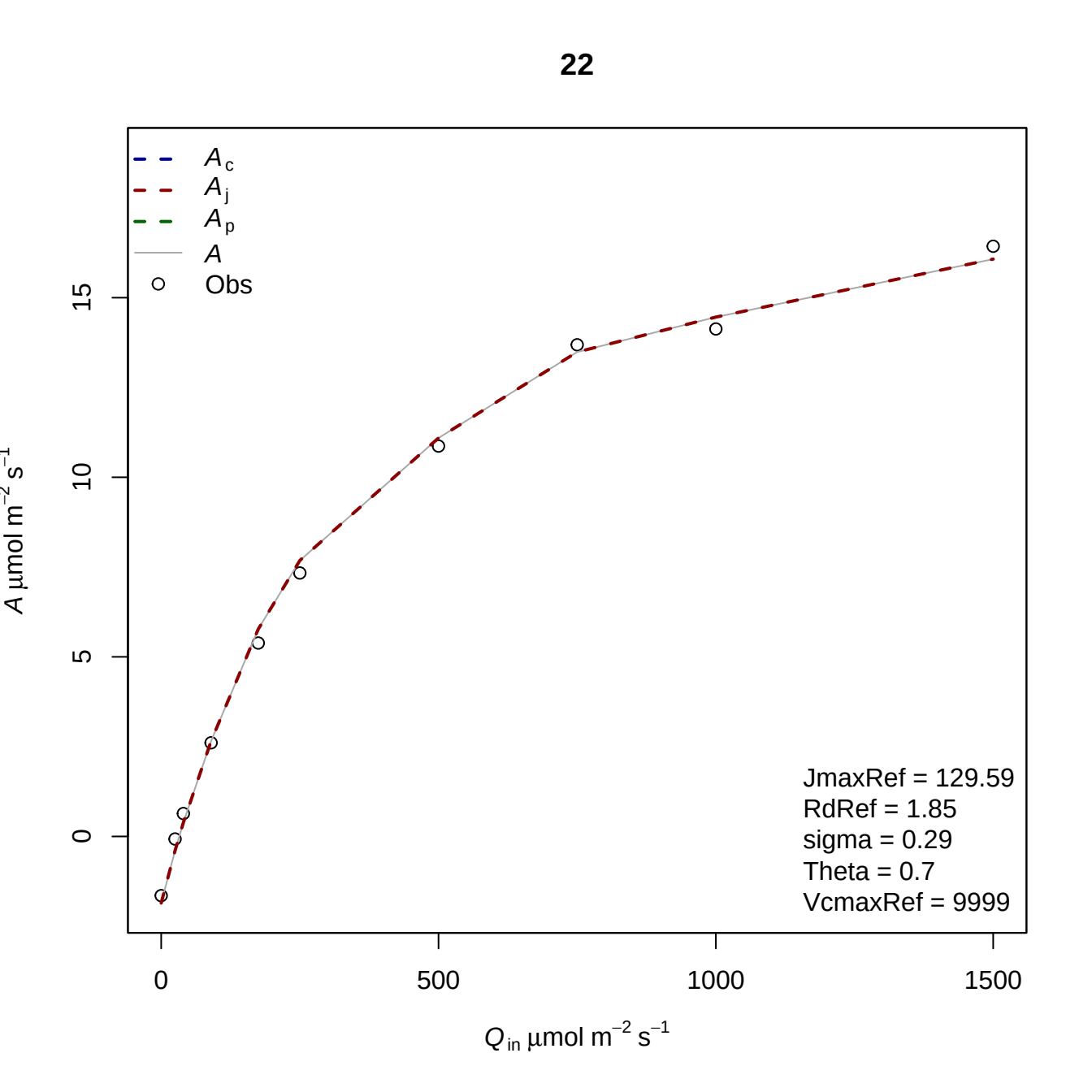

A_c
A_j
A_p
A
Obs
JmaxRef = 187.98
RdRef = 3.31
sigma = 0.46
Theta = 0.7
VcmaxRef = 151.93
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

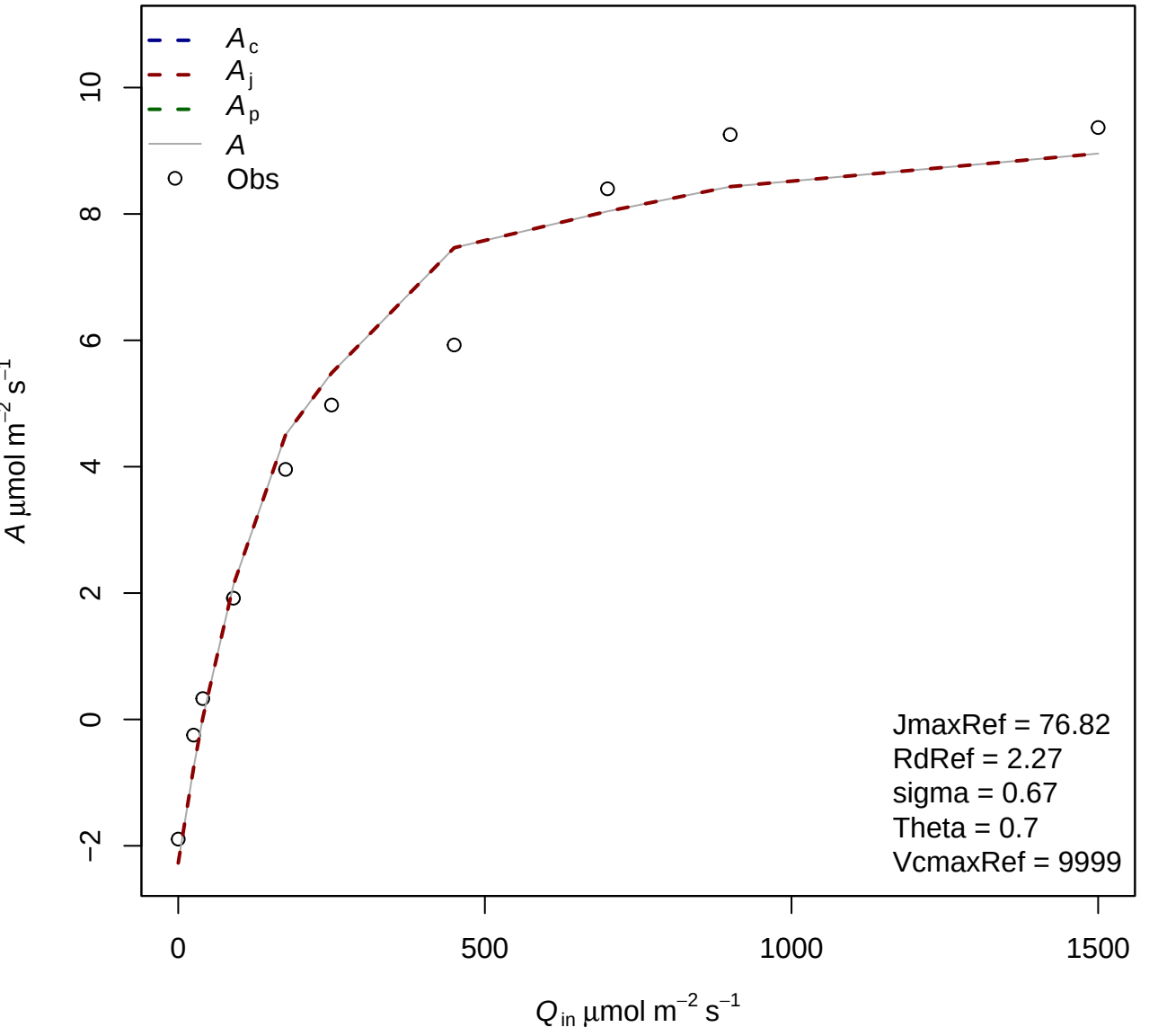

18
A_c
A_j
A_p
A
Obs
JmaxRef = 297.83
RdRef = 1.64
sigma = 0.2
Theta = 0.7
VcmaxRef = 218.44
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

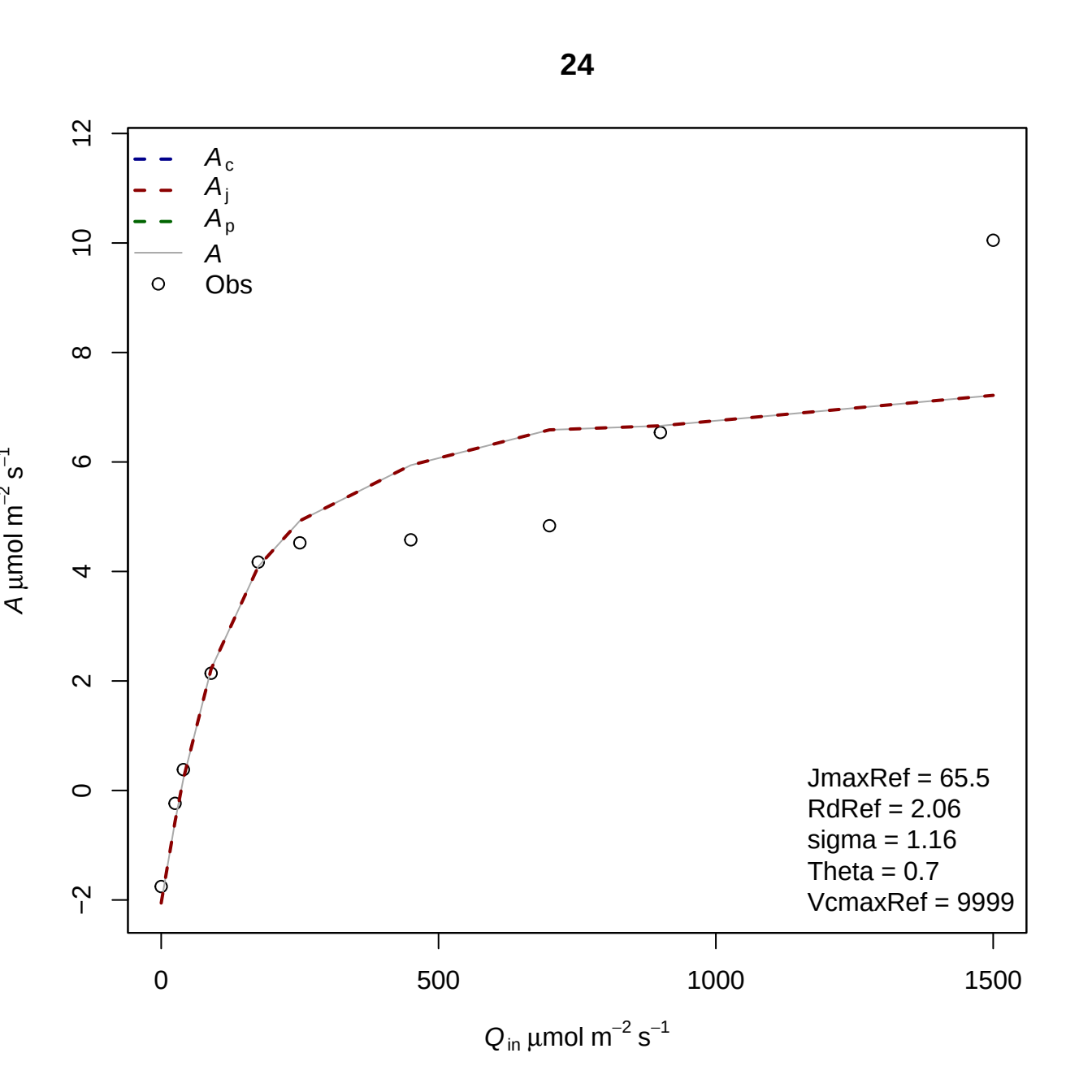

A_c
A_j
A_p
A
Obs
JmaxRef = 81.73
RdRef = 1.34
sigma = 0.28
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

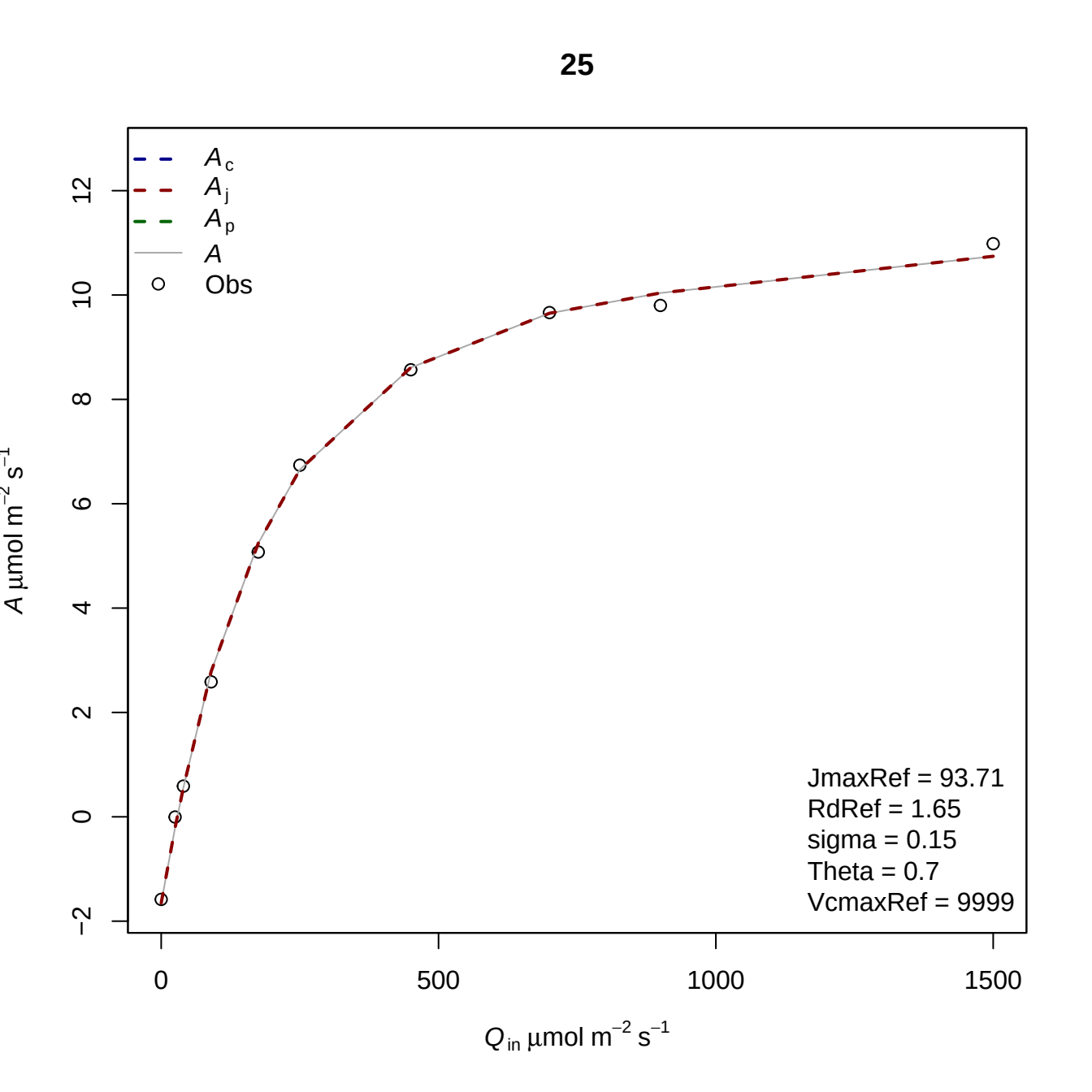

20
A_c
A_j
A_p
A
Obs
JmaxRef = 75.88
RdRef = 1.08
sigma = 0.11
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

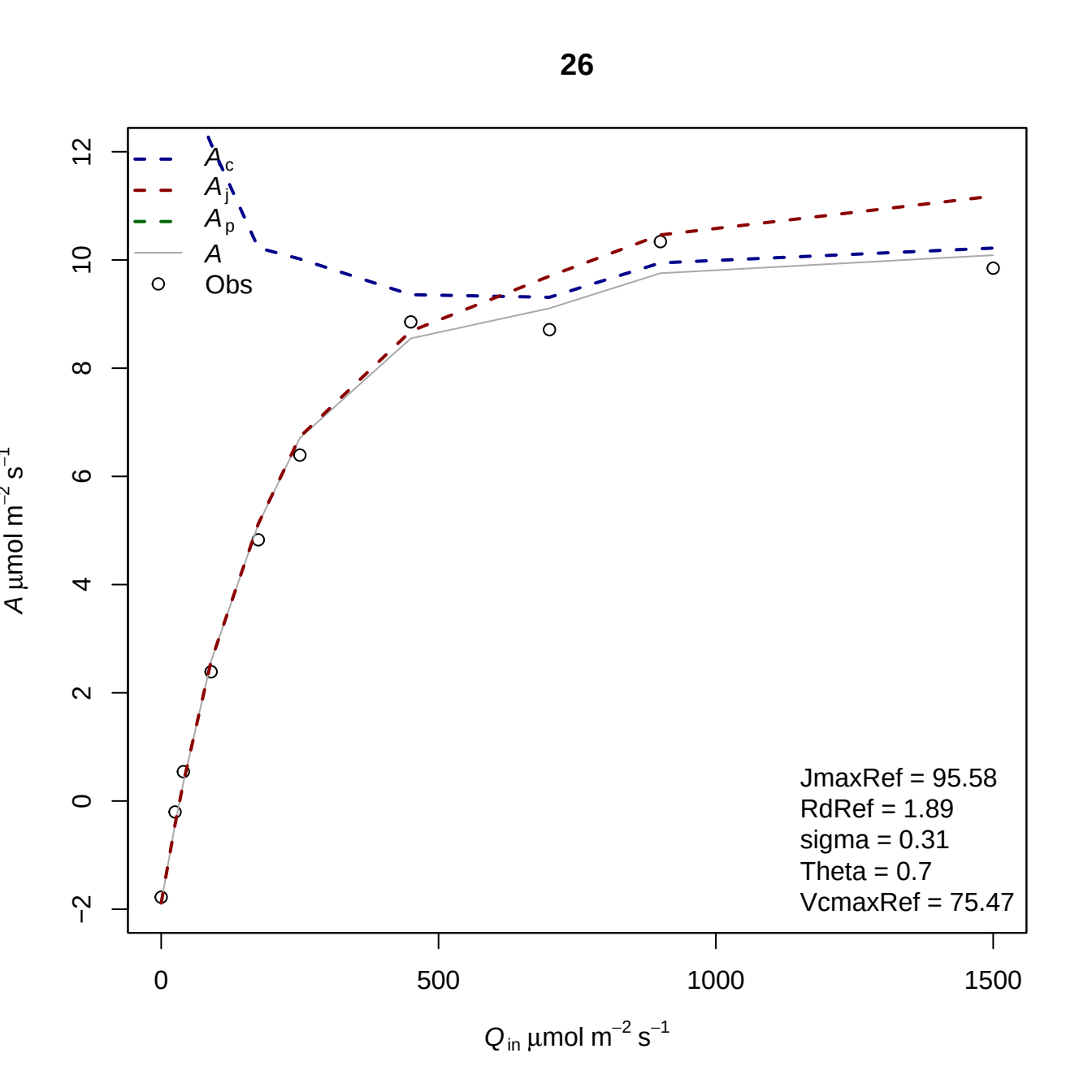

21
A_c
A_j
A_p
A
Obs
JmaxRef = 122.47
RdRef = 1.94
sigma = 0.54
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

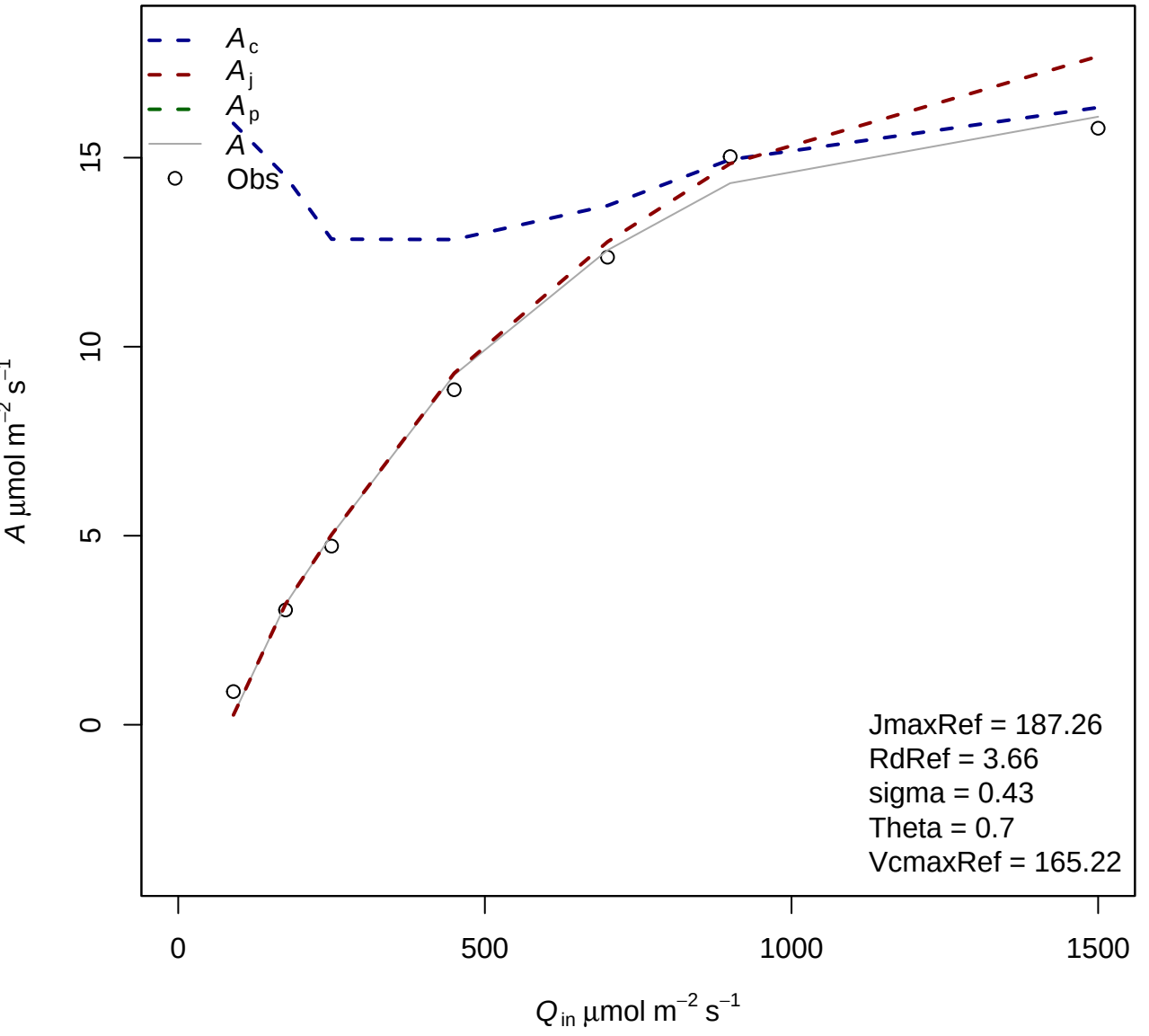

22
A_c
A_j
A_p
A
Obs
JmaxRef = 129.59
RdRef = 1.85
sigma = 0.29
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

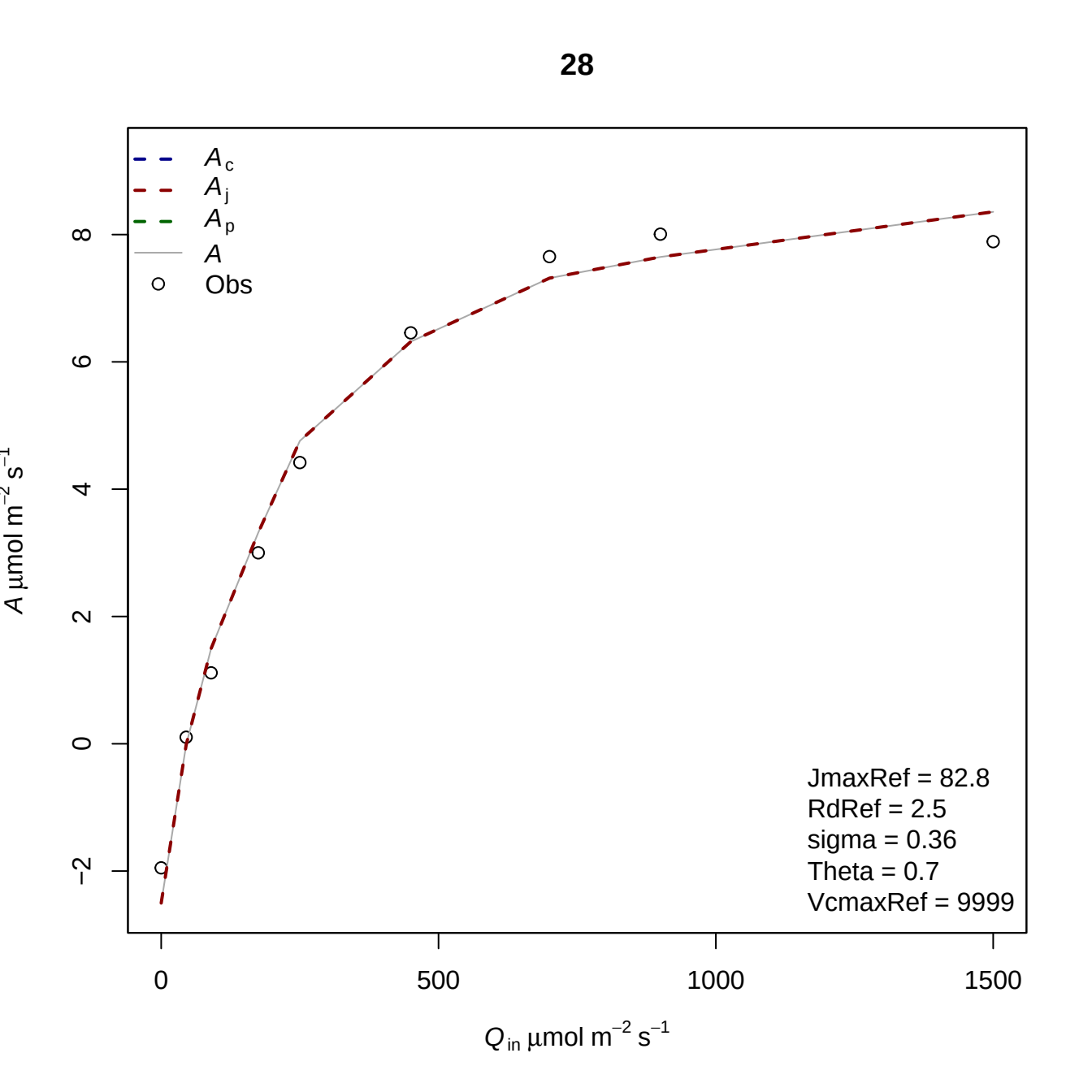

A_c
A_j
A_p
A
Obs
JmaxRef = 76.82
RdRef = 2.27
sigma = 0.67
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

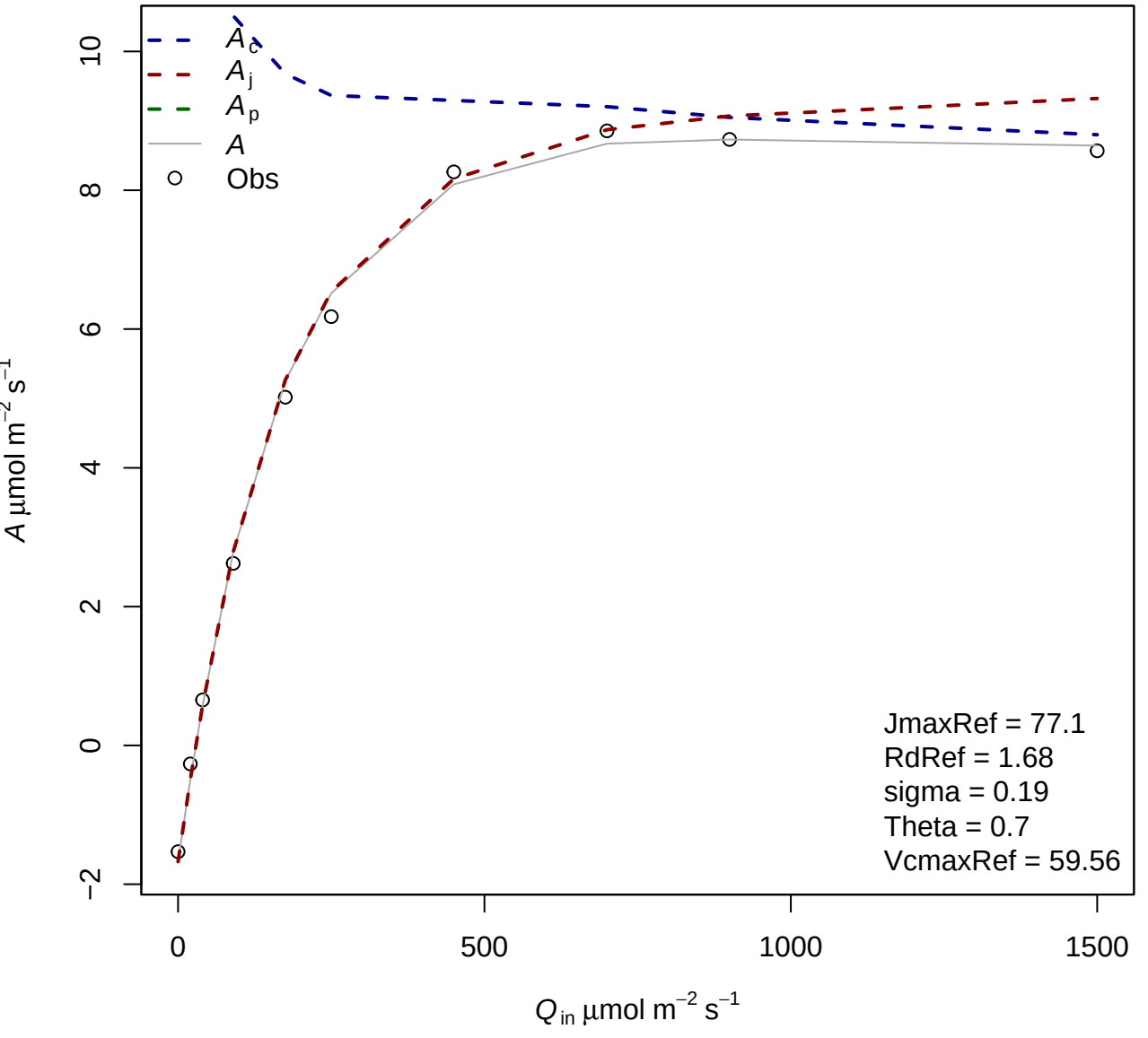

24
A_c
A_j
A_p
A
Obs
JmaxRef = 65.5
RdRef = 2.06
sigma = 1.16
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

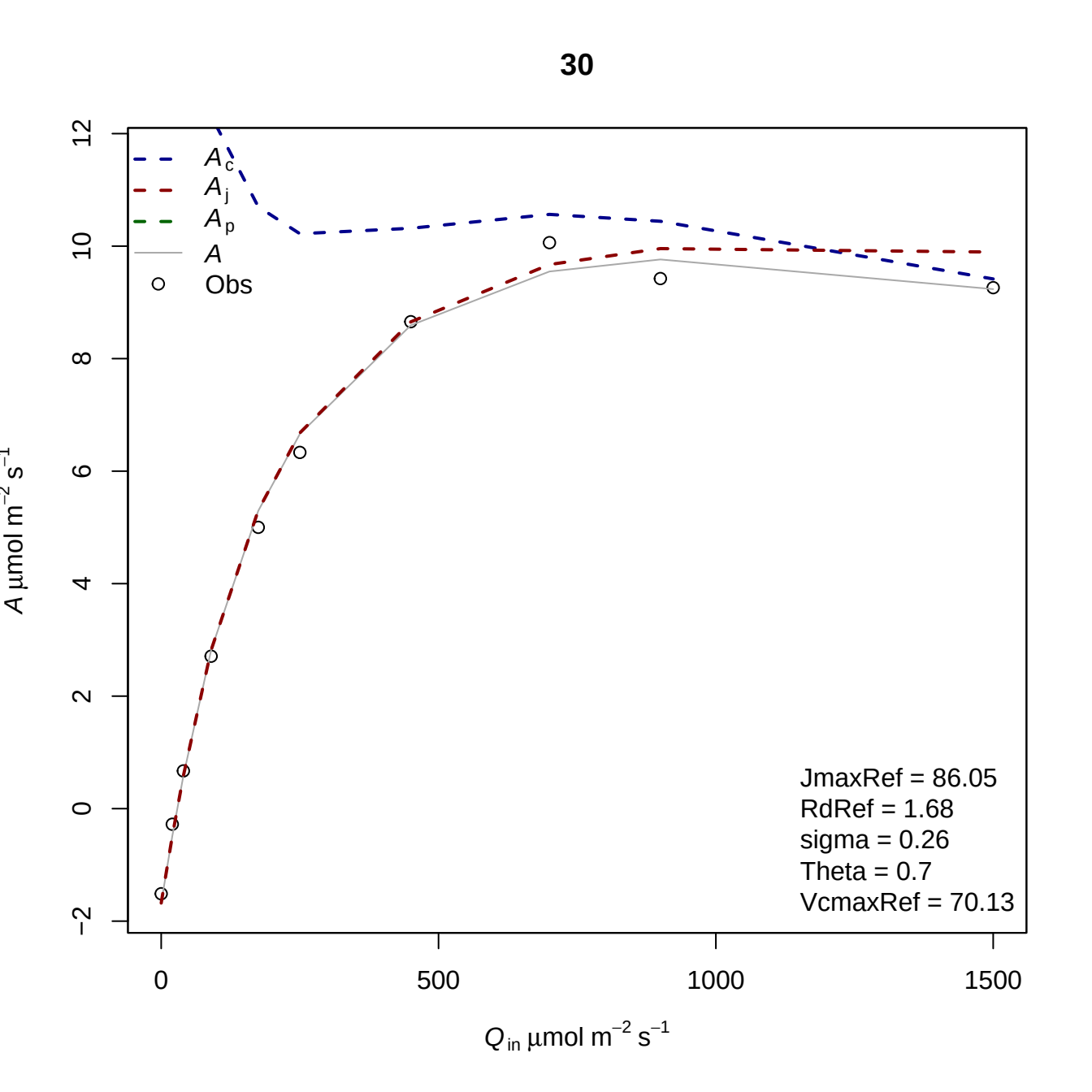

25
A_c
A_j
A_p
A
Obs
JmaxRef = 93.71
RdRef = 1.65
sigma = 0.15
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

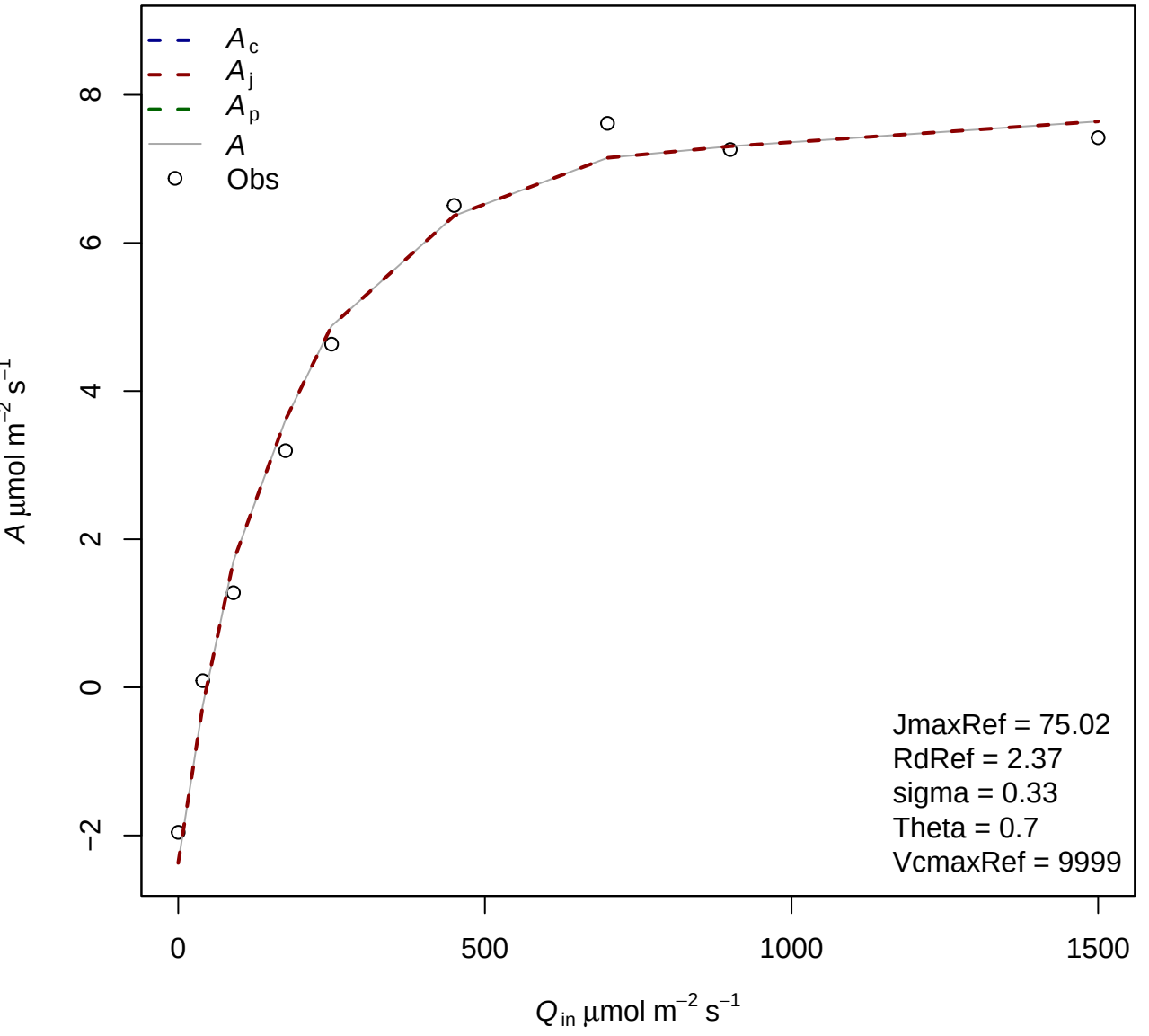

26
A_c
A_j
A_p
A
Obs
JmaxRef = 95.58
RdRef = 1.89
sigma = 0.31
Theta = 0.7
VcmaxRef = 75.47
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

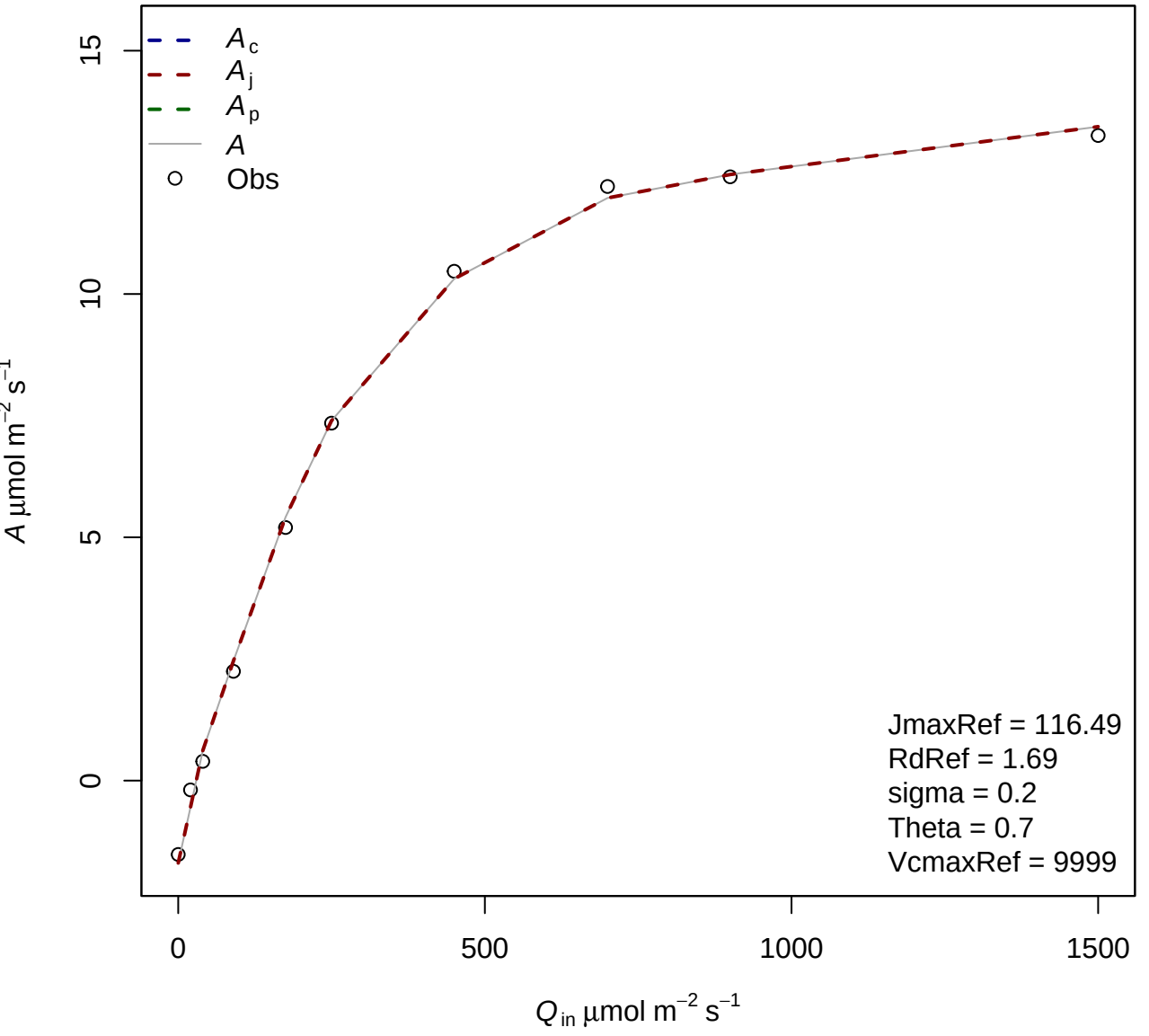

A_c
A_j
A_p
A
Obs
JmaxRef = 187.26
RdRef = 3.66
sigma = 0.43
Theta = 0.7
VcmaxRef = 165.22
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

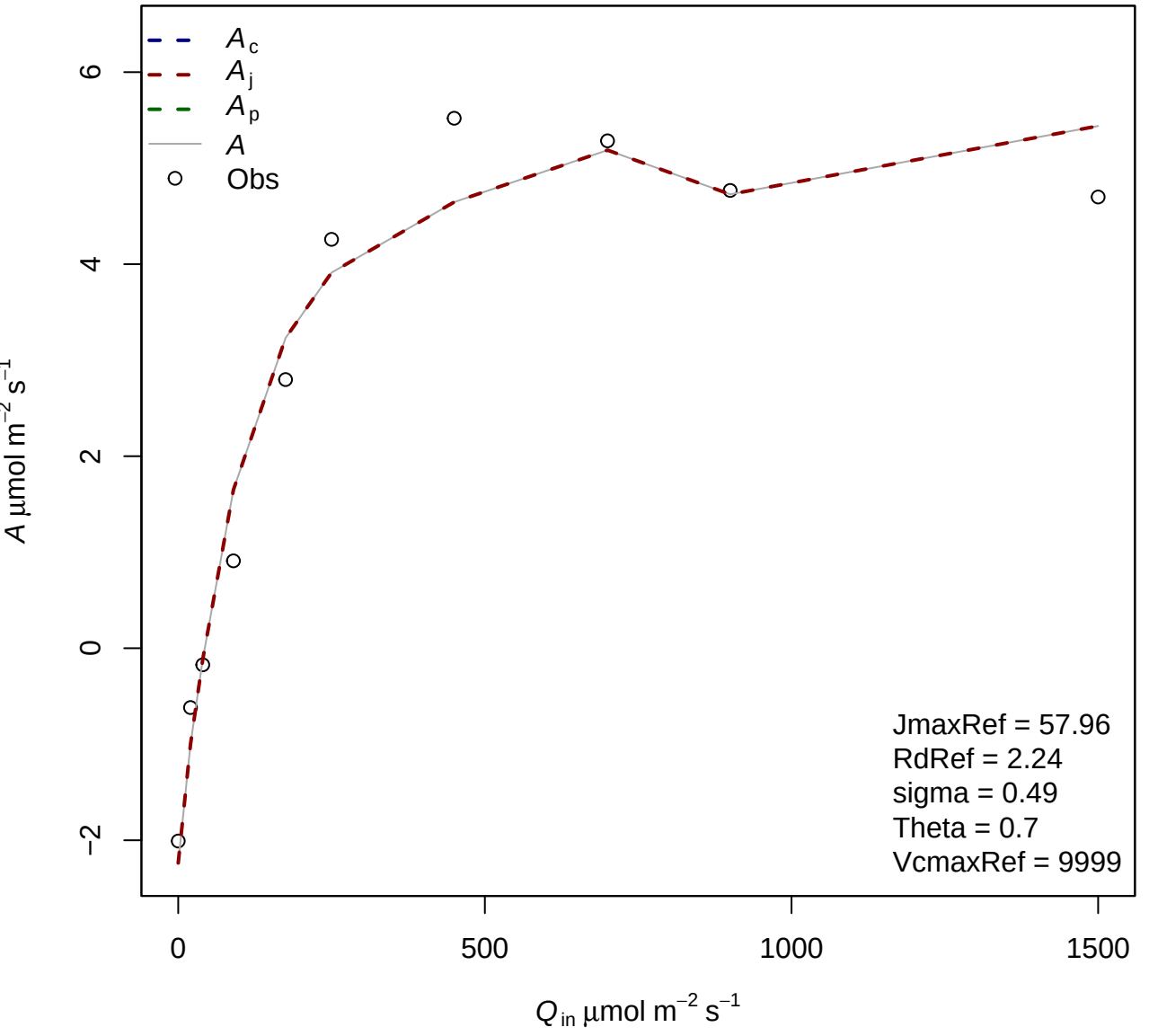

28
A_c
A_j
A_p
A
Obs
JmaxRef = 82.8
RdRef = 2.5
sigma = 0.36
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

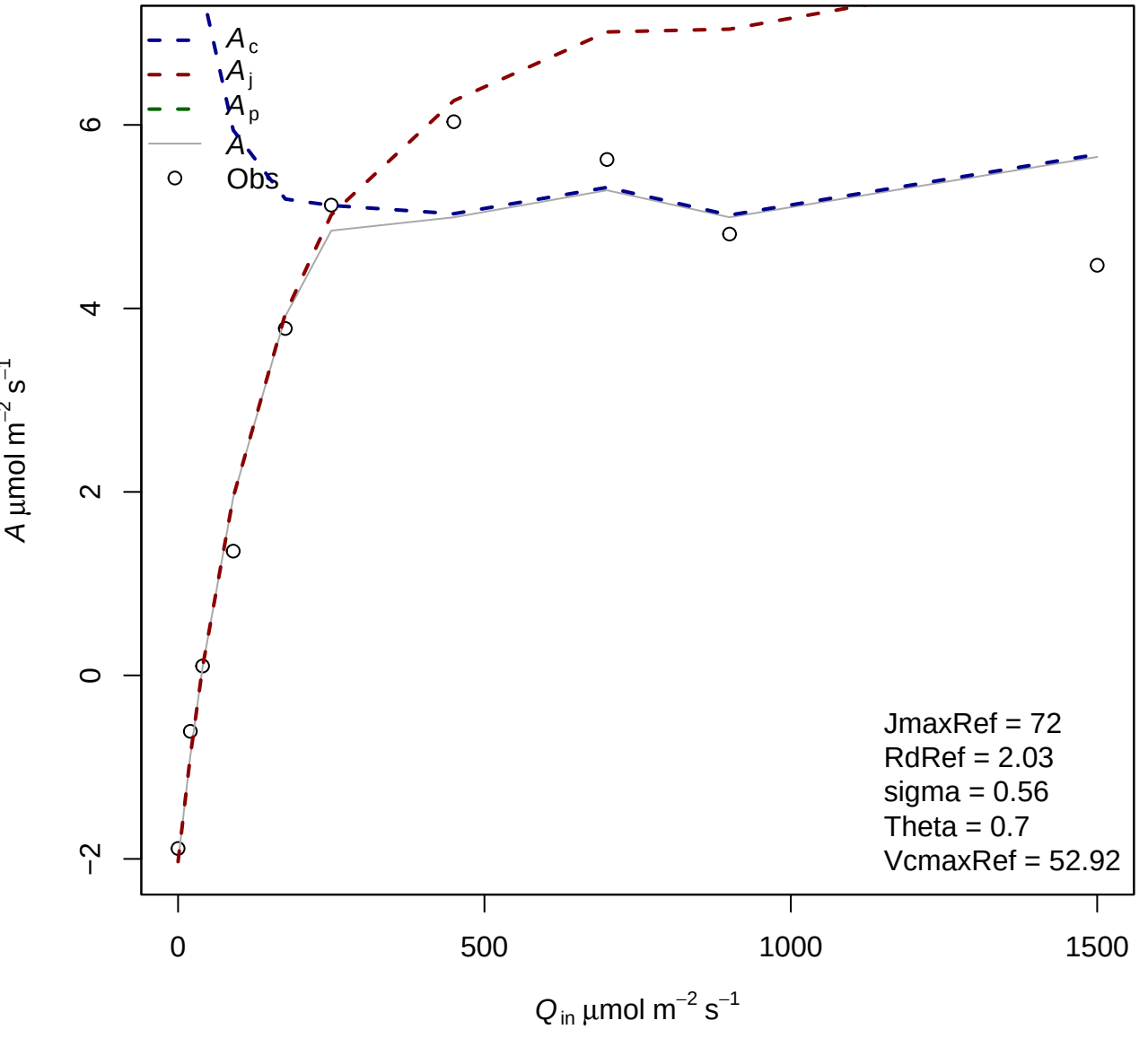

A_c
A_j
A_p
A
Obs
JmaxRef = 77.1
RdRef = 1.68
sigma = 0.19
Theta = 0.7
VcmaxRef = 59.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

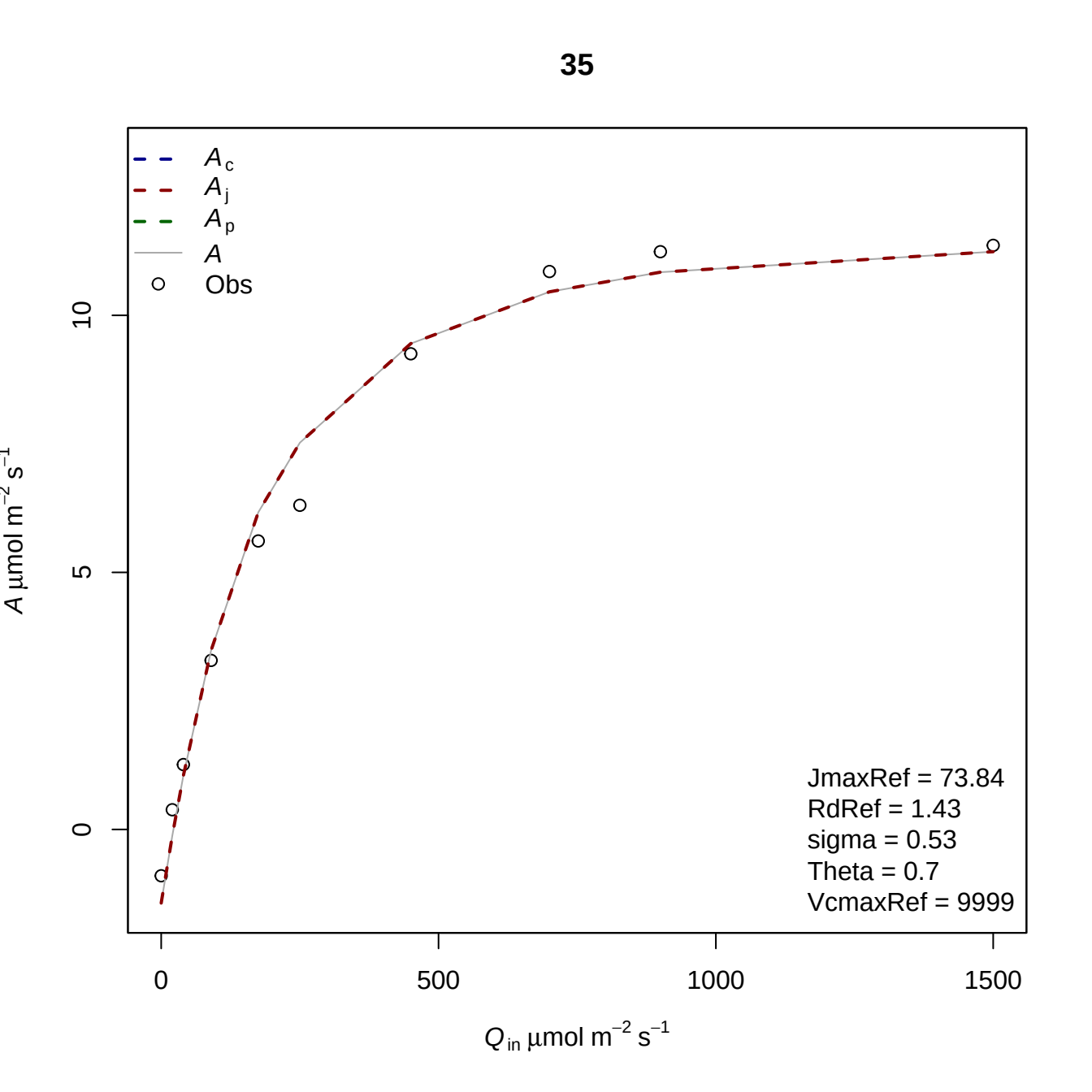

30
A_c
A_j
A_p
A
Obs
JmaxRef = 86.05
RdRef = 1.68
sigma = 0.26
Theta = 0.7
VcmaxRef = 70.13
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

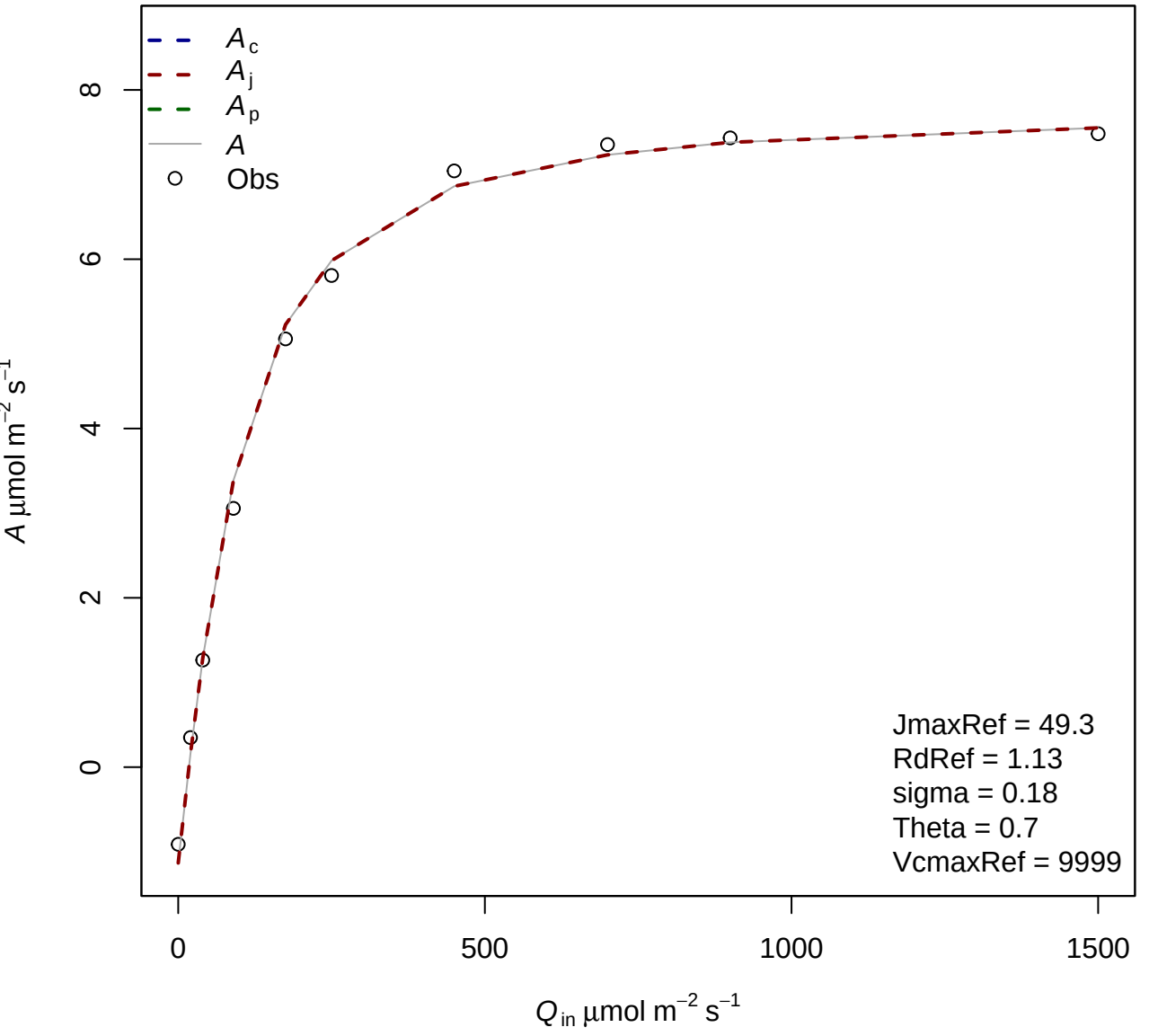

A_c
A_j
A_p
A
Obs
JmaxRef = 75.02
RdRef = 2.37
sigma = 0.33
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

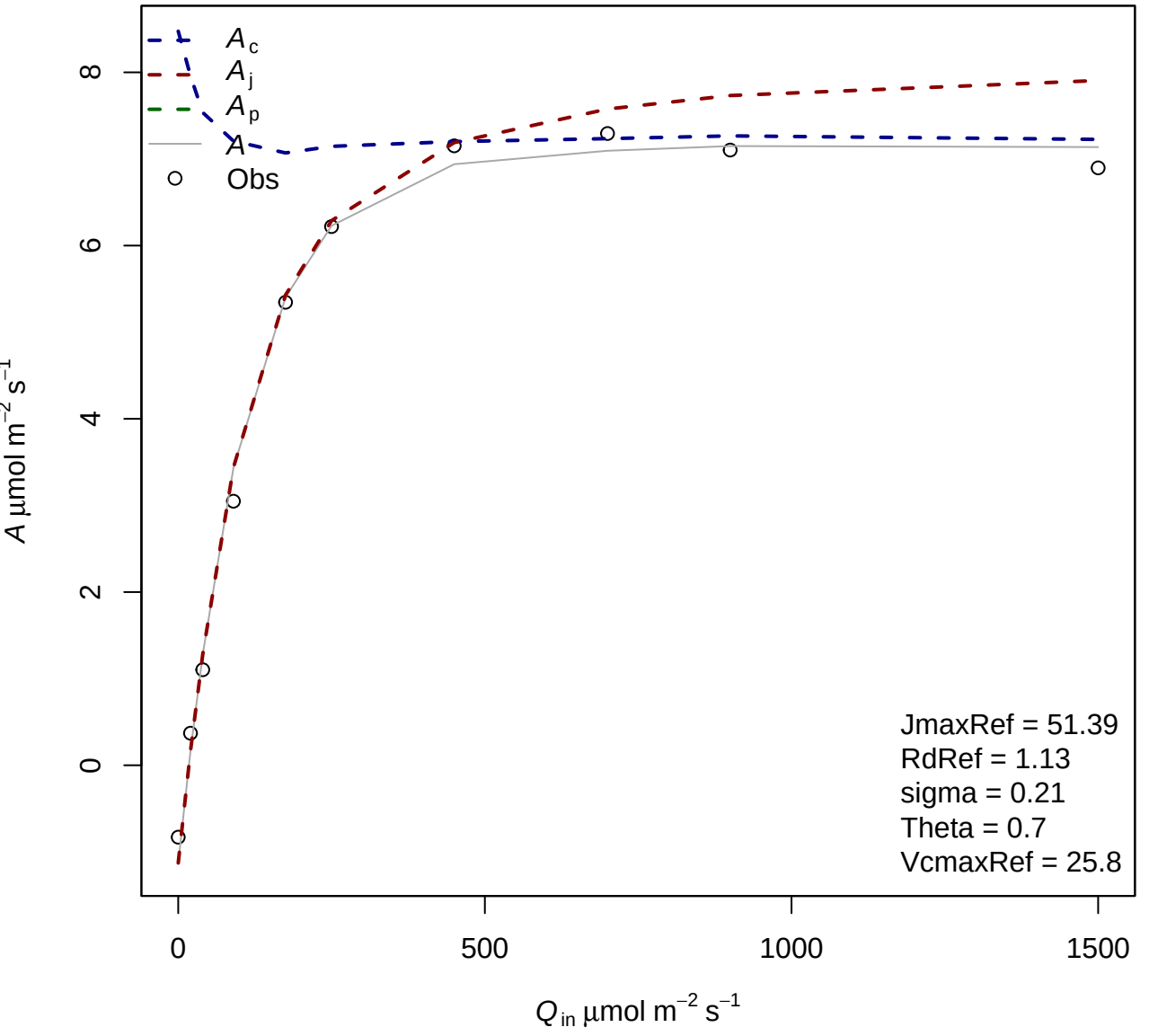

A_c
A_j
A_p
A
Obs
JmaxRef = 116.49
RdRef = 1.69
sigma = 0.2
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

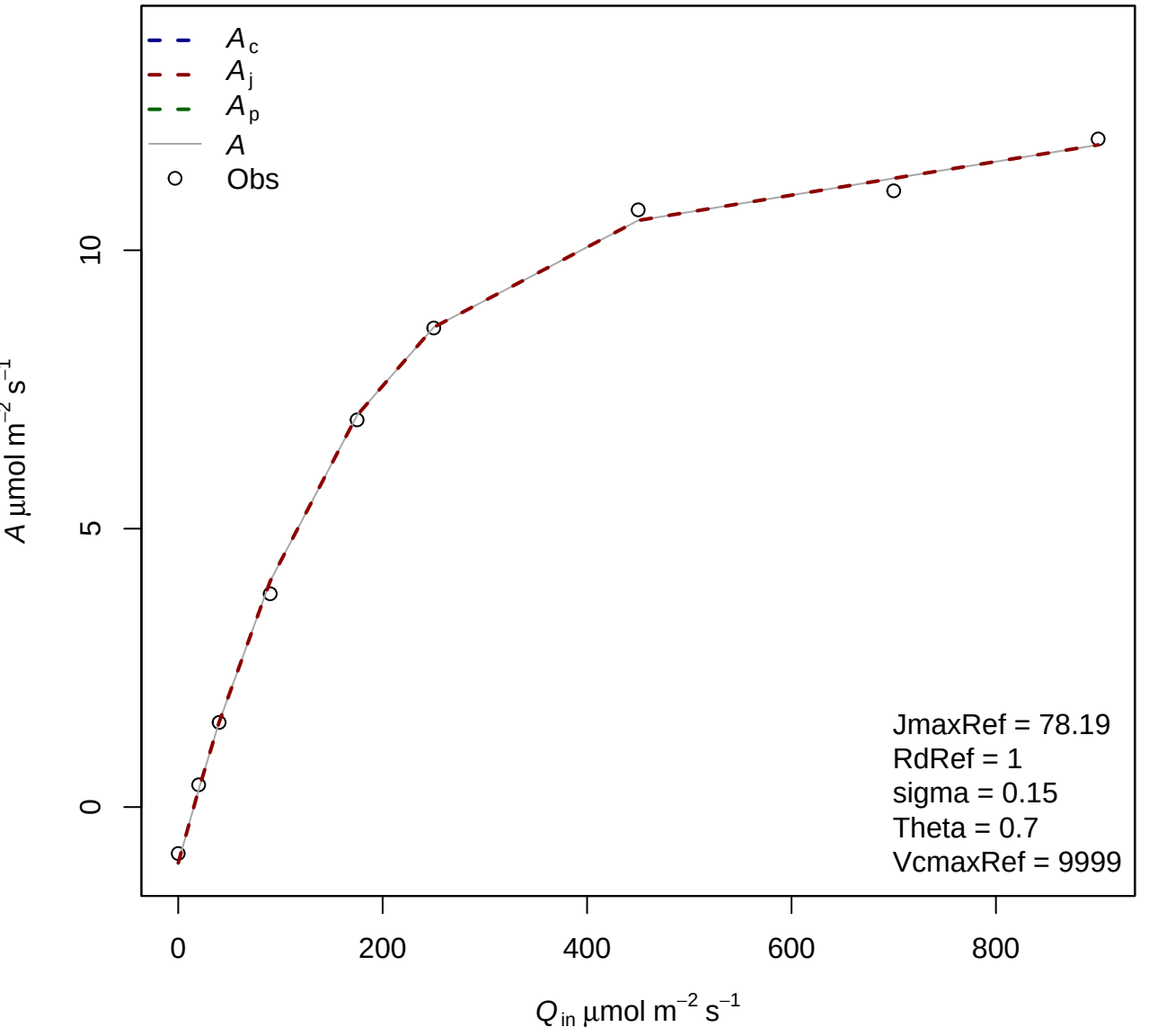

A_c
A_j
A_p
A
Obs
JmaxRef = 57.96
RdRef = 2.24
sigma = 0.49
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

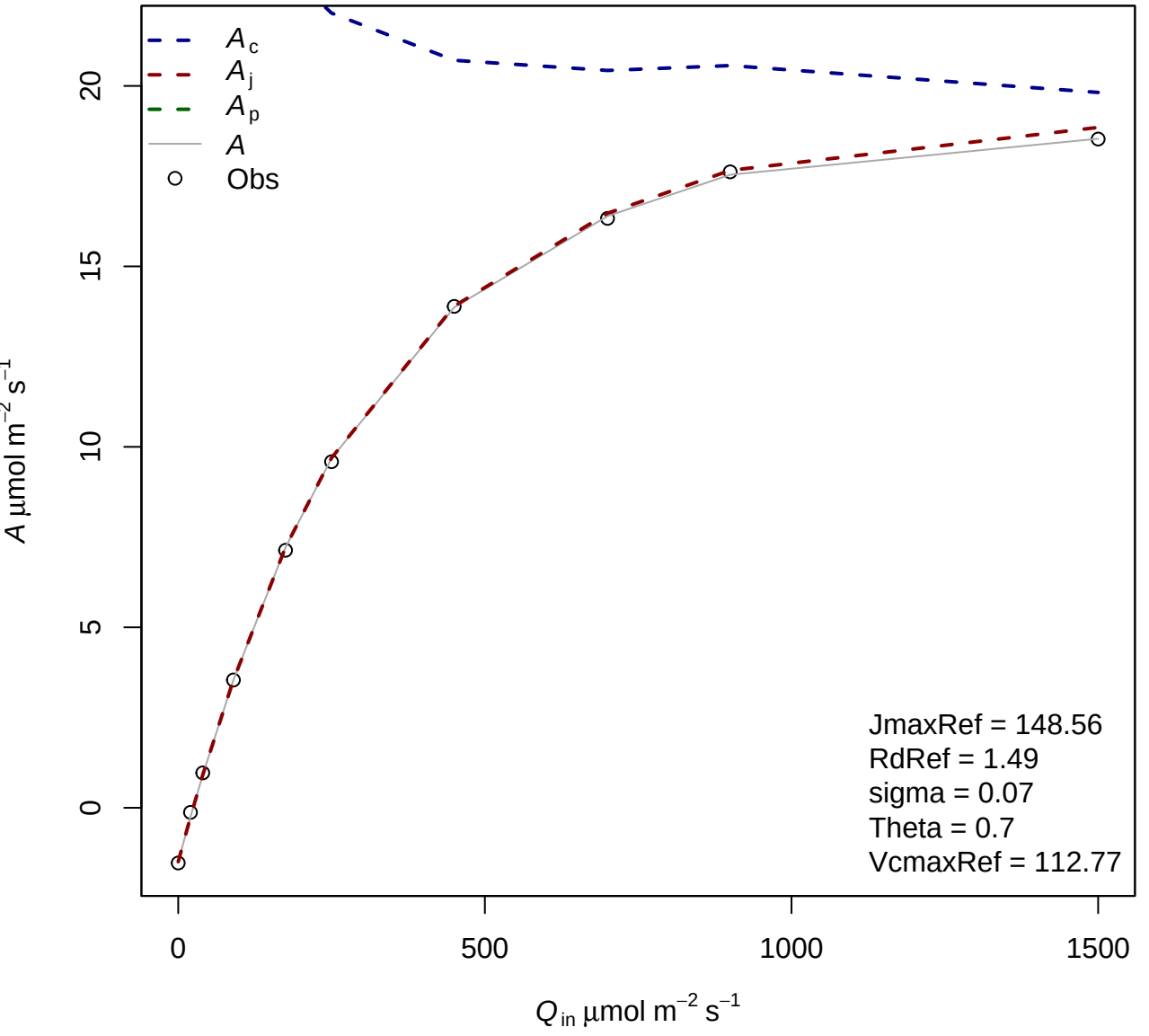

A_c
A_j
A_p
A
Obs
JmaxRef = 72
RdRef = 2.03
sigma = 0.56
Theta = 0.7
VcmaxRef = 52.92
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

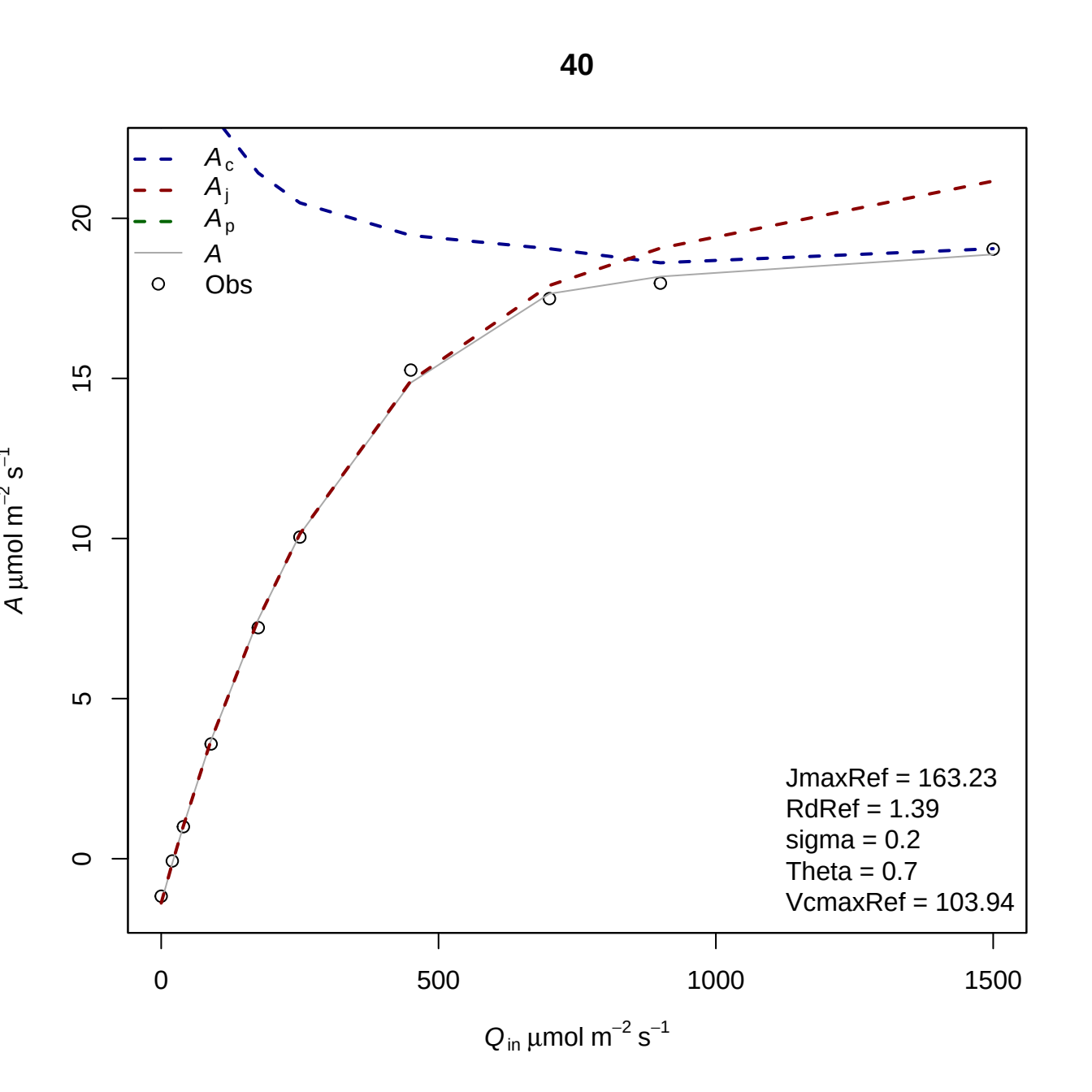

35
A_c
A_j
A_p
A
Obs
JmaxRef = 73.84
RdRef = 1.43
sigma = 0.53
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

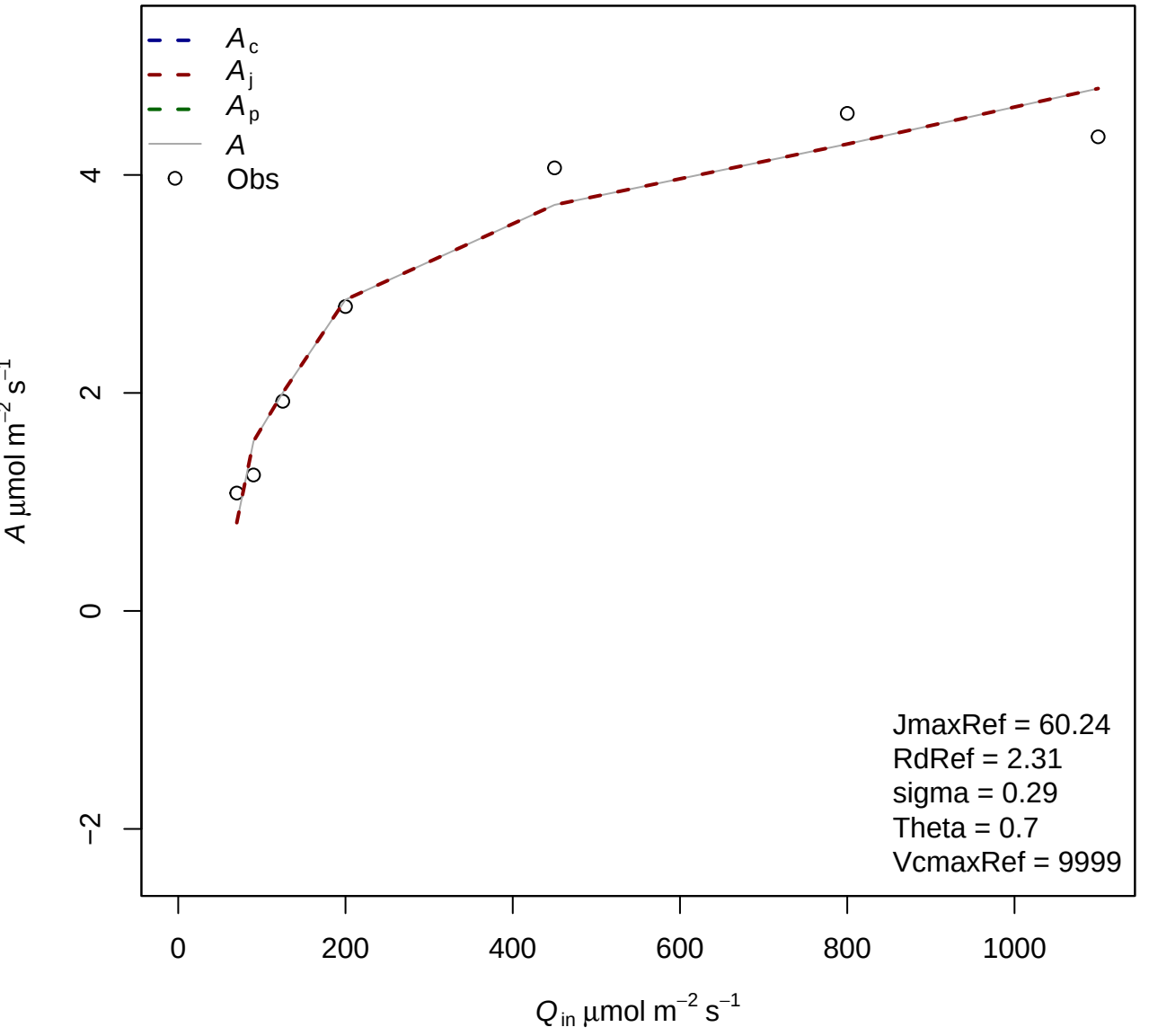

A_c
A_j
A_p
A
Obs
JmaxRef = 49.3
RdRef = 1.13
sigma = 0.18
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

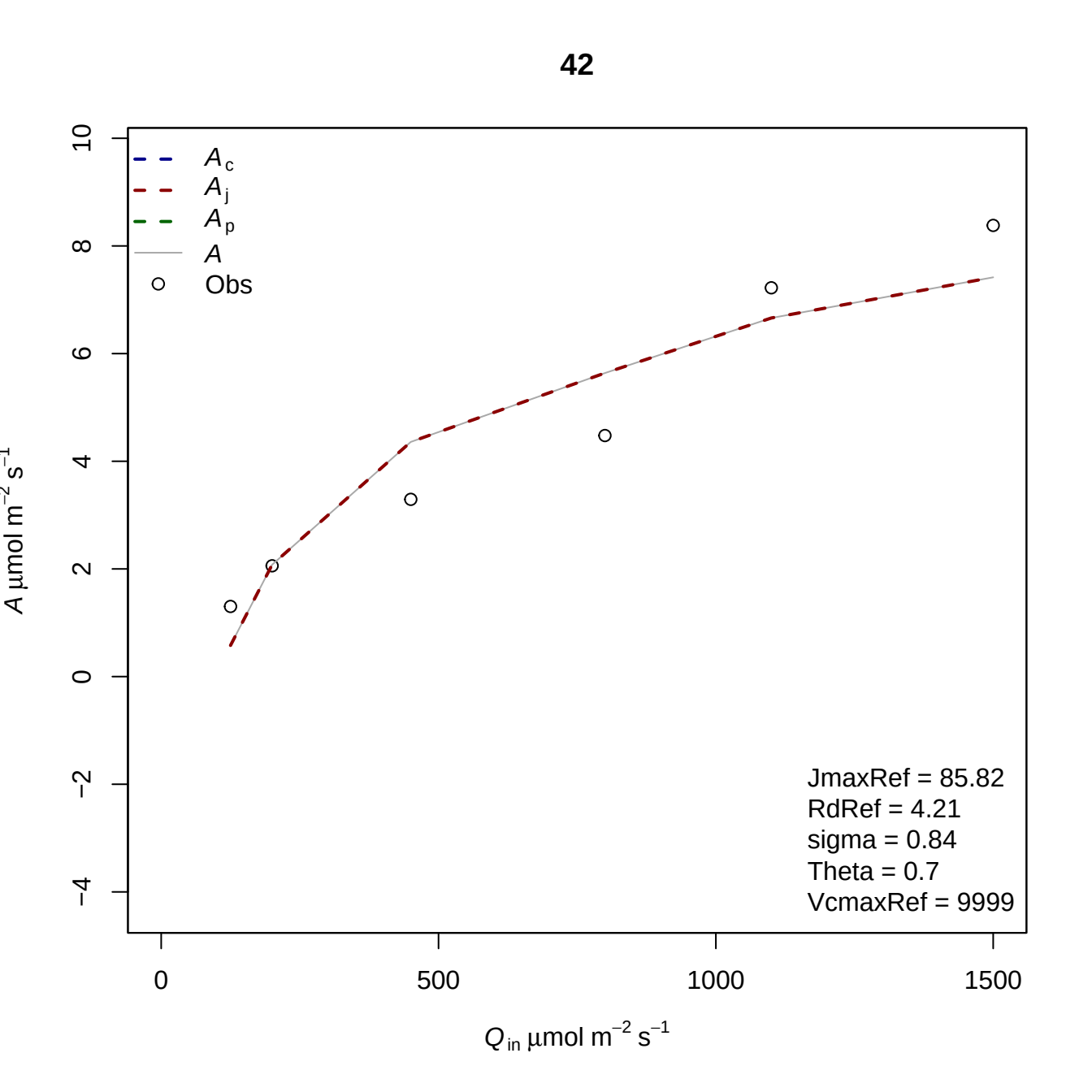

A_c
A_j
A_p
A
Obs
JmaxRef = 51.39
RdRef = 1.13
sigma = 0.21
Theta = 0.7
VcmaxRef = 25.8
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

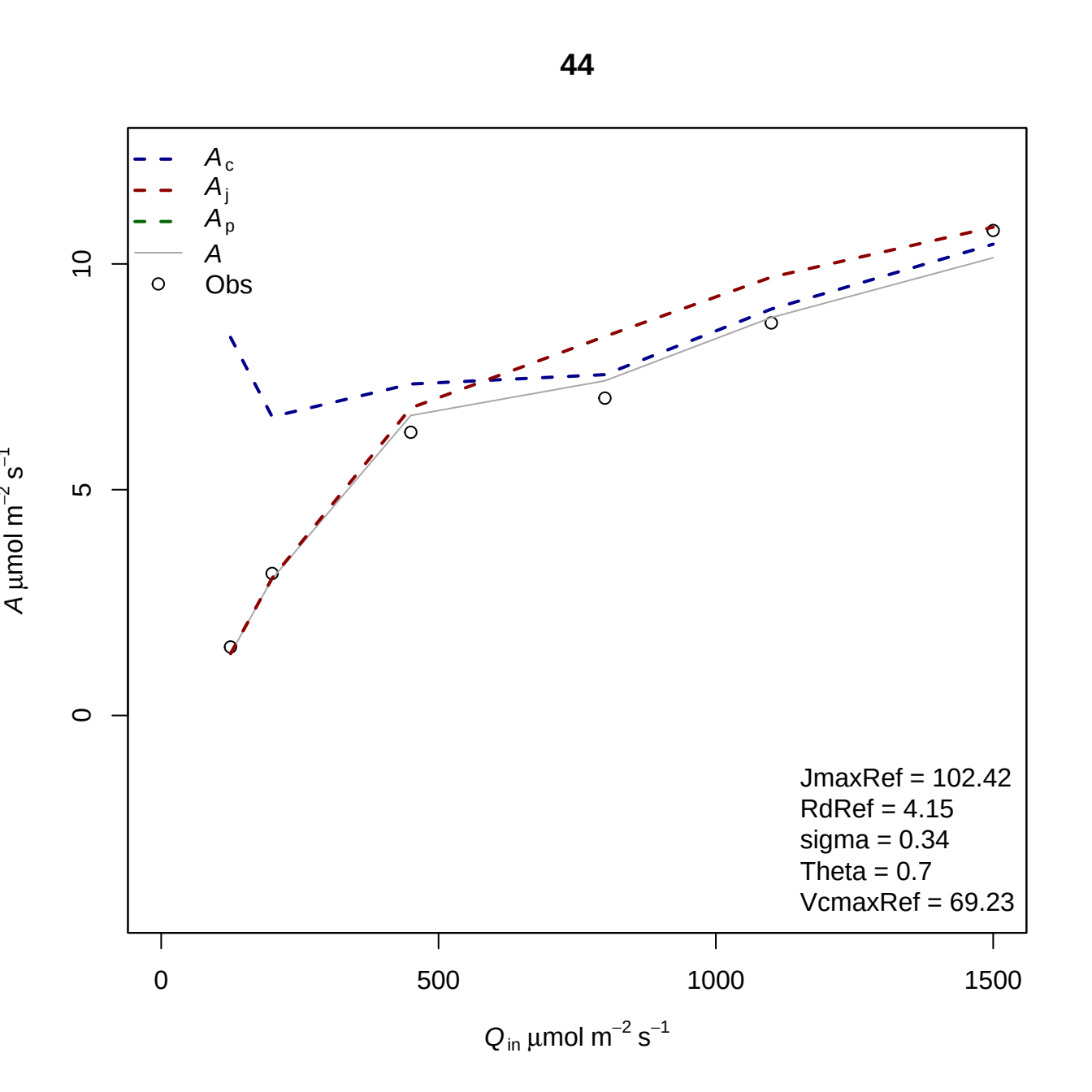

A_c
A_j
A_p
A
Obs
JmaxRef = 78.19
RdRef = 1
sigma = 0.15
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
200
400
600
800
Q_{in} µmol m^{-2} s^{-1}

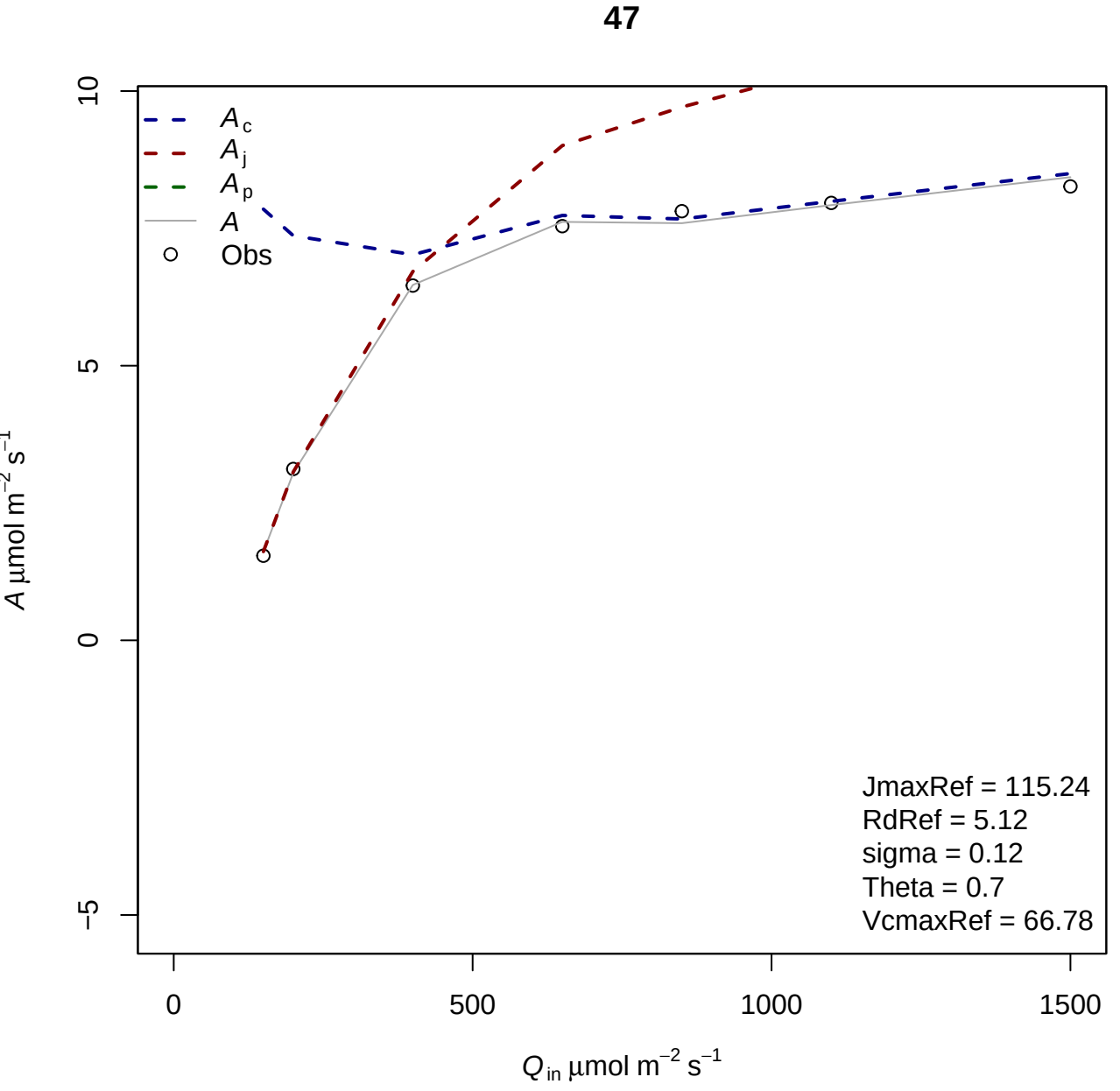

A_c
A_j
A_p
A
Obs
JmaxRef = 148.56
RdRef = 1.49
sigma = 0.07
Theta = 0.7
VcmaxRef = 112.77
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

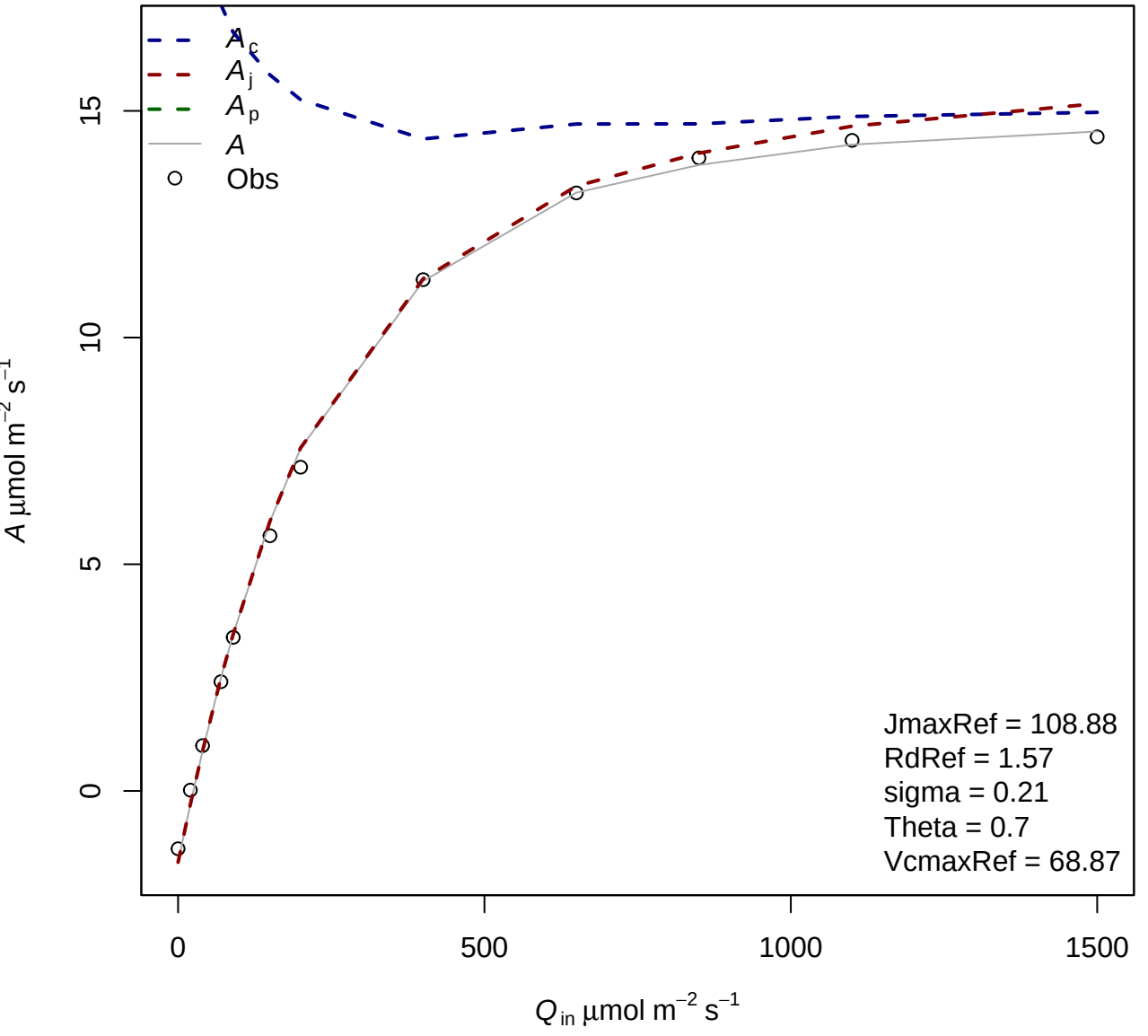

40
A_c
A_j
A_p
A
Obs
JmaxRef = 163.23
RdRef = 1.39
sigma = 0.2
Theta = 0.7
VcmaxRef = 103.94
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

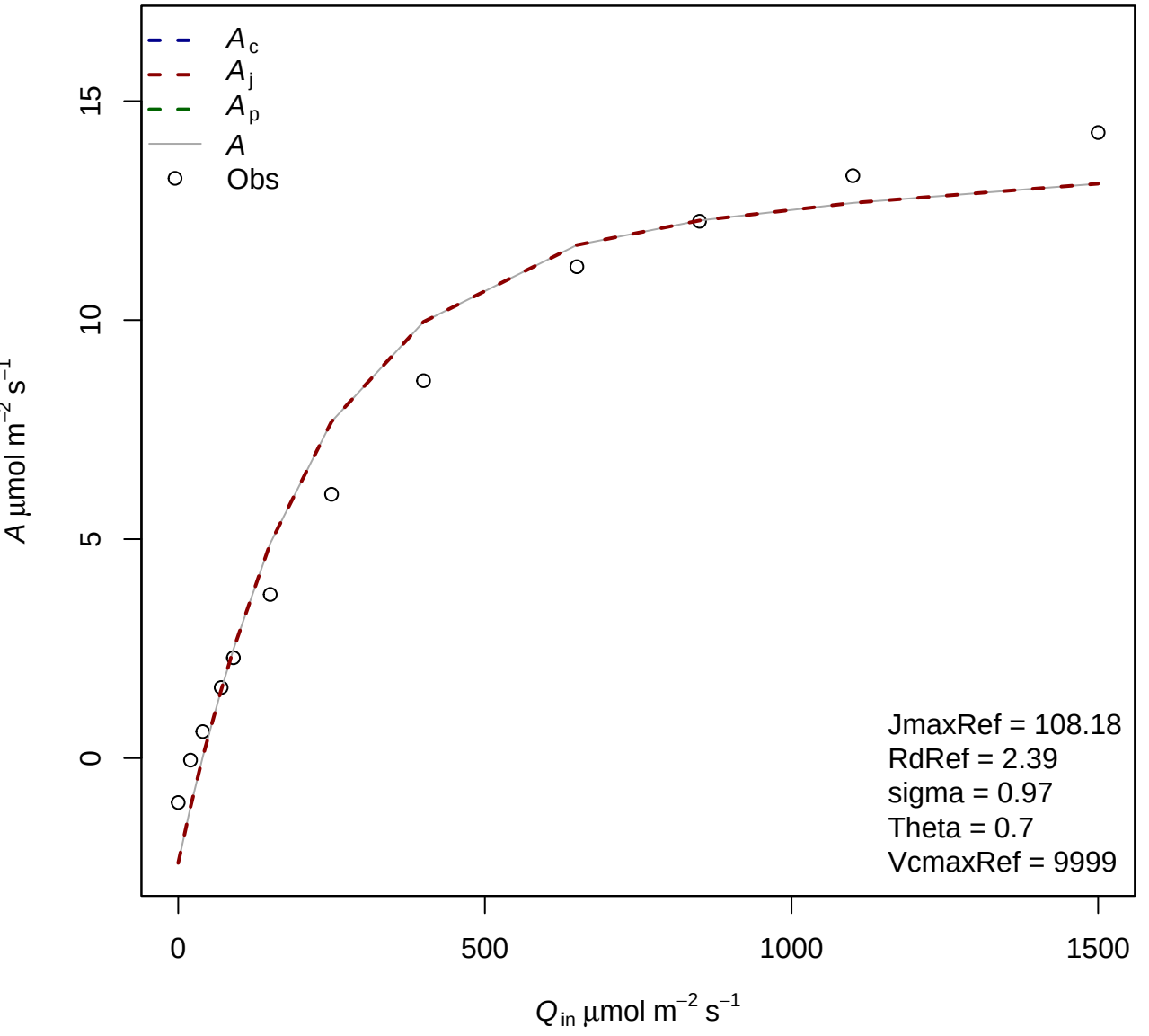

A_c
A_j
A_p
A
Obs
JmaxRef = 60.24
RdRef = 2.31
sigma = 0.29
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
200
400
600
800
1000

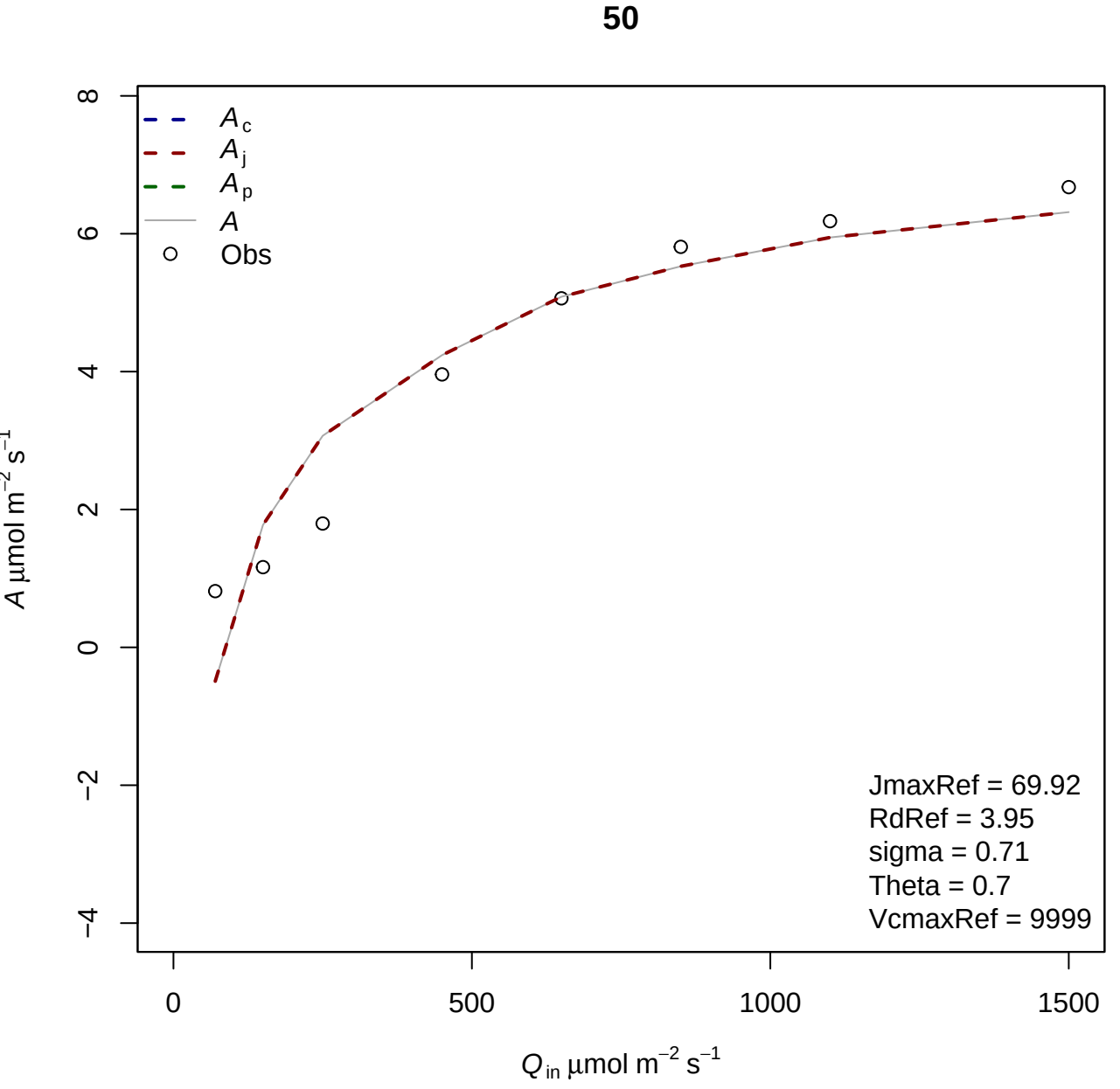

42
A_c
A_j
A_p
A
Obs
JmaxRef = 85.82
RdRef = 4.21
sigma = 0.84
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500

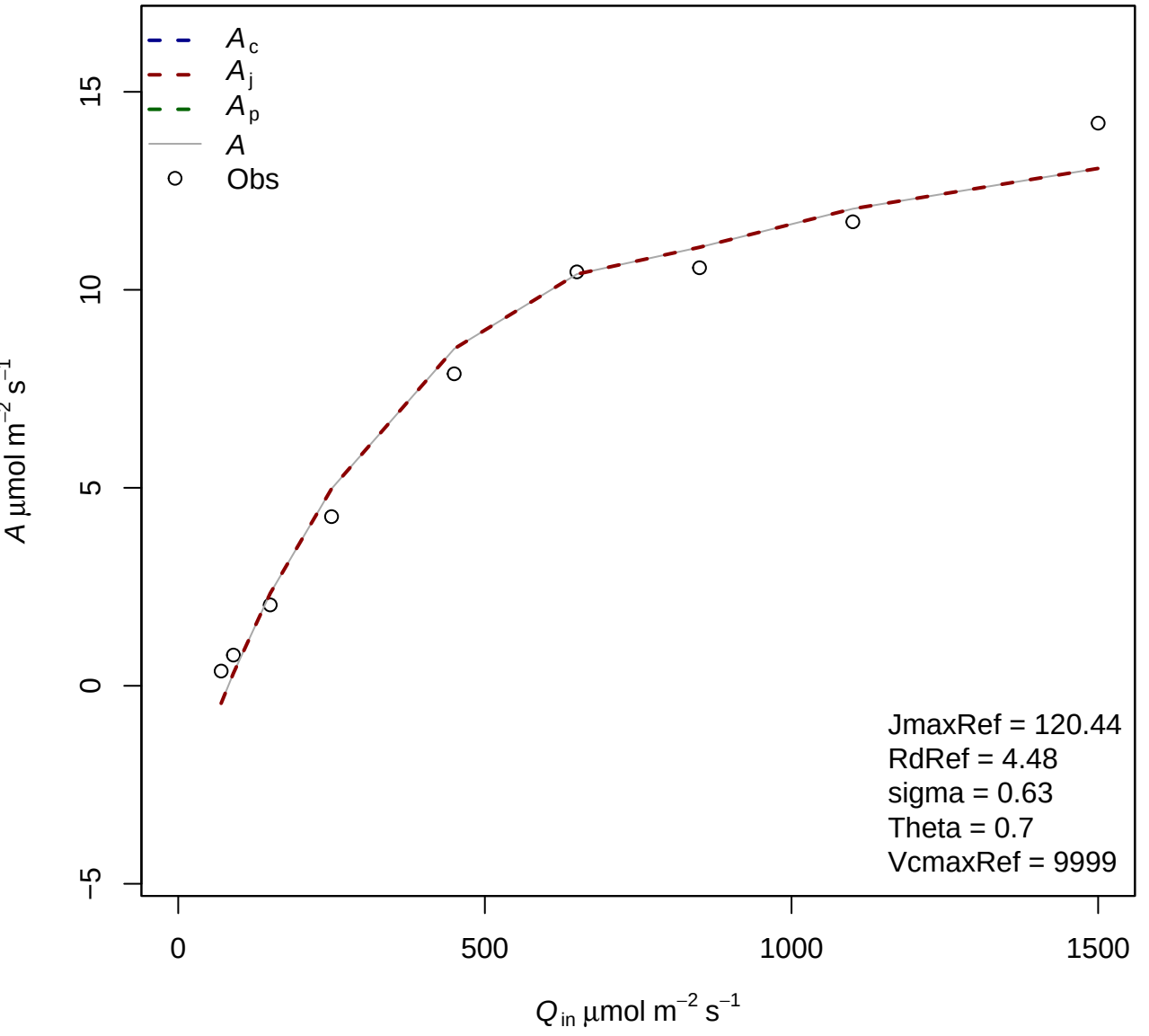

44
A_c
A_j
A_p
A
Obs
JmaxRef = 102.42
RdRef = 4.15
sigma = 0.34
Theta = 0.7
VcmaxRef = 69.23
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

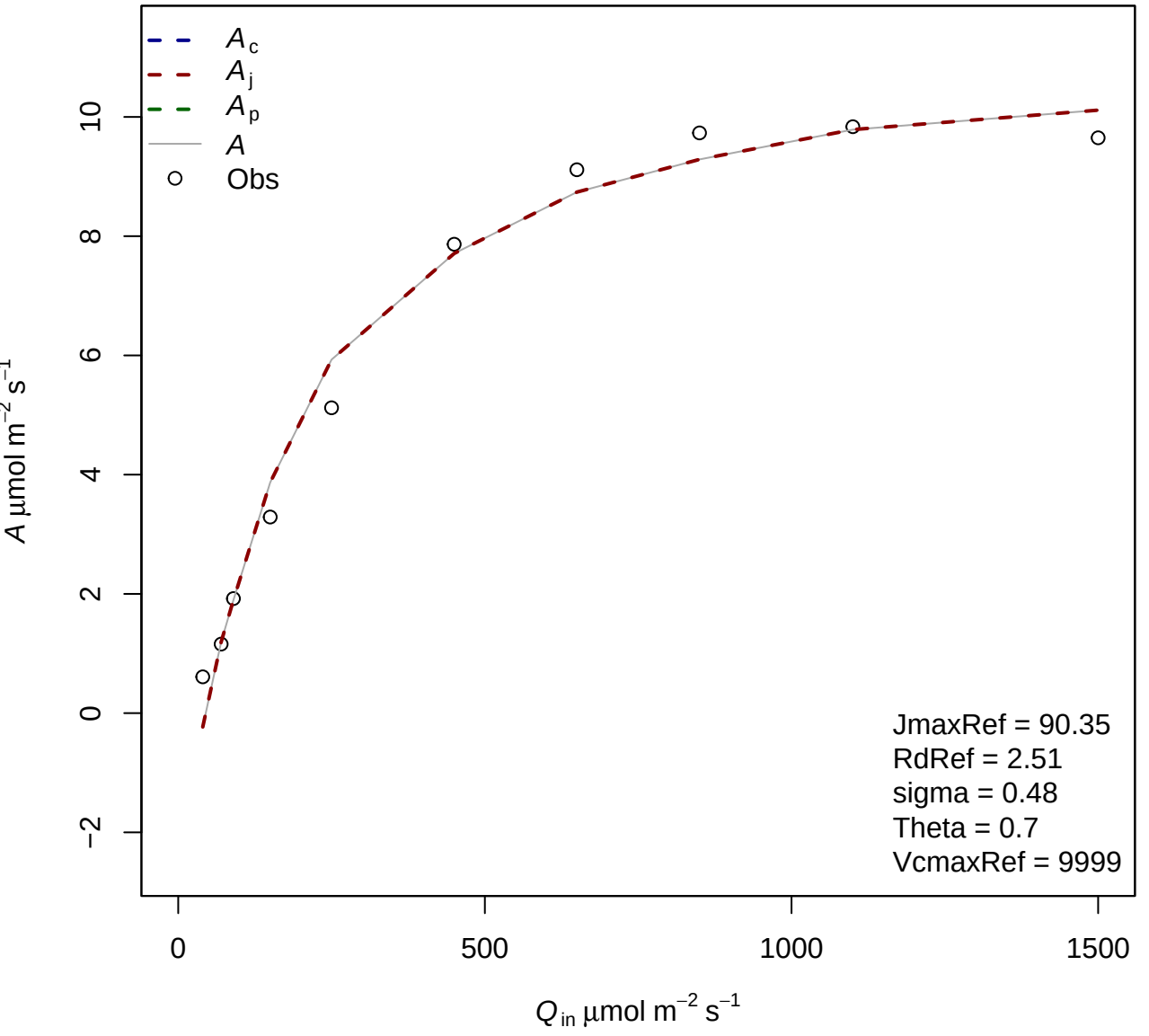

47
A_c
A_j
A_p
A
Obs
10
5
0
−5
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}
JmaxRef = 115.24
RdRef = 5.12
sigma = 0.12
Theta = 0.7
VcmaxRef = 66.78

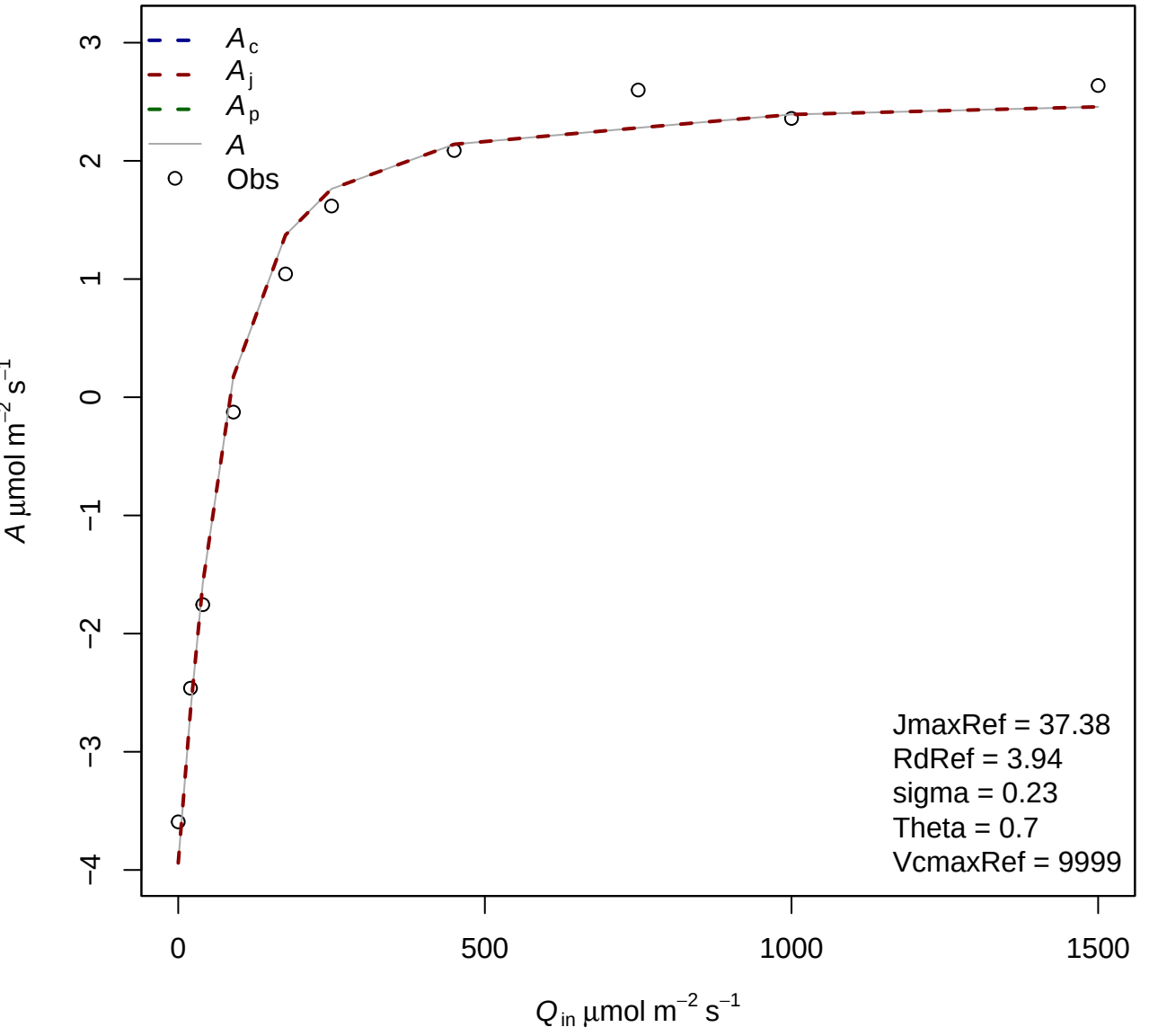

A_c
A_j
A_p
A
Obs
JmaxRef = 108.88
RdRef = 1.57
sigma = 0.21
Theta = 0.7
VcmaxRef = 68.87
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

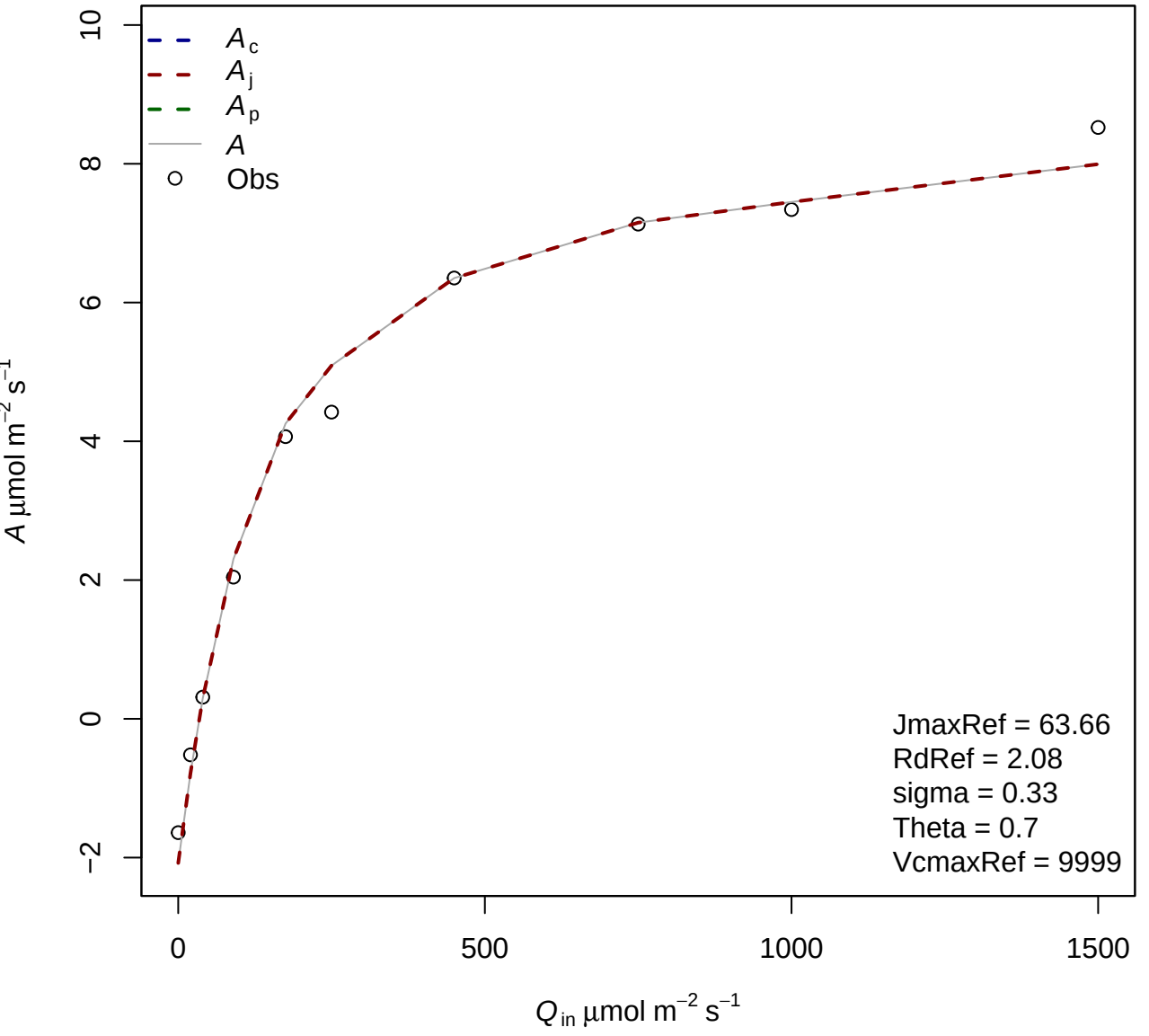

A_c
A_j
A_p
A
Obs
JmaxRef = 108.18
RdRef = 2.39
sigma = 0.97
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

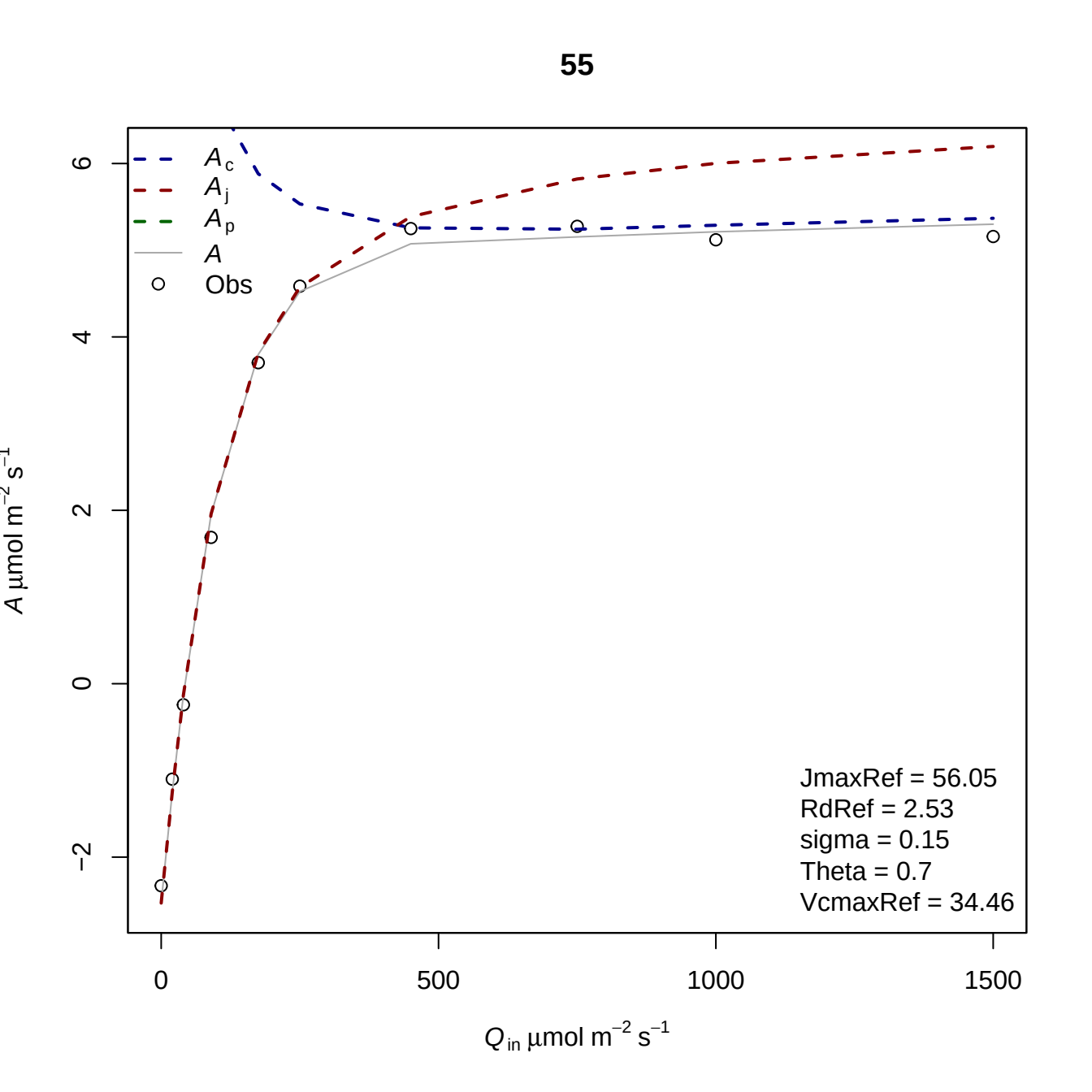

50
A_c
A_j
A_p
A
Obs
JmaxRef = 69.92
RdRef = 3.95
sigma = 0.71
Theta = 0.7
VcmaxRef = 9999
8
6
4
2
0
−2
−4
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

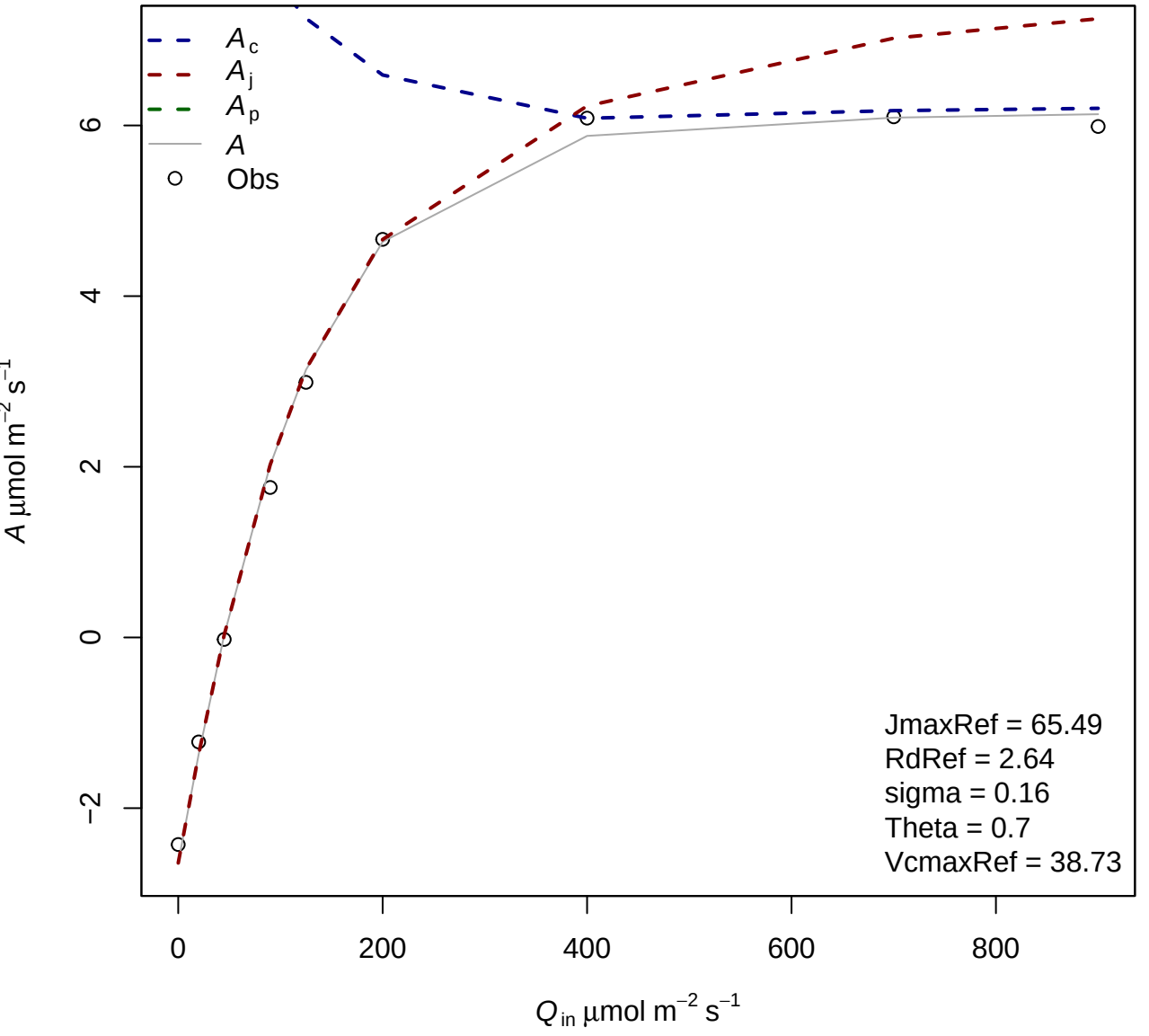

A_c
A_j
A_p
A
Obs
JmaxRef = 120.44
RdRef = 4.48
sigma = 0.63
Theta = 0.7
VcmaxRef = 9999
15
10
5
0
−5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

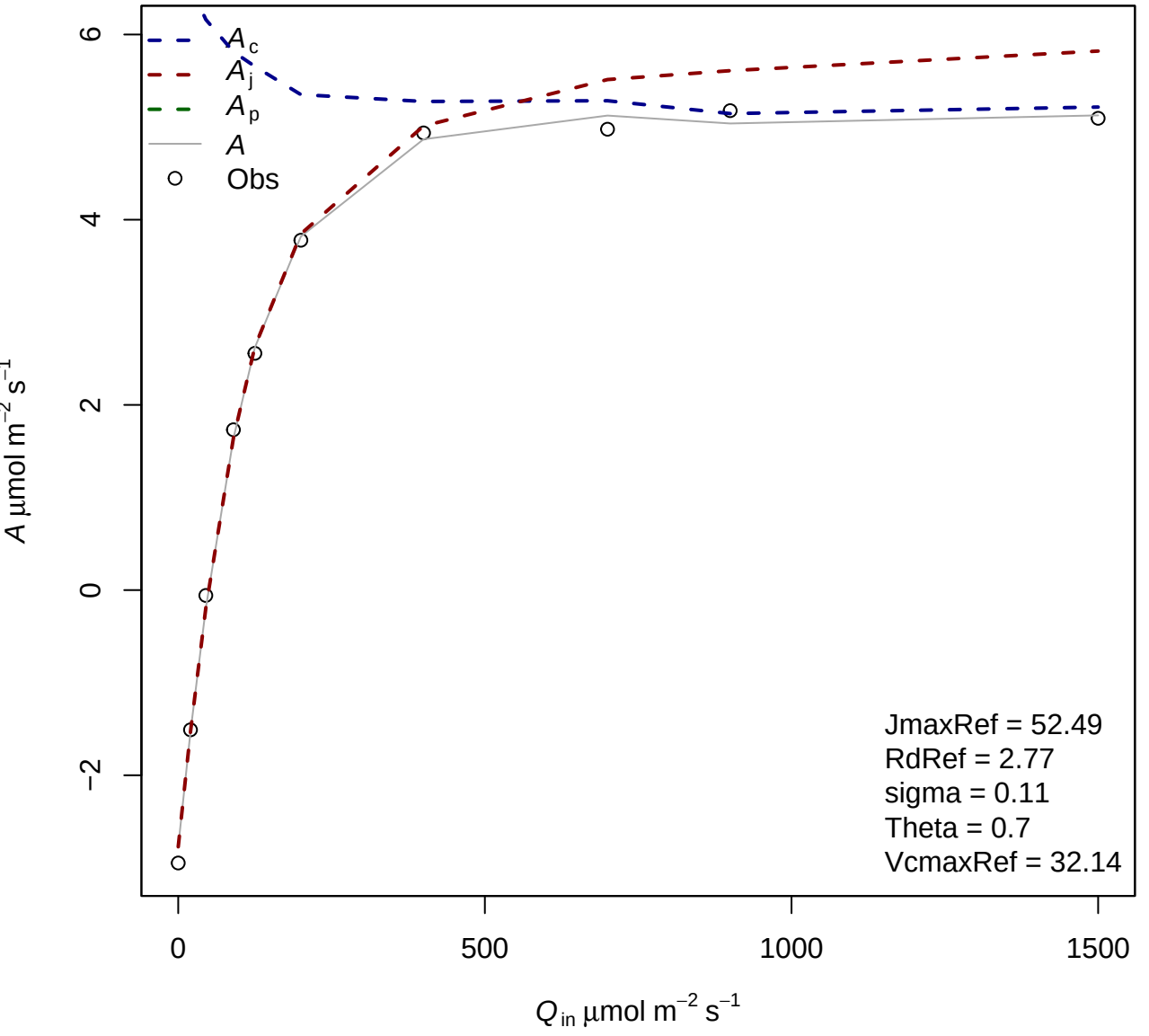

A_c
A_j
A_p
A
Obs
JmaxRef = 90.35
RdRef = 2.51
sigma = 0.48
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

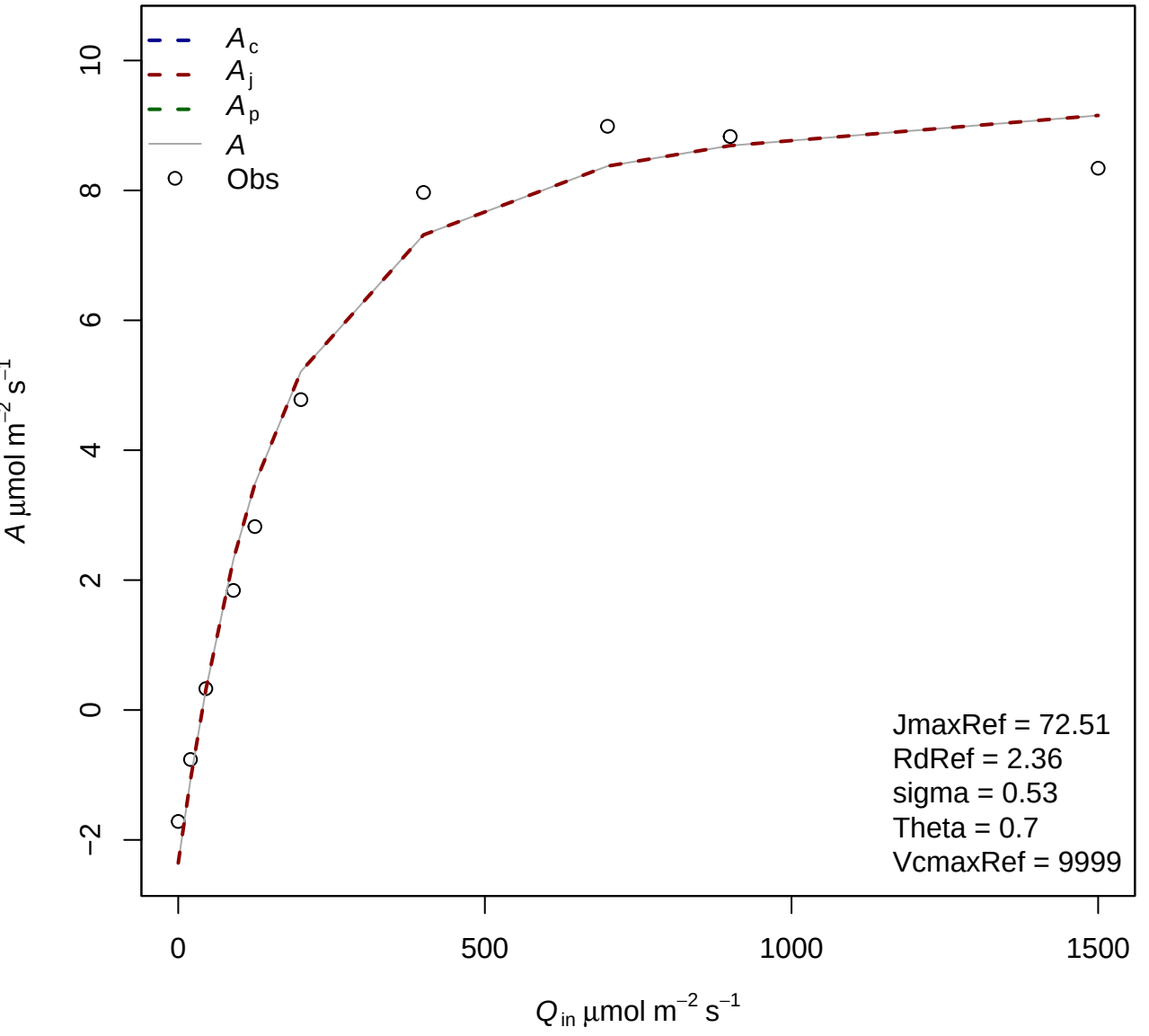

A_c
A_j
A_p
A
Obs
JmaxRef = 37.38
RdRef = 3.94
sigma = 0.23
Theta = 0.7
VcmaxRef = 9999
3
2
1
0
−1
−2
−3
−4
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

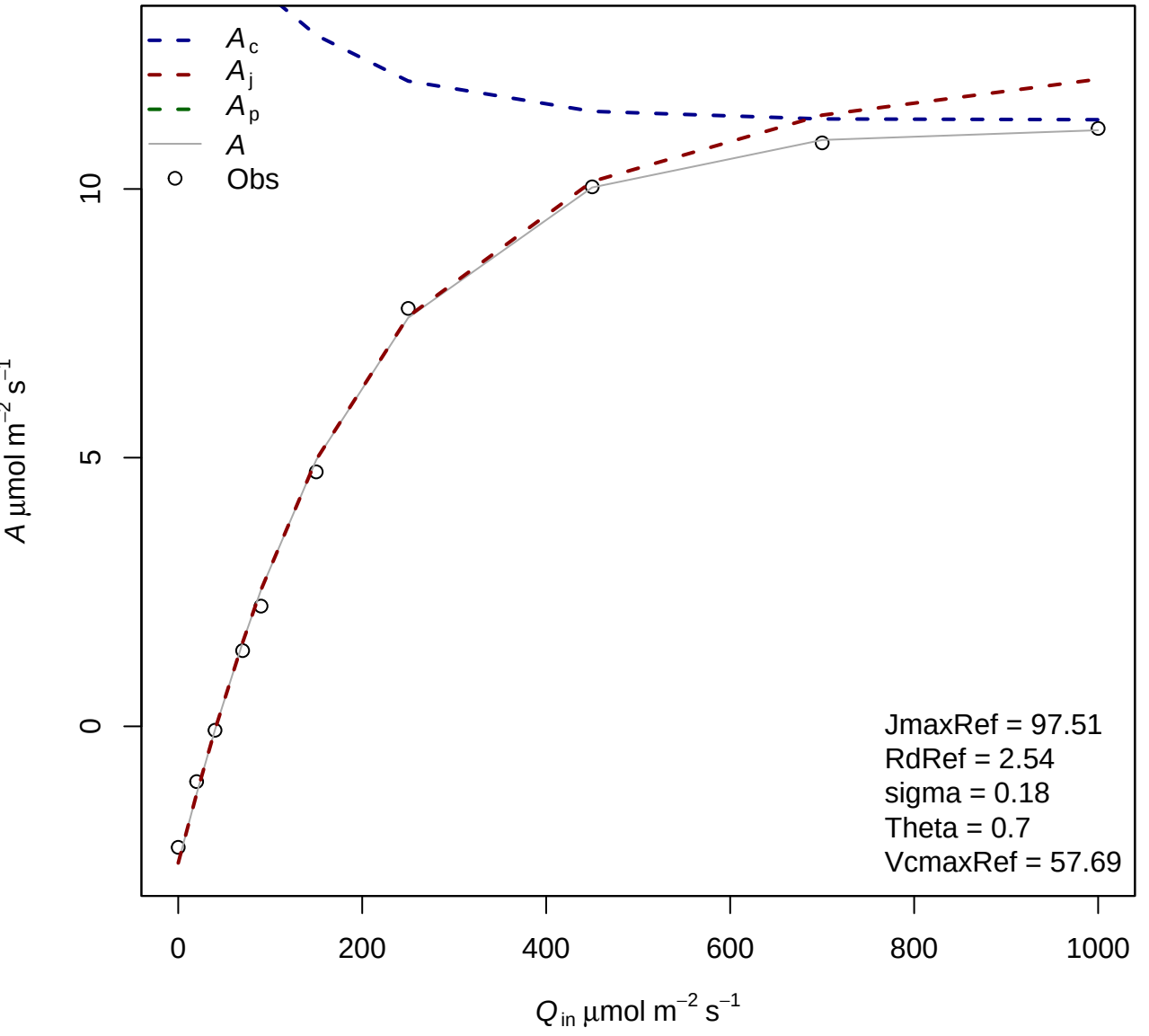

A_c
A_j
A_p
A
Obs
JmaxRef = 63.66
RdRef = 2.08
sigma = 0.33
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

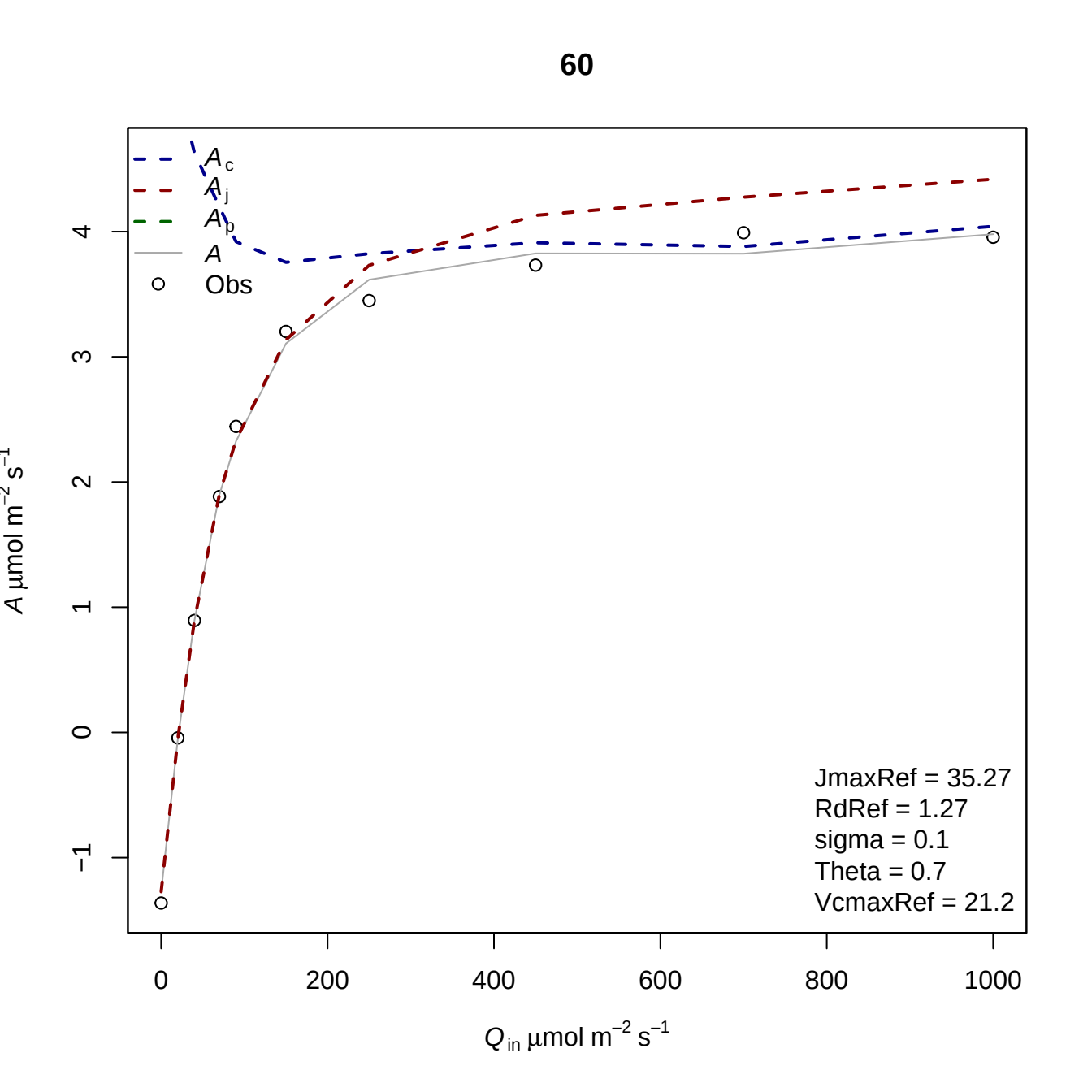

55
A_c
A_j
A_p
A
Obs
JmaxRef = 56.05
RdRef = 2.53
sigma = 0.15
Theta = 0.7
VcmaxRef = 34.46
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

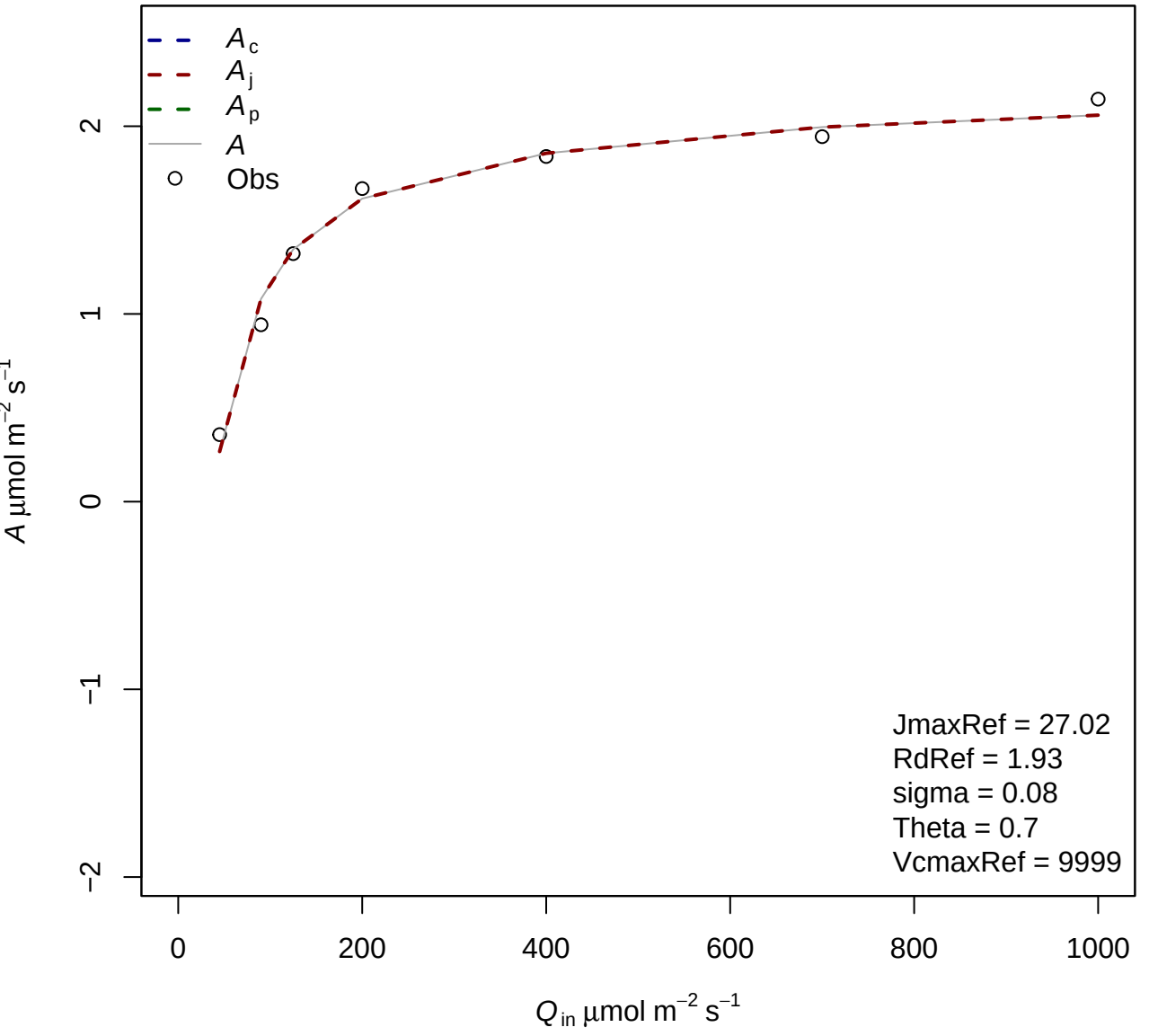

A_c
A_j
A_p
A
Obs
JmaxRef = 65.49
RdRef = 2.64
sigma = 0.16
Theta = 0.7
VcmaxRef = 38.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800

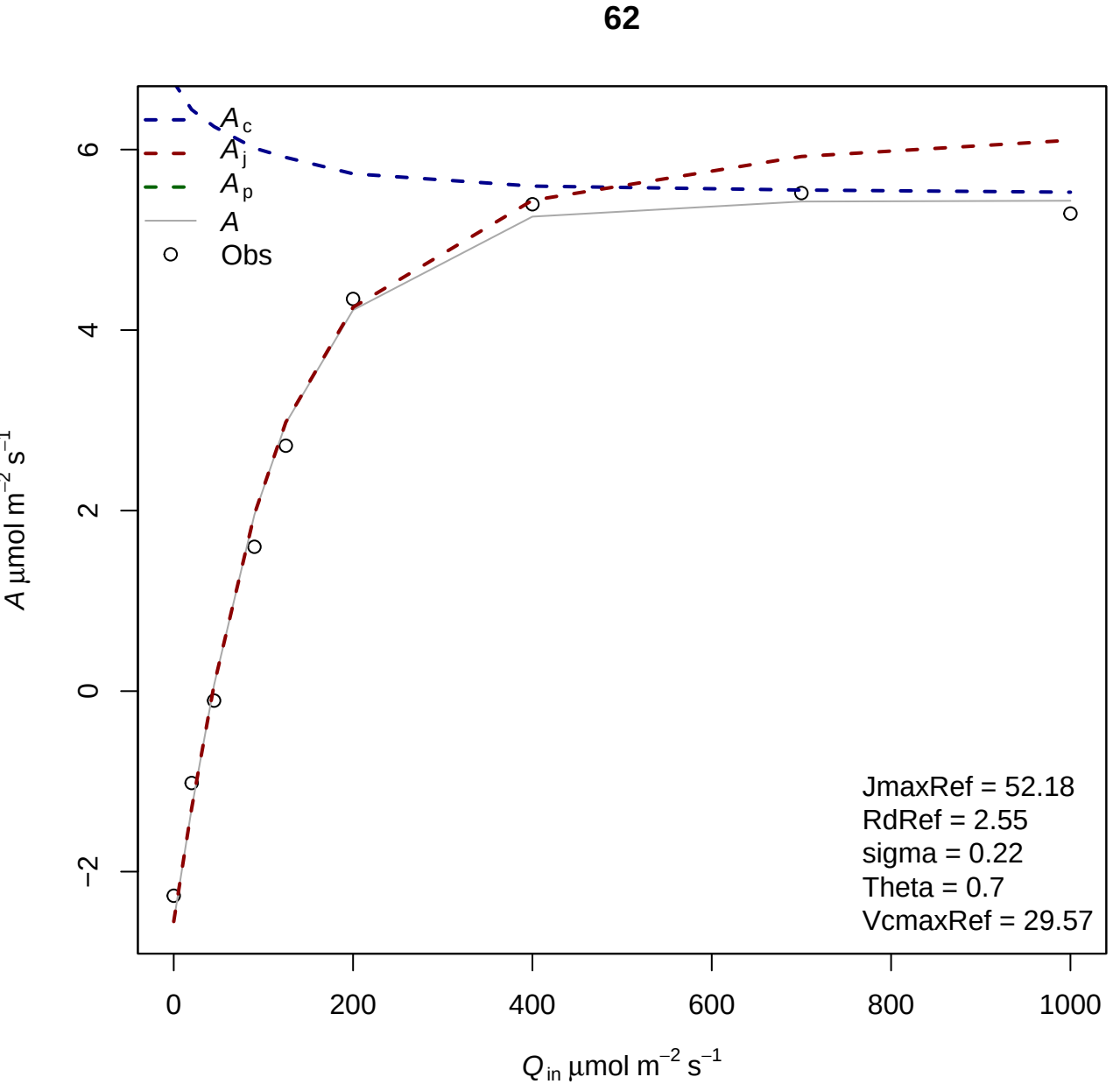

A_c
A_j
A_p
A
Obs
JmaxRef = 52.49
RdRef = 2.77
sigma = 0.11
Theta = 0.7
VcmaxRef = 32.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

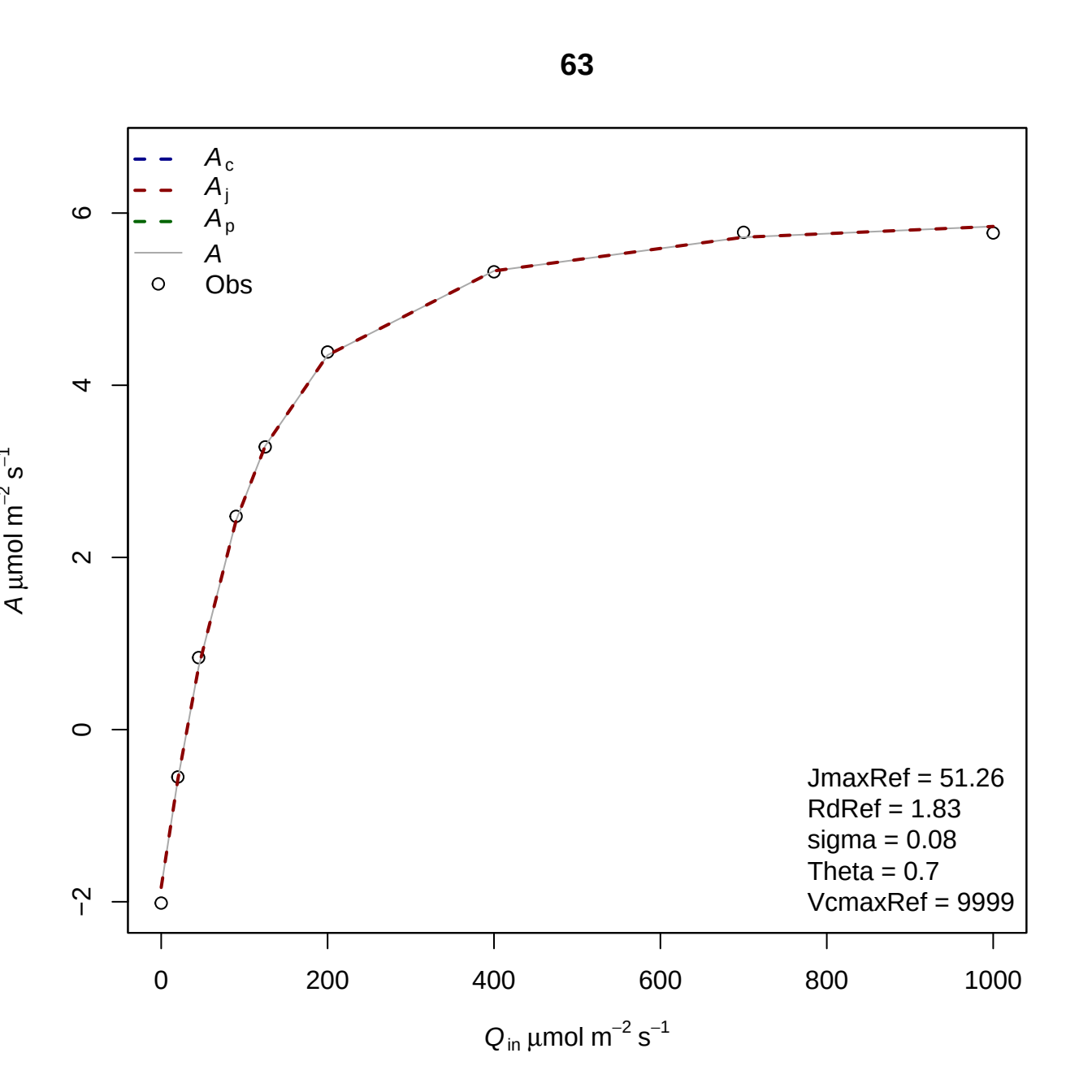

A_c
A_j
A_p
A
Obs
JmaxRef = 72.51
RdRef = 2.36
sigma = 0.53
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

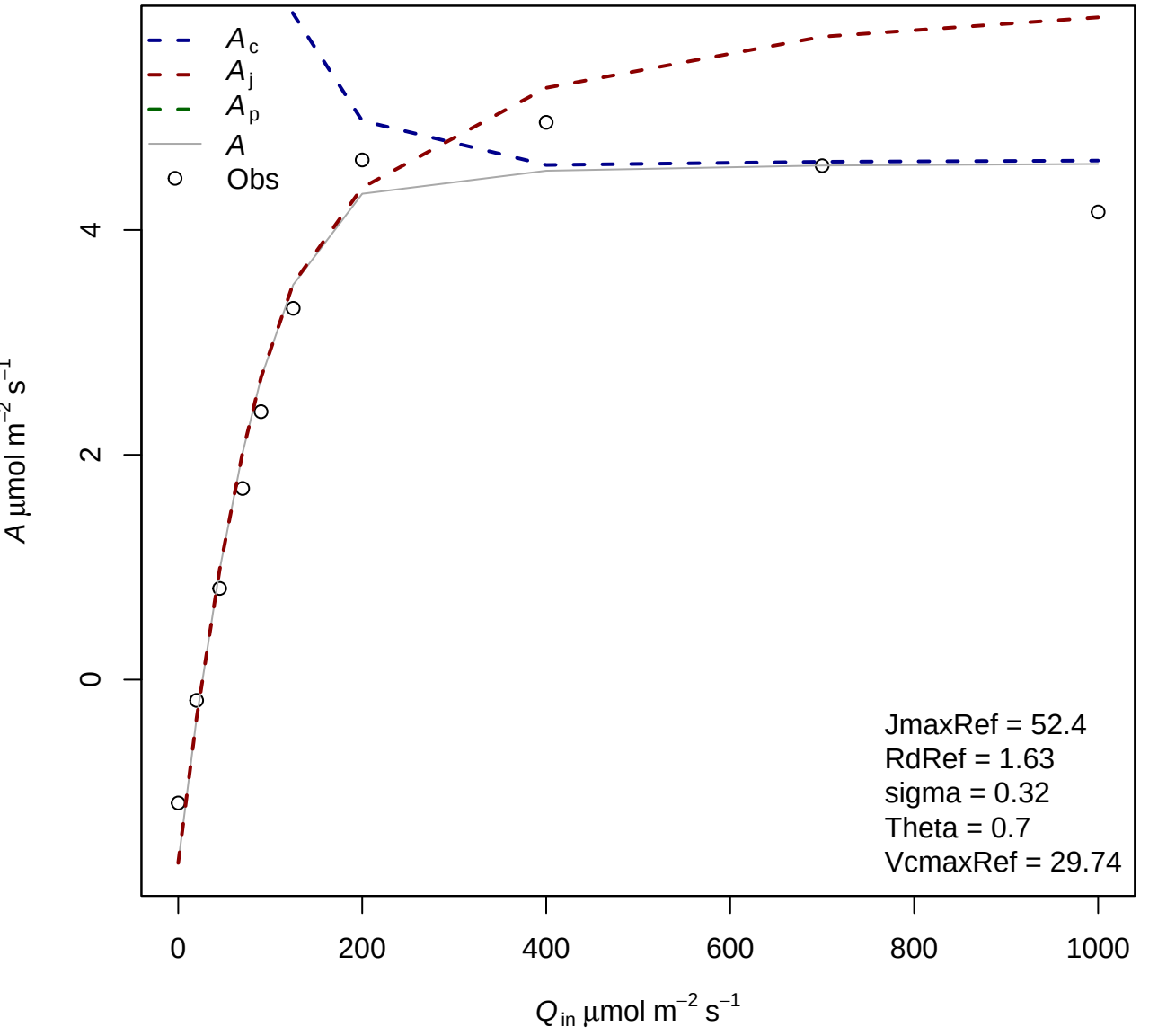

A_c
A_j
A_p
A
Obs
JmaxRef = 97.51
RdRef = 2.54
sigma = 0.18
Theta = 0.7
VcmaxRef = 57.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800
1000

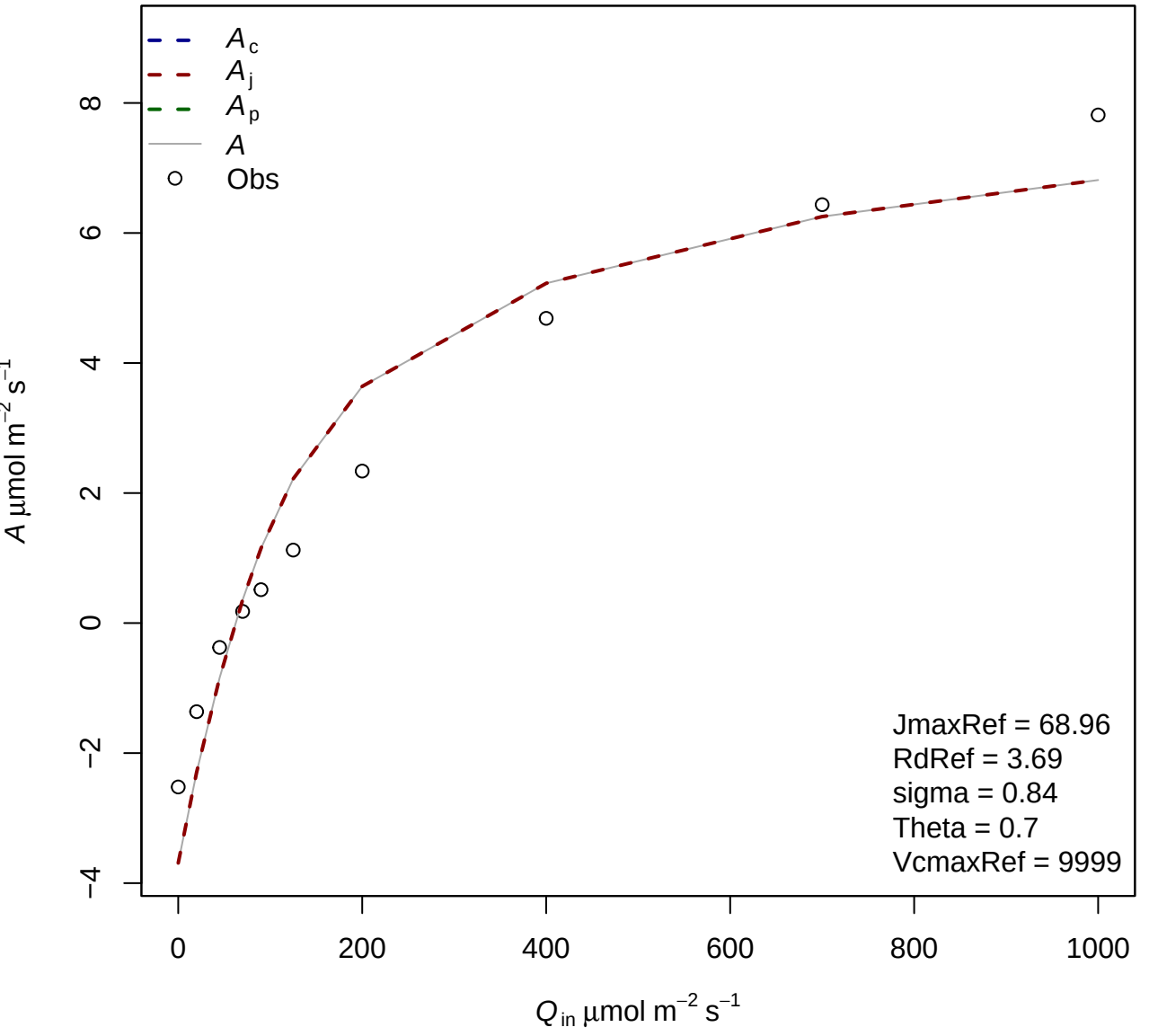

60
A_c
A_j
A_p
A
Obs
JmaxRef = 35.27
RdRef = 1.27
sigma = 0.1
Theta = 0.7
VcmaxRef = 21.2
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000

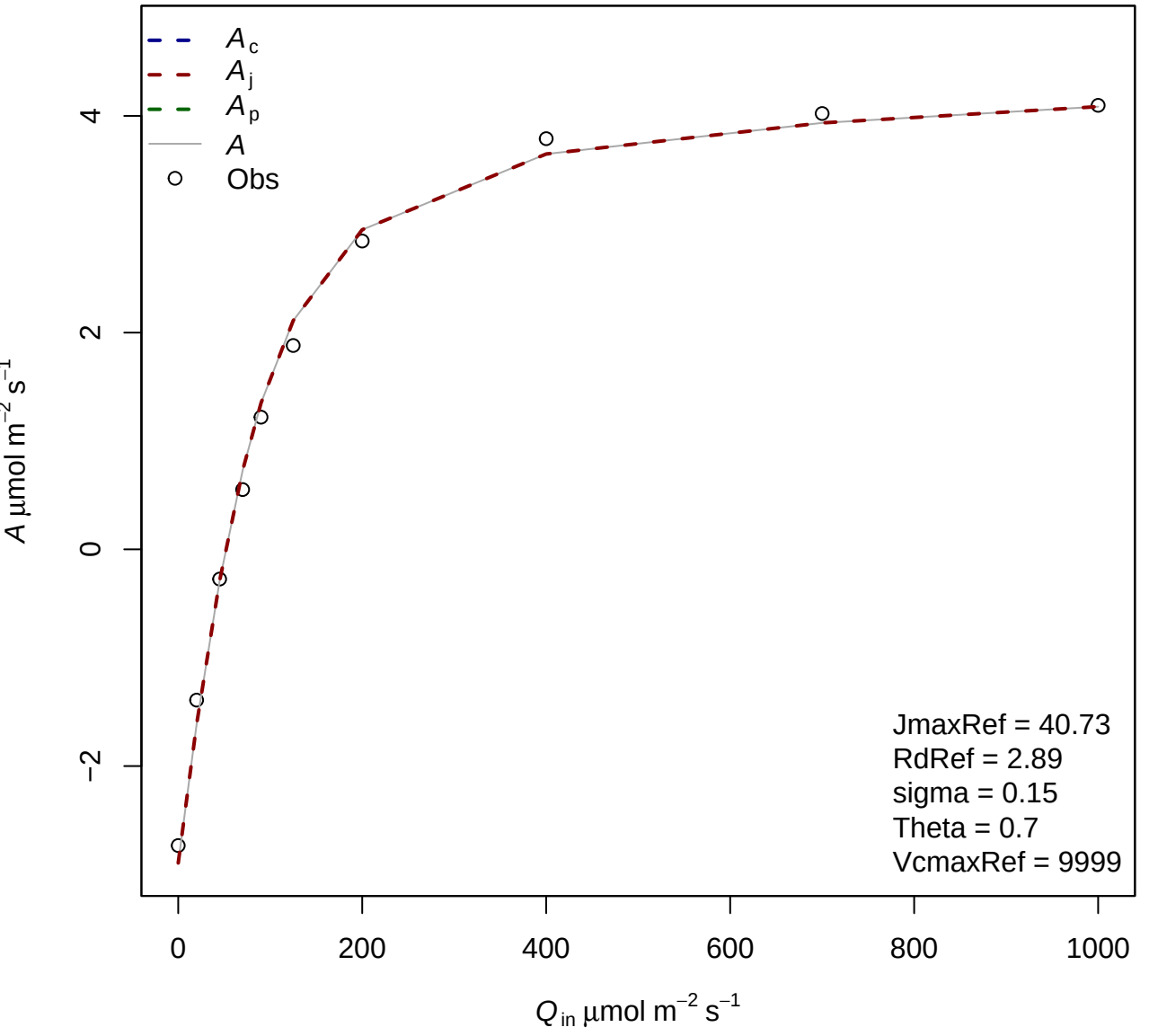

A_c
A_j
A_p
A
Obs
JmaxRef = 27.02
RdRef = 1.93
sigma = 0.08
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2
1
0
−1
−2
0
200
400
600
800
1000

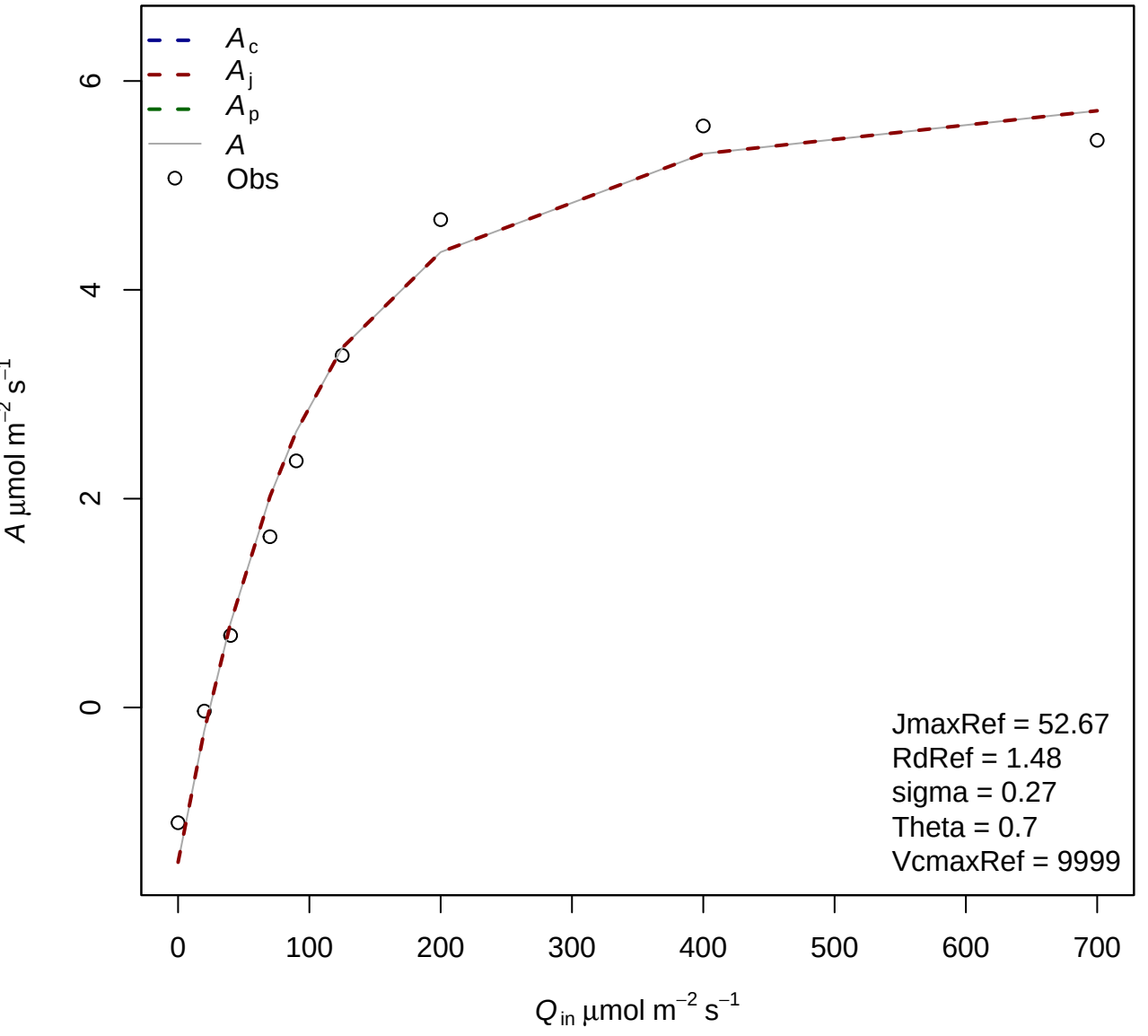

62
A_c
A_j
A_p
A
Obs
JmaxRef = 52.18
RdRef = 2.55
sigma = 0.22
Theta = 0.7
VcmaxRef = 29.57
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000

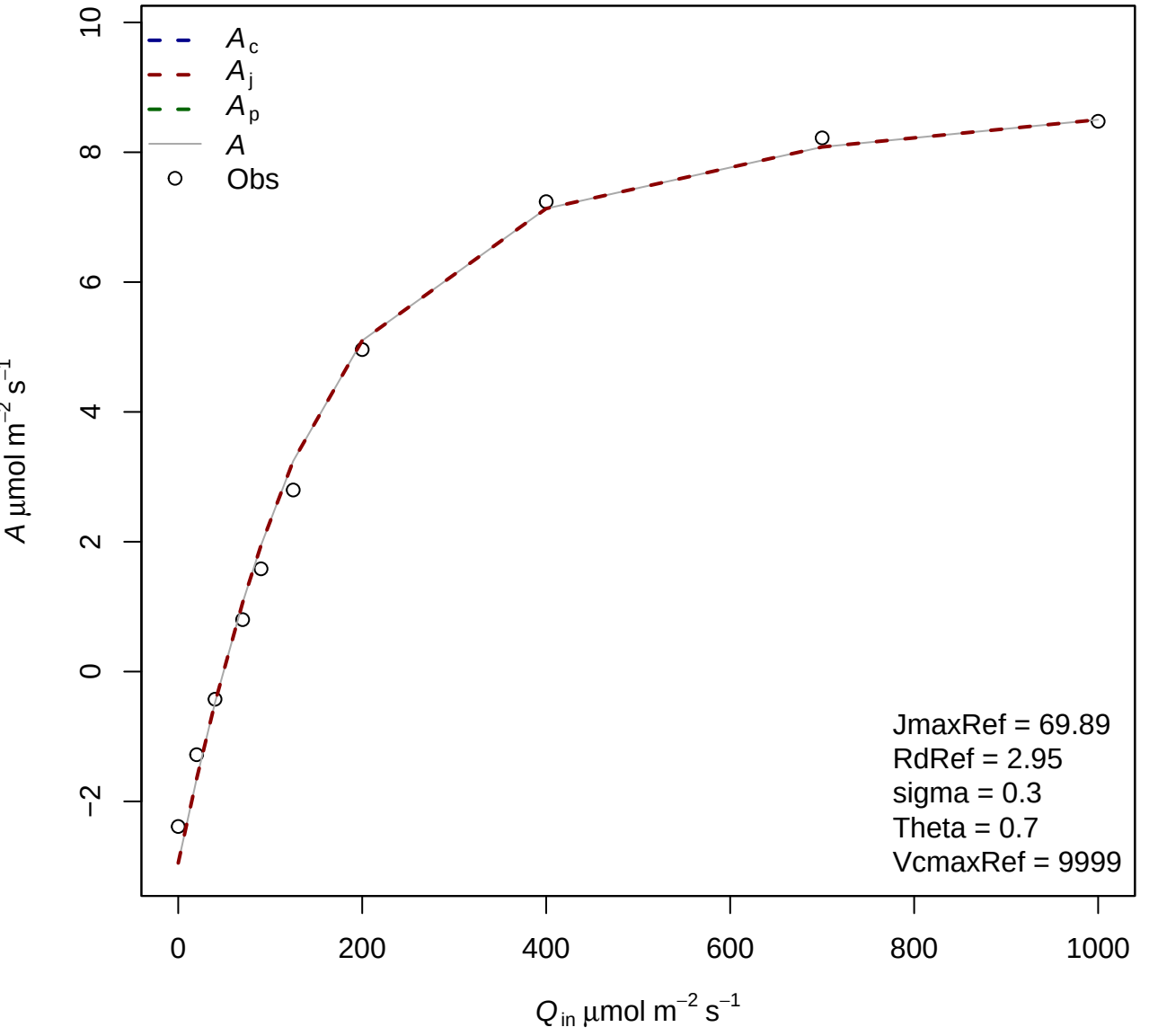

63
A_c
A_j
A_p
A
Obs
JmaxRef = 51.26
RdRef = 1.83
sigma = 0.08
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000

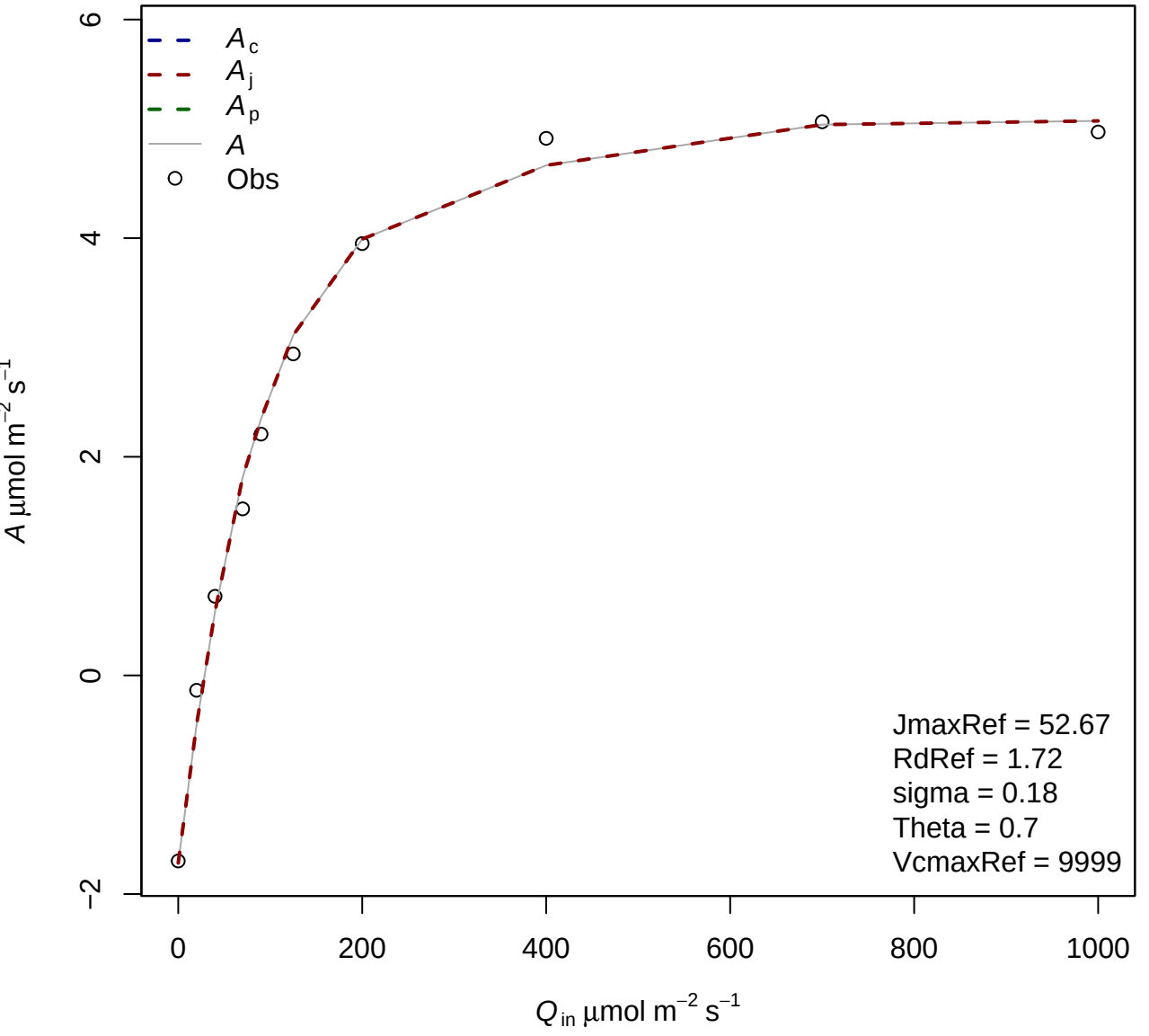

A_c
A_j
A_p
A
Obs
JmaxRef = 52.4
RdRef = 1.63
sigma = 0.32
Theta = 0.7
VcmaxRef = 29.74
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
0
200
400
600
800
1000

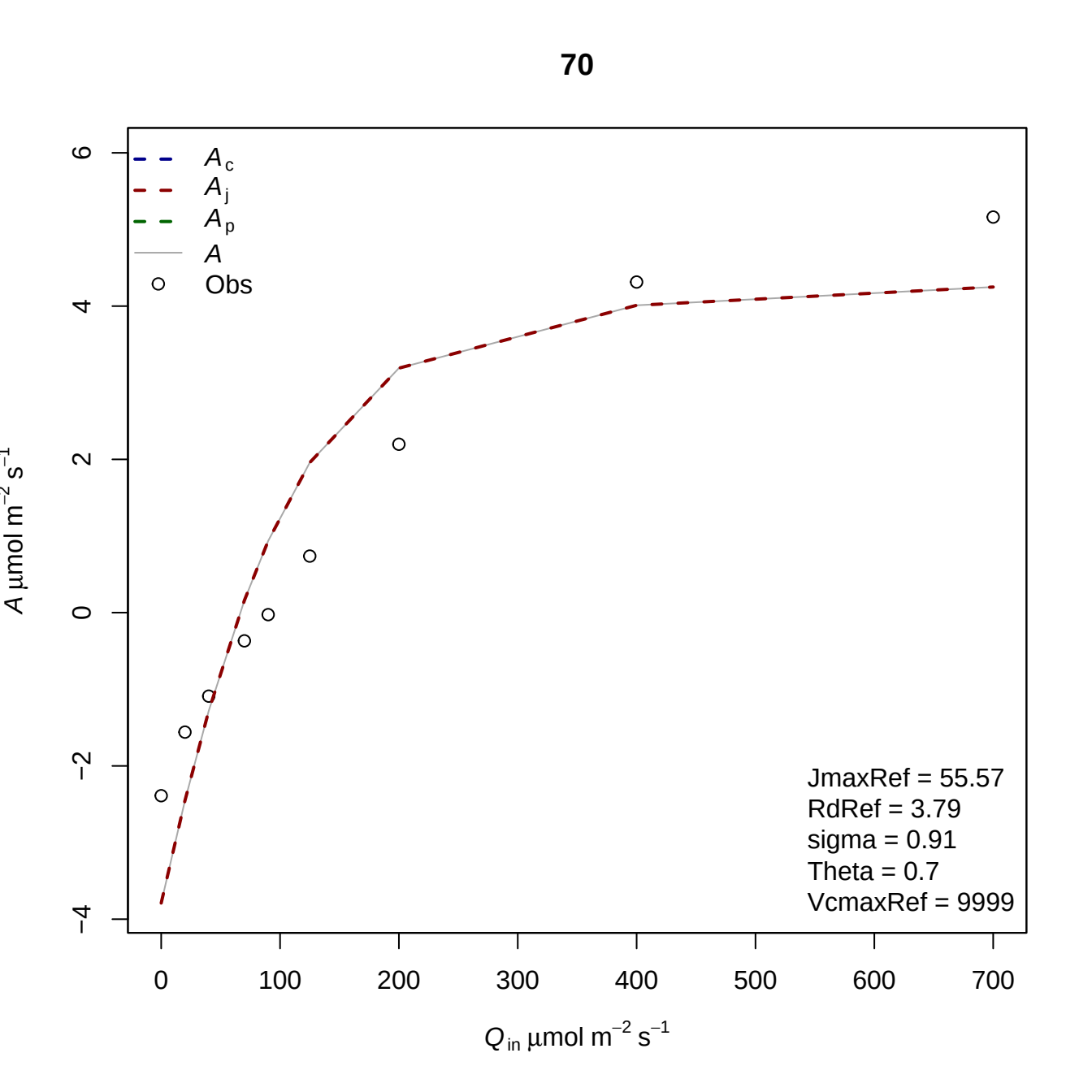

A_c
A_j
A_p
A
Obs
JmaxRef = 68.96
RdRef = 3.69
sigma = 0.84
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−4
−2
0
2
4
6
8
0
200
400
600
800
1000

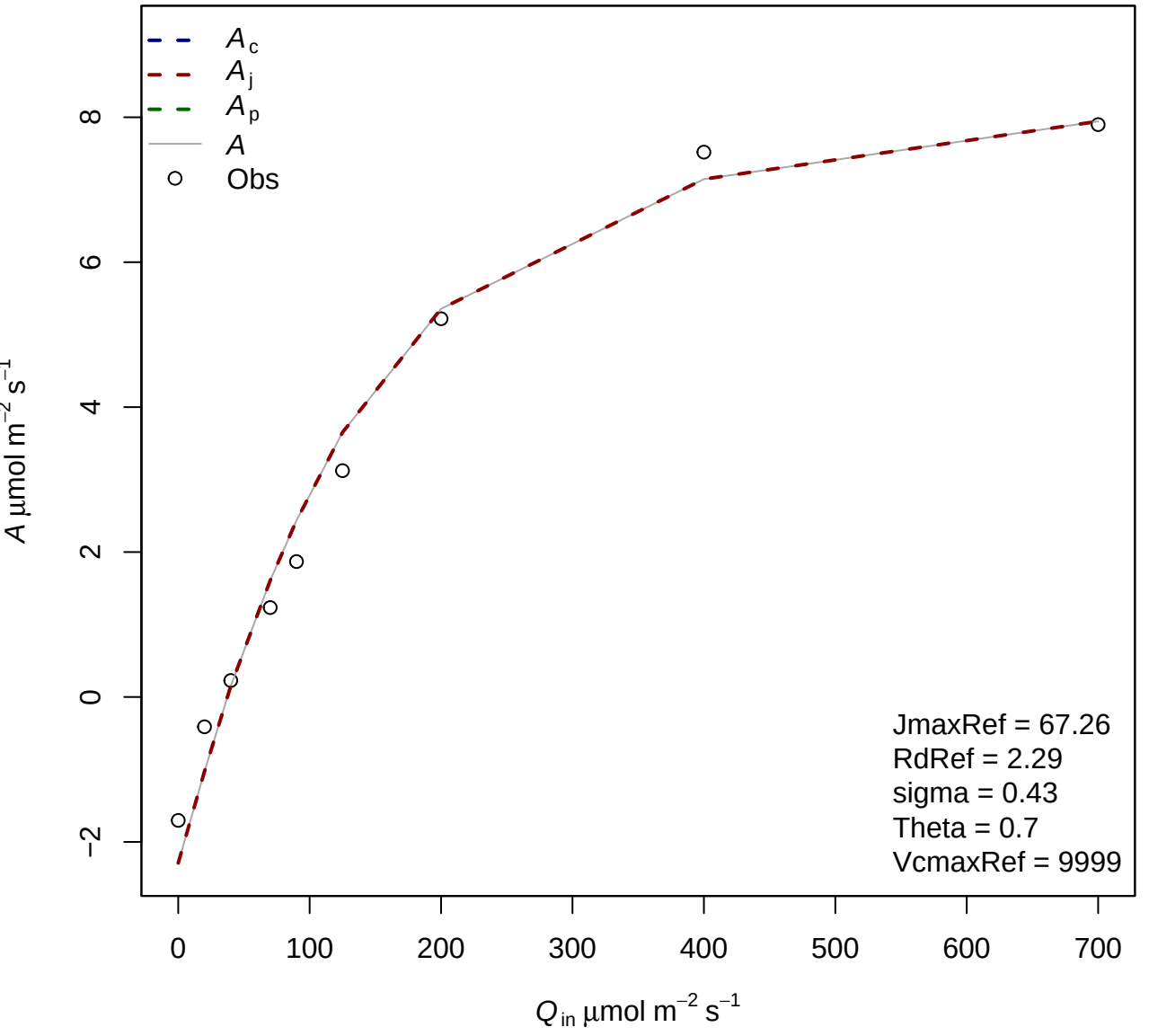

A_c
A_j
A_p
A
Obs
JmaxRef = 40.73
RdRef = 2.89
sigma = 0.15
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
200
400
600
800
1000

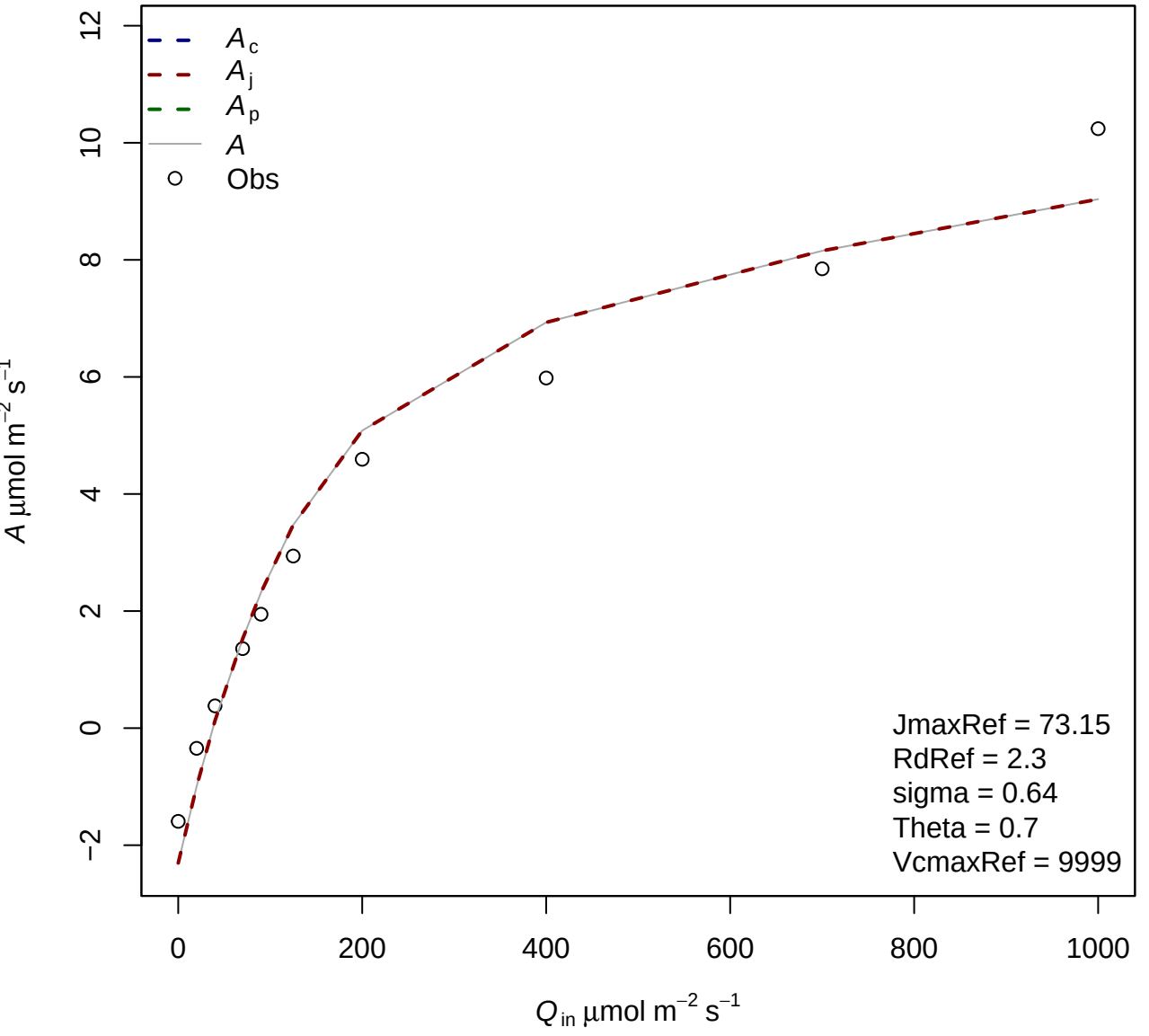

A_c
A_j
A_p
A
Obs
JmaxRef = 52.67
RdRef = 1.48
sigma = 0.27
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
100
200
300
400
500
600
700

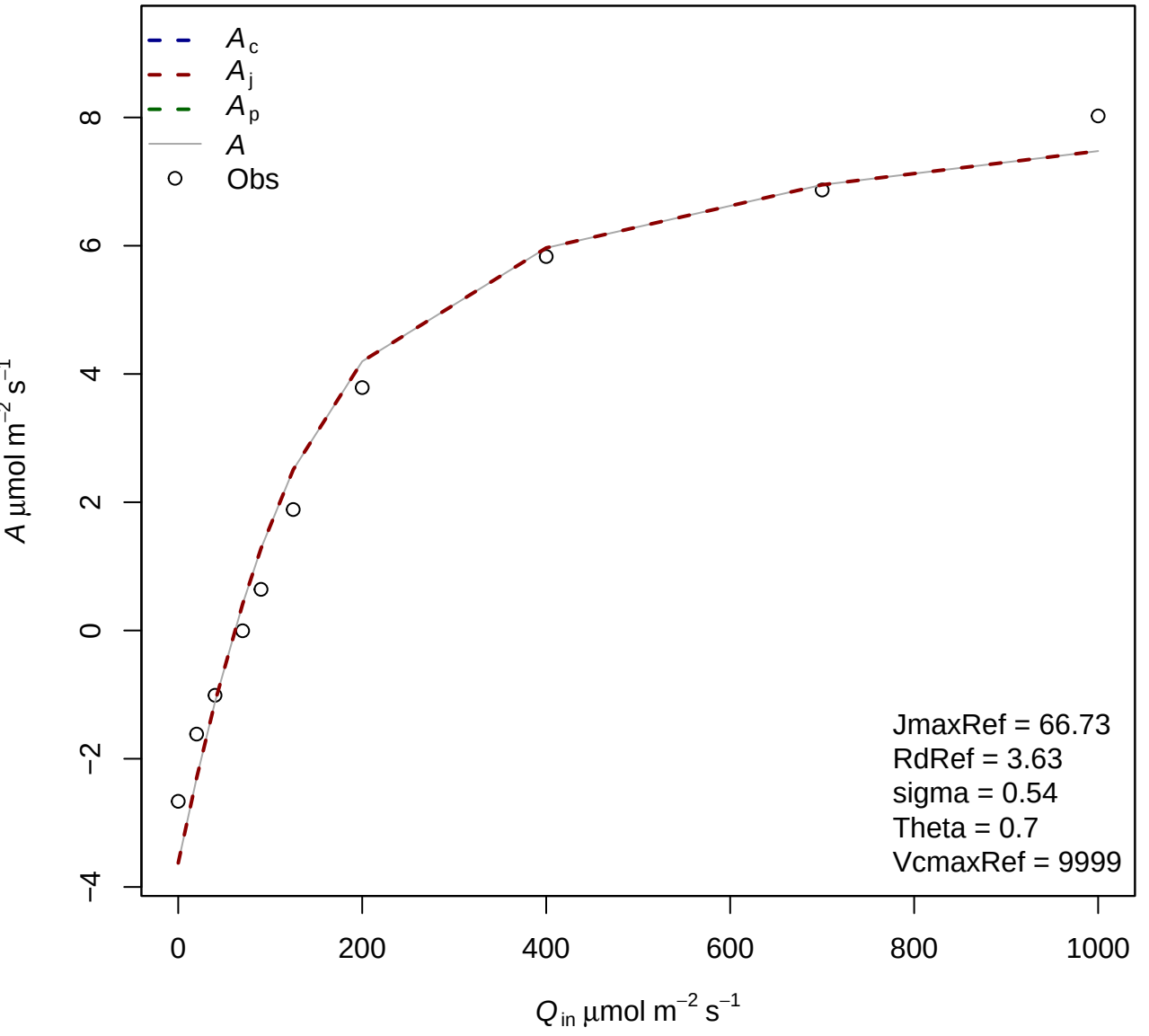

A_c
A_j
A_p
A
Obs
JmaxRef = 69.89
RdRef = 2.95
sigma = 0.3
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
200
400
600
800
1000

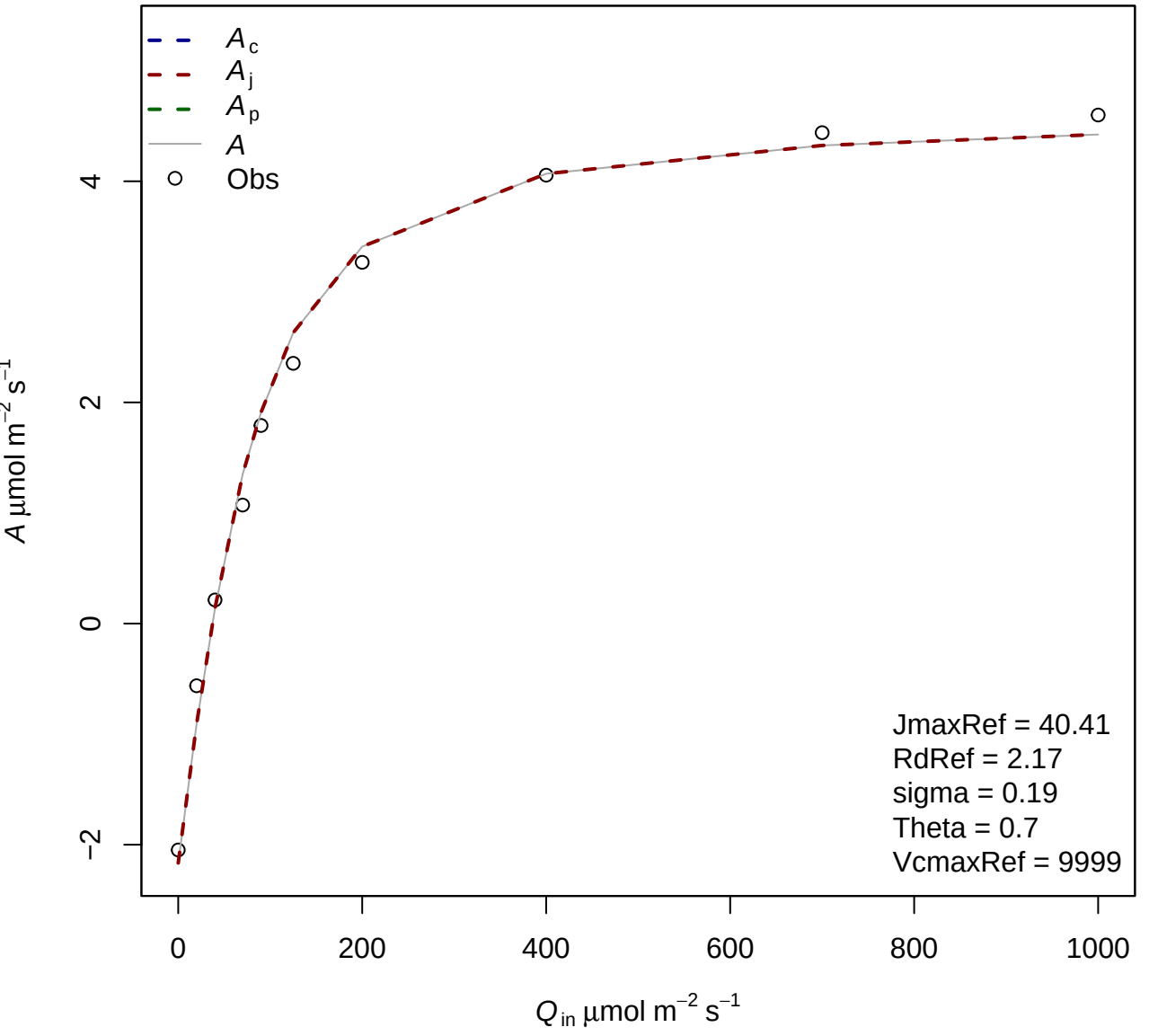

A_c
A_j
A_p
A
Obs
JmaxRef = 52.67
RdRef = 1.72
sigma = 0.18
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
200
400
600
800
1000

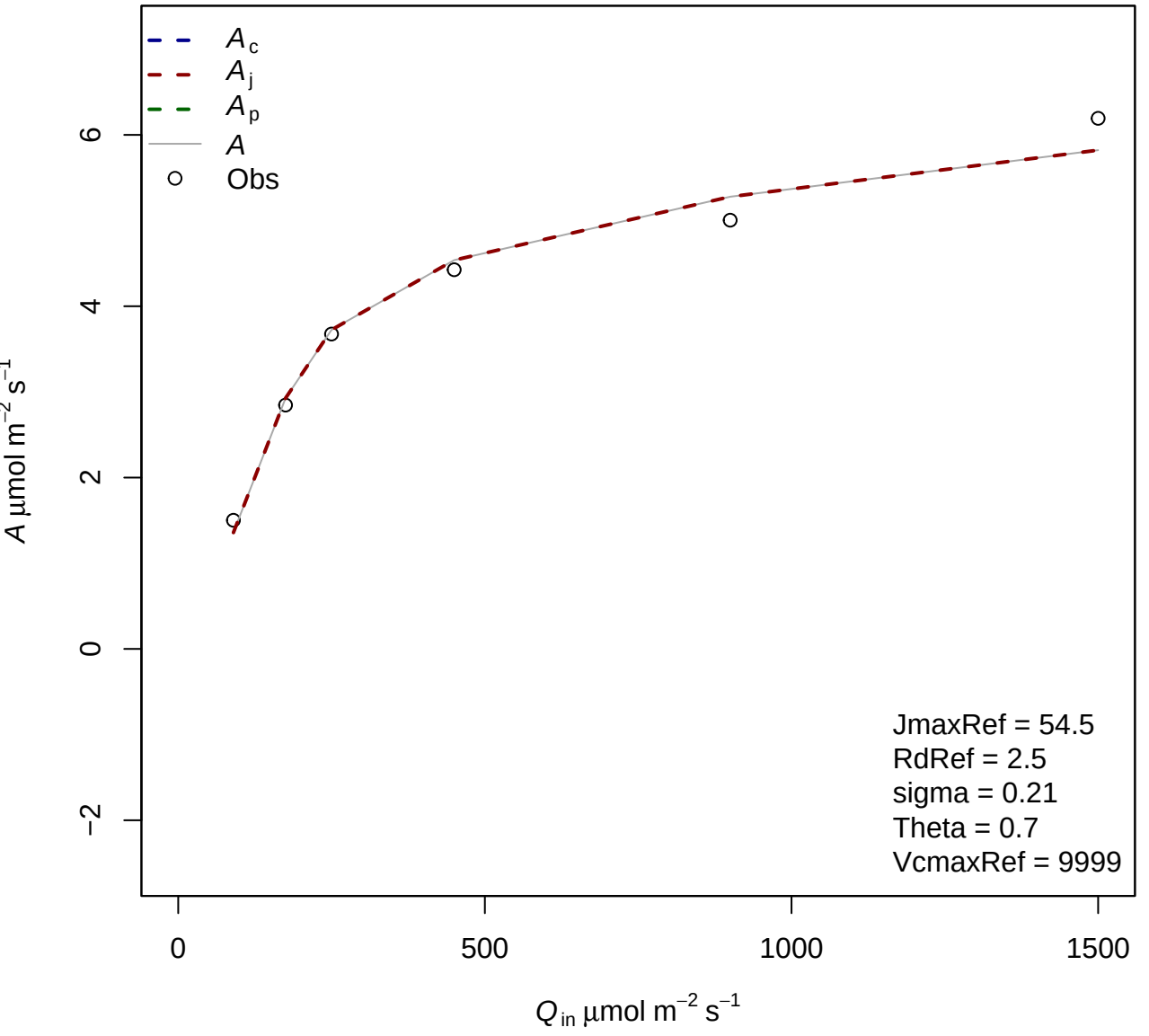

70
A_c
A_j
A_p
A
Obs
JmaxRef = 55.57
RdRef = 3.79
sigma = 0.91
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
100
200
300
400
500
600
700

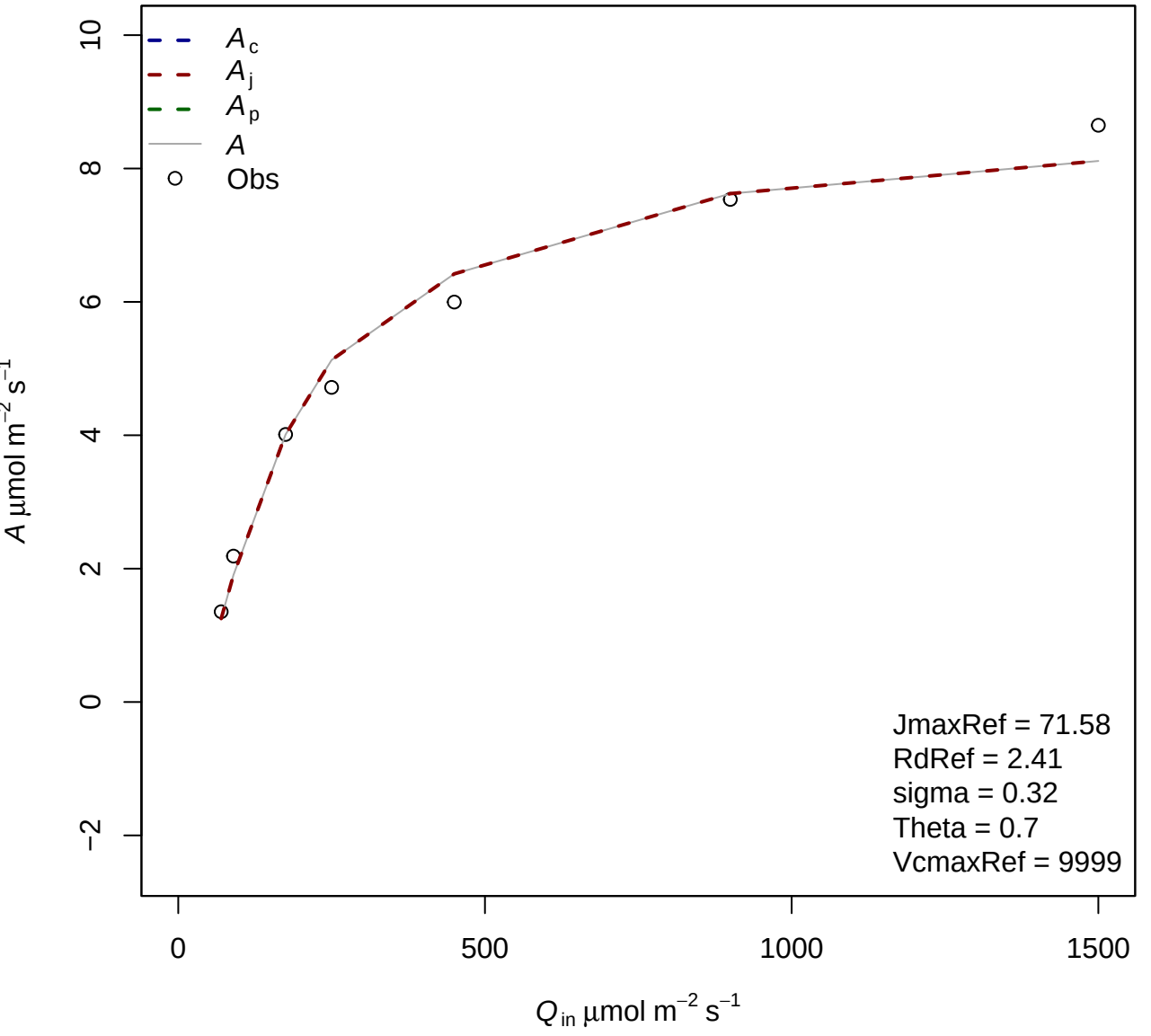

A_c
A_j
A_p
A
Obs
JmaxRef = 67.26
RdRef = 2.29
sigma = 0.43
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
100
200
300
400
500
600
700

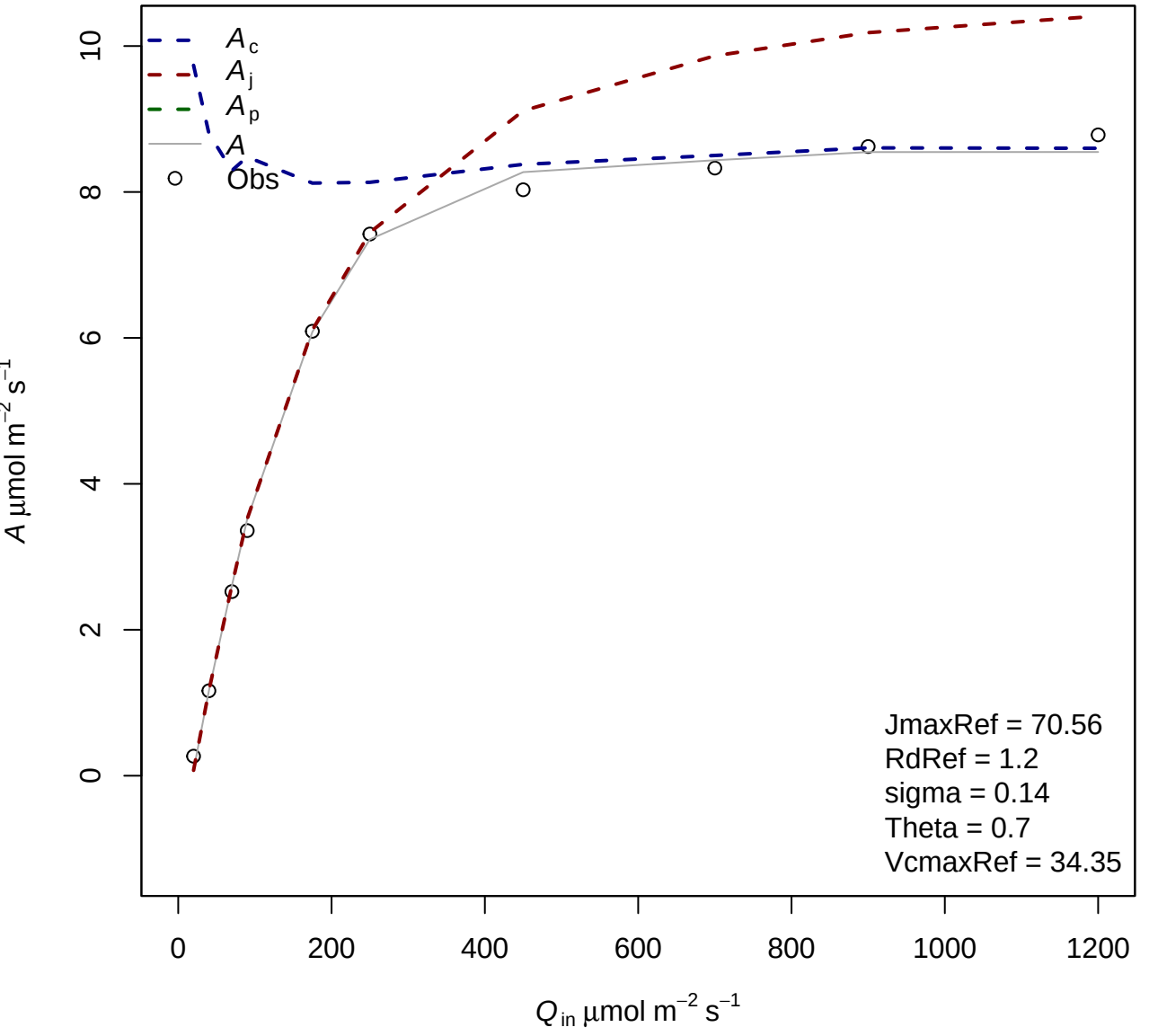

A_c
A_j
A_p
A
Obs
JmaxRef = 73.15
RdRef = 2.3
sigma = 0.64
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
200
400
600
800
1000

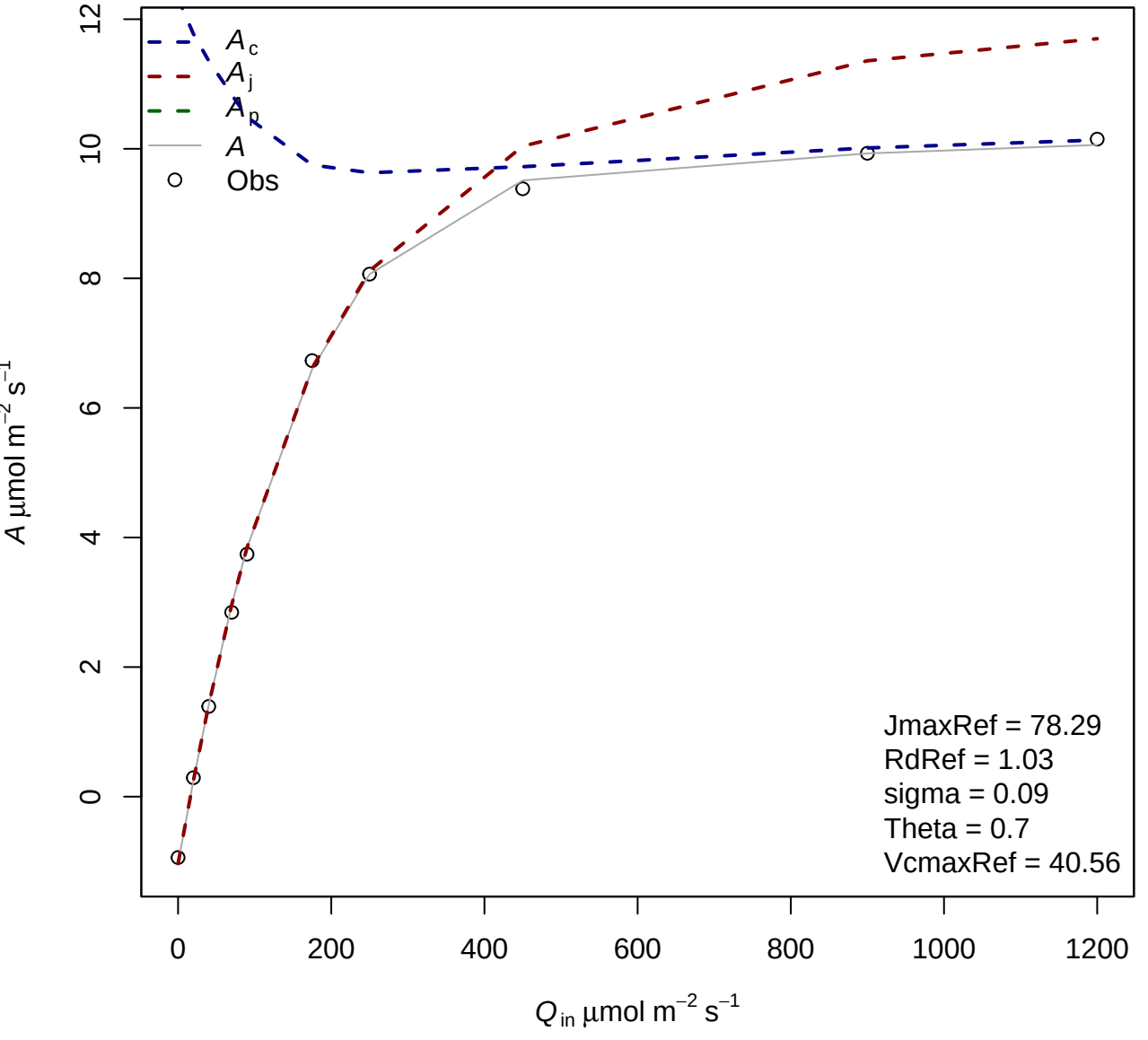

A_c
A_j
A_p
A
Obs
JmaxRef = 66.73
RdRef = 3.63
sigma = 0.54
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
200
400
600
800
1000

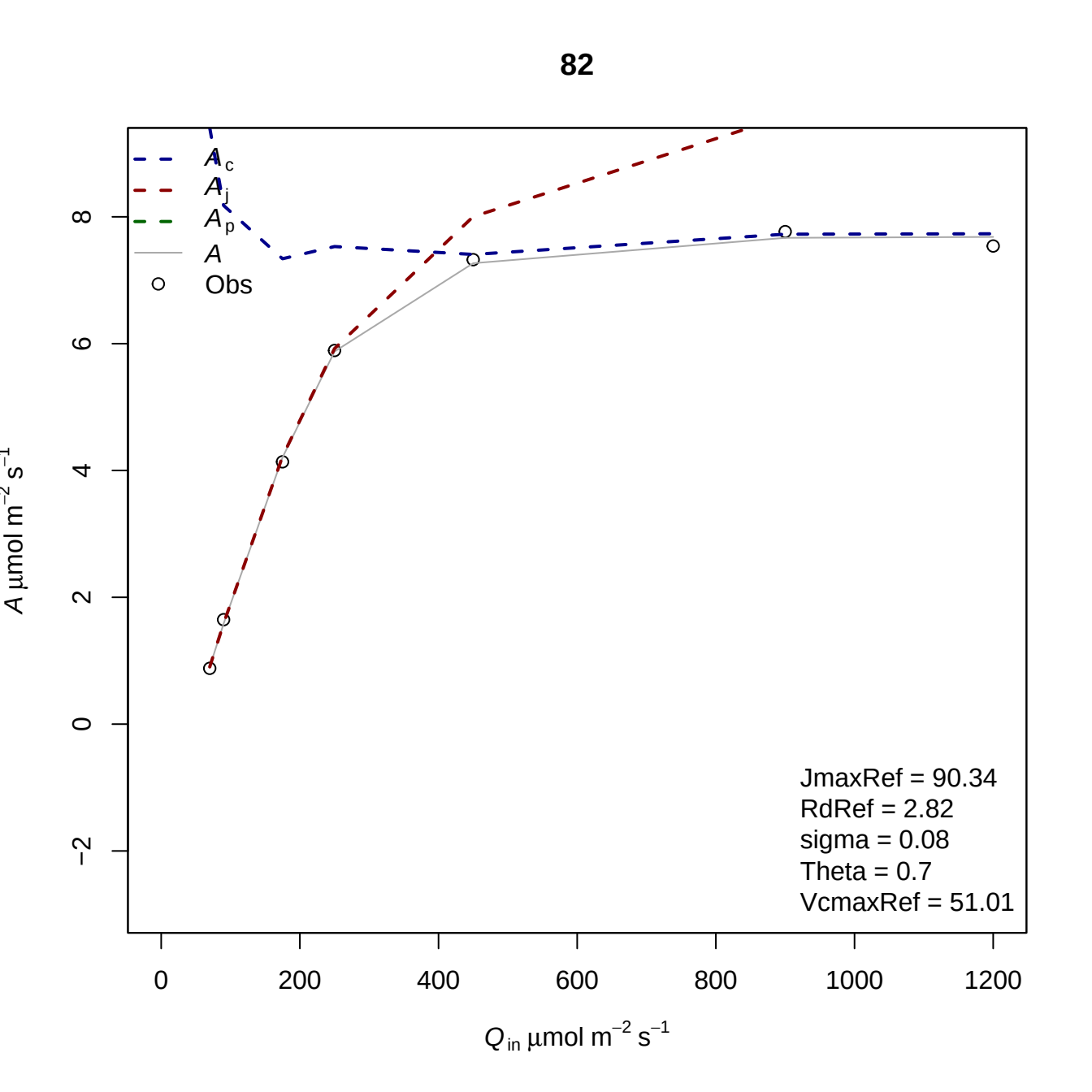

A_c
A_j
A_p
A
Obs
JmaxRef = 40.41
RdRef = 2.17
sigma = 0.19
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
200
400
600
800
1000

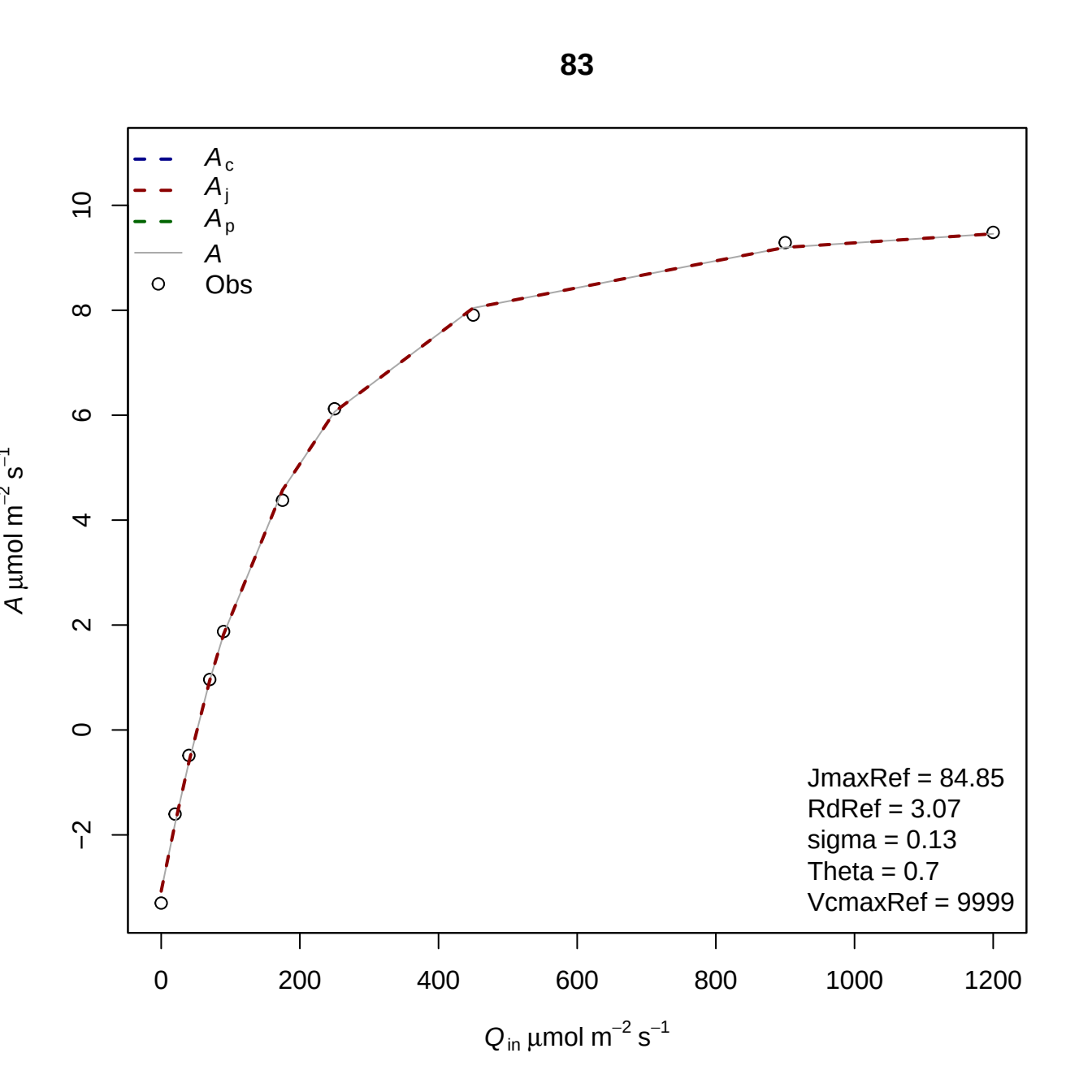

A_c
A_j
A_p
A
Obs
JmaxRef = 54.5
RdRef = 2.5
sigma = 0.21
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

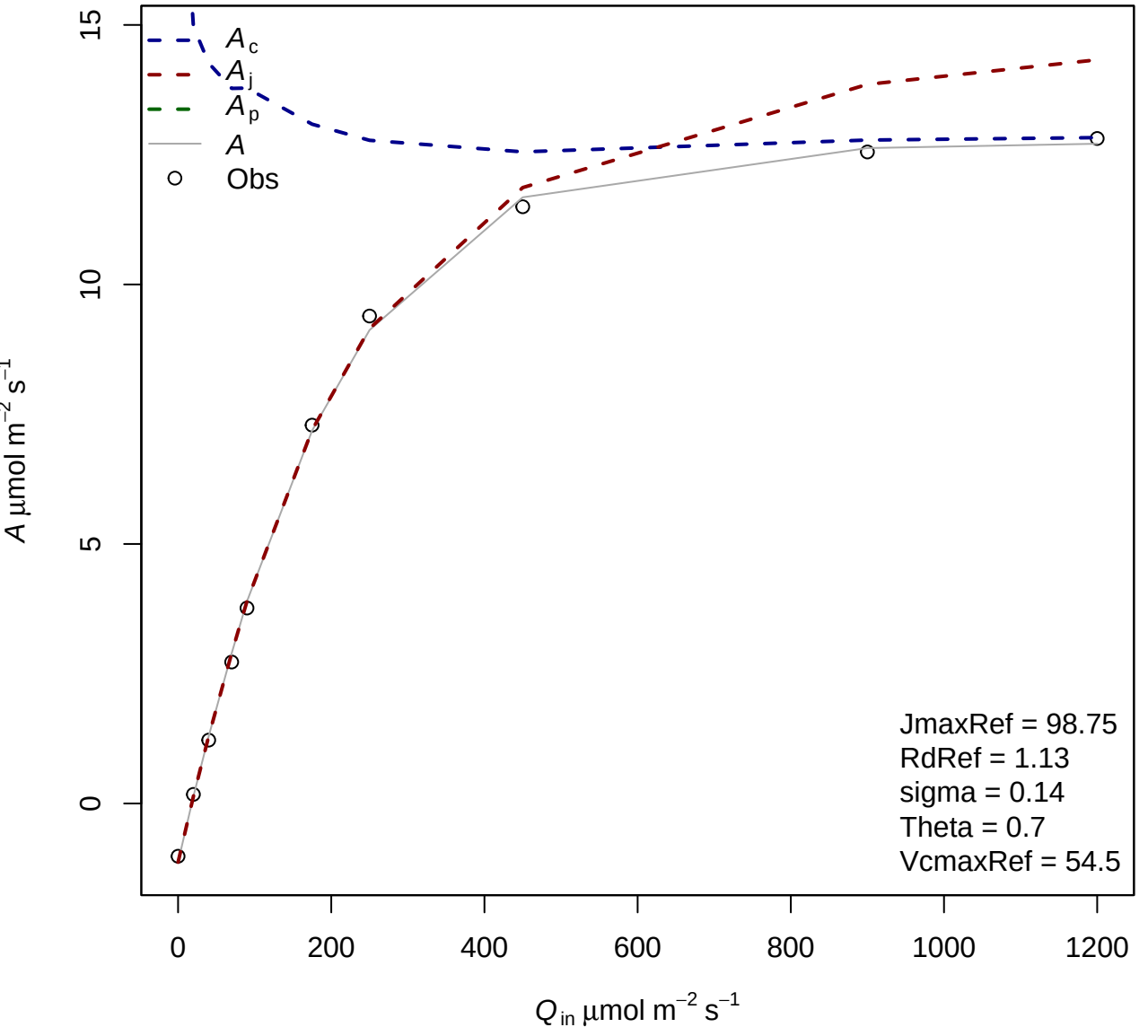

A_c
A_j
A_p
A
Obs
JmaxRef = 71.58
RdRef = 2.41
sigma = 0.32
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

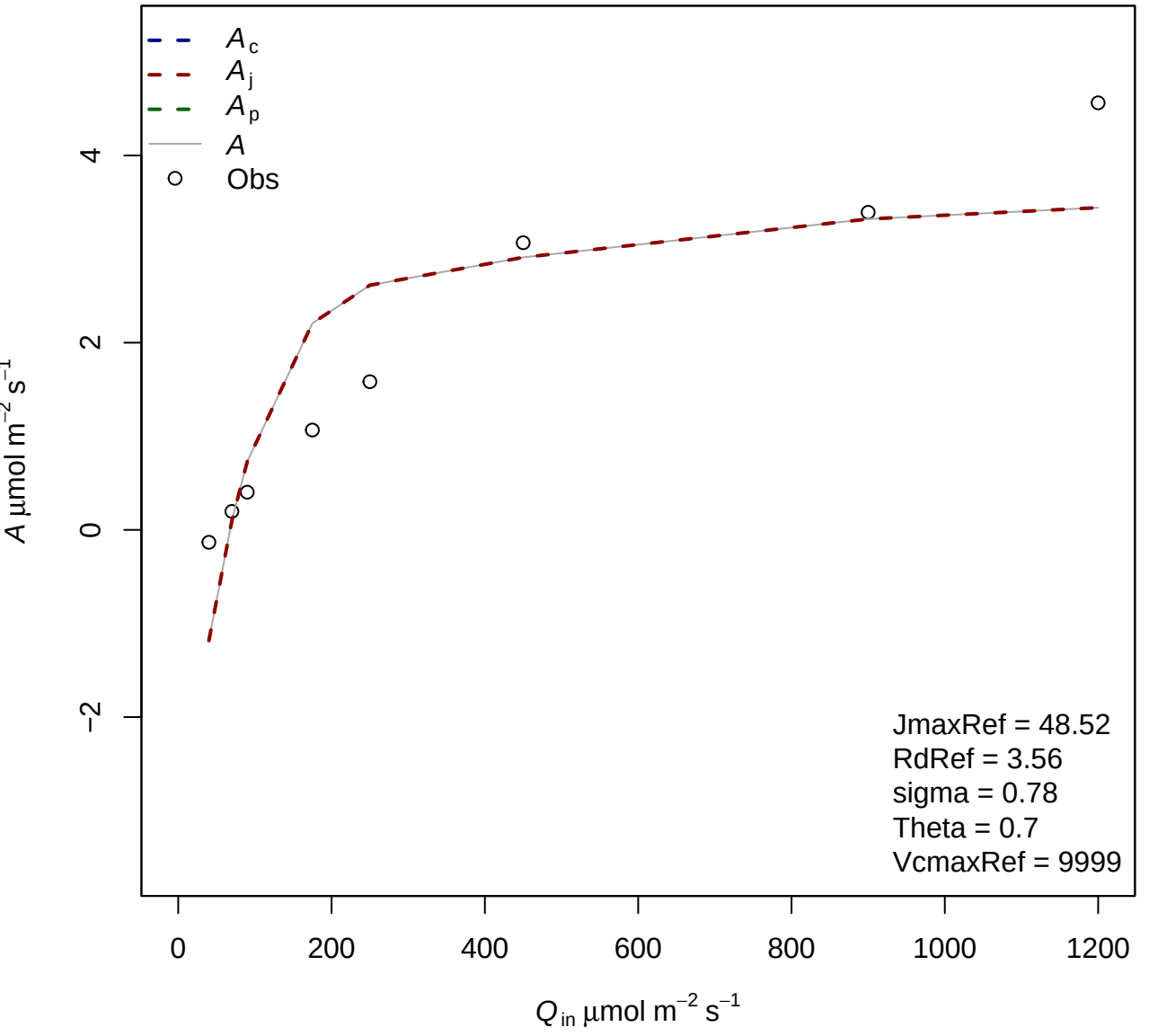

A_c
A_j
A_p
A
Obs
JmaxRef = 70.56
RdRef = 1.2
sigma = 0.14
Theta = 0.7
VcmaxRef = 34.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
200
400
600
800
1000
1200

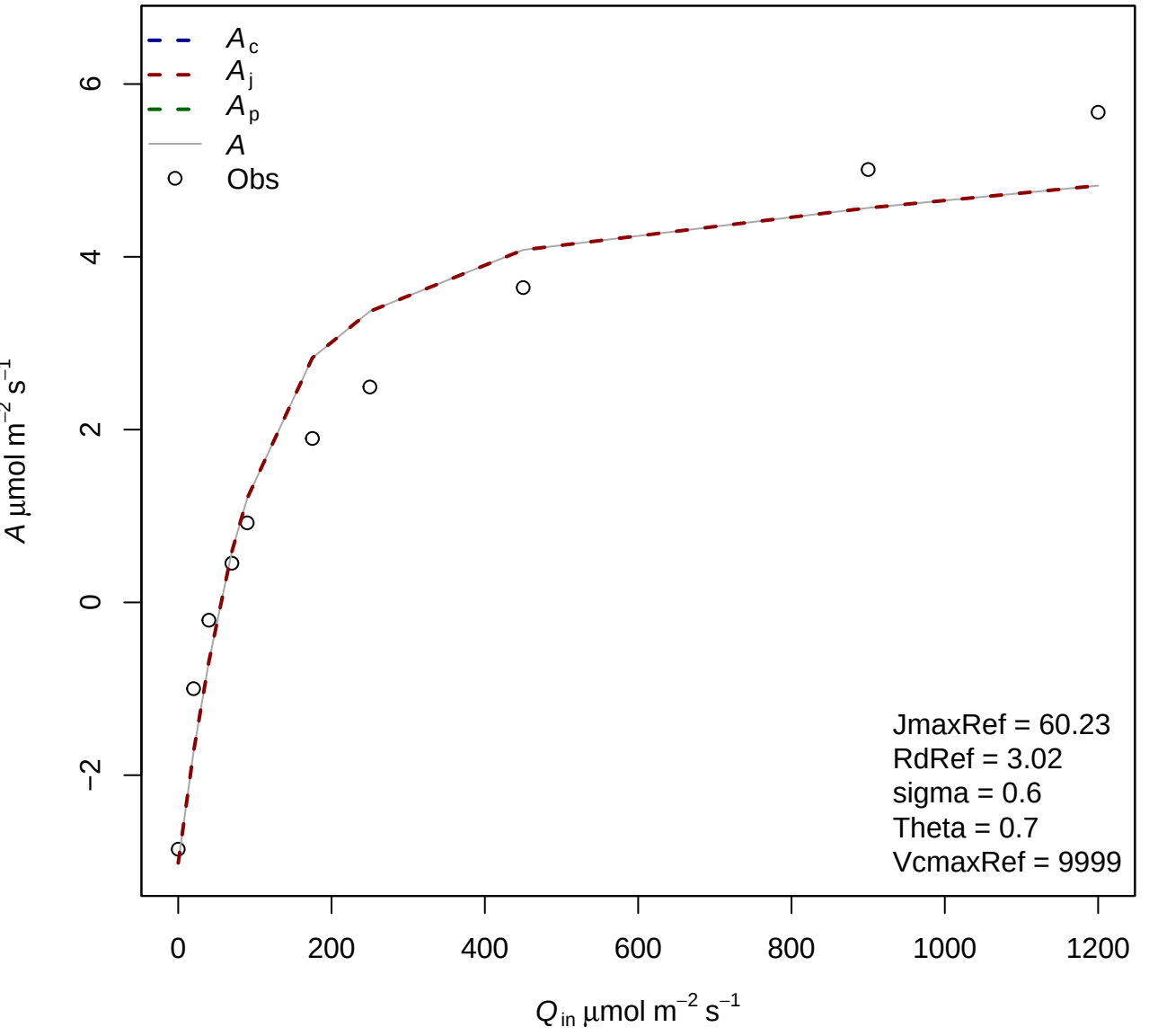

A_c
A_j
A_p
A
Obs
JmaxRef = 78.29
RdRef = 1.03
sigma = 0.09
Theta = 0.7
VcmaxRef = 40.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200

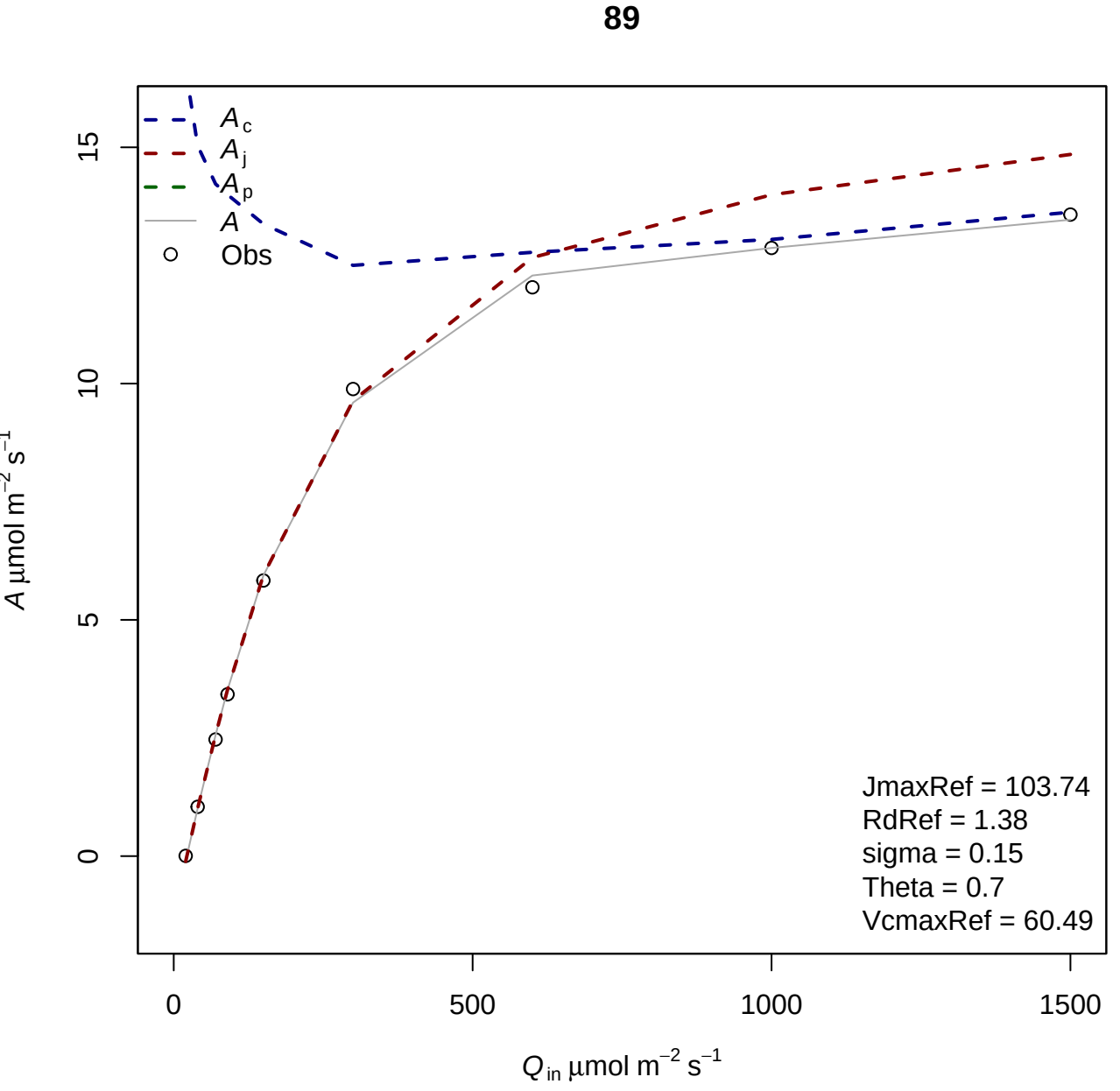

82
A_c
A_j
A_p
A
Obs
JmaxRef = 90.34
RdRef = 2.82
sigma = 0.08
Theta = 0.7
VcmaxRef = 51.01
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
200
400
600
800
1000
1200

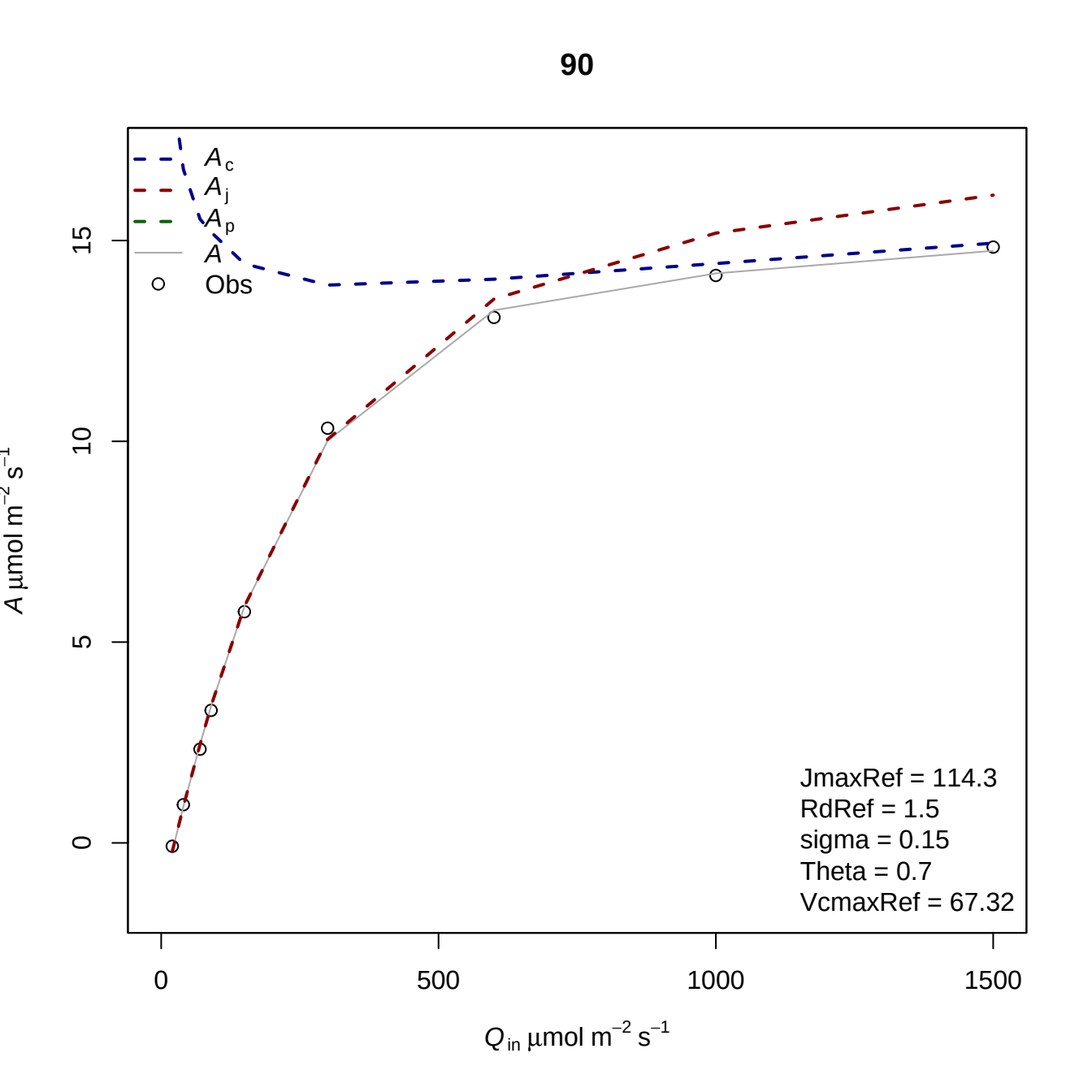

83
A_c
A_j
A_p
A
Obs
JmaxRef = 84.85
RdRef = 3.07
sigma = 0.13
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
200
400
600
800
1000
1200

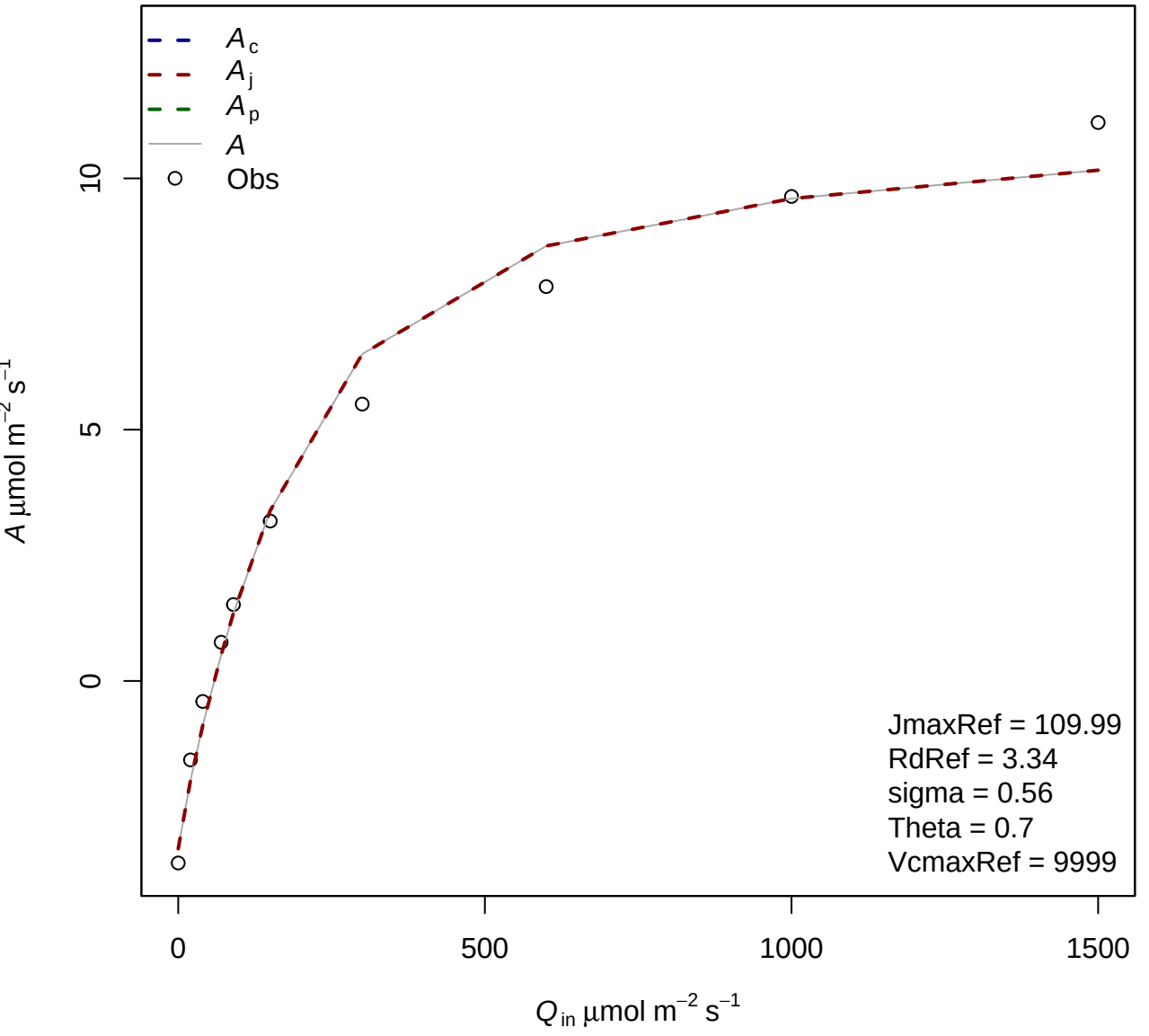

A_c
A_j
A_p
A
Obs
JmaxRef = 98.75
RdRef = 1.13
sigma = 0.14
Theta = 0.7
VcmaxRef = 54.5
0
5
10
15
A µmol m^{-2} s^{-1}
0
200
400
600
800
1000
1200
Q_{in} µmol m^{-2} s^{-1}

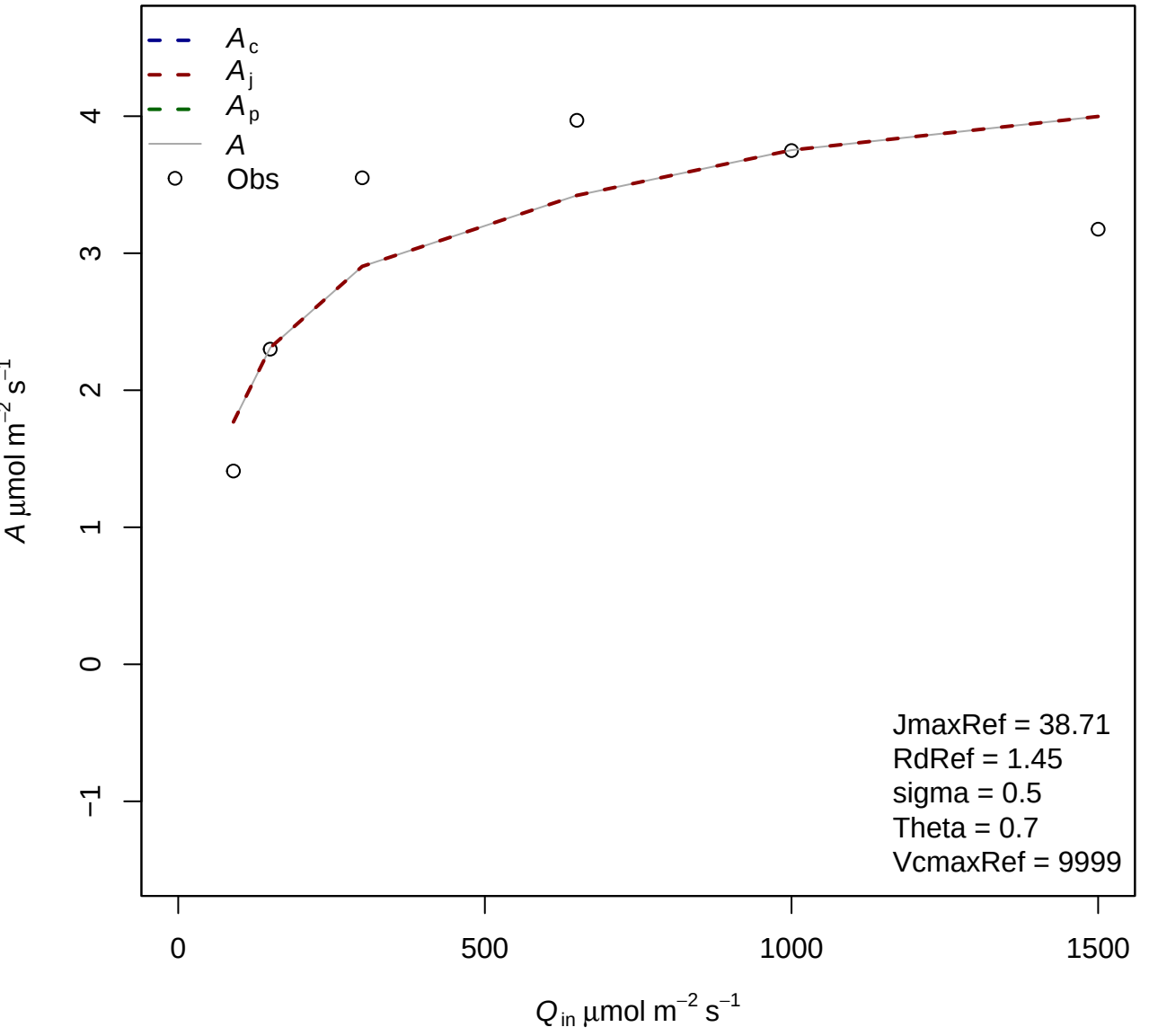

A_c
A_j
A_p
A
Obs
JmaxRef = 48.52
RdRef = 3.56
sigma = 0.78
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
200
400
600
800
1000
1200

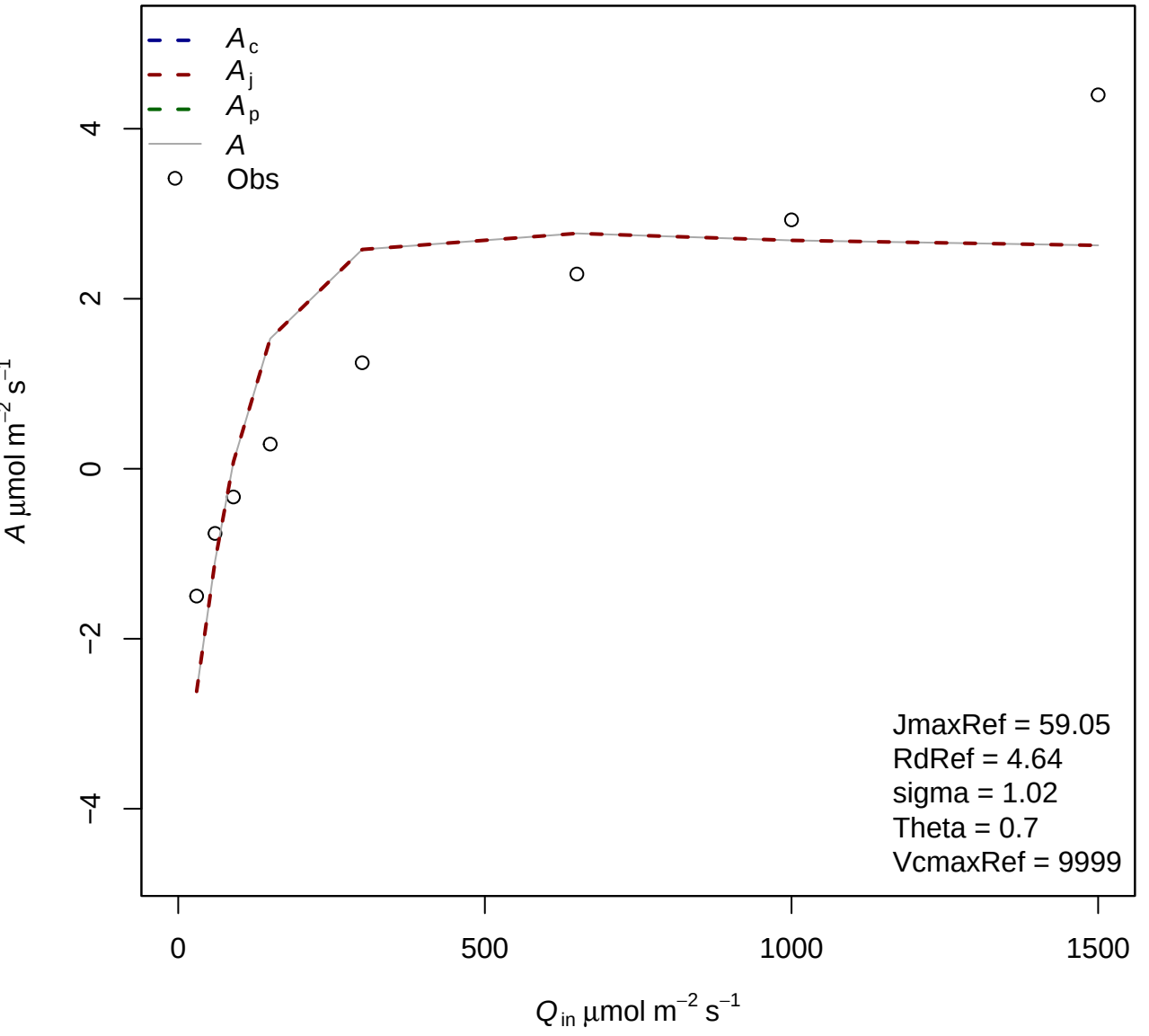

A_c
A_j
A_p
A
Obs
JmaxRef = 60.23
RdRef = 3.02
sigma = 0.6
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000
1200

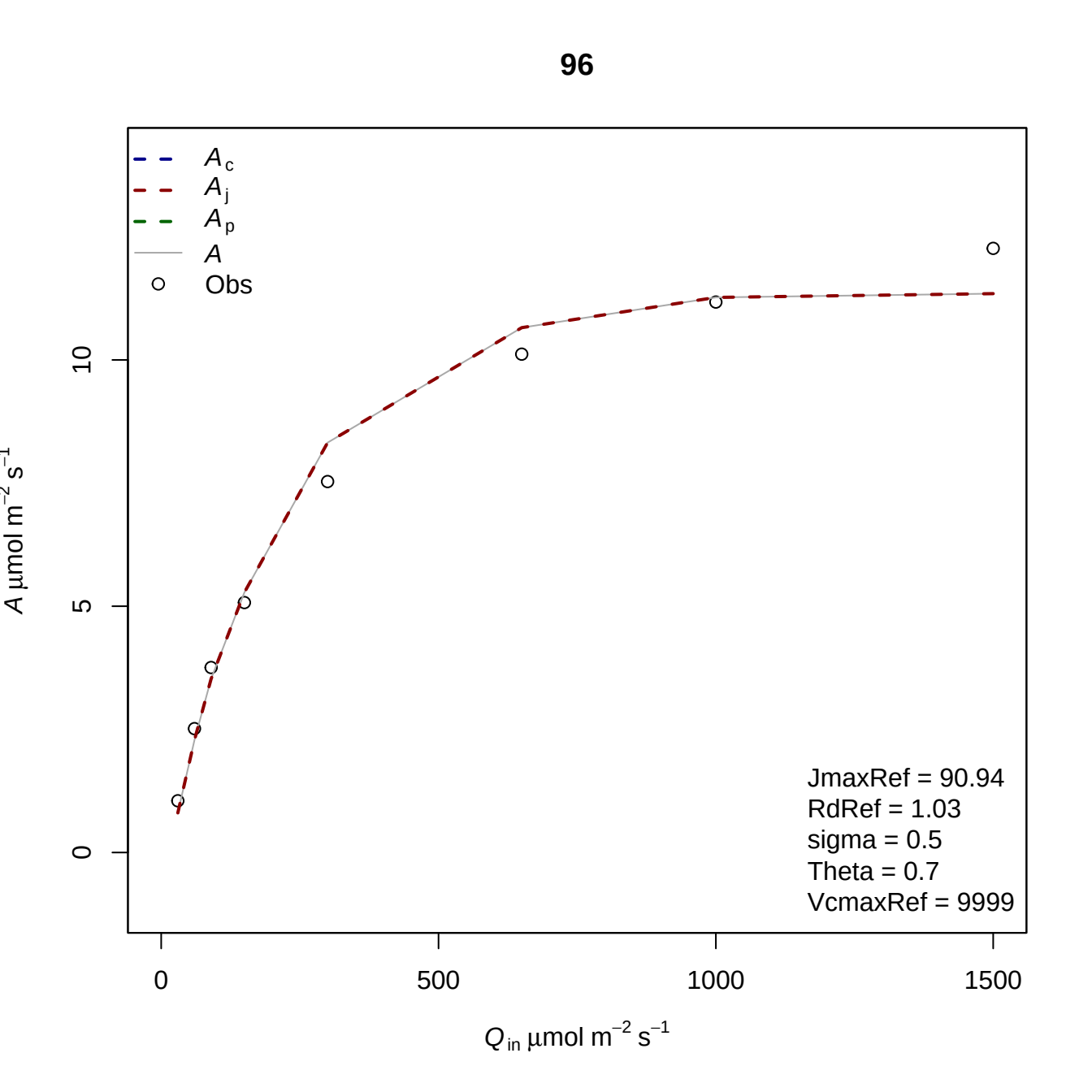

89
A_c
A_j
A_p
A
Obs
JmaxRef = 103.74
RdRef = 1.38
sigma = 0.15
Theta = 0.7
VcmaxRef = 60.49
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

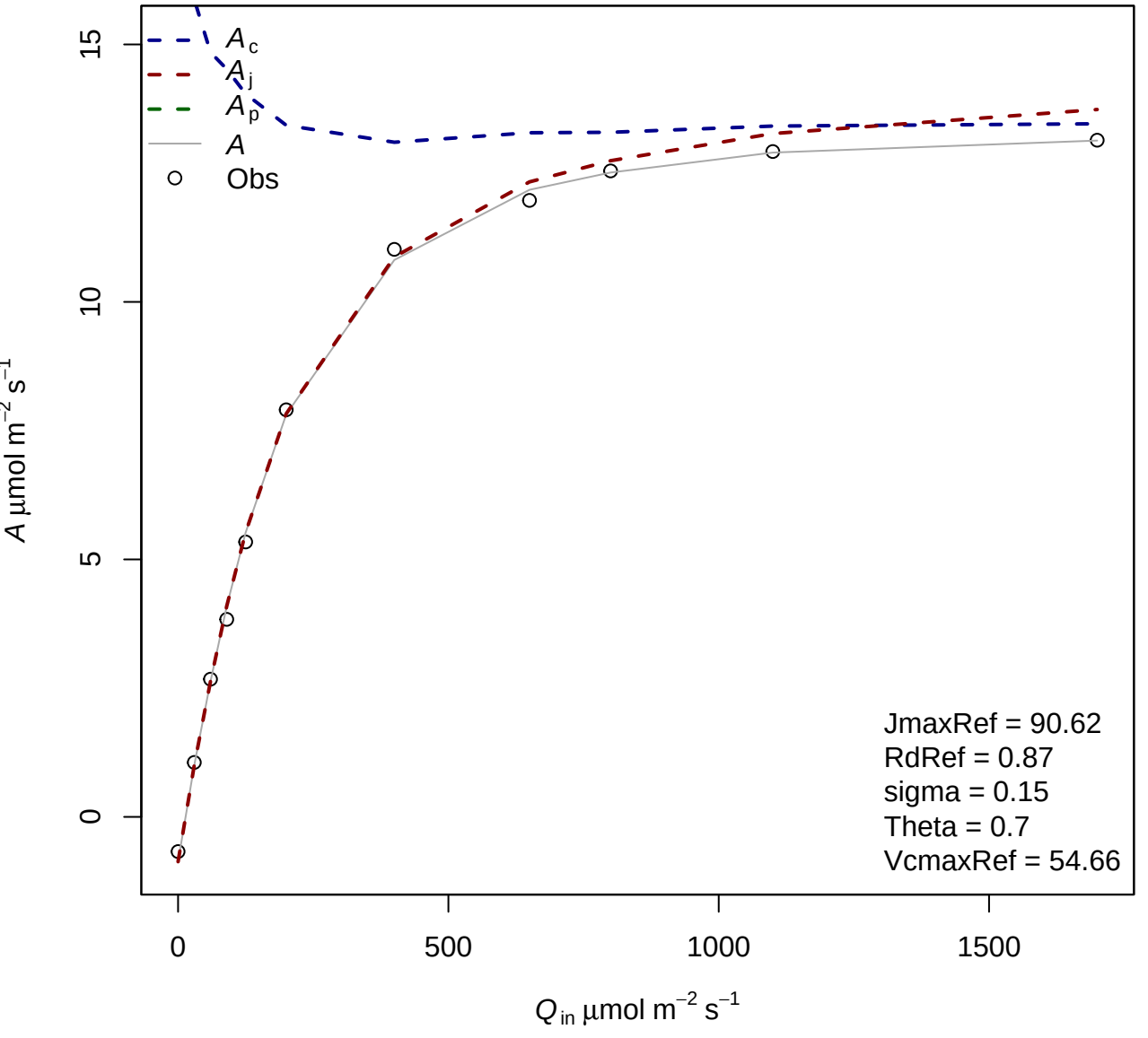

90
A_c
A_j
A_p
A
Obs
JmaxRef = 114.3
RdRef = 1.5
sigma = 0.15
Theta = 0.7
VcmaxRef = 67.32
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

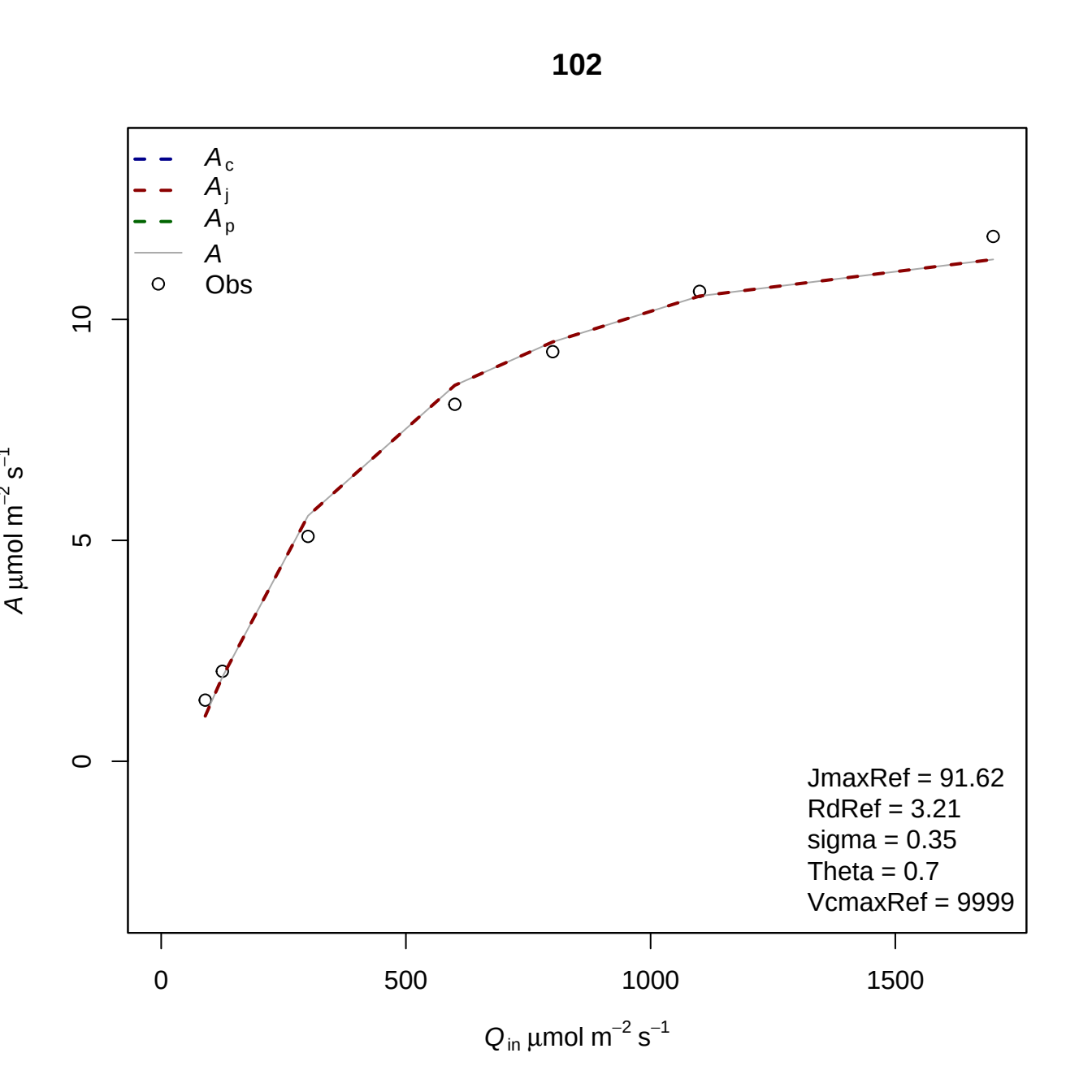

A_c
A_j
A_p
A
Obs
JmaxRef = 109.99
RdRef = 3.34
sigma = 0.56
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

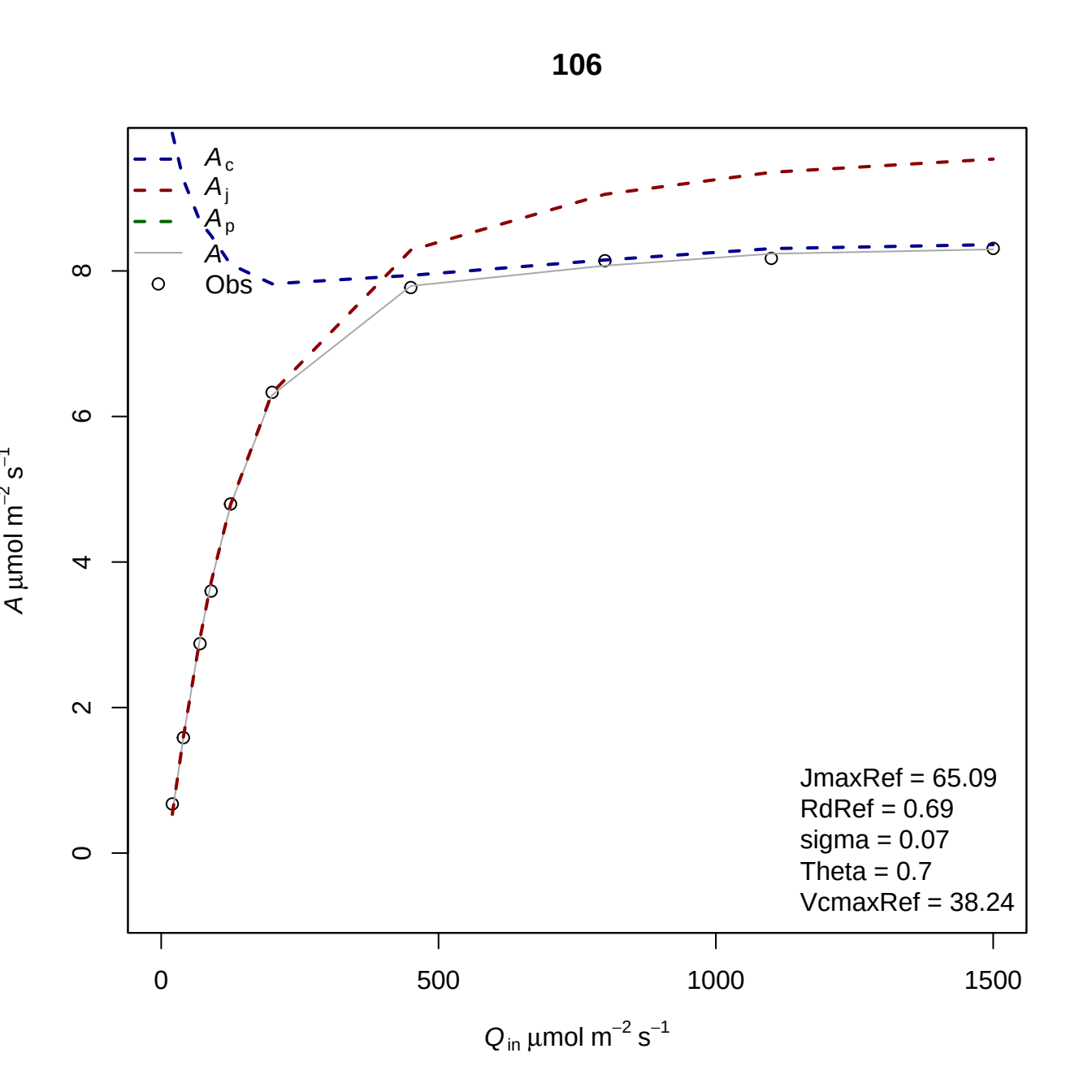

A_c
A_j
A_p
A
Obs
JmaxRef = 38.71
RdRef = 1.45
sigma = 0.5
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

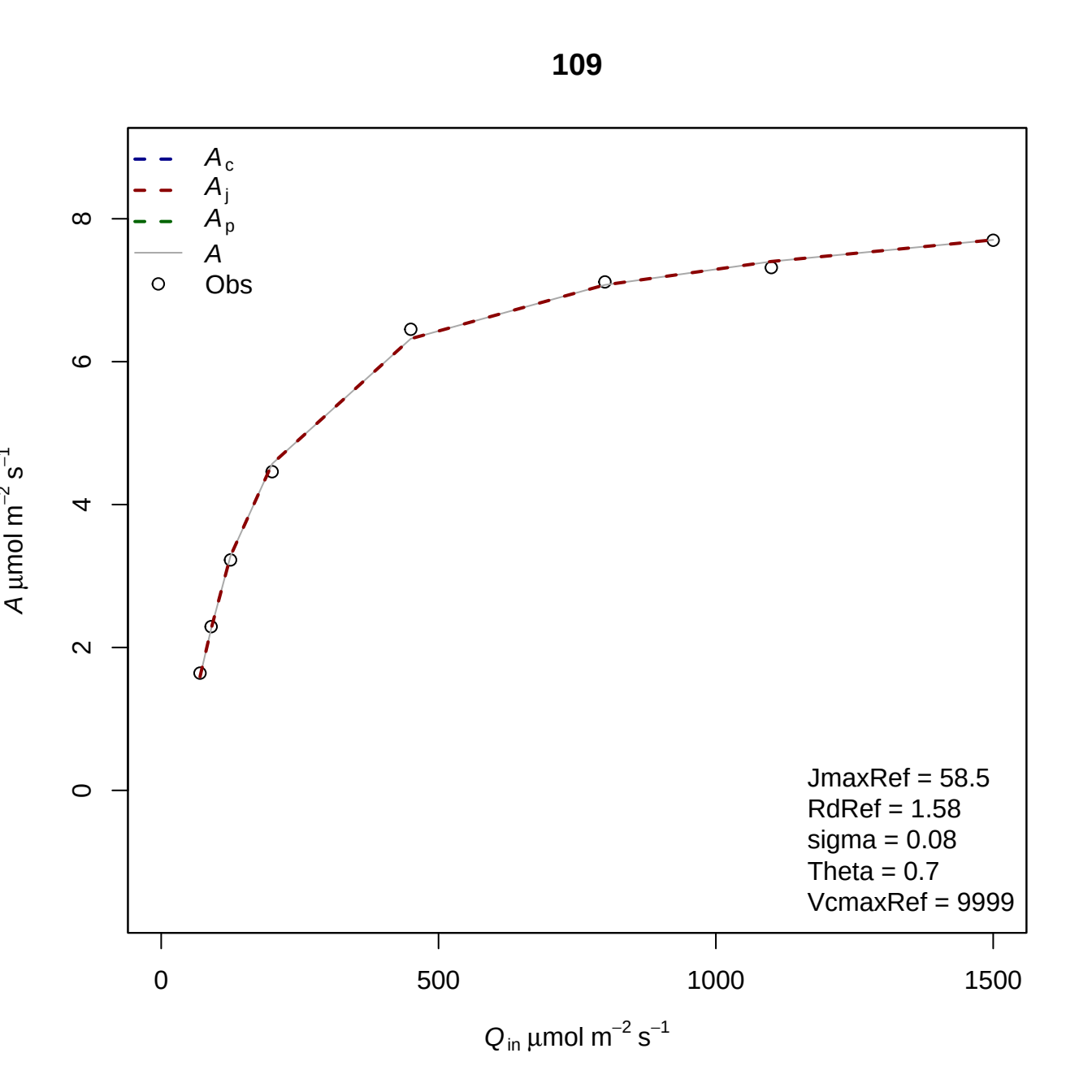

A_c
A_j
A_p
A
Obs
JmaxRef = 59.05
RdRef = 4.64
sigma = 1.02
Theta = 0.7
VcmaxRef = 9999
4
2
0
−2
−4
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

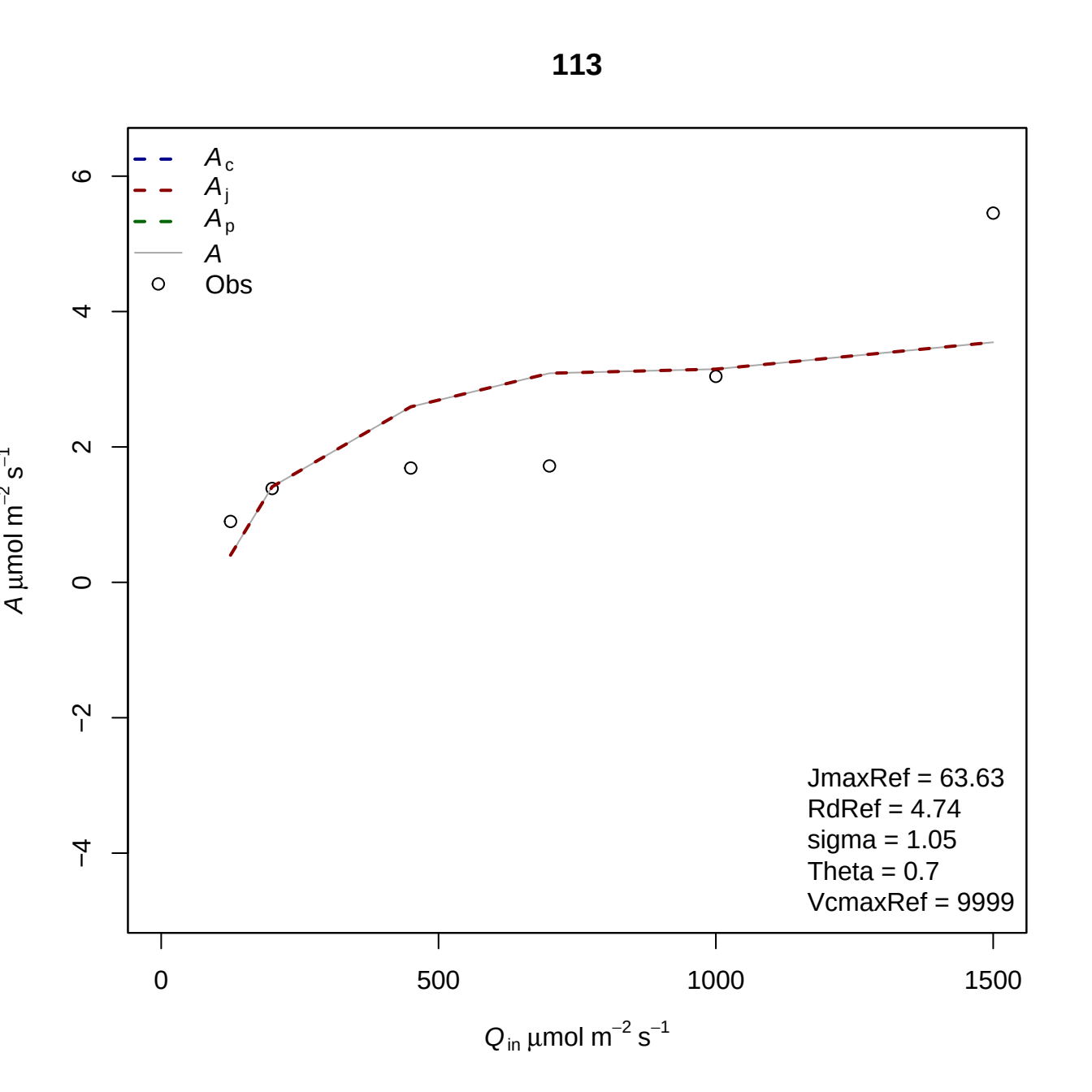

96
A_c
A_j
A_p
A
Obs
JmaxRef = 90.94
RdRef = 1.03
sigma = 0.5
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

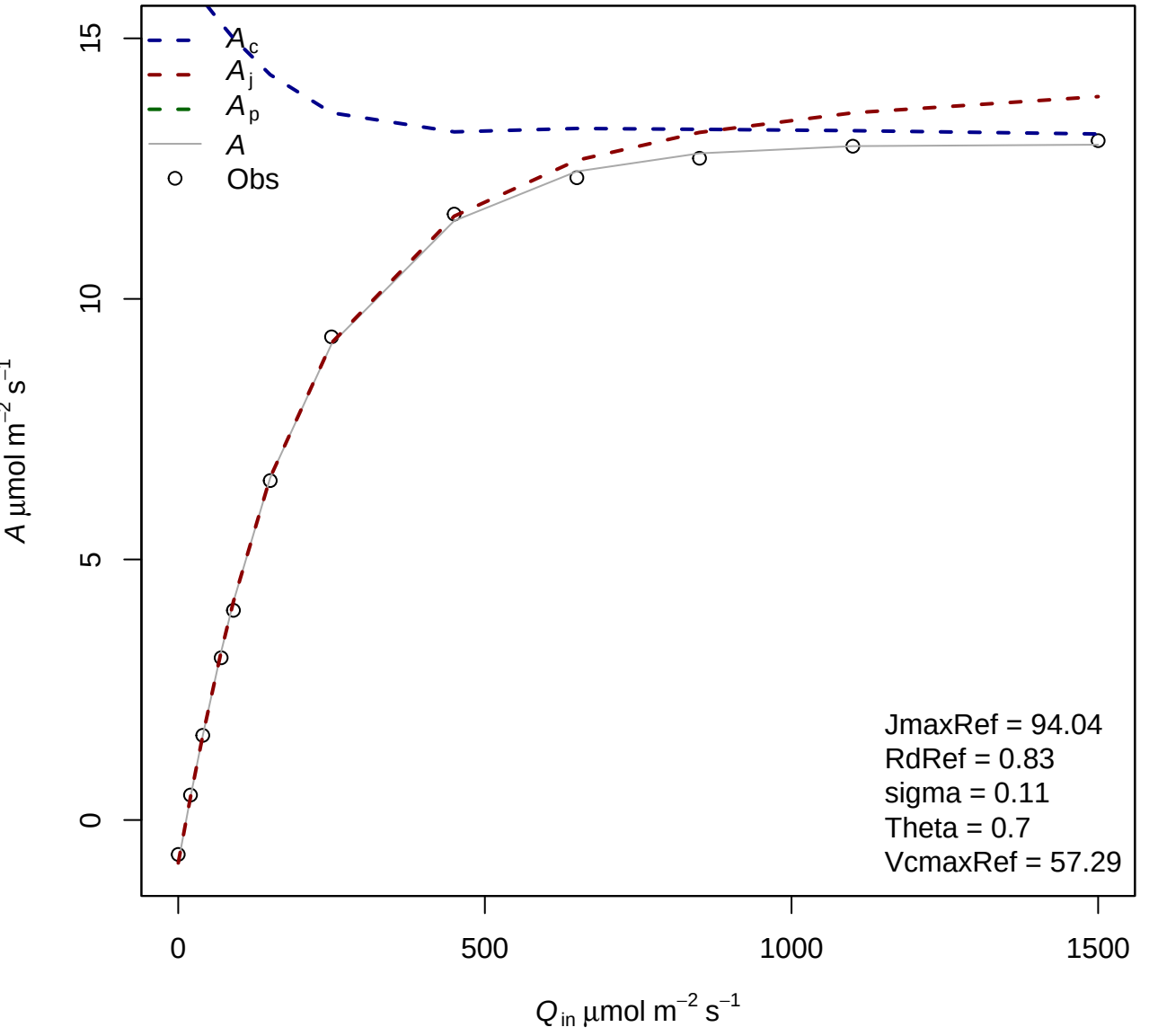

A_c
A_j
A_p
A
Obs
JmaxRef = 90.62
RdRef = 0.87
sigma = 0.15
Theta = 0.7
VcmaxRef = 54.66
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

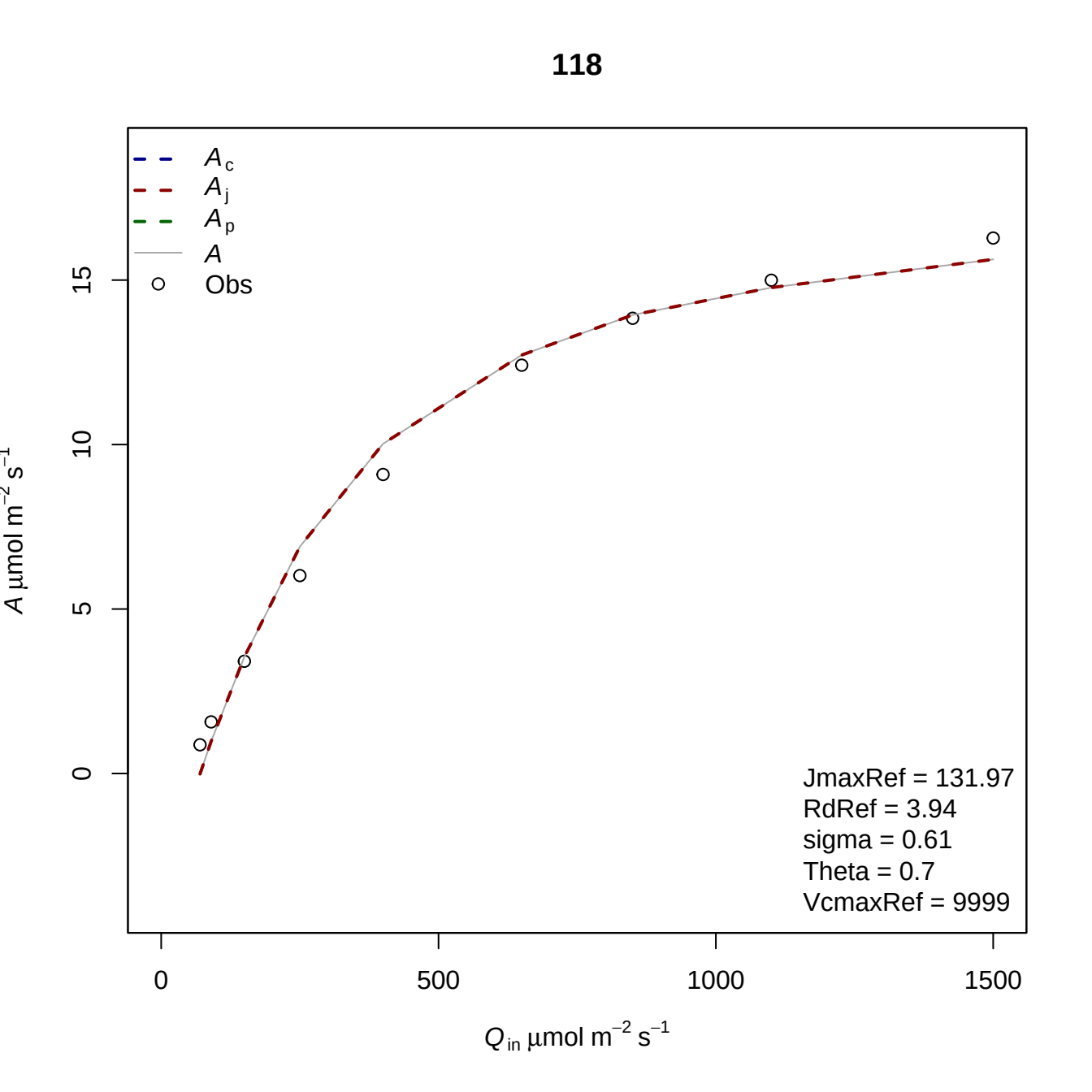

102
A_c
A_j
A_p
A
Obs
JmaxRef = 91.62
RdRef = 3.21
sigma = 0.35
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

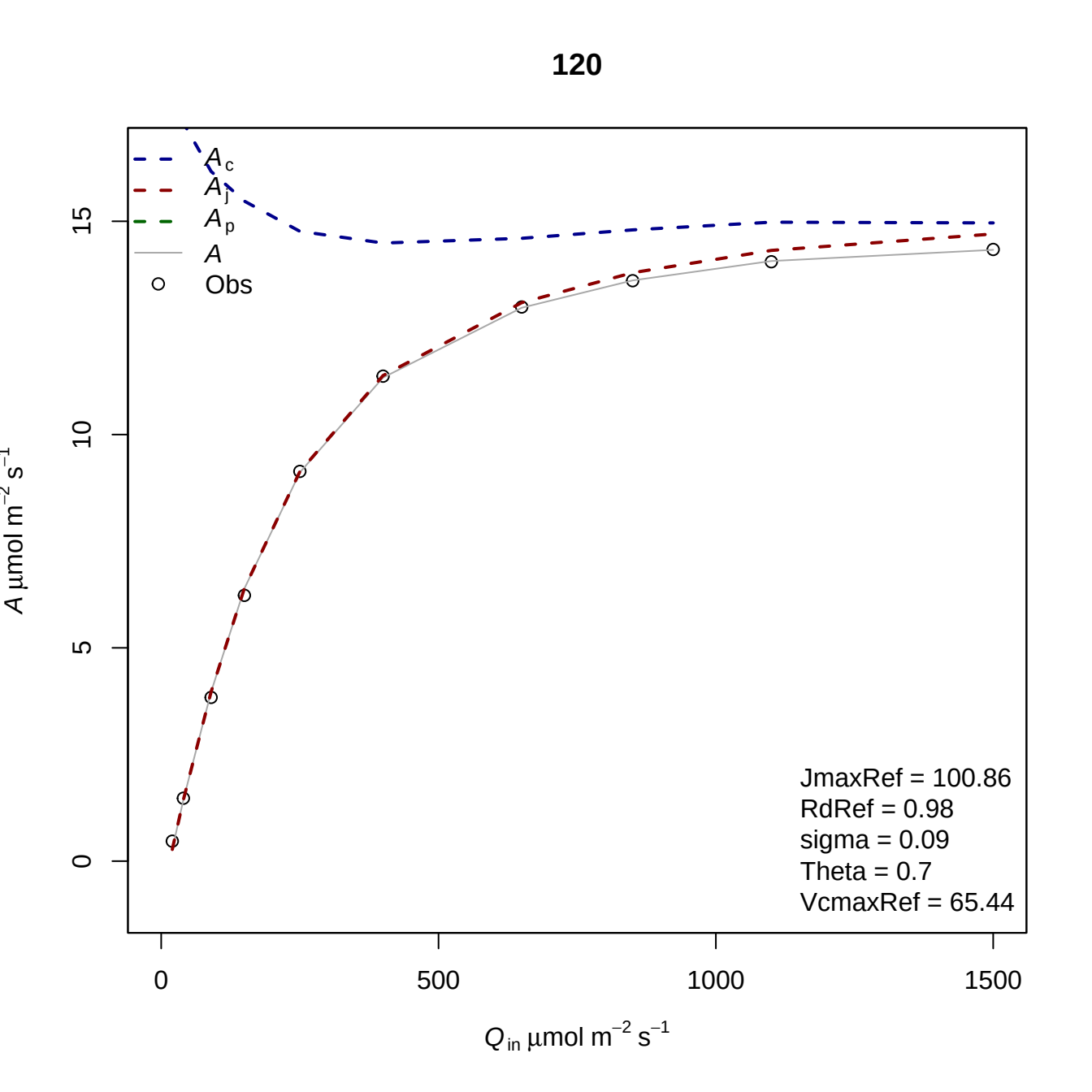

106
A_c
A_j
A_p
A
Obs
JmaxRef = 65.09
RdRef = 0.69
sigma = 0.07
Theta = 0.7
VcmaxRef = 38.24
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

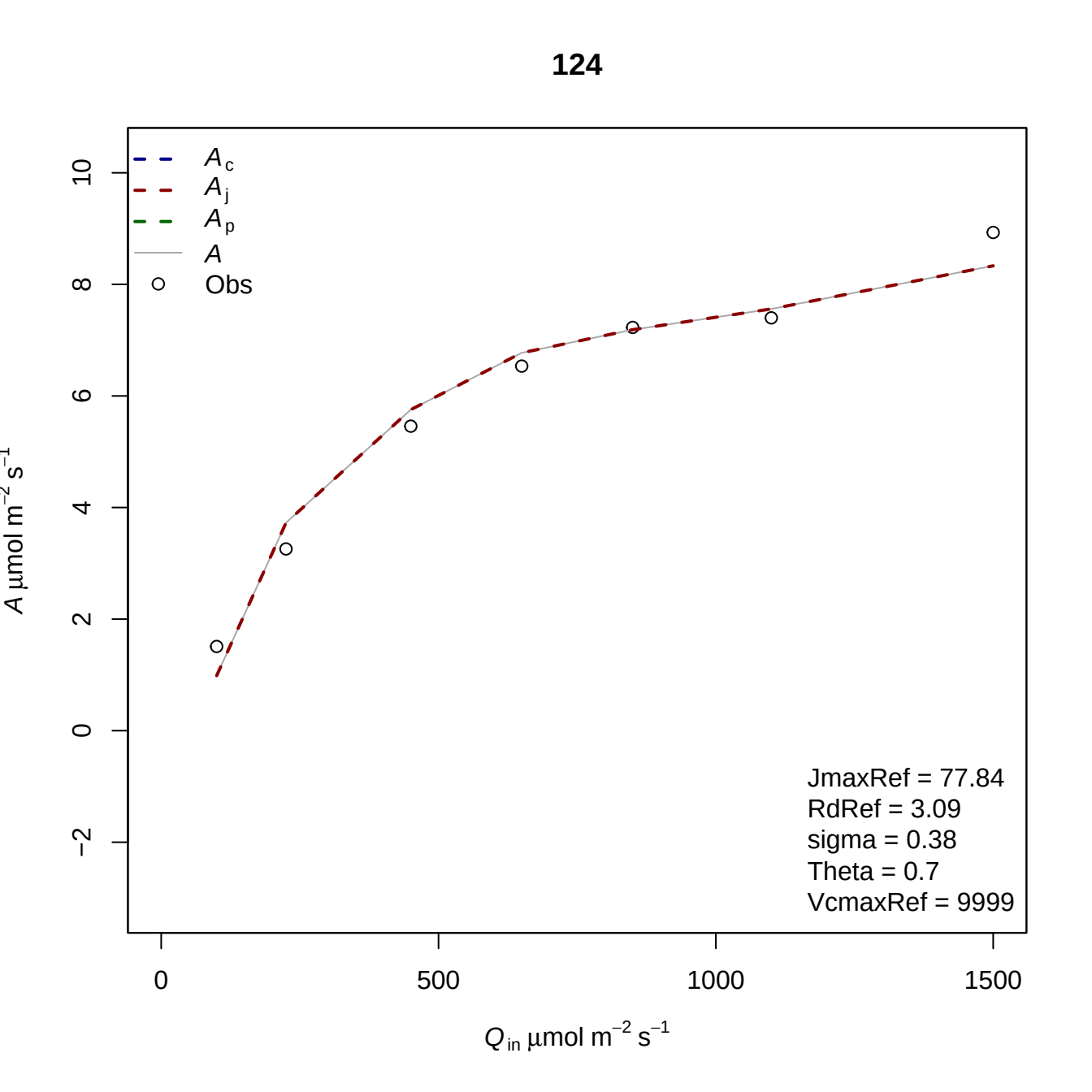

109
A_c
A_j
A_p
A
Obs
JmaxRef = 58.5
RdRef = 1.58
sigma = 0.08
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

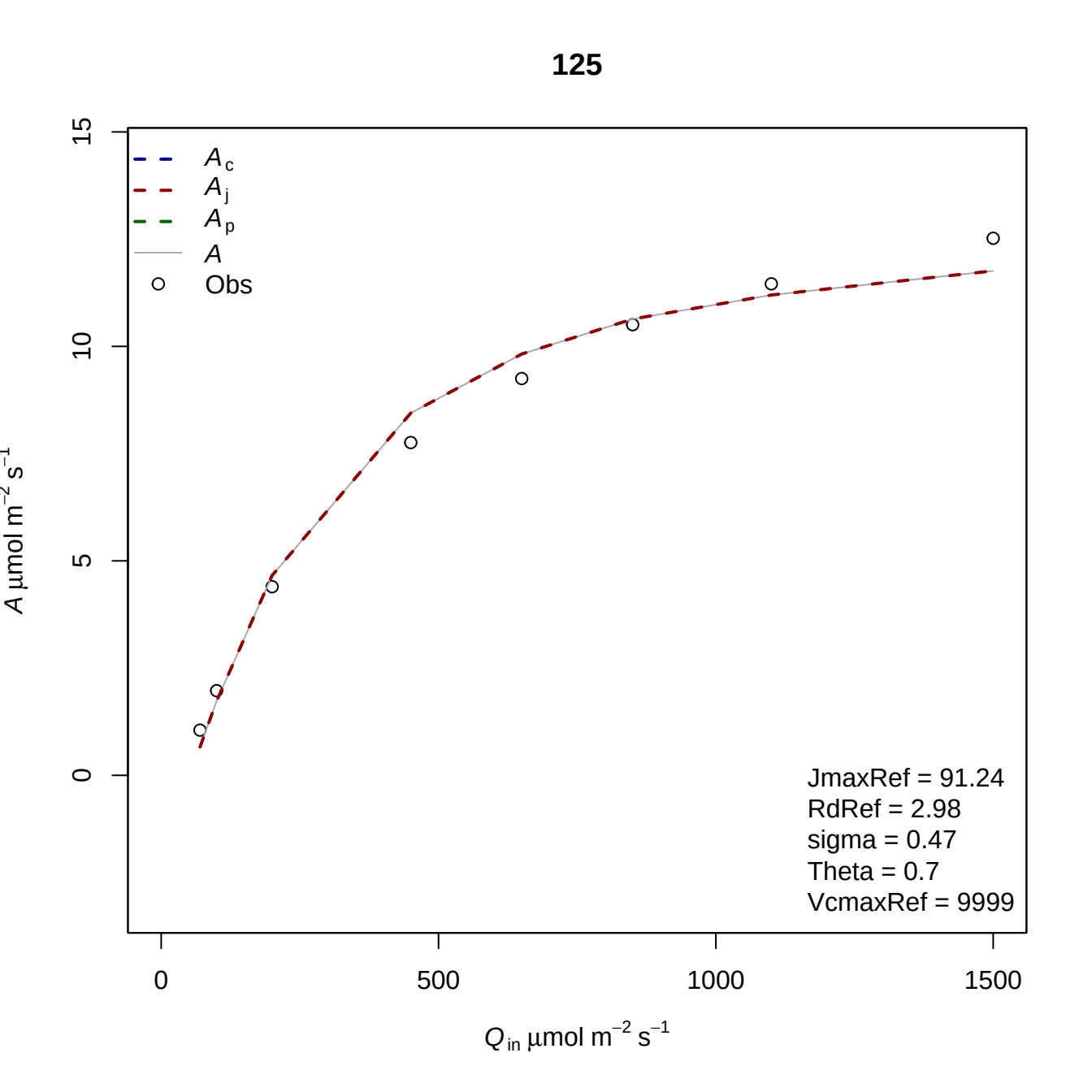

113
A_c
A_j
A_p
A
Obs
JmaxRef = 63.63
RdRef = 4.74
sigma = 1.05
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

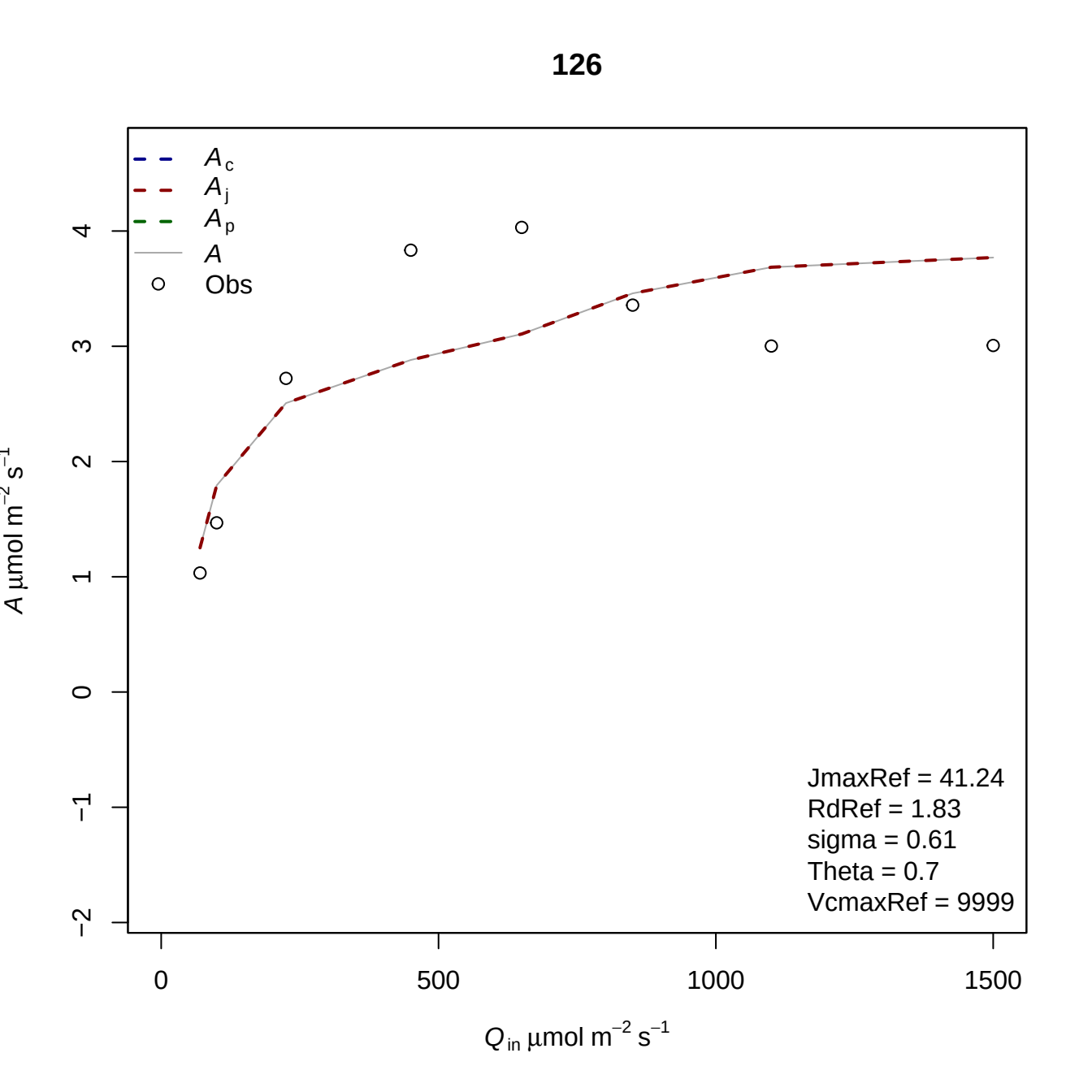

A_c
A_j
A_p
A
Obs
JmaxRef = 94.04
RdRef = 0.83
sigma = 0.11
Theta = 0.7
VcmaxRef = 57.29
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

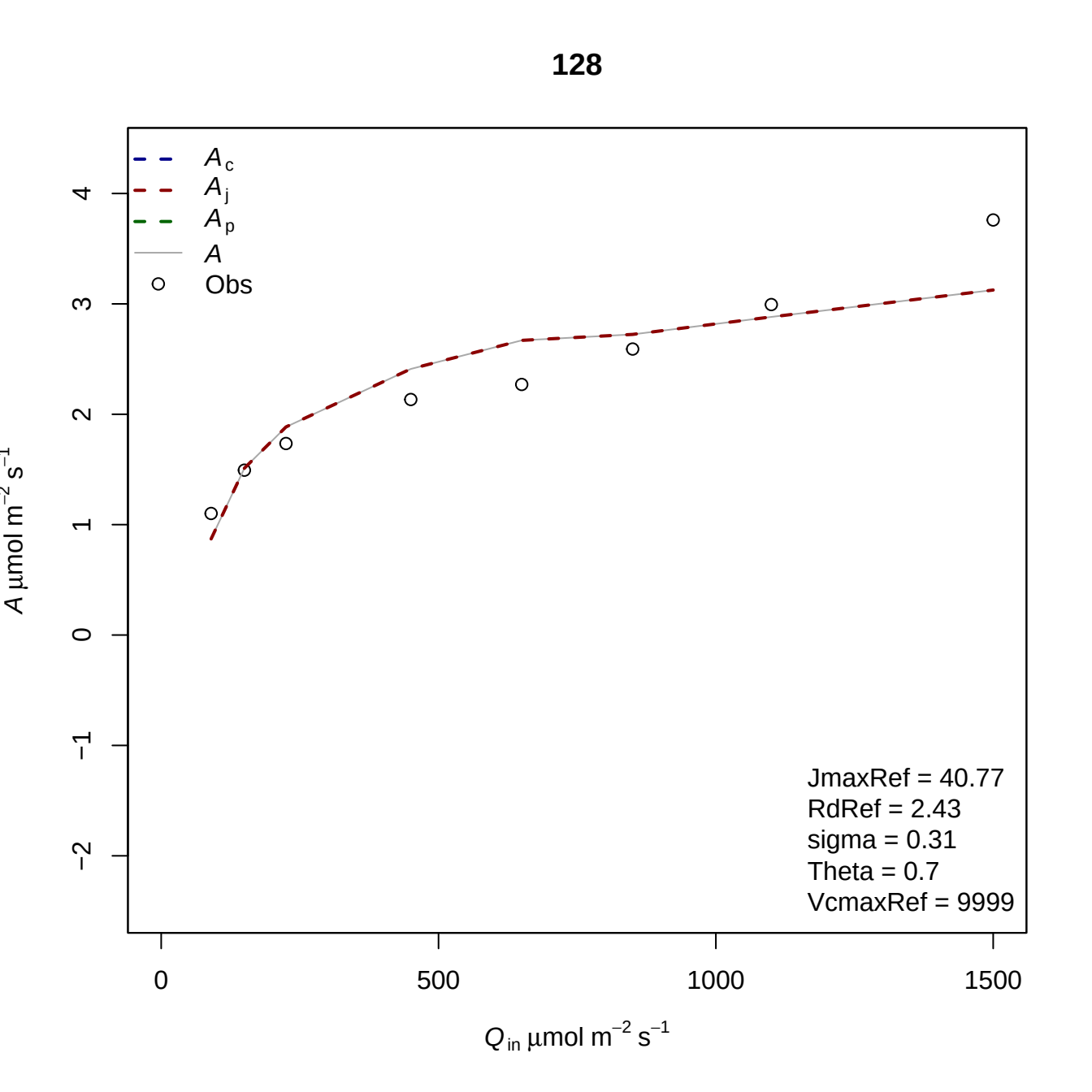

118
A_c
A_j
A_p
A
Obs
JmaxRef = 131.97
RdRef = 3.94
sigma = 0.61
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

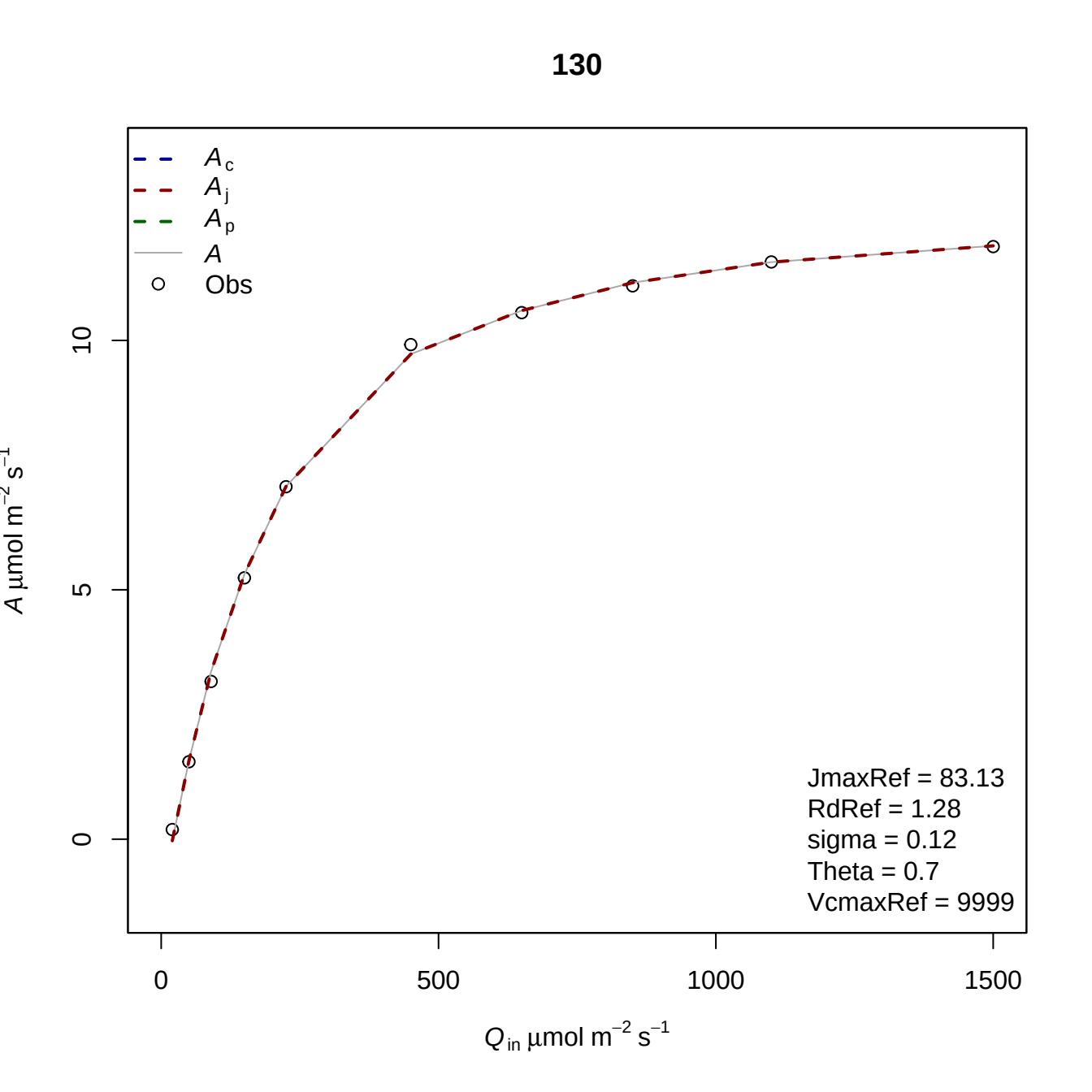

120
A_c
A_j
A_p
A
Obs
JmaxRef = 100.86
RdRef = 0.98
sigma = 0.09
Theta = 0.7
VcmaxRef = 65.44
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

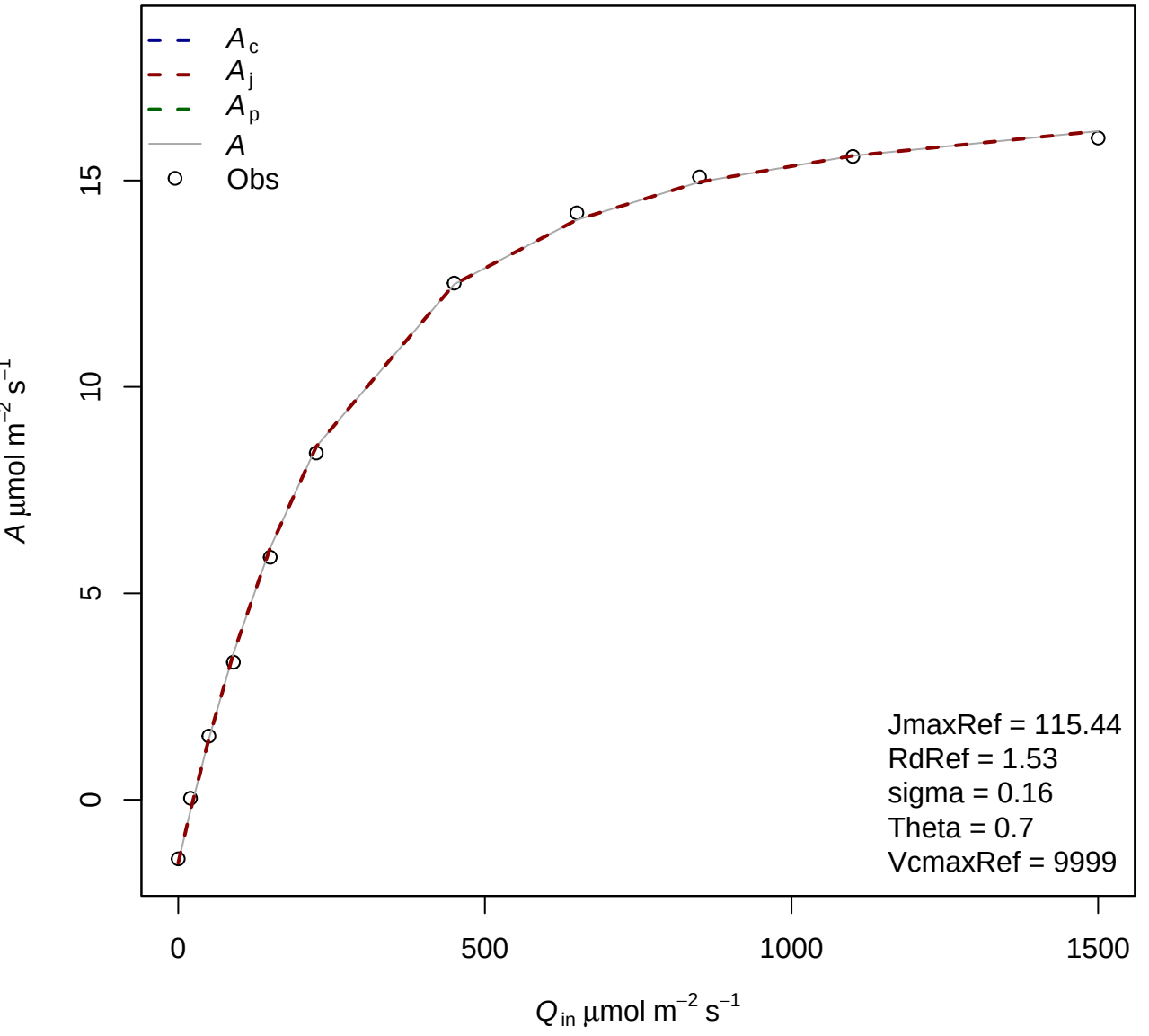

124
A_c
A_j
A_p
A
Obs
JmaxRef = 77.84
RdRef = 3.09
sigma = 0.38
Theta = 0.7
VcmaxRef = 9999
10
8
6
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

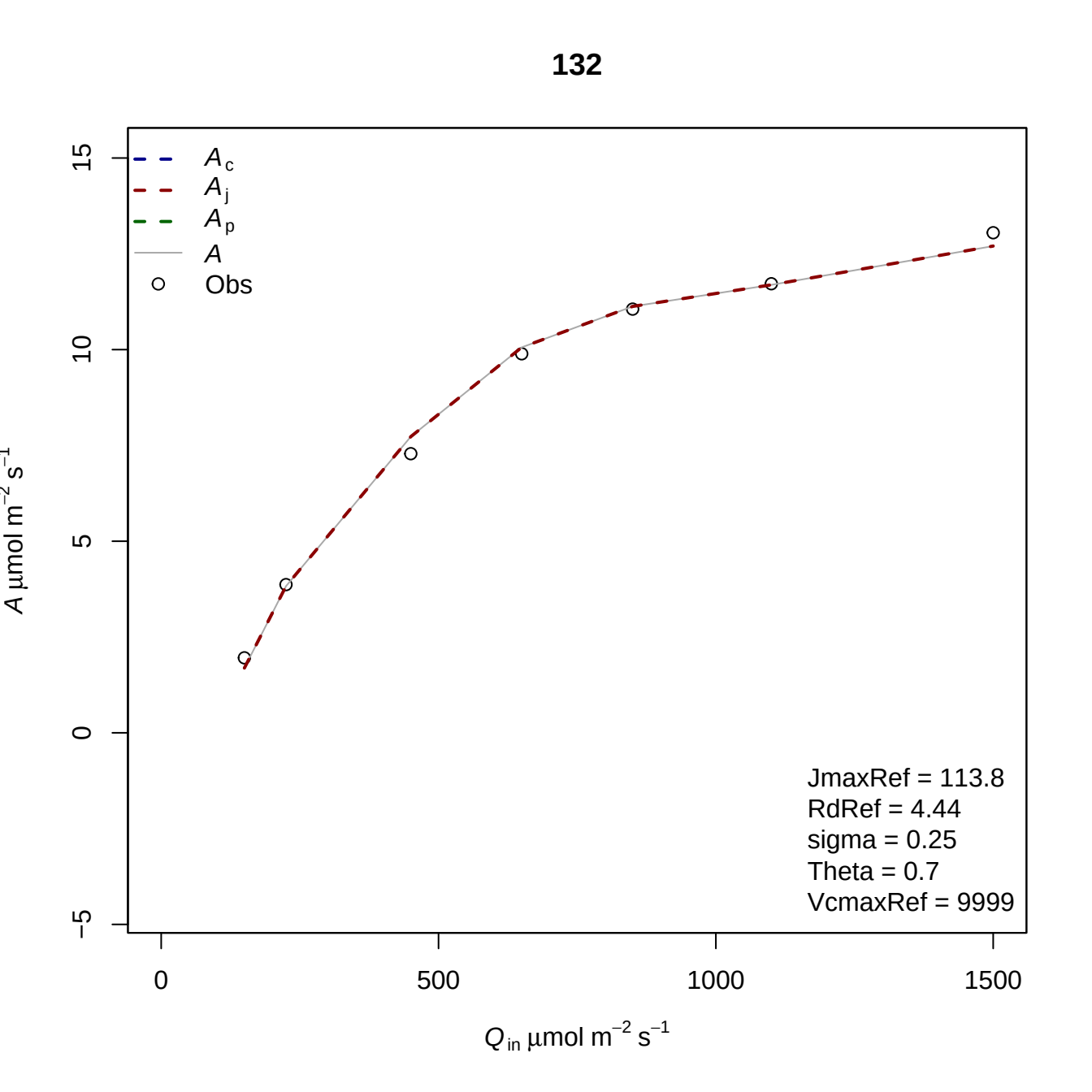

125
A_c
A_j
A_p
A
Obs
JmaxRef = 91.24
RdRef = 2.98
sigma = 0.47
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

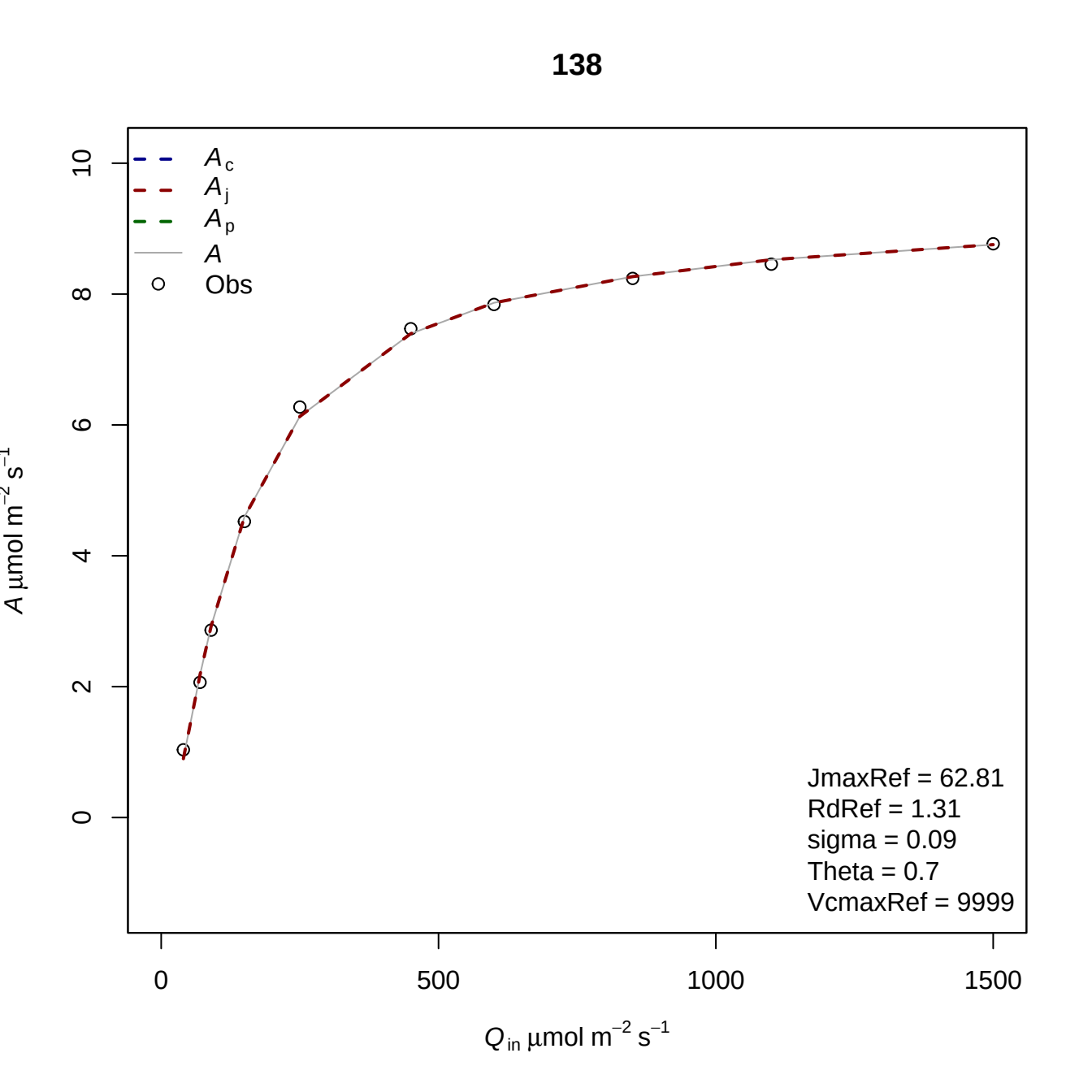

126
A_c
A_j
A_p
A
Obs
JmaxRef = 41.24
RdRef = 1.83
sigma = 0.61
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
4
0
500
1000
1500

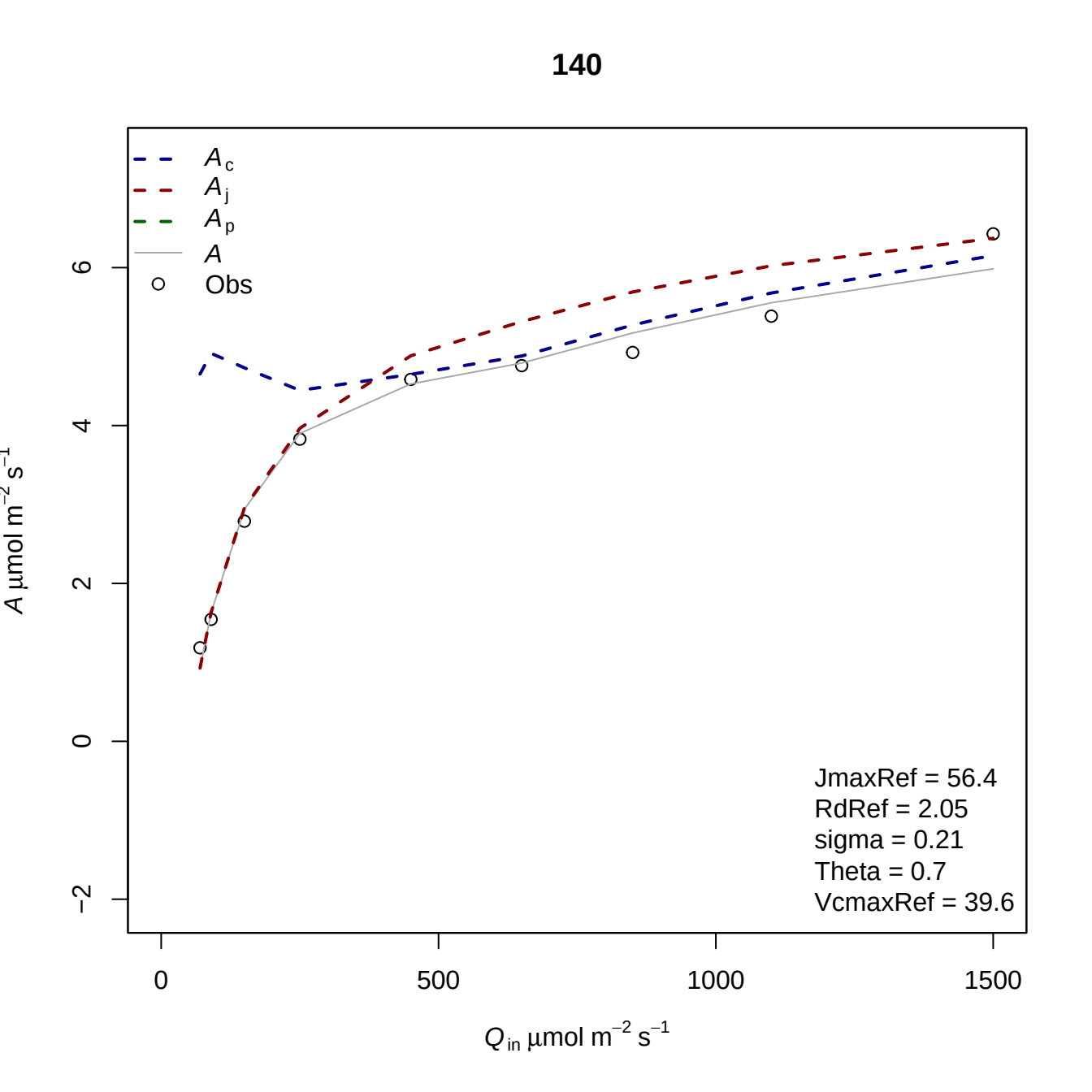

128
A_c
A_j
A_p
A
Obs
JmaxRef = 40.77
RdRef = 2.43
sigma = 0.31
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
0
500
1000
1500

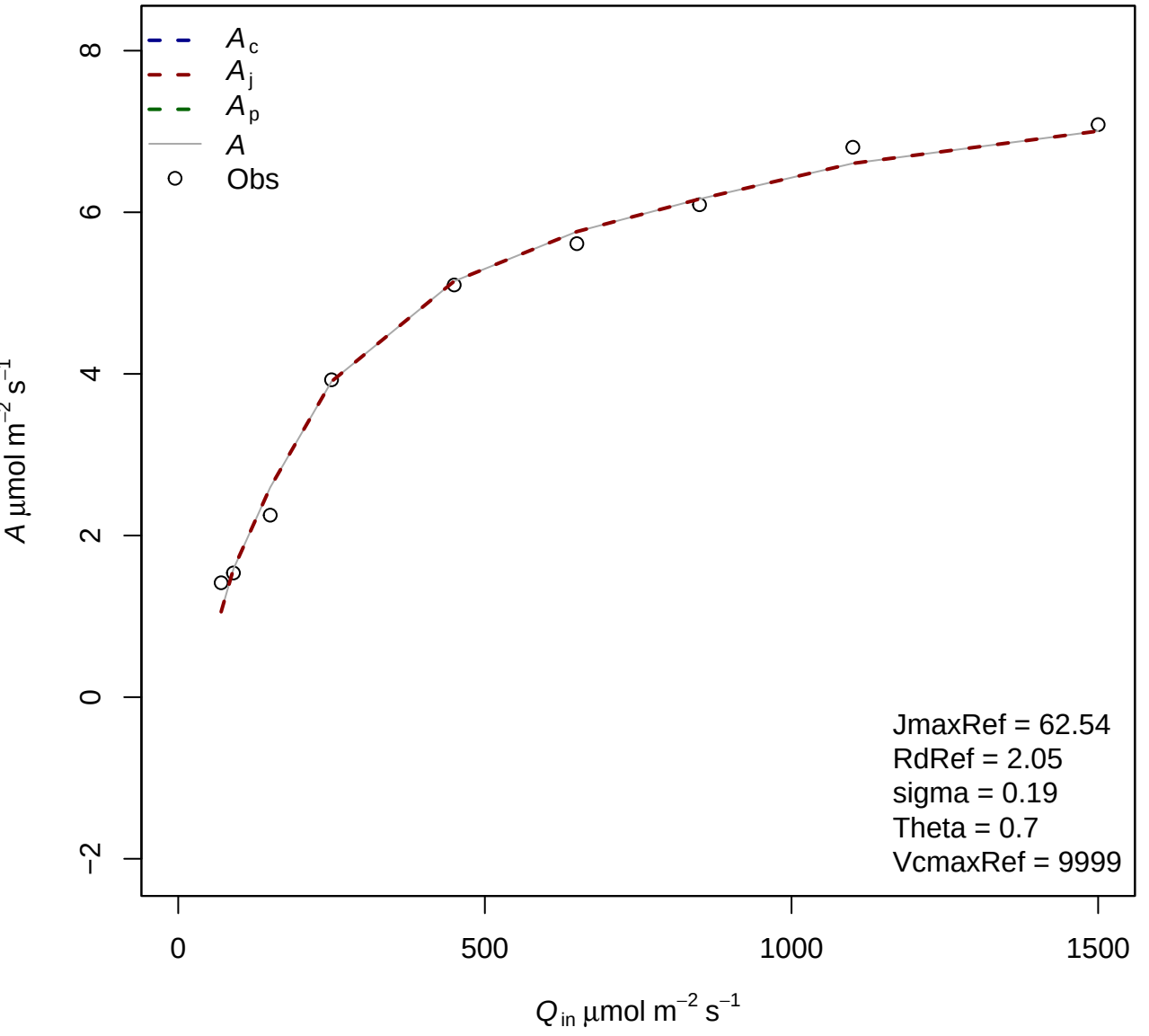

130
A_c
A_j
A_p
A
Obs
JmaxRef = 83.13
RdRef = 1.28
sigma = 0.12
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

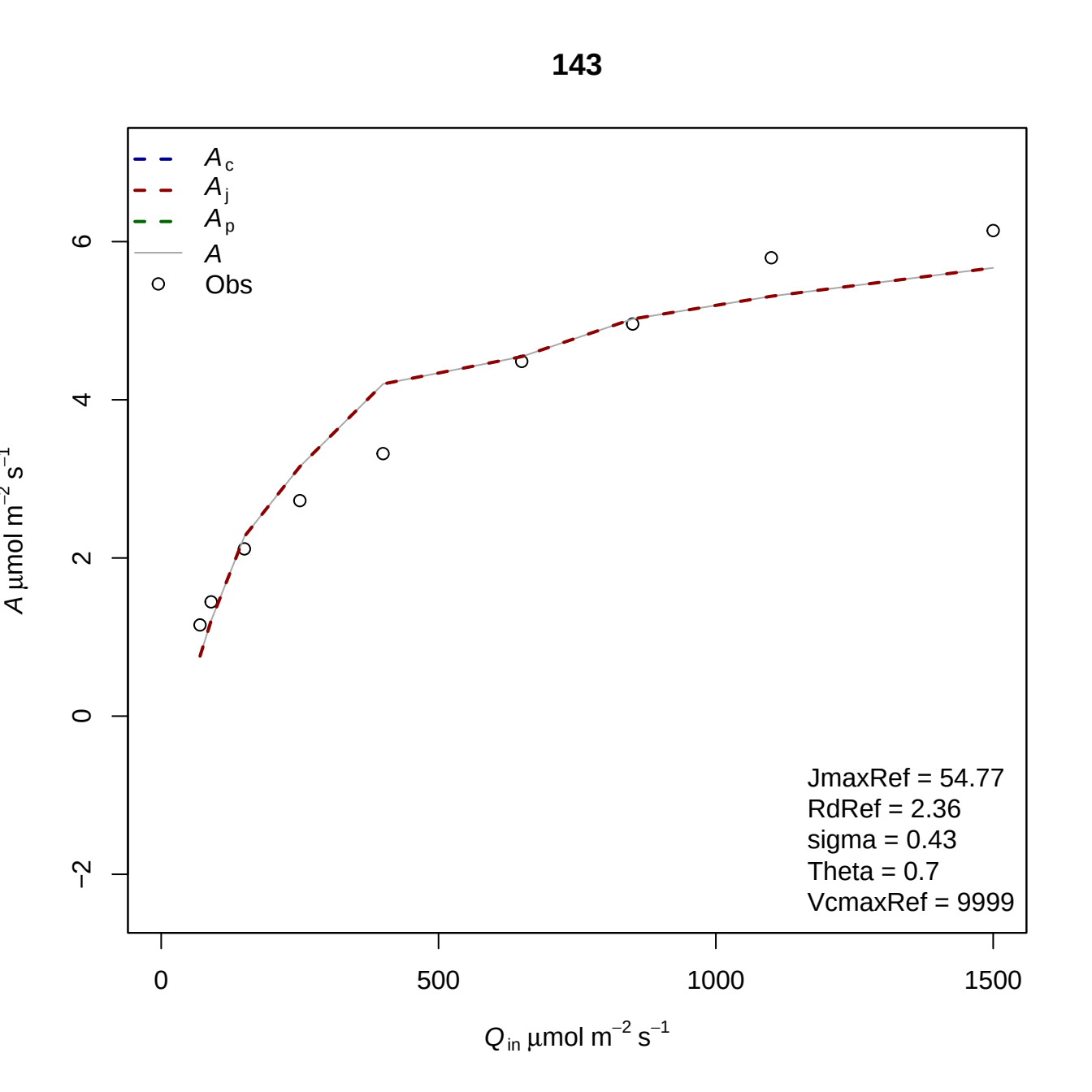

A_c
A_j
A_p
A
Obs
JmaxRef = 115.44
RdRef = 1.53
sigma = 0.16
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

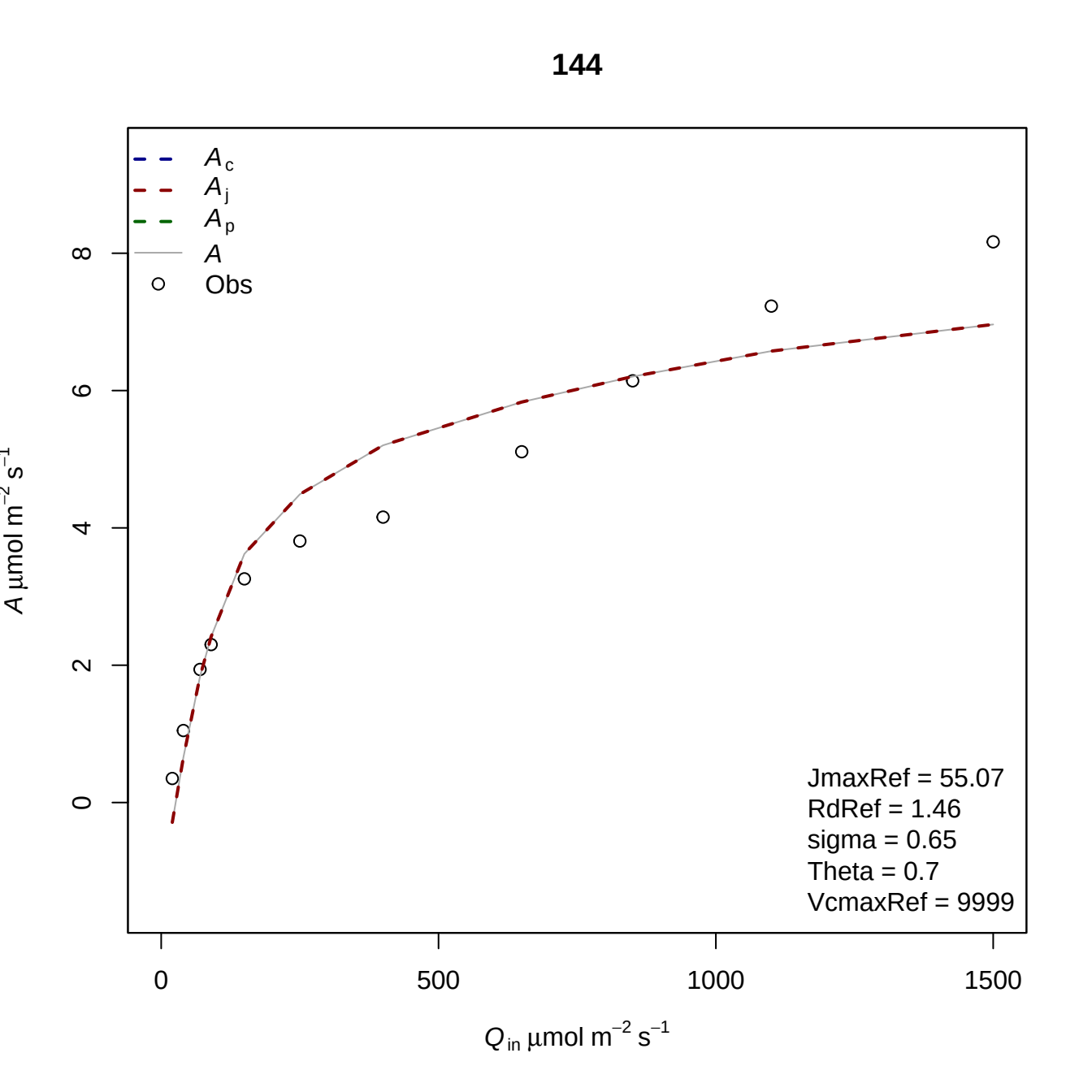

132
A_c
A_j
A_p
A
Obs
15
10
5
0
−5
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}
JmaxRef = 113.8
RdRef = 4.44
sigma = 0.25
Theta = 0.7
VcmaxRef = 9999

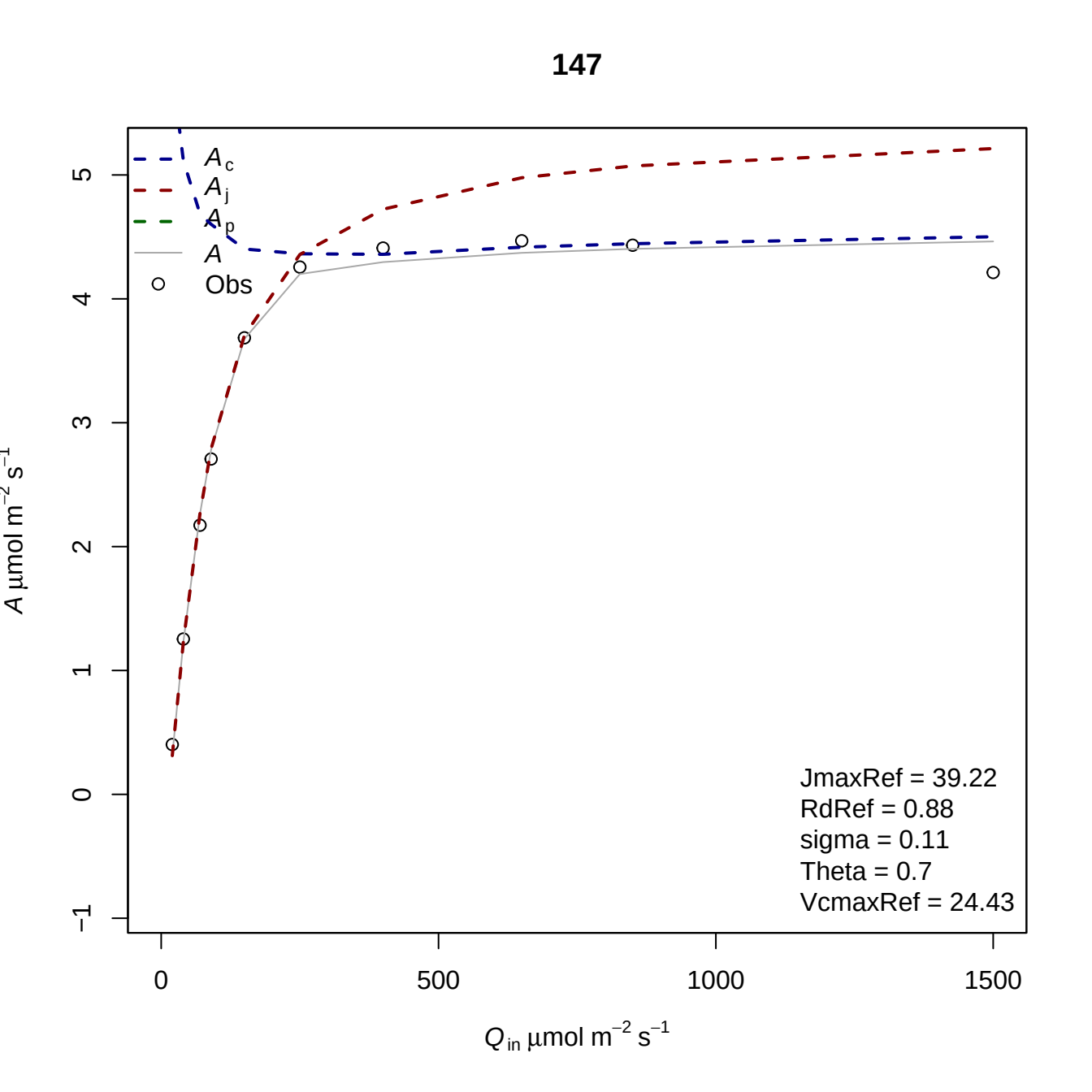

138
A_c
A_j
A_p
A
Obs
JmaxRef = 62.81
RdRef = 1.31
sigma = 0.09
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

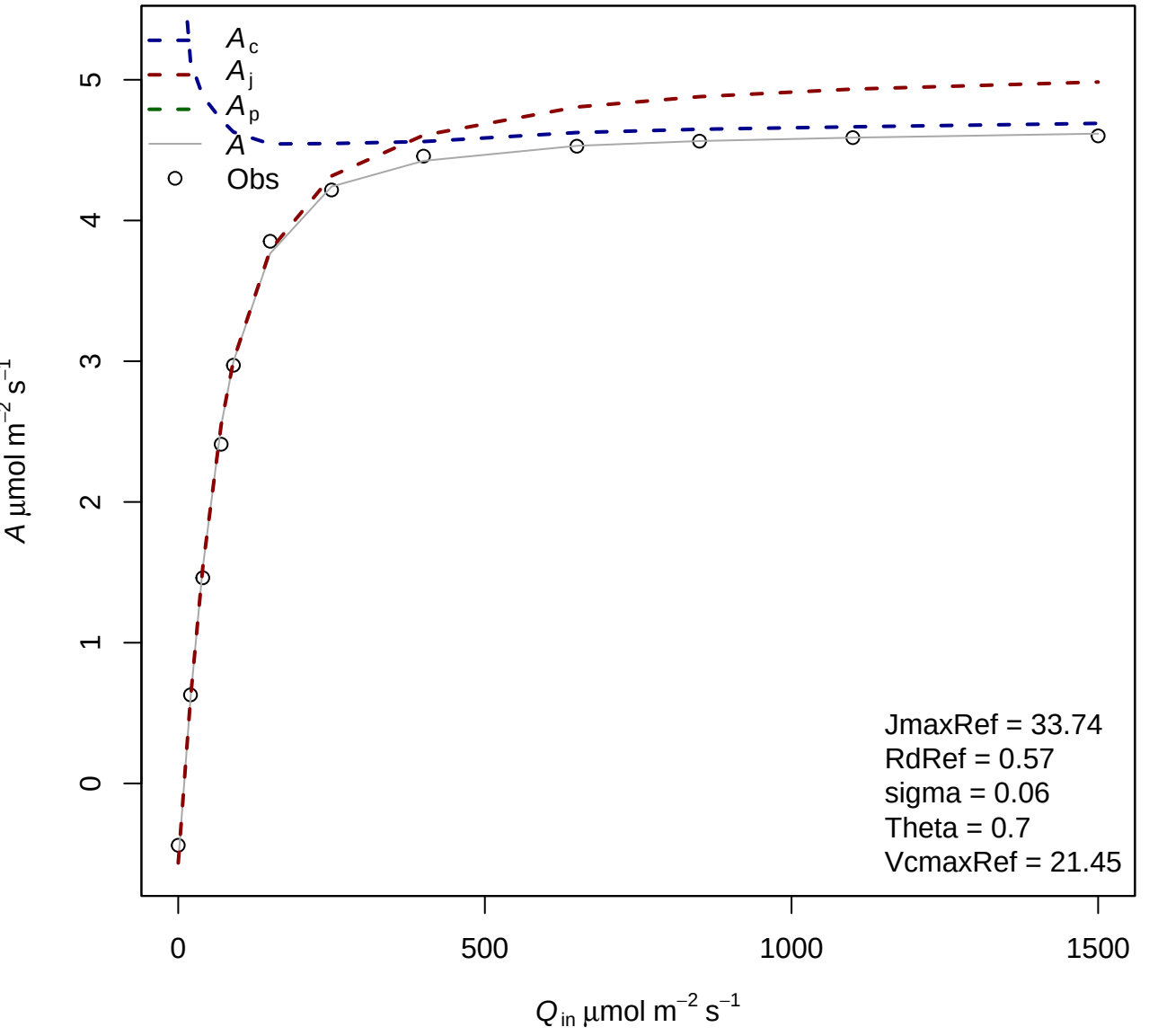

140
A_c
A_j
A_p
A
Obs
JmaxRef = 56.4
RdRef = 2.05
sigma = 0.21
Theta = 0.7
VcmaxRef = 39.6
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

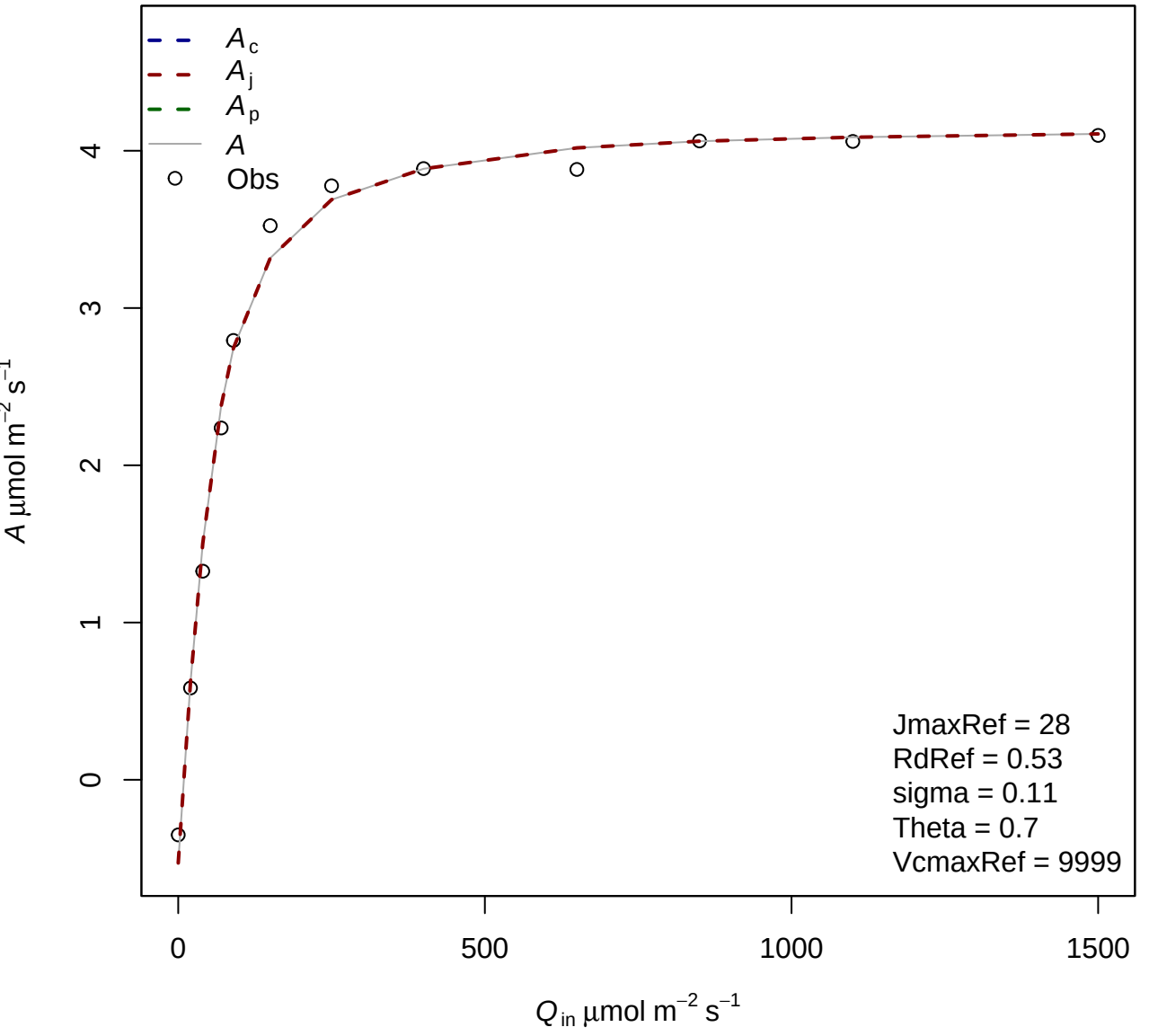

A_c
A_j
A_p
A
Obs
JmaxRef = 62.54
RdRef = 2.05
sigma = 0.19
Theta = 0.7
VcmaxRef = 9999
8
6
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

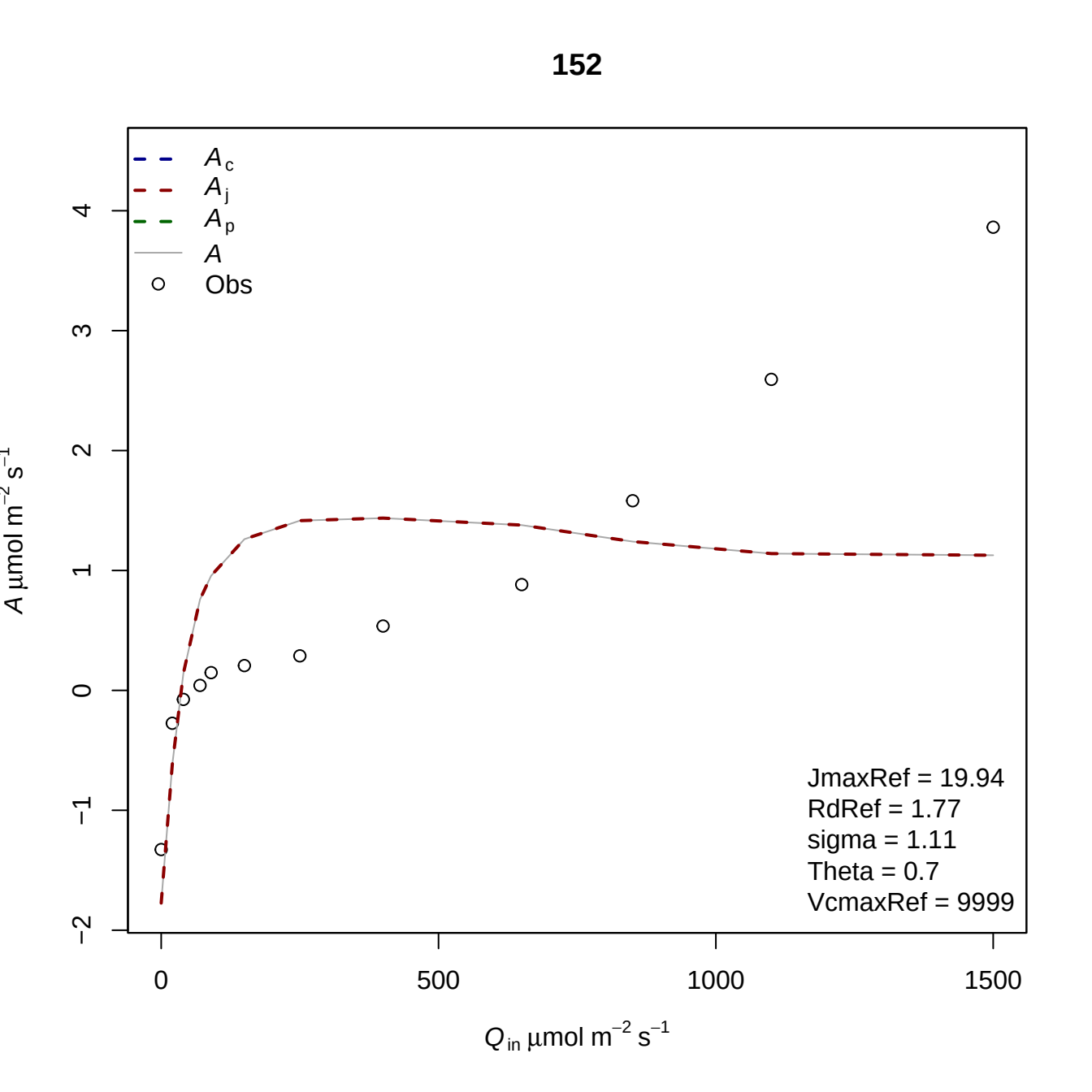

143
A_c
A_j
A_p
A
Obs
JmaxRef = 54.77
RdRef = 2.36
sigma = 0.43
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

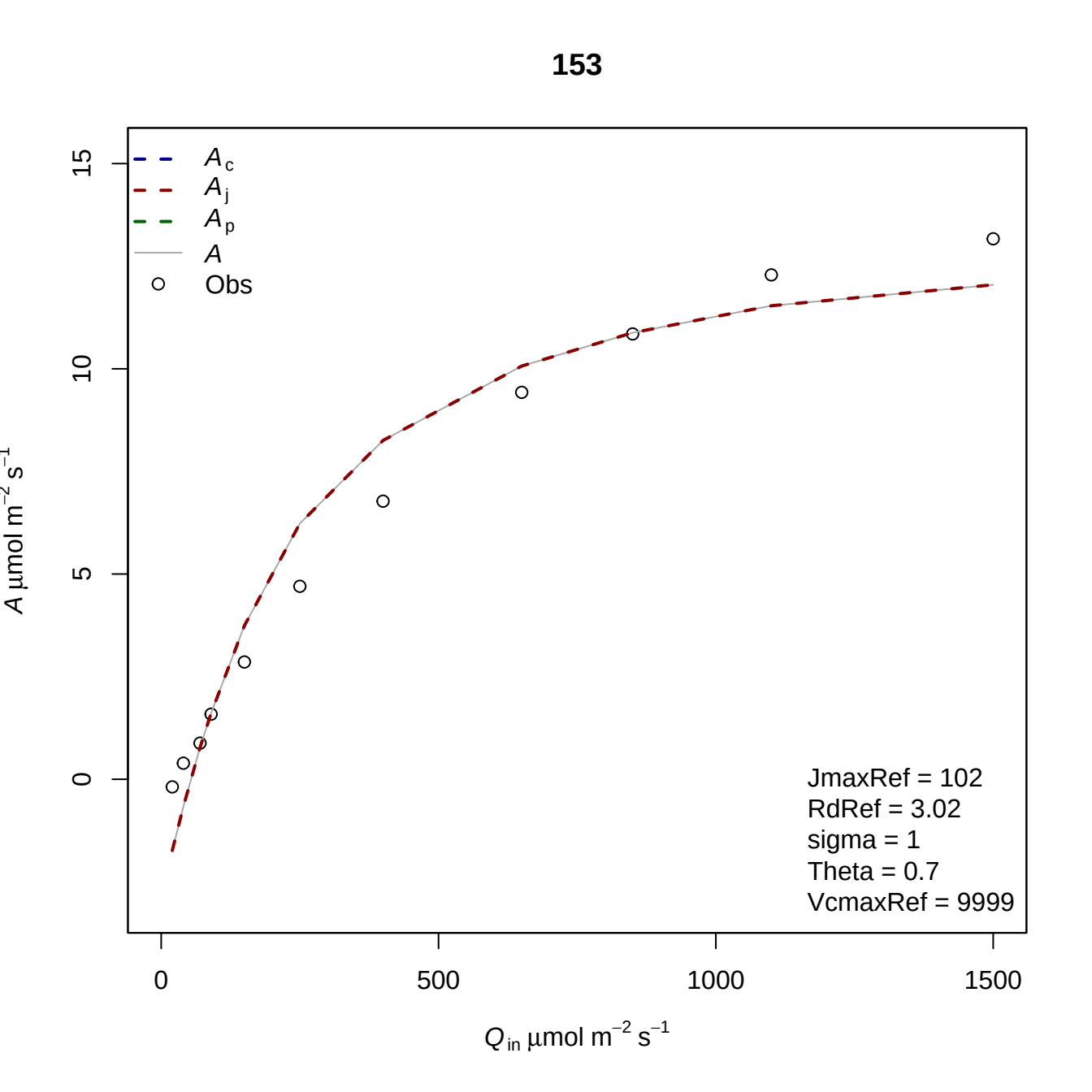

144
A_c
A_j
A_p
A
Obs
JmaxRef = 55.07
RdRef = 1.46
sigma = 0.65
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

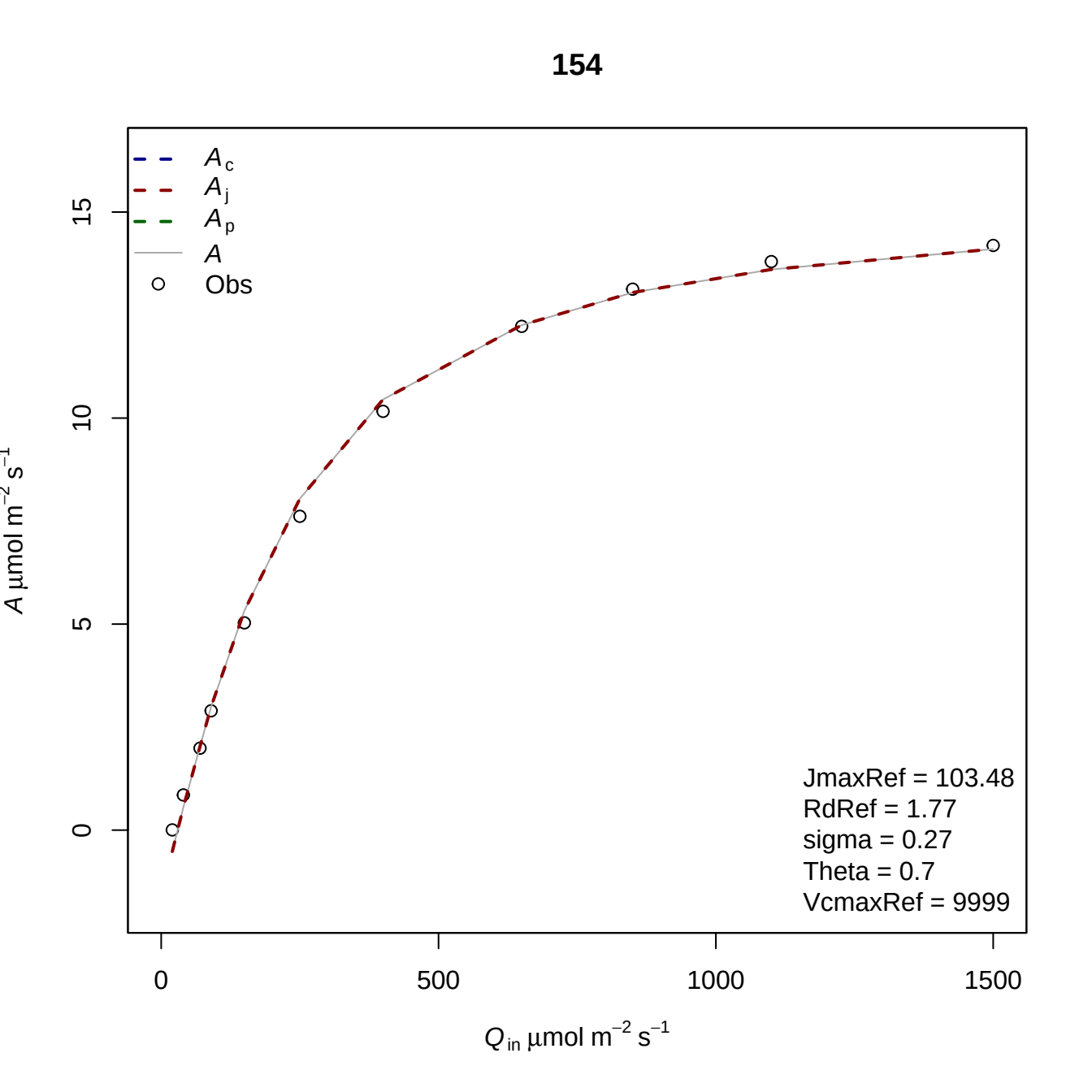

147
A_c
A_j
A_p
A
Obs
JmaxRef = 39.22
RdRef = 0.88
sigma = 0.11
Theta = 0.7
VcmaxRef = 24.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

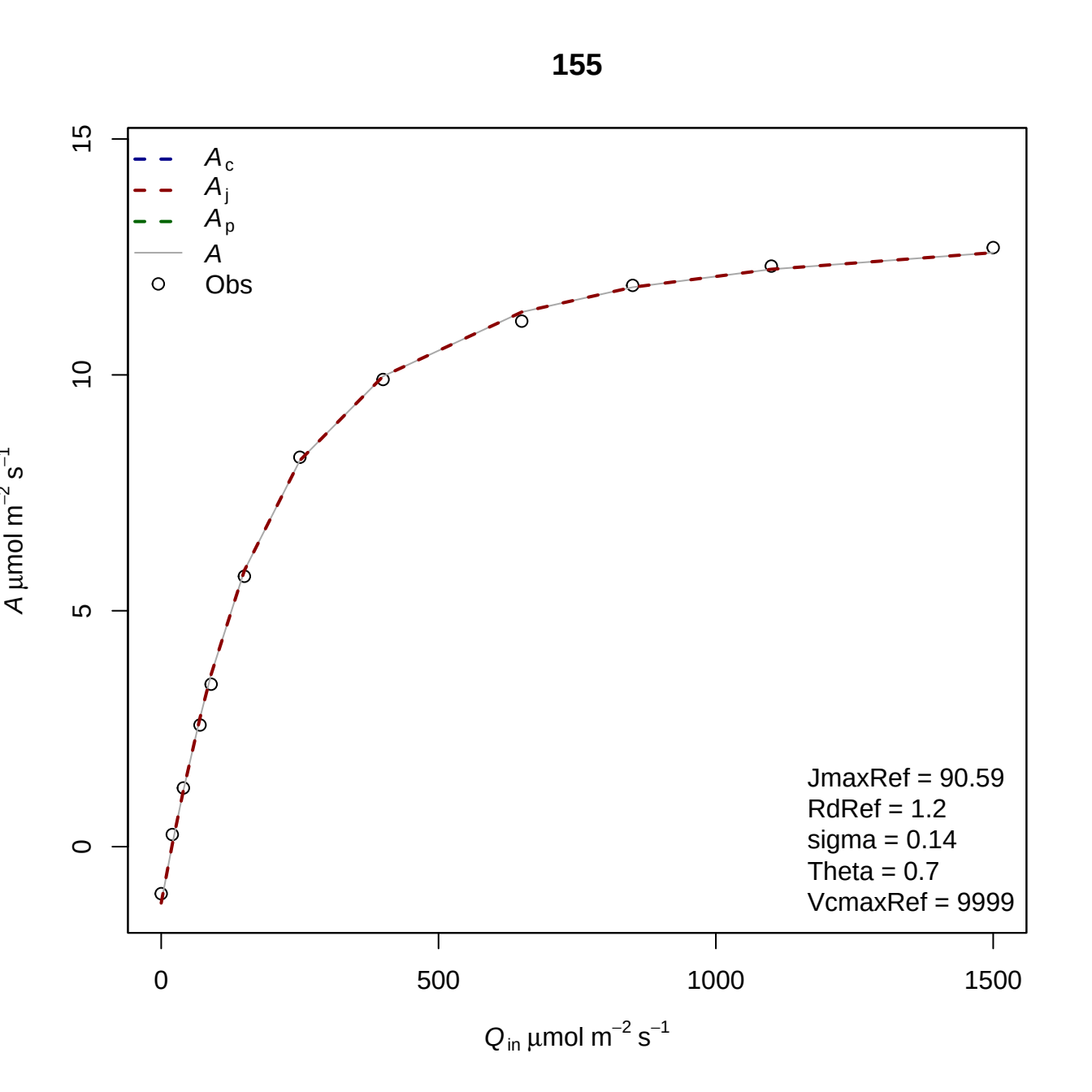

A_c
A_j
A_p
A
Obs
JmaxRef = 33.74
RdRef = 0.57
sigma = 0.06
Theta = 0.7
VcmaxRef = 21.45
0
1
2
3
4
5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

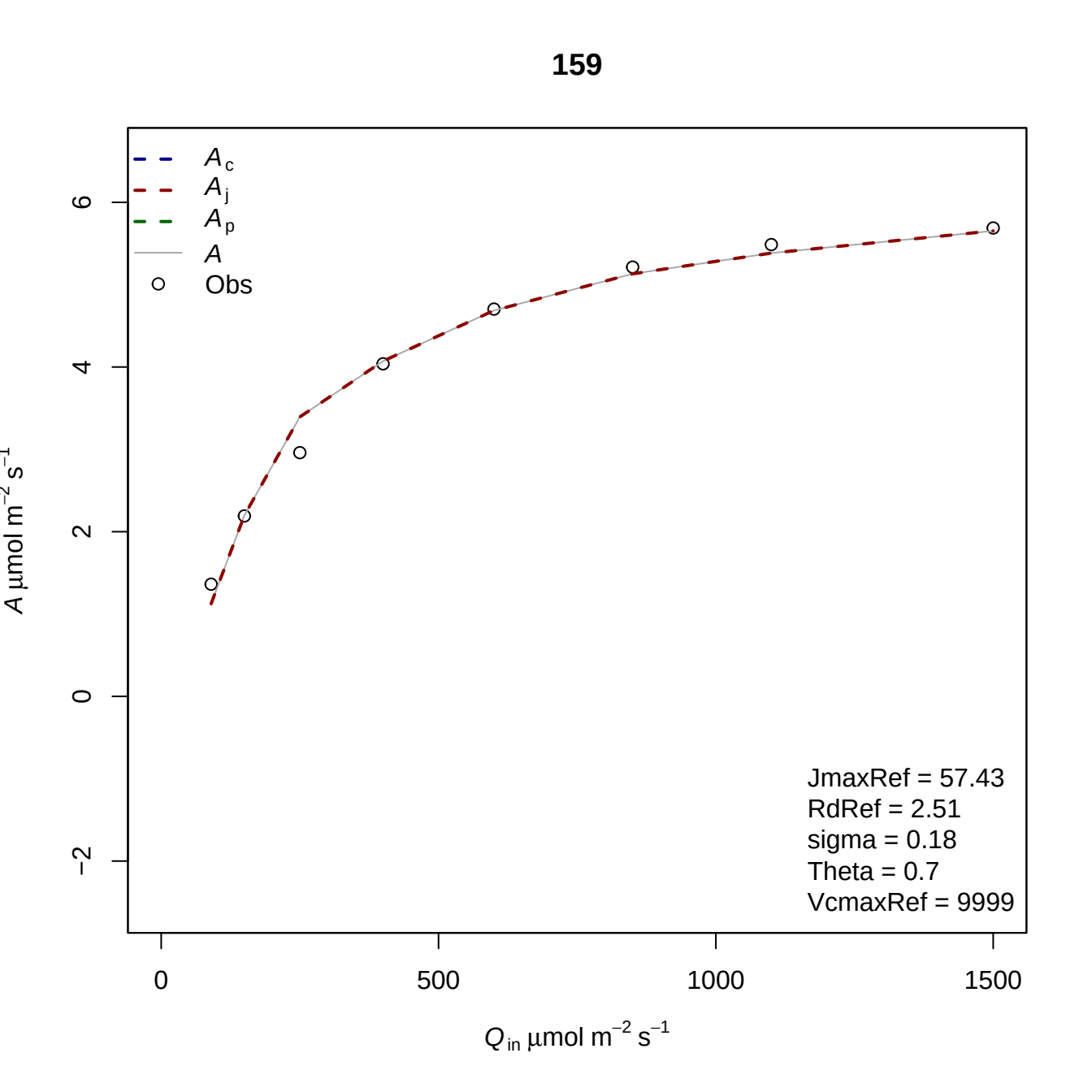

A_c
A_j
A_p
A
Obs
JmaxRef = 28
RdRef = 0.53
sigma = 0.11
Theta = 0.7
VcmaxRef = 9999
0
1
2
3
4
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

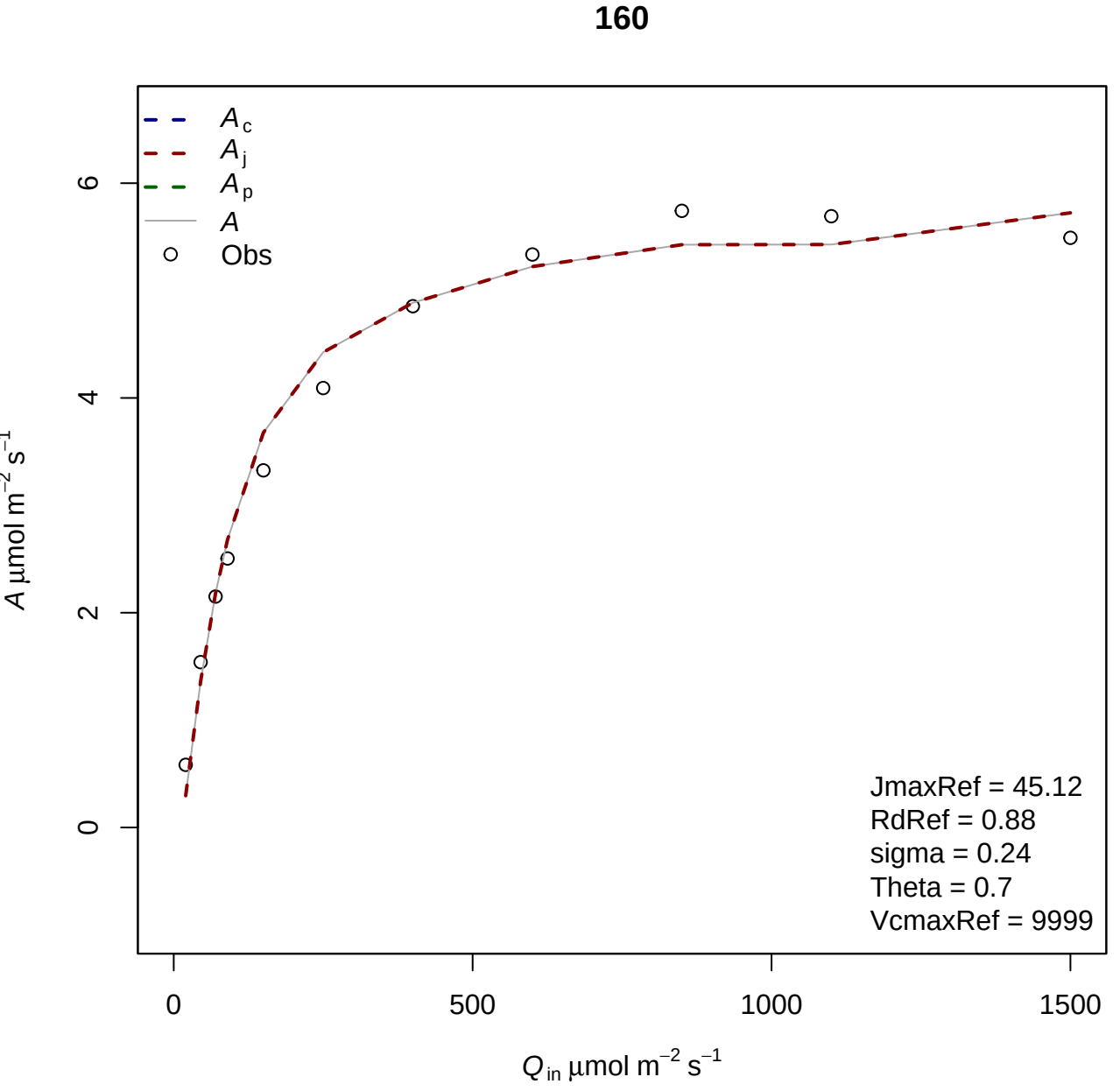

152
A_c
A_j
A_p
A
Obs
JmaxRef = 19.94
RdRef = 1.77
sigma = 1.11
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
4
0
500
1000
1500

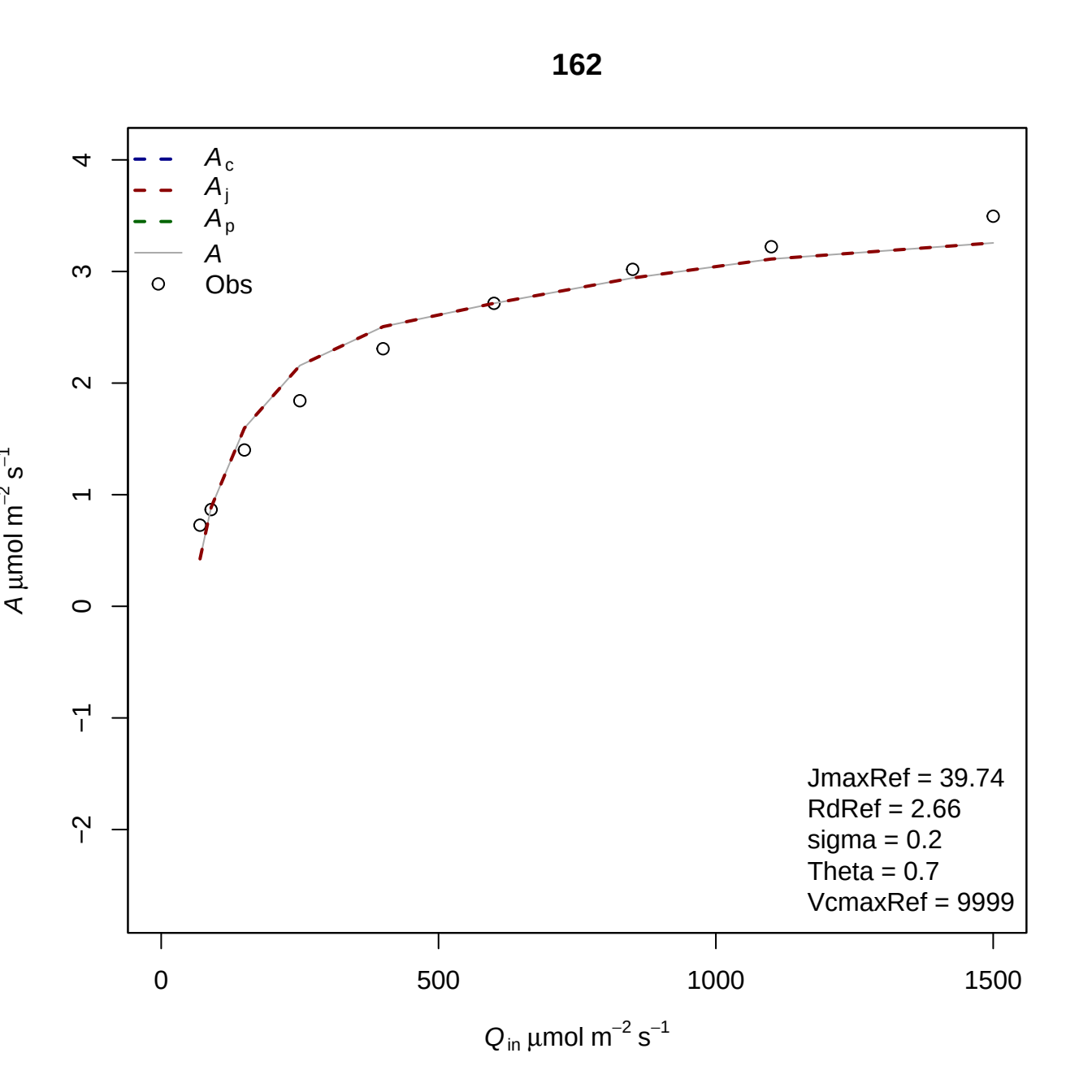

153
A_c
A_j
A_p
A
Obs
JmaxRef = 102
RdRef = 3.02
sigma = 1
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

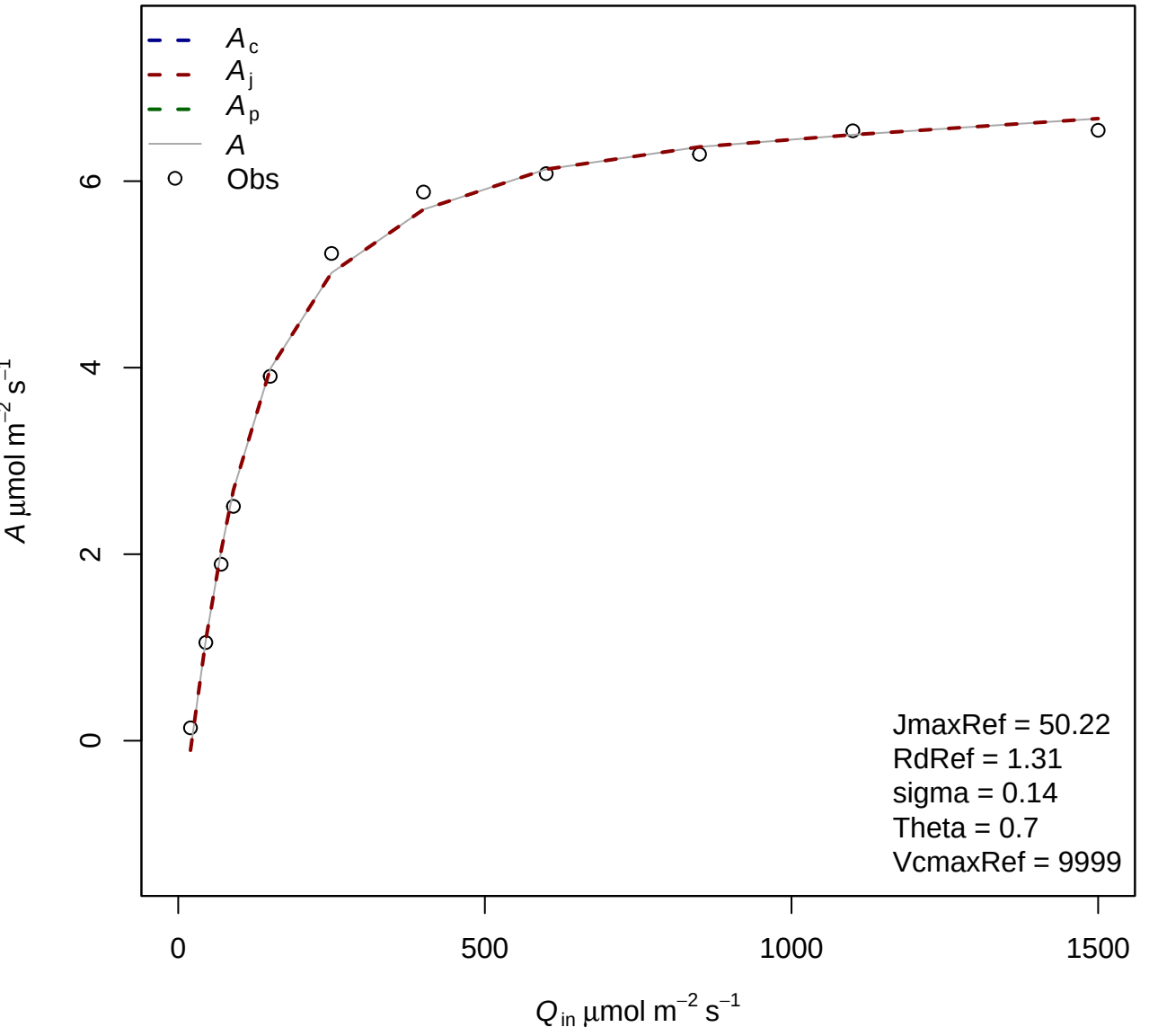

154
A_c
A_j
A_p
A
Obs
JmaxRef = 103.48
RdRef = 1.77
sigma = 0.27
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

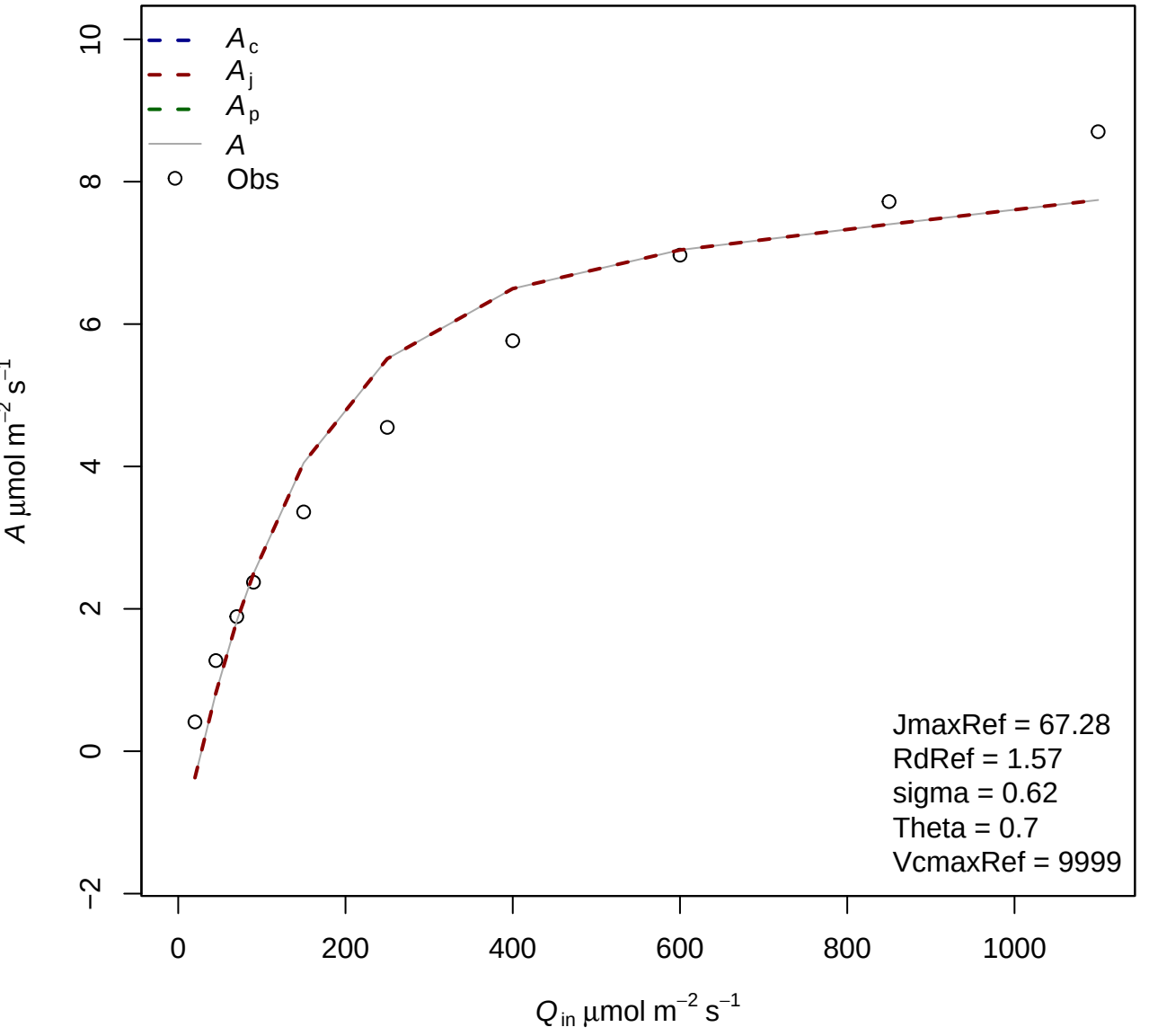

155
A_c
A_j
A_p
A
Obs
JmaxRef = 90.59
RdRef = 1.2
sigma = 0.14
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

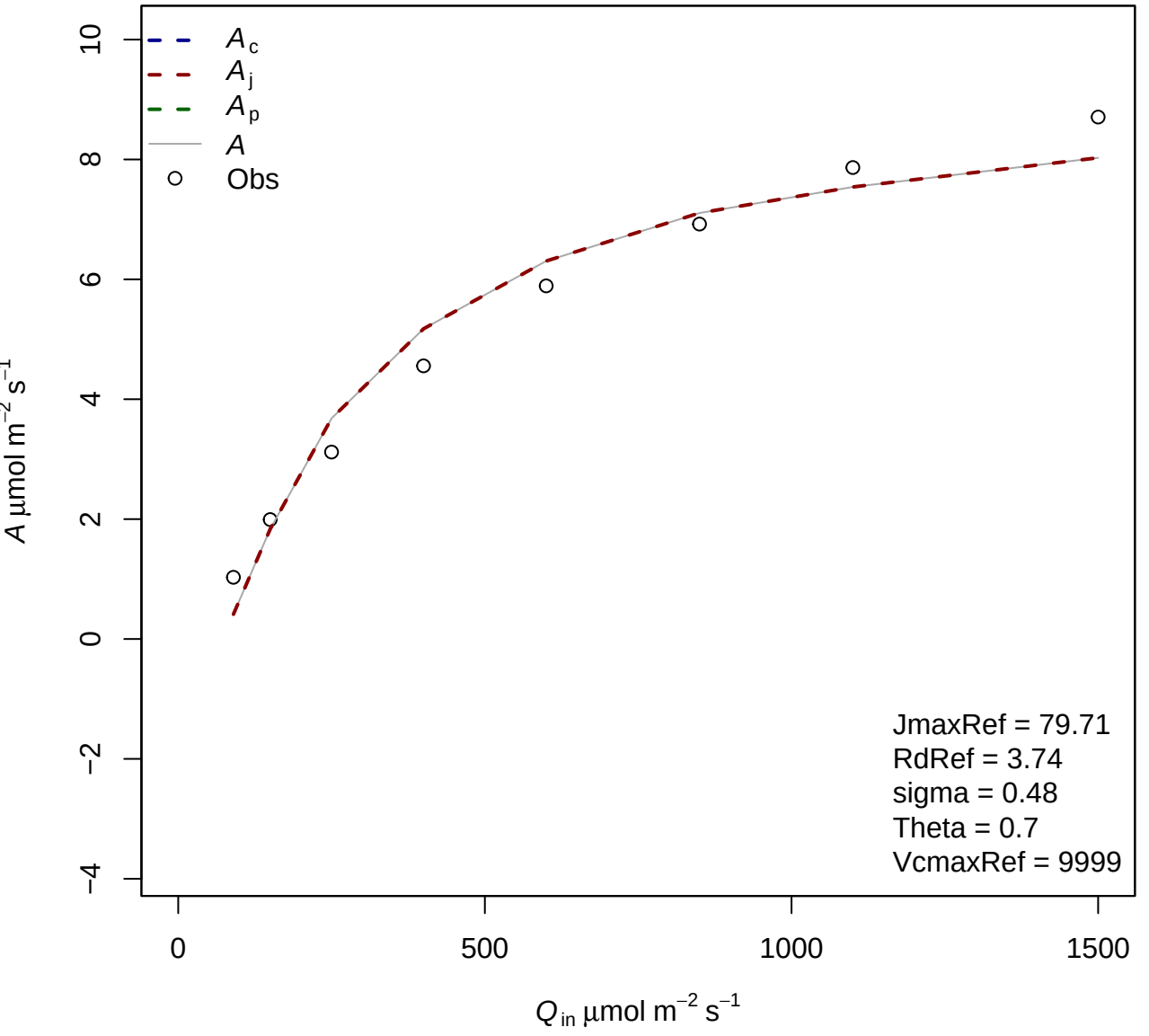

159
A_c
A_j
A_p
A
Obs
JmaxRef = 57.43
RdRef = 2.51
sigma = 0.18
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

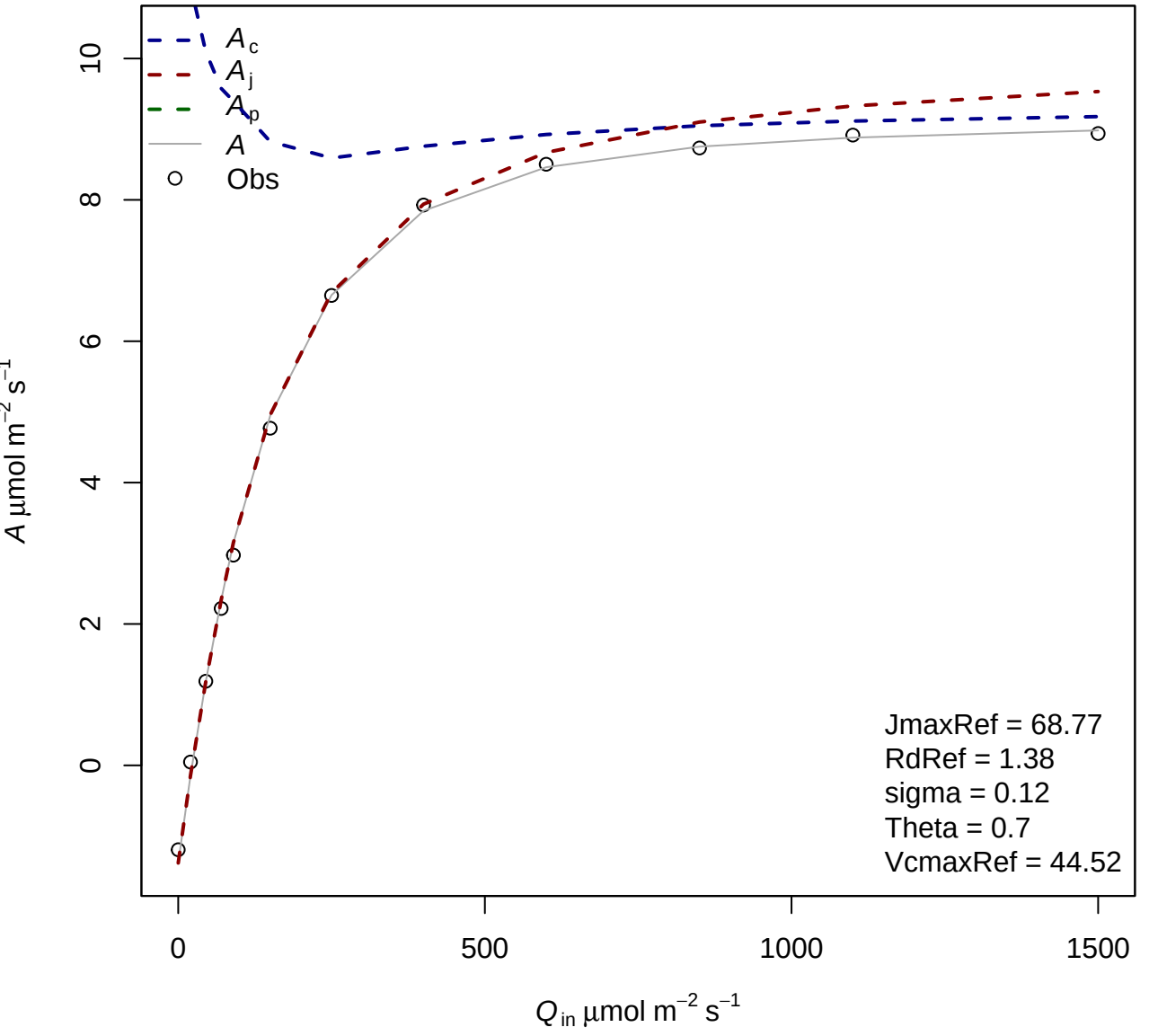

160
A_c
A_j
A_p
A
Obs
JmaxRef = 45.12
RdRef = 0.88
sigma = 0.24
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

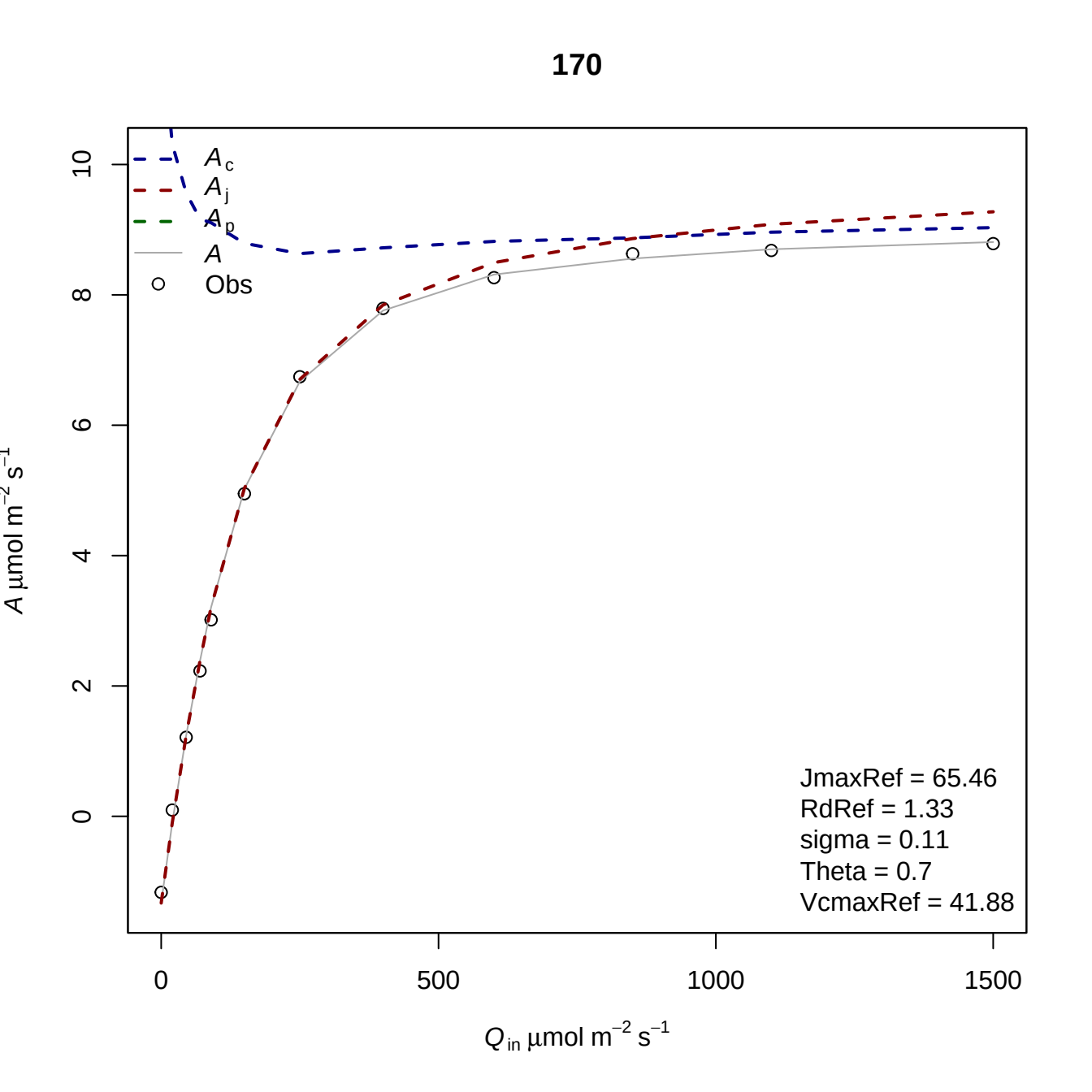

162
A_c
A_j
A_p
A
Obs
JmaxRef = 39.74
RdRef = 2.66
sigma = 0.2
Theta = 0.7
VcmaxRef = 9999
4
3
2
1
0
−1
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

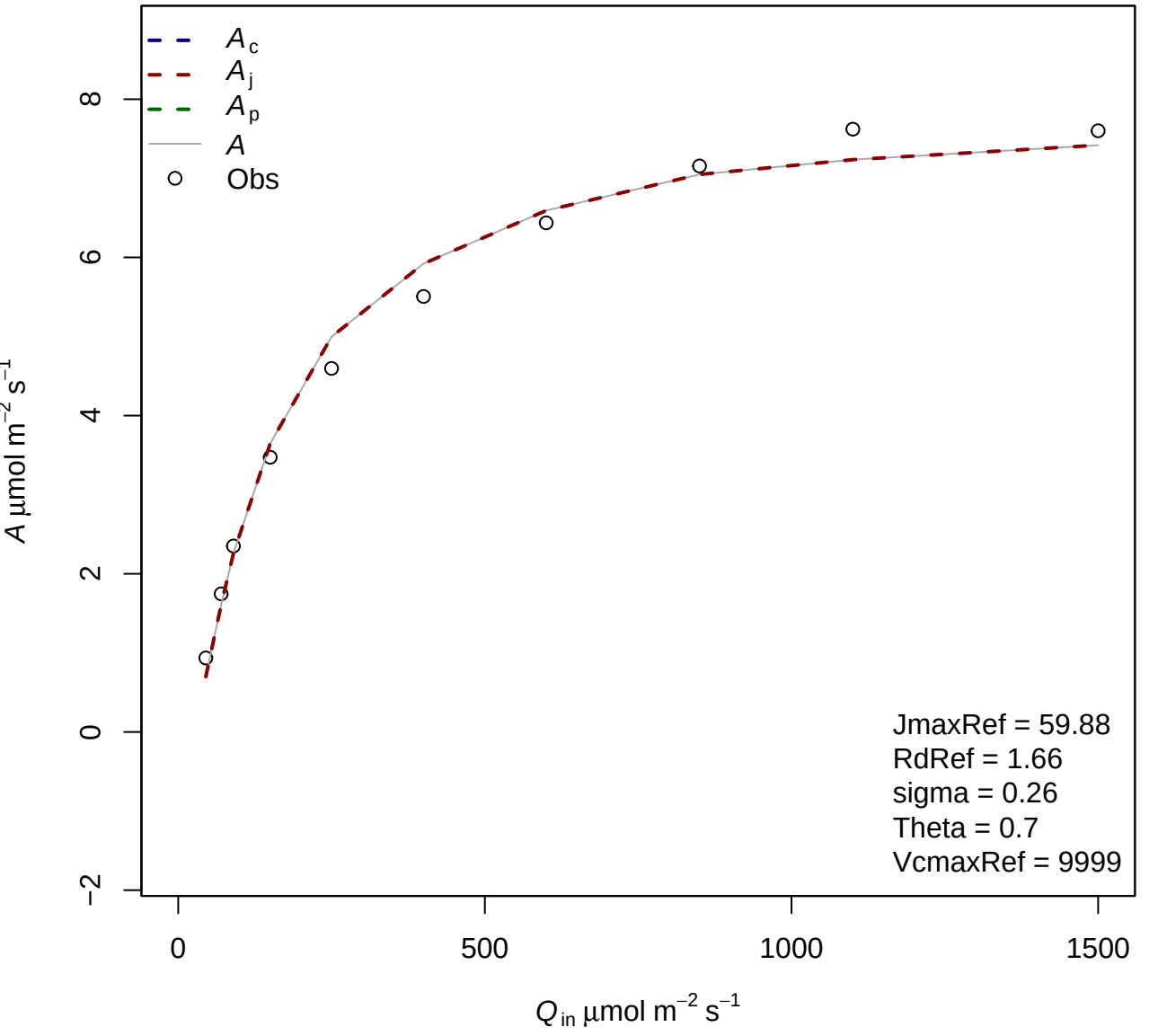

A_c
A_j
A_p
A
Obs
JmaxRef = 50.22
RdRef = 1.31
sigma = 0.14
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

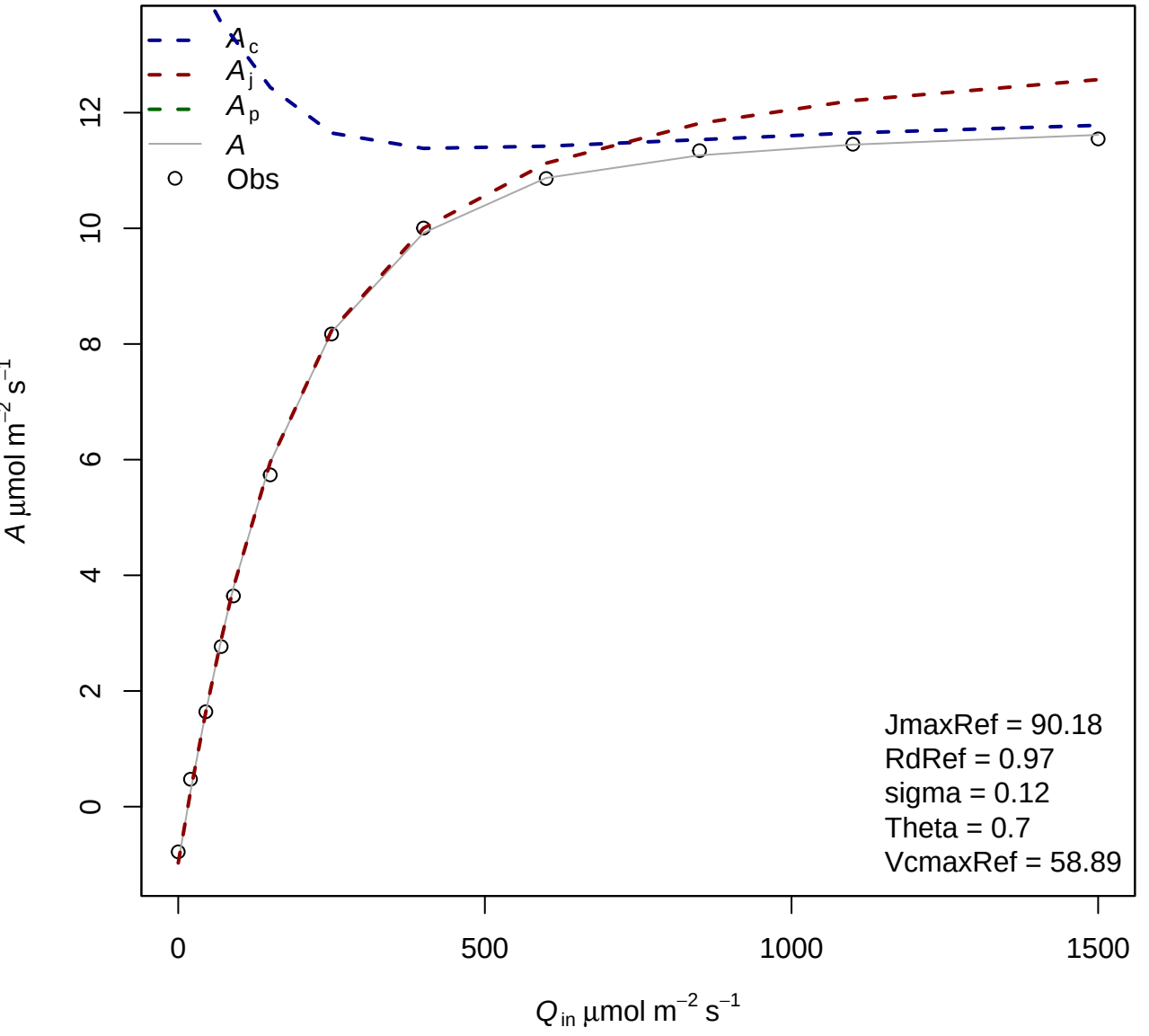

A_c
A_j
A_p
A
Obs
JmaxRef = 67.28
RdRef = 1.57
sigma = 0.62
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
200
400
600
800
1000

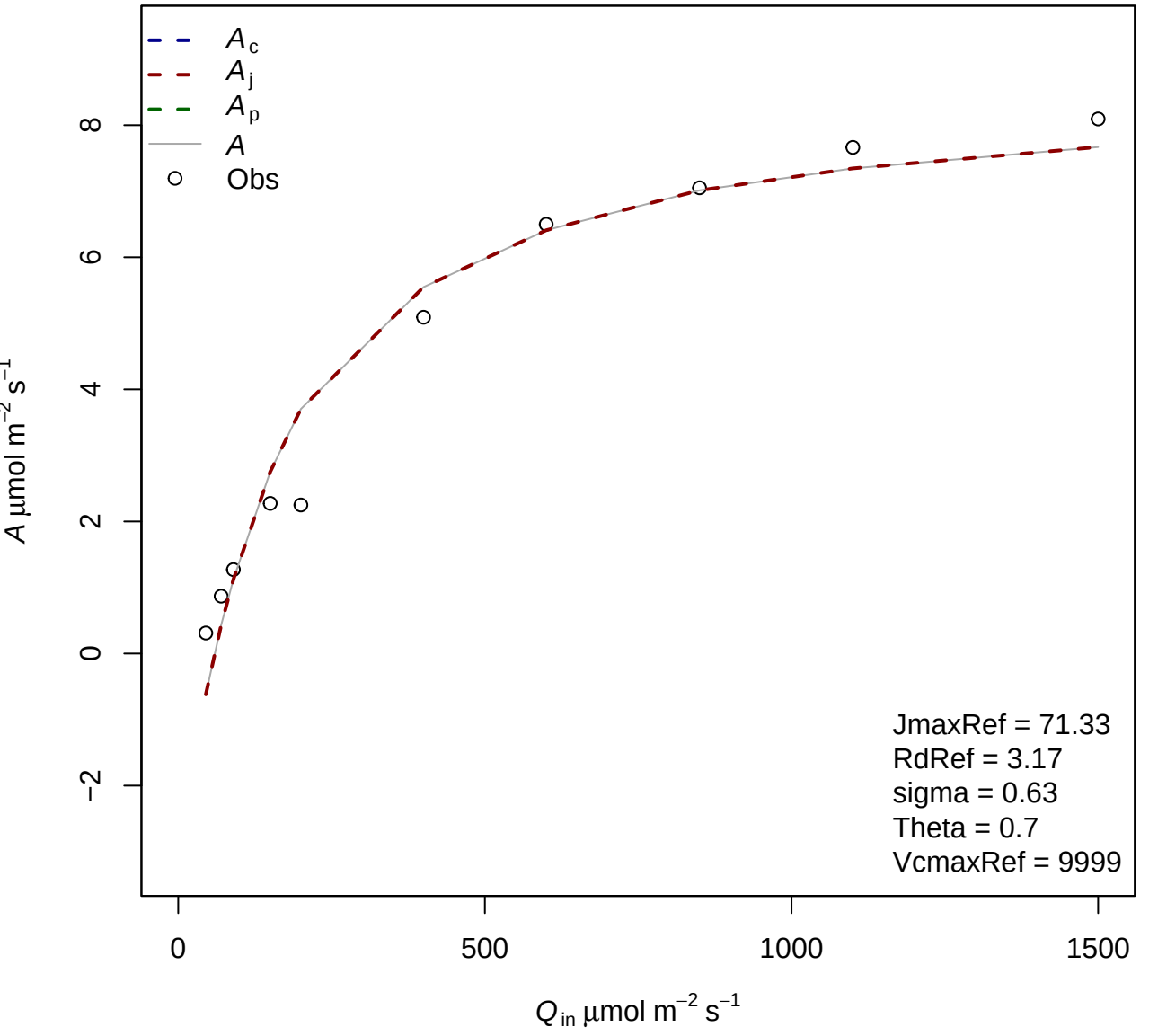

A_c
A_j
A_p
A
Obs
JmaxRef = 79.71
RdRef = 3.74
sigma = 0.48
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500

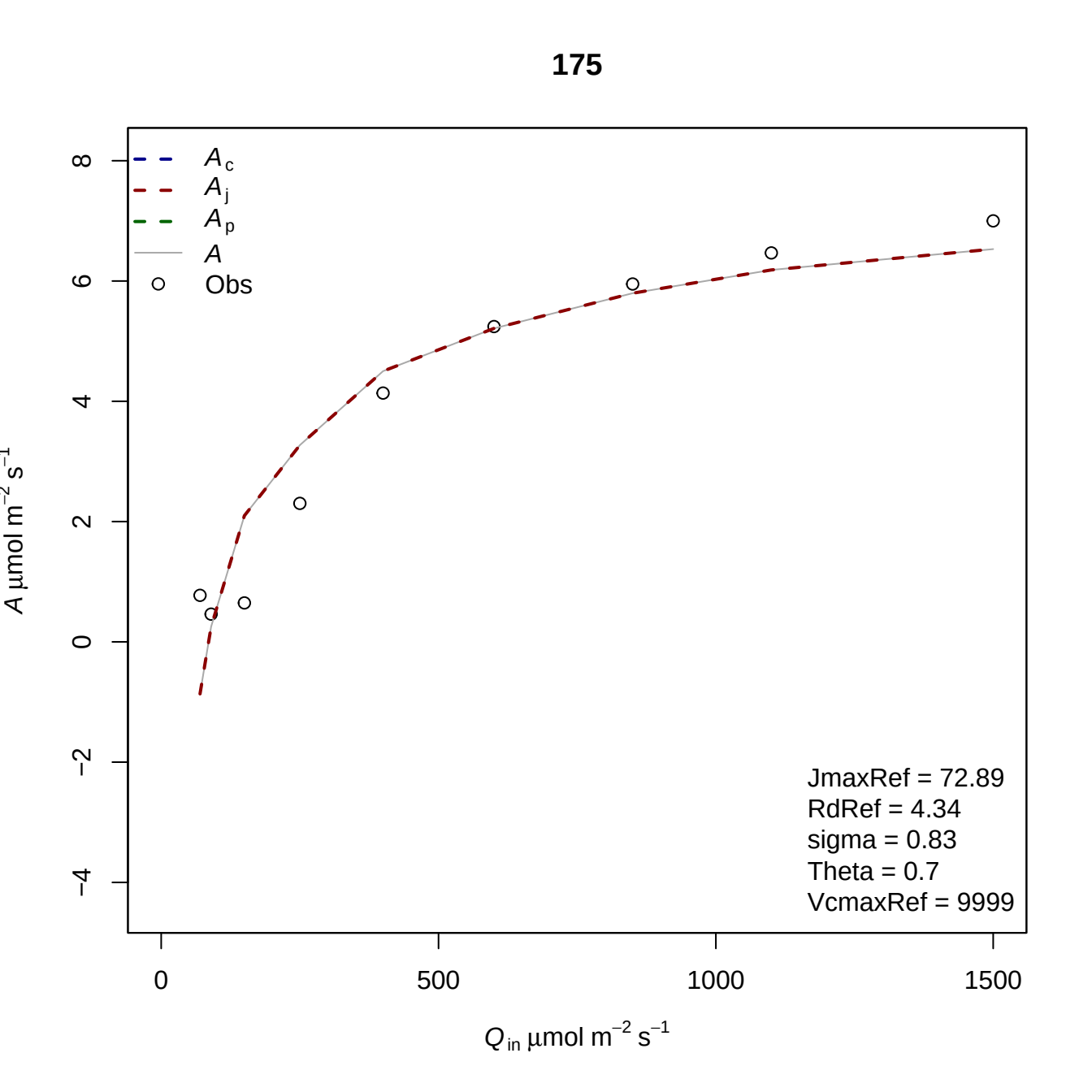

A_c
A_j
A_p
A
Obs
JmaxRef = 68.77
RdRef = 1.38
sigma = 0.12
Theta = 0.7
VcmaxRef = 44.52
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

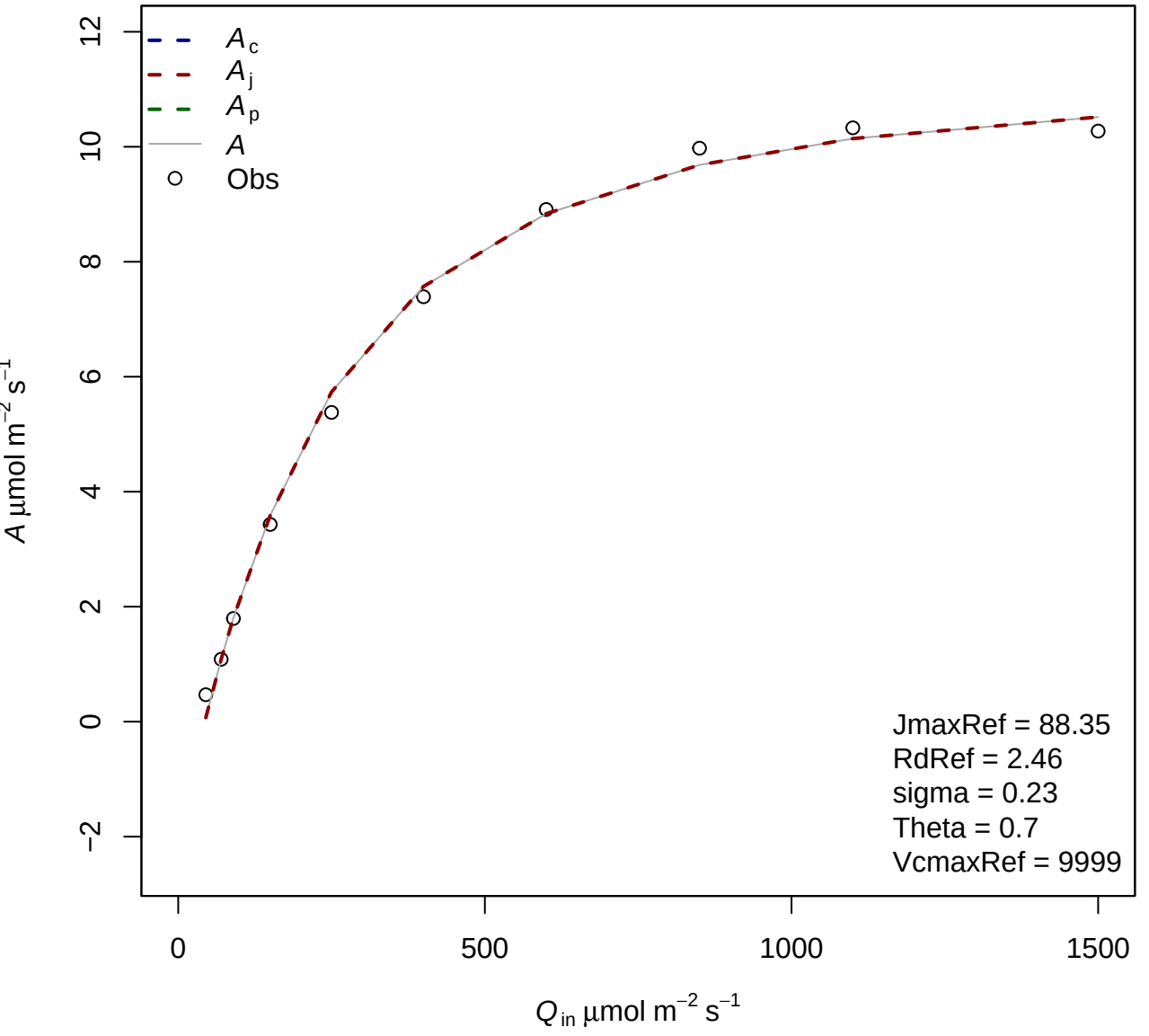

170
A_c
A_j
A_p
A
Obs
JmaxRef = 65.46
RdRef = 1.33
sigma = 0.11
Theta = 0.7
VcmaxRef = 41.88
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

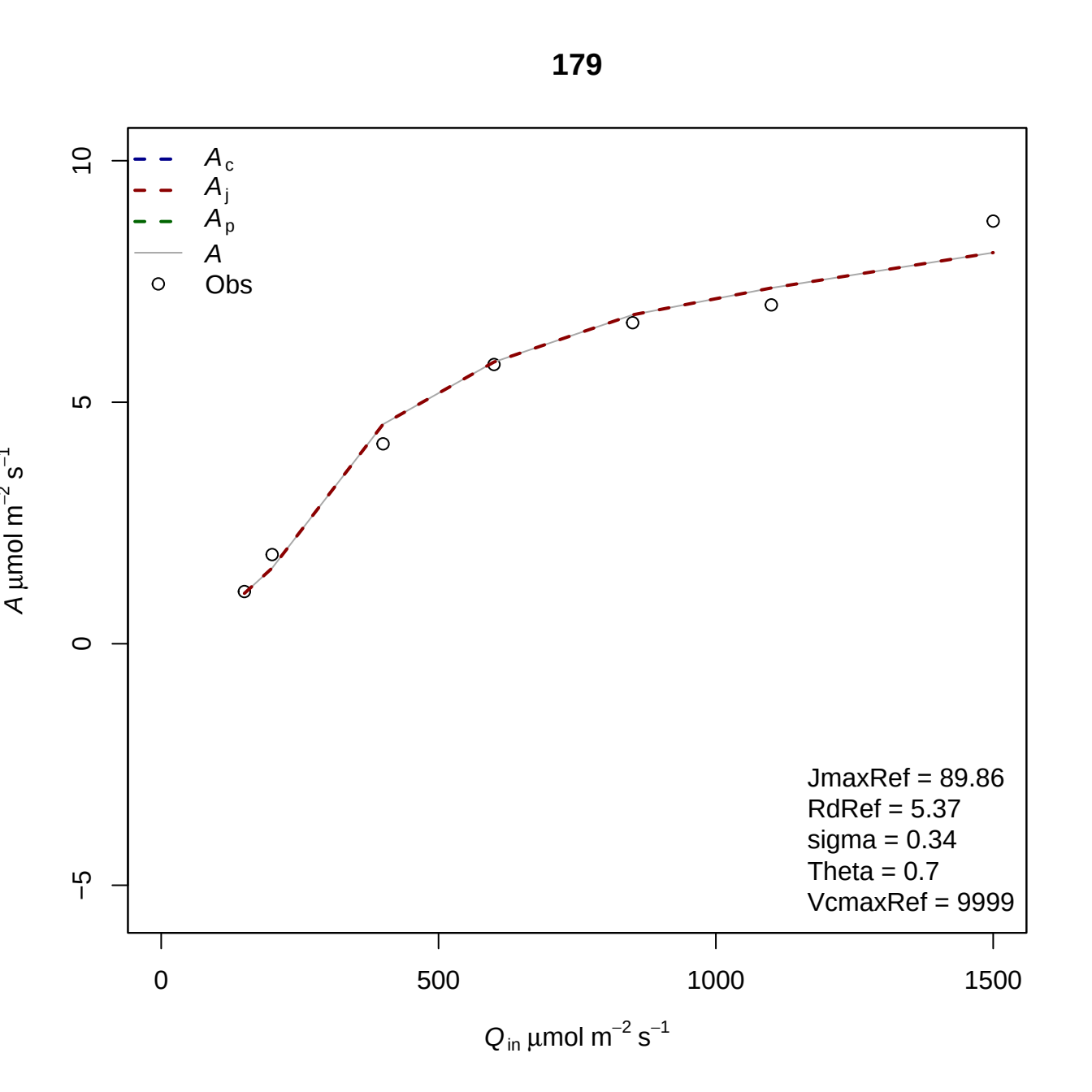

A_c
A_j
A_p
A
Obs
JmaxRef = 59.88
RdRef = 1.66
sigma = 0.26
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

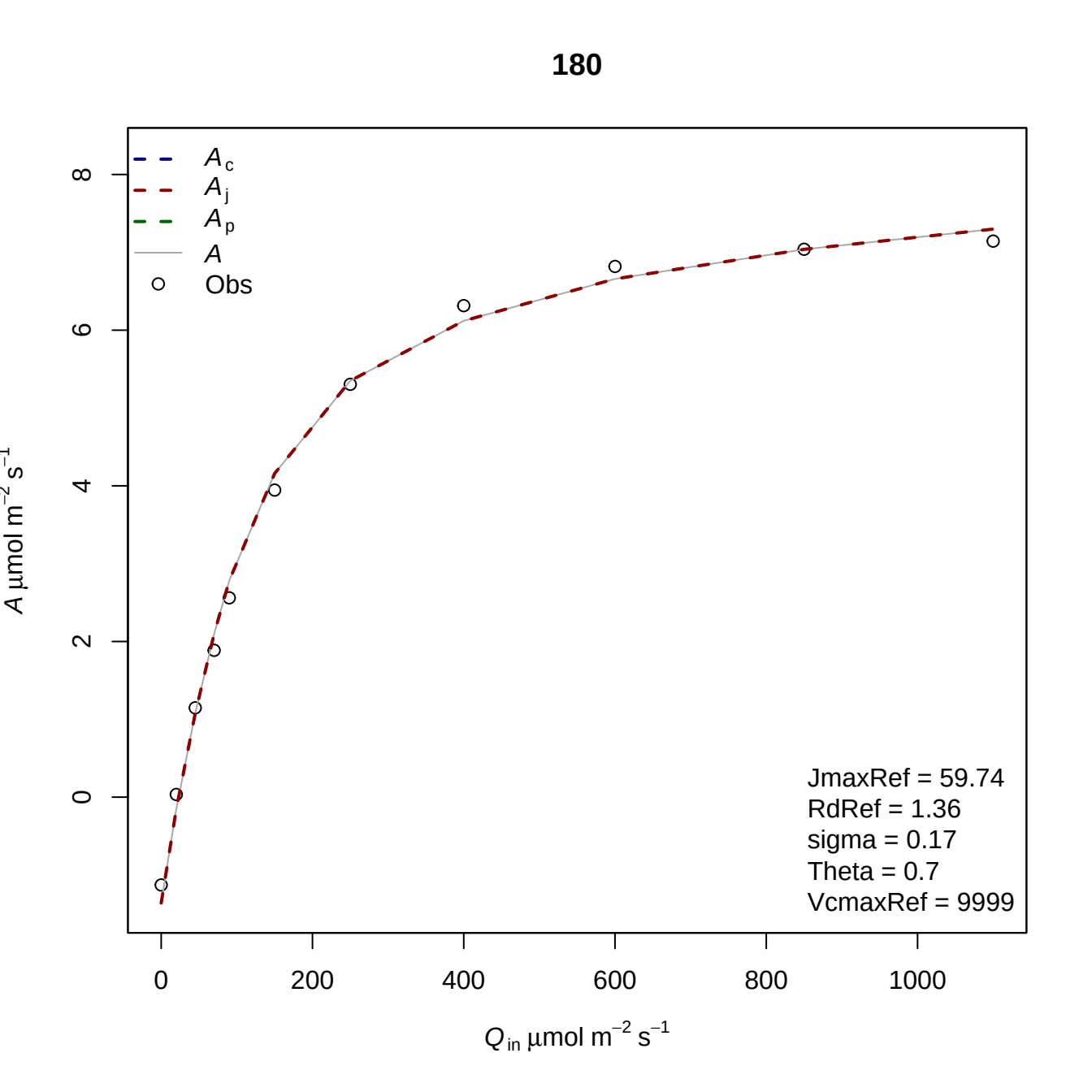

A_c
A_j
A_p
A
Obs
JmaxRef = 90.18
RdRef = 0.97
sigma = 0.12
Theta = 0.7
VcmaxRef = 58.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

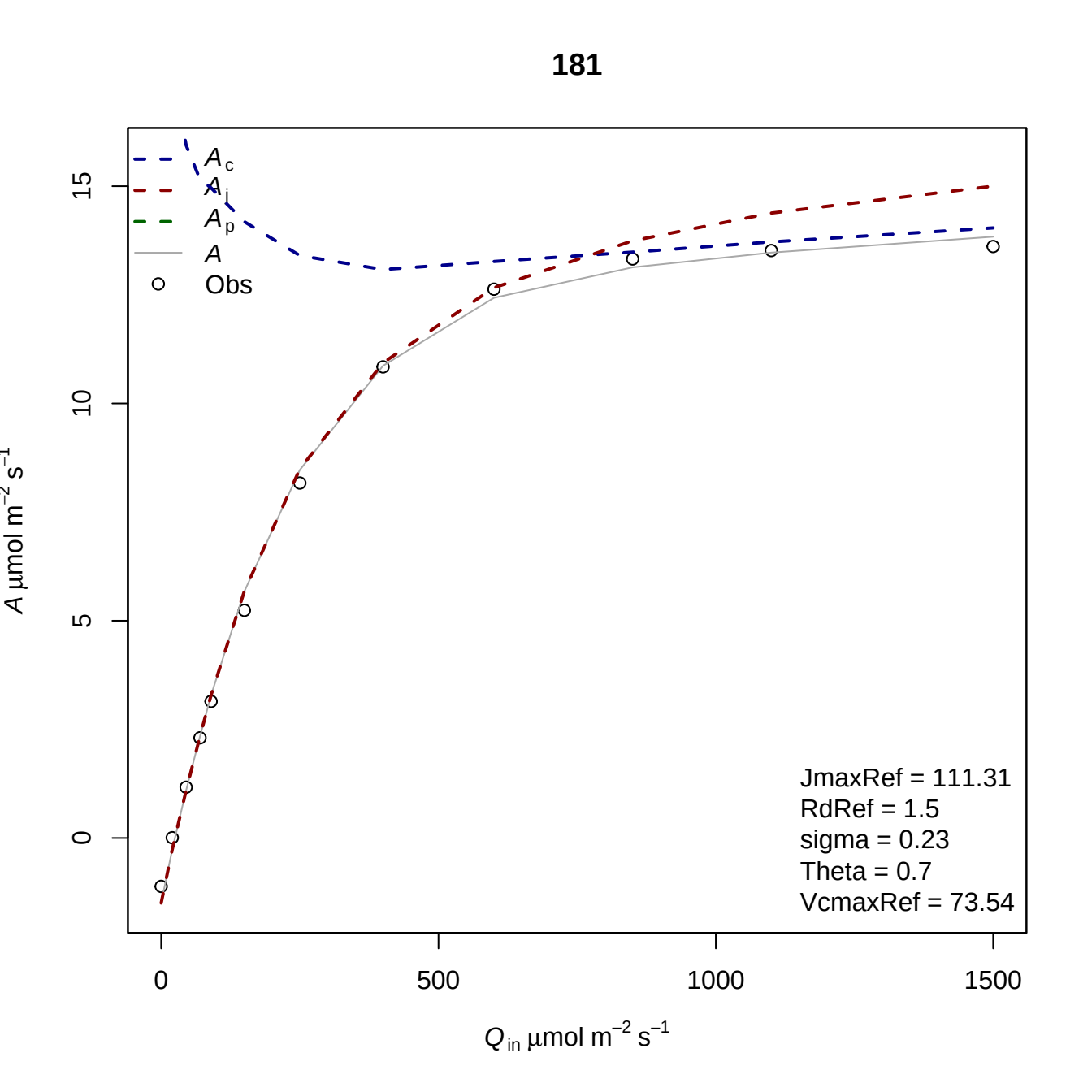

A_c
A_j
A_p
A
Obs
JmaxRef = 71.33
RdRef = 3.17
sigma = 0.63
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

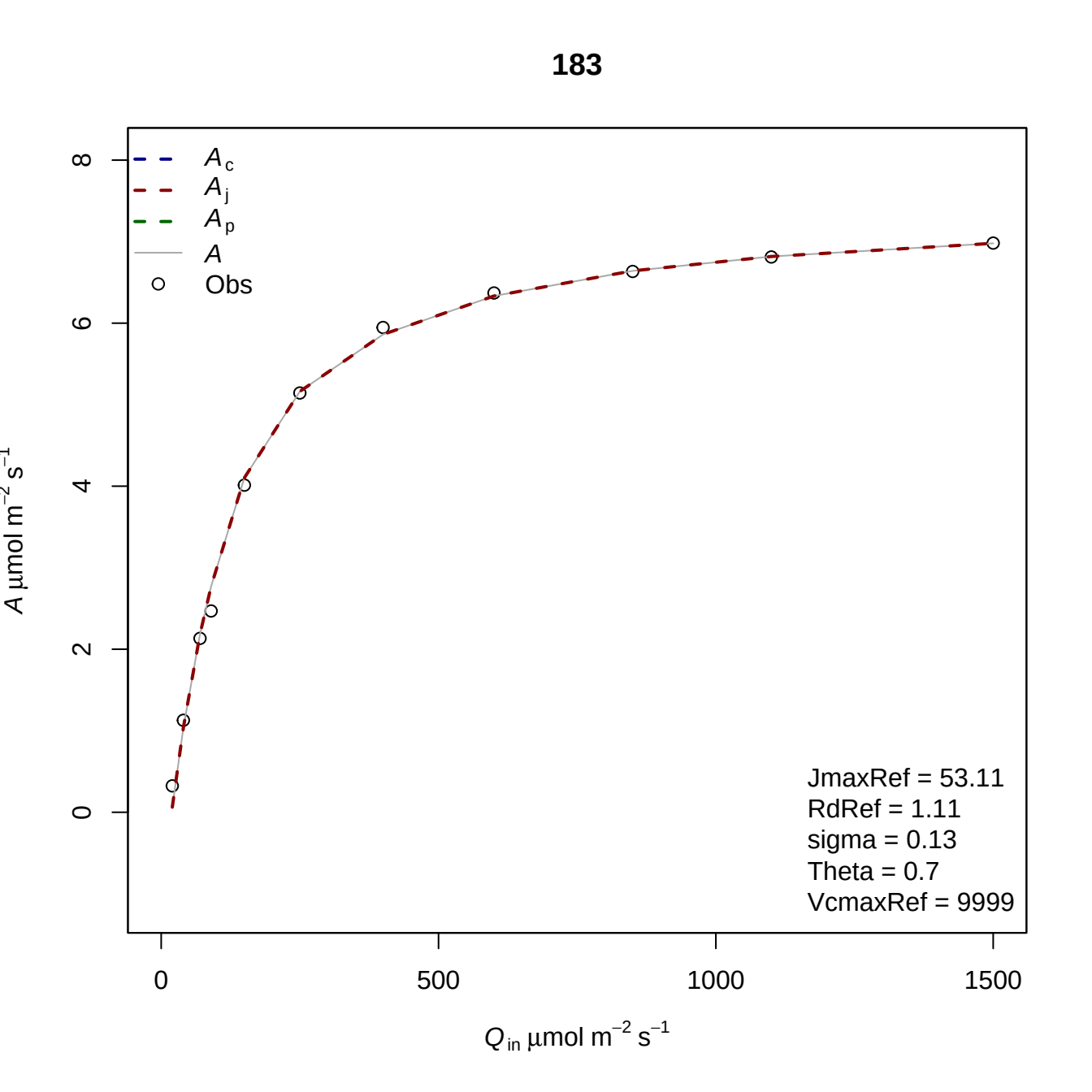

175
A_c
A_j
A_p
A
Obs
JmaxRef = 72.89
RdRef = 4.34
sigma = 0.83
Theta = 0.7
VcmaxRef = 9999
8
6
4
2
0
−2
−4
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

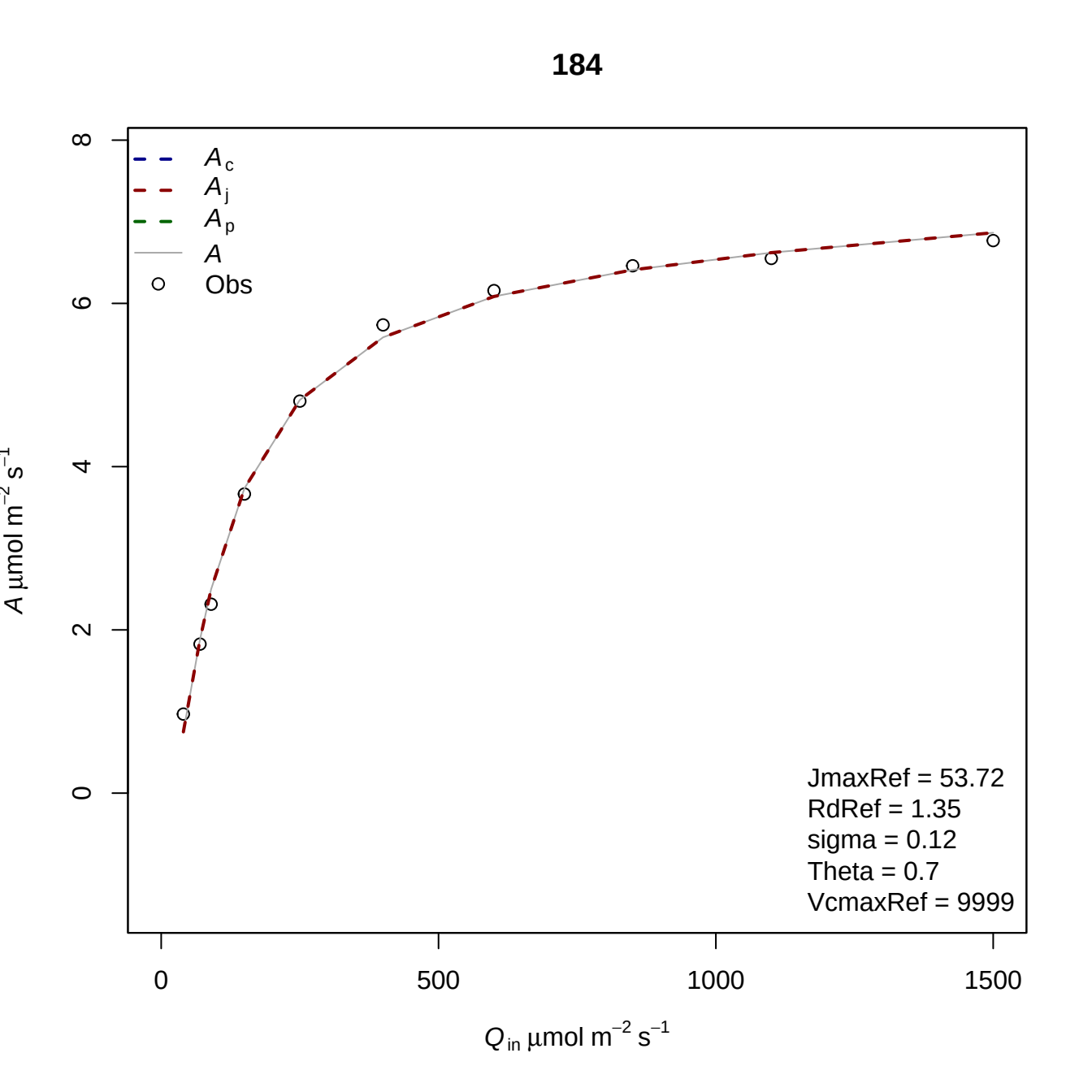

A_c
A_j
A_p
A
Obs
JmaxRef = 88.35
RdRef = 2.46
sigma = 0.23
Theta = 0.7
VcmaxRef = 9999
12
10
8
6
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

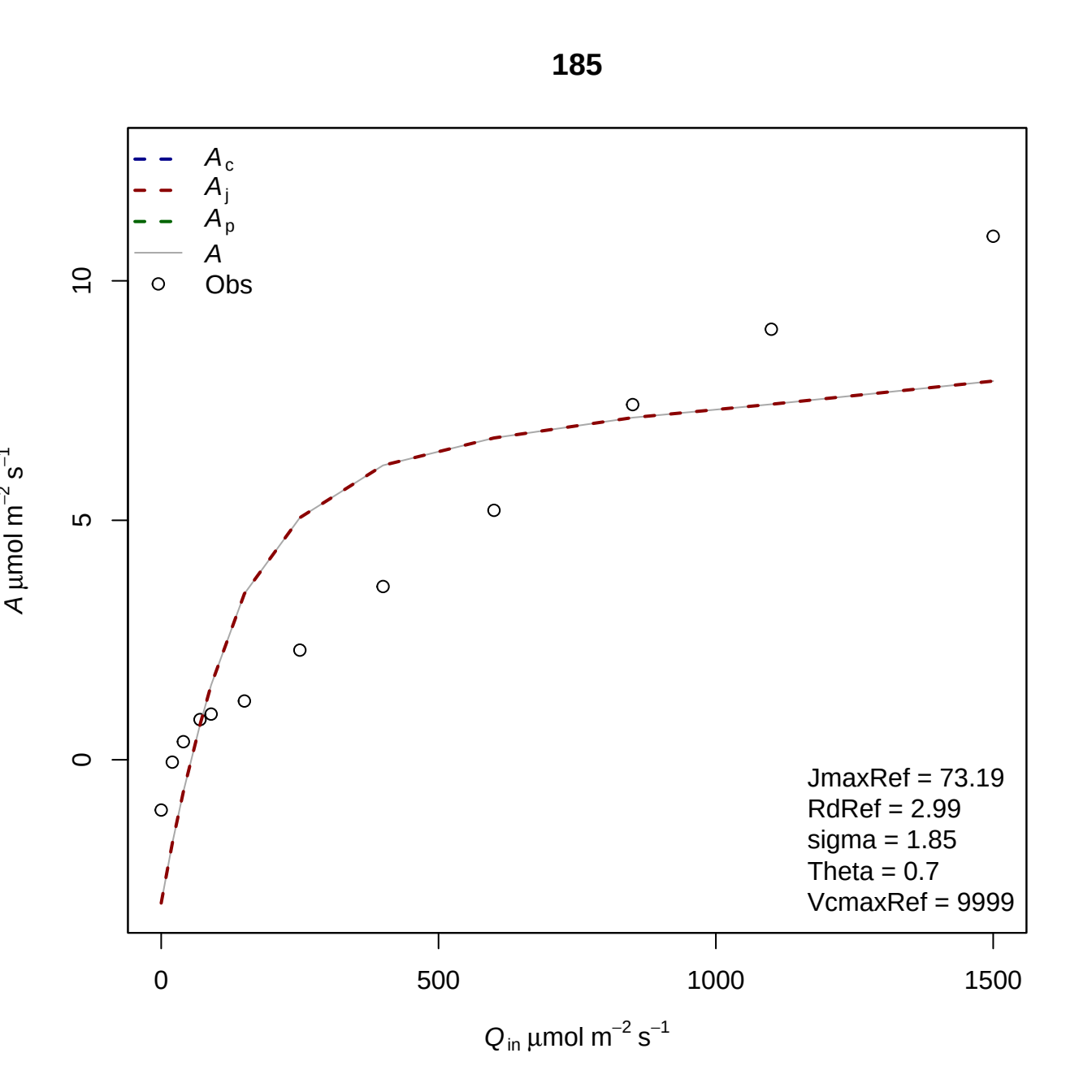

179
A_c
A_j
A_p
A
Obs
10
5
0
−5
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}
JmaxRef = 89.86
RdRef = 5.37
sigma = 0.34
Theta = 0.7
VcmaxRef = 9999

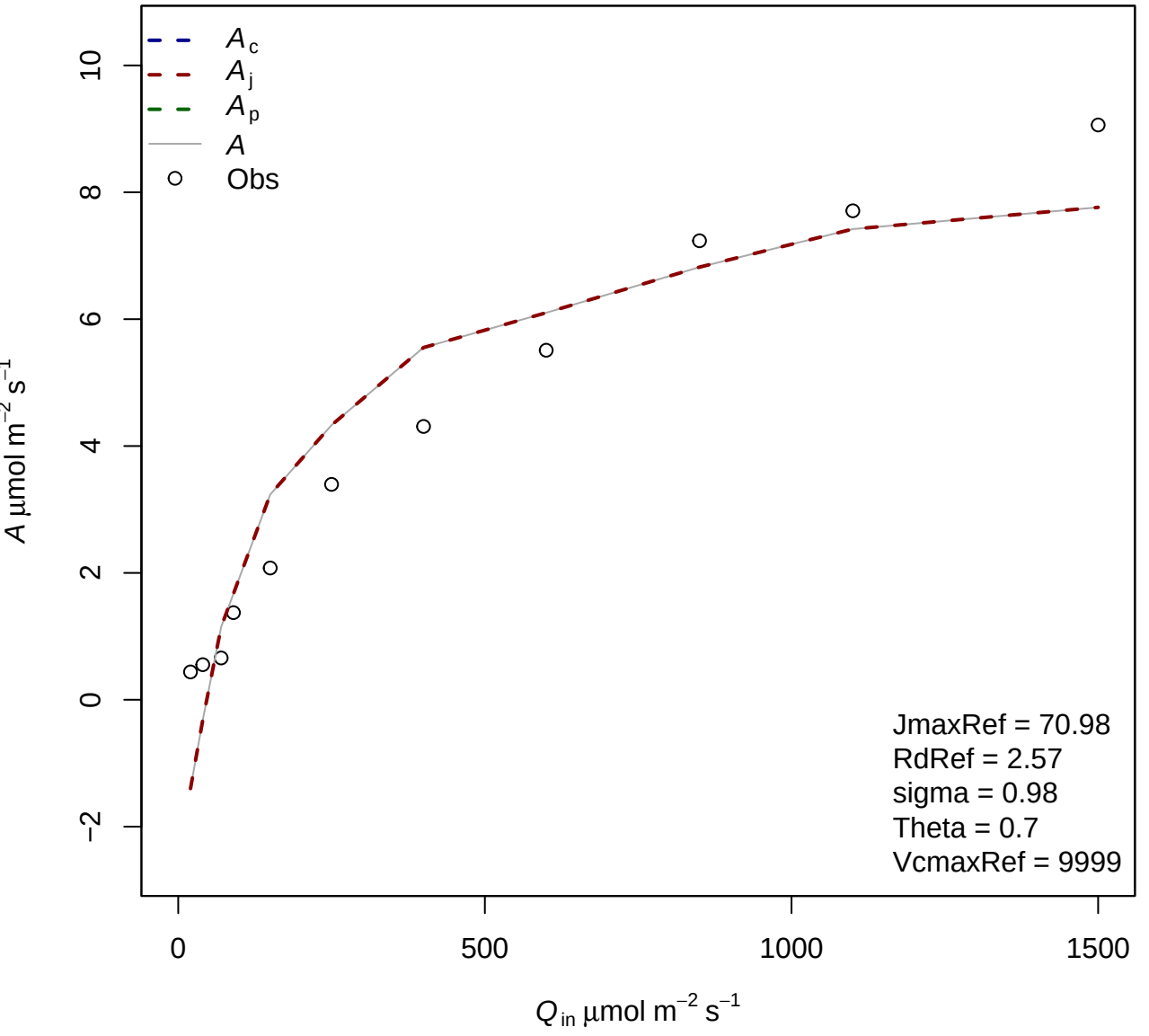

180
A_c
A_j
A_p
A
Obs
JmaxRef = 59.74
RdRef = 1.36
sigma = 0.17
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
200
400
600
800
1000

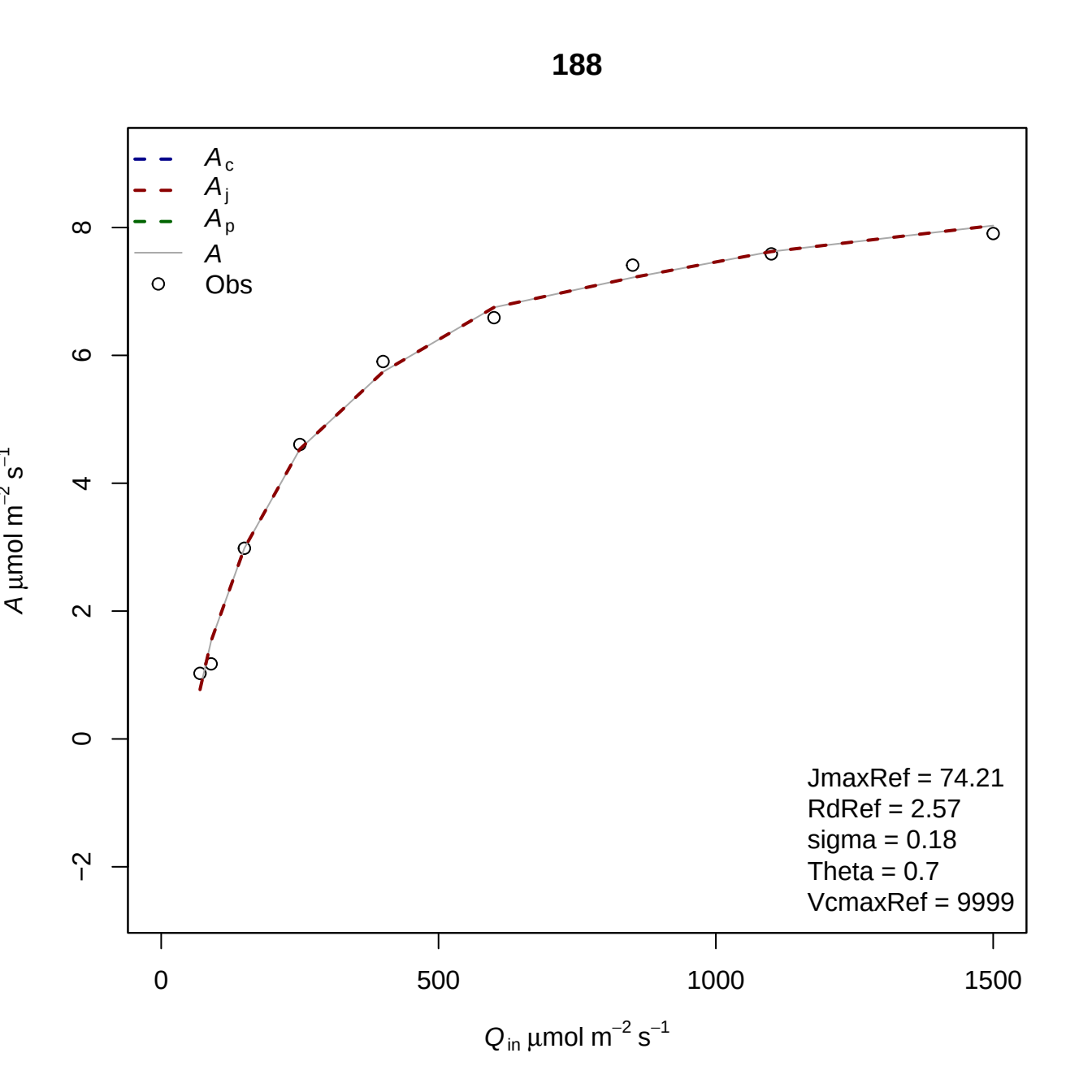

181
A_c
A_j
A_p
A
Obs
JmaxRef = 111.31
RdRef = 1.5
sigma = 0.23
Theta = 0.7
VcmaxRef = 73.54
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

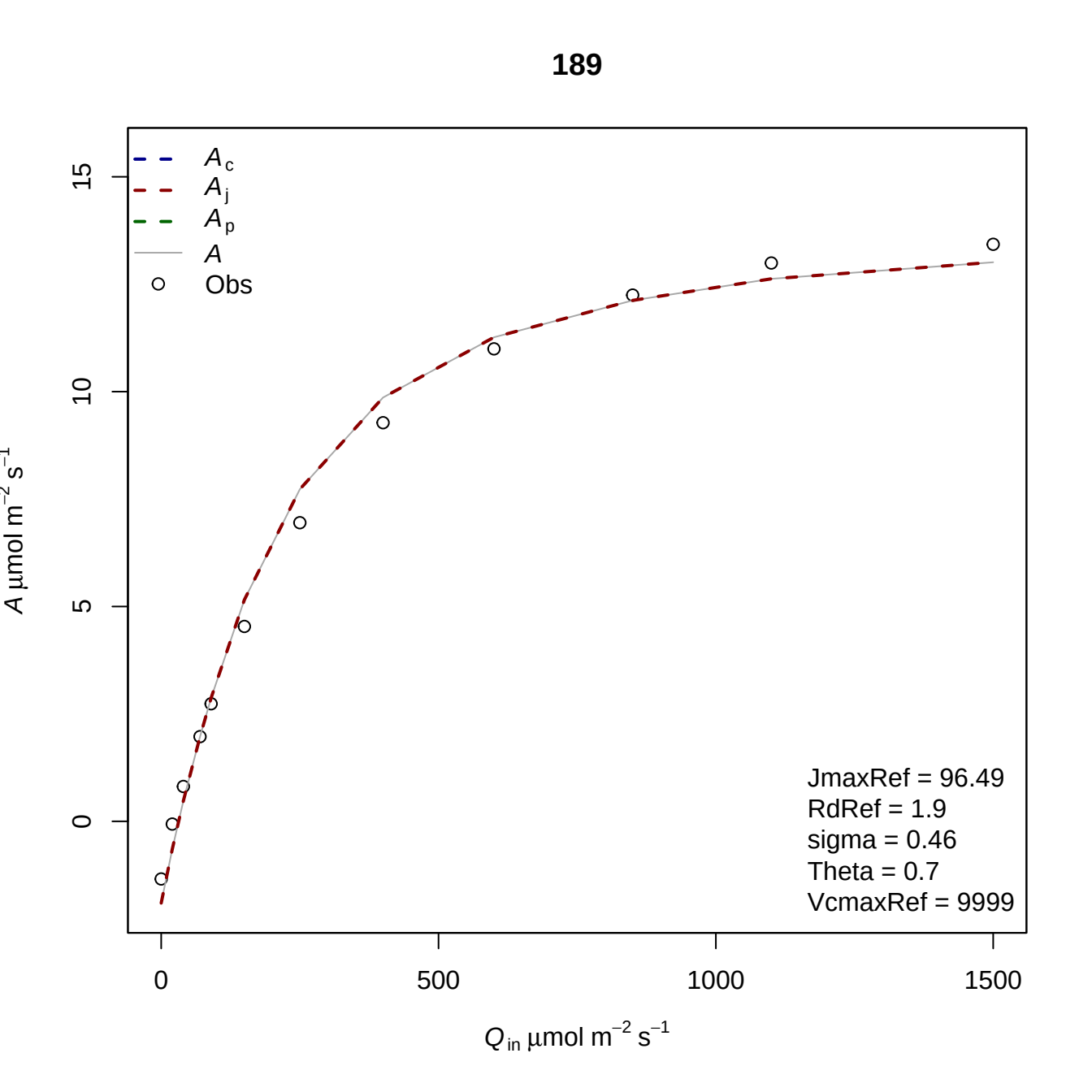

183
A_c
A_j
A_p
A
Obs
JmaxRef = 53.11
RdRef = 1.11
sigma = 0.13
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

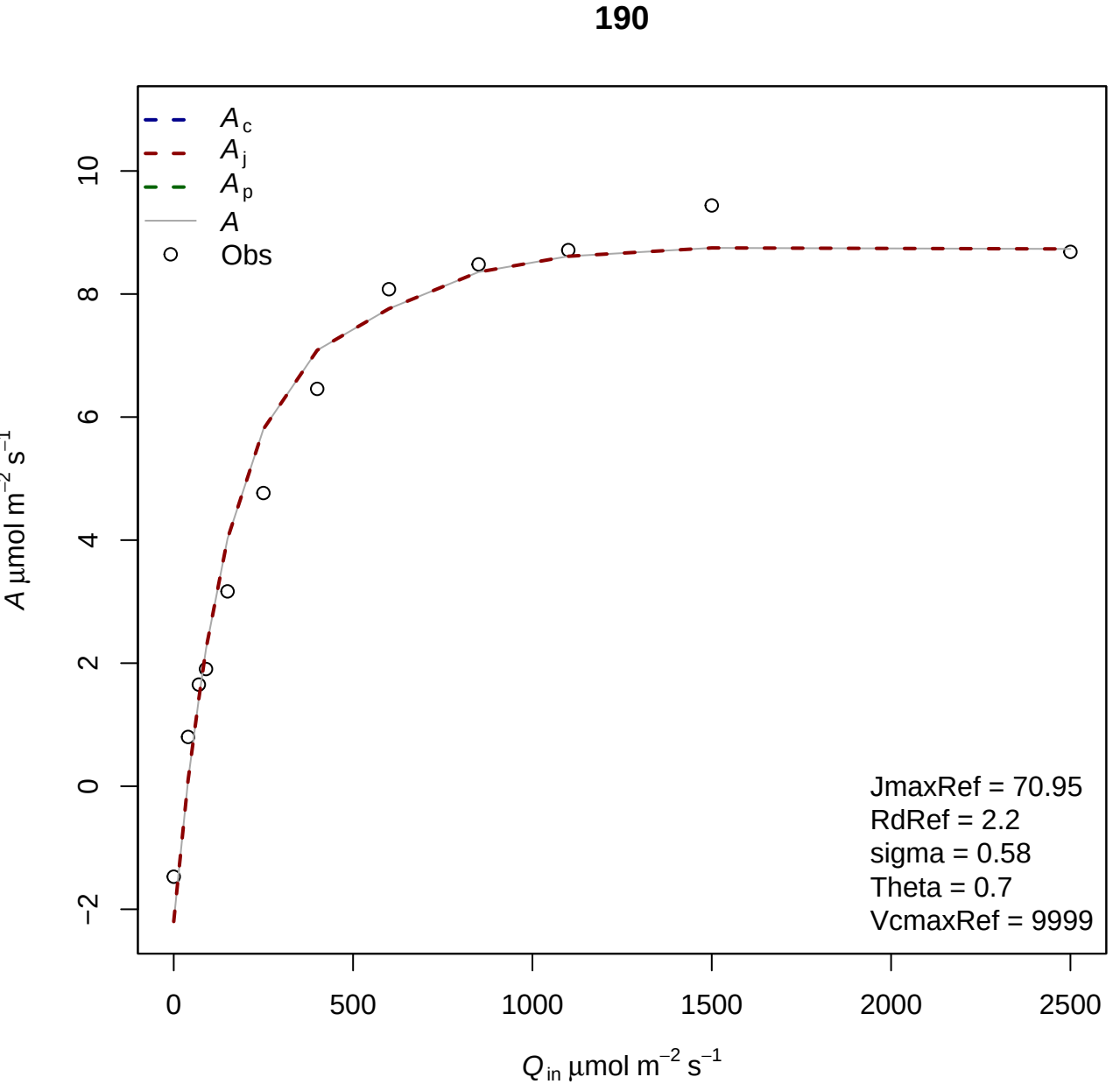

184
A_c
A_j
A_p
A
Obs
JmaxRef = 53.72
RdRef = 1.35
sigma = 0.12
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

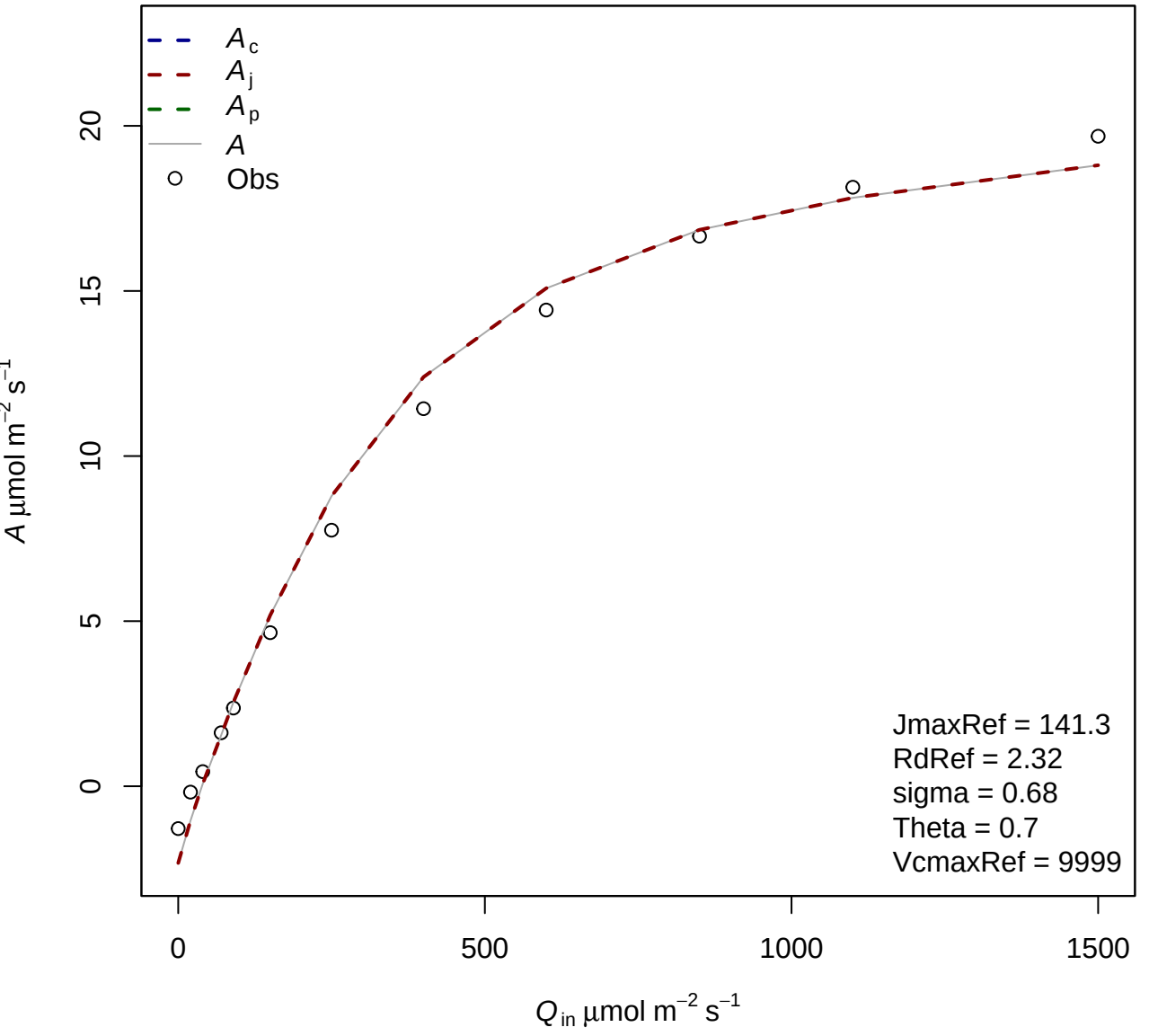

185
A_c
A_j
A_p
A
Obs
JmaxRef = 73.19
RdRef = 2.99
sigma = 1.85
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

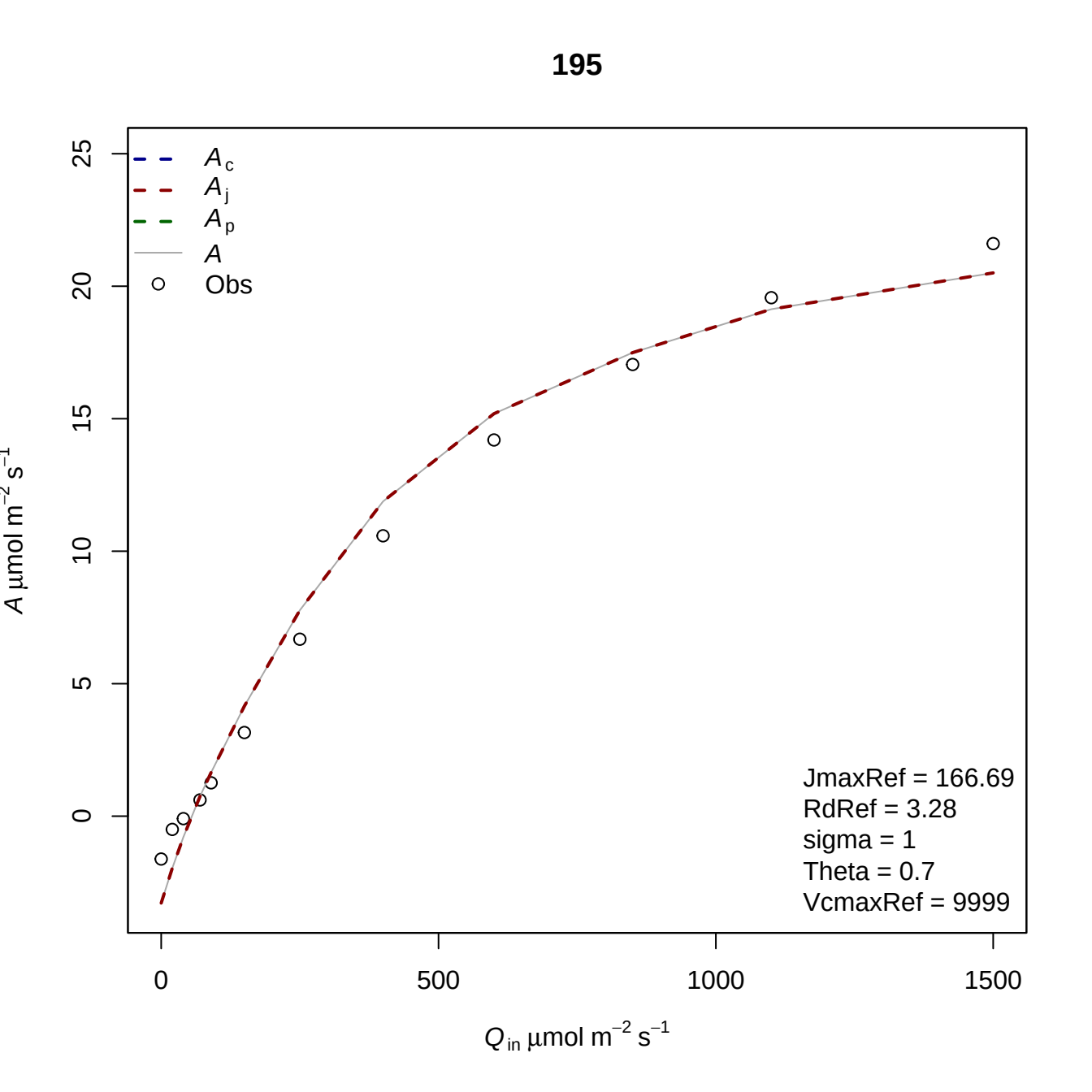

A_c
A_j
A_p
A
Obs
JmaxRef = 70.98
RdRef = 2.57
sigma = 0.98
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

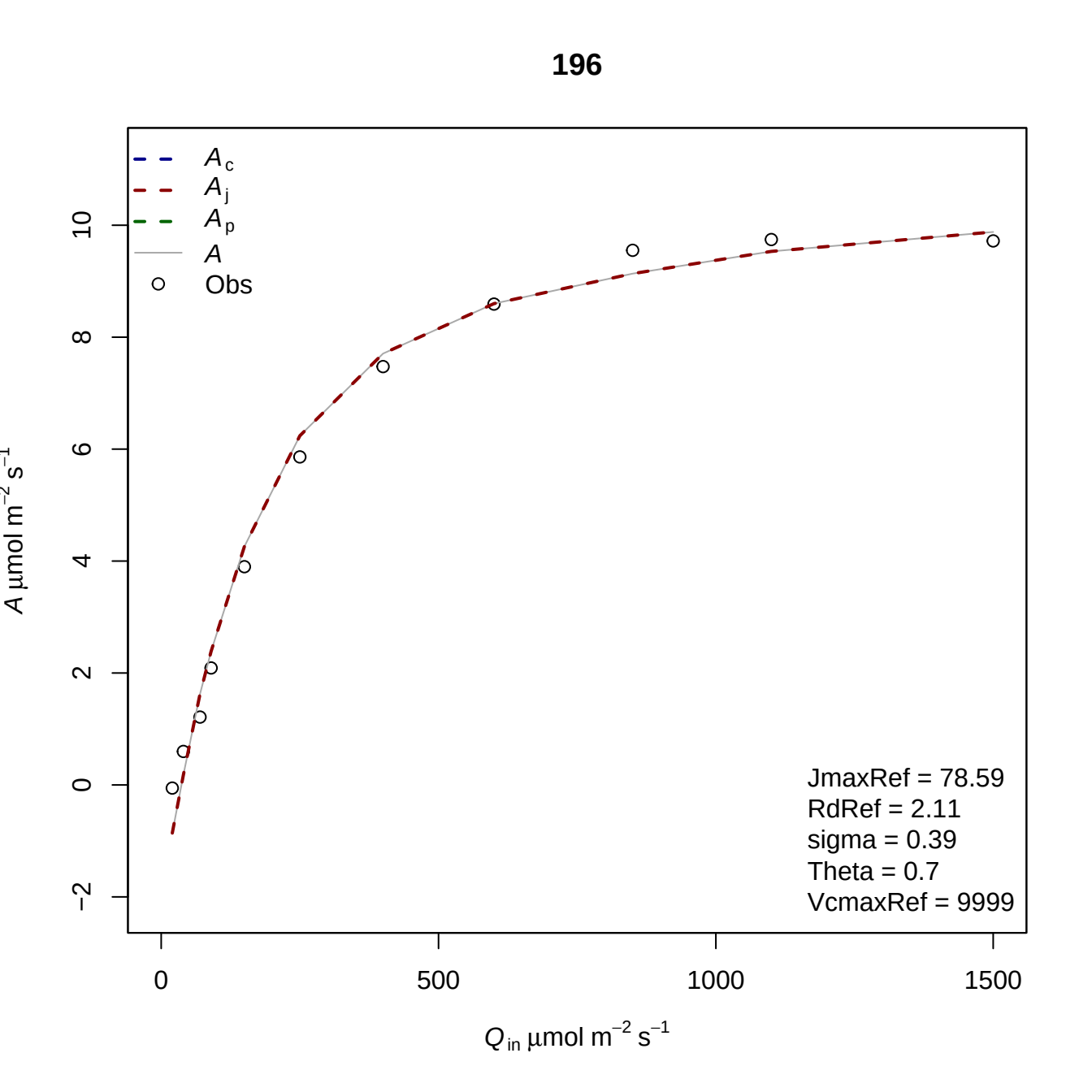

188
A_c
A_j
A_p
A
Obs
JmaxRef = 74.21
RdRef = 2.57
sigma = 0.18
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

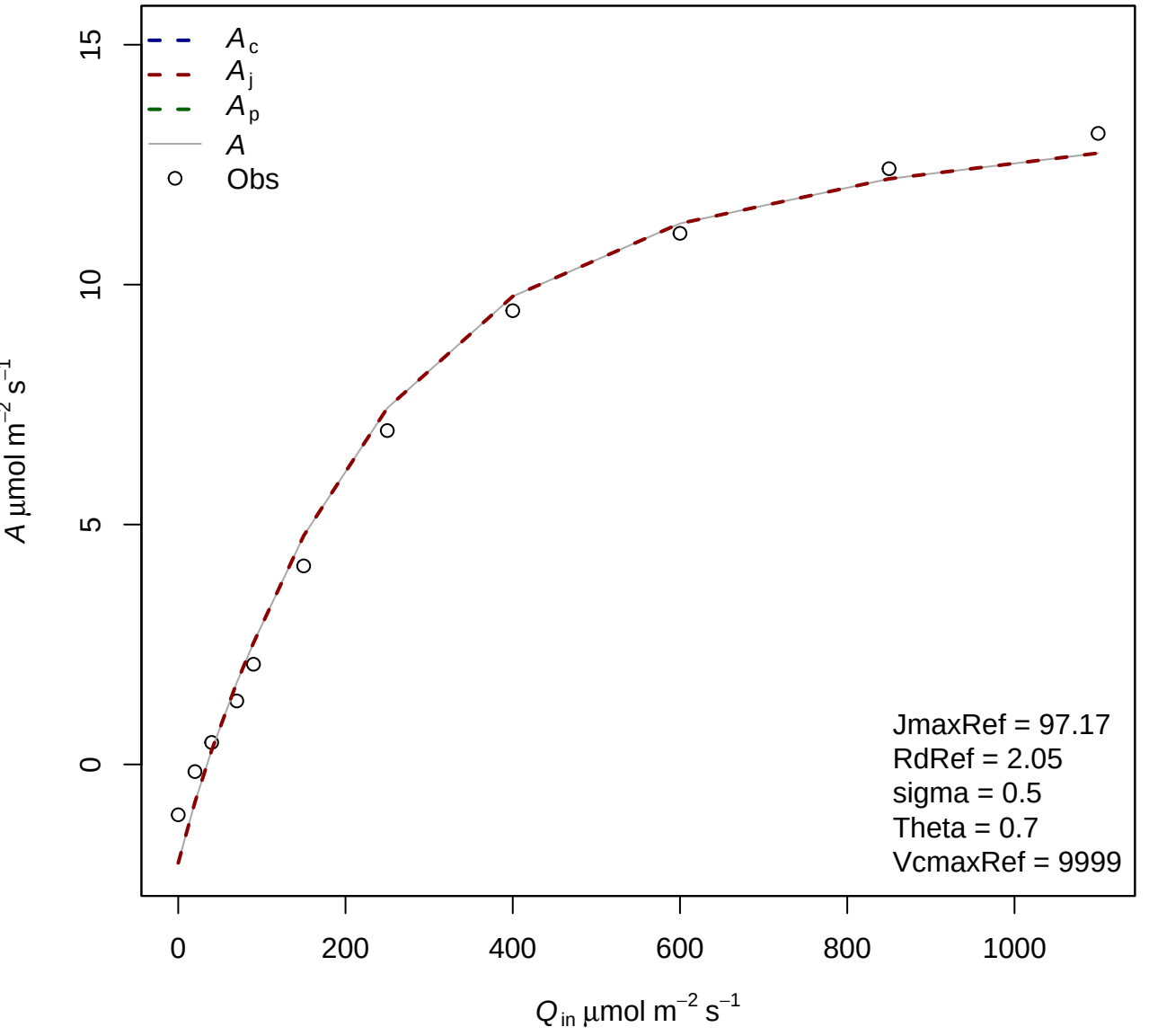

189
A_c
A_j
A_p
A
Obs
JmaxRef = 96.49
RdRef = 1.9
sigma = 0.46
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

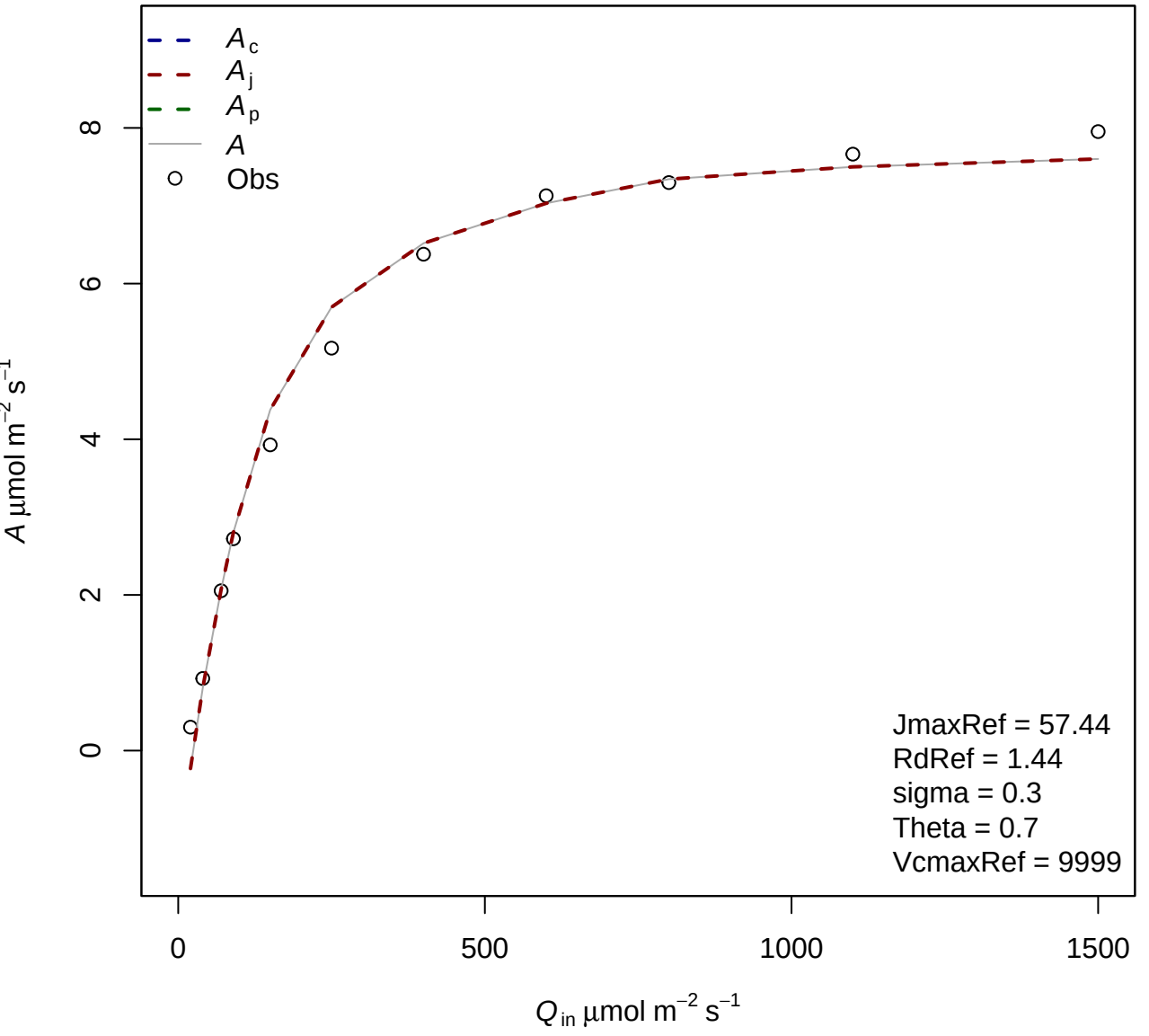

190
A_c
A_j
A_p
A
Obs
JmaxRef = 70.95
RdRef = 2.2
sigma = 0.58
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000
2500

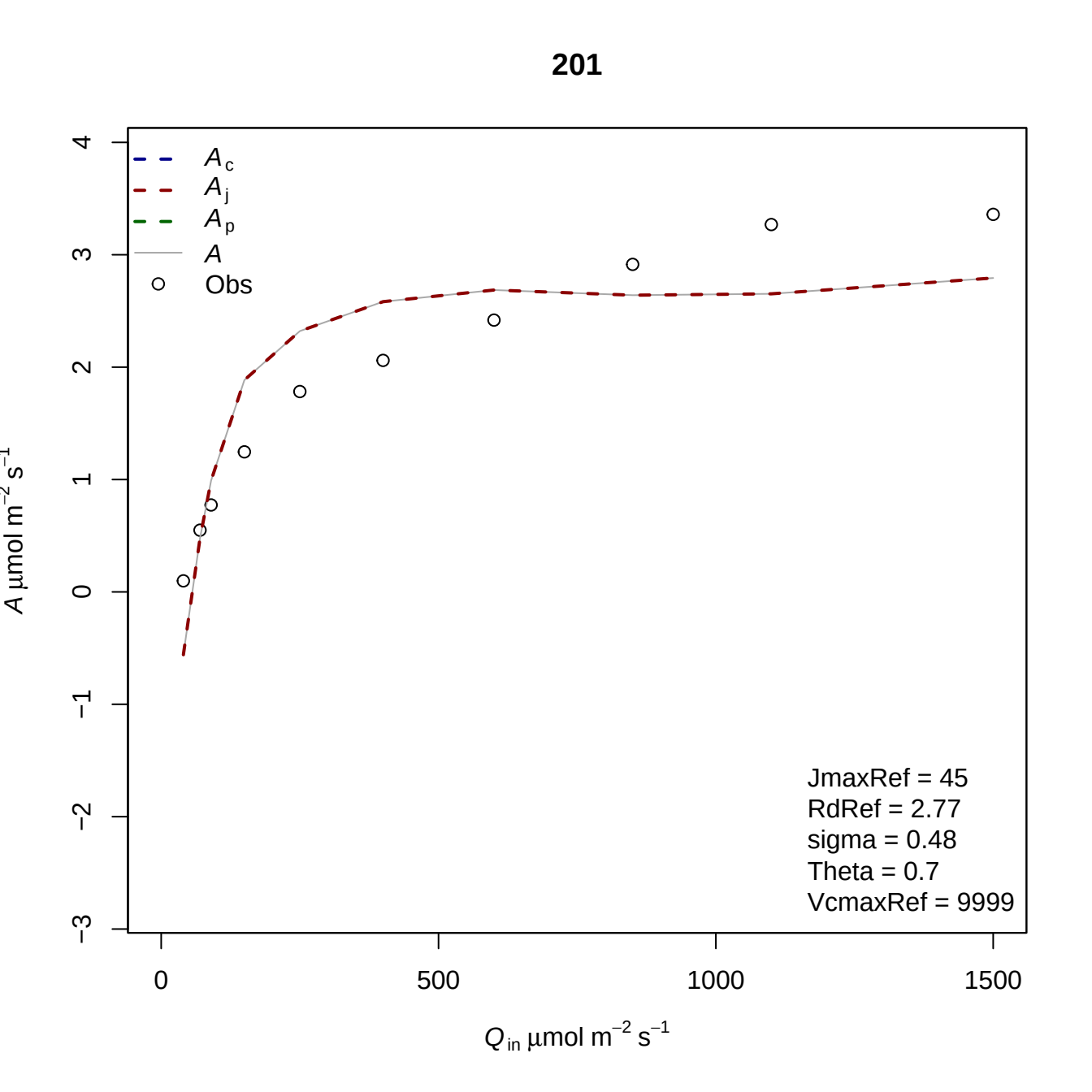

A_c
A_j
A_p
A
Obs
JmaxRef = 141.3
RdRef = 2.32
sigma = 0.68
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

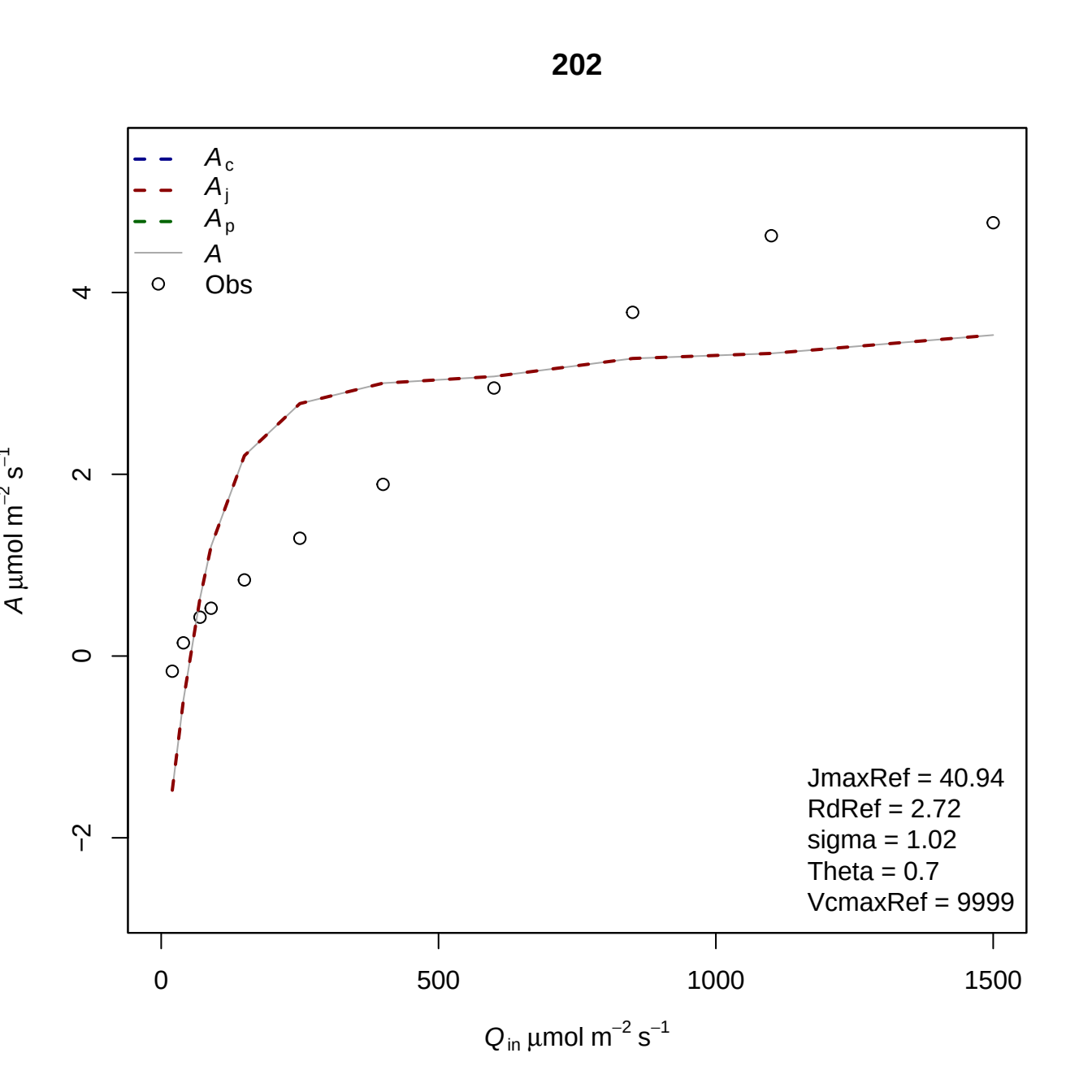

195
A_c
A_j
A_p
A
Obs
JmaxRef = 166.69
RdRef = 3.28
sigma = 1
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

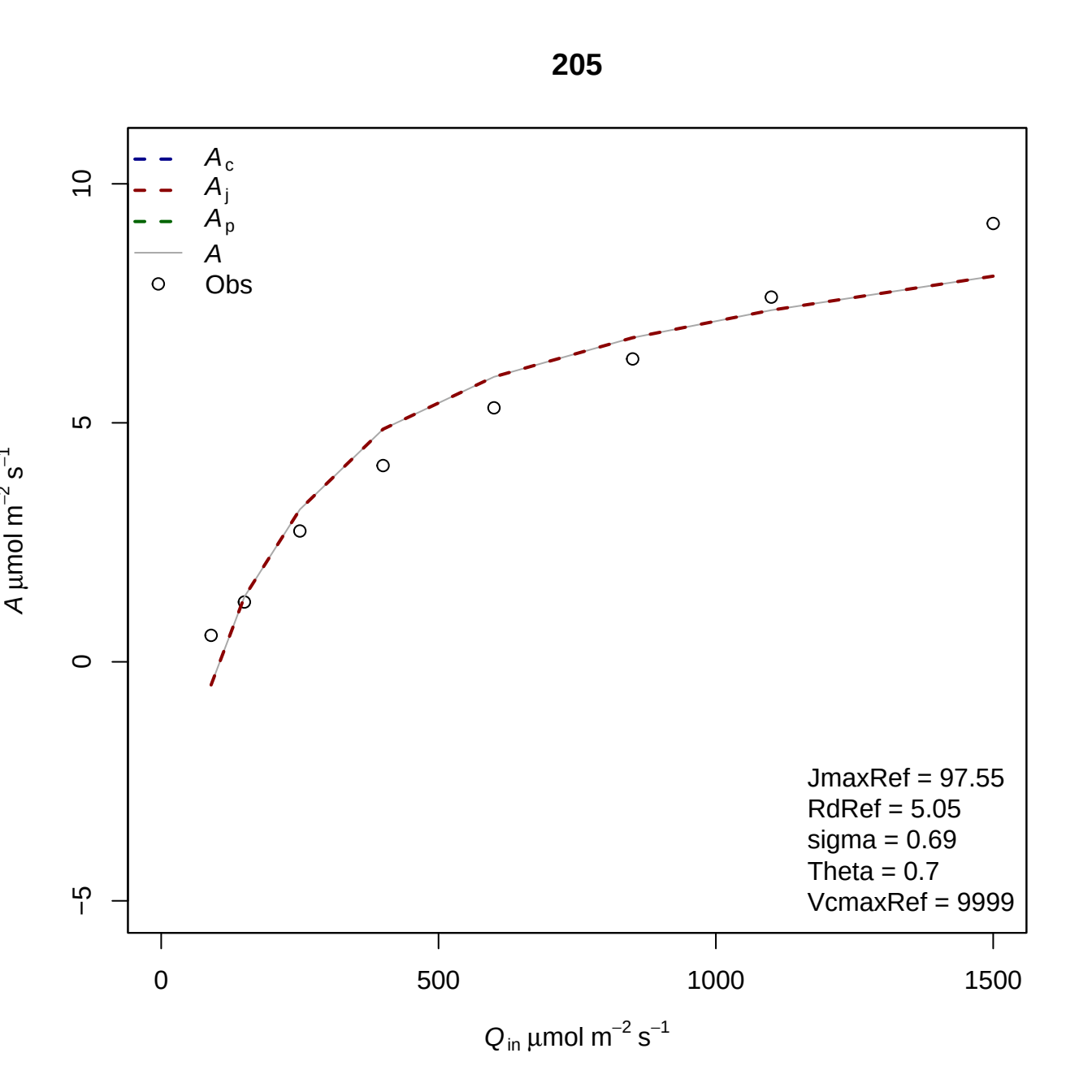

196
A_c
A_j
A_p
A
Obs
JmaxRef = 78.59
RdRef = 2.11
sigma = 0.39
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

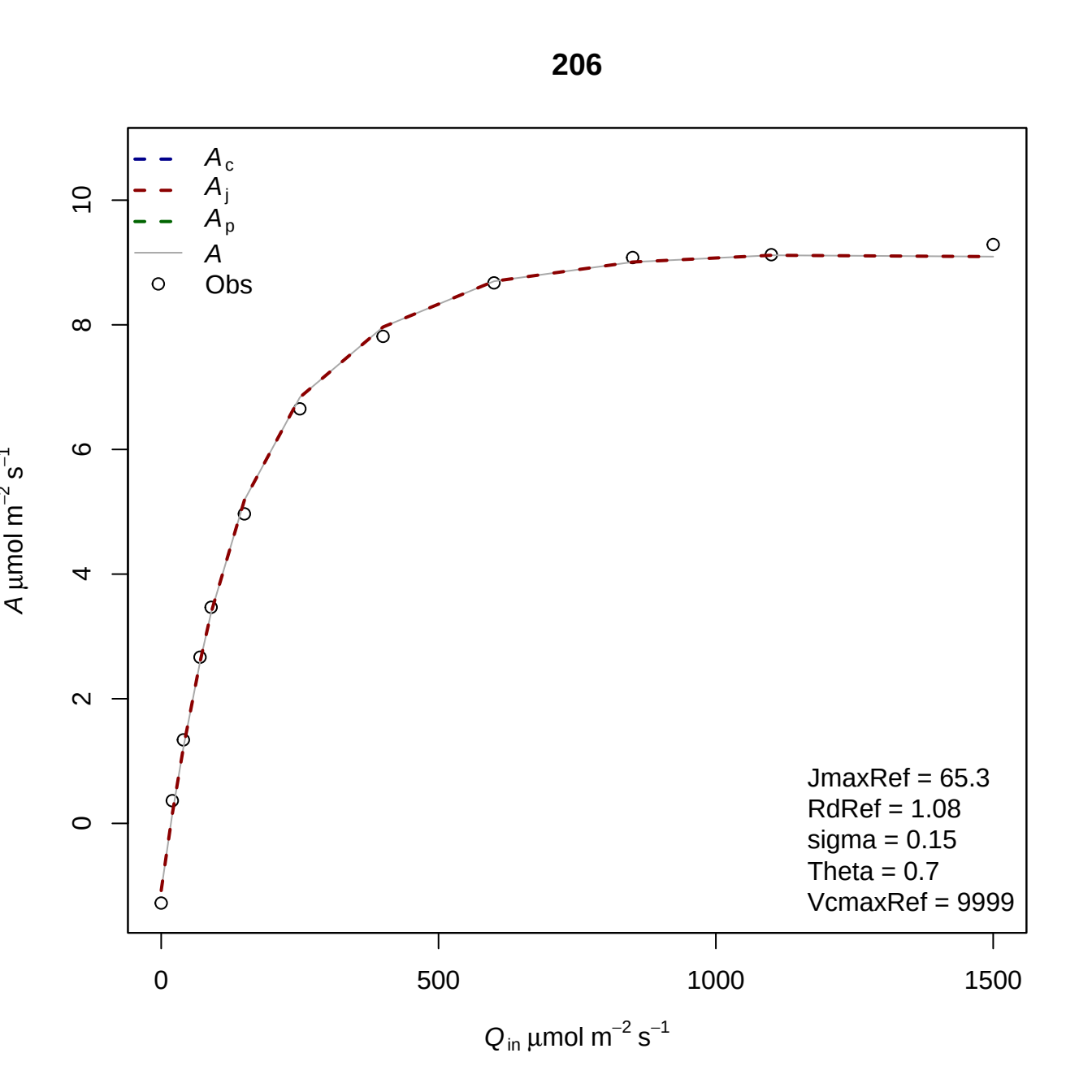

A_c
A_j
A_p
A
Obs
JmaxRef = 97.17
RdRef = 2.05
sigma = 0.5
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
200
400
600
800
1000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

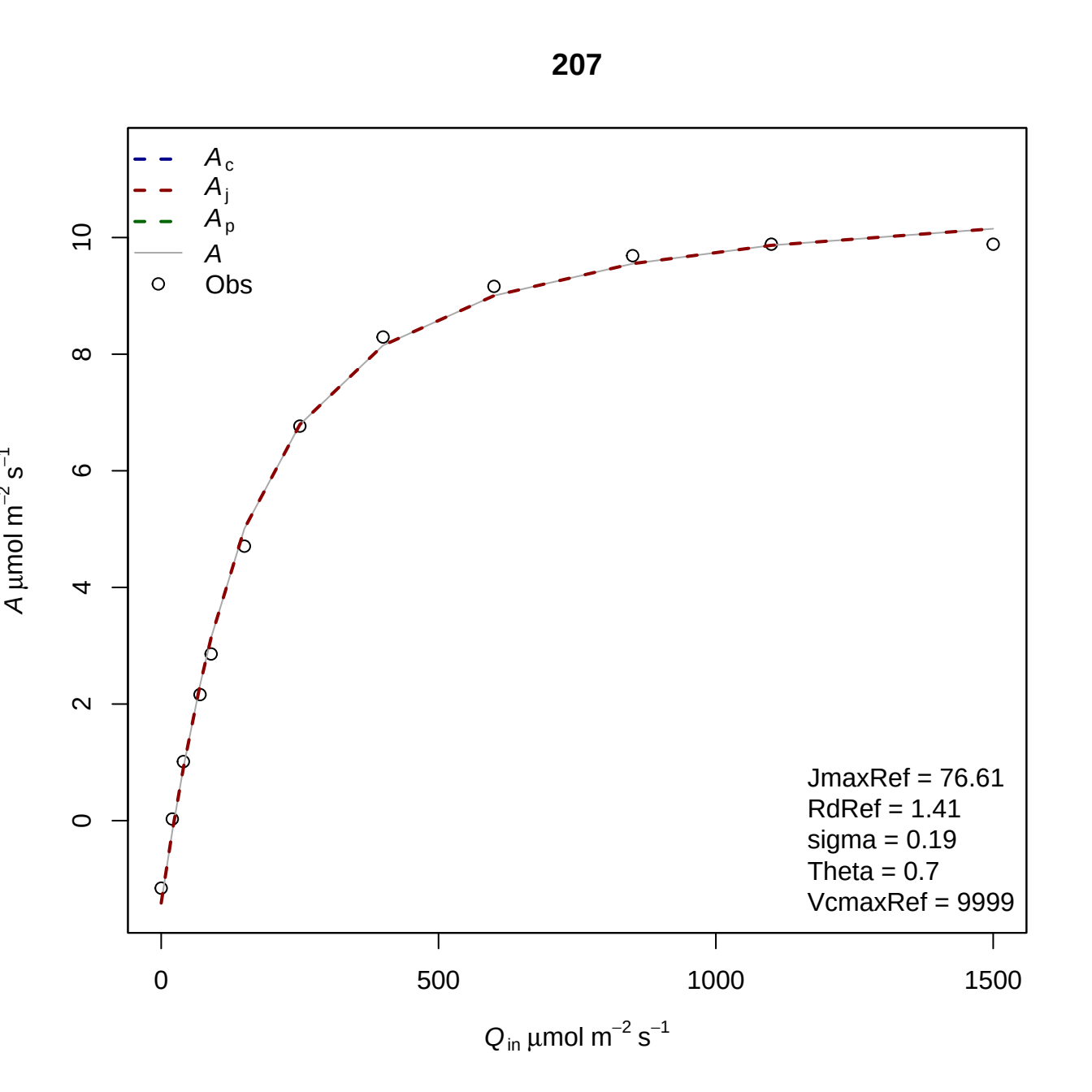

A_c
A_j
A_p
A
Obs
JmaxRef = 57.44
RdRef = 1.44
sigma = 0.3
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

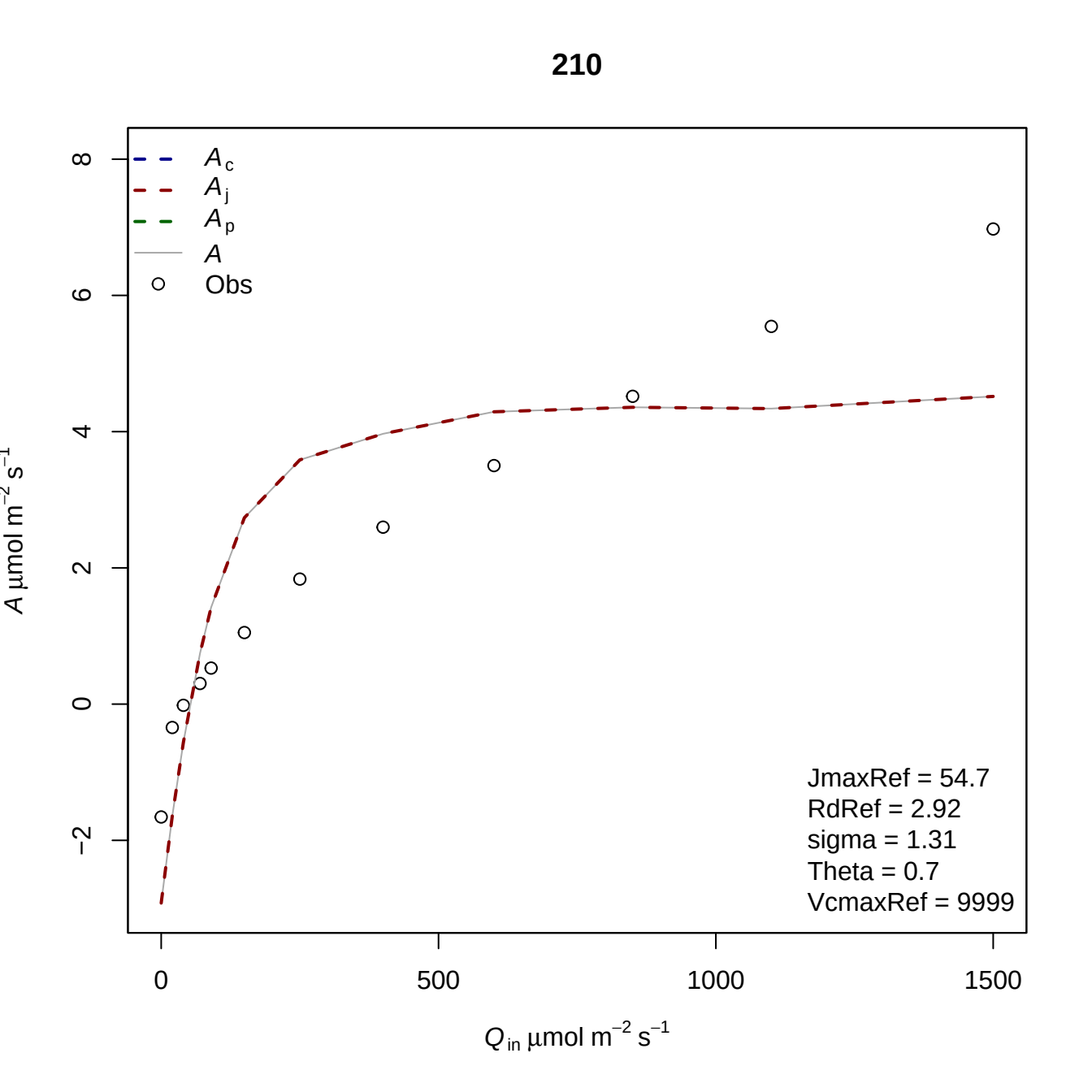

201
A_c
A_j
A_p
A
Obs
JmaxRef = 45
RdRef = 2.77
sigma = 0.48
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
−3
0
500
1000
1500

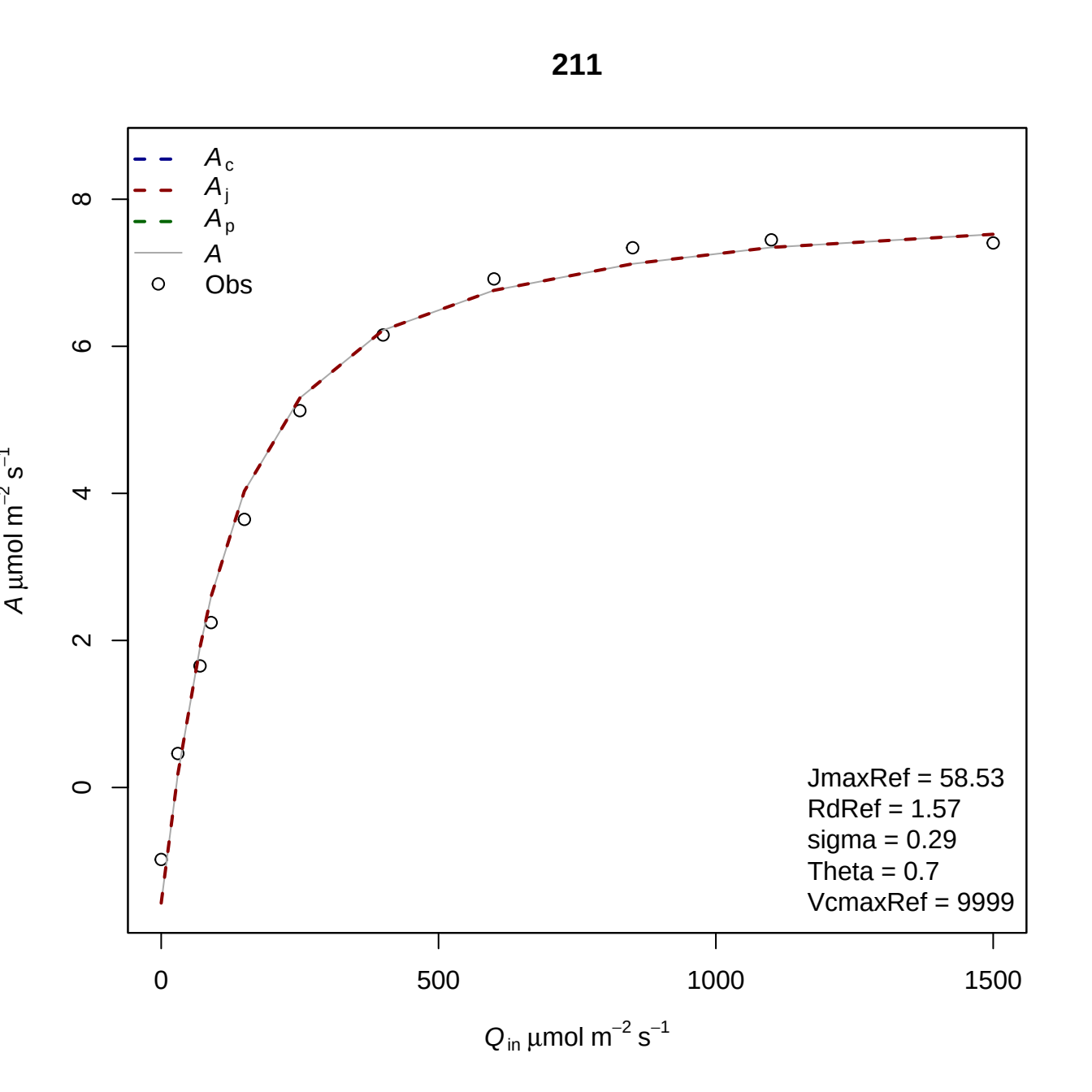

202
A_c
A_j
A_p
A
Obs
JmaxRef = 40.94
RdRef = 2.72
sigma = 1.02
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500

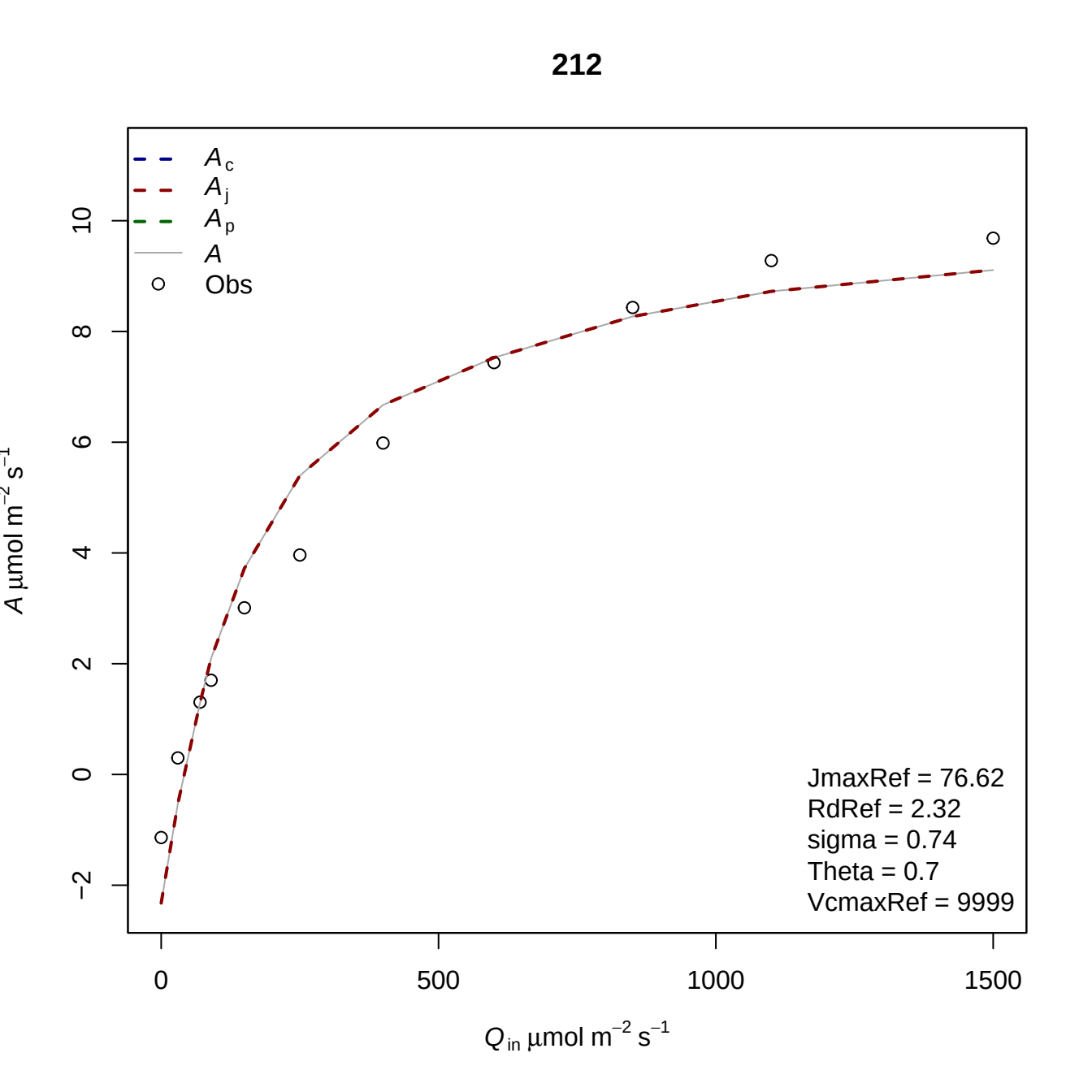

205
A_c
A_j
A_p
A
Obs
JmaxRef = 97.55
RdRef = 5.05
sigma = 0.69
Theta = 0.7
VcmaxRef = 9999
10
5
0
−5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

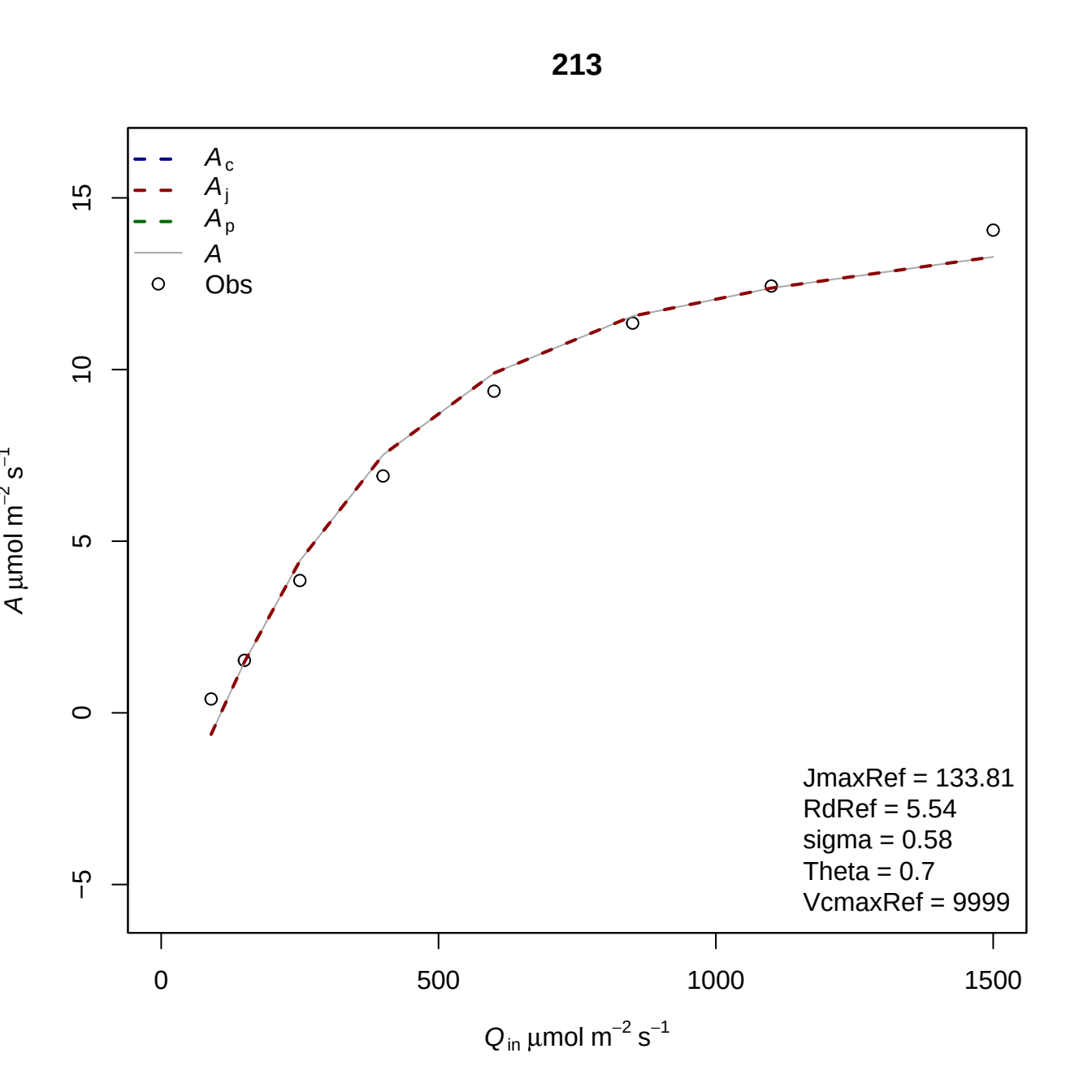

206
A_c
A_j
A_p
A
Obs
JmaxRef = 65.3
RdRef = 1.08
sigma = 0.15
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

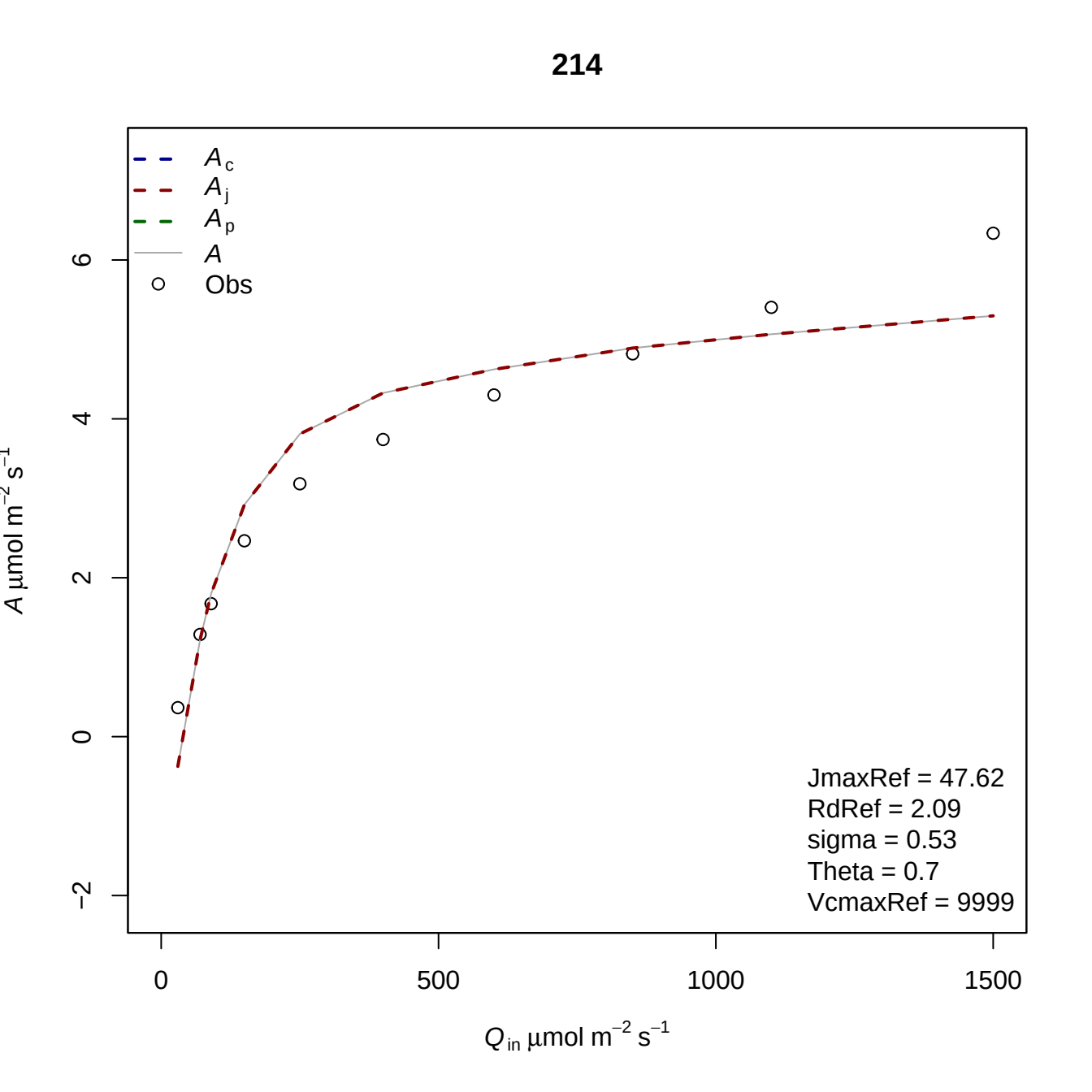

207
A_c
A_j
A_p
A
Obs
JmaxRef = 76.61
RdRef = 1.41
sigma = 0.19
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

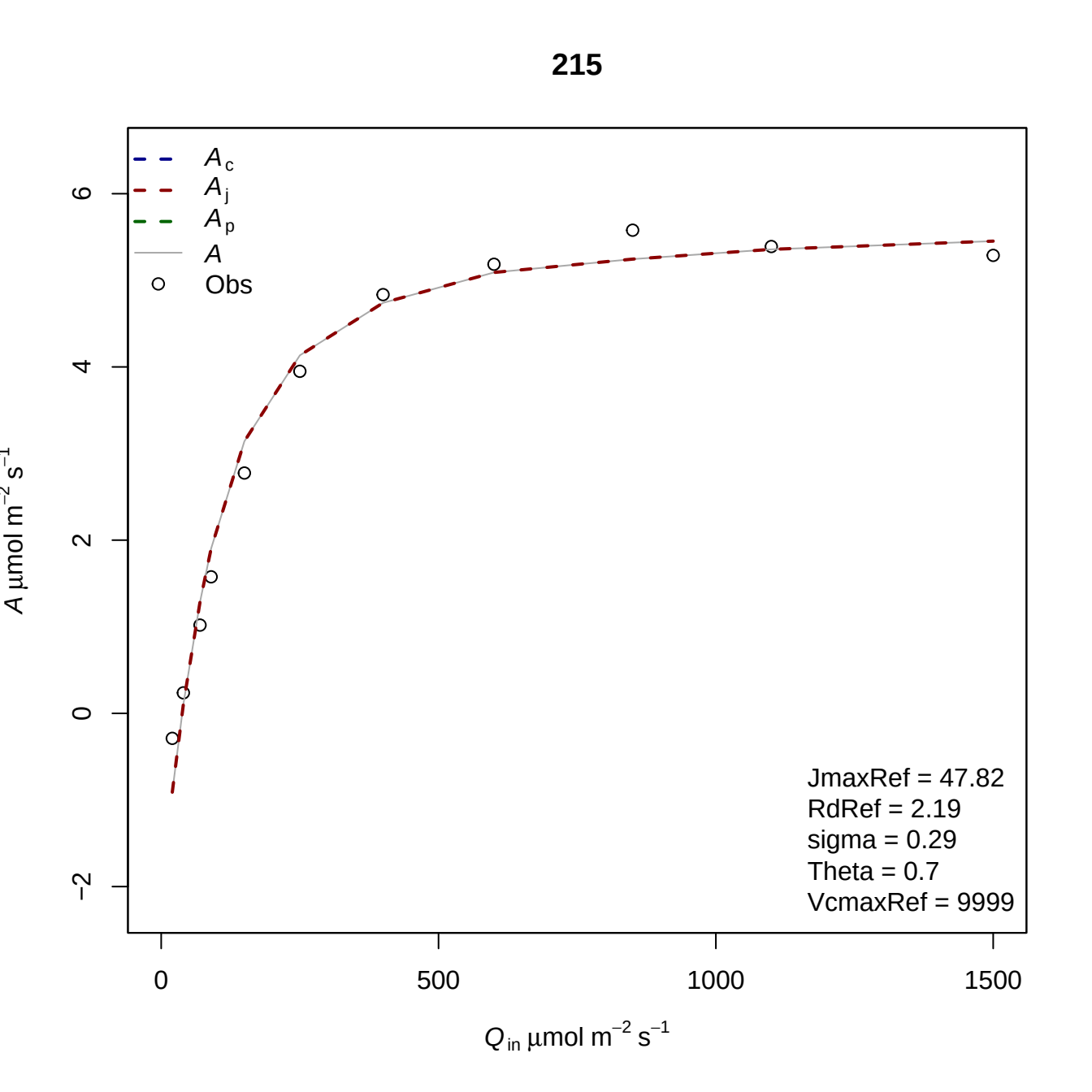

210
A_c
A_j
A_p
A
Obs
JmaxRef = 54.7
RdRef = 2.92
sigma = 1.31
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

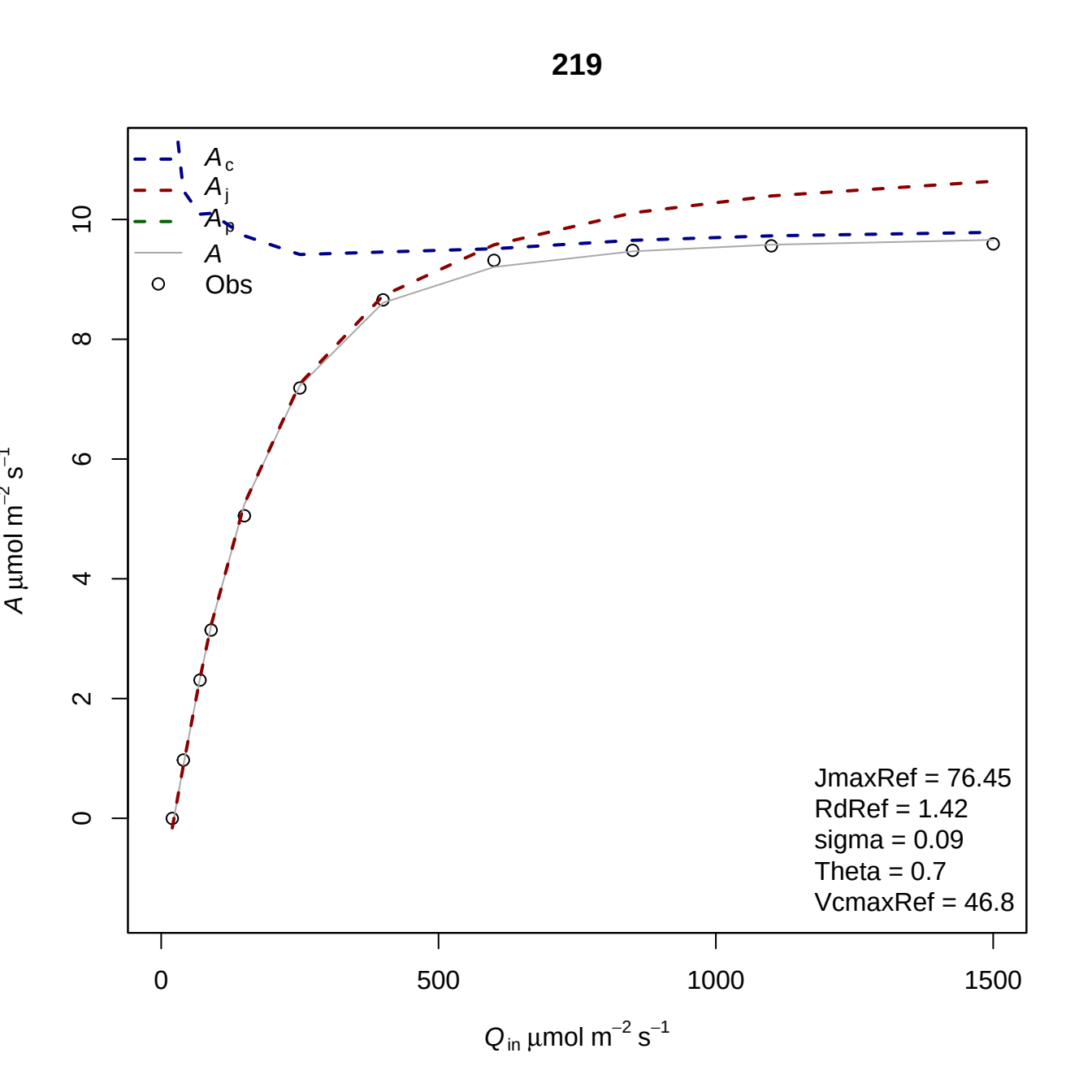

211
A_c
A_j
A_p
A
Obs
JmaxRef = 58.53
RdRef = 1.57
sigma = 0.29
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

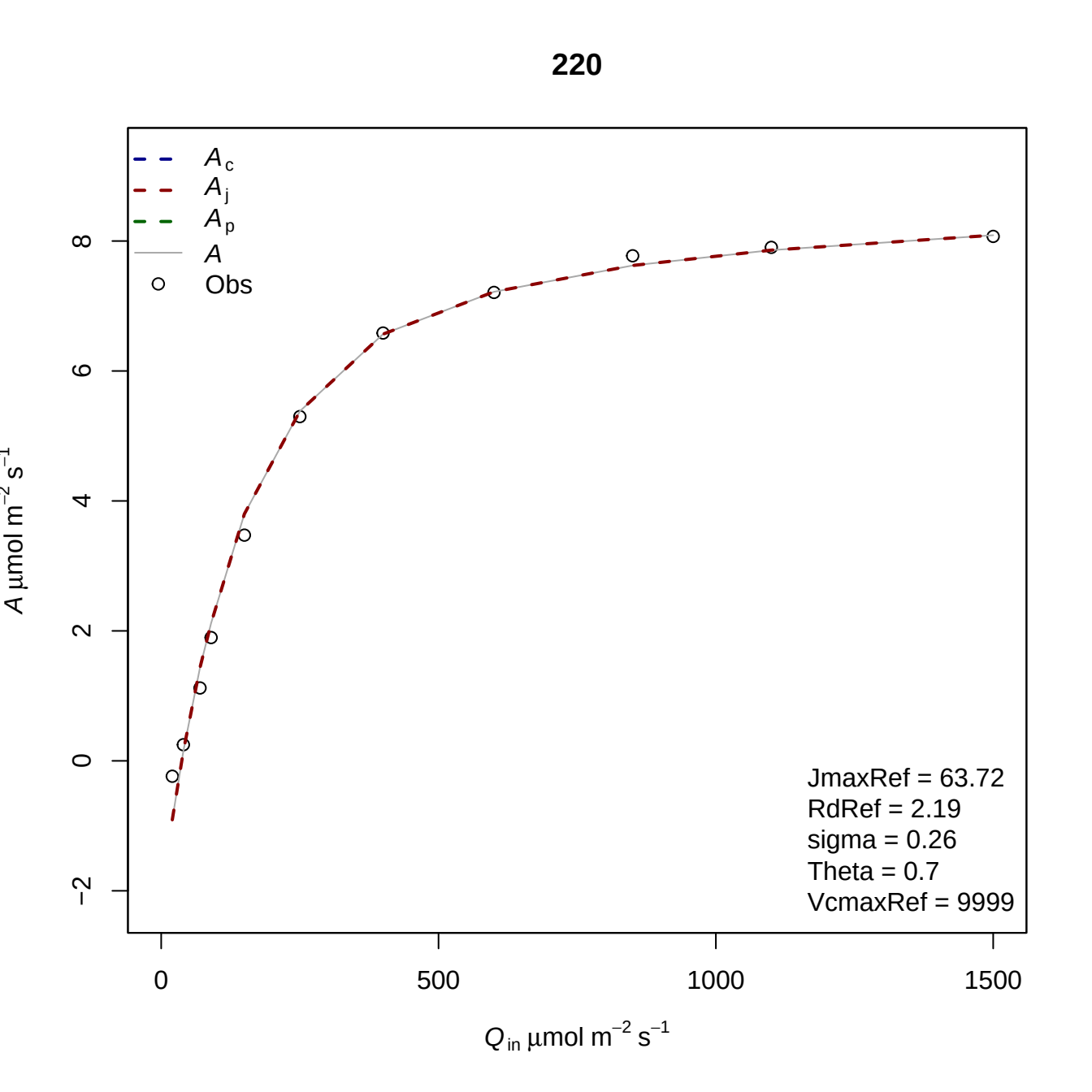

212
A_c
A_j
A_p
A
Obs
JmaxRef = 76.62
RdRef = 2.32
sigma = 0.74
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

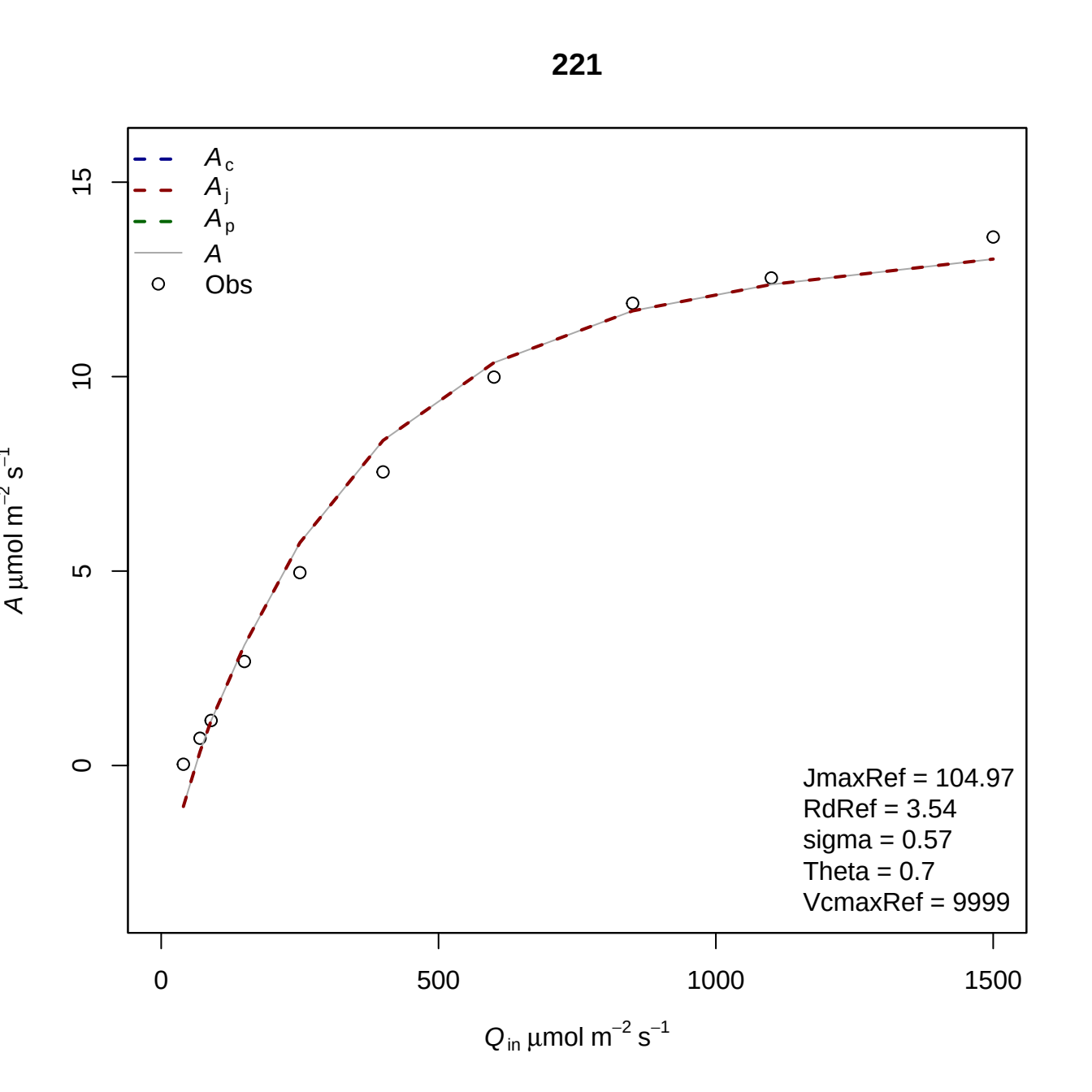

213
A_c
A_j
A_p
A
Obs
15
10
5
0
−5
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}
JmaxRef = 133.81
RdRef = 5.54
sigma = 0.58
Theta = 0.7
VcmaxRef = 9999

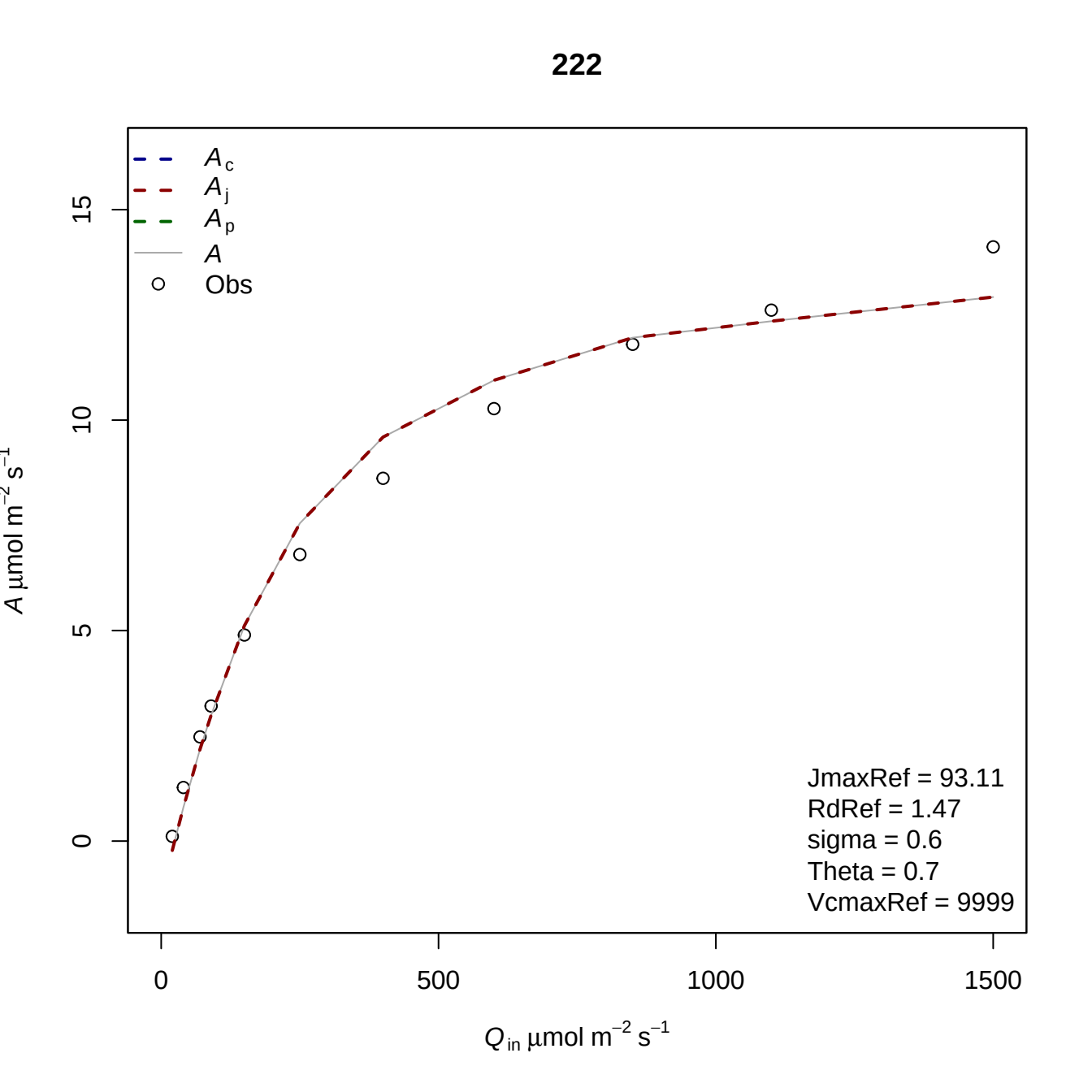

214
A_c
A_j
A_p
A
Obs
JmaxRef = 47.62
RdRef = 2.09
sigma = 0.53
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

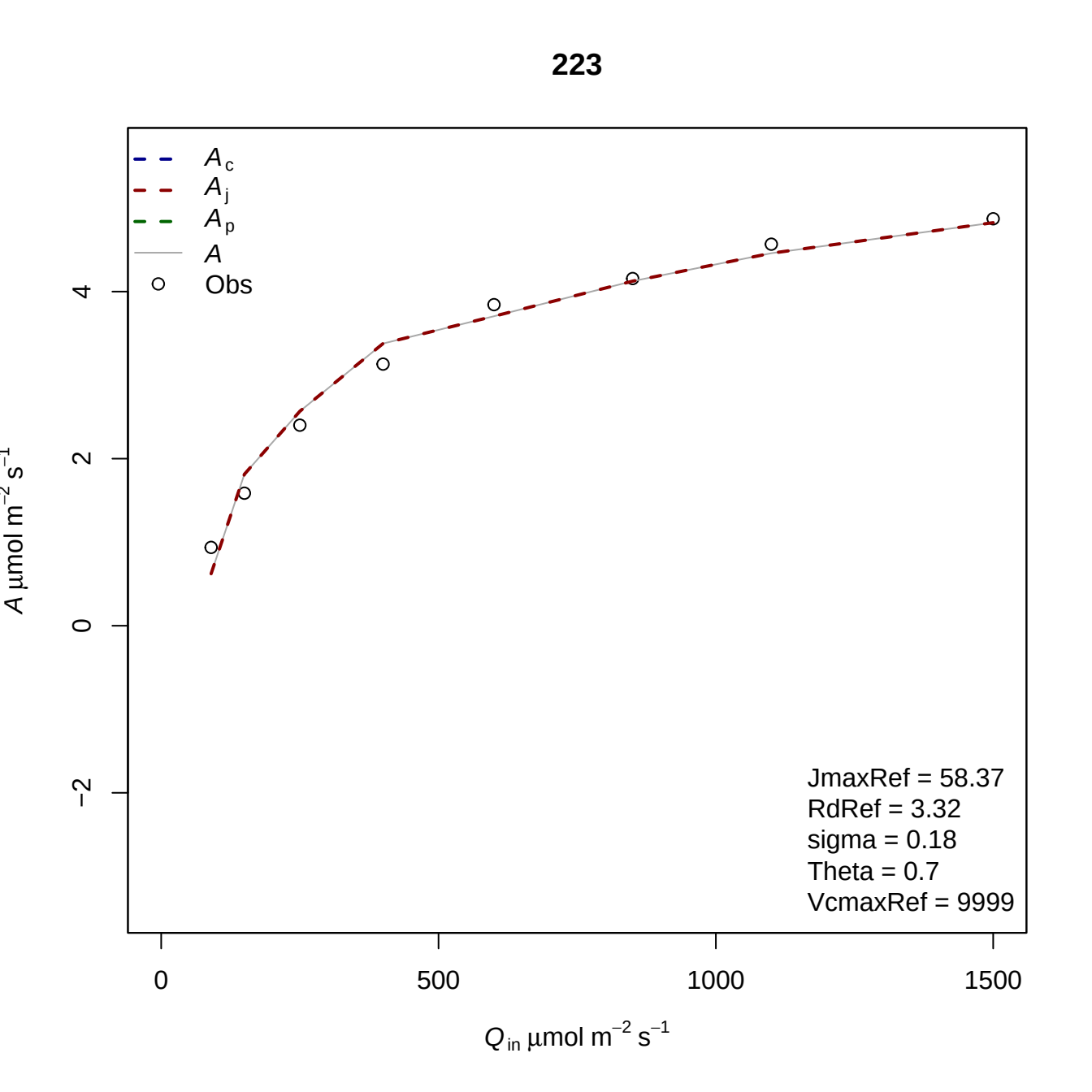

215
A_c
A_j
A_p
A
Obs
JmaxRef = 47.82
RdRef = 2.19
sigma = 0.29
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

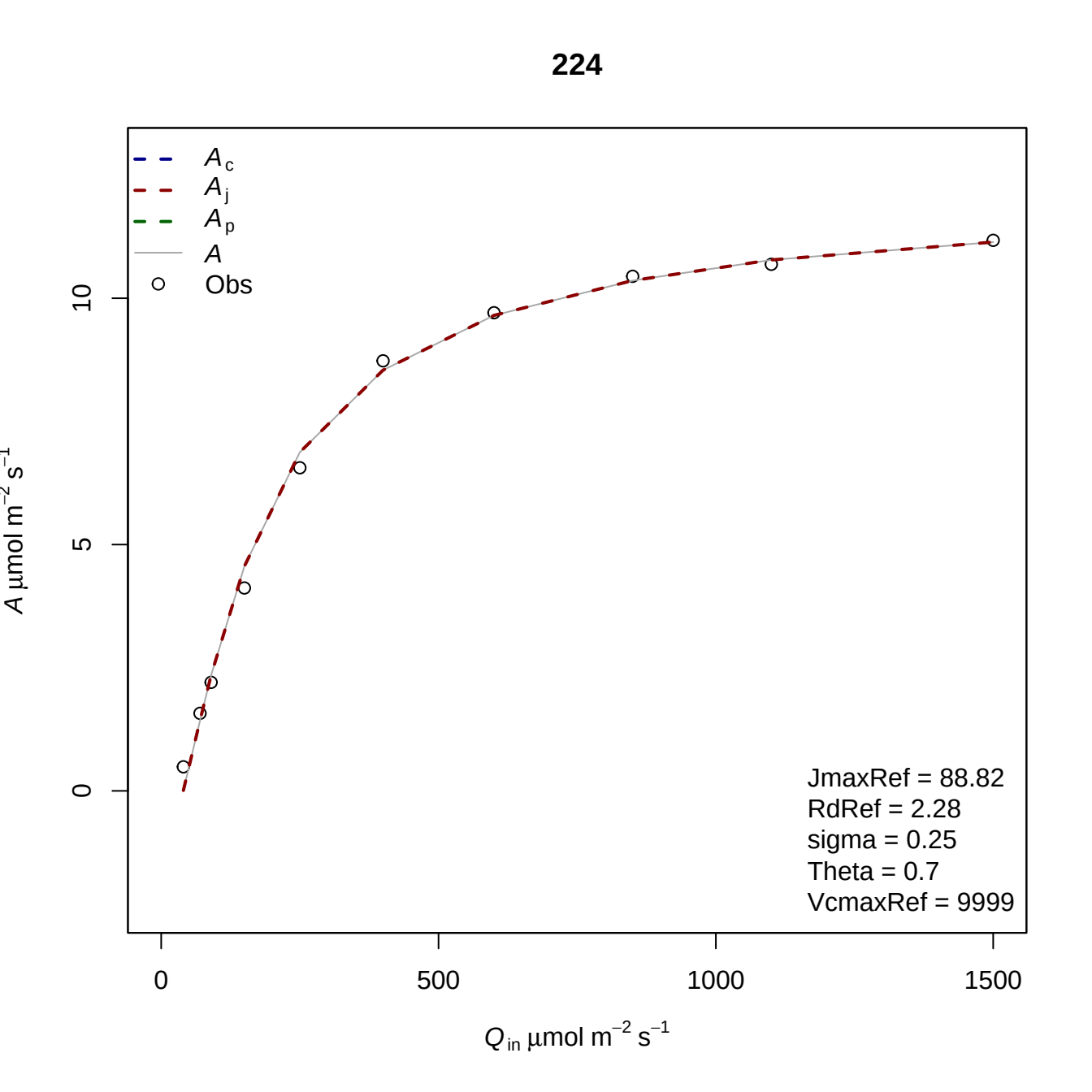

219
A_c
A_j
A_p
A
Obs
JmaxRef = 76.45
RdRef = 1.42
sigma = 0.09
Theta = 0.7
VcmaxRef = 46.8
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

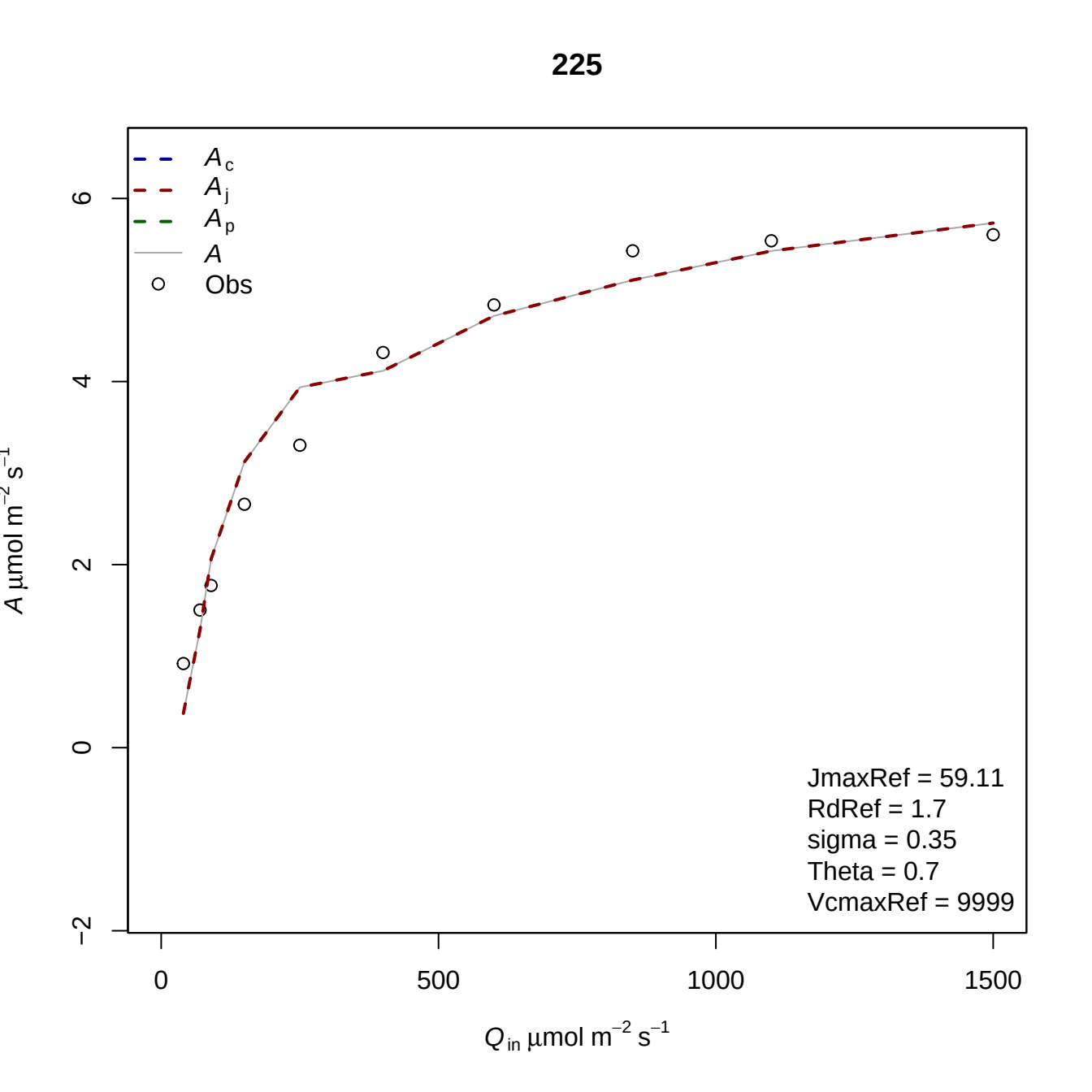

220
A_c
A_j
A_p
A
Obs
JmaxRef = 63.72
RdRef = 2.19
sigma = 0.26
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

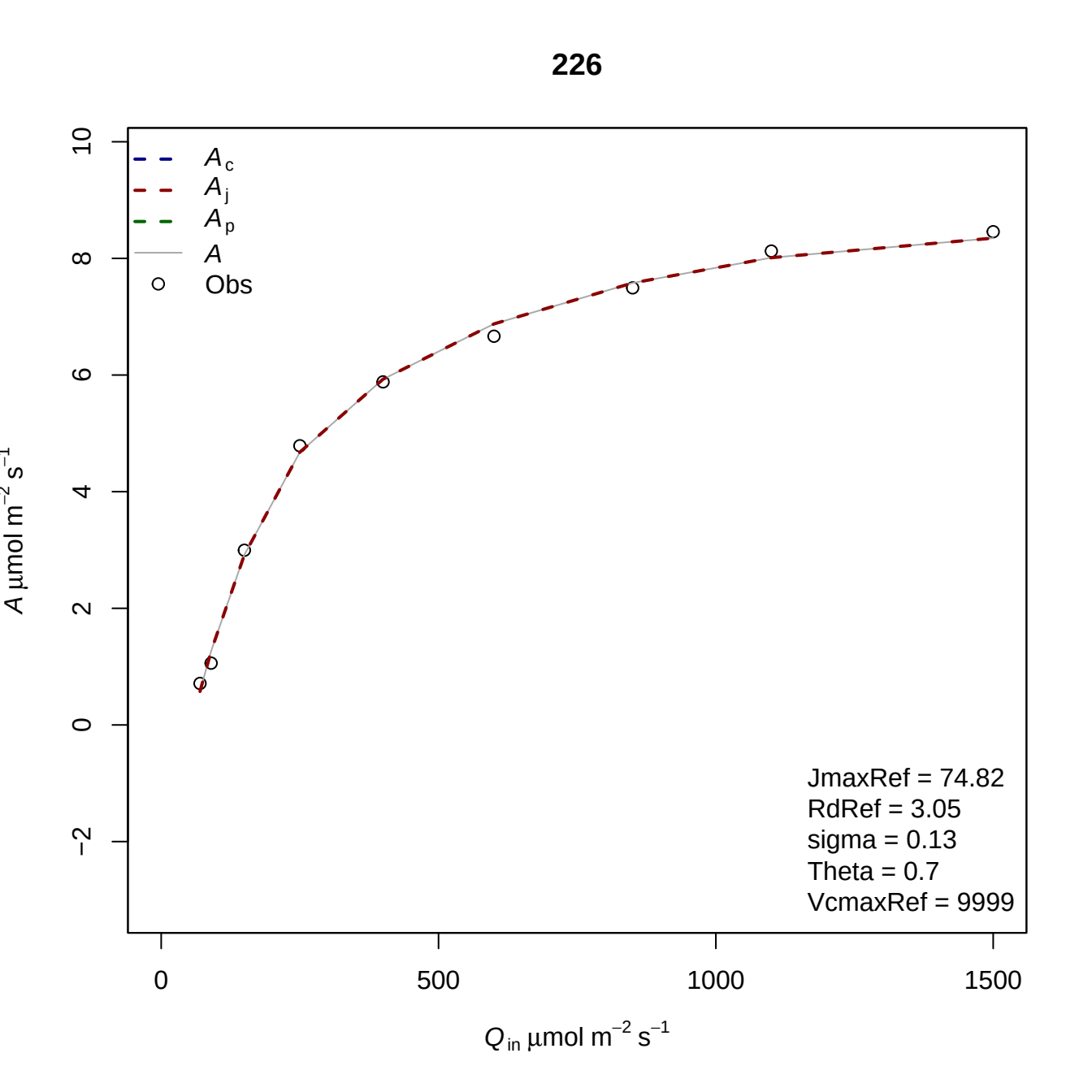

221
A_c
A_j
A_p
A
Obs
JmaxRef = 104.97
RdRef = 3.54
sigma = 0.57
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

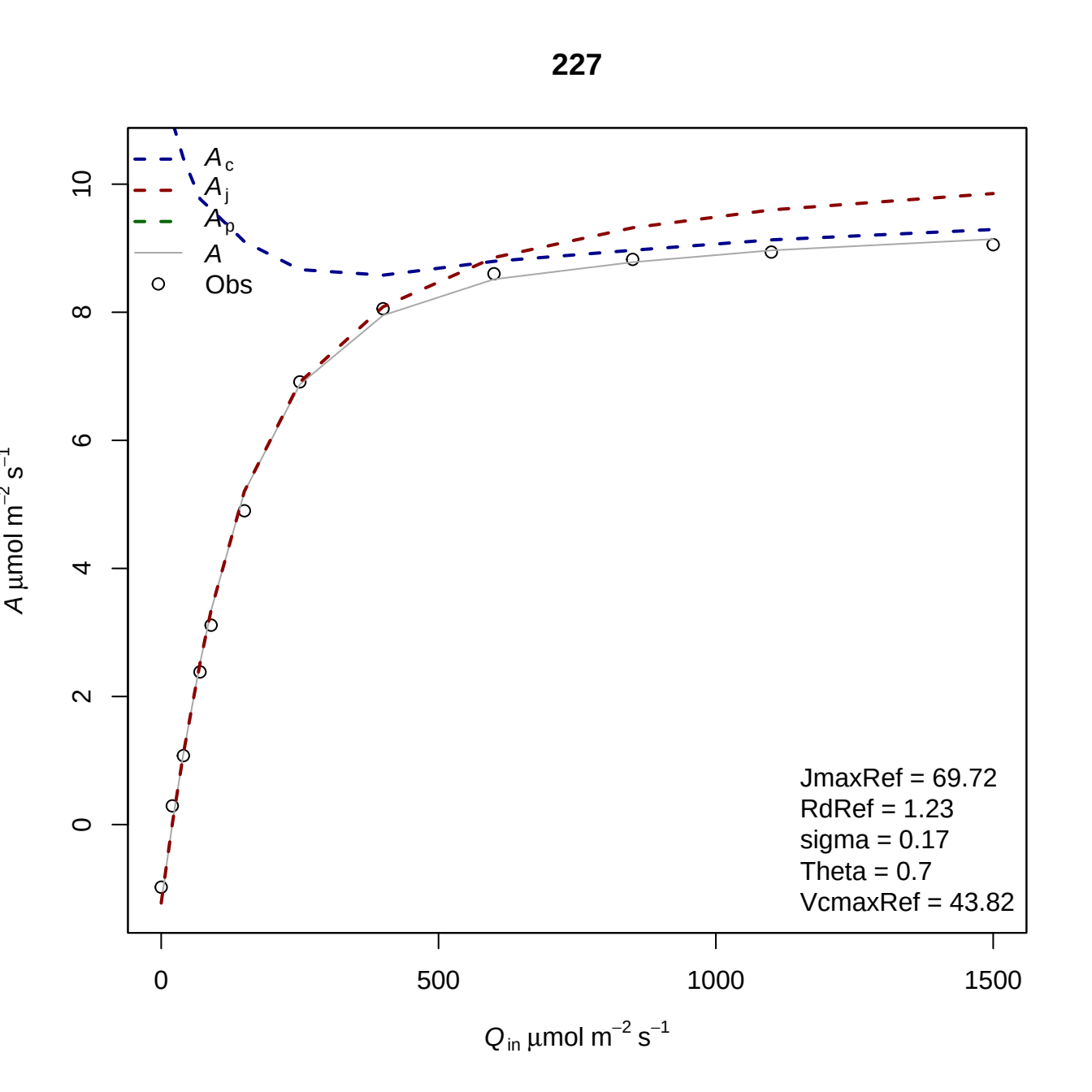

222
A_c
A_j
A_p
A
Obs
JmaxRef = 93.11
RdRef = 1.47
sigma = 0.6
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

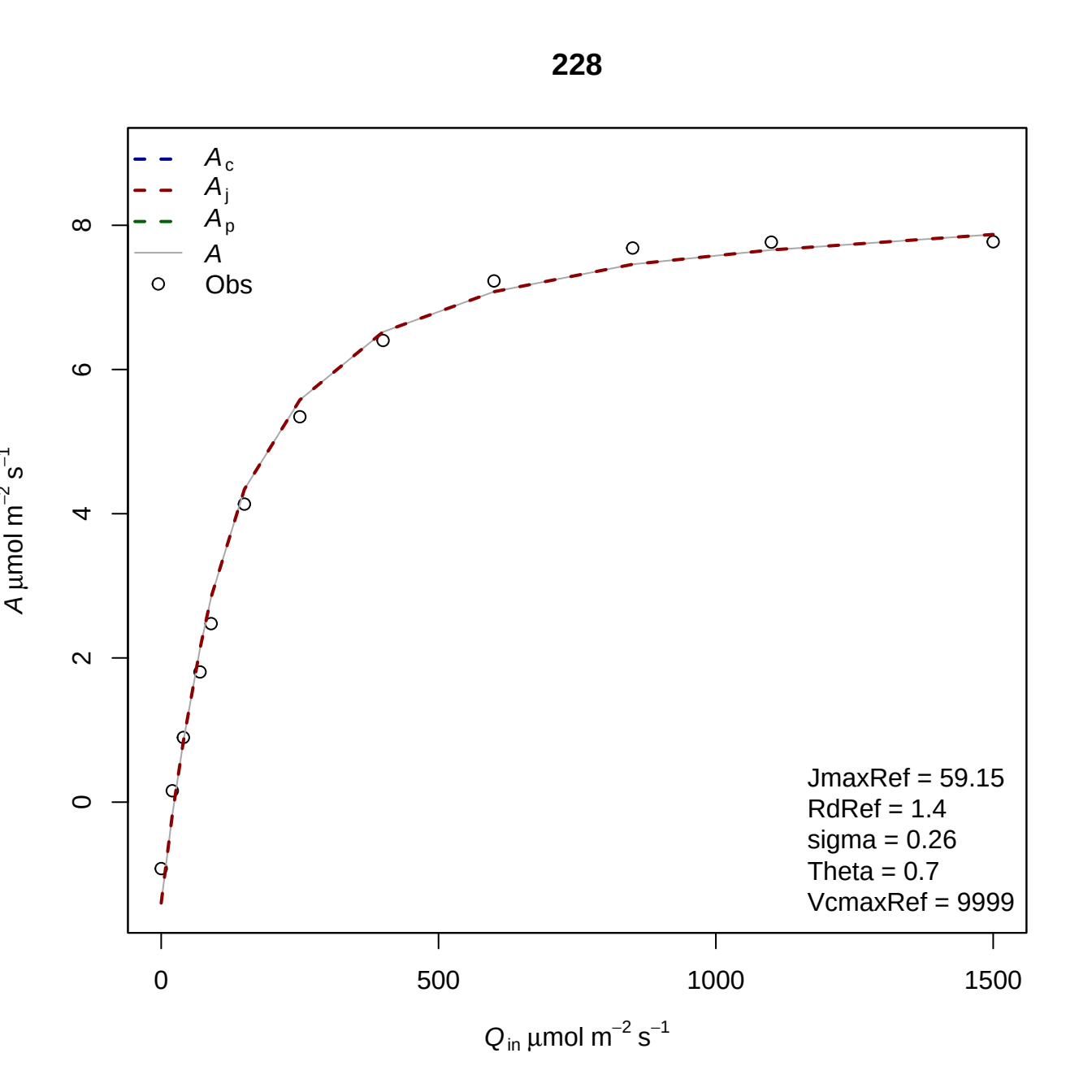

223
A_c
A_j
A_p
A
Obs
JmaxRef = 58.37
RdRef = 3.32
sigma = 0.18
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

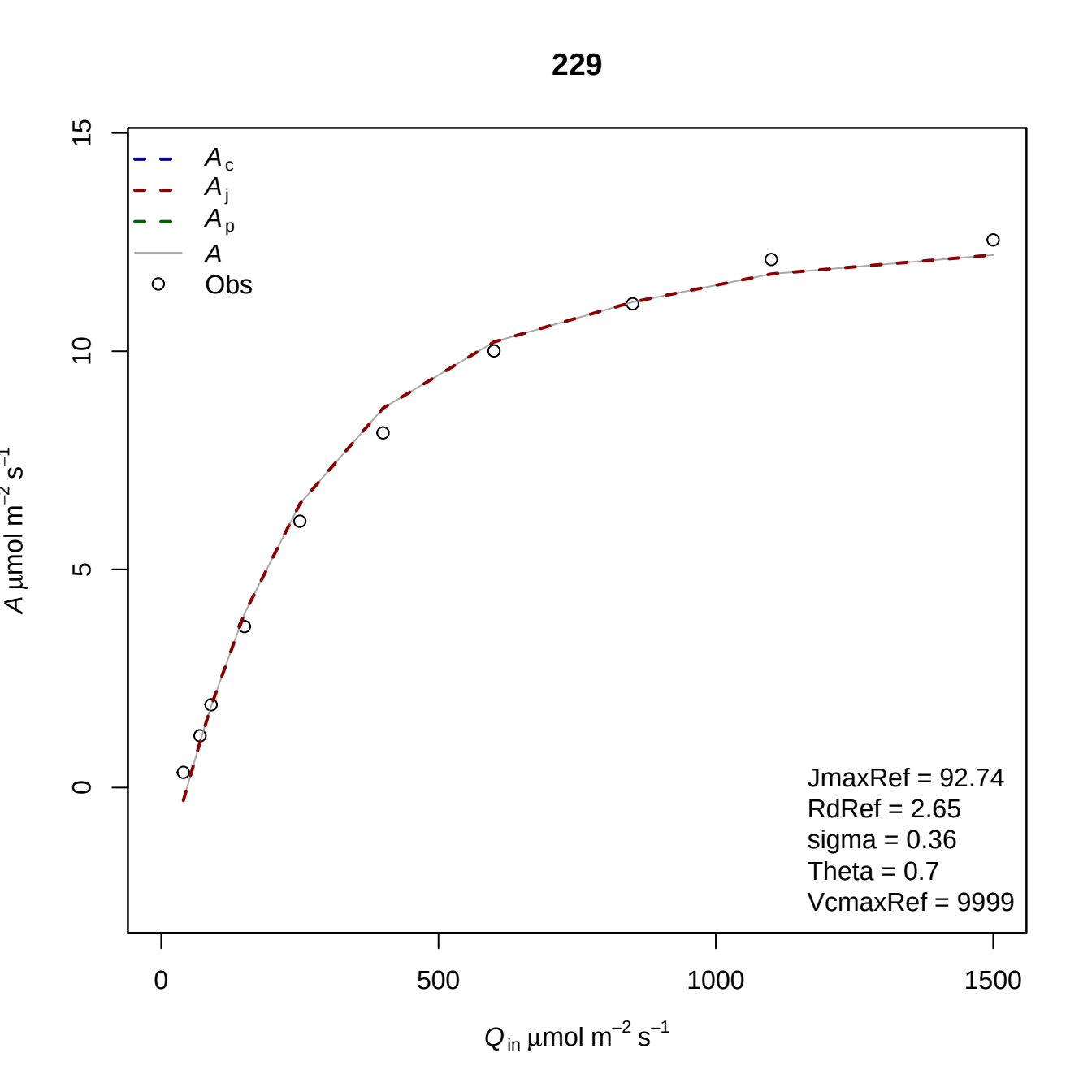

224
A_c
A_j
A_p
A
Obs
JmaxRef = 88.82
RdRef = 2.28
sigma = 0.25
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

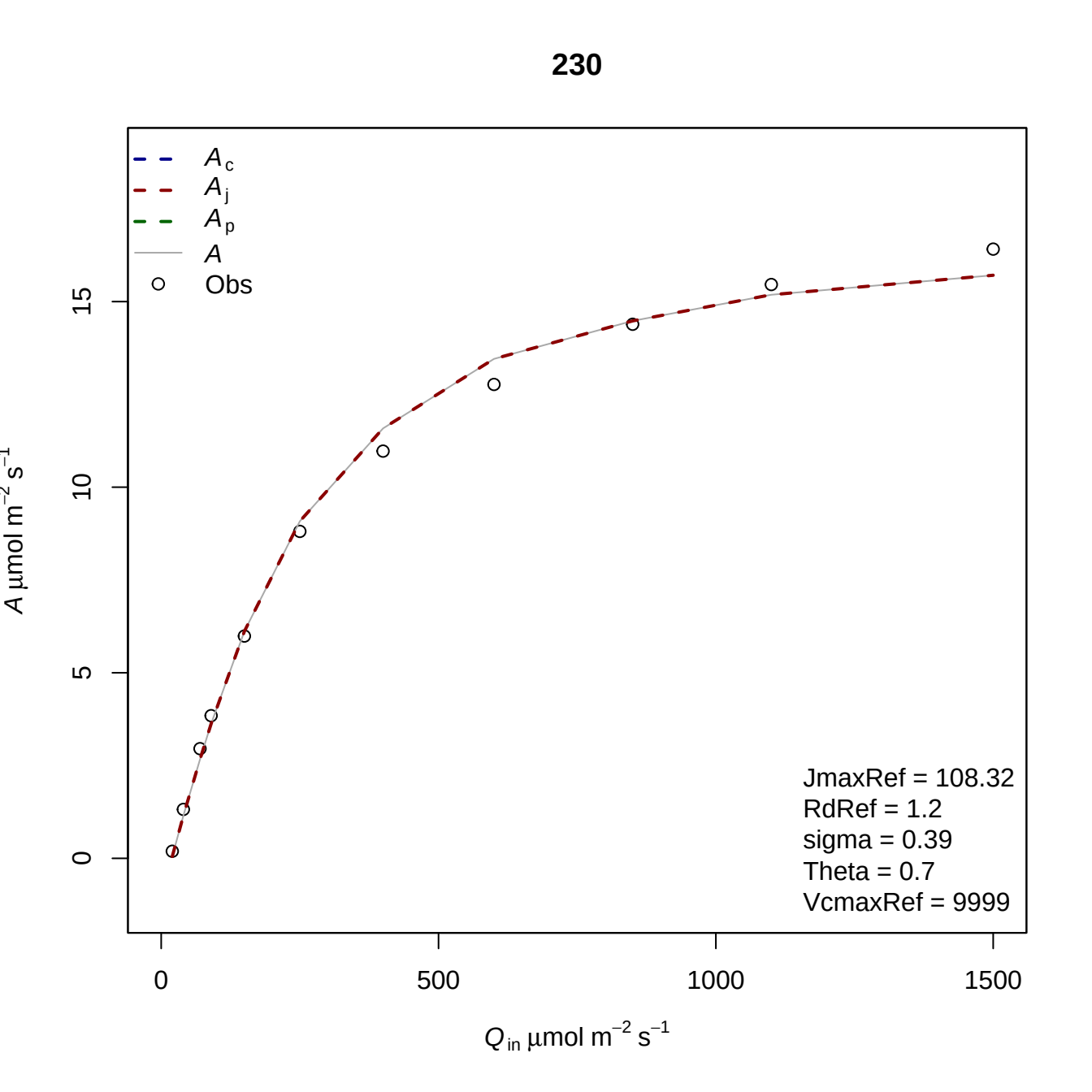

225
A_c
A_j
A_p
A
Obs
JmaxRef = 59.11
RdRef = 1.7
sigma = 0.35
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

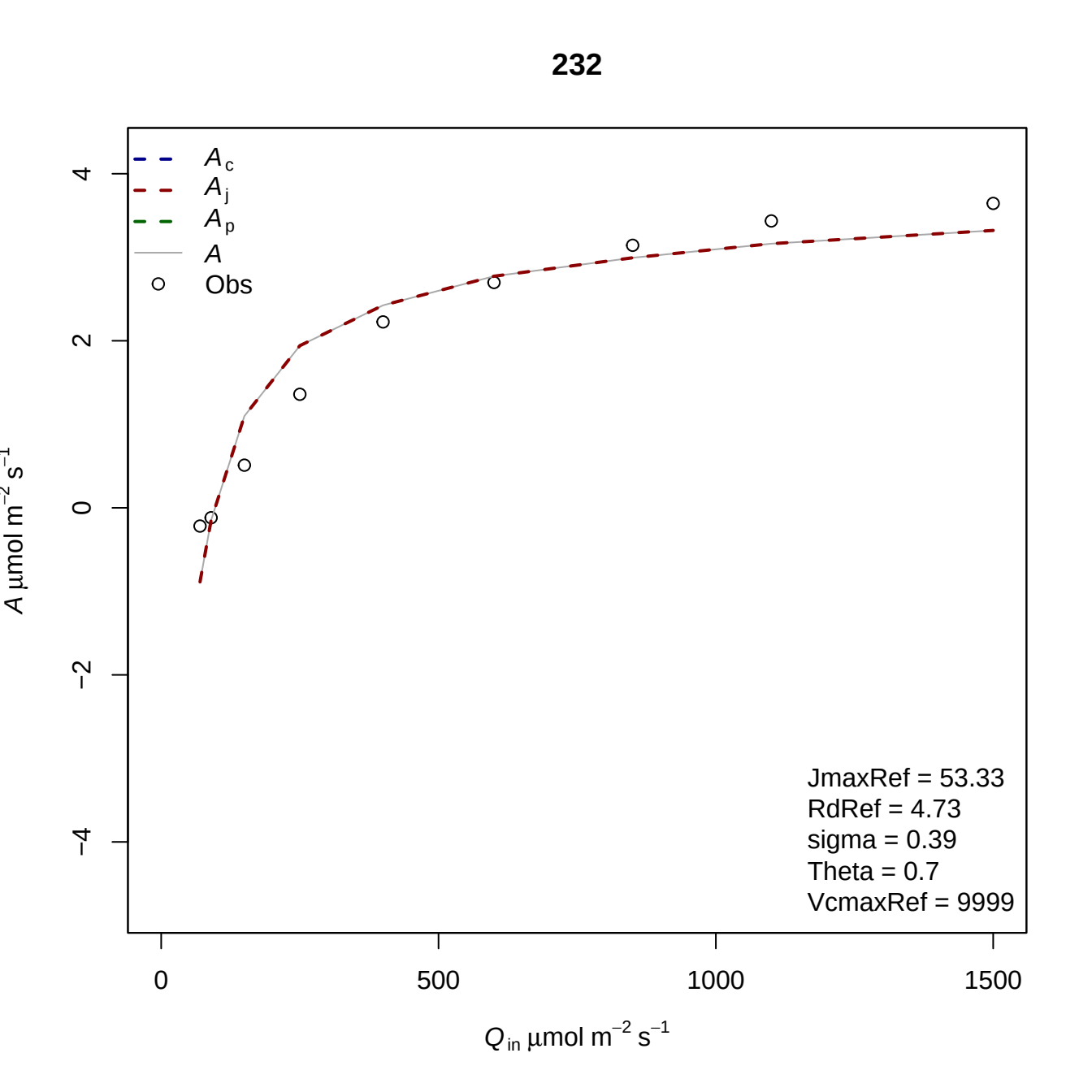

226
A_c
A_j
A_p
A
Obs
JmaxRef = 74.82
RdRef = 3.05
sigma = 0.13
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

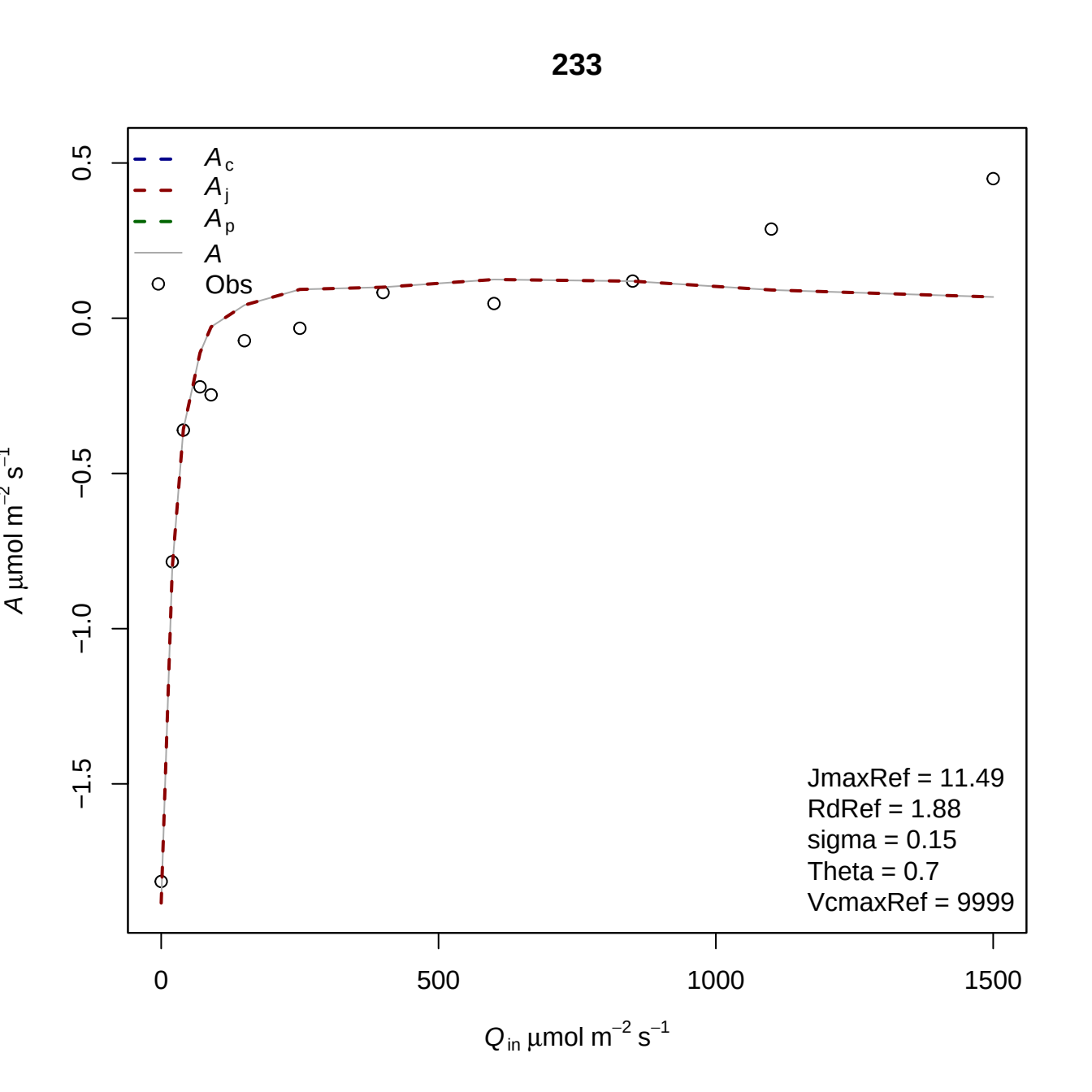

227
A_c
A_j
A_p
A
Obs
JmaxRef = 69.72
RdRef = 1.23
sigma = 0.17
Theta = 0.7
VcmaxRef = 43.82
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

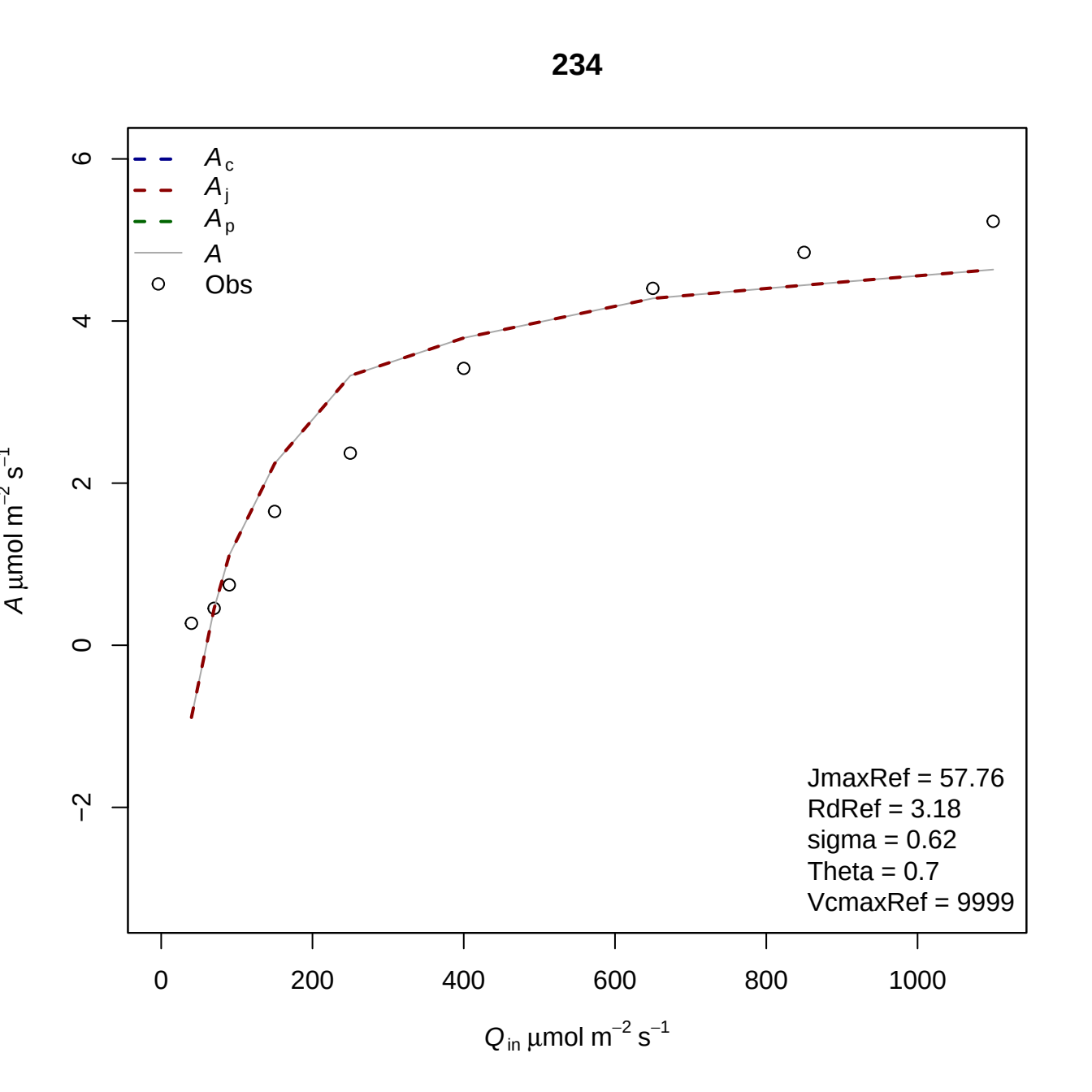

228
A_c
A_j
A_p
A
Obs
JmaxRef = 59.15
RdRef = 1.4
sigma = 0.26
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

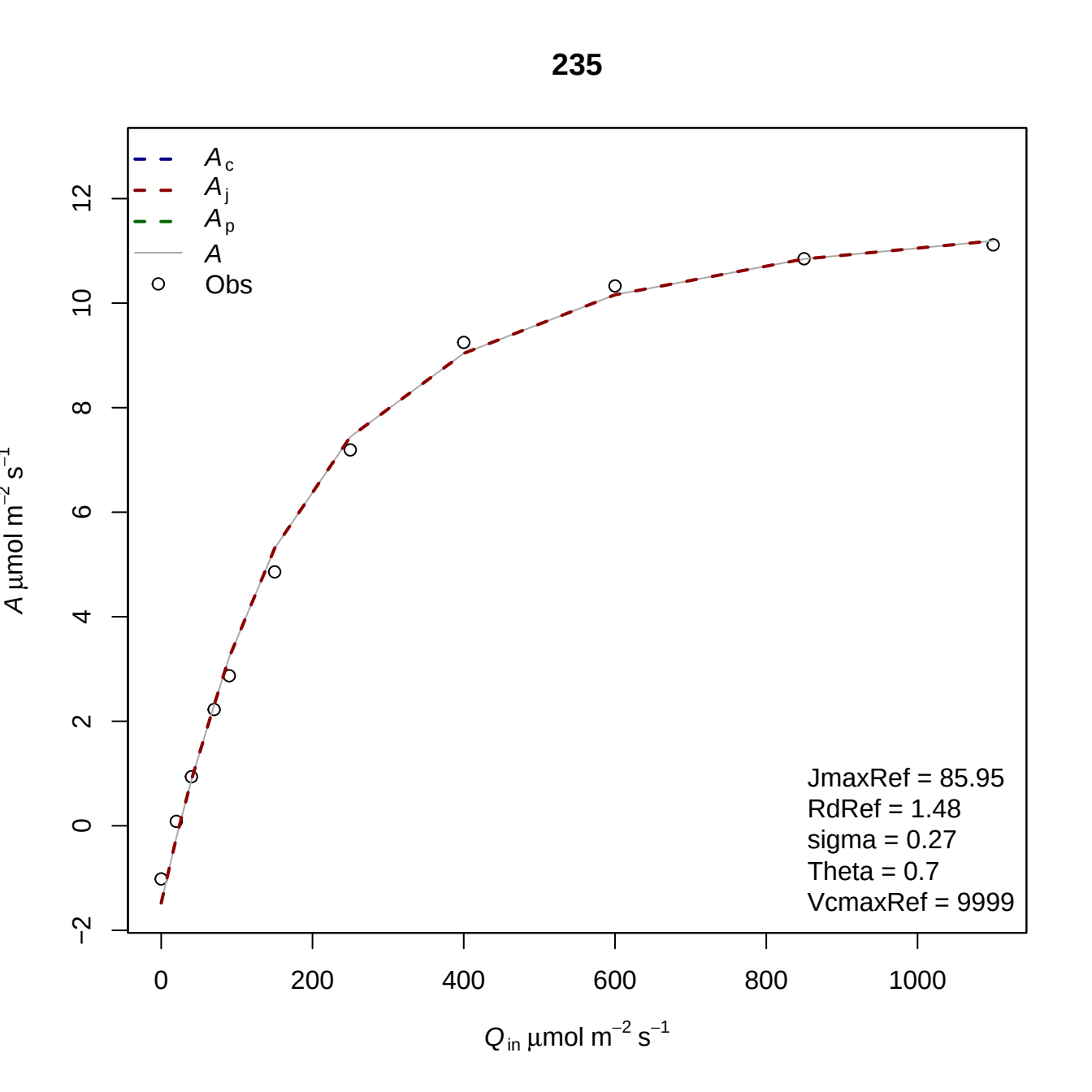

229
A_c
A_j
A_p
A
Obs
JmaxRef = 92.74
RdRef = 2.65
sigma = 0.36
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

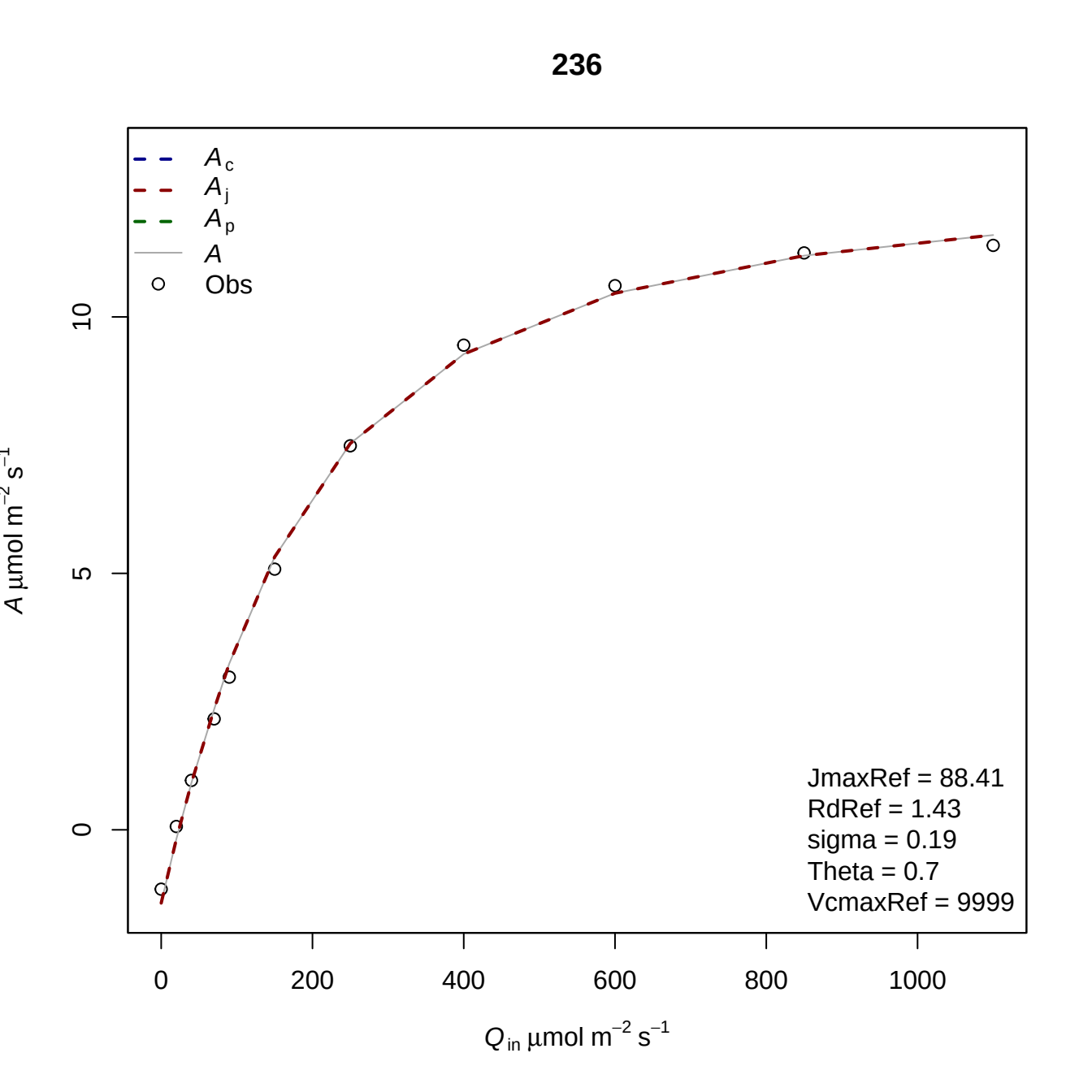

230
A_c
A_j
A_p
A
Obs
JmaxRef = 108.32
RdRef = 1.2
sigma = 0.39
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

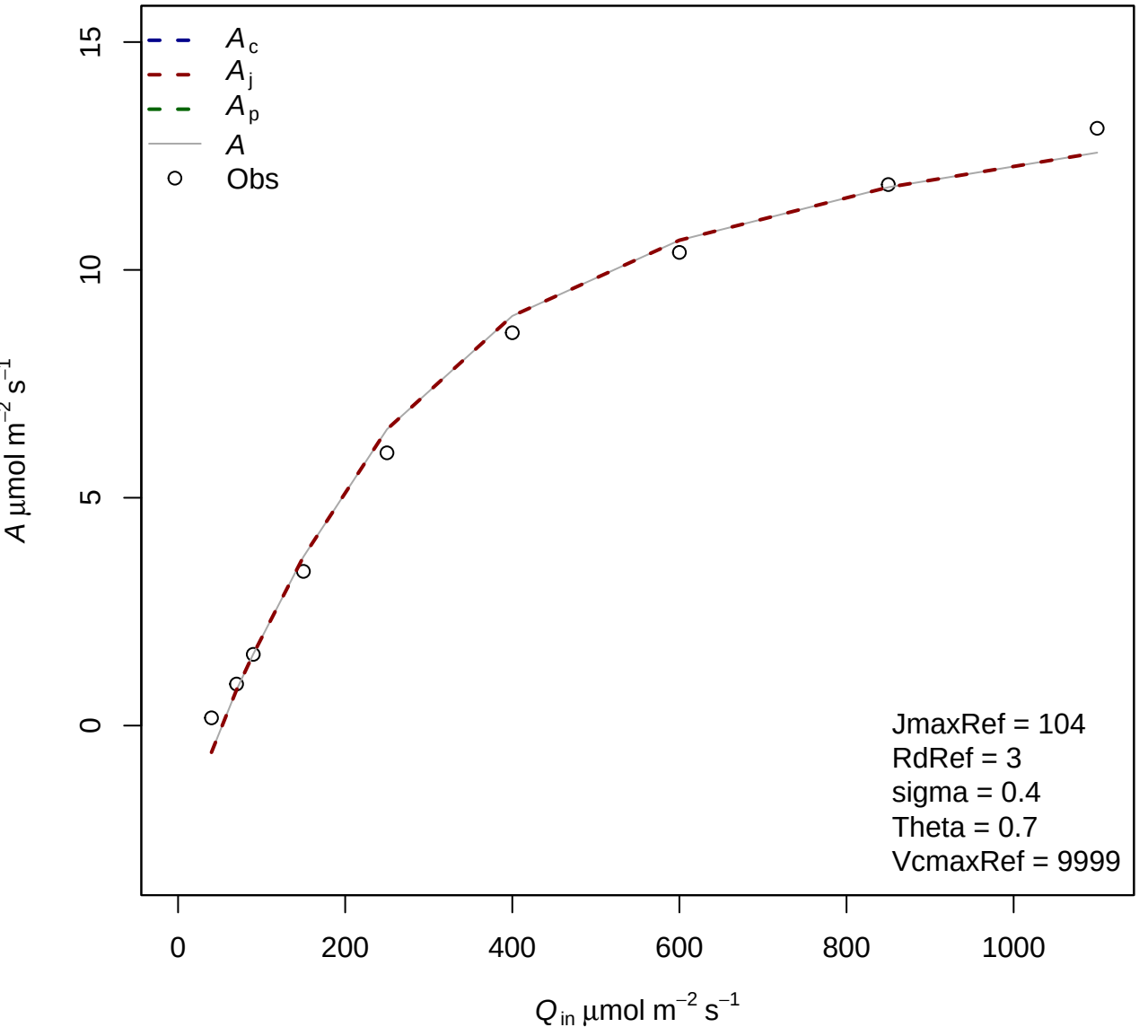

232
A_c
A_j
A_p
A
Obs
JmaxRef = 53.33
RdRef = 4.73
sigma = 0.39
Theta = 0.7
VcmaxRef = 9999
4
2
0
−2
−4
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

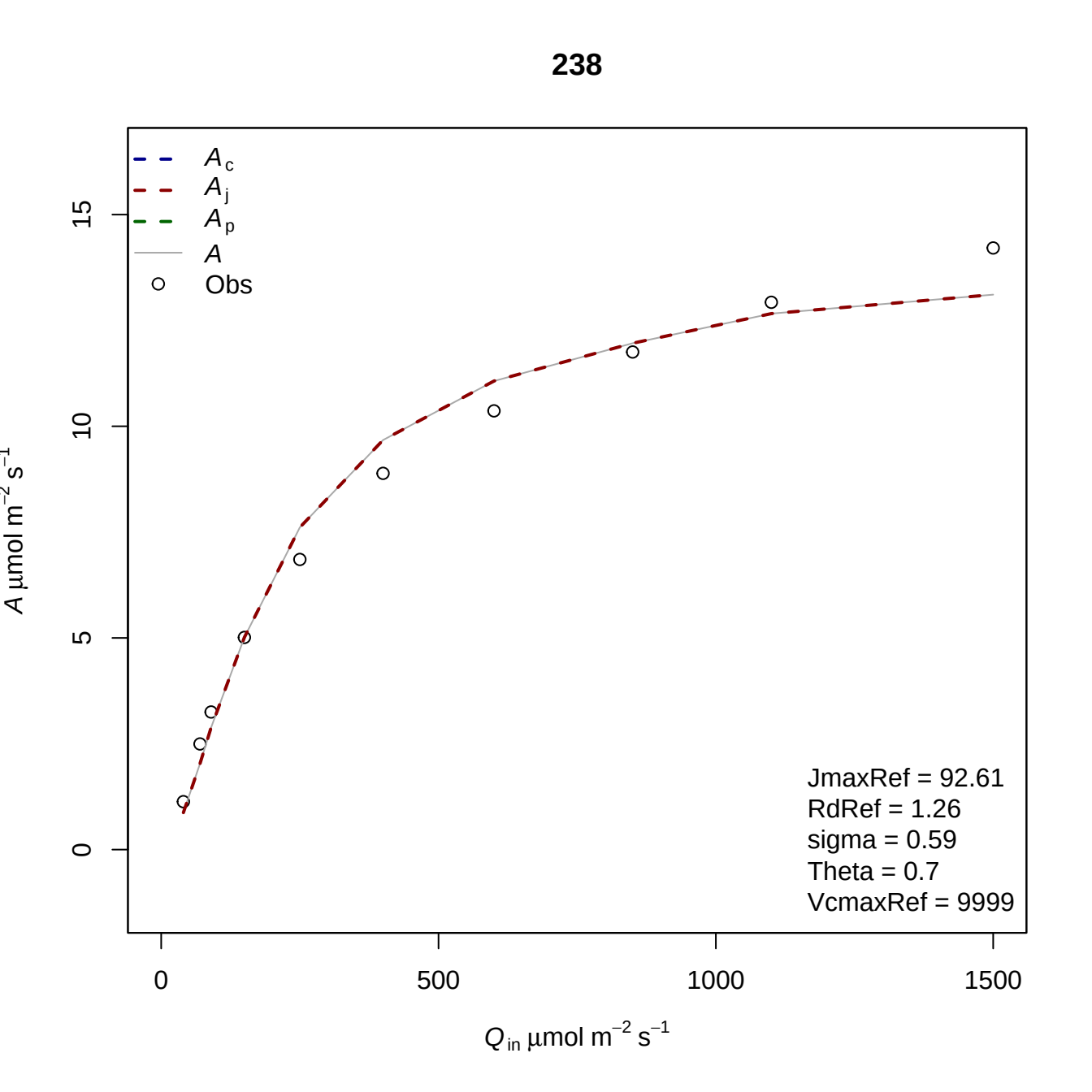

233
A_c
A_j
A_p
A
Obs
JmaxRef = 11.49
RdRef = 1.88
sigma = 0.15
Theta = 0.7
VcmaxRef = 9999
0.5
0.0
−0.5
−1.0
−1.5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

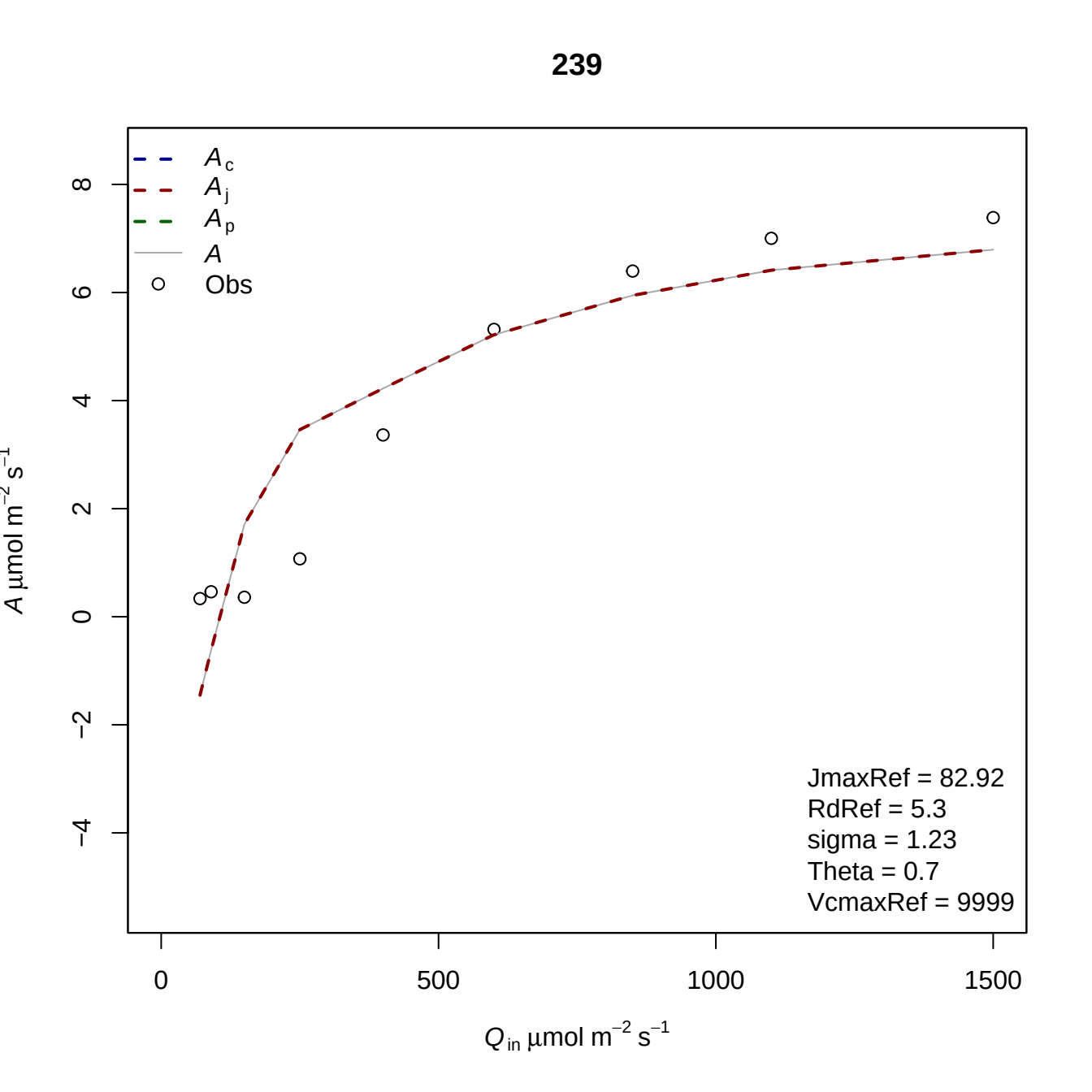

234
A_c
A_j
A_p
A
Obs
JmaxRef = 57.76
RdRef = 3.18
sigma = 0.62
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
200
400
600
800
1000

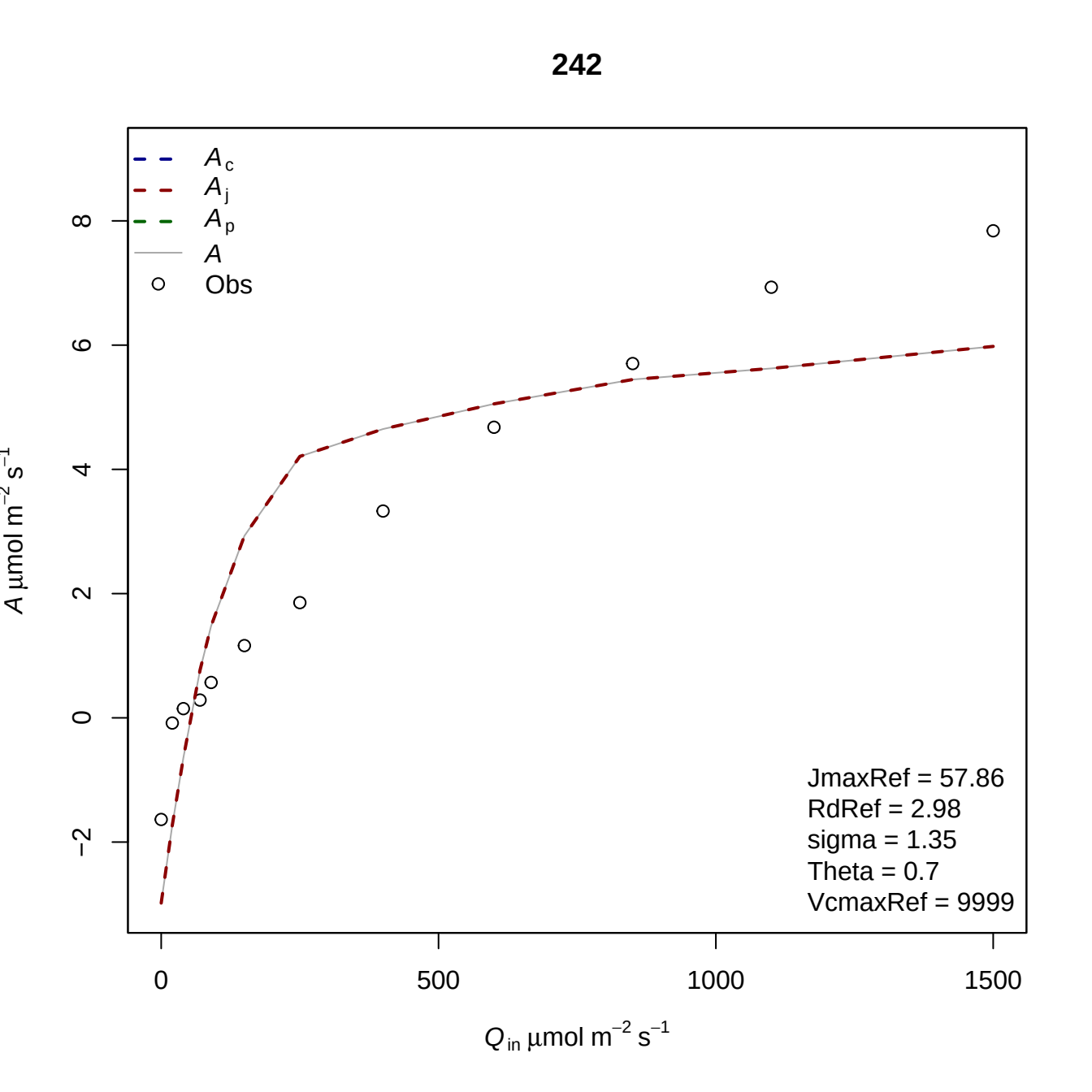

235
A_c
A_j
A_p
A
Obs
JmaxRef = 85.95
RdRef = 1.48
sigma = 0.27
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
200
400
600
800
1000

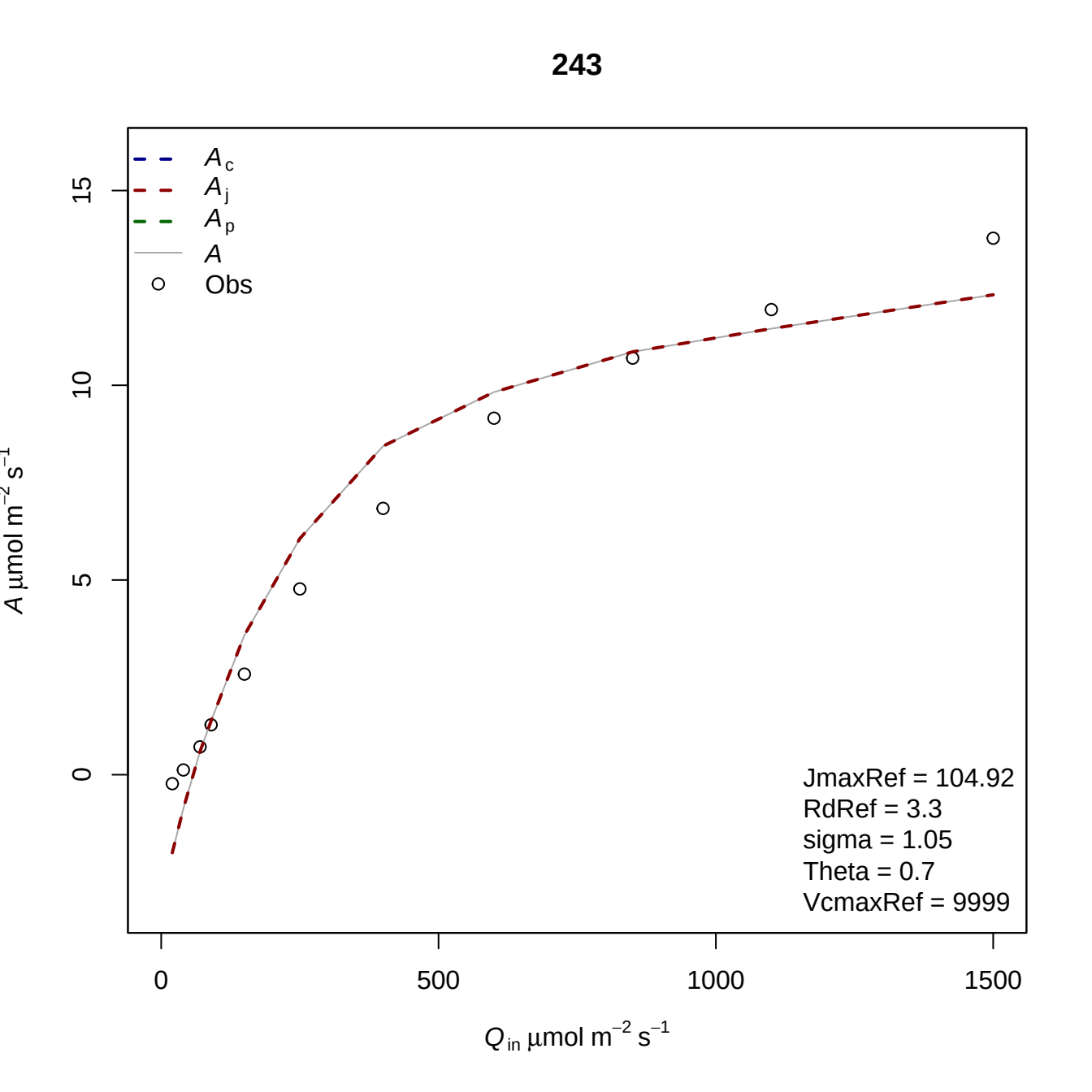

236
A_c
A_j
A_p
A
Obs
JmaxRef = 88.41
RdRef = 1.43
sigma = 0.19
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800
1000

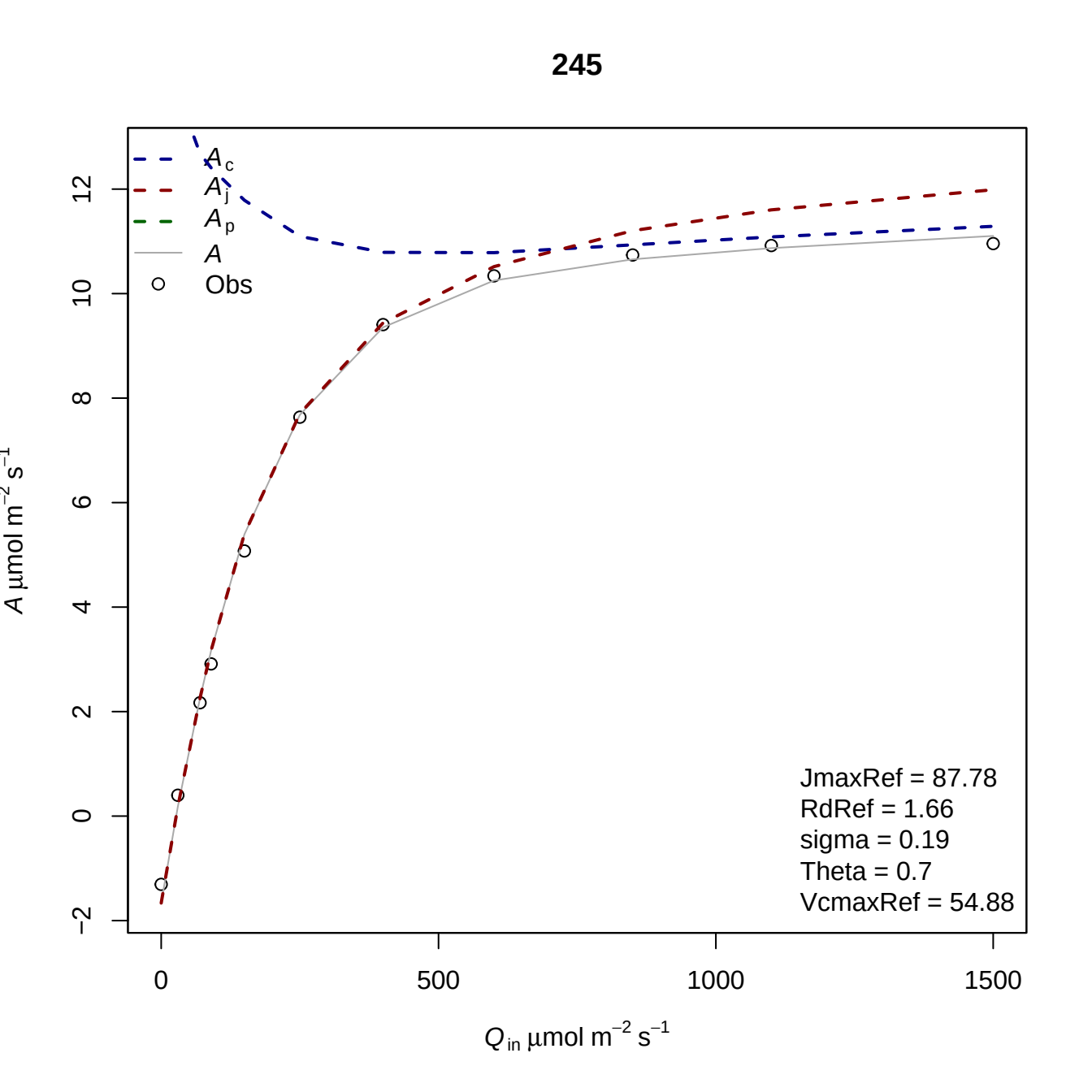

A_c
A_j
A_p
A
Obs
JmaxRef = 104
RdRef = 3
sigma = 0.4
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
200
400
600
800
1000
Q_{in} µmol m^{-2} s^{-1}

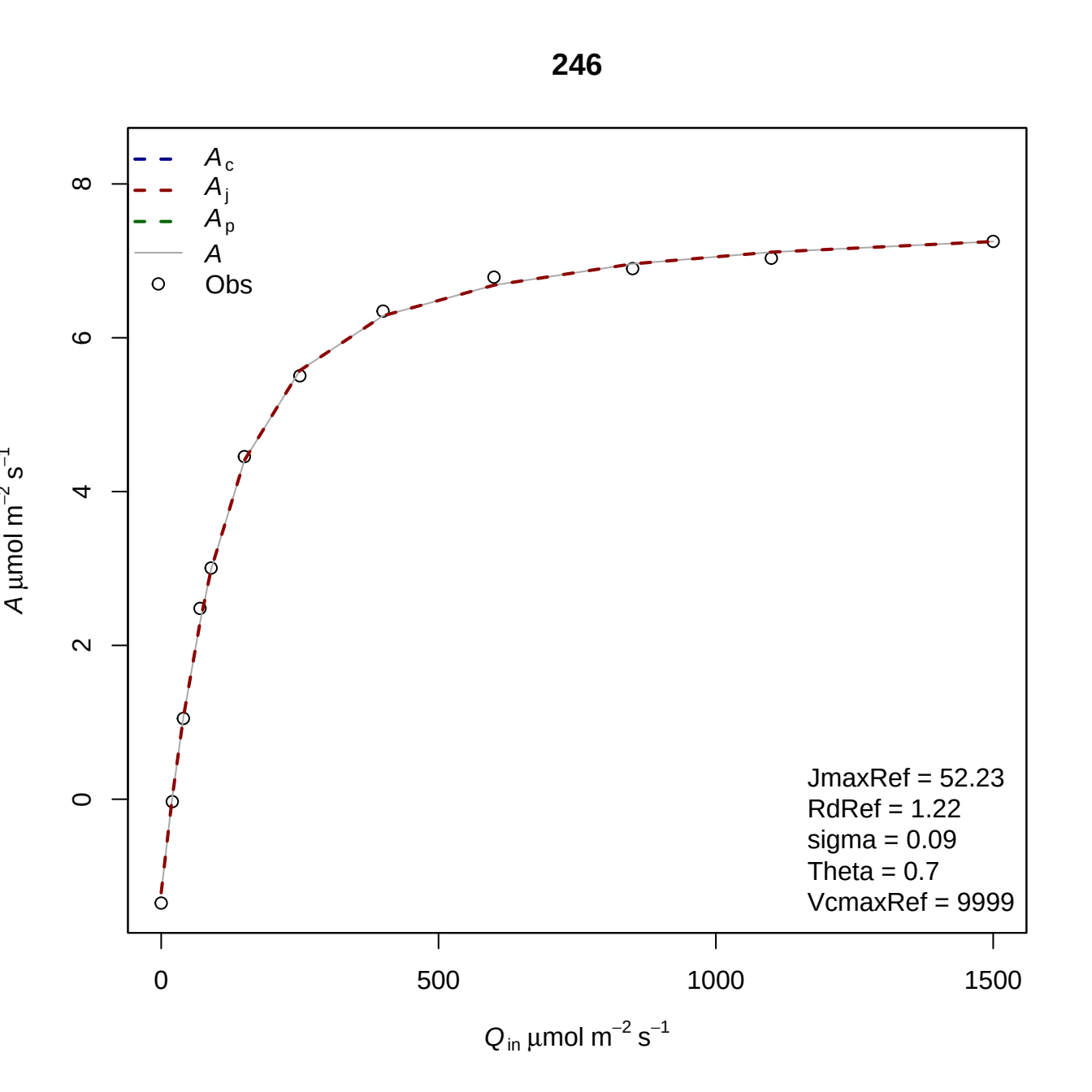

238
A_c
A_j
A_p
A
Obs
JmaxRef = 92.61
RdRef = 1.26
sigma = 0.59
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

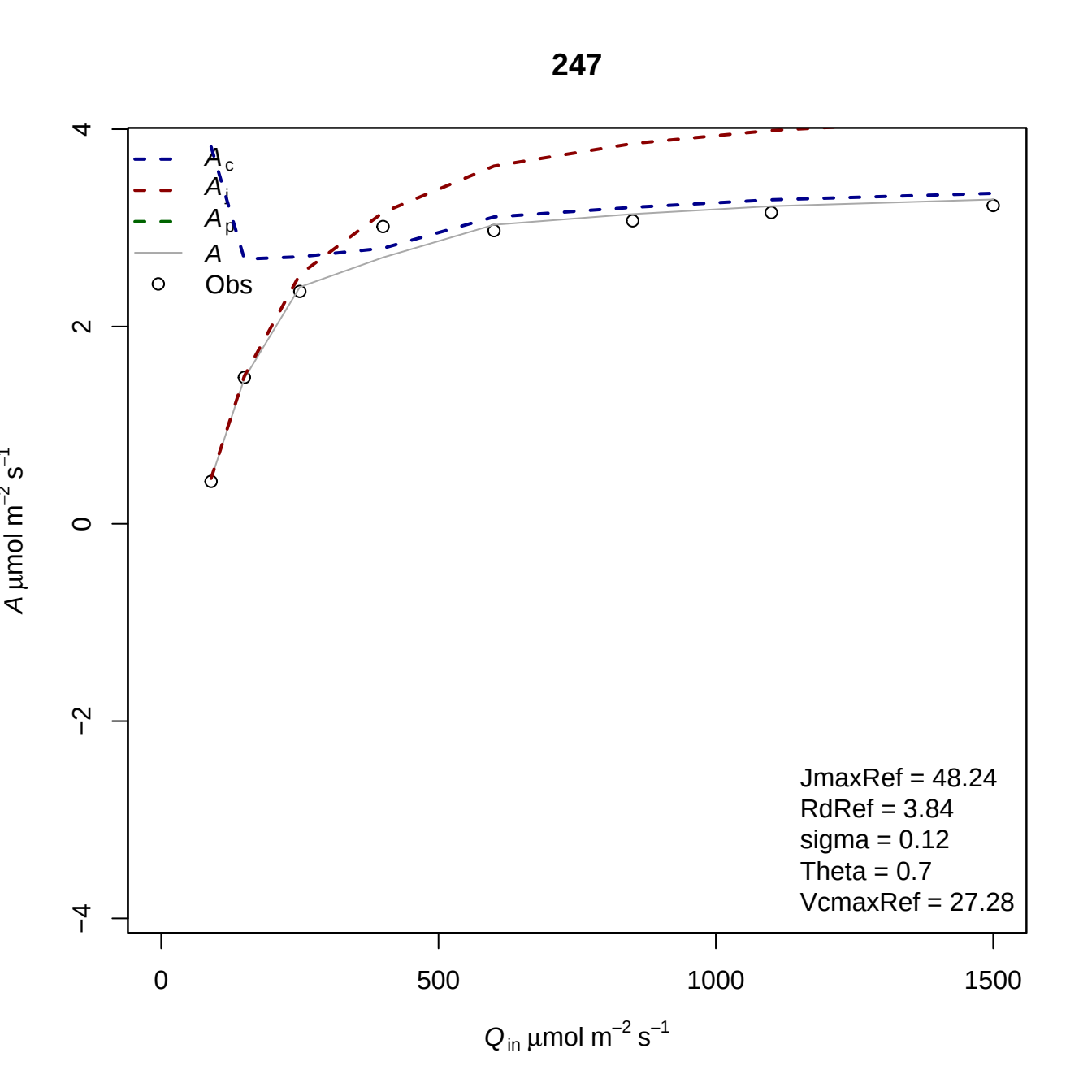

239
A_c
A_j
A_p
A
Obs
JmaxRef = 82.92
RdRef = 5.3
sigma = 1.23
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

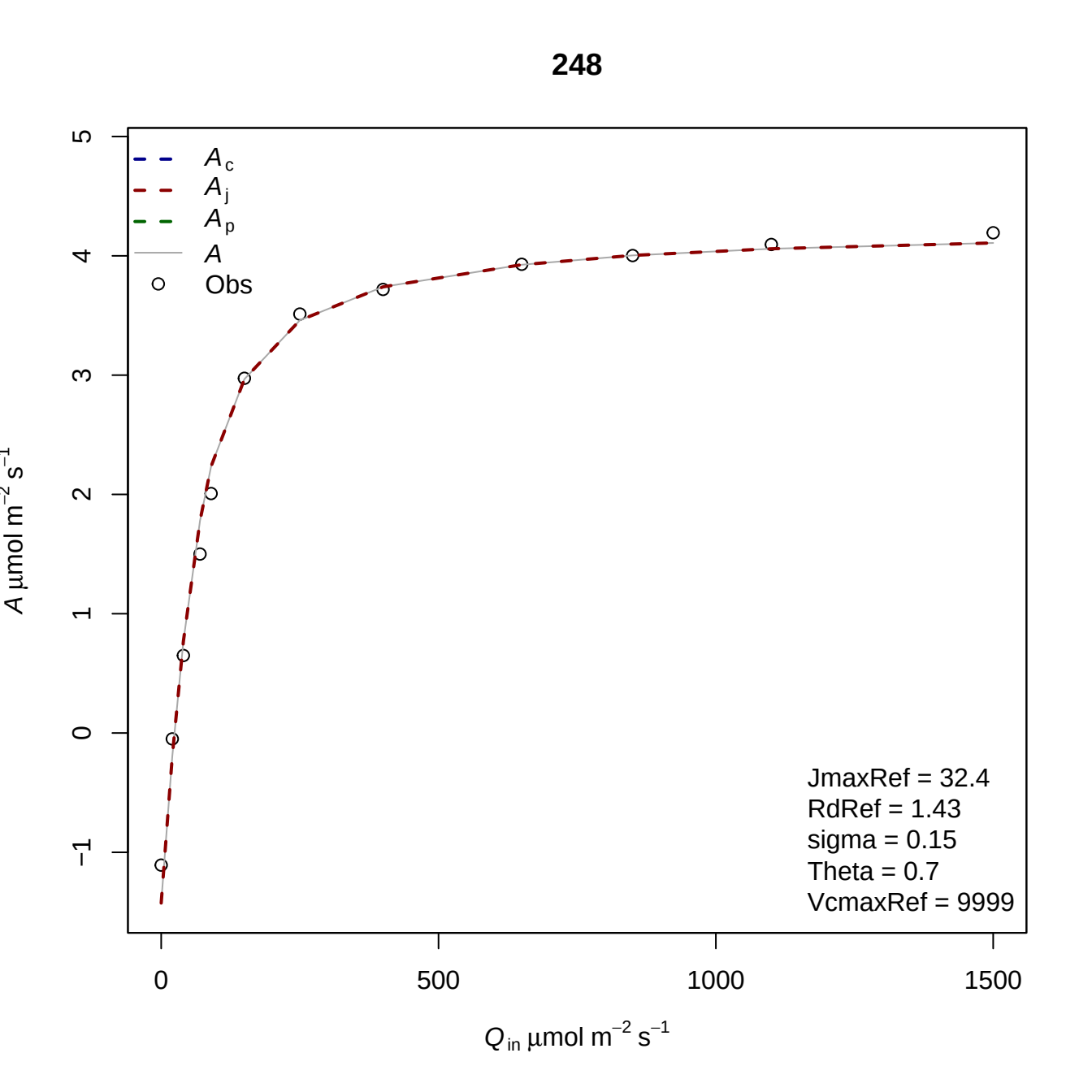

242
A_c
A_j
A_p
A
Obs
JmaxRef = 57.86
RdRef = 2.98
sigma = 1.35
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

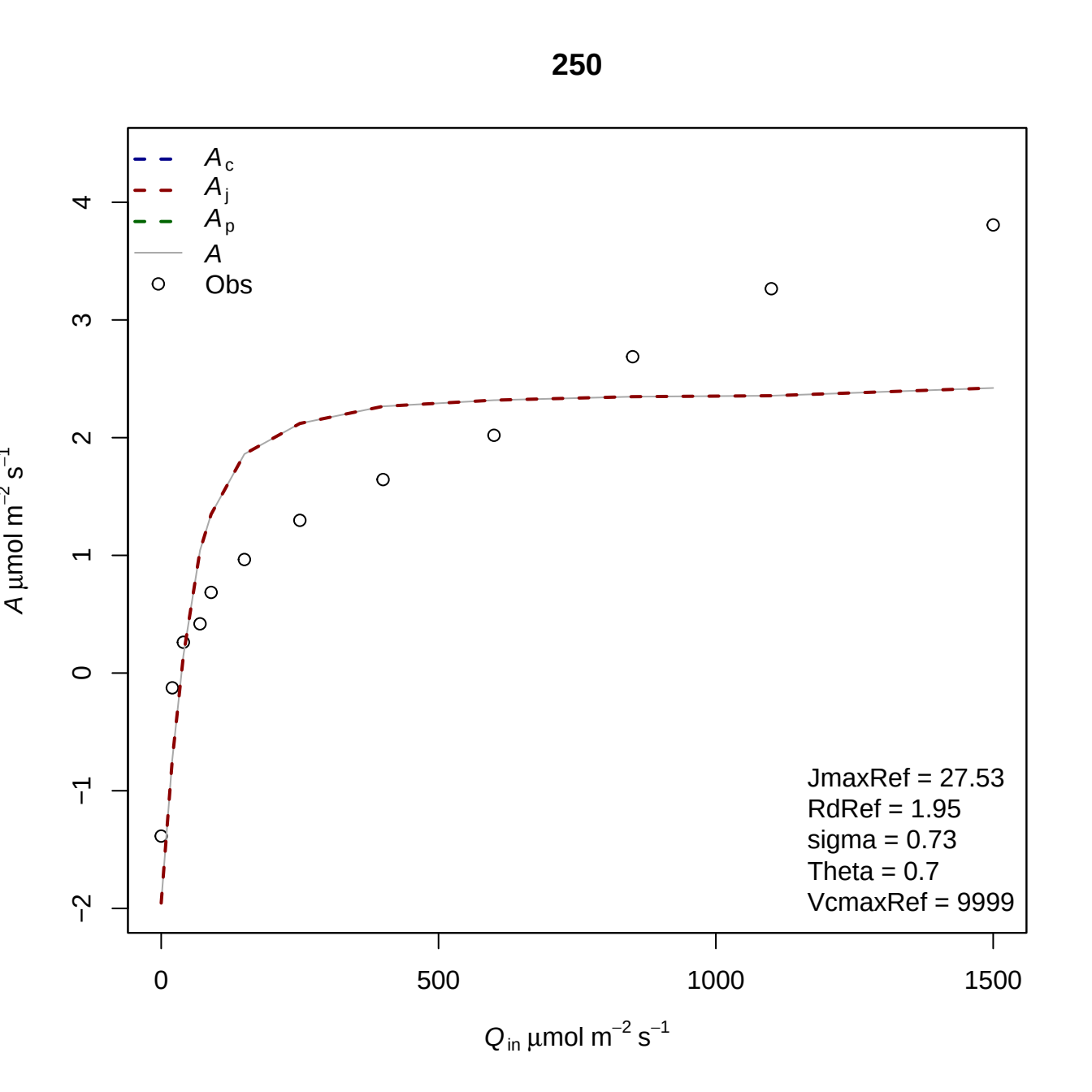

243
A_c
A_j
A_p
A
Obs
JmaxRef = 104.92
RdRef = 3.3
sigma = 1.05
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

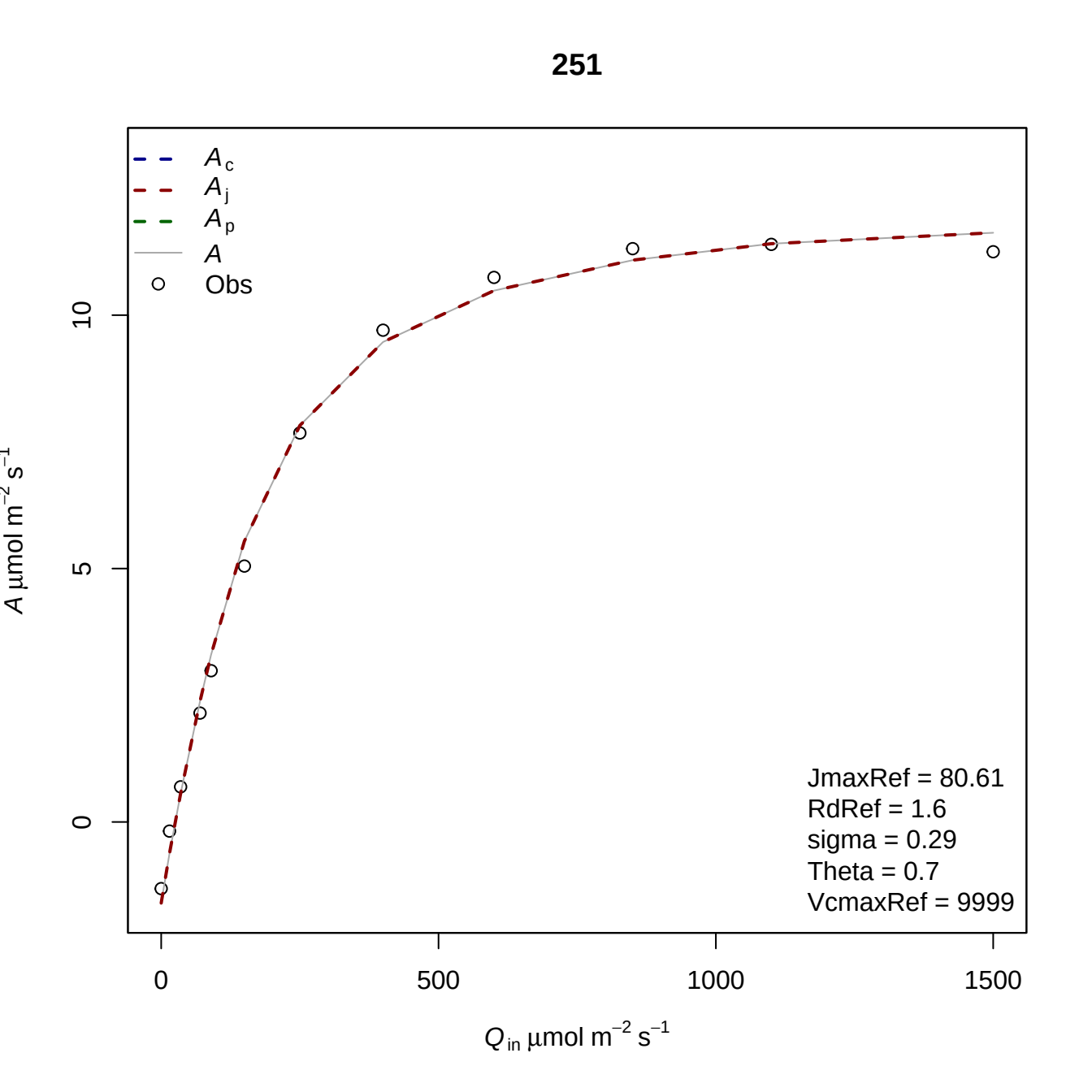

245
A_c
A_j
A_p
A
Obs
JmaxRef = 87.78
RdRef = 1.66
sigma = 0.19
Theta = 0.7
VcmaxRef = 54.88
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

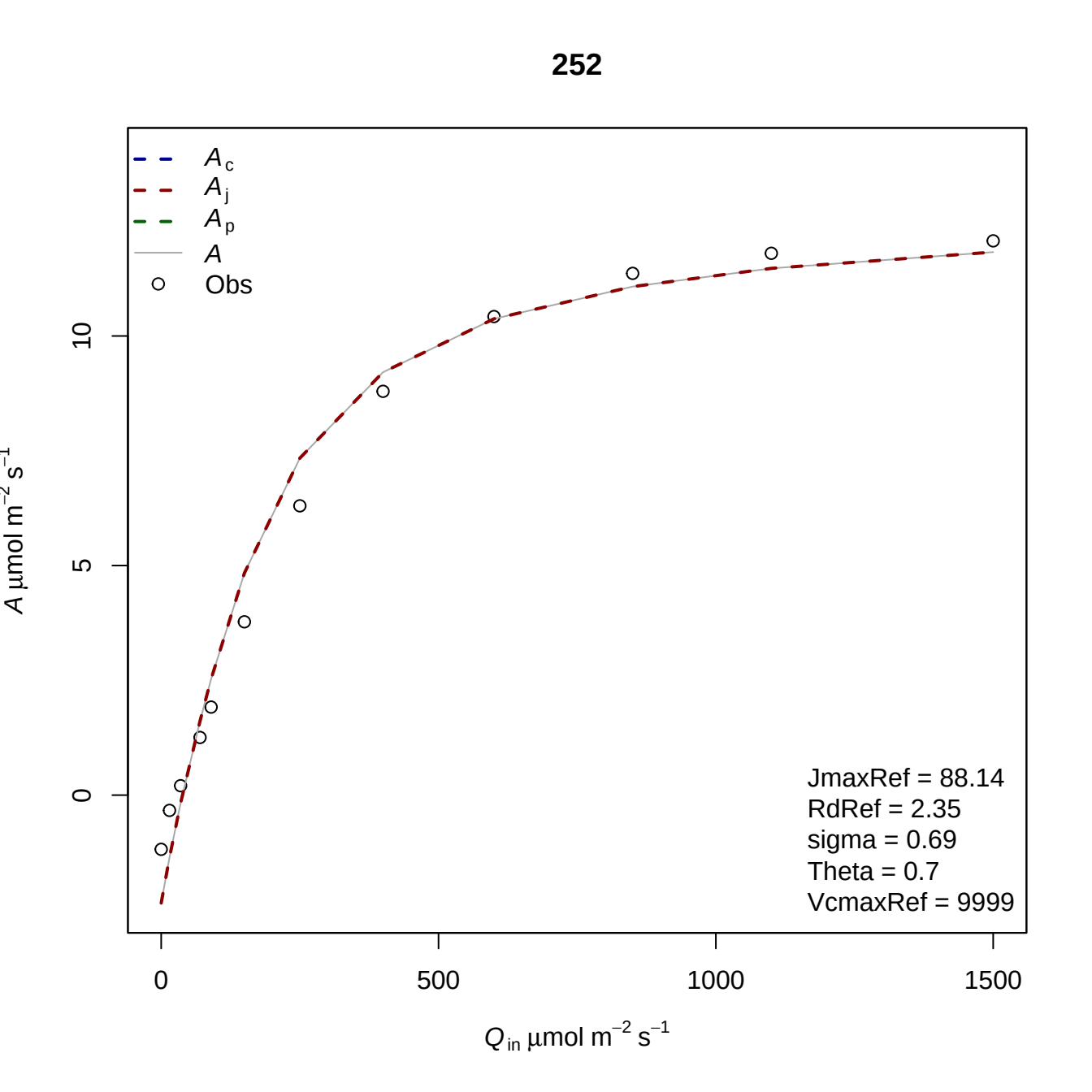

246
A_c
A_j
A_p
A
Obs
JmaxRef = 52.23
RdRef = 1.22
sigma = 0.09
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

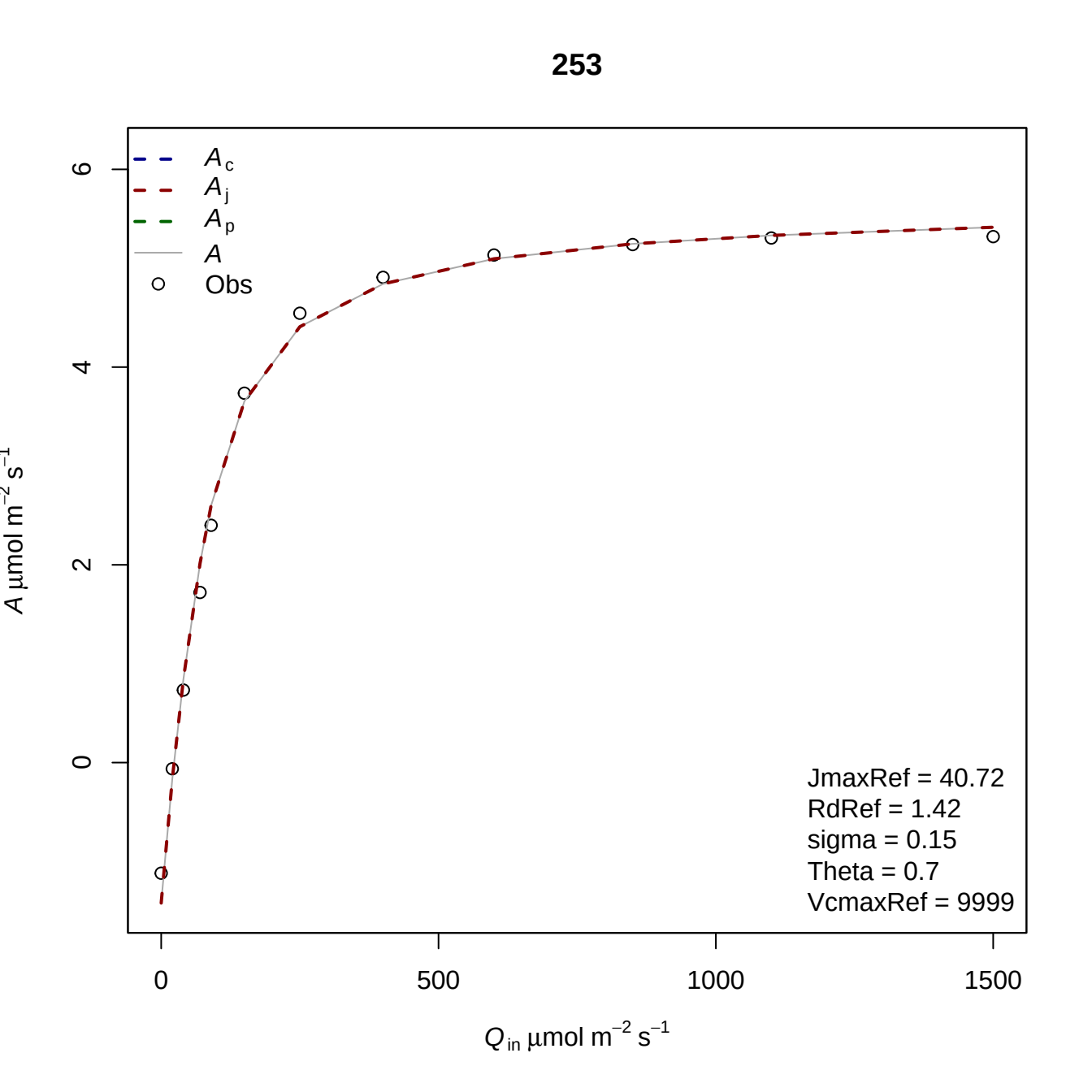

247
A_c
A_j
A_p
A
Obs
JmaxRef = 48.24
RdRef = 3.84
sigma = 0.12
Theta = 0.7
VcmaxRef = 27.28
4
2
0
−2
−4
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

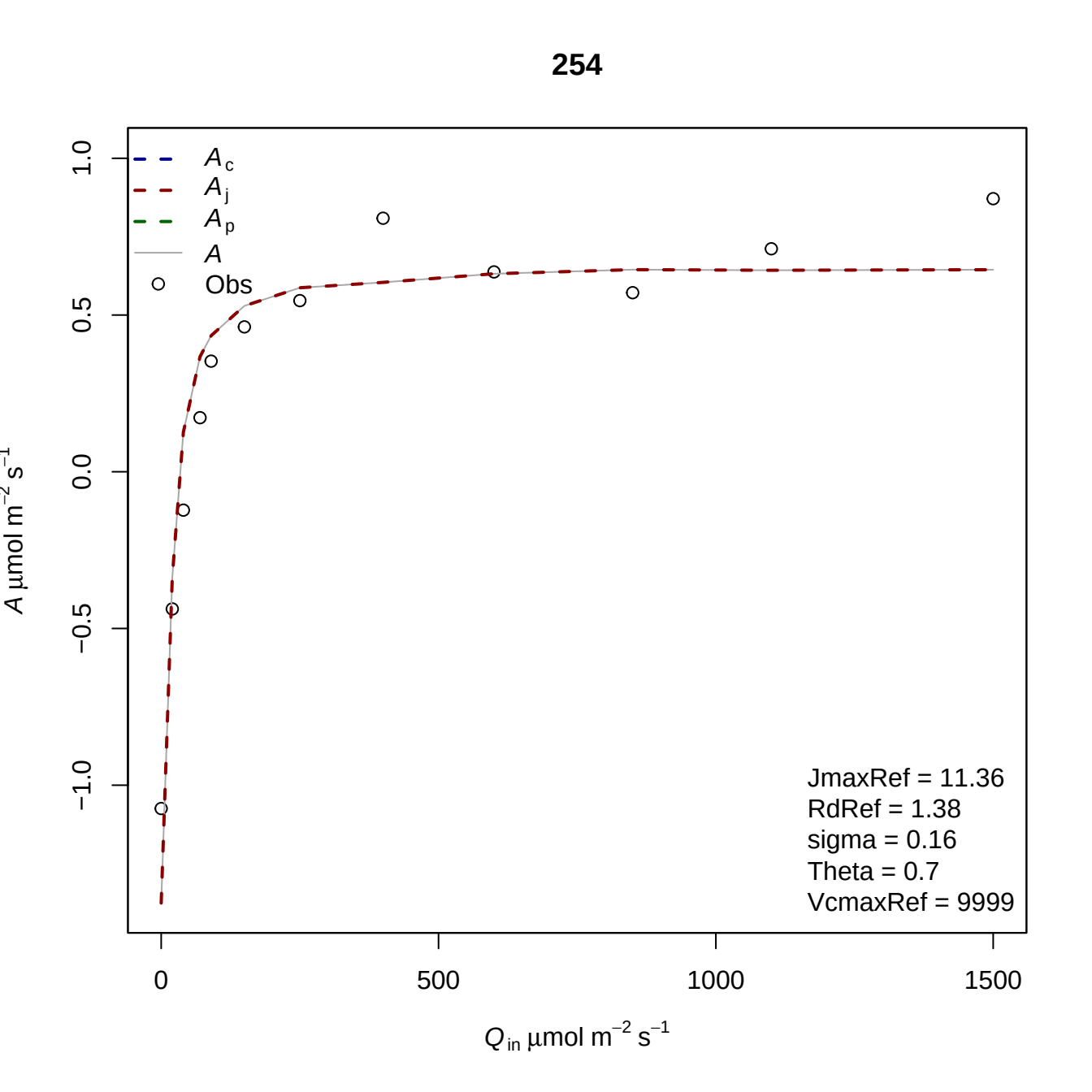

248
A_c
A_j
A_p
A
Obs
JmaxRef = 32.4
RdRef = 1.43
sigma = 0.15
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
5
4
3
2
1
0
−1
0
500
1000
1500

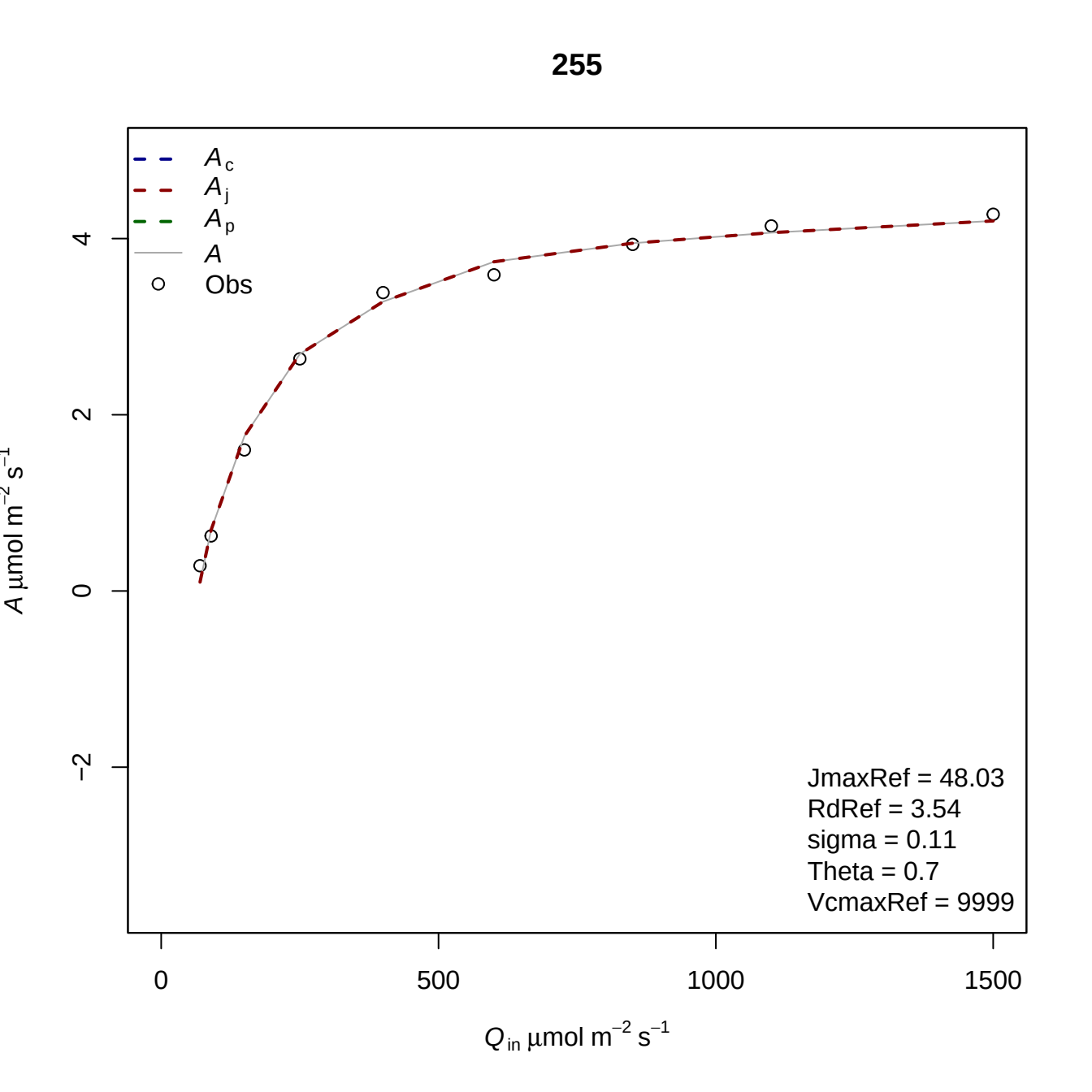

250
A_c
A_j
A_p
A
Obs
JmaxRef = 27.53
RdRef = 1.95
sigma = 0.73
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
4
0
500
1000
1500

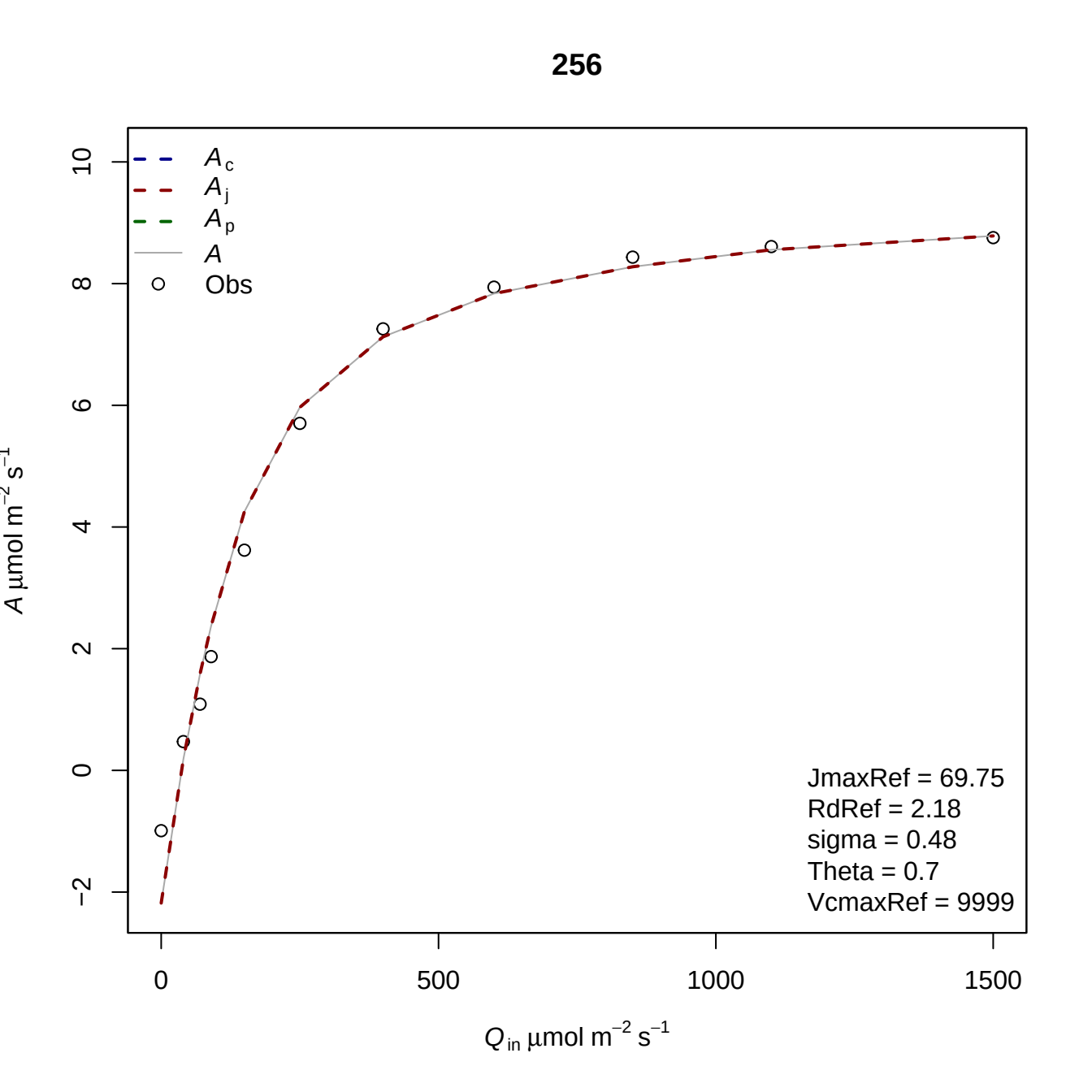

251
A_c
A_j
A_p
A
Obs
JmaxRef = 80.61
RdRef = 1.6
sigma = 0.29
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

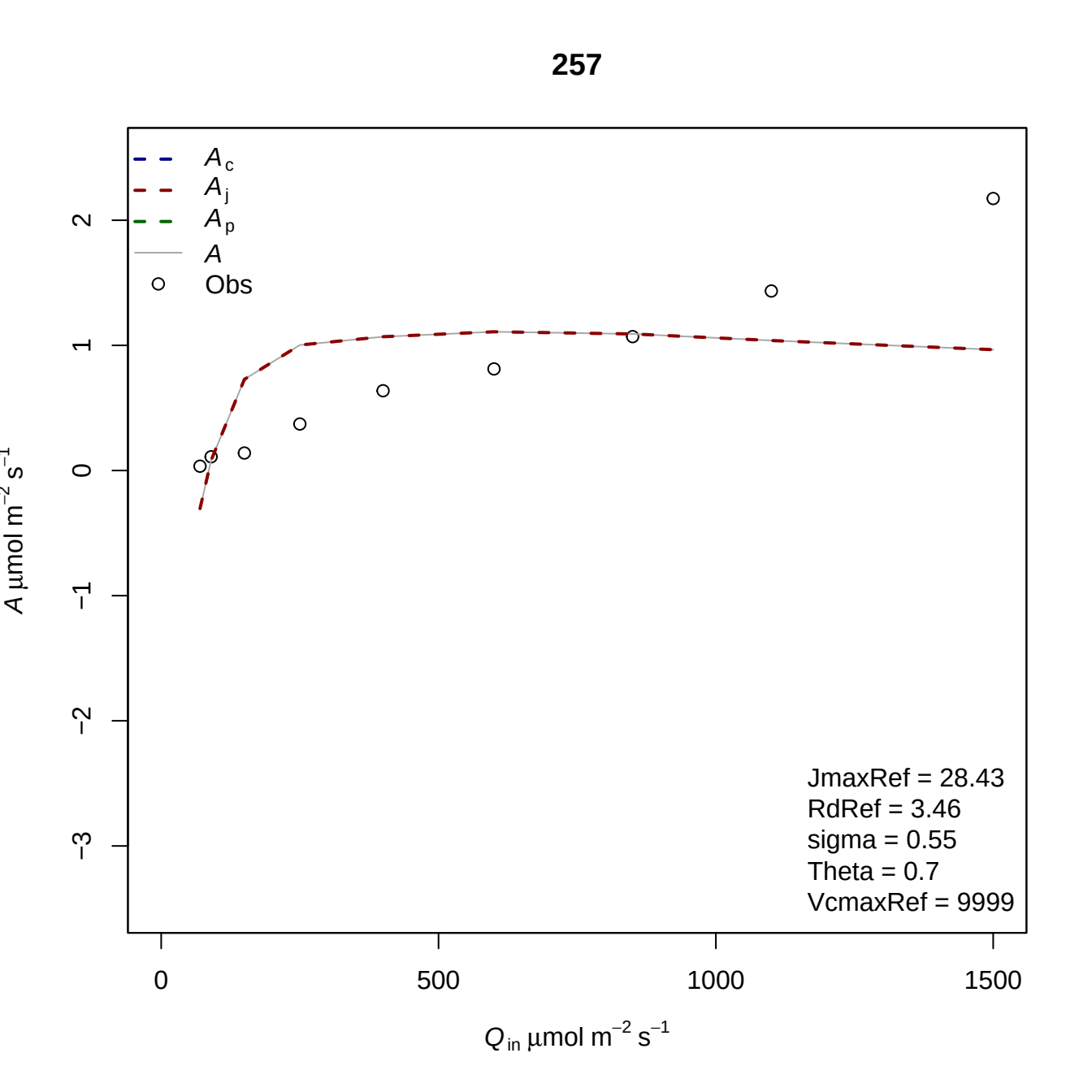

252
A_c
A_j
A_p
A
Obs
JmaxRef = 88.14
RdRef = 2.35
sigma = 0.69
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

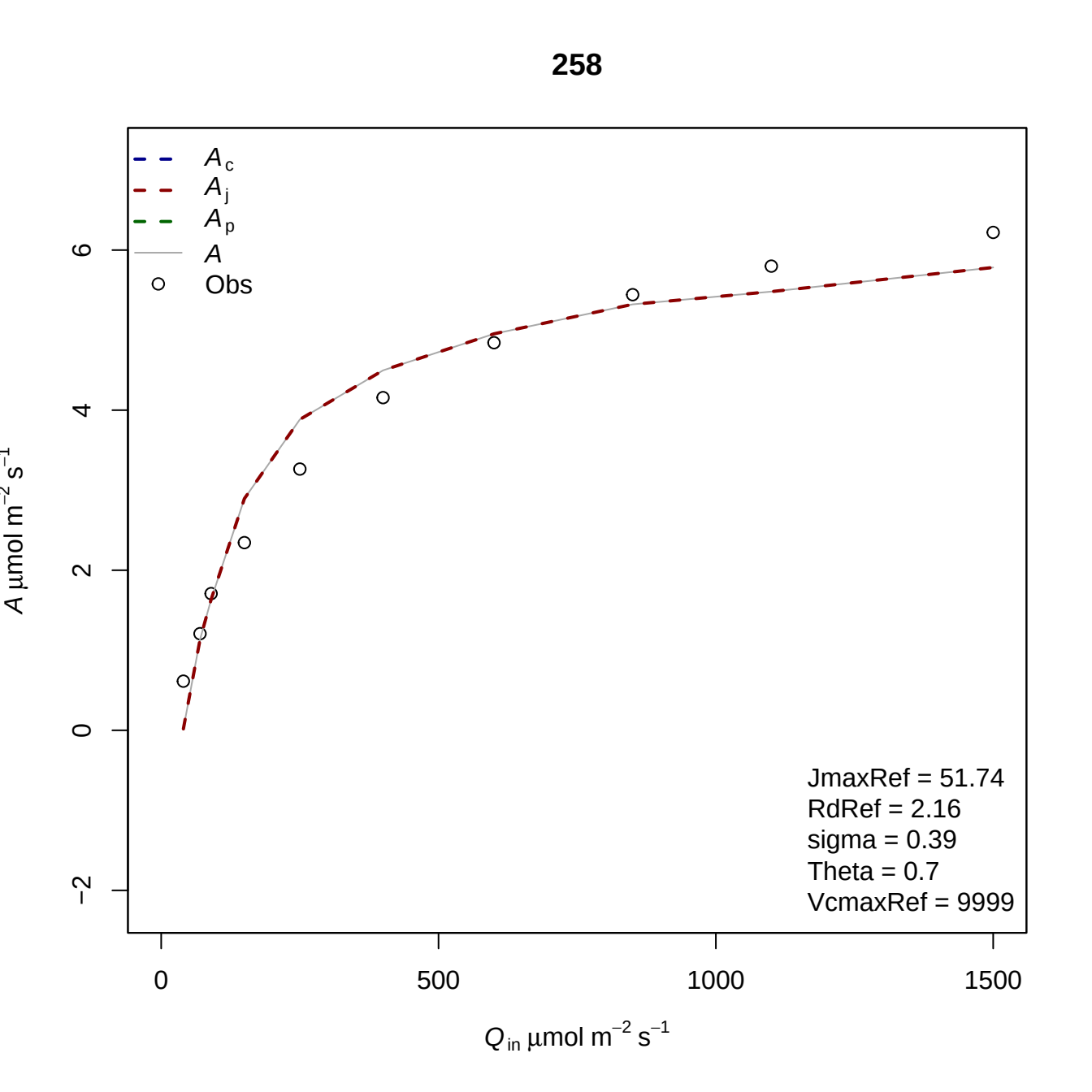

253
A_c
A_j
A_p
A
Obs
JmaxRef = 40.72
RdRef = 1.42
sigma = 0.15
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

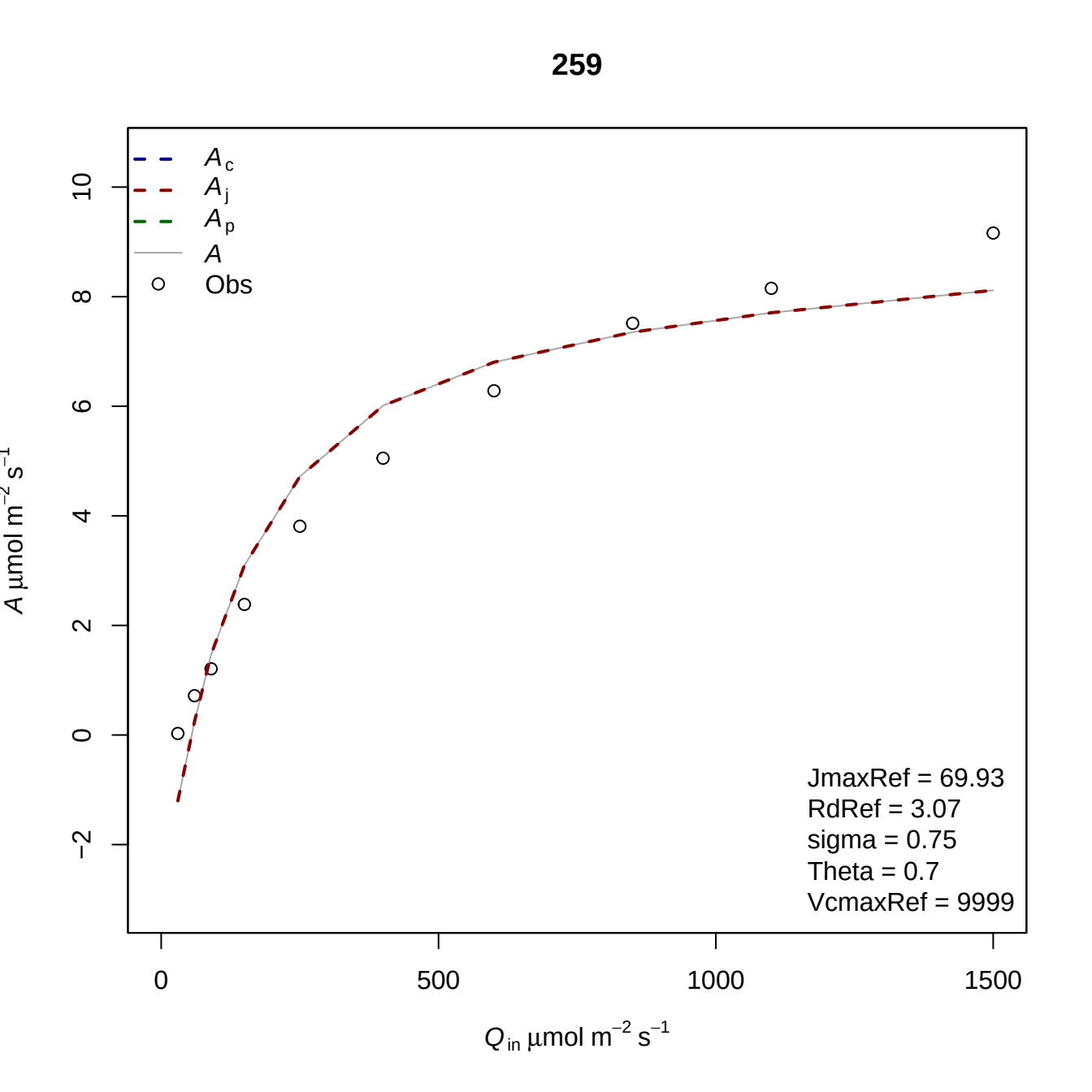

254
A_c
A_j
A_p
A
Obs
JmaxRef = 11.36
RdRef = 1.38
sigma = 0.16
Theta = 0.7
VcmaxRef = 9999
1.0
0.5
0.0
−0.5
−1.0
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

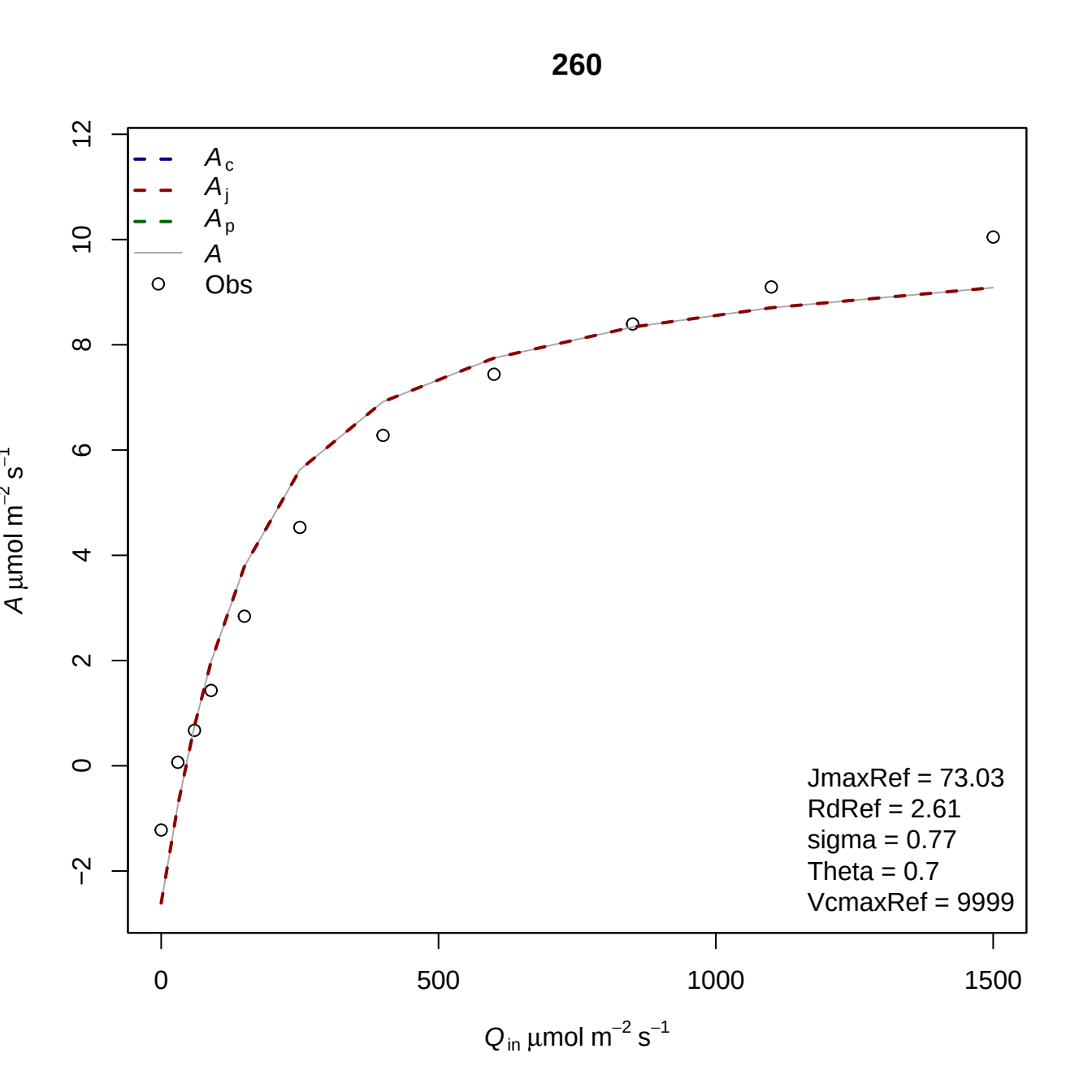

255
A_c
A_j
A_p
A
Obs
JmaxRef = 48.03
RdRef = 3.54
sigma = 0.11
Theta = 0.7
VcmaxRef = 9999
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

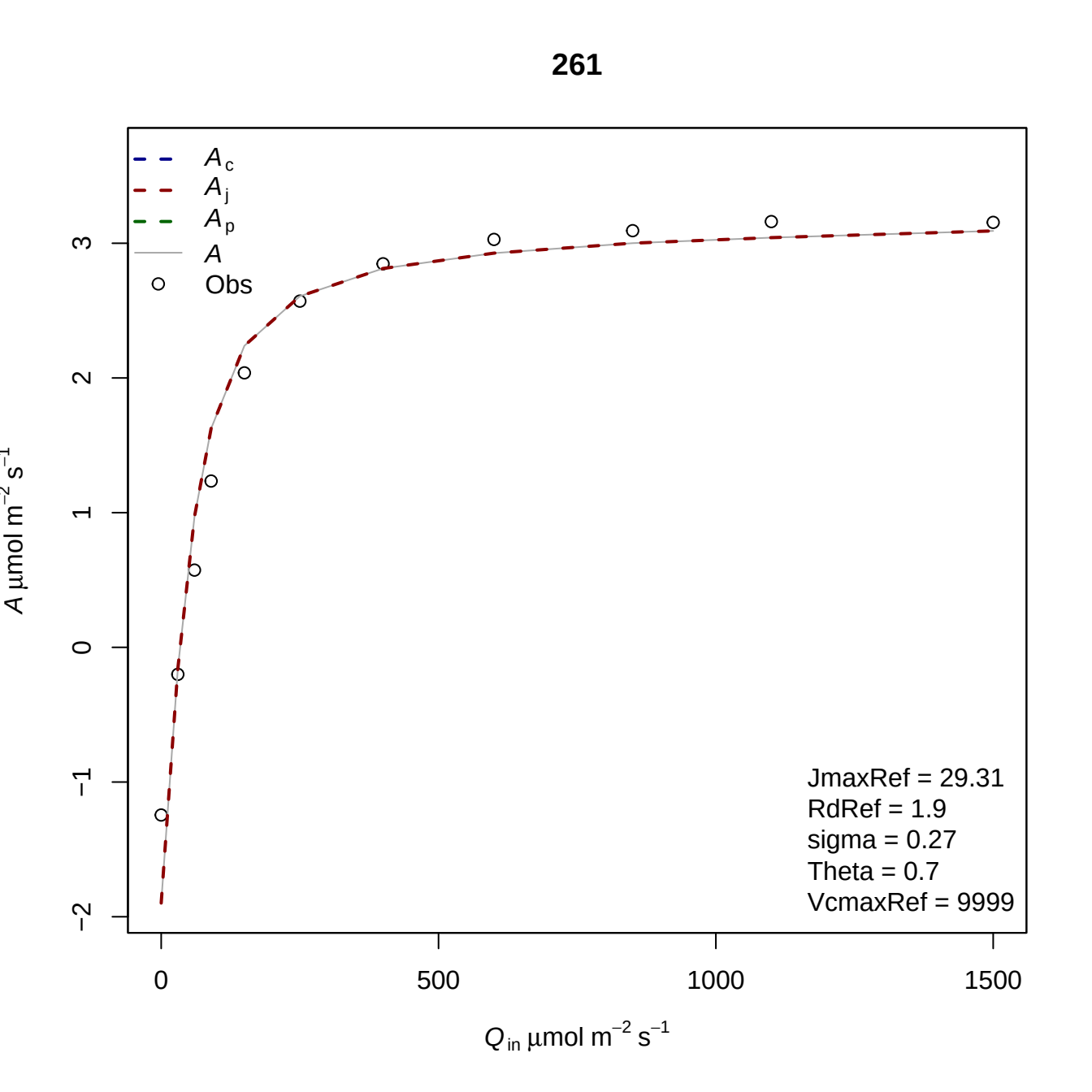

256
A_c
A_j
A_p
A
Obs
JmaxRef = 69.75
RdRef = 2.18
sigma = 0.48
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

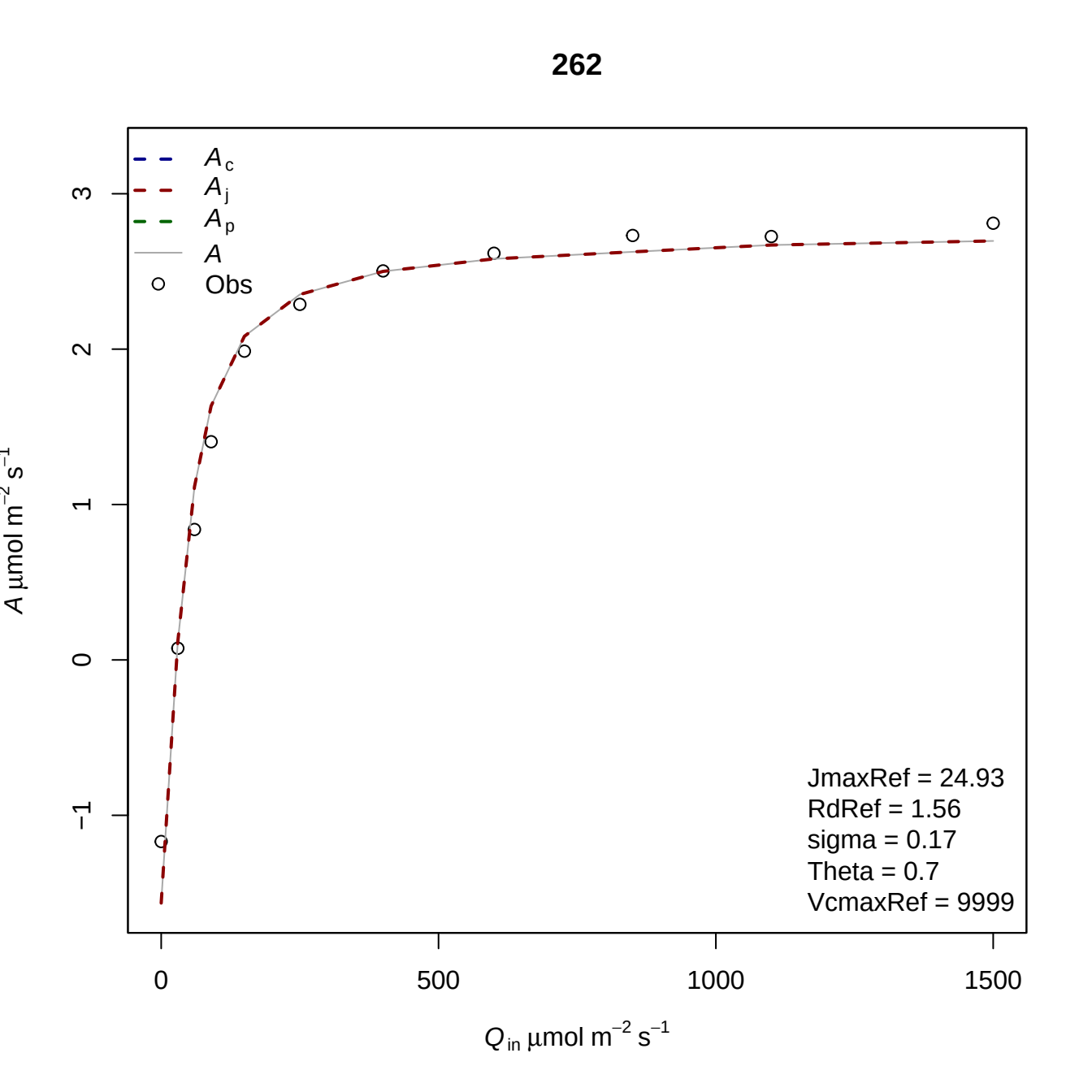

257
A_c
A_j
A_p
A
Obs
JmaxRef = 28.43
RdRef = 3.46
sigma = 0.55
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2
1
0
−1
−2
−3
0
500
1000
1500

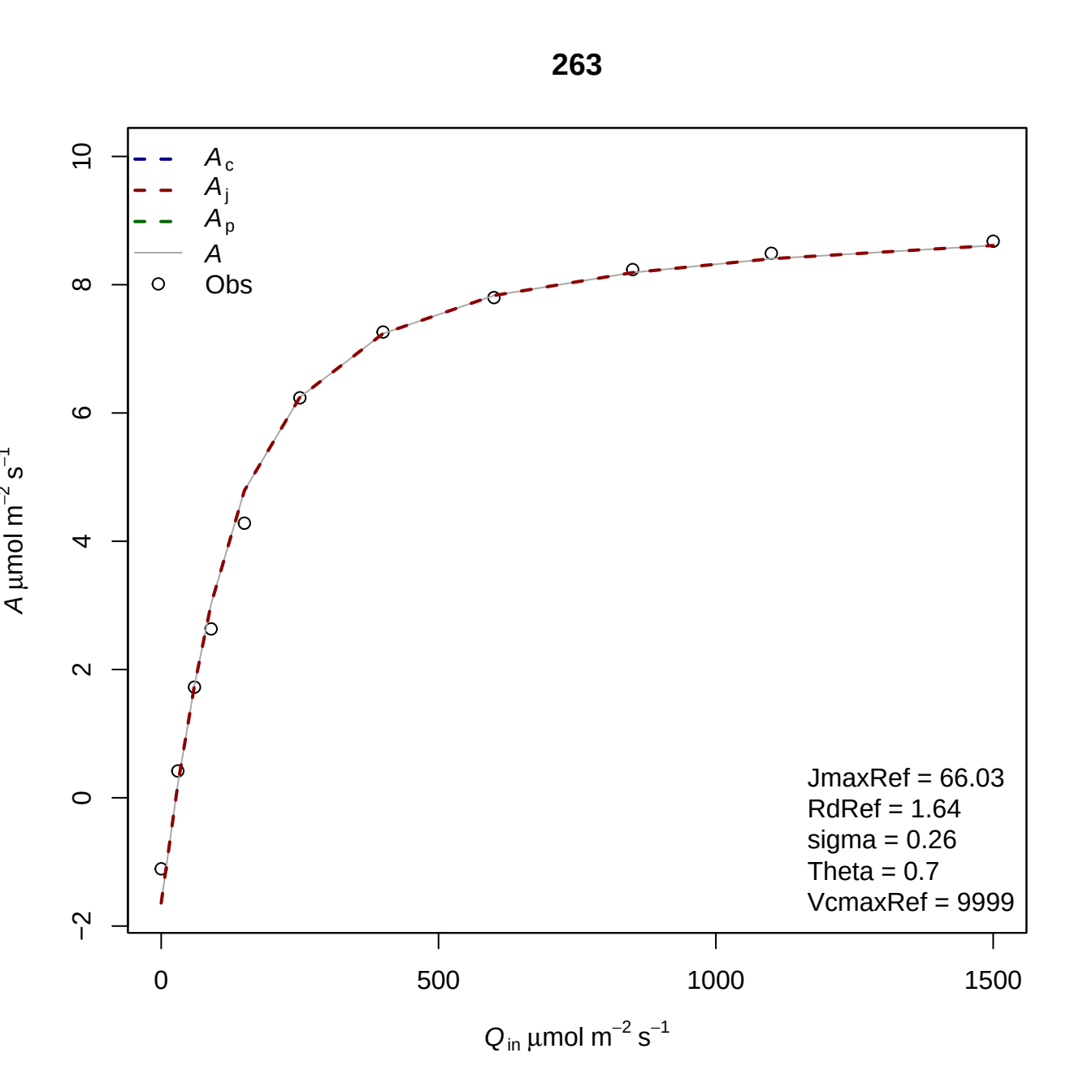

258
A_c
A_j
A_p
A
Obs
JmaxRef = 51.74
RdRef = 2.16
sigma = 0.39
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

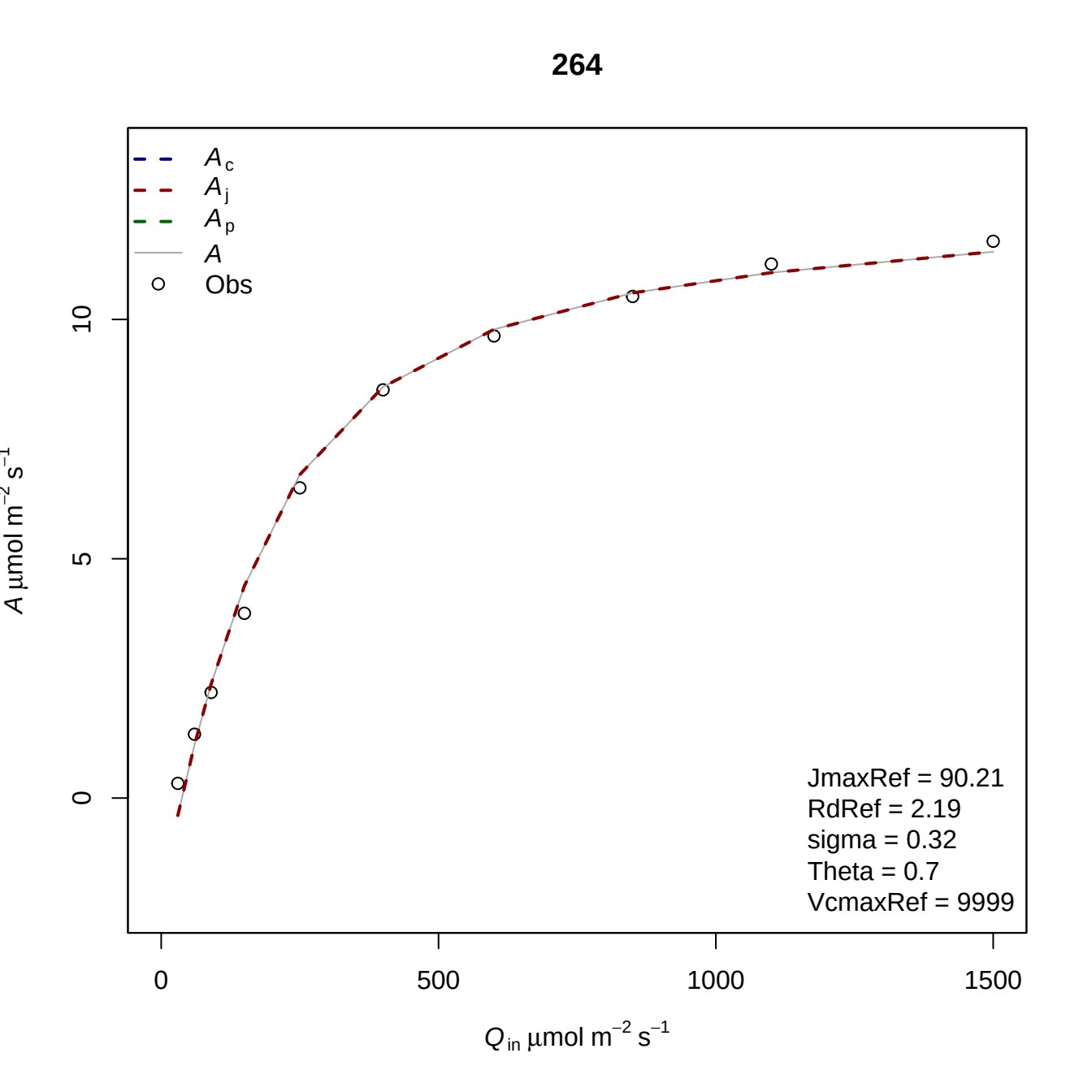

259
A_c
A_j
A_p
A
Obs
JmaxRef = 69.93
RdRef = 3.07
sigma = 0.75
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

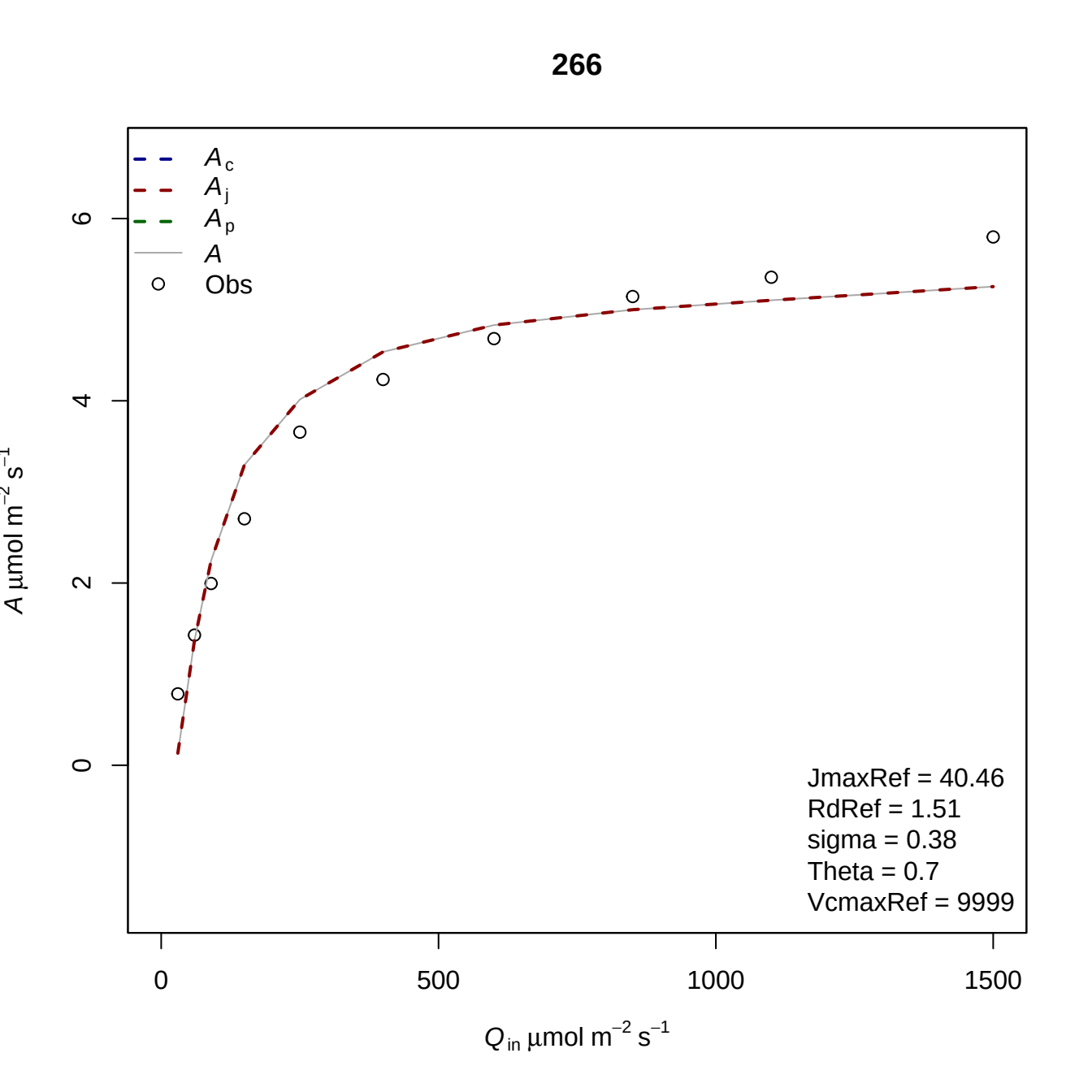

260
A_c
A_j
A_p
A
Obs
JmaxRef = 73.03
RdRef = 2.61
sigma = 0.77
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

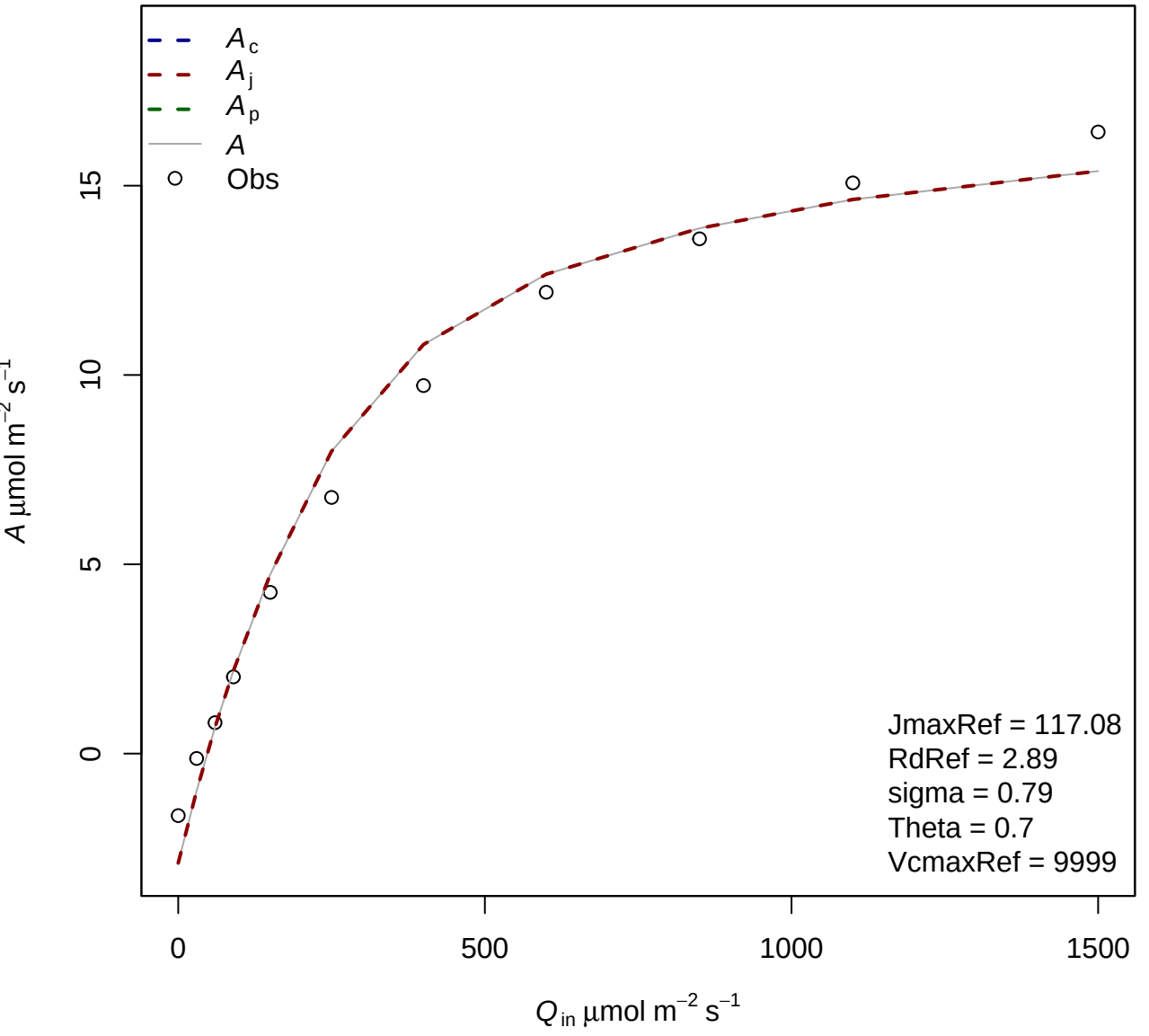

261
A_c
A_j
A_p
A
Obs
JmaxRef = 29.31
RdRef = 1.9
sigma = 0.27
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
−2
0
500
1000
1500

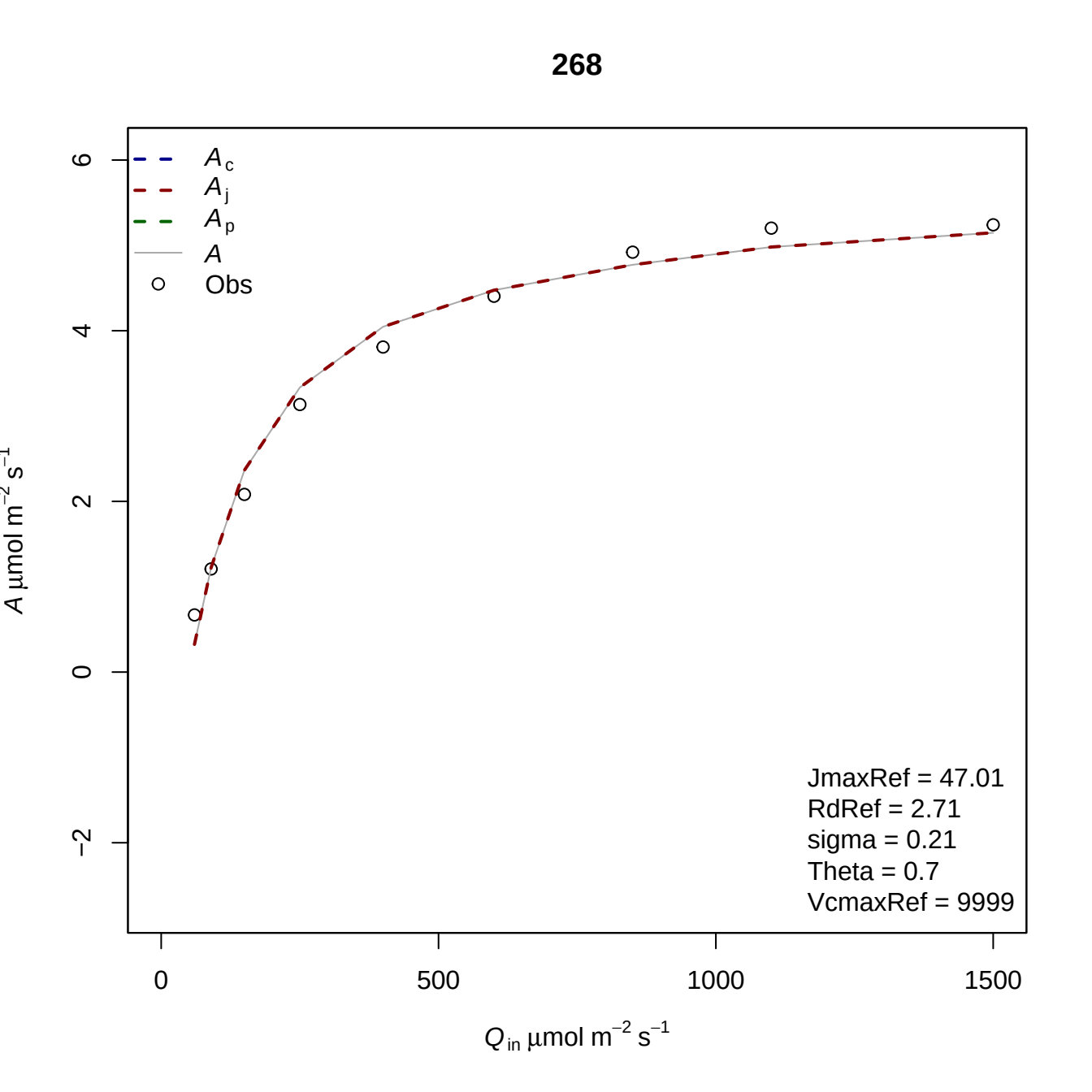

262
A_c
A_j
A_p
A
Obs
JmaxRef = 24.93
RdRef = 1.56
sigma = 0.17
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
0
500
1000
1500

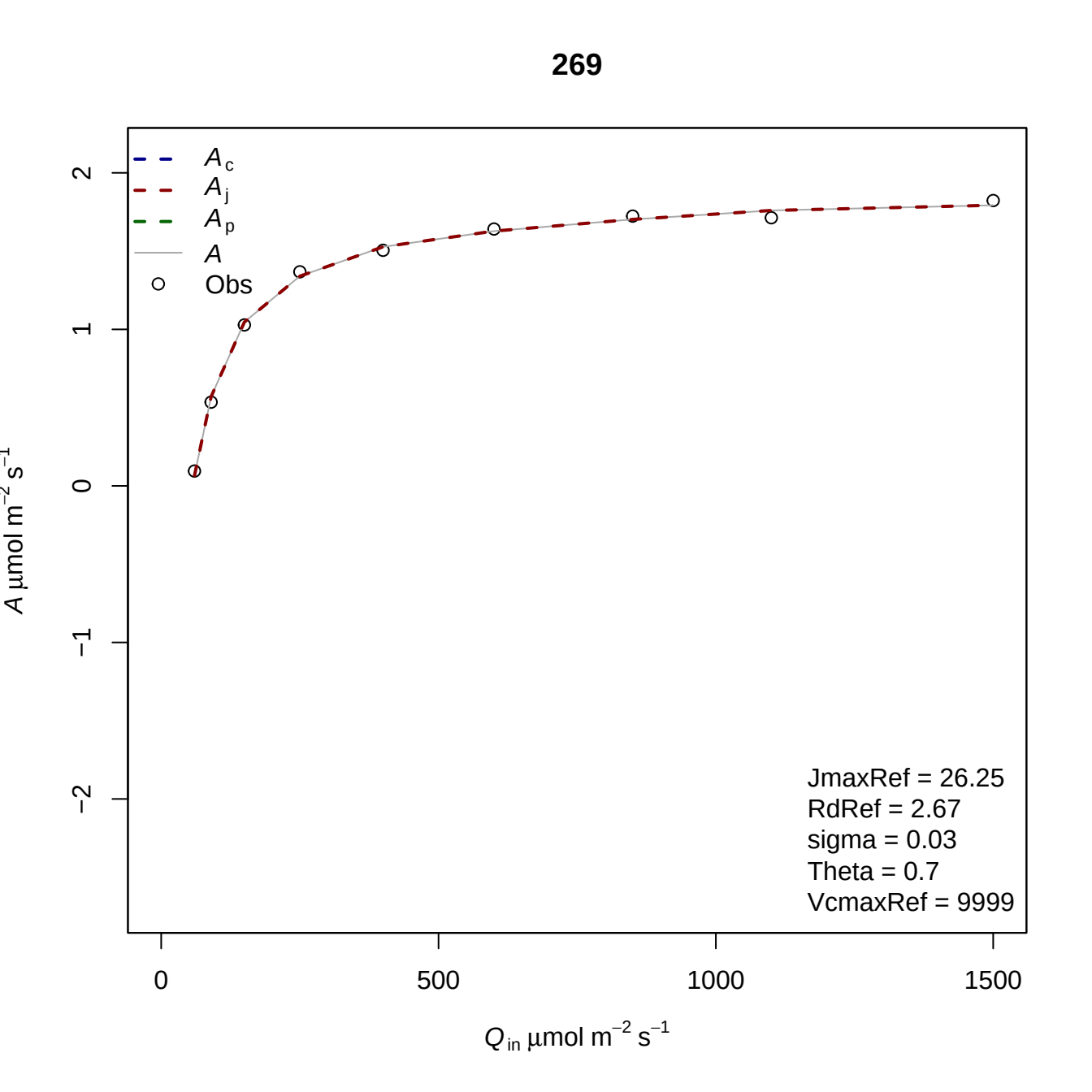

263
A_c
A_j
A_p
A
Obs
JmaxRef = 66.03
RdRef = 1.64
sigma = 0.26
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

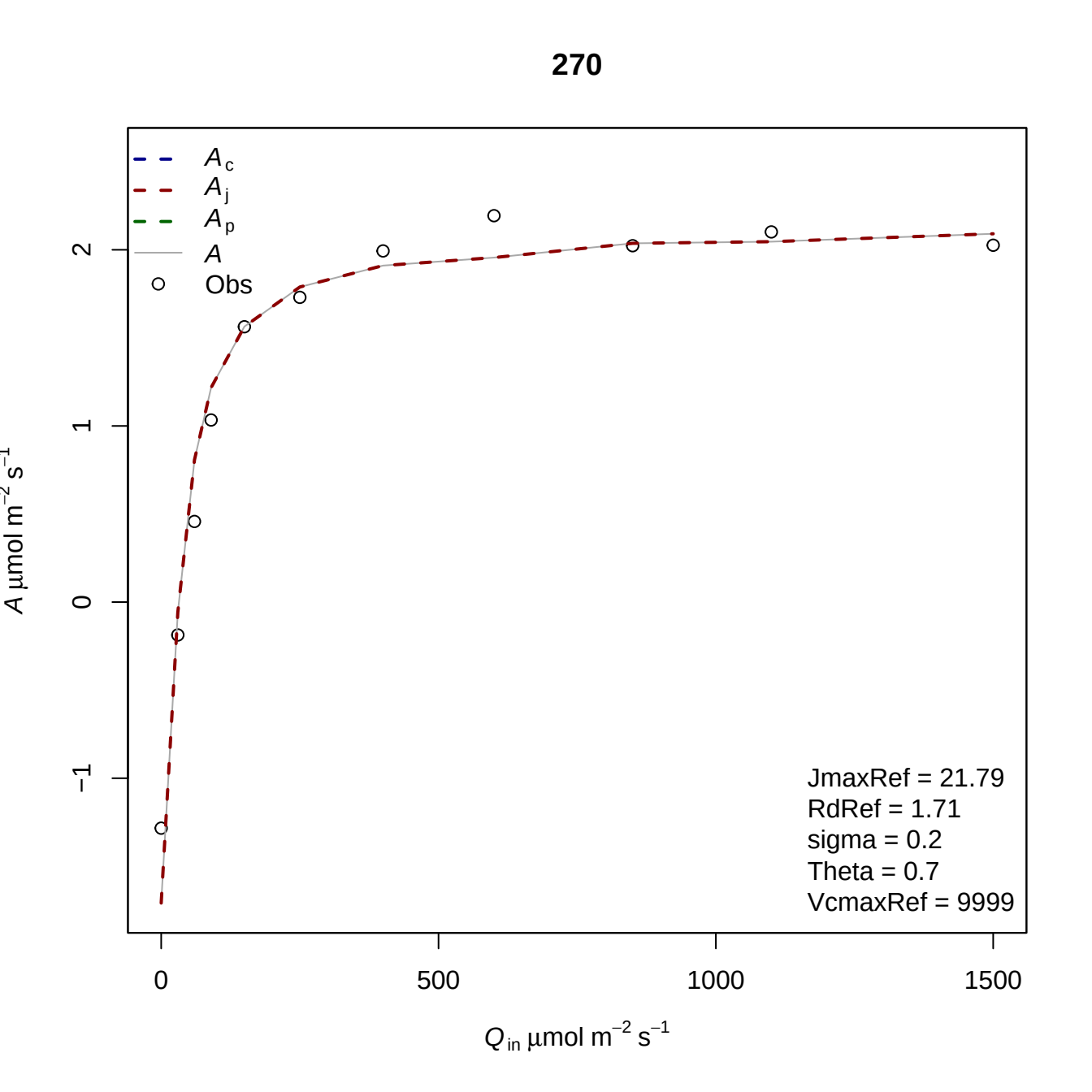

264
A_c
A_j
A_p
A
Obs
JmaxRef = 90.21
RdRef = 2.19
sigma = 0.32
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

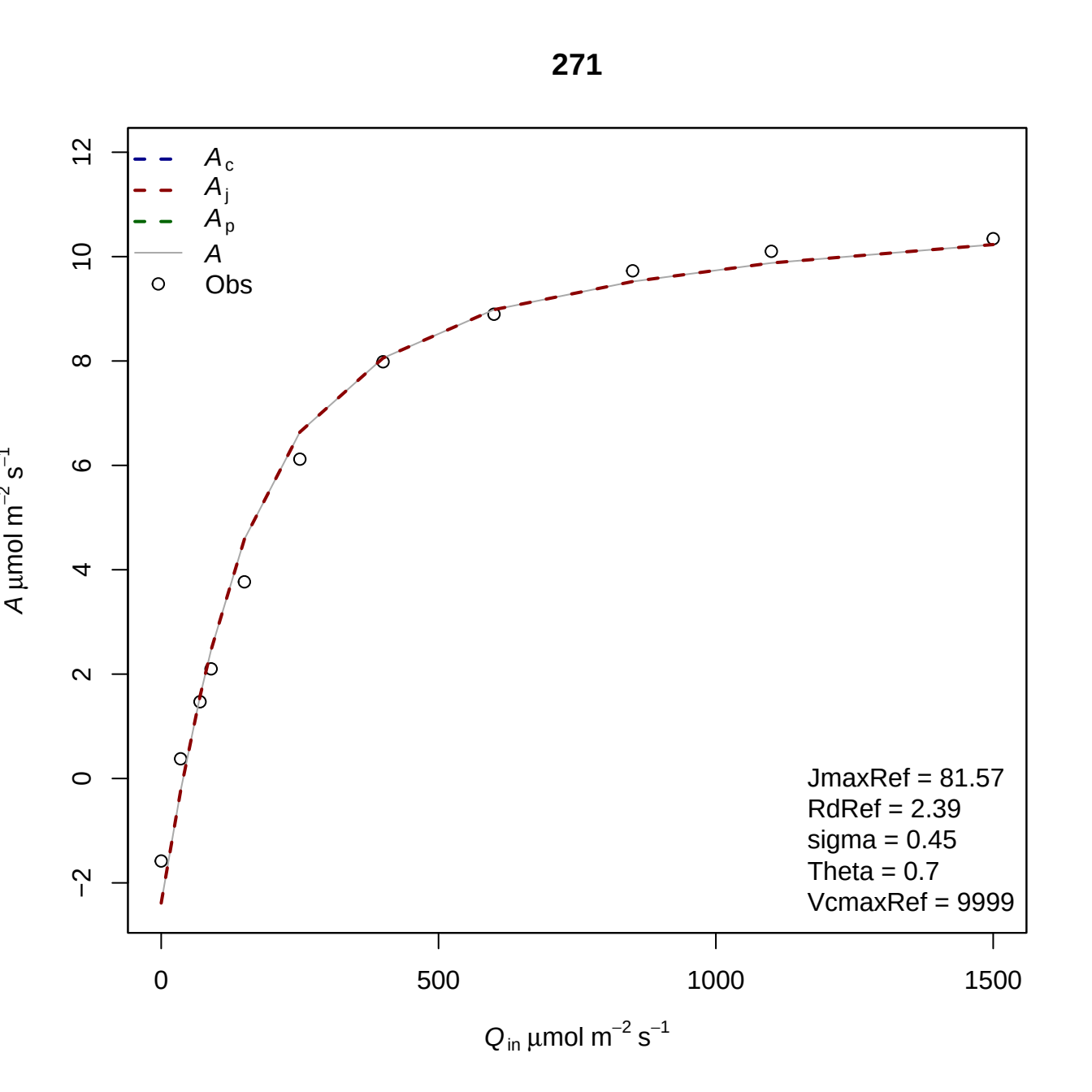

266
A_c
A_j
A_p
A
Obs
JmaxRef = 40.46
RdRef = 1.51
sigma = 0.38
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

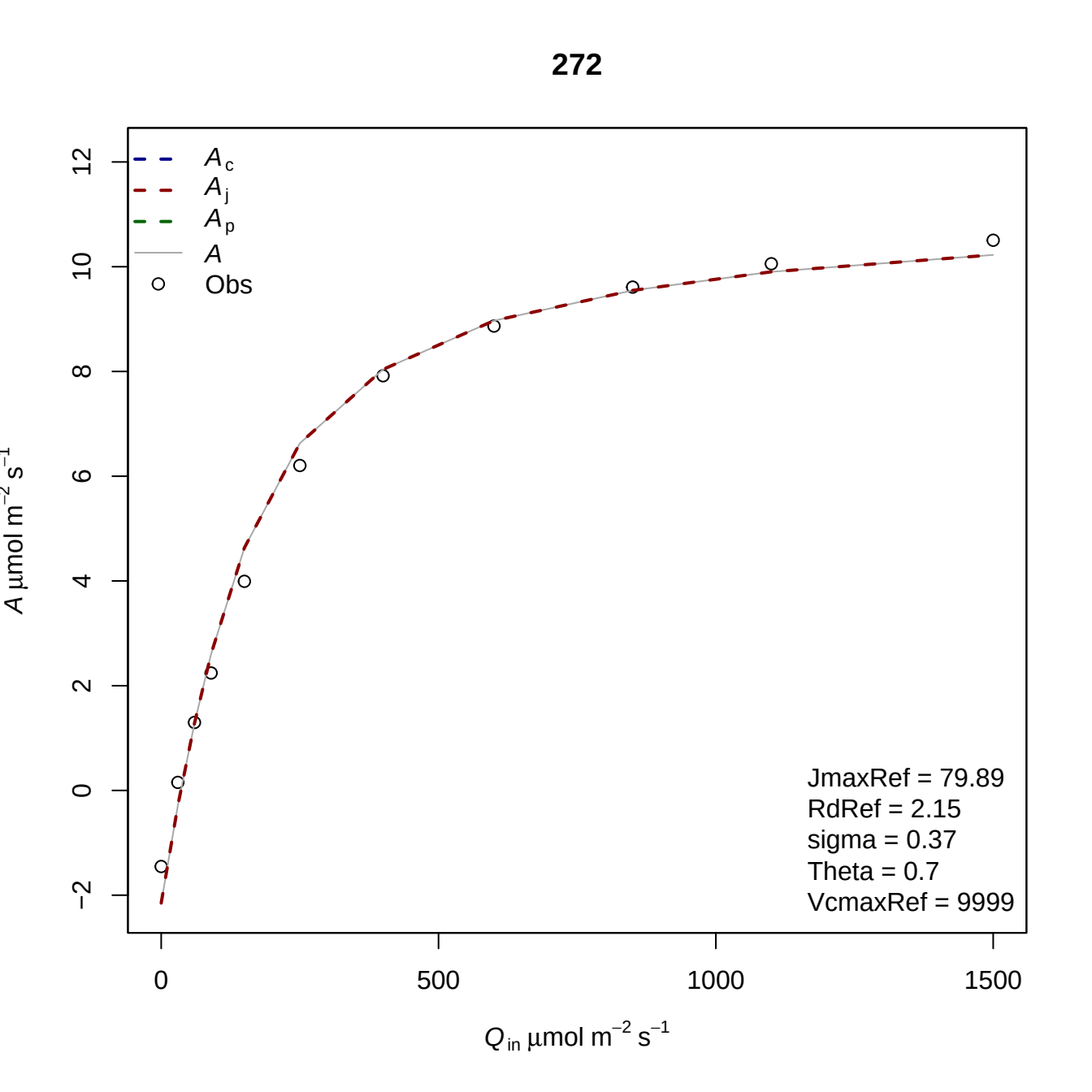

A_c
A_j
A_p
A
Obs
JmaxRef = 117.08
RdRef = 2.89
sigma = 0.79
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

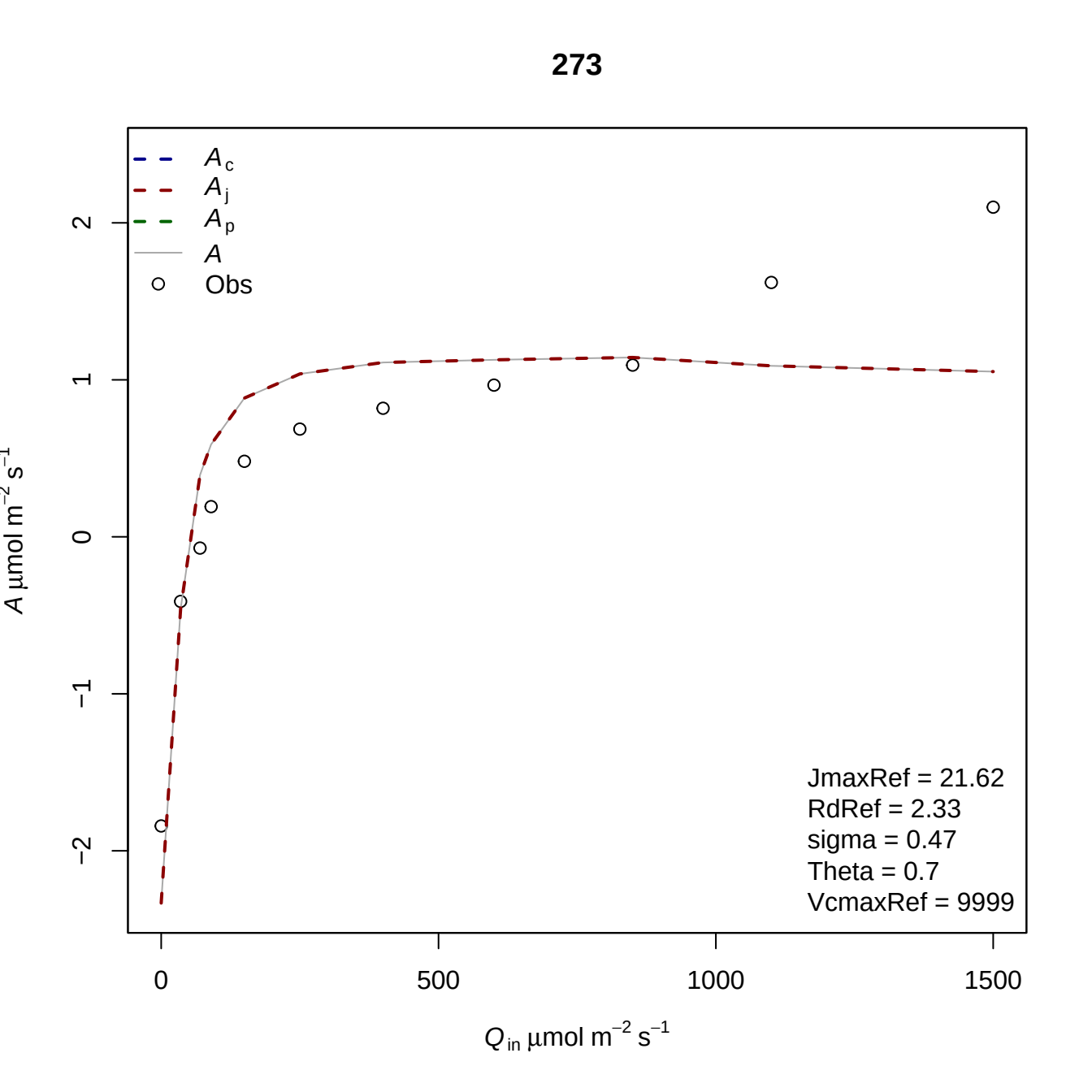

268
A_c
A_j
A_p
A
Obs
JmaxRef = 47.01
RdRef = 2.71
sigma = 0.21
Theta = 0.7
VcmaxRef = 9999
6
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

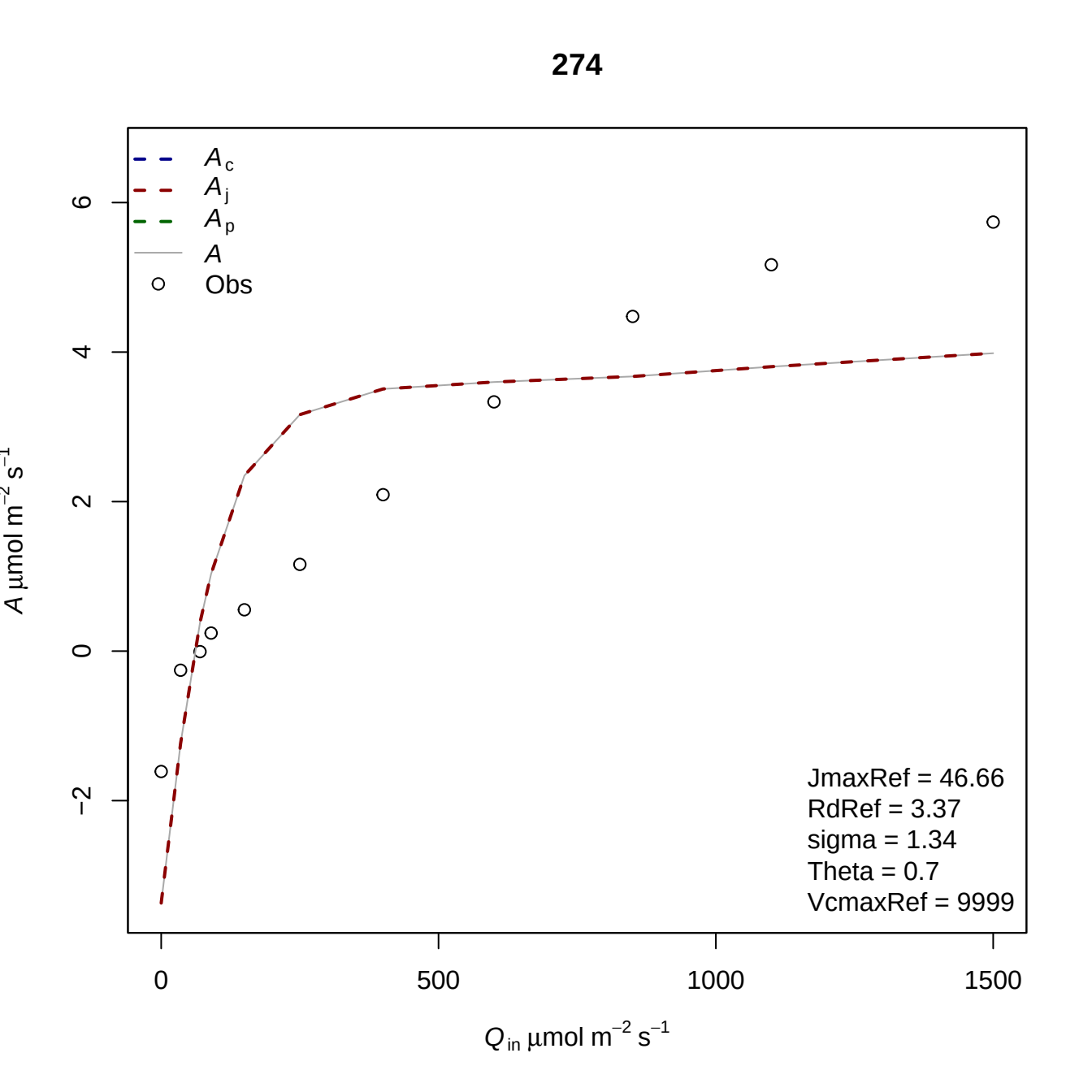

269
A_c
A_j
A_p
A
Obs
JmaxRef = 26.25
RdRef = 2.67
sigma = 0.03
Theta = 0.7
VcmaxRef = 9999
2
1
0
−1
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

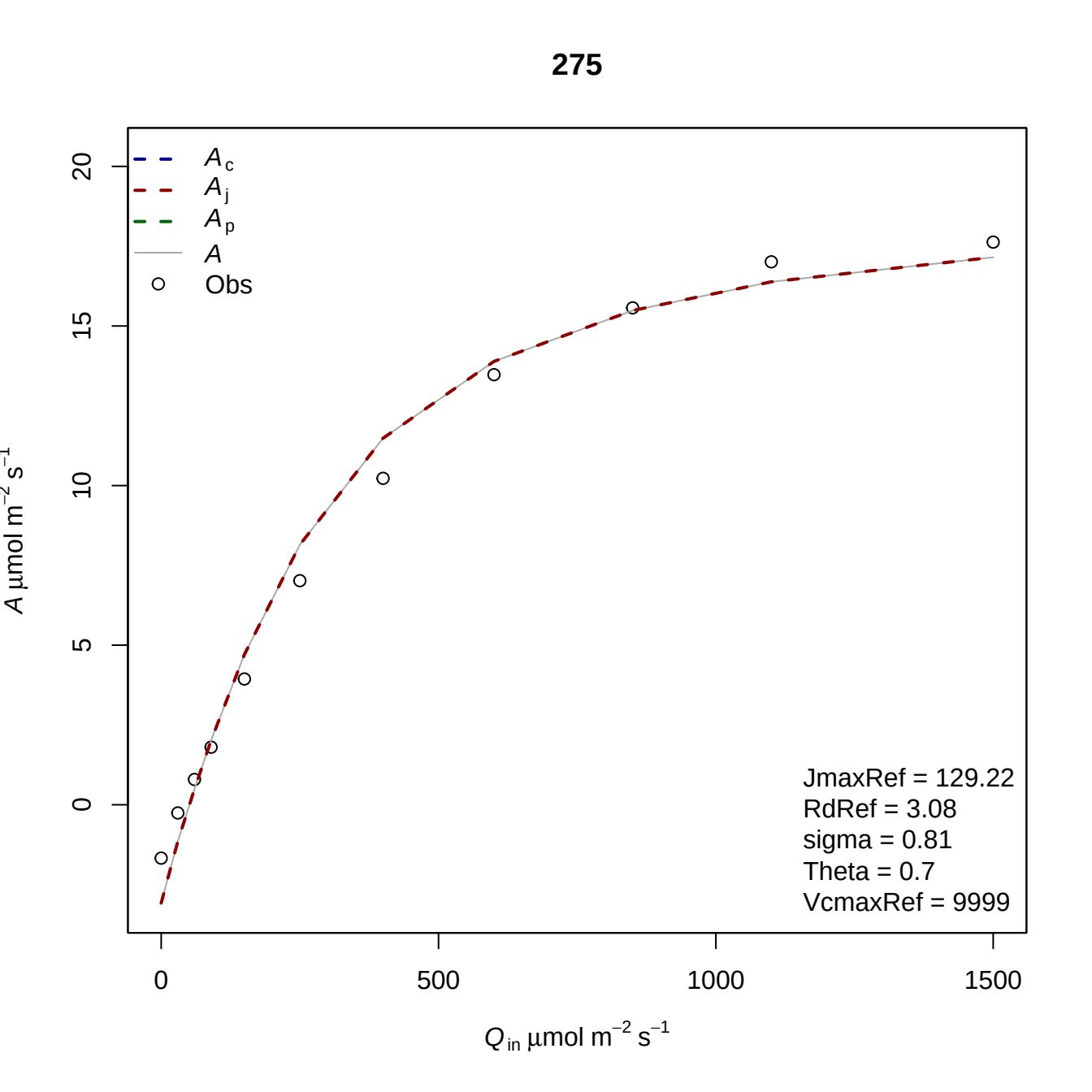

270
A_c
A_j
A_p
A
Obs
JmaxRef = 21.79
RdRef = 1.71
sigma = 0.2
Theta = 0.7
VcmaxRef = 9999
2
1
0
−1
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

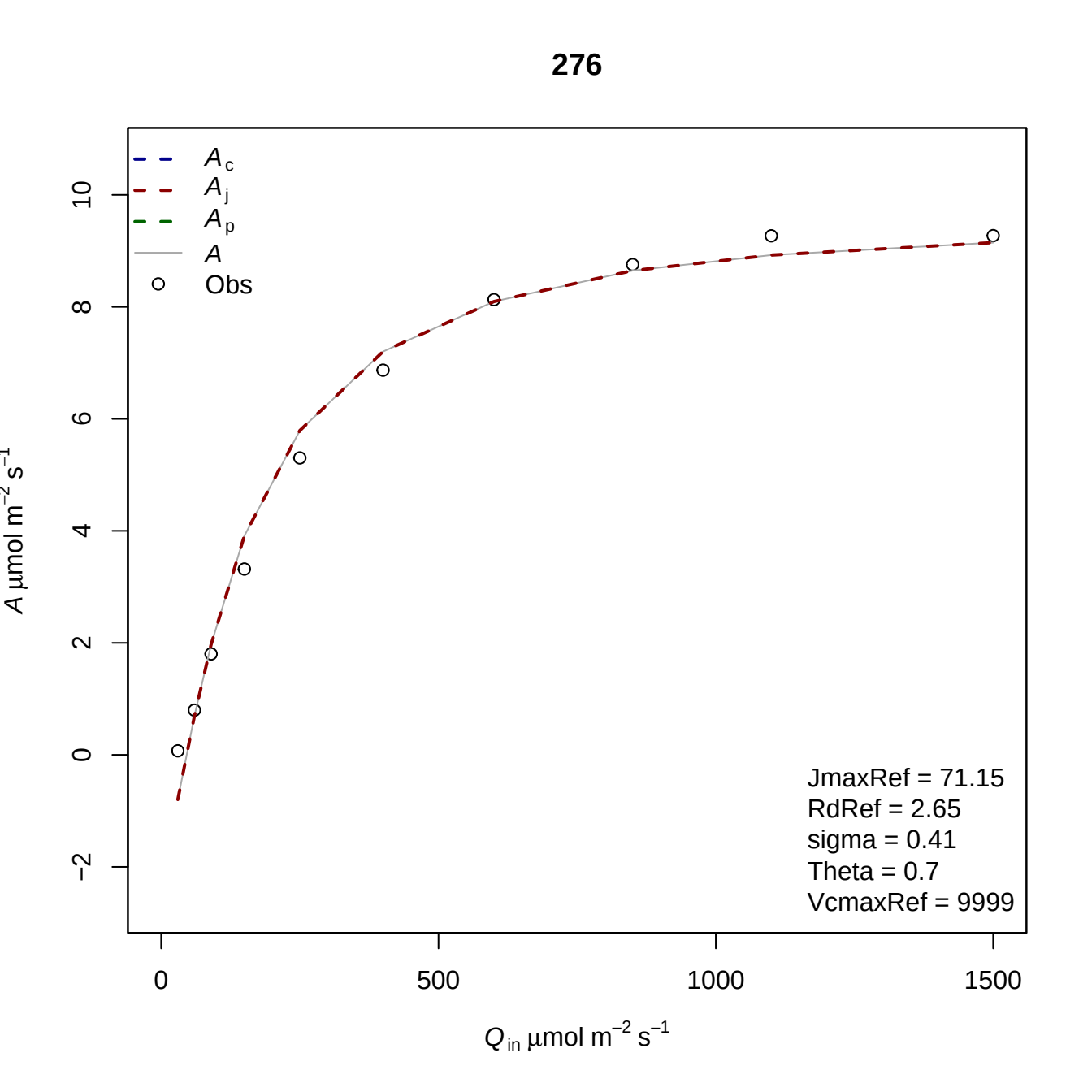

271
A_c
A_j
A_p
A
Obs
JmaxRef = 81.57
RdRef = 2.39
sigma = 0.45
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

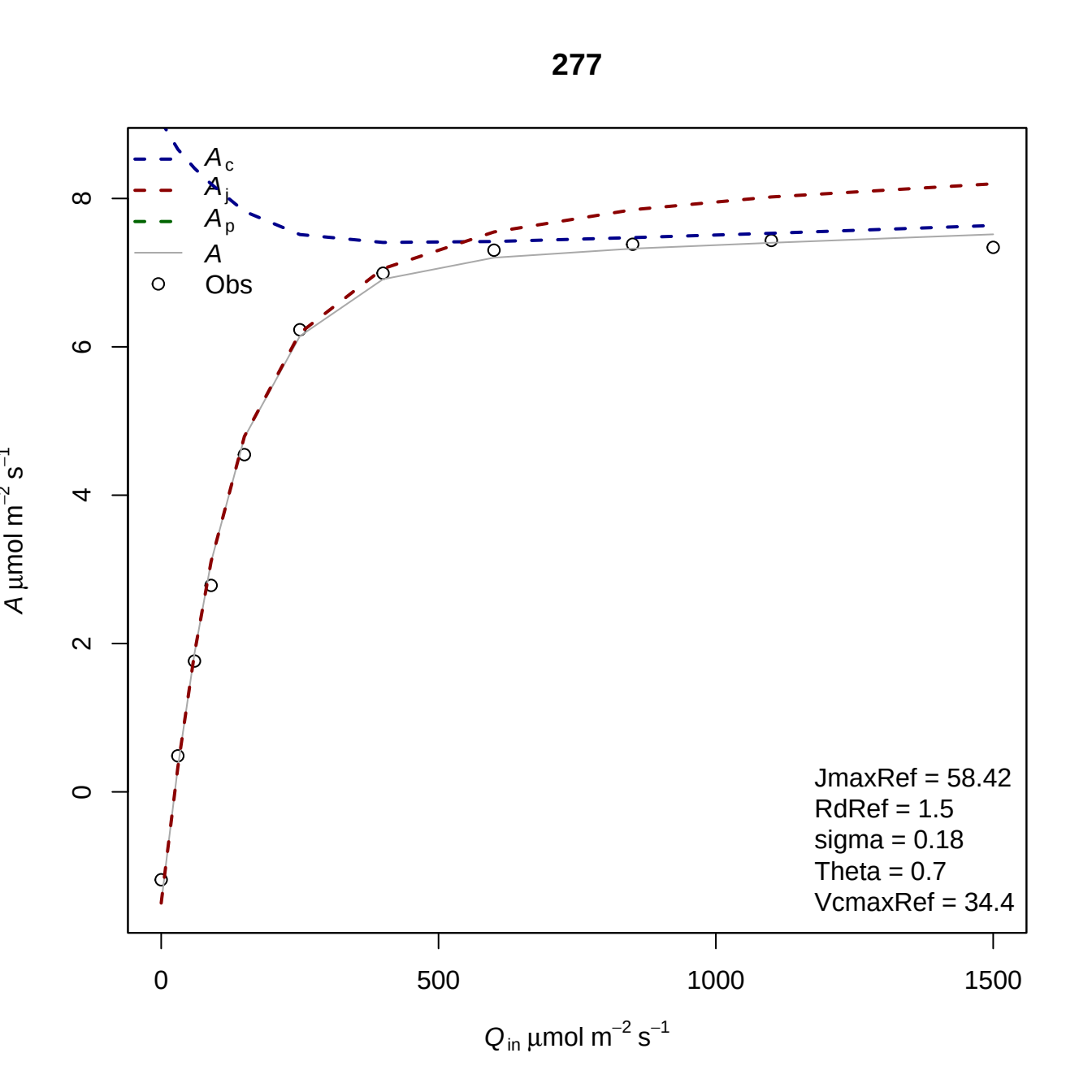

272
A_c
A_j
A_p
A
Obs
JmaxRef = 79.89
RdRef = 2.15
sigma = 0.37
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

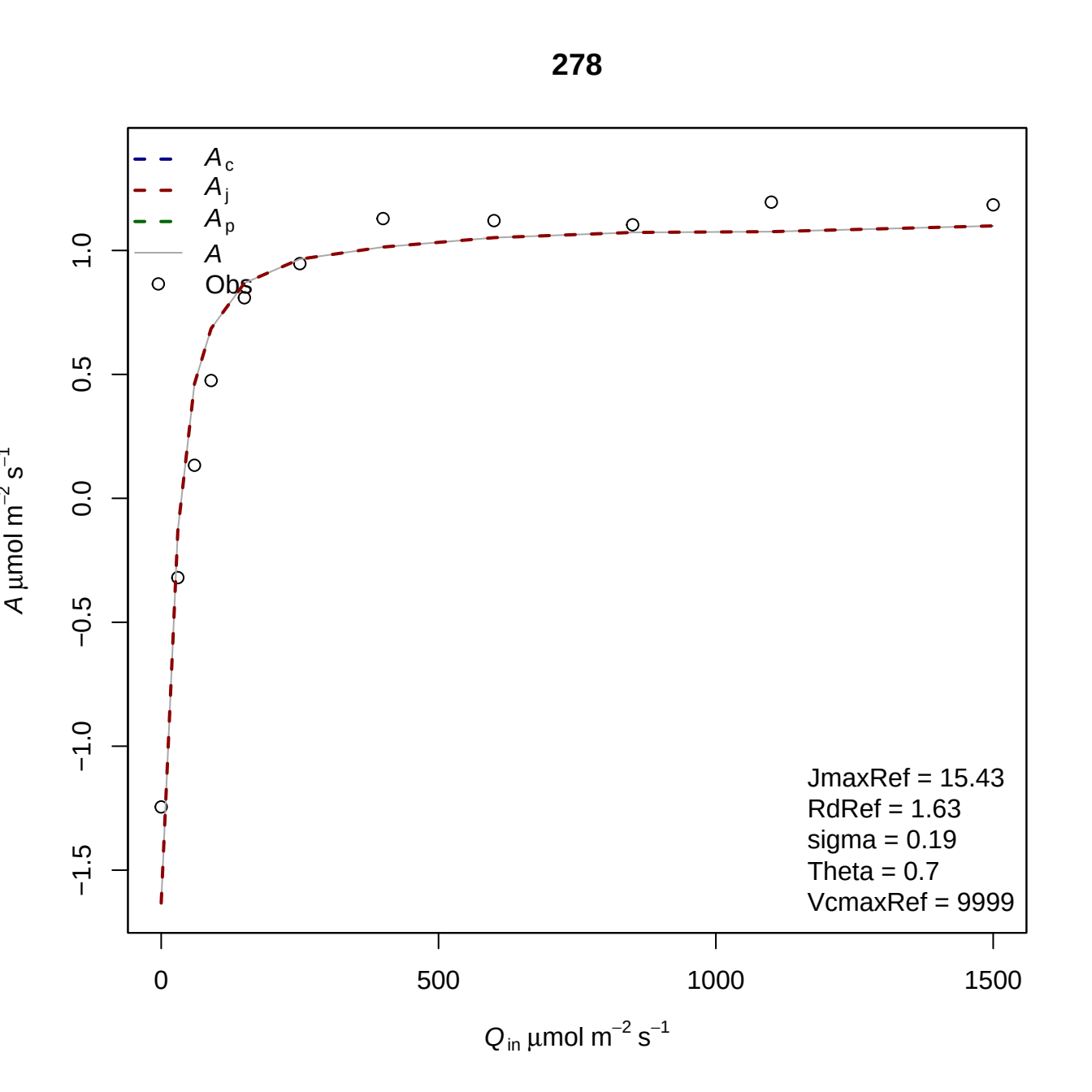

273
A_c
A_j
A_p
A
Obs
JmaxRef = 21.62
RdRef = 2.33
sigma = 0.47
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2
1
0
−1
−2
0
500
1000
1500

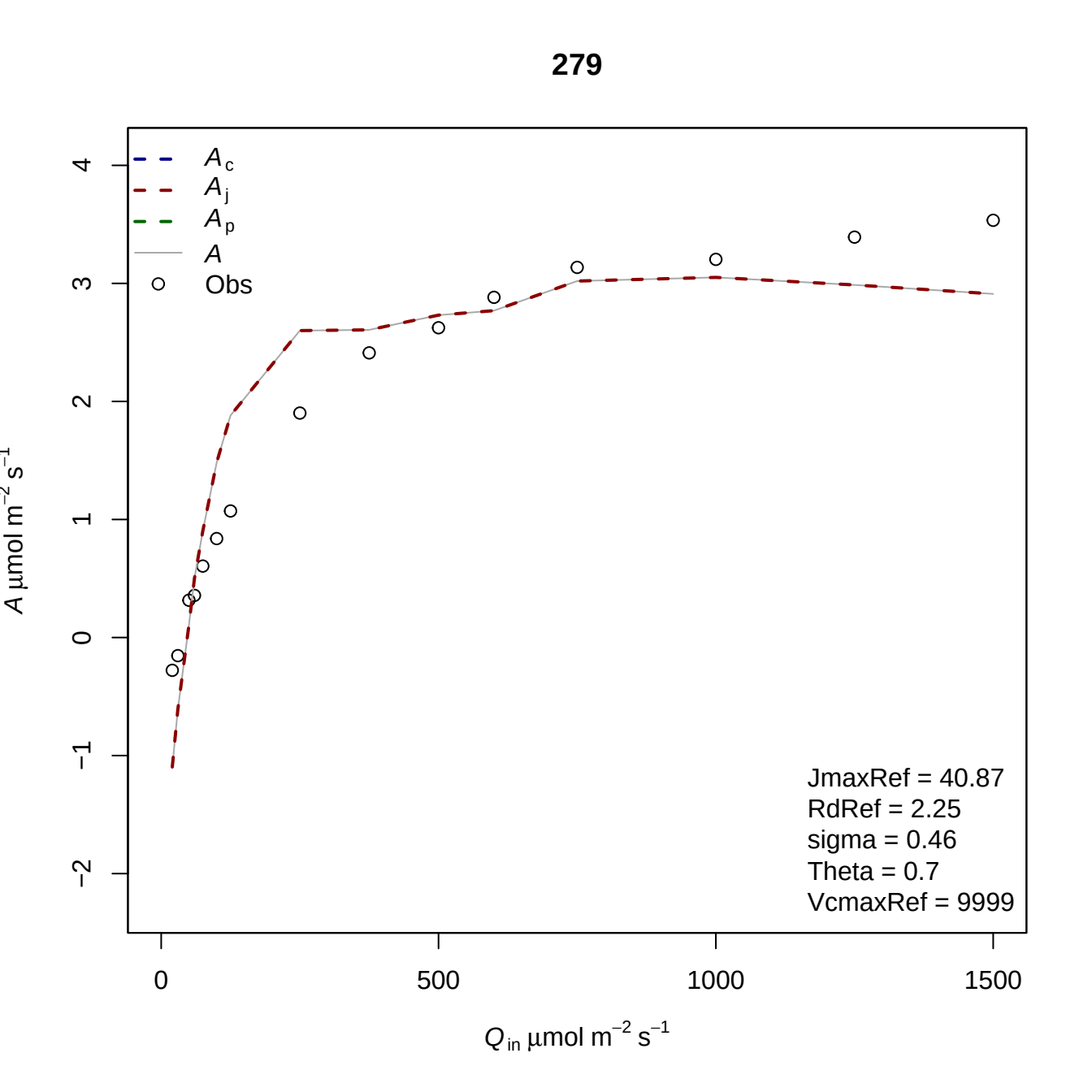

274
A_c
A_j
A_p
A
Obs
JmaxRef = 46.66
RdRef = 3.37
sigma = 1.34
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

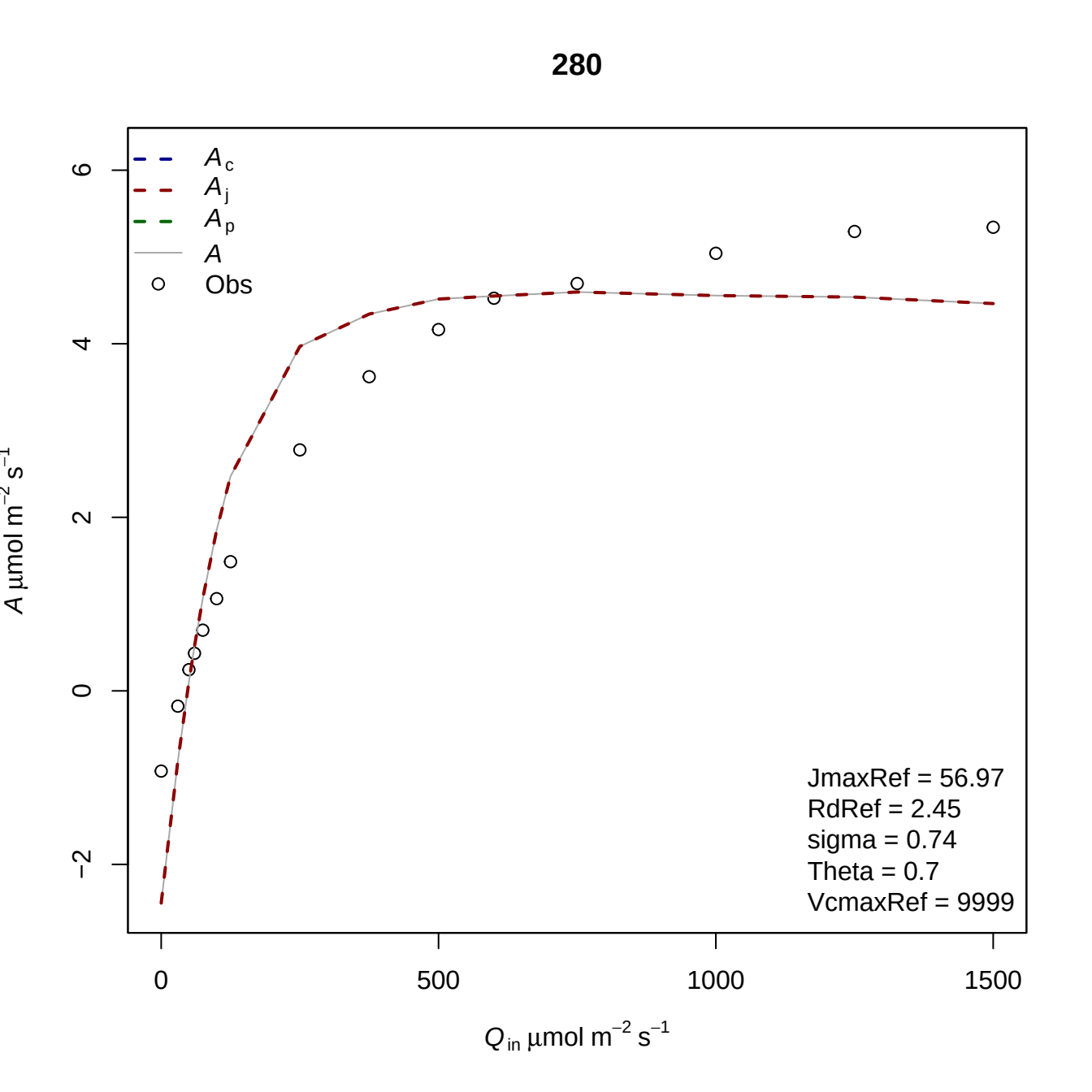

275
A_c
A_j
A_p
A
Obs
JmaxRef = 129.22
RdRef = 3.08
sigma = 0.81
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

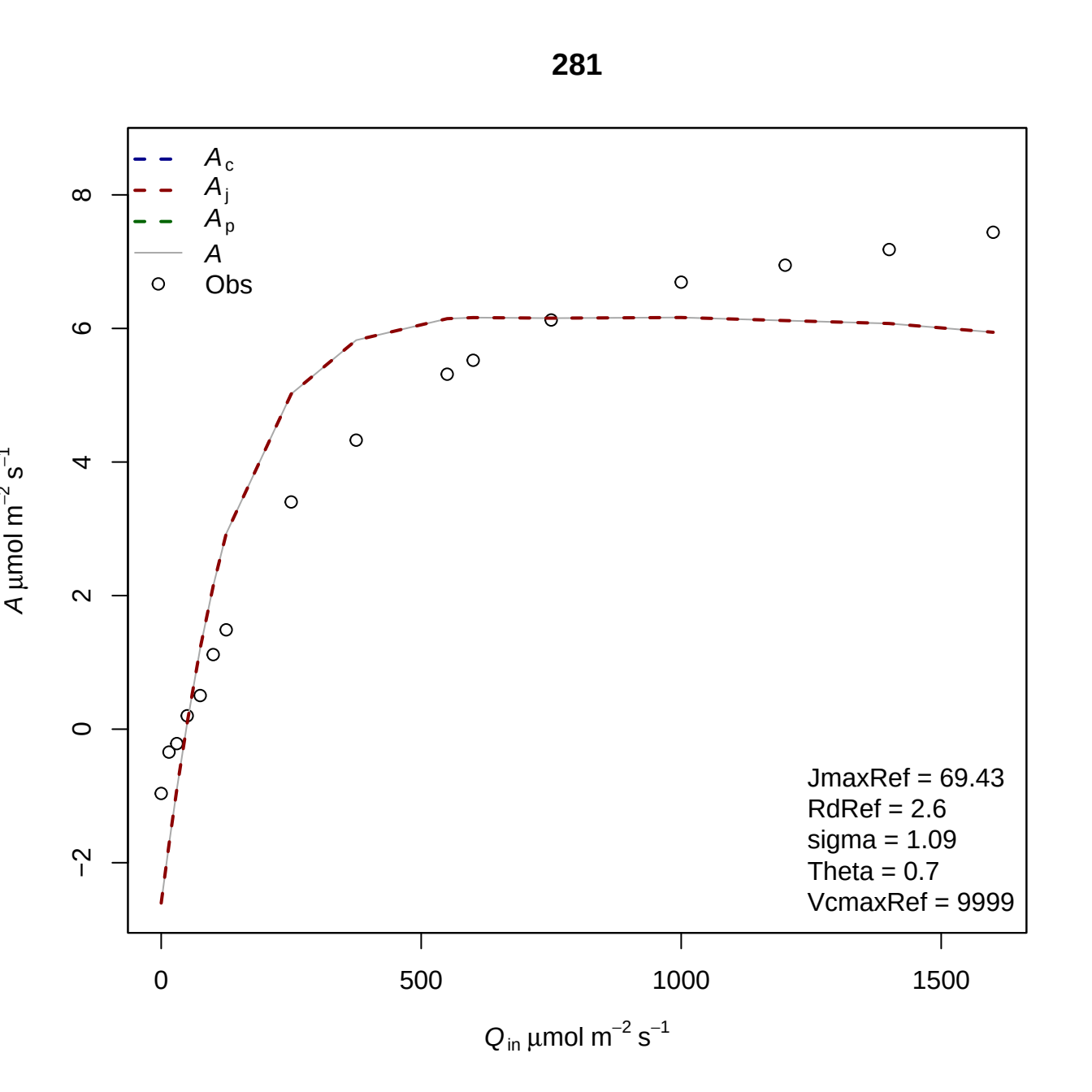

276
A_c
A_j
A_p
A
Obs
JmaxRef = 71.15
RdRef = 2.65
sigma = 0.41
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

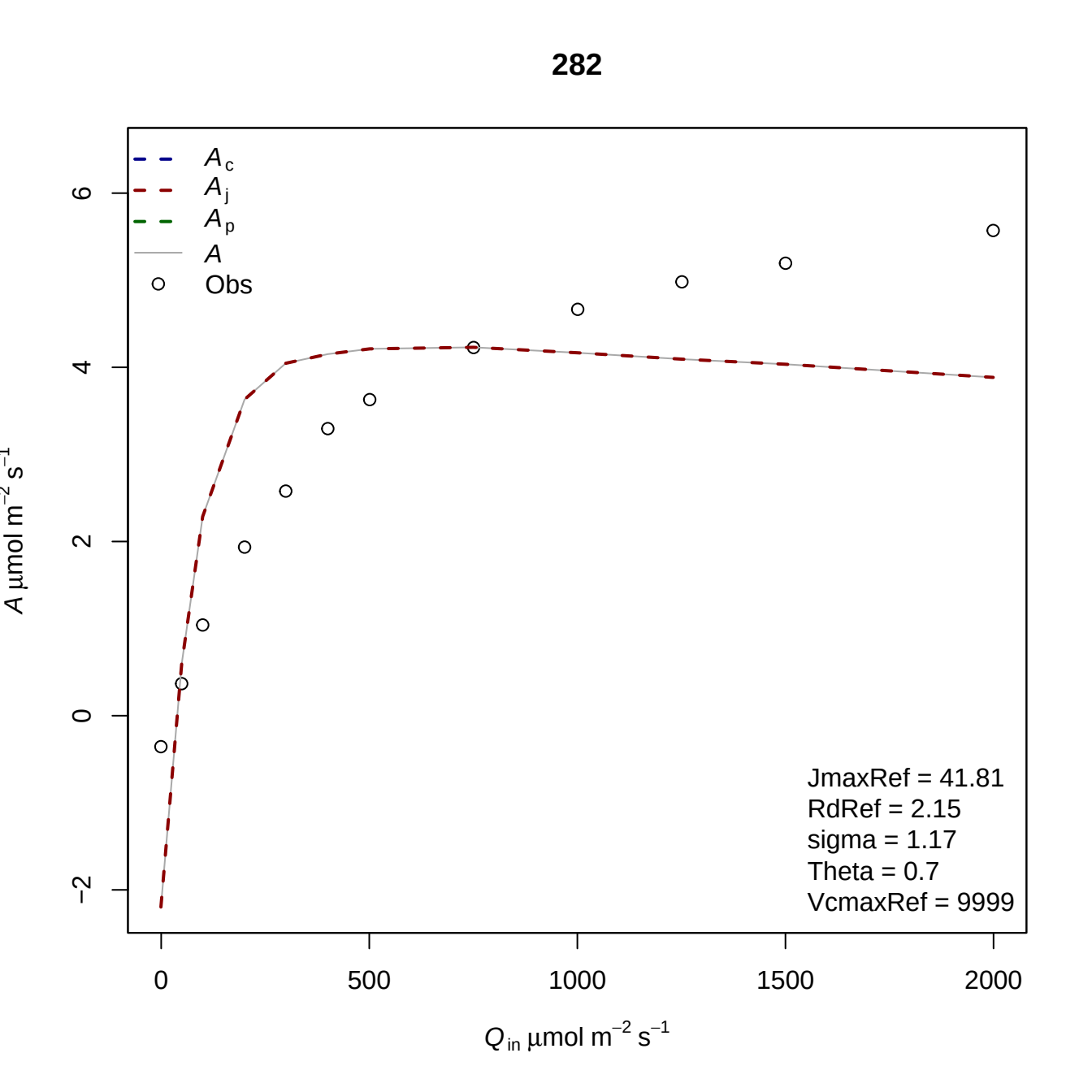

277
A_c
A_j
A_p
A
Obs
JmaxRef = 58.42
RdRef = 1.5
sigma = 0.18
Theta = 0.7
VcmaxRef = 34.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

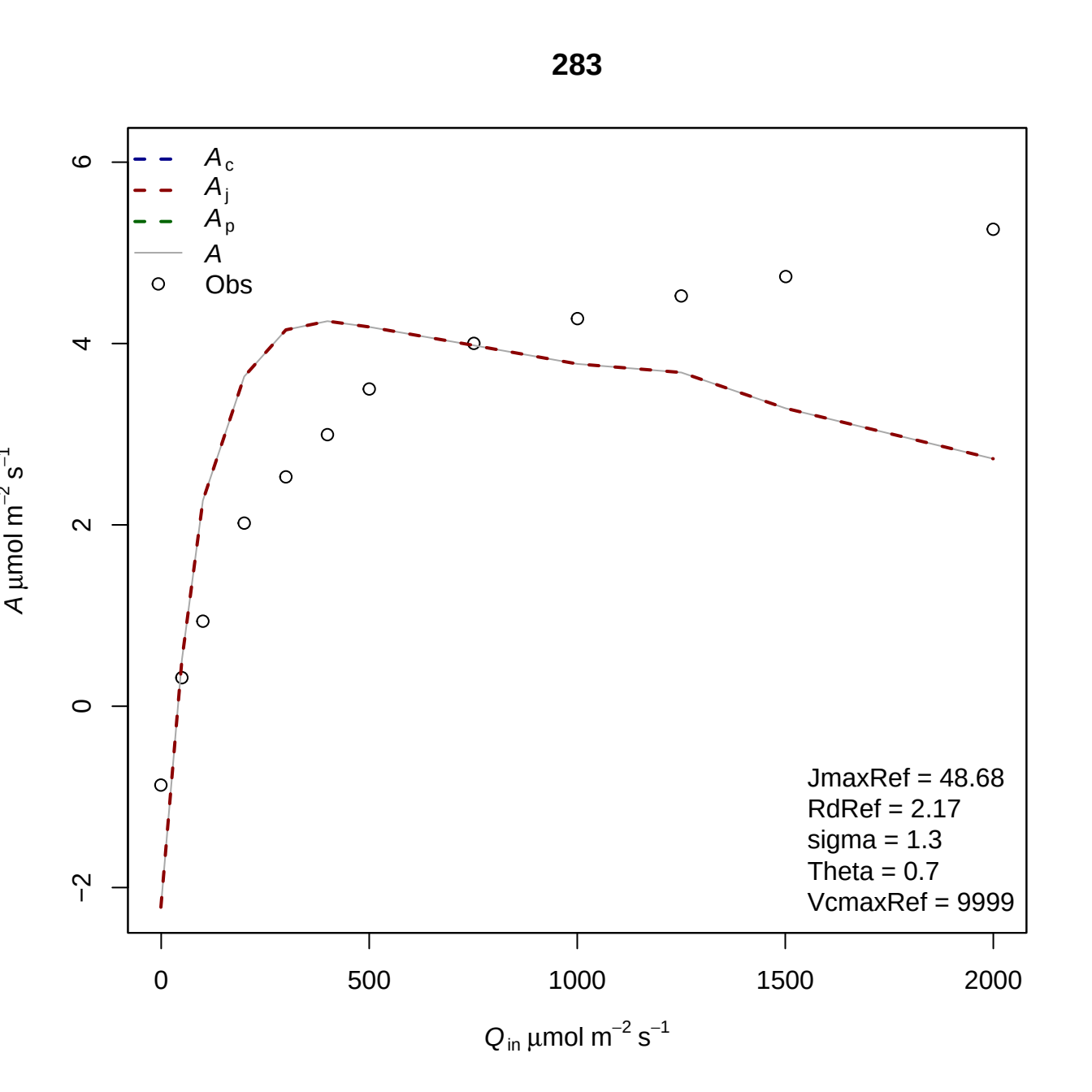

278
A_c
A_j
A_p
A
Obs
JmaxRef = 15.43
RdRef = 1.63
sigma = 0.19
Theta = 0.7
VcmaxRef = 9999
1.0
0.5
0.0
−0.5
−1.0
−1.5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

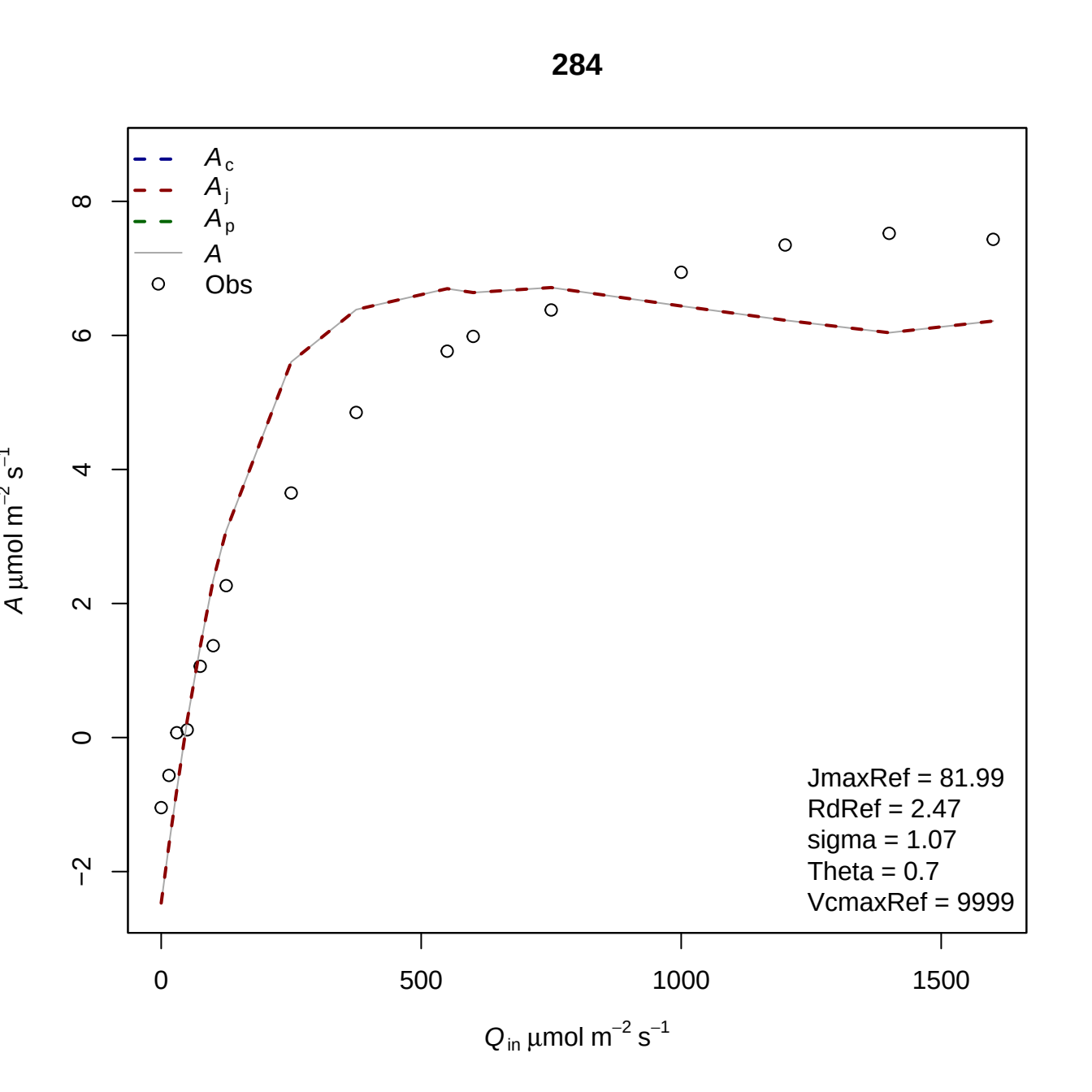

279
A_c
A_j
A_p
A
Obs
JmaxRef = 40.87
RdRef = 2.25
sigma = 0.46
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
0
500
1000
1500

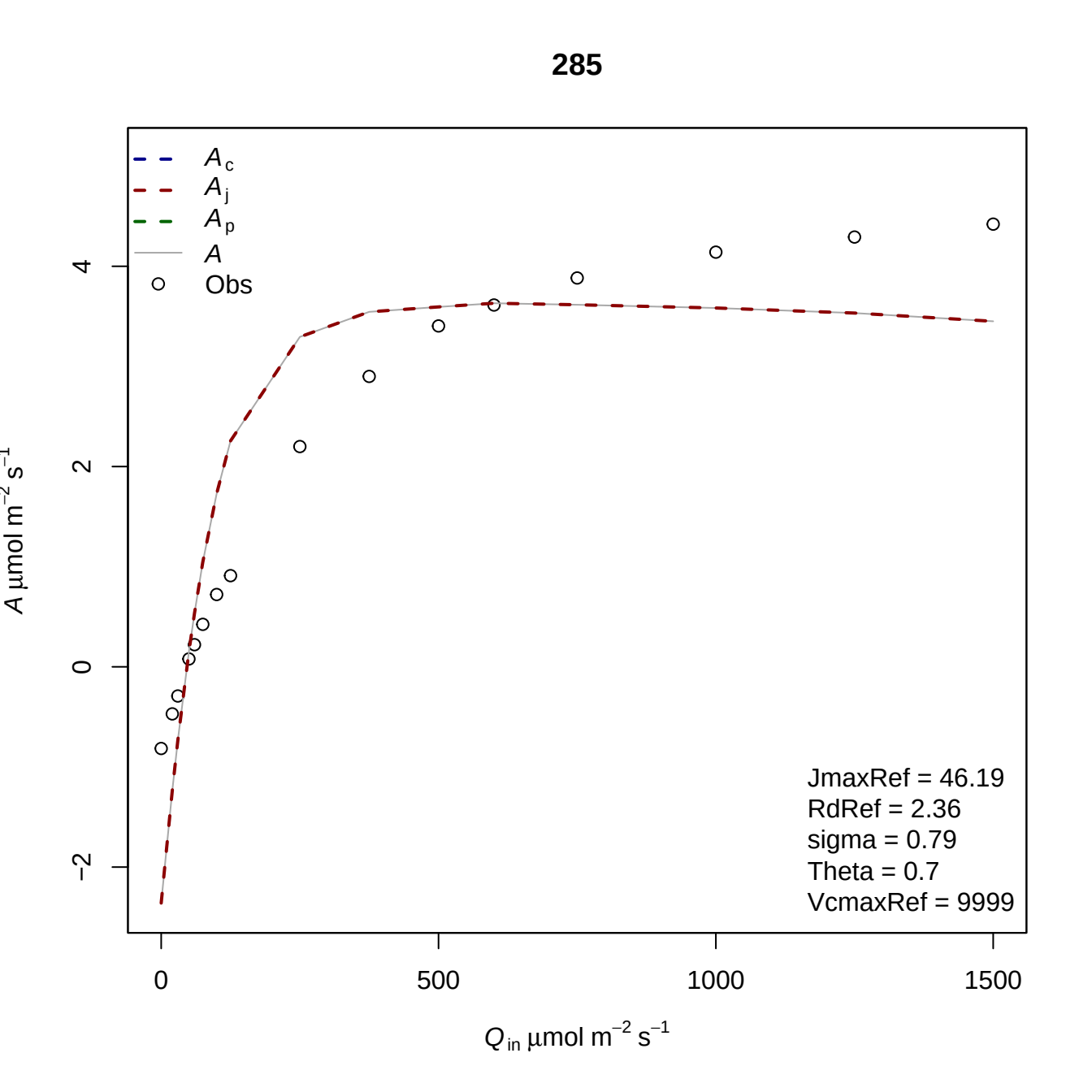

280
A_c
A_j
A_p
A
Obs
JmaxRef = 56.97
RdRef = 2.45
sigma = 0.74
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

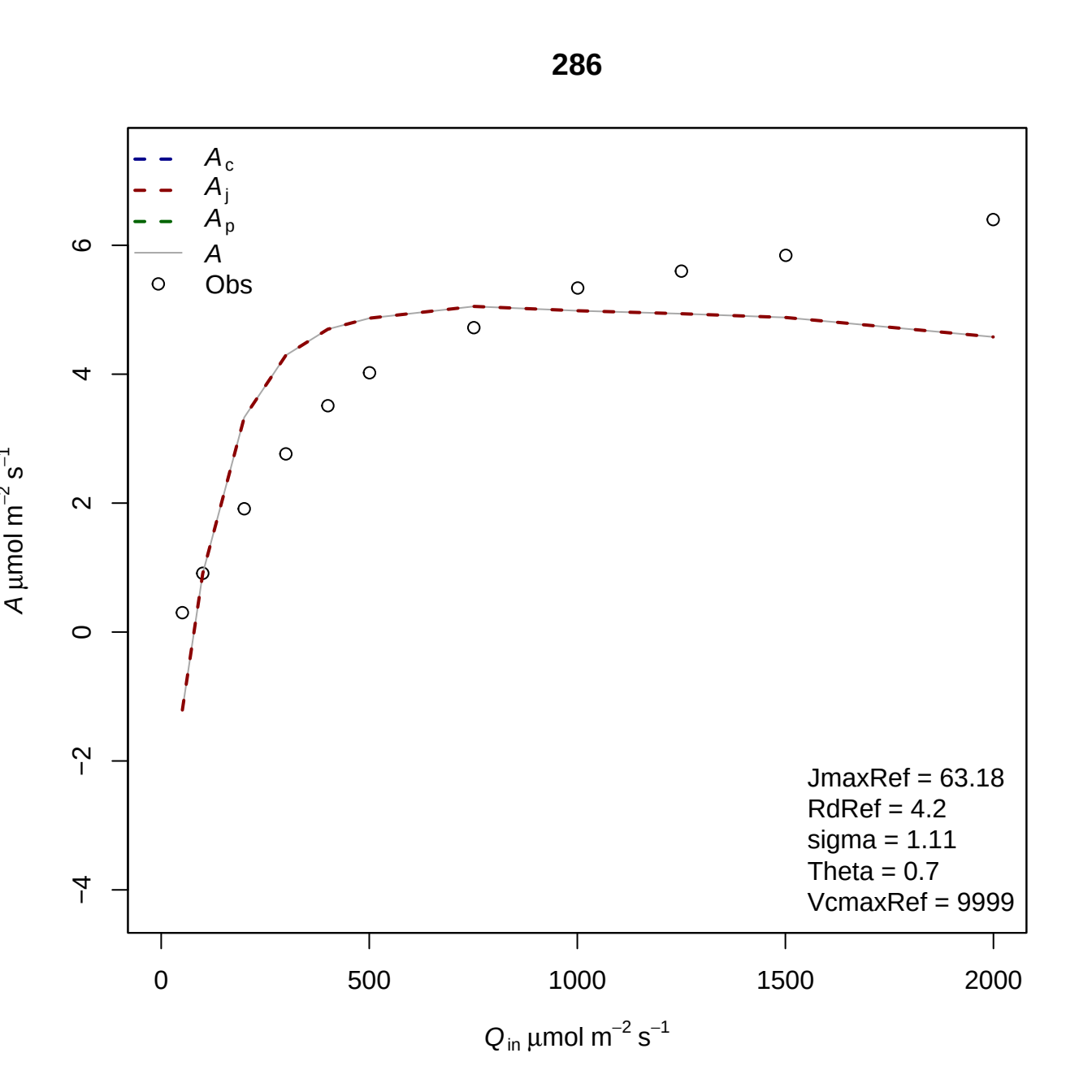

281
A_c
A_j
A_p
A
Obs
JmaxRef = 69.43
RdRef = 2.6
sigma = 1.09
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

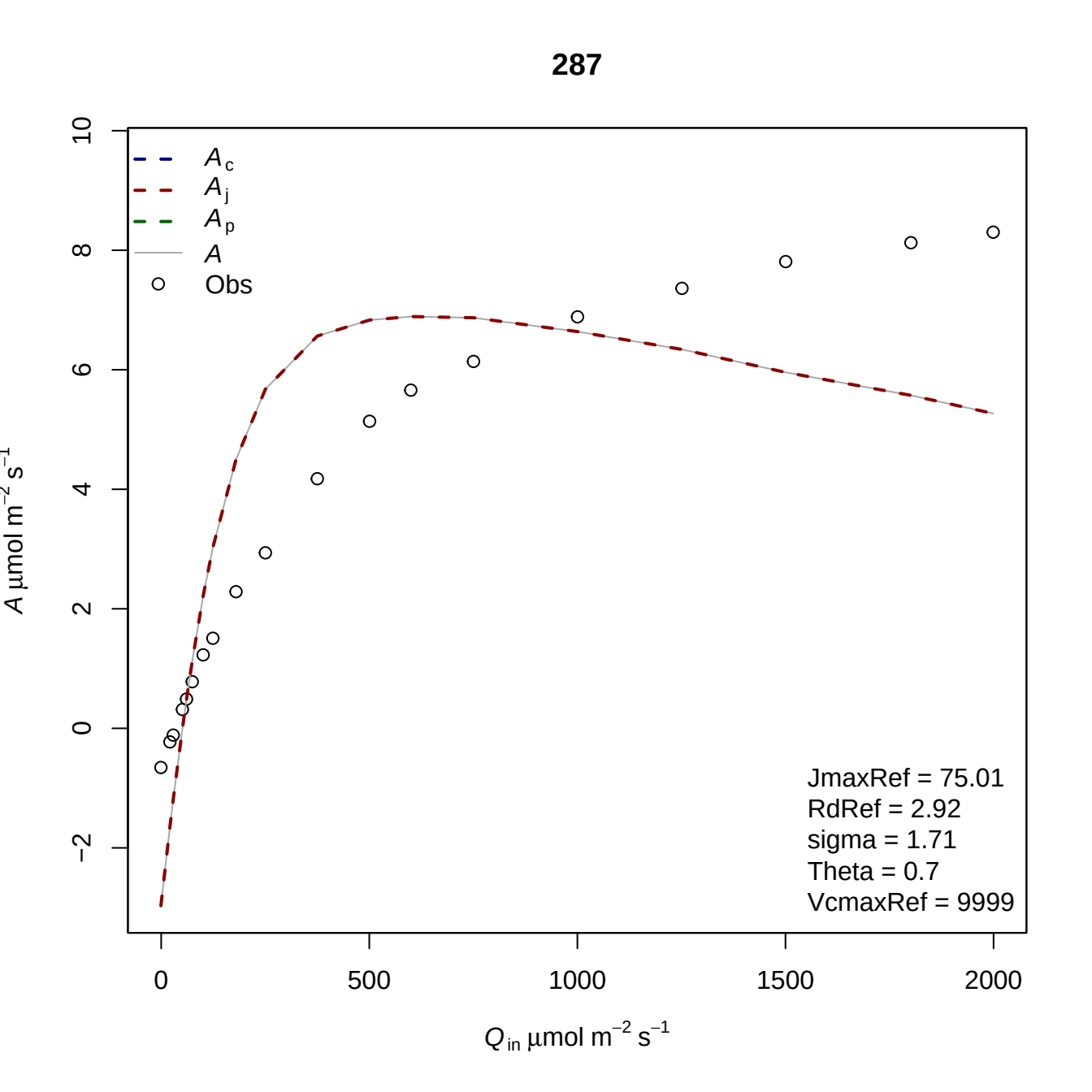

282
A_c
A_j
A_p
A
Obs
JmaxRef = 41.81
RdRef = 2.15
sigma = 1.17
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

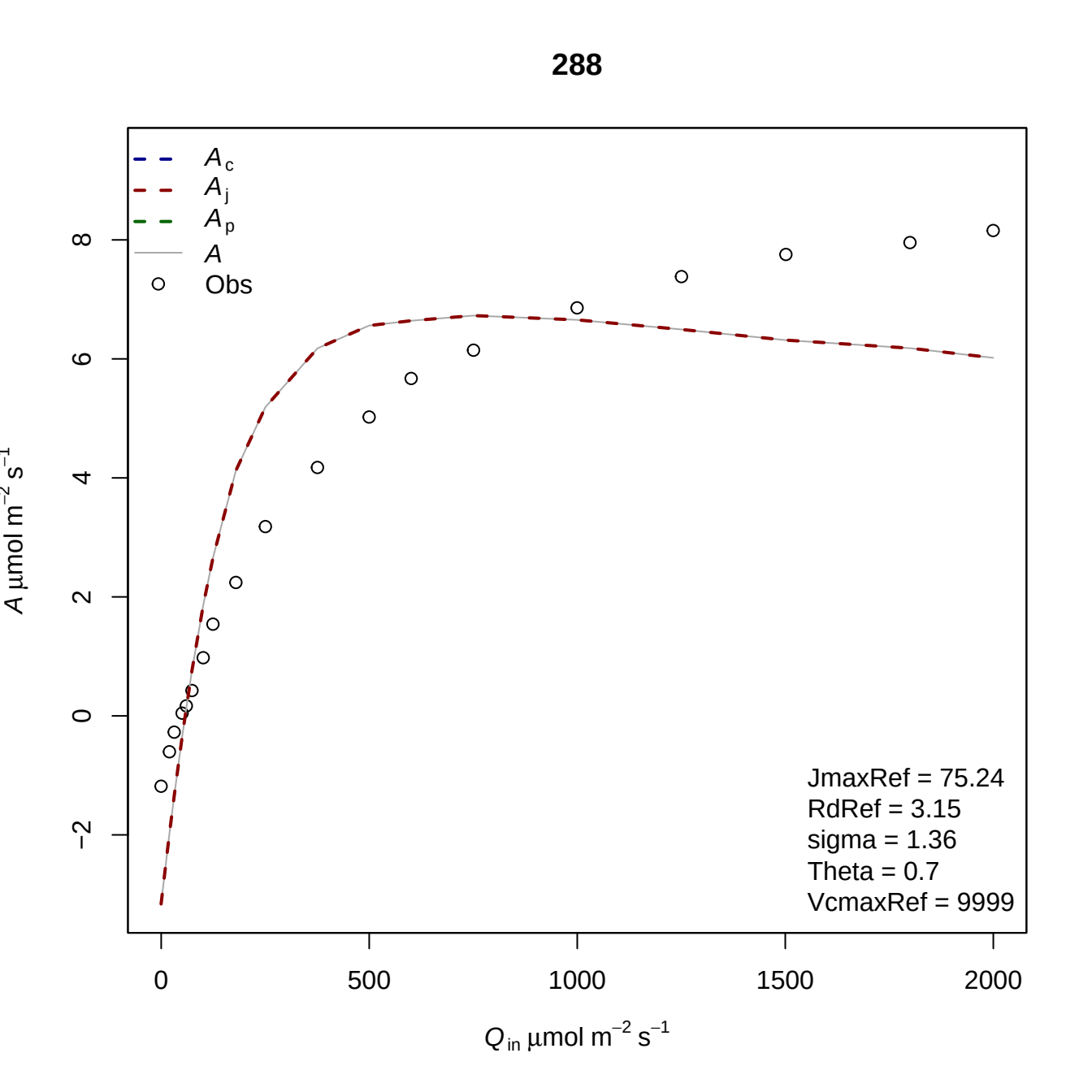

283
A_c
A_j
A_p
A
Obs
JmaxRef = 48.68
RdRef = 2.17
sigma = 1.3
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

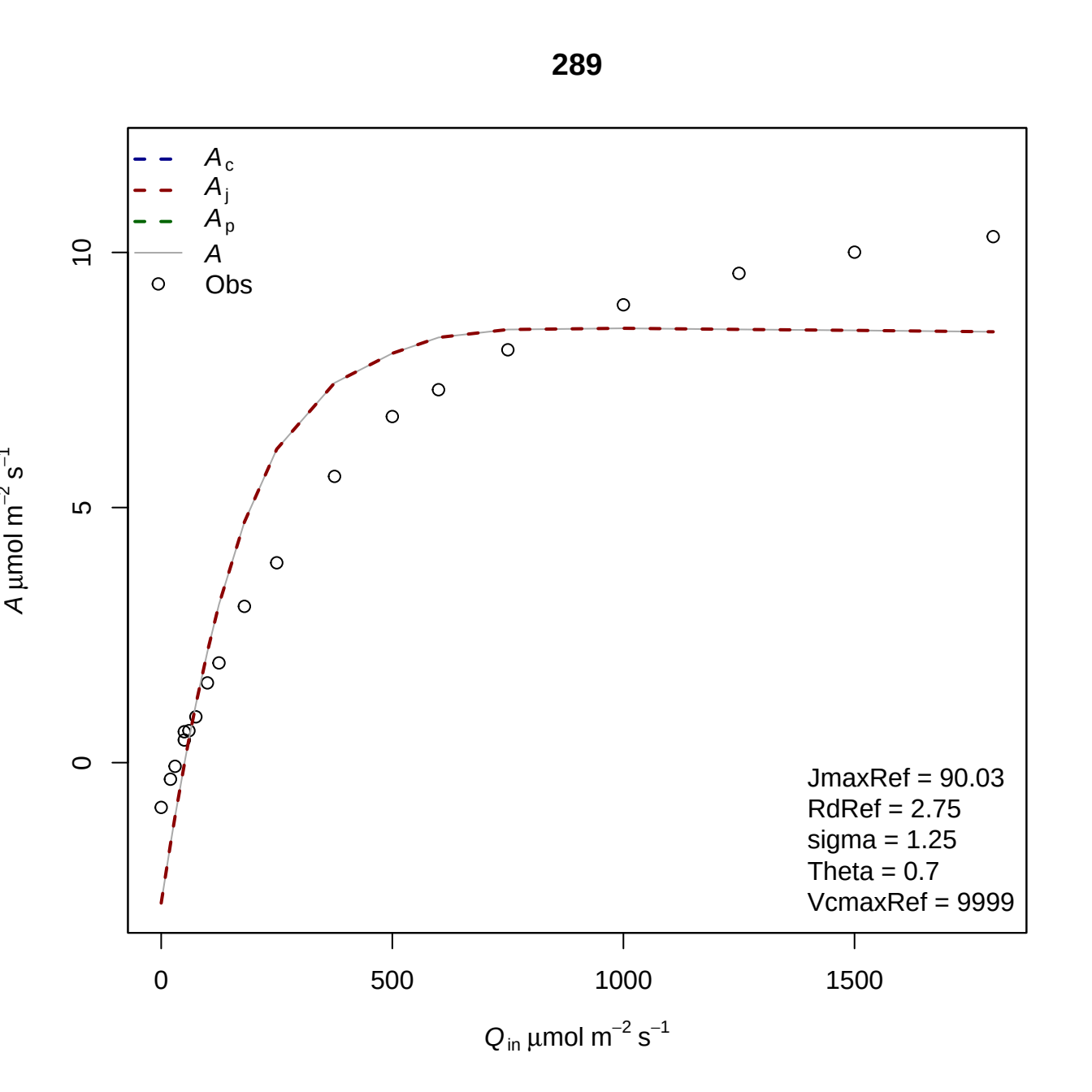

284
A_c
A_j
A_p
A
Obs
JmaxRef = 81.99
RdRef = 2.47
sigma = 1.07
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

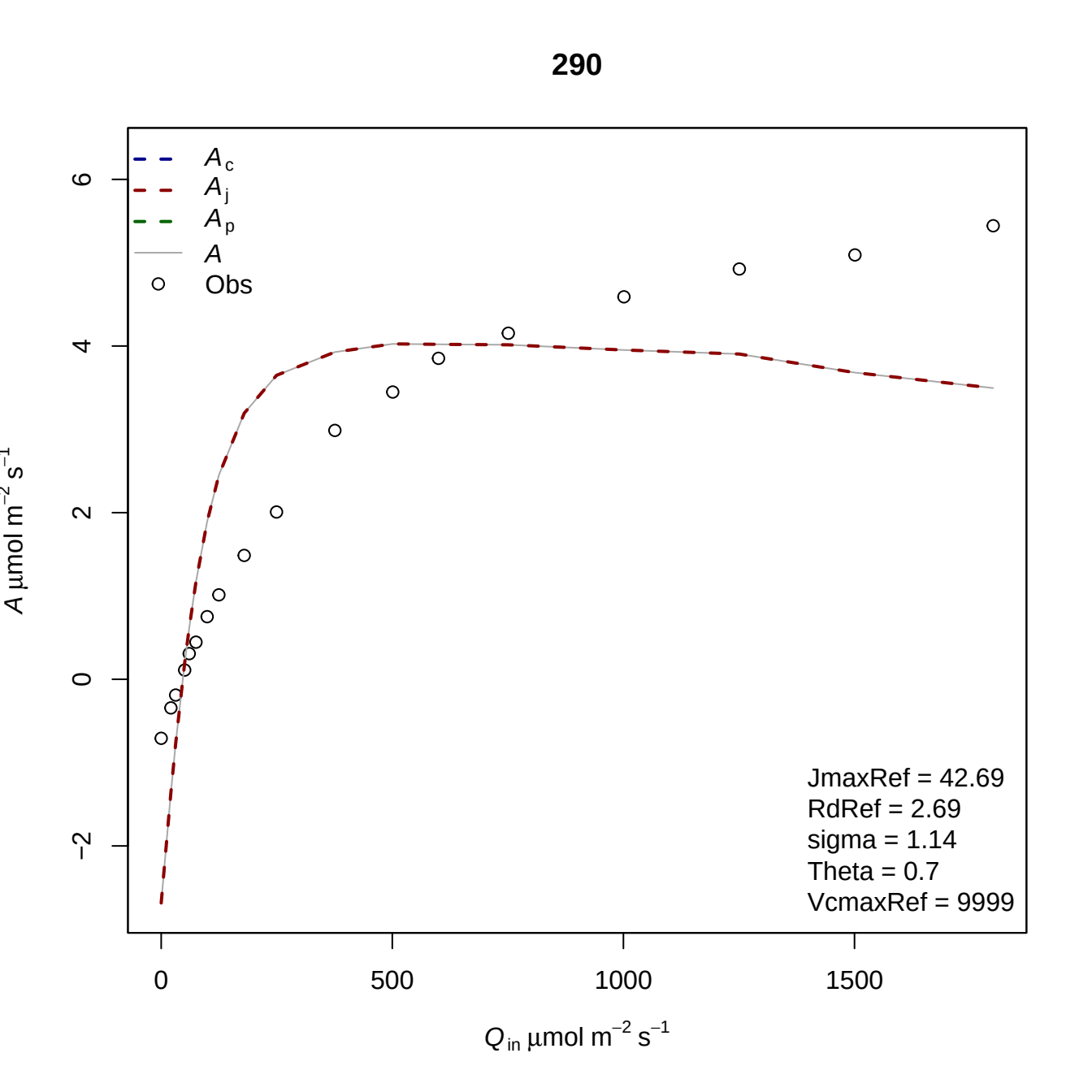

285
A_c
A_j
A_p
A
Obs
JmaxRef = 46.19
RdRef = 2.36
sigma = 0.79
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500

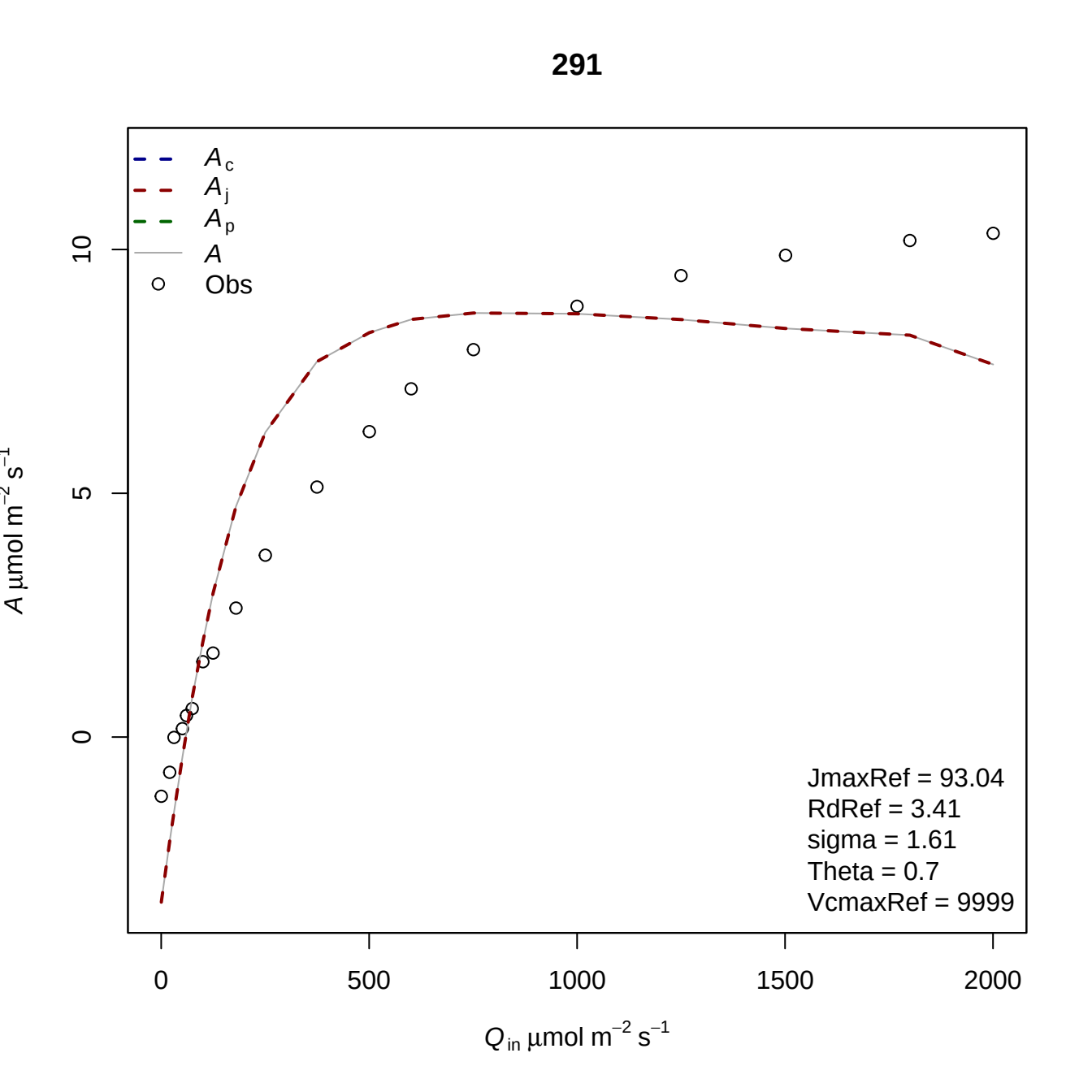

286
A_c
A_j
A_p
A
Obs
JmaxRef = 63.18
RdRef = 4.2
sigma = 1.11
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500
2000

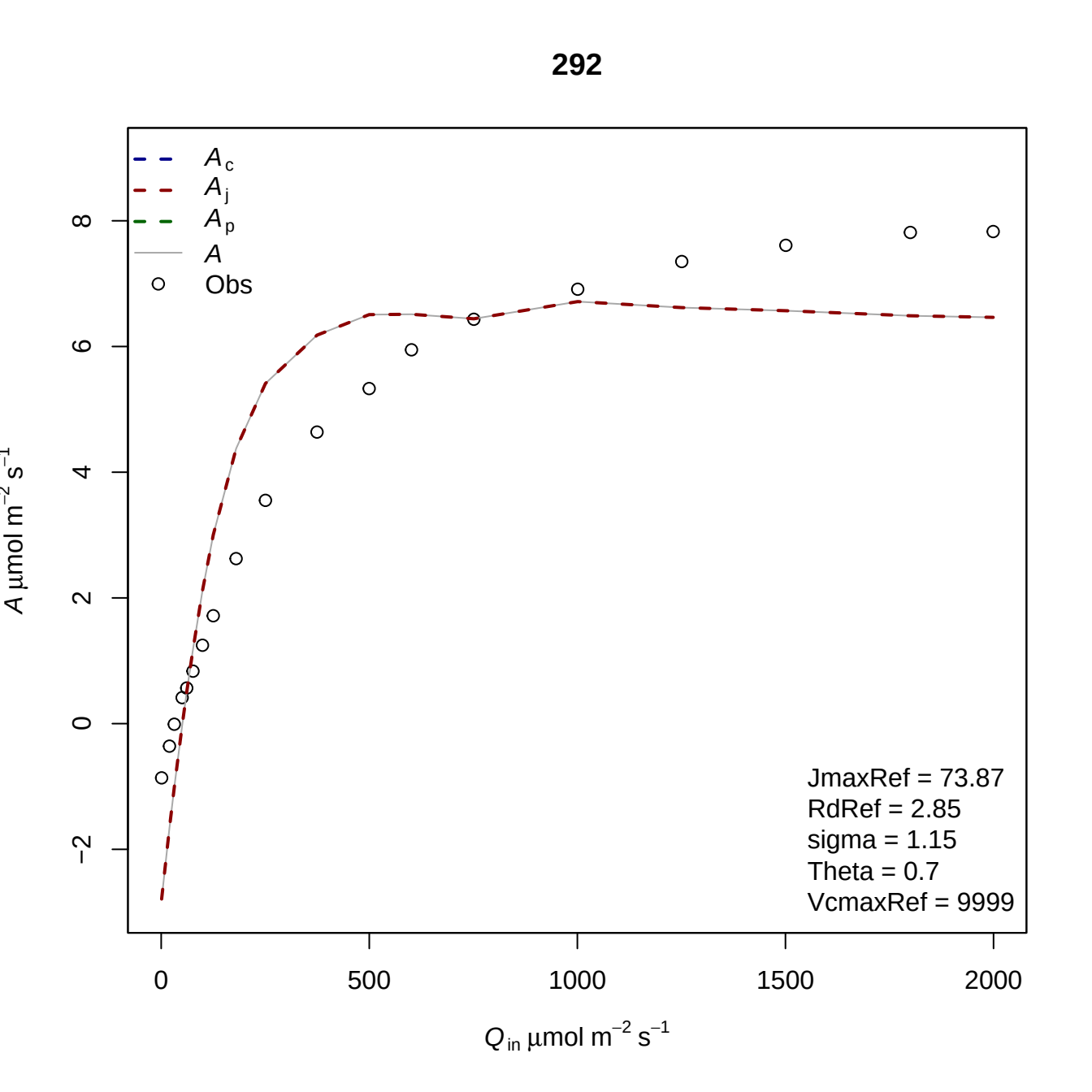

287
A_c
A_j
A_p
A
Obs
JmaxRef = 75.01
RdRef = 2.92
sigma = 1.71
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

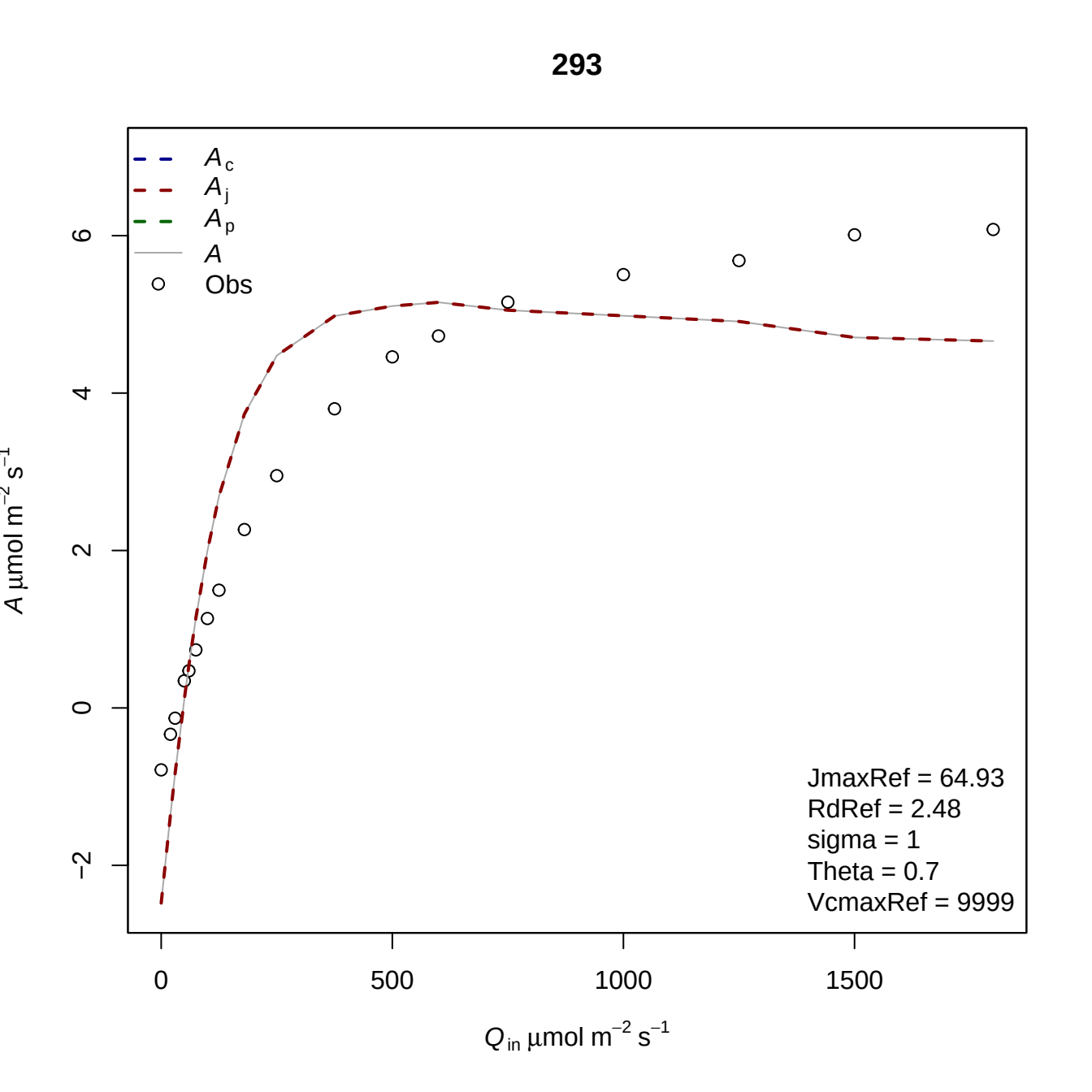

288
A_c
A_j
A_p
A
Obs
JmaxRef = 75.24
RdRef = 3.15
sigma = 1.36
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

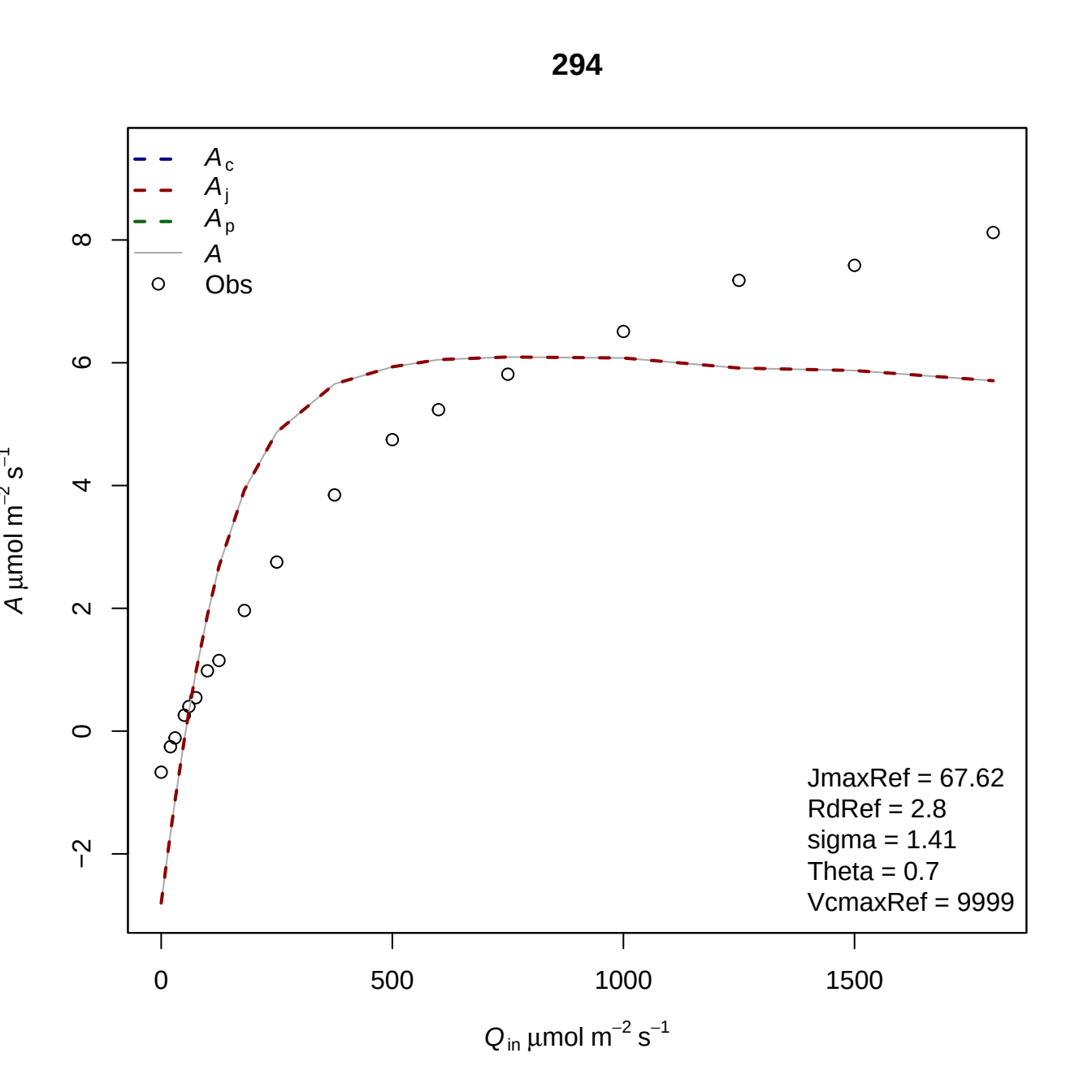

289
A_c
A_j
A_p
A
Obs
JmaxRef = 90.03
RdRef = 2.75
sigma = 1.25
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

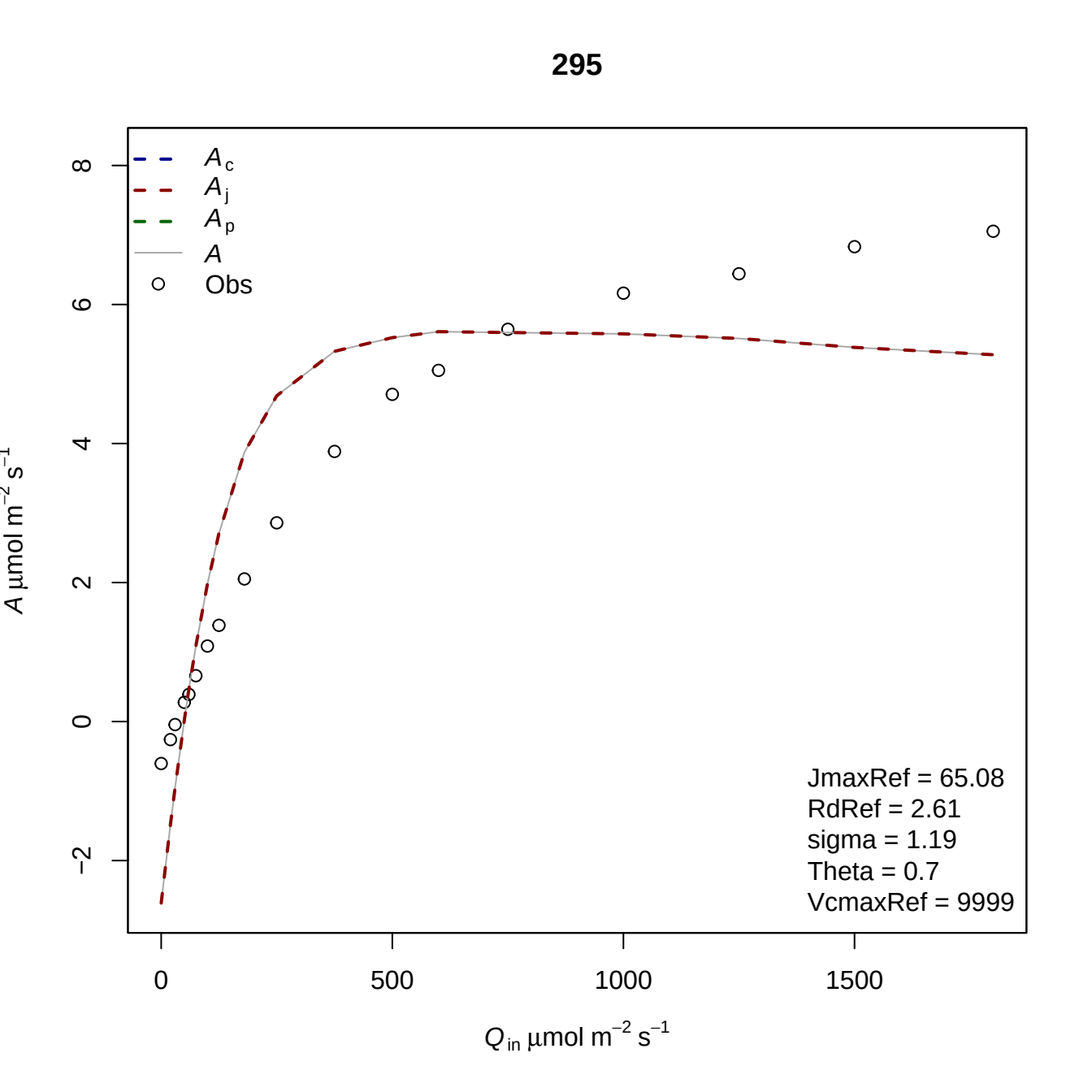

290
A_c
A_j
A_p
A
Obs
JmaxRef = 42.69
RdRef = 2.69
sigma = 1.14
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

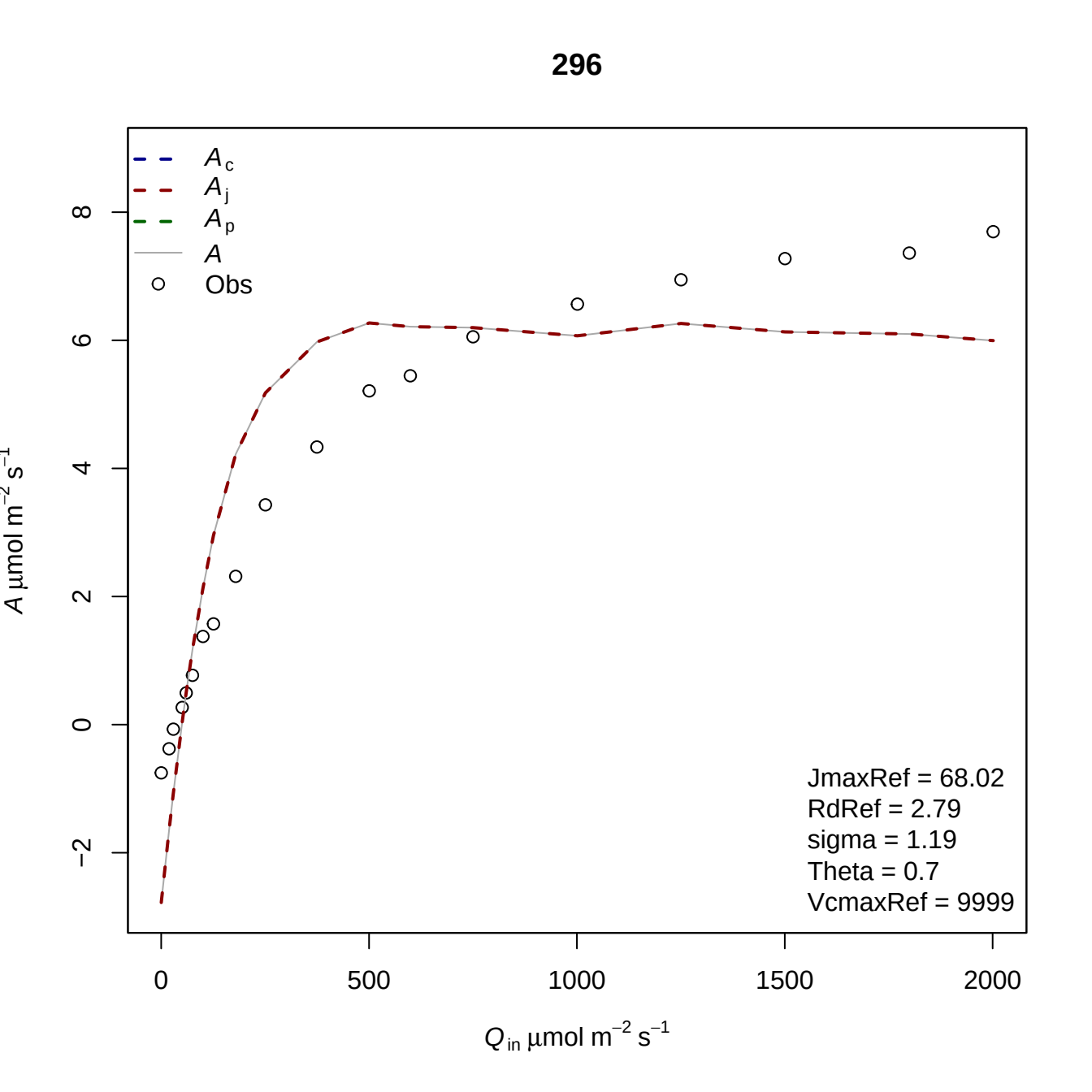

291
A_c
A_j
A_p
A
Obs
JmaxRef = 93.04
RdRef = 3.41
sigma = 1.61
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

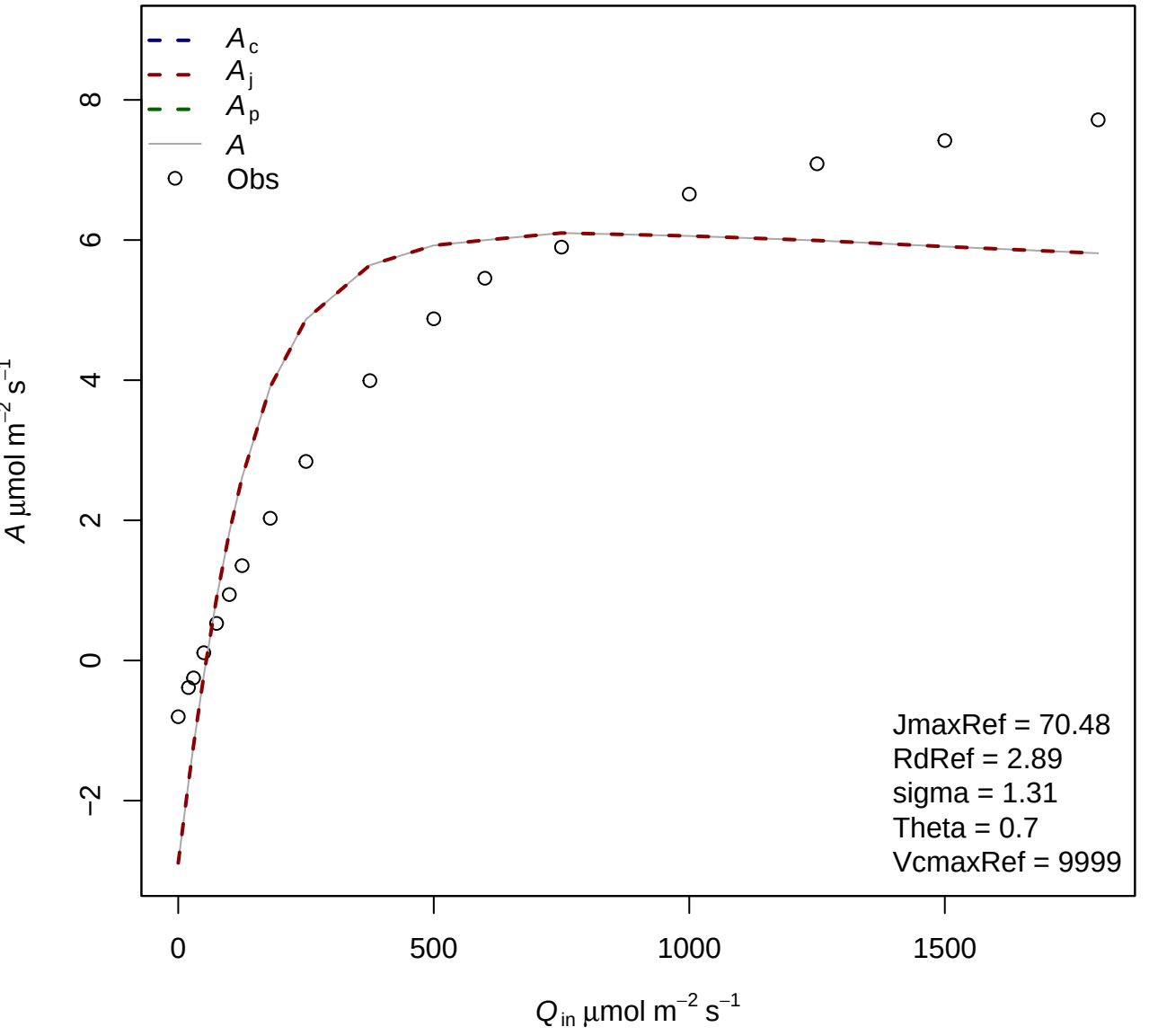

292
A_c
A_j
A_p
A
Obs
JmaxRef = 73.87
RdRef = 2.85
sigma = 1.15
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

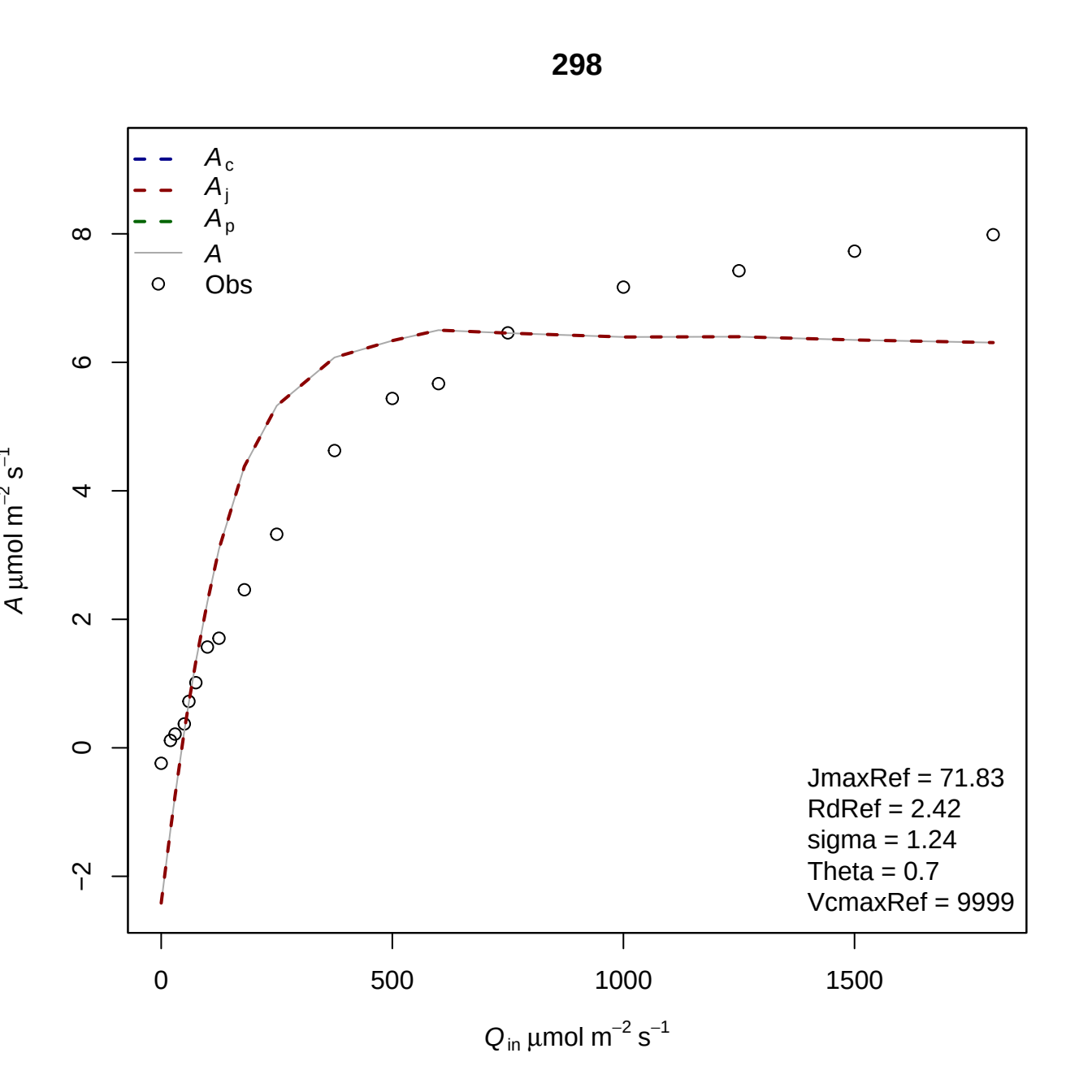

293
A_c
A_j
A_p
A
Obs
JmaxRef = 64.93
RdRef = 2.48
sigma = 1
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

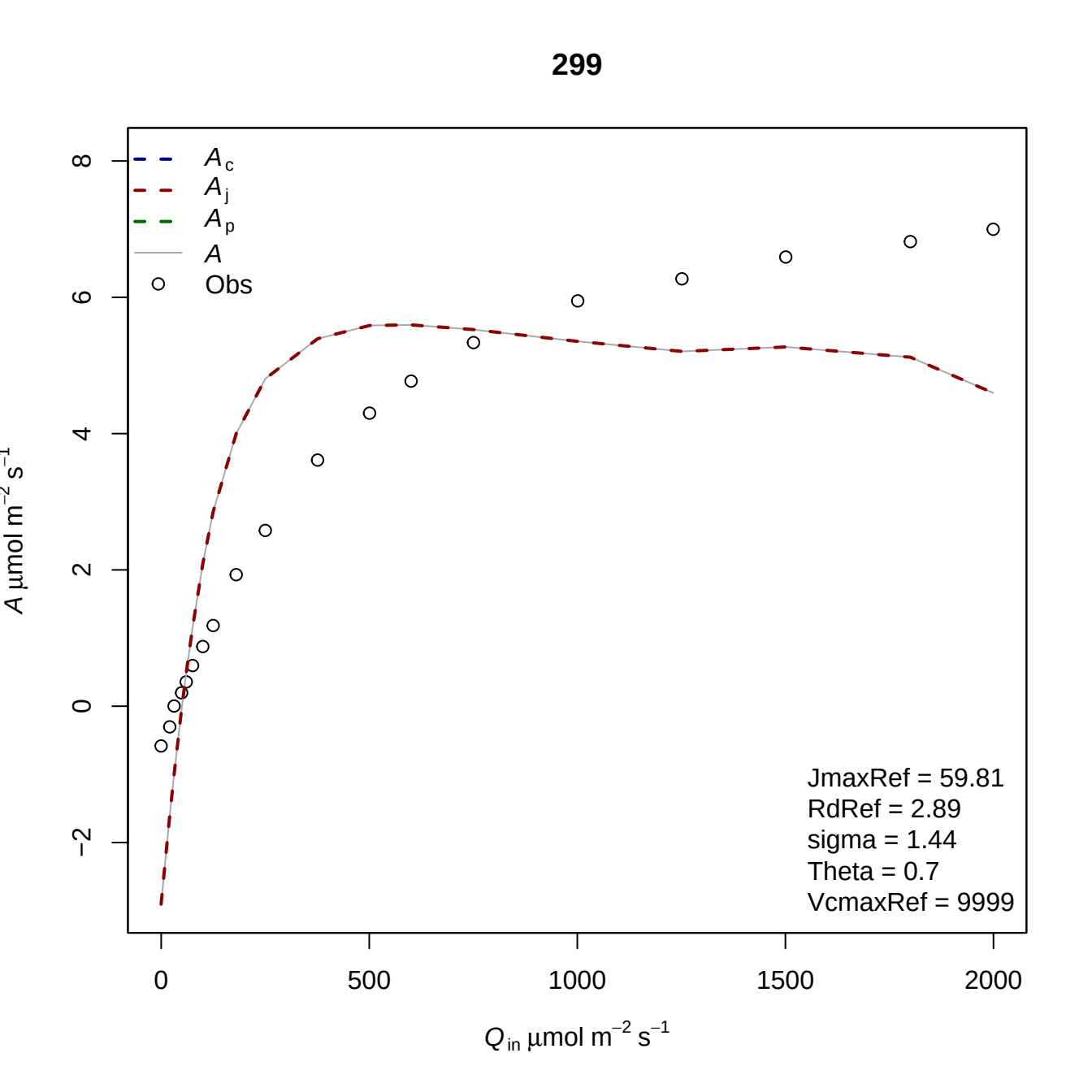

294
A_c
A_j
A_p
A
Obs
JmaxRef = 67.62
RdRef = 2.8
sigma = 1.41
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

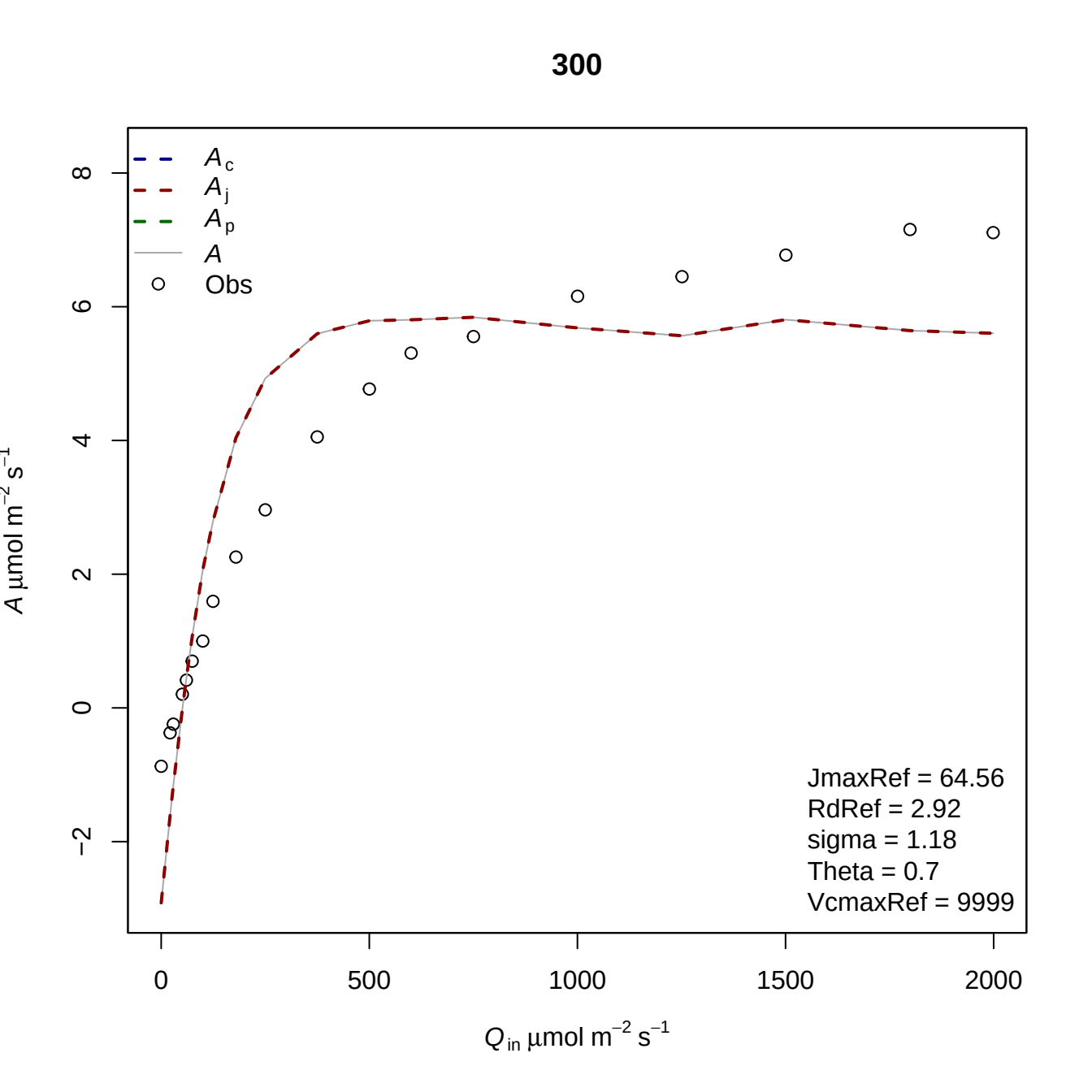

295
A_c
A_j
A_p
A
Obs
JmaxRef = 65.08
RdRef = 2.61
sigma = 1.19
Theta = 0.7
VcmaxRef = 9999
8
6
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

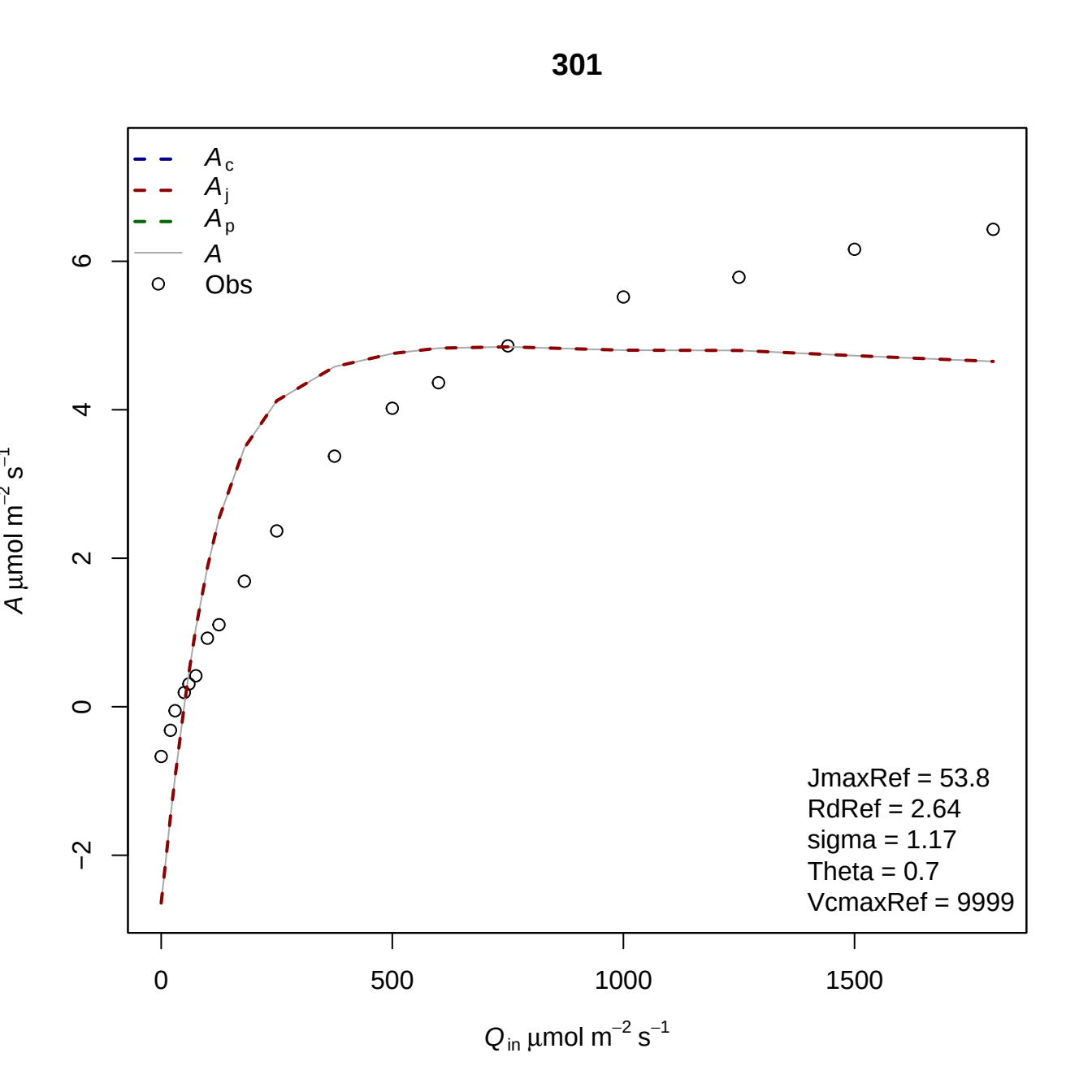

296
A_c
A_j
A_p
A
Obs
JmaxRef = 68.02
RdRef = 2.79
sigma = 1.19
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

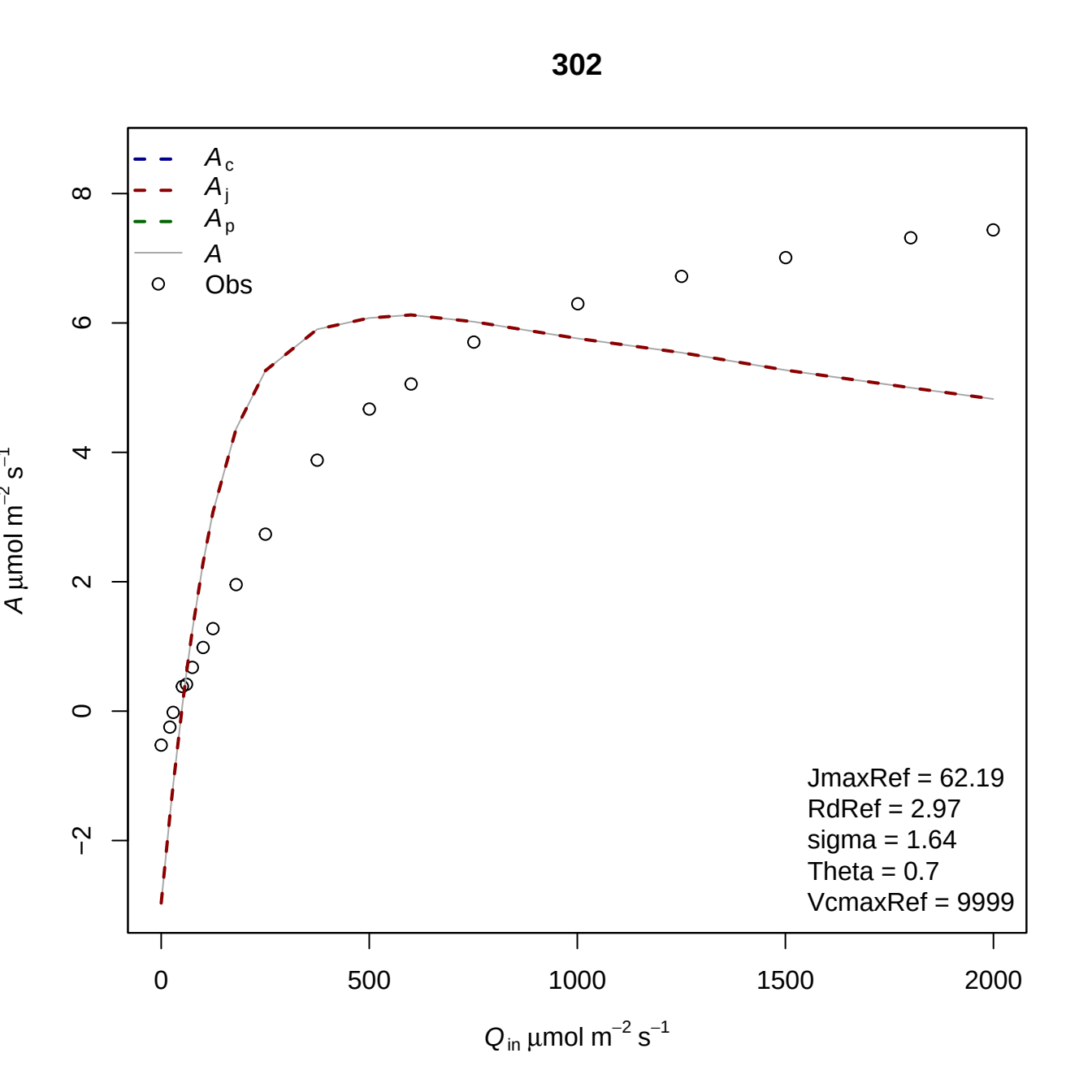

A_c
A_j
A_p
A
Obs
JmaxRef = 70.48
RdRef = 2.89
sigma = 1.31
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

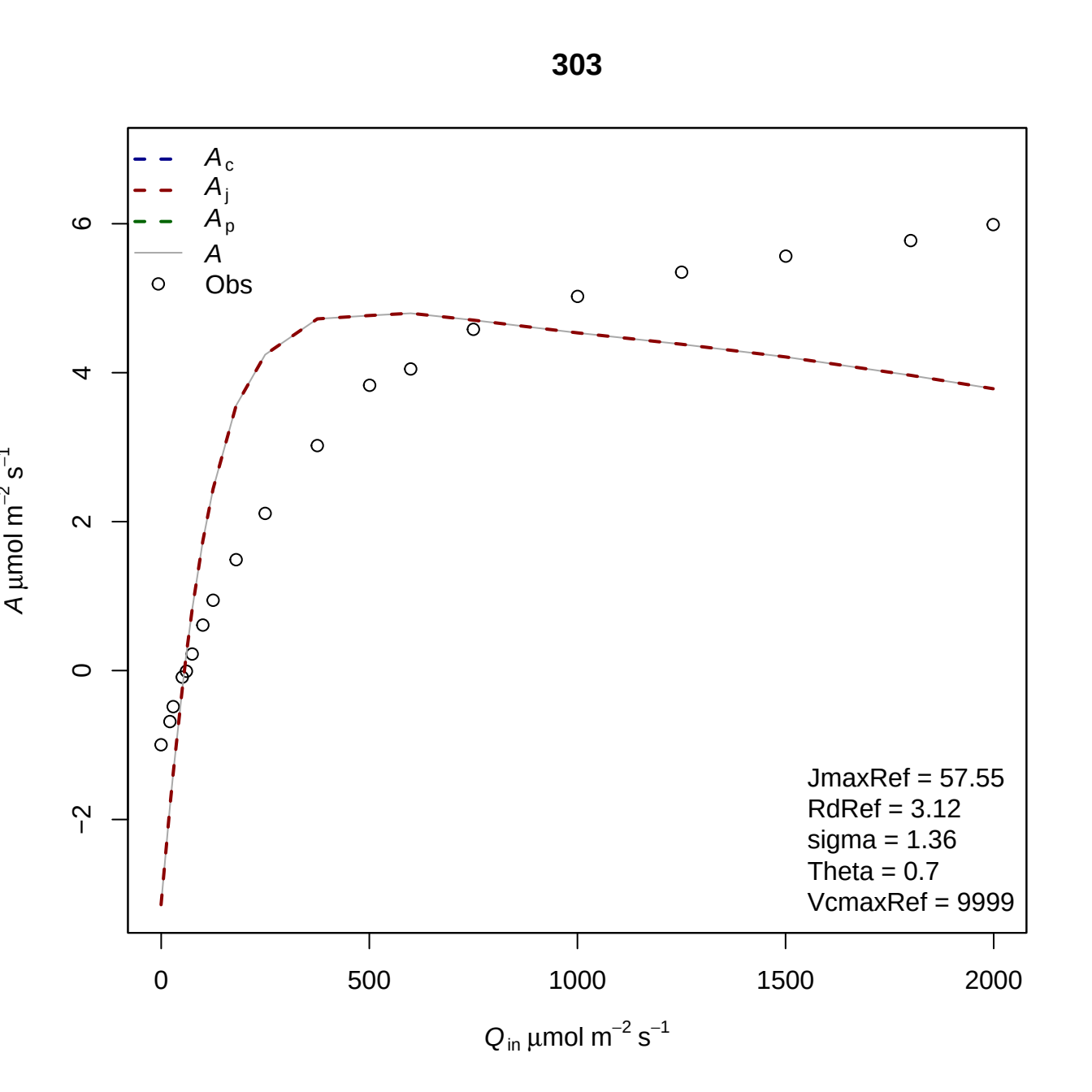

298
A_c
A_j
A_p
A
Obs
JmaxRef = 71.83
RdRef = 2.42
sigma = 1.24
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

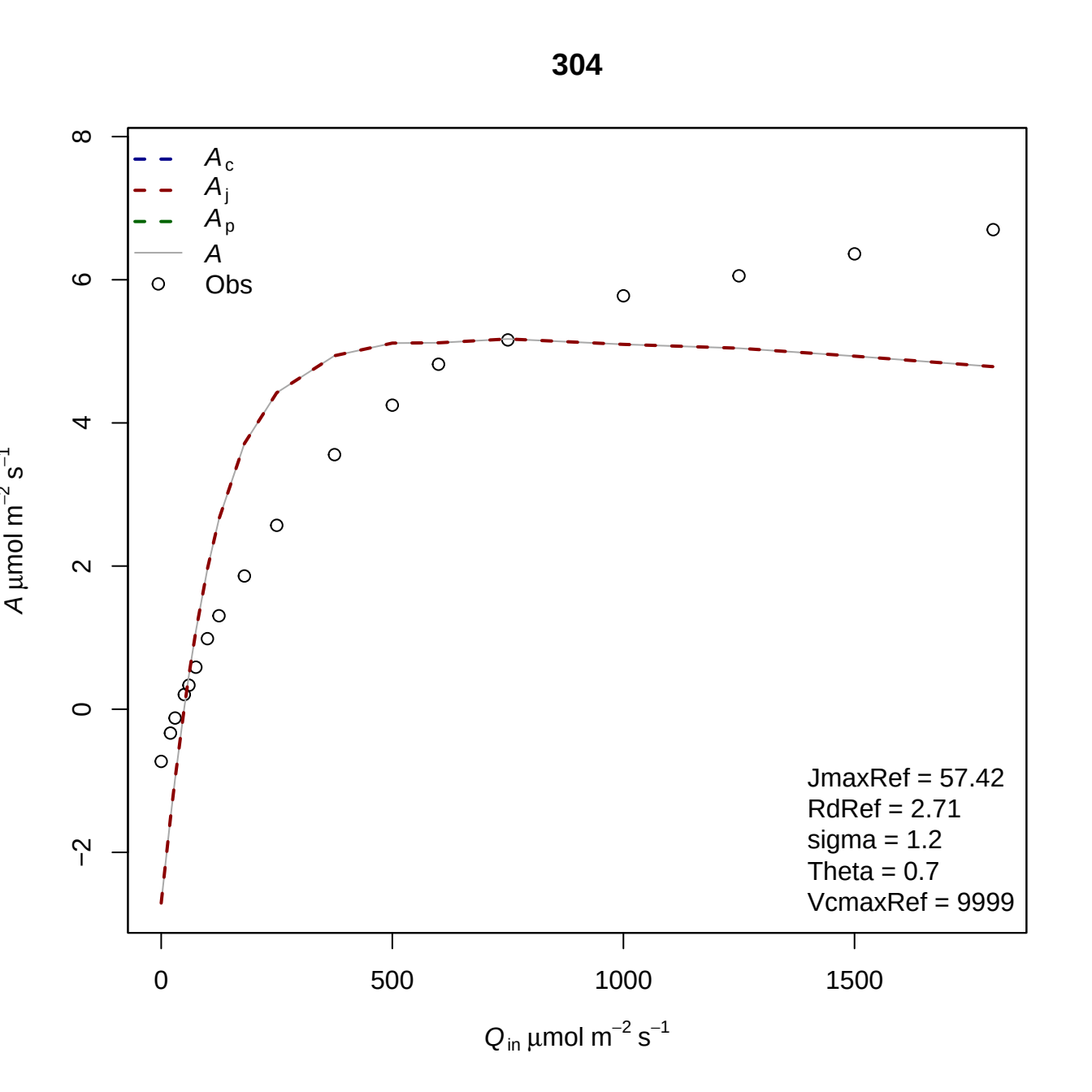

299
A_c
A_j
A_p
A
Obs
JmaxRef = 59.81
RdRef = 2.89
sigma = 1.44
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

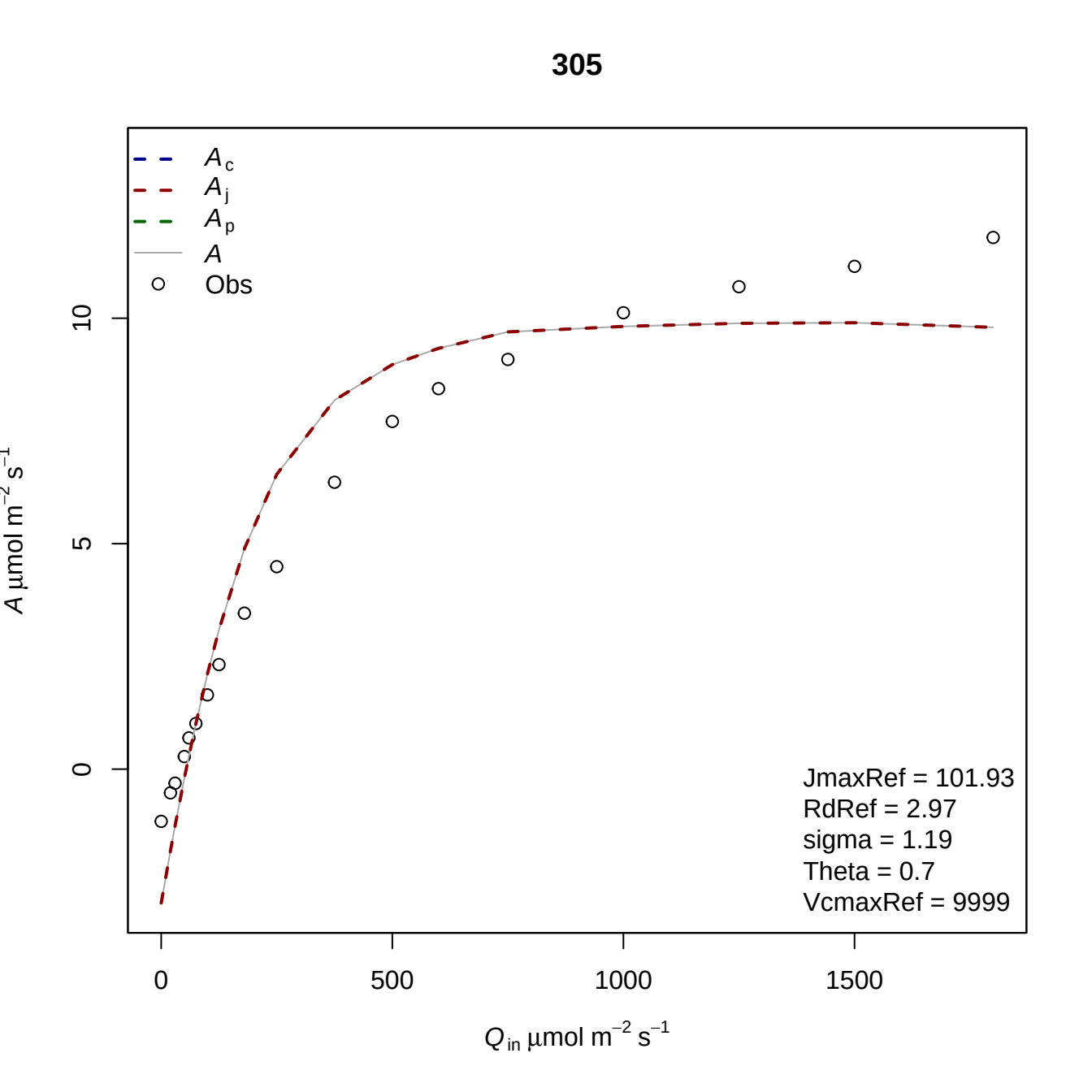

300
A_c
A_j
A_p
A
Obs
JmaxRef = 64.56
RdRef = 2.92
sigma = 1.18
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

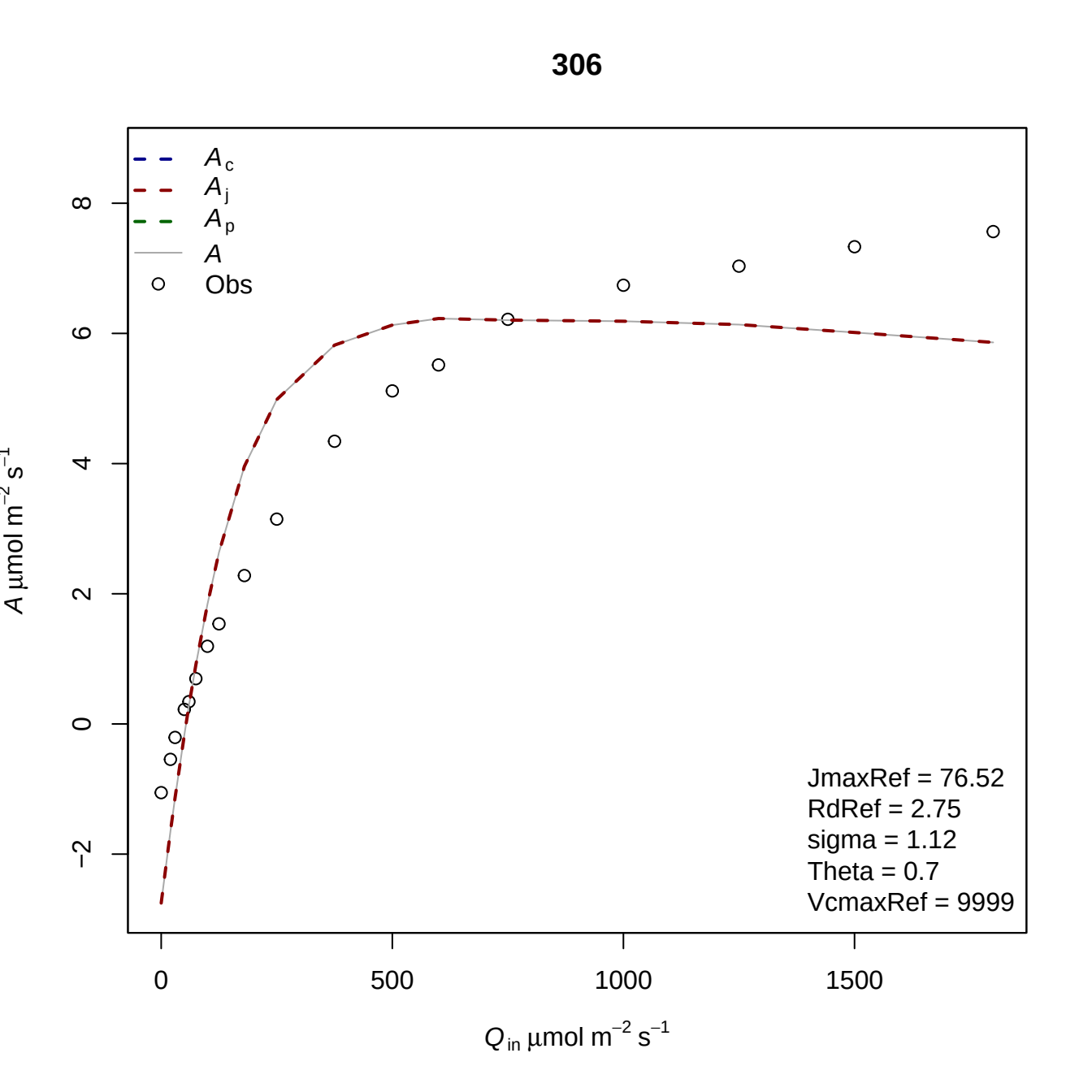

301
A_c
A_j
A_p
A
Obs
JmaxRef = 53.8
RdRef = 2.64
sigma = 1.17
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

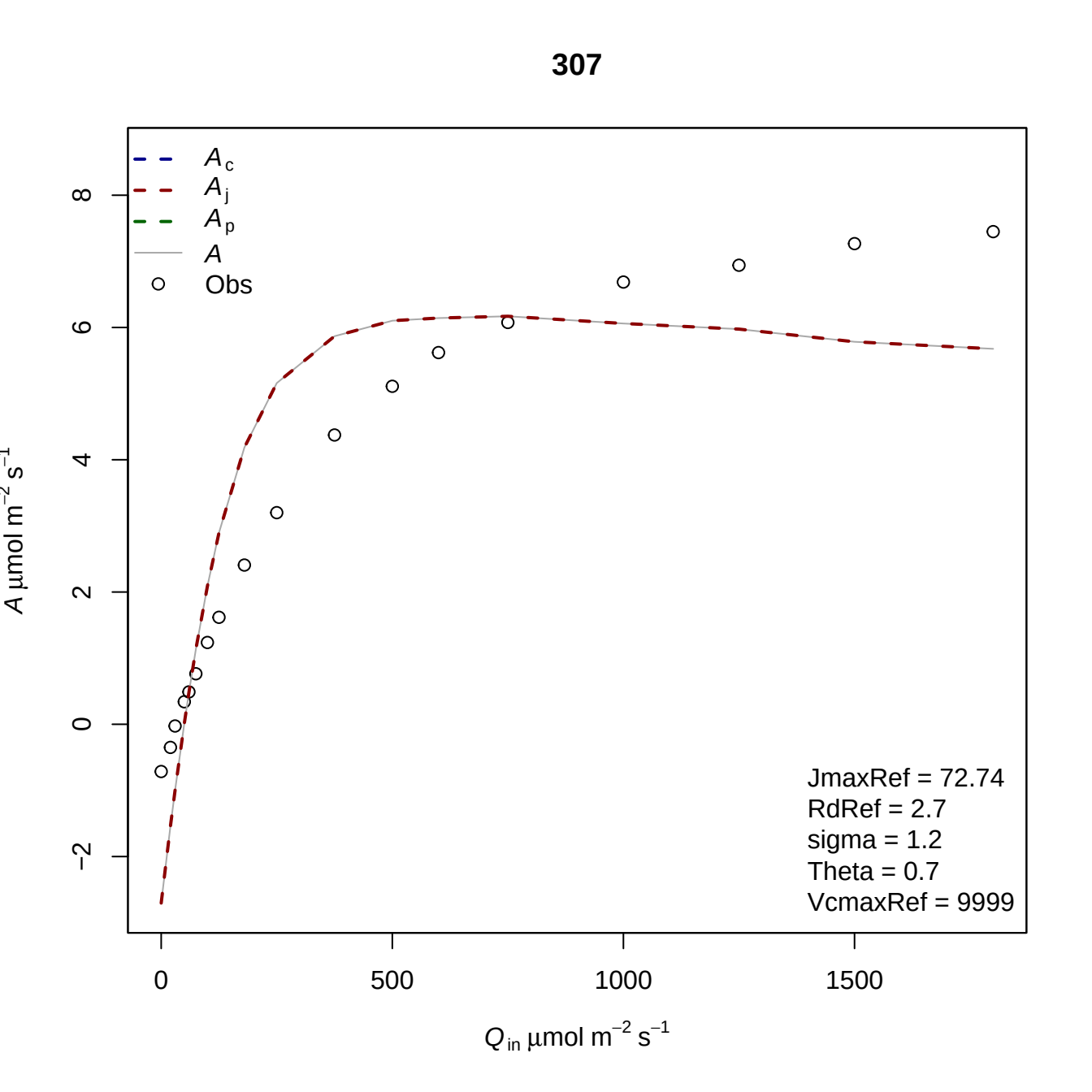

302
A_c
A_j
A_p
A
Obs
JmaxRef = 62.19
RdRef = 2.97
sigma = 1.64
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

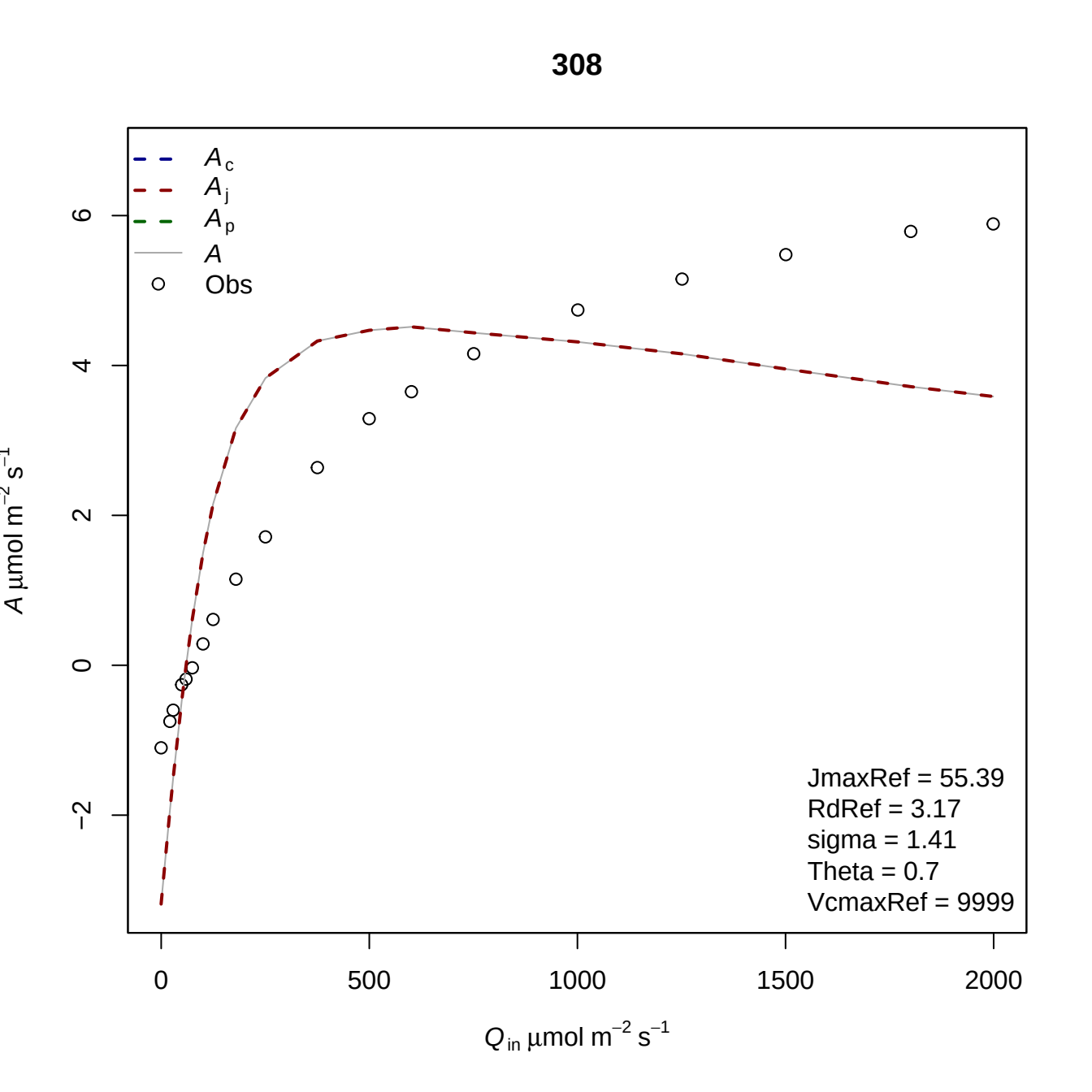

303
A_c
A_j
A_p
A
Obs
JmaxRef = 57.55
RdRef = 3.12
sigma = 1.36
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

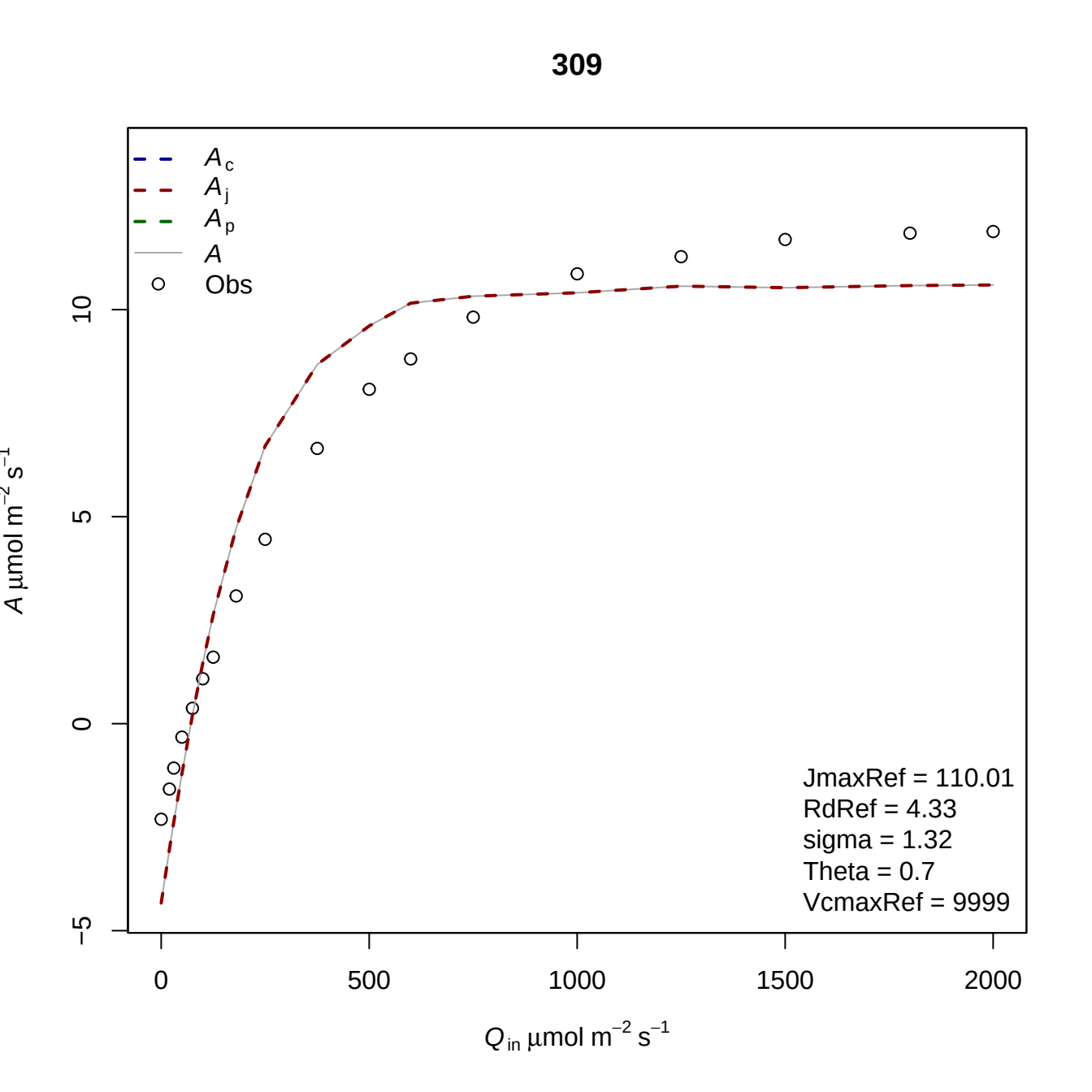

304
A_c
A_j
A_p
A
Obs
JmaxRef = 57.42
RdRef = 2.71
sigma = 1.2
Theta = 0.7
VcmaxRef = 9999
8
6
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

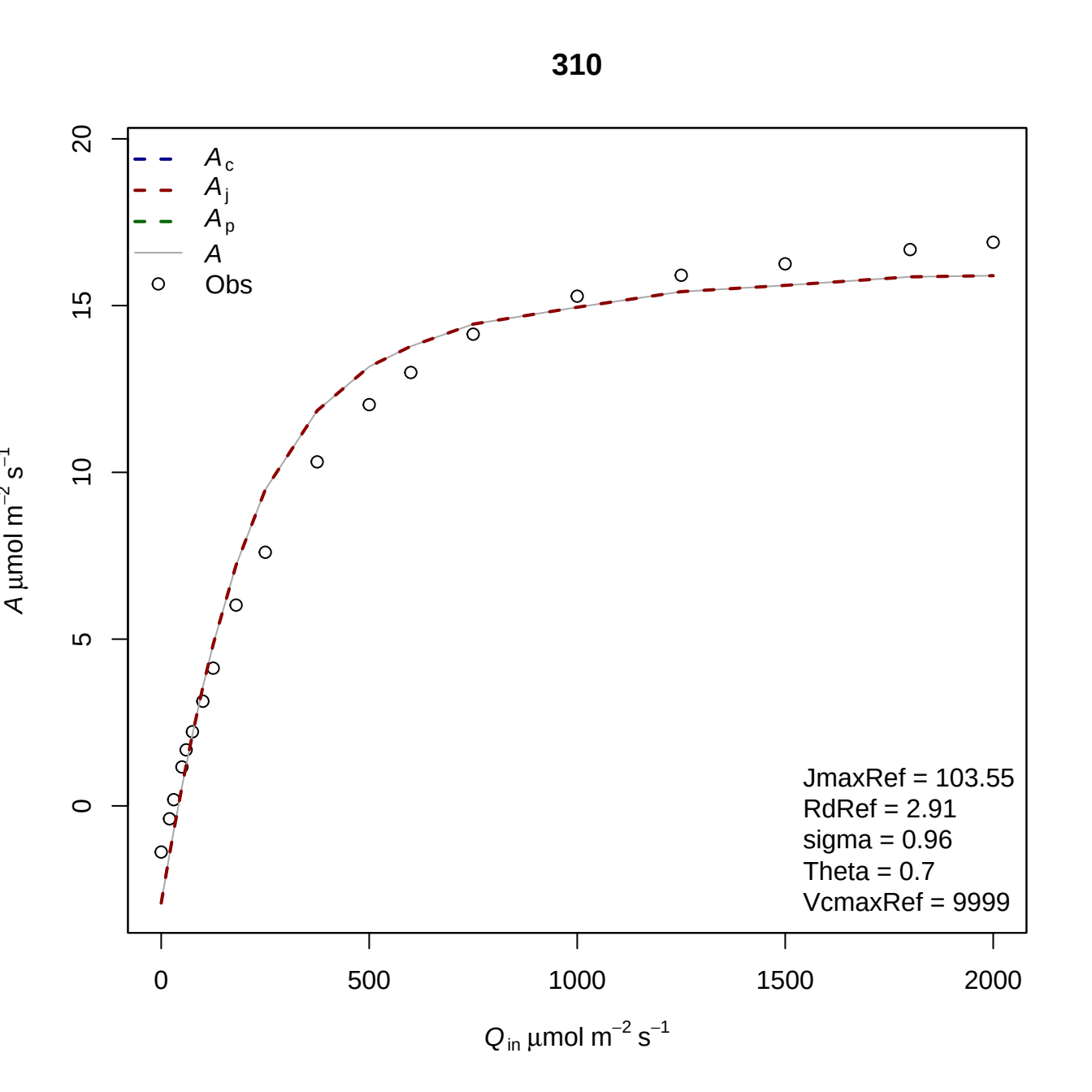

305
A_c
A_j
A_p
A
Obs
JmaxRef = 101.93
RdRef = 2.97
sigma = 1.19
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

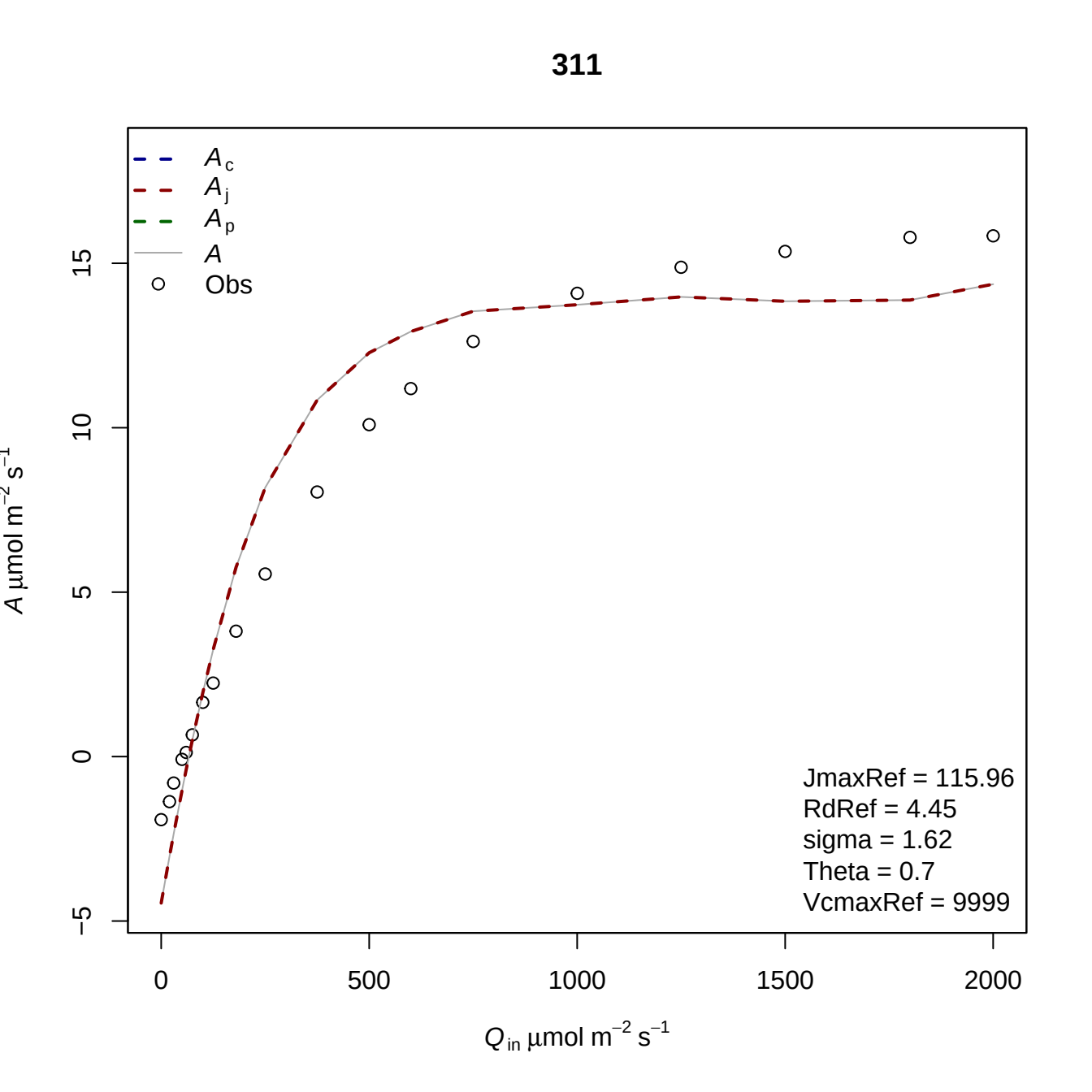

306
A_c
A_j
A_p
A
Obs
JmaxRef = 76.52
RdRef = 2.75
sigma = 1.12
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

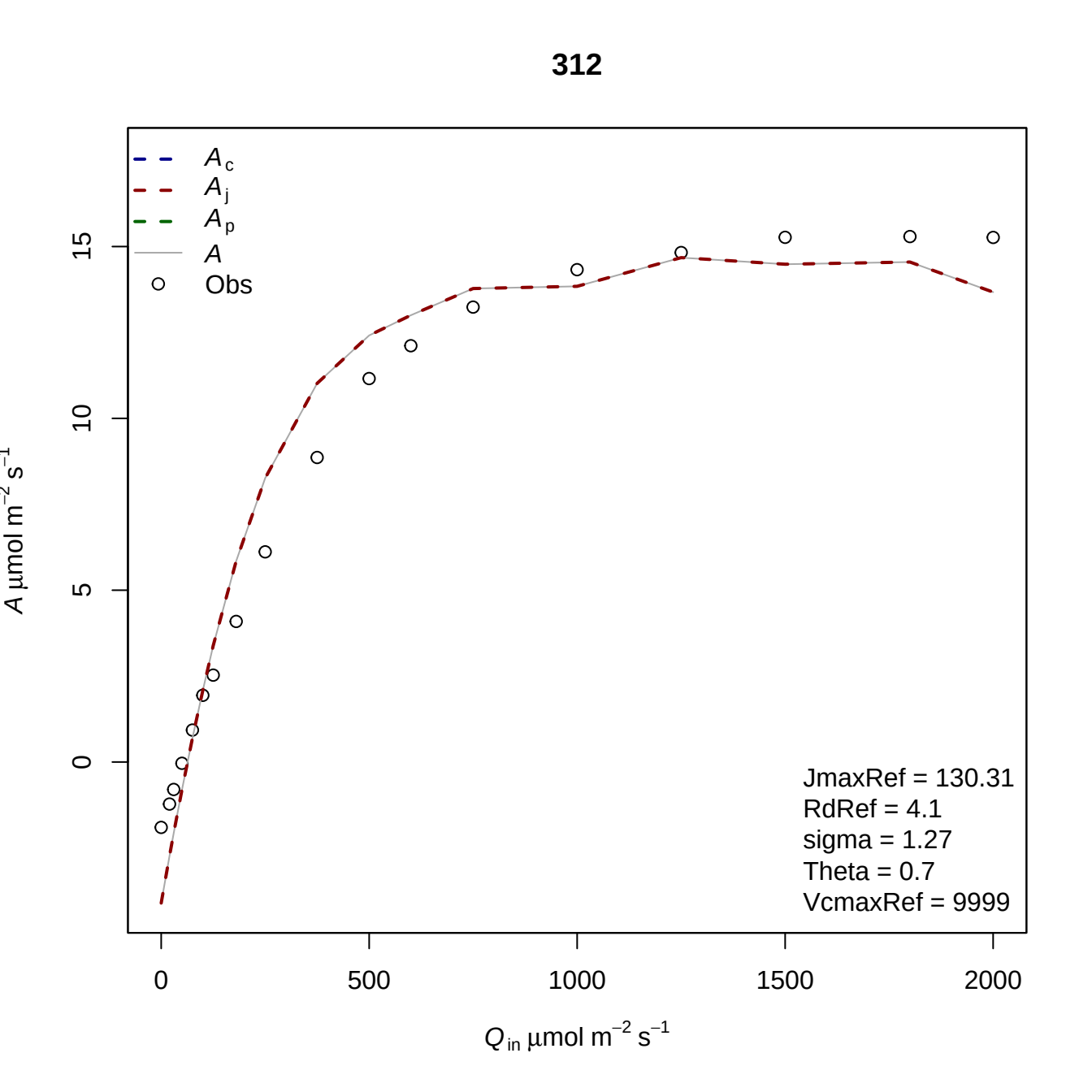

307
A_c
A_j
A_p
A
Obs
JmaxRef = 72.74
RdRef = 2.7
sigma = 1.2
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

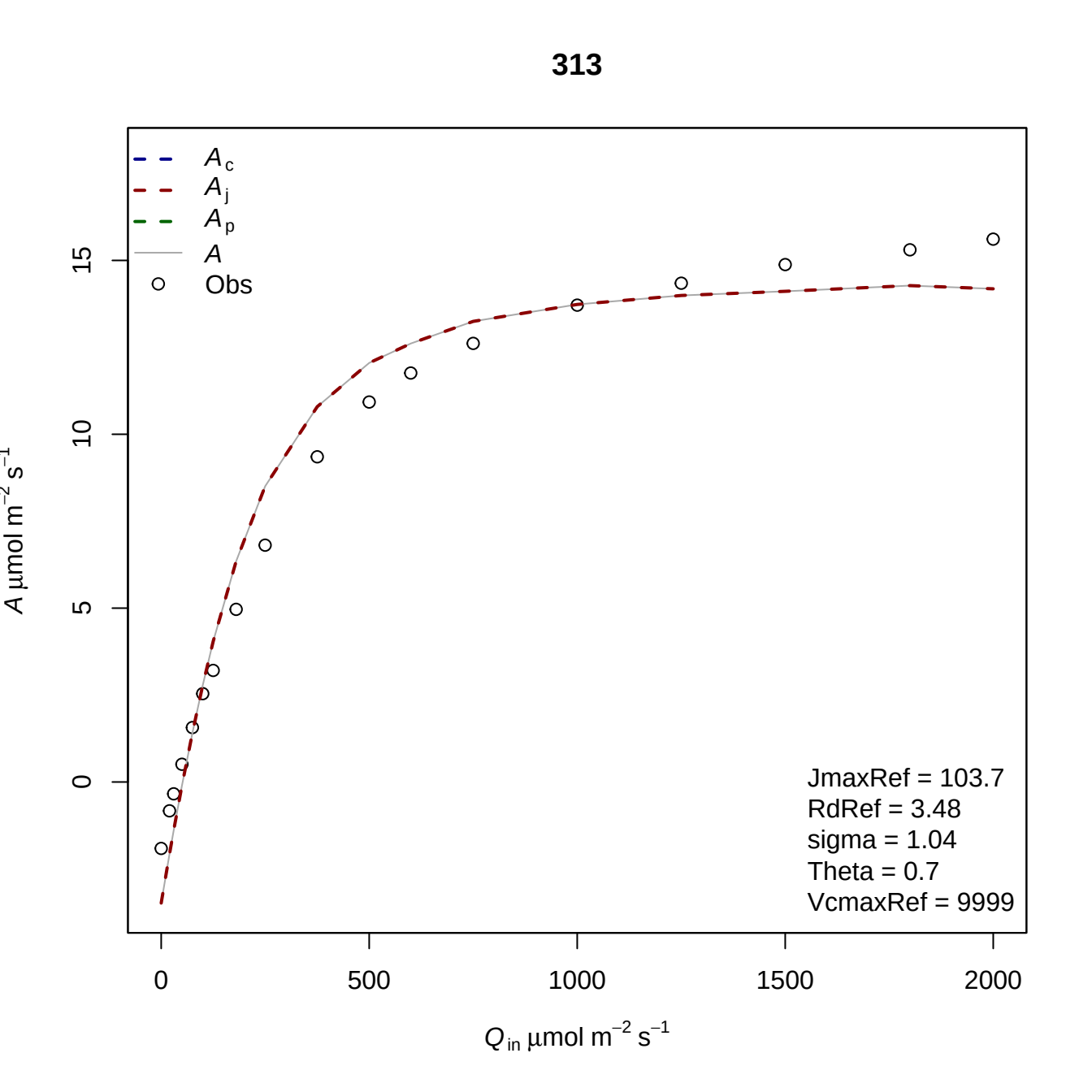

308
A_c
A_j
A_p
A
Obs
JmaxRef = 55.39
RdRef = 3.17
sigma = 1.41
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

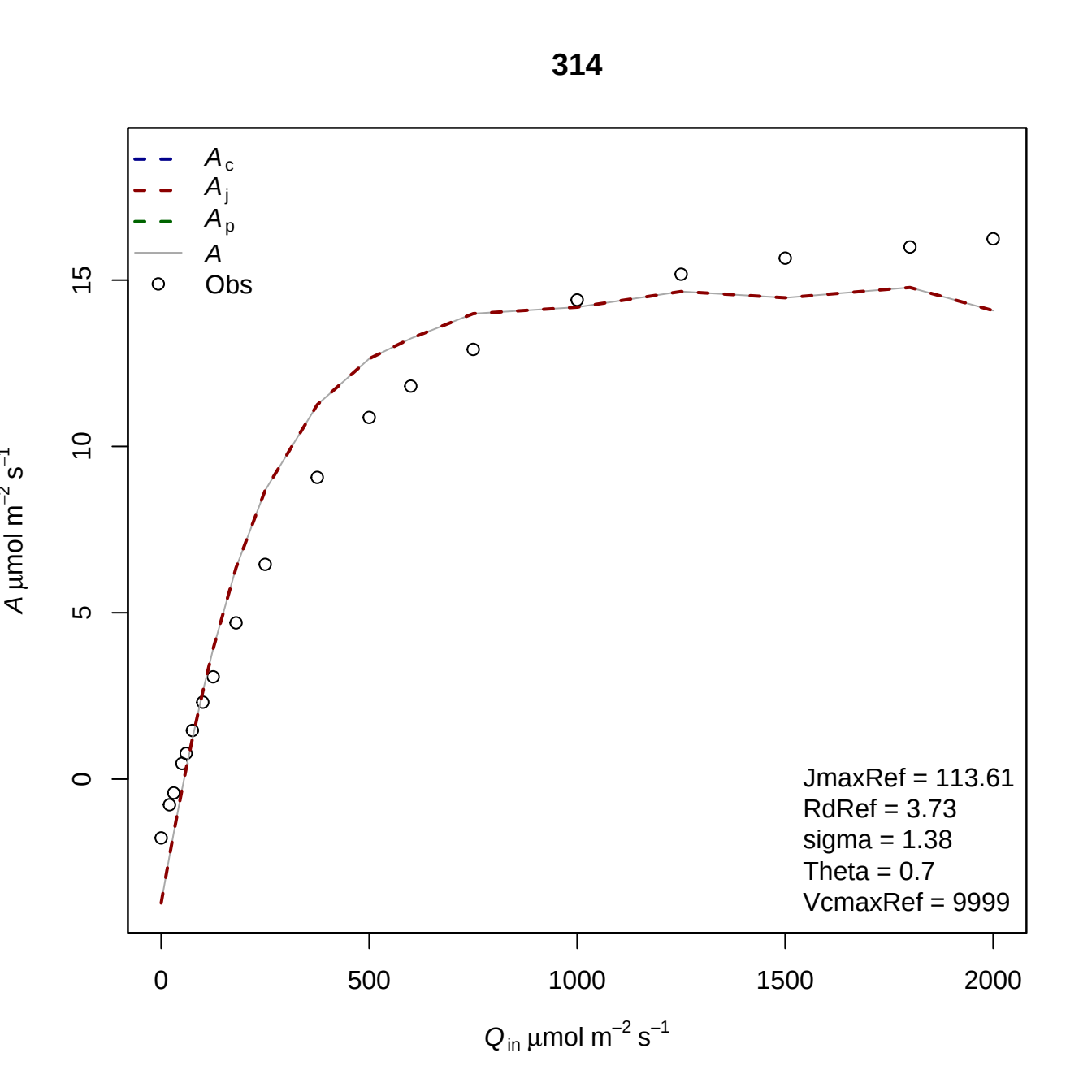

309
A_c
A_j
A_p
A
Obs
JmaxRef = 110.01
RdRef = 4.33
sigma = 1.32
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

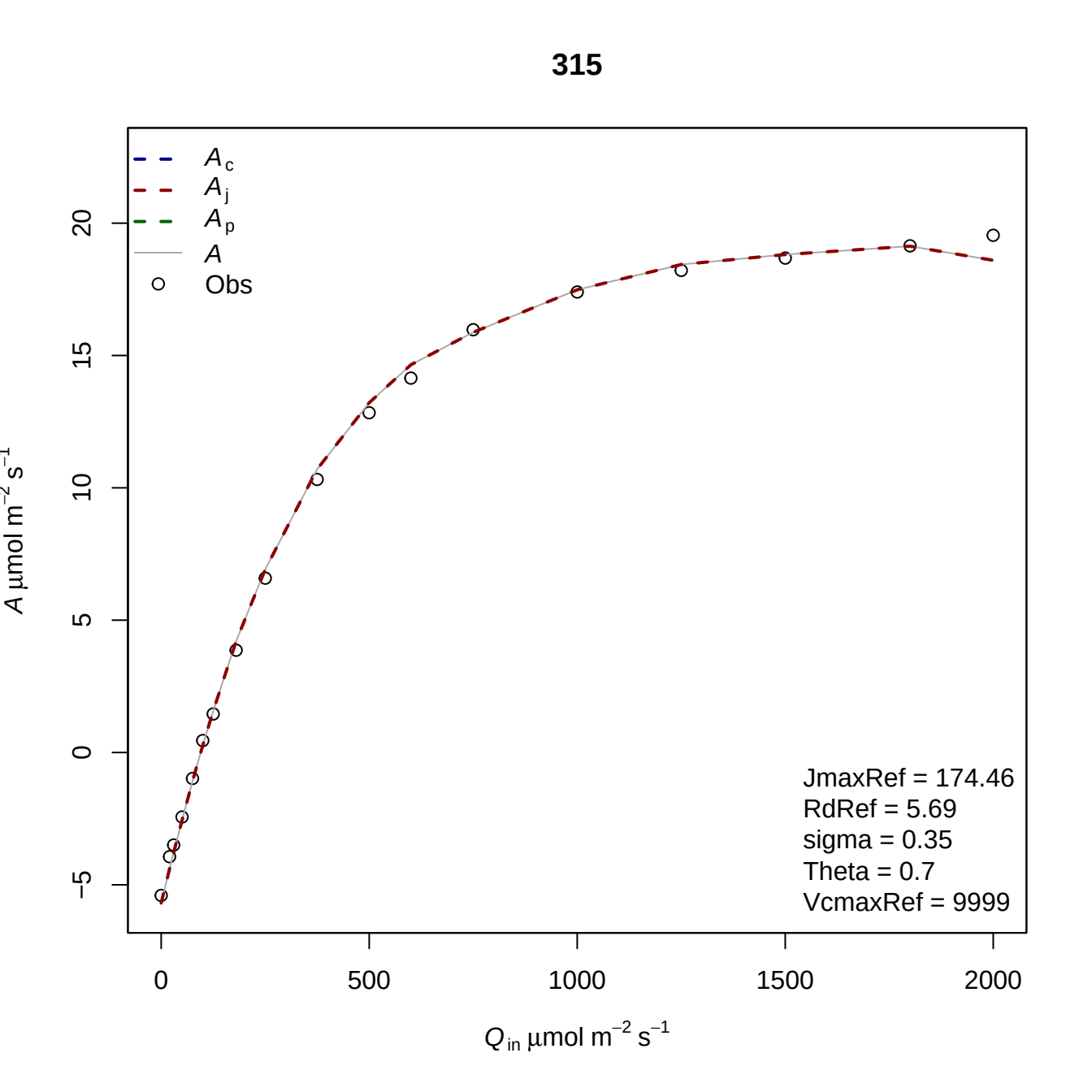

310
A_c
A_j
A_p
A
Obs
JmaxRef = 103.55
RdRef = 2.91
sigma = 0.96
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

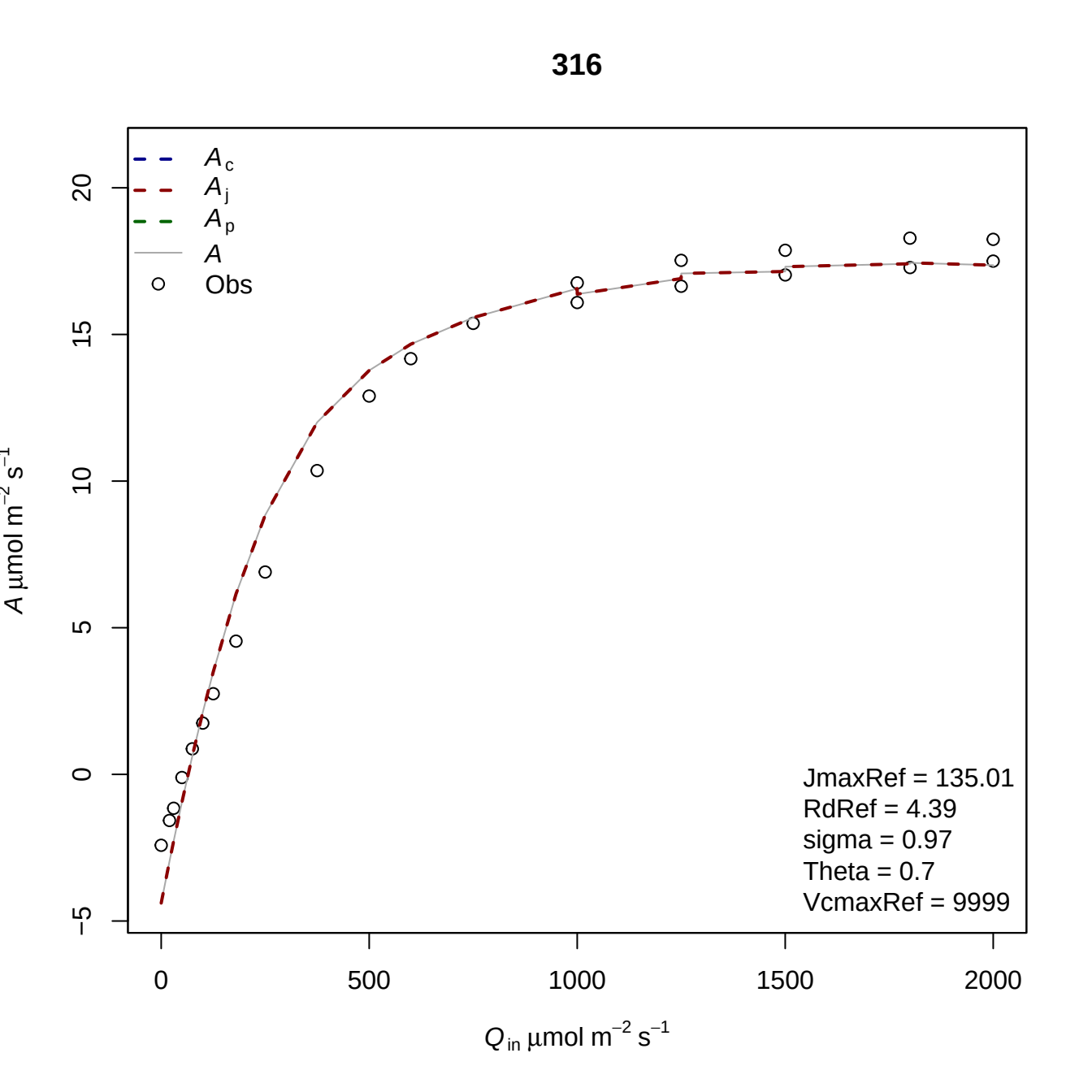

311
A_c
A_j
A_p
A
Obs
JmaxRef = 115.96
RdRef = 4.45
sigma = 1.62
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

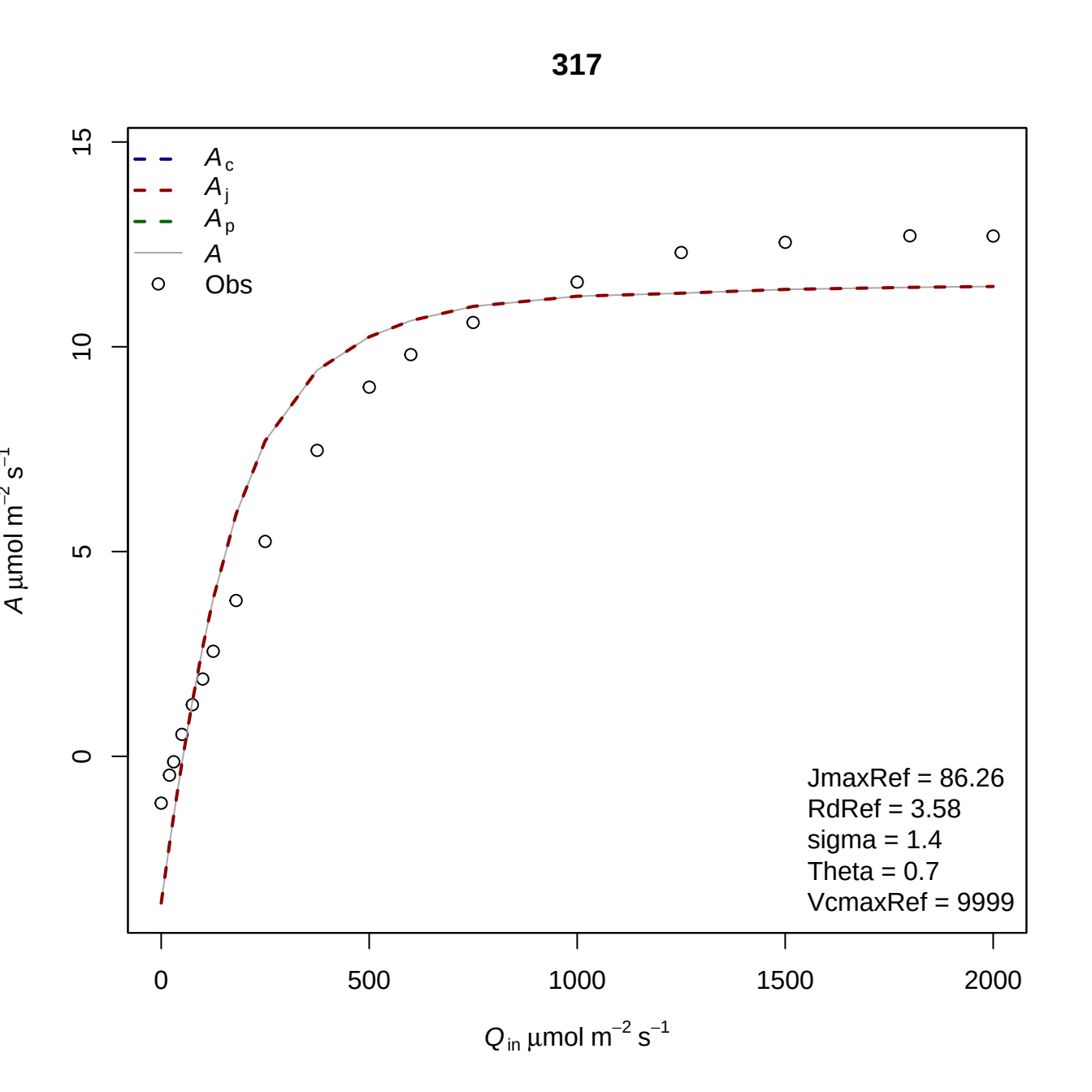

312
A_c
A_j
A_p
A
Obs
JmaxRef = 130.31
RdRef = 4.1
sigma = 1.27
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

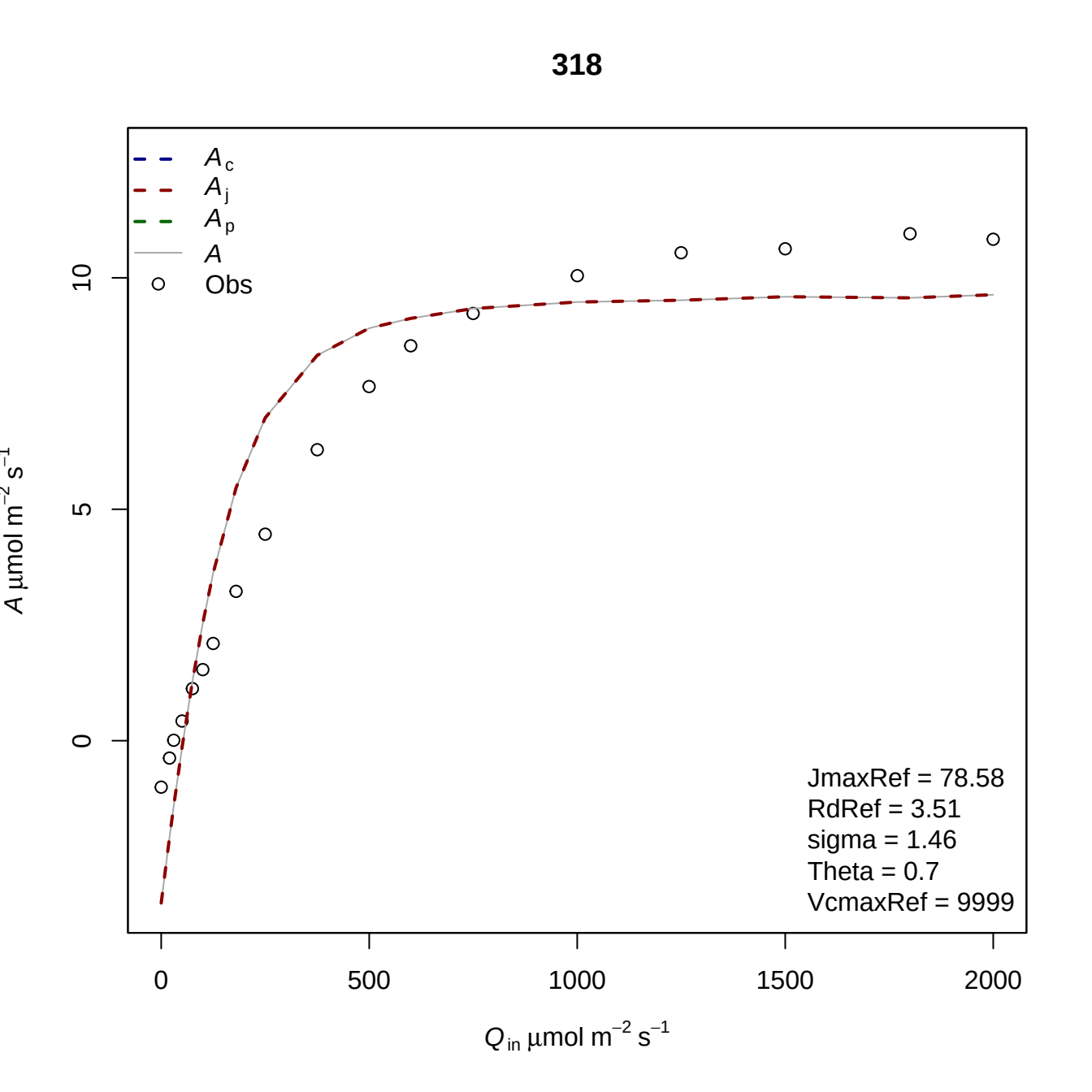

313
A_c
A_j
A_p
A
Obs
JmaxRef = 103.7
RdRef = 3.48
sigma = 1.04
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

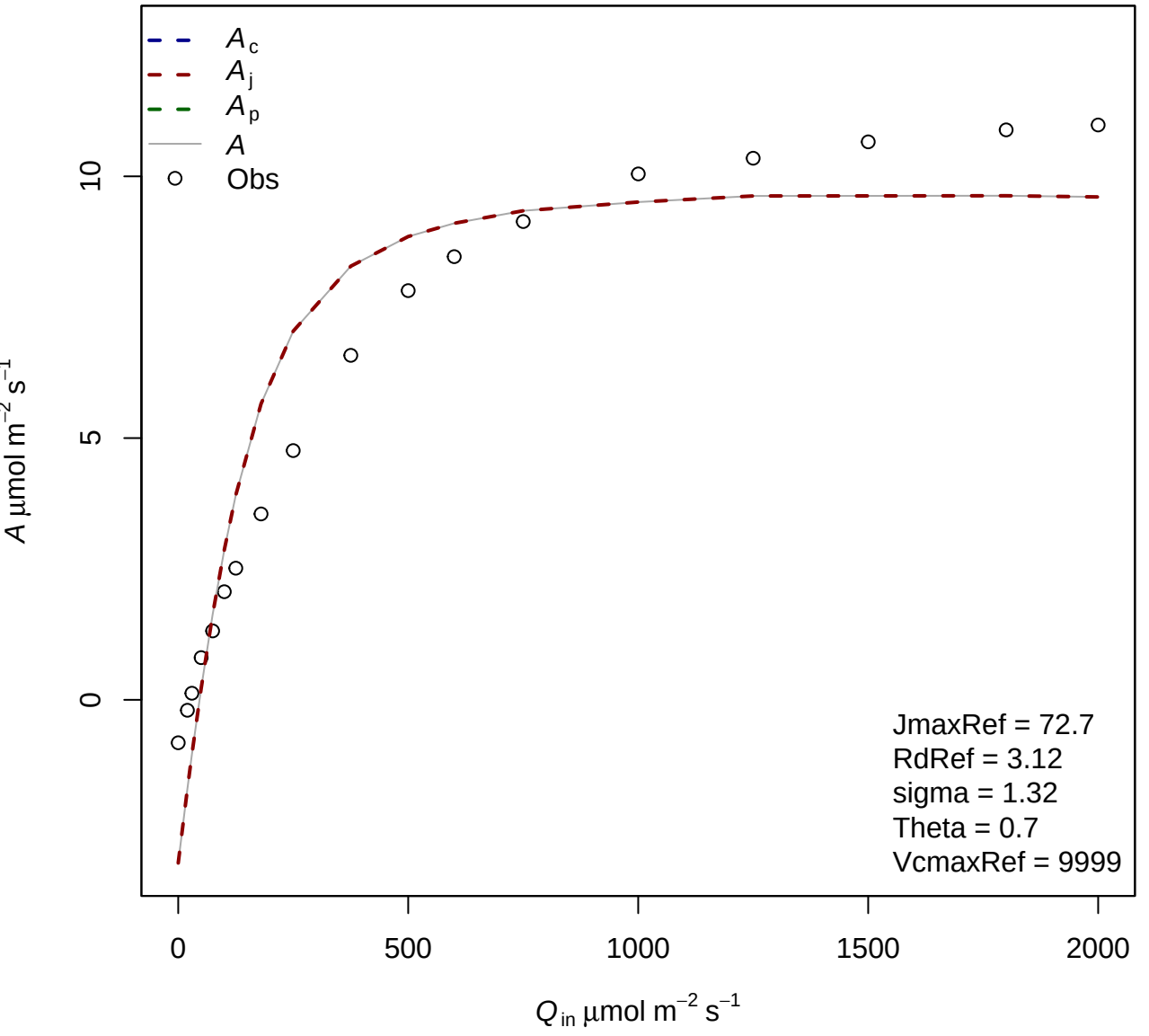

314
A_c
A_j
A_p
A
Obs
JmaxRef = 113.61
RdRef = 3.73
sigma = 1.38
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

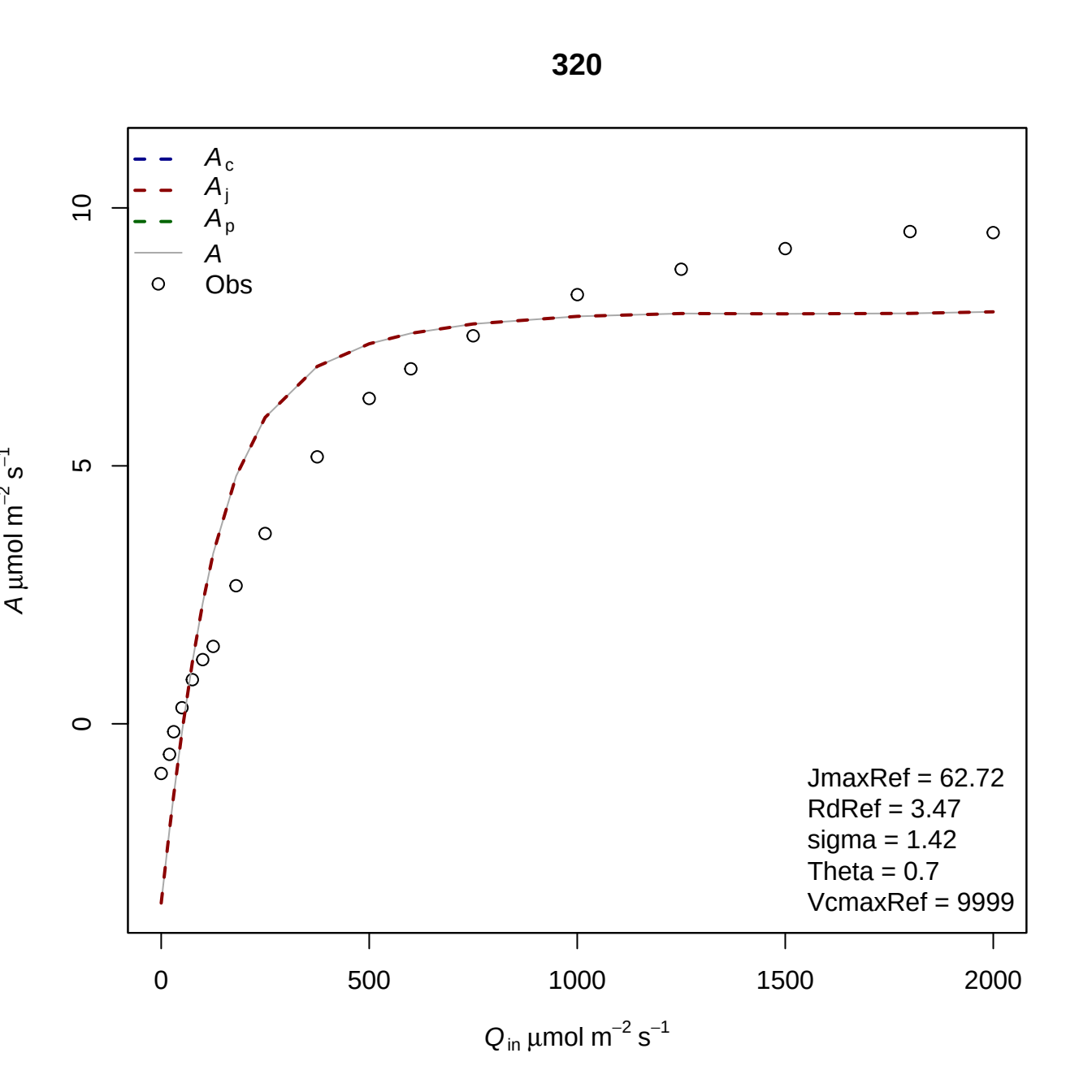

315
A_c
A_j
A_p
A
Obs
JmaxRef = 174.46
RdRef = 5.69
sigma = 0.35
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
20
15
10
5
0
−5
0
500
1000
1500
2000

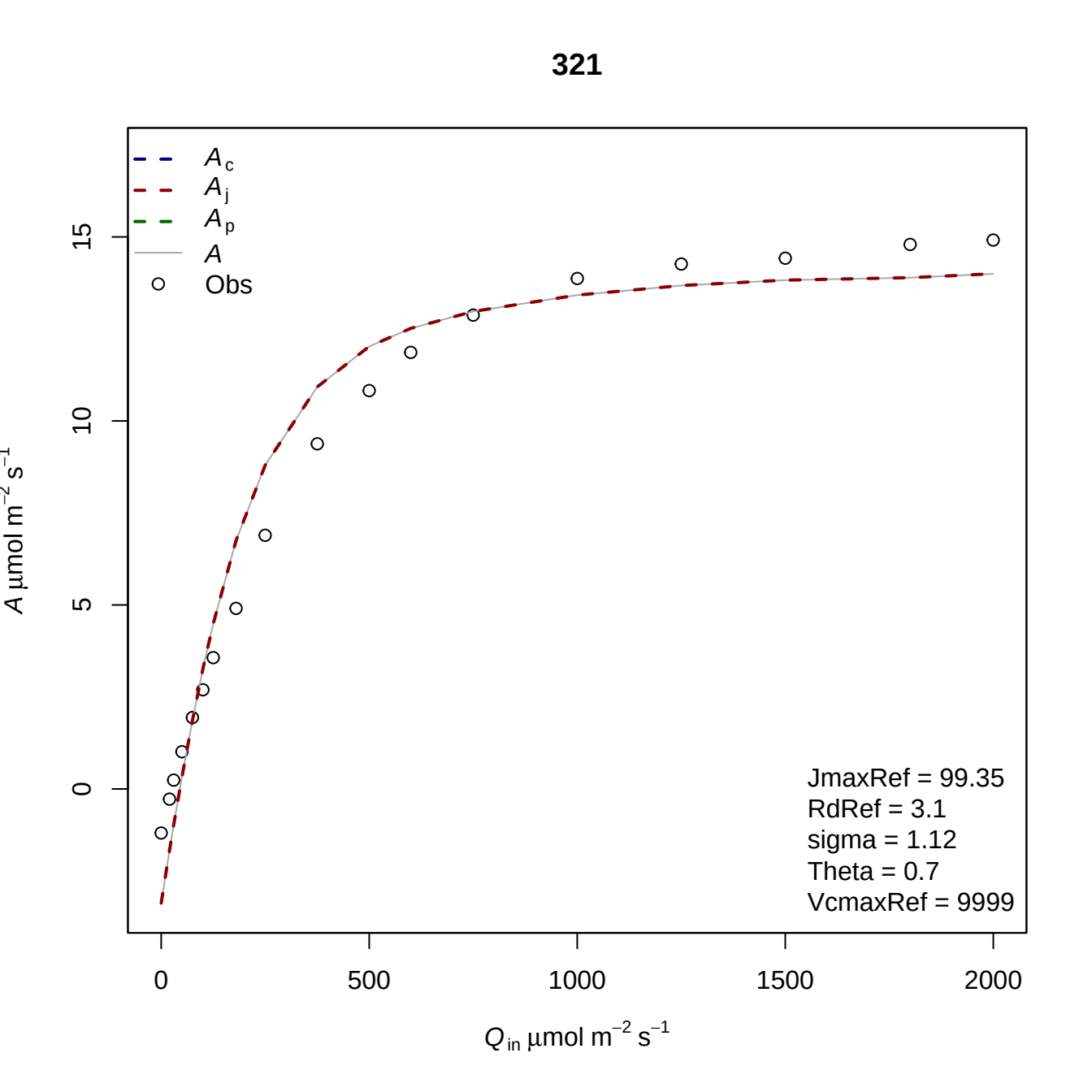

316
A_c
A_j
A_p
A
Obs
JmaxRef = 135.01
RdRef = 4.39
sigma = 0.97
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

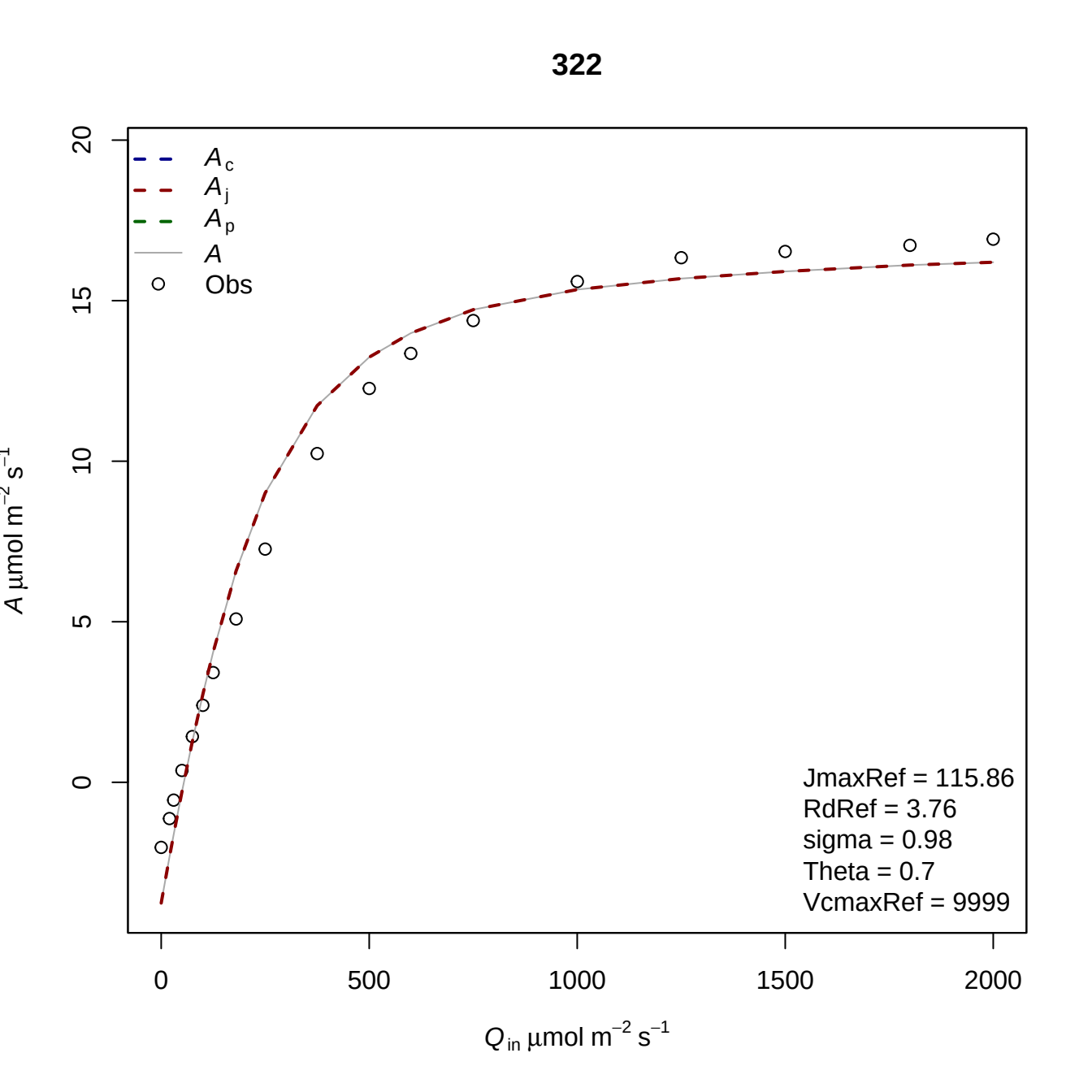

317
A_c
A_j
A_p
A
Obs
JmaxRef = 86.26
RdRef = 3.58
sigma = 1.4
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

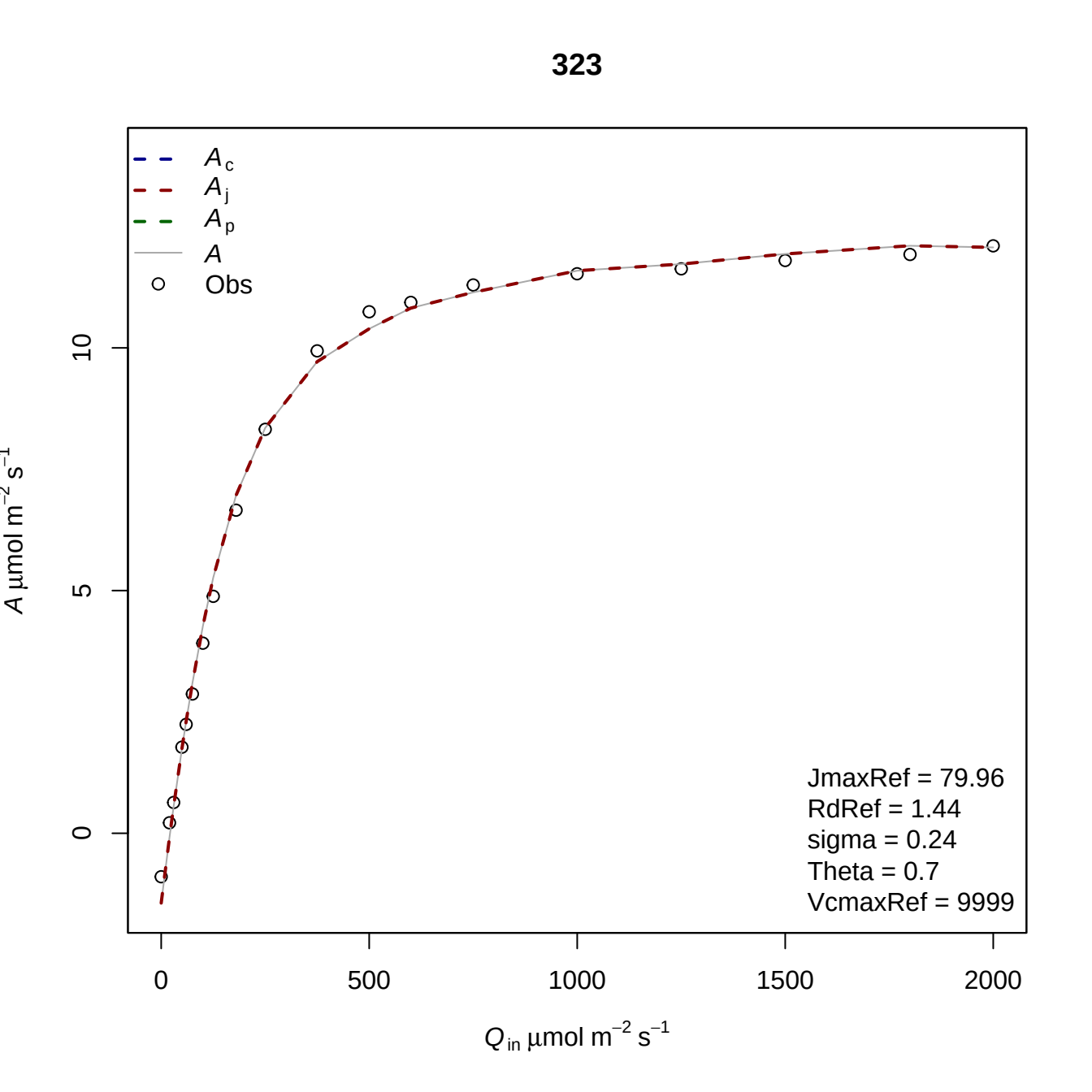

318
A_c
A_j
A_p
A
Obs
JmaxRef = 78.58
RdRef = 3.51
sigma = 1.46
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

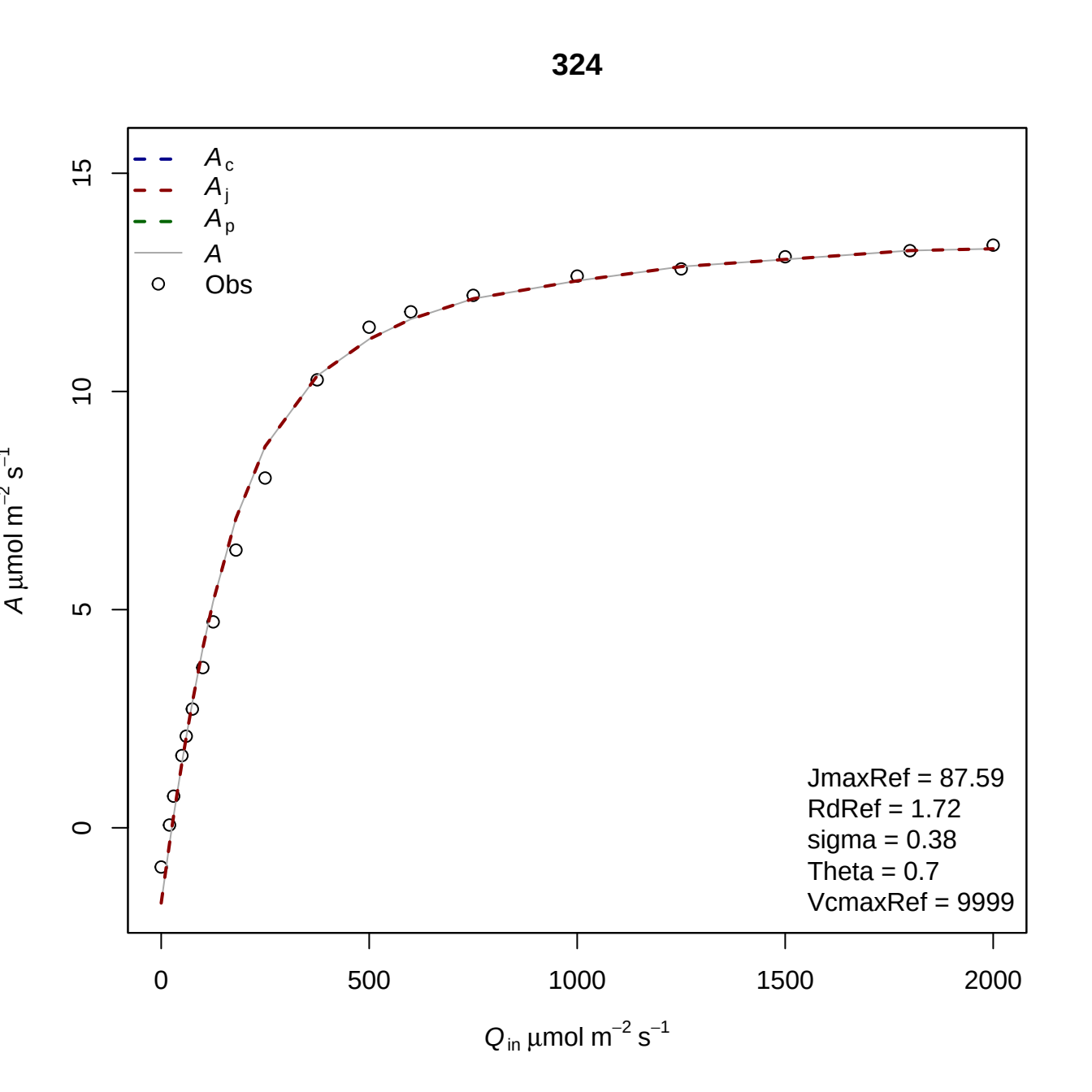

A_c
A_j
A_p
A
Obs
JmaxRef = 72.7
RdRef = 3.12
sigma = 1.32
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

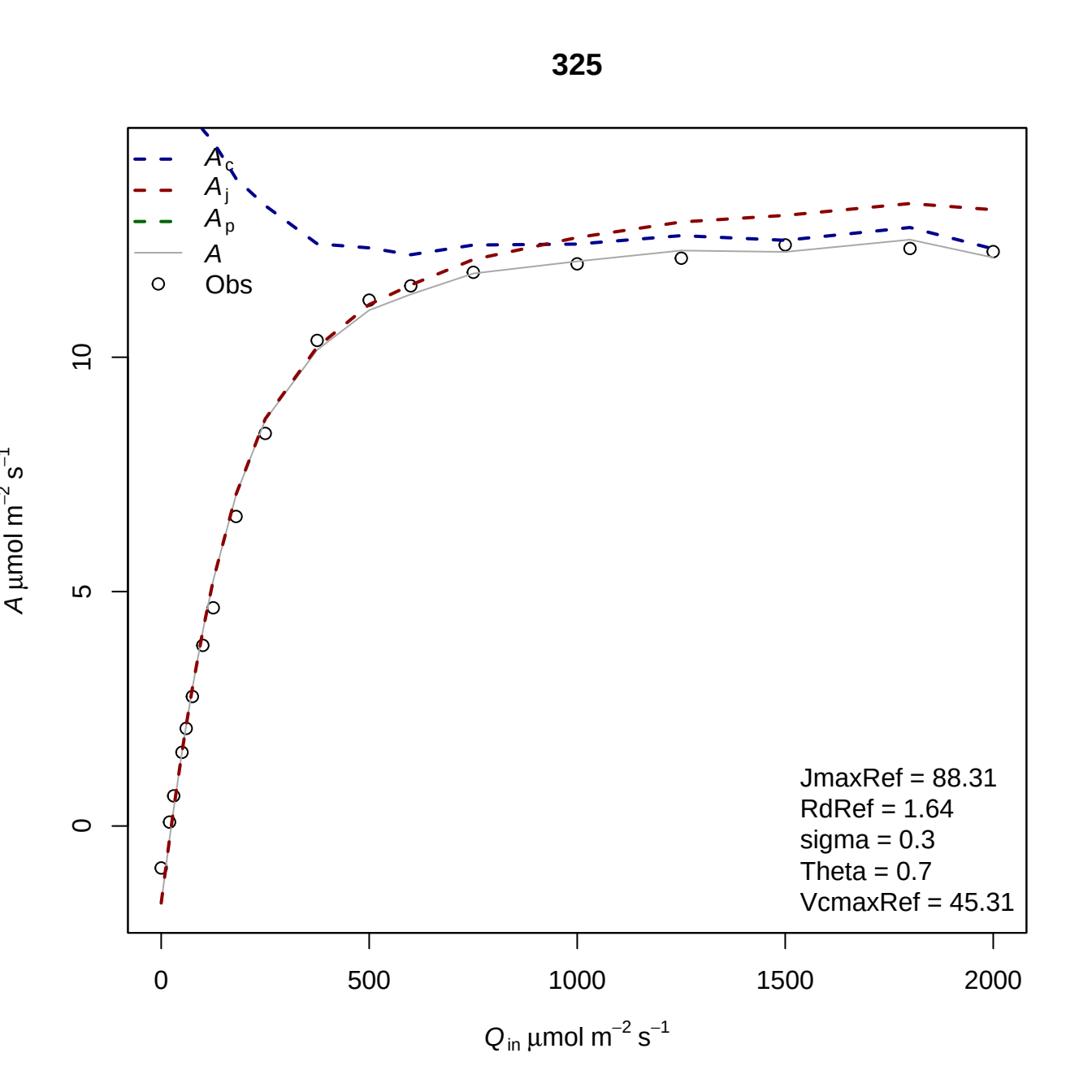

320
A_c
A_j
A_p
A
Obs
JmaxRef = 62.72
RdRef = 3.47
sigma = 1.42
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

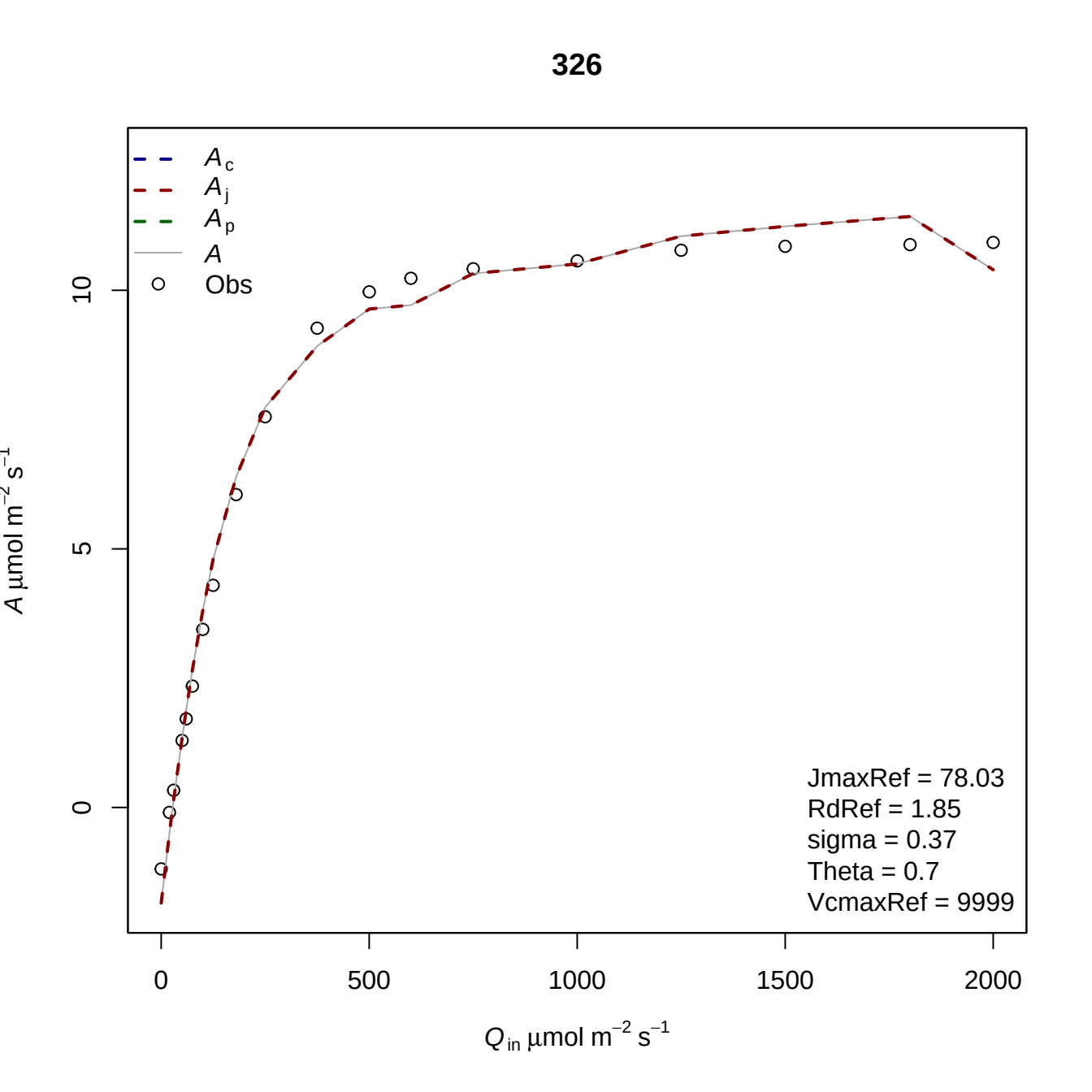

321
A_c
A_j
A_p
A
Obs
JmaxRef = 99.35
RdRef = 3.1
sigma = 1.12
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

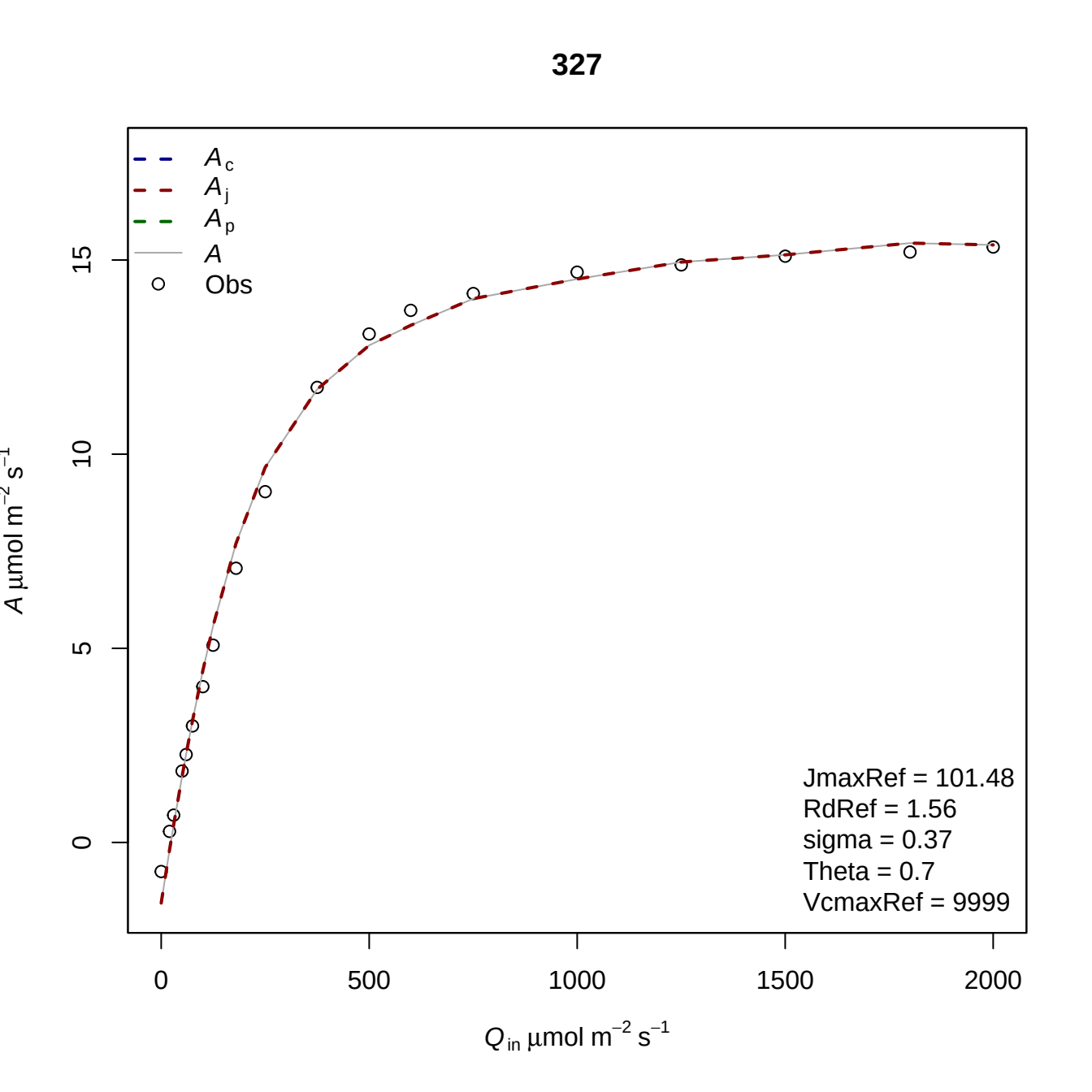

322
A_c
A_j
A_p
A
Obs
JmaxRef = 115.86
RdRef = 3.76
sigma = 0.98
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

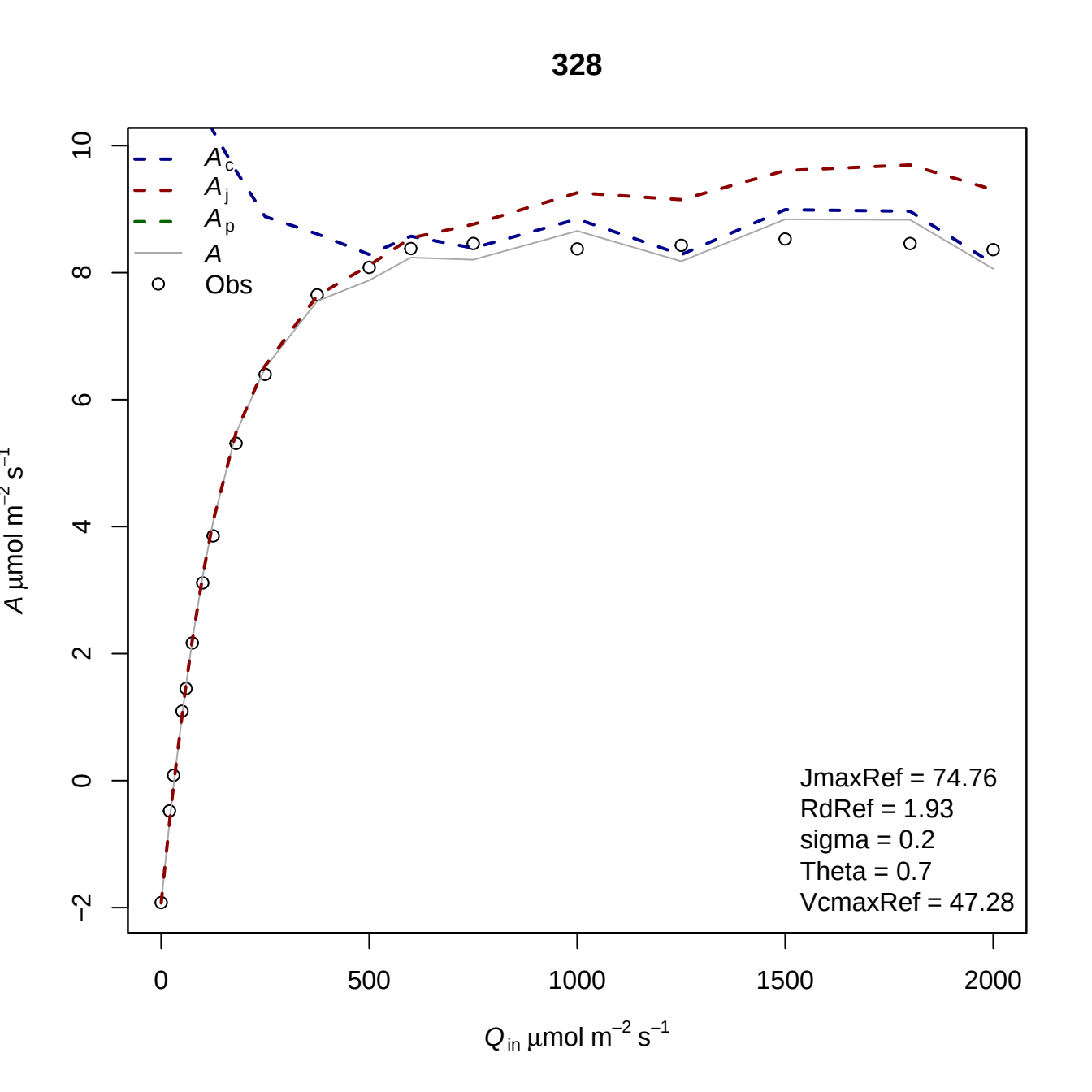

323
A_c
A_j
A_p
A
Obs
JmaxRef = 79.96
RdRef = 1.44
sigma = 0.24
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

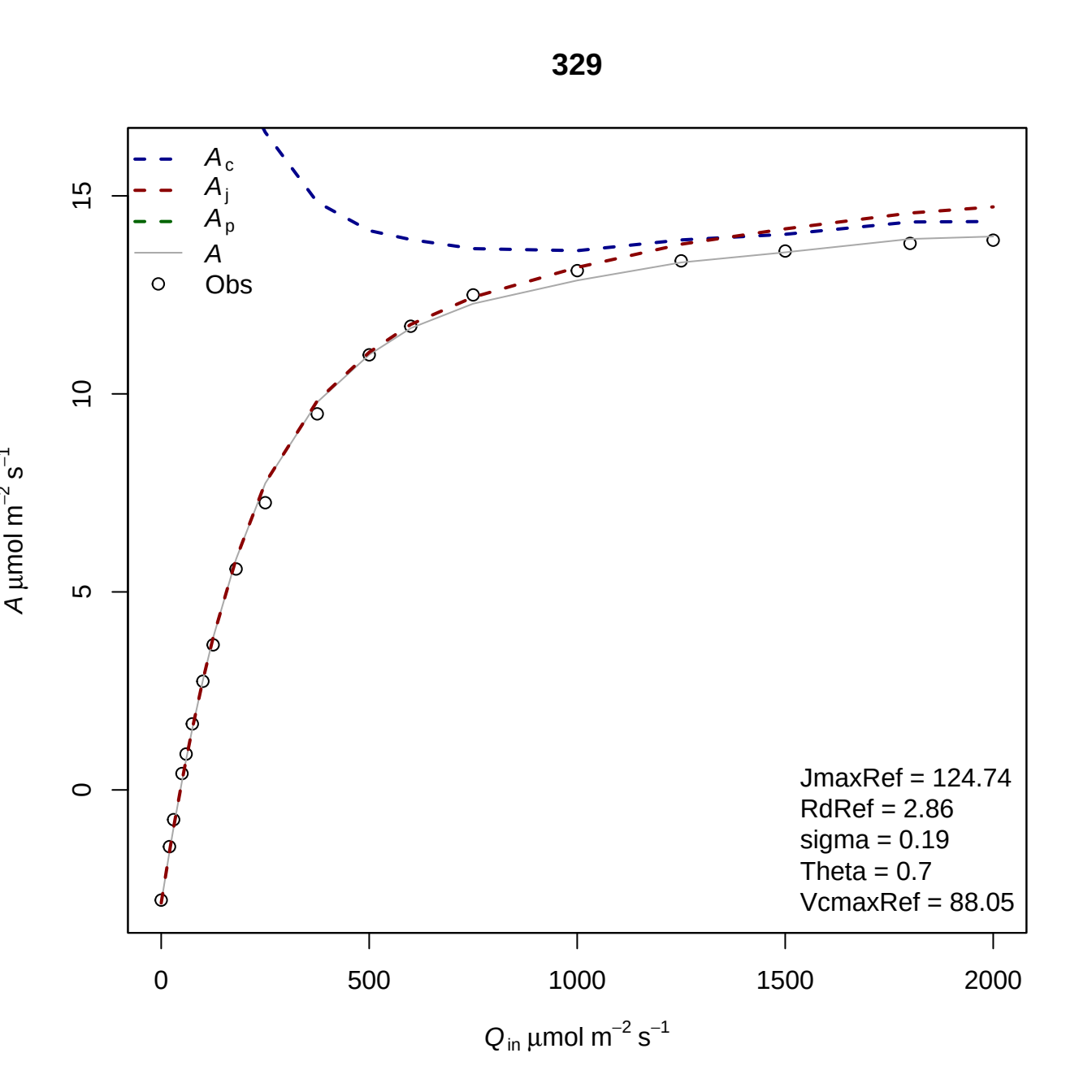

324
A_c
A_j
A_p
A
Obs
JmaxRef = 87.59
RdRef = 1.72
sigma = 0.38
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

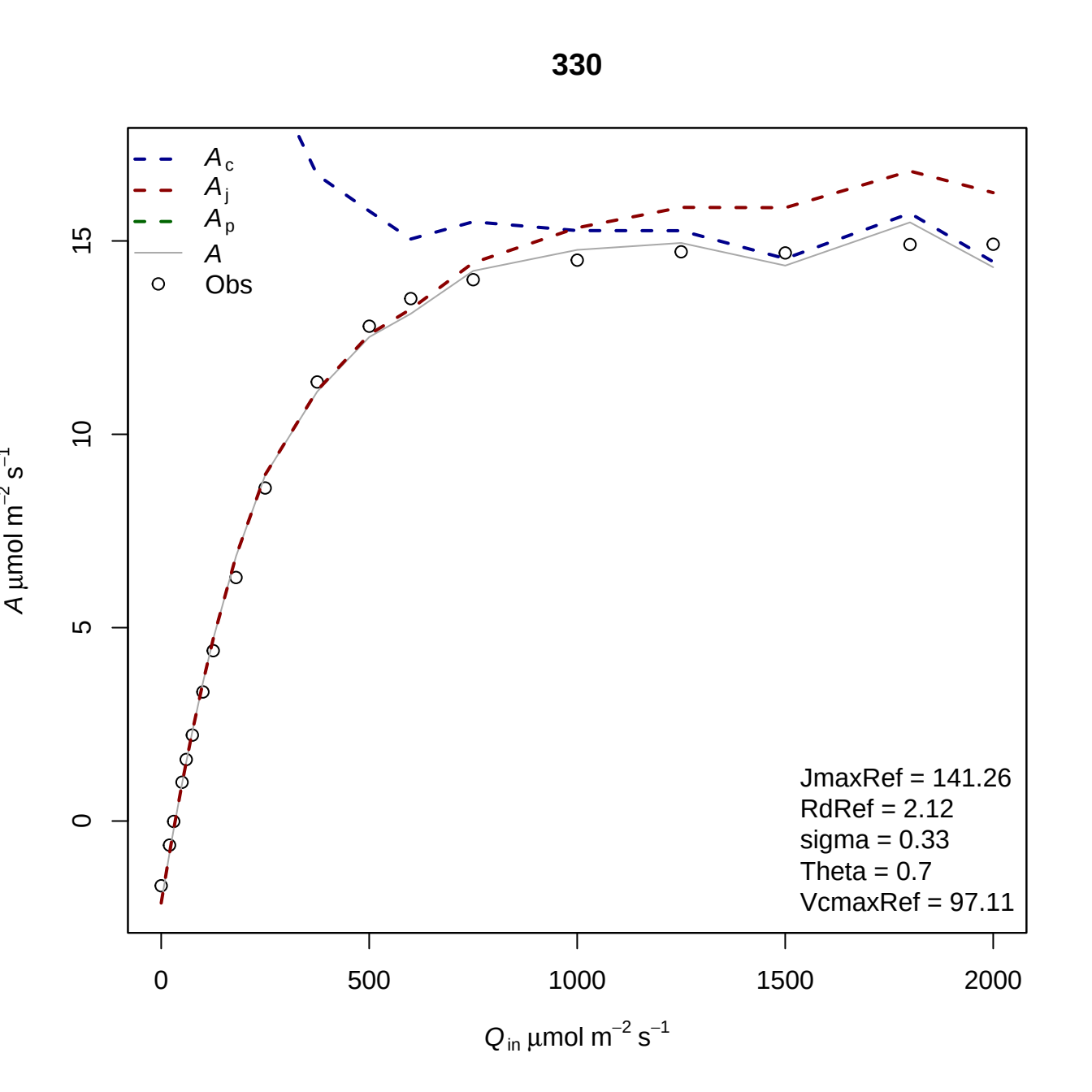

325
A_c
A_j
A_p
A
Obs
JmaxRef = 88.31
RdRef = 1.64
sigma = 0.3
Theta = 0.7
VcmaxRef = 45.31
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

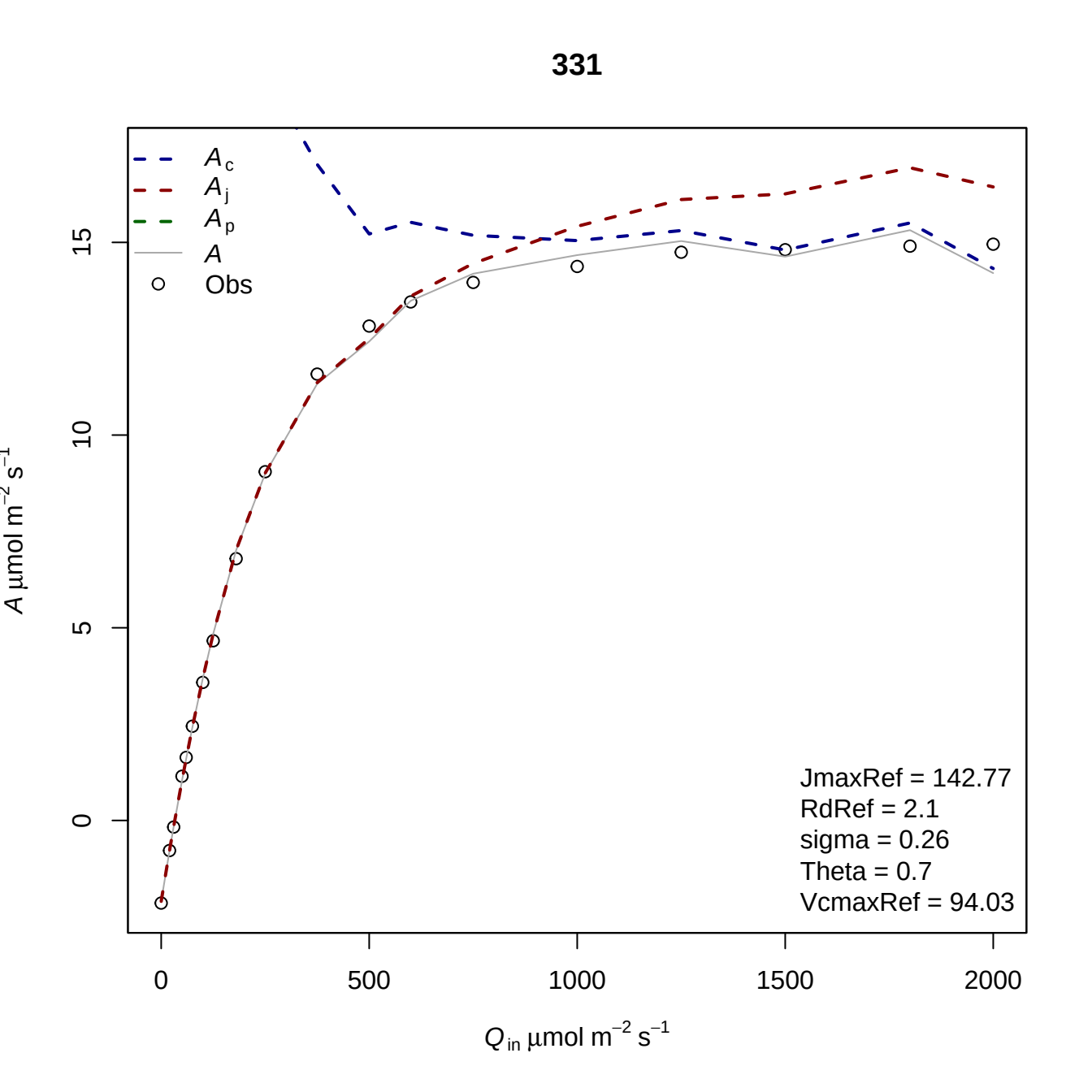

326
A_c
A_j
A_p
A
Obs
JmaxRef = 78.03
RdRef = 1.85
sigma = 0.37
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

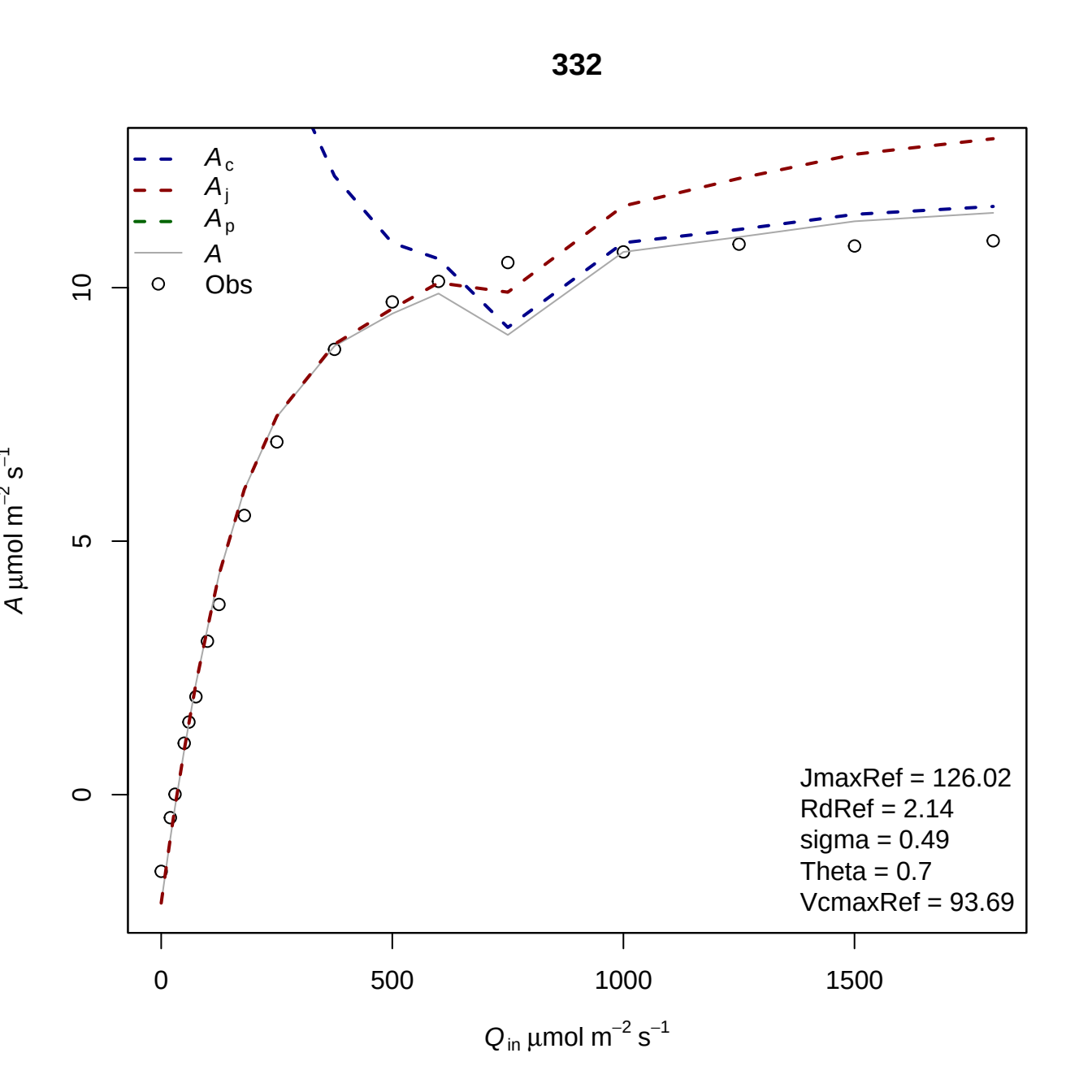

327
A_c
A_j
A_p
A
Obs
JmaxRef = 101.48
RdRef = 1.56
sigma = 0.37
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

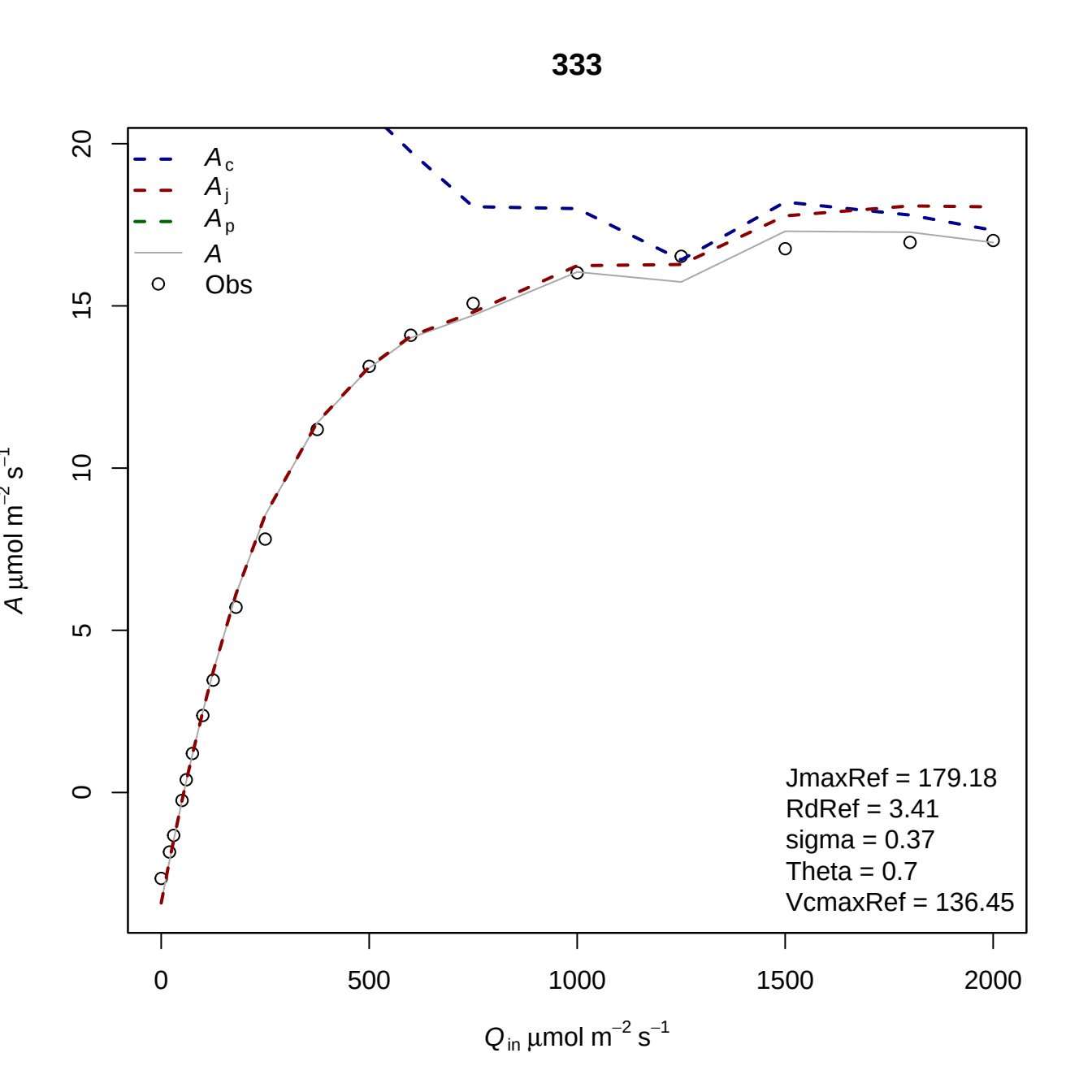

328
A_c
A_j
A_p
A
Obs
JmaxRef = 74.76
RdRef = 1.93
sigma = 0.2
Theta = 0.7
VcmaxRef = 47.28
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

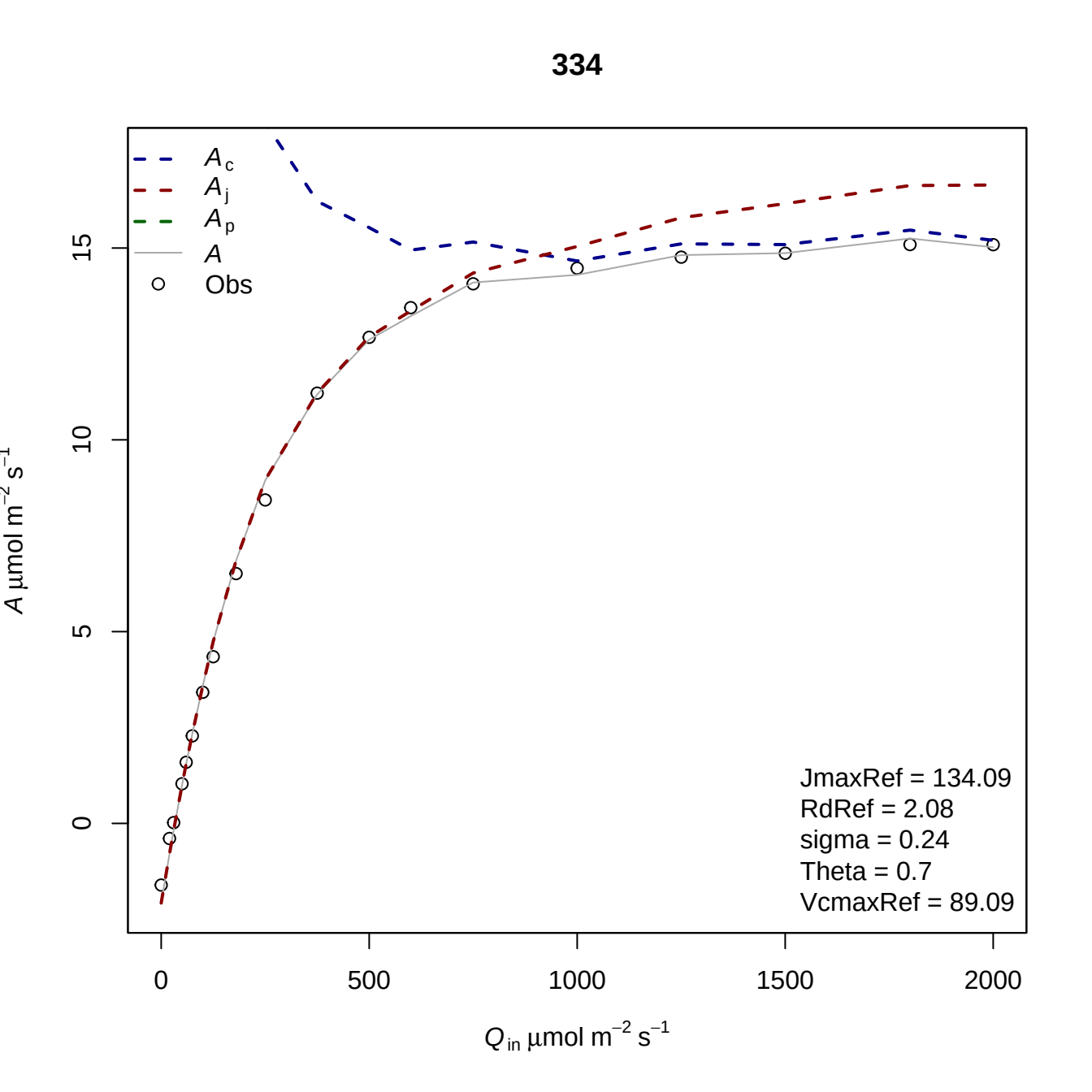

329
A_c
A_j
A_p
A
Obs
JmaxRef = 124.74
RdRef = 2.86
sigma = 0.19
Theta = 0.7
VcmaxRef = 88.05
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

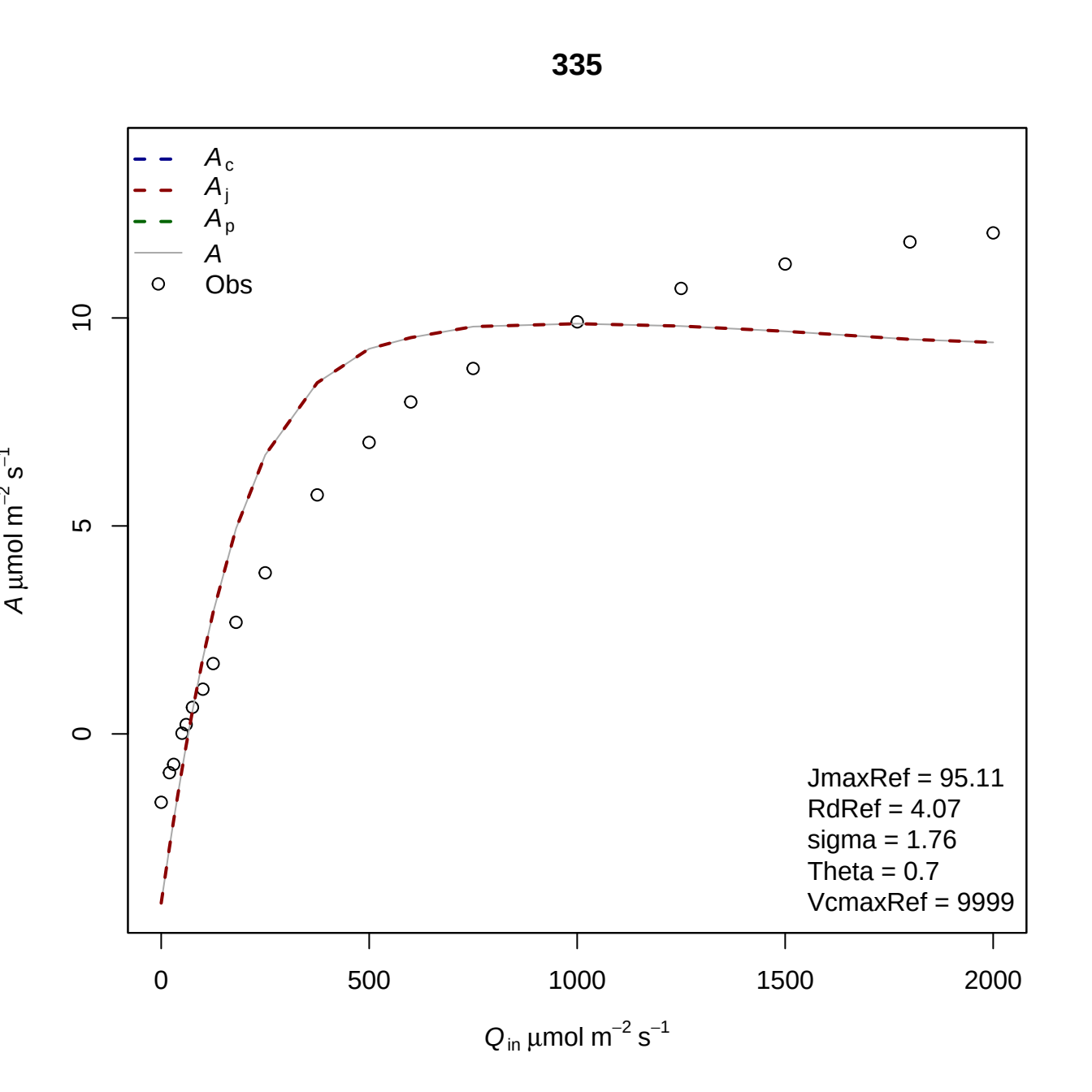

330
A_c
A_j
A_p
A
Obs
JmaxRef = 141.26
RdRef = 2.12
sigma = 0.33
Theta = 0.7
VcmaxRef = 97.11
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

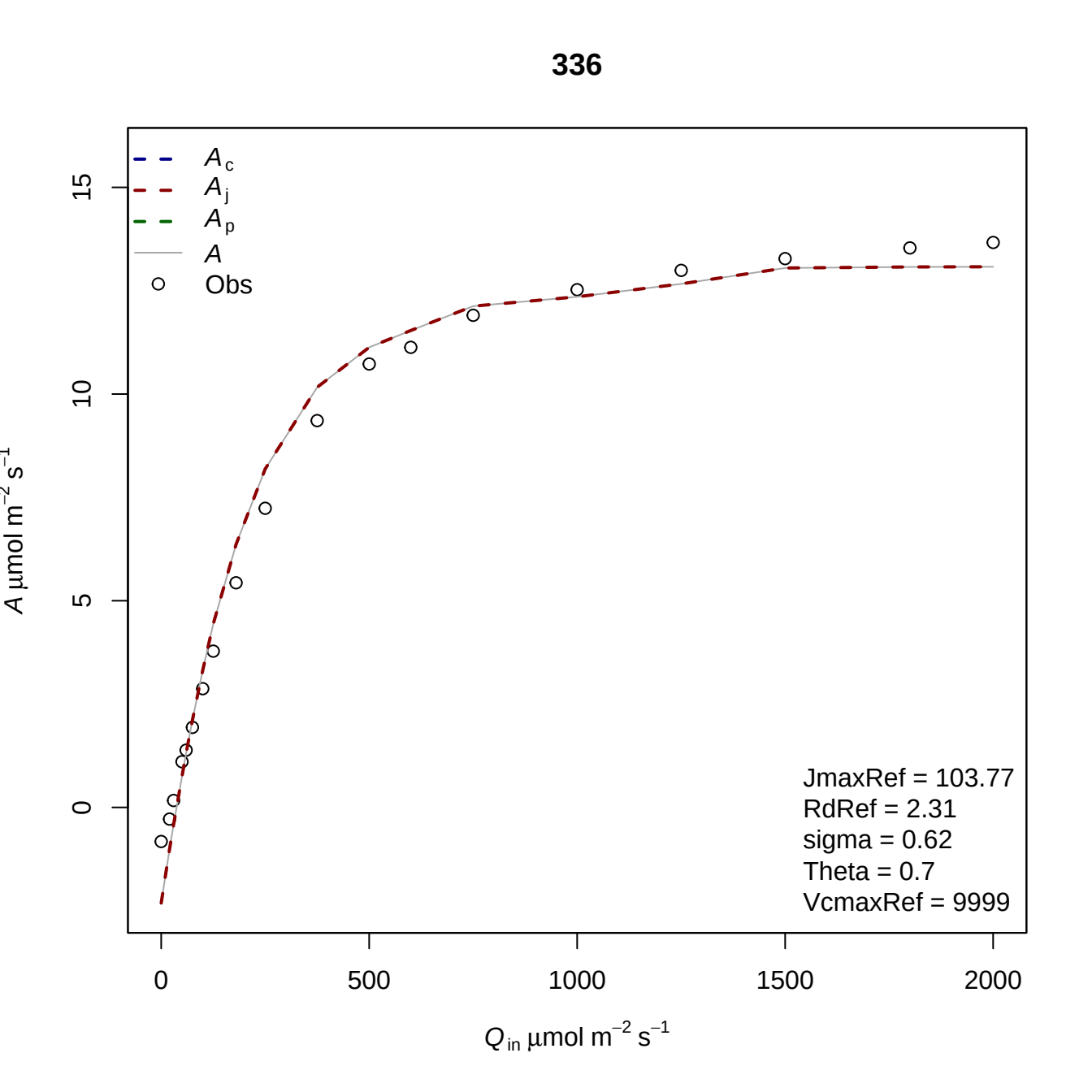

331
A_c
A_j
A_p
A
Obs
JmaxRef = 142.77
RdRef = 2.1
sigma = 0.26
Theta = 0.7
VcmaxRef = 94.03
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

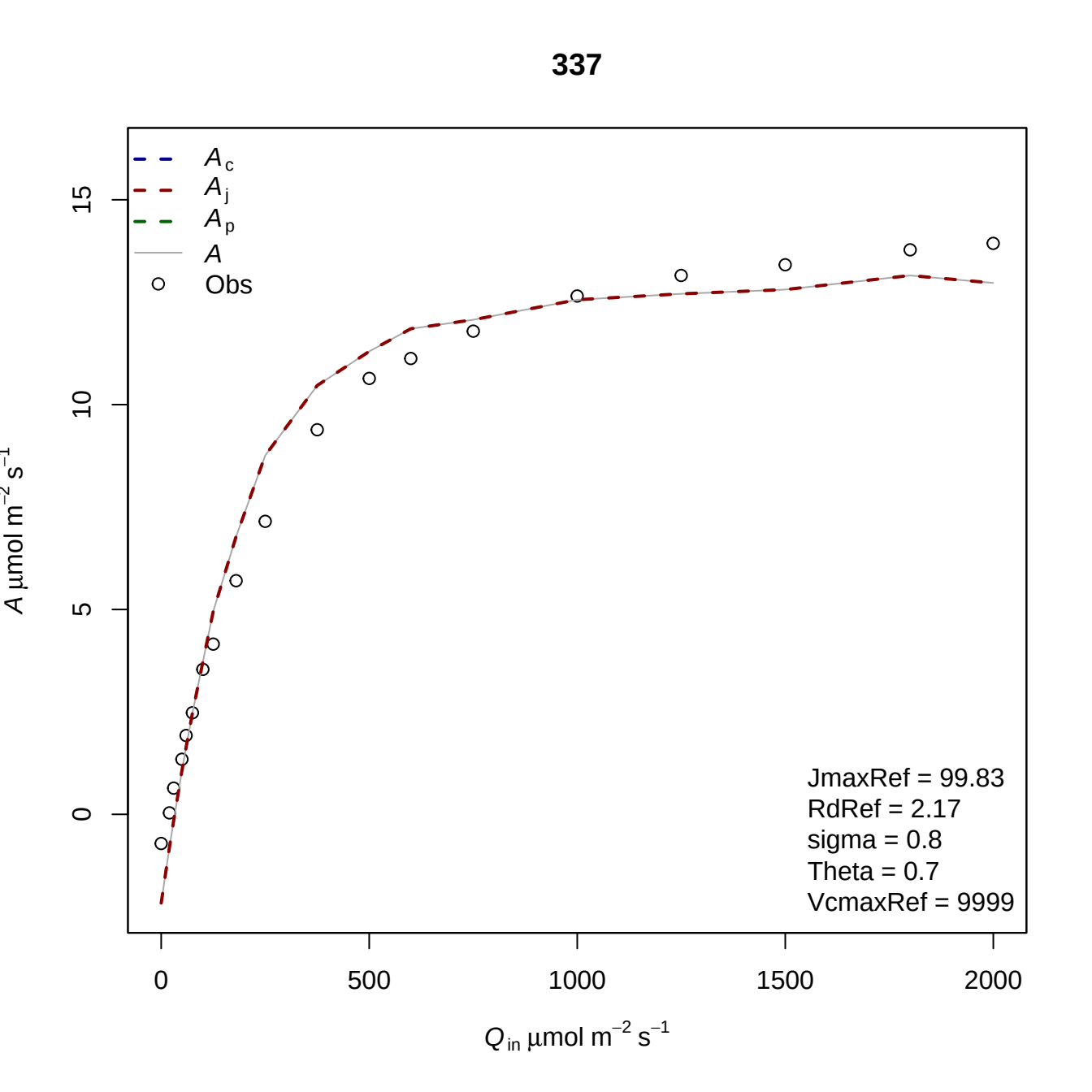

332
A_c
A_j
A_p
A
Obs
JmaxRef = 126.02
RdRef = 2.14
sigma = 0.49
Theta = 0.7
VcmaxRef = 93.69
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

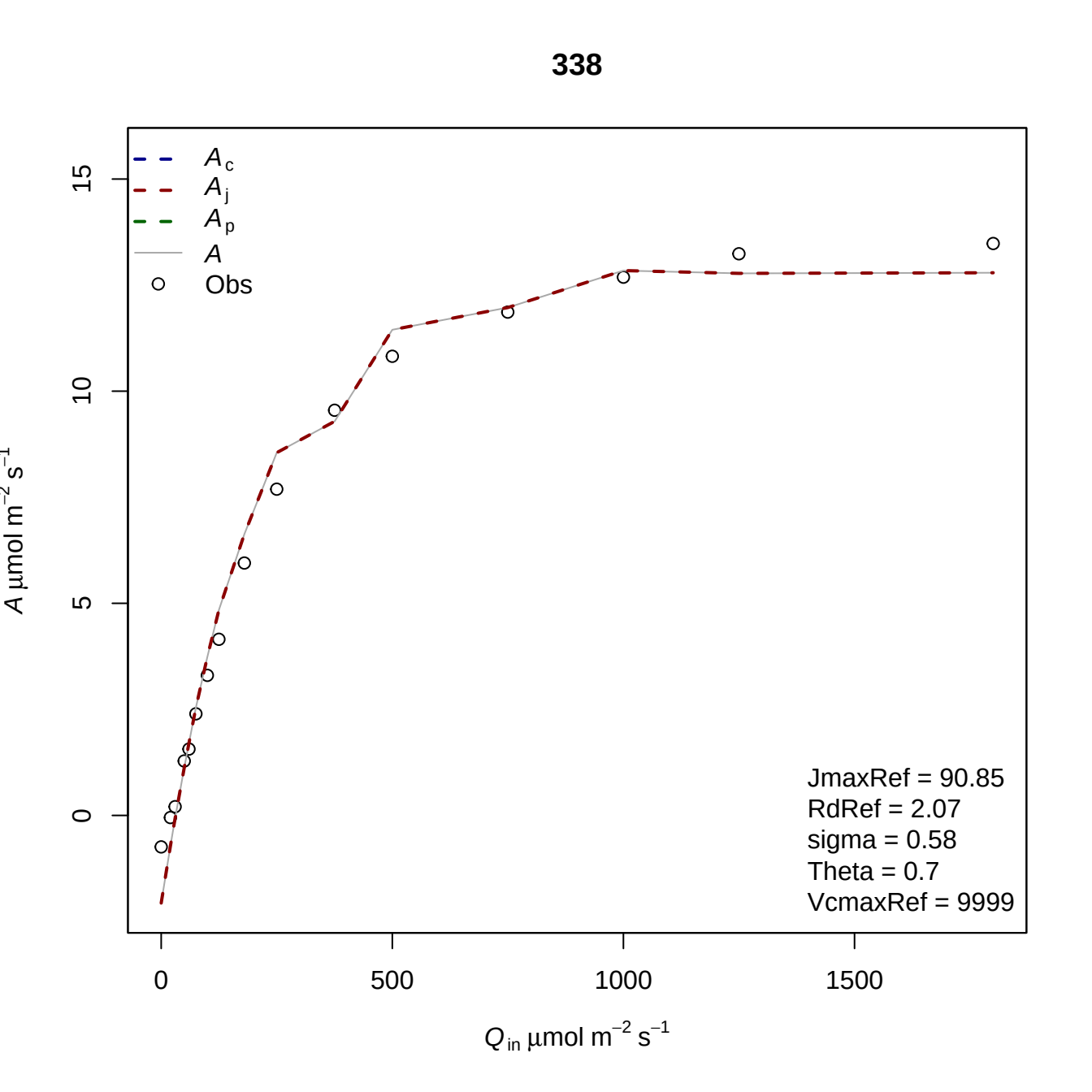

333
A_c
A_j
A_p
A
Obs
JmaxRef = 179.18
RdRef = 3.41
sigma = 0.37
Theta = 0.7
VcmaxRef = 136.45
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

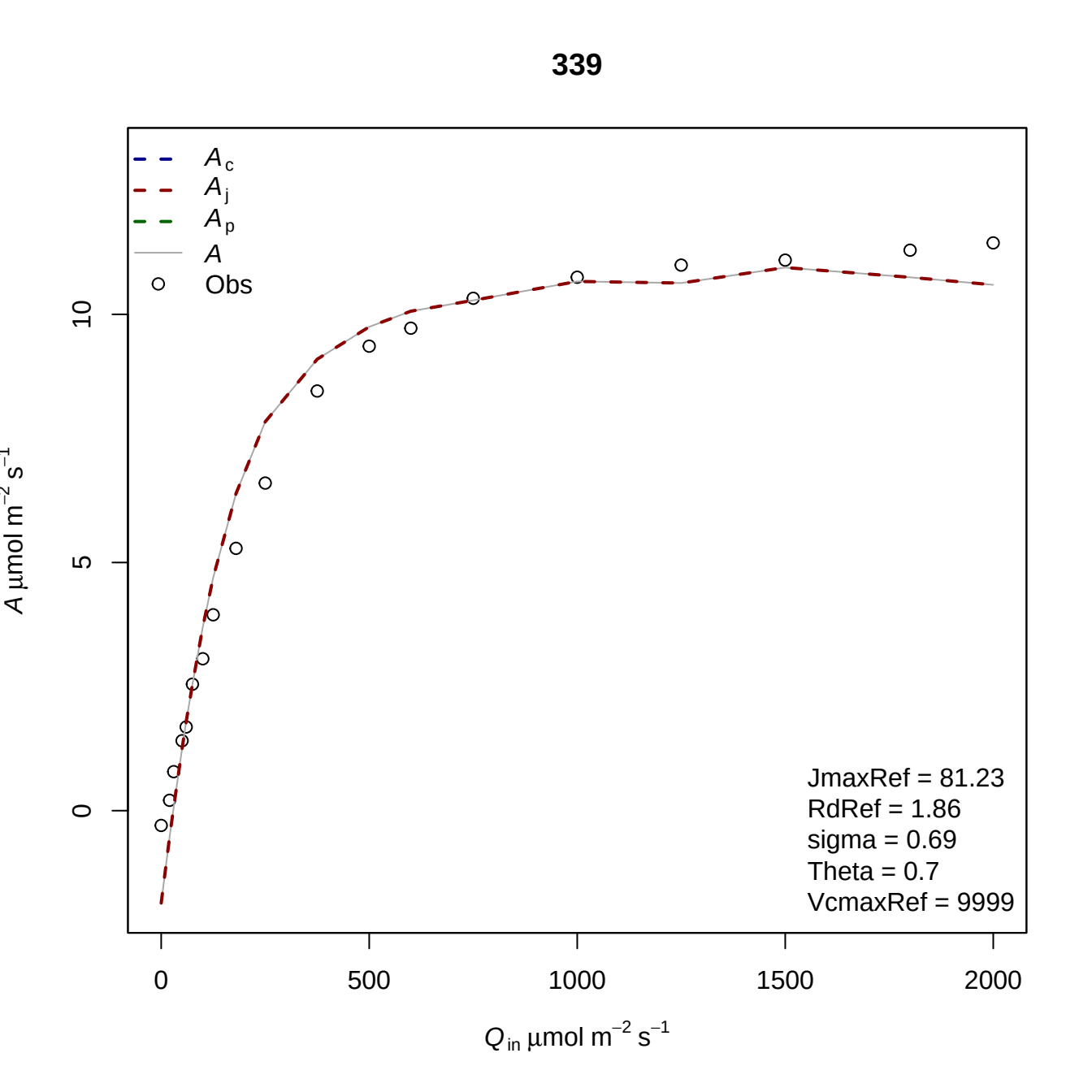

334
A_c
A_j
A_p
A
Obs
JmaxRef = 134.09
RdRef = 2.08
sigma = 0.24
Theta = 0.7
VcmaxRef = 89.09
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

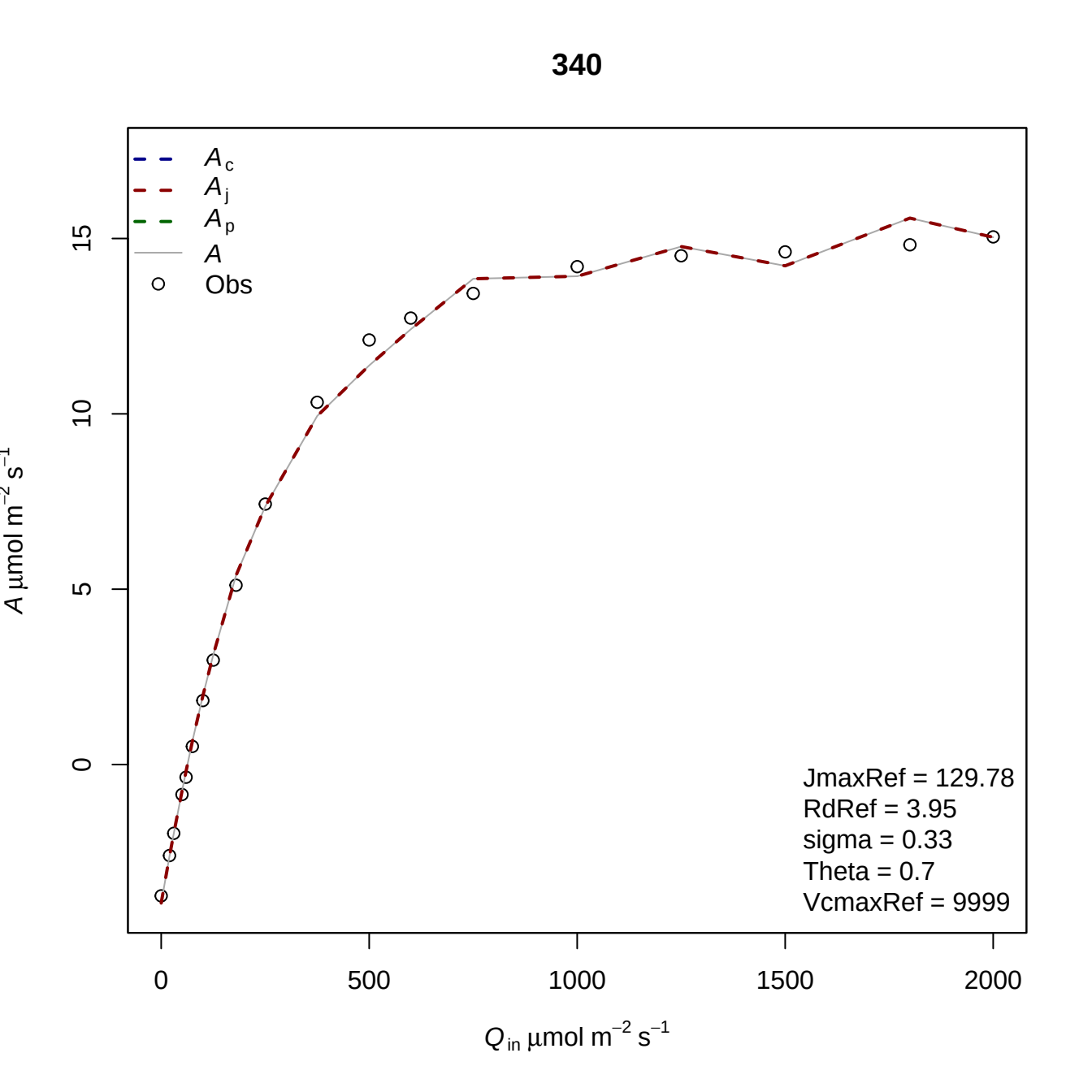

335
A_c
A_j
A_p
A
Obs
JmaxRef = 95.11
RdRef = 4.07
sigma = 1.76
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

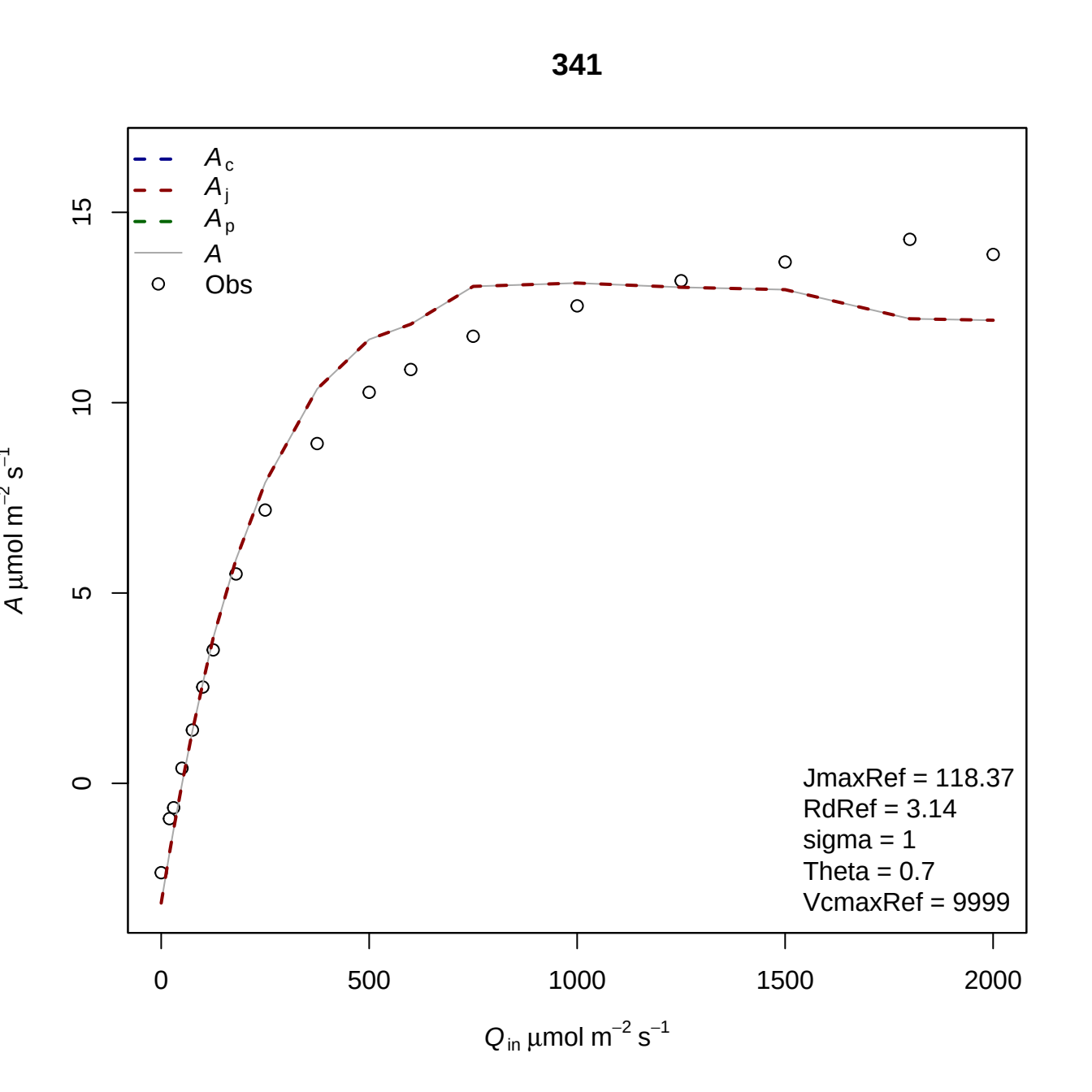

336
A_c
A_j
A_p
A
Obs
JmaxRef = 103.77
RdRef = 2.31
sigma = 0.62
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

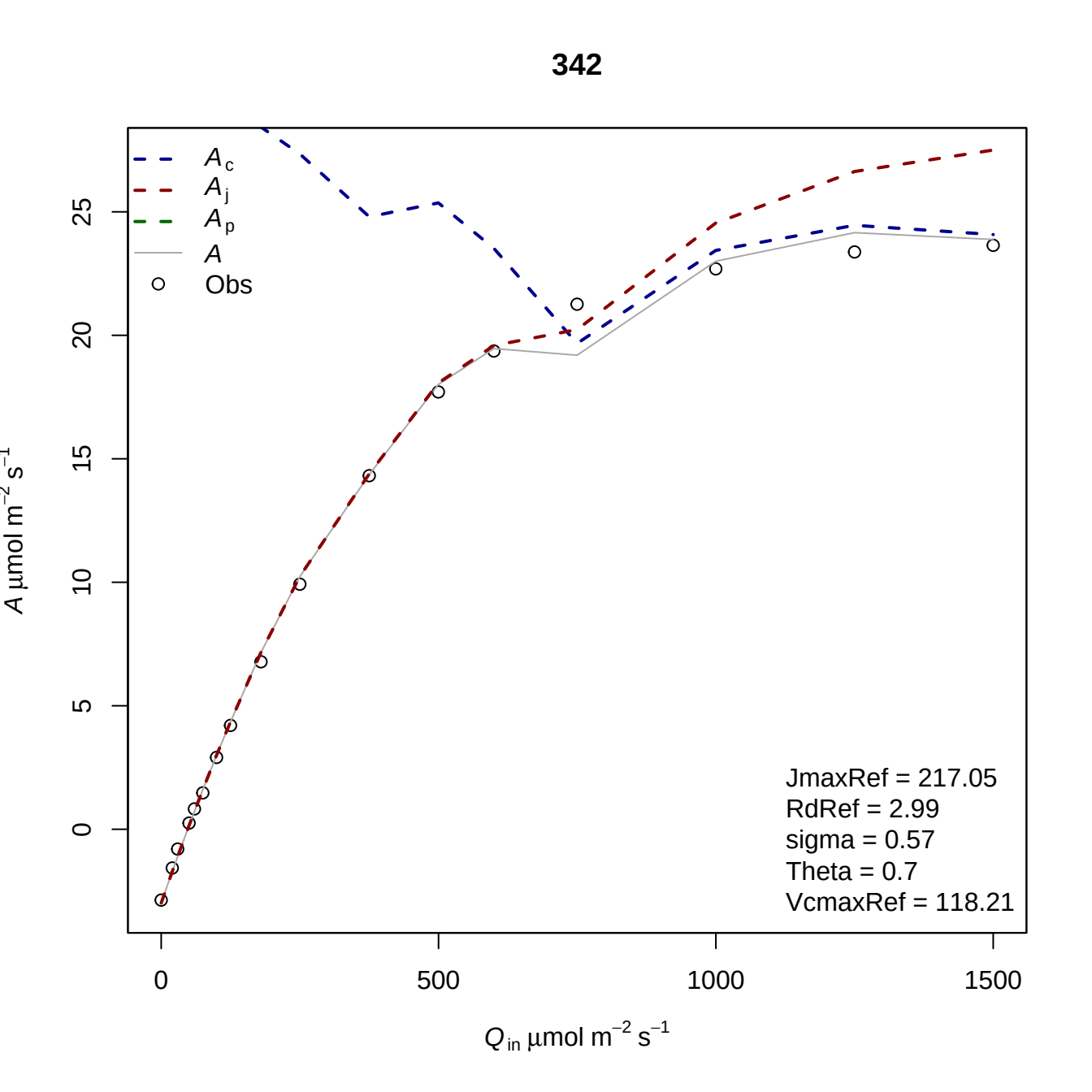

337
A_c
A_j
A_p
A
Obs
JmaxRef = 99.83
RdRef = 2.17
sigma = 0.8
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

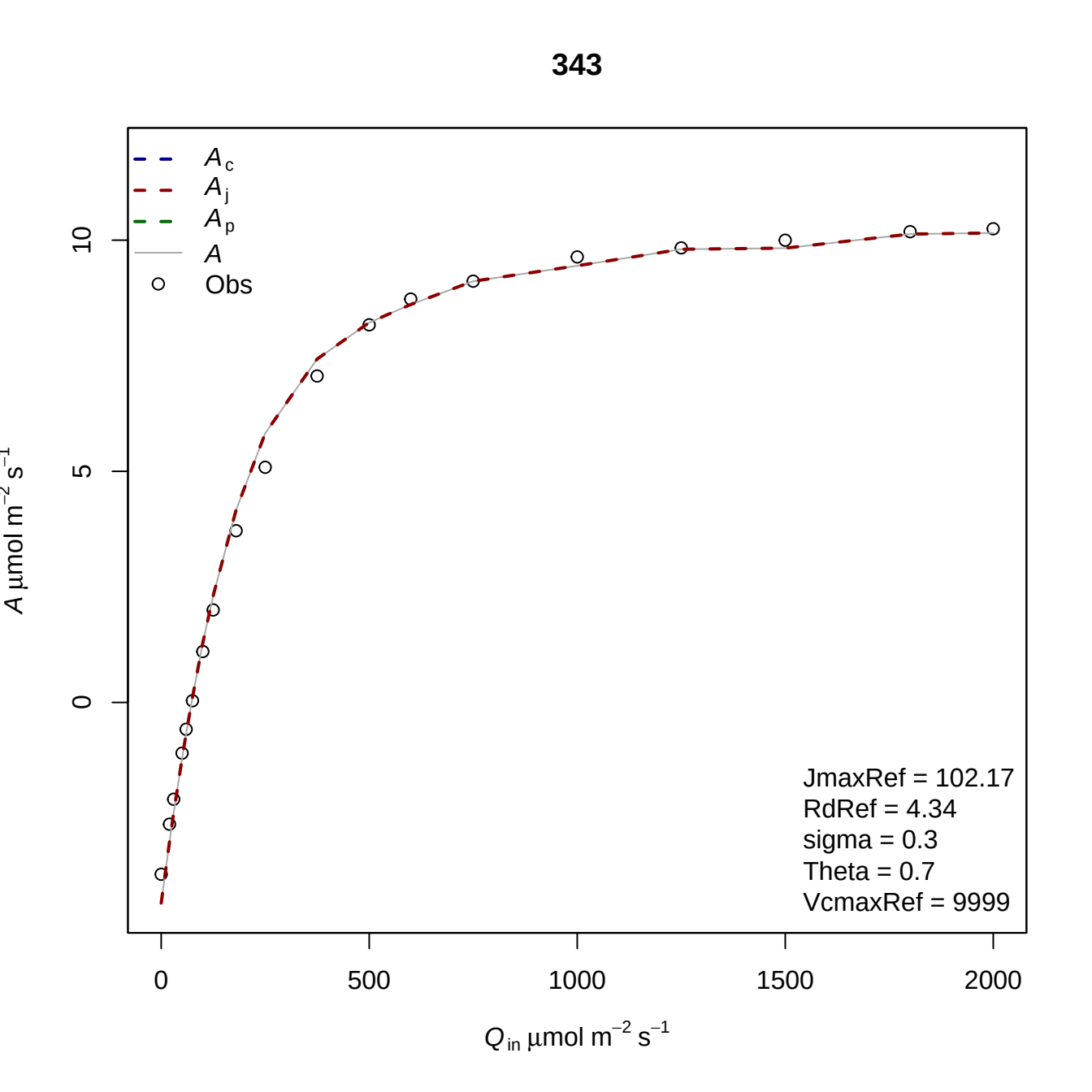

338
A_c
A_j
A_p
A
Obs
JmaxRef = 90.85
RdRef = 2.07
sigma = 0.58
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

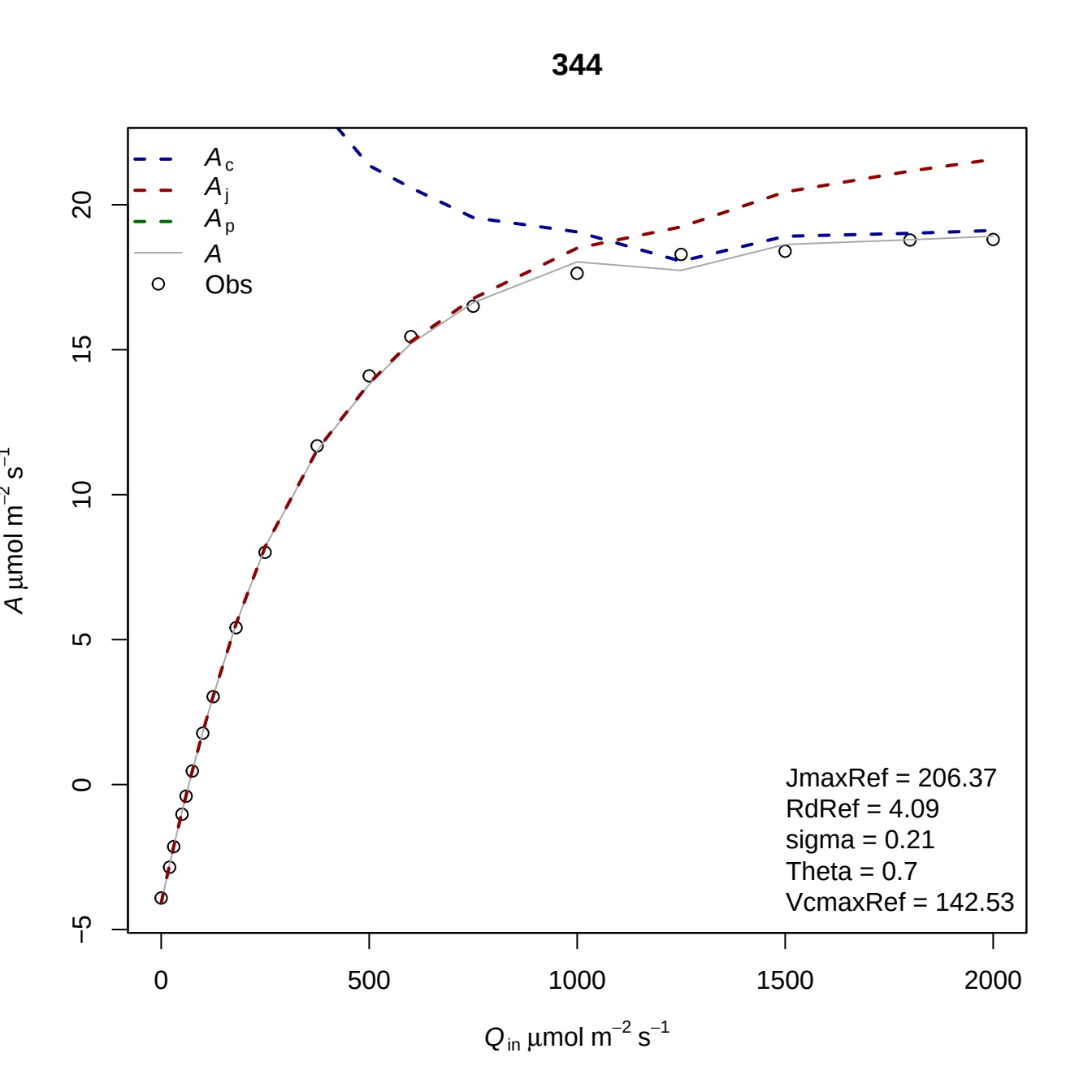

339
A_c
A_j
A_p
A
Obs
JmaxRef = 81.23
RdRef = 1.86
sigma = 0.69
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

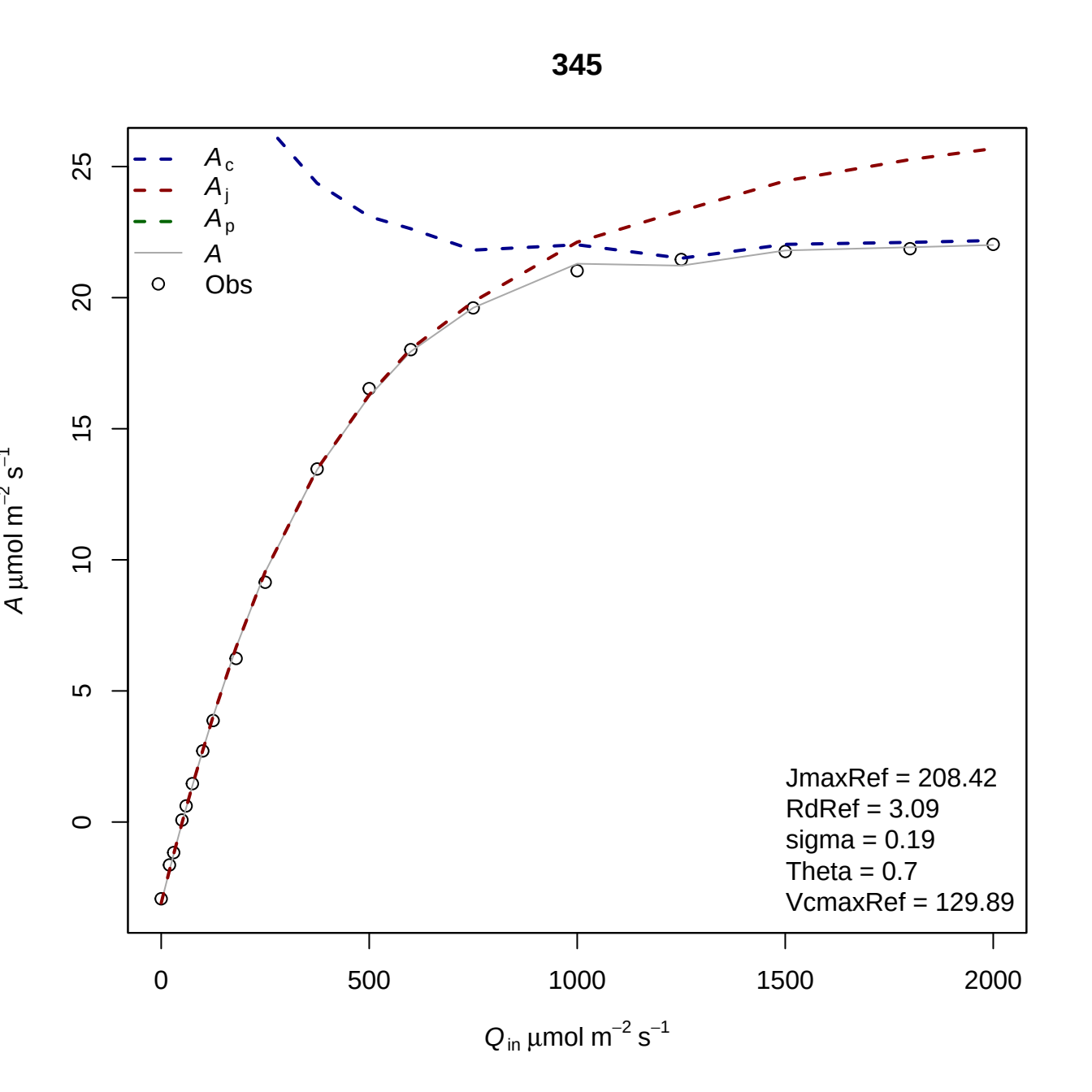

340
A_c
A_j
A_p
A
Obs
JmaxRef = 129.78
RdRef = 3.95
sigma = 0.33
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

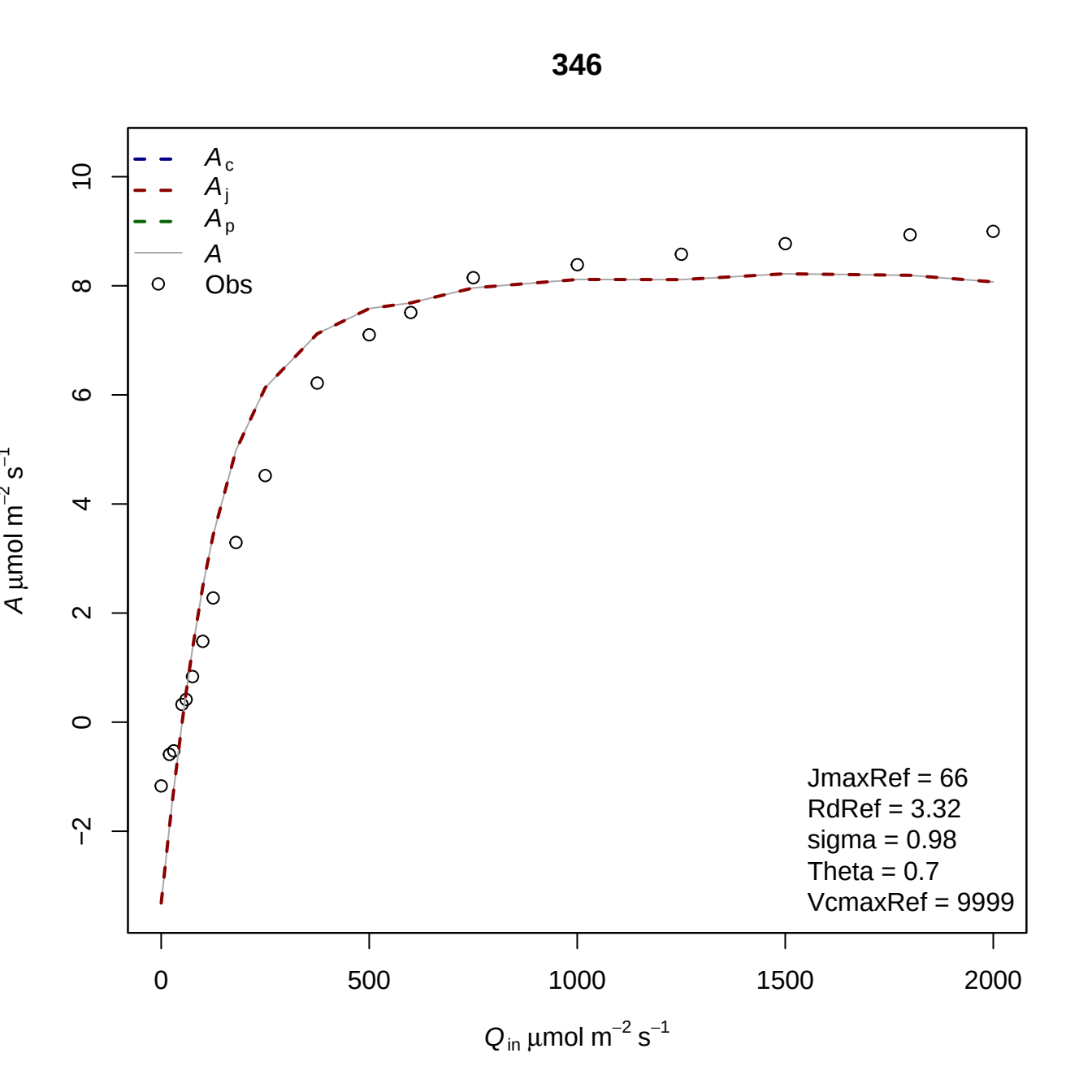

341
A_c
A_j
A_p
A
Obs
JmaxRef = 118.37
RdRef = 3.14
sigma = 1
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

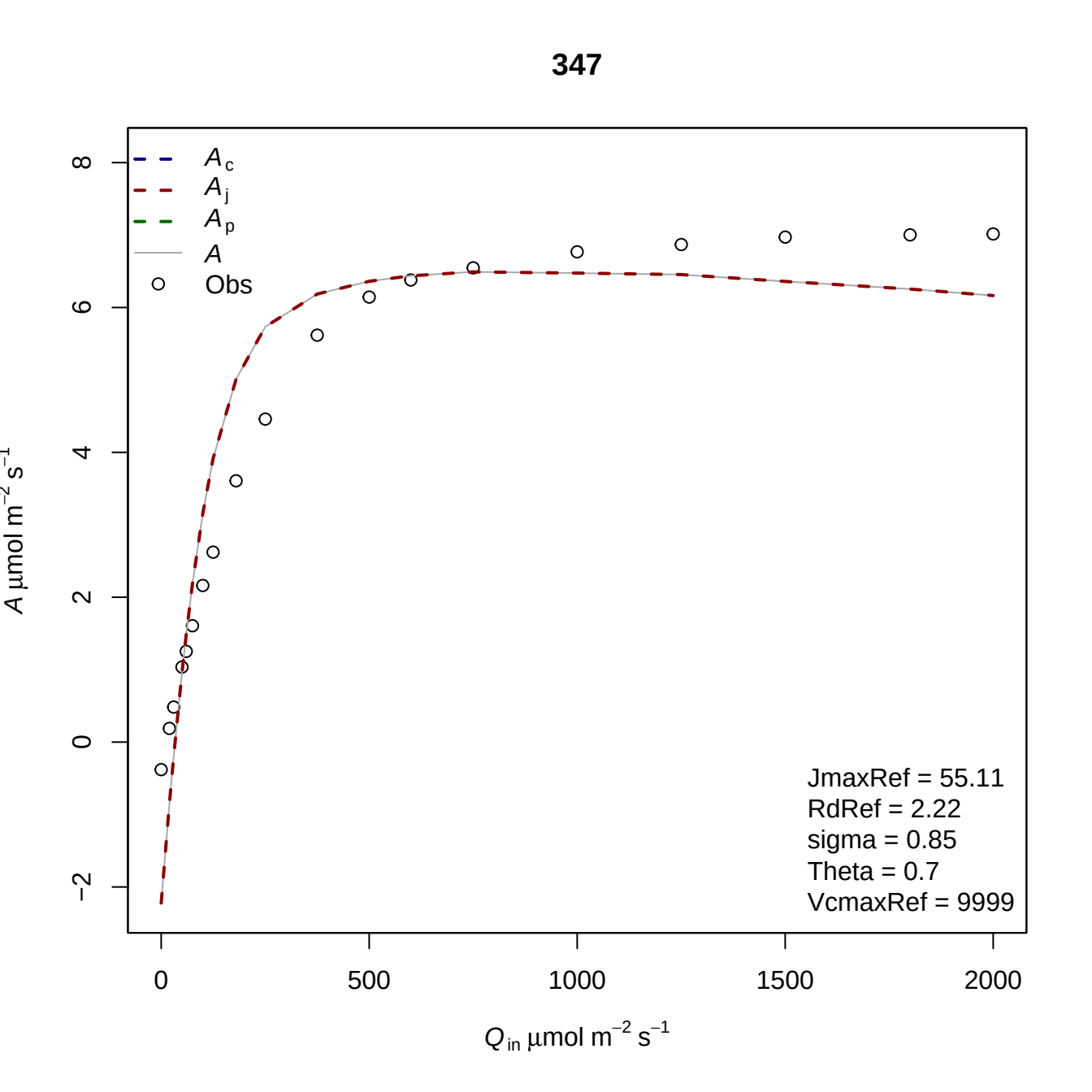

342
A_c
A_j
A_p
A
Obs
JmaxRef = 217.05
RdRef = 2.99
sigma = 0.57
Theta = 0.7
VcmaxRef = 118.21
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

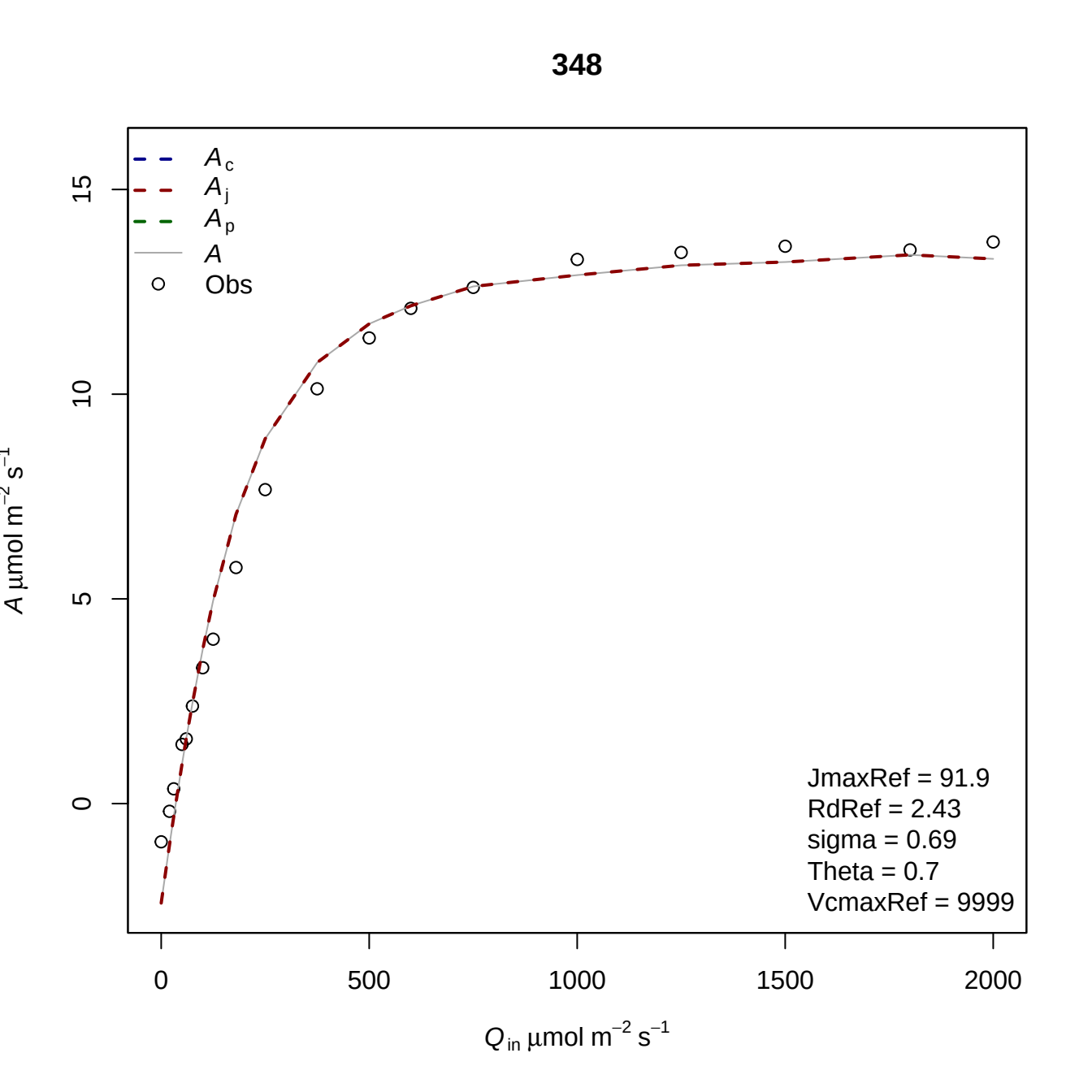

343
A_c
A_j
A_p
A
Obs
JmaxRef = 102.17
RdRef = 4.34
sigma = 0.3
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

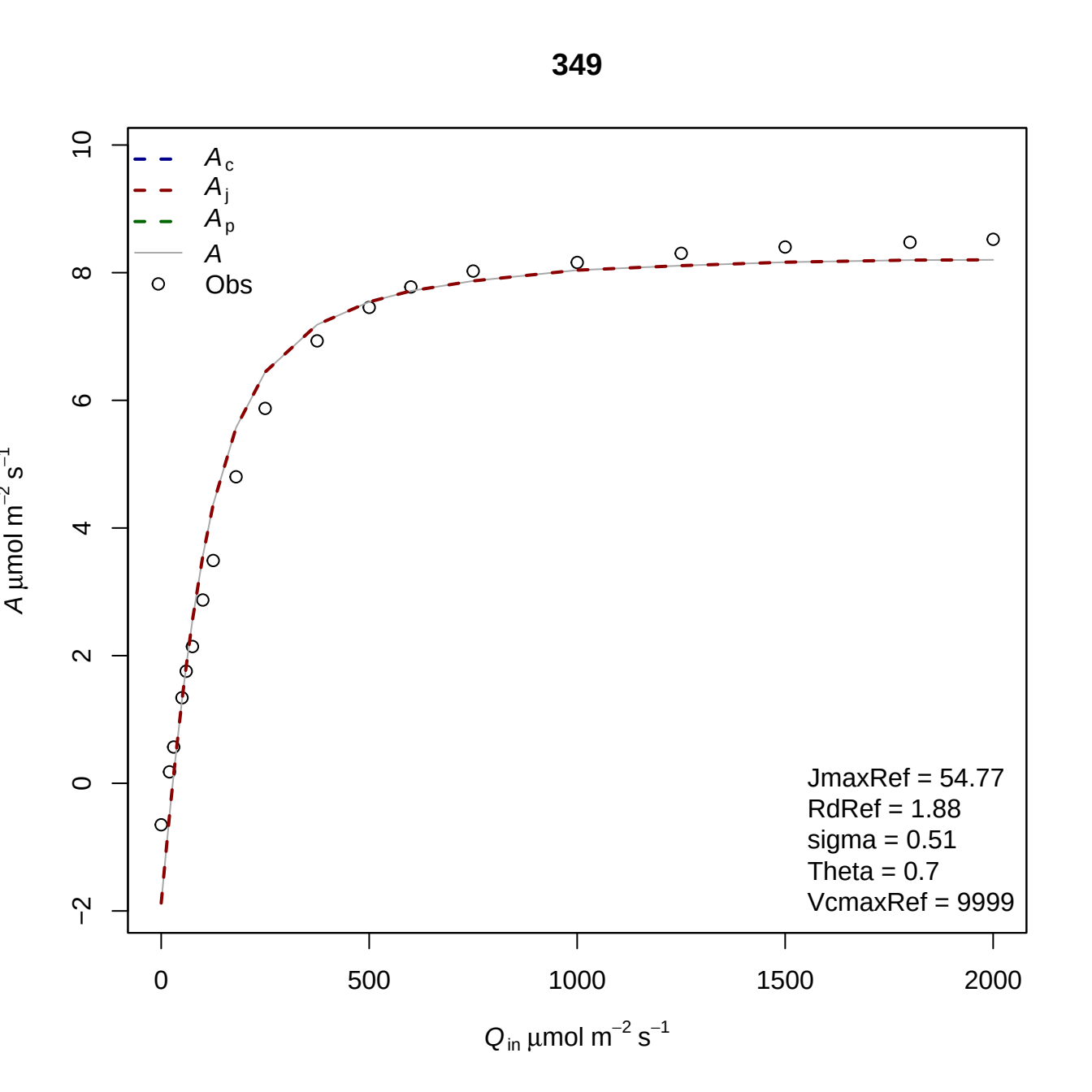

344
A_c
A_j
A_p
A
Obs
JmaxRef = 206.37
RdRef = 4.09
sigma = 0.21
Theta = 0.7
VcmaxRef = 142.53
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

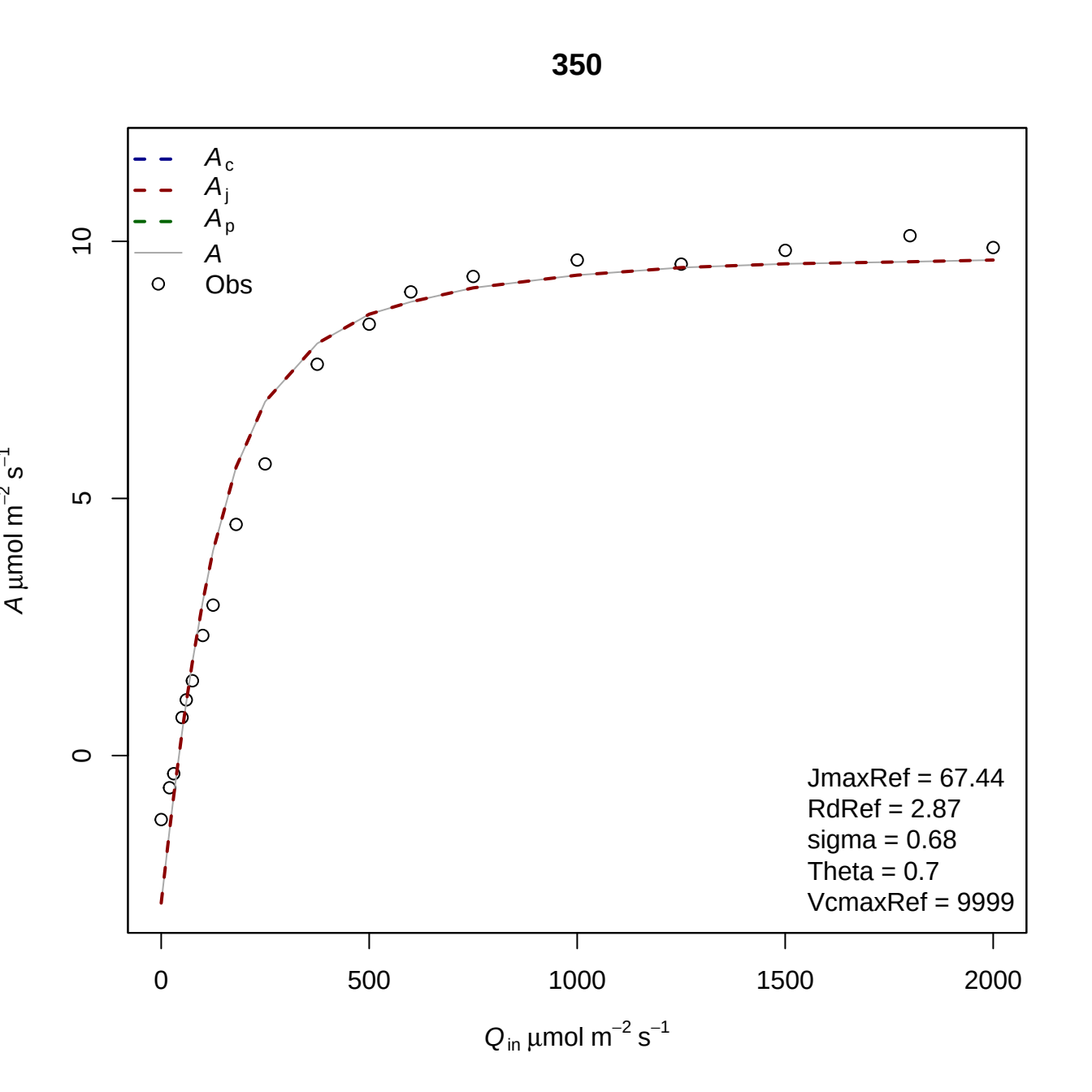

345
A_c
A_j
A_p
A
Obs
JmaxRef = 208.42
RdRef = 3.09
sigma = 0.19
Theta = 0.7
VcmaxRef = 129.89
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

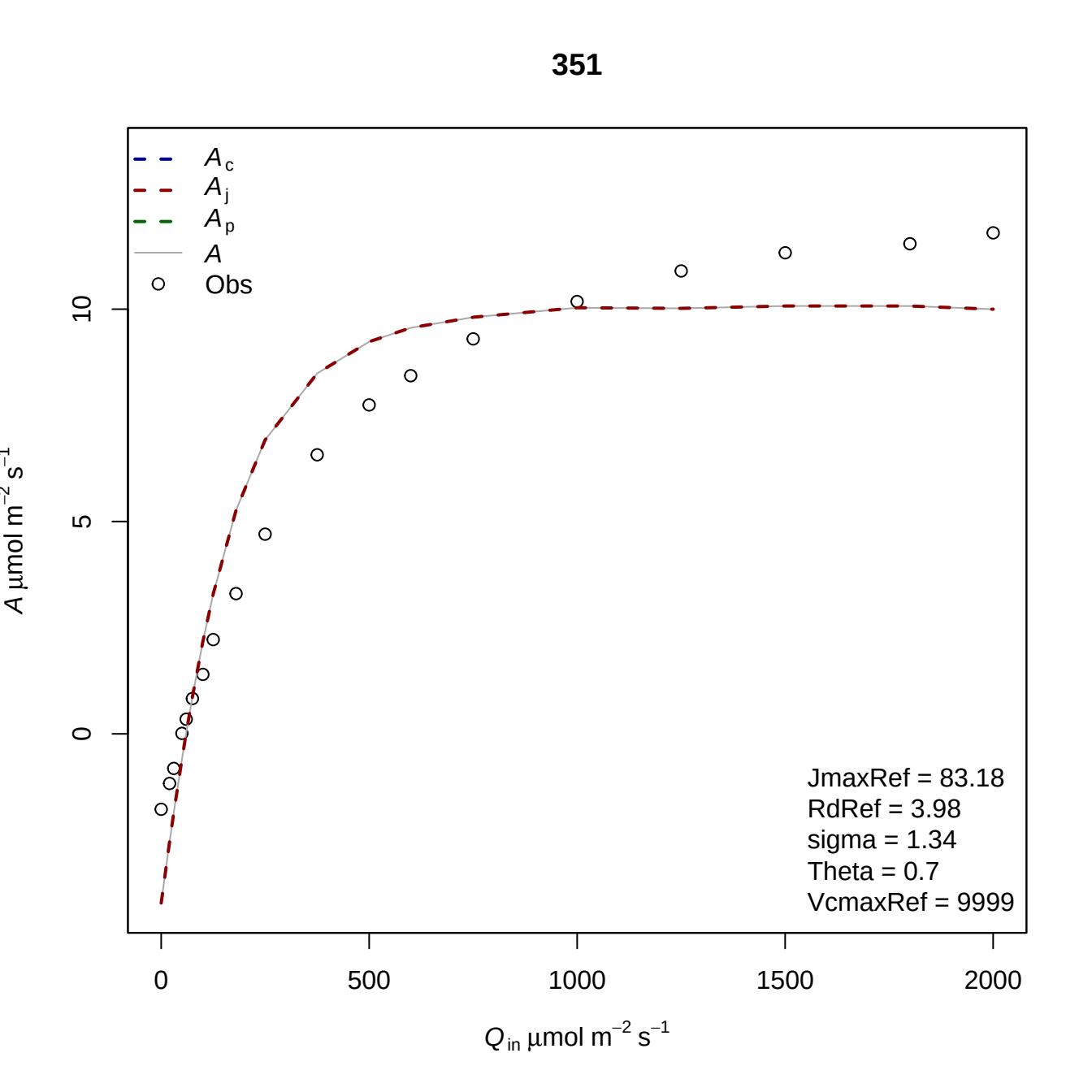

346
A_c
A_j
A_p
A
Obs
JmaxRef = 66
RdRef = 3.32
sigma = 0.98
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

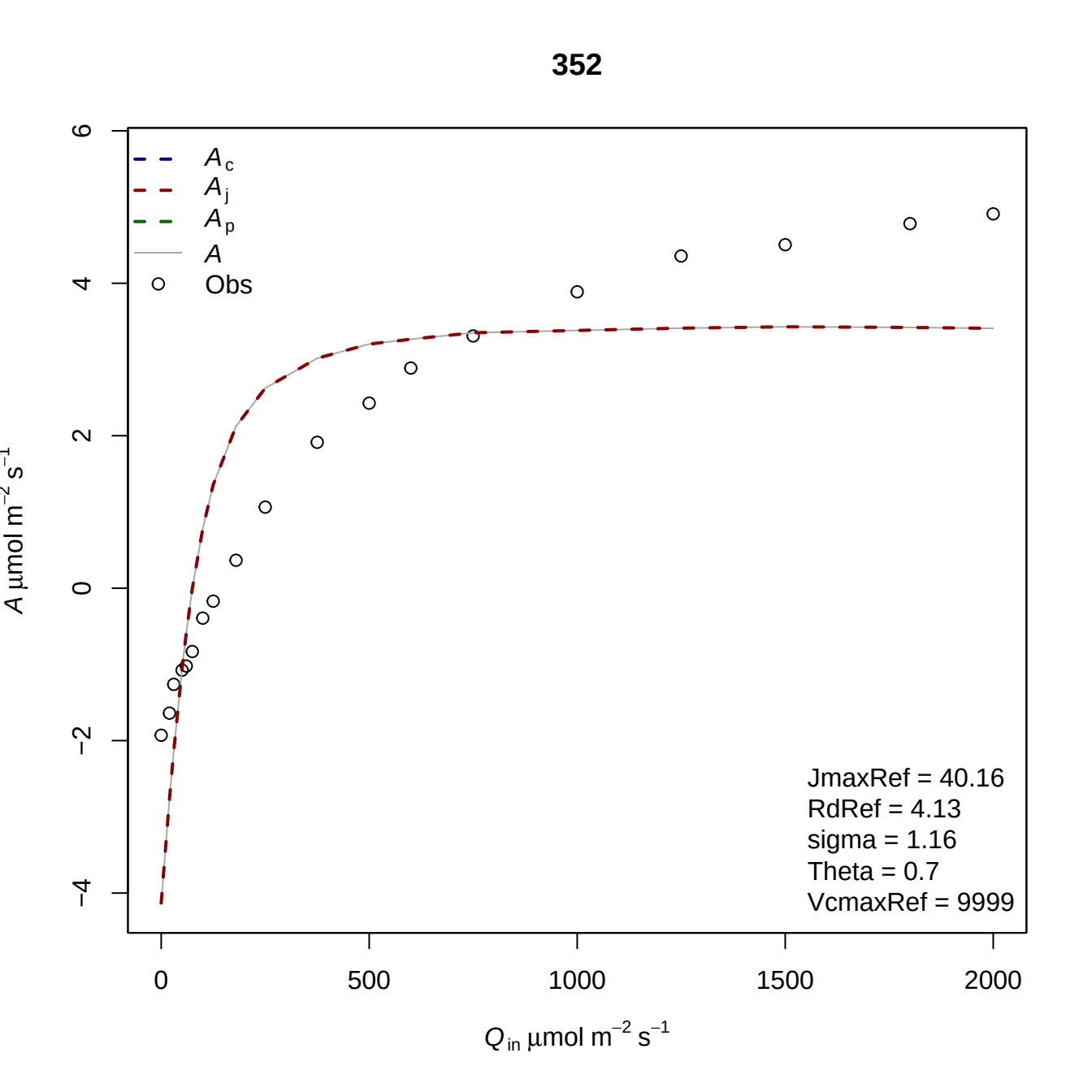

347
A_c
A_j
A_p
A
Obs
JmaxRef = 55.11
RdRef = 2.22
sigma = 0.85
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

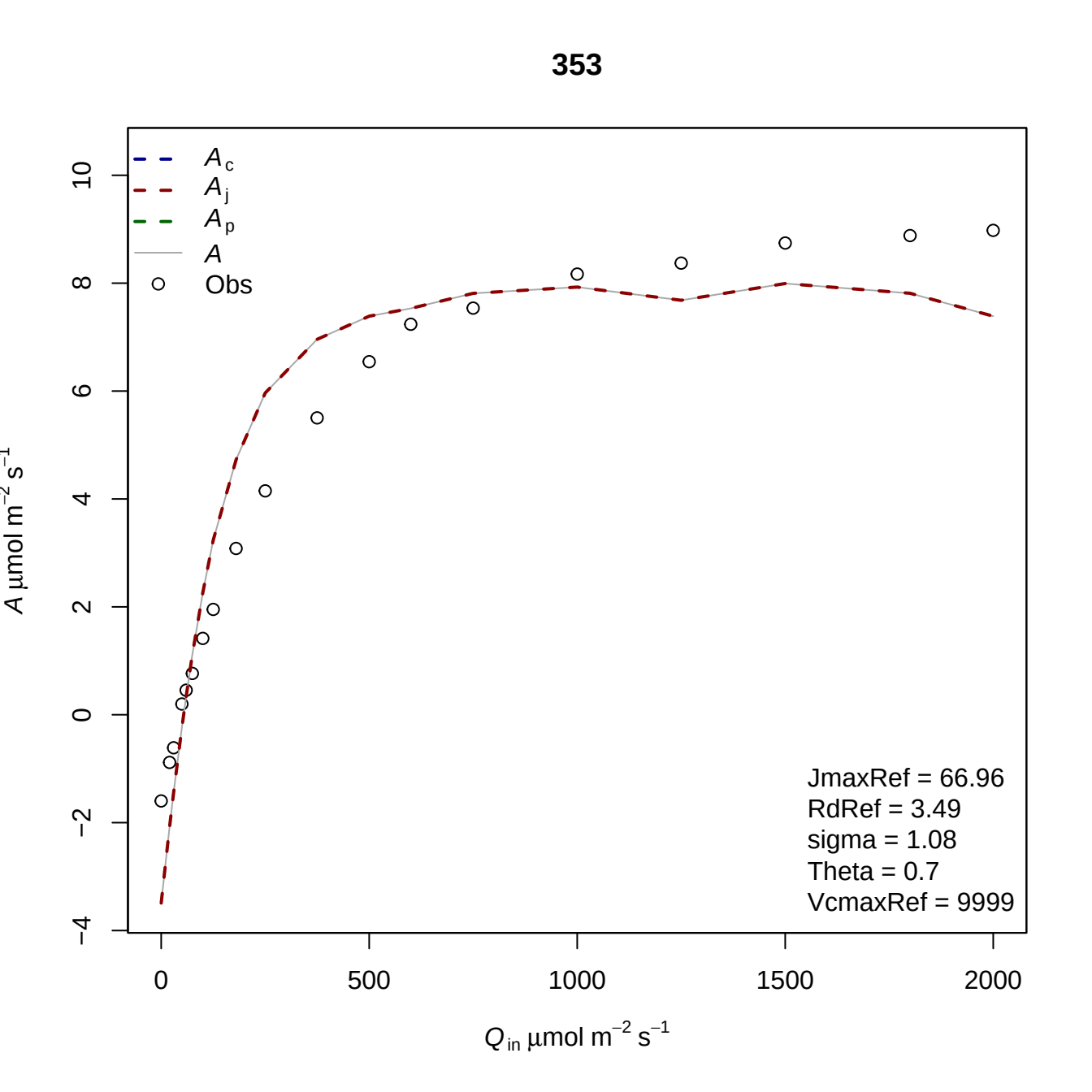

348
A_c
A_j
A_p
A
Obs
JmaxRef = 91.9
RdRef = 2.43
sigma = 0.69
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

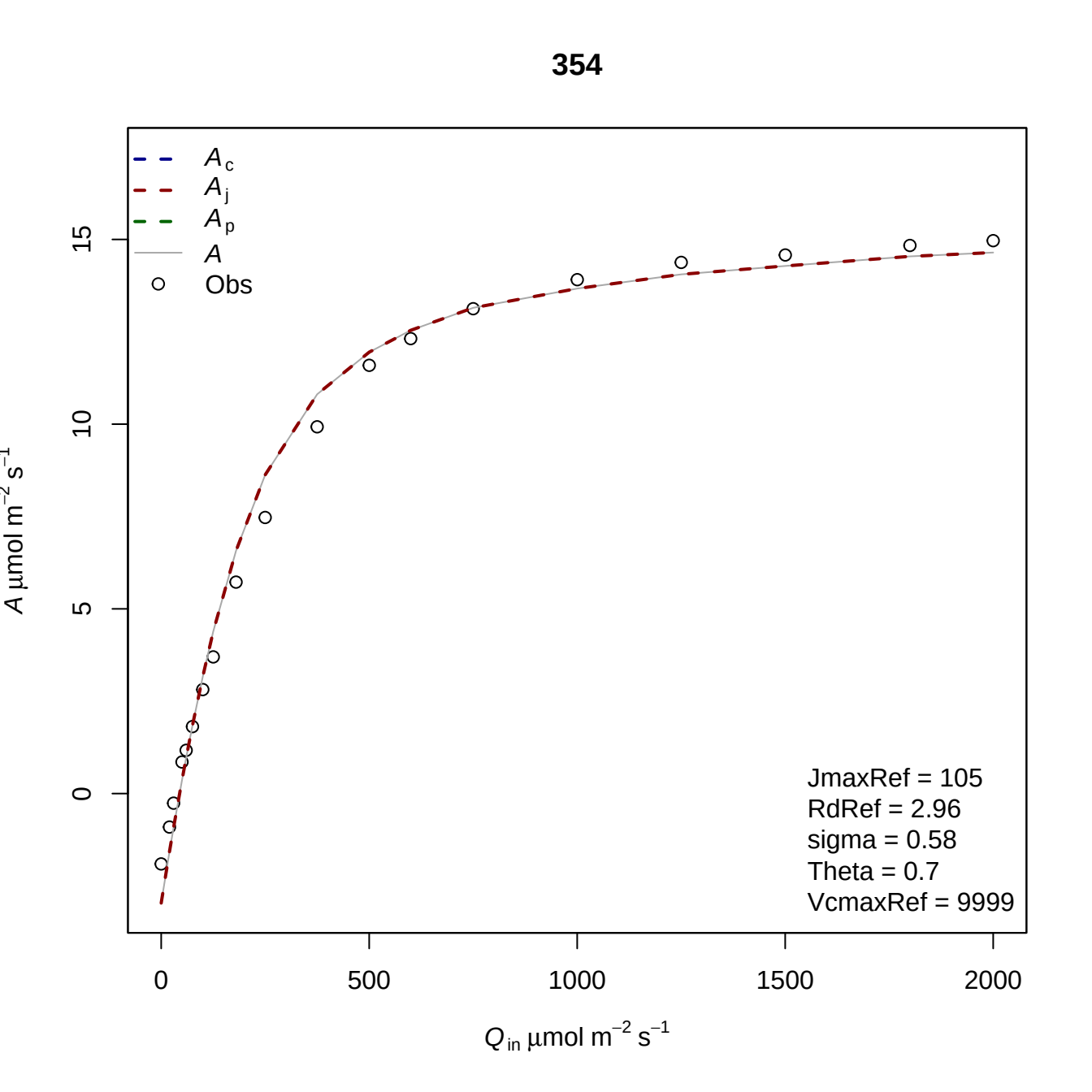

349
A_c
A_j
A_p
A
Obs
JmaxRef = 54.77
RdRef = 1.88
sigma = 0.51
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

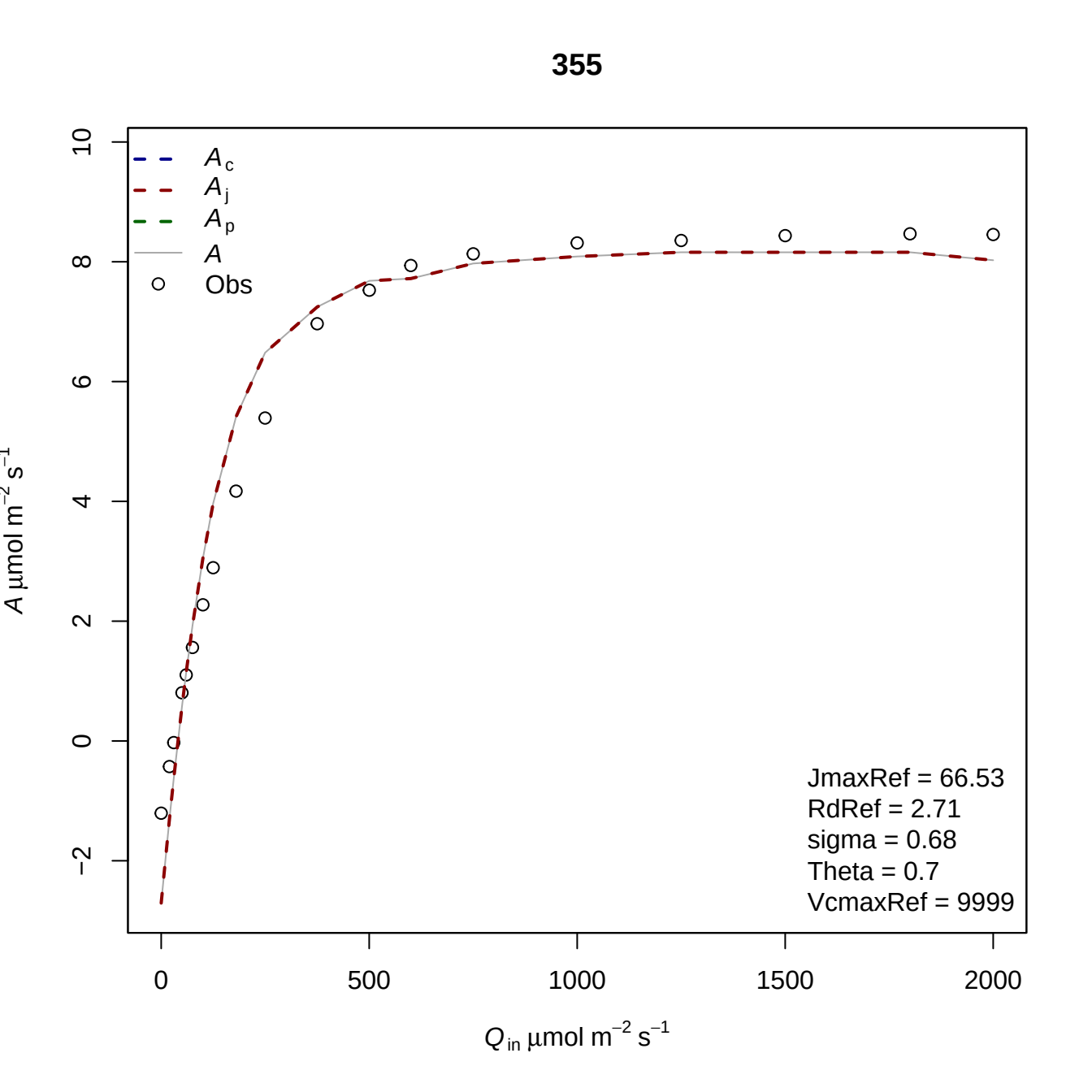

350
A_c
A_j
A_p
A
Obs
JmaxRef = 67.44
RdRef = 2.87
sigma = 0.68
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

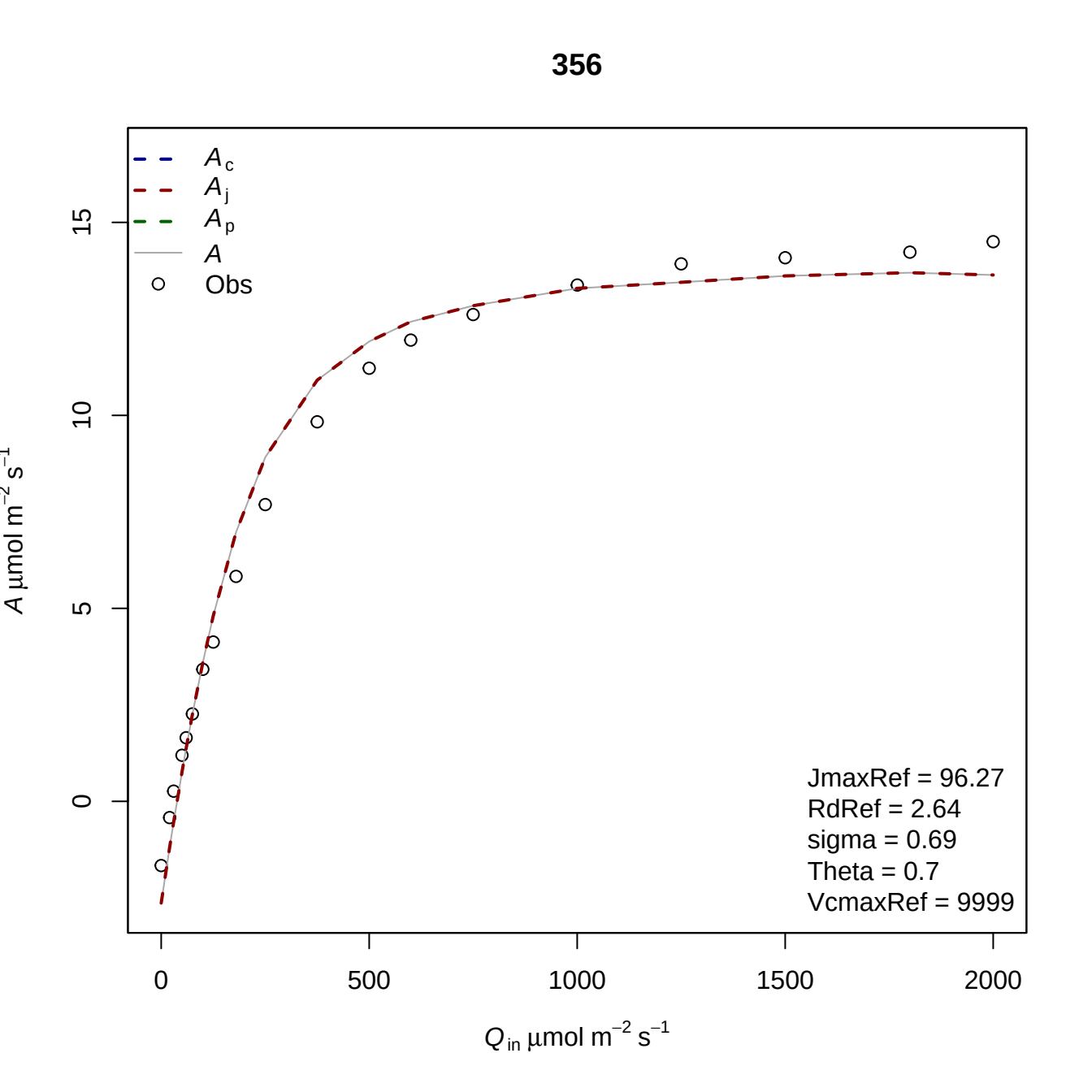

351
A_c
A_j
A_p
A
Obs
JmaxRef = 83.18
RdRef = 3.98
sigma = 1.34
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

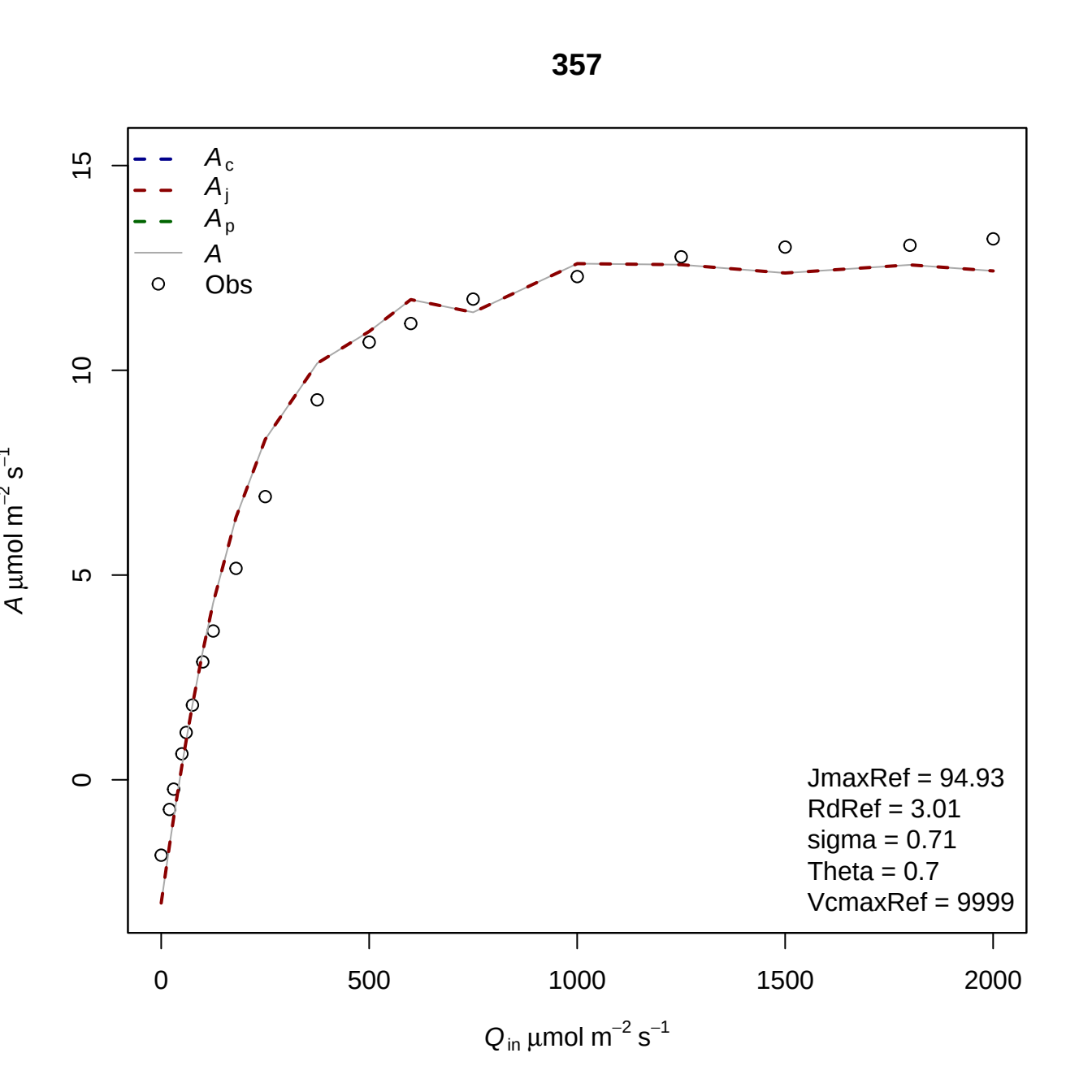

352
A_c
A_j
A_p
A
Obs
JmaxRef = 40.16
RdRef = 4.13
sigma = 1.16
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500
2000

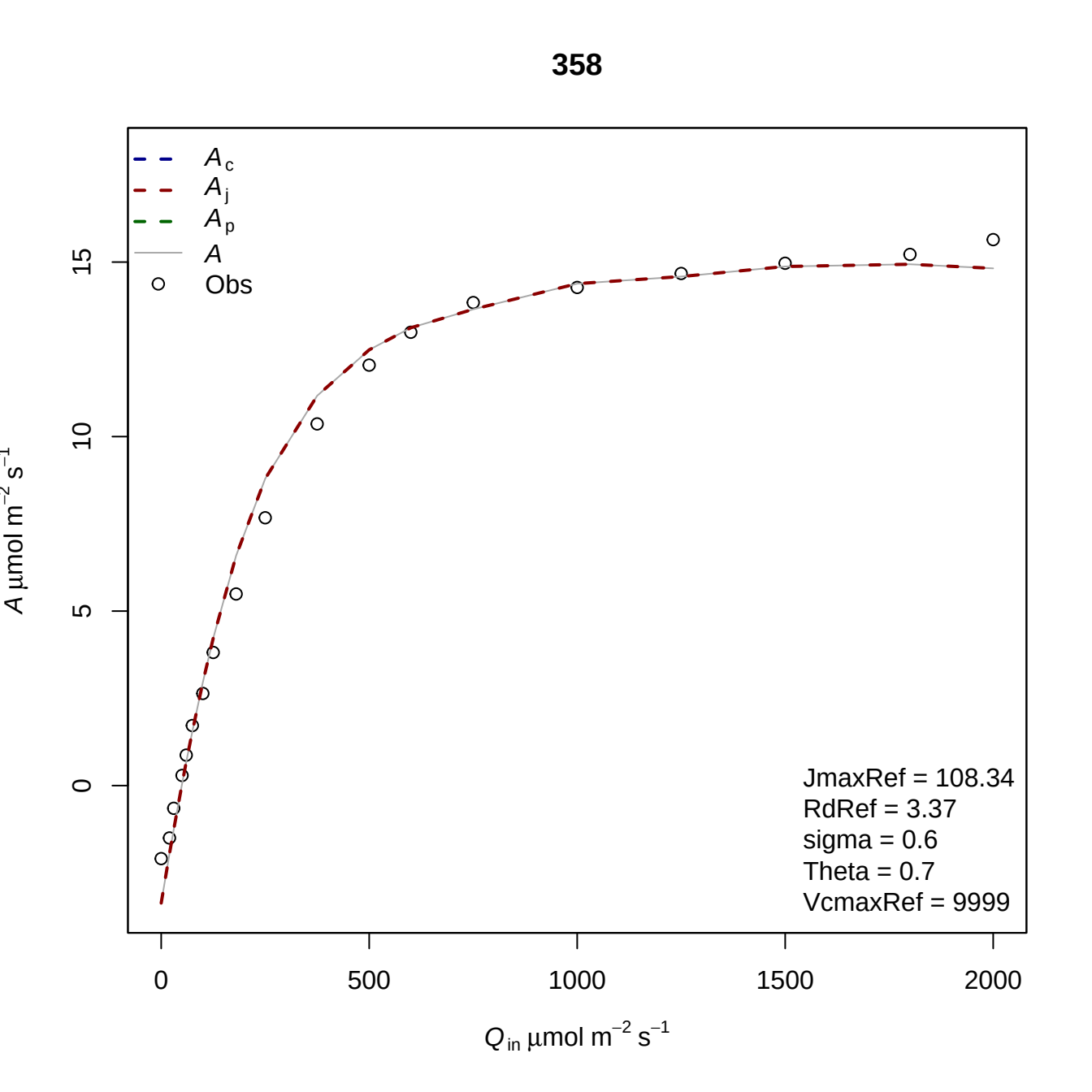

353
A_c
A_j
A_p
A
Obs
JmaxRef = 66.96
RdRef = 3.49
sigma = 1.08
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

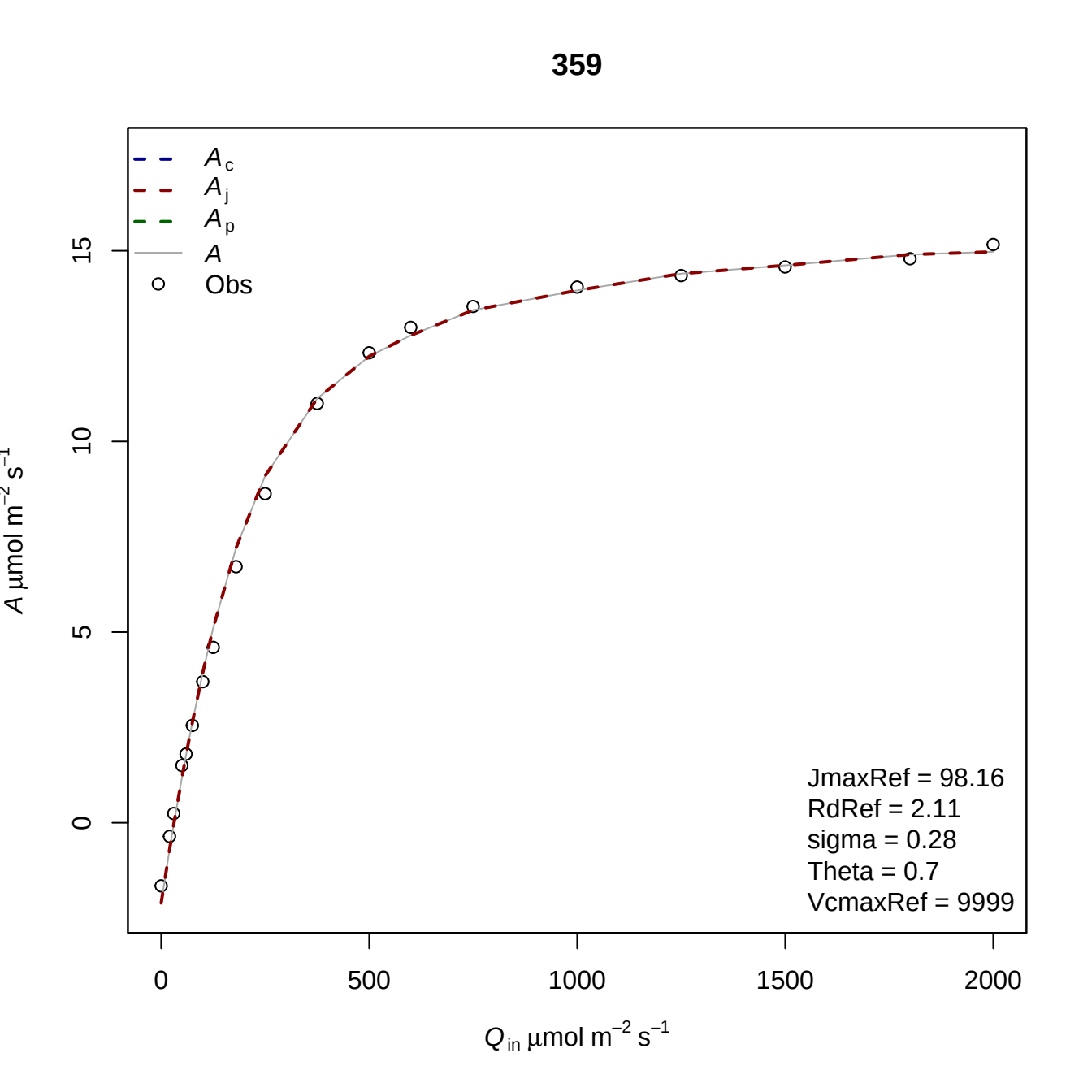

354
A_c
A_j
A_p
A
Obs
JmaxRef = 105
RdRef = 2.96
sigma = 0.58
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

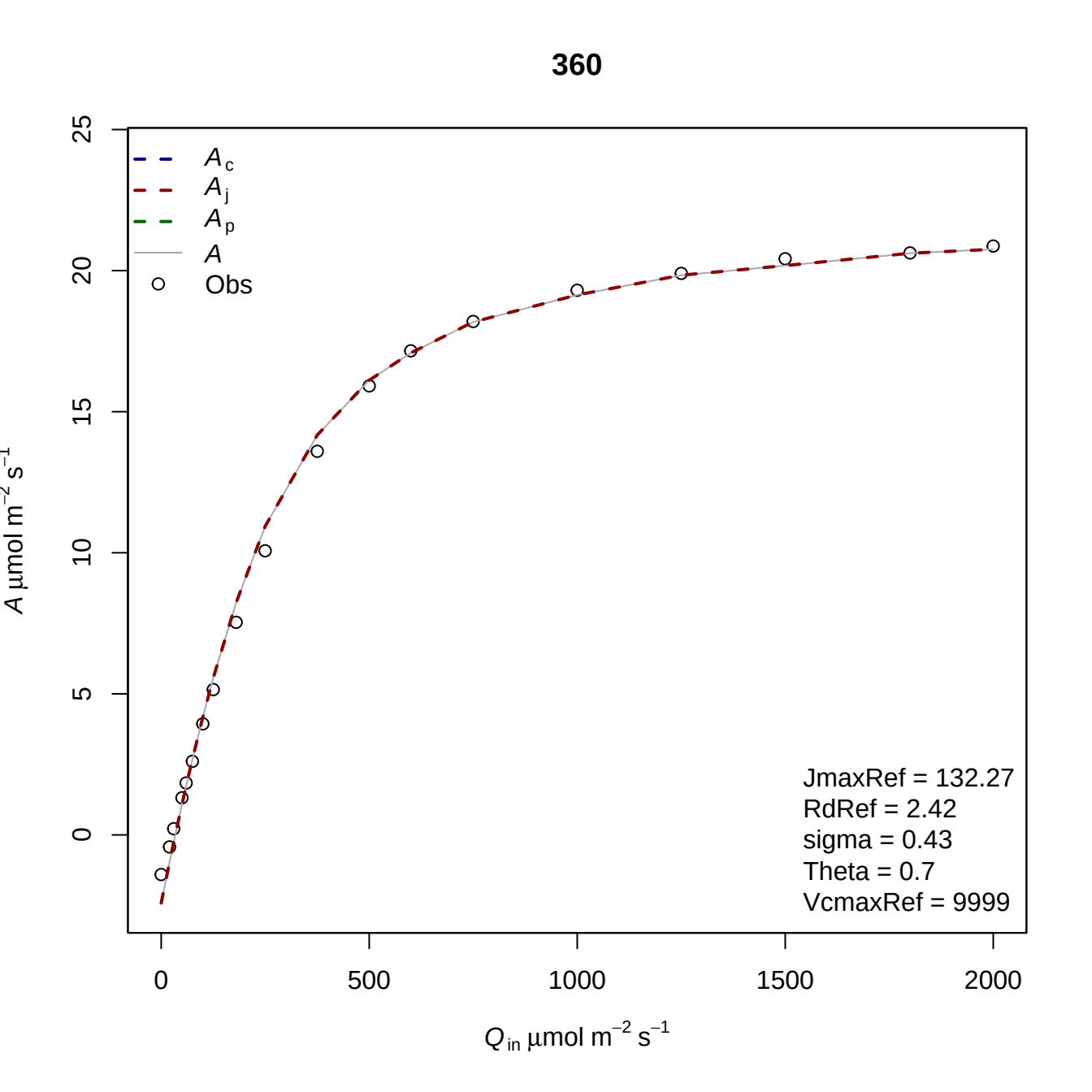

355
A_c
A_j
A_p
A
Obs
JmaxRef = 66.53
RdRef = 2.71
sigma = 0.68
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

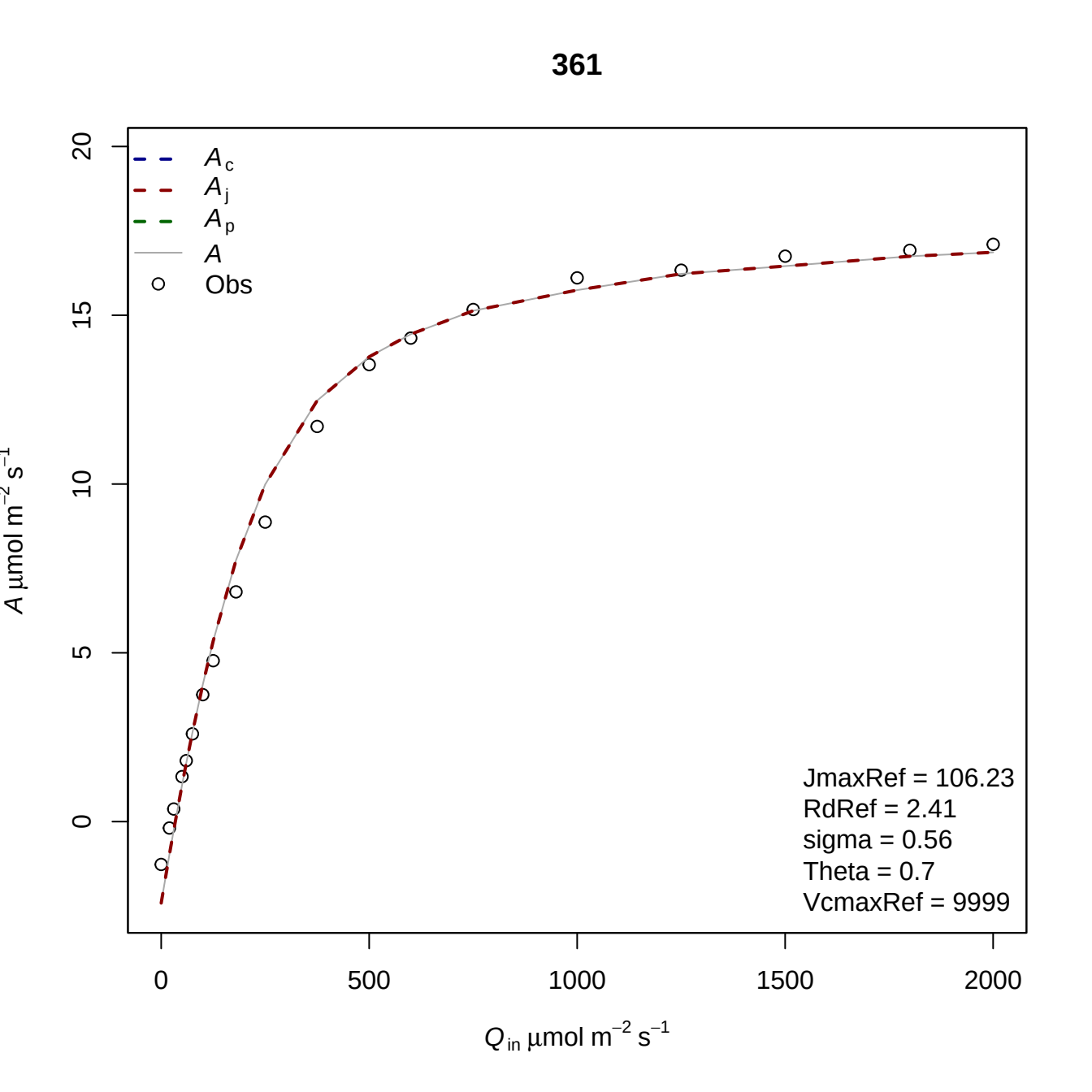

356
A_c
A_j
A_p
A
Obs
JmaxRef = 96.27
RdRef = 2.64
sigma = 0.69
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

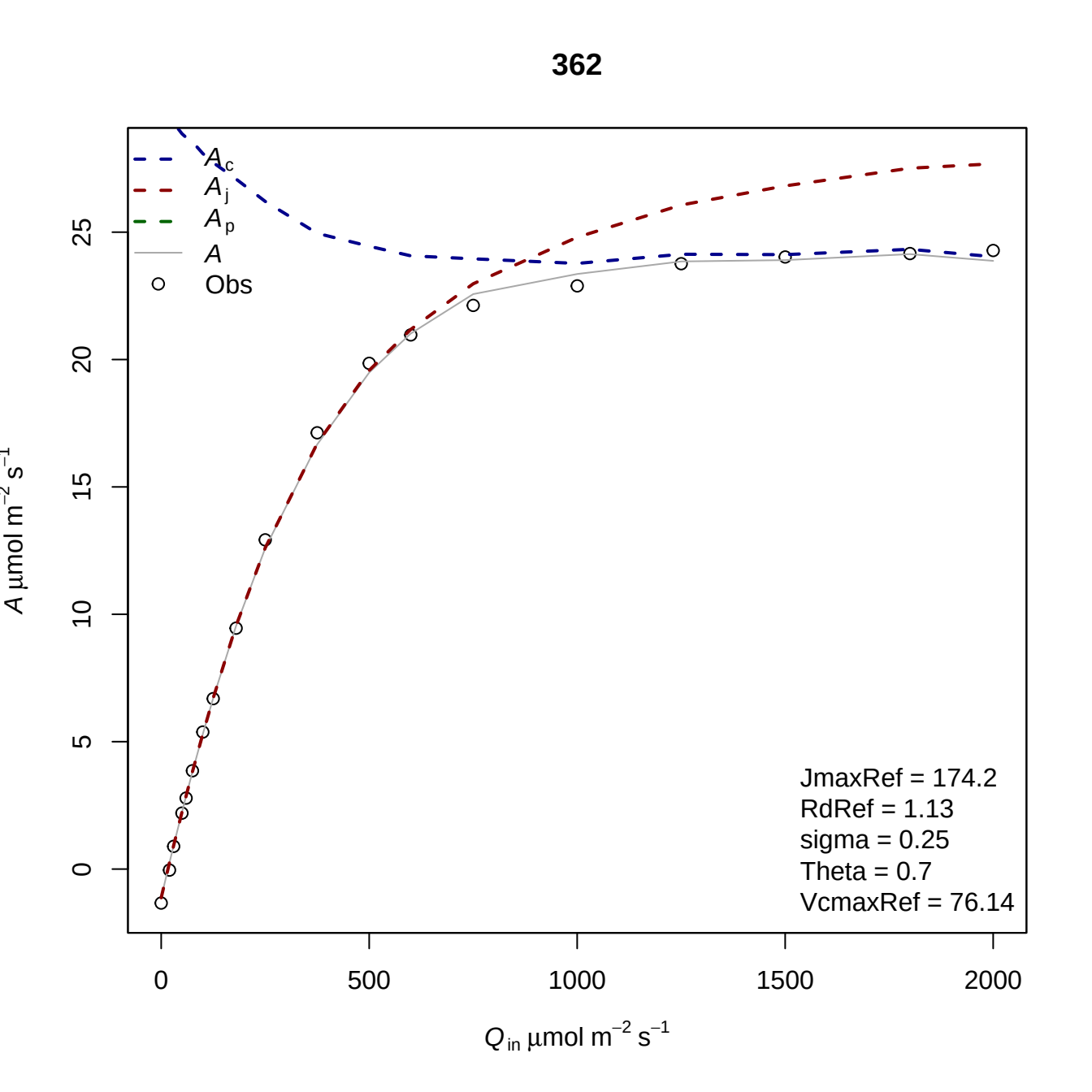

357
A_c
A_j
A_p
A
Obs
JmaxRef = 94.93
RdRef = 3.01
sigma = 0.71
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

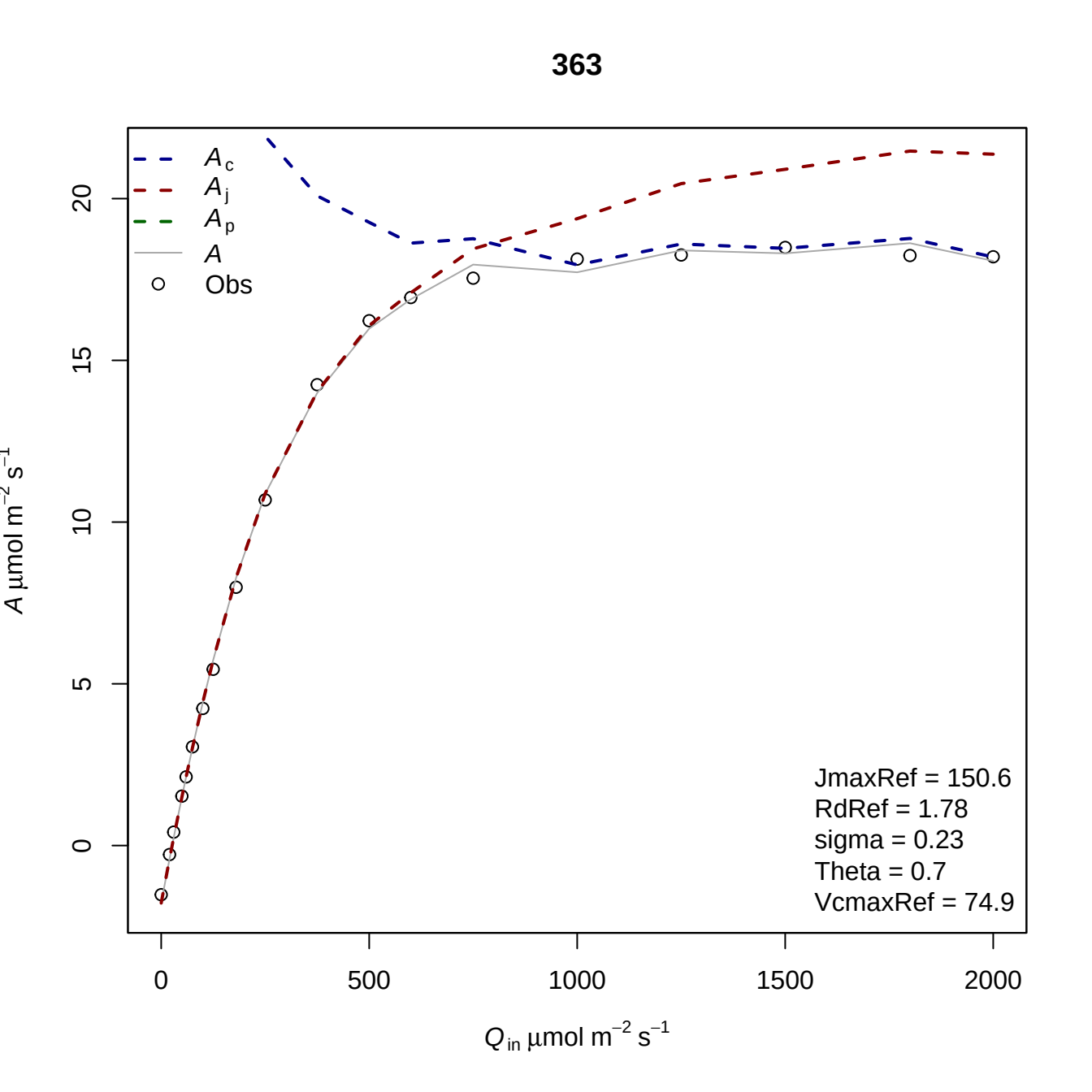

358
A_c
A_j
A_p
A
Obs
JmaxRef = 108.34
RdRef = 3.37
sigma = 0.6
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

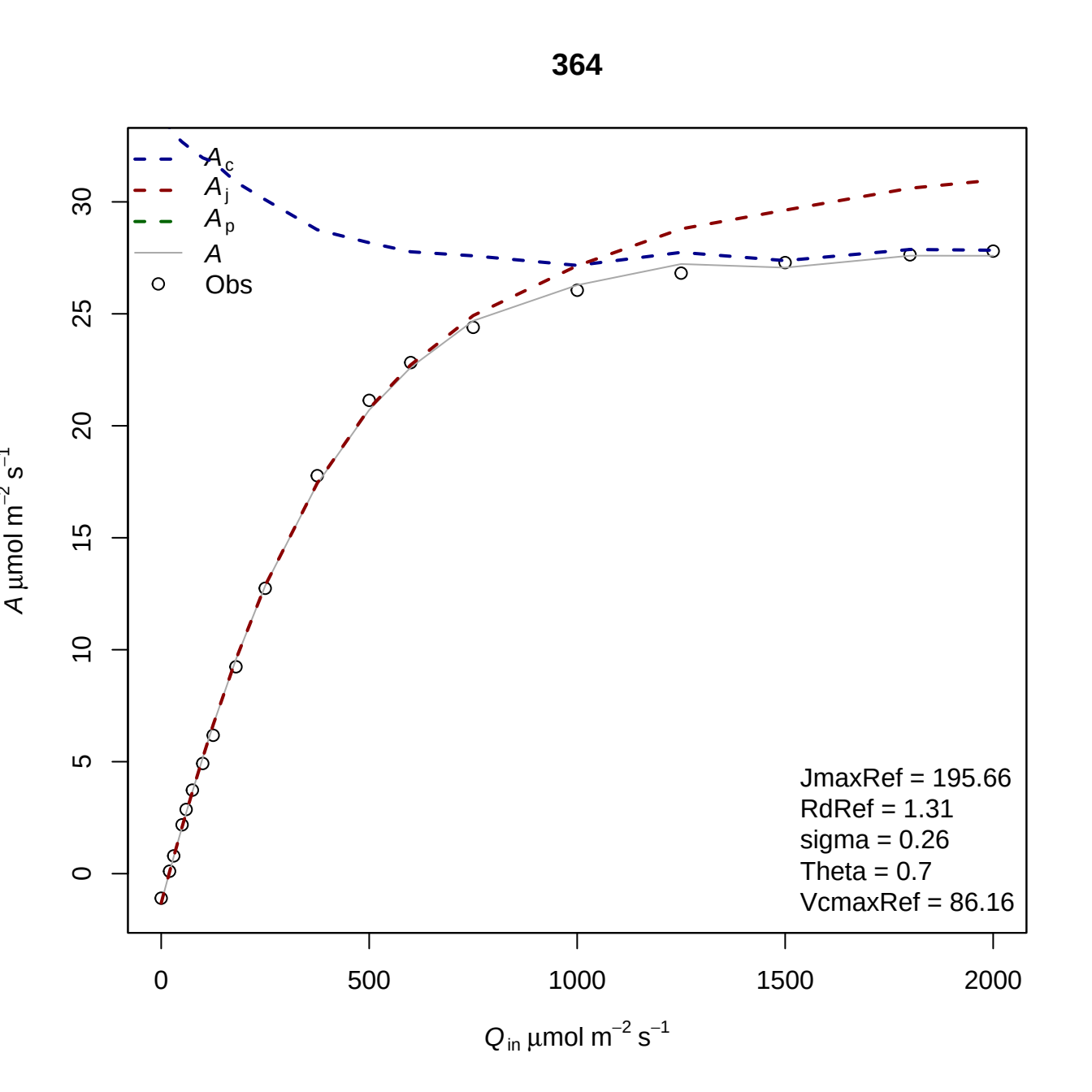

359
A_c
A_j
A_p
A
Obs
JmaxRef = 98.16
RdRef = 2.11
sigma = 0.28
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

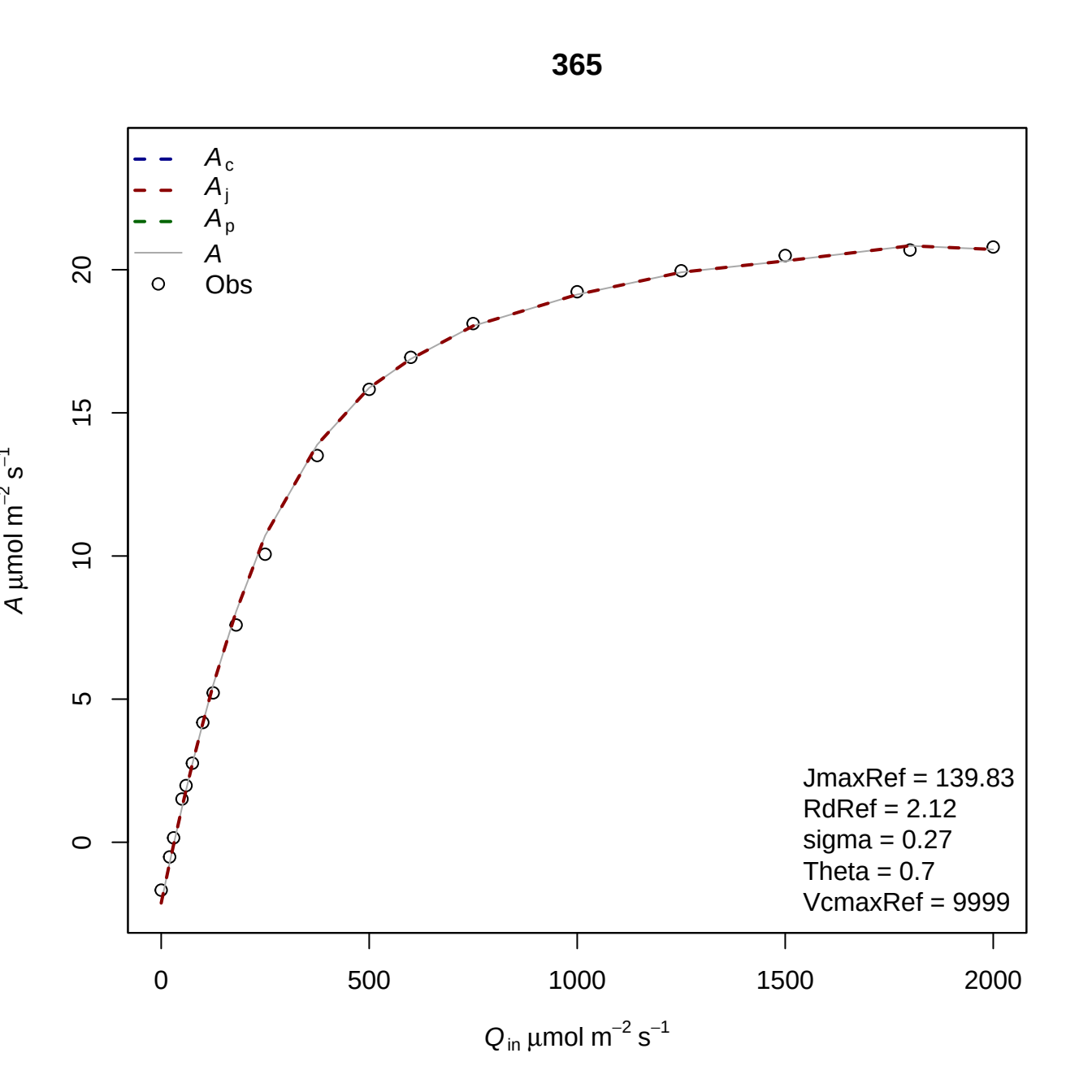

360
A_c
A_j
A_p
A
Obs
JmaxRef = 132.27
RdRef = 2.42
sigma = 0.43
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

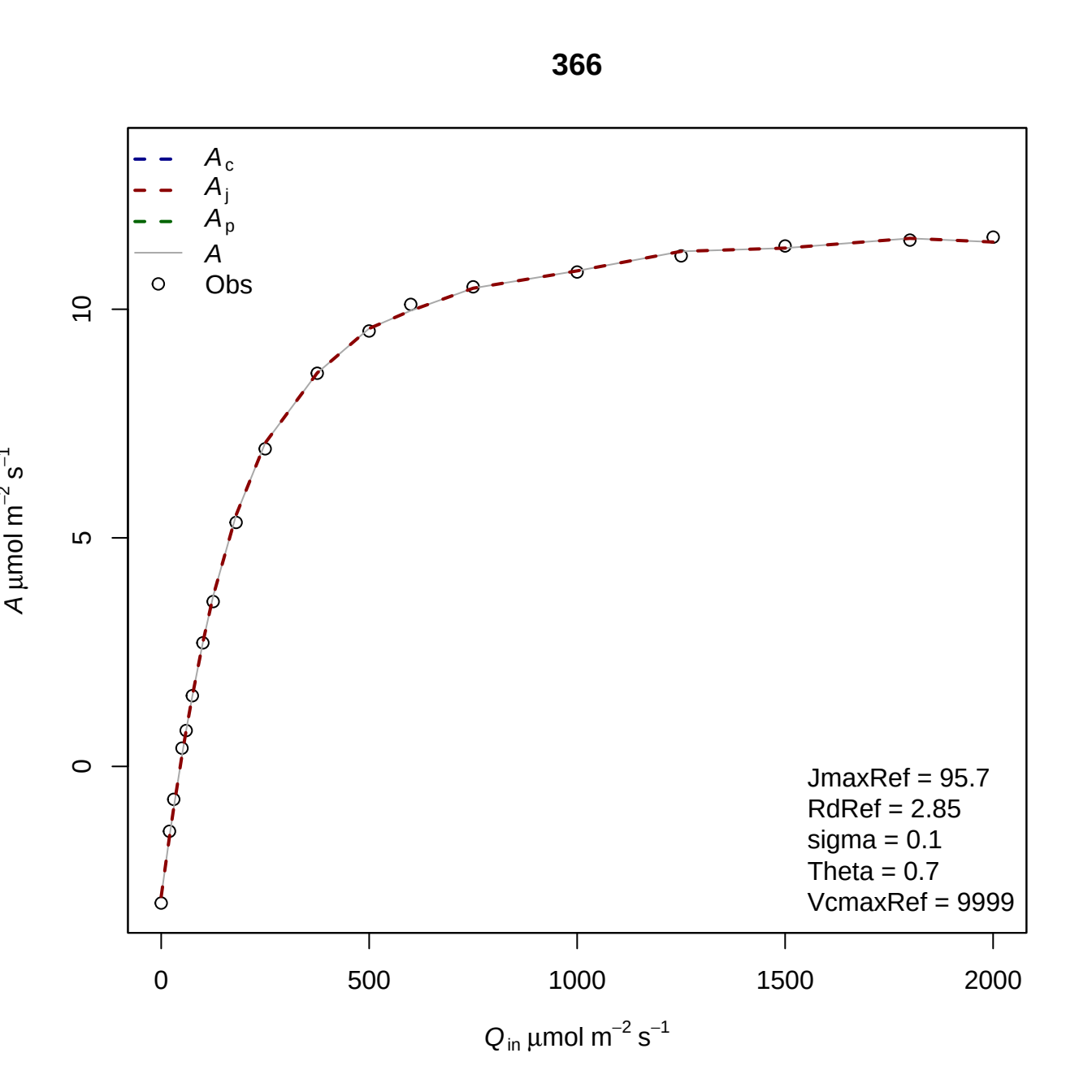

361
A_c
A_j
A_p
A
Obs
JmaxRef = 106.23
RdRef = 2.41
sigma = 0.56
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

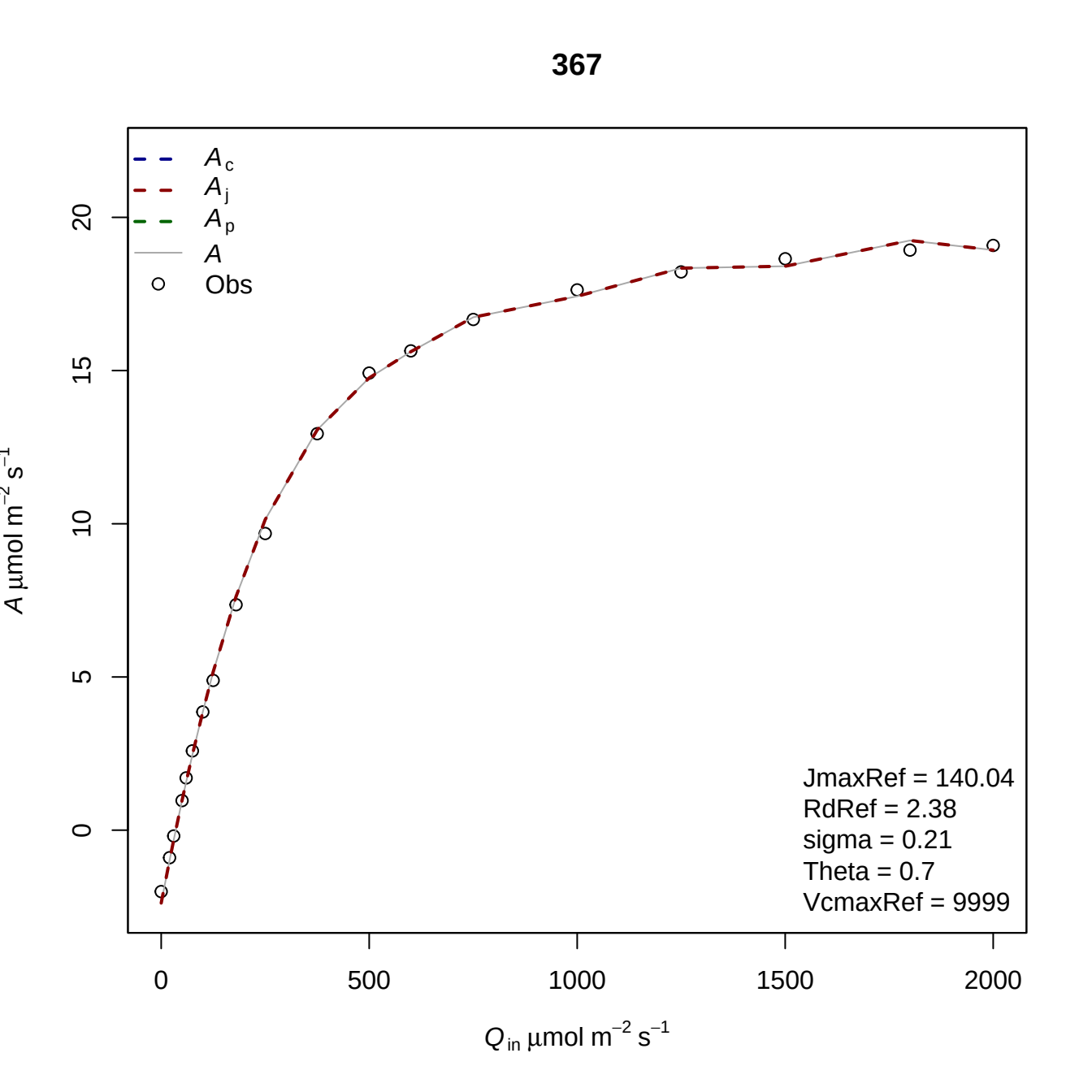

362
A_c
A_j
A_p
A
Obs
JmaxRef = 174.2
RdRef = 1.13
sigma = 0.25
Theta = 0.7
VcmaxRef = 76.14
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

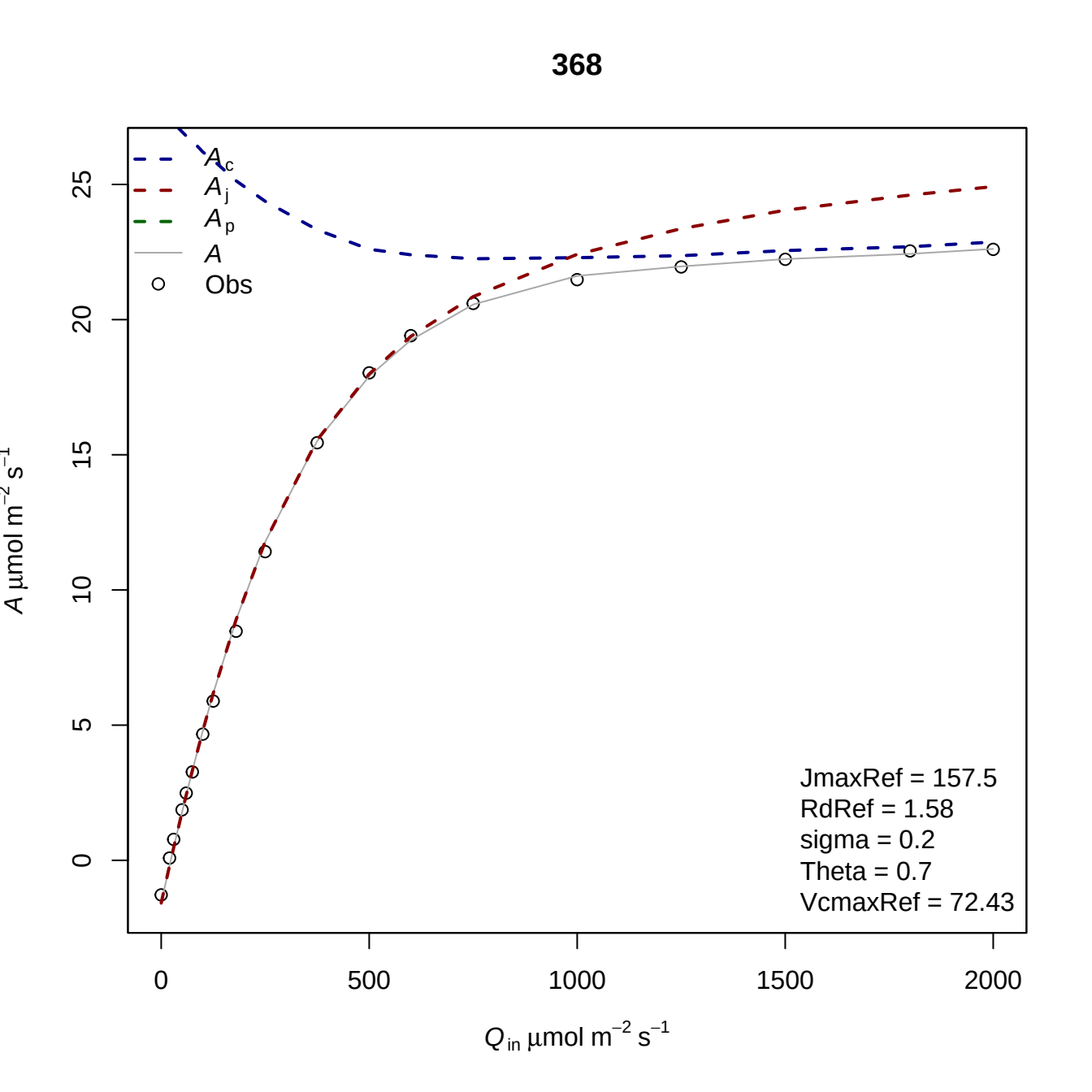

363
A_c
A_j
A_p
A
Obs
JmaxRef = 150.6
RdRef = 1.78
sigma = 0.23
Theta = 0.7
VcmaxRef = 74.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

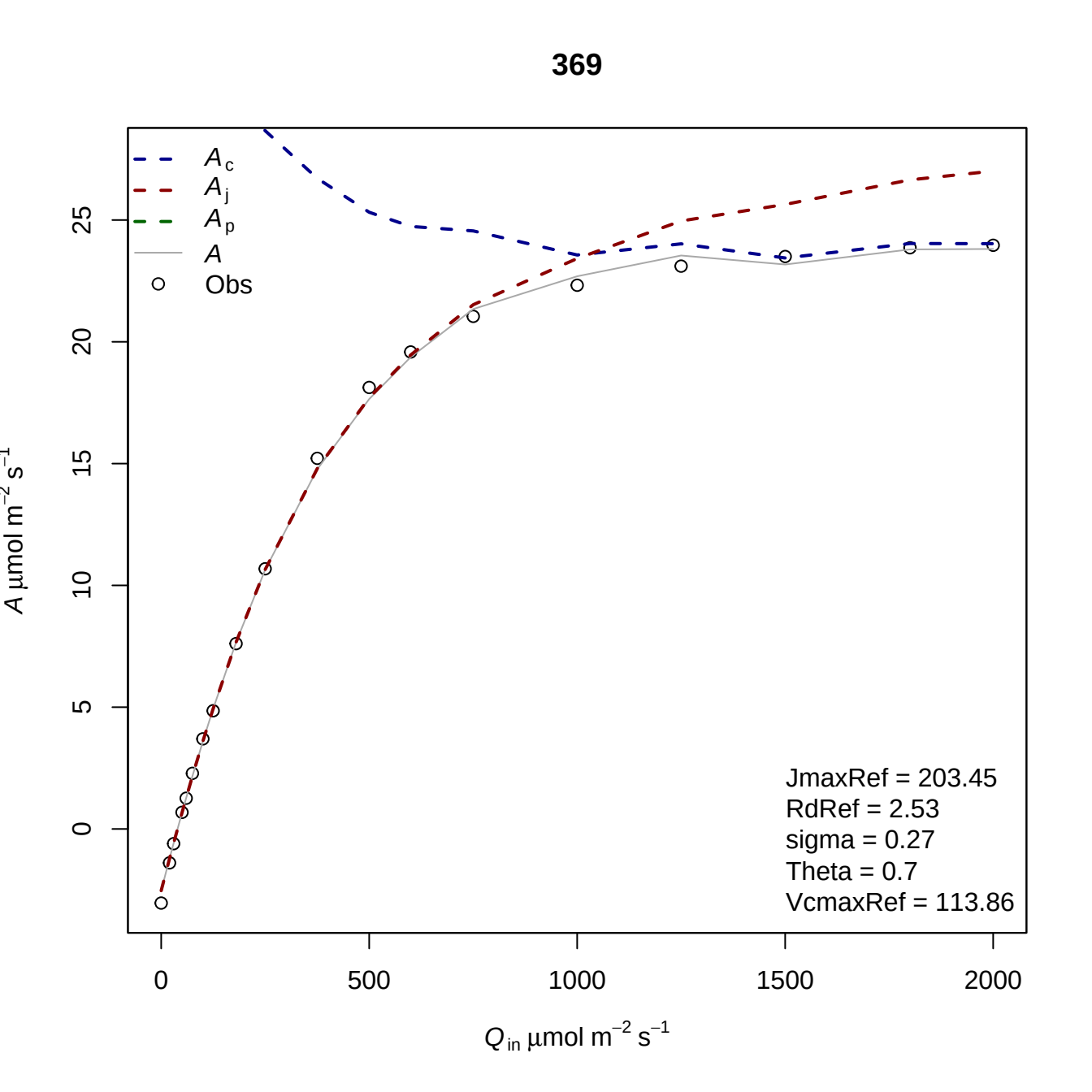

364
A_c
A_j
A_p
A
Obs
JmaxRef = 195.66
RdRef = 1.31
sigma = 0.26
Theta = 0.7
VcmaxRef = 86.16
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

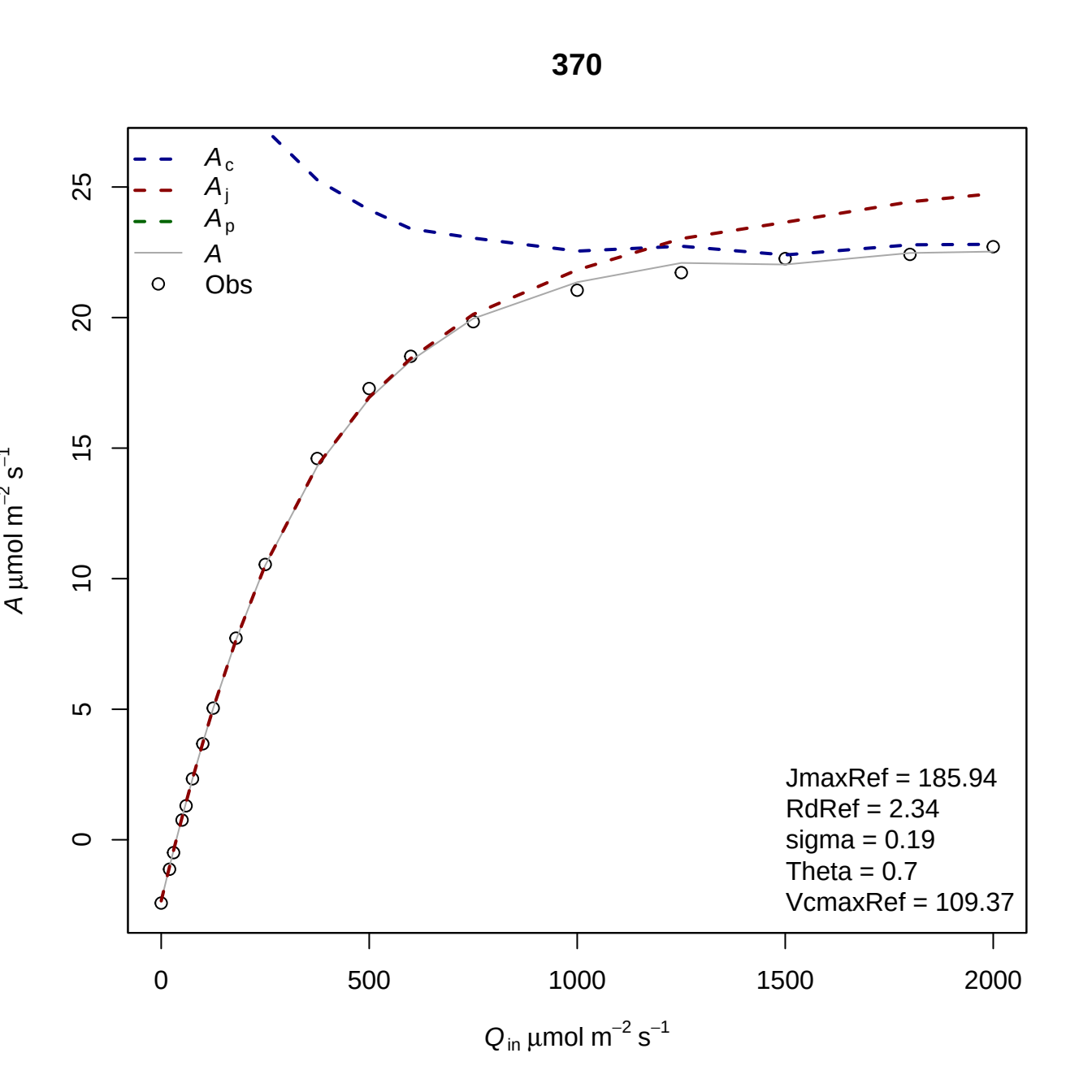

365
A_c
A_j
A_p
A
Obs
JmaxRef = 139.83
RdRef = 2.12
sigma = 0.27
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

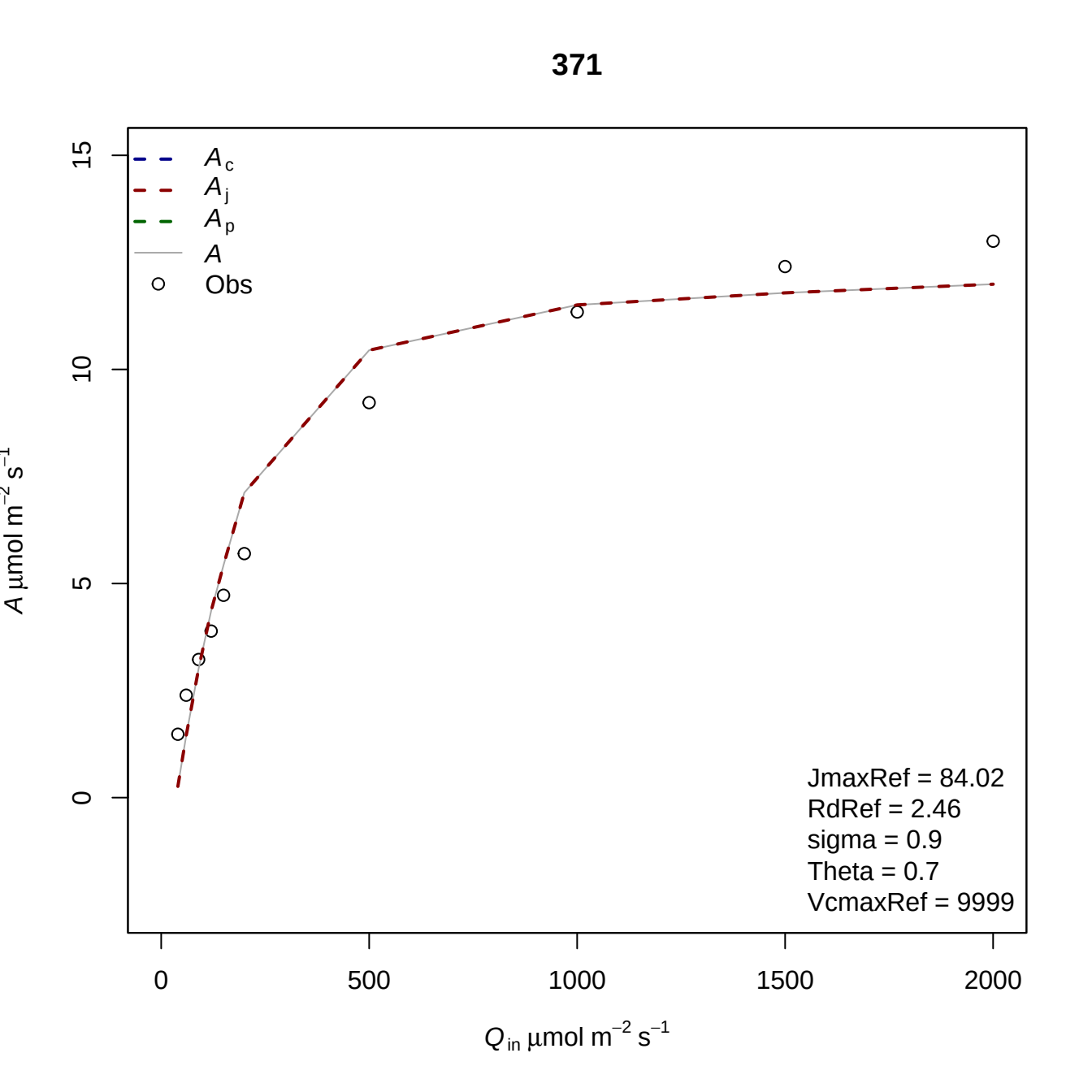

366
A_c
A_j
A_p
A
Obs
JmaxRef = 95.7
RdRef = 2.85
sigma = 0.1
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

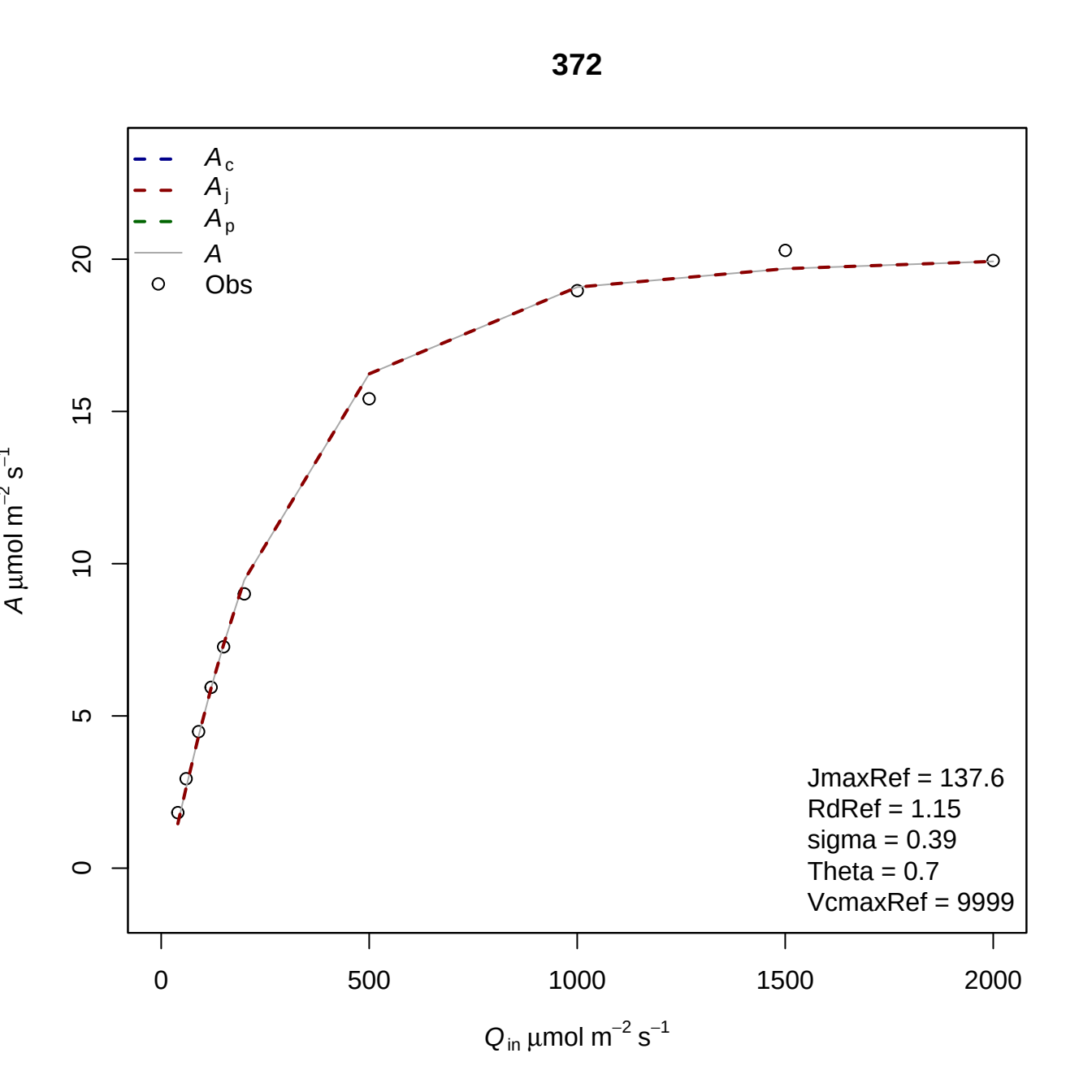

367
A_c
A_j
A_p
A
Obs
JmaxRef = 140.04
RdRef = 2.38
sigma = 0.21
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

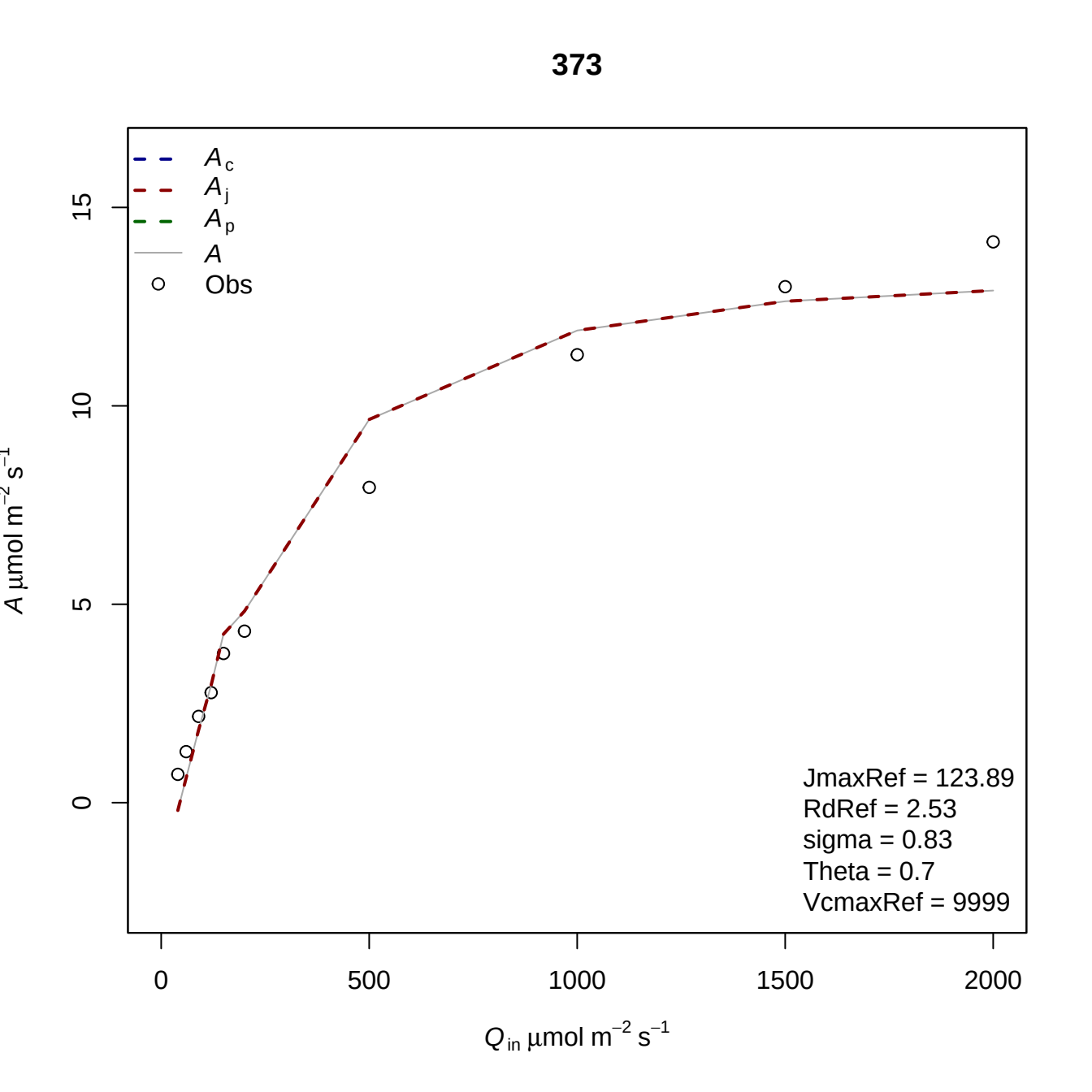

368
A_c
A_j
A_p
A
Obs
JmaxRef = 157.5
RdRef = 1.58
sigma = 0.2
Theta = 0.7
VcmaxRef = 72.43
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

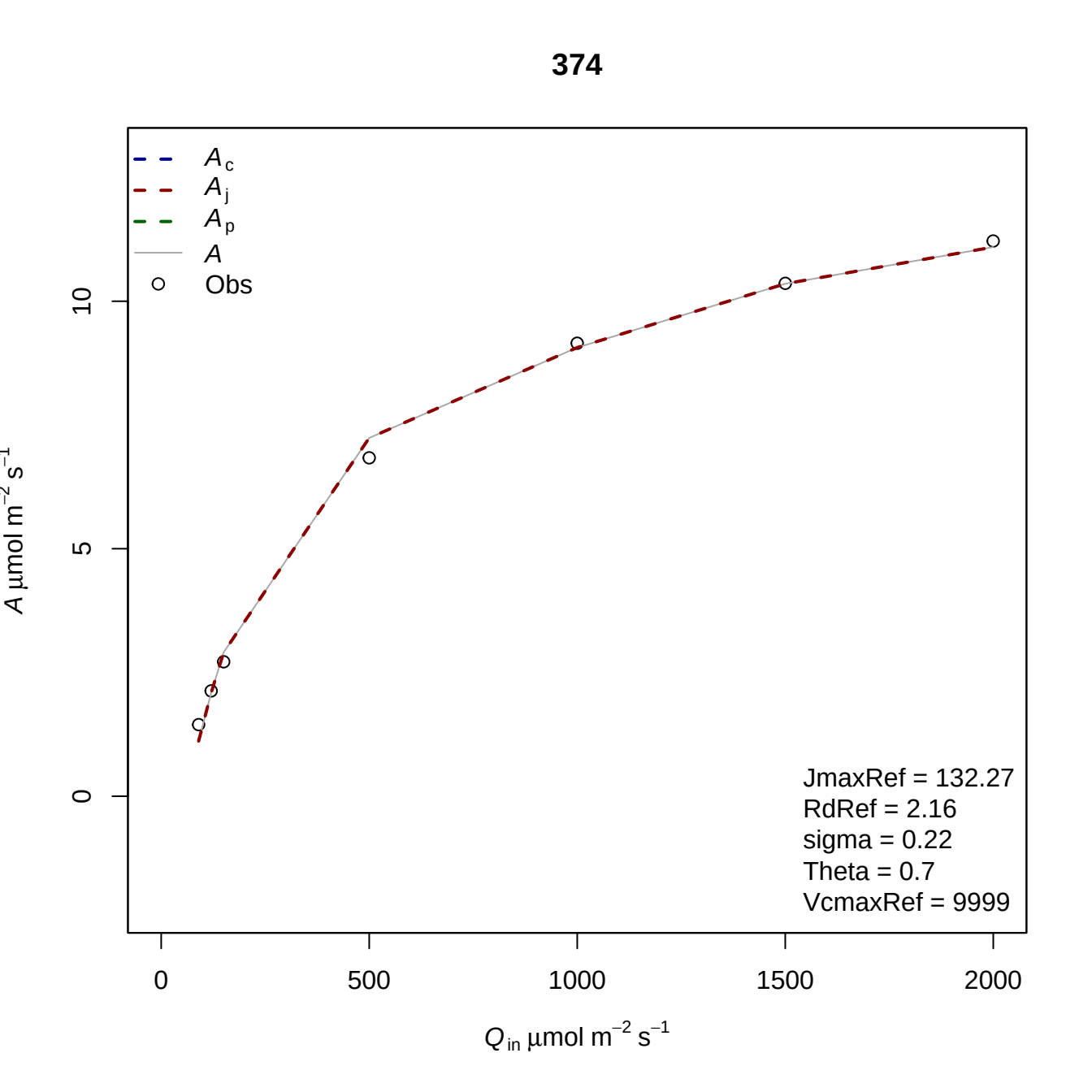

369
A_c
A_j
A_p
A
Obs
JmaxRef = 203.45
RdRef = 2.53
sigma = 0.27
Theta = 0.7
VcmaxRef = 113.86
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

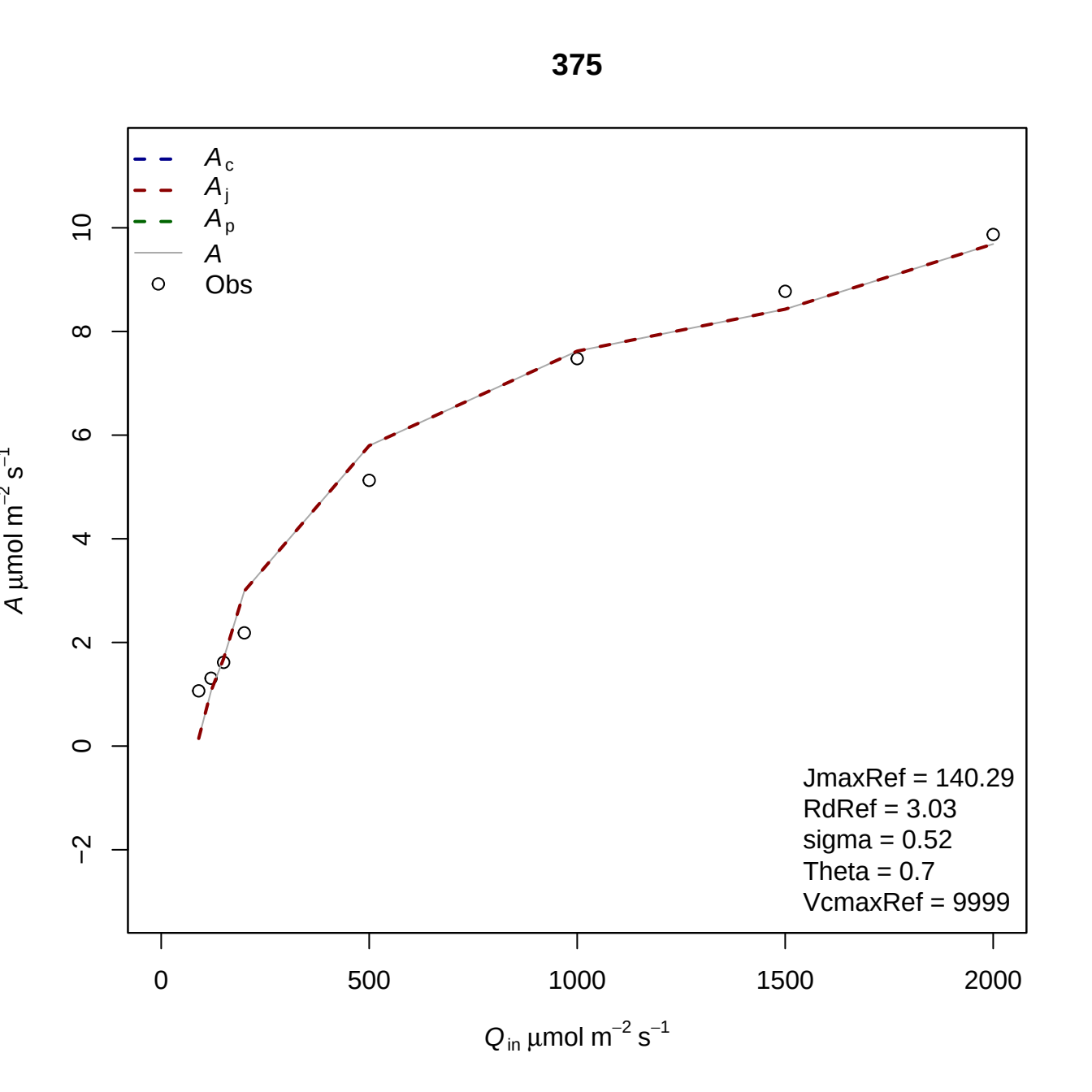

370
A_c
A_j
A_p
A
Obs
JmaxRef = 185.94
RdRef = 2.34
sigma = 0.19
Theta = 0.7
VcmaxRef = 109.37
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

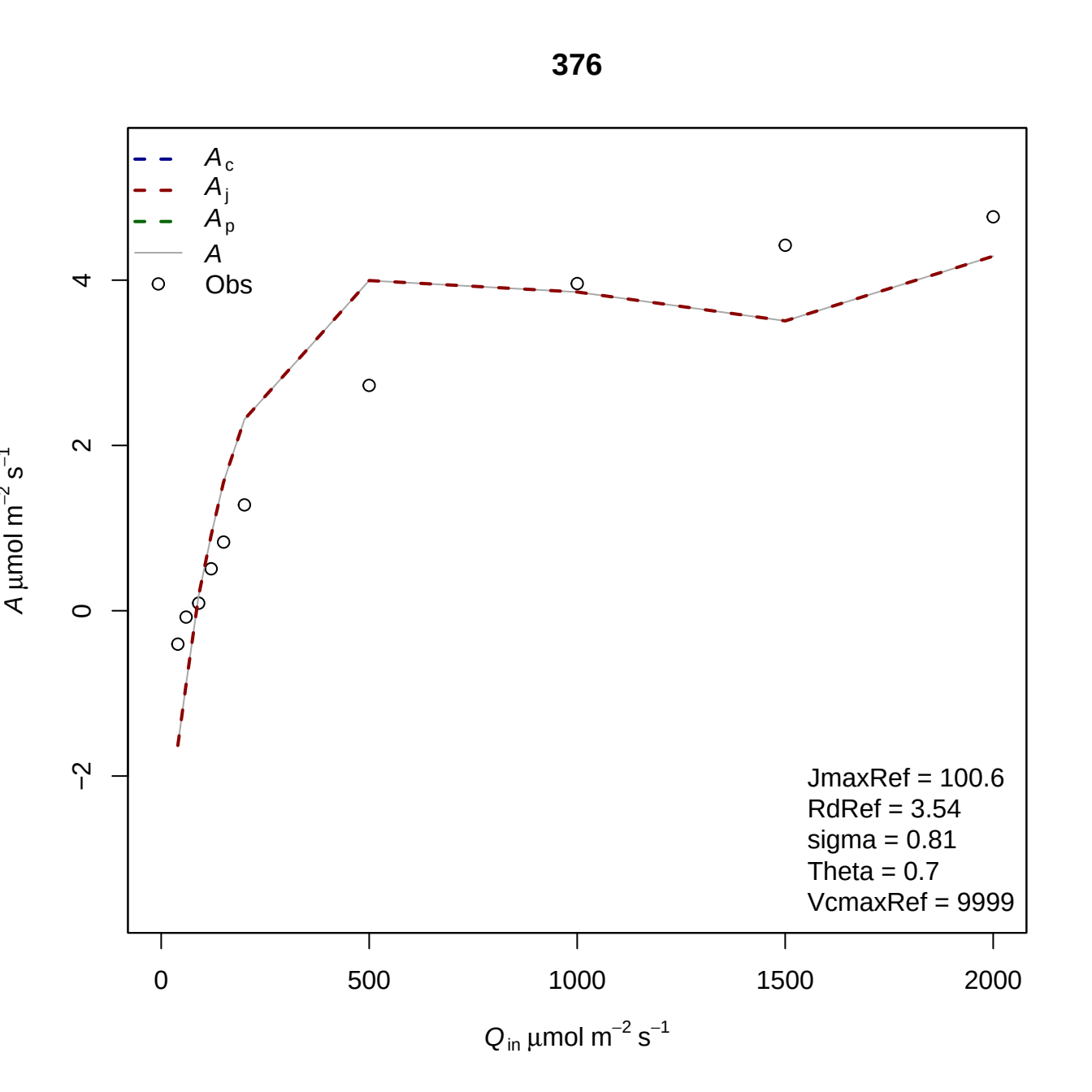

371
A_c
A_j
A_p
A
Obs
JmaxRef = 84.02
RdRef = 2.46
sigma = 0.9
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

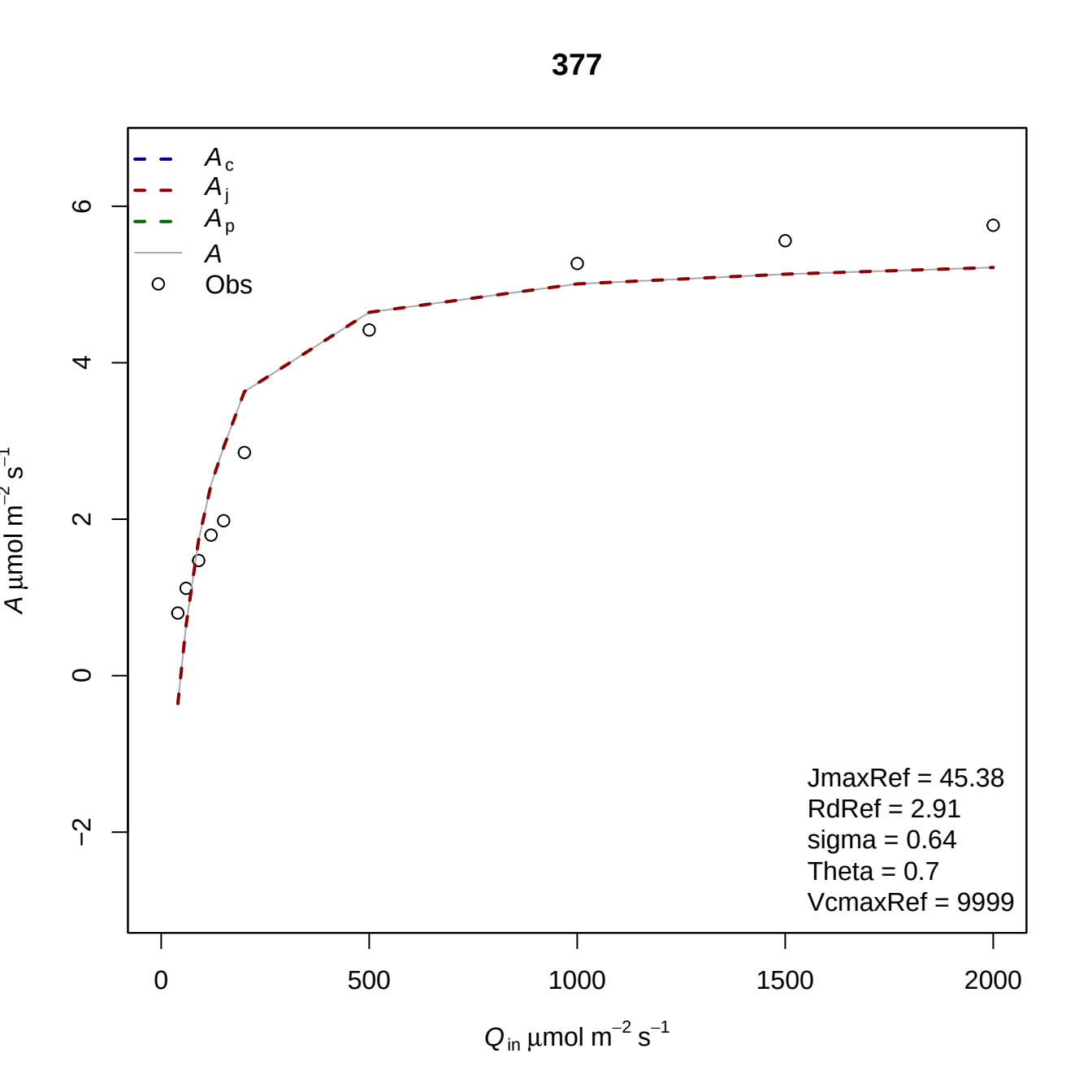

372
A_c
A_j
A_p
A
Obs
JmaxRef = 137.6
RdRef = 1.15
sigma = 0.39
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

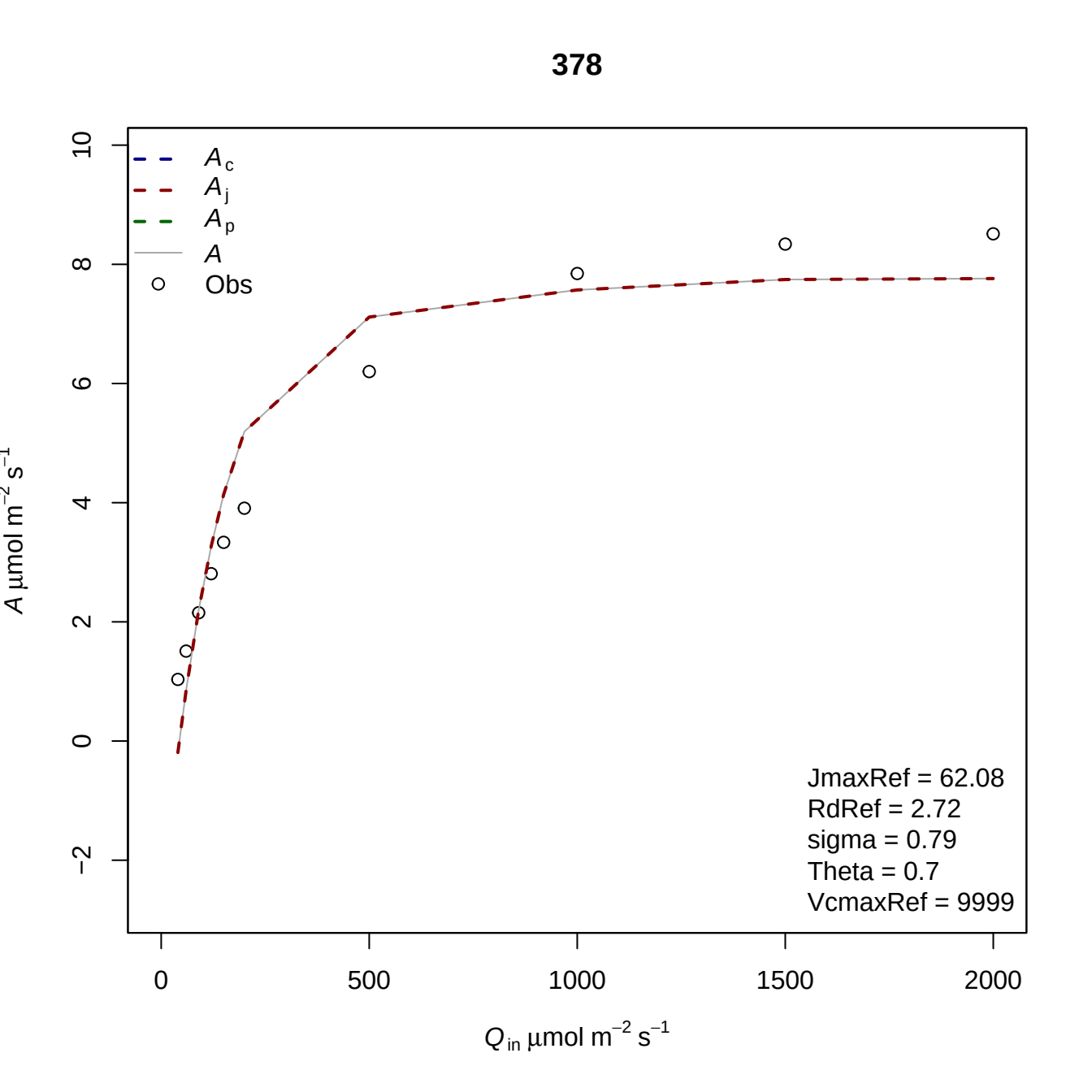

373
A_c
A_j
A_p
A
Obs
JmaxRef = 123.89
RdRef = 2.53
sigma = 0.83
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

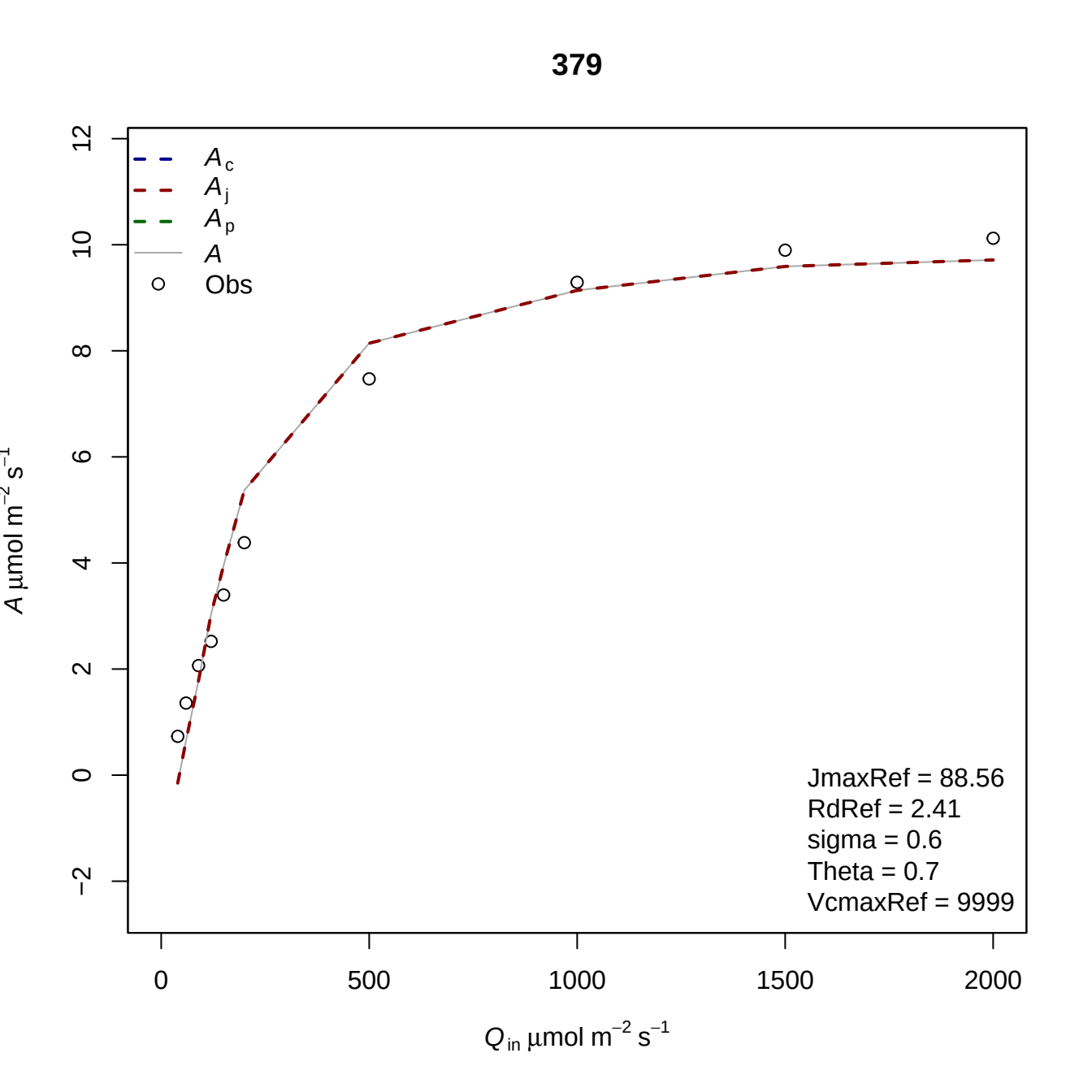

374
A_c
A_j
A_p
A
Obs
JmaxRef = 132.27
RdRef = 2.16
sigma = 0.22
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

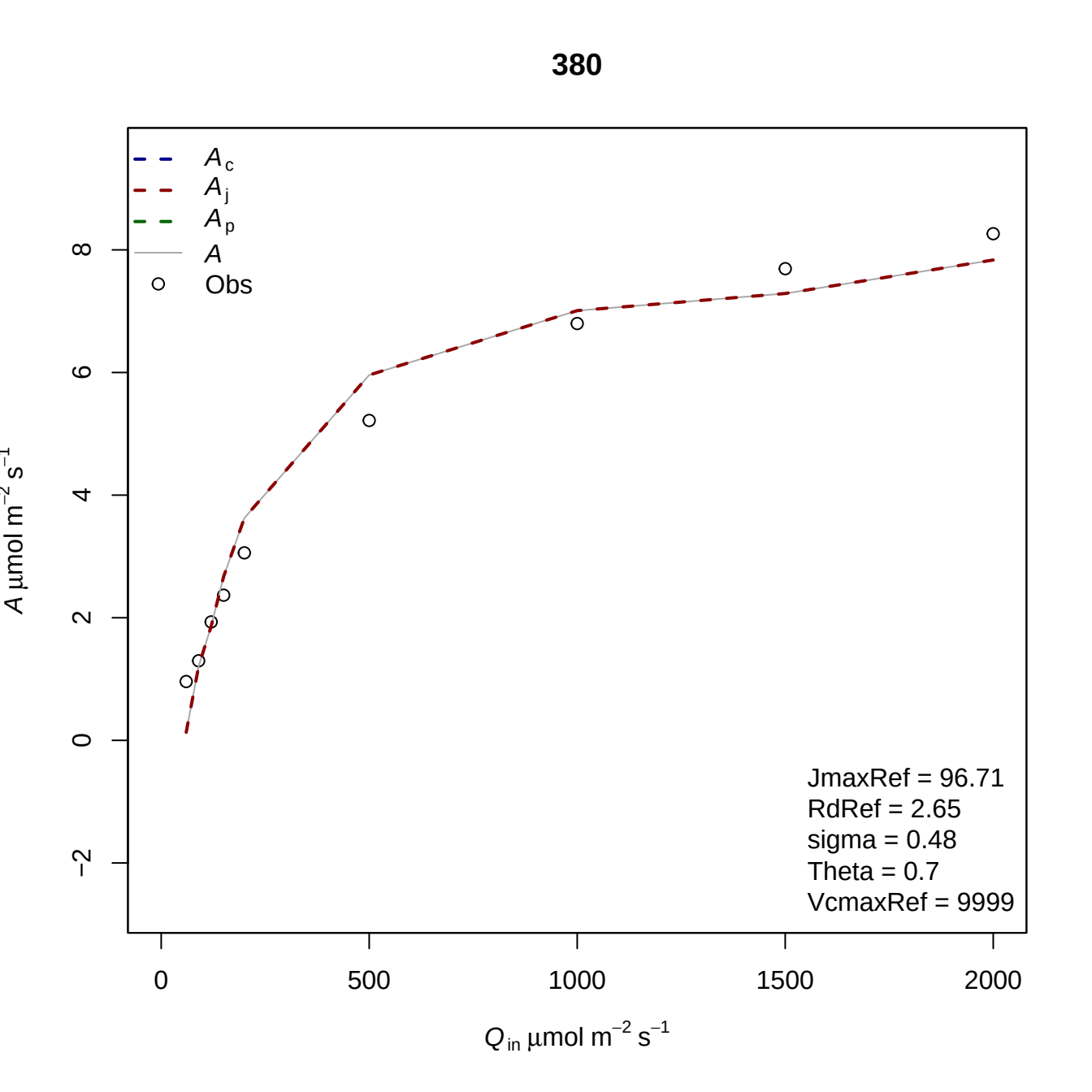

375
A_c
A_j
A_p
A
Obs
JmaxRef = 140.29
RdRef = 3.03
sigma = 0.52
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

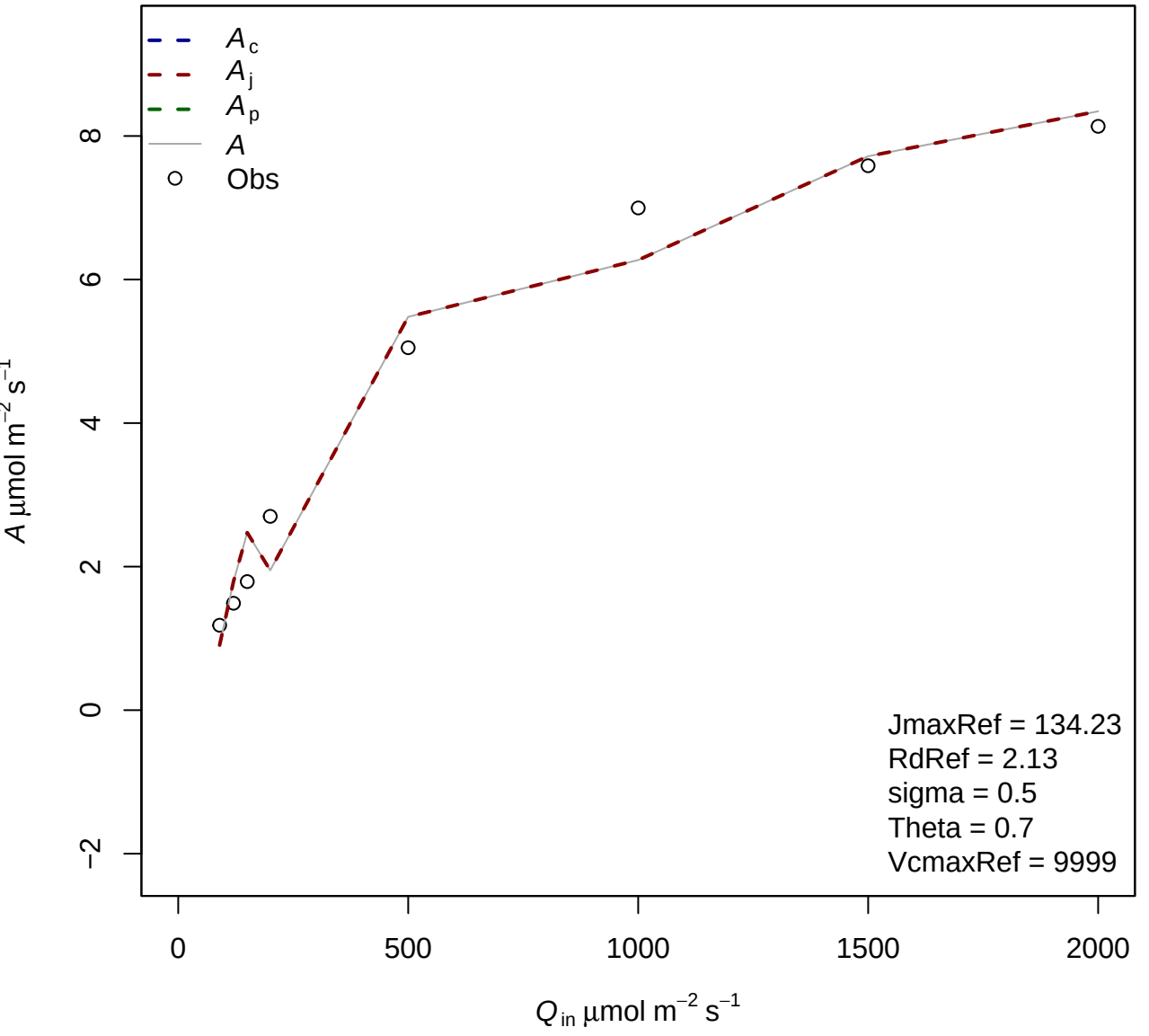

376
A_c
A_j
A_p
A
Obs
JmaxRef = 100.6
RdRef = 3.54
sigma = 0.81
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

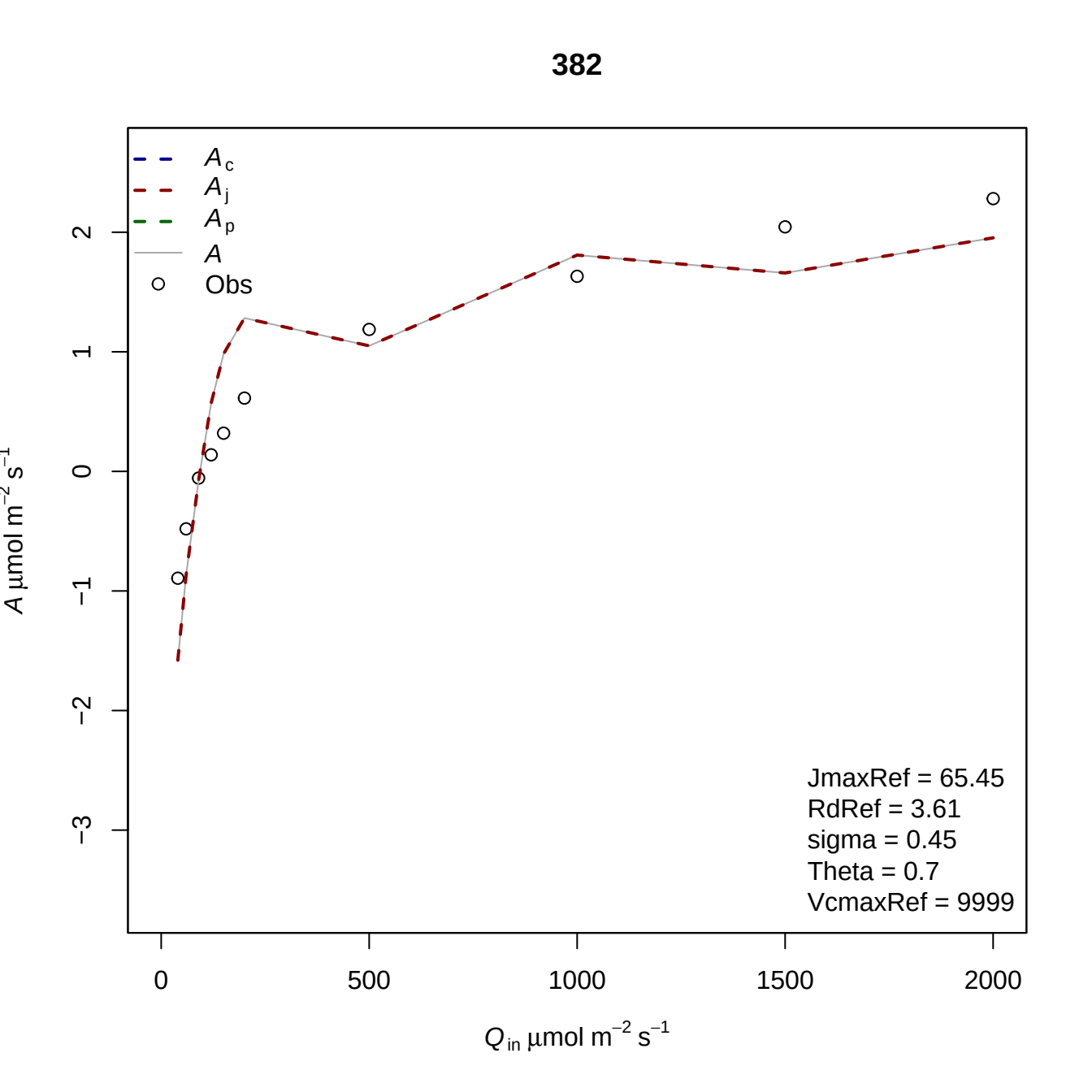

377
A_c
A_j
A_p
A
Obs
JmaxRef = 45.38
RdRef = 2.91
sigma = 0.64
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

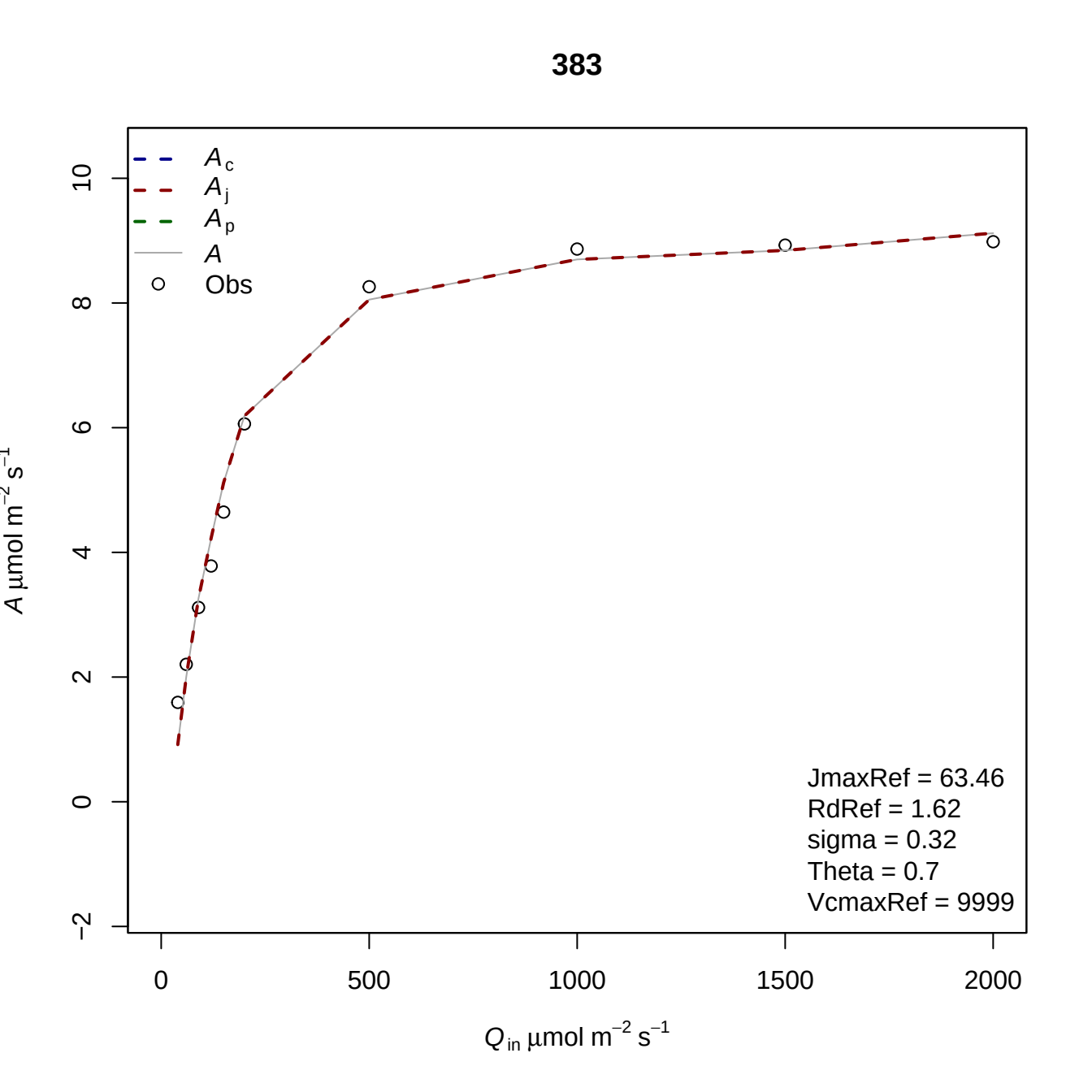

378
A_c
A_j
A_p
A
Obs
JmaxRef = 62.08
RdRef = 2.72
sigma = 0.79
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

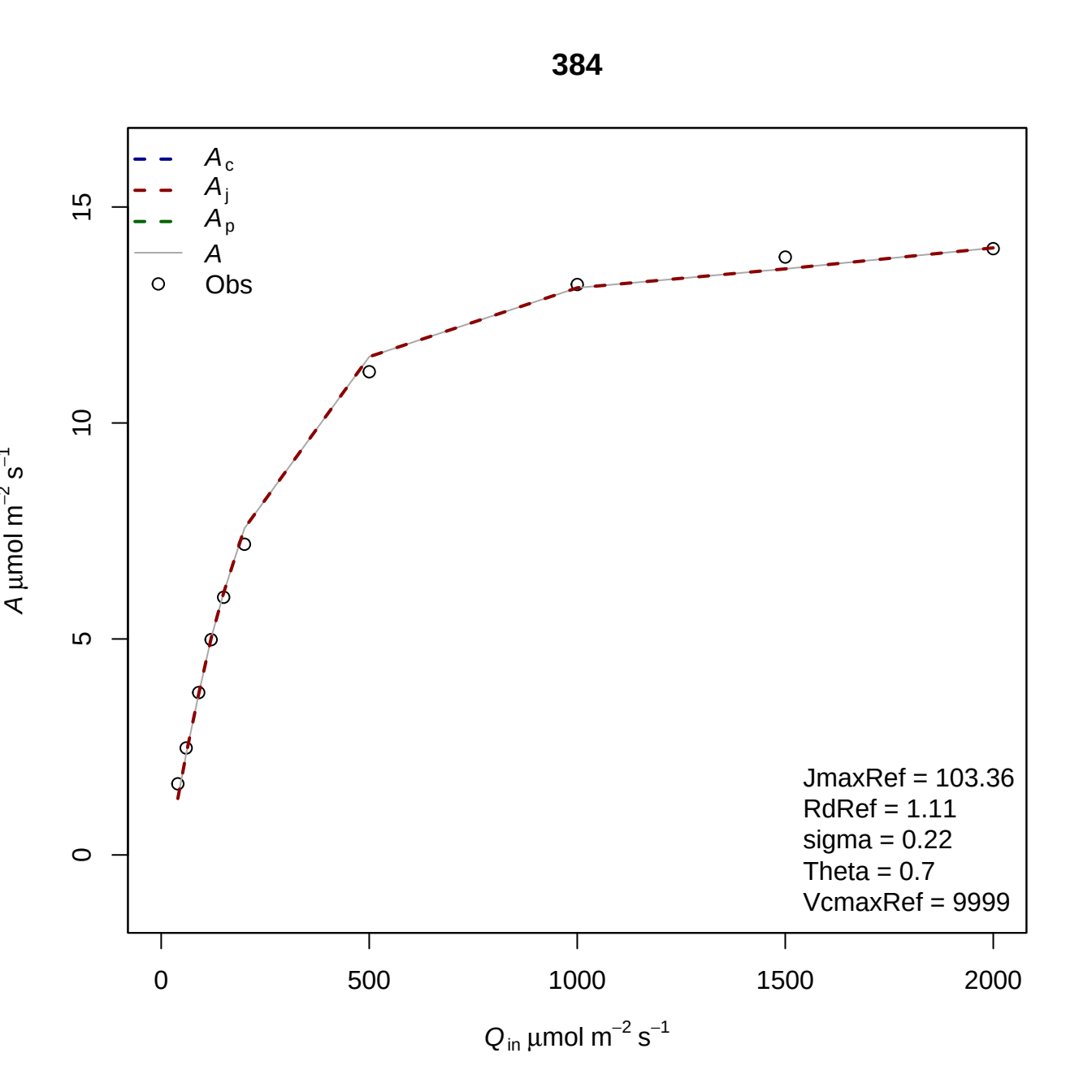

379
A_c
A_j
A_p
A
Obs
JmaxRef = 88.56
RdRef = 2.41
sigma = 0.6
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

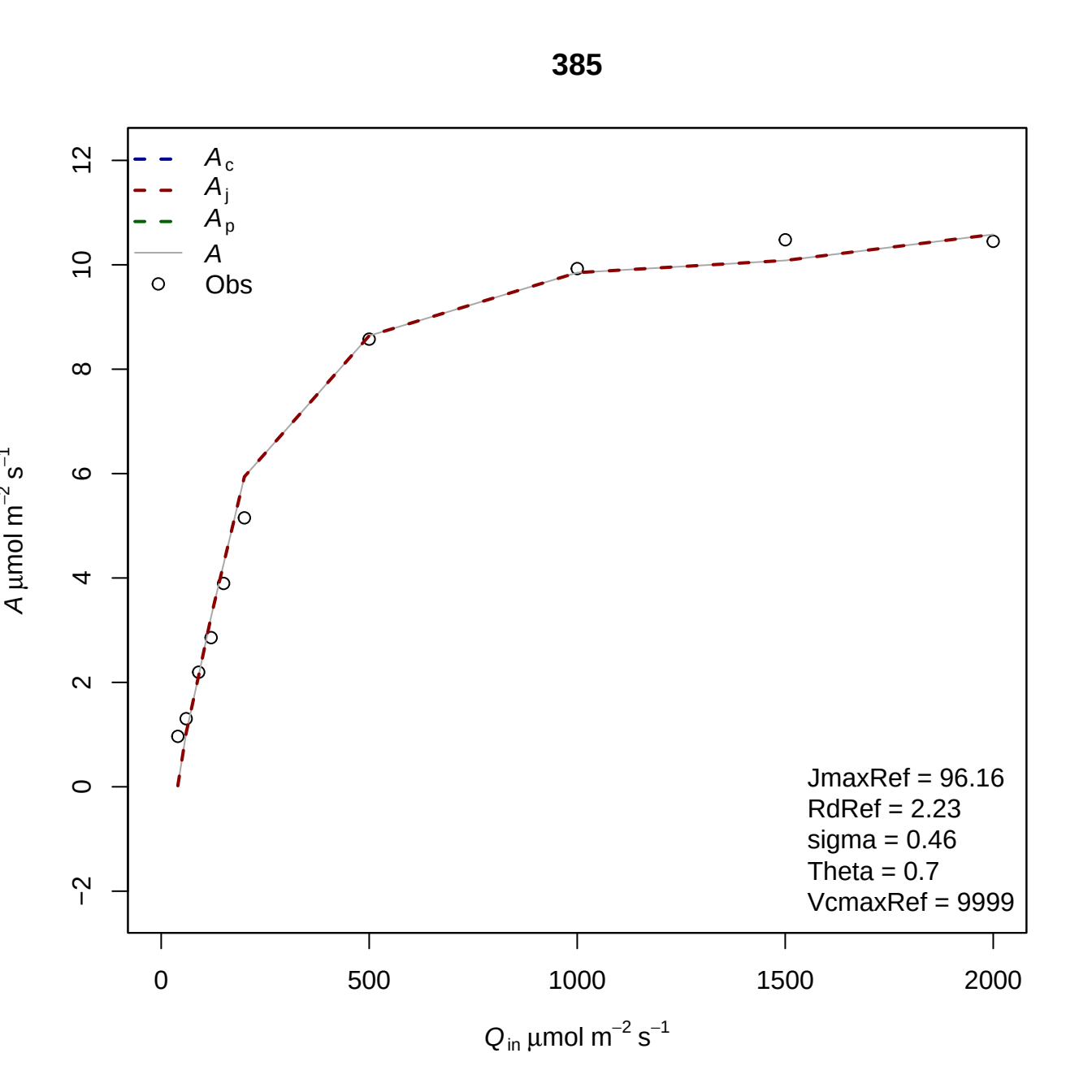

380
A_c
A_j
A_p
A
Obs
JmaxRef = 96.71
RdRef = 2.65
sigma = 0.48
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

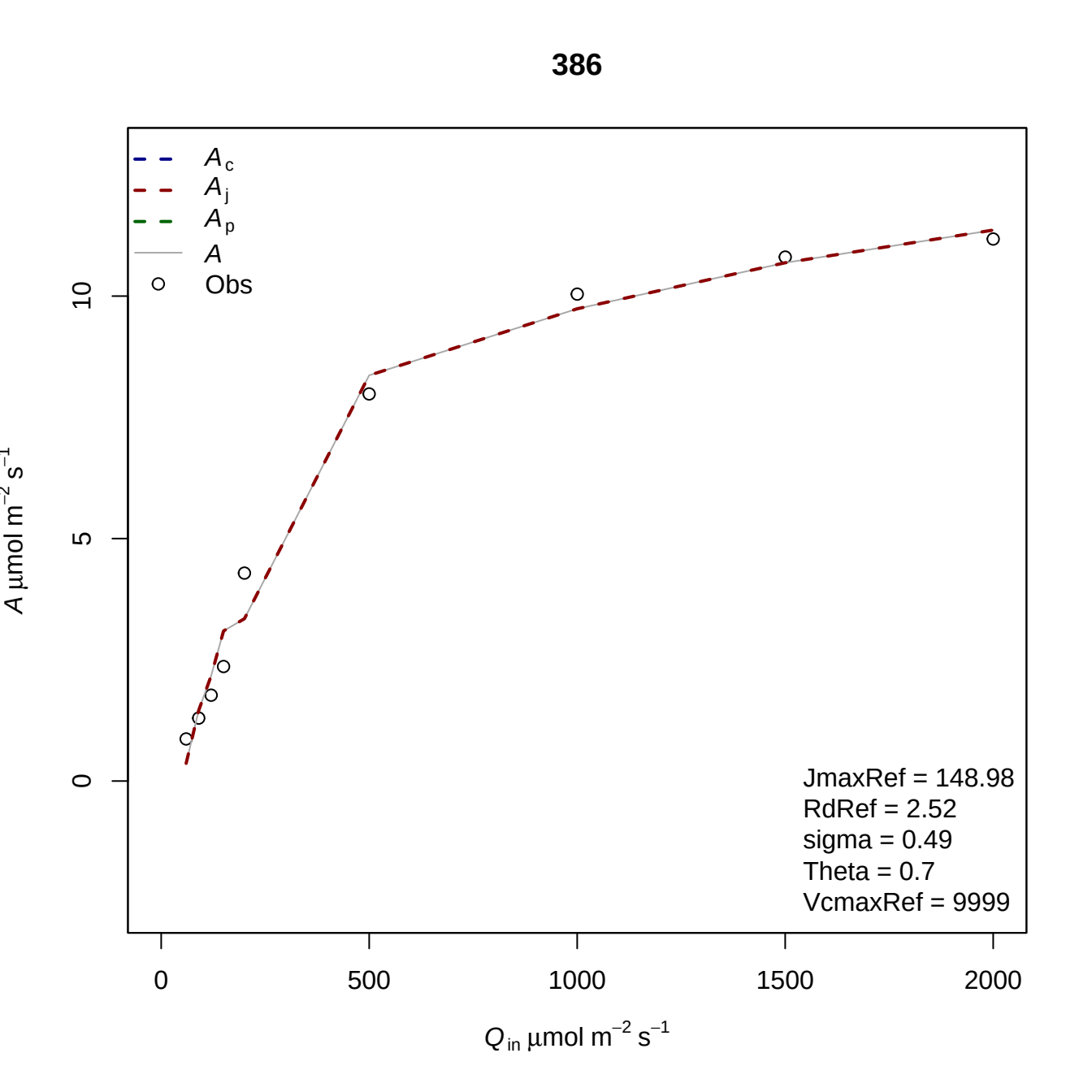

A_c
A_j
A_p
A
Obs
JmaxRef = 134.23
RdRef = 2.13
sigma = 0.5
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

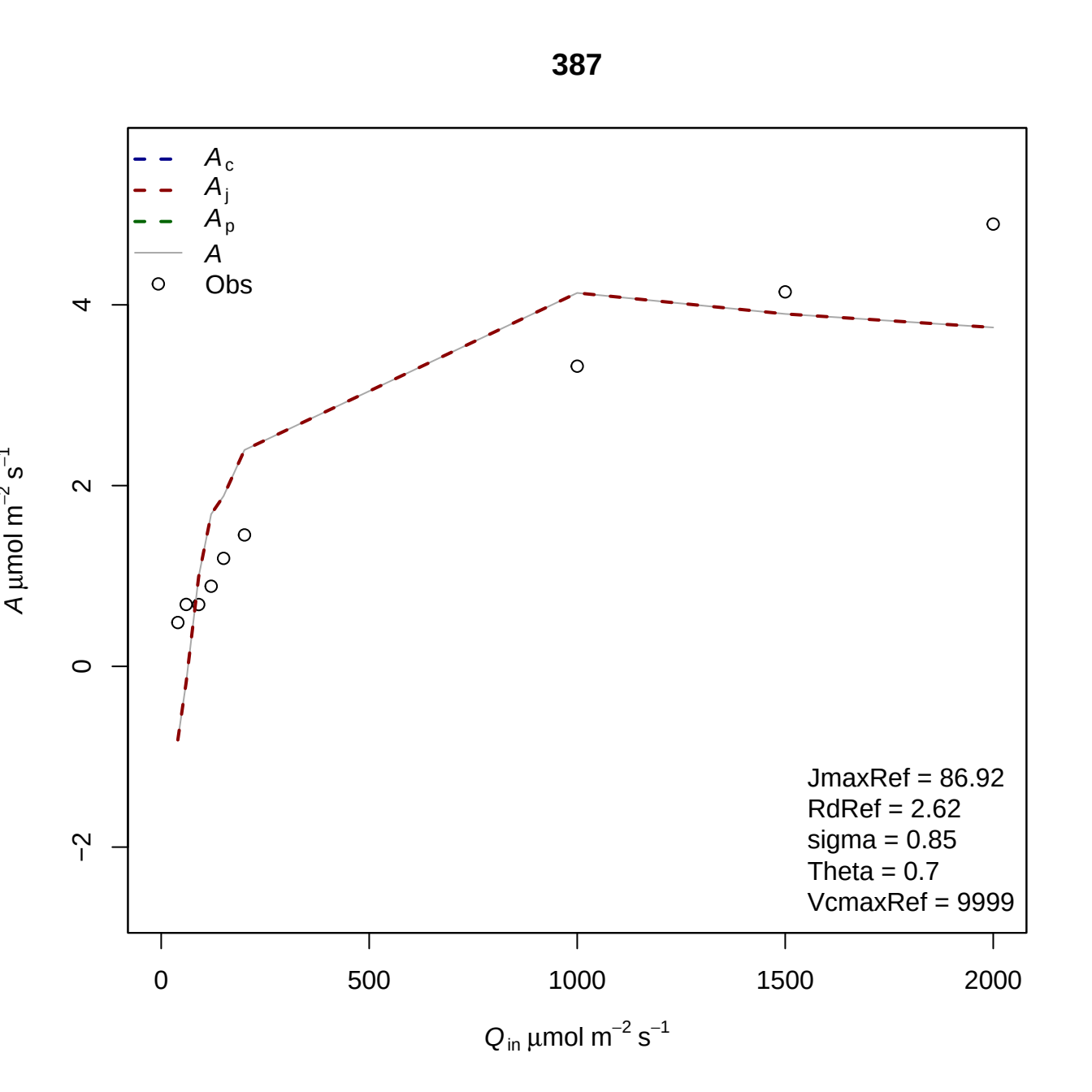

382
A_c
A_j
A_p
A
Obs
JmaxRef = 65.45
RdRef = 3.61
sigma = 0.45
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2
1
0
−1
−2
−3
0
500
1000
1500
2000

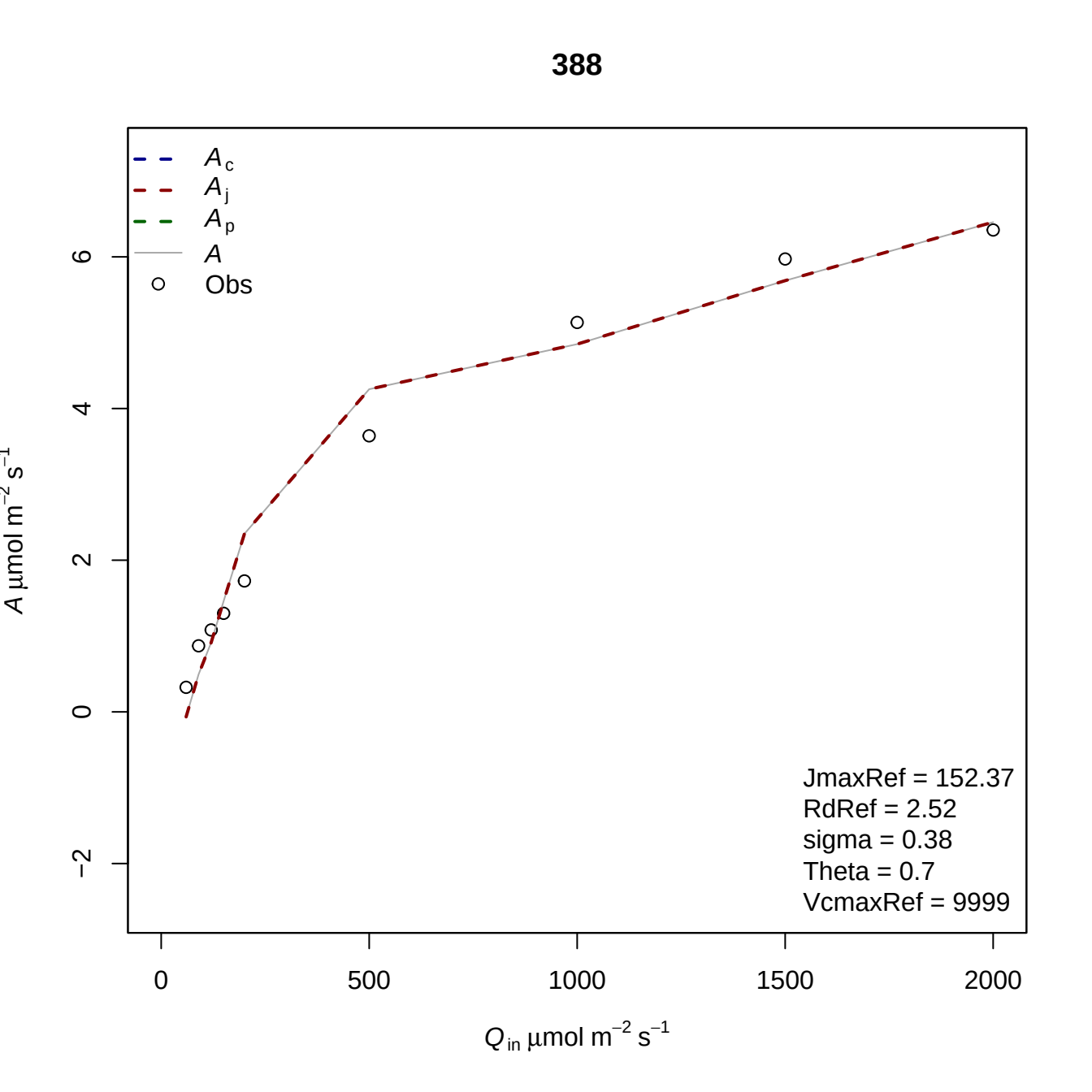

383
A_c
A_j
A_p
A
Obs
JmaxRef = 63.46
RdRef = 1.62
sigma = 0.32
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

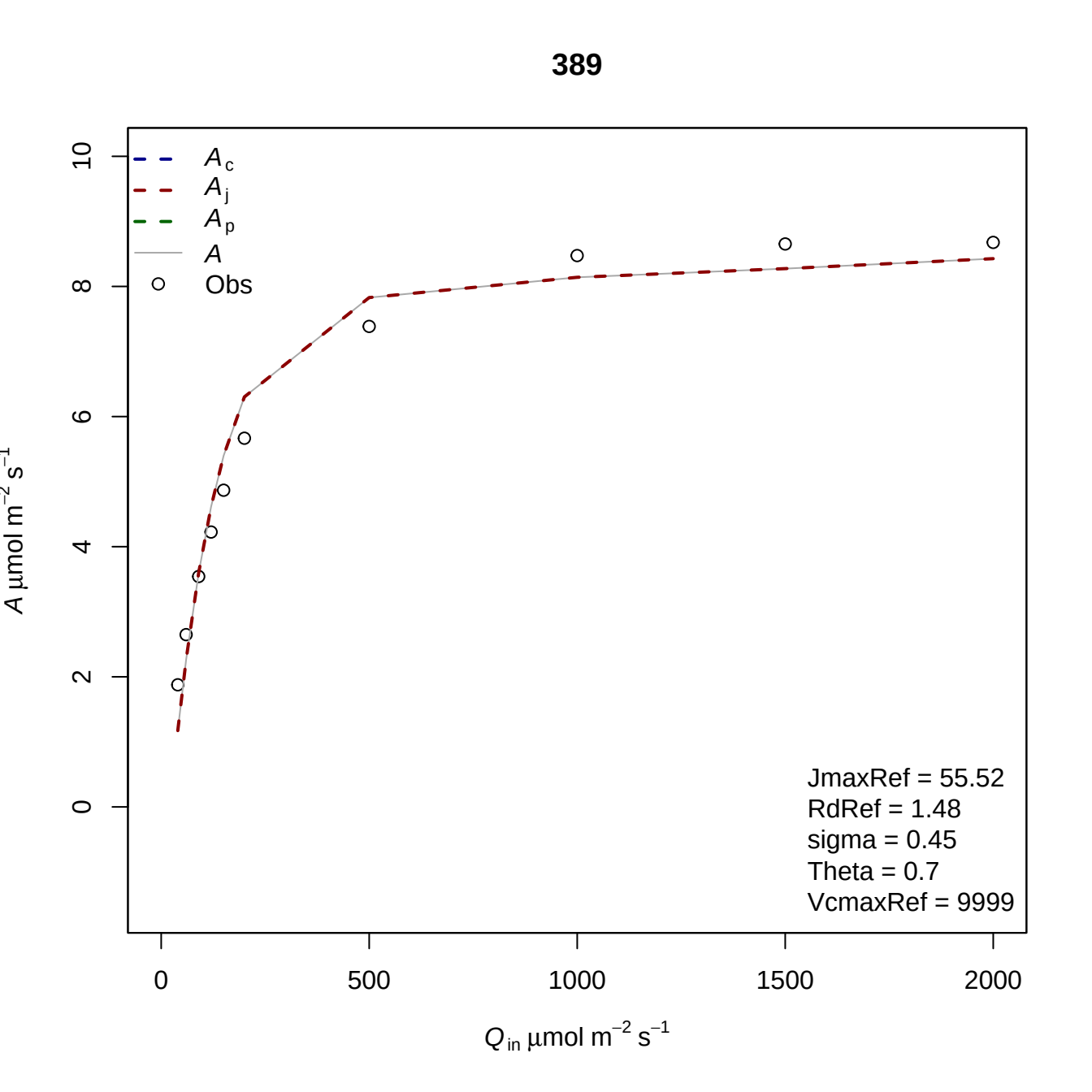

384
A_c
A_j
A_p
A
Obs
JmaxRef = 103.36
RdRef = 1.11
sigma = 0.22
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

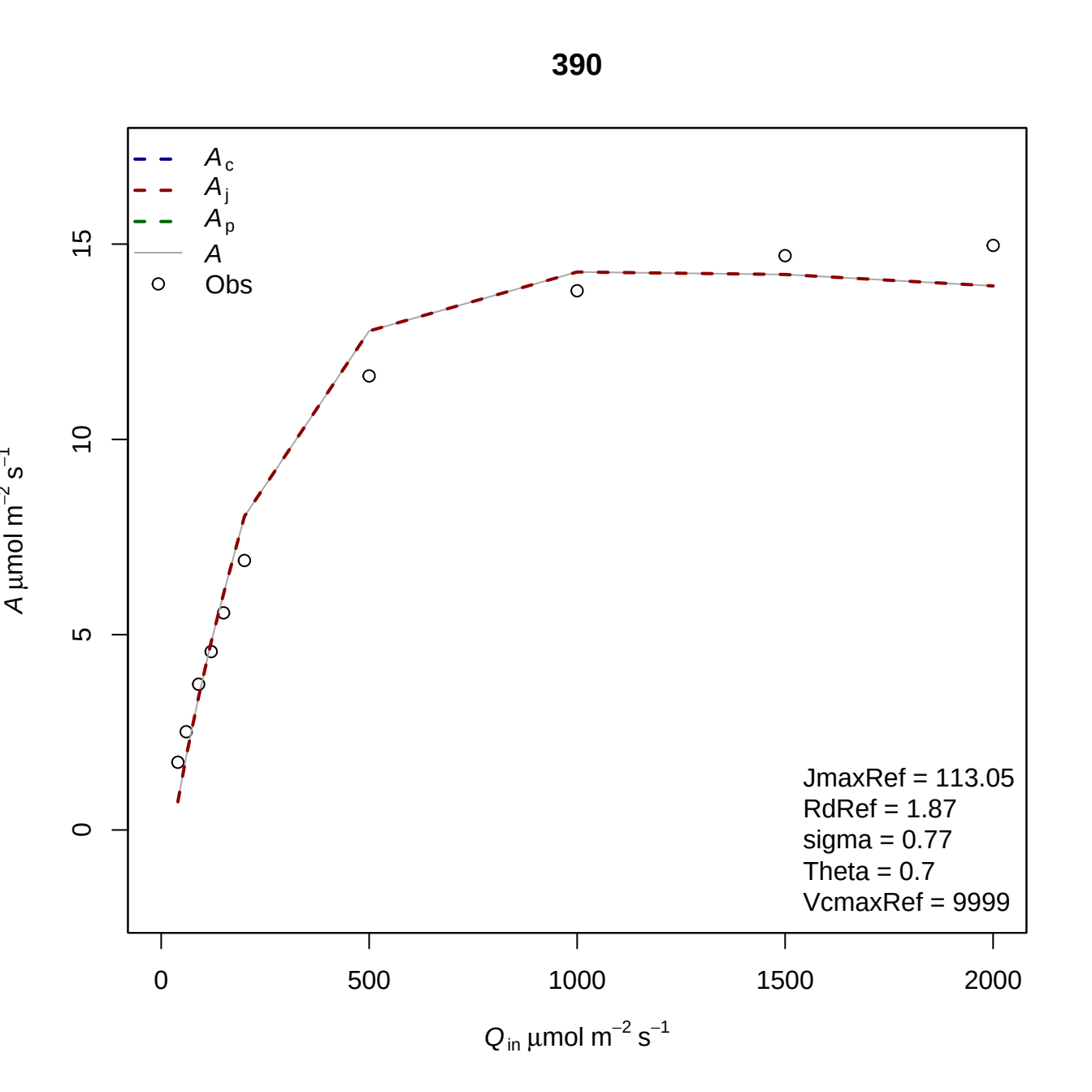

385
A_c
A_j
A_p
A
Obs
JmaxRef = 96.16
RdRef = 2.23
sigma = 0.46
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

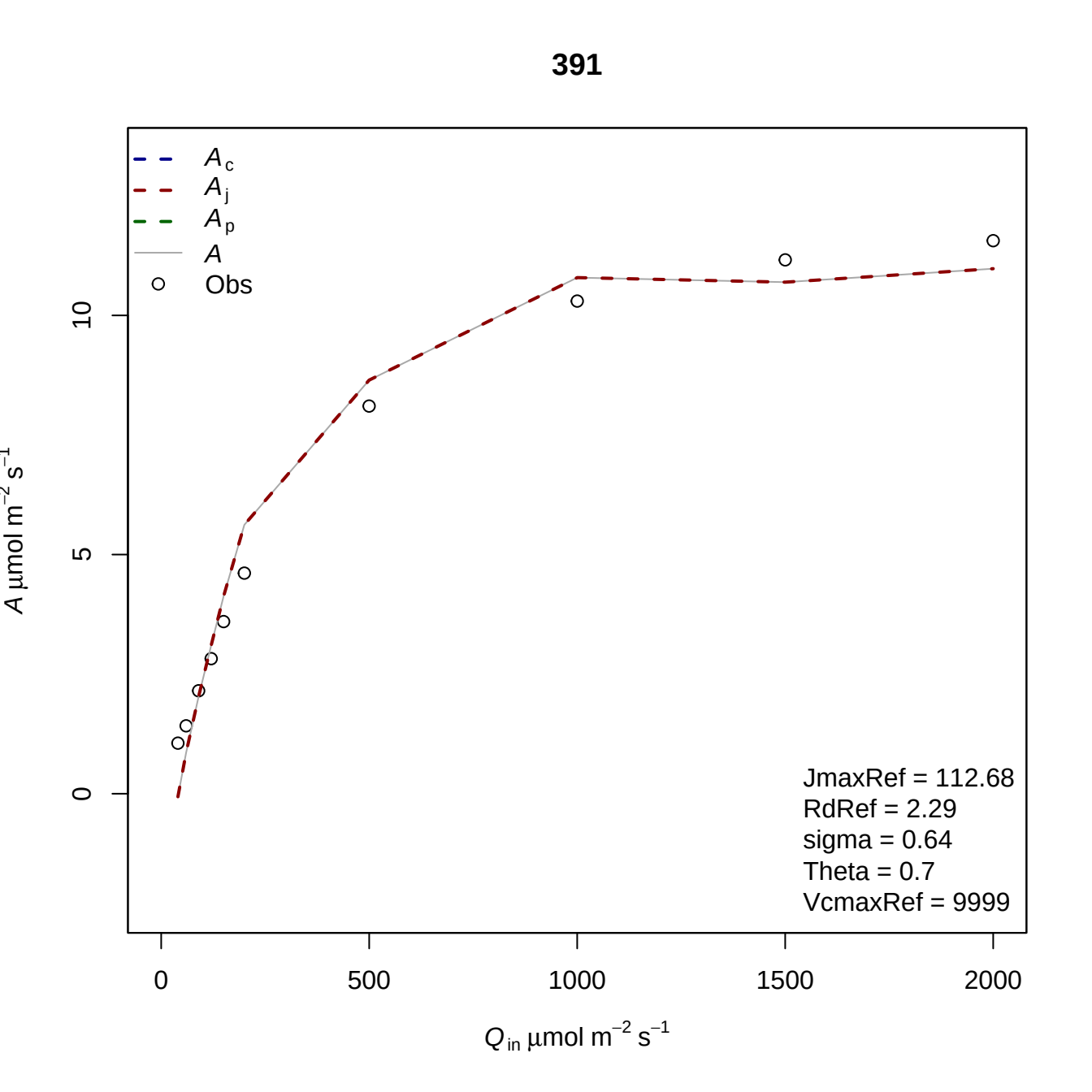

386
A_c
A_j
A_p
A
Obs
JmaxRef = 148.98
RdRef = 2.52
sigma = 0.49
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

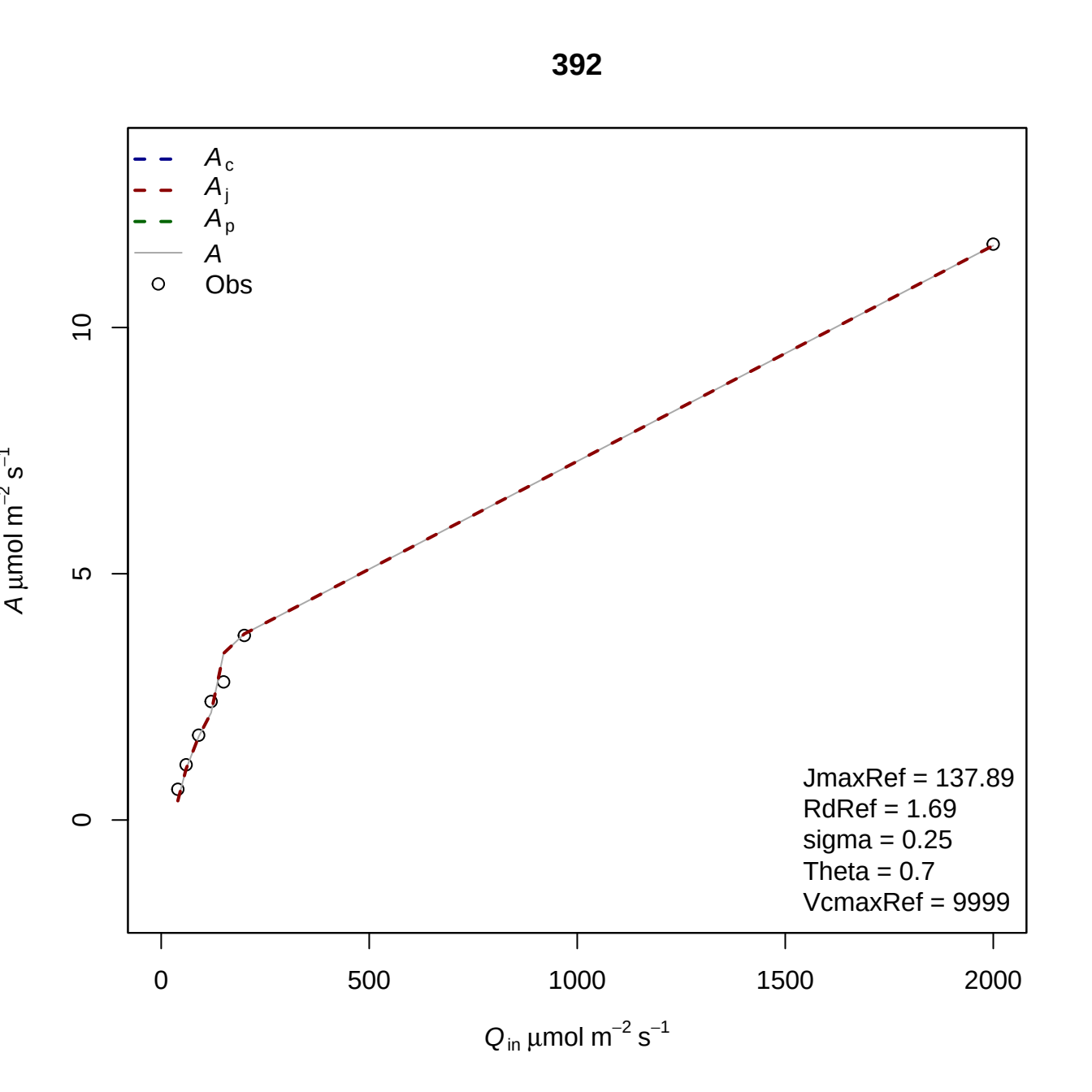

387
A_c
A_j
A_p
A
Obs
JmaxRef = 86.92
RdRef = 2.62
sigma = 0.85
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

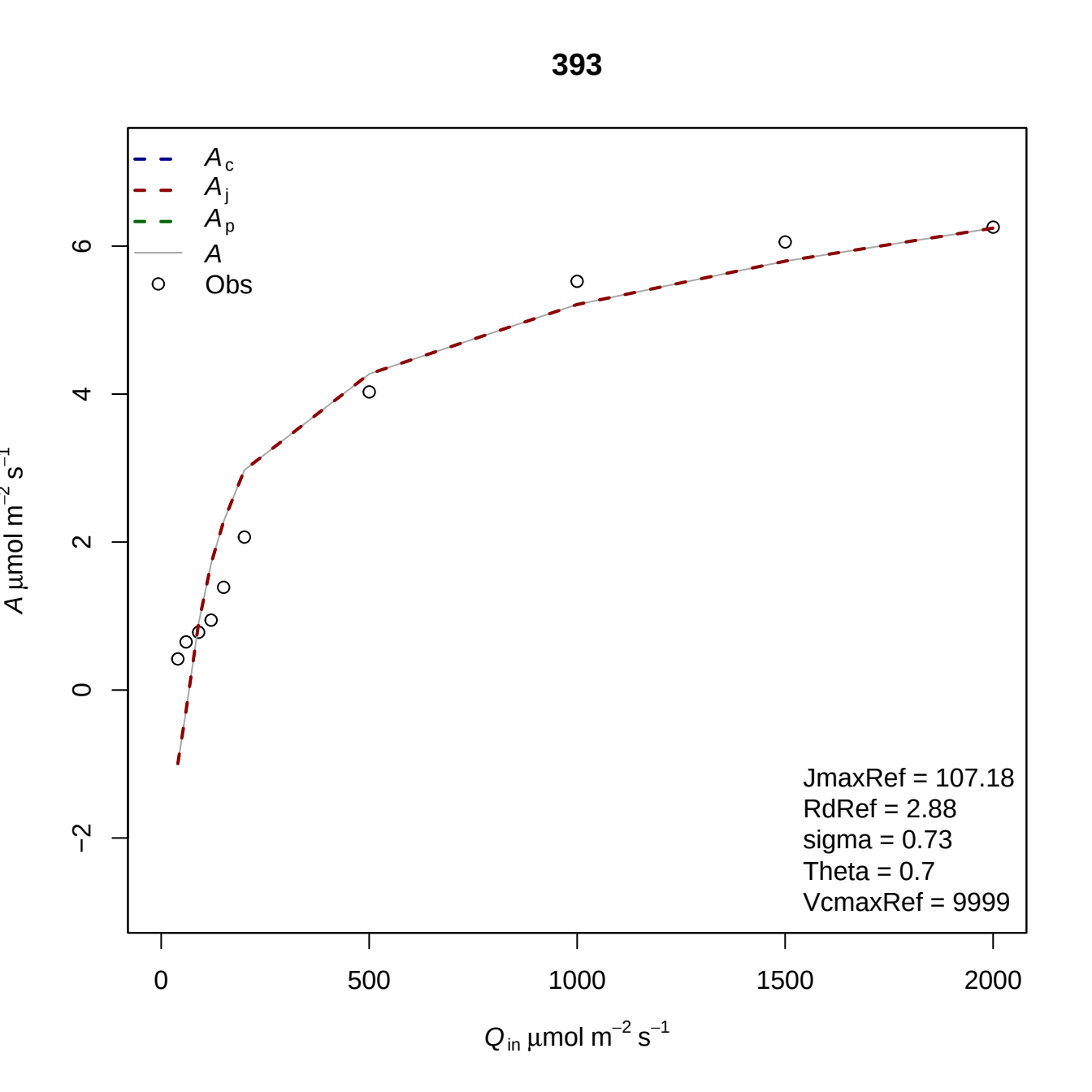

388
A_c
A_j
A_p
A
Obs
JmaxRef = 152.37
RdRef = 2.52
sigma = 0.38
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

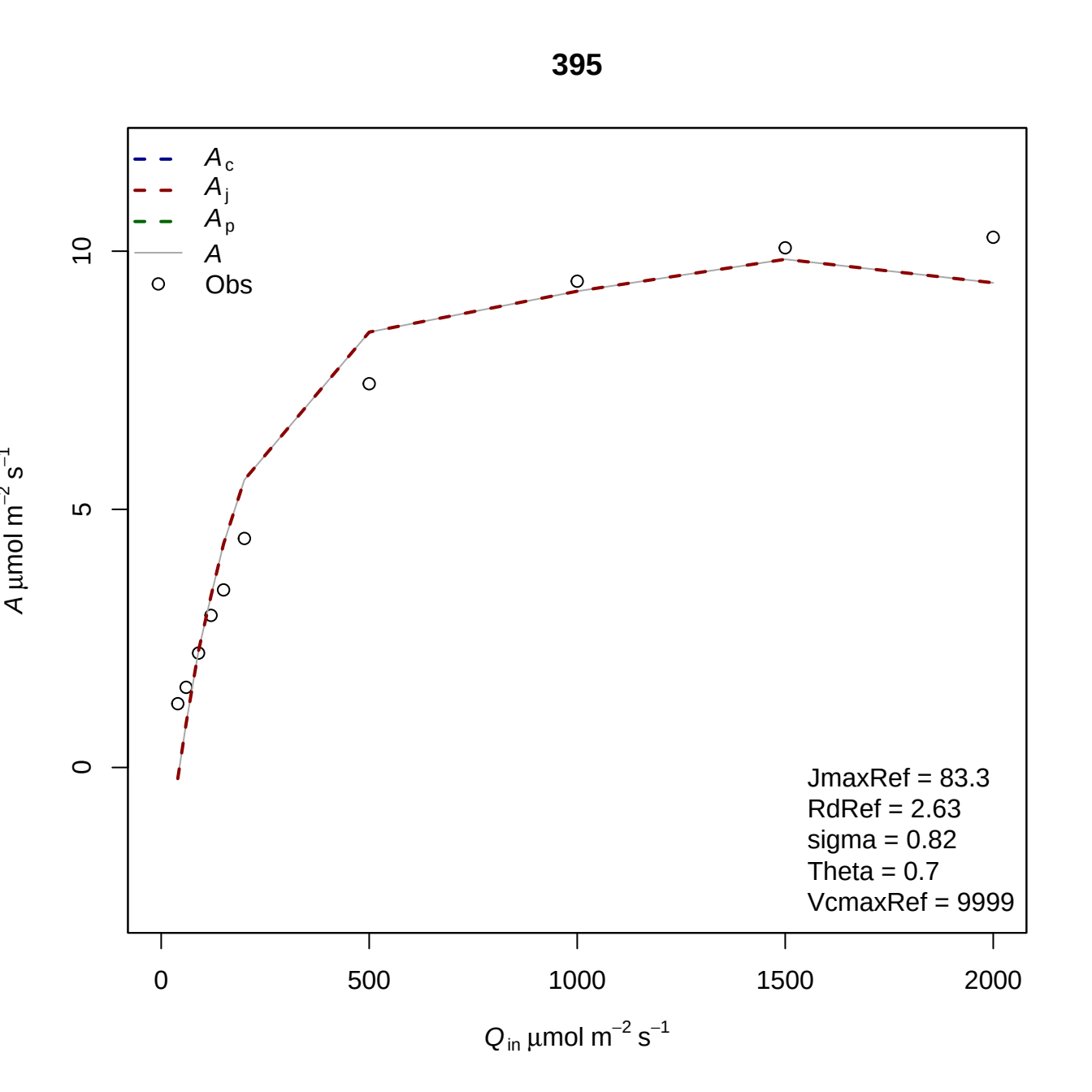

389
A_c
A_j
A_p
A
Obs
JmaxRef = 55.52
RdRef = 1.48
sigma = 0.45
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

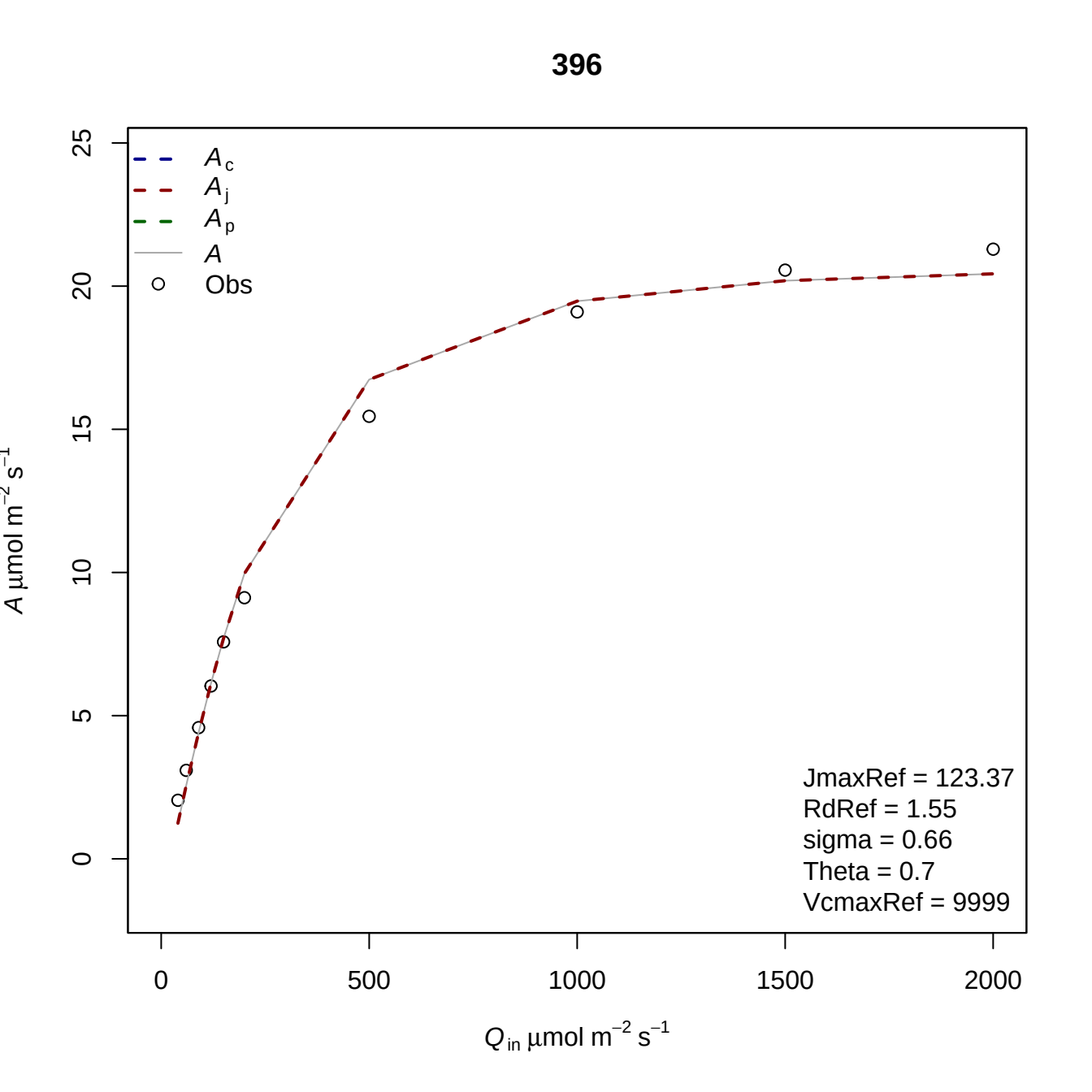

390
A_c
A_j
A_p
A
Obs
JmaxRef = 113.05
RdRef = 1.87
sigma = 0.77
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

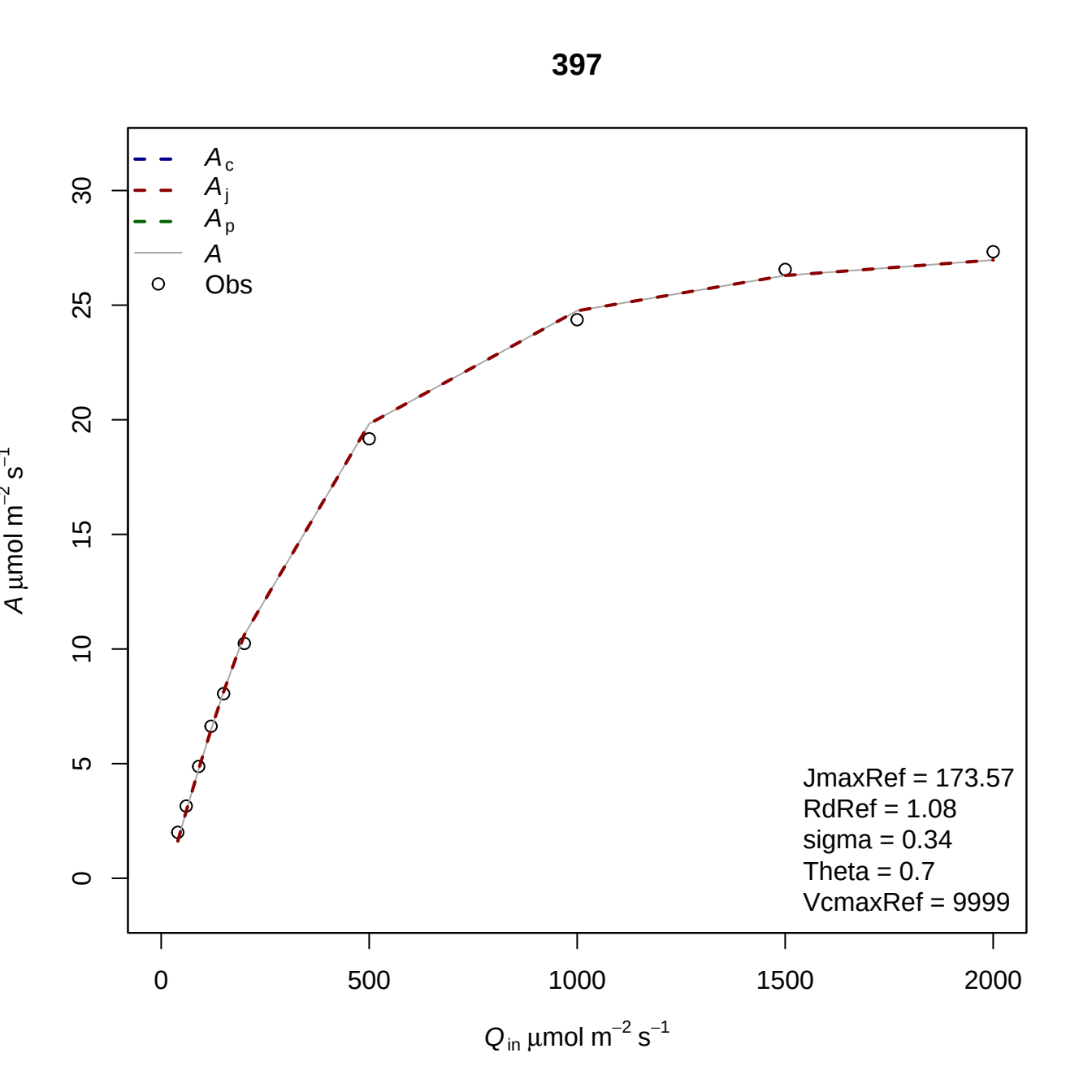

391
A_c
A_j
A_p
A
Obs
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}
JmaxRef = 112.68
RdRef = 2.29
sigma = 0.64
Theta = 0.7
VcmaxRef = 9999

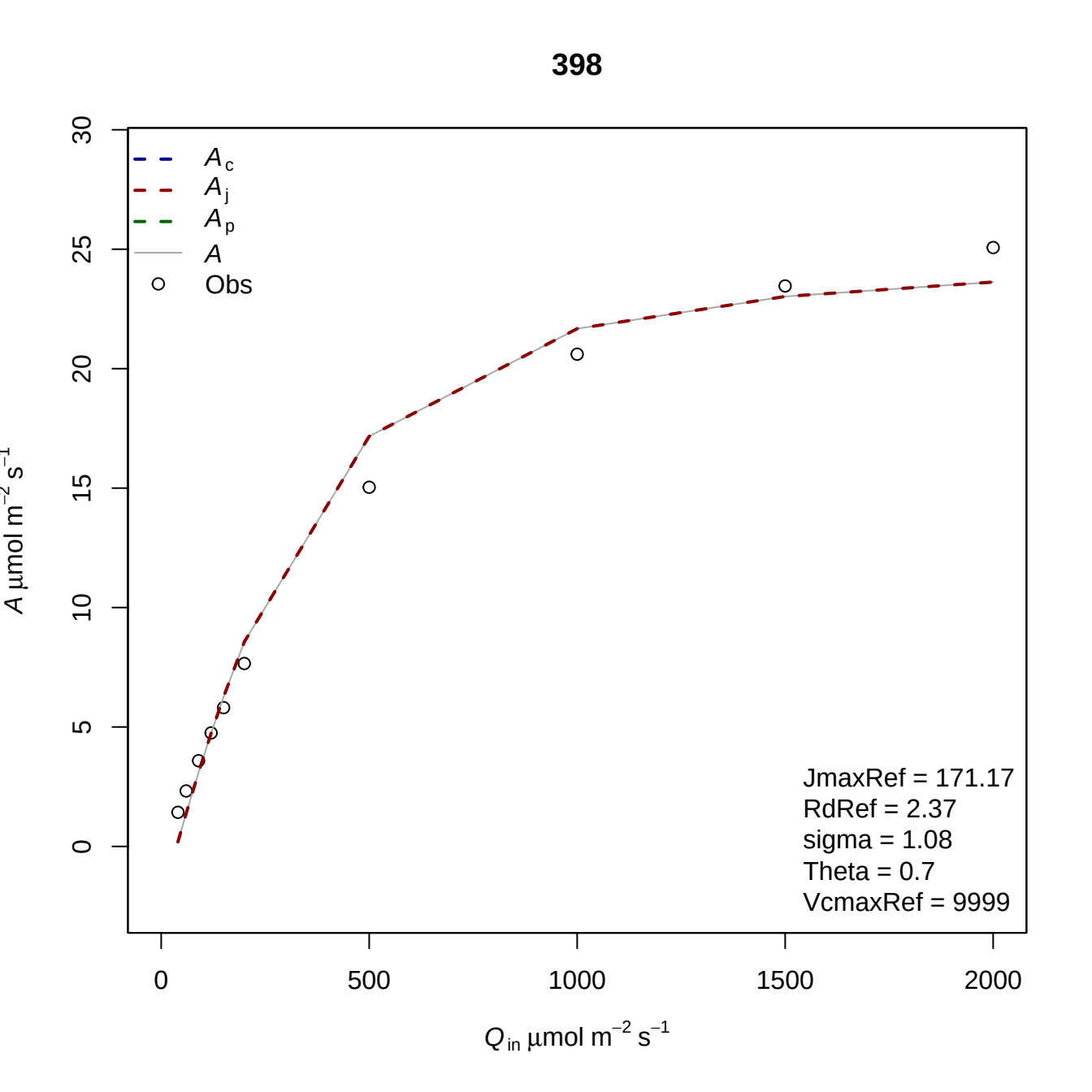

392
A_c
A_j
A_p
A
Obs
JmaxRef = 137.89
RdRef = 1.69
sigma = 0.25
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

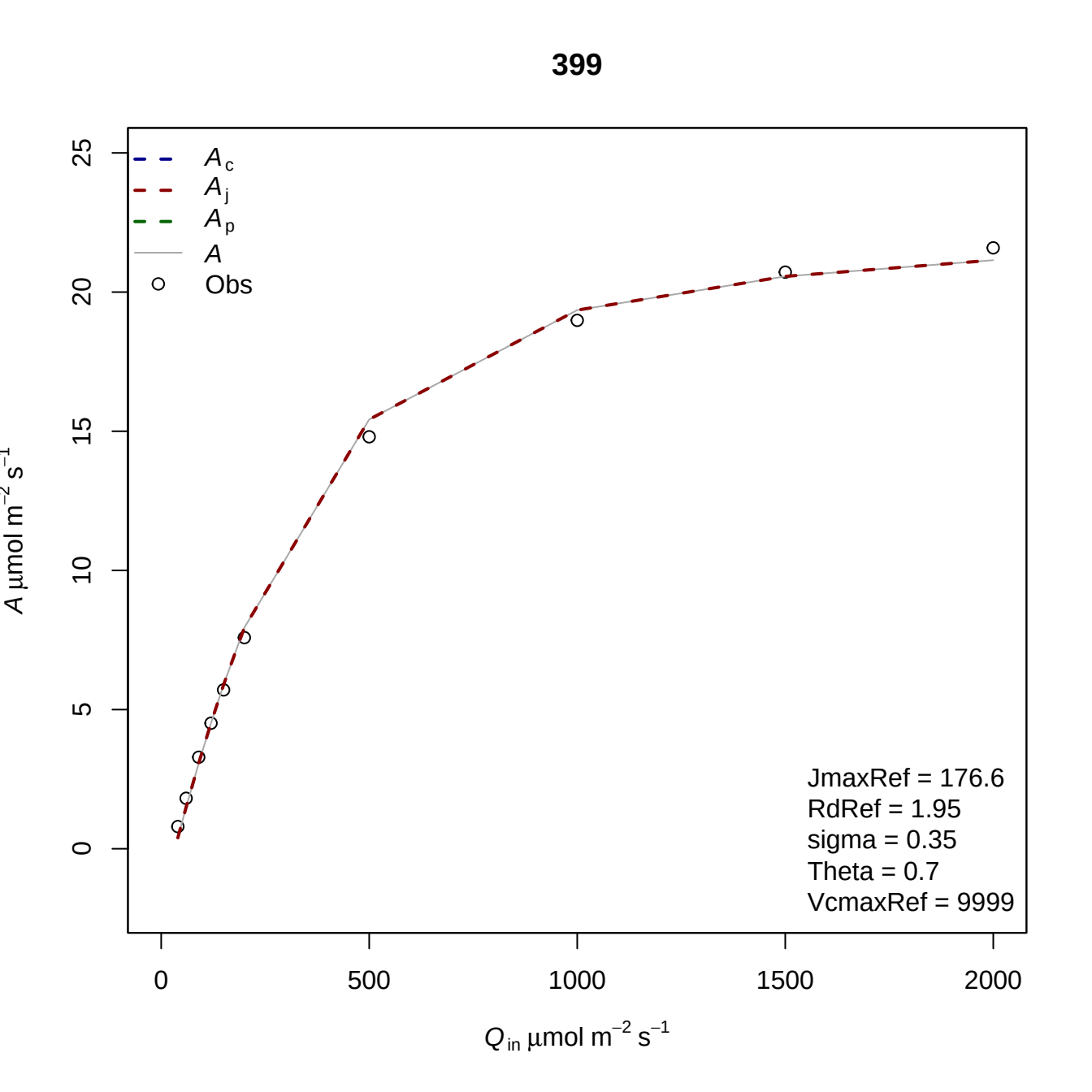

393
A_c
A_j
A_p
A
Obs
JmaxRef = 107.18
RdRef = 2.88
sigma = 0.73
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

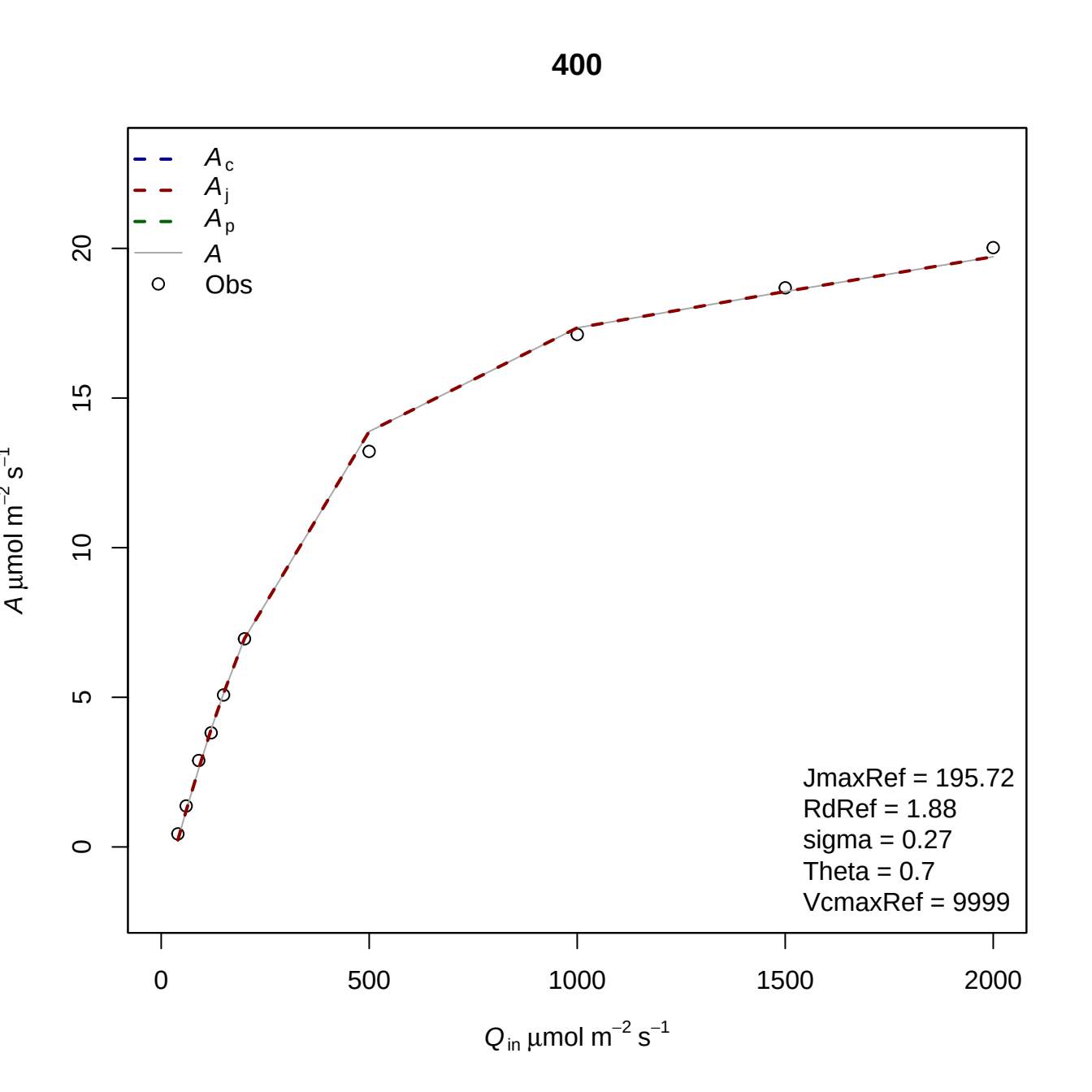

395
A_c
A_j
A_p
A
Obs
JmaxRef = 83.3
RdRef = 2.63
sigma = 0.82
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

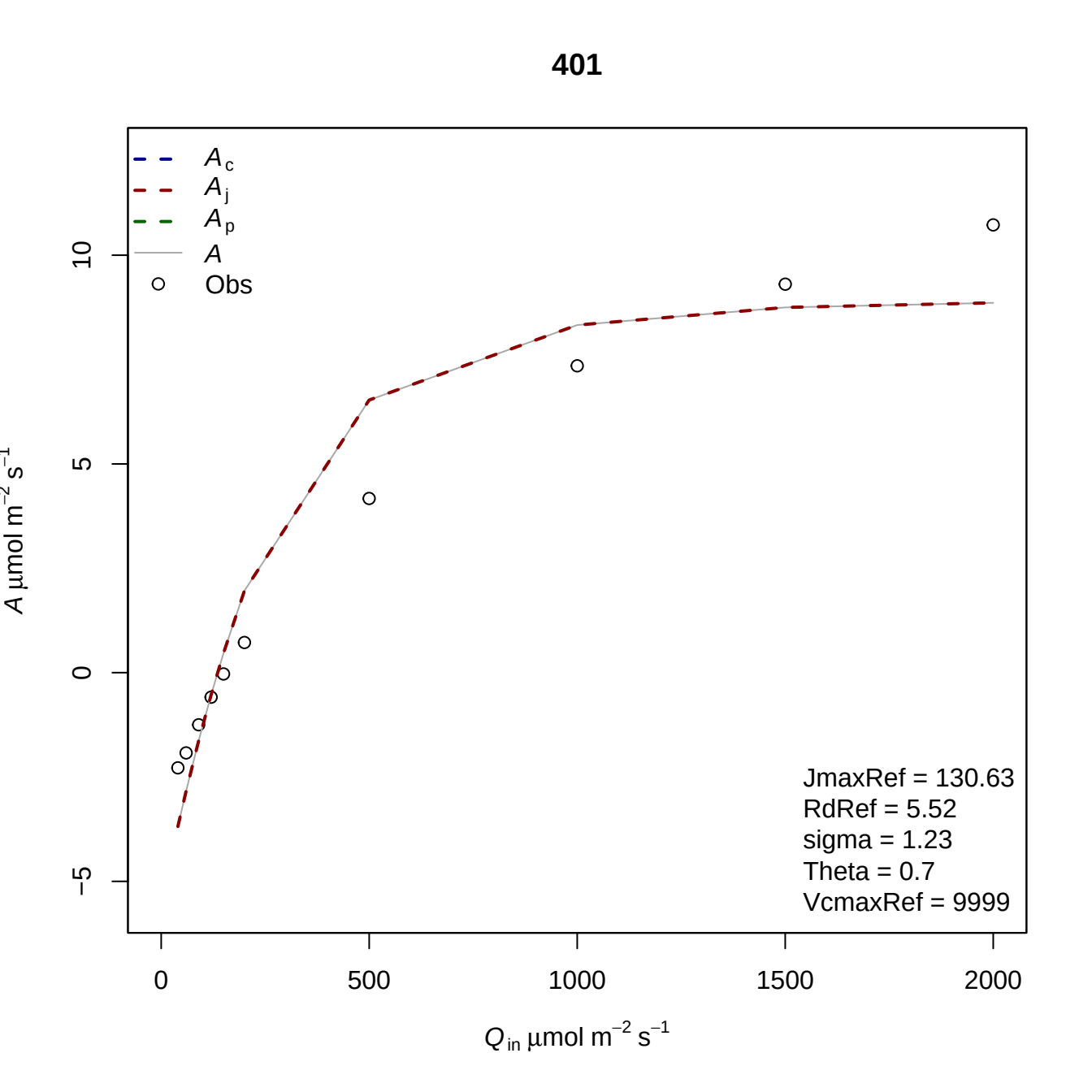

396
A_c
A_j
A_p
A
Obs
JmaxRef = 123.37
RdRef = 1.55
sigma = 0.66
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

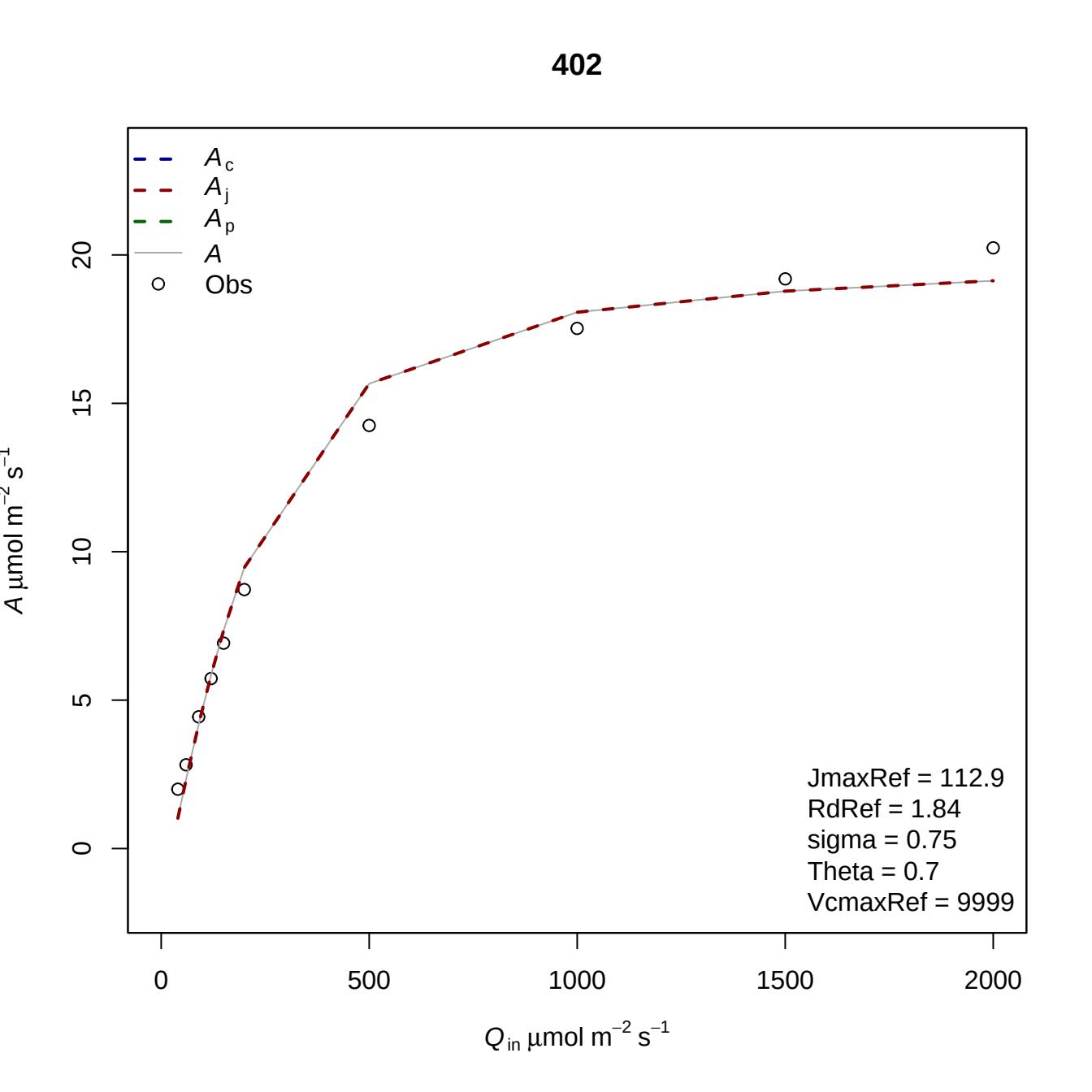

397
A_c
A_j
A_p
A
Obs
JmaxRef = 173.57
RdRef = 1.08
sigma = 0.34
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

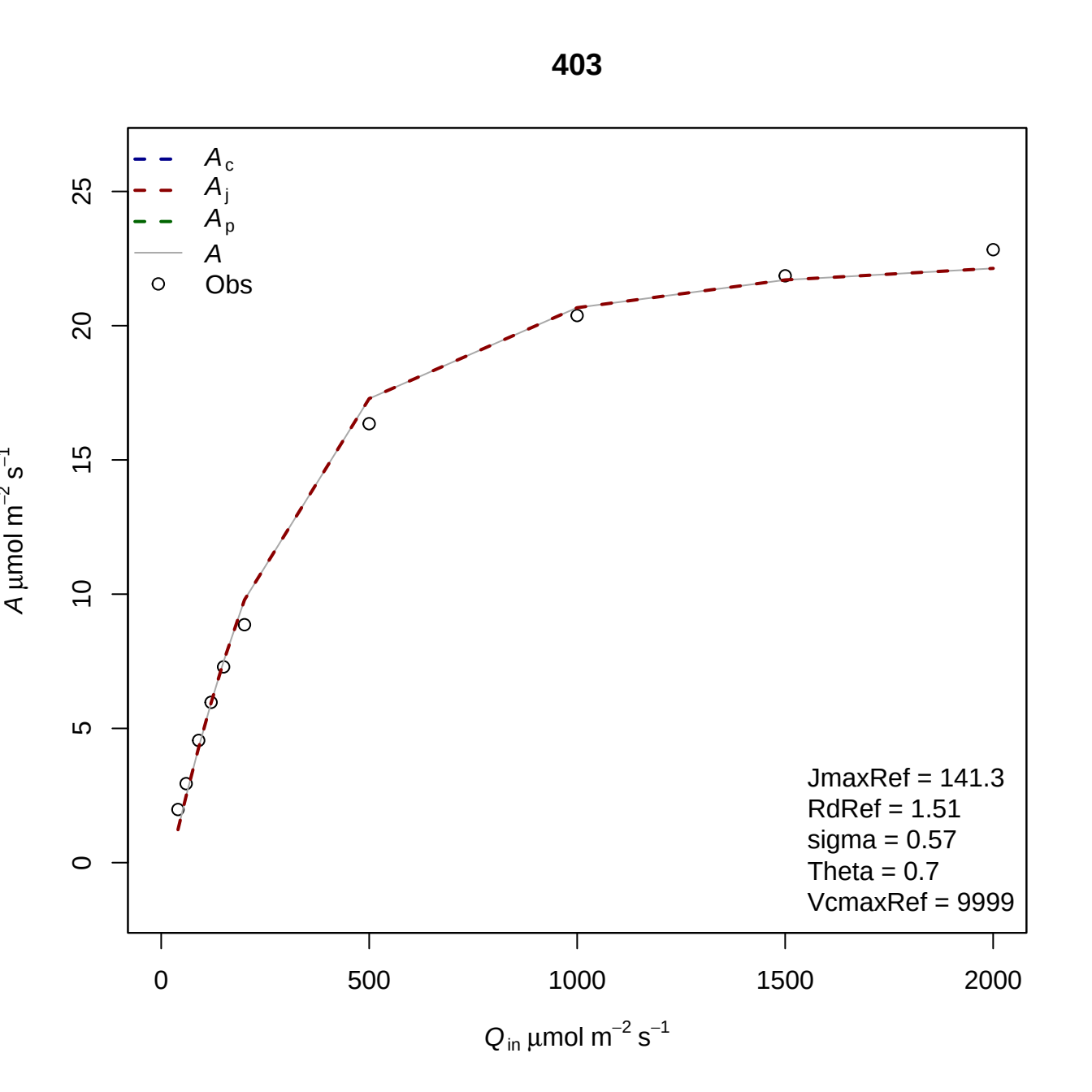

398
A_c
A_j
A_p
A
Obs
JmaxRef = 171.17
RdRef = 2.37
sigma = 1.08
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

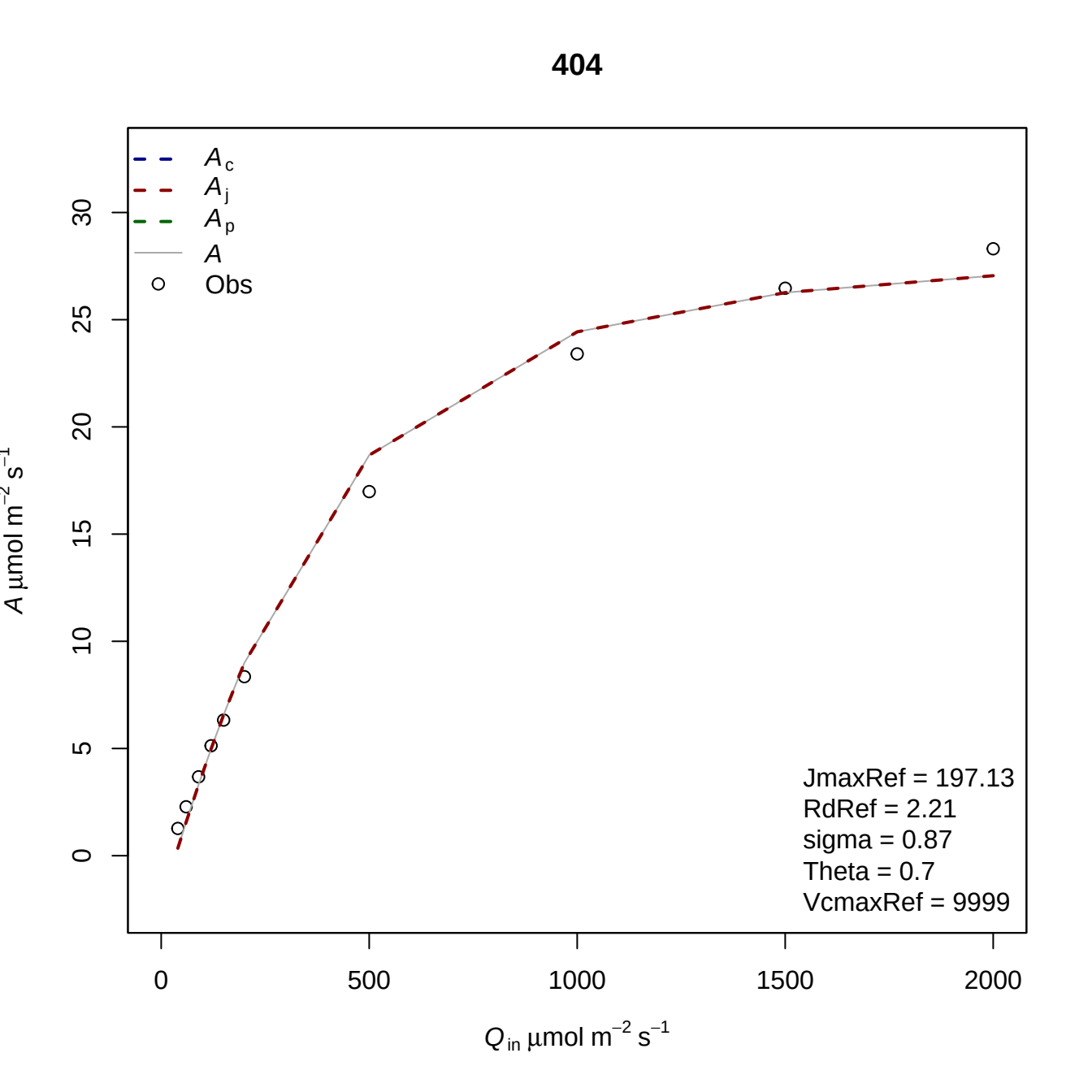

399
A_c
A_j
A_p
A
Obs
JmaxRef = 176.6
RdRef = 1.95
sigma = 0.35
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

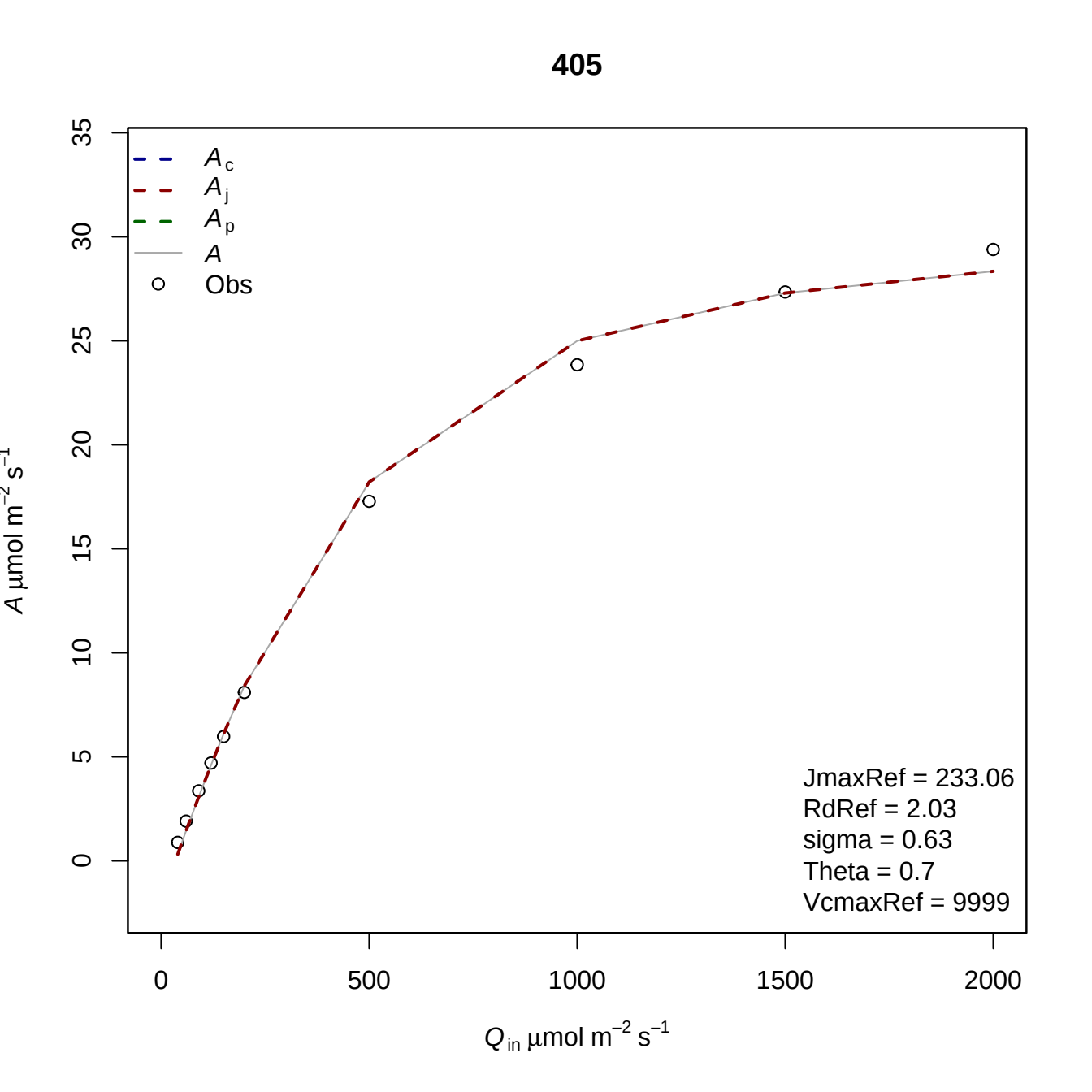

400
A_c
A_j
A_p
A
Obs
JmaxRef = 195.72
RdRef = 1.88
sigma = 0.27
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

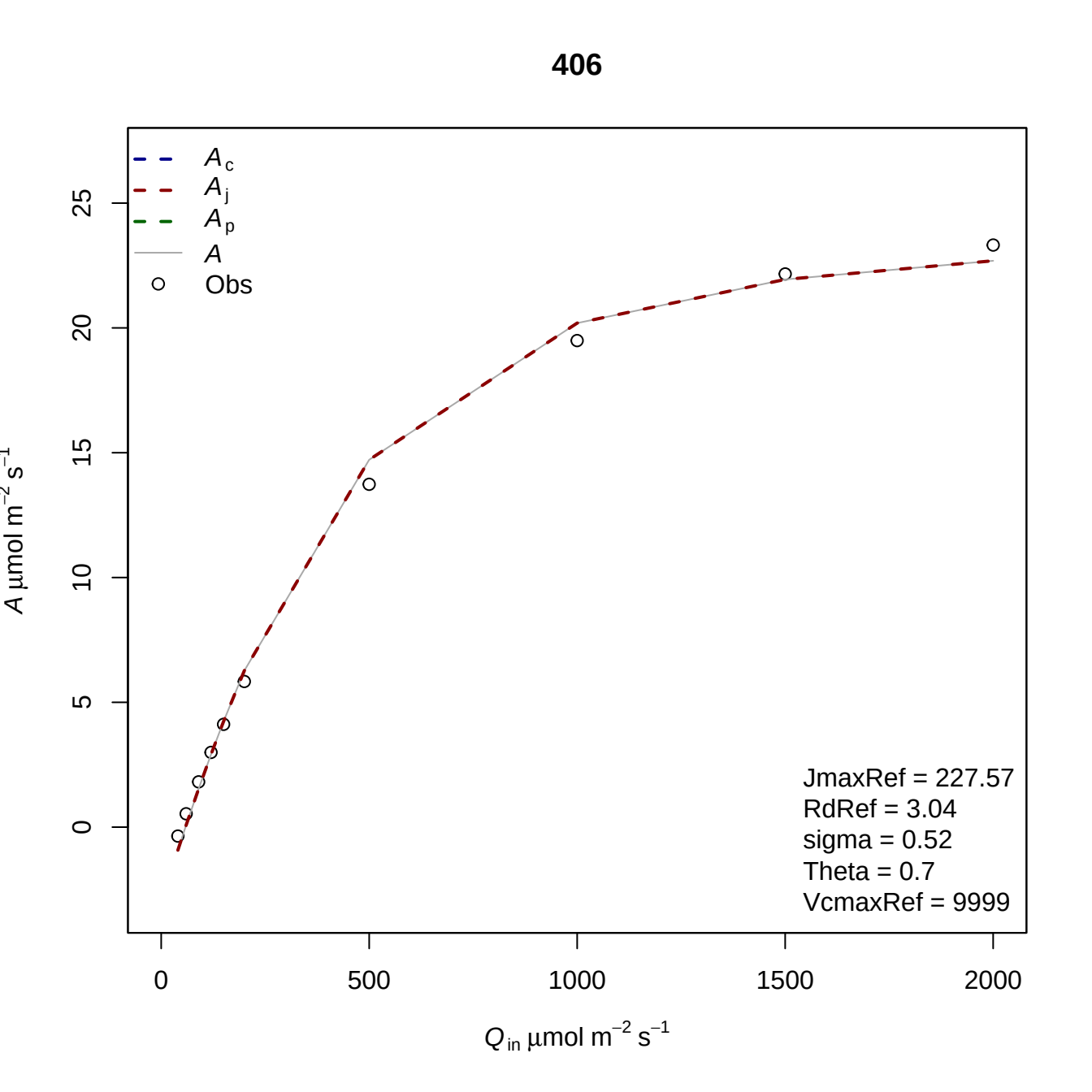

401
A_c
A_j
A_p
A
Obs
JmaxRef = 130.63
RdRef = 5.52
sigma = 1.23
Theta = 0.7
VcmaxRef = 9999
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

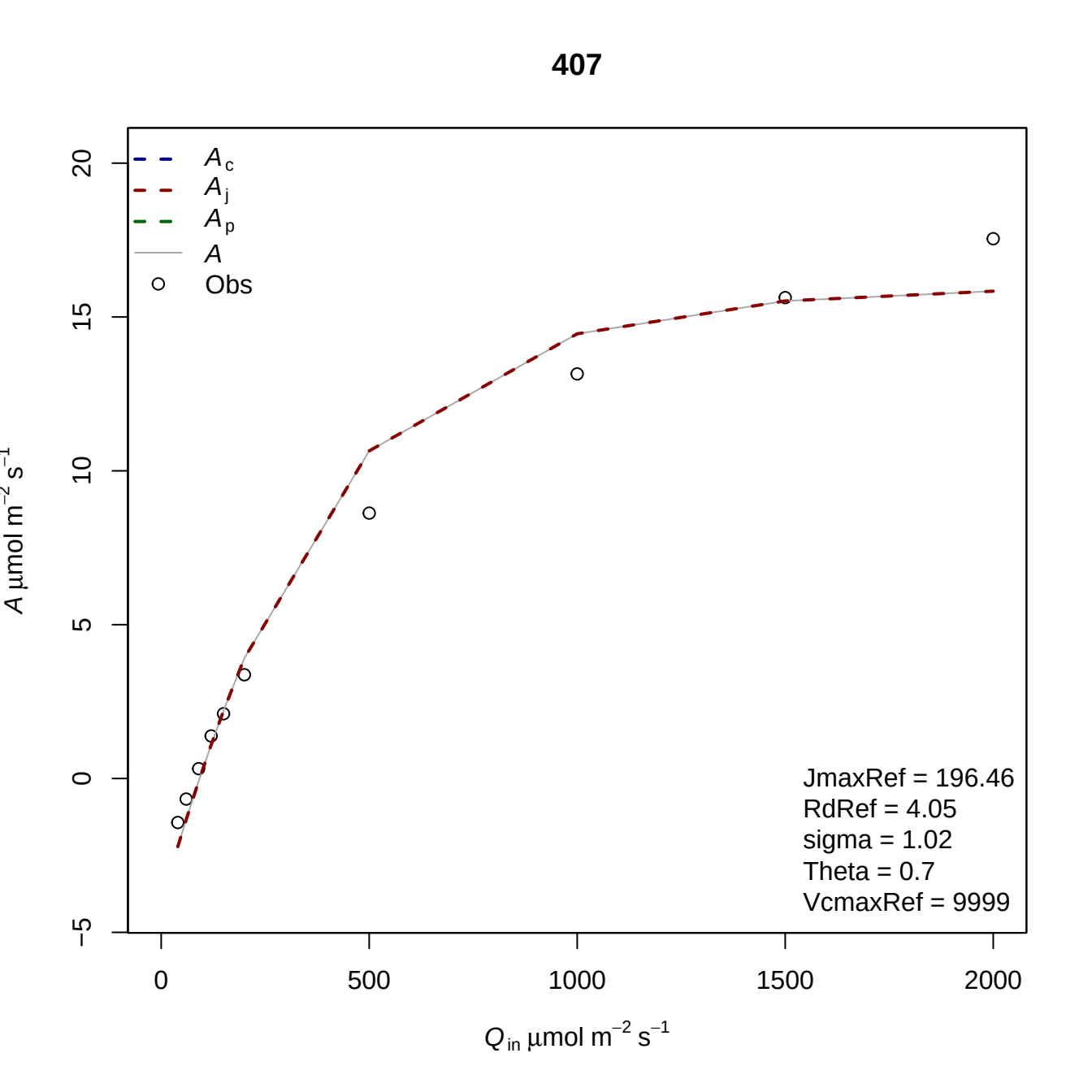

402
A_c
A_j
A_p
A
Obs
JmaxRef = 112.9
RdRef = 1.84
sigma = 0.75
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

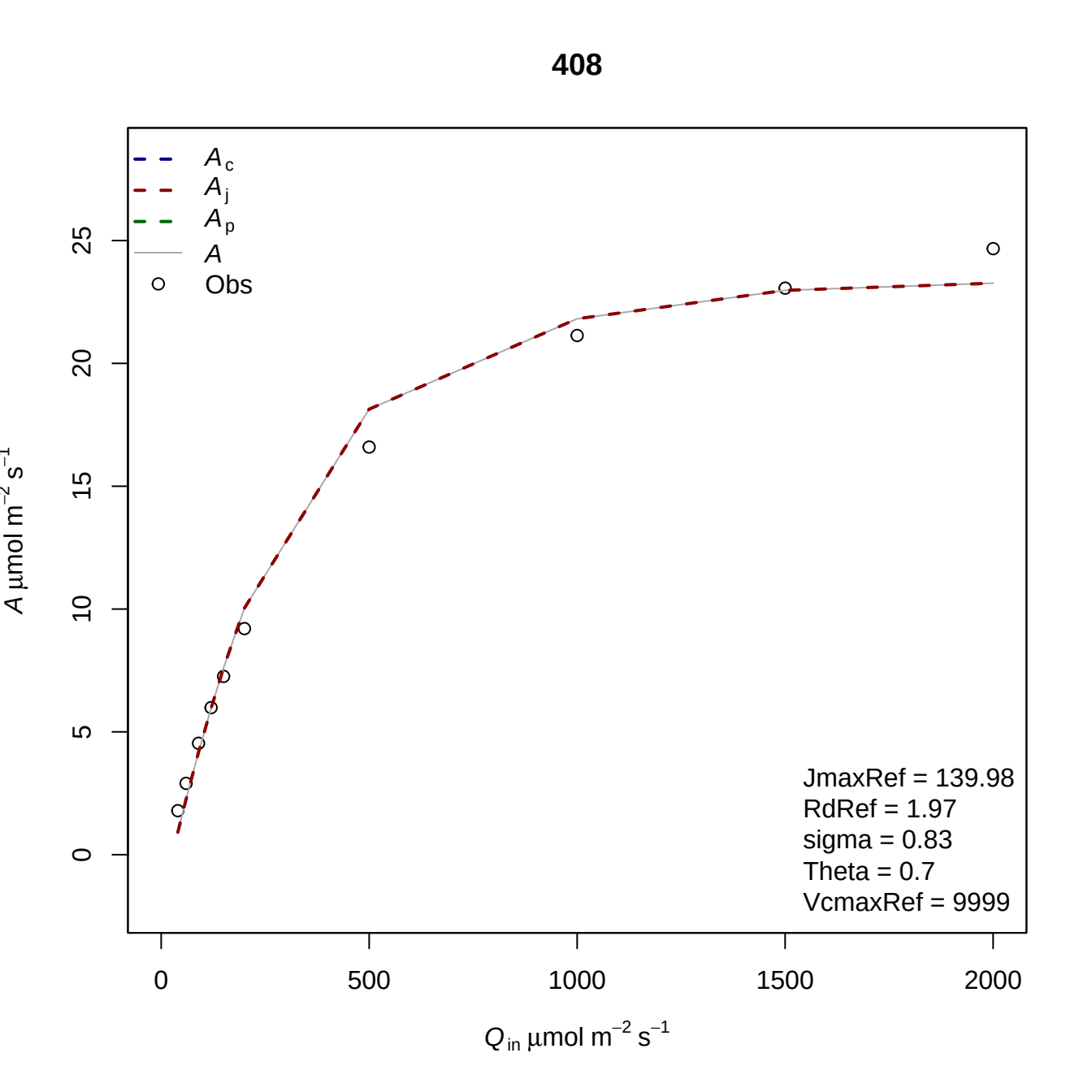

403
A_c
A_j
A_p
A
Obs
JmaxRef = 141.3
RdRef = 1.51
sigma = 0.57
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

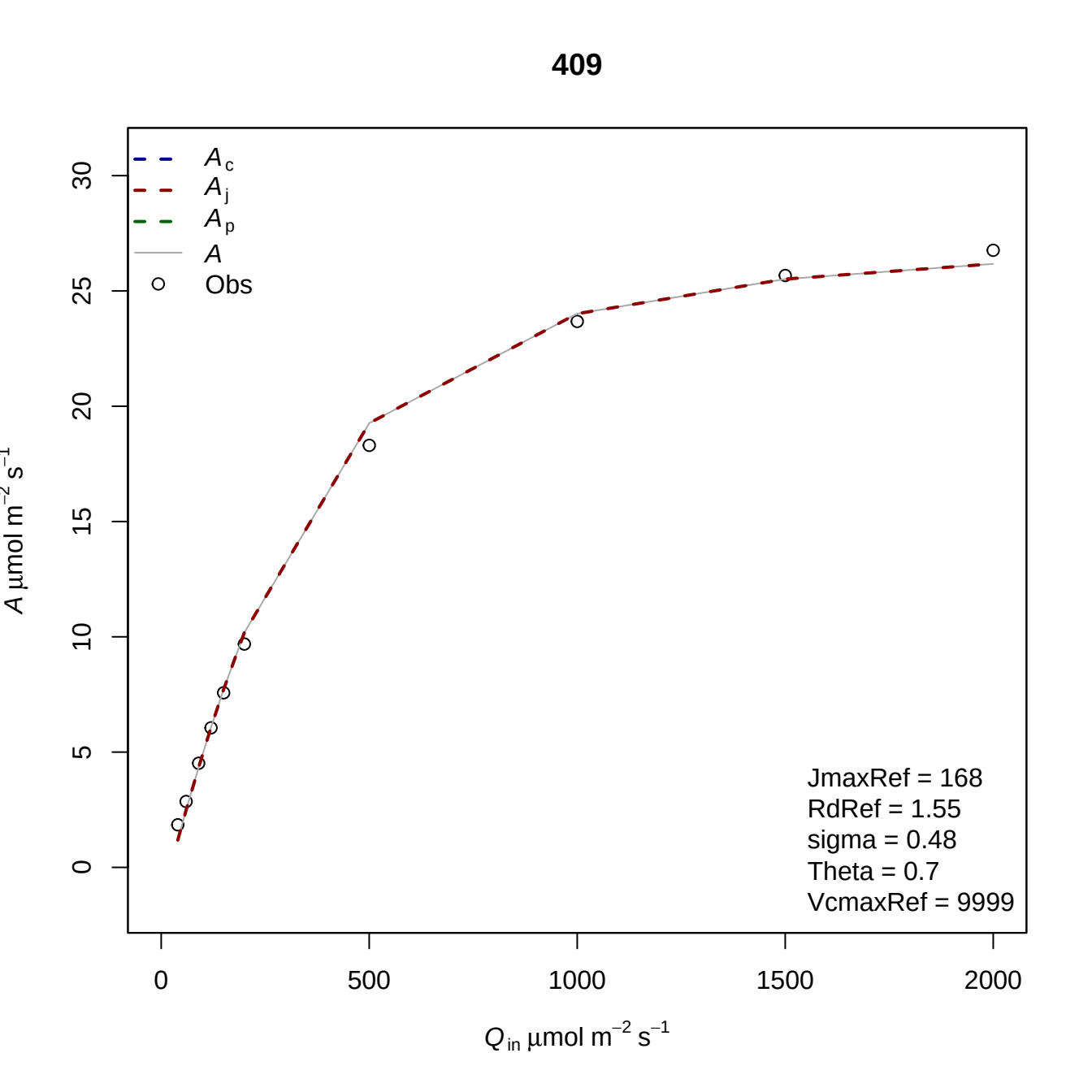

404
A_c
A_j
A_p
A
Obs
JmaxRef = 197.13
RdRef = 2.21
sigma = 0.87
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

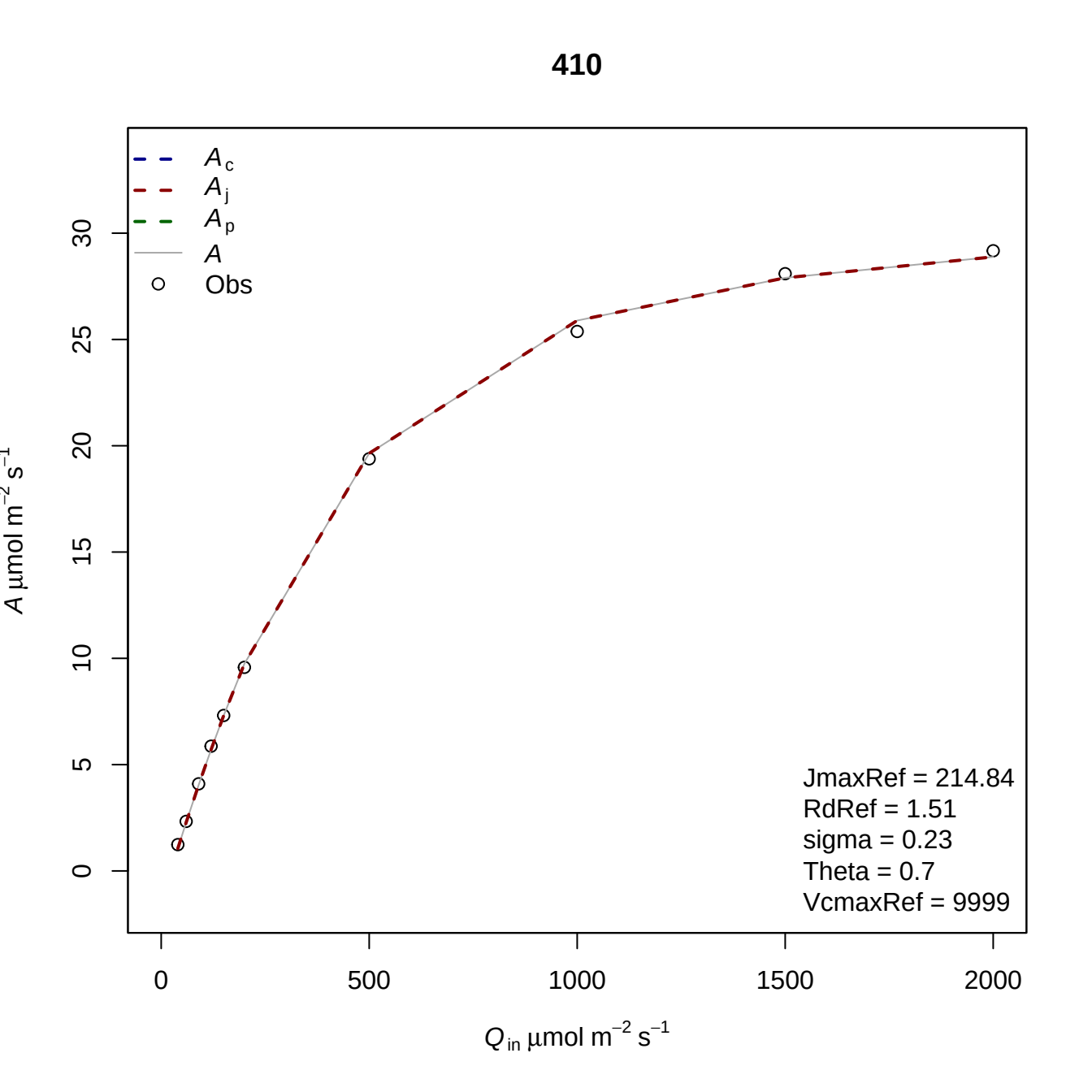

405
A_c
A_j
A_p
A
Obs
JmaxRef = 233.06
RdRef = 2.03
sigma = 0.63
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

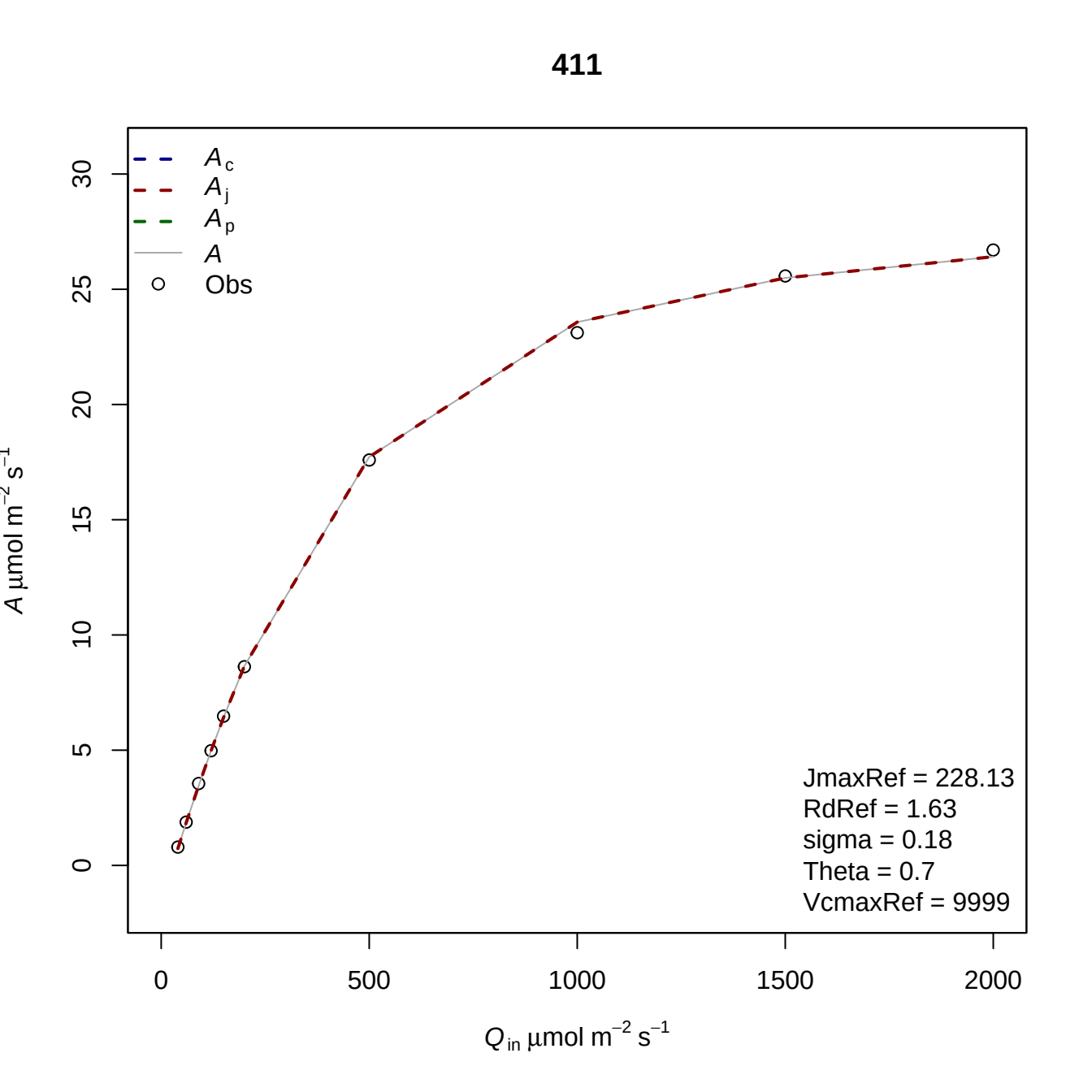

406
A_c
A_j
A_p
A
Obs
JmaxRef = 227.57
RdRef = 3.04
sigma = 0.52
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

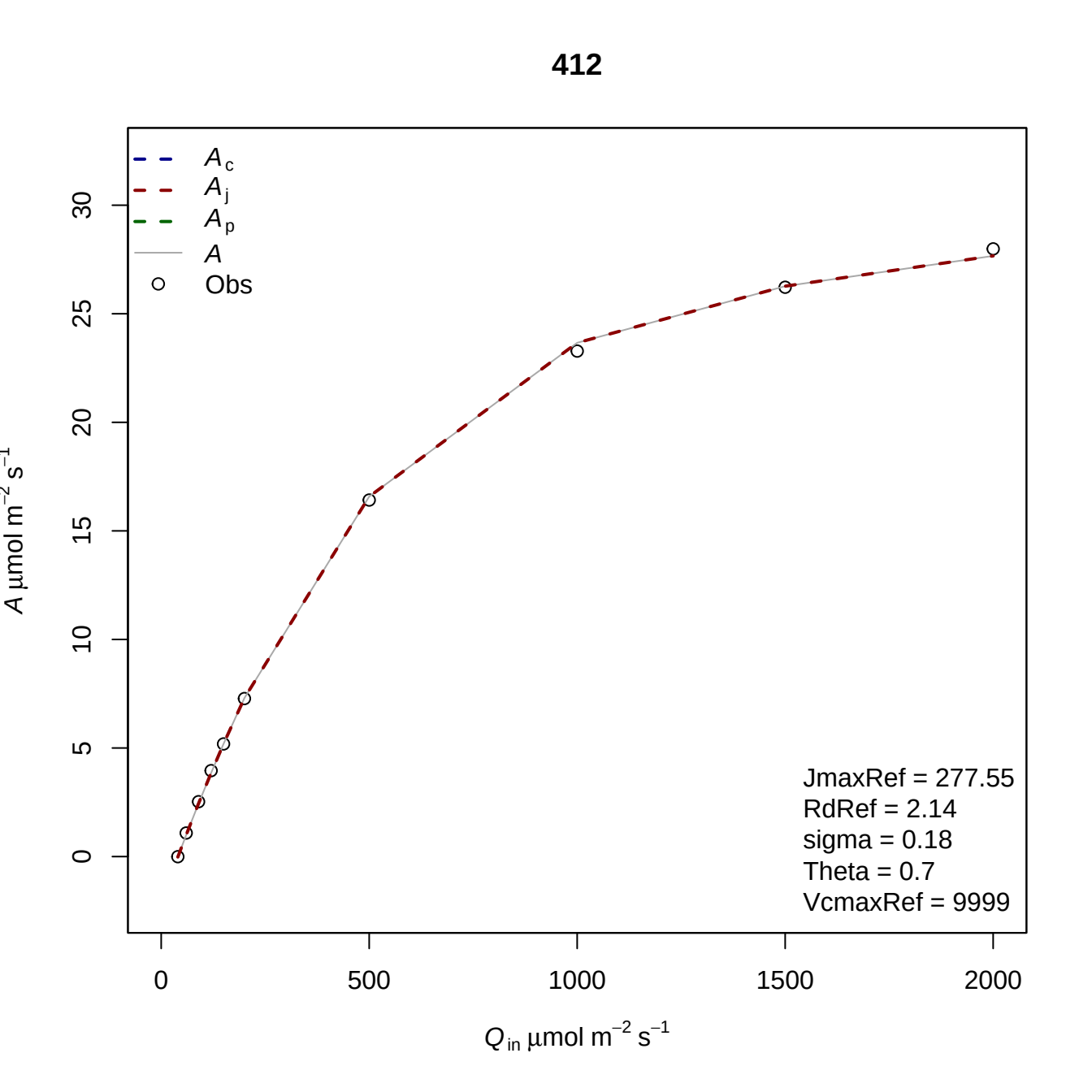

407
A_c
A_j
A_p
A
Obs
JmaxRef = 196.46
RdRef = 4.05
sigma = 1.02
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
20
15
10
5
0
−5
0
500
1000
1500
2000

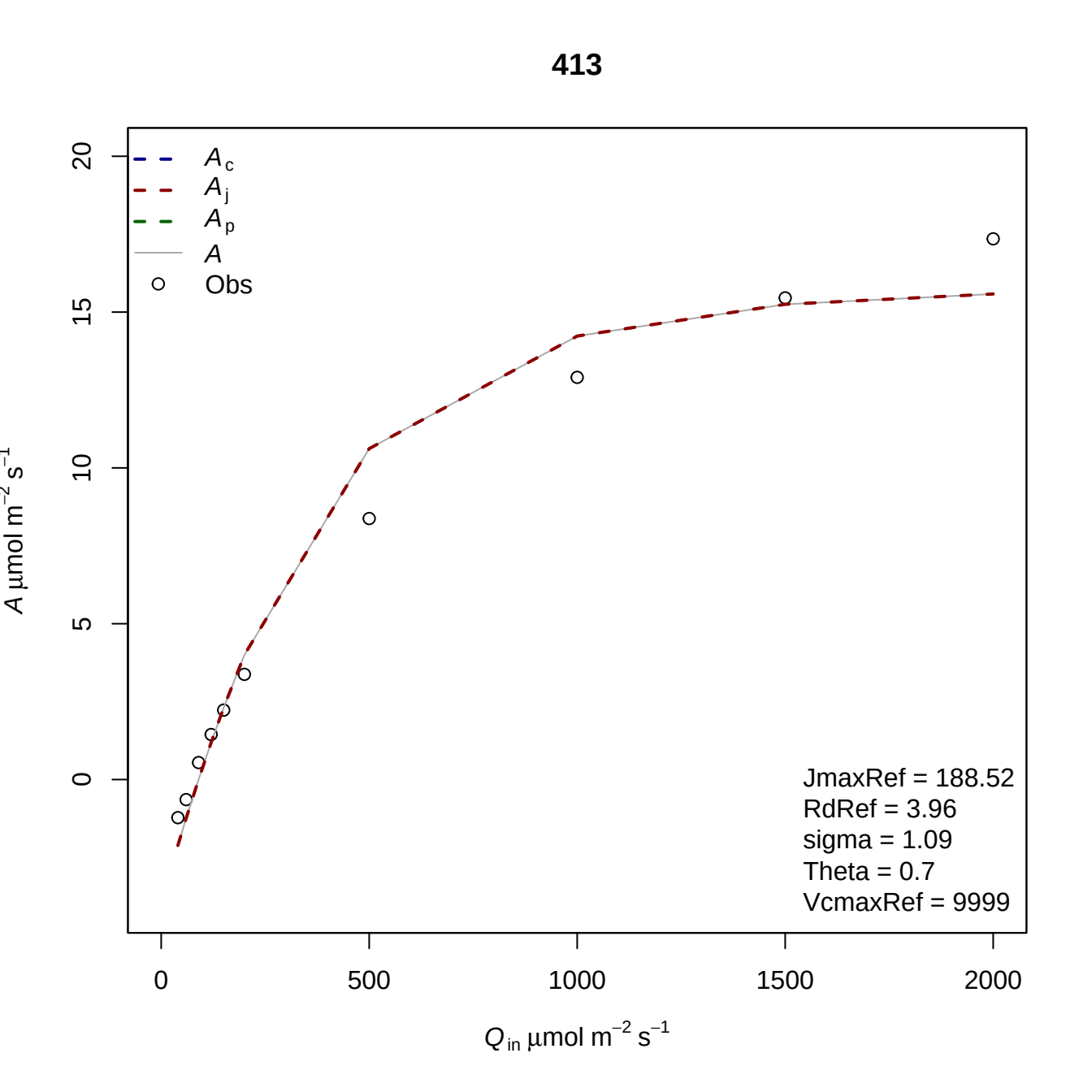

408
A_c
A_j
A_p
A
Obs
JmaxRef = 139.98
RdRef = 1.97
sigma = 0.83
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

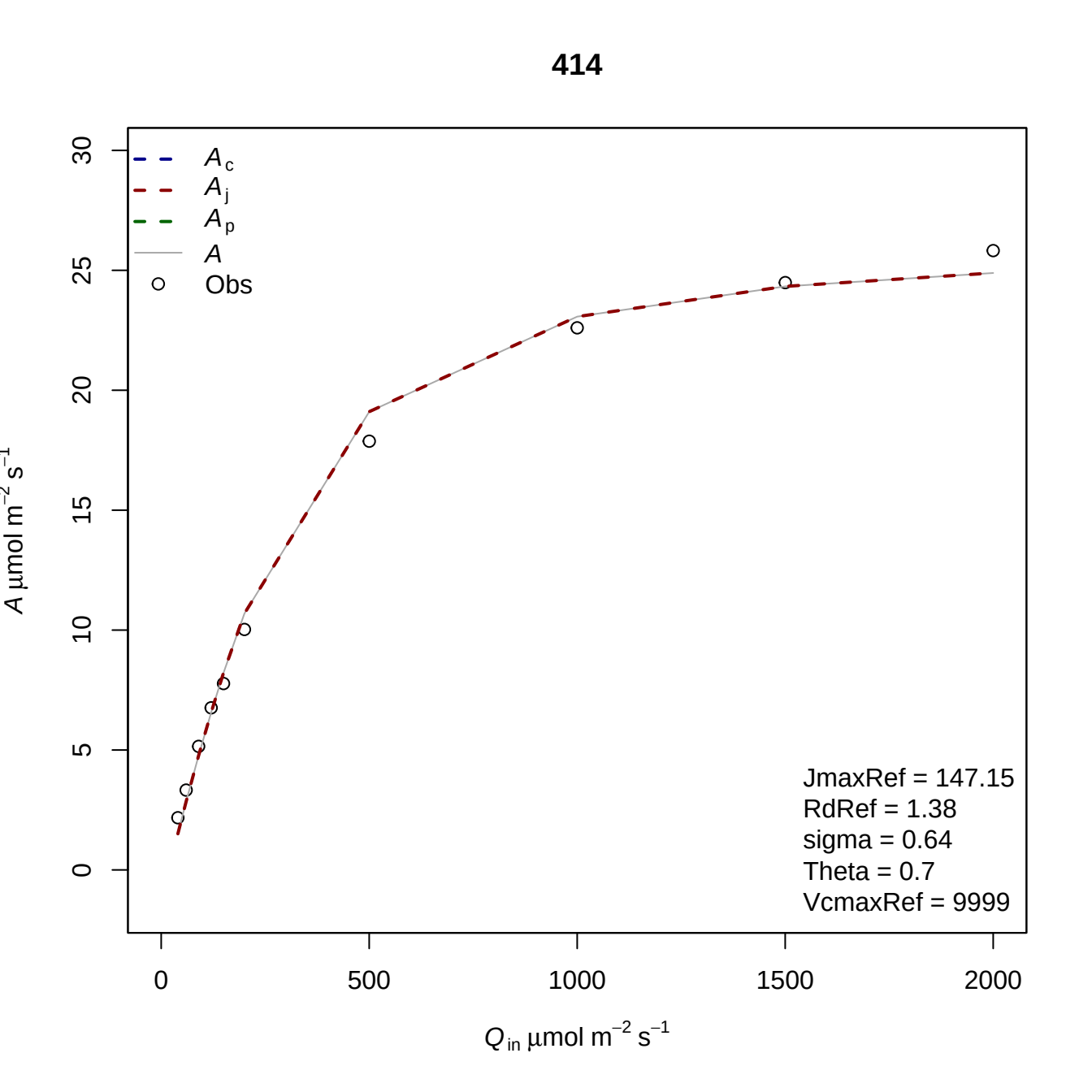

409
A_c
A_j
A_p
A
Obs
JmaxRef = 168
RdRef = 1.55
sigma = 0.48
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

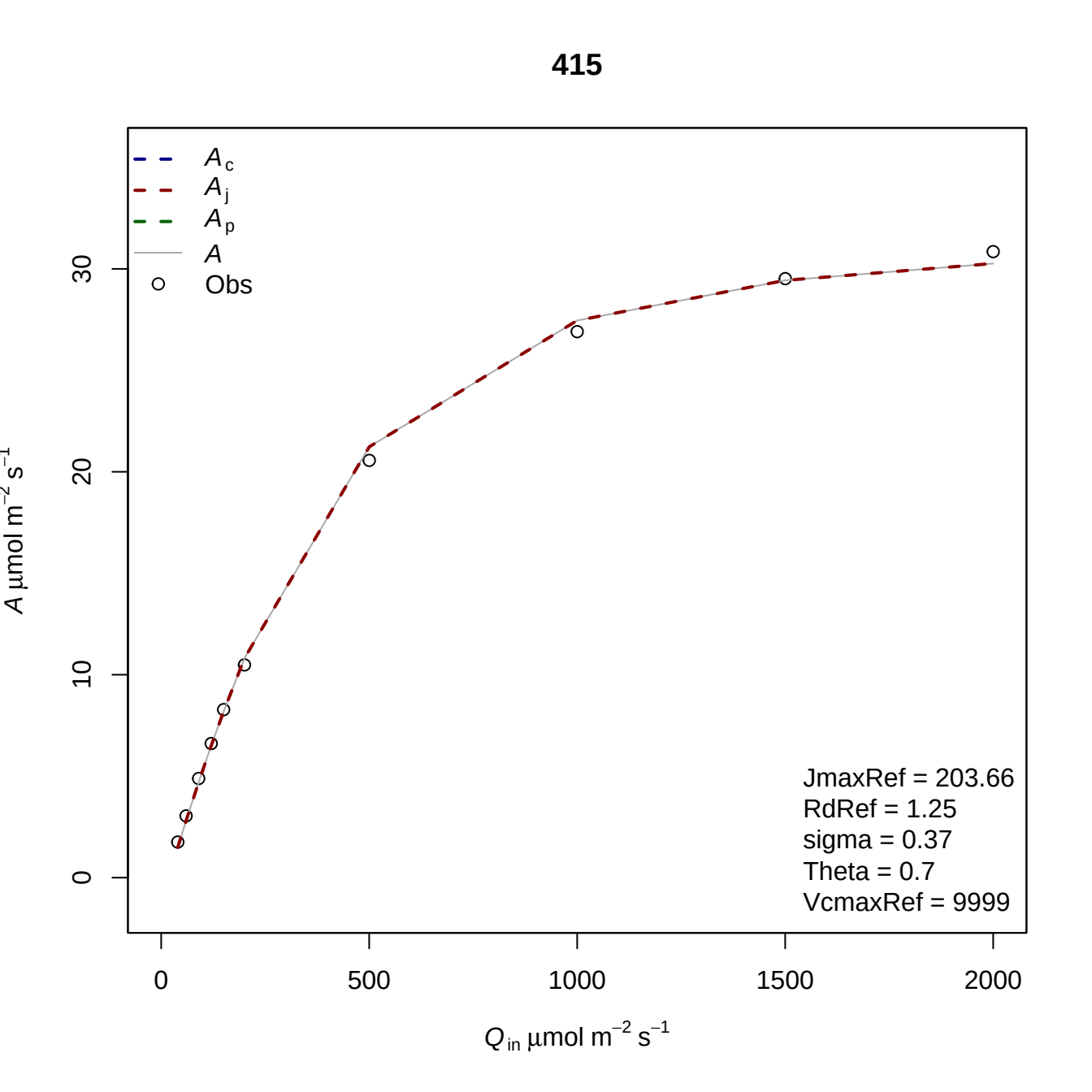

410
A_c
A_j
A_p
A
Obs
JmaxRef = 214.84
RdRef = 1.51
sigma = 0.23
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

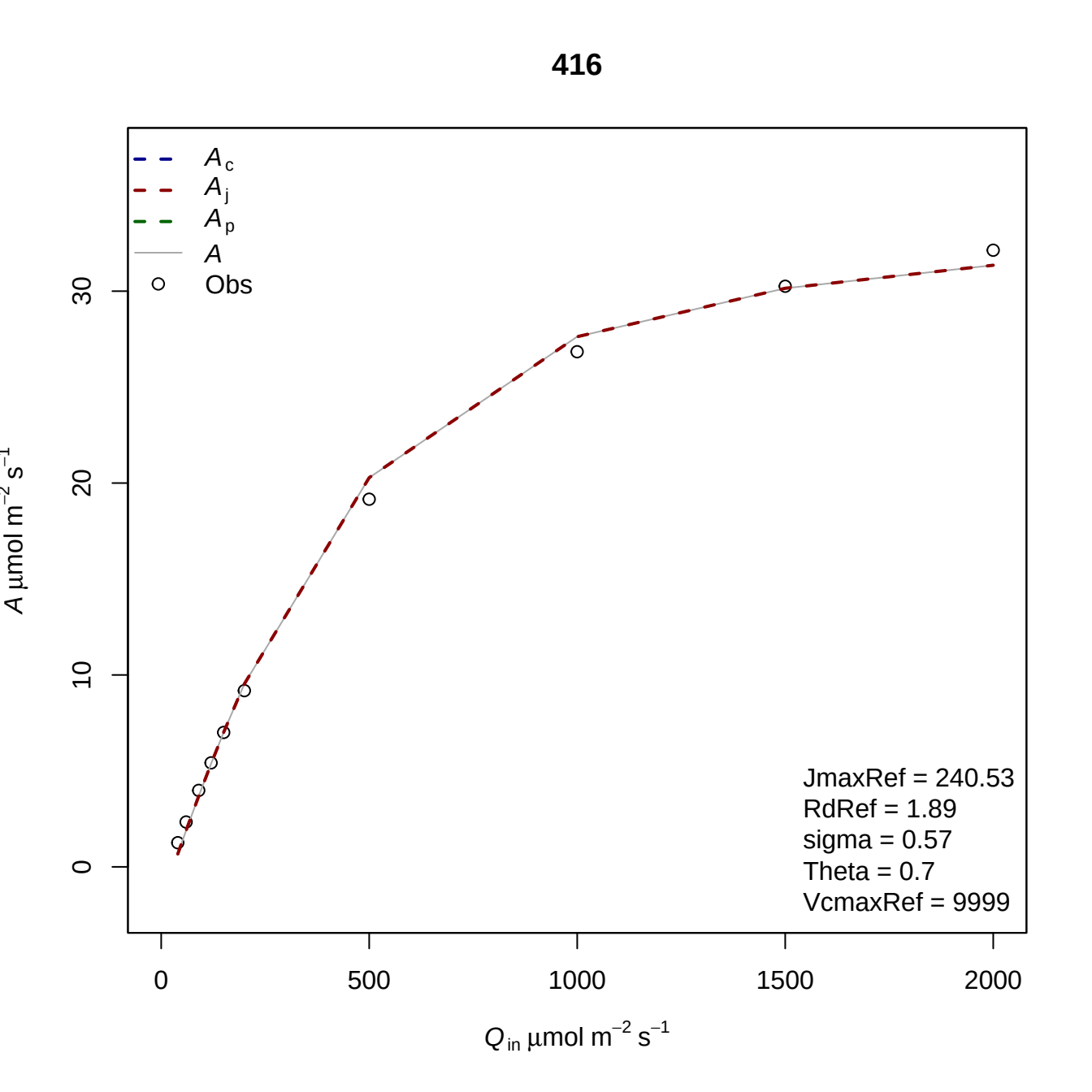

411
A_c
A_j
A_p
A
Obs
JmaxRef = 228.13
RdRef = 1.63
sigma = 0.18
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

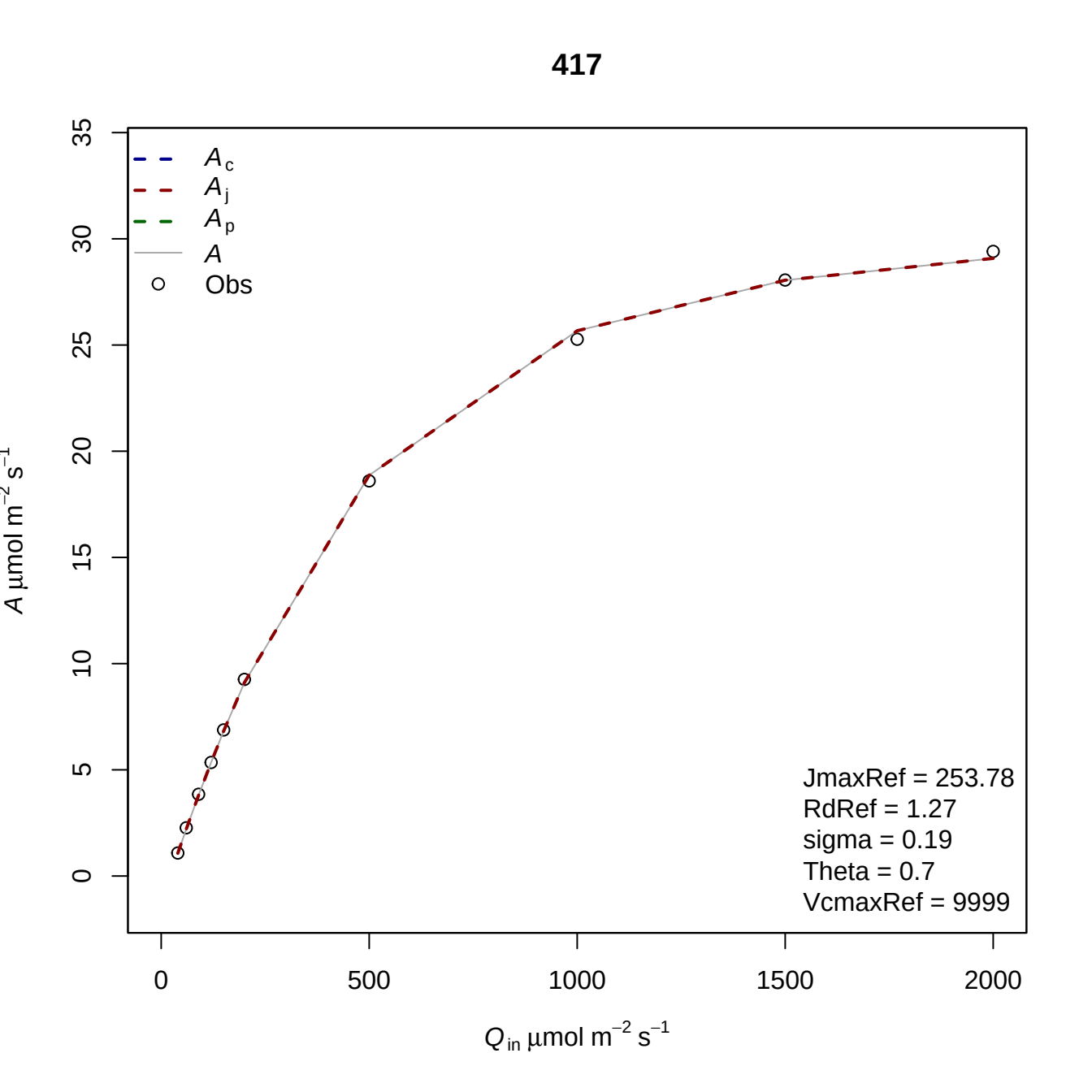

412
A_c
A_j
A_p
A
Obs
JmaxRef = 277.55
RdRef = 2.14
sigma = 0.18
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

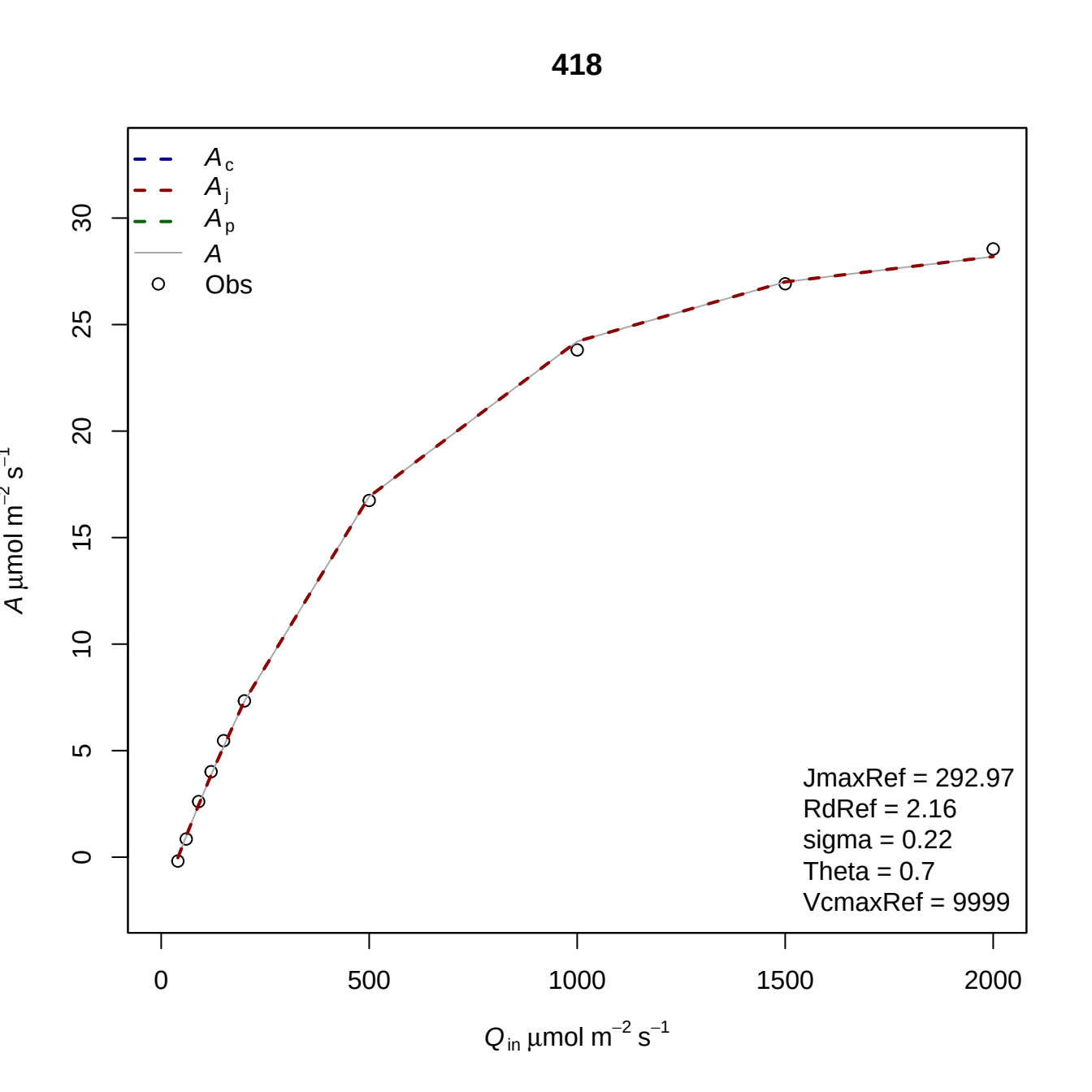

413
A_c
A_j
A_p
A
Obs
JmaxRef = 188.52
RdRef = 3.96
sigma = 1.09
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

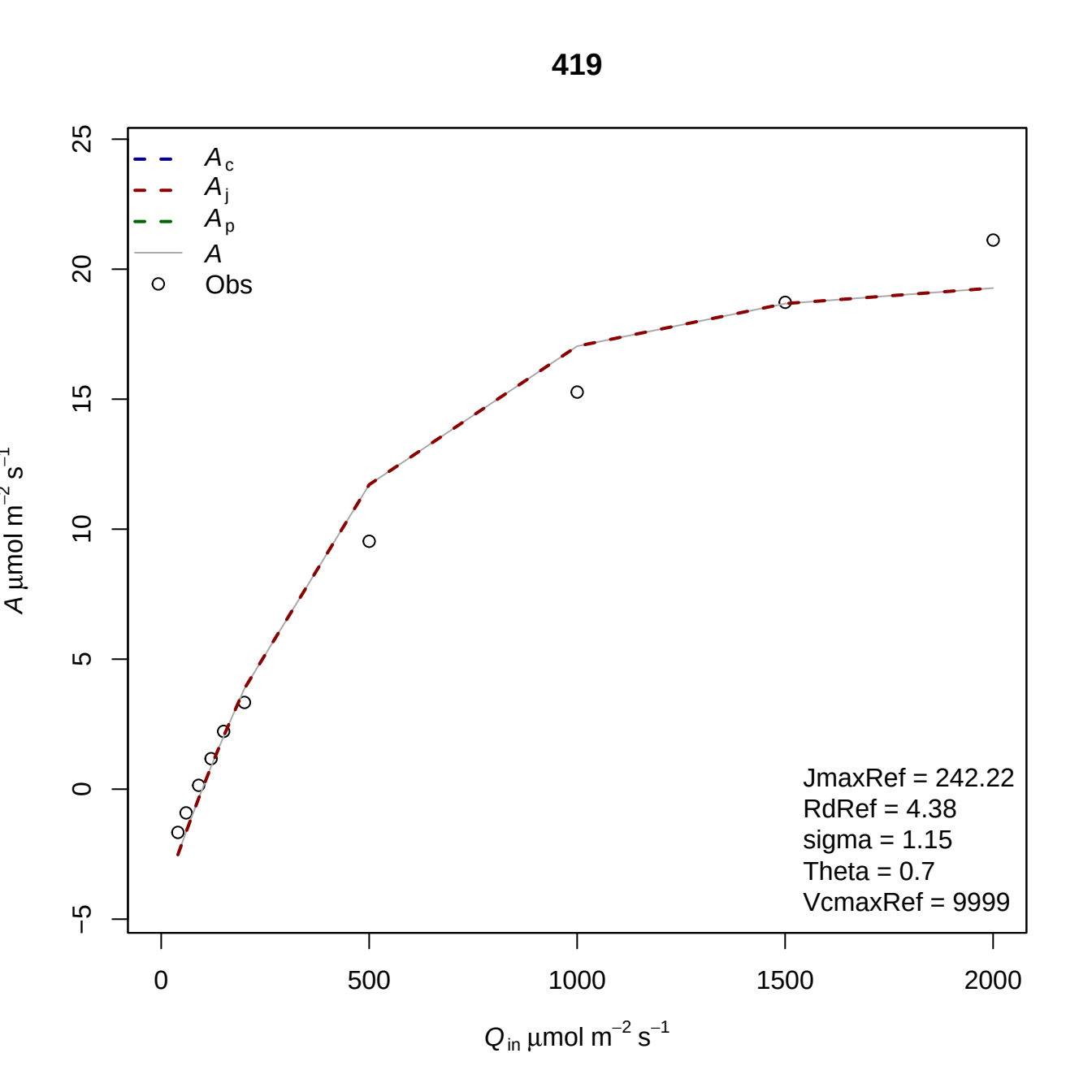

414
A_c
A_j
A_p
A
Obs
JmaxRef = 147.15
RdRef = 1.38
sigma = 0.64
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

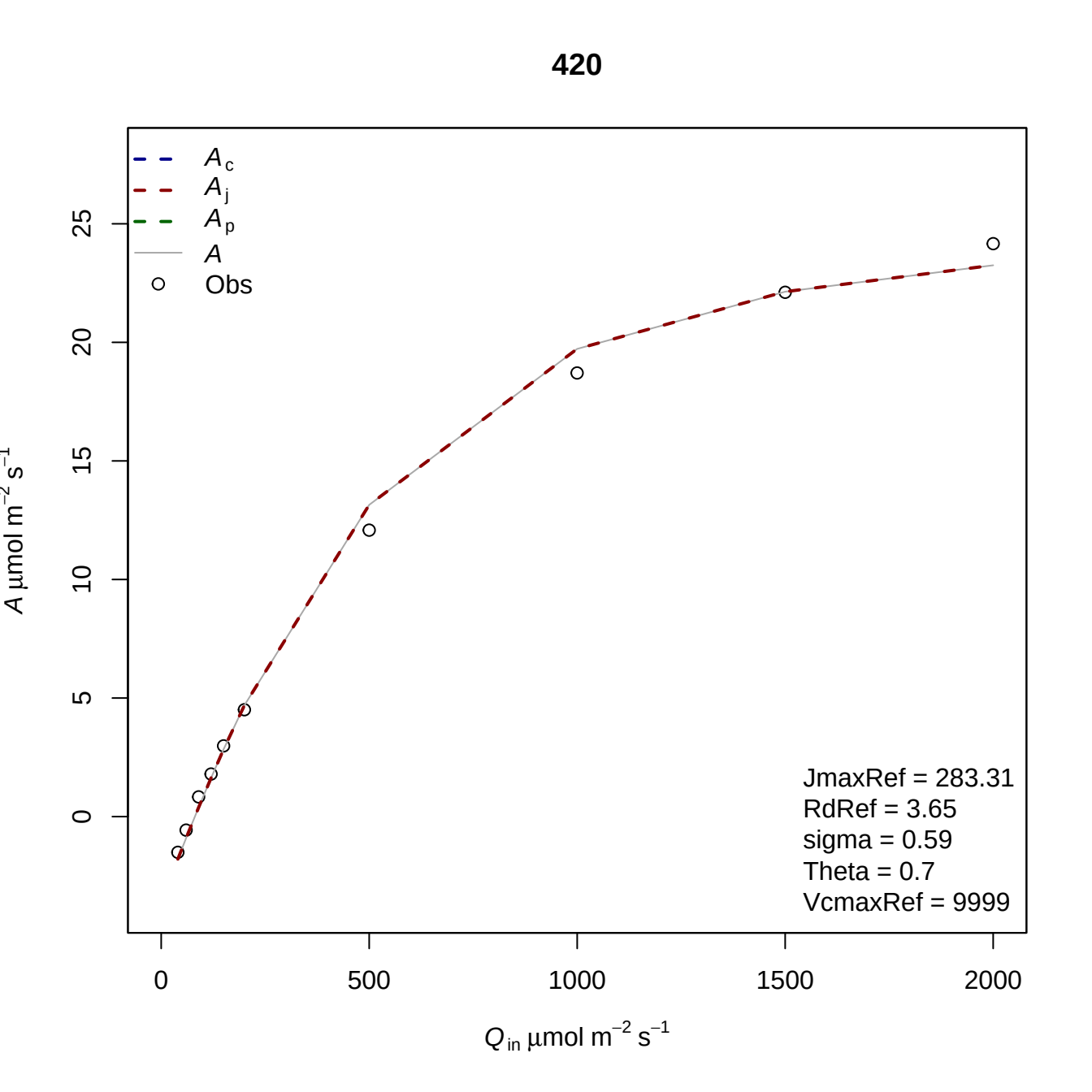

415
A_c
A_j
A_p
A
Obs
JmaxRef = 203.66
RdRef = 1.25
sigma = 0.37
Theta = 0.7
VcmaxRef = 9999
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

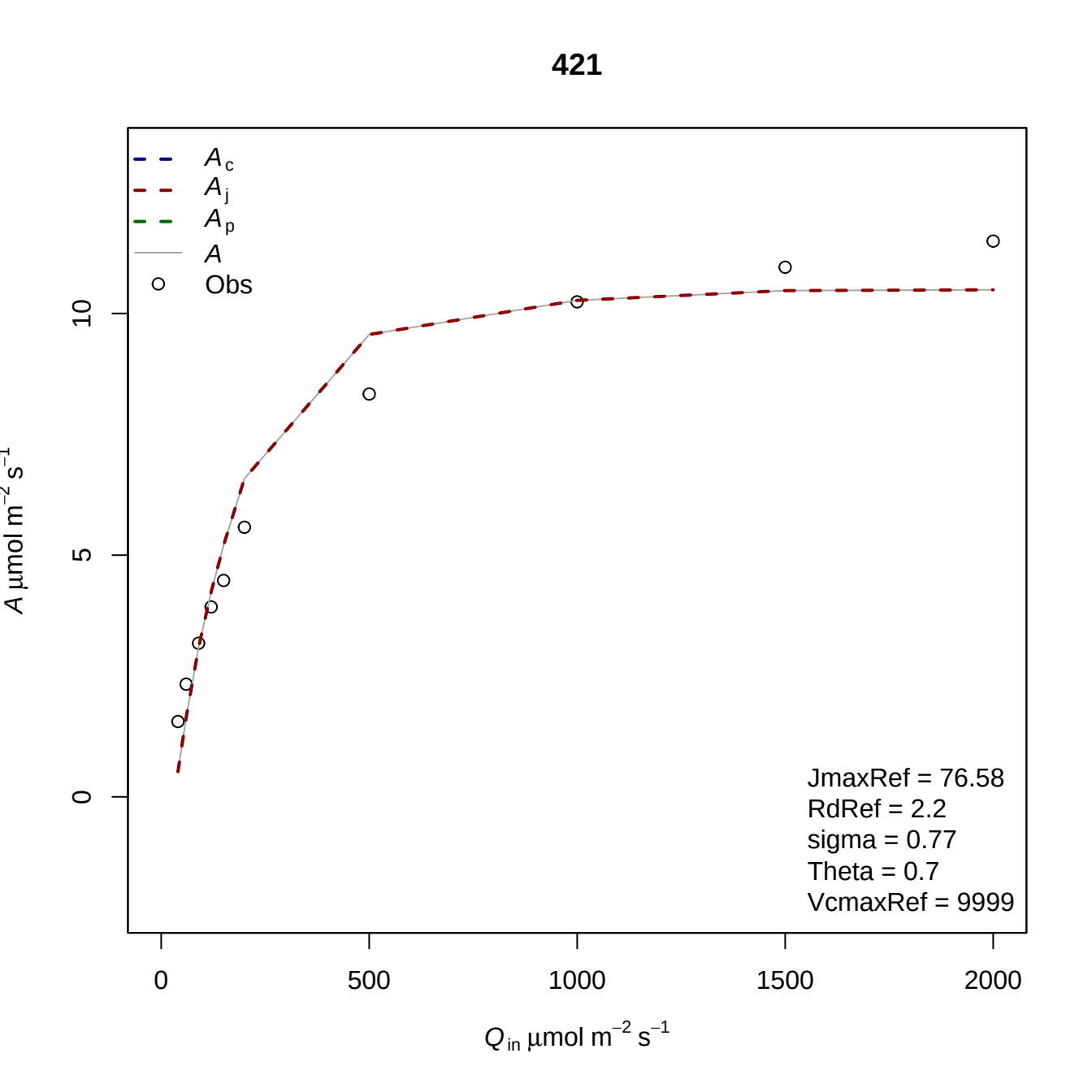

416
A_c
A_j
A_p
A
Obs
JmaxRef = 240.53
RdRef = 1.89
sigma = 0.57
Theta = 0.7
VcmaxRef = 9999
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

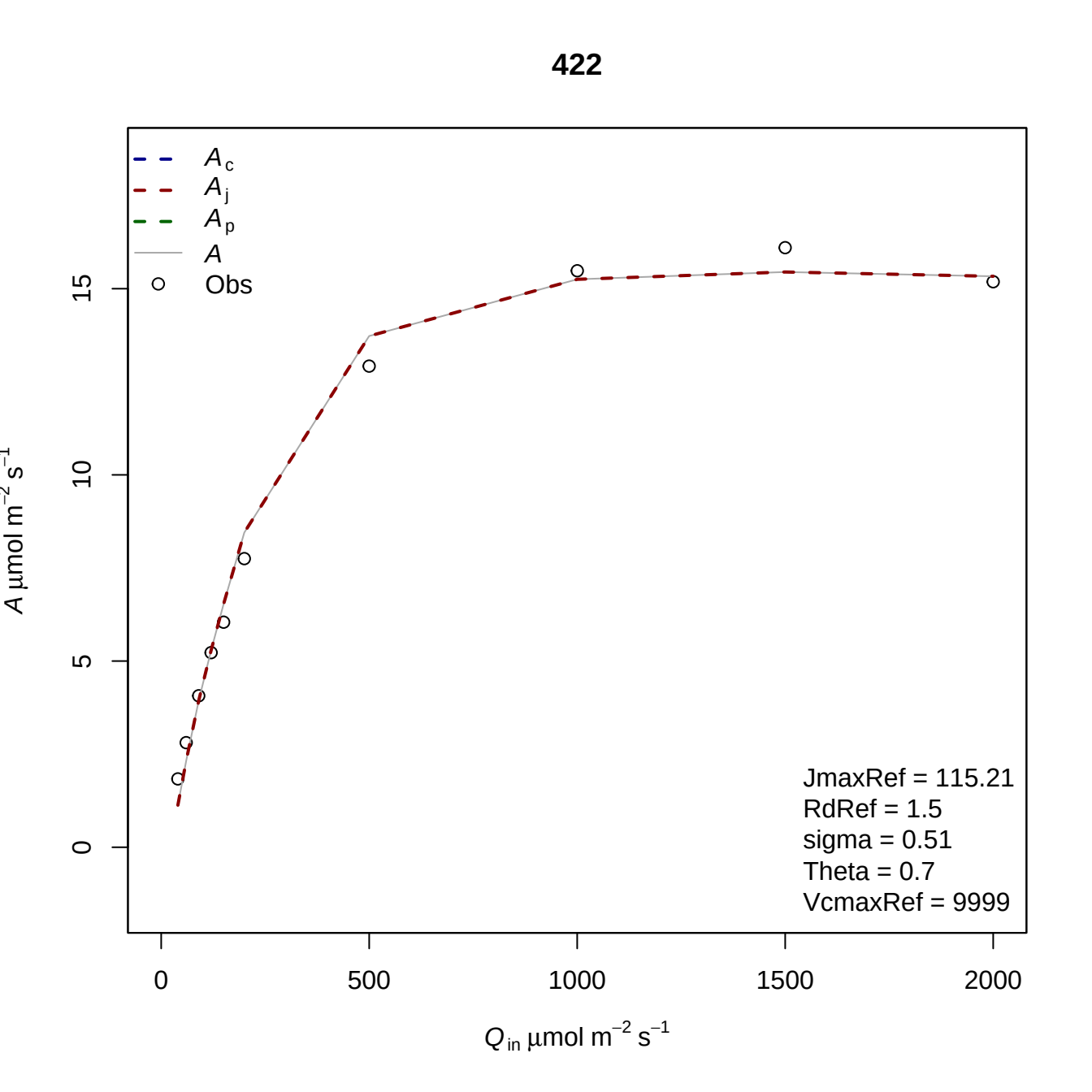

417
A_c
A_j
A_p
A
Obs
JmaxRef = 253.78
RdRef = 1.27
sigma = 0.19
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
25
30
35
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

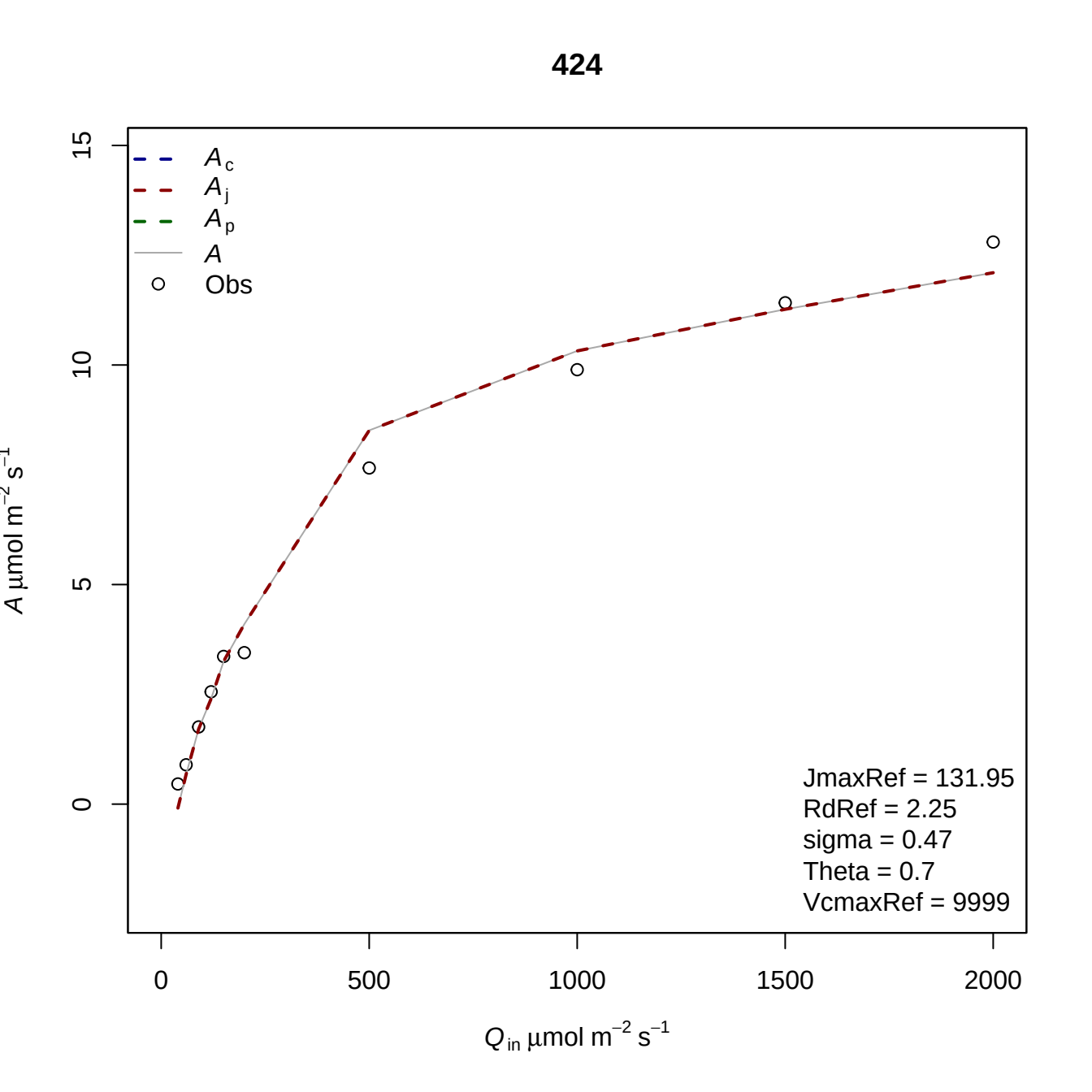

418
A_c
A_j
A_p
A
Obs
JmaxRef = 292.97
RdRef = 2.16
sigma = 0.22
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

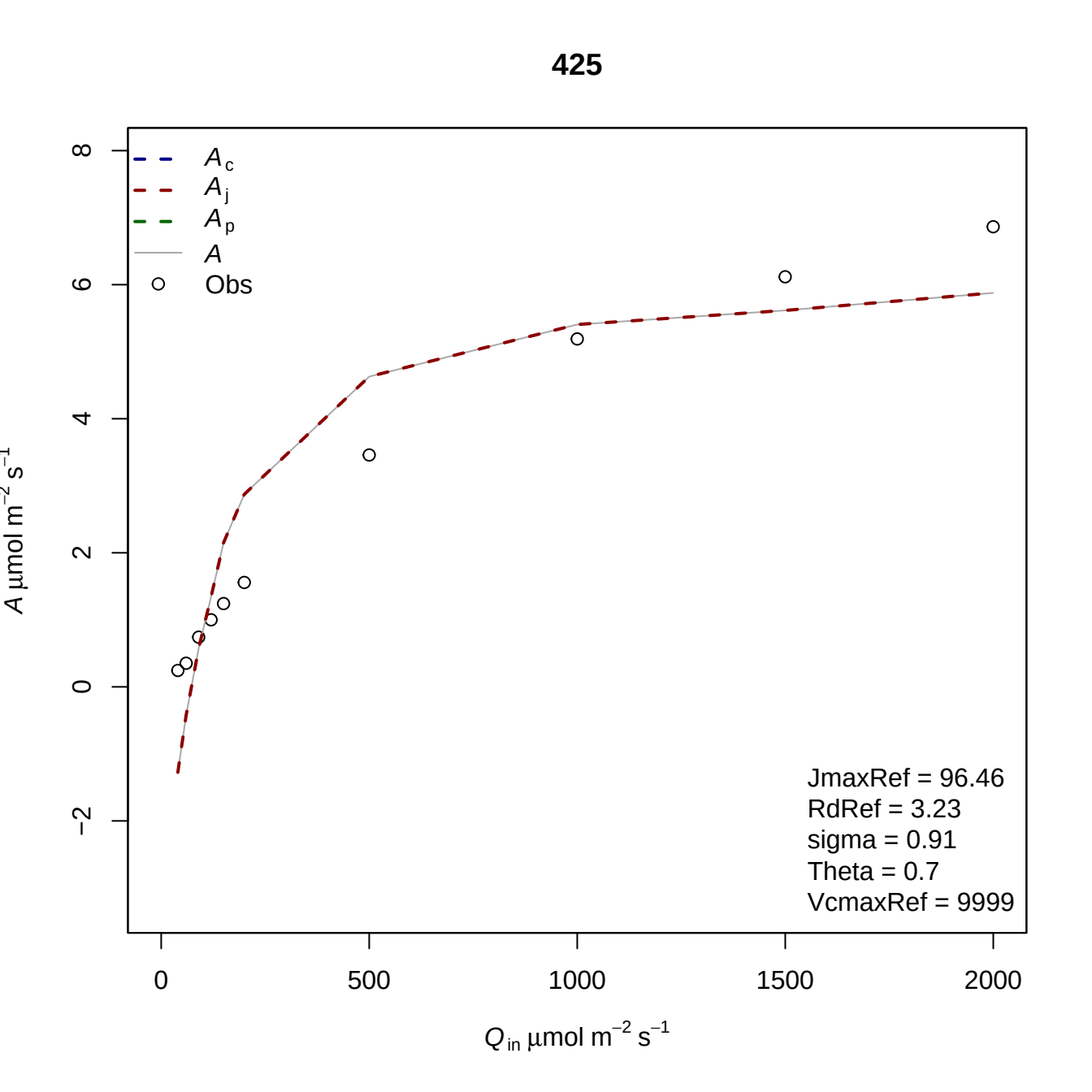

419
A_c
A_j
A_p
A
Obs
JmaxRef = 242.22
RdRef = 4.38
sigma = 1.15
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
25
20
15
10
5
0
−5
0
500
1000
1500
2000

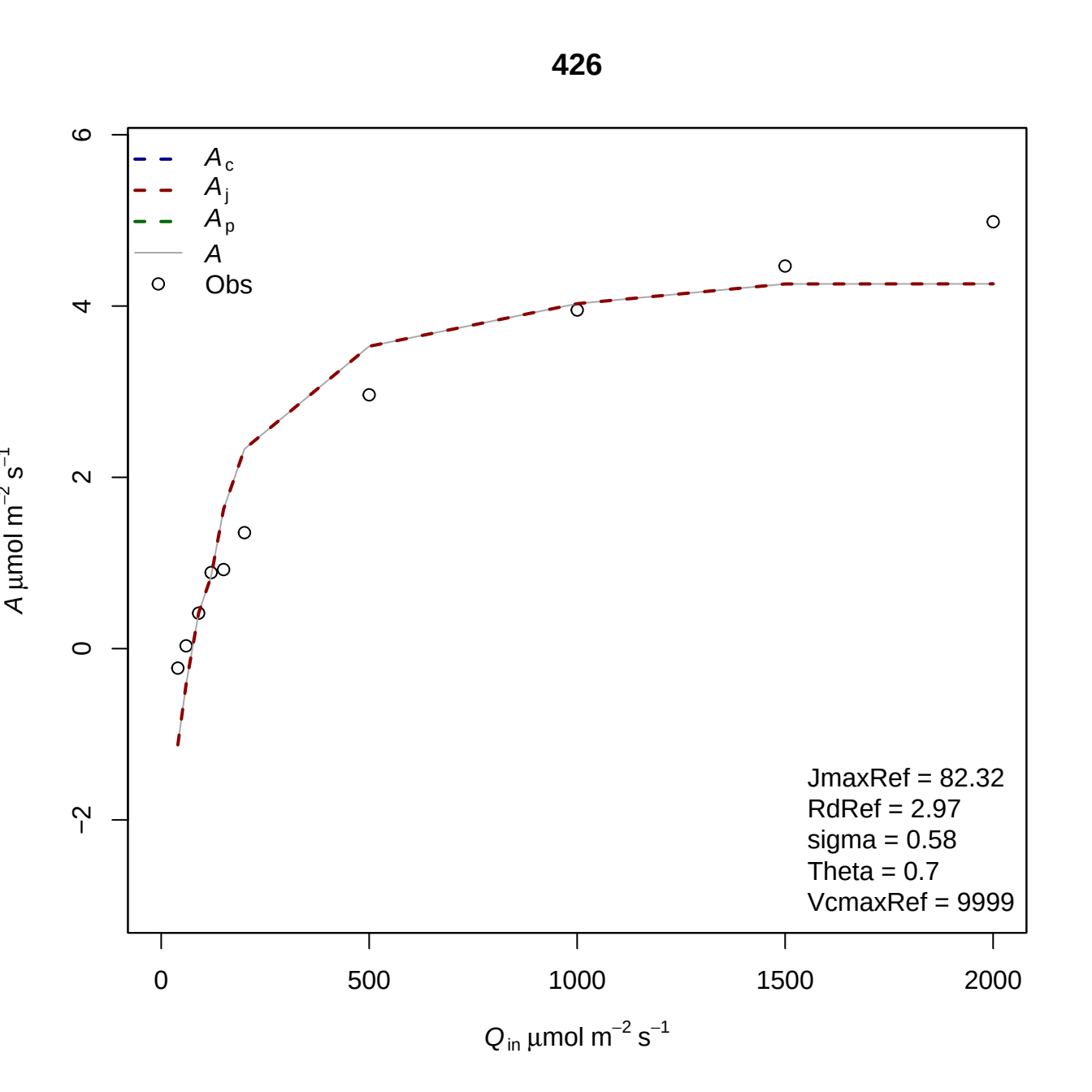

420
A_c
A_j
A_p
A
Obs
JmaxRef = 283.31
RdRef = 3.65
sigma = 0.59
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

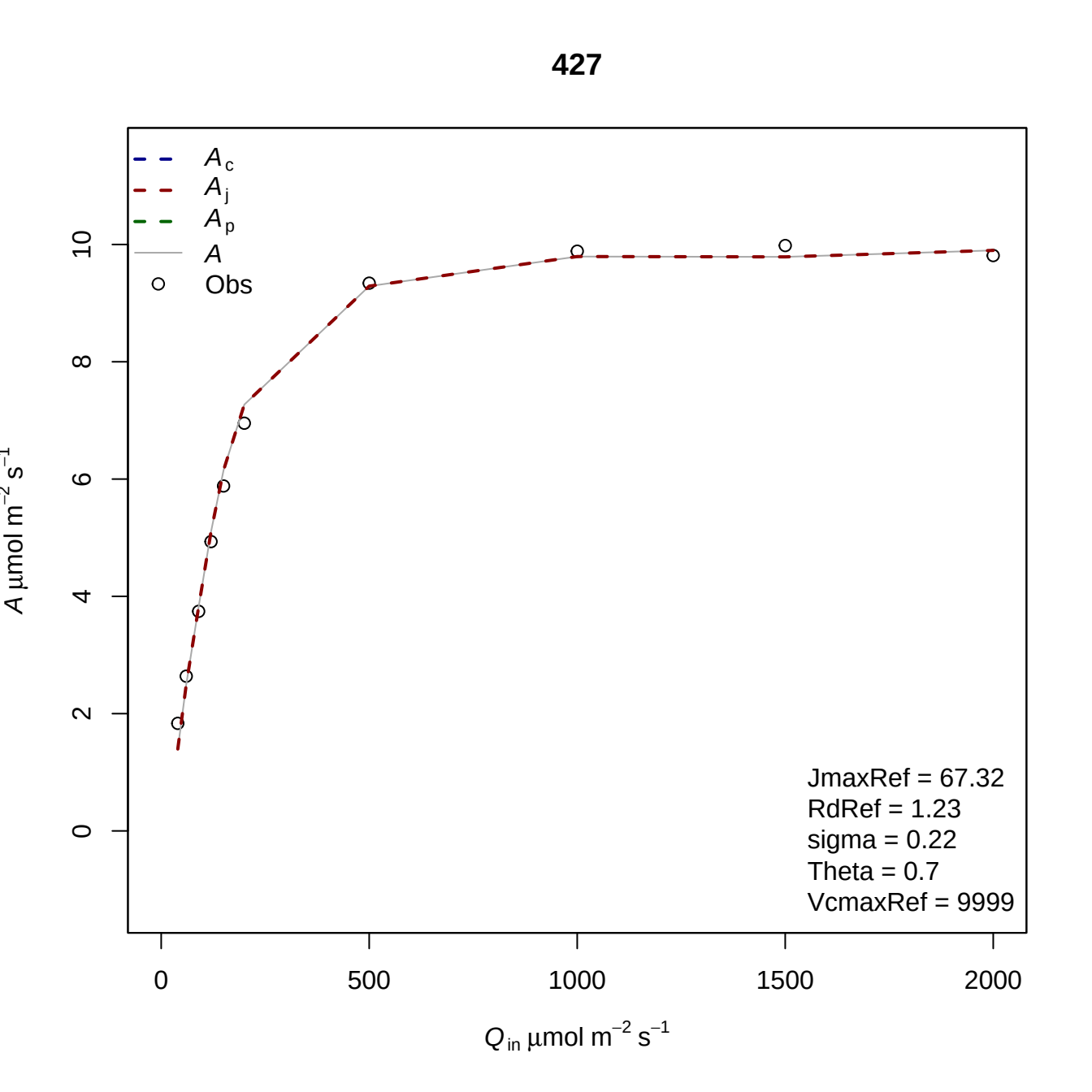

421
A_c
A_j
A_p
A
Obs
JmaxRef = 76.58
RdRef = 2.2
sigma = 0.77
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

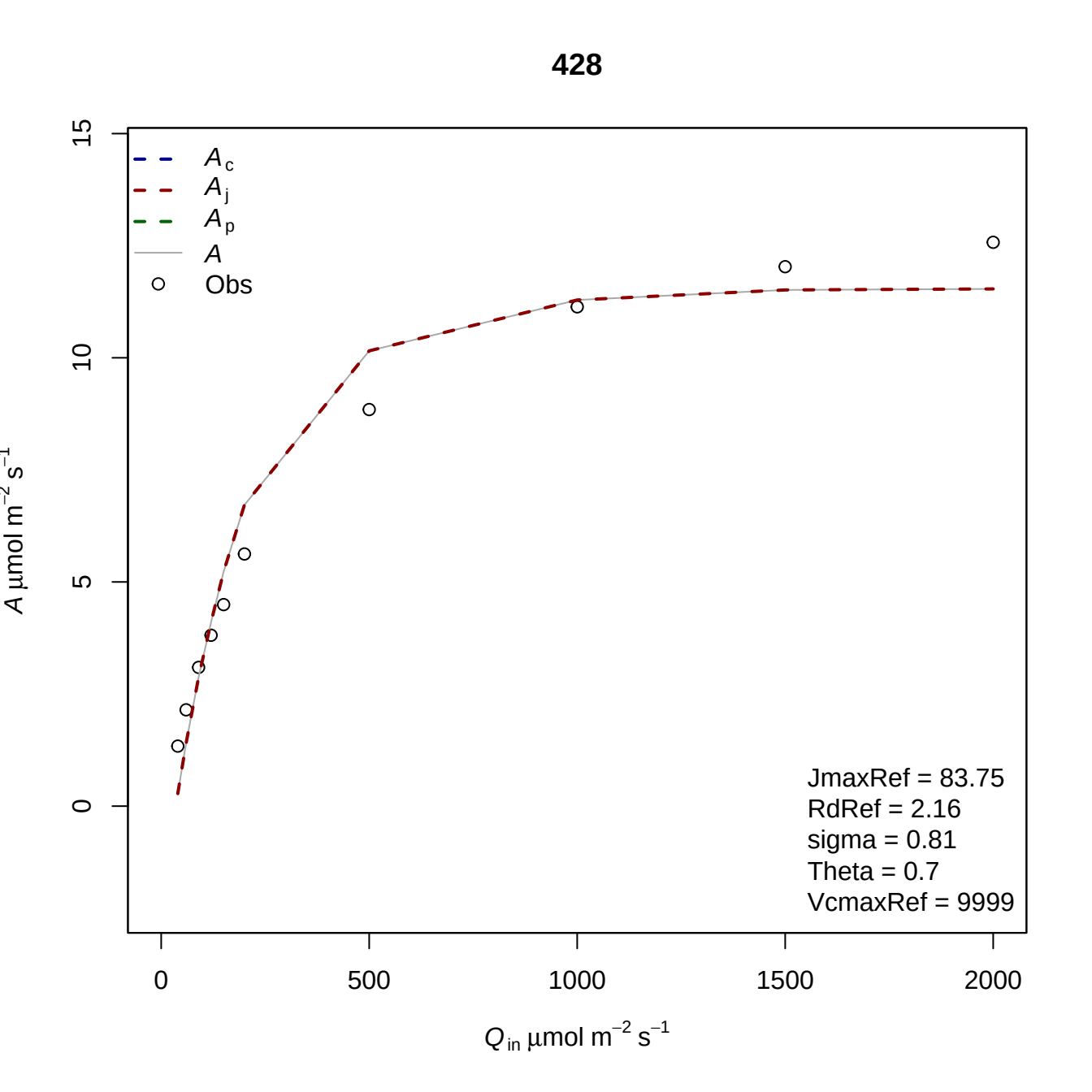

422
A_c
A_j
A_p
A
Obs
JmaxRef = 115.21
RdRef = 1.5
sigma = 0.51
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

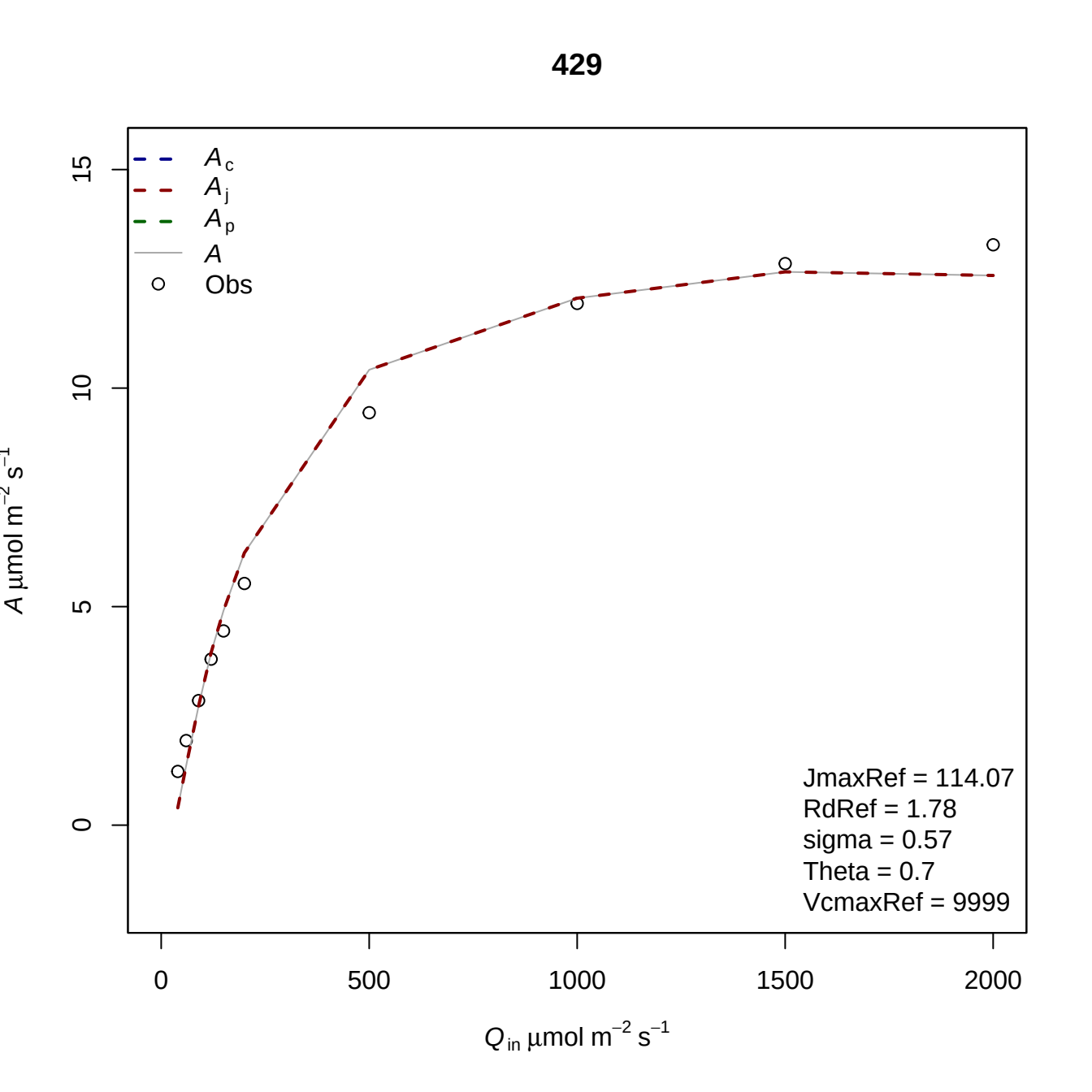

424
A_c
A_j
A_p
A
Obs
JmaxRef = 131.95
RdRef = 2.25
sigma = 0.47
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

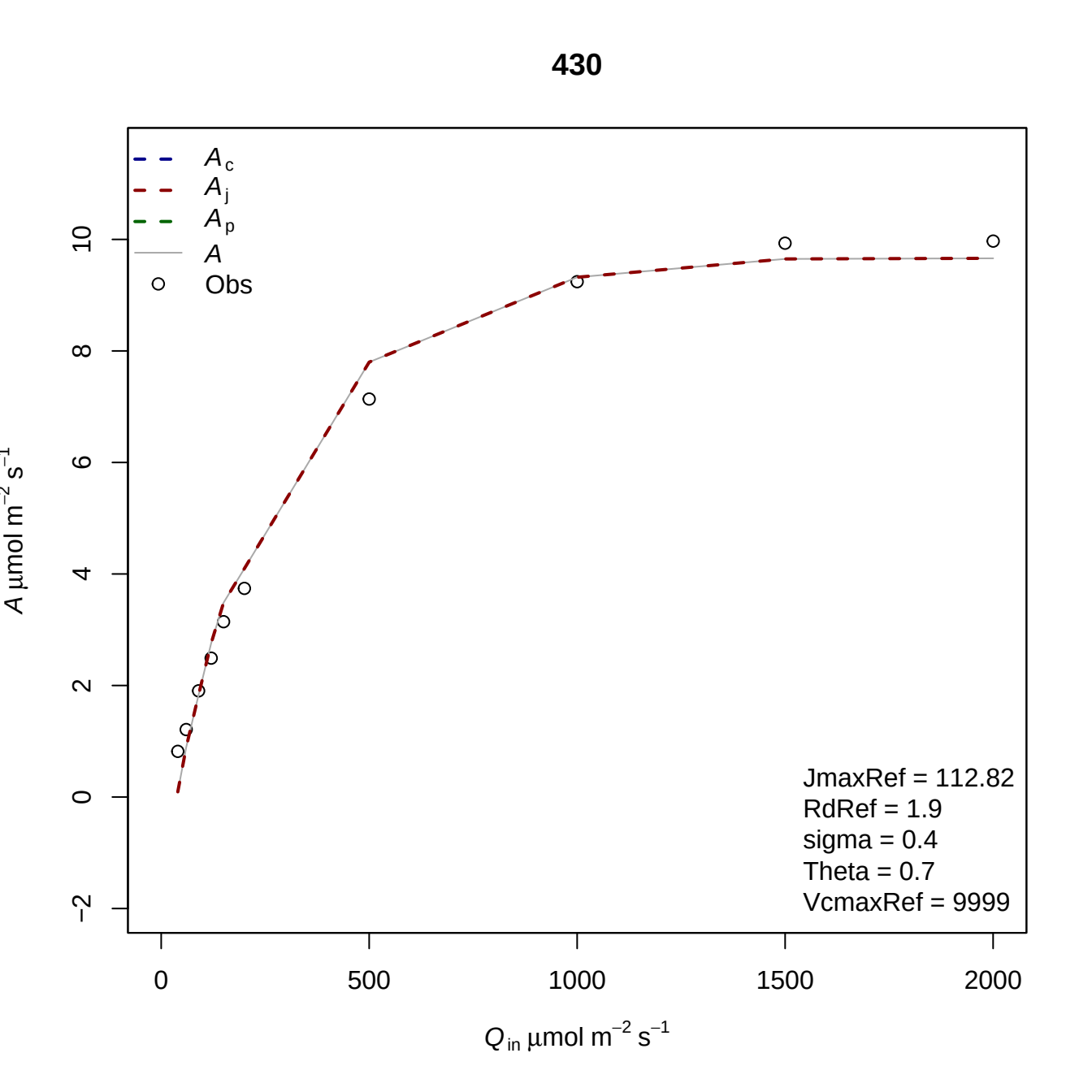

425
A_c
A_j
A_p
A
Obs
JmaxRef = 96.46
RdRef = 3.23
sigma = 0.91
Theta = 0.7
VcmaxRef = 9999
8
6
4
2
0
−2
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

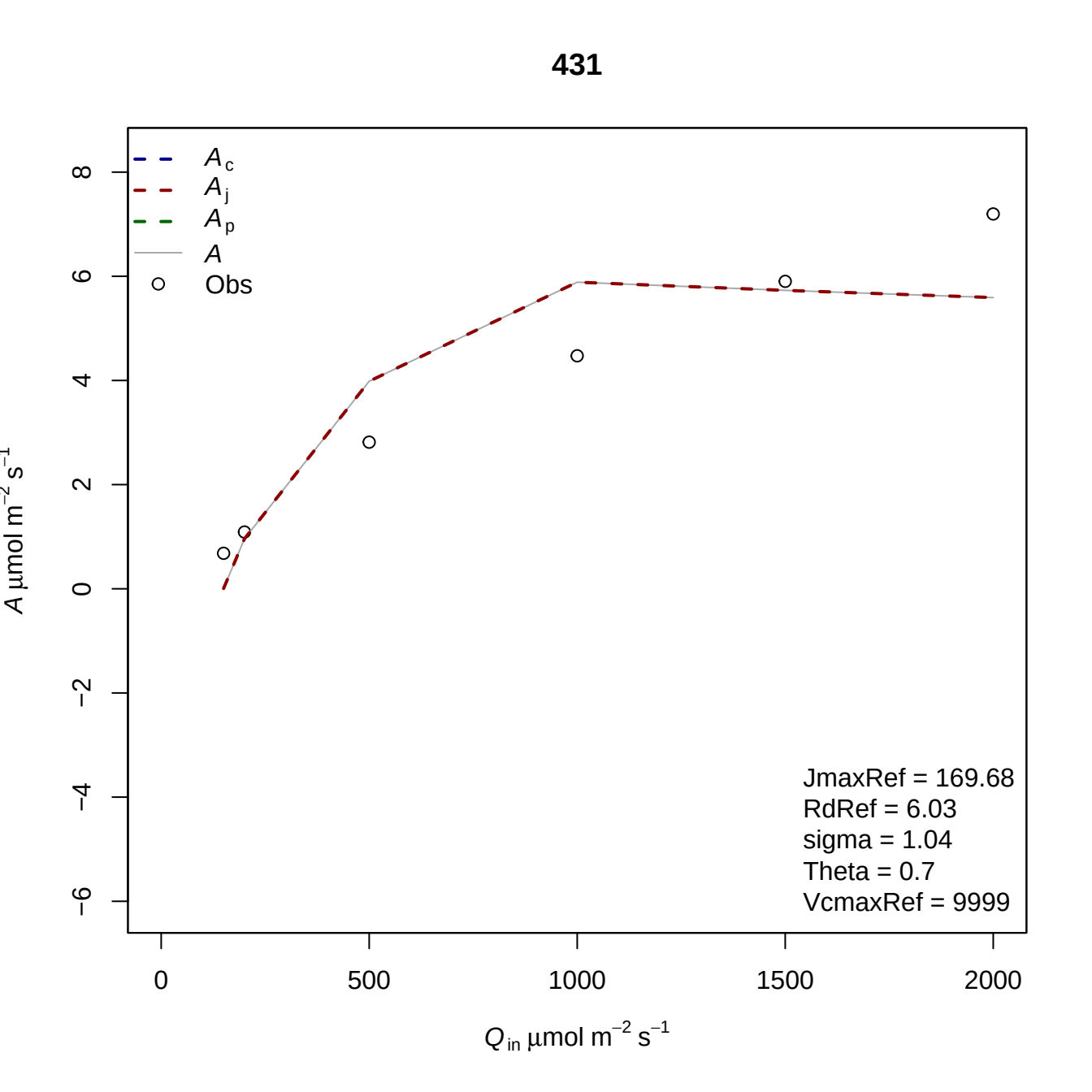

426
A_c
A_j
A_p
A
Obs
JmaxRef = 82.32
RdRef = 2.97
sigma = 0.58
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

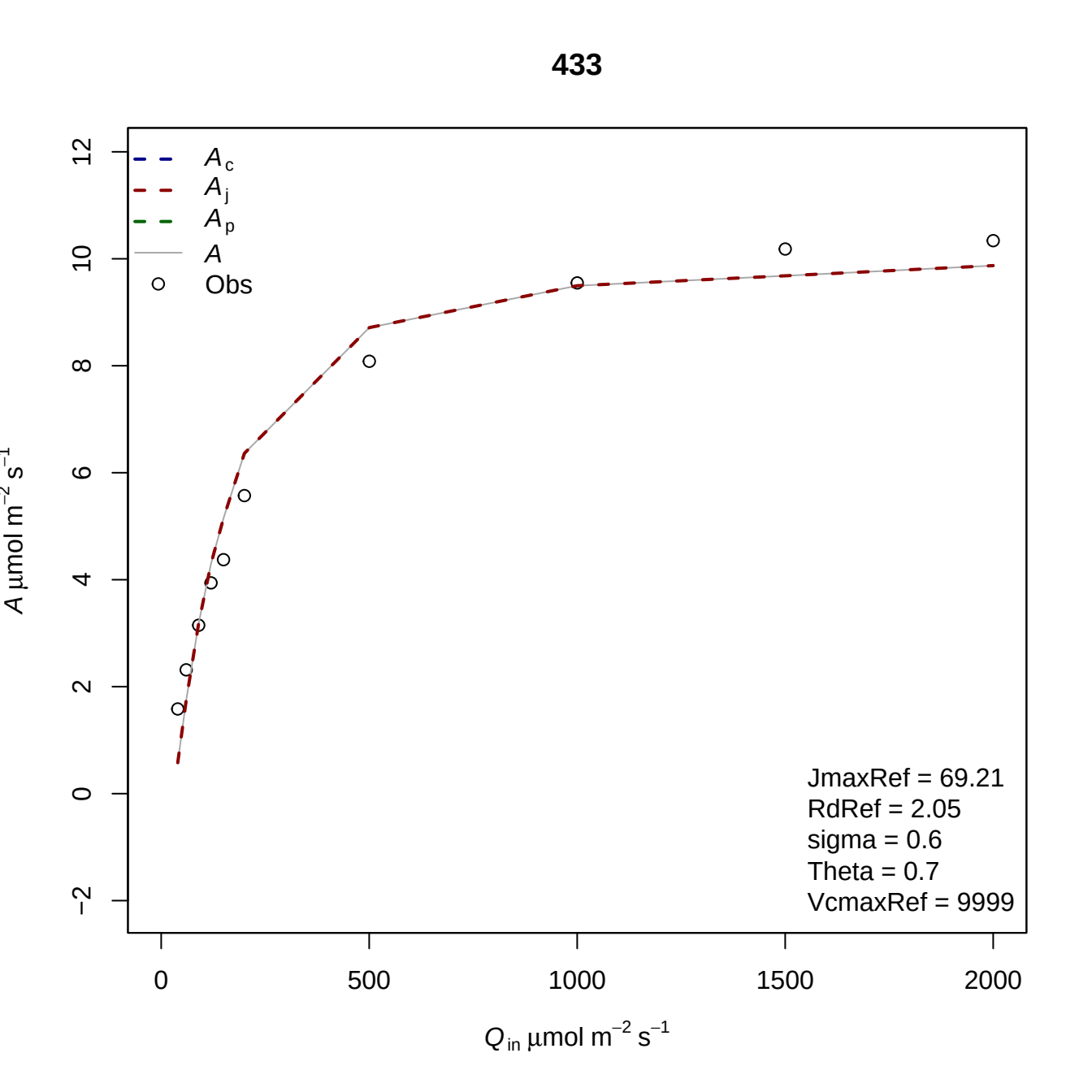

427
A_c
A_j
A_p
A
Obs
JmaxRef = 67.32
RdRef = 1.23
sigma = 0.22
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

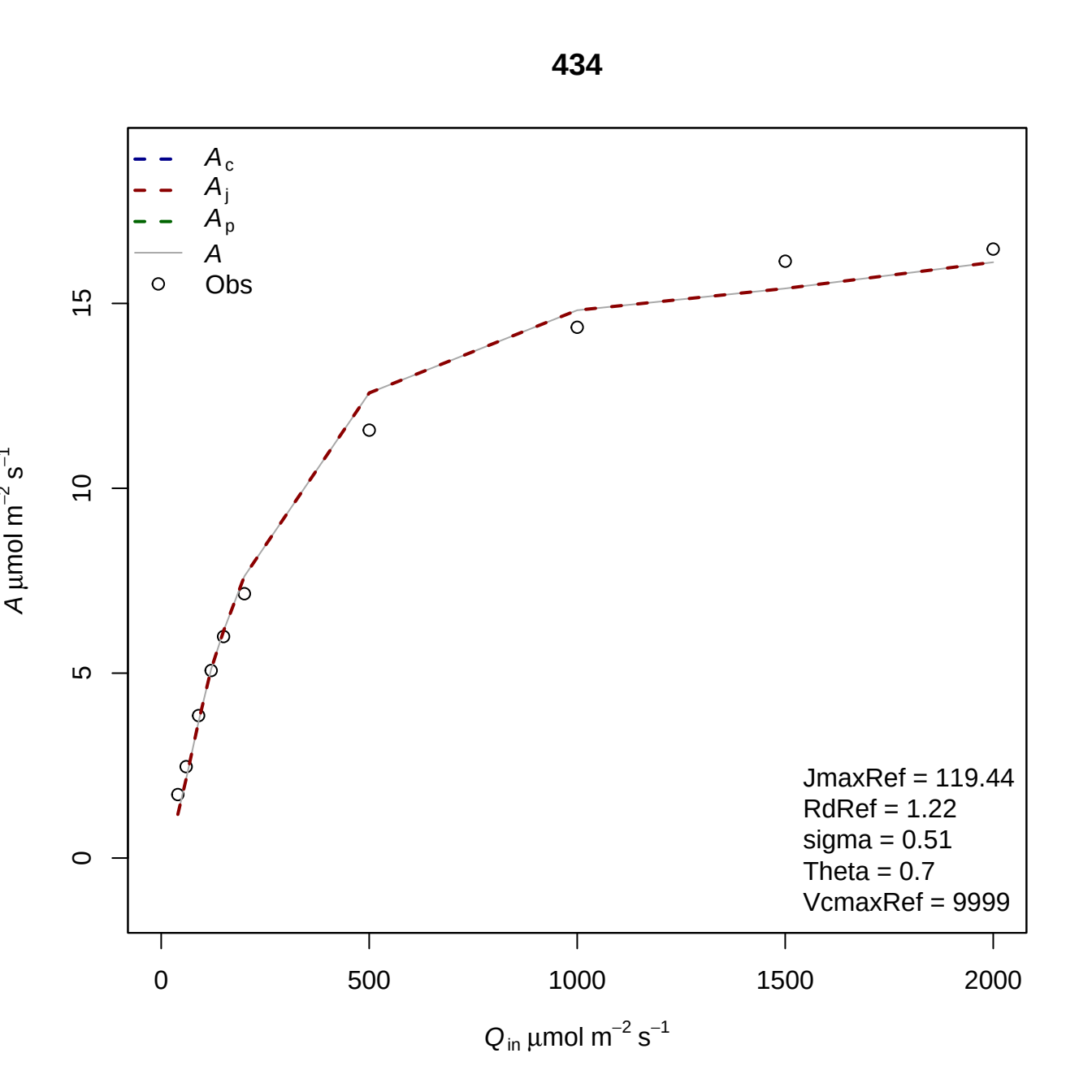

428
A_c
A_j
A_p
A
Obs
JmaxRef = 83.75
RdRef = 2.16
sigma = 0.81
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

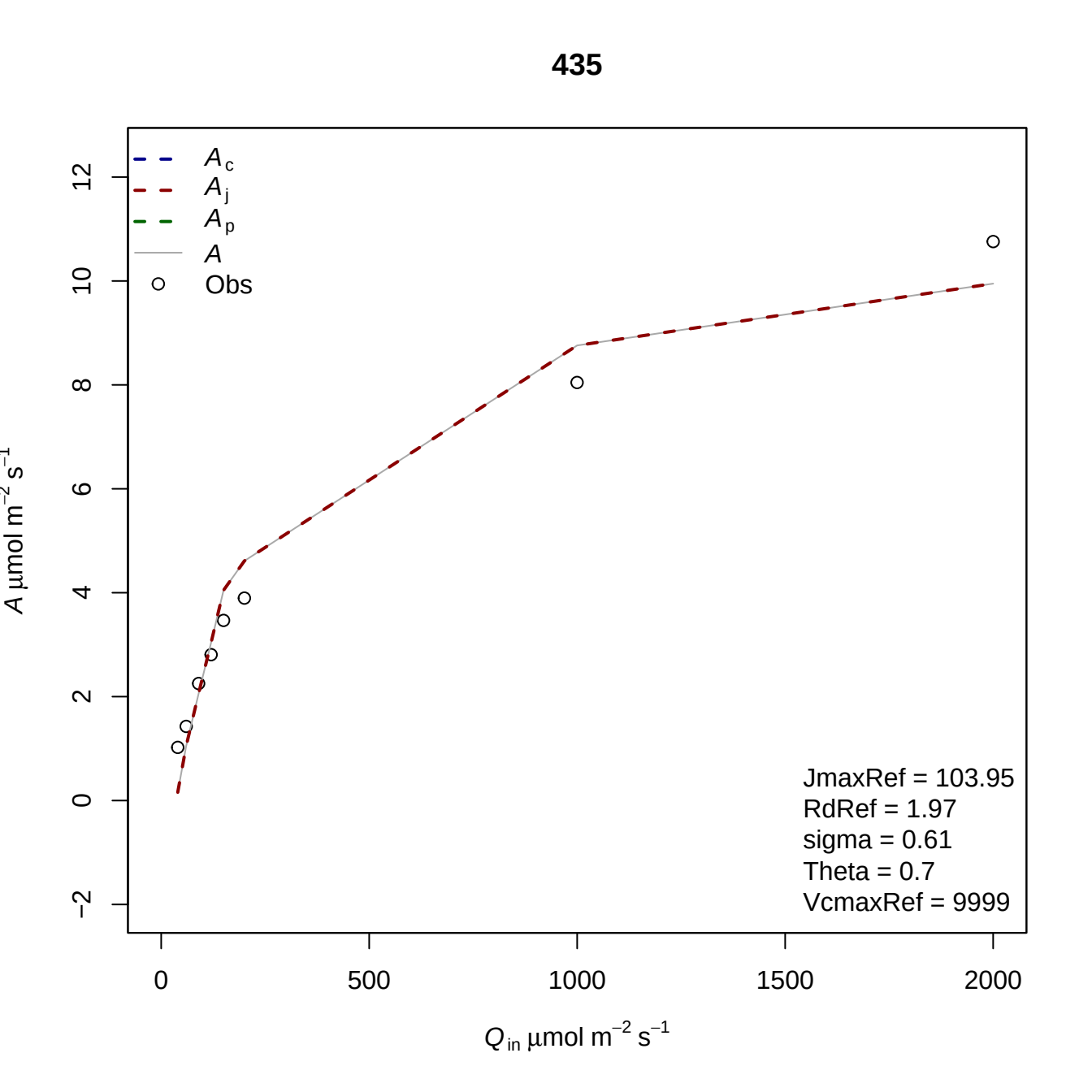

429
A_c
A_j
A_p
A
Obs
JmaxRef = 114.07
RdRef = 1.78
sigma = 0.57
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

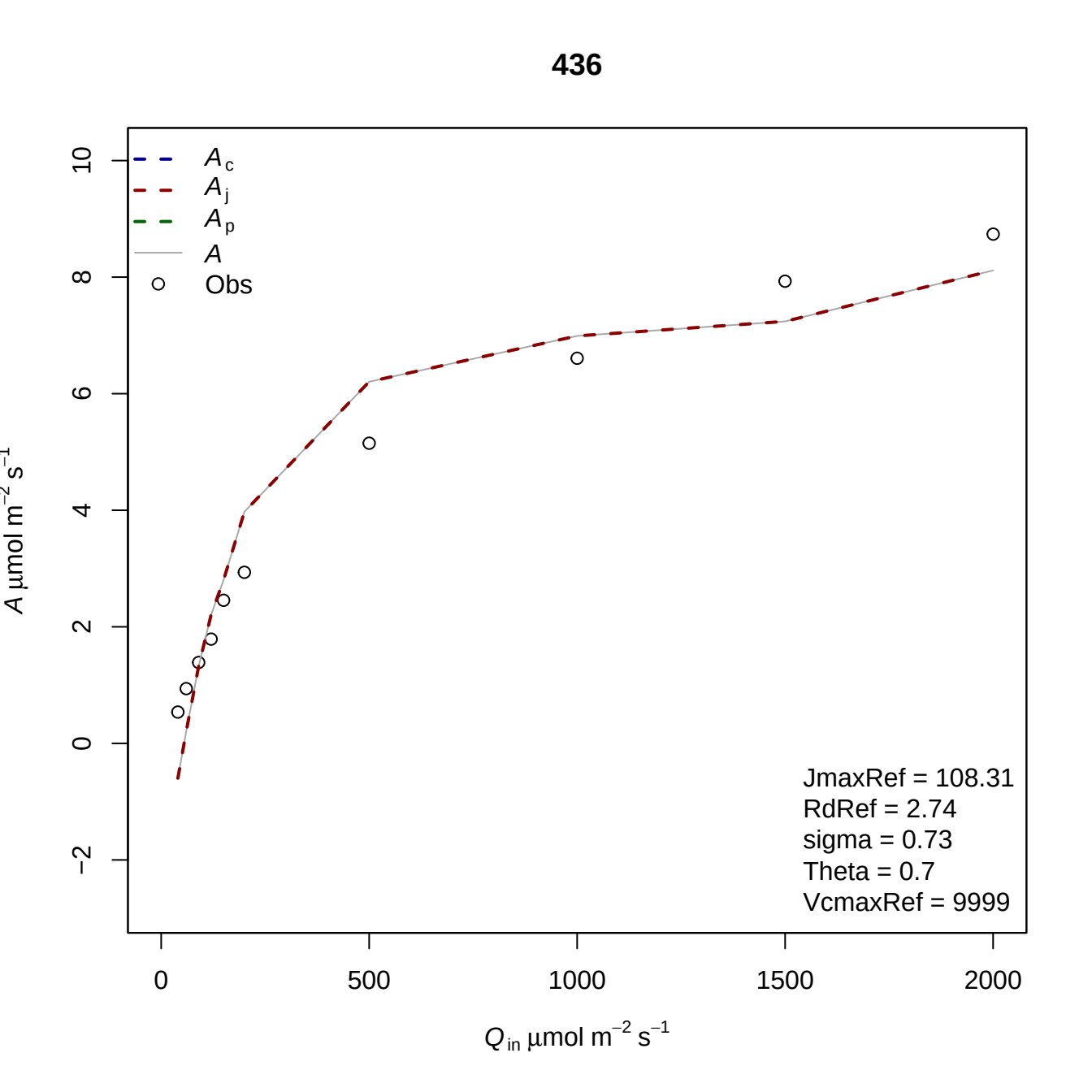

430
A_c
A_j
A_p
A
Obs
JmaxRef = 112.82
RdRef = 1.9
sigma = 0.4
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

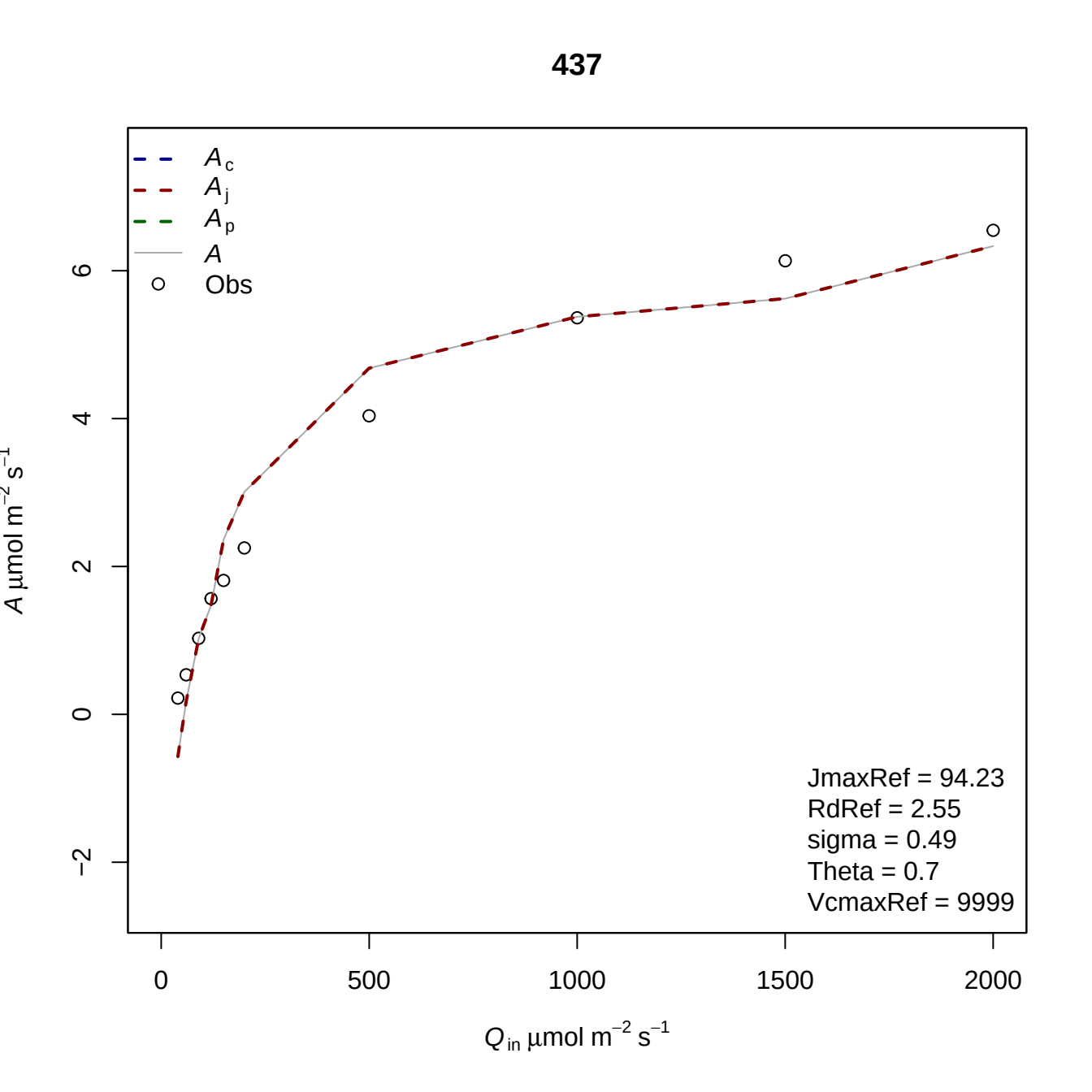

431
A_c
A_j
A_p
A
Obs
JmaxRef = 169.68
RdRef = 6.03
sigma = 1.04
Theta = 0.7
VcmaxRef = 9999
8
6
4
2
0
−2
−4
−6
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

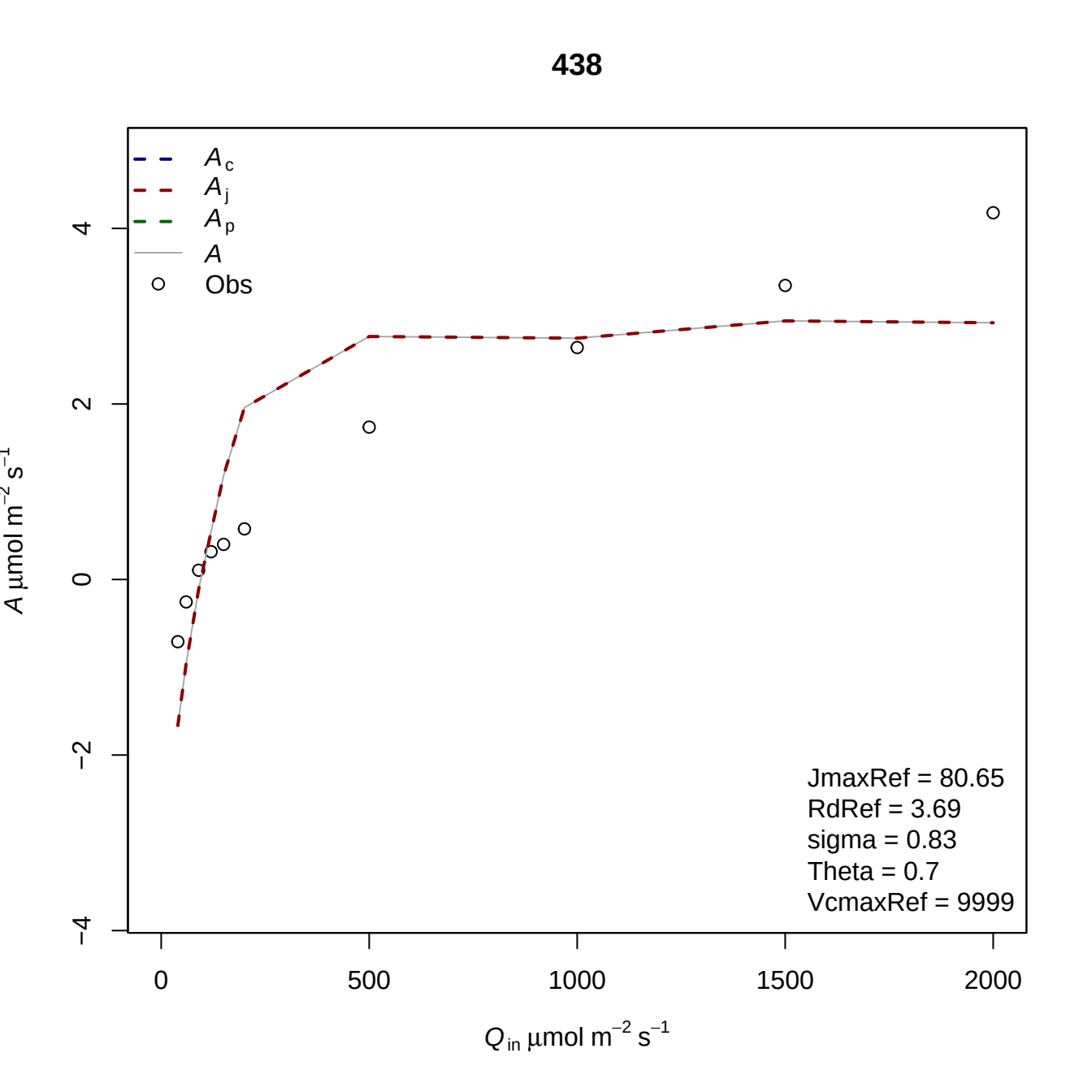

433
A_c
A_j
A_p
A
Obs
JmaxRef = 69.21
RdRef = 2.05
sigma = 0.6
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

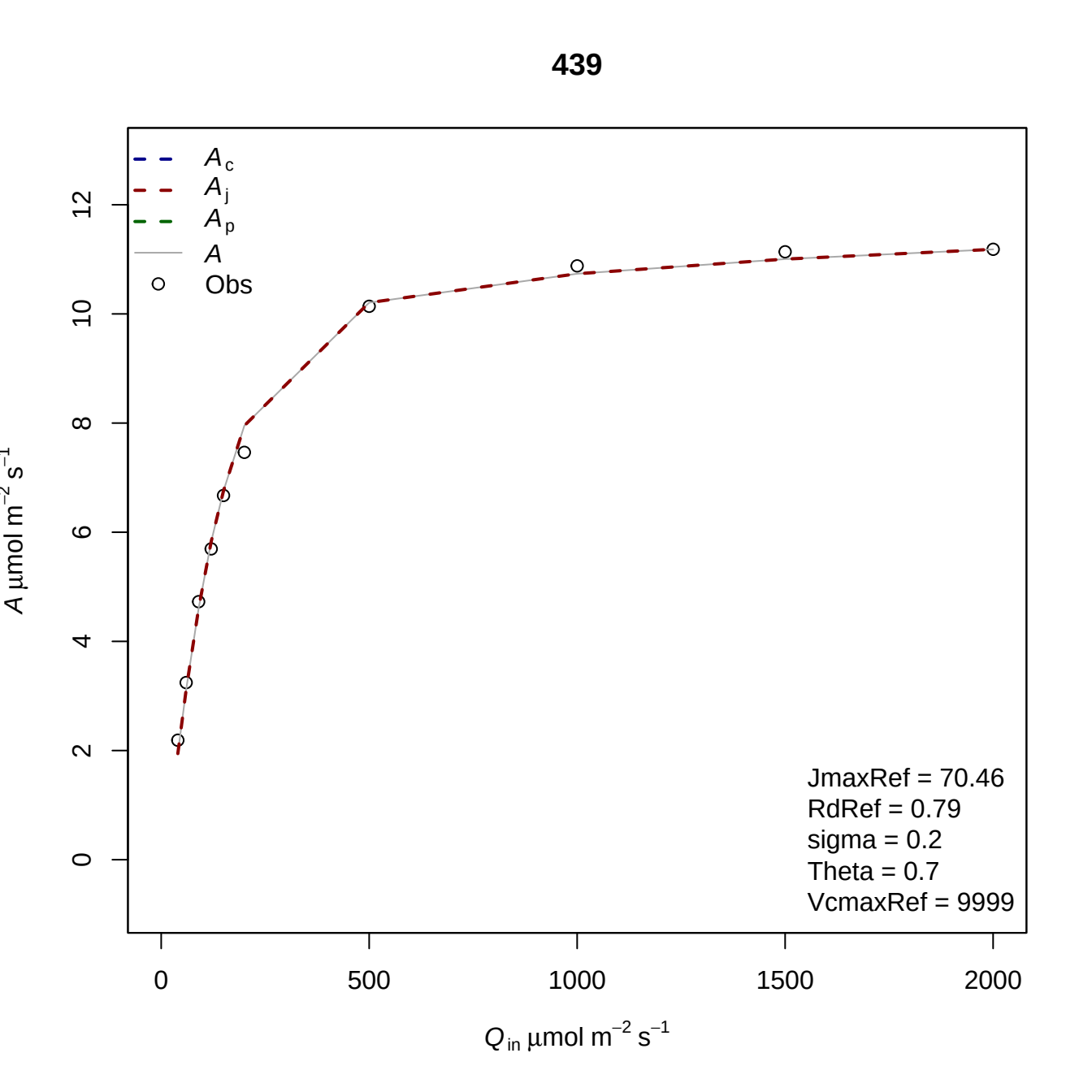

434
A_c
A_j
A_p
A
Obs
JmaxRef = 119.44
RdRef = 1.22
sigma = 0.51
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

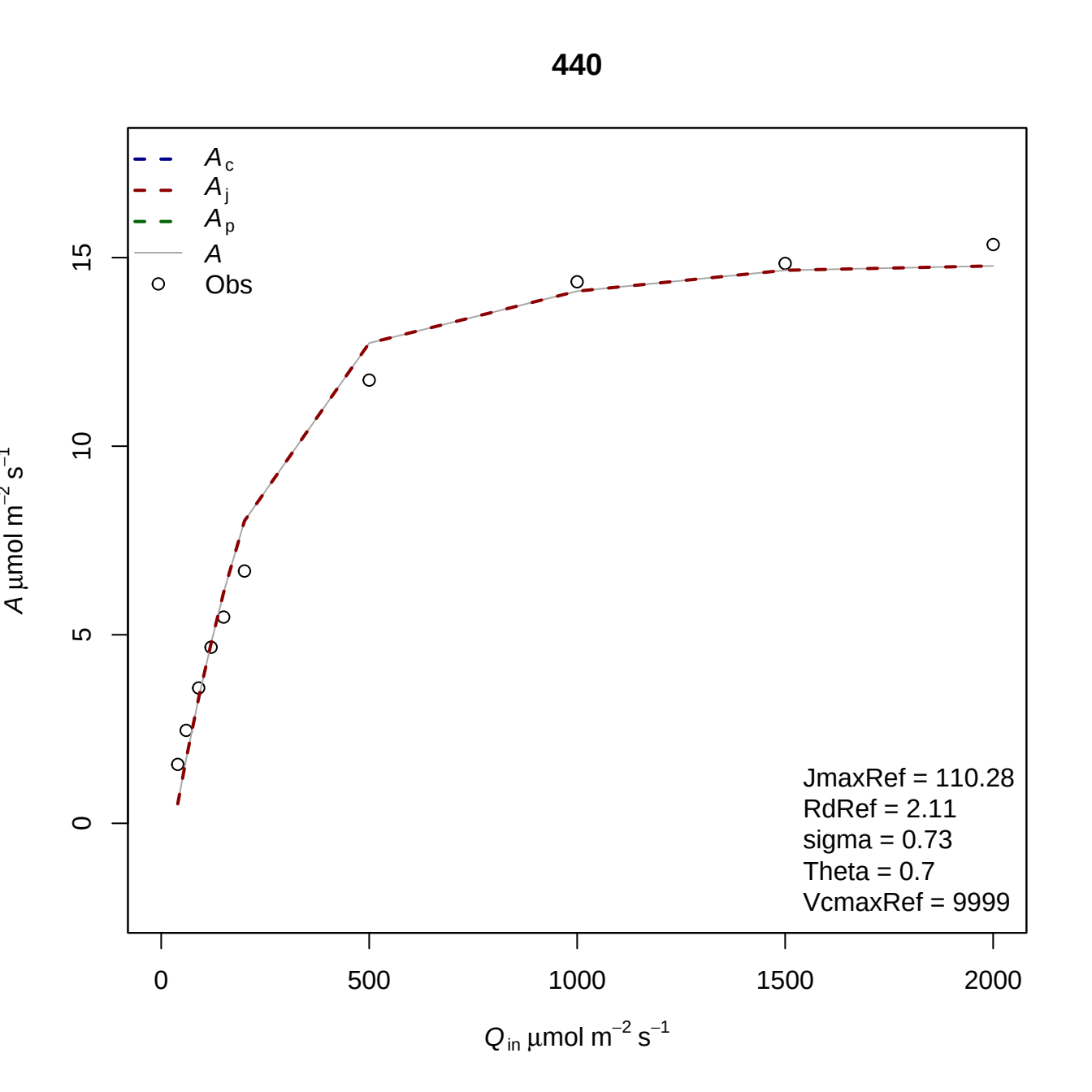

435
A_c
A_j
A_p
A
Obs
JmaxRef = 103.95
RdRef = 1.97
sigma = 0.61
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

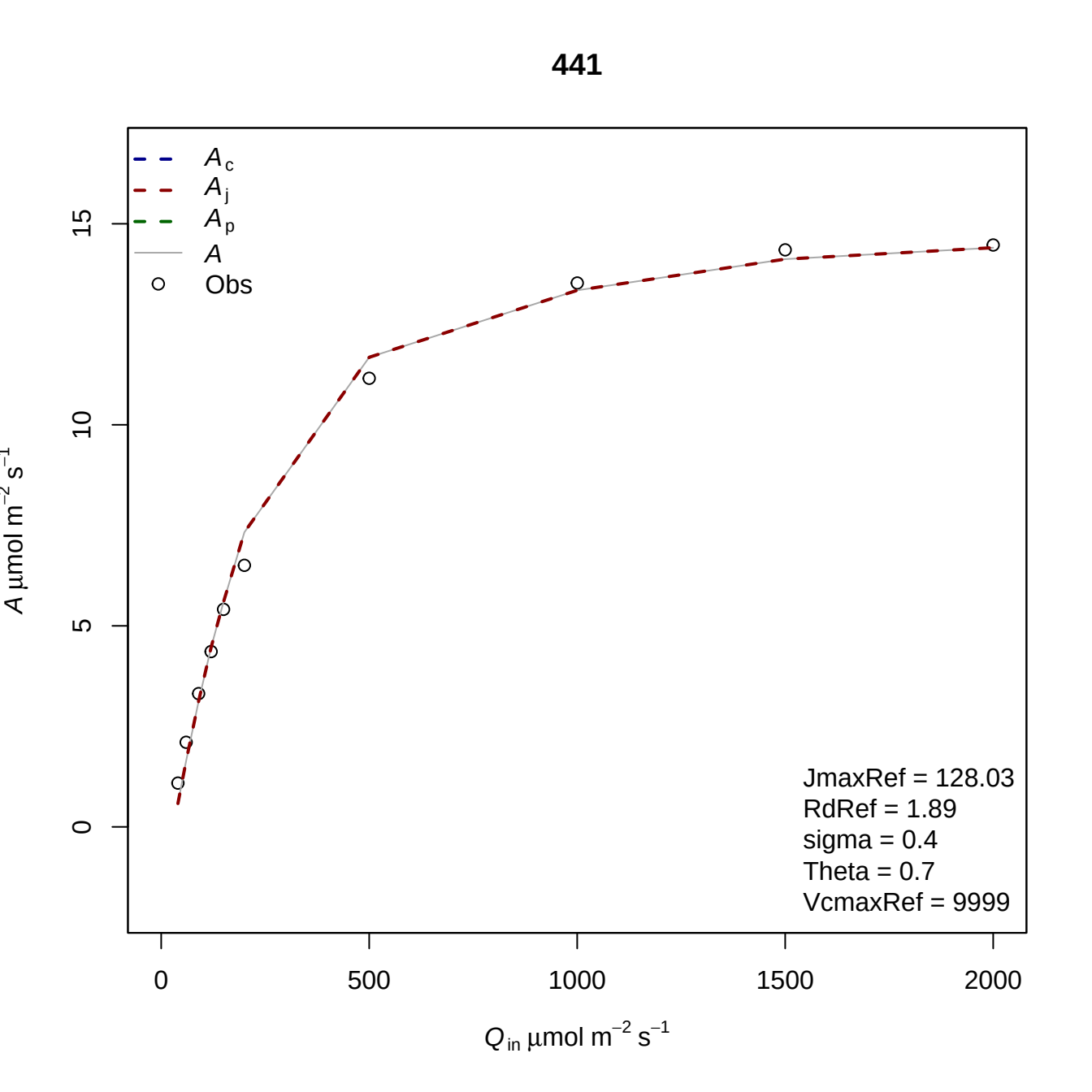

436
A_c
A_j
A_p
A
Obs
JmaxRef = 108.31
RdRef = 2.74
sigma = 0.73
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

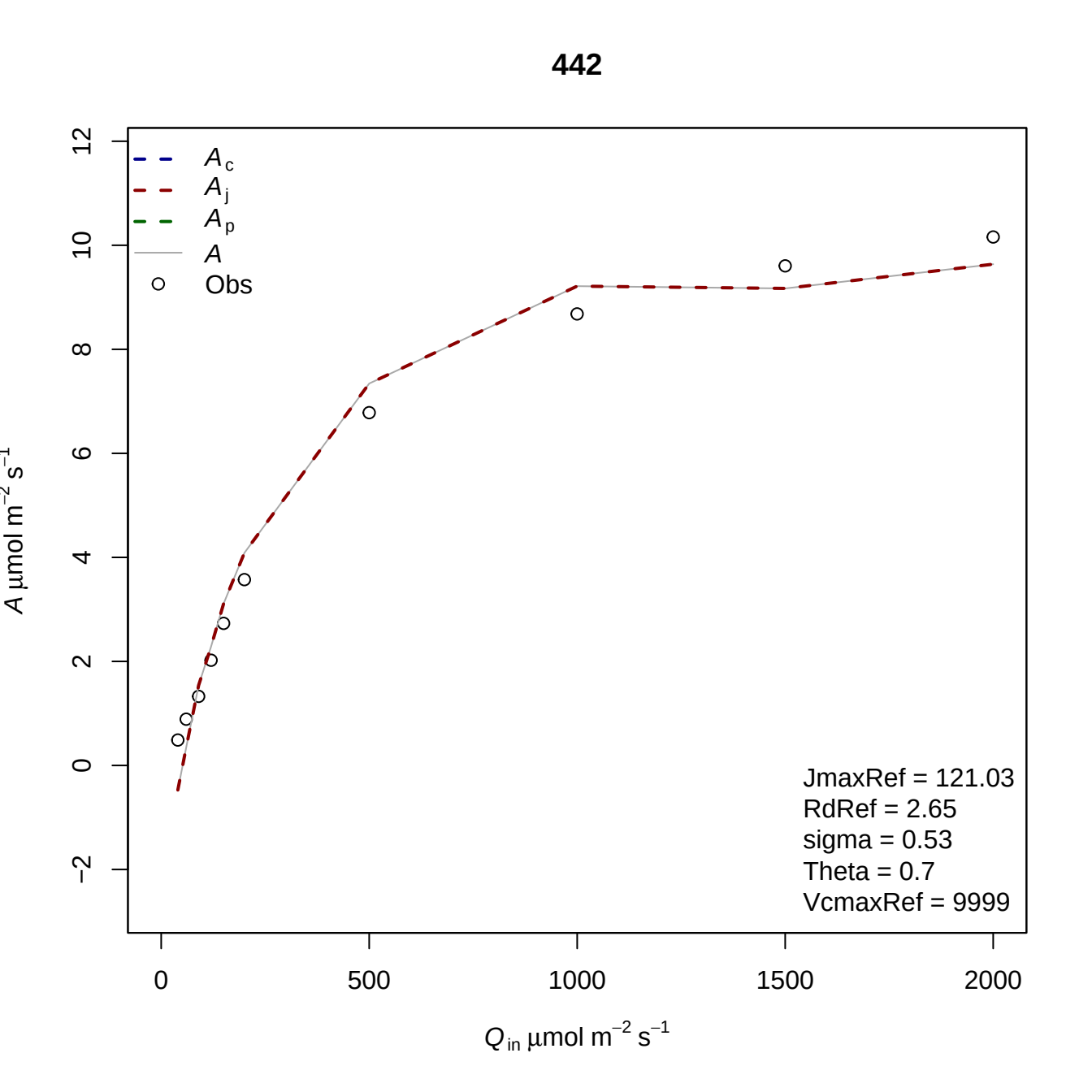

437
A_c
A_j
A_p
A
Obs
JmaxRef = 94.23
RdRef = 2.55
sigma = 0.49
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

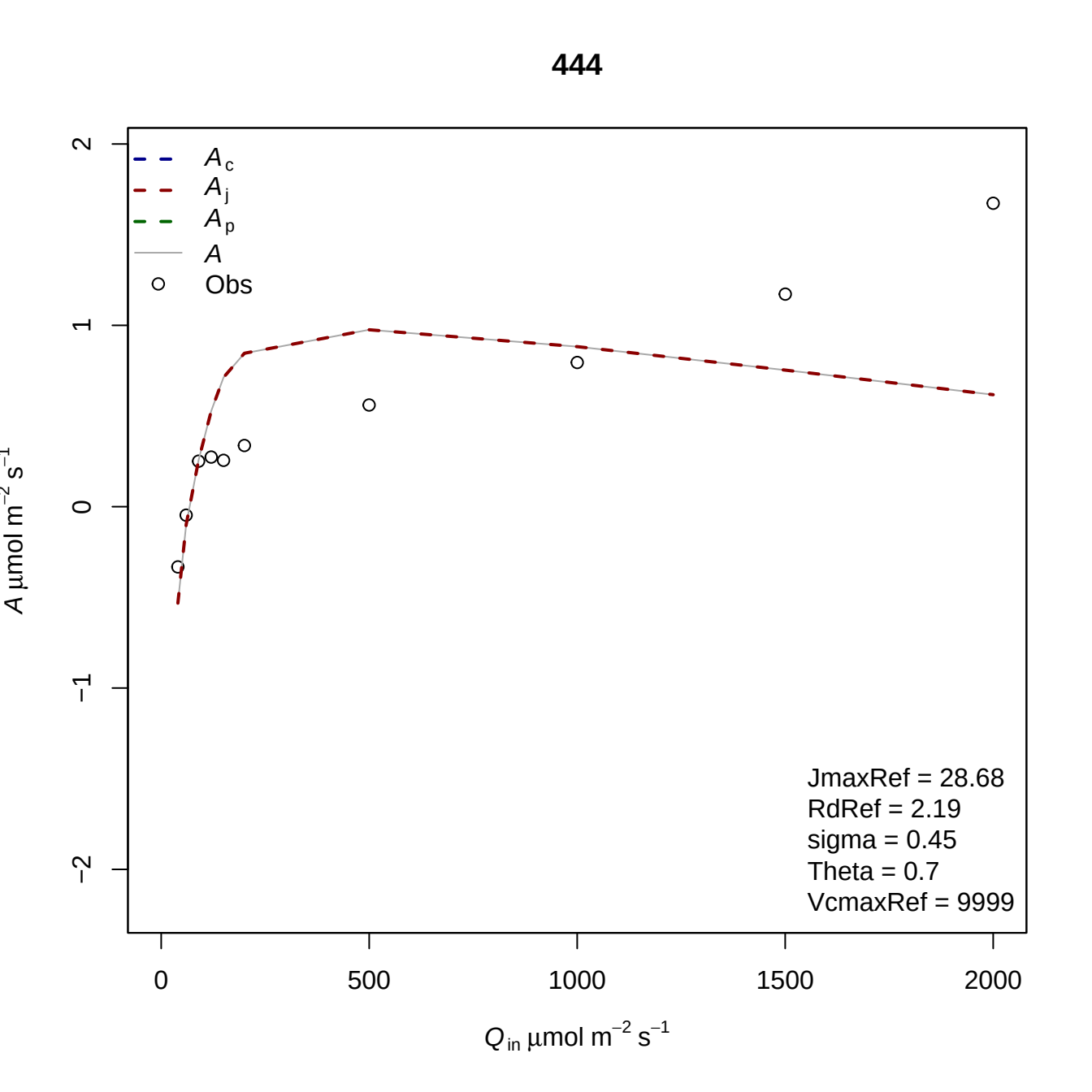

438
A_c
A_j
A_p
A
Obs
JmaxRef = 80.65
RdRef = 3.69
sigma = 0.83
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
−4
0
500
1000
1500
2000

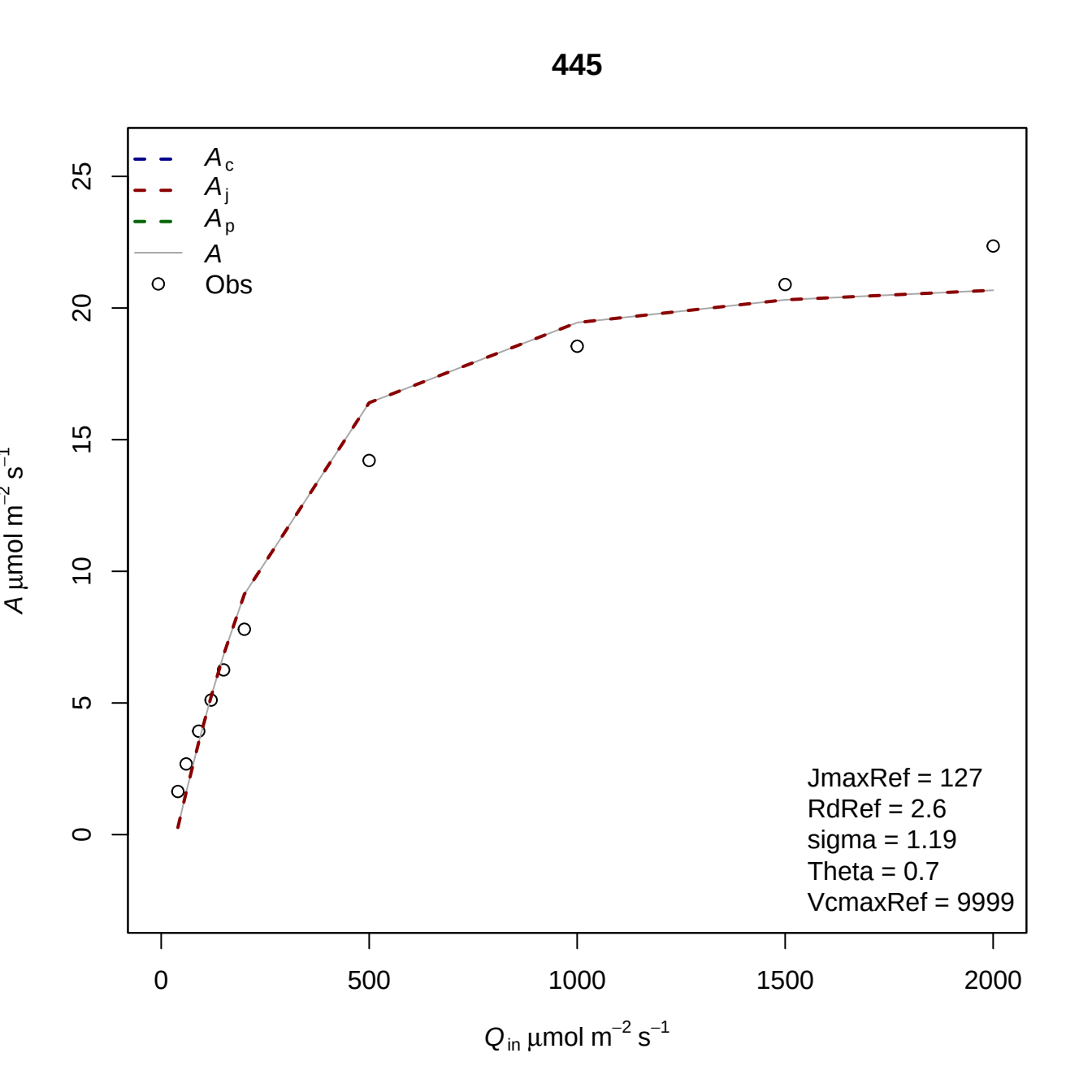

439
A_c
A_j
A_p
A
Obs
JmaxRef = 70.46
RdRef = 0.79
sigma = 0.2
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

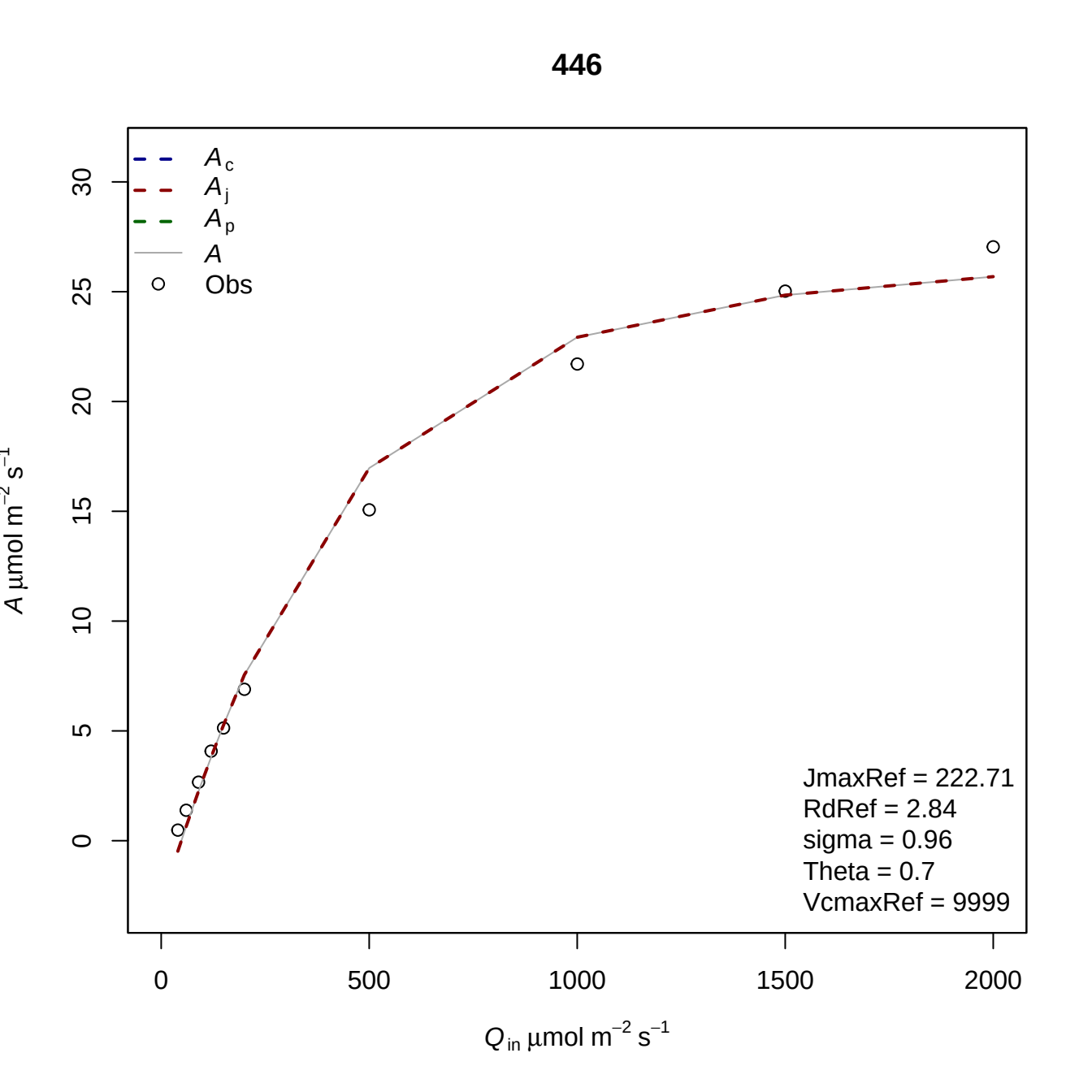

440
A_c
A_j
A_p
A
Obs
JmaxRef = 110.28
RdRef = 2.11
sigma = 0.73
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

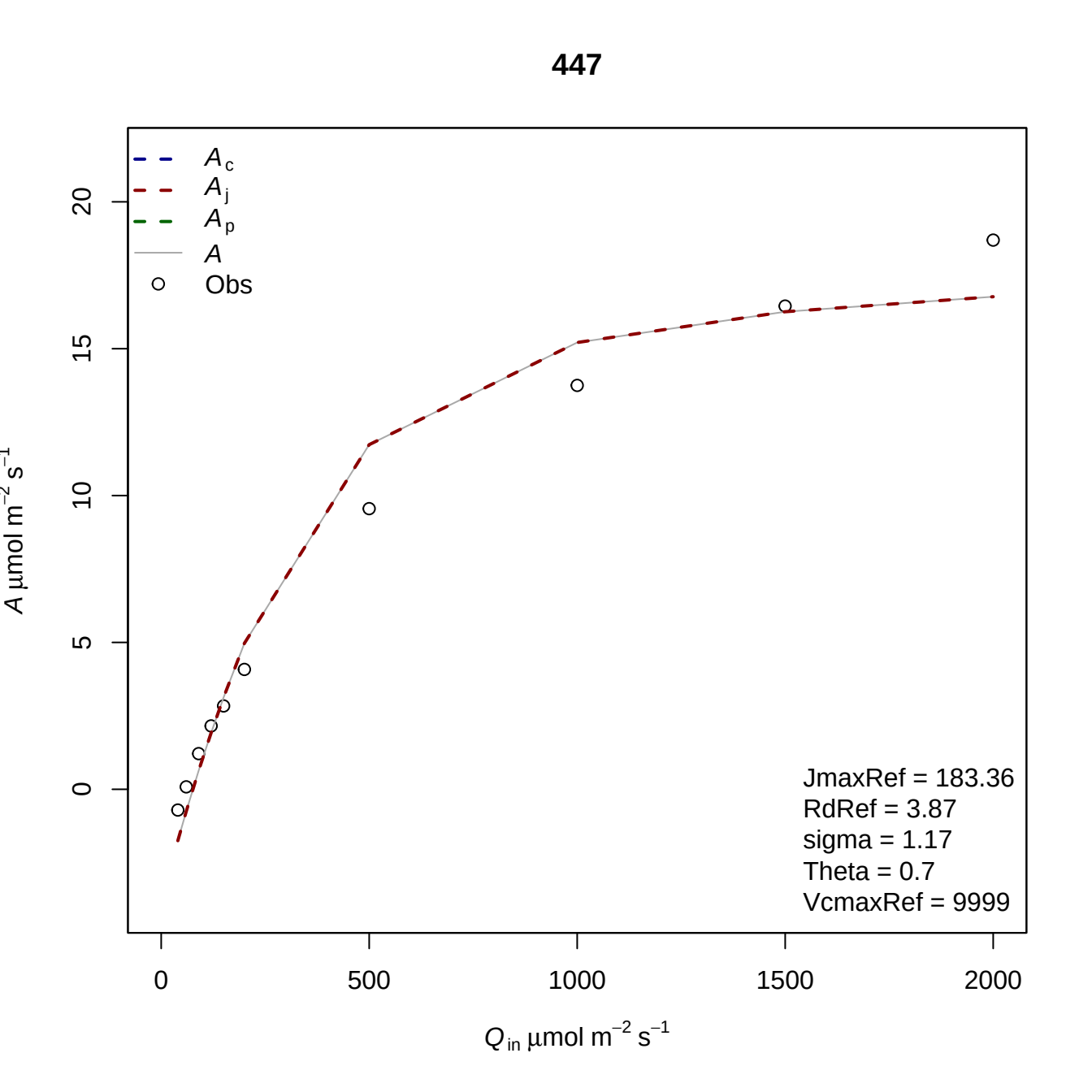

441
A_c
A_j
A_p
A
Obs
JmaxRef = 128.03
RdRef = 1.89
sigma = 0.4
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

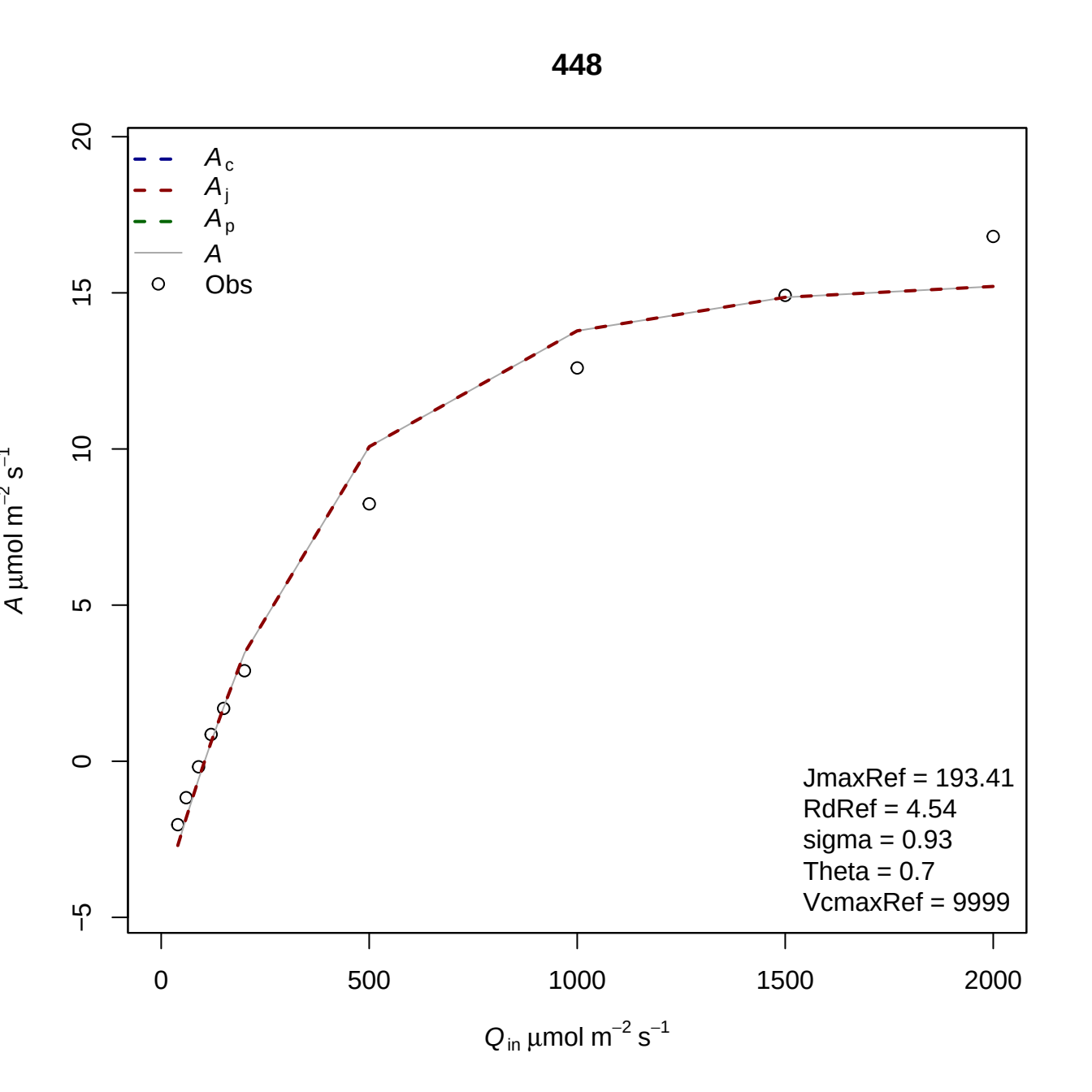

442
A_c
A_j
A_p
A
Obs
JmaxRef = 121.03
RdRef = 2.65
sigma = 0.53
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

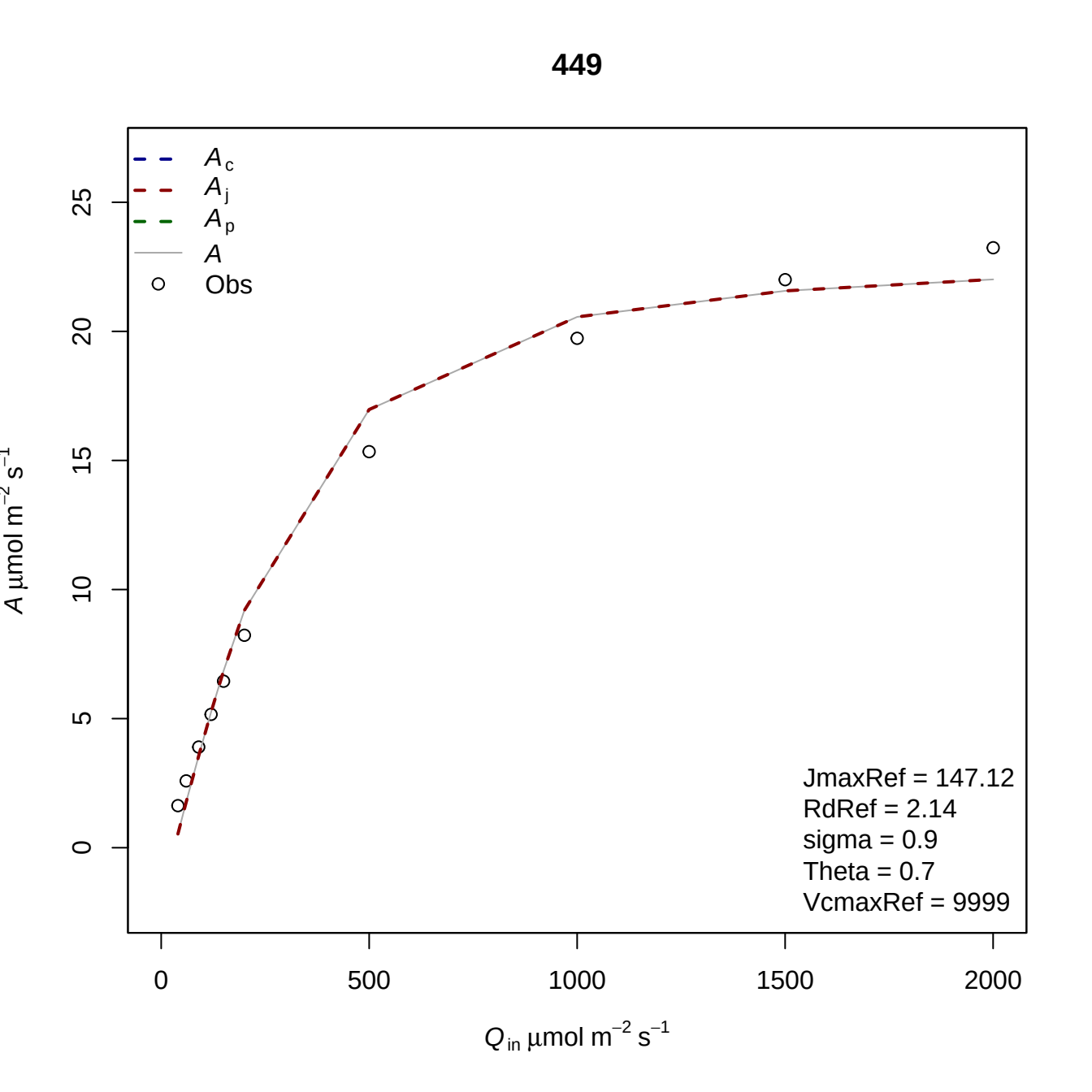

444
A_c
A_j
A_p
A
Obs
JmaxRef = 28.68
RdRef = 2.19
sigma = 0.45
Theta = 0.7
VcmaxRef = 9999
2
1
0
−1
−2
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

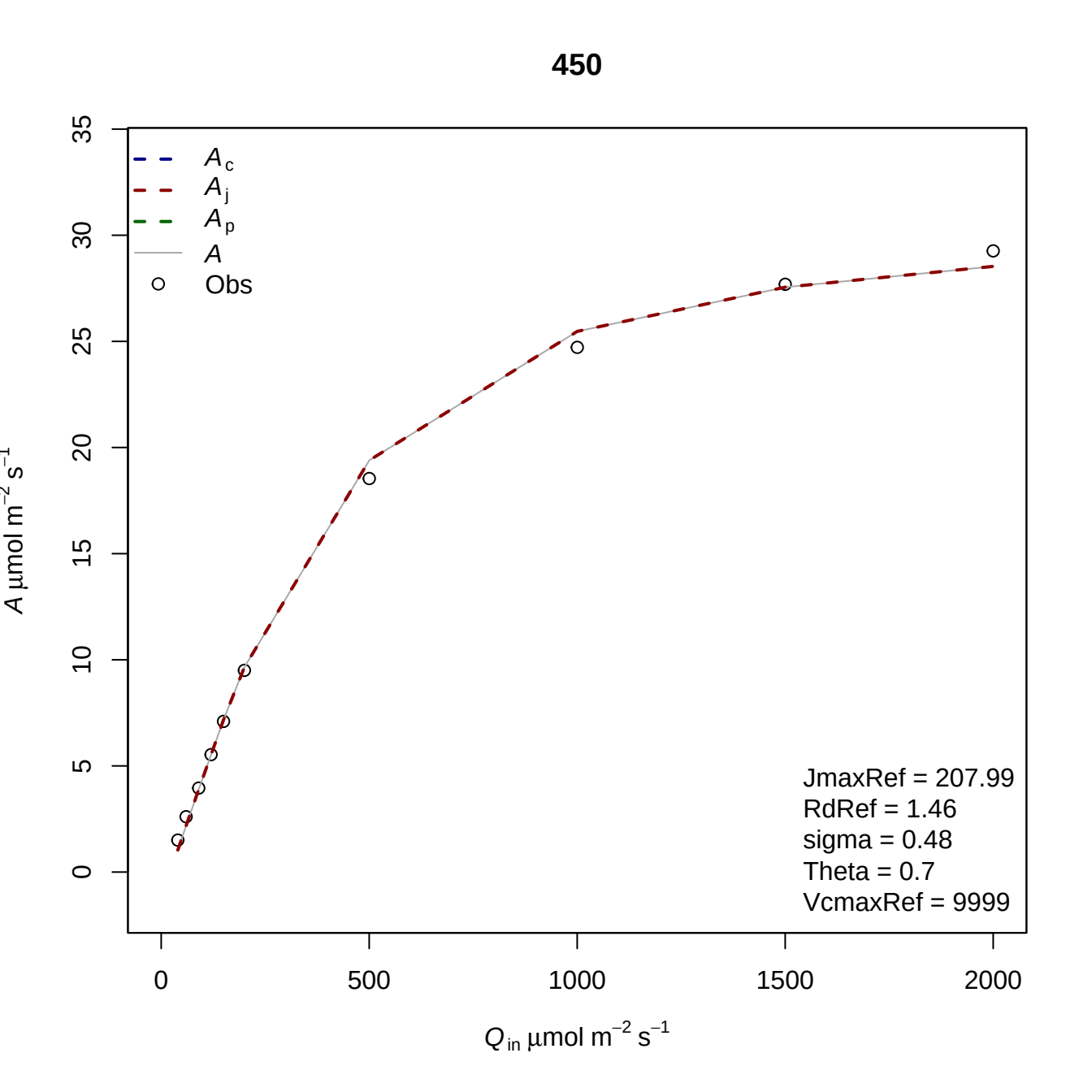

445
A_c
A_j
A_p
A
Obs
JmaxRef = 127
RdRef = 2.6
sigma = 1.19
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
25
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

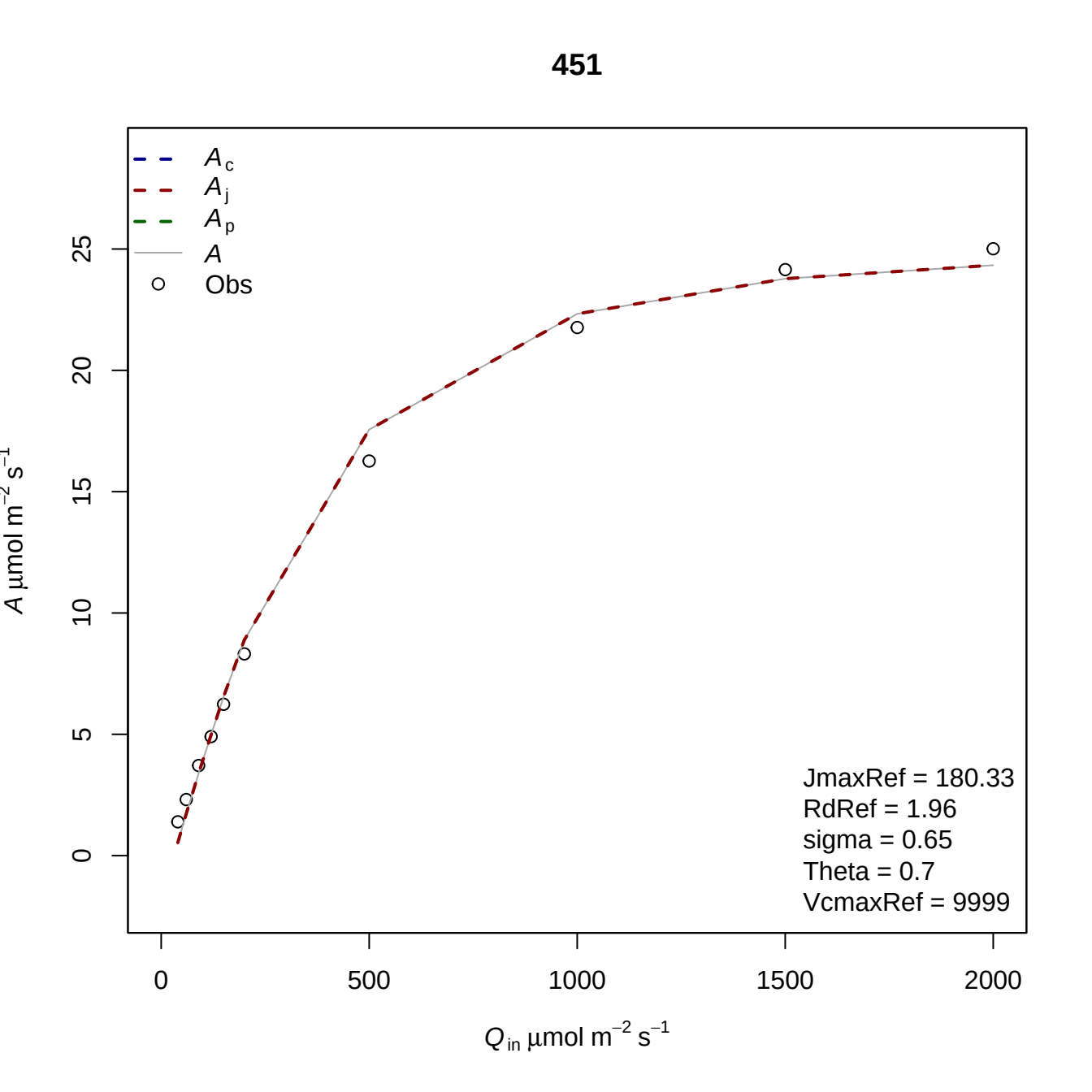

446
A_c
A_j
A_p
A
Obs
JmaxRef = 222.71
RdRef = 2.84
sigma = 0.96
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

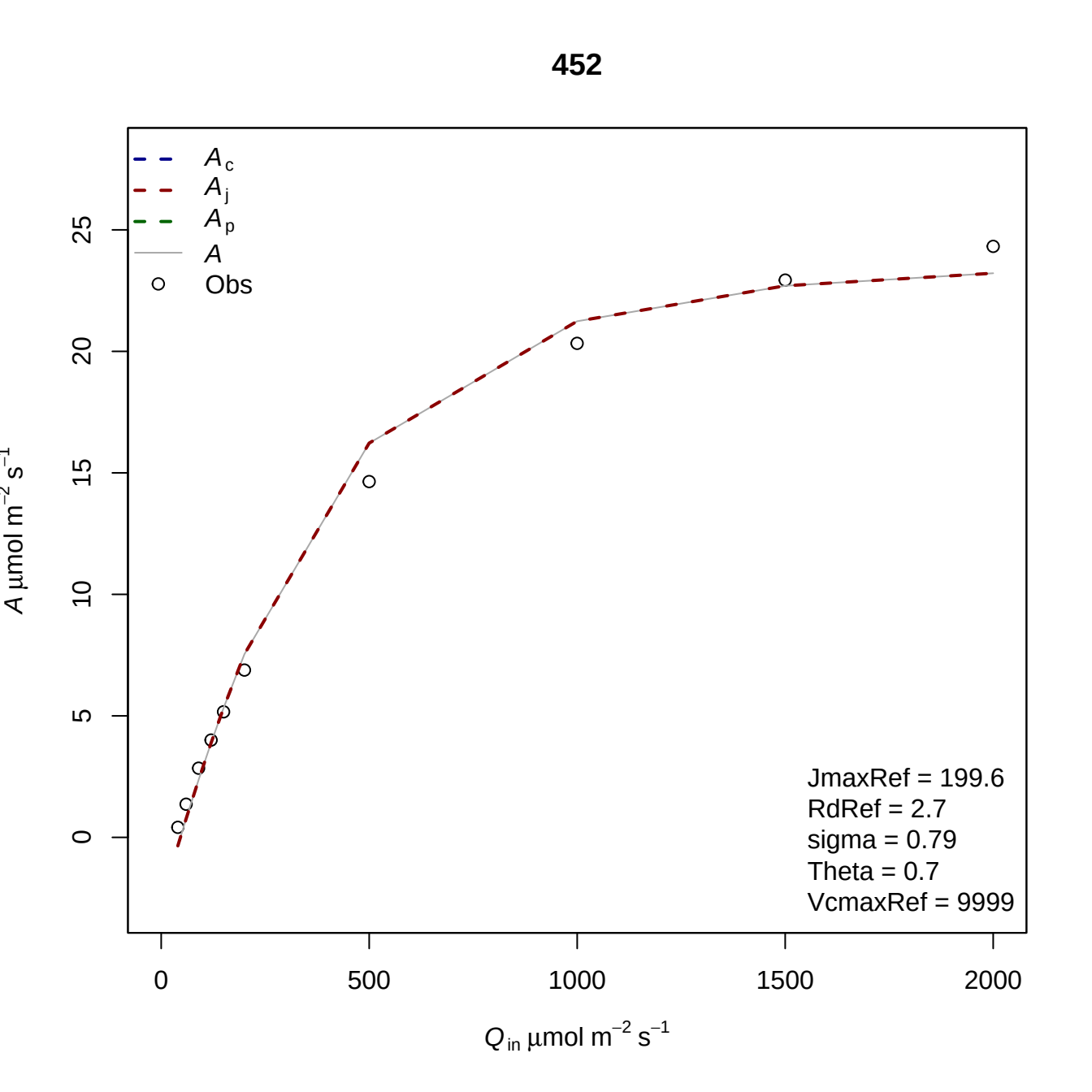

447
A_c
A_j
A_p
A
Obs
JmaxRef = 183.36
RdRef = 3.87
sigma = 1.17
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

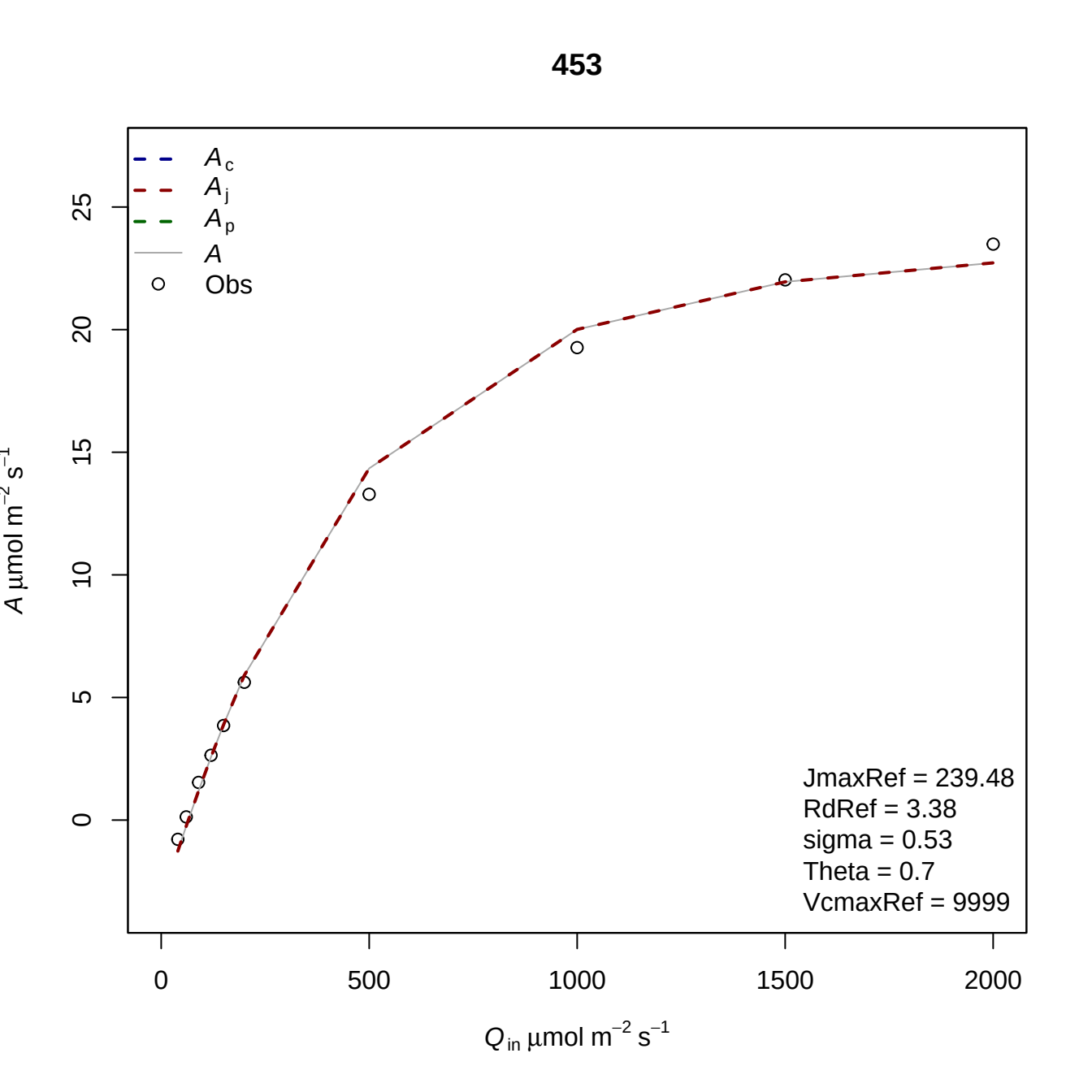

448
A_c
A_j
A_p
A
Obs
JmaxRef = 193.41
RdRef = 4.54
sigma = 0.93
Theta = 0.7
VcmaxRef = 9999
20
15
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

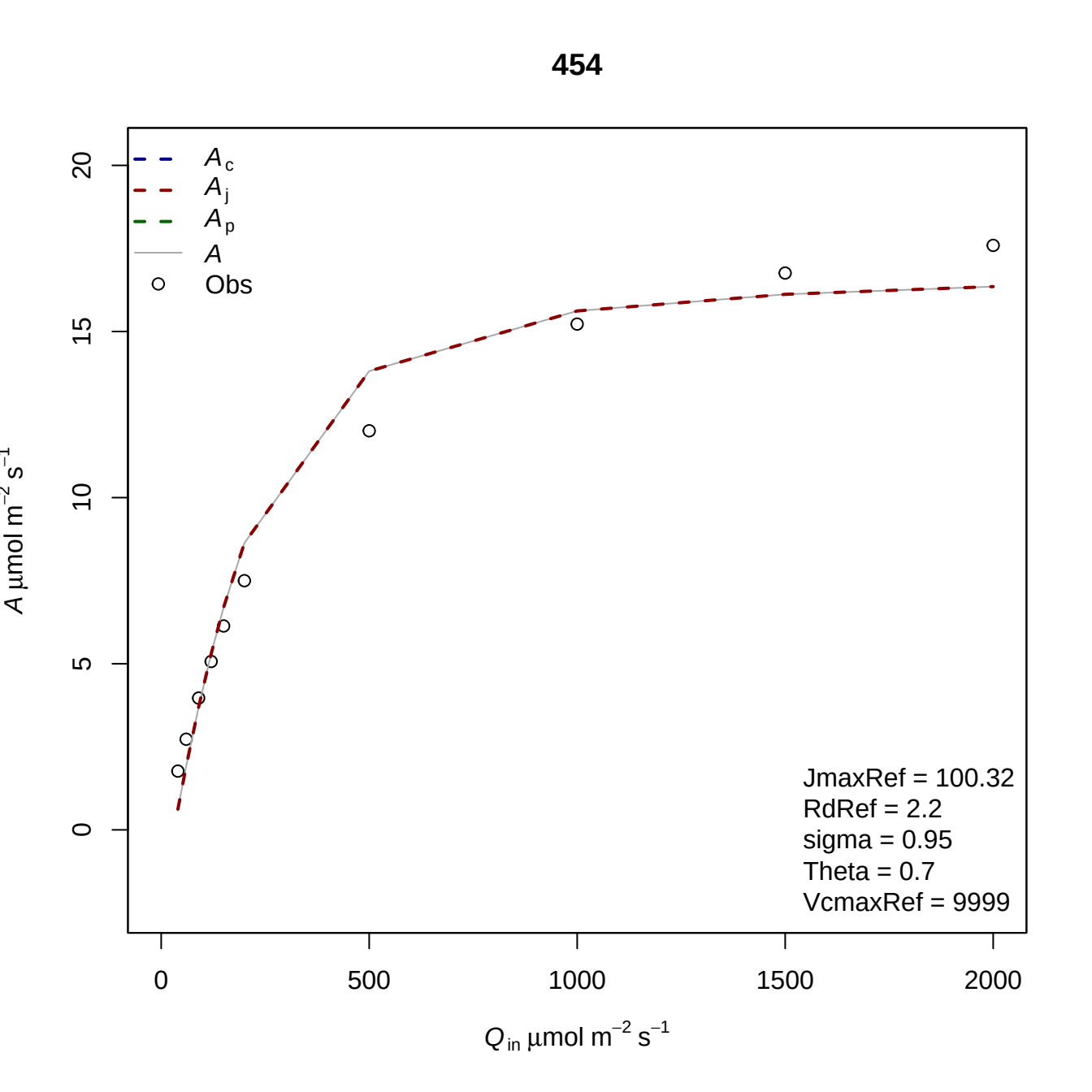

449
A_c
A_j
A_p
A
Obs
JmaxRef = 147.12
RdRef = 2.14
sigma = 0.9
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

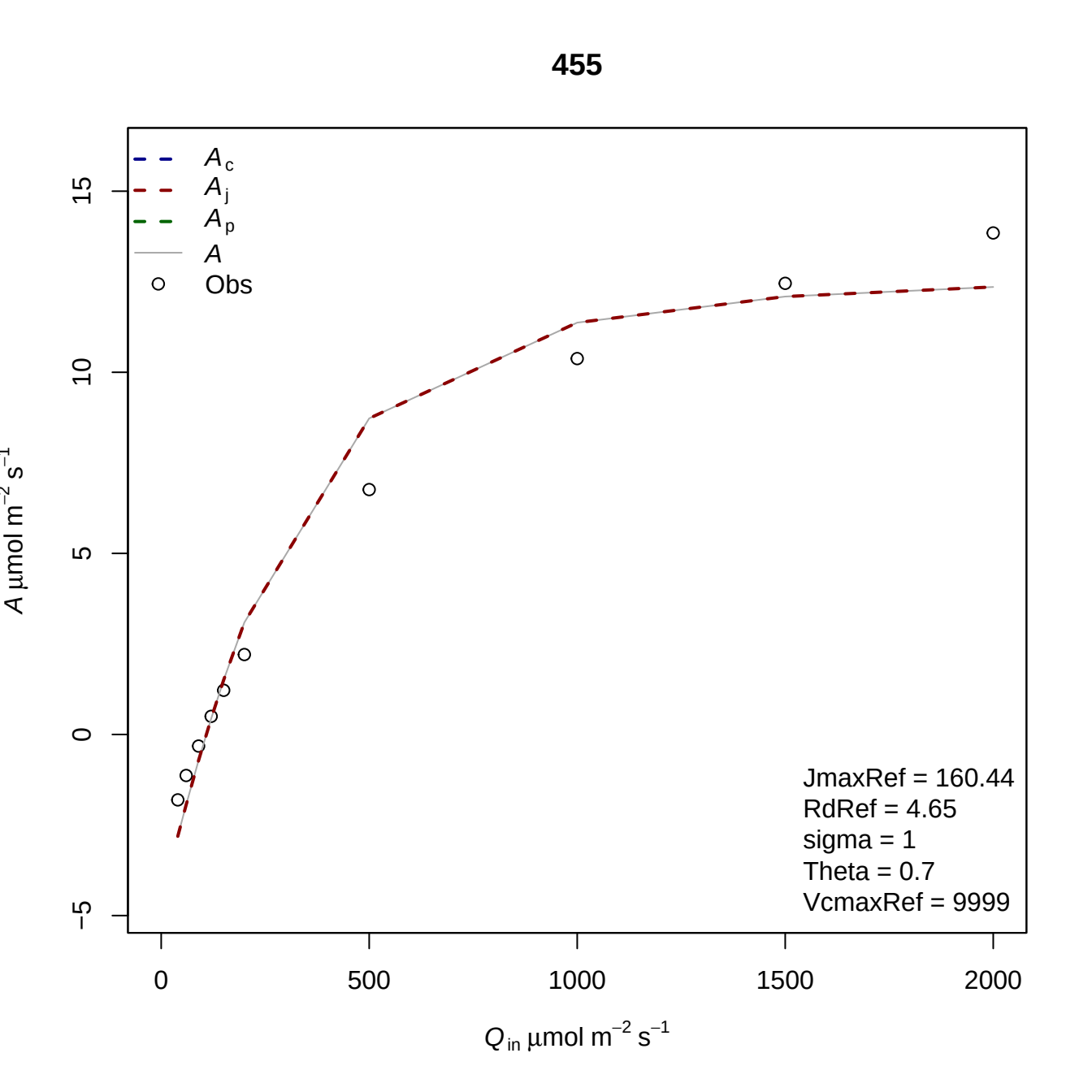

450
A_c
A_j
A_p
A
Obs
JmaxRef = 207.99
RdRef = 1.46
sigma = 0.48
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

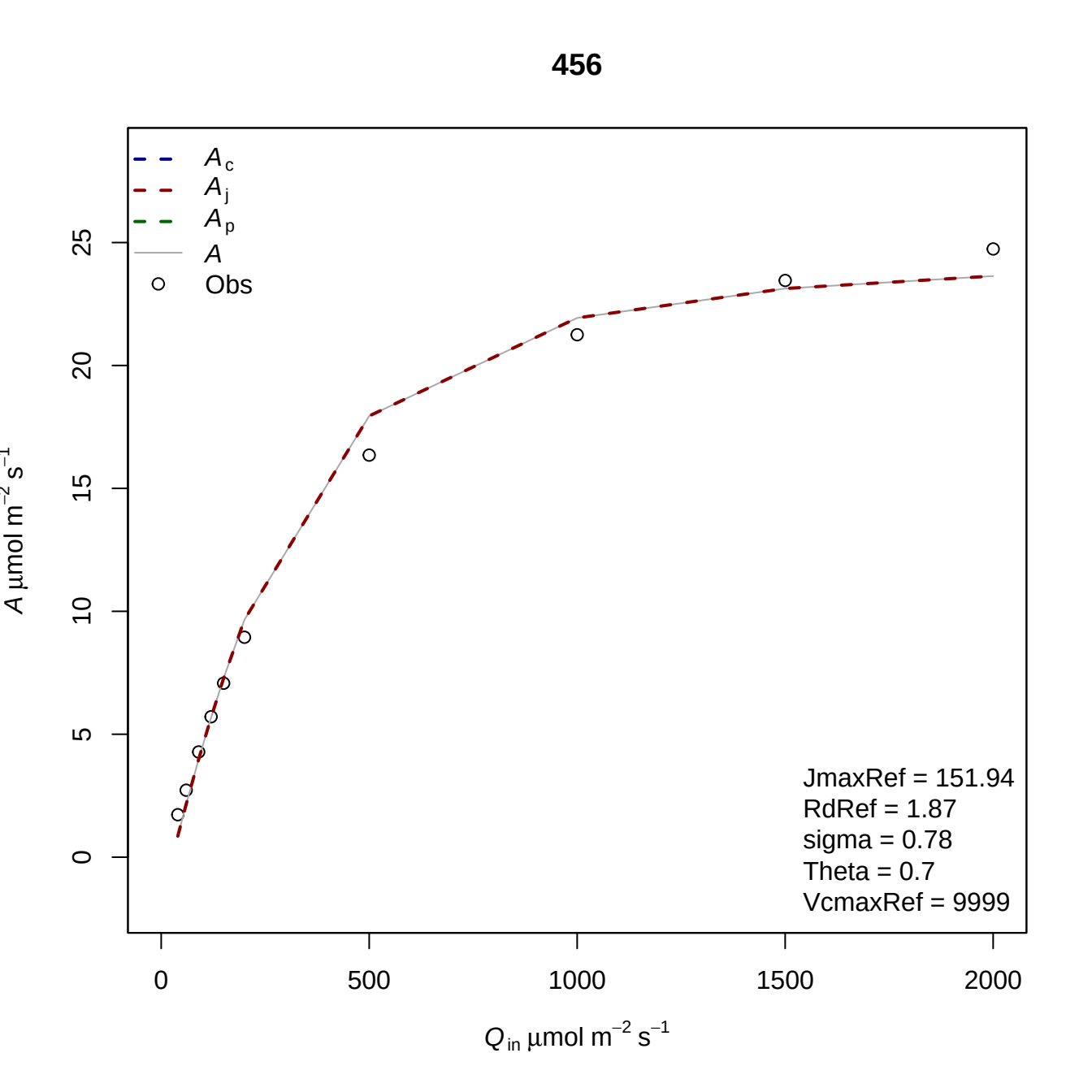

451
A_c
A_j
A_p
A
Obs
JmaxRef = 180.33
RdRef = 1.96
sigma = 0.65
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
25
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

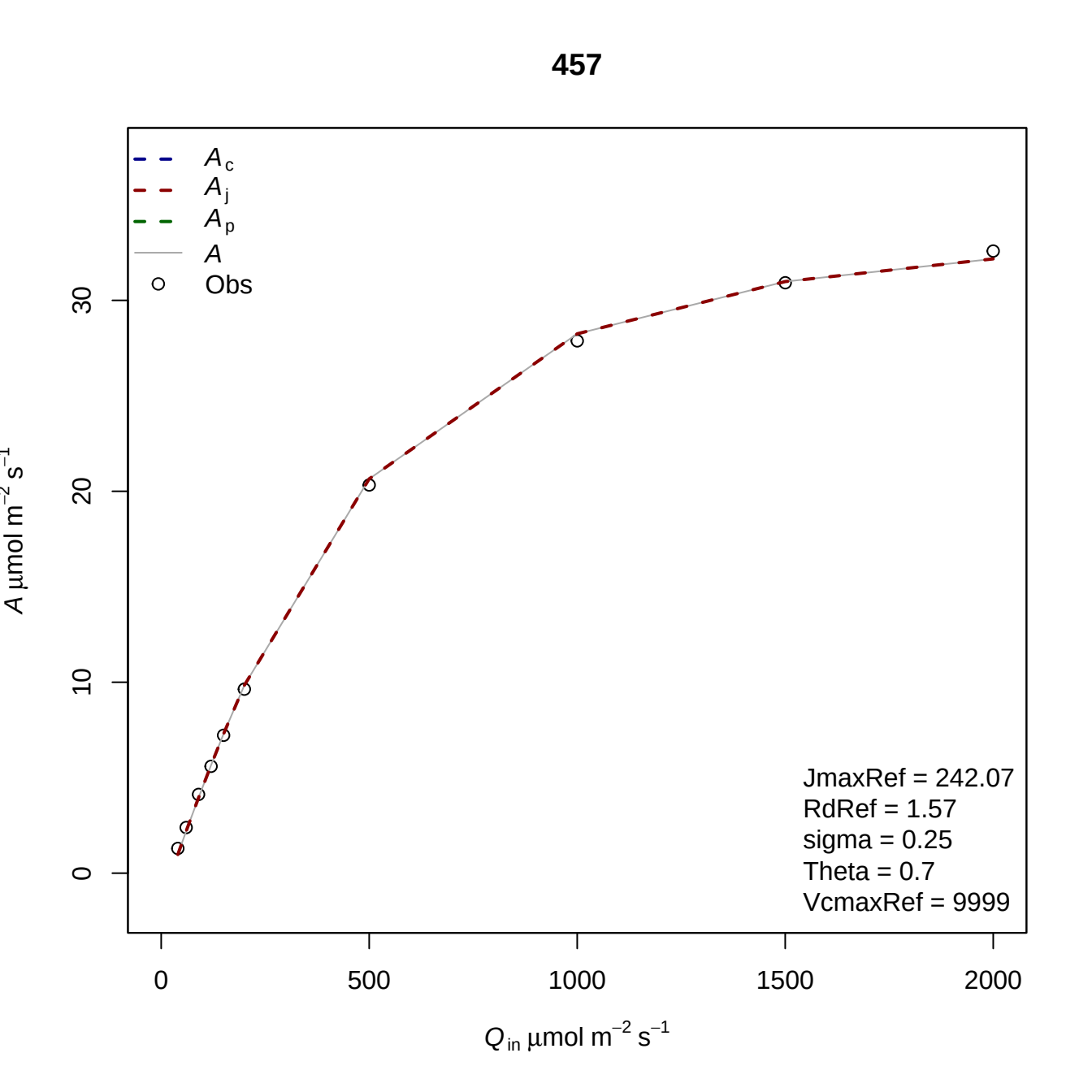

452
A_c
A_j
A_p
A
Obs
JmaxRef = 199.6
RdRef = 2.7
sigma = 0.79
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

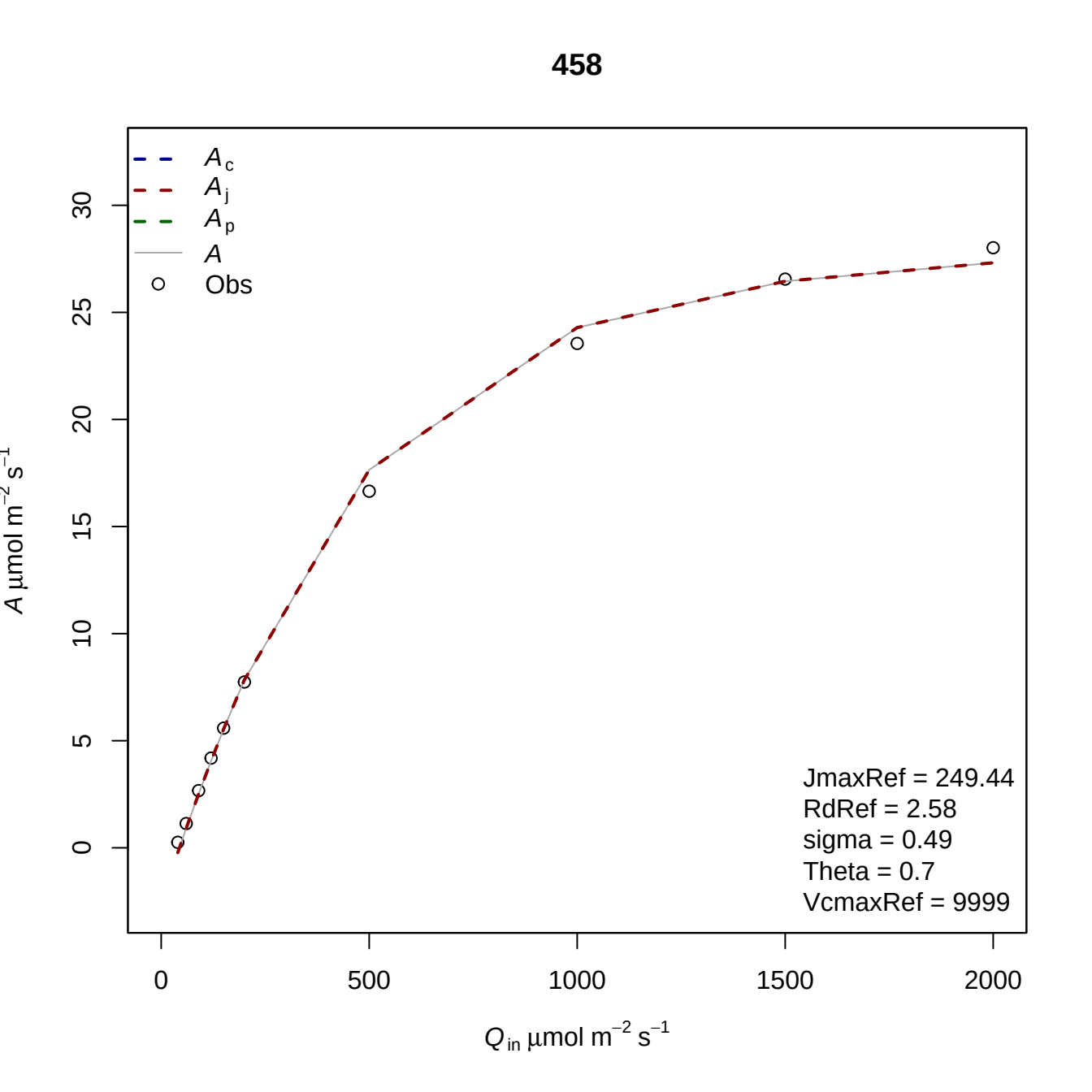

453
A_c
A_j
A_p
A
Obs
JmaxRef = 239.48
RdRef = 3.38
sigma = 0.53
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

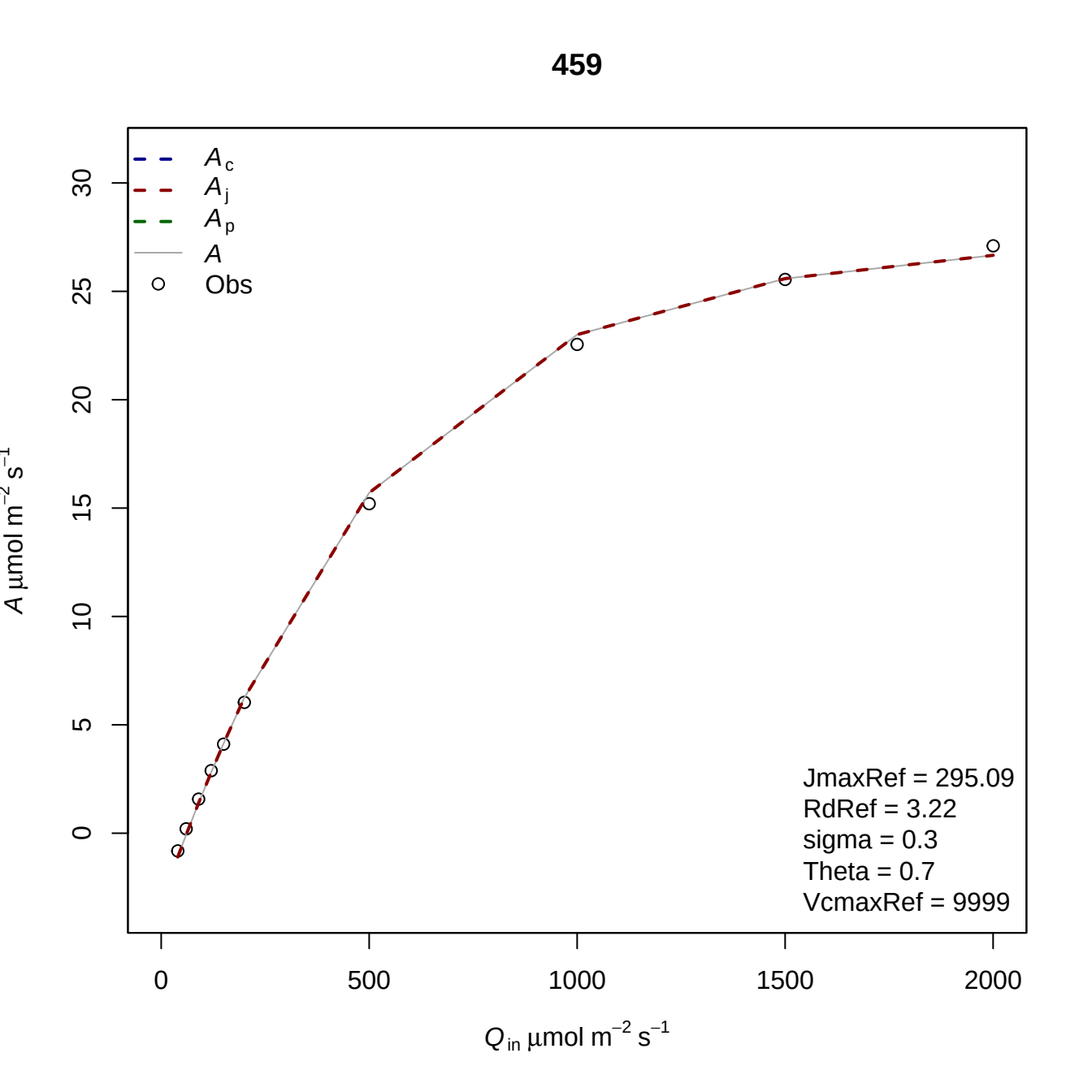

454
A_c
A_j
A_p
A
Obs
JmaxRef = 100.32
RdRef = 2.2
sigma = 0.95
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

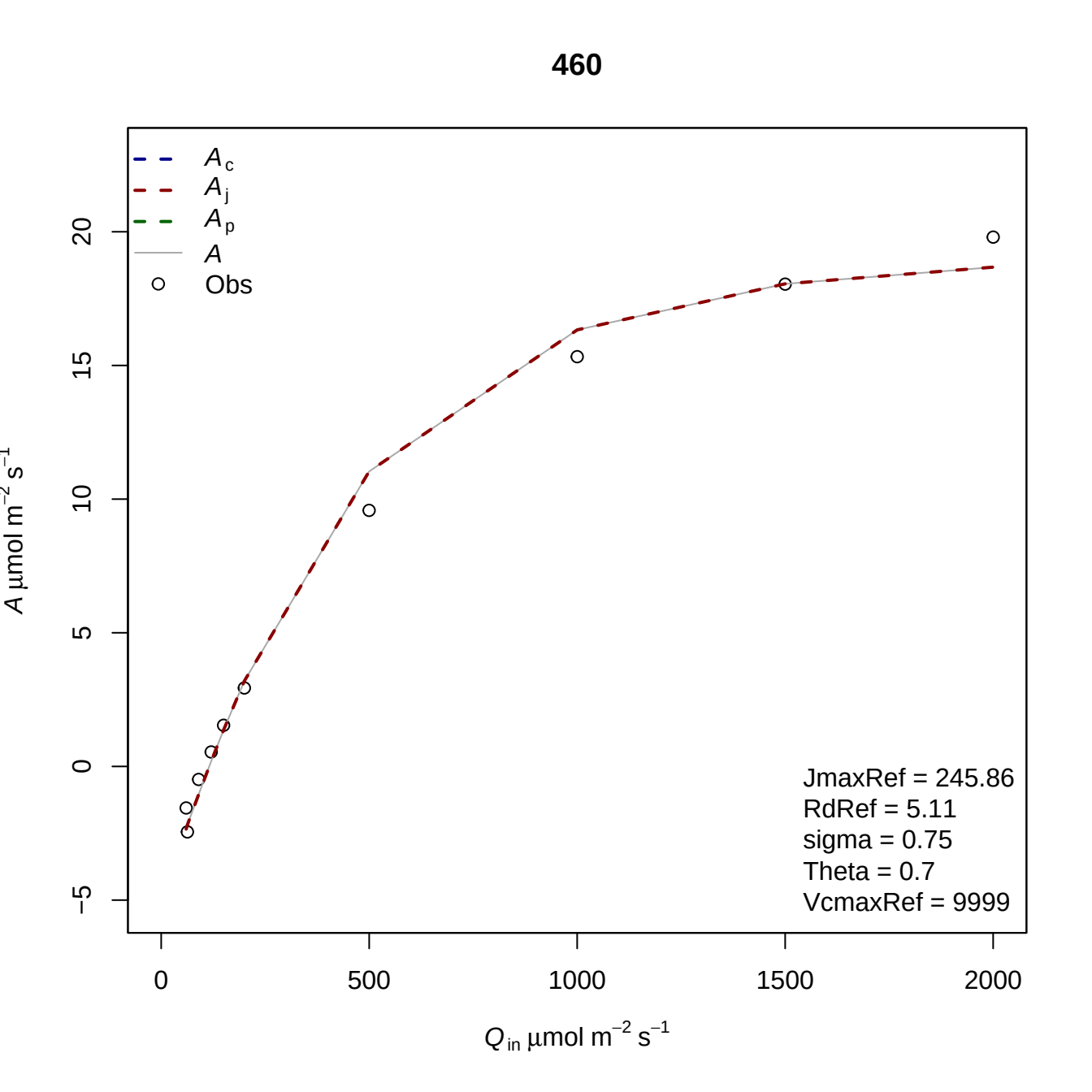

455
A_c
A_j
A_p
A
Obs
JmaxRef = 160.44
RdRef = 4.65
sigma = 1
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
15
10
5
0
−5
0
500
1000
1500
2000

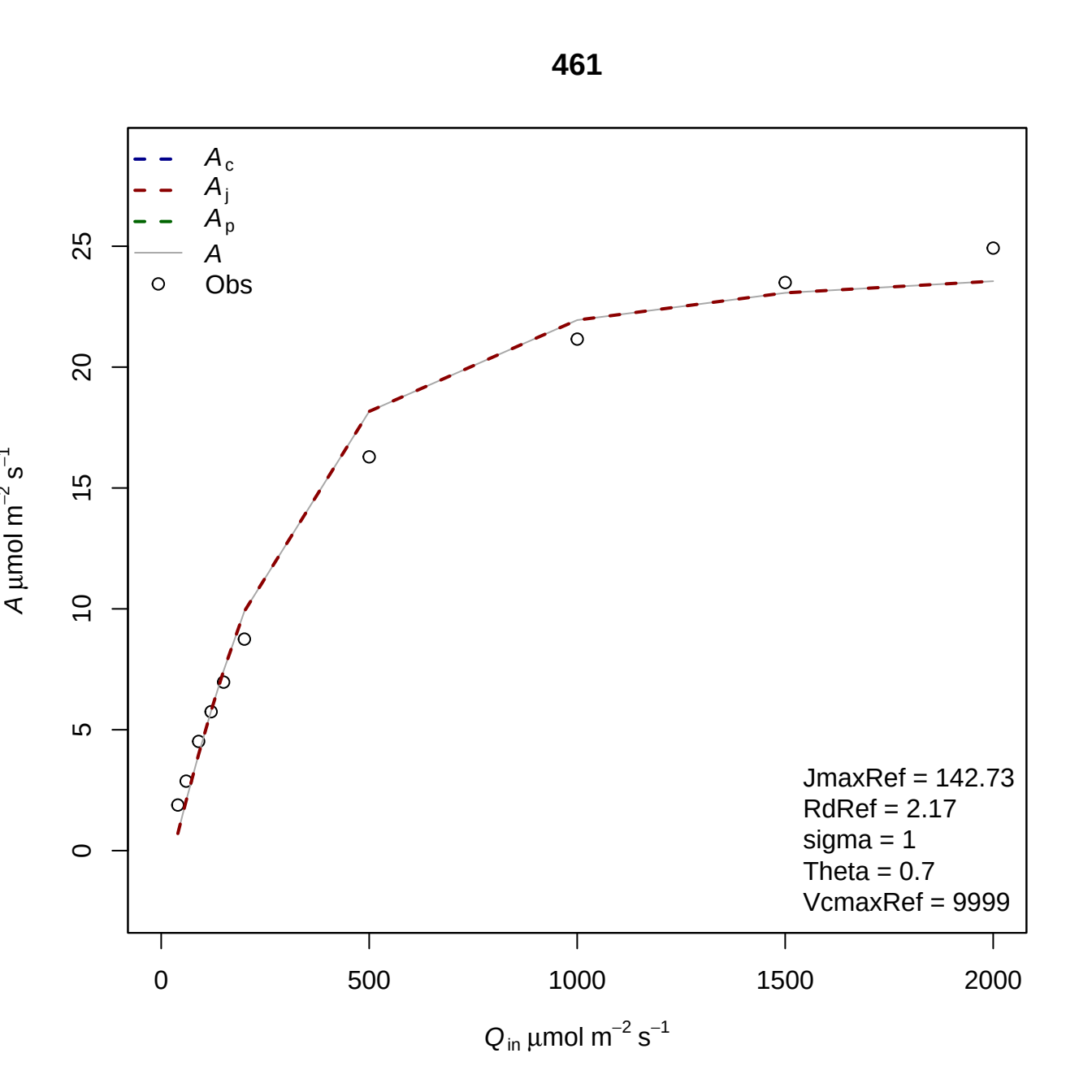

456
A_c
A_j
A_p
A
Obs
JmaxRef = 151.94
RdRef = 1.87
sigma = 0.78
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
25
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

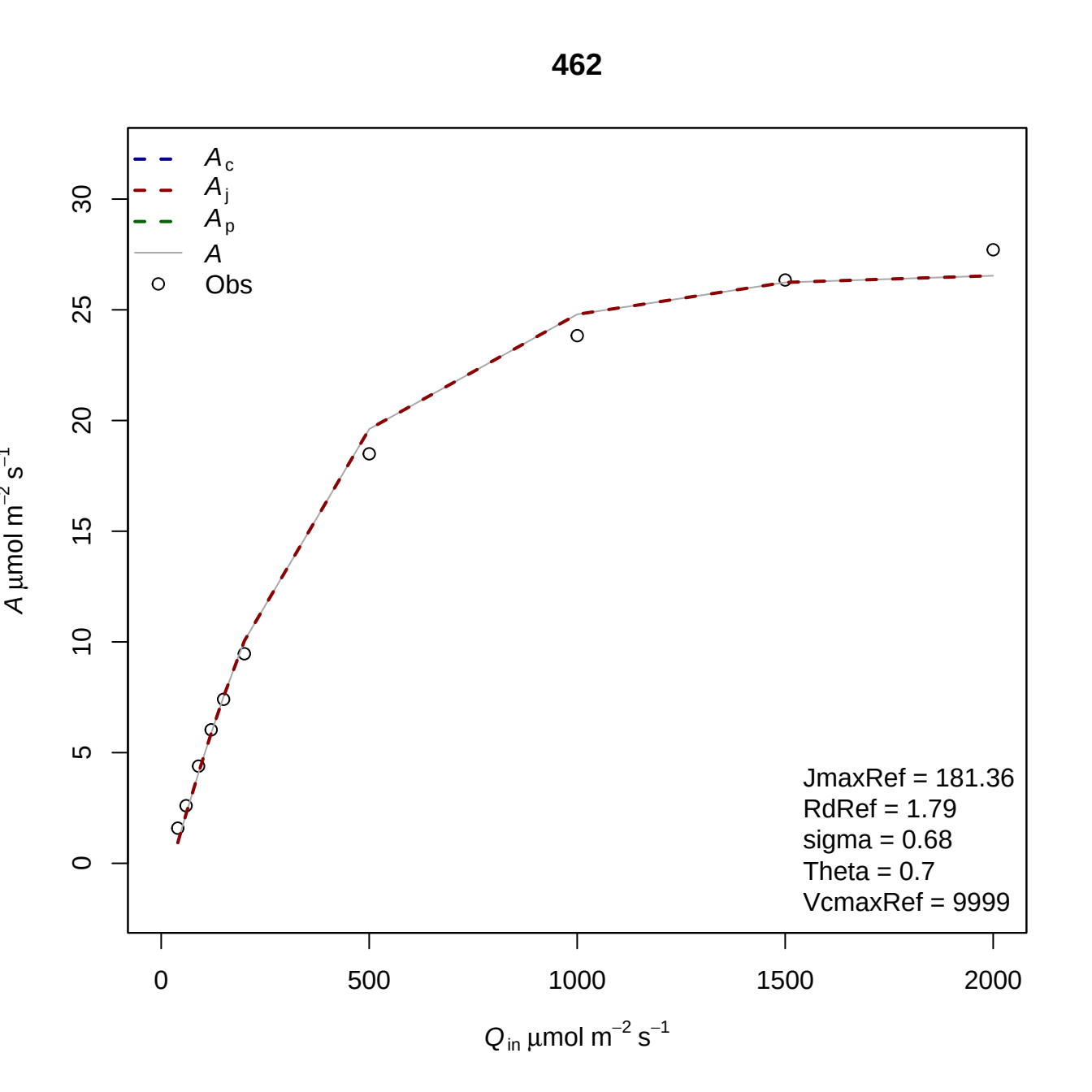

457
A_c
A_j
A_p
A
Obs
JmaxRef = 242.07
RdRef = 1.57
sigma = 0.25
Theta = 0.7
VcmaxRef = 9999
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

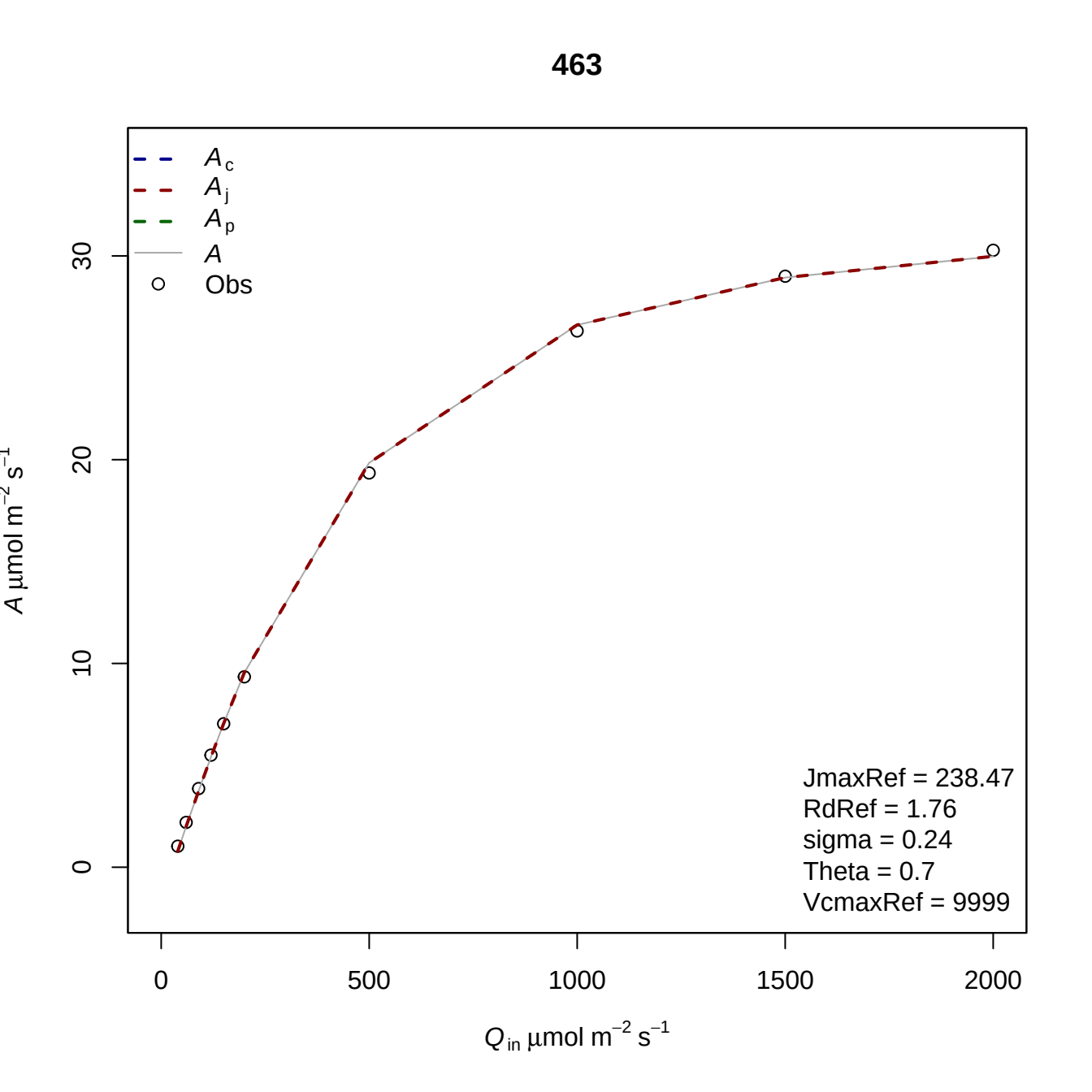

458
A_c
A_j
A_p
A
Obs
JmaxRef = 249.44
RdRef = 2.58
sigma = 0.49
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

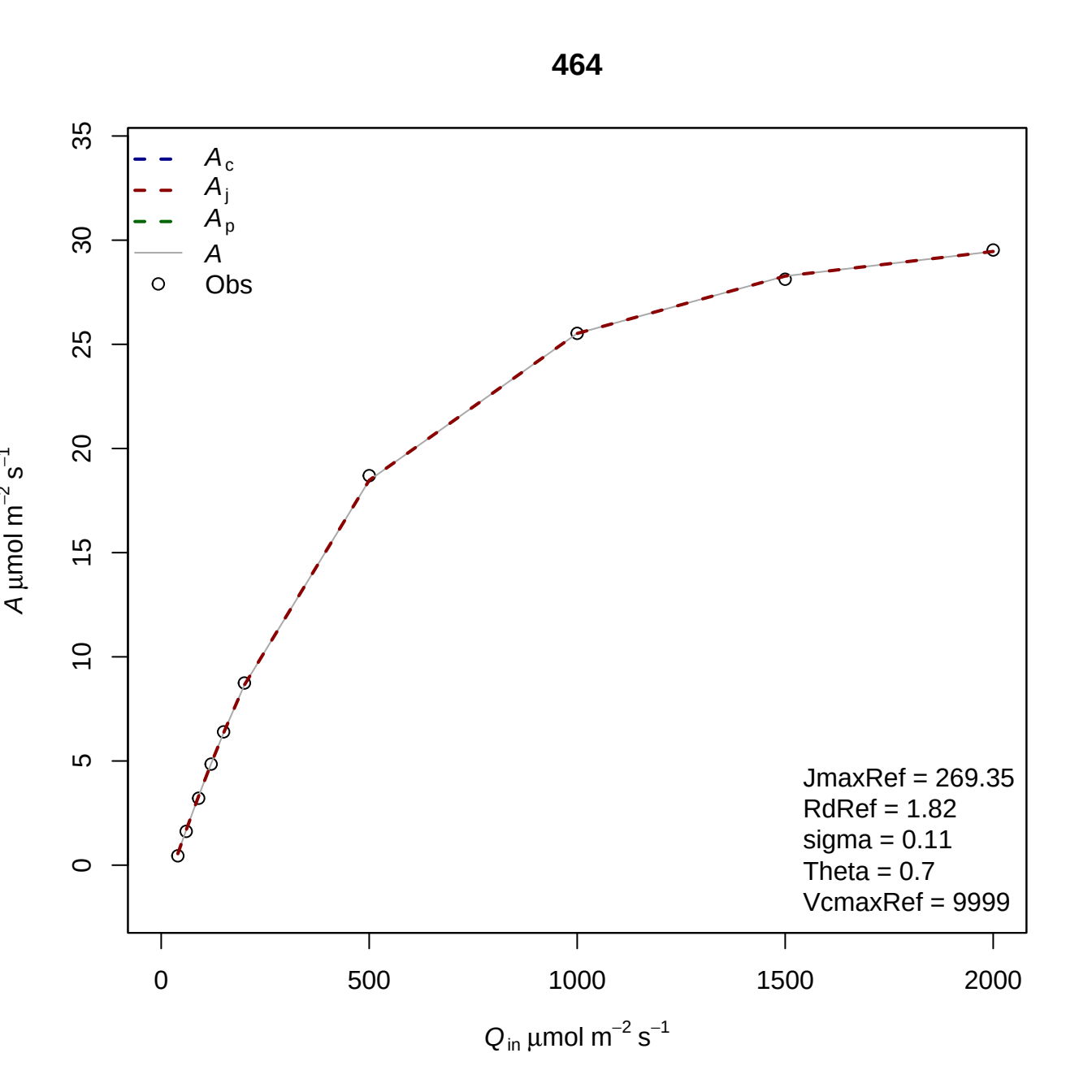

459
A_c
A_j
A_p
A
Obs
JmaxRef = 295.09
RdRef = 3.22
sigma = 0.3
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

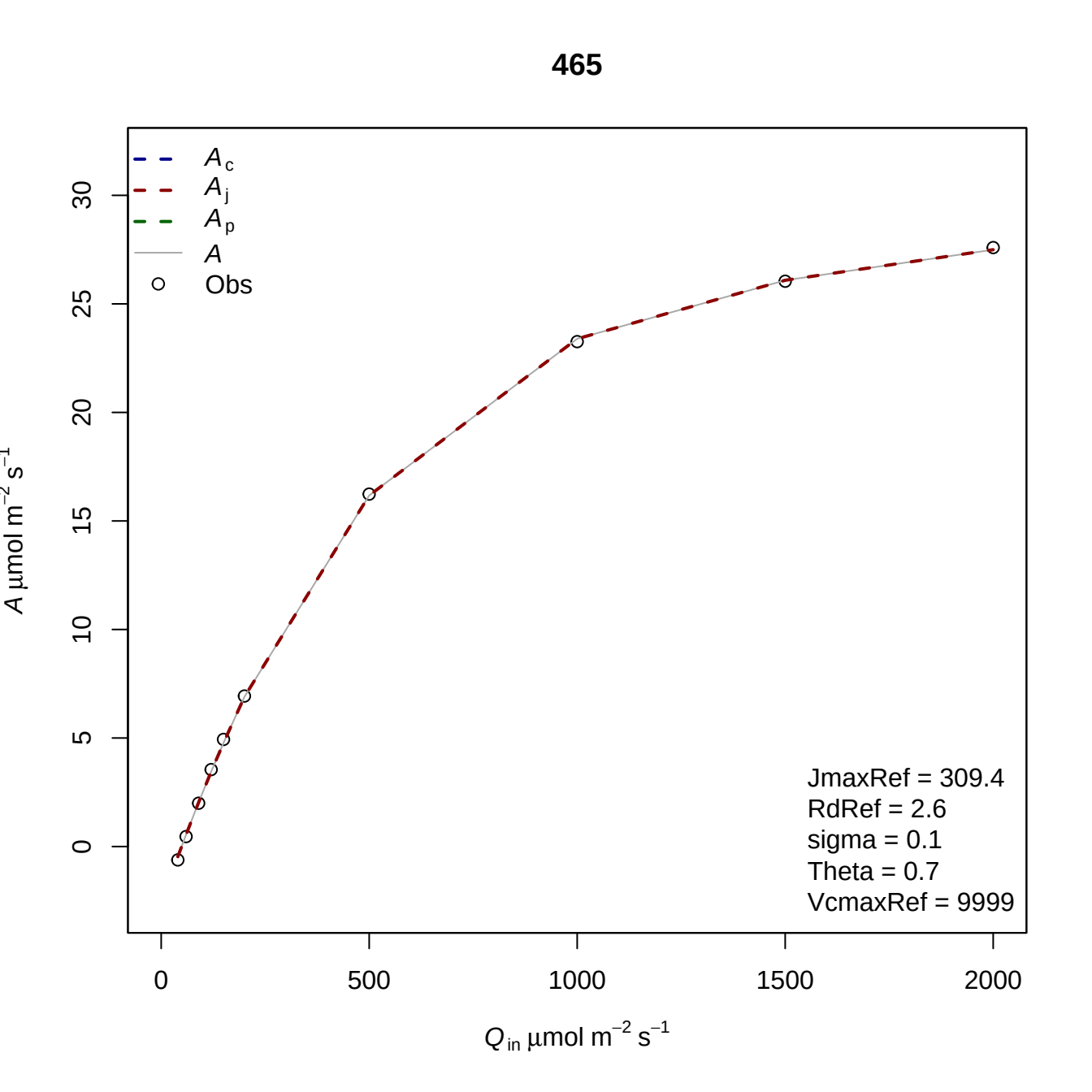

460
A_c
A_j
A_p
A
Obs
JmaxRef = 245.86
RdRef = 5.11
sigma = 0.75
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

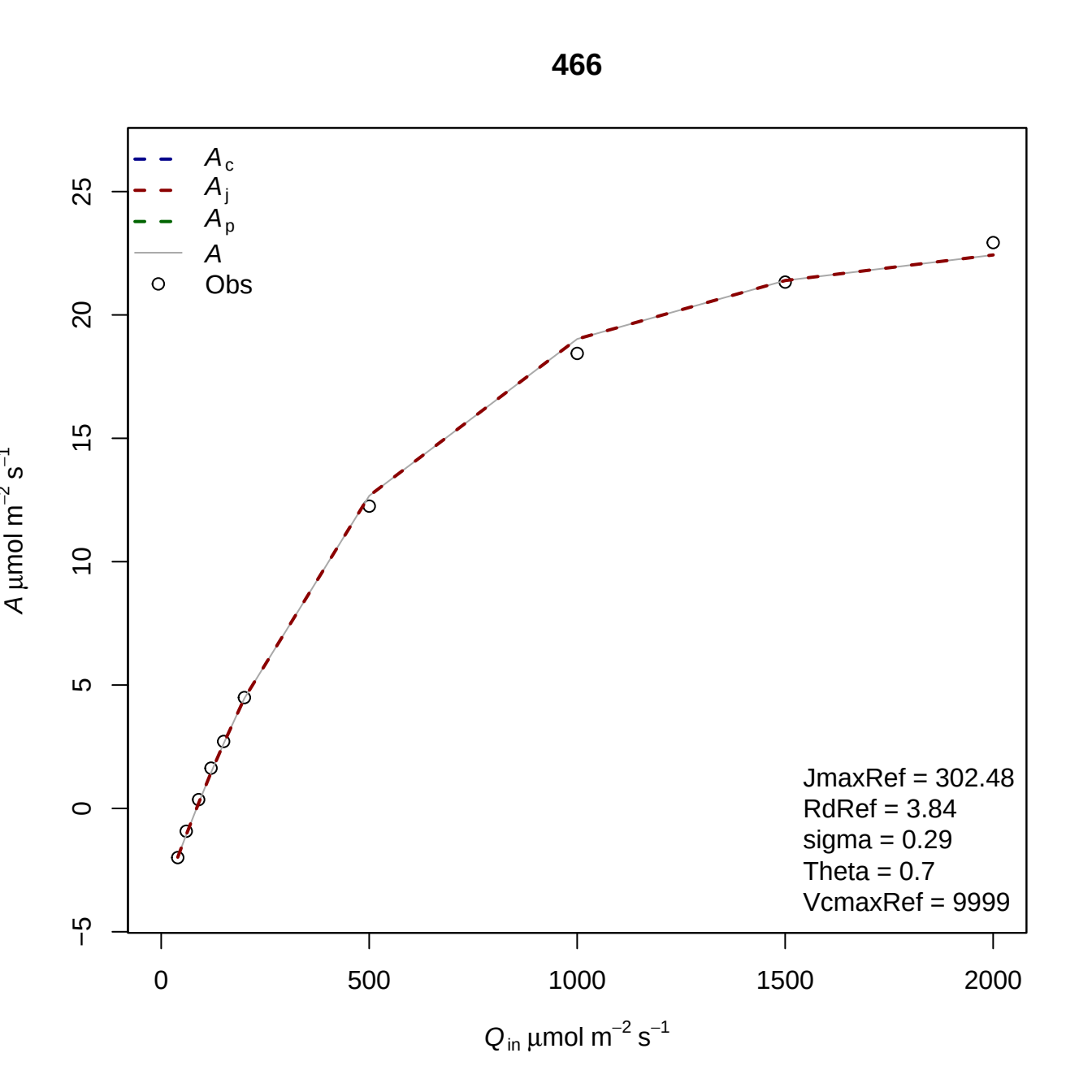

461
A_c
A_j
A_p
A
Obs
JmaxRef = 142.73
RdRef = 2.17
sigma = 1
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

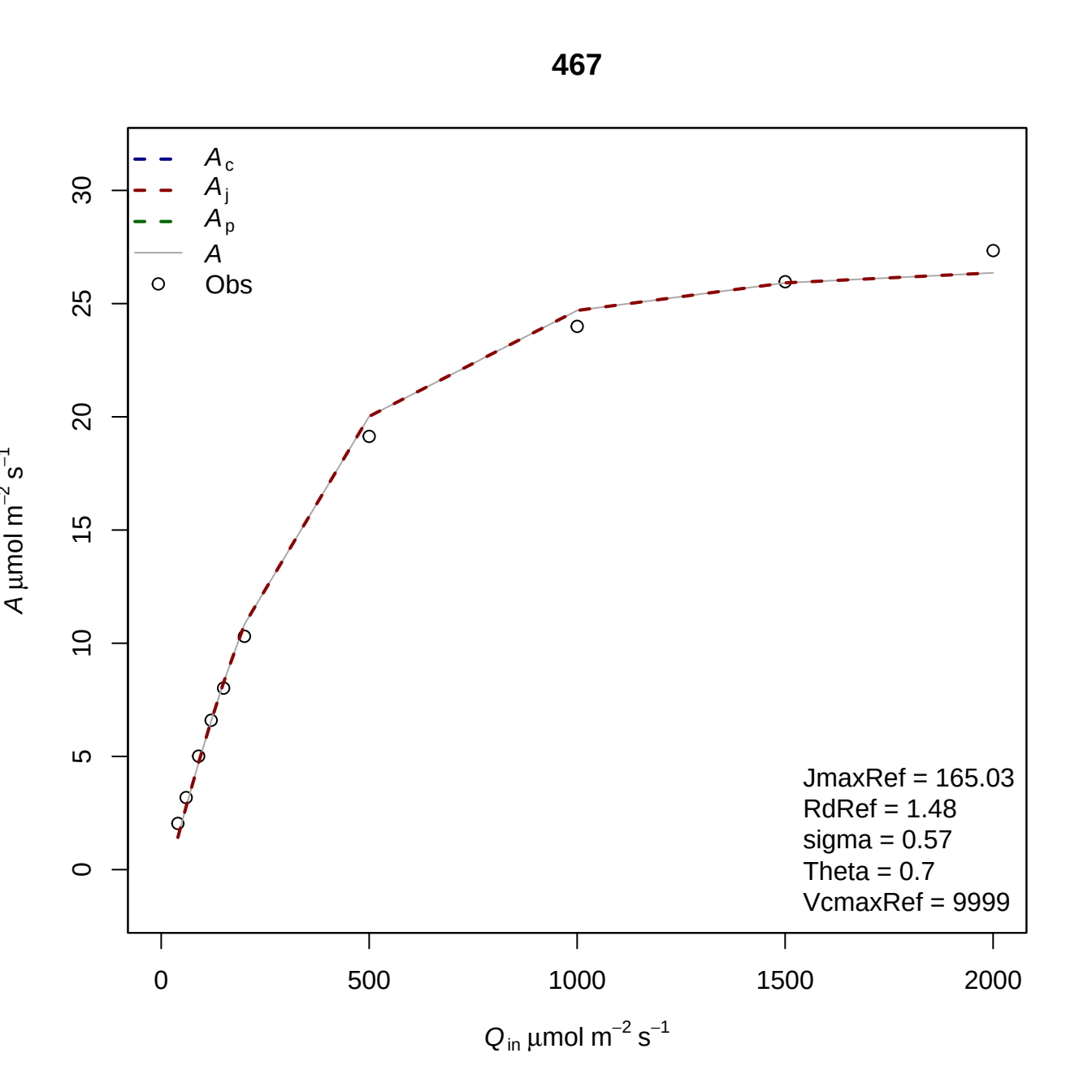

462
A_c
A_j
A_p
A
Obs
JmaxRef = 181.36
RdRef = 1.79
sigma = 0.68
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
25
30
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

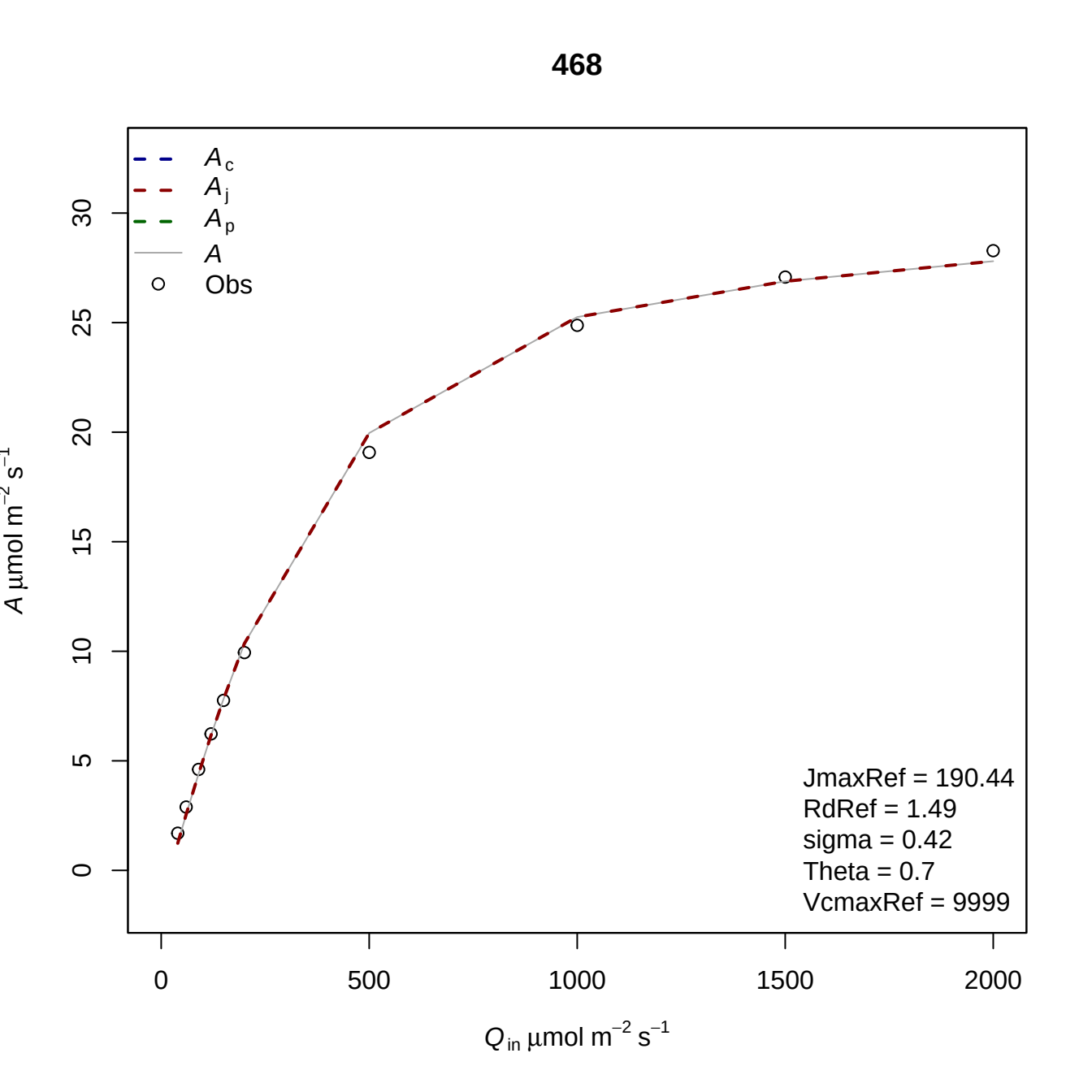

463
A_c
A_j
A_p
A
Obs
JmaxRef = 238.47
RdRef = 1.76
sigma = 0.24
Theta = 0.7
VcmaxRef = 9999
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

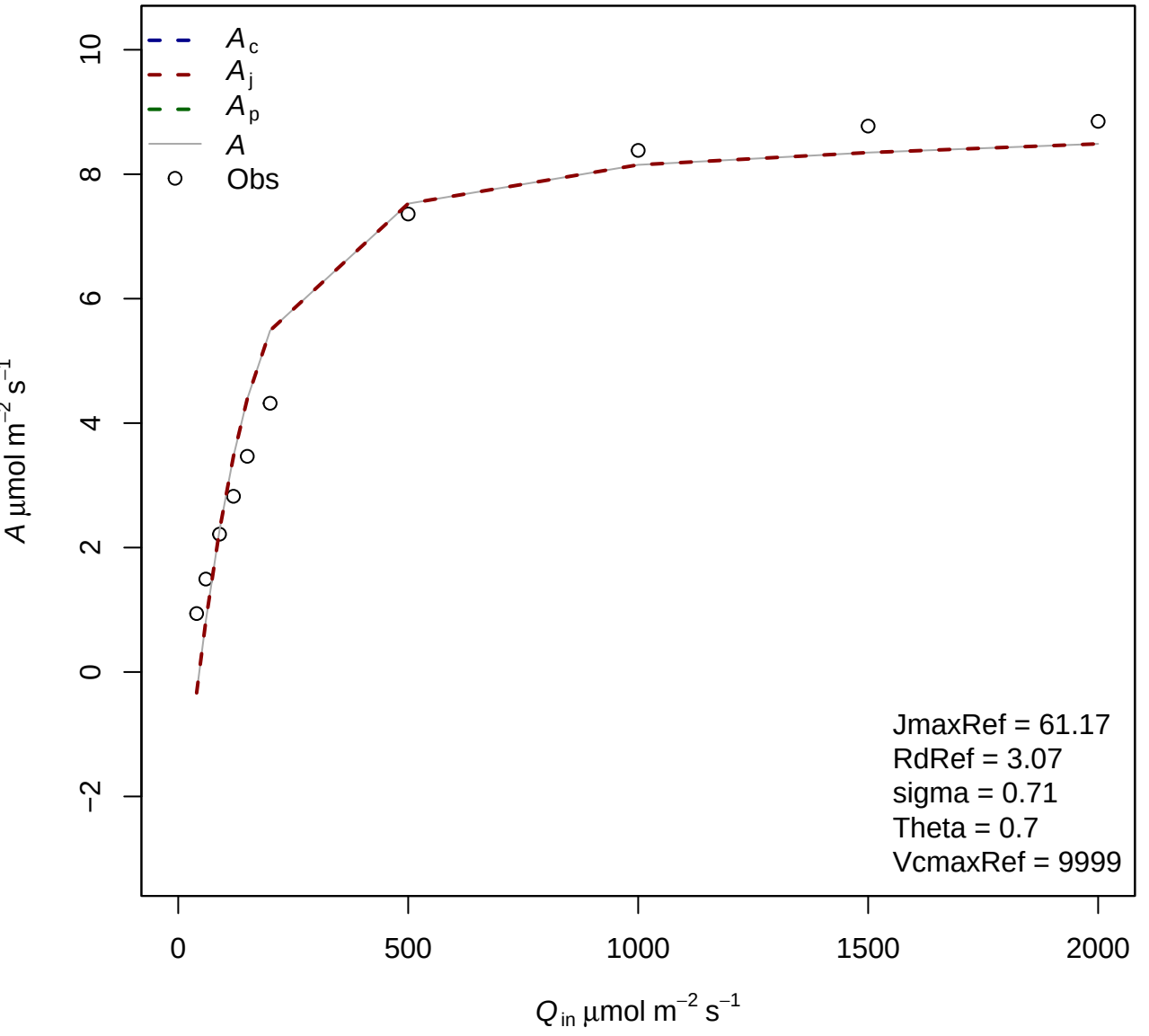

464
A_c
A_j
A_p
A
Obs
JmaxRef = 269.35
RdRef = 1.82
sigma = 0.11
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
35
0
500
1000
1500
2000

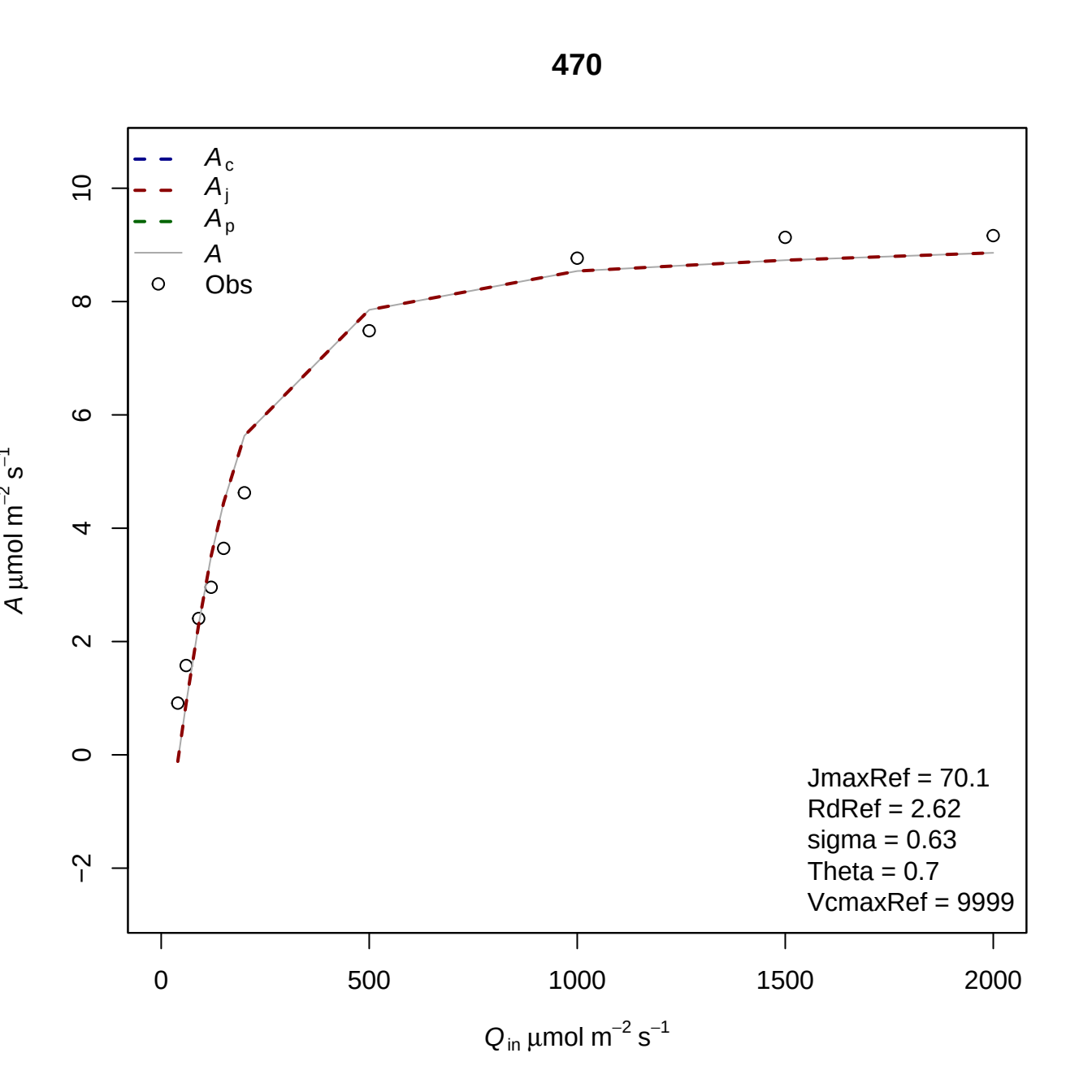

465
A_c
A_j
A_p
A
Obs
JmaxRef = 309.4
RdRef = 2.6
sigma = 0.1
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

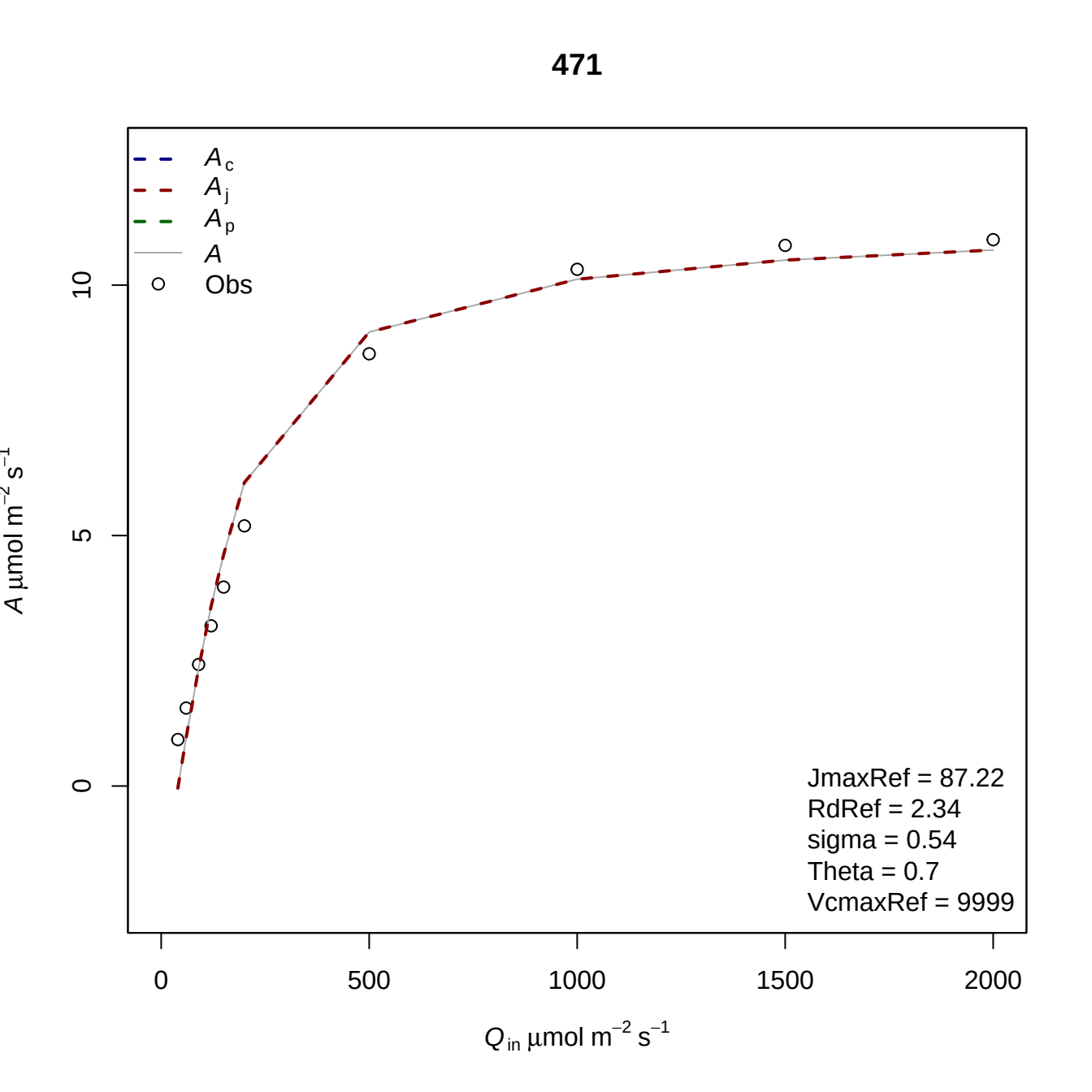

466
A_c
A_j
A_p
A
Obs
JmaxRef = 302.48
RdRef = 3.84
sigma = 0.29
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
25
0
500
1000
1500
2000

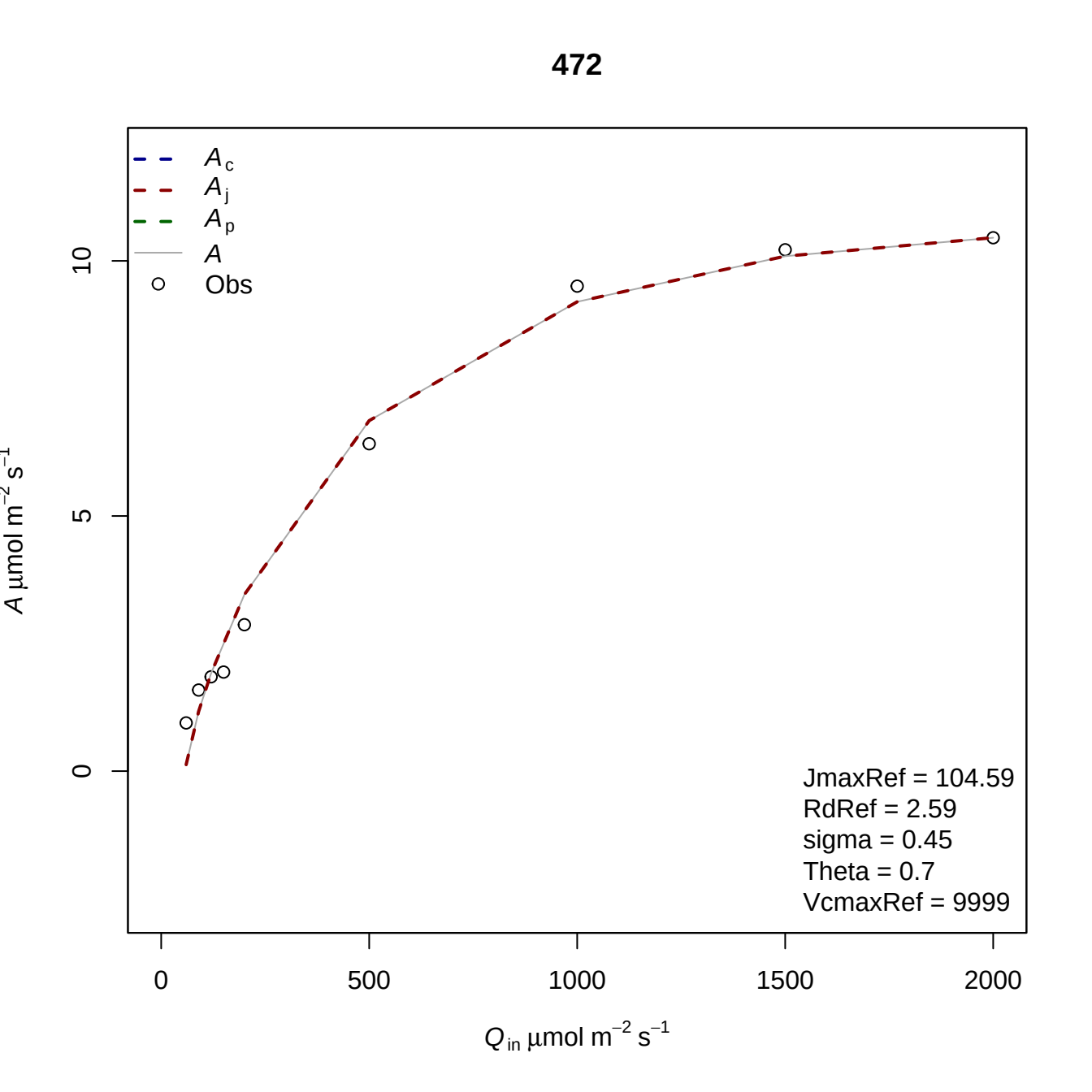

467
A_c
A_j
A_p
A
Obs
JmaxRef = 165.03
RdRef = 1.48
sigma = 0.57
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

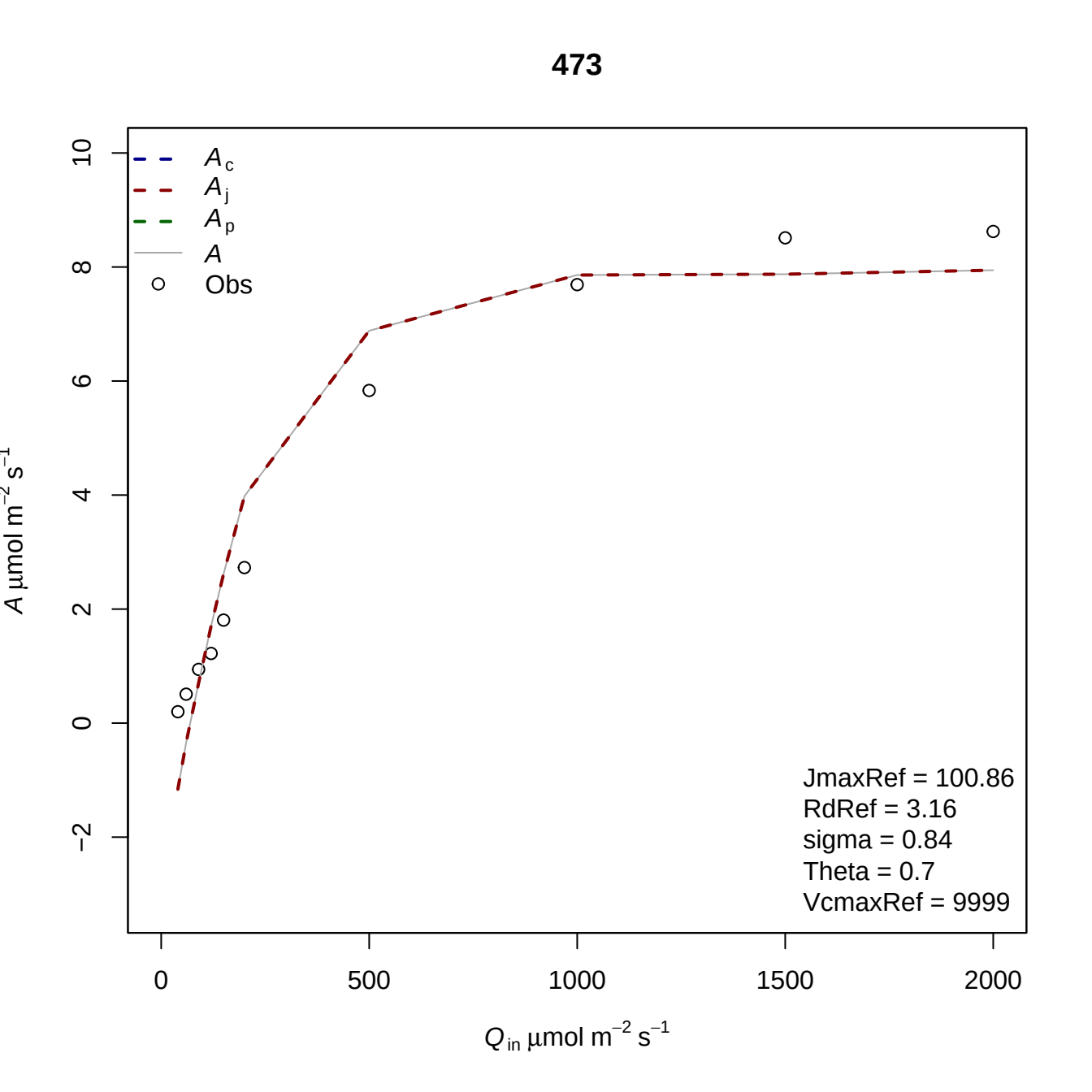

468
A_c
A_j
A_p
A
Obs
JmaxRef = 190.44
RdRef = 1.49
sigma = 0.42
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

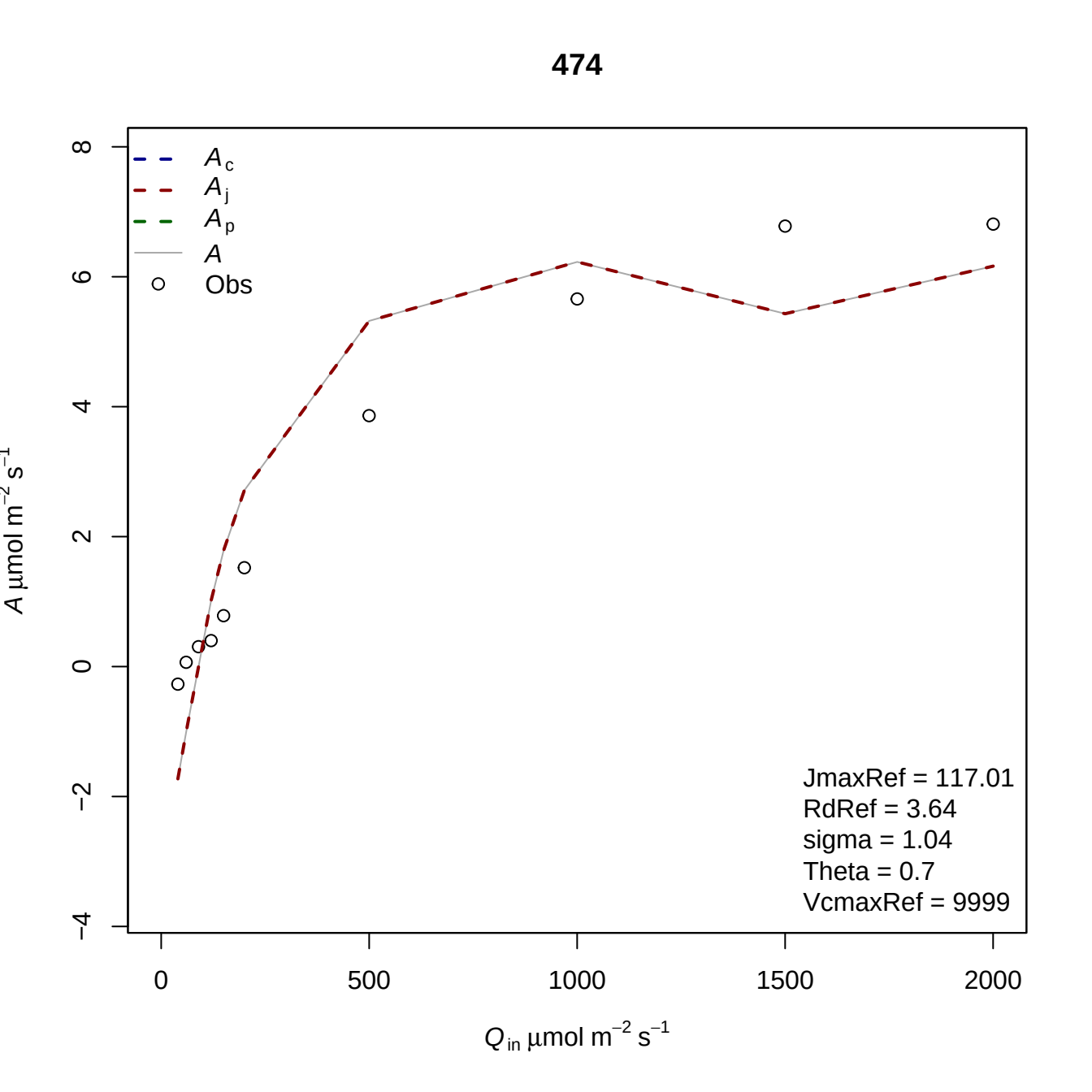

A_c
A_j
A_p
A
Obs
JmaxRef = 61.17
RdRef = 3.07
sigma = 0.71
Theta = 0.7
VcmaxRef = 9999
10
8
6
4
2
0
−2
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

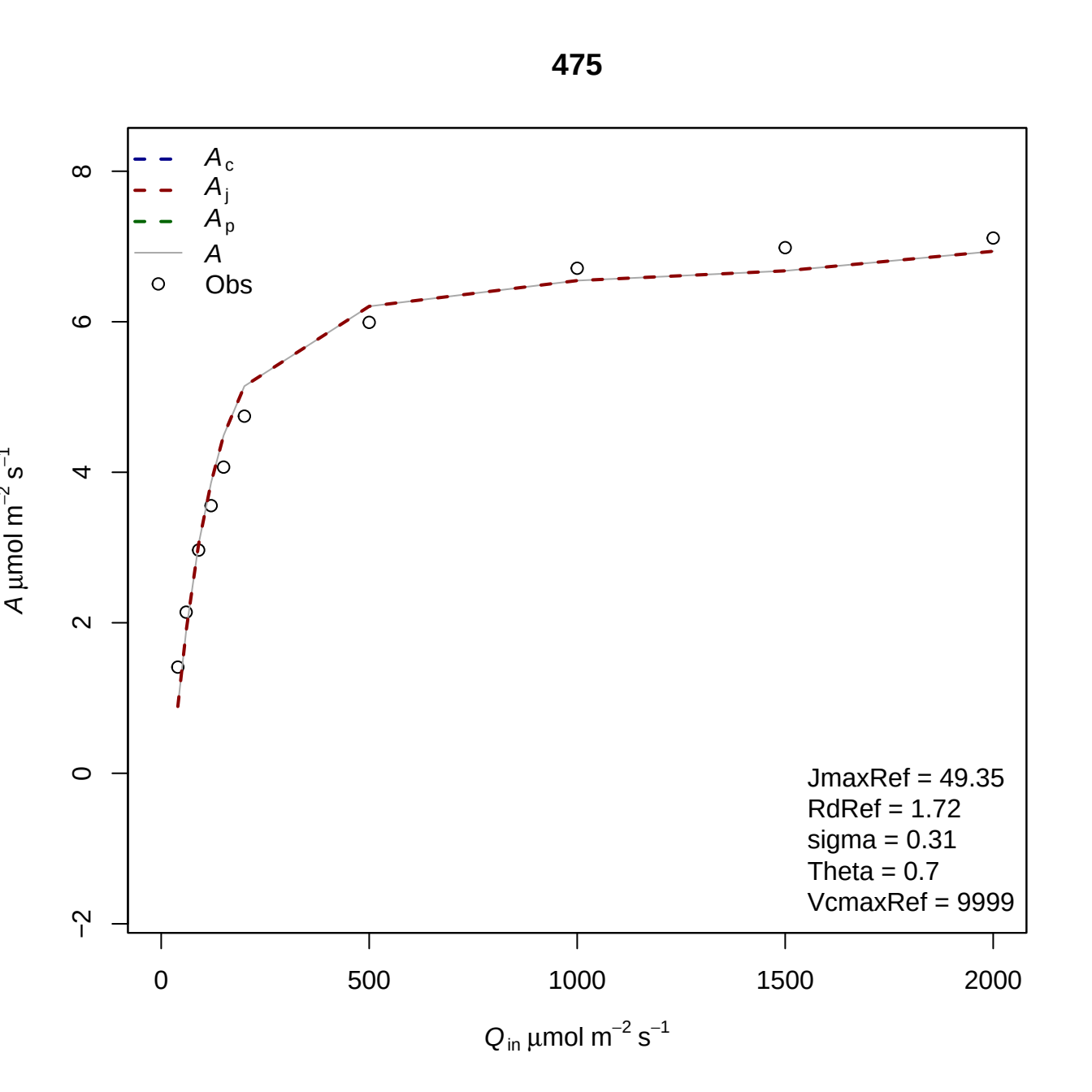

470
A_c
A_j
A_p
A
Obs
JmaxRef = 70.1
RdRef = 2.62
sigma = 0.63
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

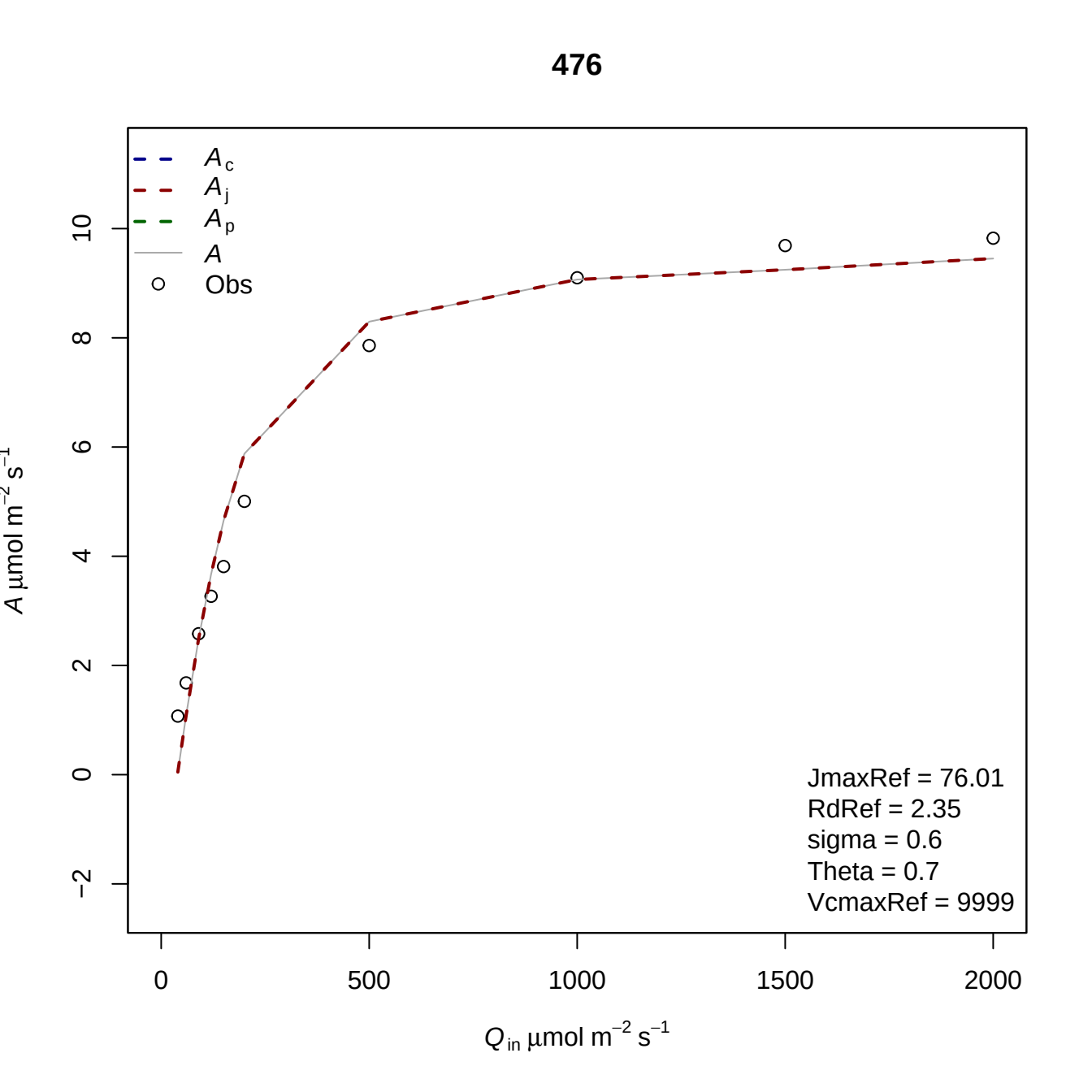

471
A_c
A_j
A_p
A
Obs
JmaxRef = 87.22
RdRef = 2.34
sigma = 0.54
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

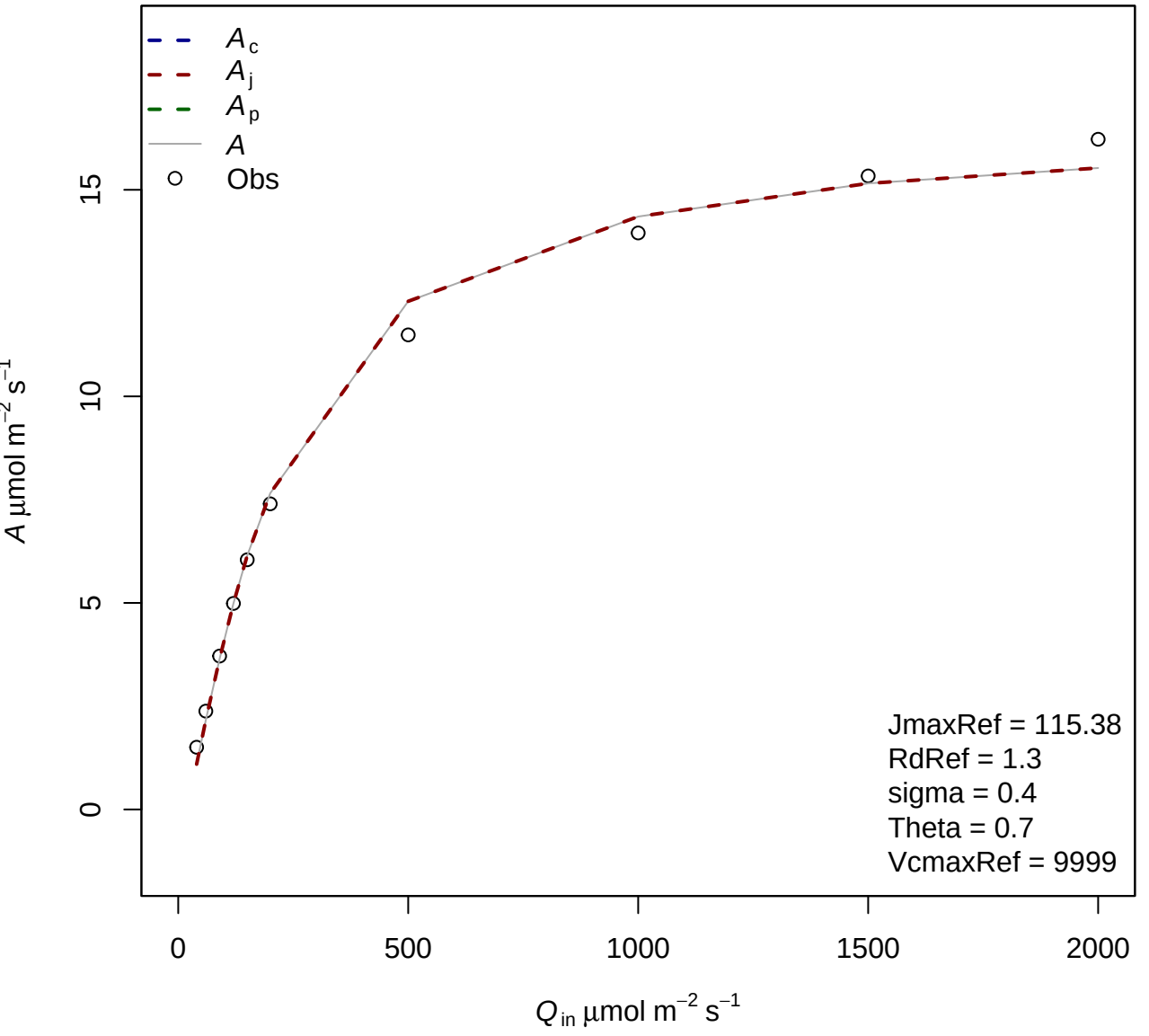

472
A_c
A_j
A_p
A
Obs
JmaxRef = 104.59
RdRef = 2.59
sigma = 0.45
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

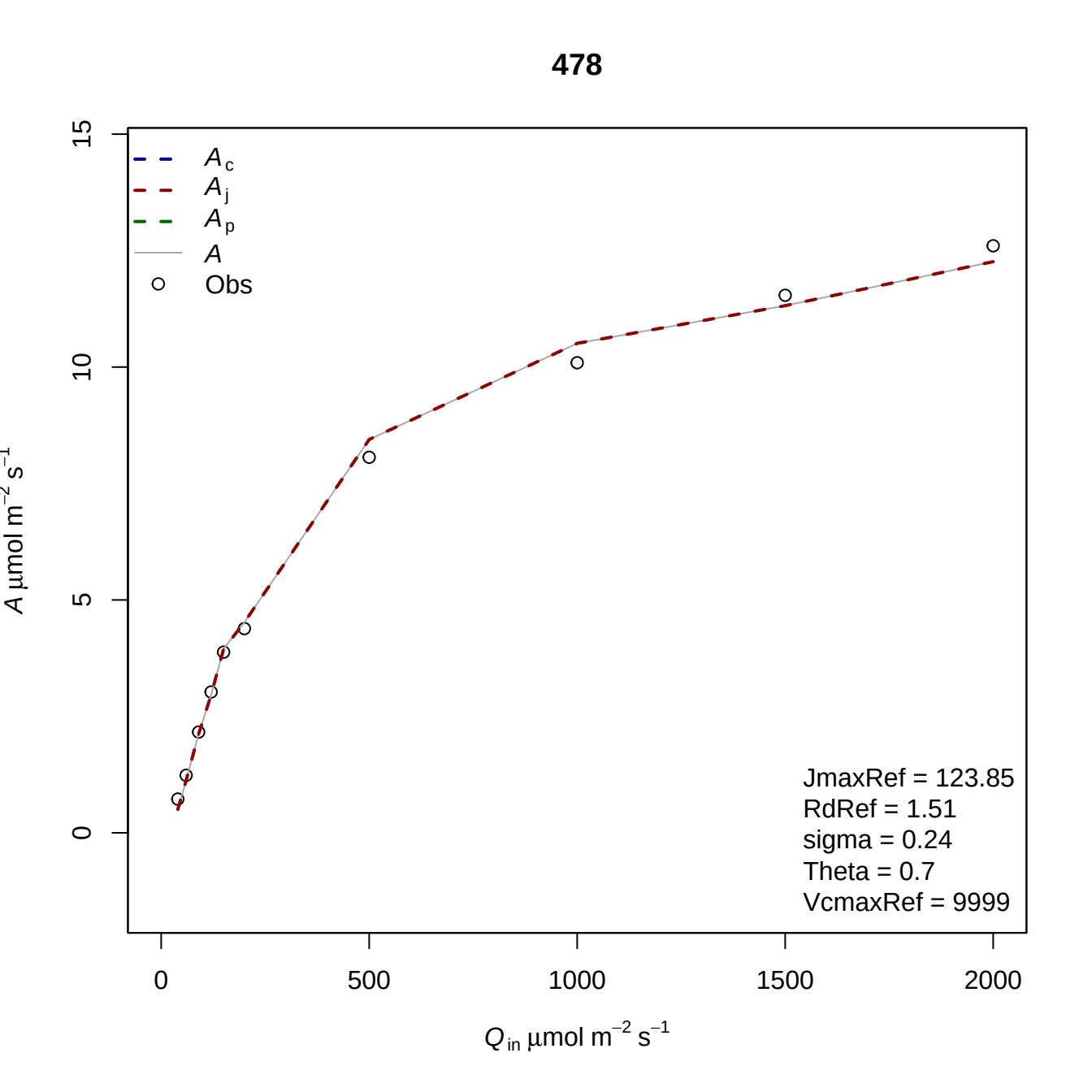

473
A_c
A_j
A_p
A
Obs
JmaxRef = 100.86
RdRef = 3.16
sigma = 0.84
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

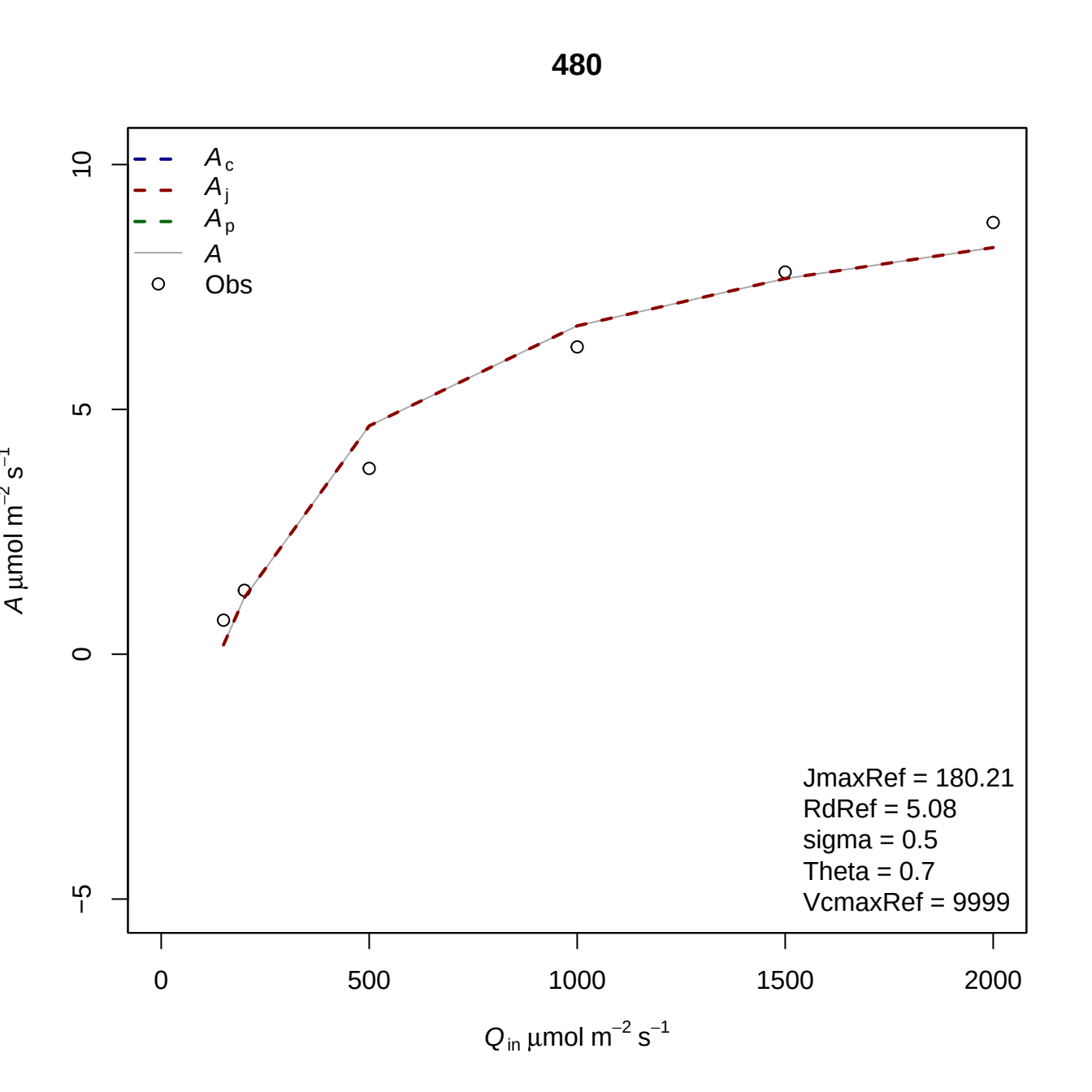

474
A_c
A_j
A_p
A
Obs
JmaxRef = 117.01
RdRef = 3.64
sigma = 1.04
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

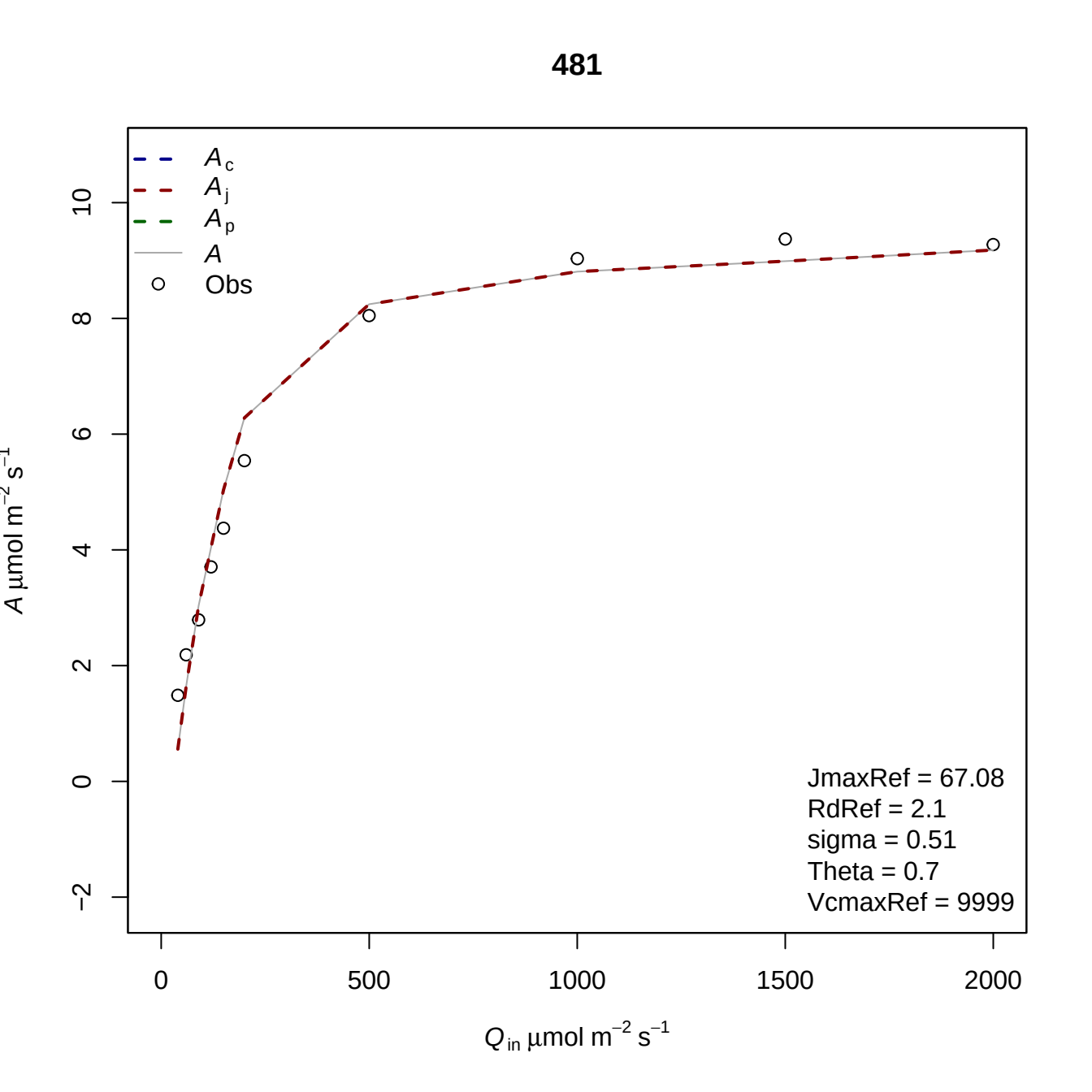

475
A_c
A_j
A_p
A
Obs
JmaxRef = 49.35
RdRef = 1.72
sigma = 0.31
Theta = 0.7
VcmaxRef = 9999
8
6
4
2
0
−2
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

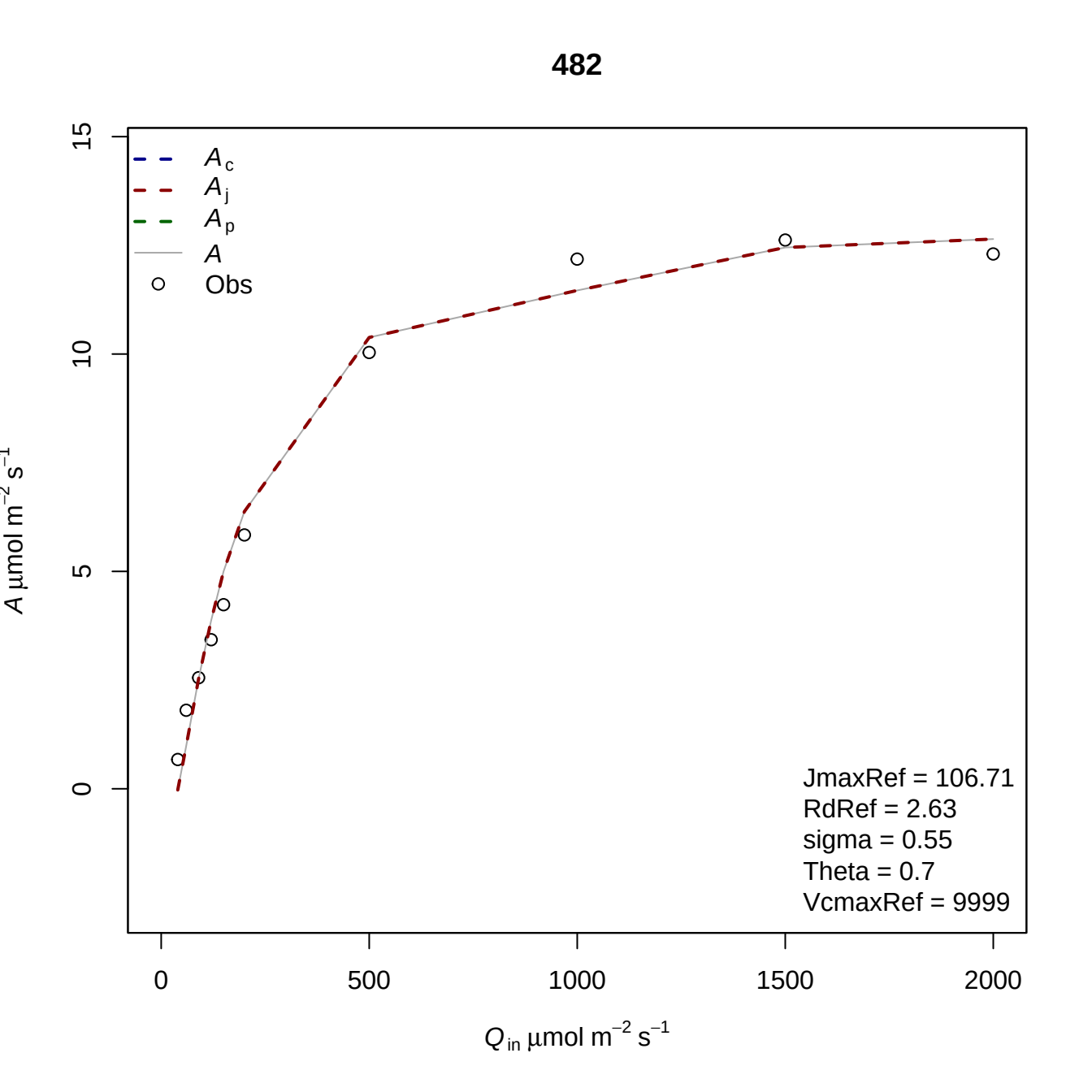

476
A_c
A_j
A_p
A
Obs
JmaxRef = 76.01
RdRef = 2.35
sigma = 0.6
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

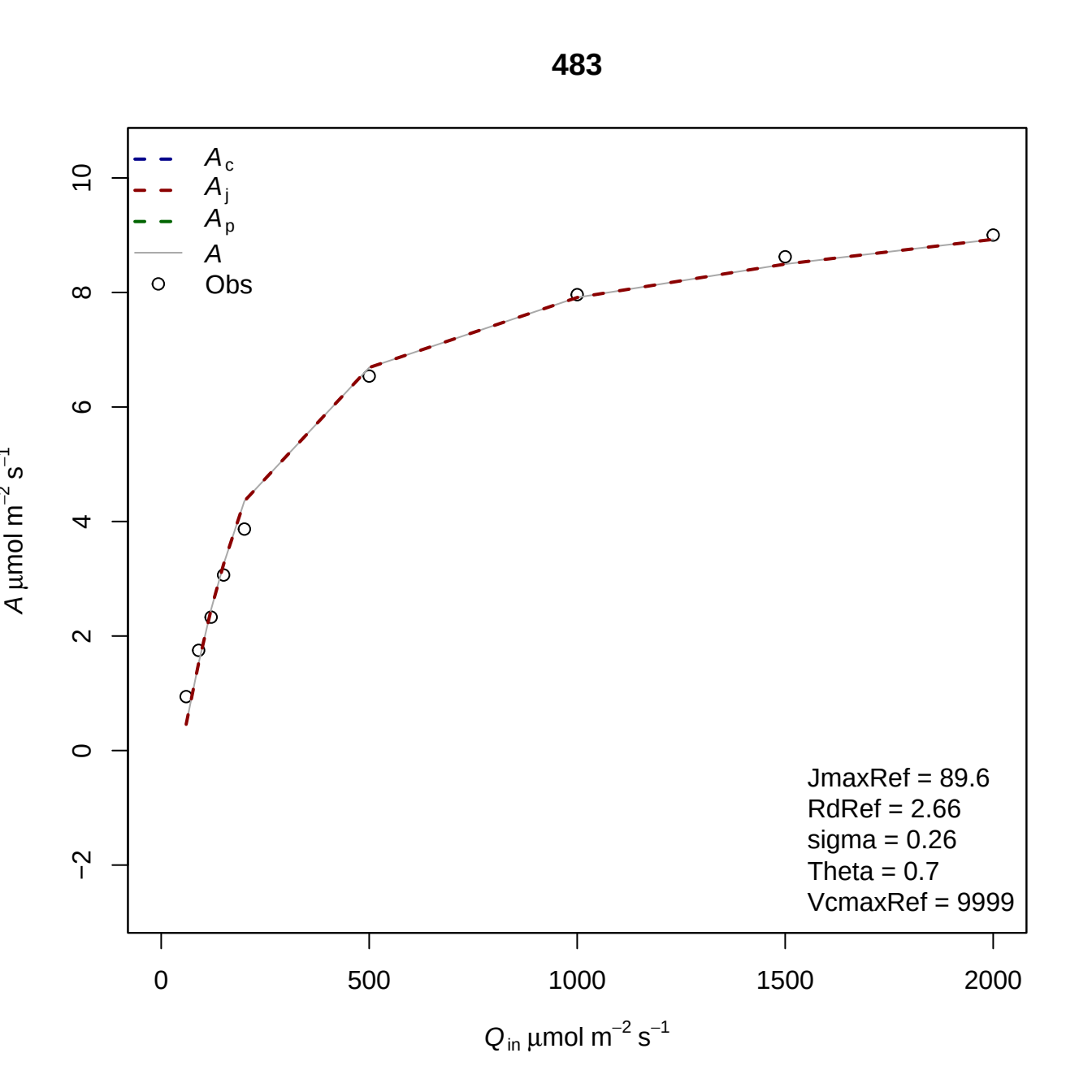

A_c
A_j
A_p
A
Obs
JmaxRef = 115.38
RdRef = 1.3
sigma = 0.4
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

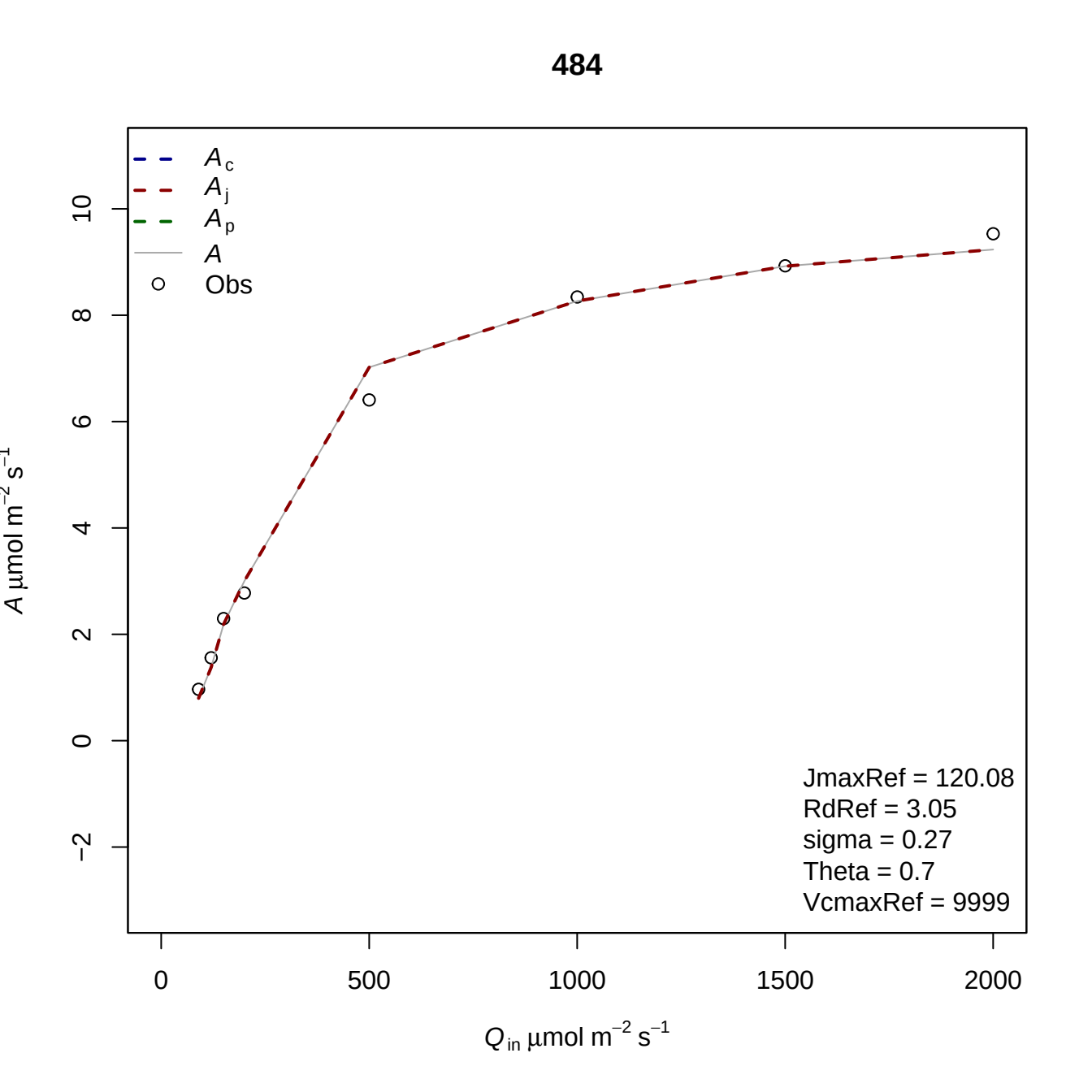

478
A_c
A_j
A_p
A
Obs
JmaxRef = 123.85
RdRef = 1.51
sigma = 0.24
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

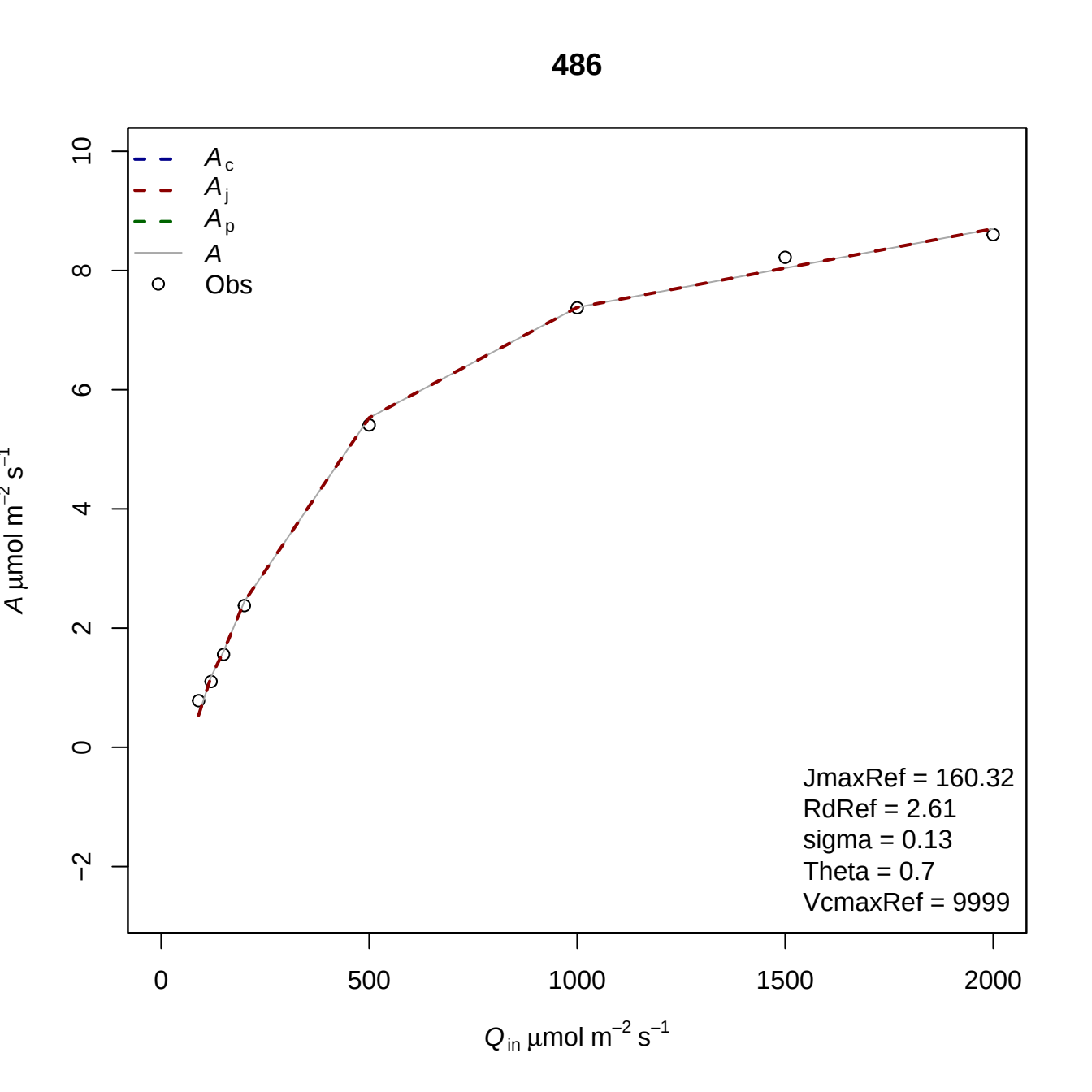

480
A_c
A_j
A_p
A
Obs
10
5
0
−5
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}
JmaxRef = 180.21
RdRef = 5.08
sigma = 0.5
Theta = 0.7
VcmaxRef = 9999

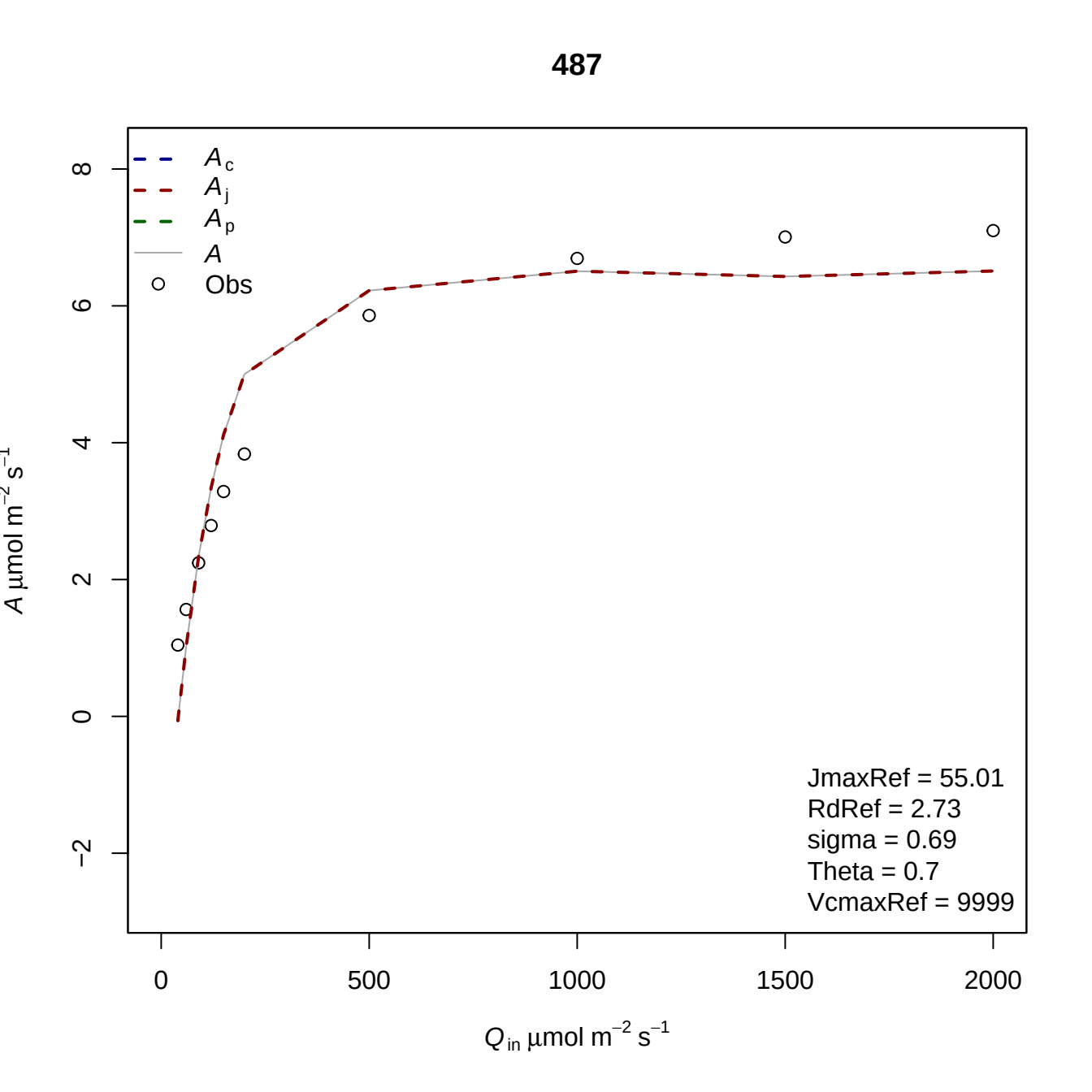

481
A_c
A_j
A_p
A
Obs
JmaxRef = 67.08
RdRef = 2.1
sigma = 0.51
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

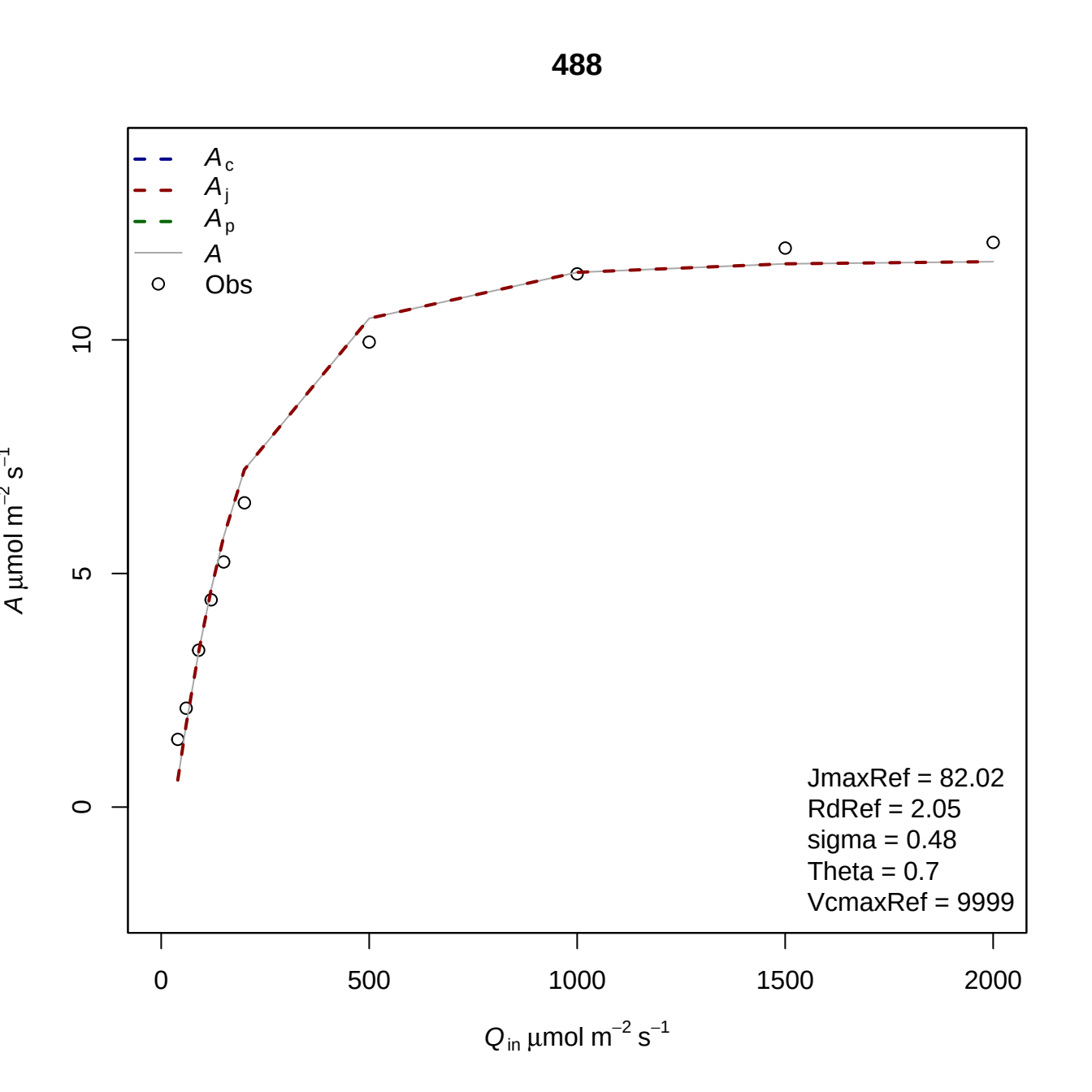

482
A_c
A_j
A_p
A
Obs
JmaxRef = 106.71
RdRef = 2.63
sigma = 0.55
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

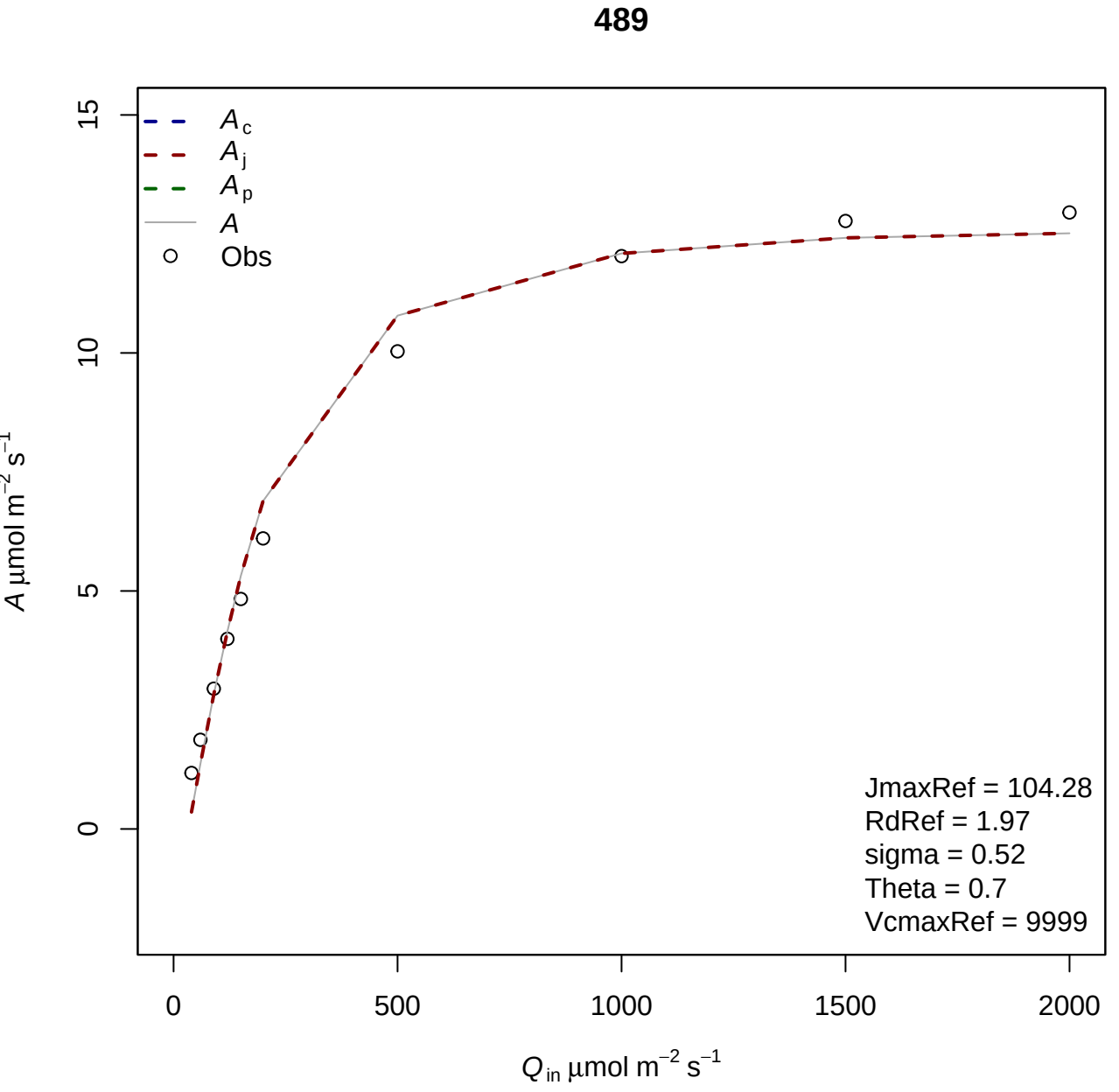

483
A_c
A_j
A_p
A
Obs
JmaxRef = 89.6
RdRef = 2.66
sigma = 0.26
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

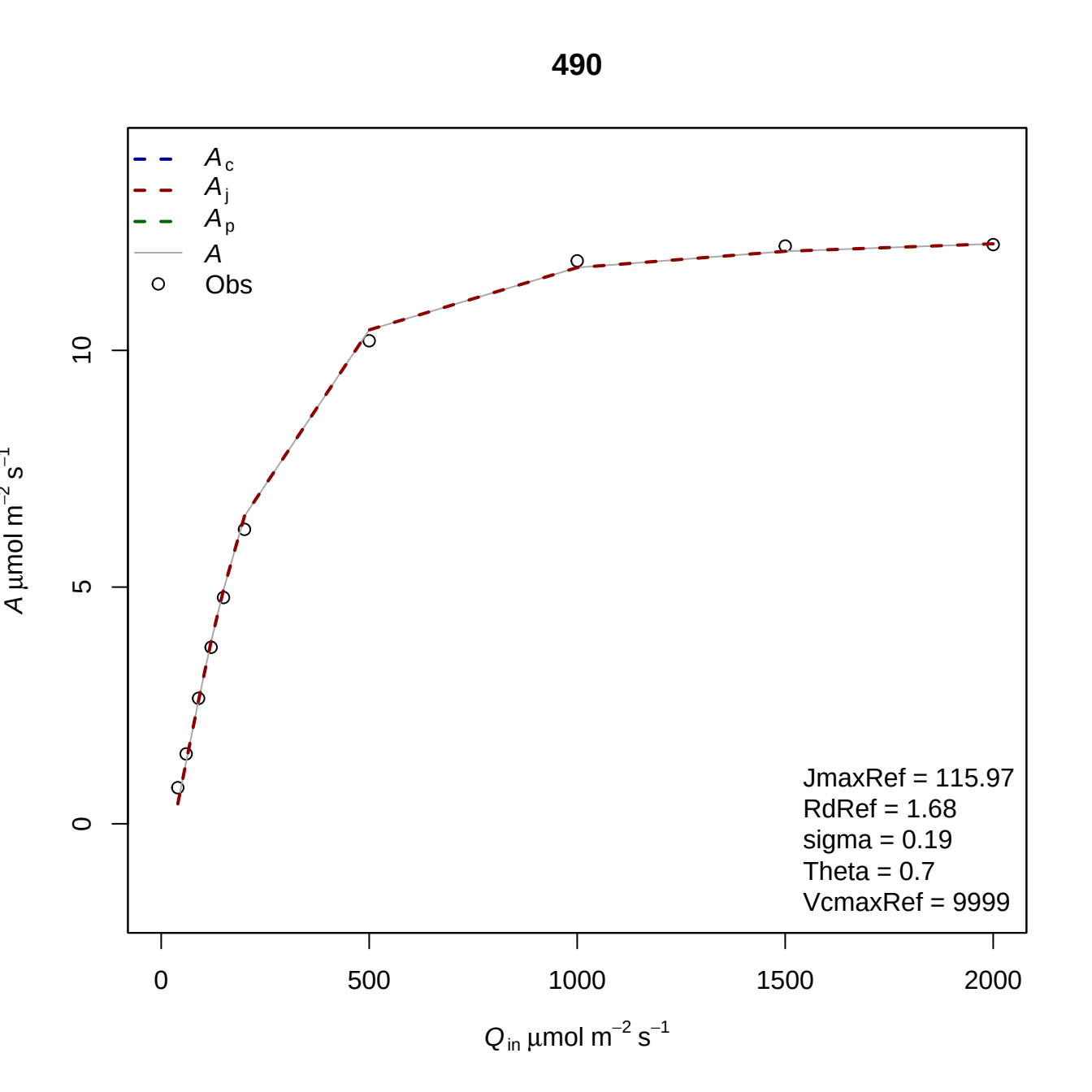

484
A_c
A_j
A_p
A
Obs
JmaxRef = 120.08
RdRef = 3.05
sigma = 0.27
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

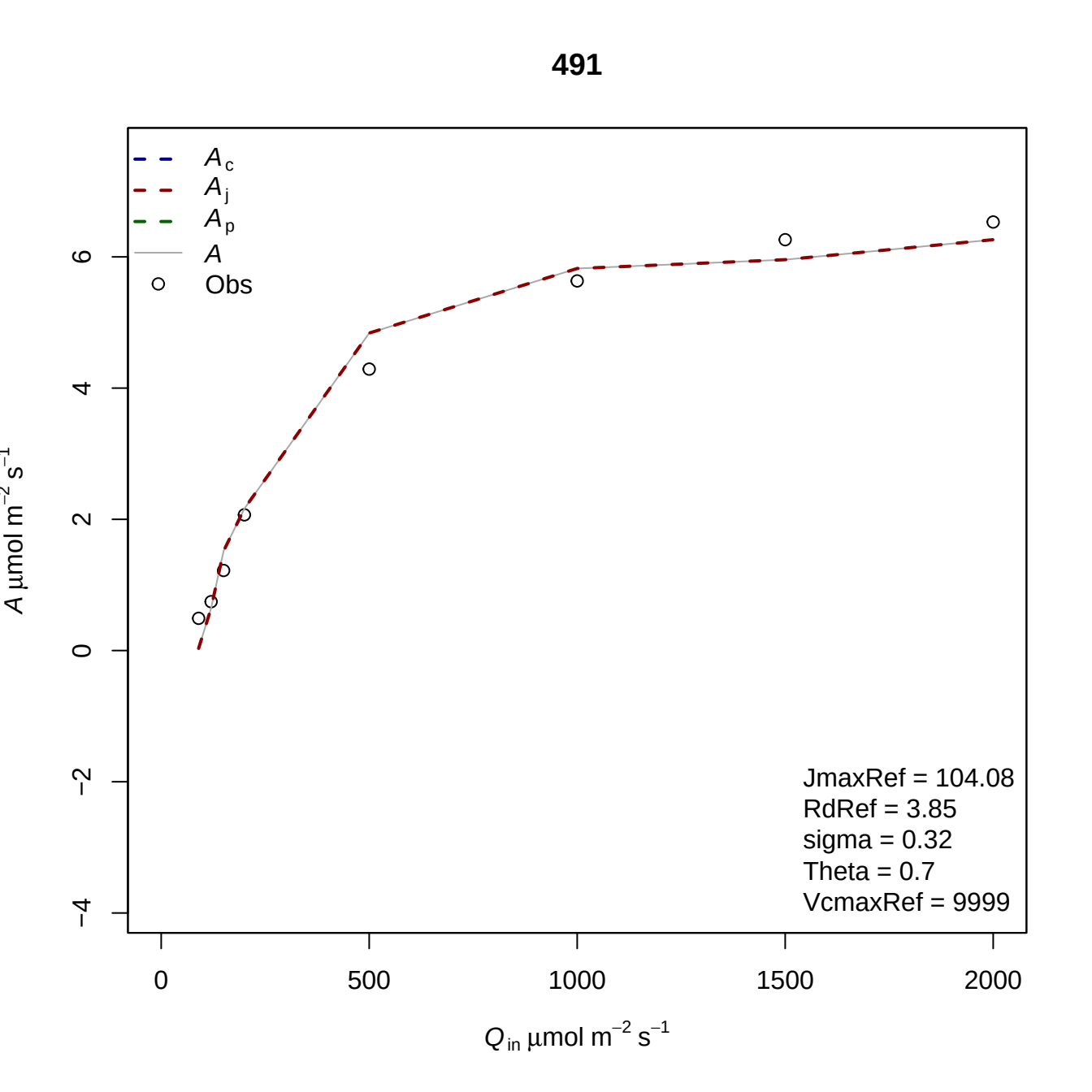

486
A_c
A_j
A_p
A
Obs
JmaxRef = 160.32
RdRef = 2.61
sigma = 0.13
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

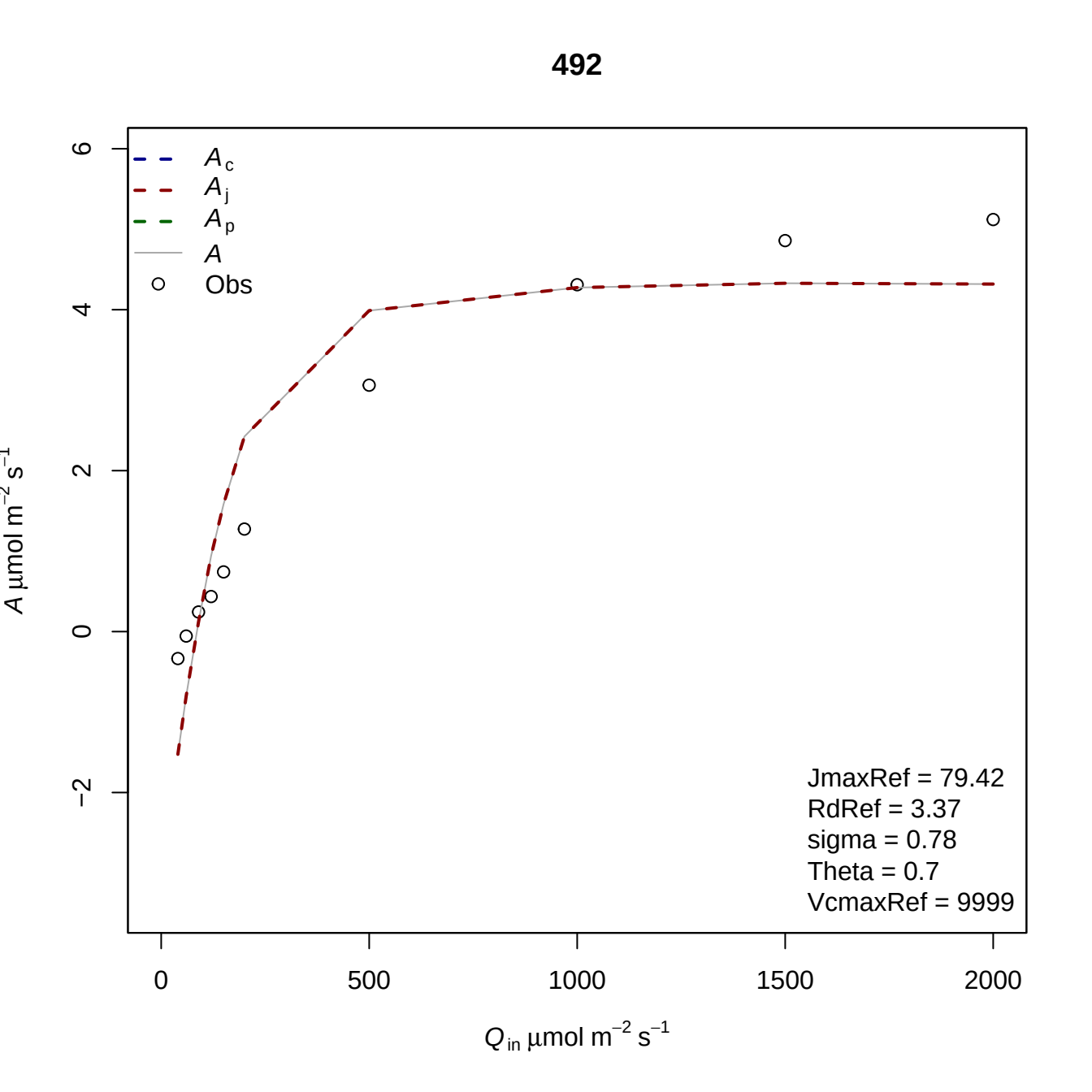

487
A_c
A_j
A_p
A
Obs
JmaxRef = 55.01
RdRef = 2.73
sigma = 0.69
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

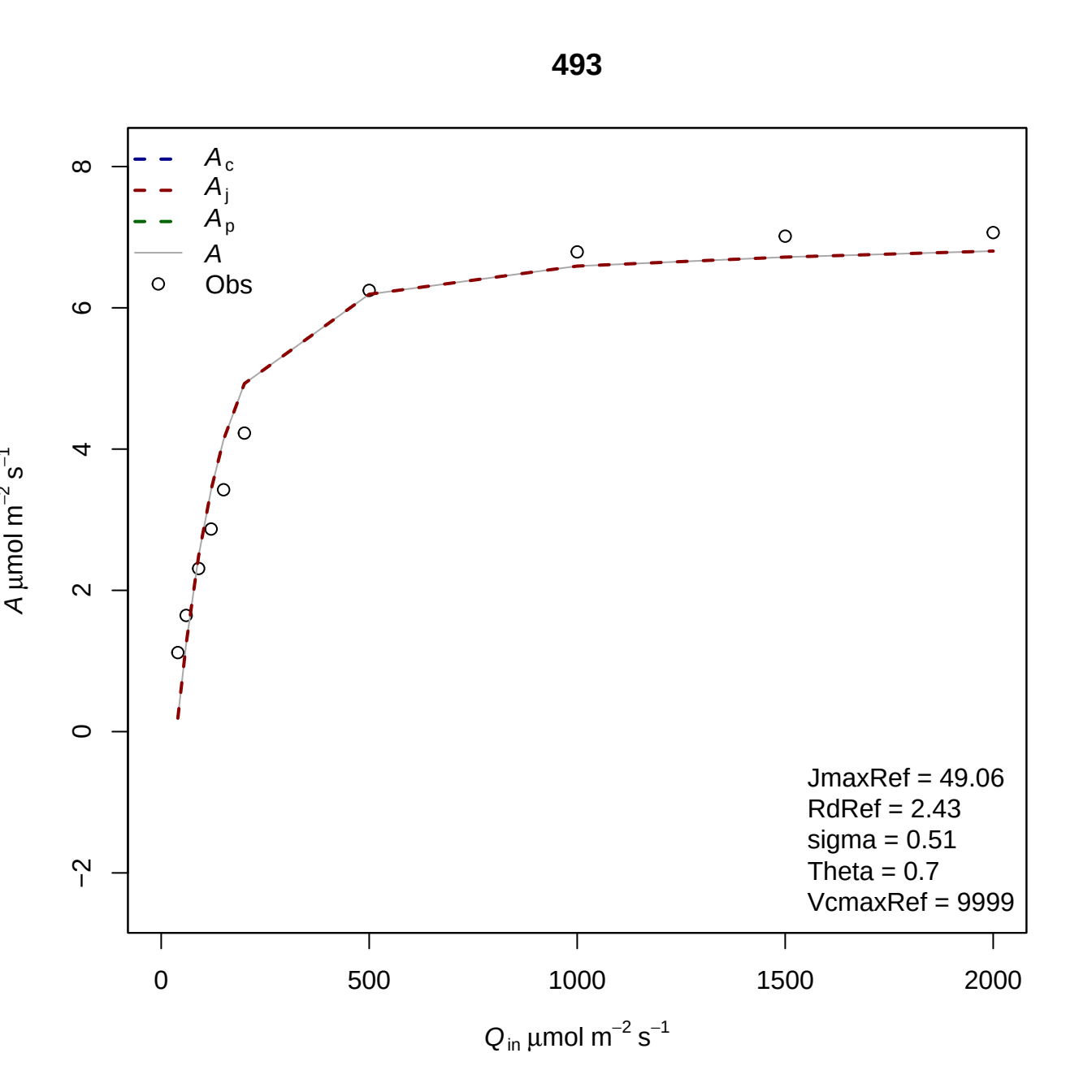

488
A_c
A_j
A_p
A
Obs
JmaxRef = 82.02
RdRef = 2.05
sigma = 0.48
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

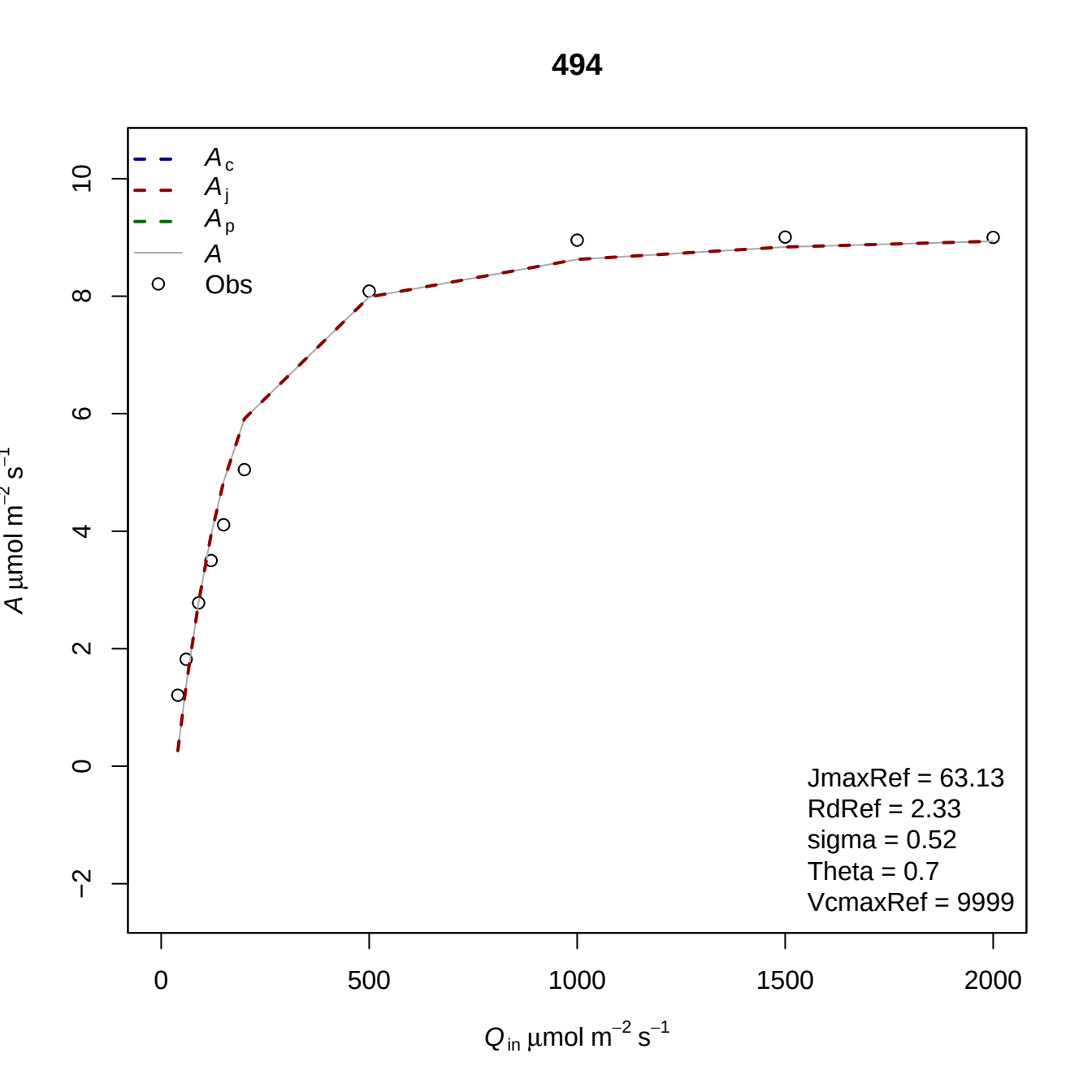

489
A_c
A_j
A_p
A
Obs
JmaxRef = 104.28
RdRef = 1.97
sigma = 0.52
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

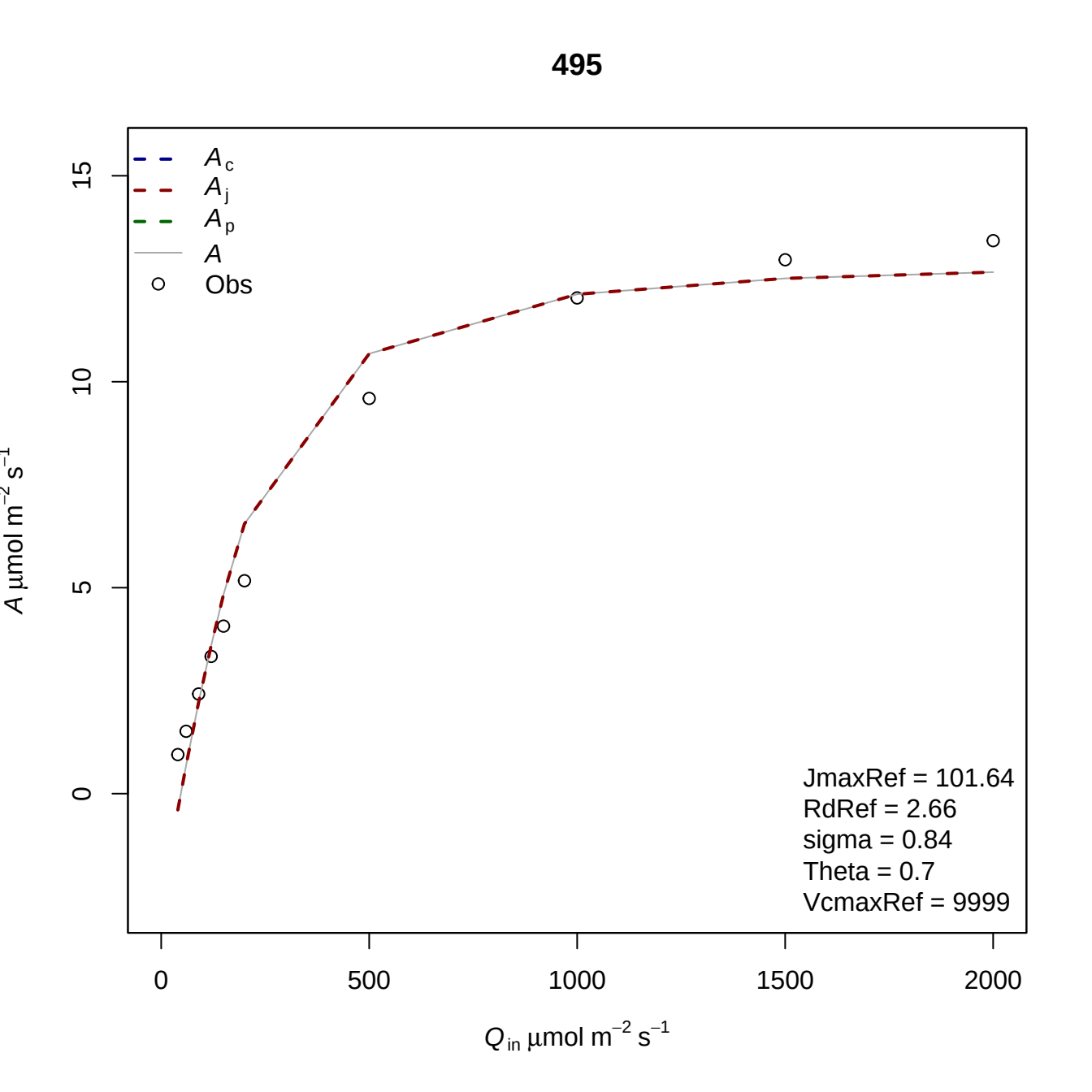

490
A_c
A_j
A_p
A
Obs
JmaxRef = 115.97
RdRef = 1.68
sigma = 0.19
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

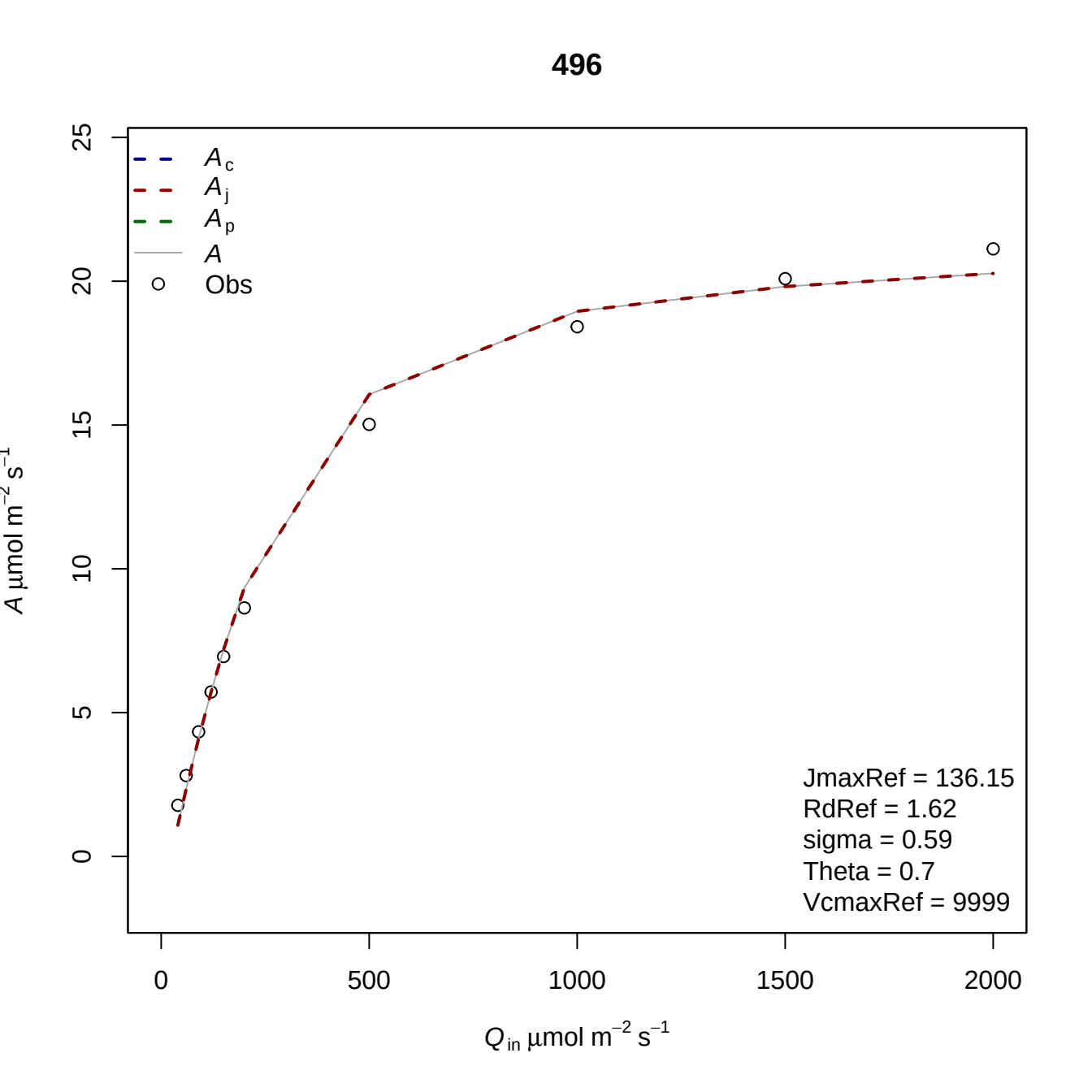

491
A_c
A_j
A_p
A
Obs
JmaxRef = 104.08
RdRef = 3.85
sigma = 0.32
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500
2000

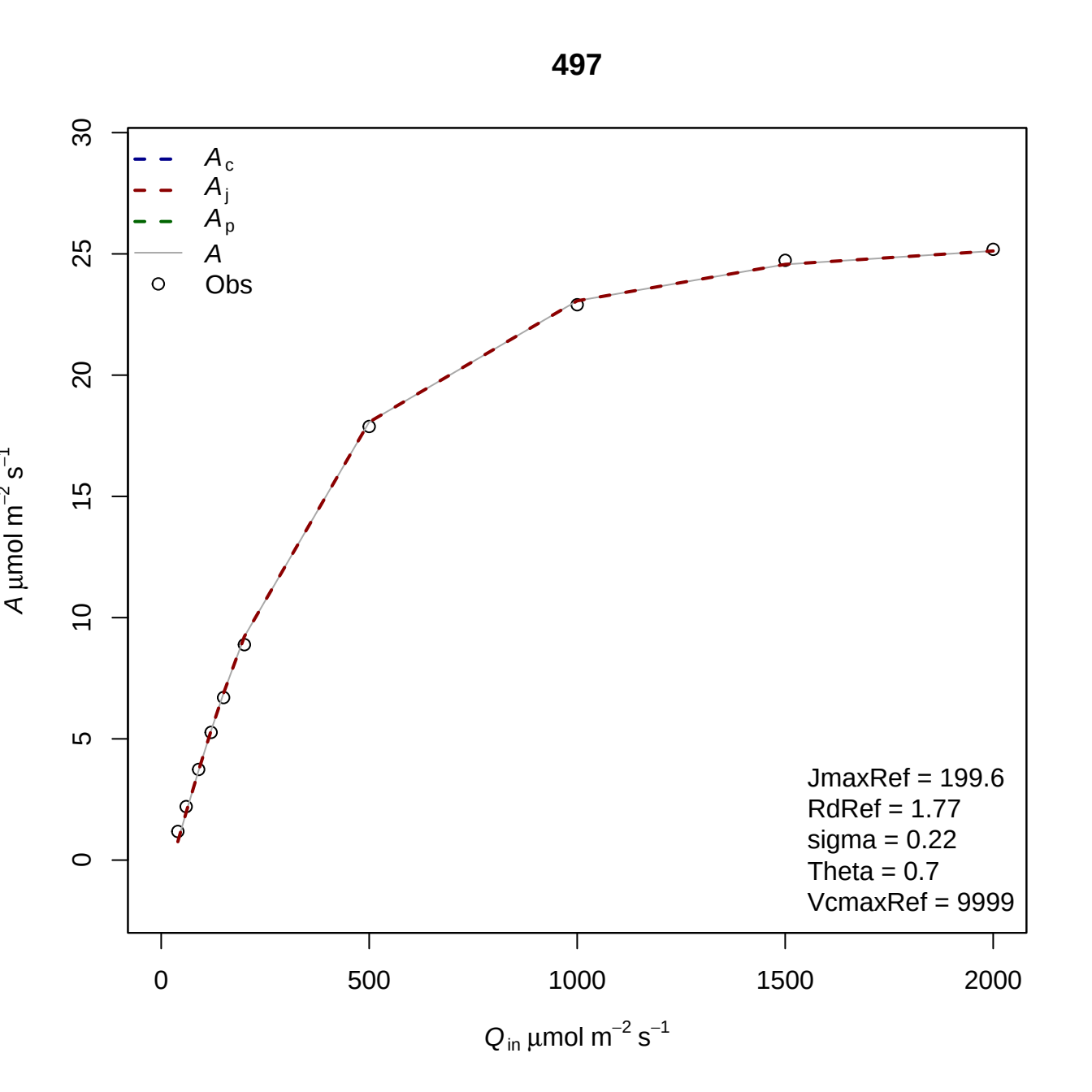

492
A_c
A_j
A_p
A
Obs
JmaxRef = 79.42
RdRef = 3.37
sigma = 0.78
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

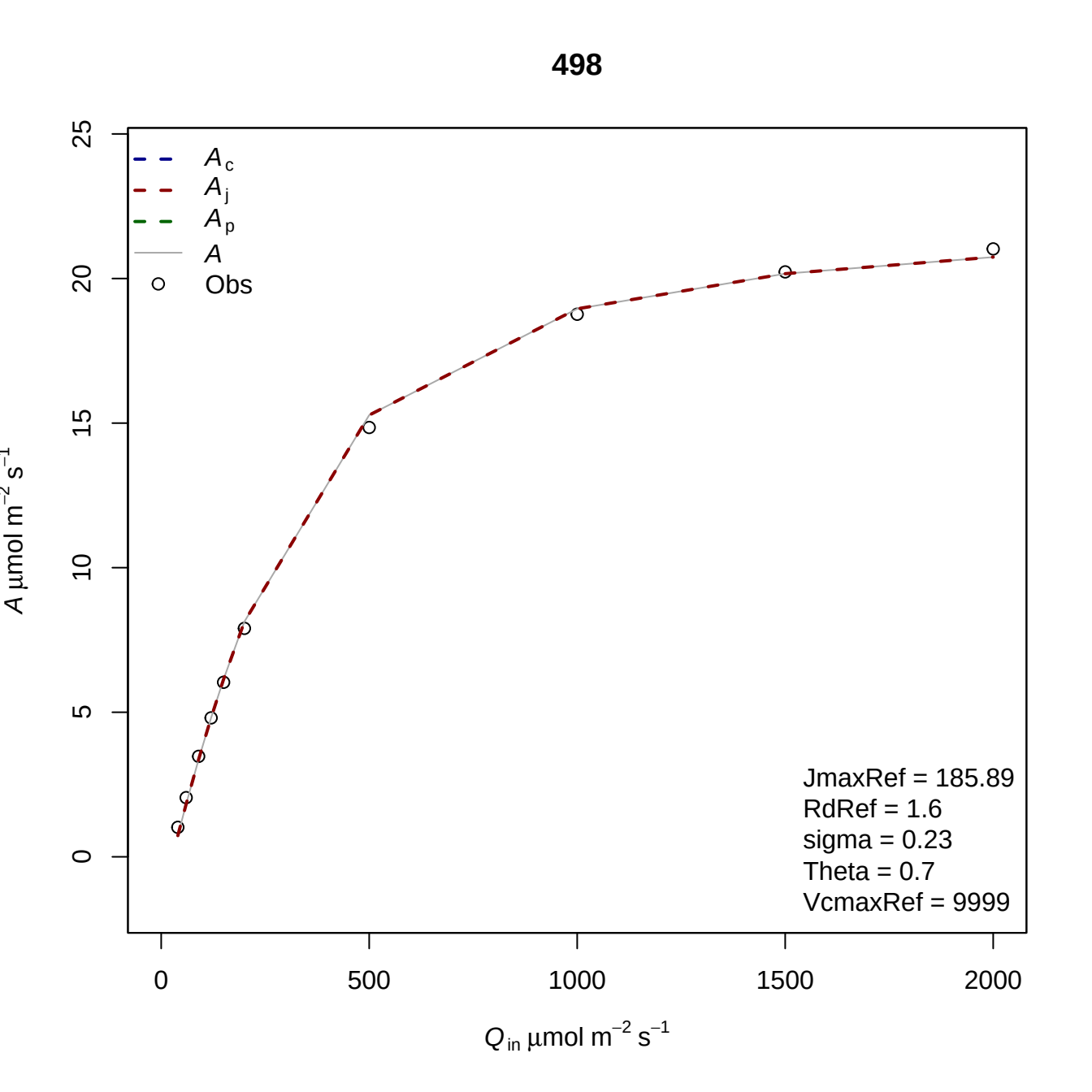

493
A_c
A_j
A_p
A
Obs
JmaxRef = 49.06
RdRef = 2.43
sigma = 0.51
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

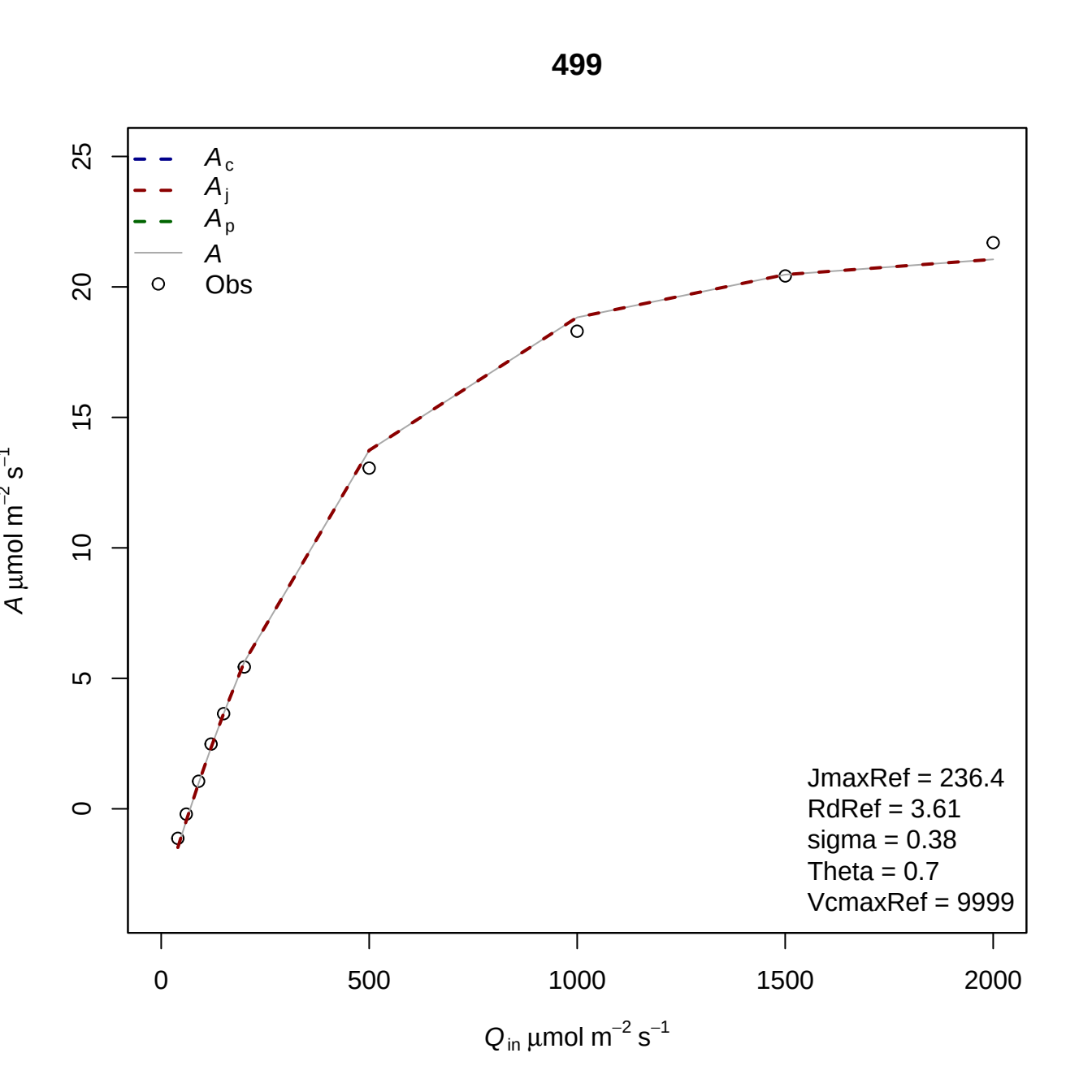

494
A_c
A_j
A_p
A
Obs
JmaxRef = 63.13
RdRef = 2.33
sigma = 0.52
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

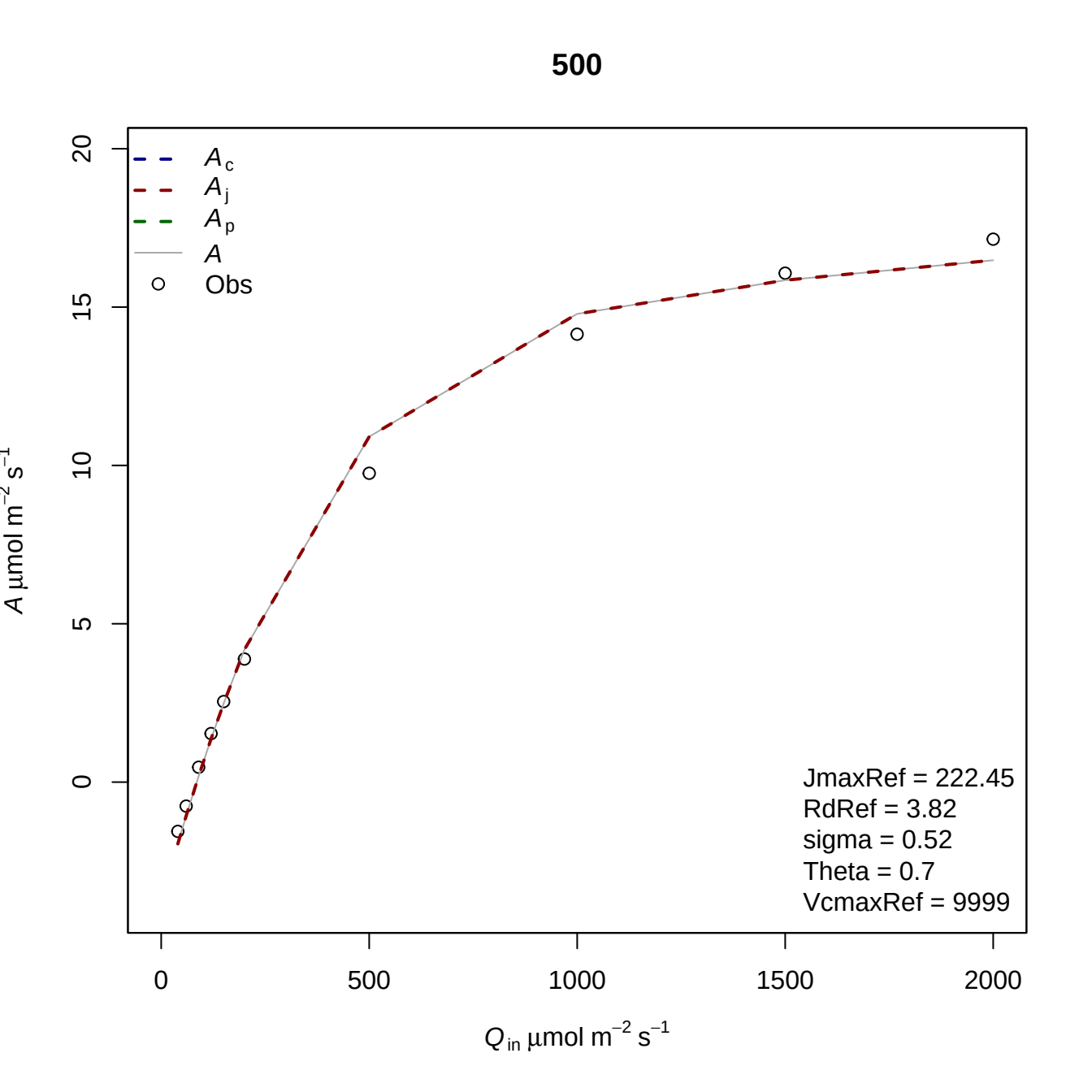

495
A_c
A_j
A_p
A
Obs
JmaxRef = 101.64
RdRef = 2.66
sigma = 0.84
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

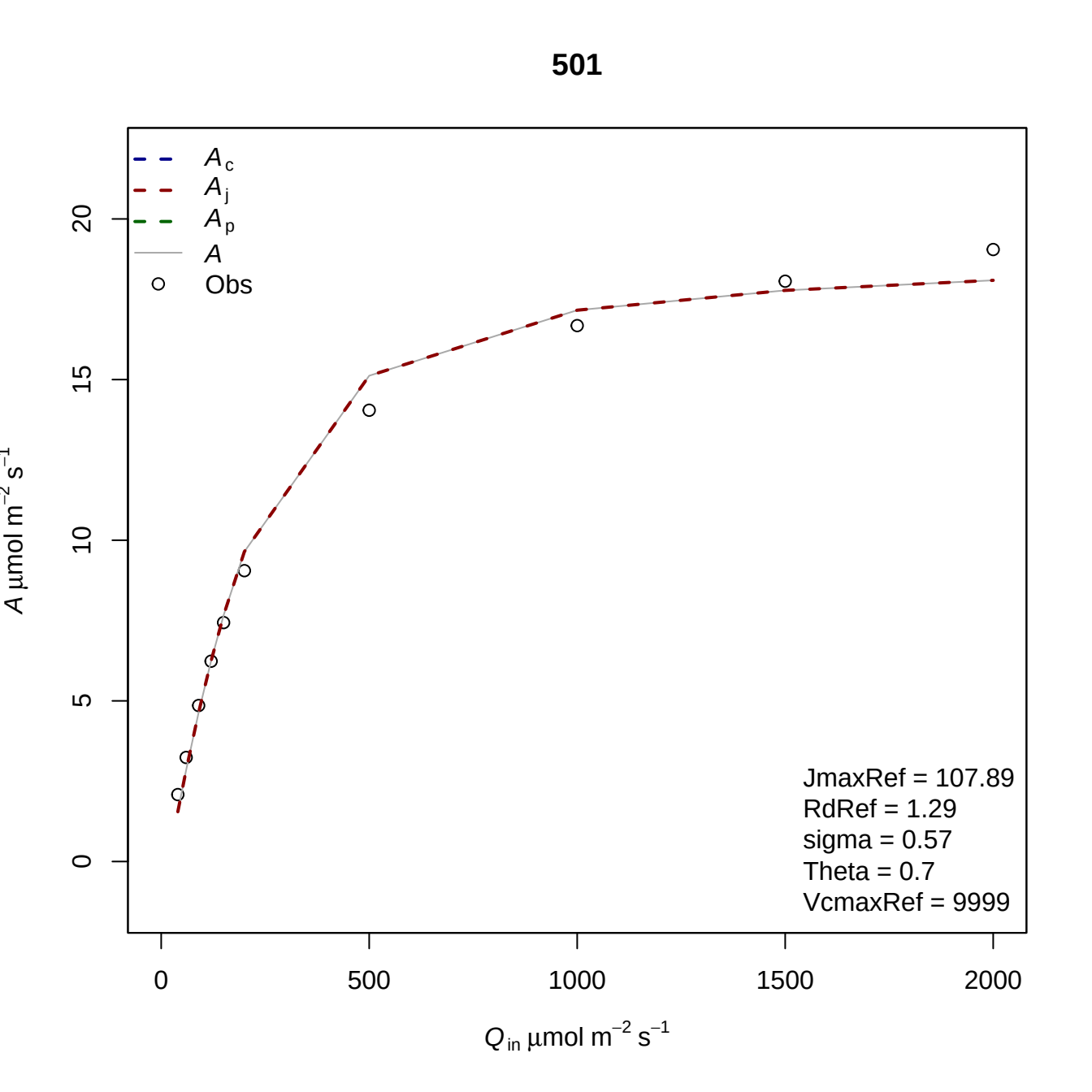

496
A_c
A_j
A_p
A
Obs
JmaxRef = 136.15
RdRef = 1.62
sigma = 0.59
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

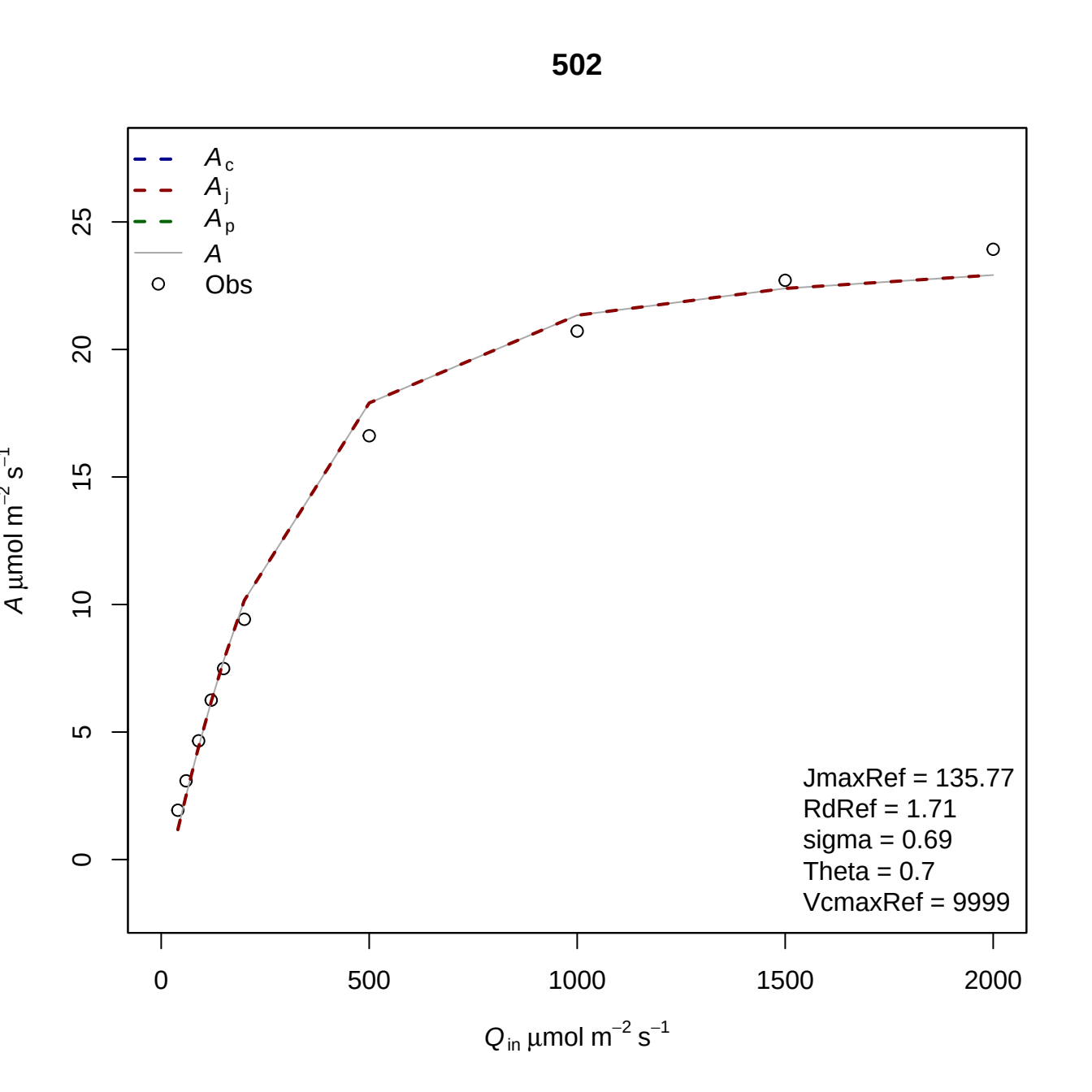

497
A_c
A_j
A_p
A
Obs
JmaxRef = 199.6
RdRef = 1.77
sigma = 0.22
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

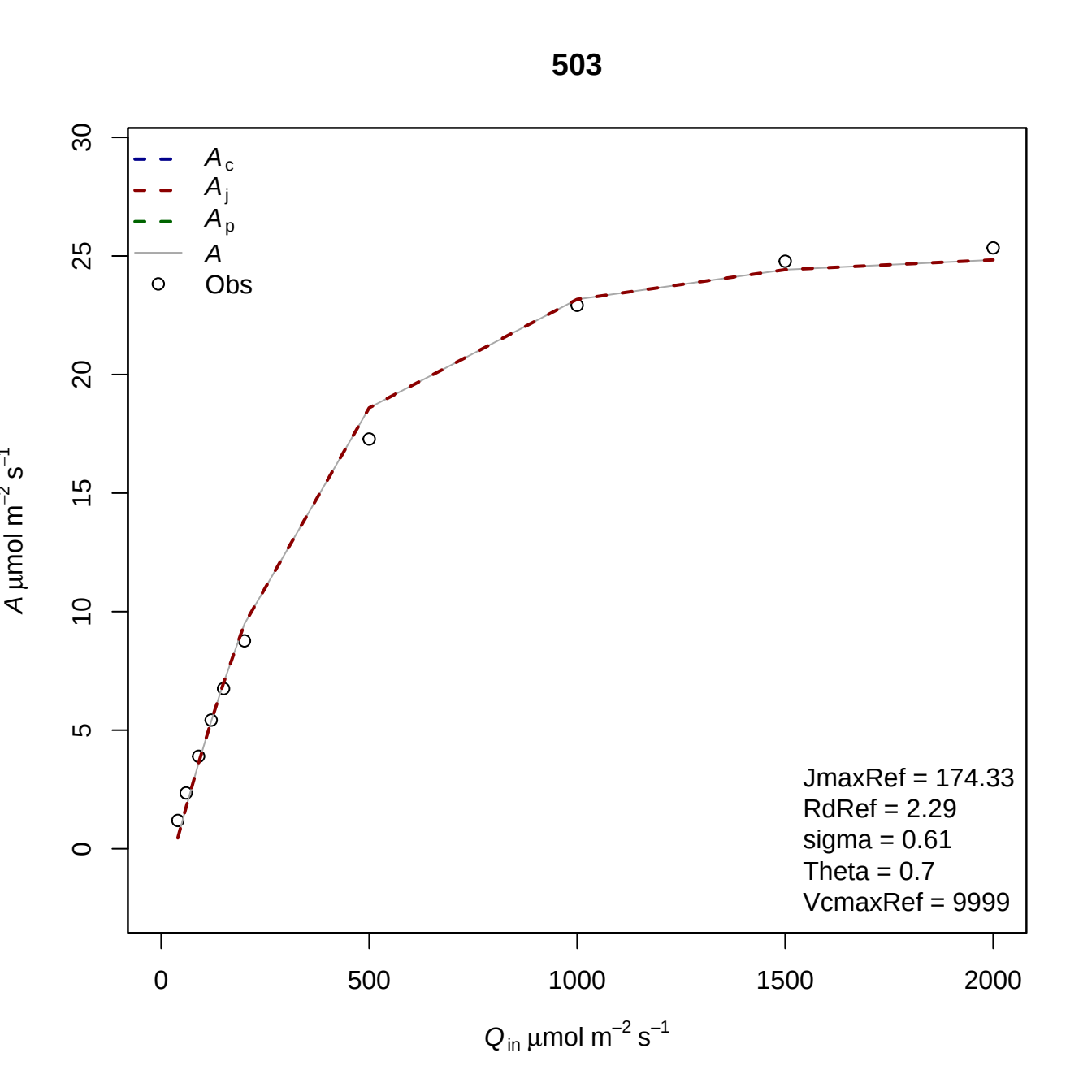

498
A_c
A_j
A_p
A
Obs
JmaxRef = 185.89
RdRef = 1.6
sigma = 0.23
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

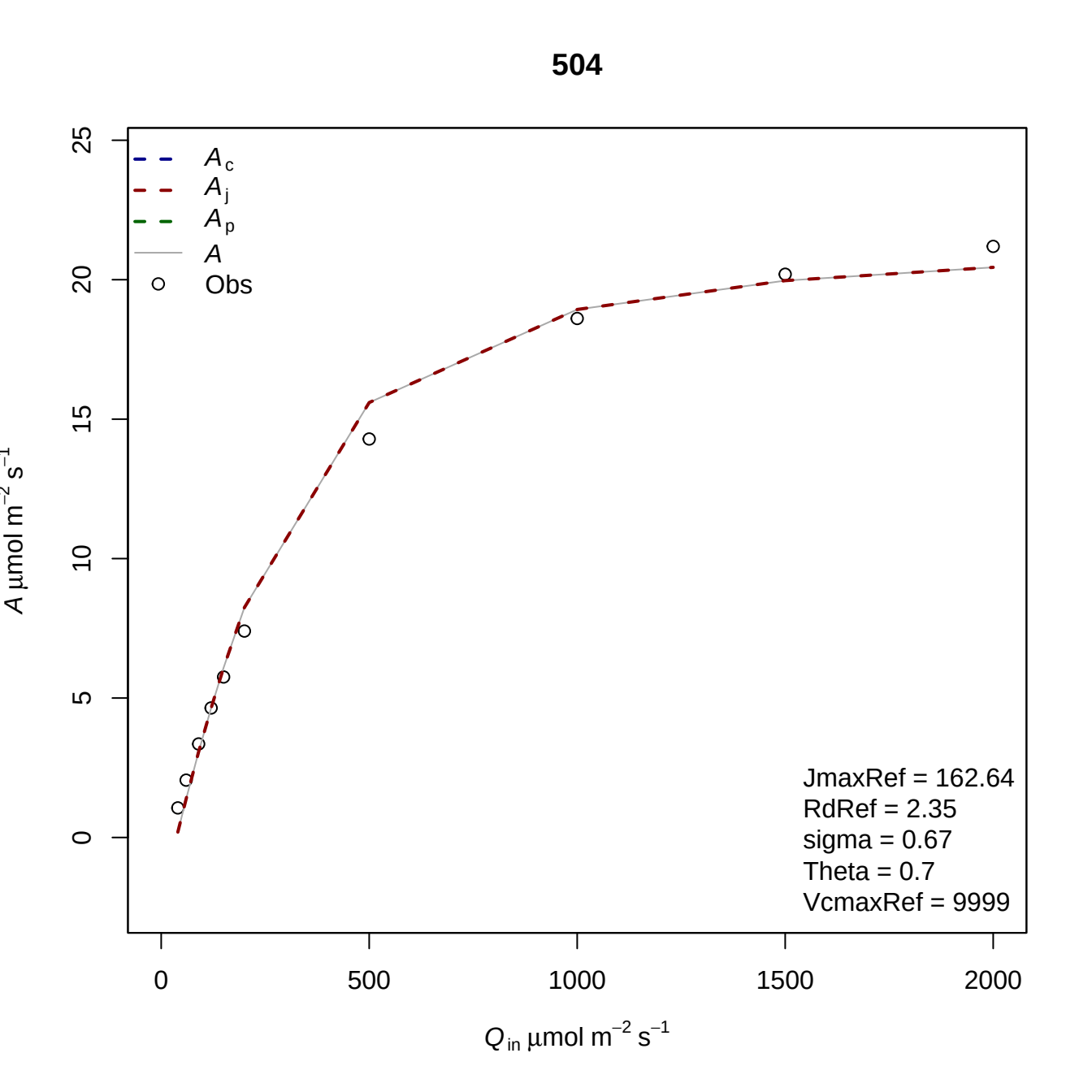

499
A_c
A_j
A_p
A
Obs
JmaxRef = 236.4
RdRef = 3.61
sigma = 0.38
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

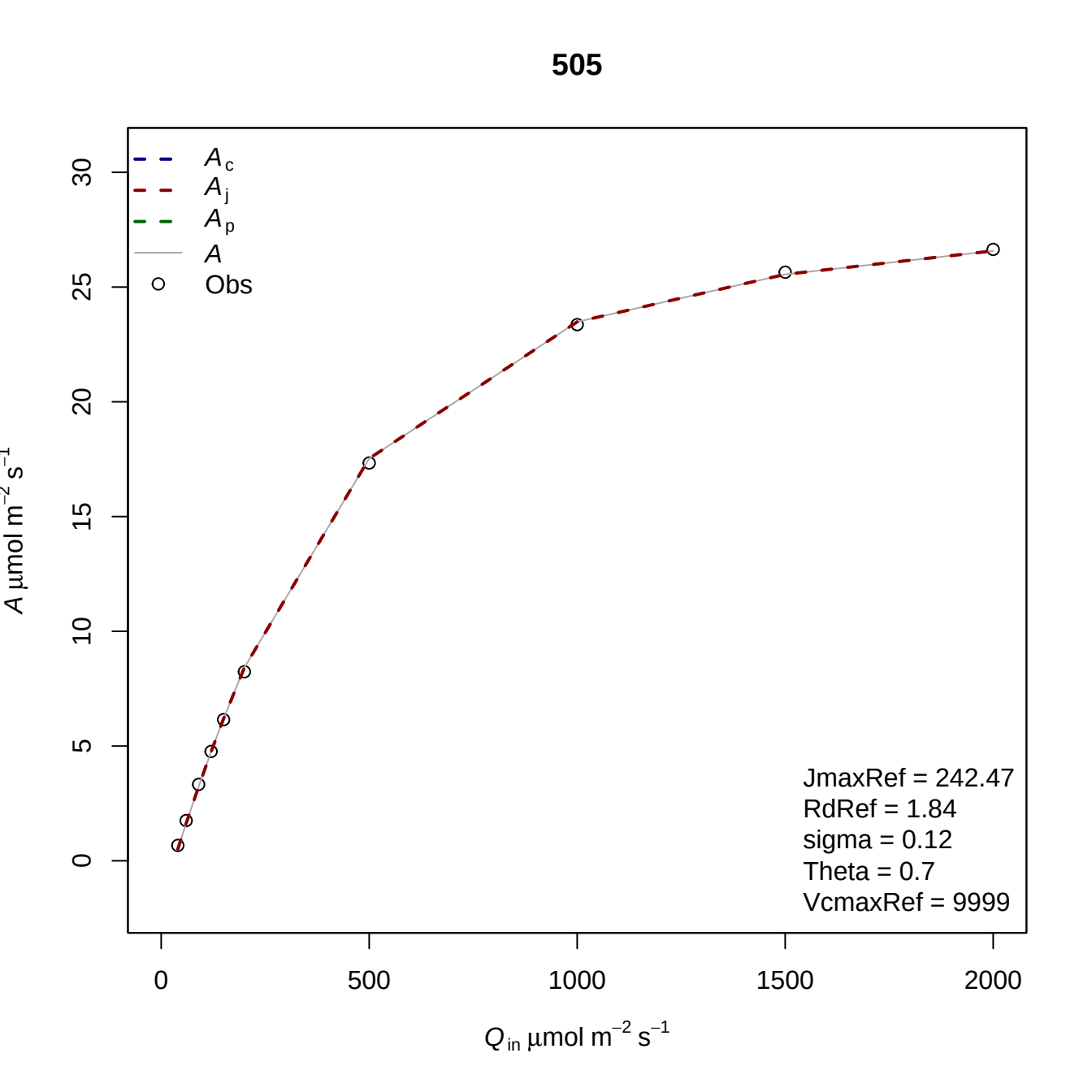

500
A_c
A_j
A_p
A
Obs
JmaxRef = 222.45
RdRef = 3.82
sigma = 0.52
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

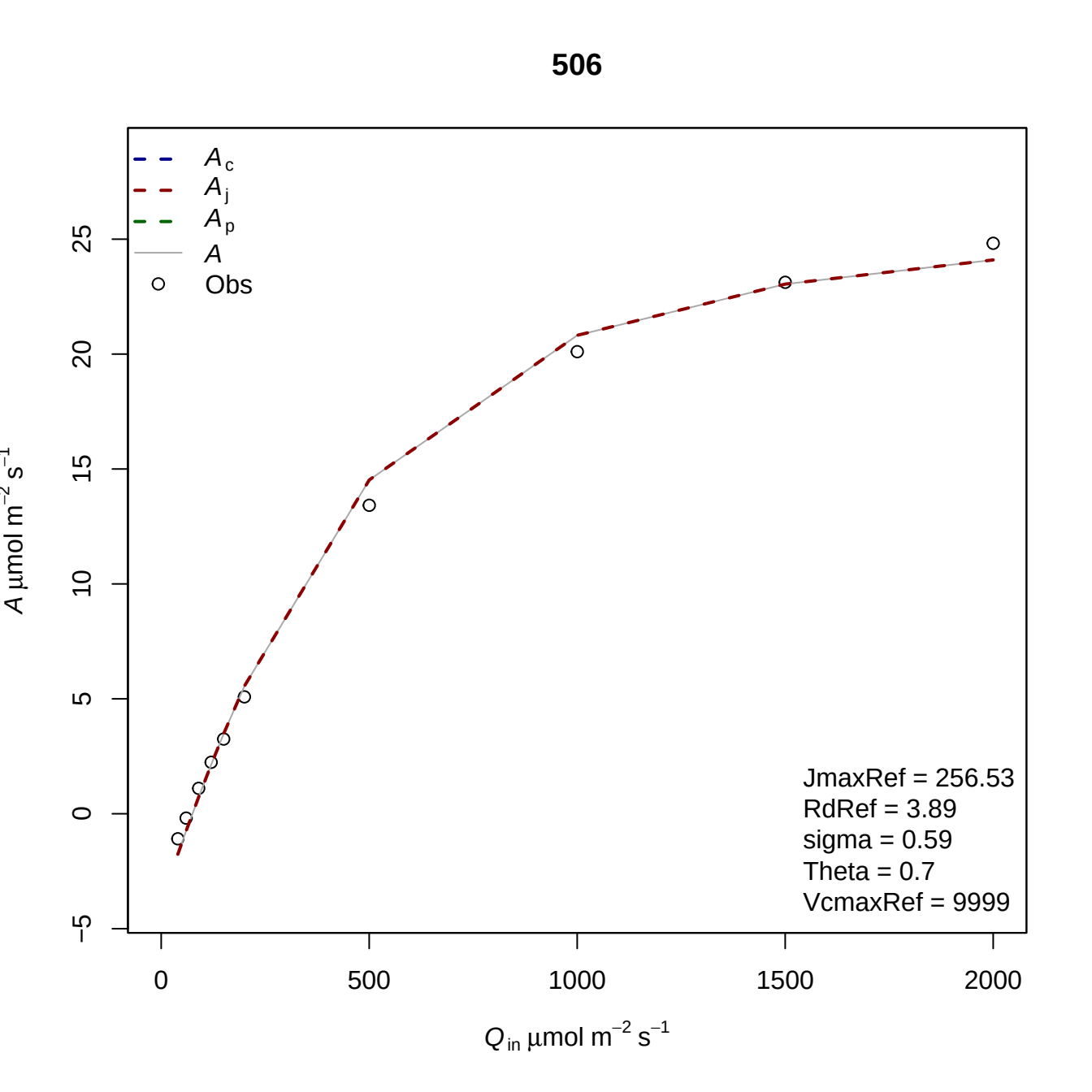

501
A_c
A_j
A_p
A
Obs
JmaxRef = 107.89
RdRef = 1.29
sigma = 0.57
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

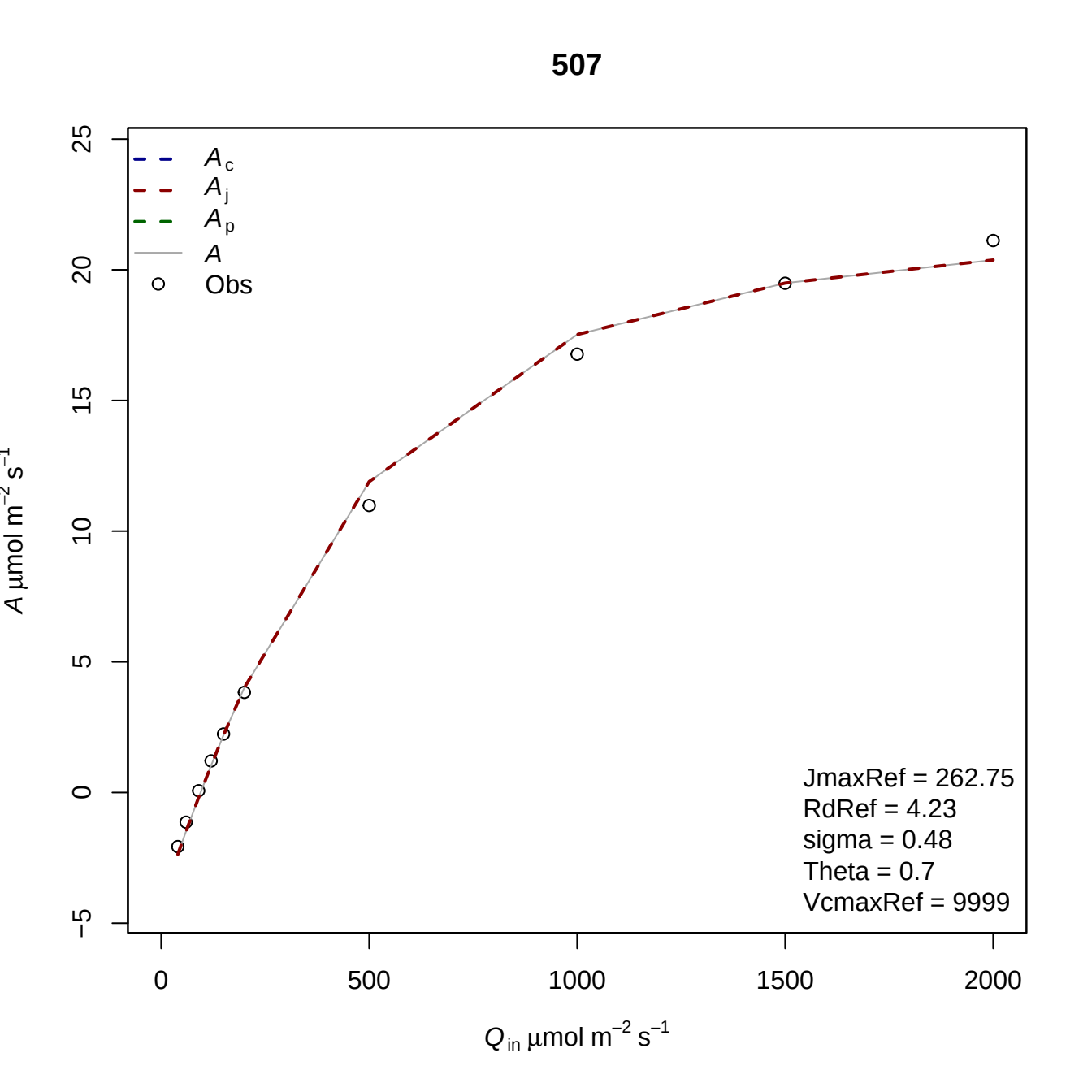

502
A_c
A_j
A_p
A
Obs
JmaxRef = 135.77
RdRef = 1.71
sigma = 0.69
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
25
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

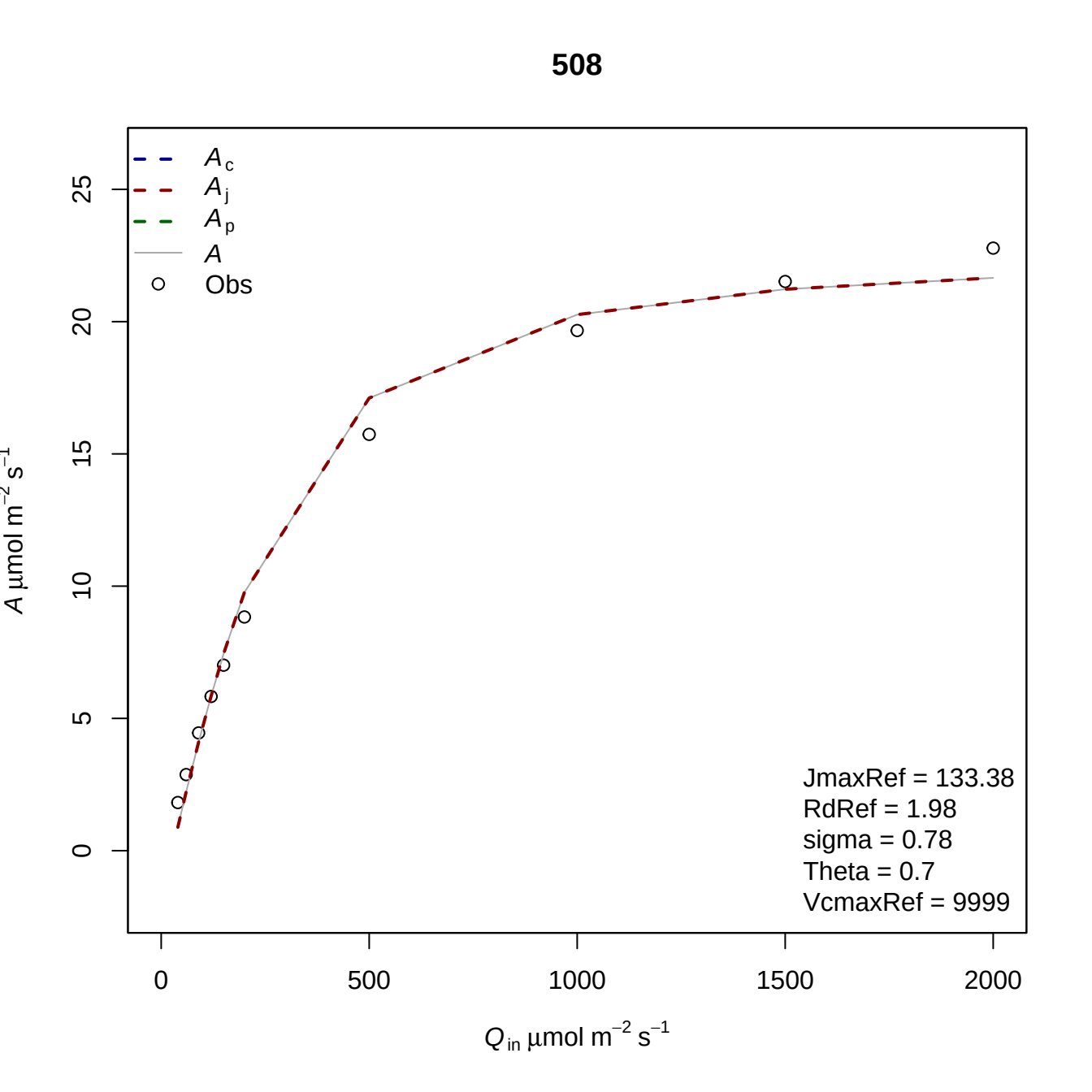

503
A_c
A_j
A_p
A
Obs
JmaxRef = 174.33
RdRef = 2.29
sigma = 0.61
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

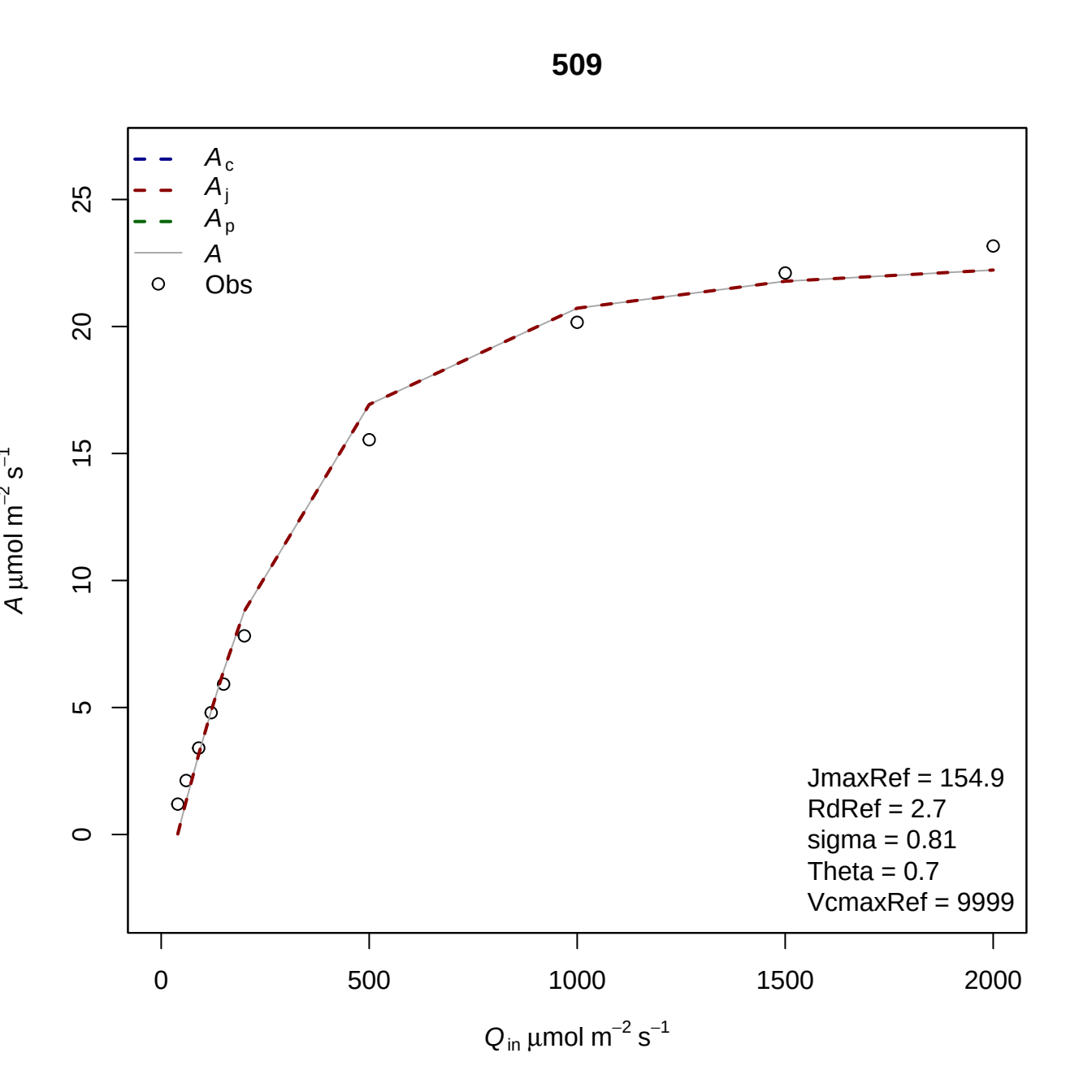

504
A_c
A_j
A_p
A
Obs
JmaxRef = 162.64
RdRef = 2.35
sigma = 0.67
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

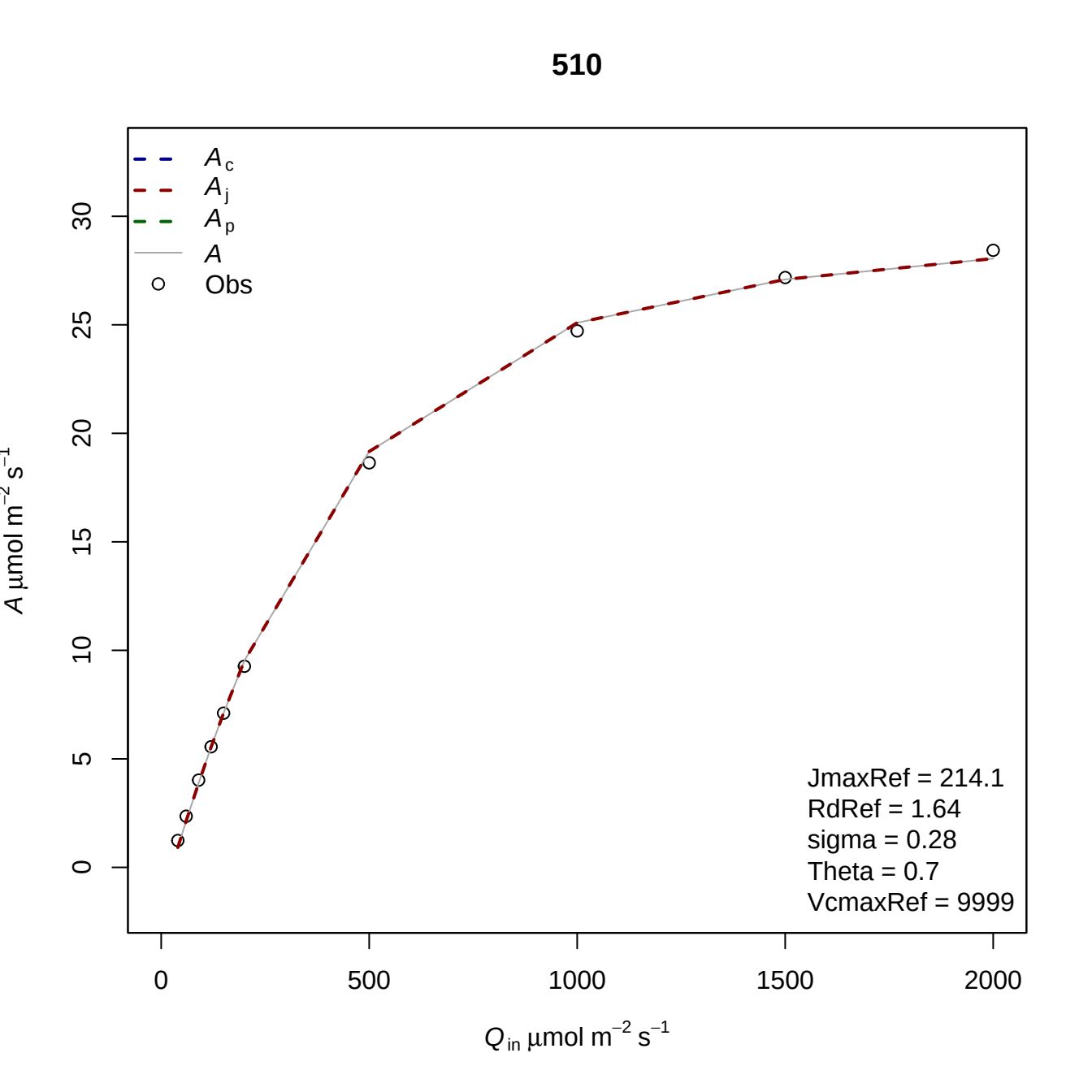

505
A_c
A_j
A_p
A
Obs
JmaxRef = 242.47
RdRef = 1.84
sigma = 0.12
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

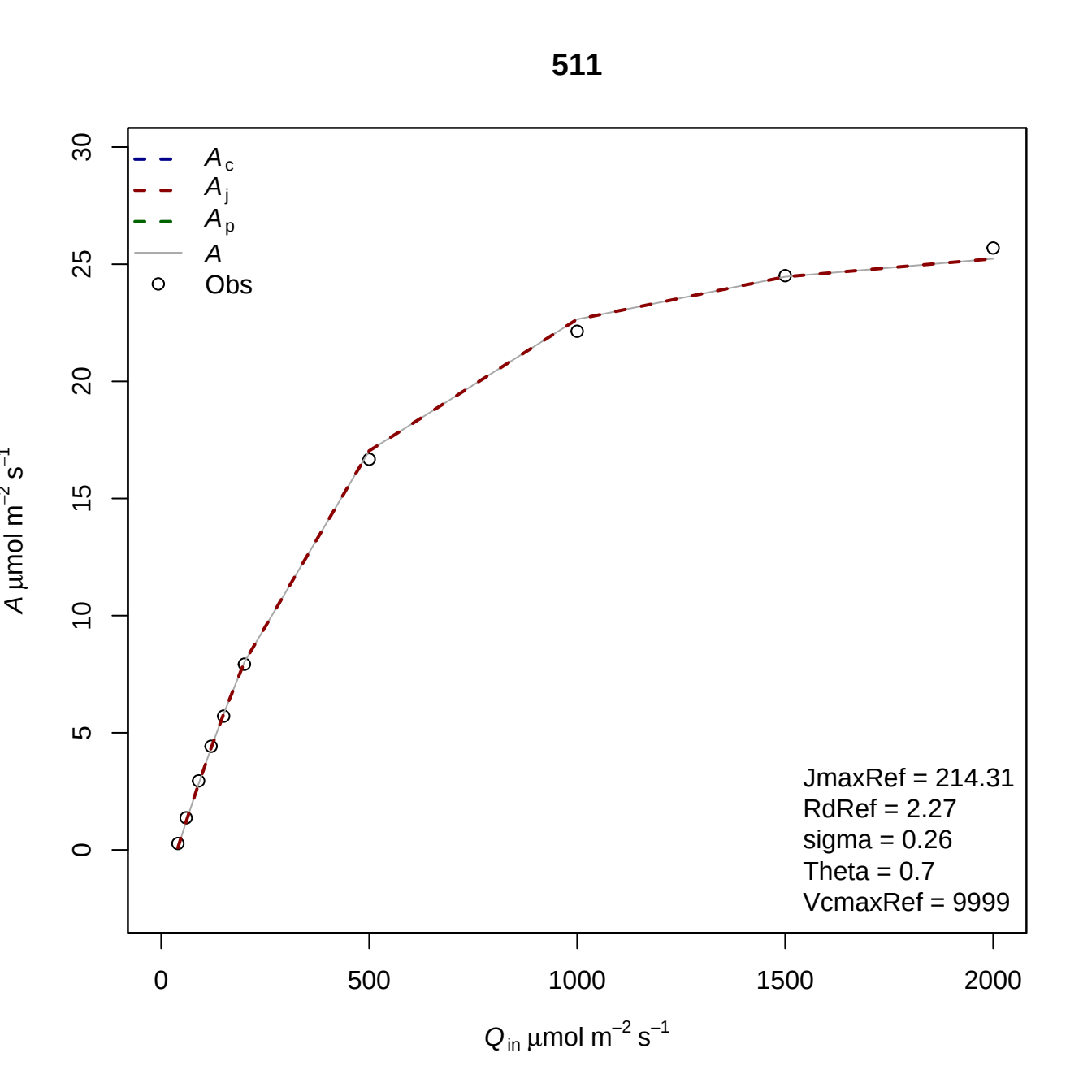

506
A_c
A_j
A_p
A
Obs
JmaxRef = 256.53
RdRef = 3.89
sigma = 0.59
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
25
0
500
1000
1500
2000

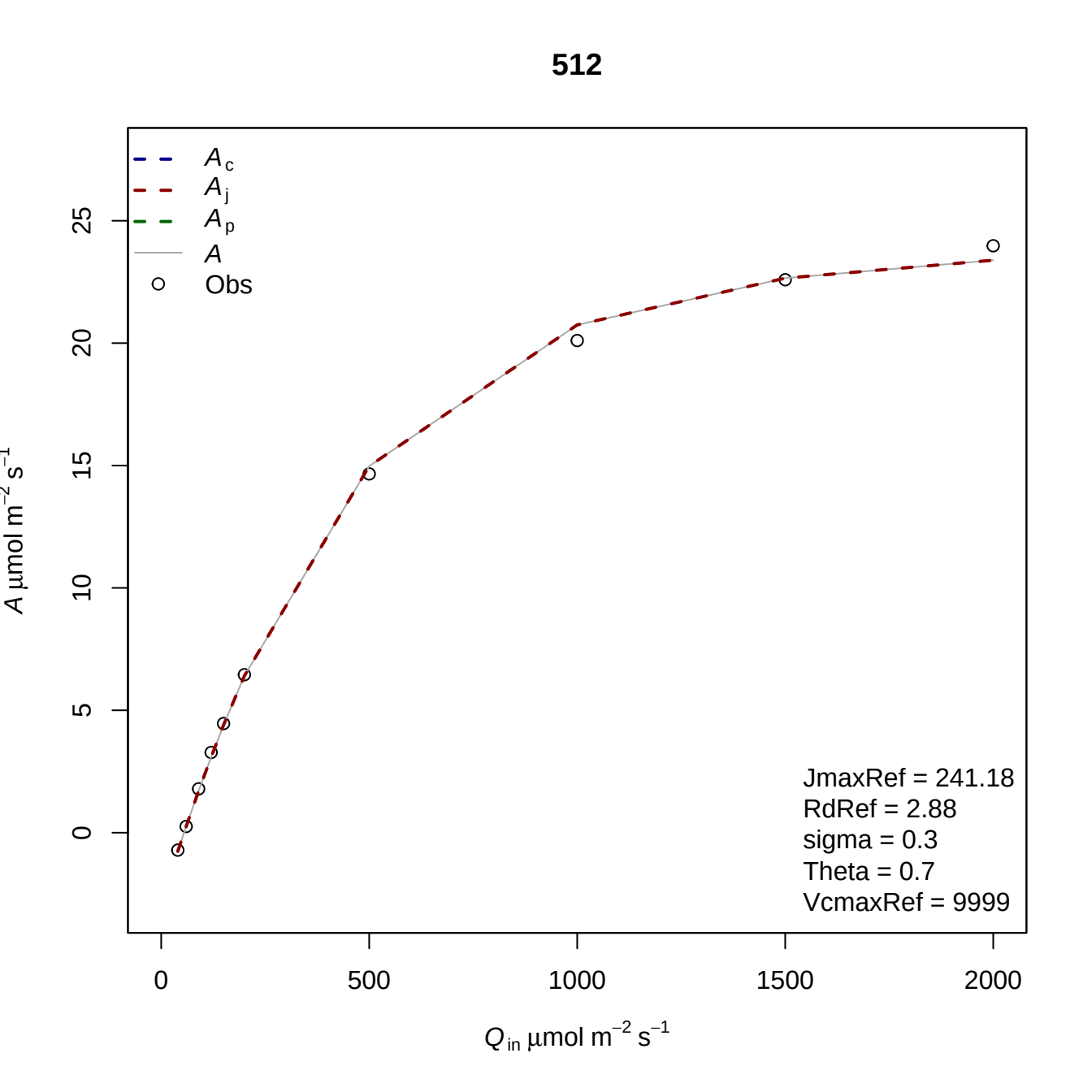

507
A_c
A_j
A_p
A
Obs
JmaxRef = 262.75
RdRef = 4.23
sigma = 0.48
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
25
20
15
10
5
0
−5
0
500
1000
1500
2000

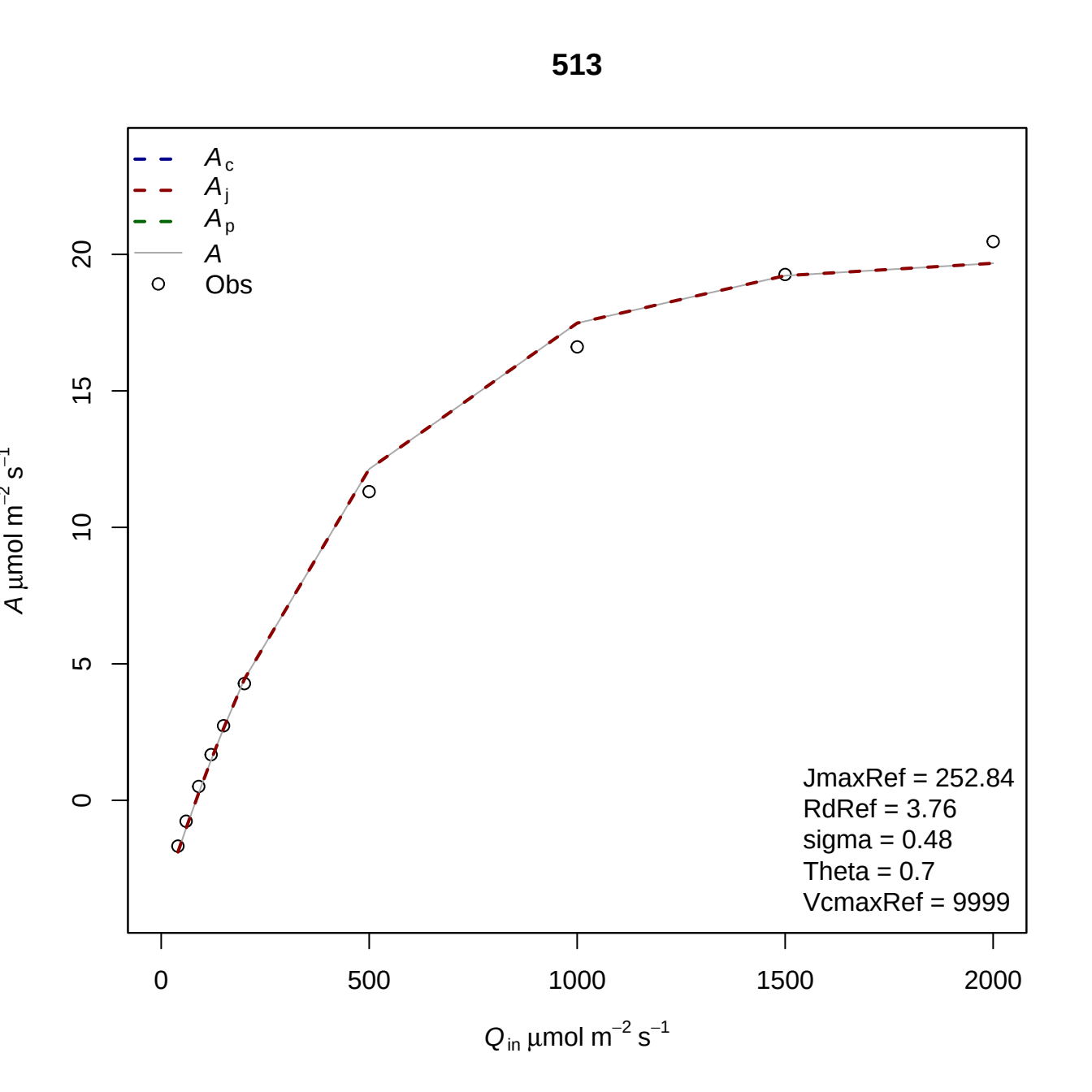

508
A_c
A_j
A_p
A
Obs
JmaxRef = 133.38
RdRef = 1.98
sigma = 0.78
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

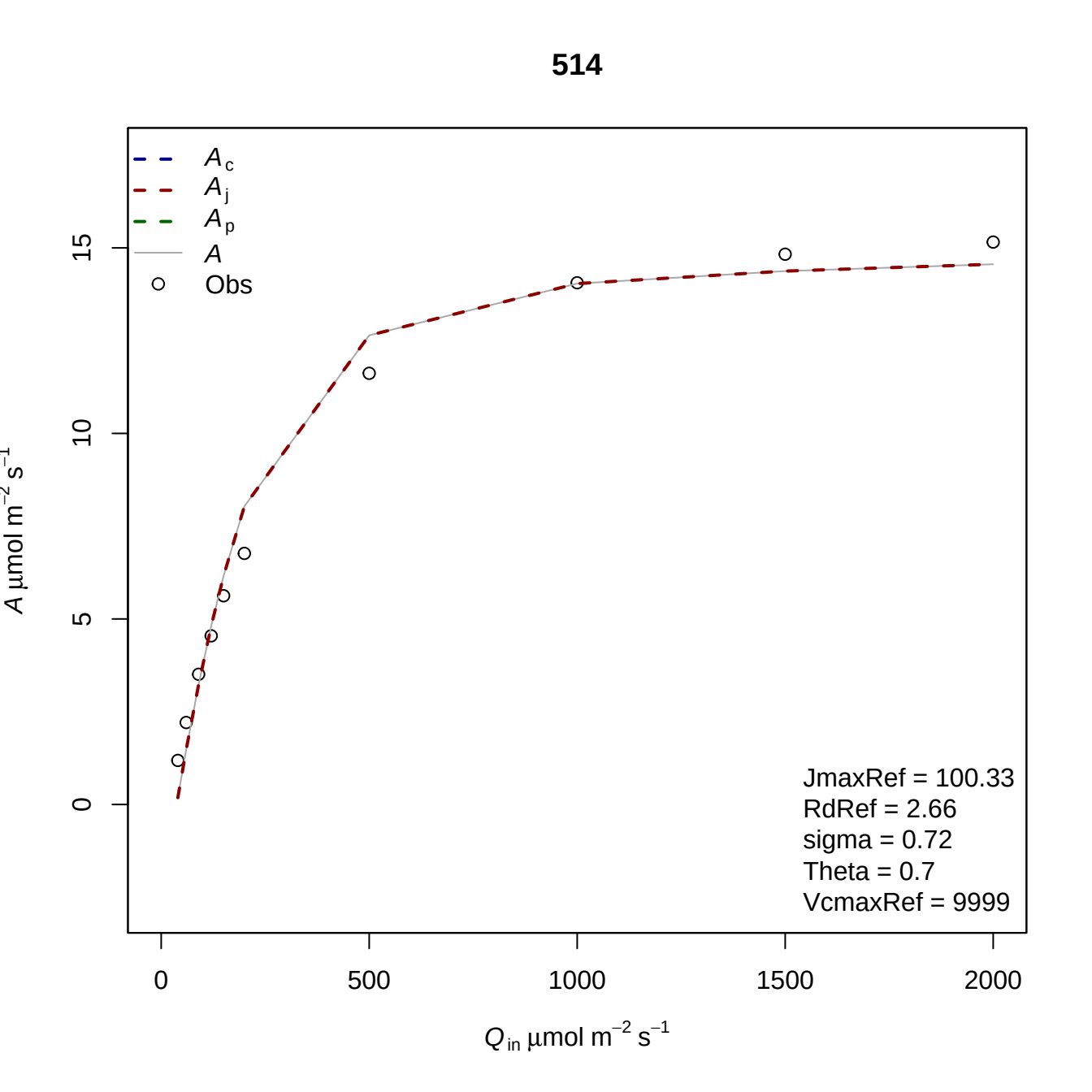

509
A_c
A_j
A_p
A
Obs
JmaxRef = 154.9
RdRef = 2.7
sigma = 0.81
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

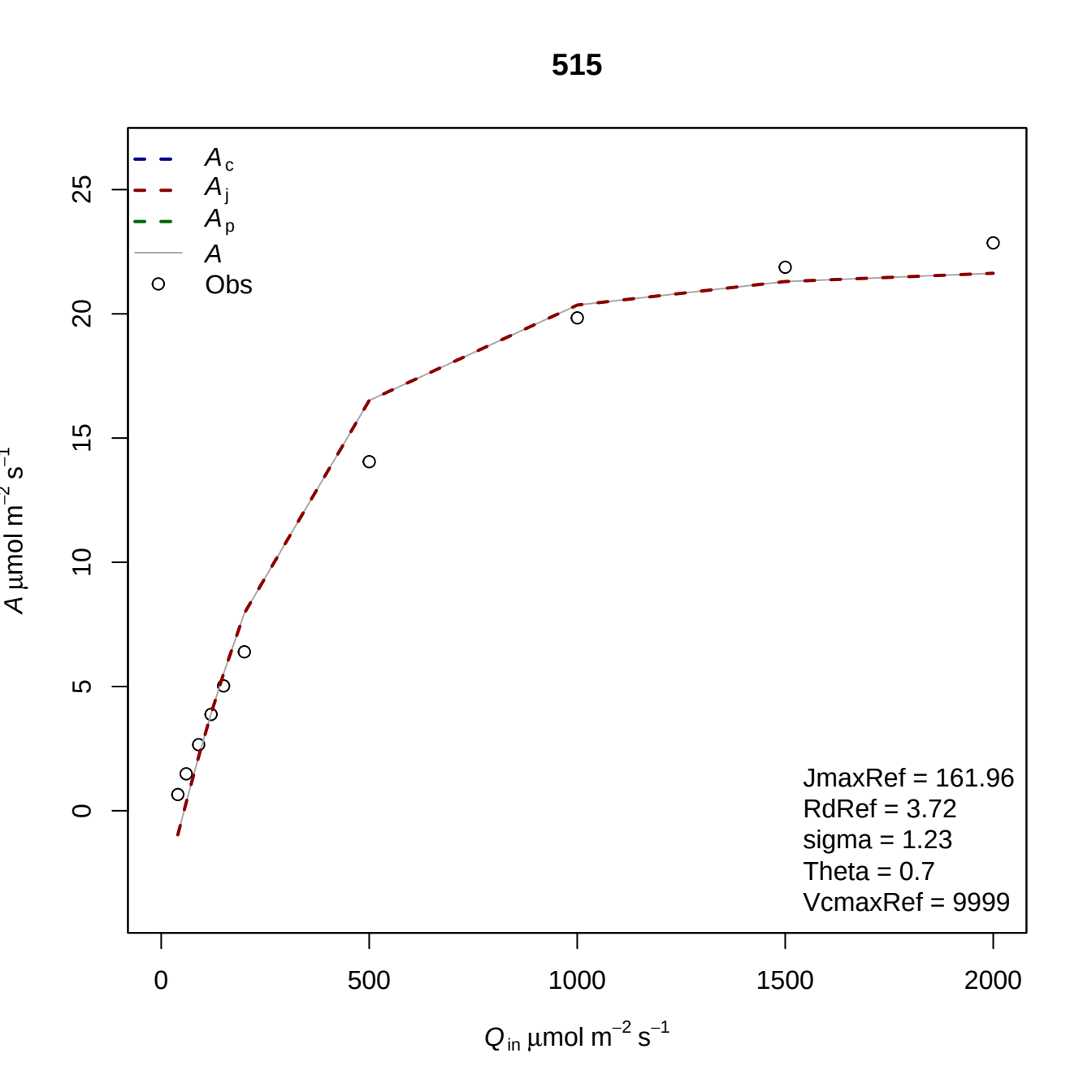

510
A_c
A_j
A_p
A
Obs
JmaxRef = 214.1
RdRef = 1.64
sigma = 0.28
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

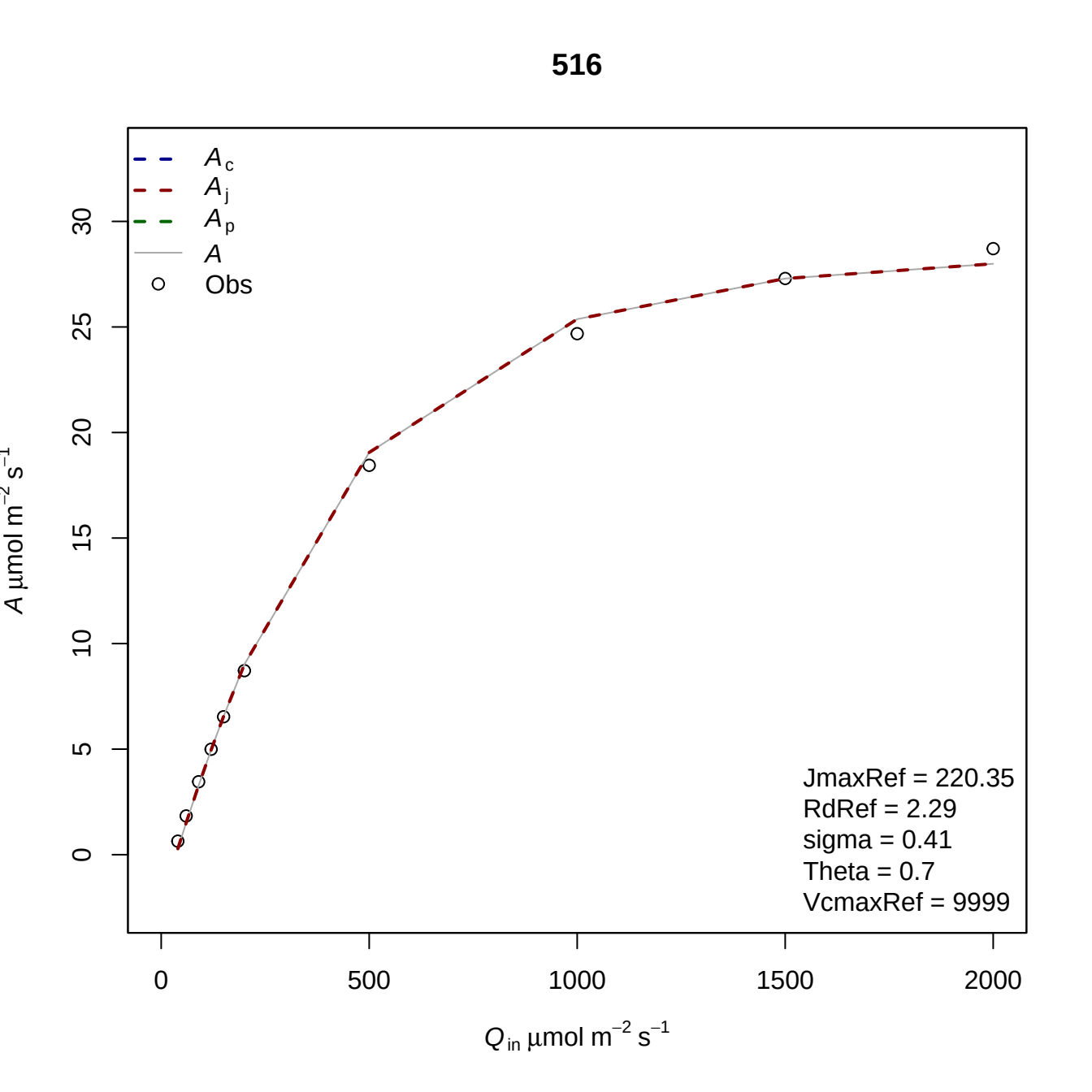

511
A_c
A_j
A_p
A
Obs
JmaxRef = 214.31
RdRef = 2.27
sigma = 0.26
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
25
30
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

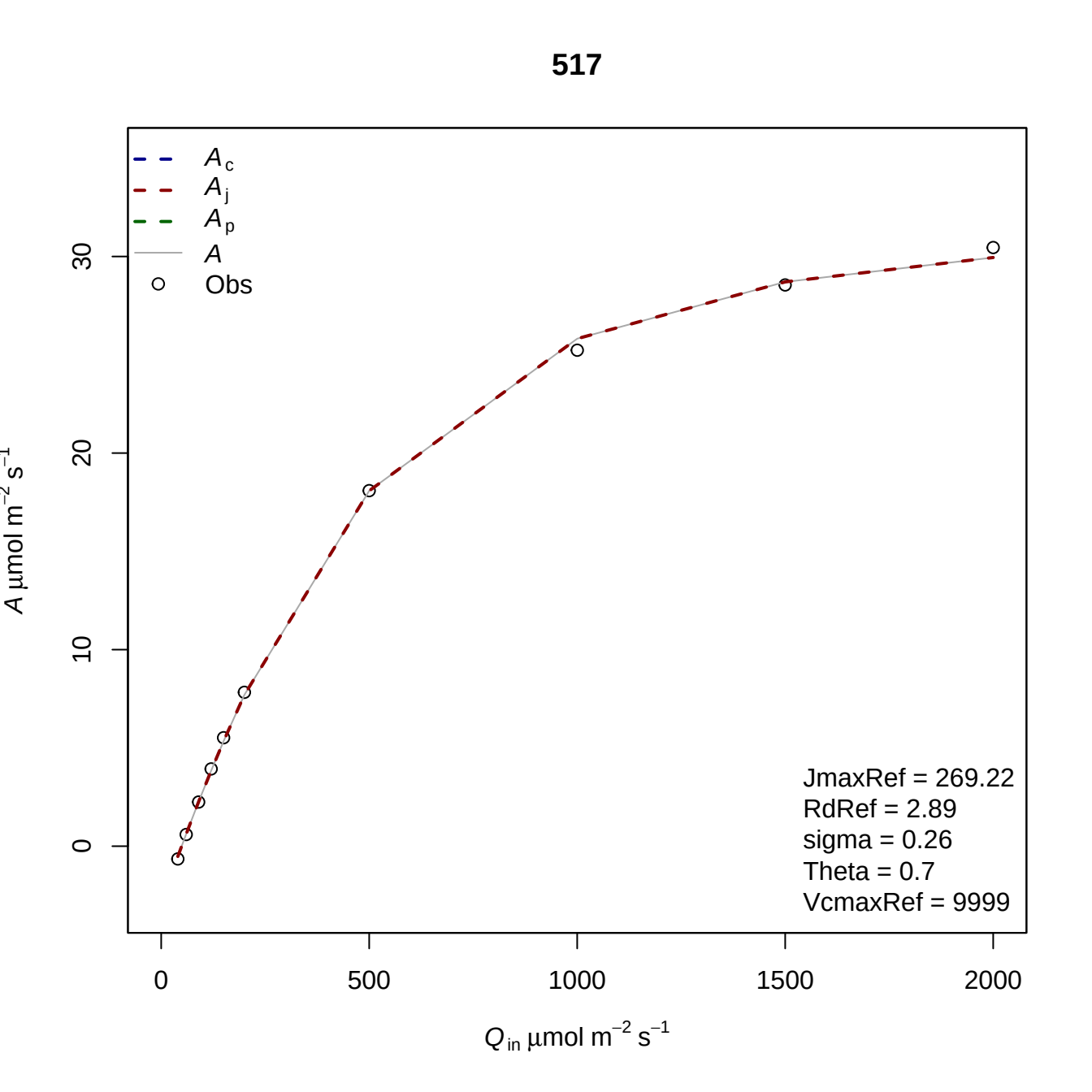

512
A_c
A_j
A_p
A
Obs
JmaxRef = 241.18
RdRef = 2.88
sigma = 0.3
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

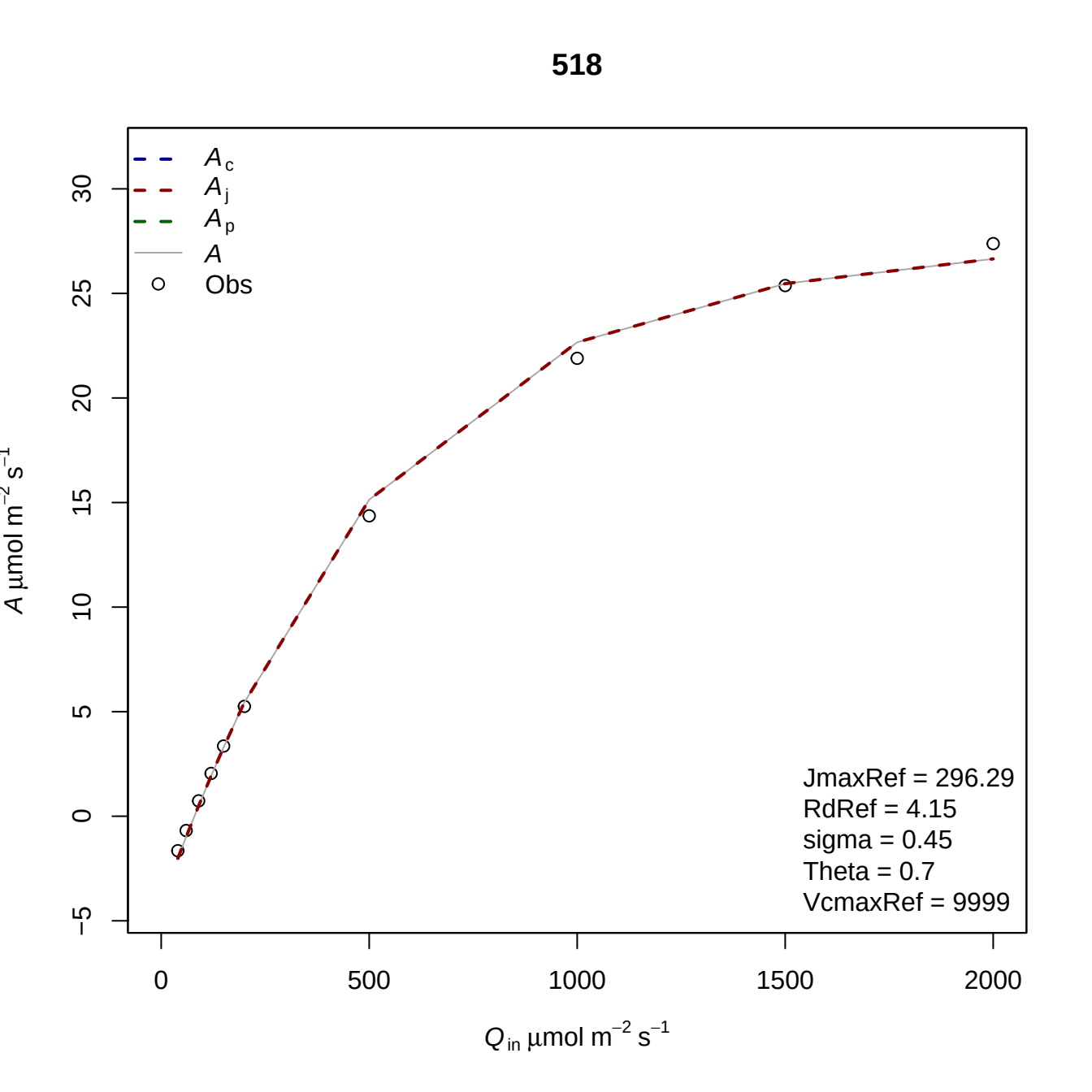

513
A_c
A_j
A_p
A
Obs
JmaxRef = 252.84
RdRef = 3.76
sigma = 0.48
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

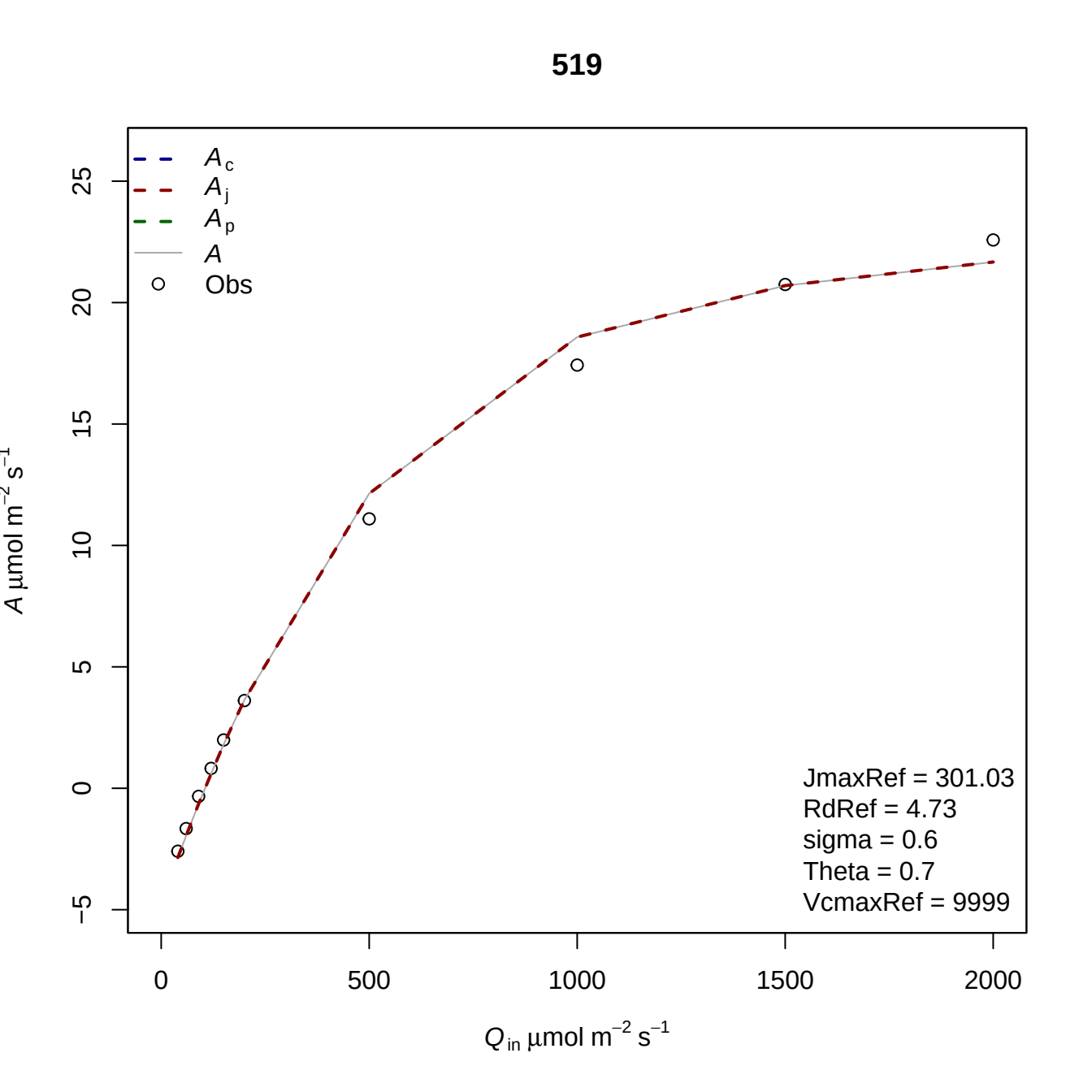

514
A_c
A_j
A_p
A
Obs
JmaxRef = 100.33
RdRef = 2.66
sigma = 0.72
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

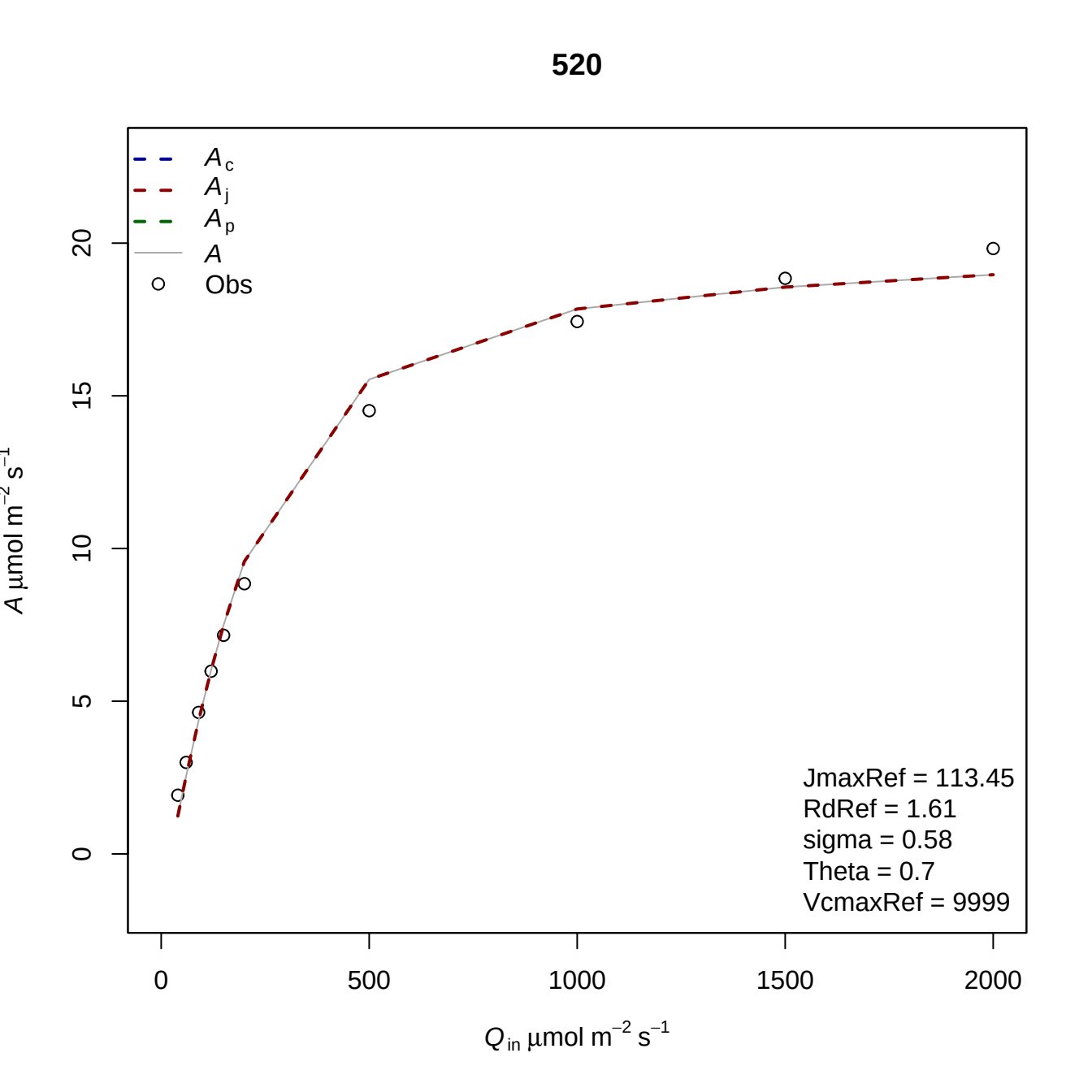

515
A_c
A_j
A_p
A
Obs
JmaxRef = 161.96
RdRef = 3.72
sigma = 1.23
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
25
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

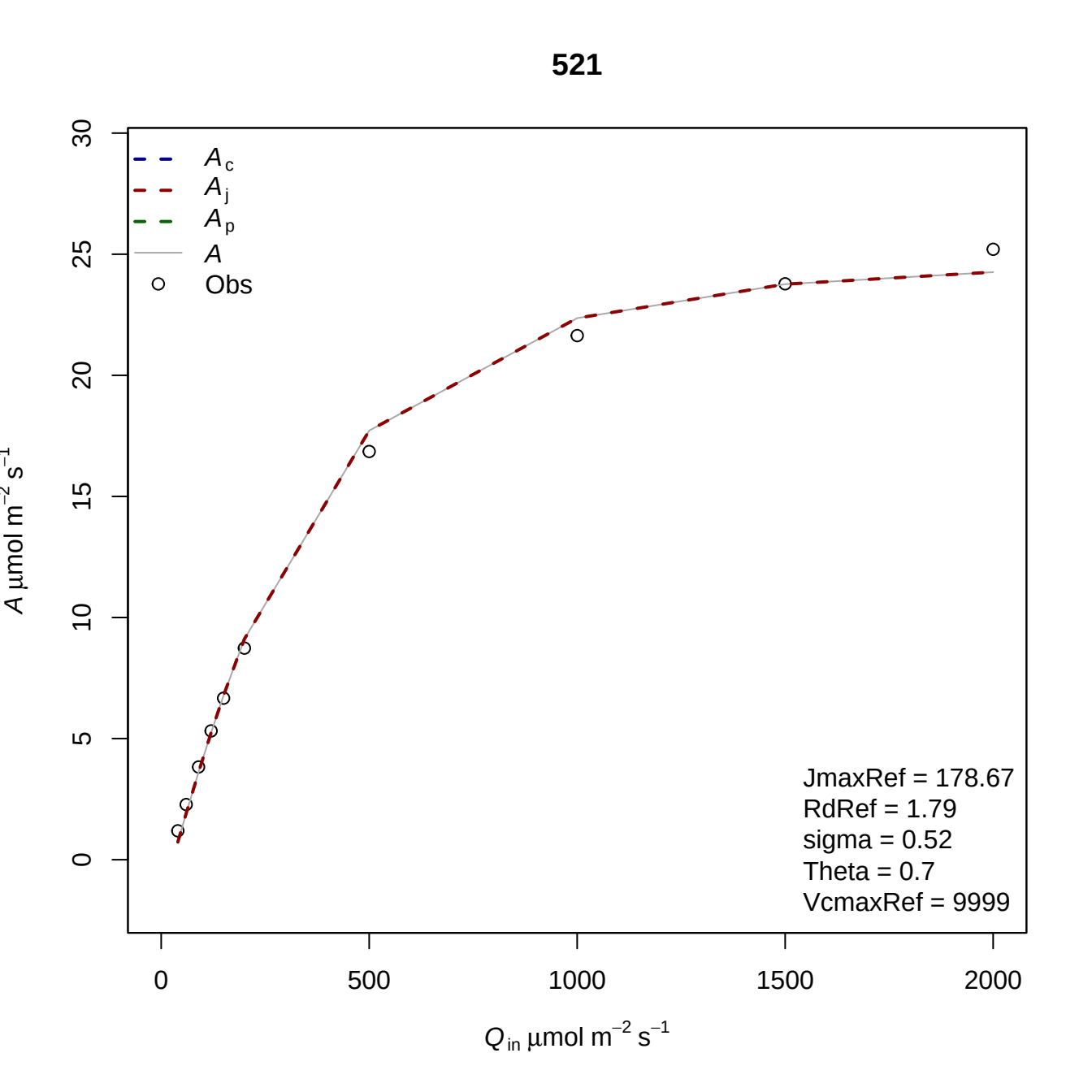

516
A_c
A_j
A_p
A
Obs
JmaxRef = 220.35
RdRef = 2.29
sigma = 0.41
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

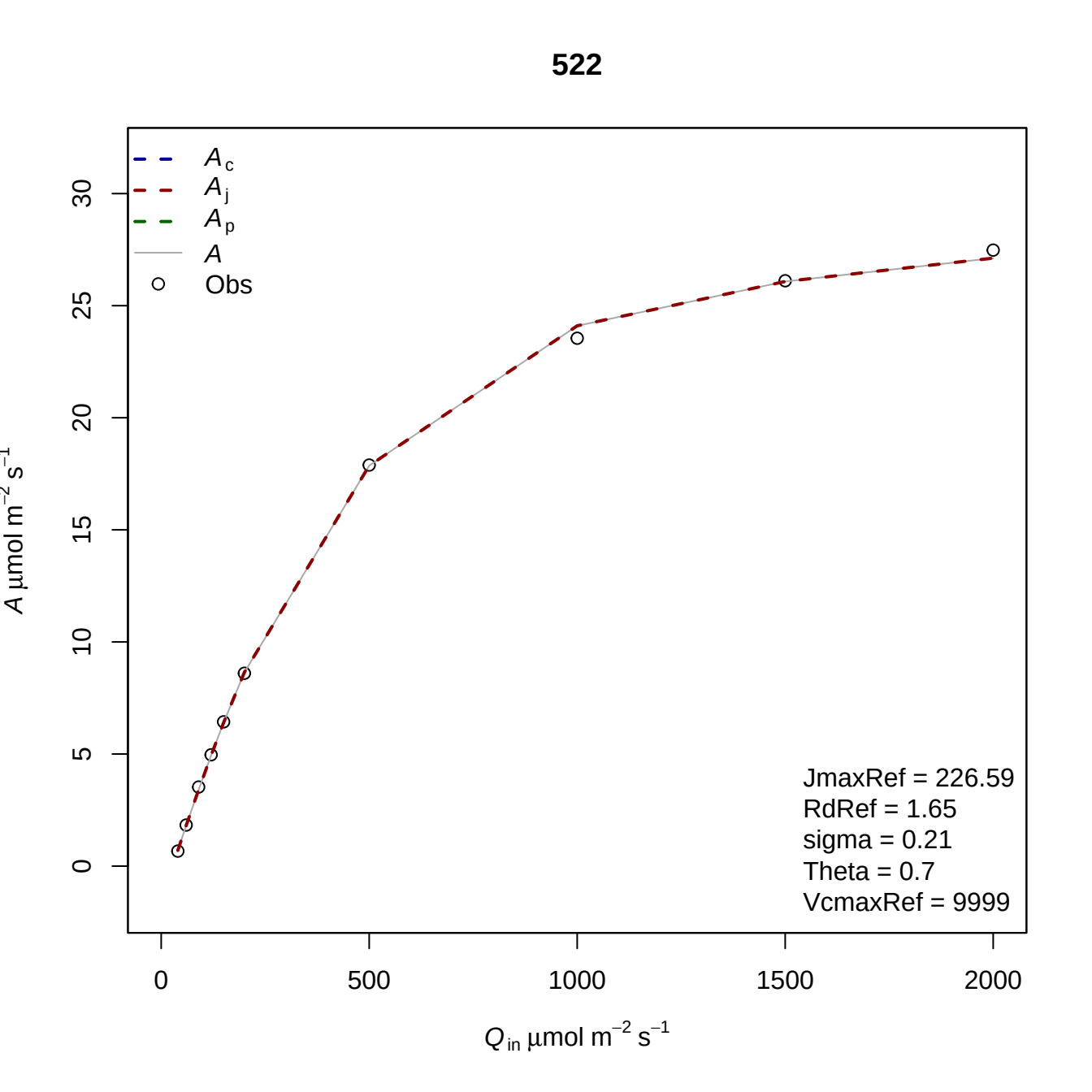

517
A_c
A_j
A_p
A
Obs
JmaxRef = 269.22
RdRef = 2.89
sigma = 0.26
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
0
500
1000
1500
2000

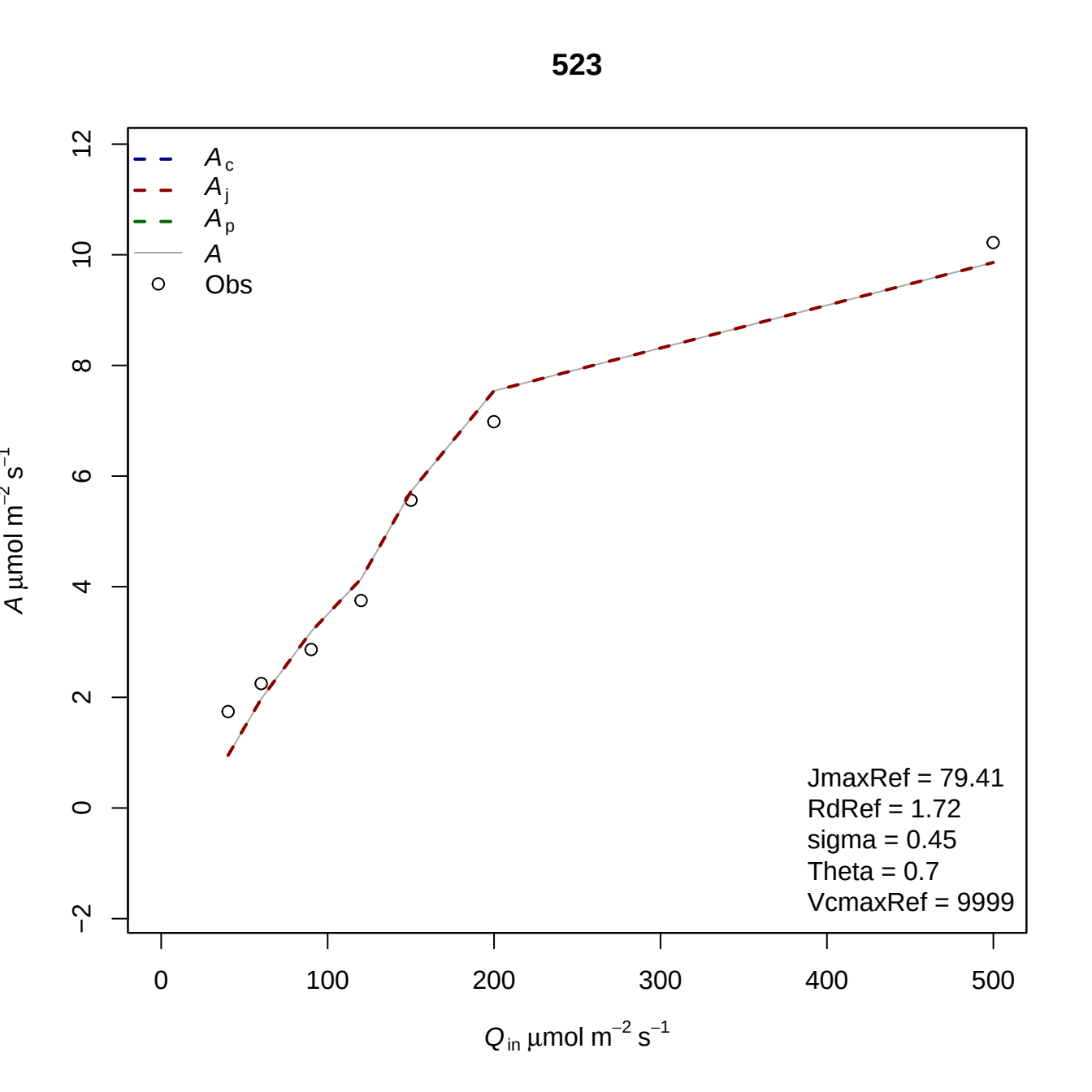

518
A_c
A_j
A_p
A
Obs
JmaxRef = 296.29
RdRef = 4.15
sigma = 0.45
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
25
30
0
500
1000
1500
2000

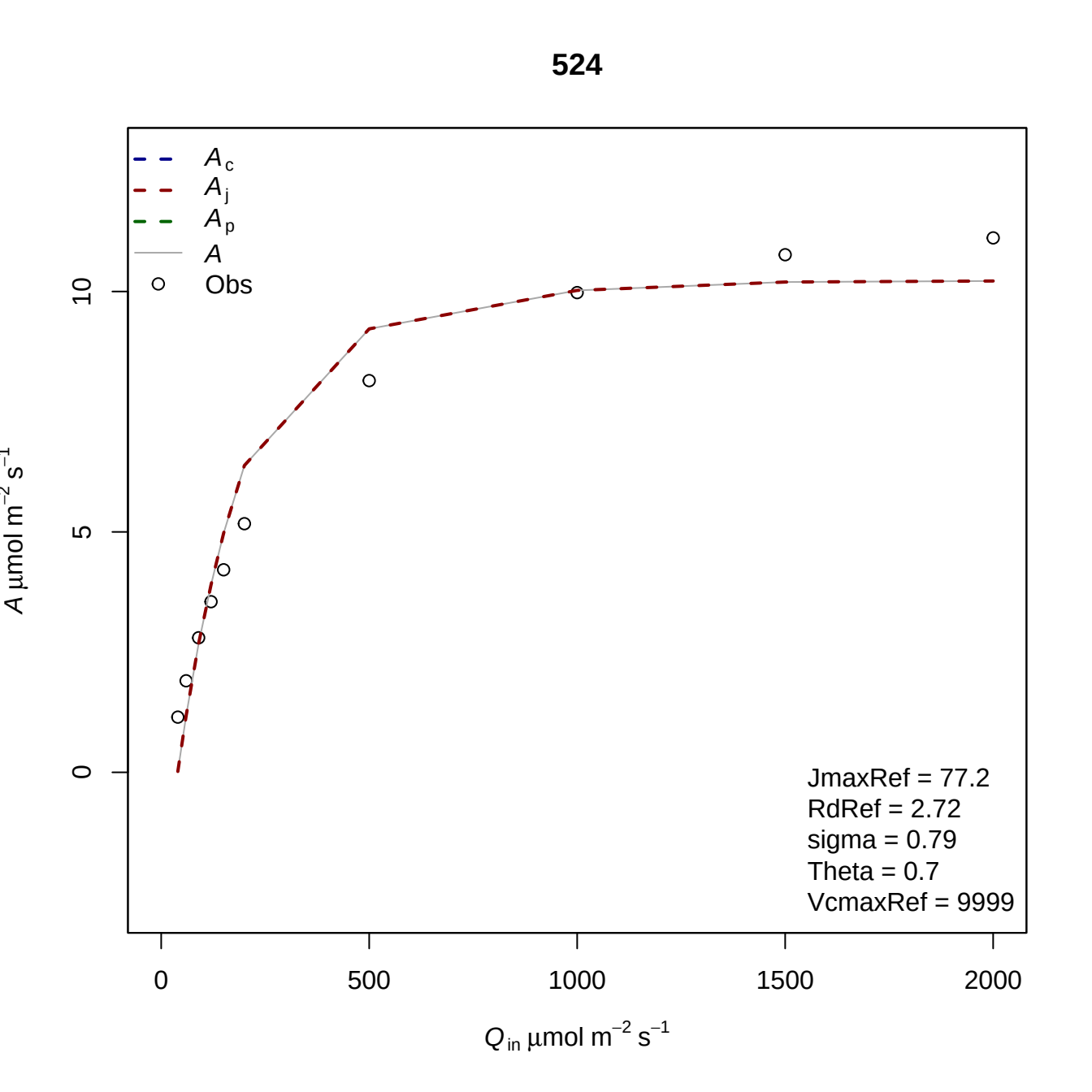

519
A_c
A_j
A_p
A
Obs
JmaxRef = 301.03
RdRef = 4.73
sigma = 0.6
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
25
20
15
10
5
0
−5
0
500
1000
1500
2000

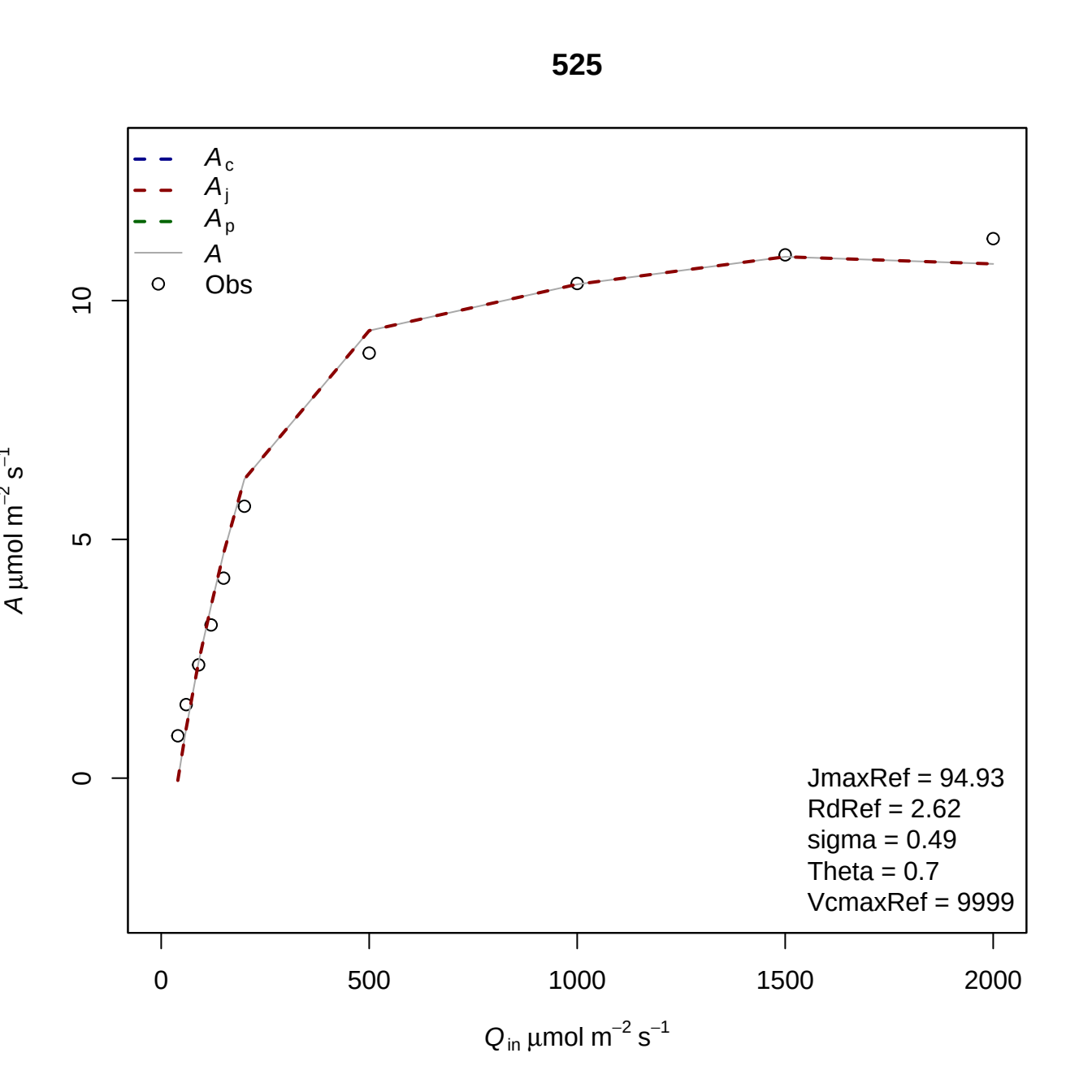

520
A_c
A_j
A_p
A
Obs
JmaxRef = 113.45
RdRef = 1.61
sigma = 0.58
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

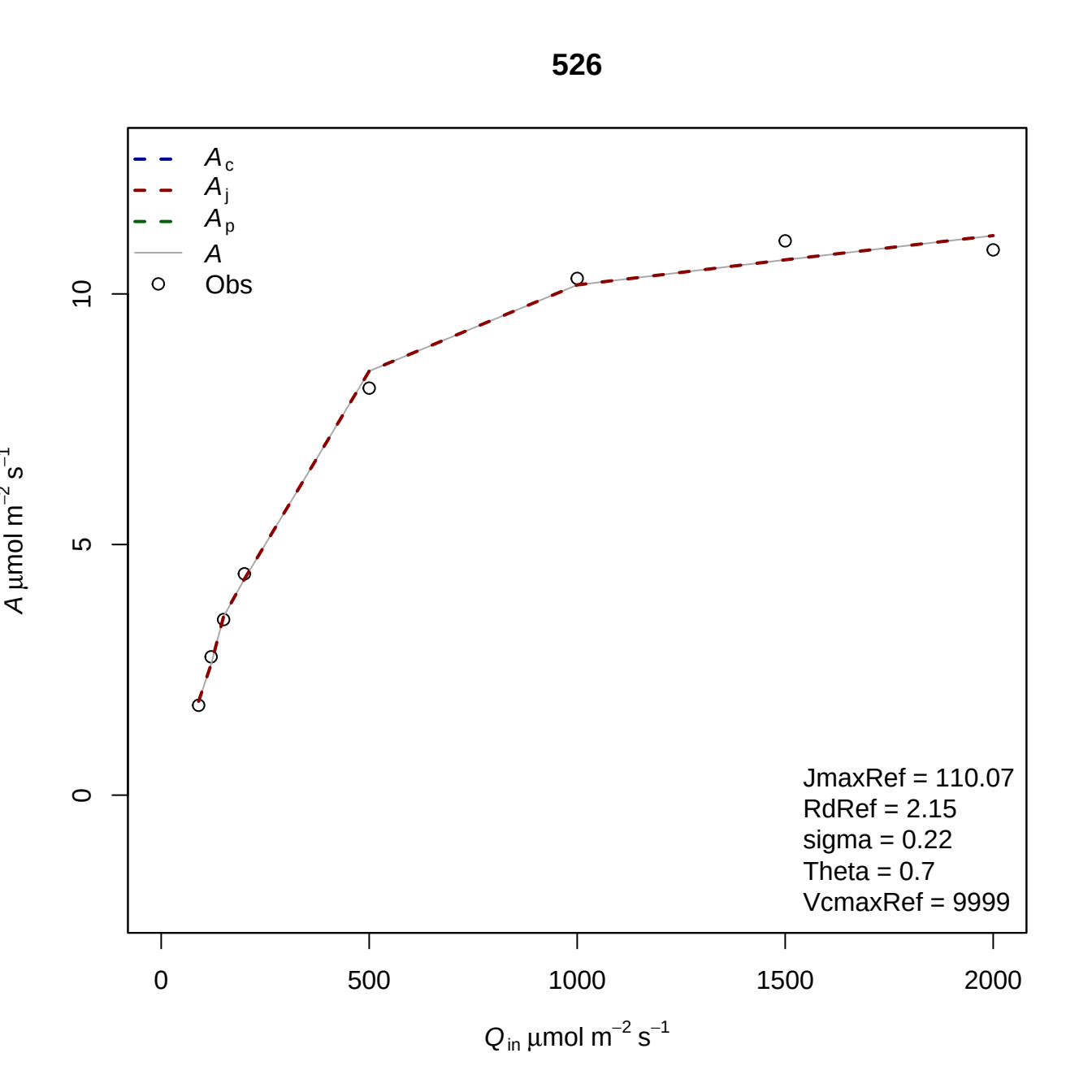

521
A_c
A_j
A_p
A
Obs
JmaxRef = 178.67
RdRef = 1.79
sigma = 0.52
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

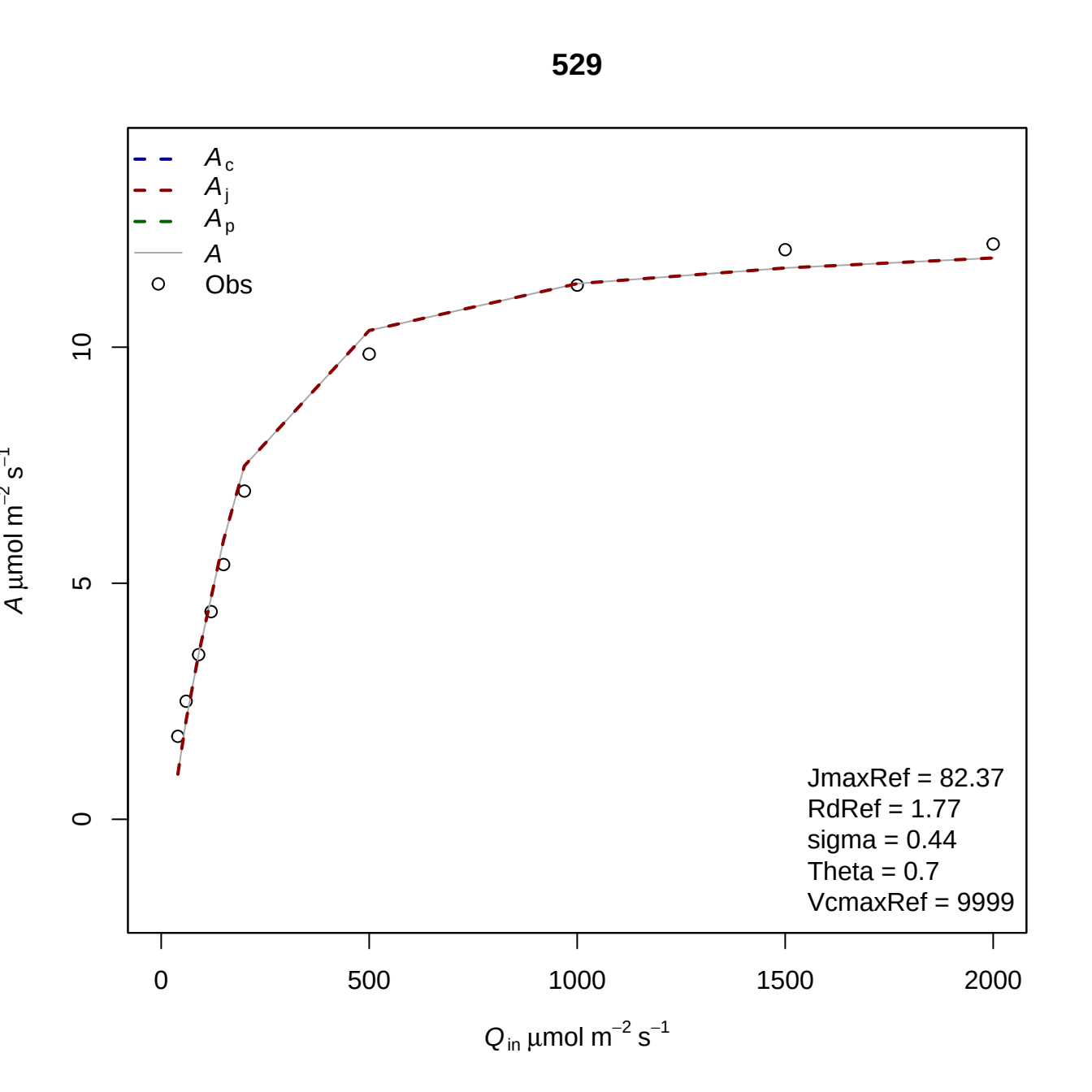

522
A_c
A_j
A_p
A
Obs
JmaxRef = 226.59
RdRef = 1.65
sigma = 0.21
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

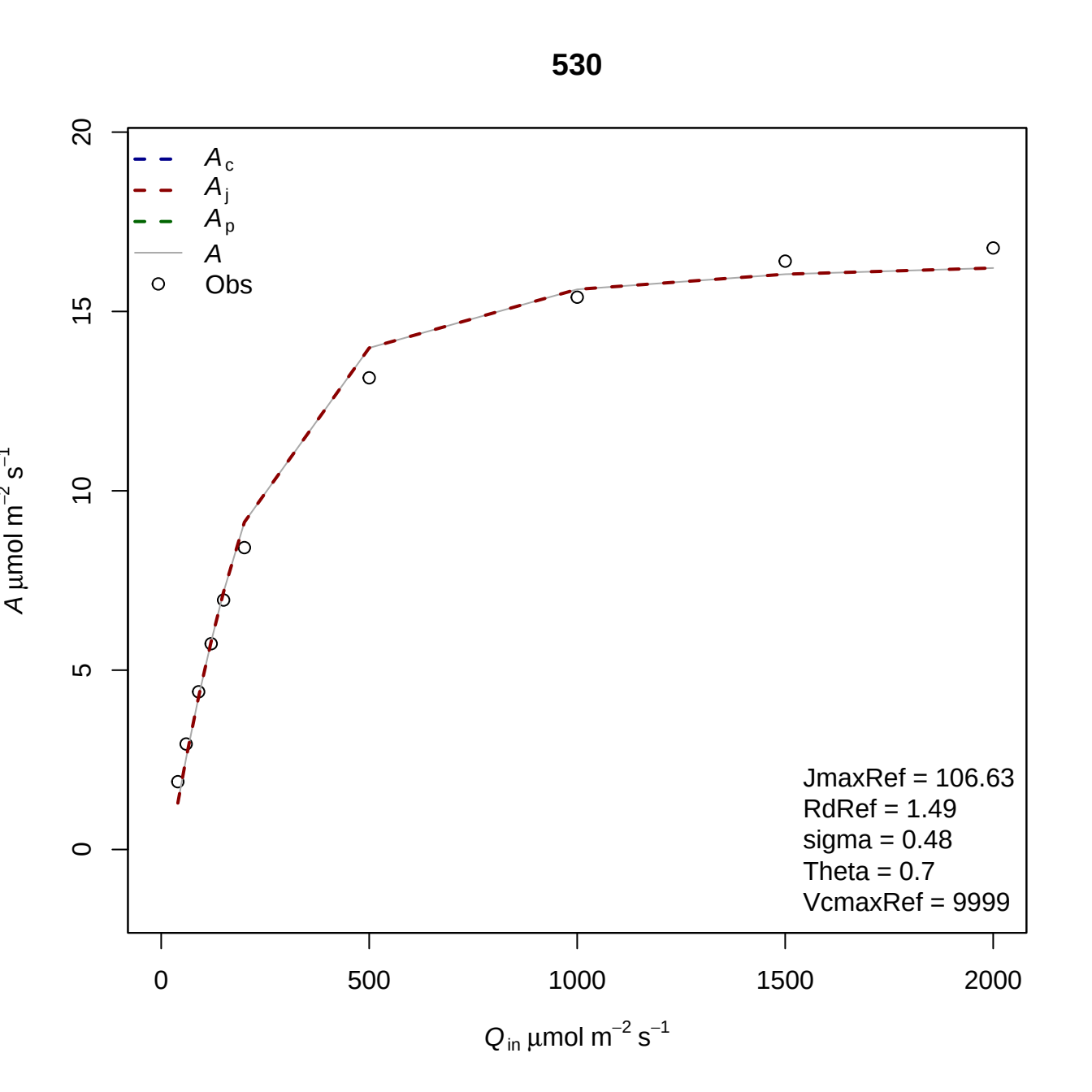

523
A_c
A_j
A_p
A
Obs
JmaxRef = 79.41
RdRef = 1.72
sigma = 0.45
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
100
200
300
400
500

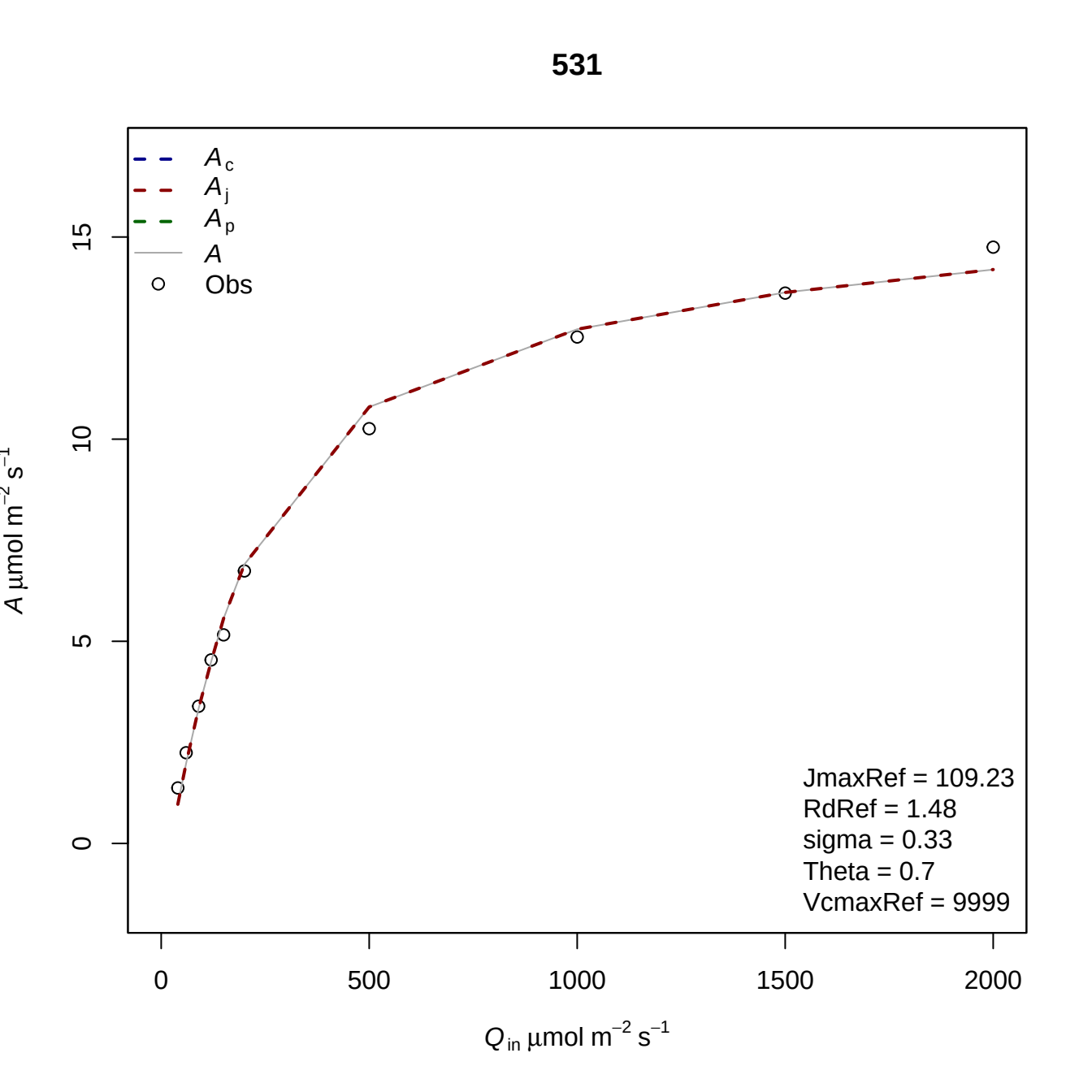

524
A_c
A_j
A_p
A
Obs
JmaxRef = 77.2
RdRef = 2.72
sigma = 0.79
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

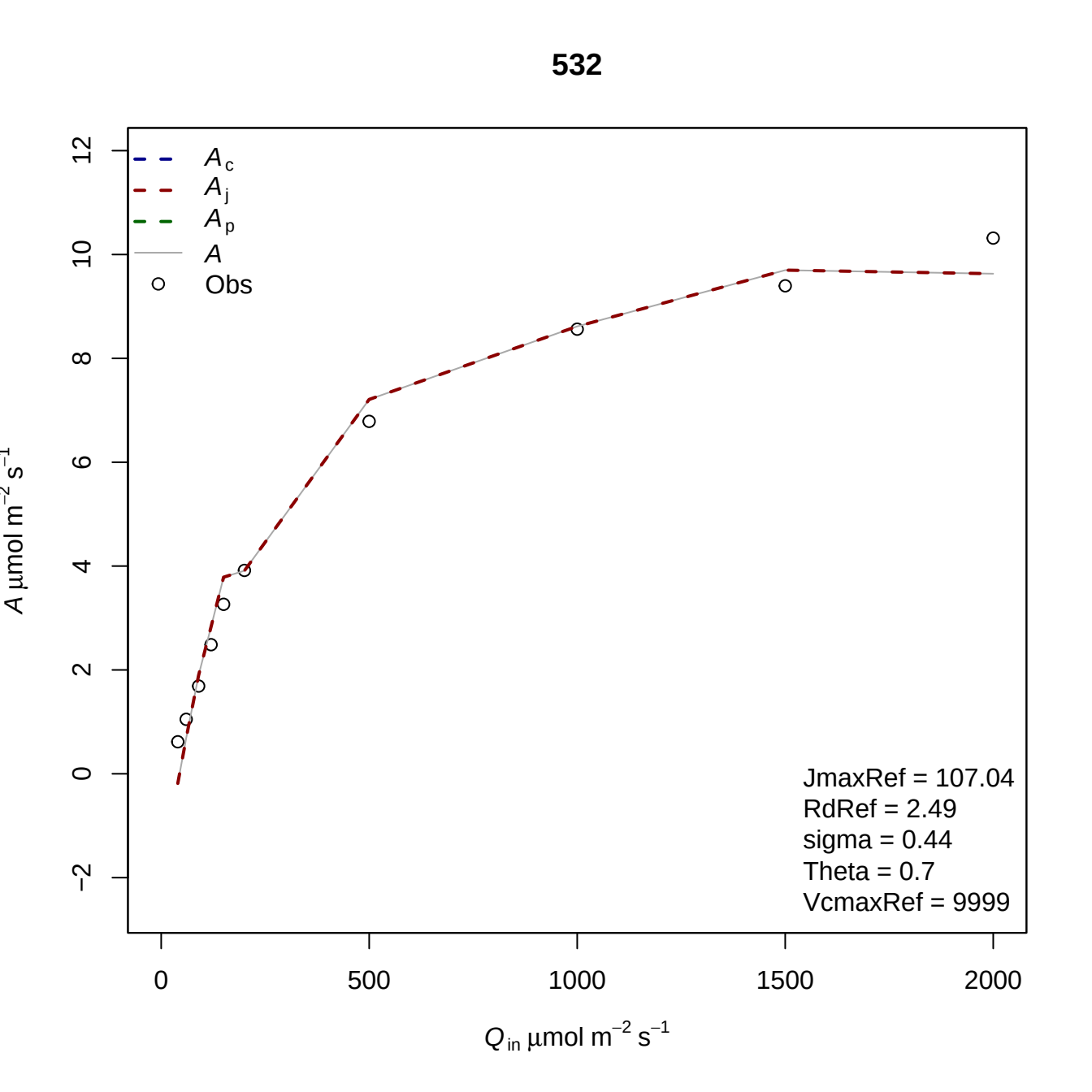

525
A_c
A_j
A_p
A
Obs
JmaxRef = 94.93
RdRef = 2.62
sigma = 0.49
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

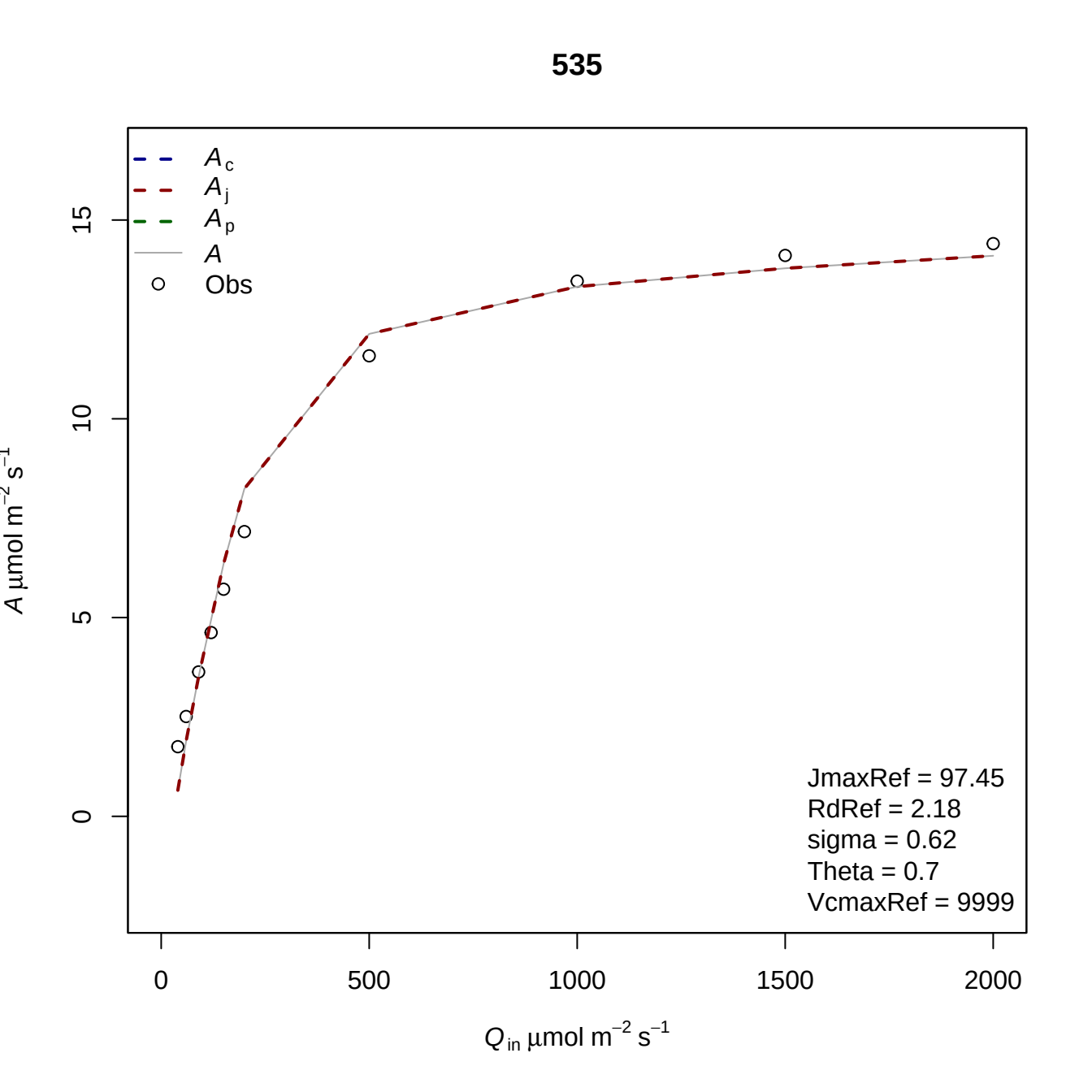

526
A_c
A_j
A_p
A
Obs
JmaxRef = 110.07
RdRef = 2.15
sigma = 0.22
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

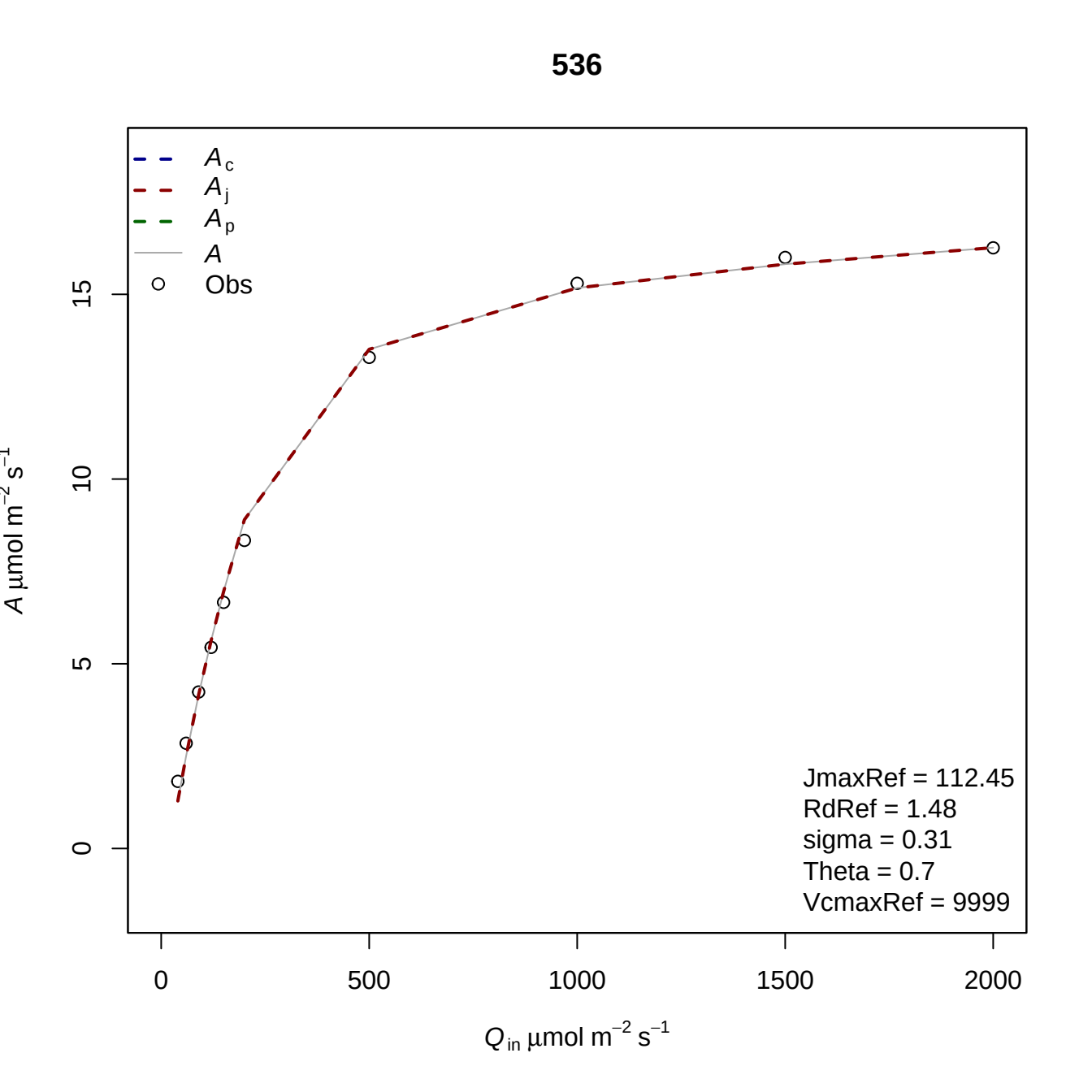

529
A_c
A_j
A_p
A
Obs
JmaxRef = 82.37
RdRef = 1.77
sigma = 0.44
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

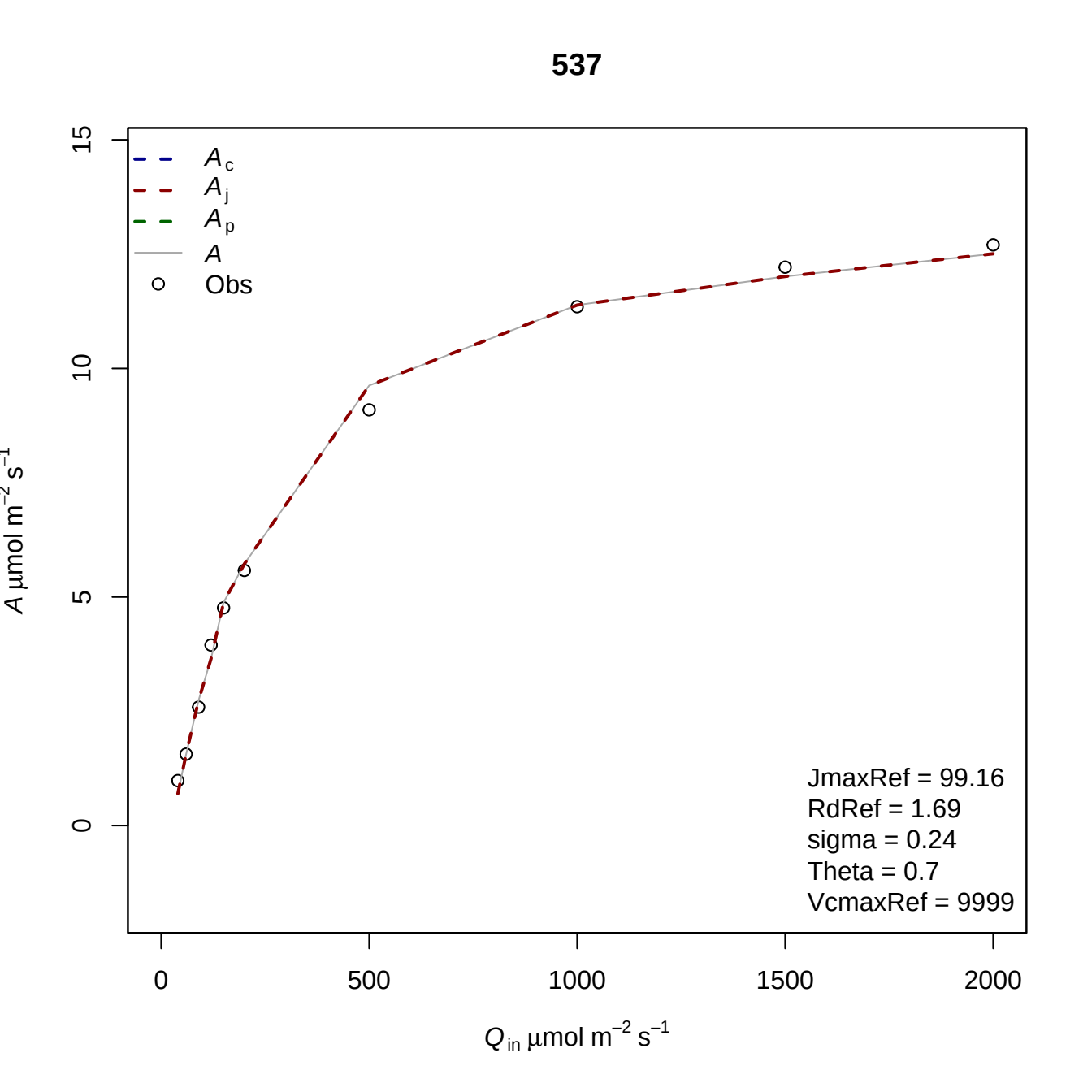

530
A_c
A_j
A_p
A
Obs
JmaxRef = 106.63
RdRef = 1.49
sigma = 0.48
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

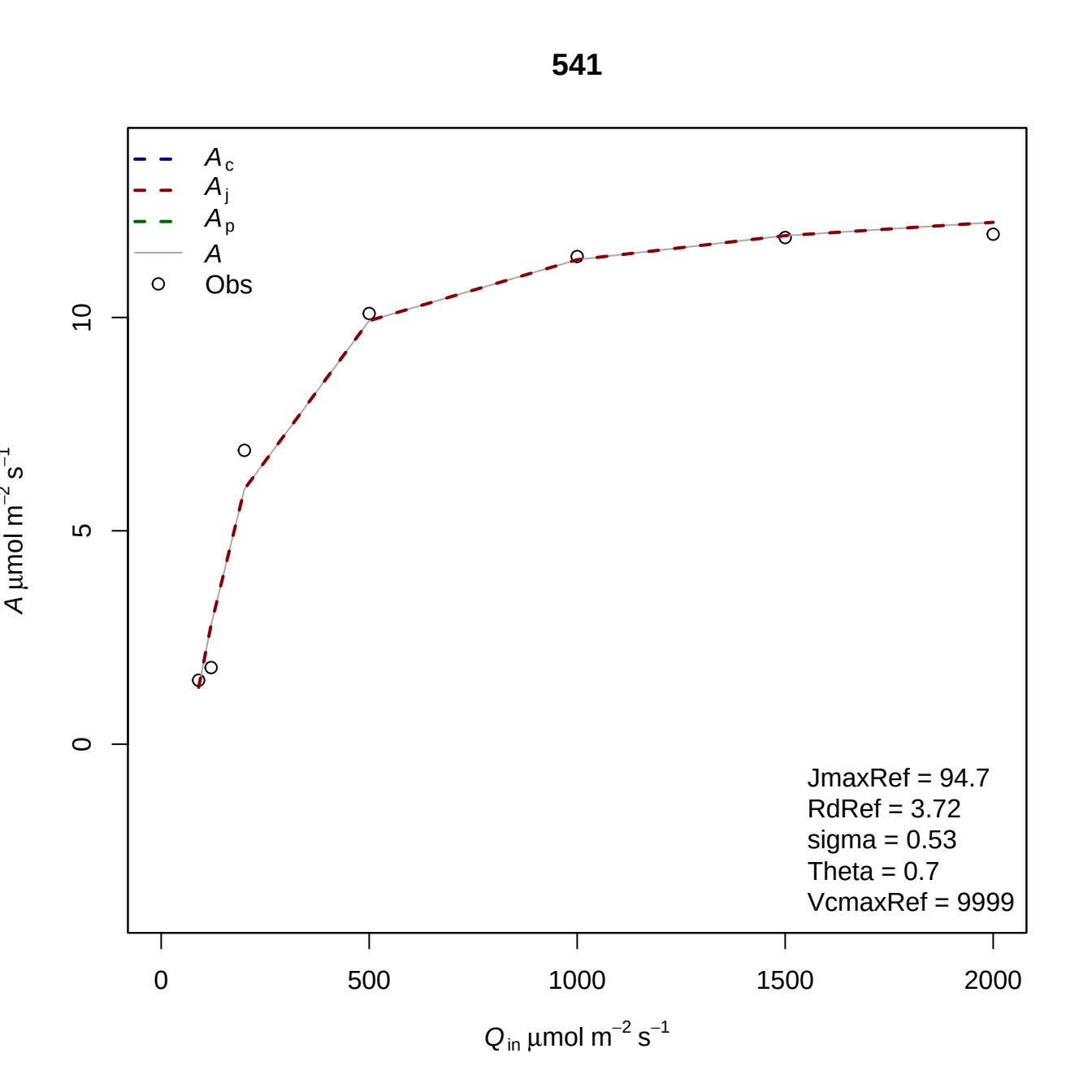

531
A_c
A_j
A_p
A
Obs
JmaxRef = 109.23
RdRef = 1.48
sigma = 0.33
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

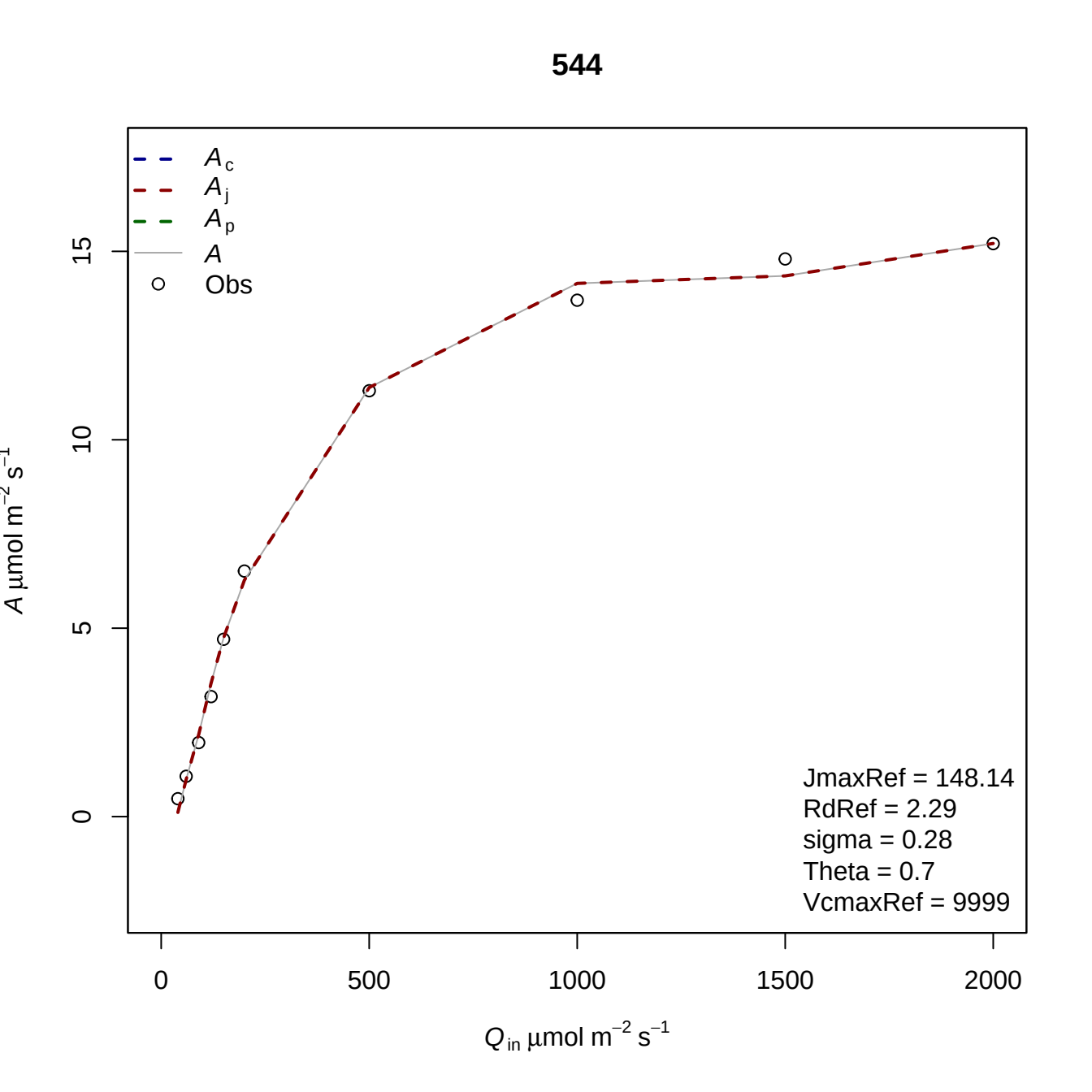

532
A_c
A_j
A_p
A
Obs
JmaxRef = 107.04
RdRef = 2.49
sigma = 0.44
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

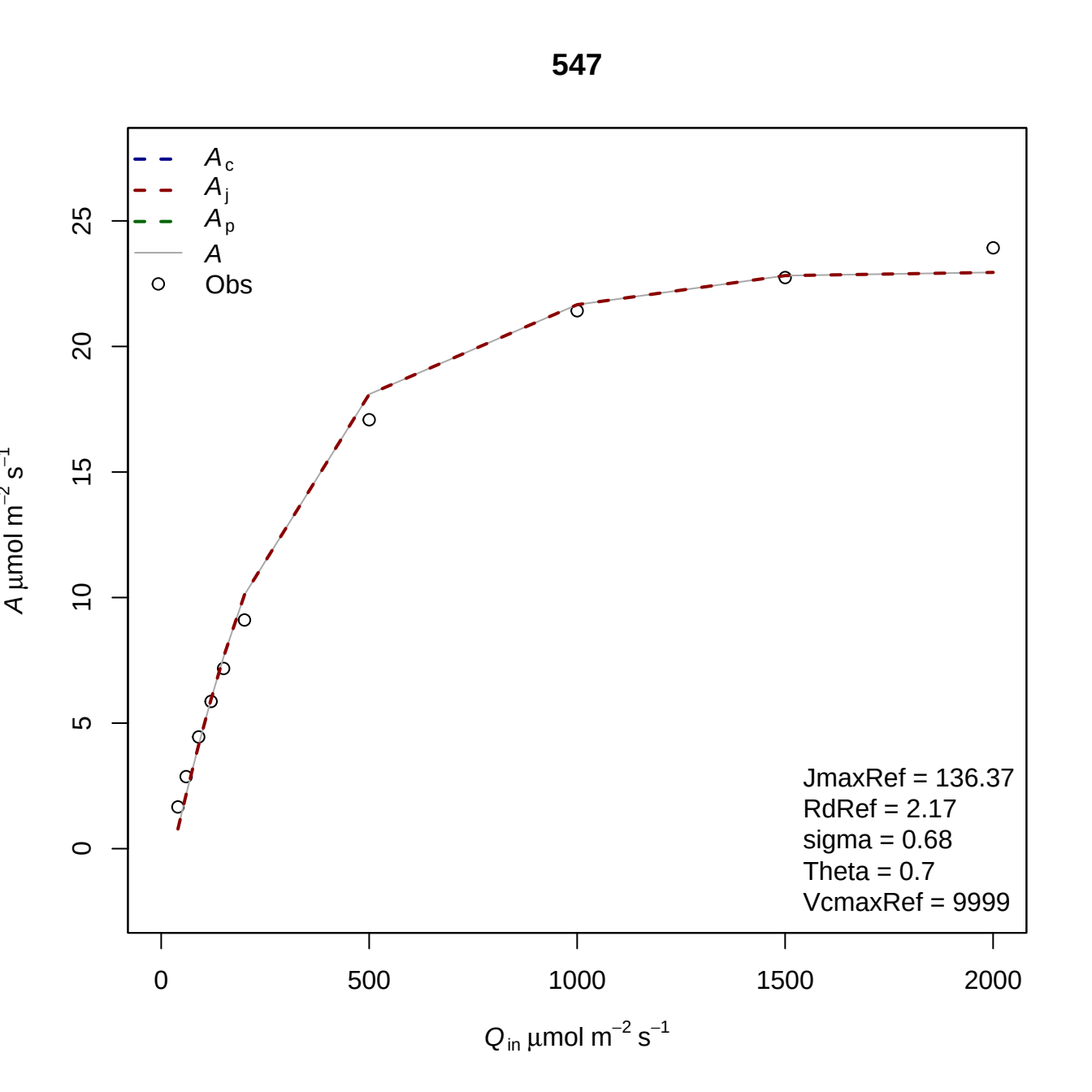

535
A_c
A_j
A_p
A
Obs
JmaxRef = 97.45
RdRef = 2.18
sigma = 0.62
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

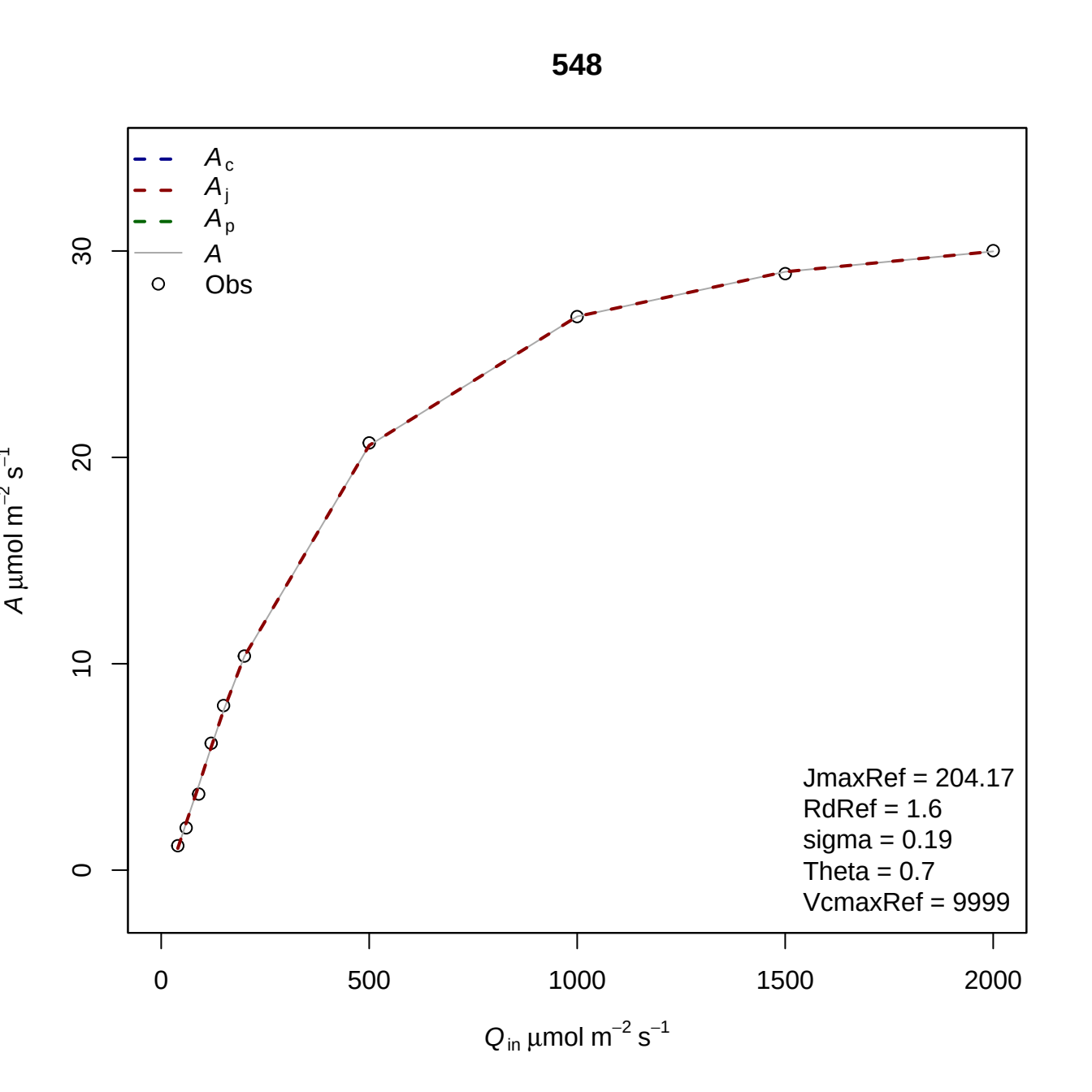

536
A_c
A_j
A_p
A
Obs
JmaxRef = 112.45
RdRef = 1.48
sigma = 0.31
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

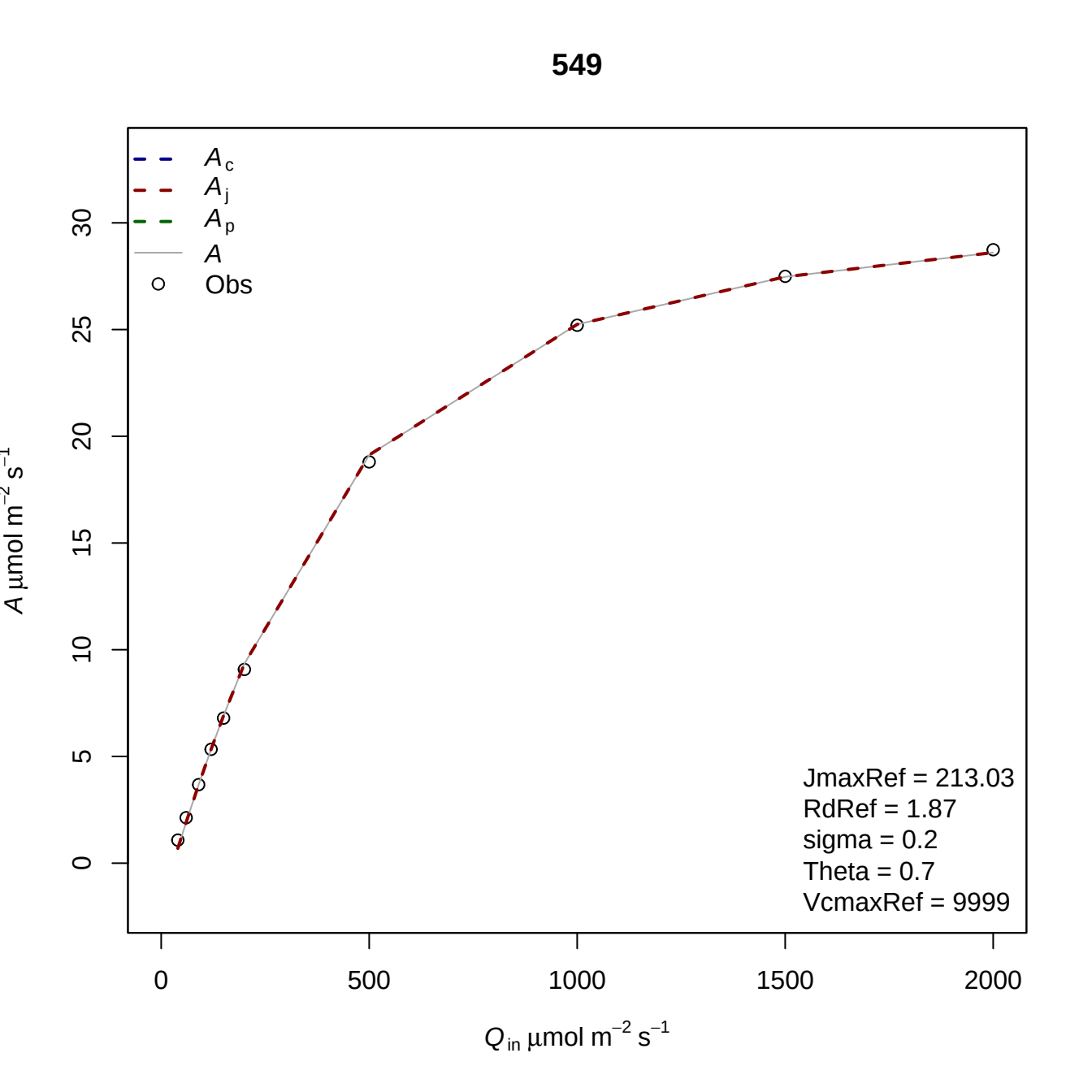

537
A_c
A_j
A_p
A
Obs
JmaxRef = 99.16
RdRef = 1.69
sigma = 0.24
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

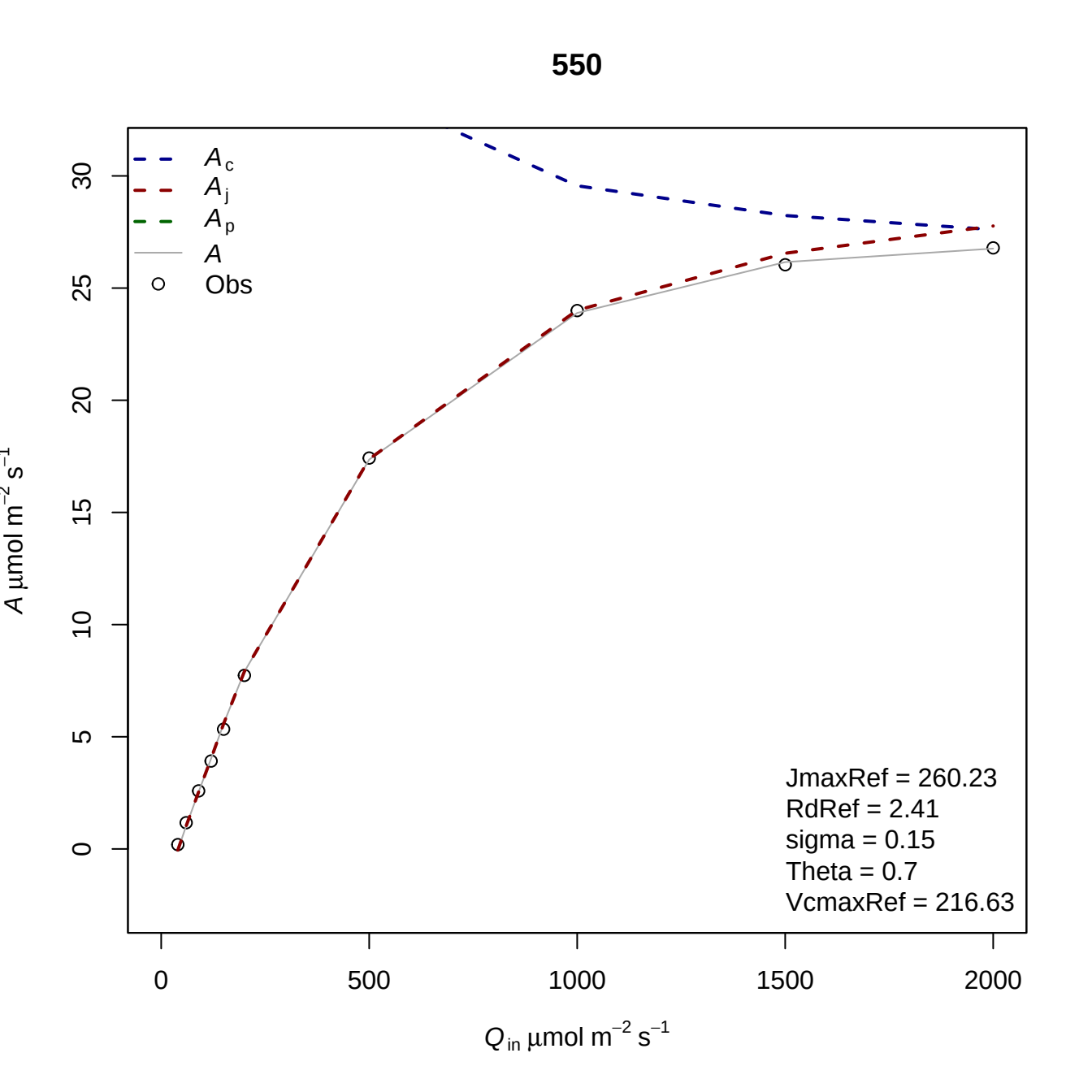

541
A_c
A_j
A_p
A
Obs
JmaxRef = 94.7
RdRef = 3.72
sigma = 0.53
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

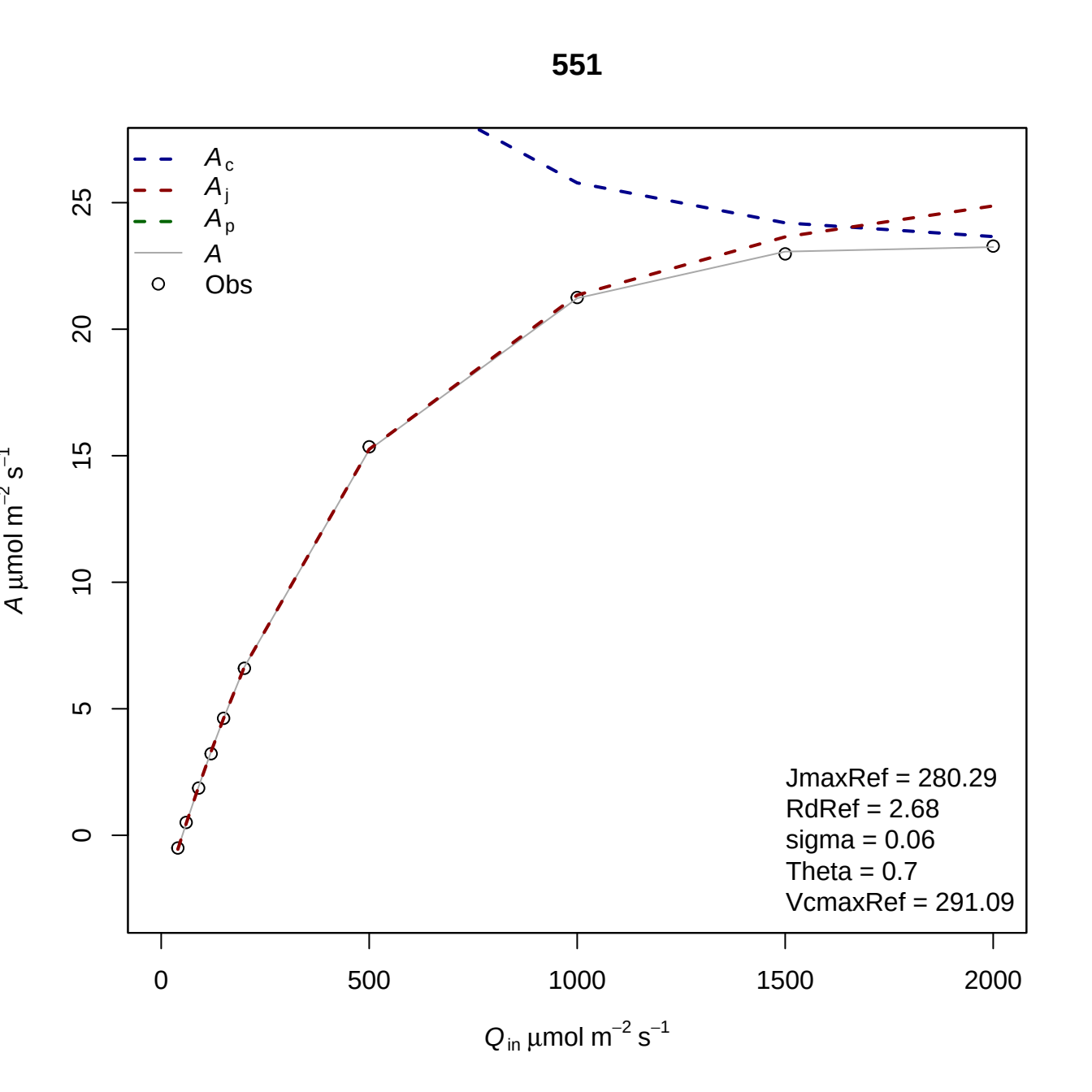

544
A_c
A_j
A_p
A
Obs
JmaxRef = 148.14
RdRef = 2.29
sigma = 0.28
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

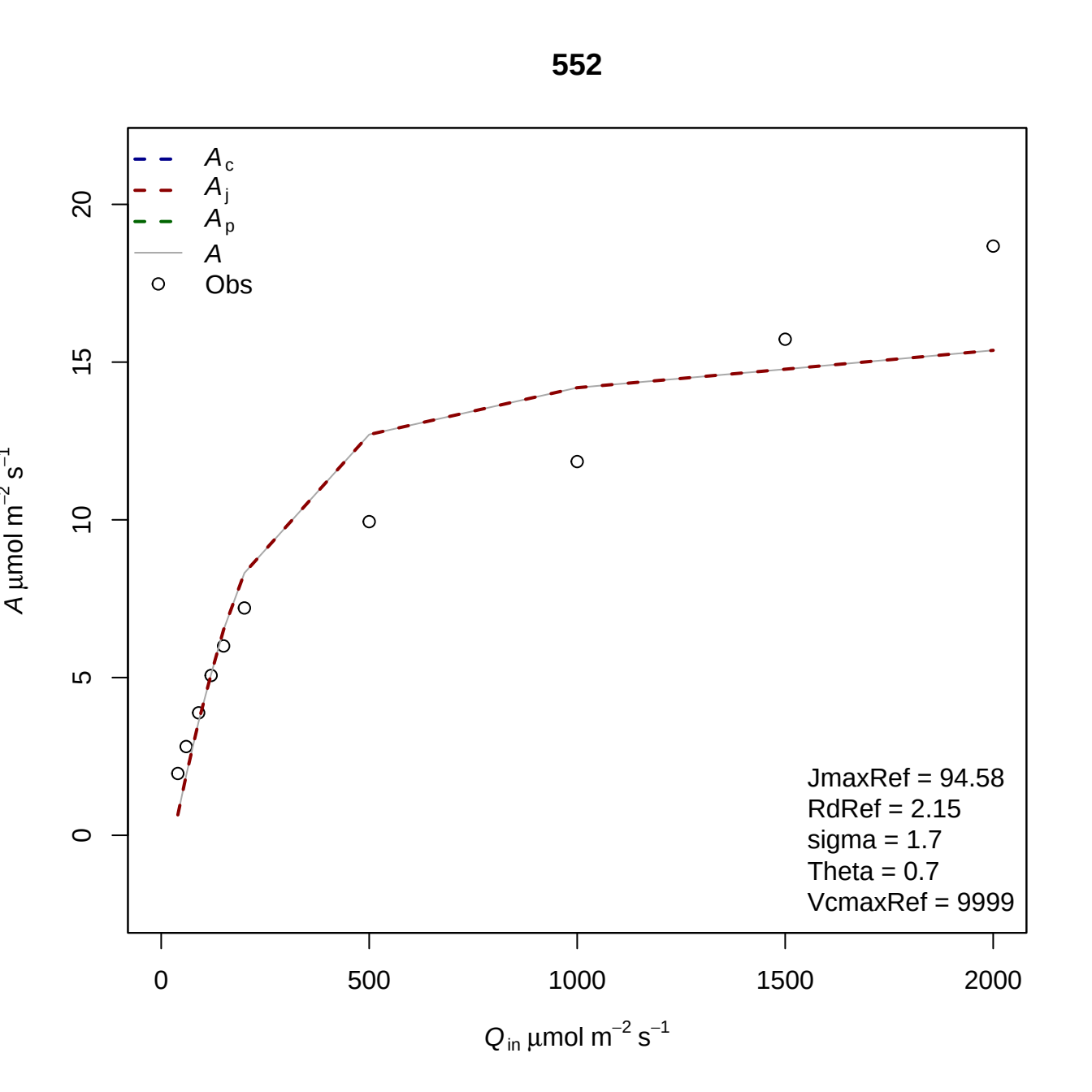

547
A_c
A_j
A_p
A
Obs
JmaxRef = 136.37
RdRef = 2.17
sigma = 0.68
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

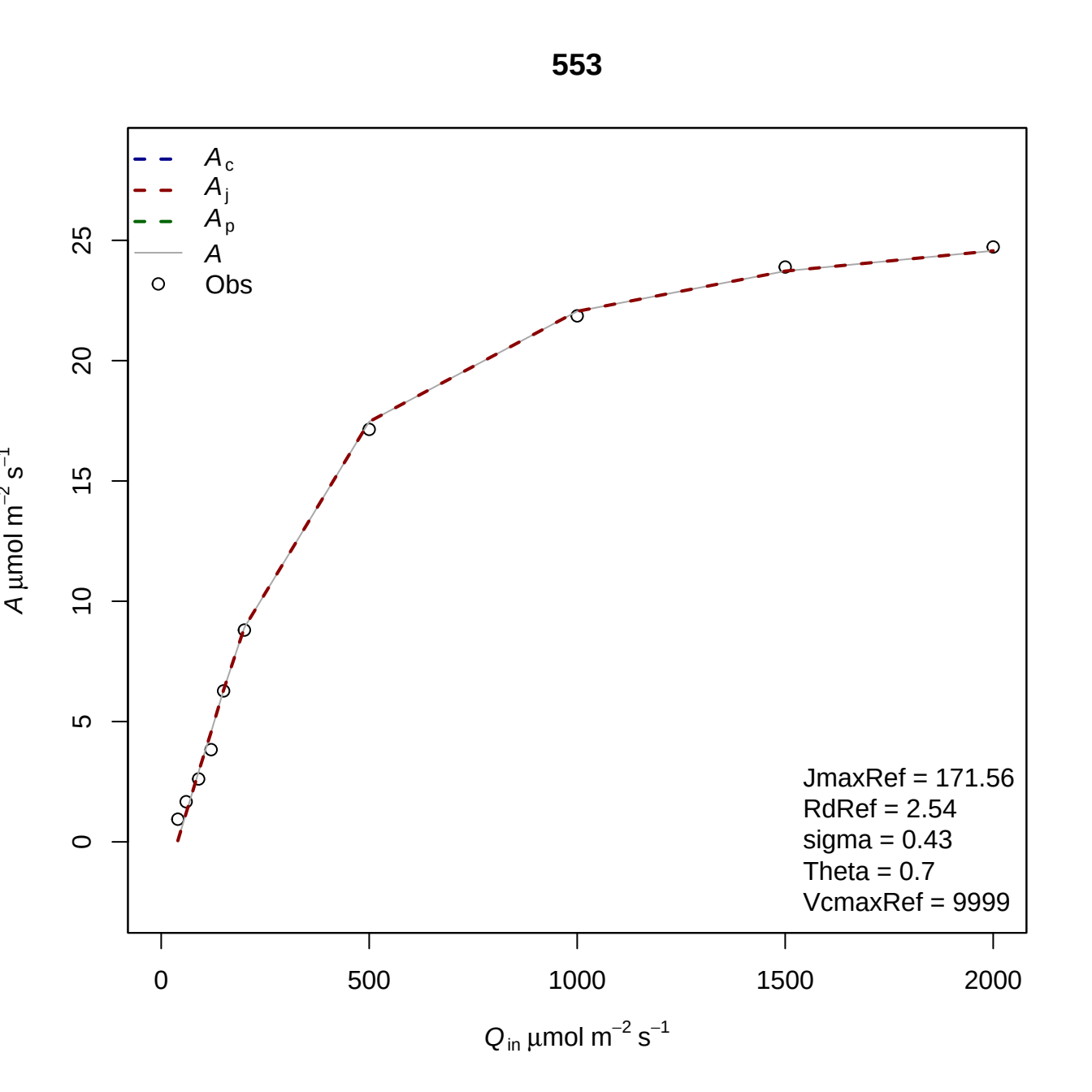

548
A_c
A_j
A_p
A
Obs
JmaxRef = 204.17
RdRef = 1.6
sigma = 0.19
Theta = 0.7
VcmaxRef = 9999
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

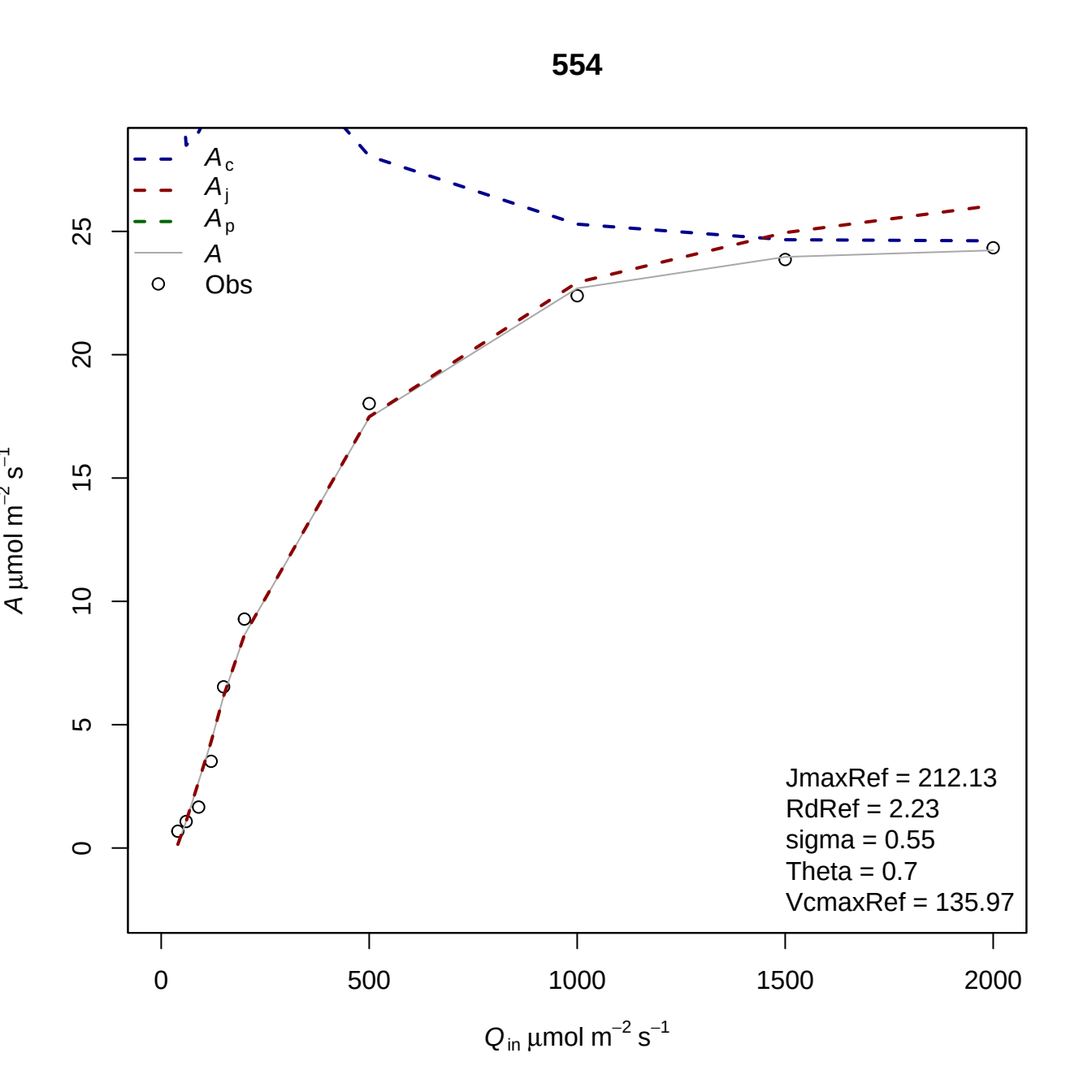

549
A_c
A_j
A_p
A
Obs
JmaxRef = 213.03
RdRef = 1.87
sigma = 0.2
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

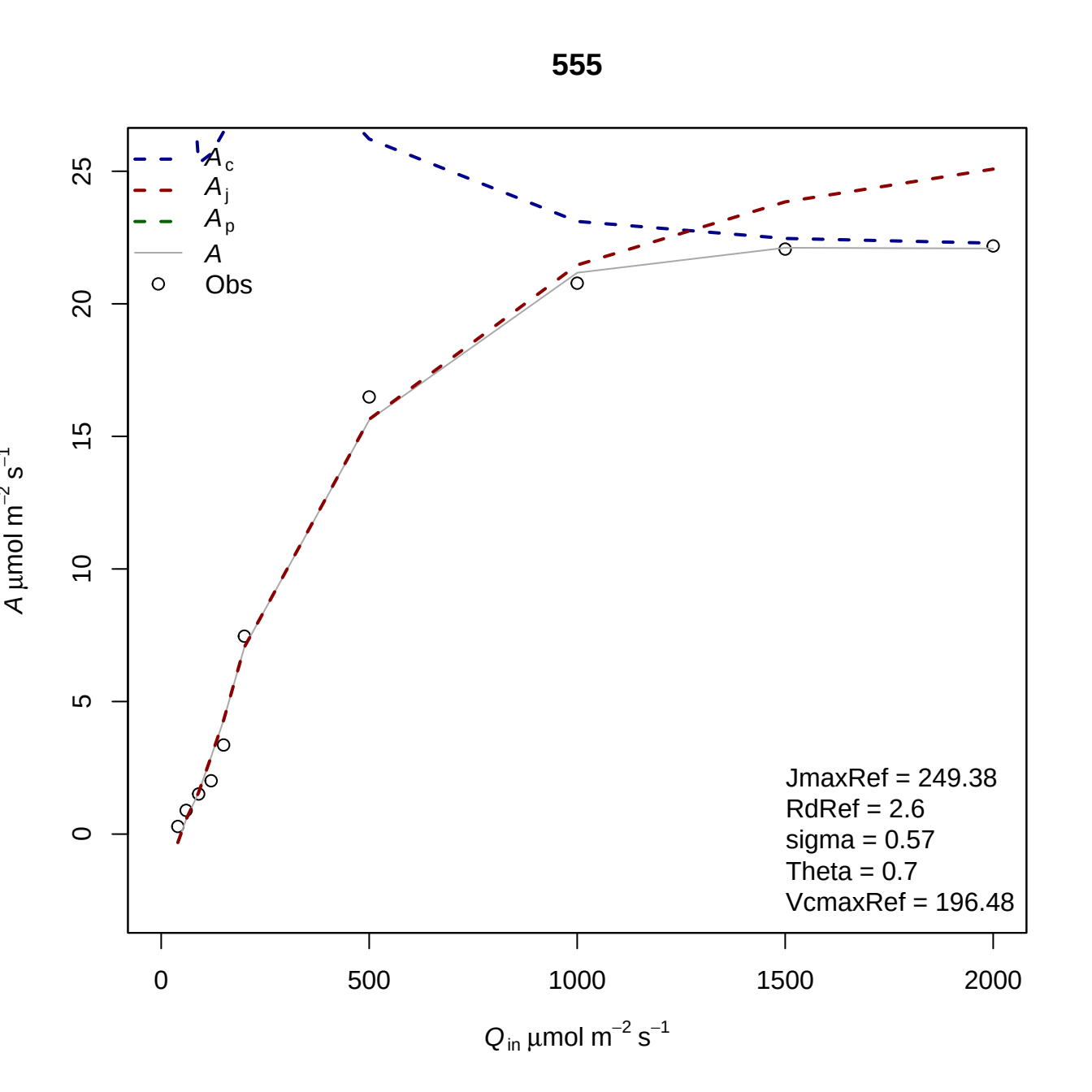

550
A_c
A_j
A_p
A
Obs
JmaxRef = 260.23
RdRef = 2.41
sigma = 0.15
Theta = 0.7
VcmaxRef = 216.63
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

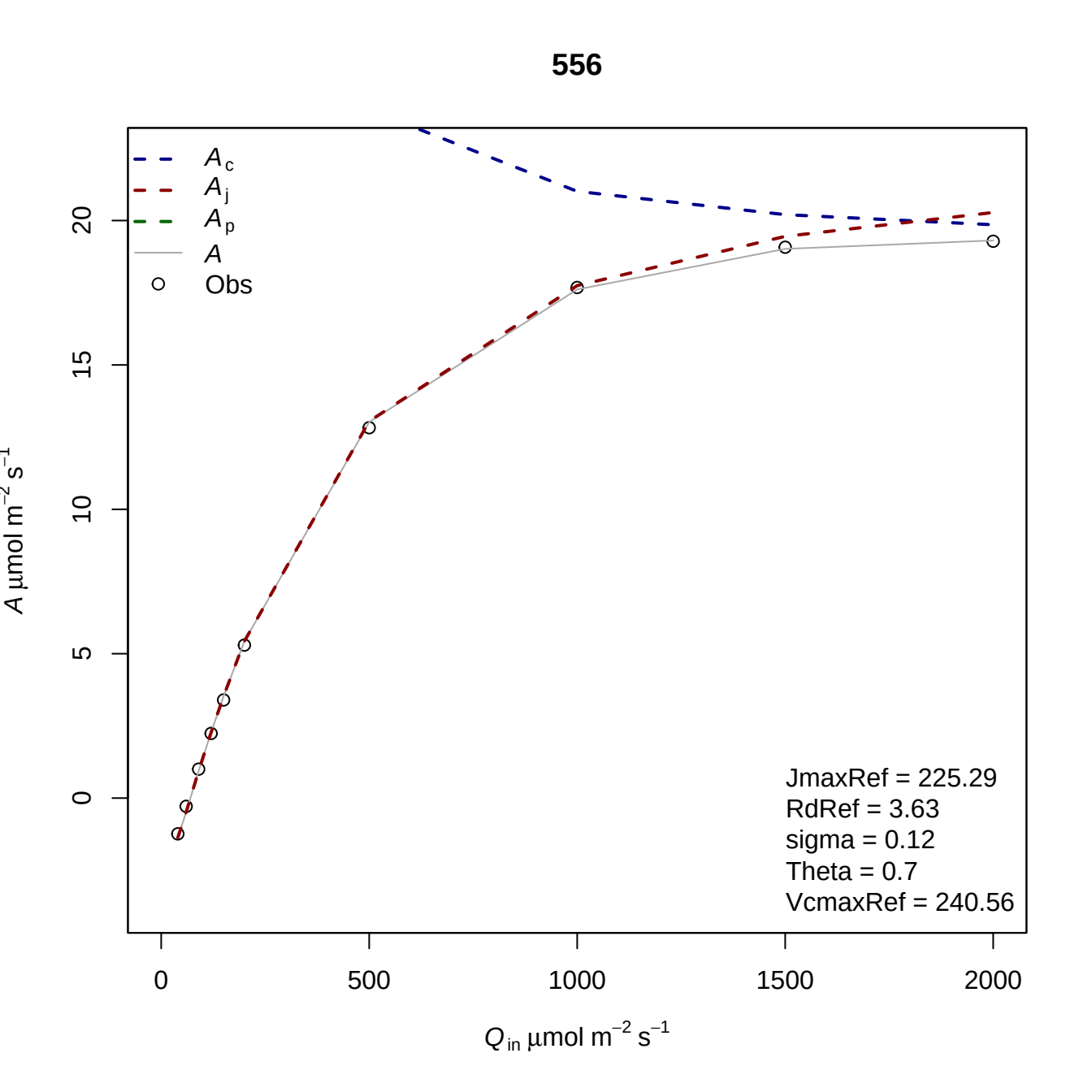

551
A_c
A_j
A_p
A
Obs
JmaxRef = 280.29
RdRef = 2.68
sigma = 0.06
Theta = 0.7
VcmaxRef = 291.09
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

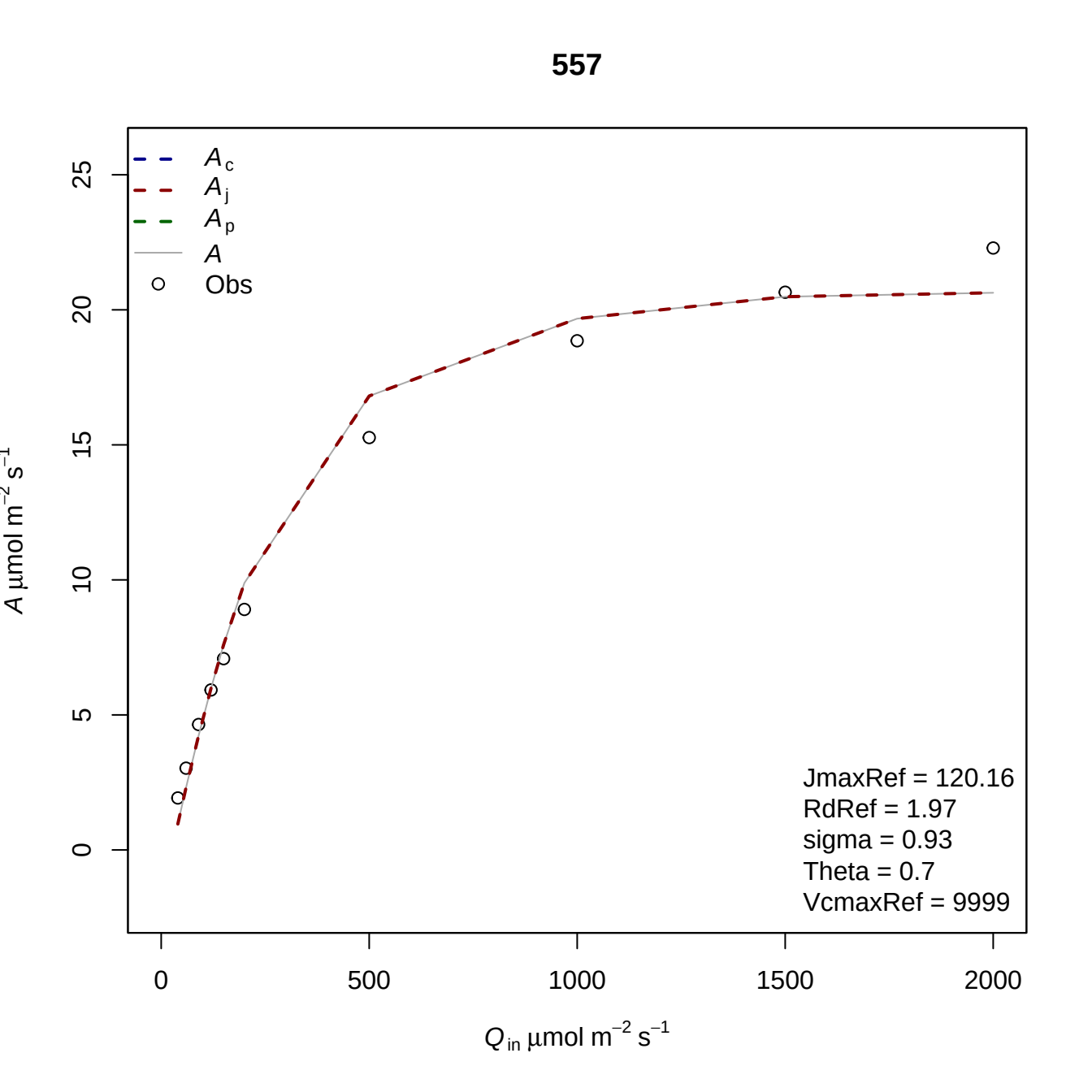

552
A_c
A_j
A_p
A
Obs
JmaxRef = 94.58
RdRef = 2.15
sigma = 1.7
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

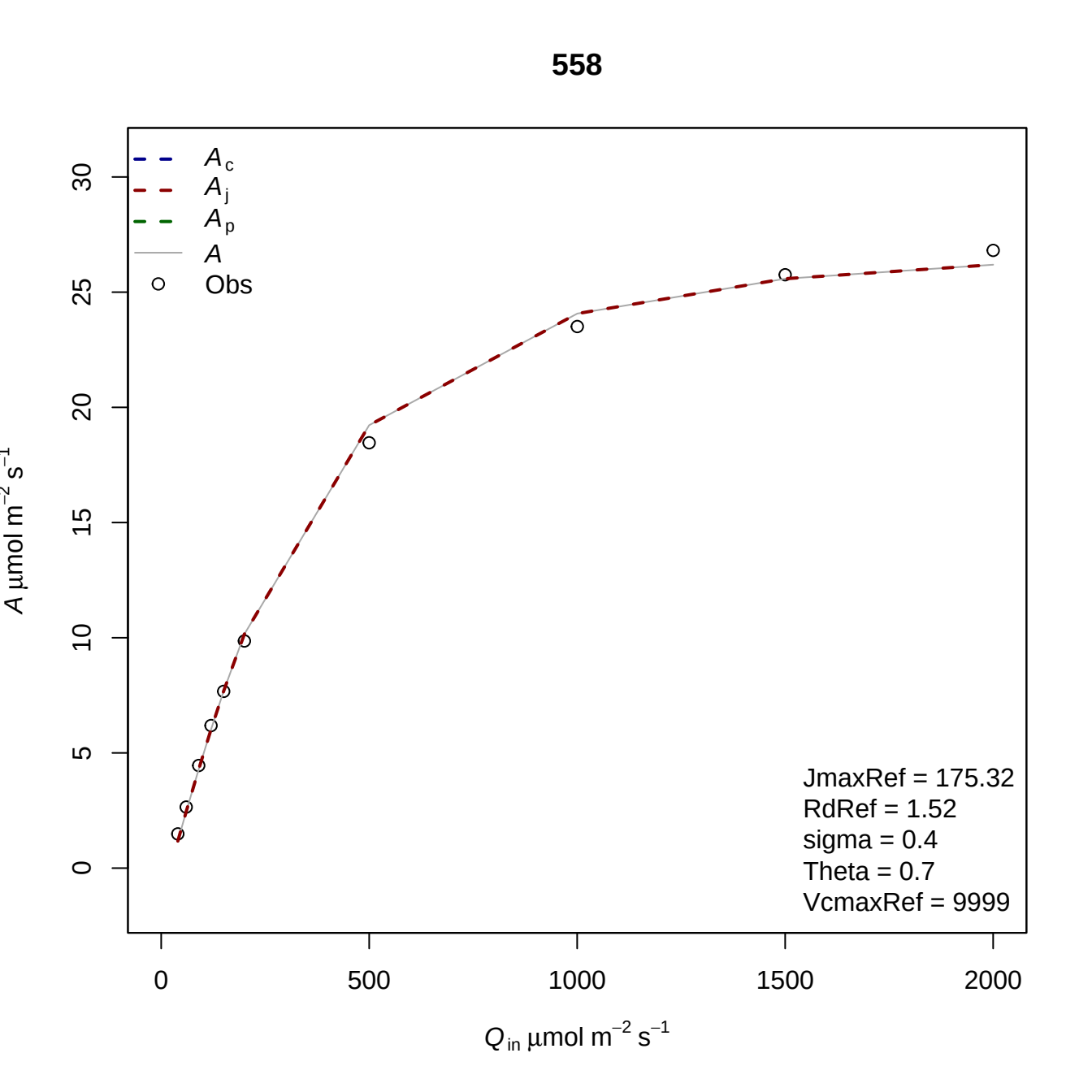

553
A_c
A_j
A_p
A
Obs
JmaxRef = 171.56
RdRef = 2.54
sigma = 0.43
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

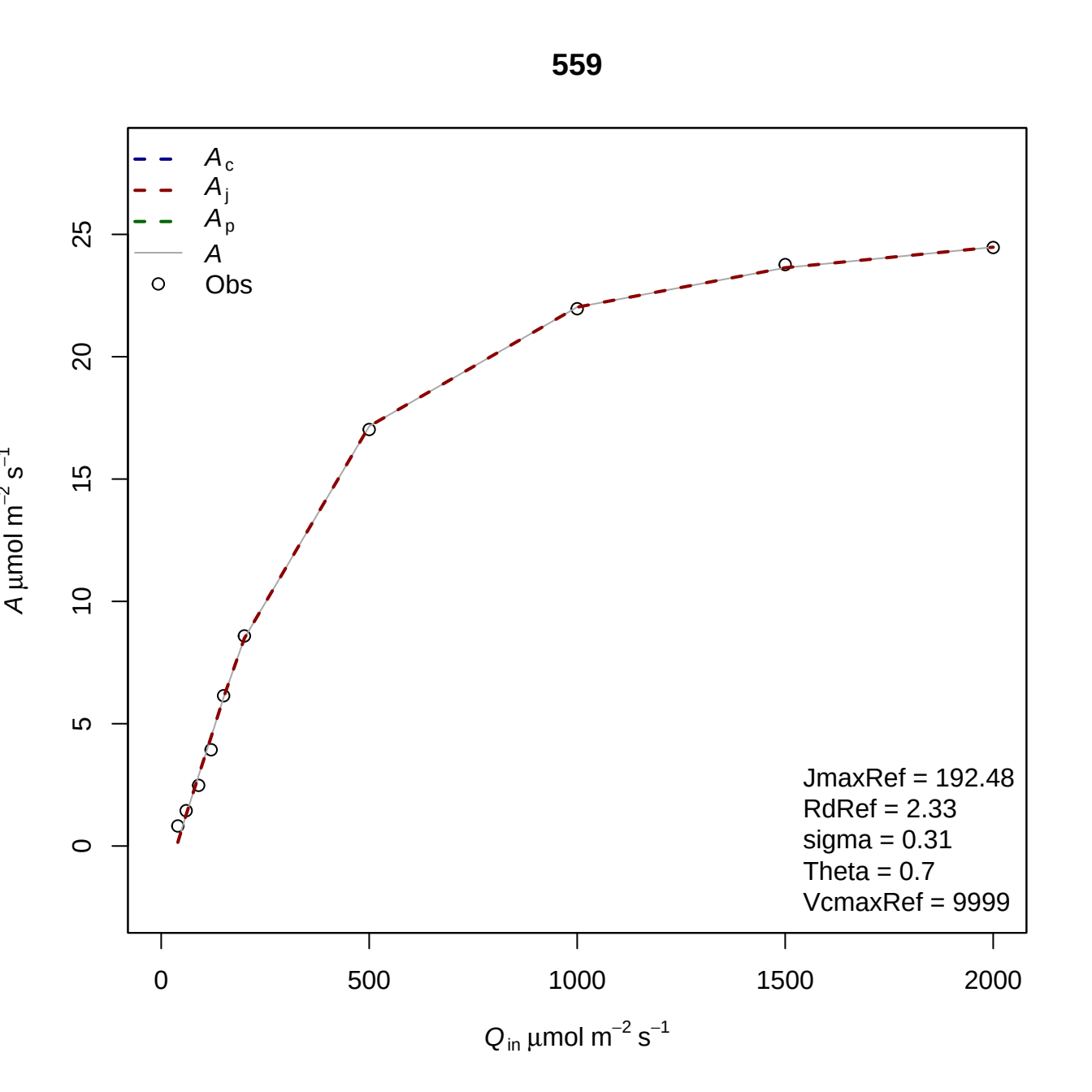

554
A_c
A_j
A_p
A
Obs
JmaxRef = 212.13
RdRef = 2.23
sigma = 0.55
Theta = 0.7
VcmaxRef = 135.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

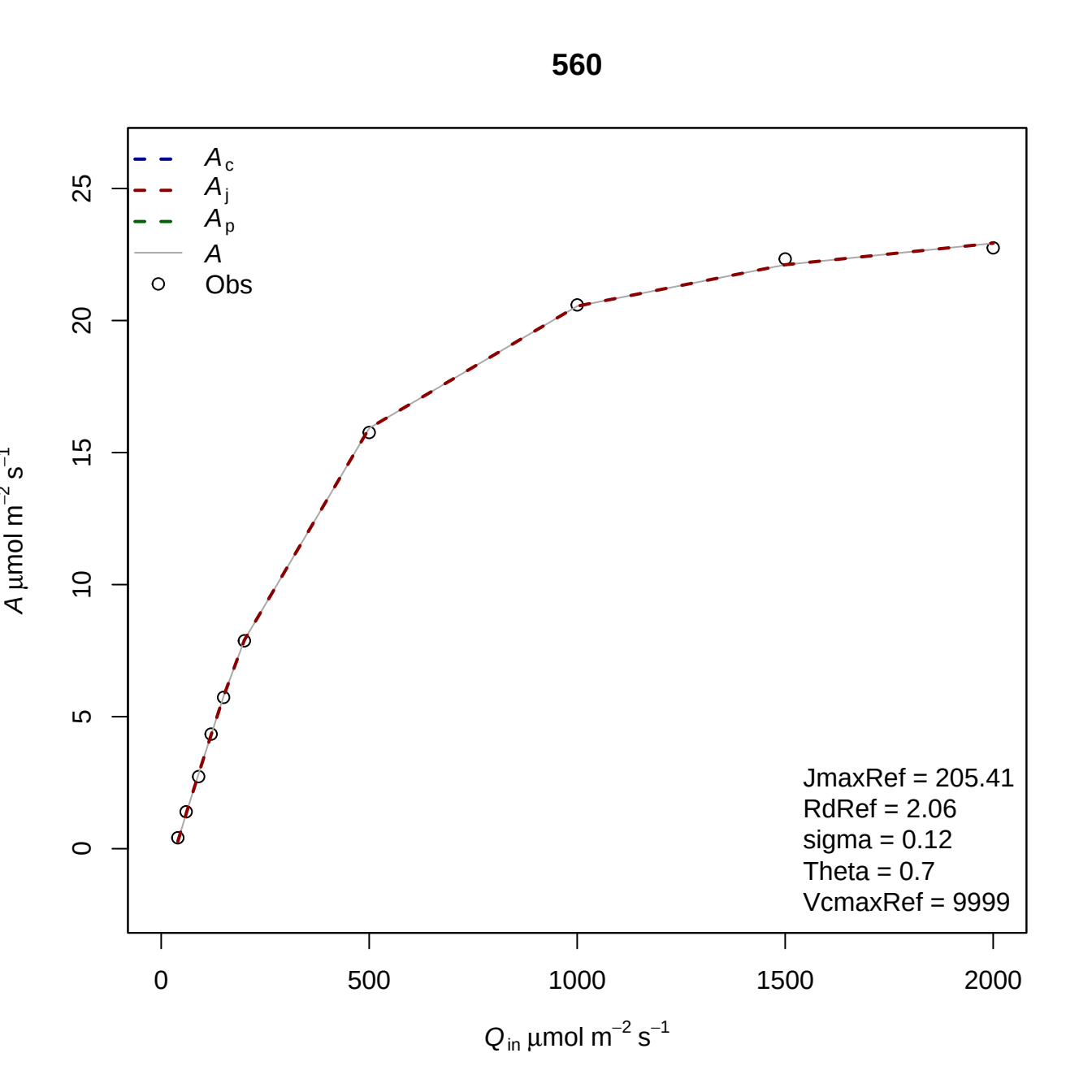

555
A_c
A_j
A_p
A
Obs
JmaxRef = 249.38
RdRef = 2.6
sigma = 0.57
Theta = 0.7
VcmaxRef = 196.48
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

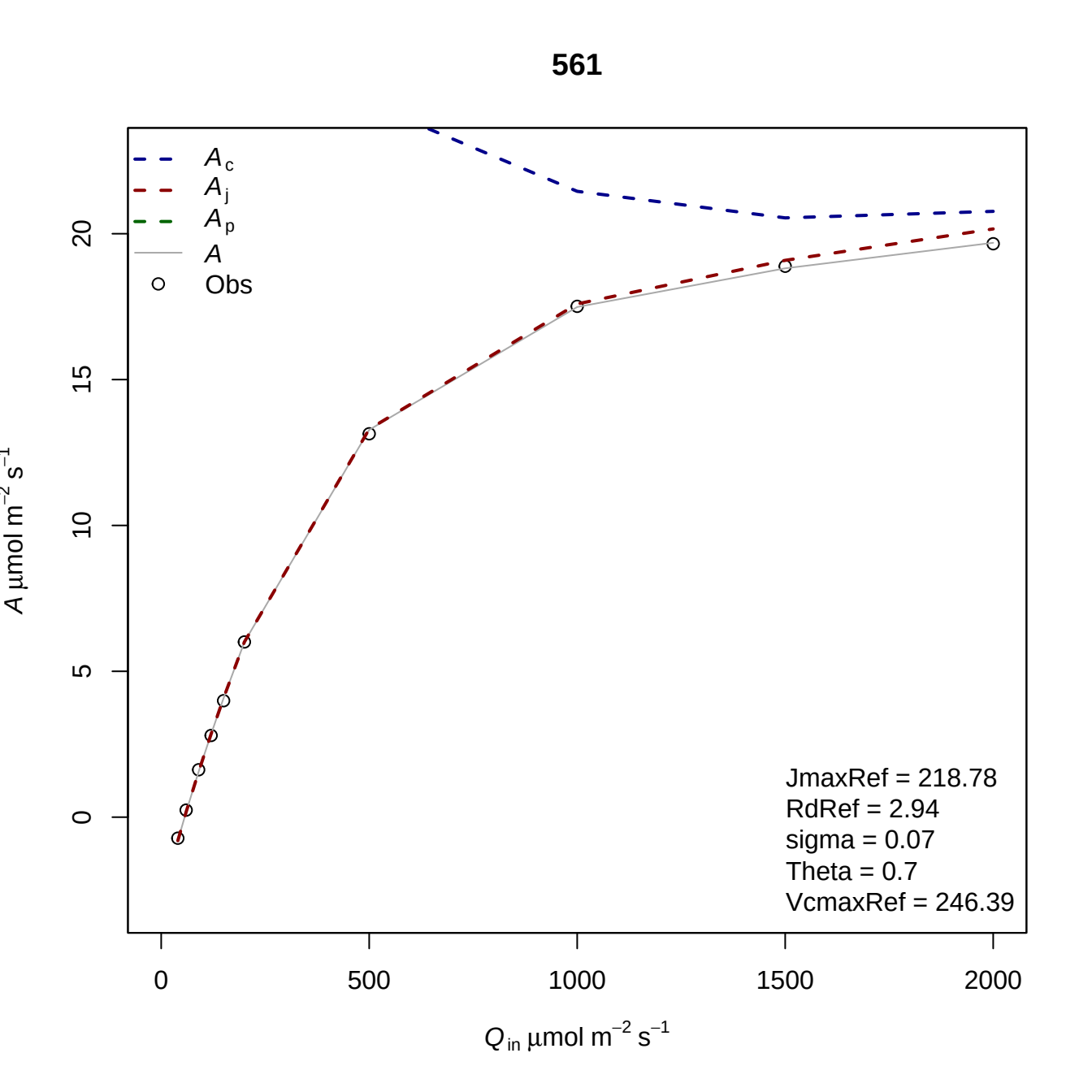

556
A_c
A_j
A_p
A
Obs
JmaxRef = 225.29
RdRef = 3.63
sigma = 0.12
Theta = 0.7
VcmaxRef = 240.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

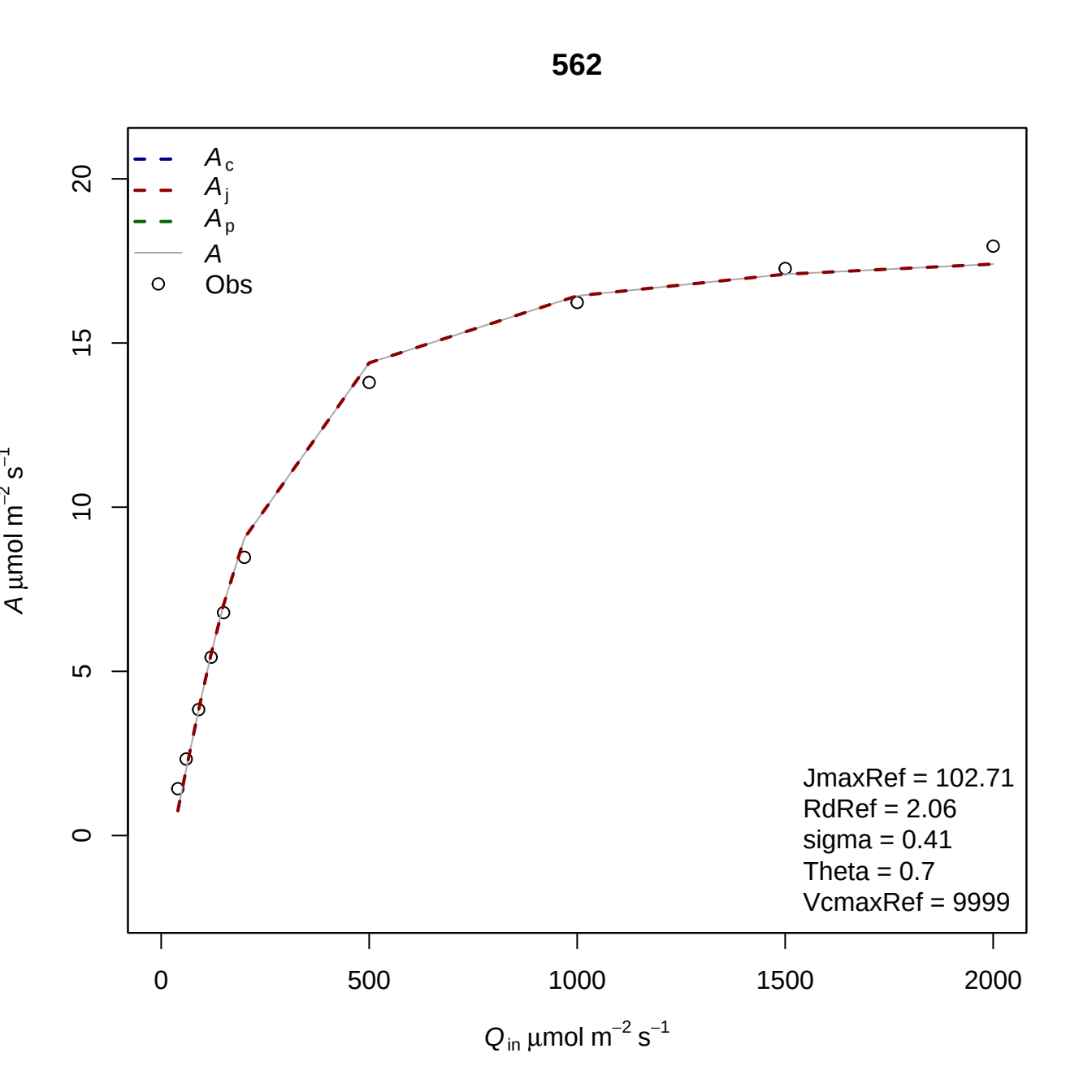

557
A_c
A_j
A_p
A
Obs
JmaxRef = 120.16
RdRef = 1.97
sigma = 0.93
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

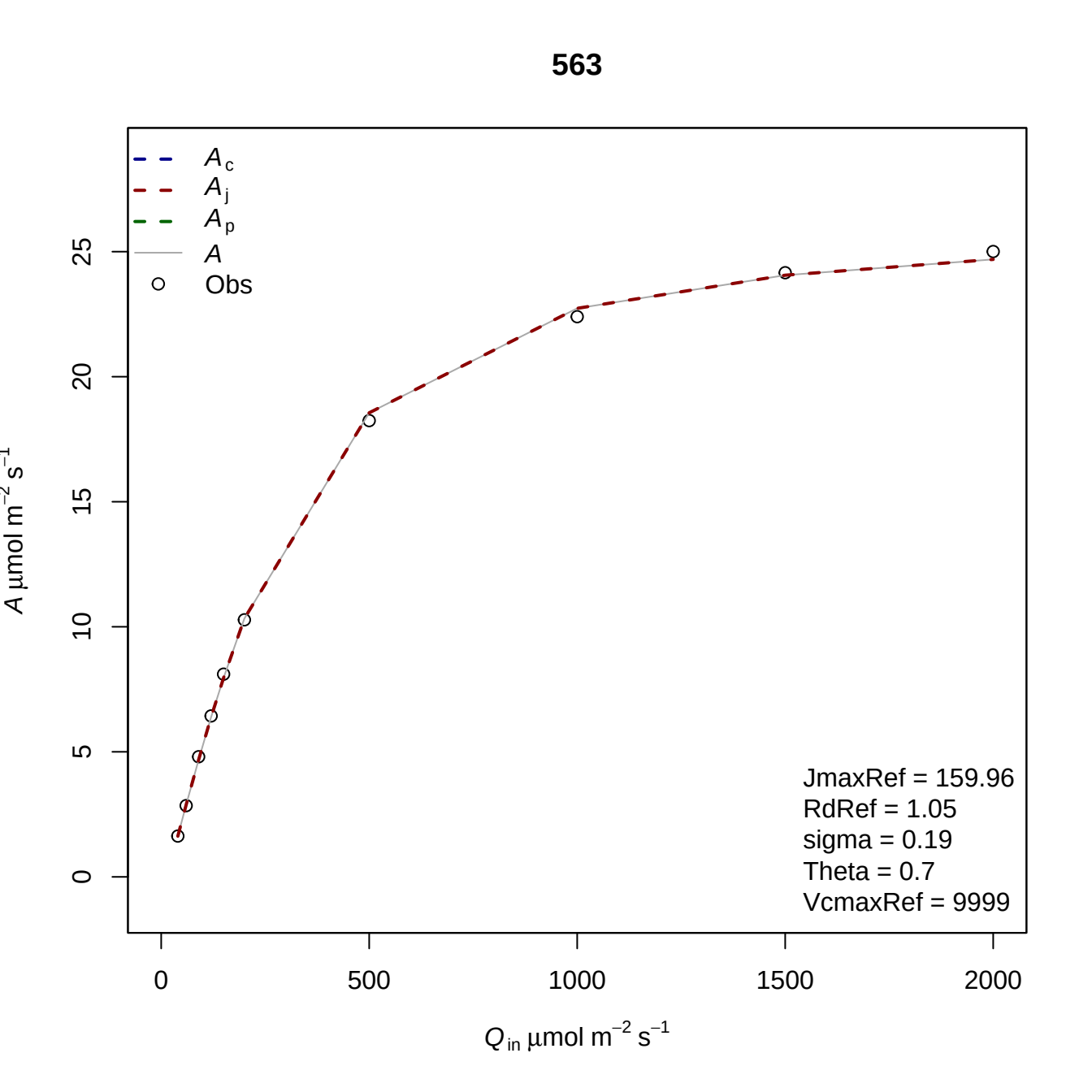

558
A_c
A_j
A_p
A
Obs
JmaxRef = 175.32
RdRef = 1.52
sigma = 0.4
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

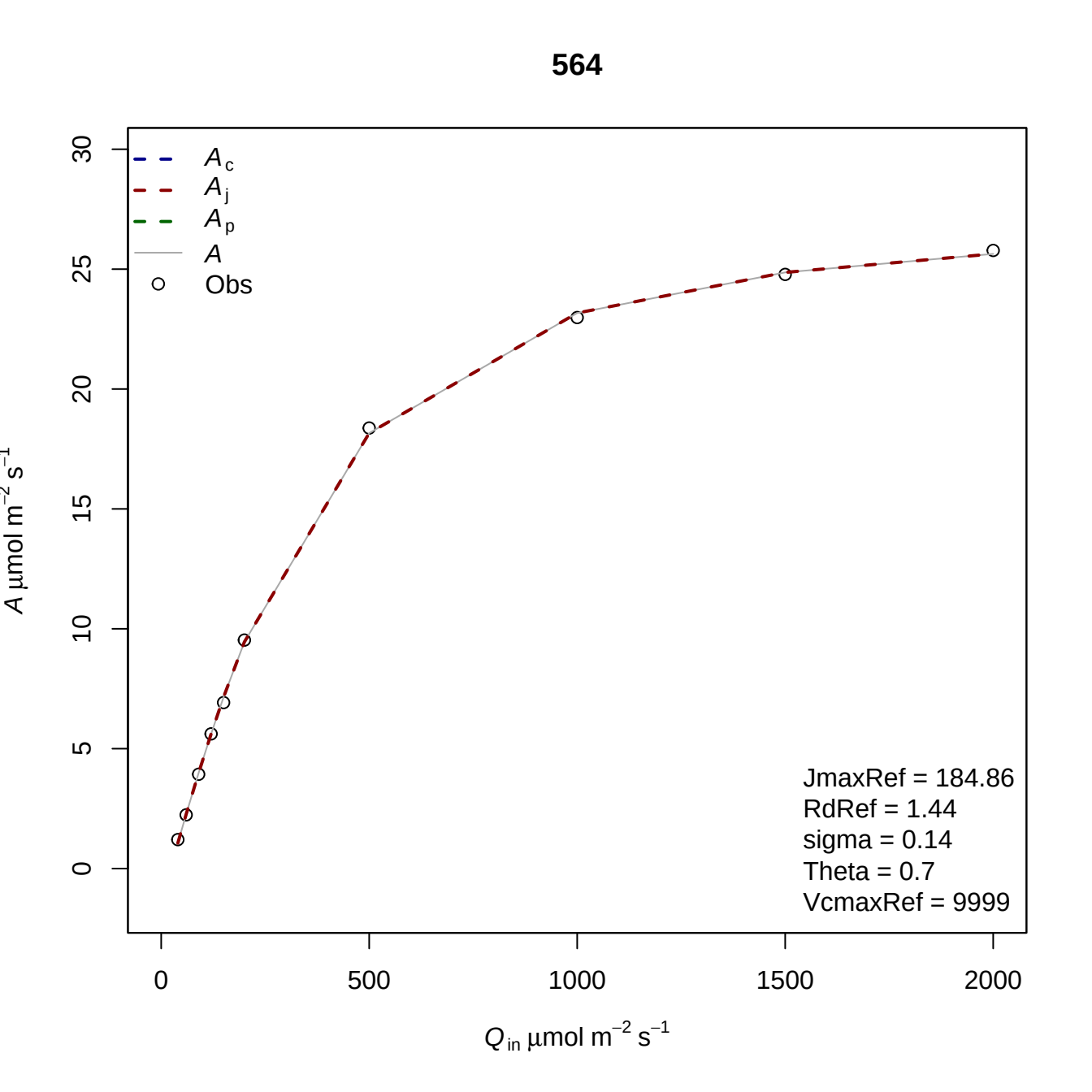

559
A_c
A_j
A_p
A
Obs
JmaxRef = 192.48
RdRef = 2.33
sigma = 0.31
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

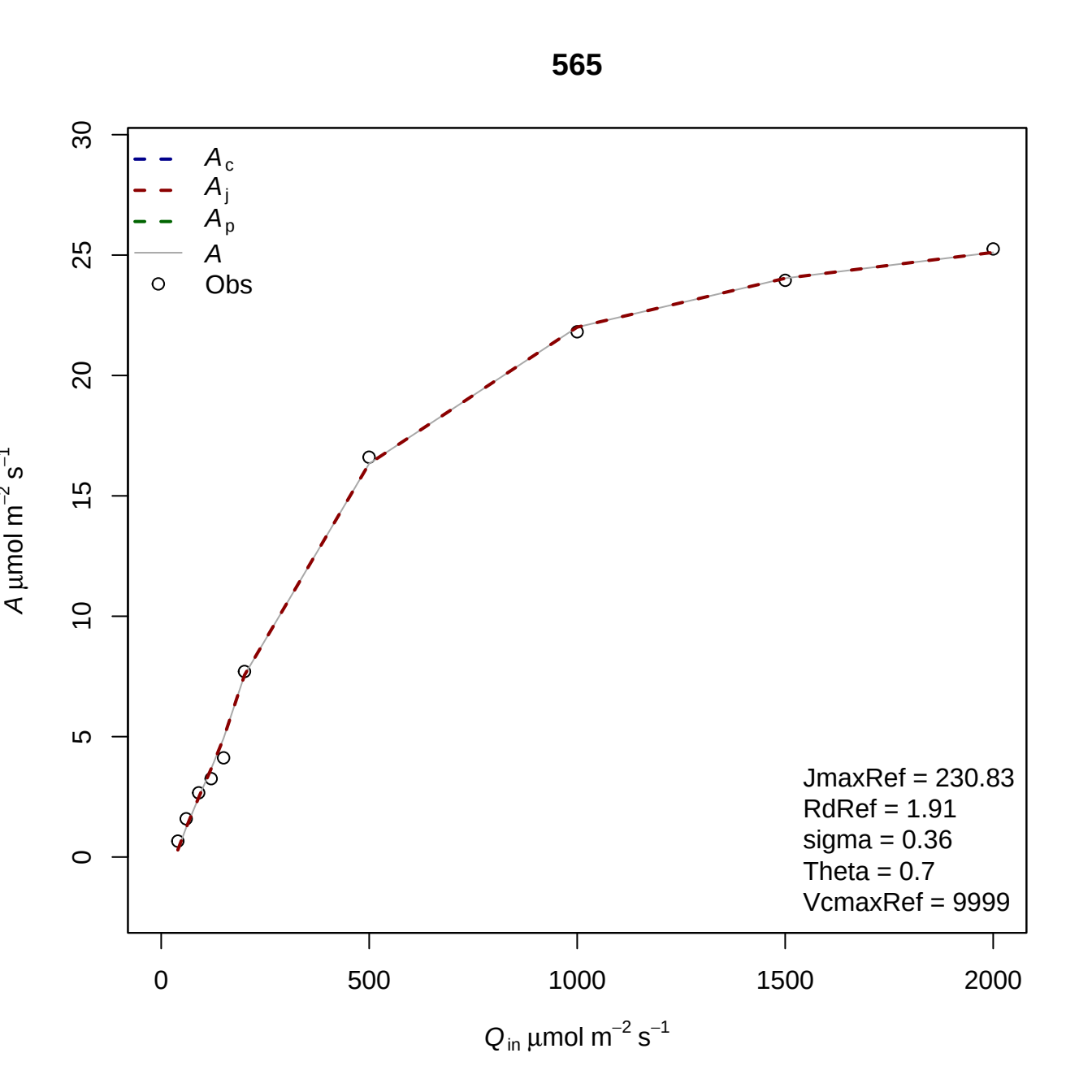

560
A_c
A_j
A_p
A
Obs
JmaxRef = 205.41
RdRef = 2.06
sigma = 0.12
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

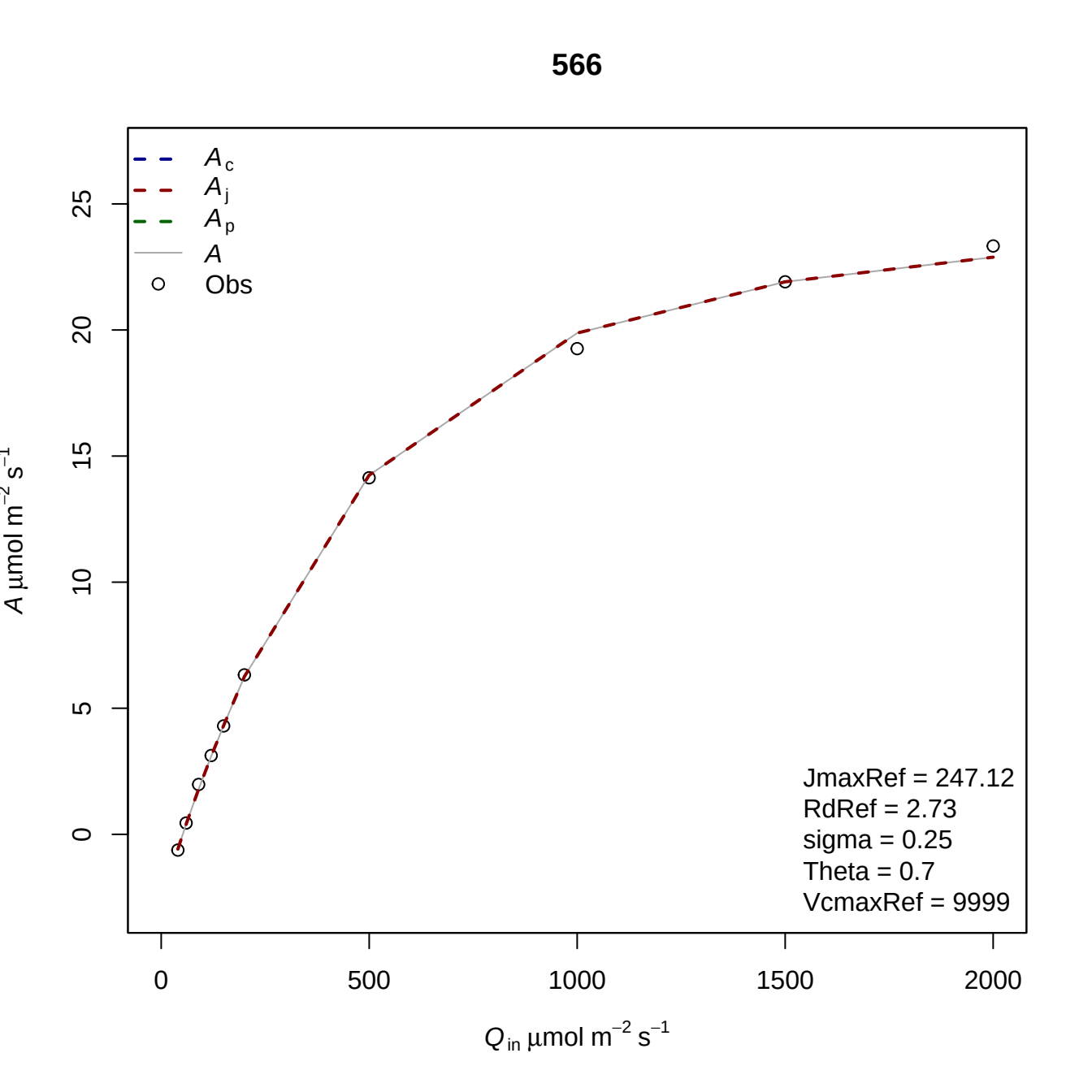

561
A_c
A_j
A_p
A
Obs
JmaxRef = 218.78
RdRef = 2.94
sigma = 0.07
Theta = 0.7
VcmaxRef = 246.39
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

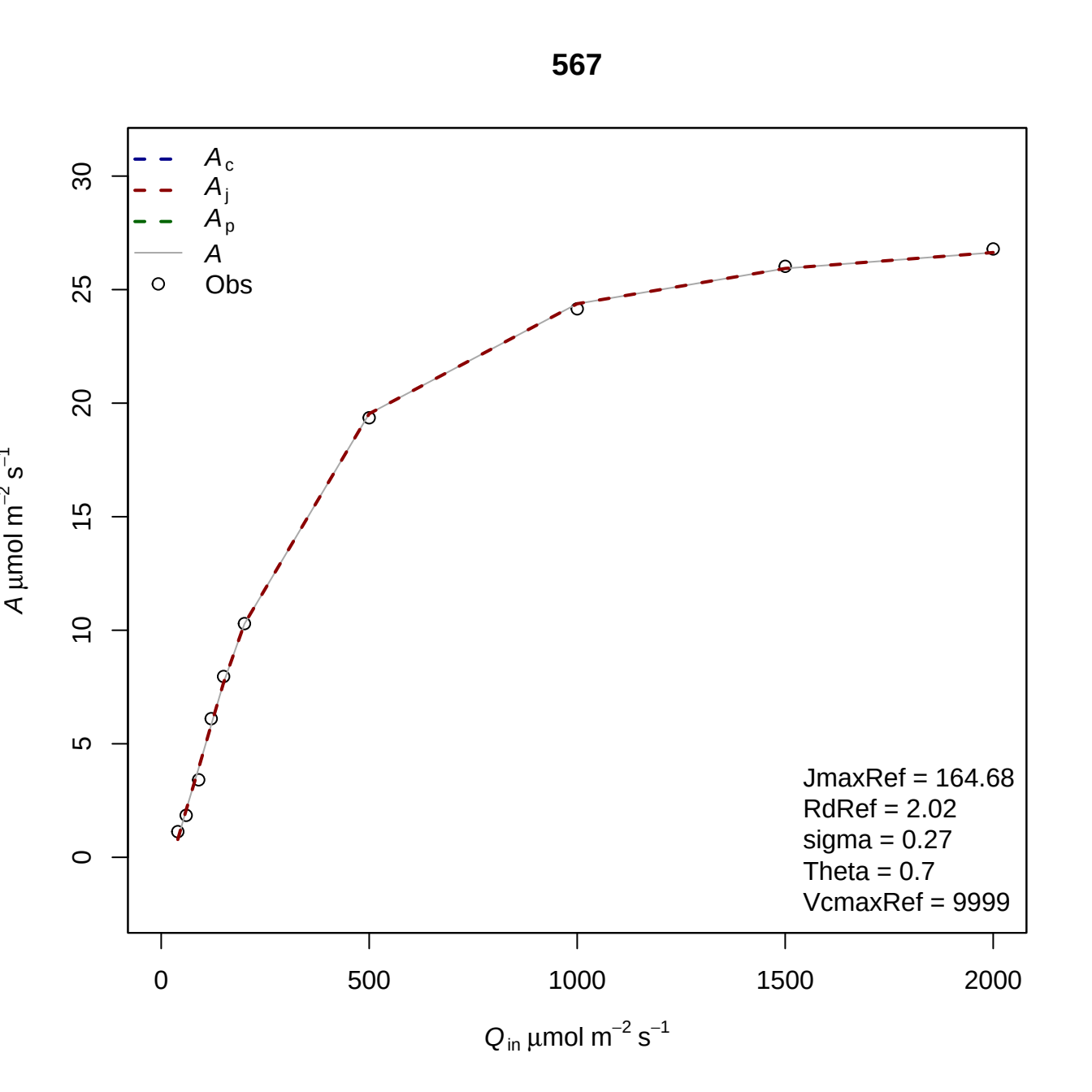

562
A_c
A_j
A_p
A
Obs
JmaxRef = 102.71
RdRef = 2.06
sigma = 0.41
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

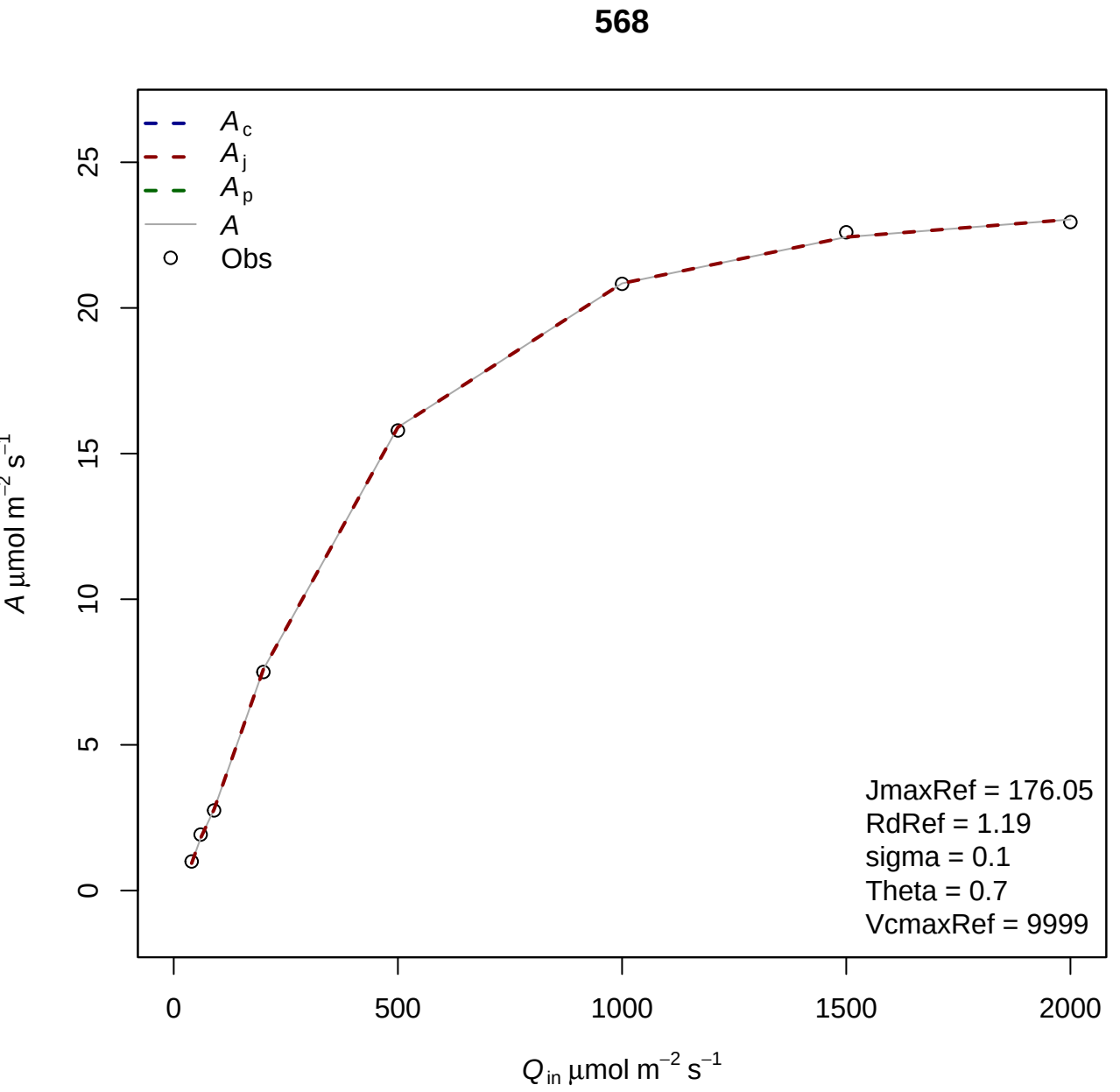

563
A_c
A_j
A_p
A
Obs
JmaxRef = 159.96
RdRef = 1.05
sigma = 0.19
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

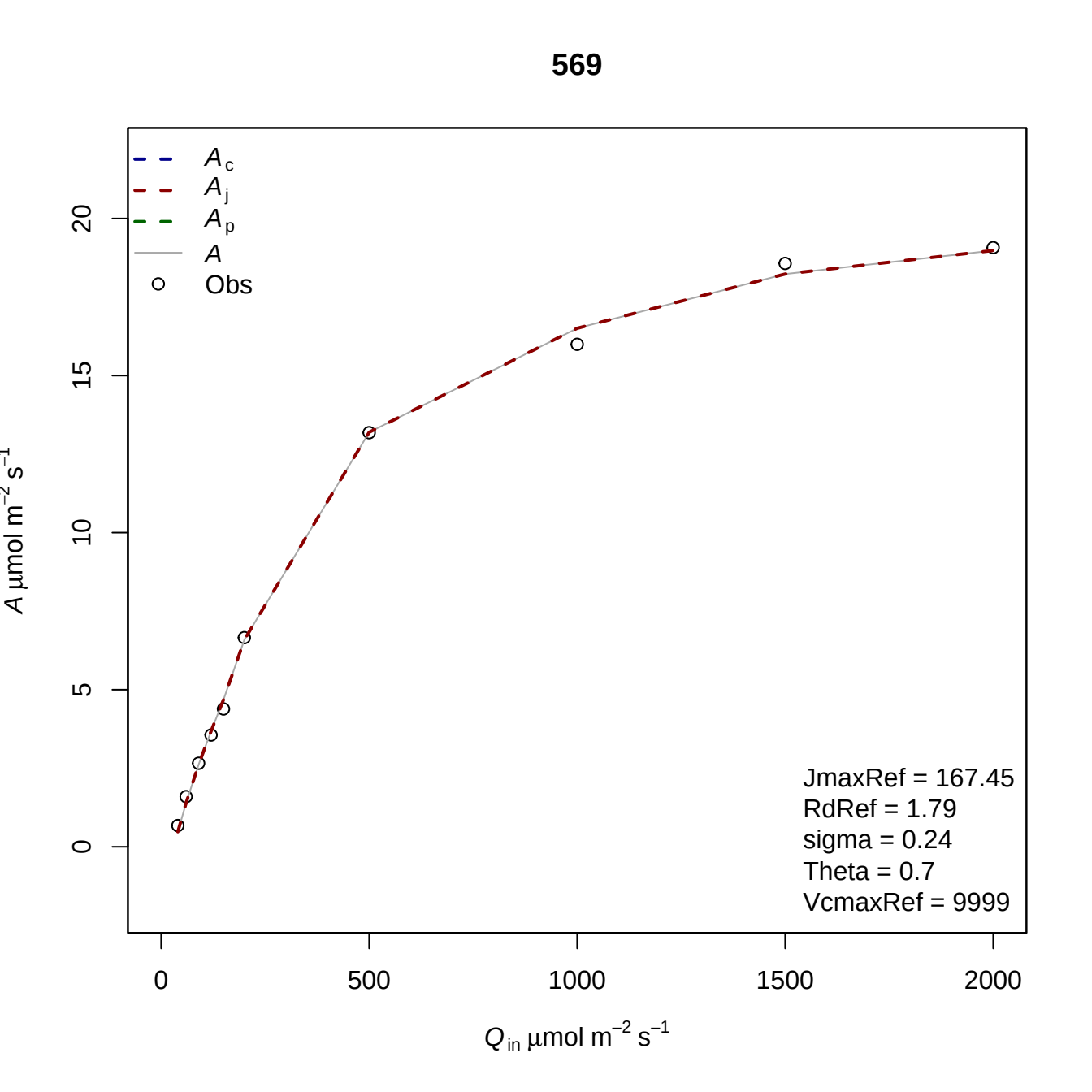

564
A_c
A_j
A_p
A
Obs
JmaxRef = 184.86
RdRef = 1.44
sigma = 0.14
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

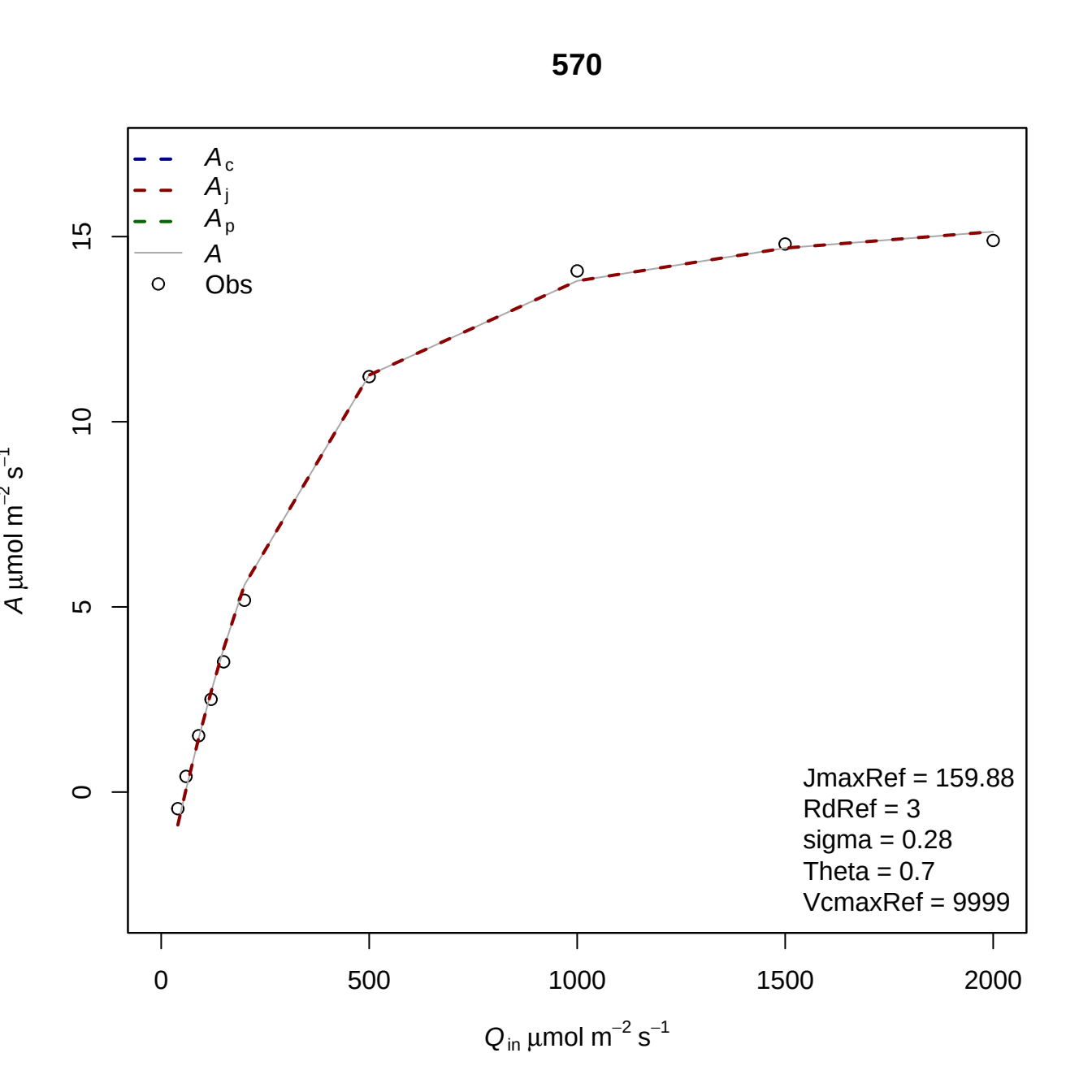

565
A_c
A_j
A_p
A
Obs
JmaxRef = 230.83
RdRef = 1.91
sigma = 0.36
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

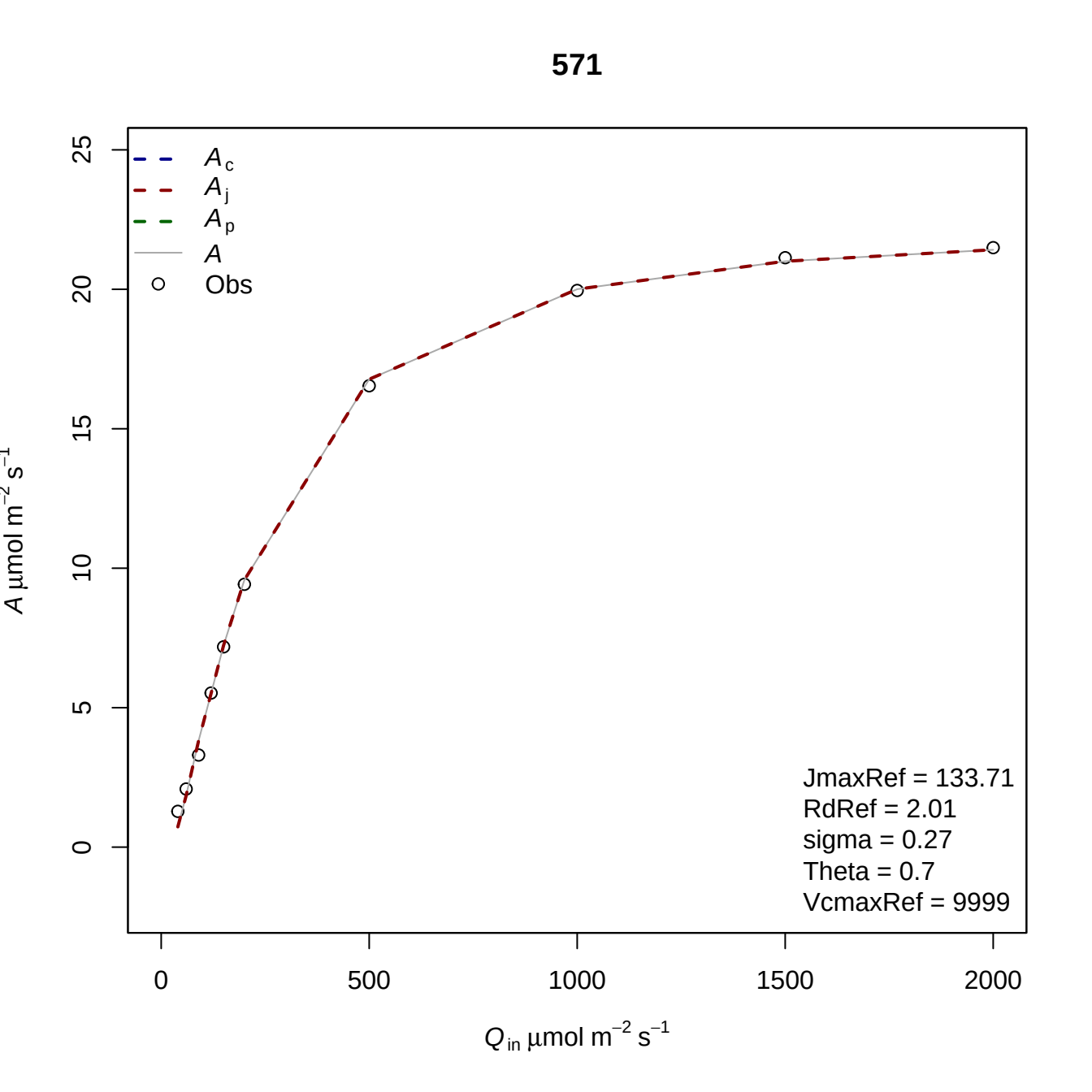

566
A_c
A_j
A_p
A
Obs
JmaxRef = 247.12
RdRef = 2.73
sigma = 0.25
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

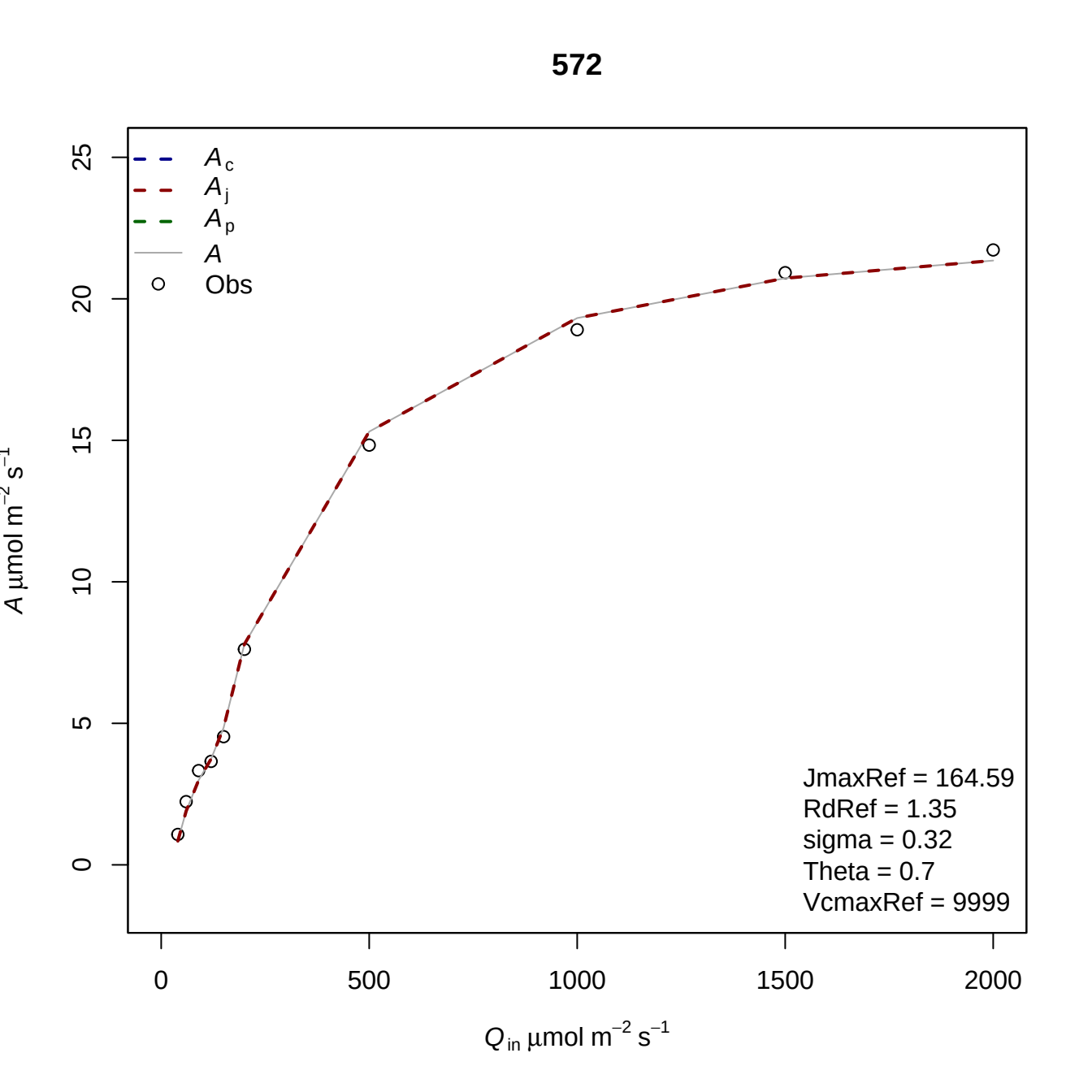

567
A_c
A_j
A_p
A
Obs
JmaxRef = 164.68
RdRef = 2.02
sigma = 0.27
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

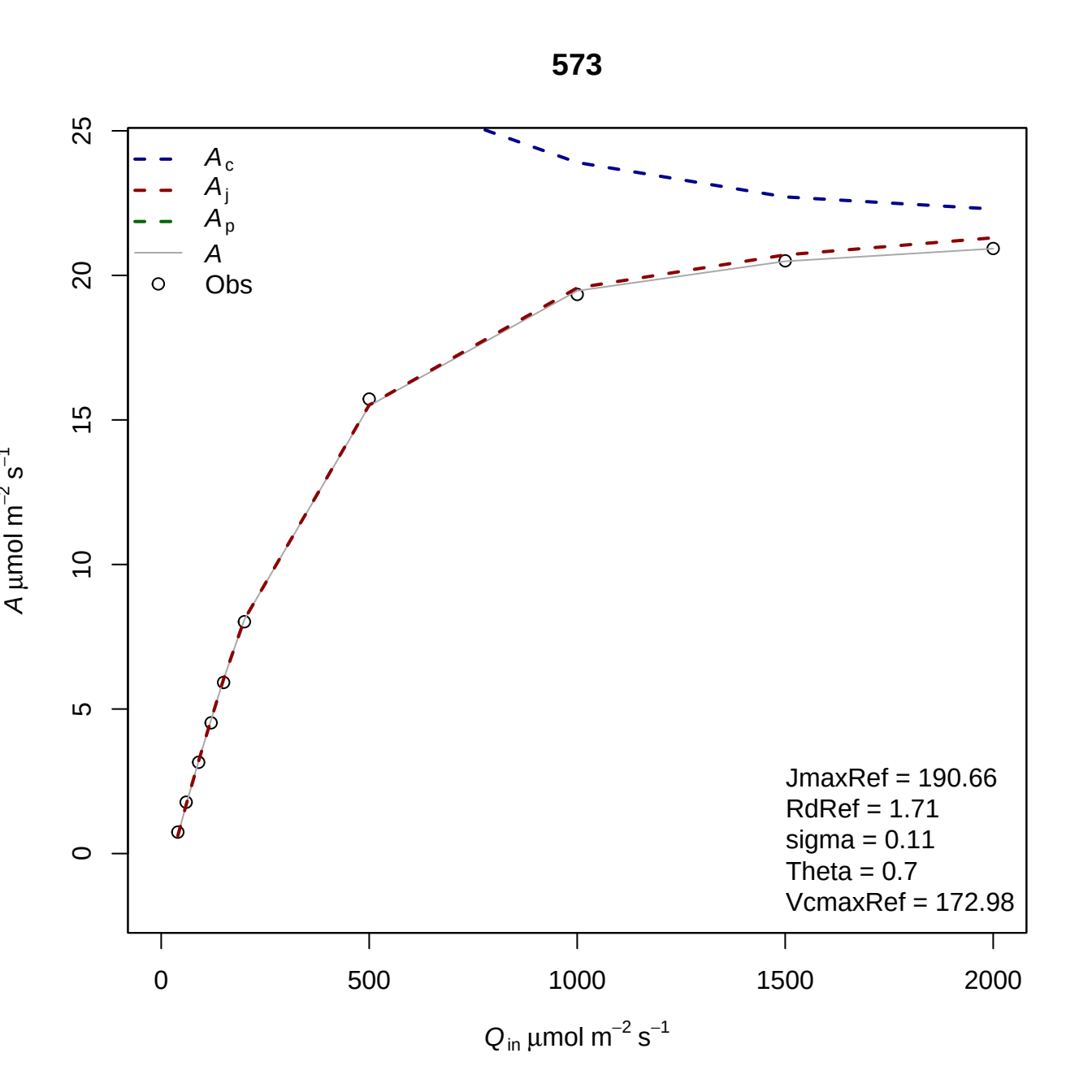

568
A_c
A_j
A_p
A
Obs
JmaxRef = 176.05
RdRef = 1.19
sigma = 0.1
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

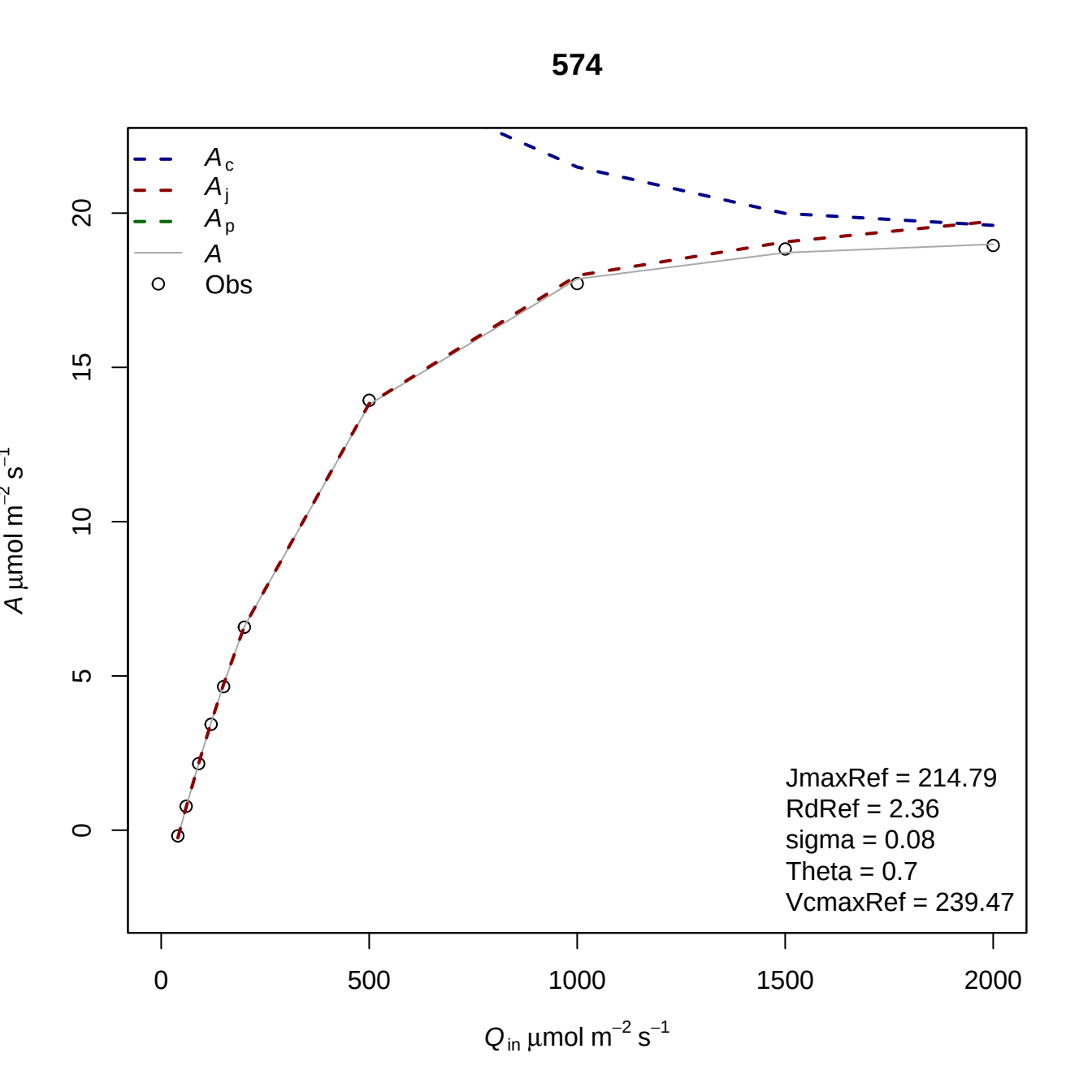

569
A_c
A_j
A_p
A
Obs
JmaxRef = 167.45
RdRef = 1.79
sigma = 0.24
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

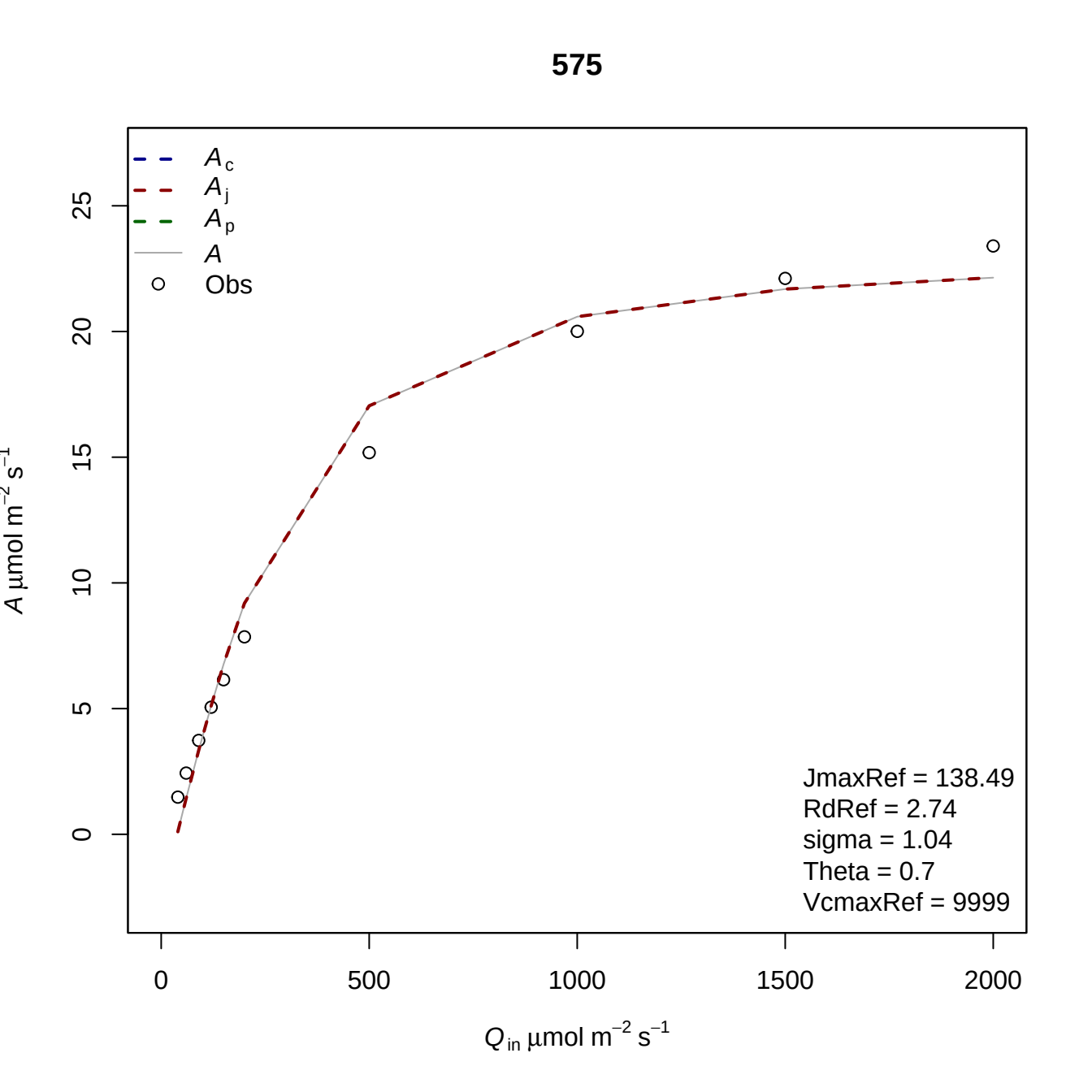

570
A_c
A_j
A_p
A
Obs
JmaxRef = 159.88
RdRef = 3
sigma = 0.28
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

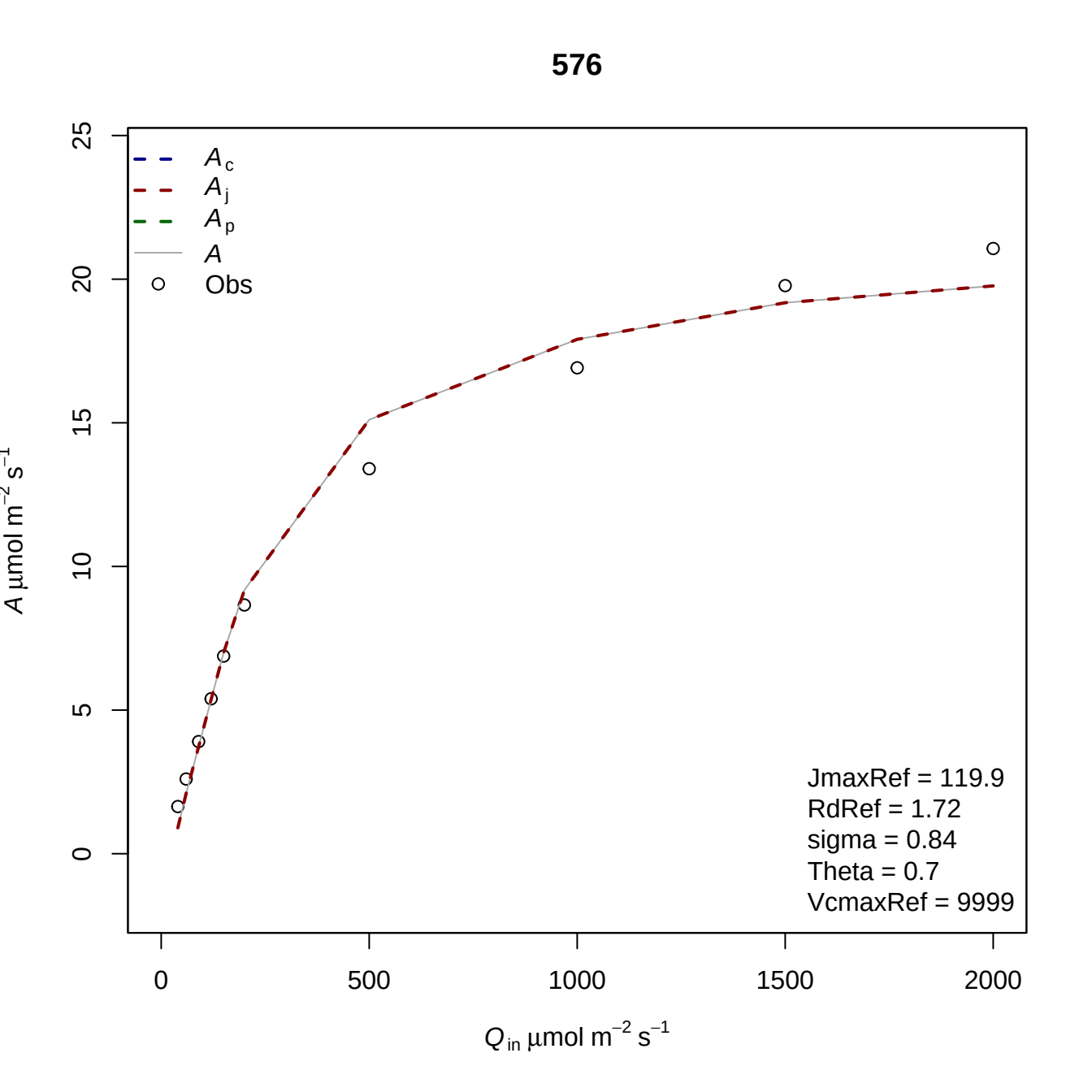

571
A_c
A_j
A_p
A
Obs
JmaxRef = 133.71
RdRef = 2.01
sigma = 0.27
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

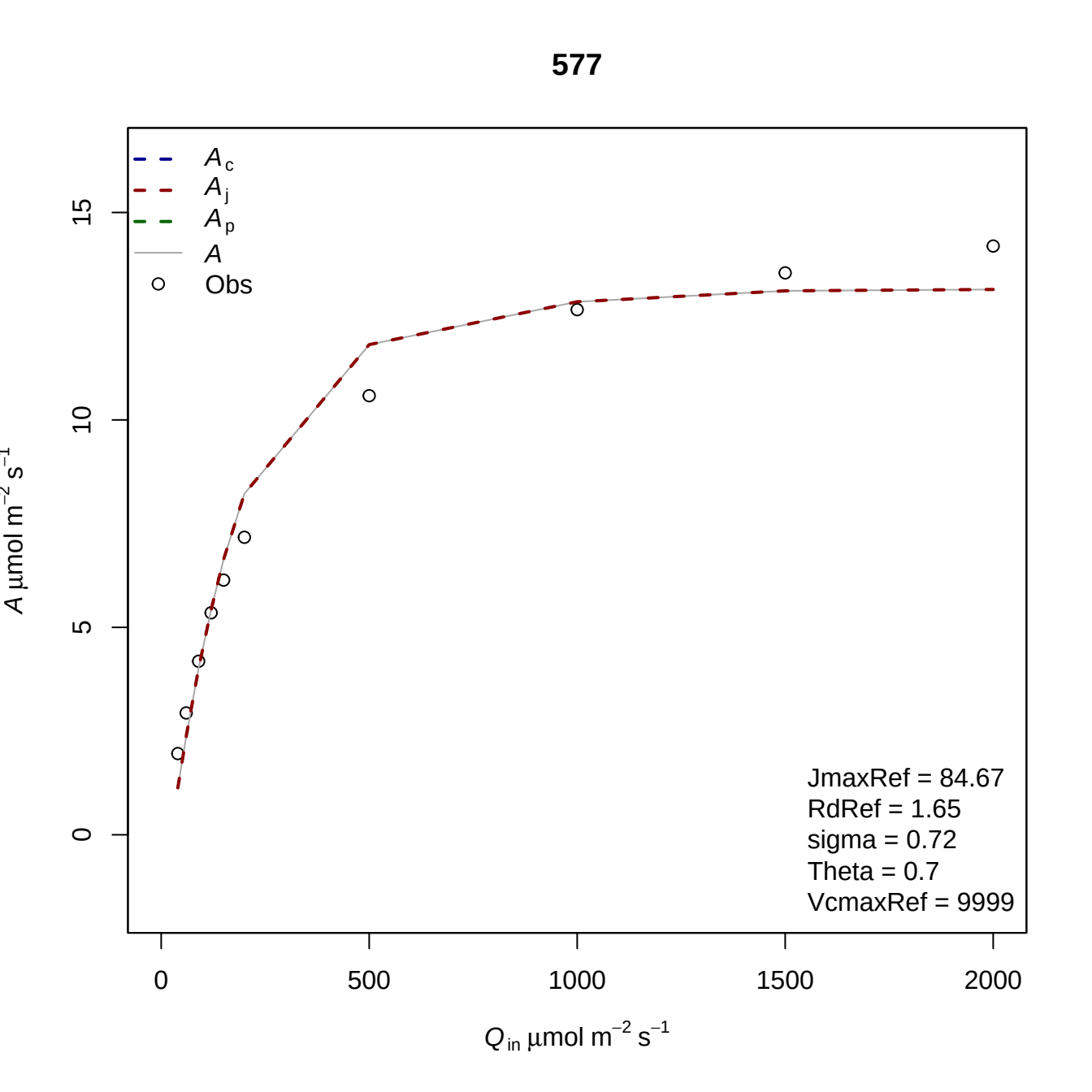

572
A_c
A_j
A_p
A
Obs
JmaxRef = 164.59
RdRef = 1.35
sigma = 0.32
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

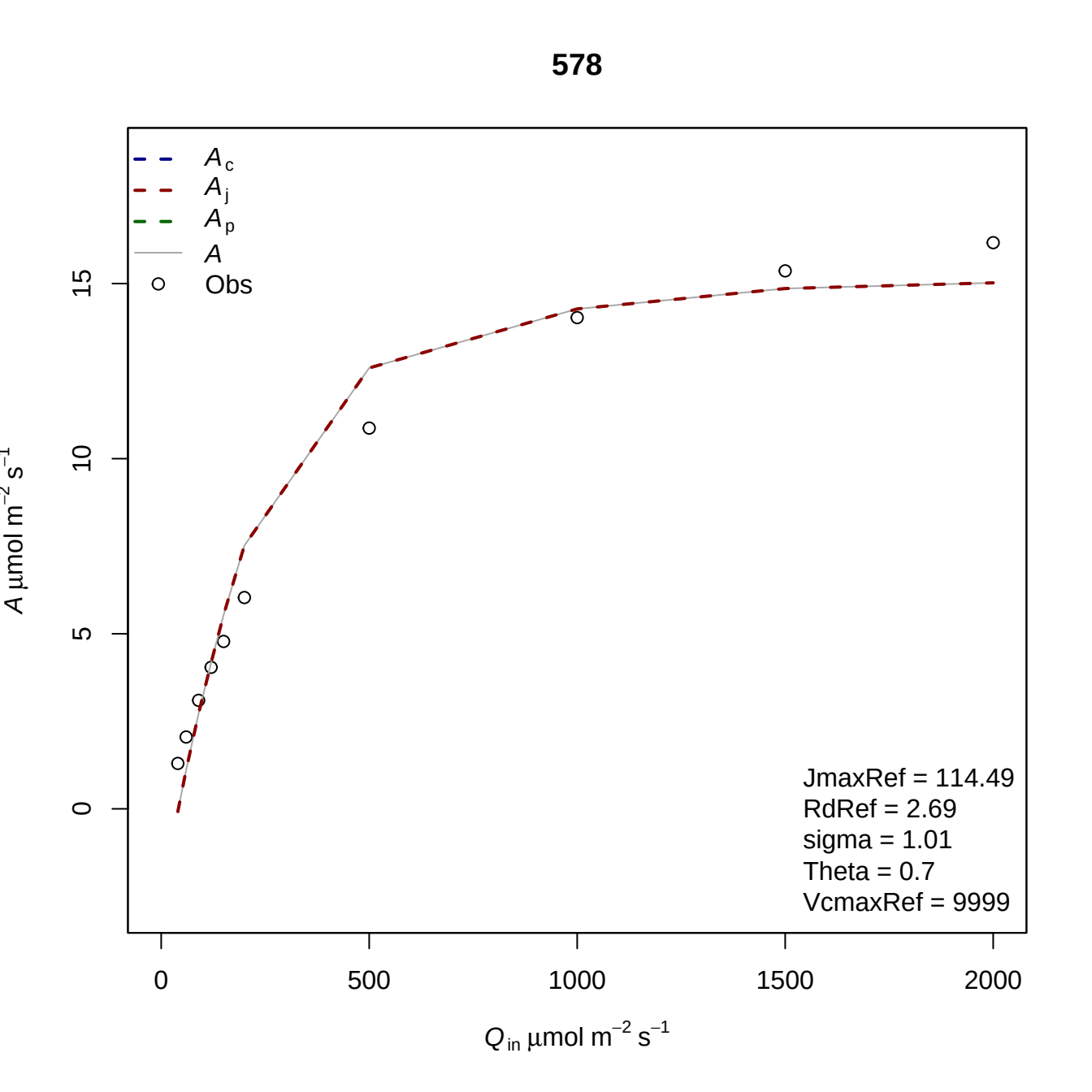

573
A_c
A_j
A_p
A
Obs
JmaxRef = 190.66
RdRef = 1.71
sigma = 0.11
Theta = 0.7
VcmaxRef = 172.98
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

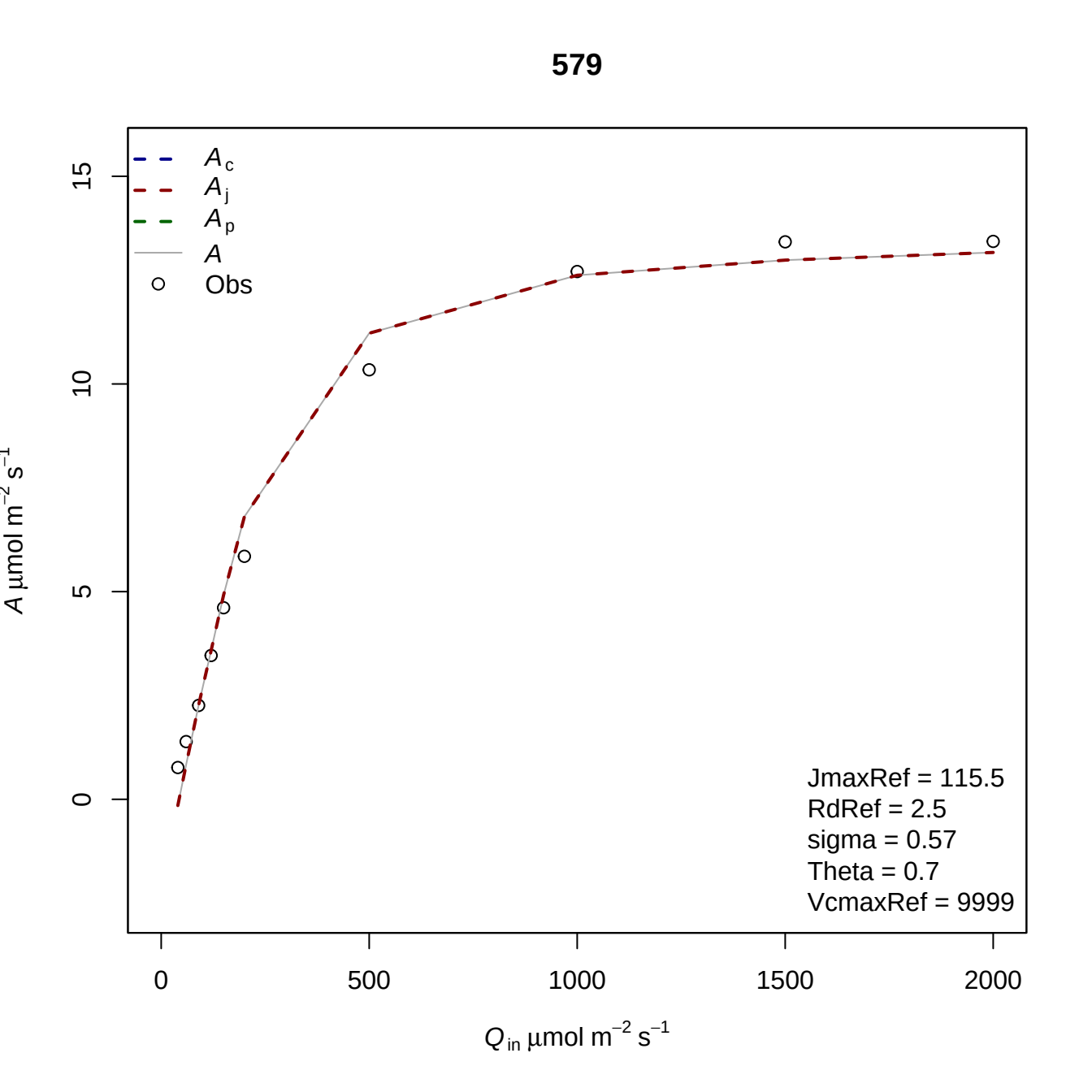

574
A_c
A_j
A_p
A
Obs
JmaxRef = 214.79
RdRef = 2.36
sigma = 0.08
Theta = 0.7
VcmaxRef = 239.47
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

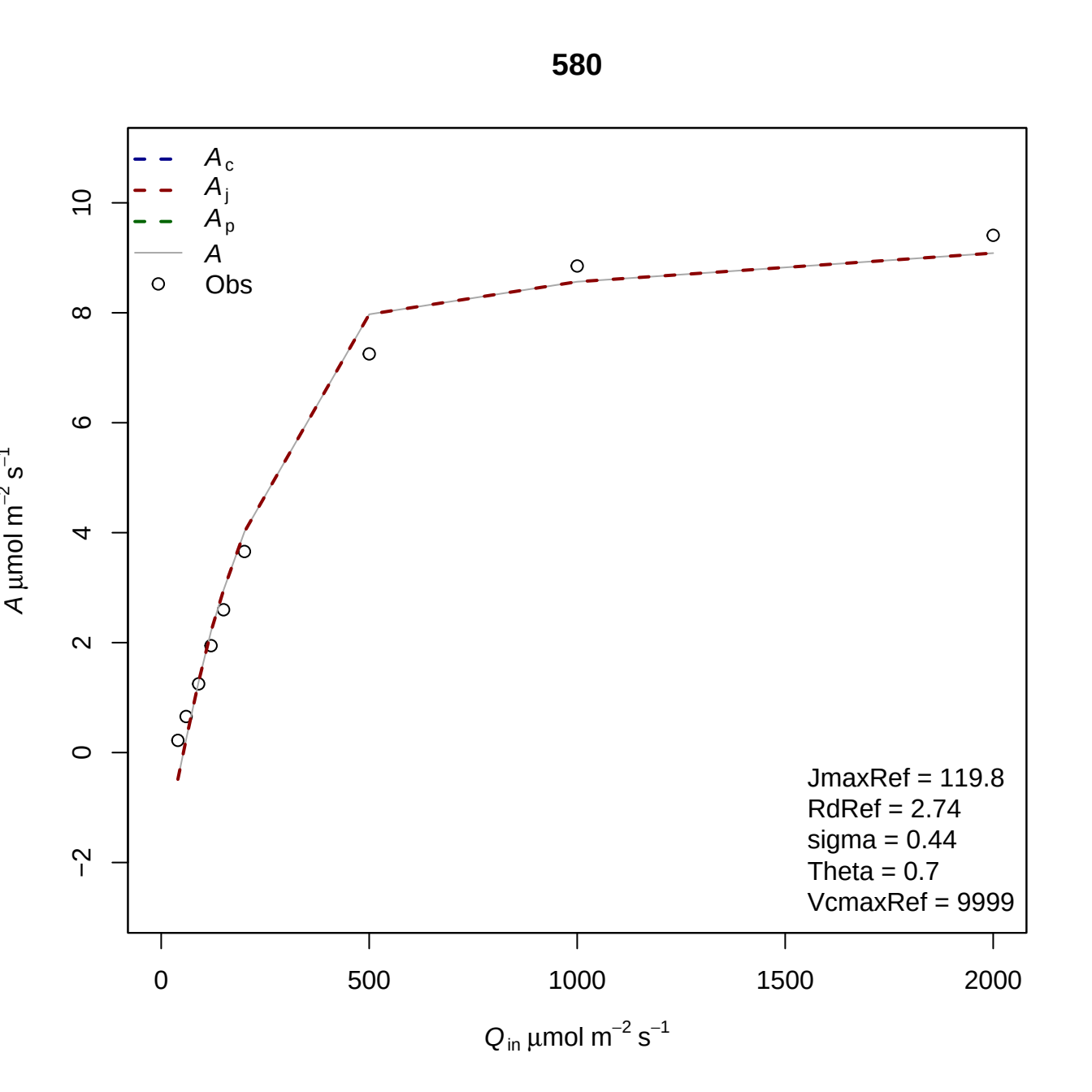

575
A_c
A_j
A_p
A
Obs
JmaxRef = 138.49
RdRef = 2.74
sigma = 1.04
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

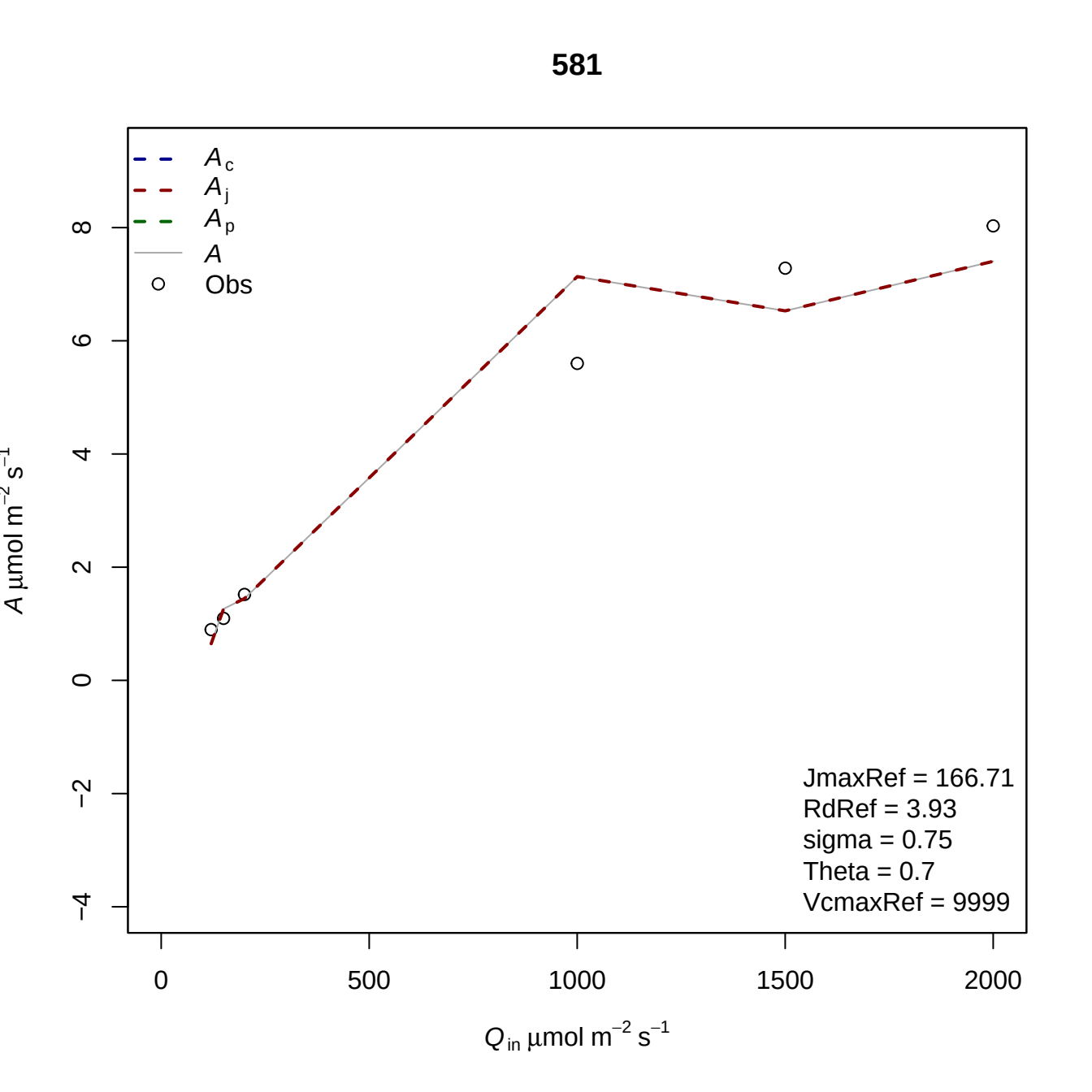

576
A_c
A_j
A_p
A
Obs
JmaxRef = 119.9
RdRef = 1.72
sigma = 0.84
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

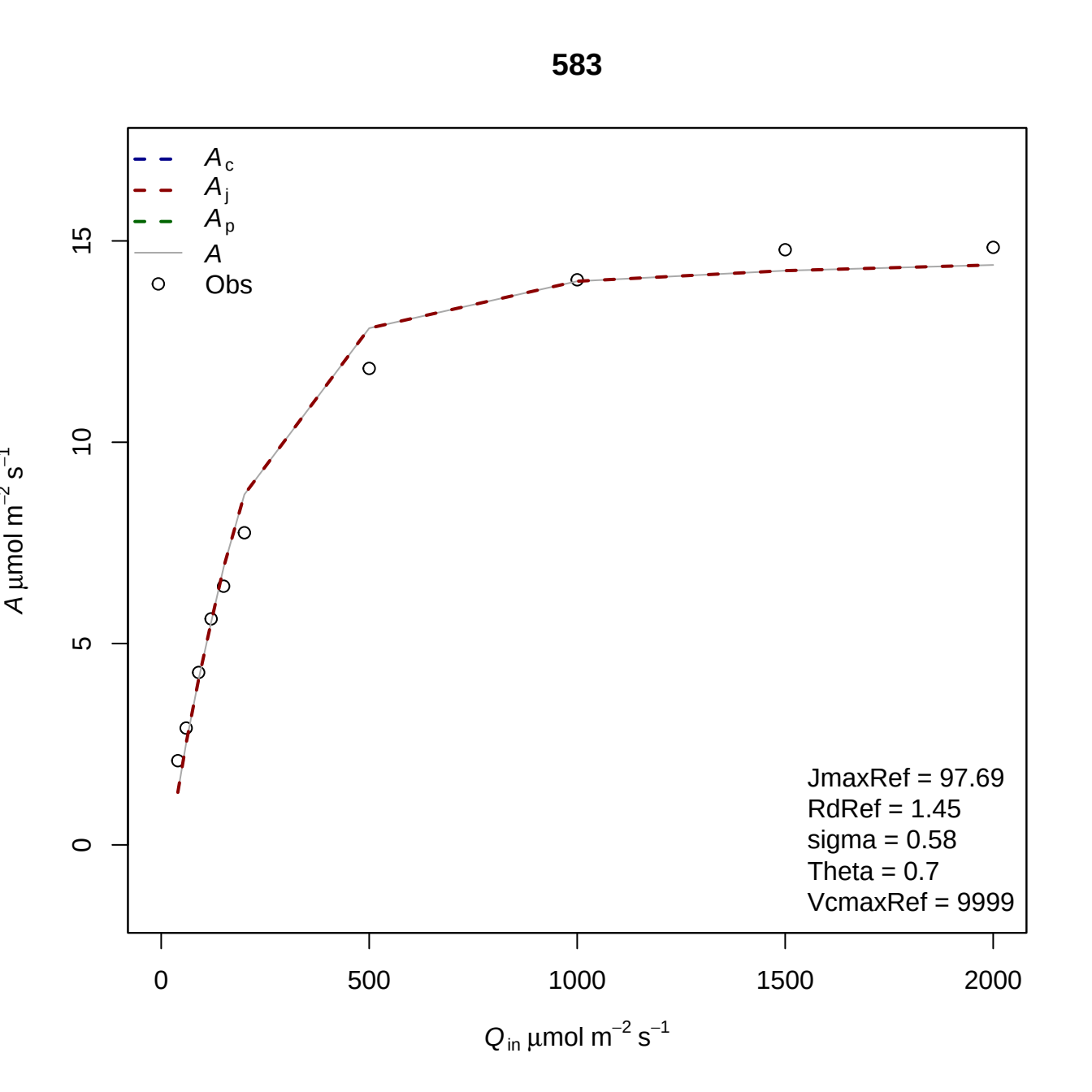

577
A_c
A_j
A_p
A
Obs
JmaxRef = 84.67
RdRef = 1.65
sigma = 0.72
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

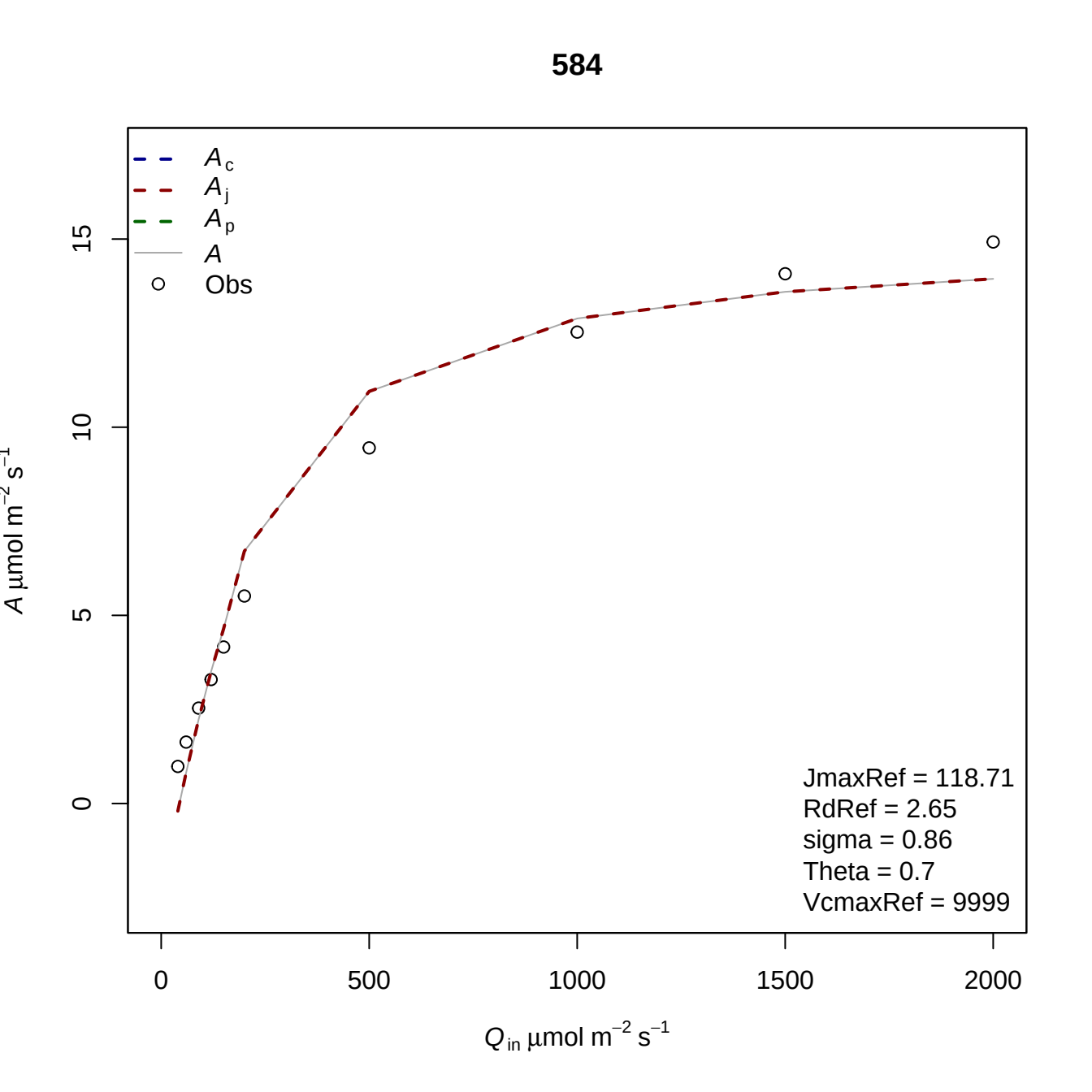

578
A_c
A_j
A_p
A
Obs
JmaxRef = 114.49
RdRef = 2.69
sigma = 1.01
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

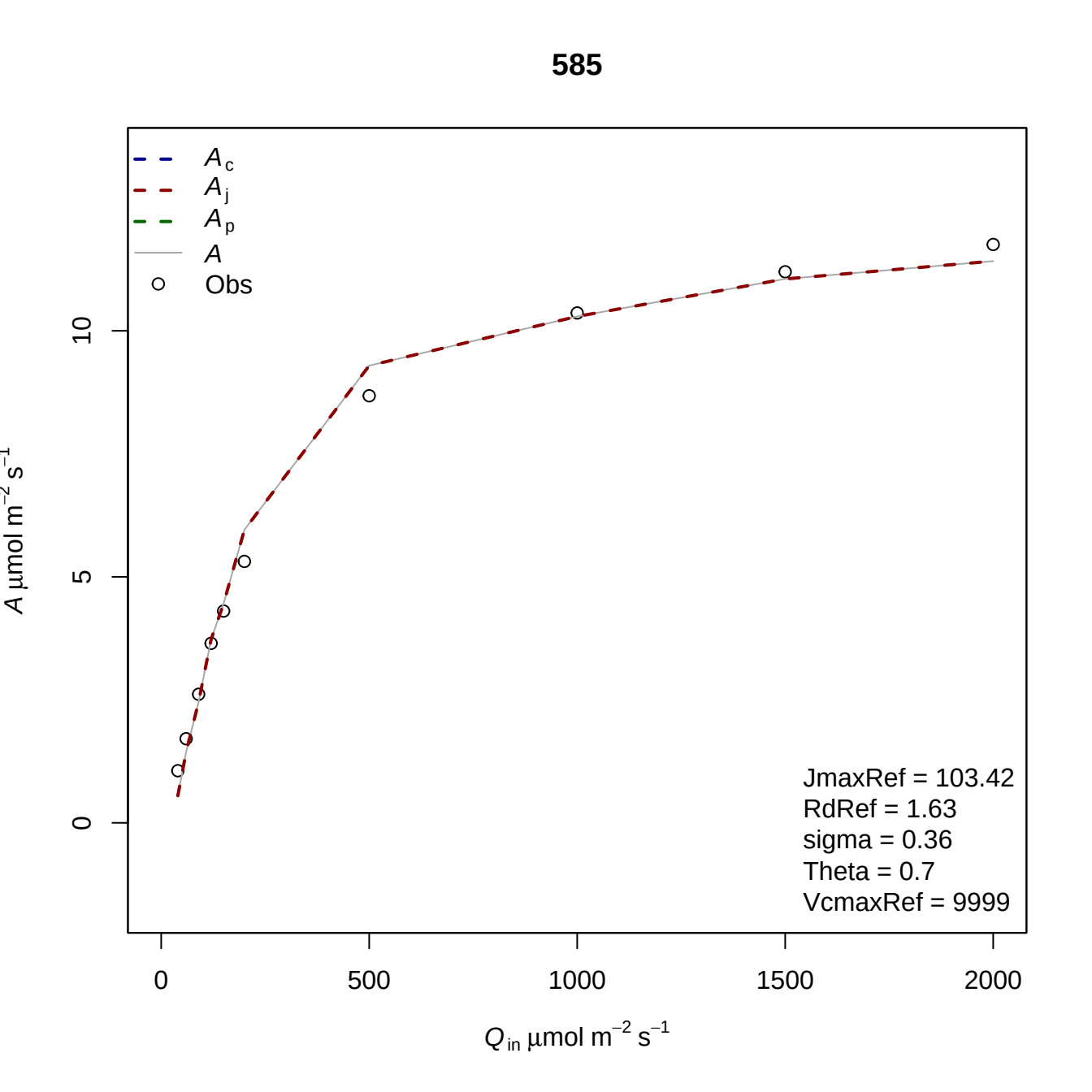

579
A_c
A_j
A_p
A
Obs
JmaxRef = 115.5
RdRef = 2.5
sigma = 0.57
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

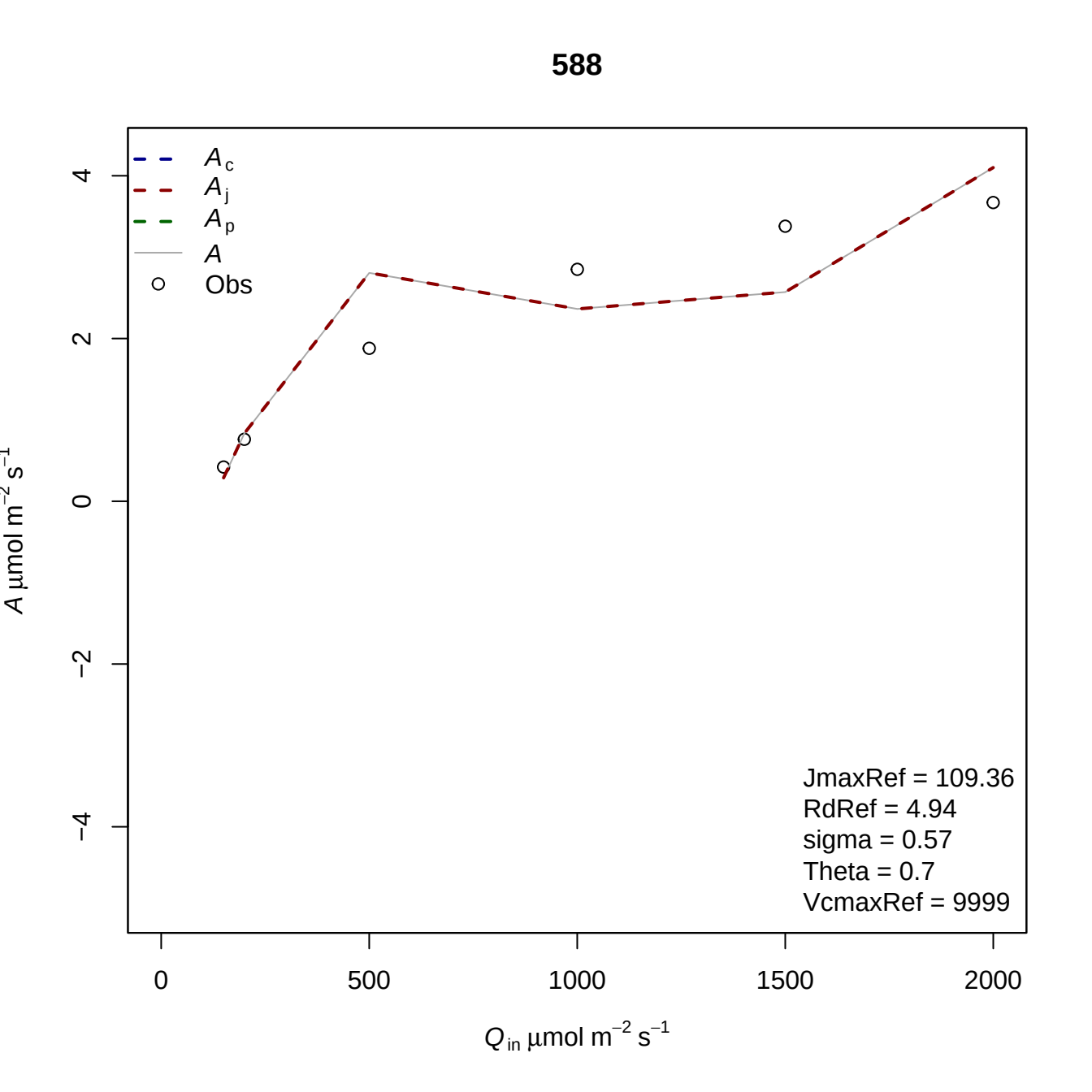

580
A_c
A_j
A_p
A
Obs
JmaxRef = 119.8
RdRef = 2.74
sigma = 0.44
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

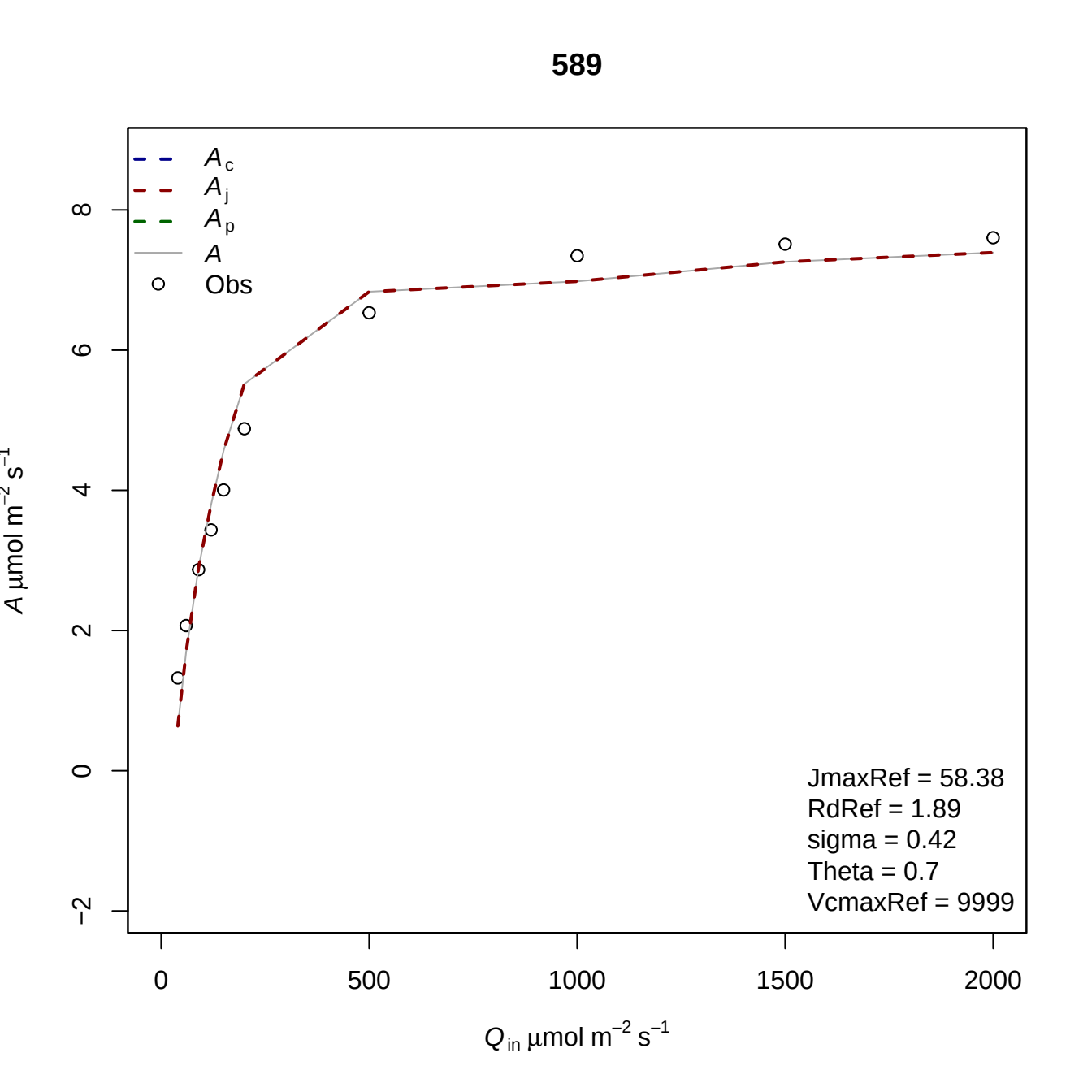

581
A_c
A_j
A_p
A
Obs
JmaxRef = 166.71
RdRef = 3.93
sigma = 0.75
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

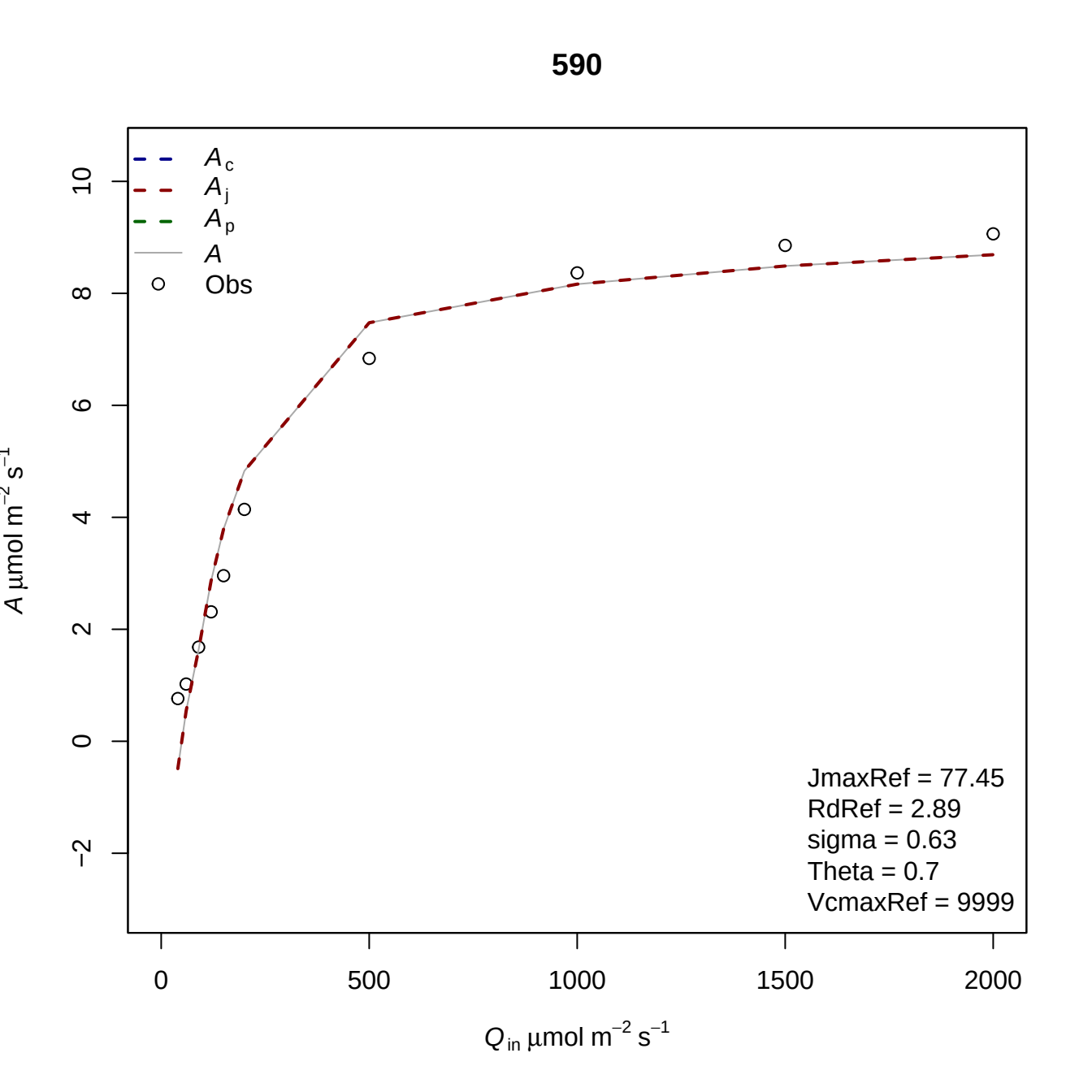

583
A_c
A_j
A_p
A
Obs
JmaxRef = 97.69
RdRef = 1.45
sigma = 0.58
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

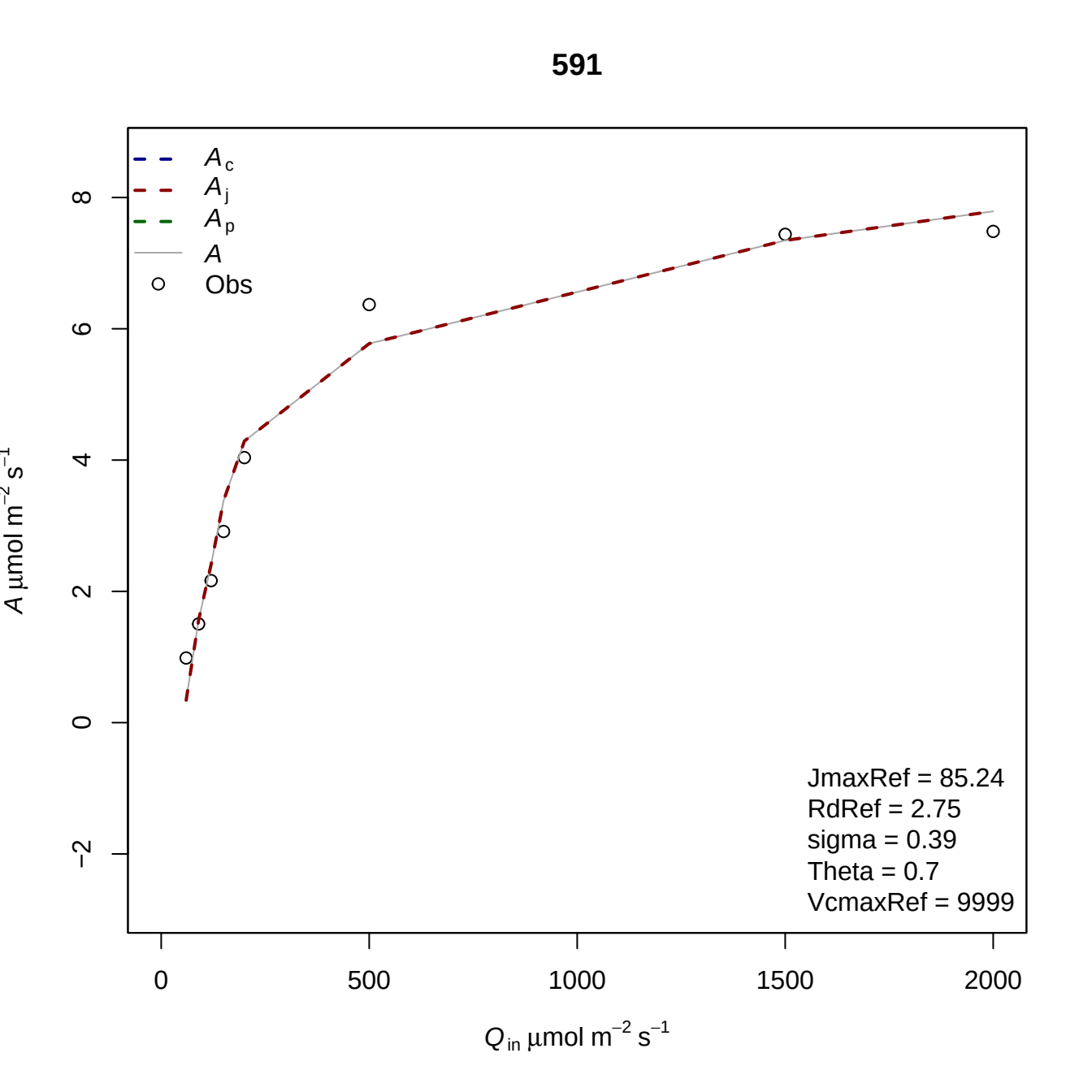

584
A_c
A_j
A_p
A
Obs
JmaxRef = 118.71
RdRef = 2.65
sigma = 0.86
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

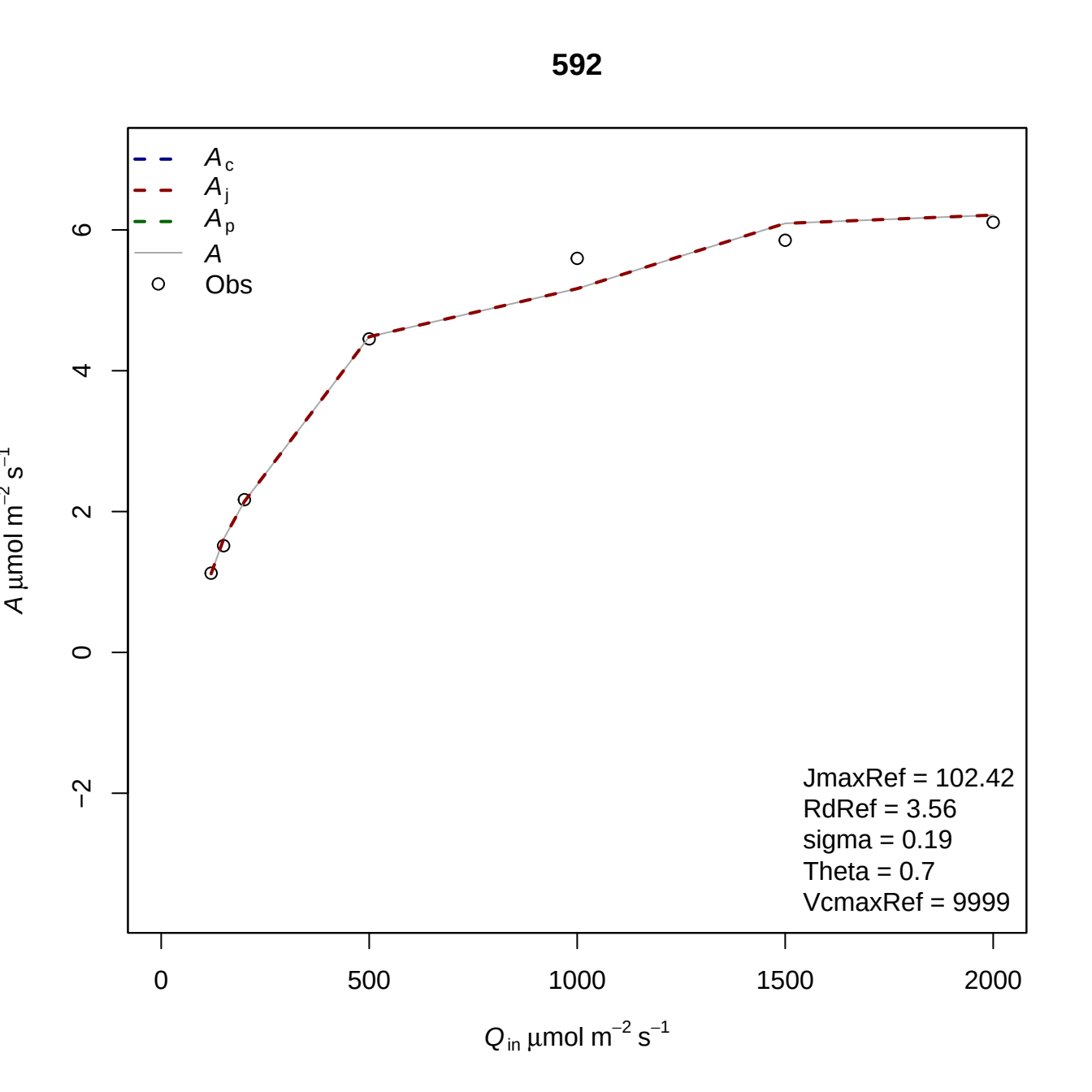

585
A_c
A_j
A_p
A
Obs
JmaxRef = 103.42
RdRef = 1.63
sigma = 0.36
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

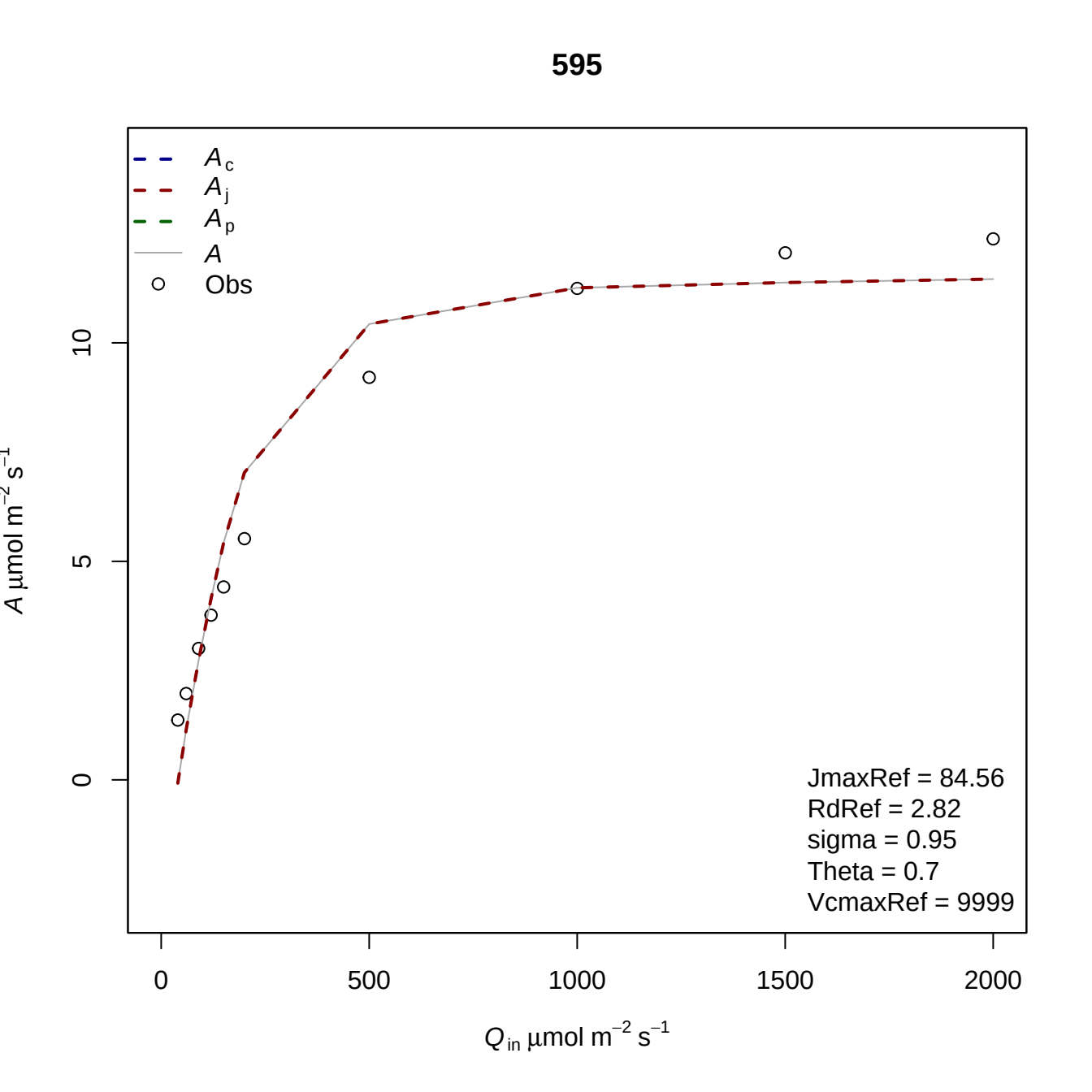

588
A_c
A_j
A_p
A
Obs
JmaxRef = 109.36
RdRef = 4.94
sigma = 0.57
Theta = 0.7
VcmaxRef = 9999
4
2
0
−2
−4
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

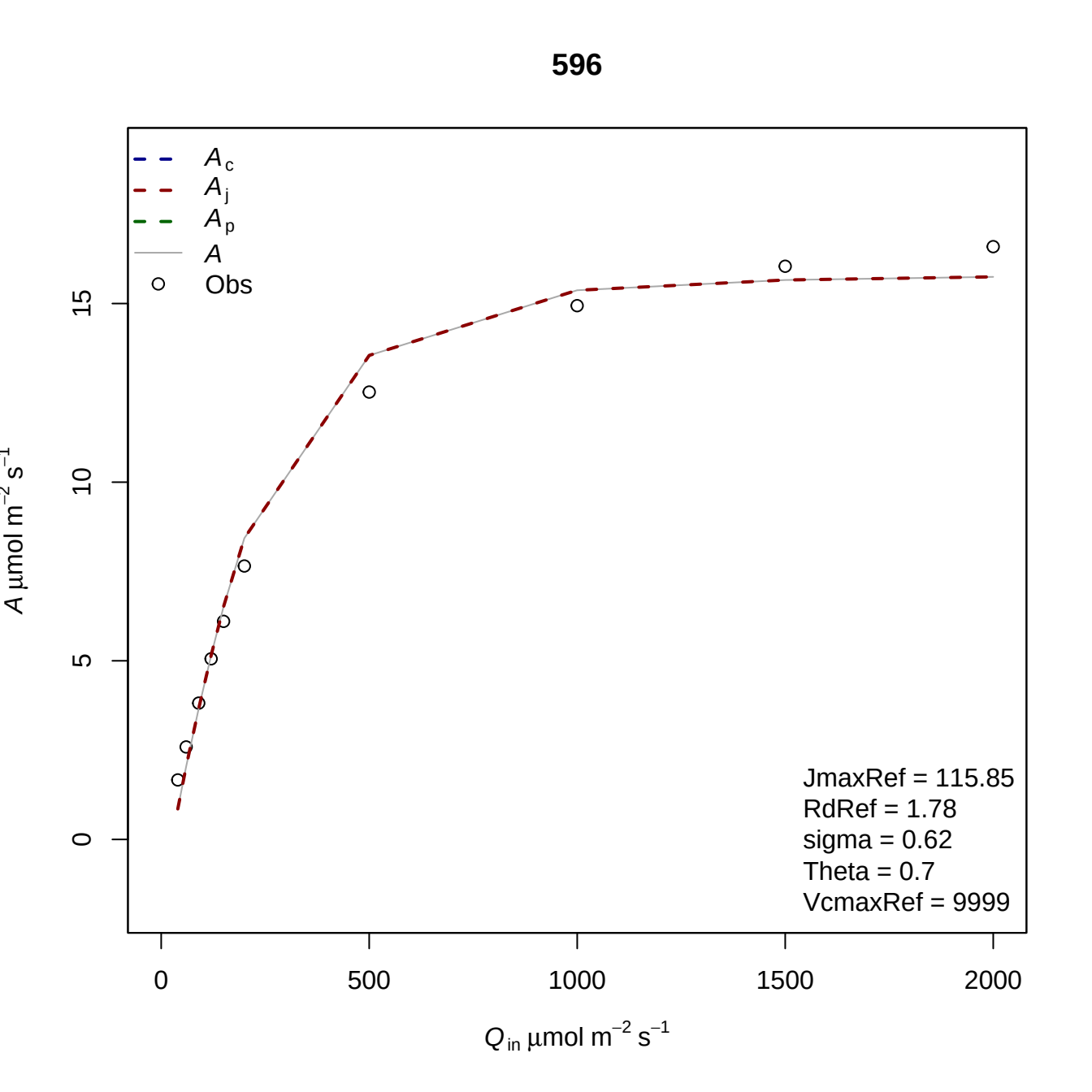

589
A_c
A_j
A_p
A
Obs
JmaxRef = 58.38
RdRef = 1.89
sigma = 0.42
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

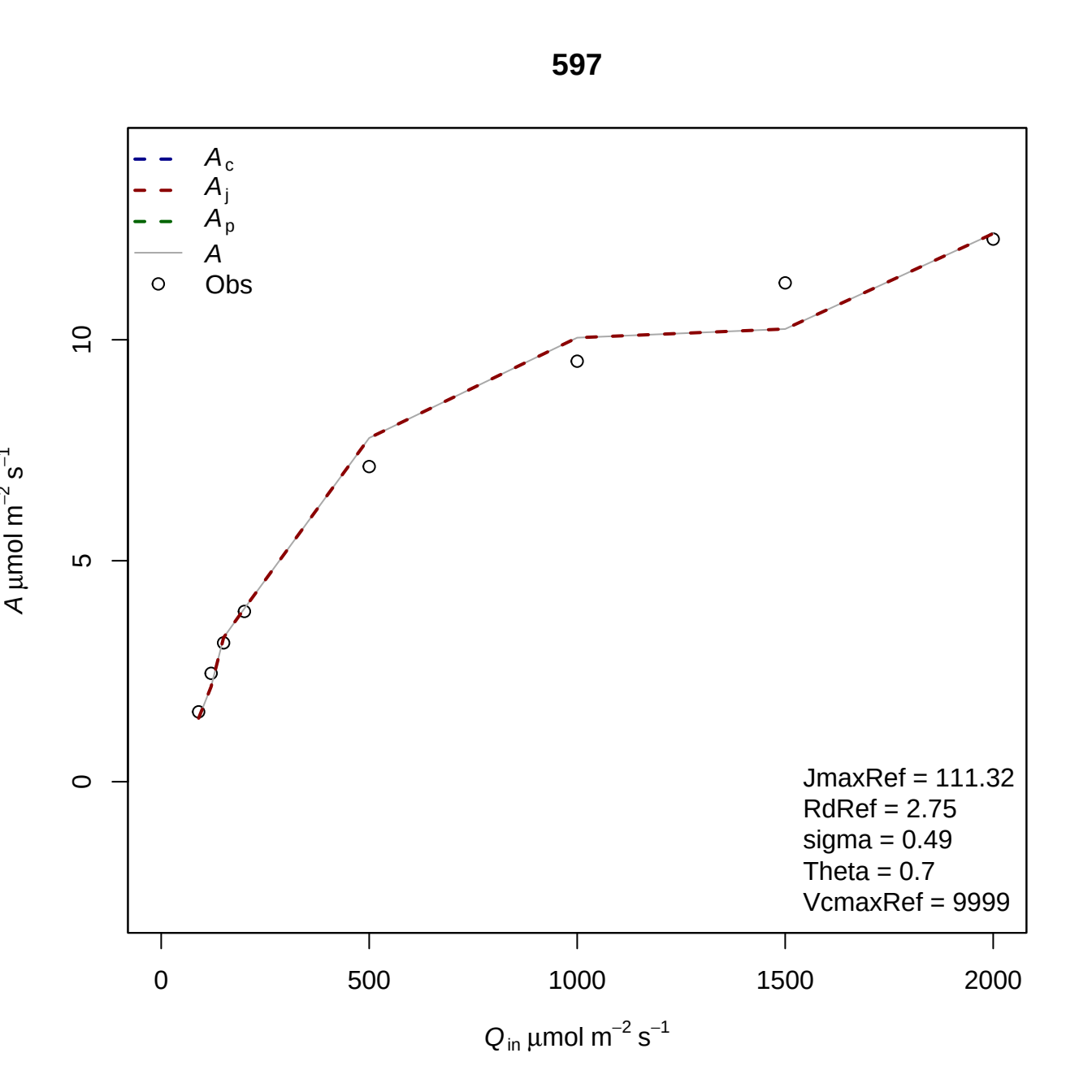

590
A_c
A_j
A_p
A
Obs
JmaxRef = 77.45
RdRef = 2.89
sigma = 0.63
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

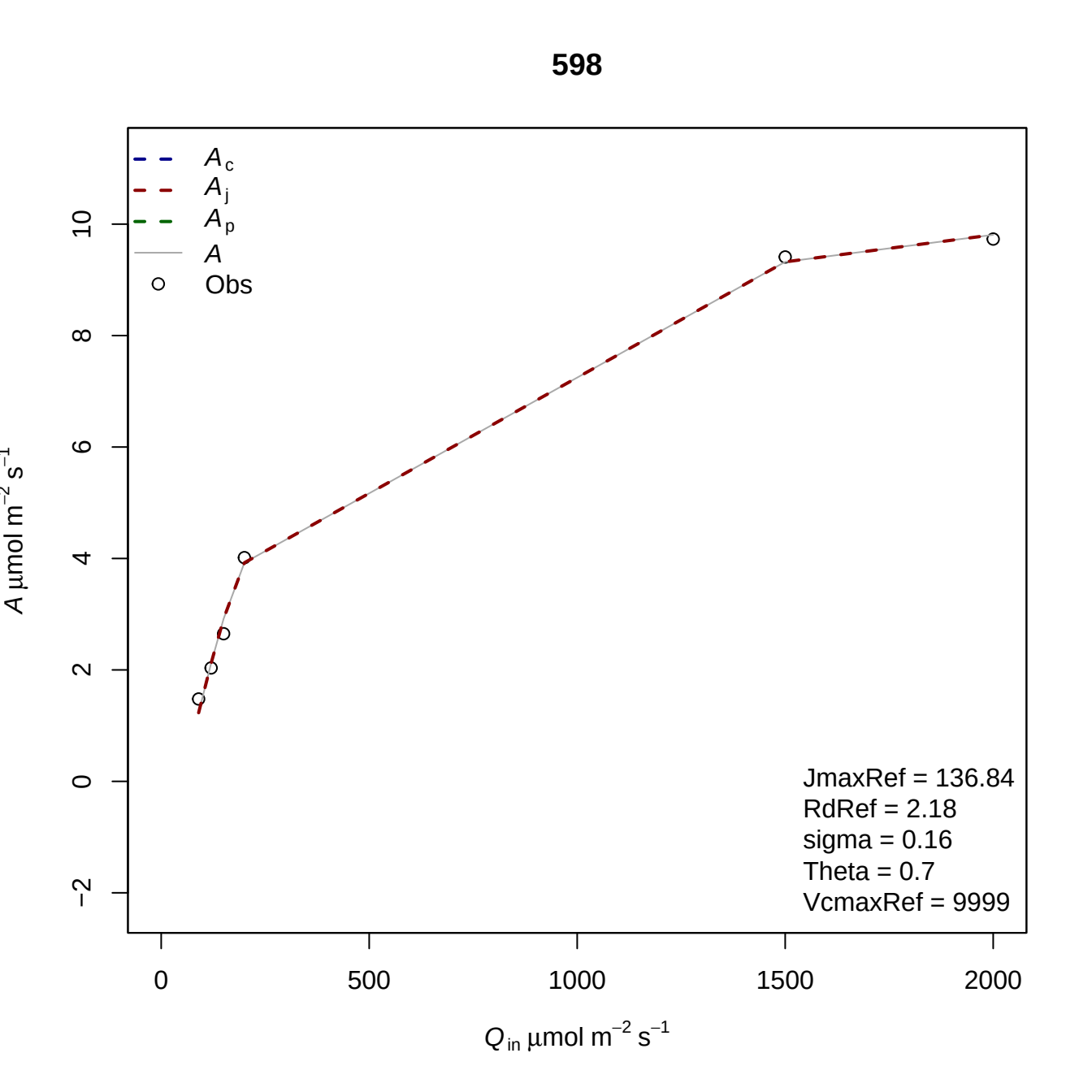

591
A_c
A_j
A_p
A
Obs
JmaxRef = 85.24
RdRef = 2.75
sigma = 0.39
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

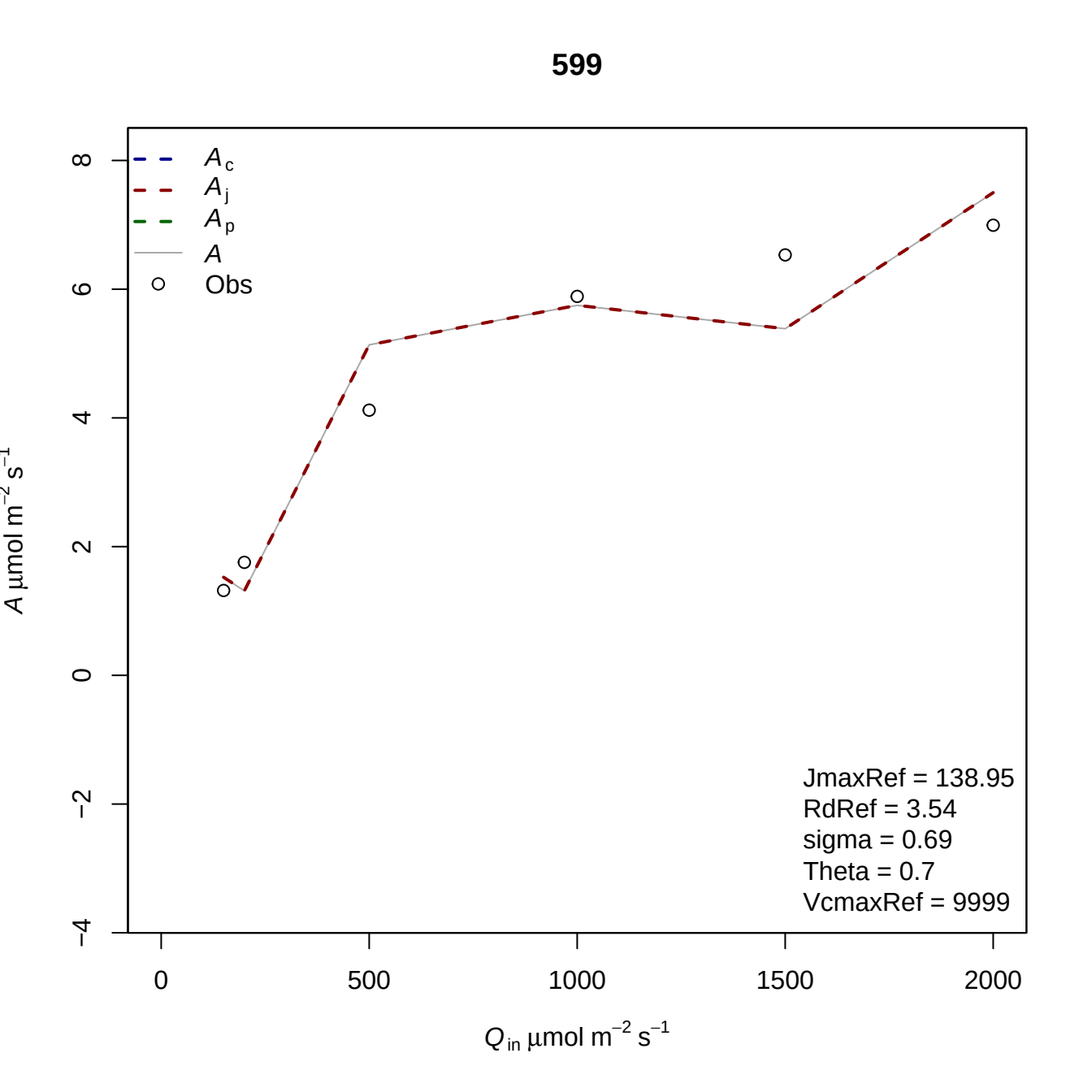

592
A_c
A_j
A_p
A
Obs
JmaxRef = 102.42
RdRef = 3.56
sigma = 0.19
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

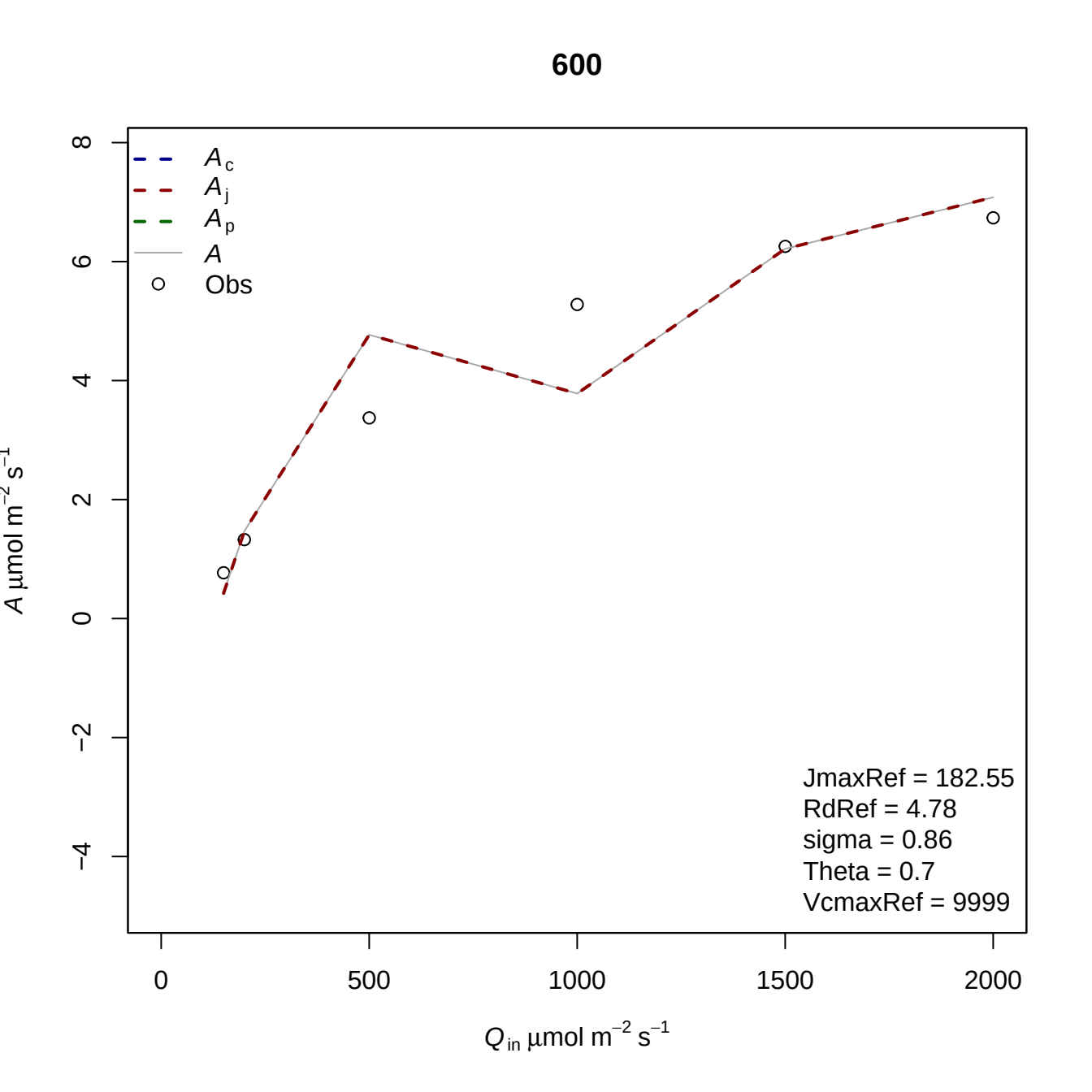

595
A_c
A_j
A_p
A
Obs
JmaxRef = 84.56
RdRef = 2.82
sigma = 0.95
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

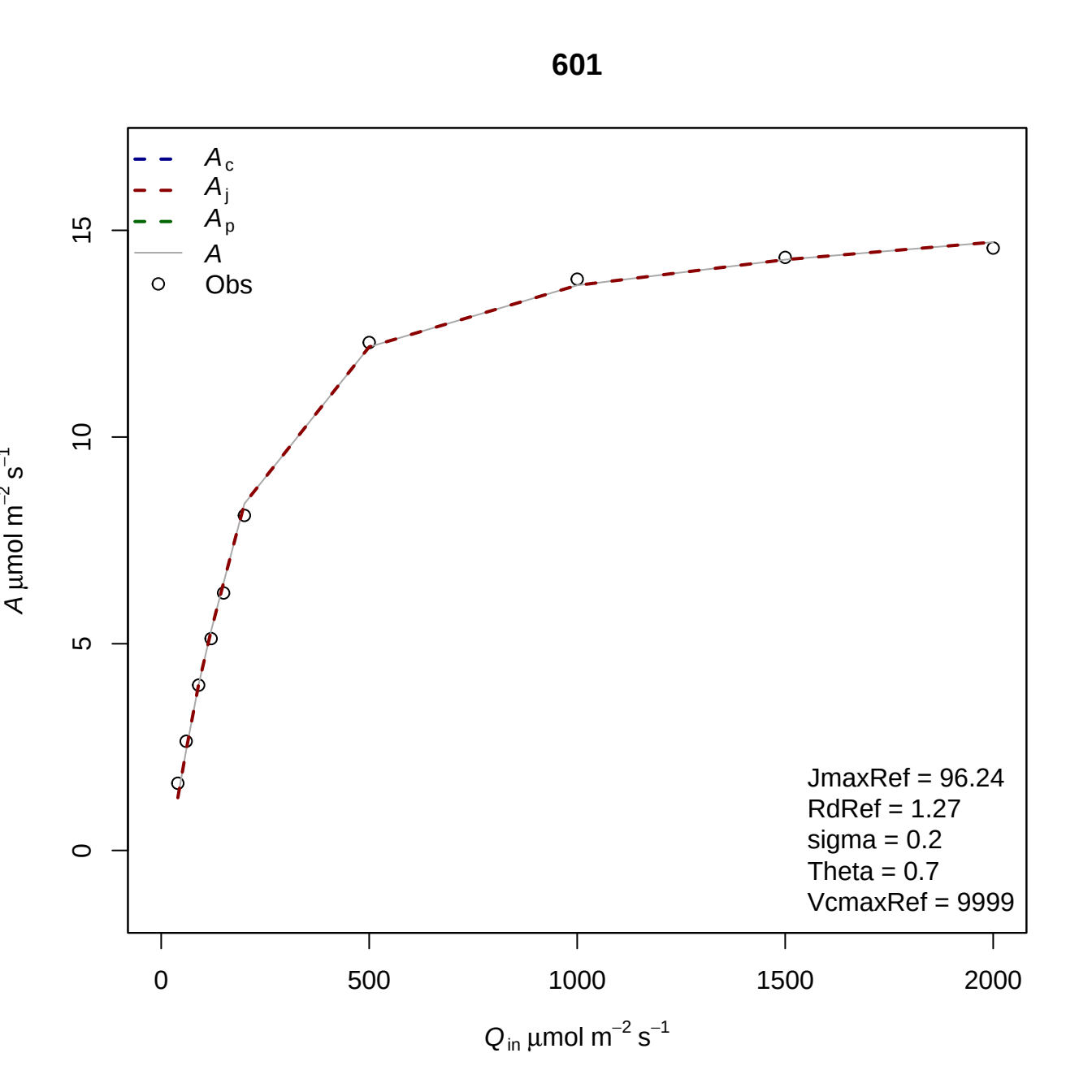

596
A_c
A_j
A_p
A
Obs
JmaxRef = 115.85
RdRef = 1.78
sigma = 0.62
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

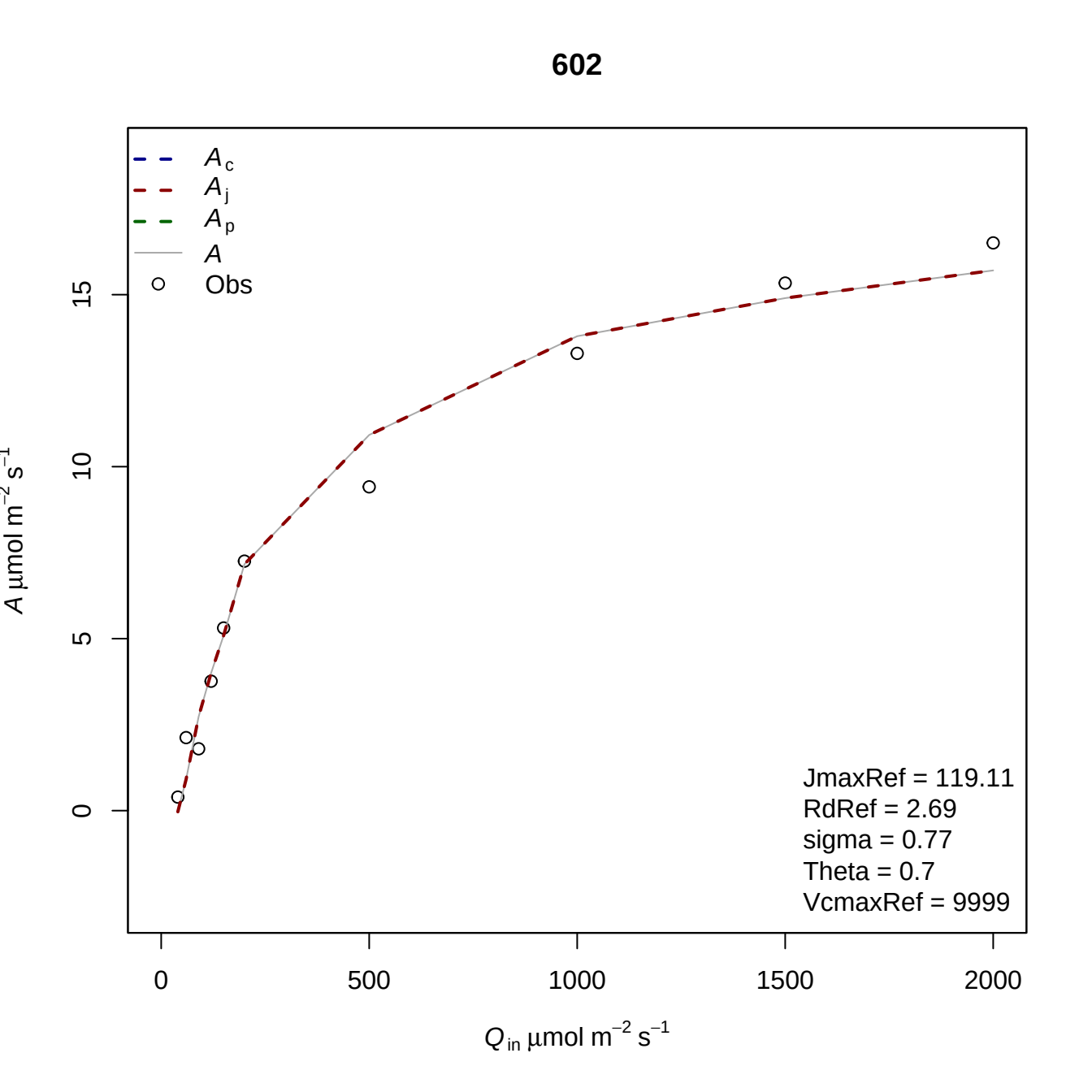

597
A_c
A_j
A_p
A
Obs
JmaxRef = 111.32
RdRef = 2.75
sigma = 0.49
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

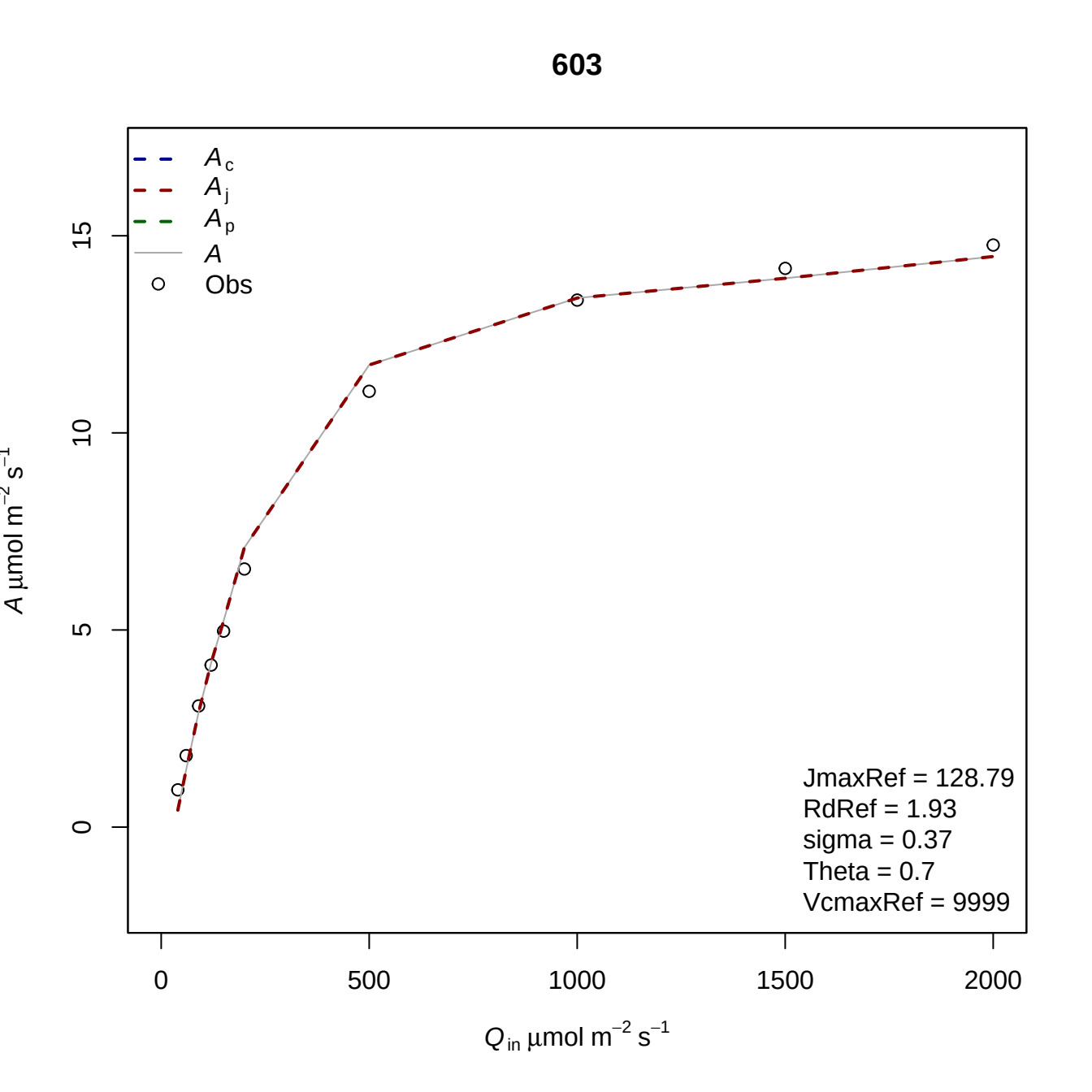

598
A_c
A_j
A_p
A
Obs
JmaxRef = 136.84
RdRef = 2.18
sigma = 0.16
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

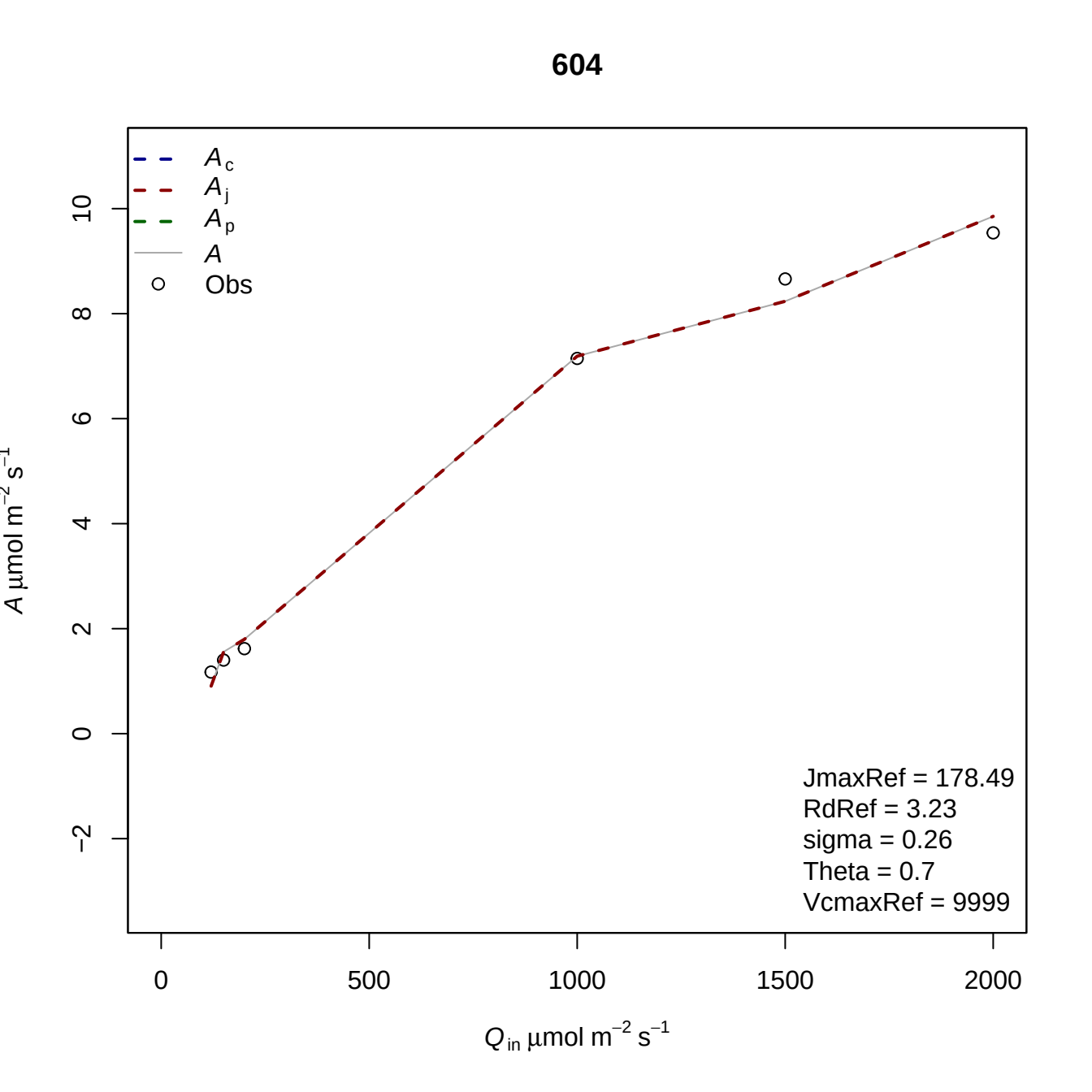

599
A_c
A_j
A_p
A
Obs
JmaxRef = 138.95
RdRef = 3.54
sigma = 0.69
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

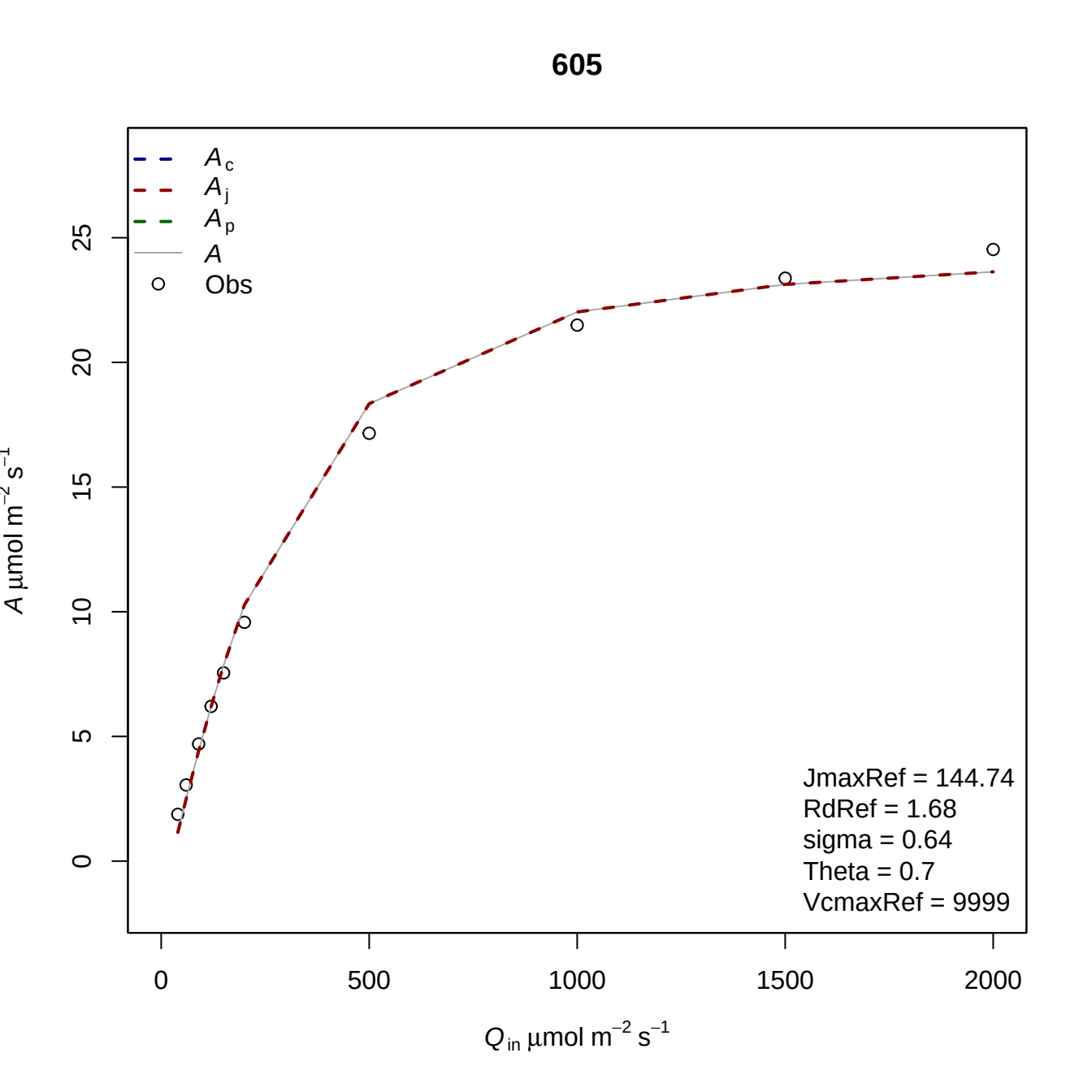

600
A_c
A_j
A_p
A
Obs
JmaxRef = 182.55
RdRef = 4.78
sigma = 0.86
Theta = 0.7
VcmaxRef = 9999
8
6
4
2
0
−2
−4
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

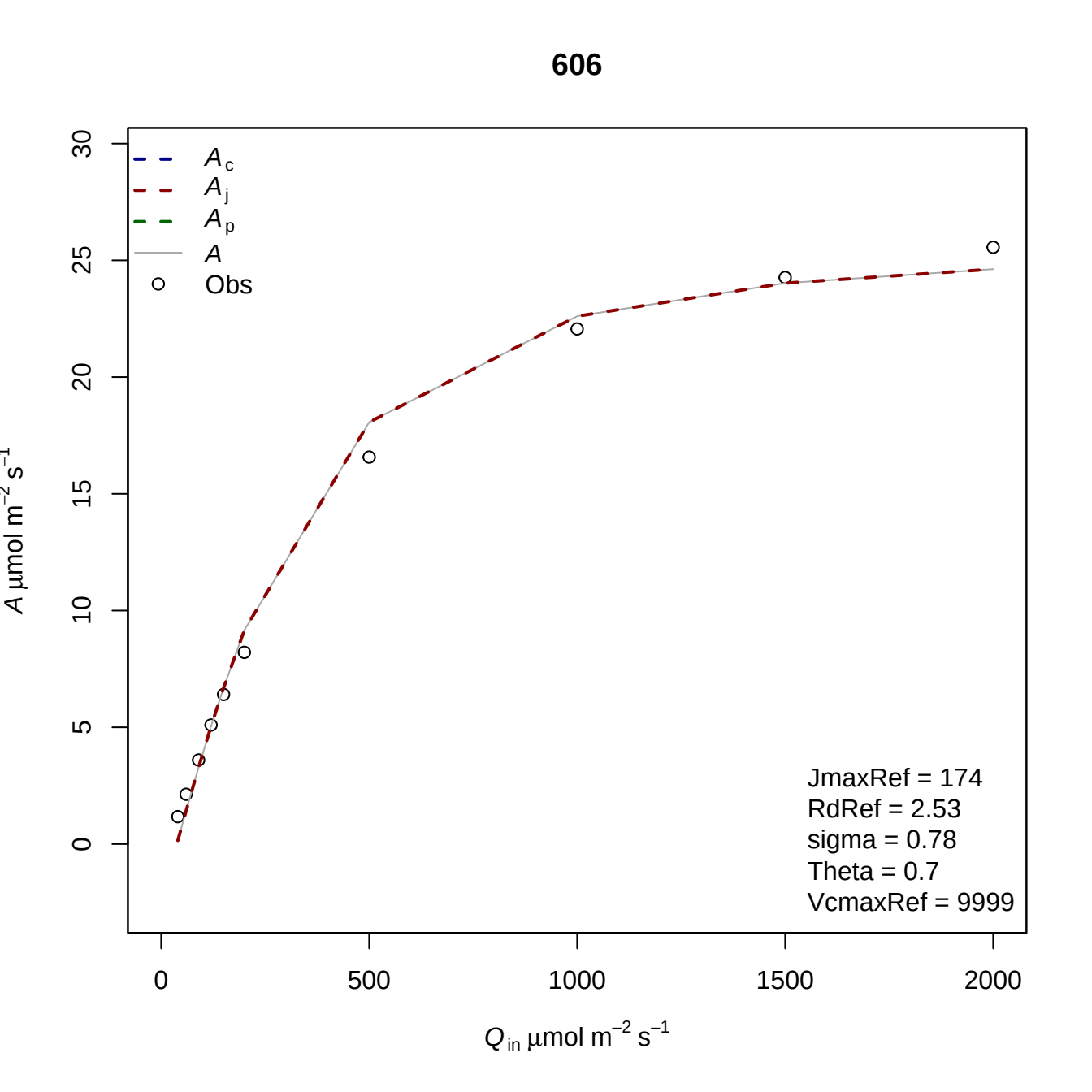

601
A_c
A_j
A_p
A
Obs
JmaxRef = 96.24
RdRef = 1.27
sigma = 0.2
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

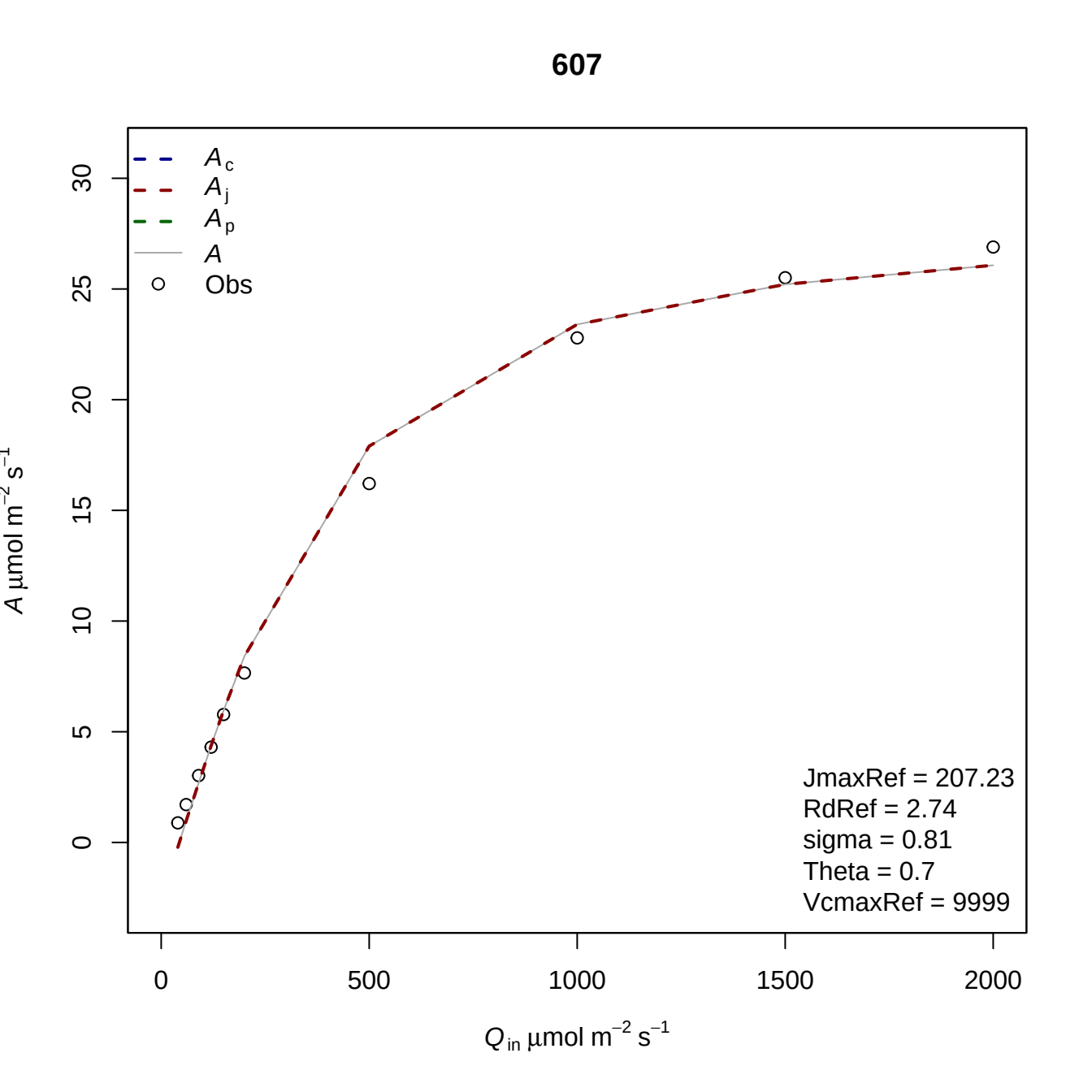

602
A_c
A_j
A_p
A
Obs
JmaxRef = 119.11
RdRef = 2.69
sigma = 0.77
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

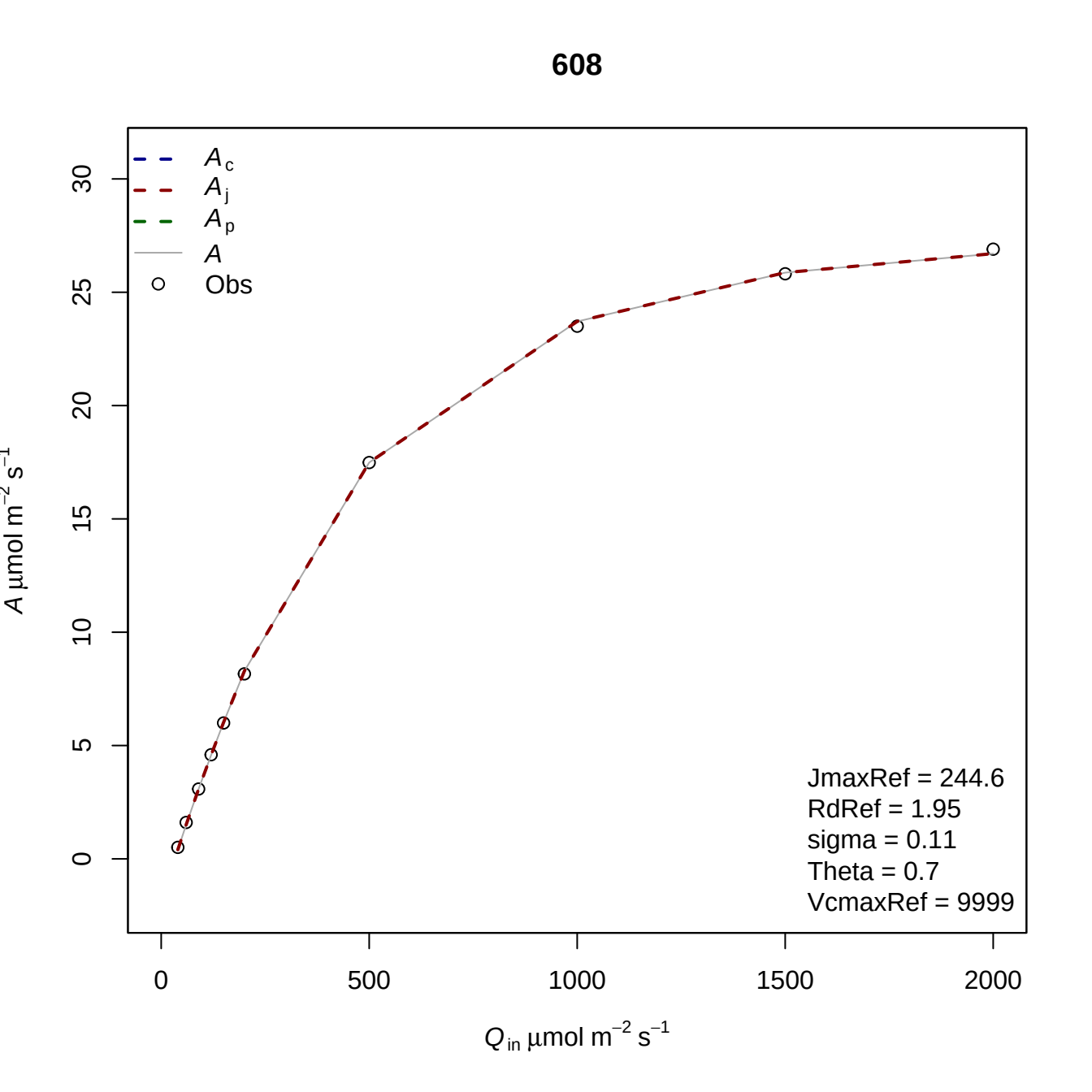

603
A_c
A_j
A_p
A
Obs
JmaxRef = 128.79
RdRef = 1.93
sigma = 0.37
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

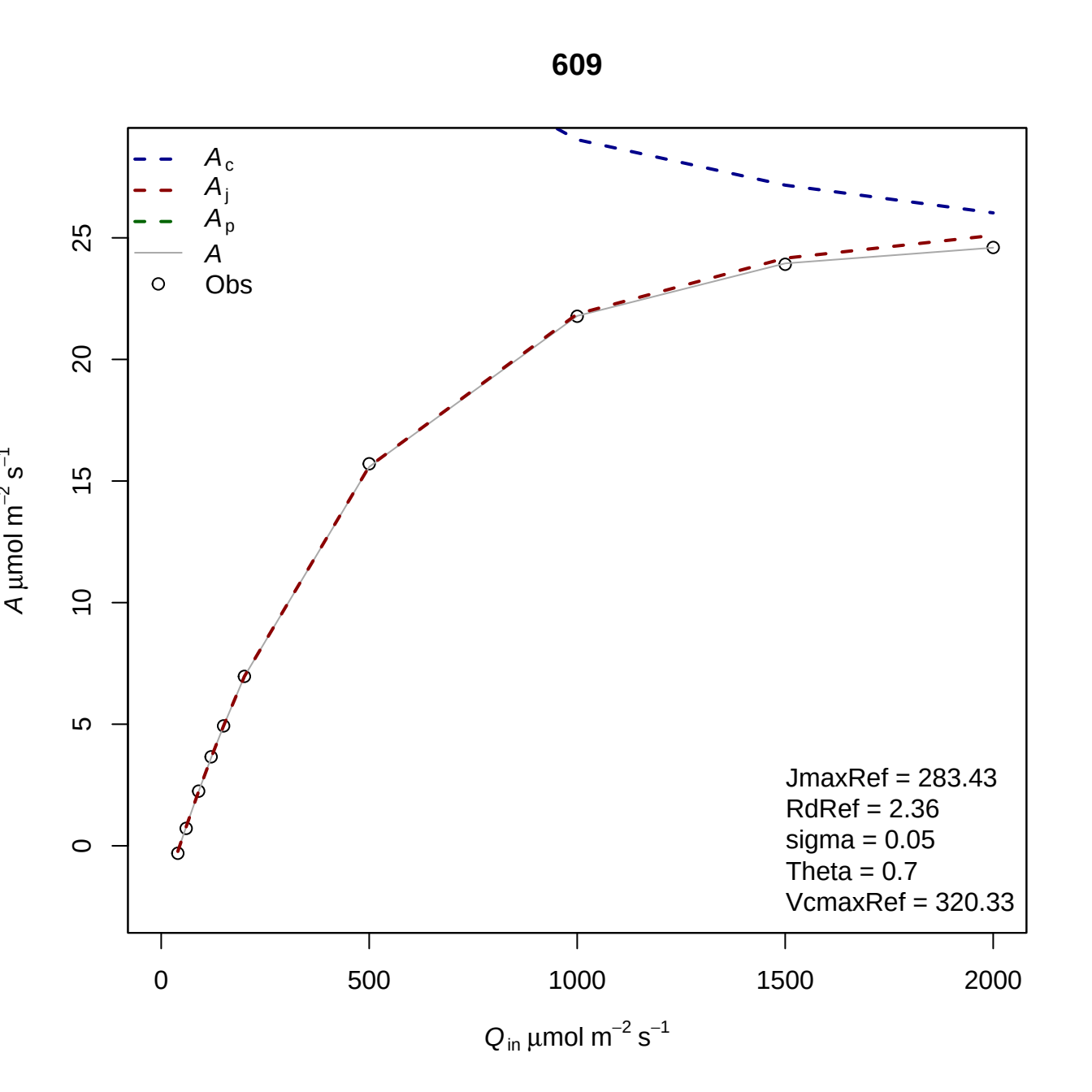

604
A_c
A_j
A_p
A
Obs
JmaxRef = 178.49
RdRef = 3.23
sigma = 0.26
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

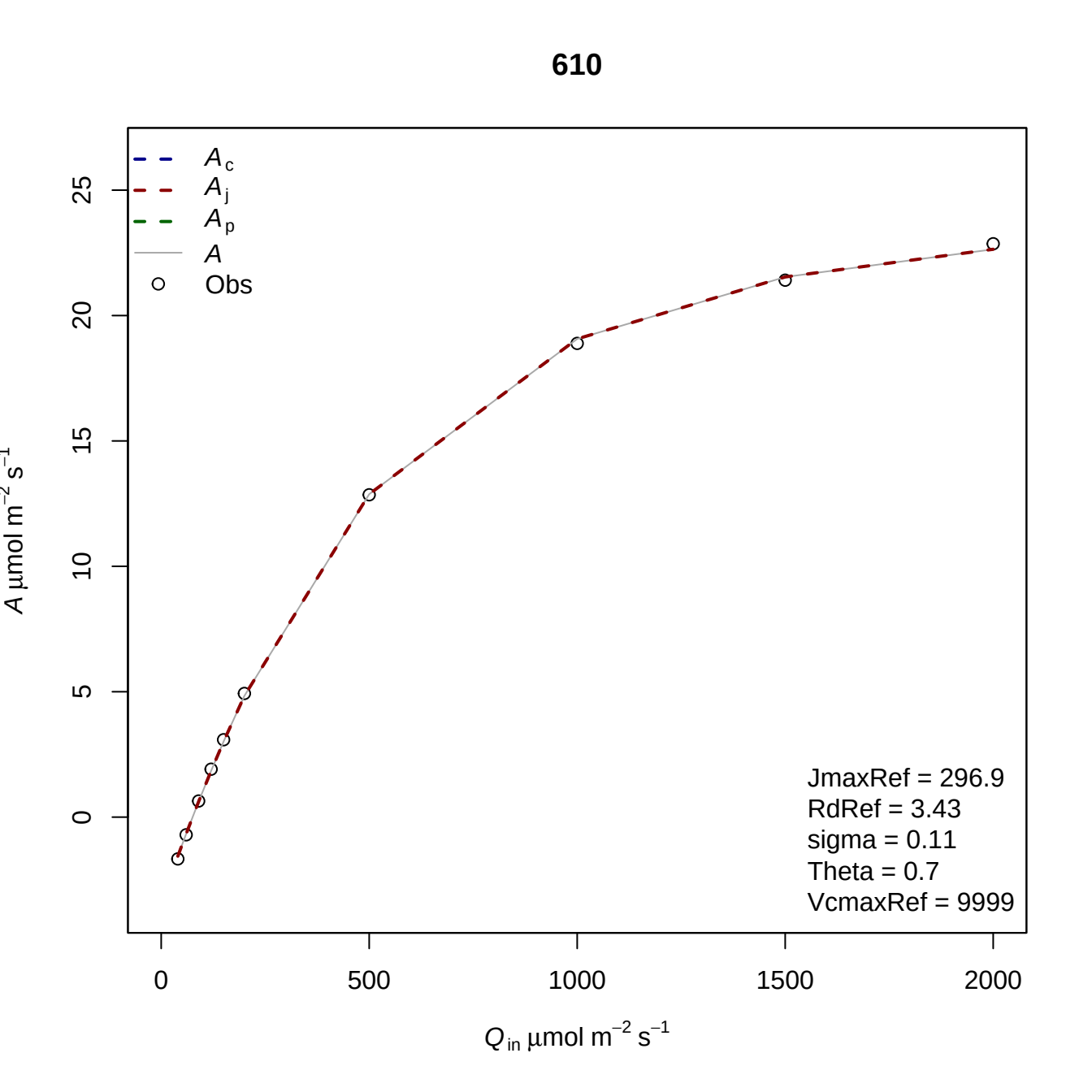

605
A_c
A_j
A_p
A
Obs
JmaxRef = 144.74
RdRef = 1.68
sigma = 0.64
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

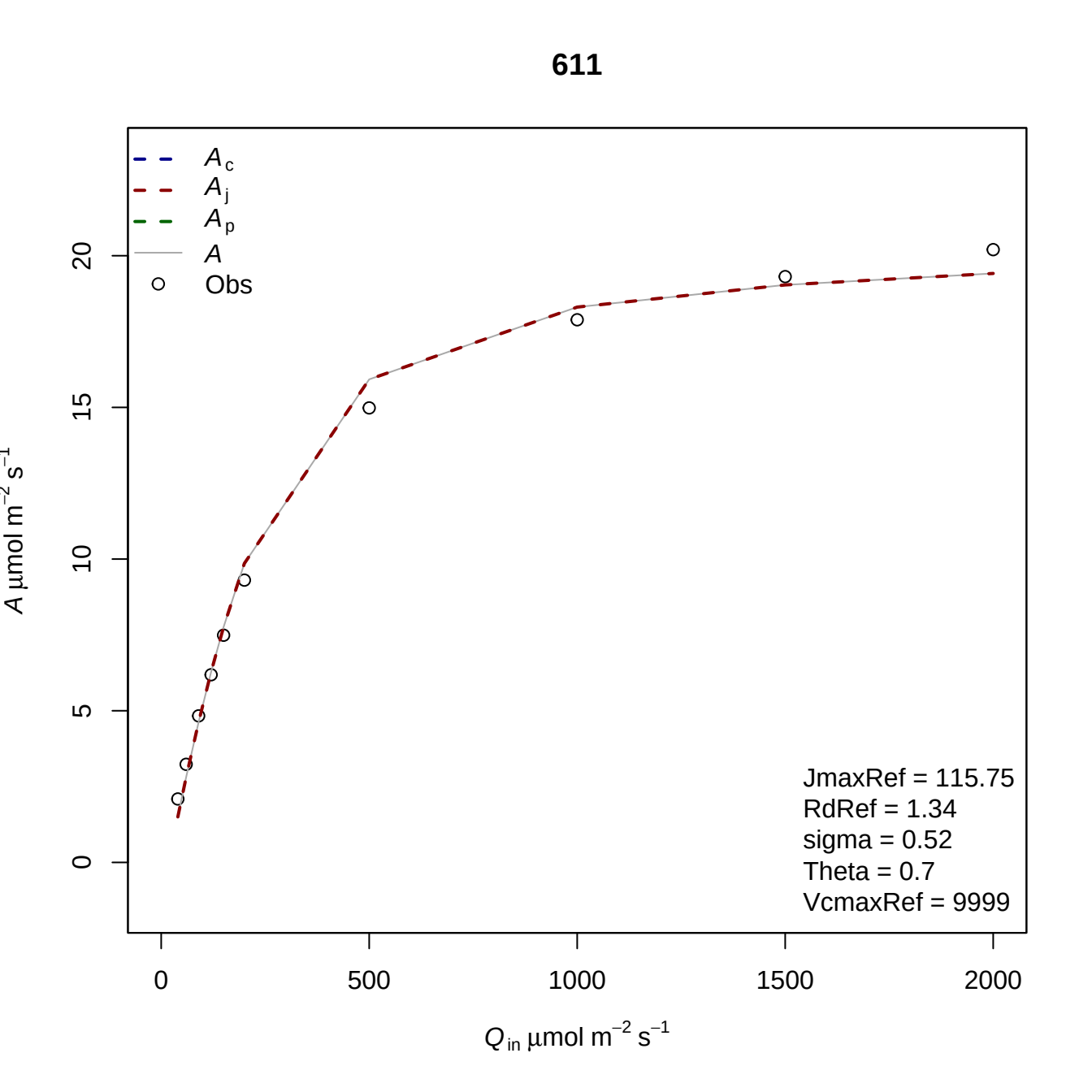

606
A_c
A_j
A_p
A
Obs
JmaxRef = 174
RdRef = 2.53
sigma = 0.78
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

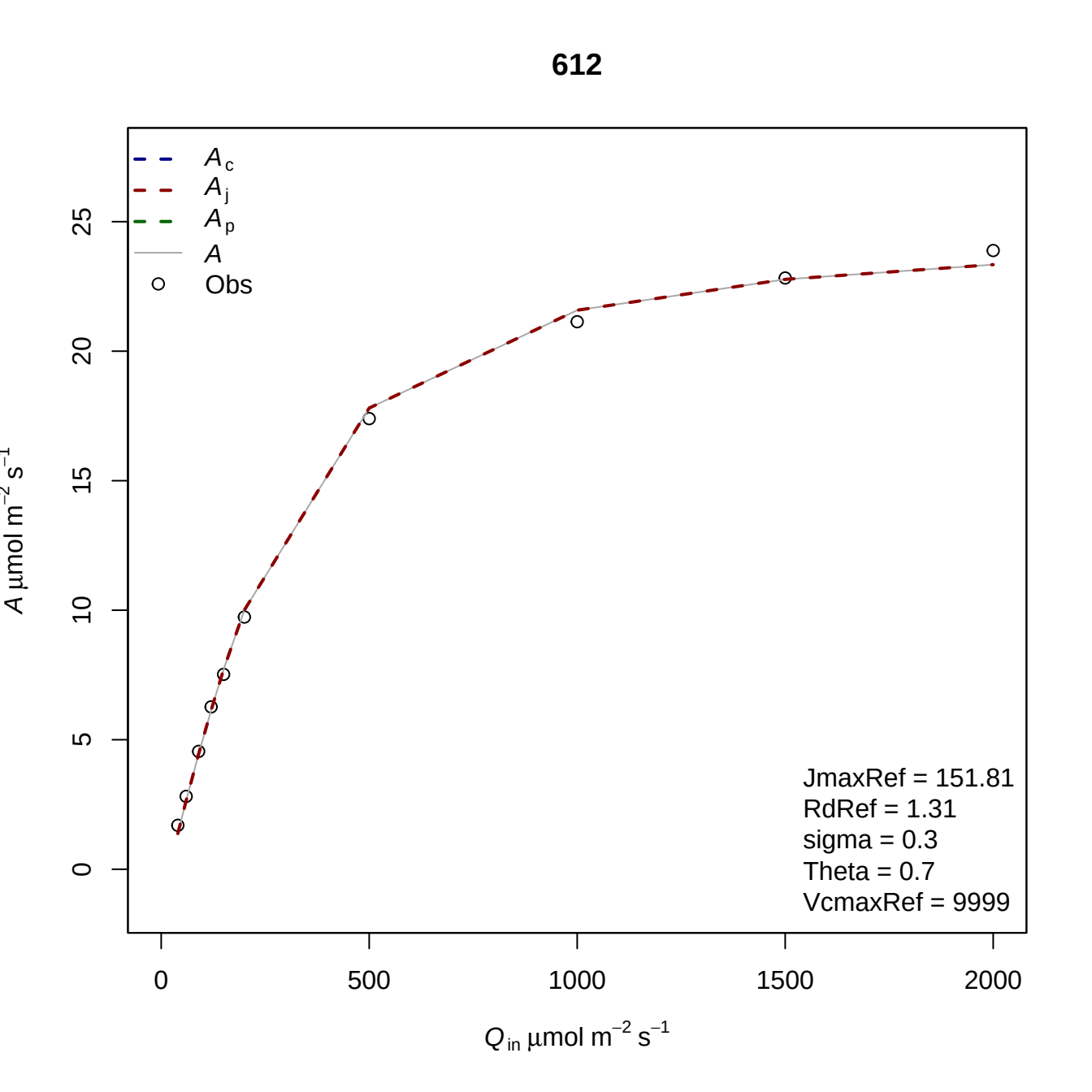

607
A_c
A_j
A_p
A
Obs
JmaxRef = 207.23
RdRef = 2.74
sigma = 0.81
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

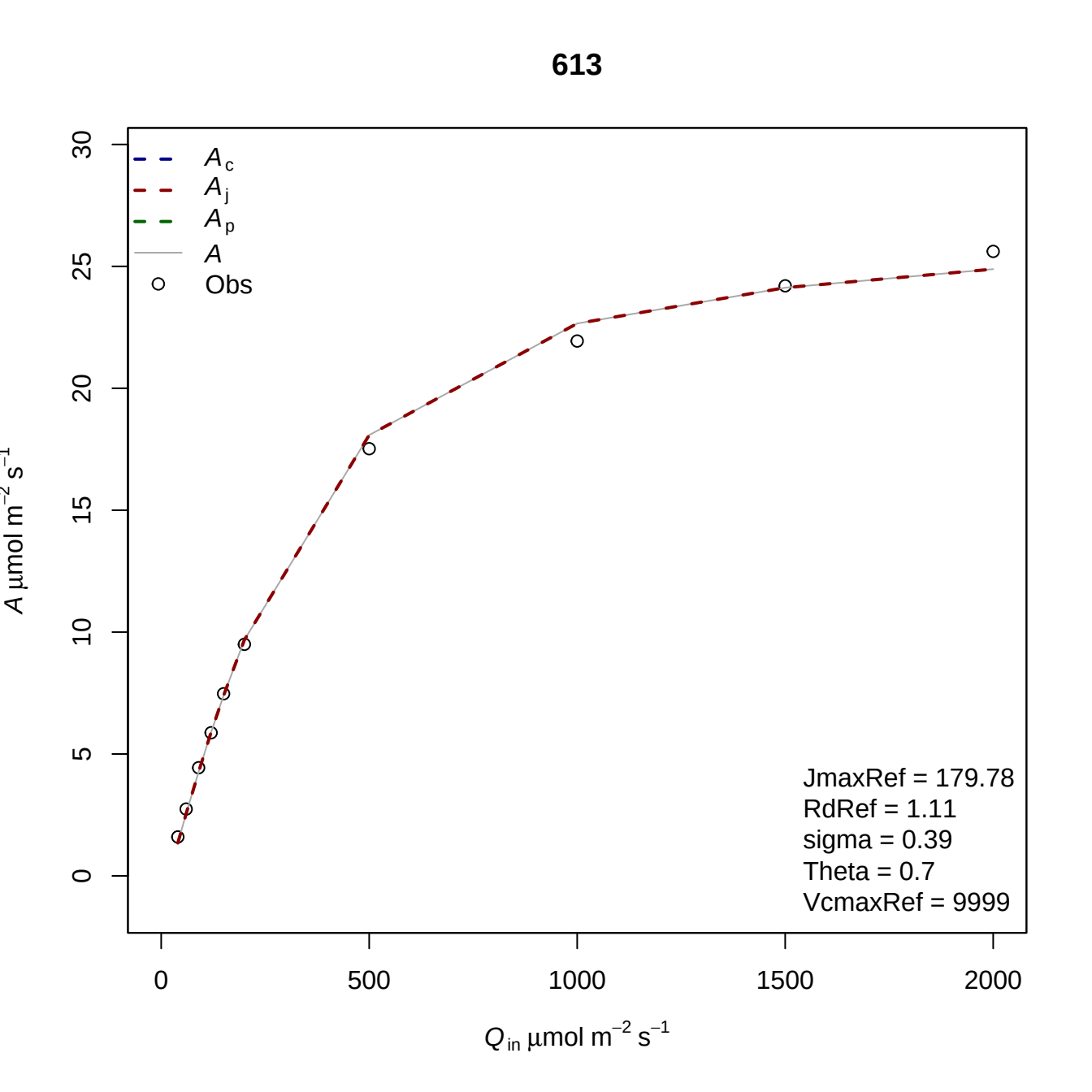

608
A_c
A_j
A_p
A
Obs
JmaxRef = 244.6
RdRef = 1.95
sigma = 0.11
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

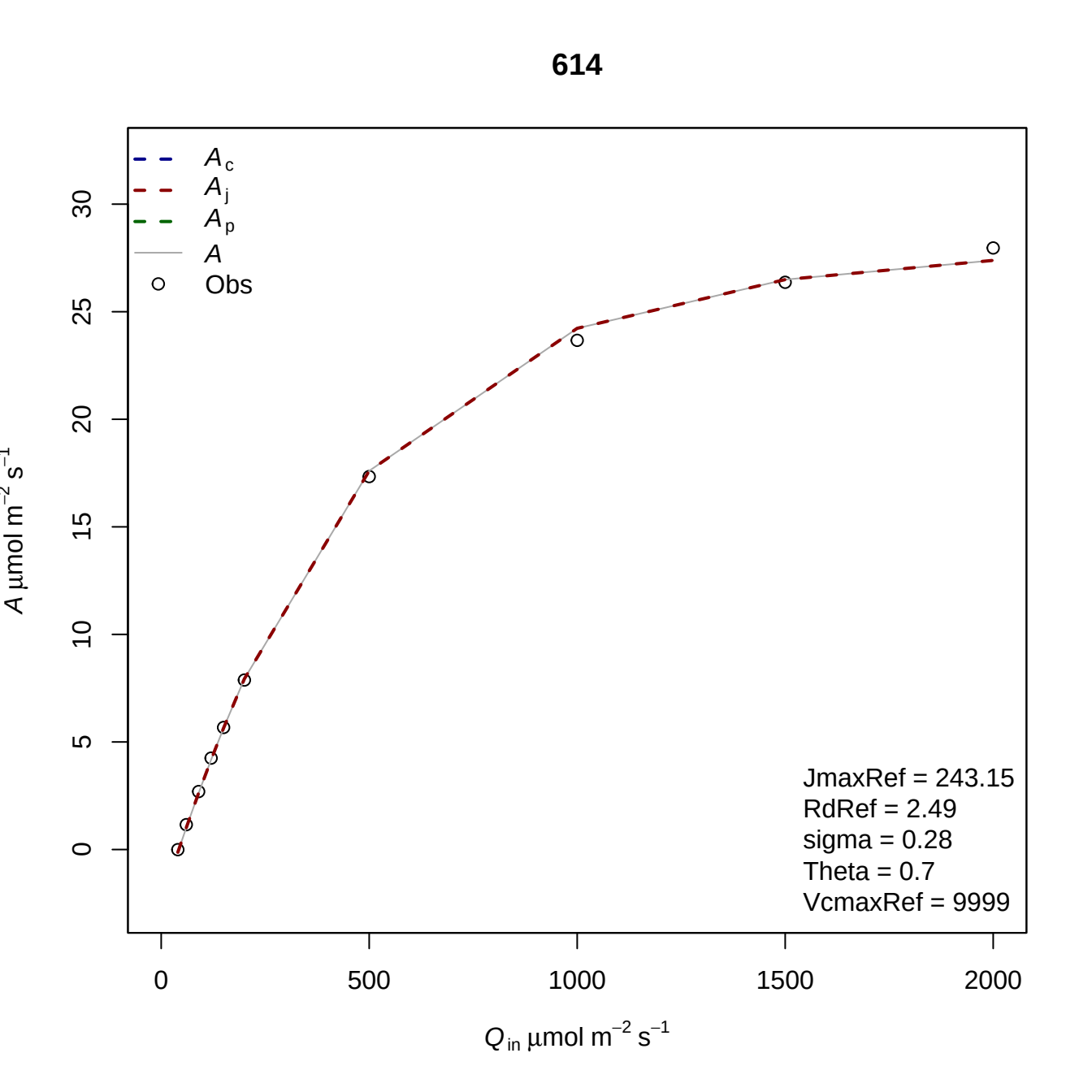

609
A_c
A_j
A_p
A
Obs
JmaxRef = 283.43
RdRef = 2.36
sigma = 0.05
Theta = 0.7
VcmaxRef = 320.33
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

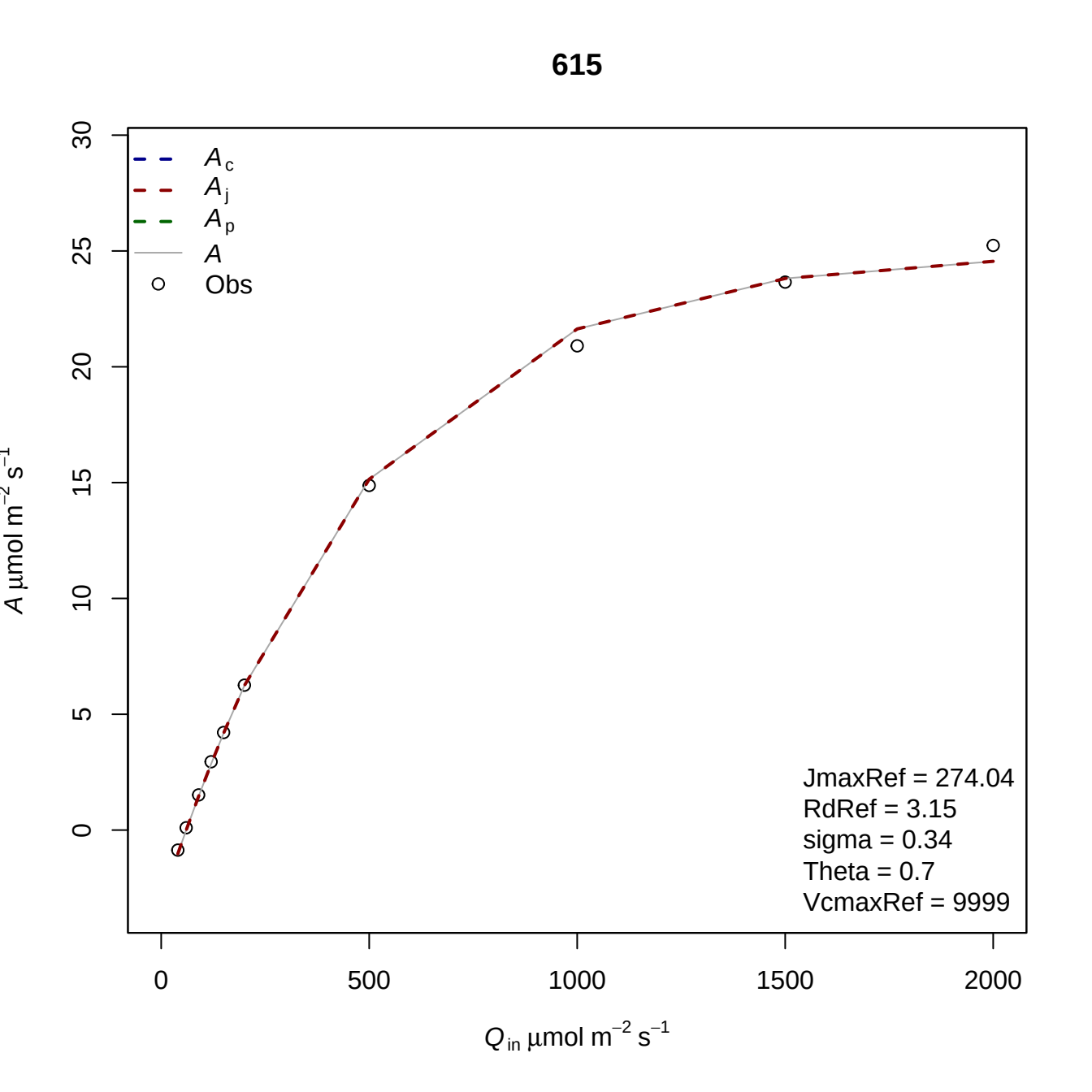

610
A_c
A_j
A_p
A
Obs
JmaxRef = 296.9
RdRef = 3.43
sigma = 0.11
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

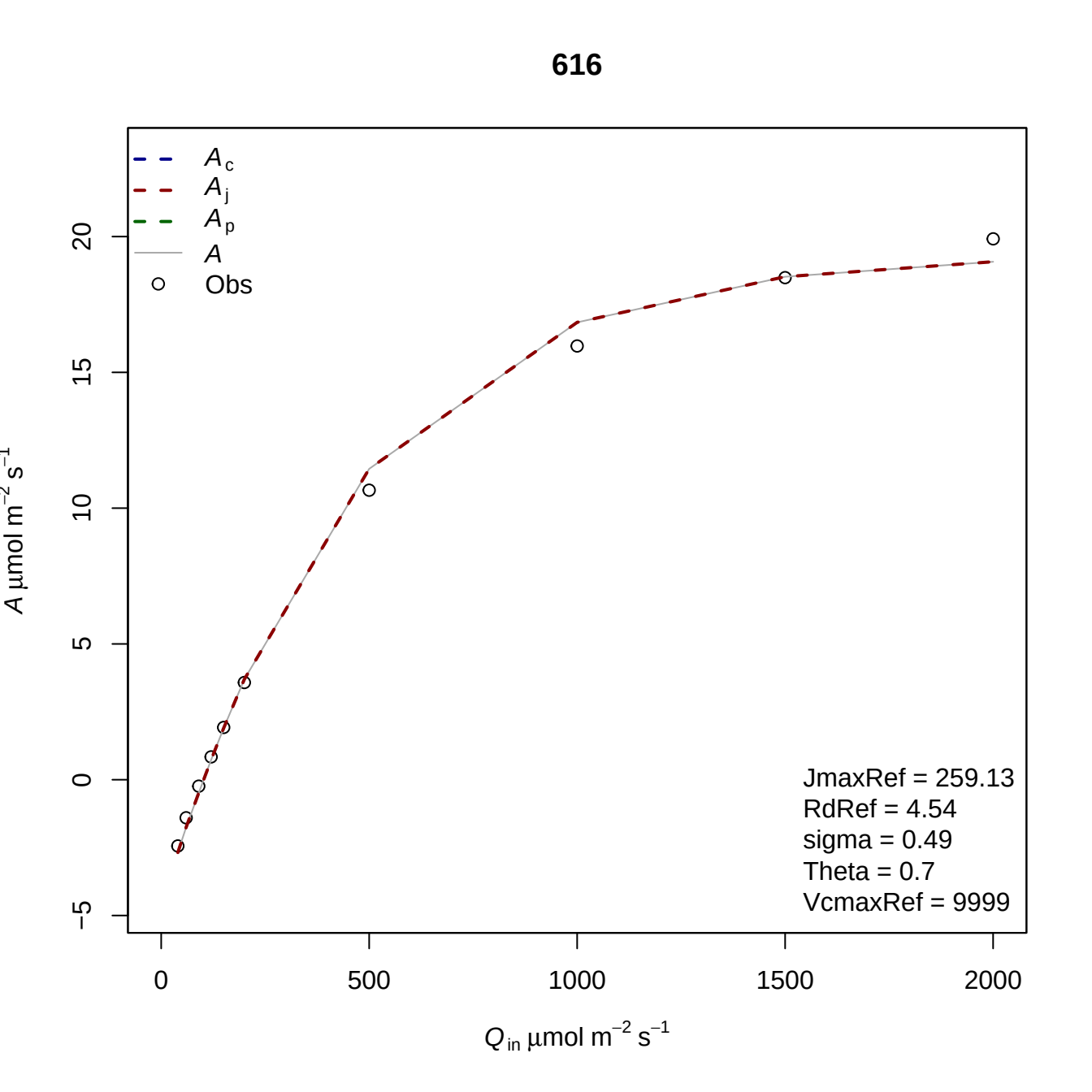

611
A_c
A_j
A_p
A
Obs
JmaxRef = 115.75
RdRef = 1.34
sigma = 0.52
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

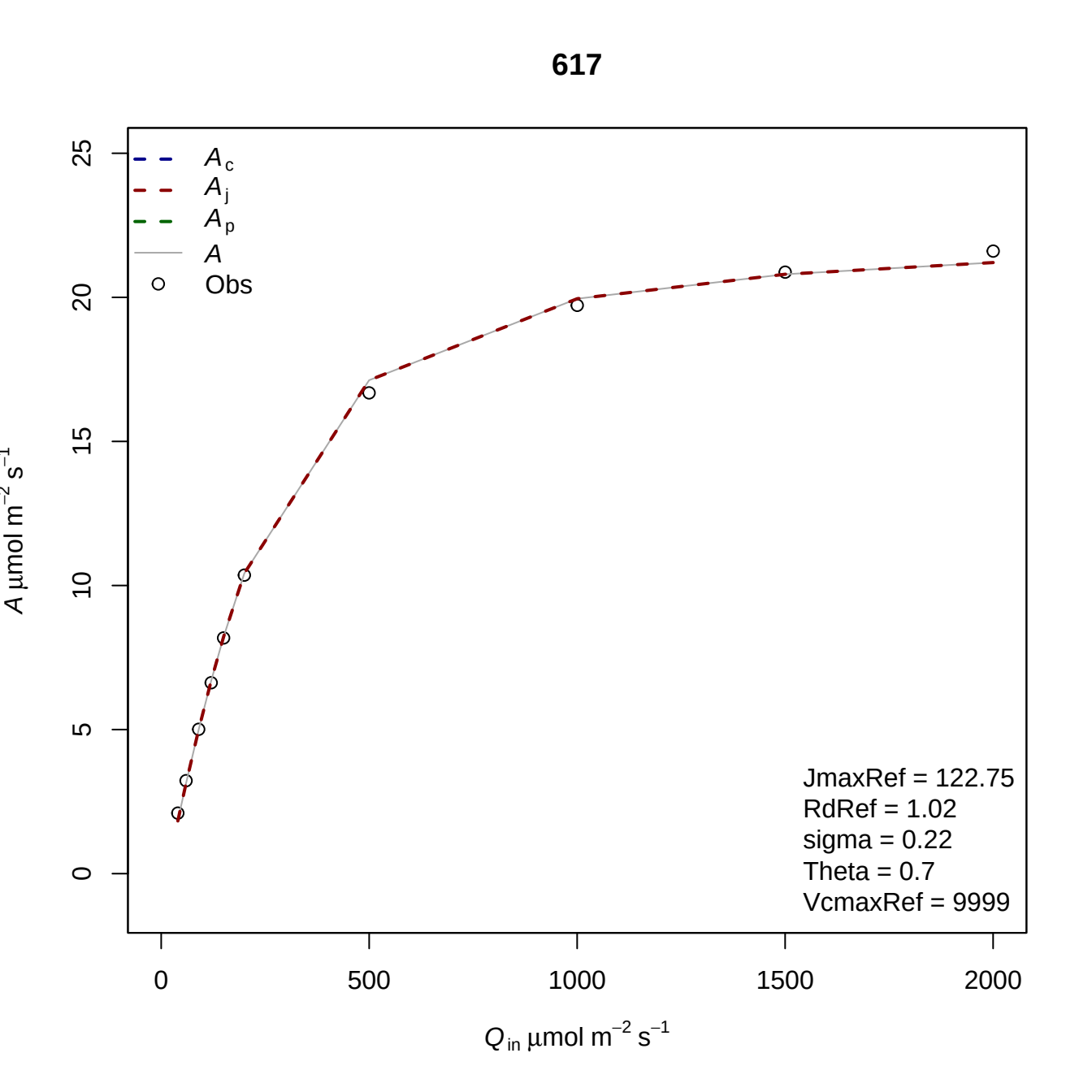

612
A_c
A_j
A_p
A
Obs
JmaxRef = 151.81
RdRef = 1.31
sigma = 0.3
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

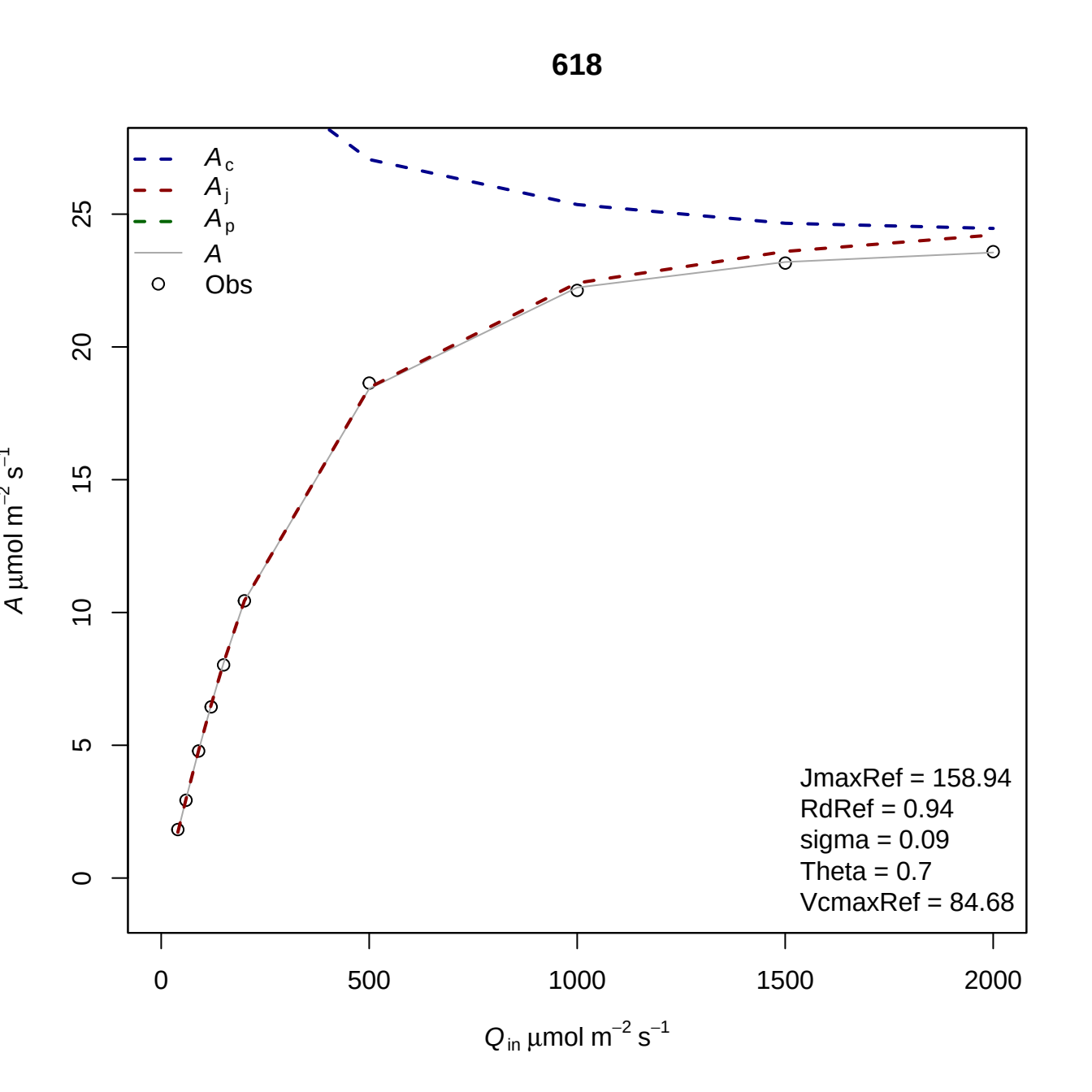

613
A_c
A_j
A_p
A
Obs
JmaxRef = 179.78
RdRef = 1.11
sigma = 0.39
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

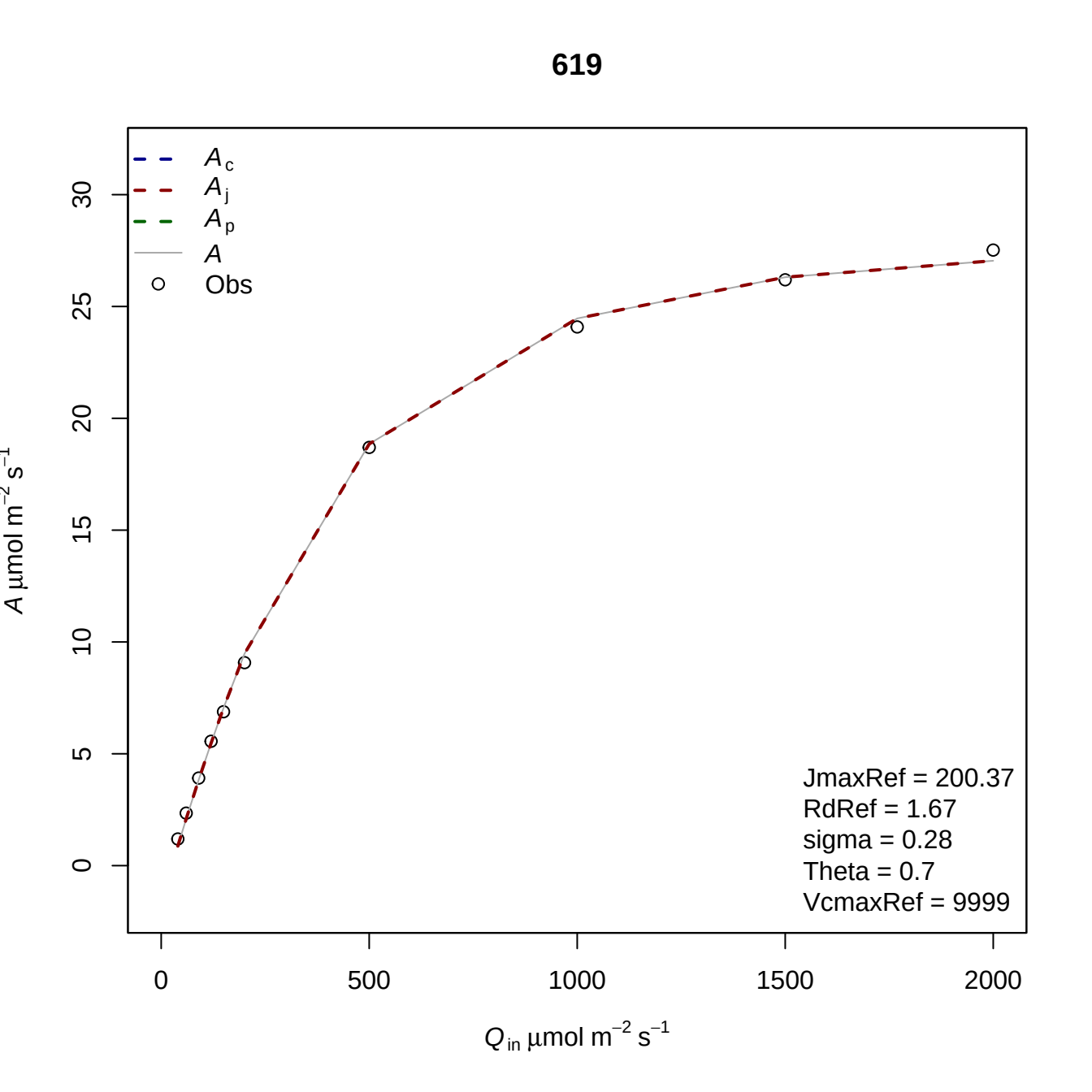

614
A_c
A_j
A_p
A
Obs
JmaxRef = 243.15
RdRef = 2.49
sigma = 0.28
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

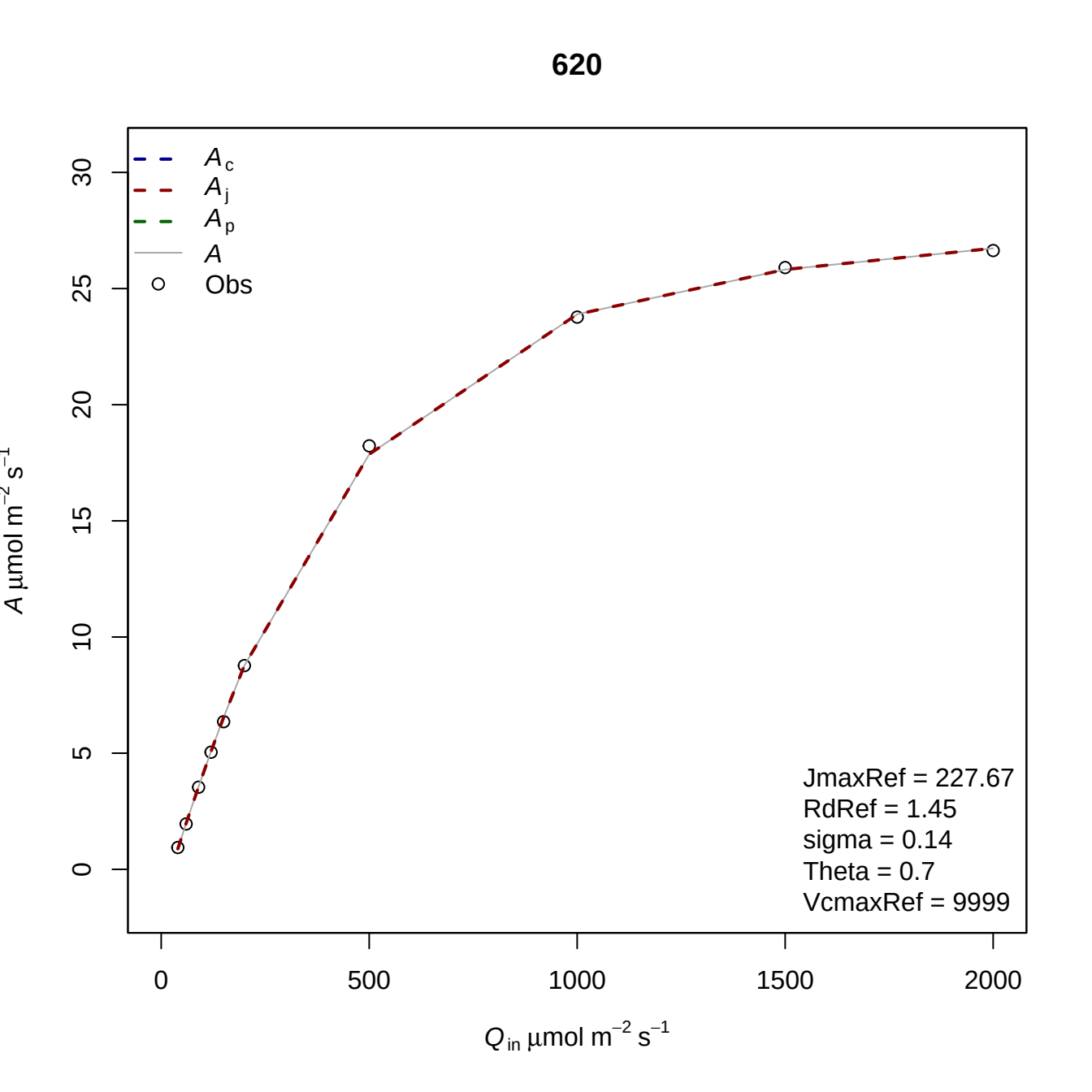

615
A_c
A_j
A_p
A
Obs
JmaxRef = 274.04
RdRef = 3.15
sigma = 0.34
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

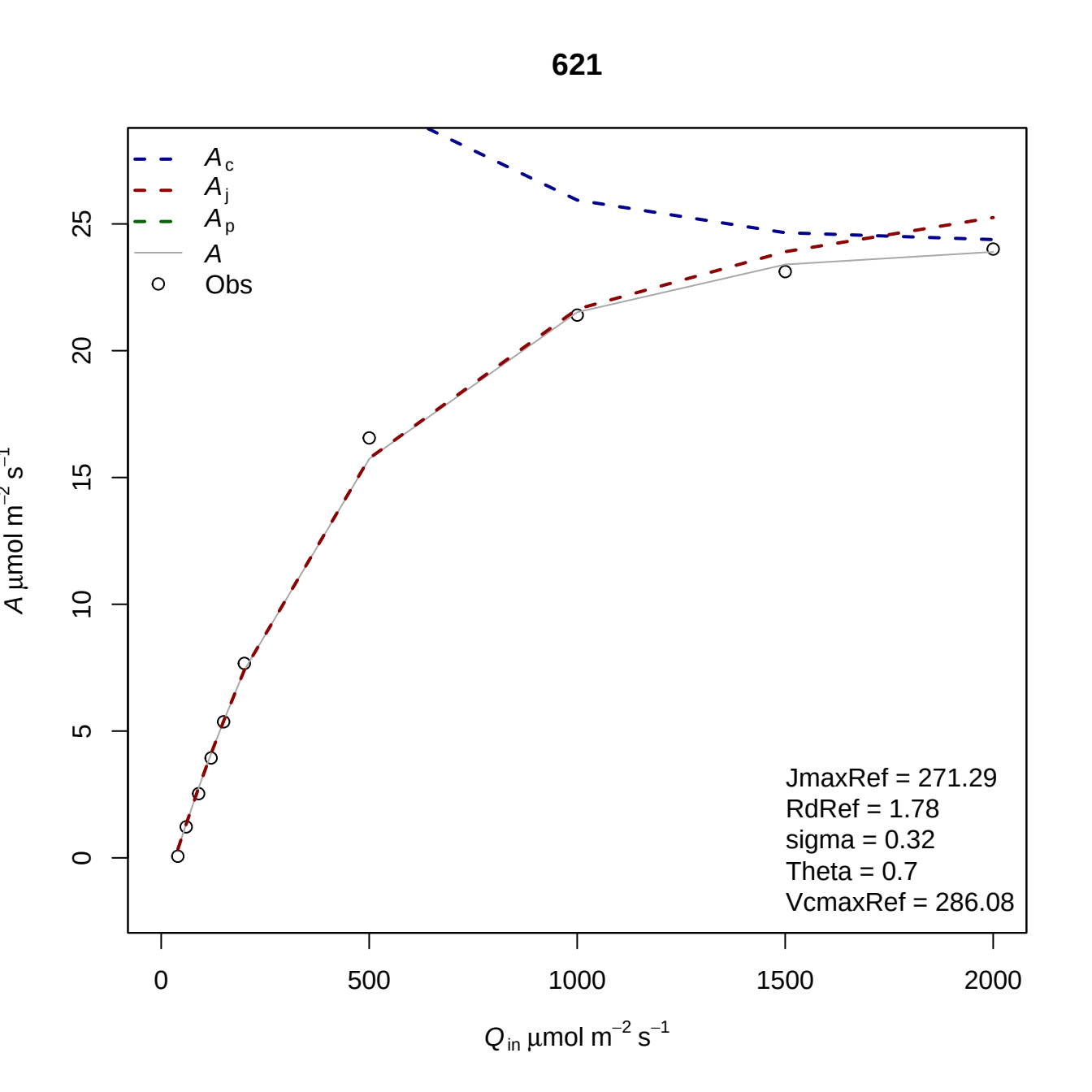

616
A_c
A_j
A_p
A
Obs
JmaxRef = 259.13
RdRef = 4.54
sigma = 0.49
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

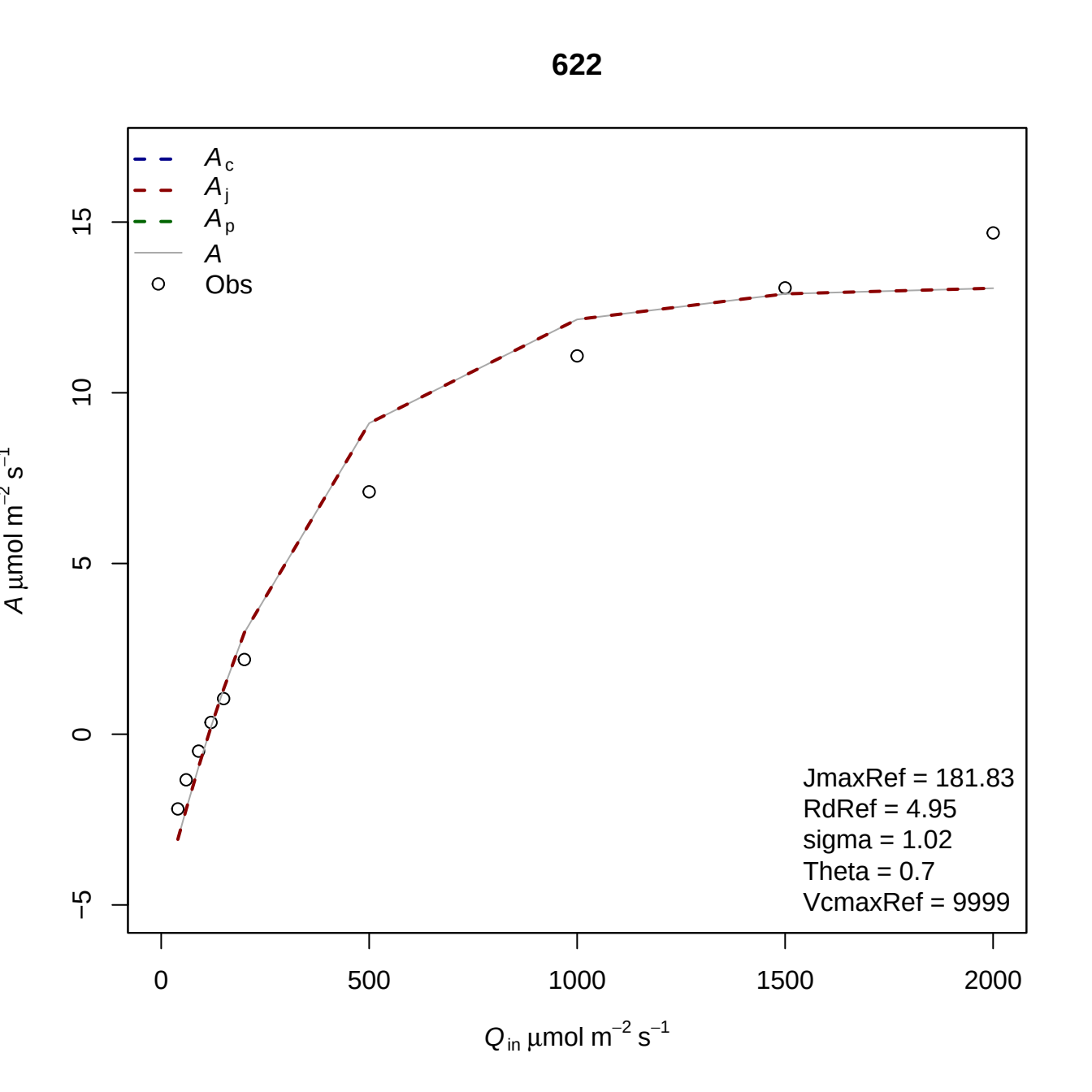

617
A_c
A_j
A_p
A
Obs
JmaxRef = 122.75
RdRef = 1.02
sigma = 0.22
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

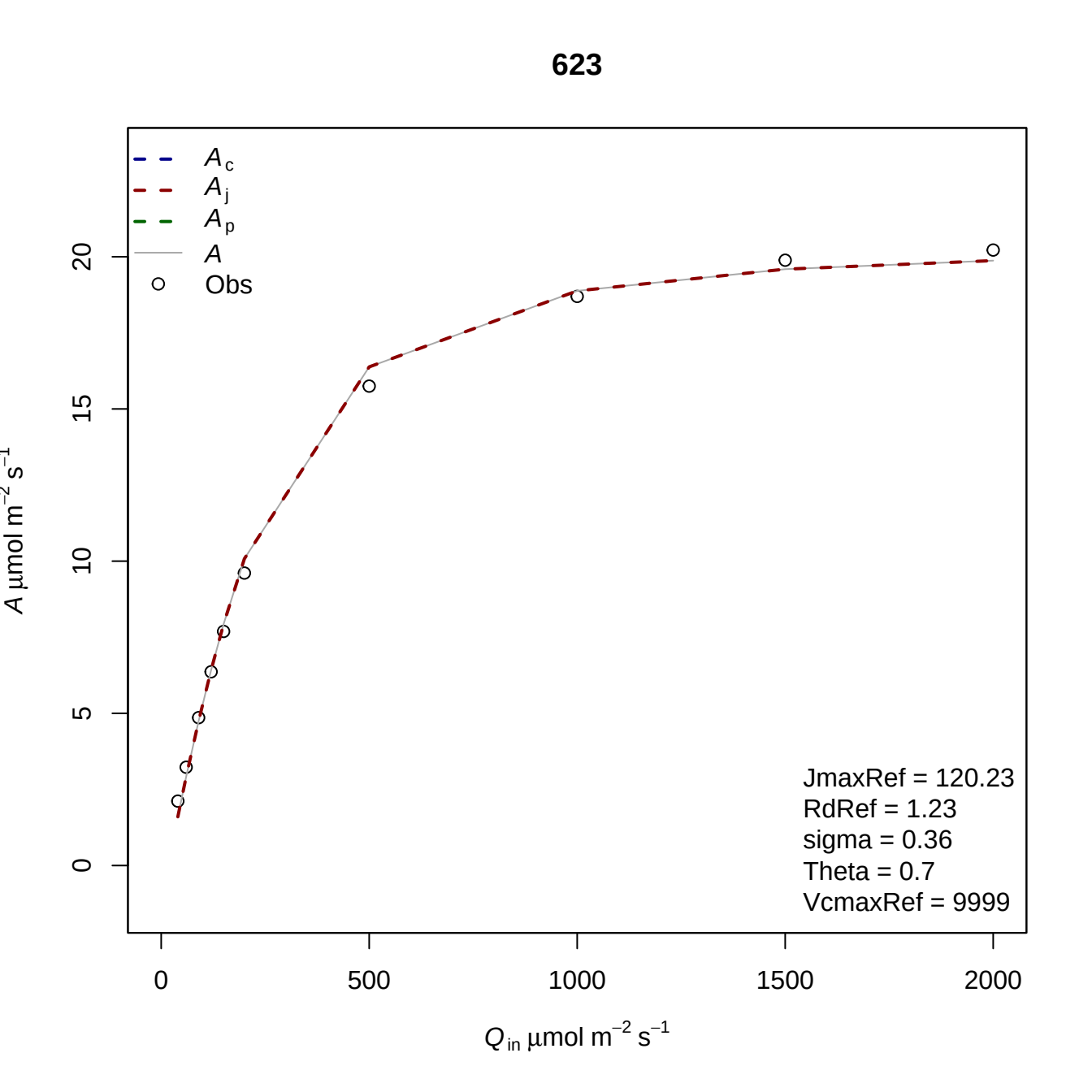

618
A_c
A_j
A_p
A
Obs
JmaxRef = 158.94
RdRef = 0.94
sigma = 0.09
Theta = 0.7
VcmaxRef = 84.68
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

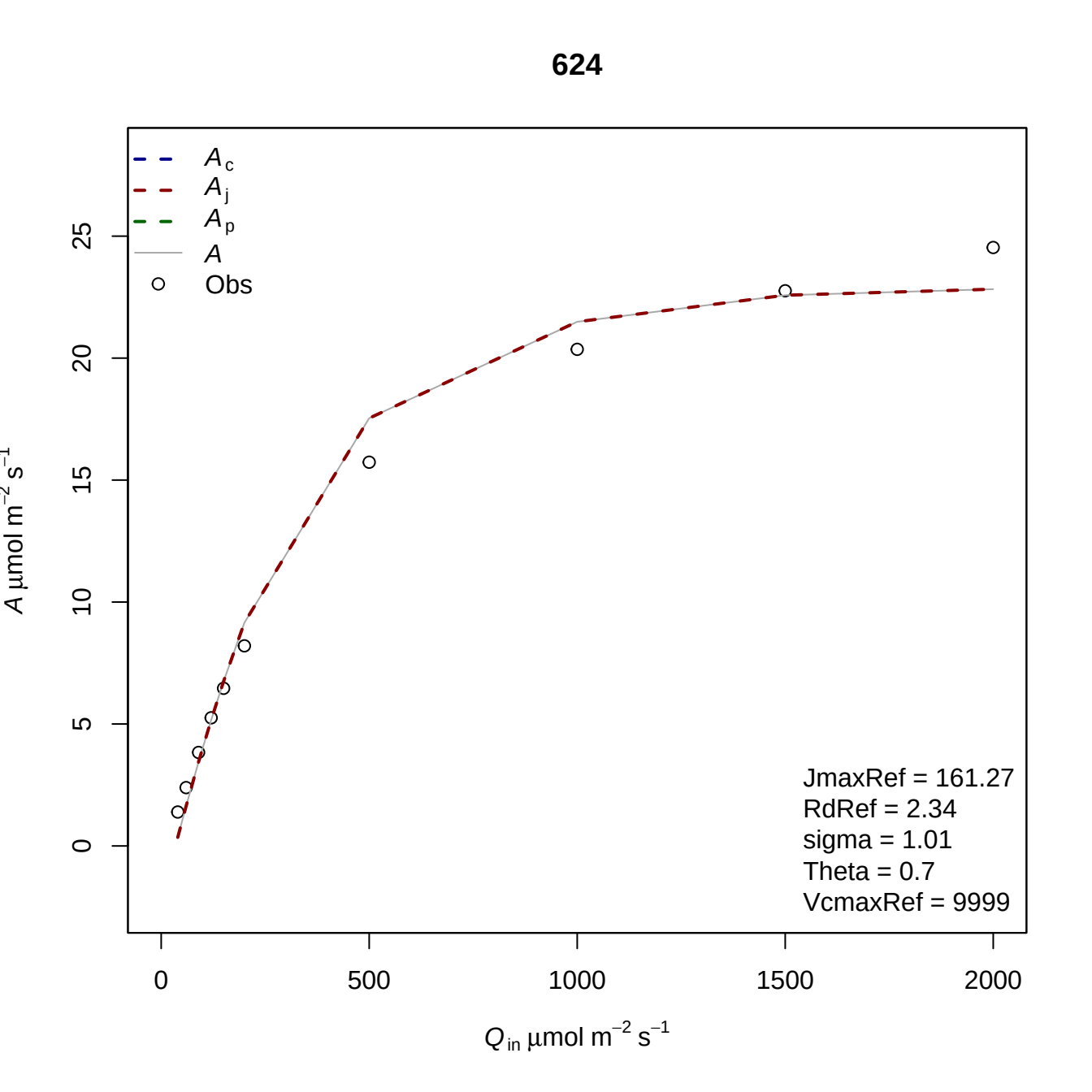

619
A_c
A_j
A_p
A
Obs
JmaxRef = 200.37
RdRef = 1.67
sigma = 0.28
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

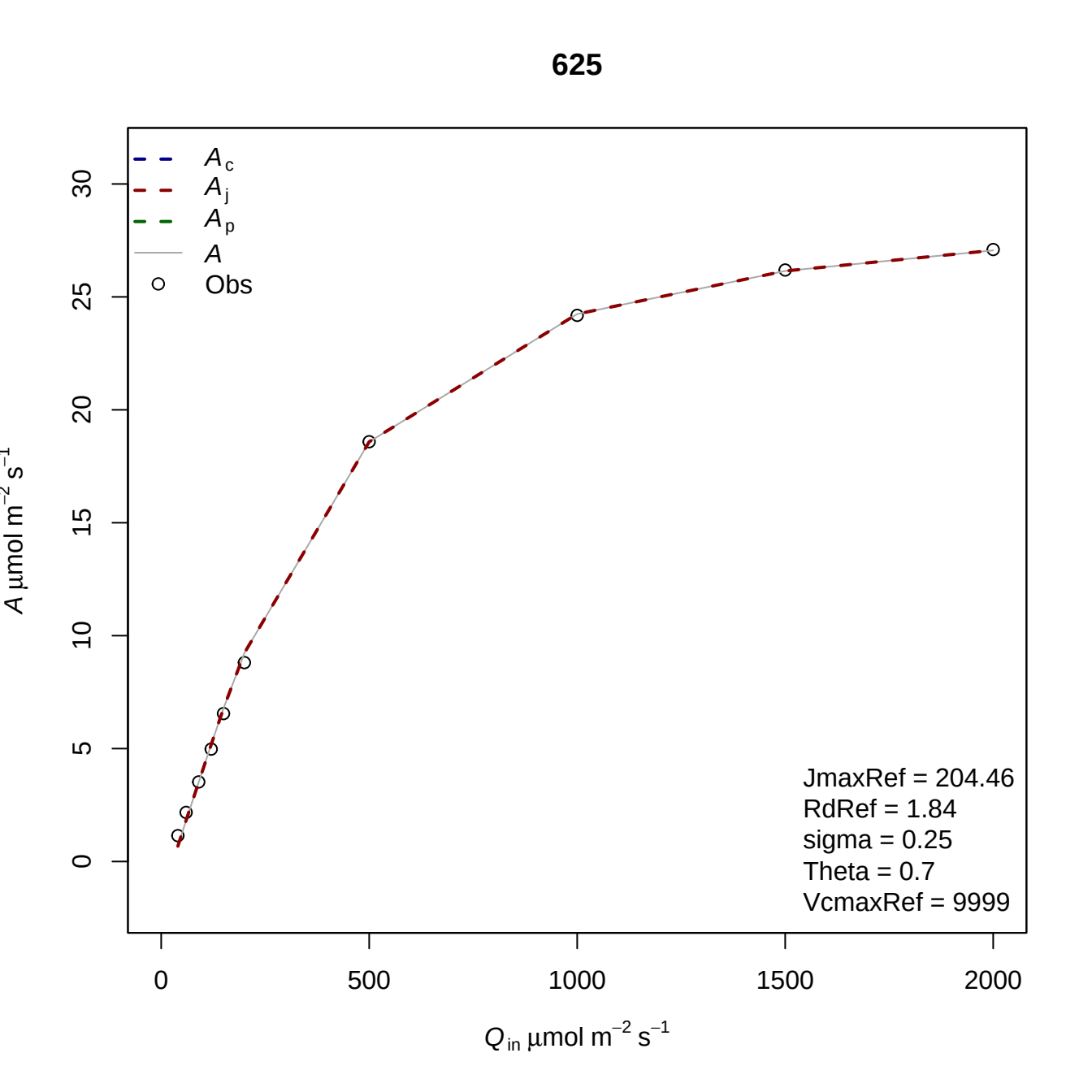

620
A_c
A_j
A_p
A
Obs
JmaxRef = 227.67
RdRef = 1.45
sigma = 0.14
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
25
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

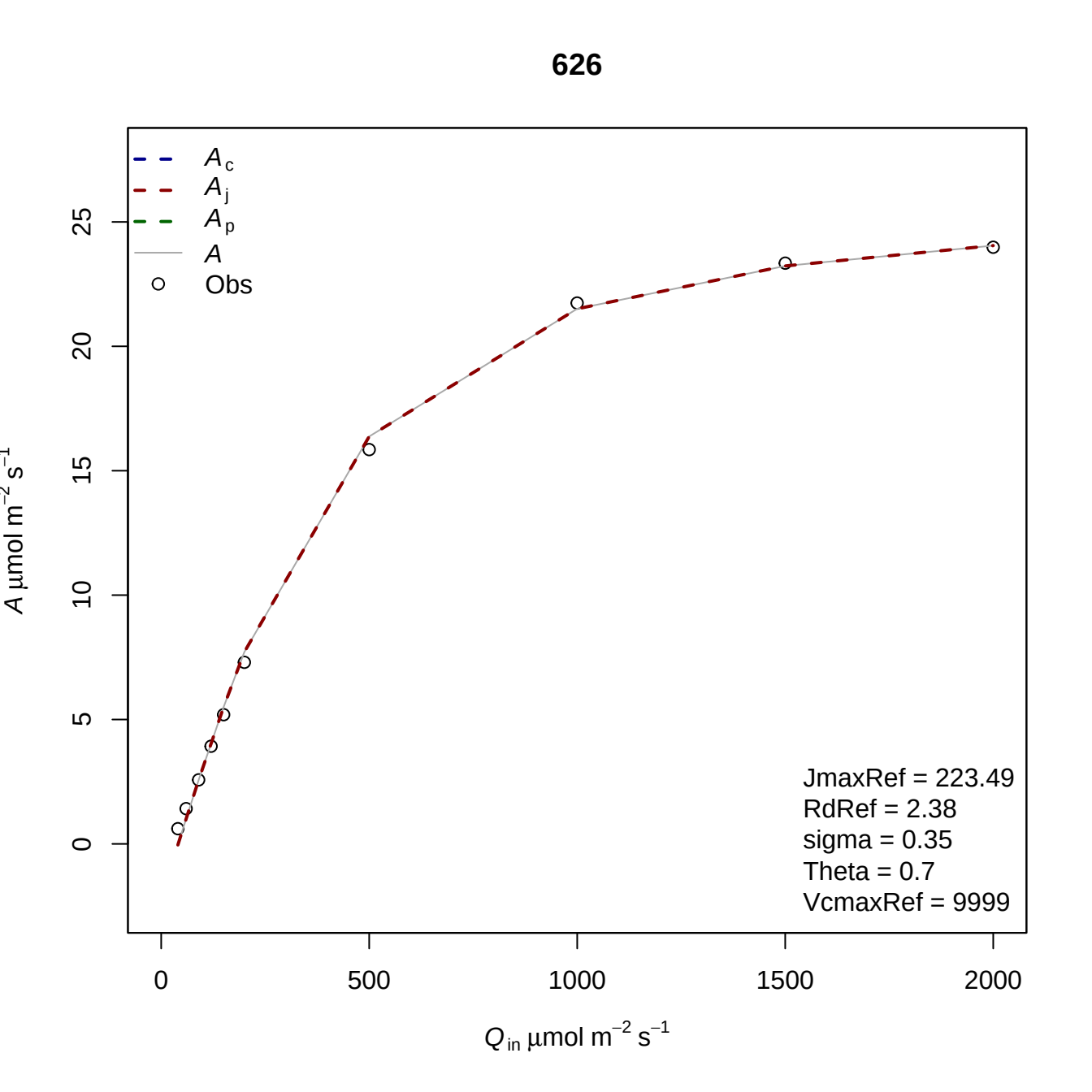

621
A_c
A_j
A_p
A
Obs
JmaxRef = 271.29
RdRef = 1.78
sigma = 0.32
Theta = 0.7
VcmaxRef = 286.08
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

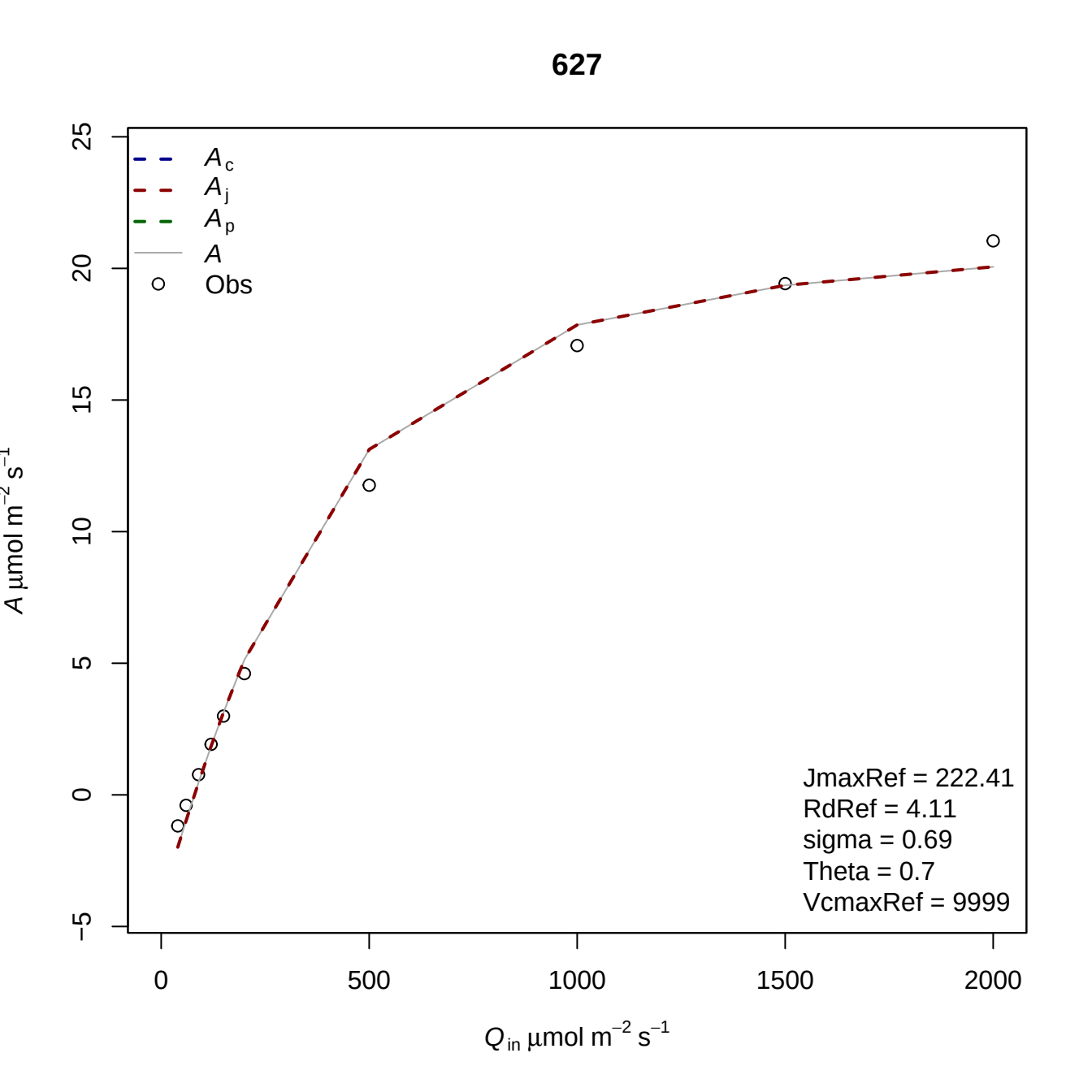

622
A_c
A_j
A_p
A
Obs
JmaxRef = 181.83
RdRef = 4.95
sigma = 1.02
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

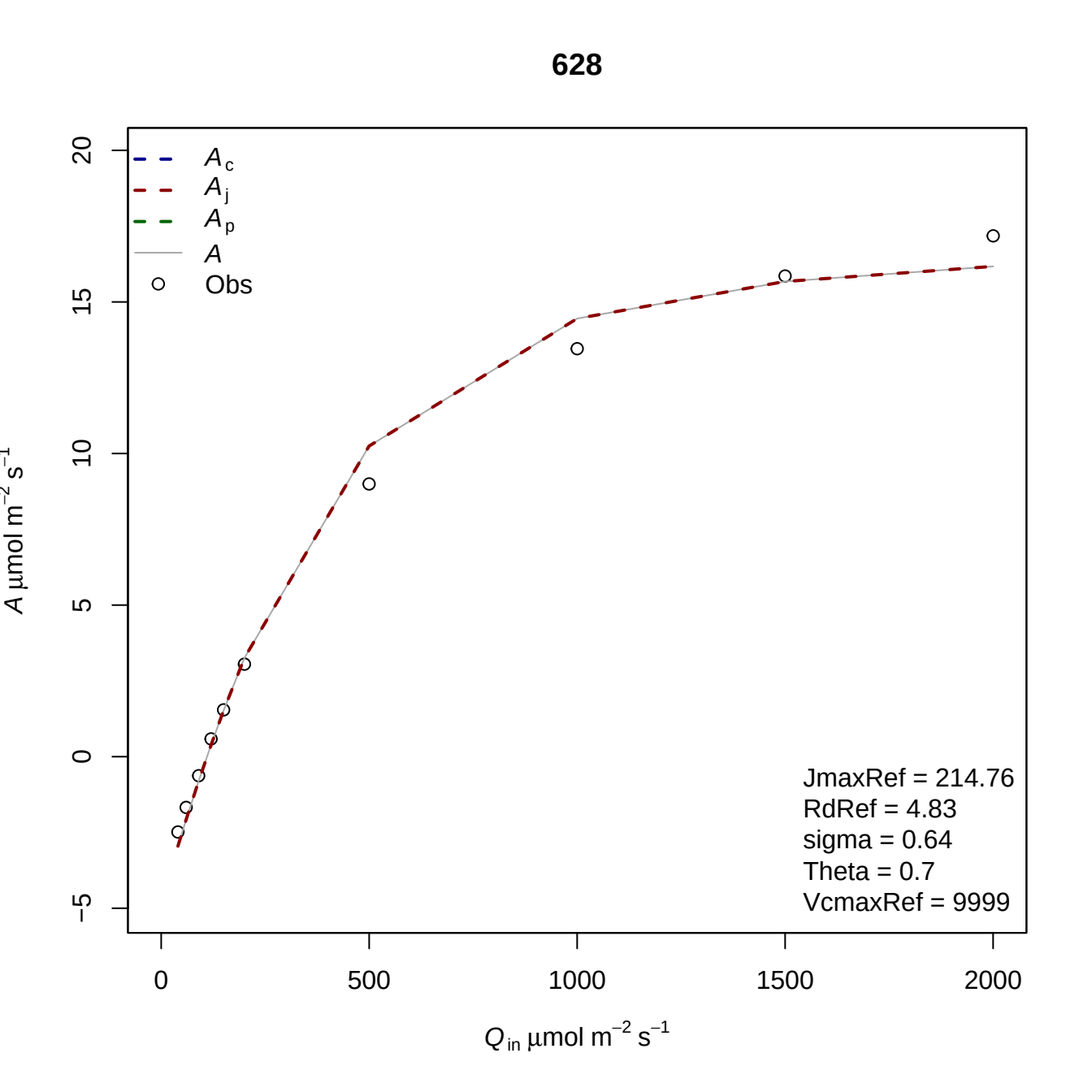

623
A_c
A_j
A_p
A
Obs
JmaxRef = 120.23
RdRef = 1.23
sigma = 0.36
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

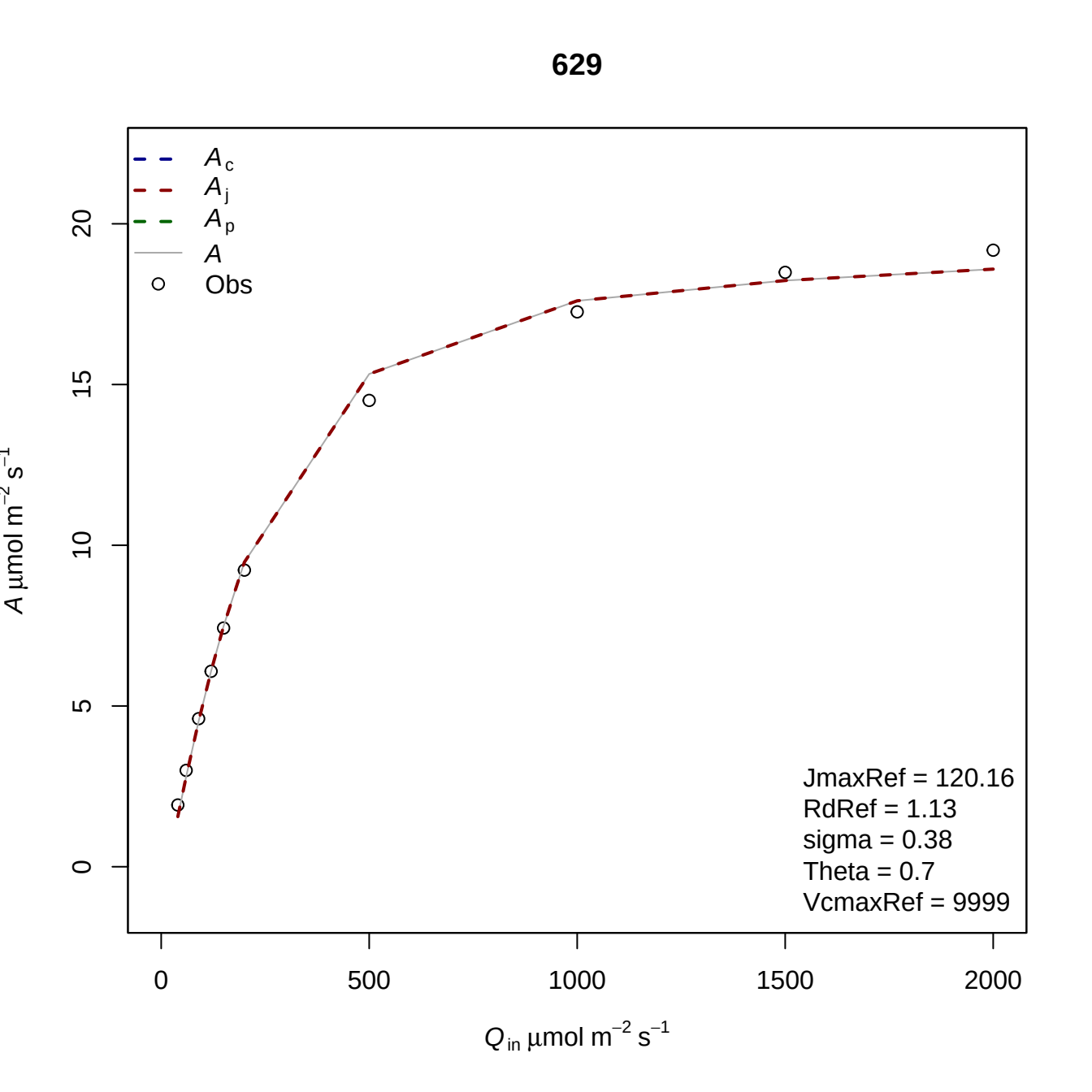

624
A_c
A_j
A_p
A
Obs
JmaxRef = 161.27
RdRef = 2.34
sigma = 1.01
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

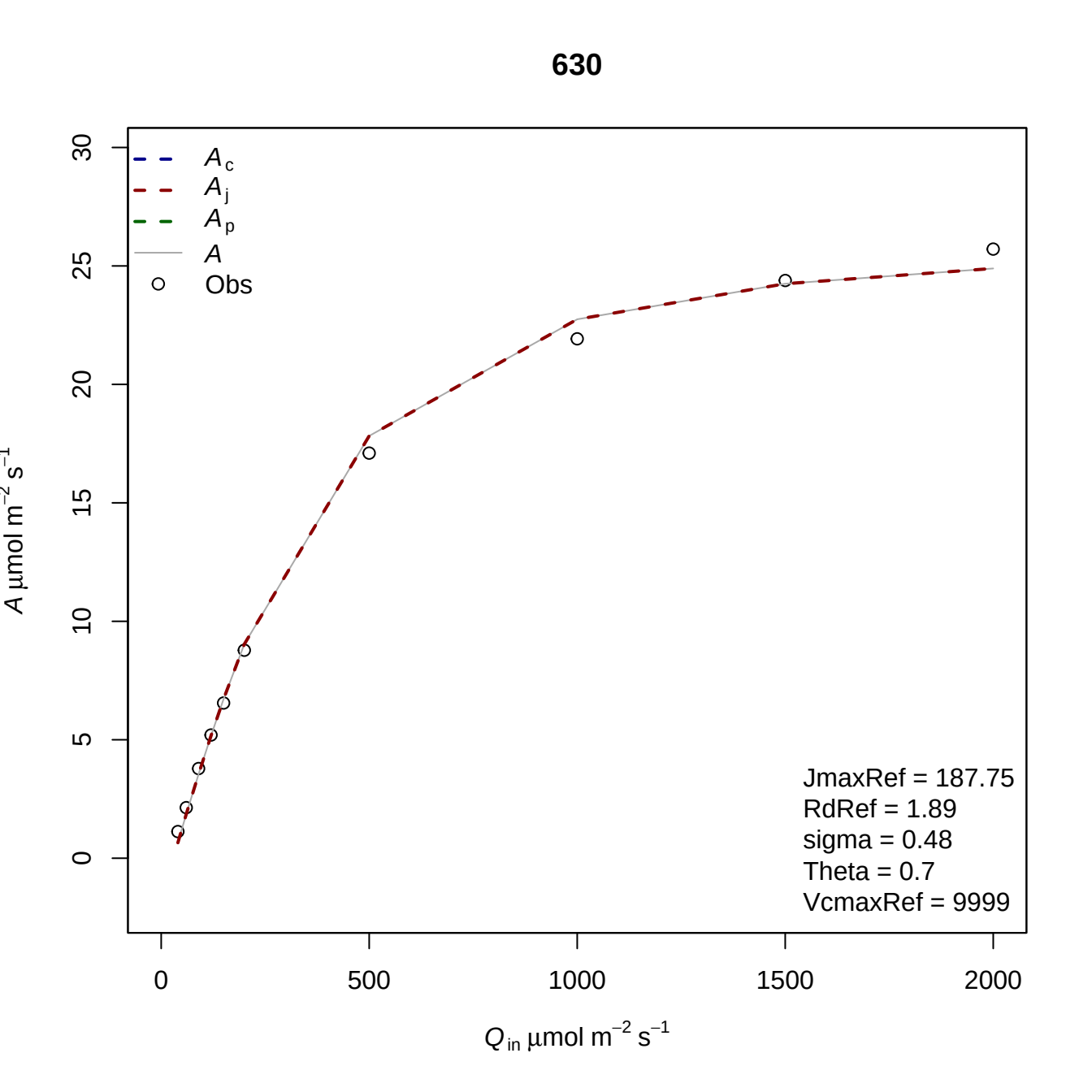

625
A_c
A_j
A_p
A
Obs
JmaxRef = 204.46
RdRef = 1.84
sigma = 0.25
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
25
30
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

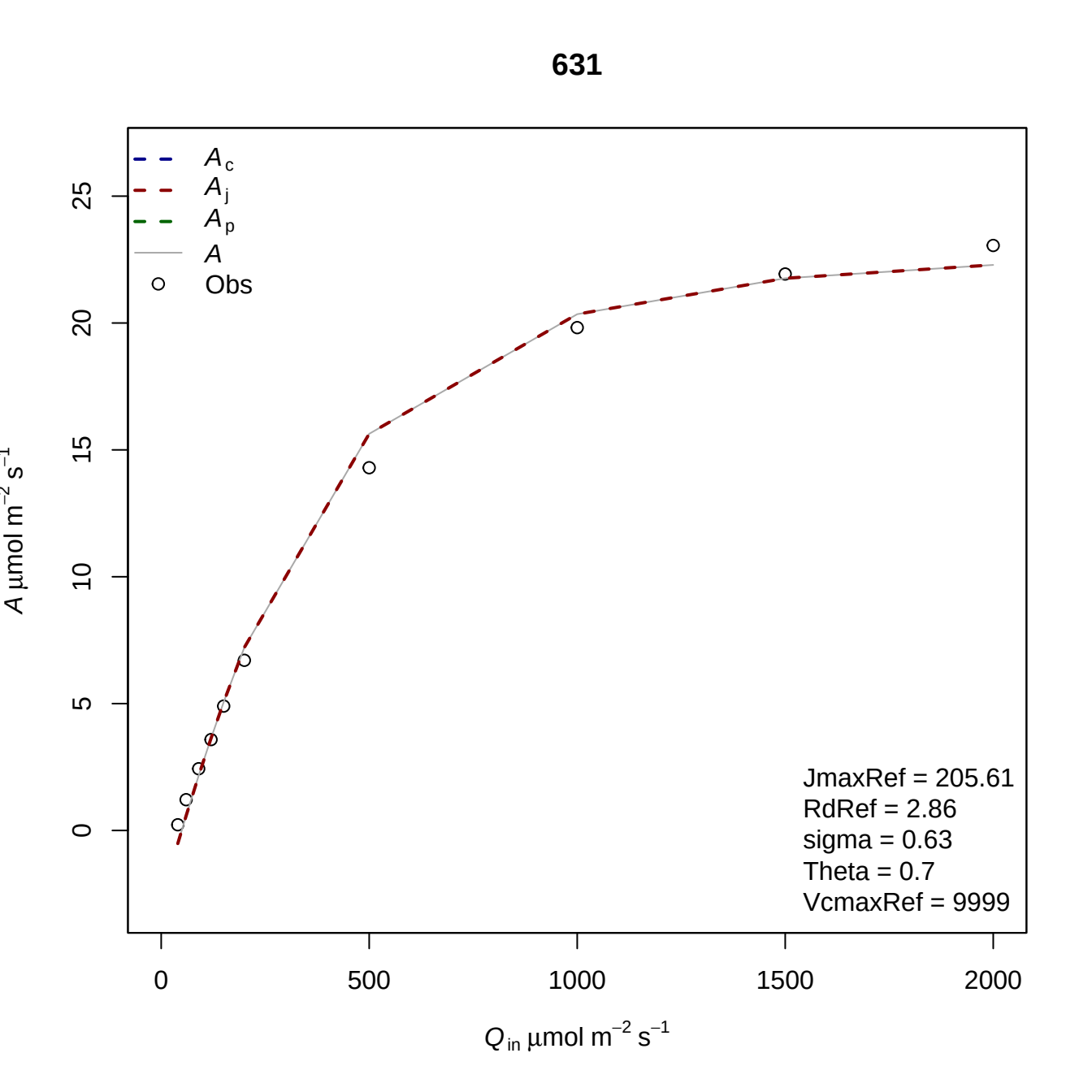

626
A_c
A_j
A_p
A
Obs
JmaxRef = 223.49
RdRef = 2.38
sigma = 0.35
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

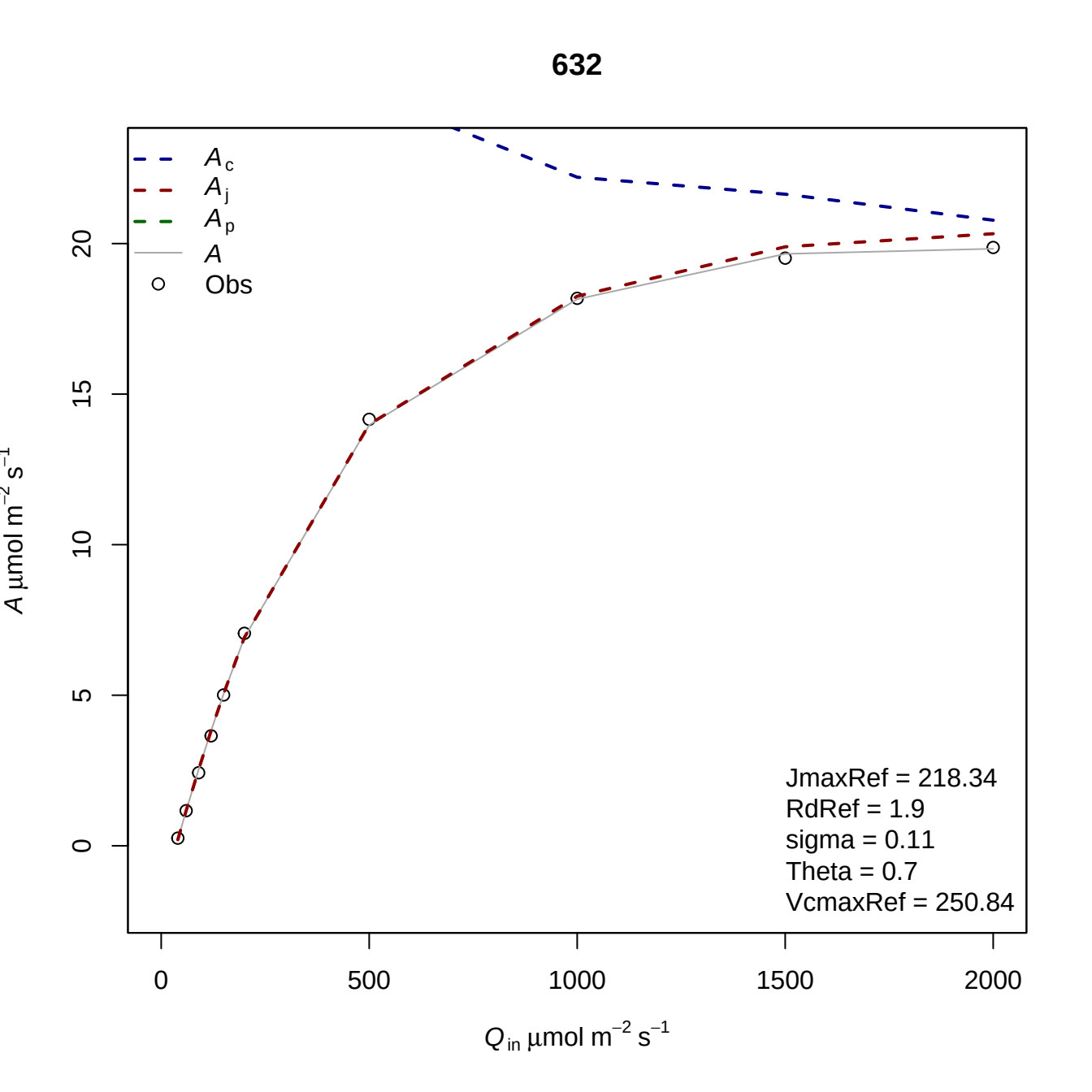

627
A_c
A_j
A_p
A
Obs
JmaxRef = 222.41
RdRef = 4.11
sigma = 0.69
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
25
20
15
10
5
0
−5
0
500
1000
1500
2000

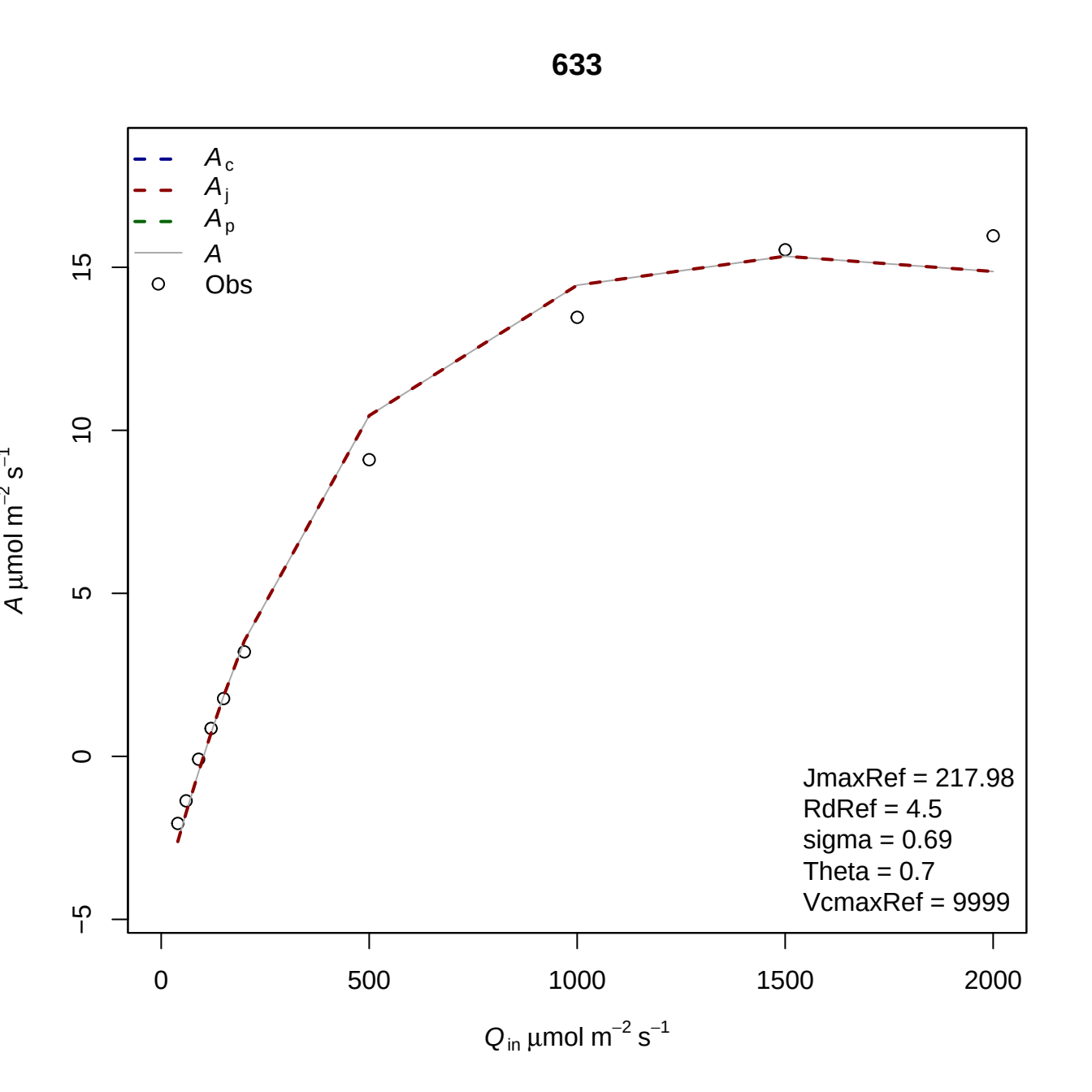

628
A_c
A_j
A_p
A
Obs
JmaxRef = 214.76
RdRef = 4.83
sigma = 0.64
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
20
15
10
5
0
−5
0
500
1000
1500
2000

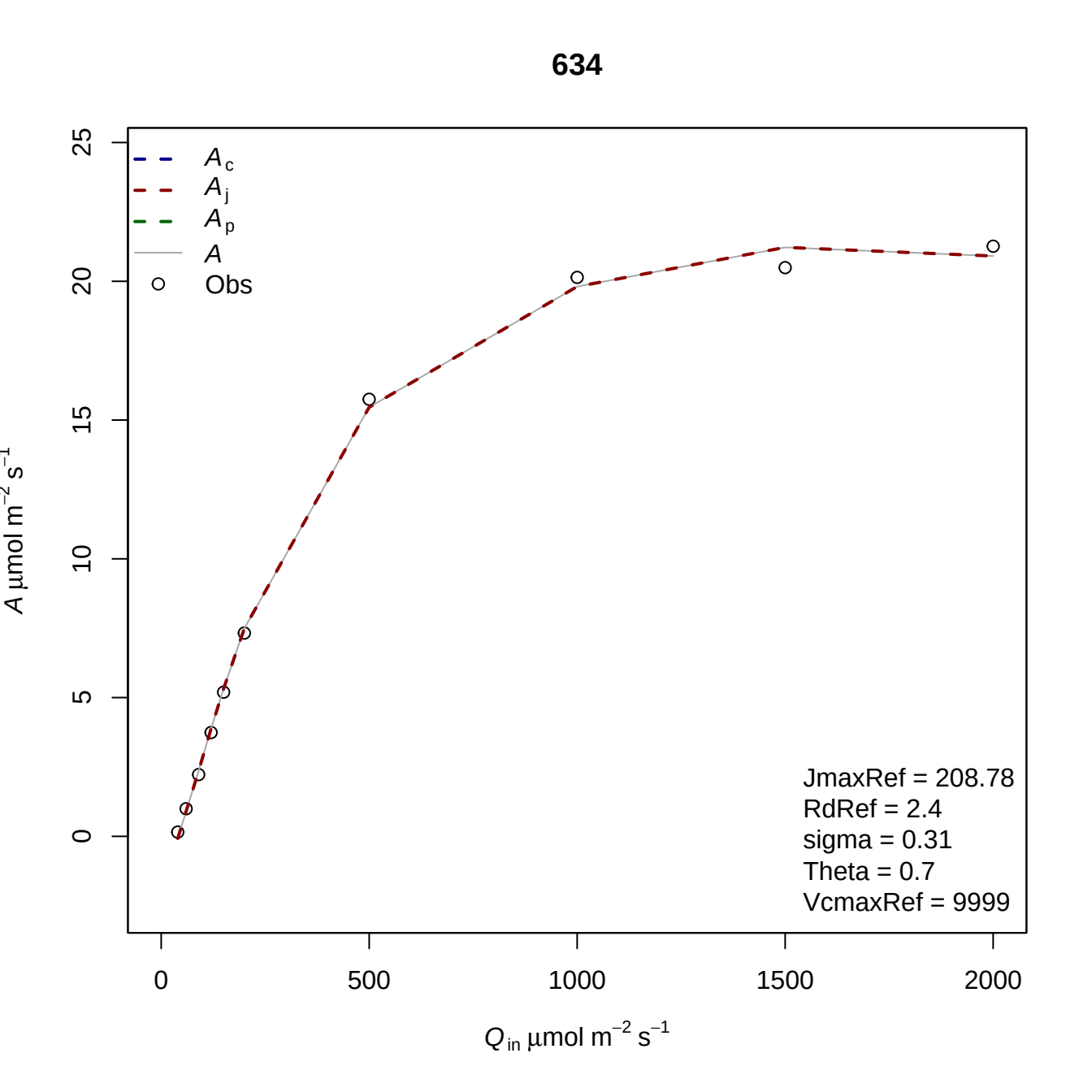

629
A_c
A_j
A_p
A
Obs
JmaxRef = 120.16
RdRef = 1.13
sigma = 0.38
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

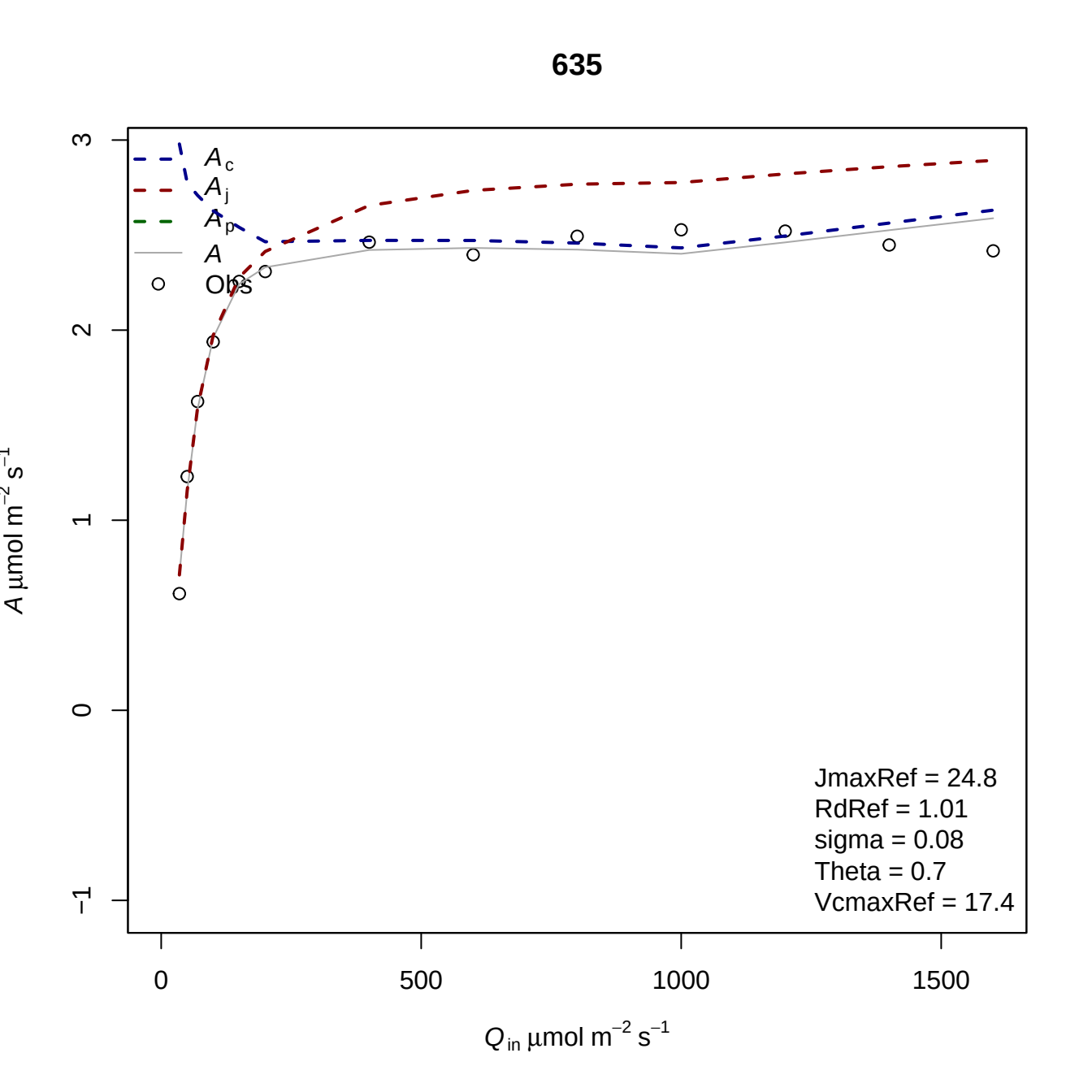

630
A_c
A_j
A_p
A
Obs
JmaxRef = 187.75
RdRef = 1.89
sigma = 0.48
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

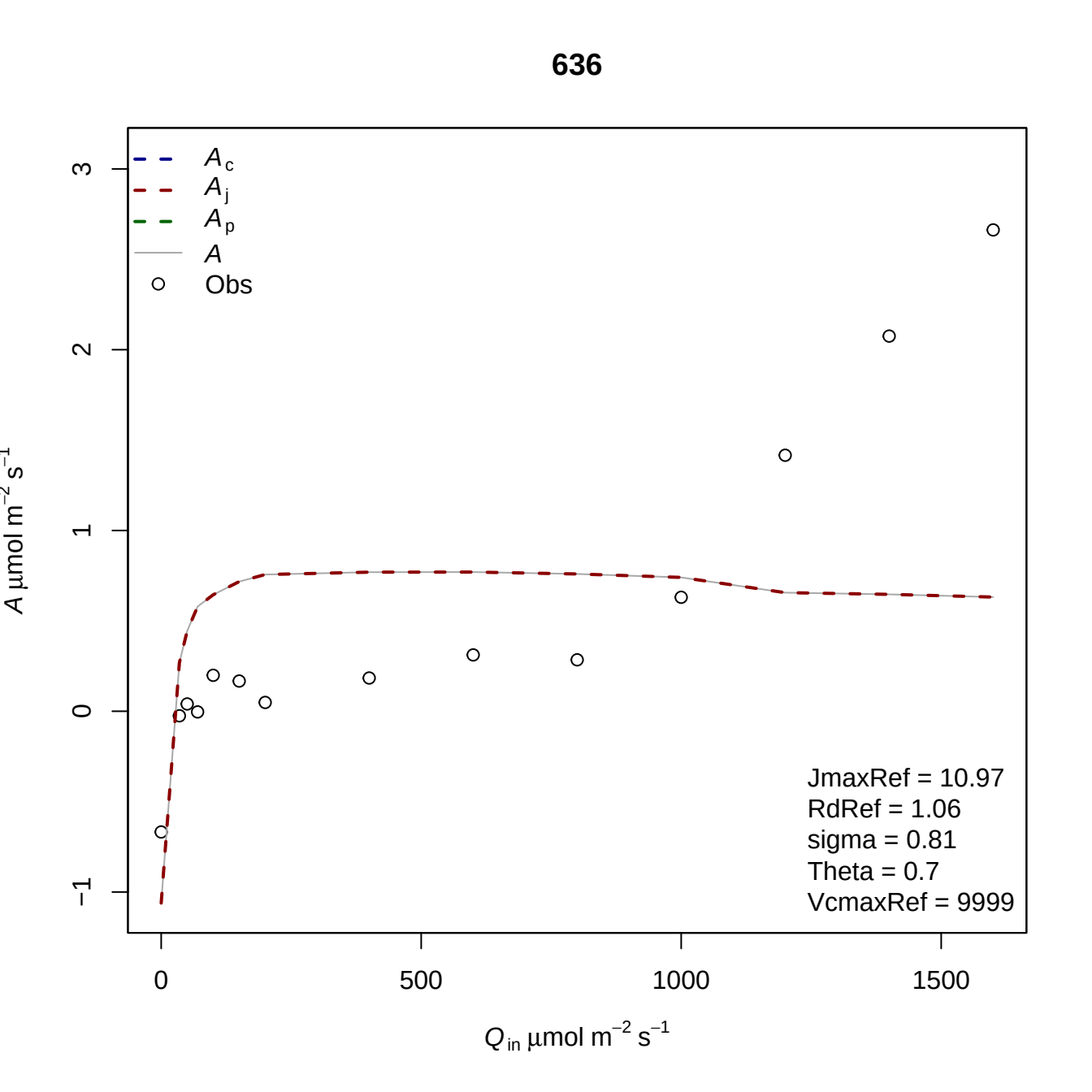

631
A_c
A_j
A_p
A
Obs
JmaxRef = 205.61
RdRef = 2.86
sigma = 0.63
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
25
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

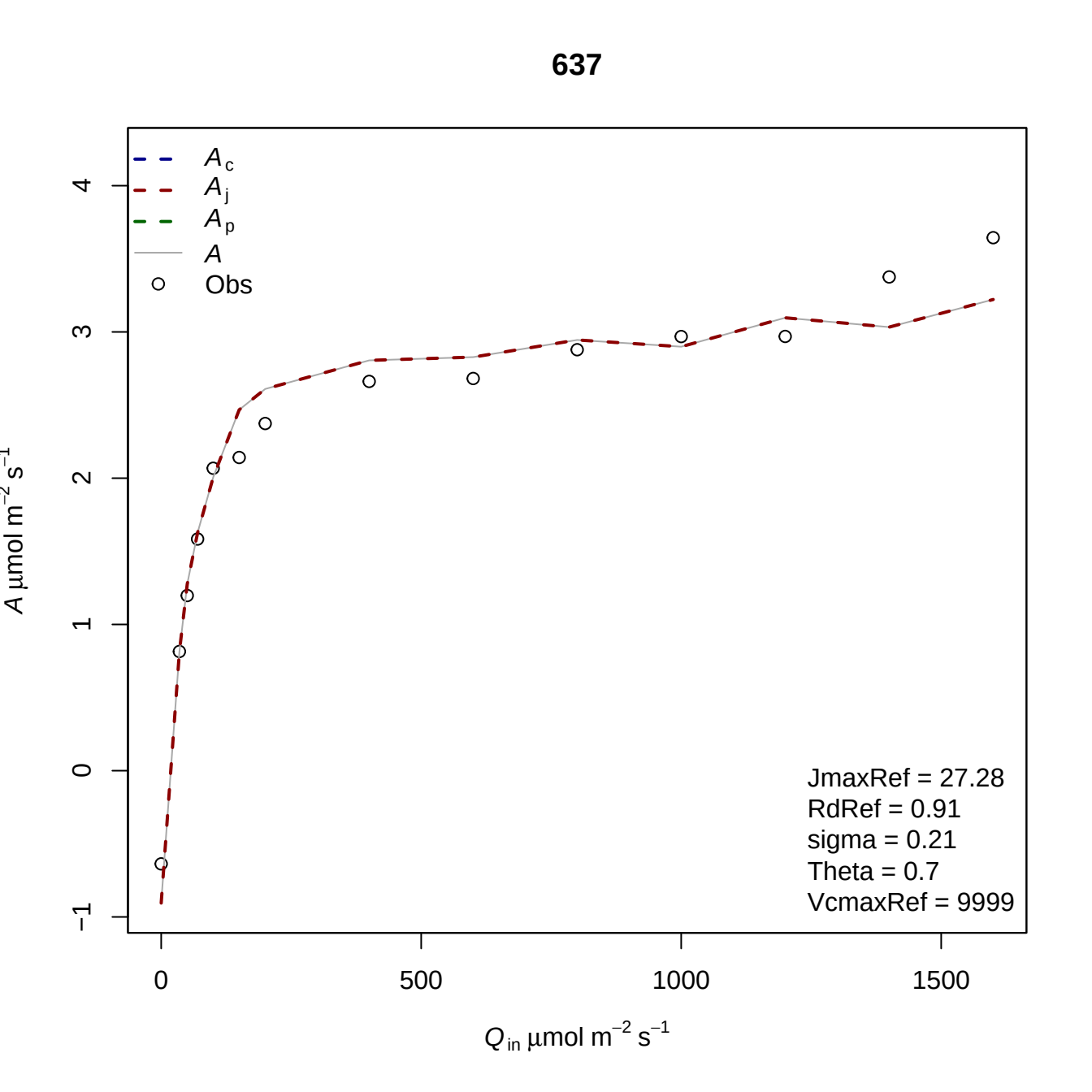

632
A_c
A_j
A_p
A
Obs
JmaxRef = 218.34
RdRef = 1.9
sigma = 0.11
Theta = 0.7
VcmaxRef = 250.84
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

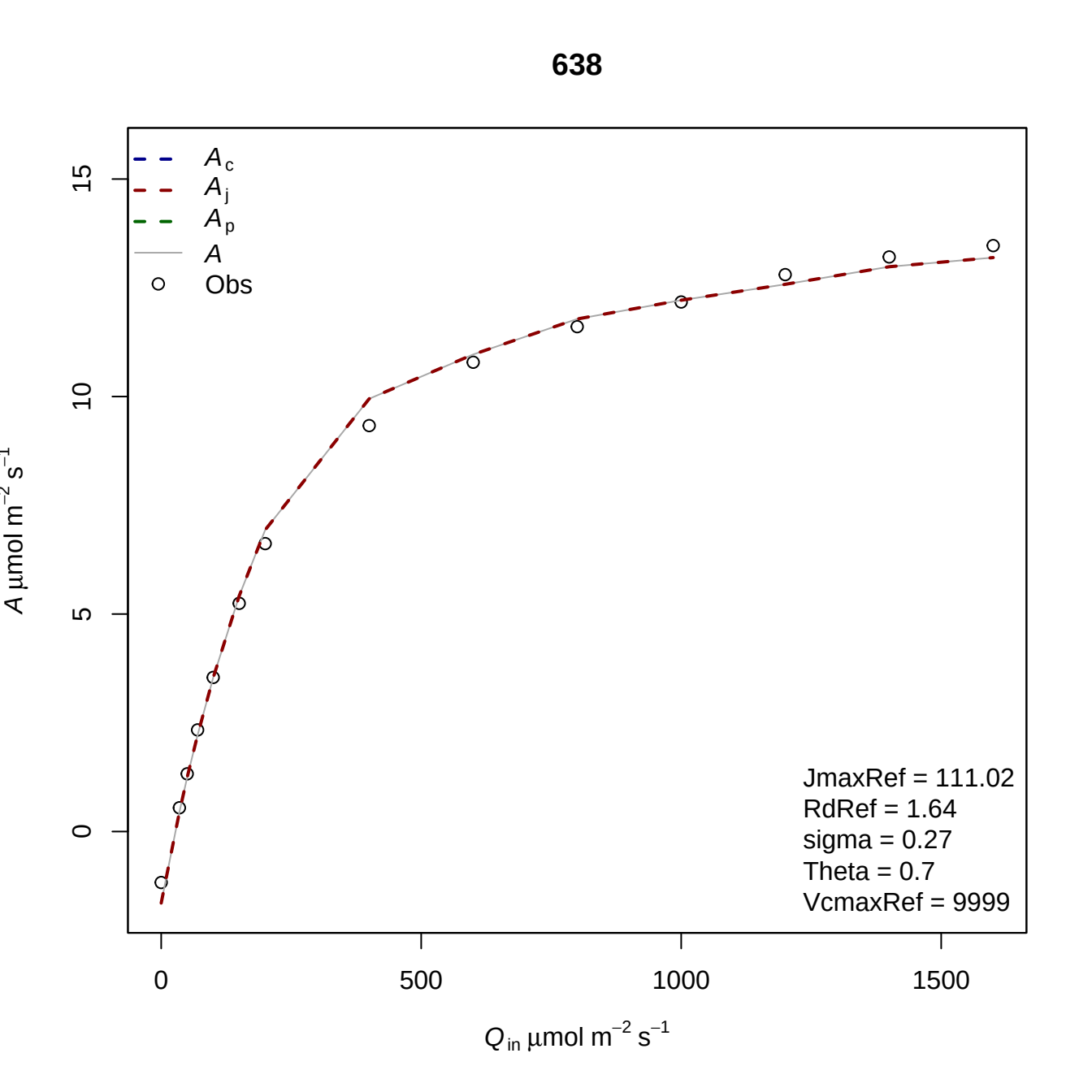

633
A_c
A_j
A_p
A
Obs
JmaxRef = 217.98
RdRef = 4.5
sigma = 0.69
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

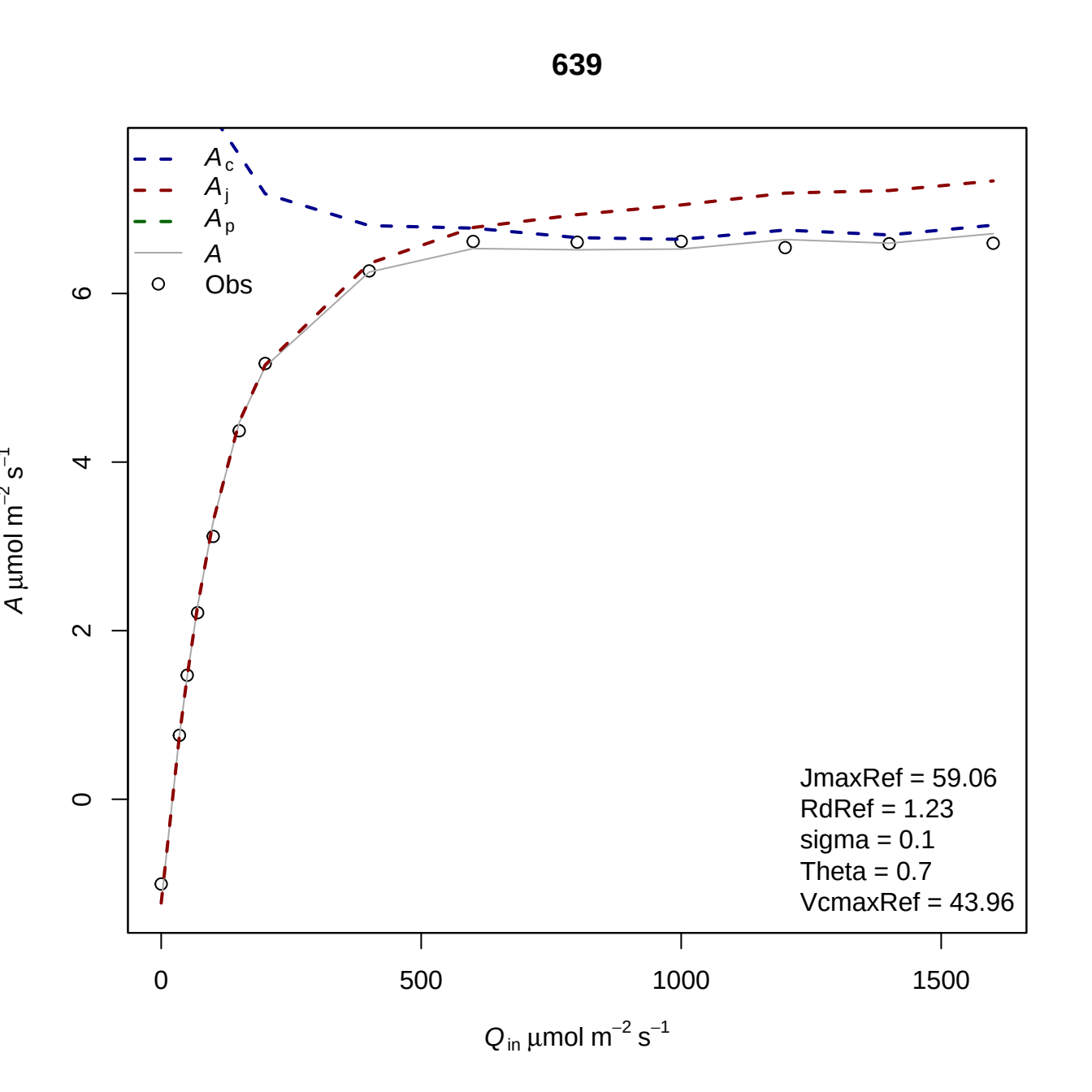

634
A_c
A_j
A_p
A
Obs
JmaxRef = 208.78
RdRef = 2.4
sigma = 0.31
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

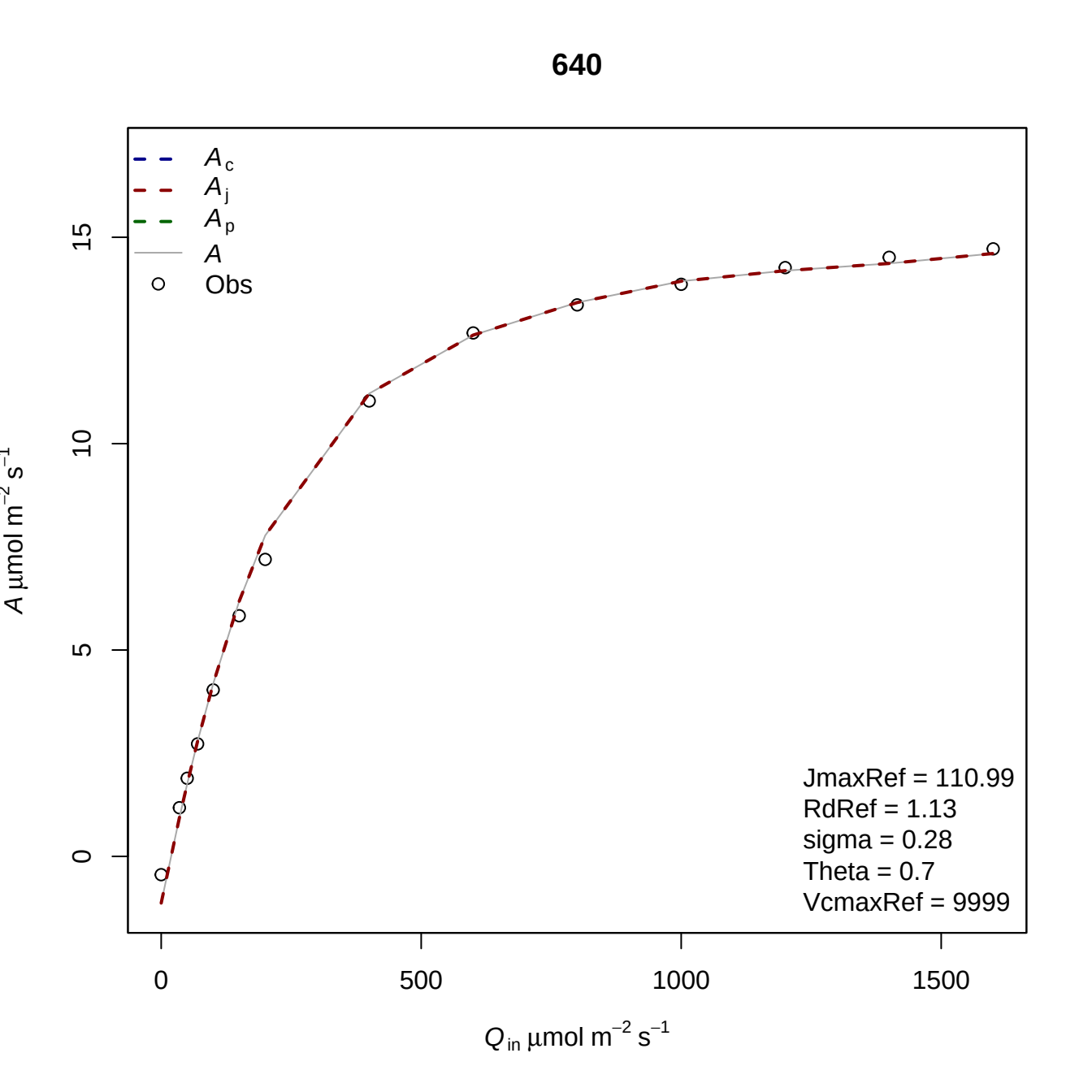

635
A_c
A_j
A_p
A
Obs
JmaxRef = 24.8
RdRef = 1.01
sigma = 0.08
Theta = 0.7
VcmaxRef = 17.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
0
500
1000
1500

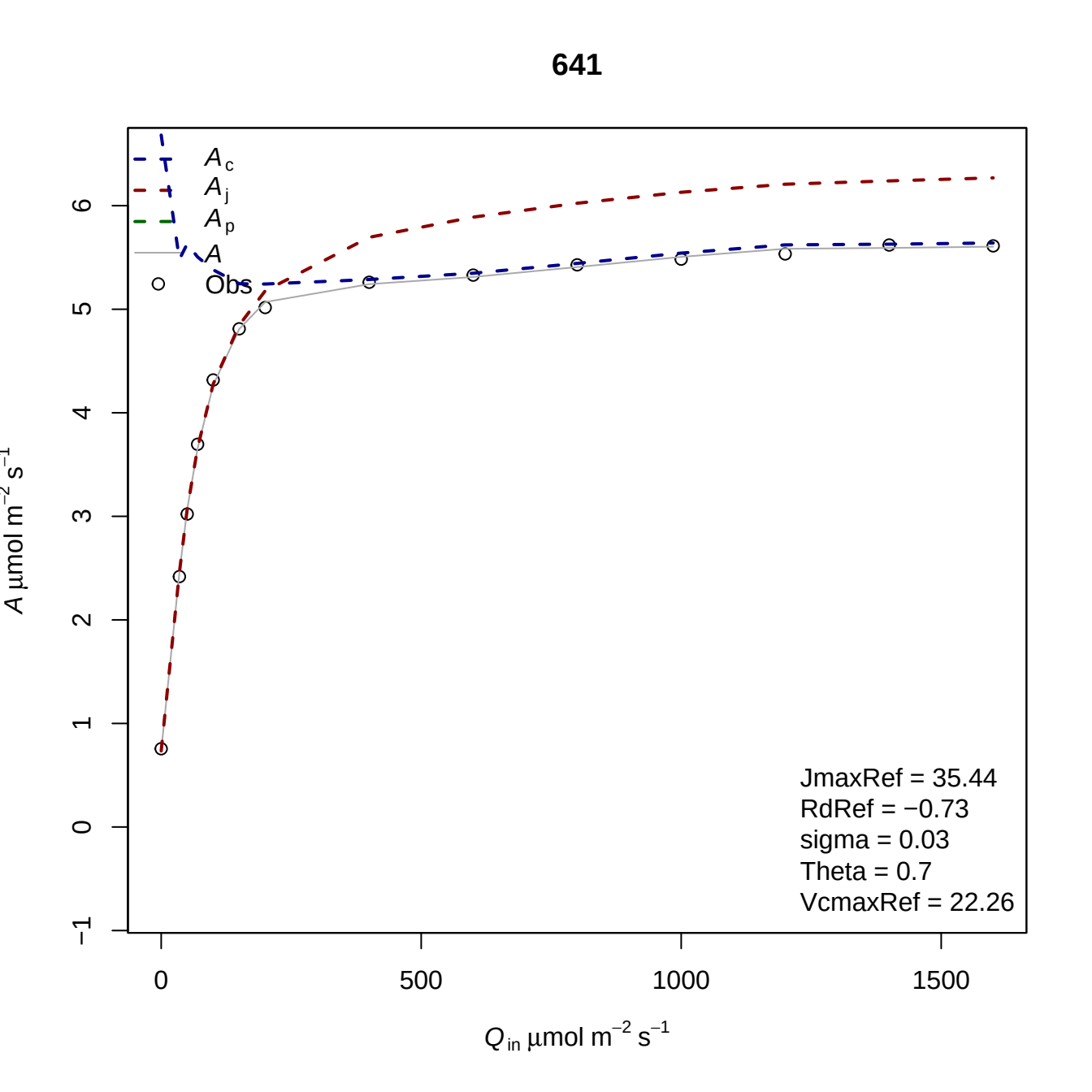

636
A_c
A_j
A_p
A
Obs
JmaxRef = 10.97
RdRef = 1.06
sigma = 0.81
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
0
500
1000
1500

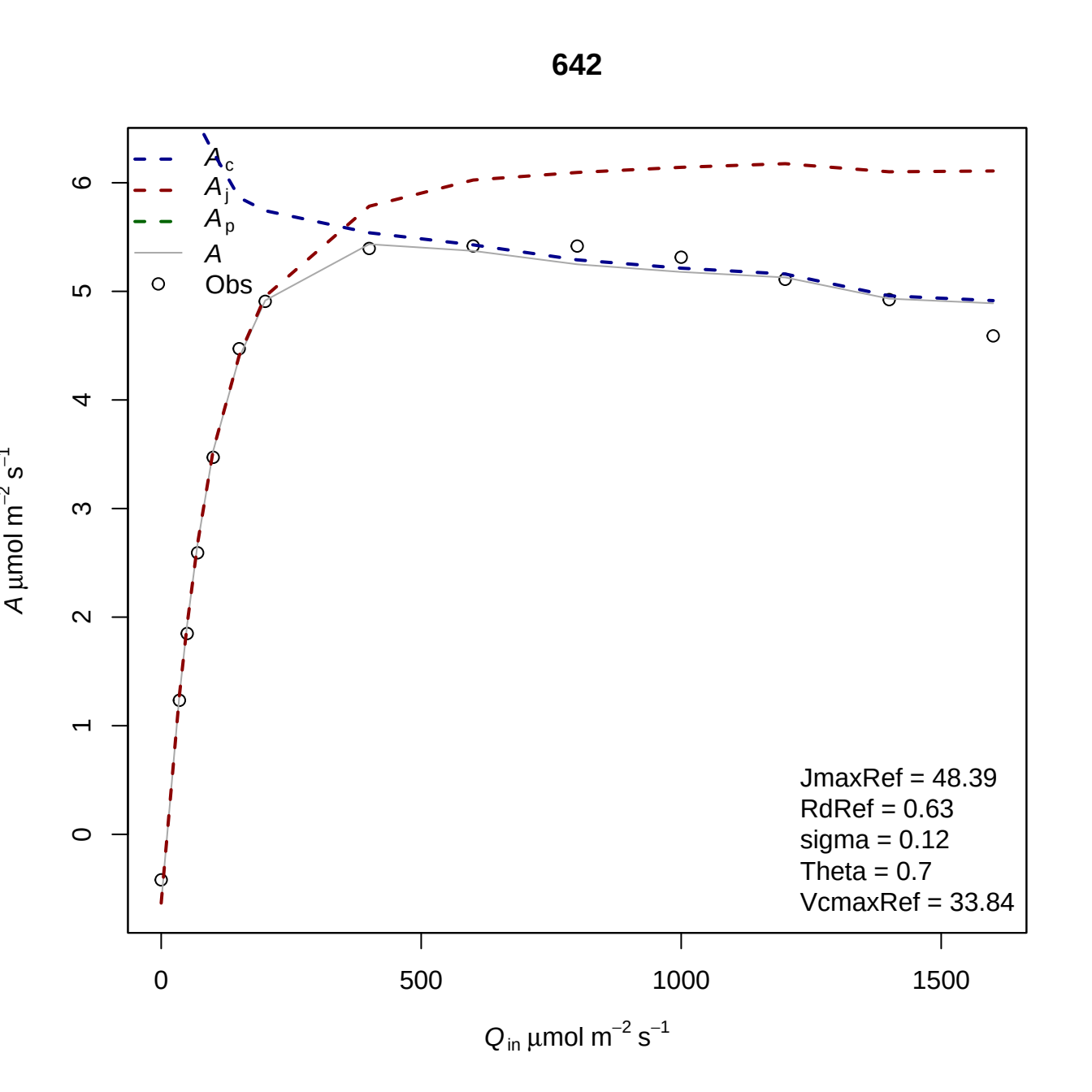

637
A_c
A_j
A_p
A
Obs
JmaxRef = 27.28
RdRef = 0.91
sigma = 0.21
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

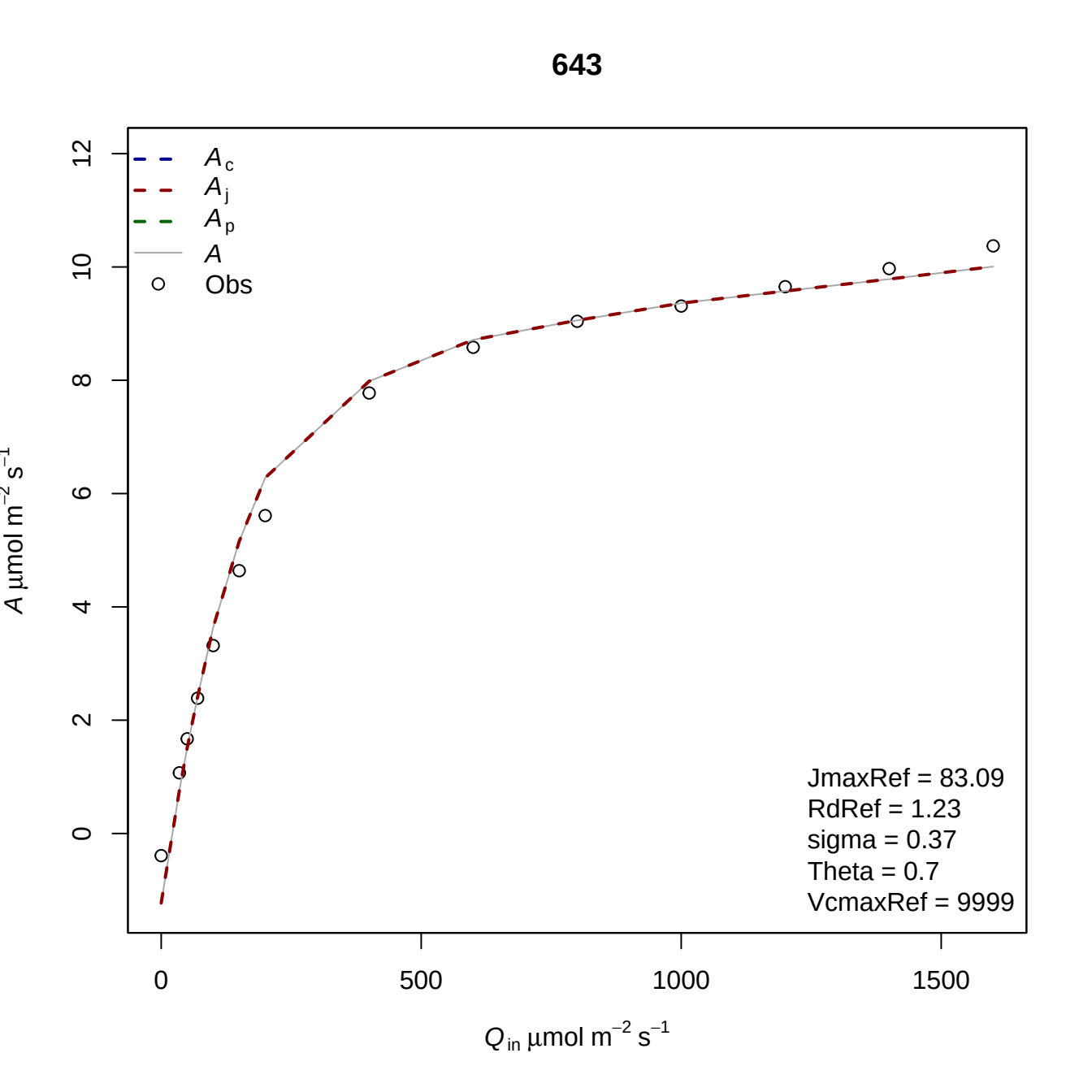

638
A_c
A_j
A_p
A
Obs
JmaxRef = 111.02
RdRef = 1.64
sigma = 0.27
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

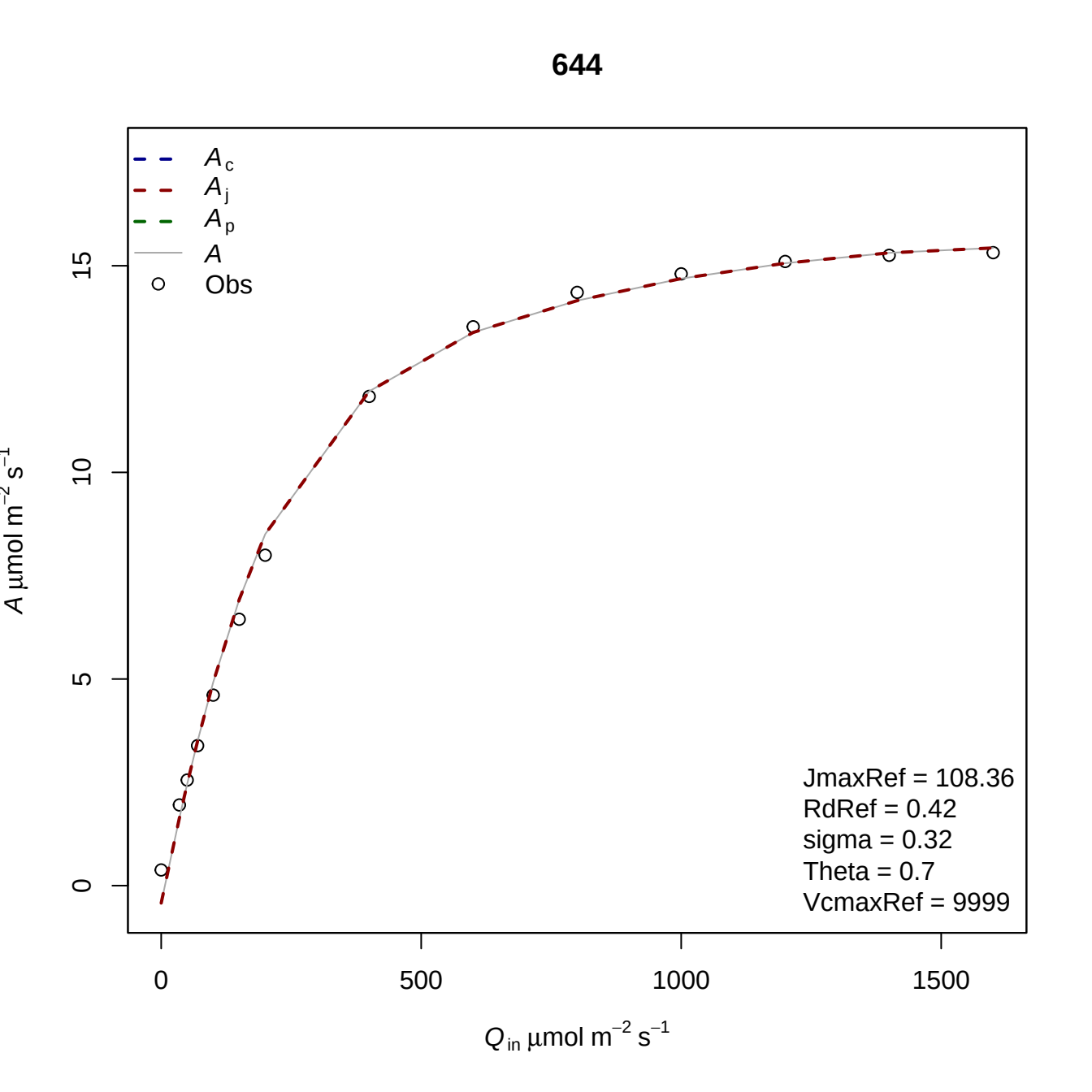

639
A_c
A_j
A_p
A
Obs
JmaxRef = 59.06
RdRef = 1.23
sigma = 0.1
Theta = 0.7
VcmaxRef = 43.96
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

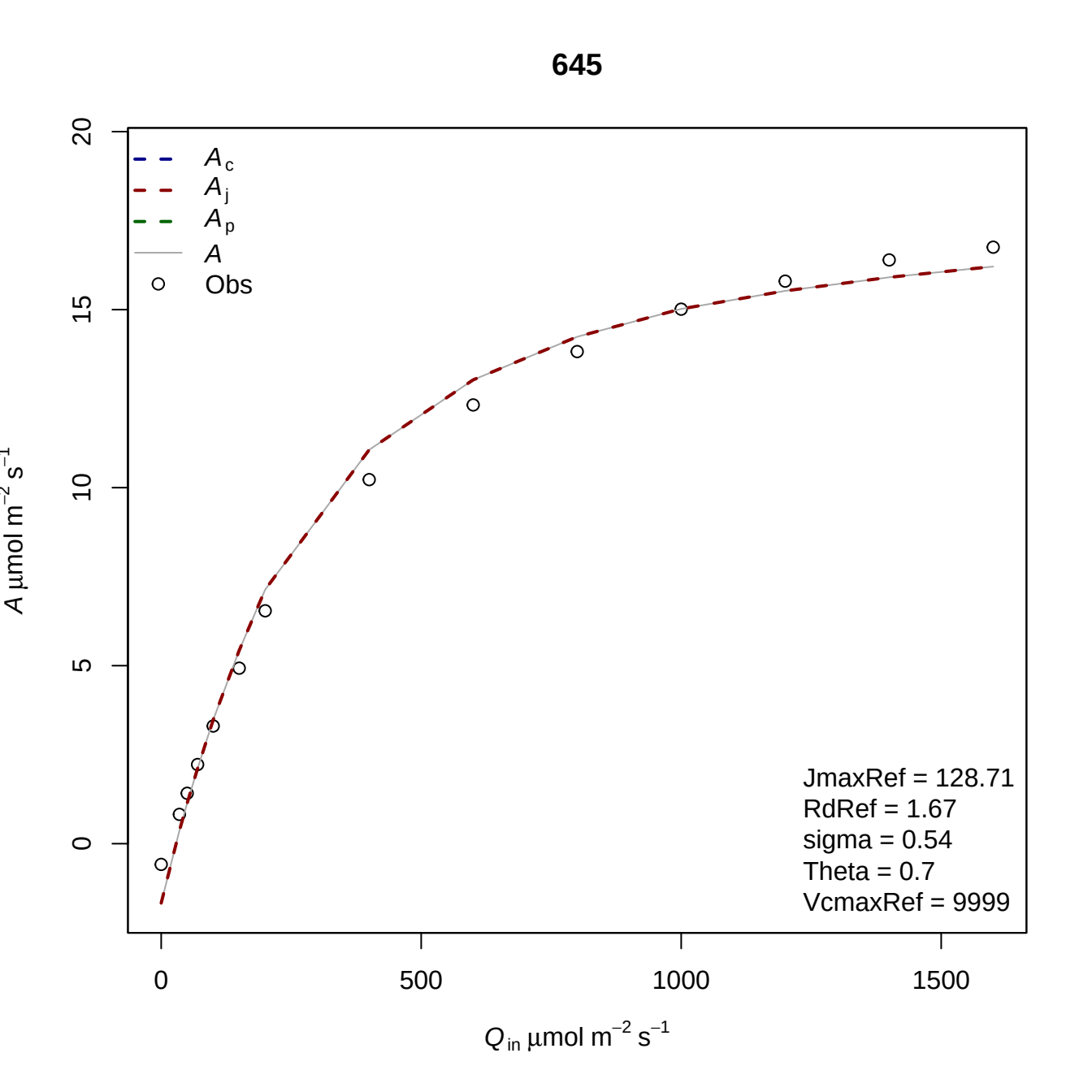

640
A_c
A_j
A_p
A
Obs
JmaxRef = 110.99
RdRef = 1.13
sigma = 0.28
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

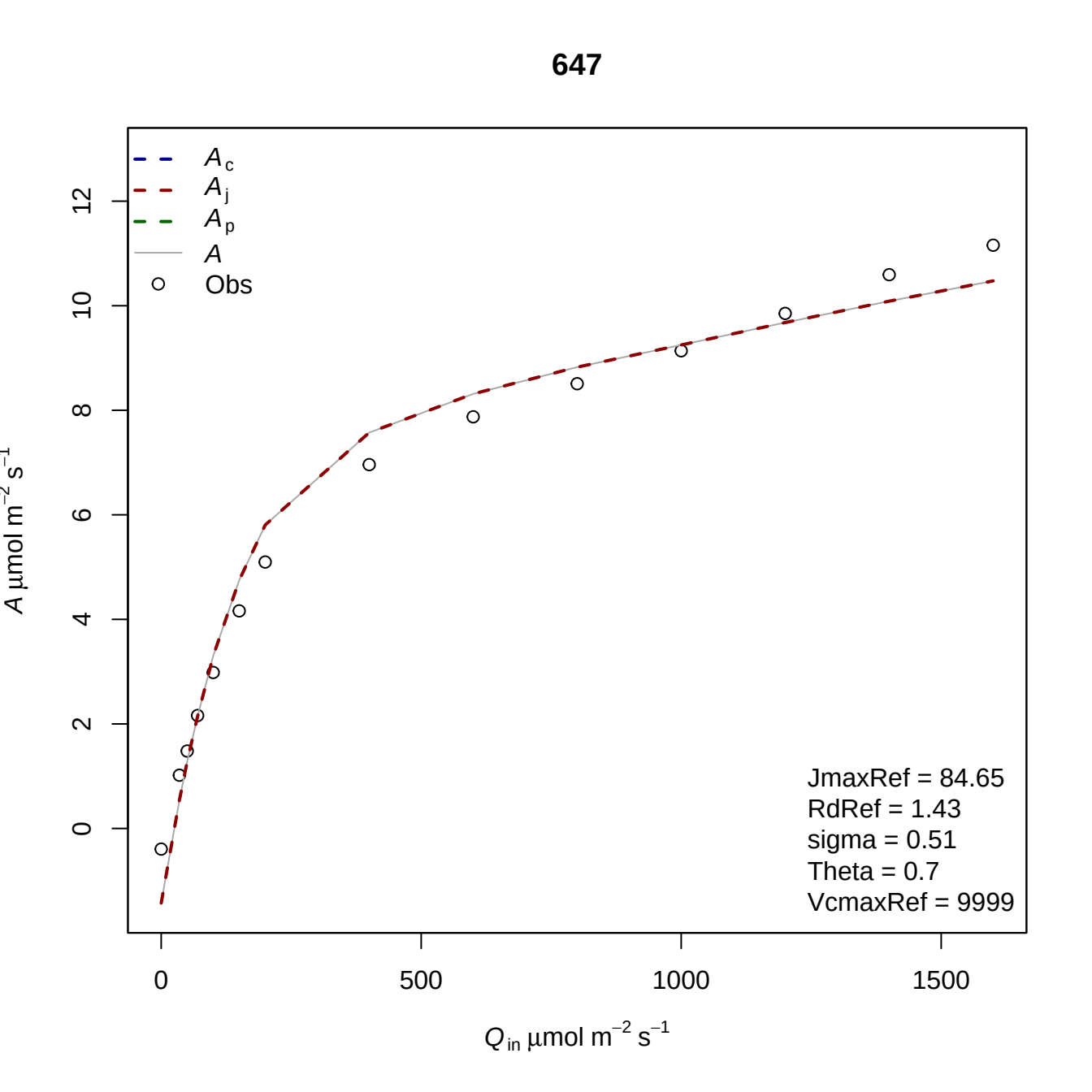

641
A_c
A_j
A_p
A
Obs
JmaxRef = 35.44
RdRef = −0.73
sigma = 0.03
Theta = 0.7
VcmaxRef = 22.26
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

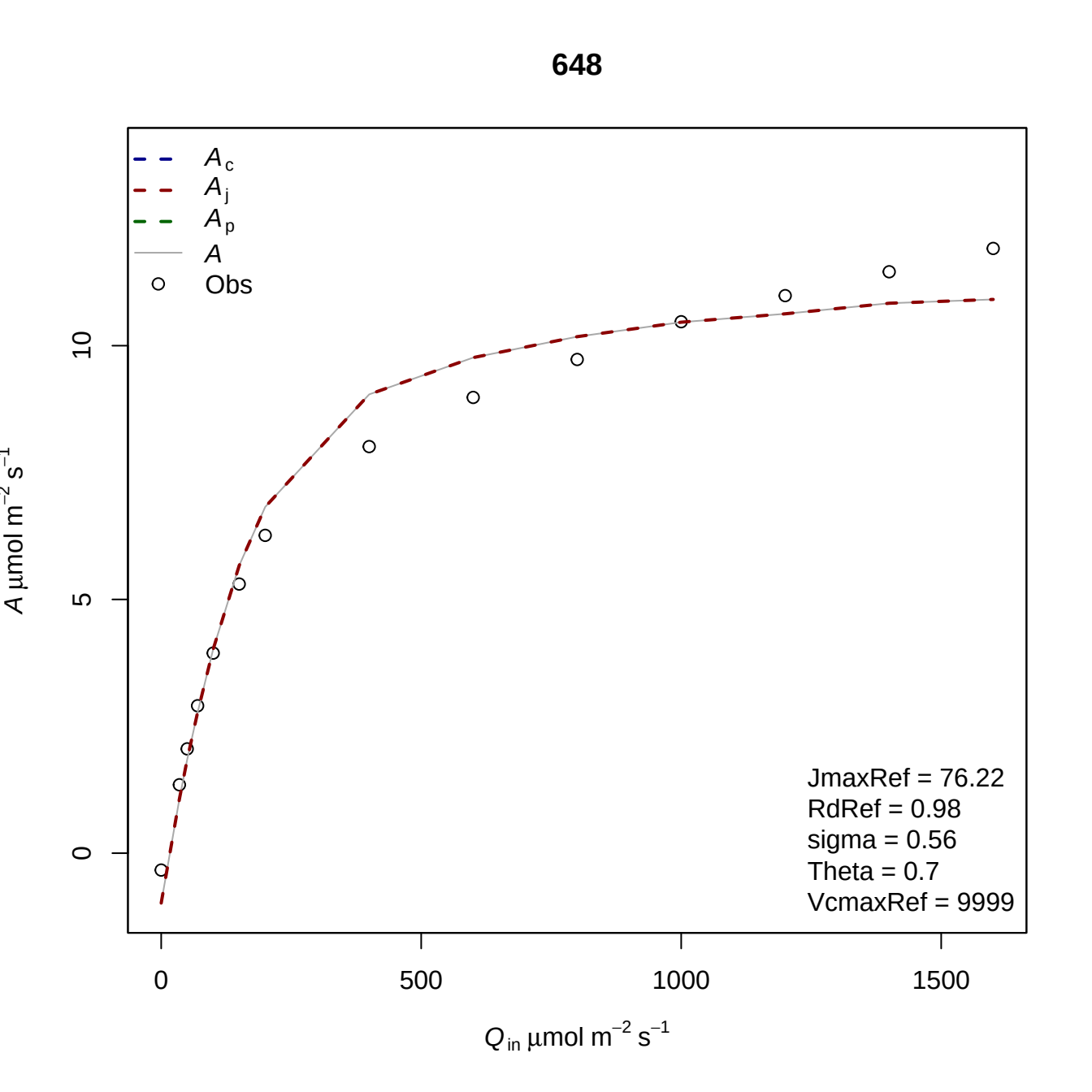

642
A_c
A_j
A_p
A
Obs
JmaxRef = 48.39
RdRef = 0.63
sigma = 0.12
Theta = 0.7
VcmaxRef = 33.84
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

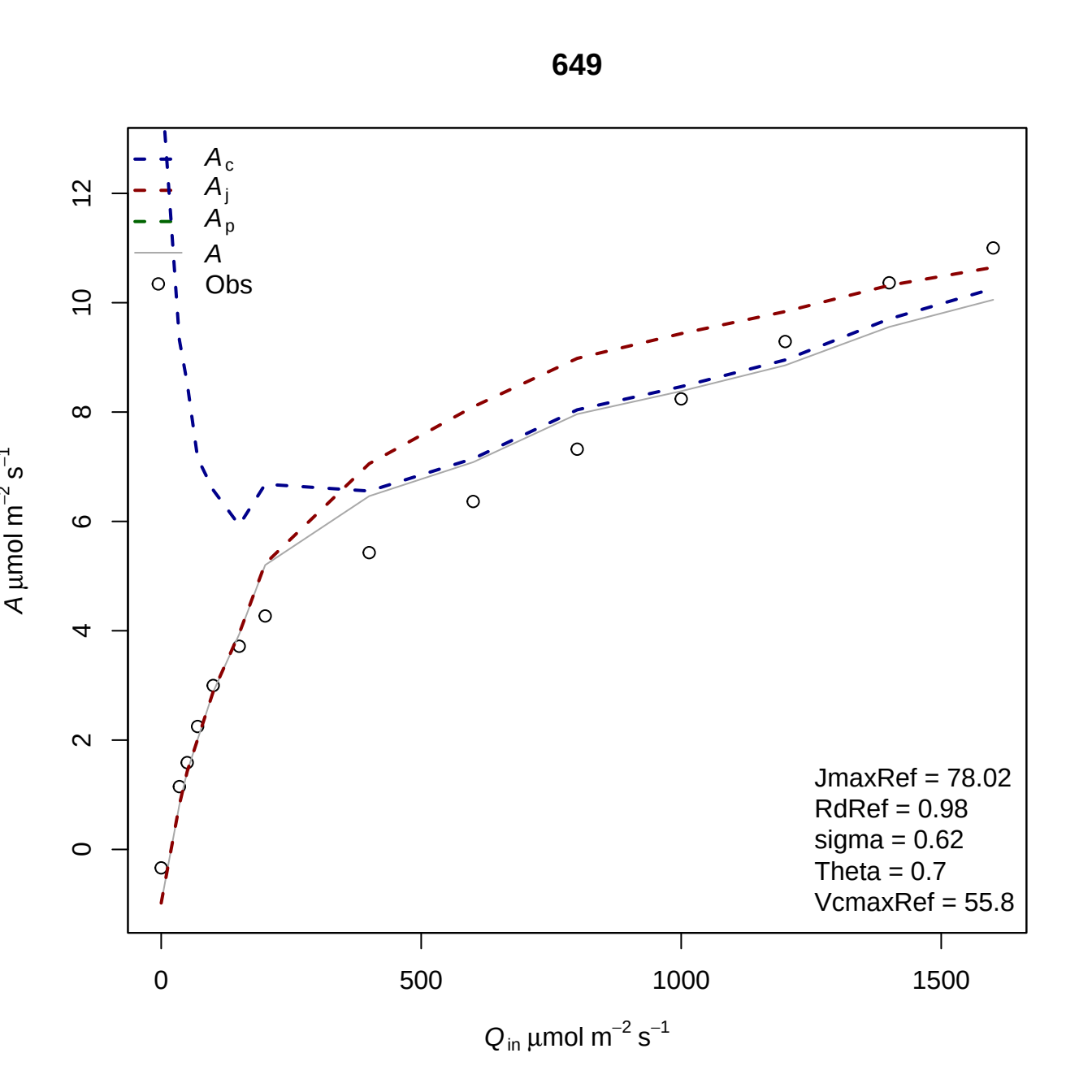

643
A_c
A_j
A_p
A
Obs
JmaxRef = 83.09
RdRef = 1.23
sigma = 0.37
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

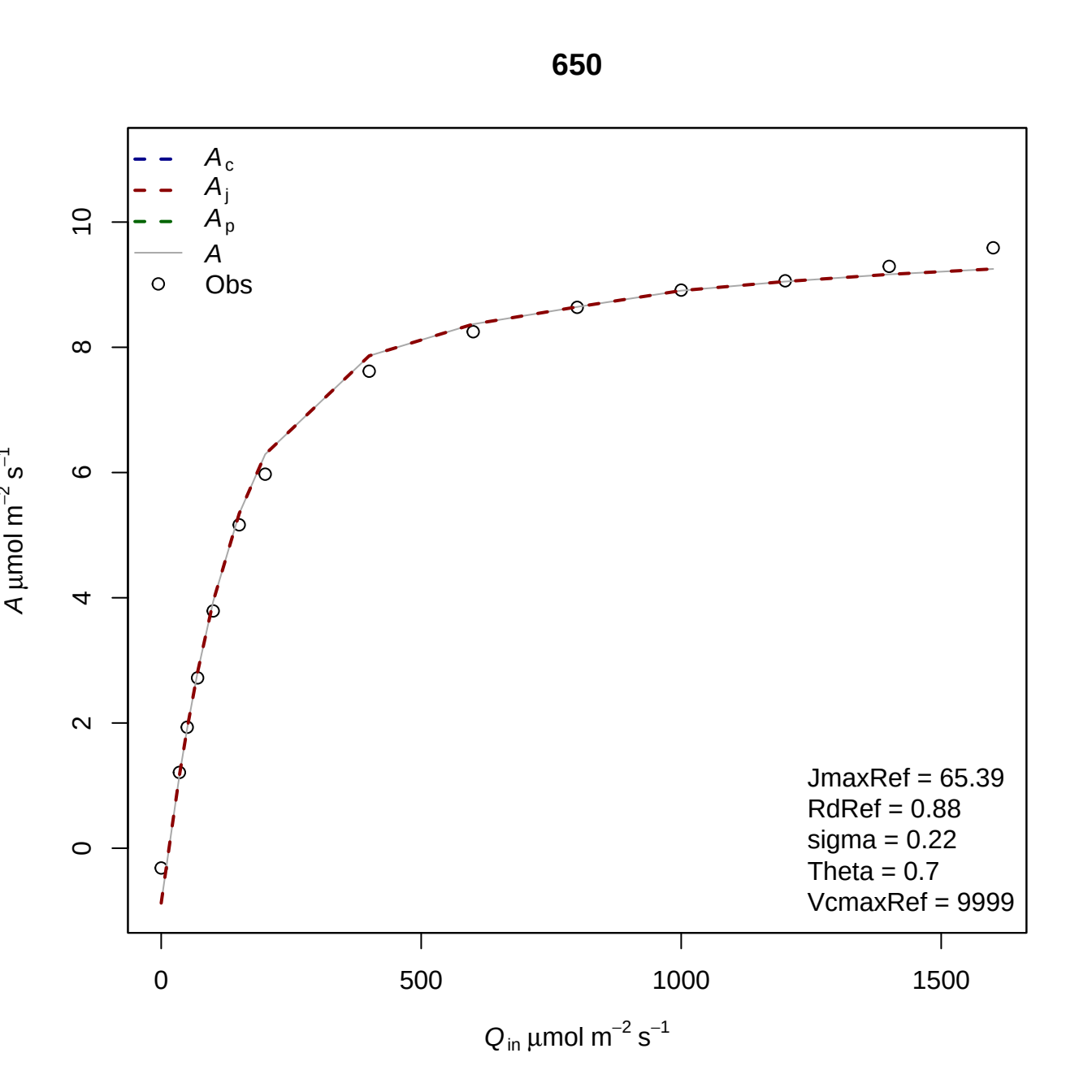

644
A_c
A_j
A_p
A
Obs
JmaxRef = 108.36
RdRef = 0.42
sigma = 0.32
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

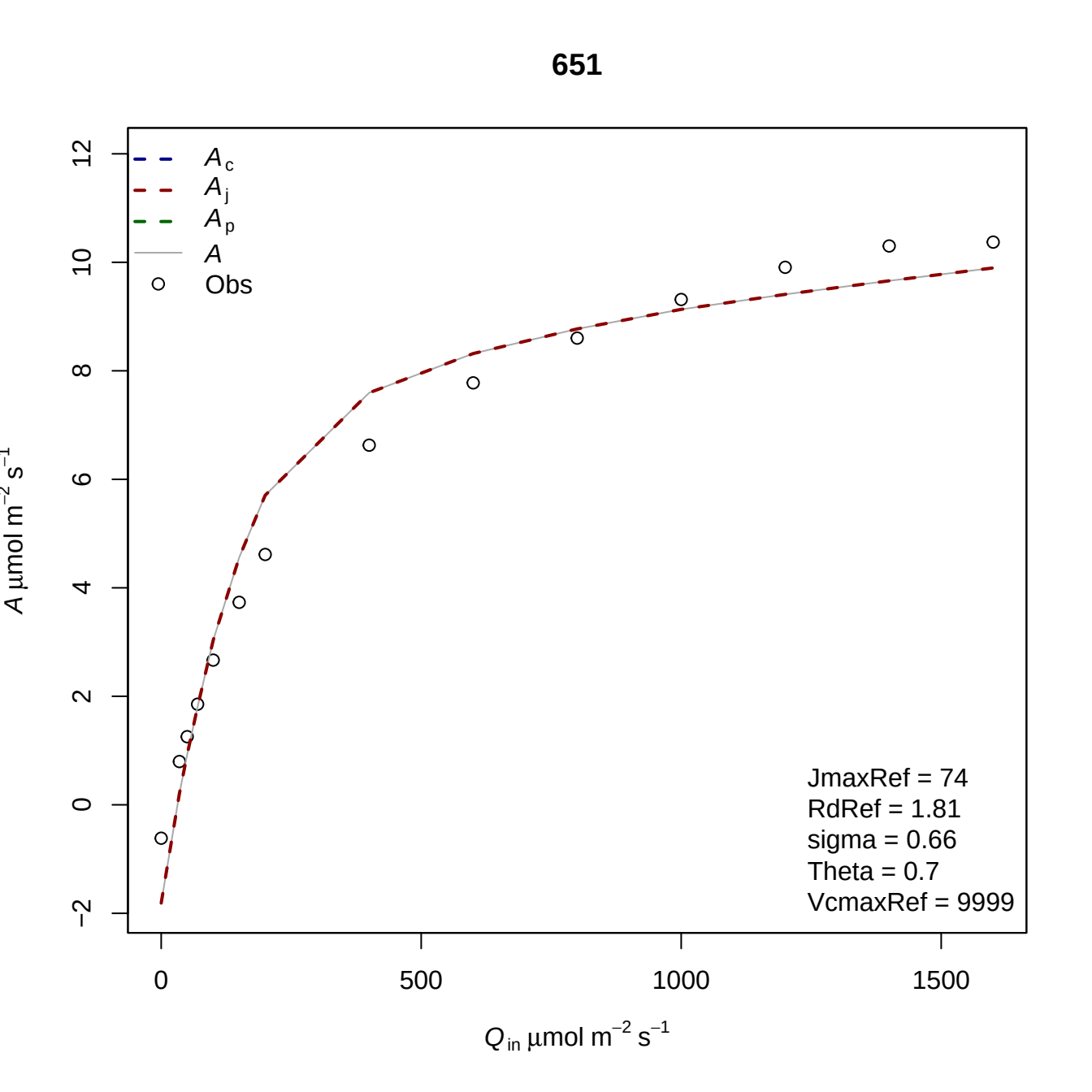

645
A_c
A_j
A_p
A
Obs
JmaxRef = 128.71
RdRef = 1.67
sigma = 0.54
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

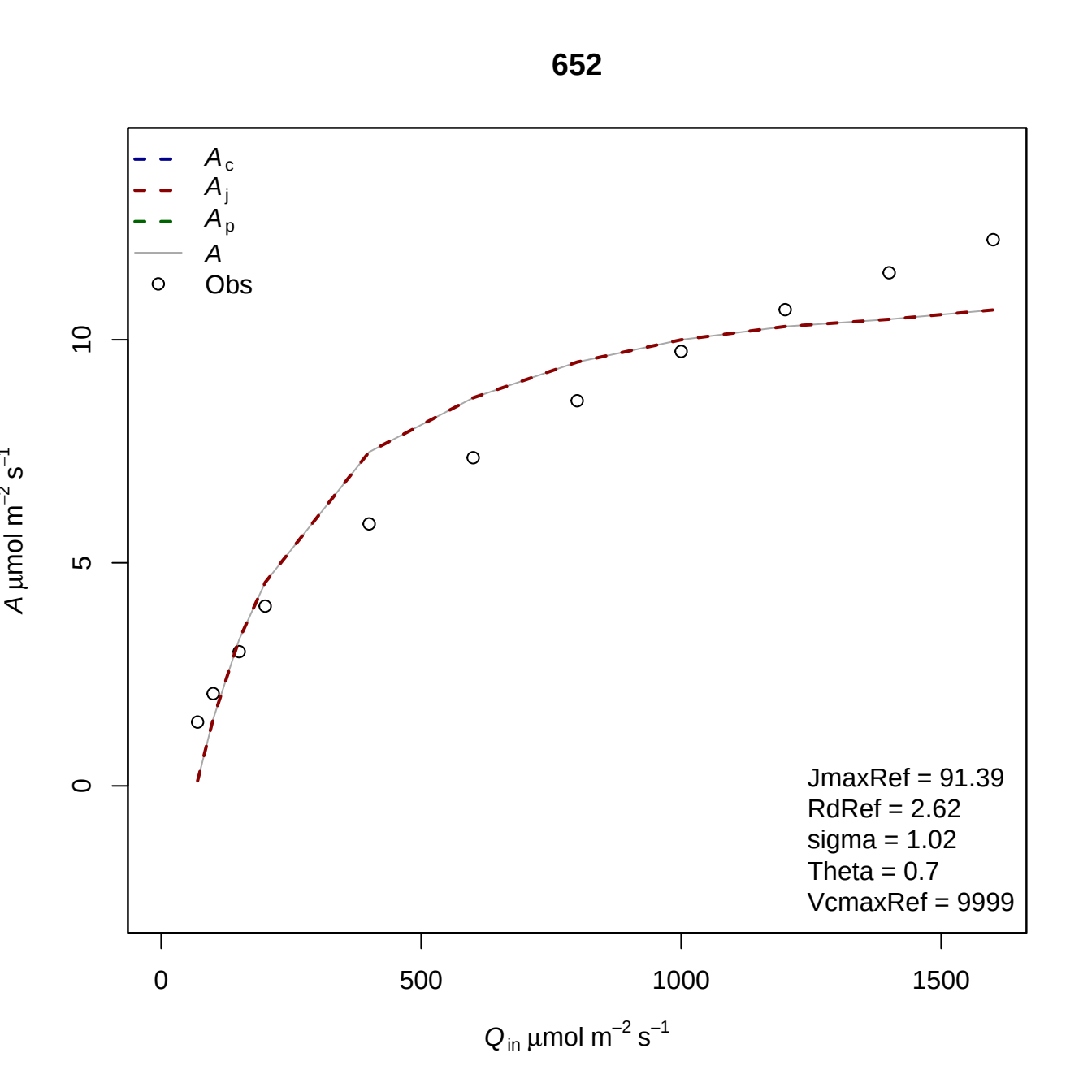

647
A_c
A_j
A_p
A
Obs
JmaxRef = 84.65
RdRef = 1.43
sigma = 0.51
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

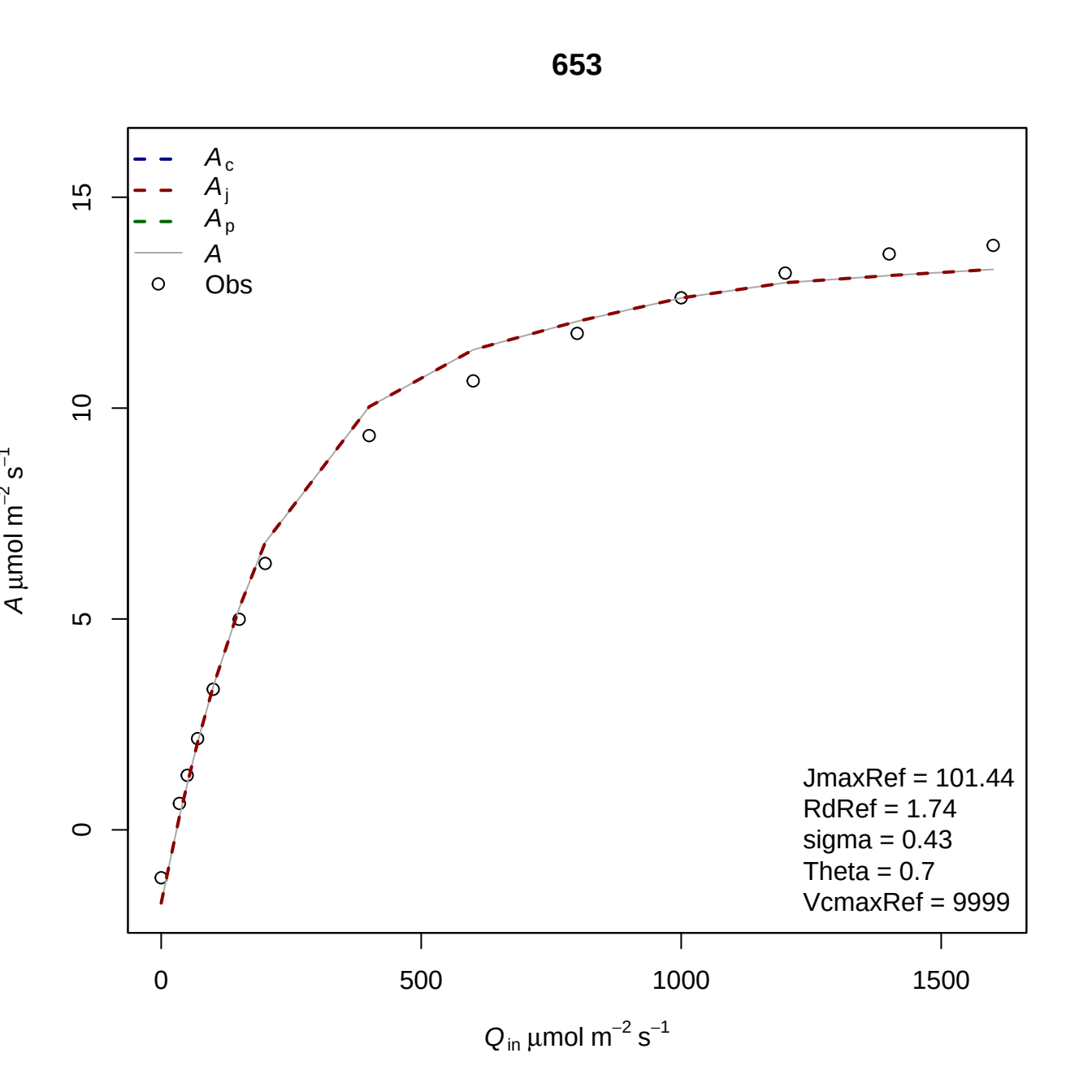

648
A_c
A_j
A_p
A
Obs
JmaxRef = 76.22
RdRef = 0.98
sigma = 0.56
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

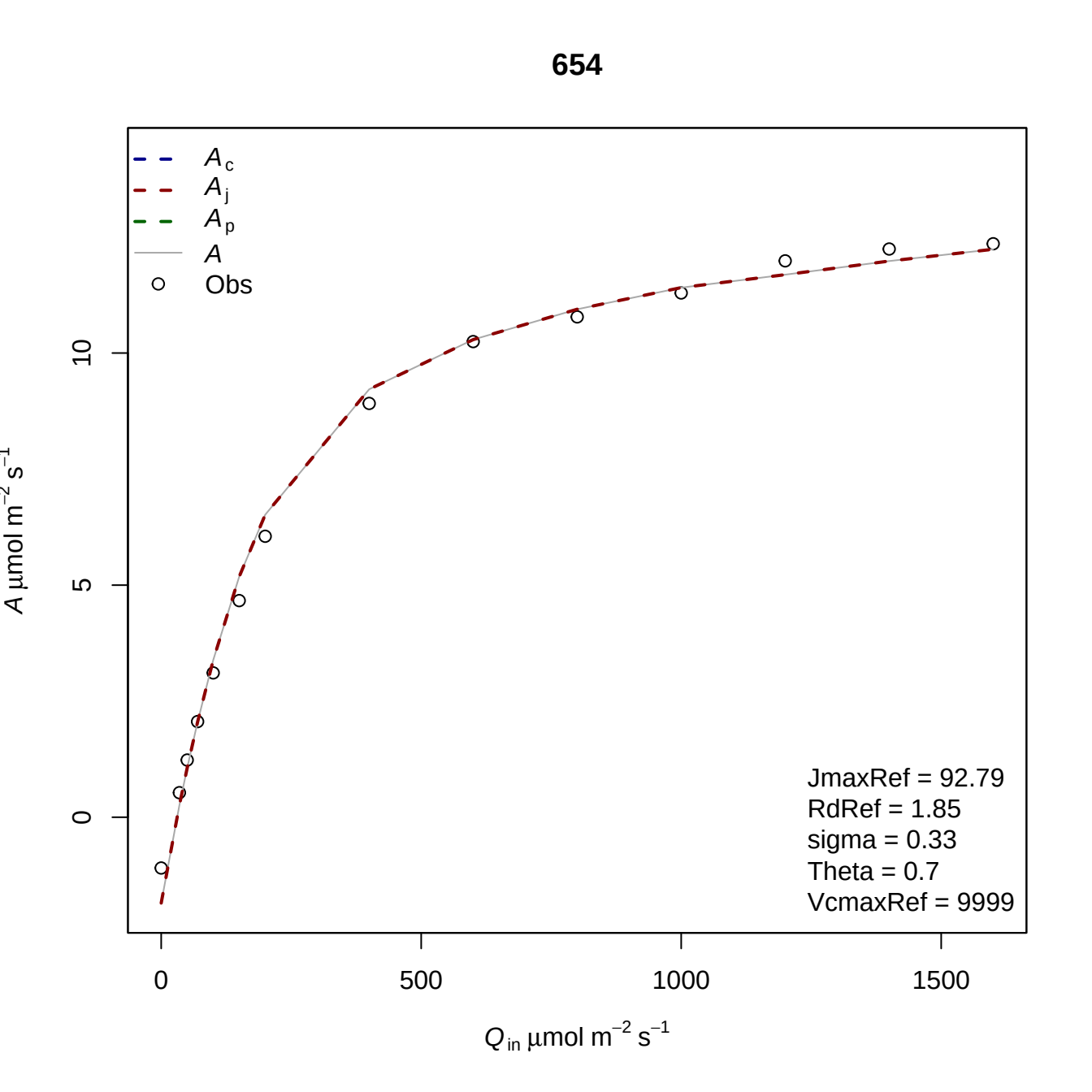

649
A_c
A_j
A_p
A
Obs
JmaxRef = 78.02
RdRef = 0.98
sigma = 0.62
Theta = 0.7
VcmaxRef = 55.8
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

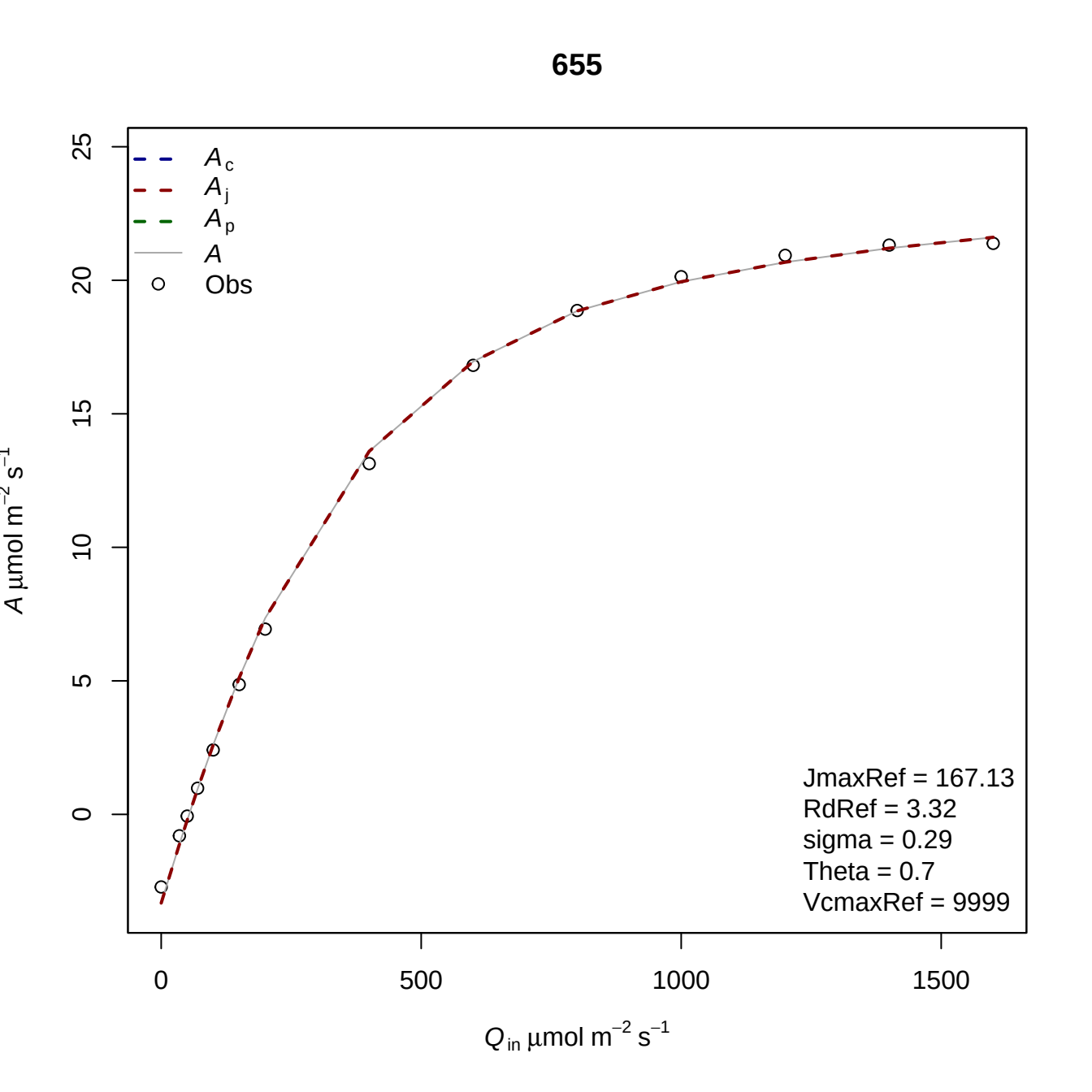

650
A_c
A_j
A_p
A
Obs
JmaxRef = 65.39
RdRef = 0.88
sigma = 0.22
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

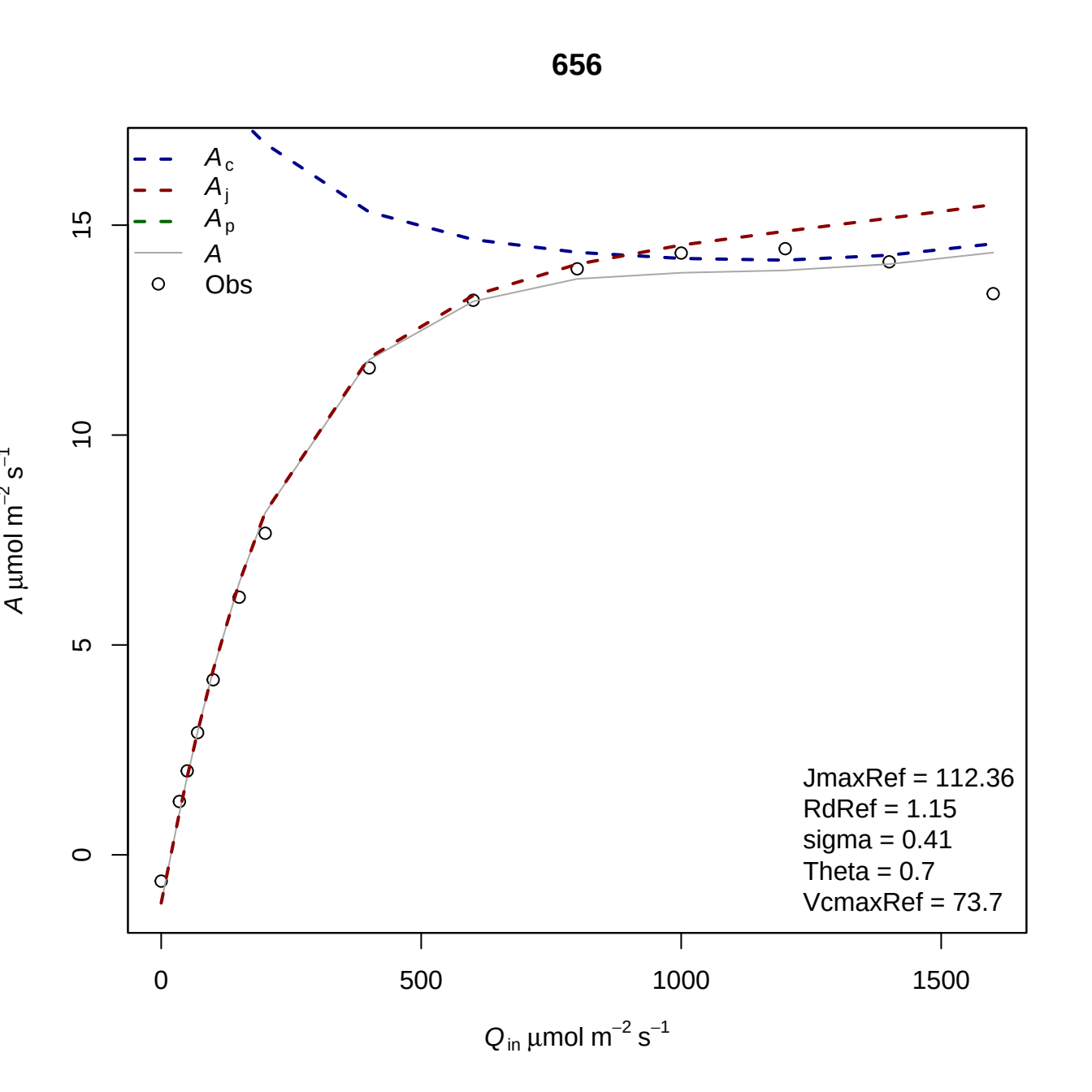

651
A_c
A_j
A_p
A
Obs
JmaxRef = 74
RdRef = 1.81
sigma = 0.66
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

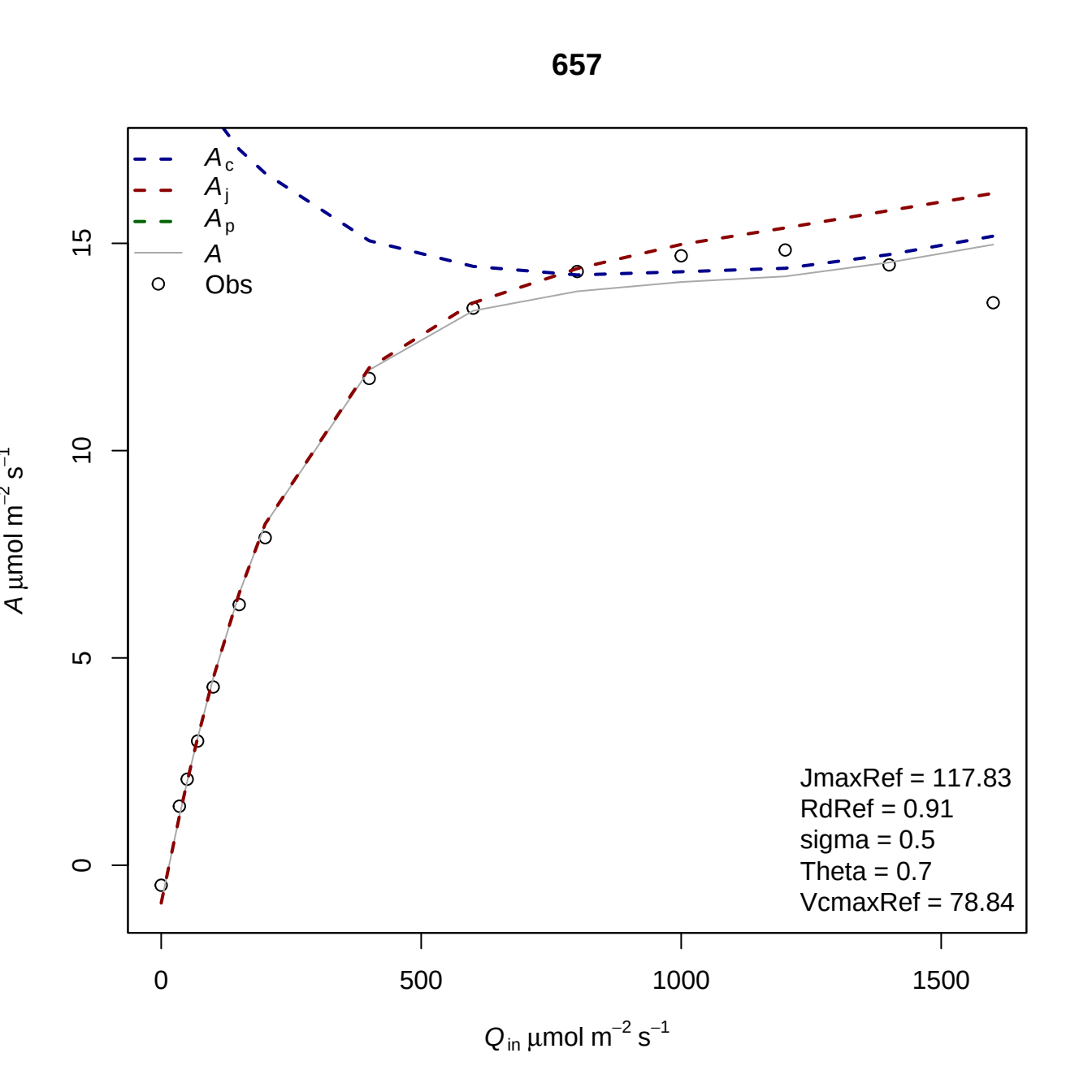

652
A_c
A_j
A_p
A
Obs
JmaxRef = 91.39
RdRef = 2.62
sigma = 1.02
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

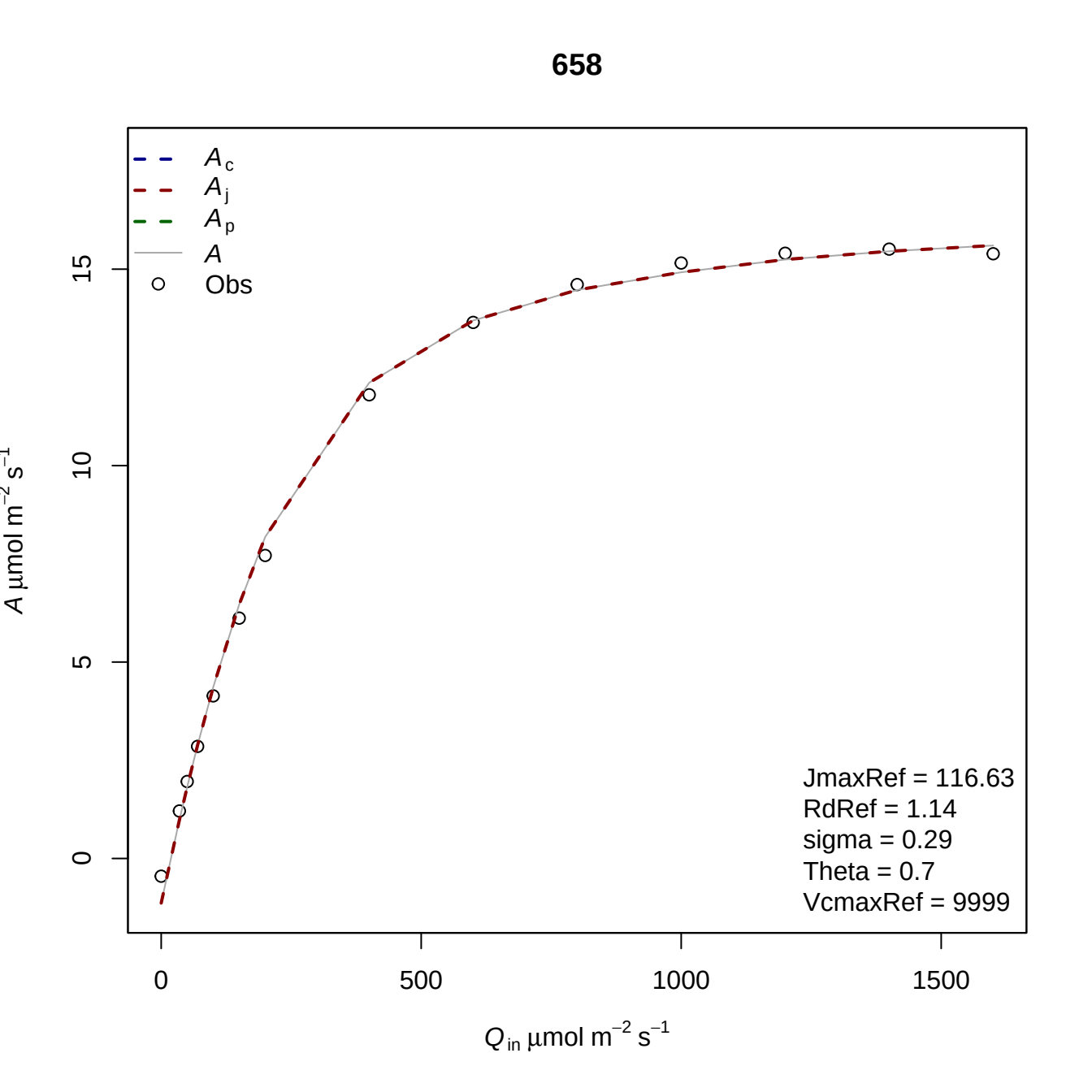

653
A_c
A_j
A_p
A
Obs
JmaxRef = 101.44
RdRef = 1.74
sigma = 0.43
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

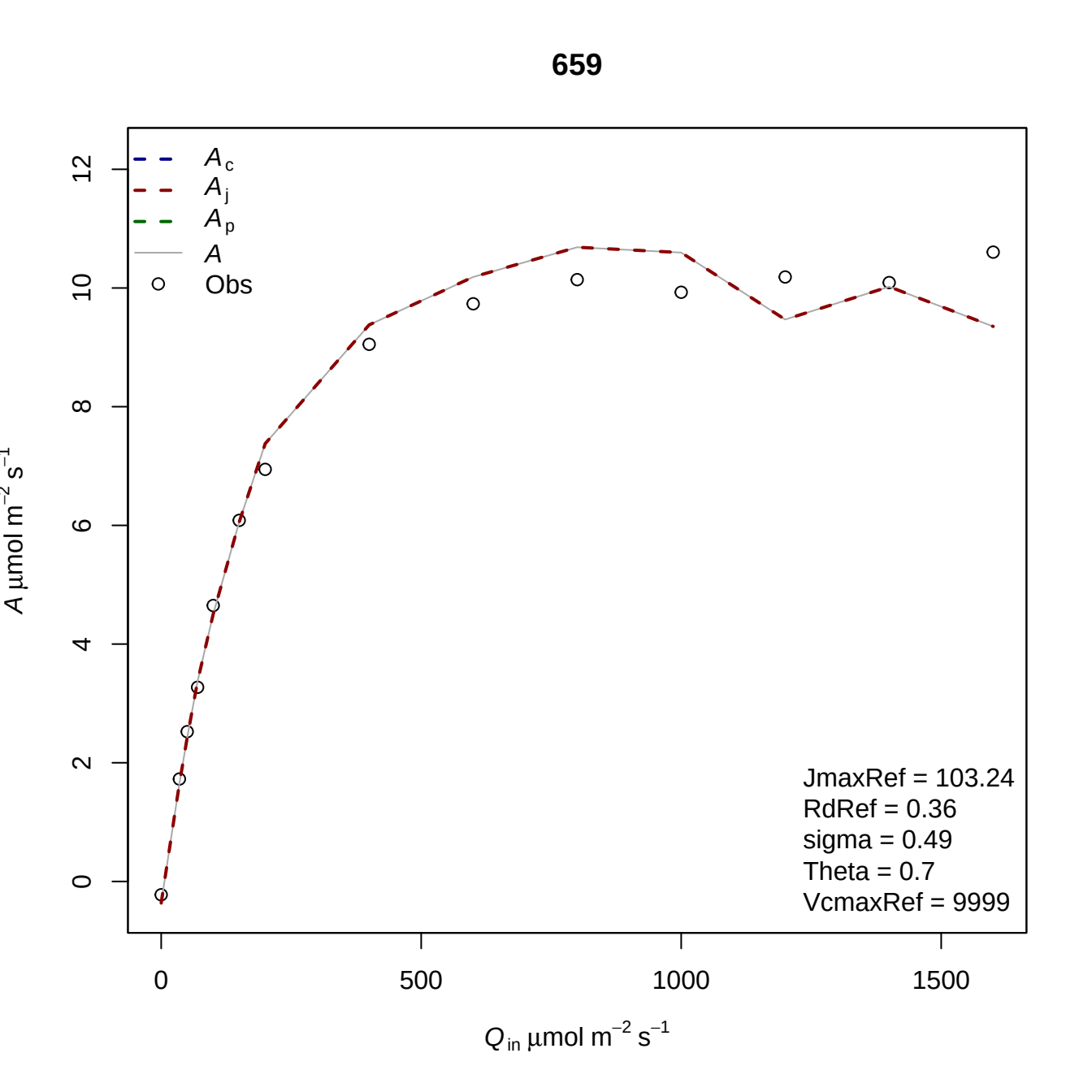

654
A_c
A_j
A_p
A
Obs
JmaxRef = 92.79
RdRef = 1.85
sigma = 0.33
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

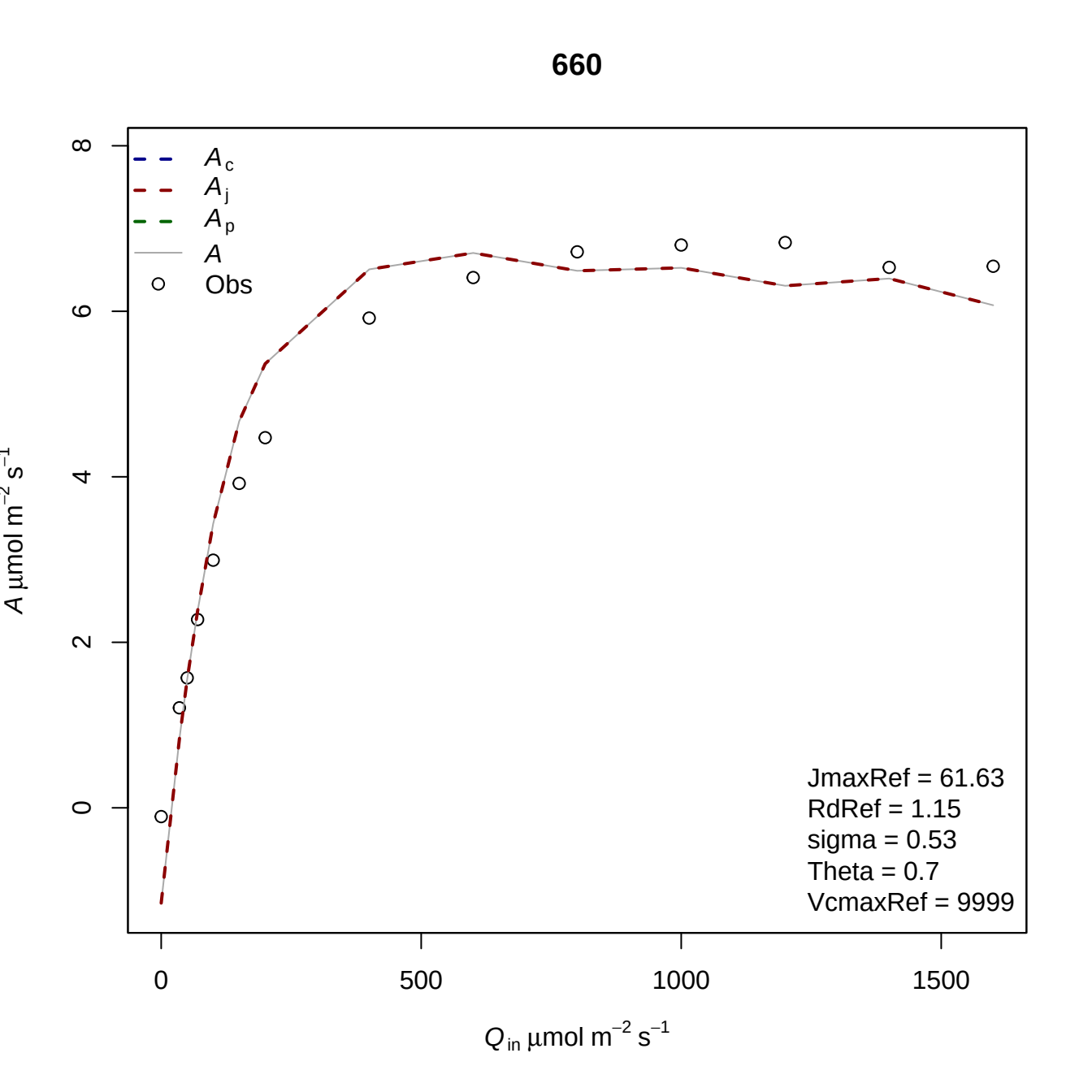

655
A_c
A_j
A_p
A
Obs
JmaxRef = 167.13
RdRef = 3.32
sigma = 0.29
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

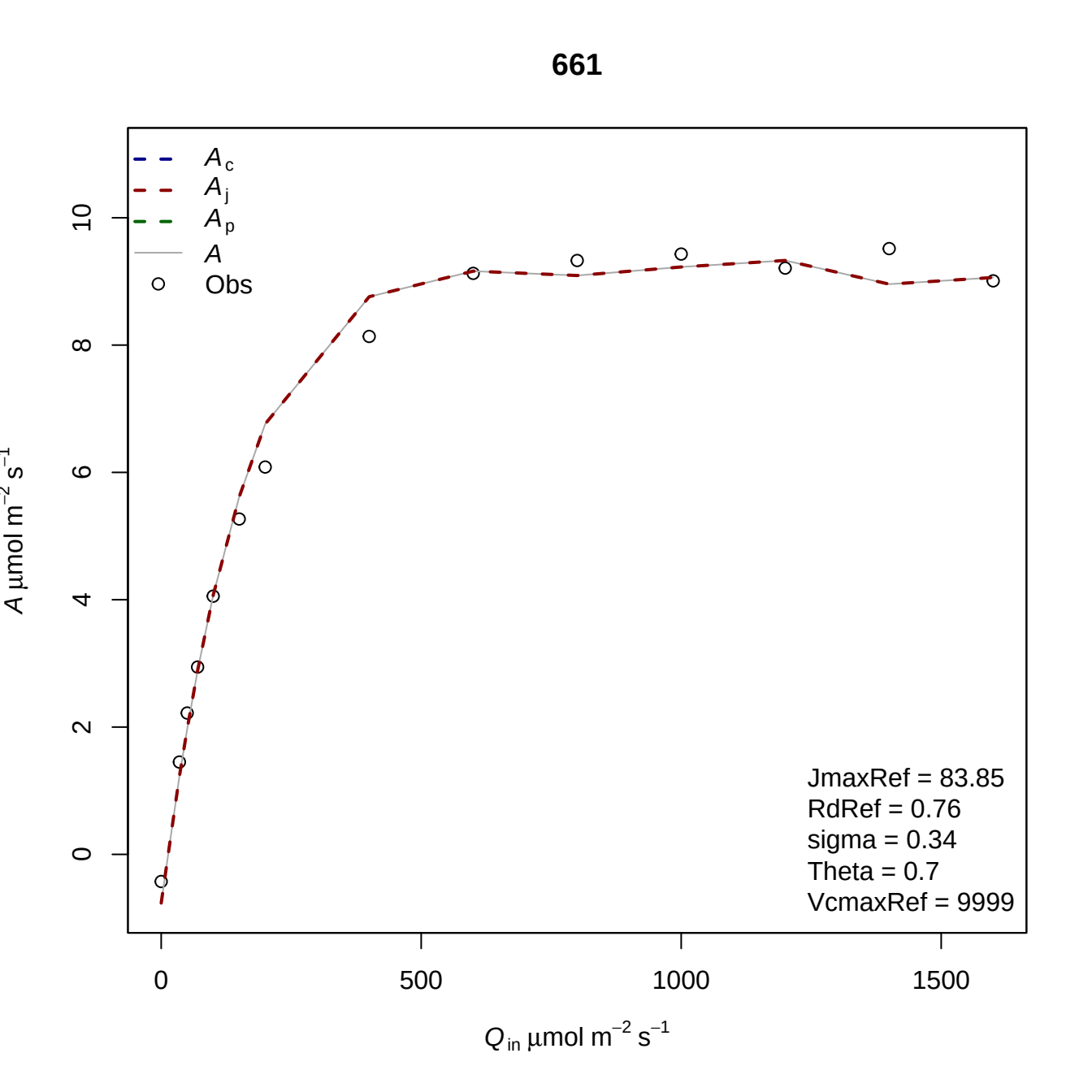

656
A_c
A_j
A_p
A
Obs
JmaxRef = 112.36
RdRef = 1.15
sigma = 0.41
Theta = 0.7
VcmaxRef = 73.7
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

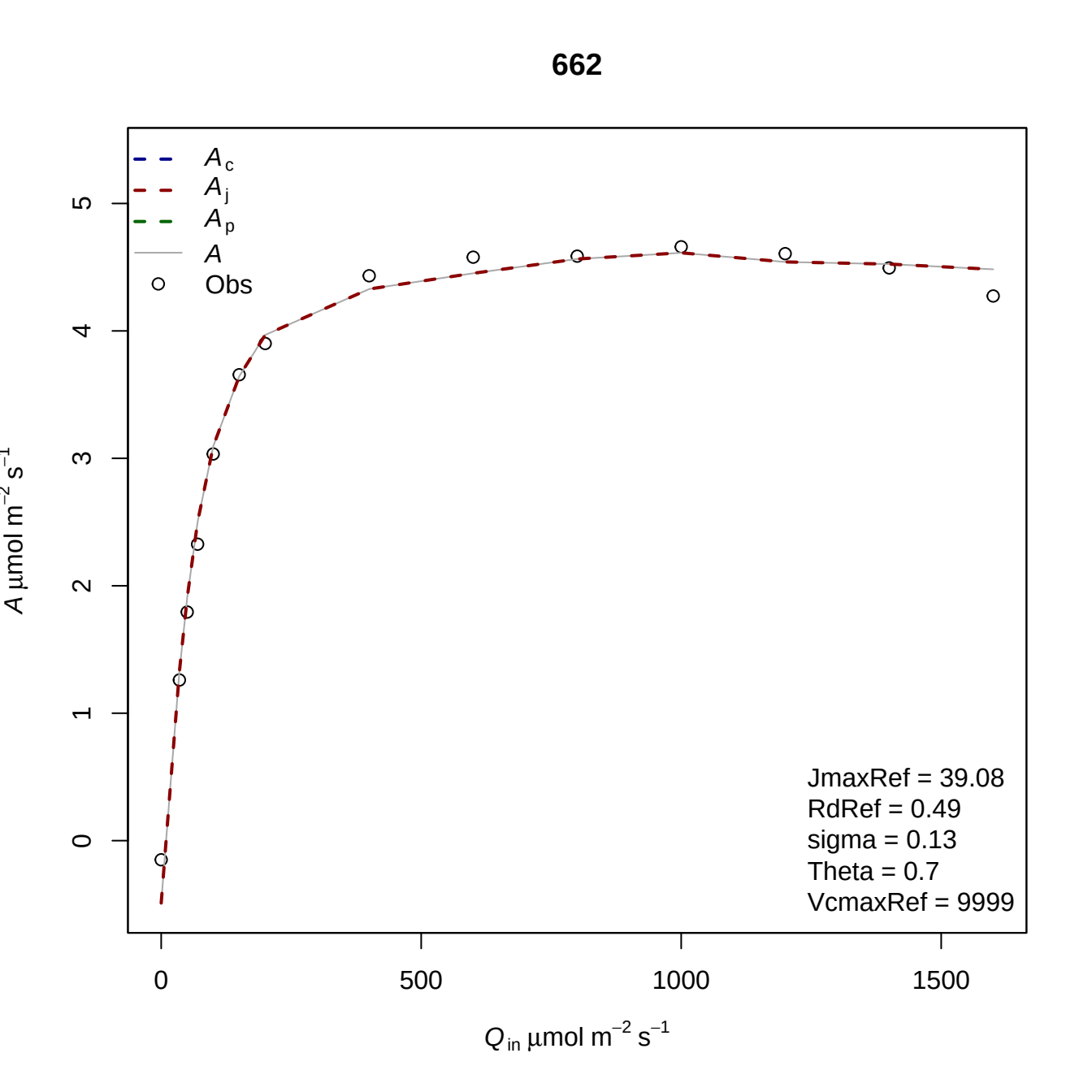

657
A_c
A_j
A_p
A
Obs
JmaxRef = 117.83
RdRef = 0.91
sigma = 0.5
Theta = 0.7
VcmaxRef = 78.84
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

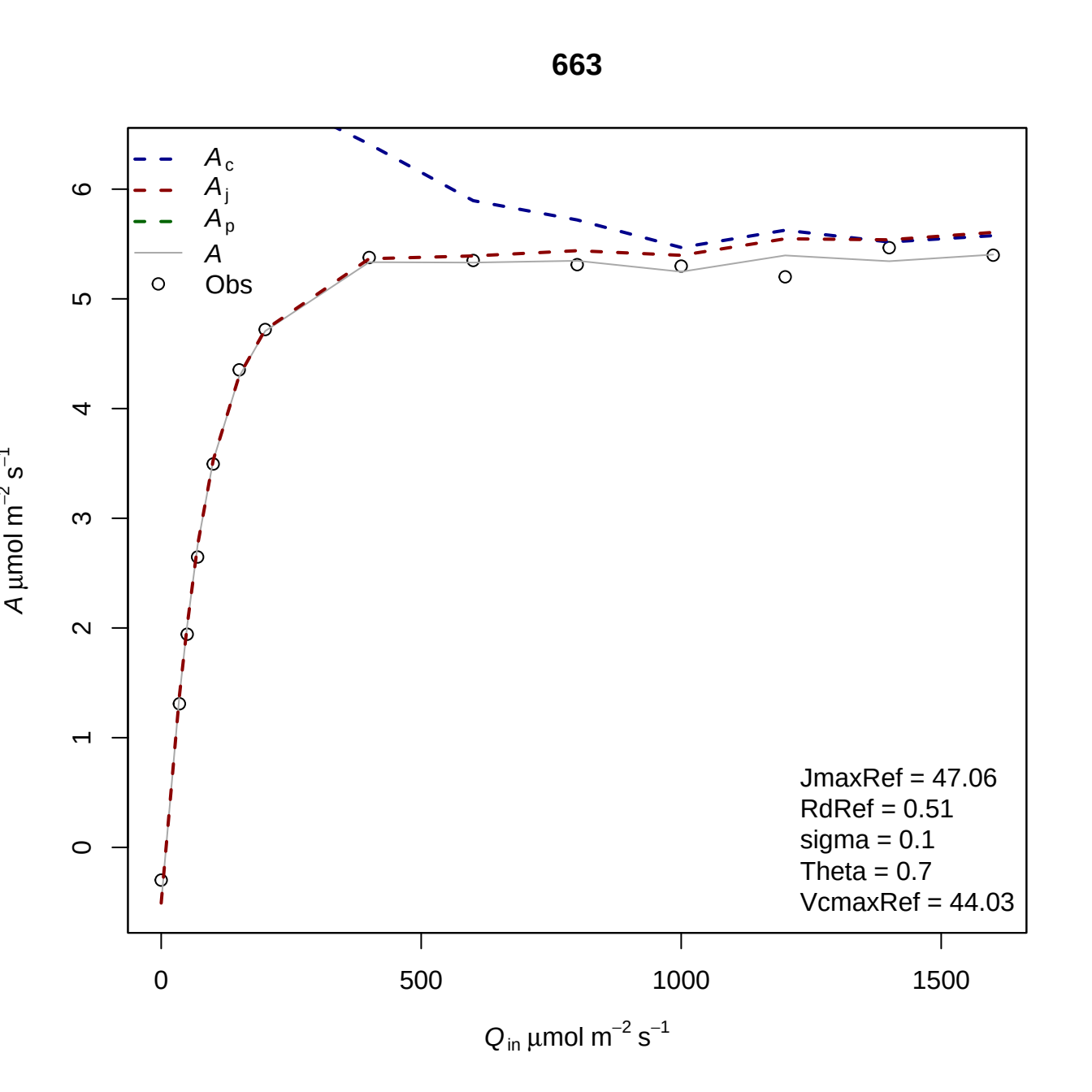

658
A_c
A_j
A_p
A
Obs
JmaxRef = 116.63
RdRef = 1.14
sigma = 0.29
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

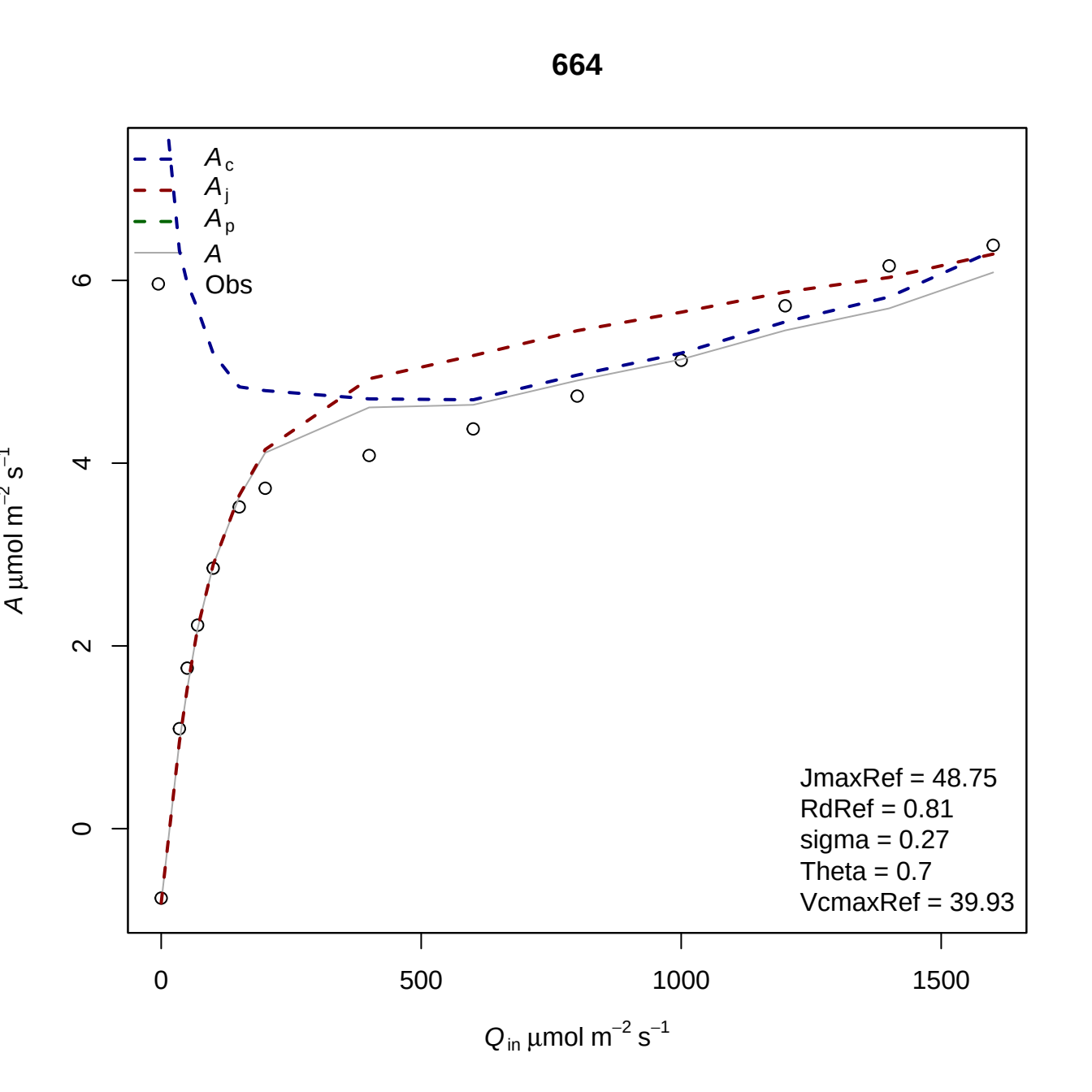

659
A_c
A_j
A_p
A
Obs
JmaxRef = 103.24
RdRef = 0.36
sigma = 0.49
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

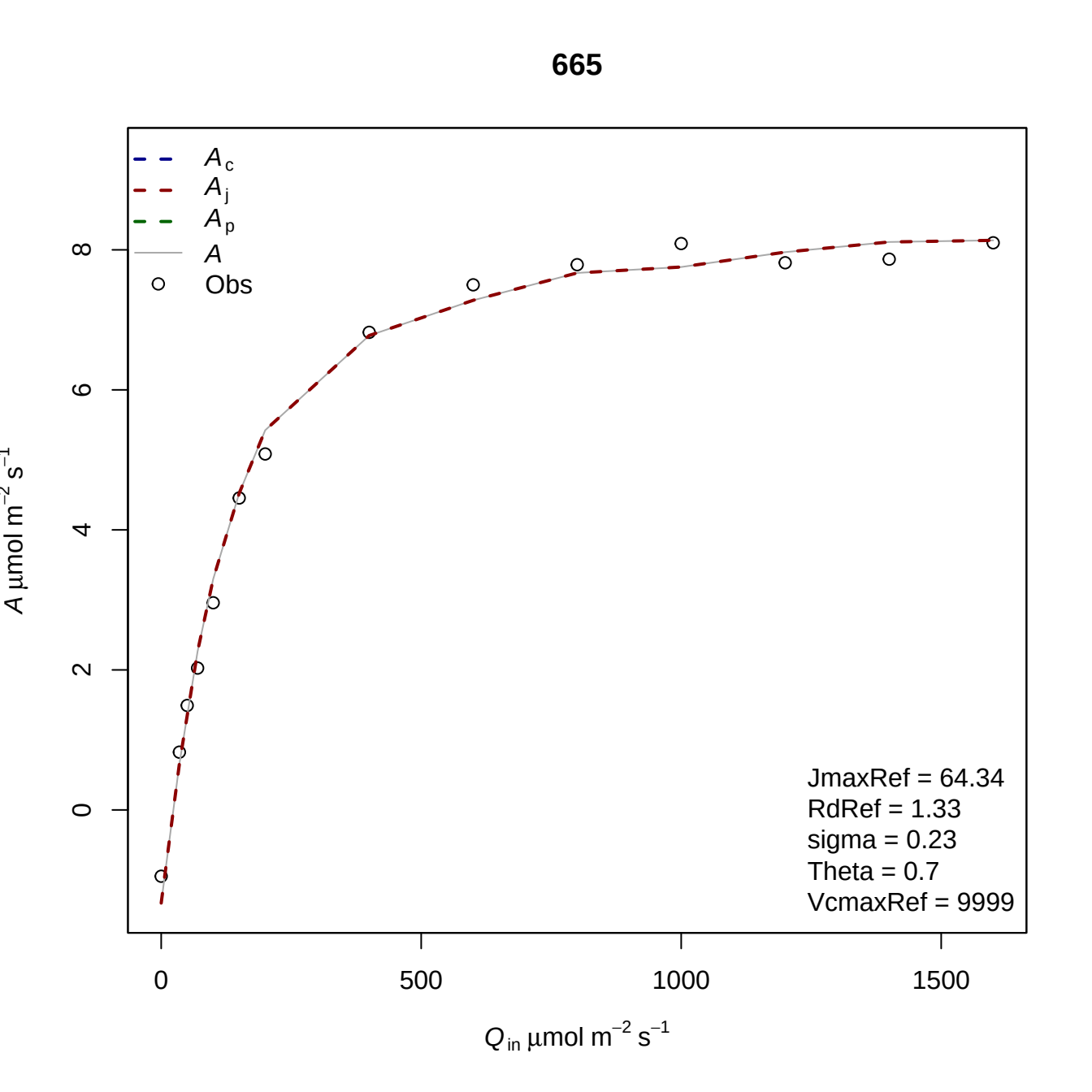

660
A_c
A_j
A_p
A
Obs
JmaxRef = 61.63
RdRef = 1.15
sigma = 0.53
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

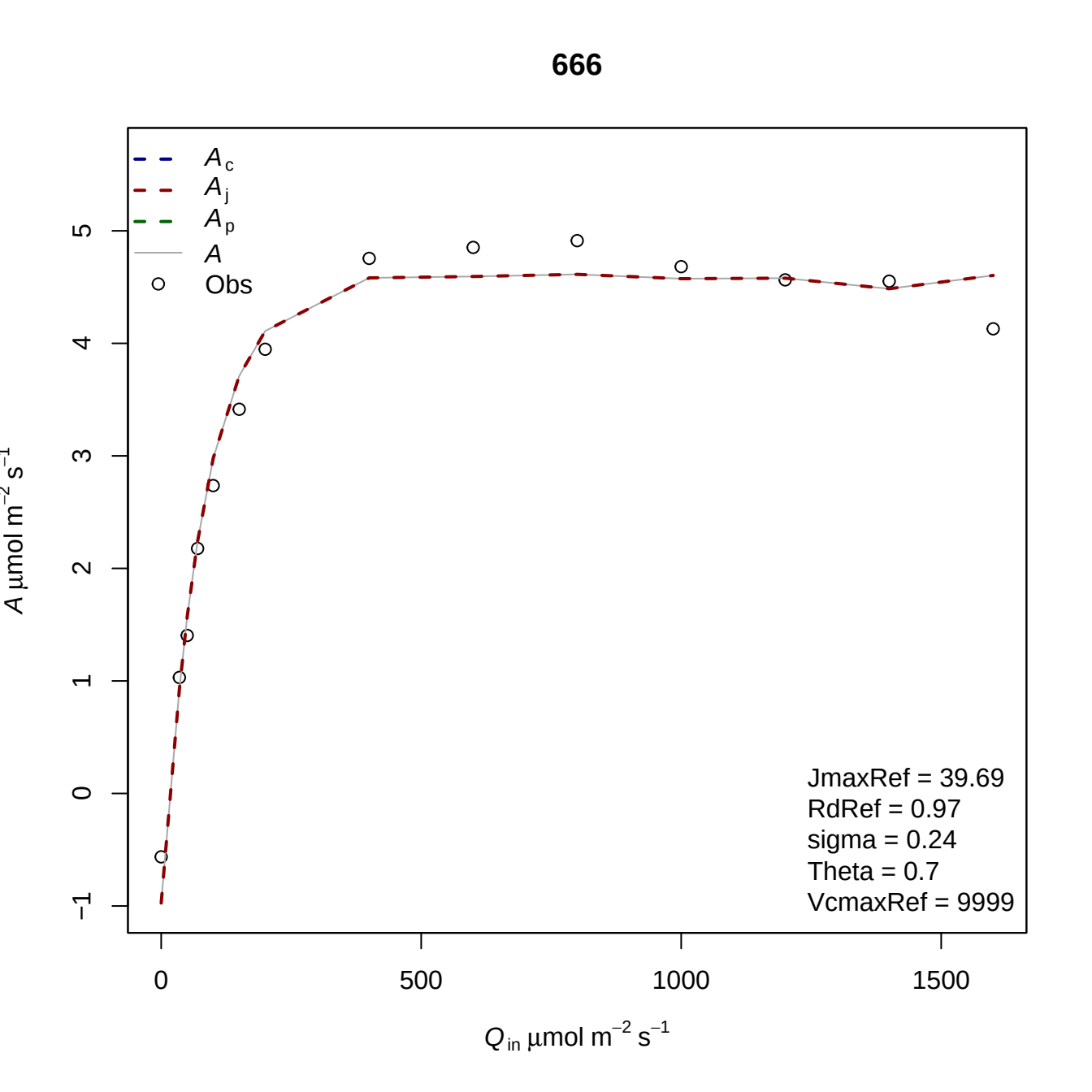

661
A_c
A_j
A_p
A
Obs
JmaxRef = 83.85
RdRef = 0.76
sigma = 0.34
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

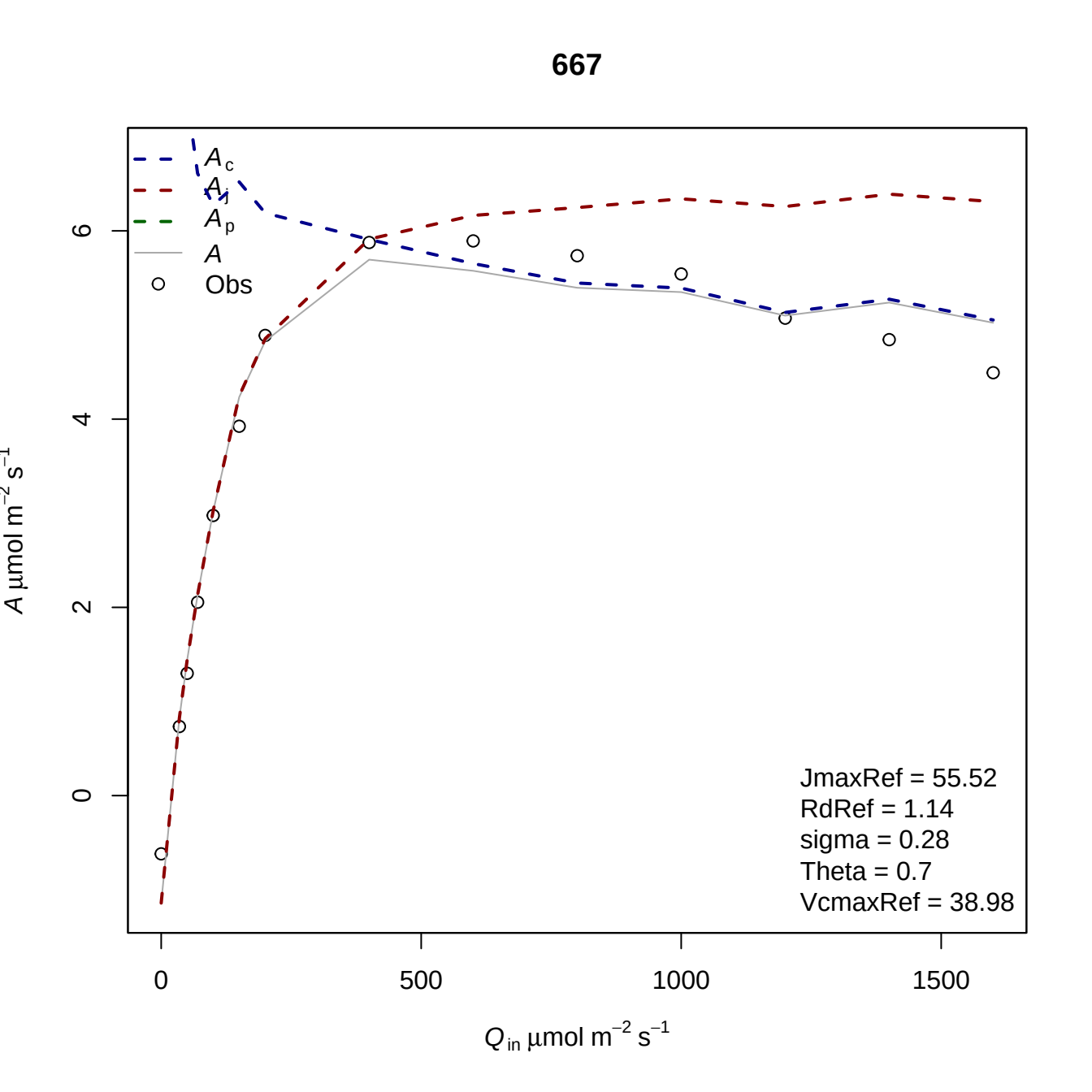

662
A_c
A_j
A_p
A
Obs
JmaxRef = 39.08
RdRef = 0.49
sigma = 0.13
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

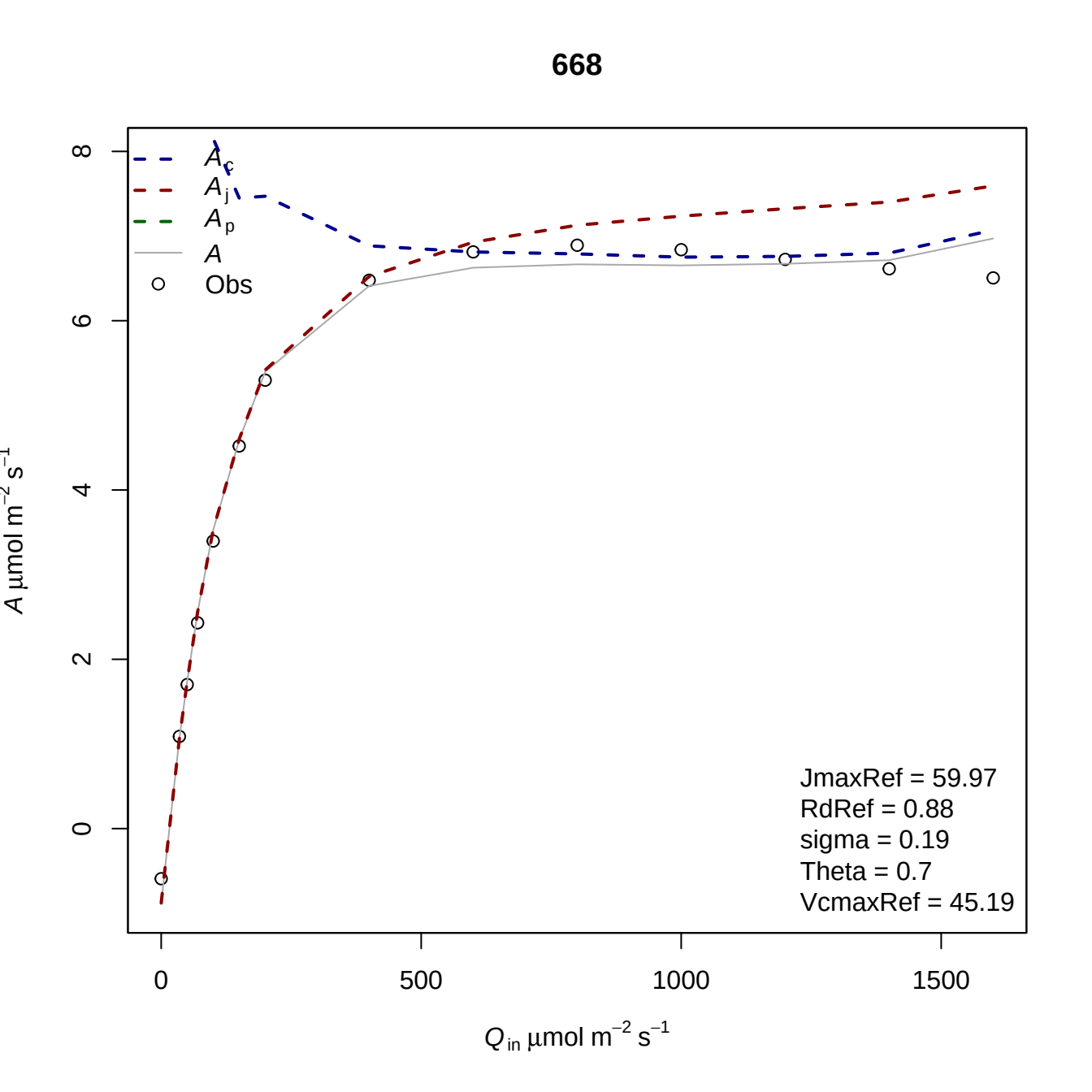

663
A_c
A_j
A_p
A
Obs
JmaxRef = 47.06
RdRef = 0.51
sigma = 0.1
Theta = 0.7
VcmaxRef = 44.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

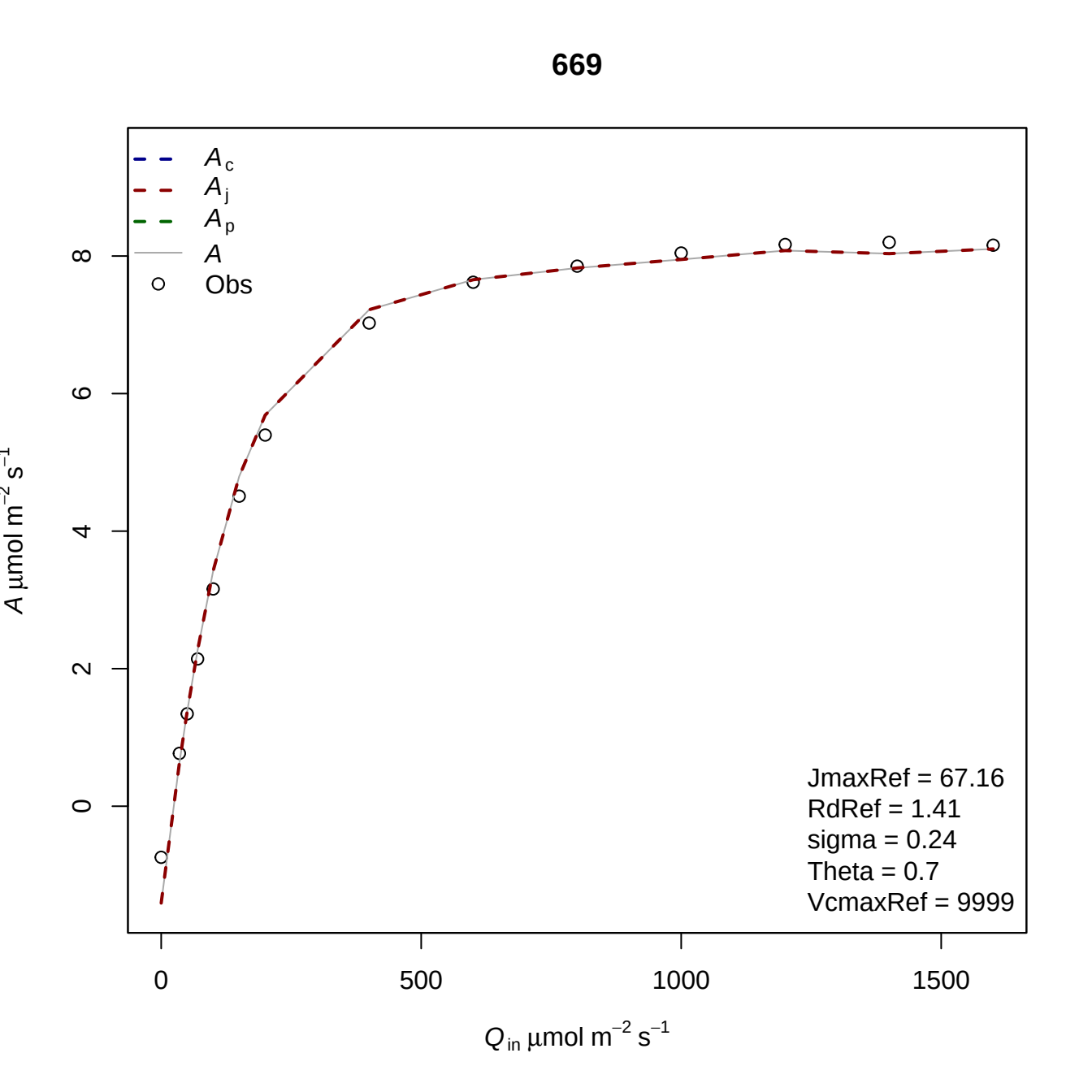

664
A_c
A_j
A_p
A
Obs
JmaxRef = 48.75
RdRef = 0.81
sigma = 0.27
Theta = 0.7
VcmaxRef = 39.93
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

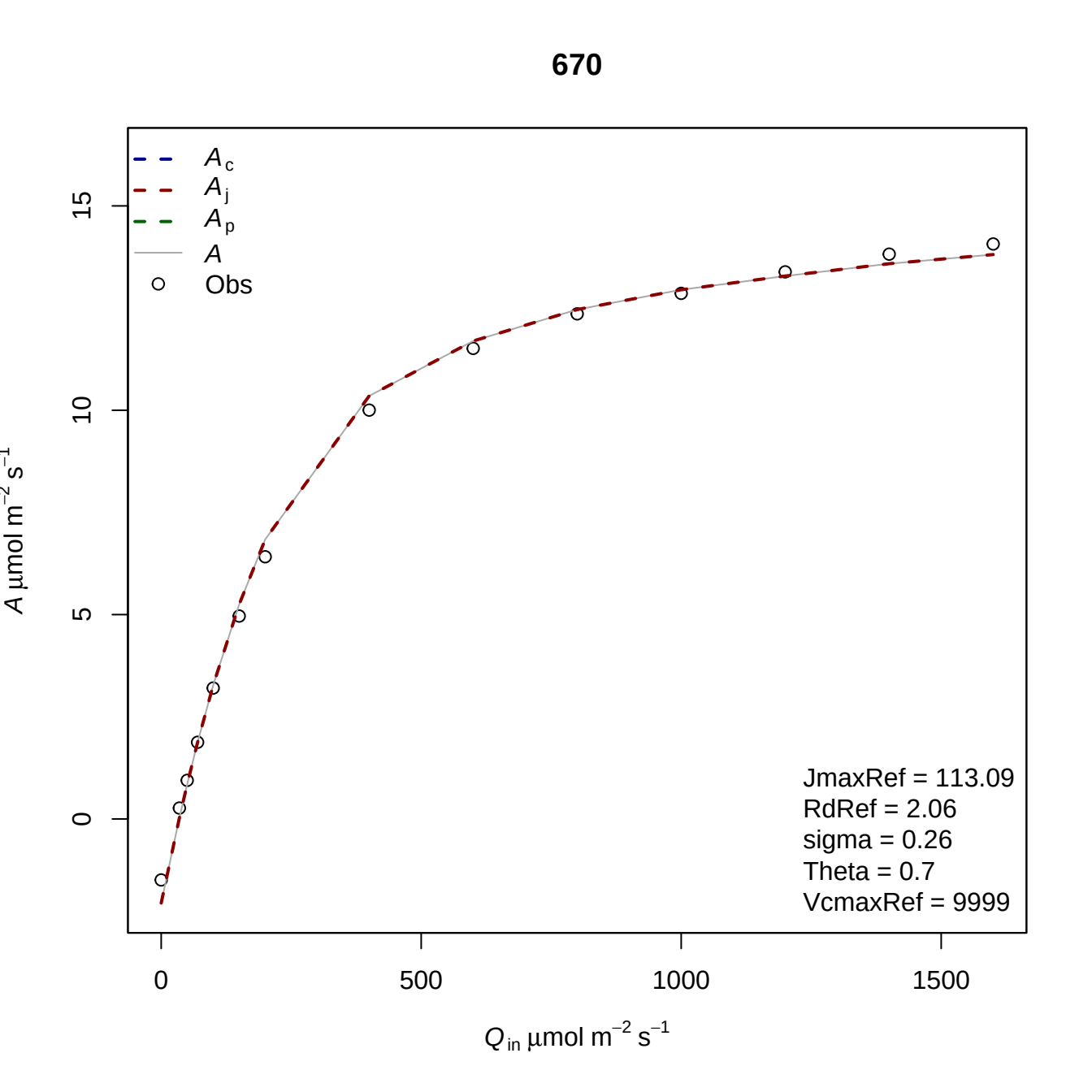

665
A_c
A_j
A_p
A
Obs
JmaxRef = 64.34
RdRef = 1.33
sigma = 0.23
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

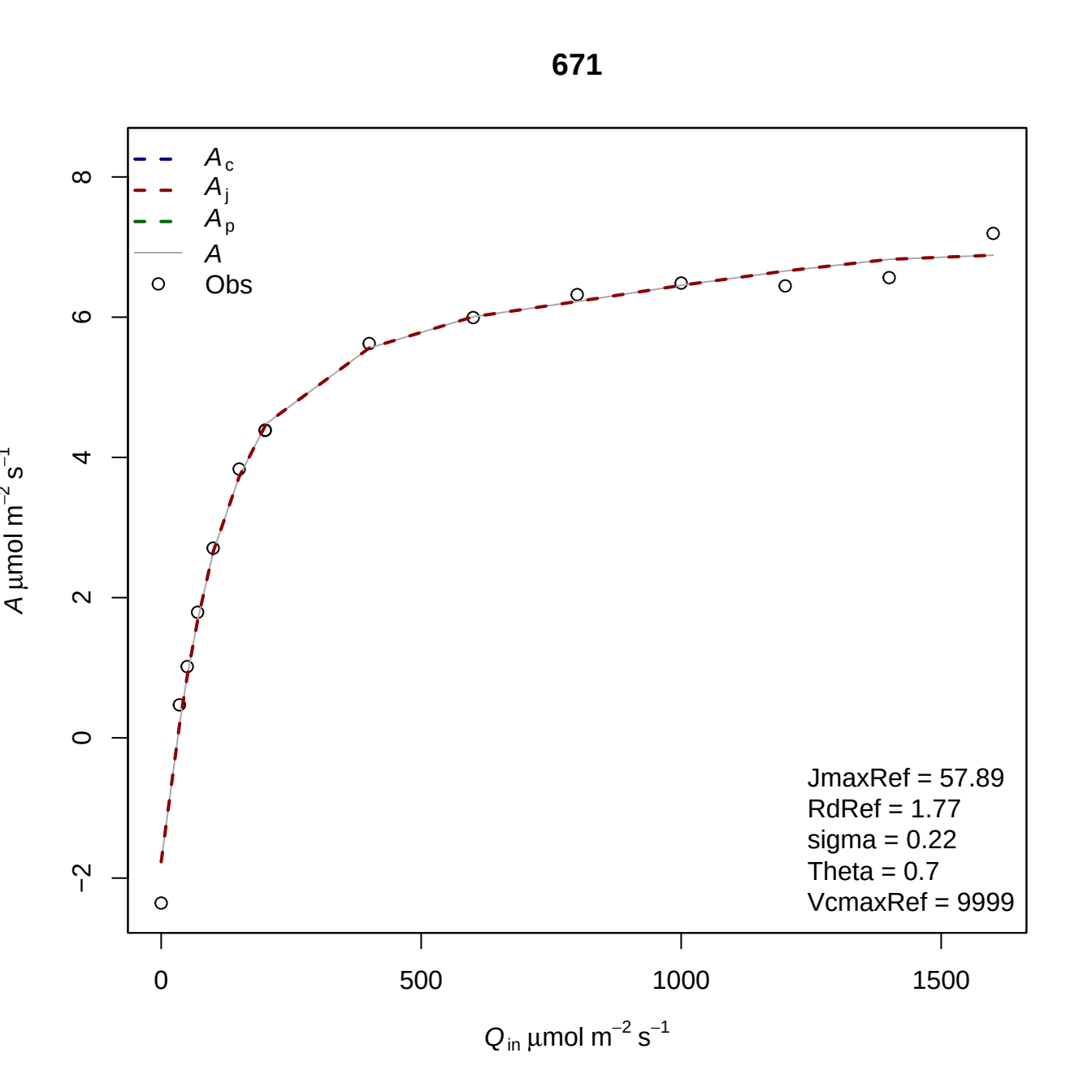

666
A_c
A_j
A_p
A
Obs
JmaxRef = 39.69
RdRef = 0.97
sigma = 0.24
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

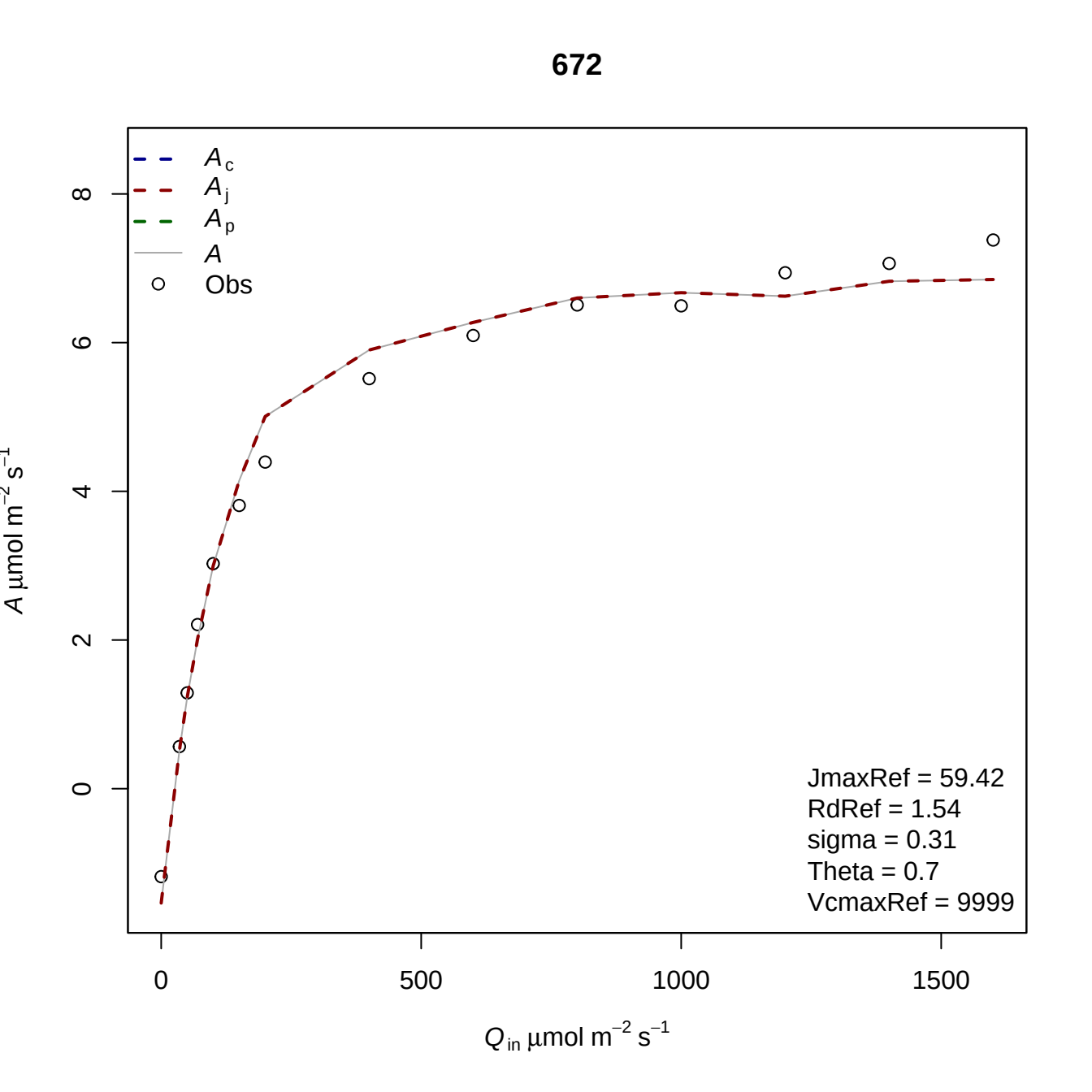

667
A_c
A_j
A_p
A
Obs
JmaxRef = 55.52
RdRef = 1.14
sigma = 0.28
Theta = 0.7
VcmaxRef = 38.98
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

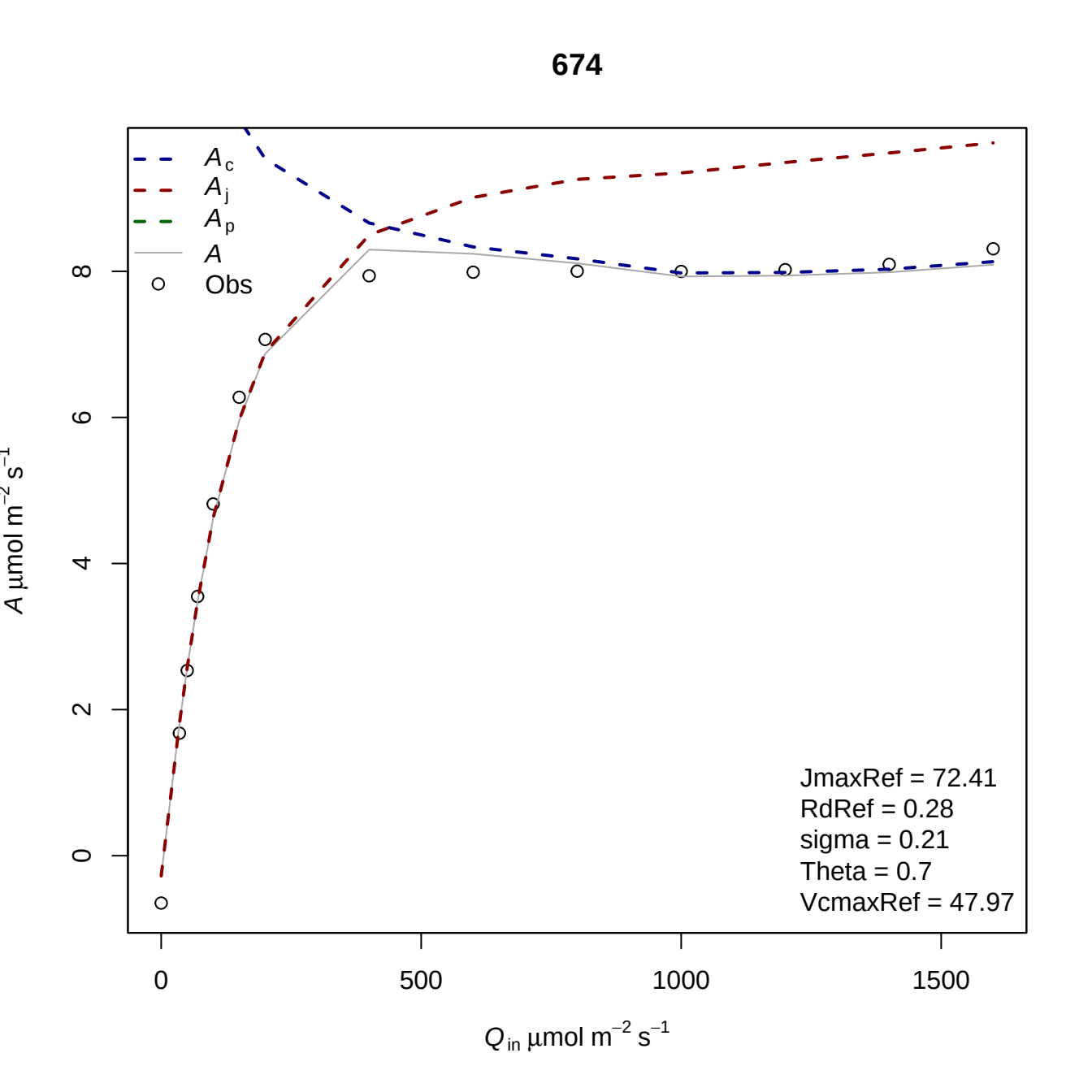

668
A_c
A_j
A_p
A
Obs
JmaxRef = 59.97
RdRef = 0.88
sigma = 0.19
Theta = 0.7
VcmaxRef = 45.19
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

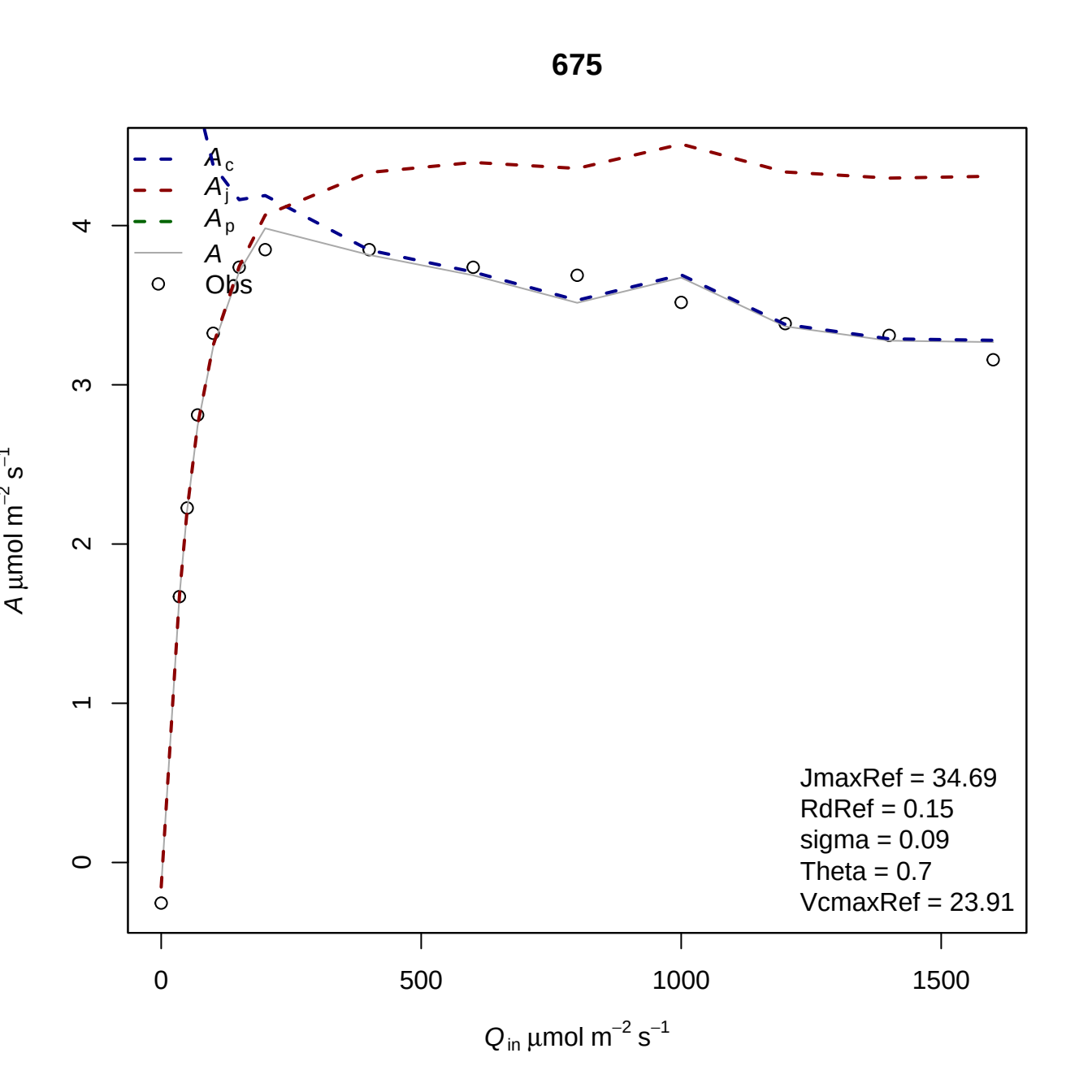

669
A_c
A_j
A_p
A
Obs
JmaxRef = 67.16
RdRef = 1.41
sigma = 0.24
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

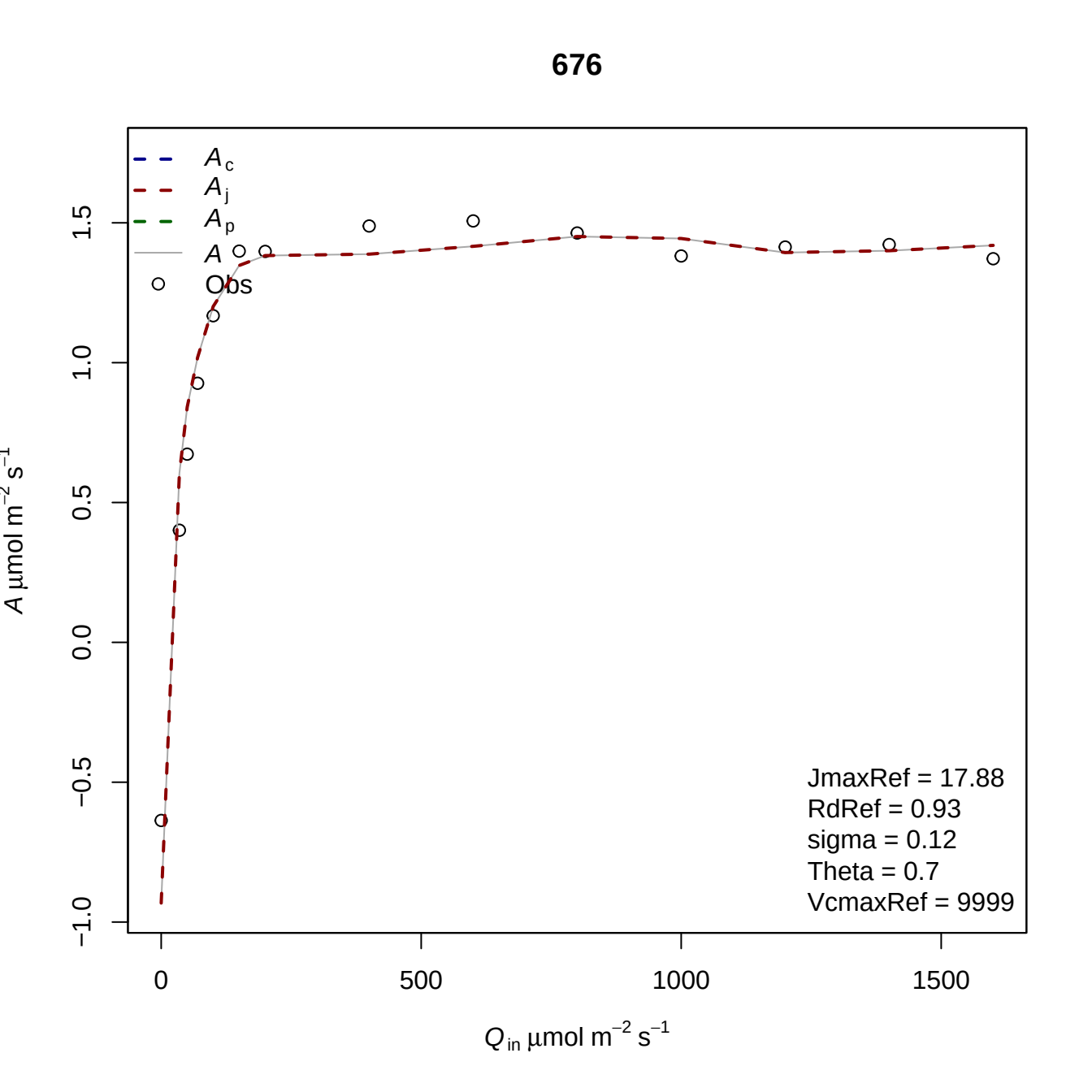

670
A_c
A_j
A_p
A
Obs
JmaxRef = 113.09
RdRef = 2.06
sigma = 0.26
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

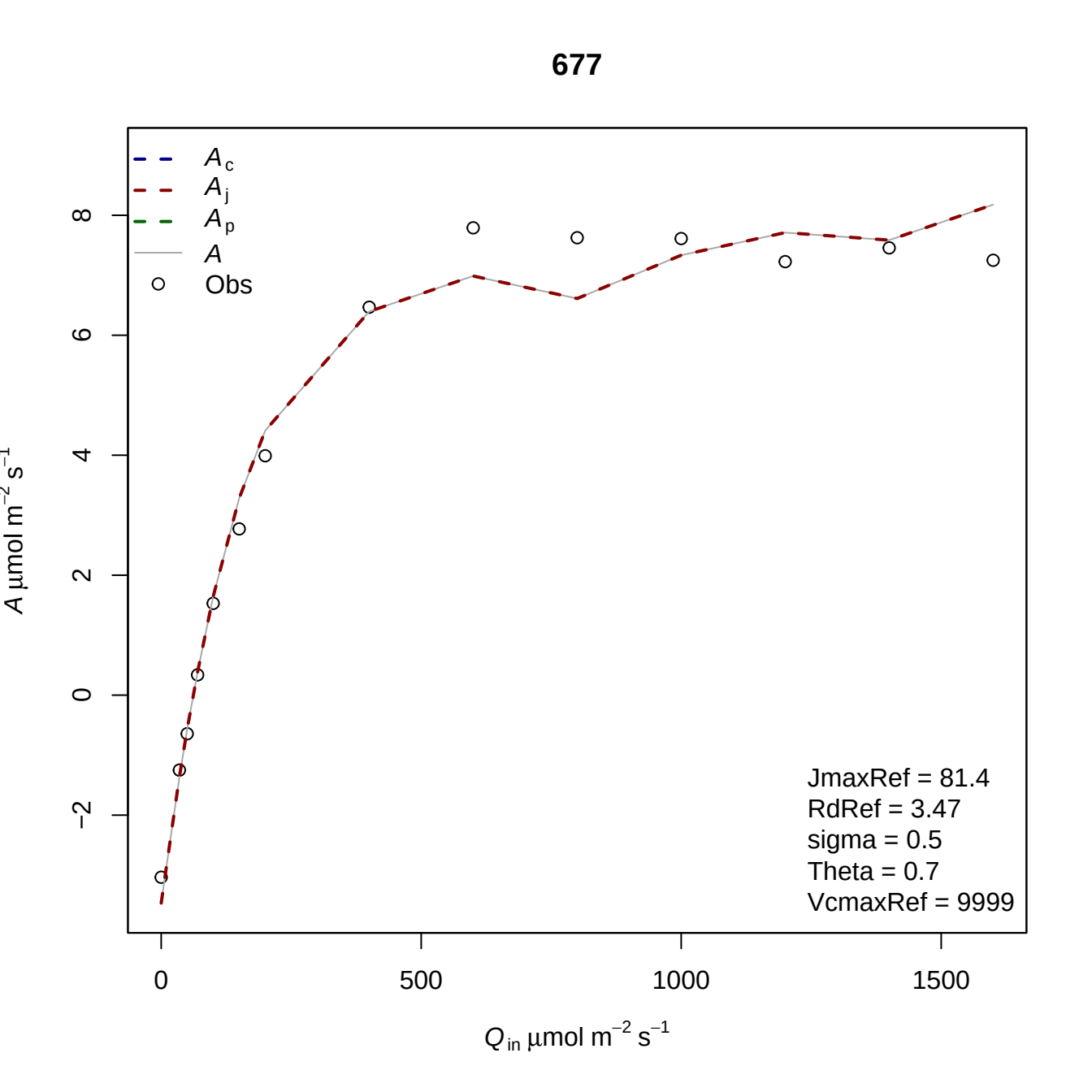

671
A_c
A_j
A_p
A
Obs
JmaxRef = 57.89
RdRef = 1.77
sigma = 0.22
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

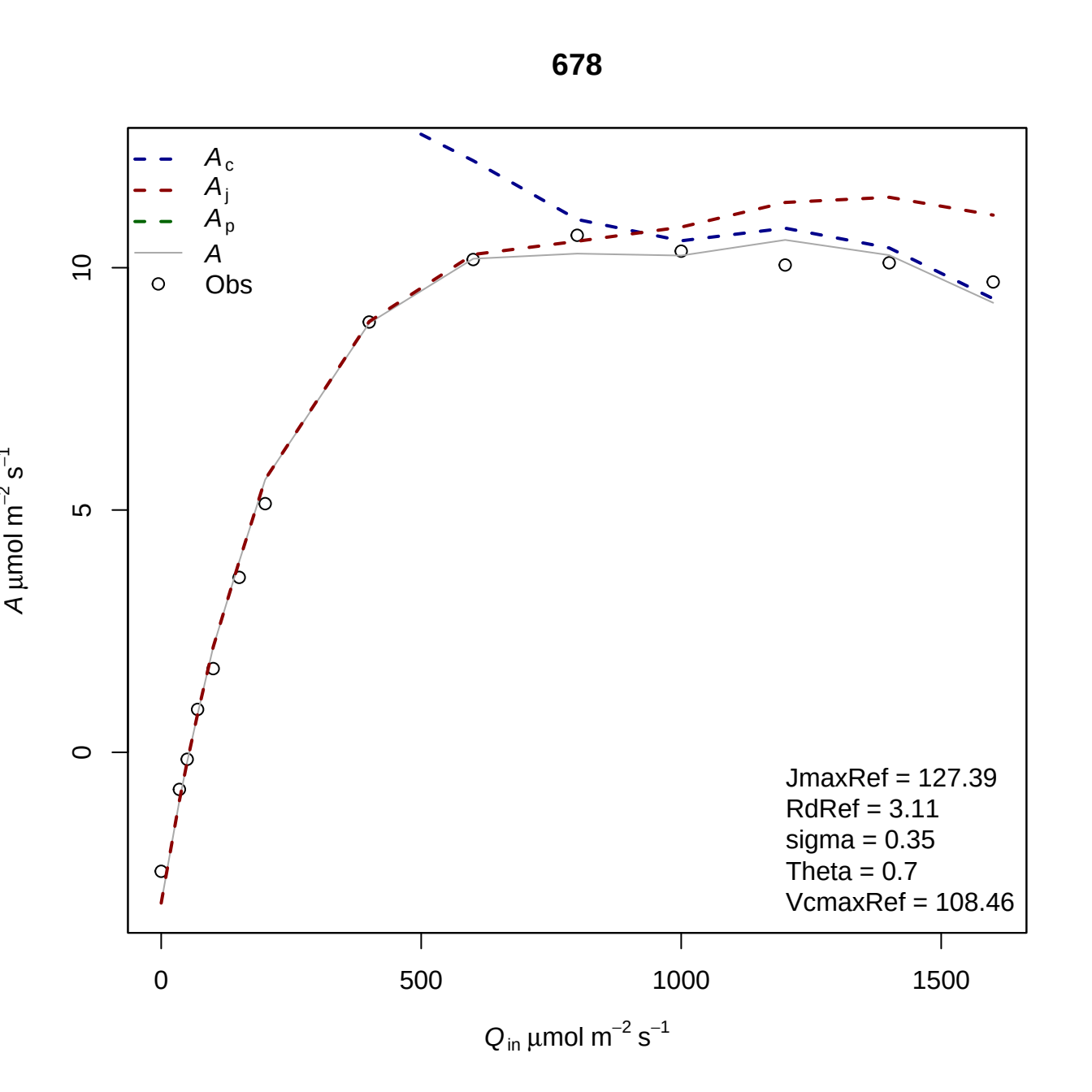

672
A_c
A_j
A_p
A
Obs
JmaxRef = 59.42
RdRef = 1.54
sigma = 0.31
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

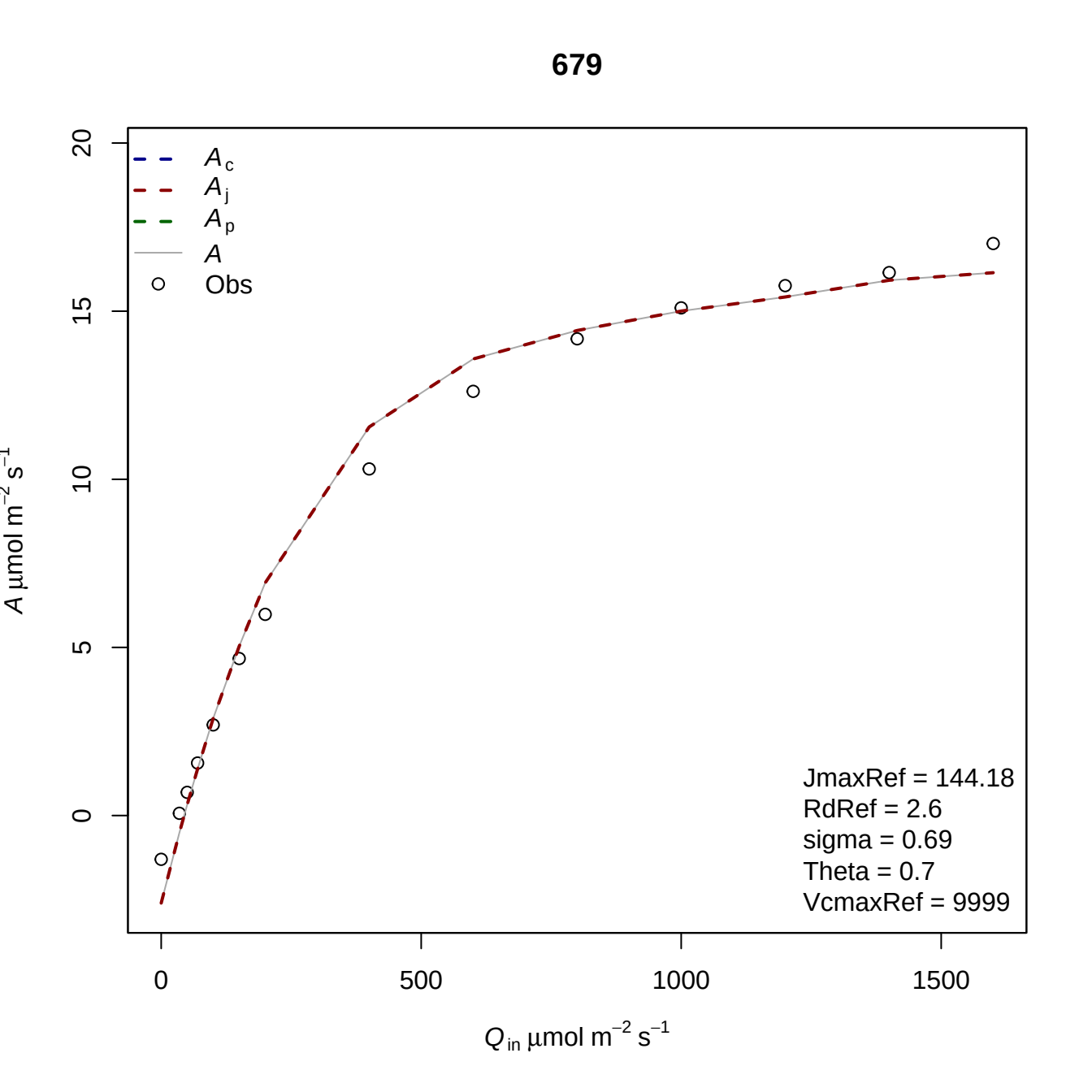

674
A_c
A_j
A_p
A
Obs
JmaxRef = 72.41
RdRef = 0.28
sigma = 0.21
Theta = 0.7
VcmaxRef = 47.97
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

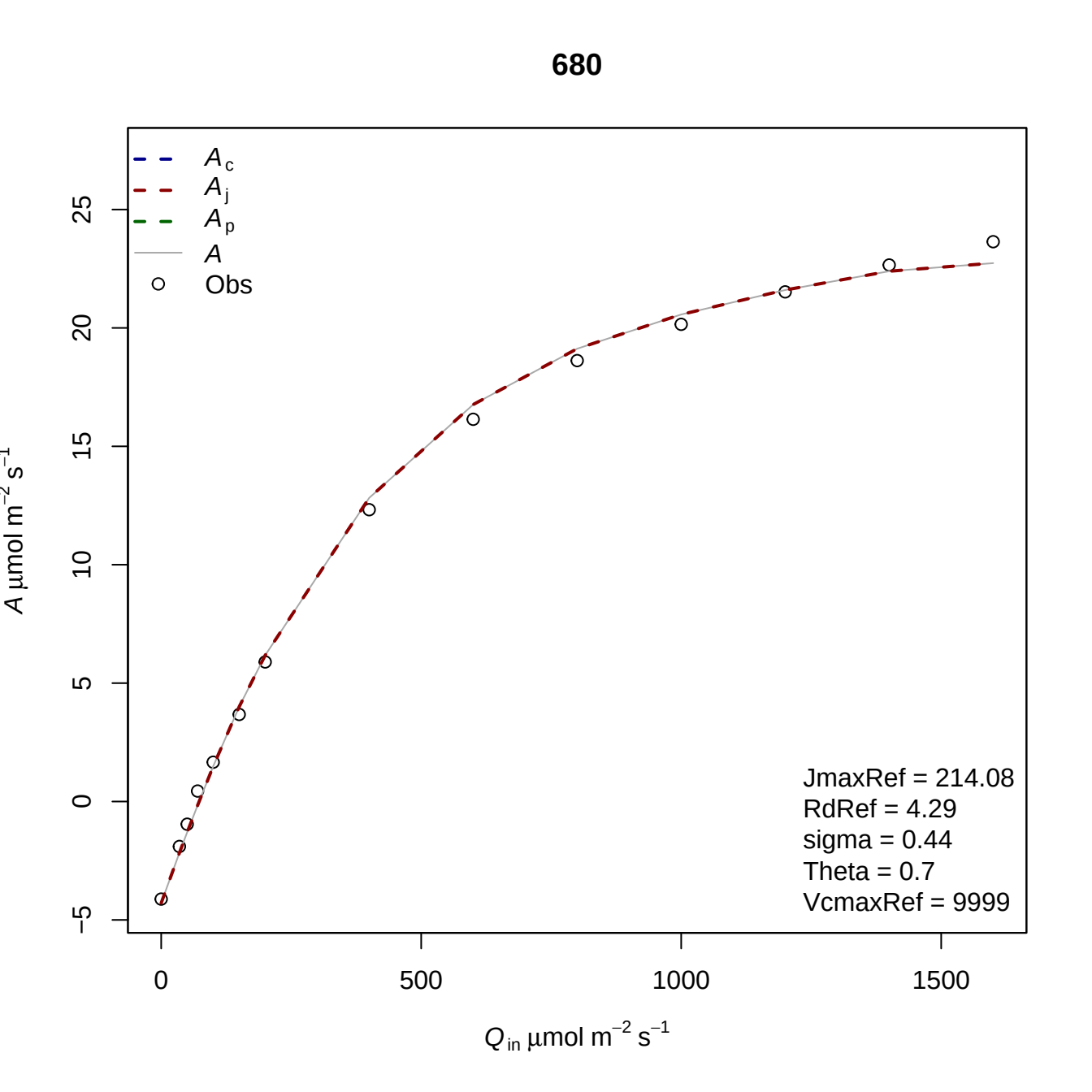

675
A_c
A_j
A_p
A
Obs
JmaxRef = 34.69
RdRef = 0.15
sigma = 0.09
Theta = 0.7
VcmaxRef = 23.91
0
1
2
3
4
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

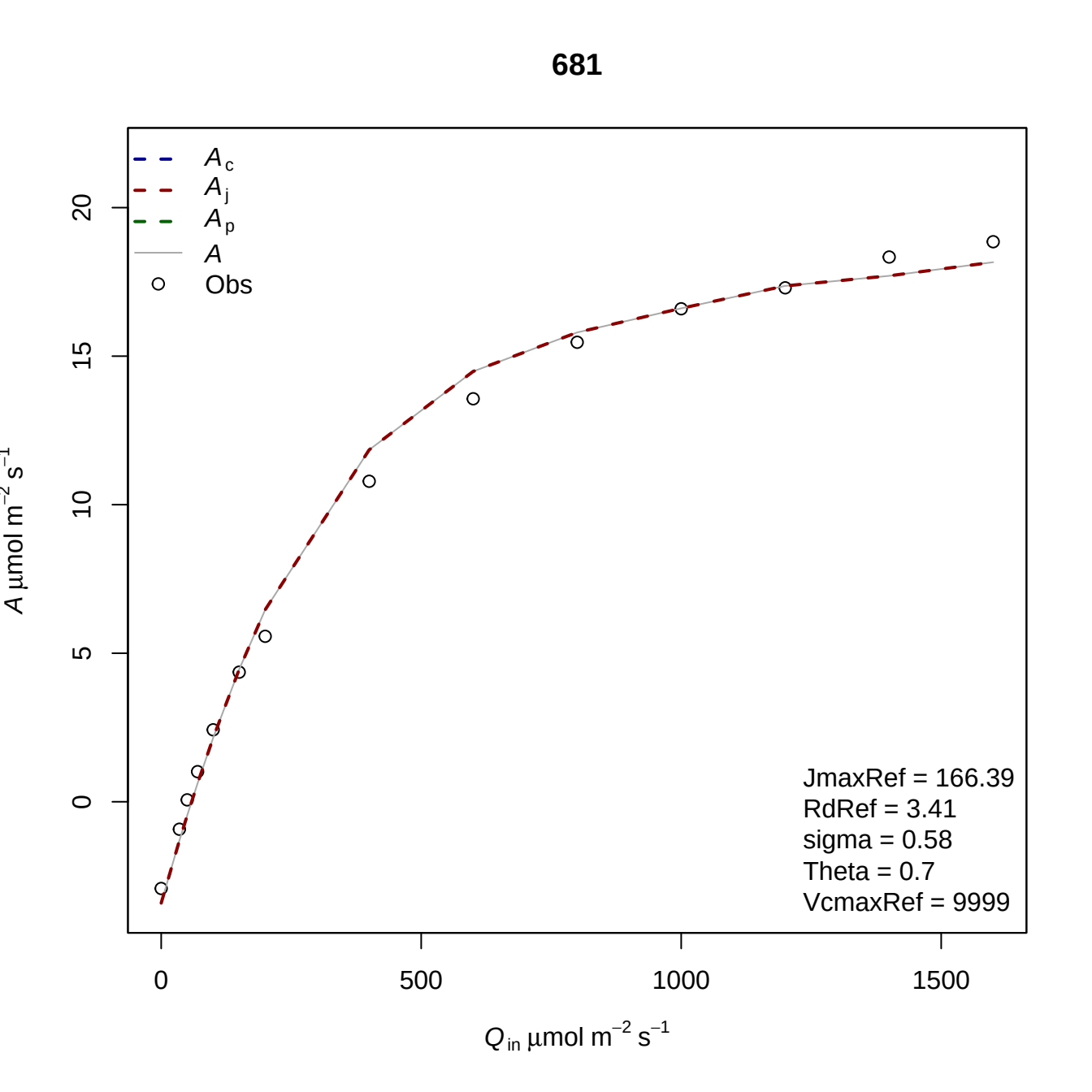

676
A_c
A_j
A_p
A
Obs
JmaxRef = 17.88
RdRef = 0.93
sigma = 0.12
Theta = 0.7
VcmaxRef = 9999
1.5
1.0
0.5
0.0
−0.5
−1.0
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

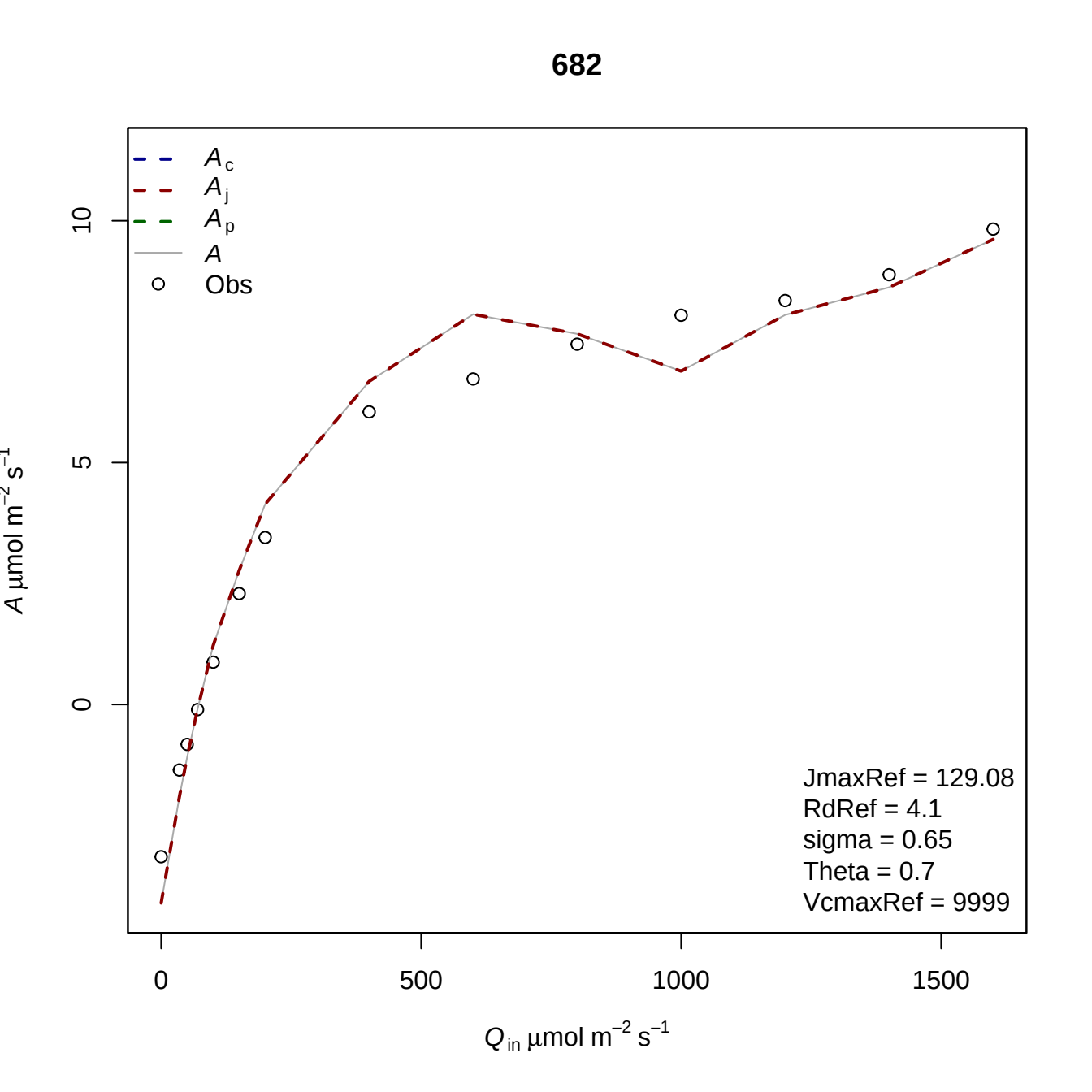

677
A_c
A_j
A_p
A
Obs
JmaxRef = 81.4
RdRef = 3.47
sigma = 0.5
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

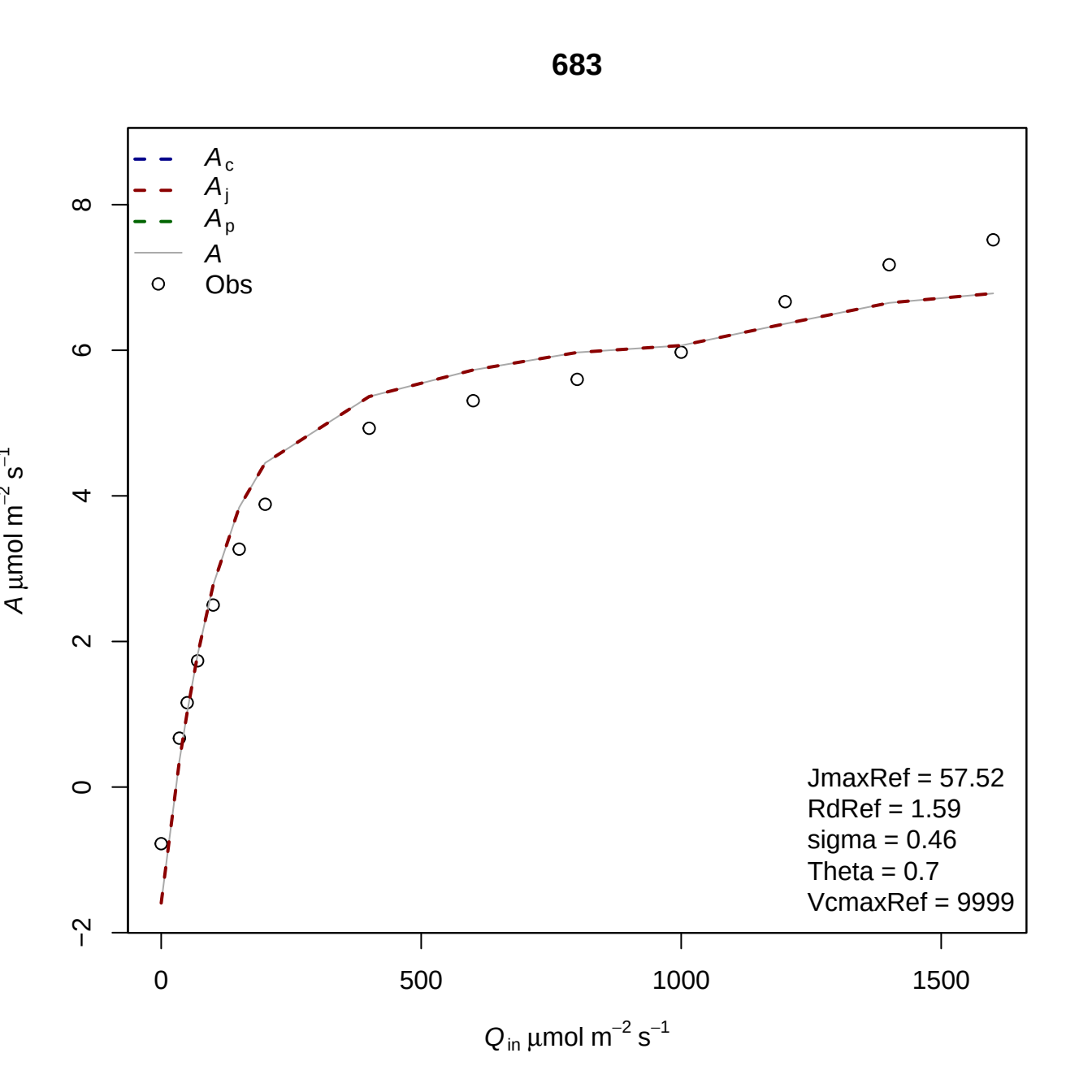

678
A_c
A_j
A_p
A
Obs
JmaxRef = 127.39
RdRef = 3.11
sigma = 0.35
Theta = 0.7
VcmaxRef = 108.46
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

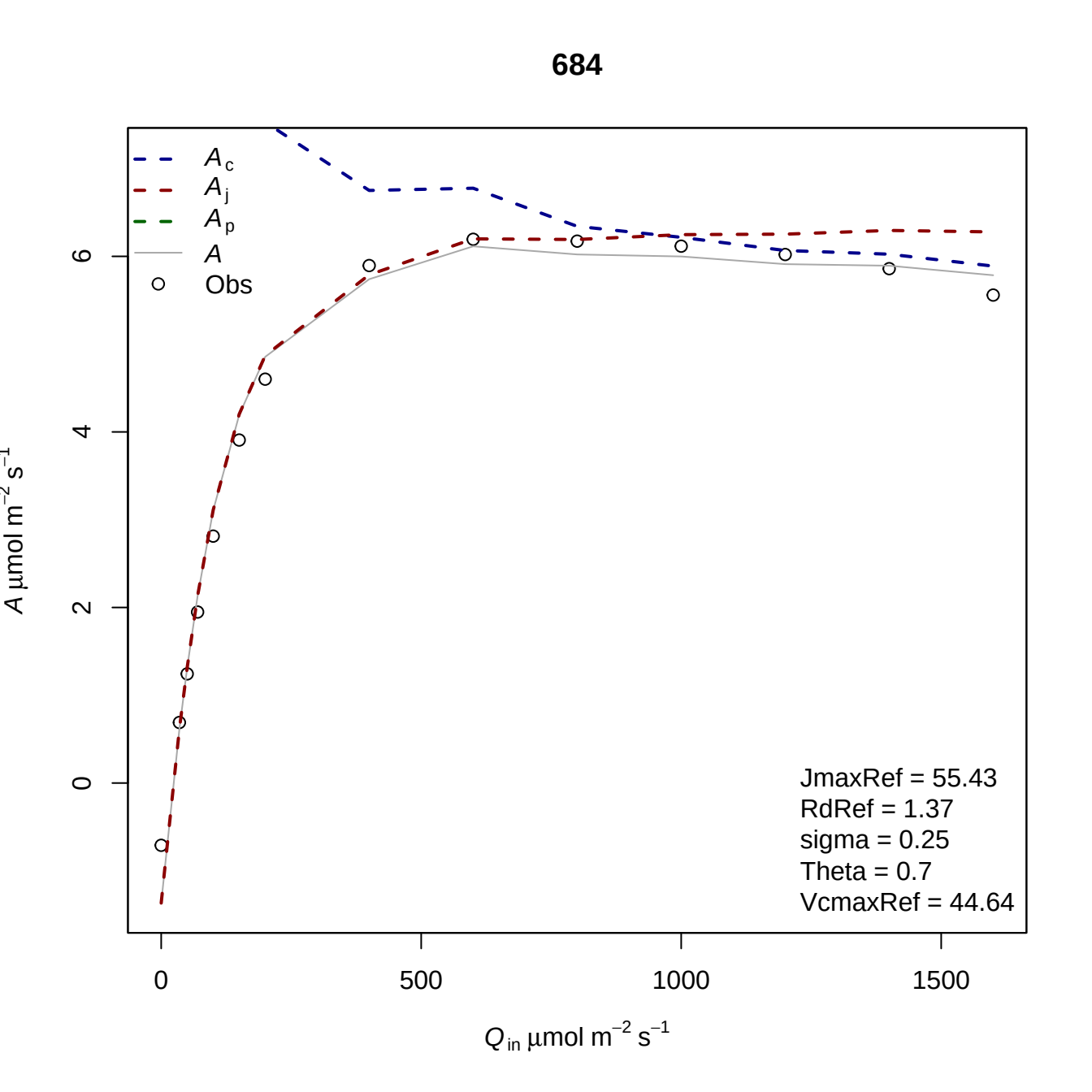

679
A_c
A_j
A_p
A
Obs
JmaxRef = 144.18
RdRef = 2.6
sigma = 0.69
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

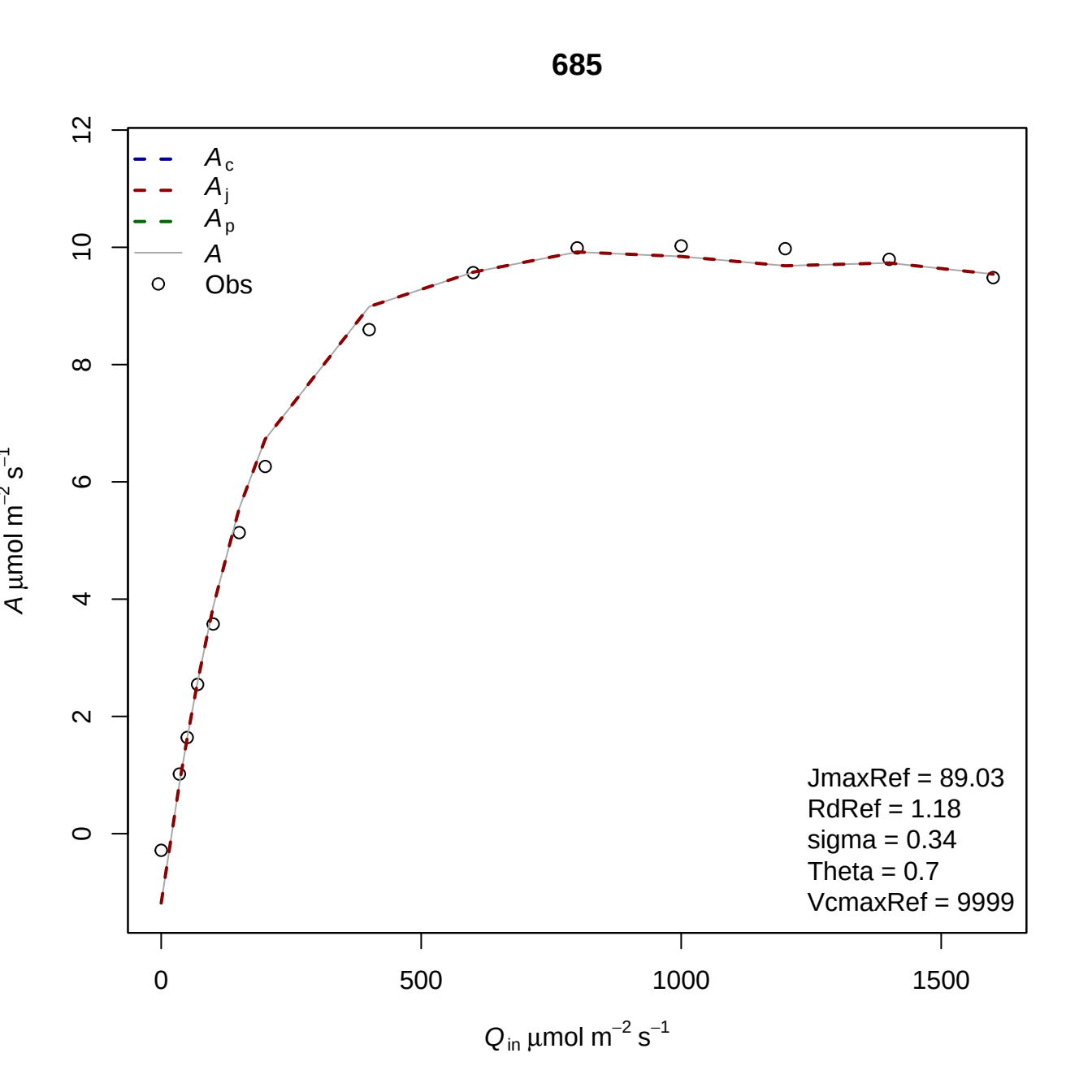

680
A_c
A_j
A_p
A
Obs
JmaxRef = 214.08
RdRef = 4.29
sigma = 0.44
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
25
0
500
1000
1500

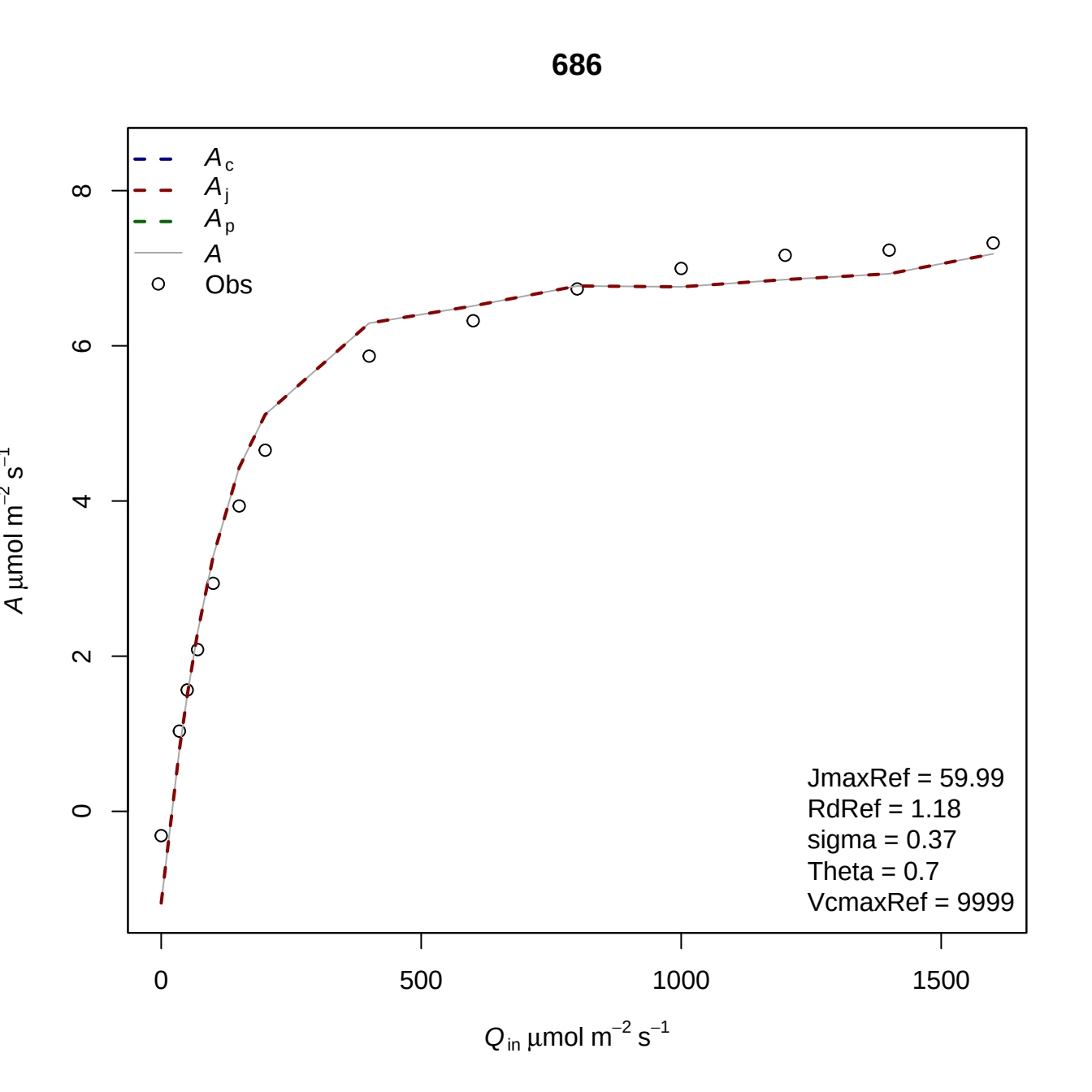

681
A_c
A_j
A_p
A
Obs
JmaxRef = 166.39
RdRef = 3.41
sigma = 0.58
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

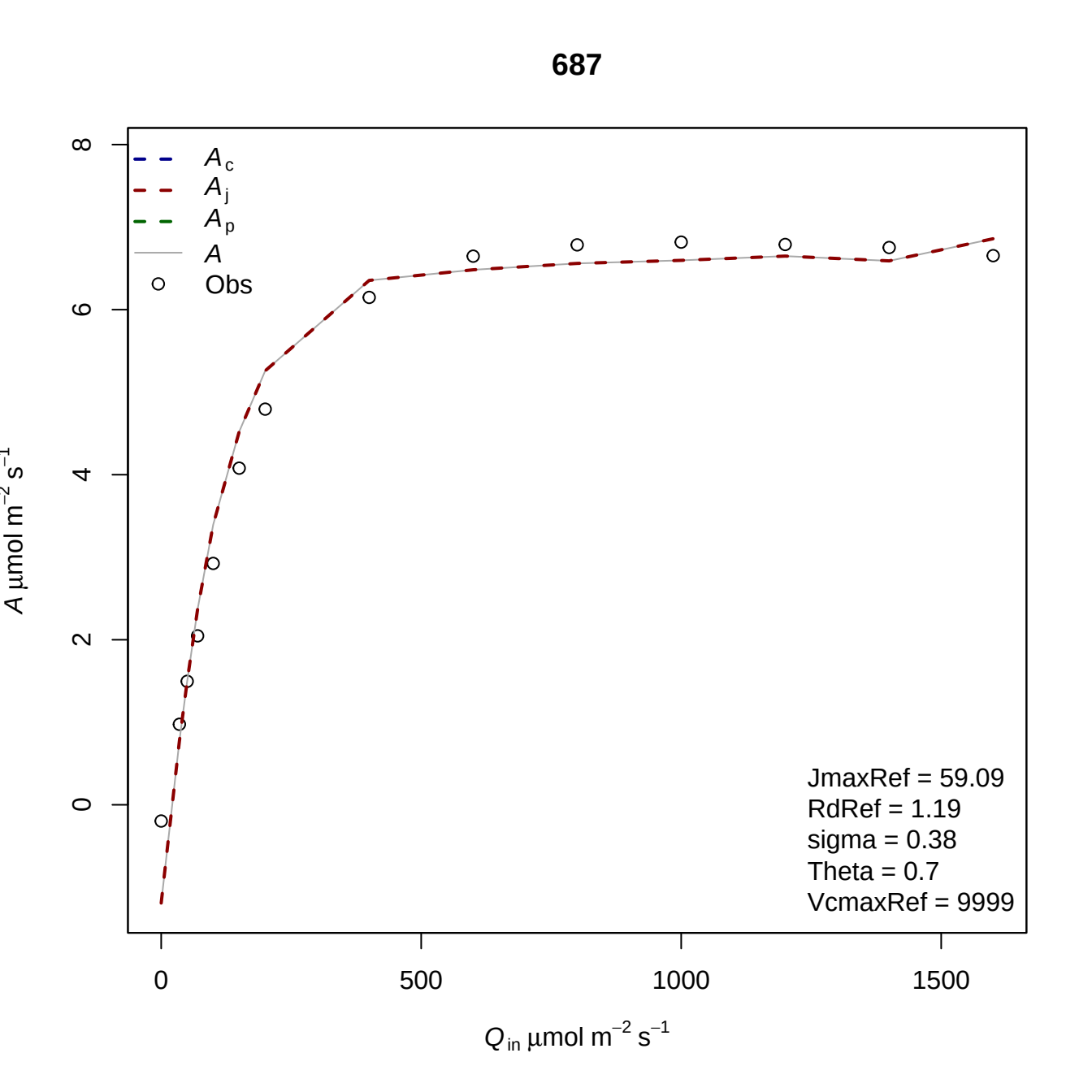

682
A_c
A_j
A_p
A
Obs
JmaxRef = 129.08
RdRef = 4.1
sigma = 0.65
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

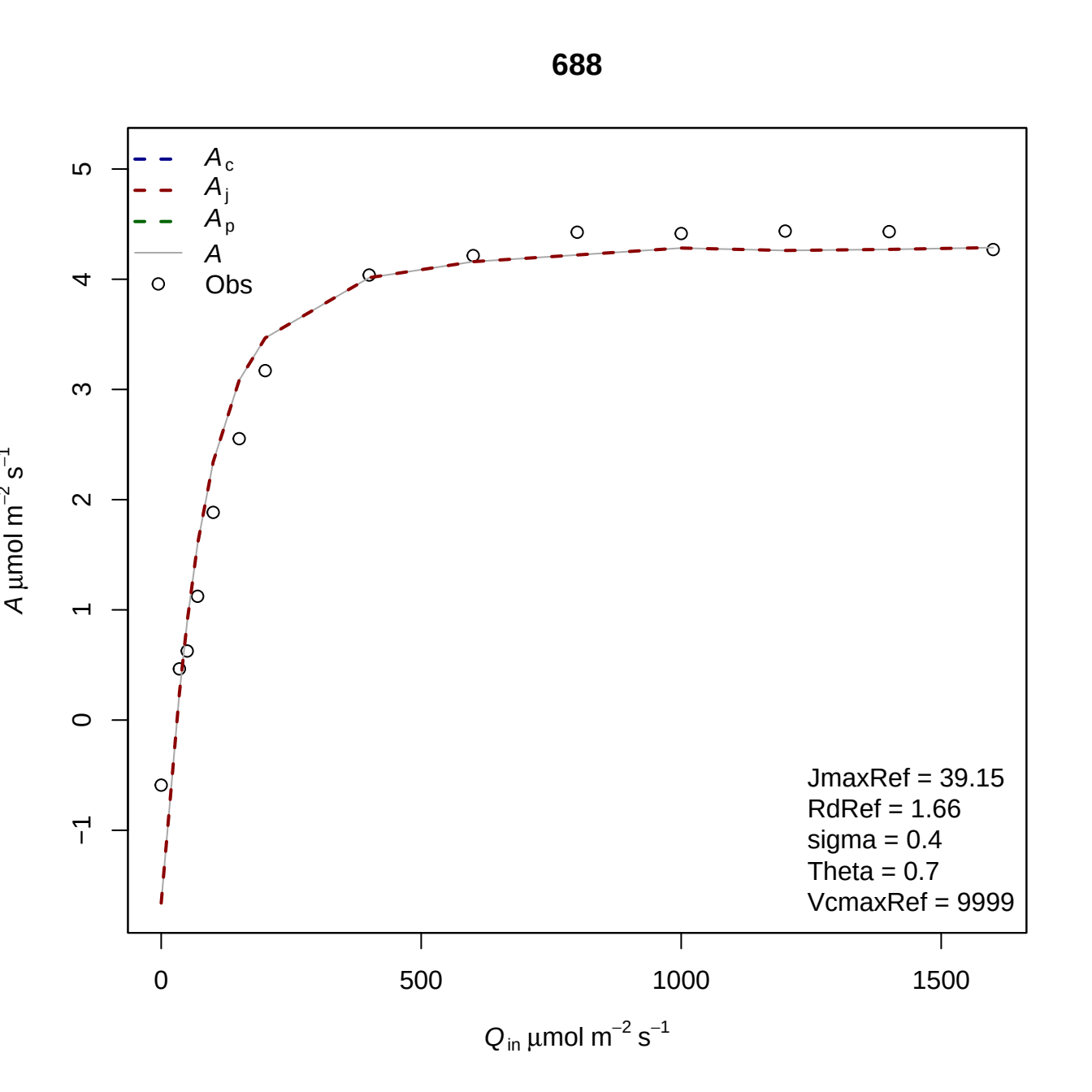

683
A_c
A_j
A_p
A
Obs
JmaxRef = 57.52
RdRef = 1.59
sigma = 0.46
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

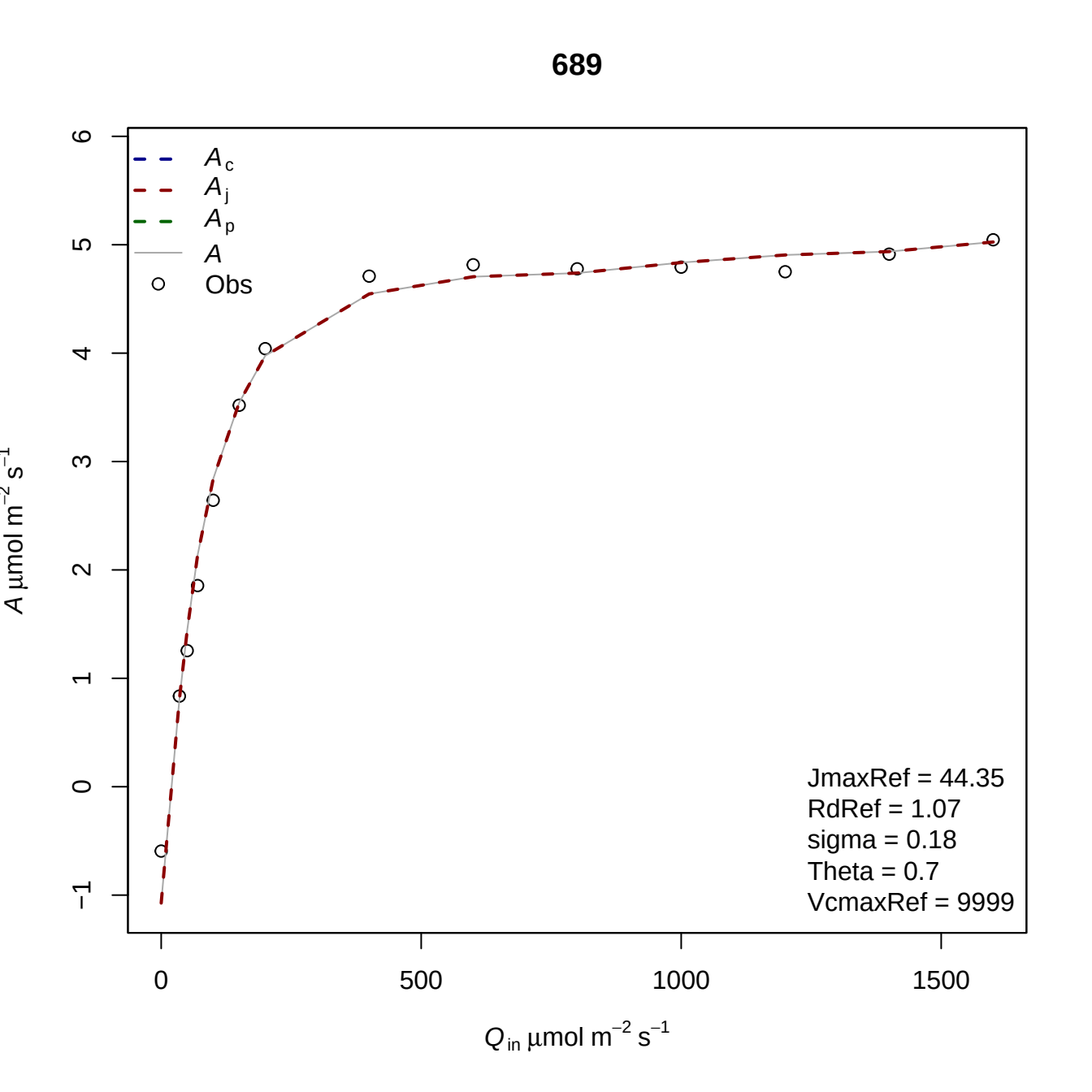

684
A_c
A_j
A_p
A
Obs
JmaxRef = 55.43
RdRef = 1.37
sigma = 0.25
Theta = 0.7
VcmaxRef = 44.64
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

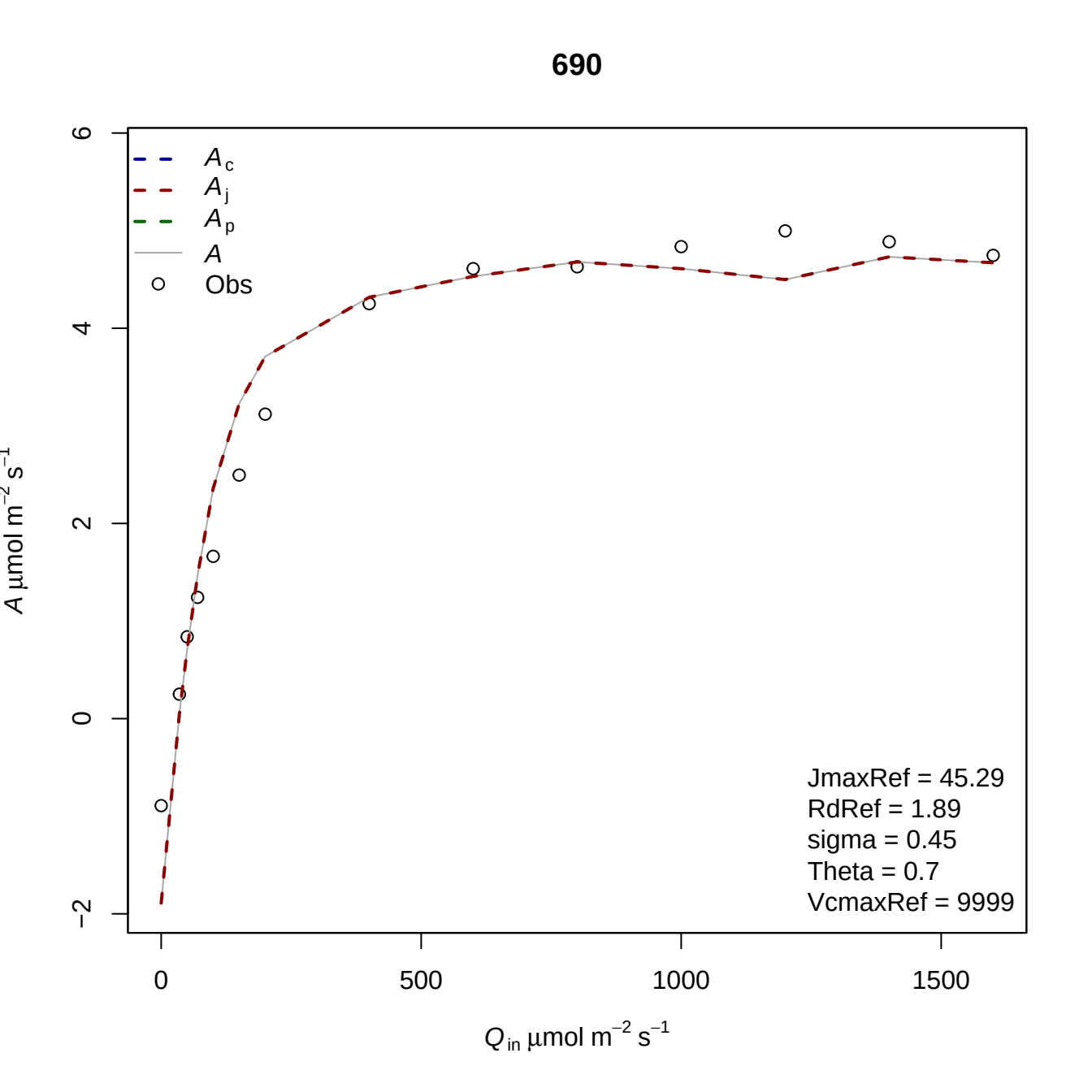

685
A_c
A_j
A_p
A
Obs
JmaxRef = 89.03
RdRef = 1.18
sigma = 0.34
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

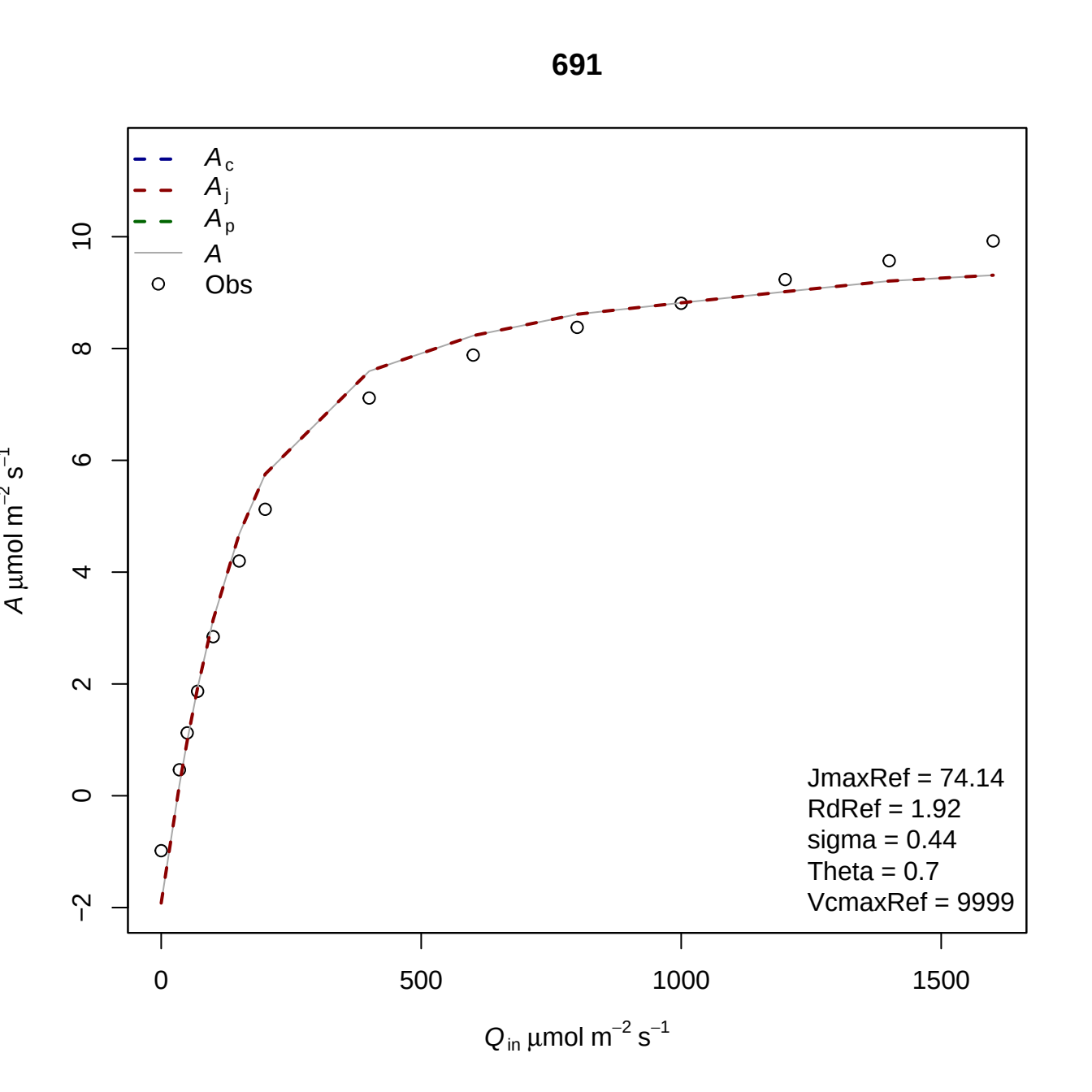

686
A_c
A_j
A_p
A
Obs
JmaxRef = 59.99
RdRef = 1.18
sigma = 0.37
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

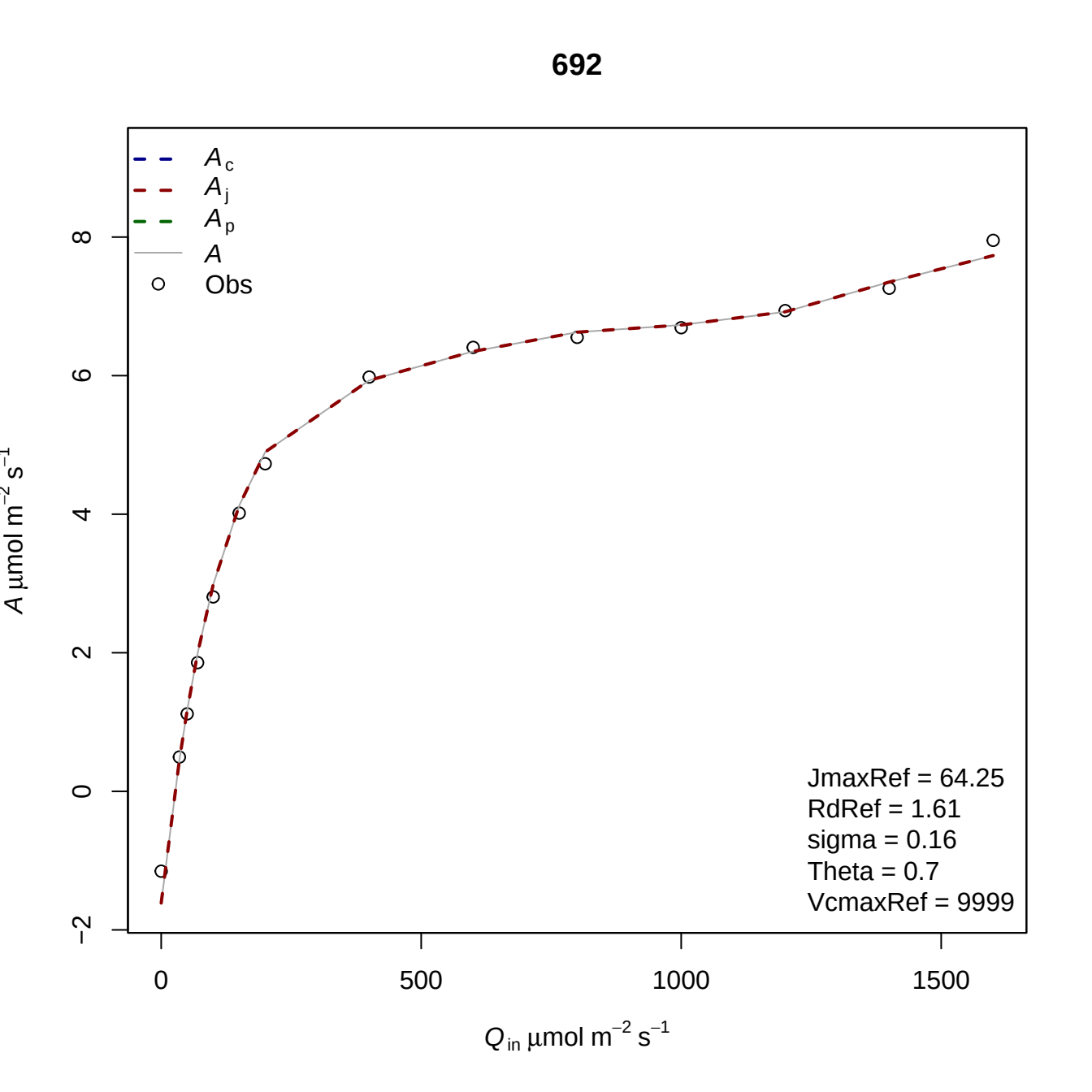

687
A_c
A_j
A_p
A
Obs
JmaxRef = 59.09
RdRef = 1.19
sigma = 0.38
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

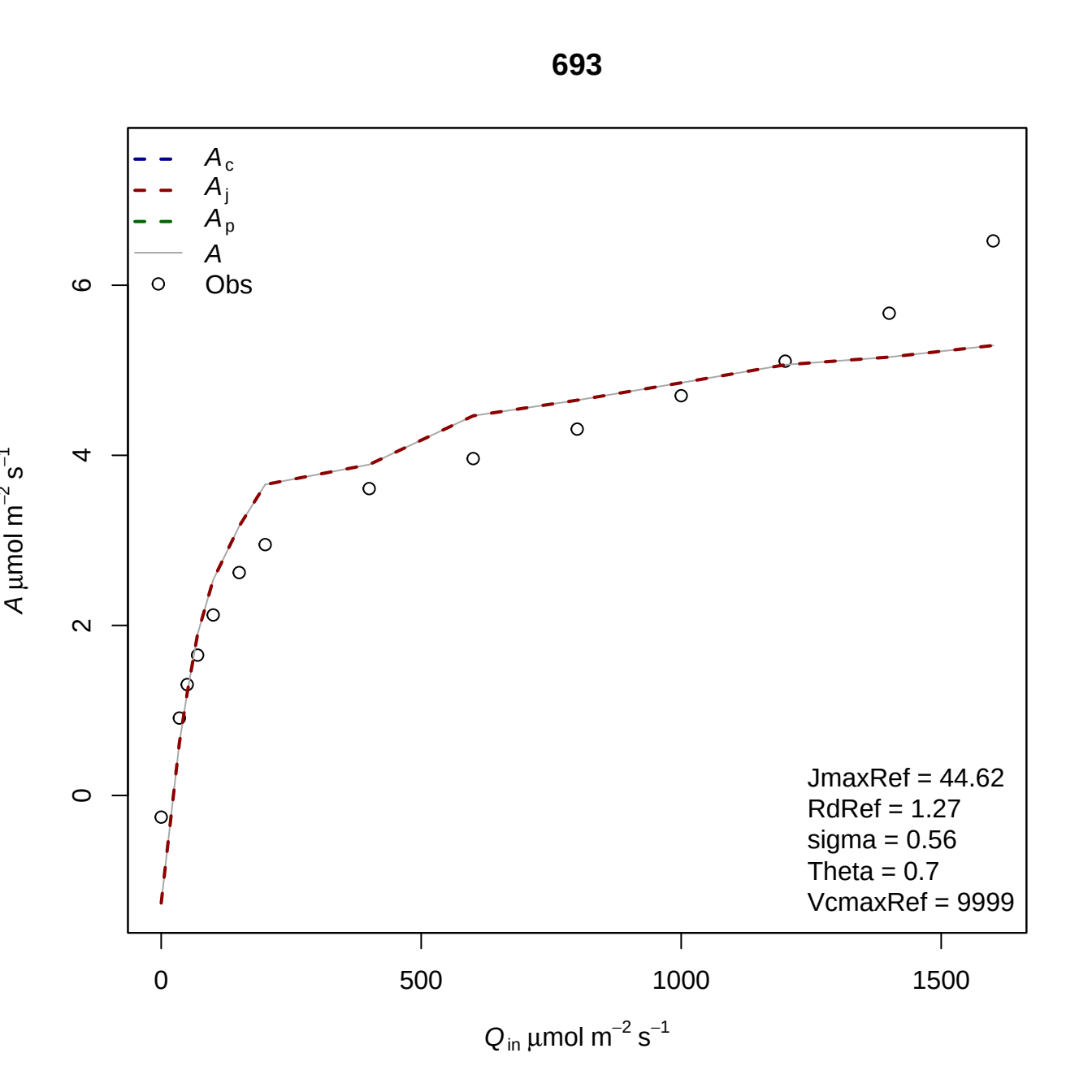

688
A_c
A_j
A_p
A
Obs
JmaxRef = 39.15
RdRef = 1.66
sigma = 0.4
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

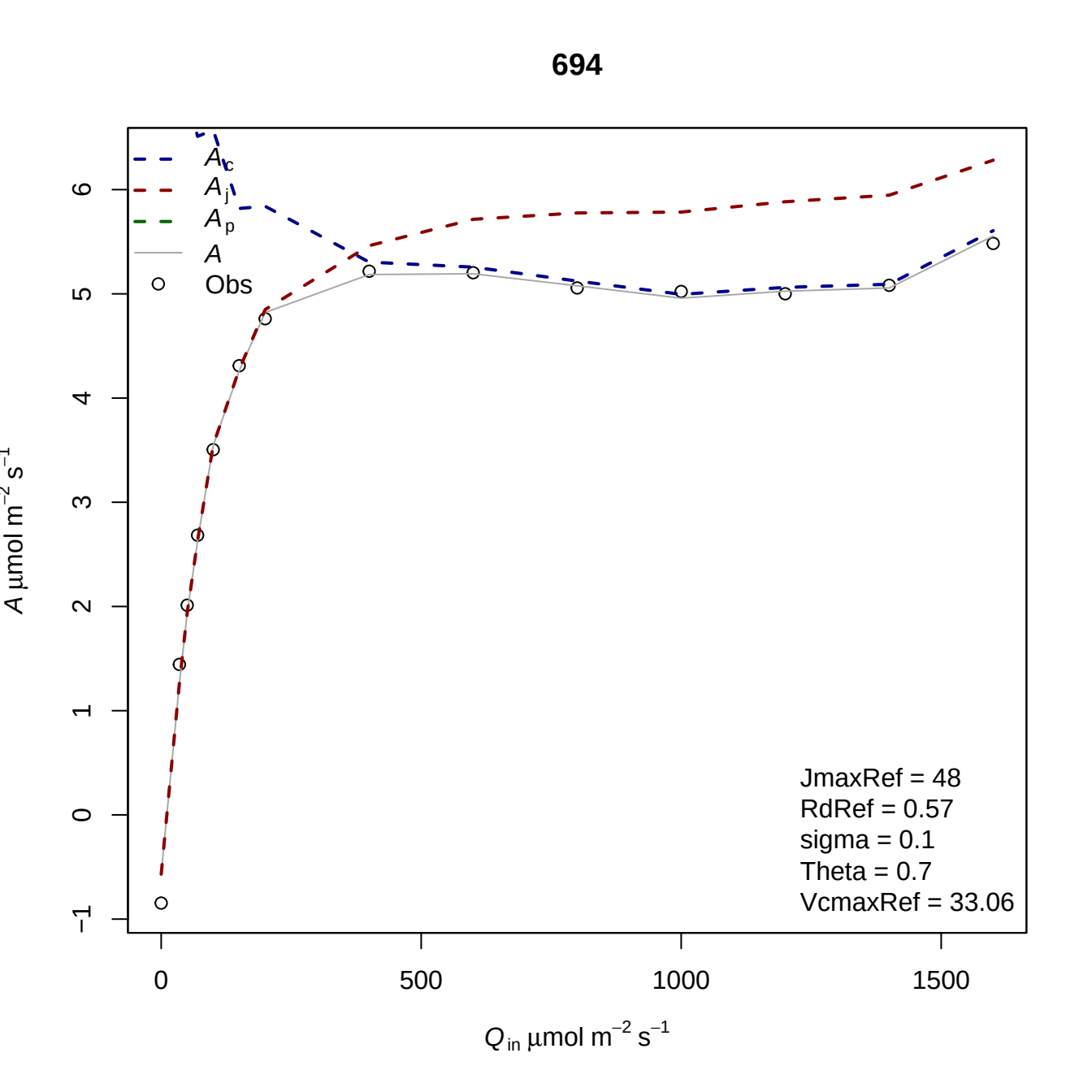

689
A_c
A_j
A_p
A
Obs
JmaxRef = 44.35
RdRef = 1.07
sigma = 0.18
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

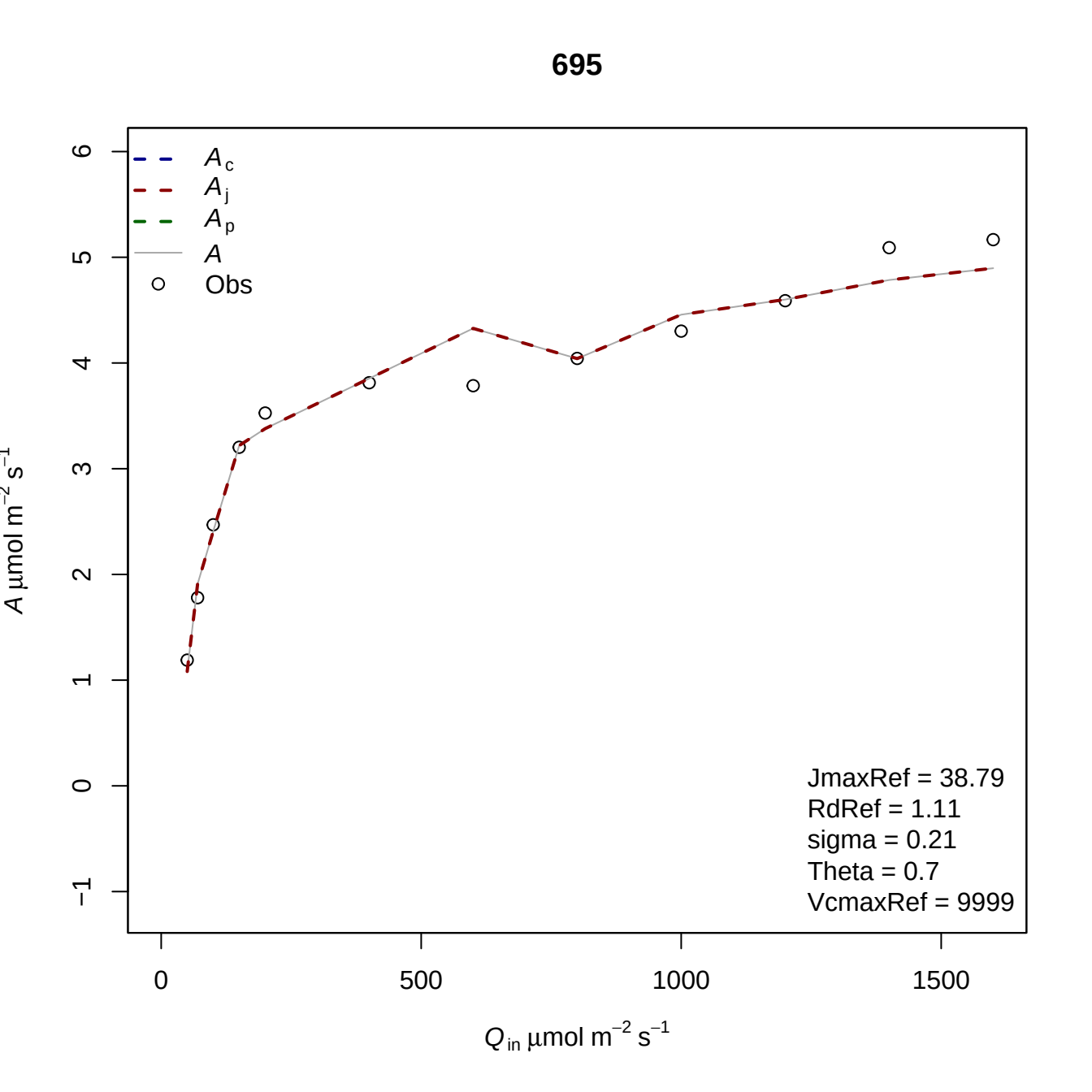

690
A_c
A_j
A_p
A
Obs
JmaxRef = 45.29
RdRef = 1.89
sigma = 0.45
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

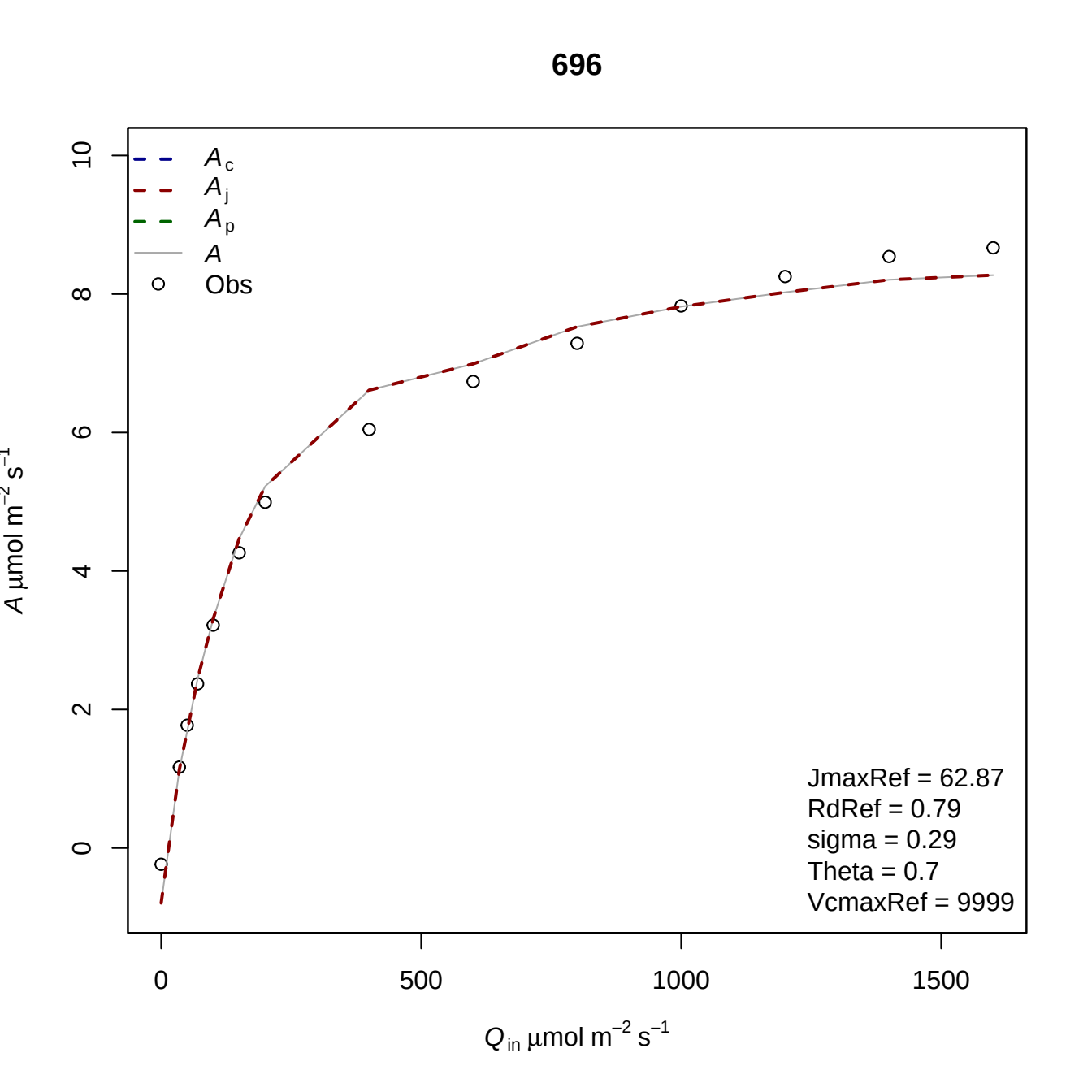

691
A_c
A_j
A_p
A
Obs
JmaxRef = 74.14
RdRef = 1.92
sigma = 0.44
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

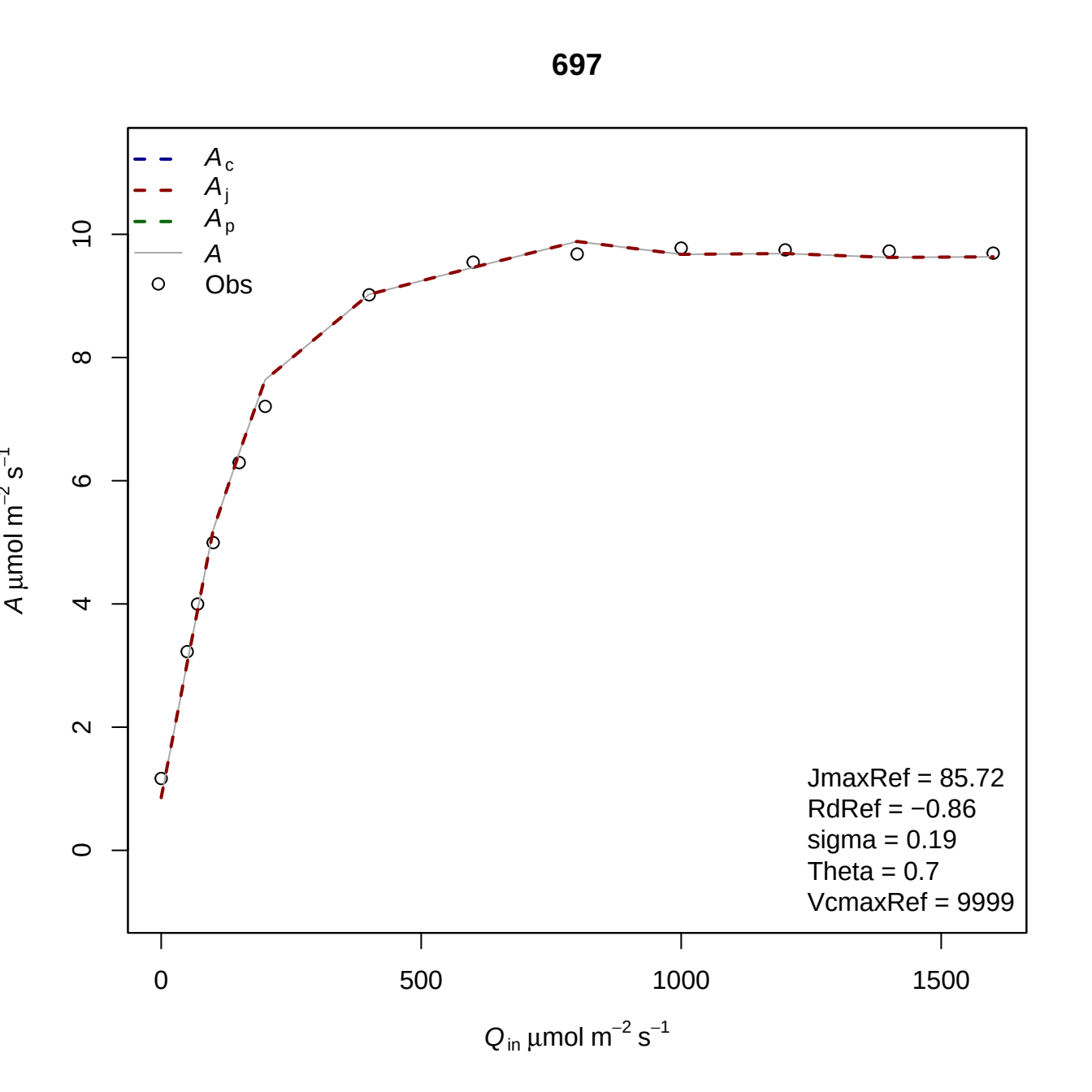

692
A_c
A_j
A_p
A
Obs
JmaxRef = 64.25
RdRef = 1.61
sigma = 0.16
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

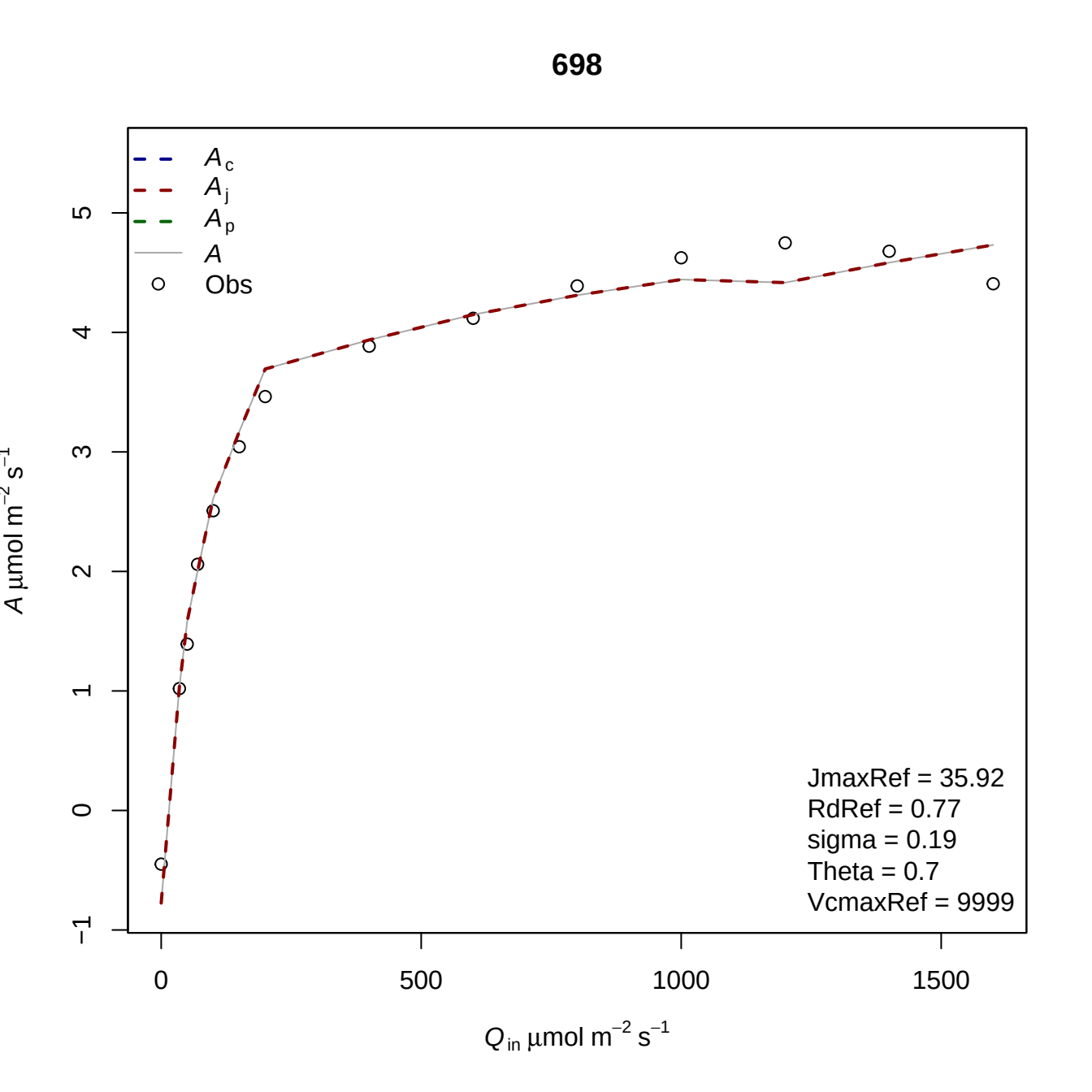

693
A_c
A_j
A_p
A
Obs
JmaxRef = 44.62
RdRef = 1.27
sigma = 0.56
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

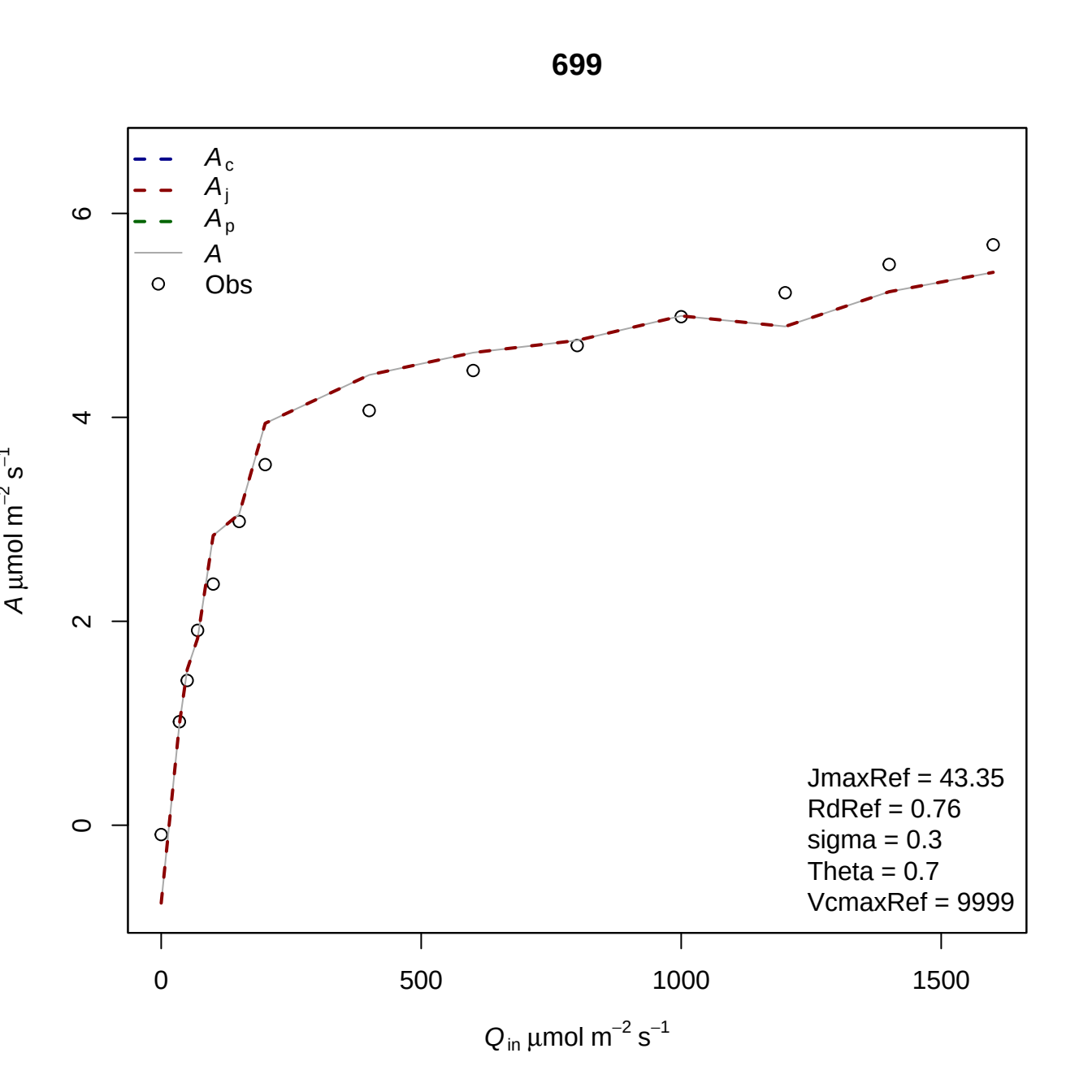

694
A_c
A_j
A_p
A
Obs
JmaxRef = 48
RdRef = 0.57
sigma = 0.1
Theta = 0.7
VcmaxRef = 33.06
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

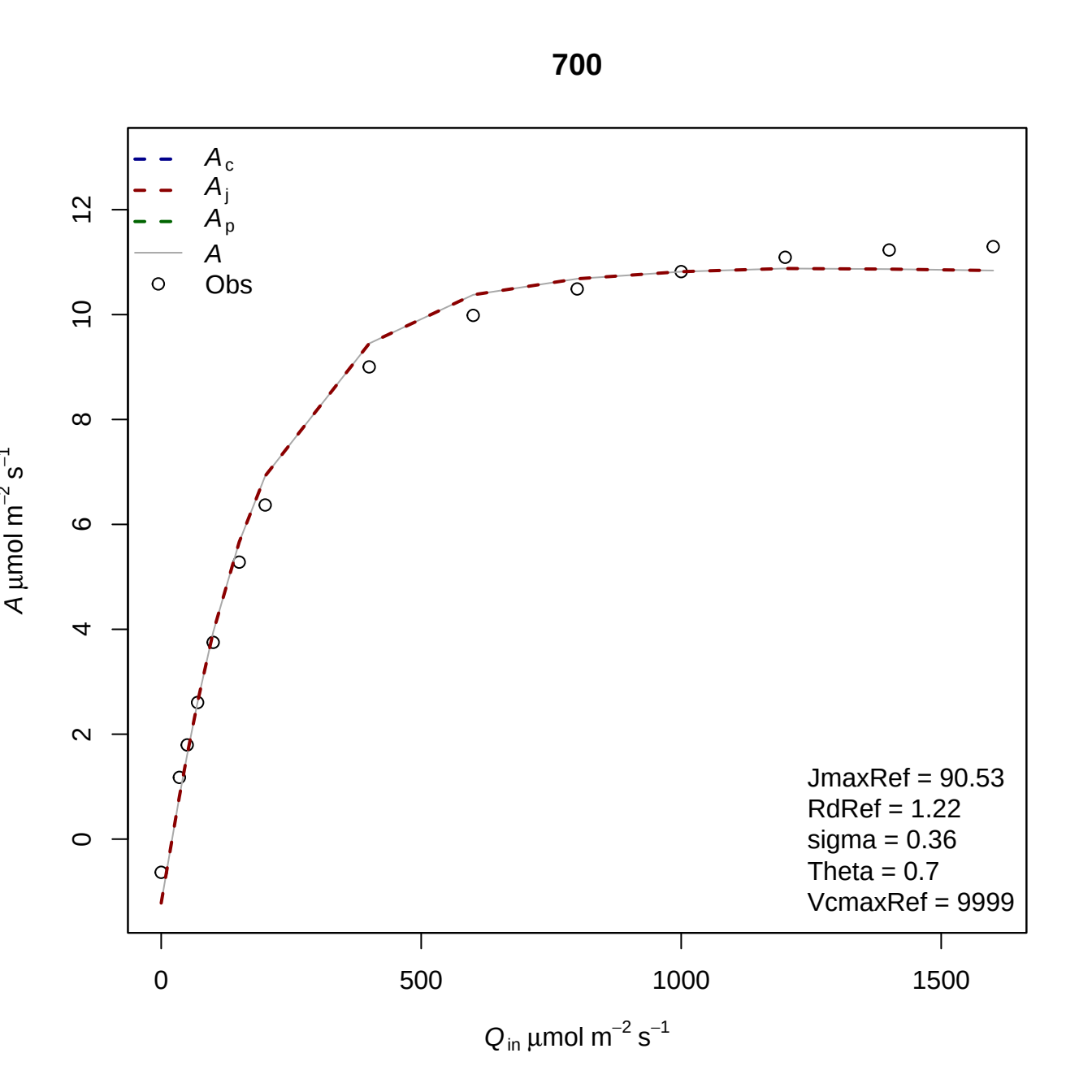

695
A_c
A_j
A_p
A
Obs
JmaxRef = 38.79
RdRef = 1.11
sigma = 0.21
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

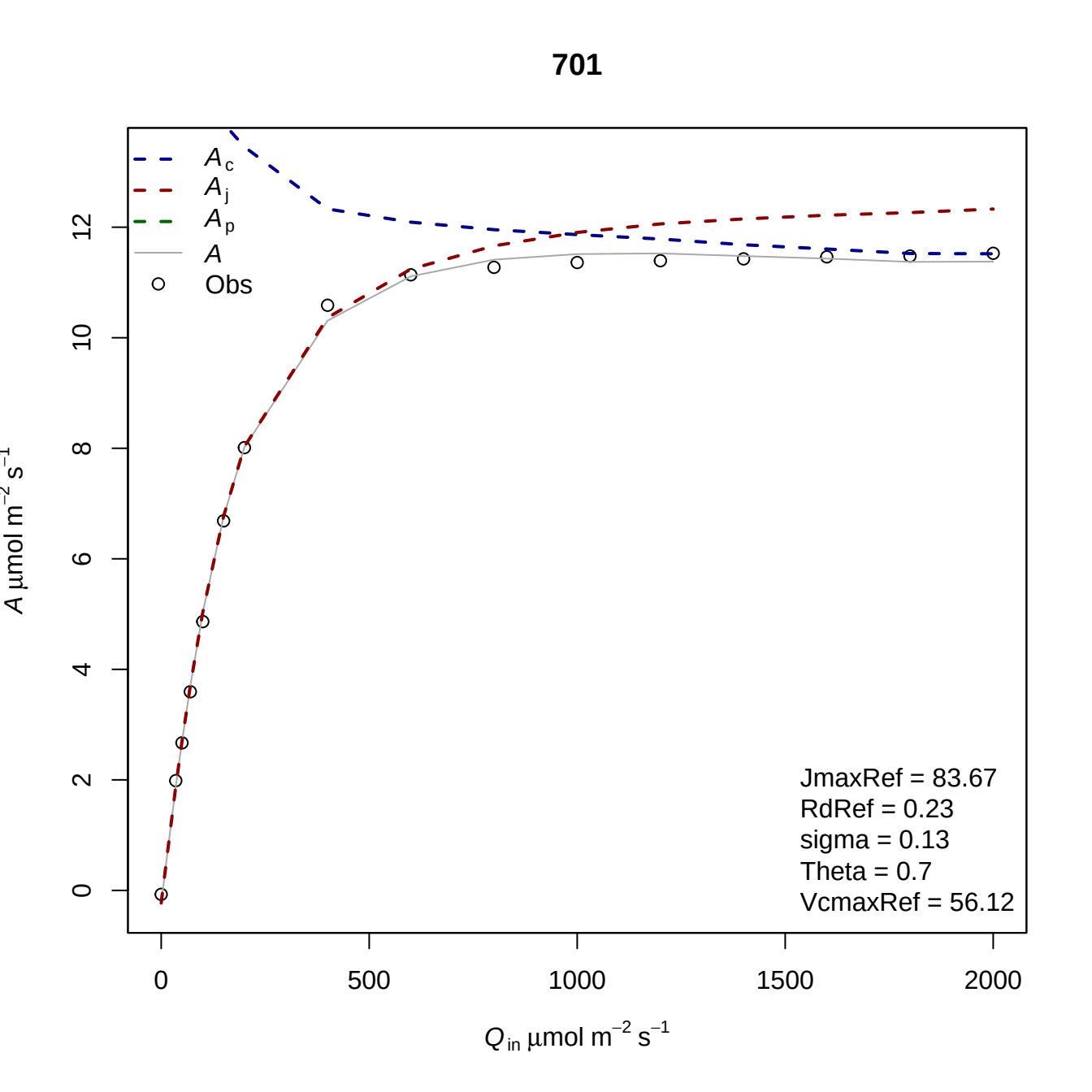

696
A_c
A_j
A_p
A
Obs
JmaxRef = 62.87
RdRef = 0.79
sigma = 0.29
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

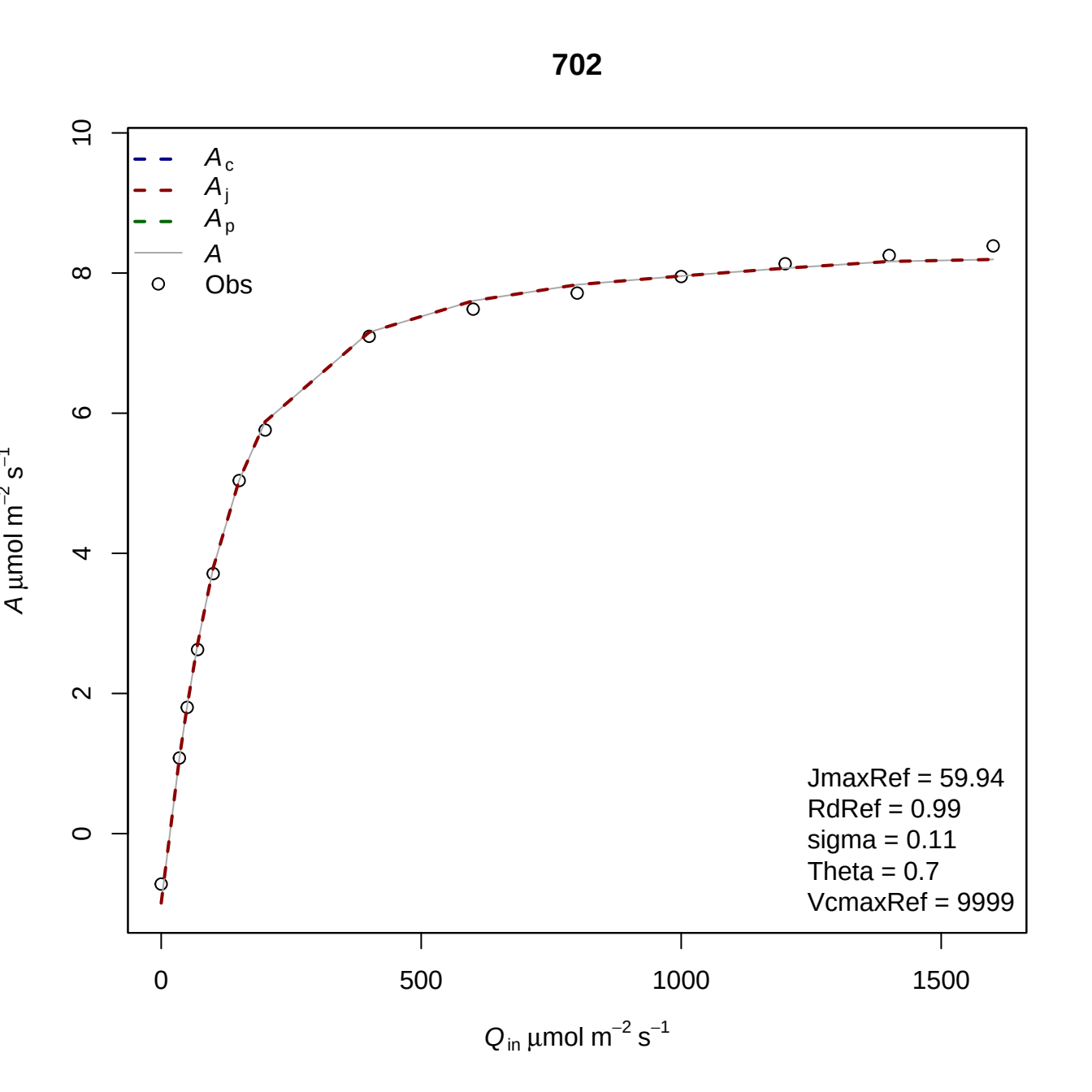

697
A_c
A_j
A_p
A
Obs
JmaxRef = 85.72
RdRef = −0.86
sigma = 0.19
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

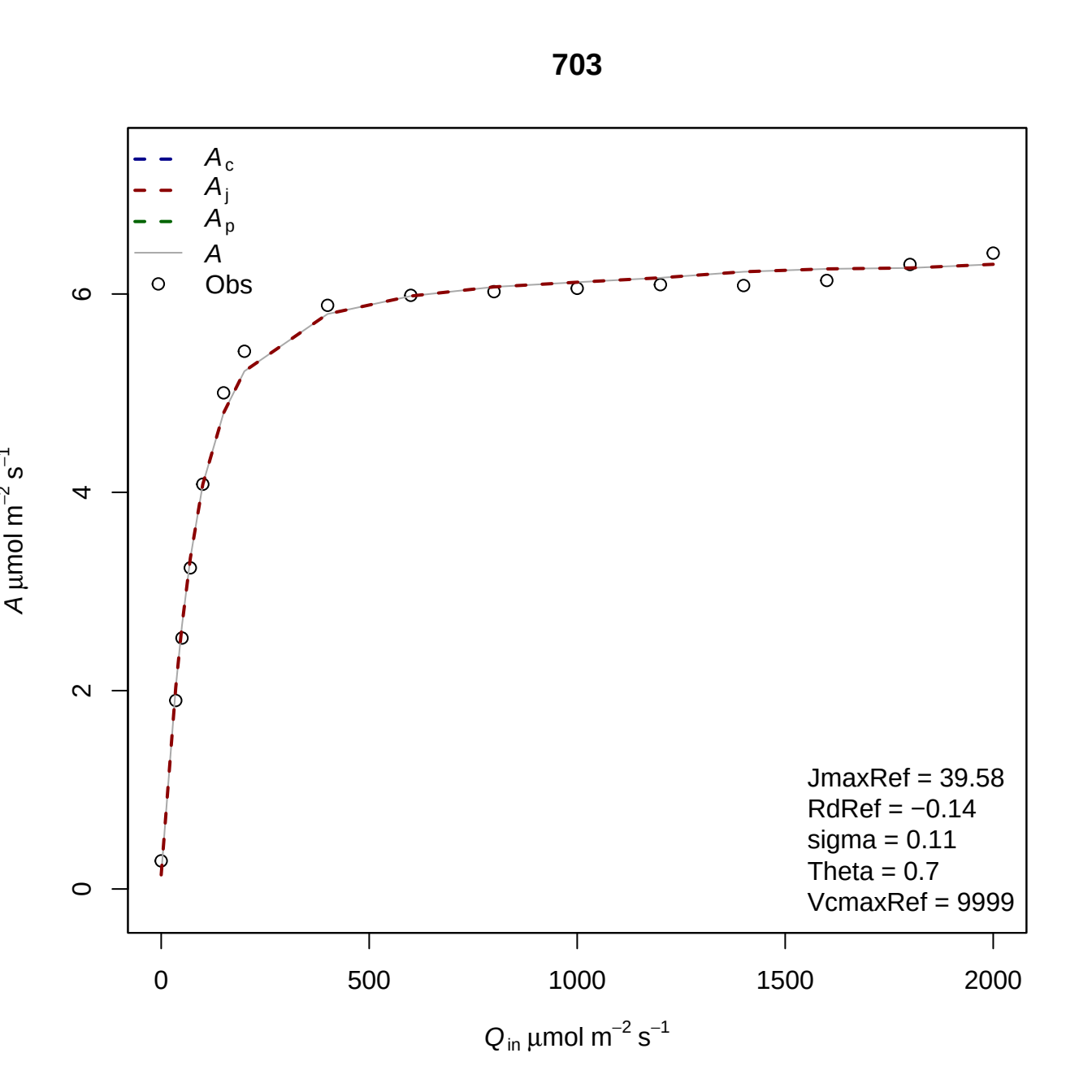

698
A_c
A_j
A_p
A
Obs
JmaxRef = 35.92
RdRef = 0.77
sigma = 0.19
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

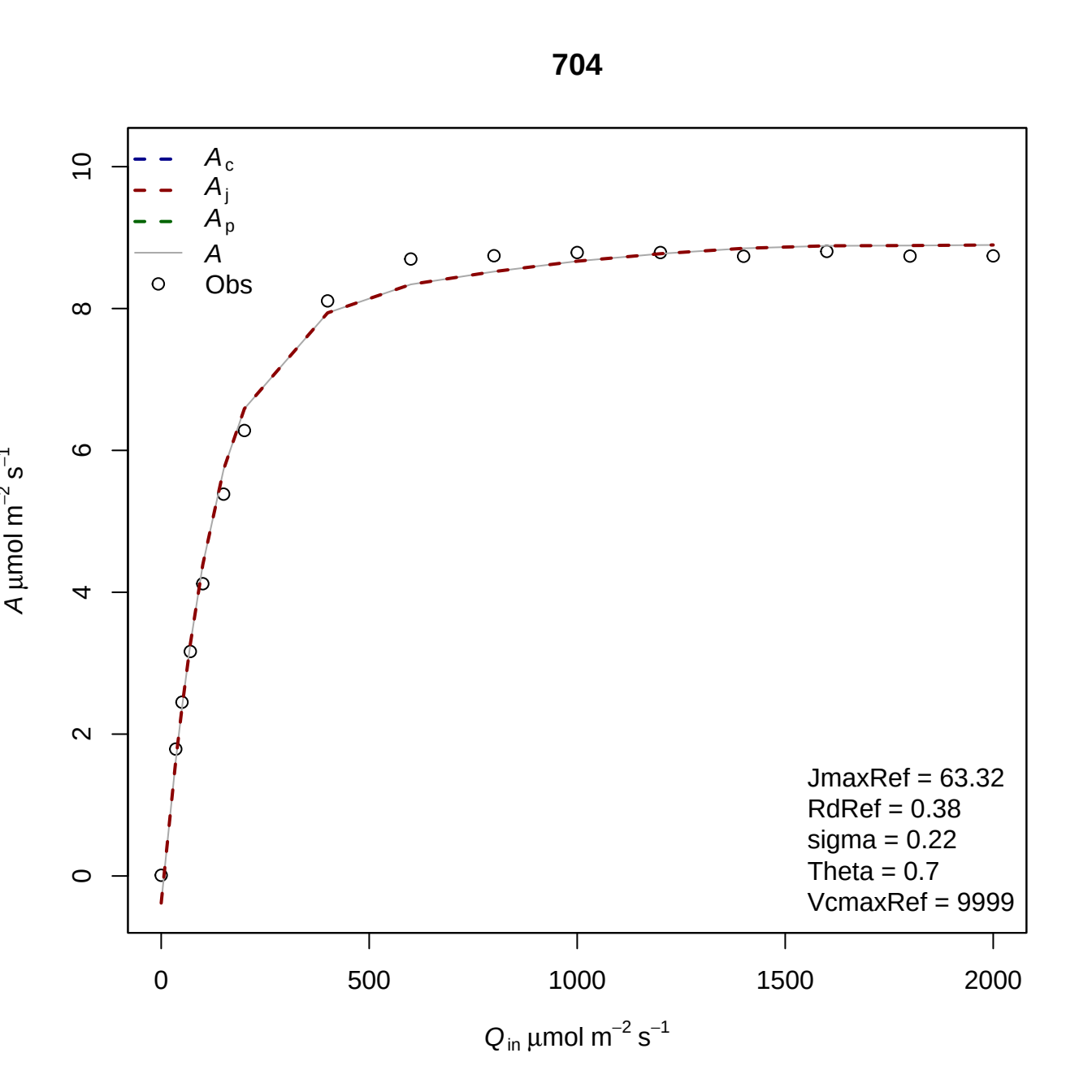

699
A_c
A_j
A_p
A
Obs
JmaxRef = 43.35
RdRef = 0.76
sigma = 0.3
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

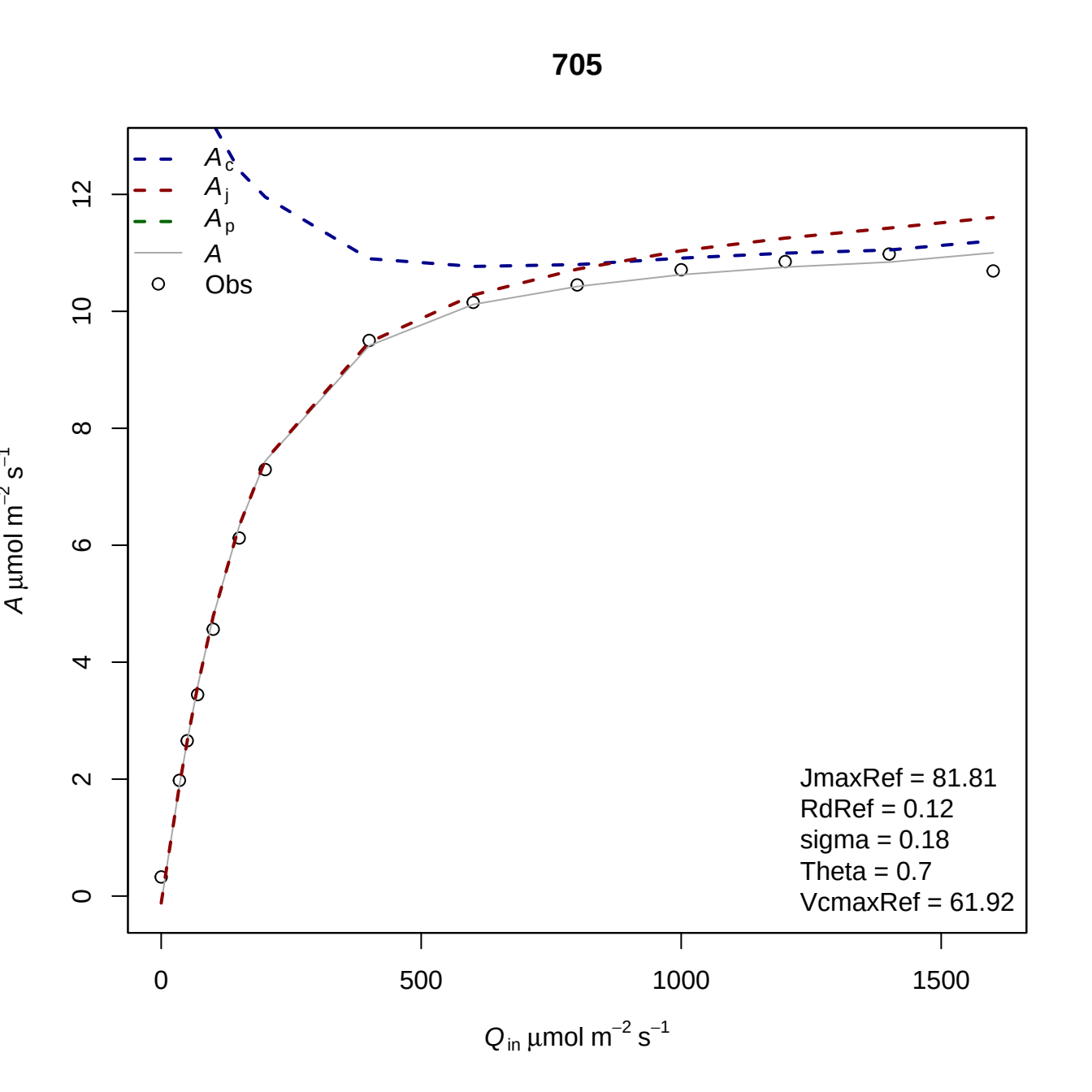

700
A_c
A_j
A_p
A
Obs
JmaxRef = 90.53
RdRef = 1.22
sigma = 0.36
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

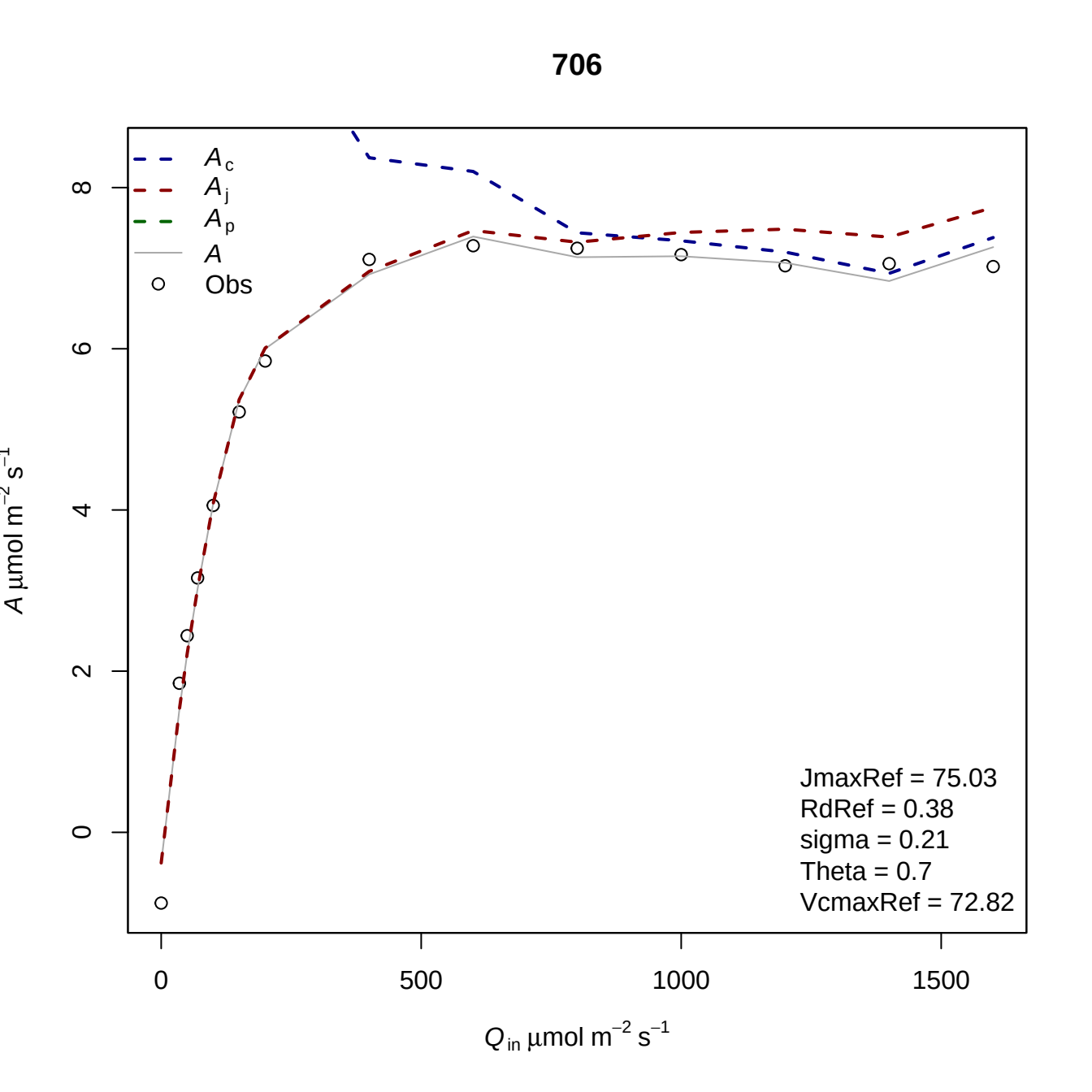

701
A_c
A_j
A_p
A
Obs
JmaxRef = 83.67
RdRef = 0.23
sigma = 0.13
Theta = 0.7
VcmaxRef = 56.12
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

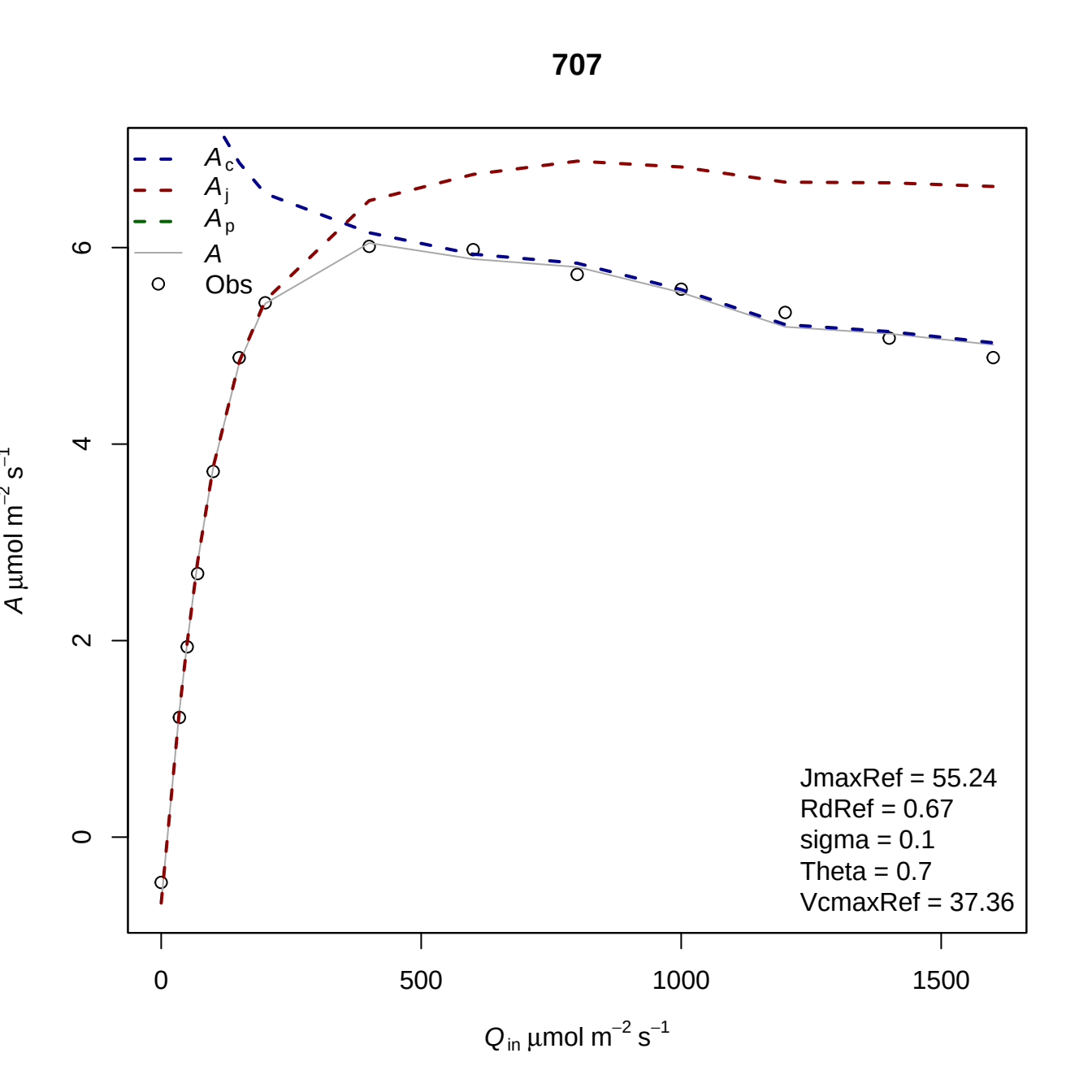

702
A_c
A_j
A_p
A
Obs
JmaxRef = 59.94
RdRef = 0.99
sigma = 0.11
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

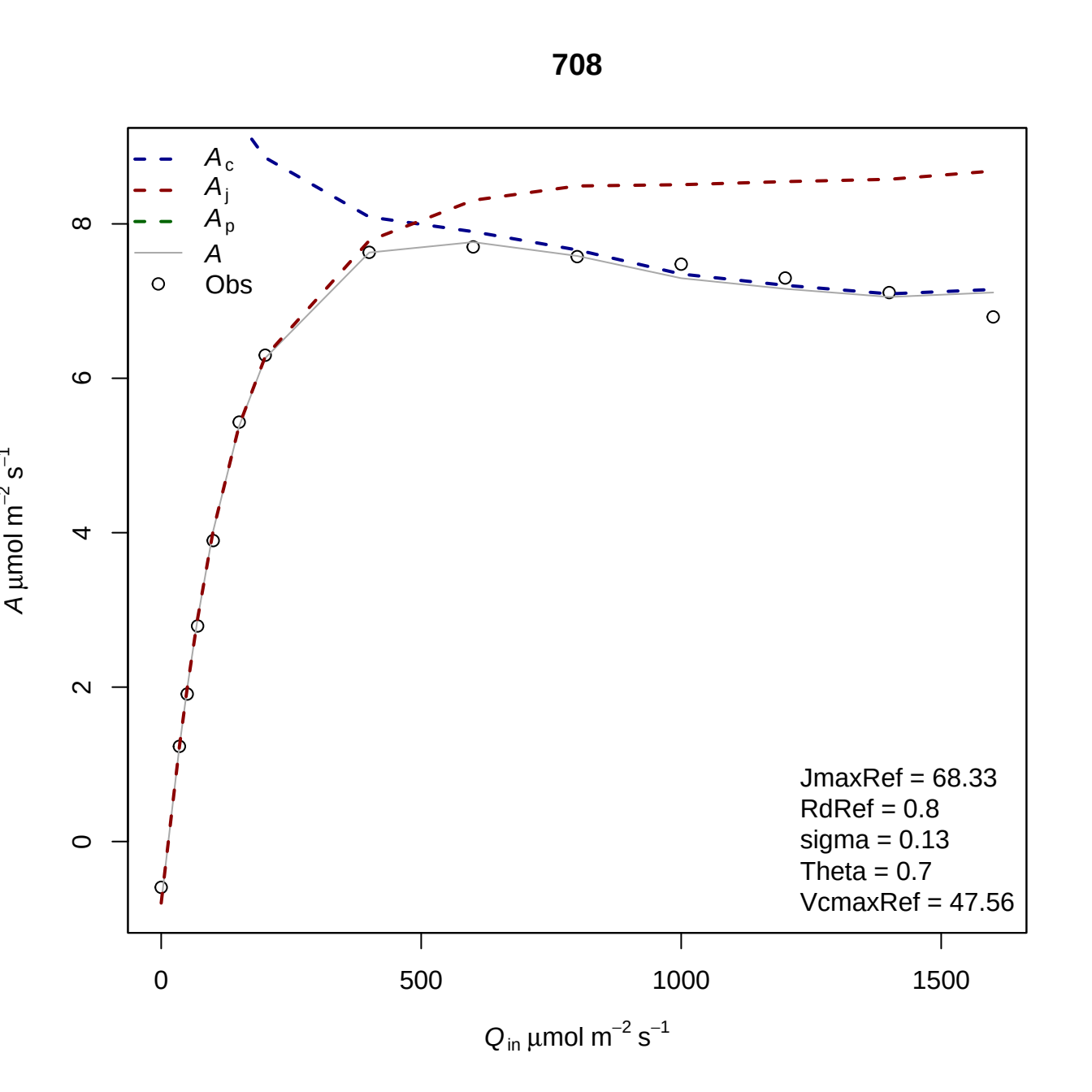

703
A_c
A_j
A_p
A
Obs
JmaxRef = 39.58
RdRef = −0.14
sigma = 0.11
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

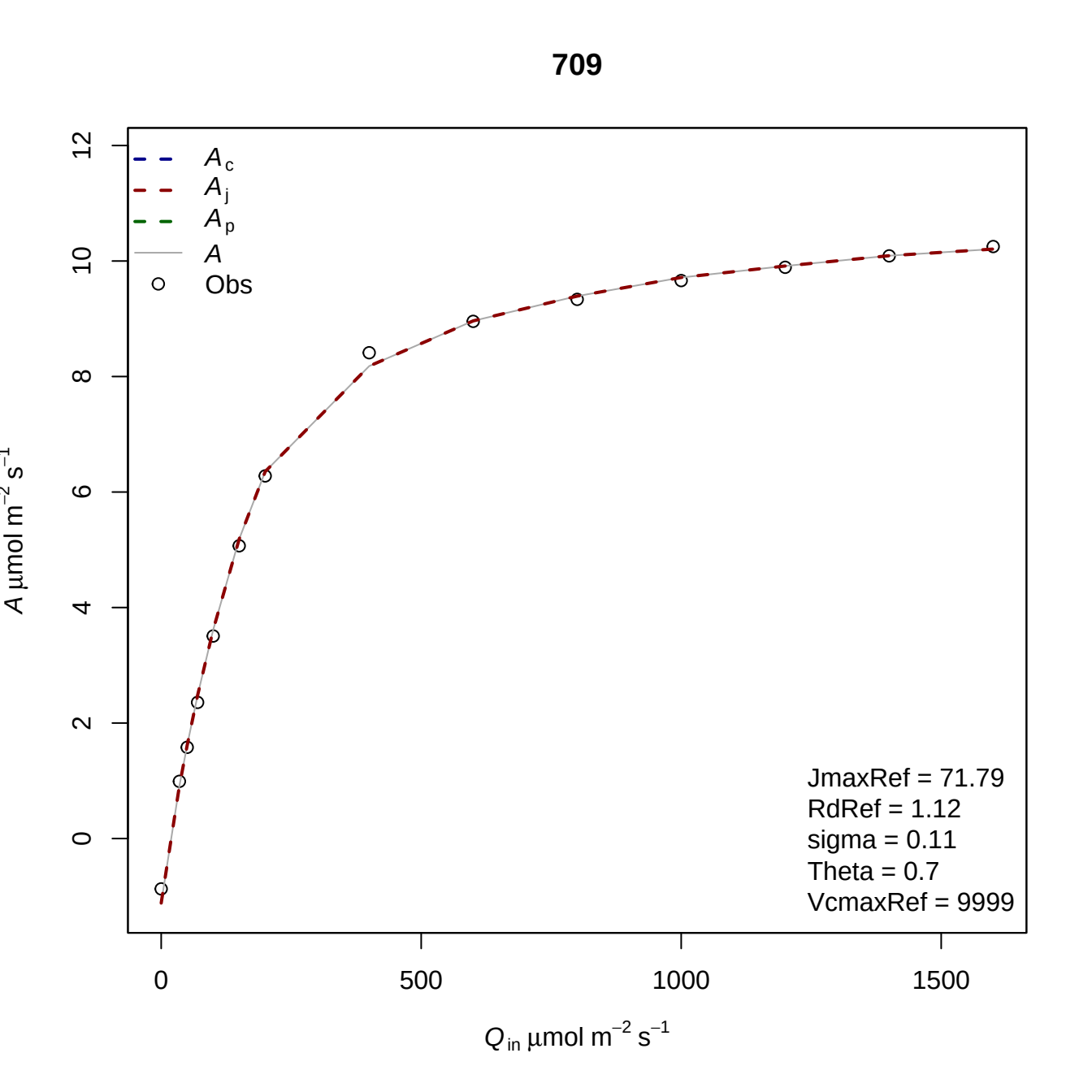

704
A_c
A_j
A_p
A
Obs
JmaxRef = 63.32
RdRef = 0.38
sigma = 0.22
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

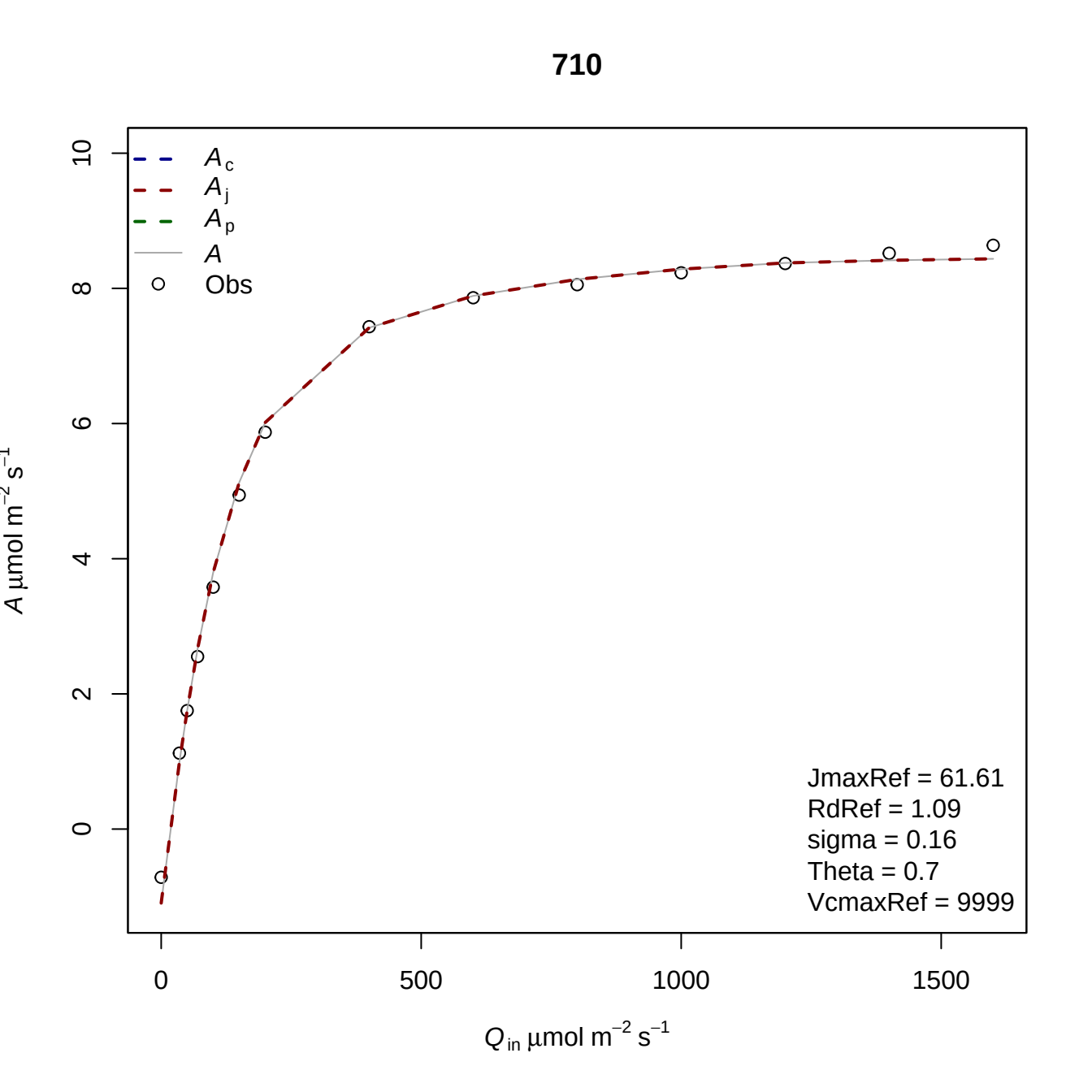

705
A_c
A_j
A_p
A
Obs
JmaxRef = 81.81
RdRef = 0.12
sigma = 0.18
Theta = 0.7
VcmaxRef = 61.92
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

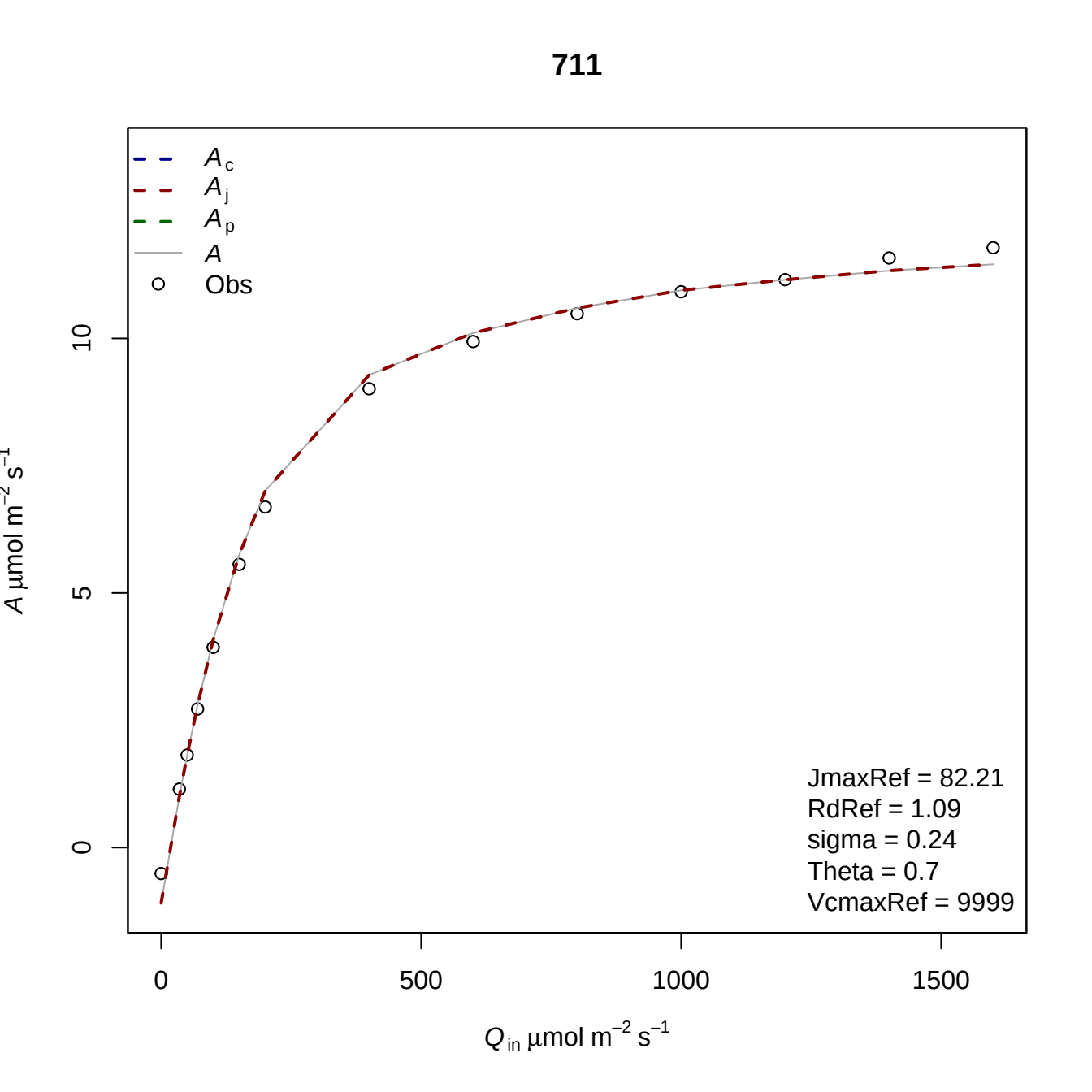

706
A_c
A_j
A_p
A
Obs
JmaxRef = 75.03
RdRef = 0.38
sigma = 0.21
Theta = 0.7
VcmaxRef = 72.82
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

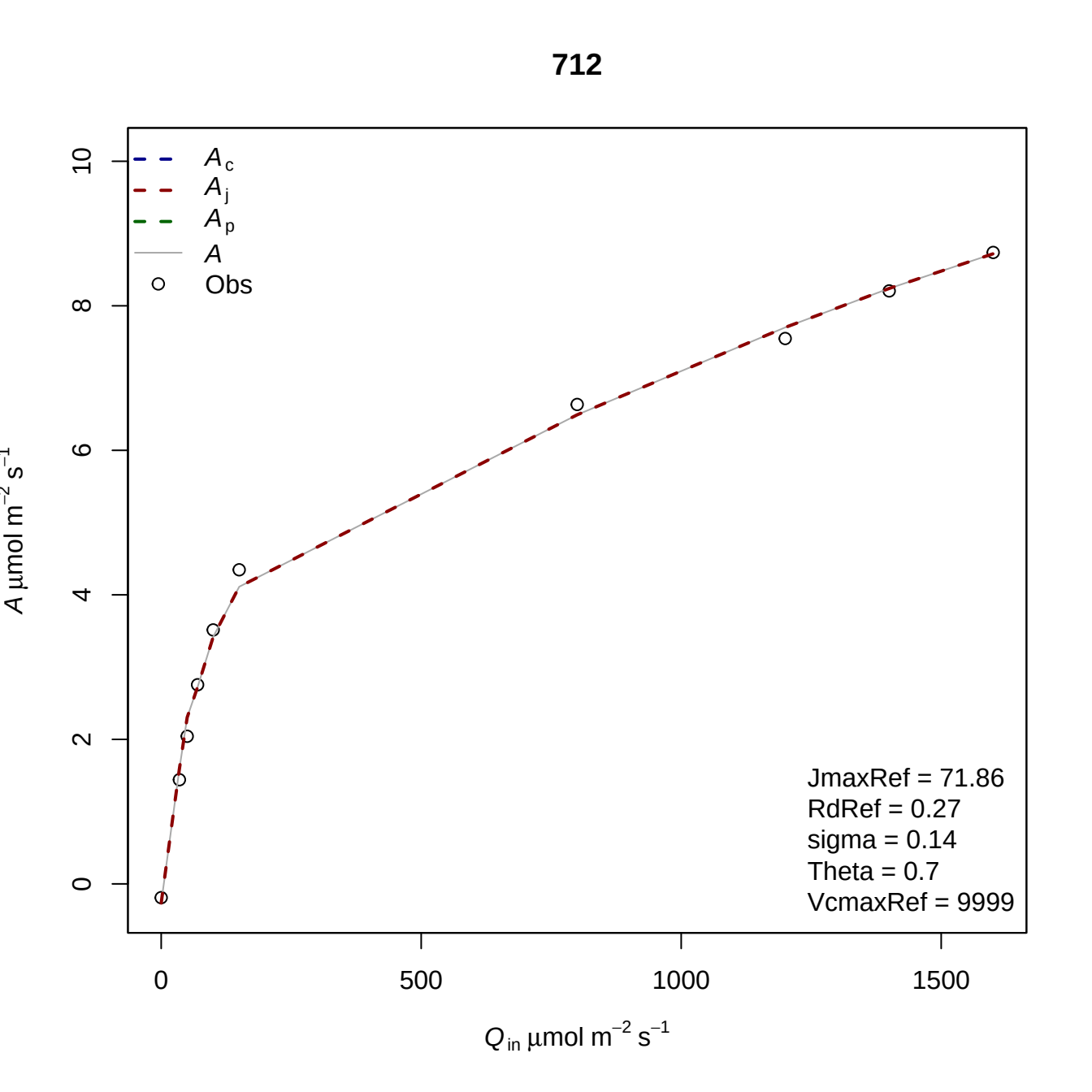

707
A_c
A_j
A_p
A
Obs
JmaxRef = 55.24
RdRef = 0.67
sigma = 0.1
Theta = 0.7
VcmaxRef = 37.36
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

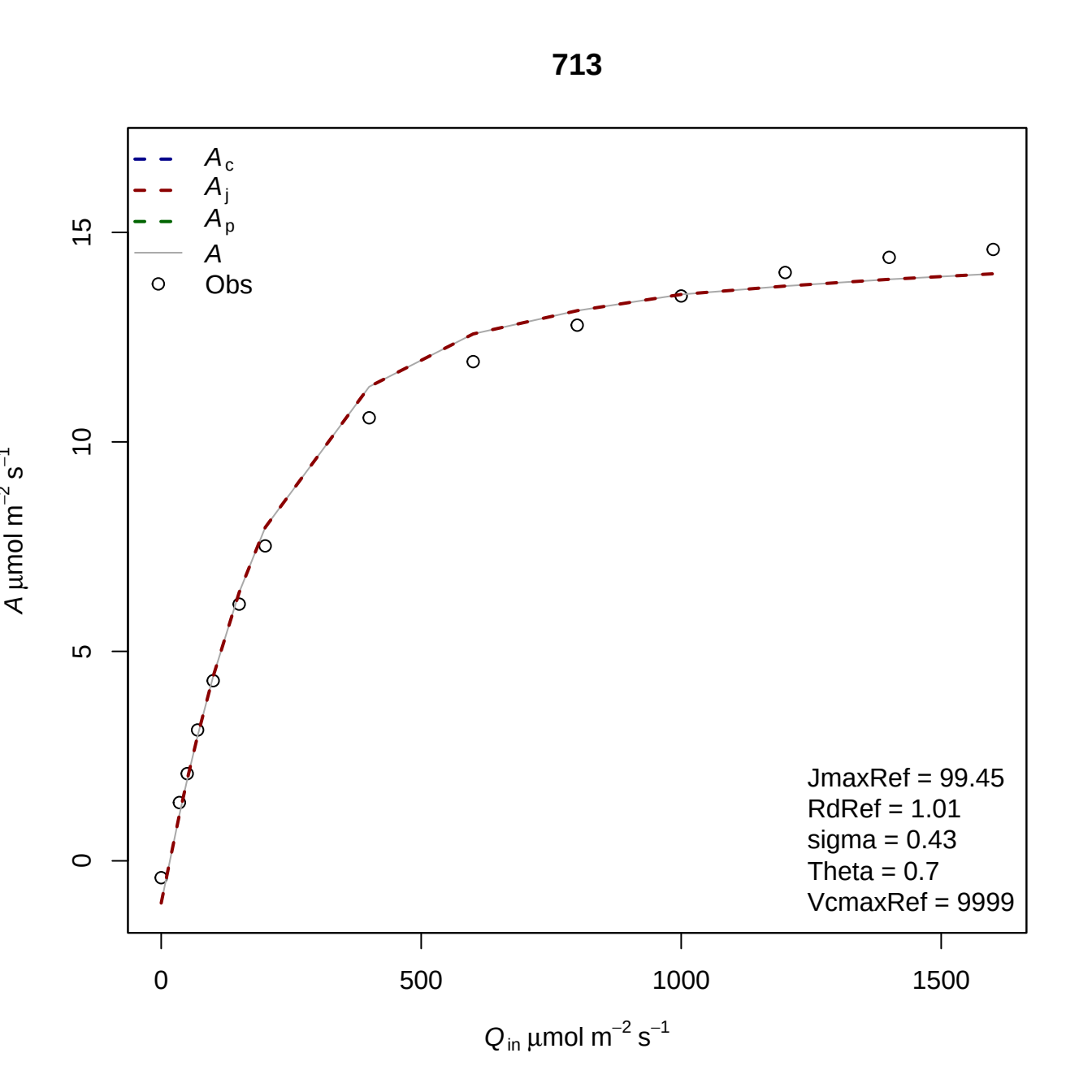

708
A_c
A_j
A_p
A
Obs
JmaxRef = 68.33
RdRef = 0.8
sigma = 0.13
Theta = 0.7
VcmaxRef = 47.56
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

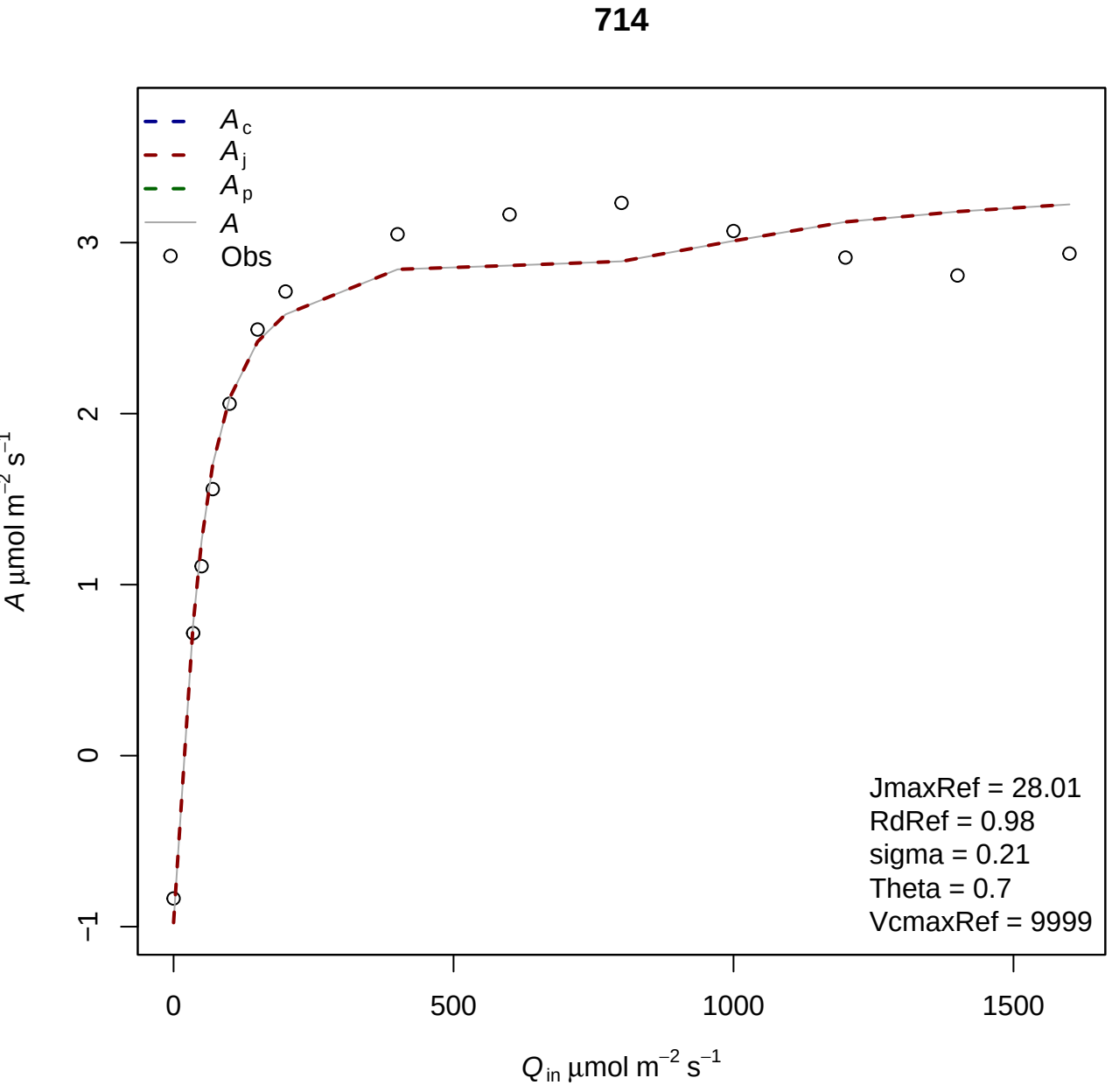

709
A_c
A_j
A_p
A
Obs
JmaxRef = 71.79
RdRef = 1.12
sigma = 0.11
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

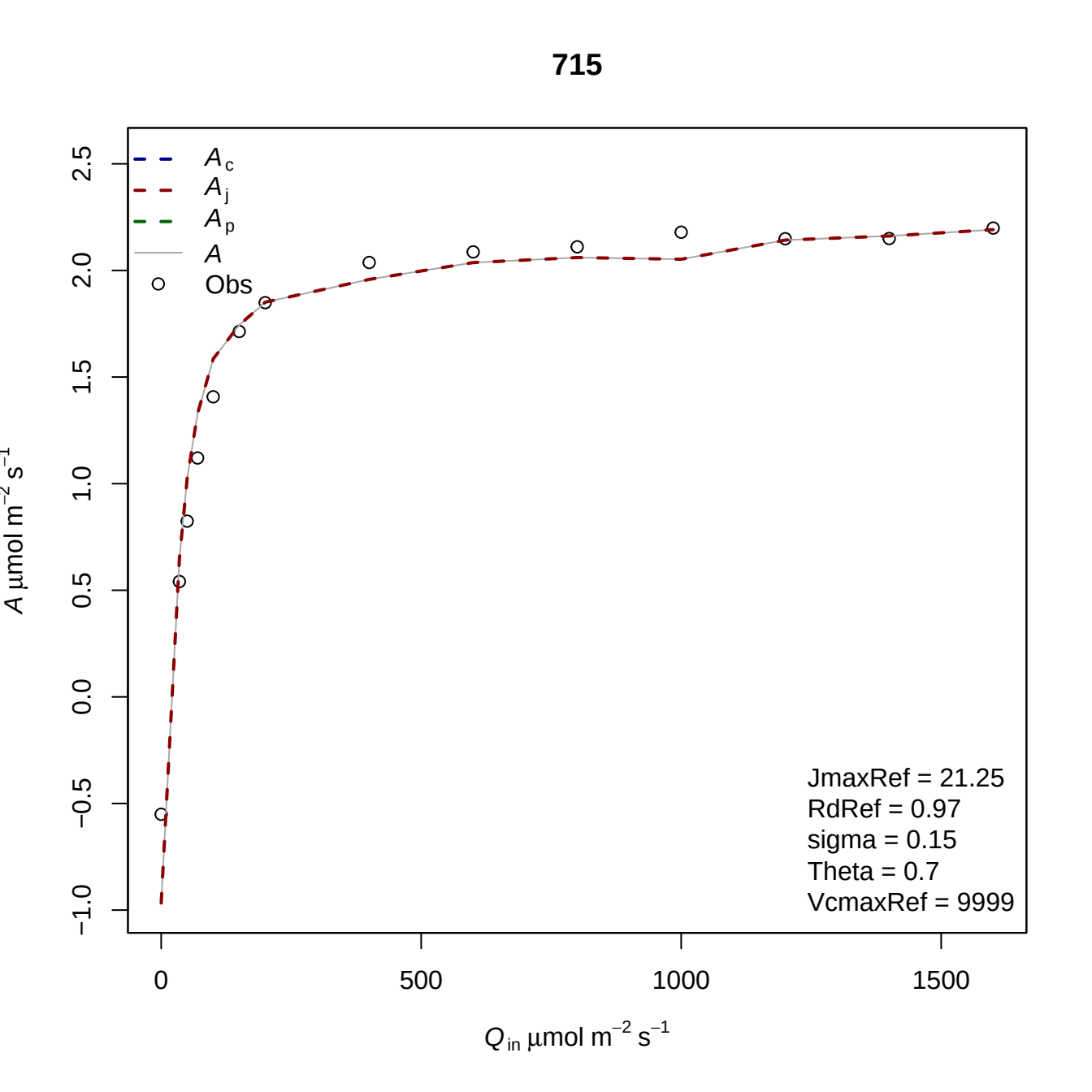

710
A_c
A_j
A_p
A
Obs
JmaxRef = 61.61
RdRef = 1.09
sigma = 0.16
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

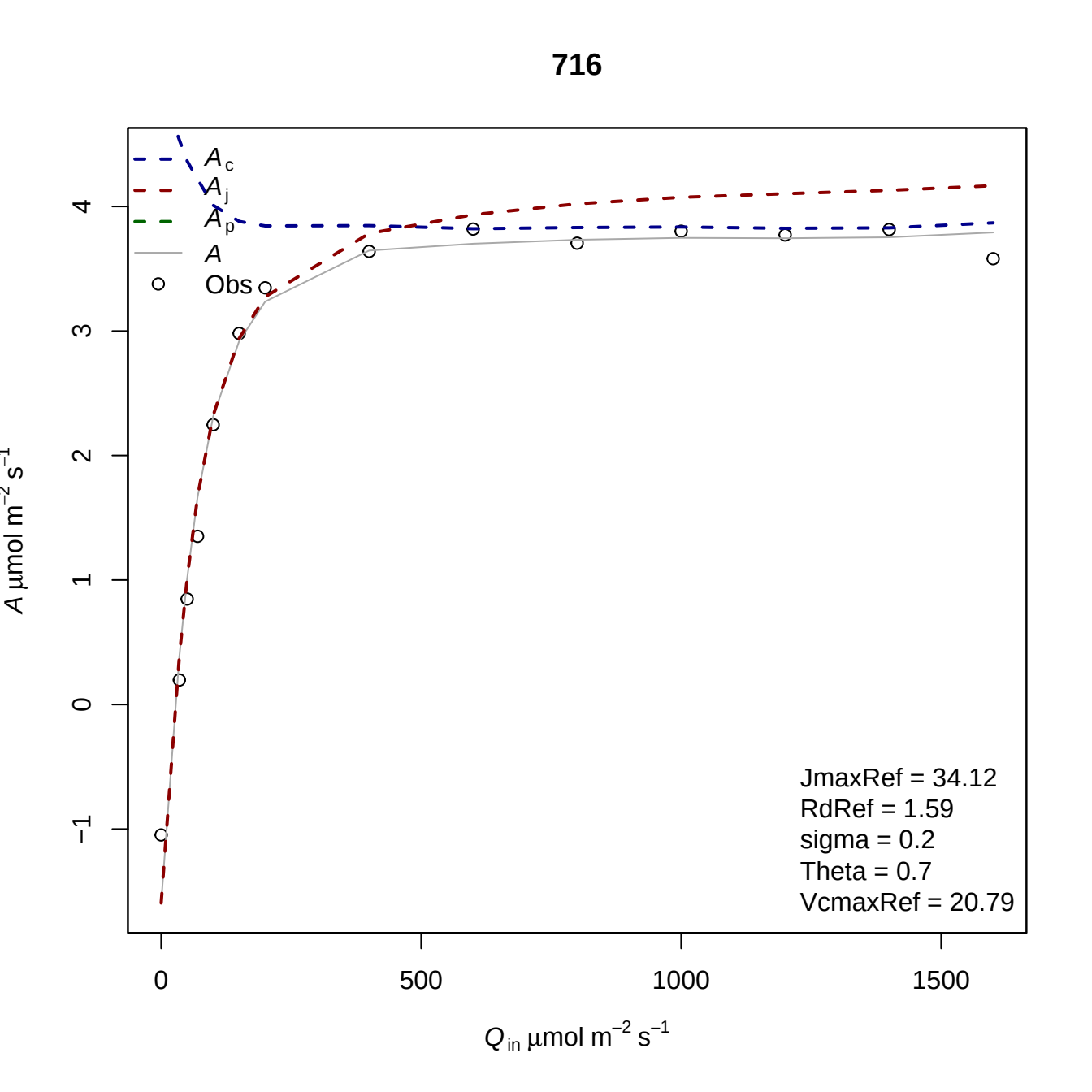

711
A_c
A_j
A_p
A
Obs
JmaxRef = 82.21
RdRef = 1.09
sigma = 0.24
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

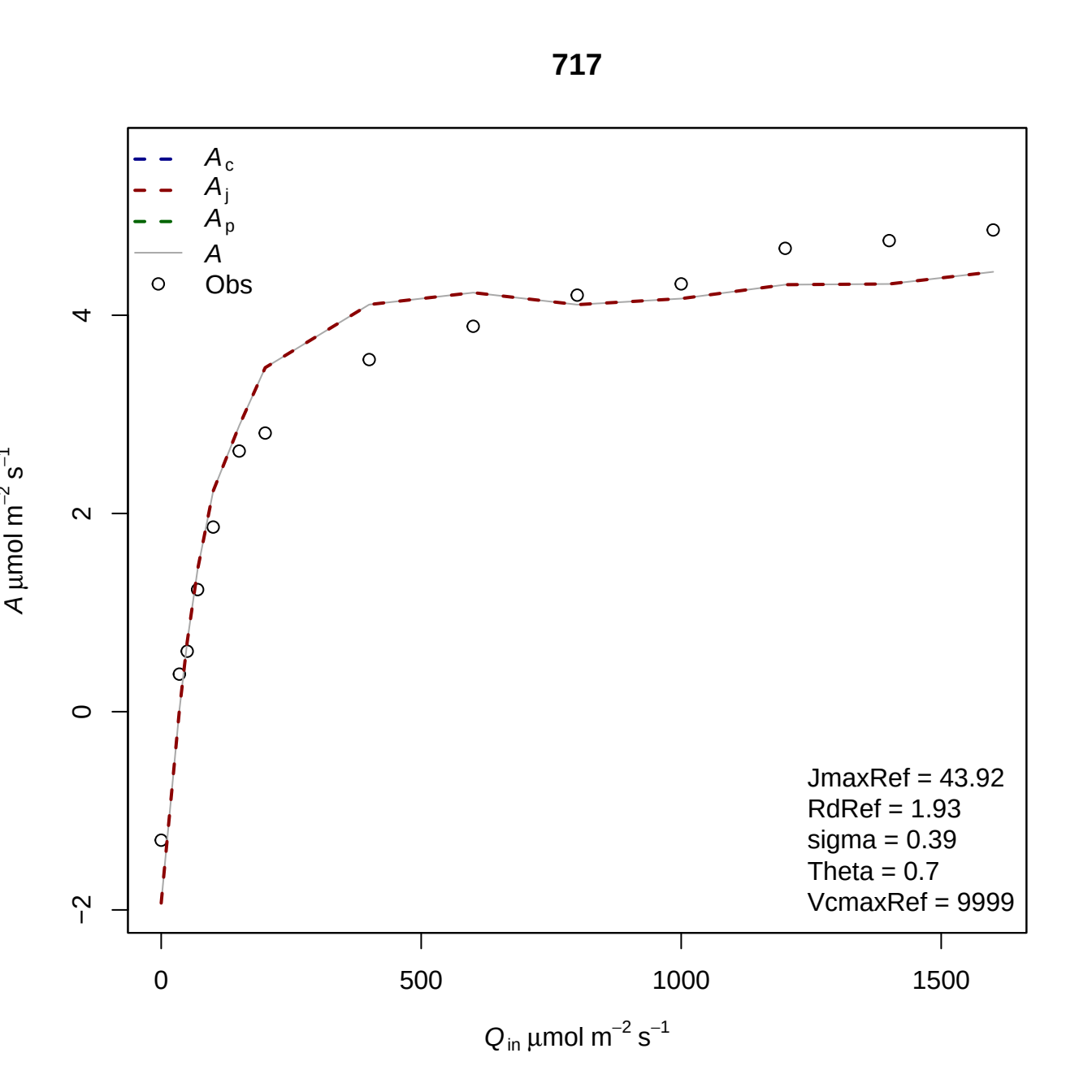

712
A_c
A_j
A_p
A
Obs
JmaxRef = 71.86
RdRef = 0.27
sigma = 0.14
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

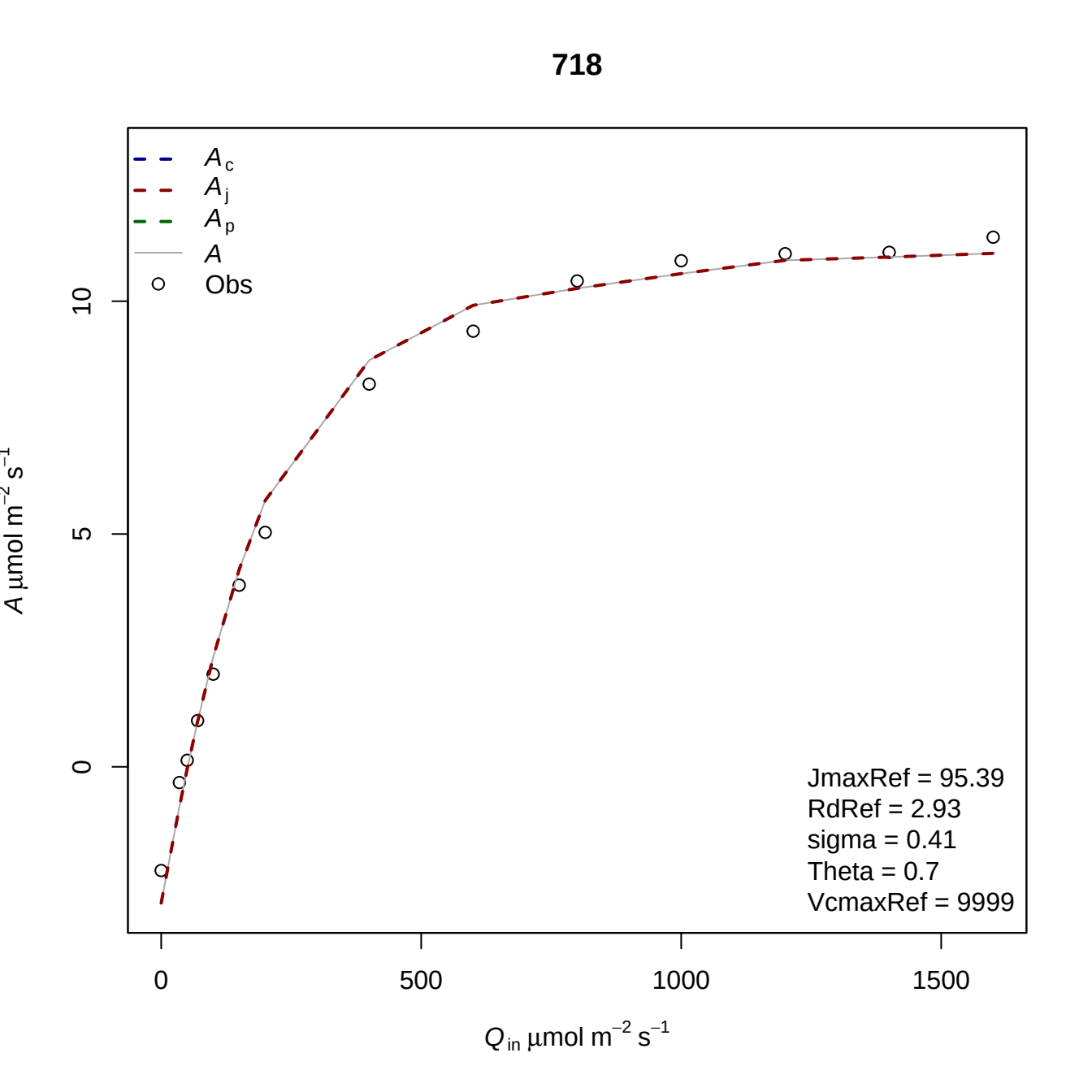

713
A_c
A_j
A_p
A
Obs
JmaxRef = 99.45
RdRef = 1.01
sigma = 0.43
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

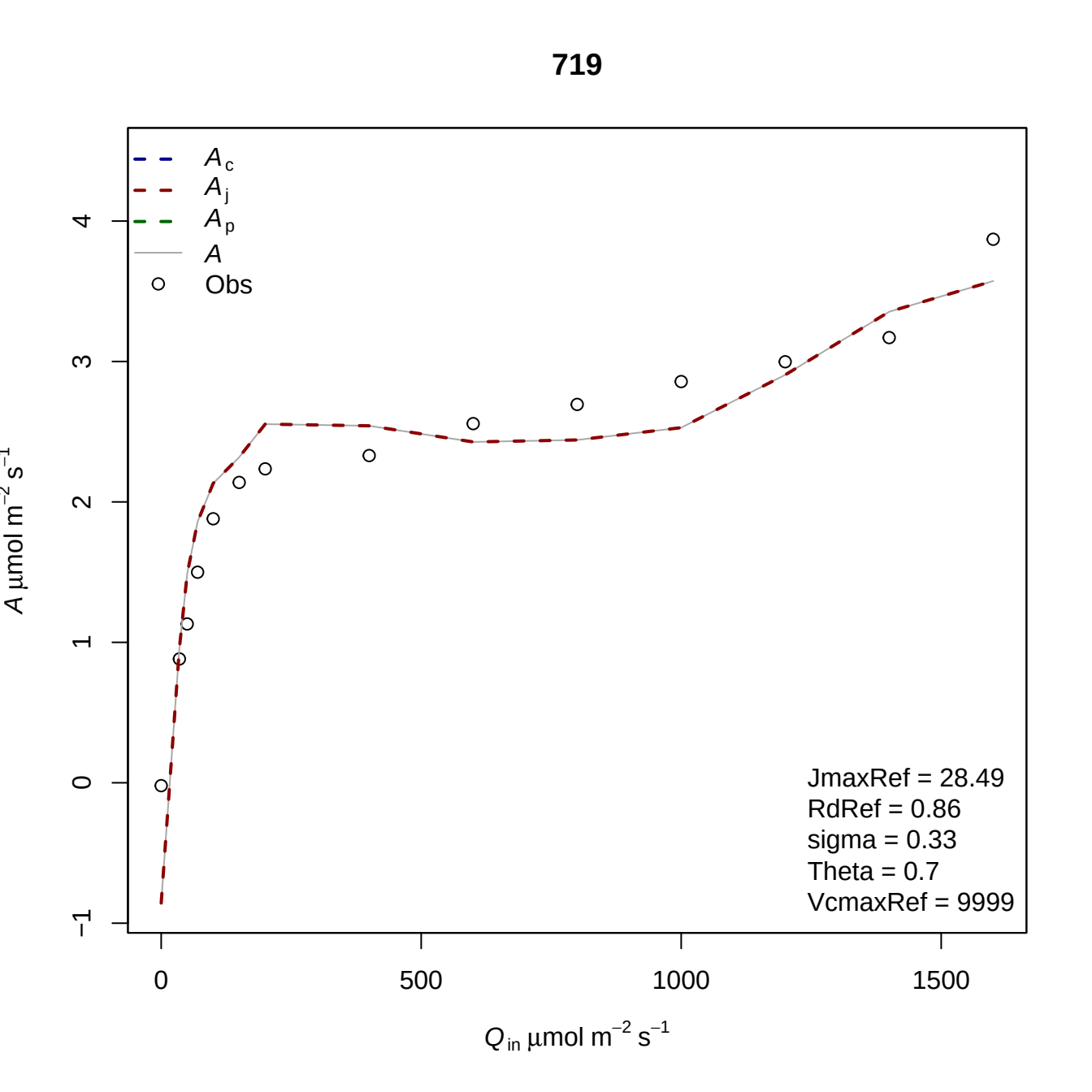

714
A_c
A_j
A_p
A
Obs
JmaxRef = 28.01
RdRef = 0.98
sigma = 0.21
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

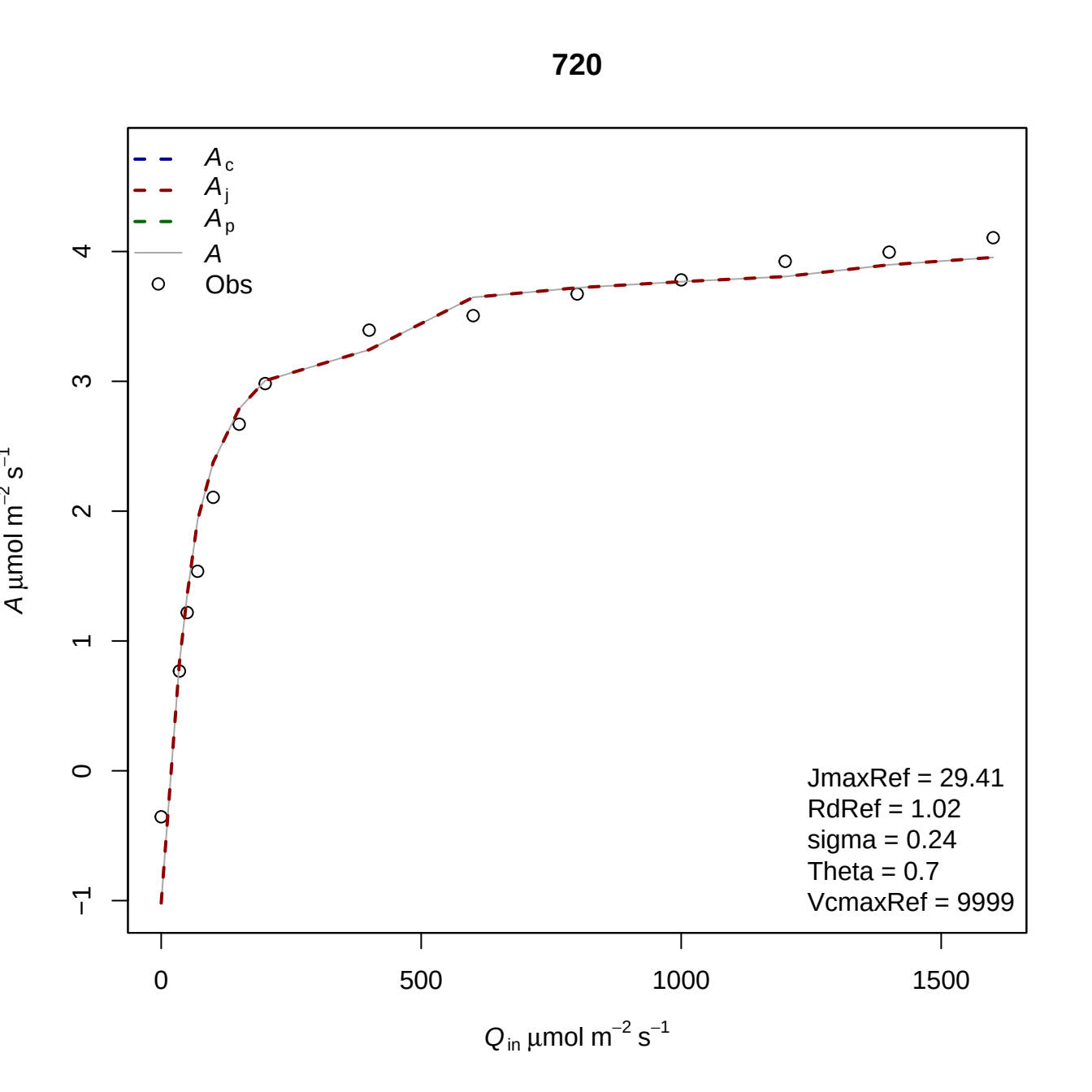

715
A_c
A_j
A_p
A
Obs
JmaxRef = 21.25
RdRef = 0.97
sigma = 0.15
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
0
500
1000
1500

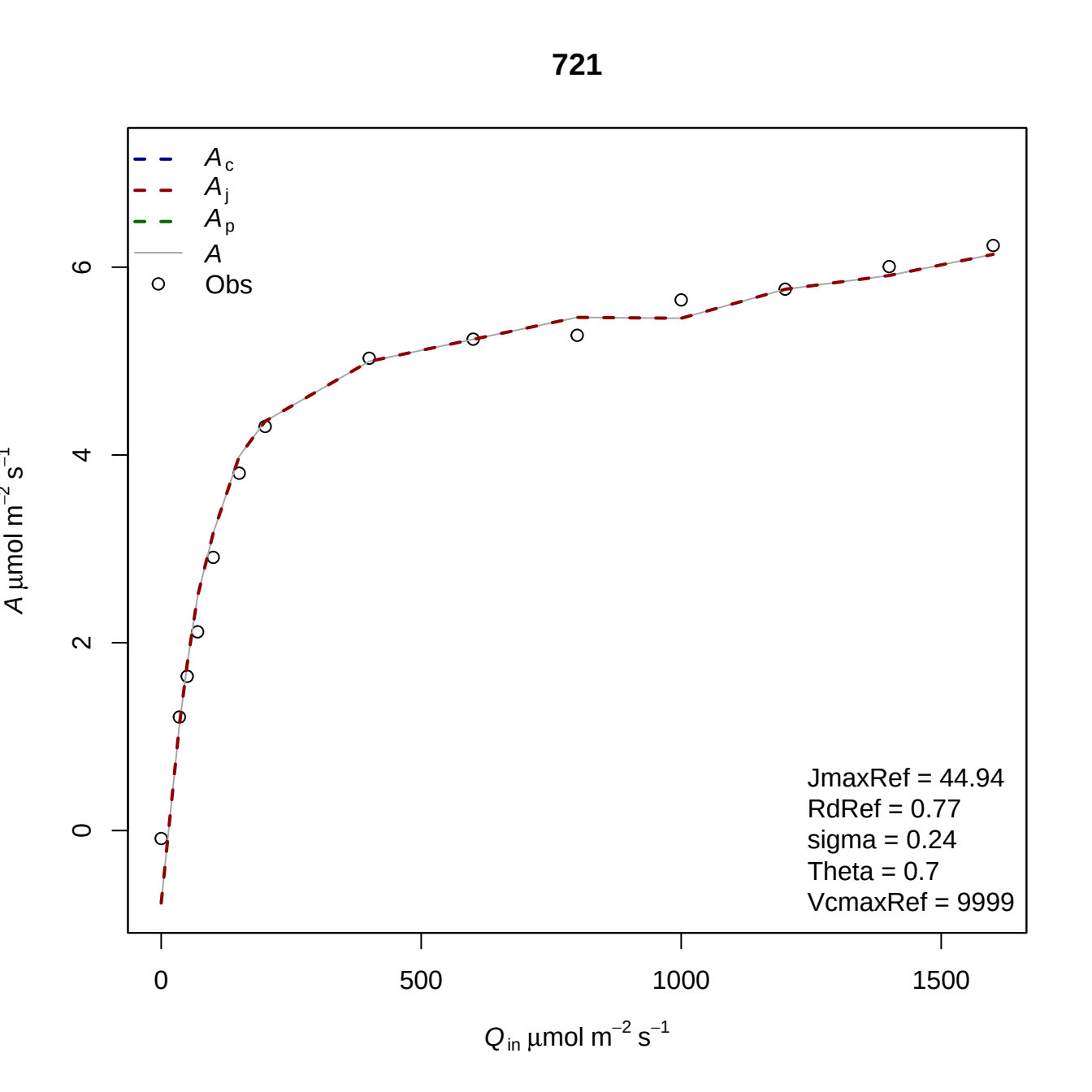

716
A_c
A_j
A_p
A
Obs
JmaxRef = 34.12
RdRef = 1.59
sigma = 0.2
Theta = 0.7
VcmaxRef = 20.79
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

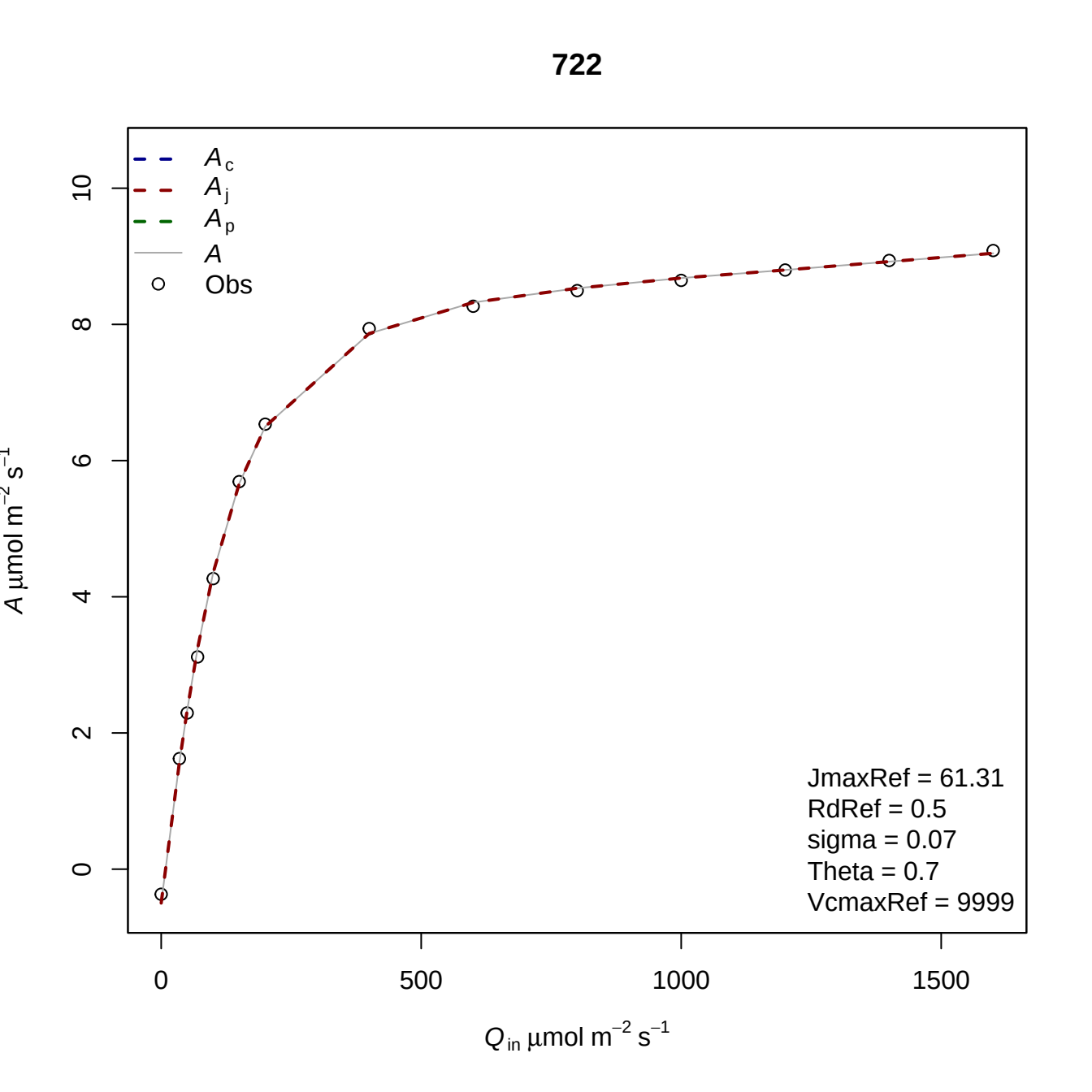

717
A_c
A_j
A_p
A
Obs
JmaxRef = 43.92
RdRef = 1.93
sigma = 0.39
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

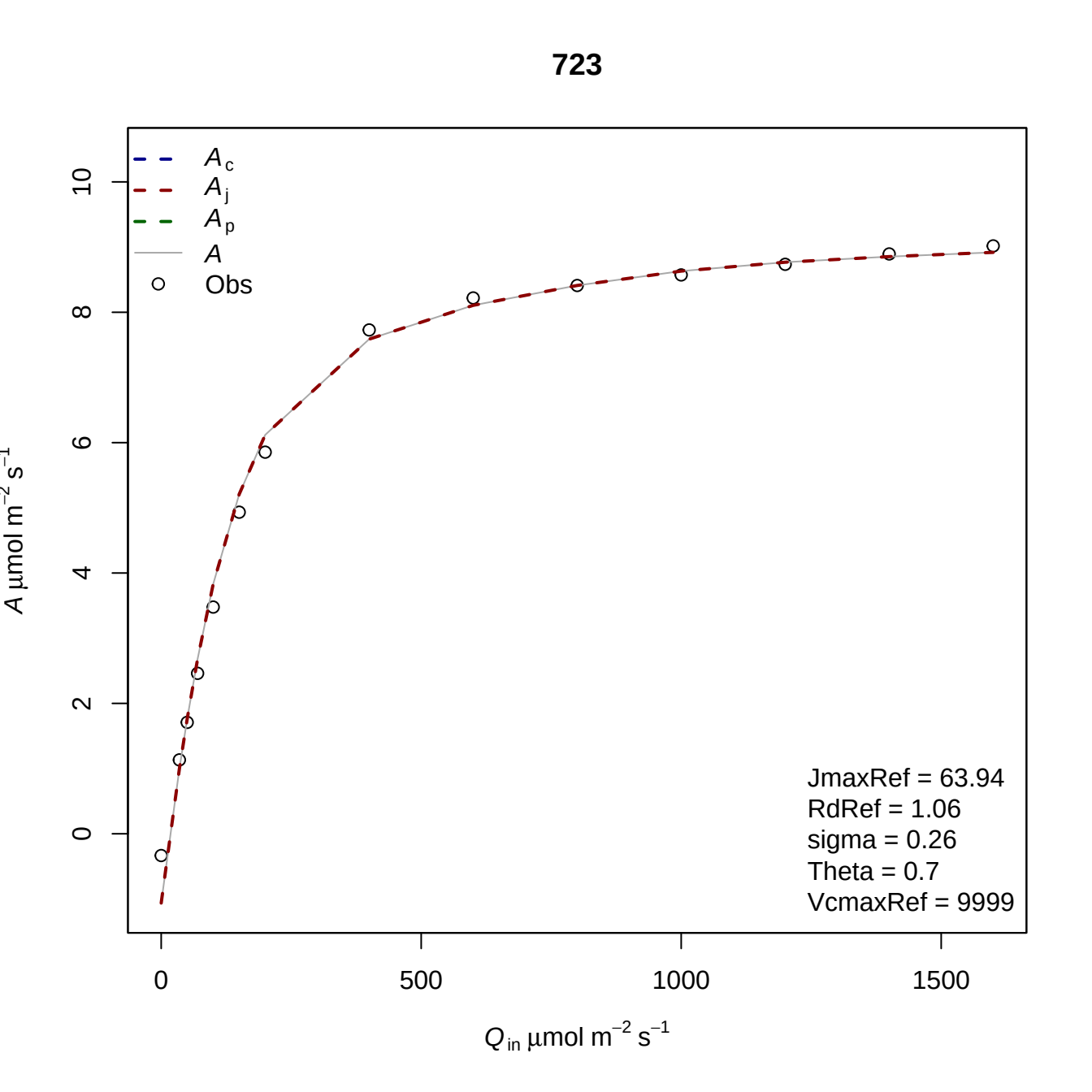

718
A_c
A_j
A_p
A
Obs
JmaxRef = 95.39
RdRef = 2.93
sigma = 0.41
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

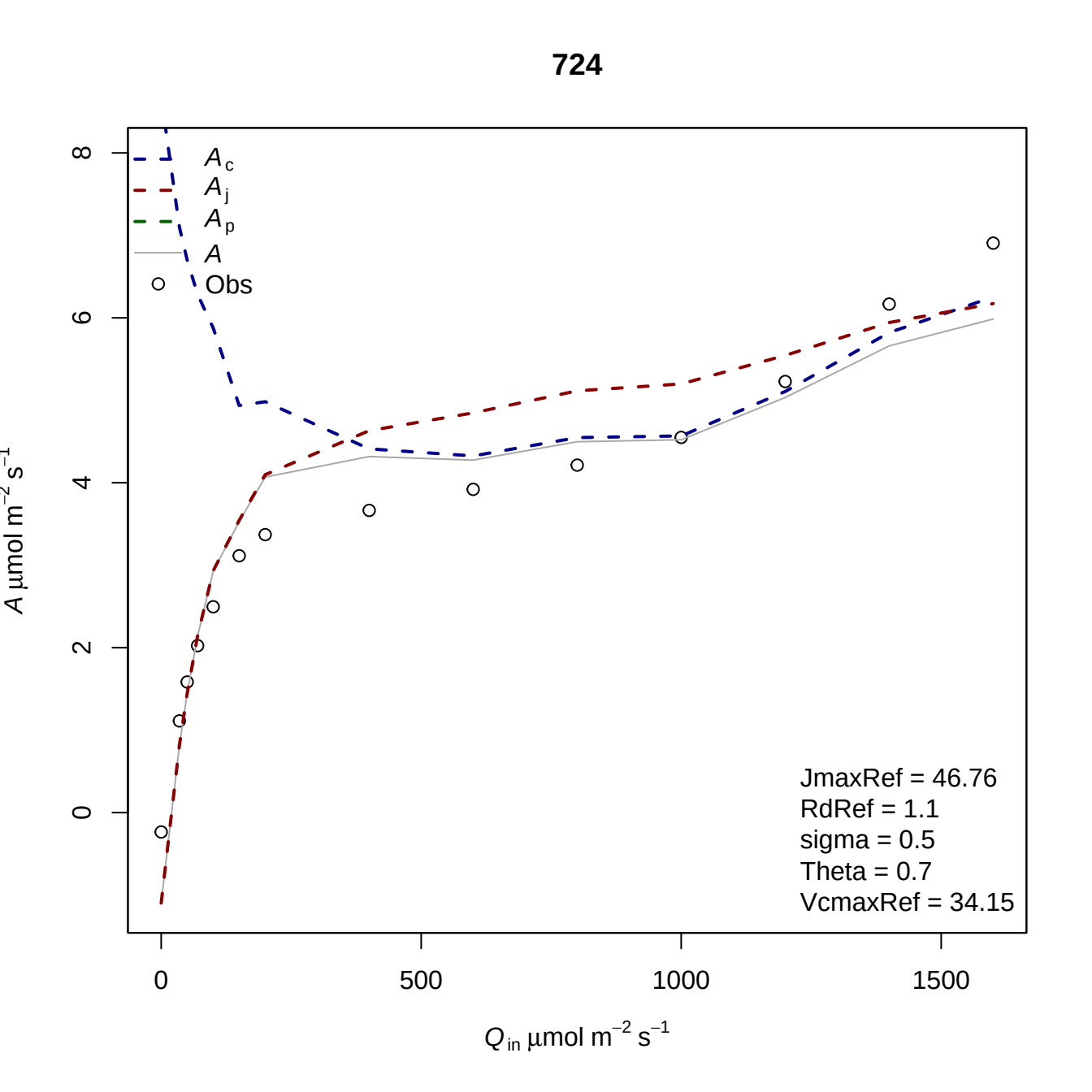

719
A_c
A_j
A_p
A
Obs
JmaxRef = 28.49
RdRef = 0.86
sigma = 0.33
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

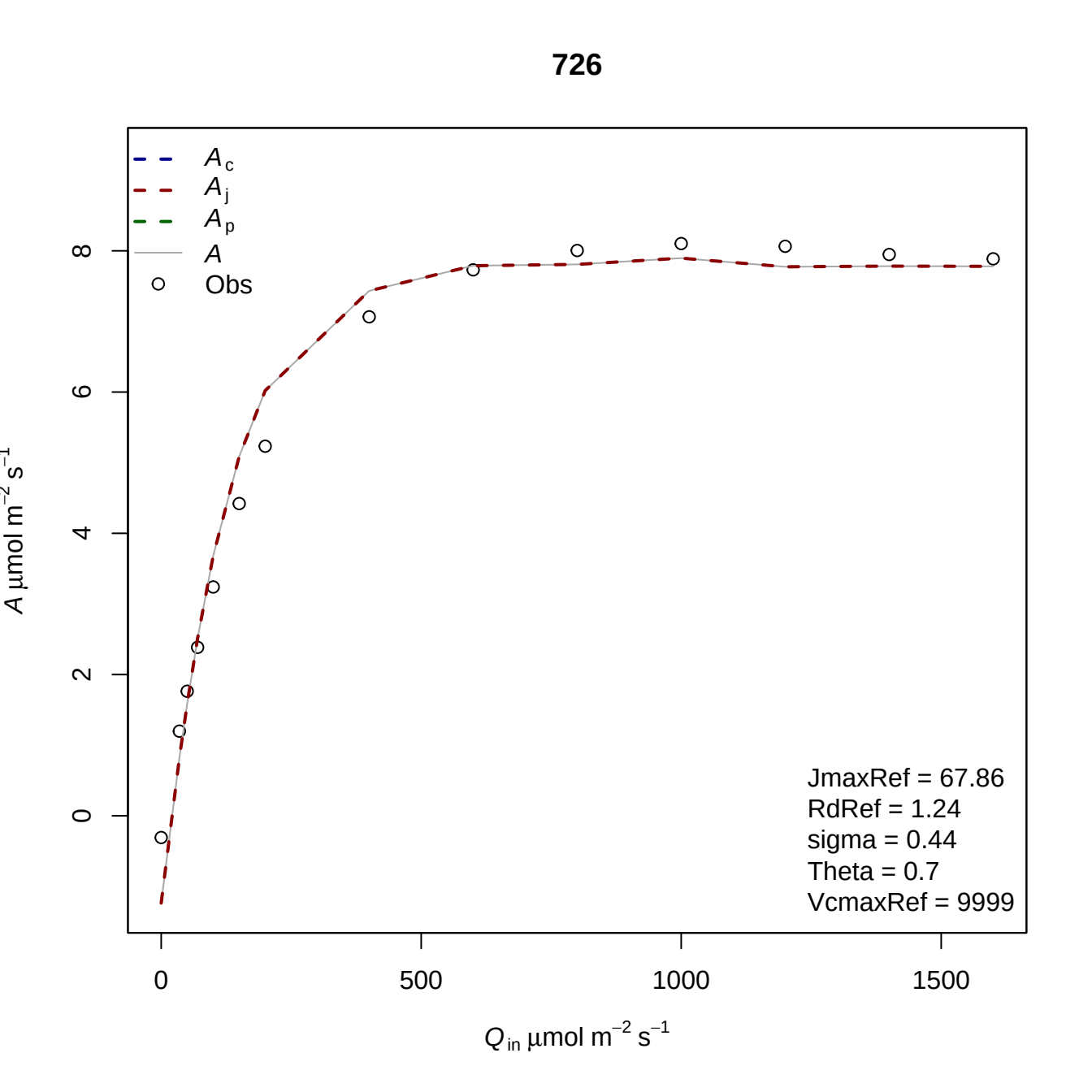

720
A_c
A_j
A_p
A
Obs
JmaxRef = 29.41
RdRef = 1.02
sigma = 0.24
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

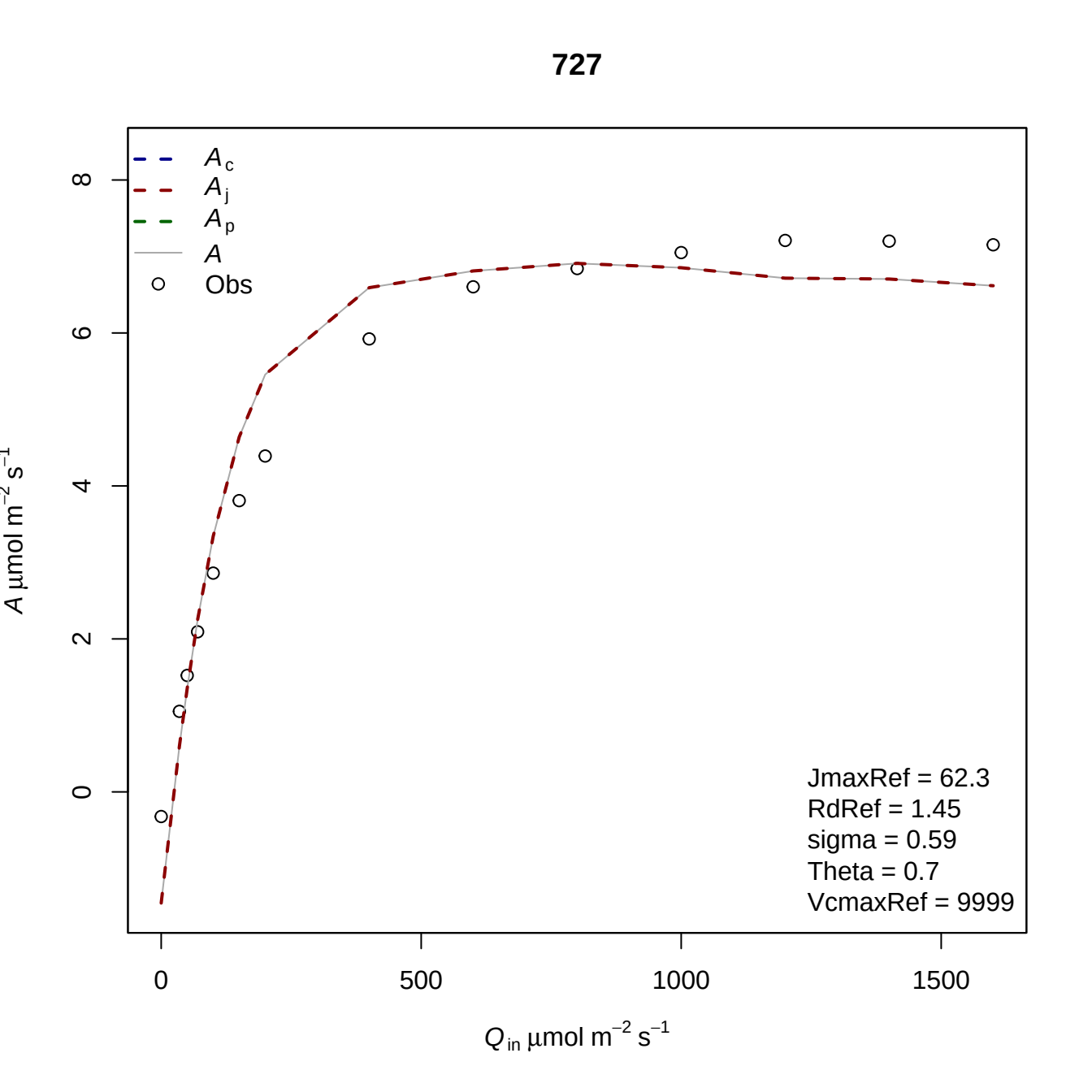

721
A_c
A_j
A_p
A
Obs
JmaxRef = 44.94
RdRef = 0.77
sigma = 0.24
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

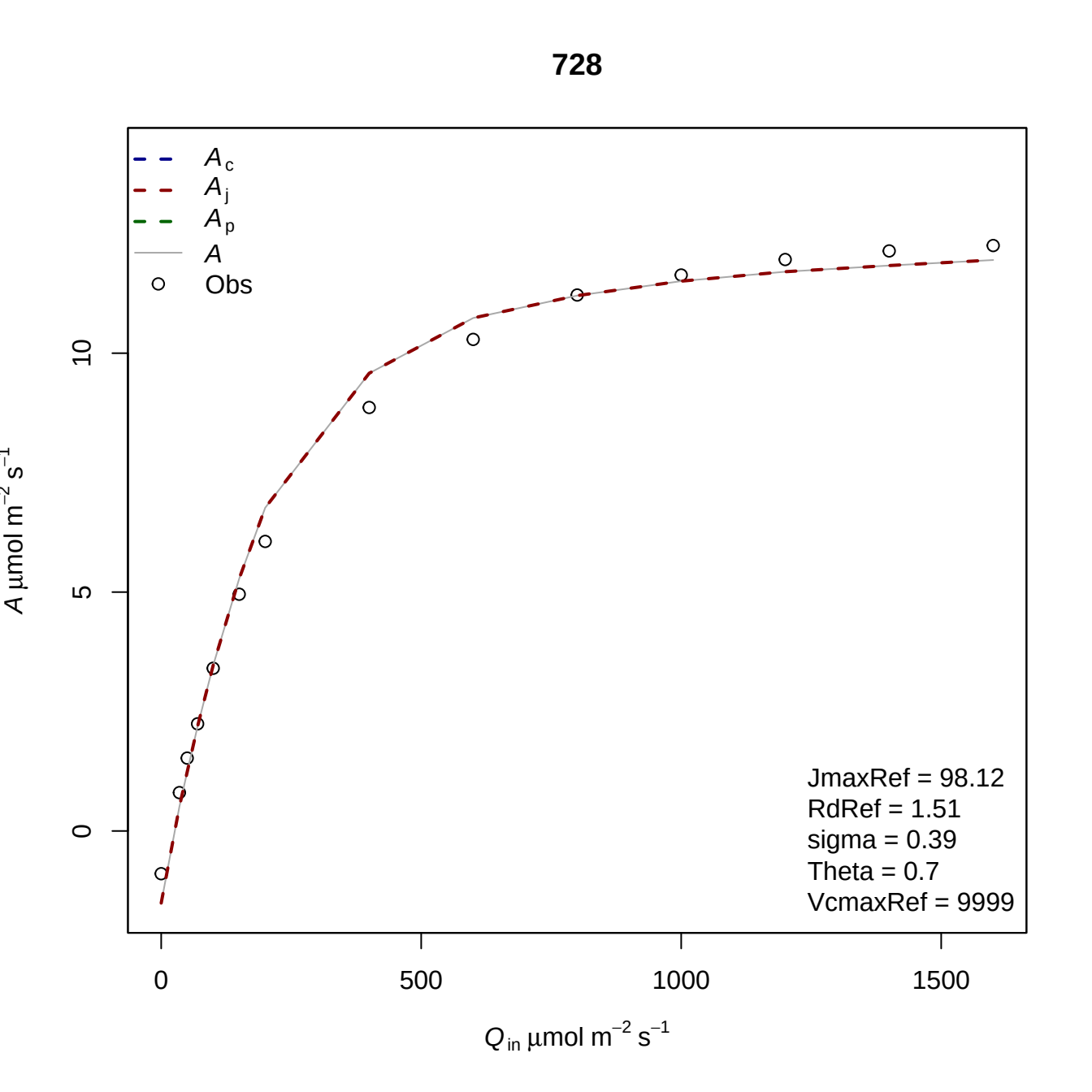

722
A_c
A_j
A_p
A
Obs
JmaxRef = 61.31
RdRef = 0.5
sigma = 0.07
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

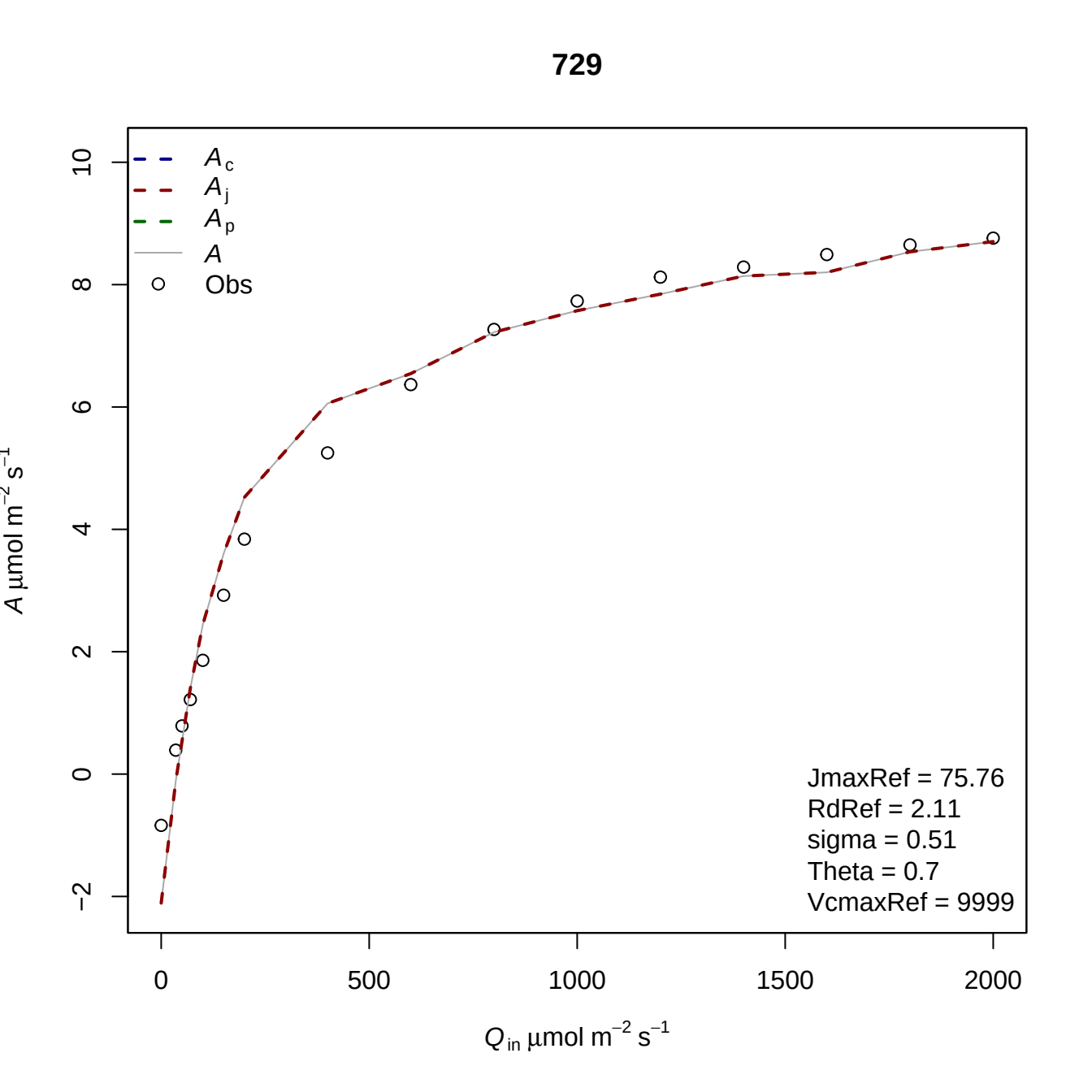

723
A_c
A_j
A_p
A
Obs
JmaxRef = 63.94
RdRef = 1.06
sigma = 0.26
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

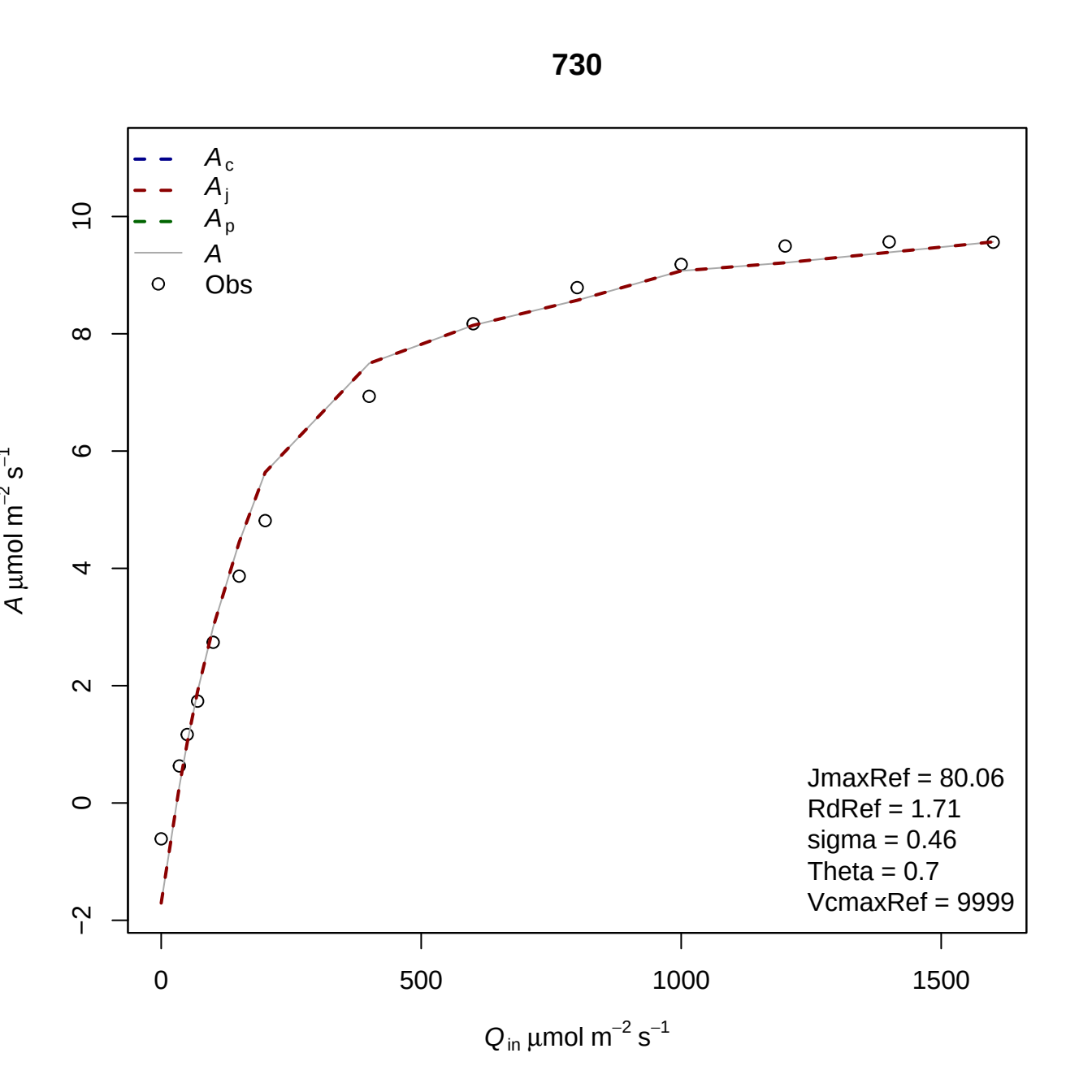

724
A_c
A_j
A_p
A
Obs
JmaxRef = 46.76
RdRef = 1.1
sigma = 0.5
Theta = 0.7
VcmaxRef = 34.15
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

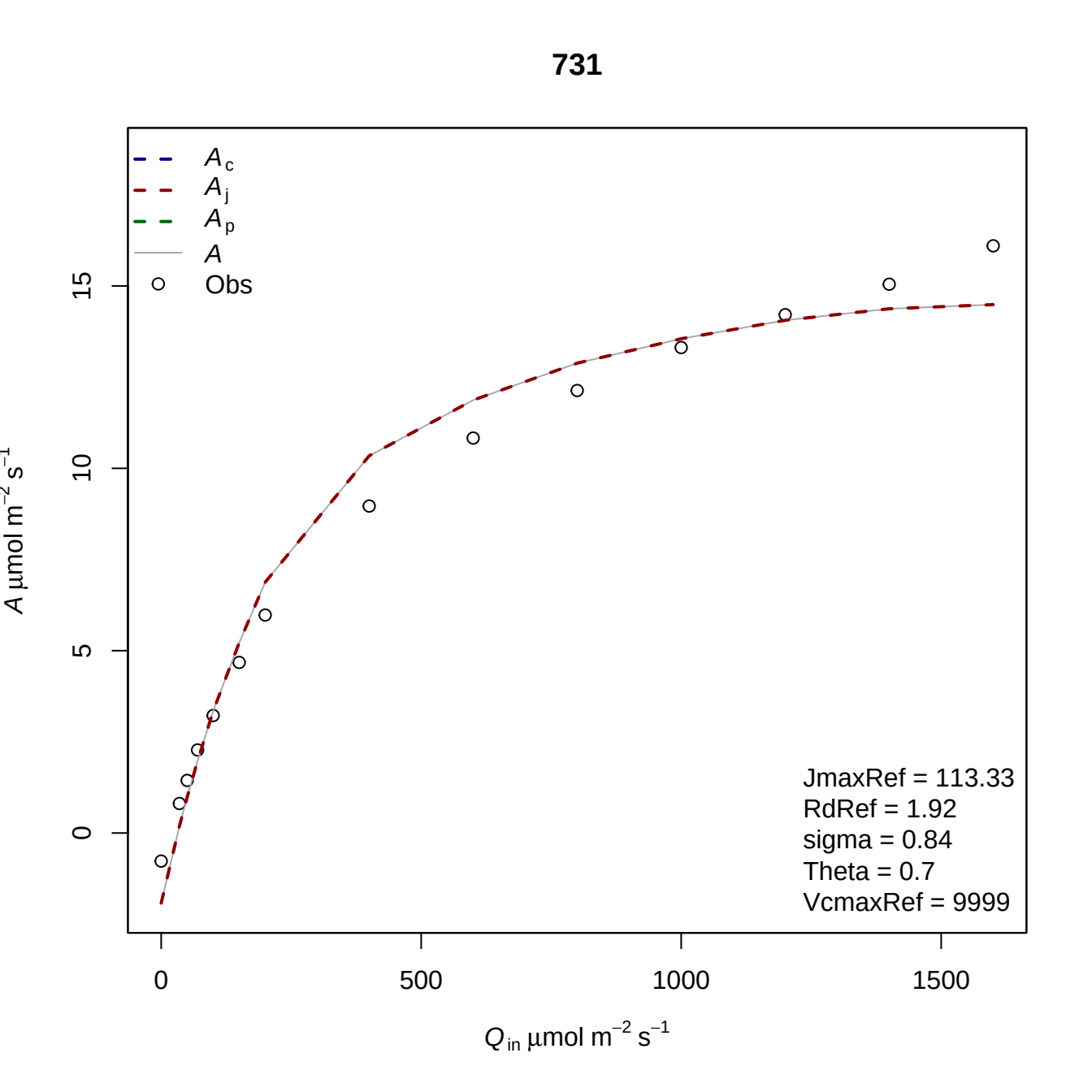

726
A_c
A_j
A_p
A
Obs
JmaxRef = 67.86
RdRef = 1.24
sigma = 0.44
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

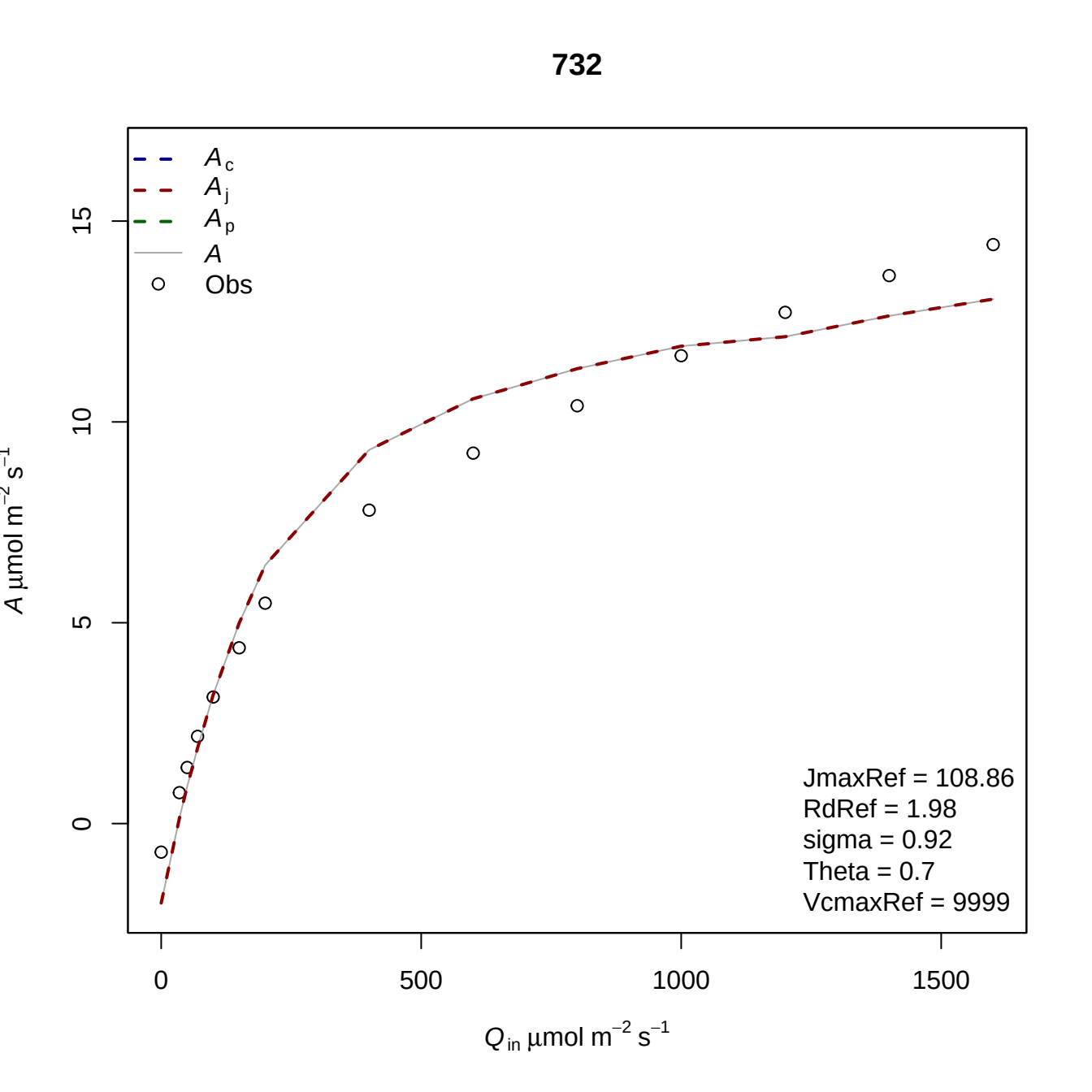

727
A_c
A_j
A_p
A
Obs
JmaxRef = 62.3
RdRef = 1.45
sigma = 0.59
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

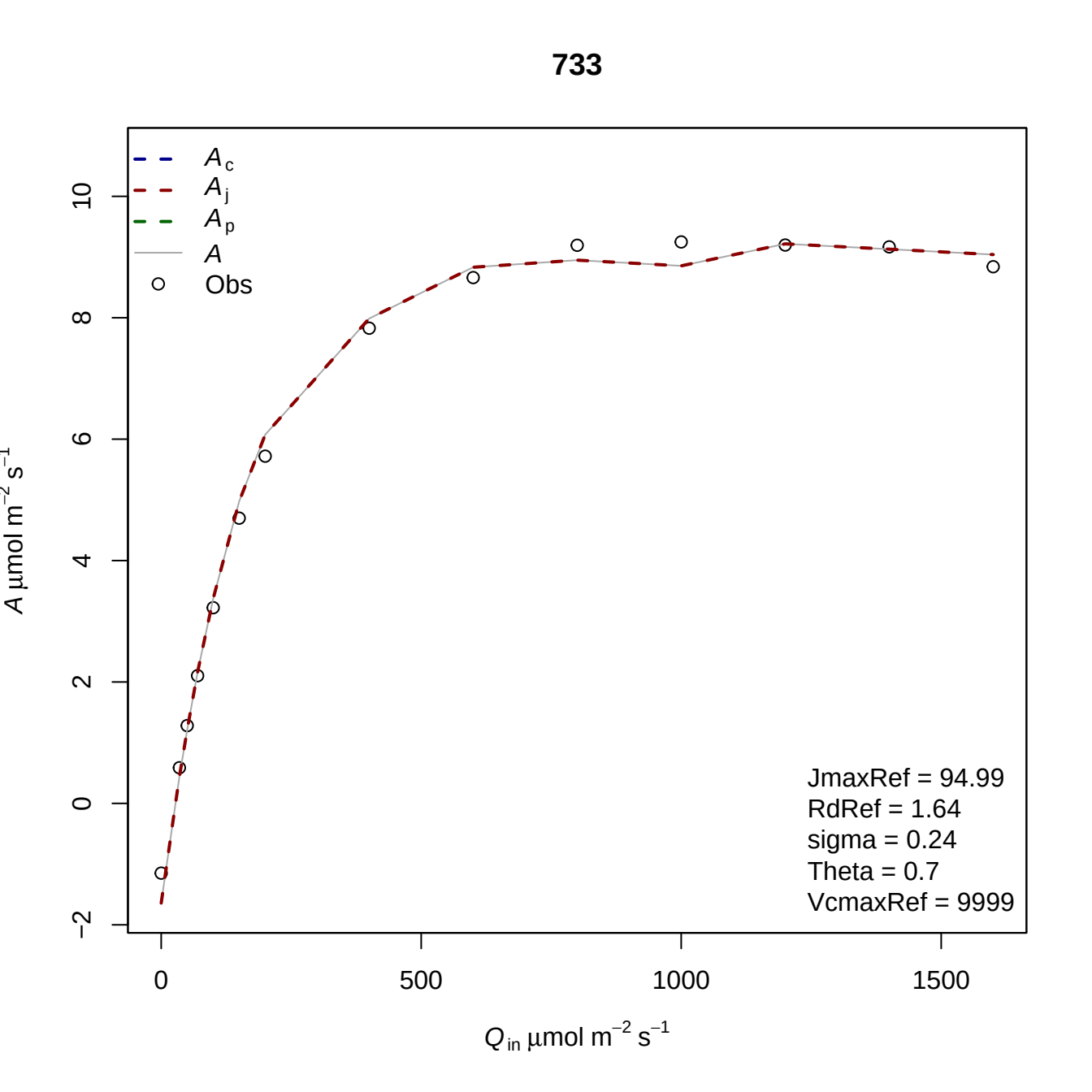

728
A_c
A_j
A_p
A
Obs
JmaxRef = 98.12
RdRef = 1.51
sigma = 0.39
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

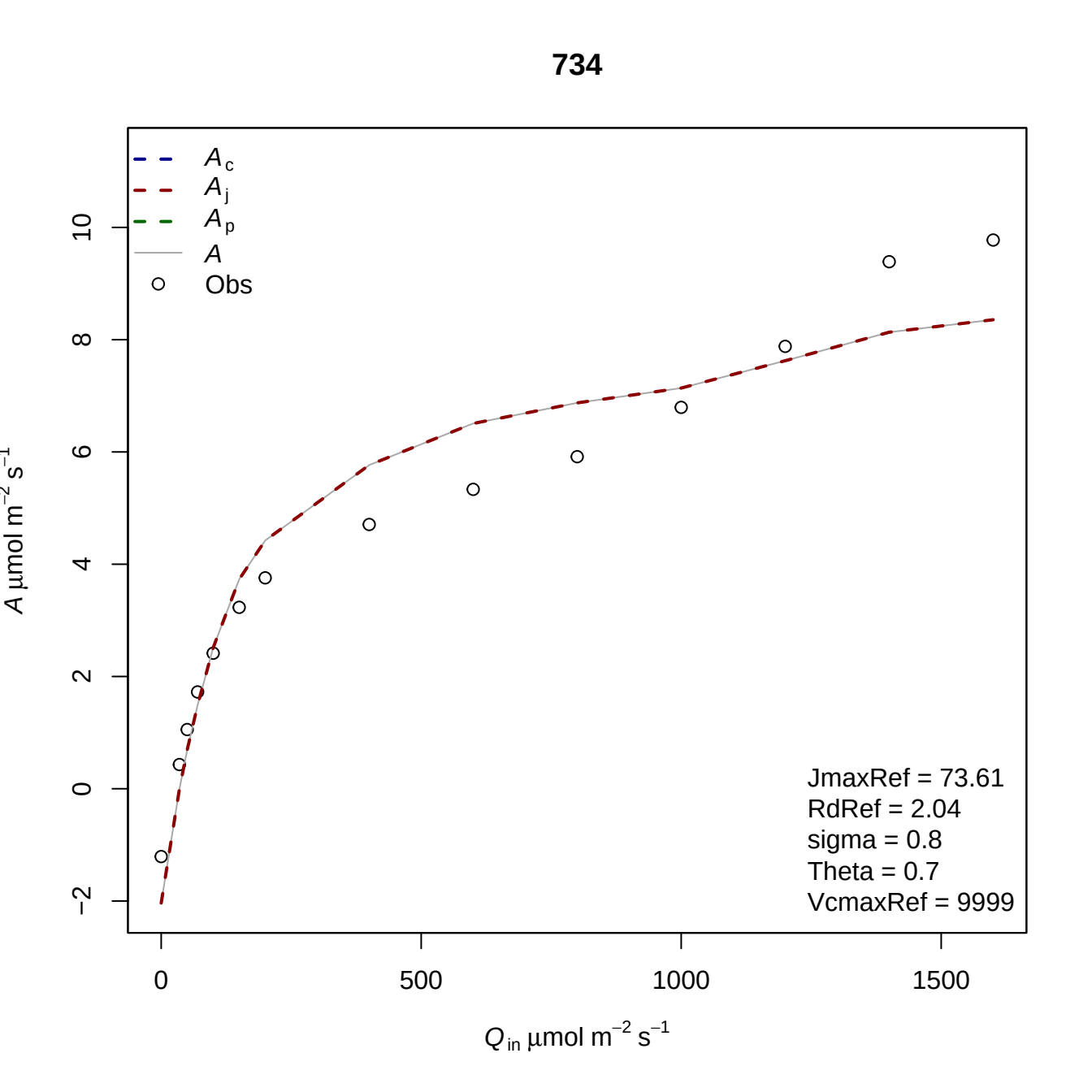

729
A_c
A_j
A_p
A
Obs
JmaxRef = 75.76
RdRef = 2.11
sigma = 0.51
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

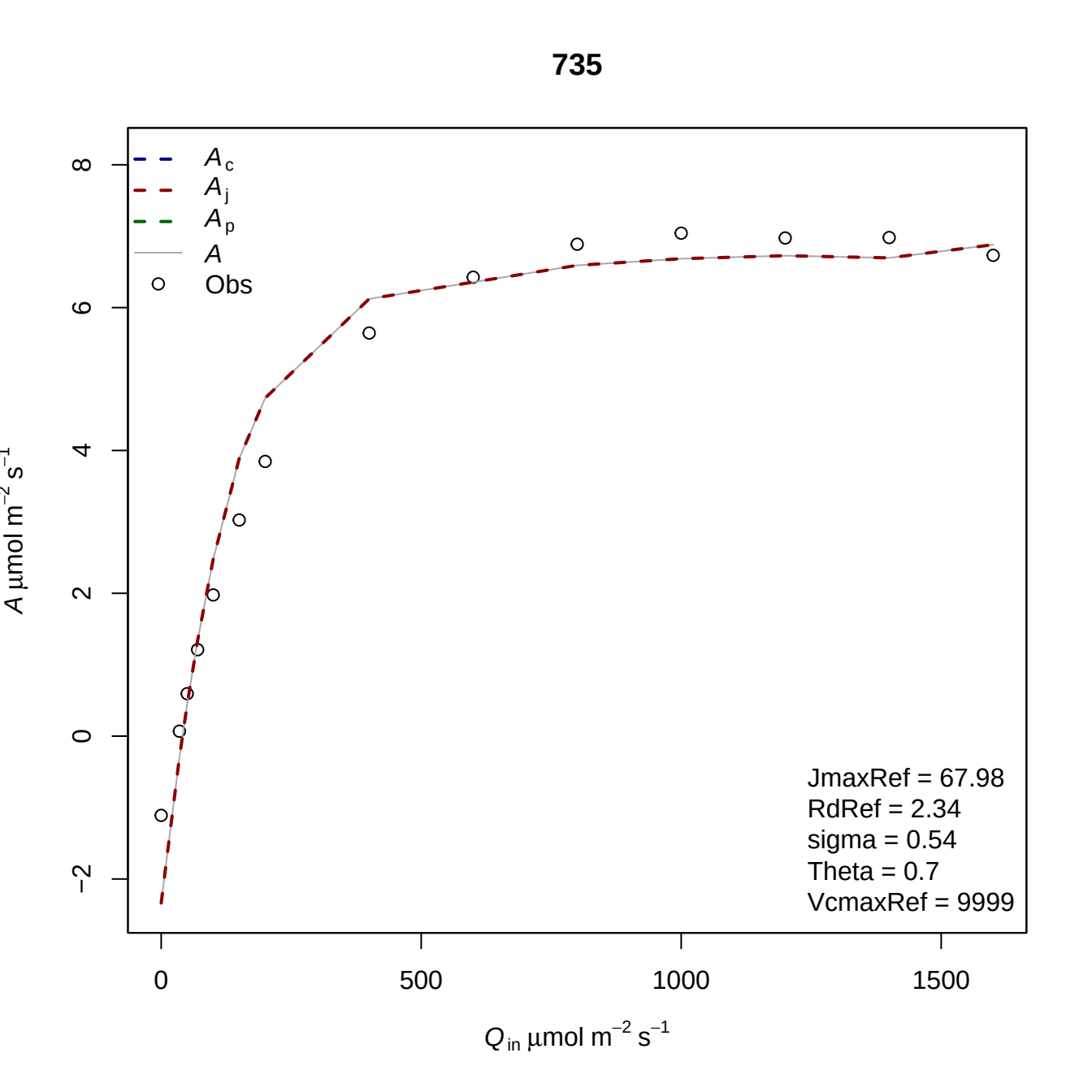

730
A_c
A_j
A_p
A
Obs
JmaxRef = 80.06
RdRef = 1.71
sigma = 0.46
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

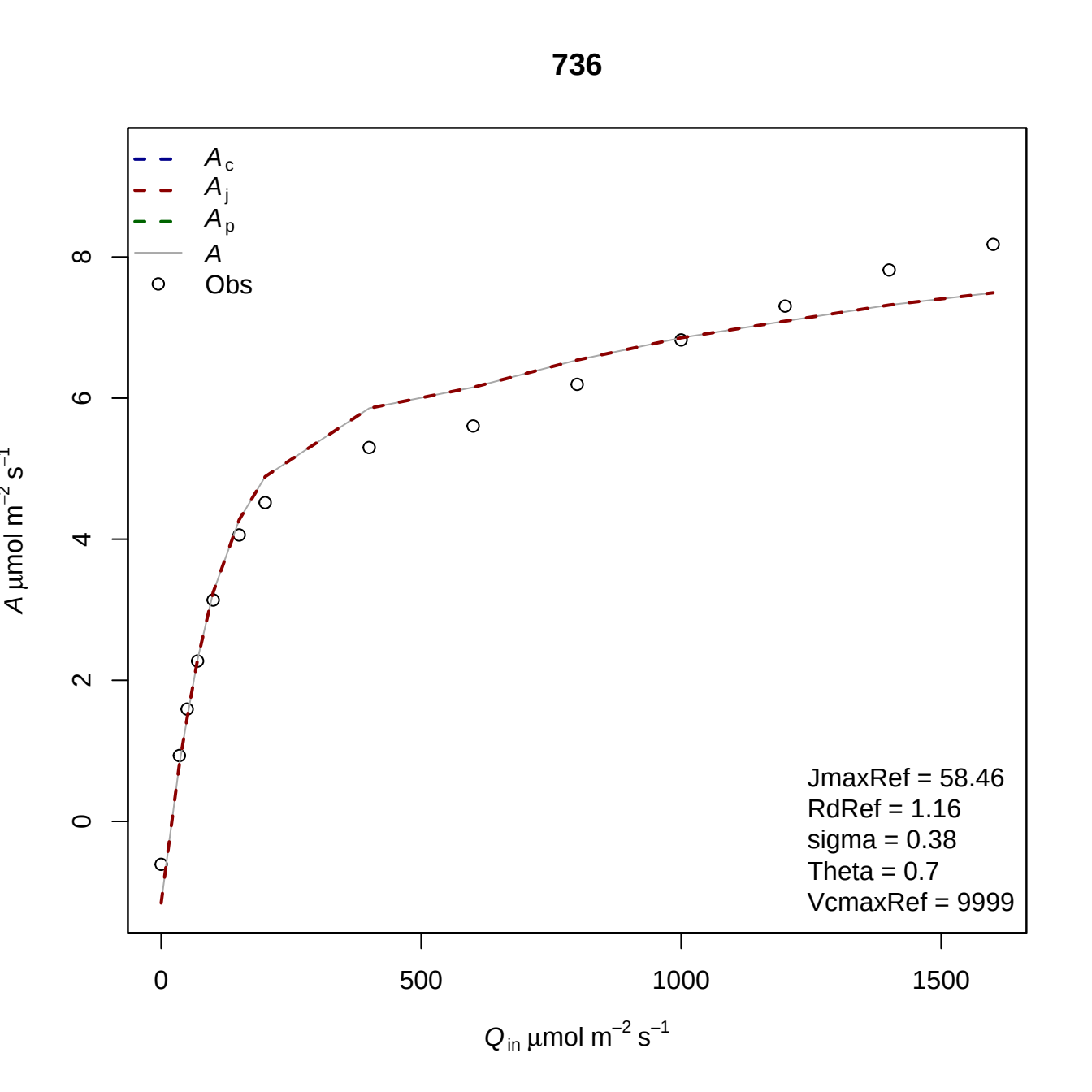

731
A_c
A_j
A_p
A
Obs
JmaxRef = 113.33
RdRef = 1.92
sigma = 0.84
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

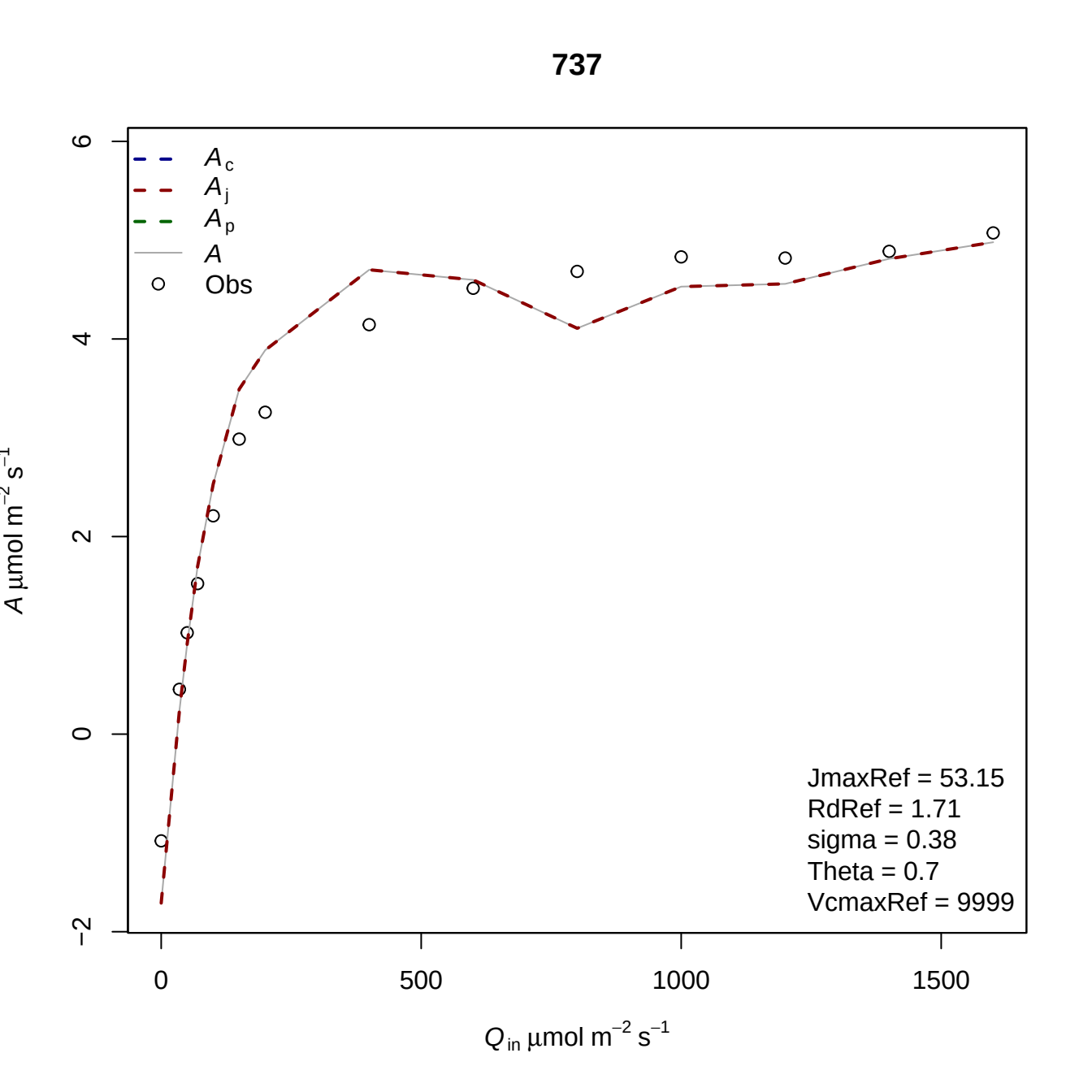

732
A_c
A_j
A_p
A
Obs
JmaxRef = 108.86
RdRef = 1.98
sigma = 0.92
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

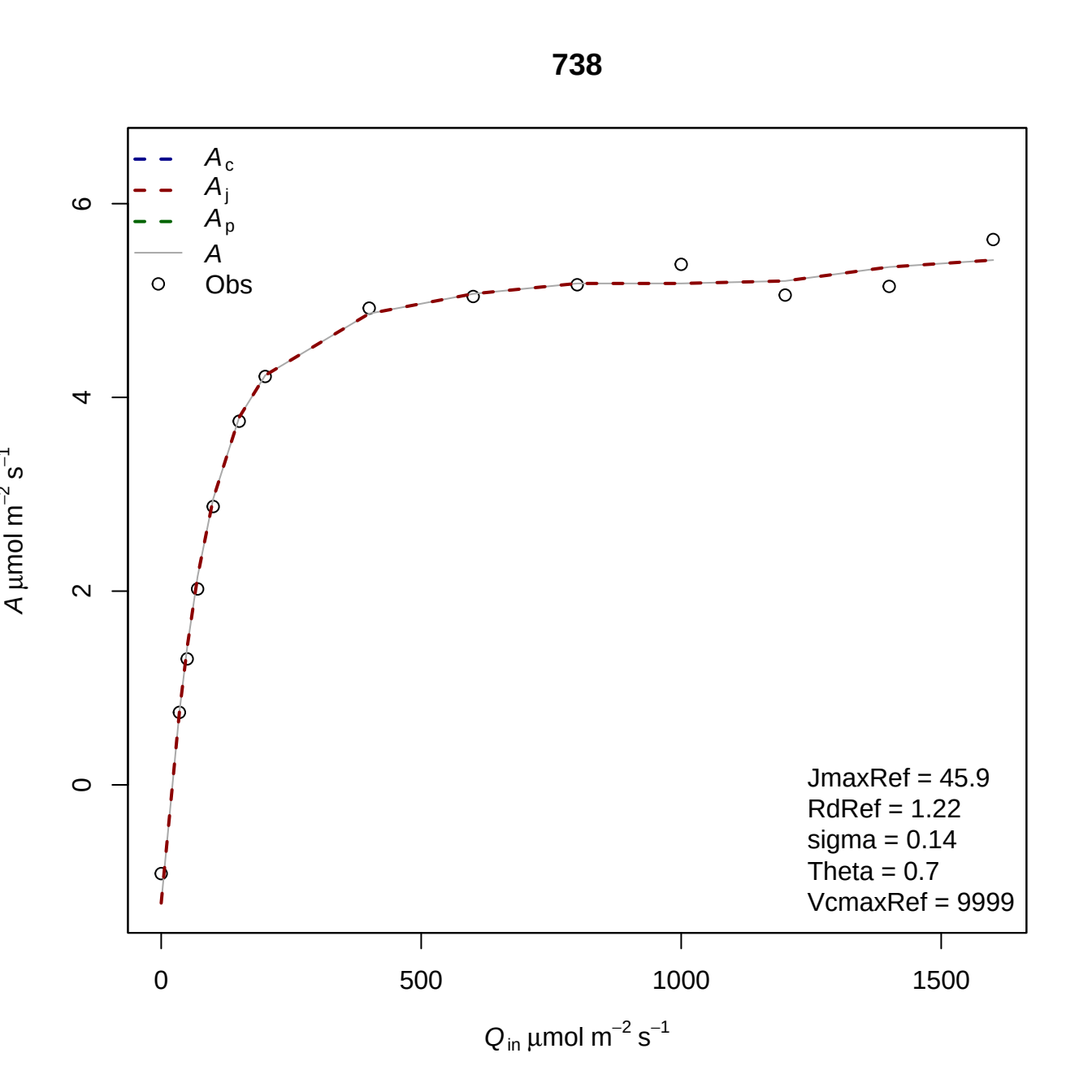

733
A_c
A_j
A_p
A
Obs
JmaxRef = 94.99
RdRef = 1.64
sigma = 0.24
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

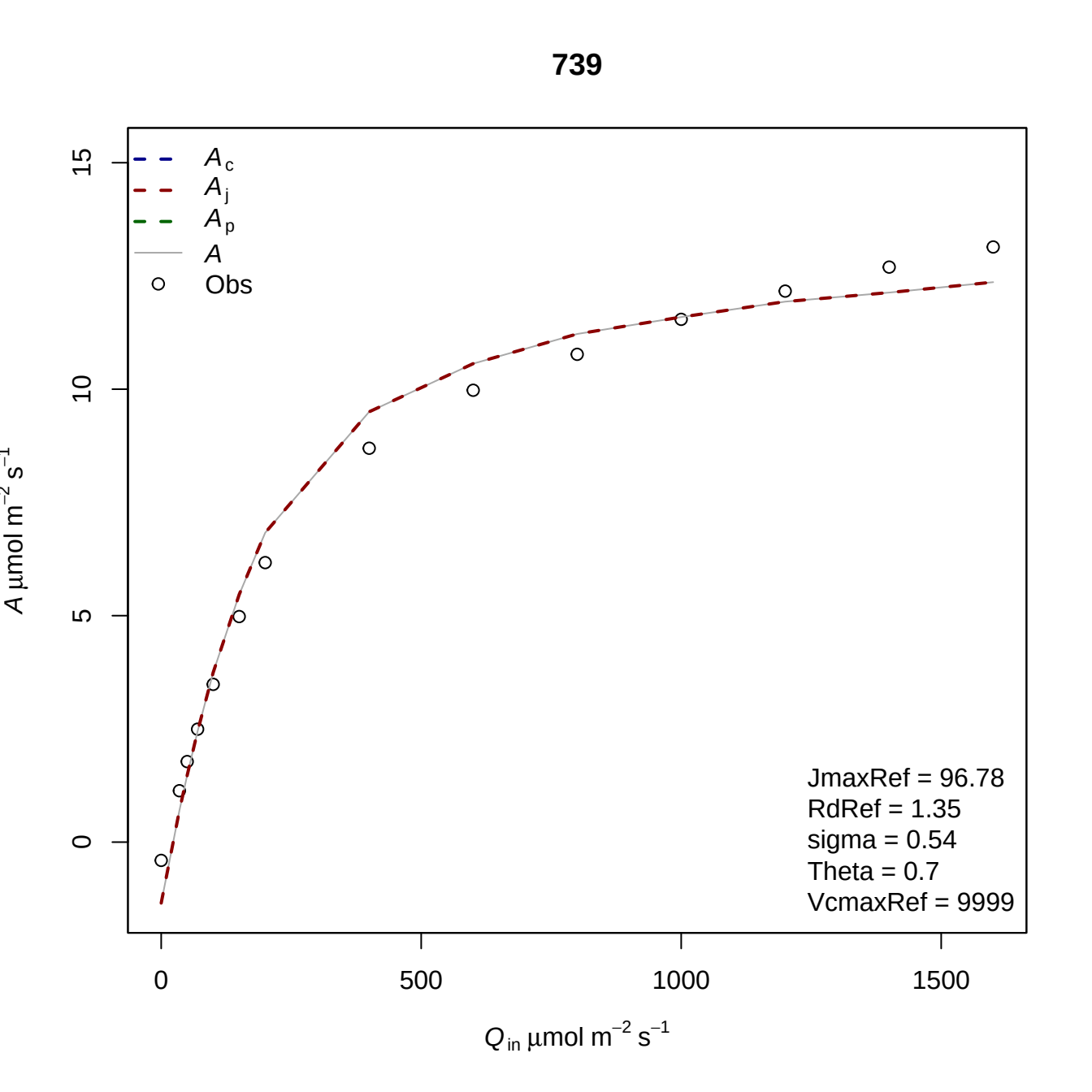

734
A_c
A_j
A_p
A
Obs
JmaxRef = 73.61
RdRef = 2.04
sigma = 0.8
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

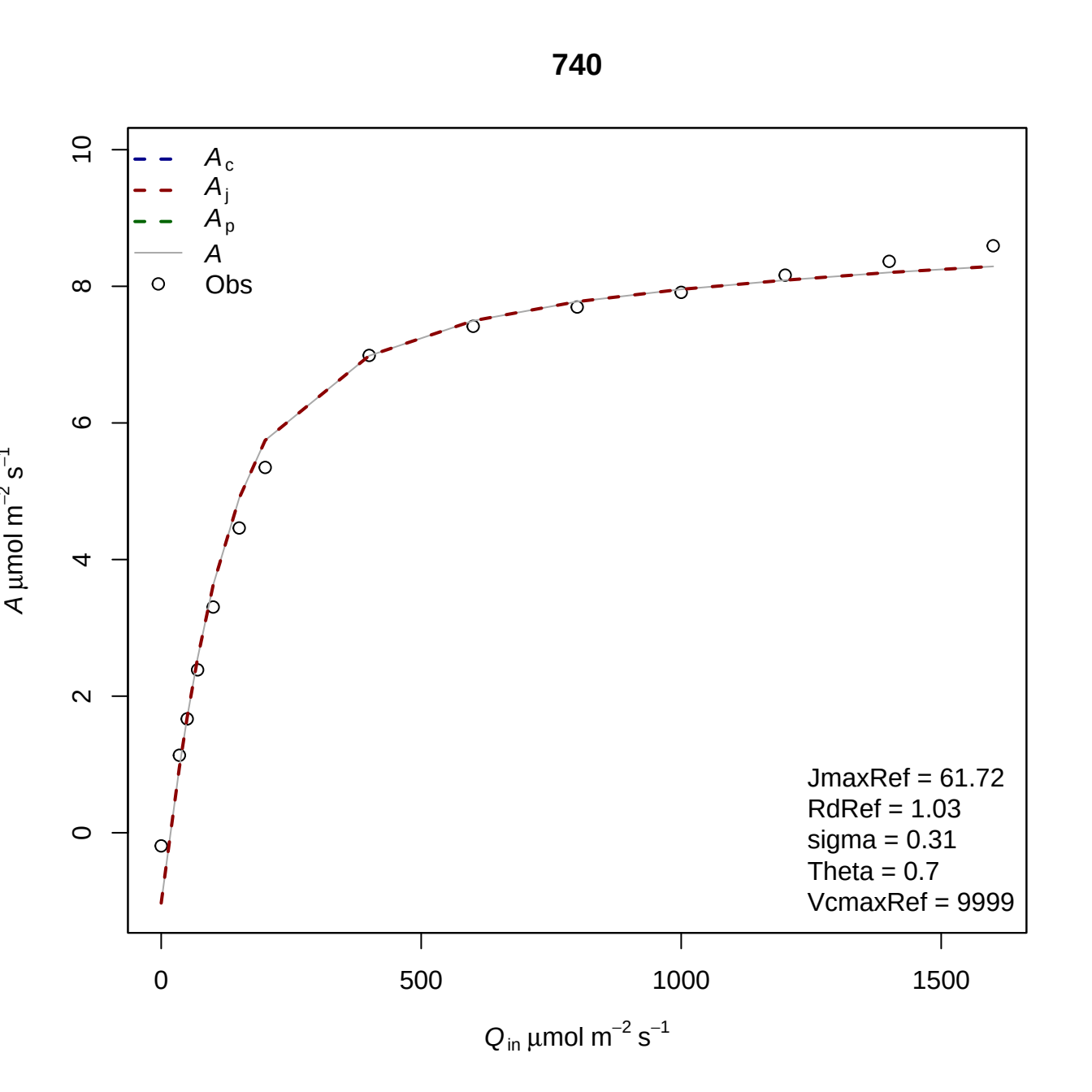

735
A_c
A_j
A_p
A
Obs
JmaxRef = 67.98
RdRef = 2.34
sigma = 0.54
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

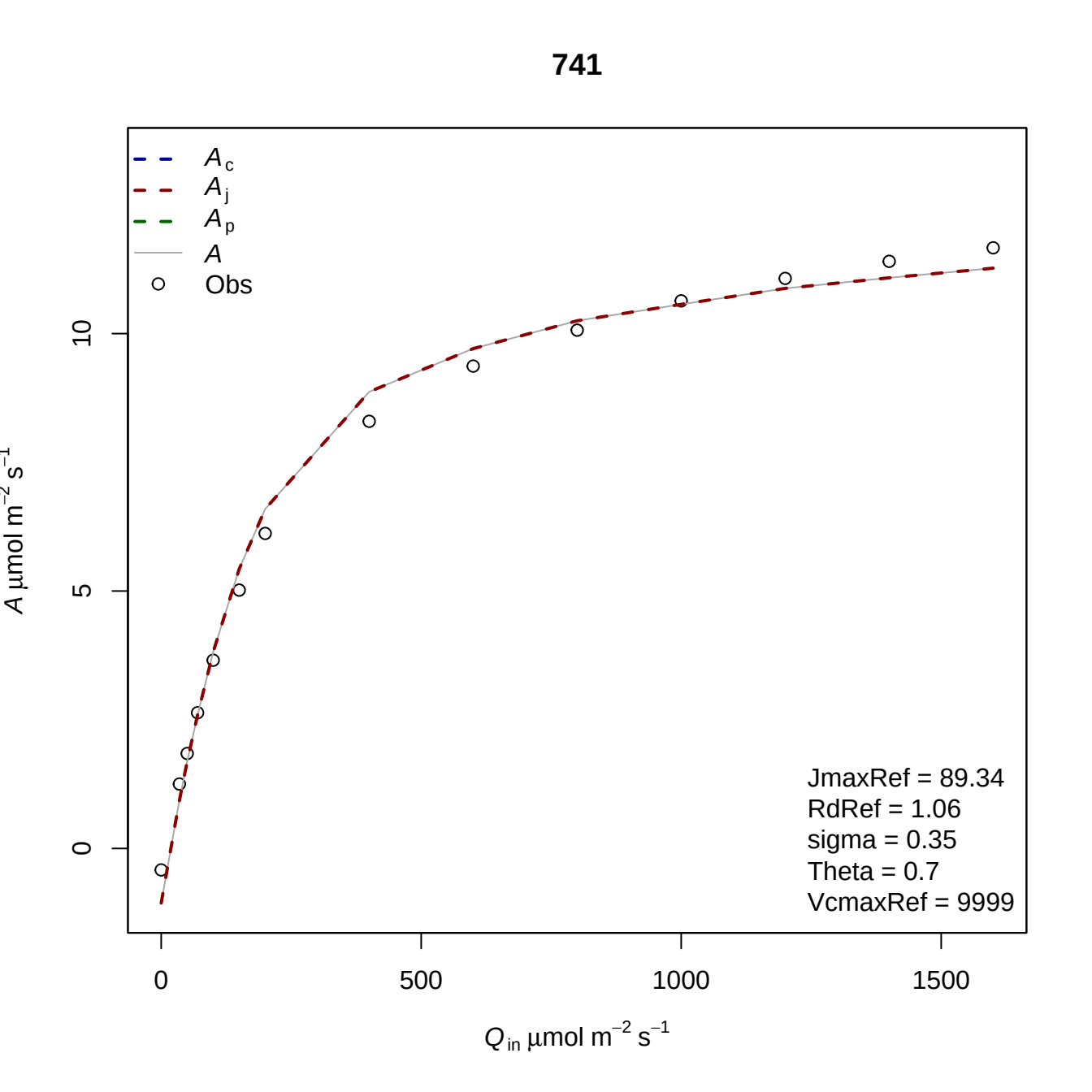

736
A_c
A_j
A_p
A
Obs
JmaxRef = 58.46
RdRef = 1.16
sigma = 0.38
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

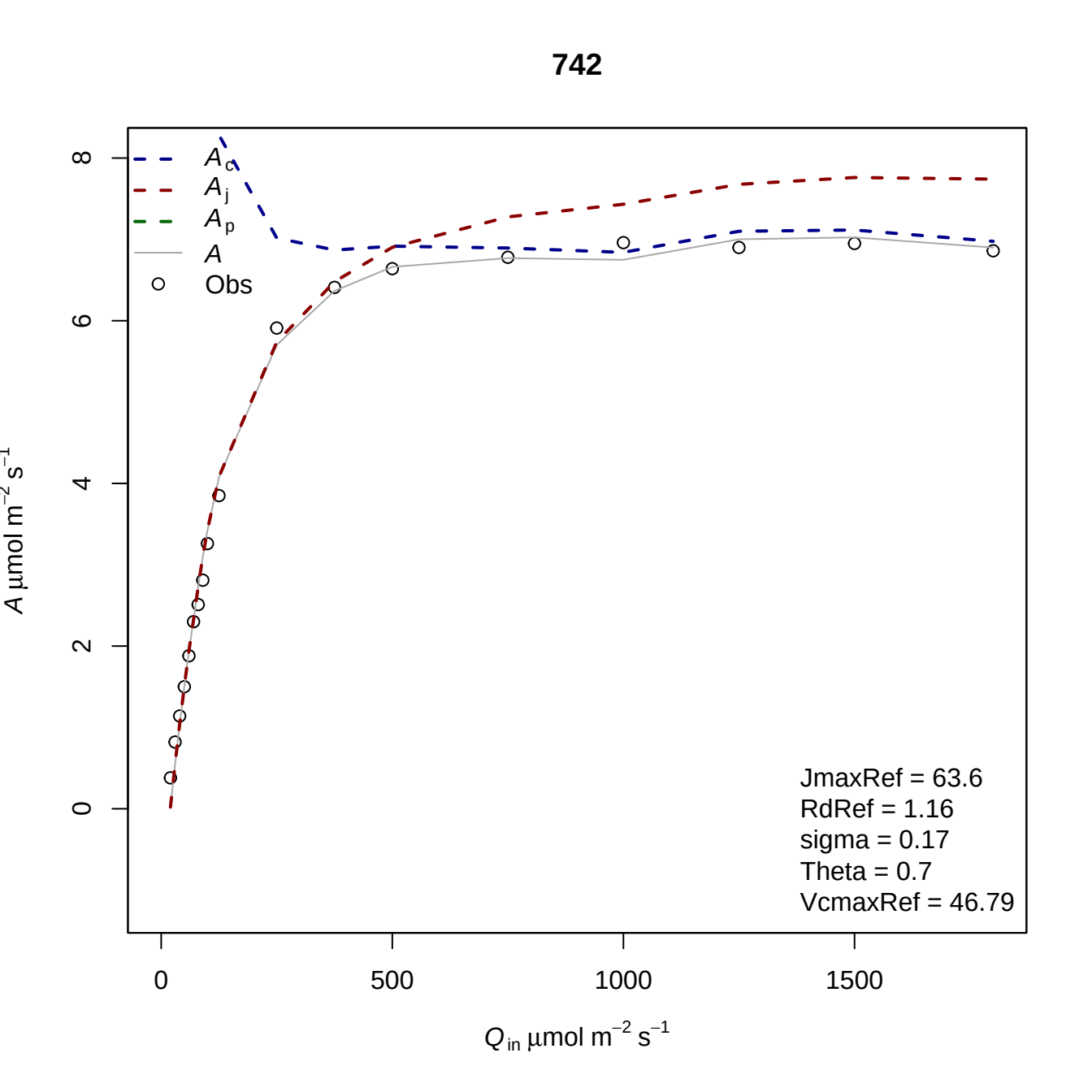

737
A_c
A_j
A_p
A
Obs
JmaxRef = 53.15
RdRef = 1.71
sigma = 0.38
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

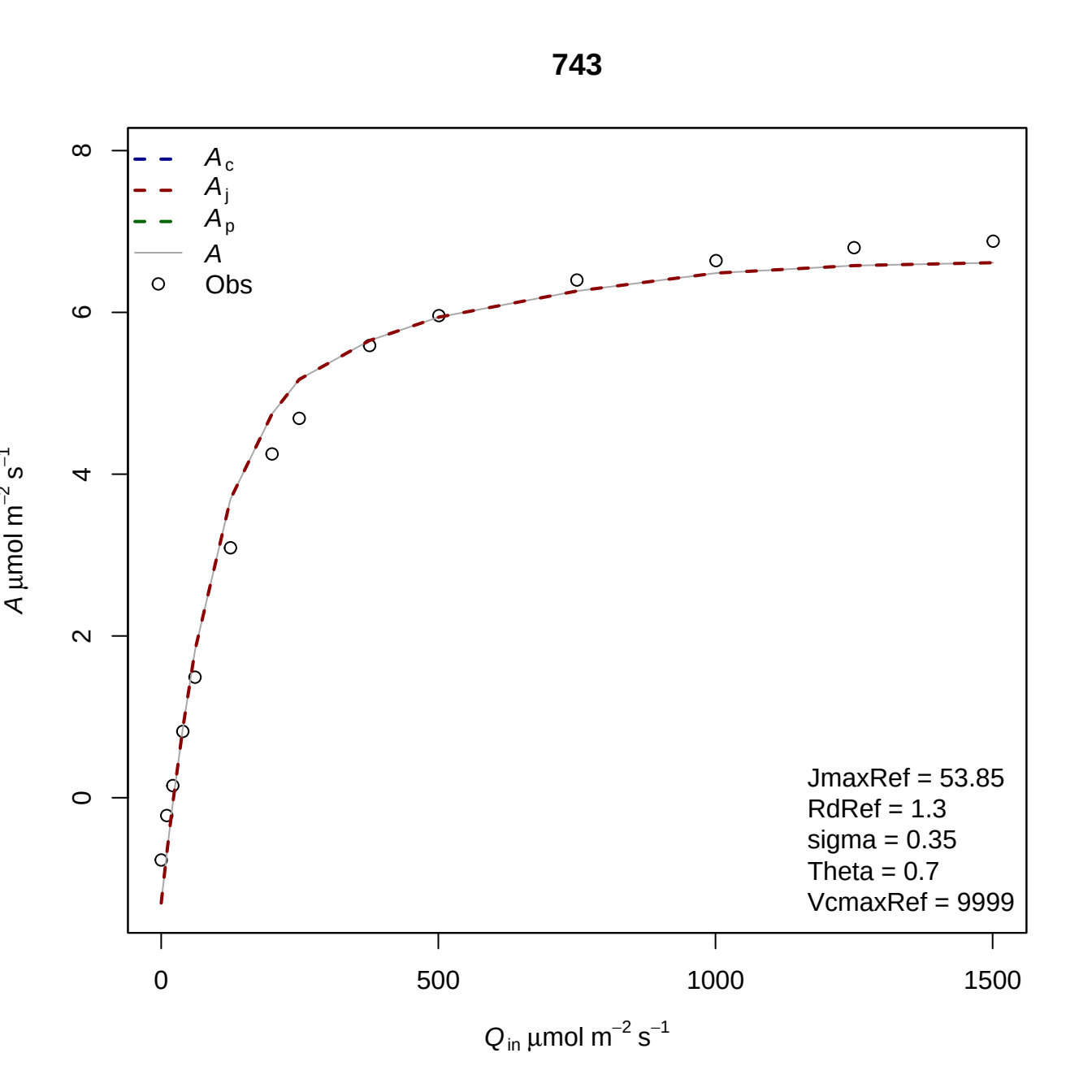

738
A_c
A_j
A_p
A
Obs
JmaxRef = 45.9
RdRef = 1.22
sigma = 0.14
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

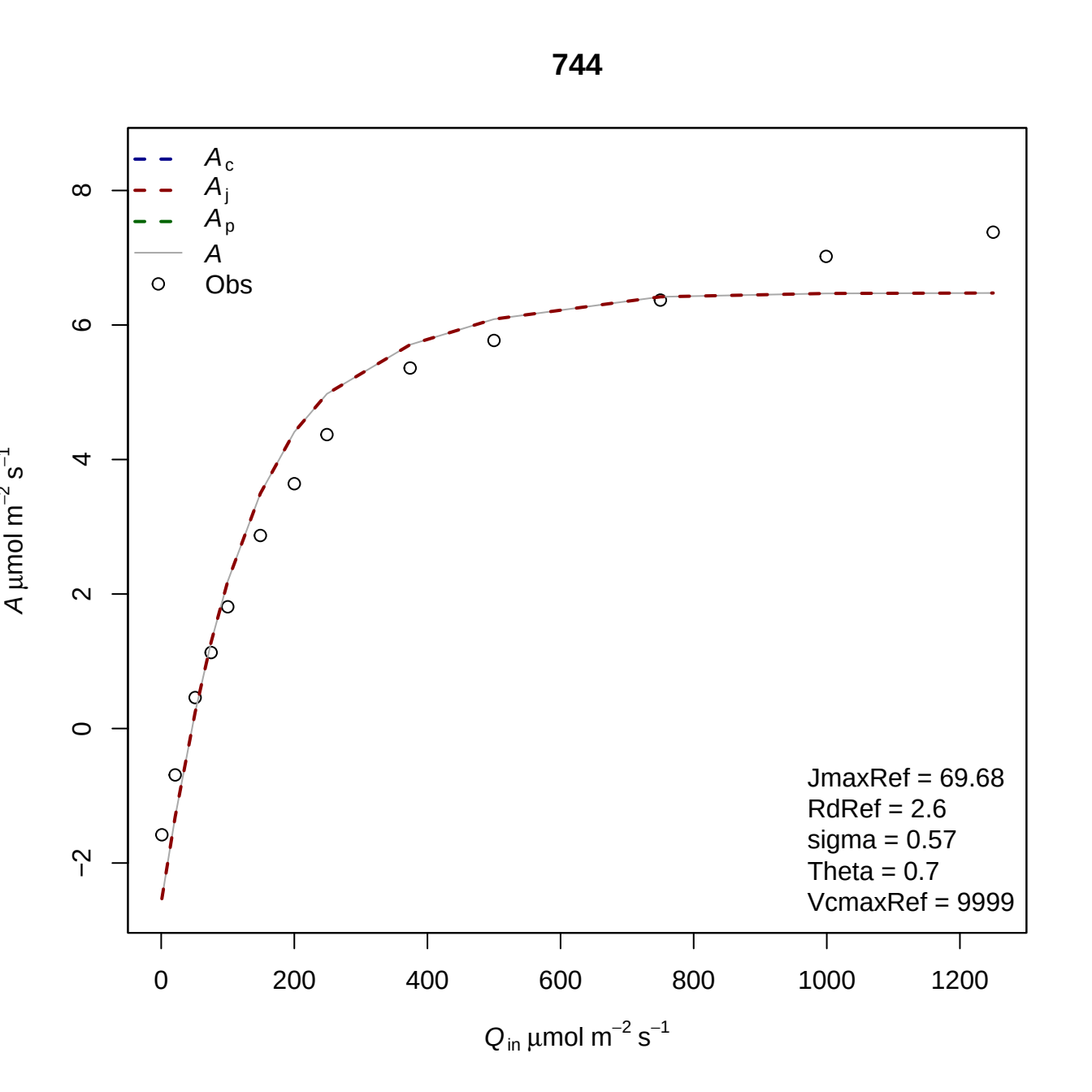

739
A_c
A_j
A_p
A
Obs
JmaxRef = 96.78
RdRef = 1.35
sigma = 0.54
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

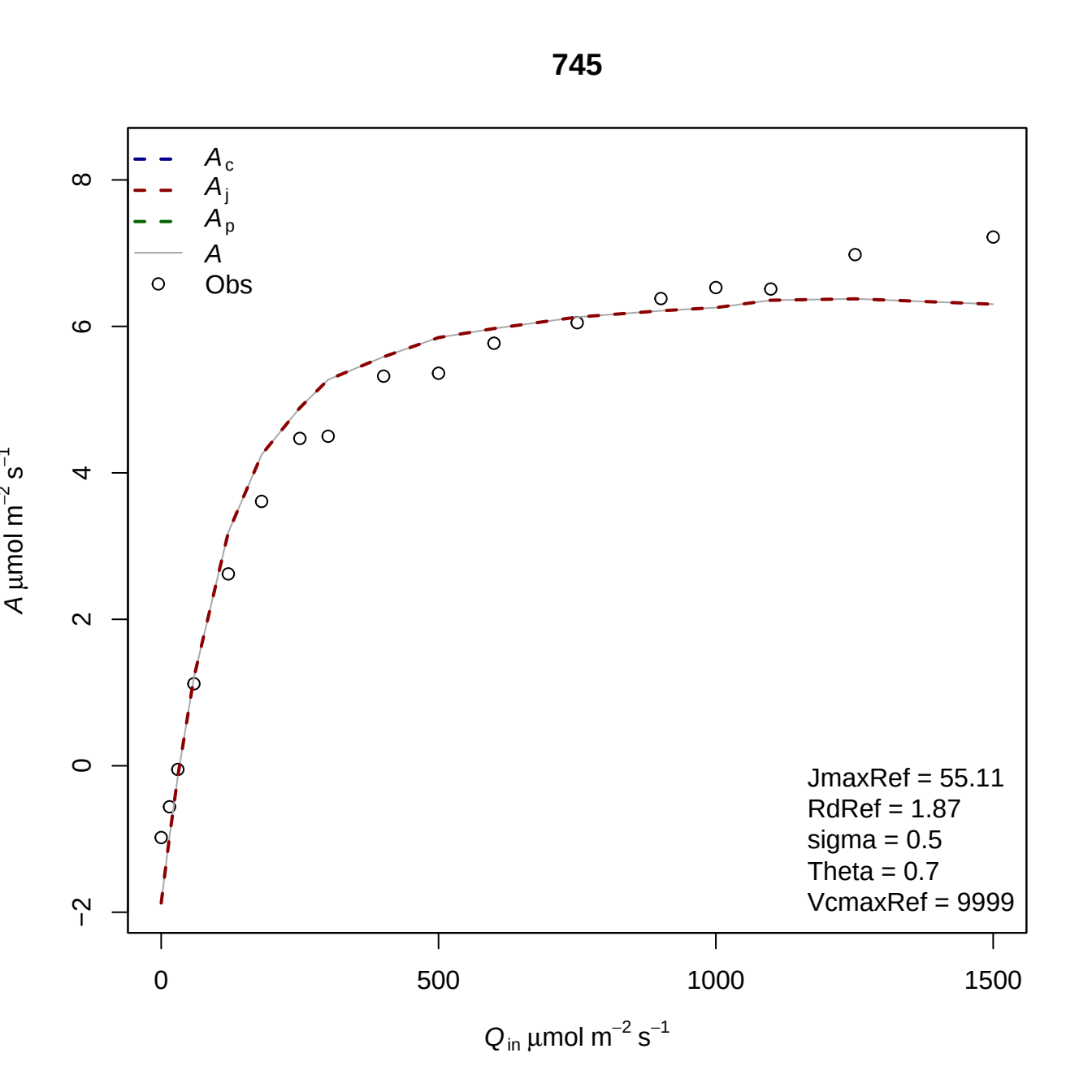

740
A_c
A_j
A_p
A
Obs
JmaxRef = 61.72
RdRef = 1.03
sigma = 0.31
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

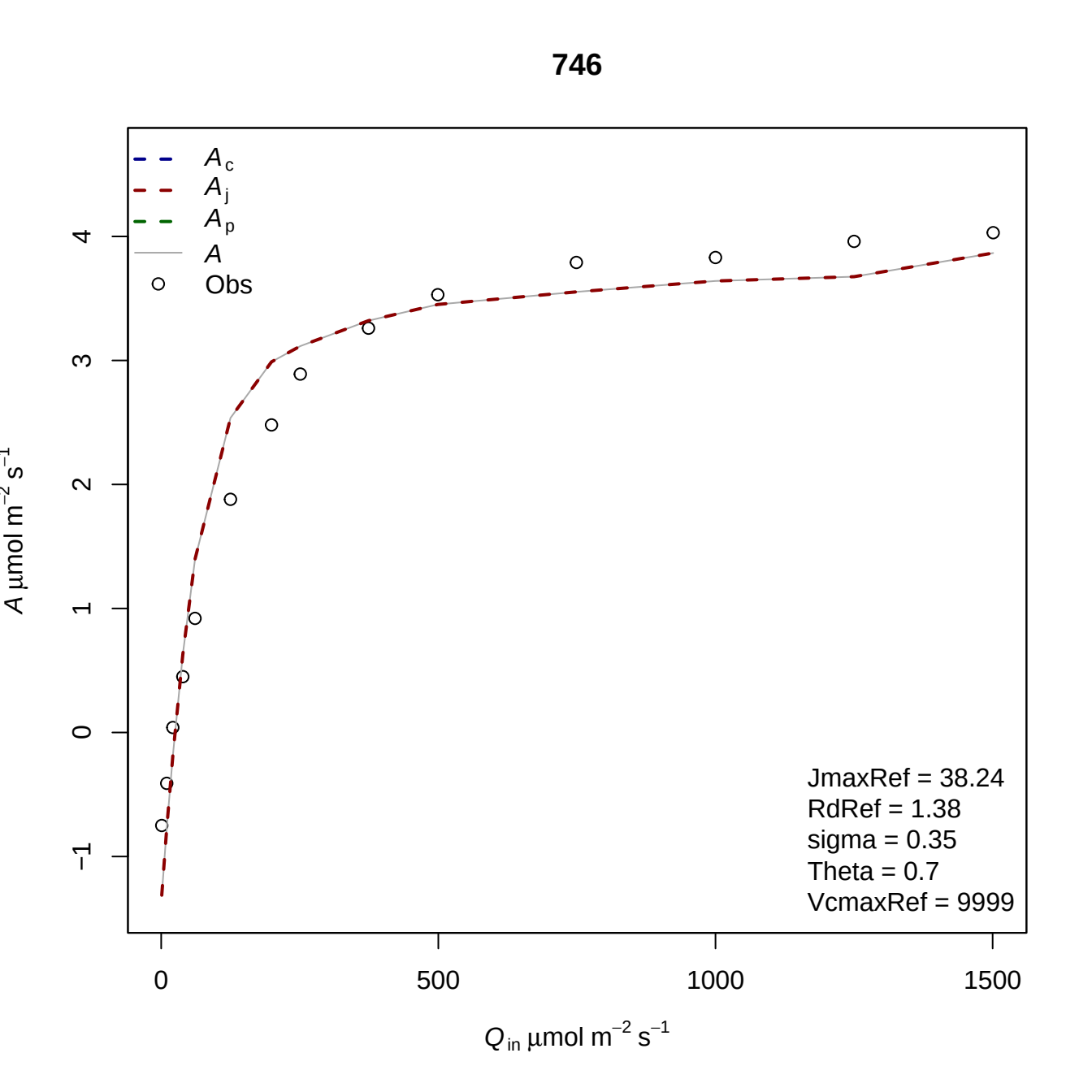

741
A_c
A_j
A_p
A
Obs
JmaxRef = 89.34
RdRef = 1.06
sigma = 0.35
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

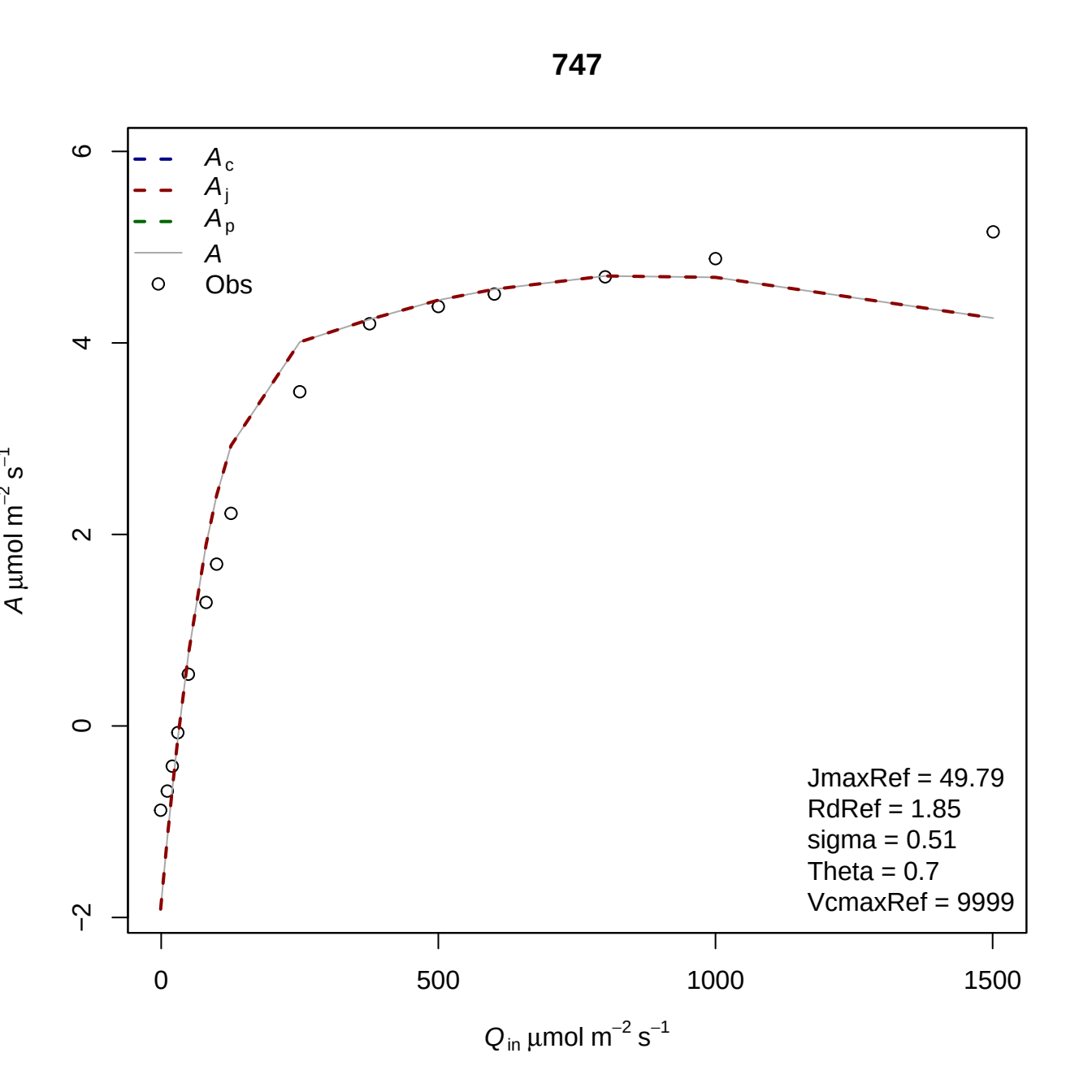

742
A_c
A_j
A_p
A
Obs
JmaxRef = 63.6
RdRef = 1.16
sigma = 0.17
Theta = 0.7
VcmaxRef = 46.79
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

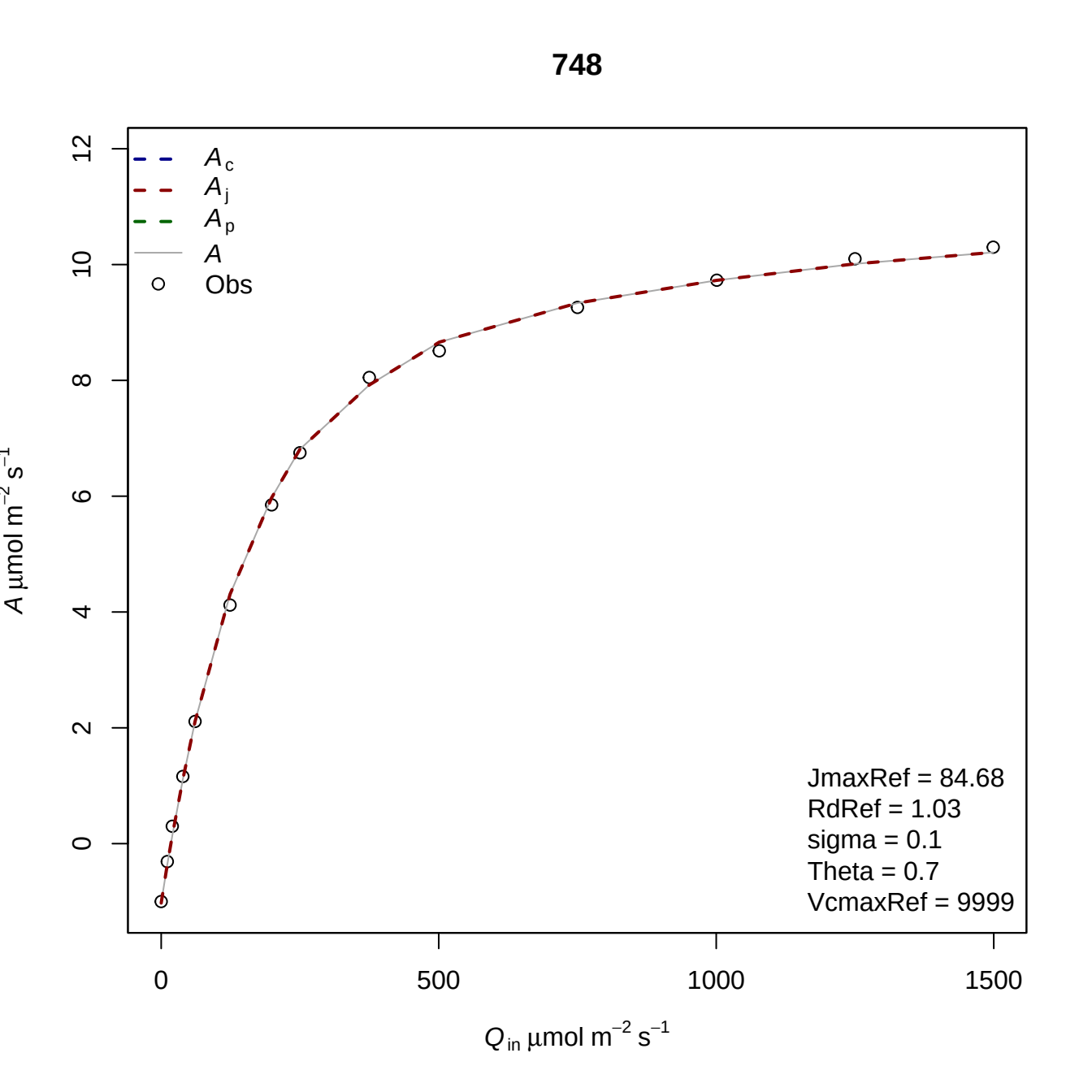

743
A_c
A_j
A_p
A
Obs
JmaxRef = 53.85
RdRef = 1.3
sigma = 0.35
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

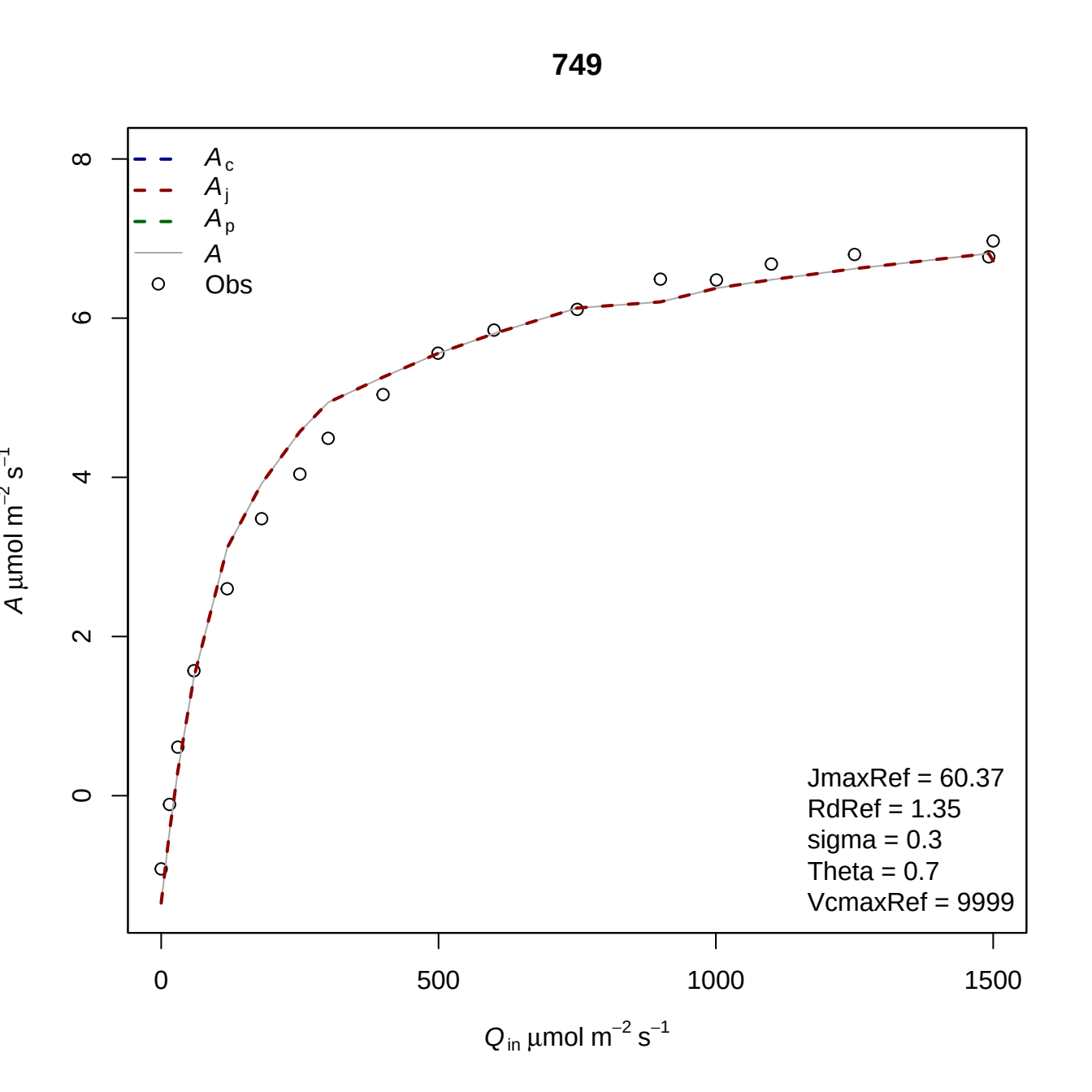

744
A_c
A_j
A_p
A
Obs
JmaxRef = 69.68
RdRef = 2.6
sigma = 0.57
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
200
400
600
800
1000
1200

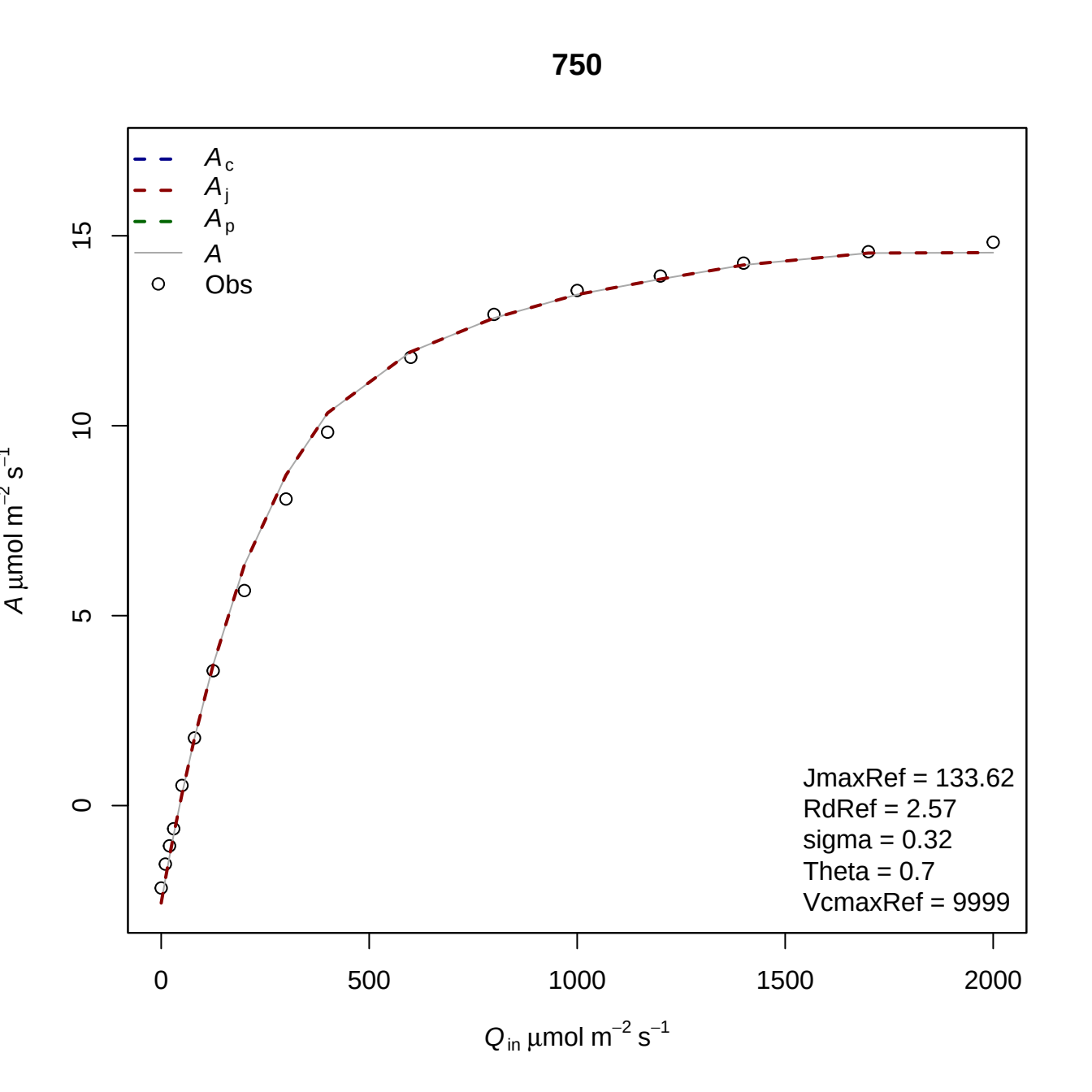

745
A_c
A_j
A_p
A
Obs
JmaxRef = 55.11
RdRef = 1.87
sigma = 0.5
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

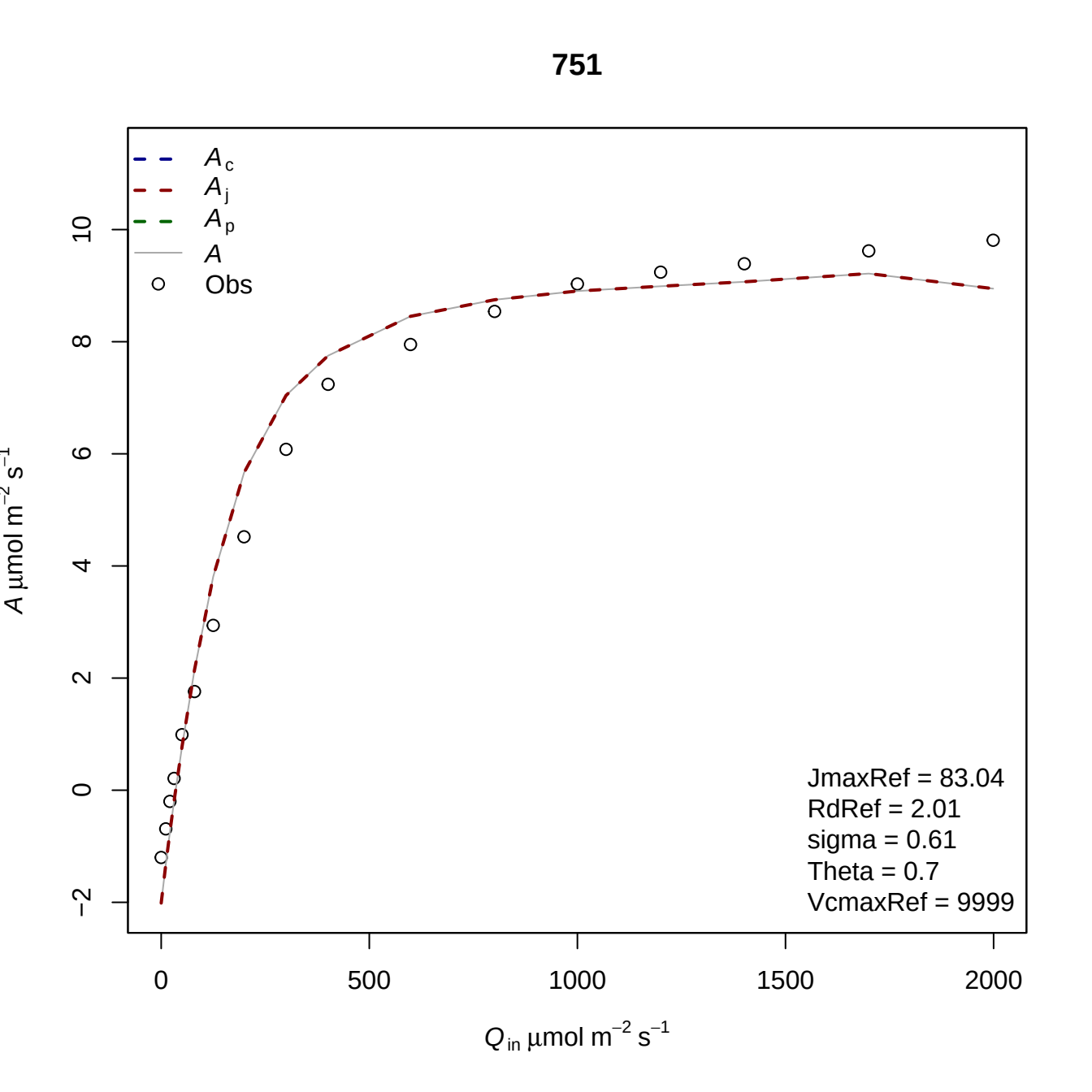

746
A_c
A_j
A_p
A
Obs
JmaxRef = 38.24
RdRef = 1.38
sigma = 0.35
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

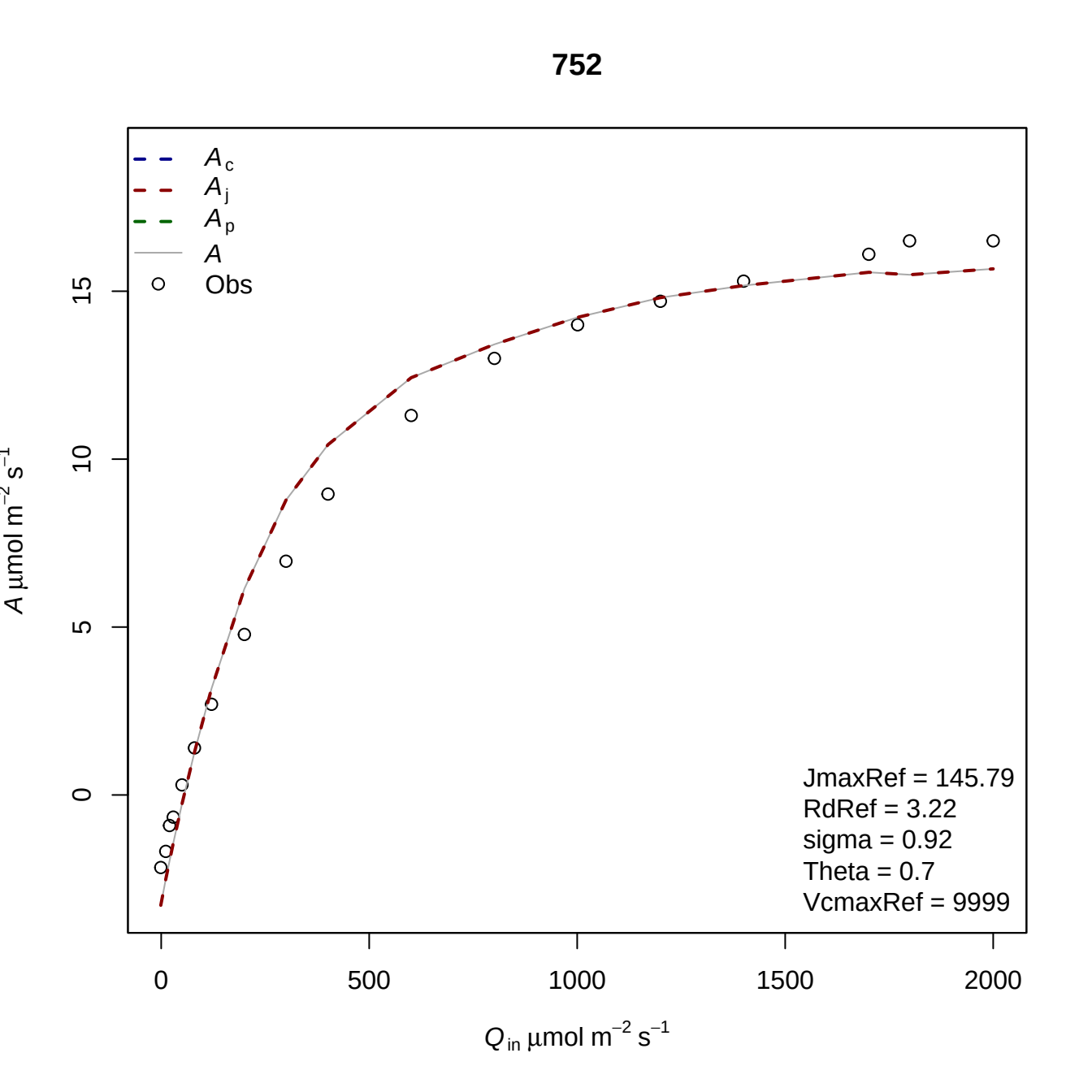

747
A_c
A_j
A_p
A
Obs
JmaxRef = 49.79
RdRef = 1.85
sigma = 0.51
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

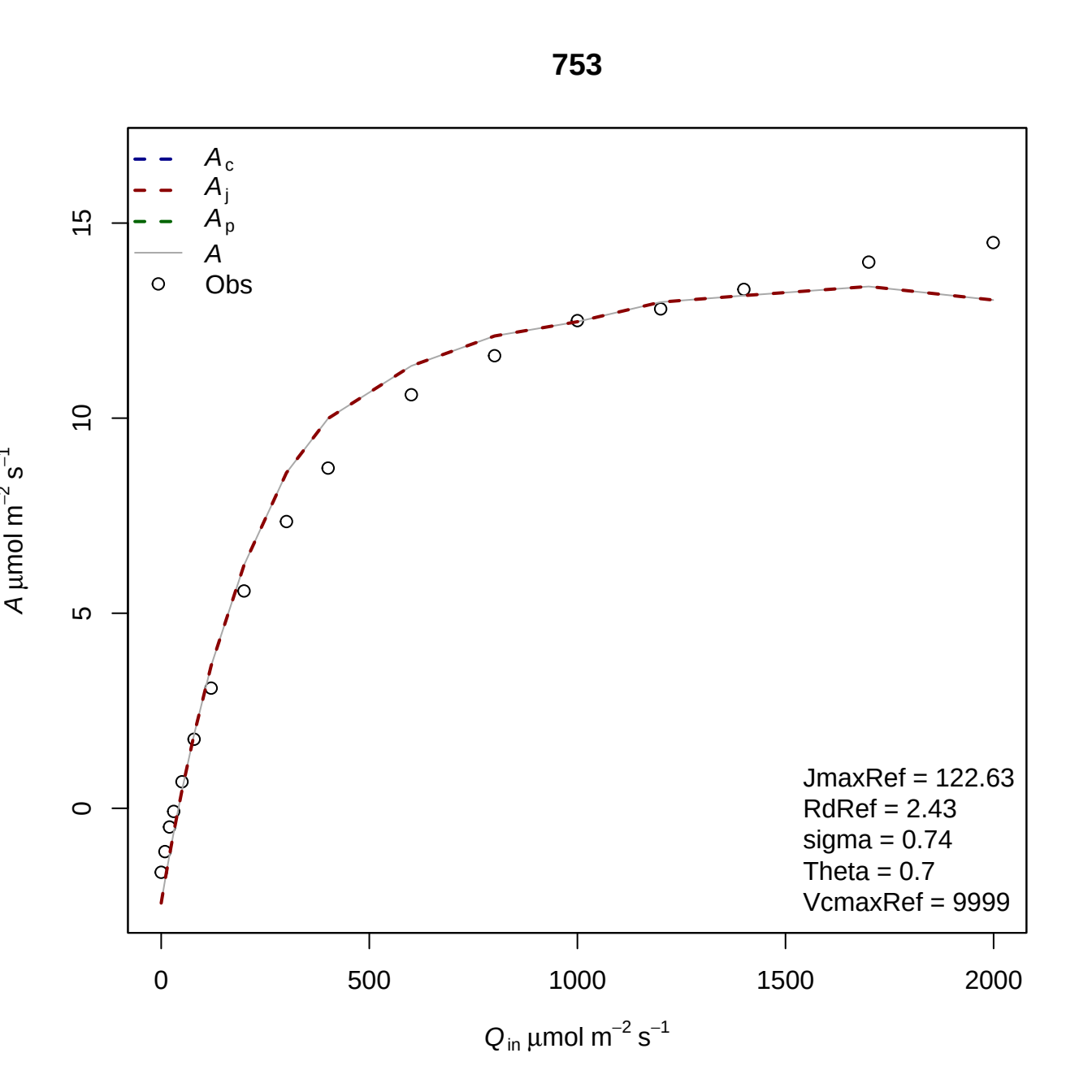

748
A_c
A_j
A_p
A
Obs
JmaxRef = 84.68
RdRef = 1.03
sigma = 0.1
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

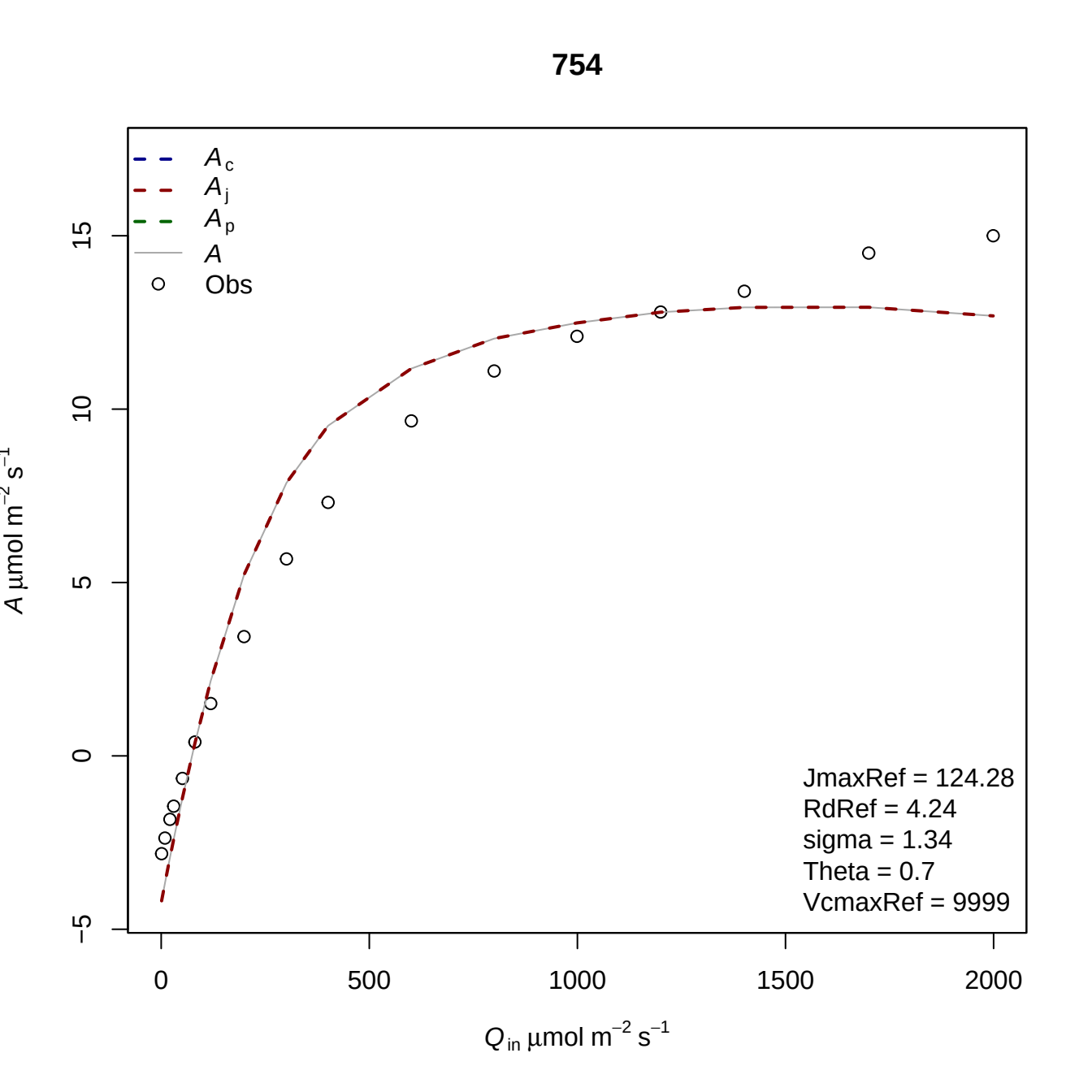

749
A_c
A_j
A_p
A
Obs
JmaxRef = 60.37
RdRef = 1.35
sigma = 0.3
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

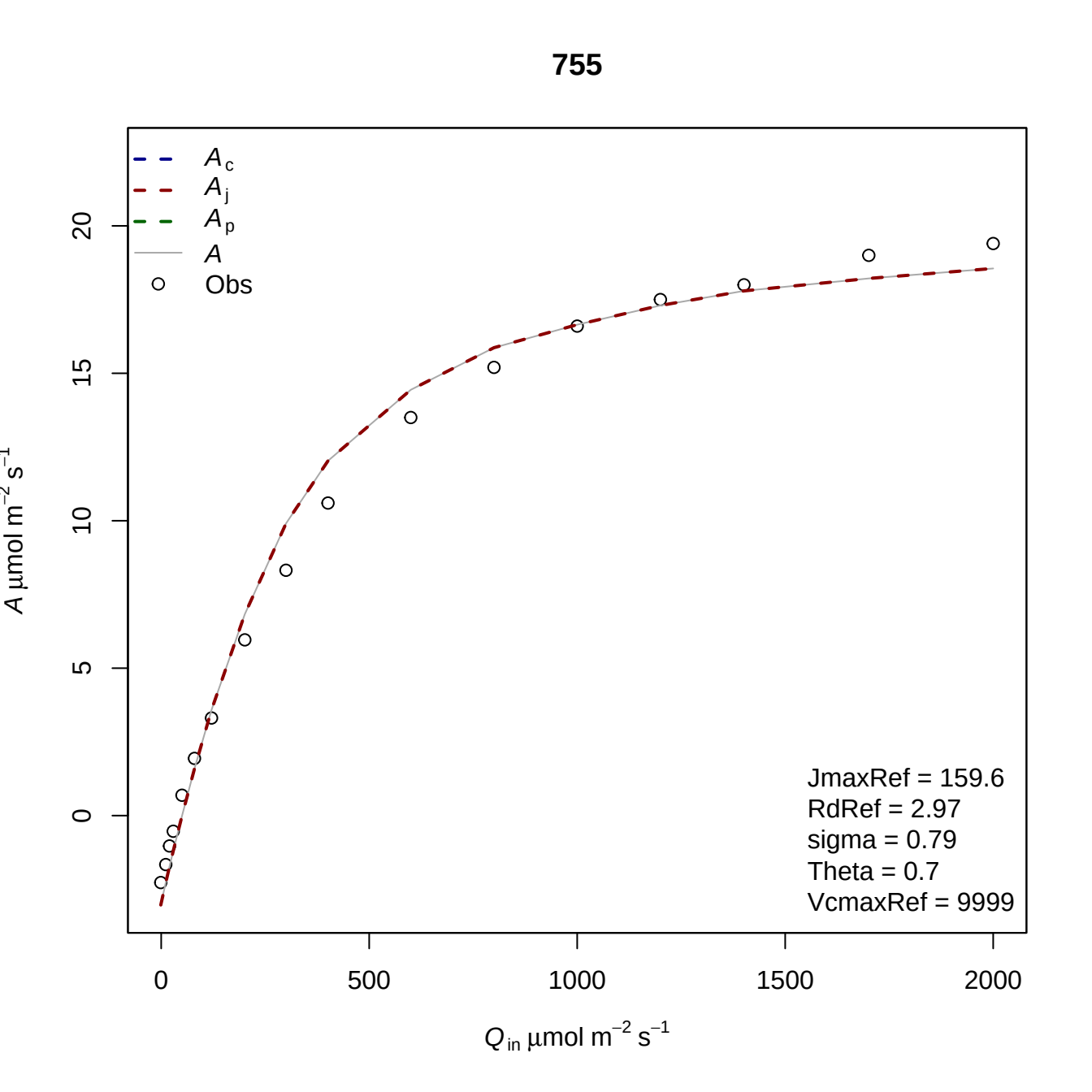

750
A_c
A_j
A_p
A
Obs
JmaxRef = 133.62
RdRef = 2.57
sigma = 0.32
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

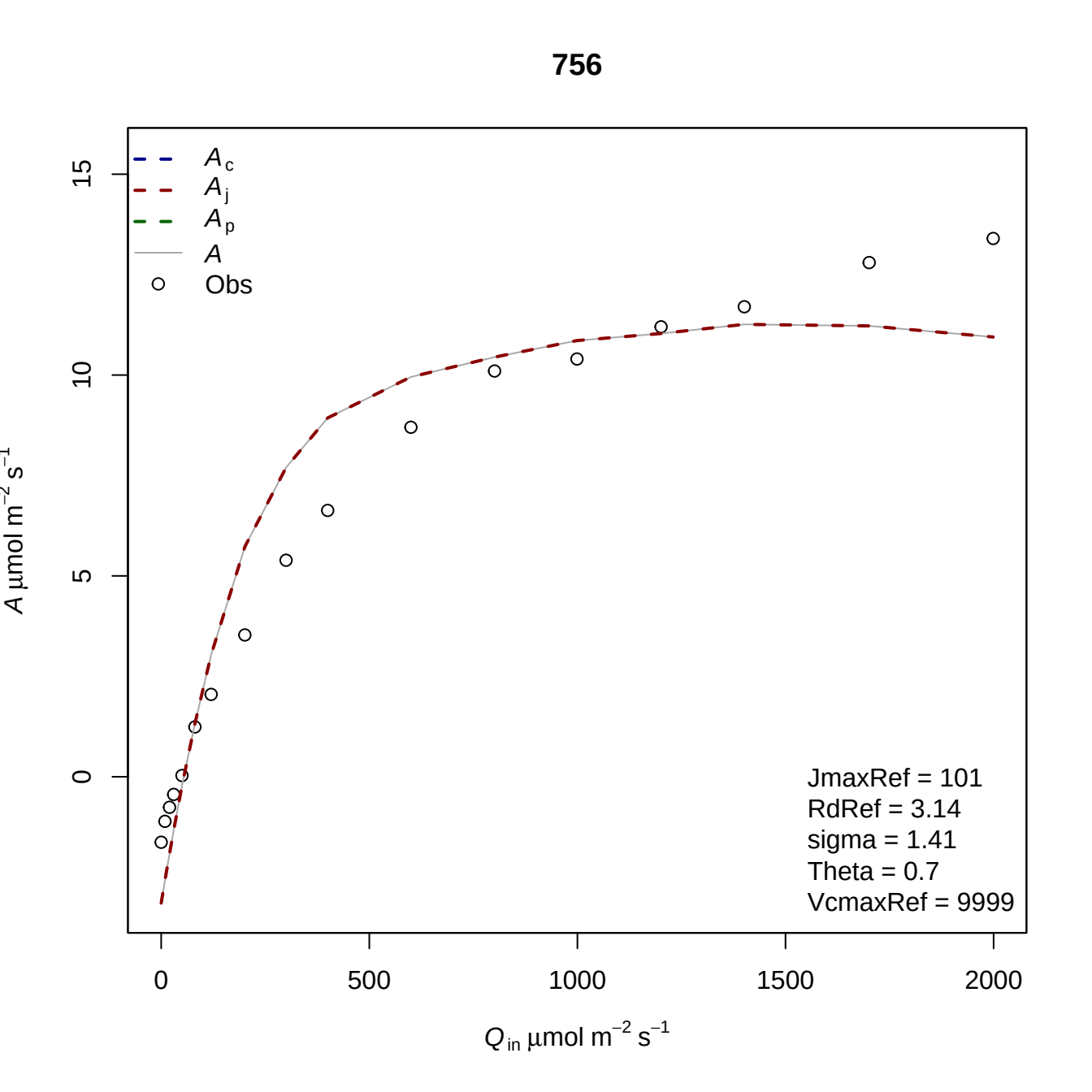

751
A_c
A_j
A_p
A
Obs
JmaxRef = 83.04
RdRef = 2.01
sigma = 0.61
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

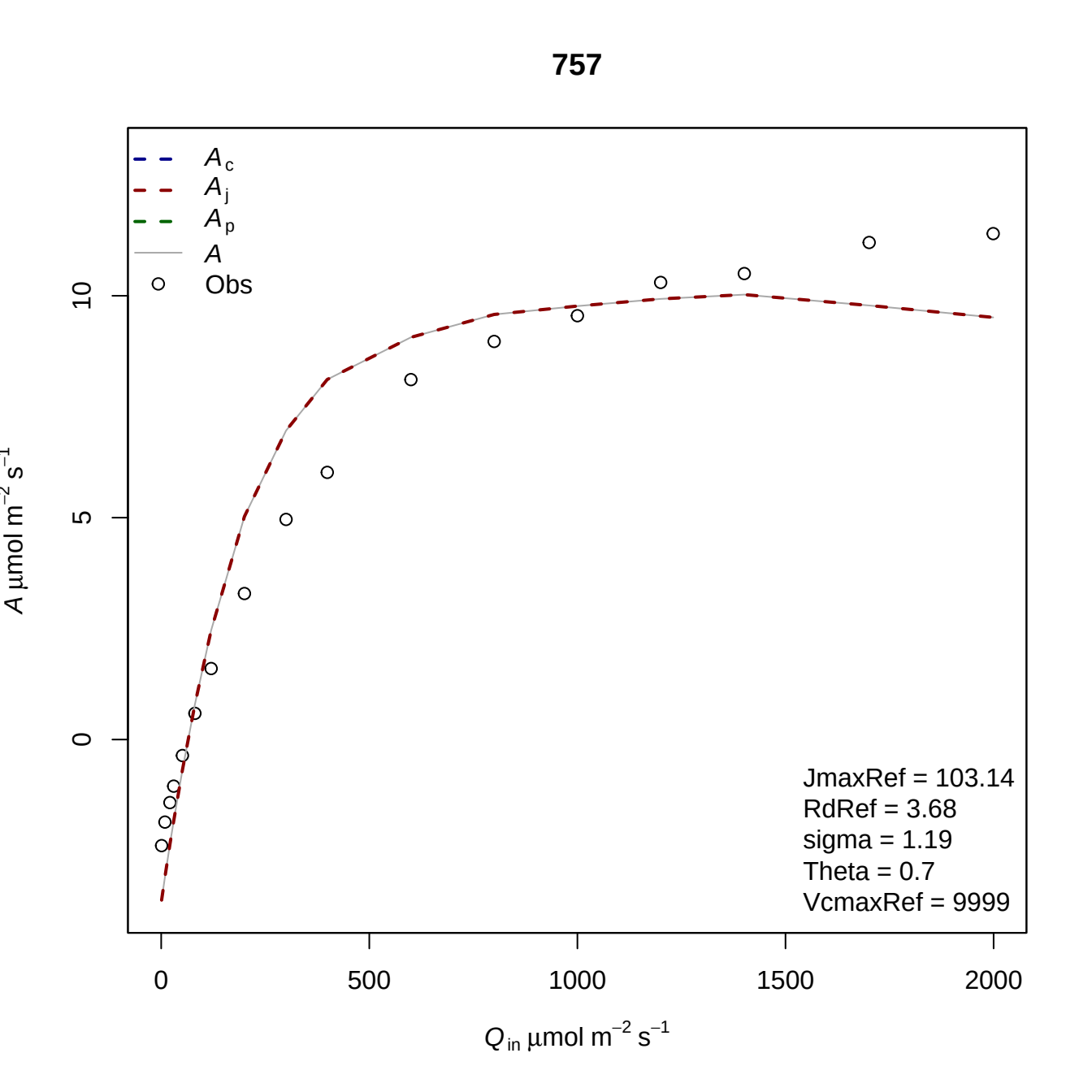

752
A_c
A_j
A_p
A
Obs
JmaxRef = 145.79
RdRef = 3.22
sigma = 0.92
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

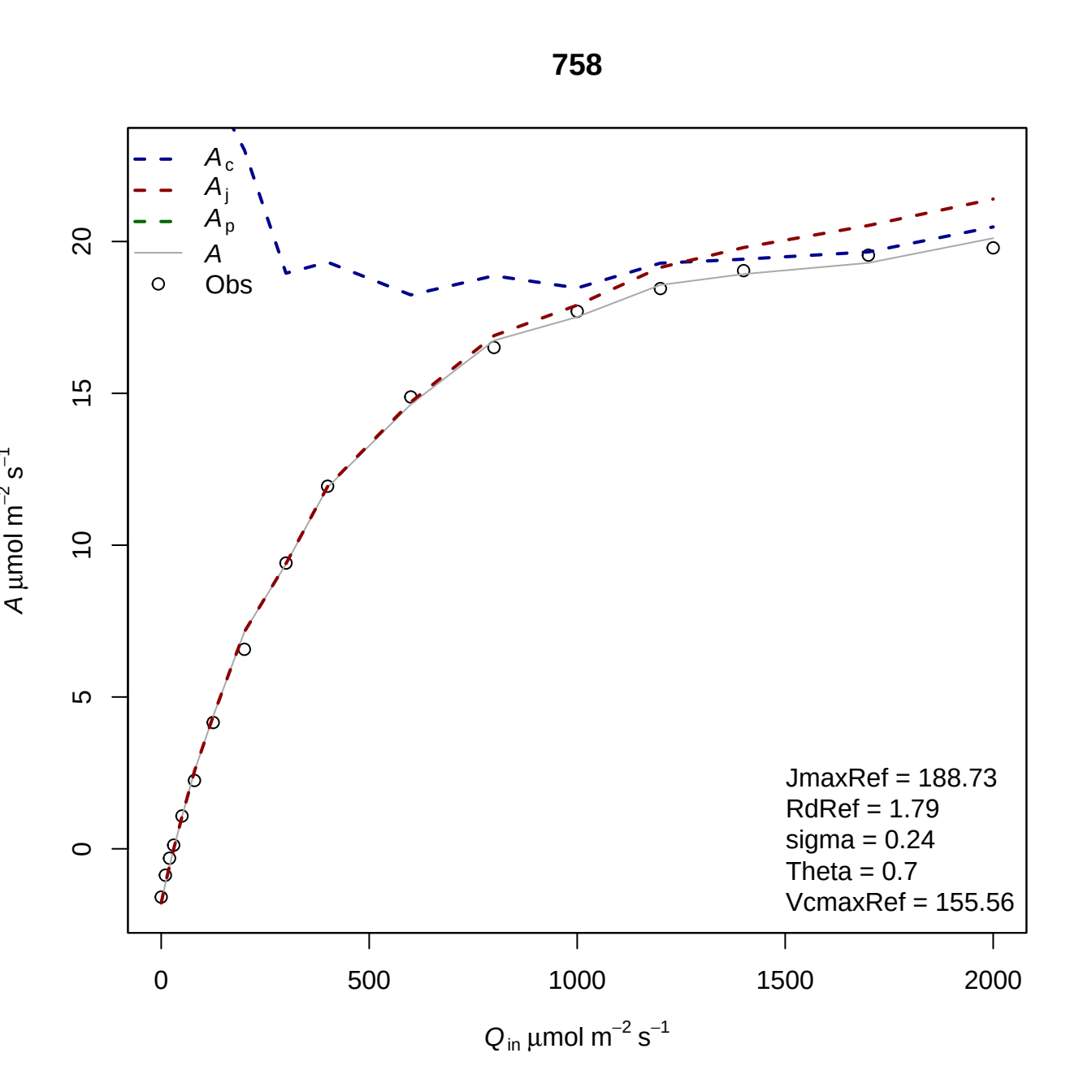

753
A_c
A_j
A_p
A
Obs
JmaxRef = 122.63
RdRef = 2.43
sigma = 0.74
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

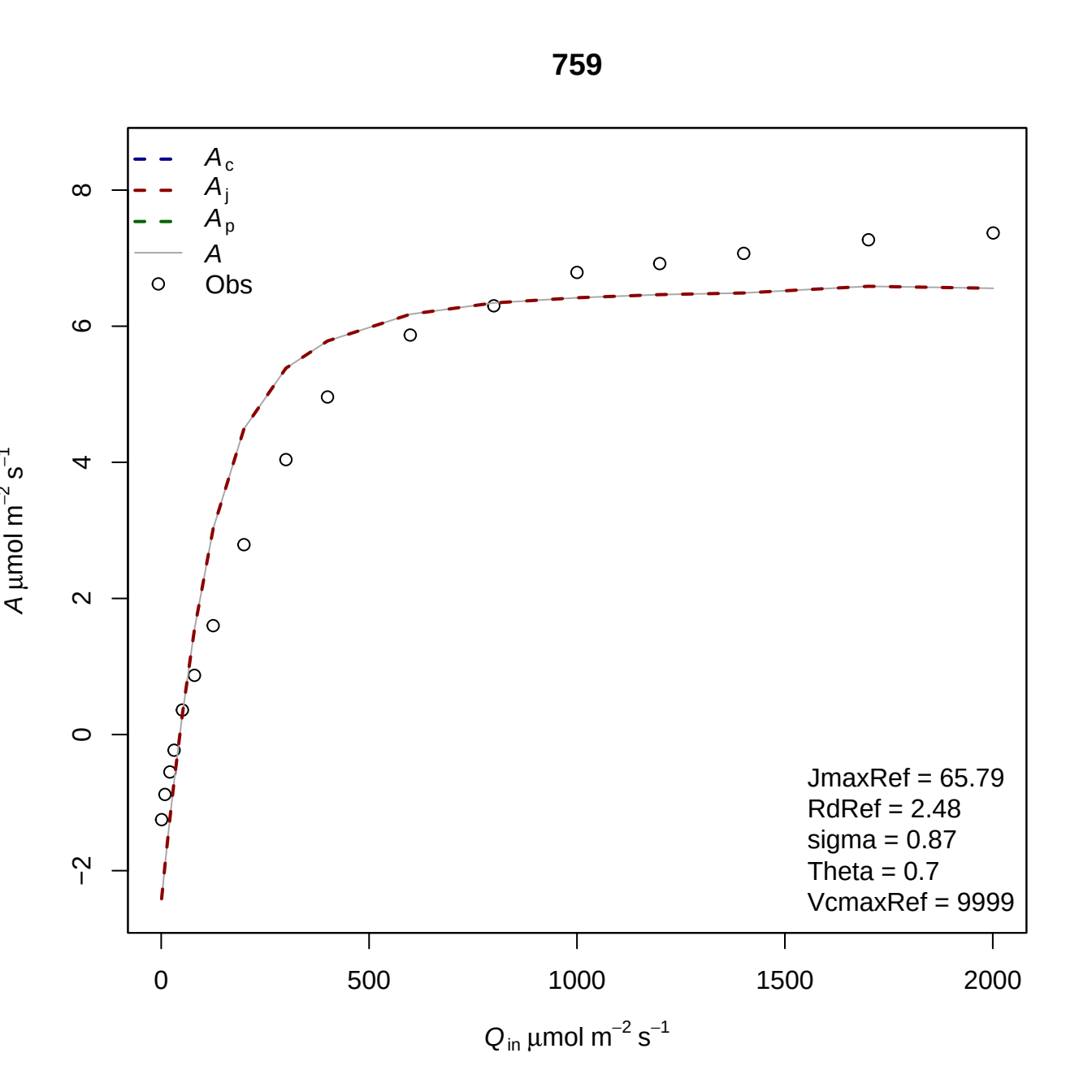

754
A_c
A_j
A_p
A
Obs
JmaxRef = 124.28
RdRef = 4.24
sigma = 1.34
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

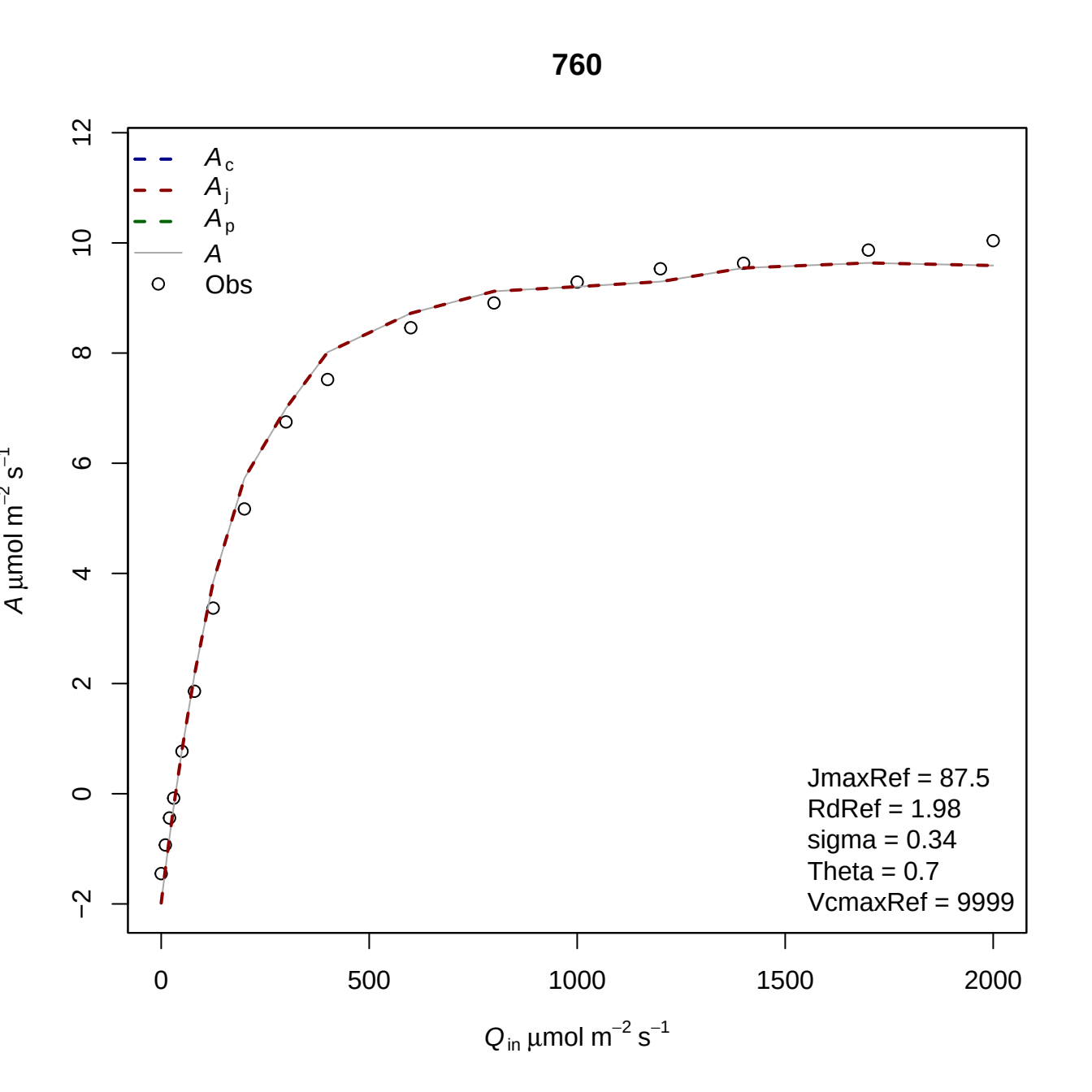

755
A_c
A_j
A_p
A
Obs
JmaxRef = 159.6
RdRef = 2.97
sigma = 0.79
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

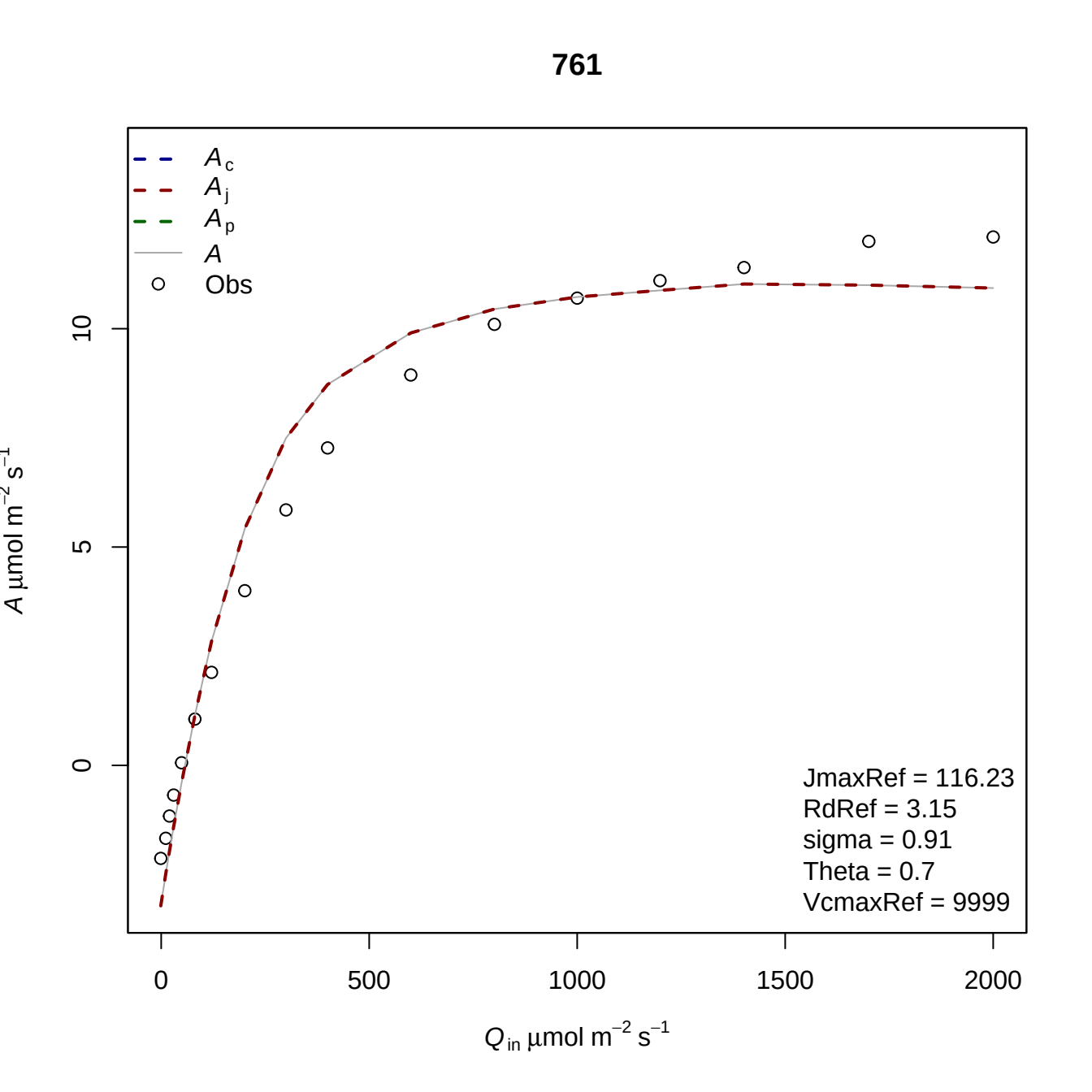

756
A_c
A_j
A_p
A
Obs
JmaxRef = 101
RdRef = 3.14
sigma = 1.41
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

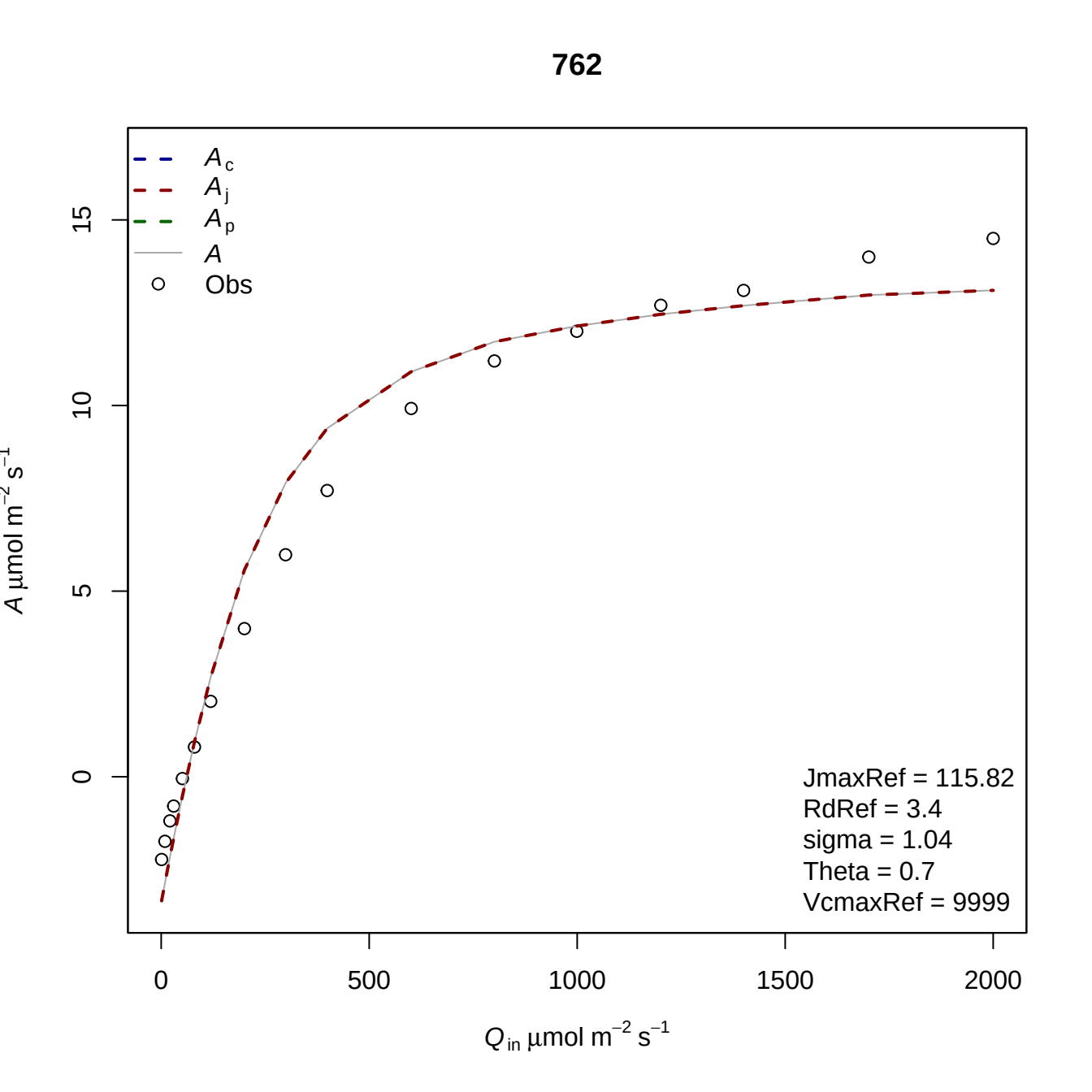

757
A_c
A_j
A_p
A
Obs
JmaxRef = 103.14
RdRef = 3.68
sigma = 1.19
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

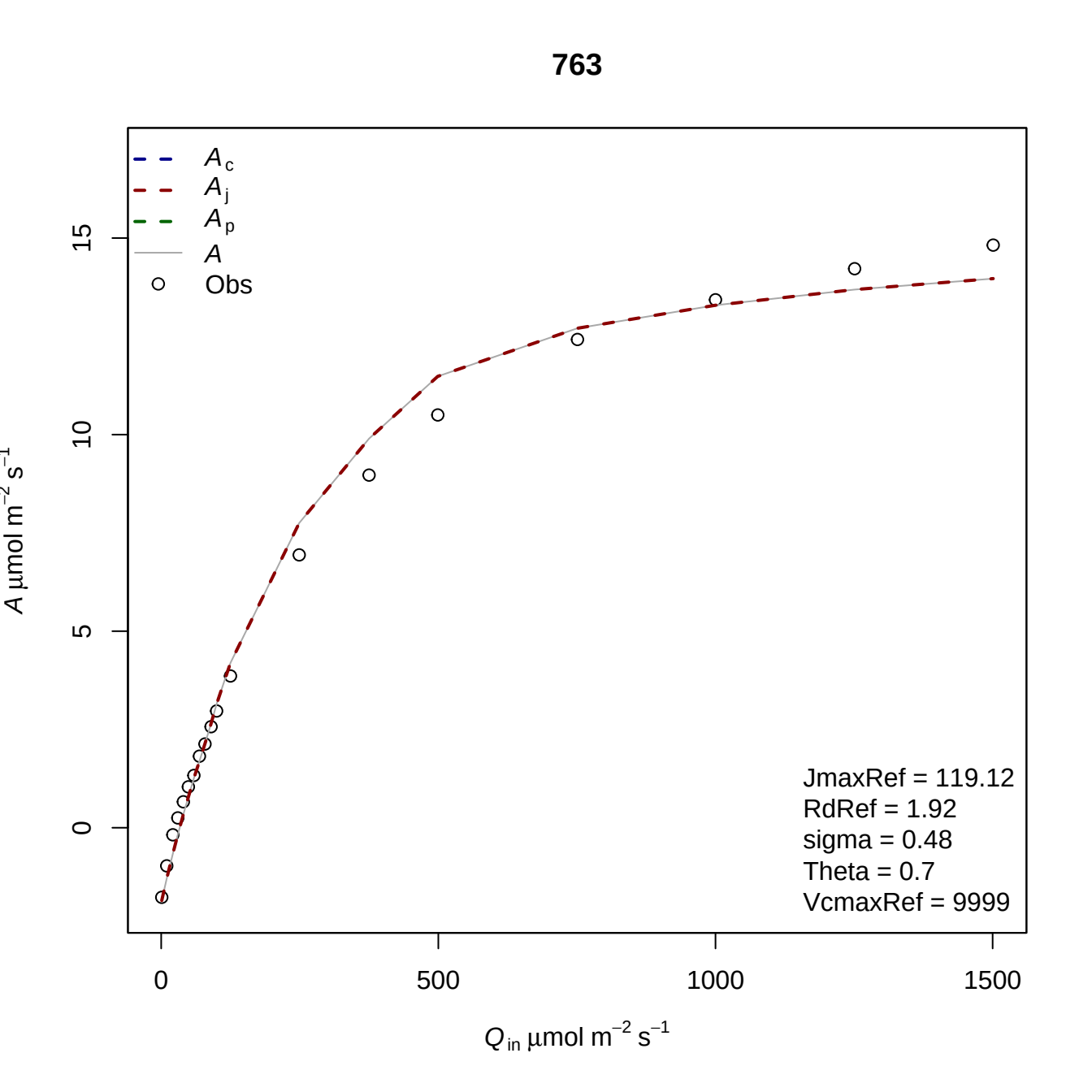

758
A_c
A_j
A_p
A
Obs
JmaxRef = 188.73
RdRef = 1.79
sigma = 0.24
Theta = 0.7
VcmaxRef = 155.56
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

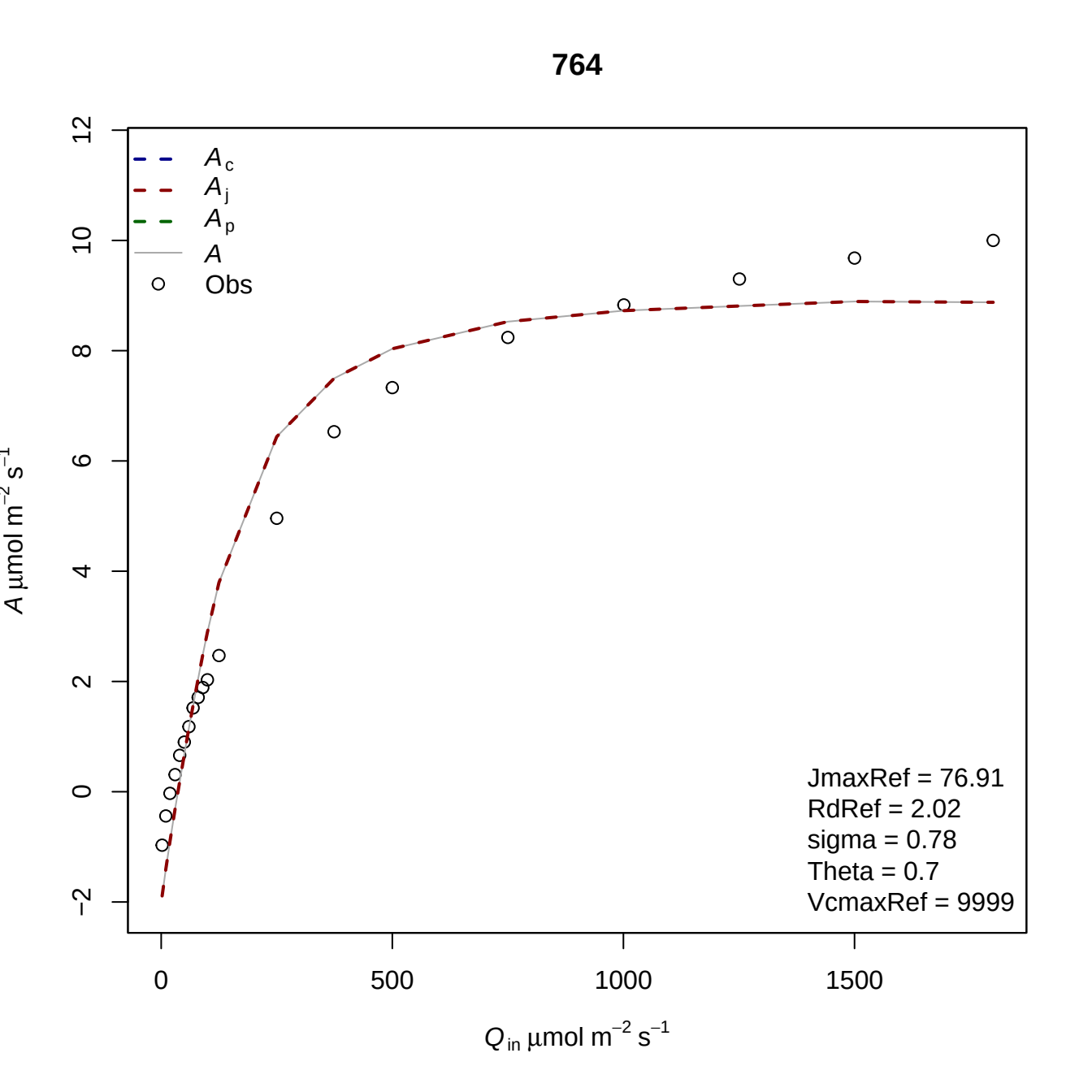

759
A_c
A_j
A_p
A
Obs
JmaxRef = 65.79
RdRef = 2.48
sigma = 0.87
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

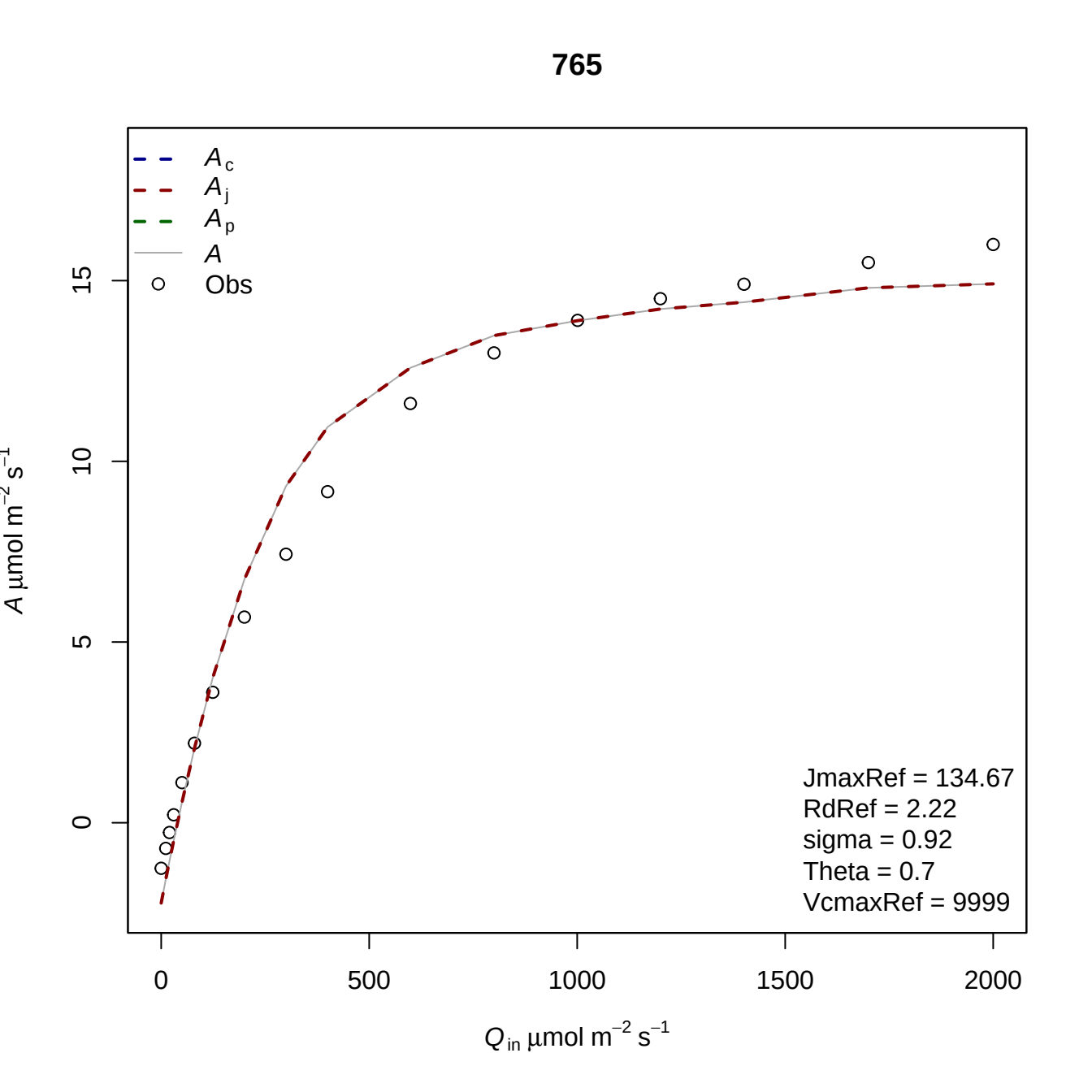

760
A_c
A_j
A_p
A
Obs
JmaxRef = 87.5
RdRef = 1.98
sigma = 0.34
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

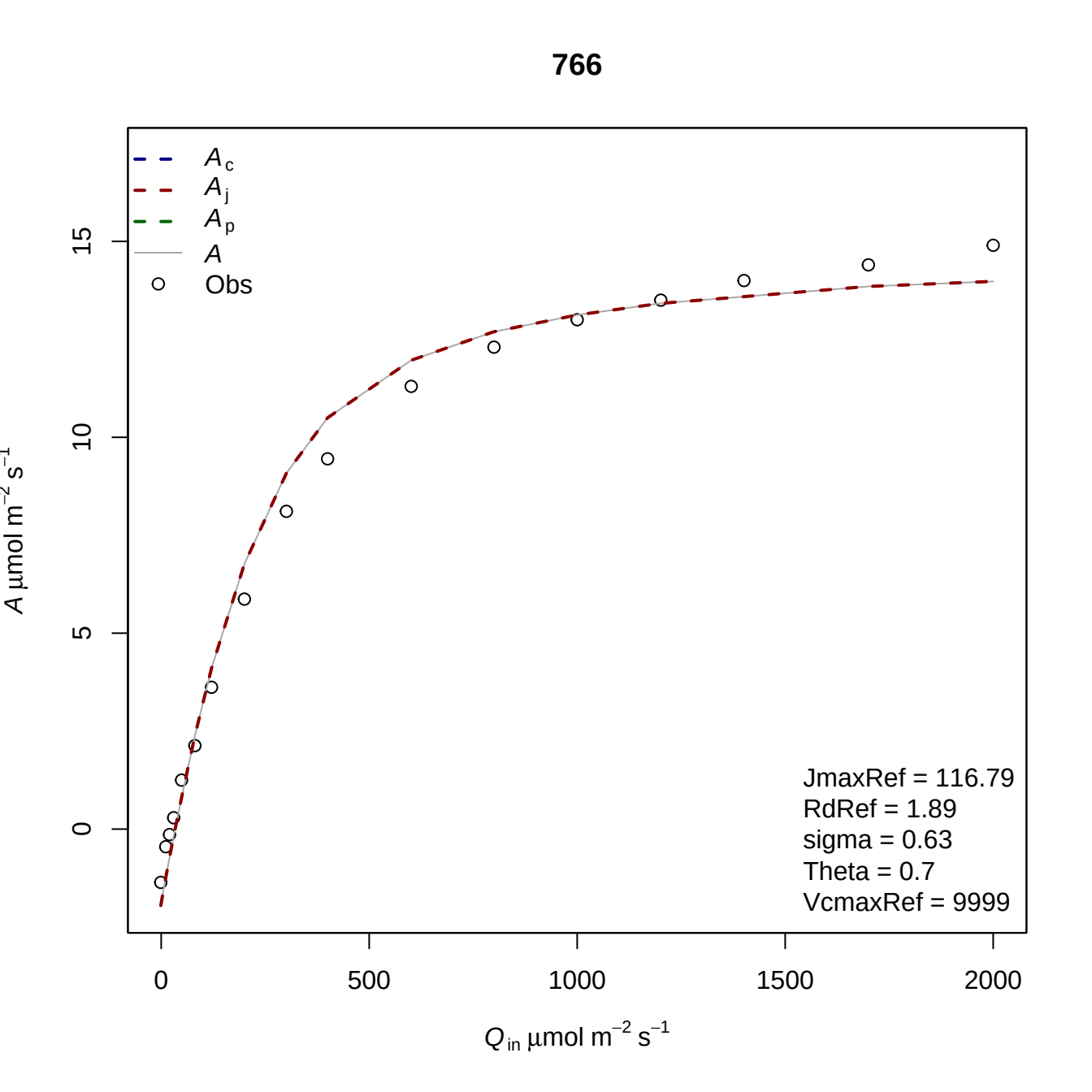

761
A_c
A_j
A_p
A
Obs
JmaxRef = 116.23
RdRef = 3.15
sigma = 0.91
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

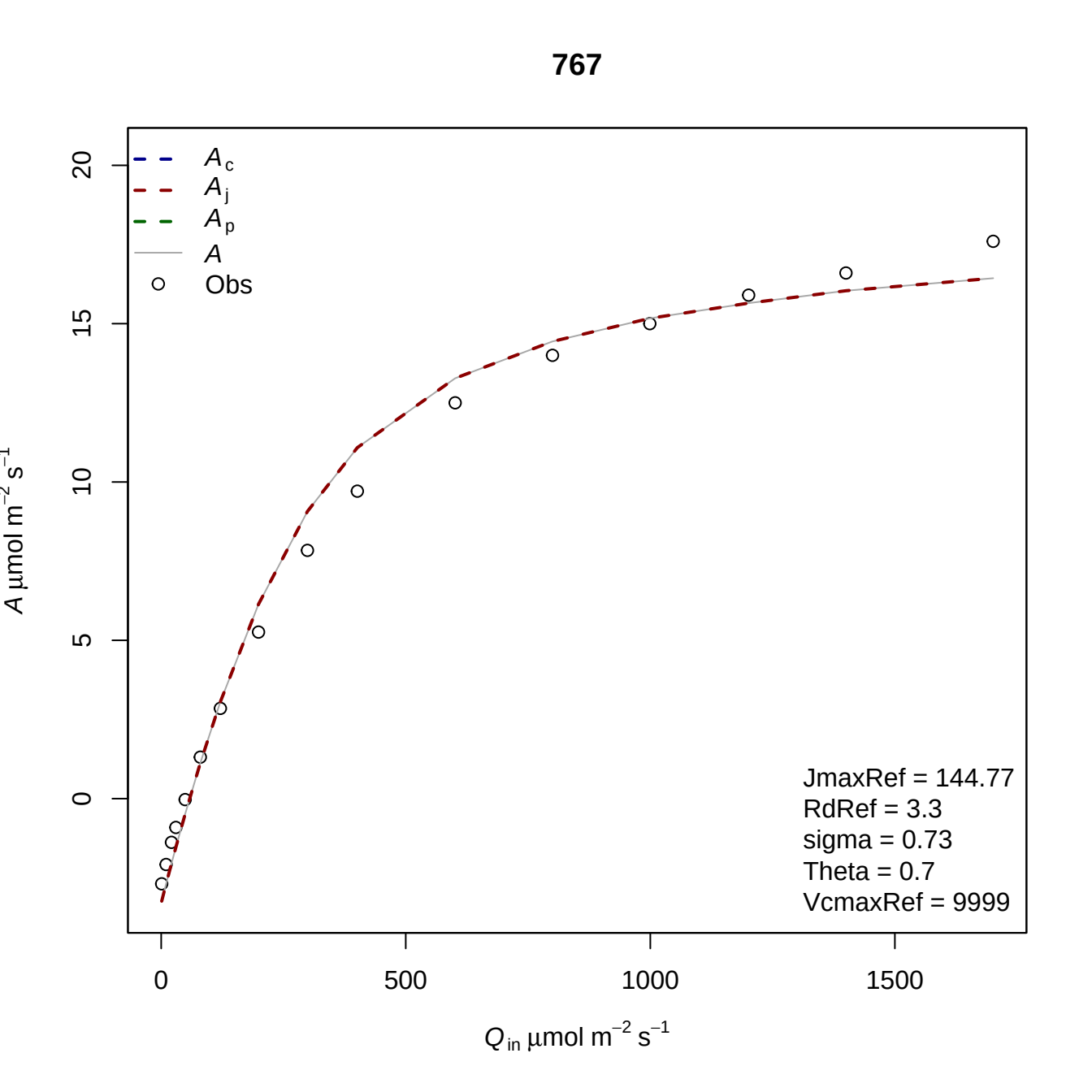

762
A_c
A_j
A_p
A
Obs
JmaxRef = 115.82
RdRef = 3.4
sigma = 1.04
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

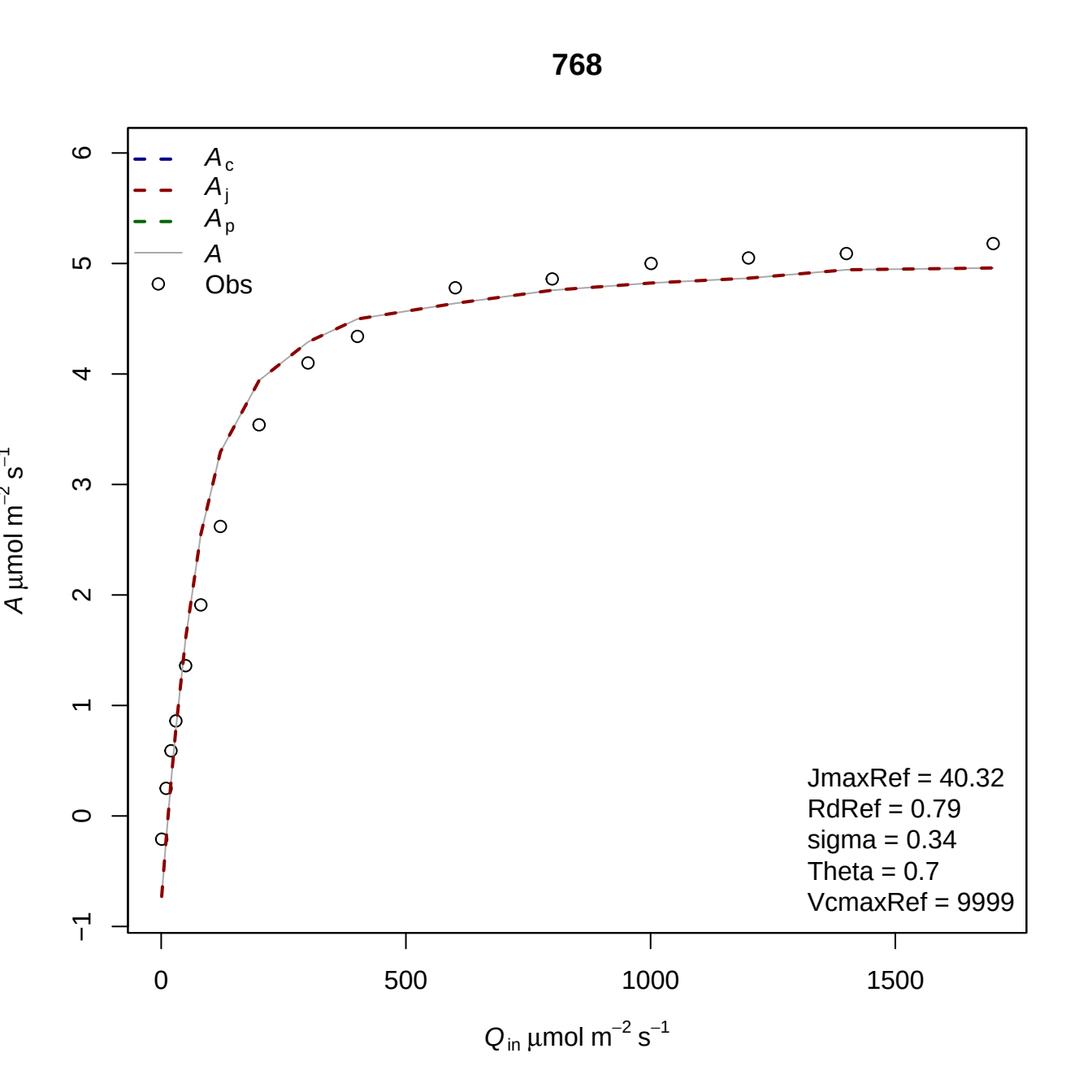

763
A_c
A_j
A_p
A
Obs
JmaxRef = 119.12
RdRef = 1.92
sigma = 0.48
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

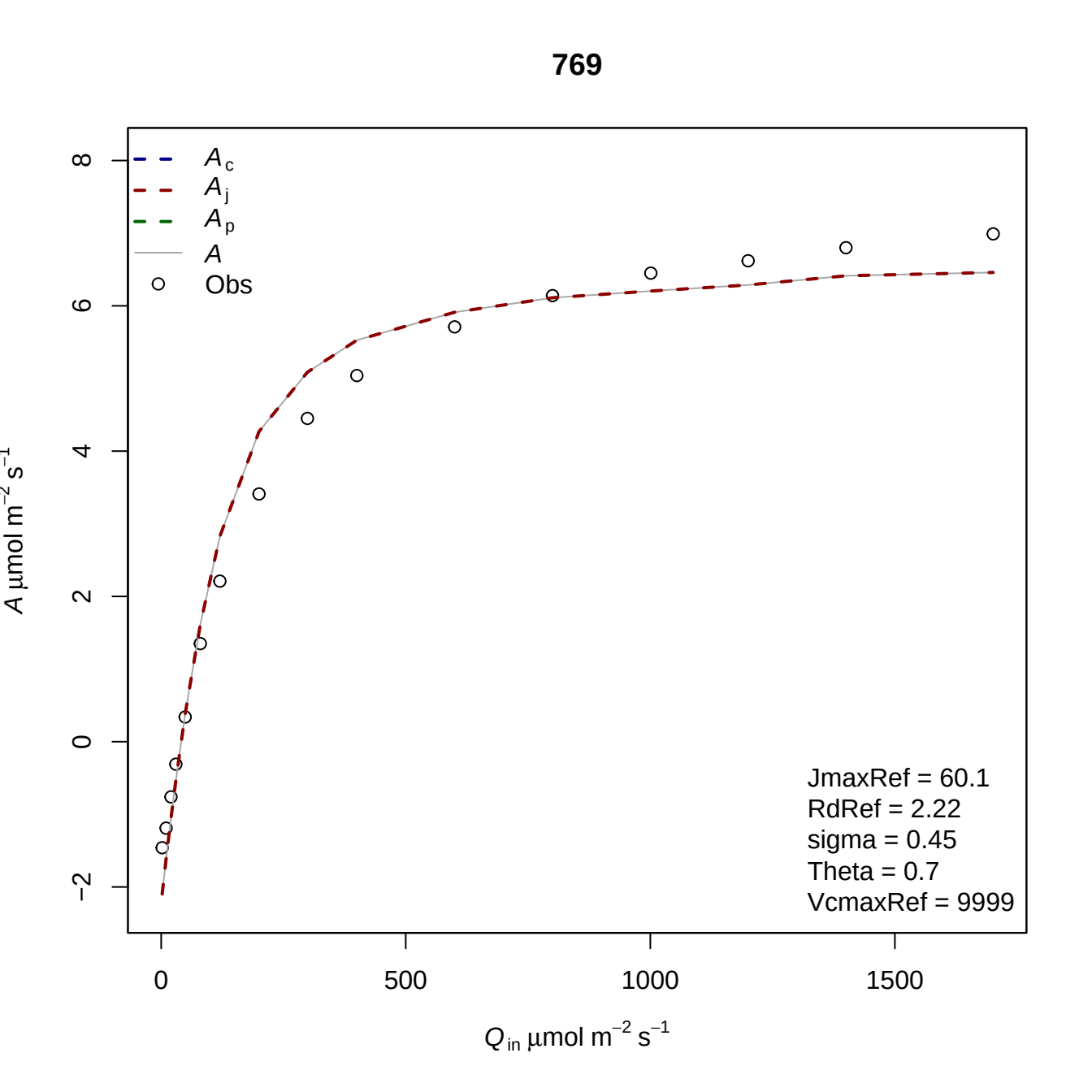

764
A_c
A_j
A_p
A
Obs
JmaxRef = 76.91
RdRef = 2.02
sigma = 0.78
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

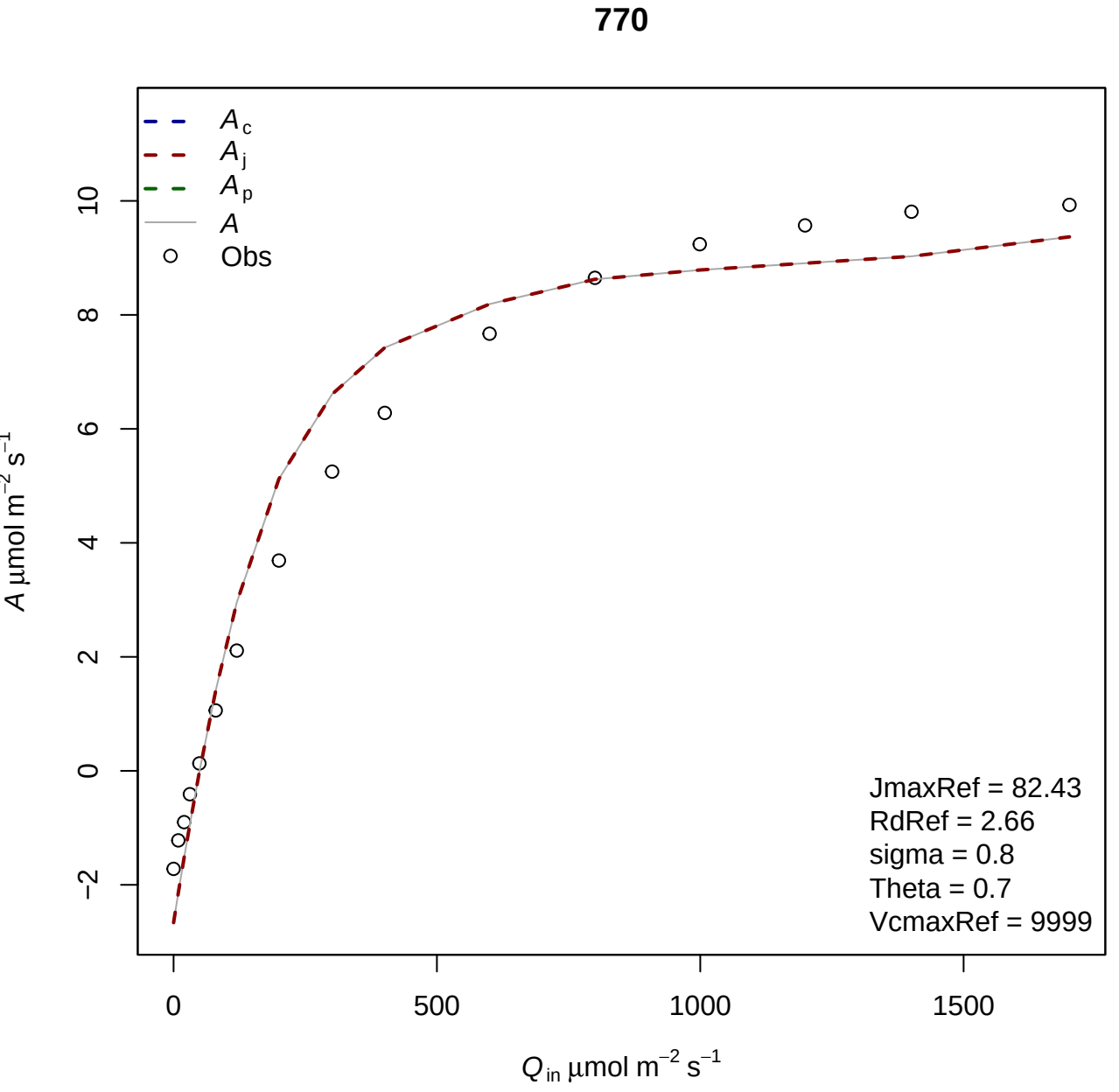

765
A_c
A_j
A_p
A
Obs
JmaxRef = 134.67
RdRef = 2.22
sigma = 0.92
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

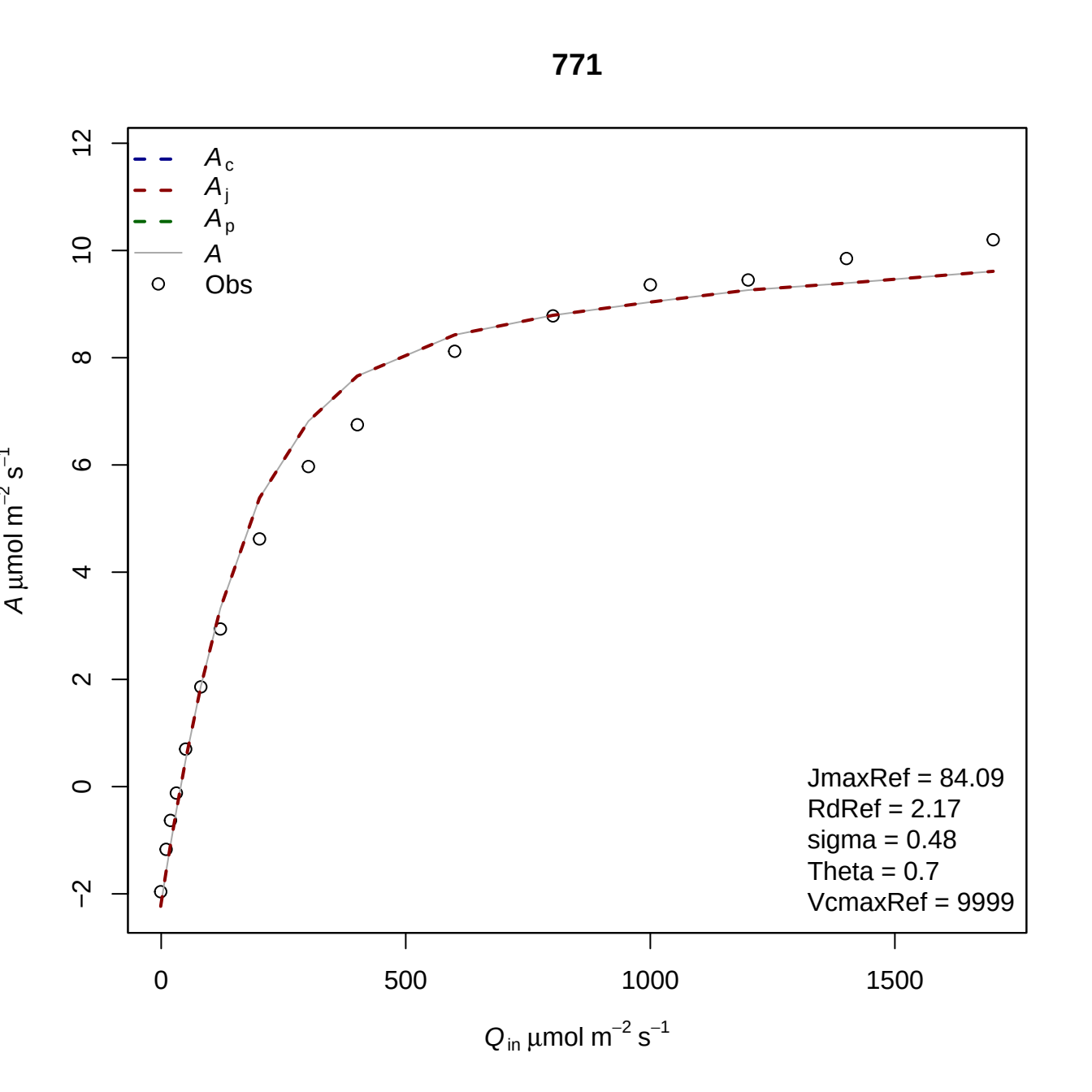

766
A_c
A_j
A_p
A
Obs
JmaxRef = 116.79
RdRef = 1.89
sigma = 0.63
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

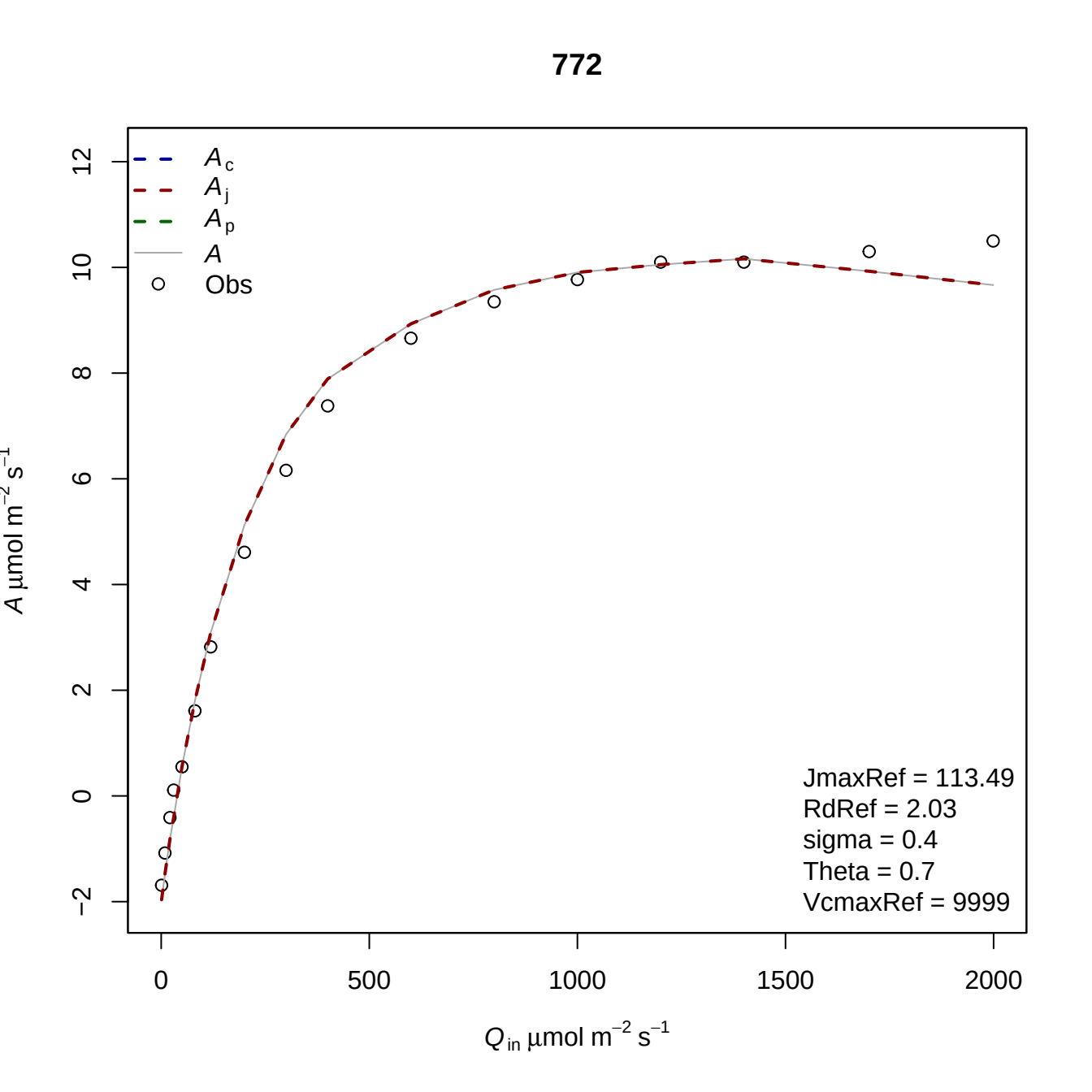

767
A_c
A_j
A_p
A
Obs
JmaxRef = 144.77
RdRef = 3.3
sigma = 0.73
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

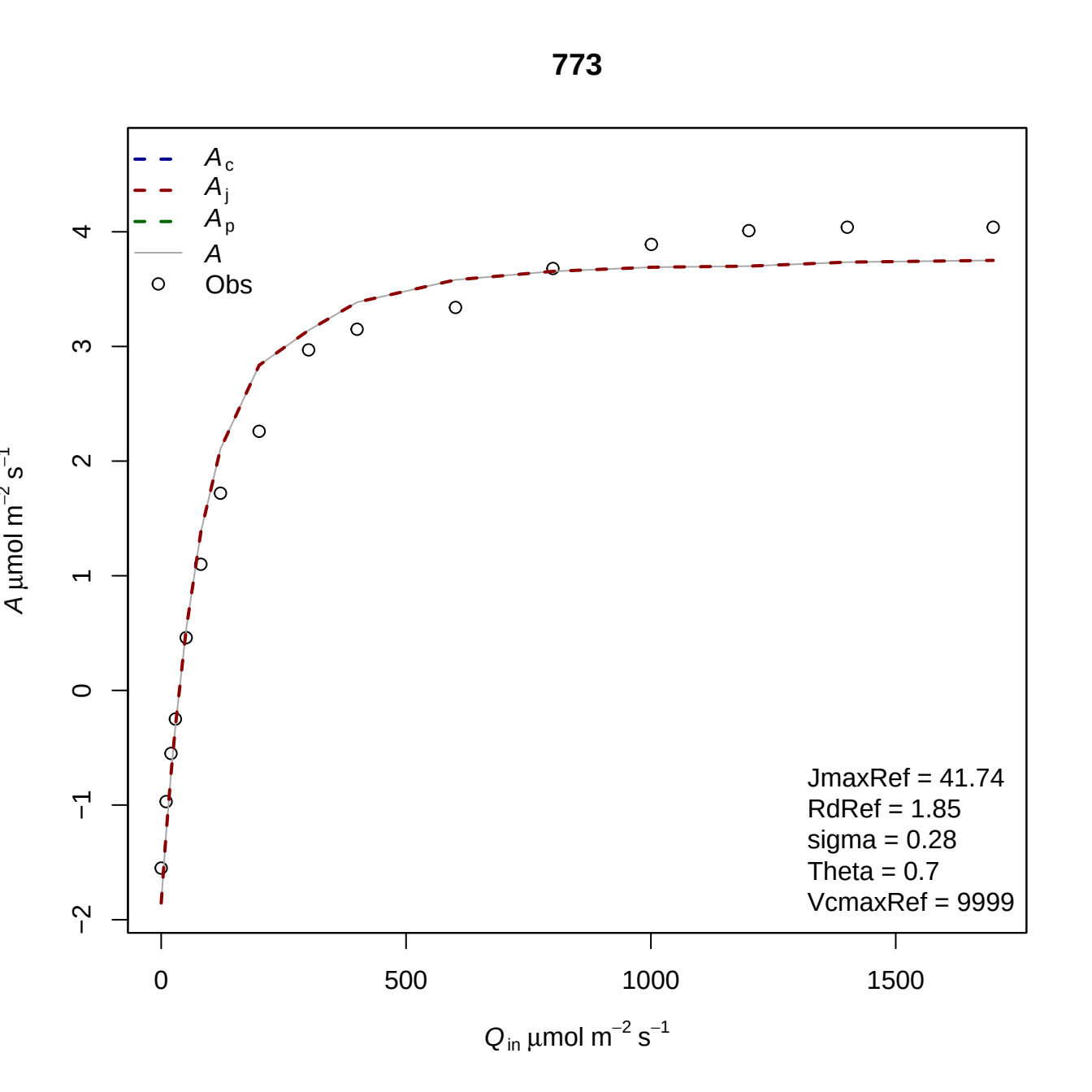

768
A_c
A_j
A_p
A
Obs
JmaxRef = 40.32
RdRef = 0.79
sigma = 0.34
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

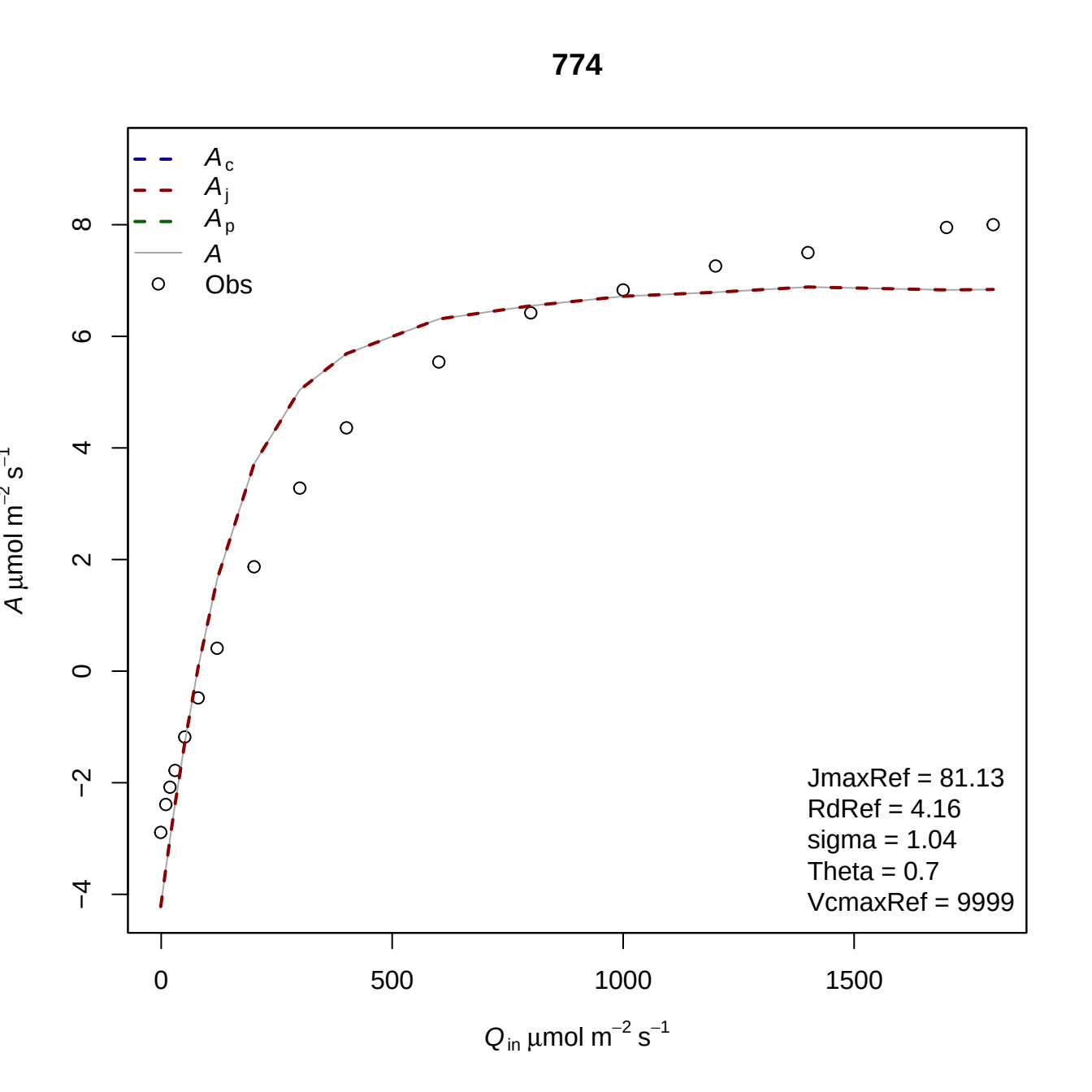

769
A_c
A_j
A_p
A
Obs
JmaxRef = 60.1
RdRef = 2.22
sigma = 0.45
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

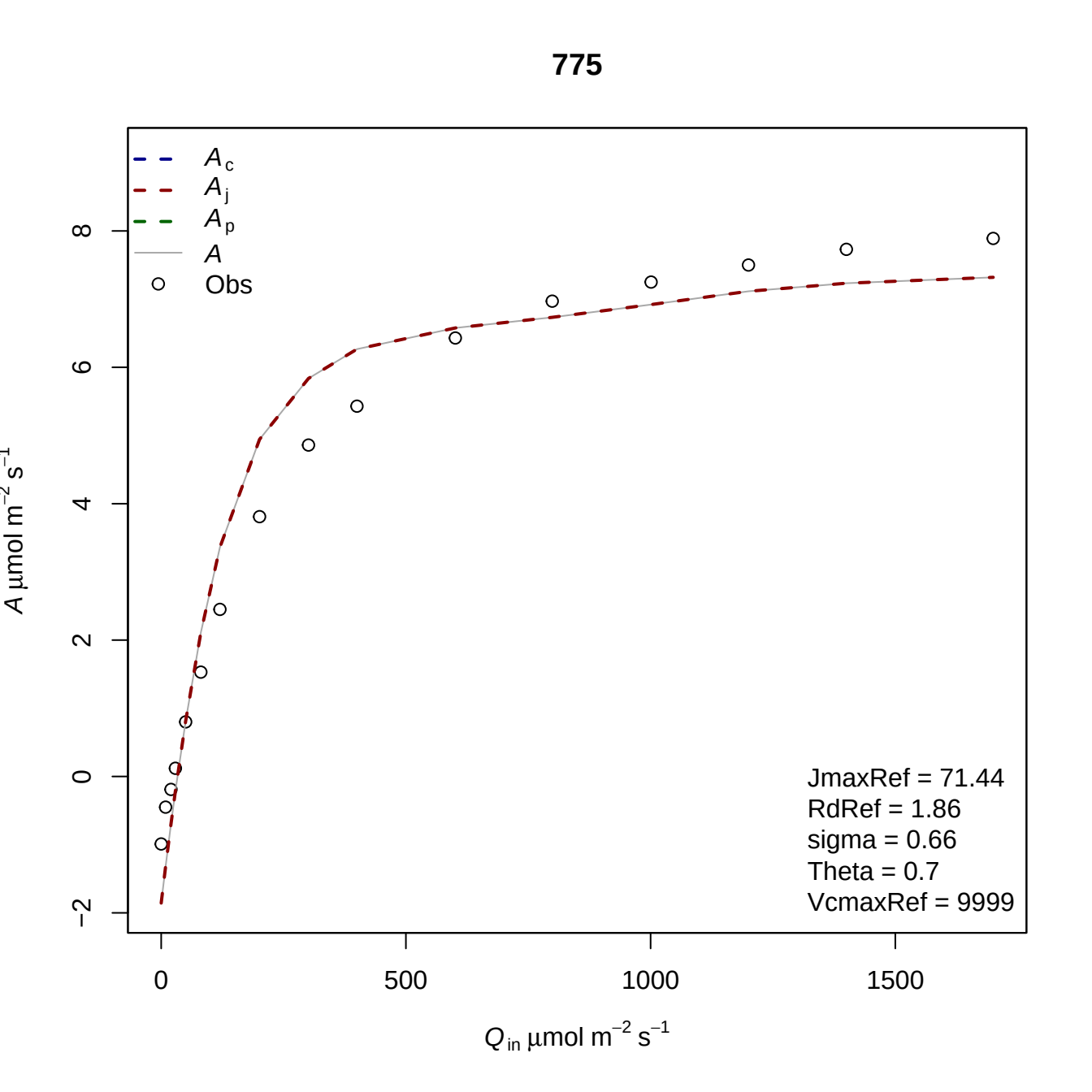

770
A_c
A_j
A_p
A
Obs
JmaxRef = 82.43
RdRef = 2.66
sigma = 0.8
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

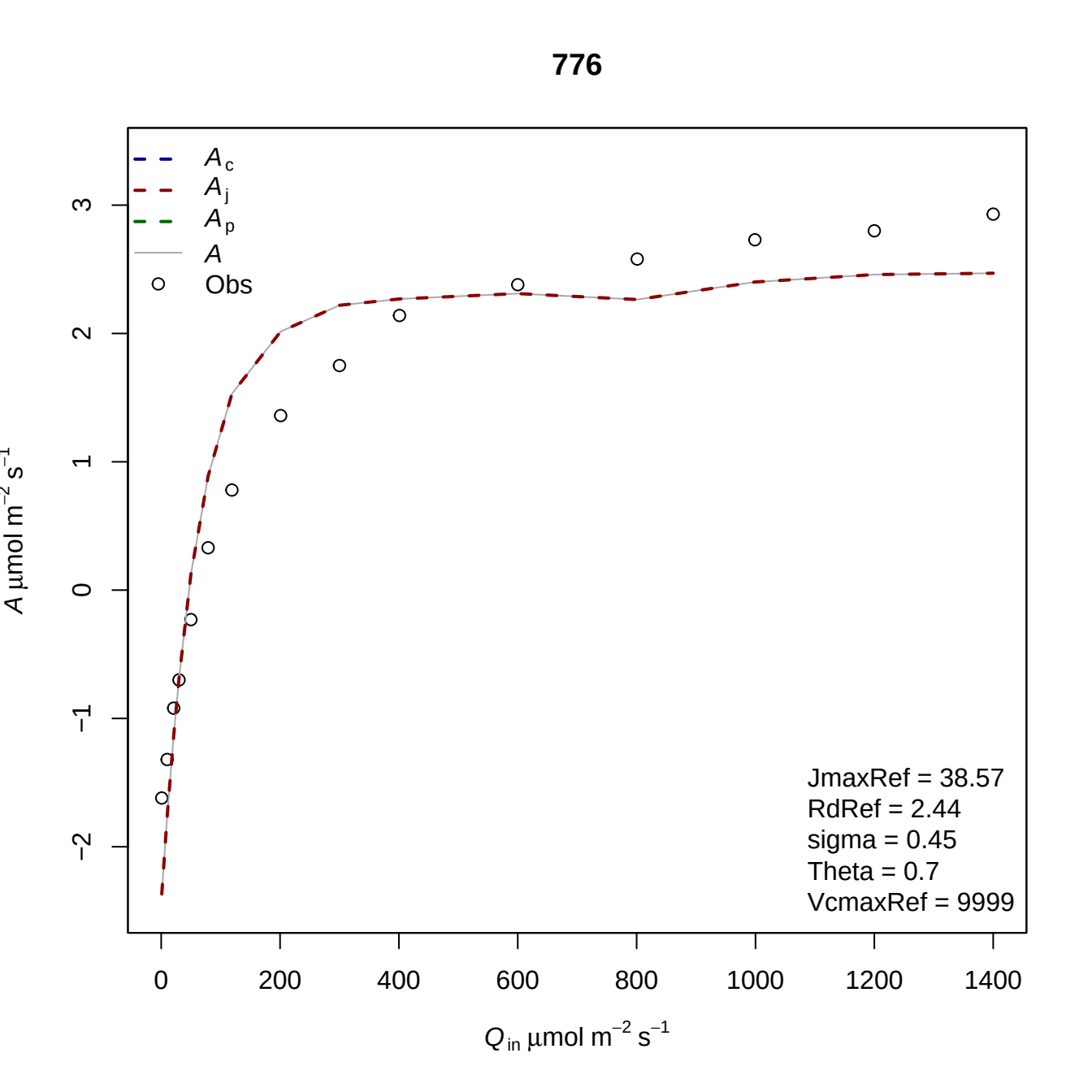

771
A_c
A_j
A_p
A
Obs
JmaxRef = 84.09
RdRef = 2.17
sigma = 0.48
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

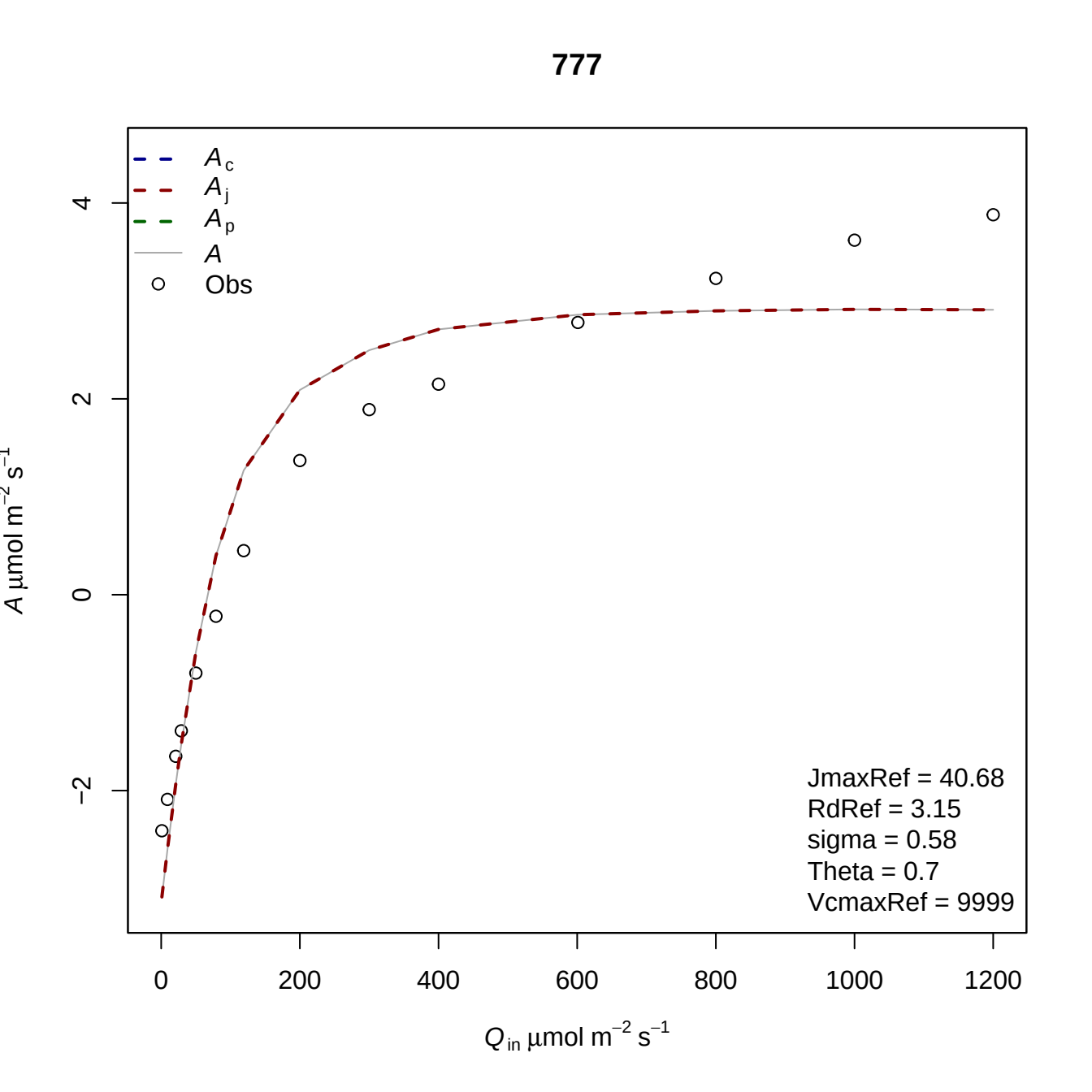

772
A_c
A_j
A_p
A
Obs
JmaxRef = 113.49
RdRef = 2.03
sigma = 0.4
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

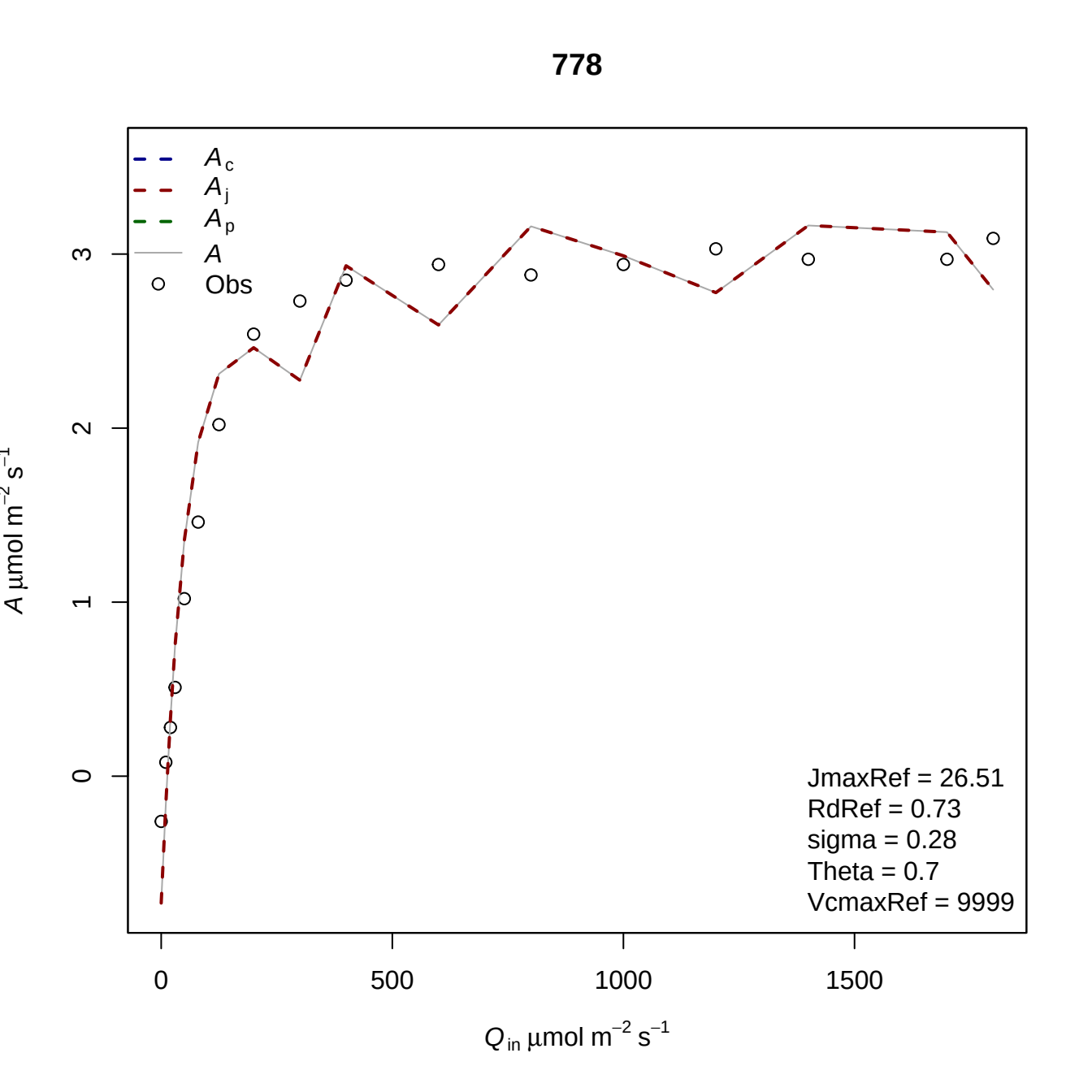

773
A_c
A_j
A_p
A
Obs
JmaxRef = 41.74
RdRef = 1.85
sigma = 0.28
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
4
0
500
1000
1500

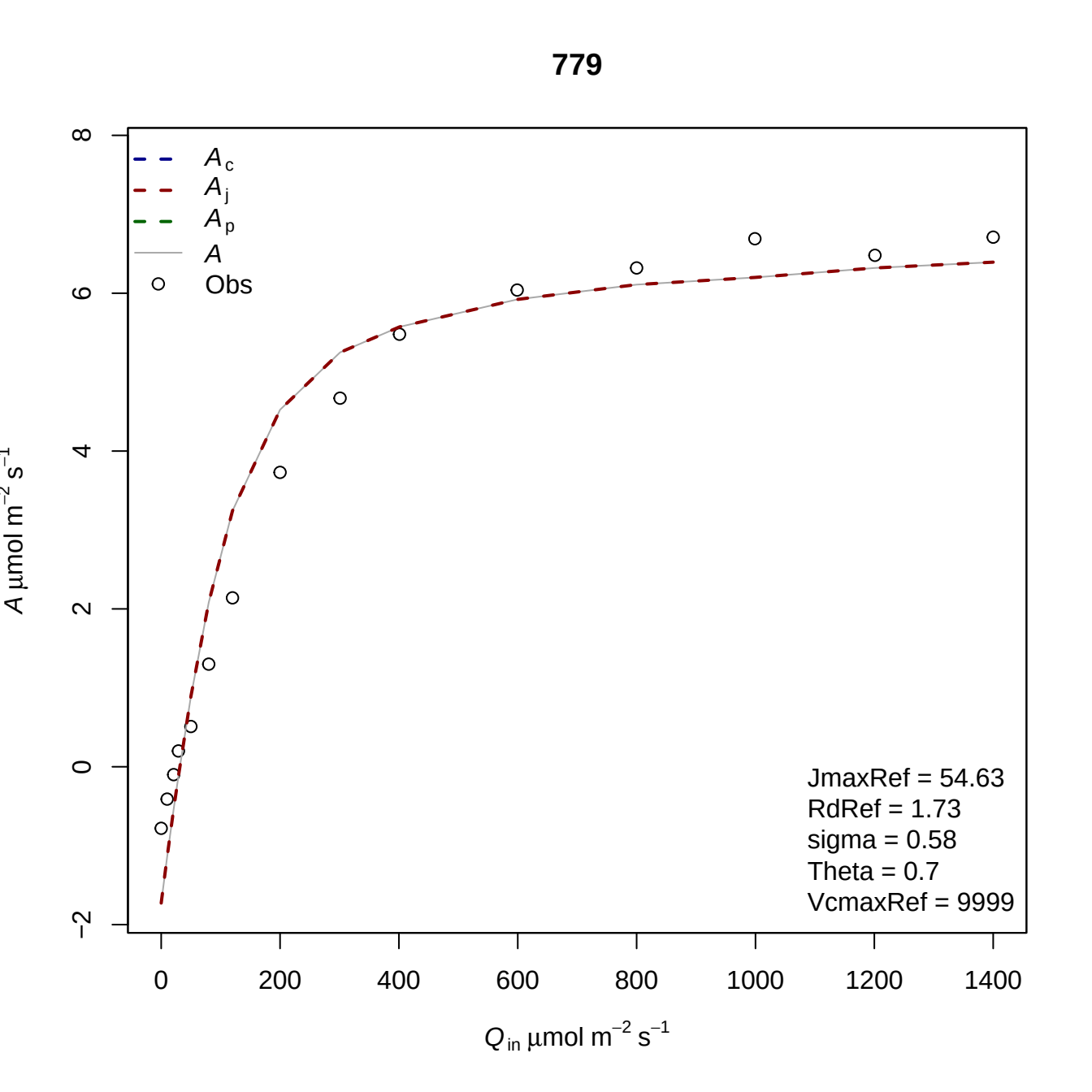

774
A_c
A_j
A_p
A
Obs
JmaxRef = 81.13
RdRef = 4.16
sigma = 1.04
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

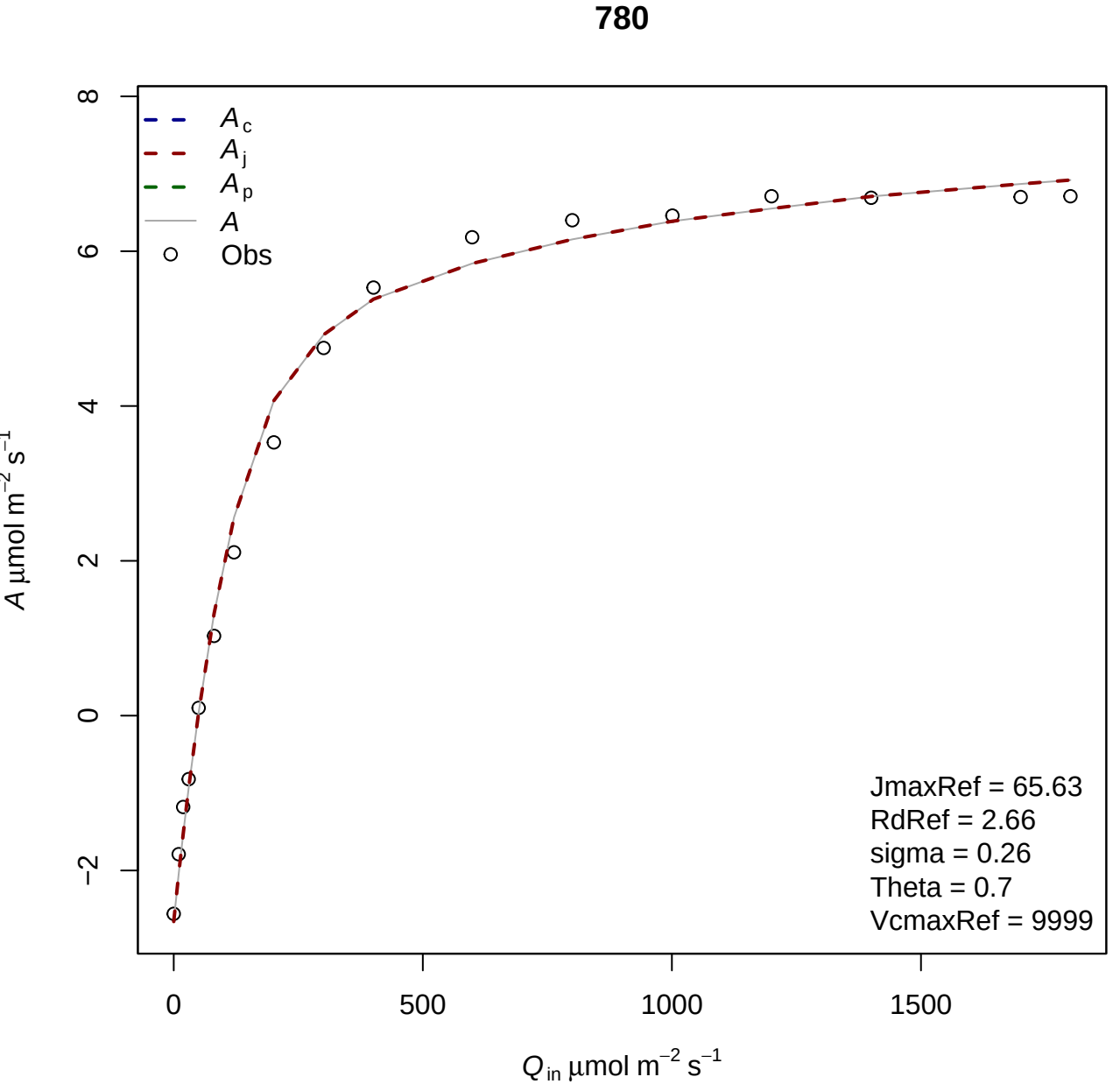

775
A_c
A_j
A_p
A
Obs
JmaxRef = 71.44
RdRef = 1.86
sigma = 0.66
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

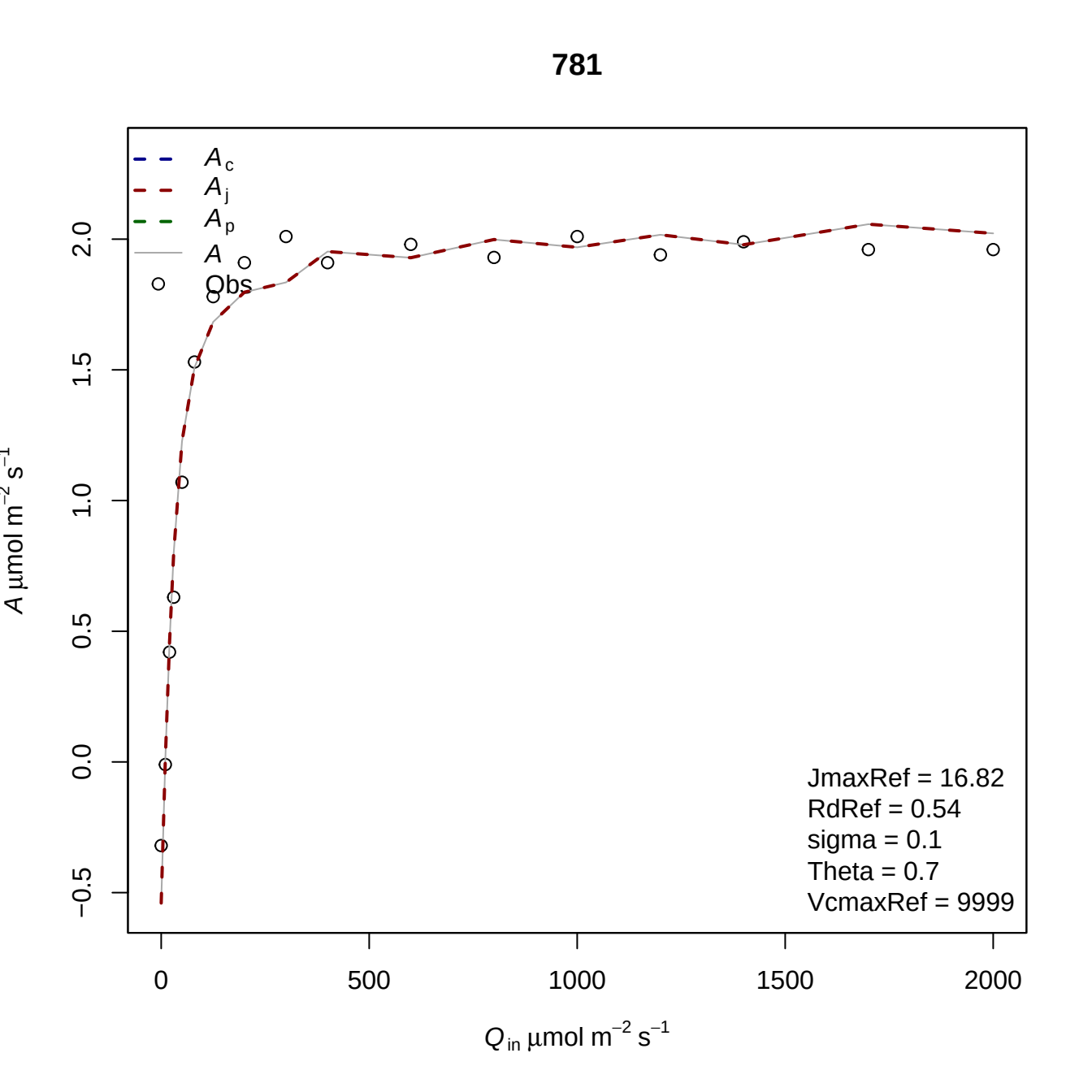

776
A_c
A_j
A_p
A
Obs
JmaxRef = 38.57
RdRef = 2.44
sigma = 0.45
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
0
200
400
600
800
1000
1200
1400

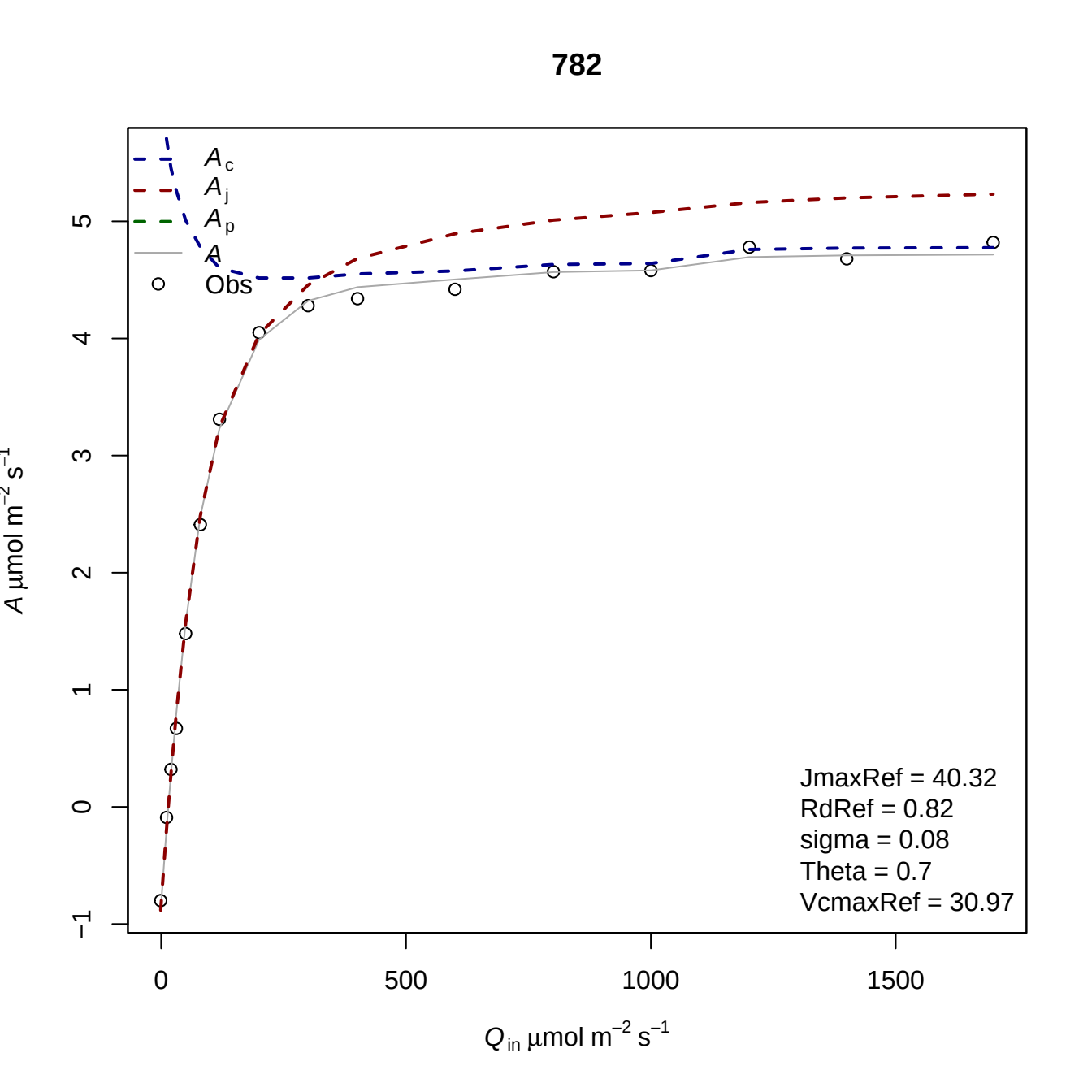

777
A_c
A_j
A_p
A
Obs
JmaxRef = 40.68
RdRef = 3.15
sigma = 0.58
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
200
400
600
800
1000
1200

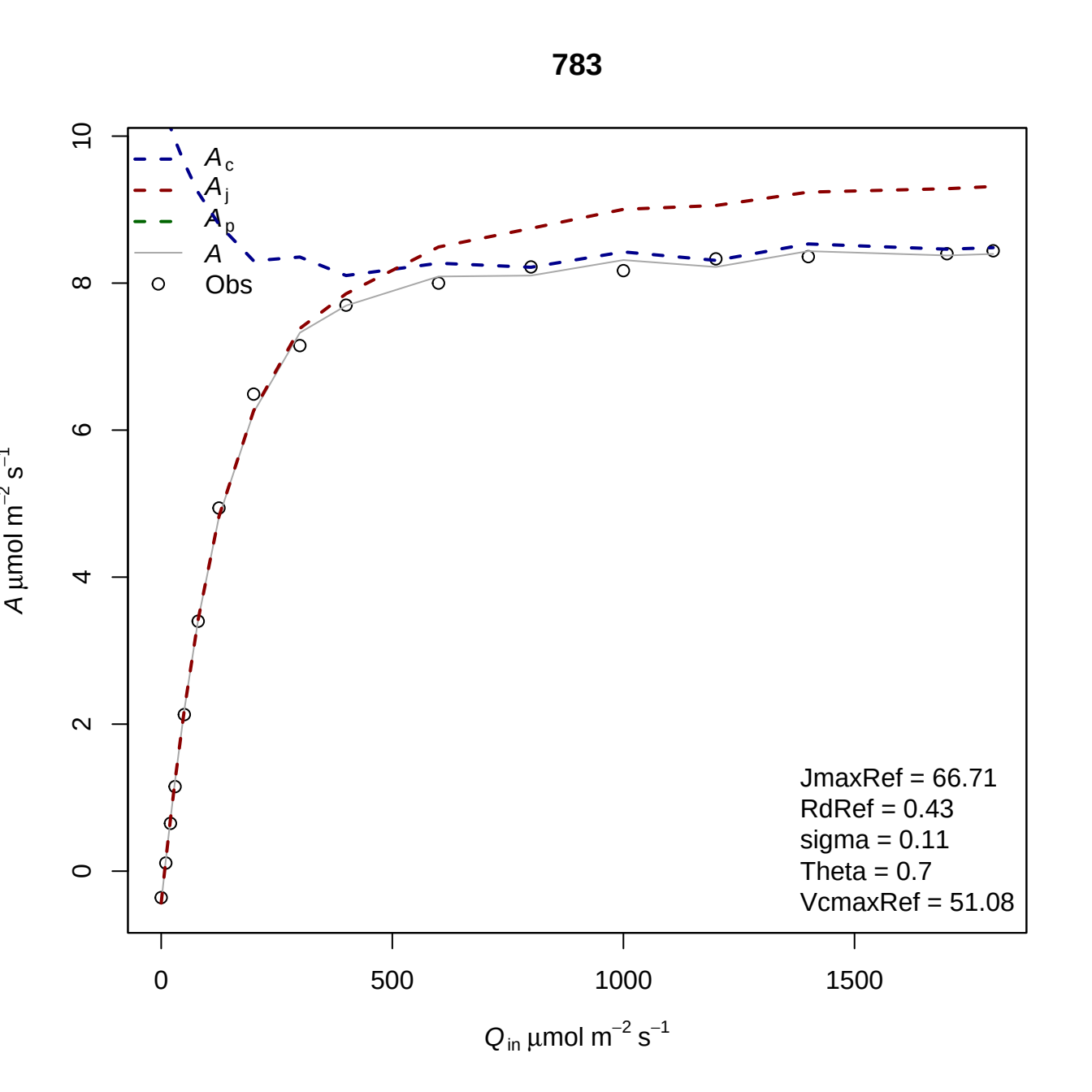

778
A_c
A_j
A_p
A
Obs
JmaxRef = 26.51
RdRef = 0.73
sigma = 0.28
Theta = 0.7
VcmaxRef = 9999
0
1
2
3
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

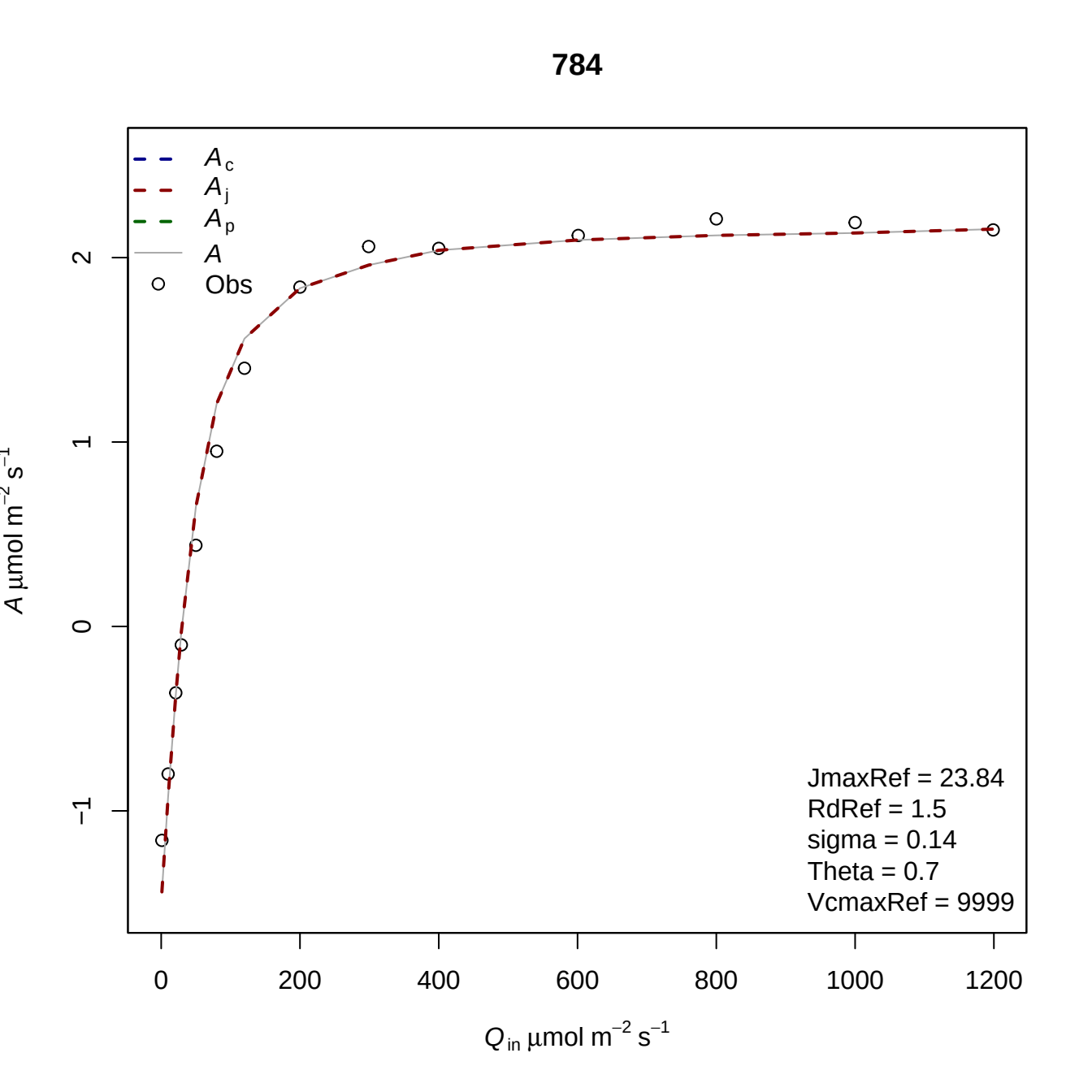

779
A_c
A_j
A_p
A
Obs
JmaxRef = 54.63
RdRef = 1.73
sigma = 0.58
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
200
400
600
800
1000
1200
1400

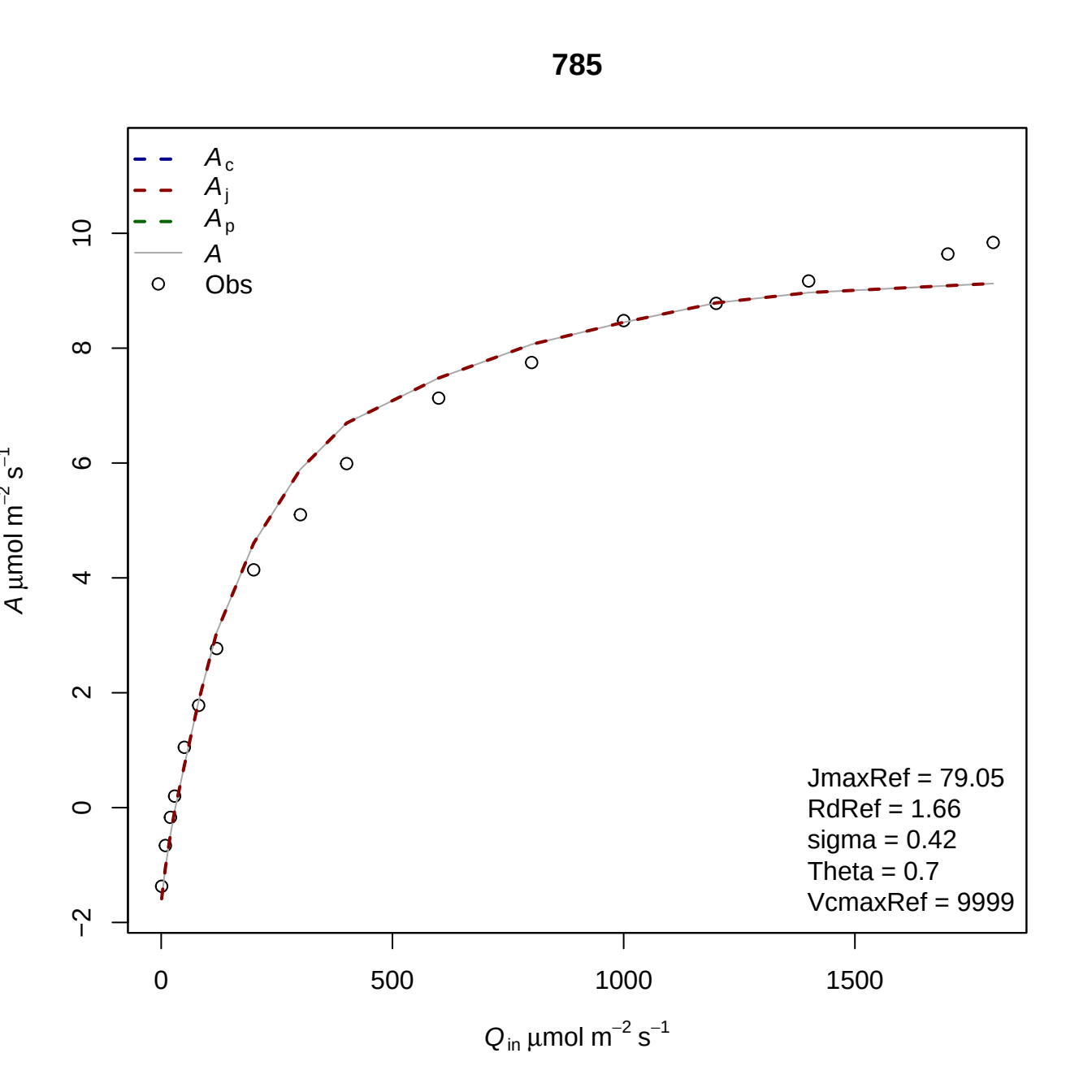

780
A_c
A_j
A_p
A
Obs
JmaxRef = 65.63
RdRef = 2.66
sigma = 0.26
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

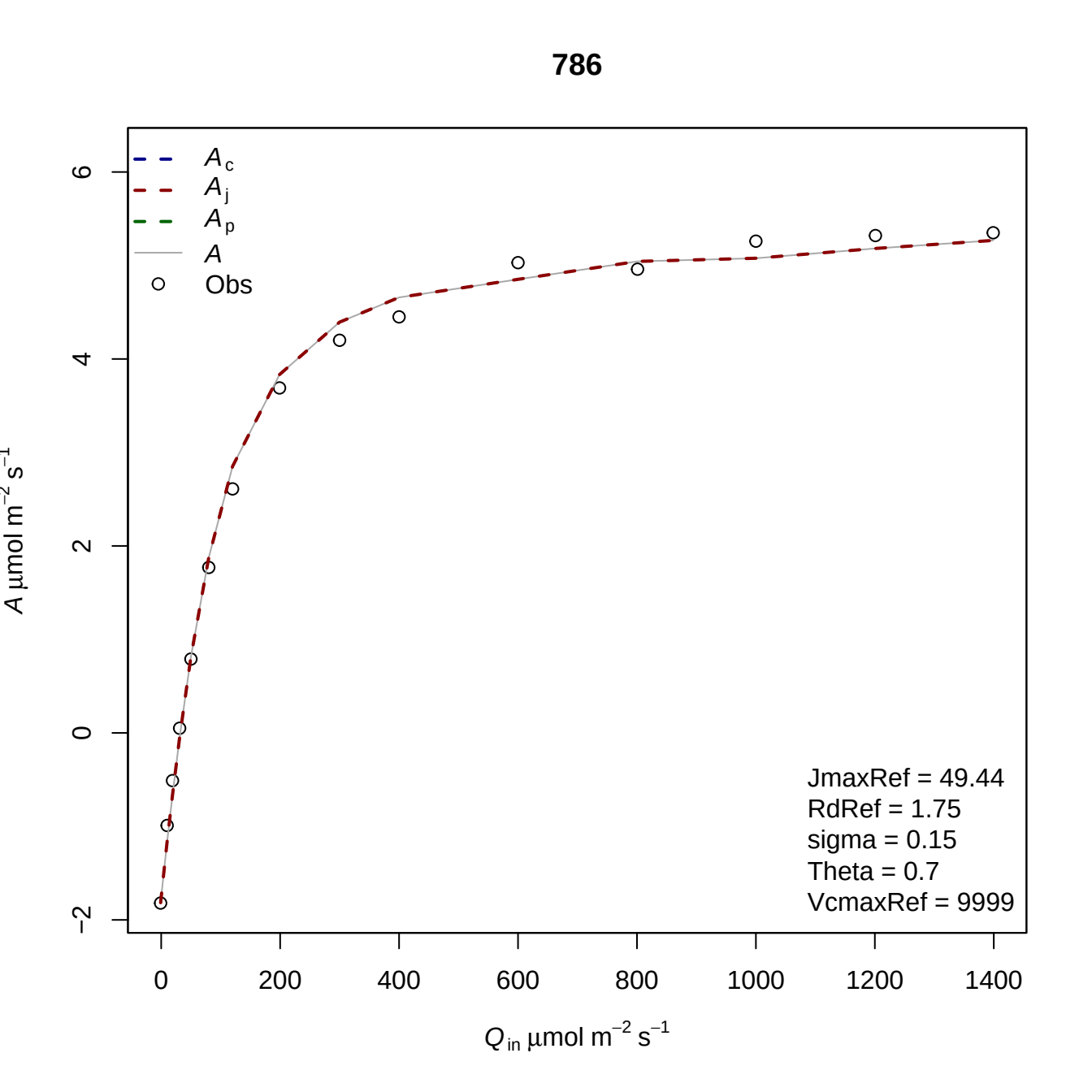

781
A_c
A_j
A_p
A
Obs
JmaxRef = 16.82
RdRef = 0.54
sigma = 0.1
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−0.5
0.0
0.5
1.0
1.5
2.0
0
500
1000
1500
2000

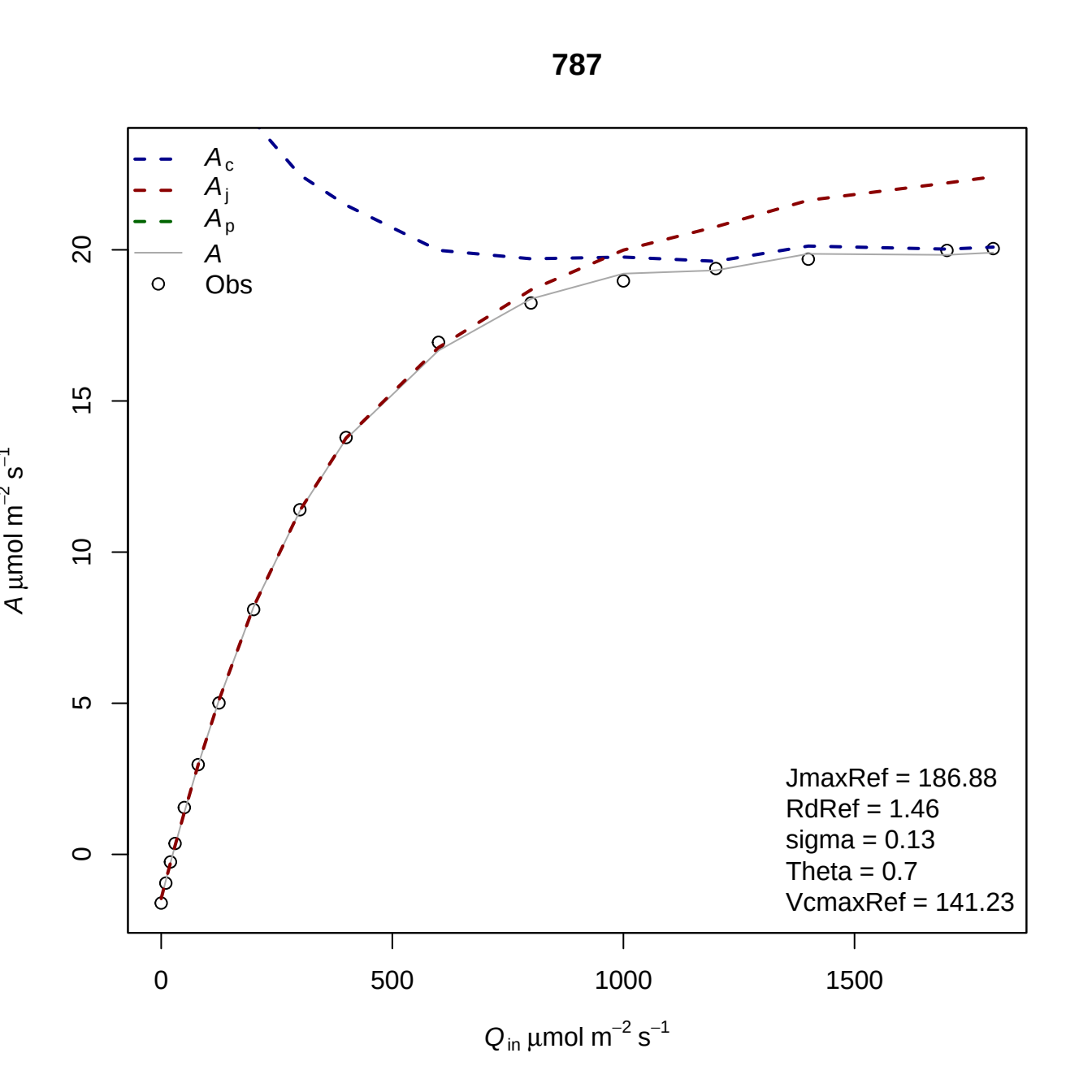

782
A_c
A_j
A_p
A
Obs
JmaxRef = 40.32
RdRef = 0.82
sigma = 0.08
Theta = 0.7
VcmaxRef = 30.97
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

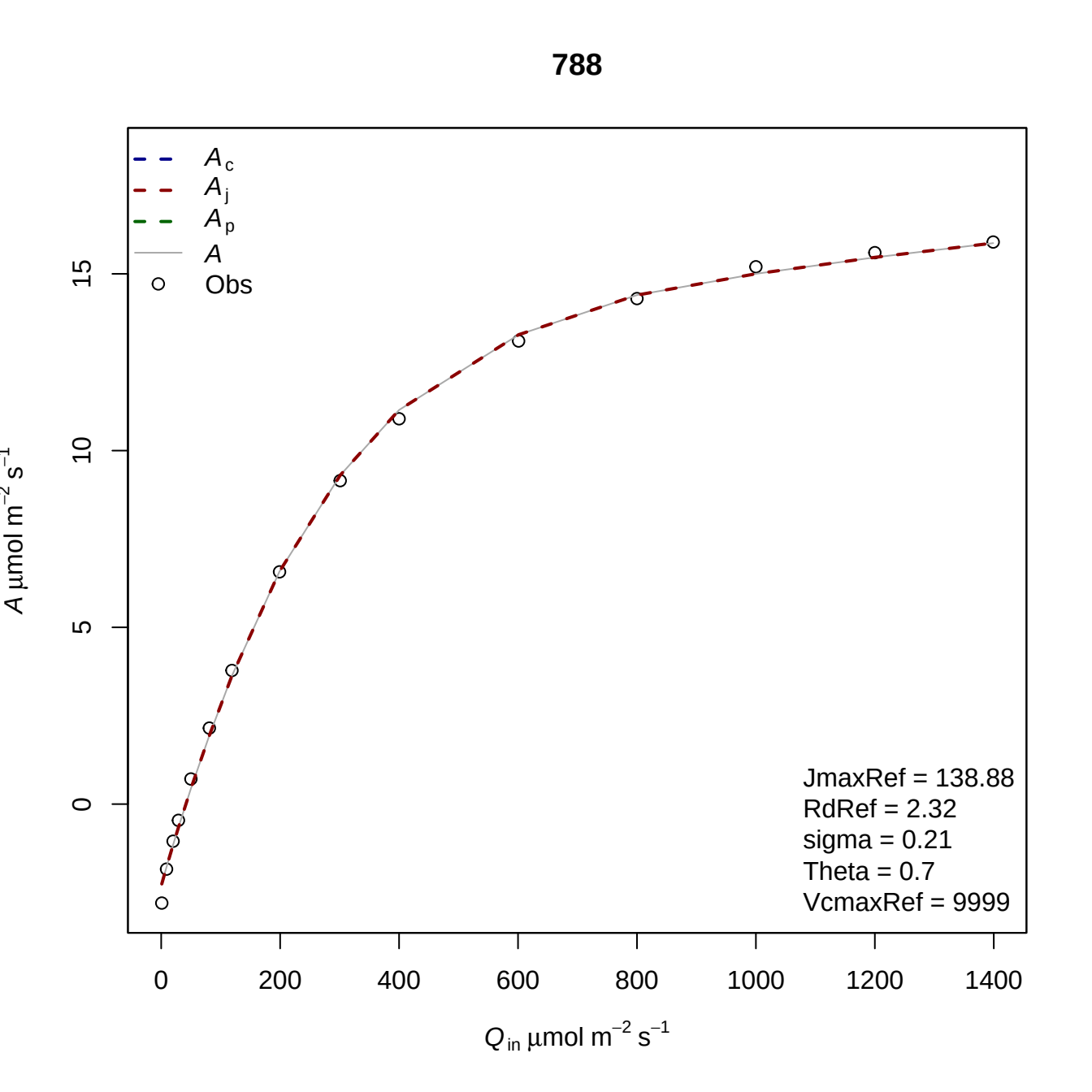

783
A_c
A_j
A_p
A
Obs
JmaxRef = 66.71
RdRef = 0.43
sigma = 0.11
Theta = 0.7
VcmaxRef = 51.08
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

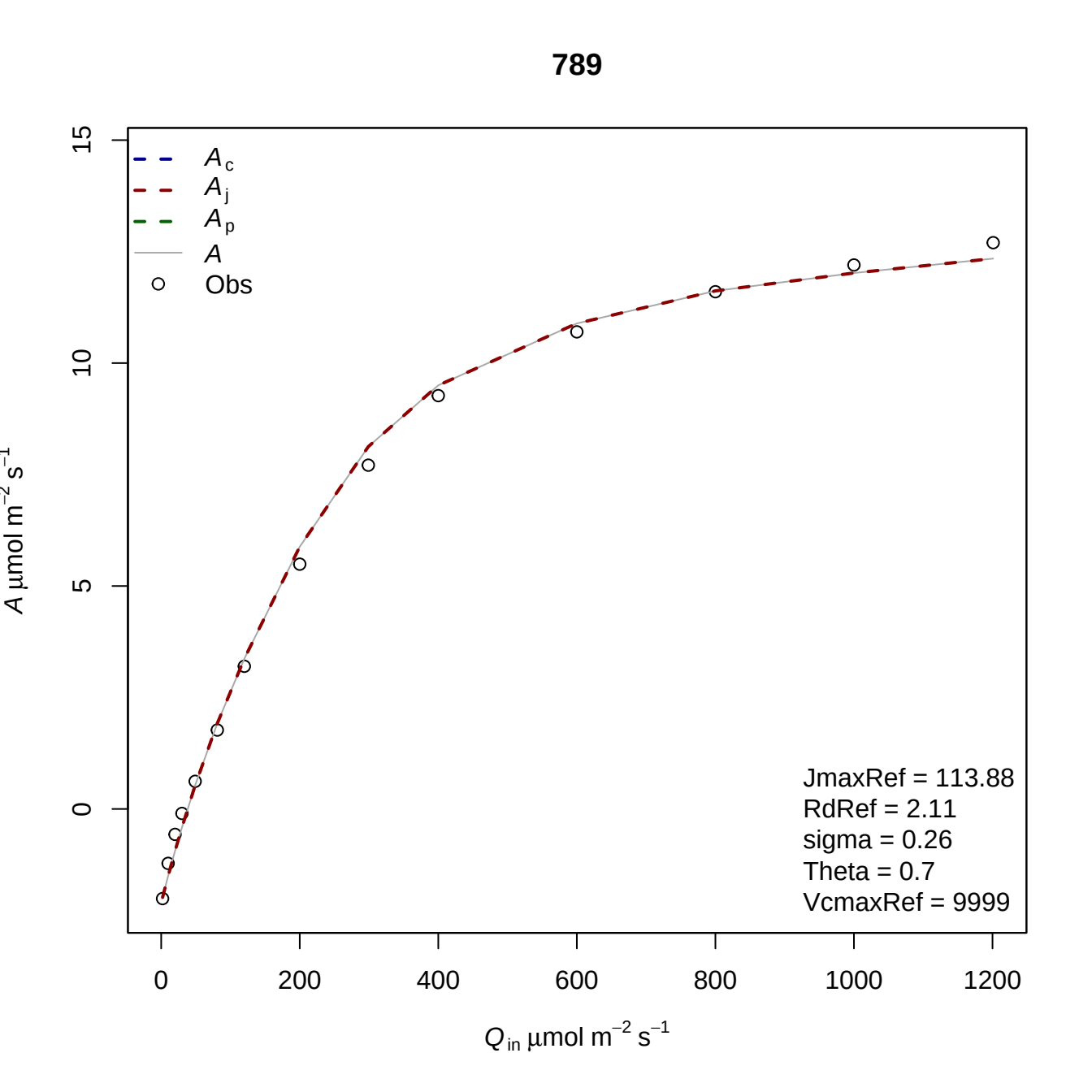

784
A_c
A_j
A_p
A
Obs
JmaxRef = 23.84
RdRef = 1.5
sigma = 0.14
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
200
400
600
800
1000
1200

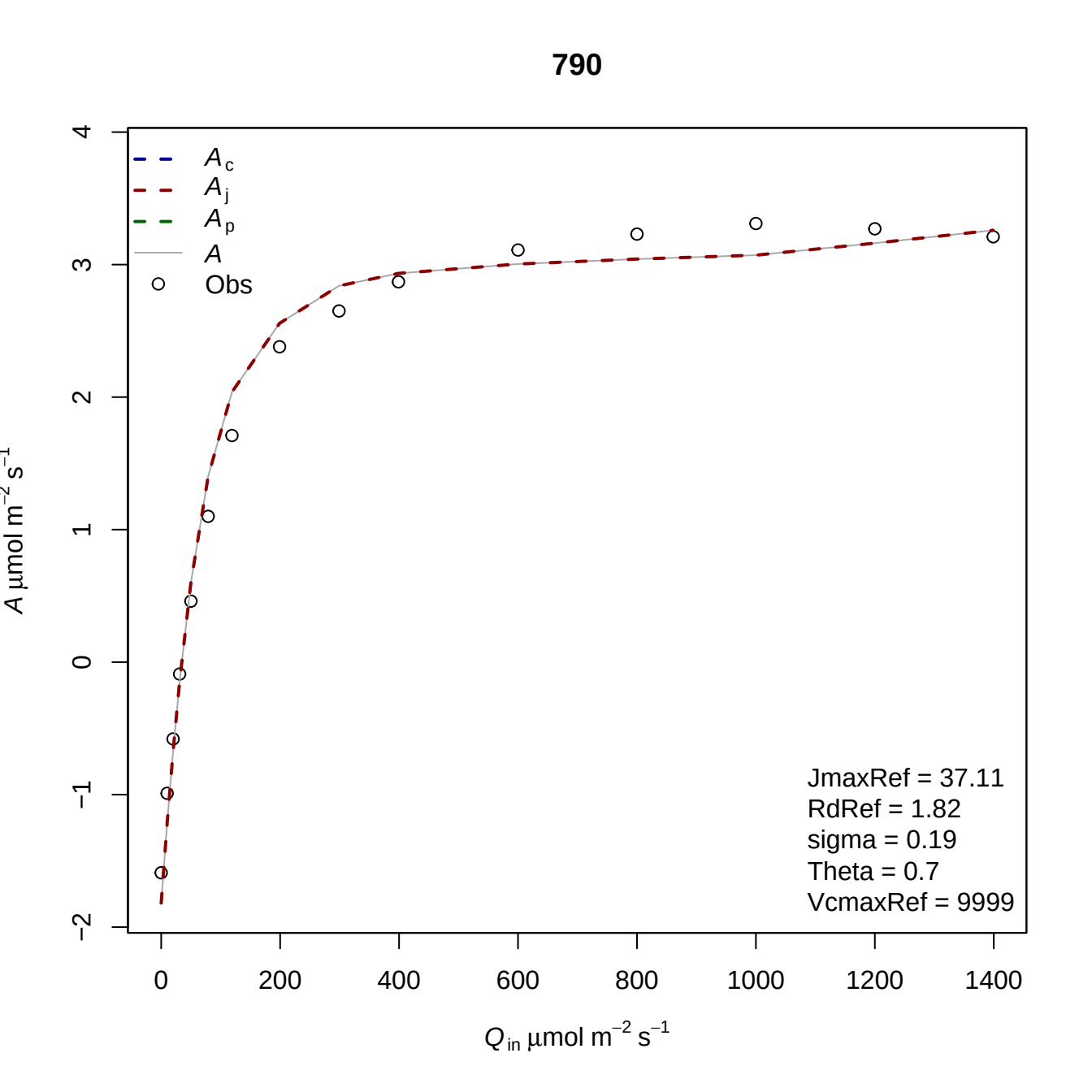

785
A_c
A_j
A_p
A
Obs
JmaxRef = 79.05
RdRef = 1.66
sigma = 0.42
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

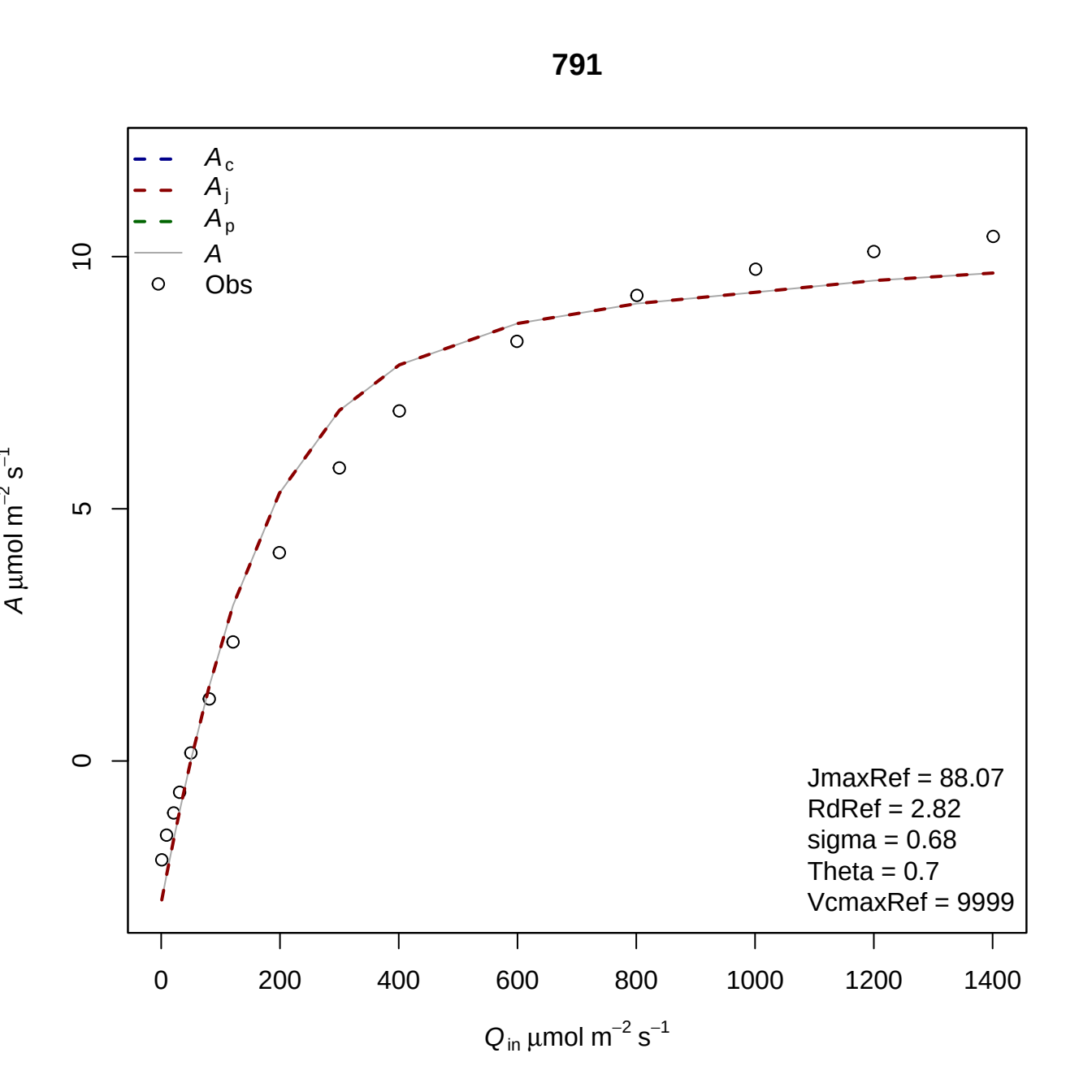

786
A_c
A_j
A_p
A
Obs
JmaxRef = 49.44
RdRef = 1.75
sigma = 0.15
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000
1200
1400

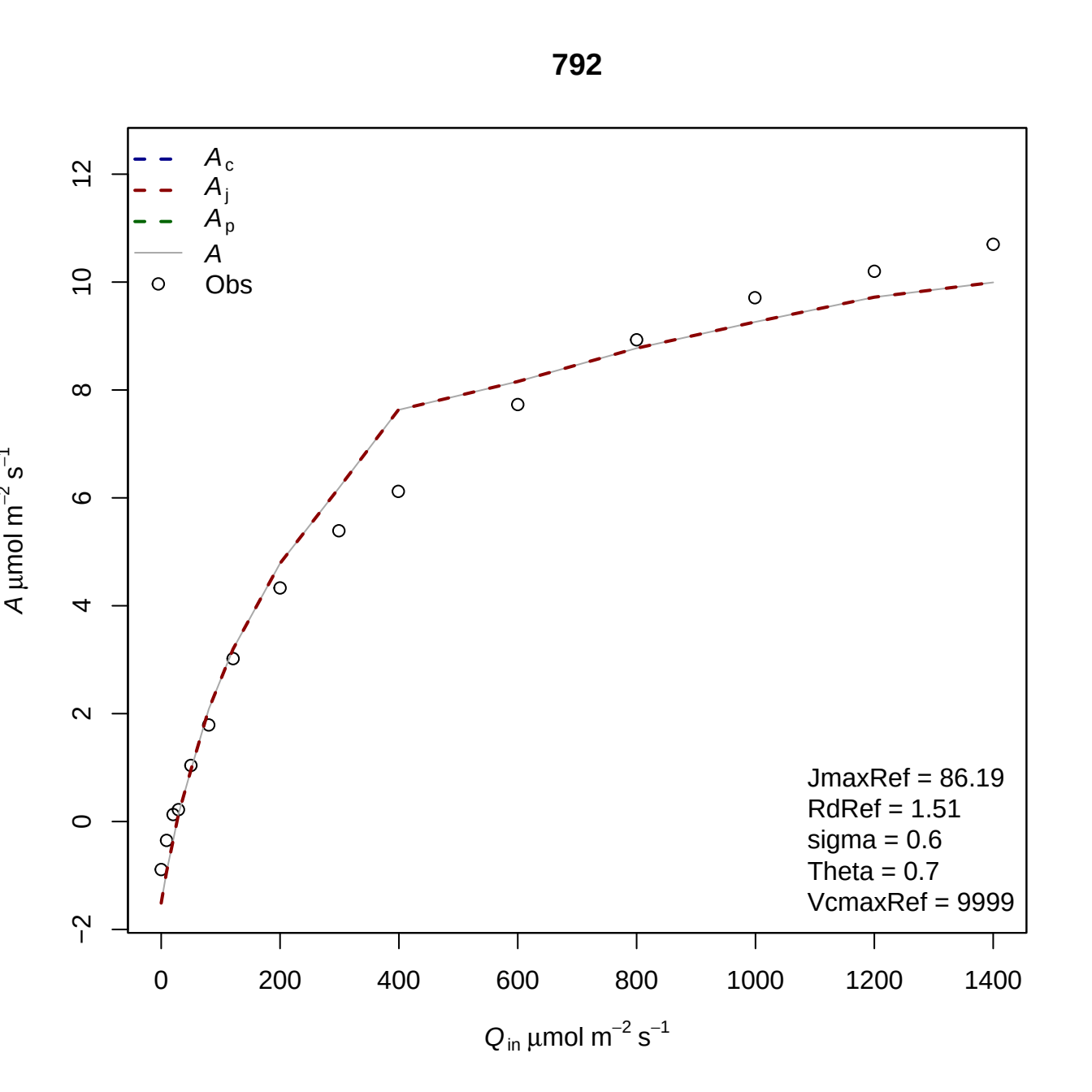

787
A_c
A_j
A_p
A
Obs
JmaxRef = 186.88
RdRef = 1.46
sigma = 0.13
Theta = 0.7
VcmaxRef = 141.23
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

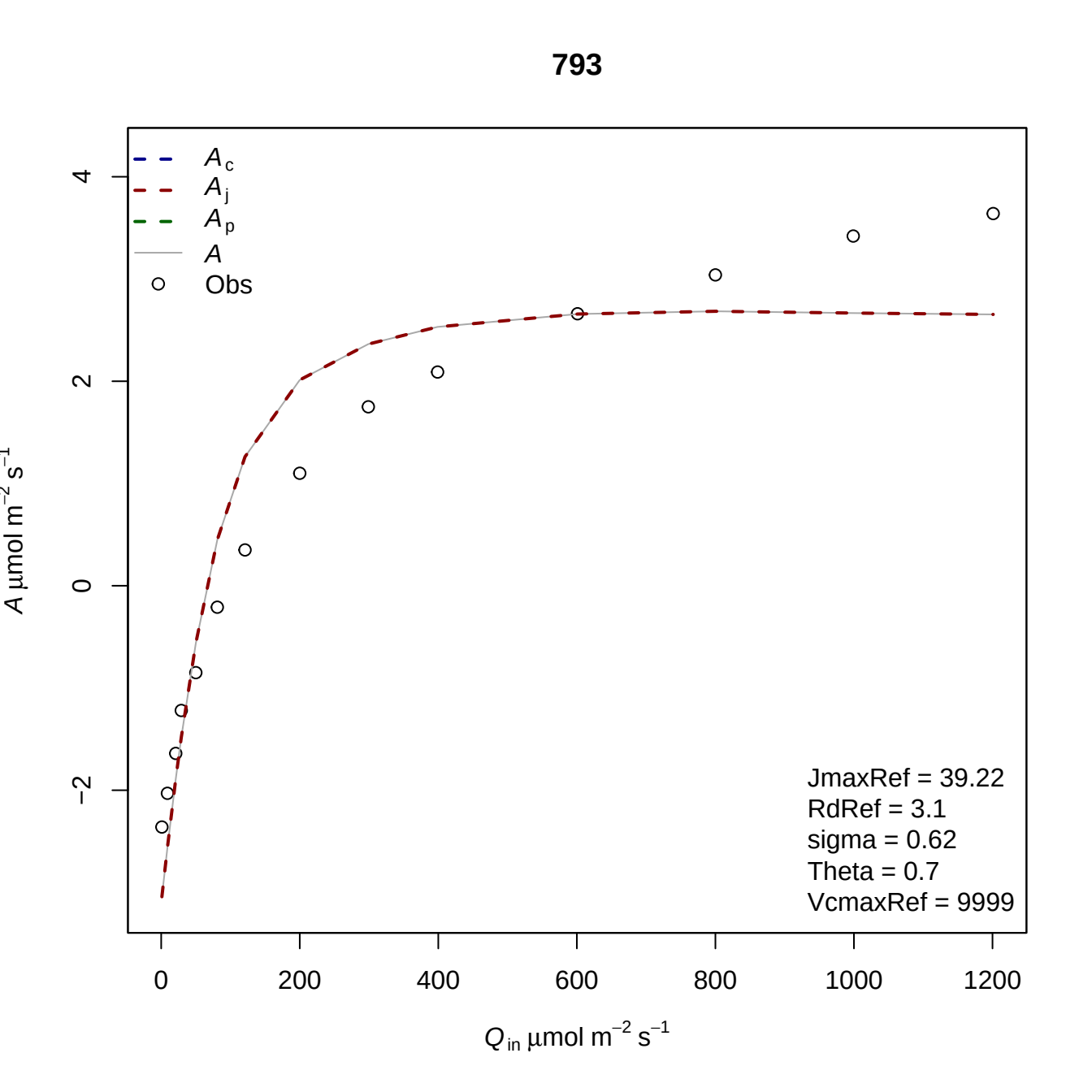

788
A_c
A_j
A_p
A
Obs
JmaxRef = 138.88
RdRef = 2.32
sigma = 0.21
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
200
400
600
800
1000
1200
1400

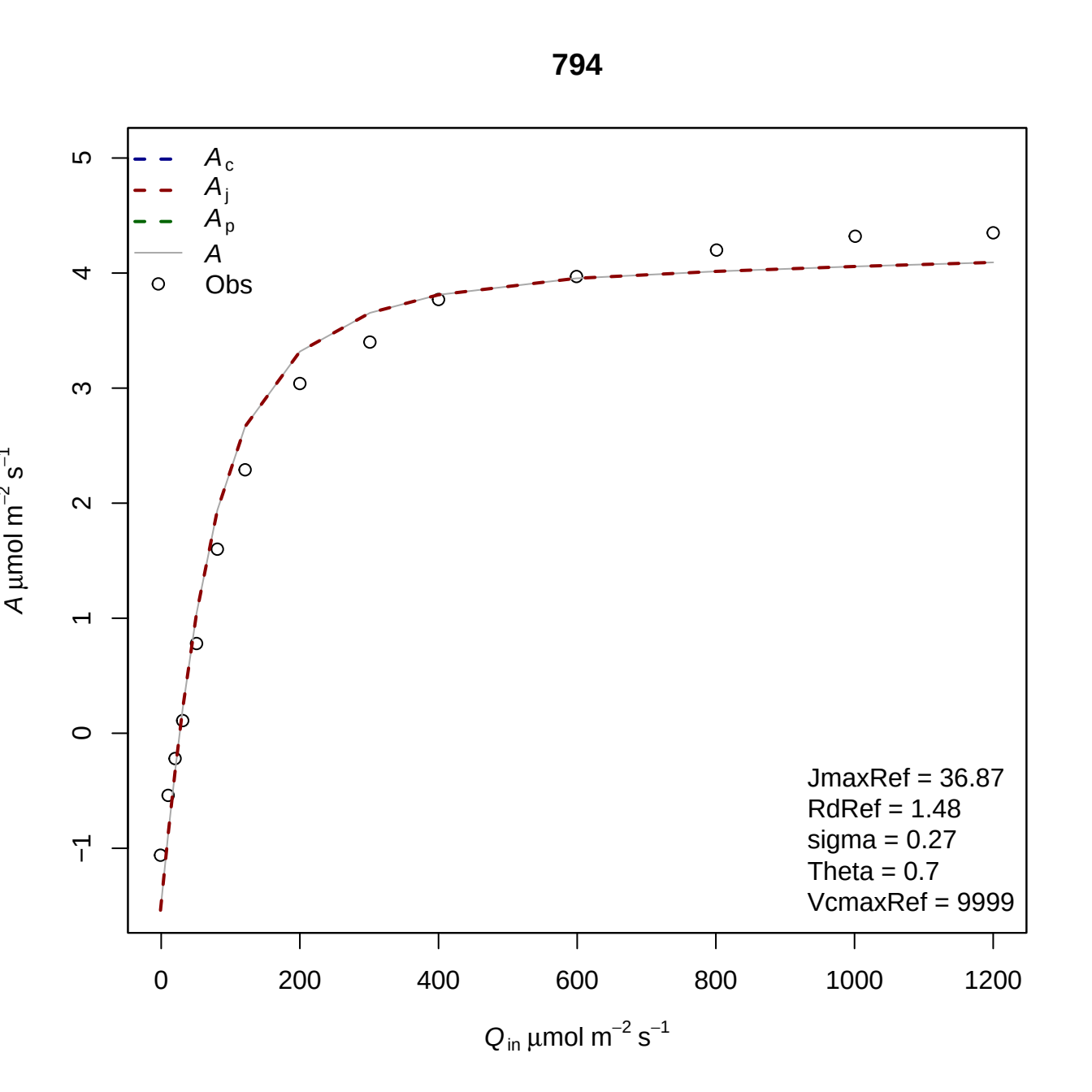

789
A_c
A_j
A_p
A
Obs
JmaxRef = 113.88
RdRef = 2.11
sigma = 0.26
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
200
400
600
800
1000
1200

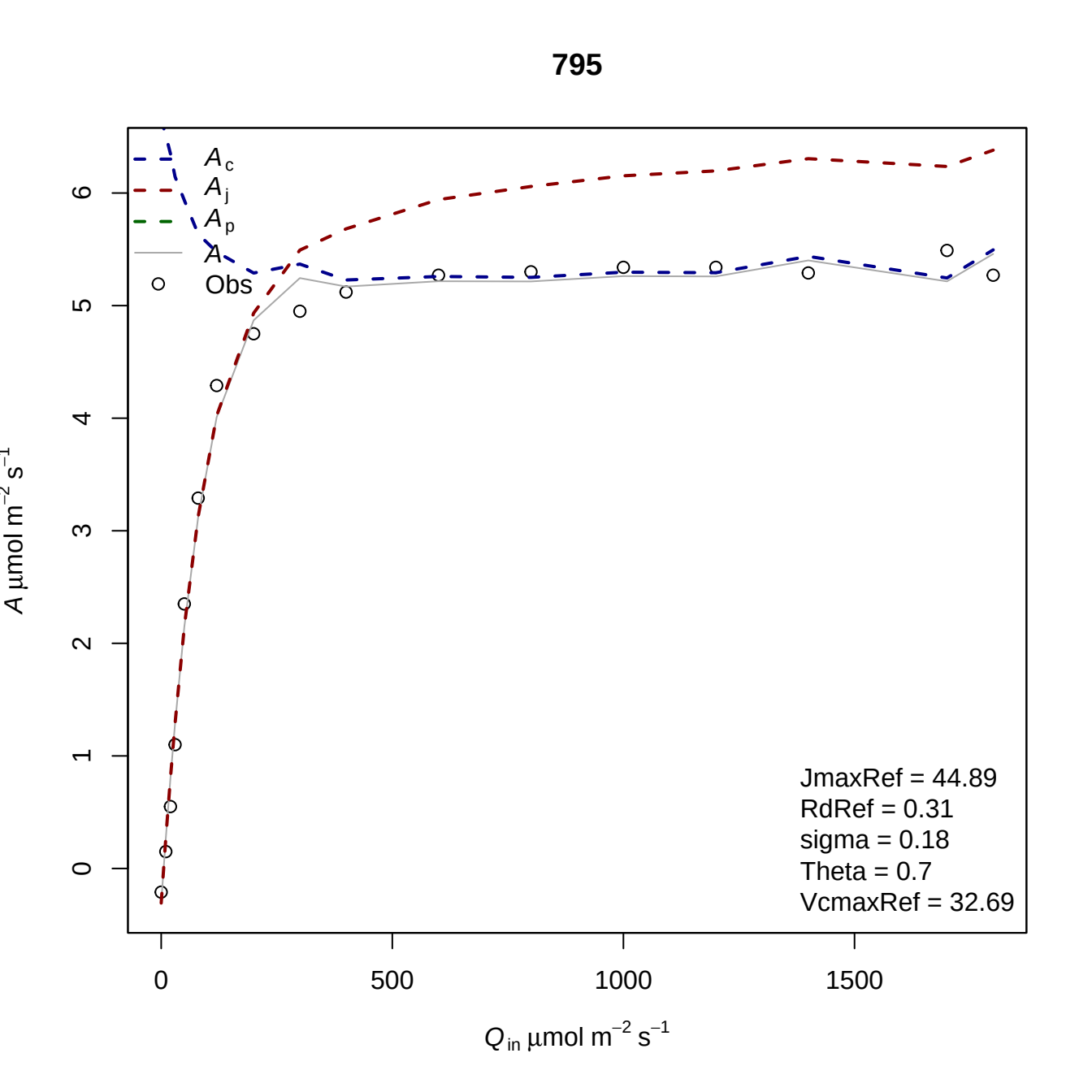

790
A_c
A_j
A_p
A
Obs
JmaxRef = 37.11
RdRef = 1.82
sigma = 0.19
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
0
200
400
600
800
1000
1200
1400

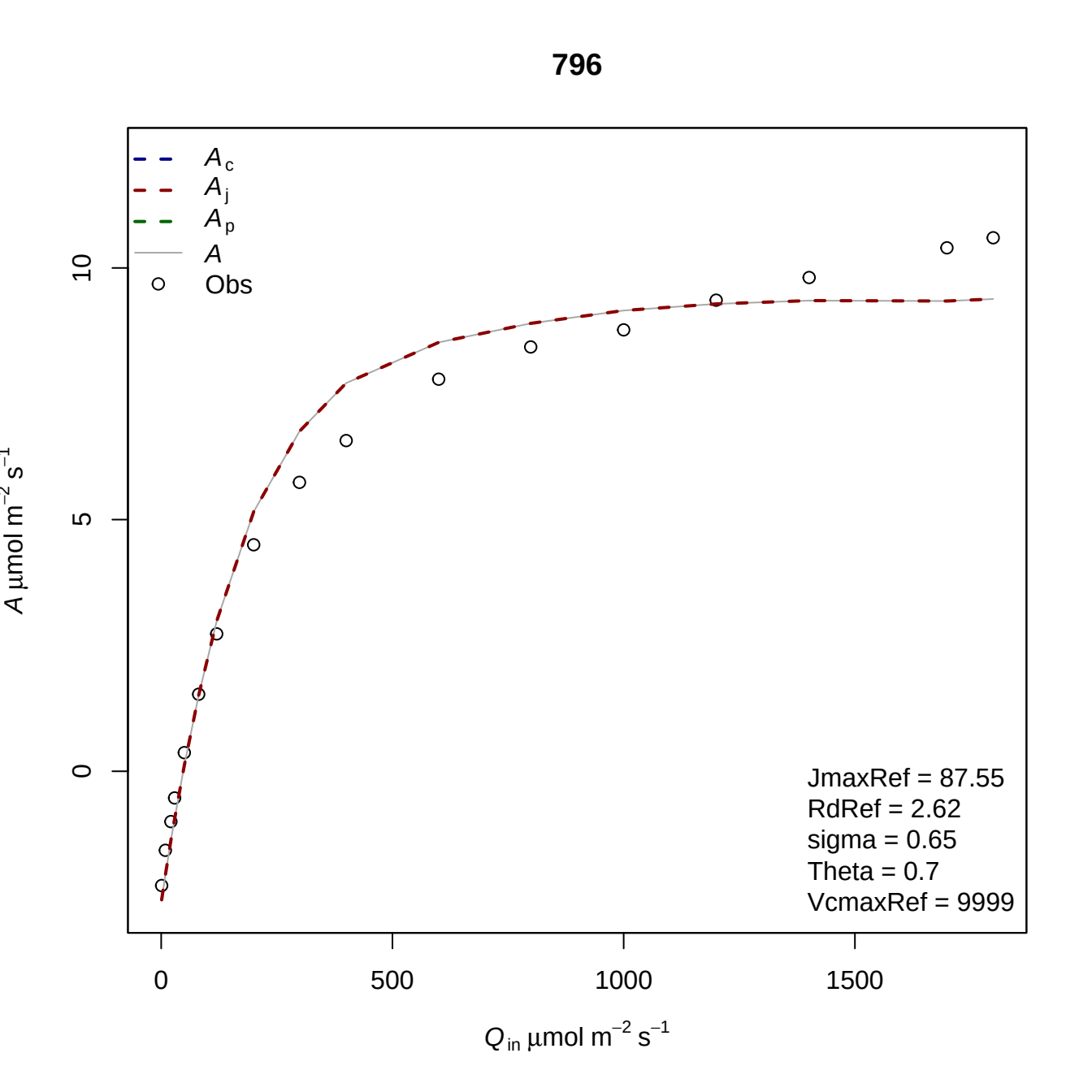

791
A_c
A_j
A_p
A
Obs
JmaxRef = 88.07
RdRef = 2.82
sigma = 0.68
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
200
400
600
800
1000
1200
1400
Q_{in} µmol m^{-2} s^{-1}

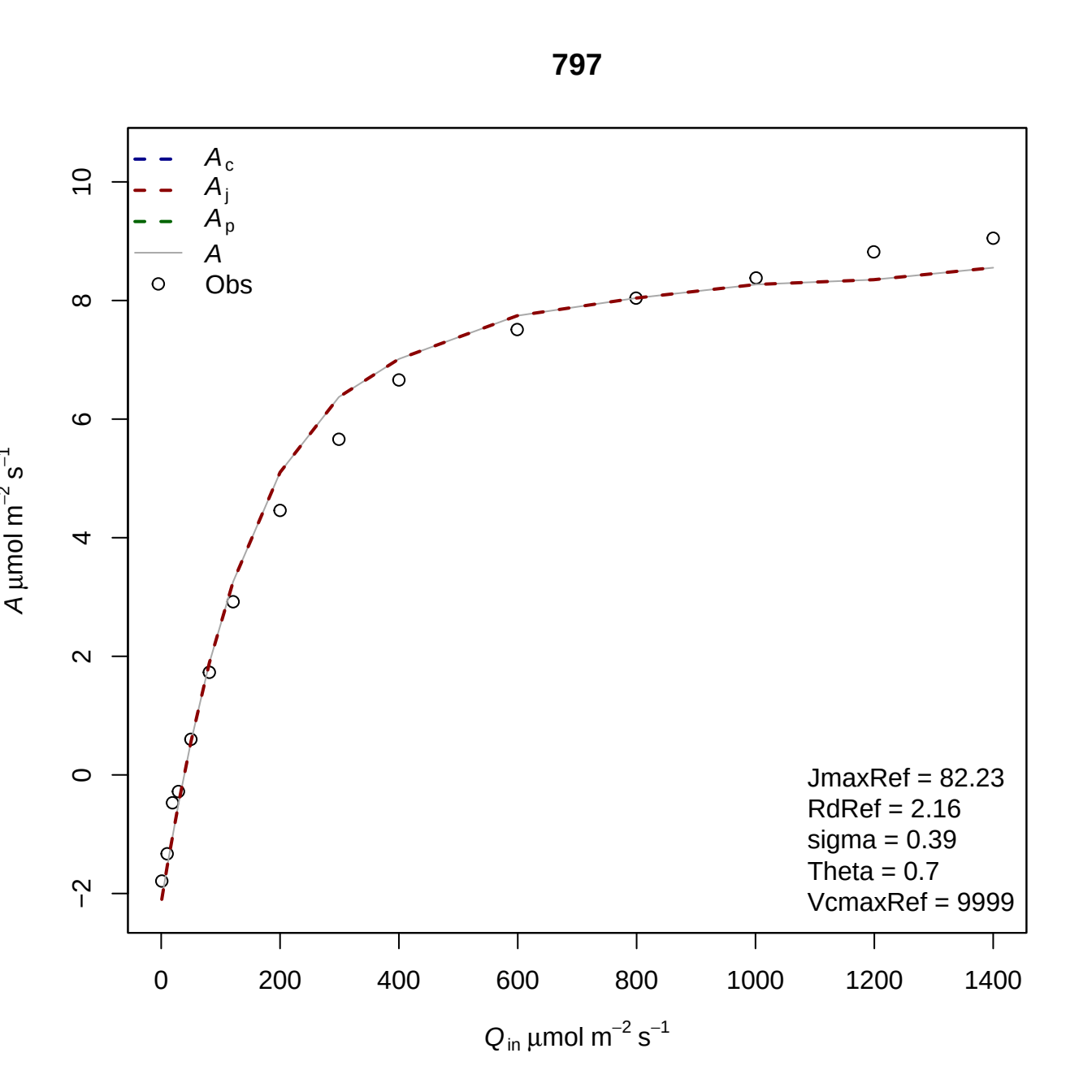

792
A_c
A_j
A_p
A
Obs
JmaxRef = 86.19
RdRef = 1.51
sigma = 0.6
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
1400

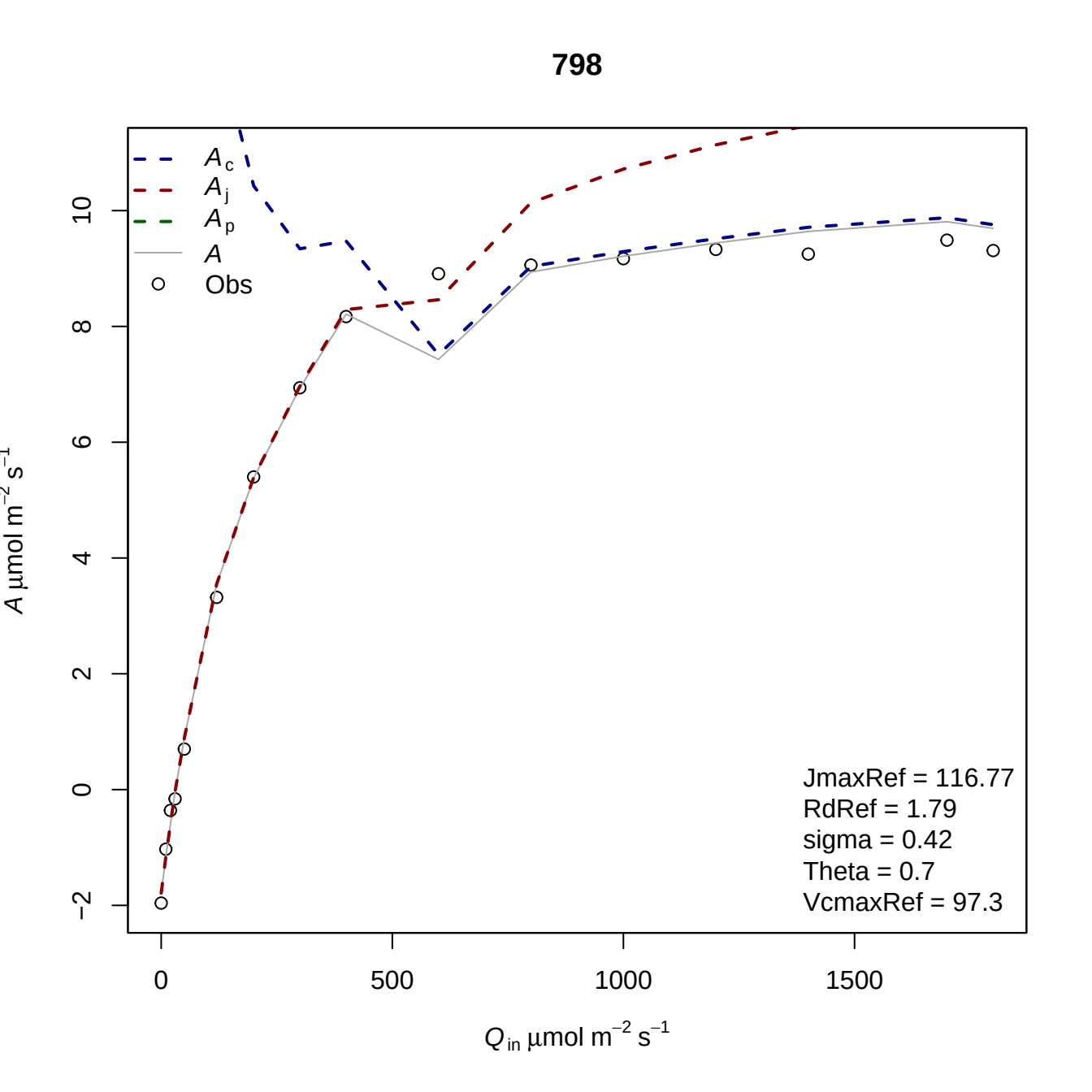

793
A_c
A_j
A_p
A
Obs
JmaxRef = 39.22
RdRef = 3.1
sigma = 0.62
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
0
200
400
600
800
1000
1200

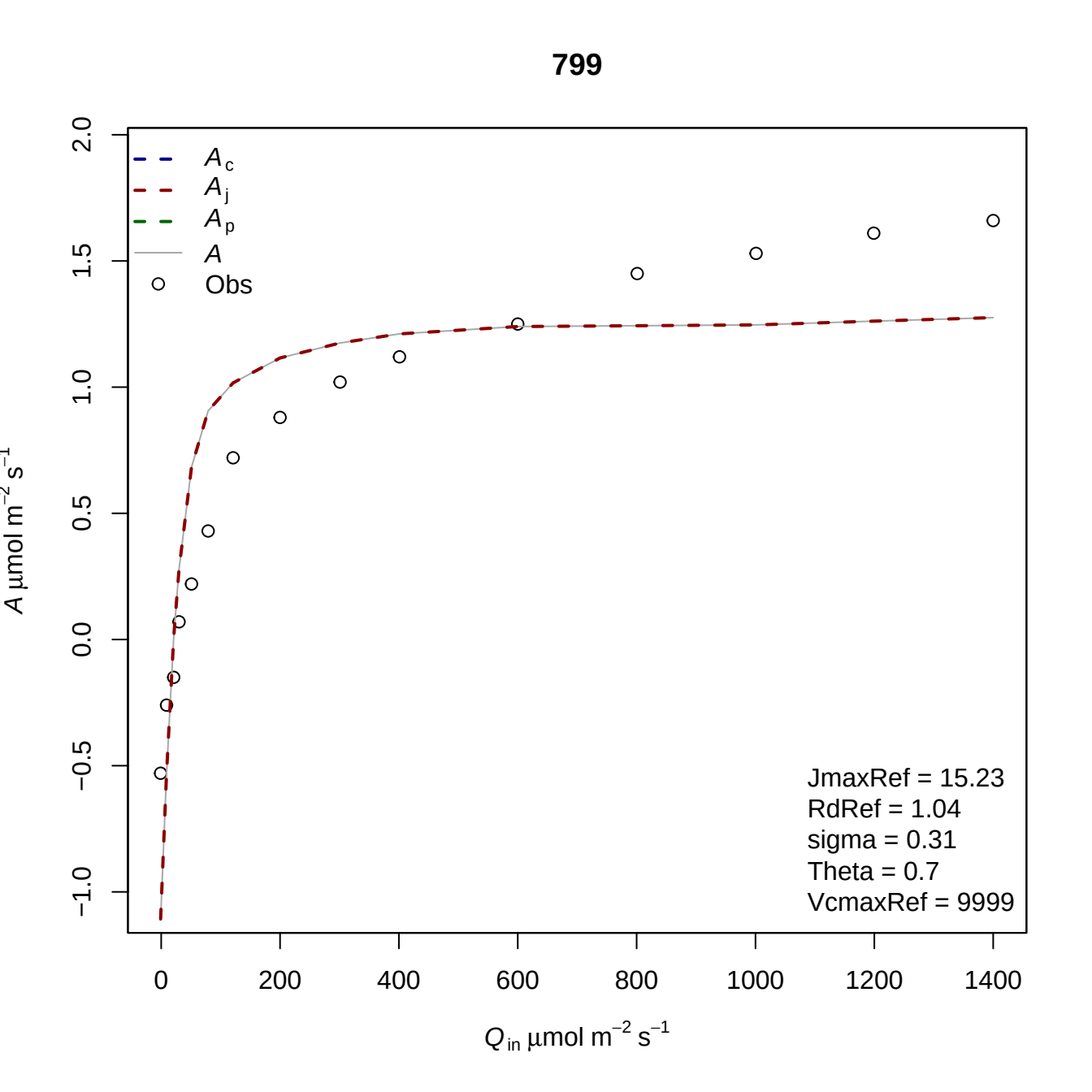

794
A_c
A_j
A_p
A
Obs
JmaxRef = 36.87
RdRef = 1.48
sigma = 0.27
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
5
4
3
2
1
0
−1
0
200
400
600
800
1000
1200

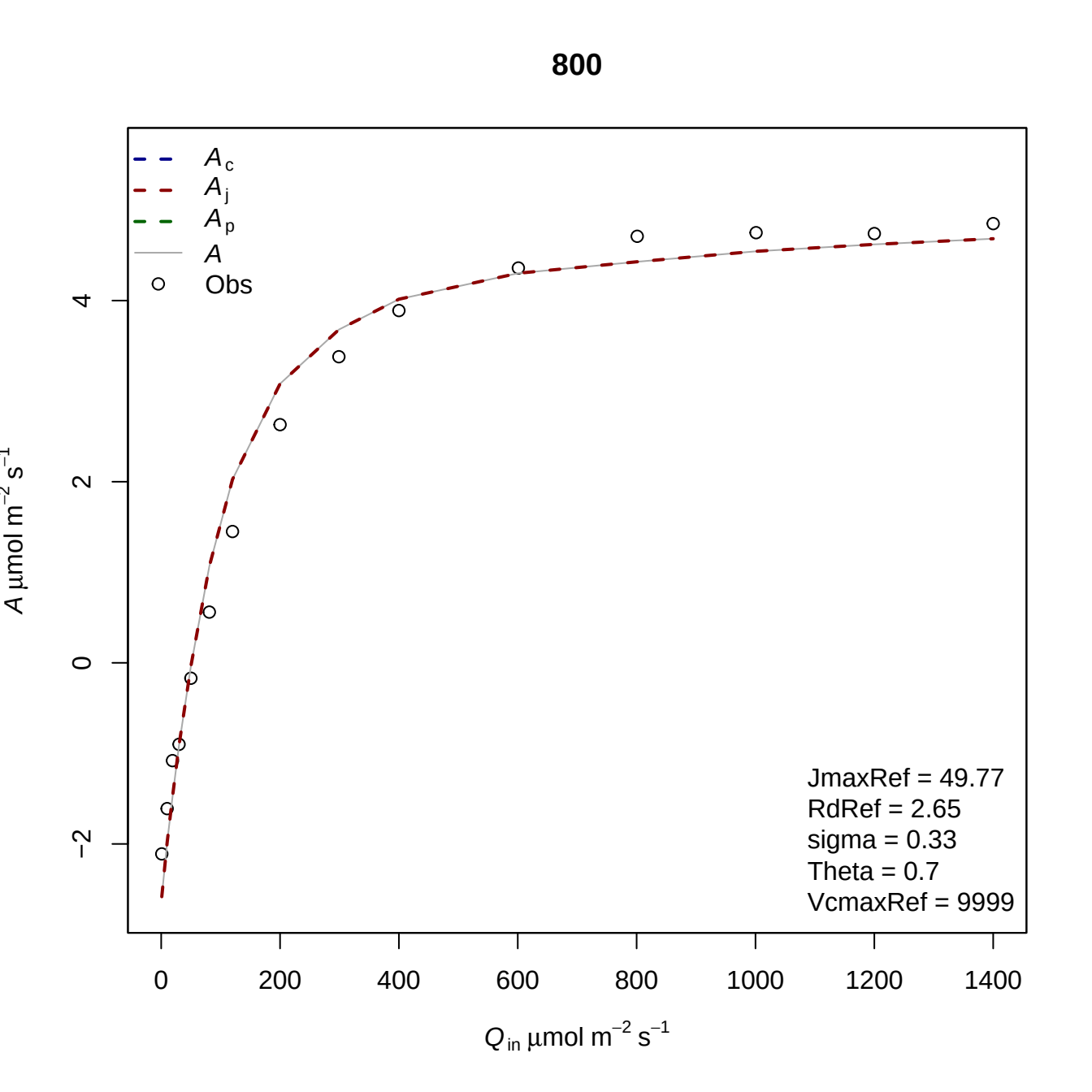

795
A_c
A_j
A_p
A
Obs
JmaxRef = 44.89
RdRef = 0.31
sigma = 0.18
Theta = 0.7
VcmaxRef = 32.69
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

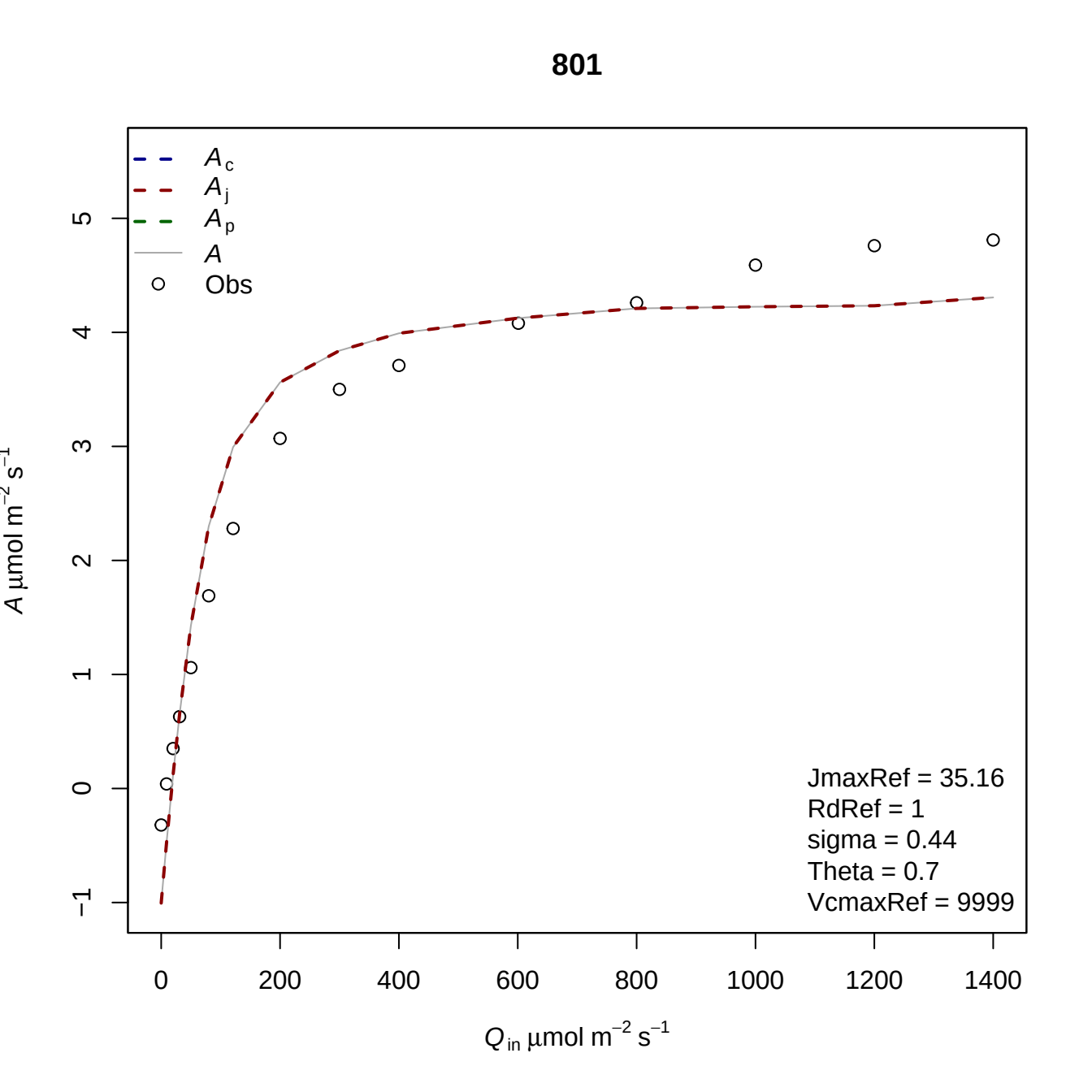

796
A_c
A_j
A_p
A
Obs
JmaxRef = 87.55
RdRef = 2.62
sigma = 0.65
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

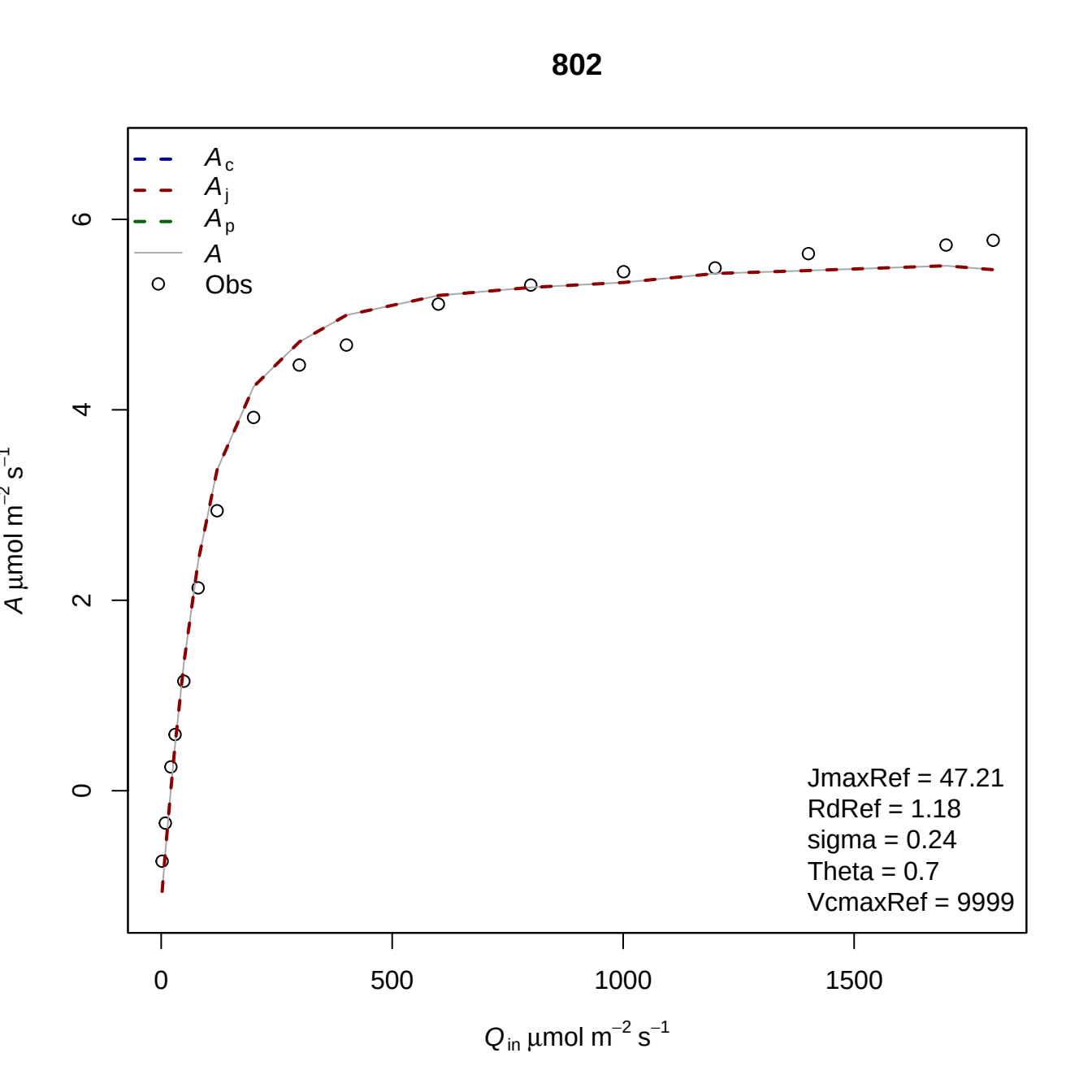

797
A_c
A_j
A_p
A
Obs
JmaxRef = 82.23
RdRef = 2.16
sigma = 0.39
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
200
400
600
800
1000
1200
1400

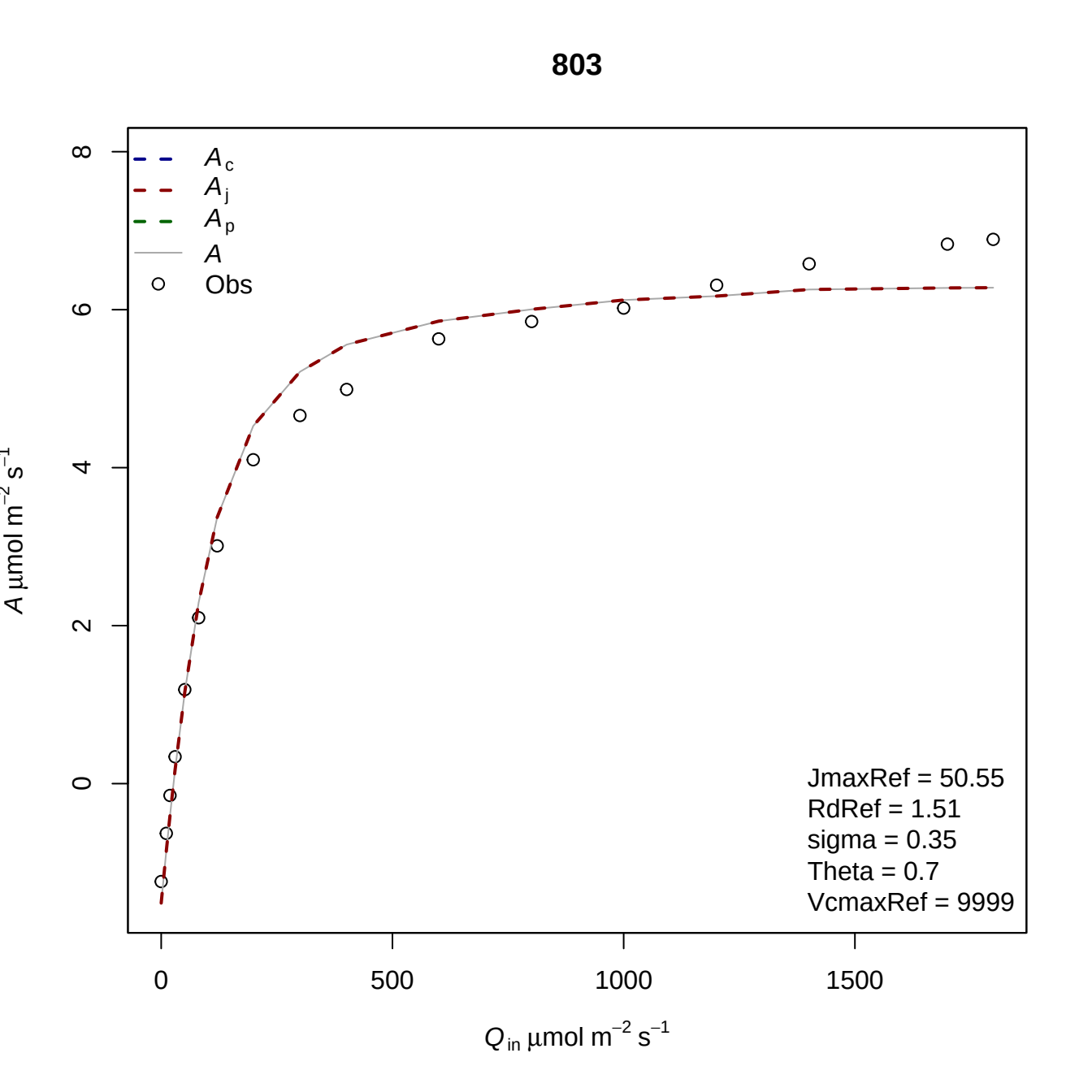

798
A_c
A_j
A_p
A
Obs
JmaxRef = 116.77
RdRef = 1.79
sigma = 0.42
Theta = 0.7
VcmaxRef = 97.3
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

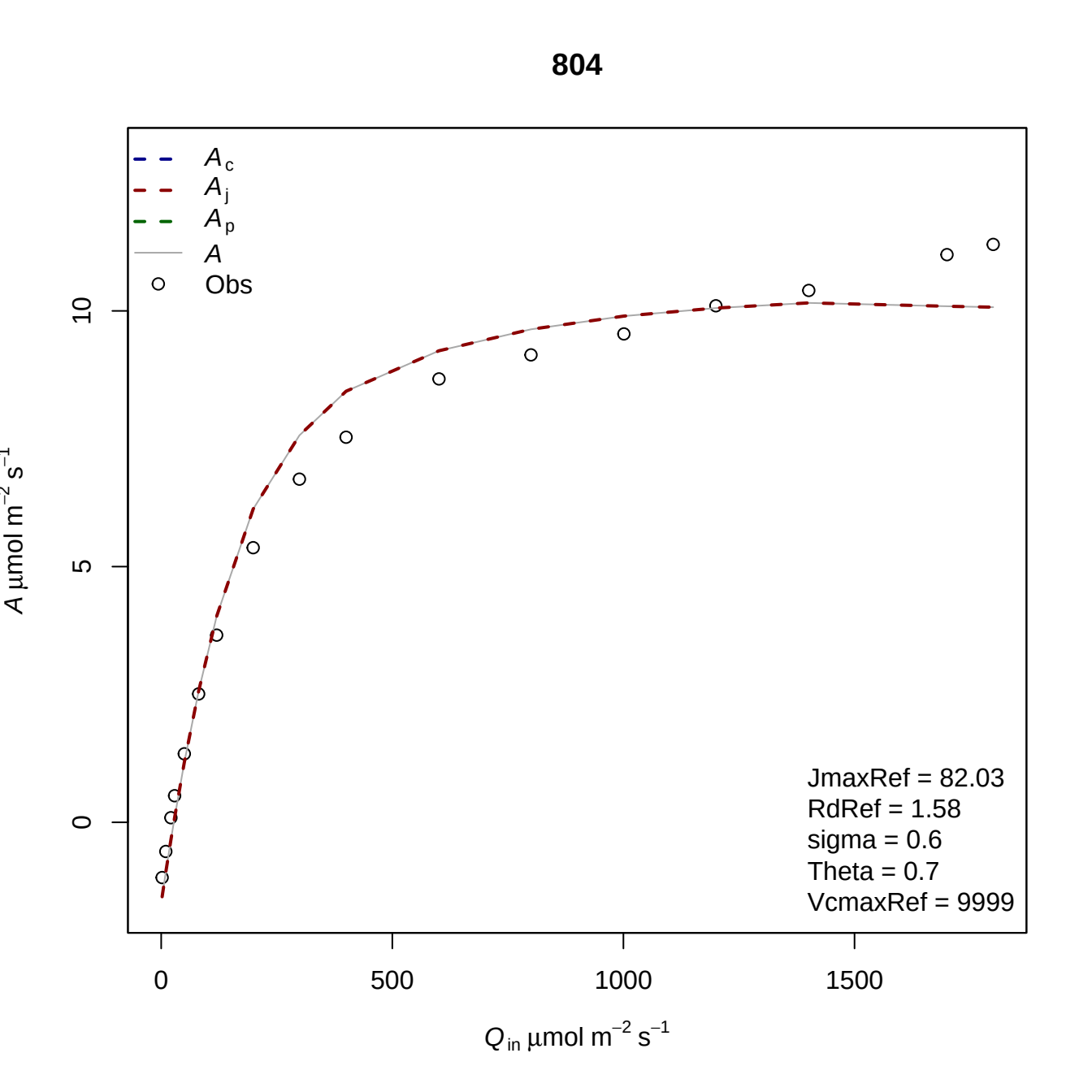

799
A_c
A_j
A_p
A
Obs
JmaxRef = 15.23
RdRef = 1.04
sigma = 0.31
Theta = 0.7
VcmaxRef = 9999
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
0
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

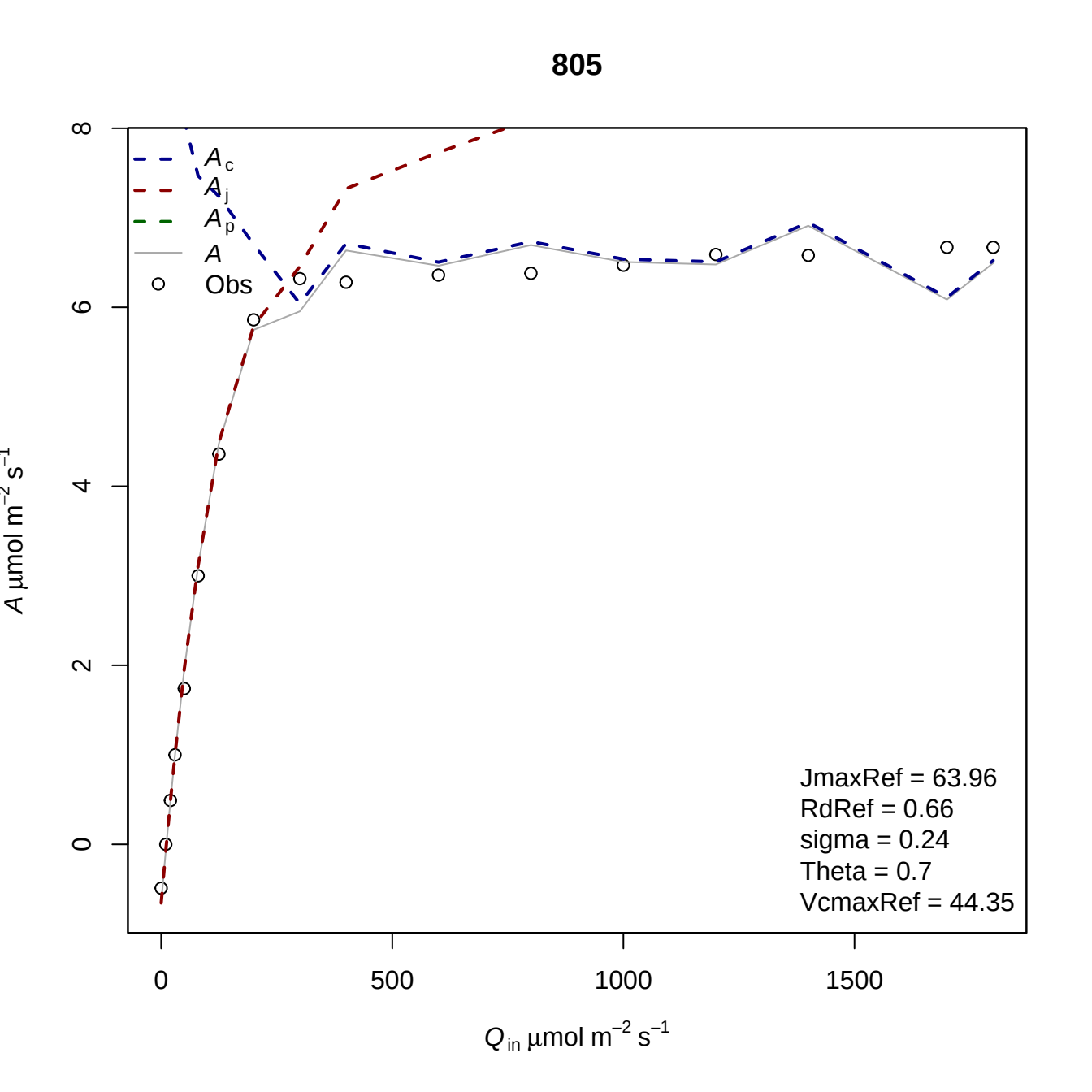

800
A_c
A_j
A_p
A
Obs
JmaxRef = 49.77
RdRef = 2.65
sigma = 0.33
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
200
400
600
800
1000
1200
1400

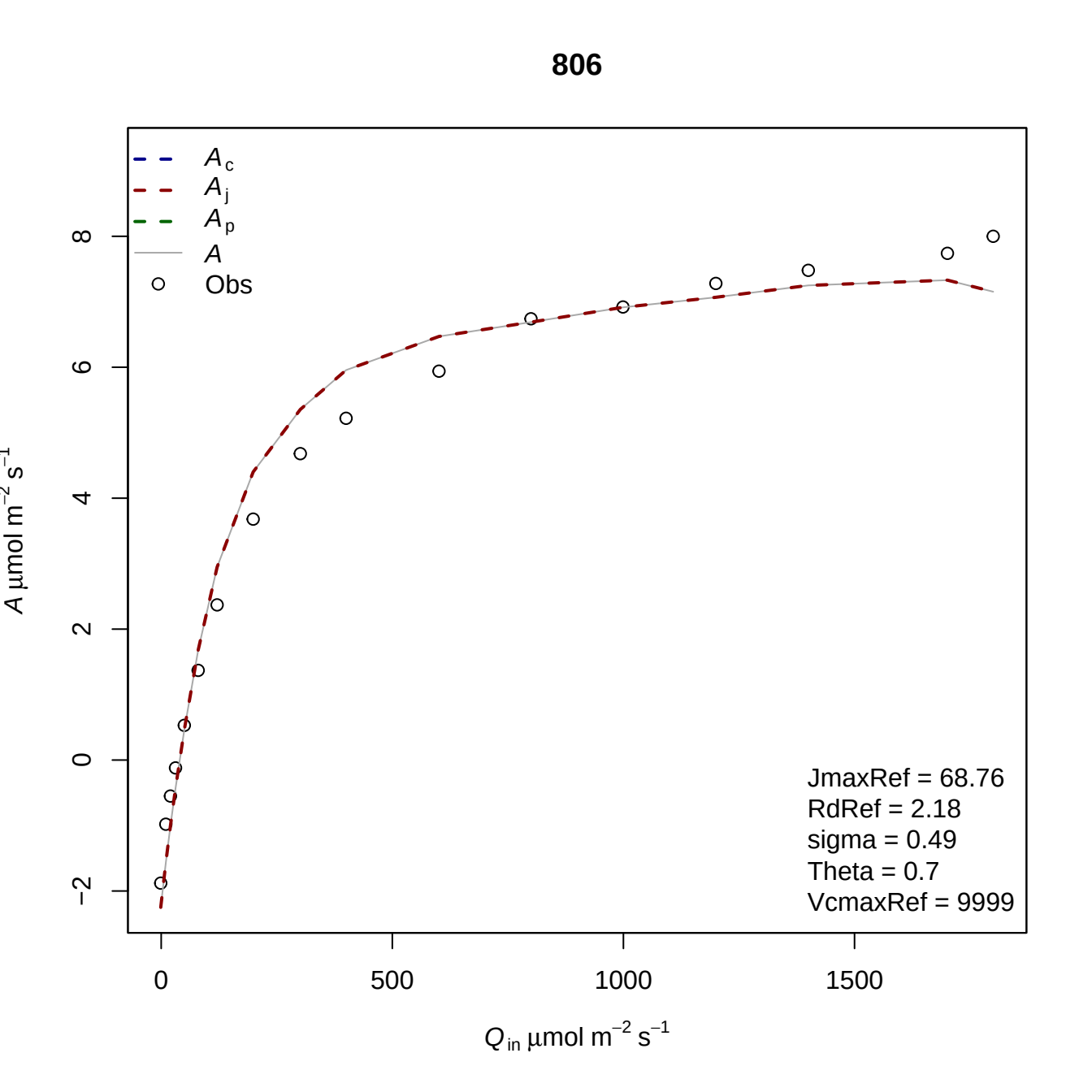

801
A_c
A_j
A_p
A
Obs
JmaxRef = 35.16
RdRef = 1
sigma = 0.44
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
200
400
600
800
1000
1200
1400

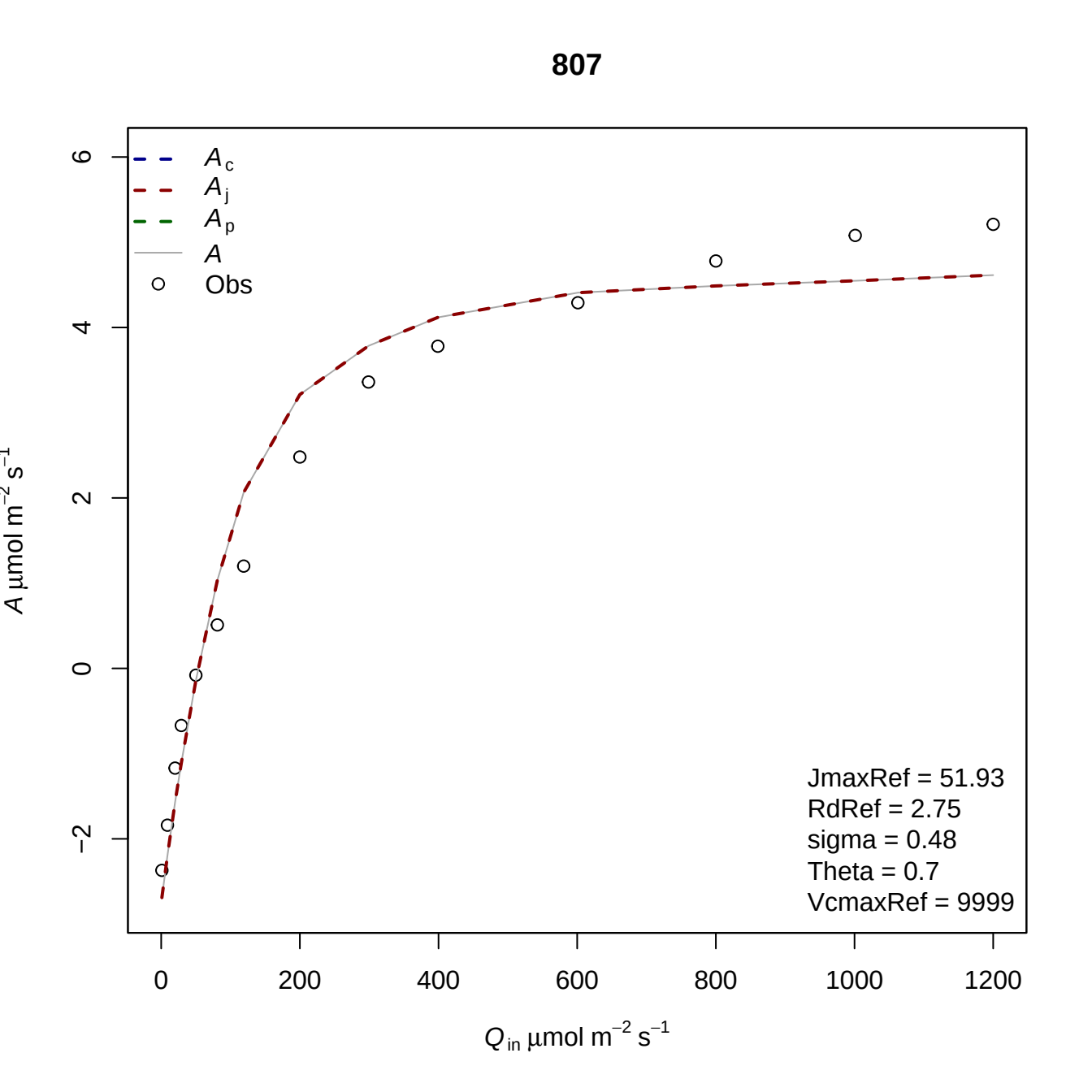

802
A_c
A_j
A_p
A
Obs
JmaxRef = 47.21
RdRef = 1.18
sigma = 0.24
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

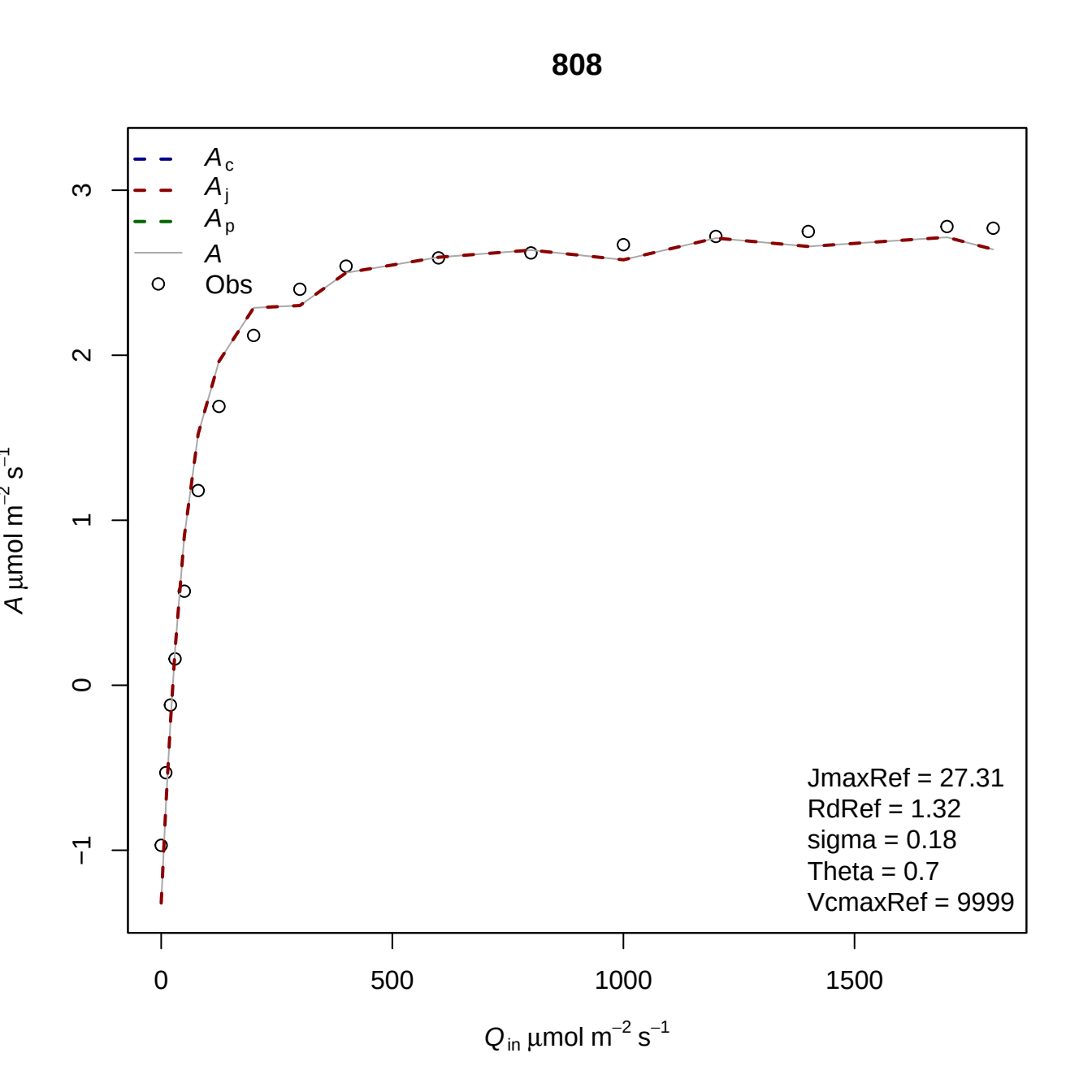

803
A_c
A_j
A_p
A
Obs
JmaxRef = 50.55
RdRef = 1.51
sigma = 0.35
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

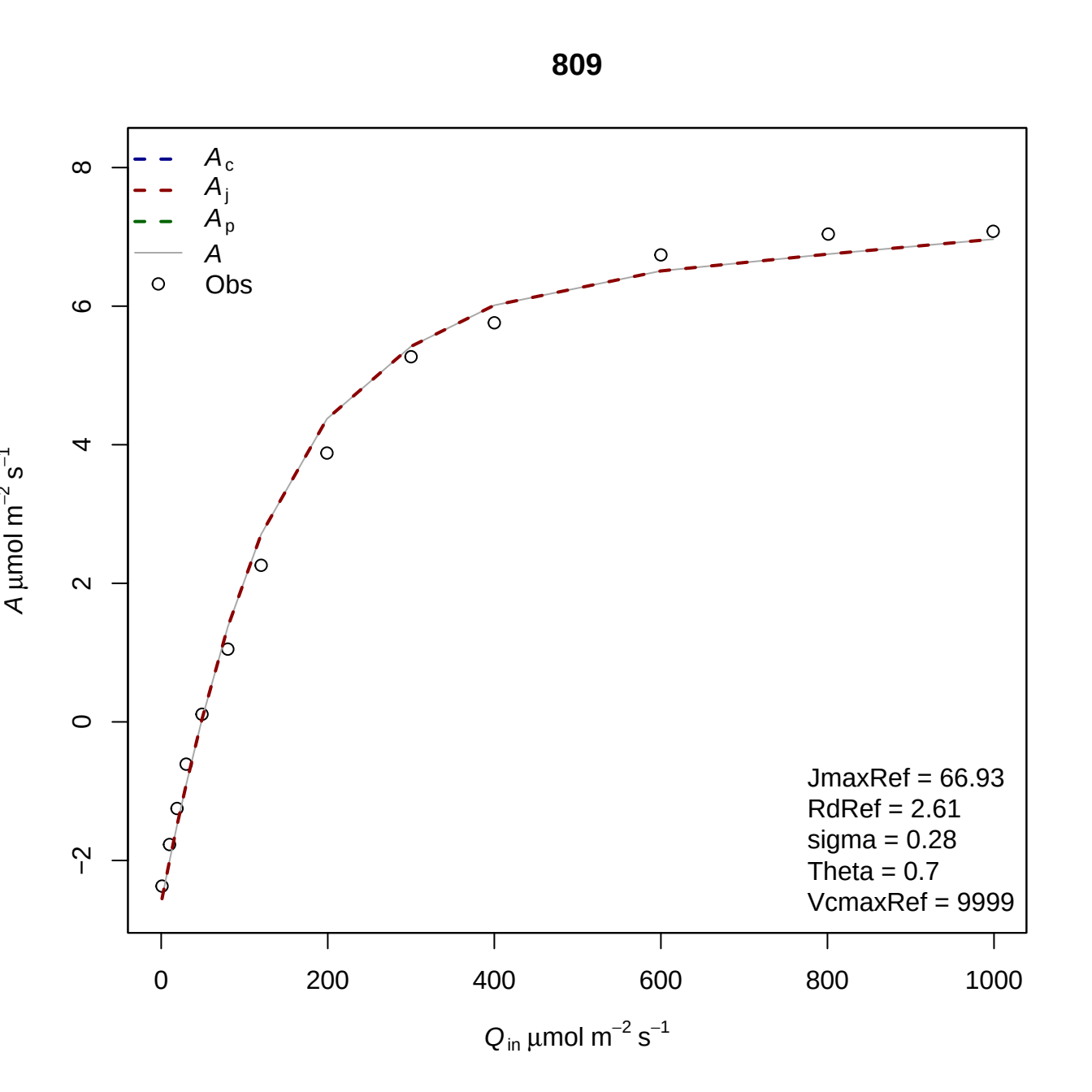

804
A_c
A_j
A_p
A
Obs
JmaxRef = 82.03
RdRef = 1.58
sigma = 0.6
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

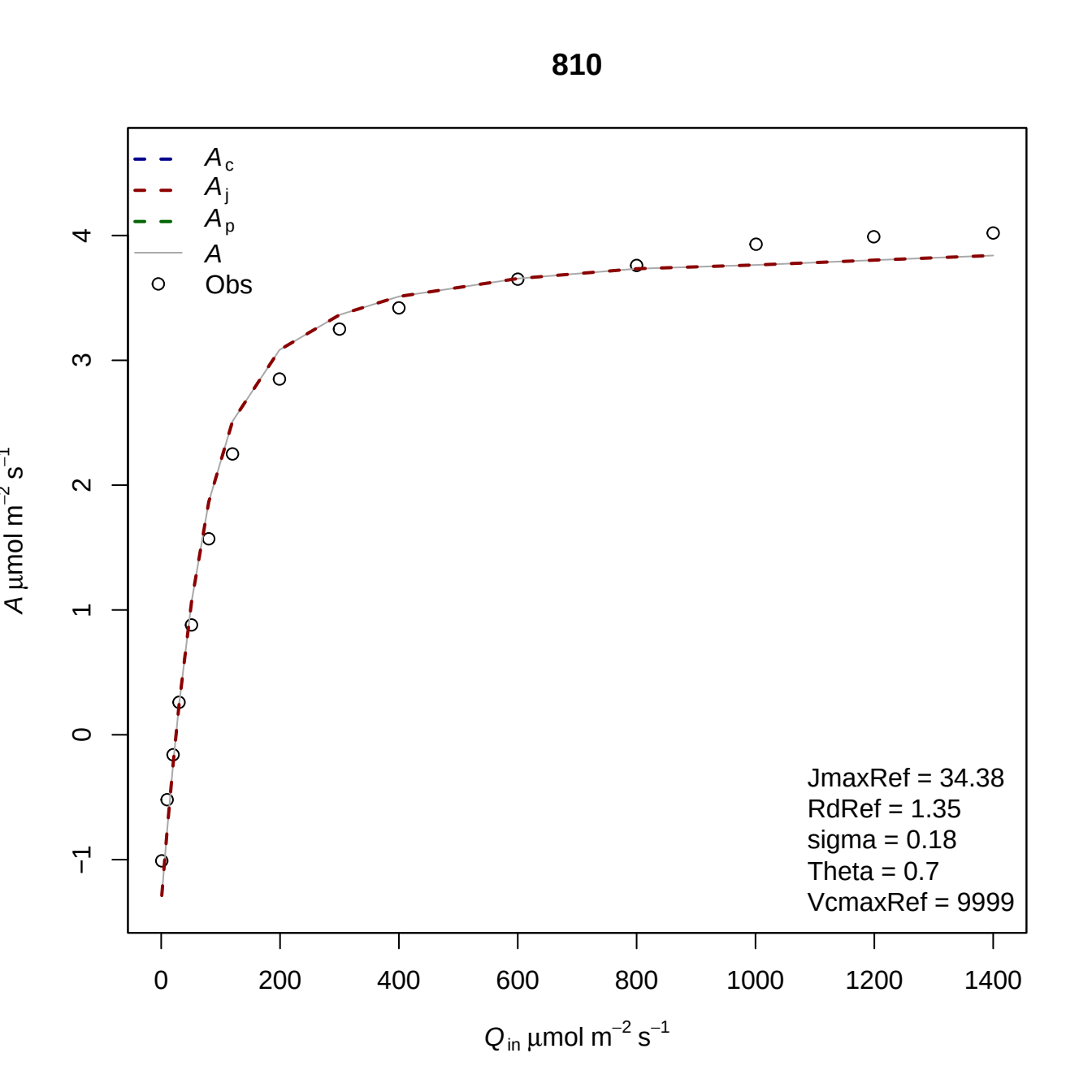

805
A_c
A_j
A_p
A
Obs
JmaxRef = 63.96
RdRef = 0.66
sigma = 0.24
Theta = 0.7
VcmaxRef = 44.35
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

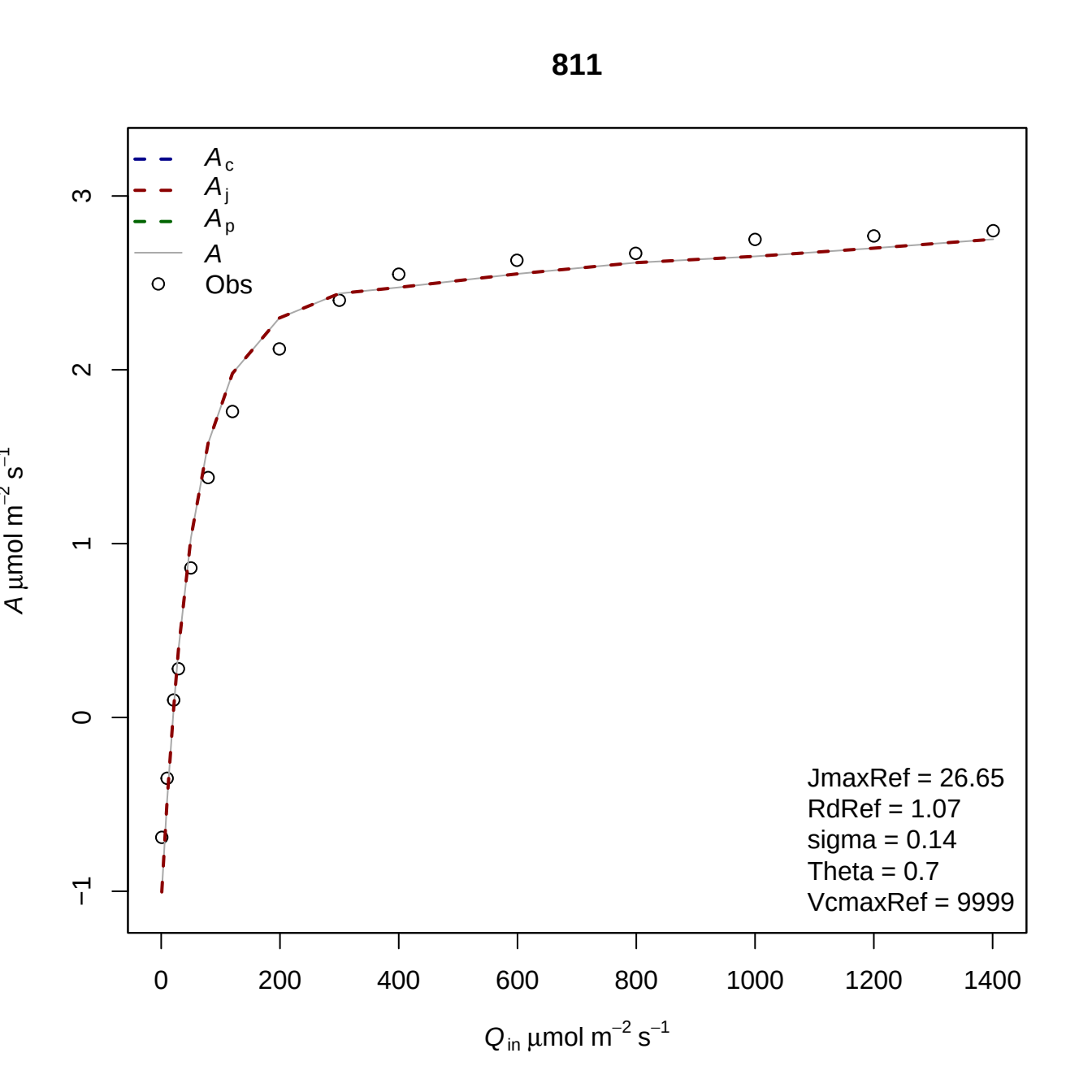

806
A_c
A_j
A_p
A
Obs
JmaxRef = 68.76
RdRef = 2.18
sigma = 0.49
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

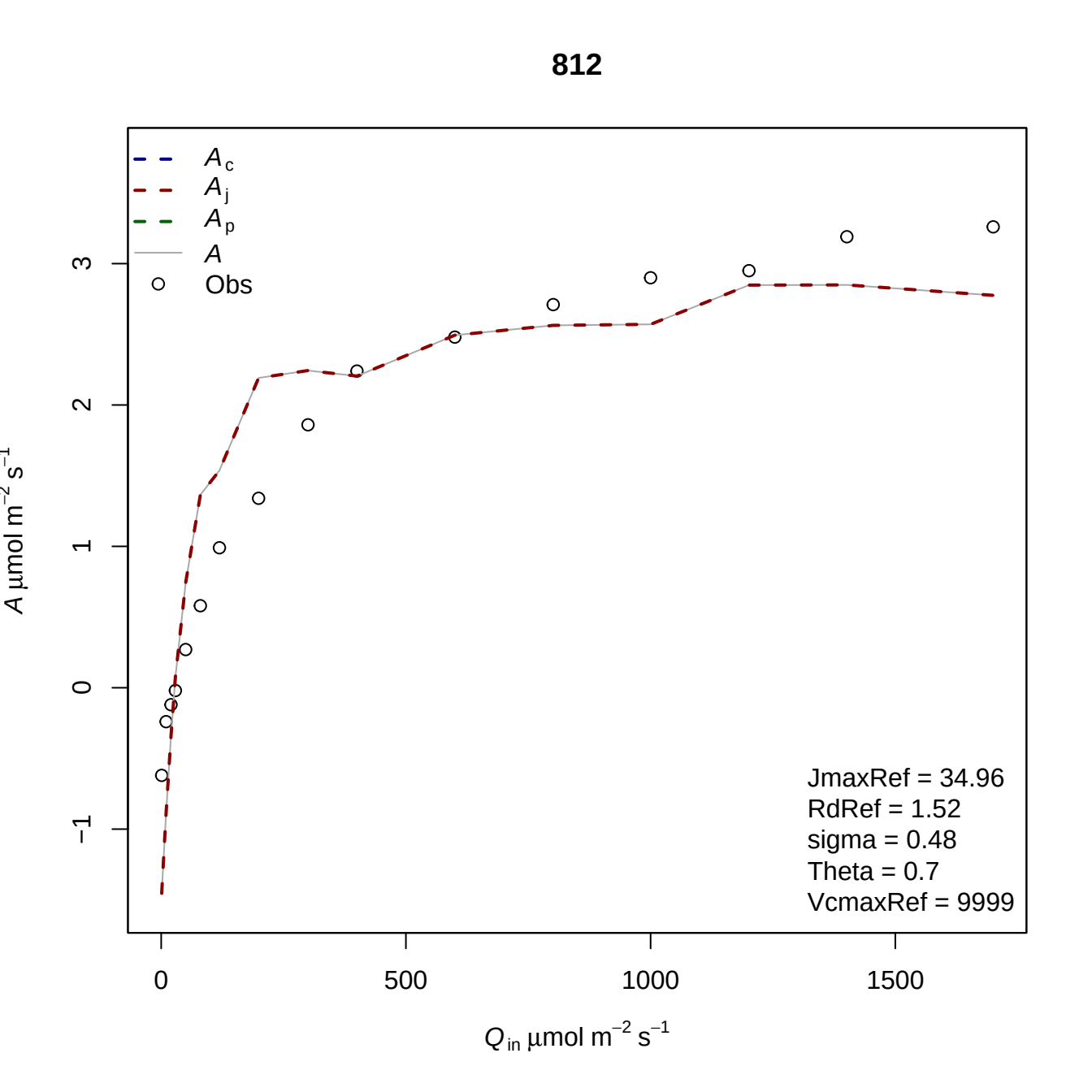

807
A_c
A_j
A_p
A
Obs
JmaxRef = 51.93
RdRef = 2.75
sigma = 0.48
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
200
400
600
800
1000
1200

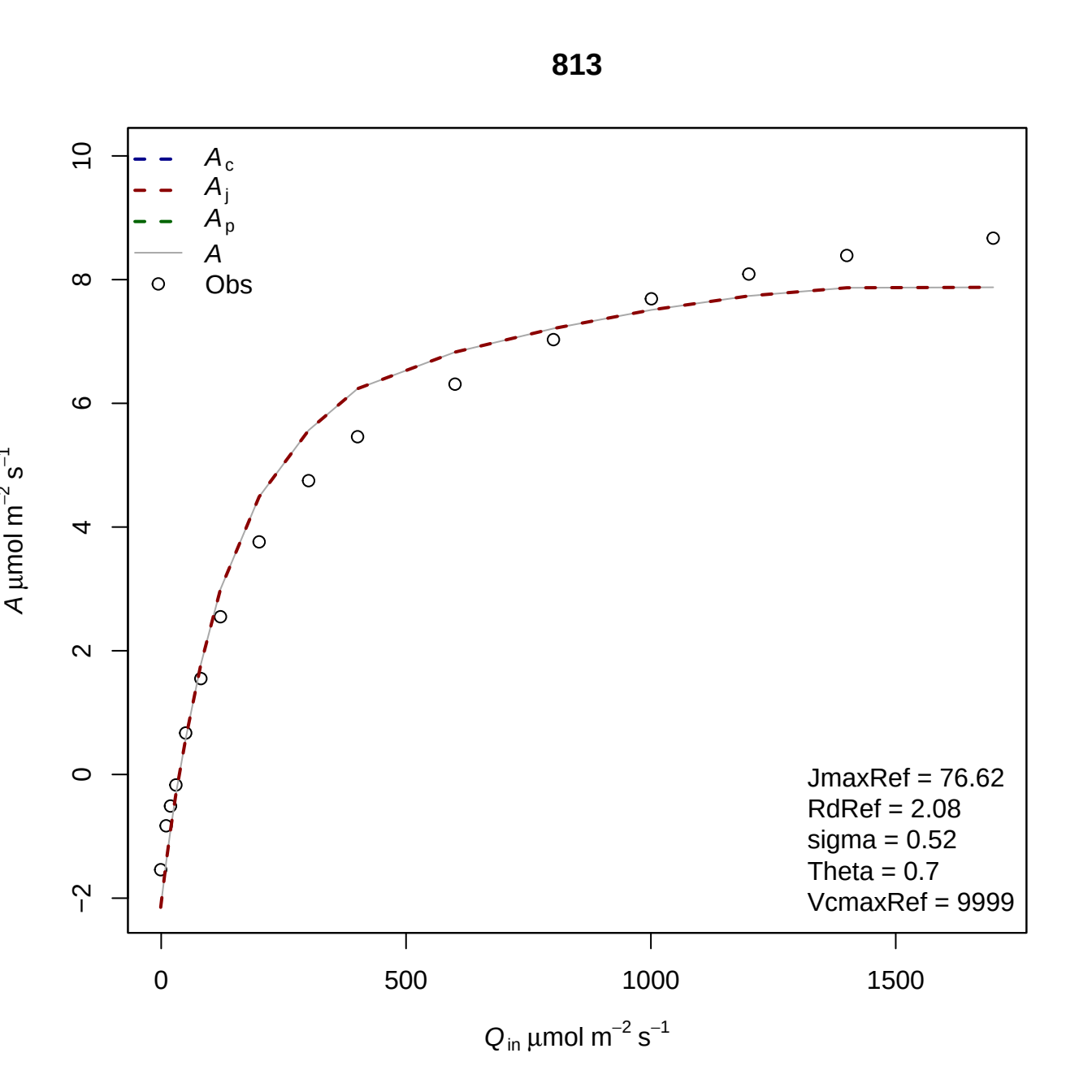

808
A_c
A_j
A_p
A
Obs
JmaxRef = 27.31
RdRef = 1.32
sigma = 0.18
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
0
500
1000
1500

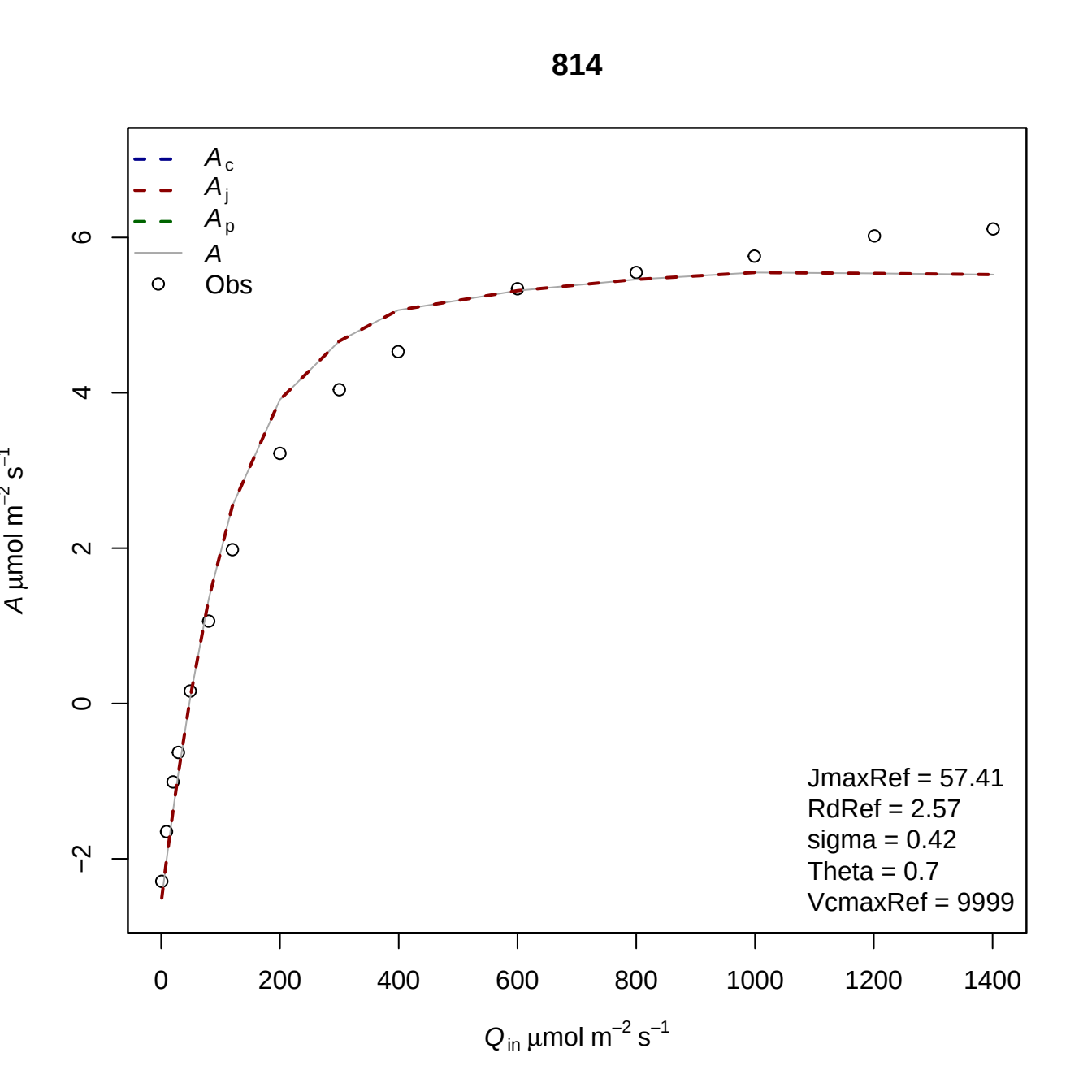

809
A_c
A_j
A_p
A
Obs
JmaxRef = 66.93
RdRef = 2.61
sigma = 0.28
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
200
400
600
800
1000

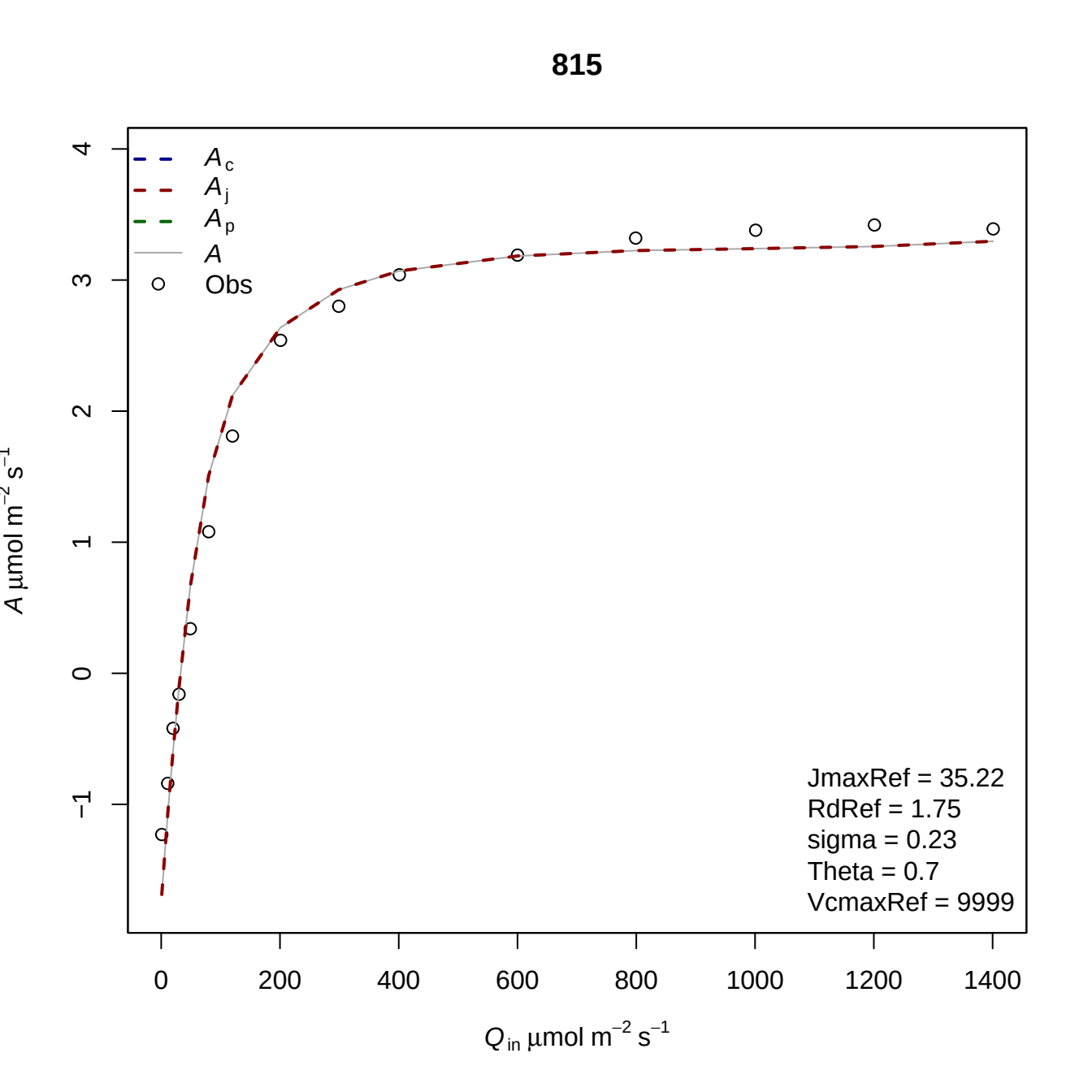

810
A_c
A_j
A_p
A
Obs
JmaxRef = 34.38
RdRef = 1.35
sigma = 0.18
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000
1200
1400

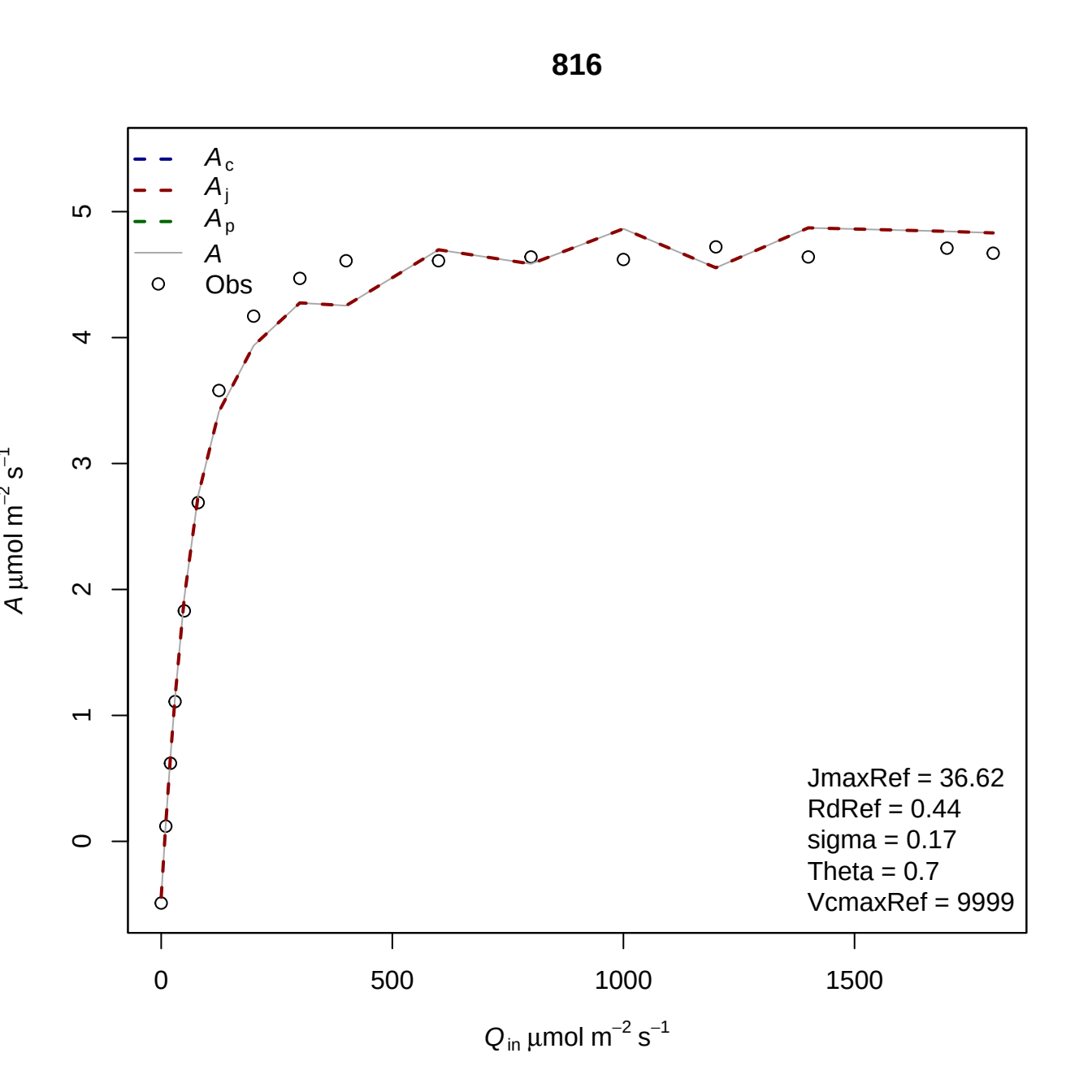

811
A_c
A_j
A_p
A
Obs
JmaxRef = 26.65
RdRef = 1.07
sigma = 0.14
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800
1000
1200
1400

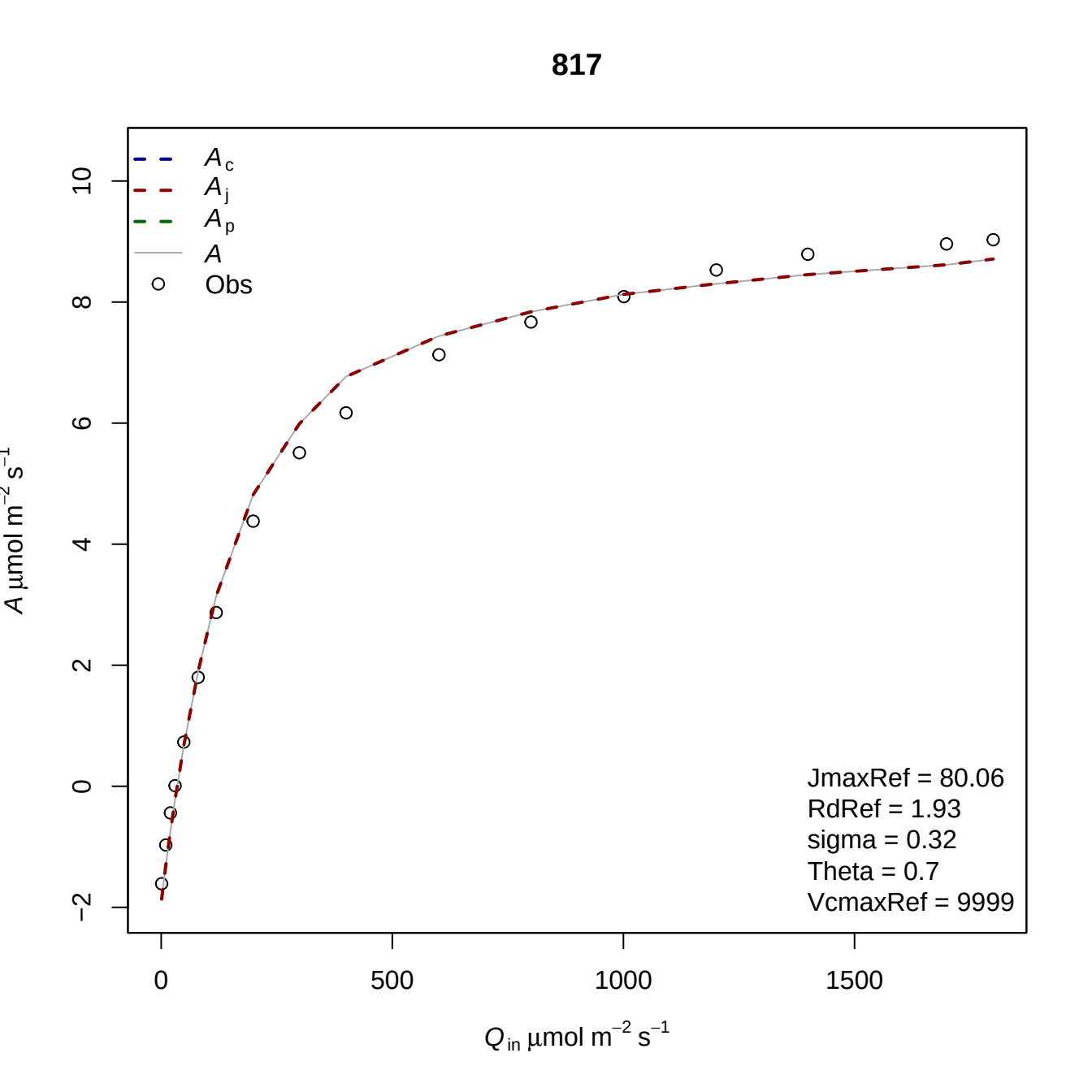

812
A_c
A_j
A_p
A
Obs
JmaxRef = 34.96
RdRef = 1.52
sigma = 0.48
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

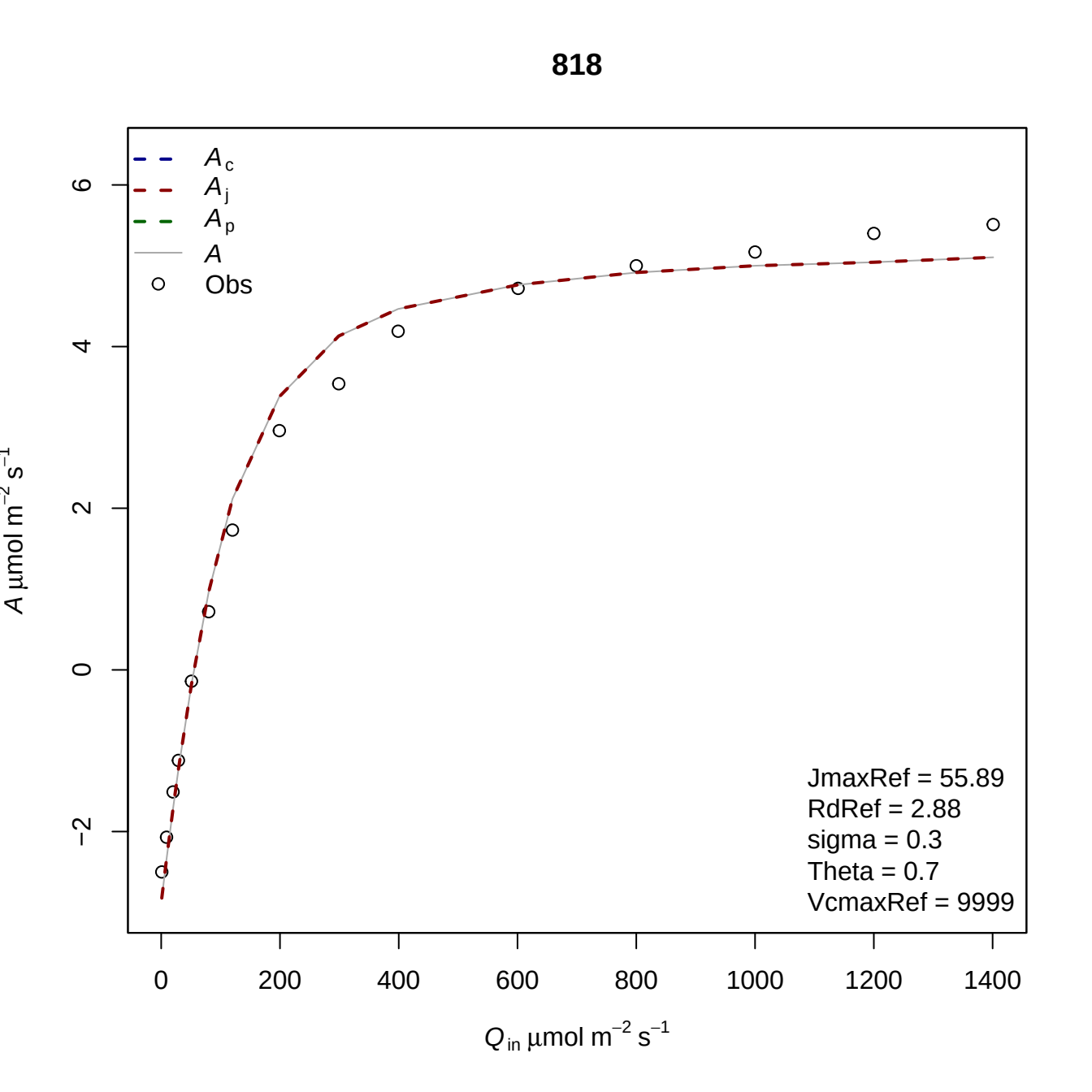

813
A_c
A_j
A_p
A
Obs
JmaxRef = 76.62
RdRef = 2.08
sigma = 0.52
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

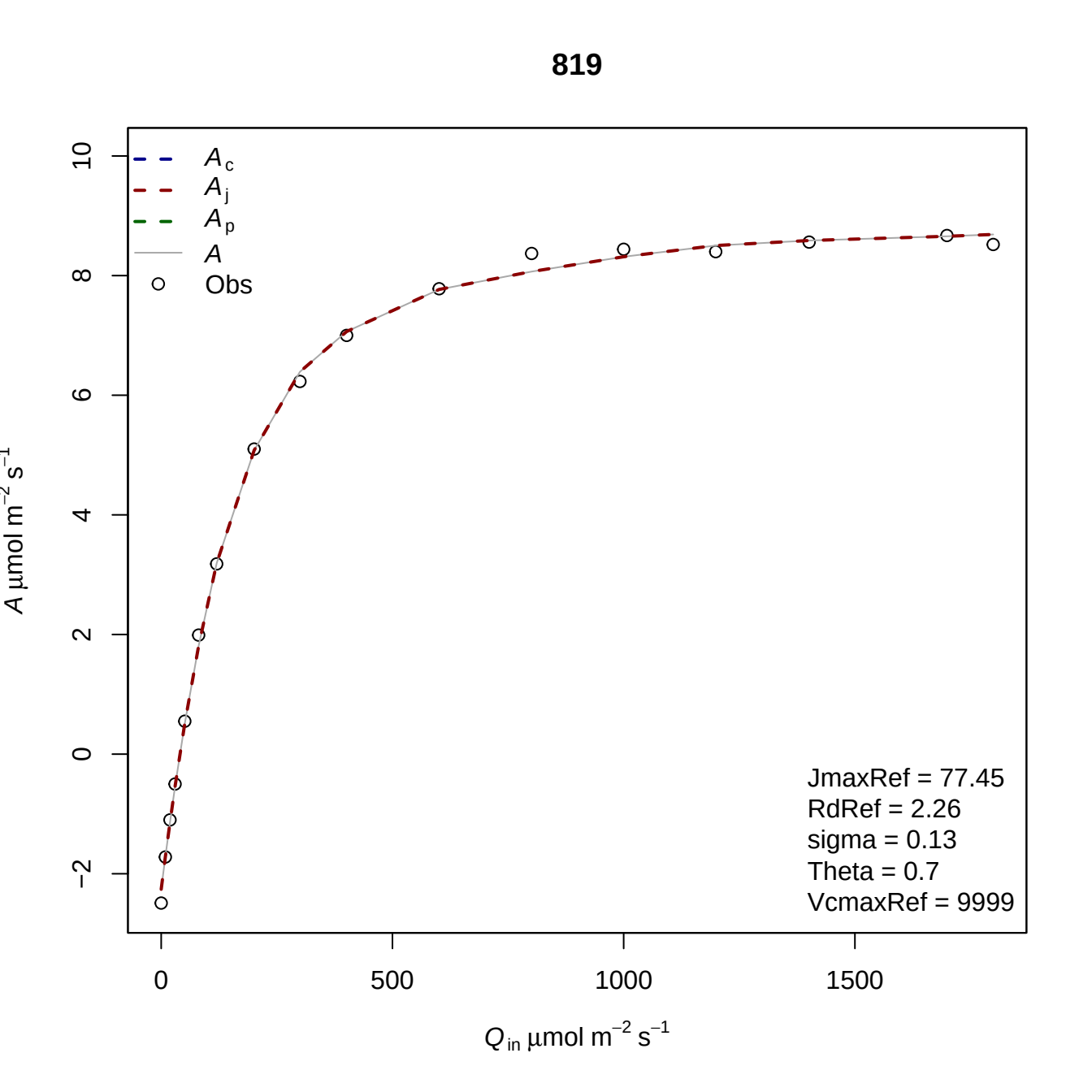

814
A_c
A_j
A_p
A
Obs
JmaxRef = 57.41
RdRef = 2.57
sigma = 0.42
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000
1200
1400

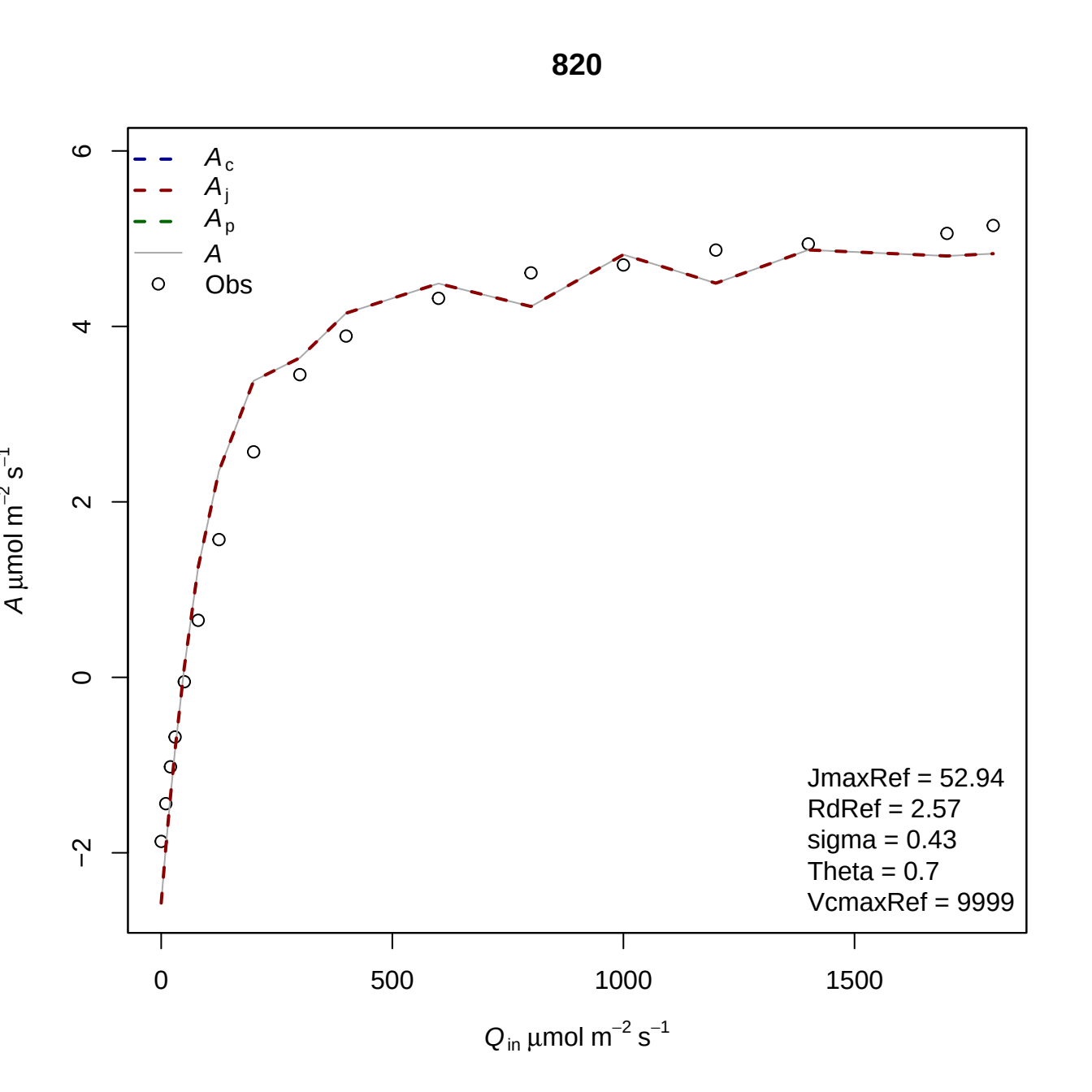

815
A_c
A_j
A_p
A
Obs
JmaxRef = 35.22
RdRef = 1.75
sigma = 0.23
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000
1200
1400

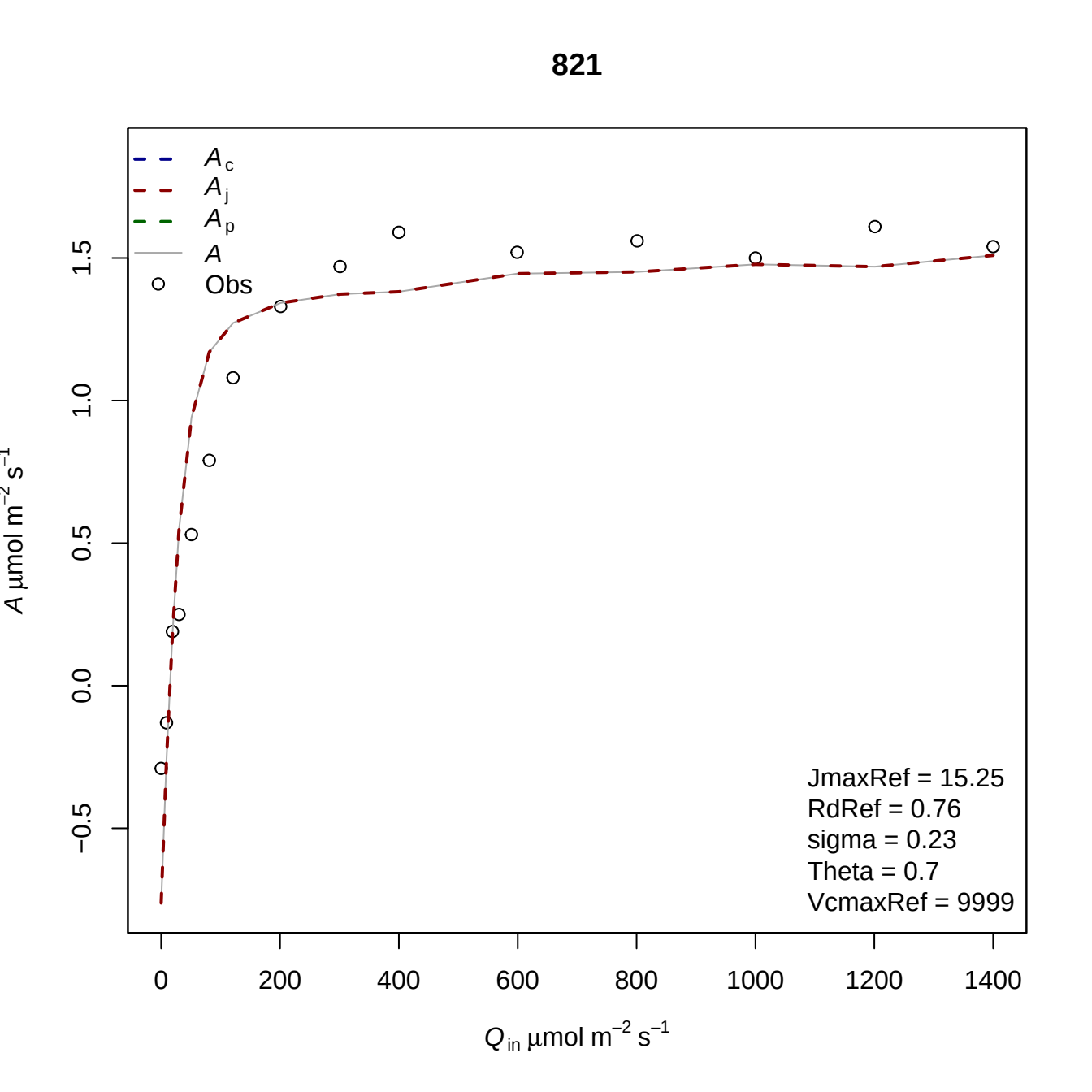

816
A_c
A_j
A_p
A
Obs
JmaxRef = 36.62
RdRef = 0.44
sigma = 0.17
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

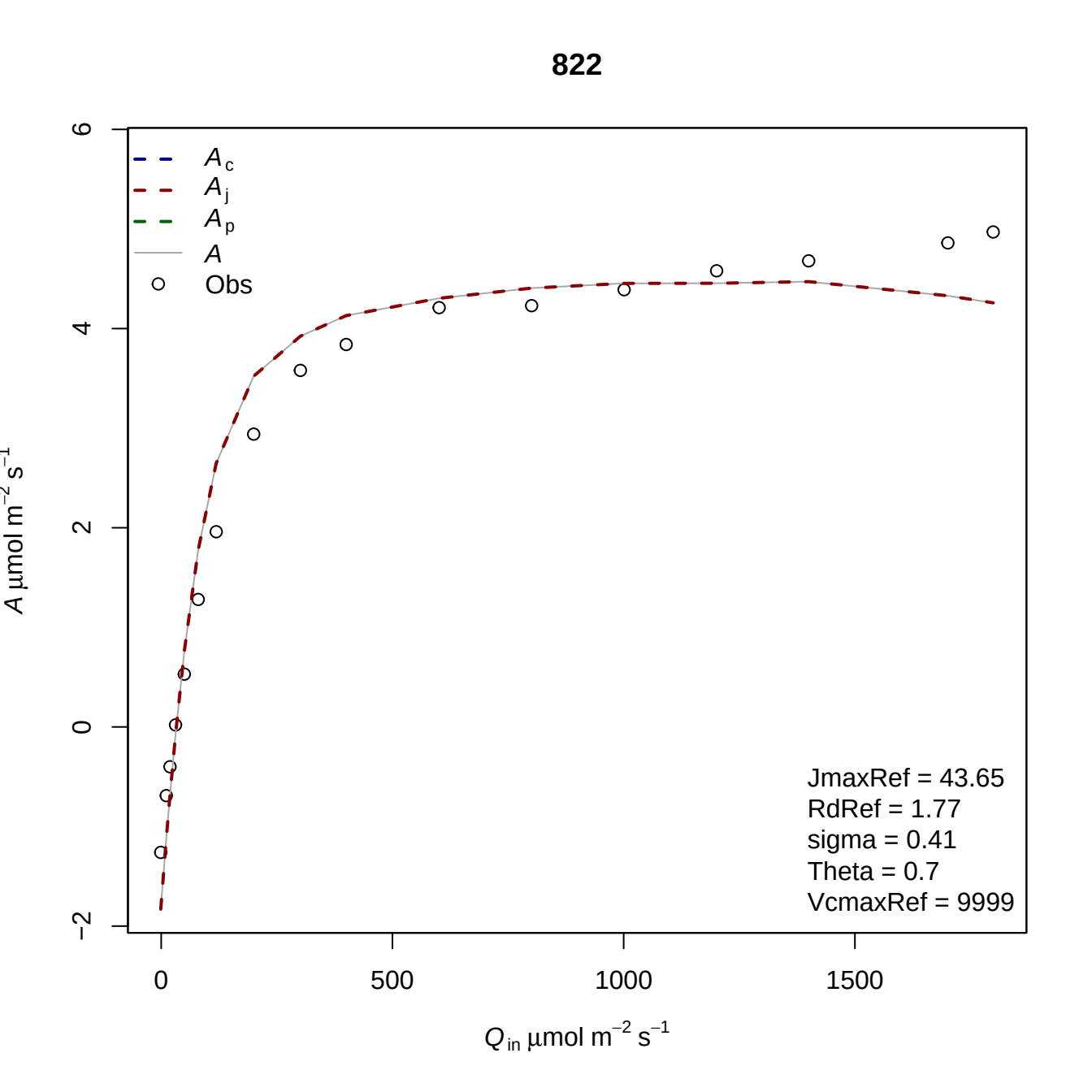

817
A_c
A_j
A_p
A
Obs
JmaxRef = 80.06
RdRef = 1.93
sigma = 0.32
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

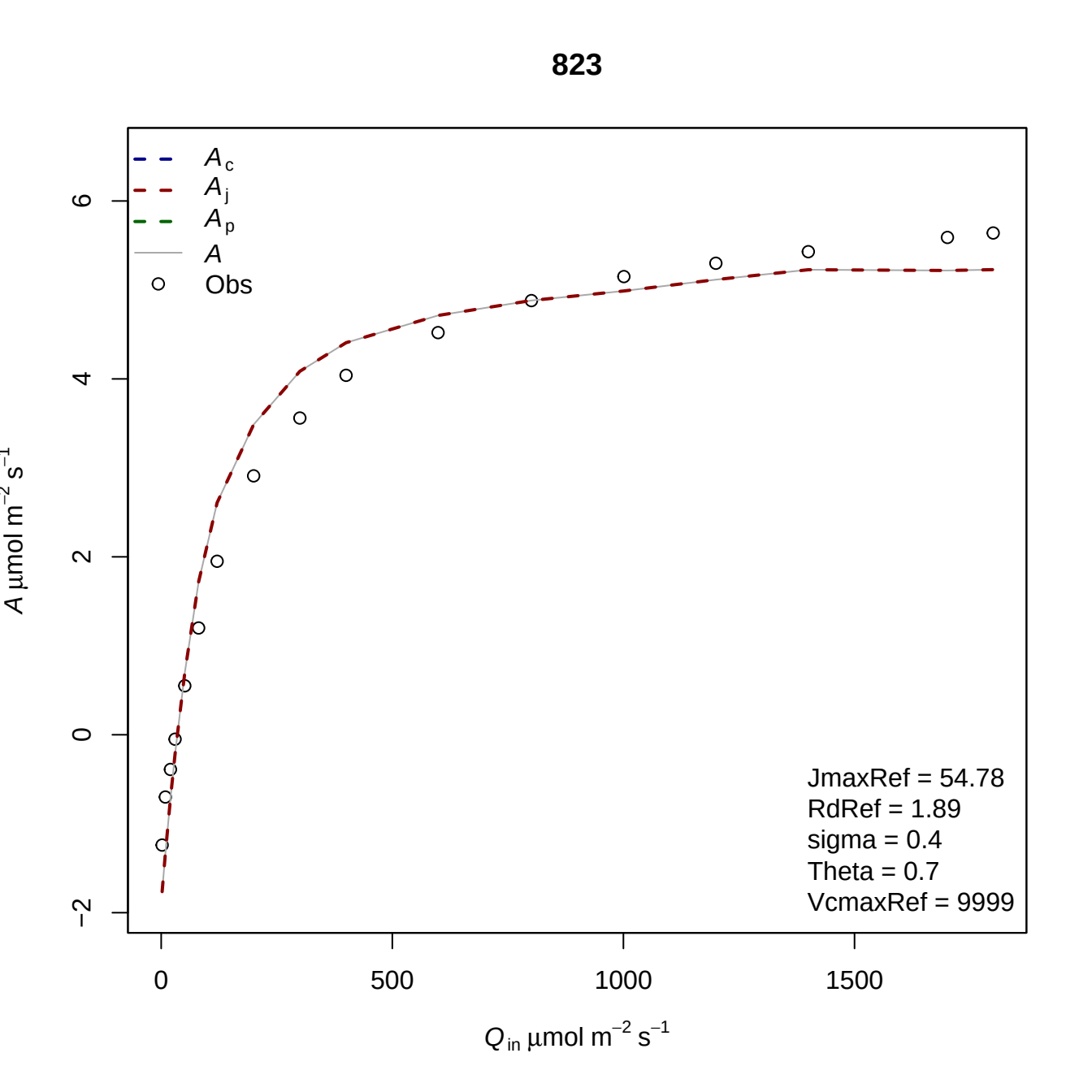

818
A_c
A_j
A_p
A
Obs
JmaxRef = 55.89
RdRef = 2.88
sigma = 0.3
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
200
400
600
800
1000
1200
1400

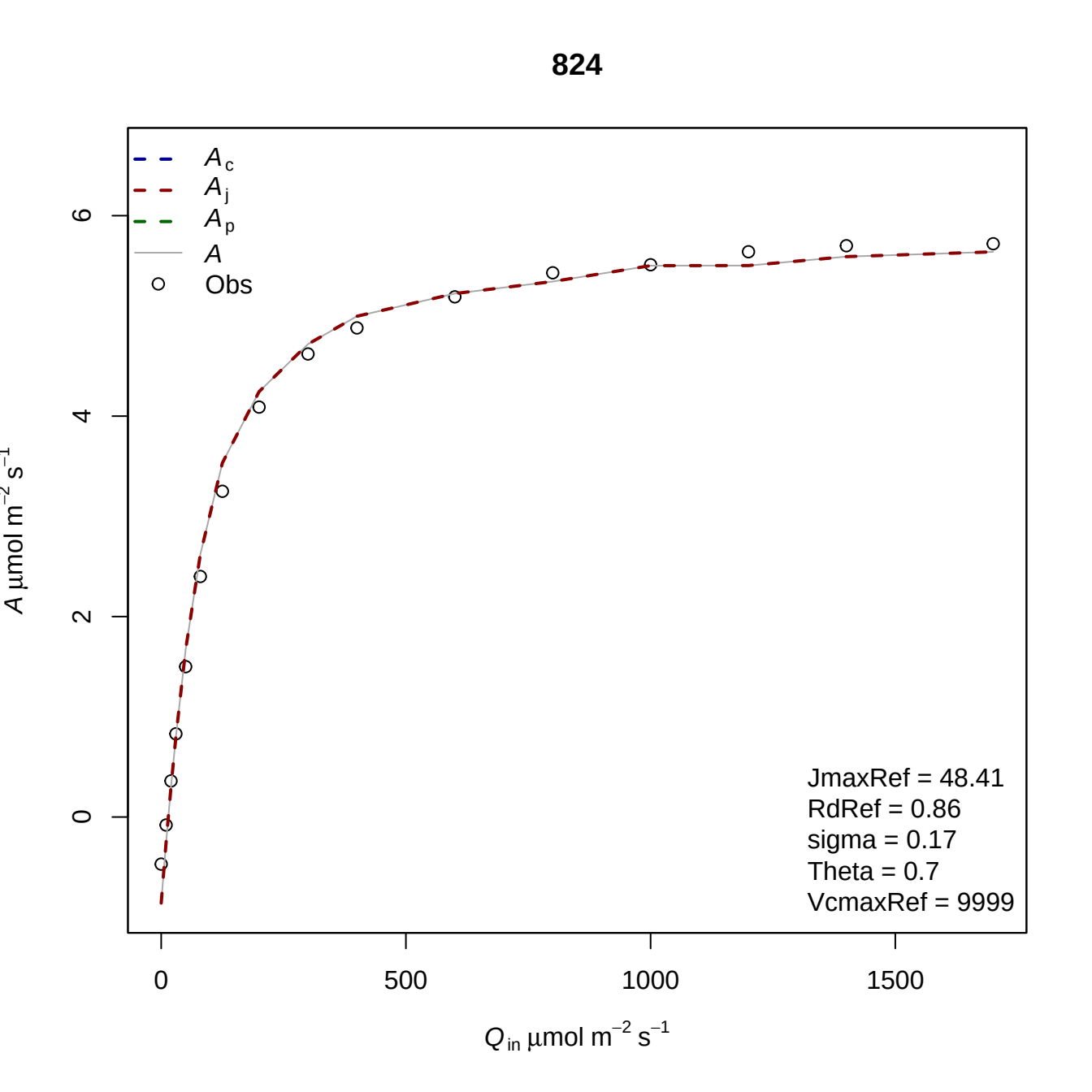

819
A_c
A_j
A_p
A
Obs
JmaxRef = 77.45
RdRef = 2.26
sigma = 0.13
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

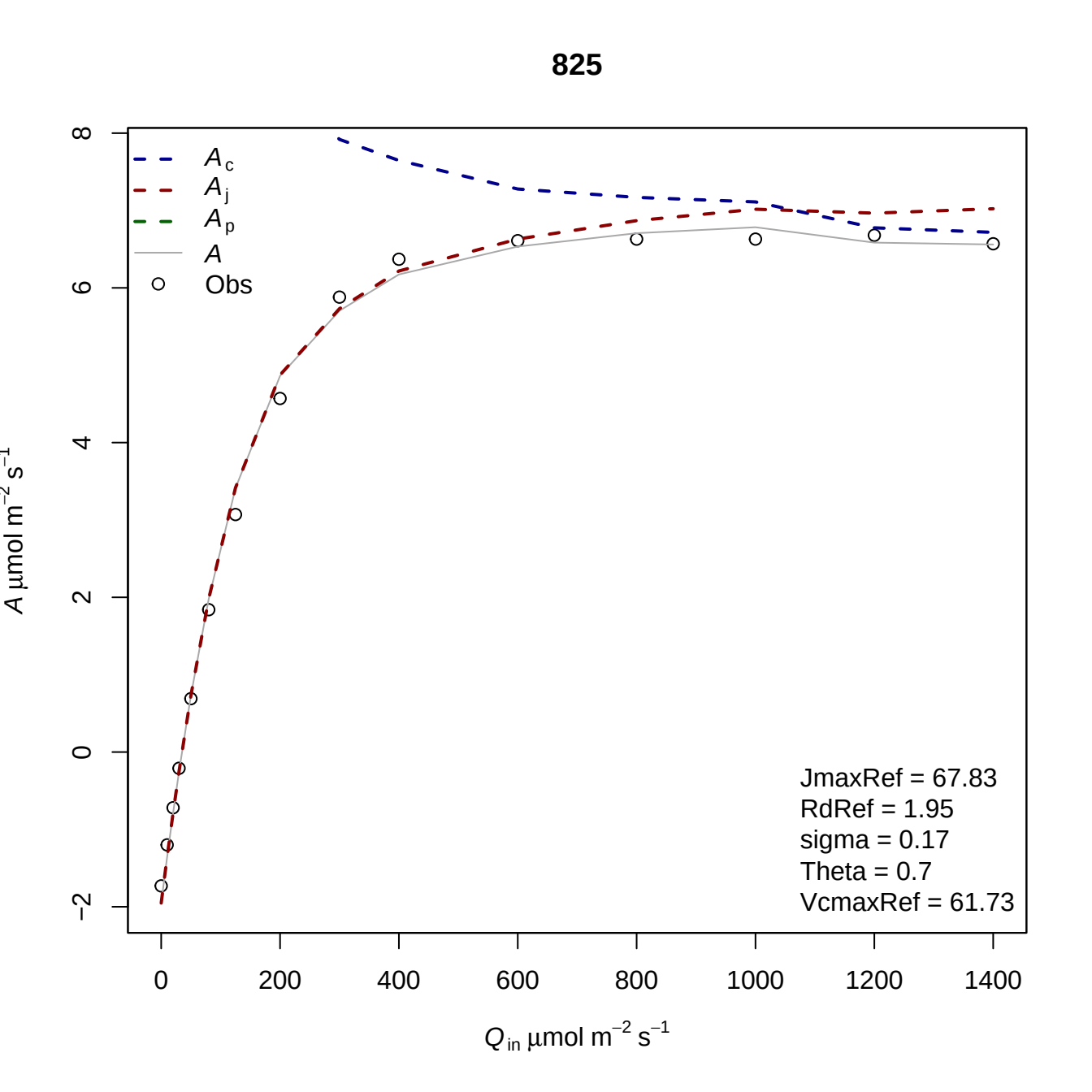

820
A_c
A_j
A_p
A
Obs
JmaxRef = 52.94
RdRef = 2.57
sigma = 0.43
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

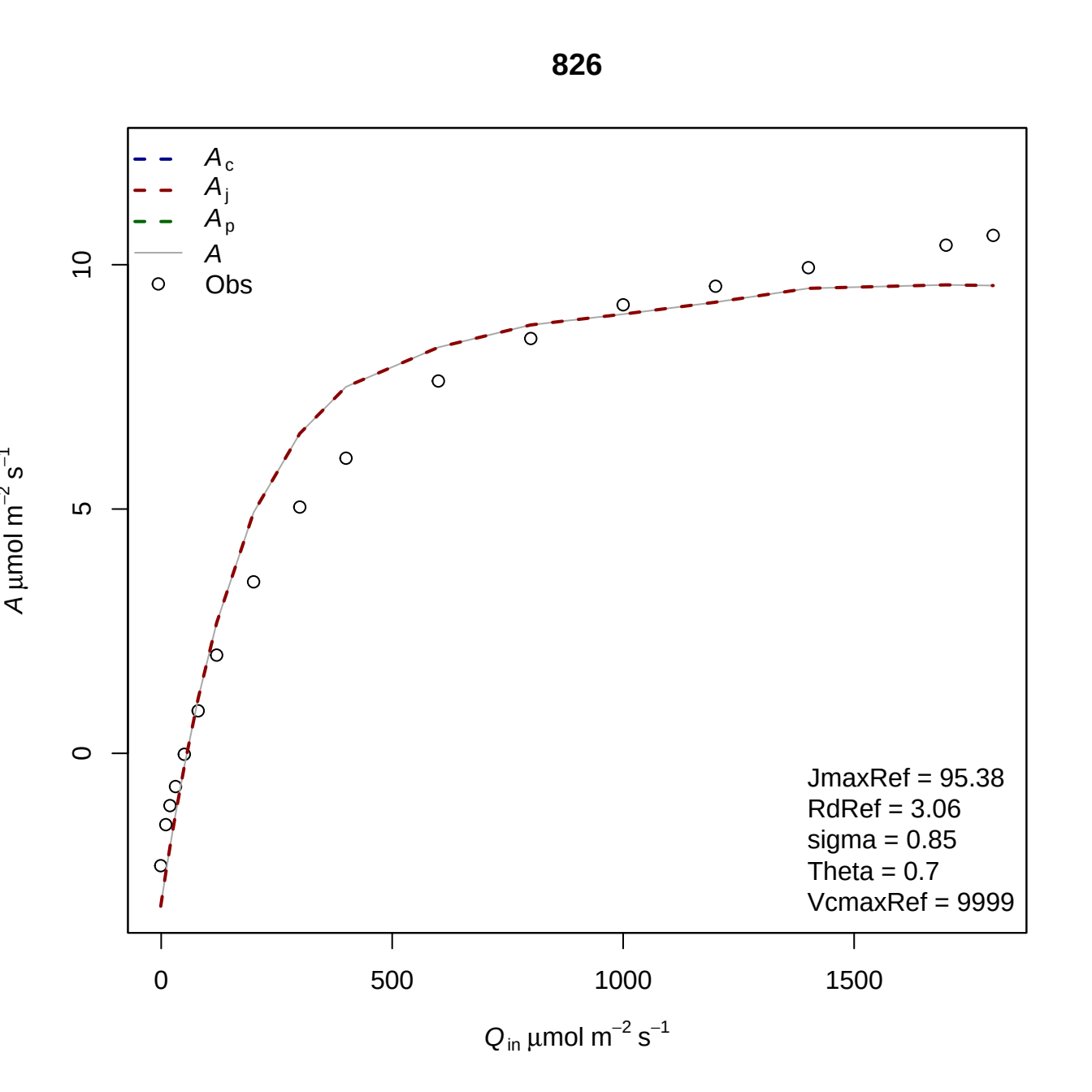

821
A_c
A_j
A_p
A
Obs
JmaxRef = 15.25
RdRef = 0.76
sigma = 0.23
Theta = 0.7
VcmaxRef = 9999
−0.5
0.0
0.5
1.0
1.5
0
200
400
600
800
1000
1200
1400
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

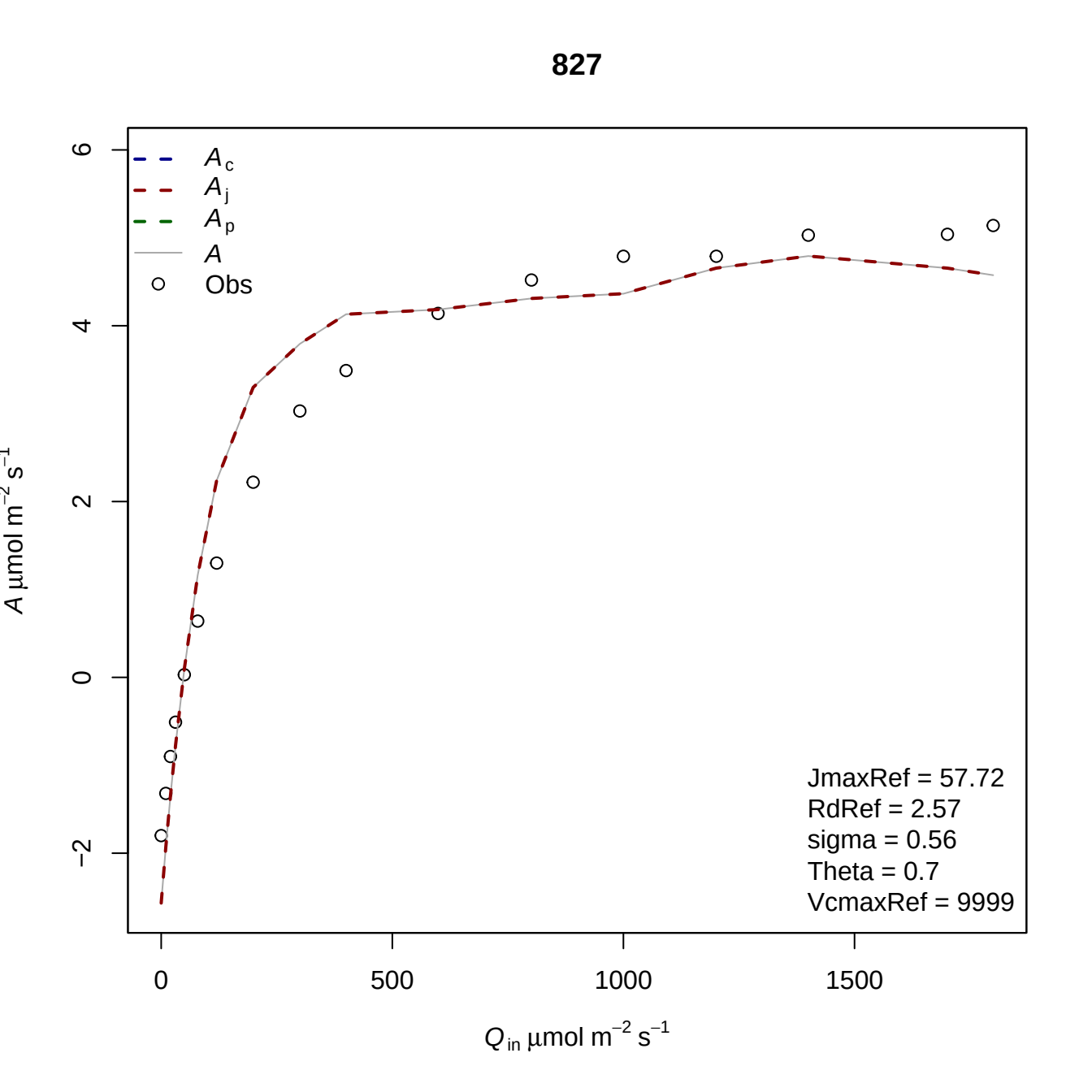

822
A_c
A_j
A_p
A
Obs
JmaxRef = 43.65
RdRef = 1.77
sigma = 0.41
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

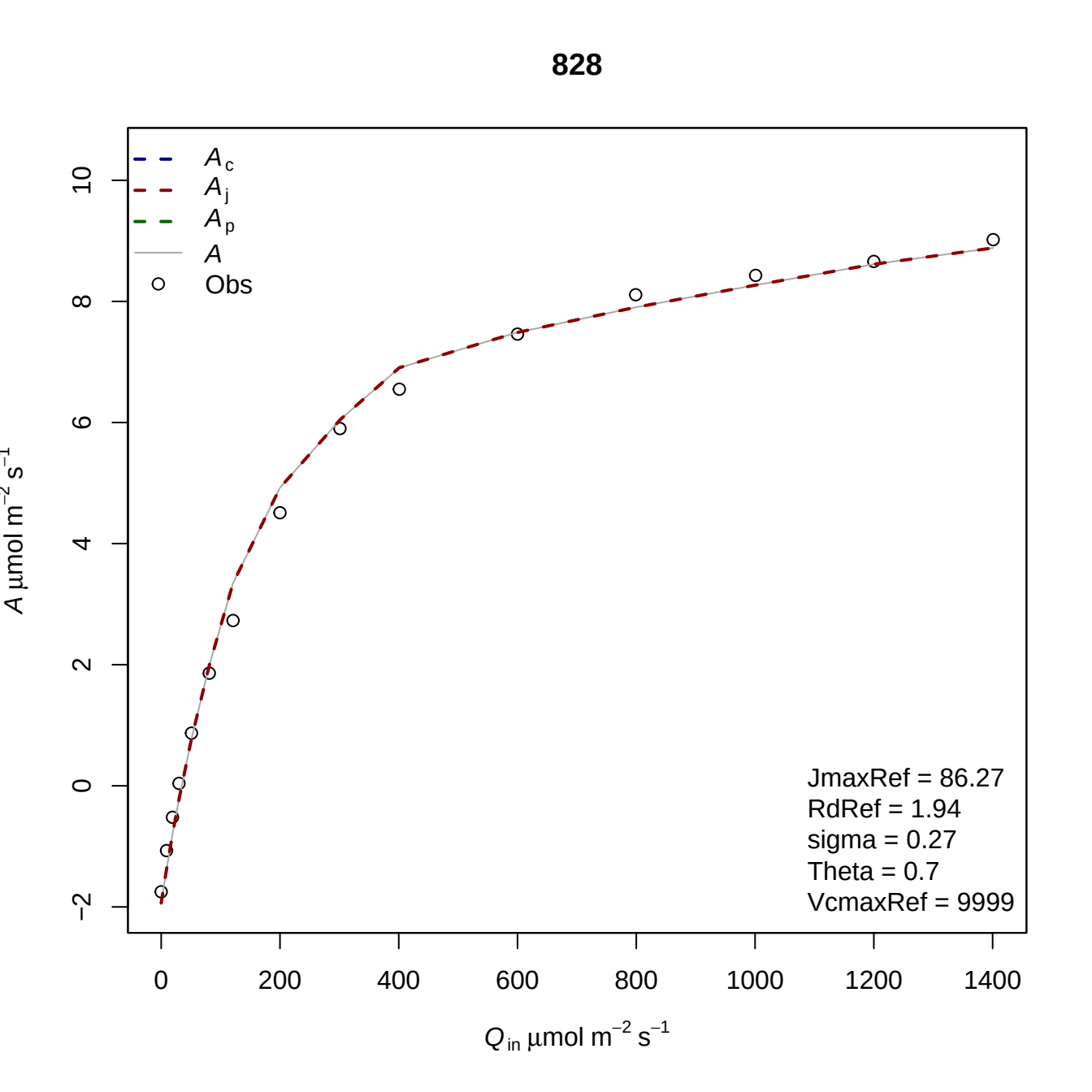

823
A_c
A_j
A_p
A
Obs
JmaxRef = 54.78
RdRef = 1.89
sigma = 0.4
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

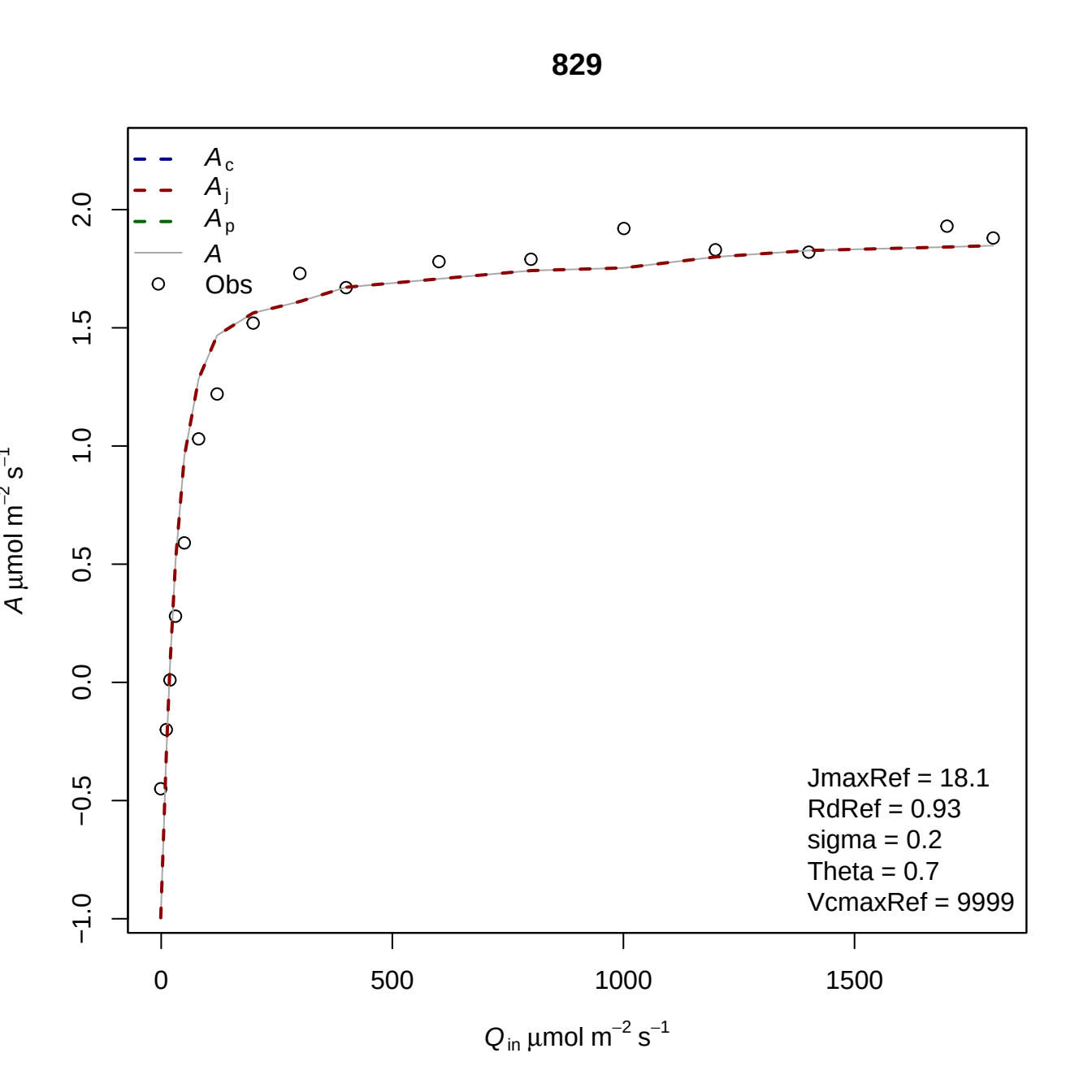

824
A_c
A_j
A_p
A
Obs
JmaxRef = 48.41
RdRef = 0.86
sigma = 0.17
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

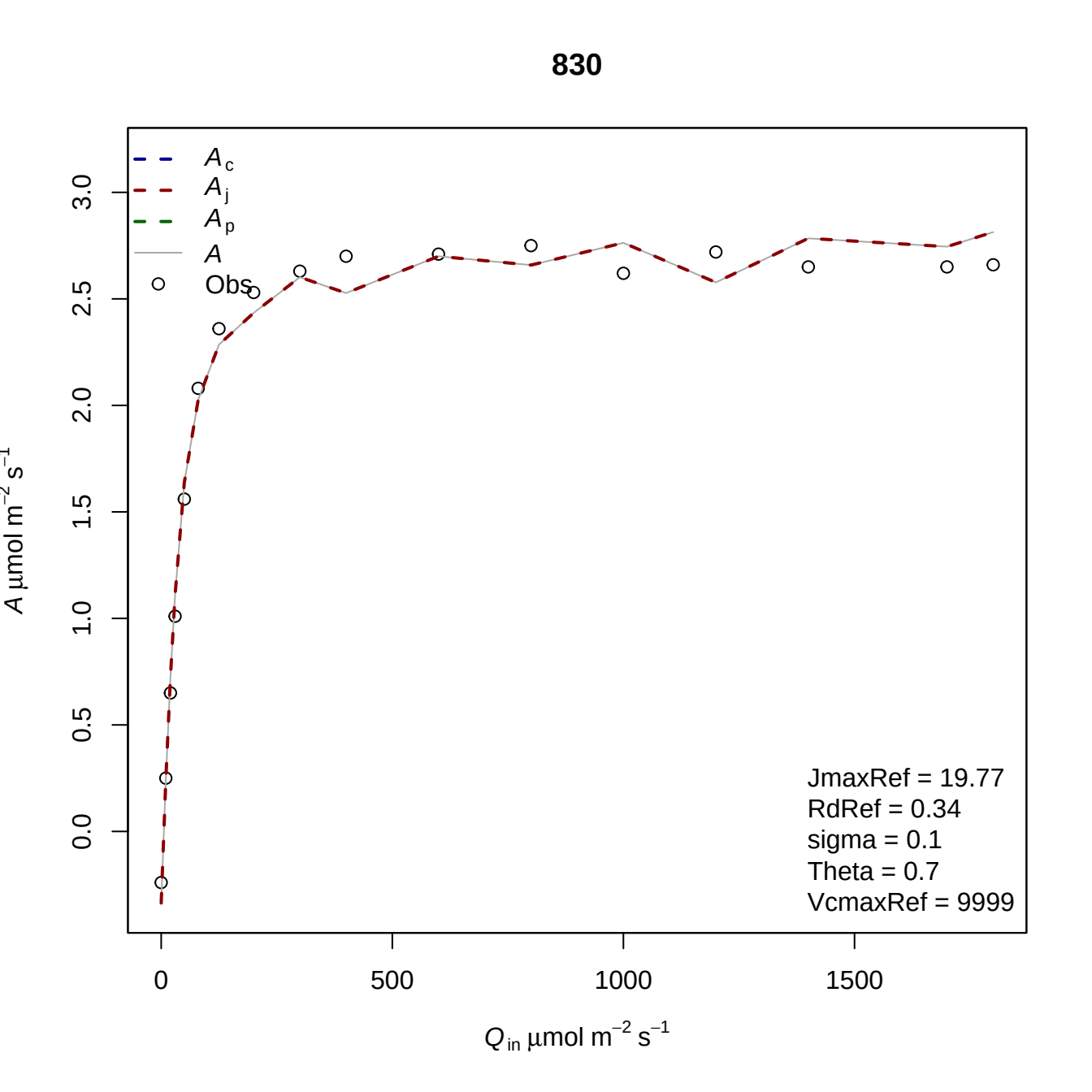

825
A_c
A_j
A_p
A
Obs
JmaxRef = 67.83
RdRef = 1.95
sigma = 0.17
Theta = 0.7
VcmaxRef = 61.73
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
200
400
600
800
1000
1200
1400

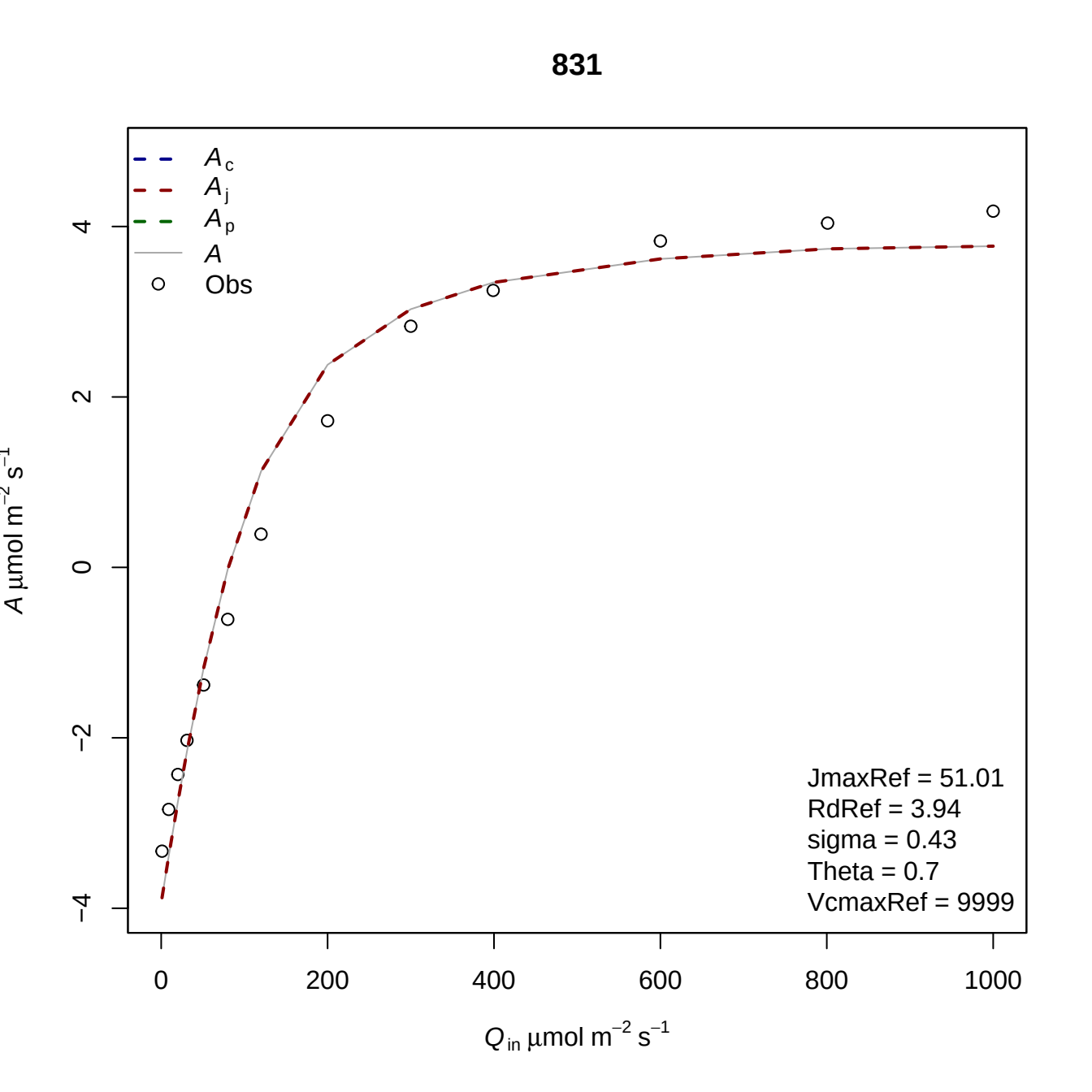

826
A_c
A_j
A_p
A
Obs
JmaxRef = 95.38
RdRef = 3.06
sigma = 0.85
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

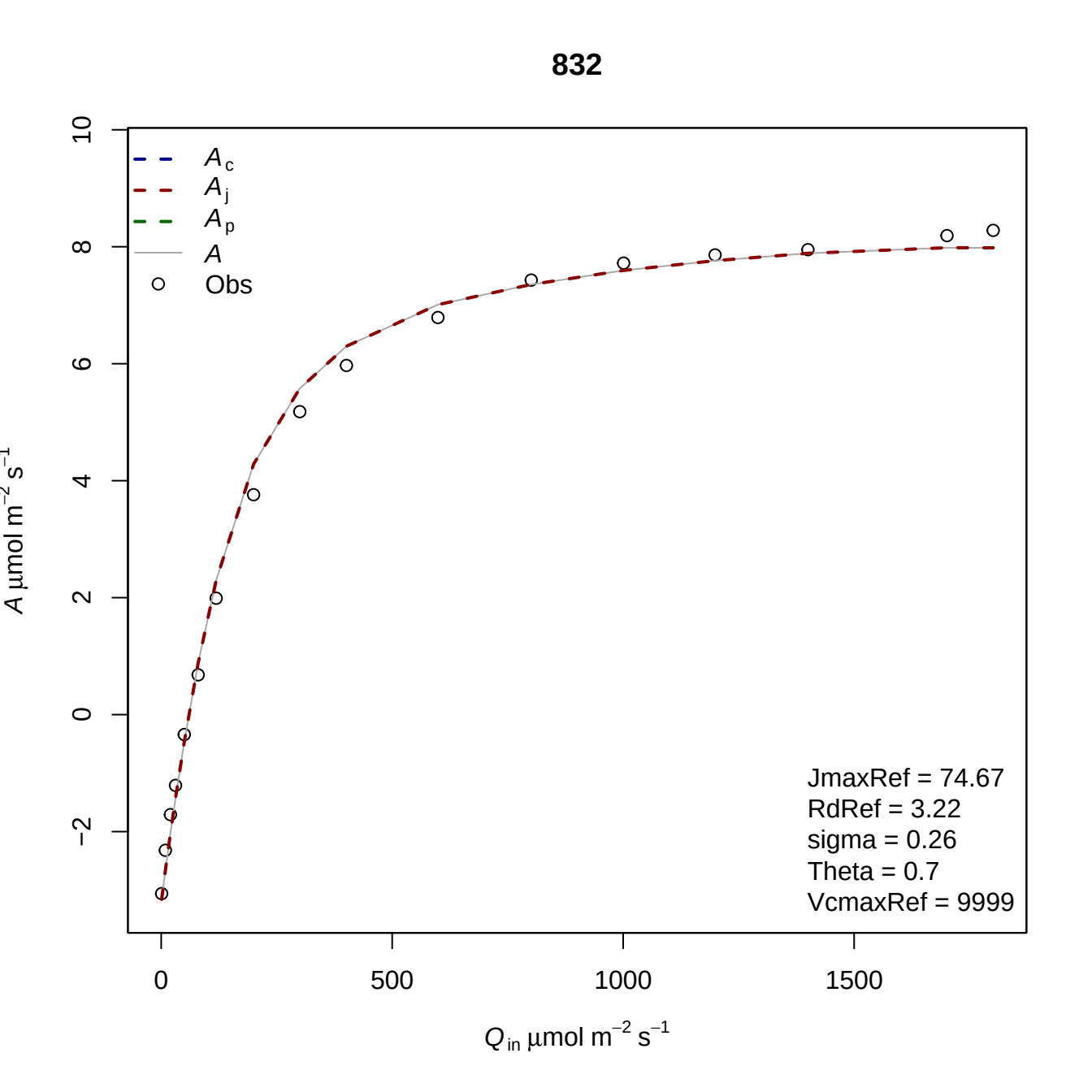

827
A_c
A_j
A_p
A
Obs
JmaxRef = 57.72
RdRef = 2.57
sigma = 0.56
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

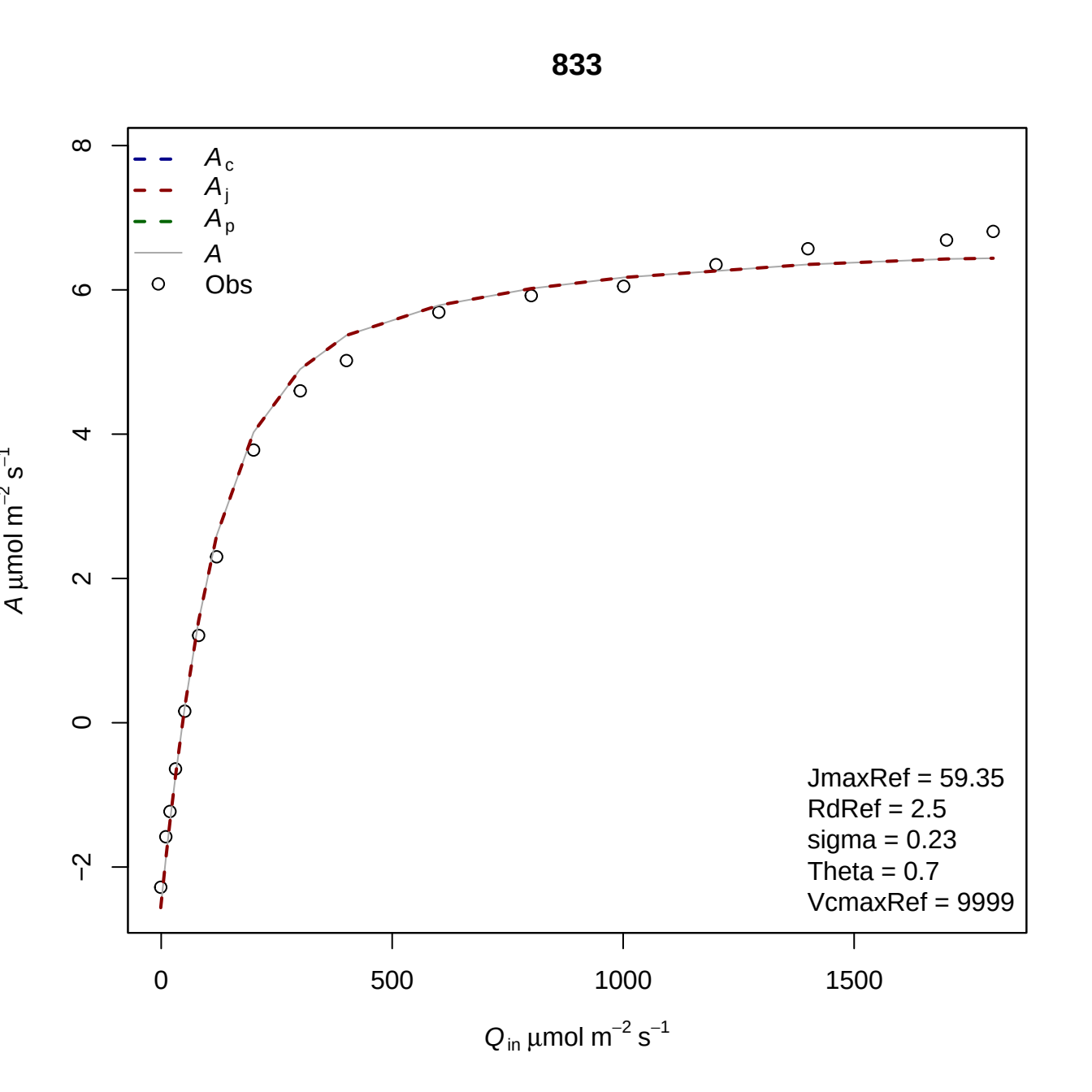

828
A_c
A_j
A_p
A
Obs
JmaxRef = 86.27
RdRef = 1.94
sigma = 0.27
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
200
400
600
800
1000
1200
1400

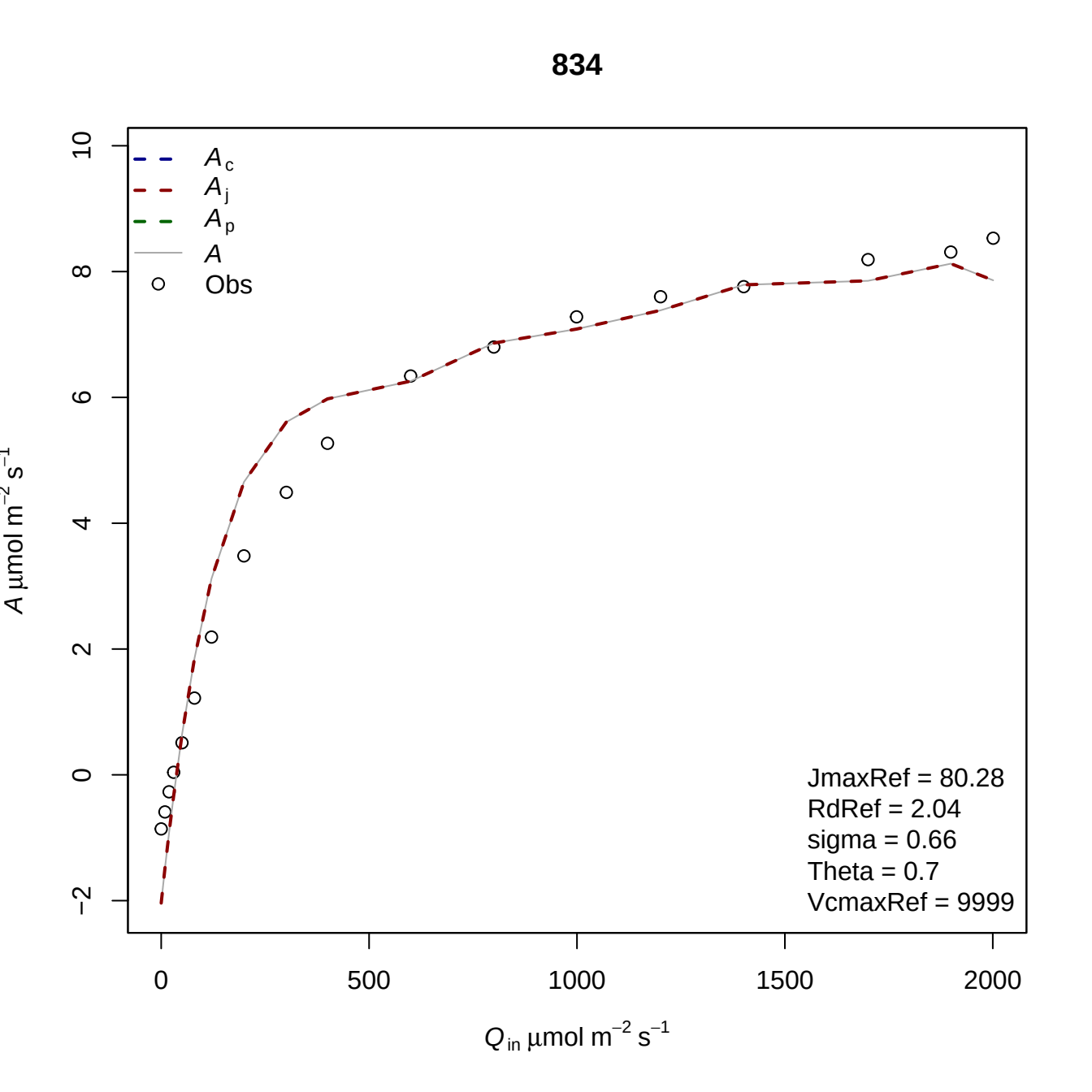

829
A_c
A_j
A_p
A
Obs
JmaxRef = 18.1
RdRef = 0.93
sigma = 0.2
Theta = 0.7
VcmaxRef = 9999
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

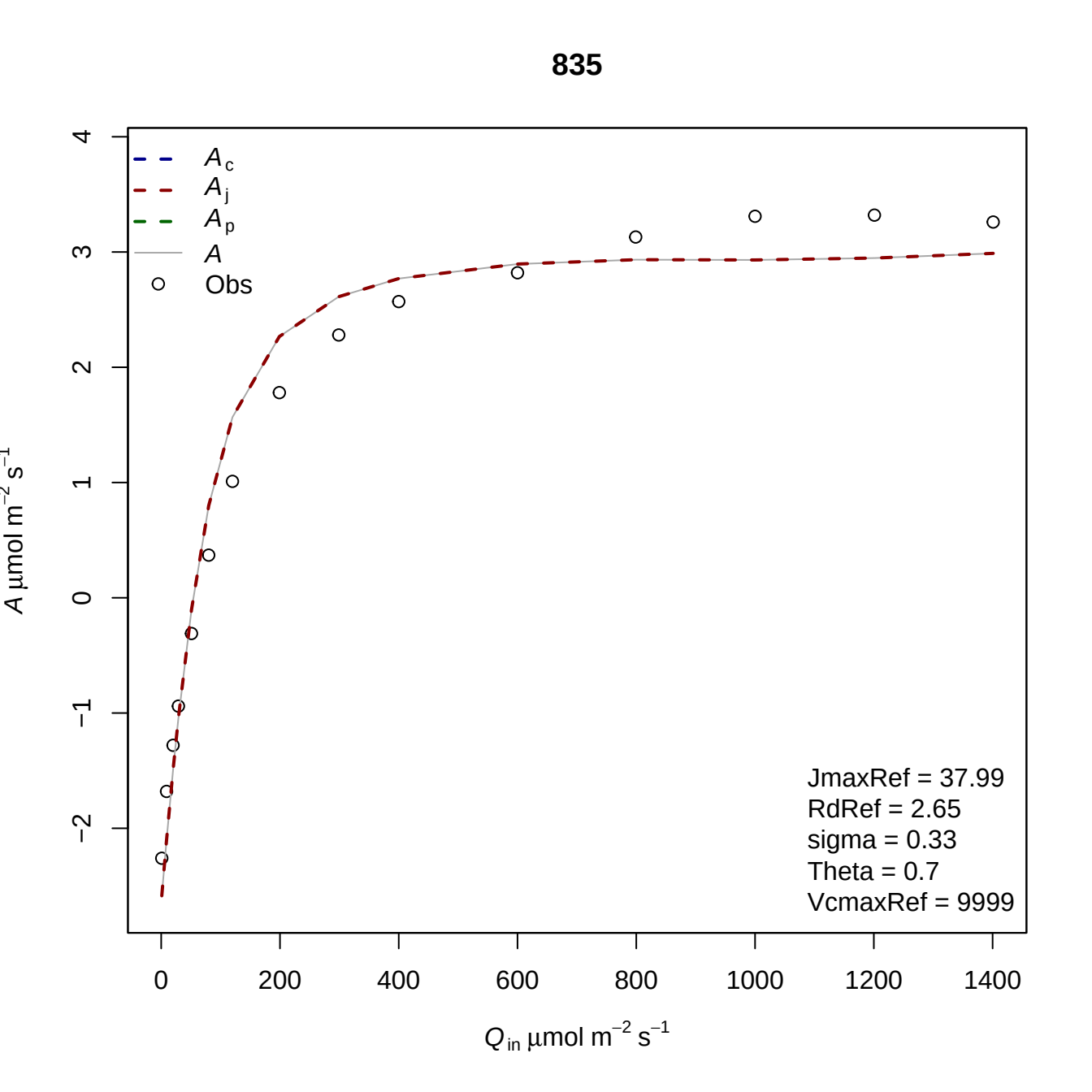

830
A_c
A_j
A_p
A
Obs
JmaxRef = 19.77
RdRef = 0.34
sigma = 0.1
Theta = 0.7
VcmaxRef = 9999
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

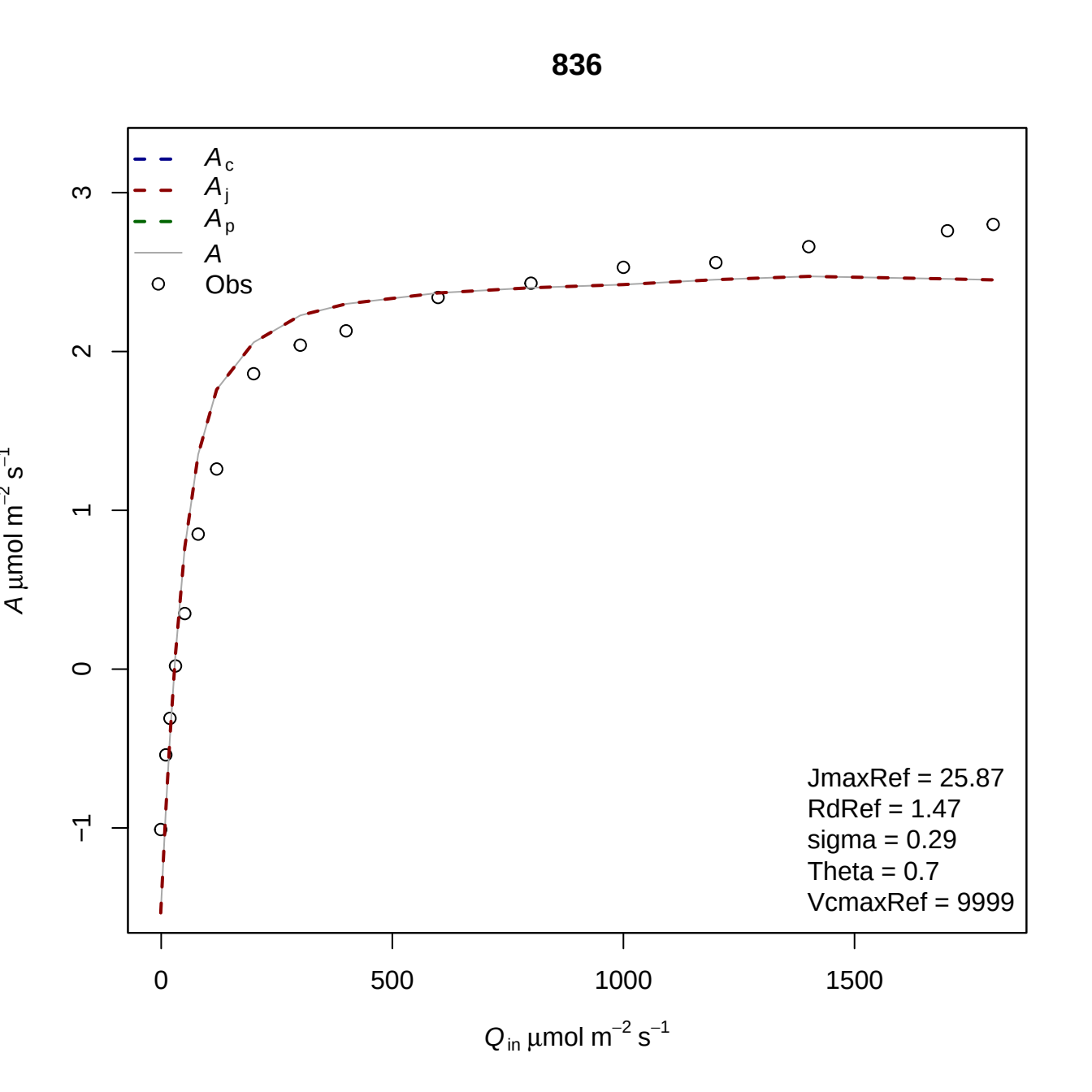

831
A_c
A_j
A_p
A
Obs
JmaxRef = 51.01
RdRef = 3.94
sigma = 0.43
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
−4
0
200
400
600
800
1000

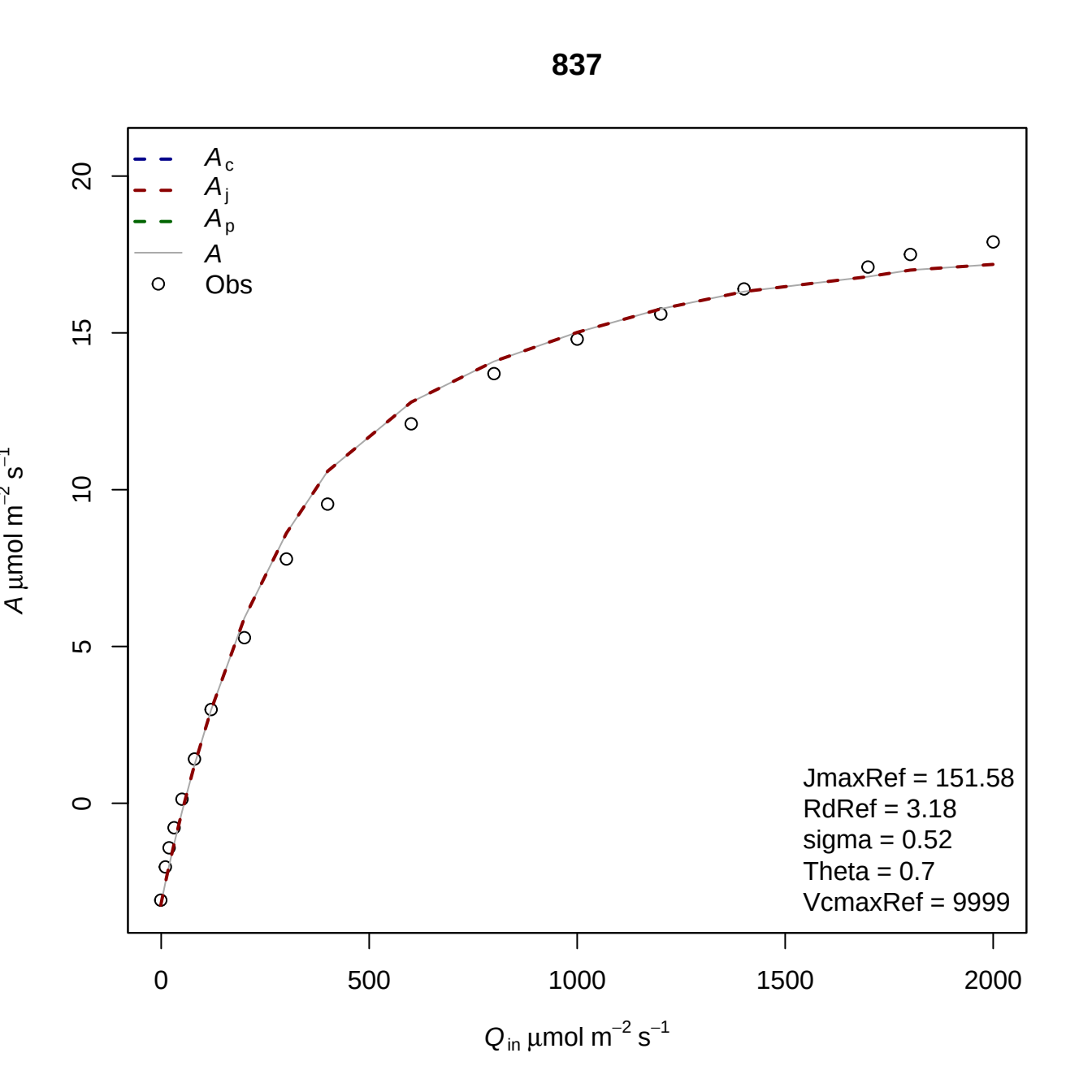

832
A_c
A_j
A_p
A
Obs
JmaxRef = 74.67
RdRef = 3.22
sigma = 0.26
Theta = 0.7
VcmaxRef = 9999
10
8
6
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

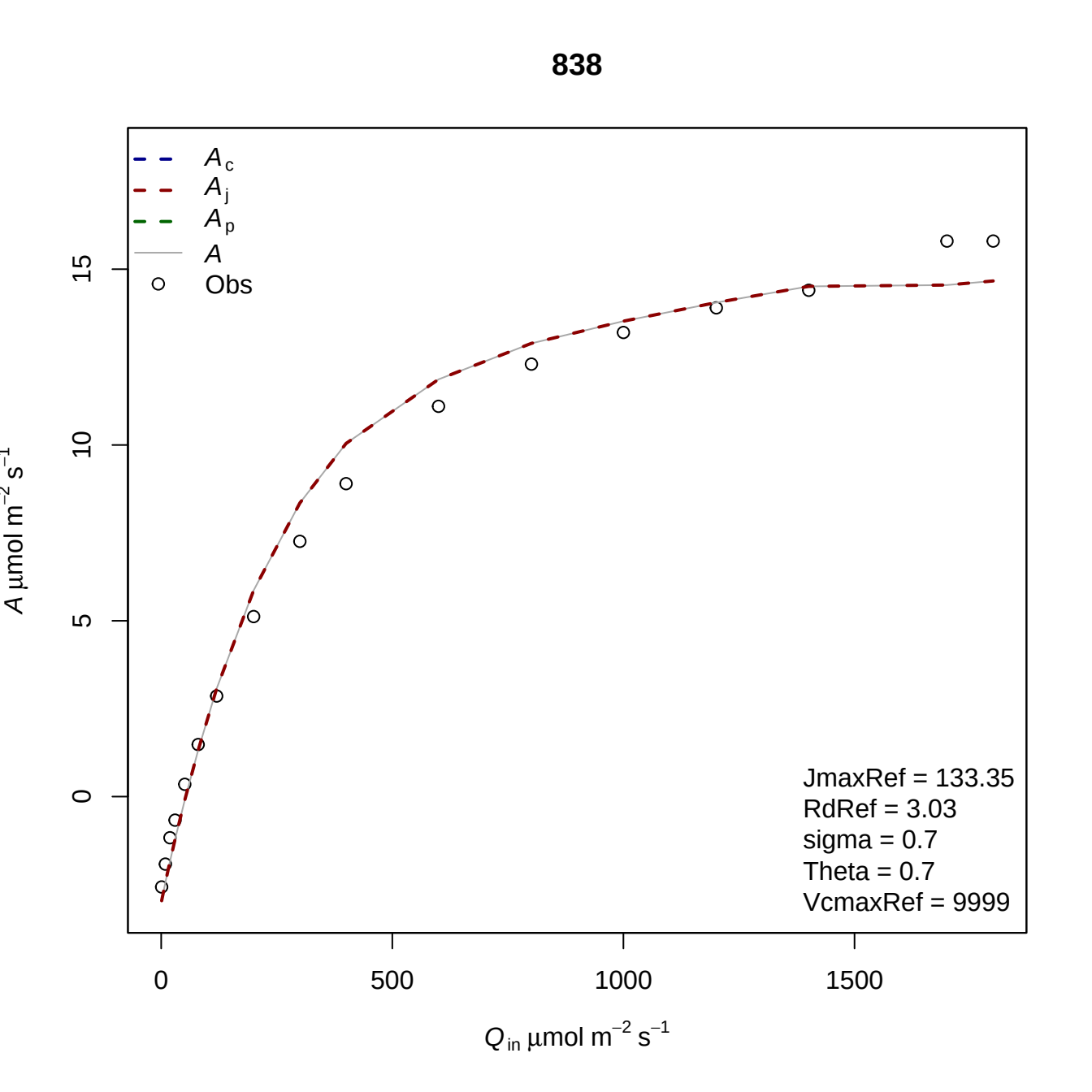

833
A_c
A_j
A_p
A
Obs
JmaxRef = 59.35
RdRef = 2.5
sigma = 0.23
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

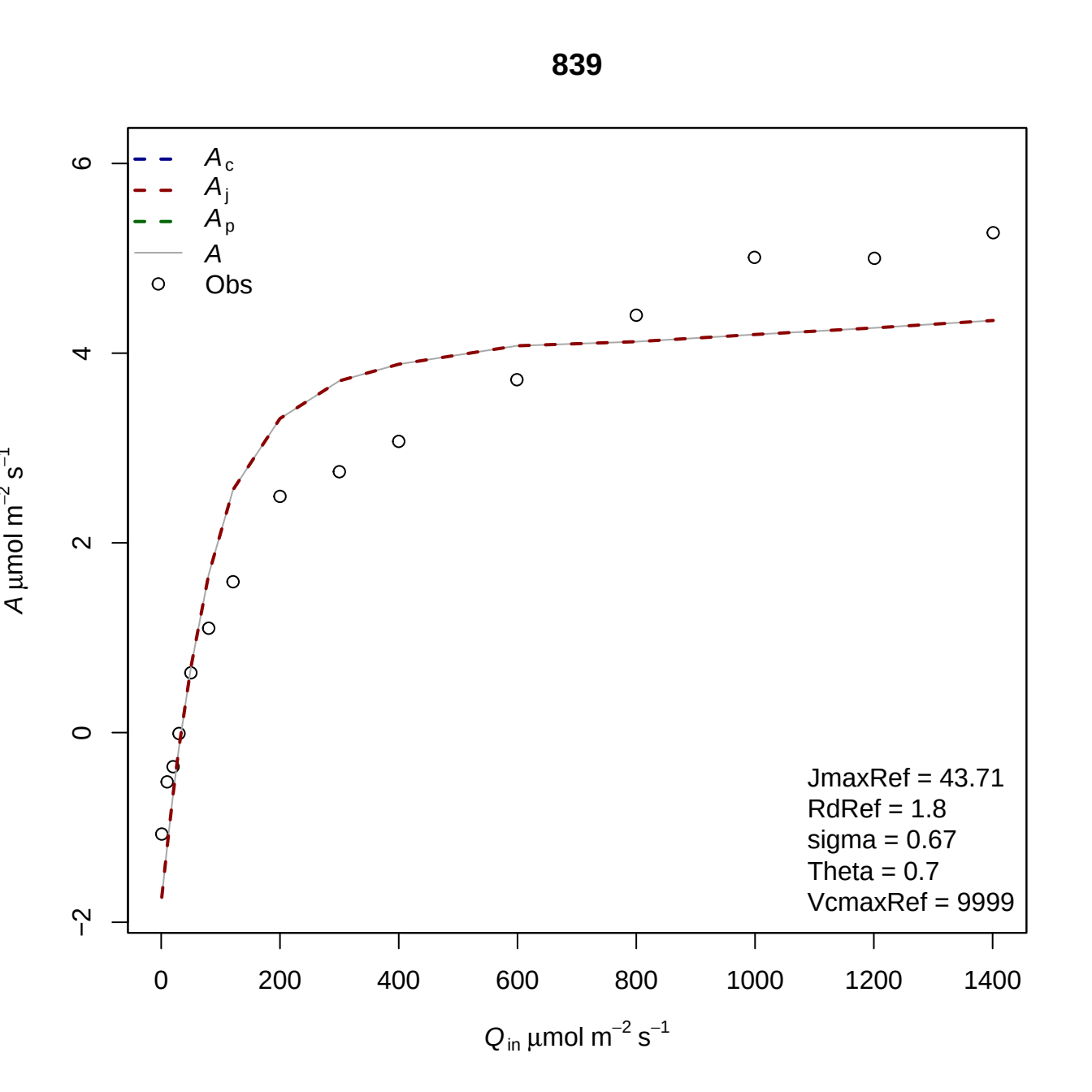

834
A_c
A_j
A_p
A
Obs
JmaxRef = 80.28
RdRef = 2.04
sigma = 0.66
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

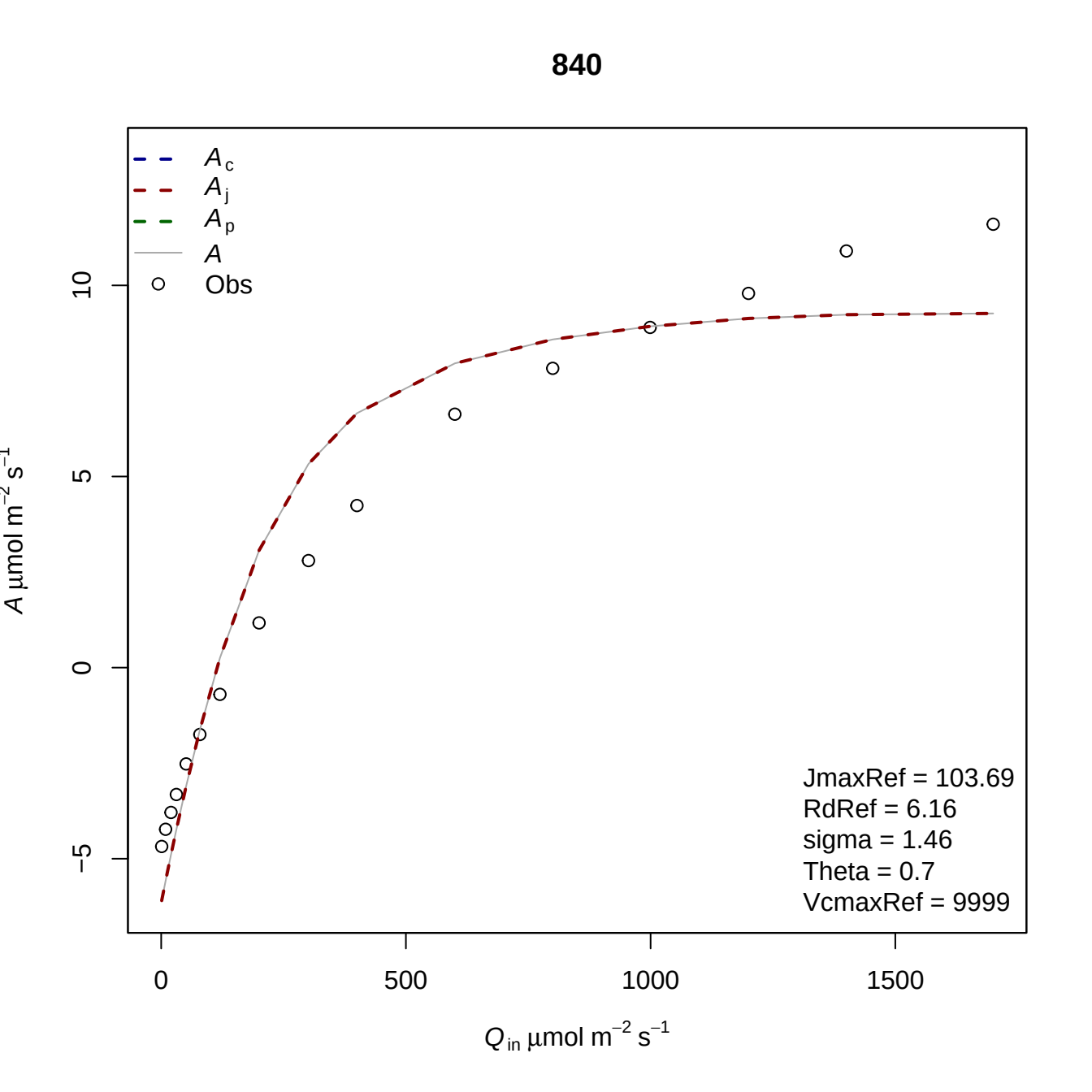

835
A_c
A_j
A_p
A
Obs
JmaxRef = 37.99
RdRef = 2.65
sigma = 0.33
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
0
200
400
600
800
1000
1200
1400

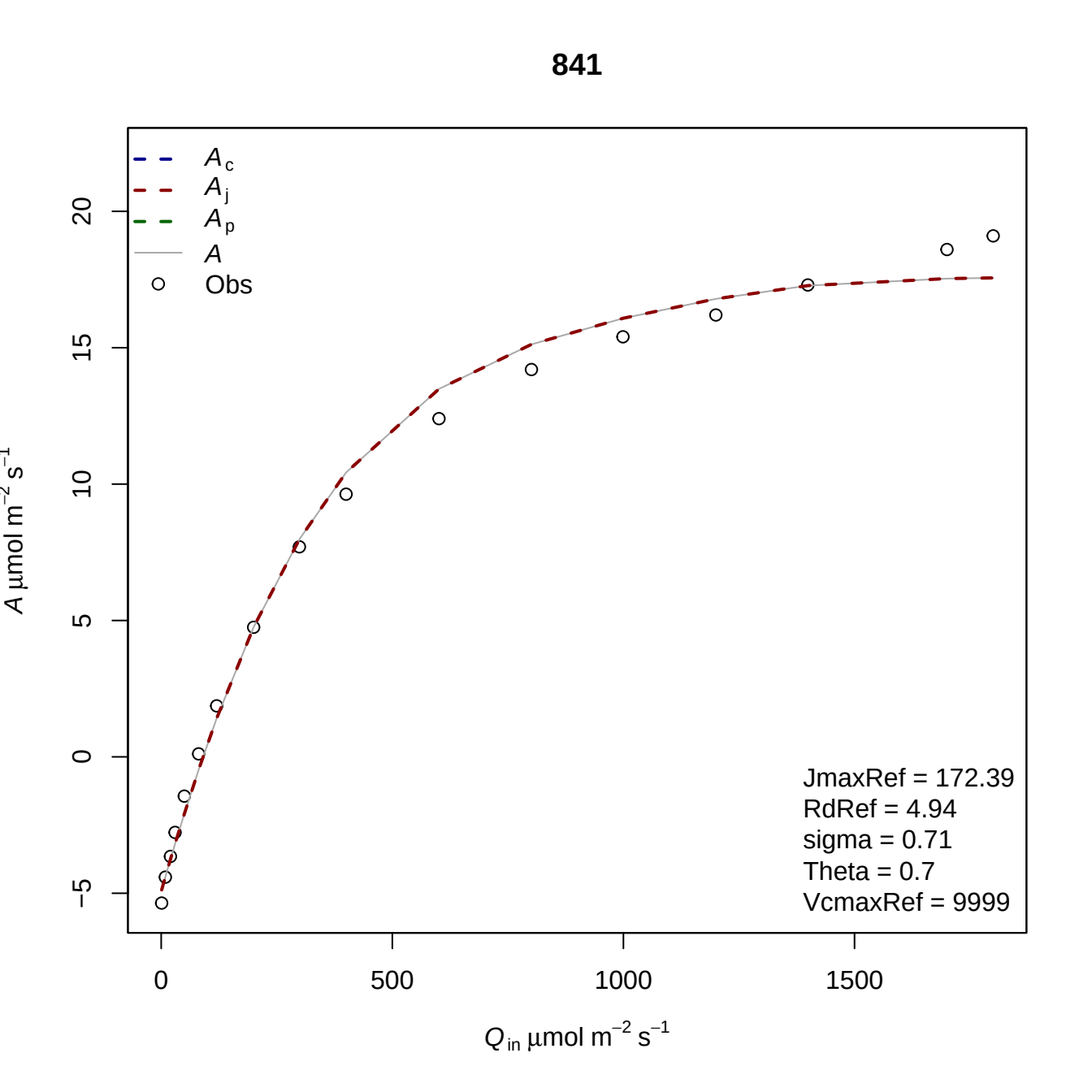

836
A_c
A_j
A_p
A
Obs
JmaxRef = 25.87
RdRef = 1.47
sigma = 0.29
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
0
500
1000
1500

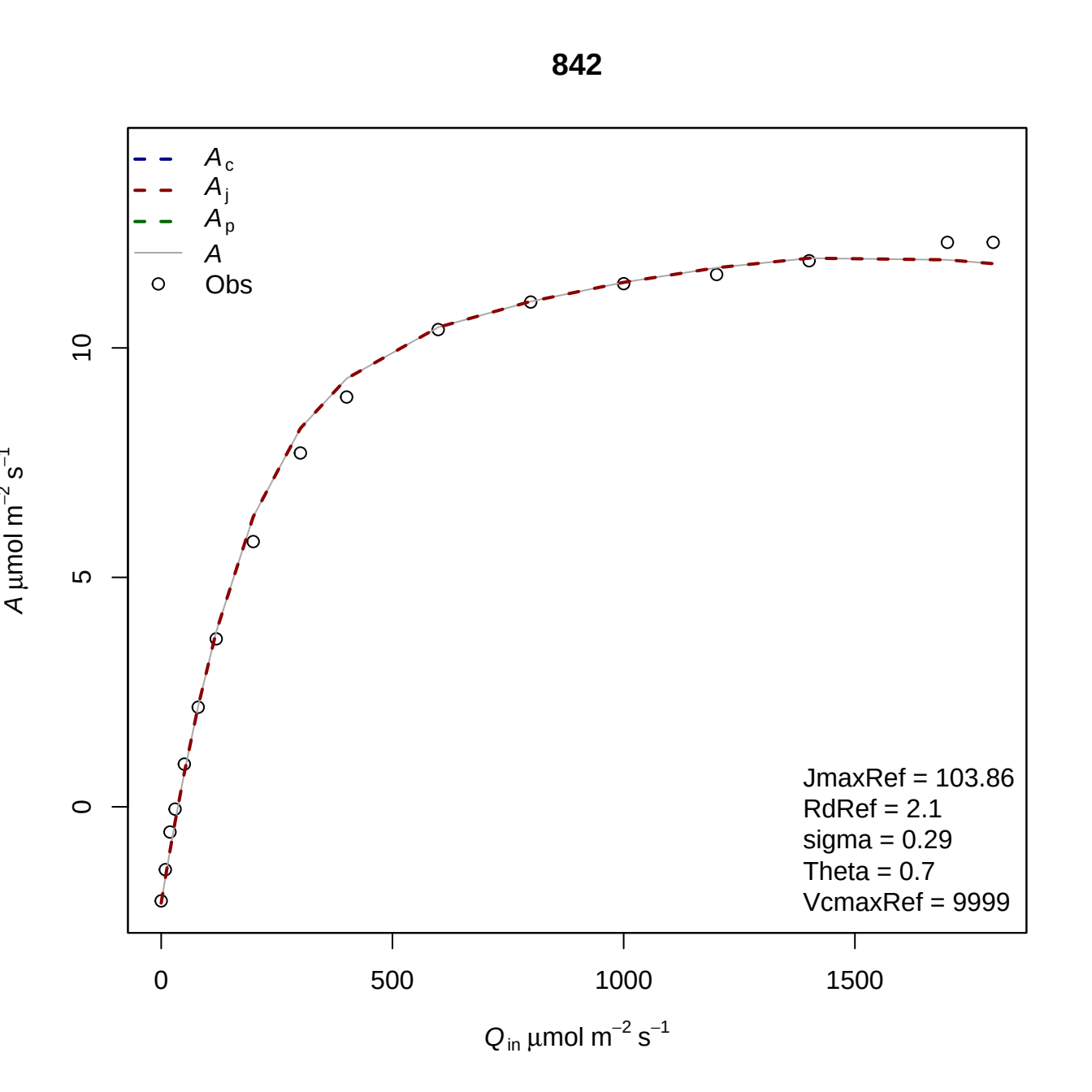

837
A_c
A_j
A_p
A
Obs
JmaxRef = 151.58
RdRef = 3.18
sigma = 0.52
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

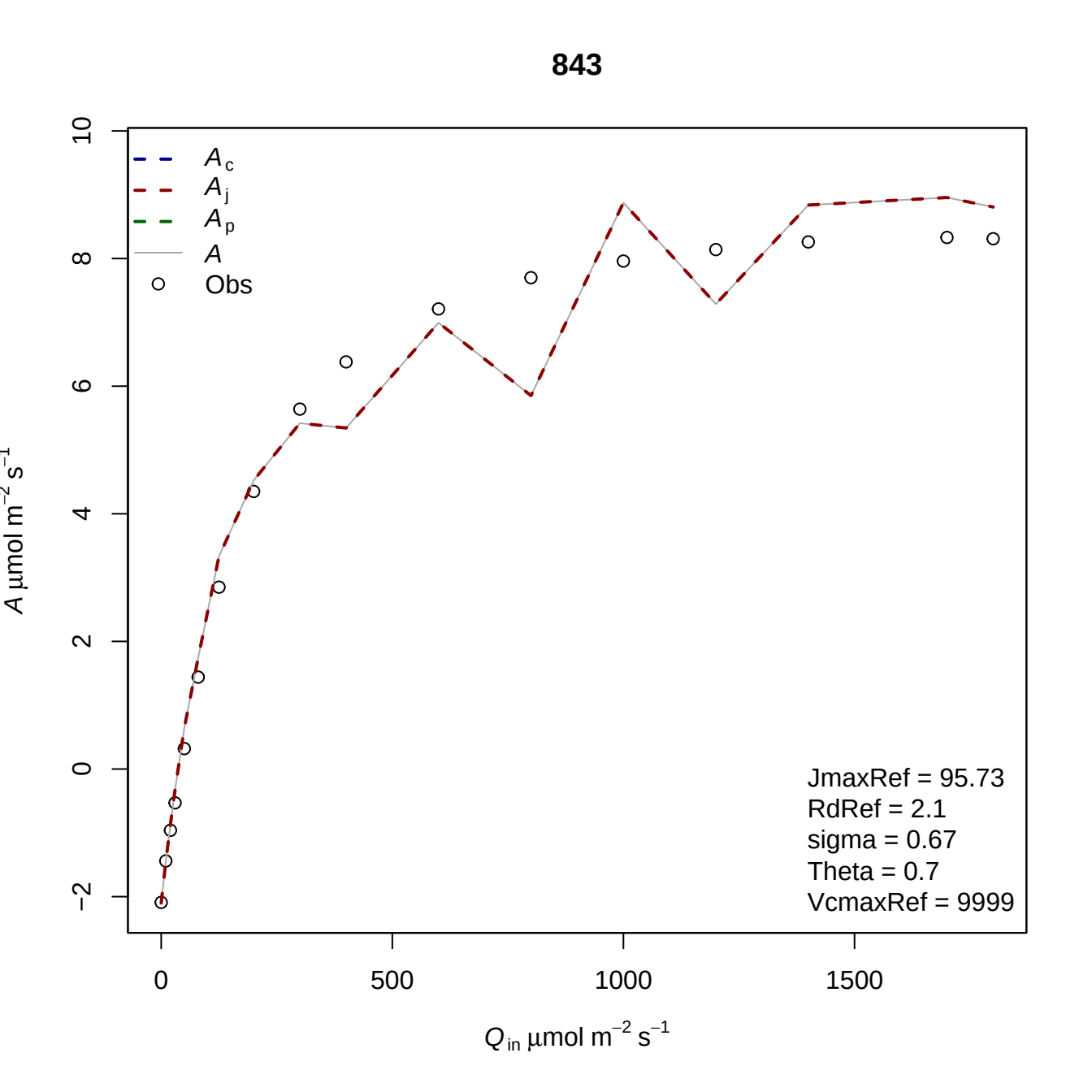

838
A_c
A_j
A_p
A
Obs
JmaxRef = 133.35
RdRef = 3.03
sigma = 0.7
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

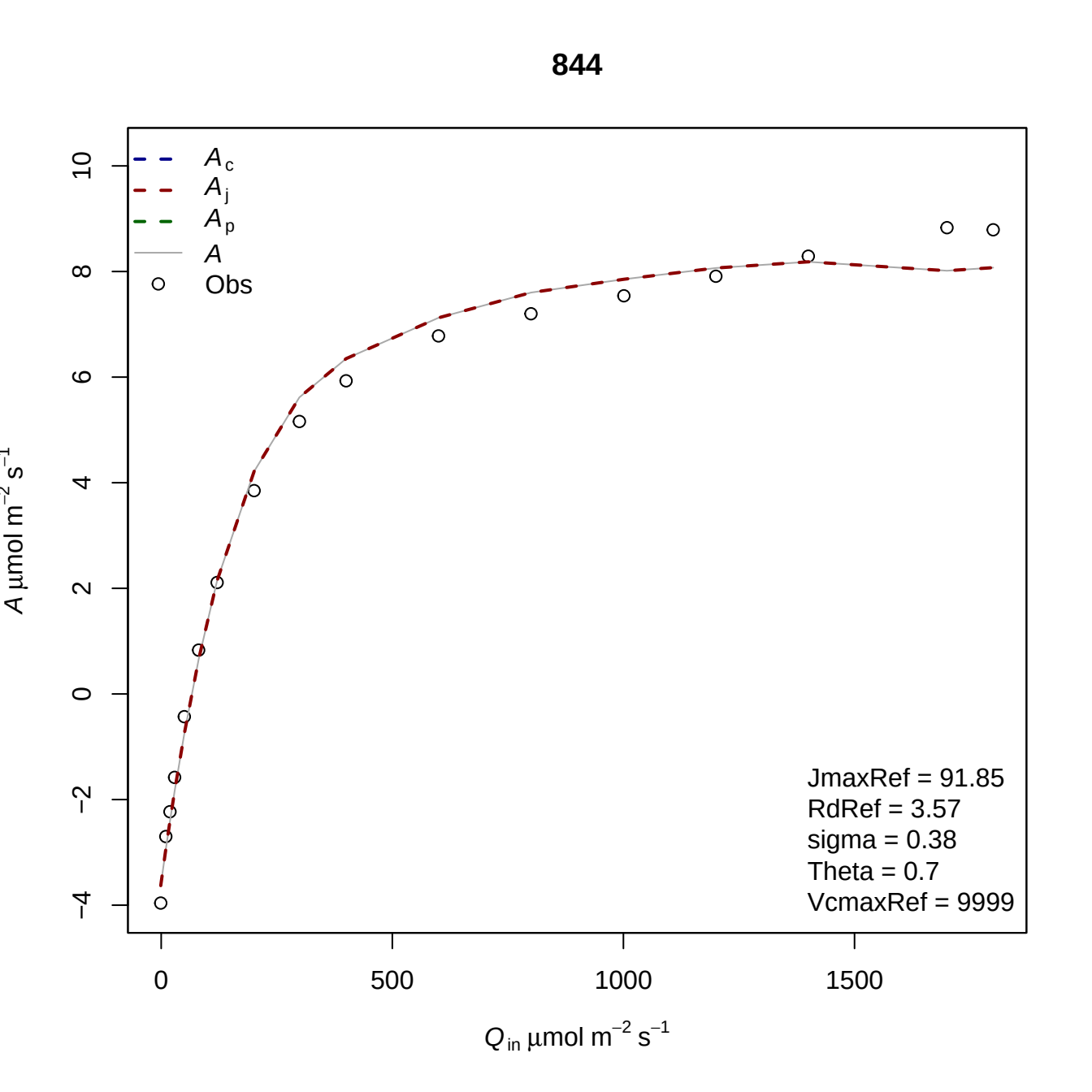

839
A_c
A_j
A_p
A
Obs
JmaxRef = 43.71
RdRef = 1.8
sigma = 0.67
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000
1200
1400

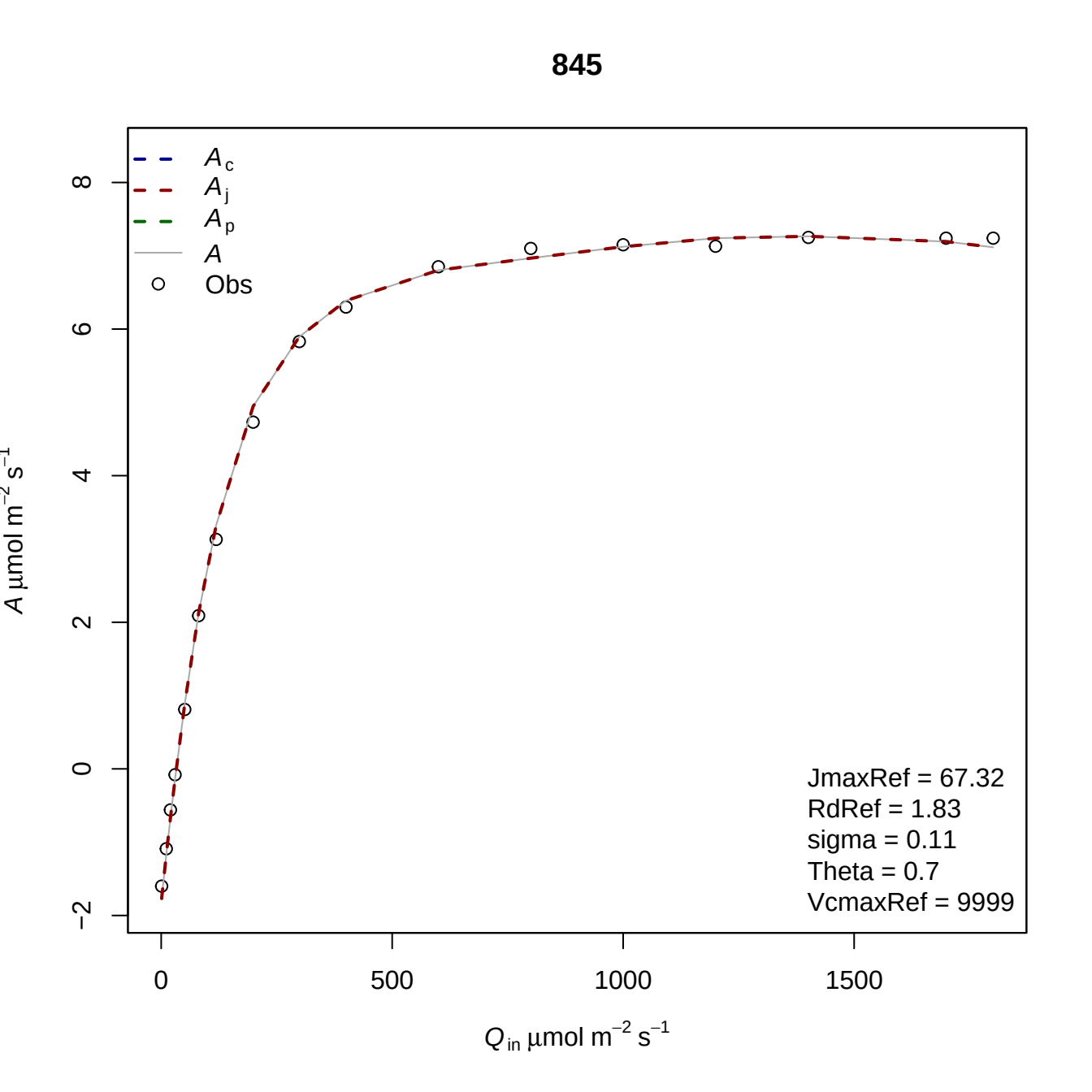

840
A_c
A_j
A_p
A
Obs
JmaxRef = 103.69
RdRef = 6.16
sigma = 1.46
Theta = 0.7
VcmaxRef = 9999
10
5
0
−5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

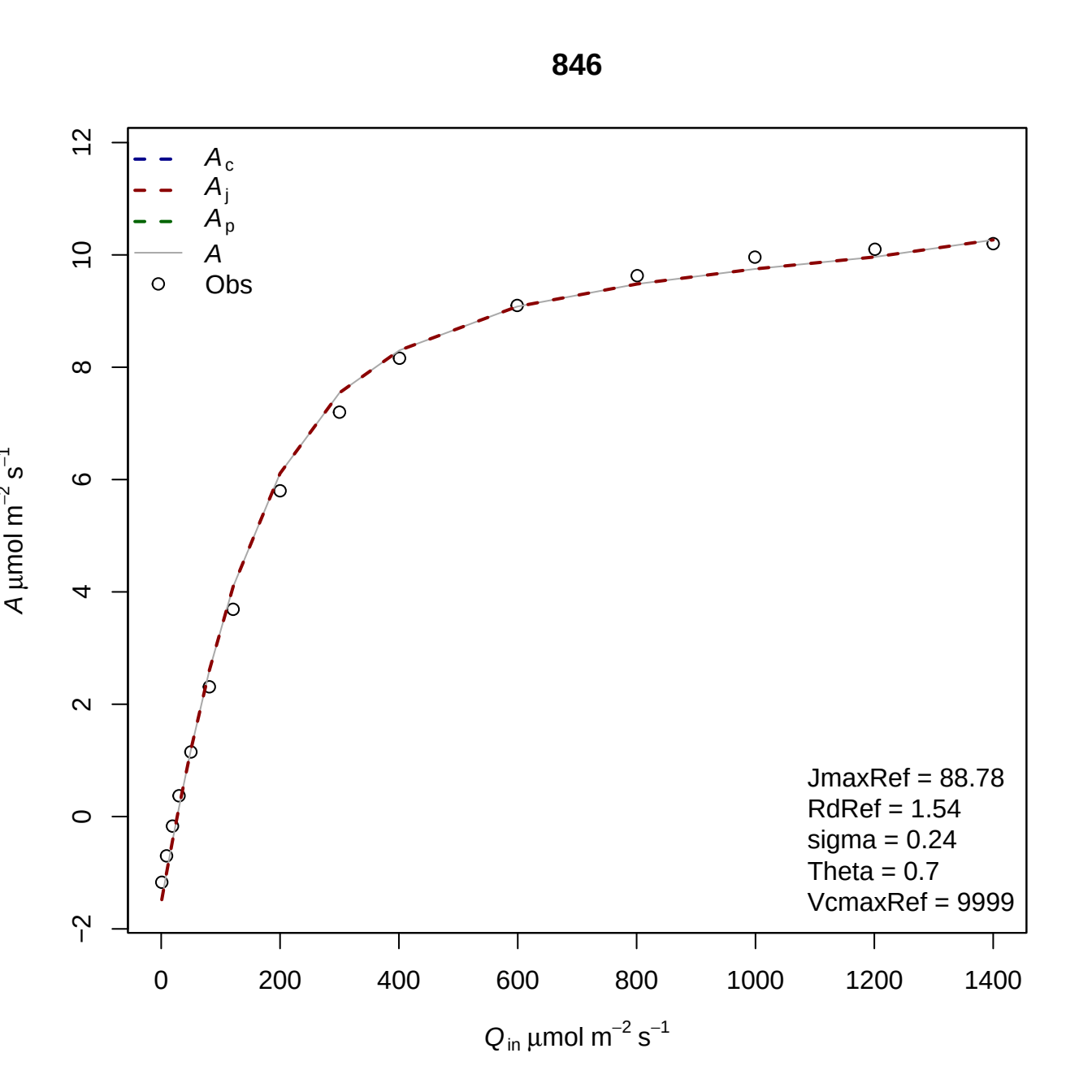

841
A_c
A_j
A_p
A
Obs
JmaxRef = 172.39
RdRef = 4.94
sigma = 0.71
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500

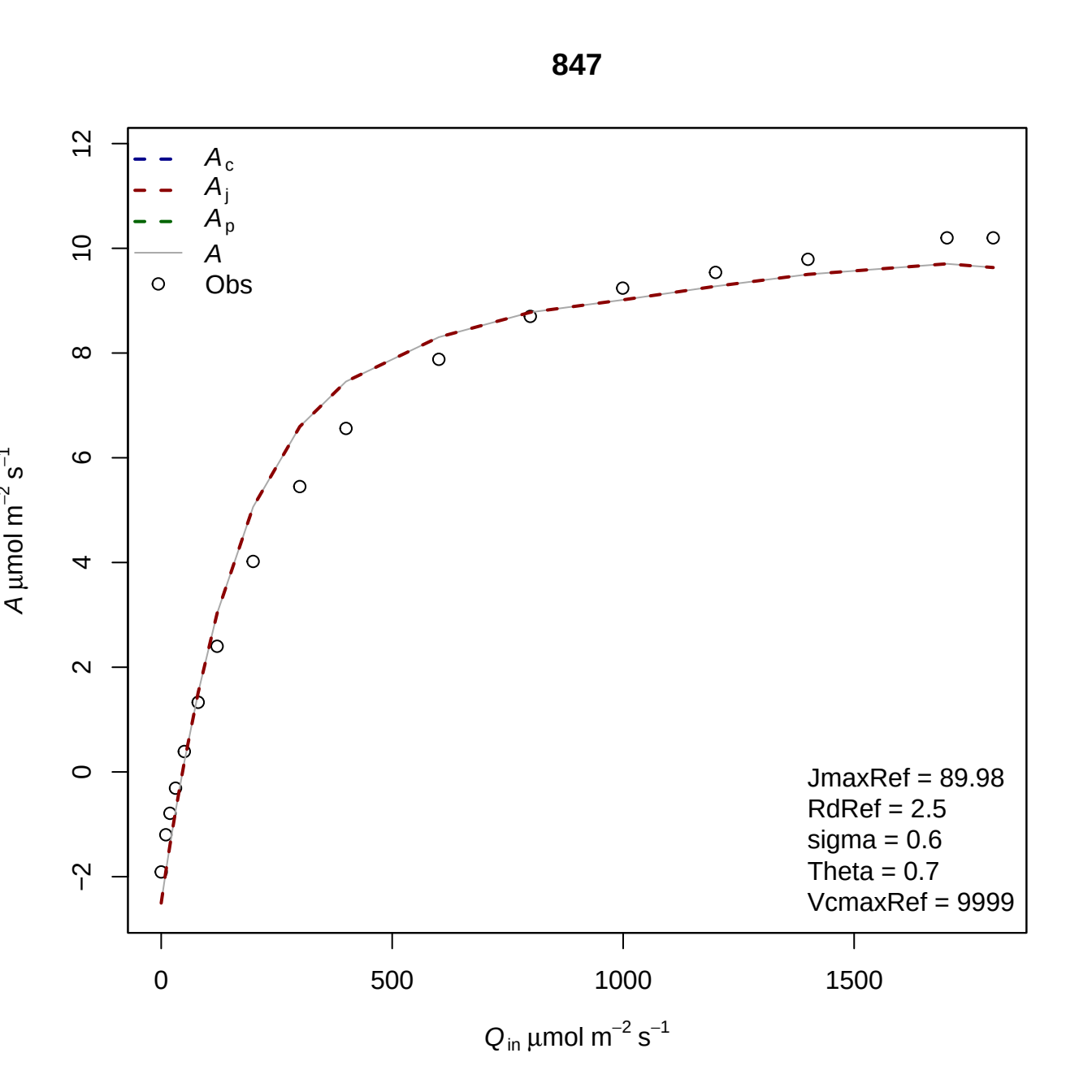

842
A_c
A_j
A_p
A
Obs
JmaxRef = 103.86
RdRef = 2.1
sigma = 0.29
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

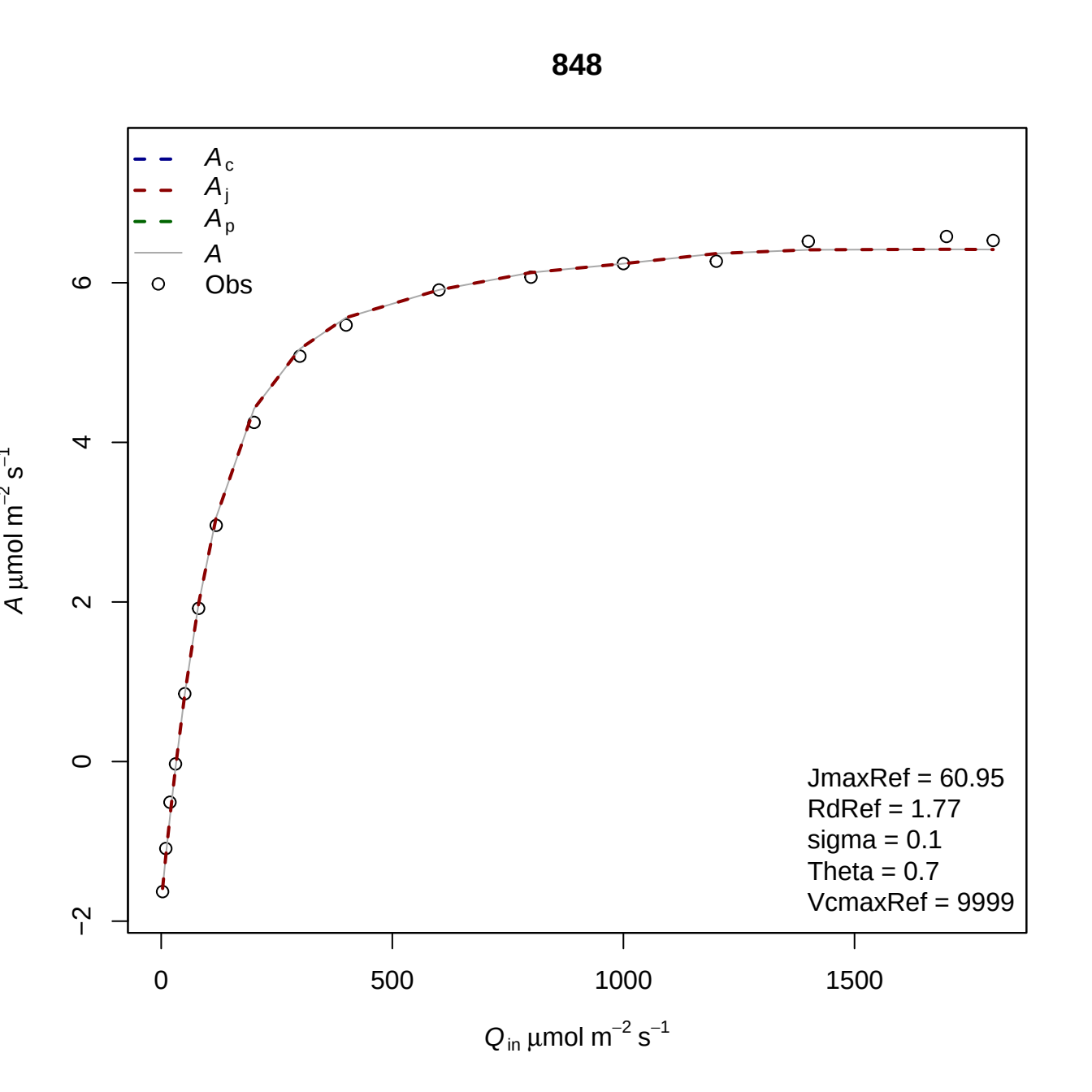

843
A_c
A_j
A_p
A
Obs
JmaxRef = 95.73
RdRef = 2.1
sigma = 0.67
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

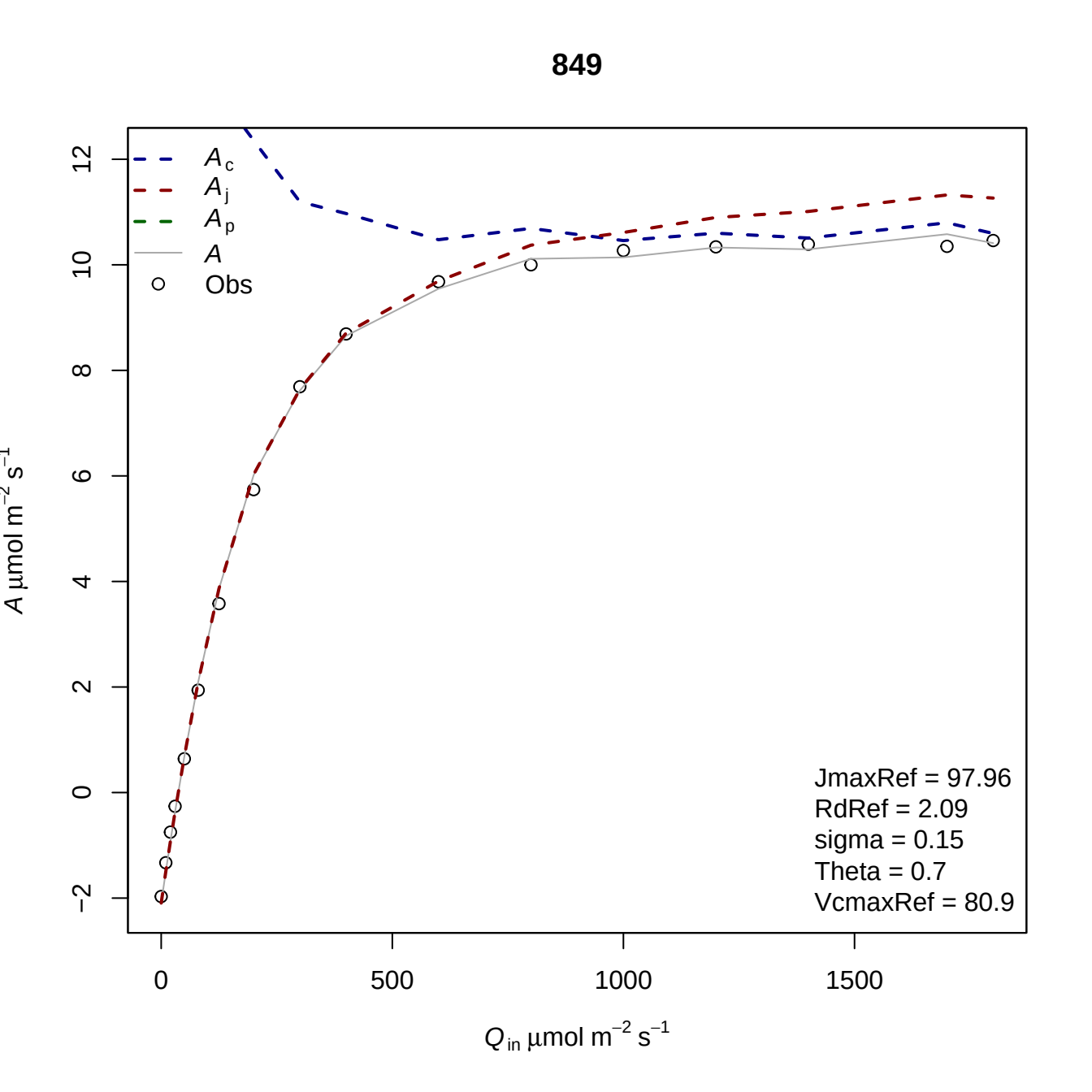

844
A_c
A_j
A_p
A
Obs
JmaxRef = 91.85
RdRef = 3.57
sigma = 0.38
Theta = 0.7
VcmaxRef = 9999
10
8
6
4
2
0
−2
−4
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

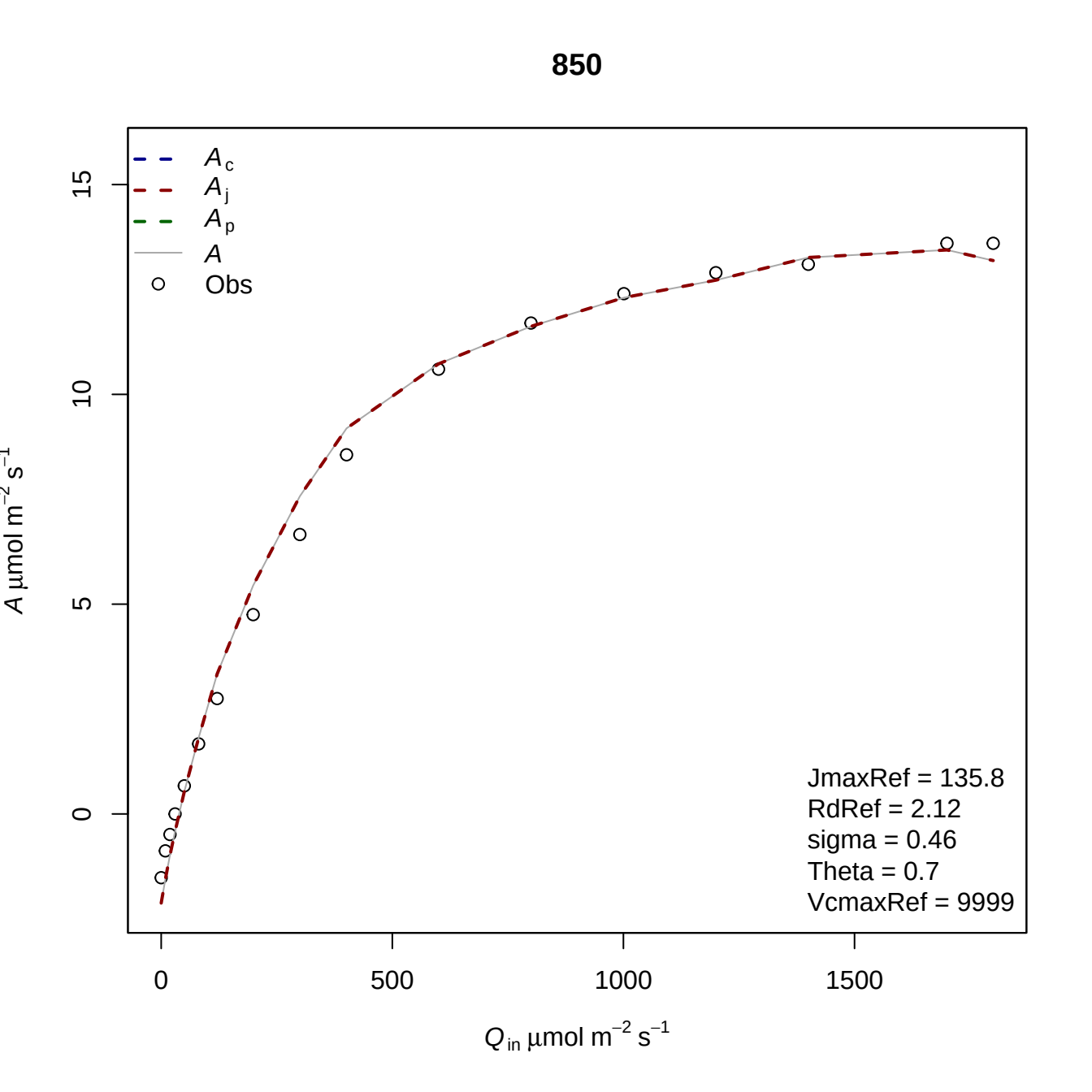

845
A_c
A_j
A_p
A
Obs
JmaxRef = 67.32
RdRef = 1.83
sigma = 0.11
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

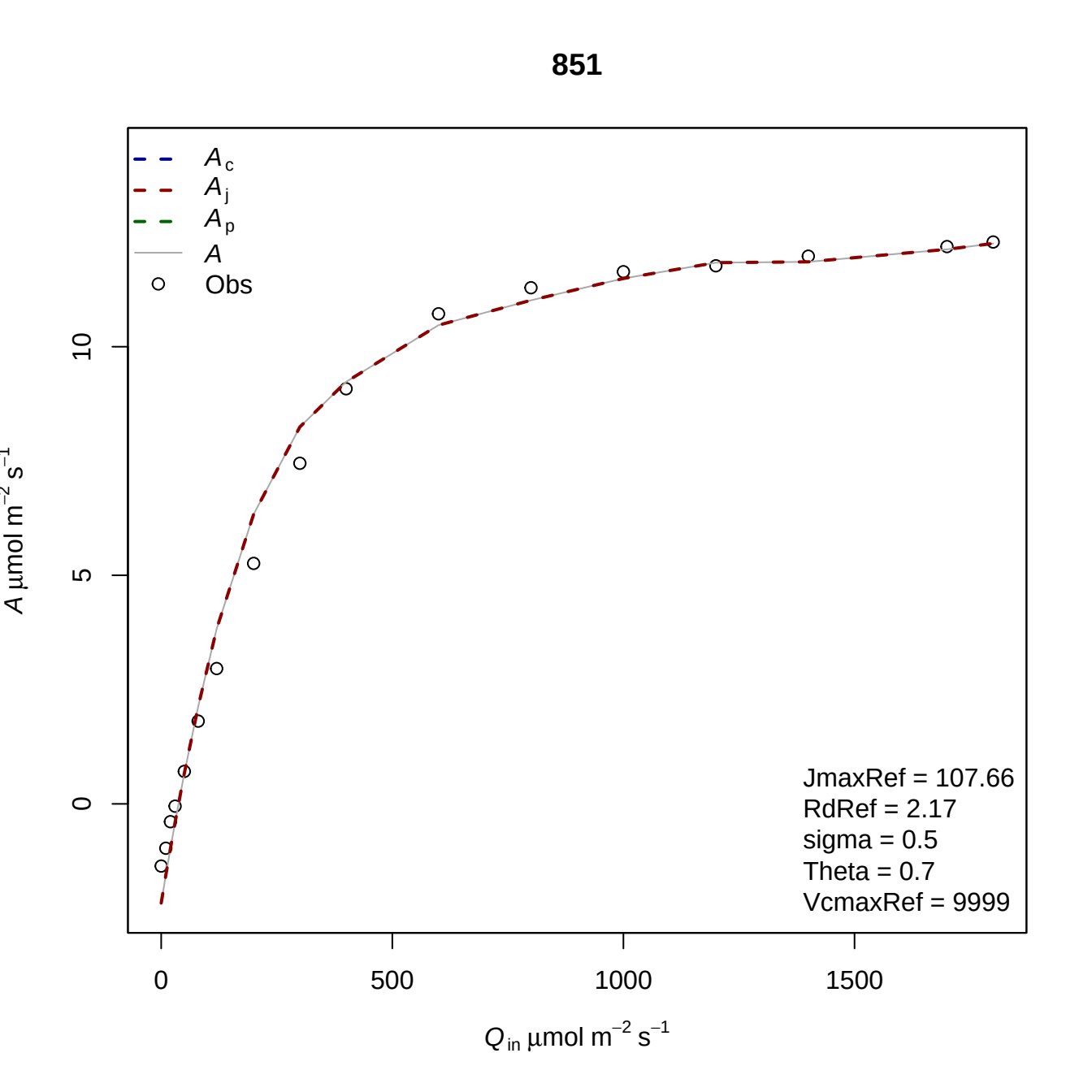

846
A_c
A_j
A_p
A
Obs
JmaxRef = 88.78
RdRef = 1.54
sigma = 0.24
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
200
400
600
800
1000
1200
1400

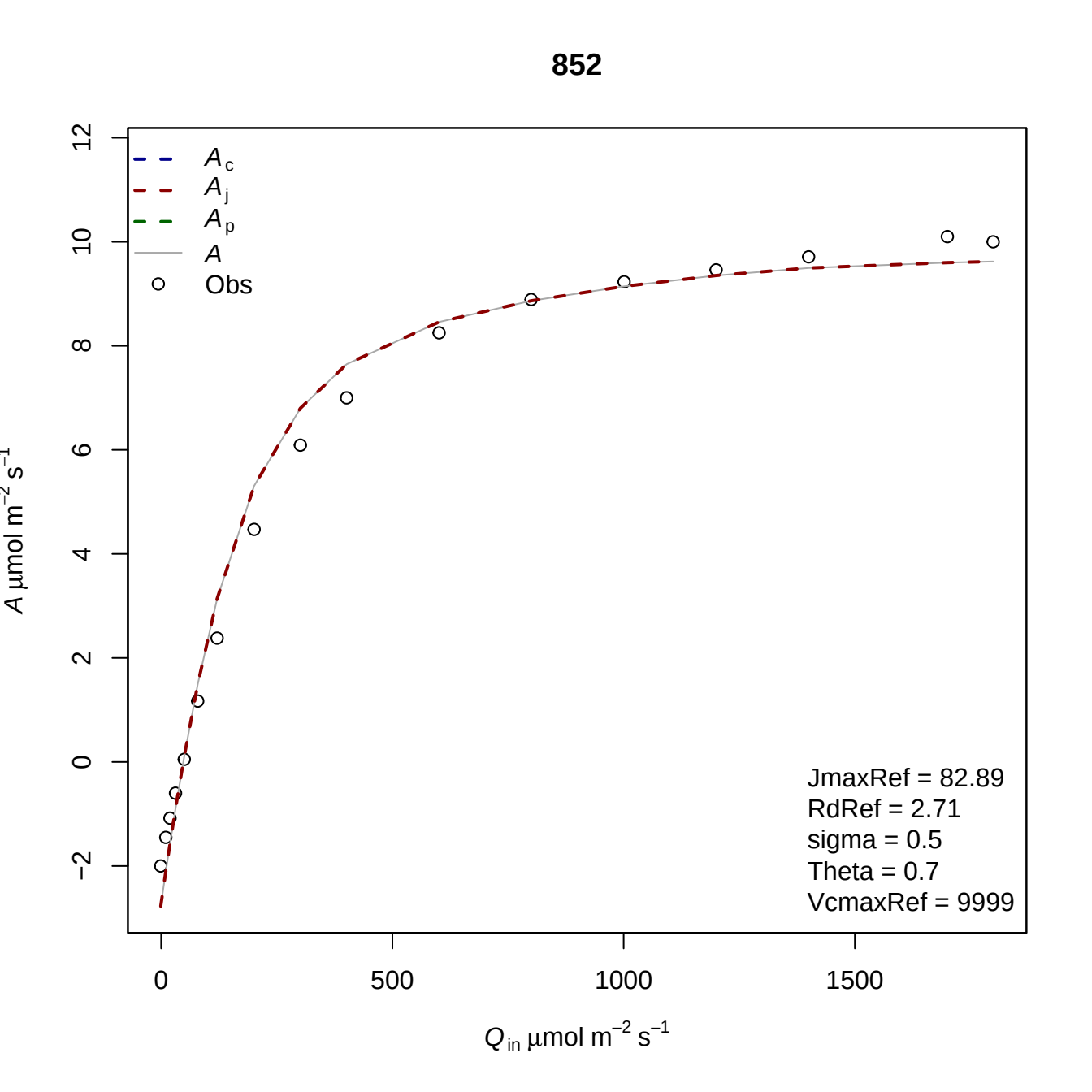

847
A_c
A_j
A_p
A
Obs
JmaxRef = 89.98
RdRef = 2.5
sigma = 0.6
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

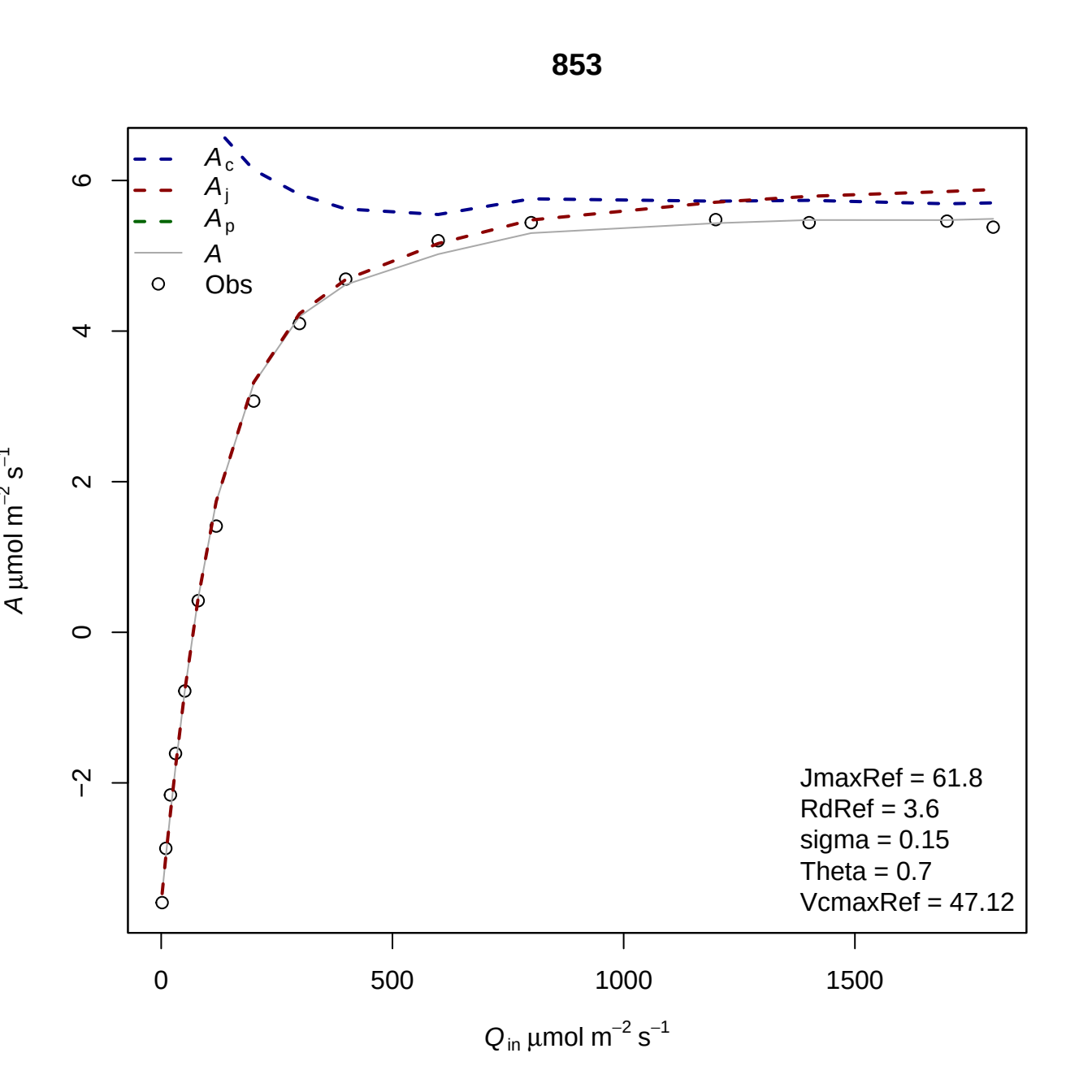

848
A_c
A_j
A_p
A
Obs
JmaxRef = 60.95
RdRef = 1.77
sigma = 0.1
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500

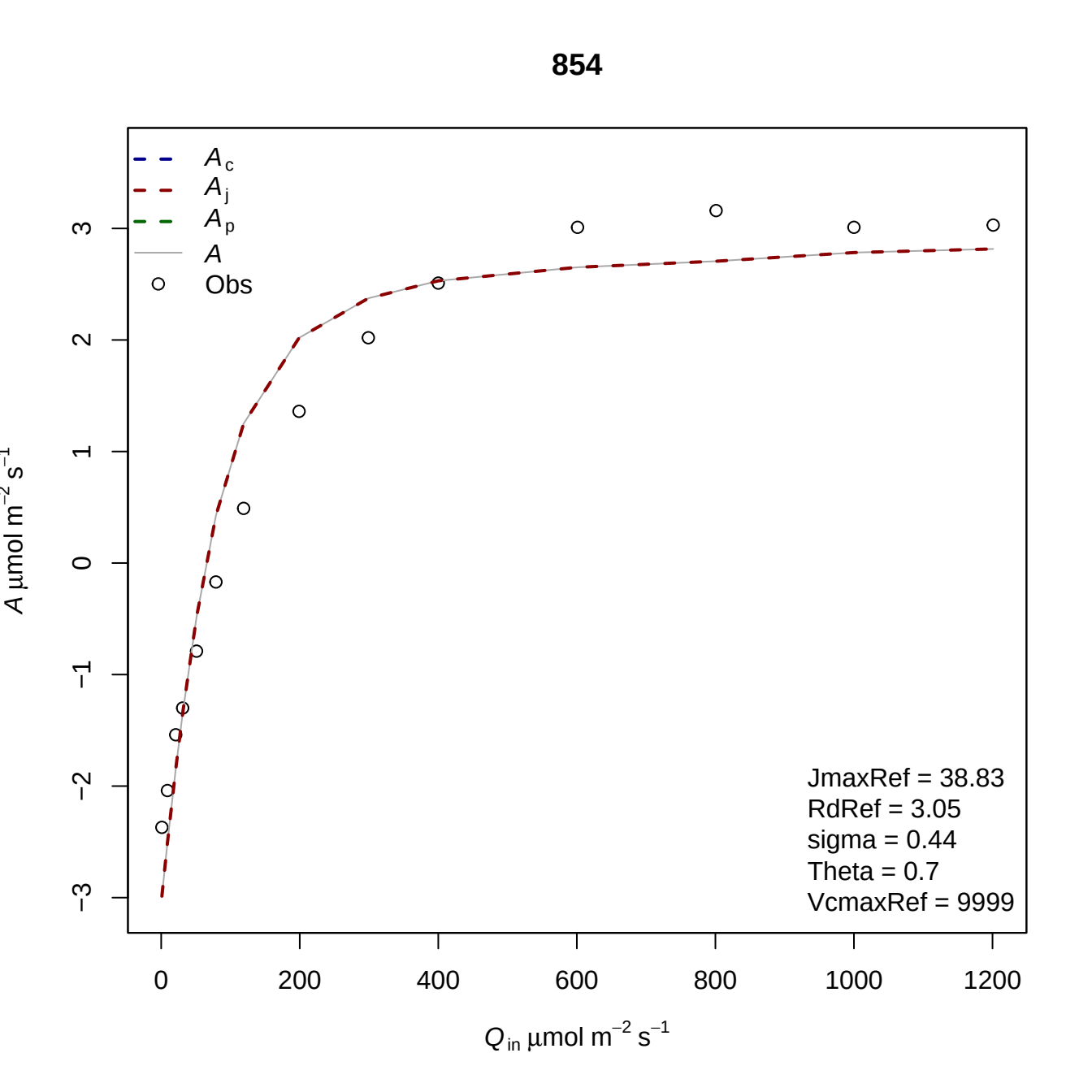

849
A_c
A_j
A_p
A
Obs
JmaxRef = 97.96
RdRef = 2.09
sigma = 0.15
Theta = 0.7
VcmaxRef = 80.9
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

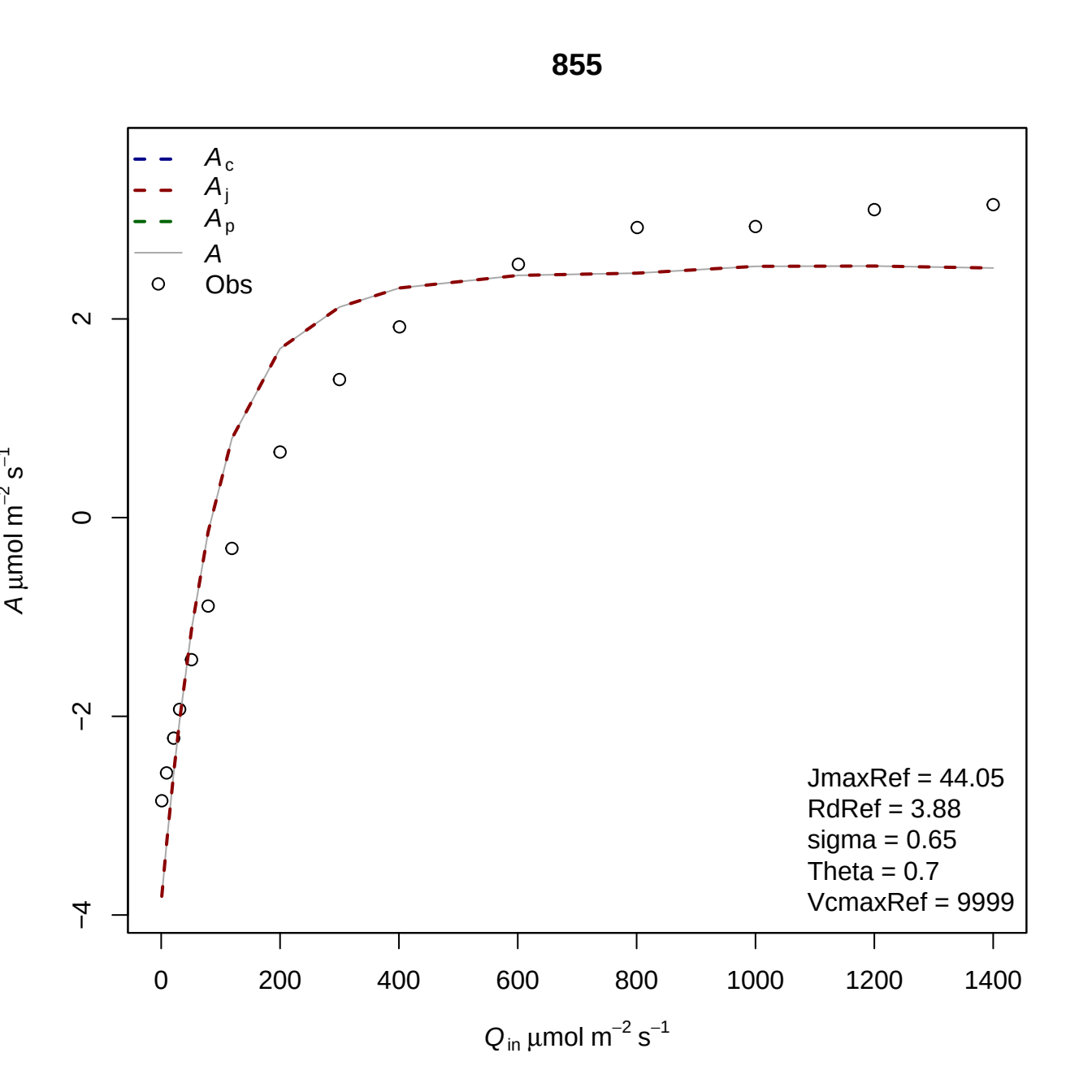

850
A_c
A_j
A_p
A
Obs
JmaxRef = 135.8
RdRef = 2.12
sigma = 0.46
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

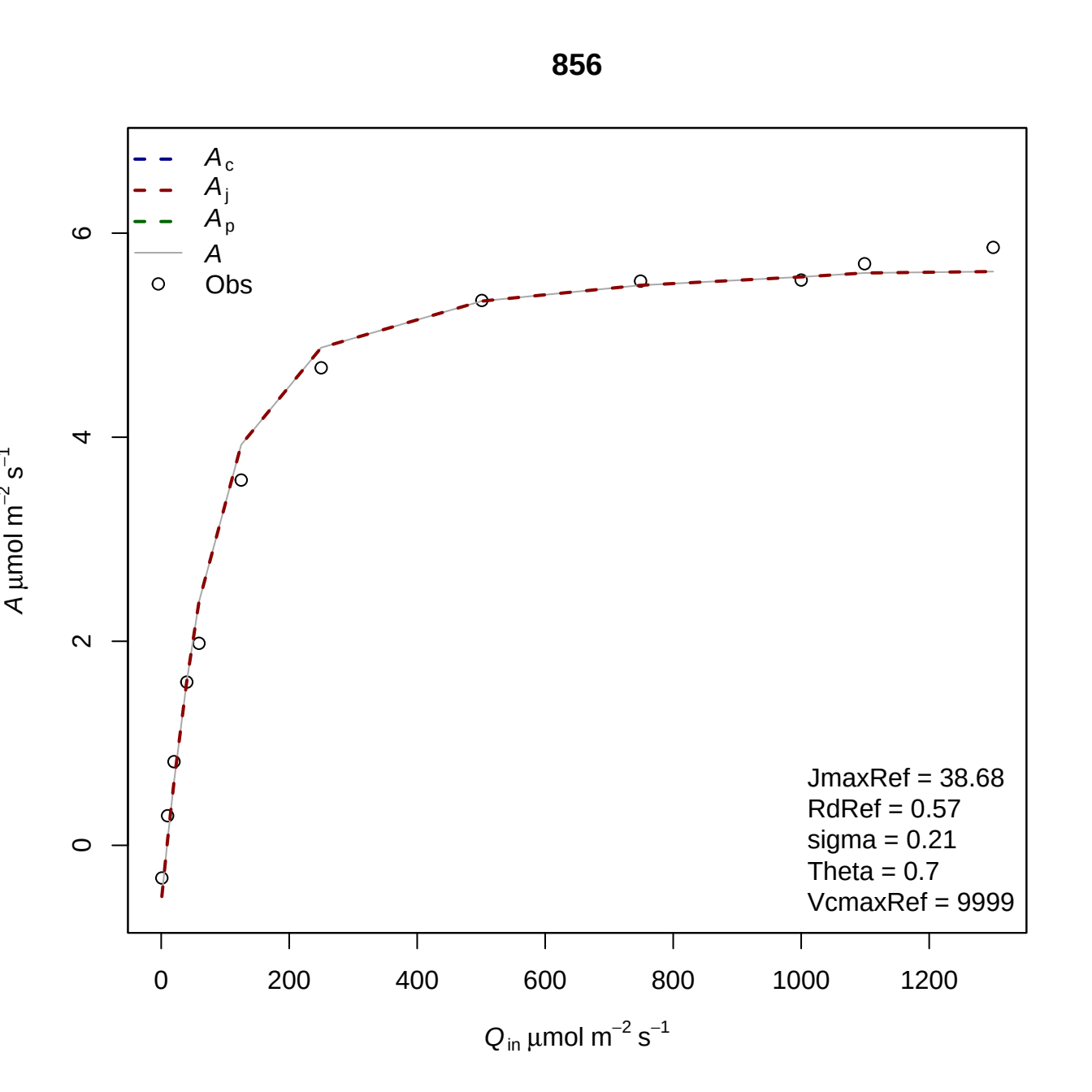

851
A_c
A_j
A_p
A
Obs
JmaxRef = 107.66
RdRef = 2.17
sigma = 0.5
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

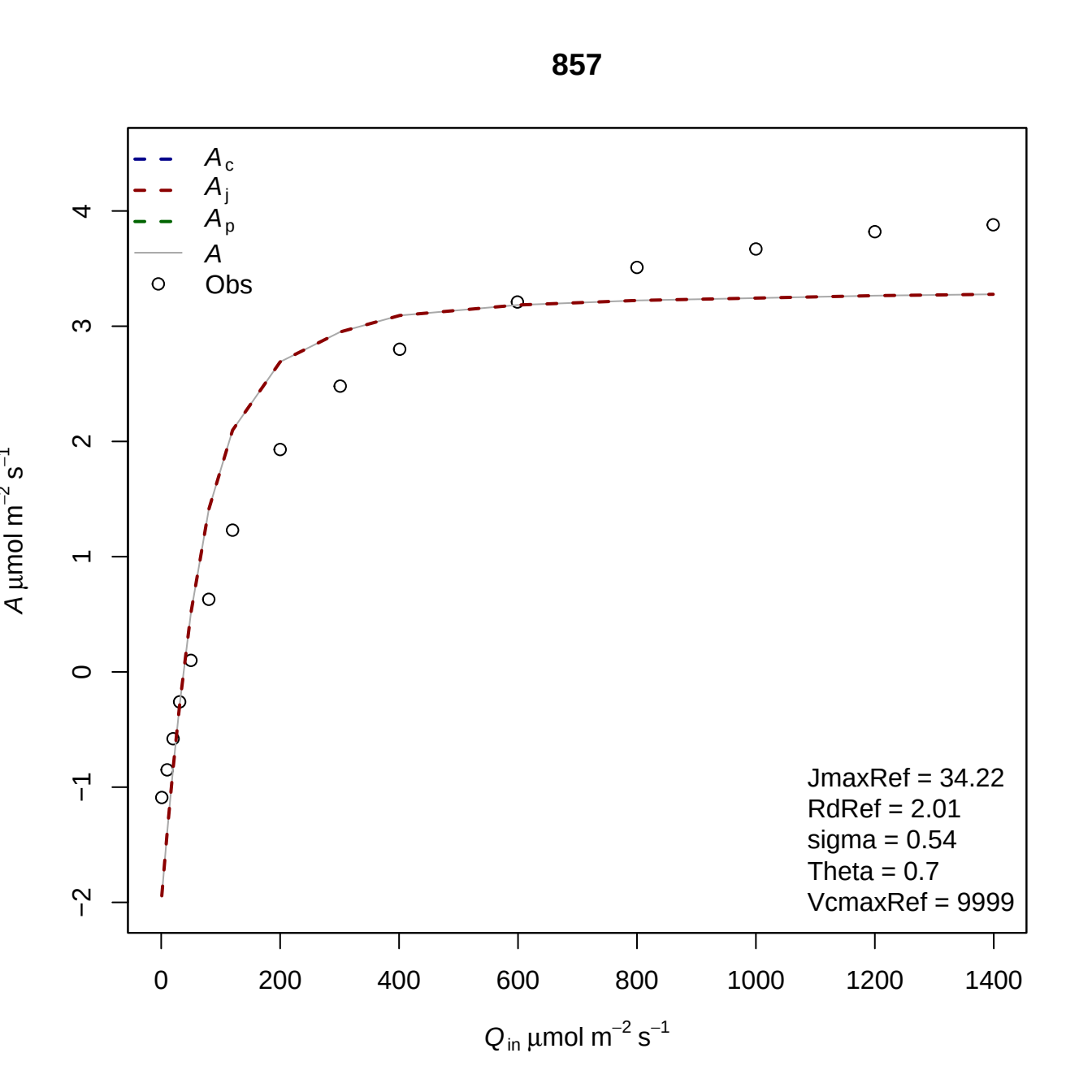

852
A_c
A_j
A_p
A
Obs
JmaxRef = 82.89
RdRef = 2.71
sigma = 0.5
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

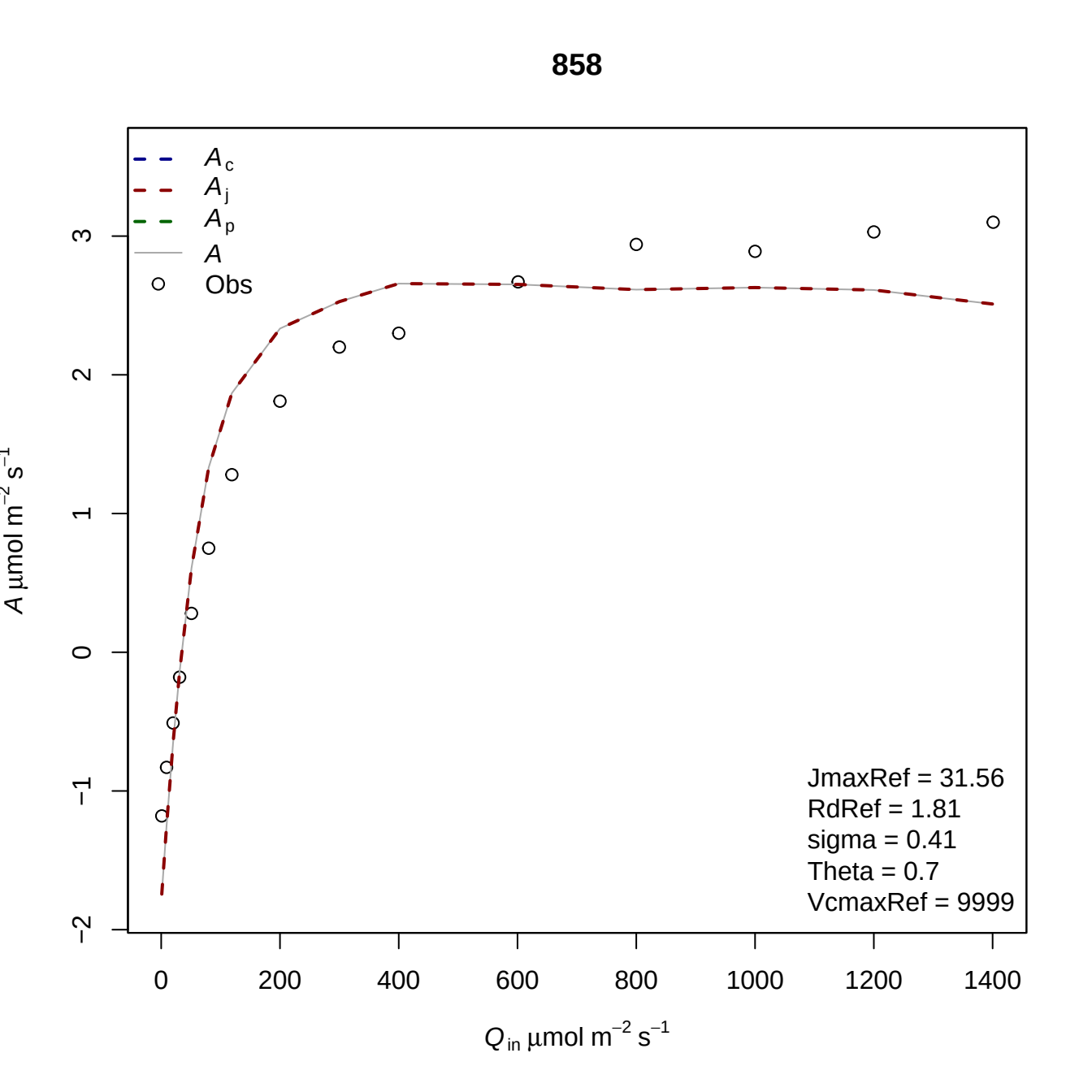

853
A_c
A_j
A_p
A
Obs
JmaxRef = 61.8
RdRef = 3.6
sigma = 0.15
Theta = 0.7
VcmaxRef = 47.12
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

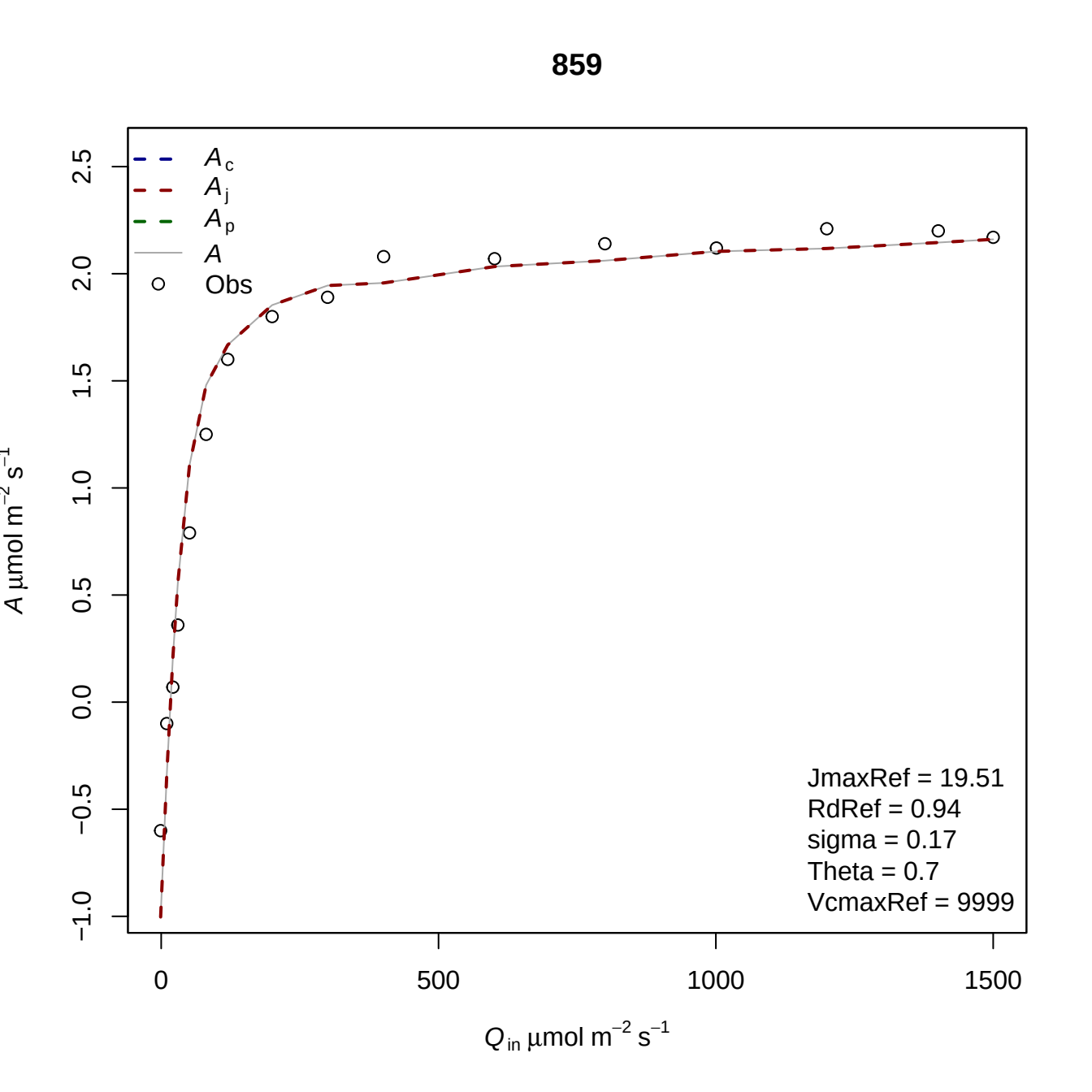

854
A_c
A_j
A_p
A
Obs
JmaxRef = 38.83
RdRef = 3.05
sigma = 0.44
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−3
−2
−1
0
1
2
3
0
200
400
600
800
1000
1200

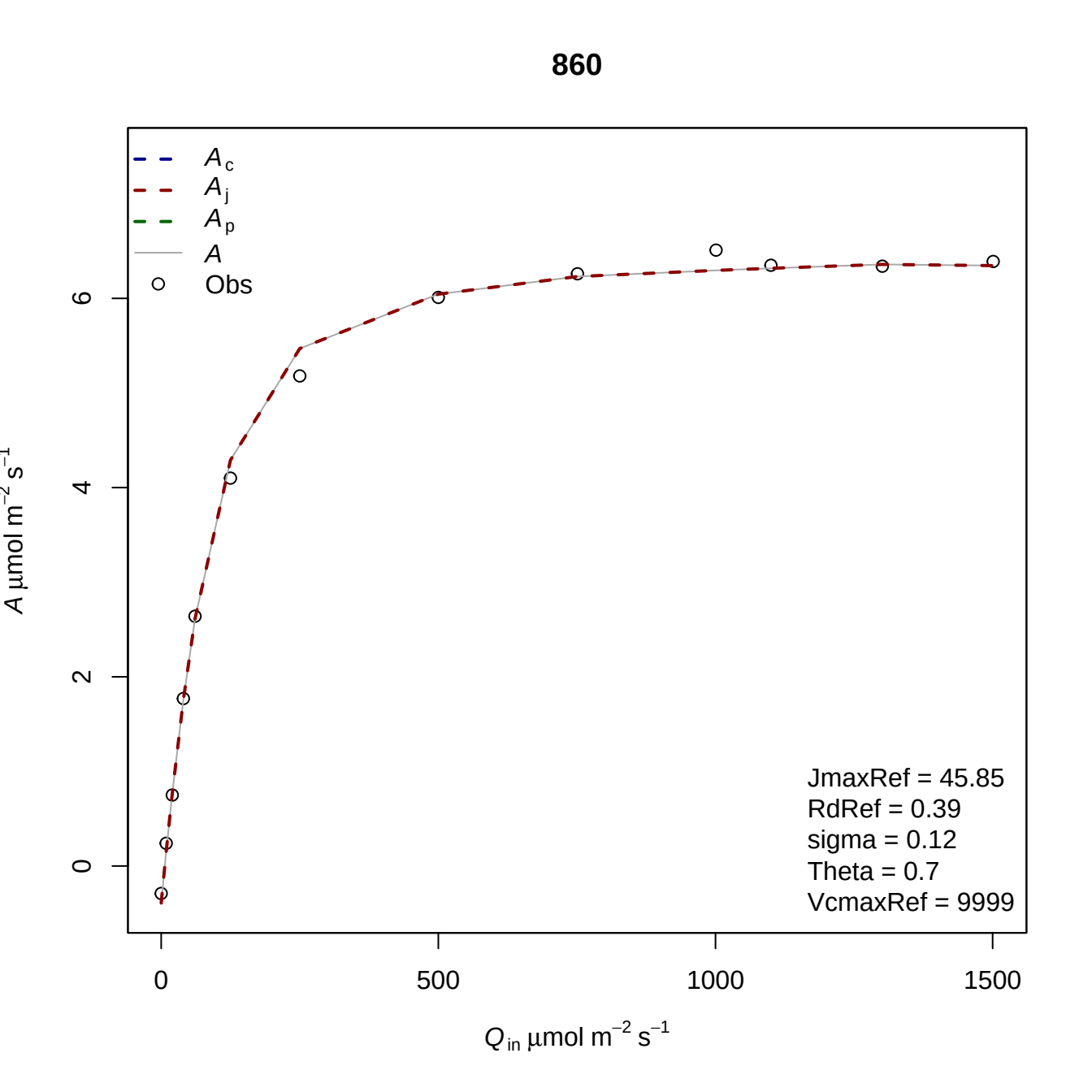

855
A_c
A_j
A_p
A
Obs
JmaxRef = 44.05
RdRef = 3.88
sigma = 0.65
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−4
−2
0
2
0
200
400
600
800
1000
1200
1400

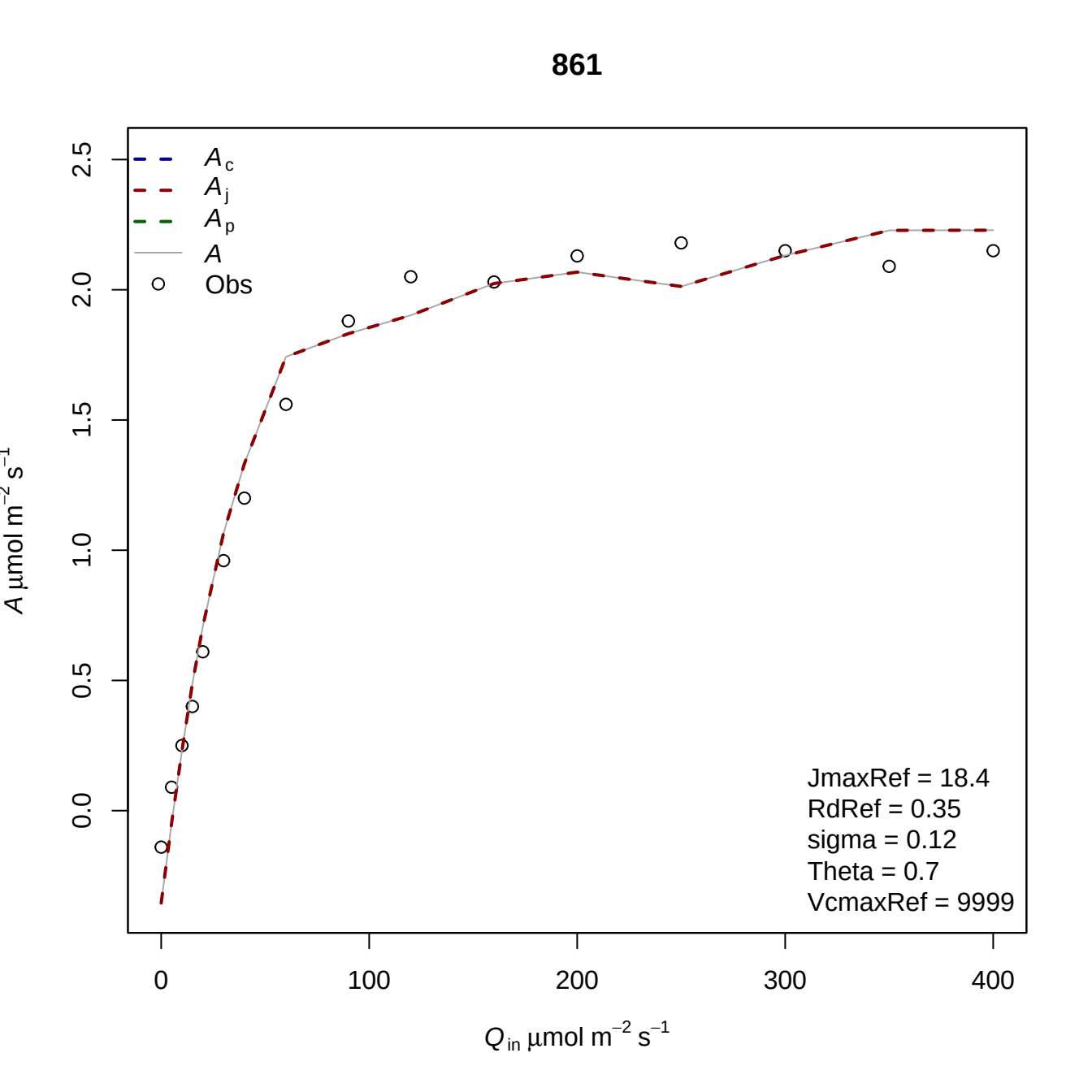

856
A_c
A_j
A_p
A
Obs
JmaxRef = 38.68
RdRef = 0.57
sigma = 0.21
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
0
200
400
600
800
1000
1200
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

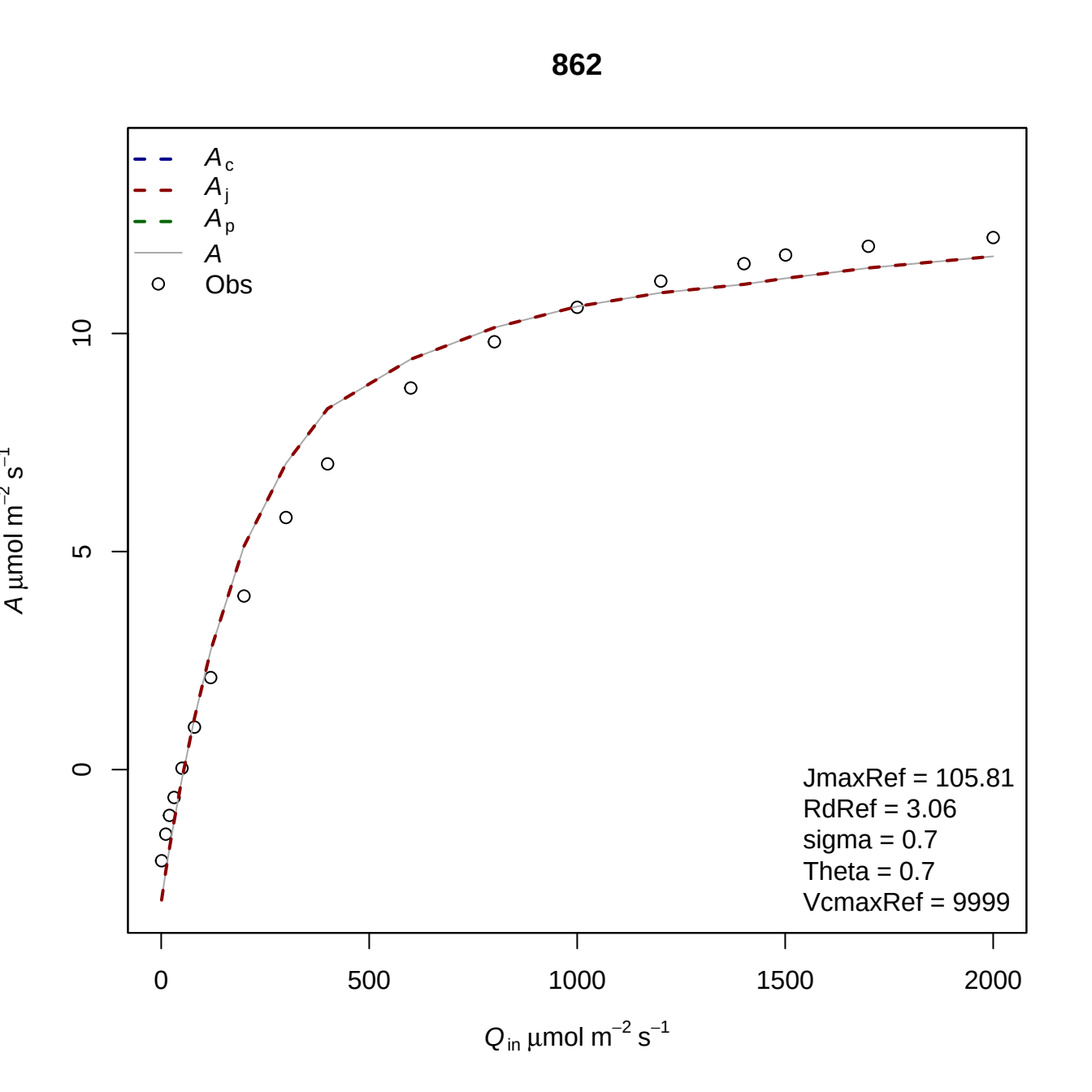

857
A_c
A_j
A_p
A
Obs
JmaxRef = 34.22
RdRef = 2.01
sigma = 0.54
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
4
0
200
400
600
800
1000
1200
1400

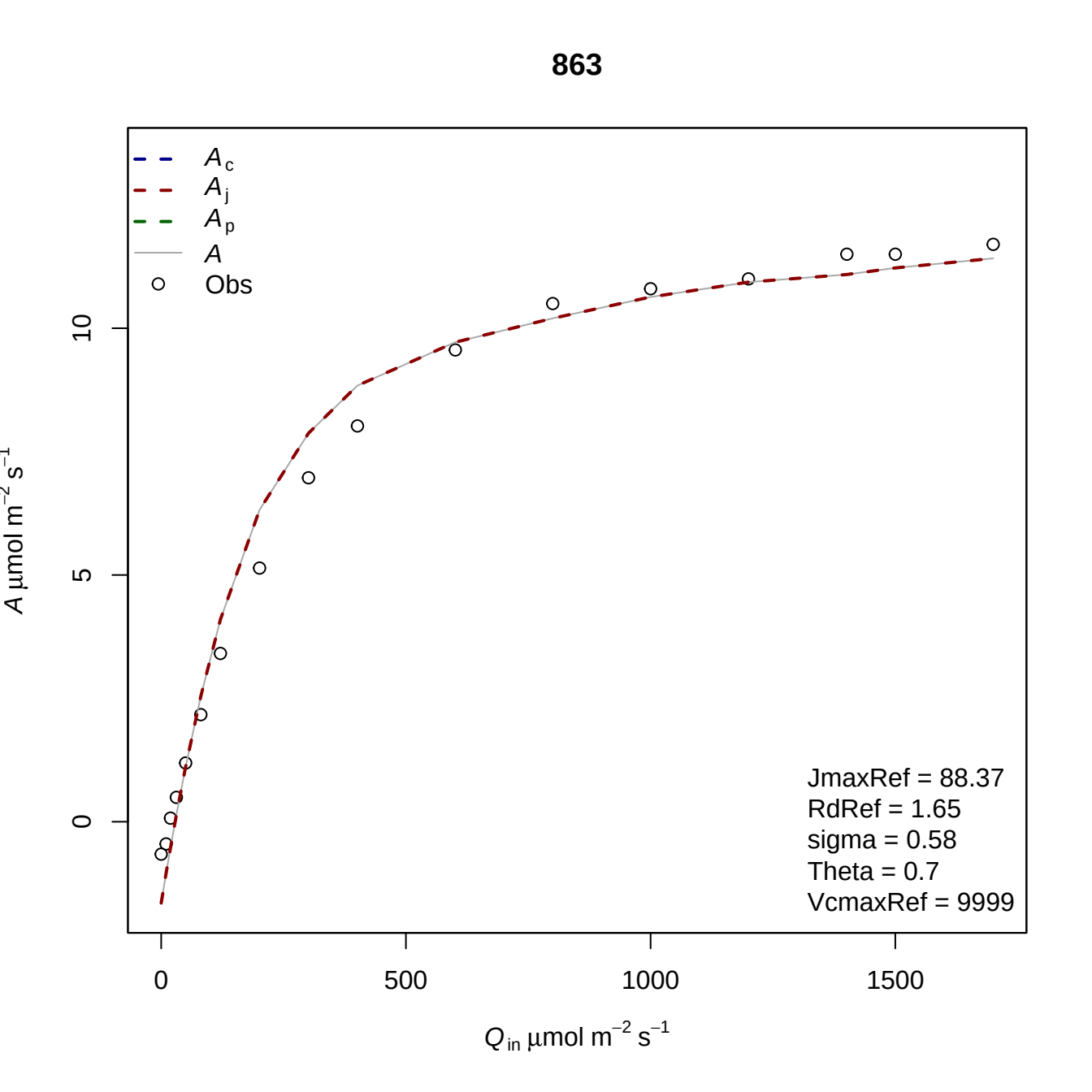

858
A_c
A_j
A_p
A
Obs
JmaxRef = 31.56
RdRef = 1.81
sigma = 0.41
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
−1
0
1
2
3
0
200
400
600
800
1000
1200
1400

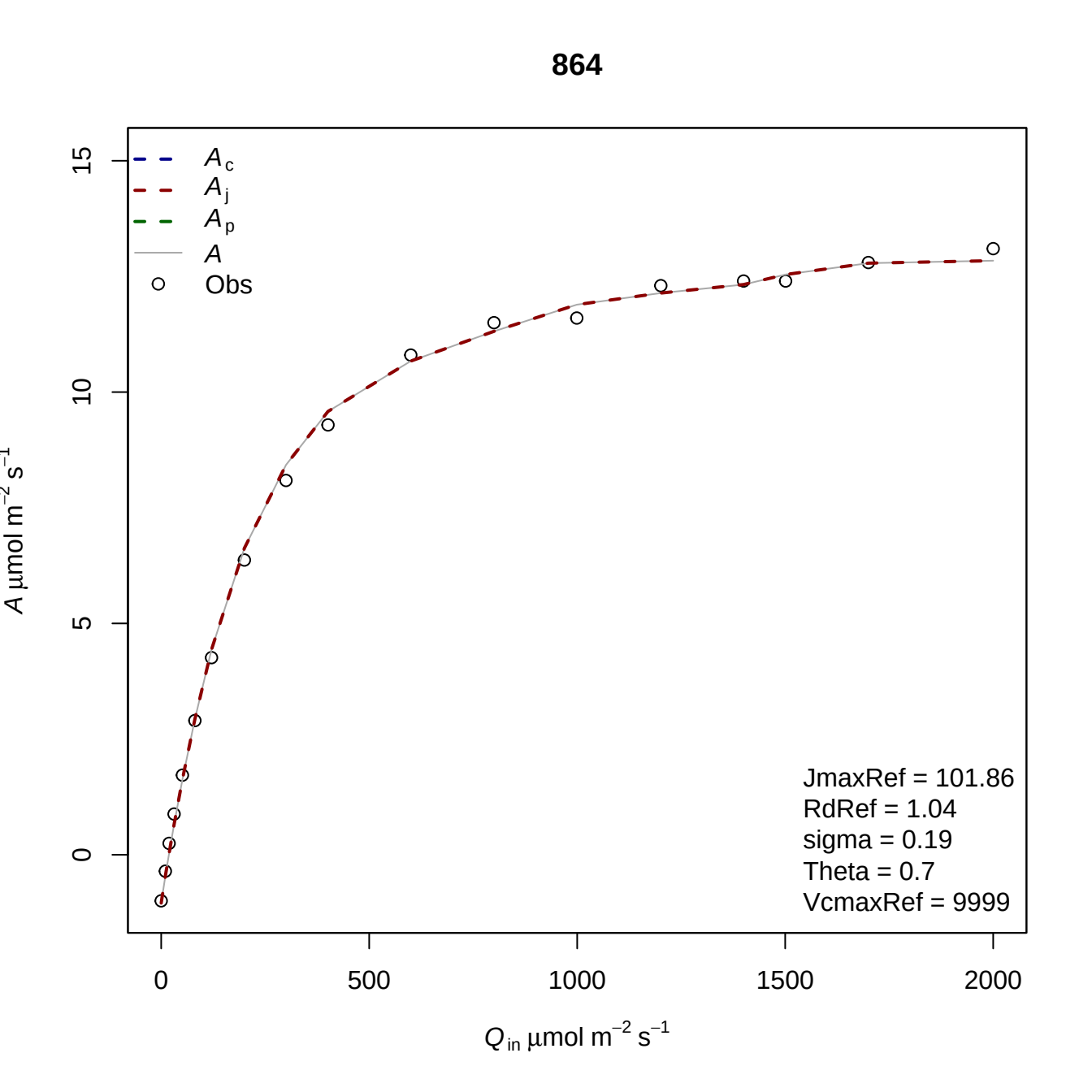

859
A_c
A_j
A_p
A
Obs
JmaxRef = 19.51
RdRef = 0.94
sigma = 0.17
Theta = 0.7
VcmaxRef = 9999
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

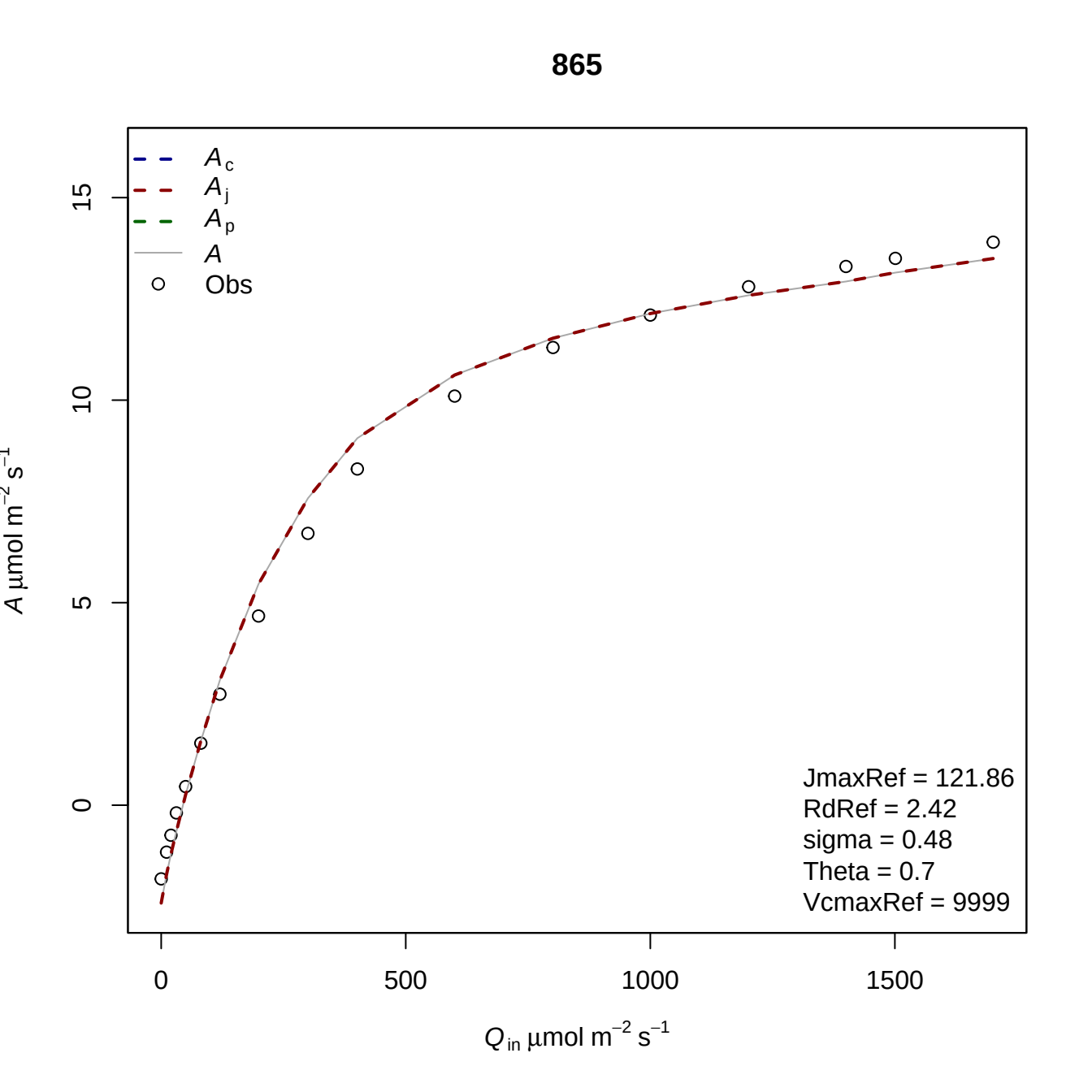

860
A_c
A_j
A_p
A
Obs
JmaxRef = 45.85
RdRef = 0.39
sigma = 0.12
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

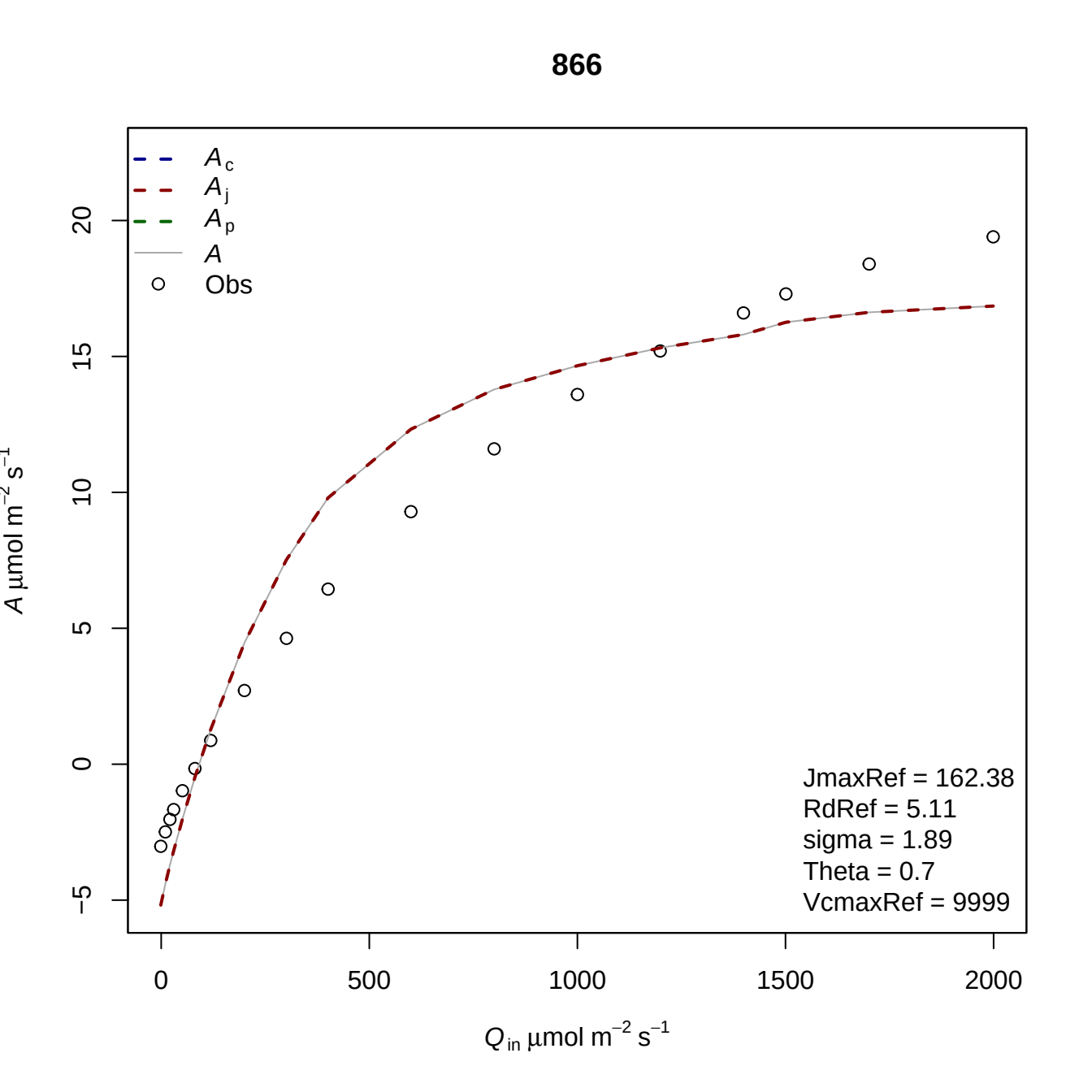

861
A_c
A_j
A_p
A
Obs
JmaxRef = 18.4
RdRef = 0.35
sigma = 0.12
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0.0
0.5
1.0
1.5
2.0
2.5
0
100
200
300
400

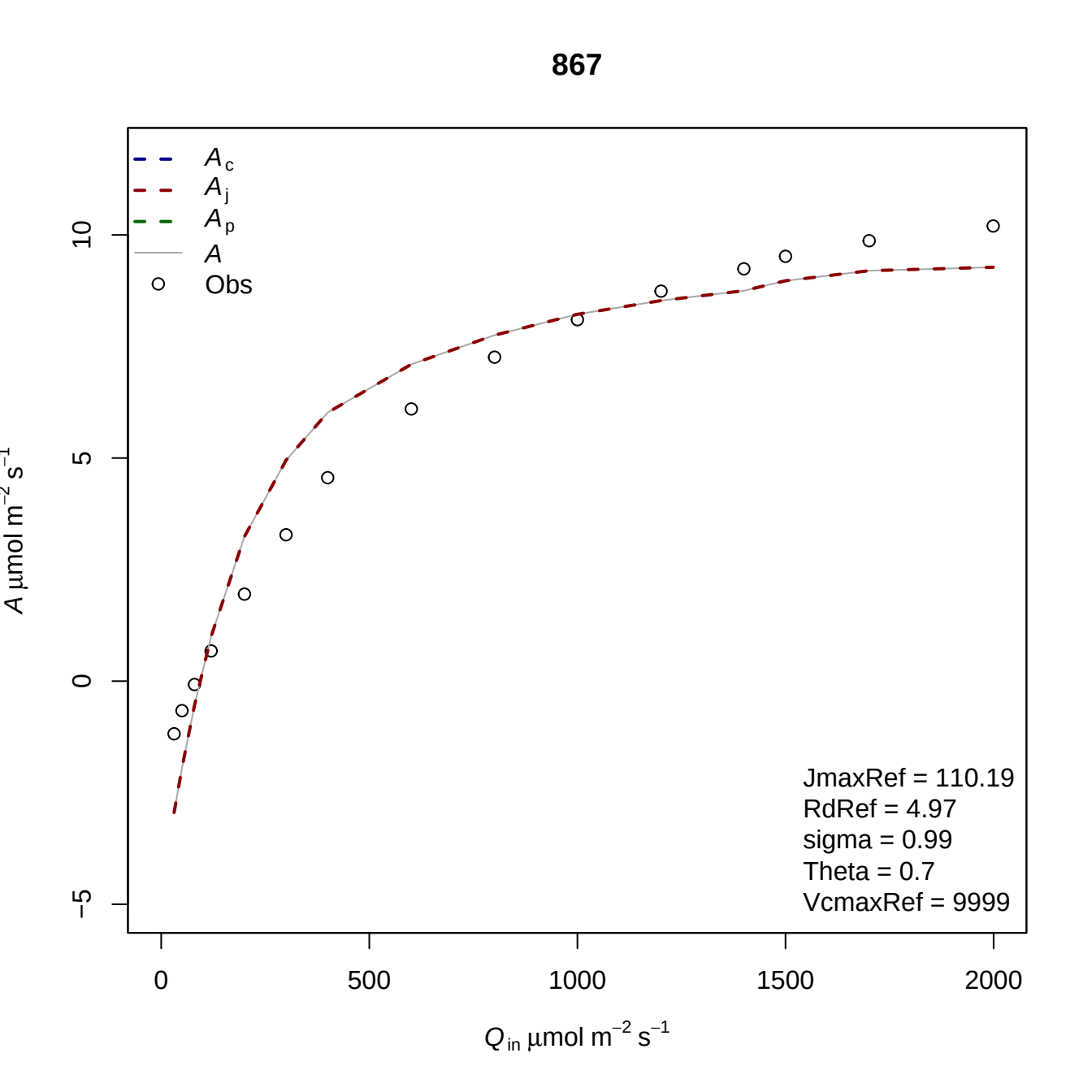

862
A_c
A_j
A_p
A
Obs
JmaxRef = 105.81
RdRef = 3.06
sigma = 0.7
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

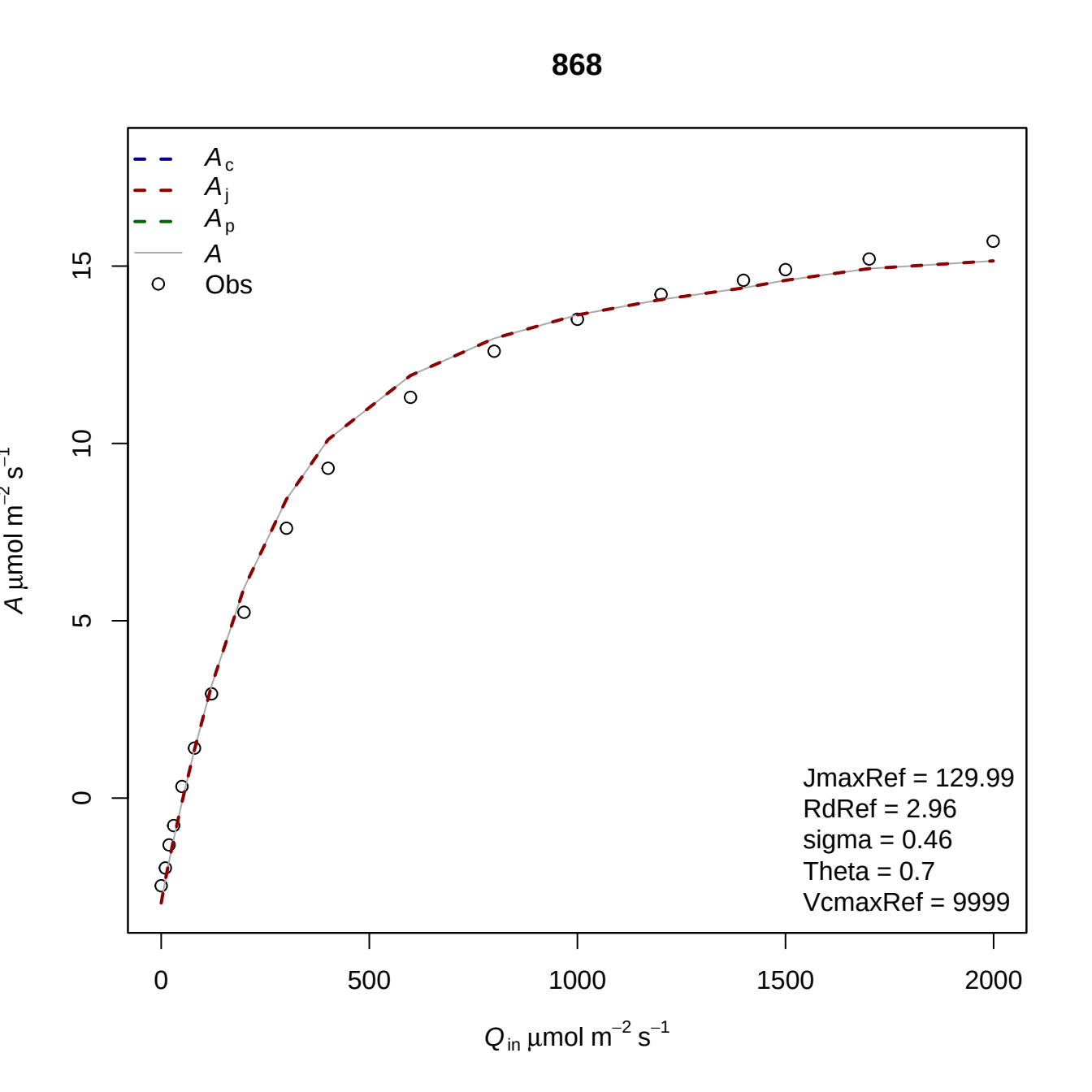

863
A_c
A_j
A_p
A
Obs
JmaxRef = 88.37
RdRef = 1.65
sigma = 0.58
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

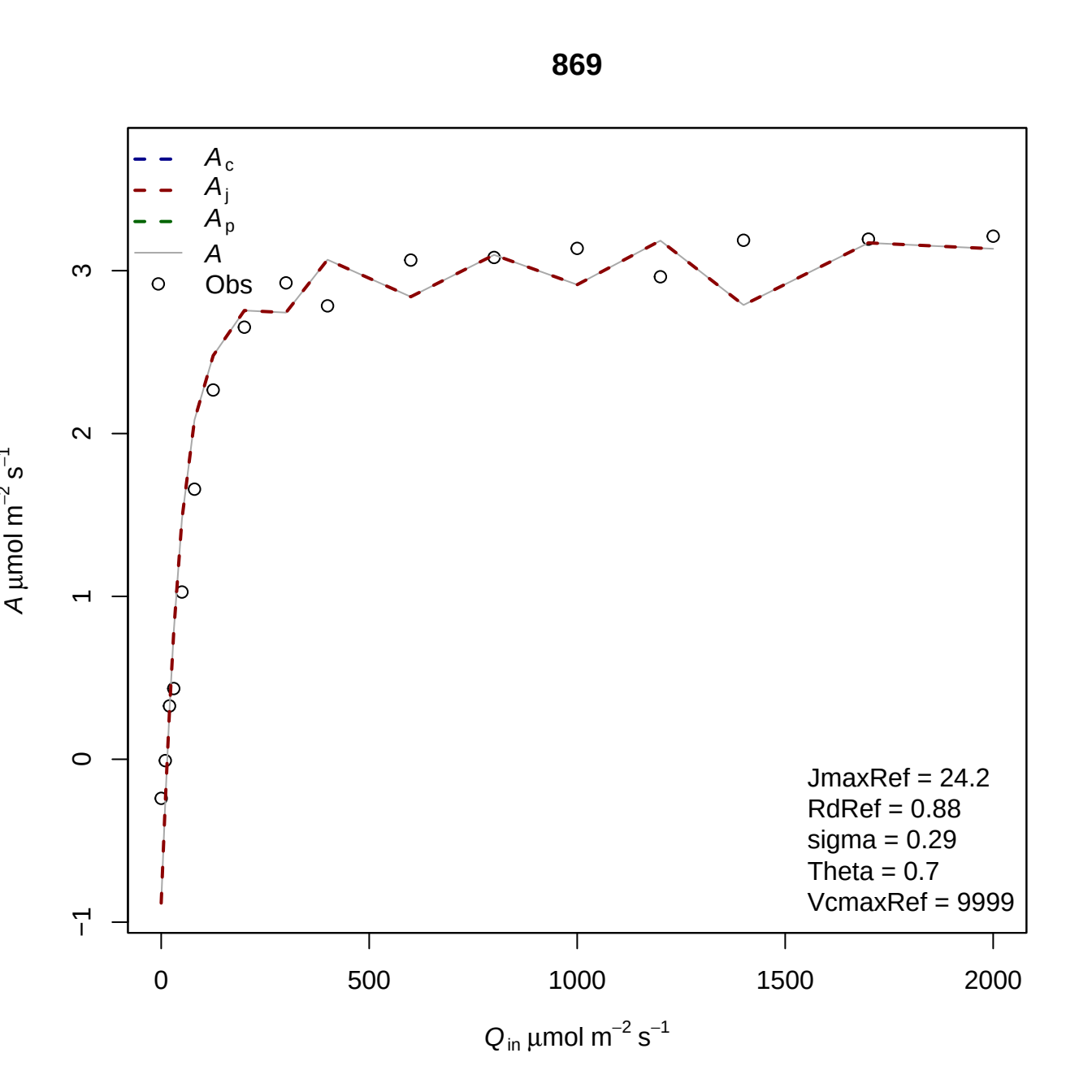

864
A_c
A_j
A_p
A
Obs
JmaxRef = 101.86
RdRef = 1.04
sigma = 0.19
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

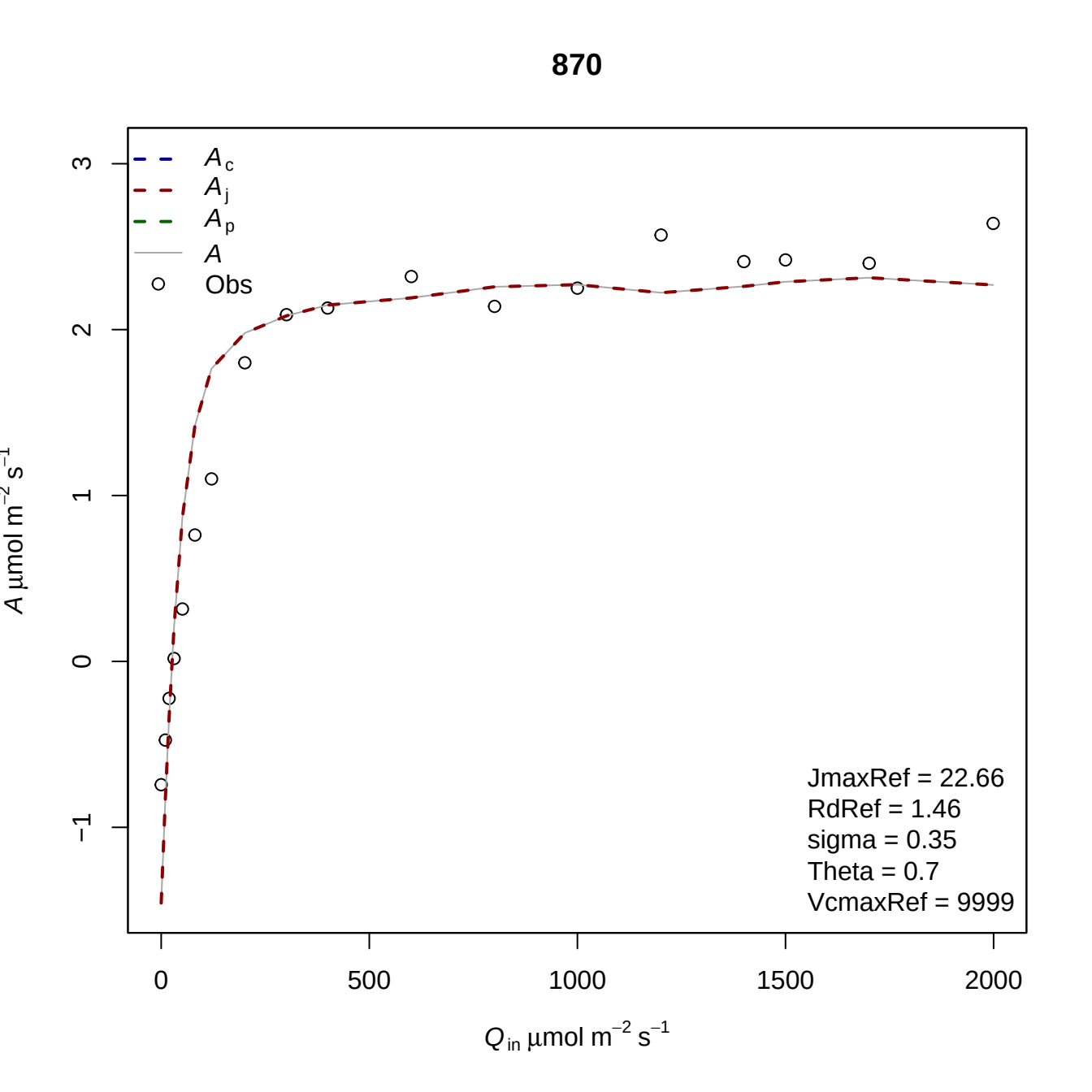

865
A_c
A_j
A_p
A
Obs
JmaxRef = 121.86
RdRef = 2.42
sigma = 0.48
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

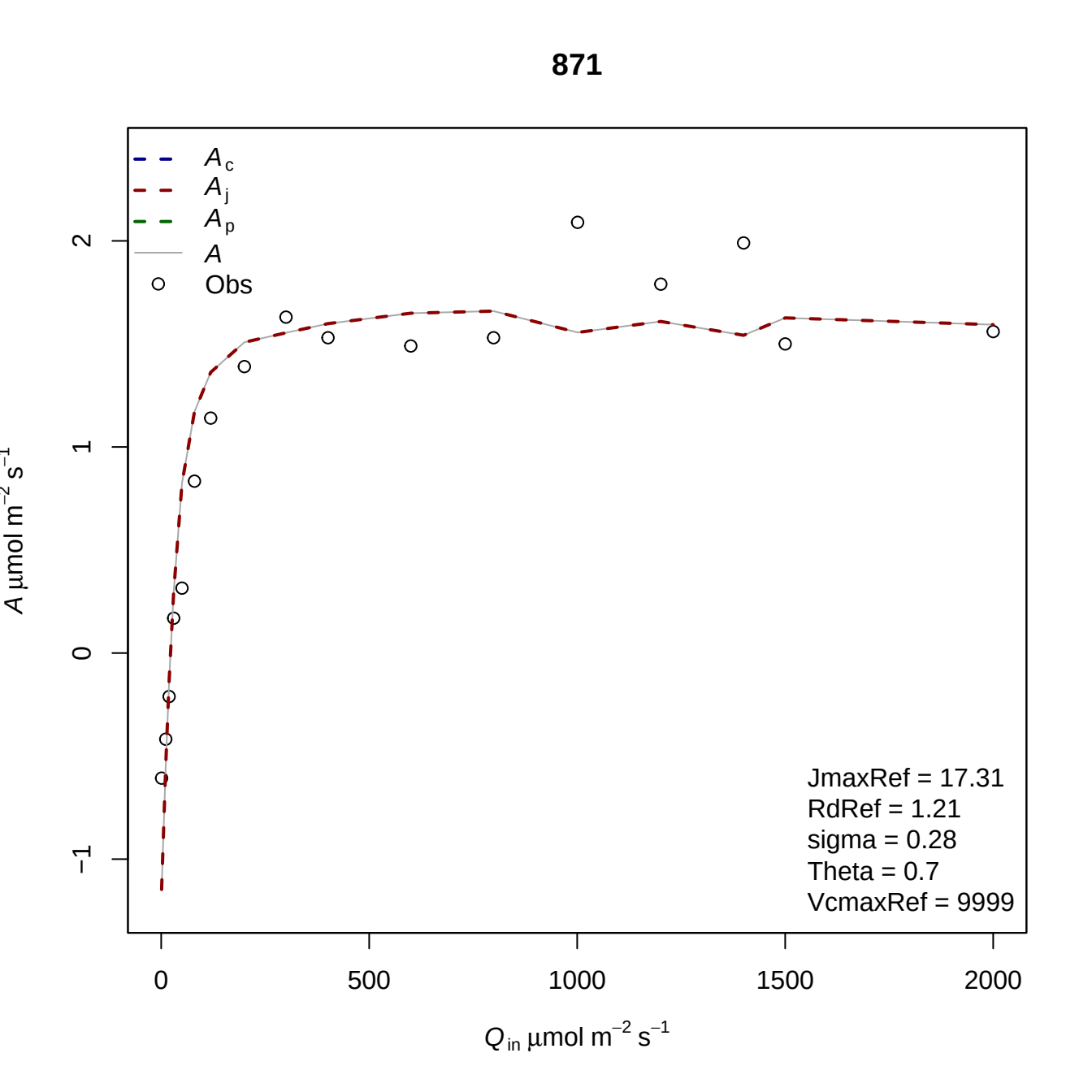

866
A_c
A_j
A_p
A
Obs
JmaxRef = 162.38
RdRef = 5.11
sigma = 1.89
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

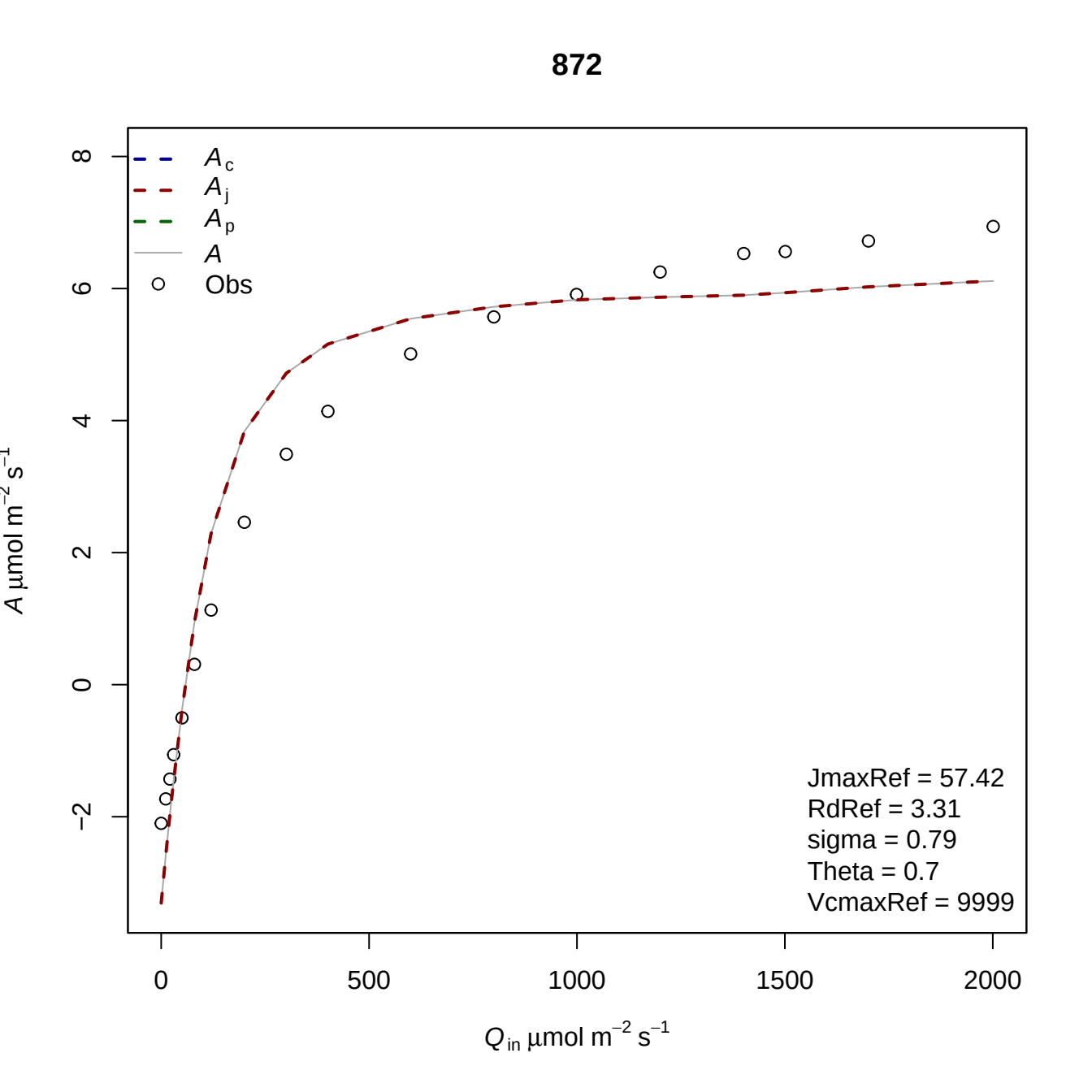

867
A_c
A_j
A_p
A
Obs
JmaxRef = 110.19
RdRef = 4.97
sigma = 0.99
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
5
0
−5
0
500
1000
1500
2000

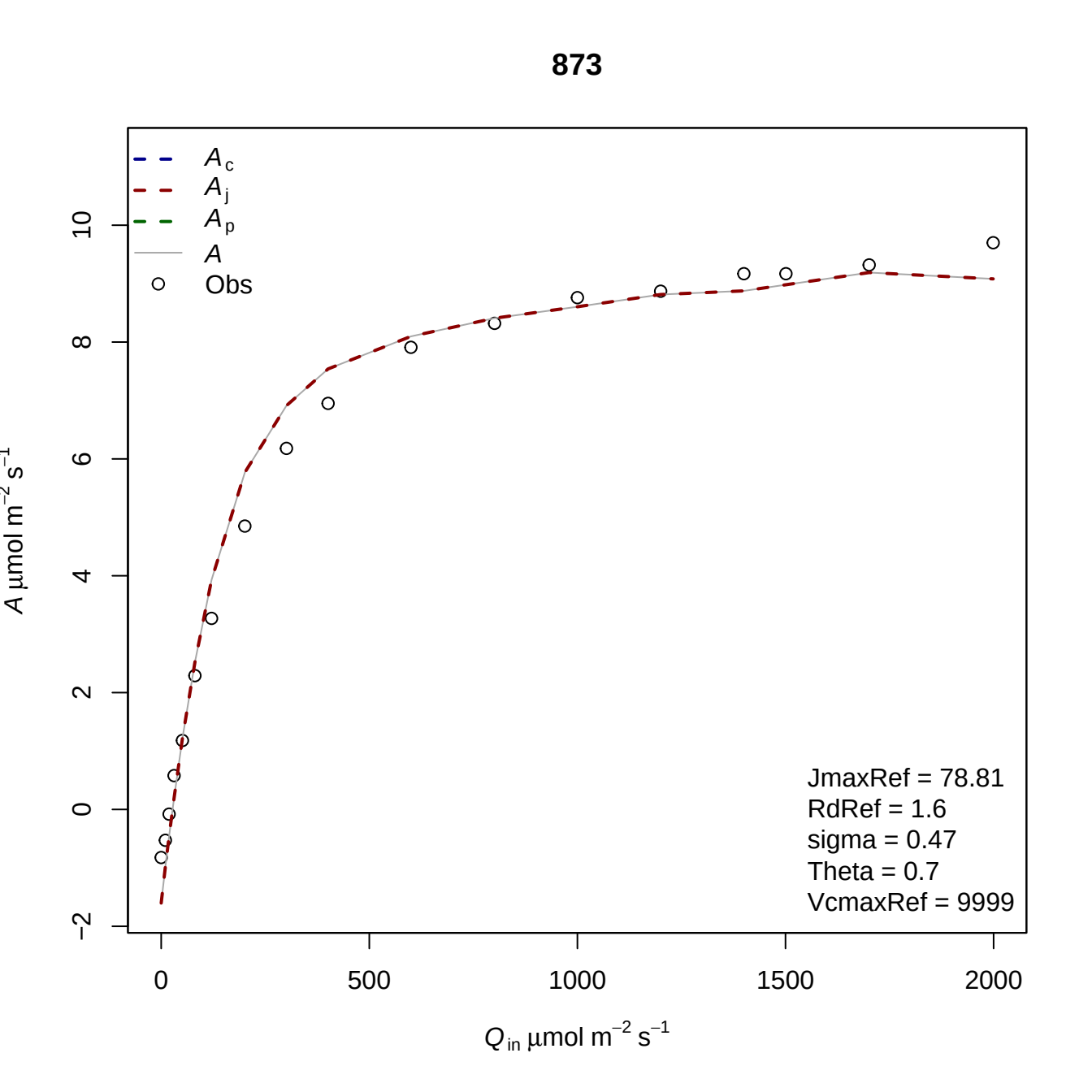

868
A_c
A_j
A_p
A
Obs
JmaxRef = 129.99
RdRef = 2.96
sigma = 0.46
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

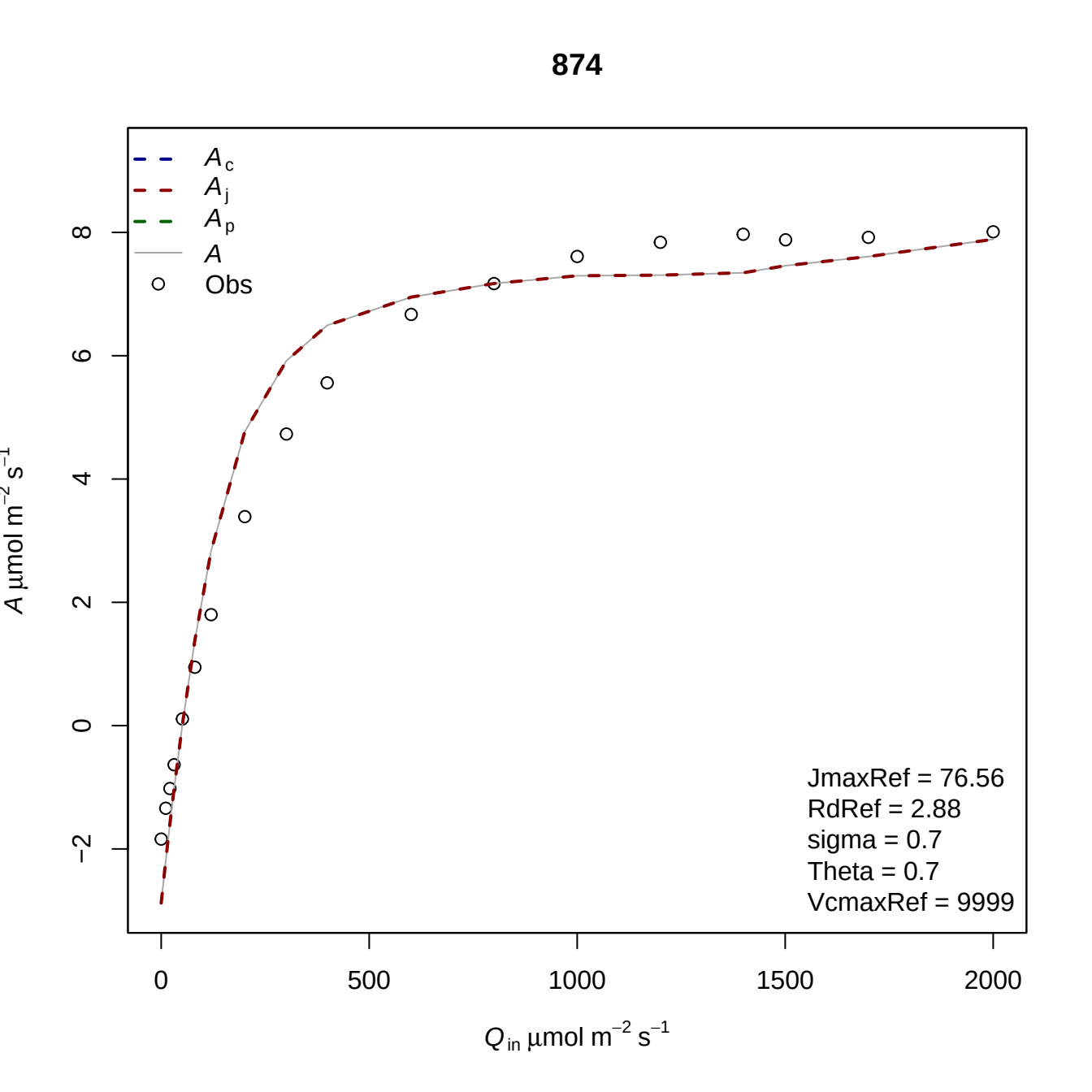

869
A_c
A_j
A_p
A
Obs
JmaxRef = 24.2
RdRef = 0.88
sigma = 0.29
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

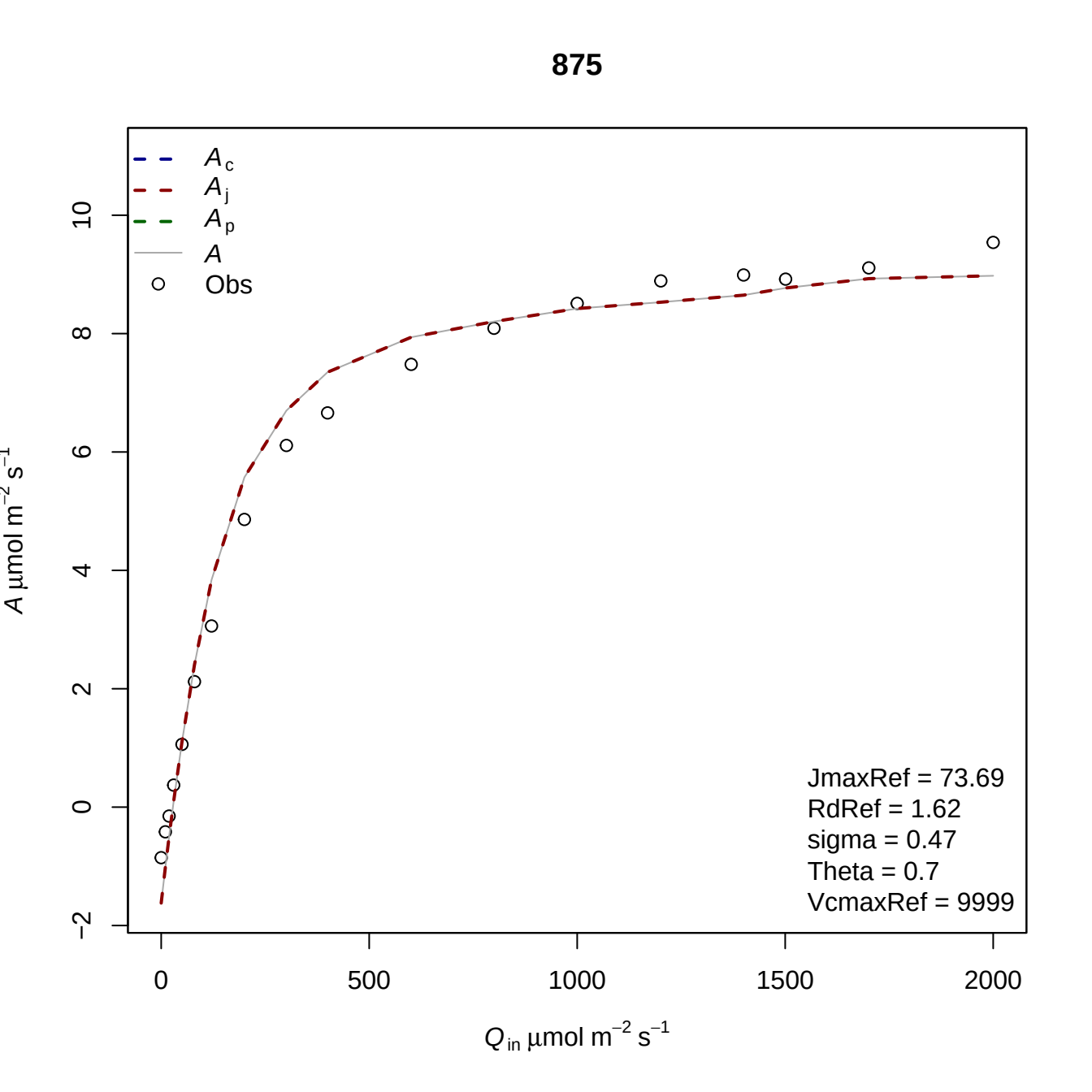

870
A_c
A_j
A_p
A
Obs
JmaxRef = 22.66
RdRef = 1.46
sigma = 0.35
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
0
500
1000
1500
2000

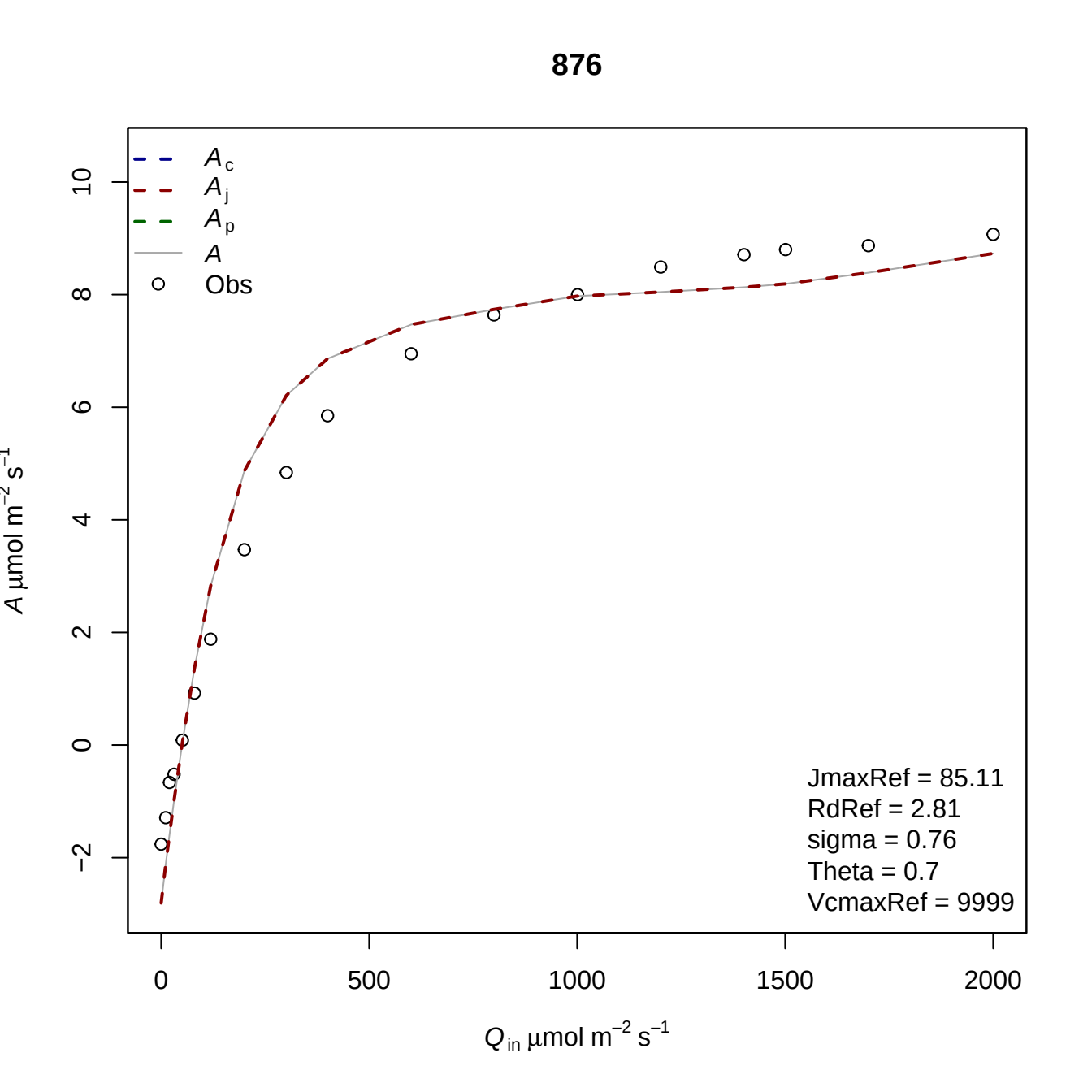

871
A_c
A_j
A_p
A
Obs
JmaxRef = 17.31
RdRef = 1.21
sigma = 0.28
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
500
1000
1500
2000

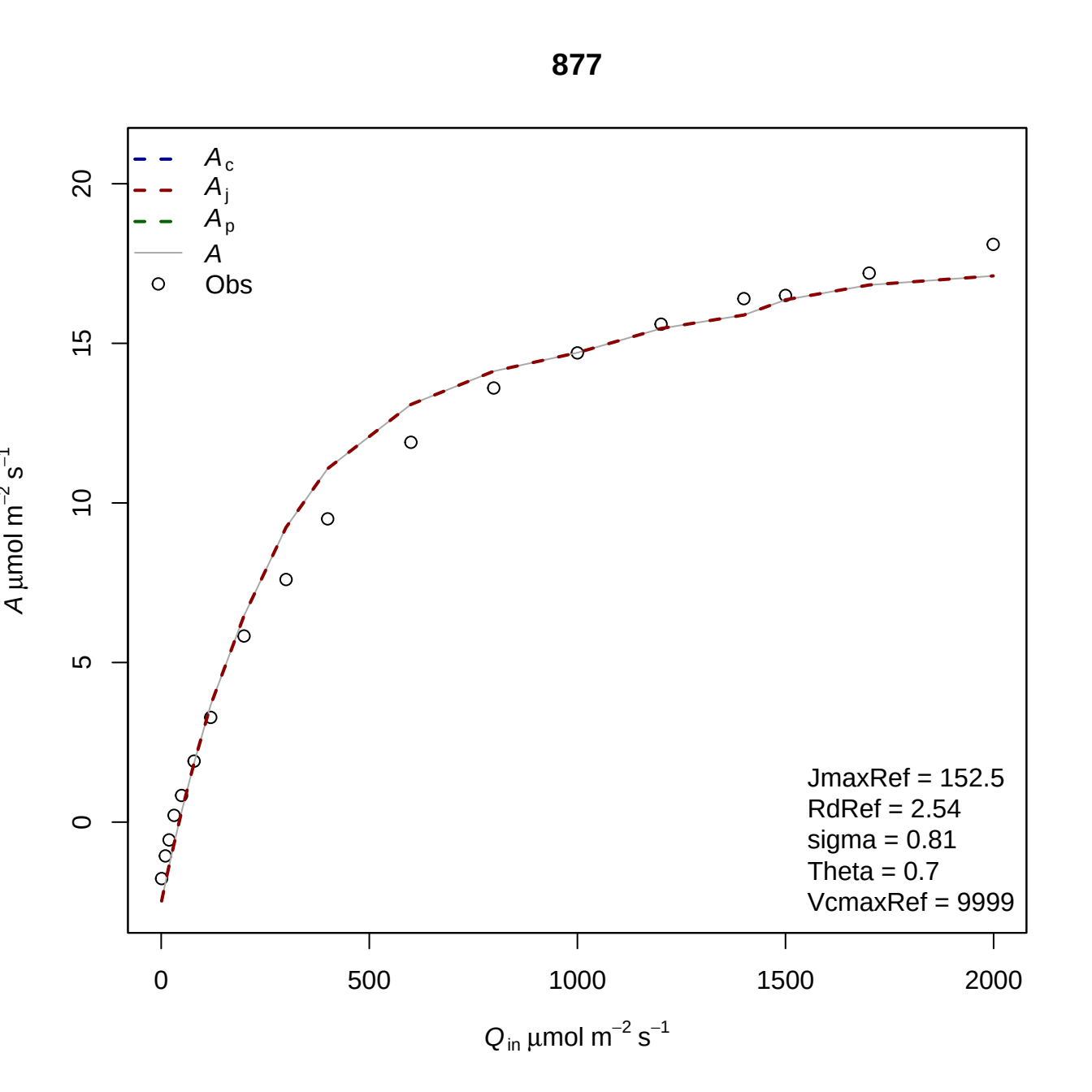

872
A_c
A_j
A_p
A
Obs
JmaxRef = 57.42
RdRef = 3.31
sigma = 0.79
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

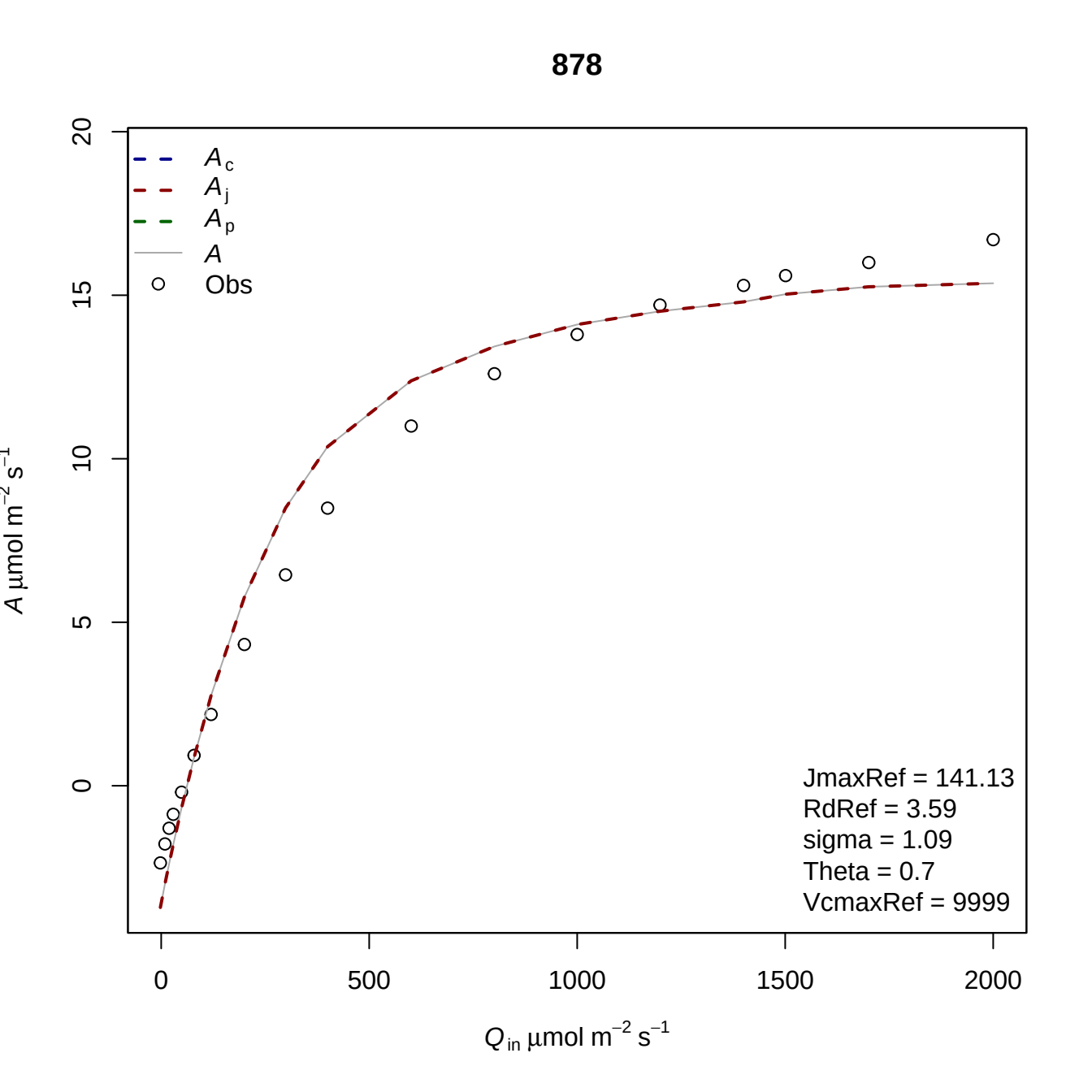

873
A_c
A_j
A_p
A
Obs
JmaxRef = 78.81
RdRef = 1.6
sigma = 0.47
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

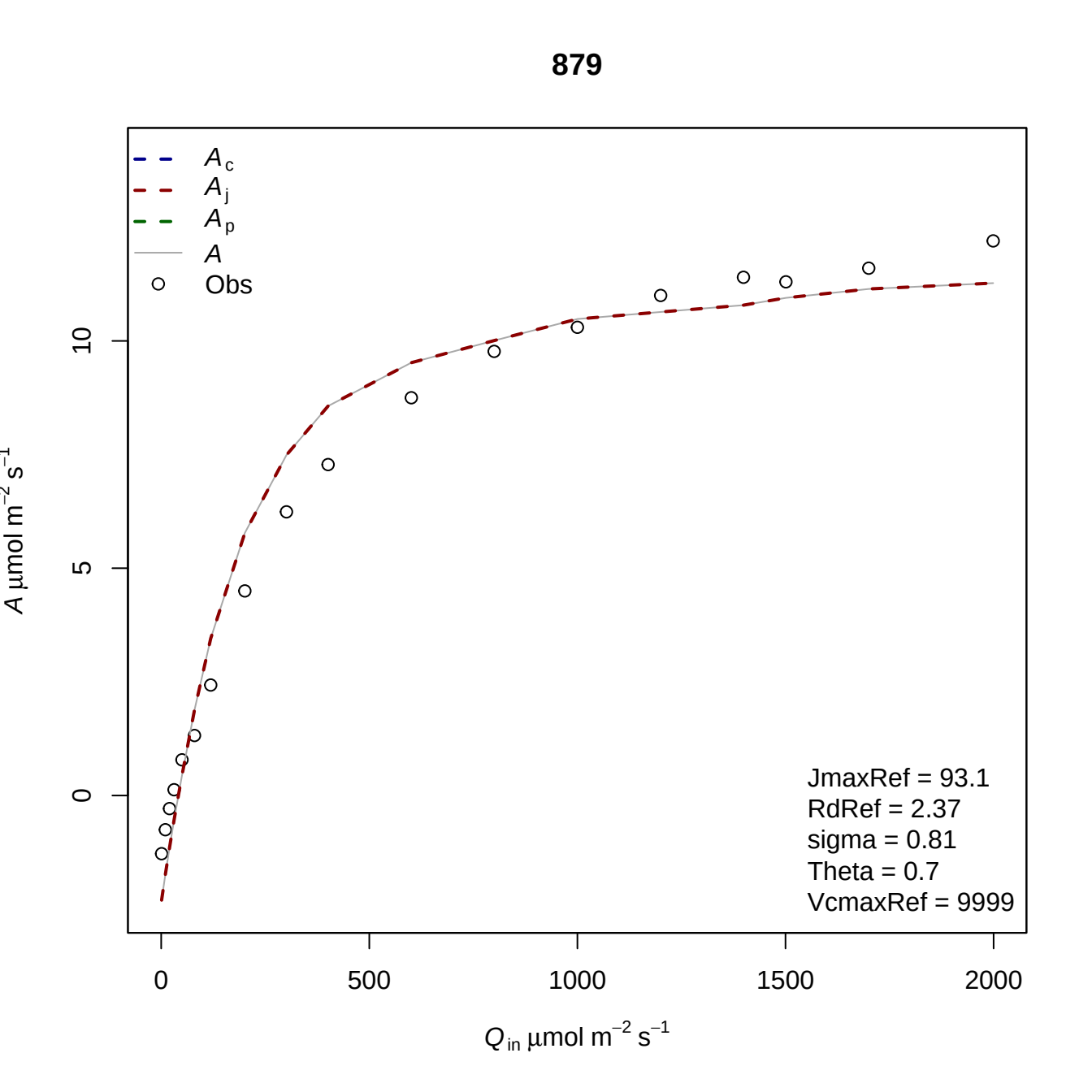

874
A_c
A_j
A_p
A
Obs
JmaxRef = 76.56
RdRef = 2.88
sigma = 0.7
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

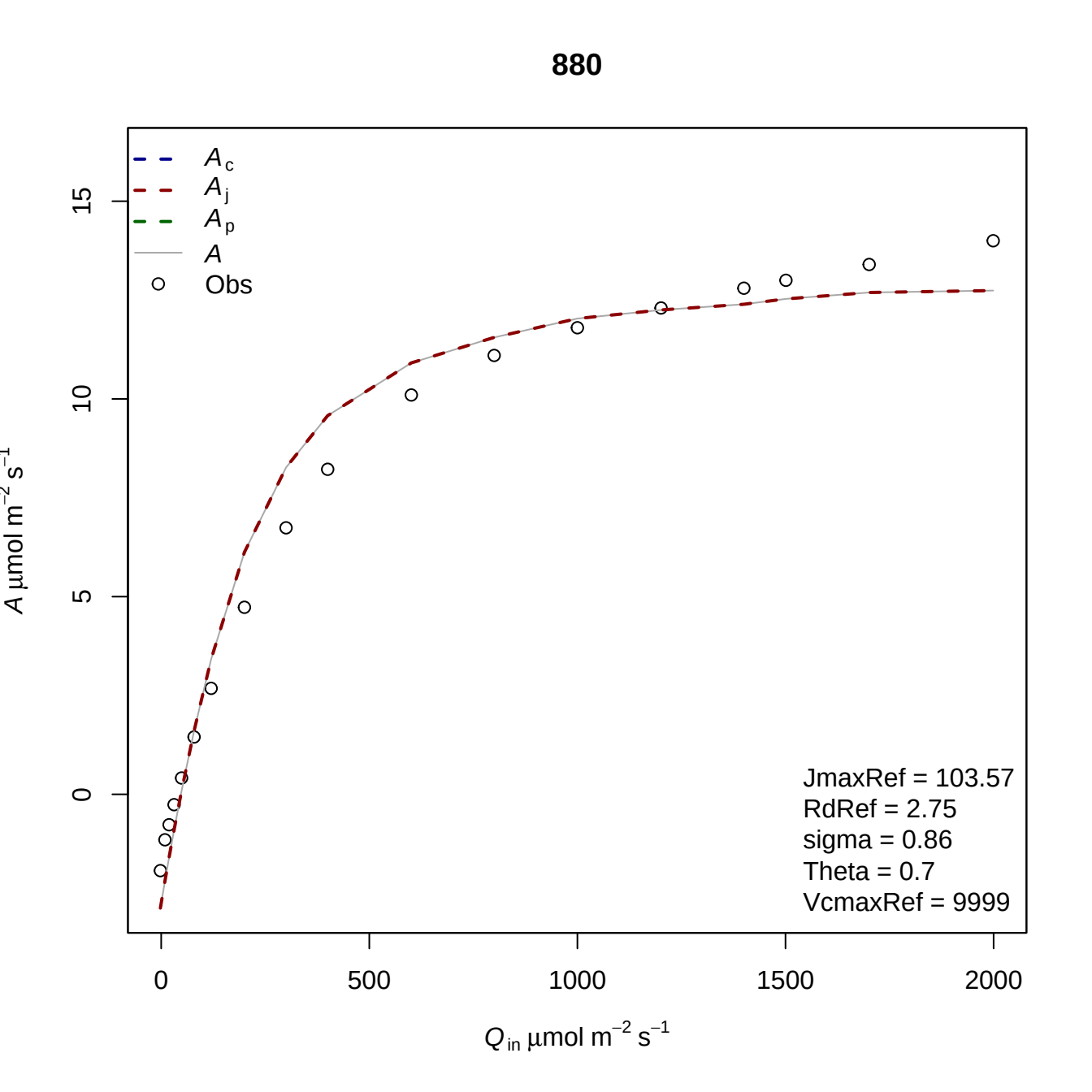

875
A_c
A_j
A_p
A
Obs
JmaxRef = 73.69
RdRef = 1.62
sigma = 0.47
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

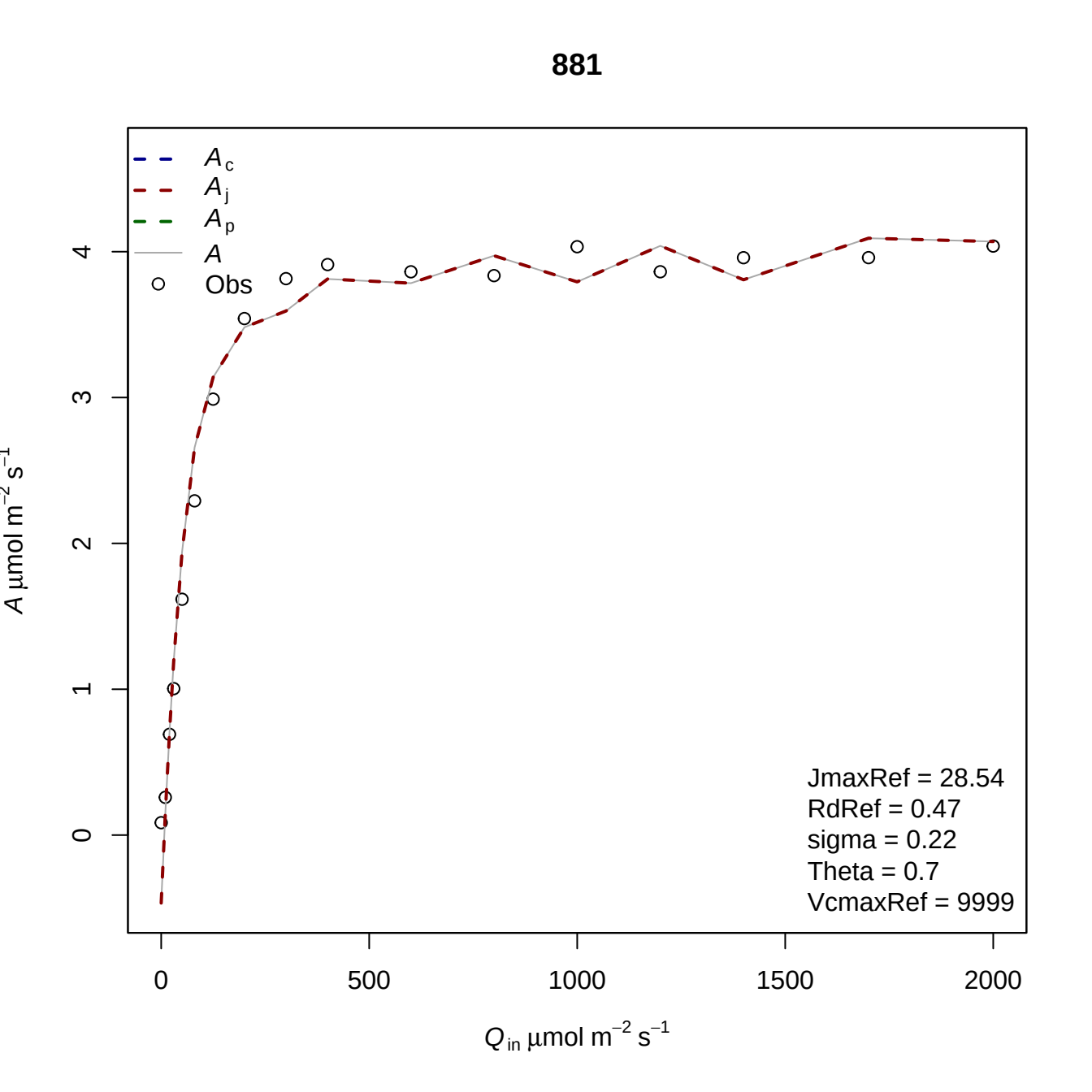

876
A_c
A_j
A_p
A
Obs
JmaxRef = 85.11
RdRef = 2.81
sigma = 0.76
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

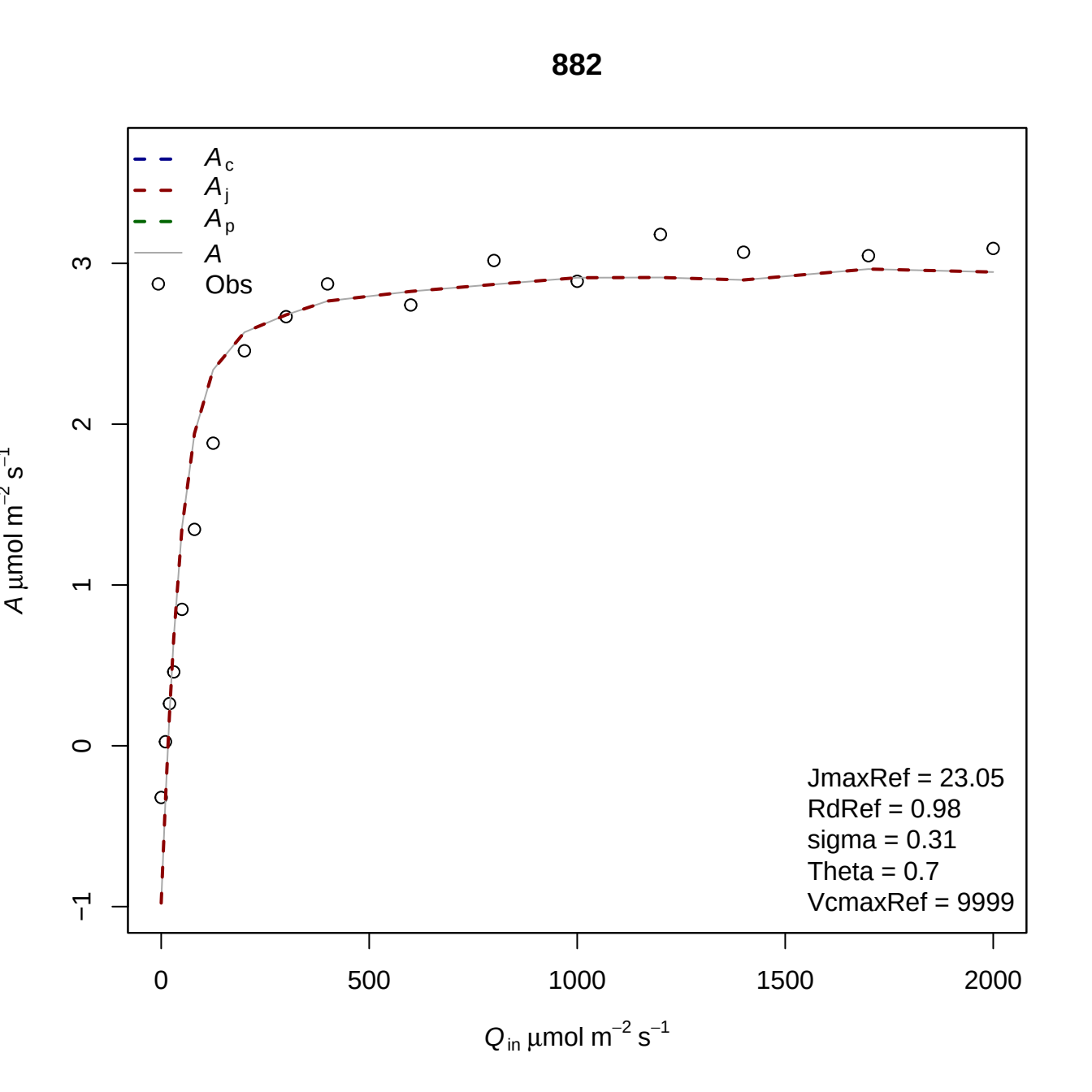

877
A_c
A_j
A_p
A
Obs
JmaxRef = 152.5
RdRef = 2.54
sigma = 0.81
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

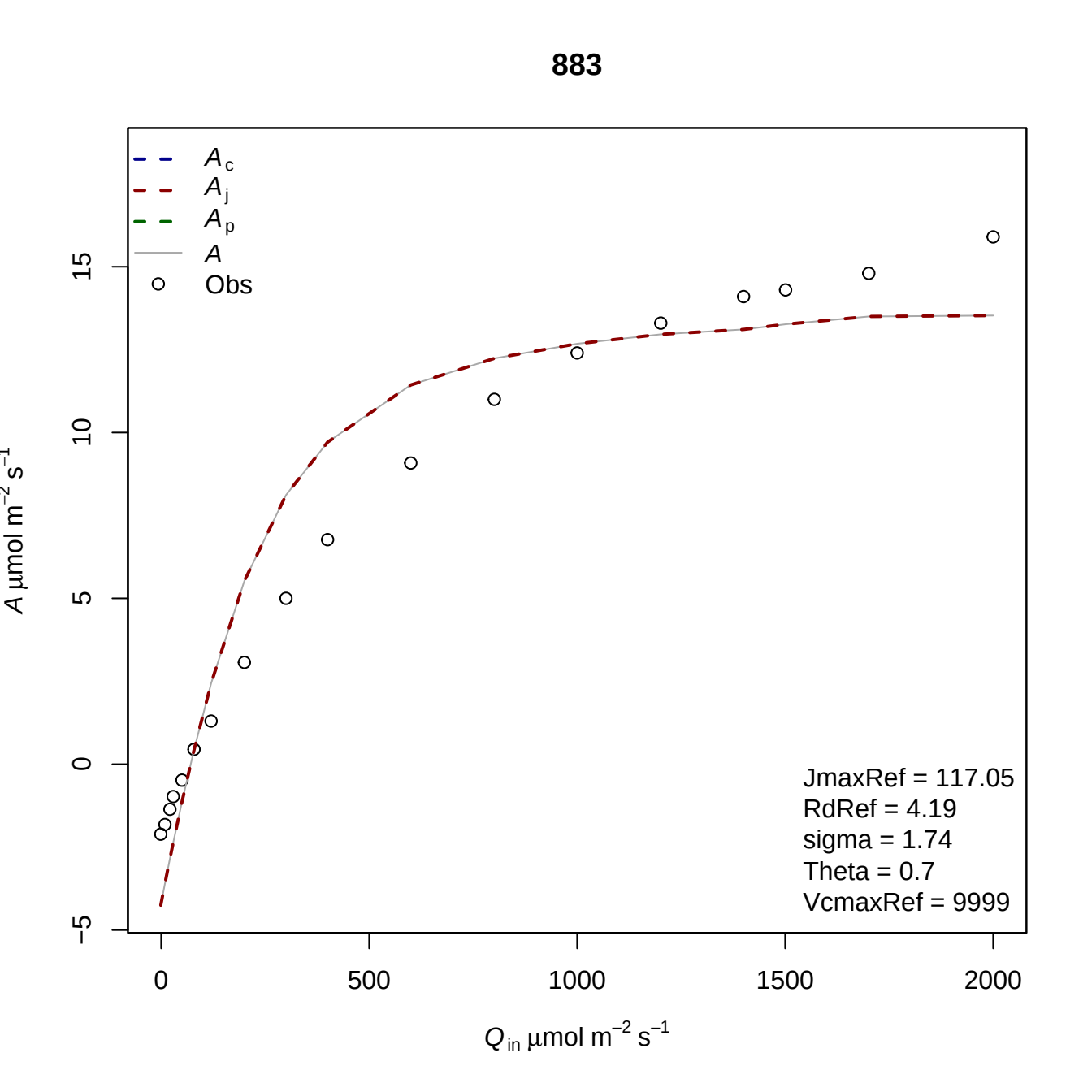

878
A_c
A_j
A_p
A
Obs
JmaxRef = 141.13
RdRef = 3.59
sigma = 1.09
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

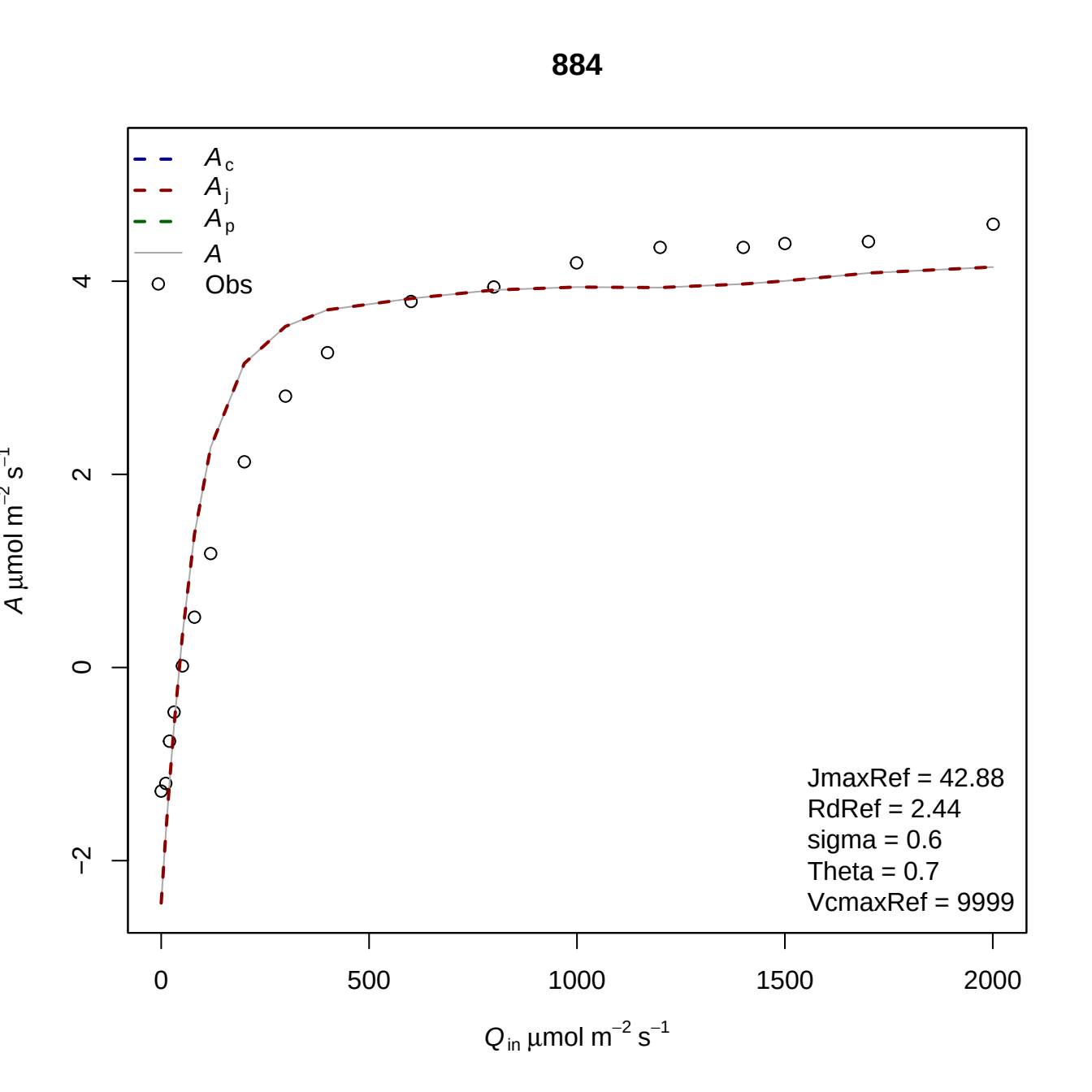

879
A_c
A_j
A_p
A
Obs
JmaxRef = 93.1
RdRef = 2.37
sigma = 0.81
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

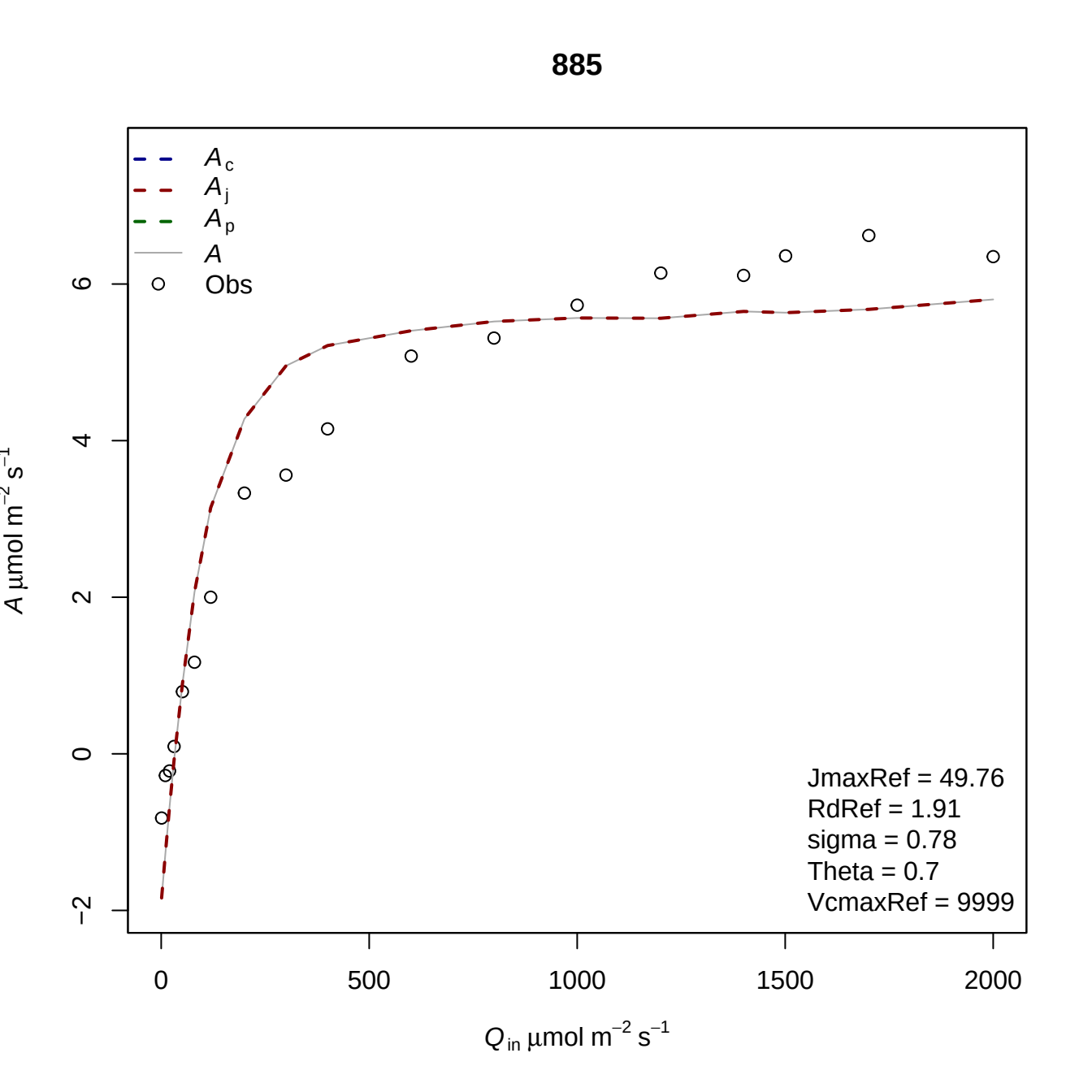

880
A_c
A_j
A_p
A
Obs
JmaxRef = 103.57
RdRef = 2.75
sigma = 0.86
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

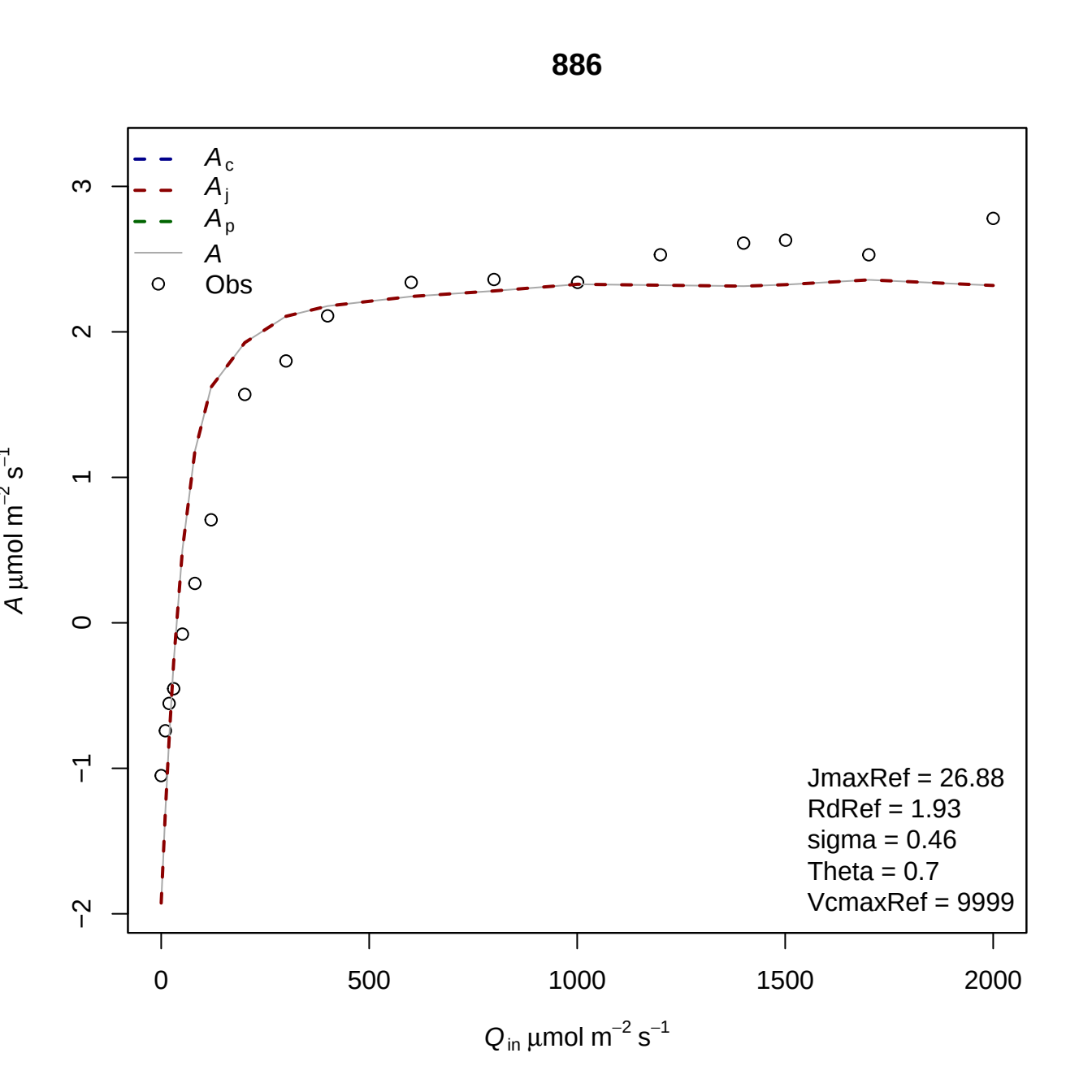

881
A_c
A_j
A_p
A
Obs
JmaxRef = 28.54
RdRef = 0.47
sigma = 0.22
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500
2000

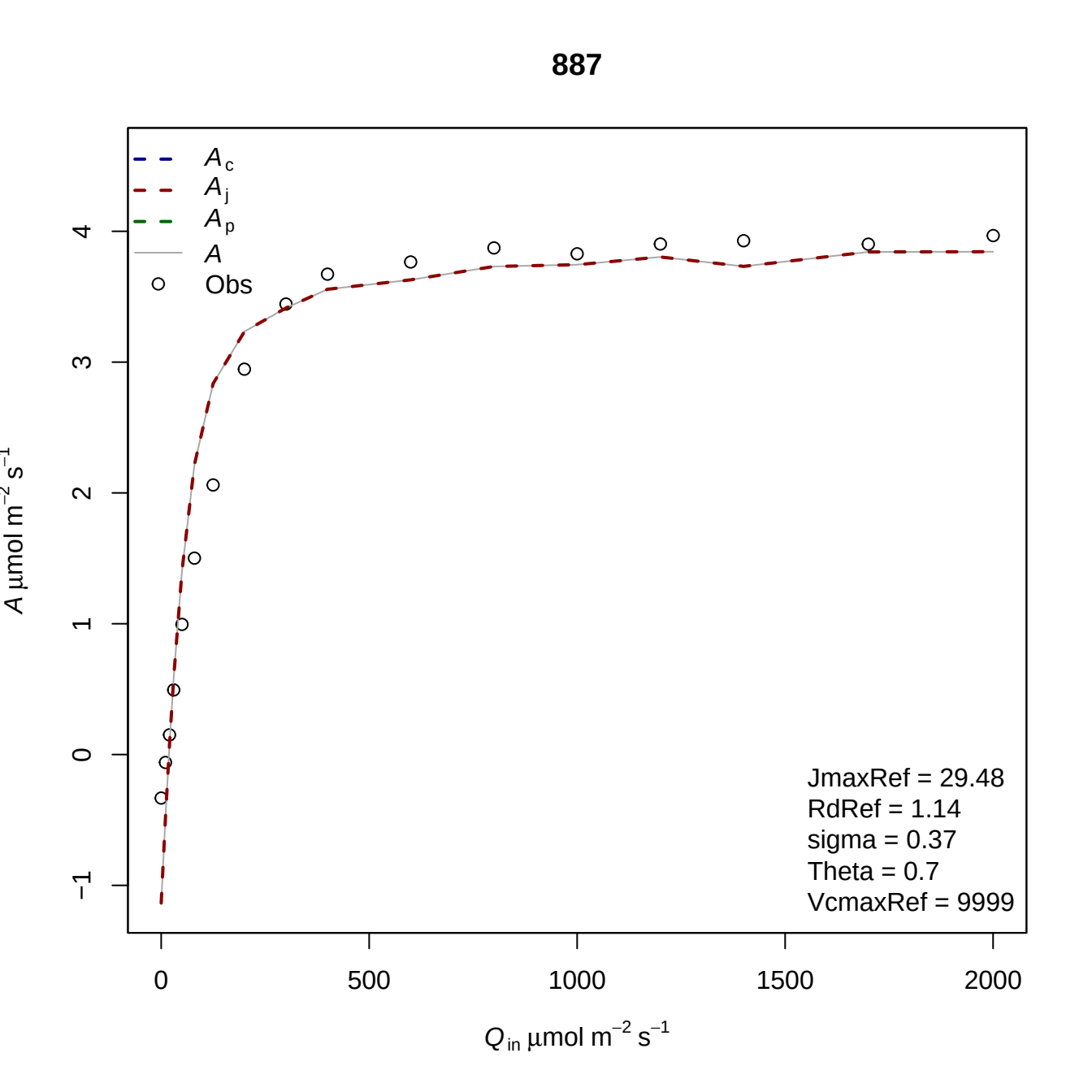

882
A_c
A_j
A_p
A
Obs
JmaxRef = 23.05
RdRef = 0.98
sigma = 0.31
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

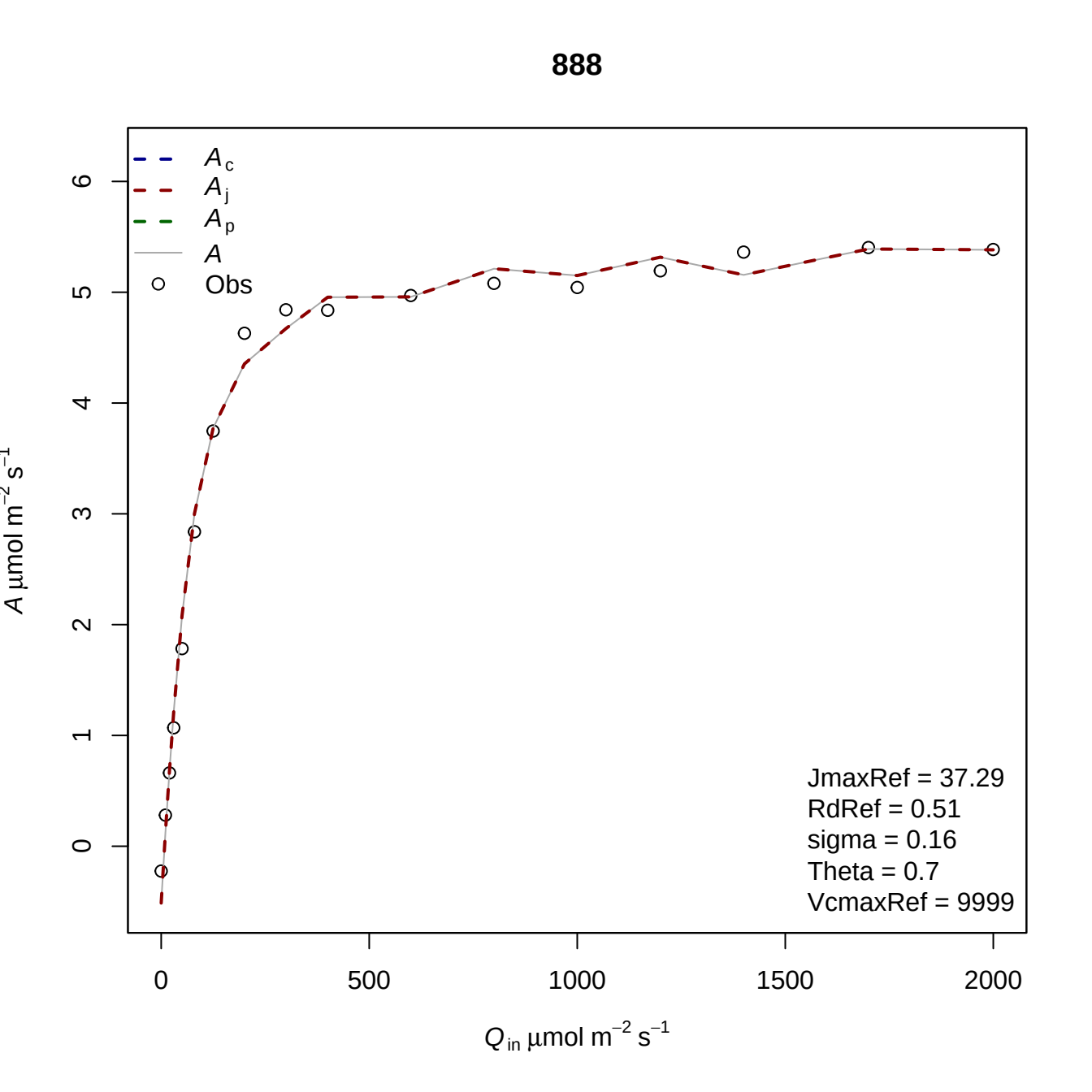

883
A_c
A_j
A_p
A
Obs
JmaxRef = 117.05
RdRef = 4.19
sigma = 1.74
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

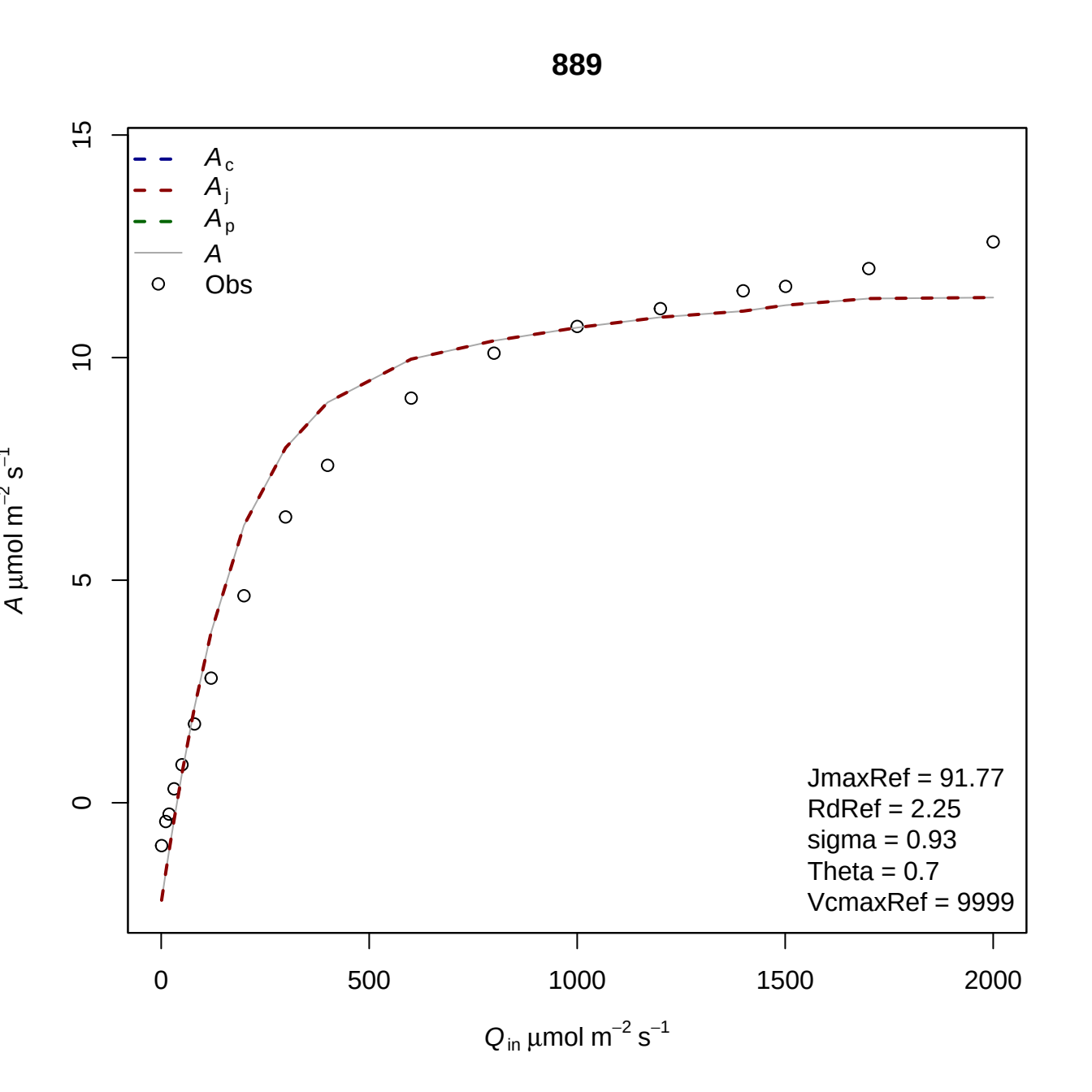

884
A_c
A_j
A_p
A
Obs
JmaxRef = 42.88
RdRef = 2.44
sigma = 0.6
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

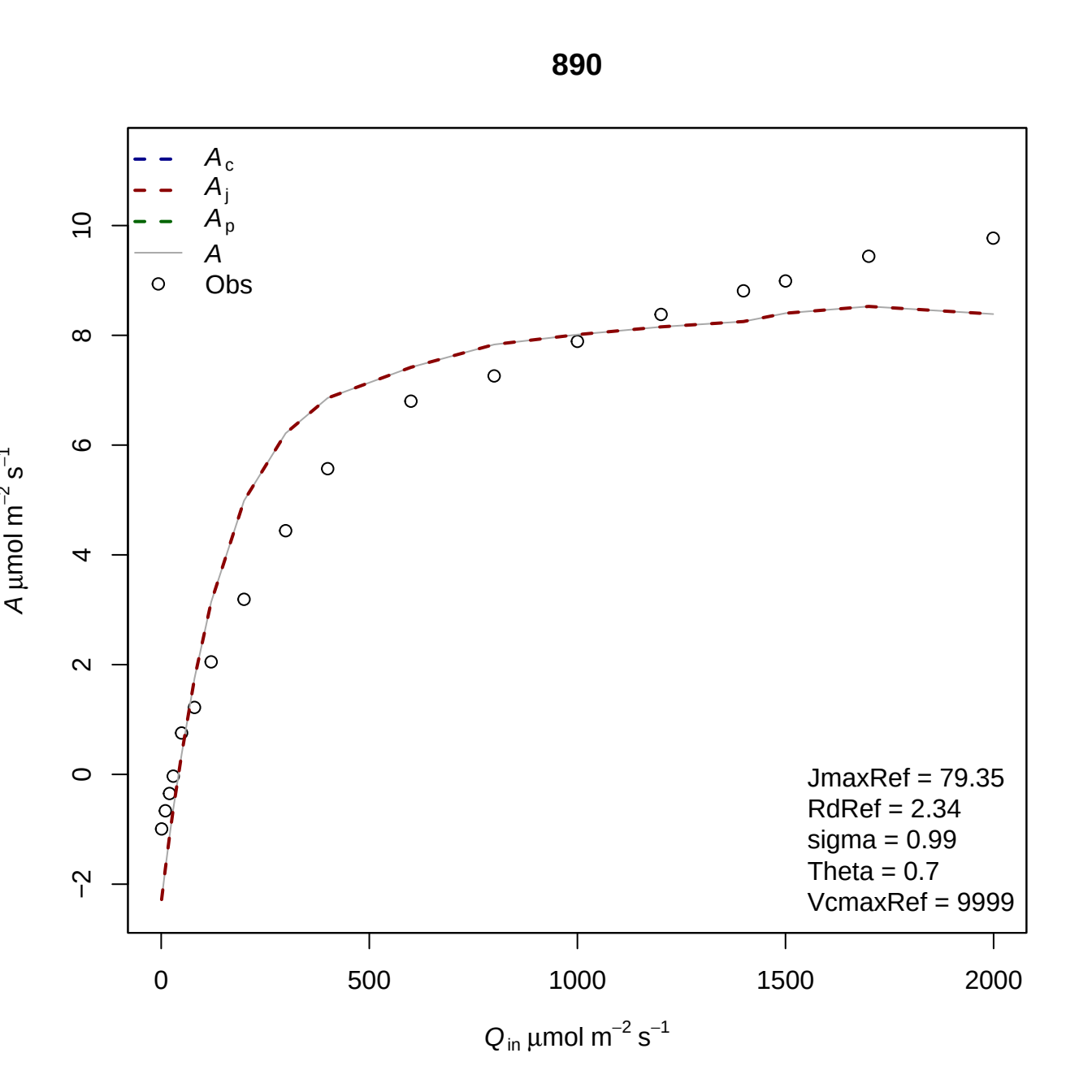

885
A_c
A_j
A_p
A
Obs
JmaxRef = 49.76
RdRef = 1.91
sigma = 0.78
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

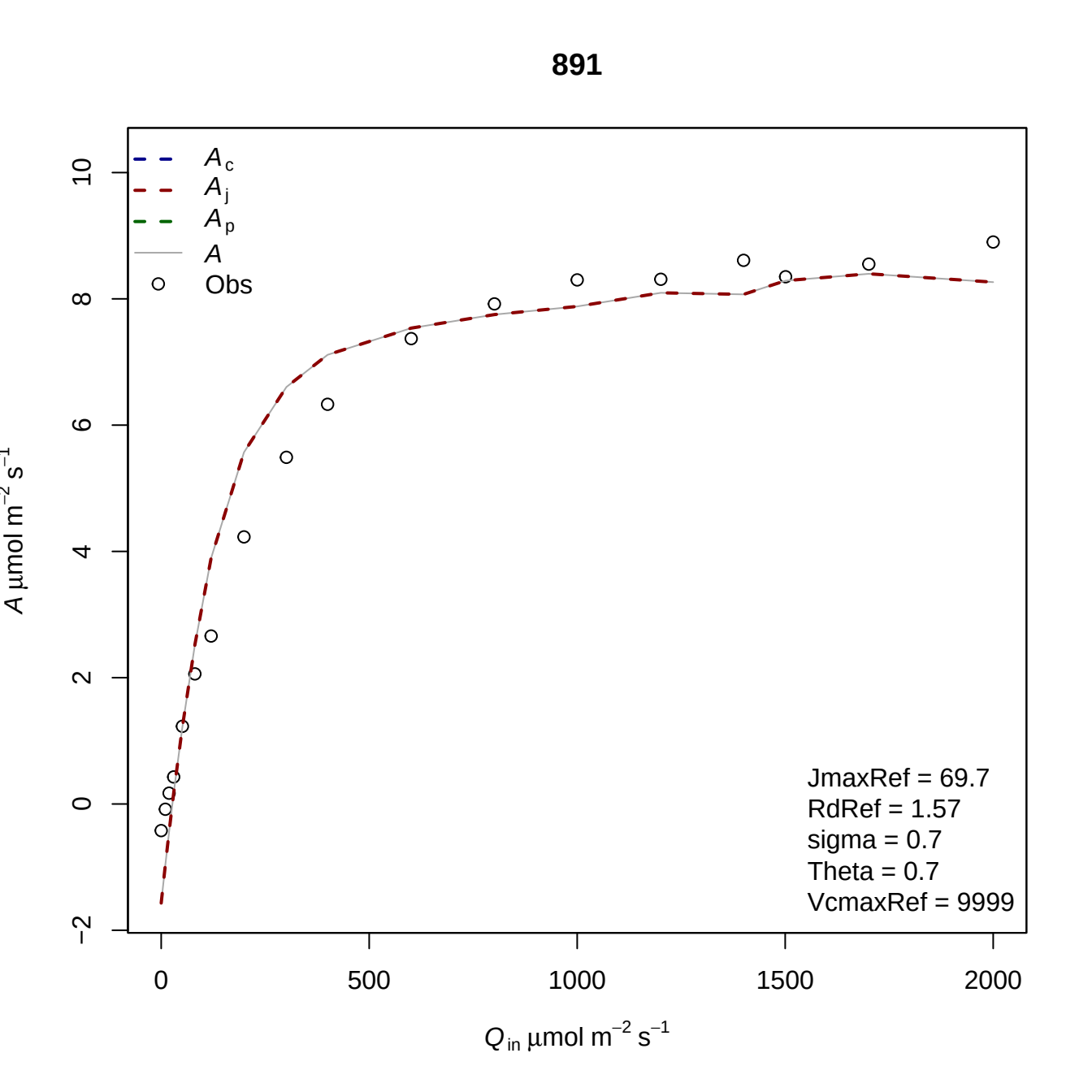

886
A_c
A_j
A_p
A
Obs
JmaxRef = 26.88
RdRef = 1.93
sigma = 0.46
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
−2
0
500
1000
1500
2000

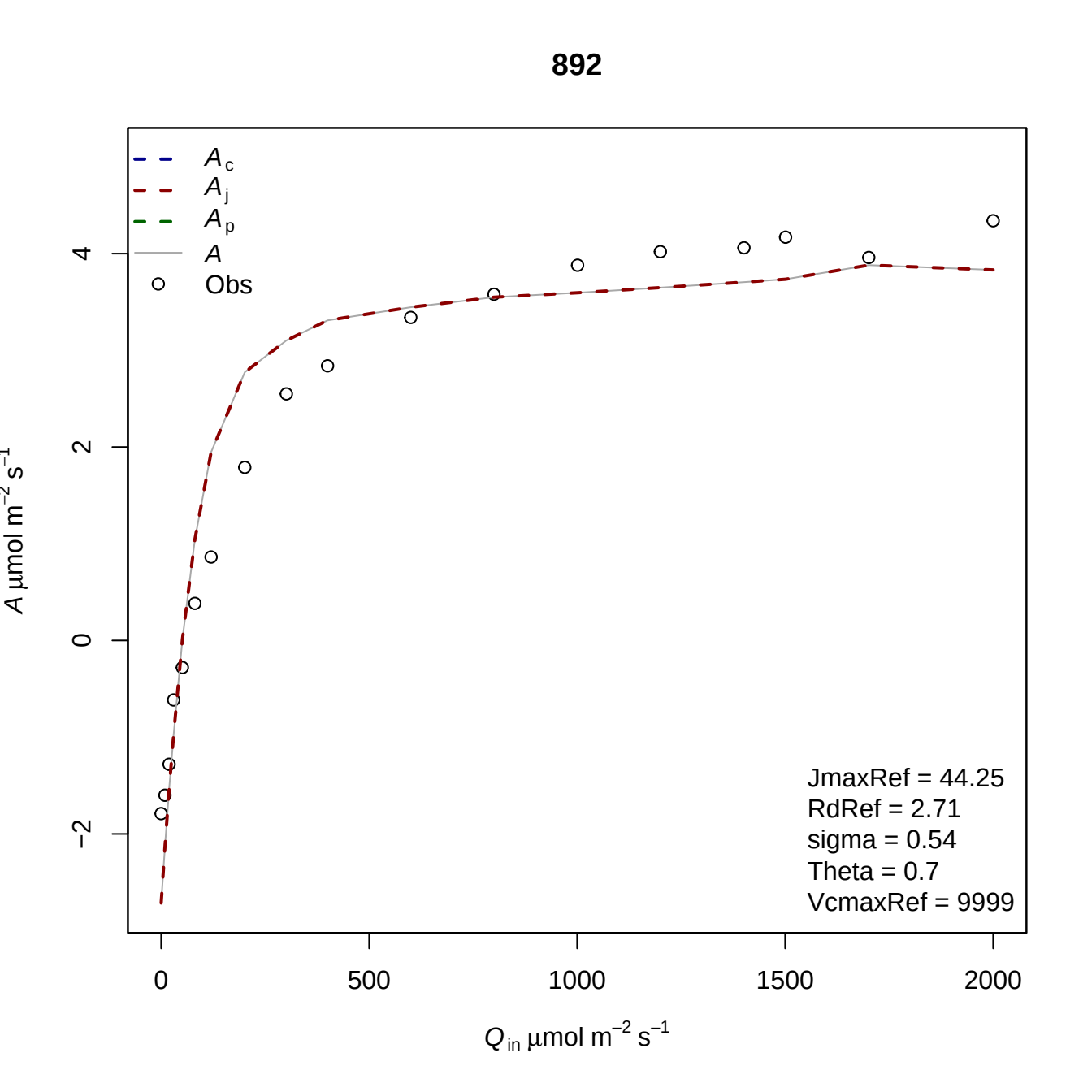

887
A_c
A_j
A_p
A
Obs
JmaxRef = 29.48
RdRef = 1.14
sigma = 0.37
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500
2000

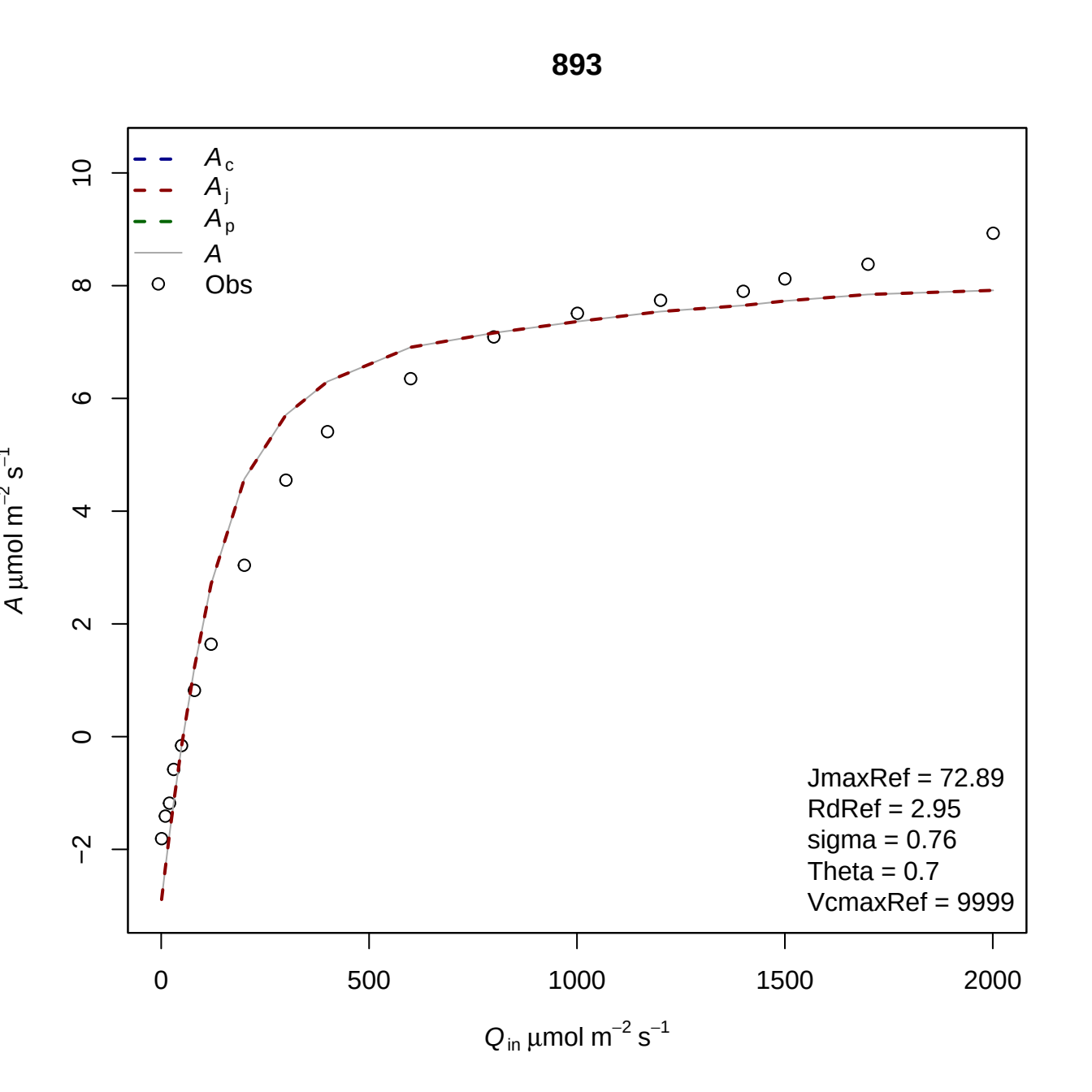

888
A_c
A_j
A_p
A
Obs
JmaxRef = 37.29
RdRef = 0.51
sigma = 0.16
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500
2000

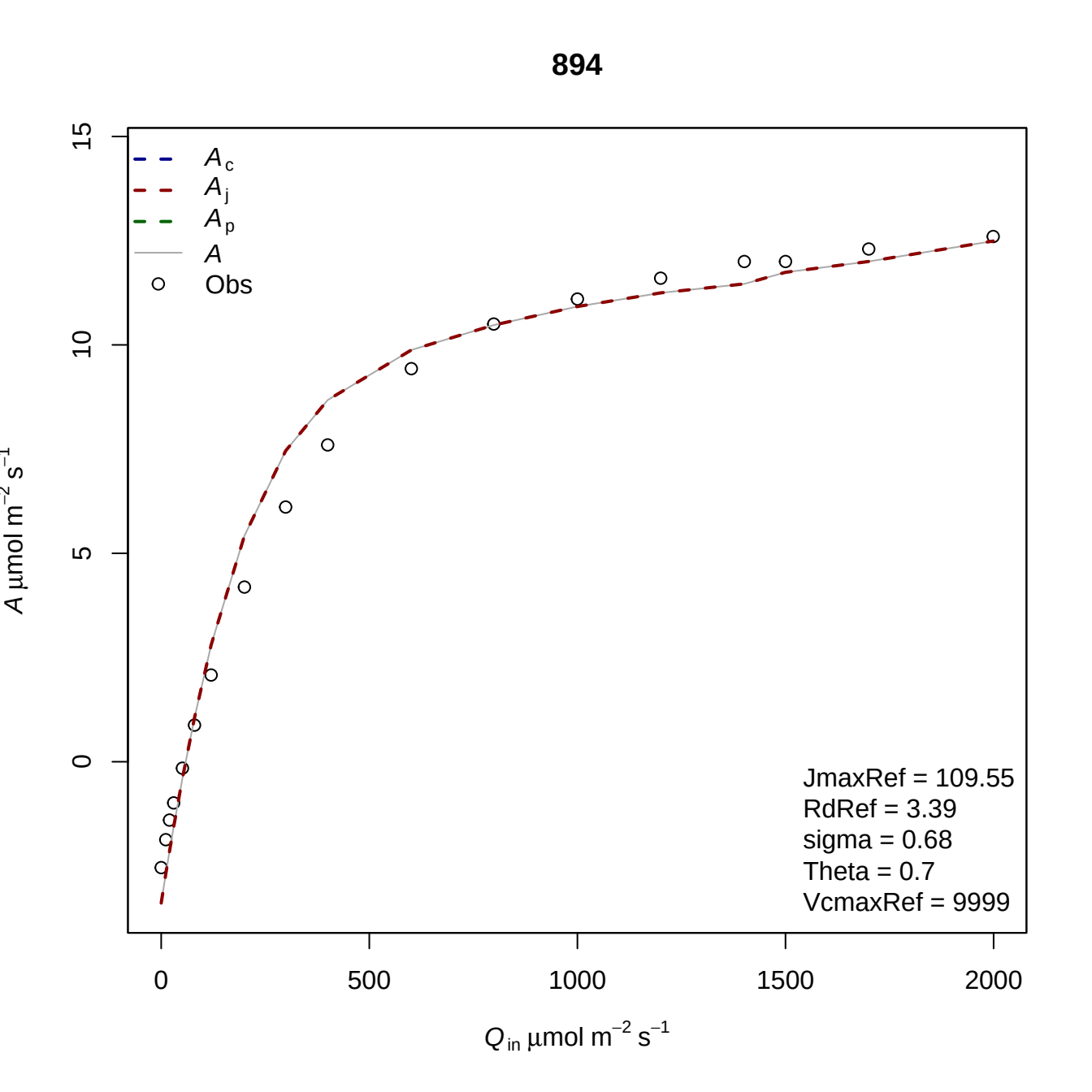

889
A_c
A_j
A_p
A
Obs
JmaxRef = 91.77
RdRef = 2.25
sigma = 0.93
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

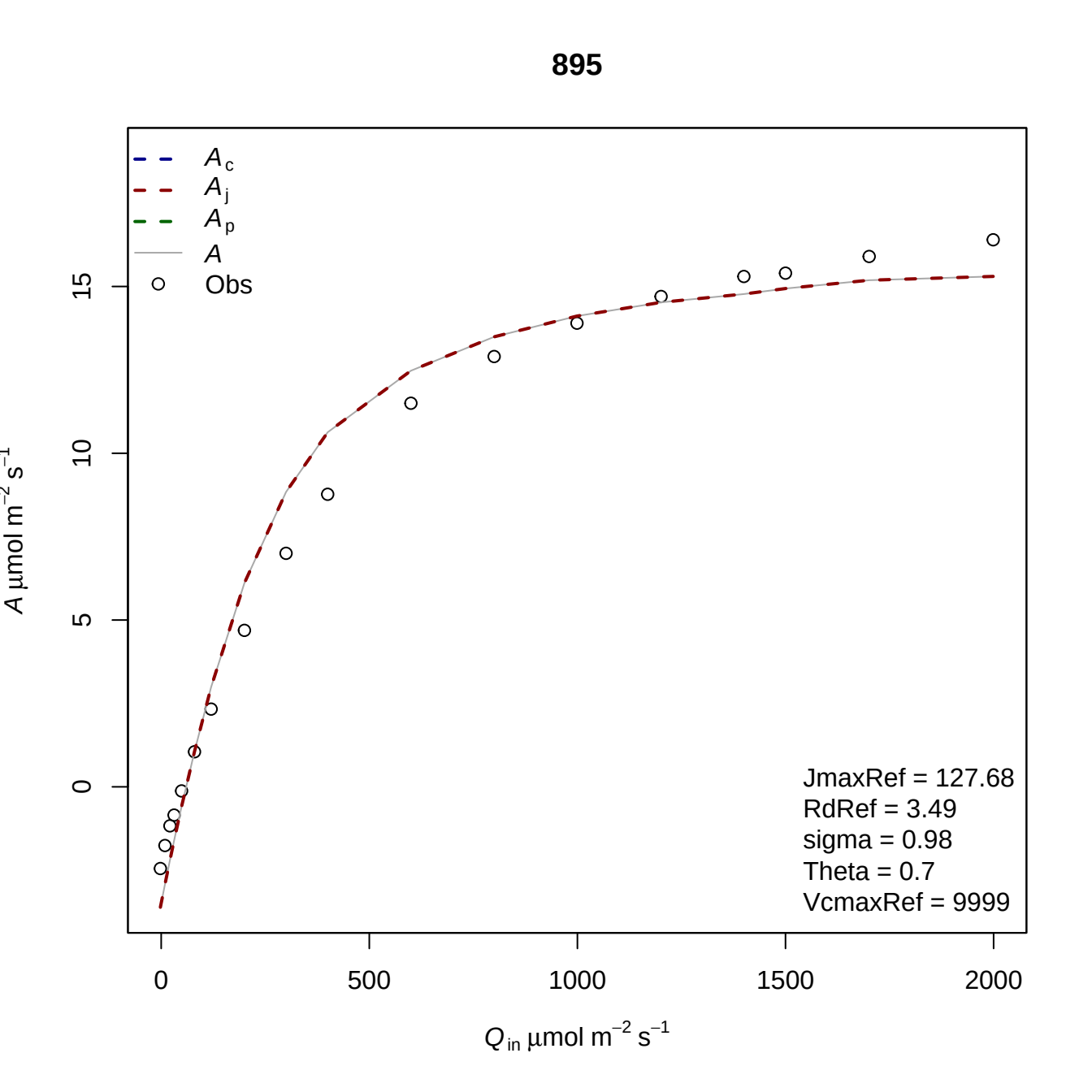

890
A_c
A_j
A_p
A
Obs
JmaxRef = 79.35
RdRef = 2.34
sigma = 0.99
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

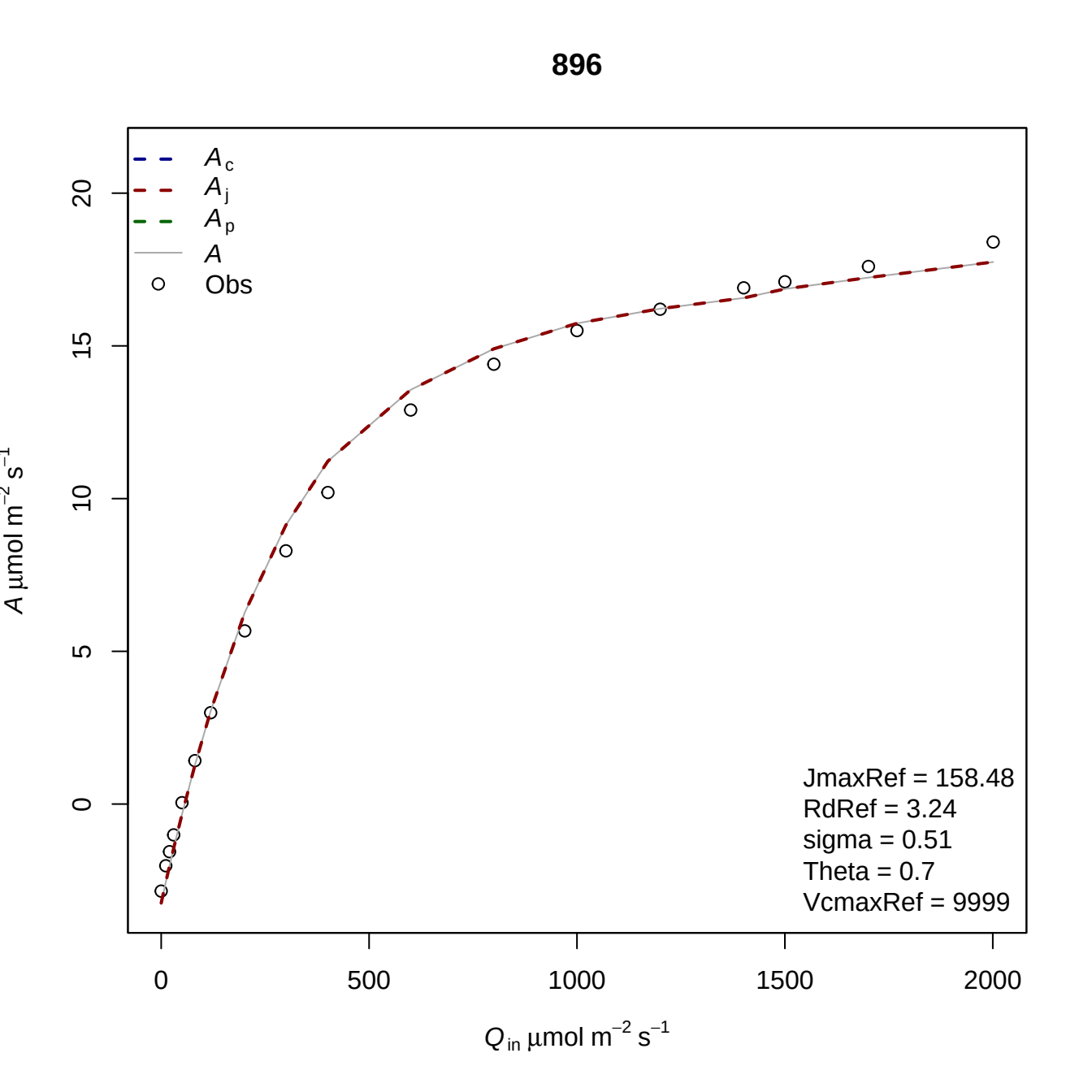

891
A_c
A_j
A_p
A
Obs
JmaxRef = 69.7
RdRef = 1.57
sigma = 0.7
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

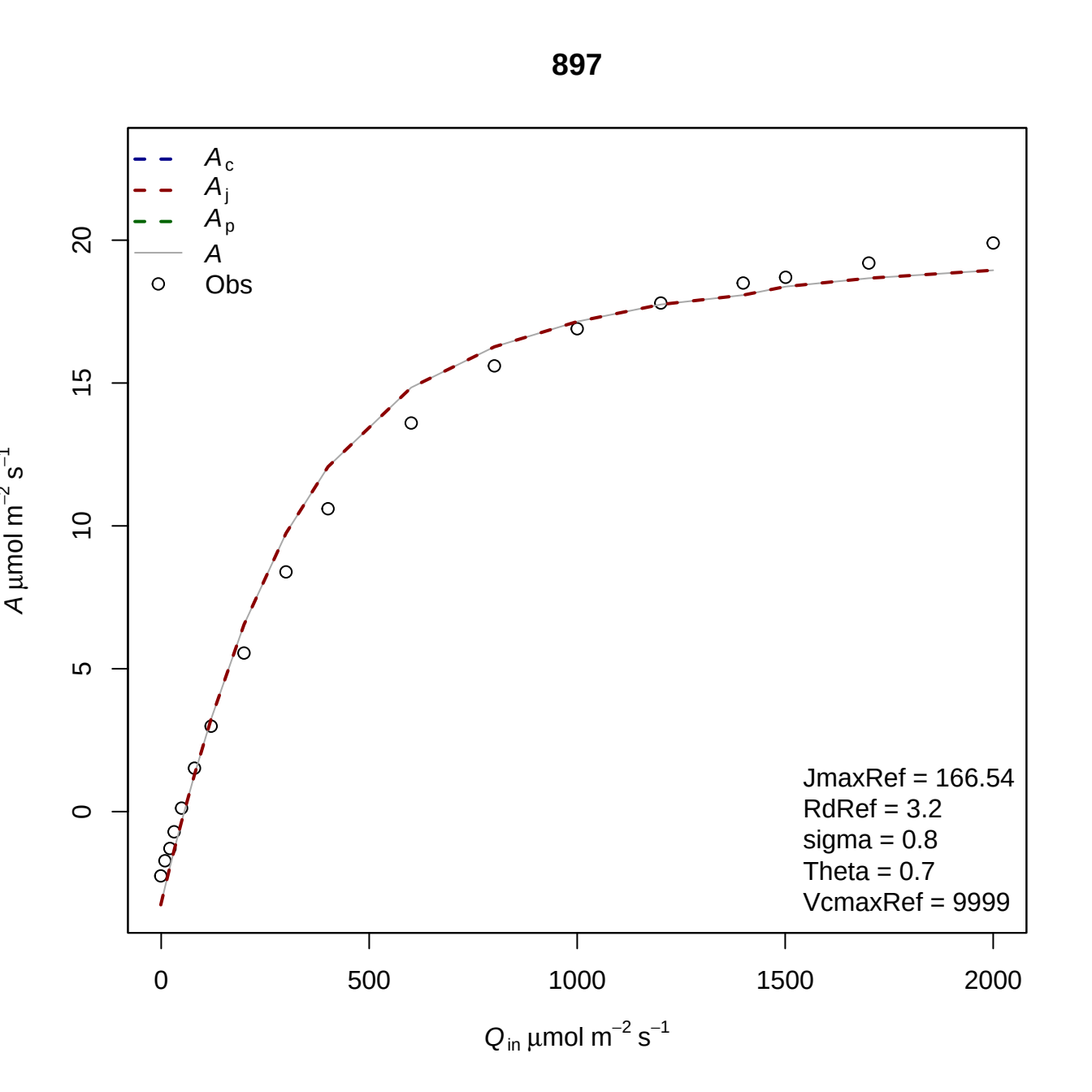

892
A_c
A_j
A_p
A
Obs
JmaxRef = 44.25
RdRef = 2.71
sigma = 0.54
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

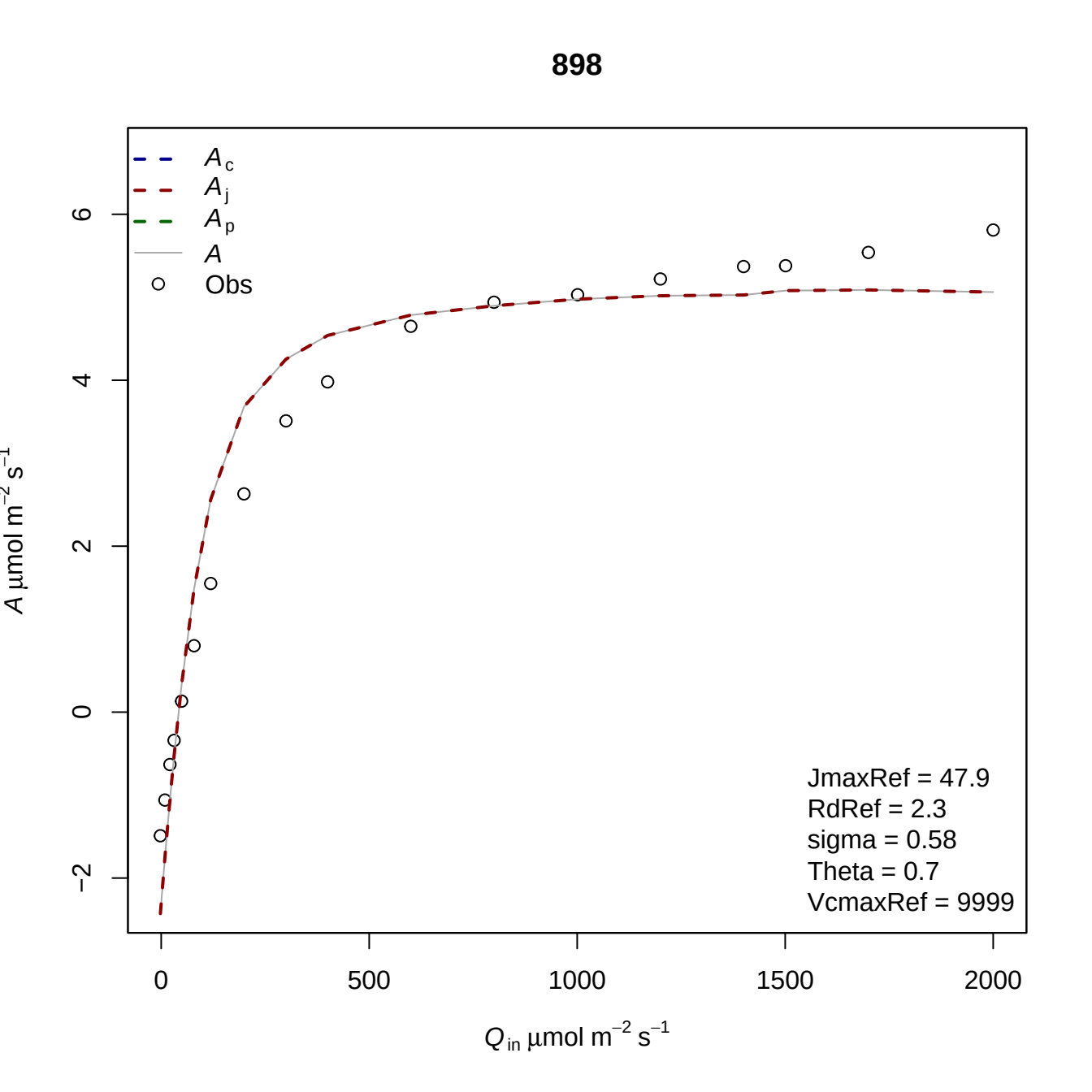

893
A_c
A_j
A_p
A
Obs
JmaxRef = 72.89
RdRef = 2.95
sigma = 0.76
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

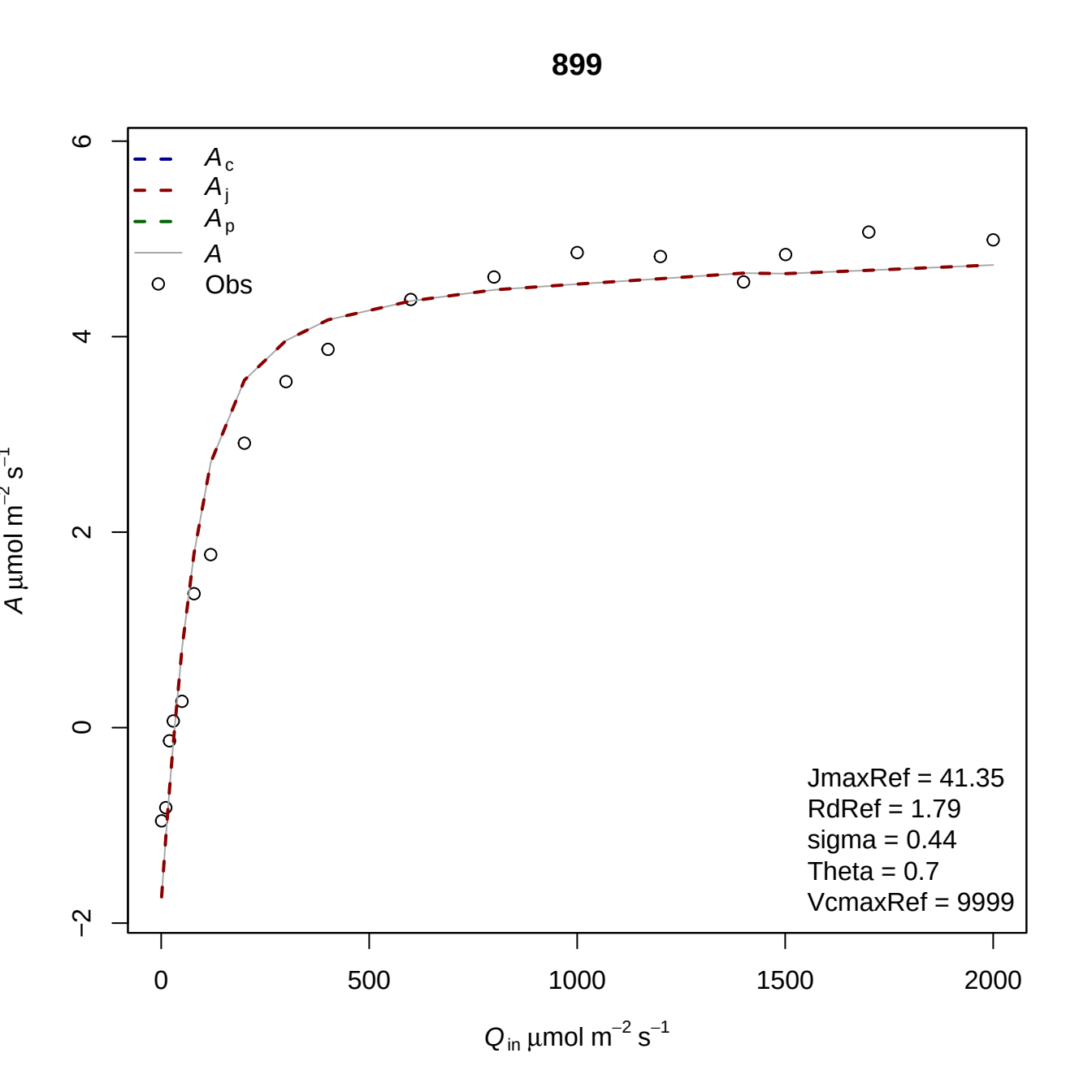

894
A_c
A_j
A_p
A
Obs
JmaxRef = 109.55
RdRef = 3.39
sigma = 0.68
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

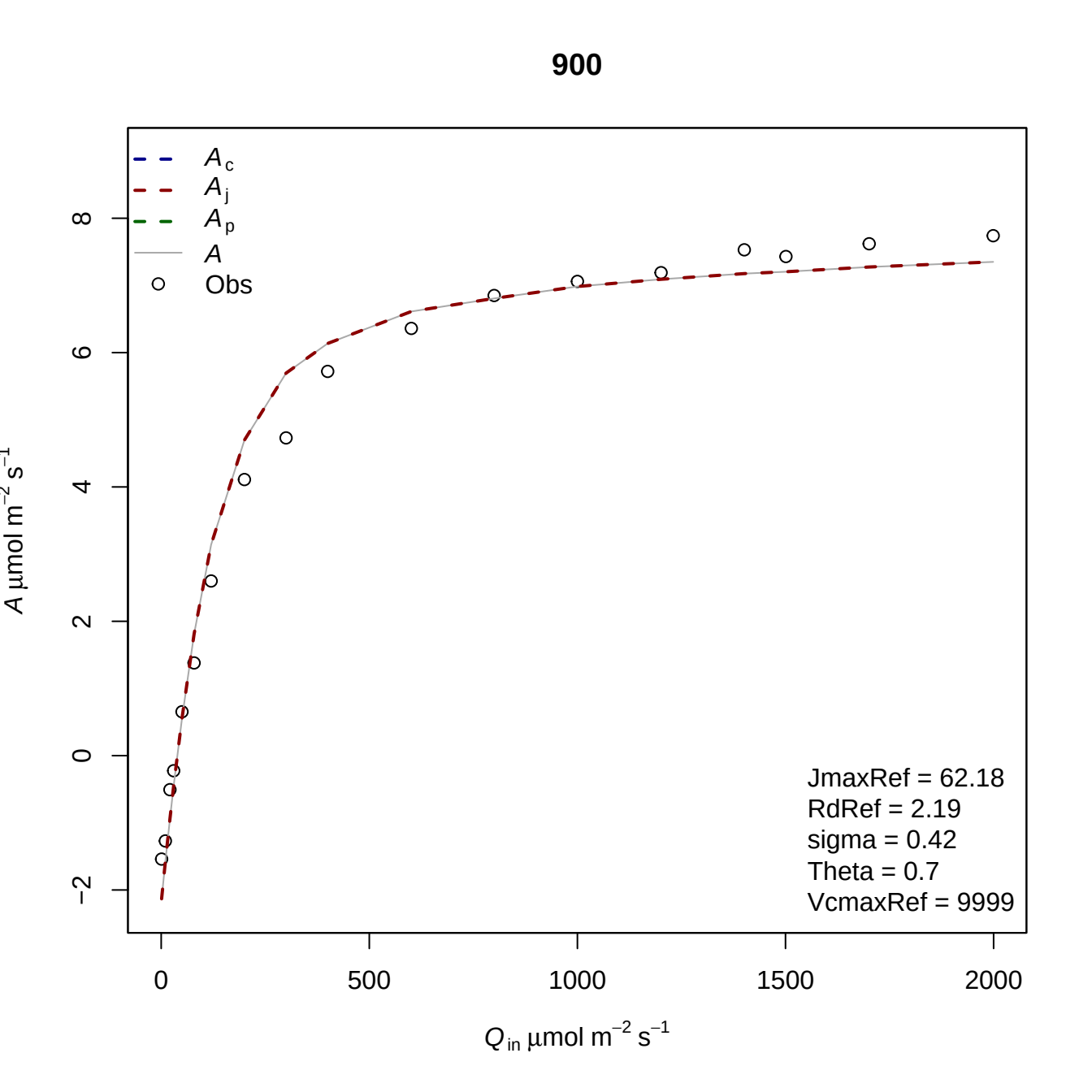

895
A_c
A_j
A_p
A
Obs
JmaxRef = 127.68
RdRef = 3.49
sigma = 0.98
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

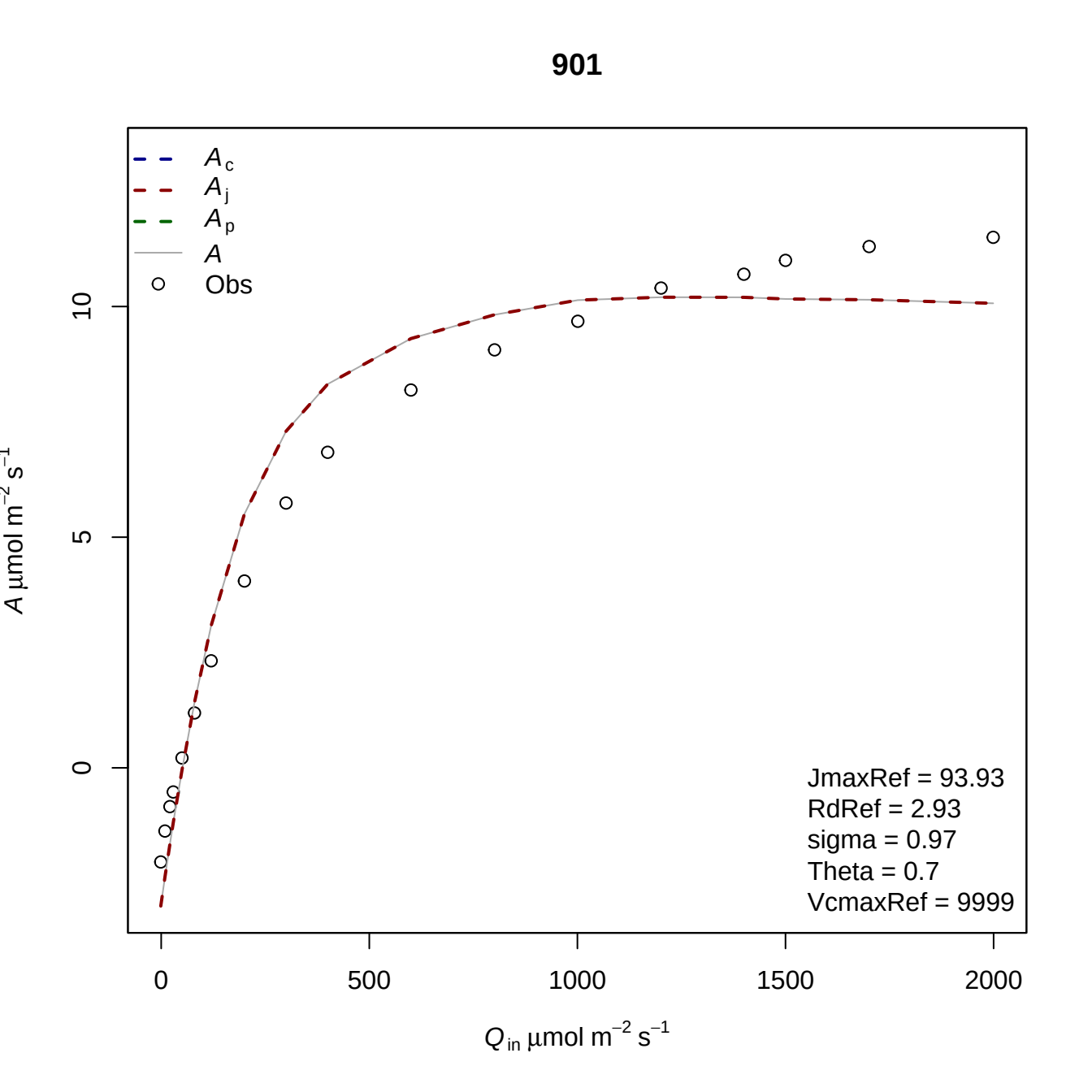

896
A_c
A_j
A_p
A
Obs
JmaxRef = 158.48
RdRef = 3.24
sigma = 0.51
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

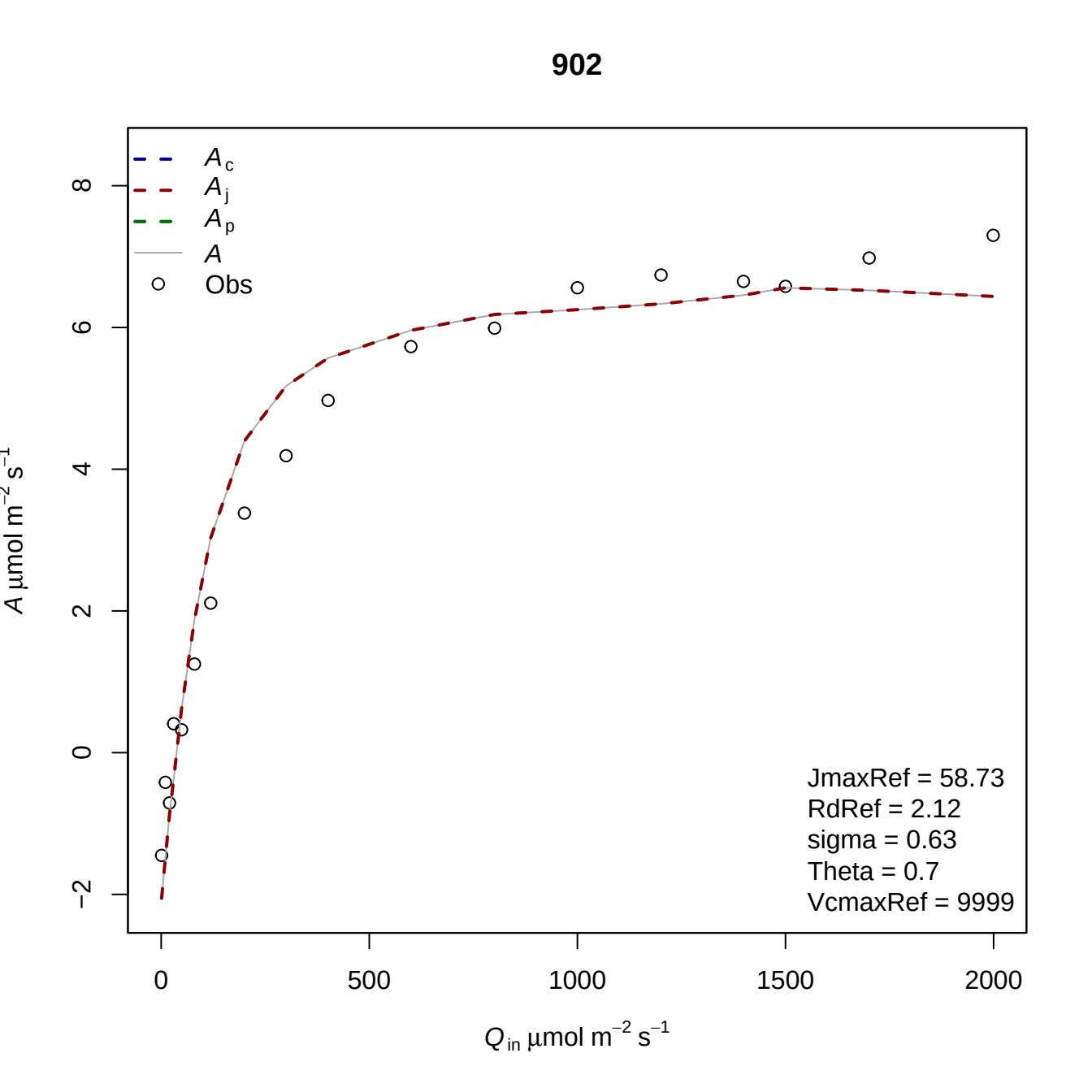

897
A_c
A_j
A_p
A
Obs
JmaxRef = 166.54
RdRef = 3.2
sigma = 0.8
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

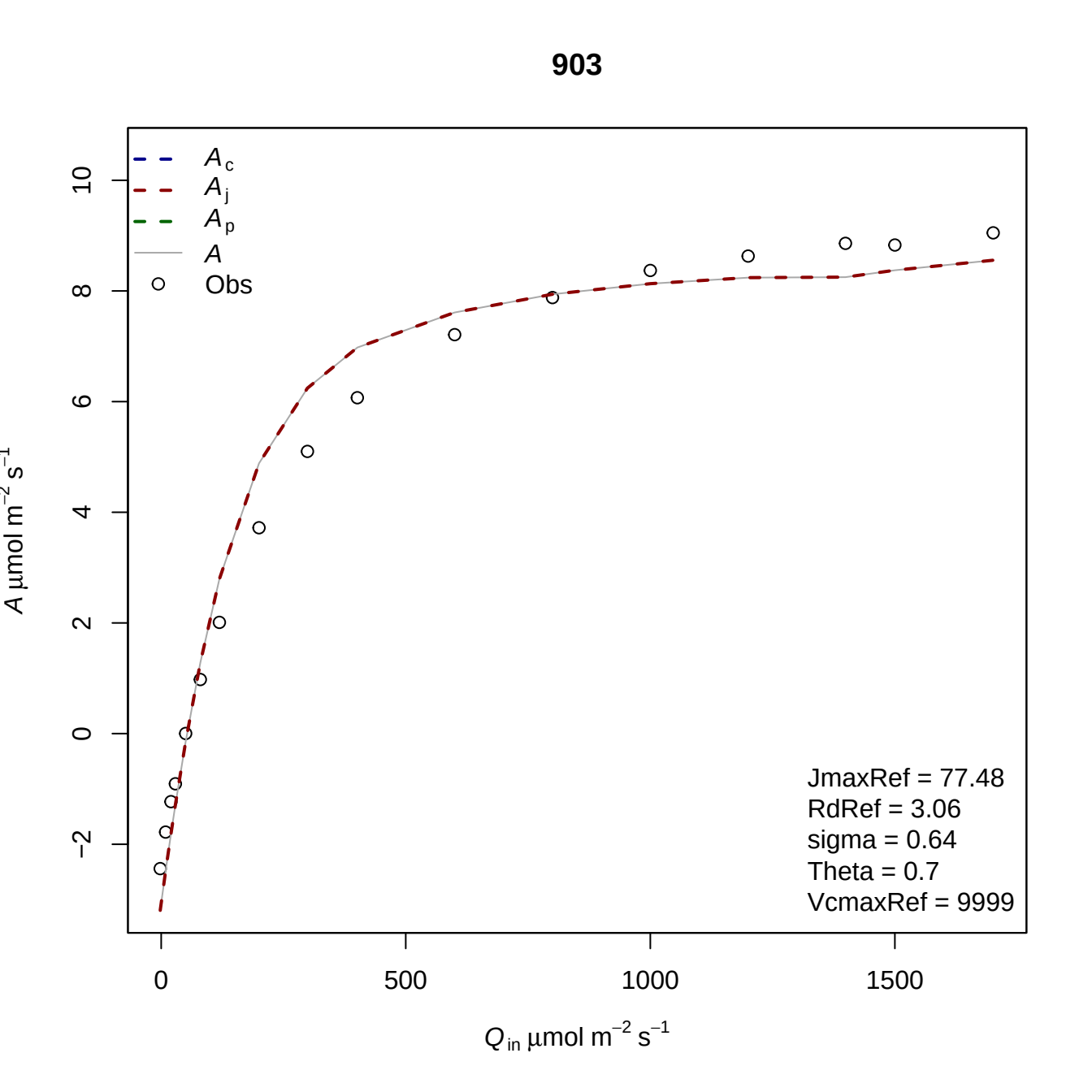

898
A_c
A_j
A_p
A
Obs
JmaxRef = 47.9
RdRef = 2.3
sigma = 0.58
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

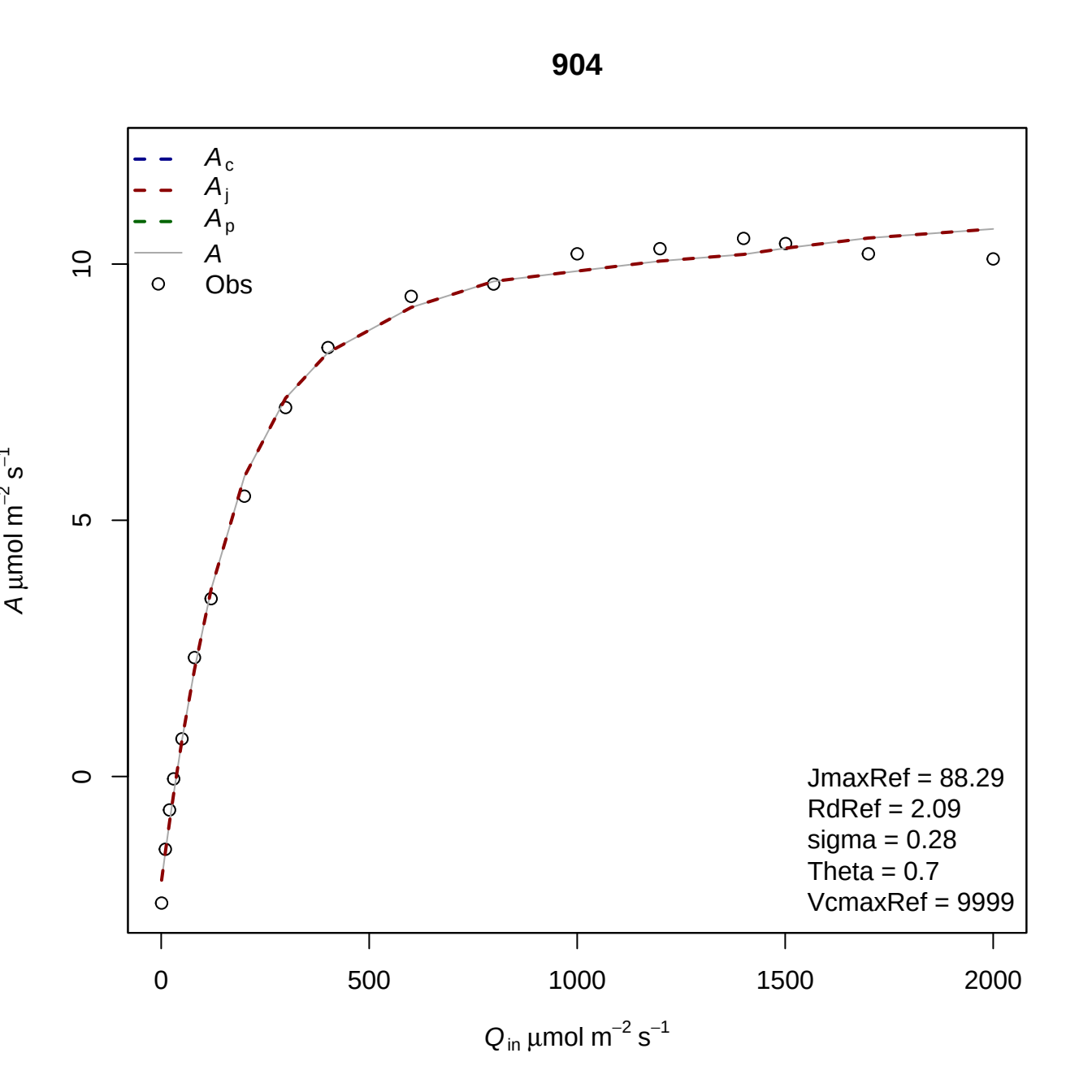

899
A_c
A_j
A_p
A
Obs
JmaxRef = 41.35
RdRef = 1.79
sigma = 0.44
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

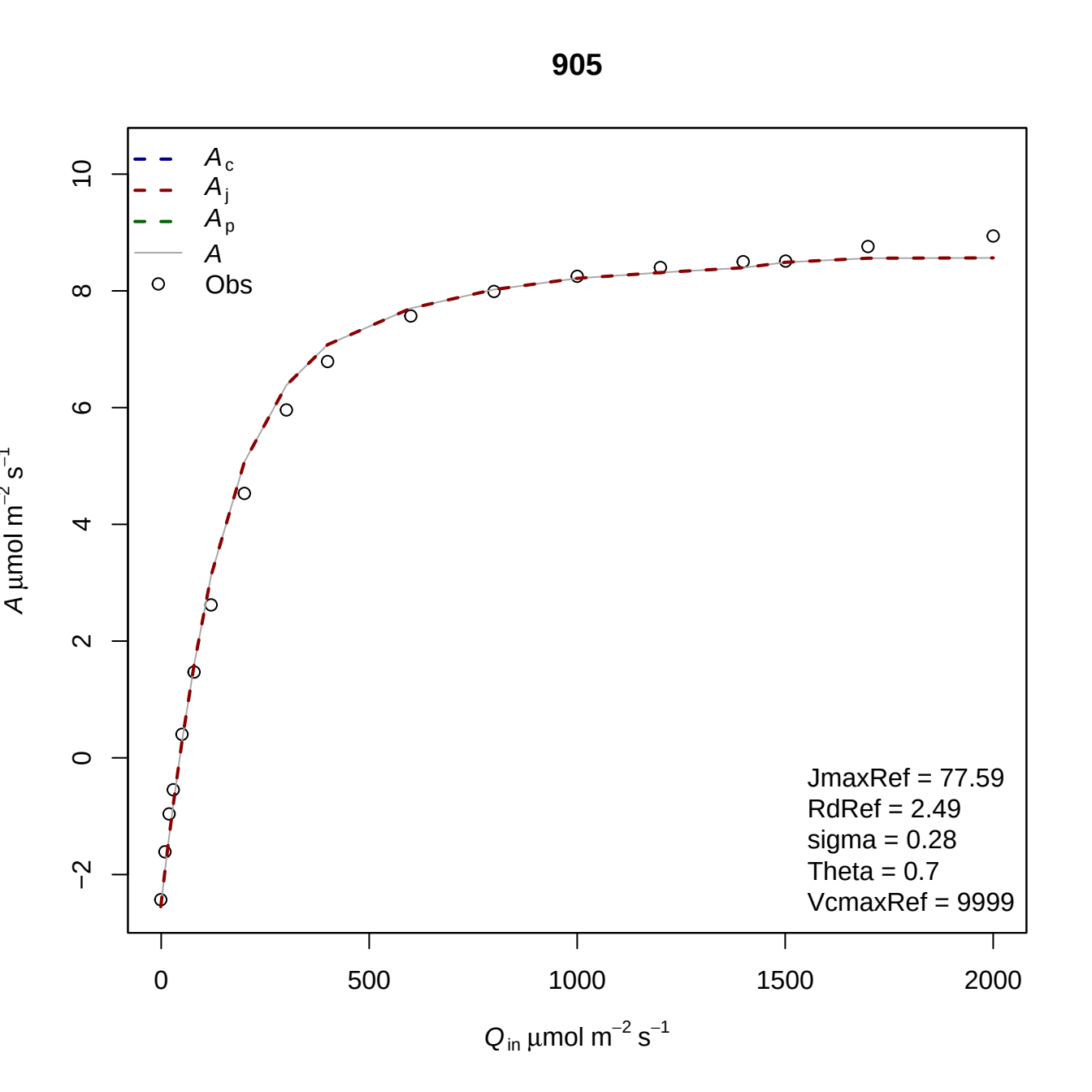

900
A_c
A_j
A_p
A
Obs
JmaxRef = 62.18
RdRef = 2.19
sigma = 0.42
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

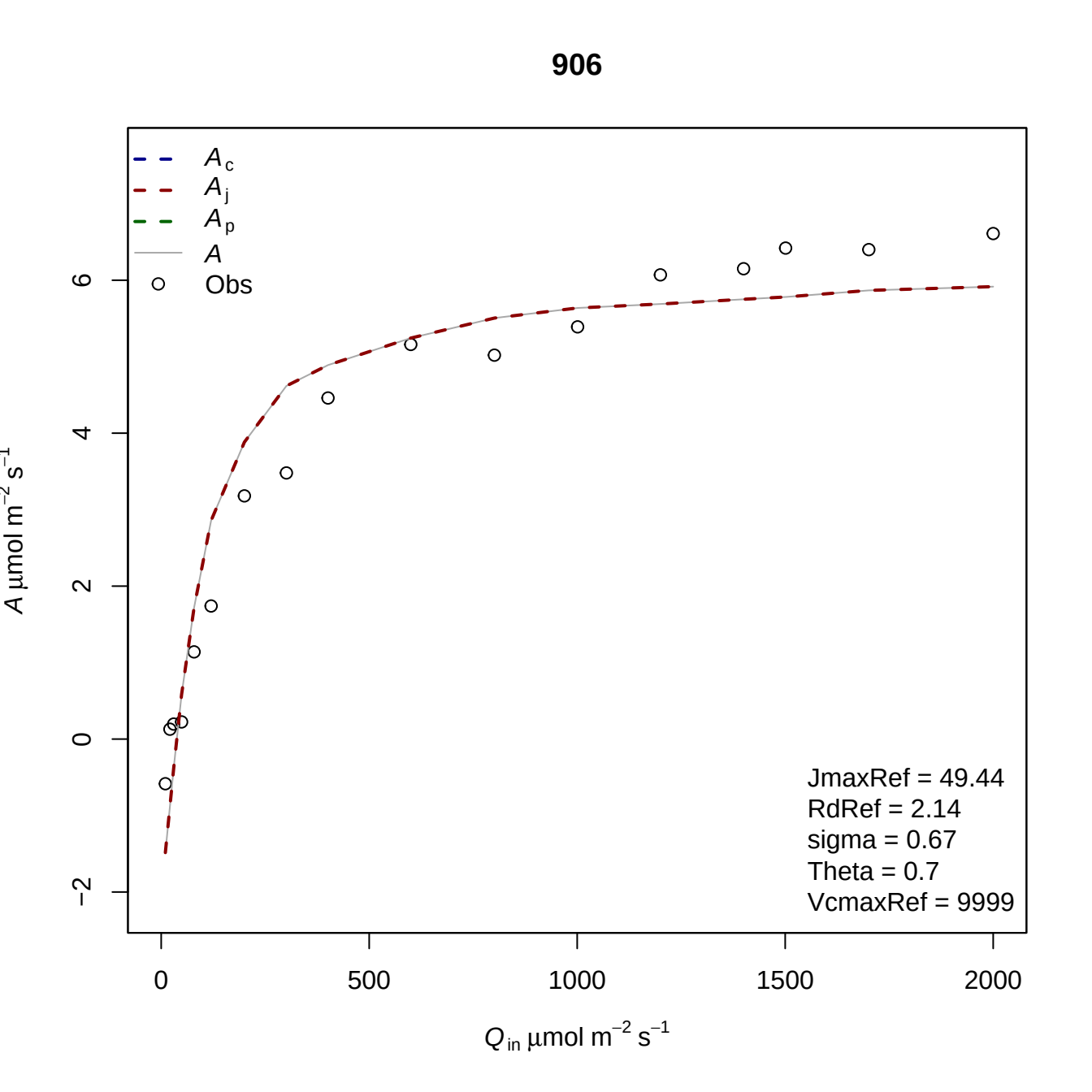

901
A_c
A_j
A_p
A
Obs
JmaxRef = 93.93
RdRef = 2.93
sigma = 0.97
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

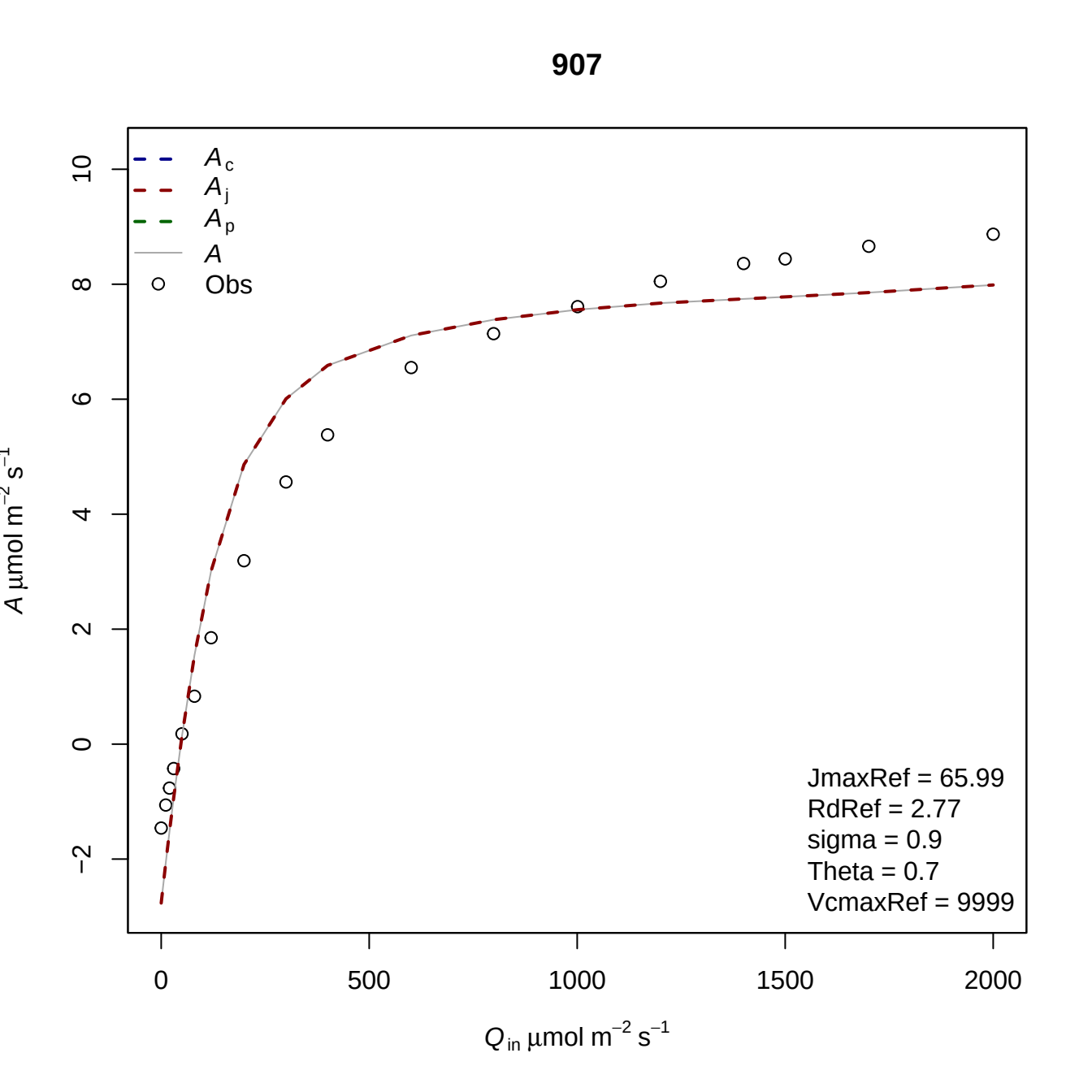

902
A_c
A_j
A_p
A
Obs
JmaxRef = 58.73
RdRef = 2.12
sigma = 0.63
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

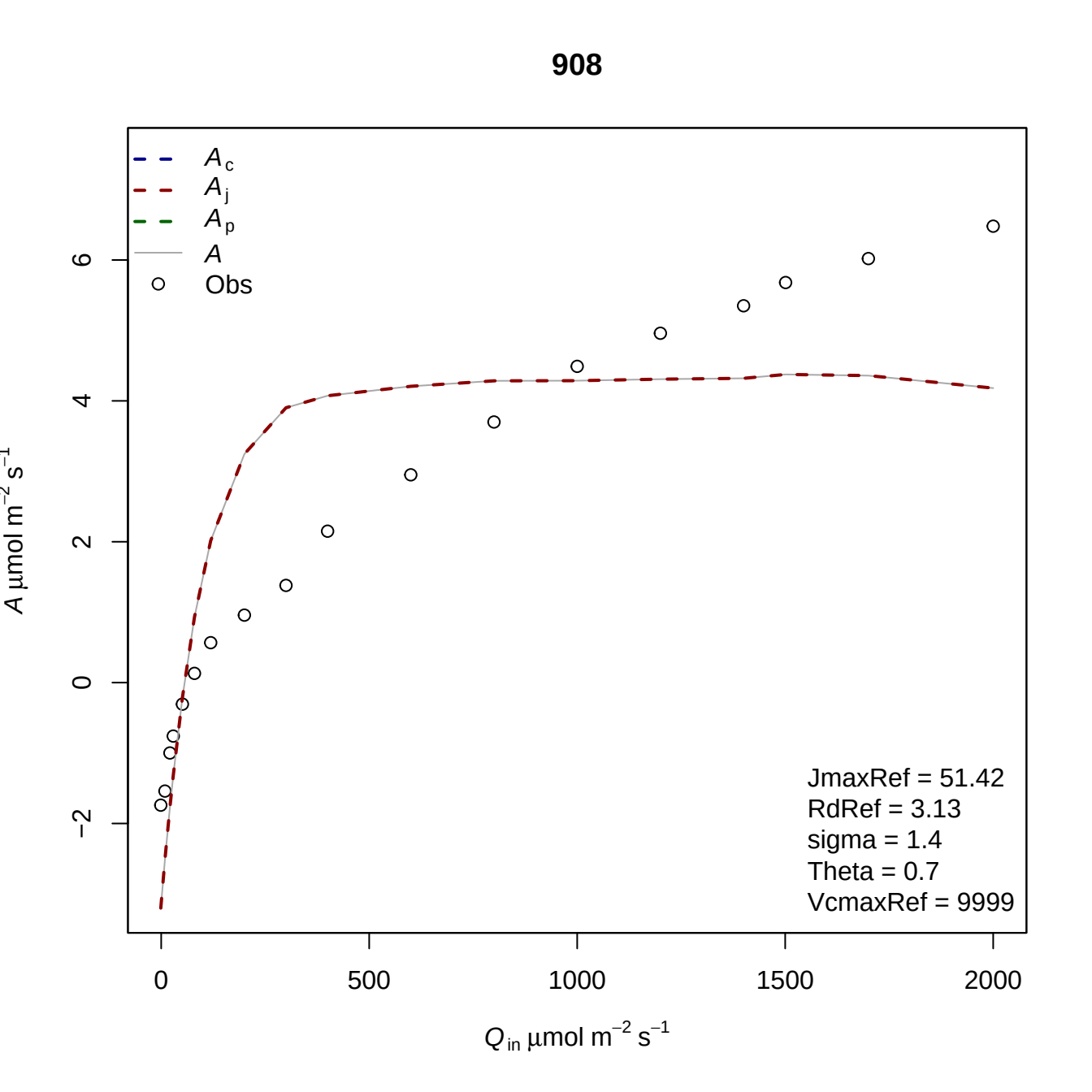

903
A_c
A_j
A_p
A
Obs
JmaxRef = 77.48
RdRef = 3.06
sigma = 0.64
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

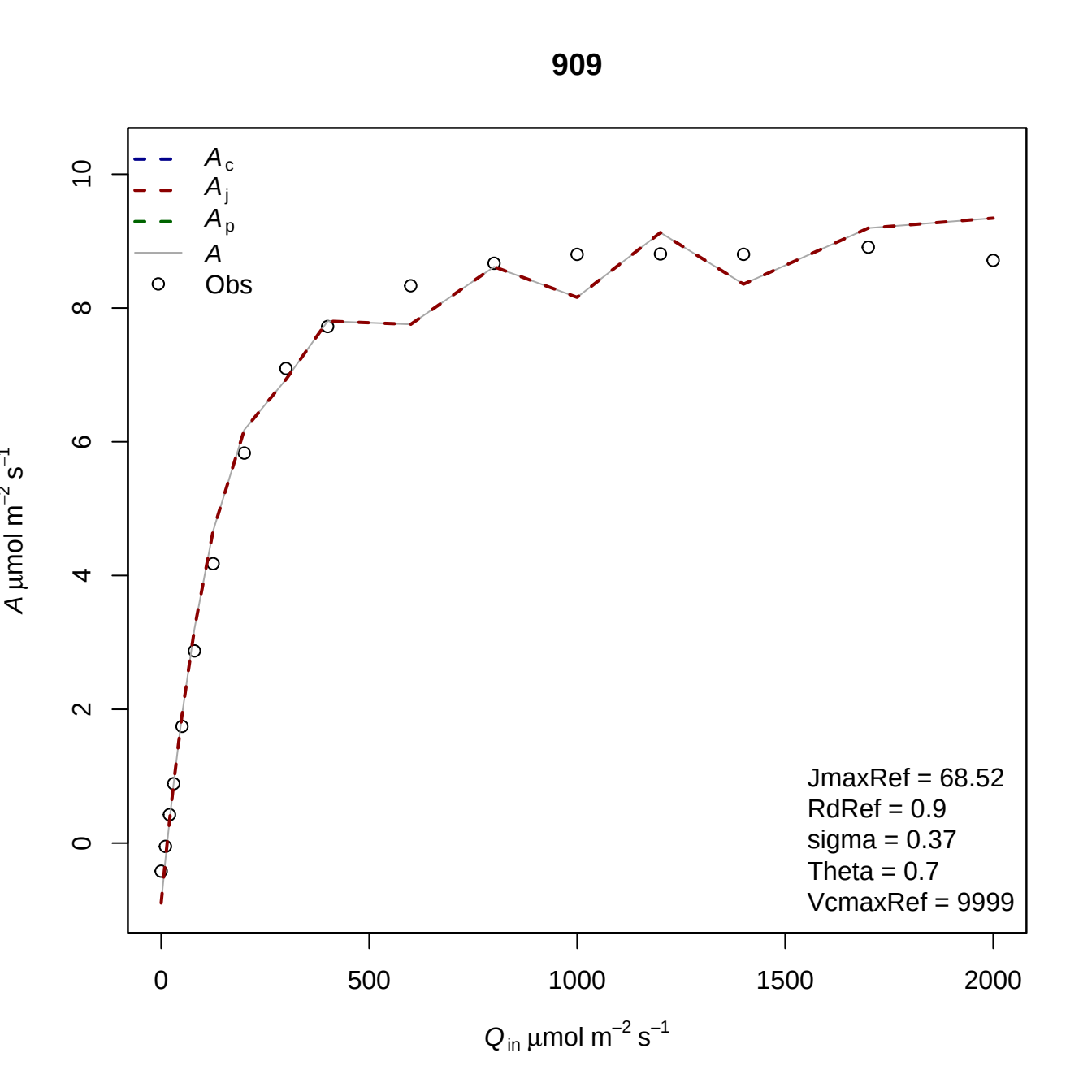

904
A_c
A_j
A_p
A
Obs
JmaxRef = 88.29
RdRef = 2.09
sigma = 0.28
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

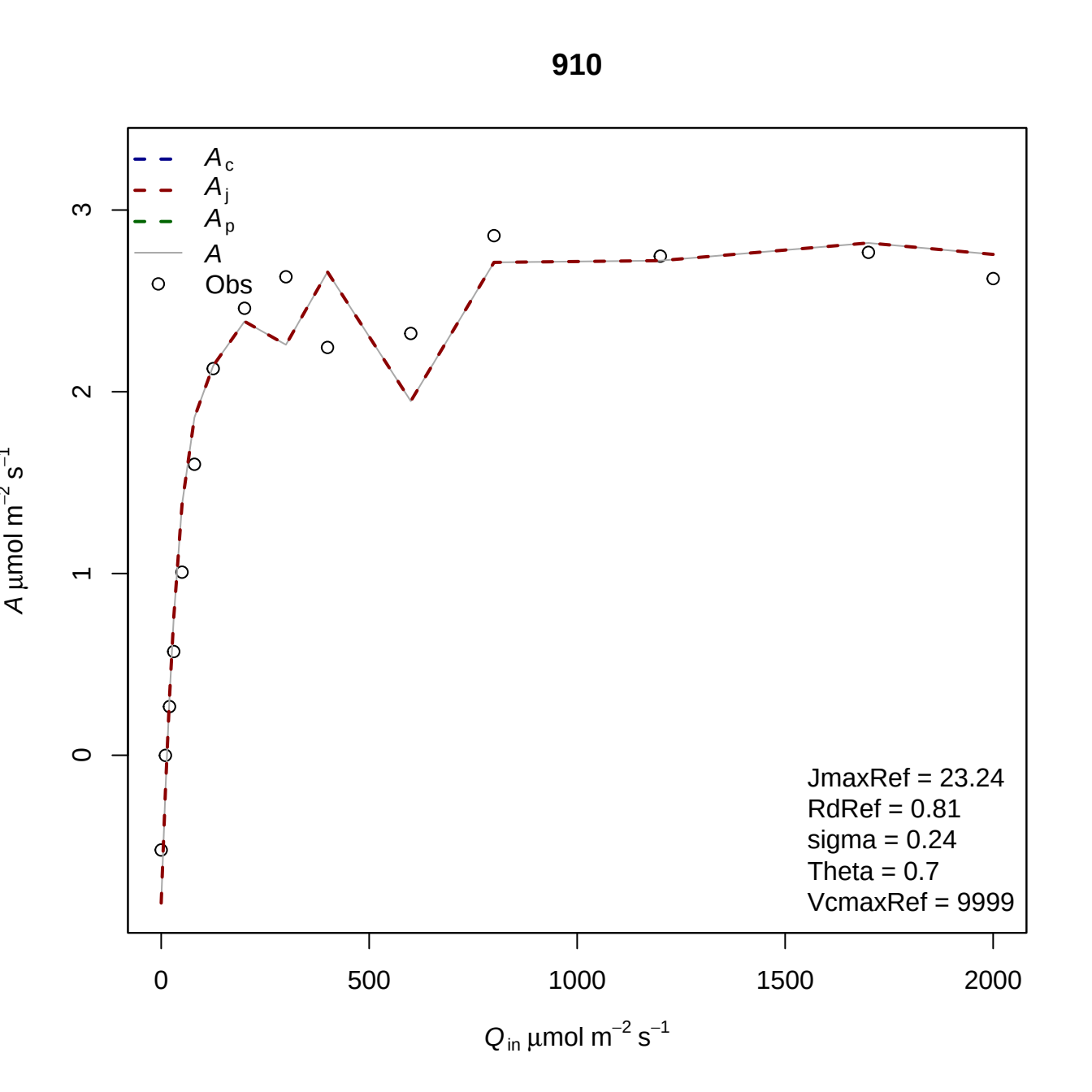

905
A_c
A_j
A_p
A
Obs
JmaxRef = 77.59
RdRef = 2.49
sigma = 0.28
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

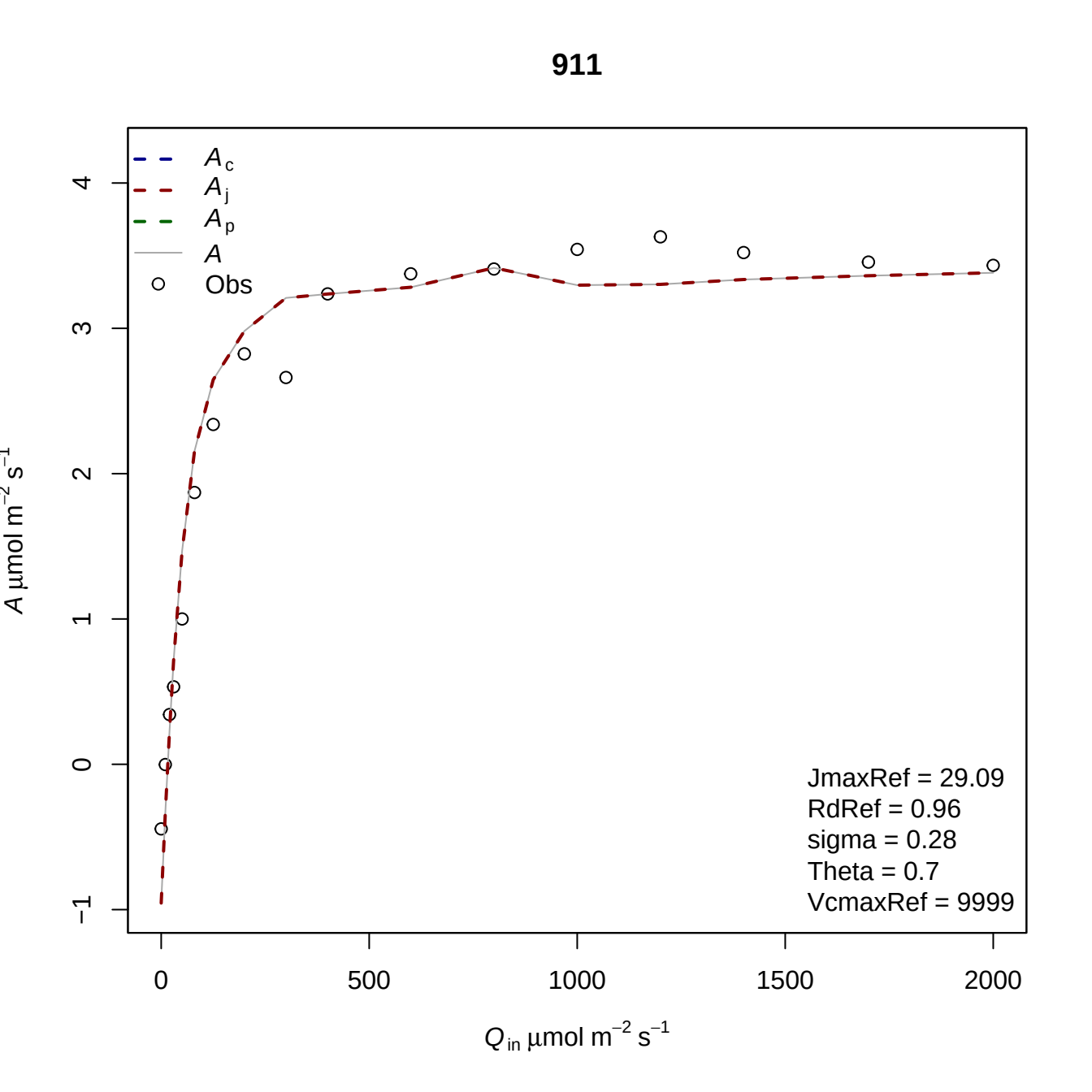

906
A_c
A_j
A_p
A
Obs
JmaxRef = 49.44
RdRef = 2.14
sigma = 0.67
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

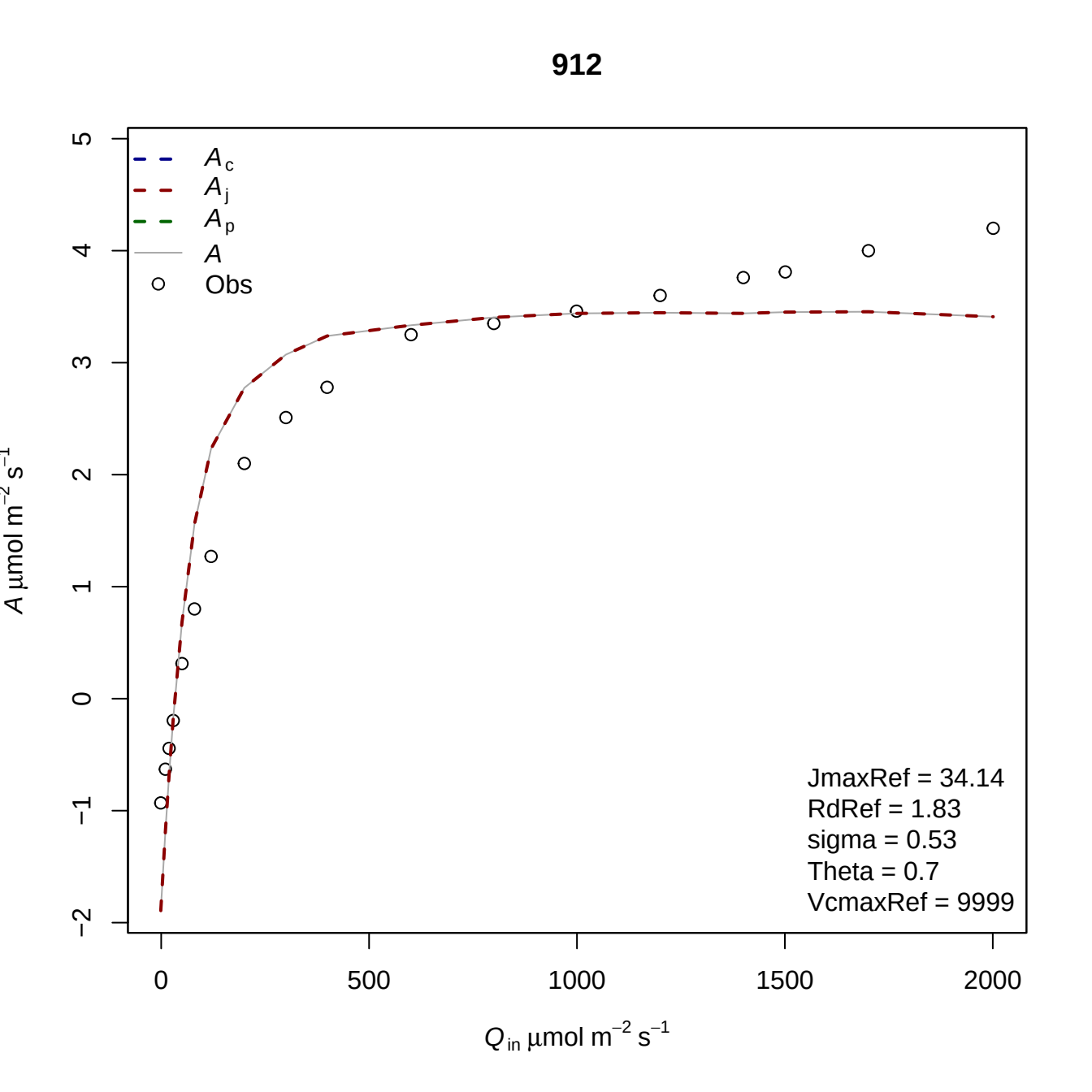

907
A_c
A_j
A_p
A
Obs
JmaxRef = 65.99
RdRef = 2.77
sigma = 0.9
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

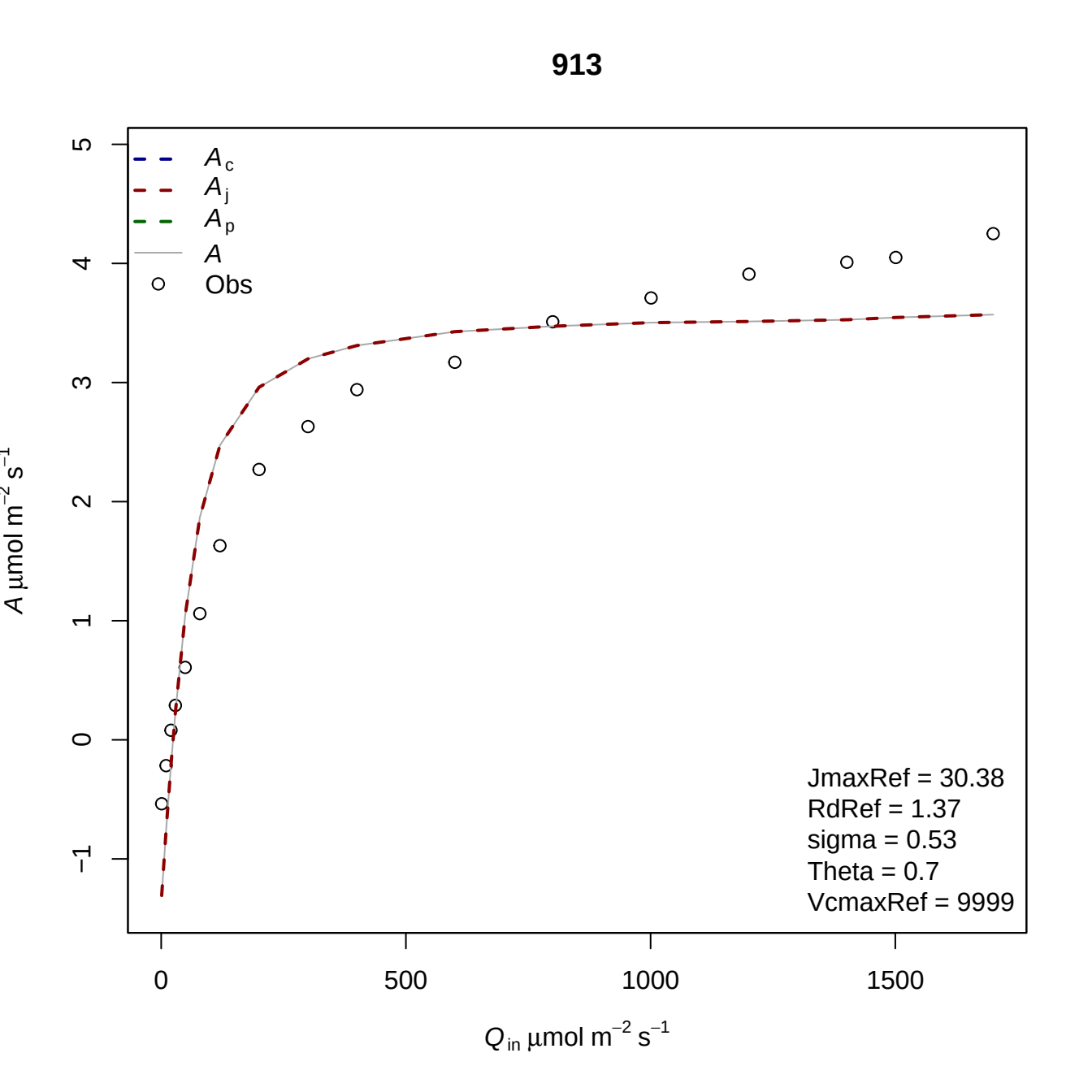

908
A_c
A_j
A_p
A
Obs
JmaxRef = 51.42
RdRef = 3.13
sigma = 1.4
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

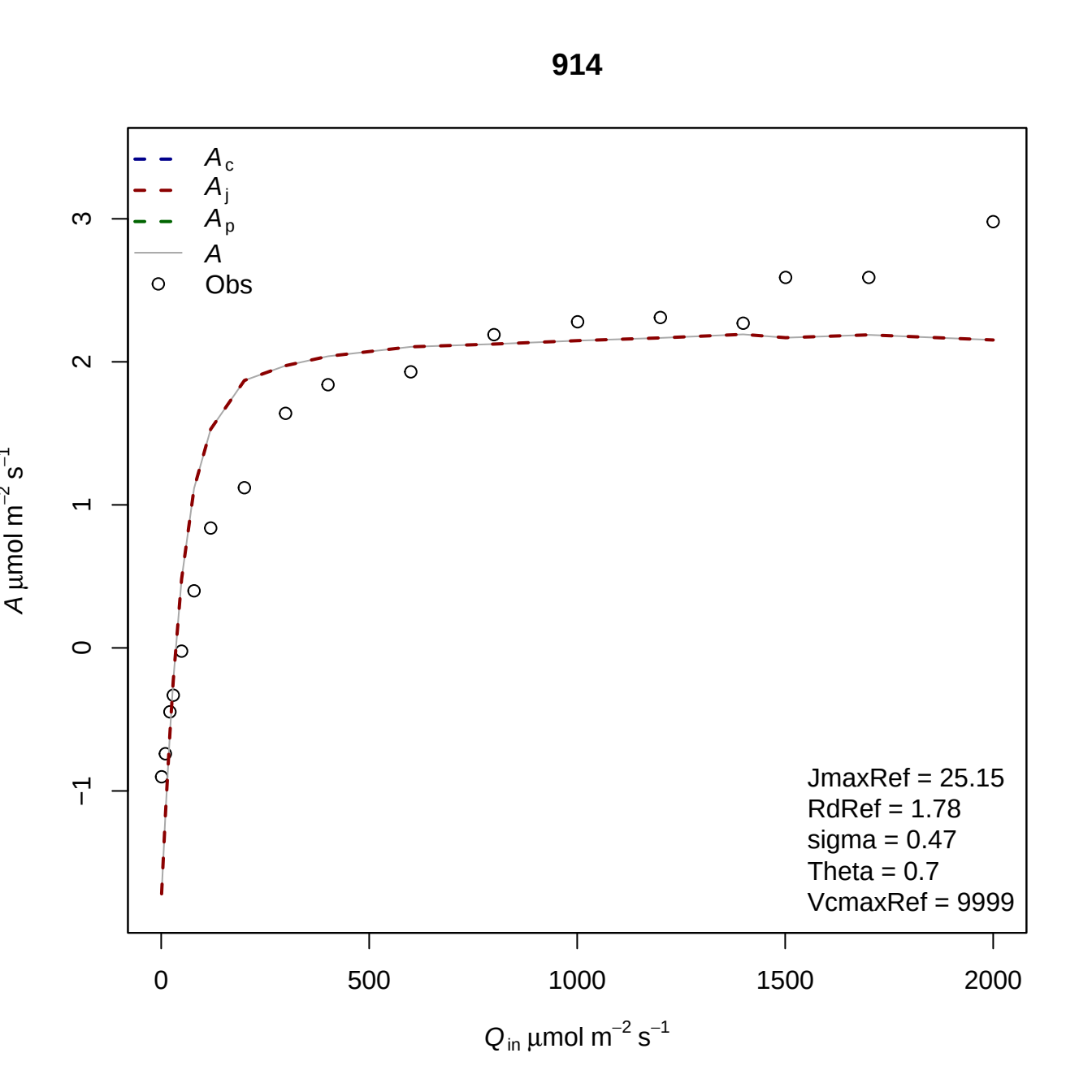

909
A_c
A_j
A_p
A
Obs
JmaxRef = 68.52
RdRef = 0.9
sigma = 0.37
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

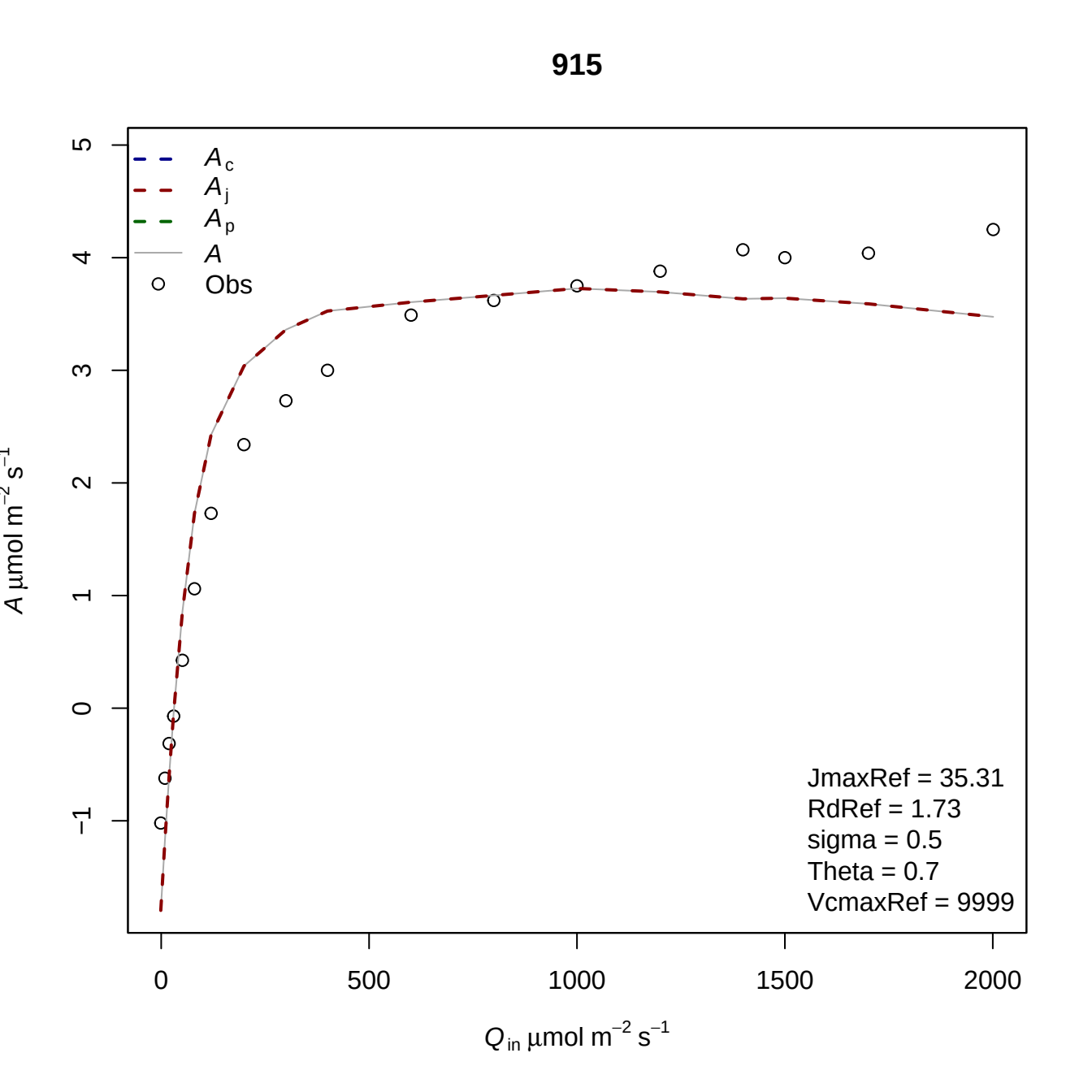

910
A_c
A_j
A_p
A
Obs
JmaxRef = 23.24
RdRef = 0.81
sigma = 0.24
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
500
1000
1500
2000

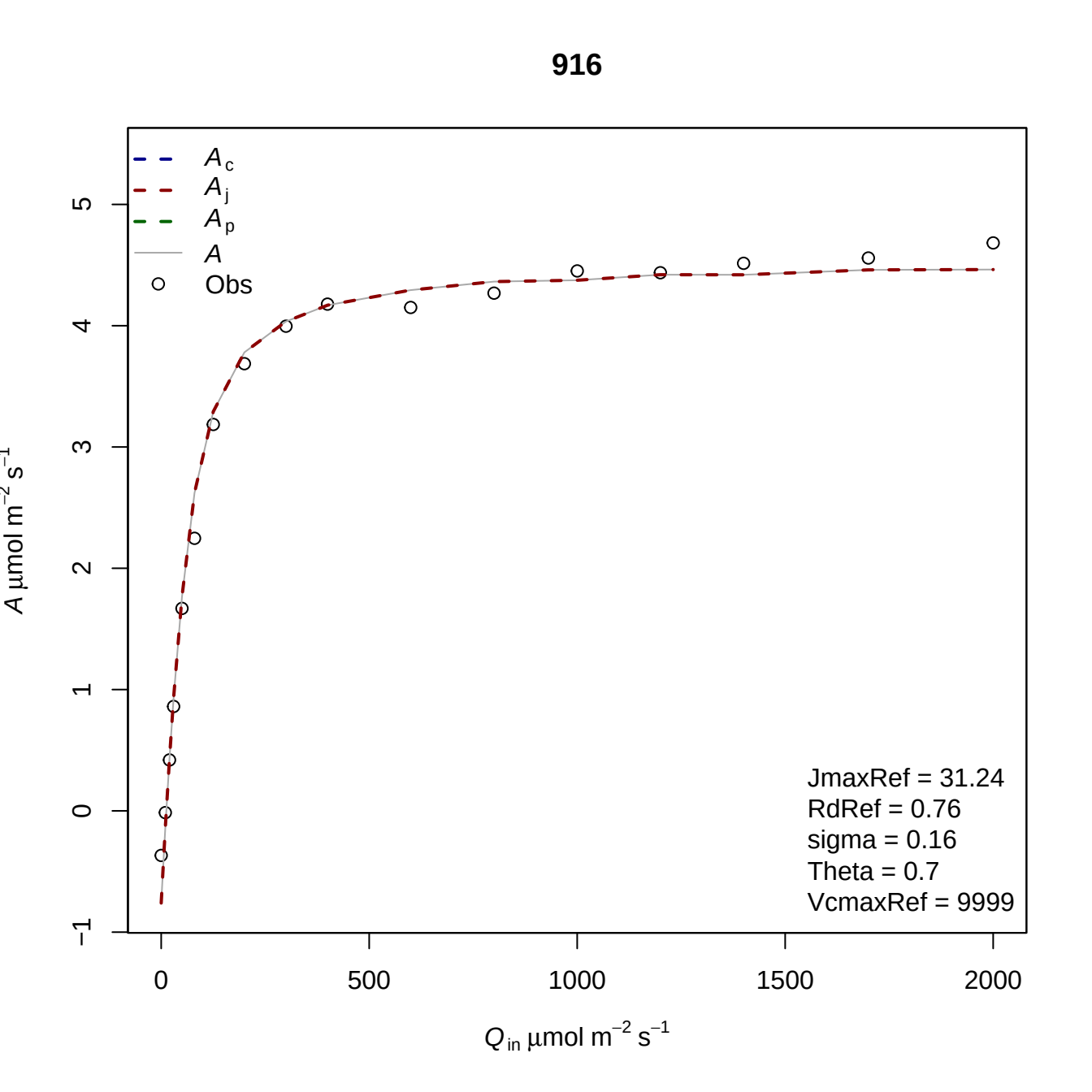

911
A_c
A_j
A_p
A
Obs
JmaxRef = 29.09
RdRef = 0.96
sigma = 0.28
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500
2000

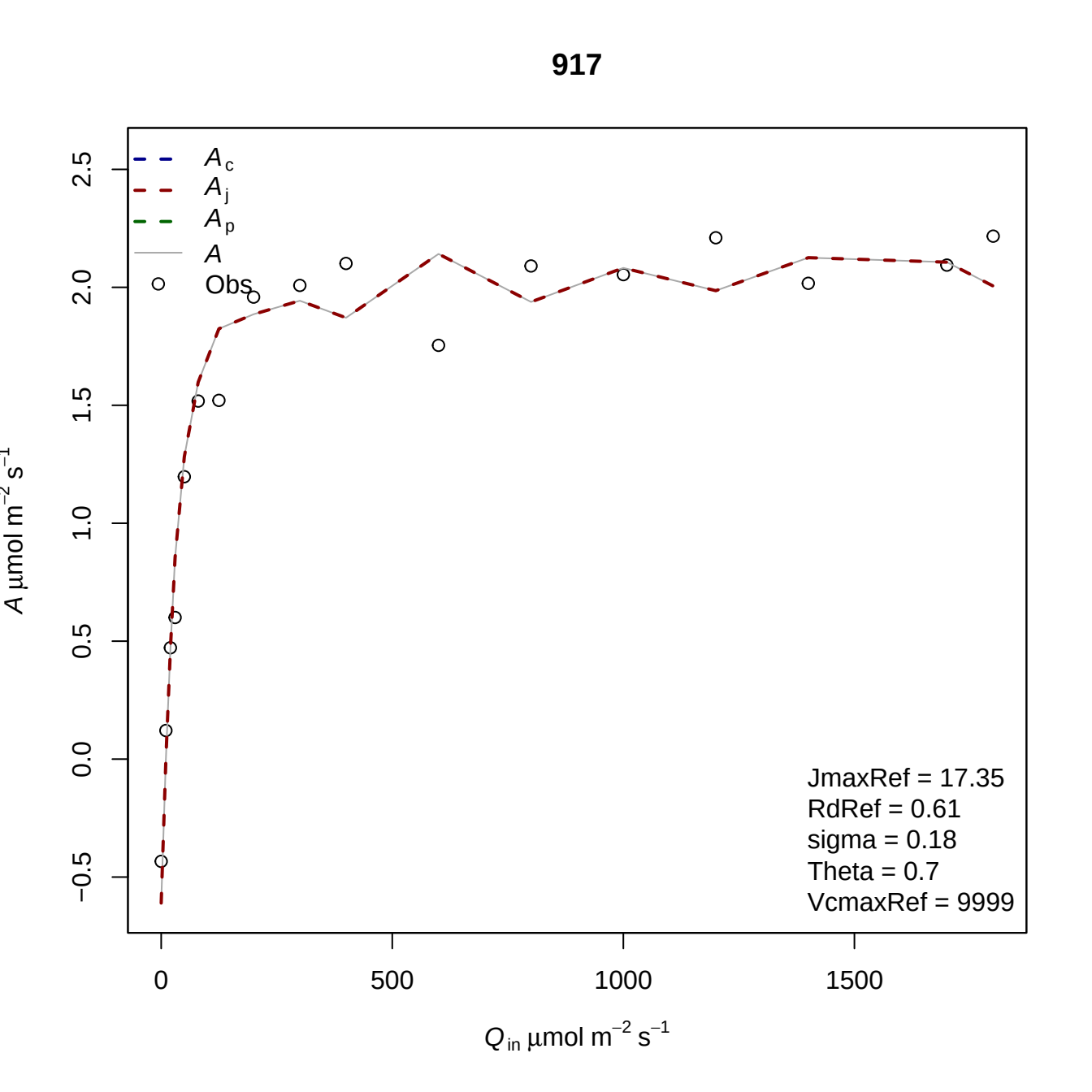

912
A_c
A_j
A_p
A
Obs
JmaxRef = 34.14
RdRef = 1.83
sigma = 0.53
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
5
4
3
2
1
0
−1
−2
0
500
1000
1500
2000

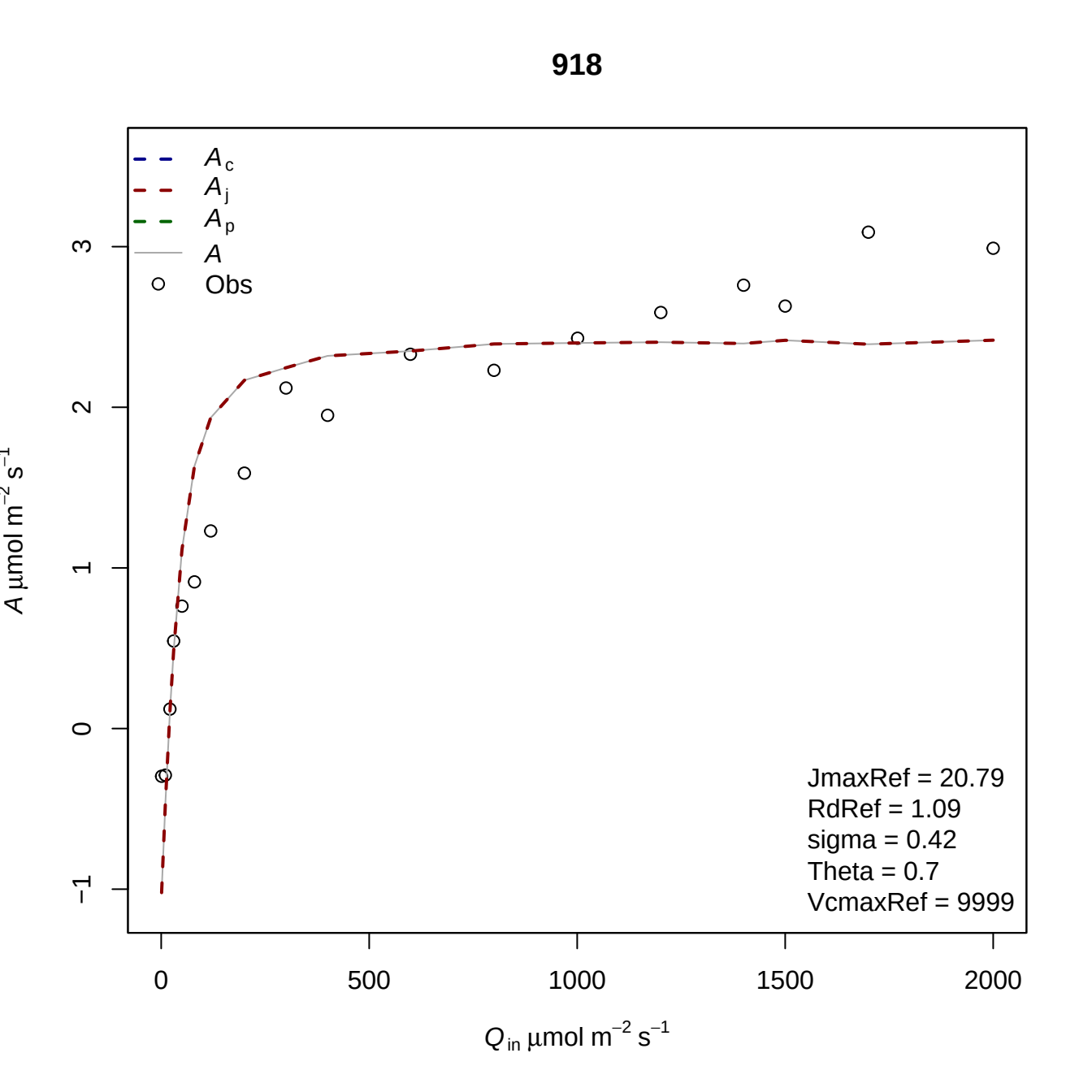

913
A_c
A_j
A_p
A
Obs
JmaxRef = 30.38
RdRef = 1.37
sigma = 0.53
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
5
4
3
2
1
0
−1
0
500
1000
1500

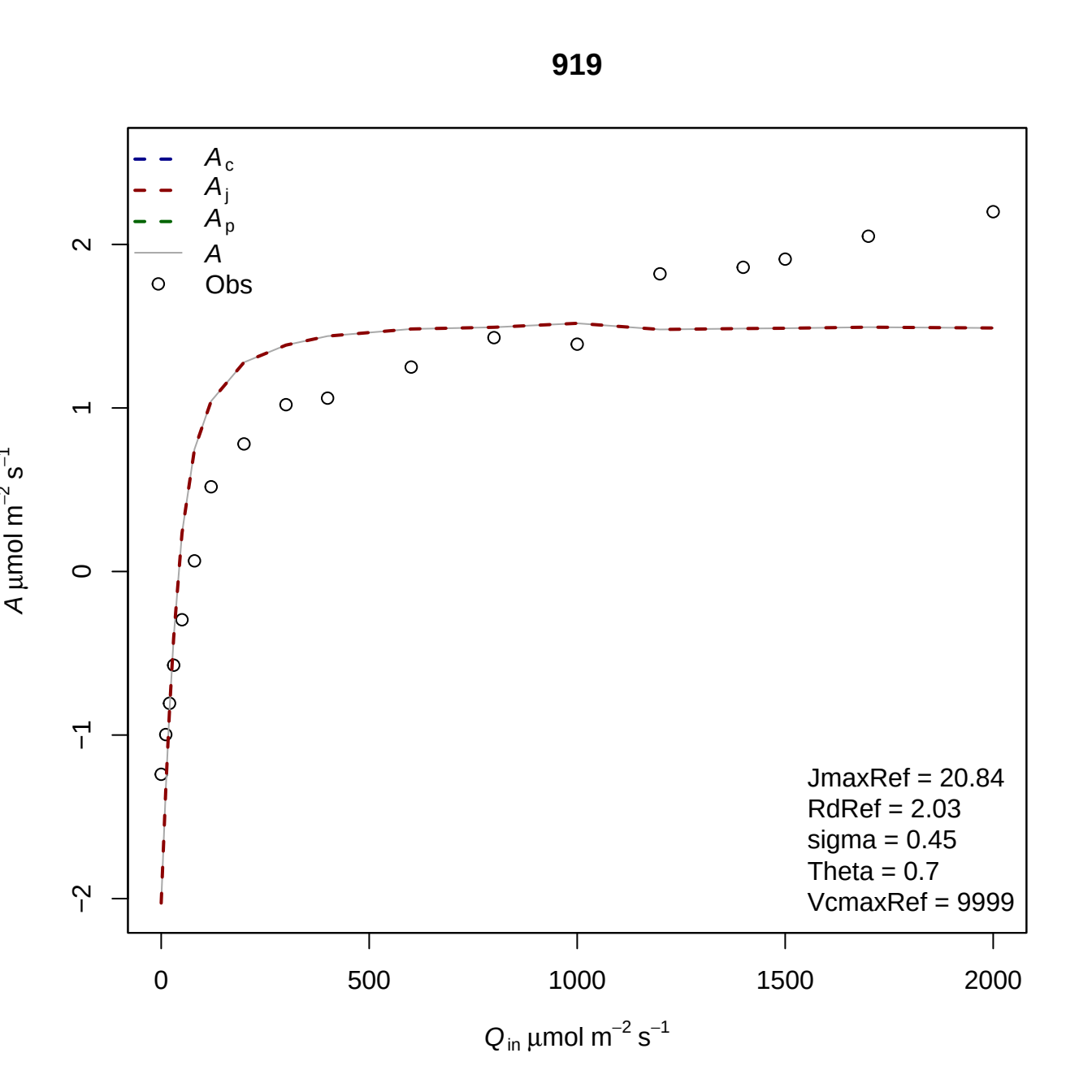

914
A_c
A_j
A_p
A
Obs
JmaxRef = 25.15
RdRef = 1.78
sigma = 0.47
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

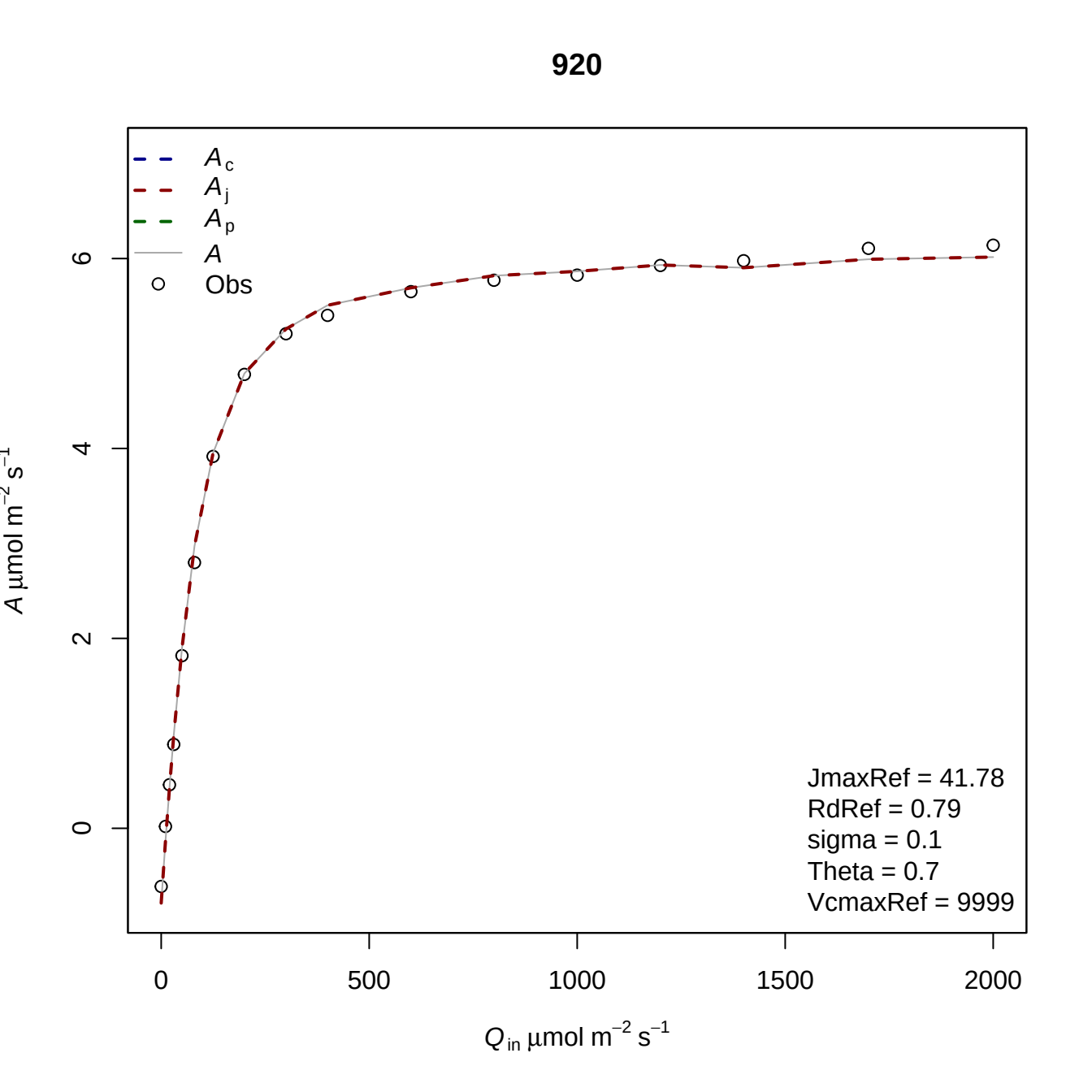

915
A_c
A_j
A_p
A
Obs
JmaxRef = 35.31
RdRef = 1.73
sigma = 0.5
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500
2000

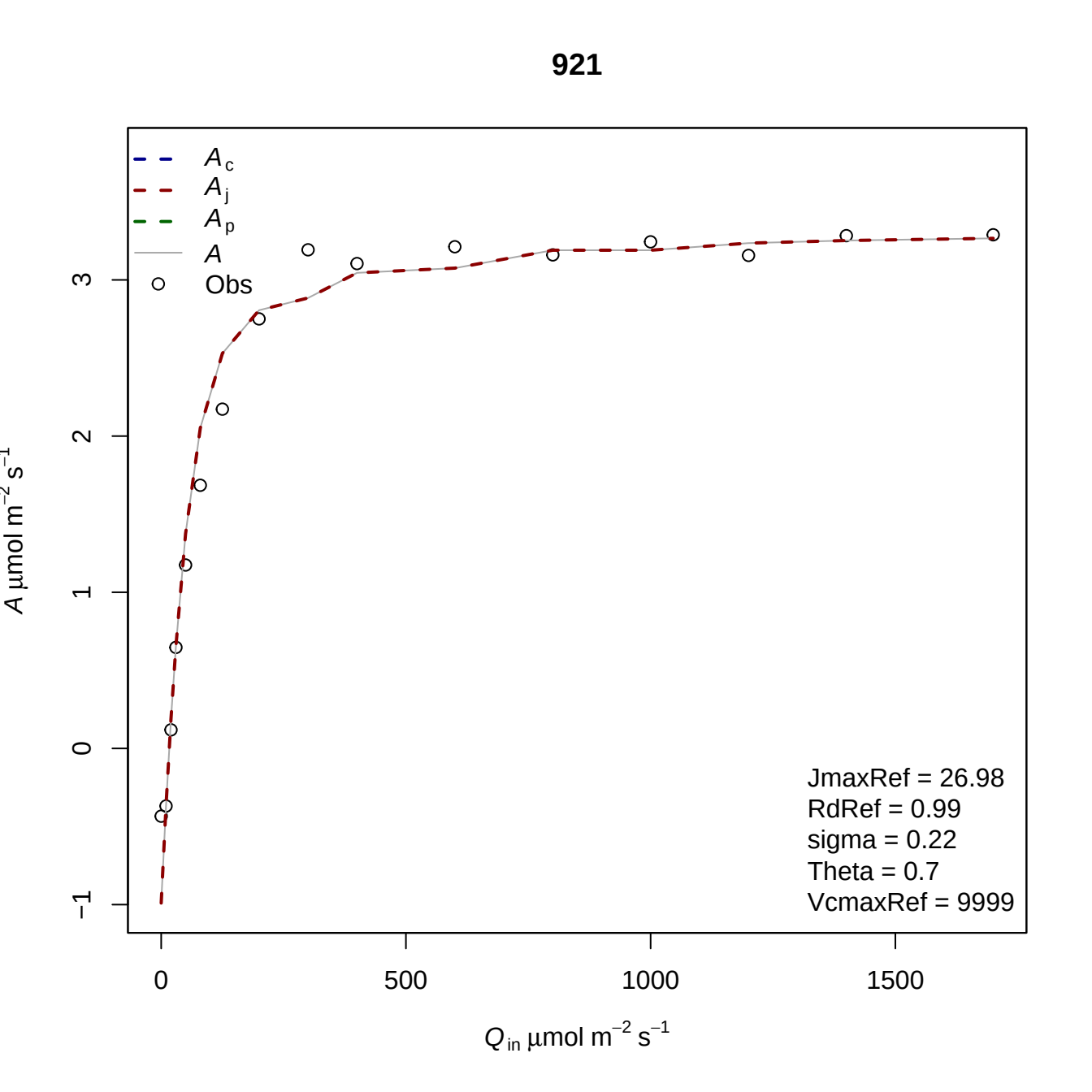

916
A_c
A_j
A_p
A
Obs
JmaxRef = 31.24
RdRef = 0.76
sigma = 0.16
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500
2000

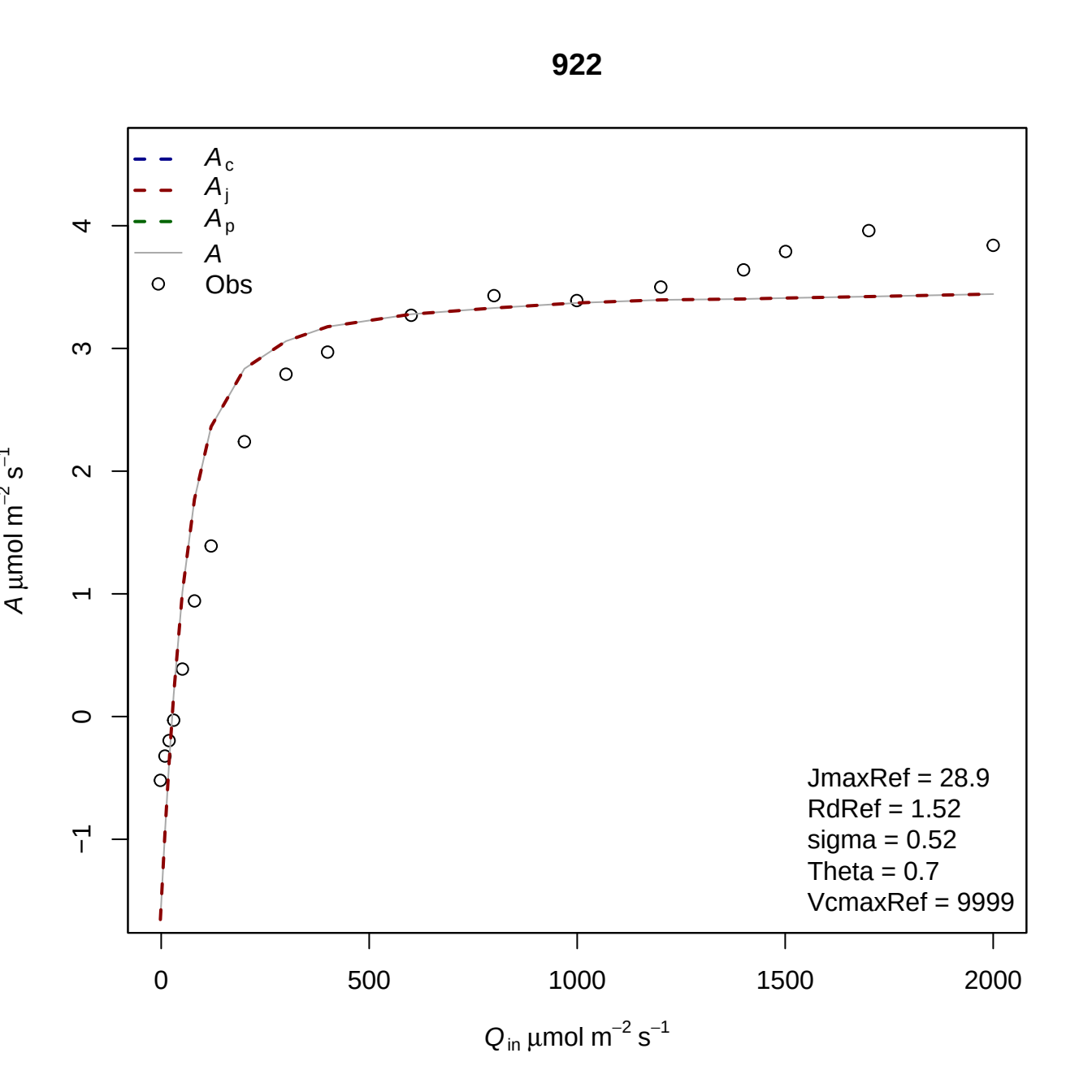

917
A_c
A_j
A_p
A
Obs
JmaxRef = 17.35
RdRef = 0.61
sigma = 0.18
Theta = 0.7
VcmaxRef = 9999
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

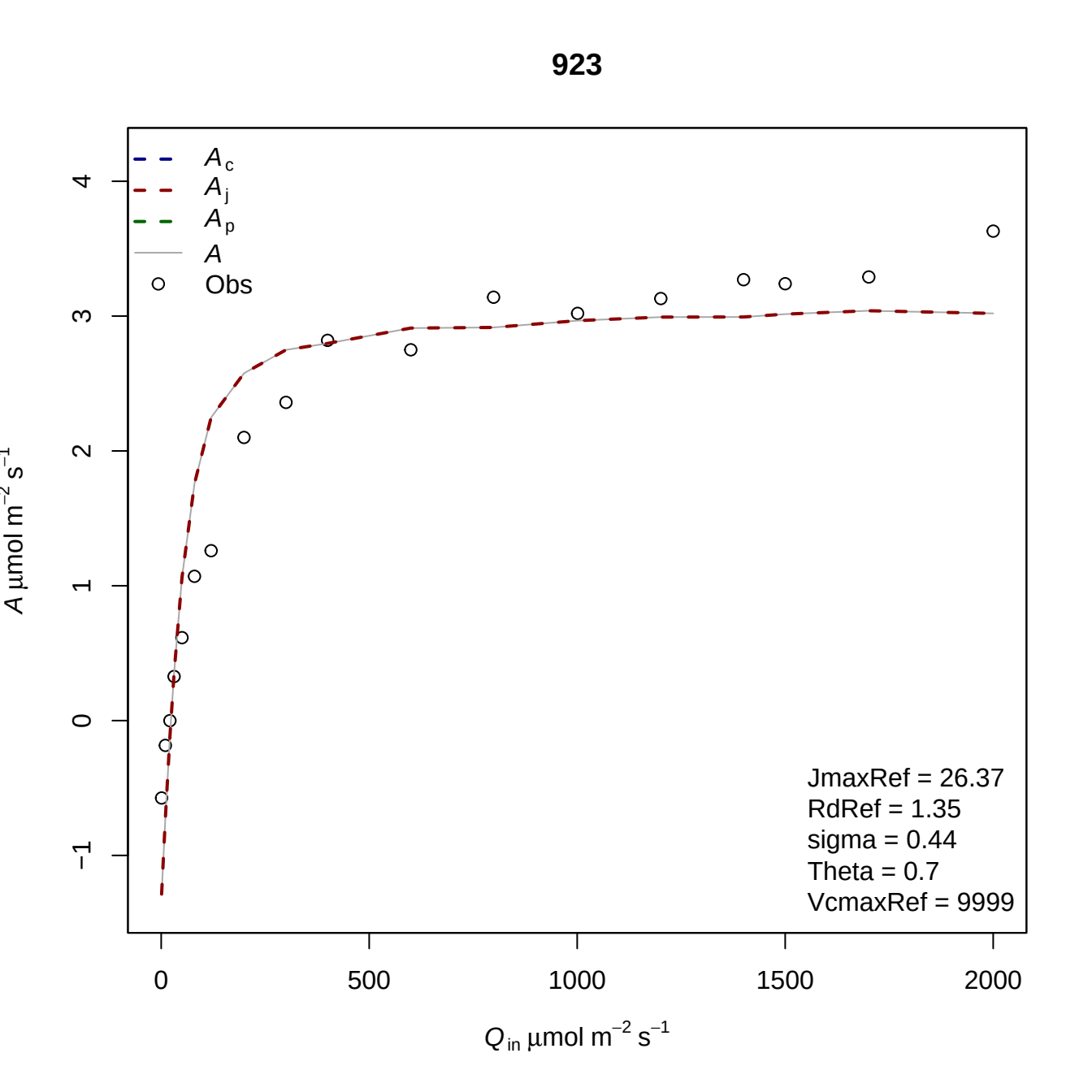

918
A_c
A_j
A_p
A
Obs
JmaxRef = 20.79
RdRef = 1.09
sigma = 0.42
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

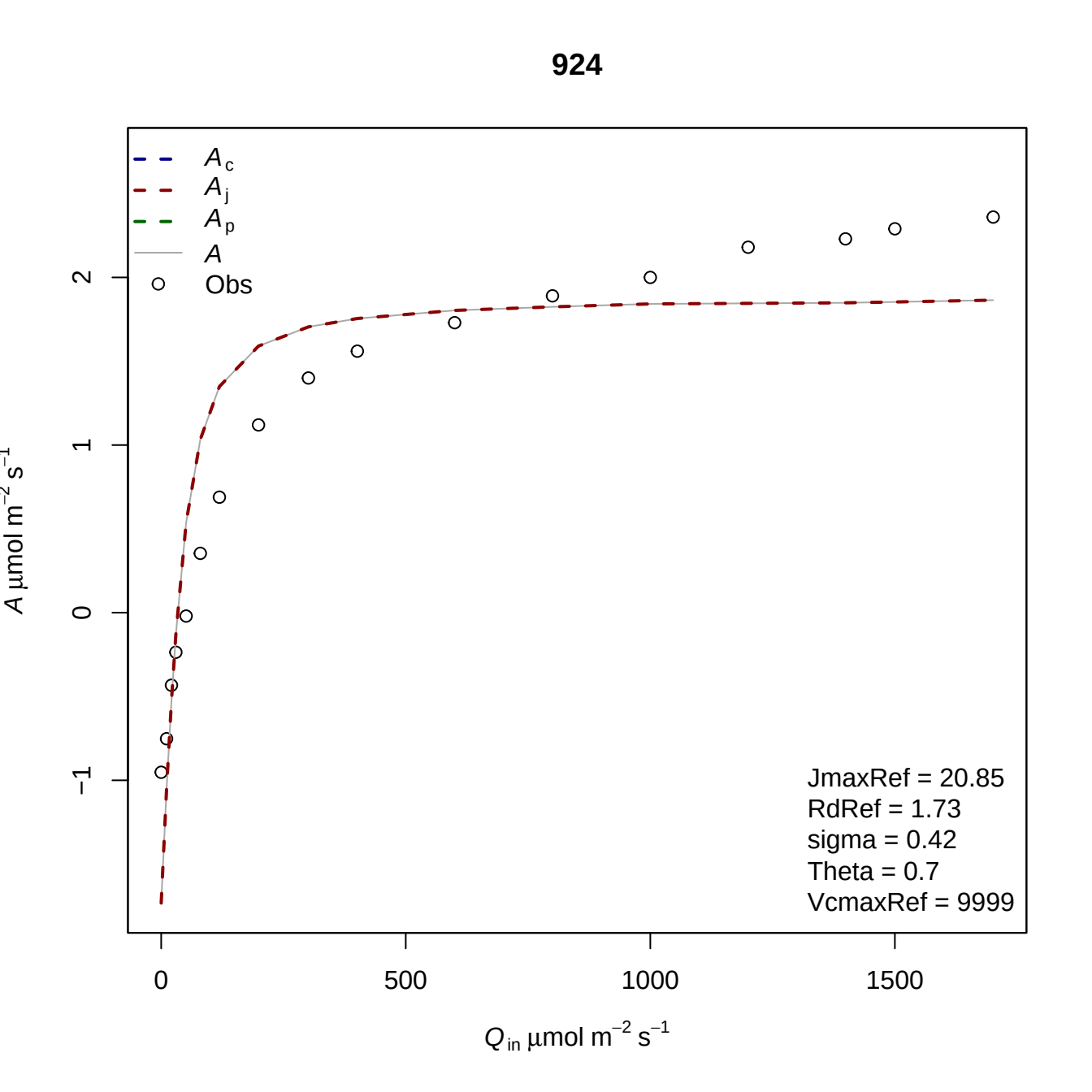

919
A_c
A_j
A_p
A
Obs
JmaxRef = 20.84
RdRef = 2.03
sigma = 0.45
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2
1
0
−1
−2
0
500
1000
1500
2000

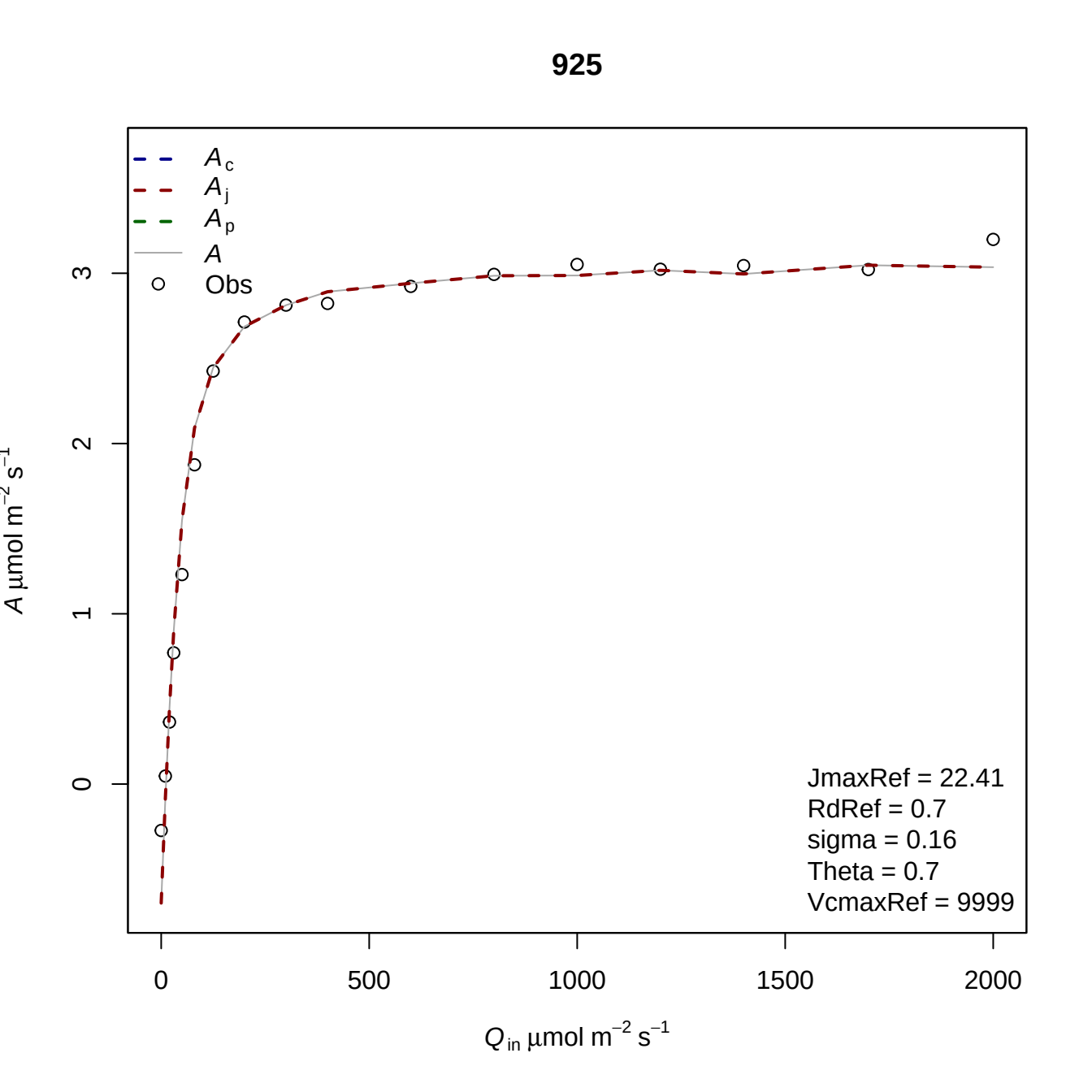

920
A_c
A_j
A_p
A
Obs
JmaxRef = 41.78
RdRef = 0.79
sigma = 0.1
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

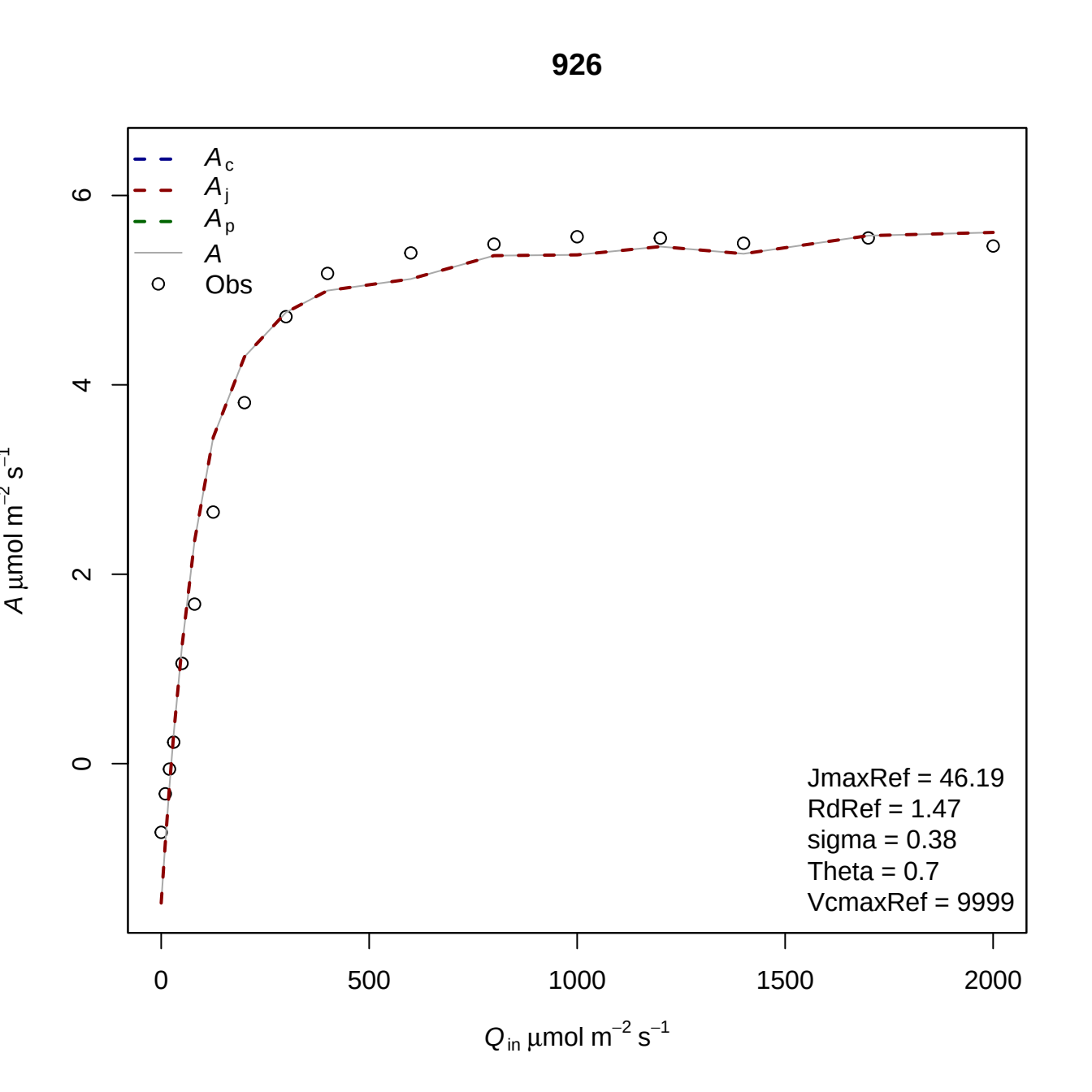

921
A_c
A_j
A_p
A
Obs
JmaxRef = 26.98
RdRef = 0.99
sigma = 0.22
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

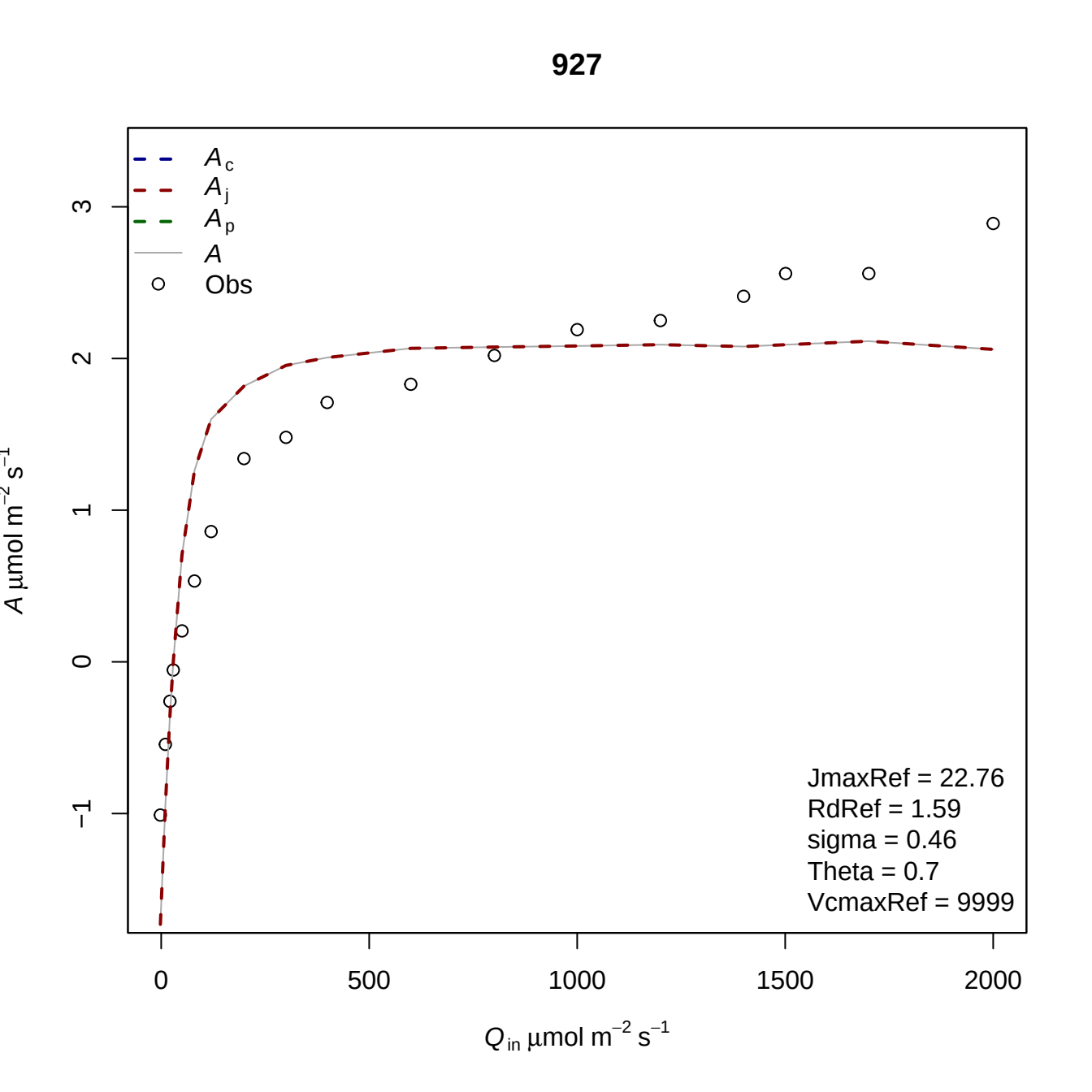

922
A_c
A_j
A_p
A
Obs
JmaxRef = 28.9
RdRef = 1.52
sigma = 0.52
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500
2000

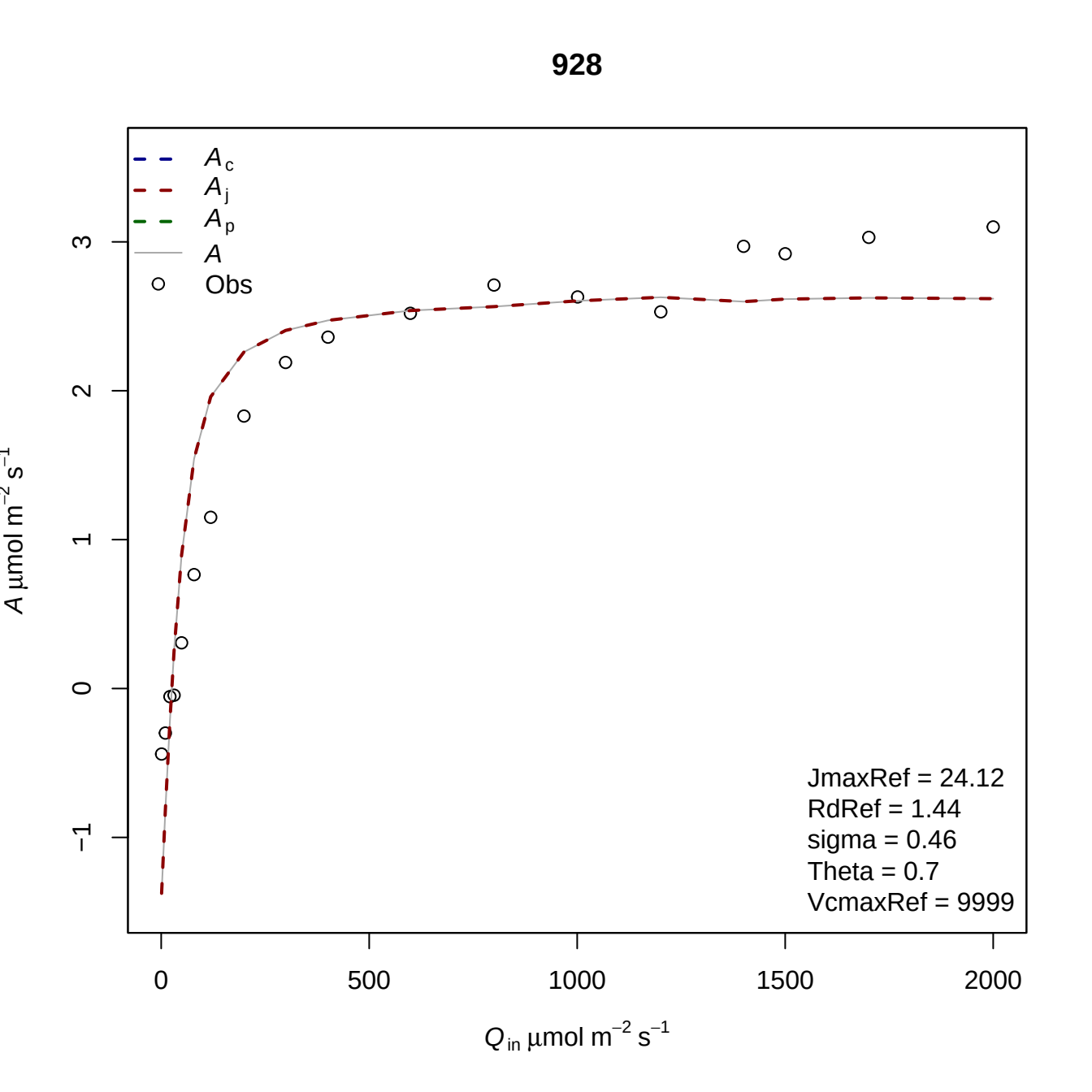

923
A_c
A_j
A_p
A
Obs
JmaxRef = 26.37
RdRef = 1.35
sigma = 0.44
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500
2000

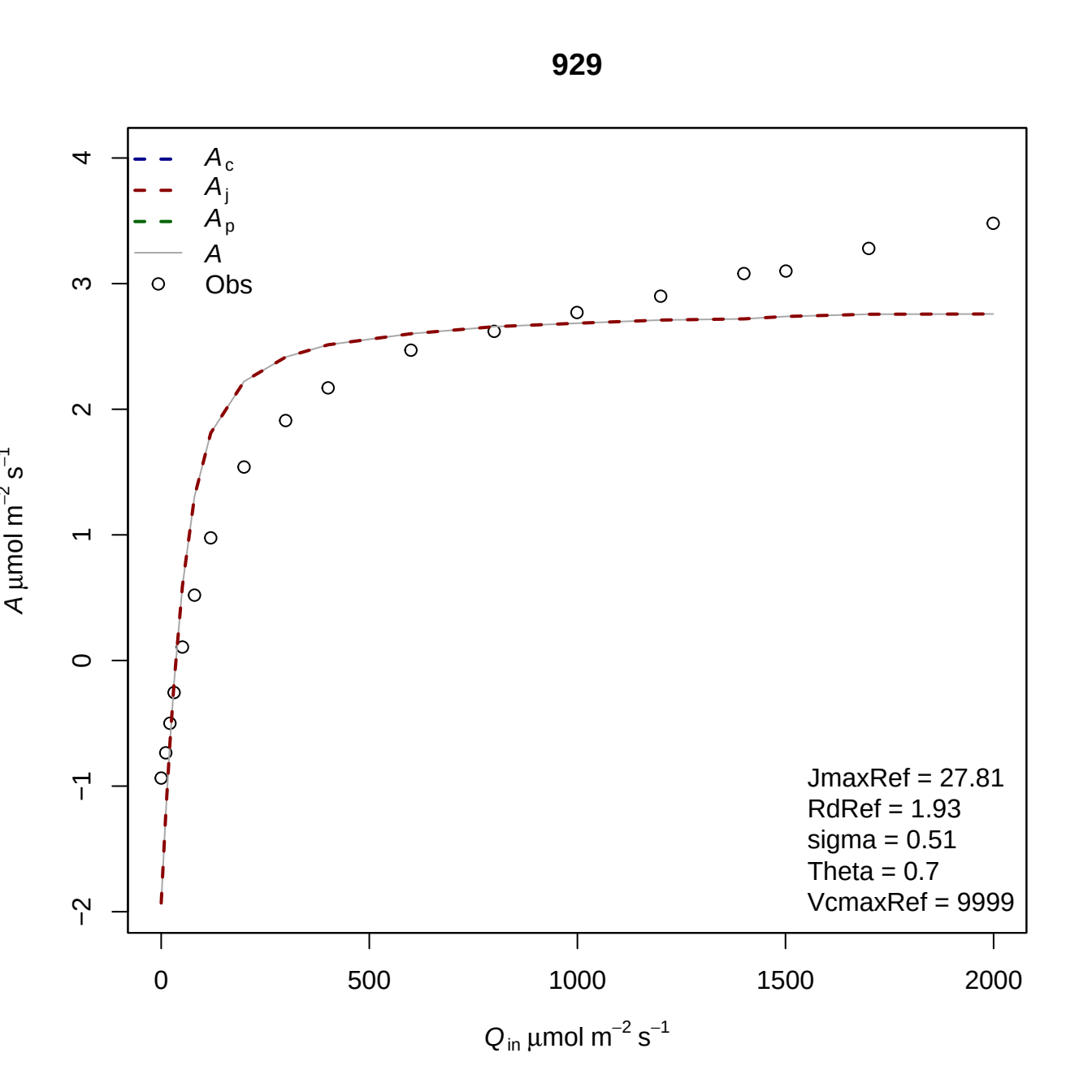

924
A_c
A_j
A_p
A
Obs
JmaxRef = 20.85
RdRef = 1.73
sigma = 0.42
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
500
1000
1500

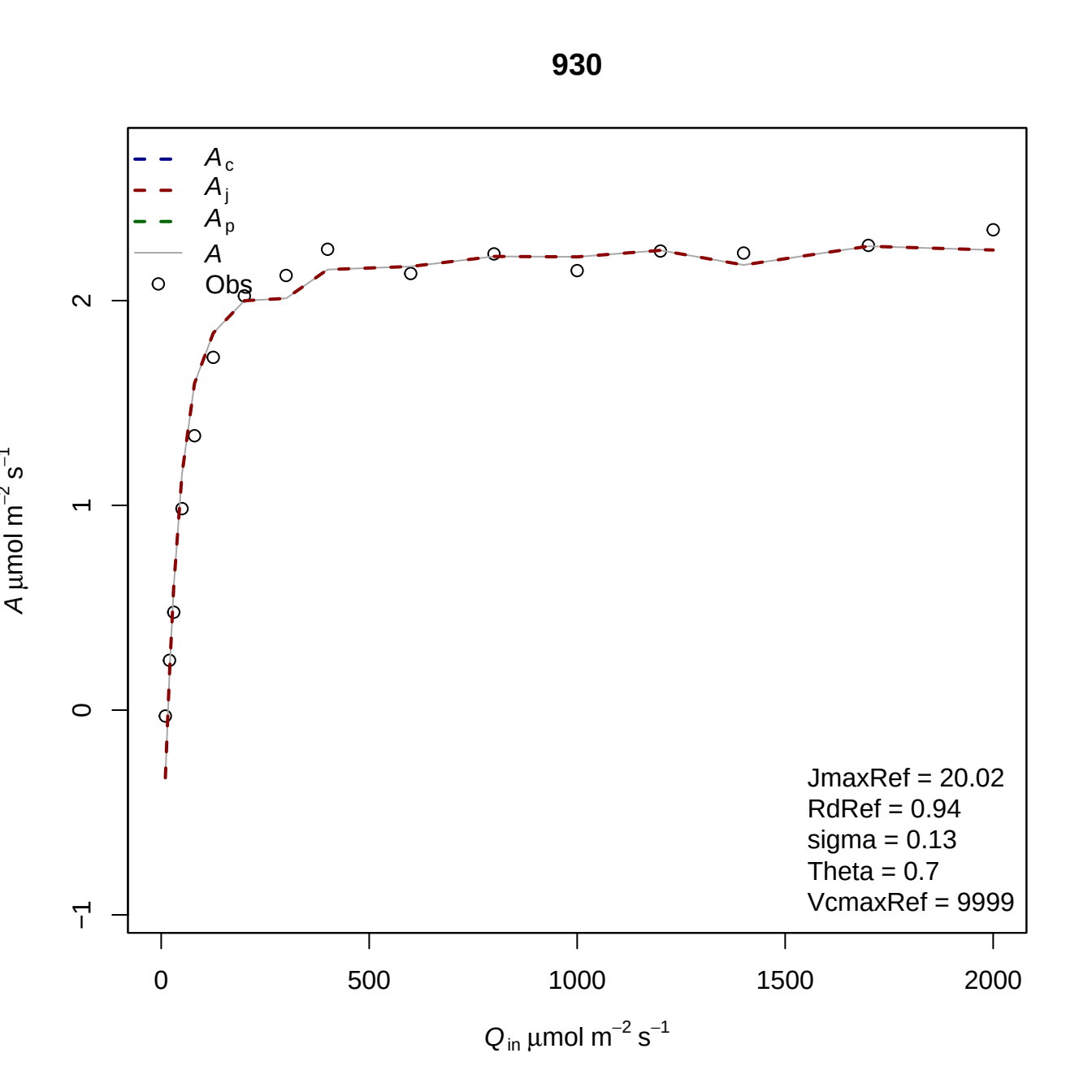

925
A_c
A_j
A_p
A
Obs
JmaxRef = 22.41
RdRef = 0.7
sigma = 0.16
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
500
1000
1500
2000

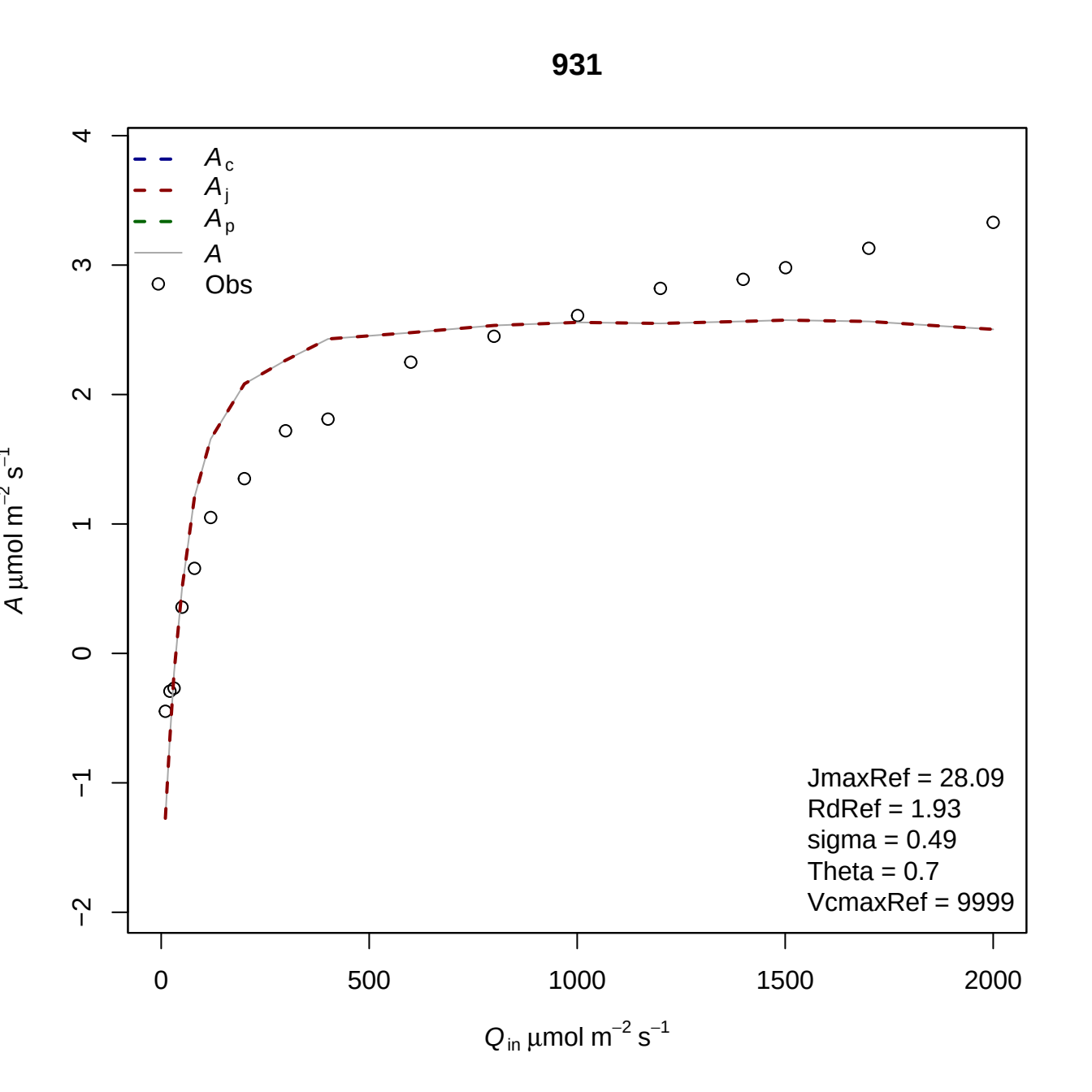

926
A_c
A_j
A_p
A
Obs
JmaxRef = 46.19
RdRef = 1.47
sigma = 0.38
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

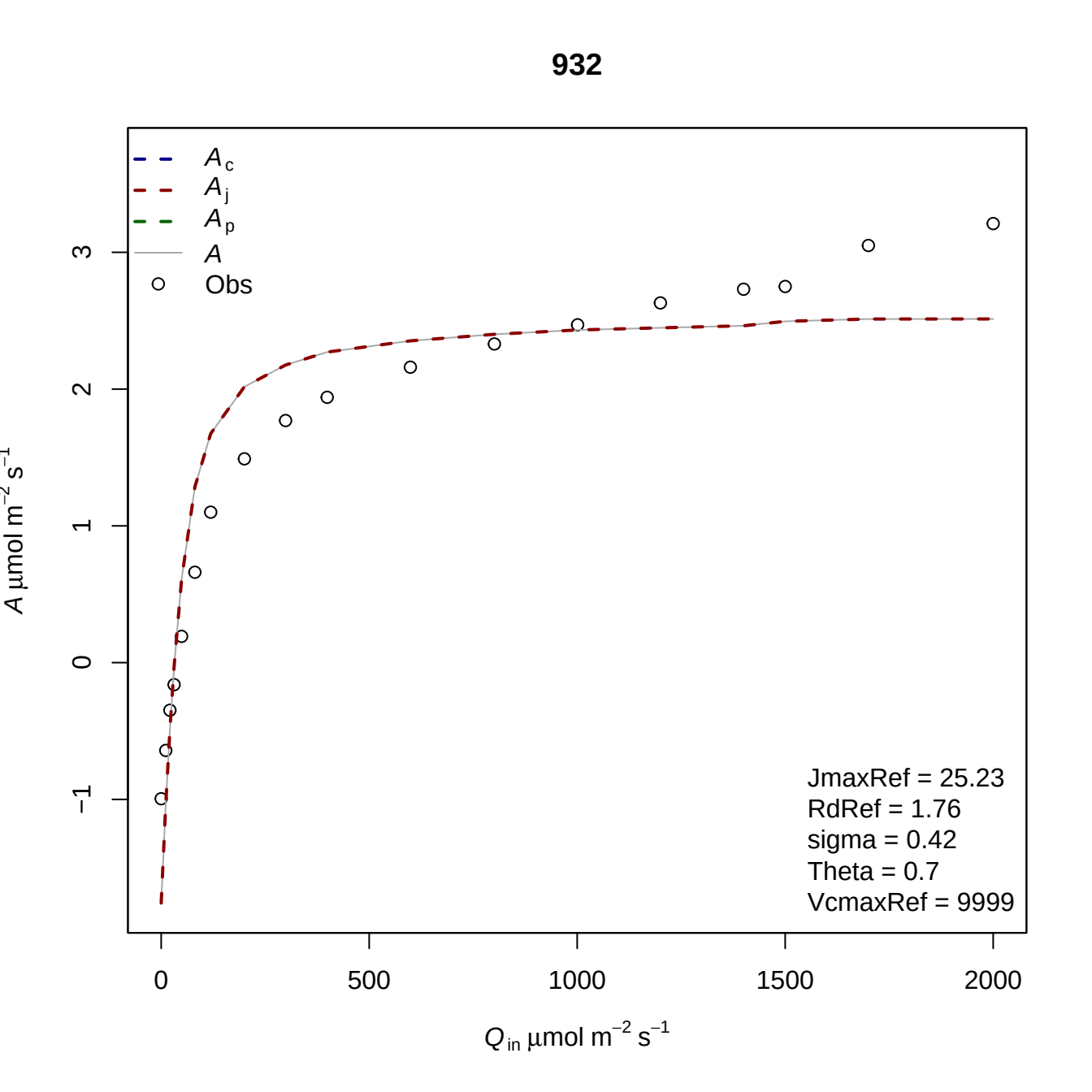

927
A_c
A_j
A_p
A
Obs
JmaxRef = 22.76
RdRef = 1.59
sigma = 0.46
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

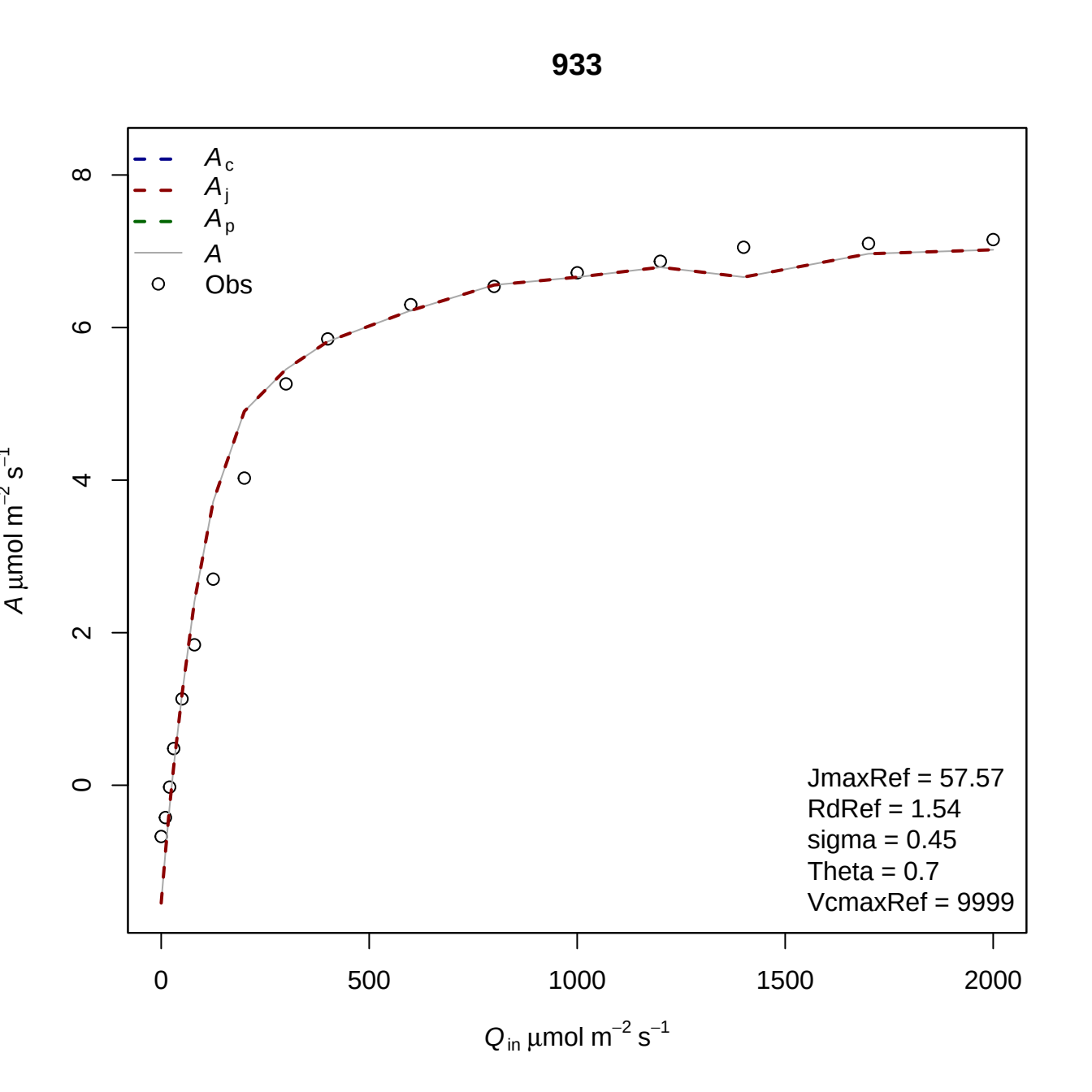

928
A_c
A_j
A_p
A
Obs
JmaxRef = 24.12
RdRef = 1.44
sigma = 0.46
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

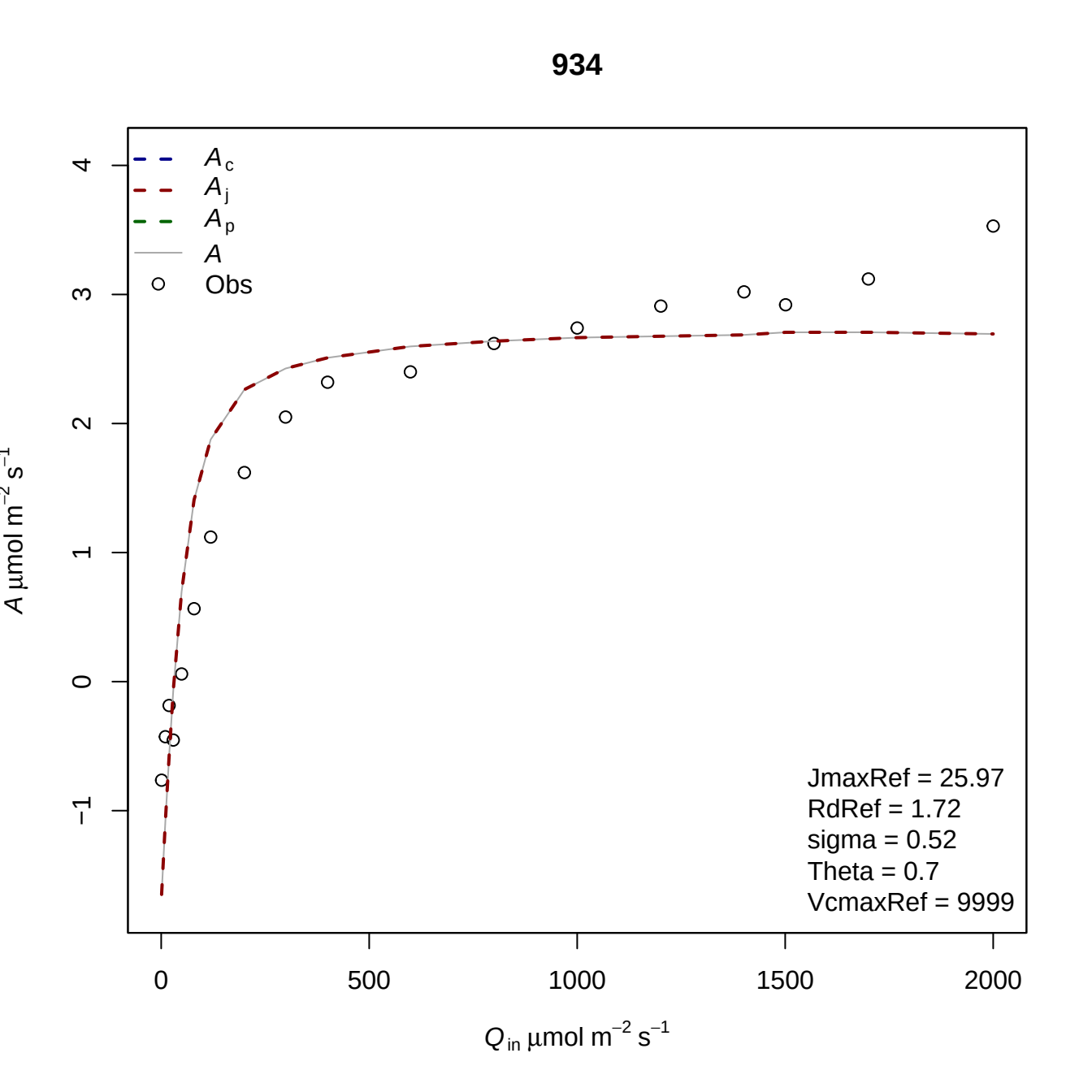

929
A_c
A_j
A_p
A
Obs
JmaxRef = 27.81
RdRef = 1.93
sigma = 0.51
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
0
500
1000
1500
2000

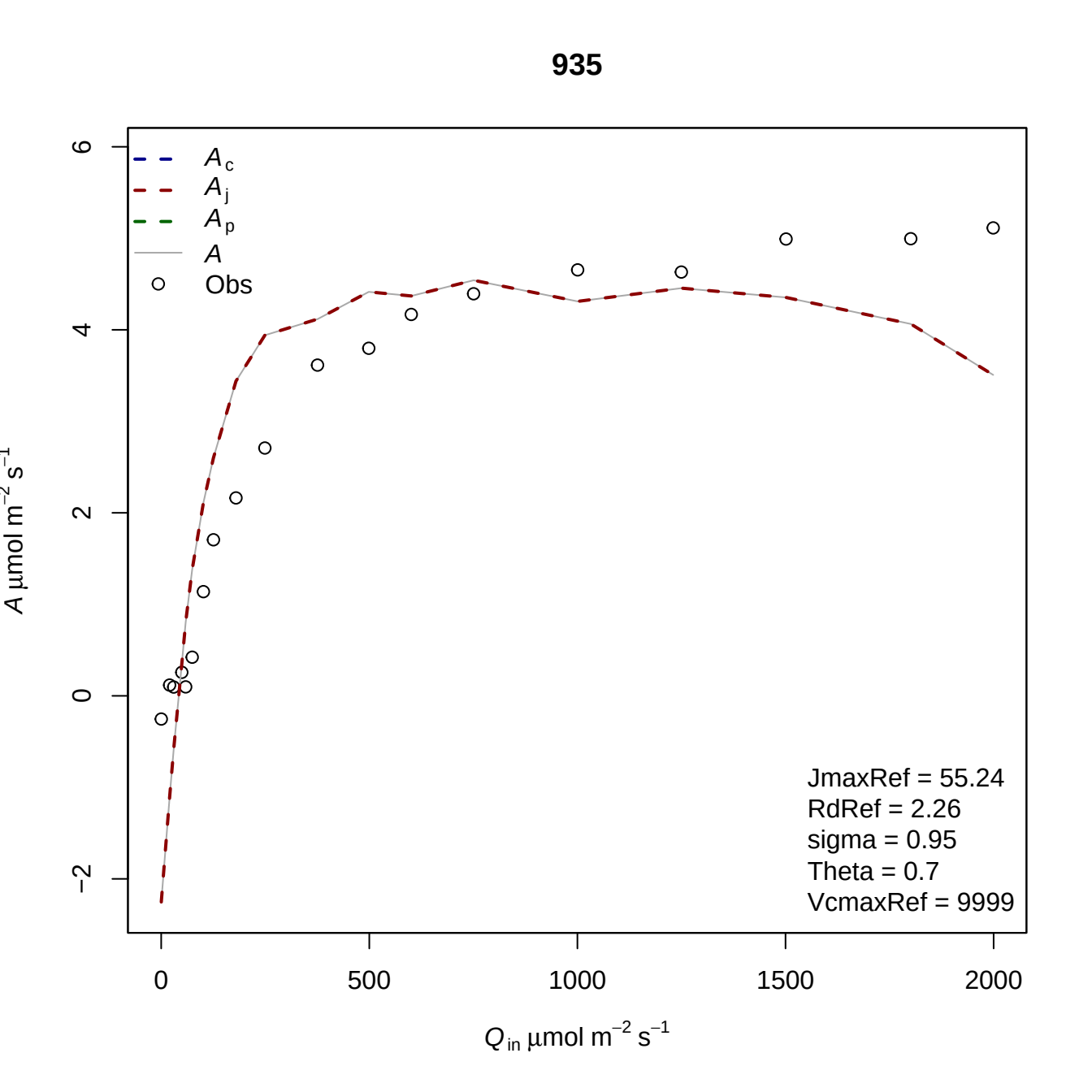

930
A_c
A_j
A_p
A
Obs
JmaxRef = 20.02
RdRef = 0.94
sigma = 0.13
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
500
1000
1500
2000

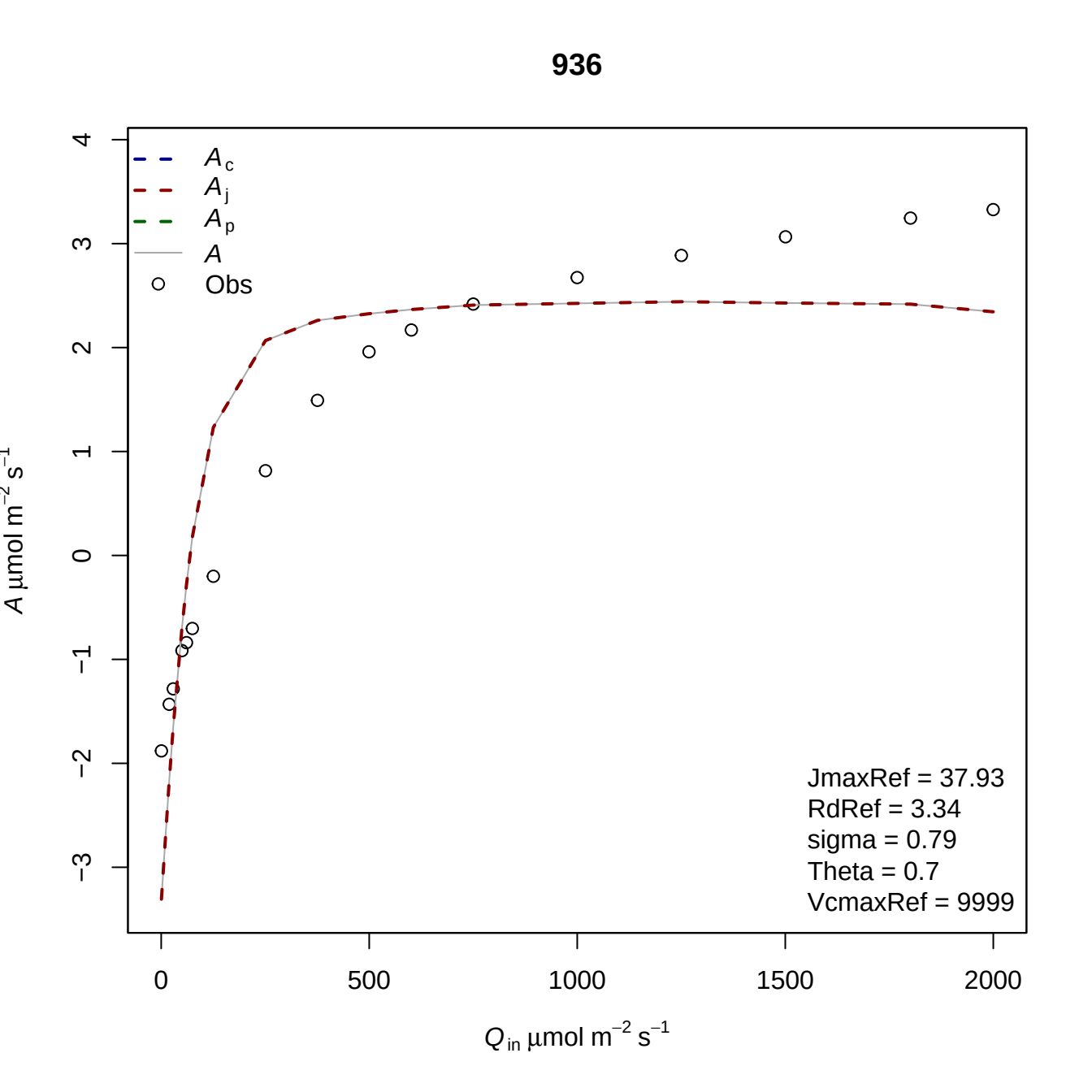

931
A_c
A_j
A_p
A
Obs
JmaxRef = 28.09
RdRef = 1.93
sigma = 0.49
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
0
500
1000
1500
2000

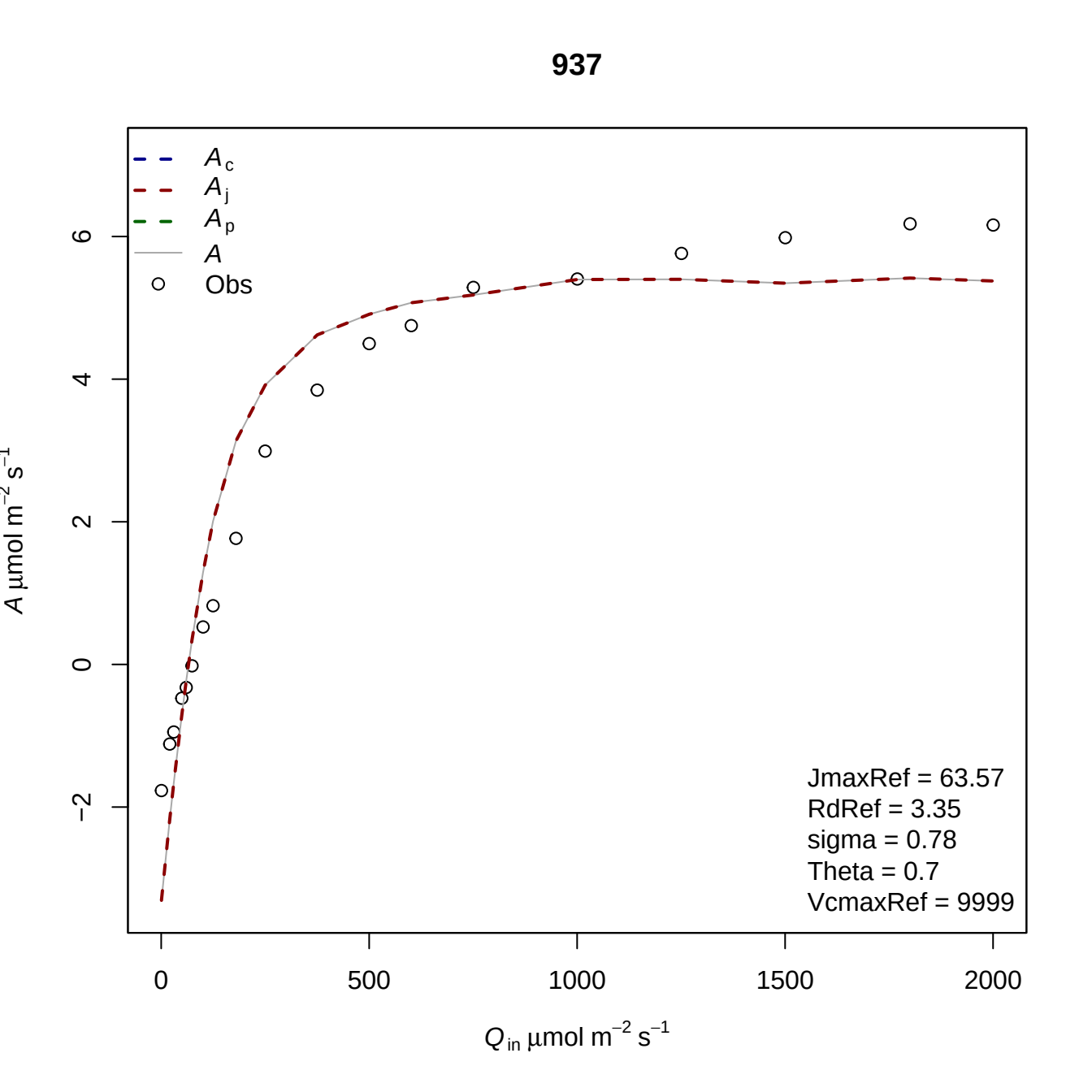

932
A_c
A_j
A_p
A
Obs
JmaxRef = 25.23
RdRef = 1.76
sigma = 0.42
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

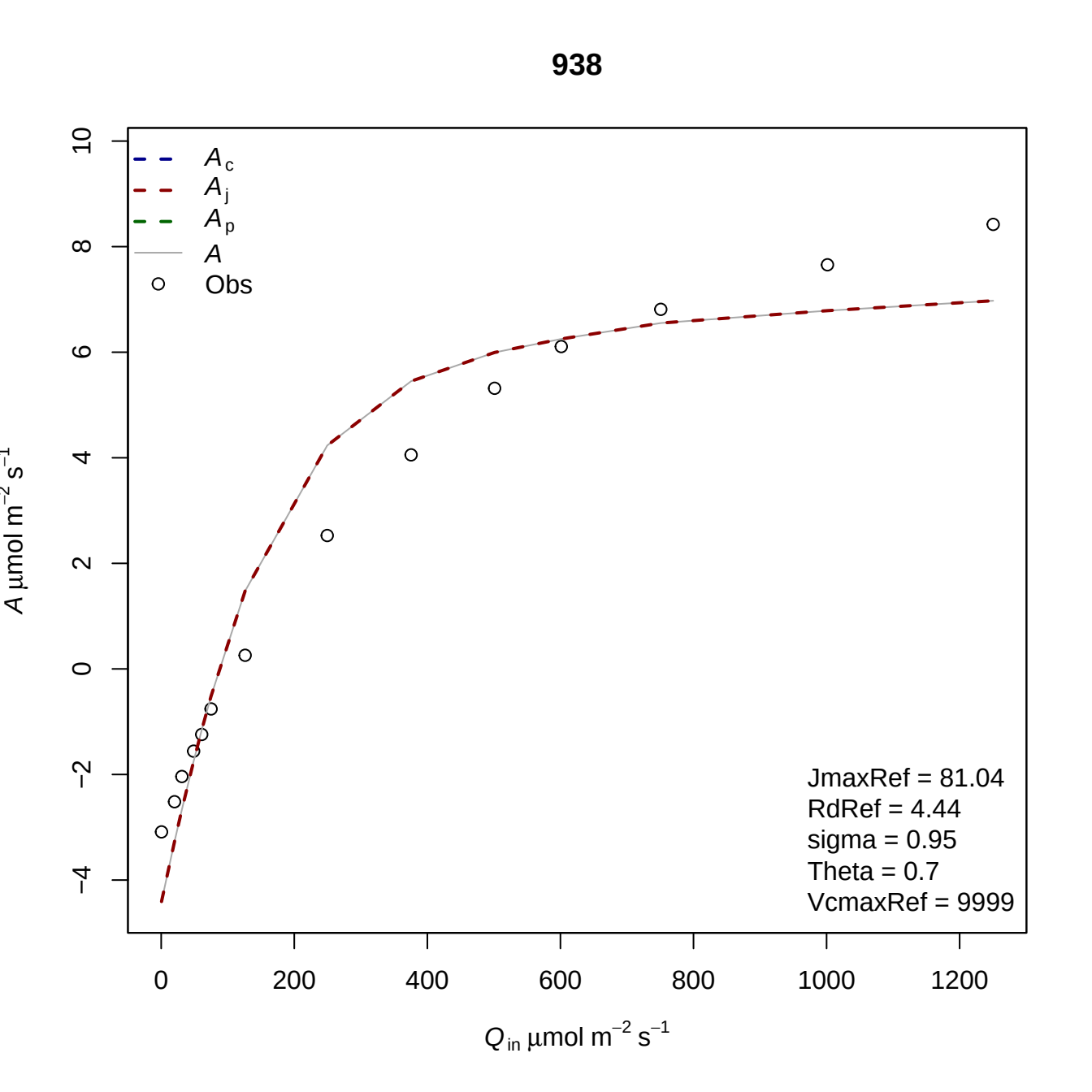

933
A_c
A_j
A_p
A
Obs
JmaxRef = 57.57
RdRef = 1.54
sigma = 0.45
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

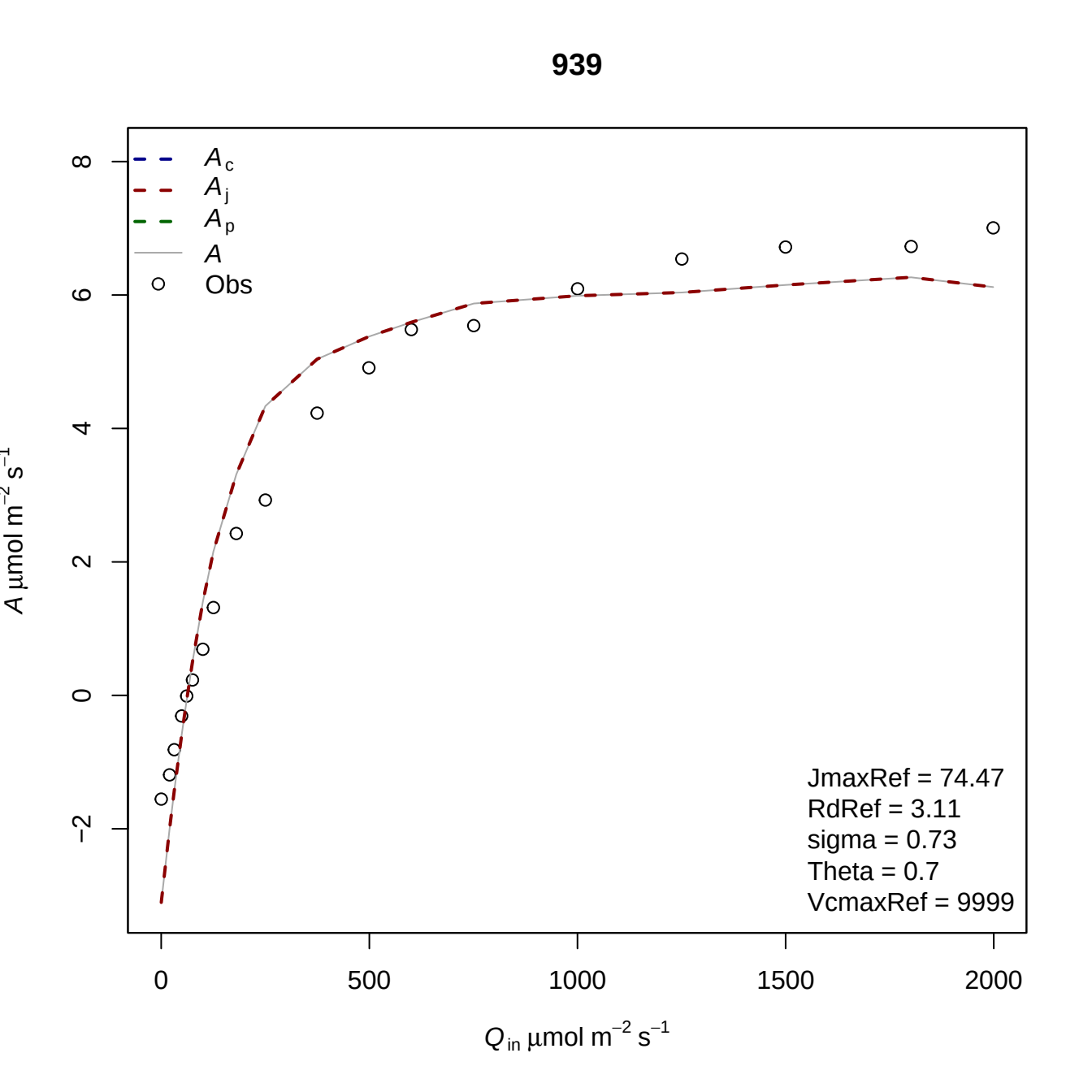

934
A_c
A_j
A_p
A
Obs
JmaxRef = 25.97
RdRef = 1.72
sigma = 0.52
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500
2000

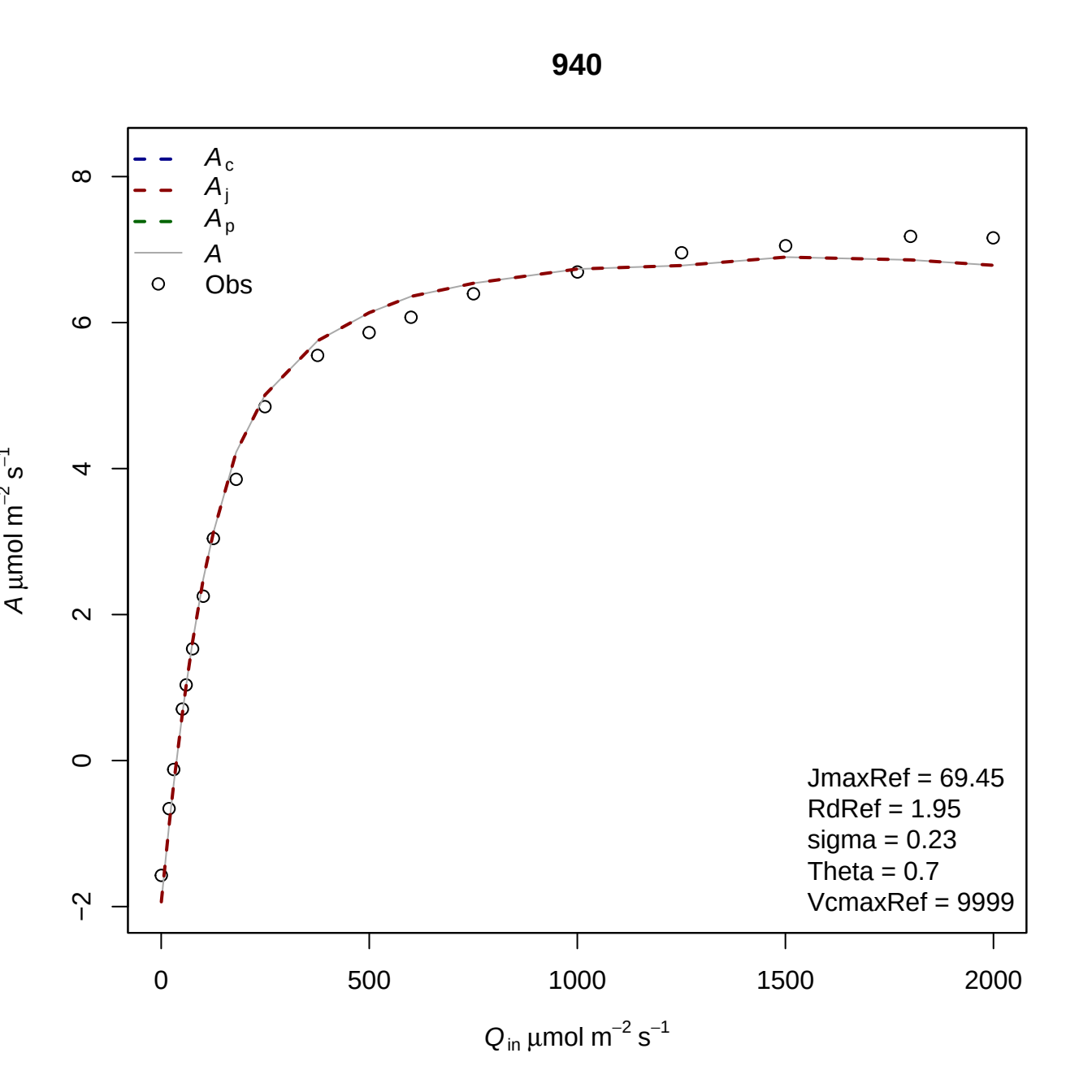

935
A_c
A_j
A_p
A
Obs
JmaxRef = 55.24
RdRef = 2.26
sigma = 0.95
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

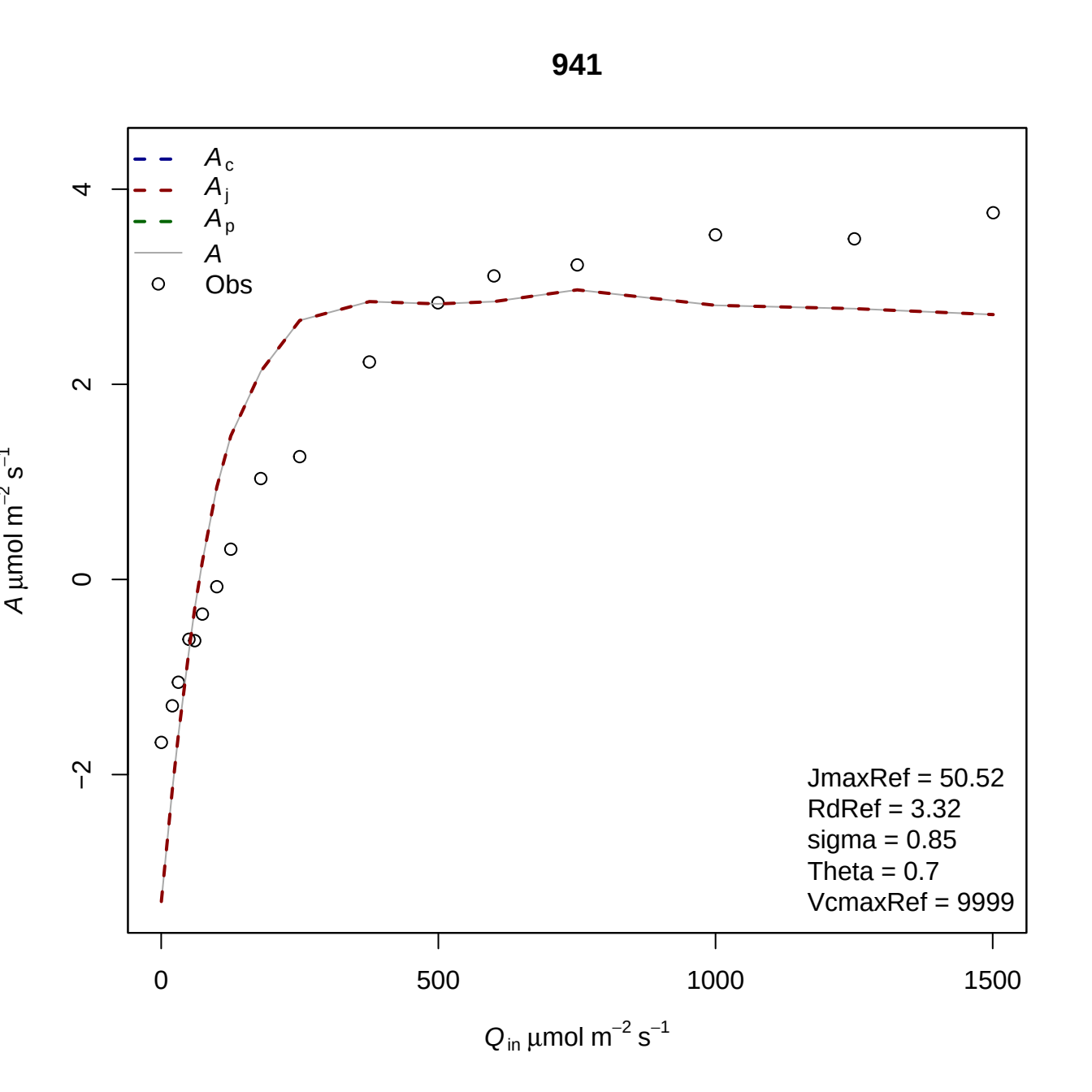

936
A_c
A_j
A_p
A
Obs
JmaxRef = 37.93
RdRef = 3.34
sigma = 0.79
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
−3
0
500
1000
1500
2000

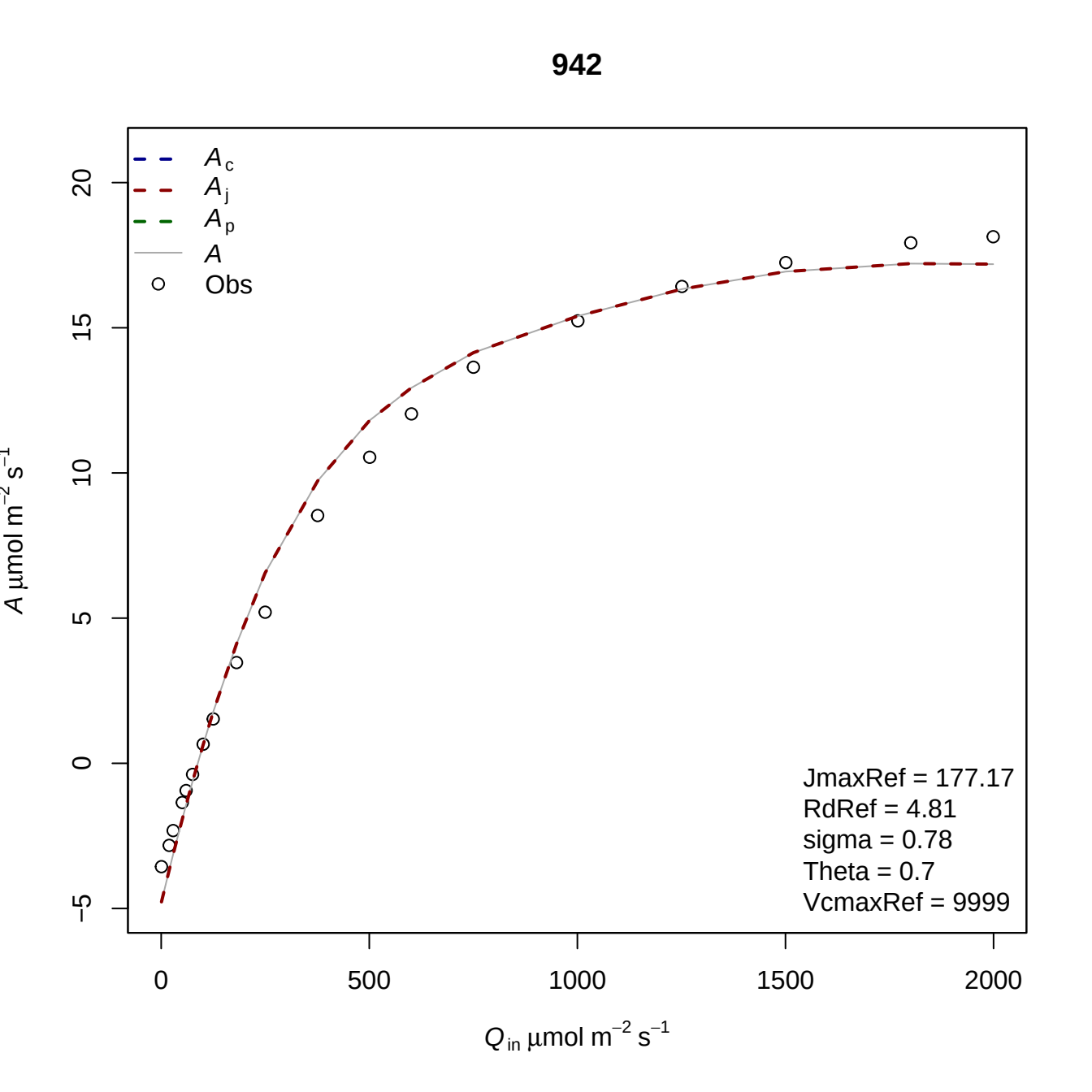

937
A_c
A_j
A_p
A
Obs
JmaxRef = 63.57
RdRef = 3.35
sigma = 0.78
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

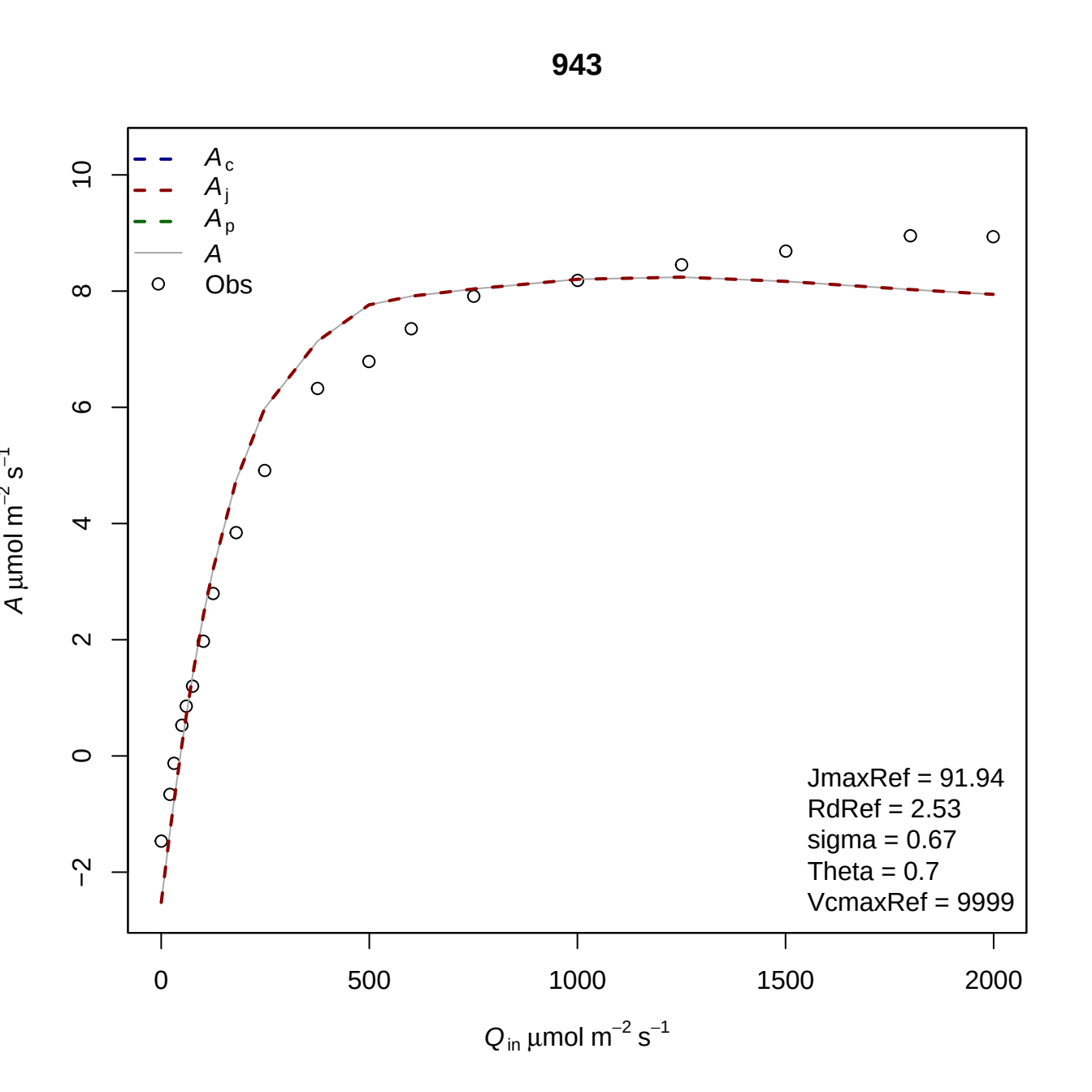

938
A_c
A_j
A_p
A
Obs
JmaxRef = 81.04
RdRef = 4.44
sigma = 0.95
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
200
400
600
800
1000
1200

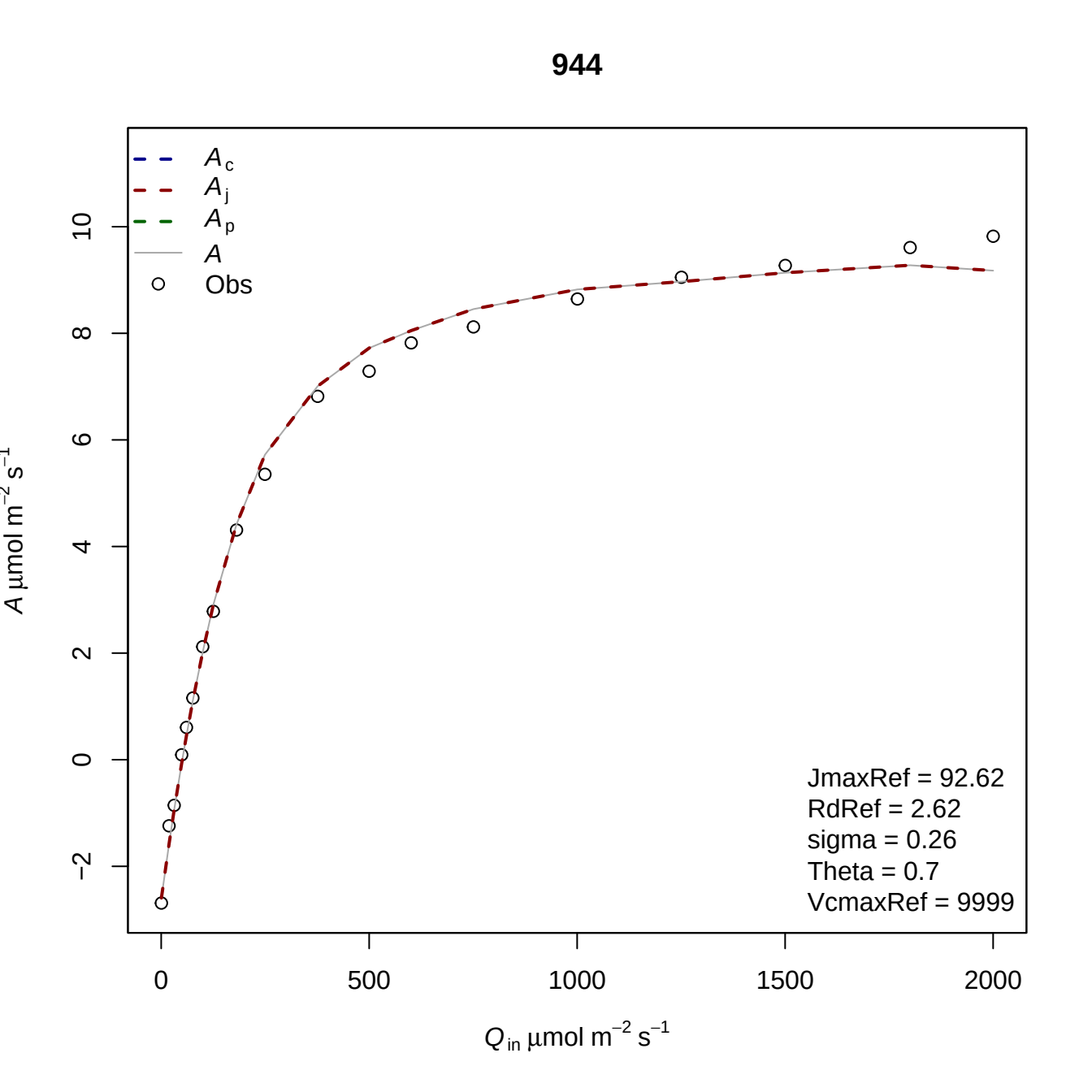

939
A_c
A_j
A_p
A
Obs
JmaxRef = 74.47
RdRef = 3.11
sigma = 0.73
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

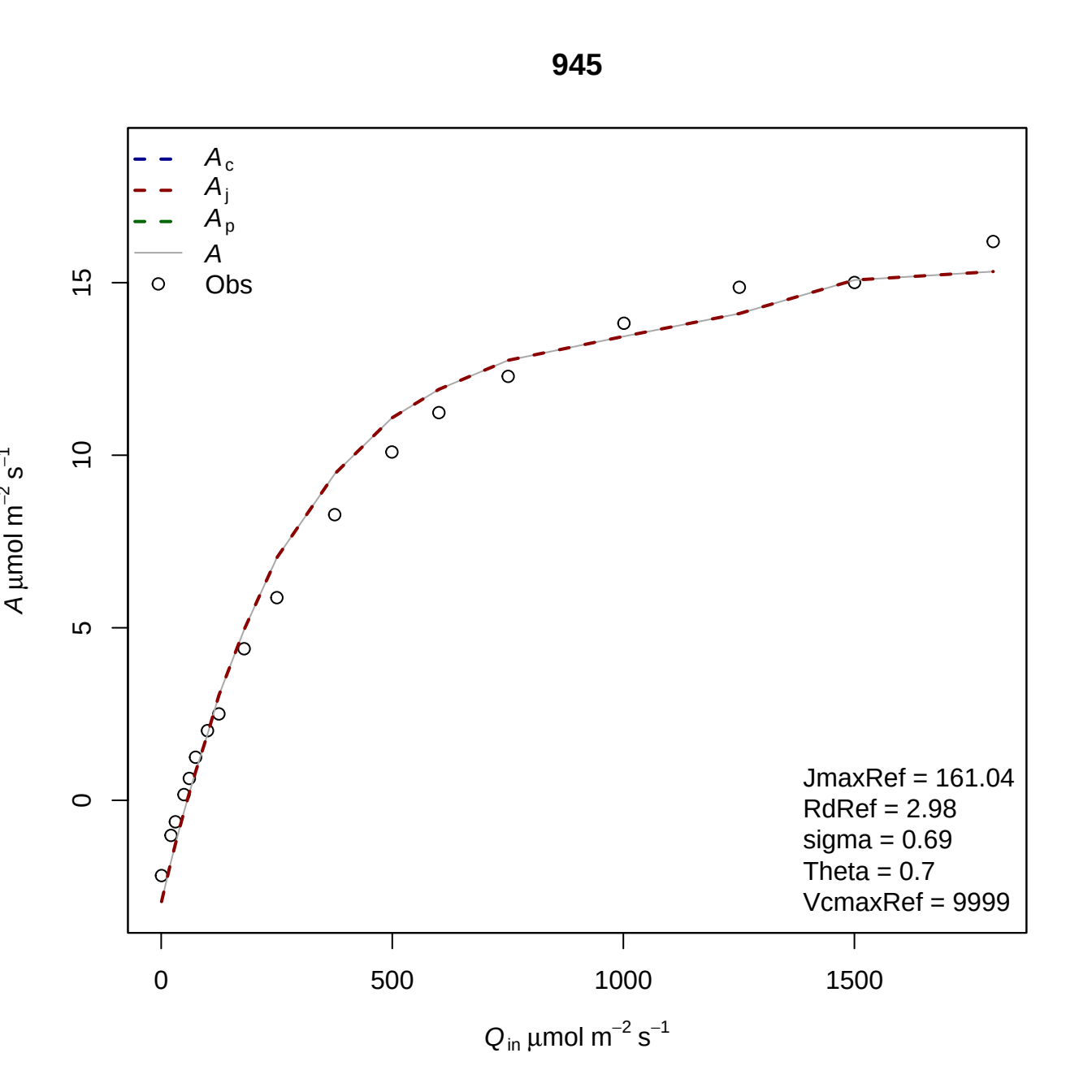

940
A_c
A_j
A_p
A
Obs
JmaxRef = 69.45
RdRef = 1.95
sigma = 0.23
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

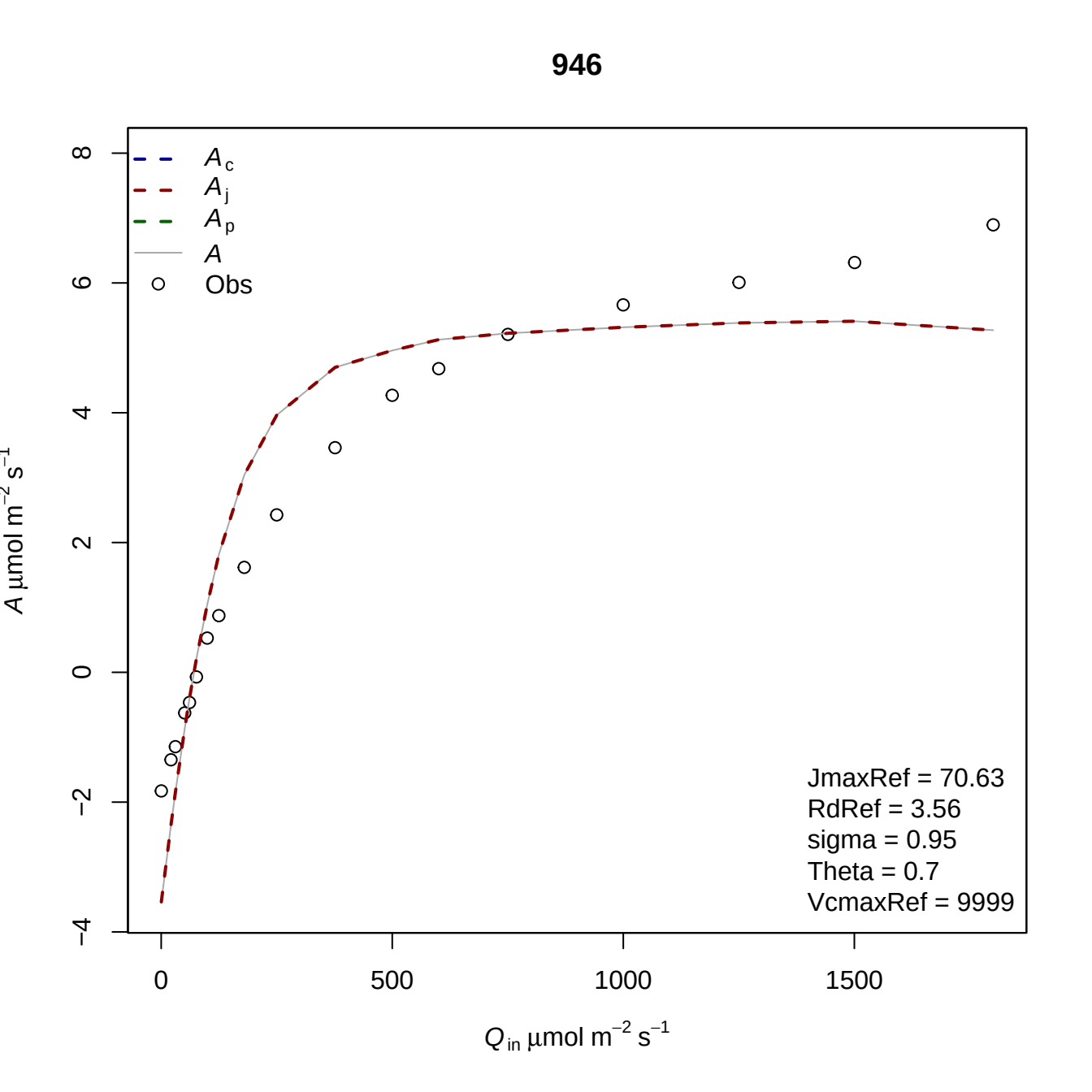

941
A_c
A_j
A_p
A
Obs
JmaxRef = 50.52
RdRef = 3.32
sigma = 0.85
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
0
500
1000
1500

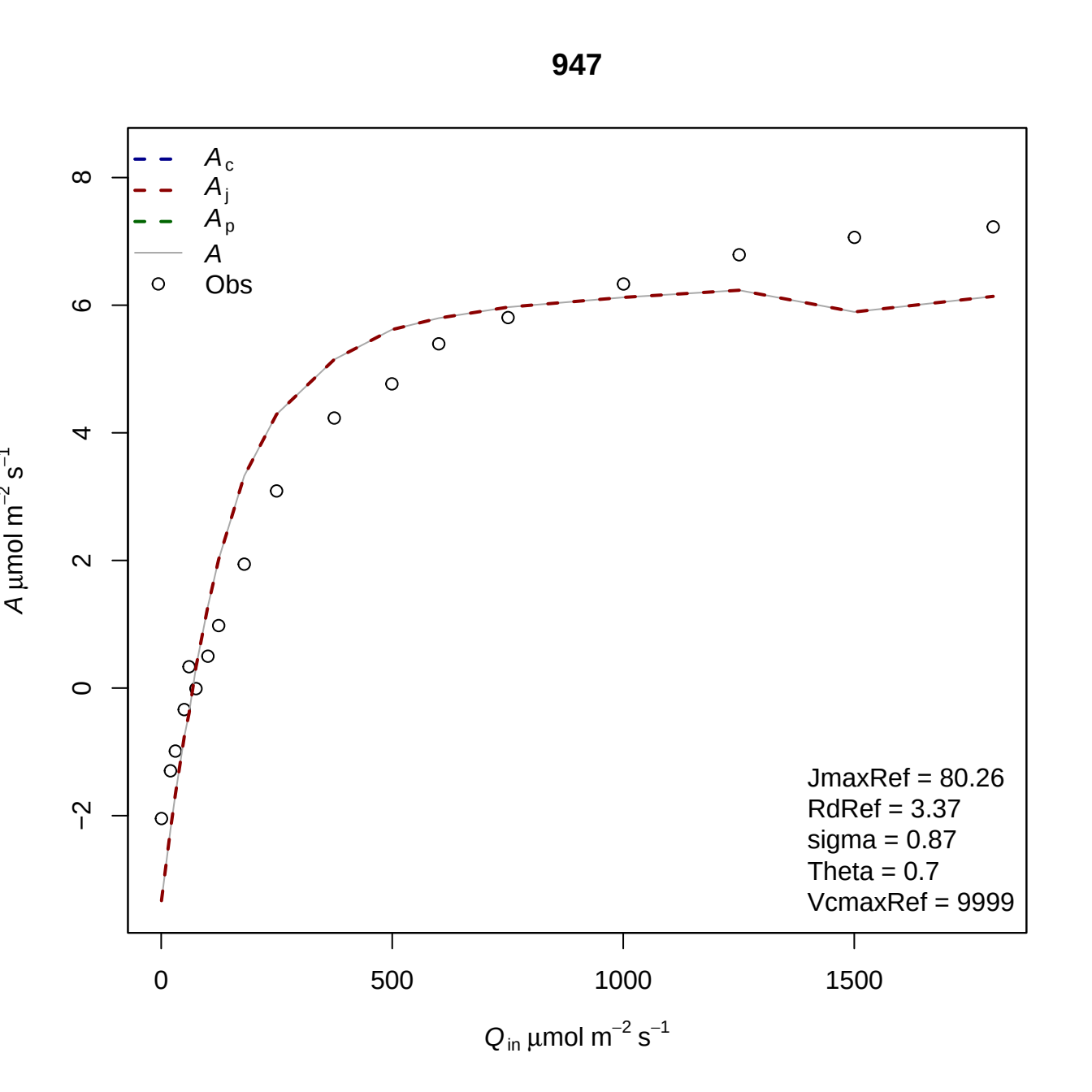

942
A_c
A_j
A_p
A
Obs
JmaxRef = 177.17
RdRef = 4.81
sigma = 0.78
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

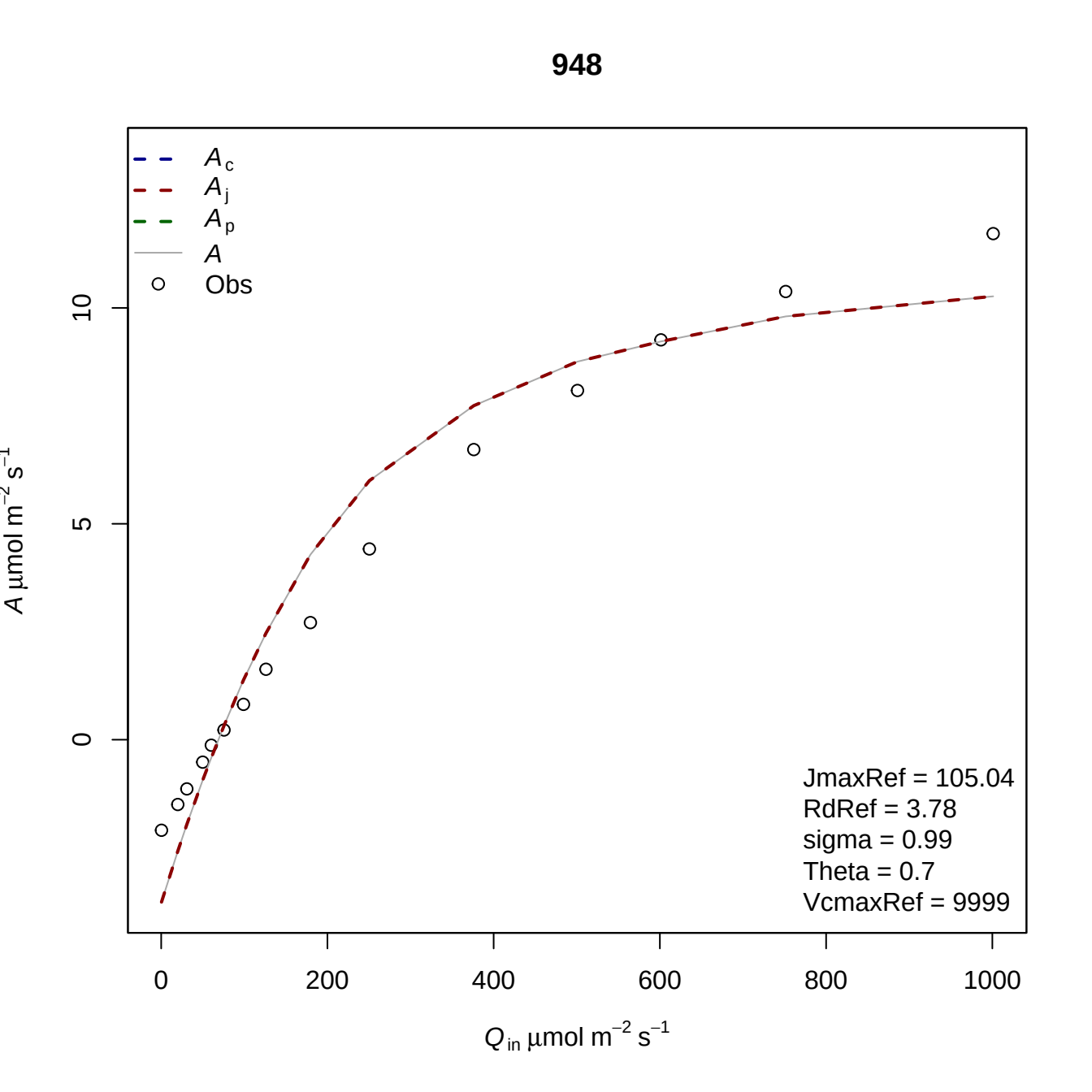

943
A_c
A_j
A_p
A
Obs
JmaxRef = 91.94
RdRef = 2.53
sigma = 0.67
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

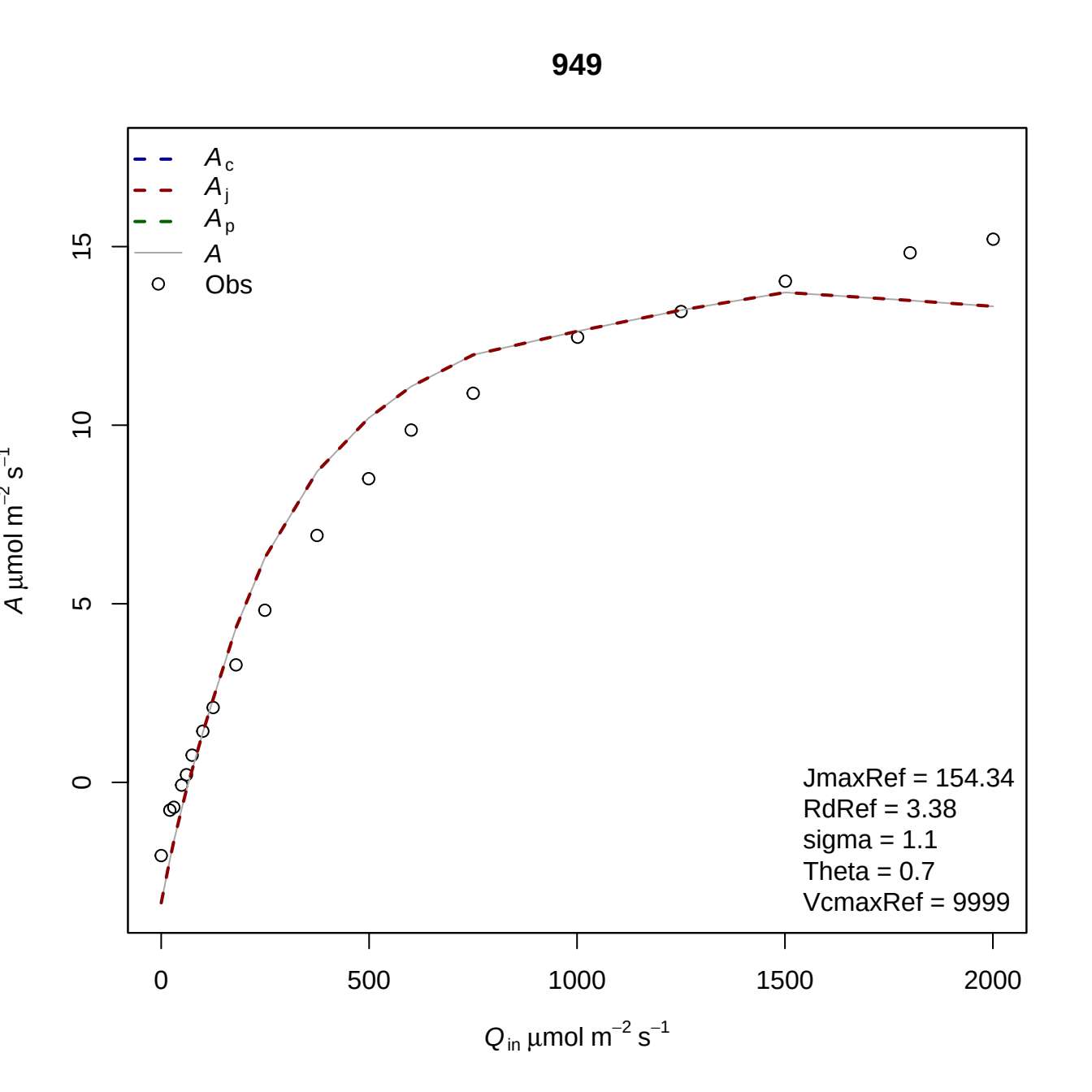

944
A_c
A_j
A_p
A
Obs
JmaxRef = 92.62
RdRef = 2.62
sigma = 0.26
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

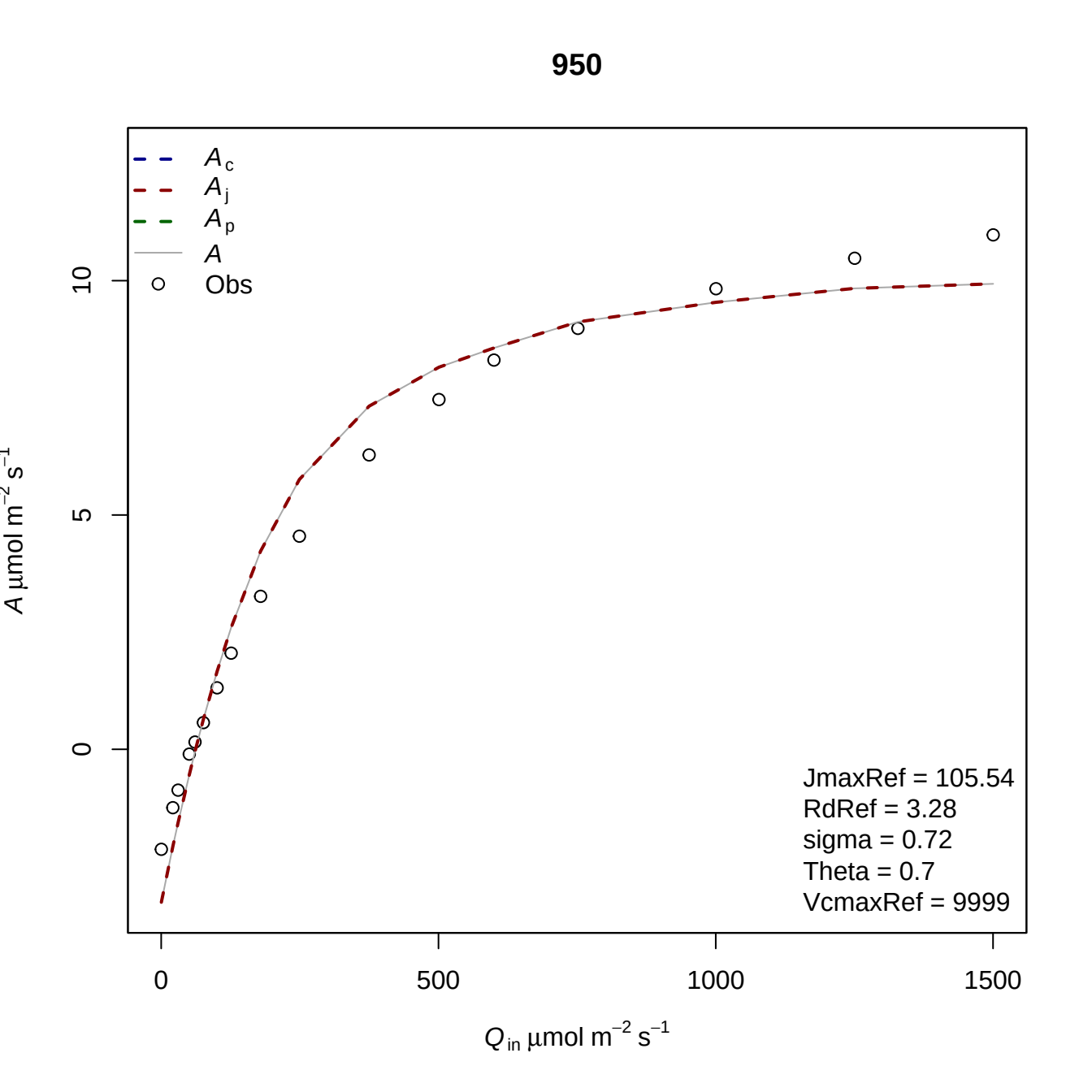

945
A_c
A_j
A_p
A
Obs
JmaxRef = 161.04
RdRef = 2.98
sigma = 0.69
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

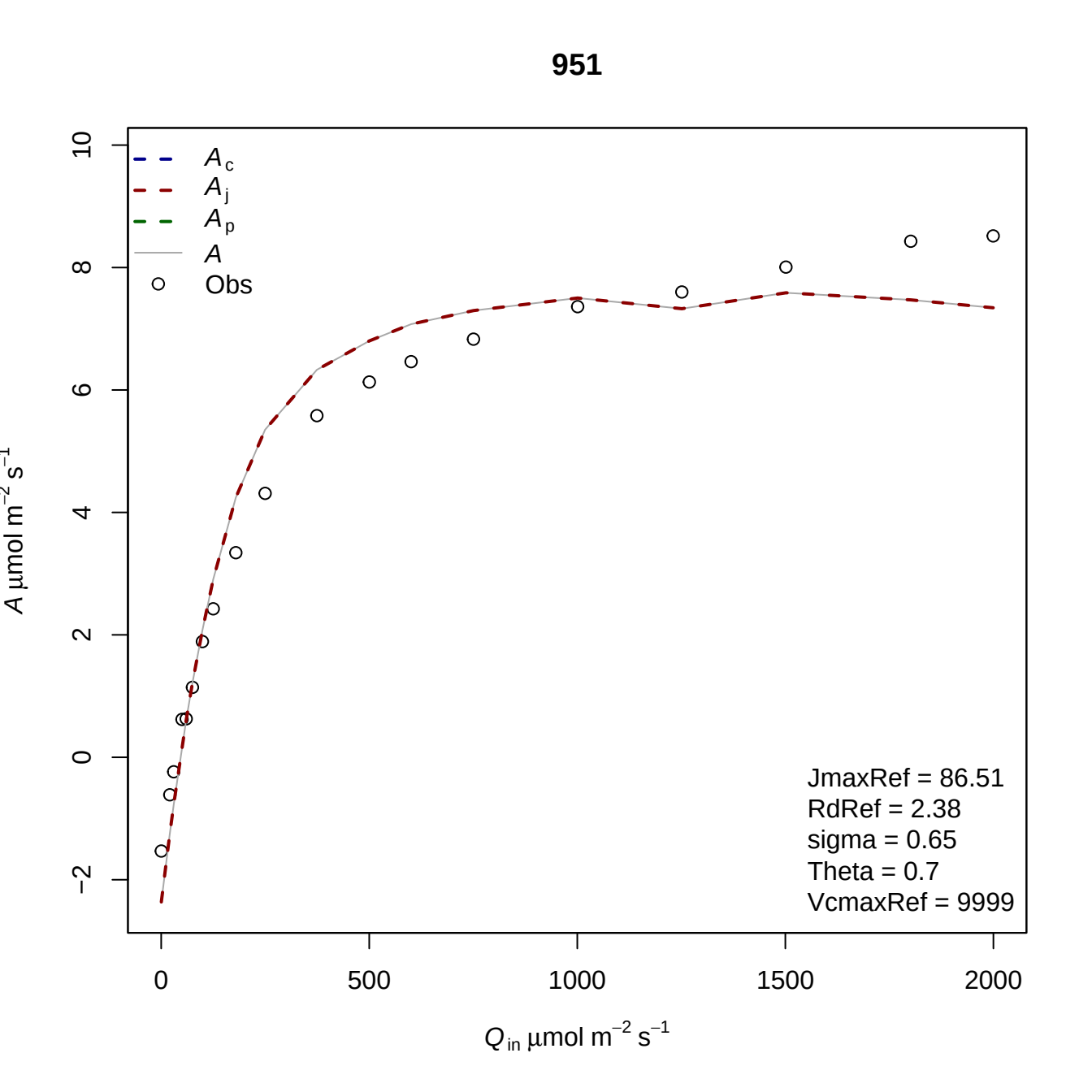

946
A_c
A_j
A_p
A
Obs
JmaxRef = 70.63
RdRef = 3.56
sigma = 0.95
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

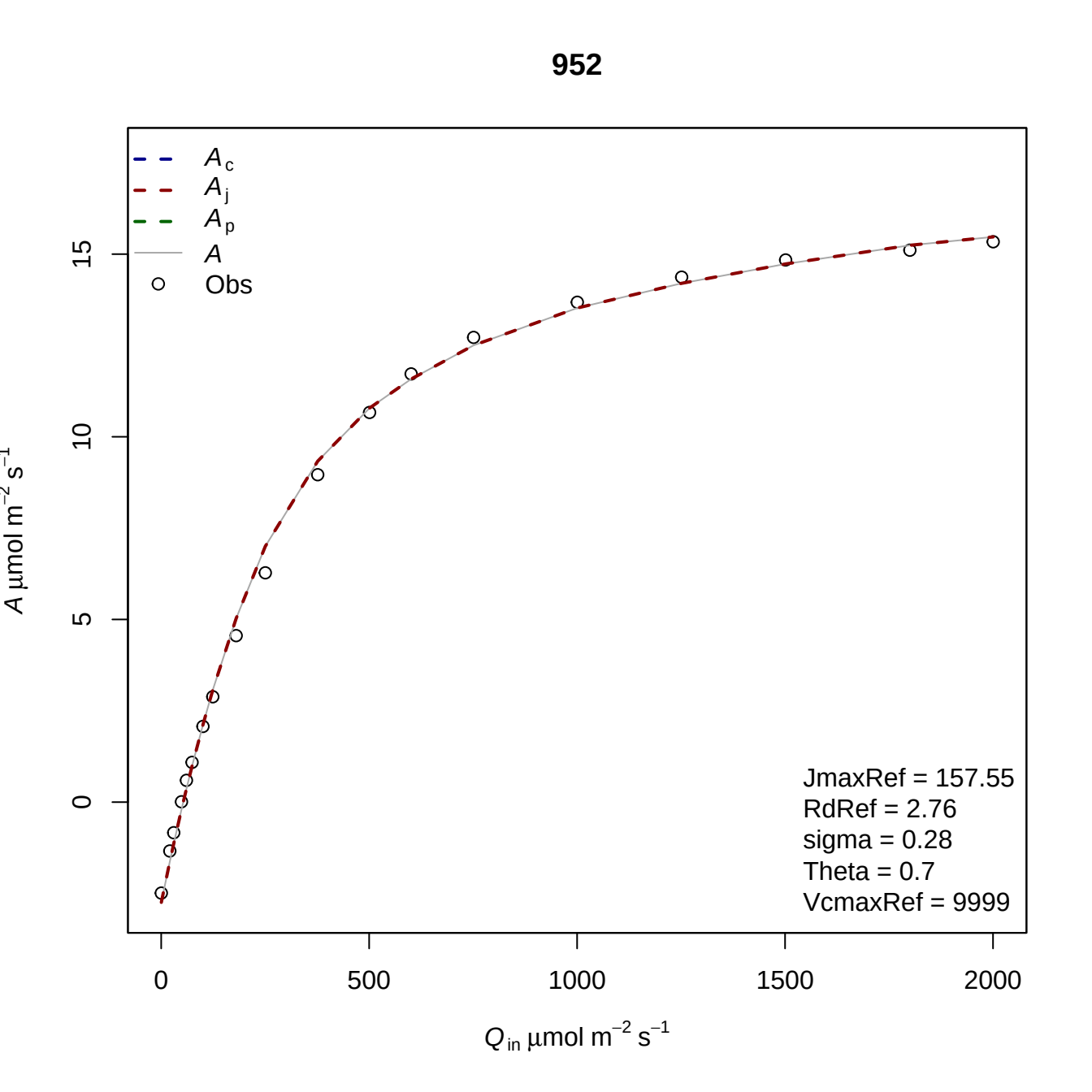

947
A_c
A_j
A_p
A
Obs
JmaxRef = 80.26
RdRef = 3.37
sigma = 0.87
Theta = 0.7
VcmaxRef = 9999
8
6
4
2
0
−2
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

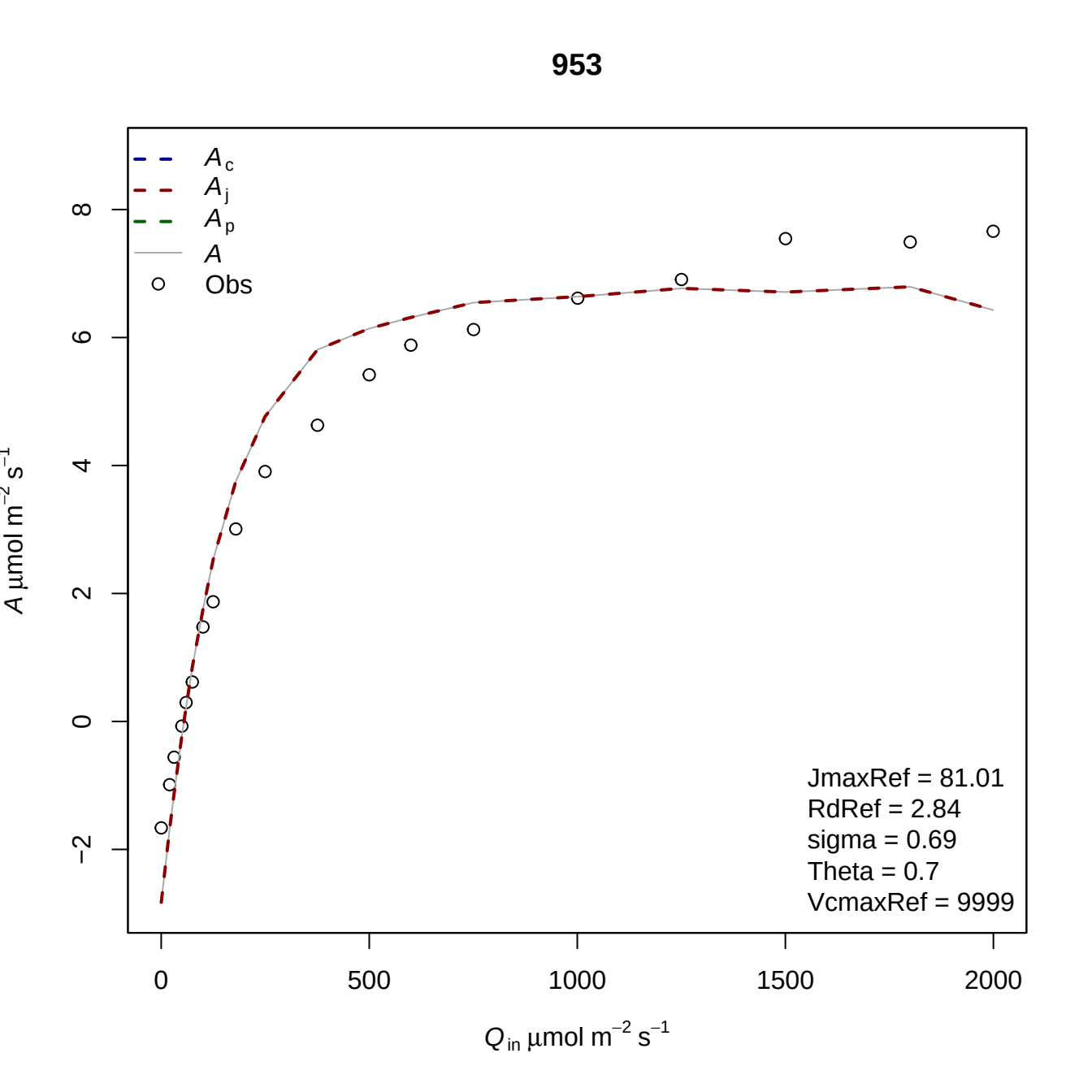

948
A_c
A_j
A_p
A
Obs
JmaxRef = 105.04
RdRef = 3.78
sigma = 0.99
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800
1000

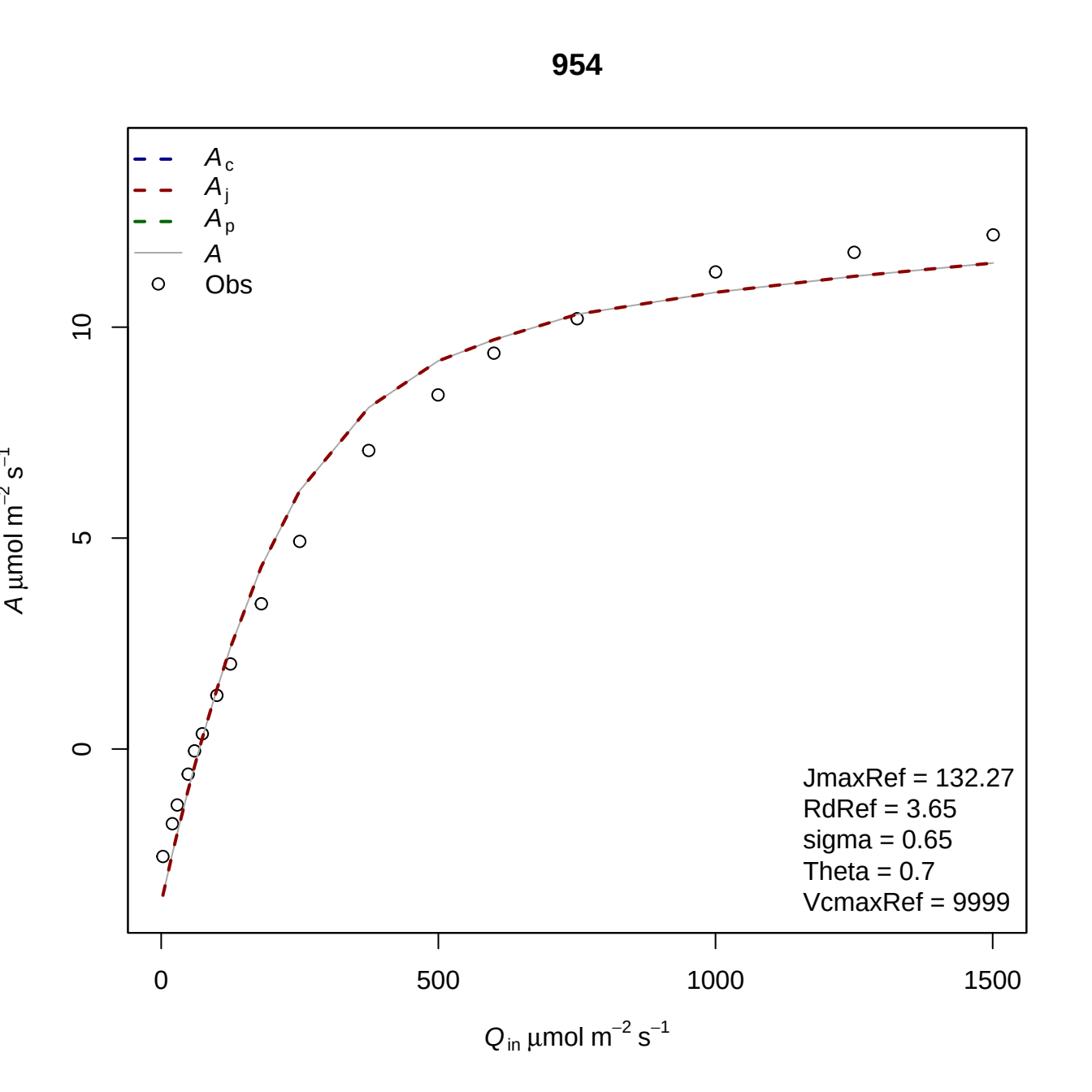

949
A_c
A_j
A_p
A
Obs
JmaxRef = 154.34
RdRef = 3.38
sigma = 1.1
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

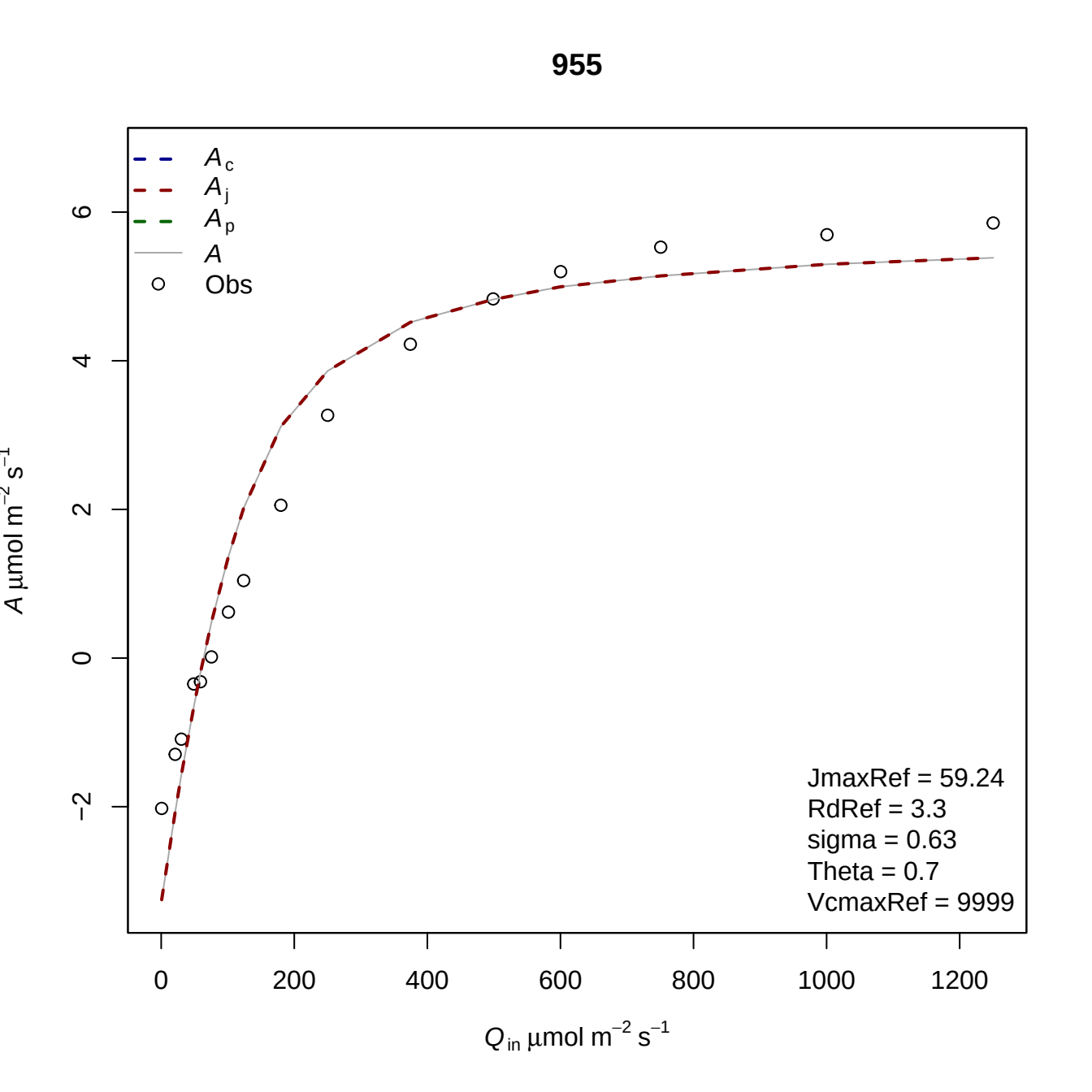

950
A_c
A_j
A_p
A
Obs
JmaxRef = 105.54
RdRef = 3.28
sigma = 0.72
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

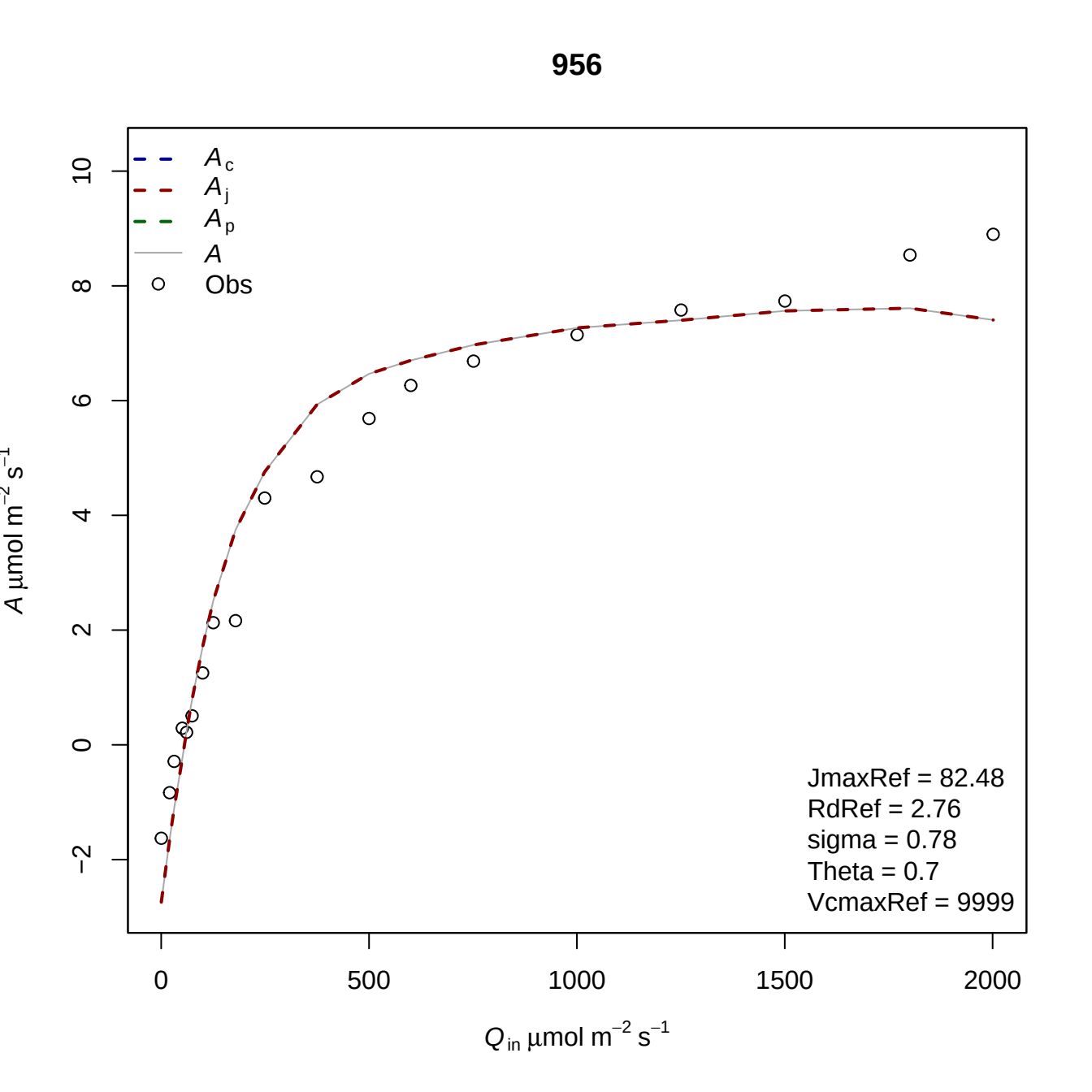

951
A_c
A_j
A_p
A
Obs
JmaxRef = 86.51
RdRef = 2.38
sigma = 0.65
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

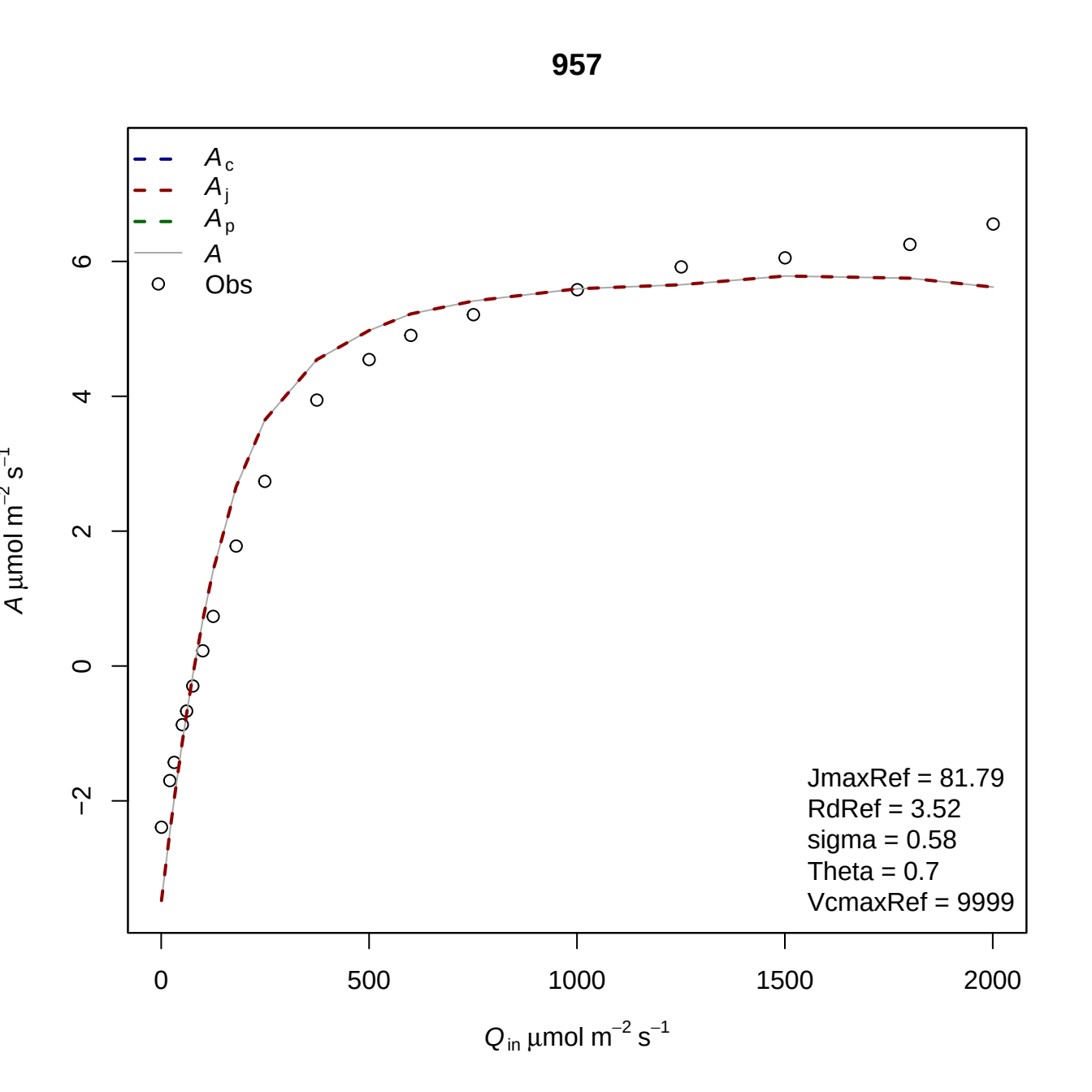

952
A_c
A_j
A_p
A
Obs
JmaxRef = 157.55
RdRef = 2.76
sigma = 0.28
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

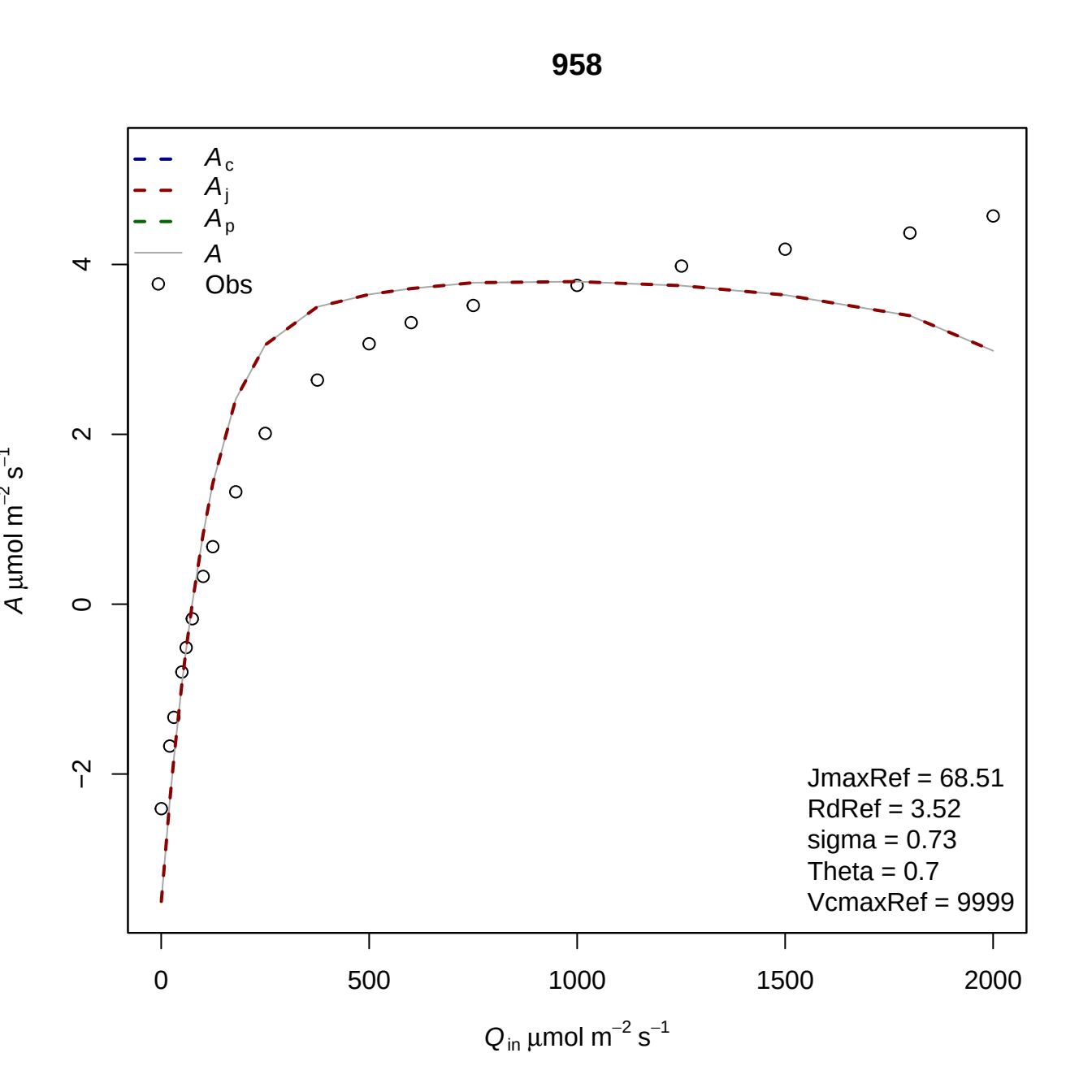

953
A_c
A_j
A_p
A
Obs
JmaxRef = 81.01
RdRef = 2.84
sigma = 0.69
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

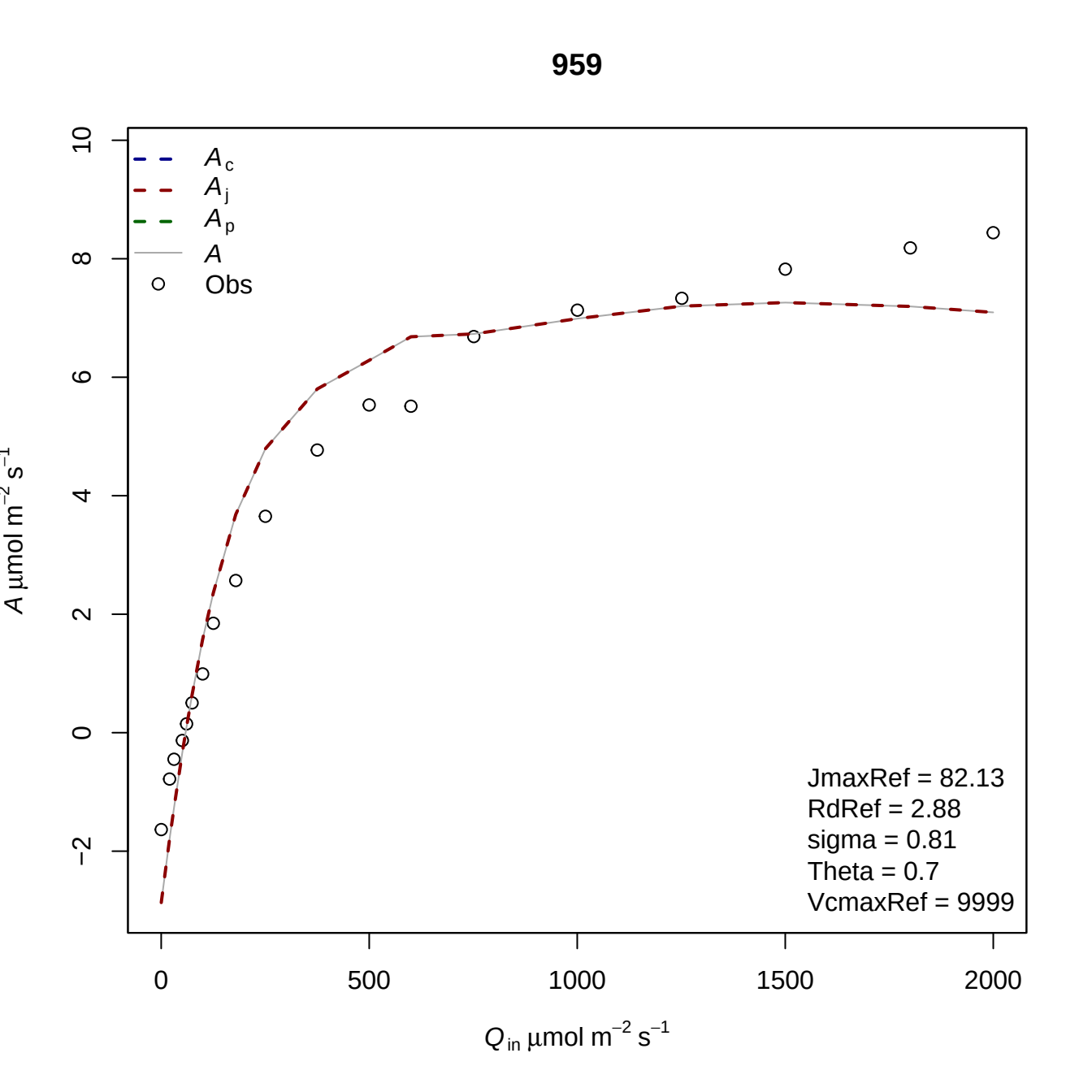

954
A_c
A_j
A_p
A
Obs
JmaxRef = 132.27
RdRef = 3.65
sigma = 0.65
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

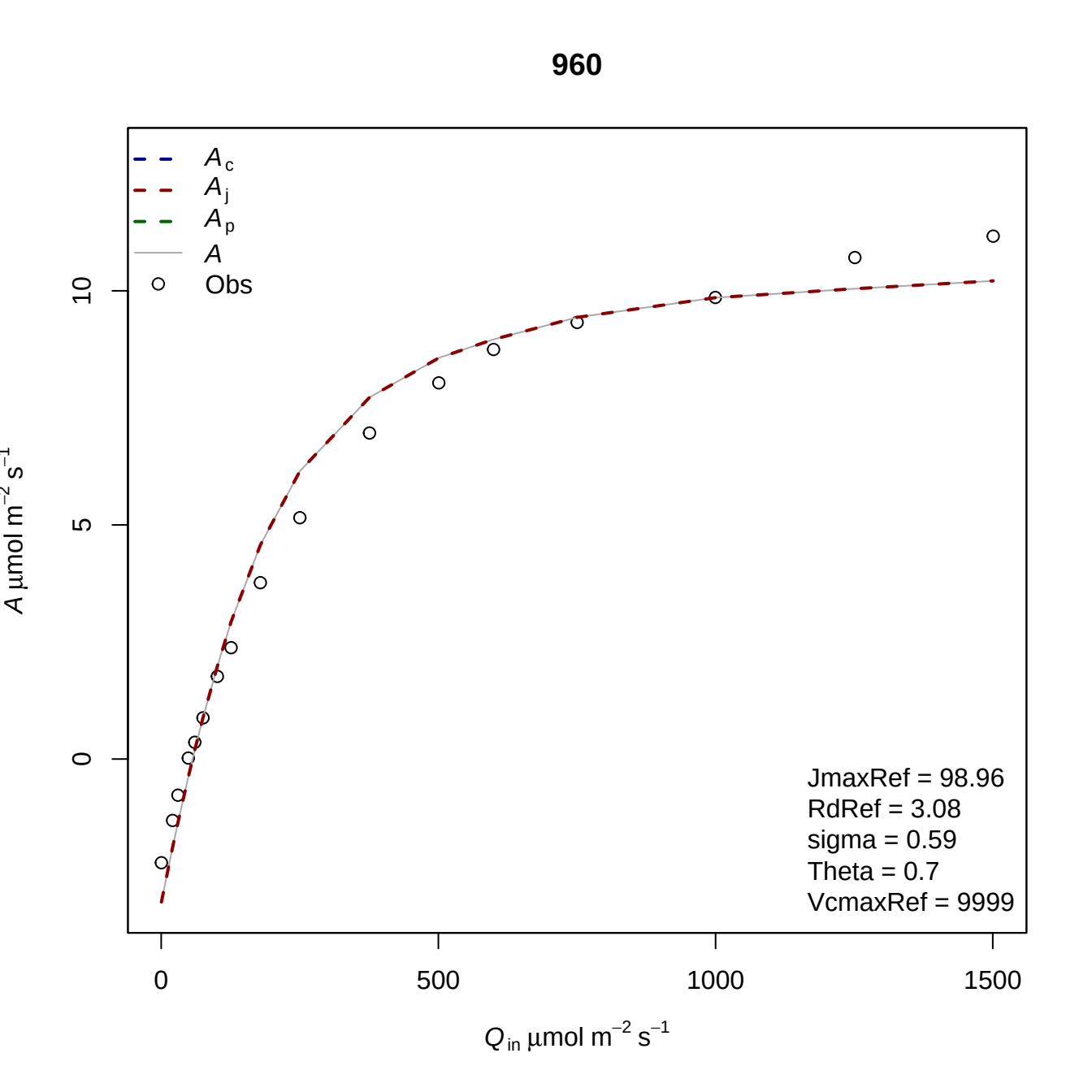

955
A_c
A_j
A_p
A
Obs
JmaxRef = 59.24
RdRef = 3.3
sigma = 0.63
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000
1200

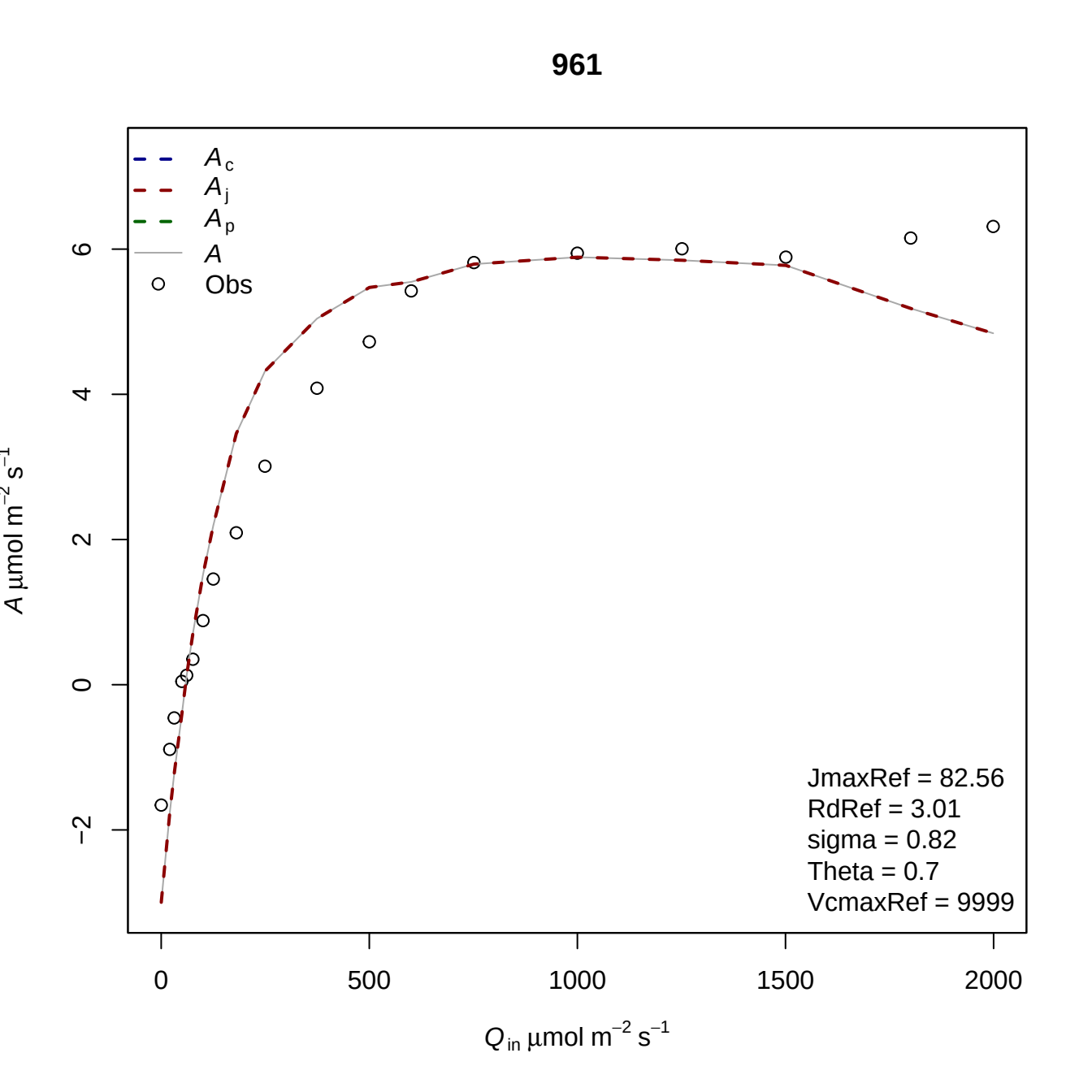

956
A_c
A_j
A_p
A
Obs
JmaxRef = 82.48
RdRef = 2.76
sigma = 0.78
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

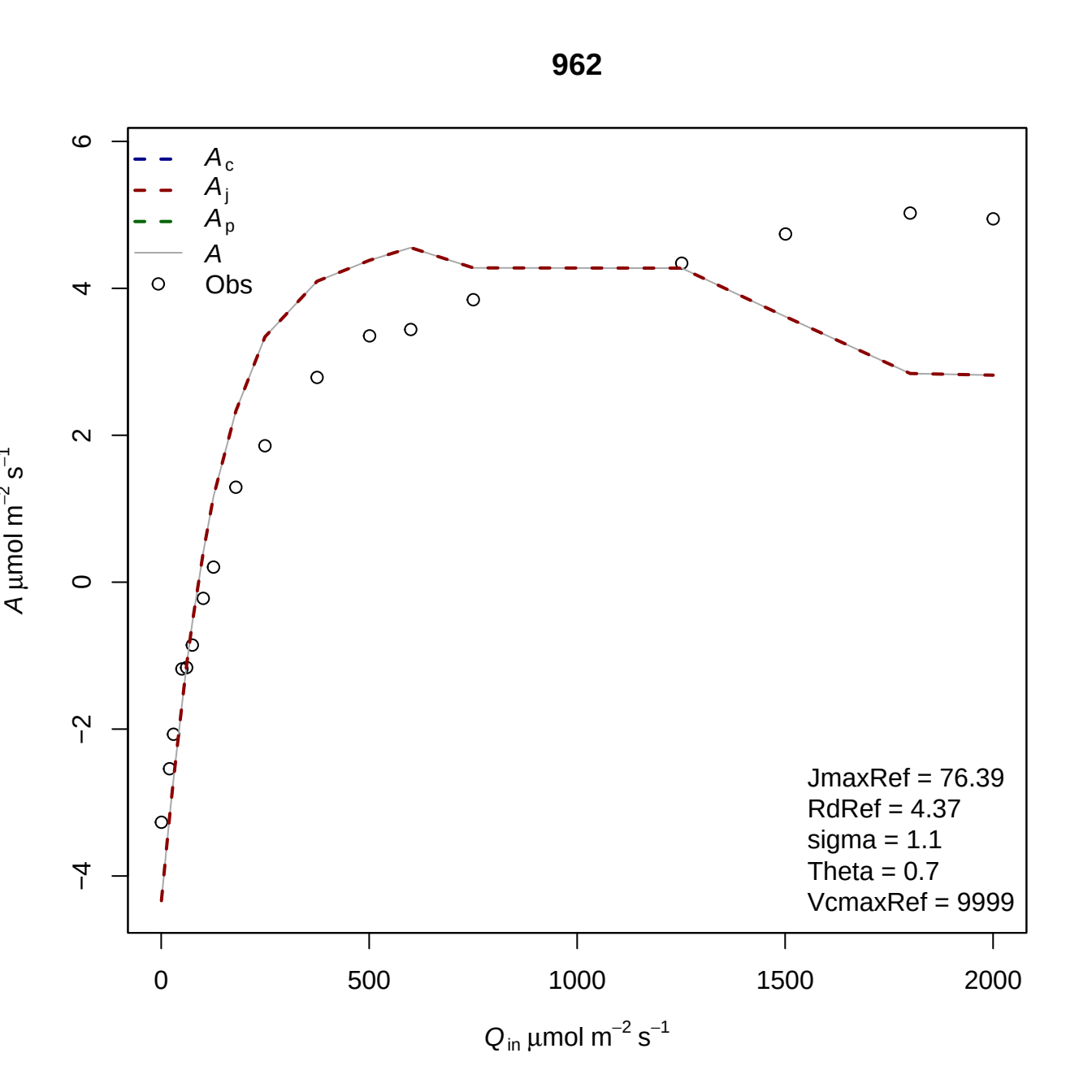

957
A_c
A_j
A_p
A
Obs
JmaxRef = 81.79
RdRef = 3.52
sigma = 0.58
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

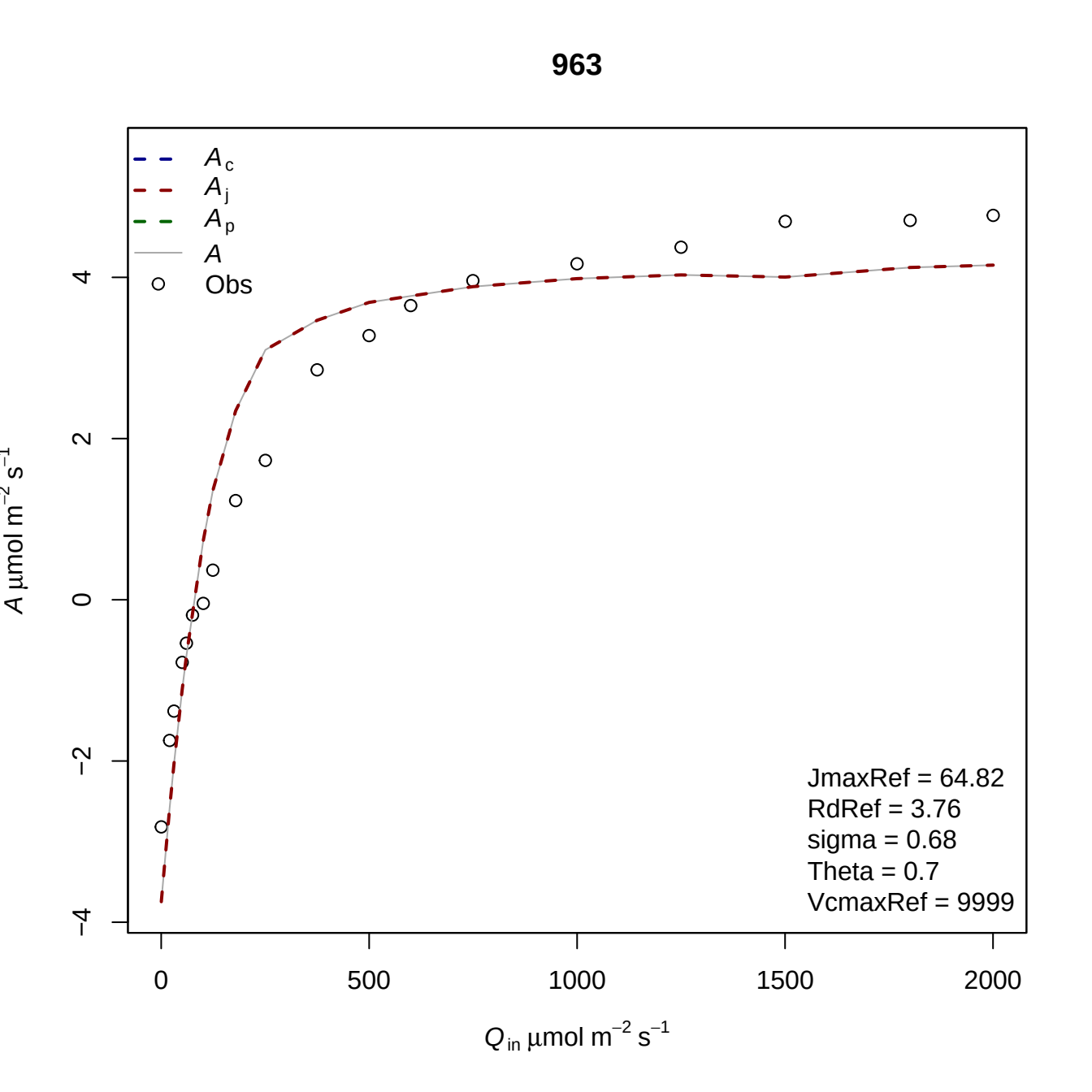

958
A_c
A_j
A_p
A
Obs
JmaxRef = 68.51
RdRef = 3.52
sigma = 0.73
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000

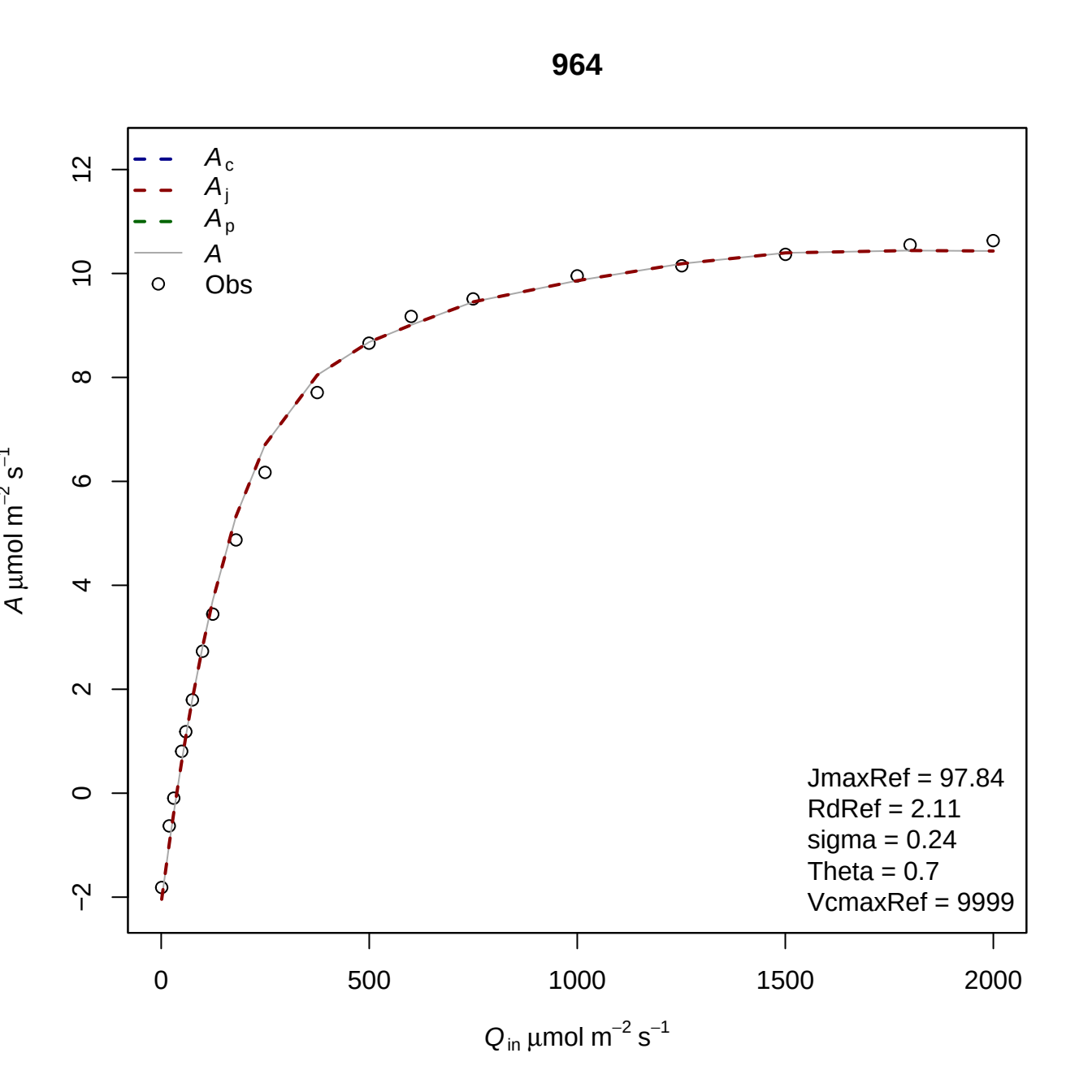

959
A_c
A_j
A_p
A
Obs
JmaxRef = 82.13
RdRef = 2.88
sigma = 0.81
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

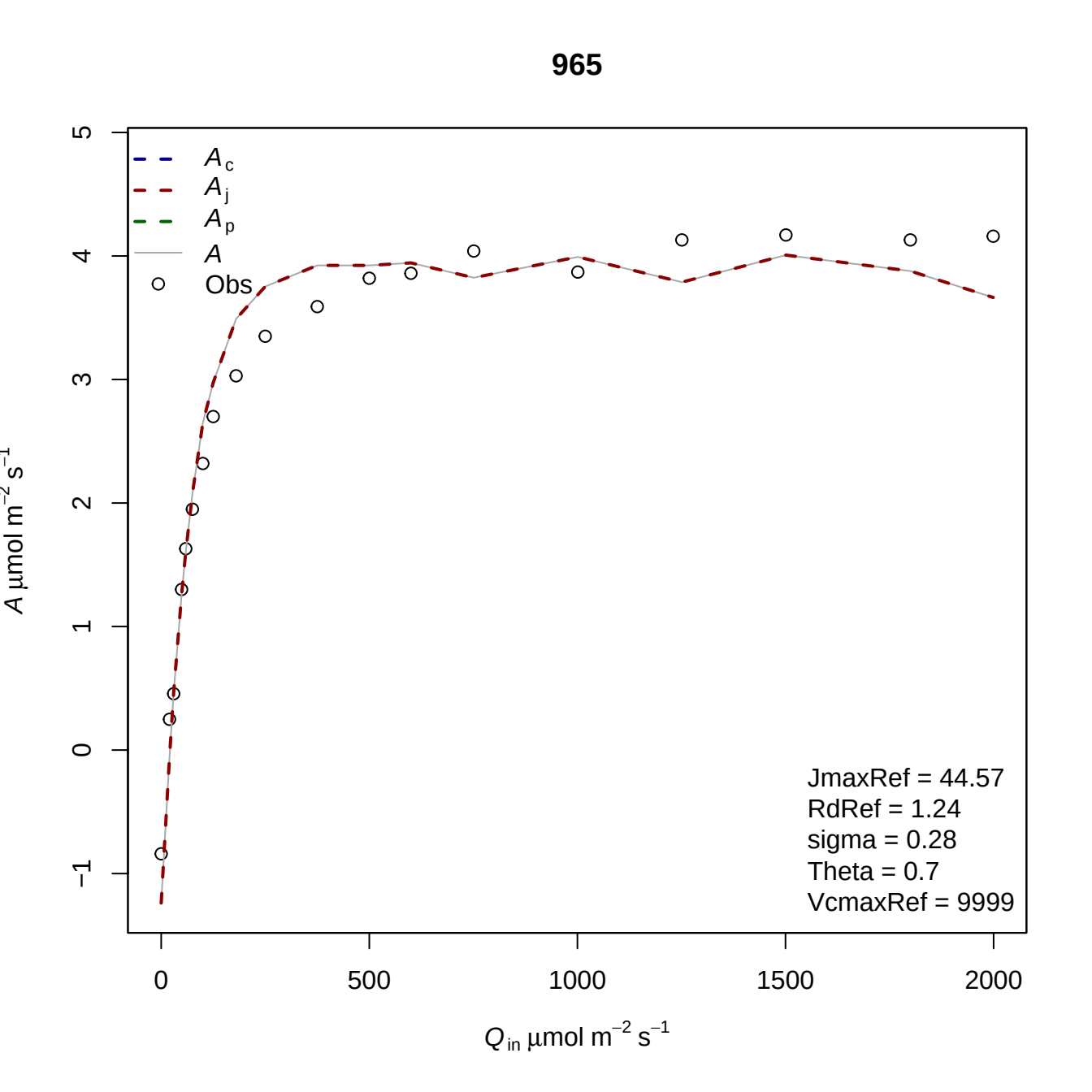

960
A_c
A_j
A_p
A
Obs
JmaxRef = 98.96
RdRef = 3.08
sigma = 0.59
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

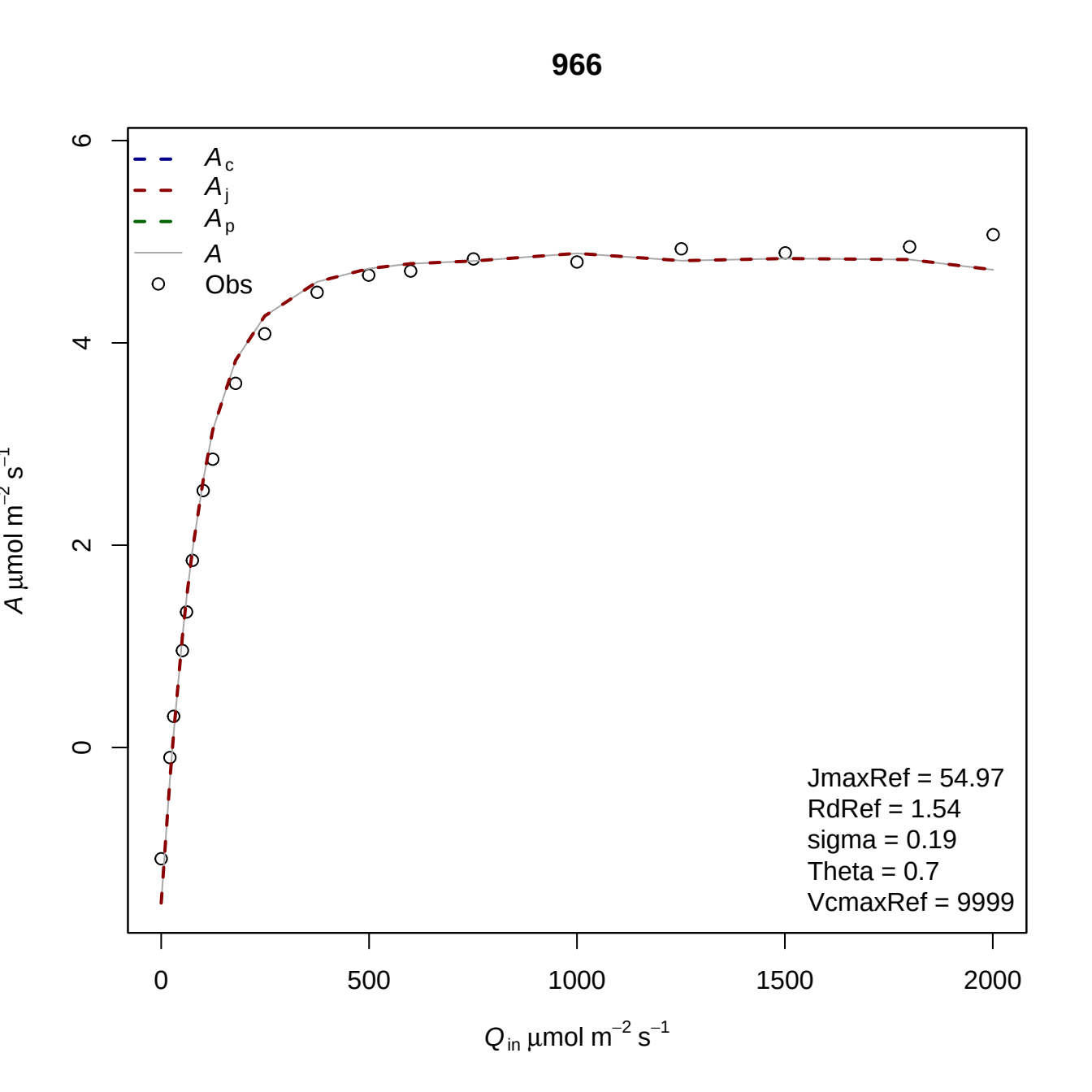

961
A_c
A_j
A_p
A
Obs
JmaxRef = 82.56
RdRef = 3.01
sigma = 0.82
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

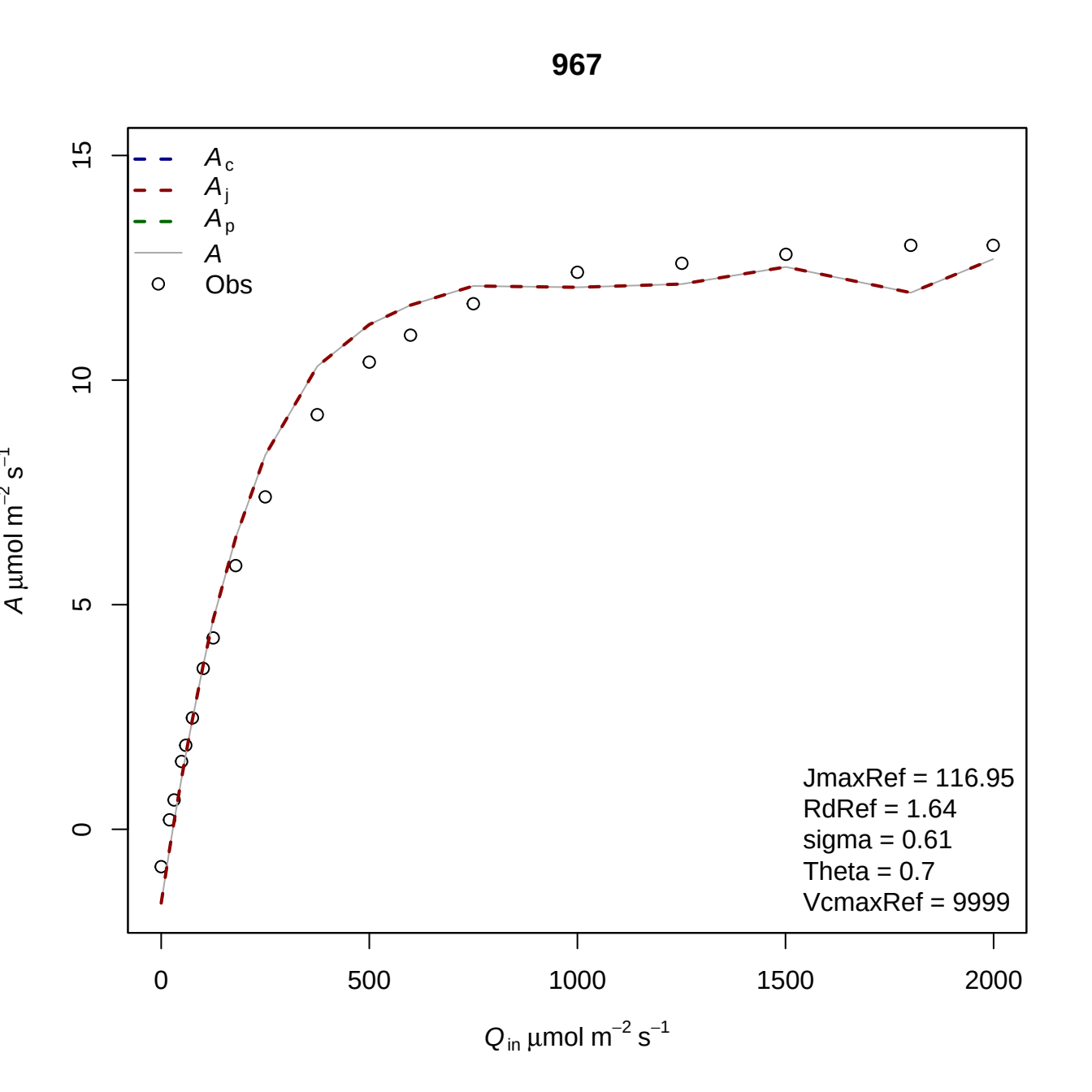

962
A_c
A_j
A_p
A
Obs
JmaxRef = 76.39
RdRef = 4.37
sigma = 1.1
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500
2000

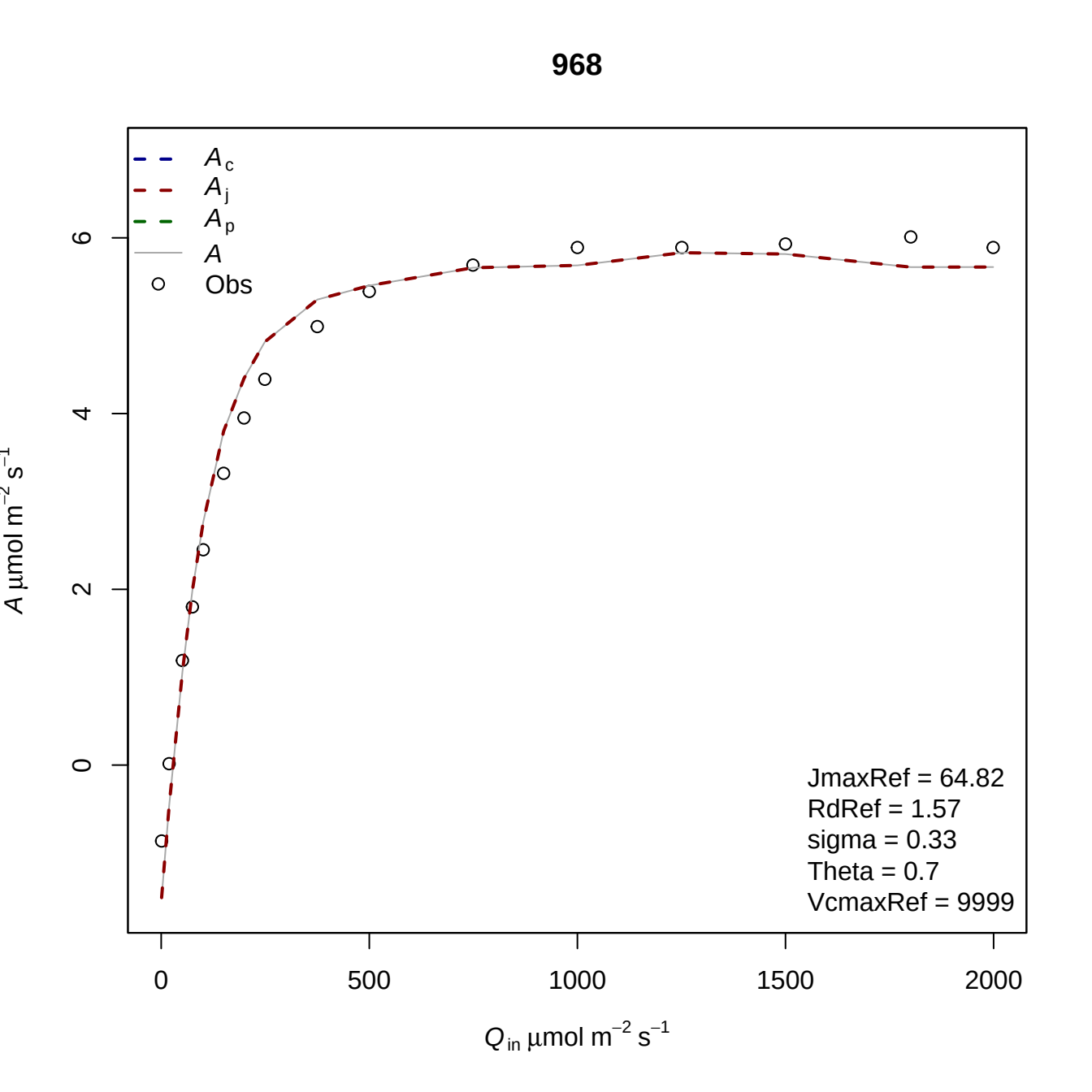

963
A_c
A_j
A_p
A
Obs
JmaxRef = 64.82
RdRef = 3.76
sigma = 0.68
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
−4
0
500
1000
1500
2000

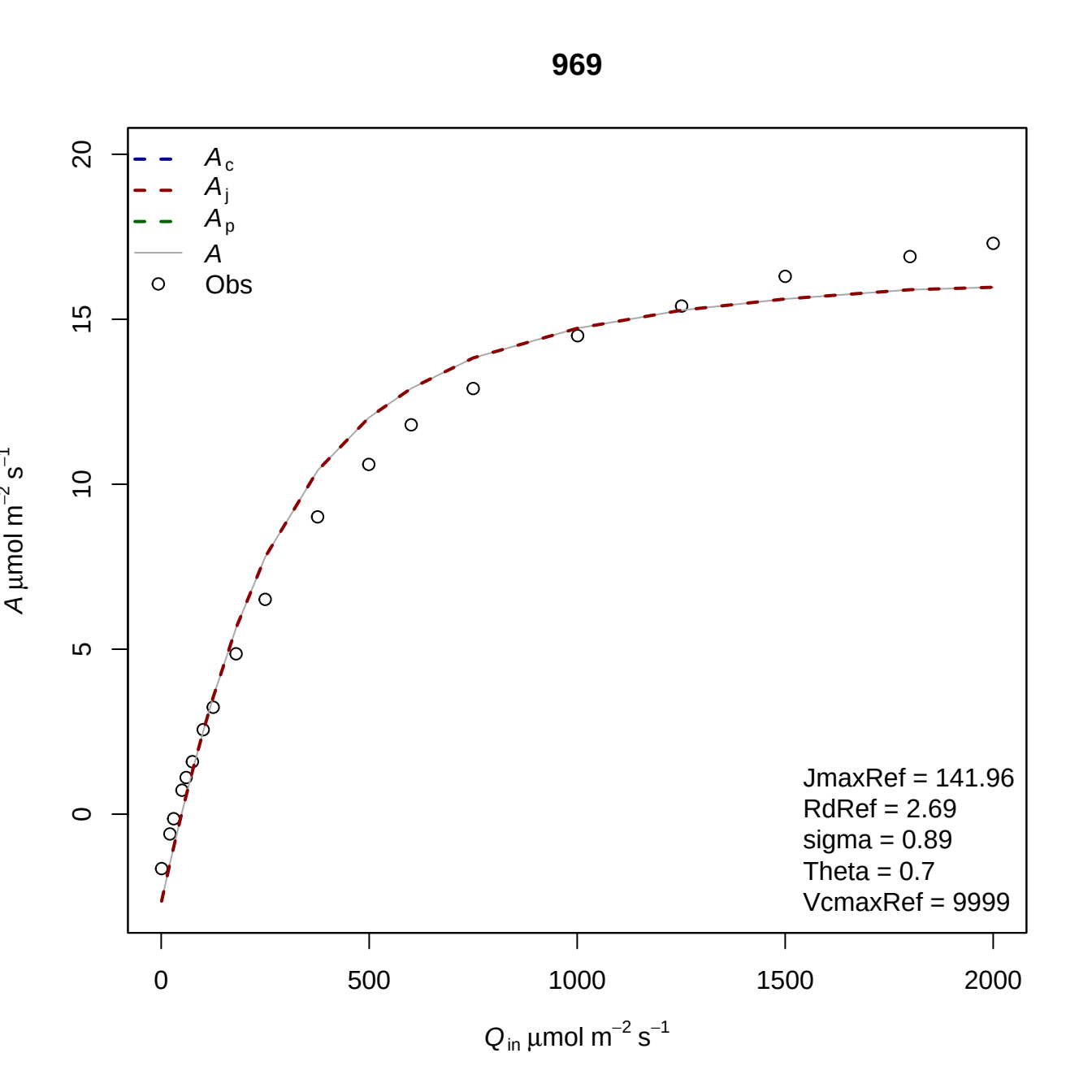

964
A_c
A_j
A_p
A
Obs
JmaxRef = 97.84
RdRef = 2.11
sigma = 0.24
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

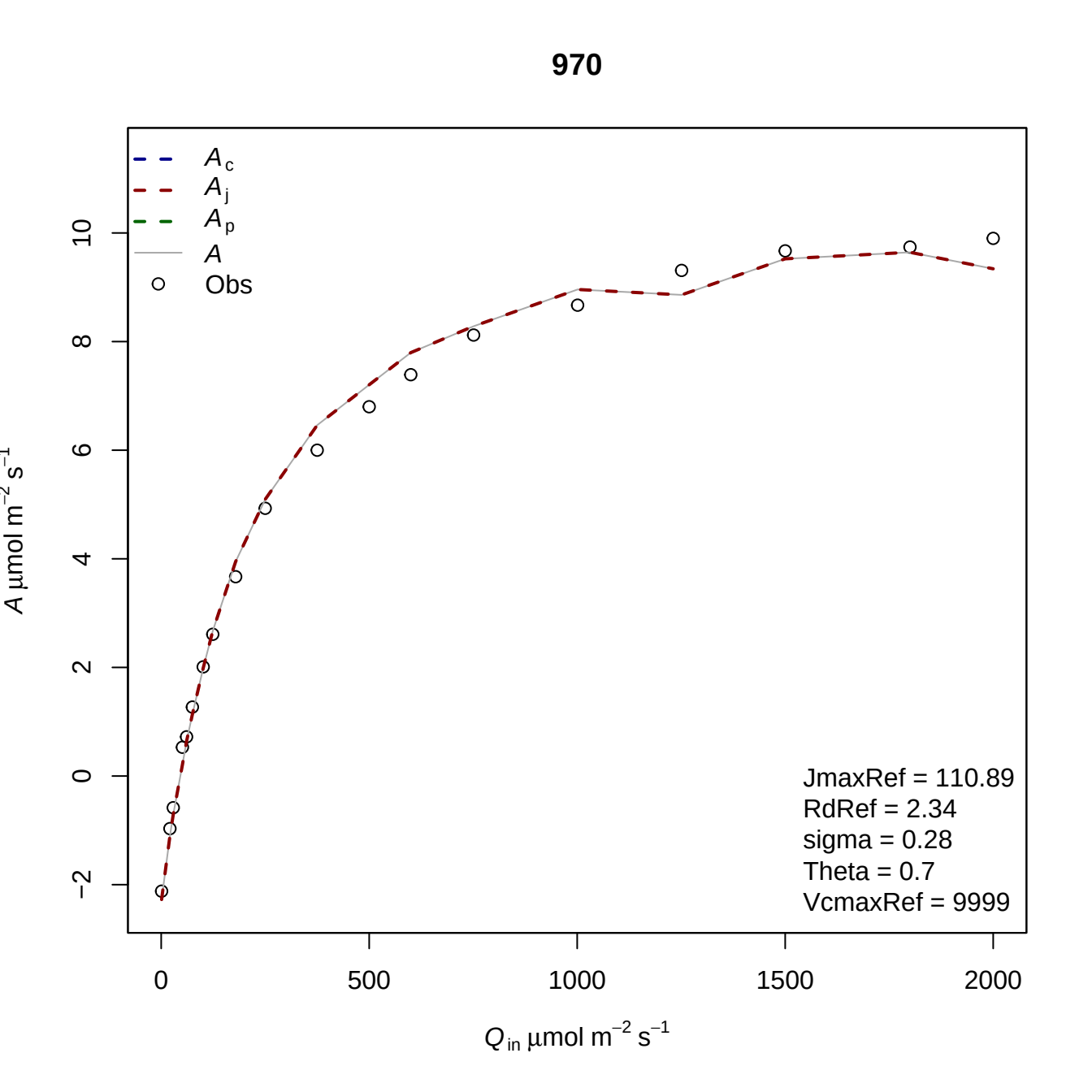

965
A_c
A_j
A_p
A
Obs
JmaxRef = 44.57
RdRef = 1.24
sigma = 0.28
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
5
4
3
2
1
0
−1
0
500
1000
1500
2000

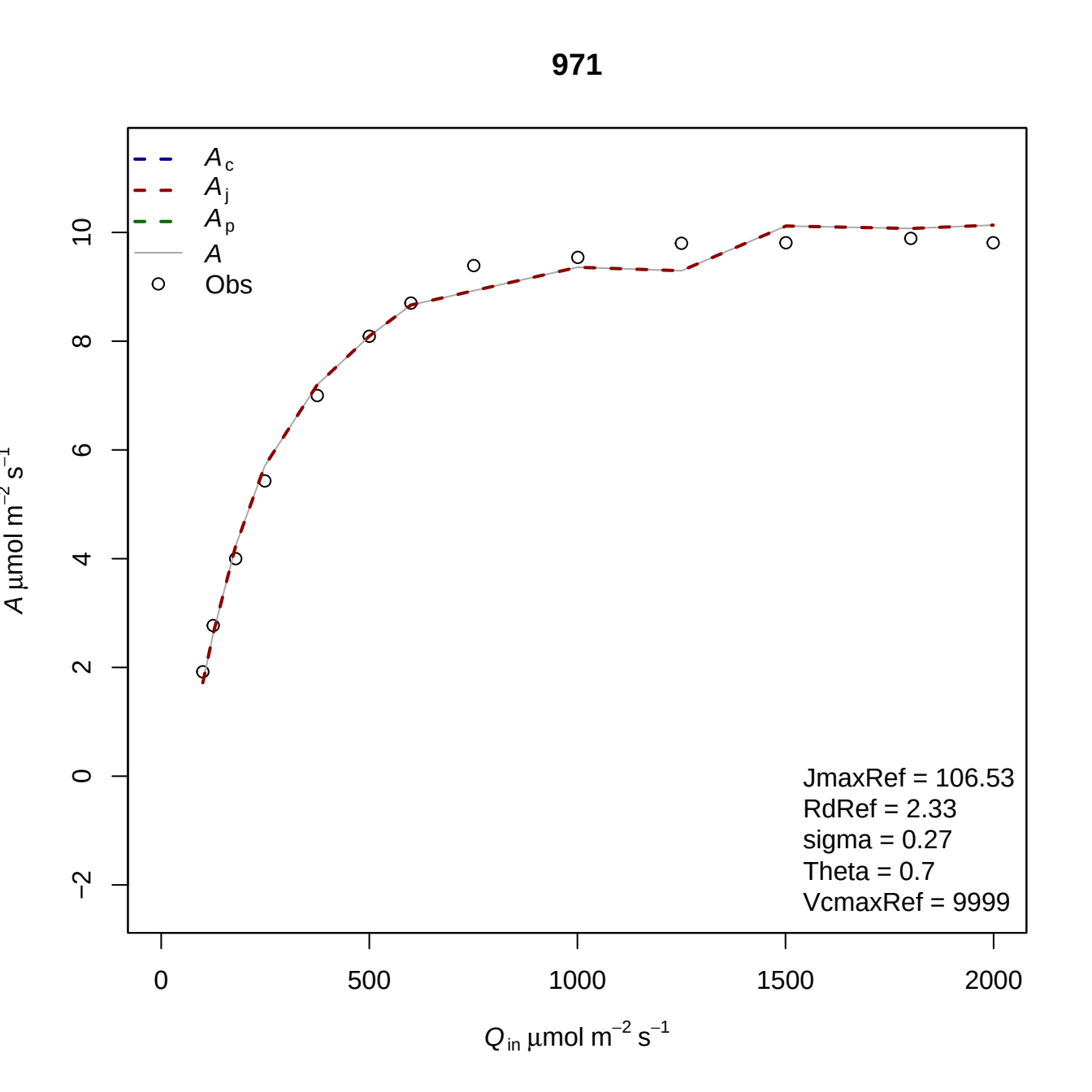

966
A_c
A_j
A_p
A
Obs
JmaxRef = 54.97
RdRef = 1.54
sigma = 0.19
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

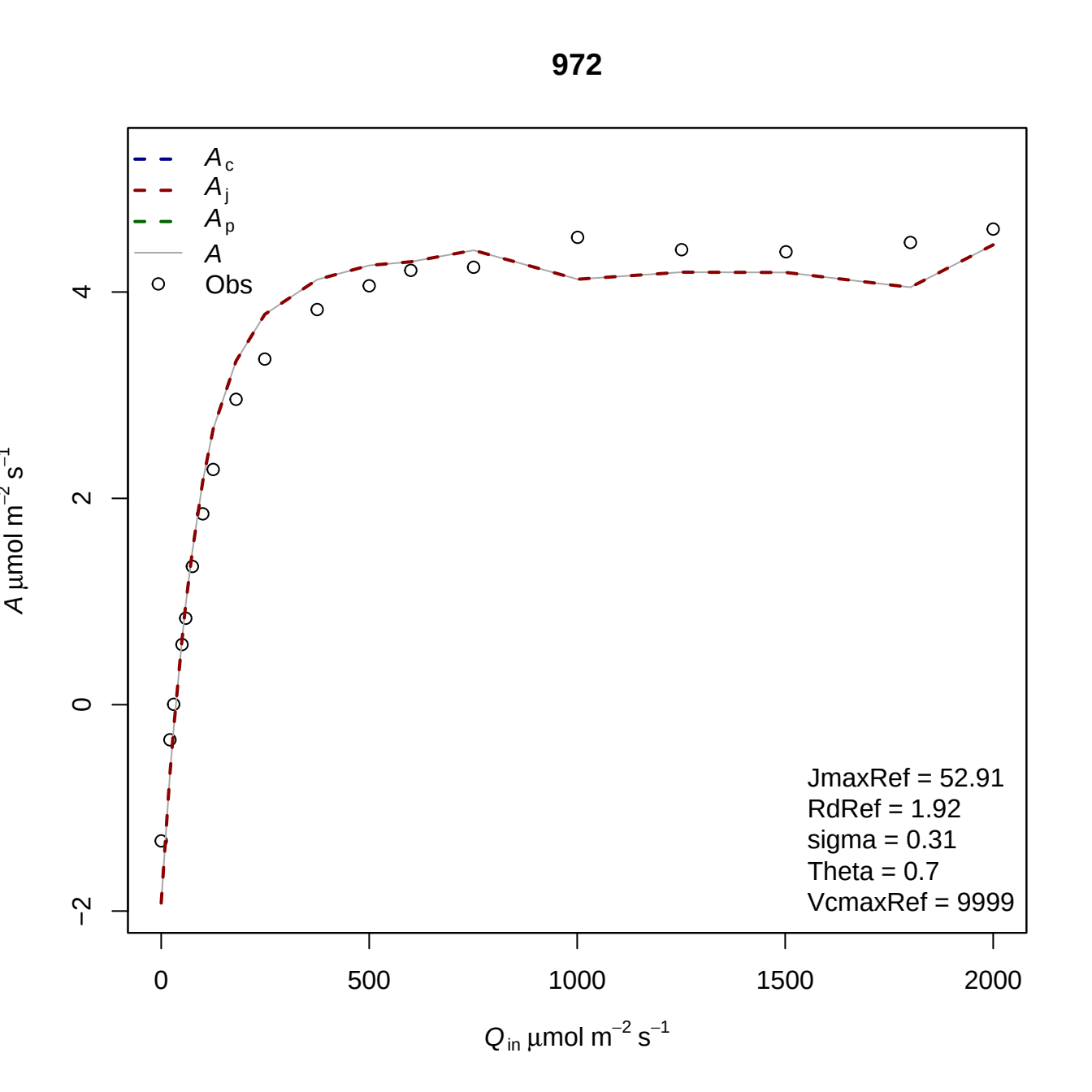

967
A_c
A_j
A_p
A
Obs
JmaxRef = 116.95
RdRef = 1.64
sigma = 0.61
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

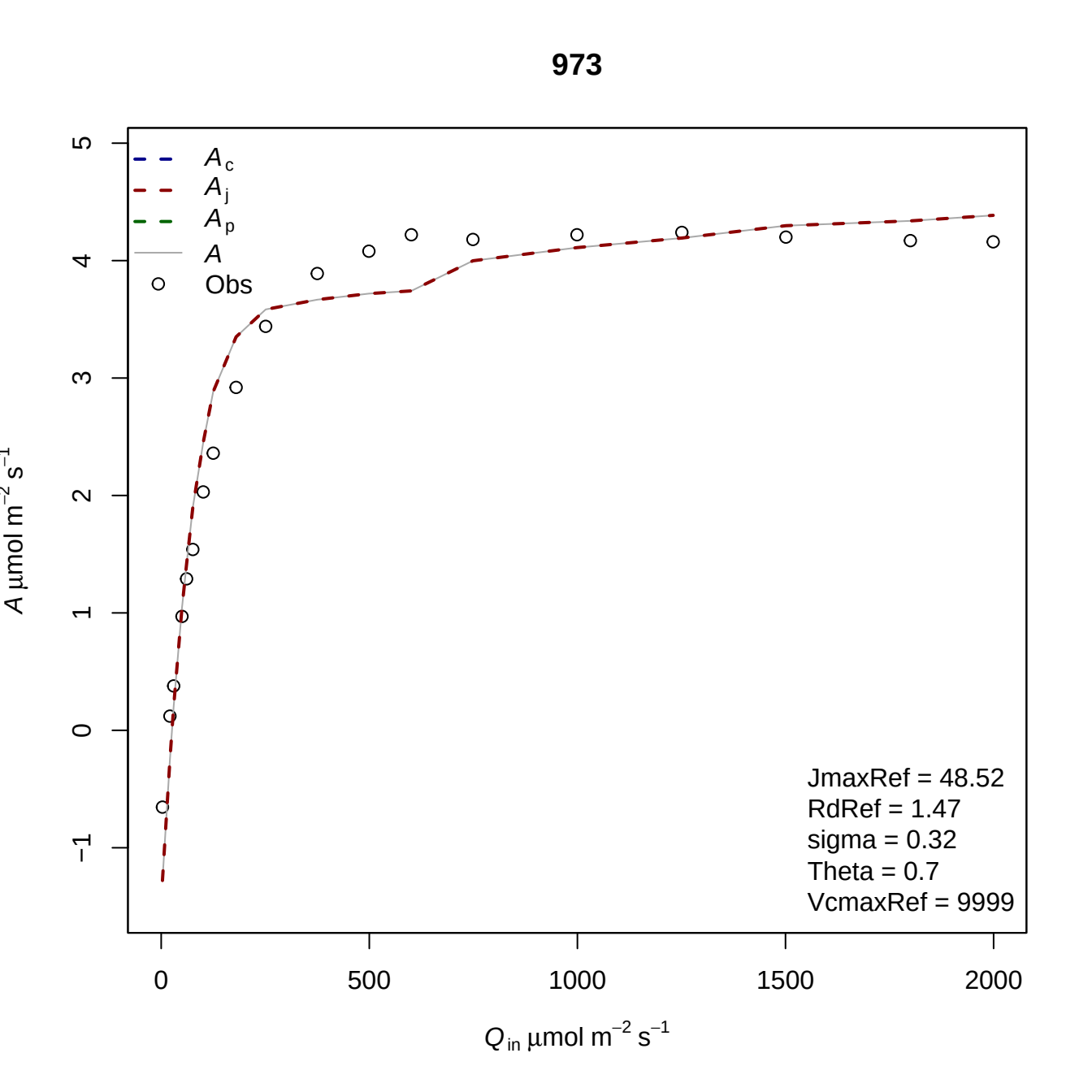

968
A_c
A_j
A_p
A
Obs
JmaxRef = 64.82
RdRef = 1.57
sigma = 0.33
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

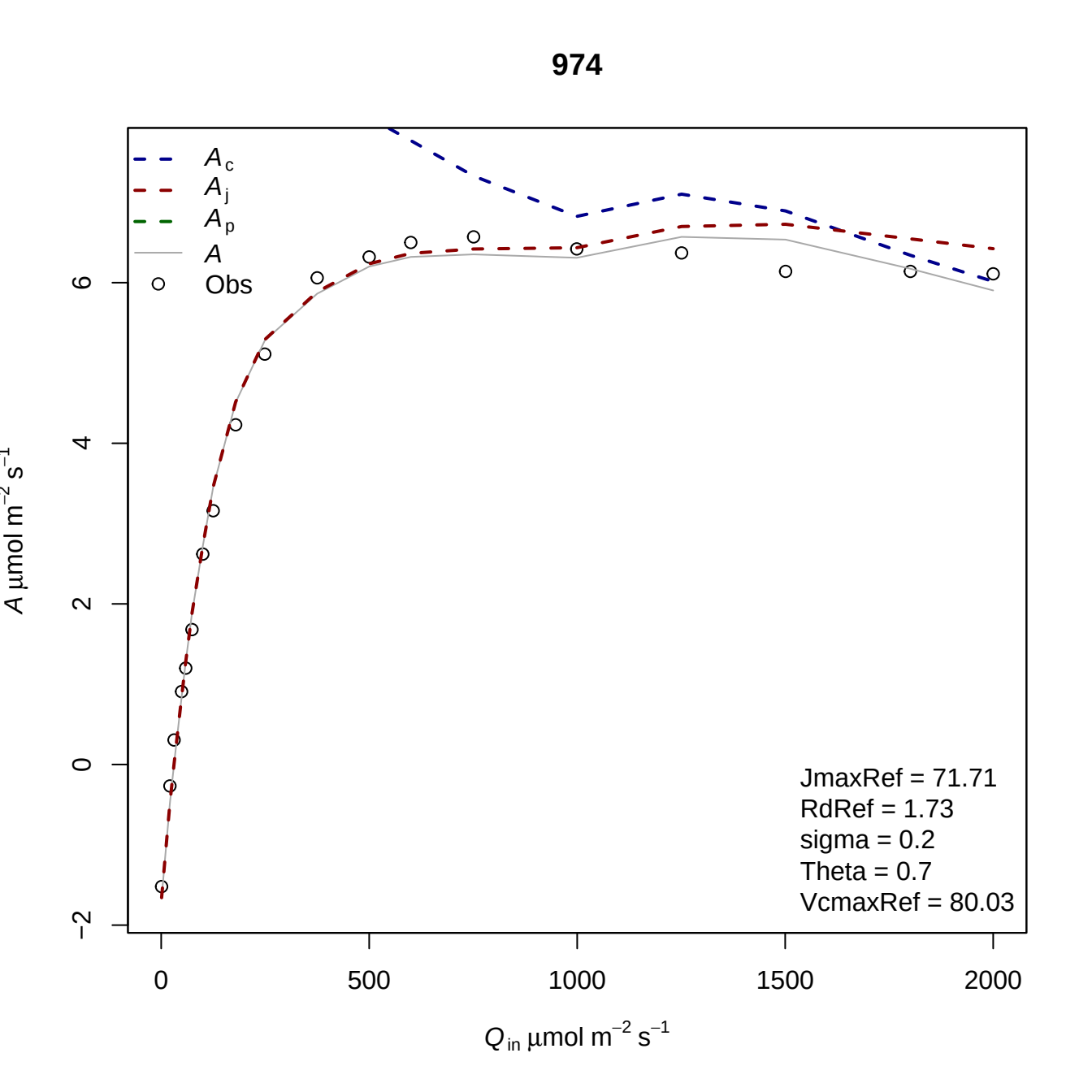

969
A_c
A_j
A_p
A
Obs
JmaxRef = 141.96
RdRef = 2.69
sigma = 0.89
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

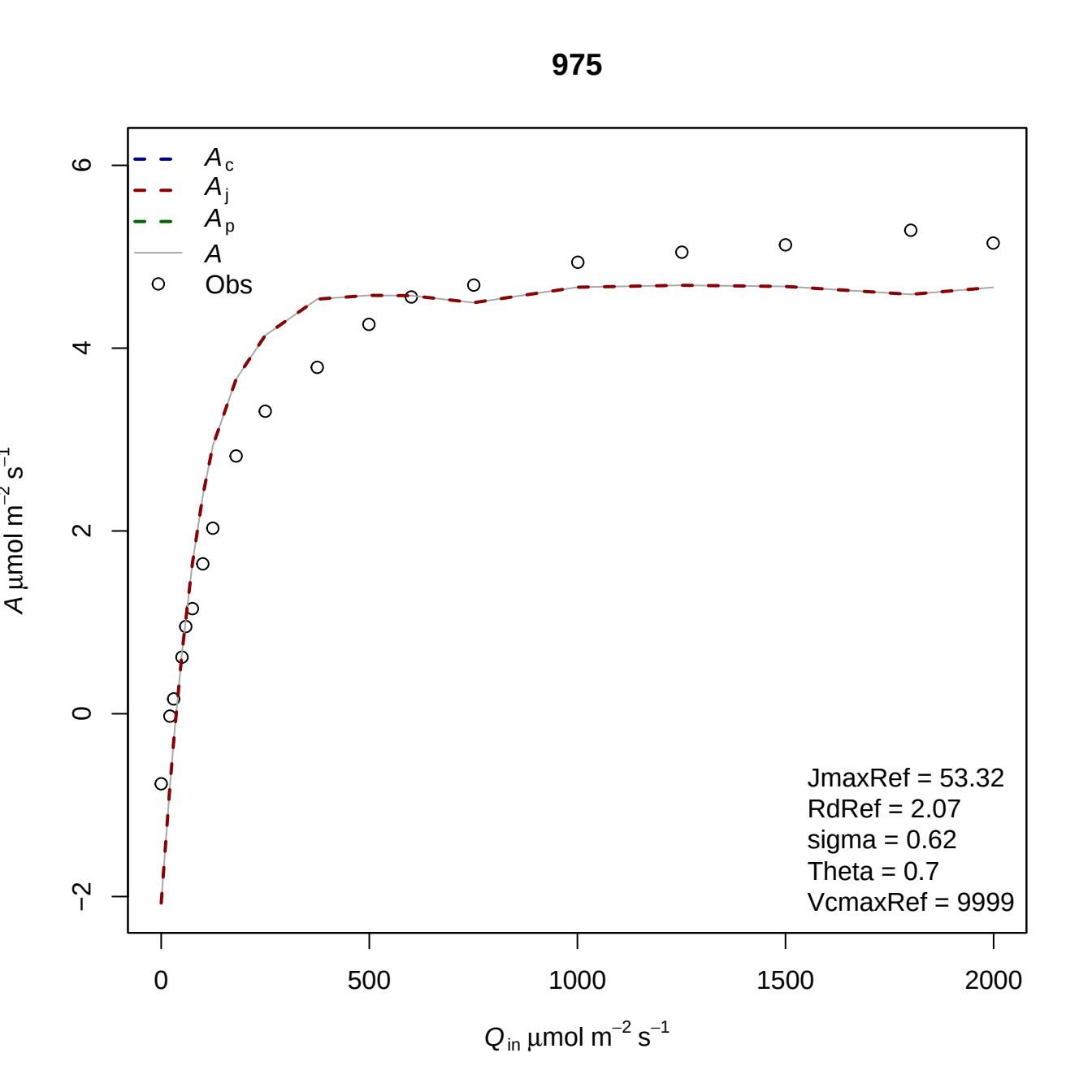

970
A_c
A_j
A_p
A
Obs
JmaxRef = 110.89
RdRef = 2.34
sigma = 0.28
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

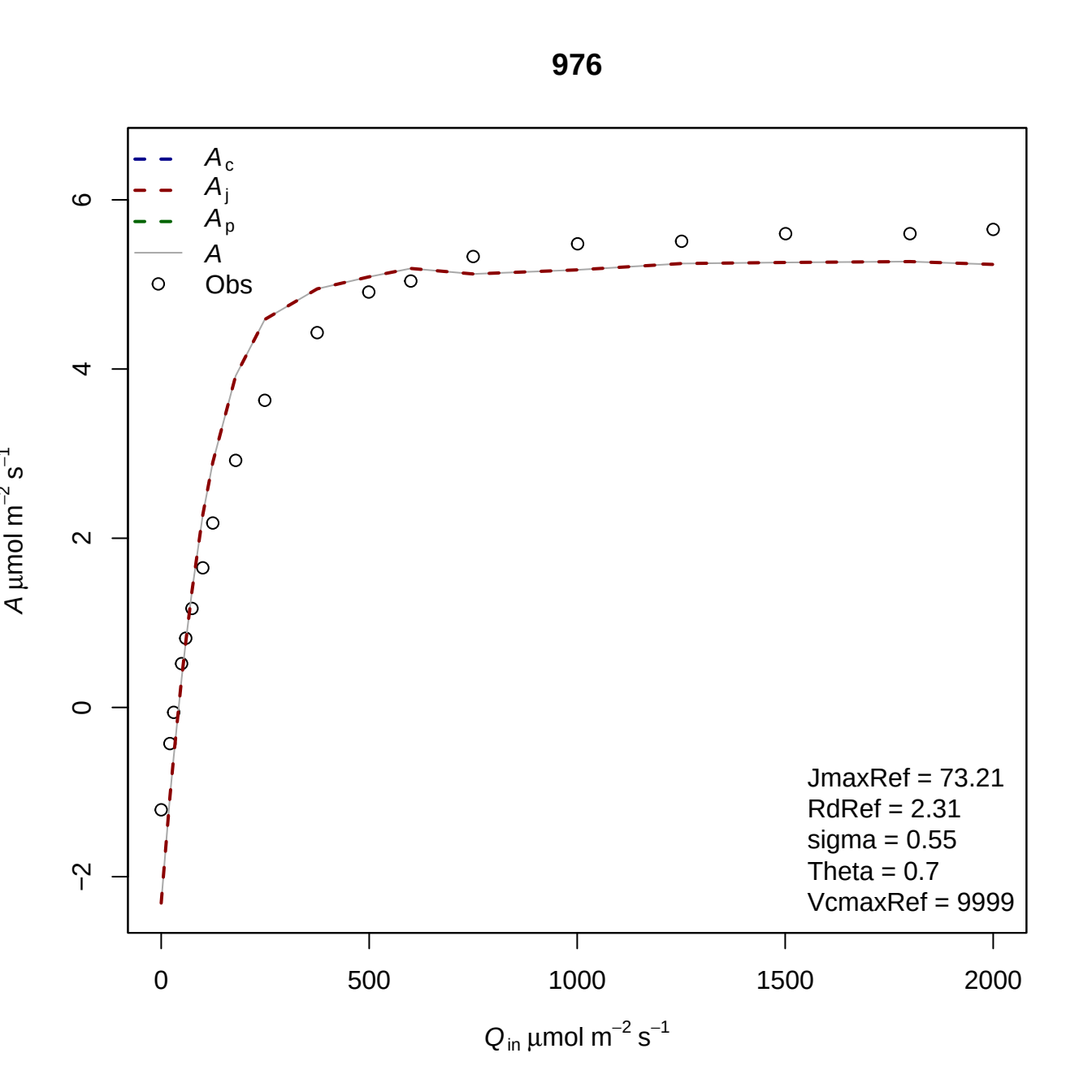

971
A_c
A_j
A_p
A
Obs
JmaxRef = 106.53
RdRef = 2.33
sigma = 0.27
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

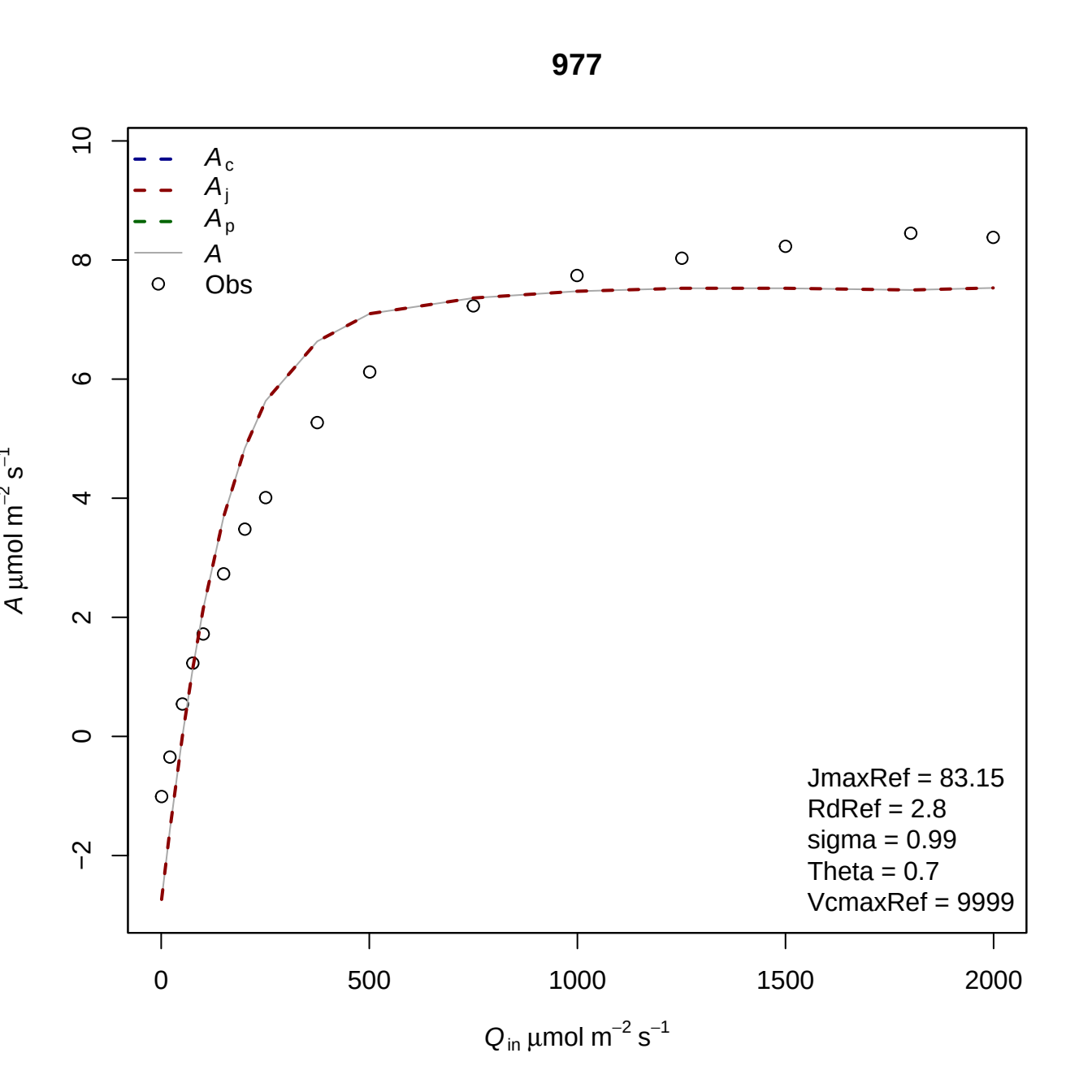

972
A_c
A_j
A_p
A
Obs
JmaxRef = 52.91
RdRef = 1.92
sigma = 0.31
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

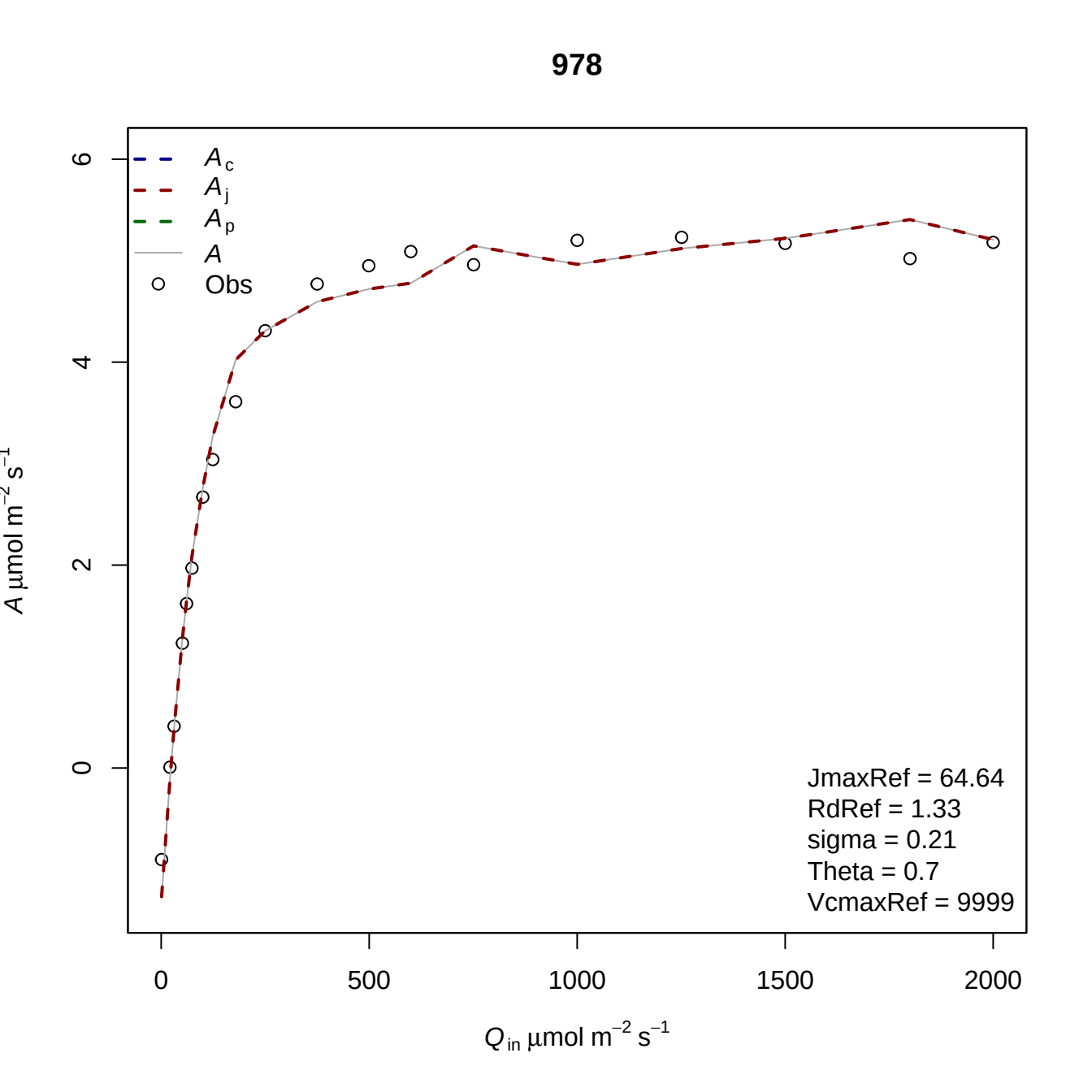

973
A_c
A_j
A_p
A
Obs
JmaxRef = 48.52
RdRef = 1.47
sigma = 0.32
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500
2000

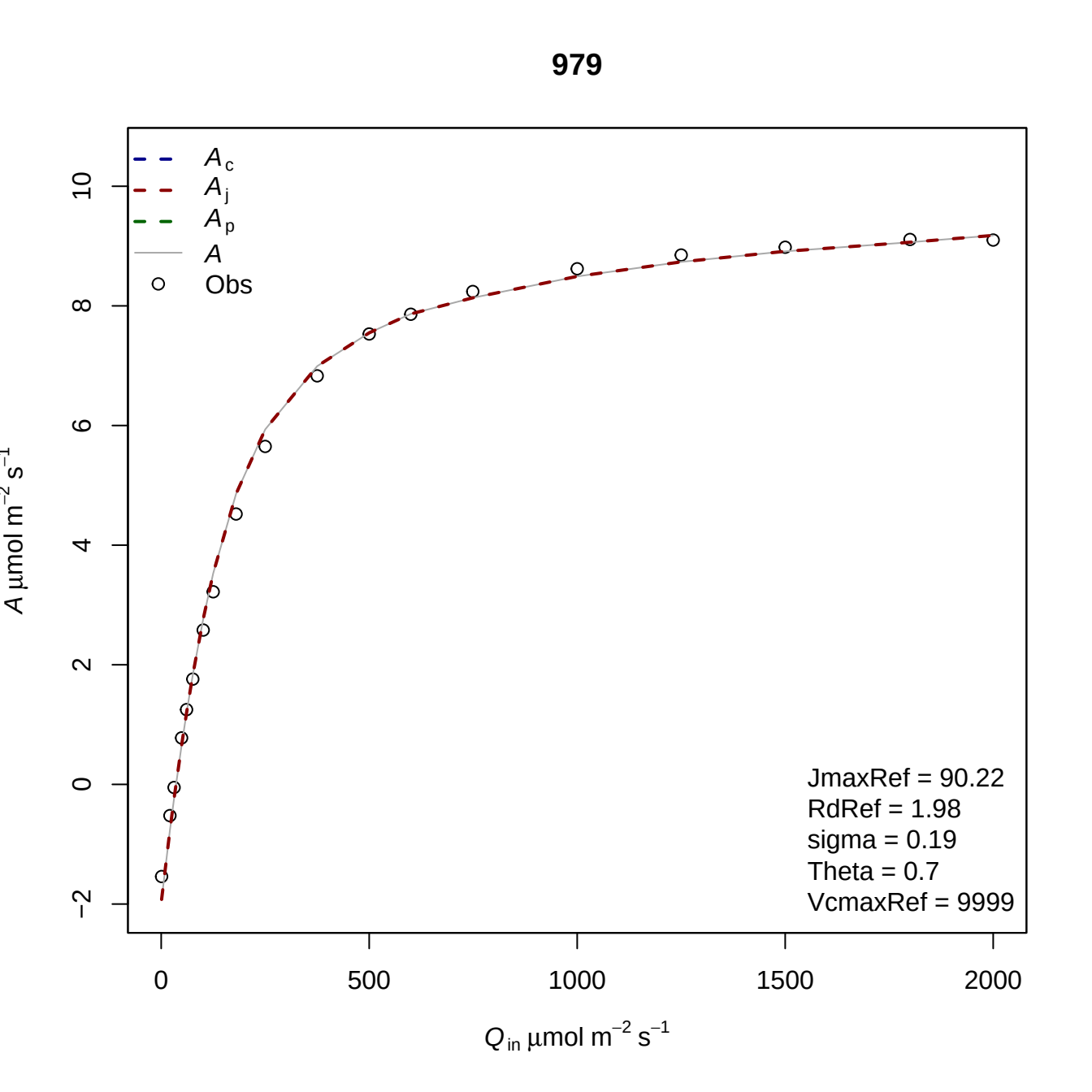

974
A_c
A_j
A_p
A
Obs
JmaxRef = 71.71
RdRef = 1.73
sigma = 0.2
Theta = 0.7
VcmaxRef = 80.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

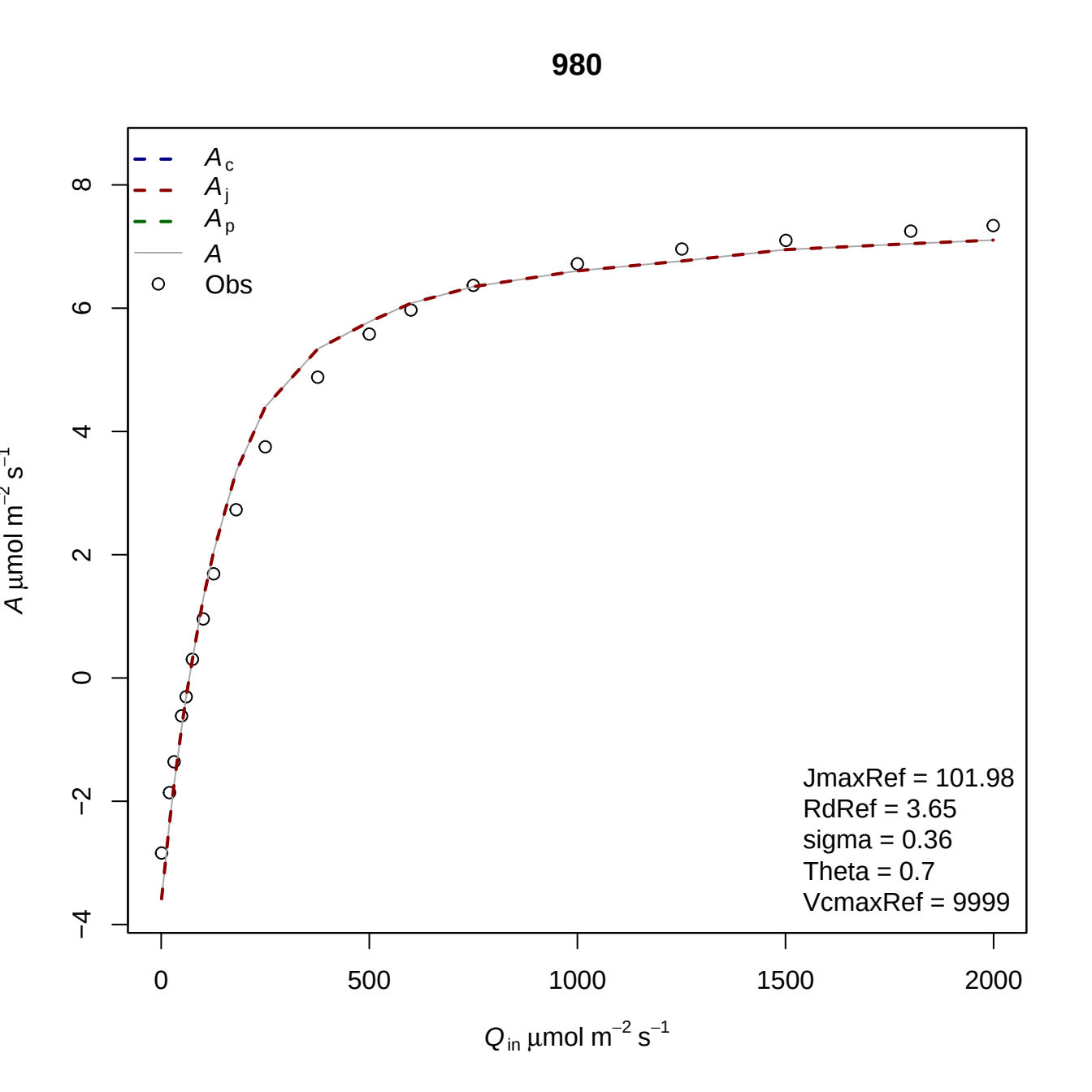

975
A_c
A_j
A_p
A
Obs
JmaxRef = 53.32
RdRef = 2.07
sigma = 0.62
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

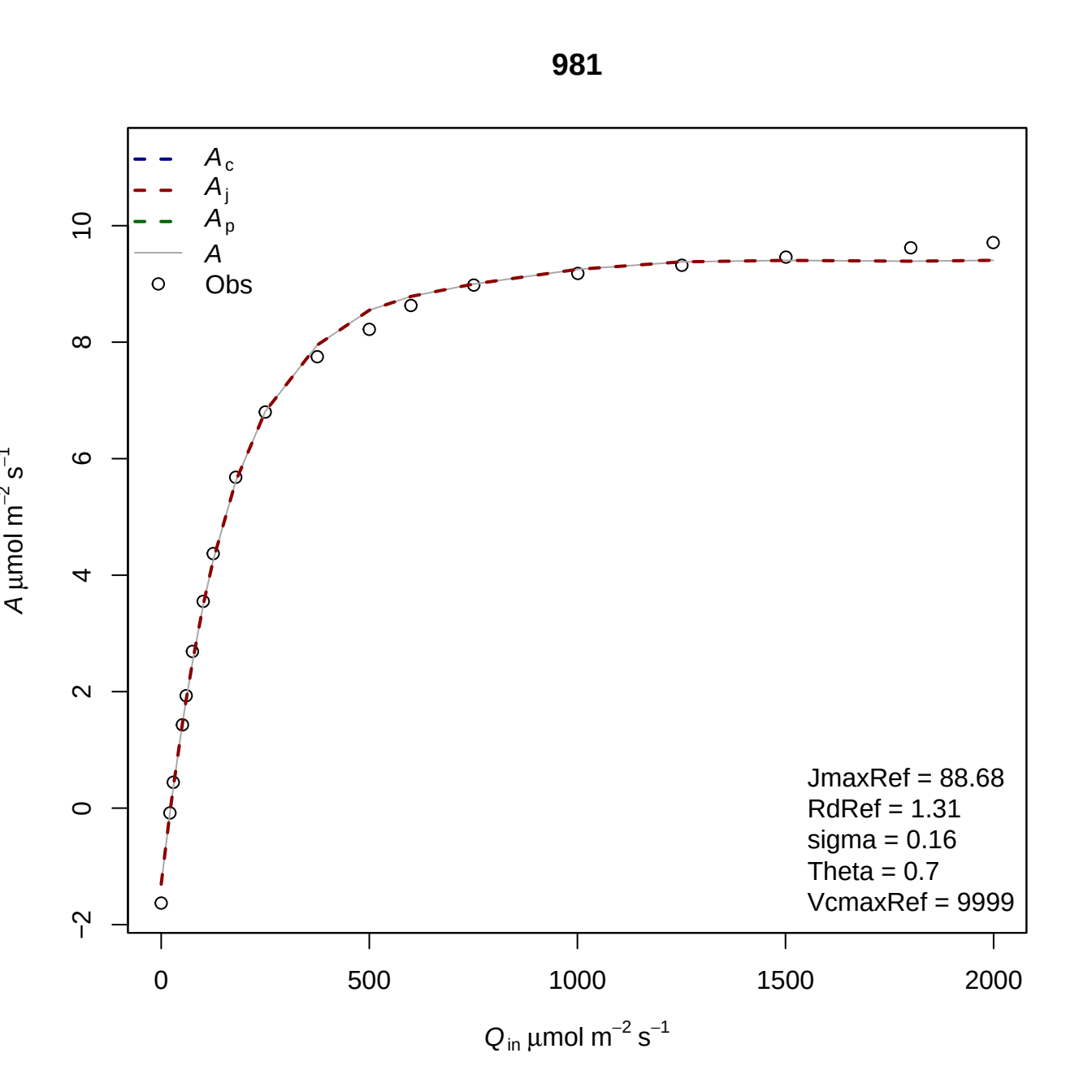

976
A_c
A_j
A_p
A
Obs
JmaxRef = 73.21
RdRef = 2.31
sigma = 0.55
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

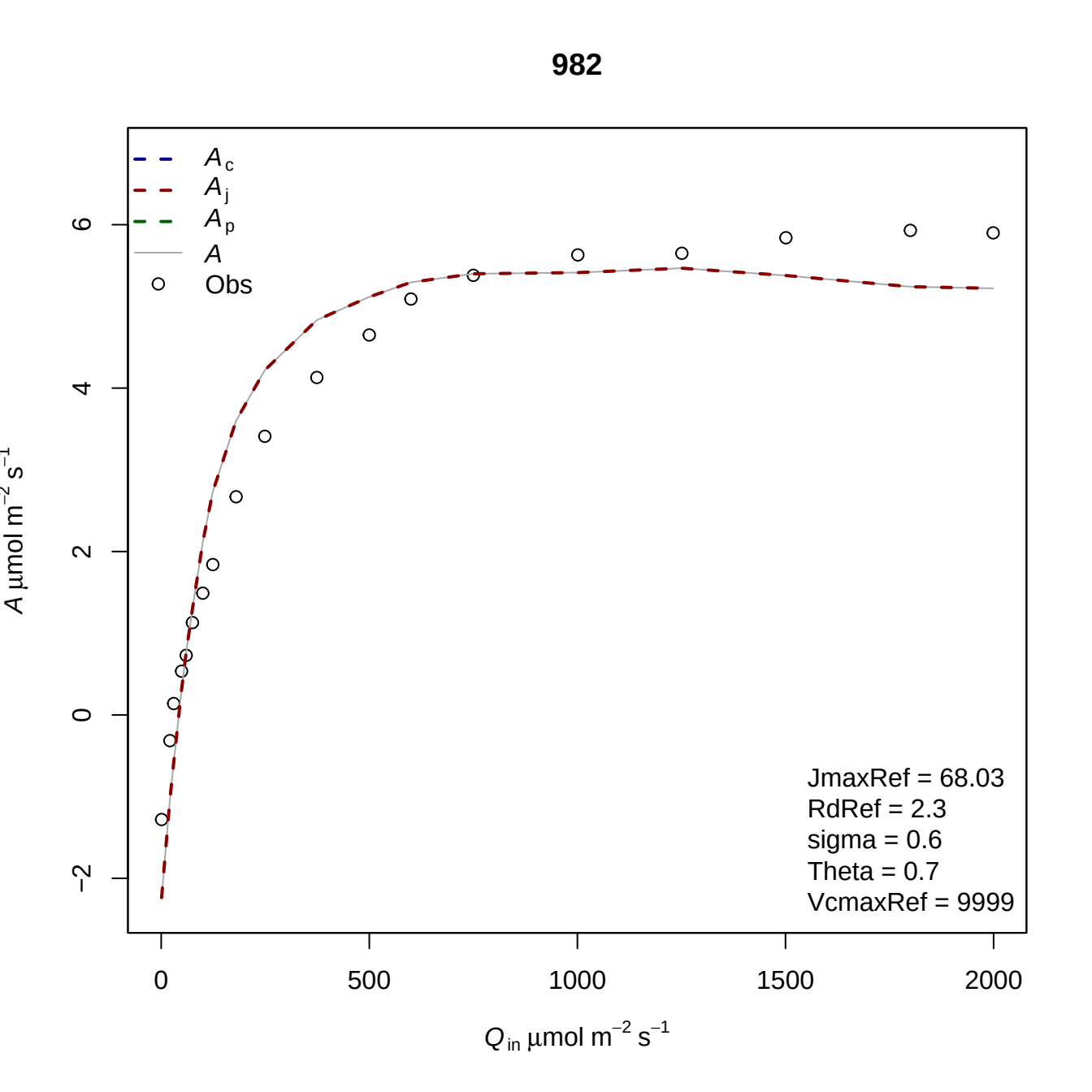

977
A_c
A_j
A_p
A
Obs
JmaxRef = 83.15
RdRef = 2.8
sigma = 0.99
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

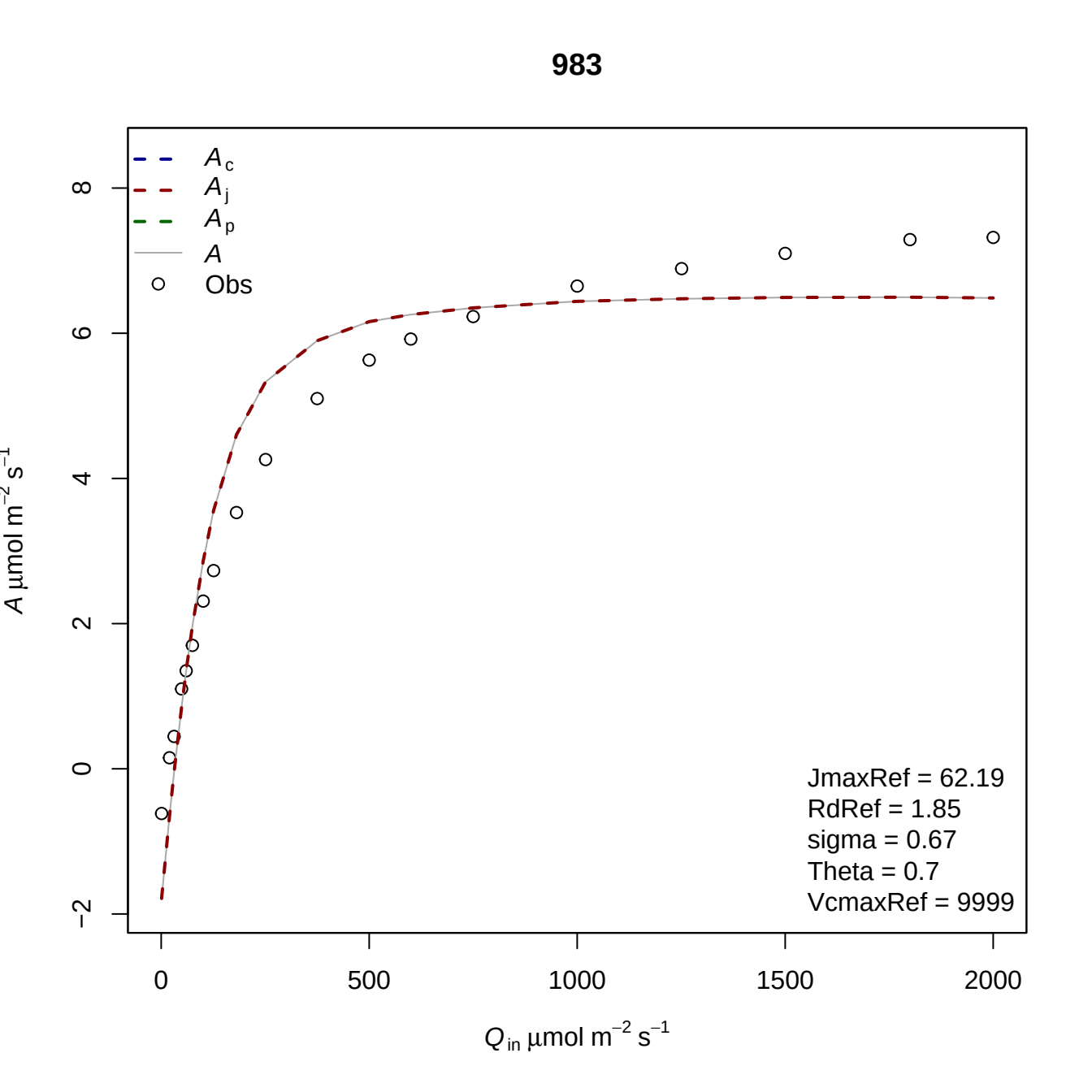

978
A_c
A_j
A_p
A
Obs
JmaxRef = 64.64
RdRef = 1.33
sigma = 0.21
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500
2000

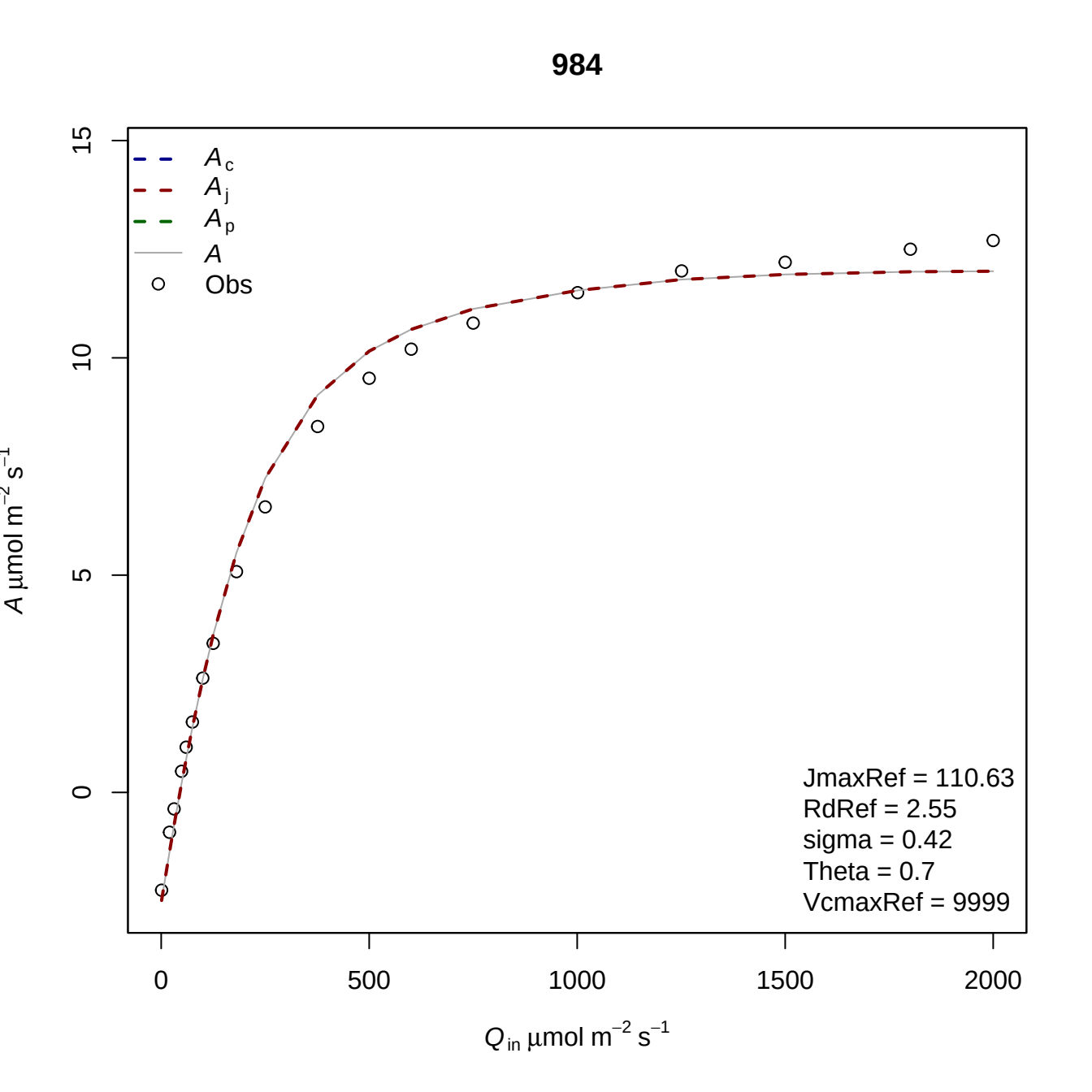

979
A_c
A_j
A_p
A
Obs
JmaxRef = 90.22
RdRef = 1.98
sigma = 0.19
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

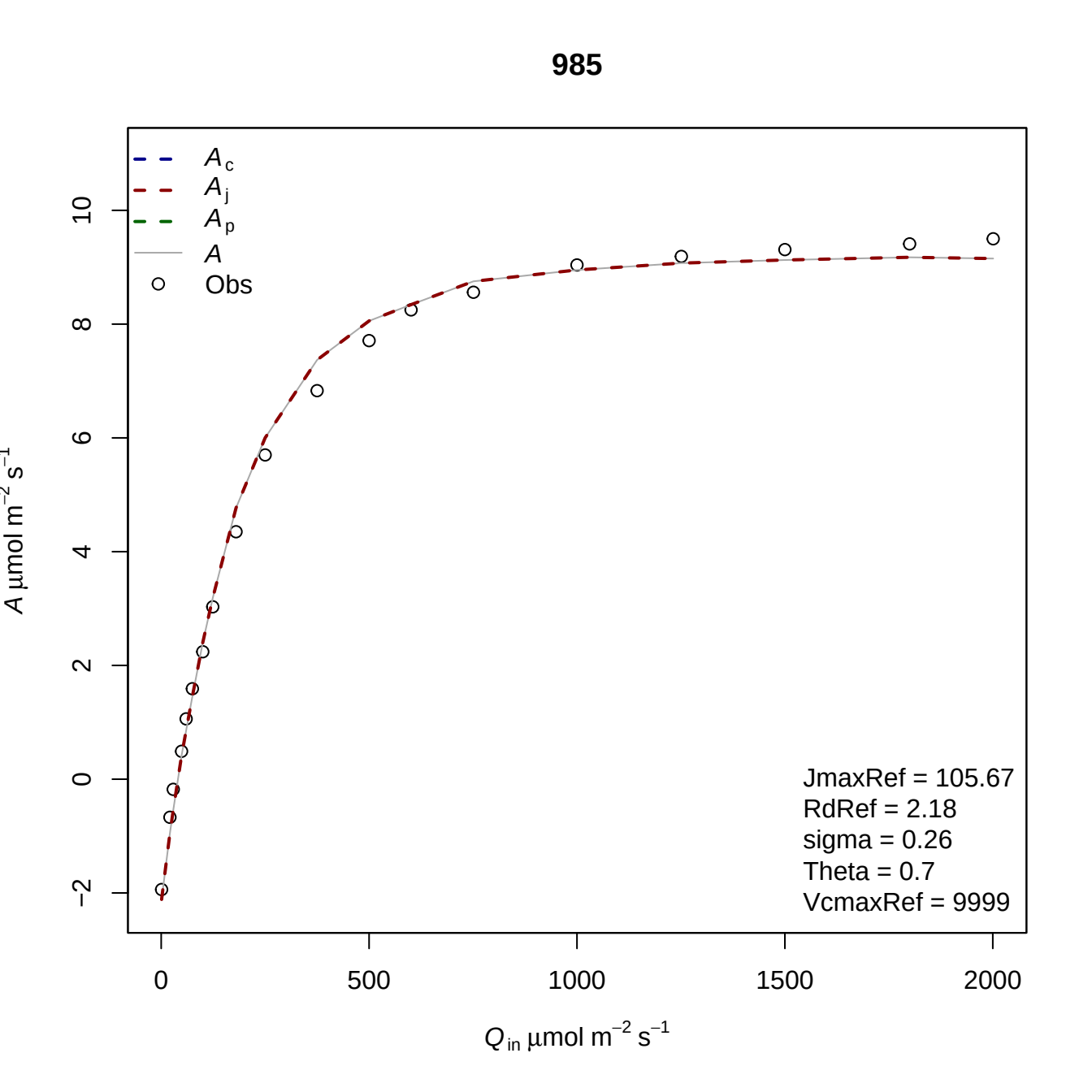

980
A_c
A_j
A_p
A
Obs
JmaxRef = 101.98
RdRef = 3.65
sigma = 0.36
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

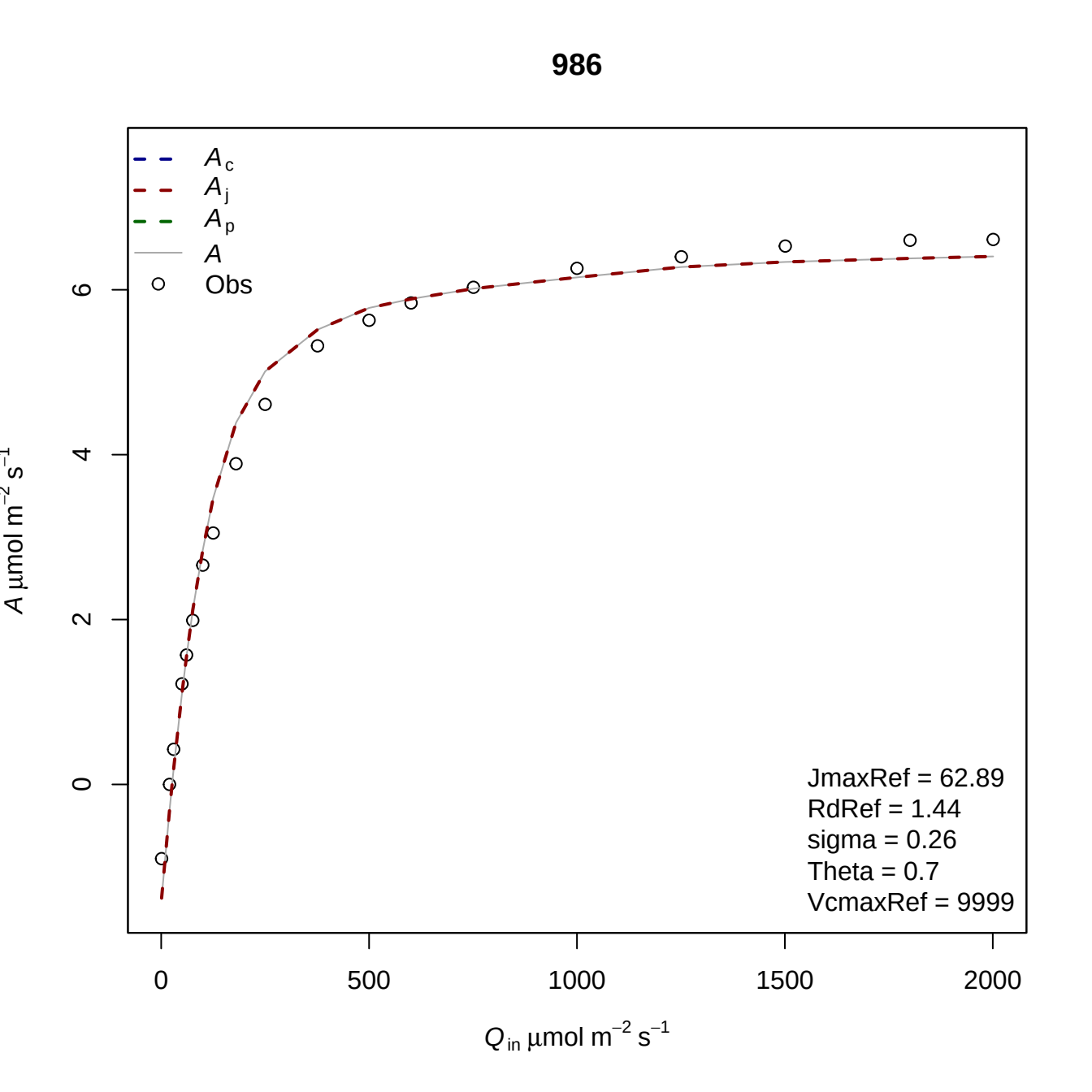

981
A_c
A_j
A_p
A
Obs
JmaxRef = 88.68
RdRef = 1.31
sigma = 0.16
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

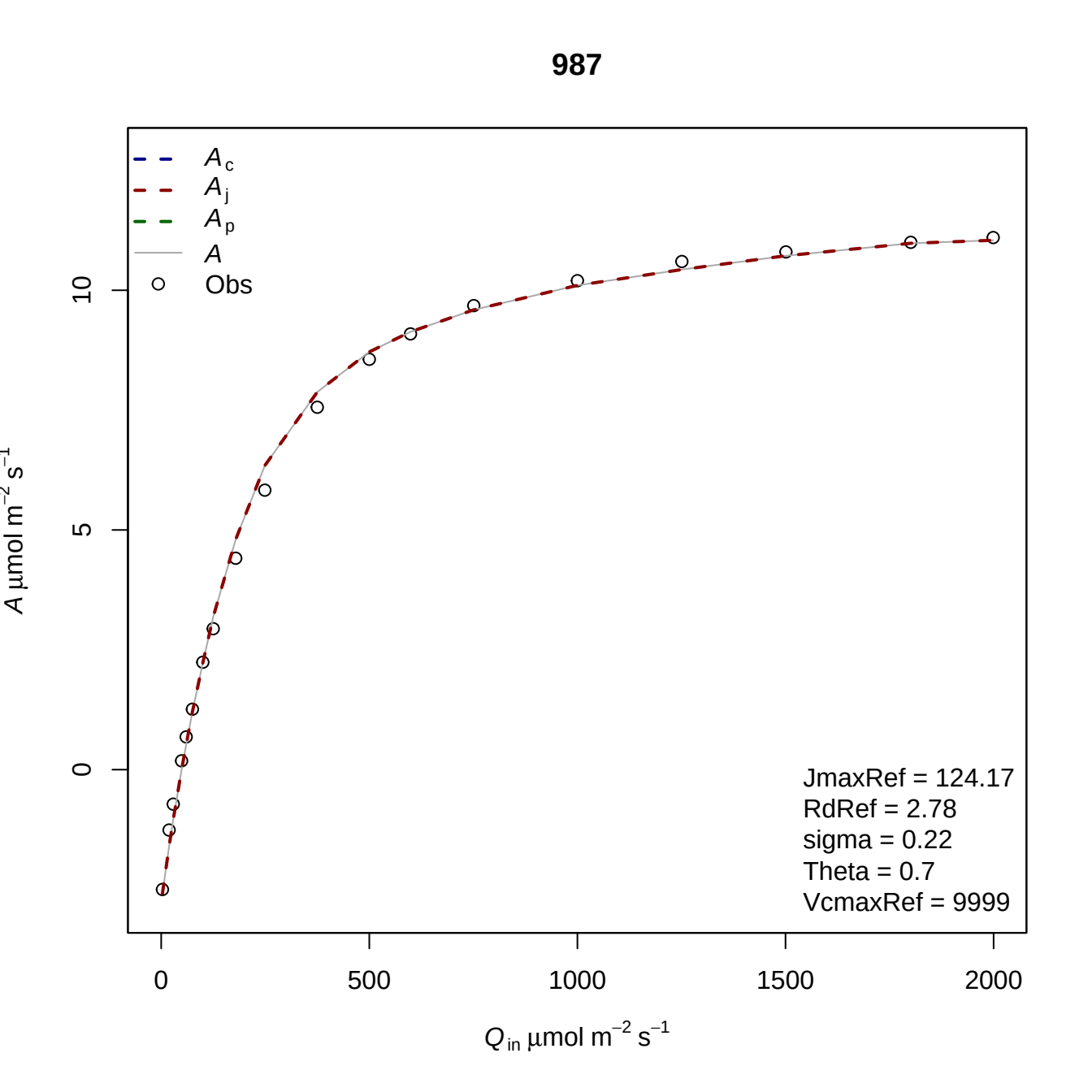

982
A_c
A_j
A_p
A
Obs
JmaxRef = 68.03
RdRef = 2.3
sigma = 0.6
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

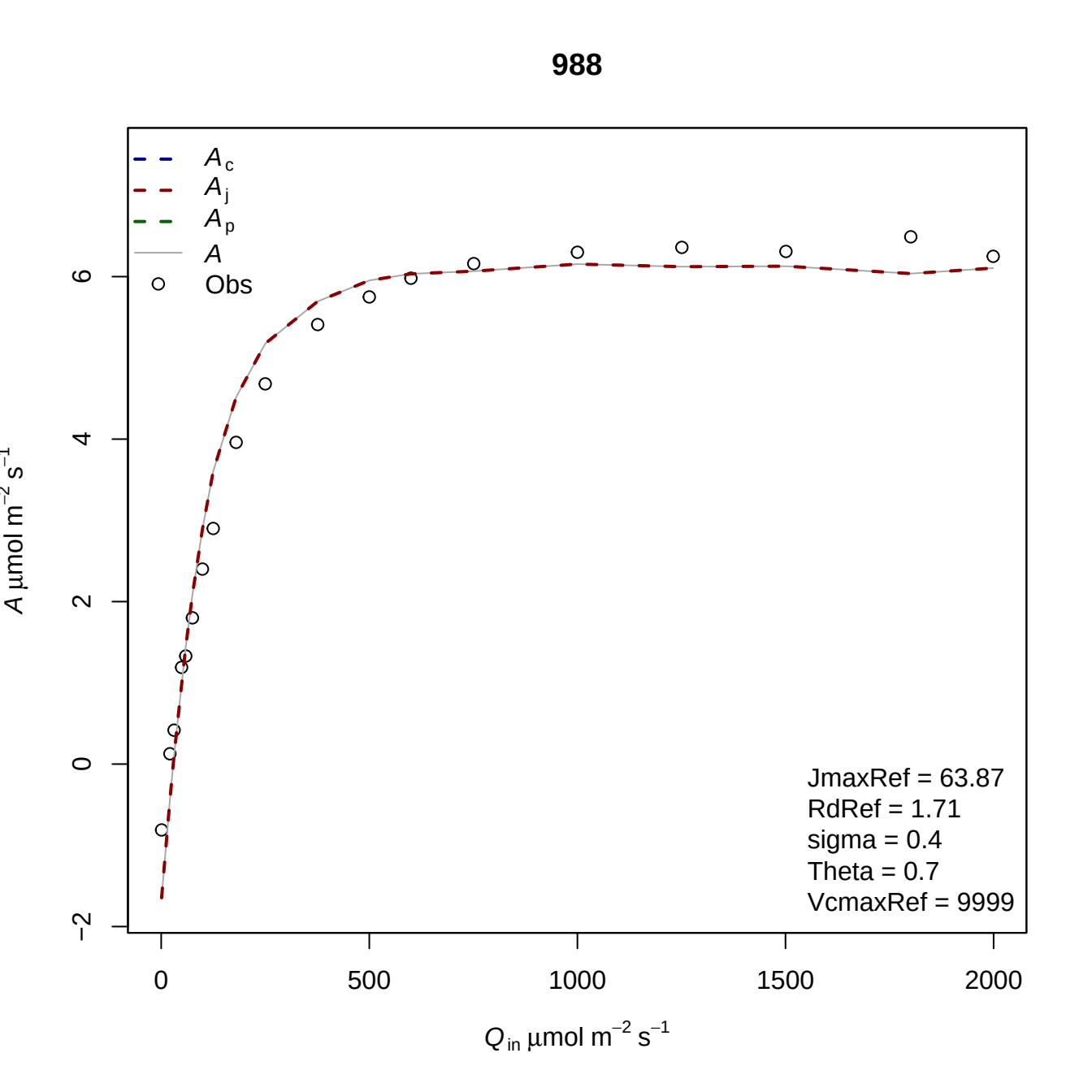

983
A_c
A_j
A_p
A
Obs
JmaxRef = 62.19
RdRef = 1.85
sigma = 0.67
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

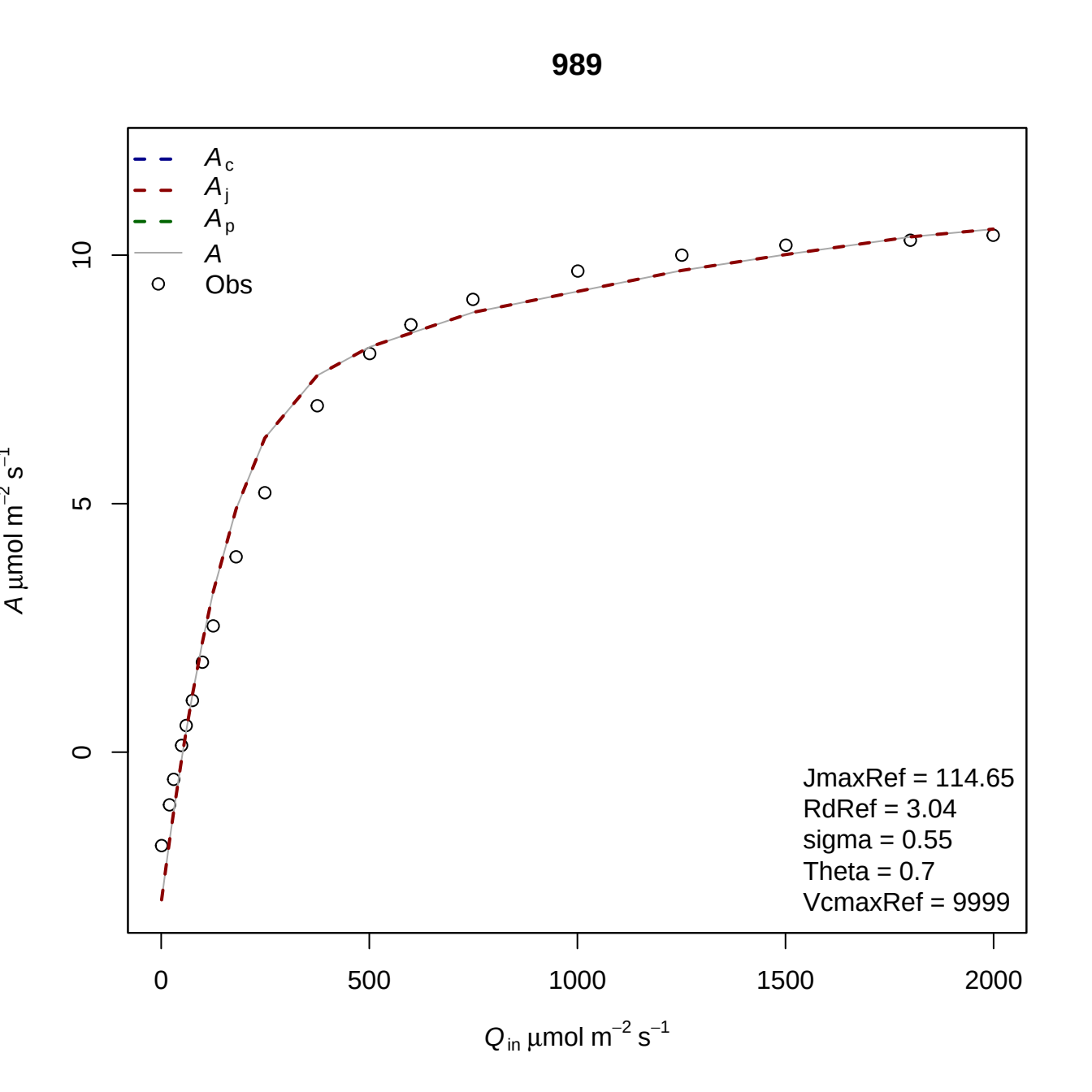

984
A_c
A_j
A_p
A
Obs
JmaxRef = 110.63
RdRef = 2.55
sigma = 0.42
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

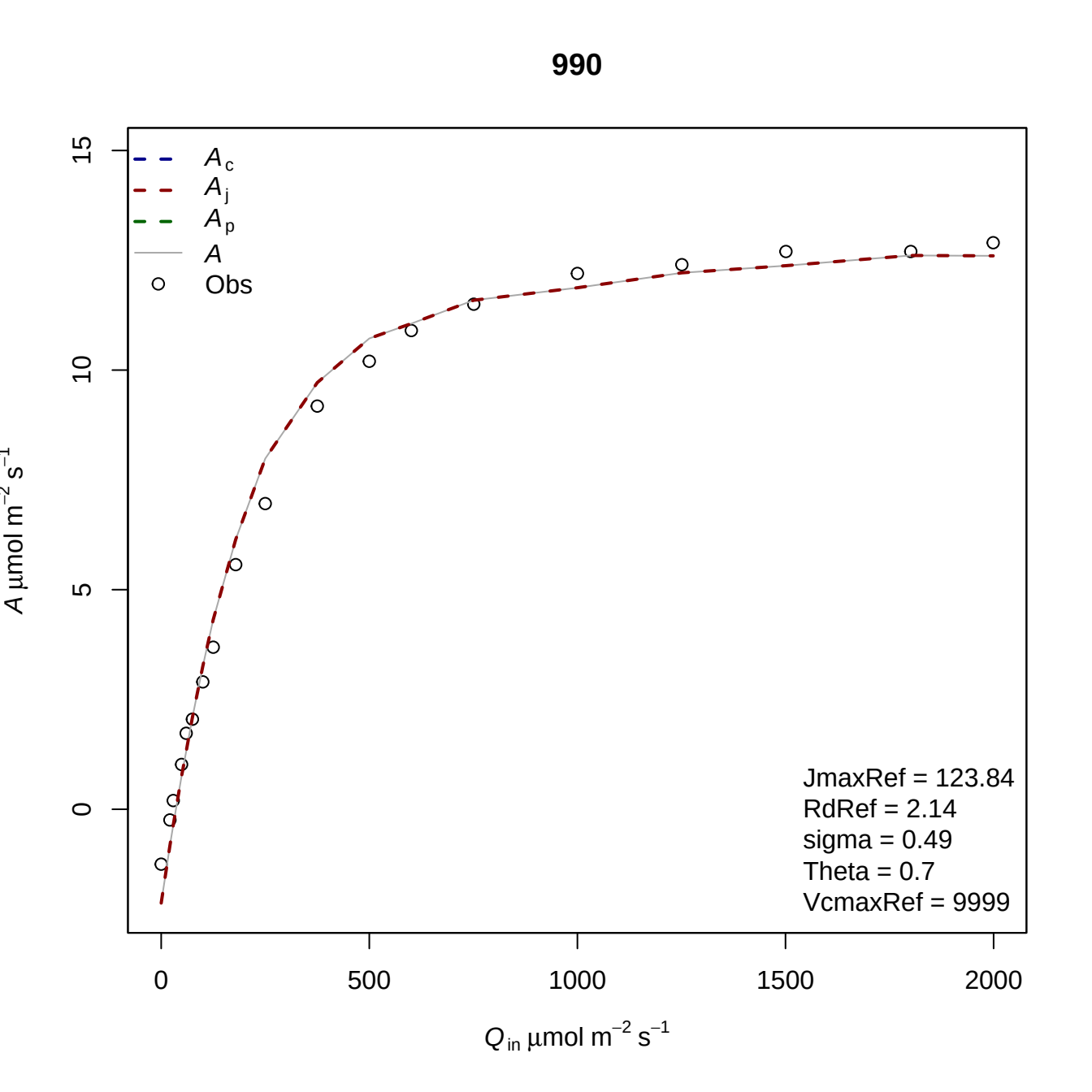

985
A_c
A_j
A_p
A
Obs
JmaxRef = 105.67
RdRef = 2.18
sigma = 0.26
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

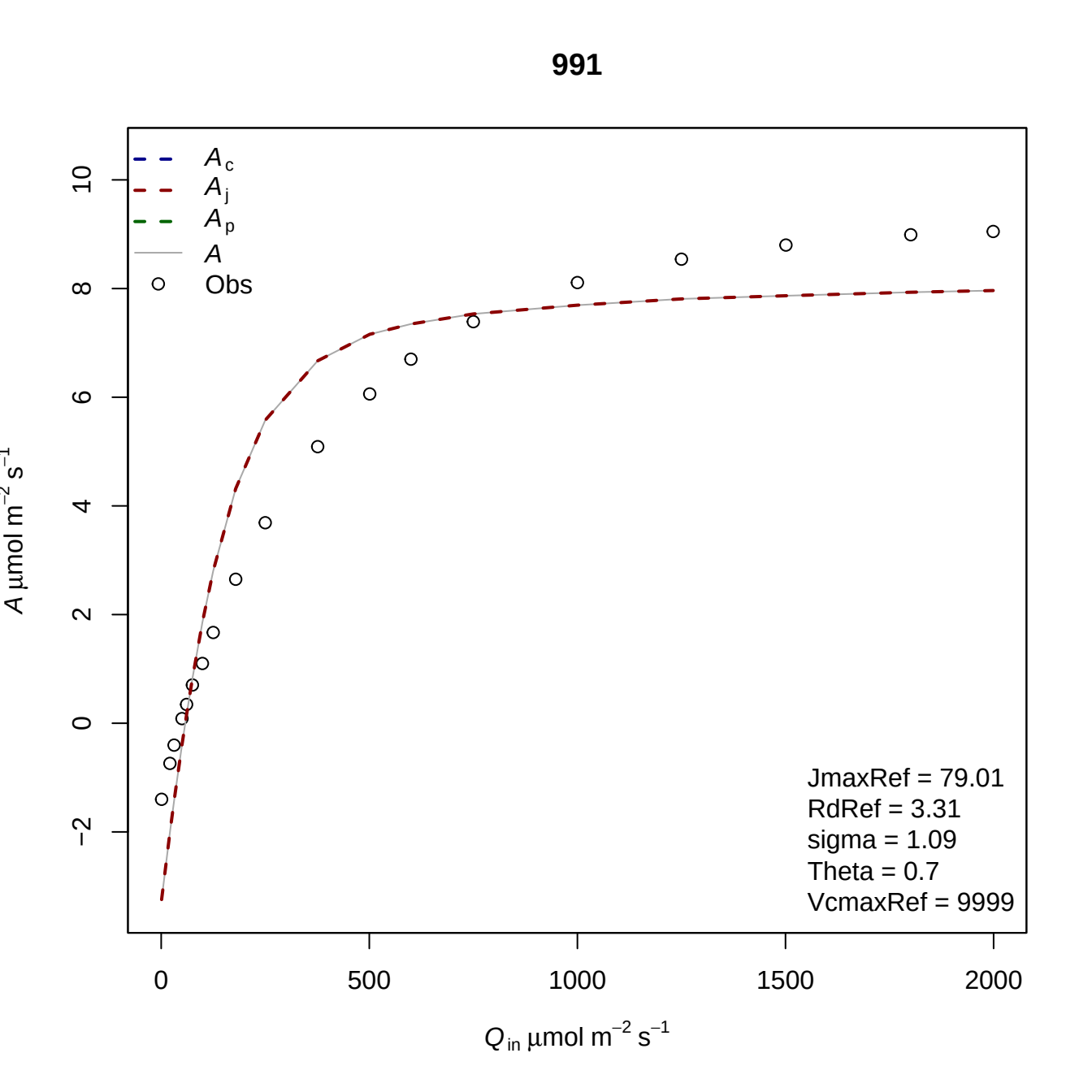

986
A_c
A_j
A_p
A
Obs
JmaxRef = 62.89
RdRef = 1.44
sigma = 0.26
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

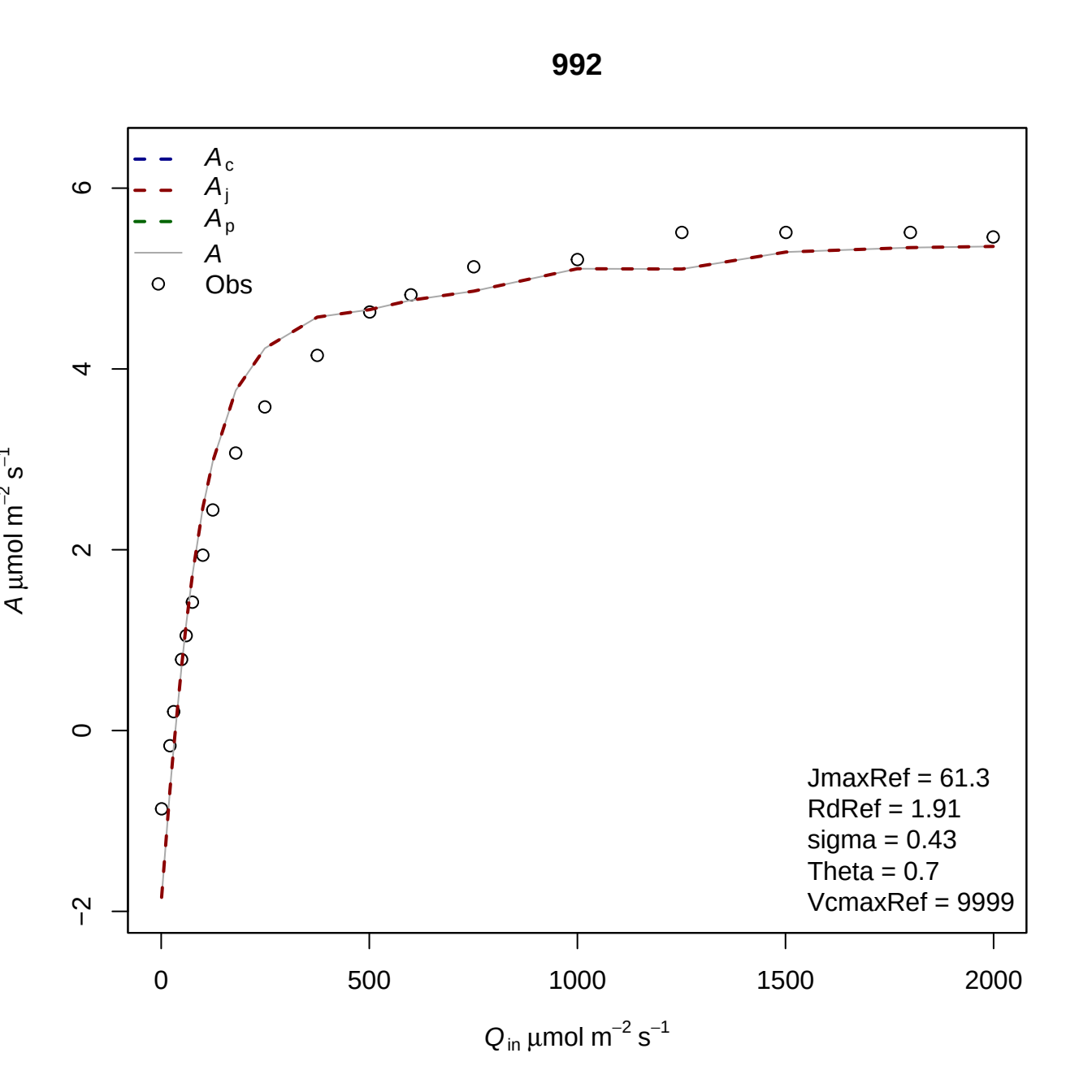

987
A_c
A_j
A_p
A
Obs
JmaxRef = 124.17
RdRef = 2.78
sigma = 0.22
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

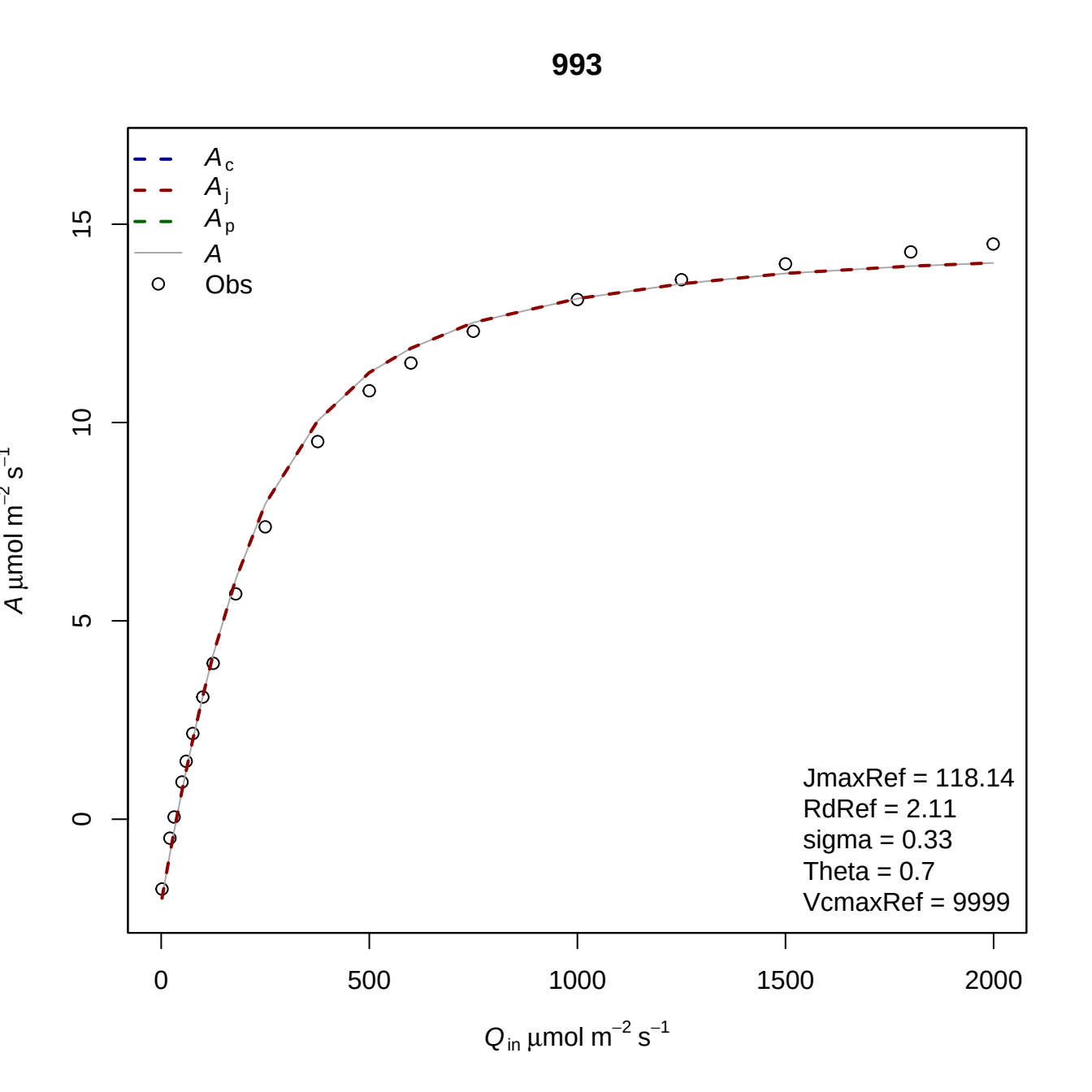

988
A_c
A_j
A_p
A
Obs
JmaxRef = 63.87
RdRef = 1.71
sigma = 0.4
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

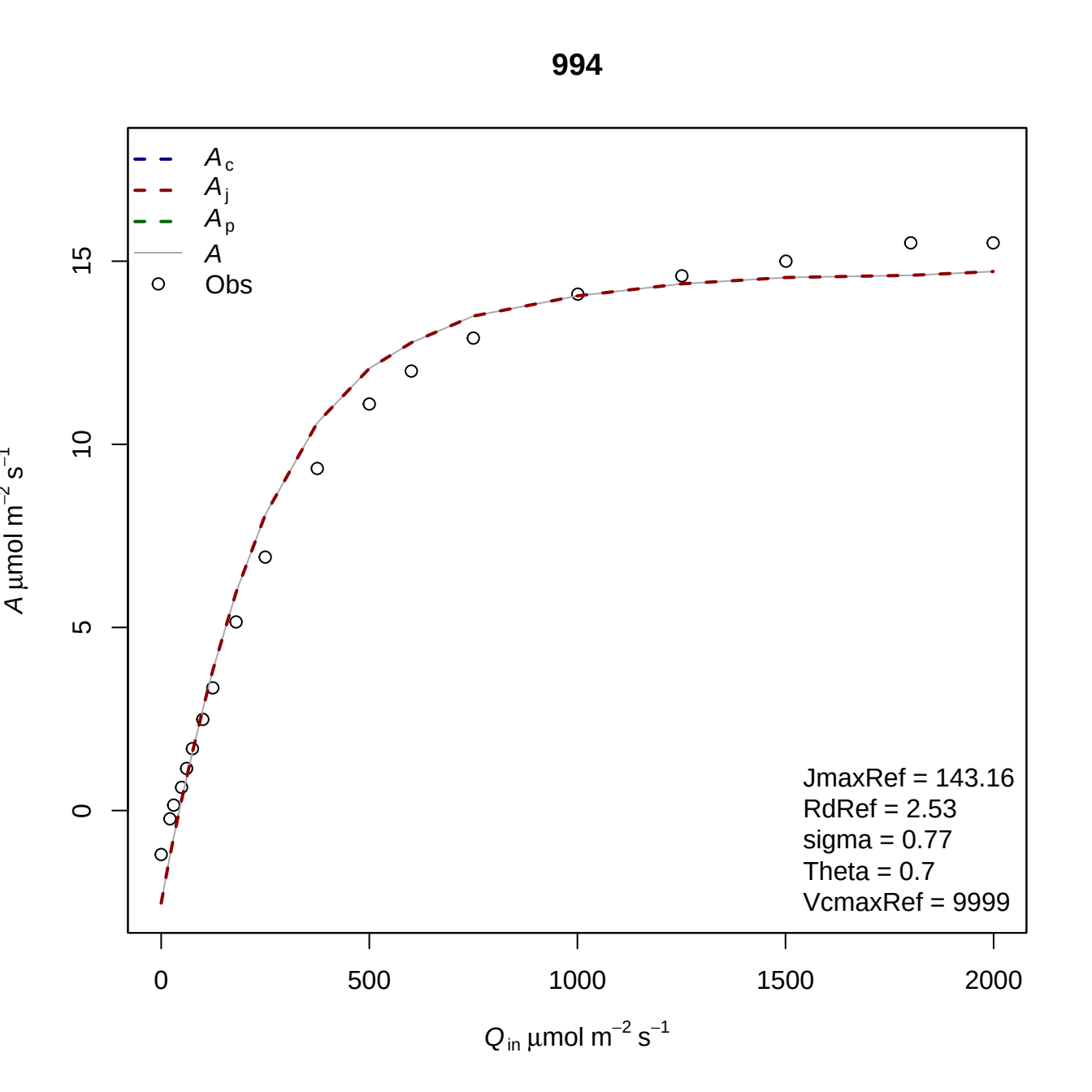

989
A_c
A_j
A_p
A
Obs
JmaxRef = 114.65
RdRef = 3.04
sigma = 0.55
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

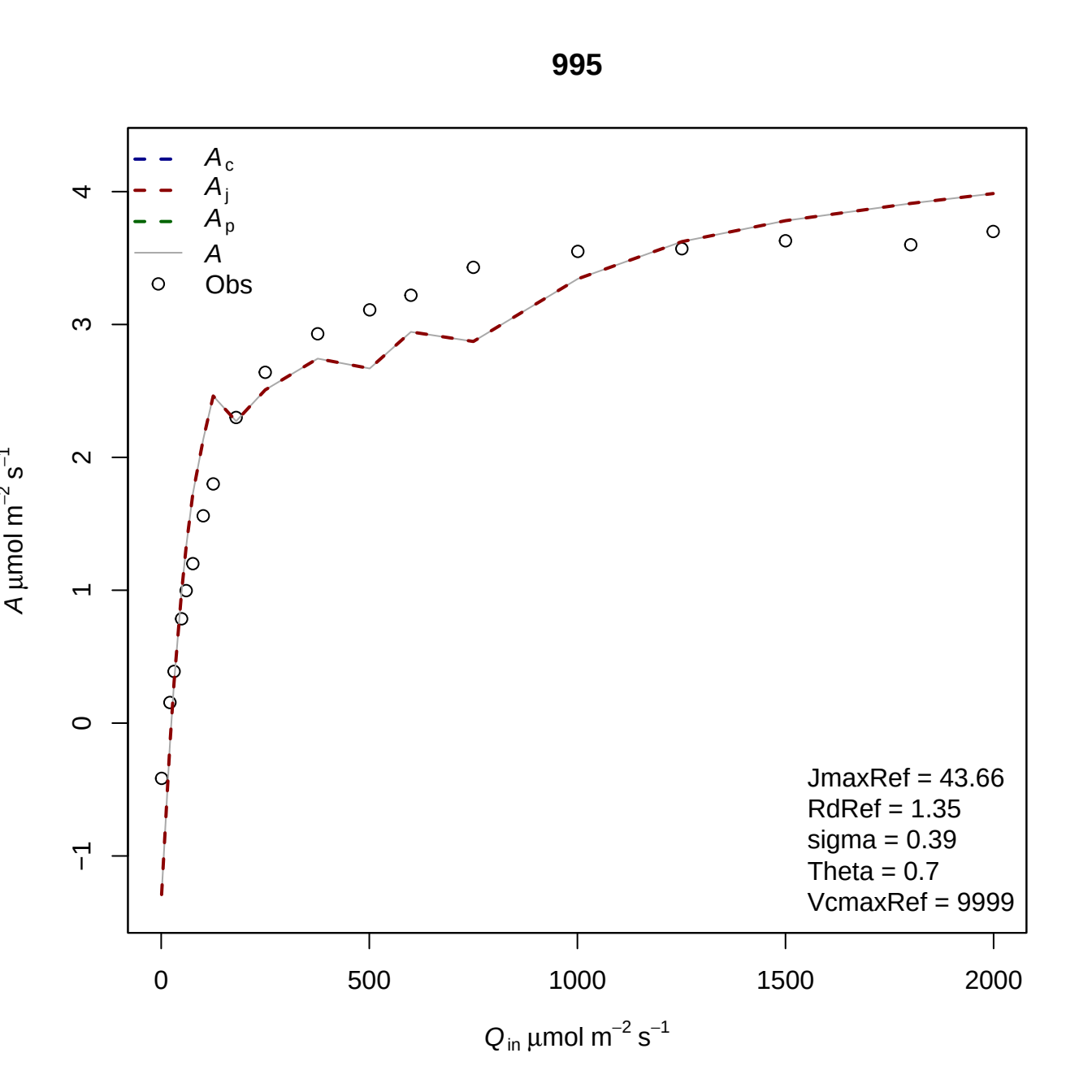

990
A_c
A_j
A_p
A
Obs
JmaxRef = 123.84
RdRef = 2.14
sigma = 0.49
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

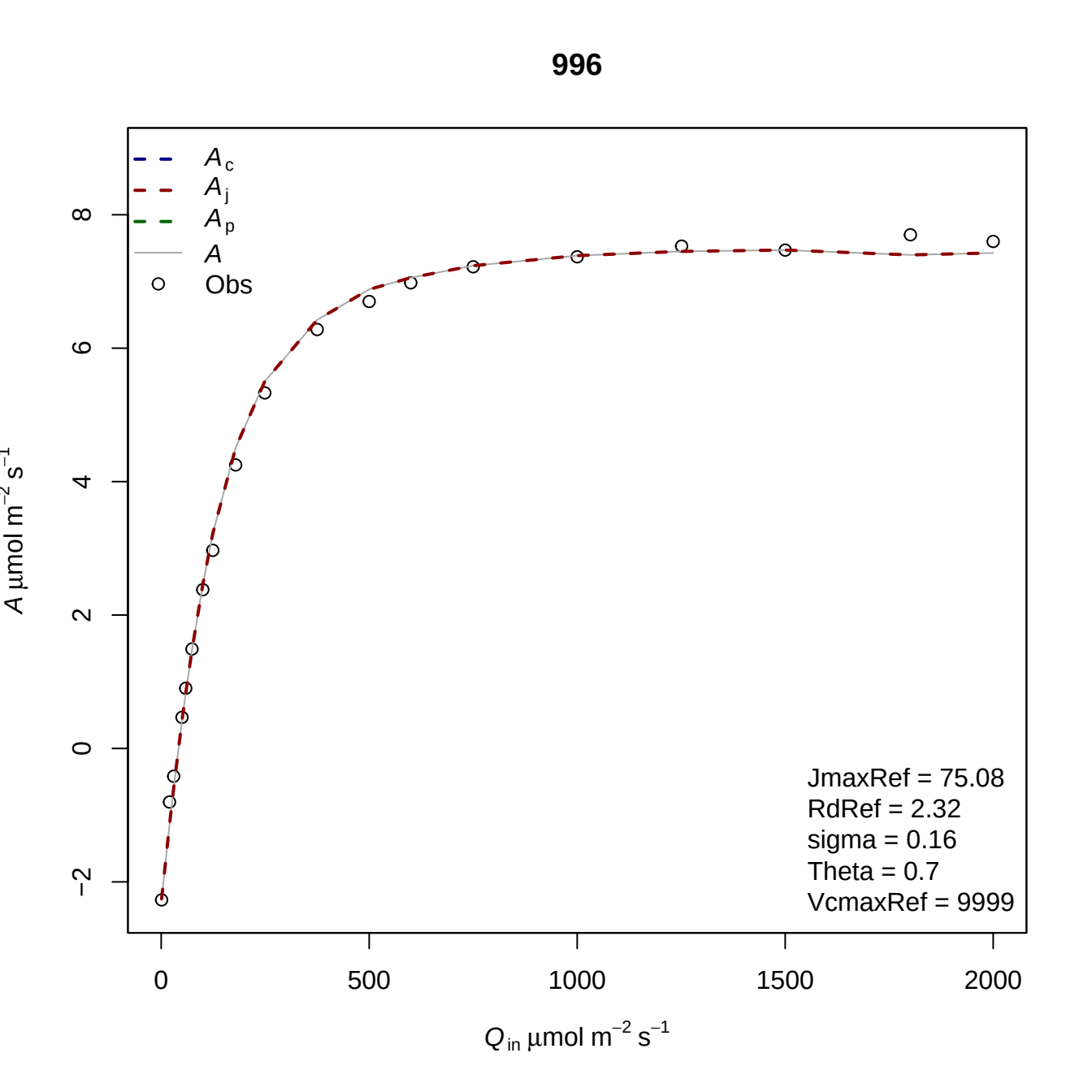

991
A_c
A_j
A_p
A
Obs
JmaxRef = 79.01
RdRef = 3.31
sigma = 1.09
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

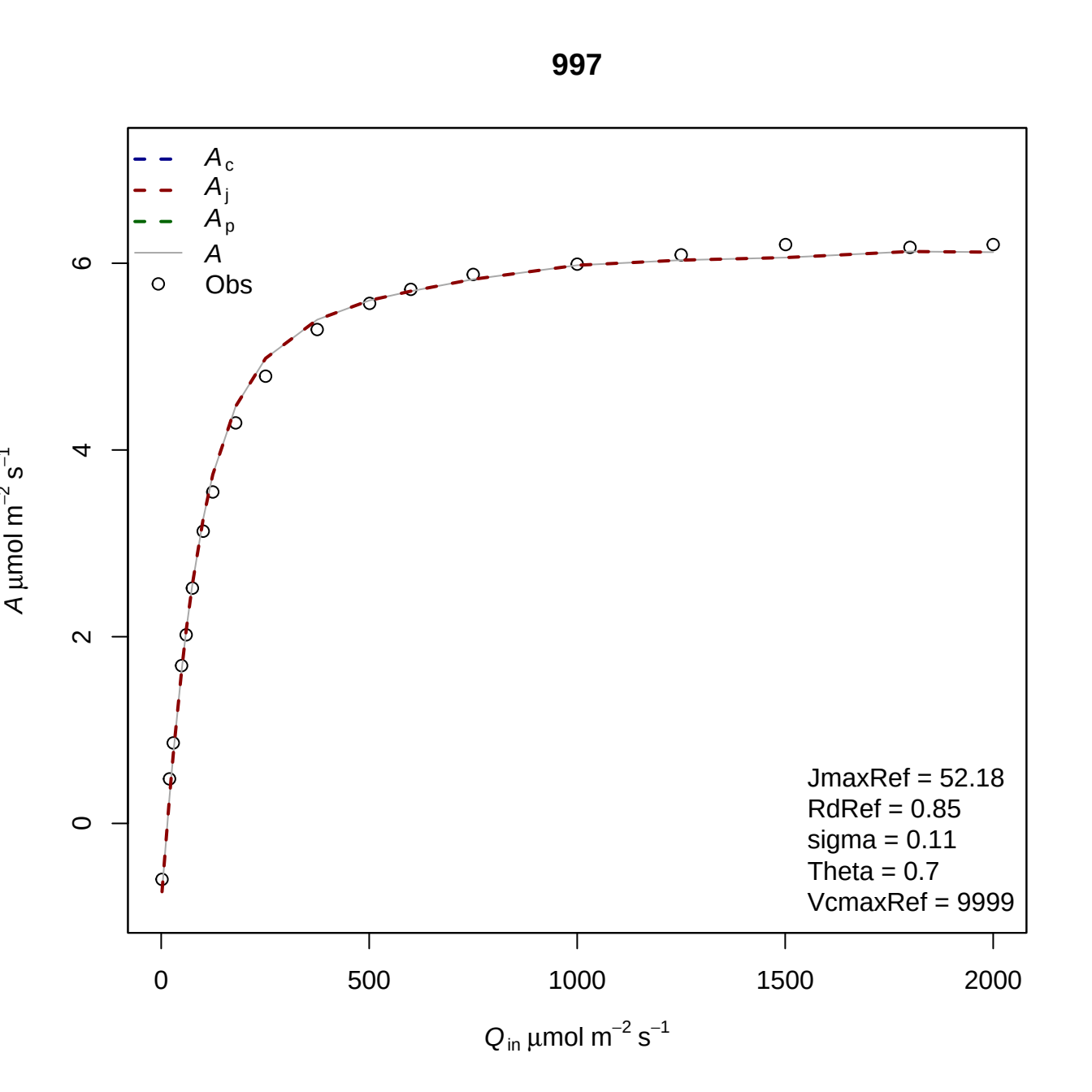

992
A_c
A_j
A_p
A
Obs
JmaxRef = 61.3
RdRef = 1.91
sigma = 0.43
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

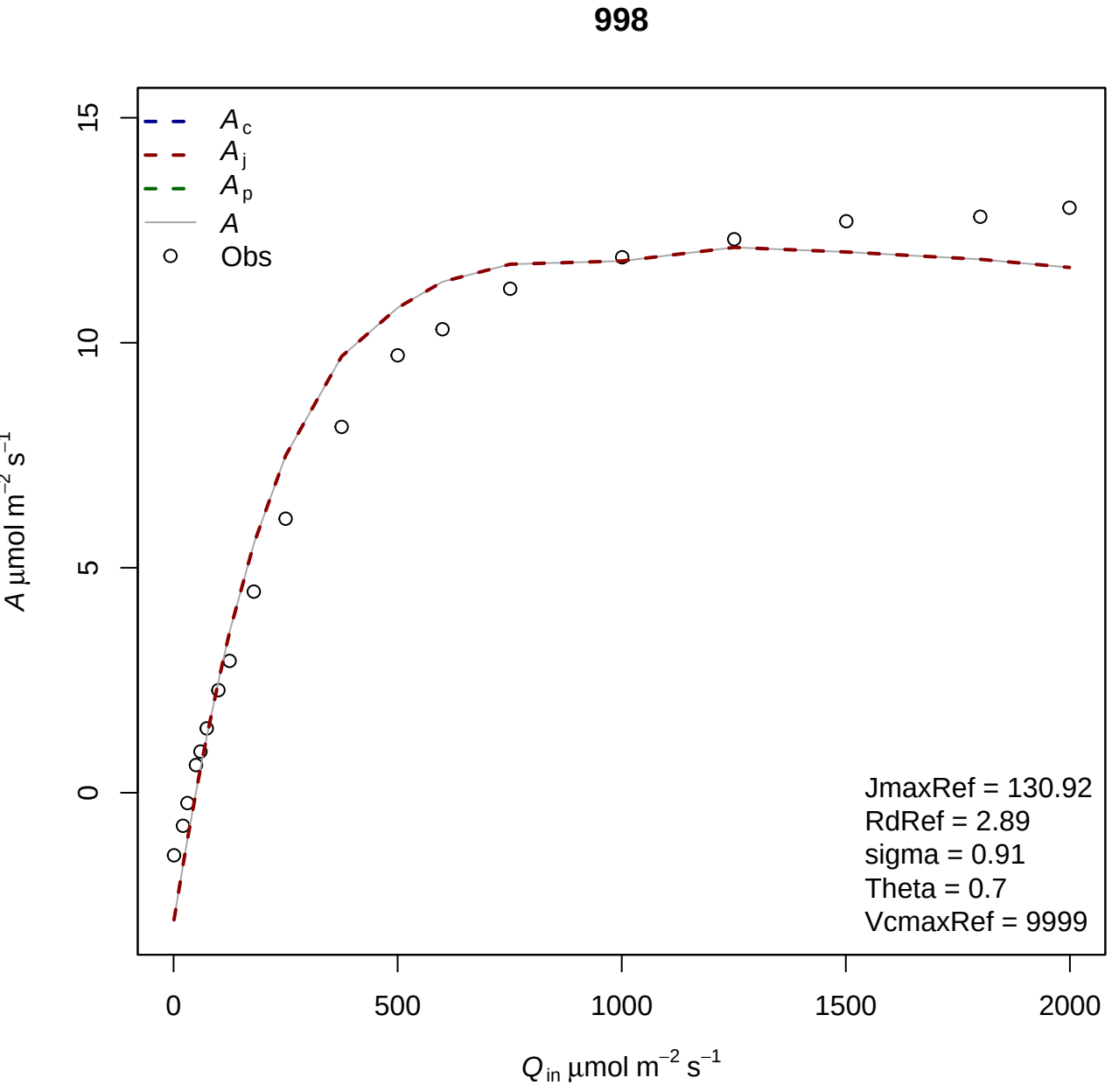

993
A_c
A_j
A_p
A
Obs
JmaxRef = 118.14
RdRef = 2.11
sigma = 0.33
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

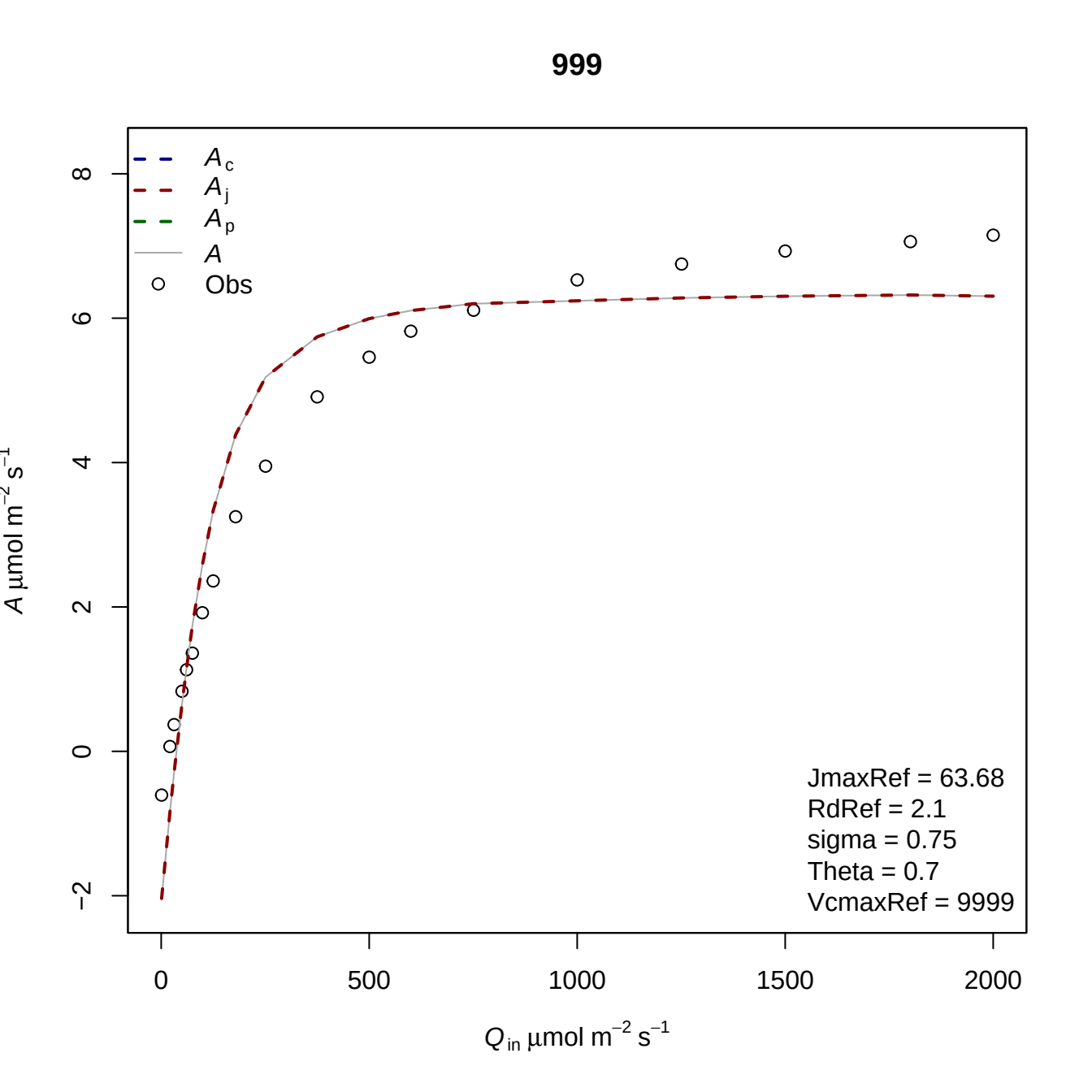

994
A_c
A_j
A_p
A
Obs
JmaxRef = 143.16
RdRef = 2.53
sigma = 0.77
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

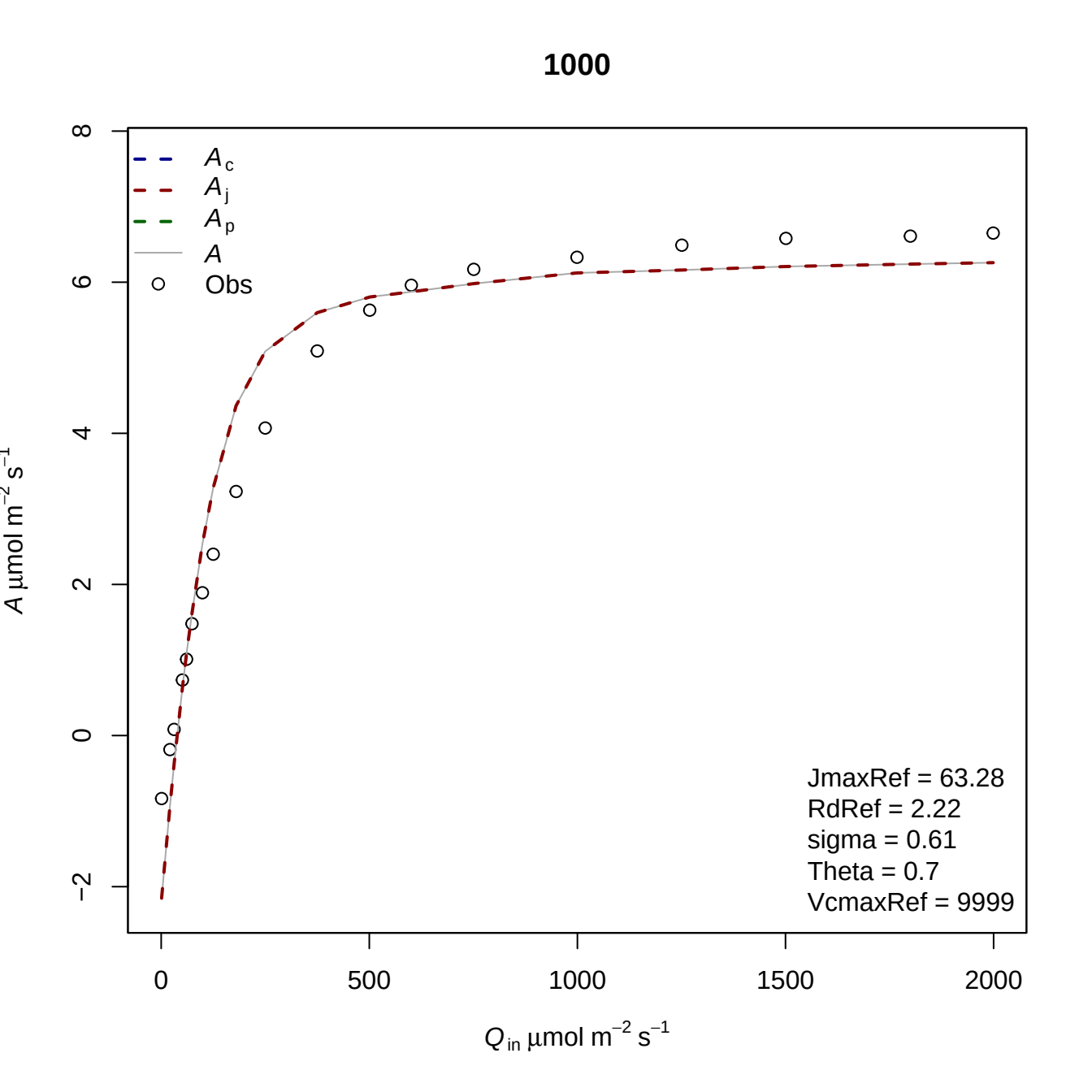

995
A_c
A_j
A_p
A
Obs
JmaxRef = 43.66
RdRef = 1.35
sigma = 0.39
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500
2000

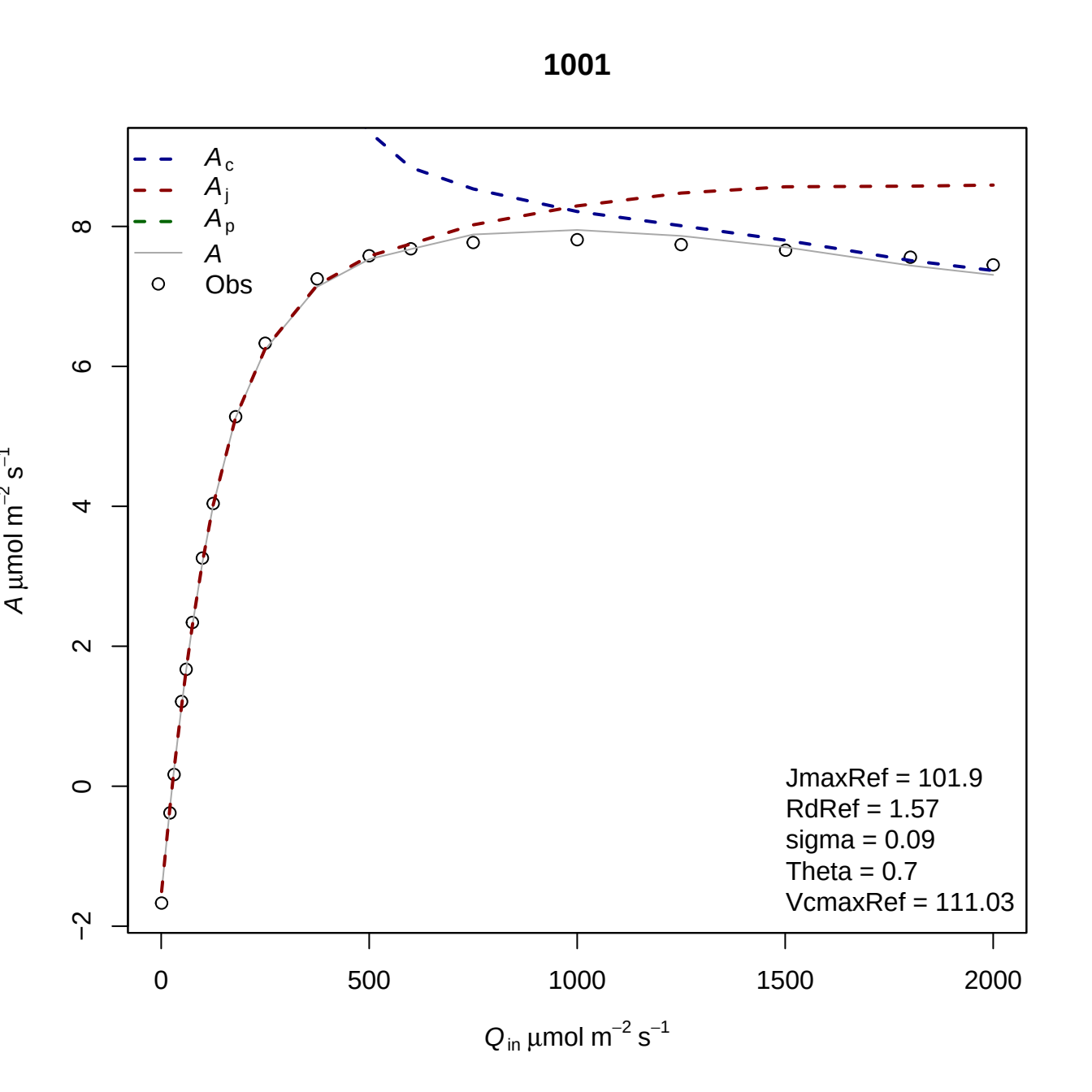

996
A_c
A_j
A_p
A
Obs
JmaxRef = 75.08
RdRef = 2.32
sigma = 0.16
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

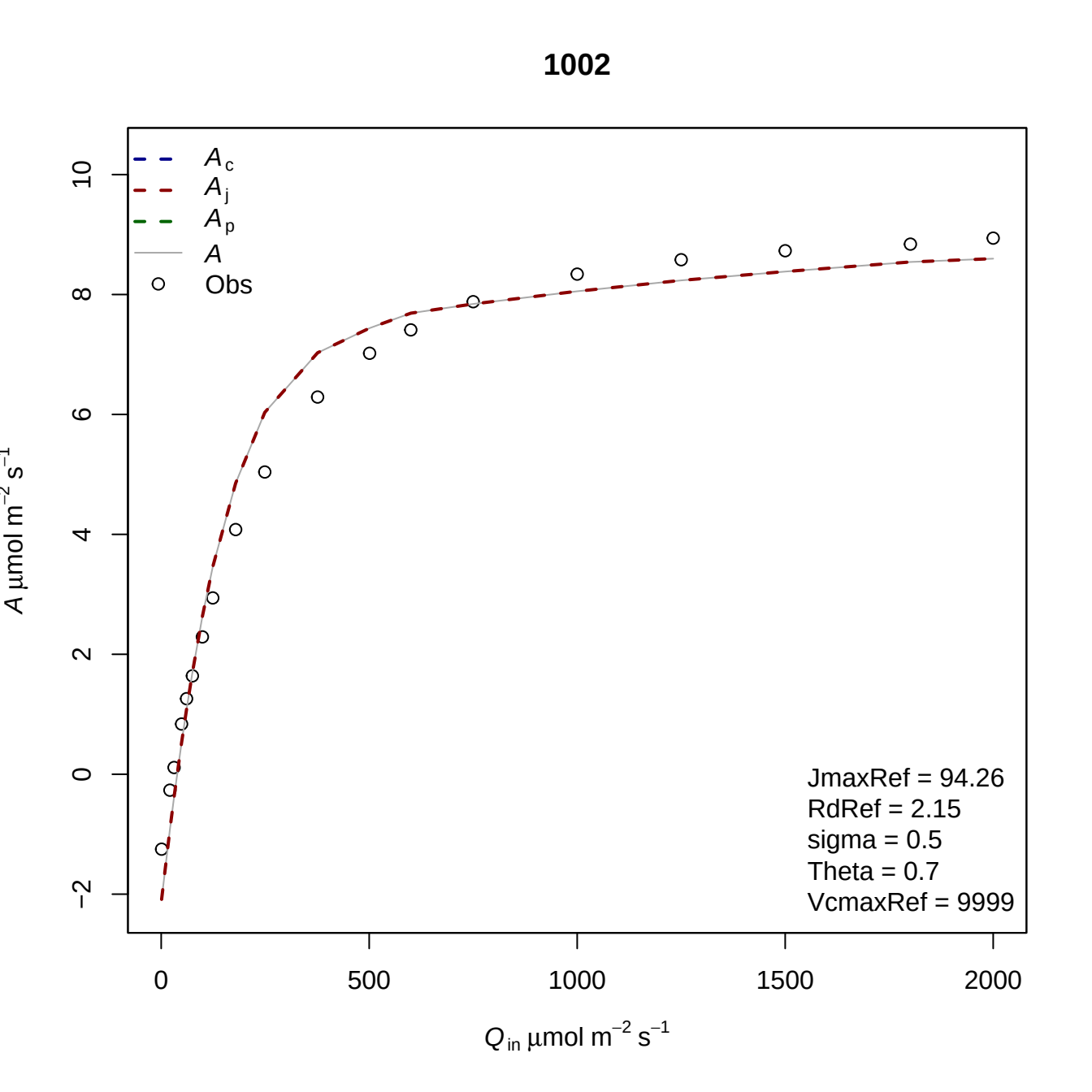

997
A_c
A_j
A_p
A
Obs
JmaxRef = 52.18
RdRef = 0.85
sigma = 0.11
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

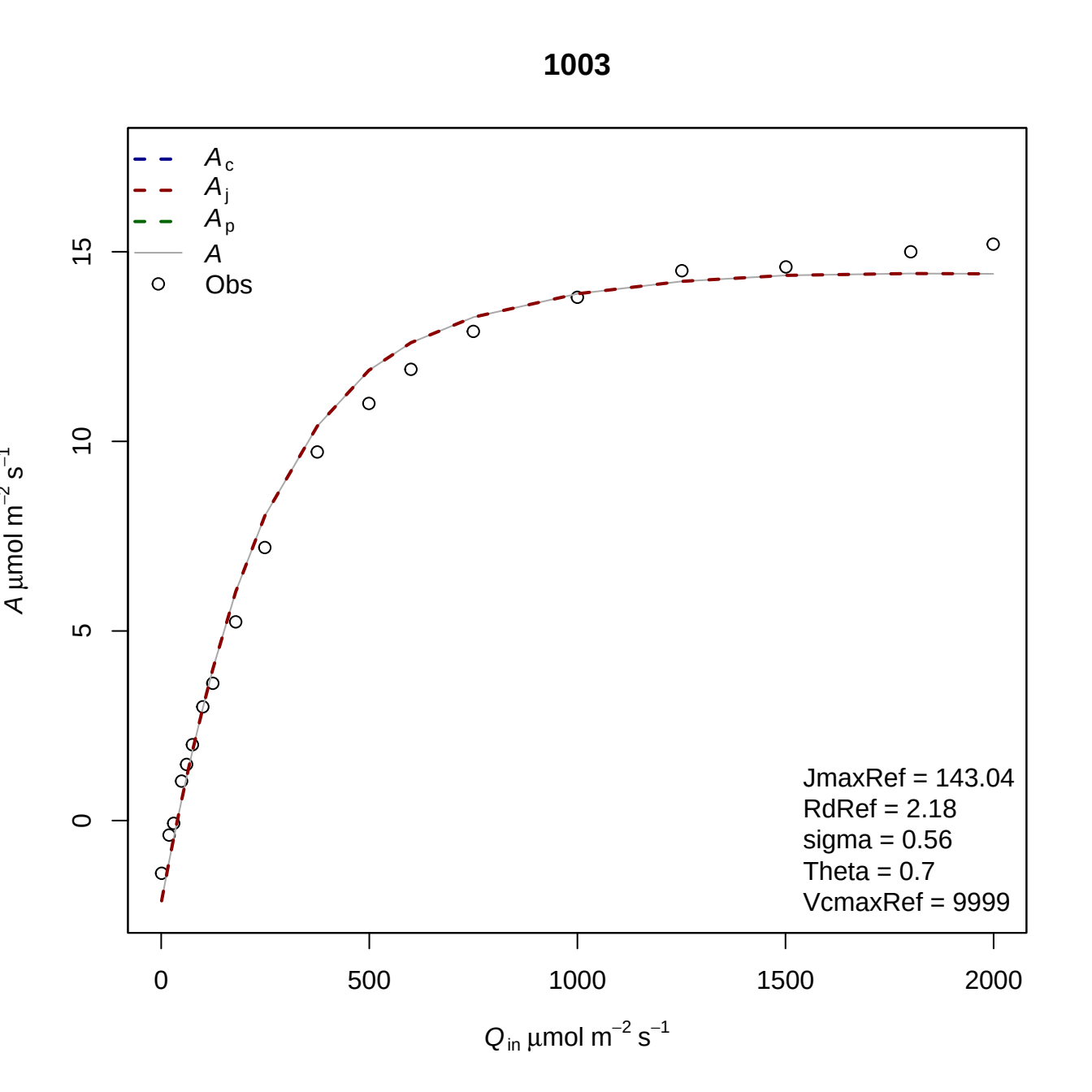

998
A_c
A_j
A_p
A
Obs
JmaxRef = 130.92
RdRef = 2.89
sigma = 0.91
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

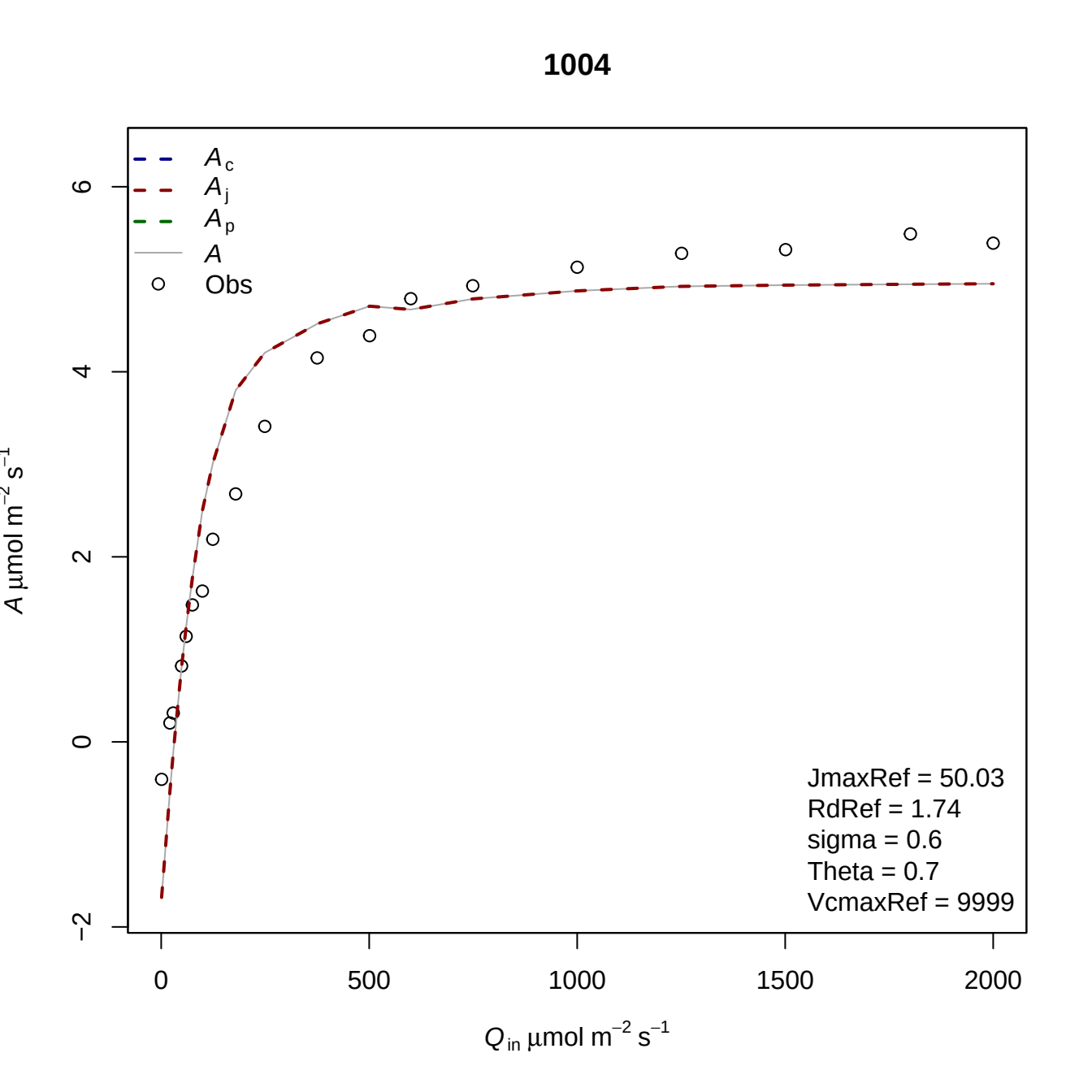

999
A_c
A_j
A_p
A
Obs
JmaxRef = 63.68
RdRef = 2.1
sigma = 0.75
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

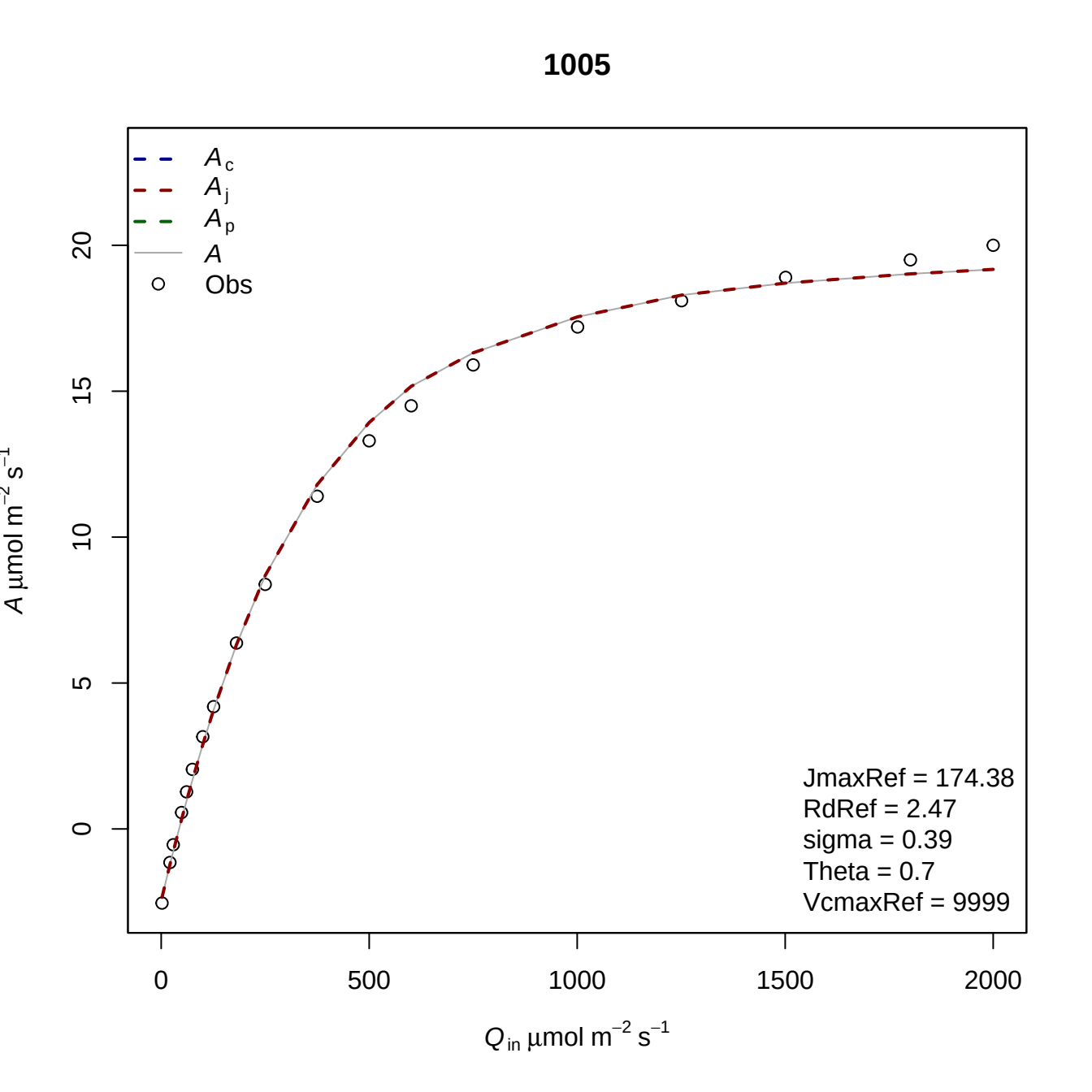

1000
A_c
A_j
A_p
A
Obs
JmaxRef = 63.28
RdRef = 2.22
sigma = 0.61
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

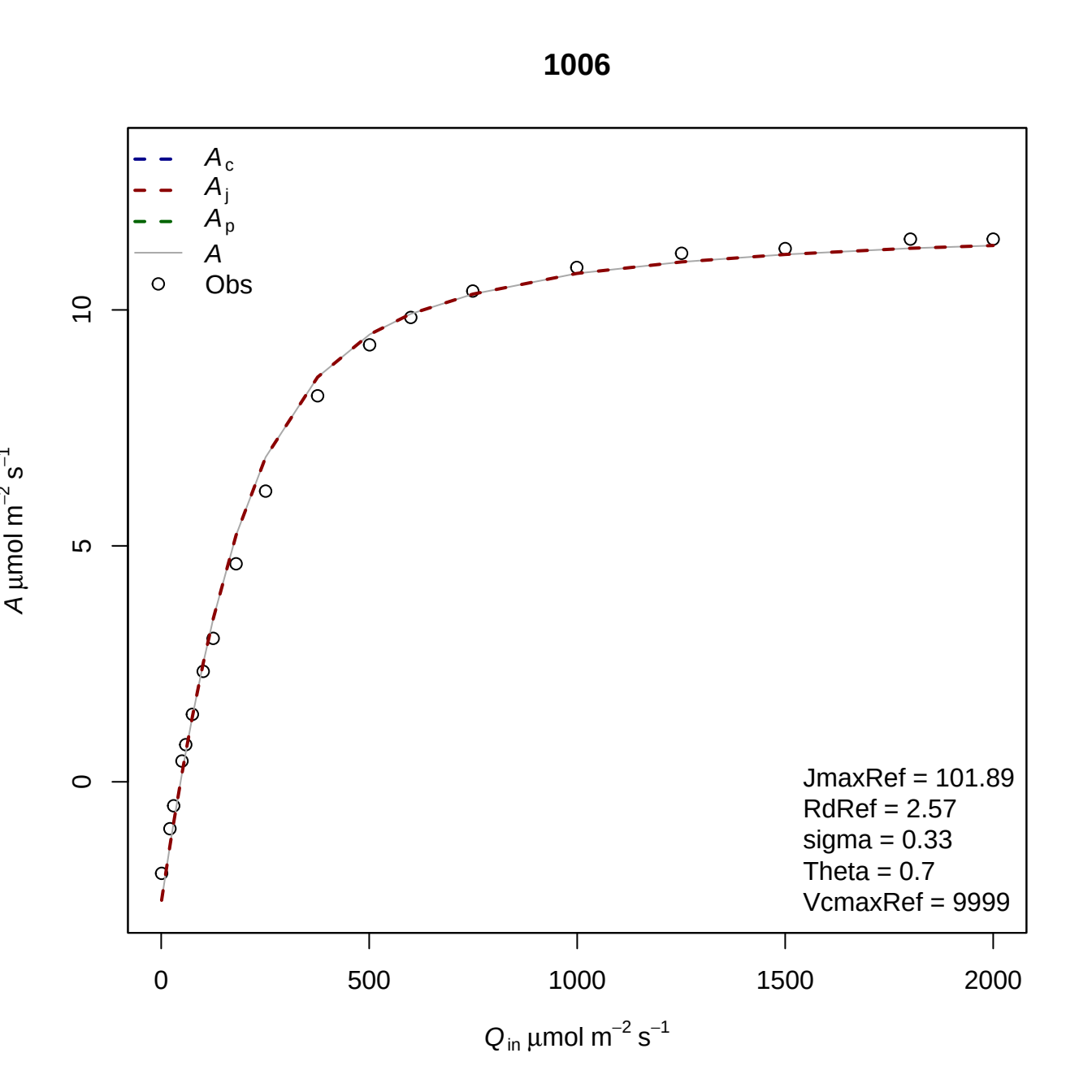

1001
A_c
A_j
A_p
A
Obs
JmaxRef = 101.9
RdRef = 1.57
sigma = 0.09
Theta = 0.7
VcmaxRef = 111.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

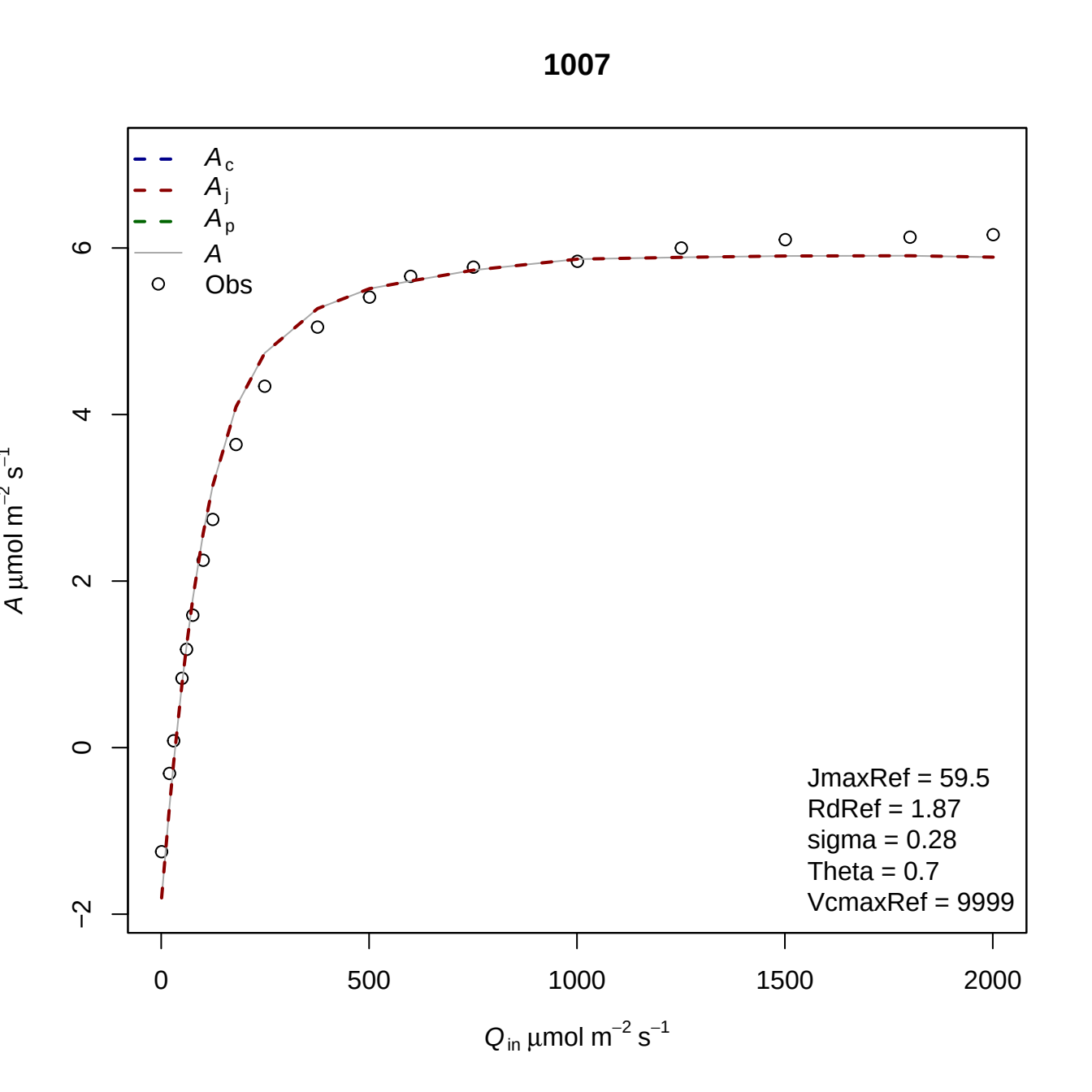

1002
A_c
A_j
A_p
A
Obs
JmaxRef = 94.26
RdRef = 2.15
sigma = 0.5
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

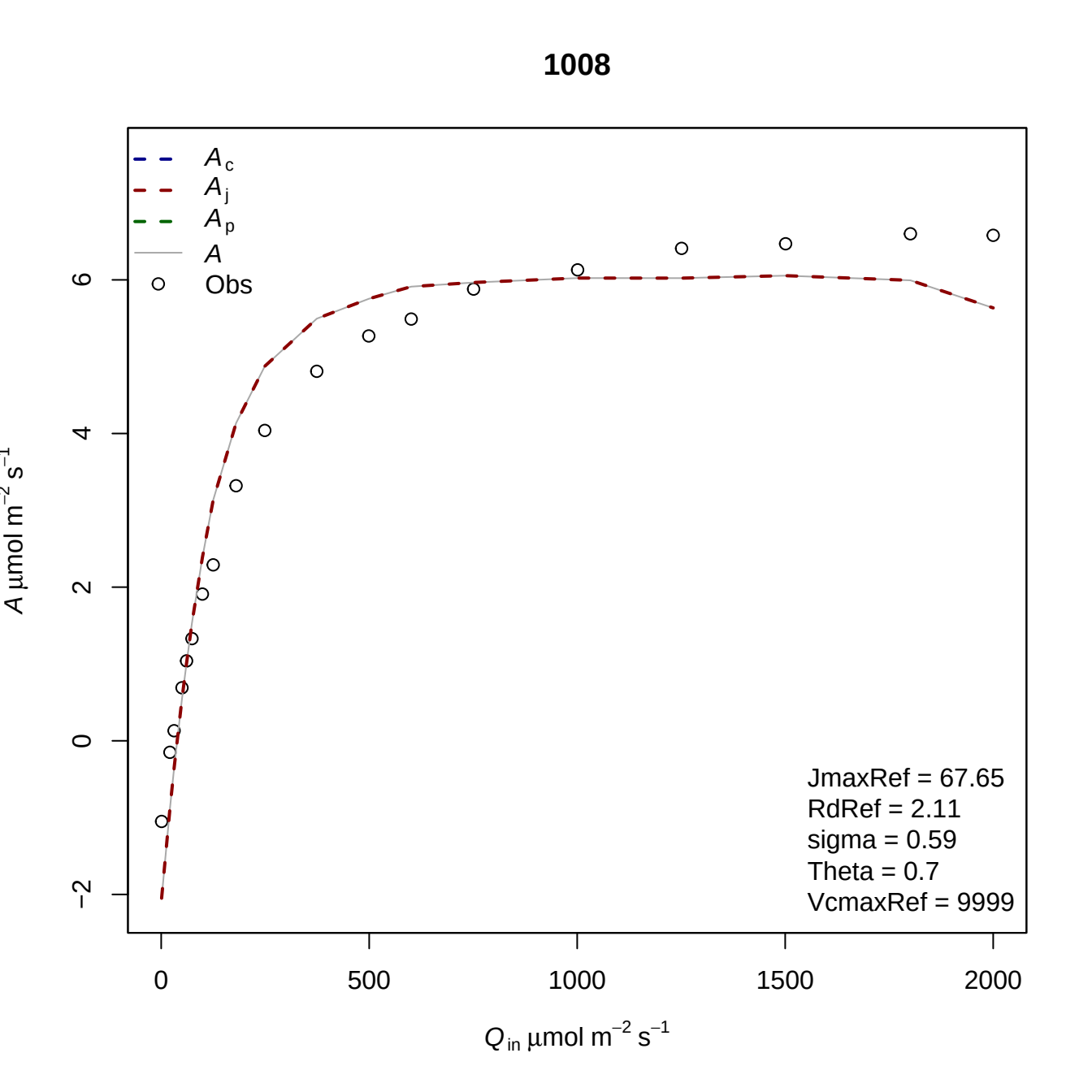

1003
A_c
A_j
A_p
A
Obs
JmaxRef = 143.04
RdRef = 2.18
sigma = 0.56
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

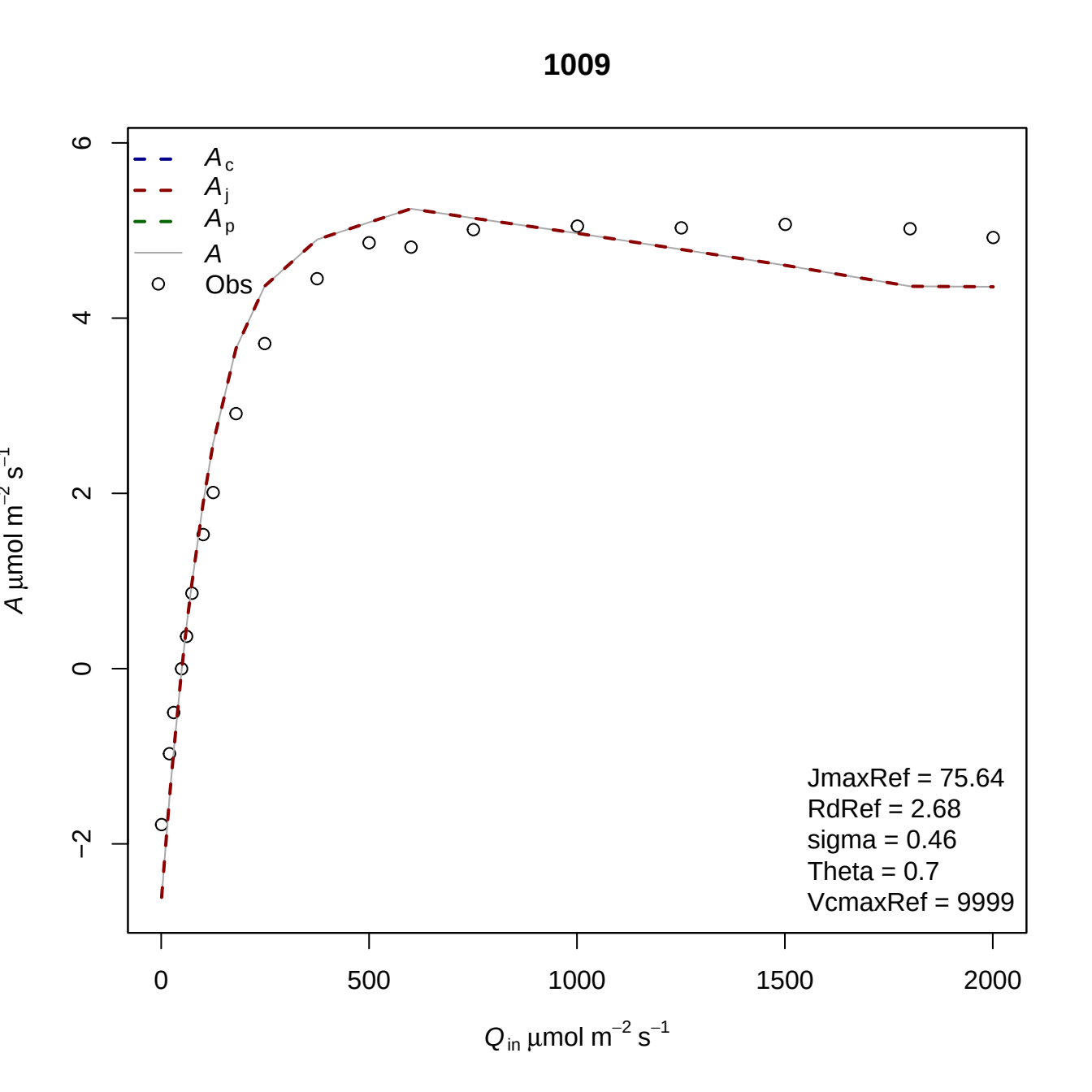

1004
A_c
A_j
A_p
A
Obs
JmaxRef = 50.03
RdRef = 1.74
sigma = 0.6
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

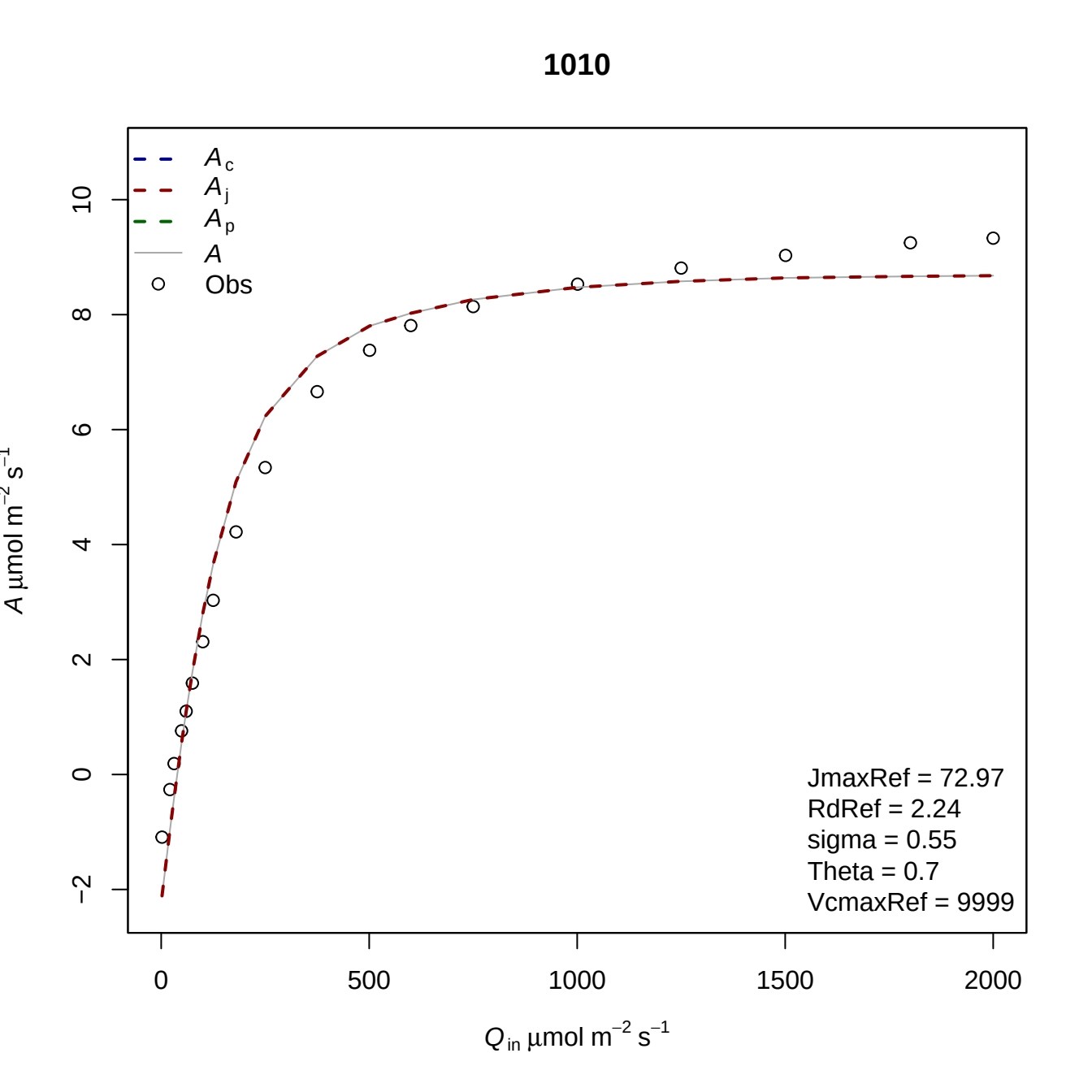

1005
A_c
A_j
A_p
A
Obs
JmaxRef = 174.38
RdRef = 2.47
sigma = 0.39
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

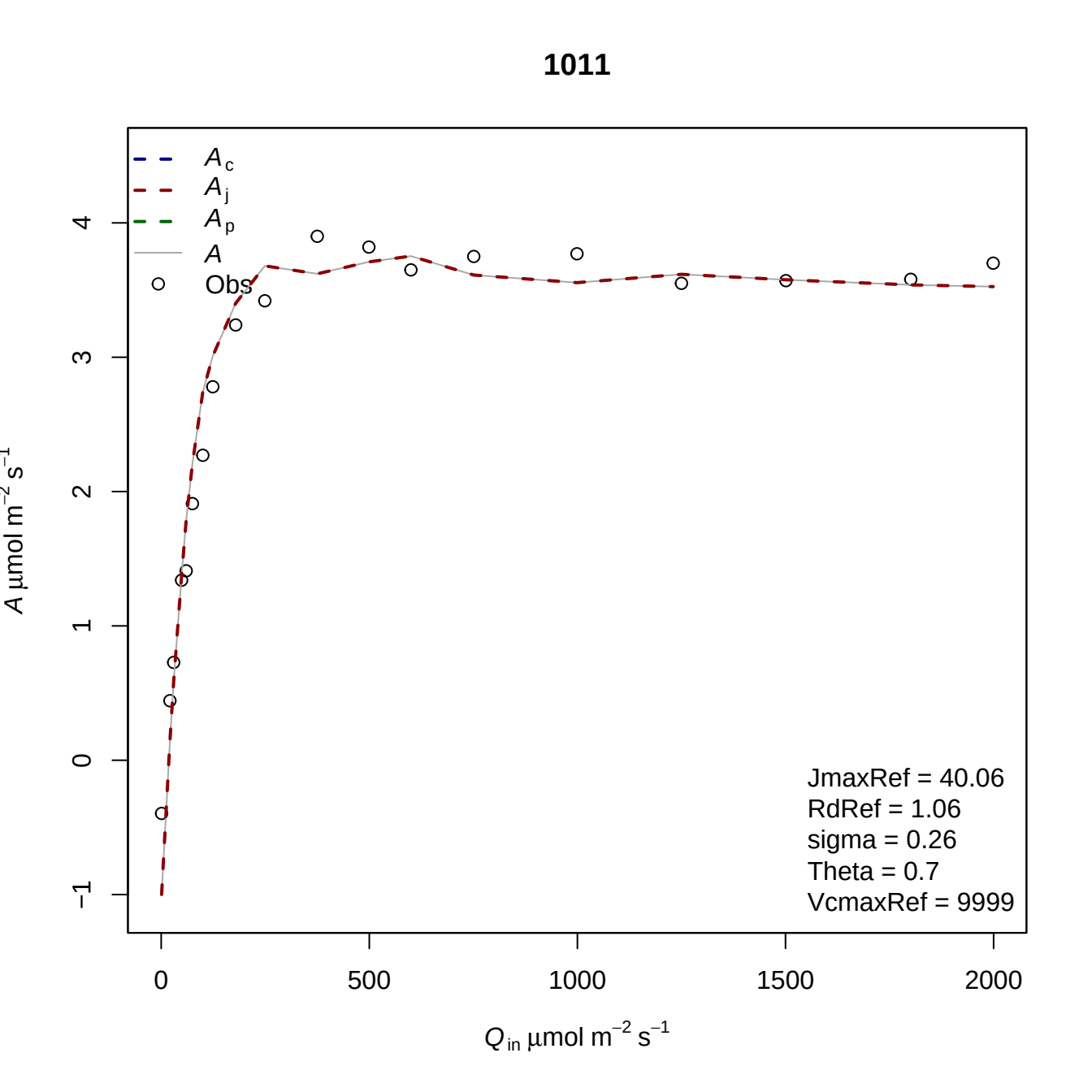

1006
A_c
A_j
A_p
A
Obs
JmaxRef = 101.89
RdRef = 2.57
sigma = 0.33
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

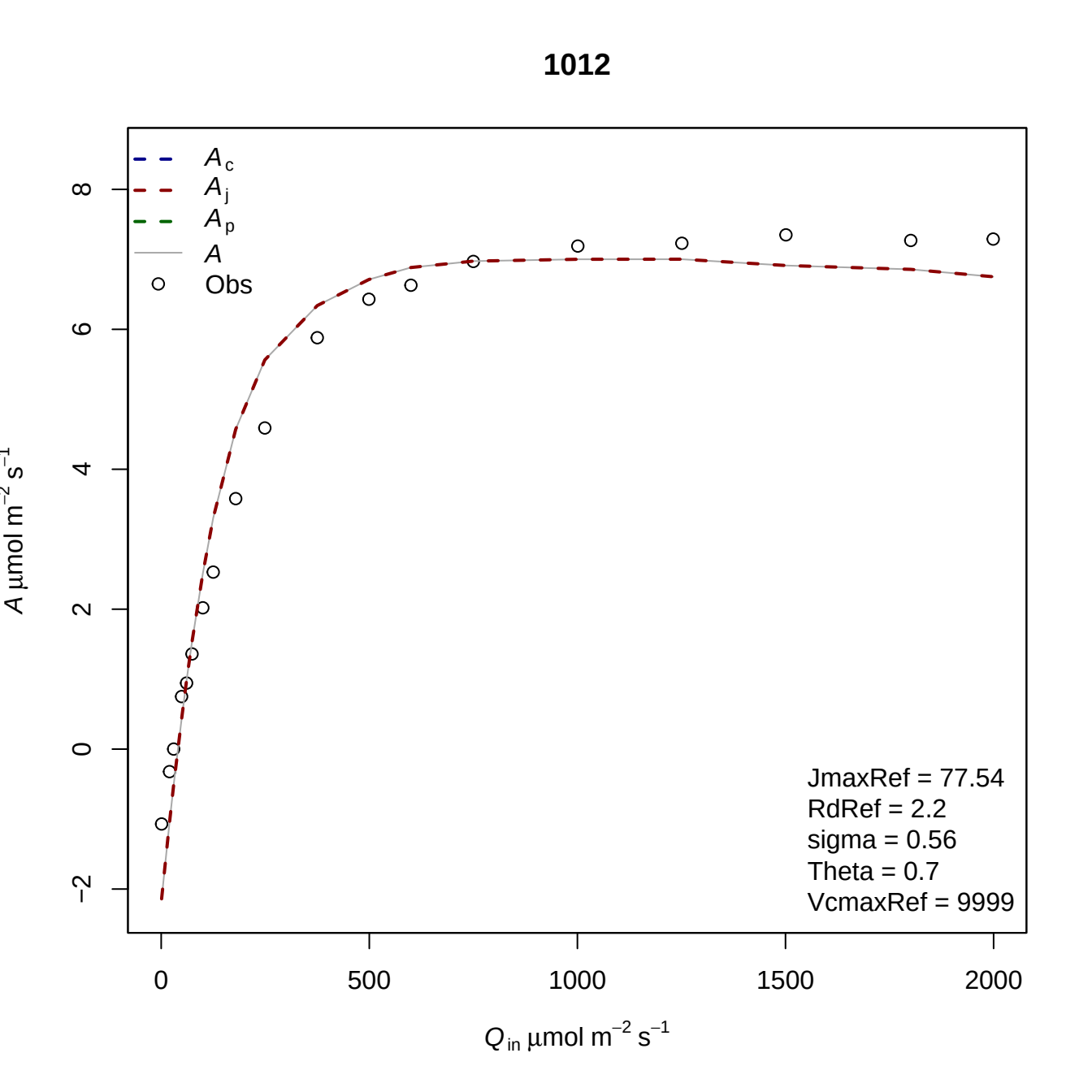

1007
A_c
A_j
A_p
A
Obs
JmaxRef = 59.5
RdRef = 1.87
sigma = 0.28
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

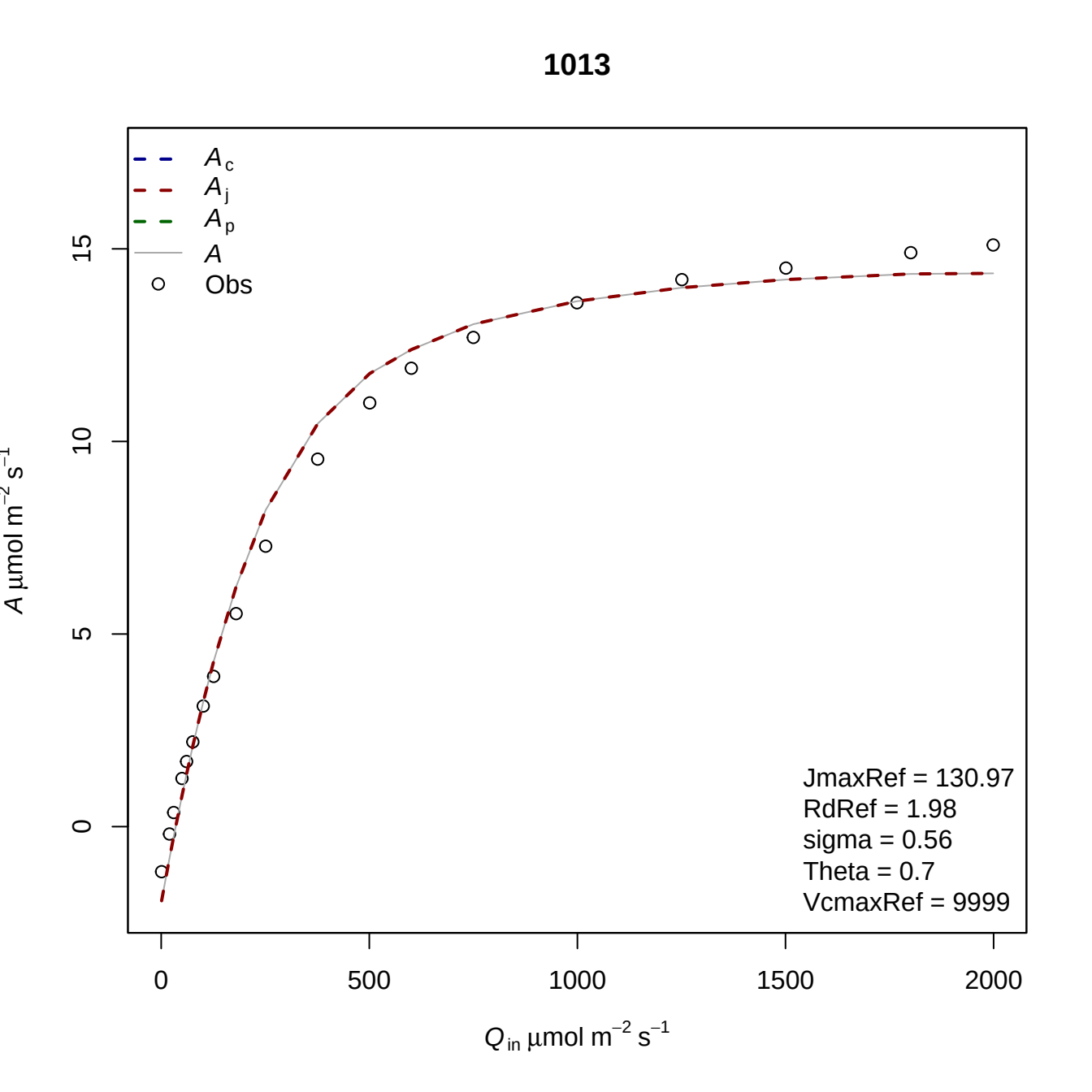

1008
A_c
A_j
A_p
A
Obs
JmaxRef = 67.65
RdRef = 2.11
sigma = 0.59
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

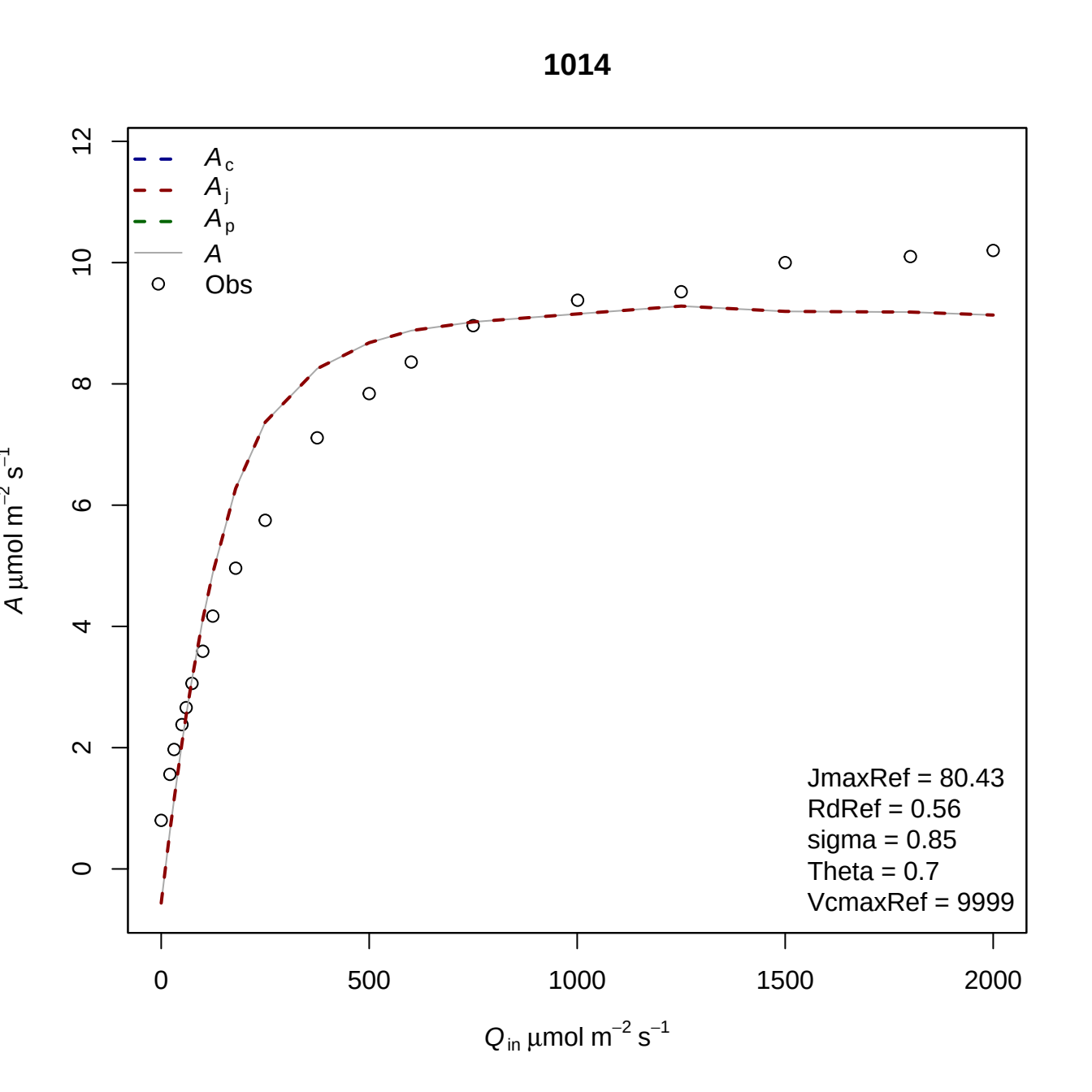

1009
A_c
A_j
A_p
A
Obs
JmaxRef = 75.64
RdRef = 2.68
sigma = 0.46
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

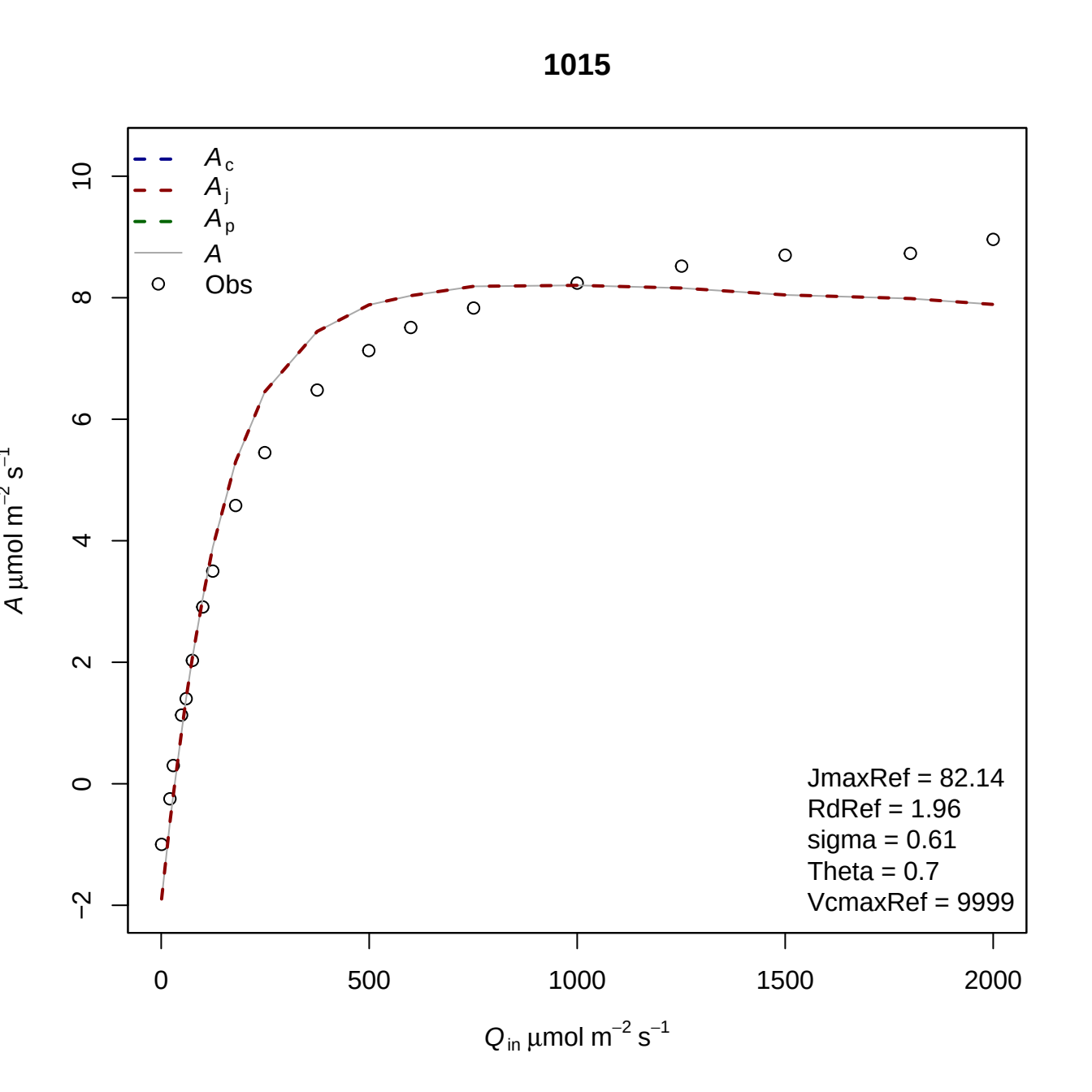

1010
A_c
A_j
A_p
A
Obs
JmaxRef = 72.97
RdRef = 2.24
sigma = 0.55
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

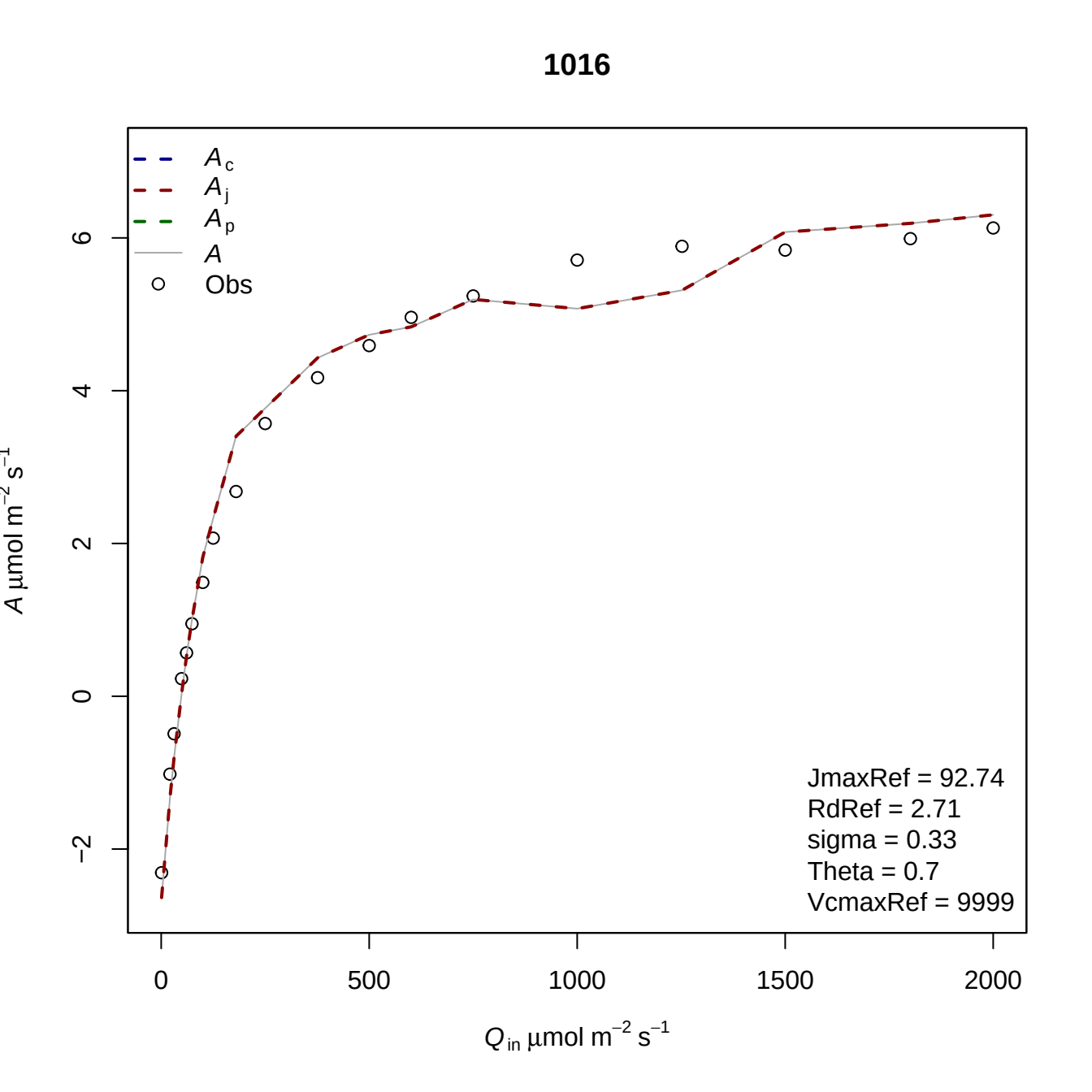

1011
A_c
A_j
A_p
A
Obs
JmaxRef = 40.06
RdRef = 1.06
sigma = 0.26
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500
2000

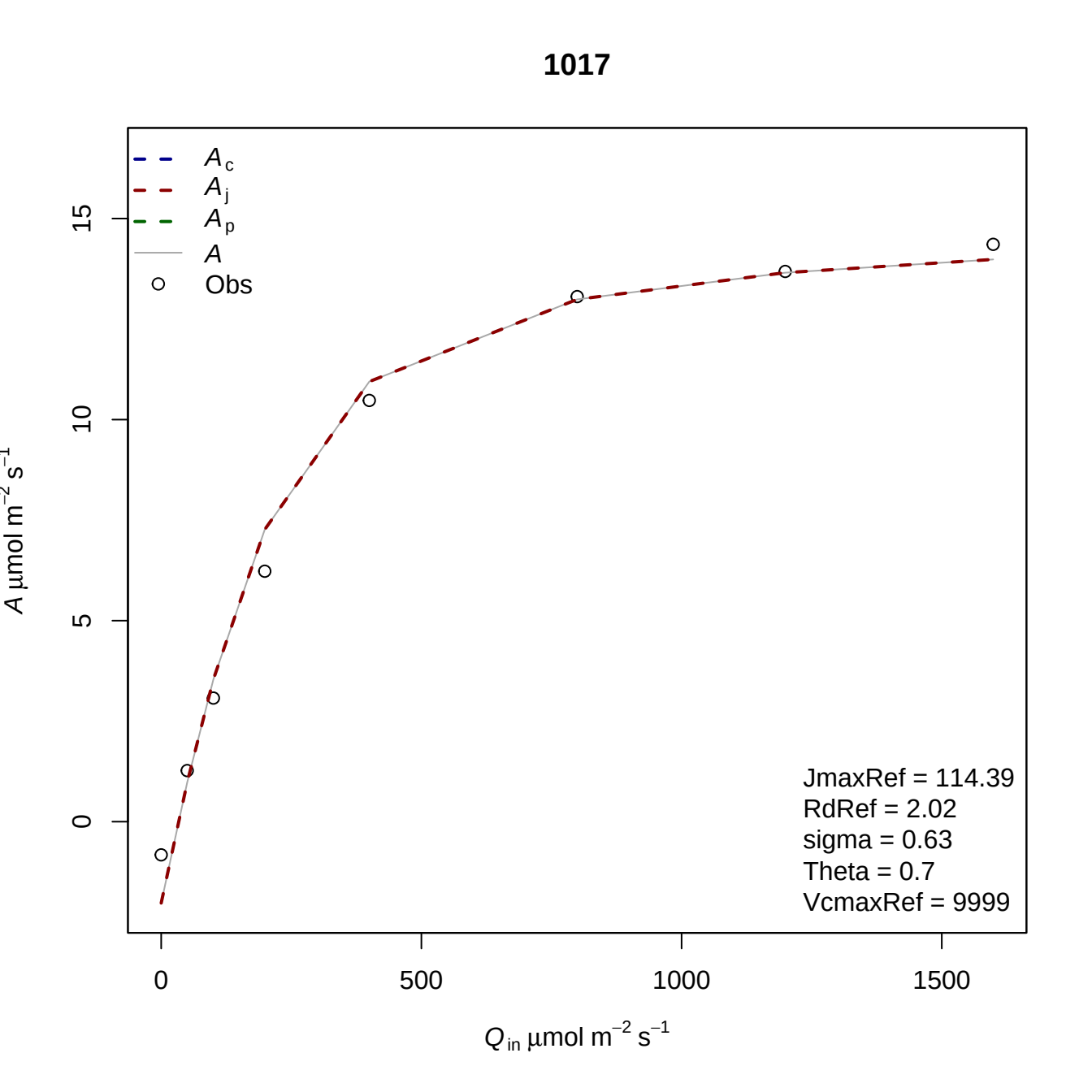

1012
A_c
A_j
A_p
A
Obs
JmaxRef = 77.54
RdRef = 2.2
sigma = 0.56
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

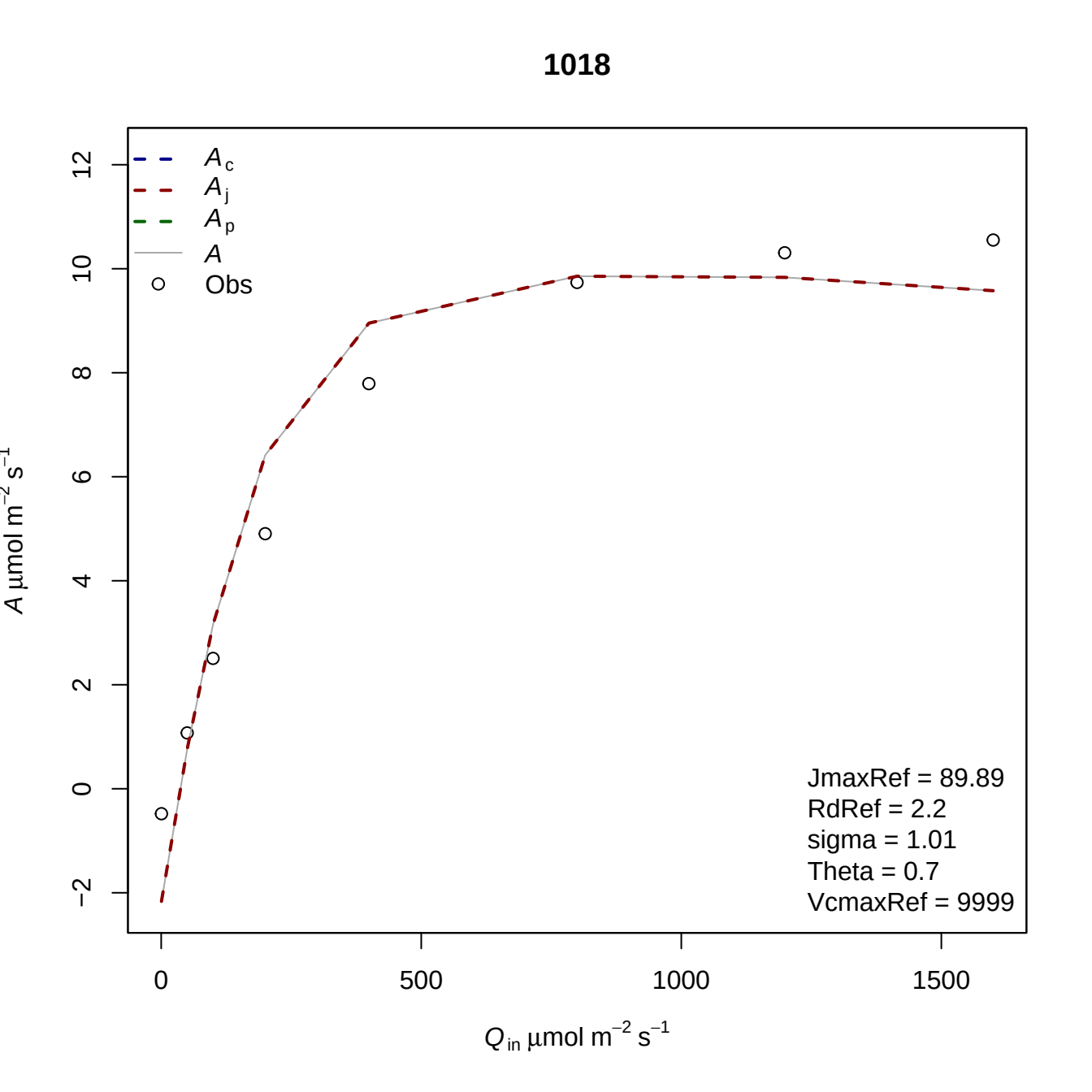

1013
A_c
A_j
A_p
A
Obs
JmaxRef = 130.97
RdRef = 1.98
sigma = 0.56
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

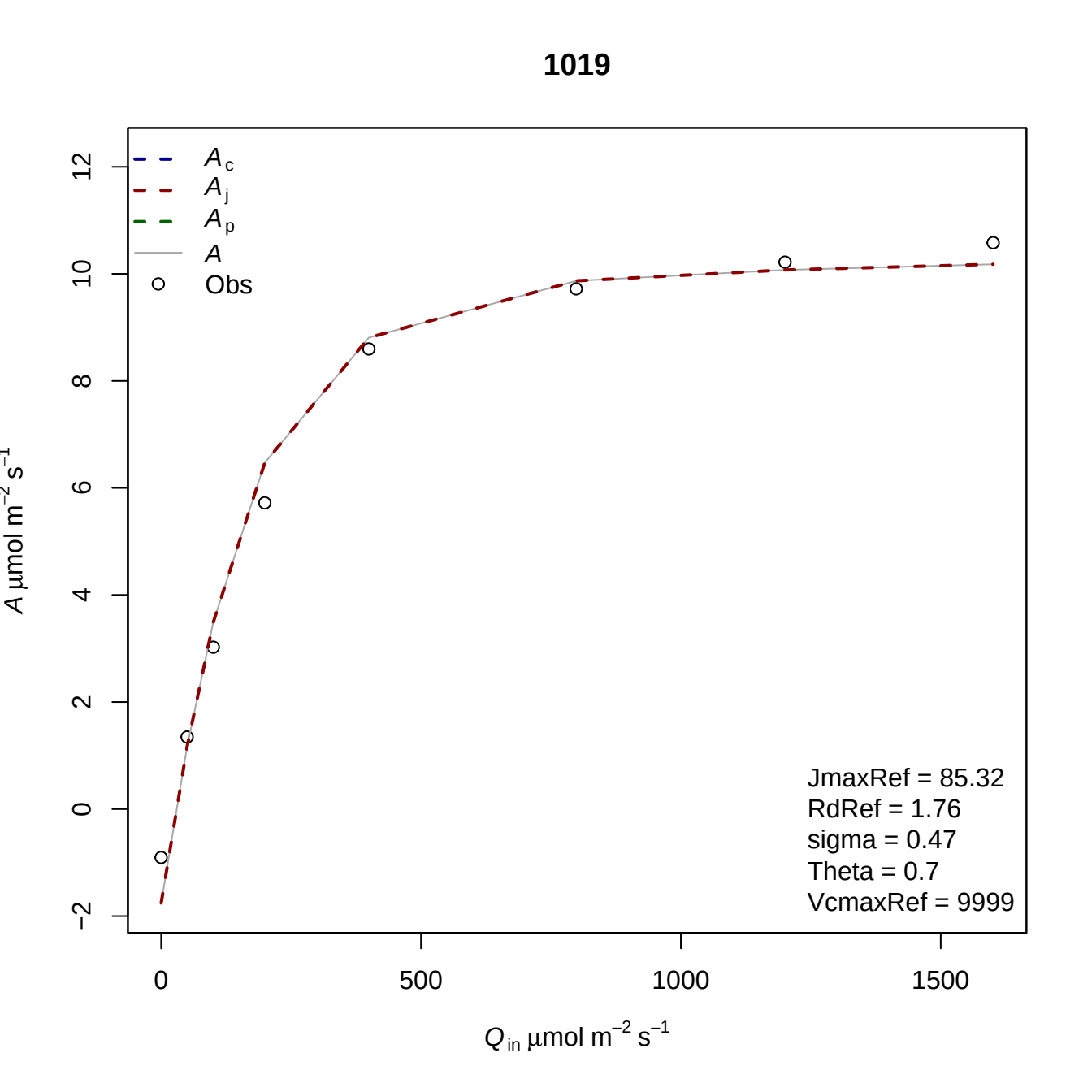

1014
A_c
A_j
A_p
A
Obs
JmaxRef = 80.43
RdRef = 0.56
sigma = 0.85
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

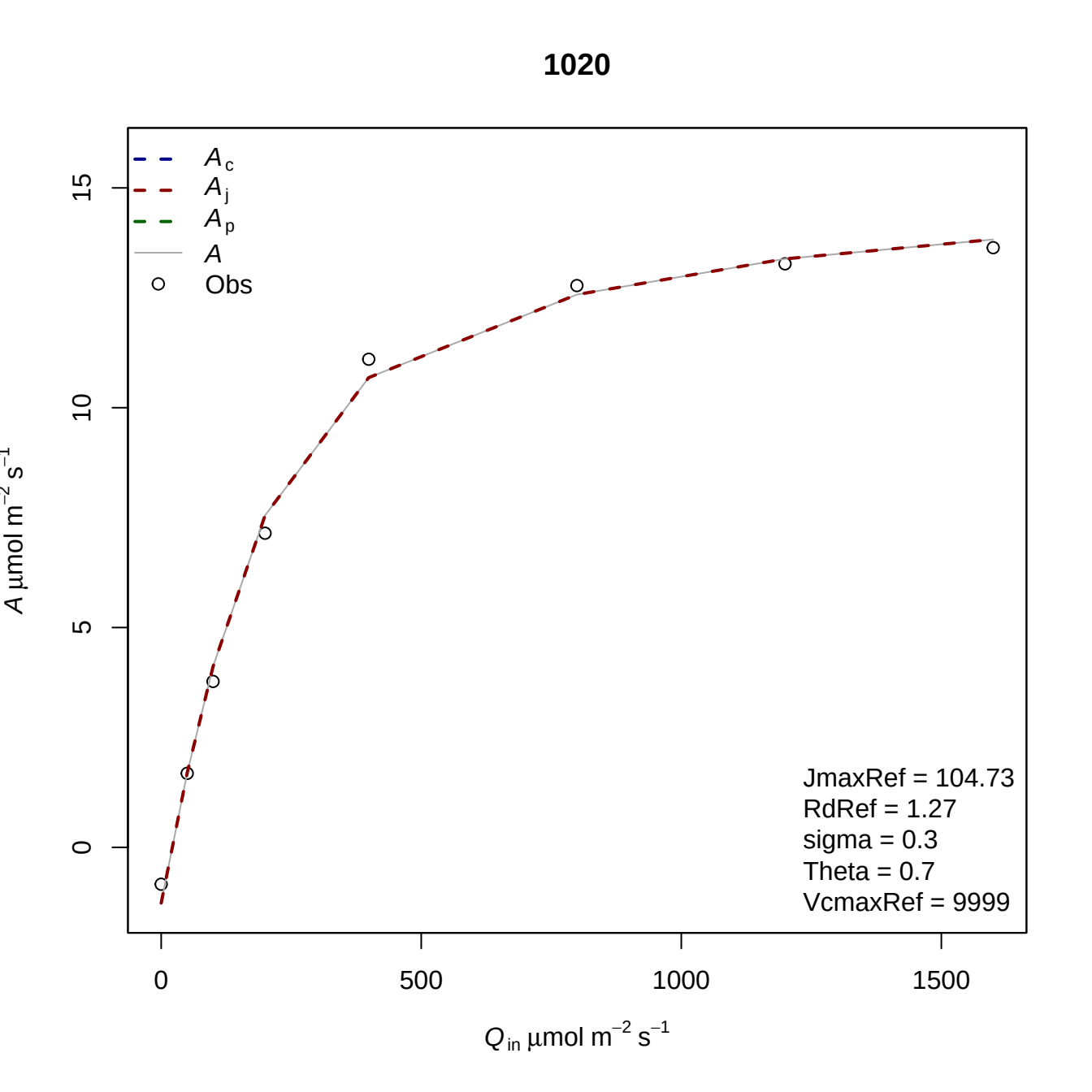

1015
A_c
A_j
A_p
A
Obs
JmaxRef = 82.14
RdRef = 1.96
sigma = 0.61
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

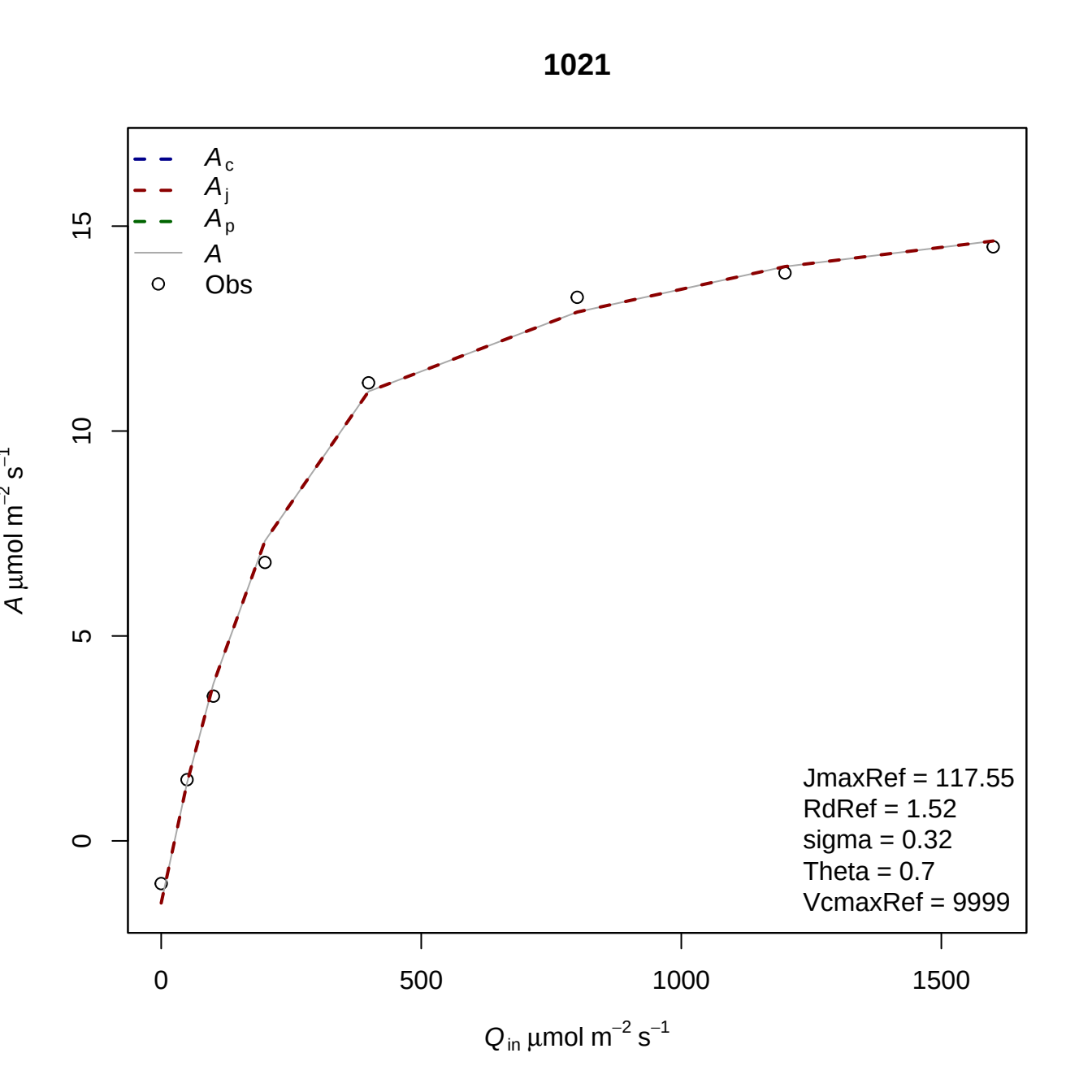

1016
A_c
A_j
A_p
A
Obs
JmaxRef = 92.74
RdRef = 2.71
sigma = 0.33
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

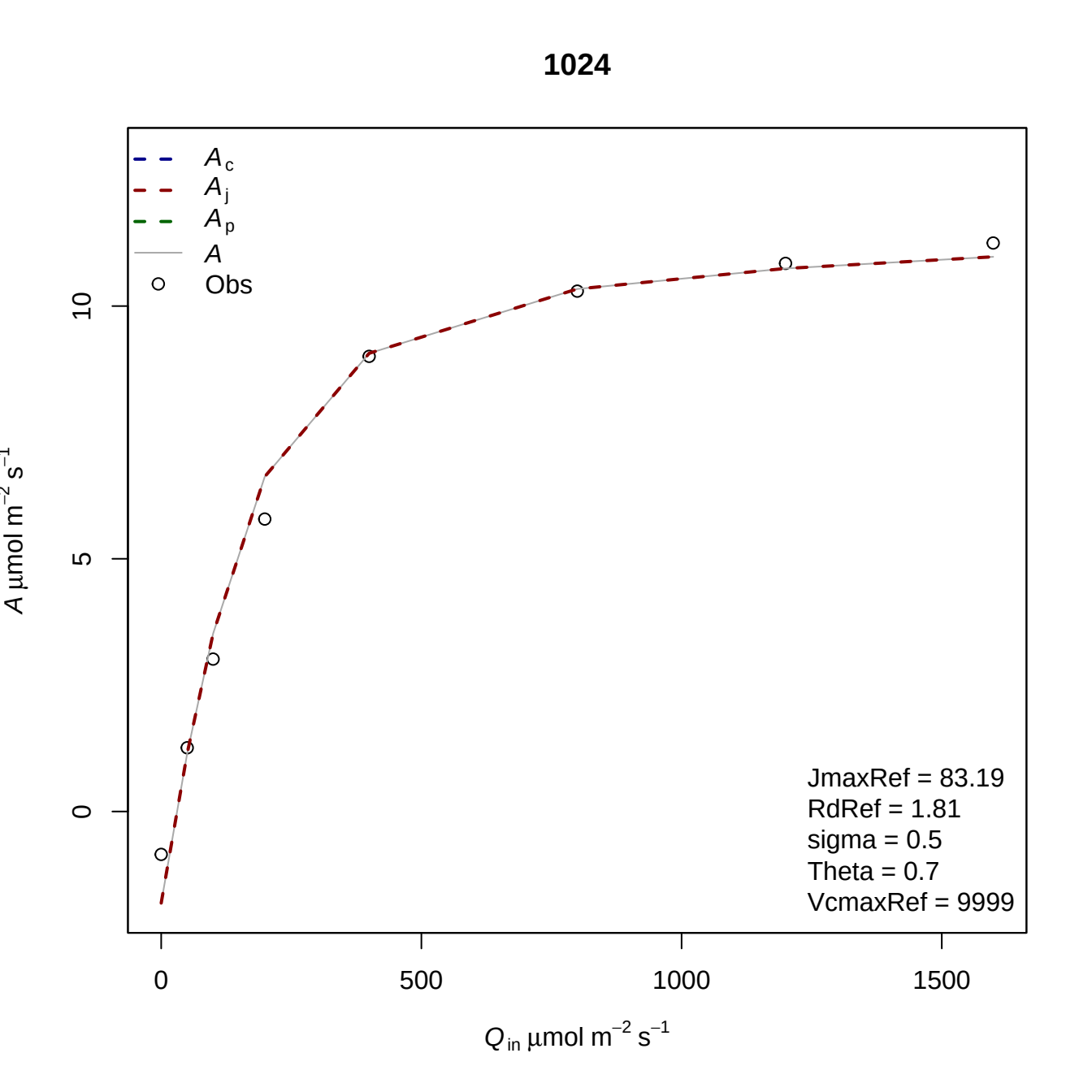

1017
A_c
A_j
A_p
A
Obs
JmaxRef = 114.39
RdRef = 2.02
sigma = 0.63
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

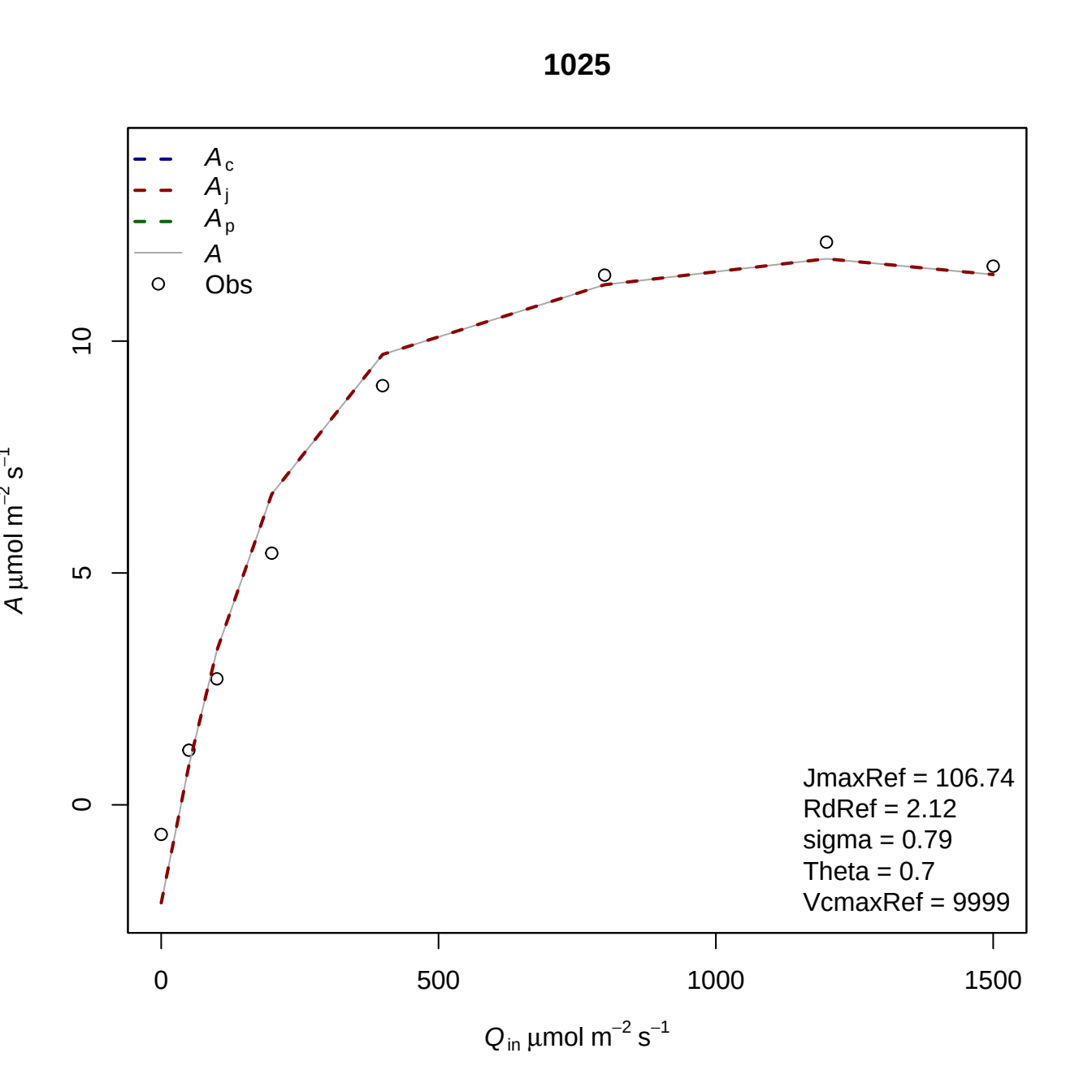

1018
A_c
A_j
A_p
A
Obs
JmaxRef = 89.89
RdRef = 2.2
sigma = 1.01
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

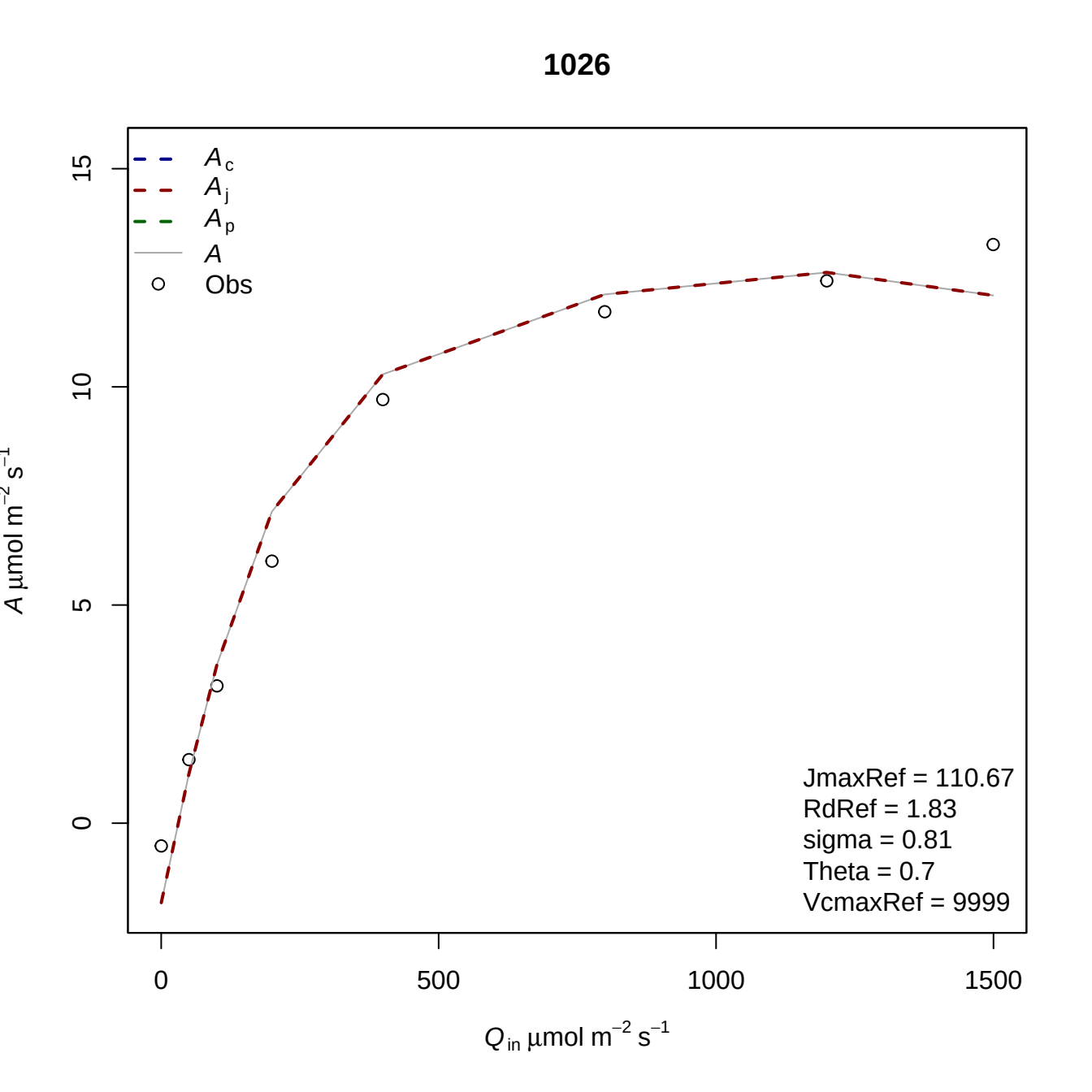

1019
A_c
A_j
A_p
A
Obs
JmaxRef = 85.32
RdRef = 1.76
sigma = 0.47
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

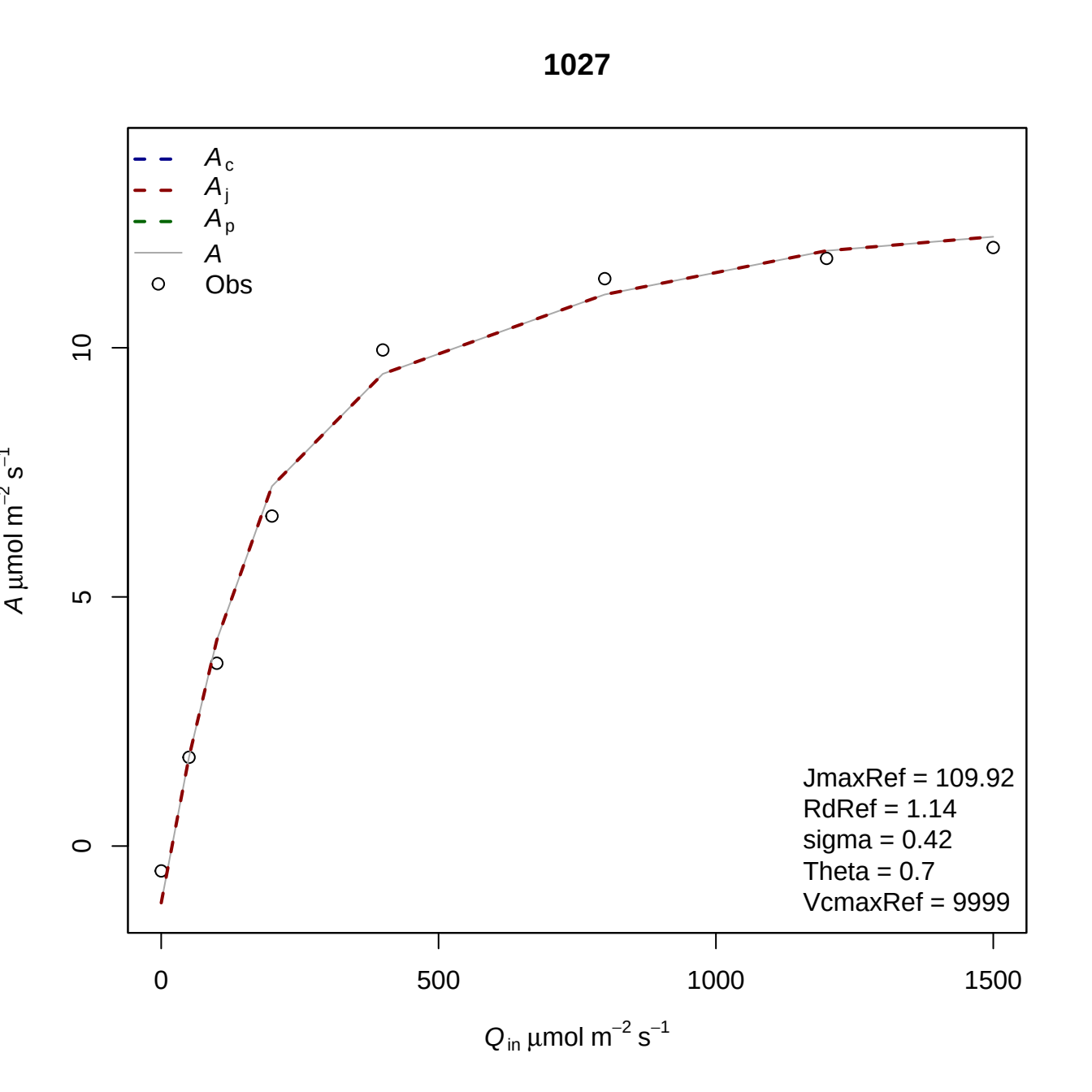

1020
A_c
A_j
A_p
A
Obs
JmaxRef = 104.73
RdRef = 1.27
sigma = 0.3
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

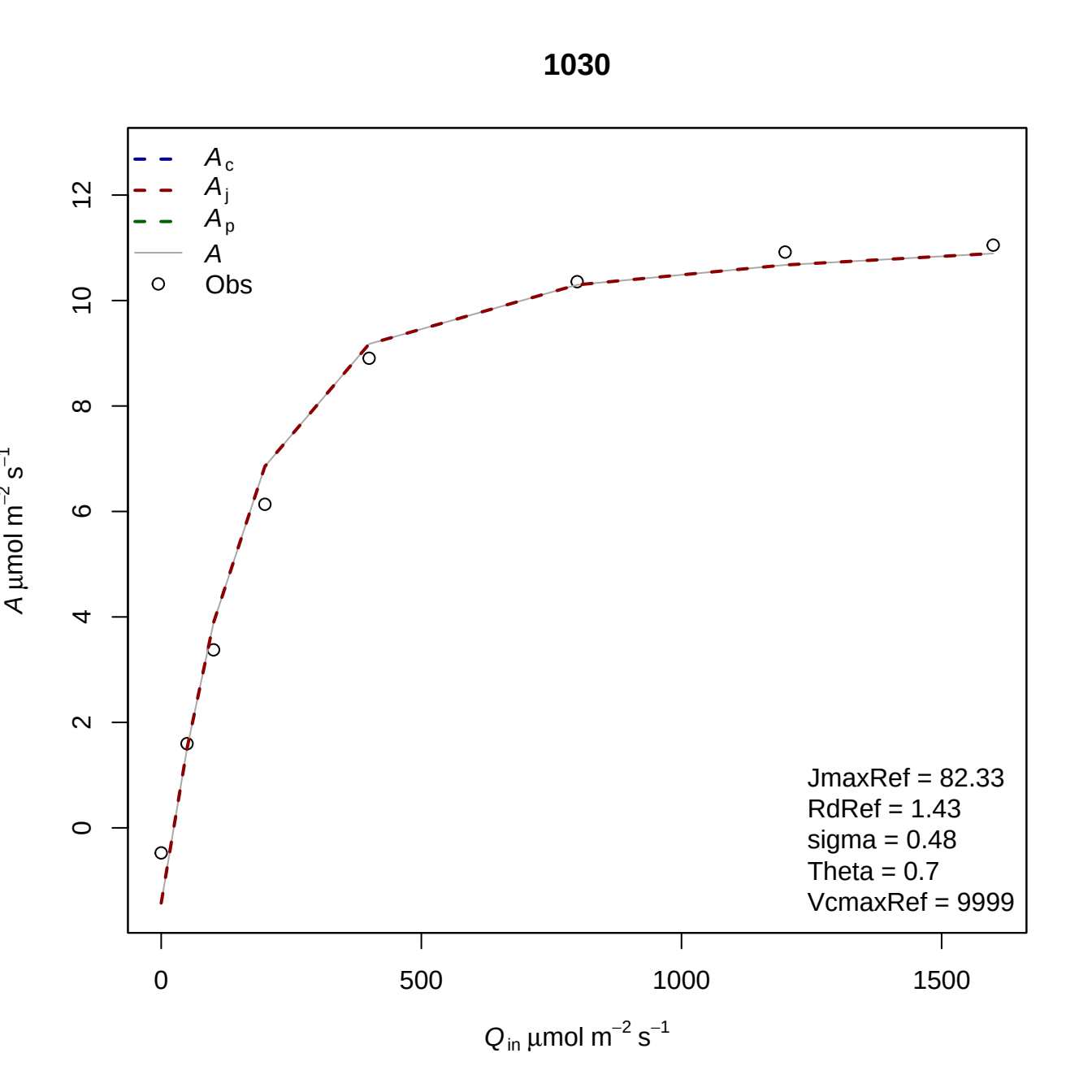

1021
A_c
A_j
A_p
A
Obs
JmaxRef = 117.55
RdRef = 1.52
sigma = 0.32
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

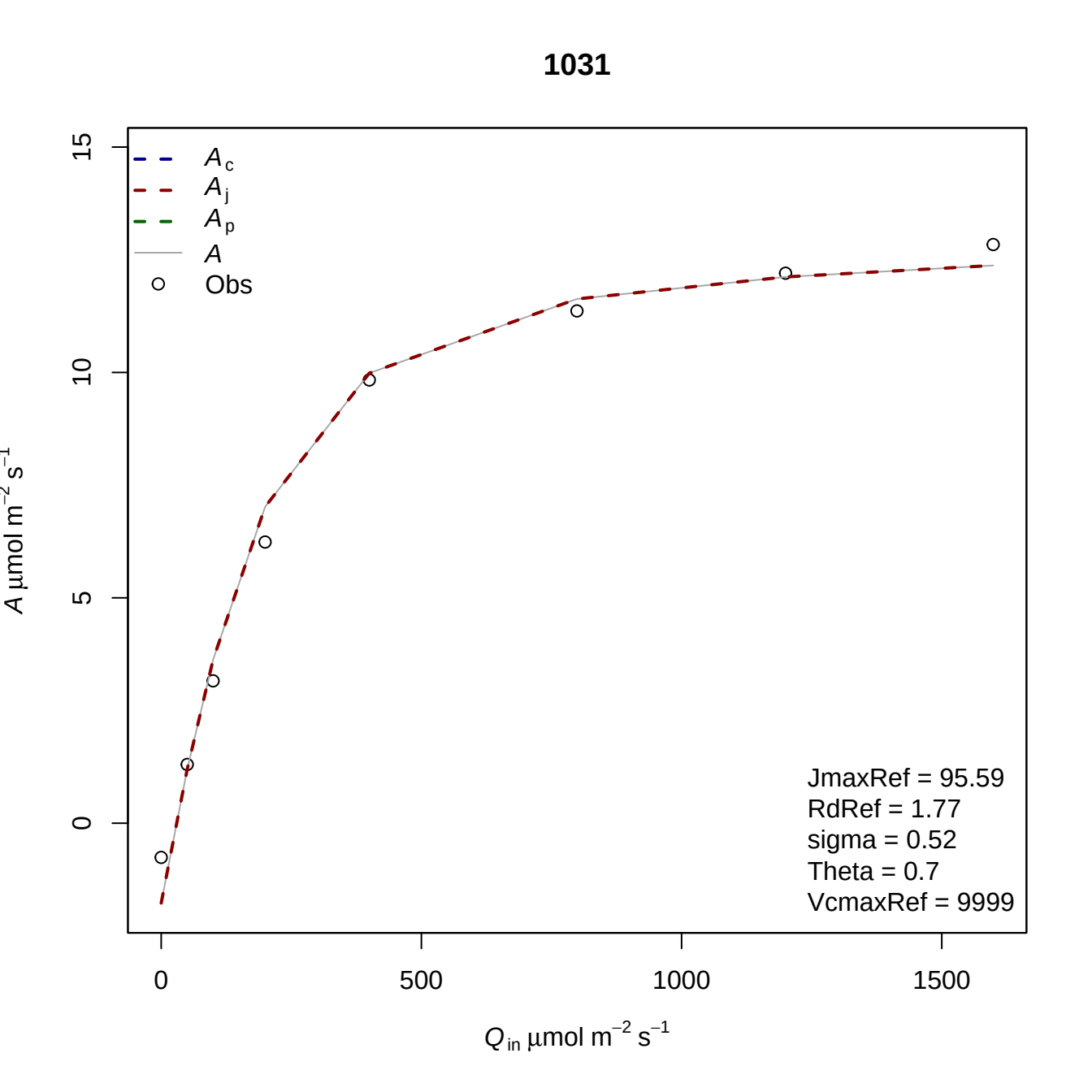

1024
A_c
A_j
A_p
A
Obs
JmaxRef = 83.19
RdRef = 1.81
sigma = 0.5
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

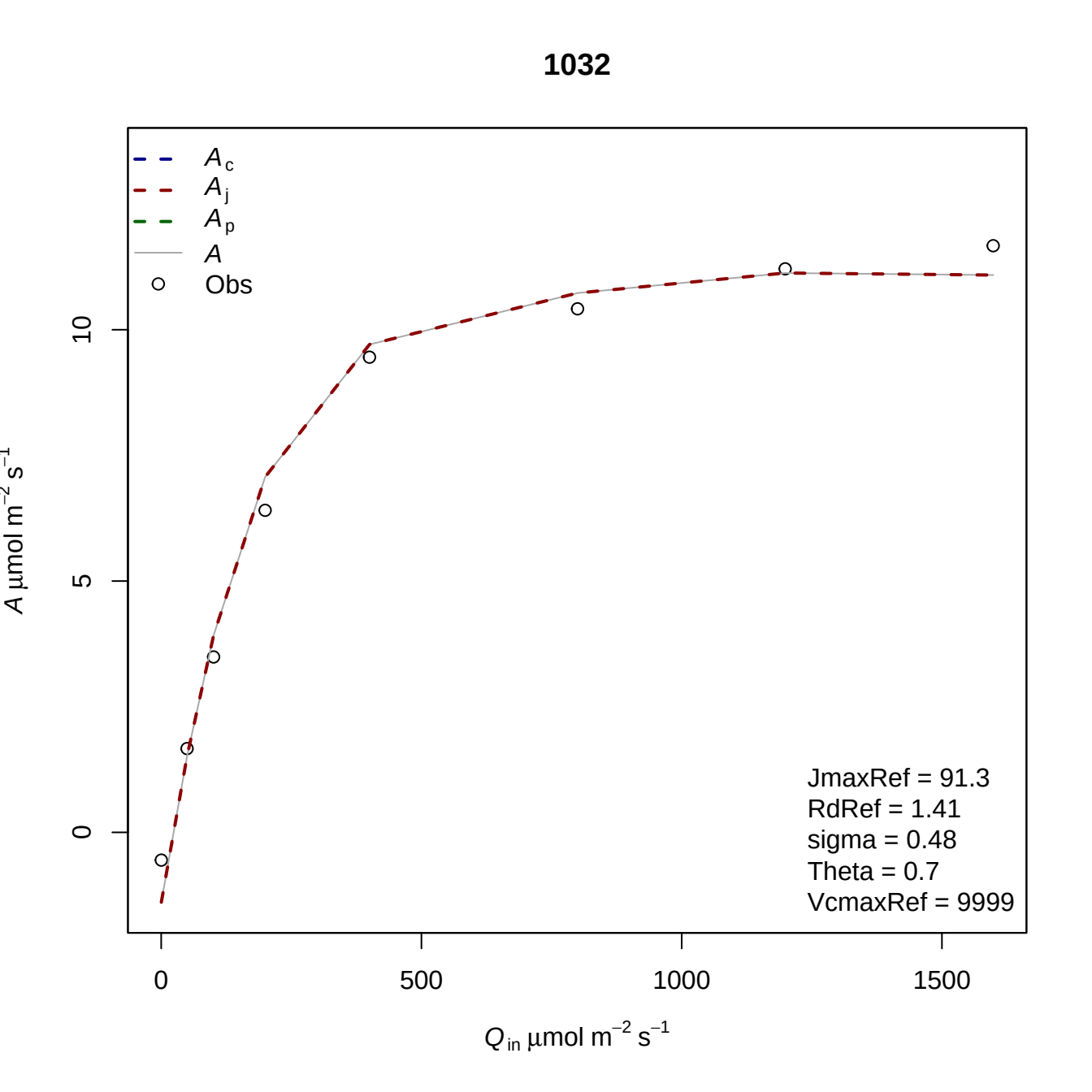

1025
A_c
A_j
A_p
A
Obs
JmaxRef = 106.74
RdRef = 2.12
sigma = 0.79
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

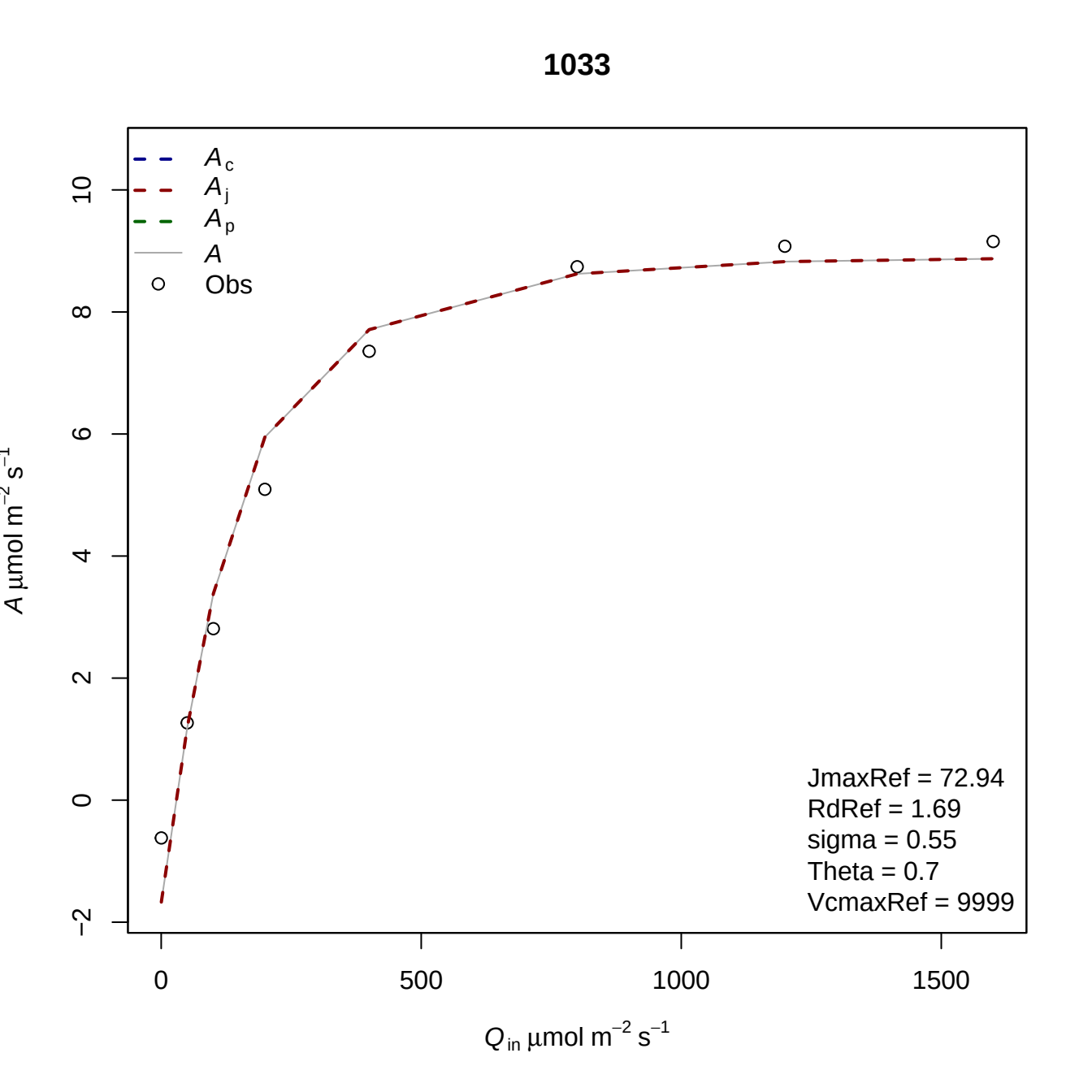

1026
A_c
A_j
A_p
A
Obs
JmaxRef = 110.67
RdRef = 1.83
sigma = 0.81
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

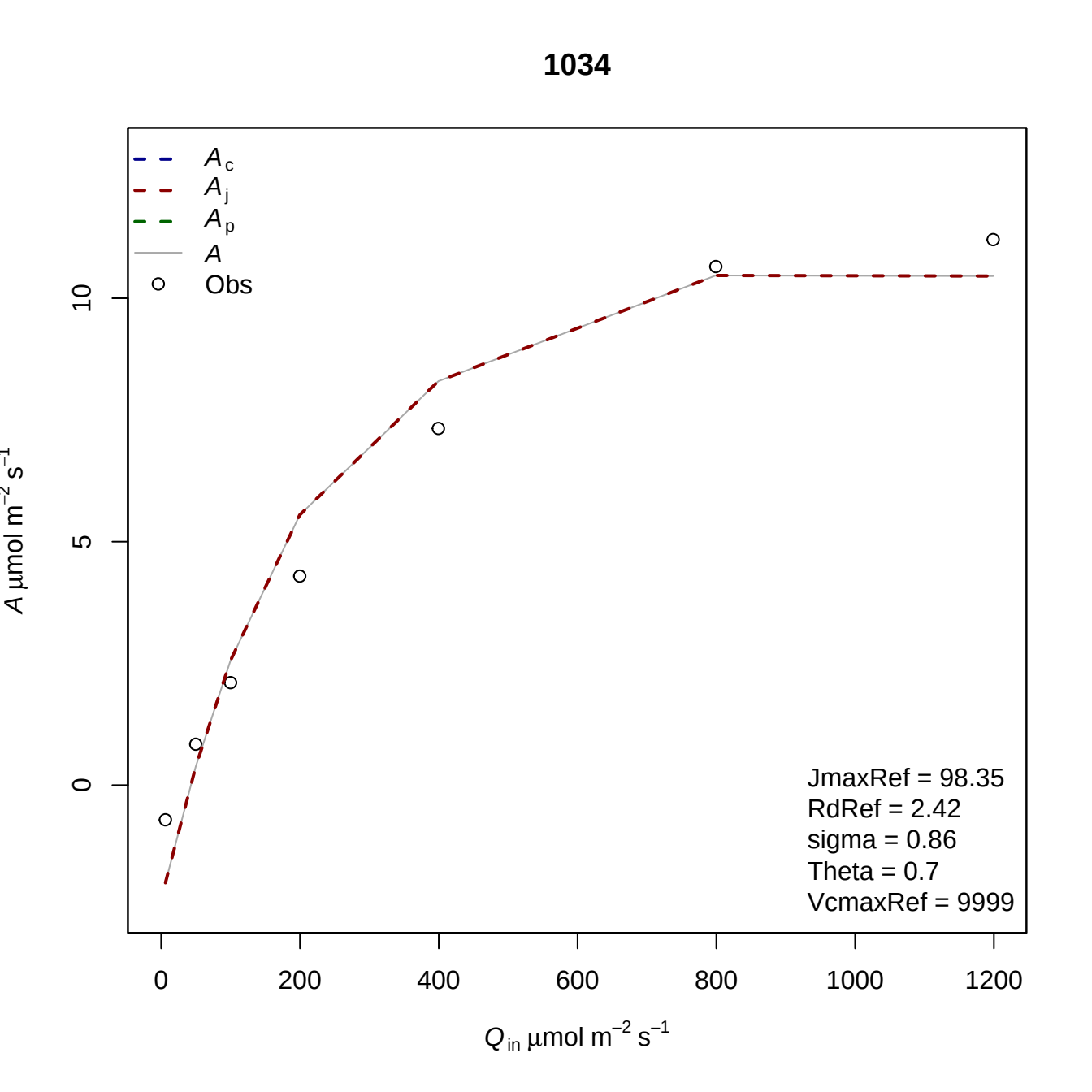

1027
A_c
A_j
A_p
A
Obs
JmaxRef = 109.92
RdRef = 1.14
sigma = 0.42
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

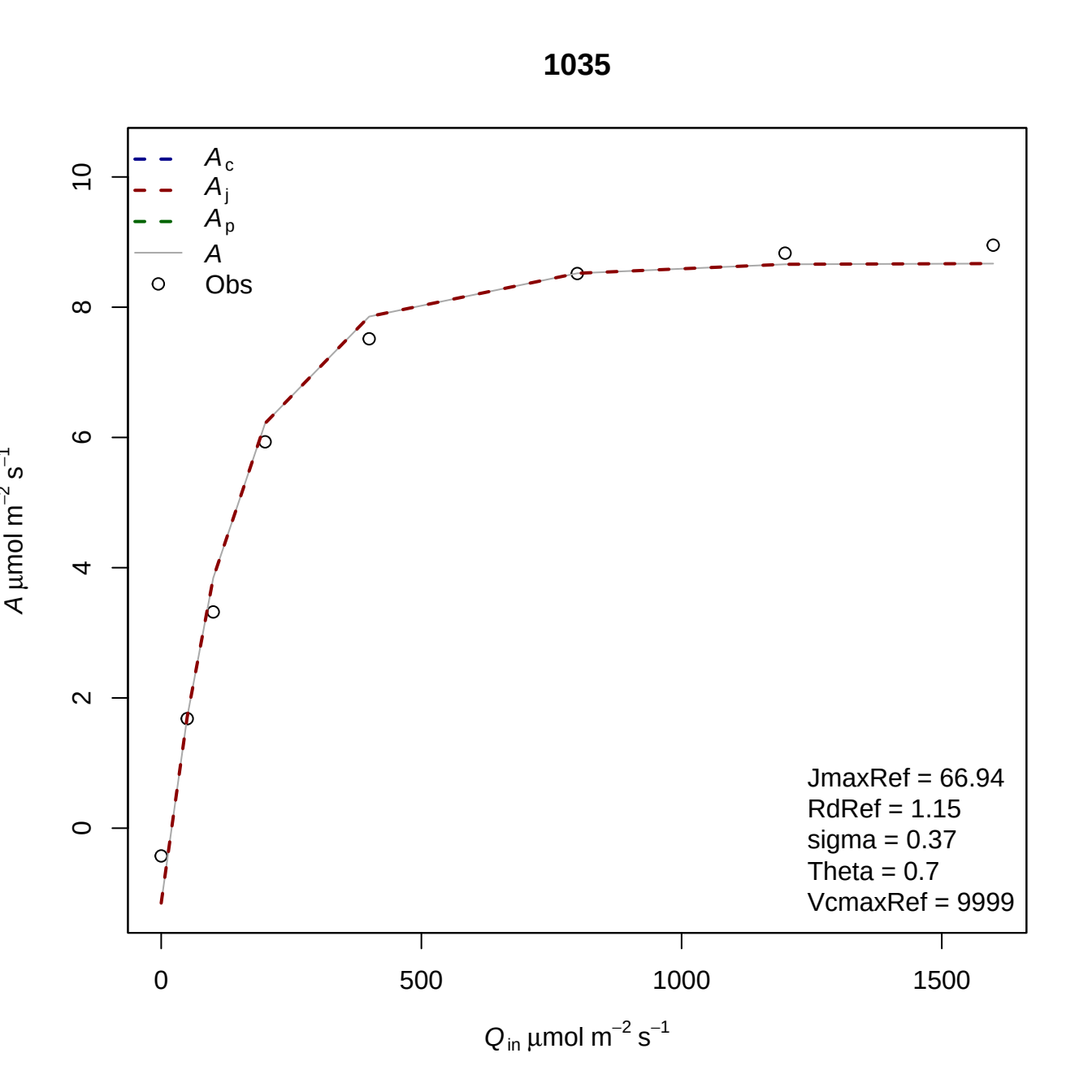

1030
A_c
A_j
A_p
A
Obs
JmaxRef = 82.33
RdRef = 1.43
sigma = 0.48
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

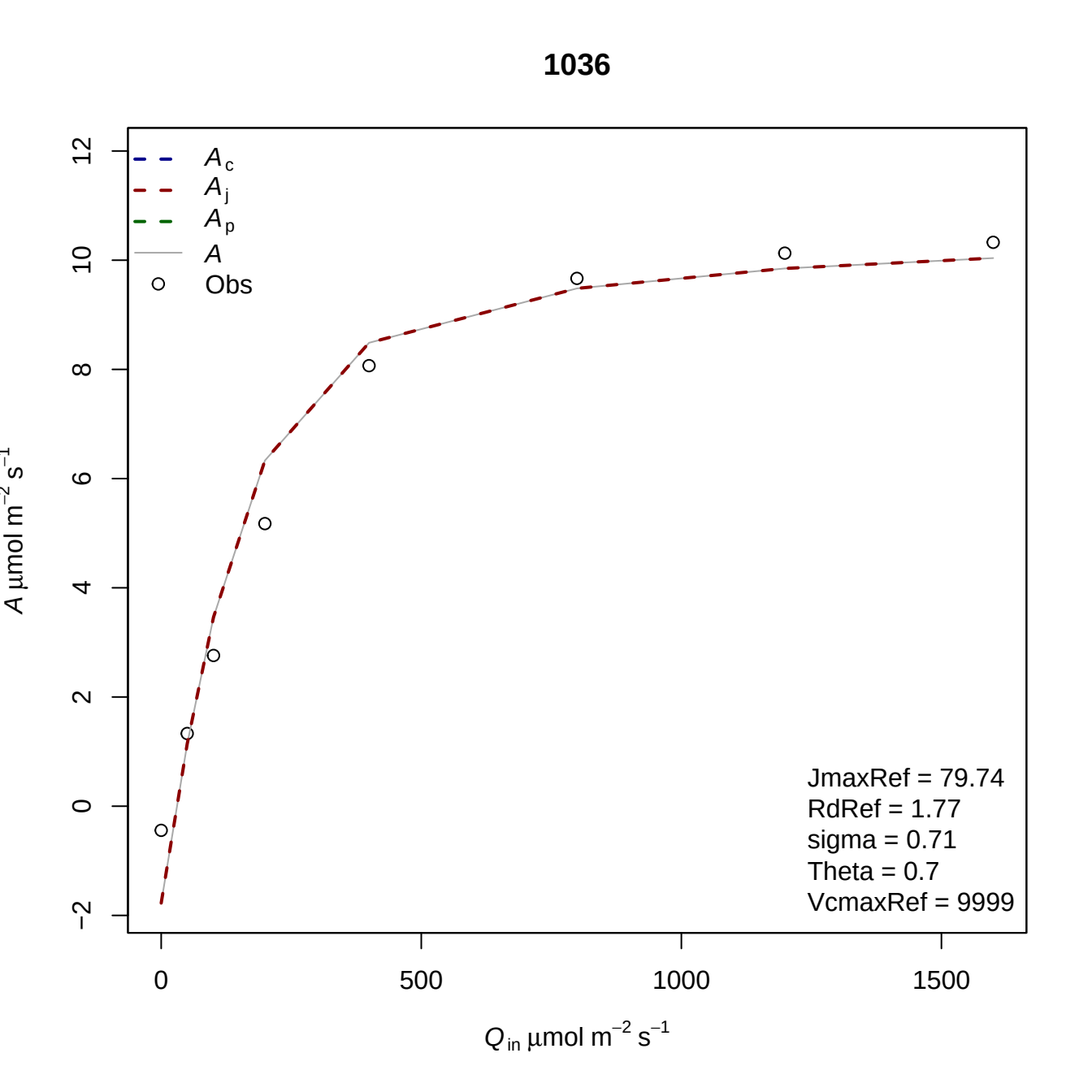

1031
A_c
A_j
A_p
A
Obs
JmaxRef = 95.59
RdRef = 1.77
sigma = 0.52
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

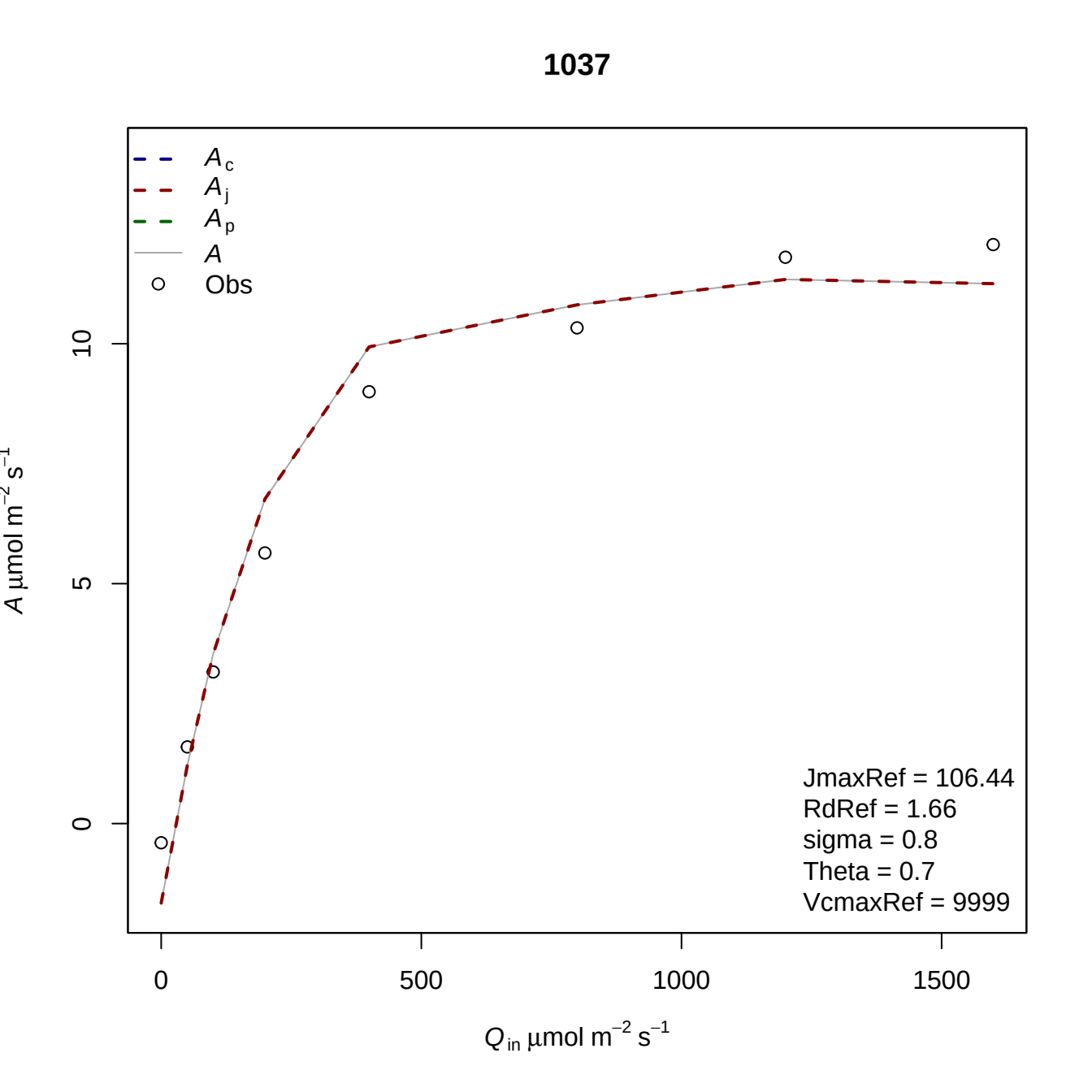

1032
A_c
A_j
A_p
A
Obs
JmaxRef = 91.3
RdRef = 1.41
sigma = 0.48
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

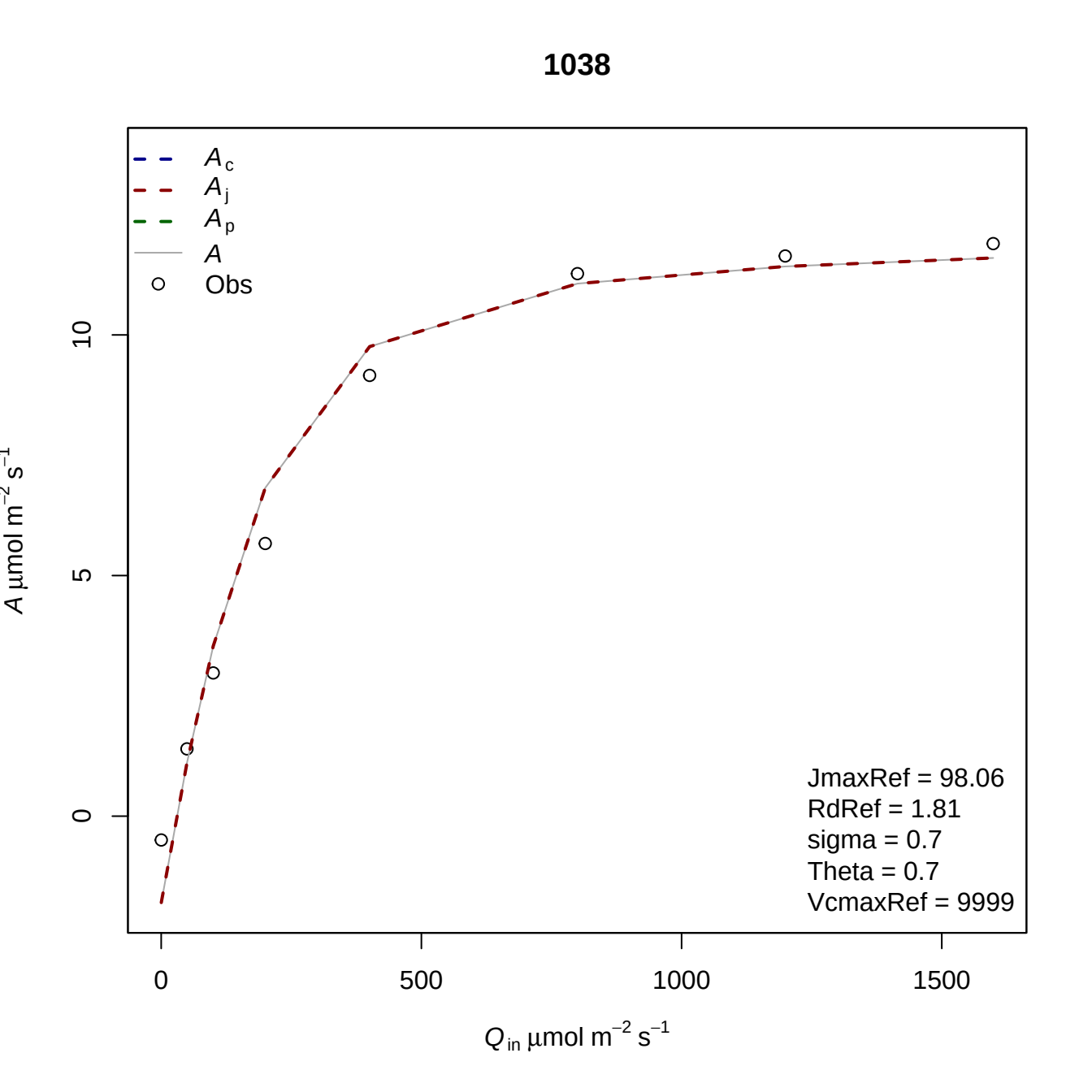

1033
A_c
A_j
A_p
A
Obs
JmaxRef = 72.94
RdRef = 1.69
sigma = 0.55
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

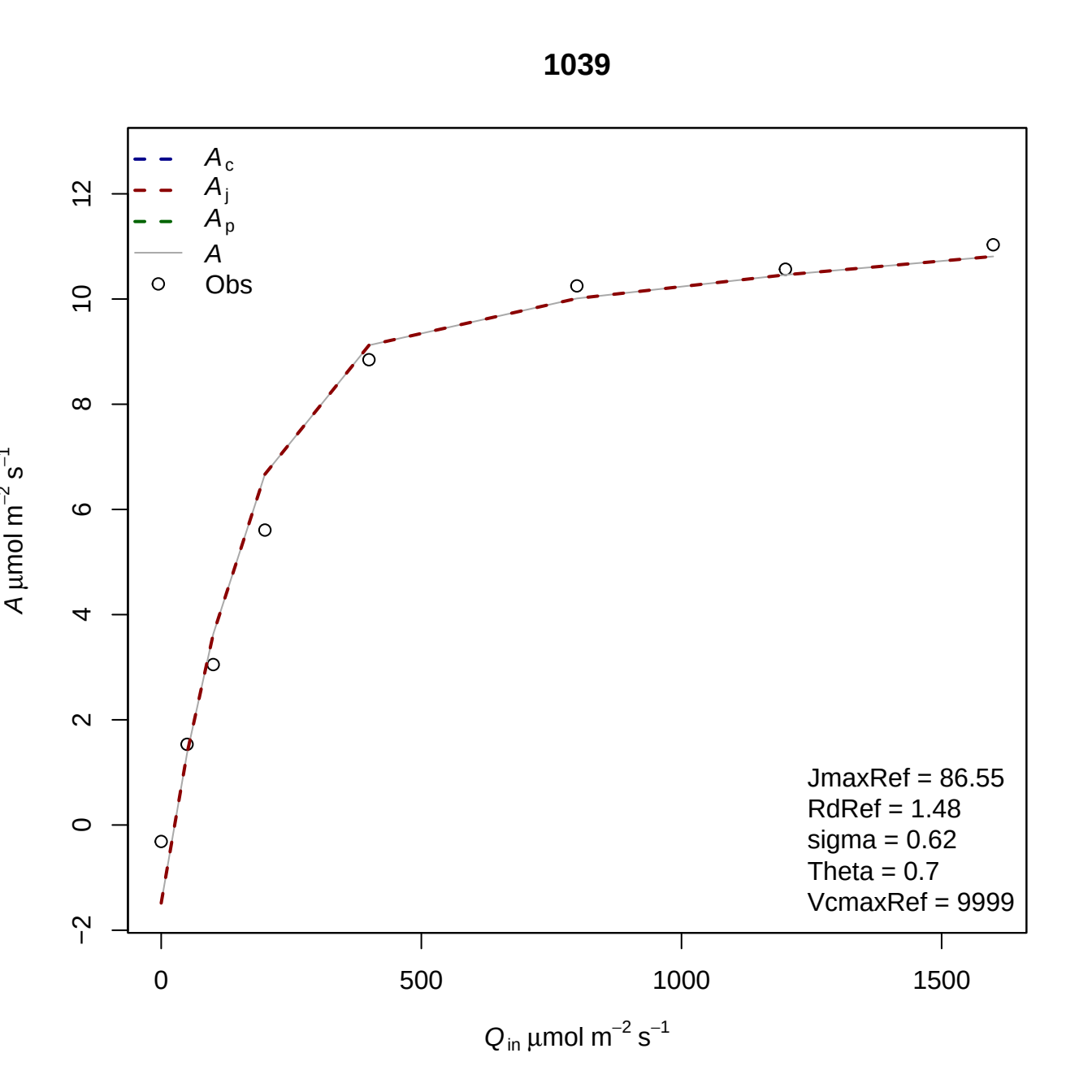

1034
A_c
A_j
A_p
A
Obs
JmaxRef = 98.35
RdRef = 2.42
sigma = 0.86
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800
1000
1200
Q_{in} µmol m^{-2} s^{-1}

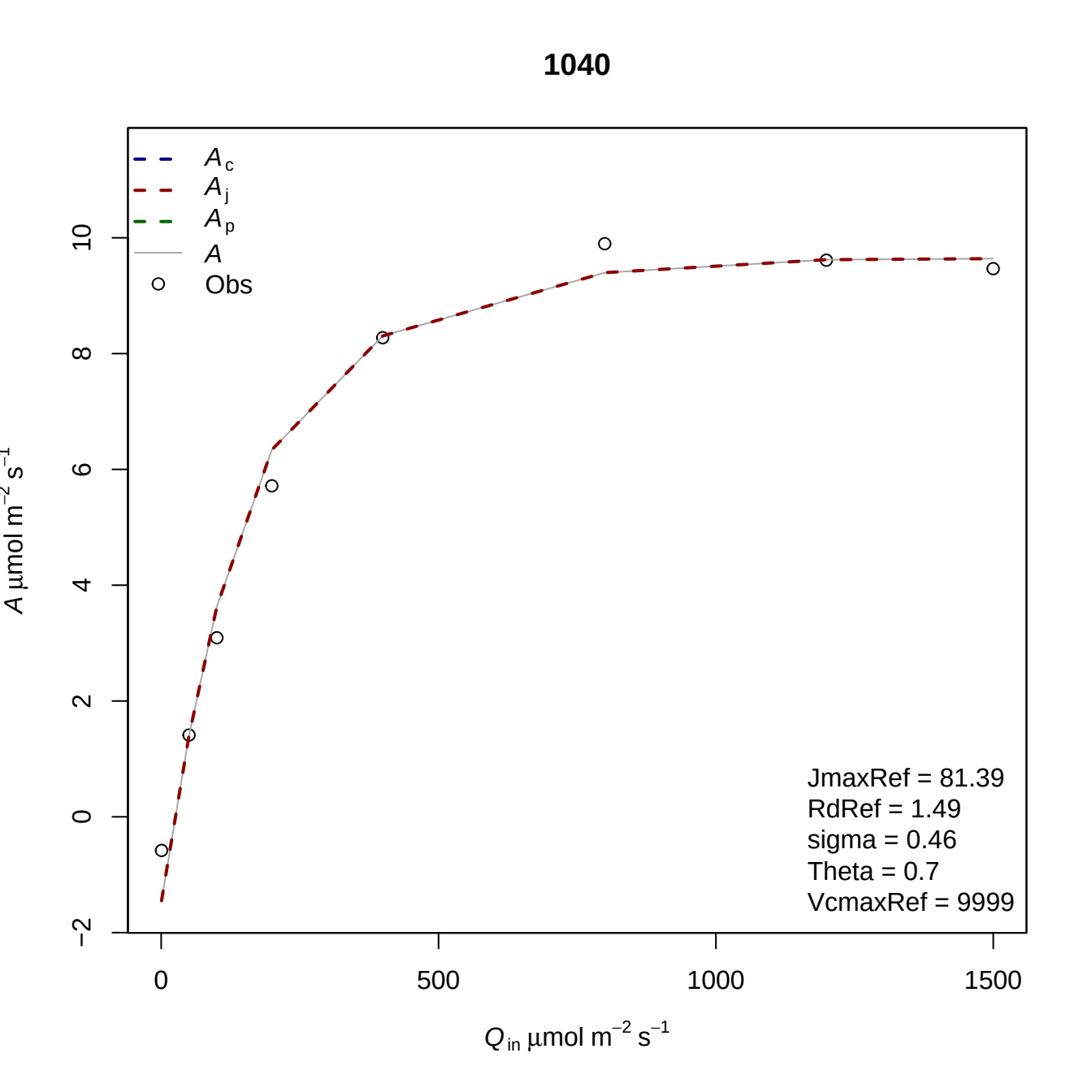

1035
A_c
A_j
A_p
A
Obs
JmaxRef = 66.94
RdRef = 1.15
sigma = 0.37
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

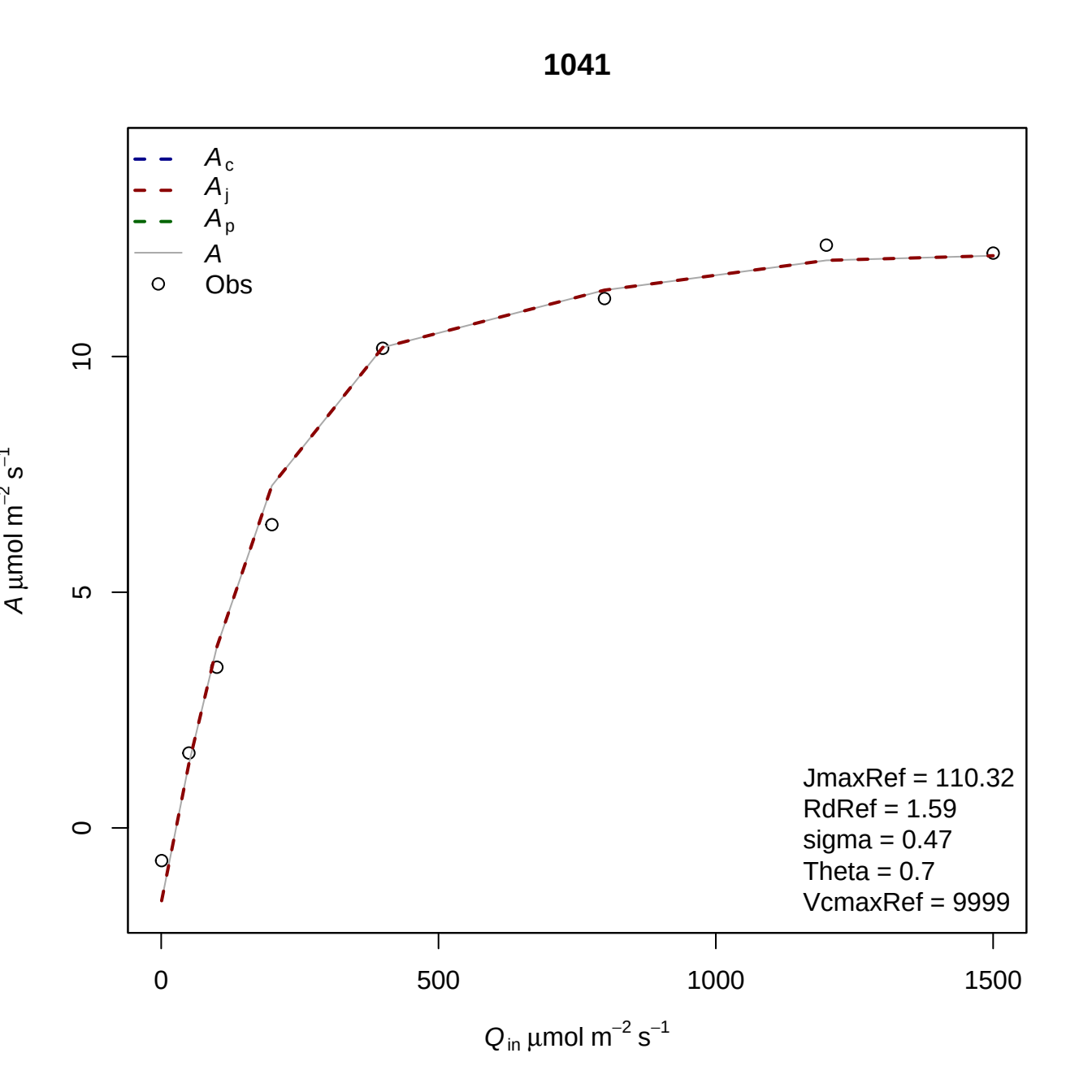

1036
A_c
A_j
A_p
A
Obs
JmaxRef = 79.74
RdRef = 1.77
sigma = 0.71
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

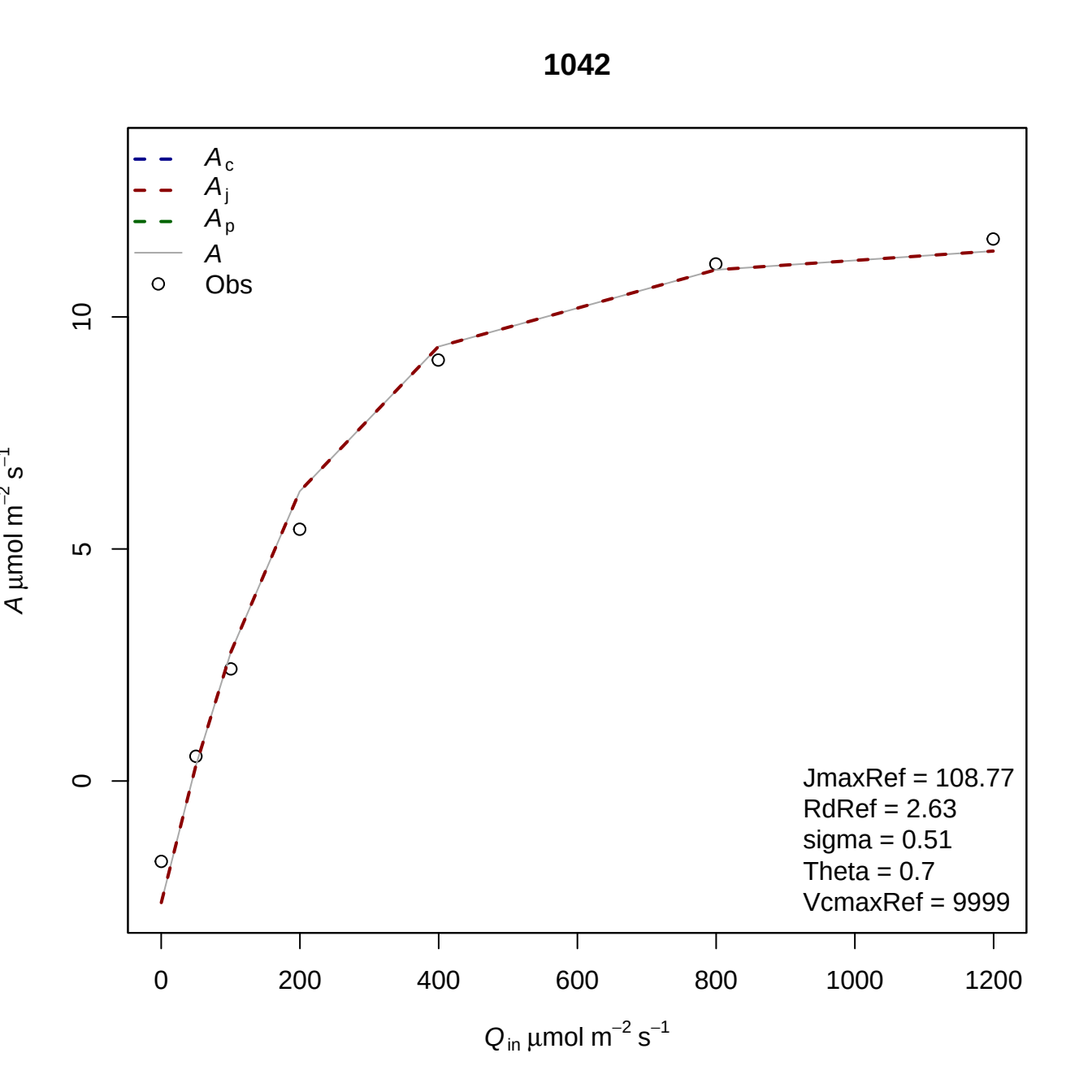

1037
A_c
A_j
A_p
A
Obs
JmaxRef = 106.44
RdRef = 1.66
sigma = 0.8
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

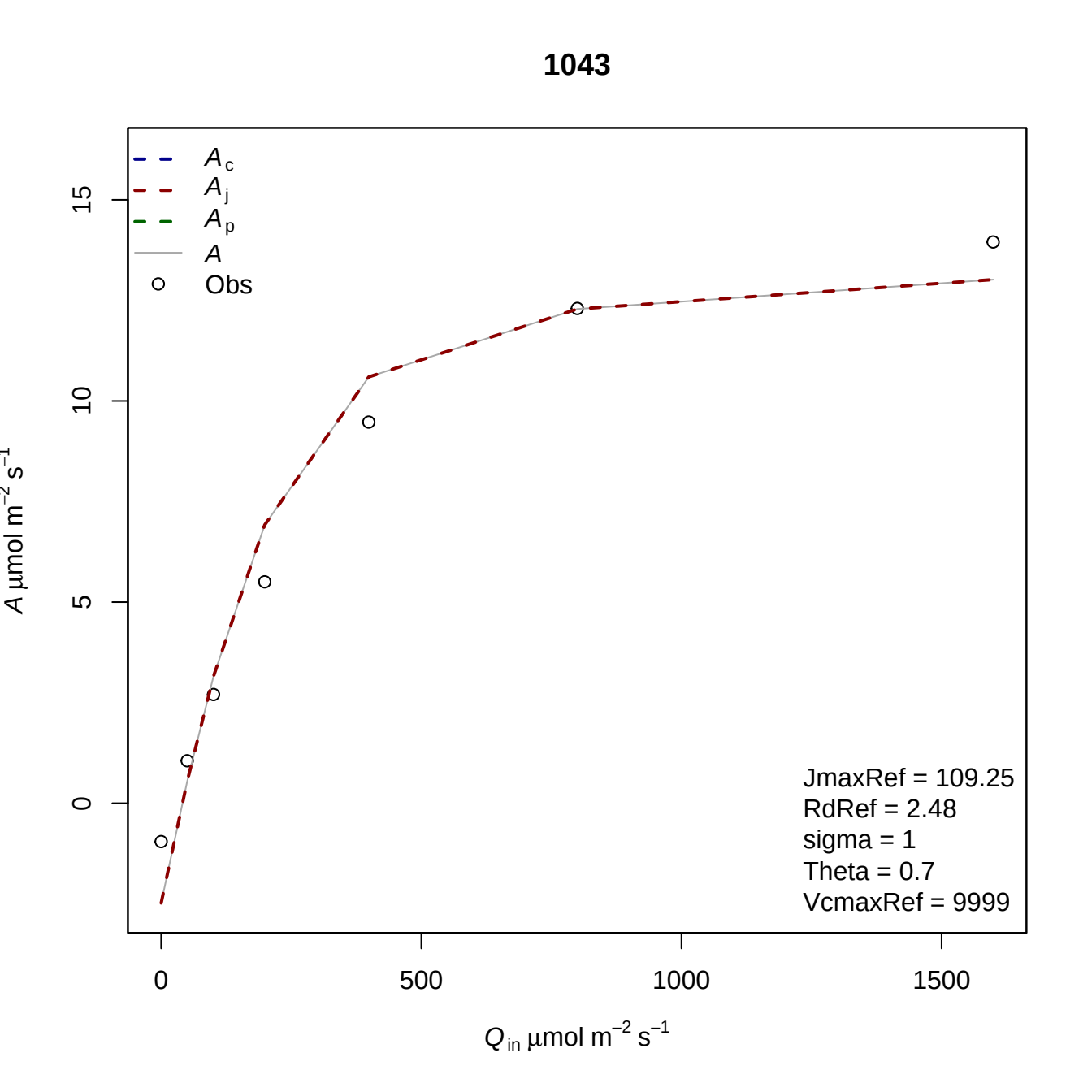

1038
A_c
A_j
A_p
A
Obs
JmaxRef = 98.06
RdRef = 1.81
sigma = 0.7
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

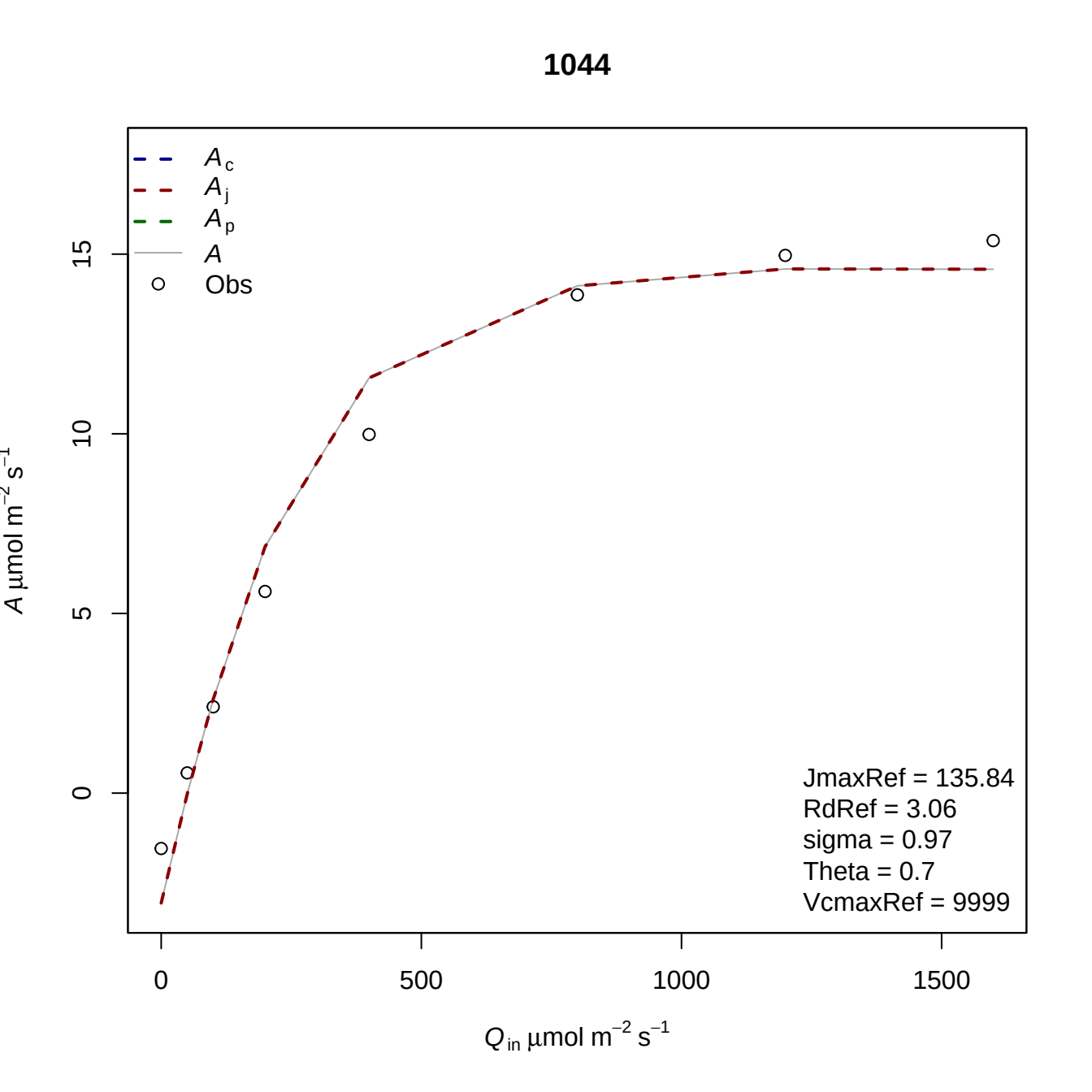

1039
A_c
A_j
A_p
A
Obs
JmaxRef = 86.55
RdRef = 1.48
sigma = 0.62
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
12
0
500
1000
1500

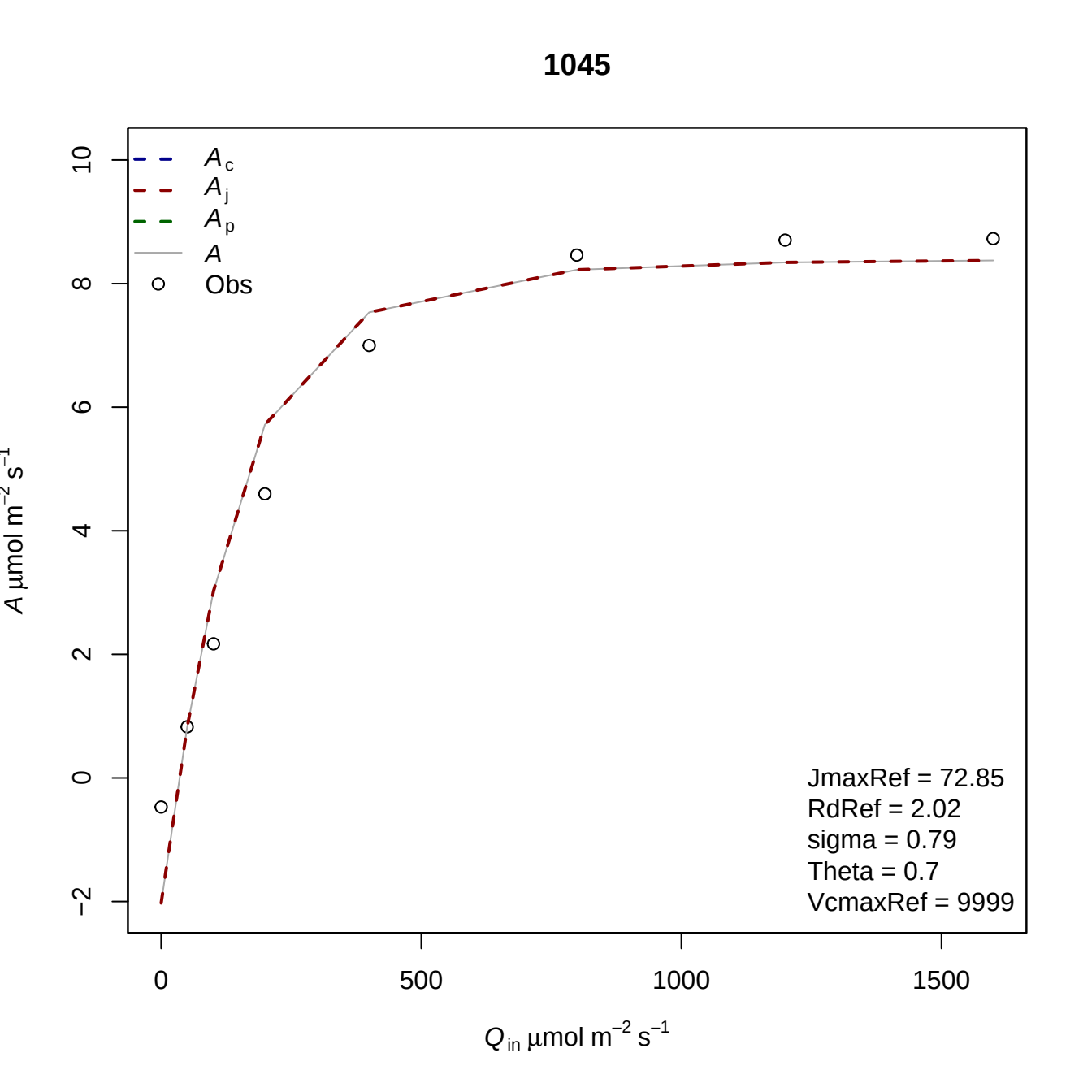

1040
A_c
A_j
A_p
A
Obs
JmaxRef = 81.39
RdRef = 1.49
sigma = 0.46
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

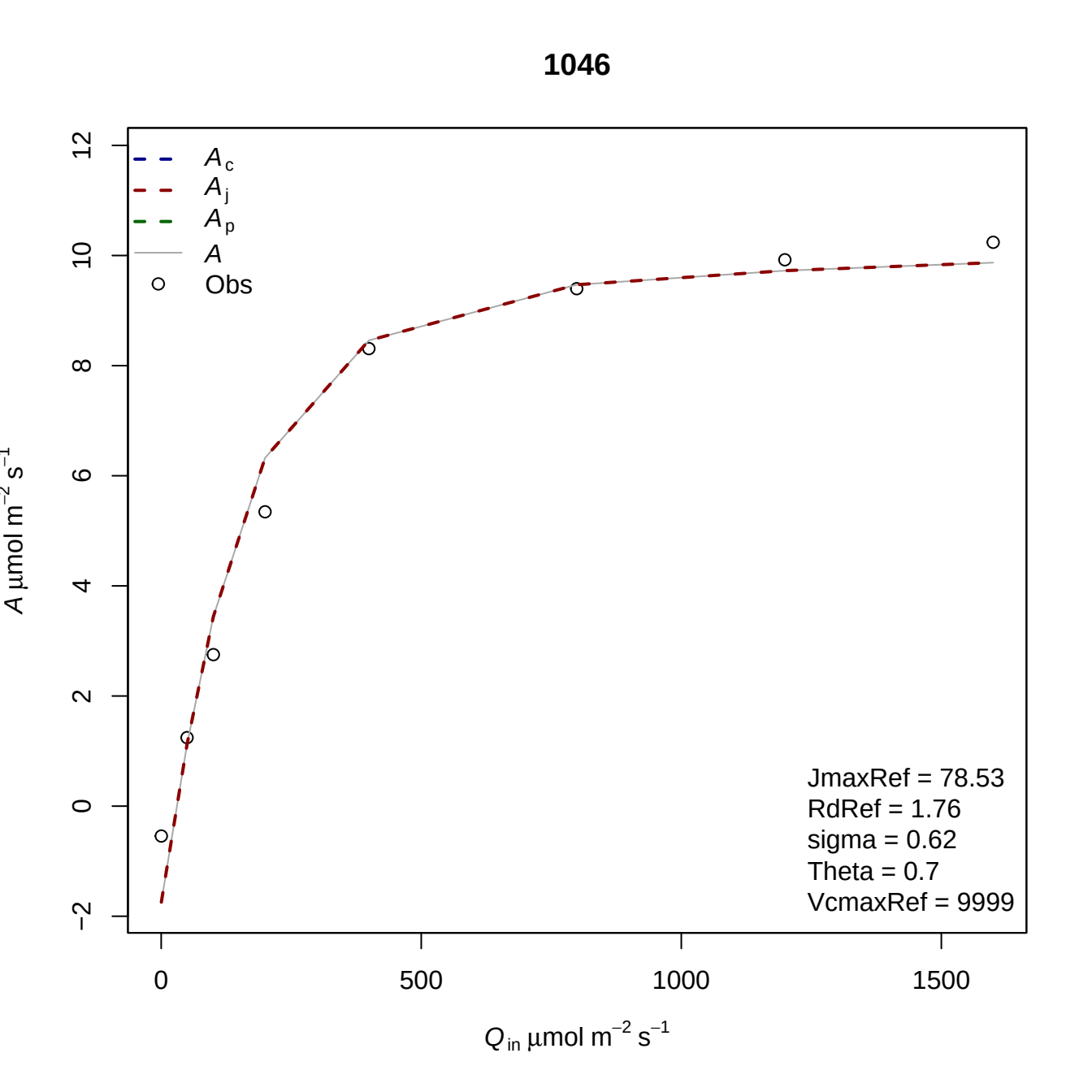

1041
A_c
A_j
A_p
A
Obs
JmaxRef = 110.32
RdRef = 1.59
sigma = 0.47
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

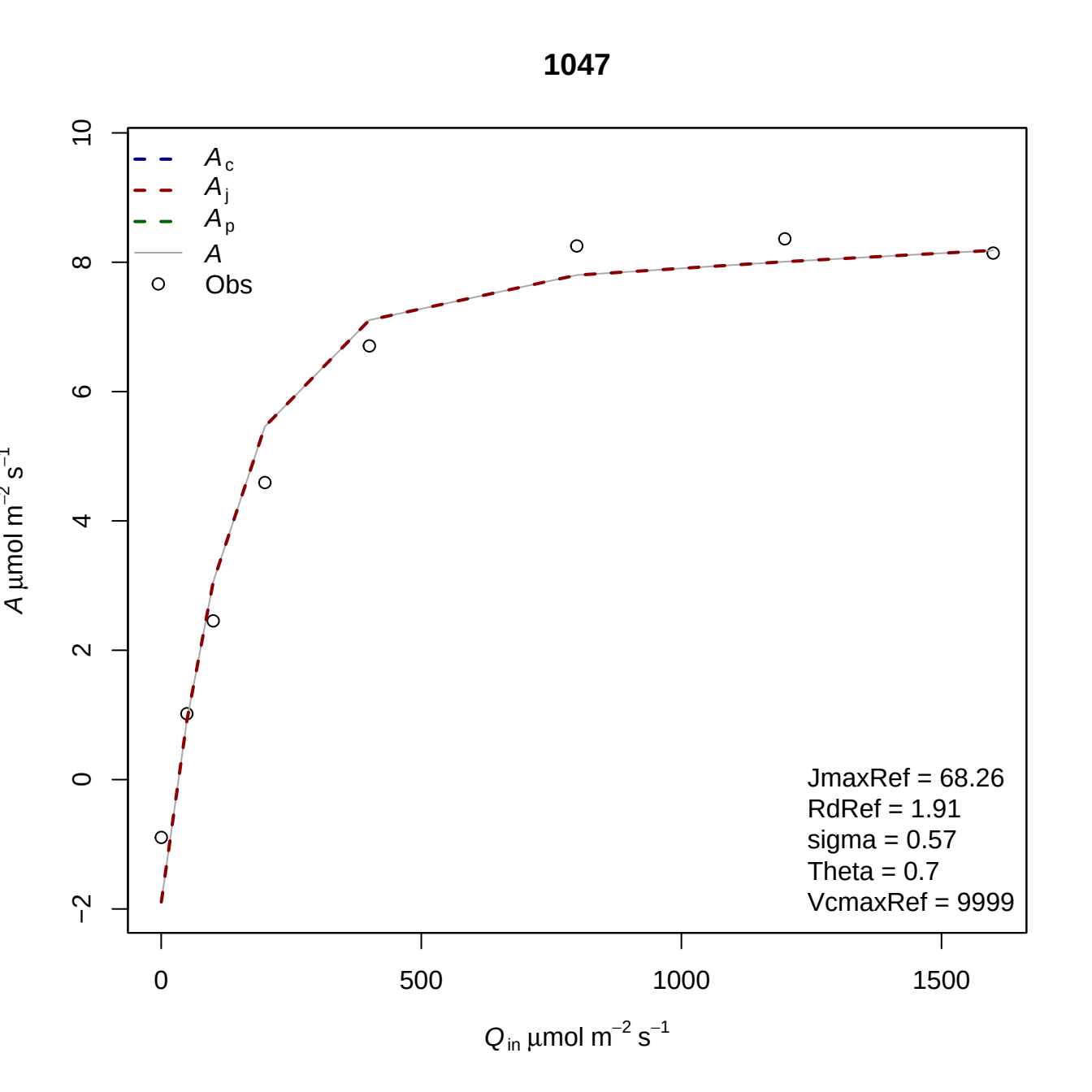

1042
A_c
A_j
A_p
A
Obs
JmaxRef = 108.77
RdRef = 2.63
sigma = 0.51
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800
1000
1200

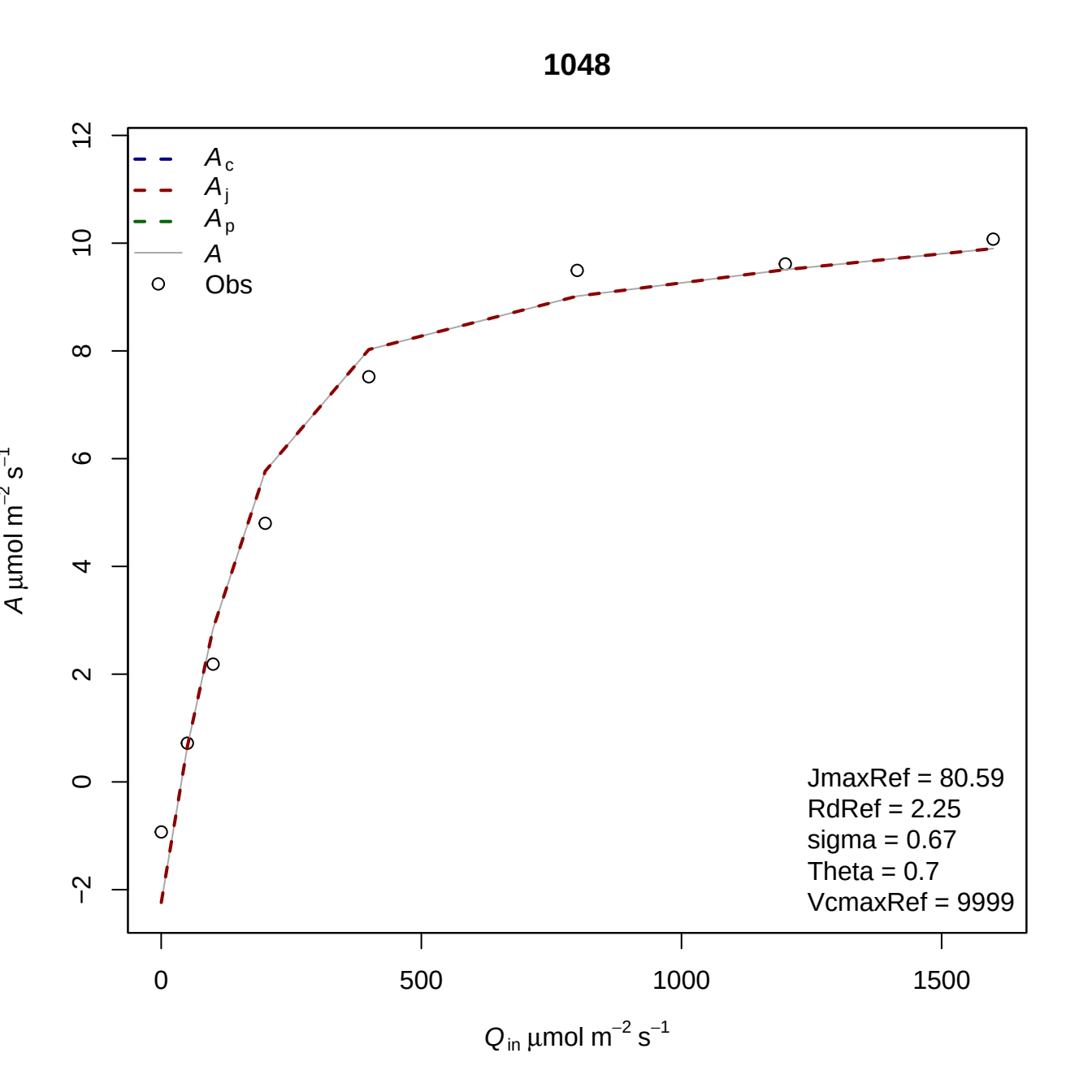

1043
A_c
A_j
A_p
A
Obs
JmaxRef = 109.25
RdRef = 2.48
sigma = 1
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

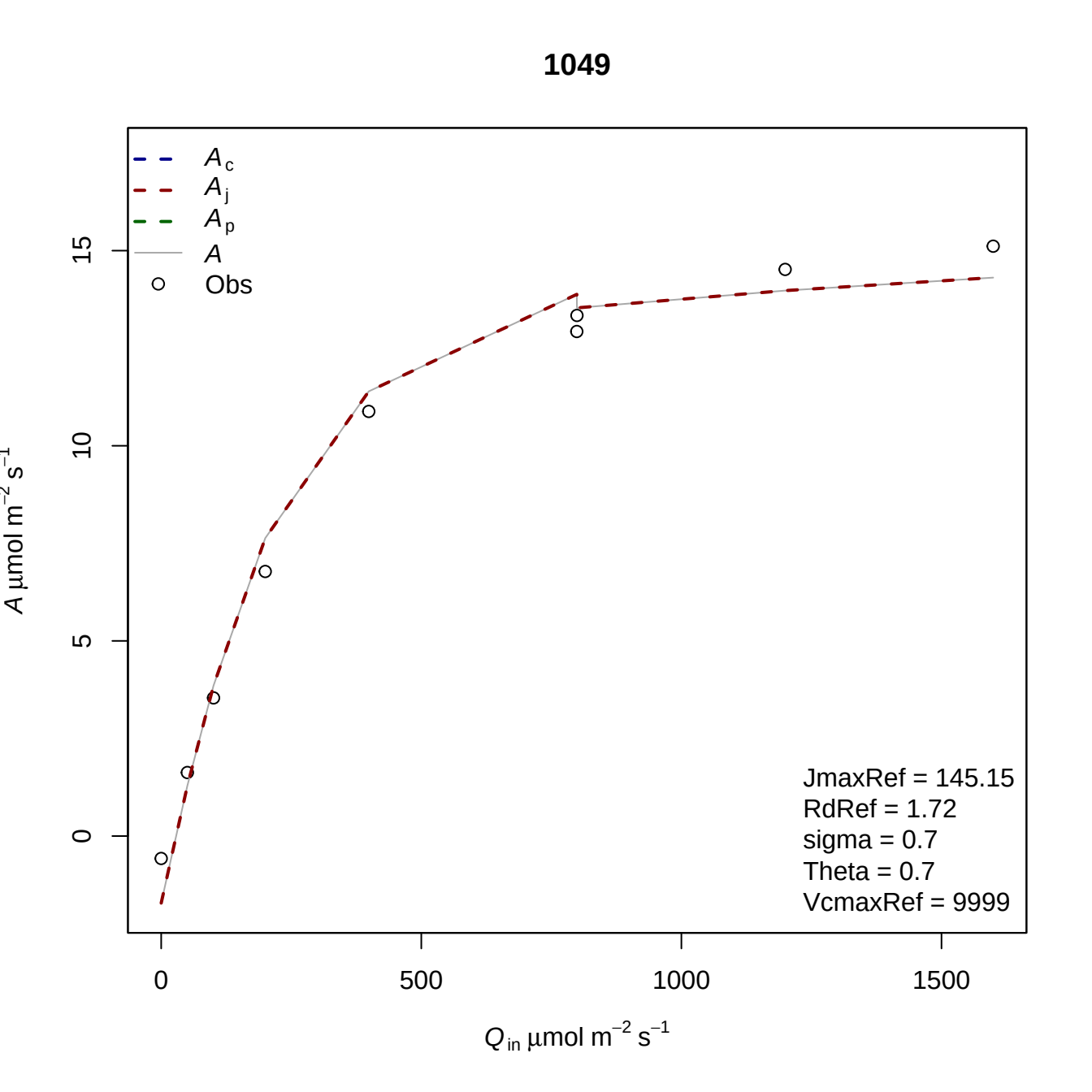

1044
A_c
A_j
A_p
A
Obs
JmaxRef = 135.84
RdRef = 3.06
sigma = 0.97
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

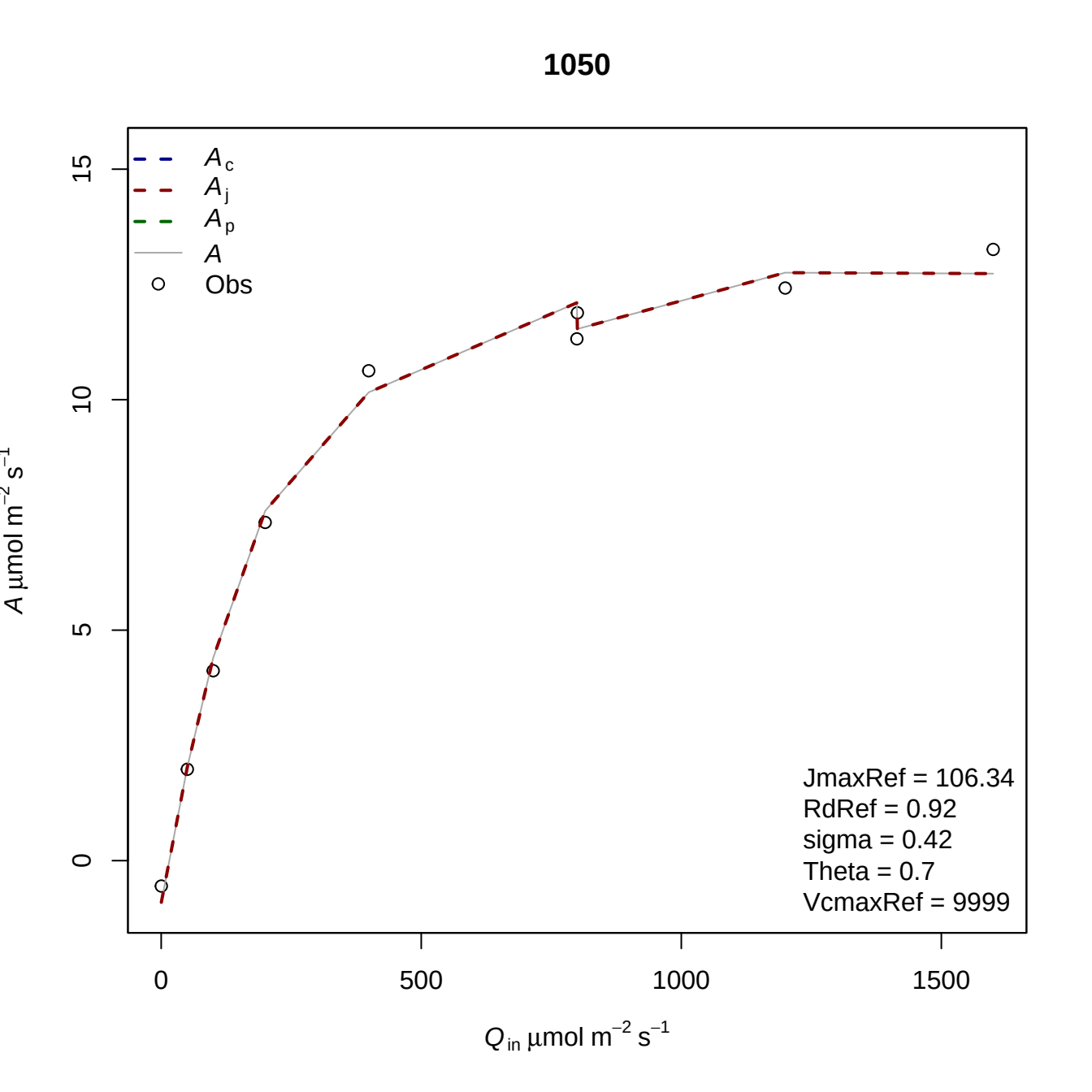

1045
A_c
A_j
A_p
A
Obs
JmaxRef = 72.85
RdRef = 2.02
sigma = 0.79
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

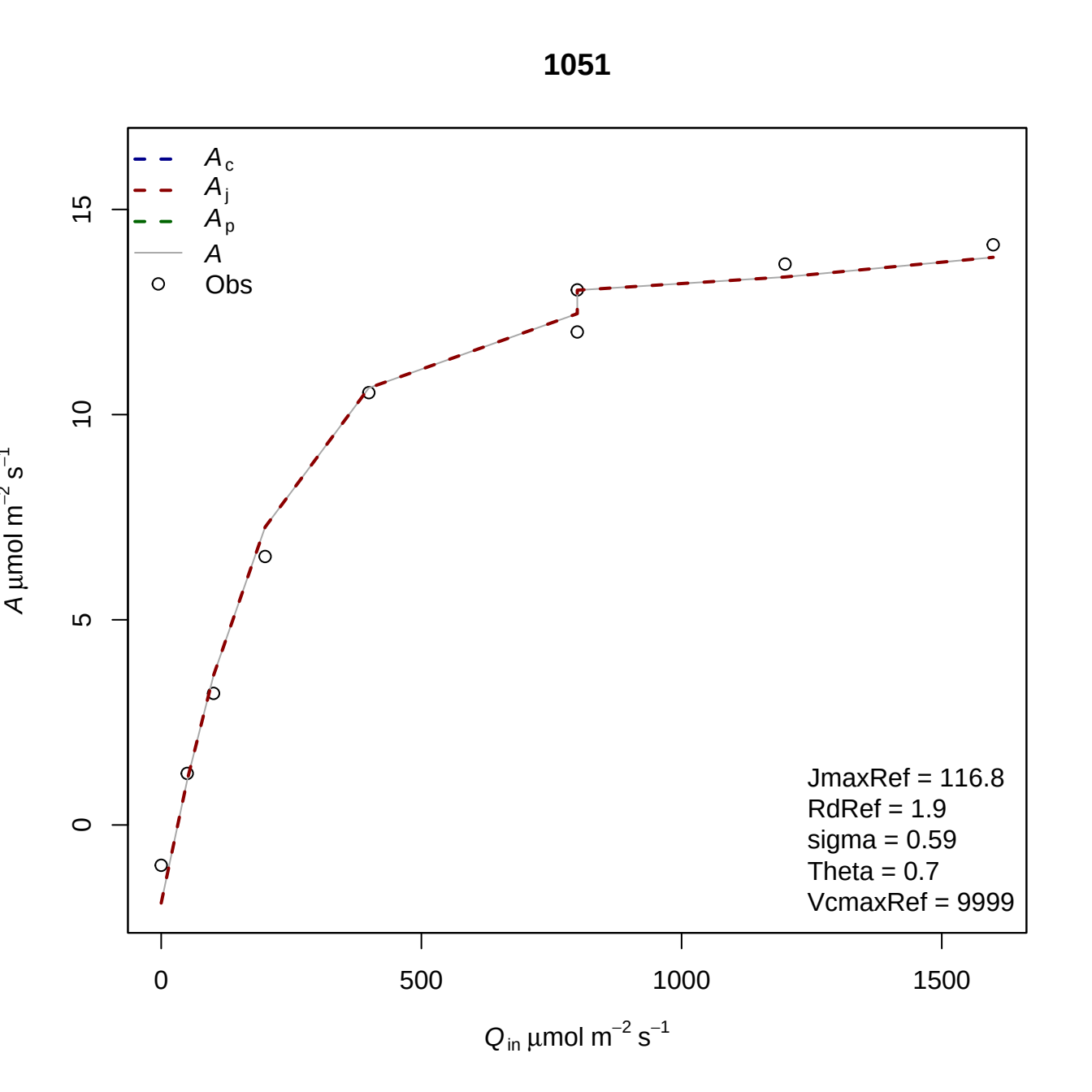

1046
A_c
A_j
A_p
A
Obs
JmaxRef = 78.53
RdRef = 1.76
sigma = 0.62
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

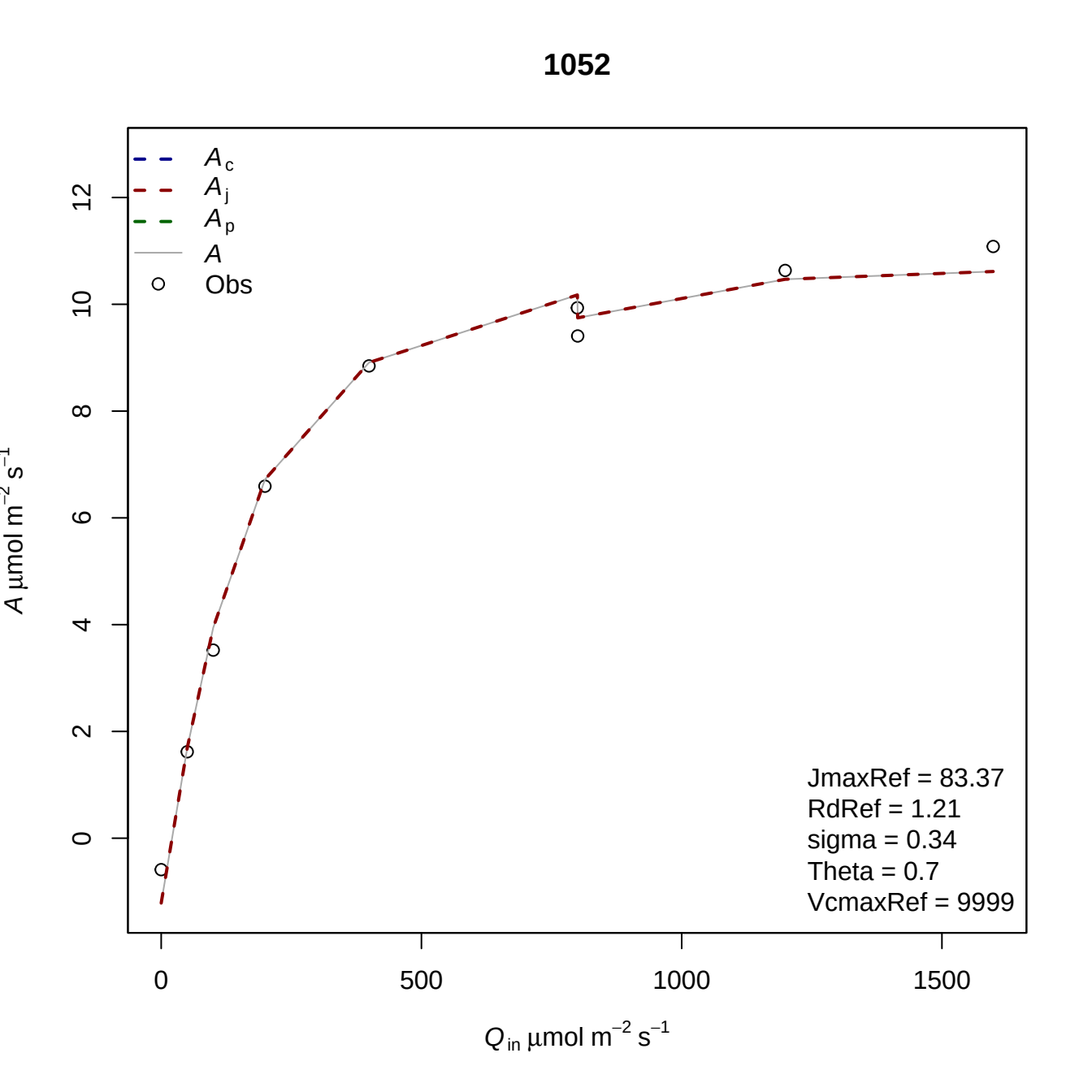

1047
A_c
A_j
A_p
A
Obs
JmaxRef = 68.26
RdRef = 1.91
sigma = 0.57
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

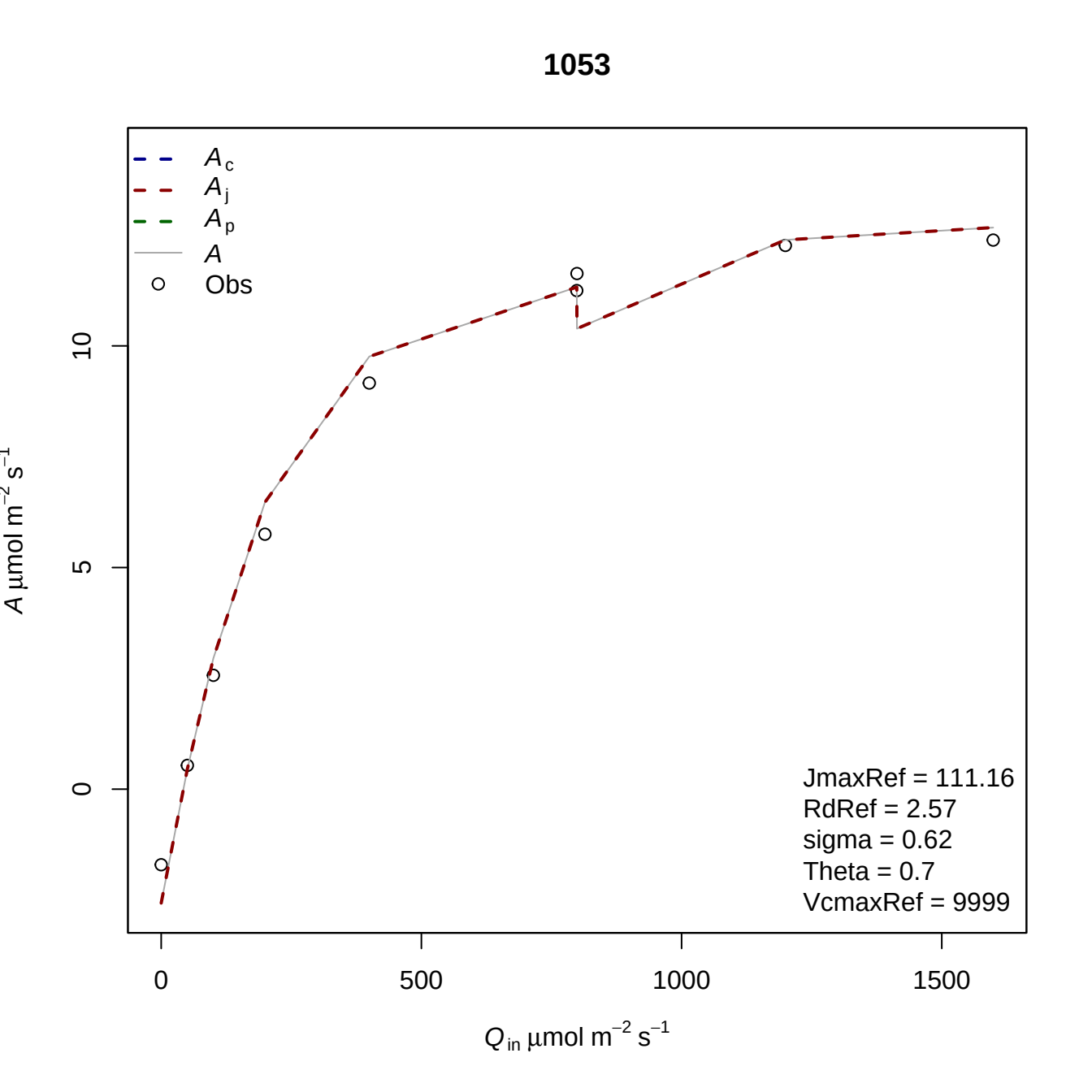

1048
A_c
A_j
A_p
A
Obs
JmaxRef = 80.59
RdRef = 2.25
sigma = 0.67
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

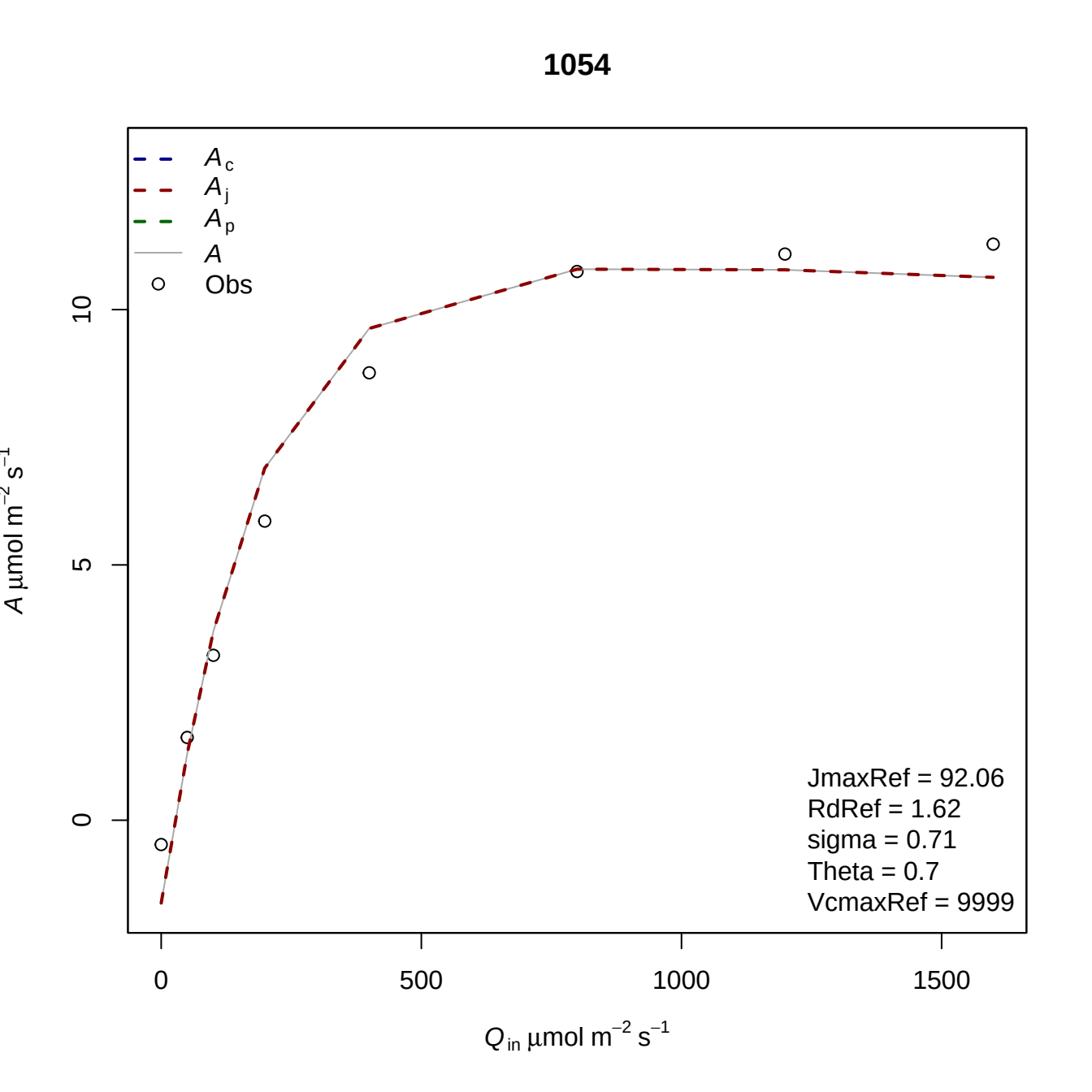

1049
A_c
A_j
A_p
A
Obs
JmaxRef = 145.15
RdRef = 1.72
sigma = 0.7
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

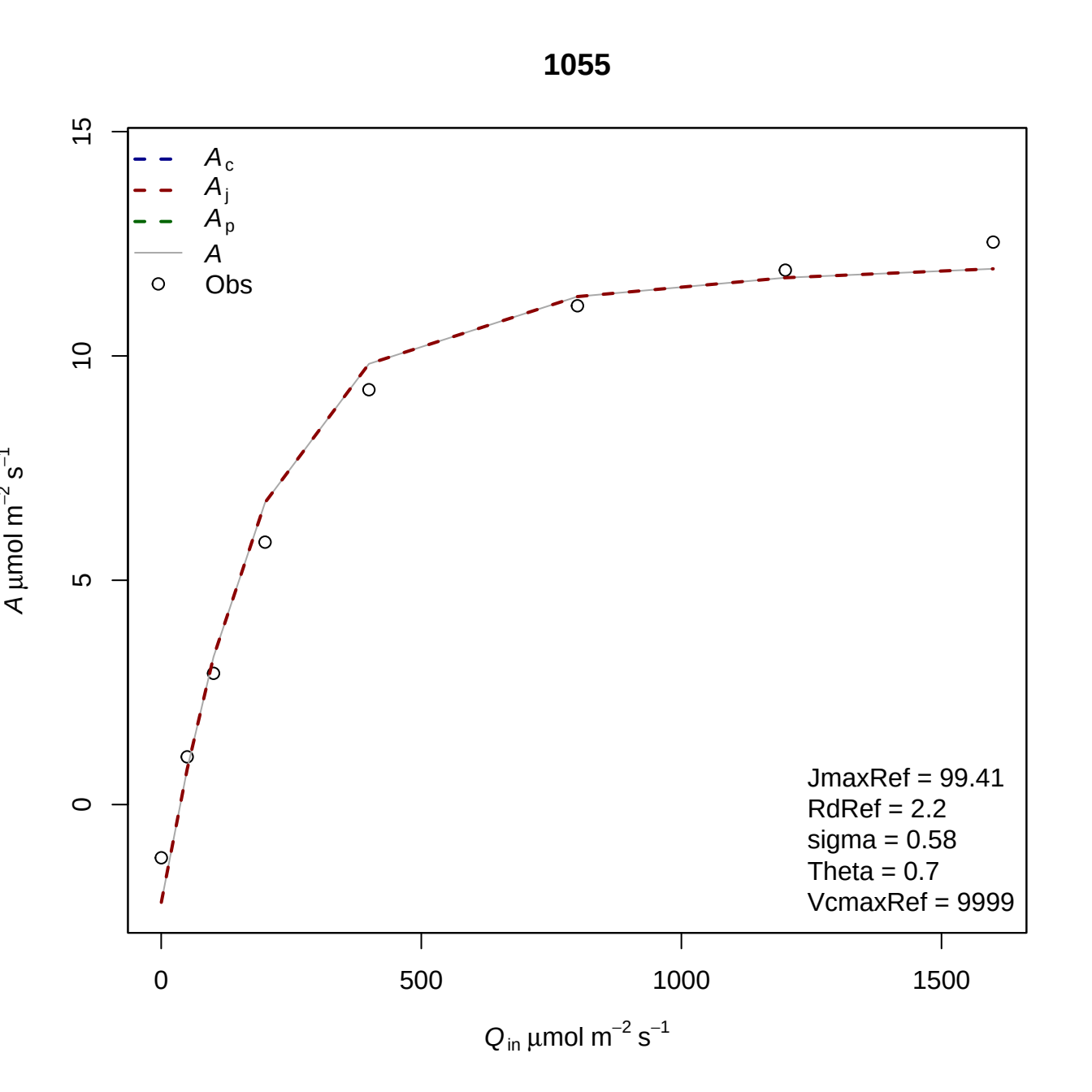

1050
A_c
A_j
A_p
A
Obs
JmaxRef = 106.34
RdRef = 0.92
sigma = 0.42
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

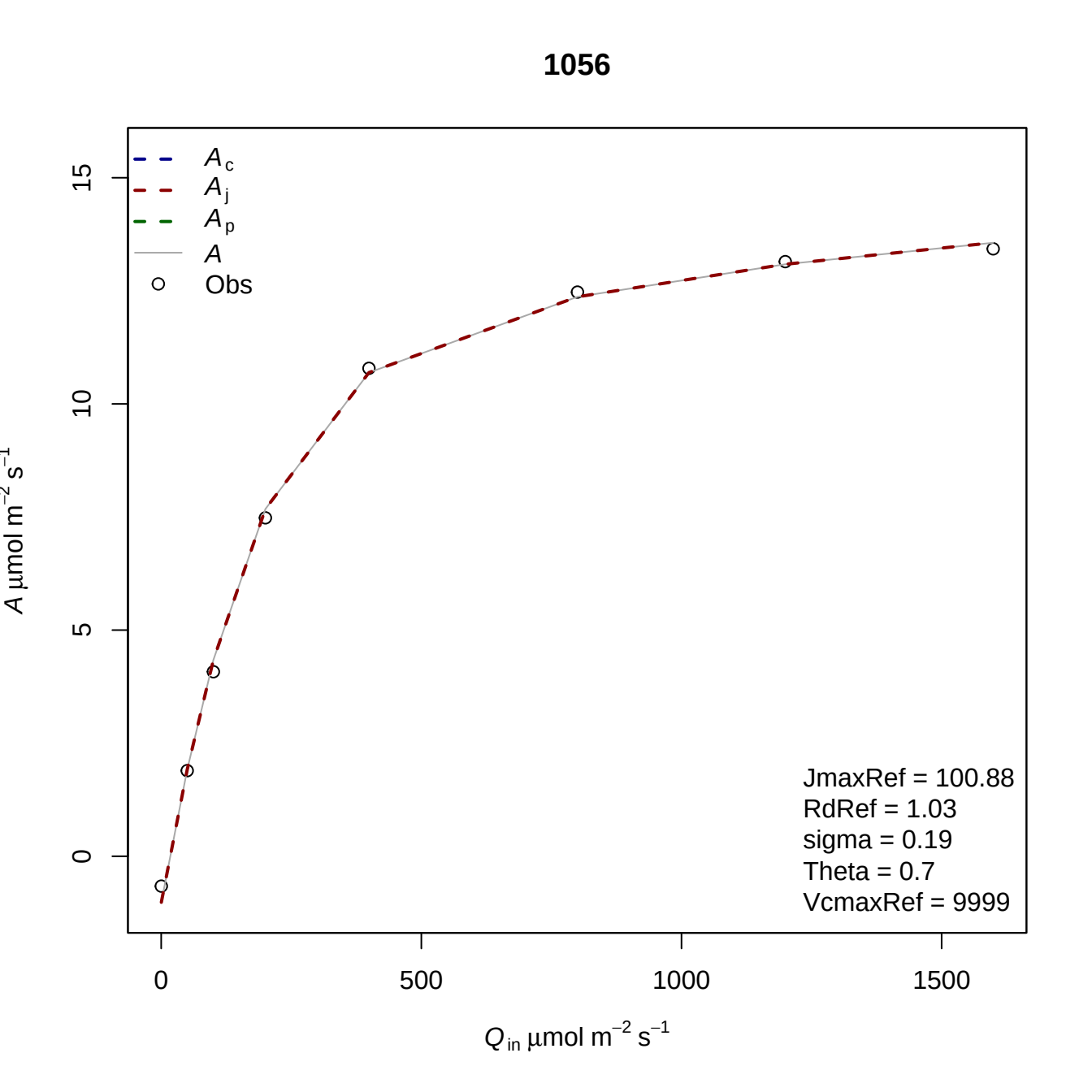

1051
A_c
A_j
A_p
A
Obs
JmaxRef = 116.8
RdRef = 1.9
sigma = 0.59
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

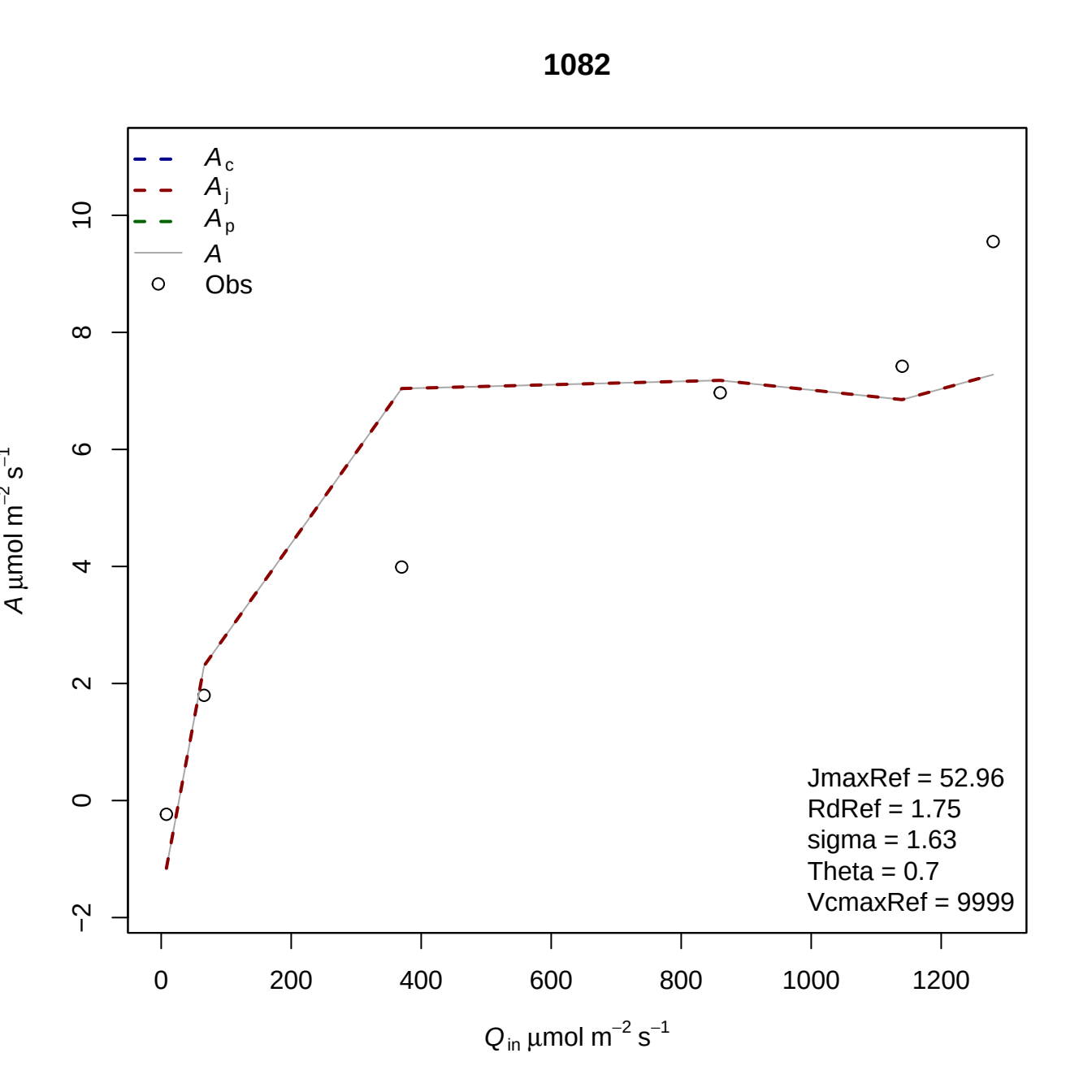

1052
A_c
A_j
A_p
A
Obs
JmaxRef = 83.37
RdRef = 1.21
sigma = 0.34
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

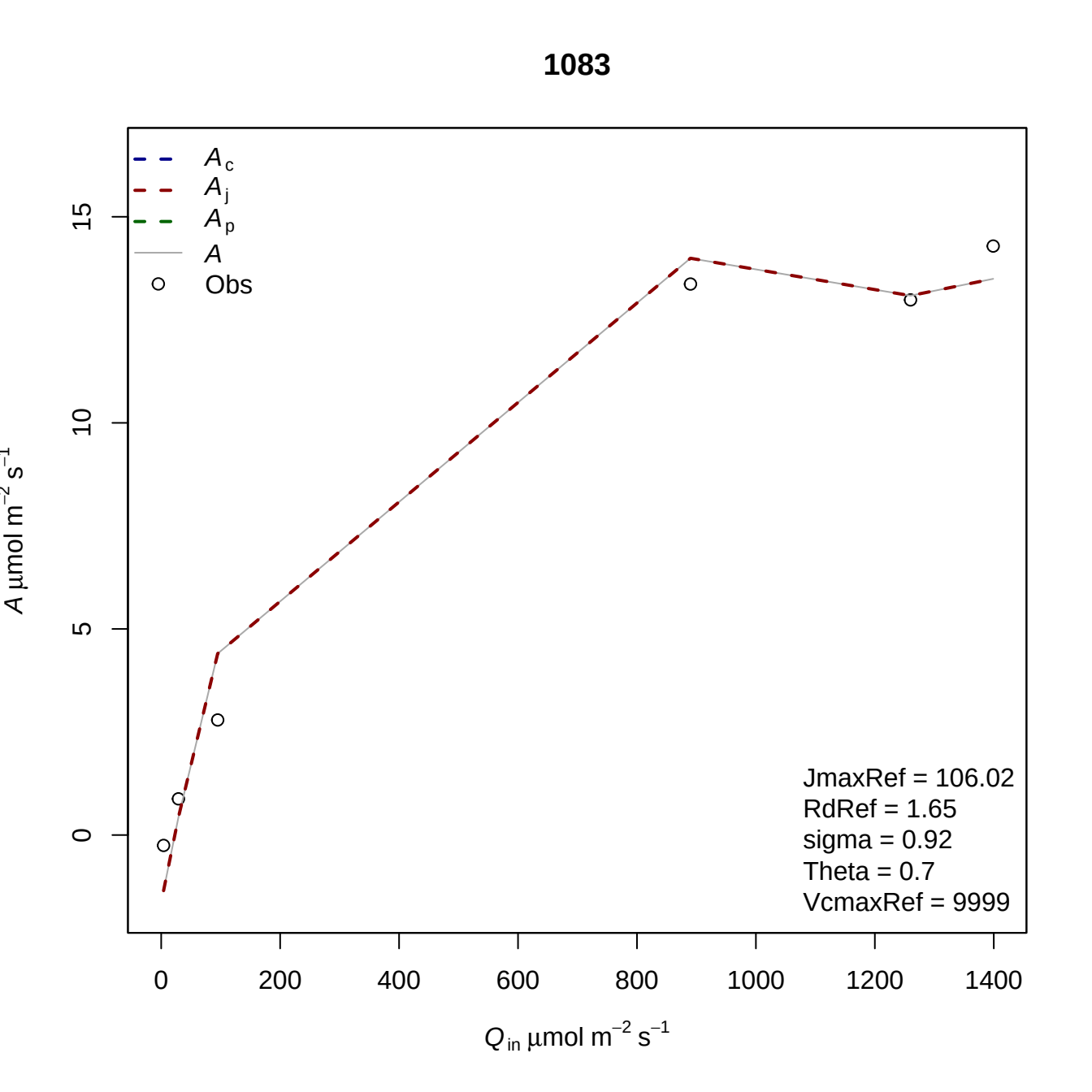

1053
A_c
A_j
A_p
A
Obs
JmaxRef = 111.16
RdRef = 2.57
sigma = 0.62
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

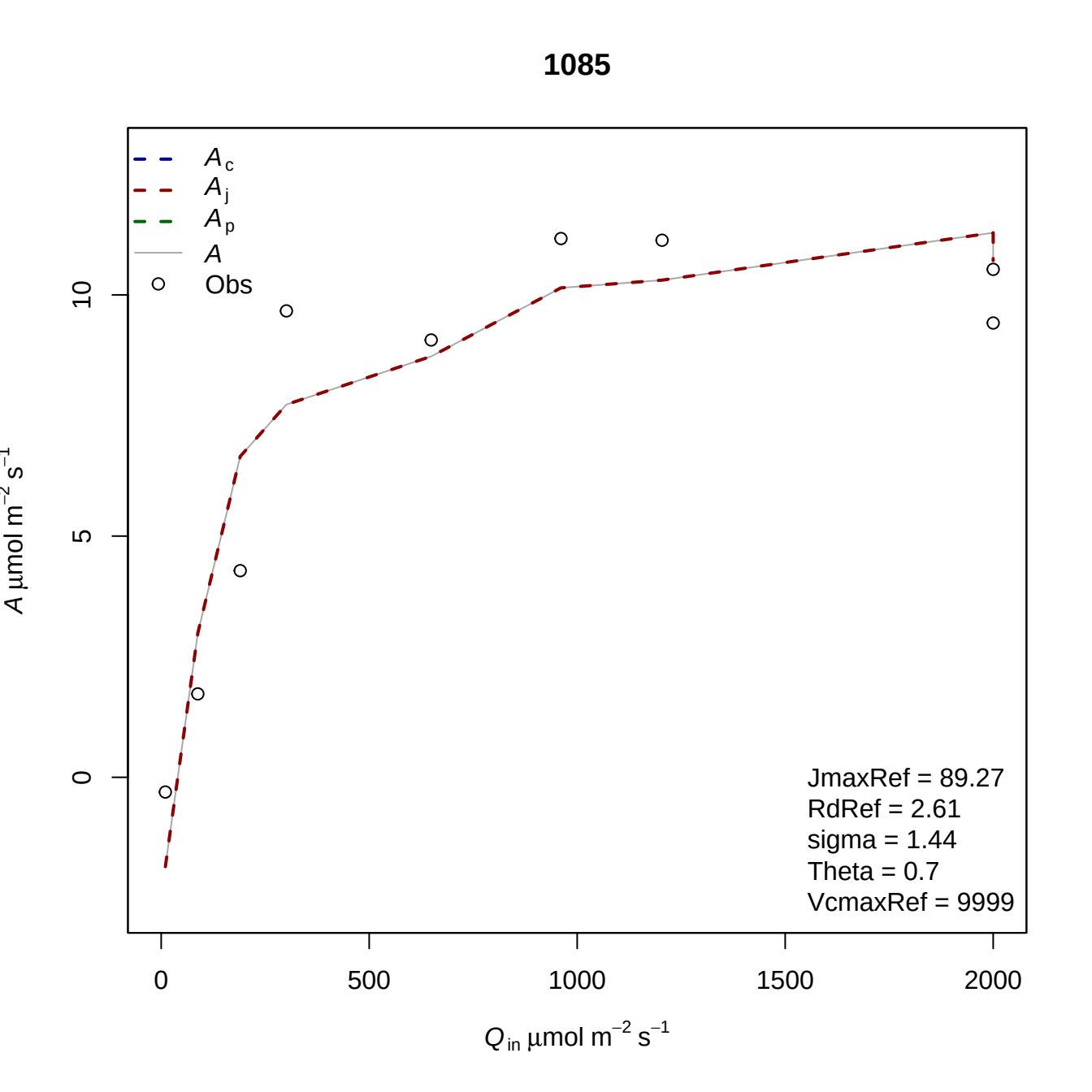

1054
A_c
A_j
A_p
A
Obs
JmaxRef = 92.06
RdRef = 1.62
sigma = 0.71
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

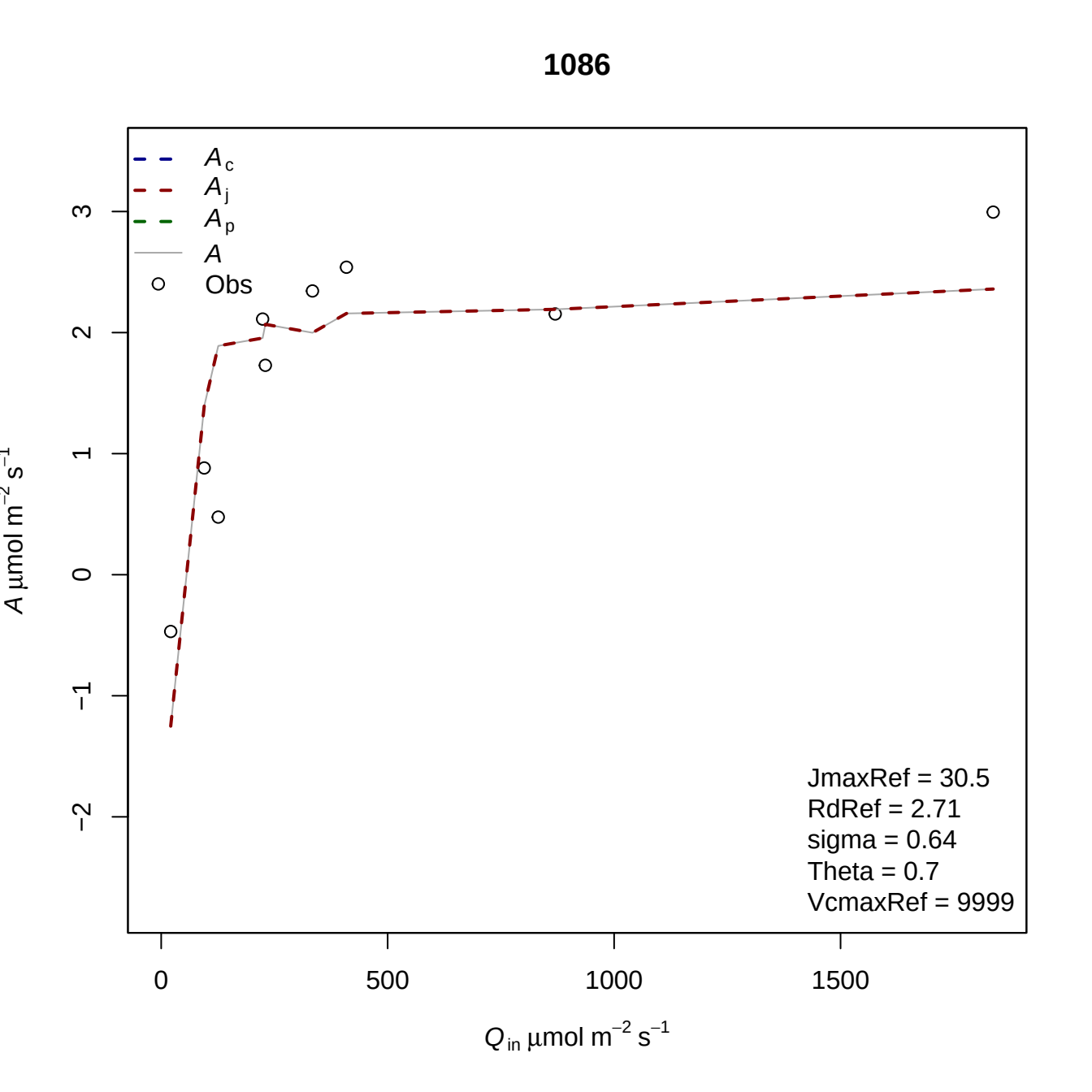

1055
A_c
A_j
A_p
A
Obs
JmaxRef = 99.41
RdRef = 2.2
sigma = 0.58
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

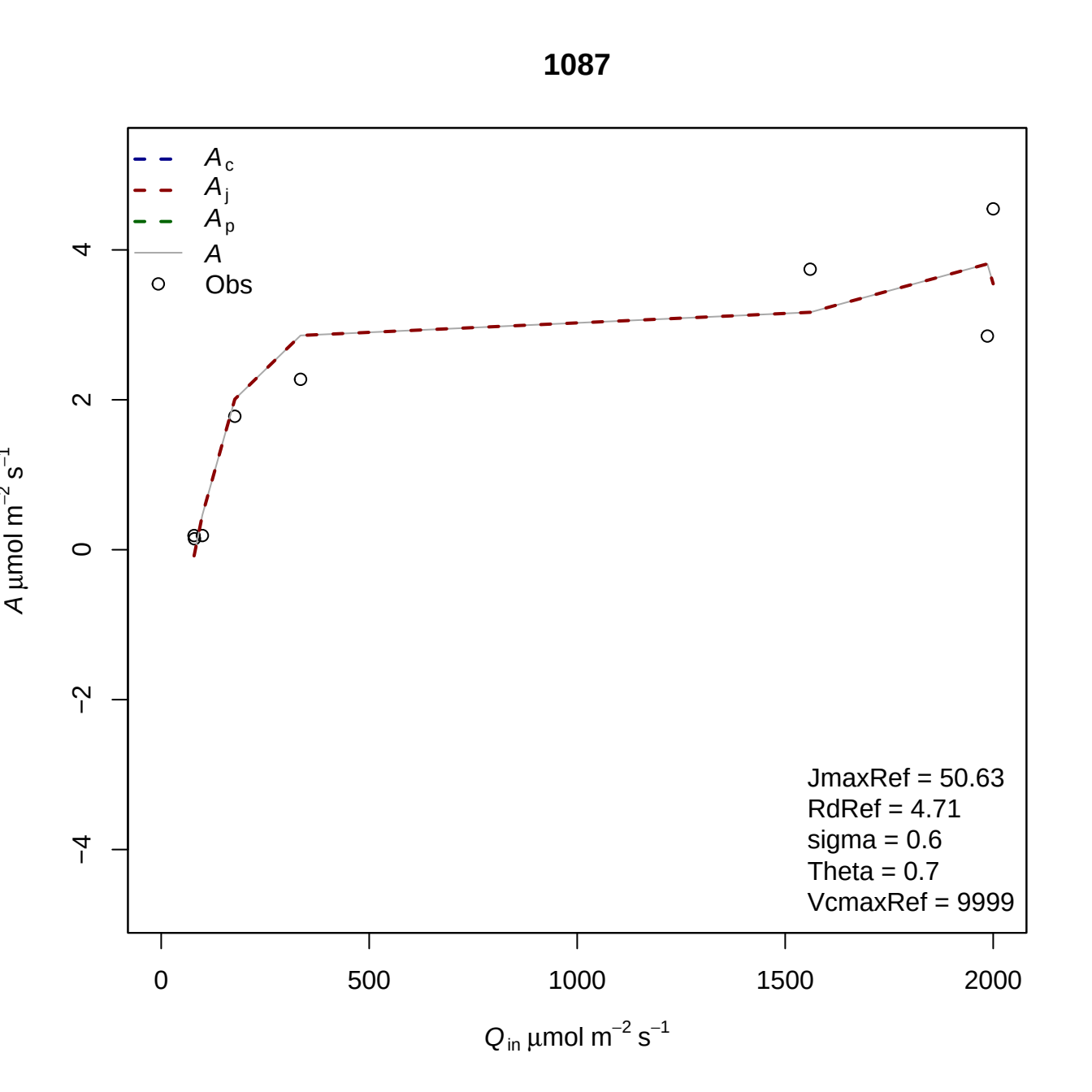

1056
A_c
A_j
A_p
A
Obs
JmaxRef = 100.88
RdRef = 1.03
sigma = 0.19
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

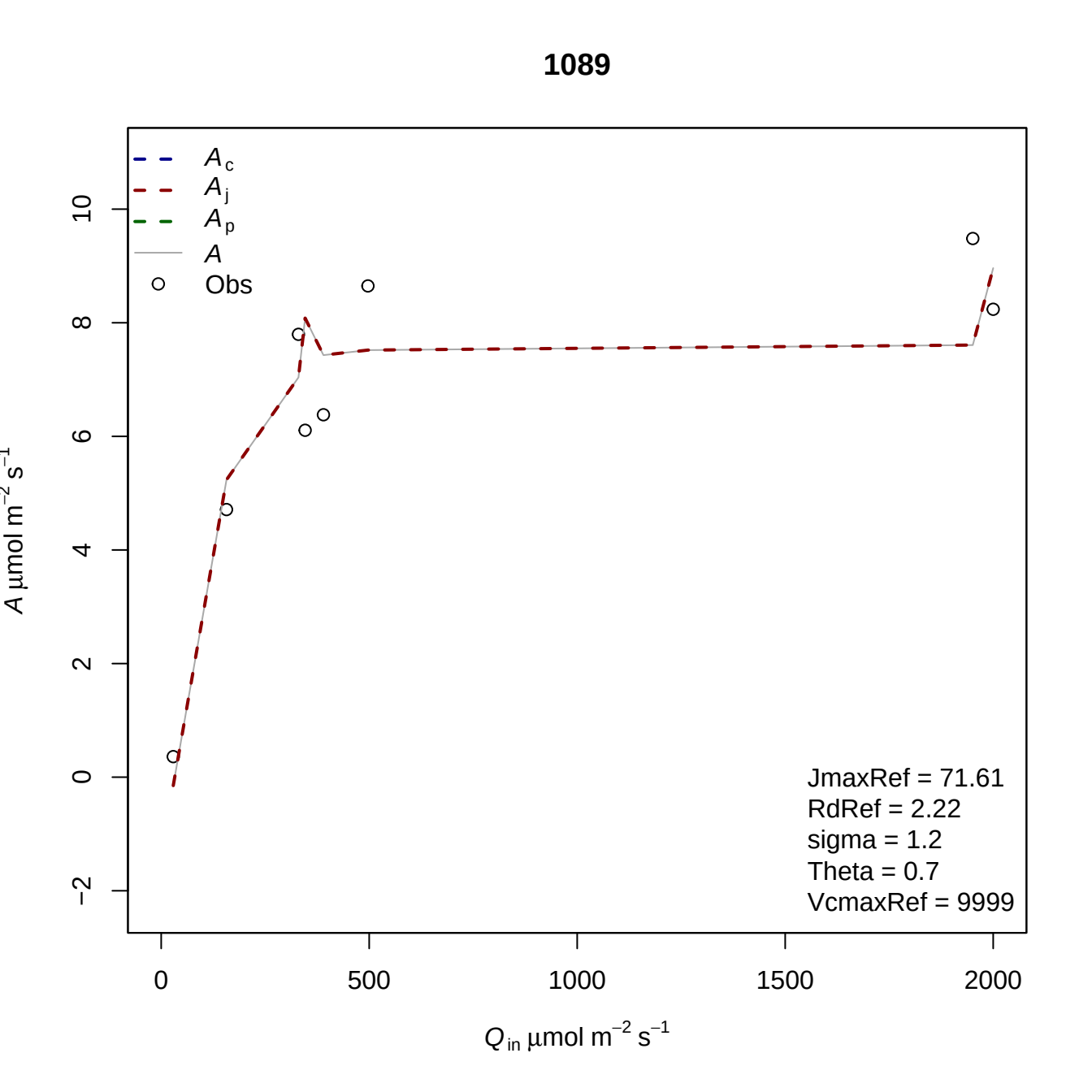

1082
A_c
A_j
A_p
A
Obs
JmaxRef = 52.96
RdRef = 1.75
sigma = 1.63
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
200
400
600
800
1000
1200

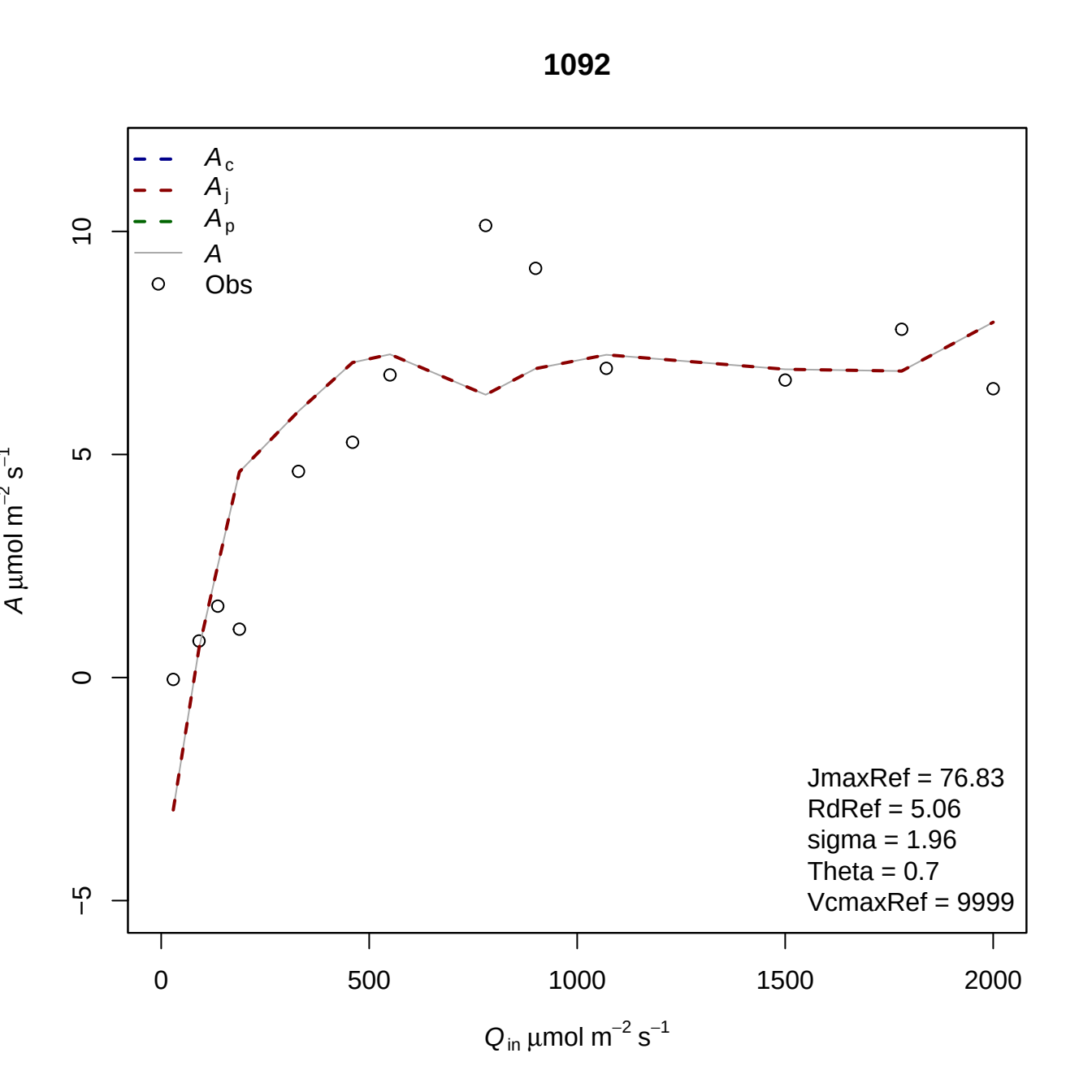

1083
A_c
A_j
A_p
A
Obs
JmaxRef = 106.02
RdRef = 1.65
sigma = 0.92
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
200
400
600
800
1000
1200
1400

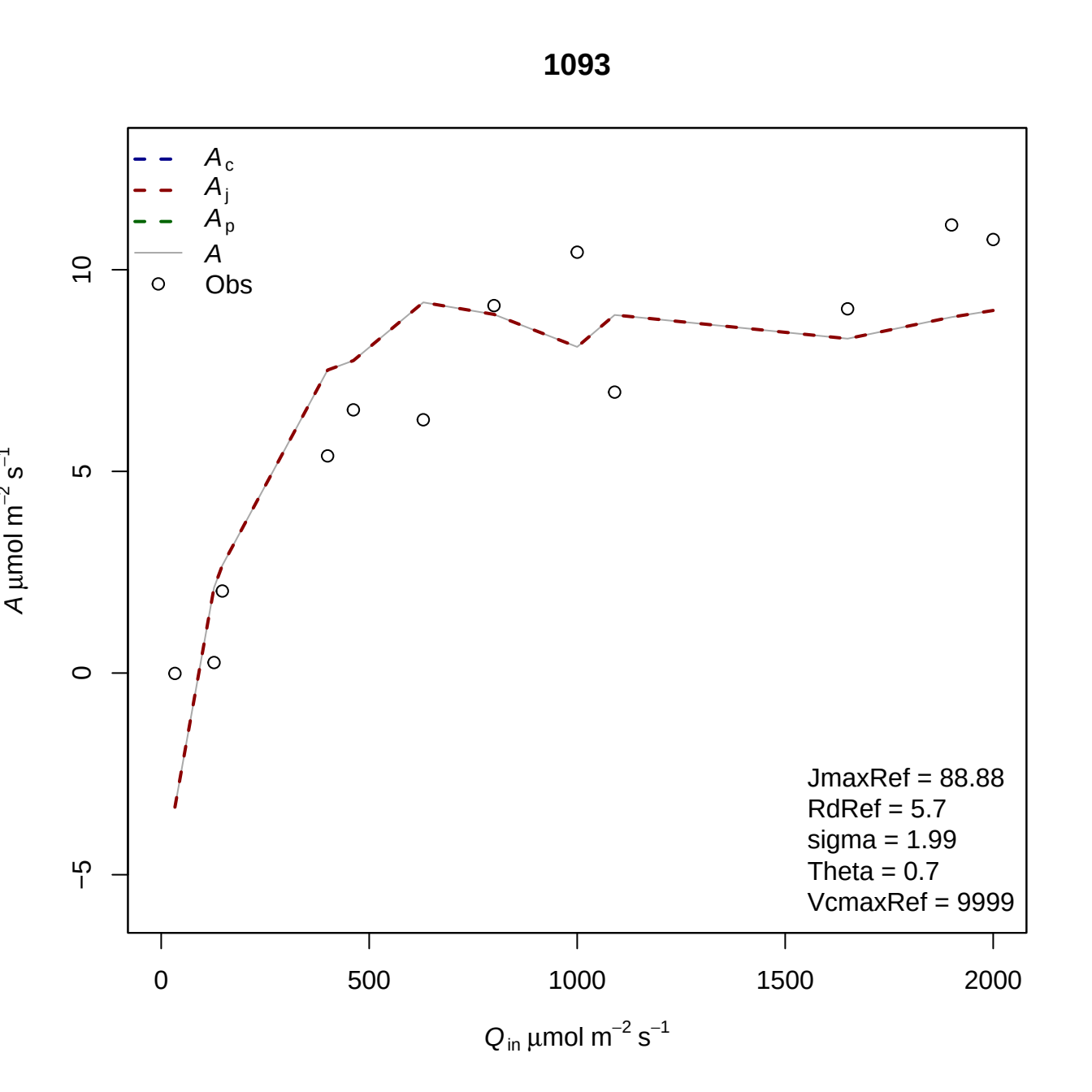

1085
A_c
A_j
A_p
A
Obs
JmaxRef = 89.27
RdRef = 2.61
sigma = 1.44
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

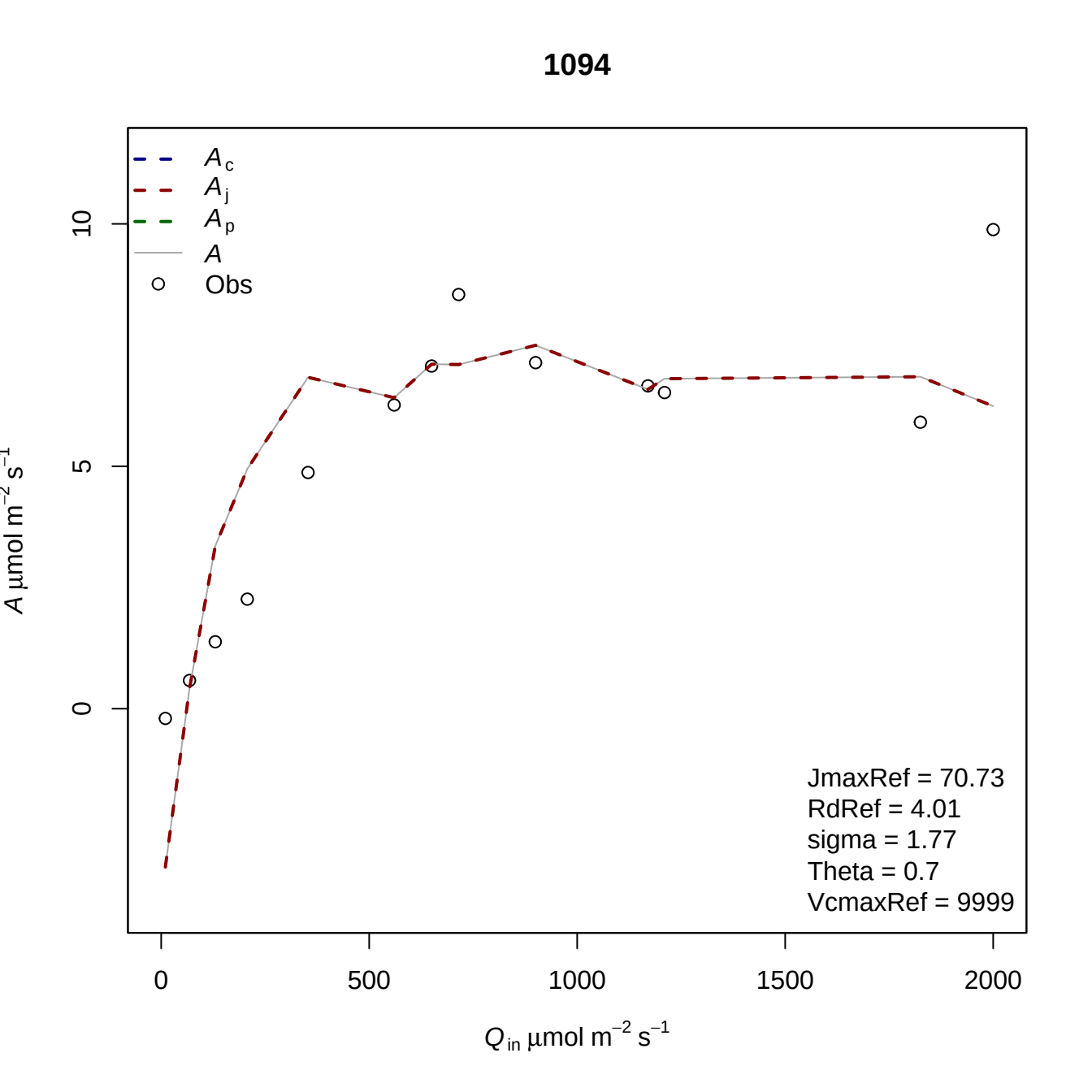

1086
A_c
A_j
A_p
A
Obs
JmaxRef = 30.5
RdRef = 2.71
sigma = 0.64
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
−2
0
500
1000
1500

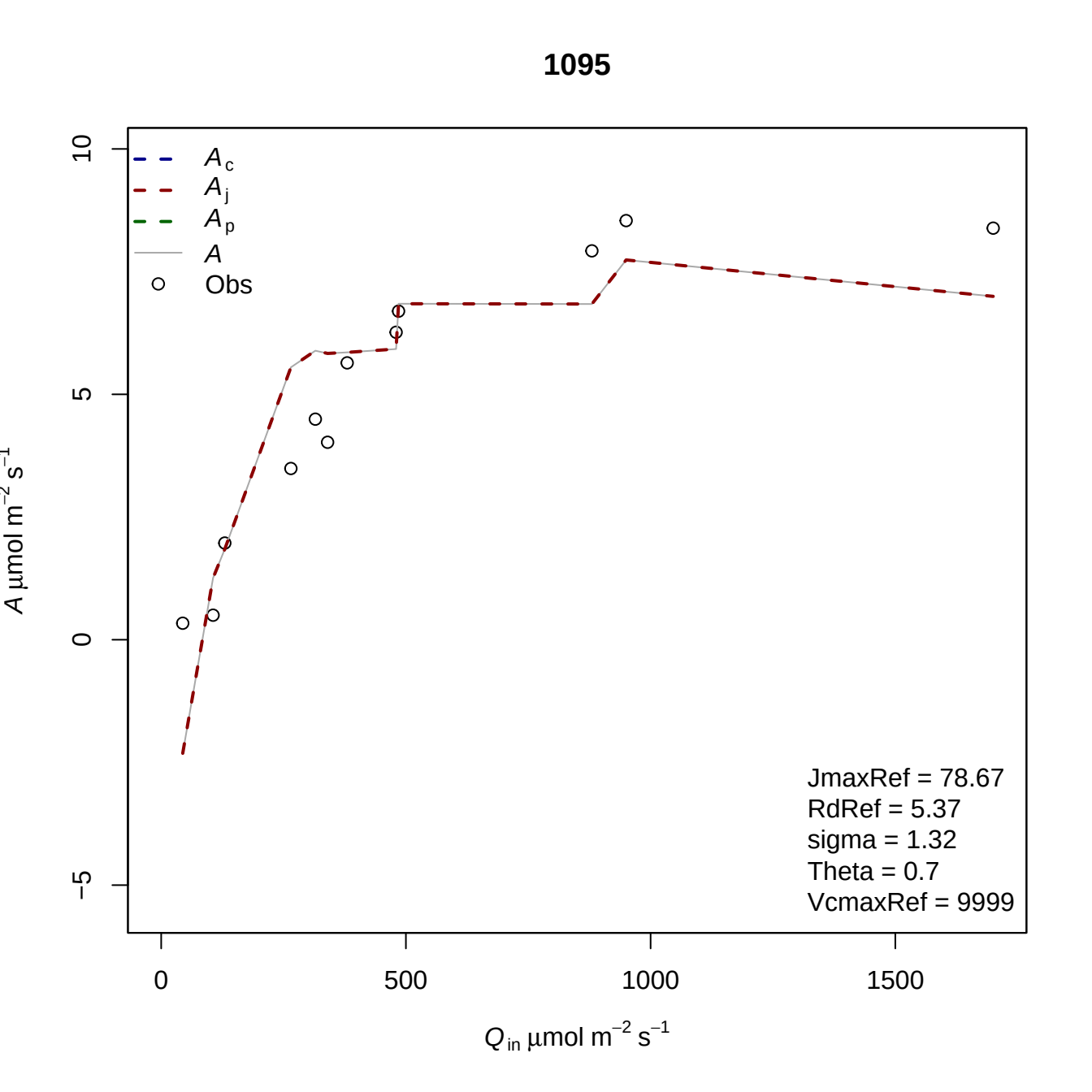

1087
A_c
A_j
A_p
A
Obs
JmaxRef = 50.63
RdRef = 4.71
sigma = 0.6
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
−4
0
500
1000
1500
2000

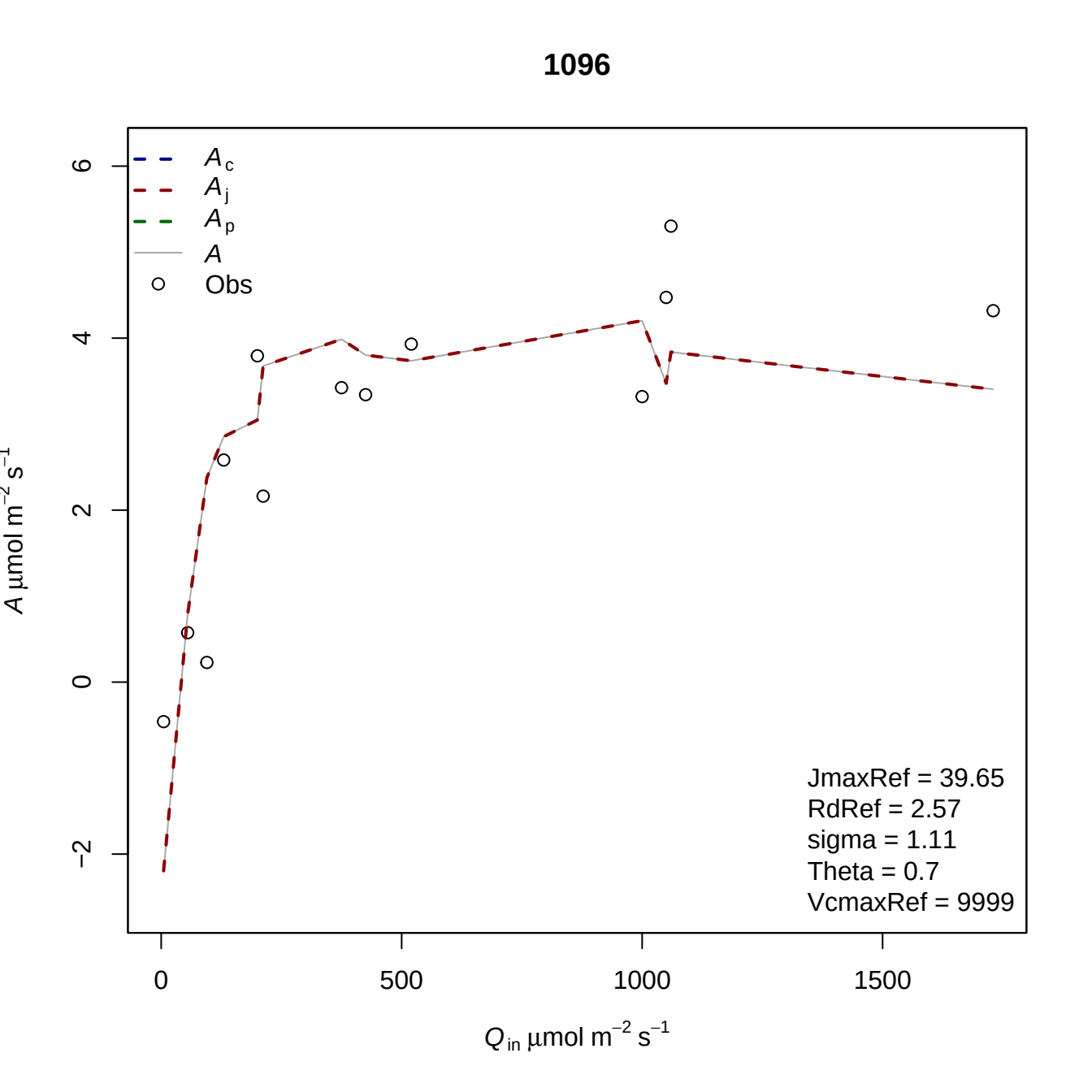

1089
A_c
A_j
A_p
A
Obs
JmaxRef = 71.61
RdRef = 2.22
sigma = 1.2
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

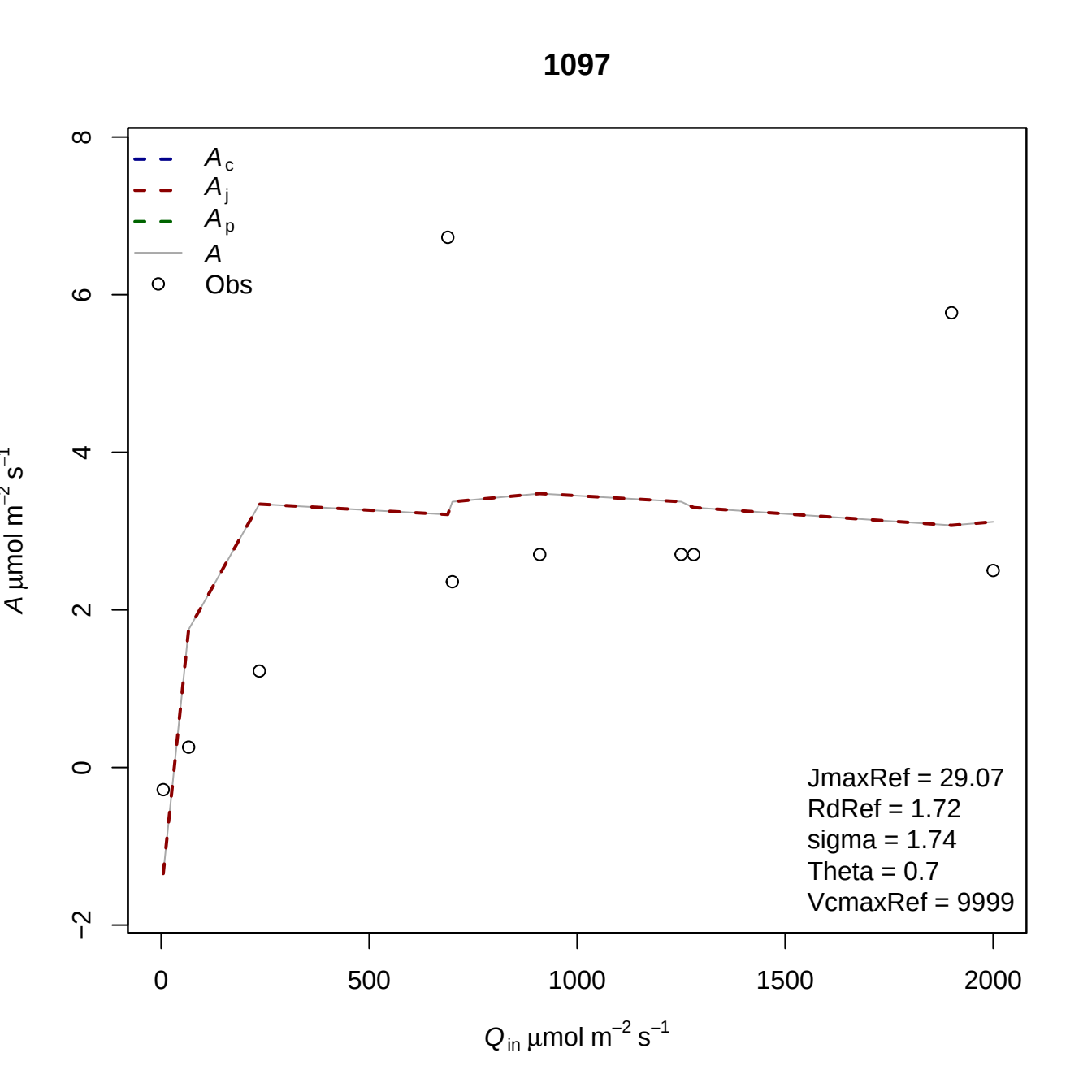

1092
A_c
A_j
A_p
A
Obs
JmaxRef = 76.83
RdRef = 5.06
sigma = 1.96
Theta = 0.7
VcmaxRef = 9999
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

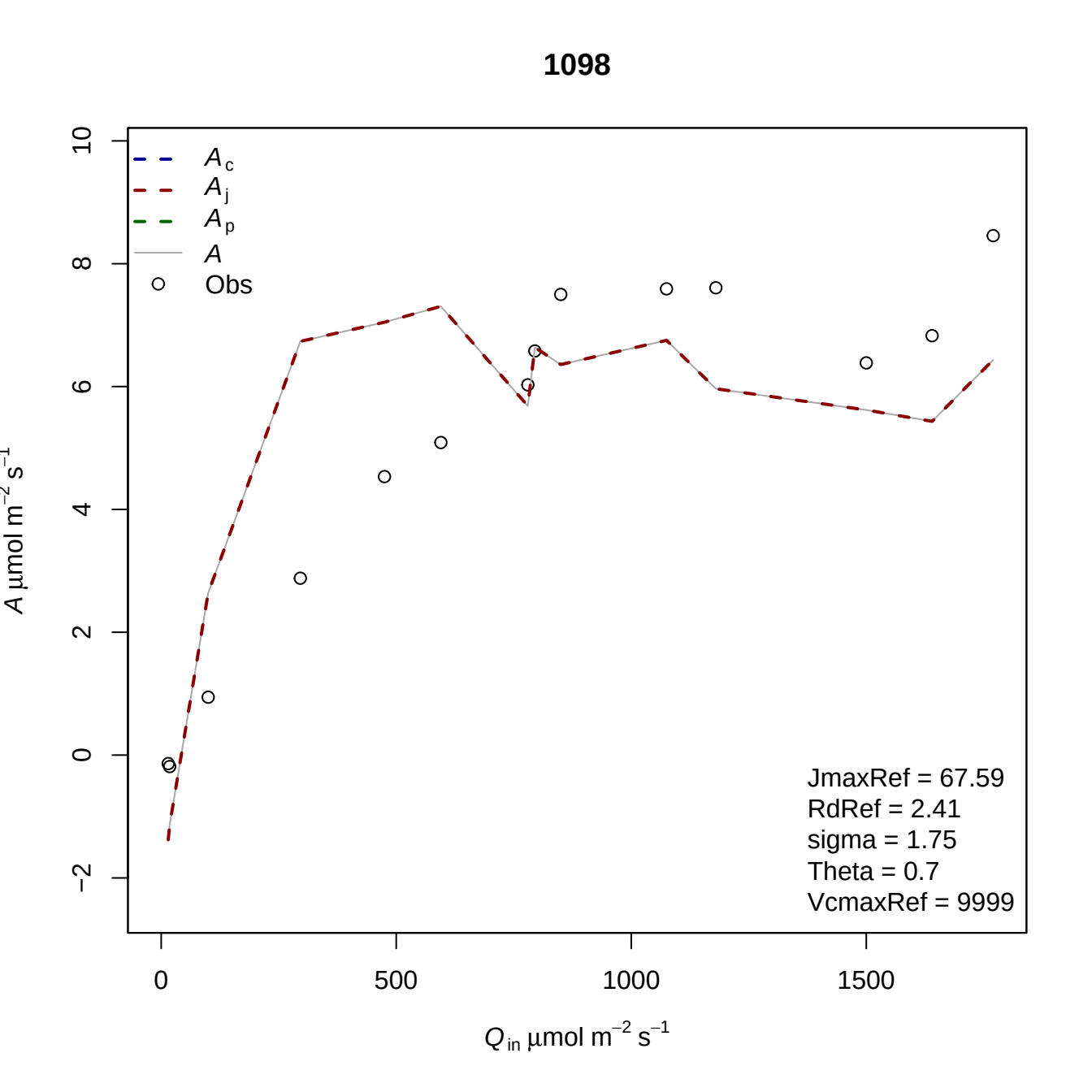

1093
A_c
A_j
A_p
A
Obs
JmaxRef = 88.88
RdRef = 5.7
sigma = 1.99
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

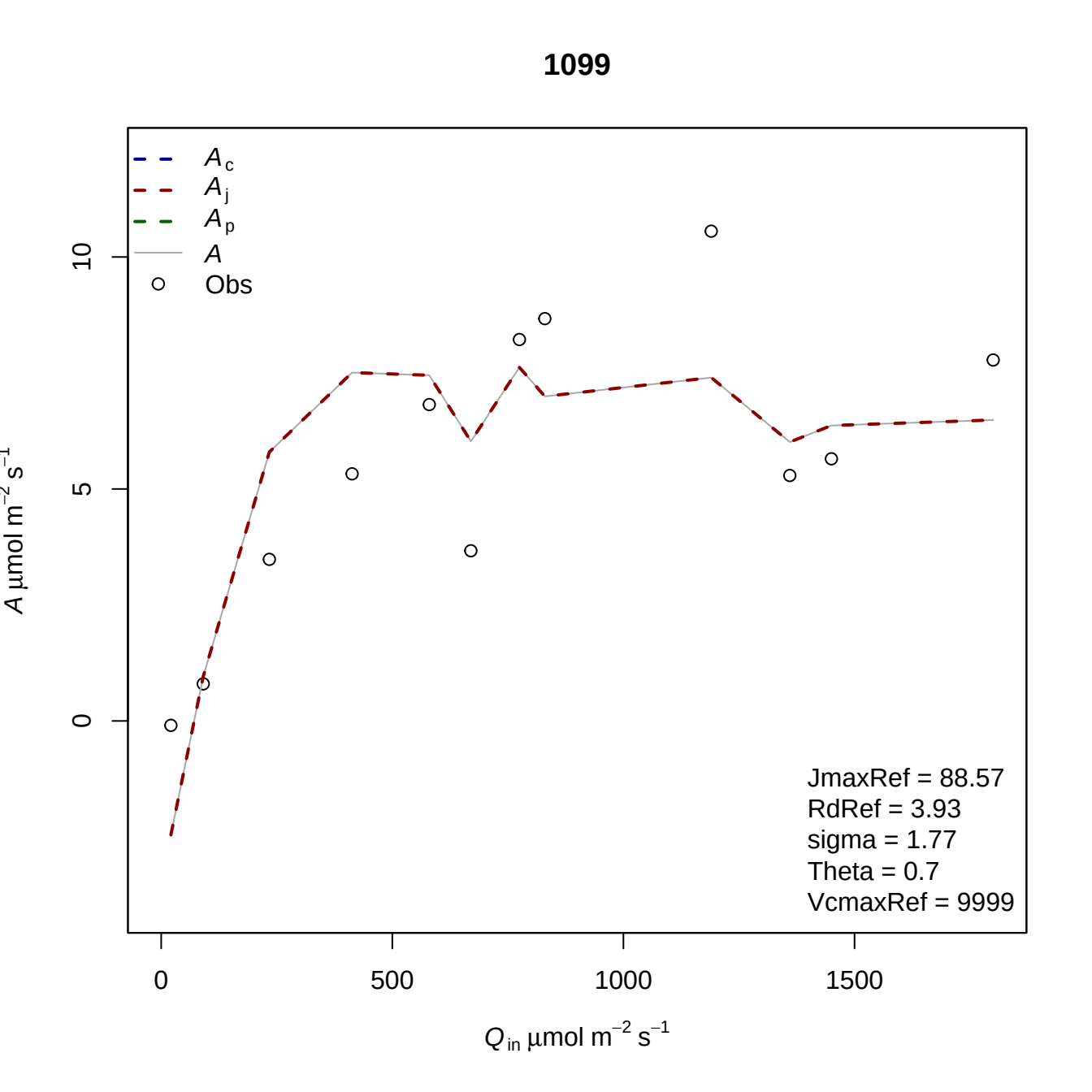

1094
A_c
A_j
A_p
A
Obs
JmaxRef = 70.73
RdRef = 4.01
sigma = 1.77
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

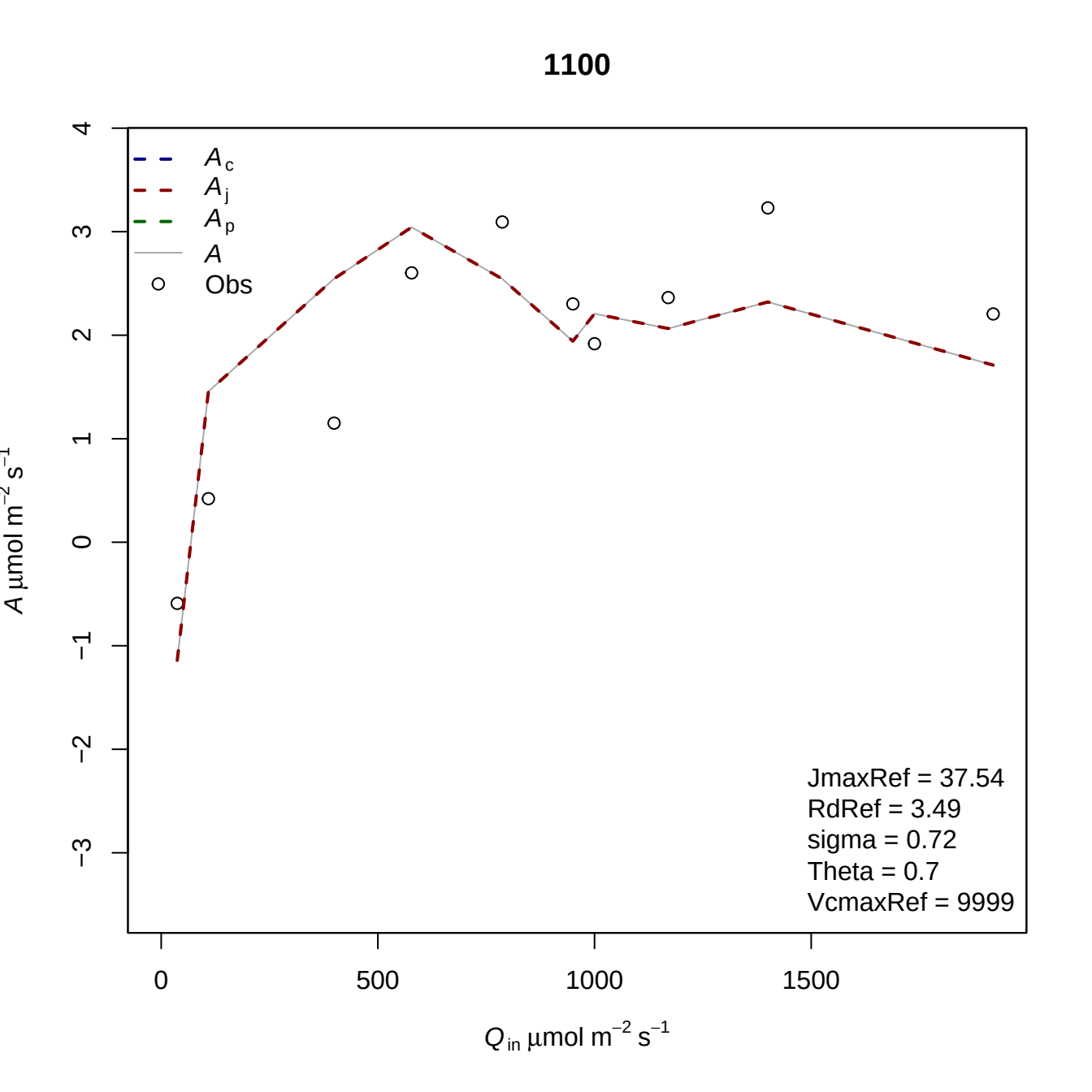

1095
A_c
A_j
A_p
A
Obs
JmaxRef = 78.67
RdRef = 5.37
sigma = 1.32
Theta = 0.7
VcmaxRef = 9999
10
5
0
−5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

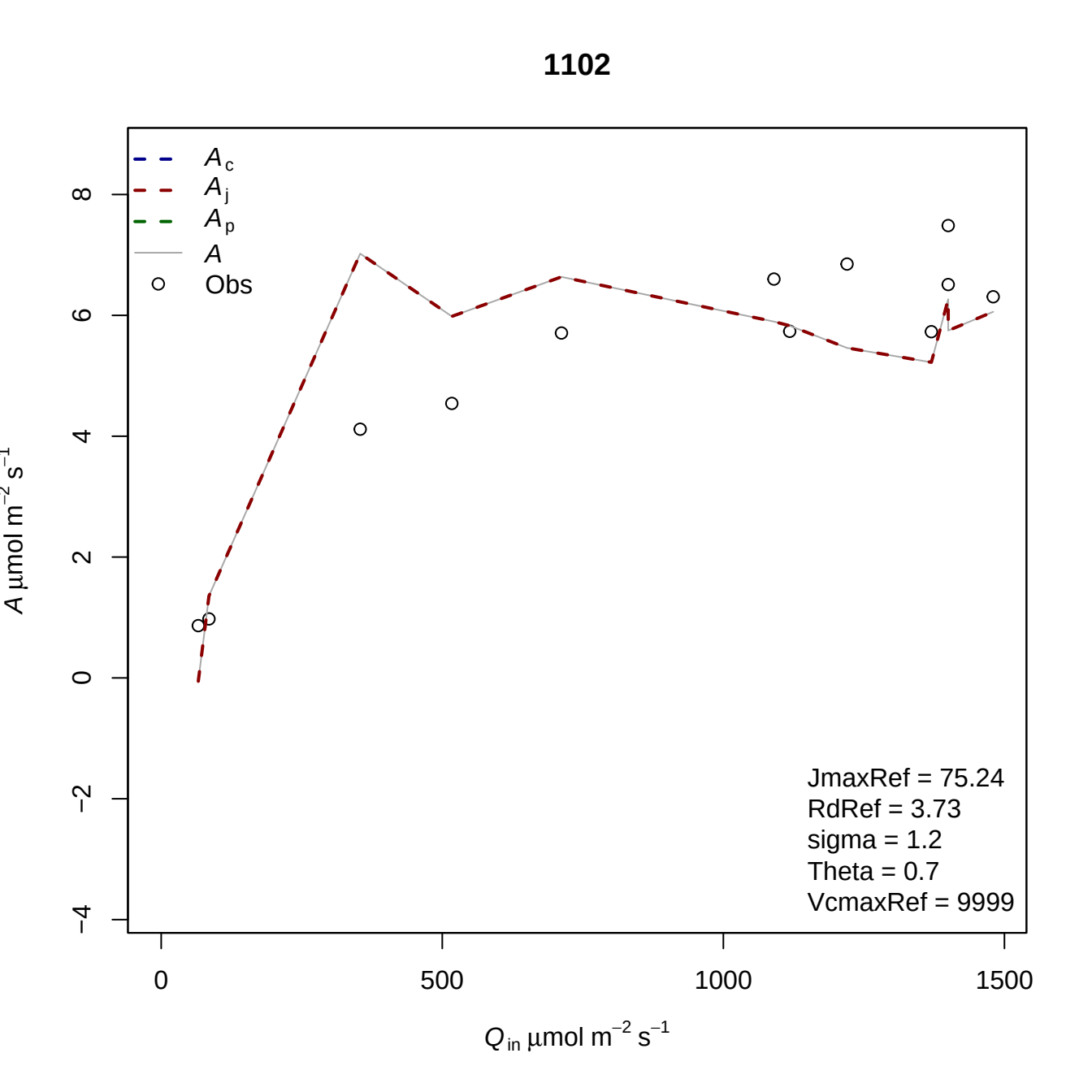

1096
A_c
A_j
A_p
A
Obs
JmaxRef = 39.65
RdRef = 2.57
sigma = 1.11
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500

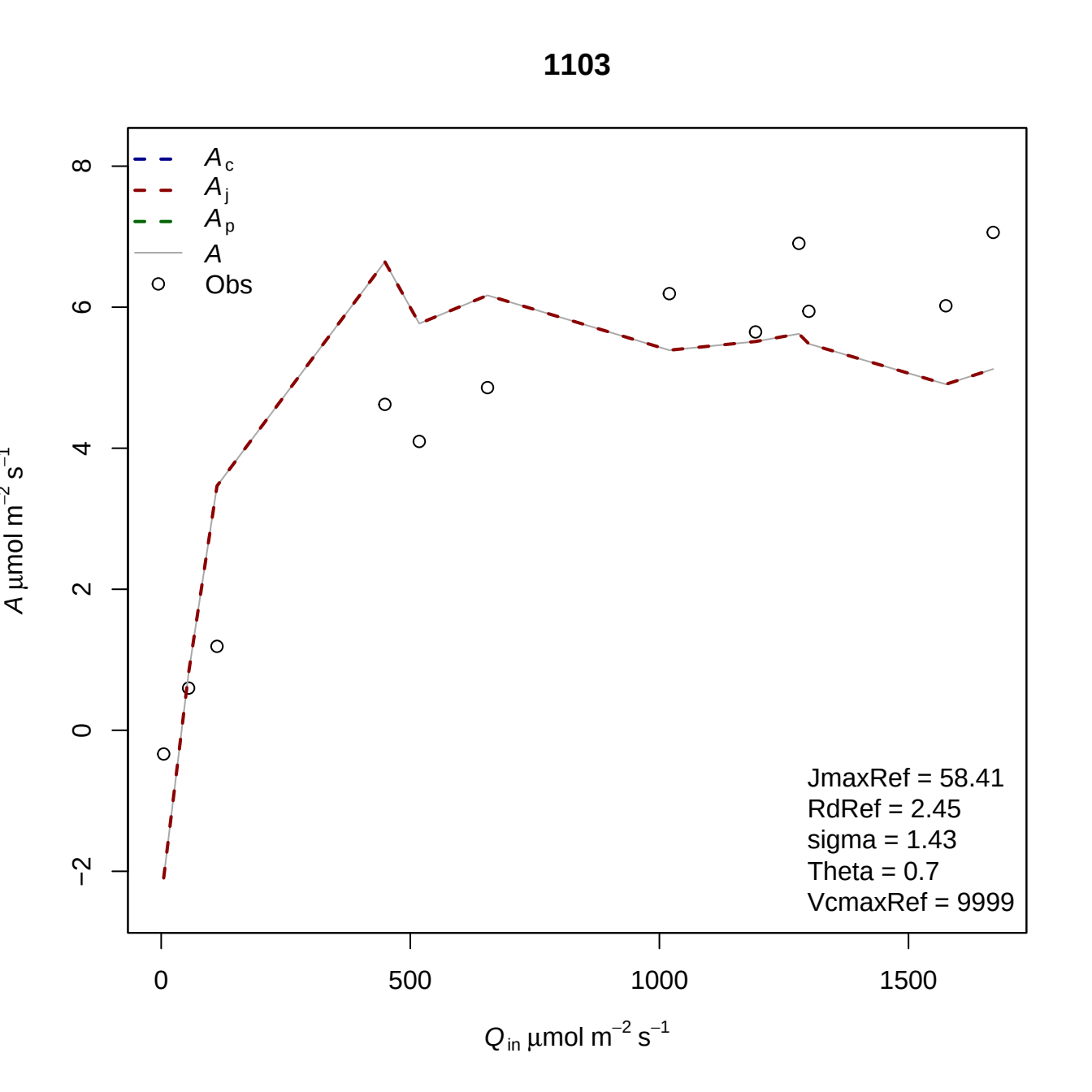

1097
A_c
A_j
A_p
A
Obs
JmaxRef = 29.07
RdRef = 1.72
sigma = 1.74
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

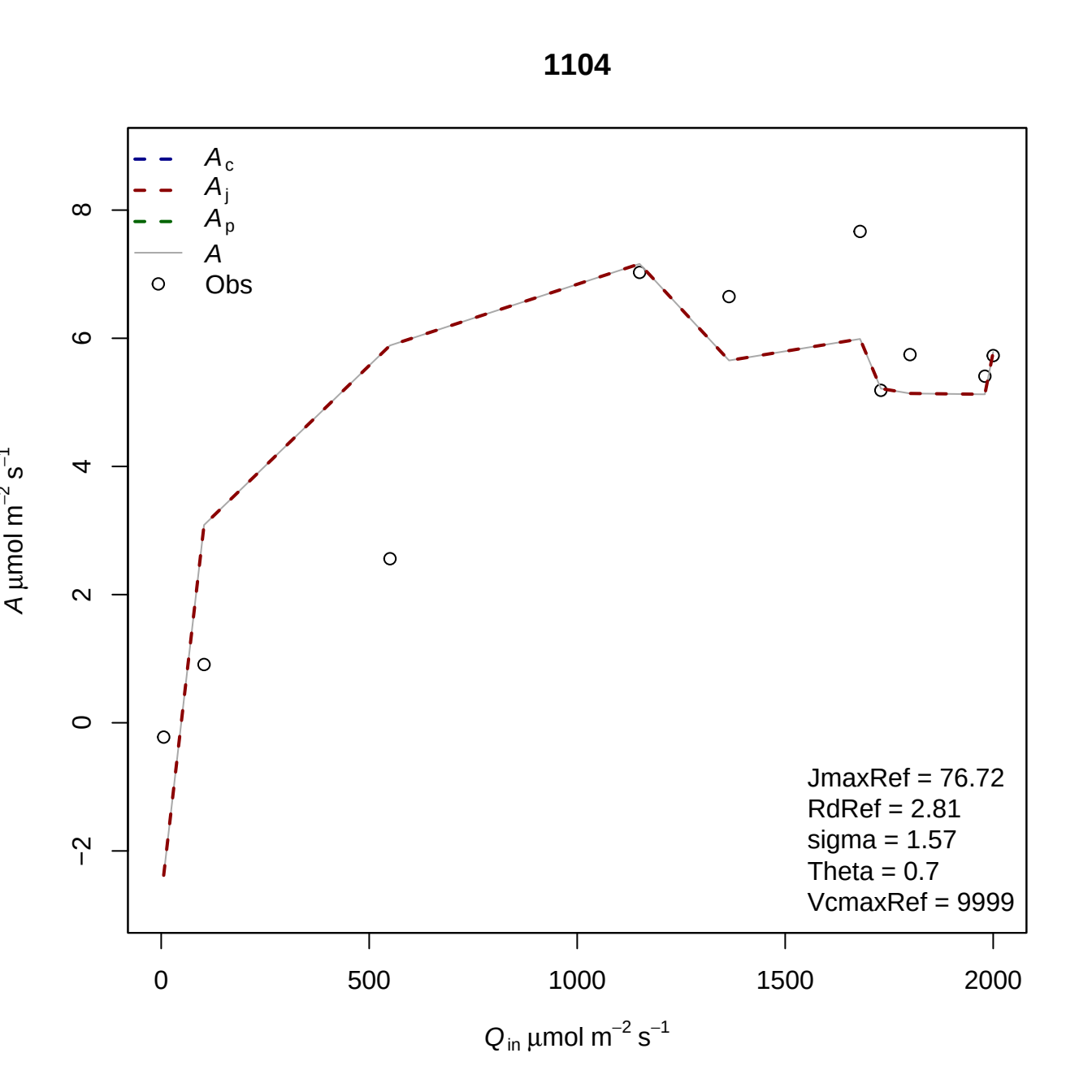

1098
A_c
A_j
A_p
A
Obs
JmaxRef = 67.59
RdRef = 2.41
sigma = 1.75
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

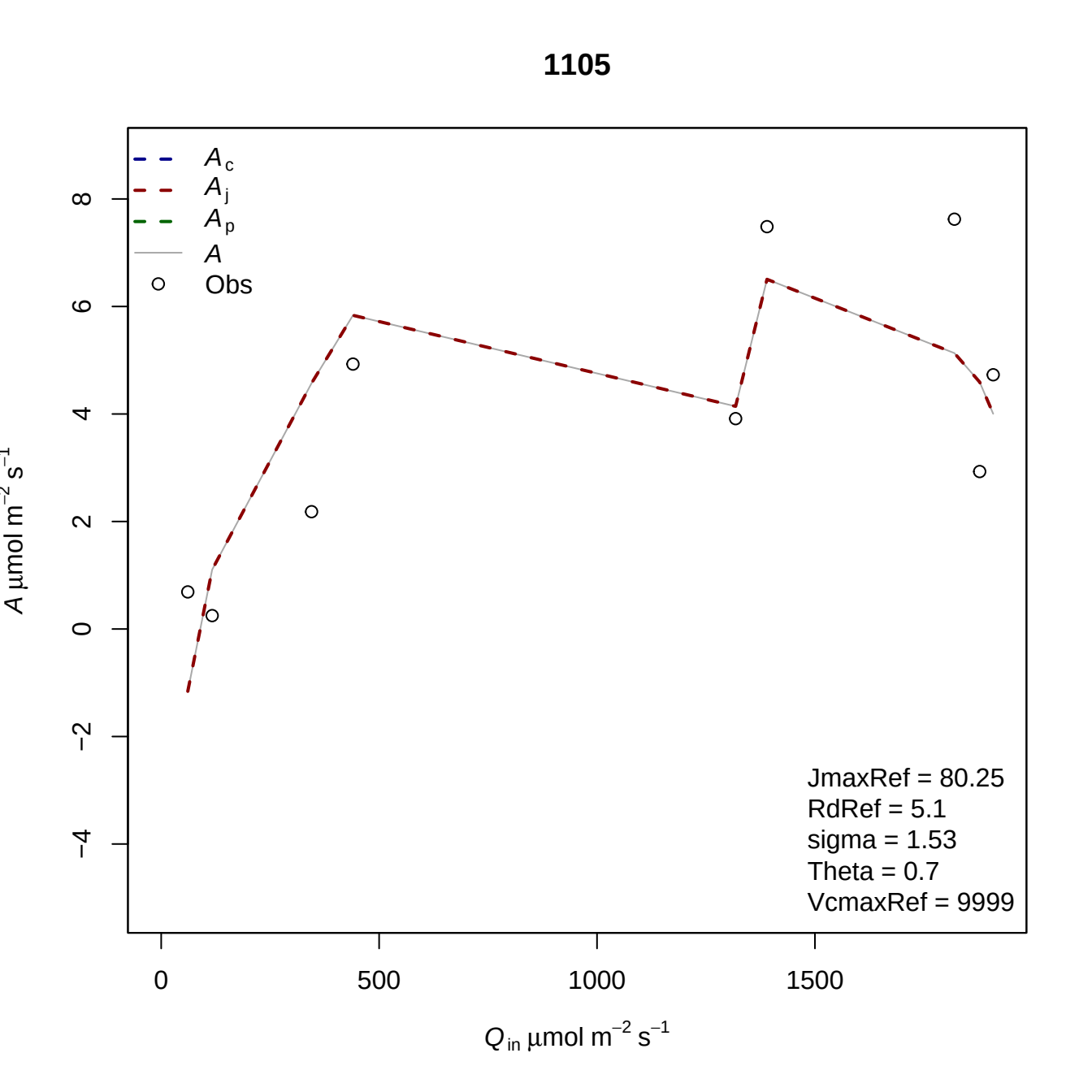

1099
A_c
A_j
A_p
A
Obs
JmaxRef = 88.57
RdRef = 3.93
sigma = 1.77
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

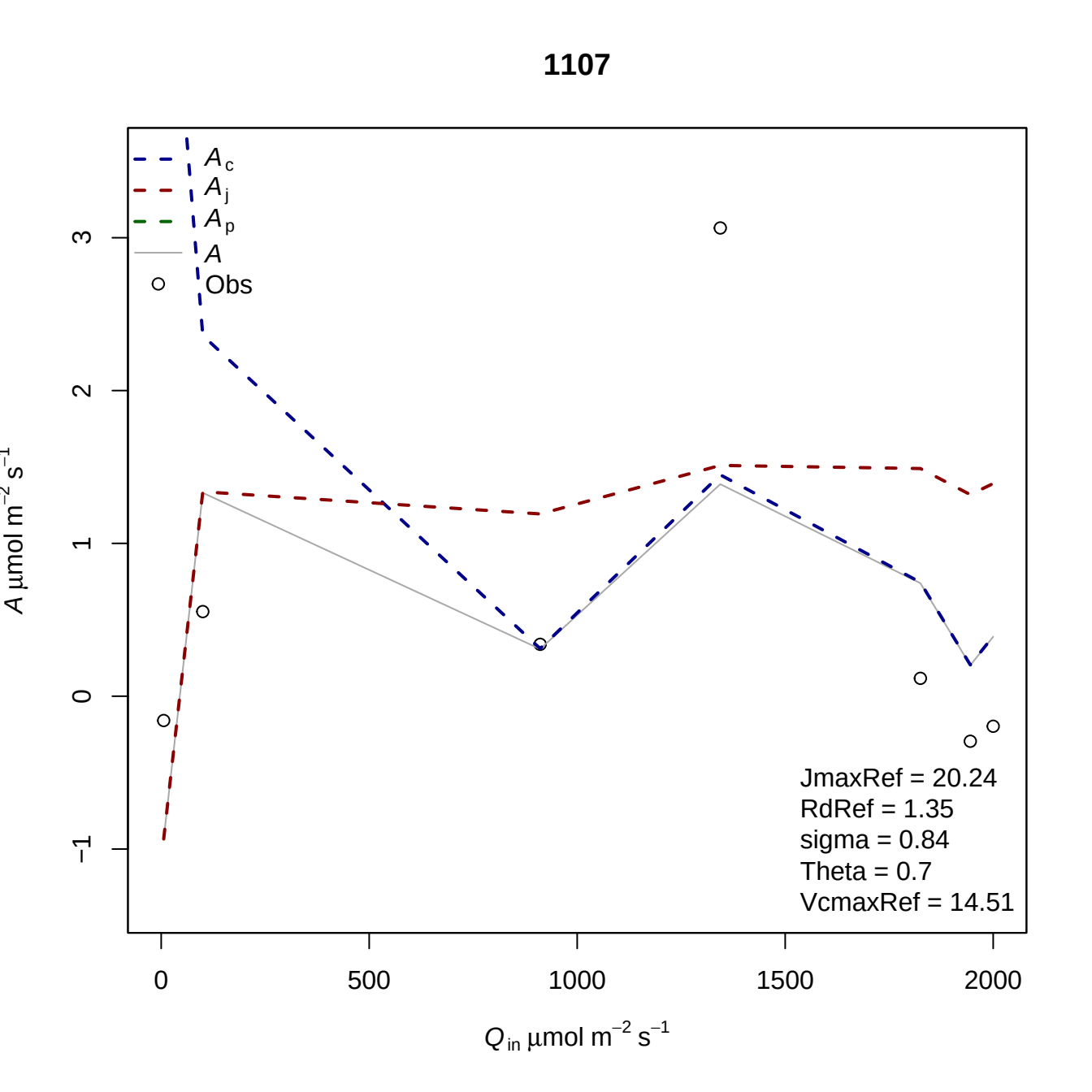

1100
A_c
A_j
A_p
A
Obs
JmaxRef = 37.54
RdRef = 3.49
sigma = 0.72
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
−3
0
500
1000
1500

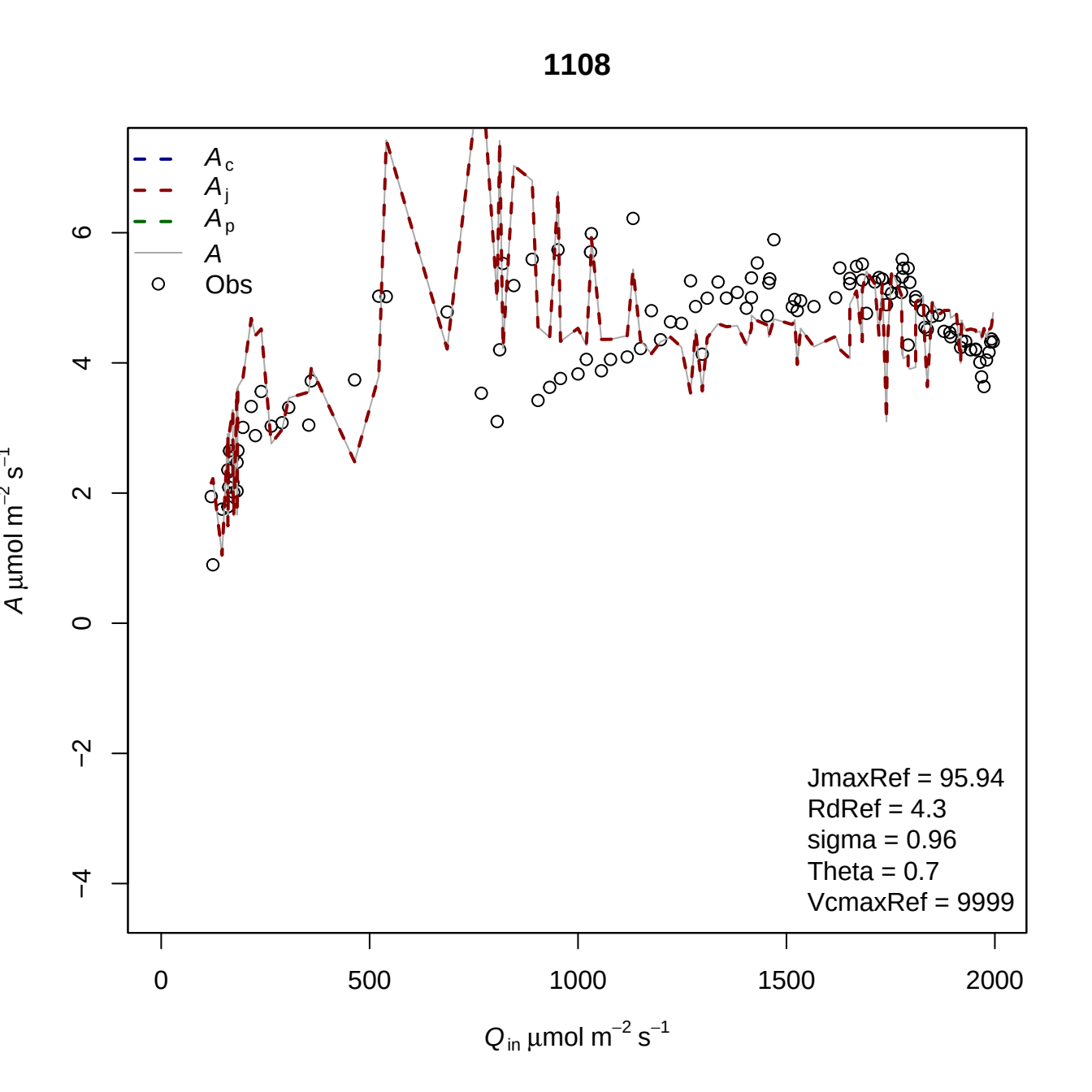

1102
A_c
A_j
A_p
A
Obs
JmaxRef = 75.24
RdRef = 3.73
sigma = 1.2
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

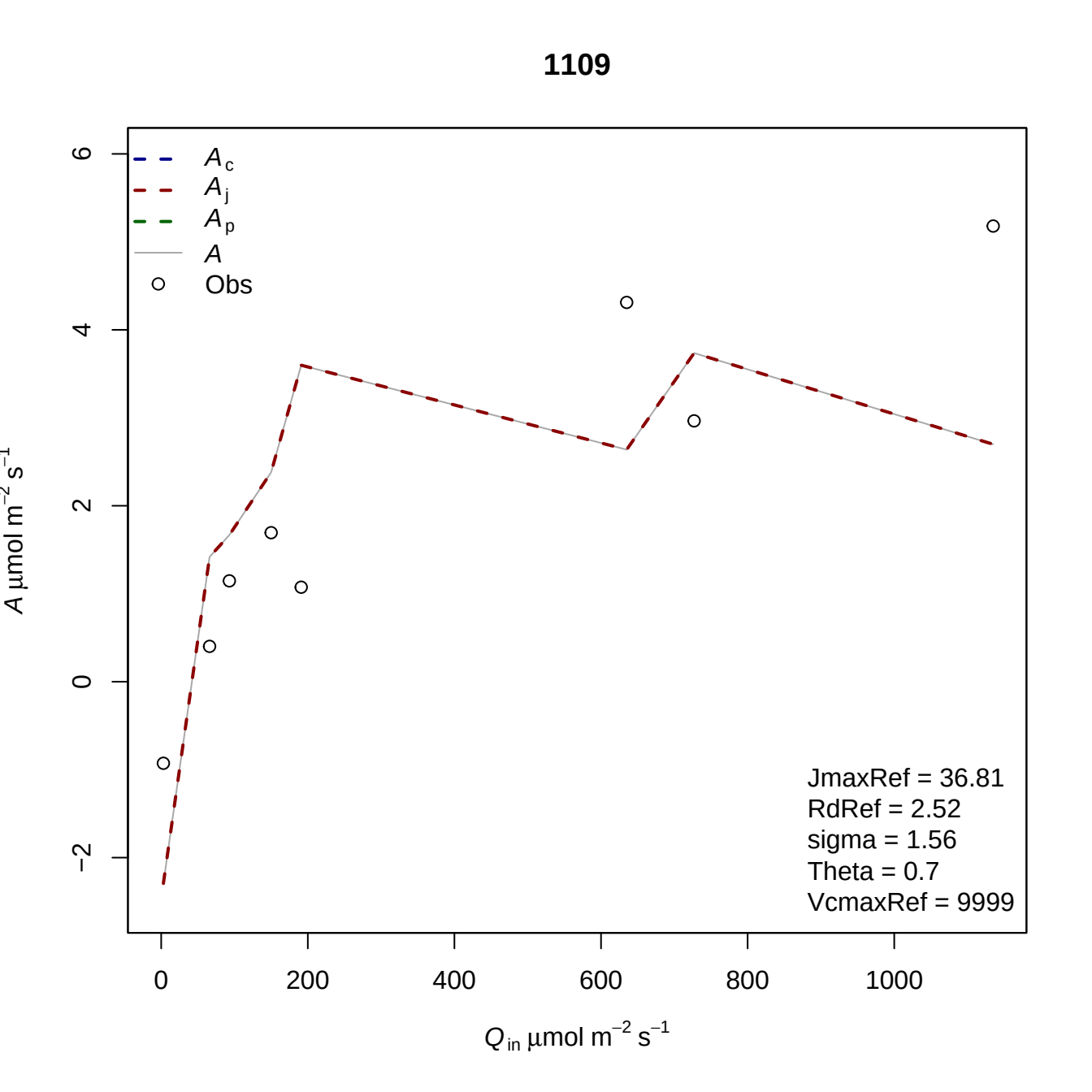

1103
A_c
A_j
A_p
A
Obs
JmaxRef = 58.41
RdRef = 2.45
sigma = 1.43
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

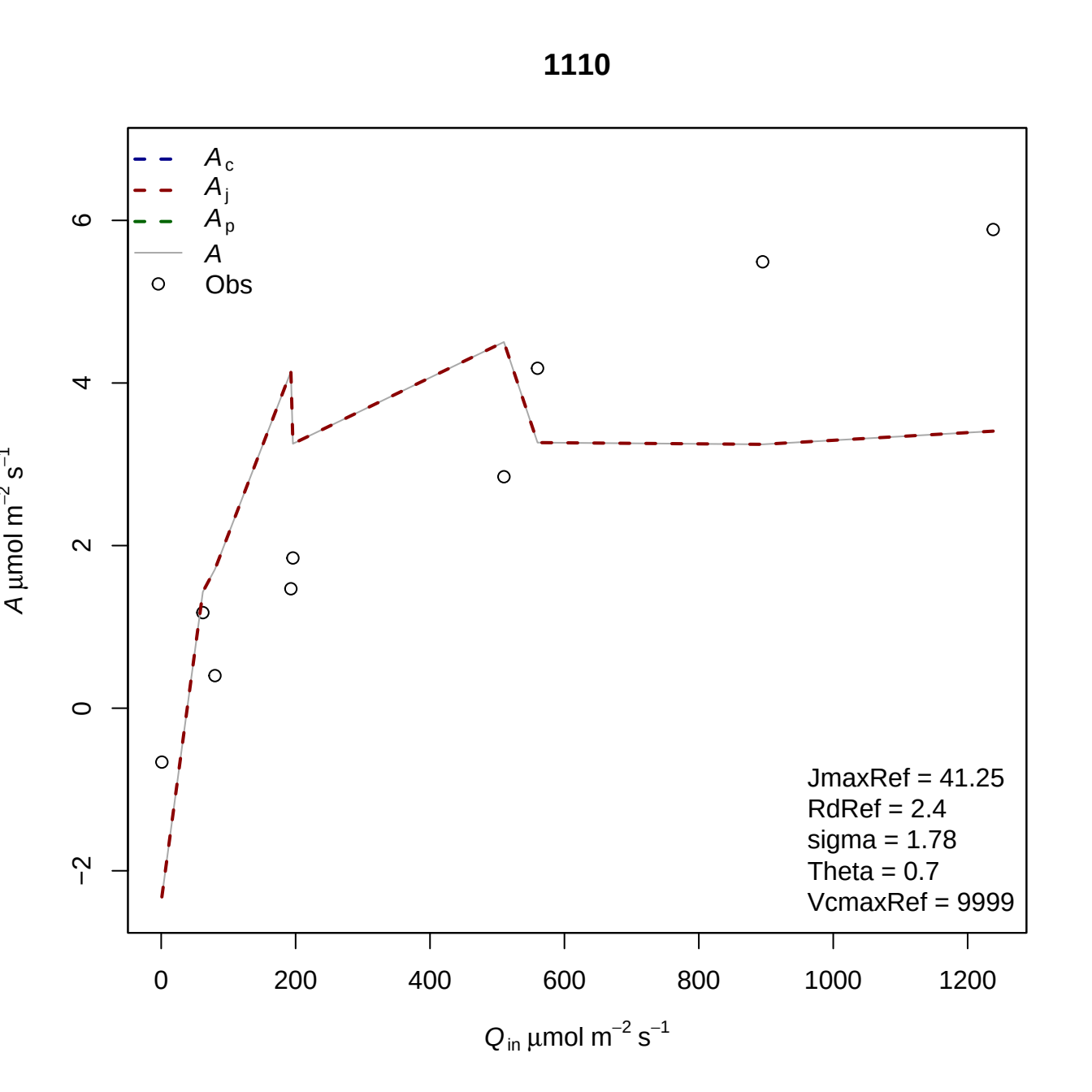

1104
A_c
A_j
A_p
A
Obs
JmaxRef = 76.72
RdRef = 2.81
sigma = 1.57
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

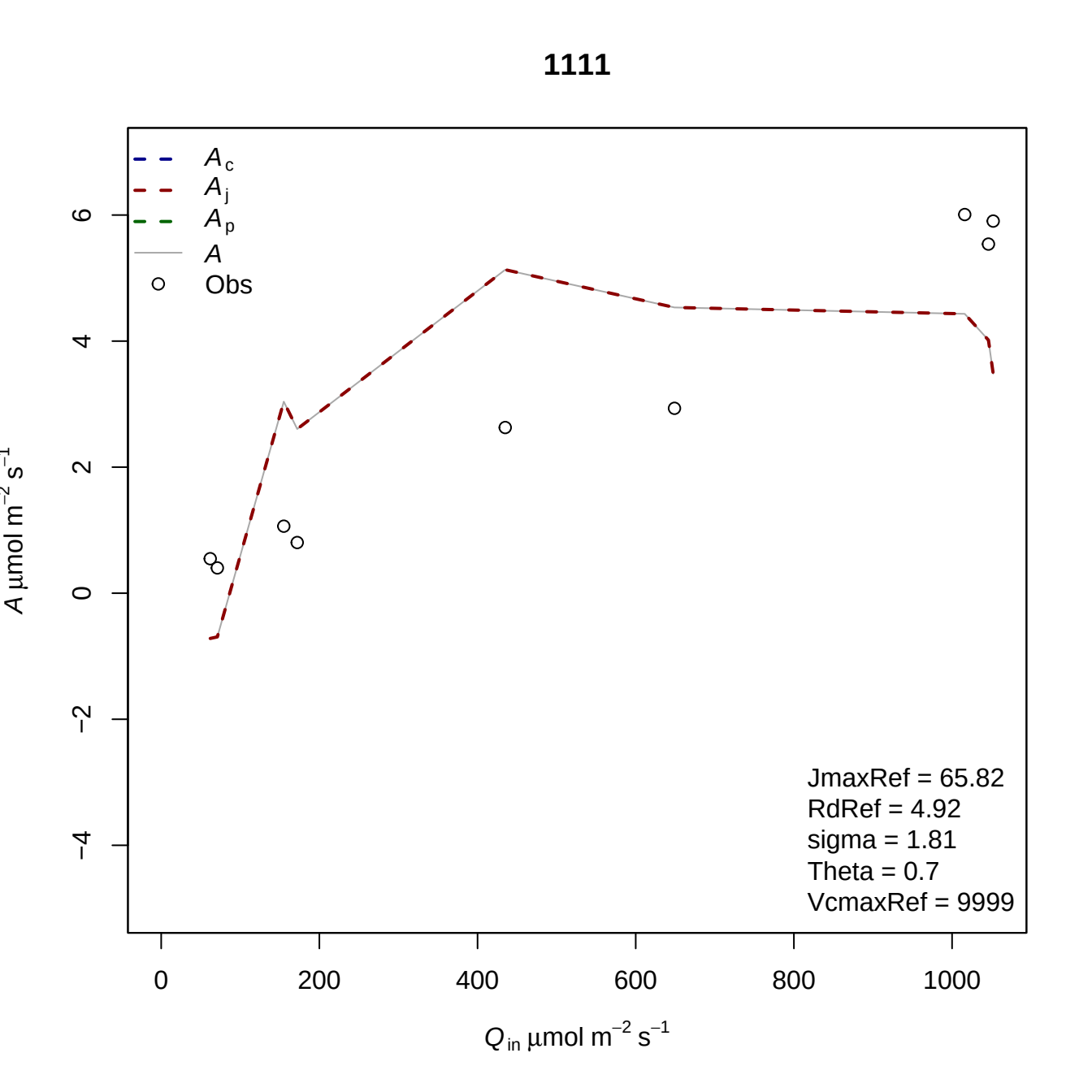

1105
A_c
A_j
A_p
A
Obs
JmaxRef = 80.25
RdRef = 5.1
sigma = 1.53
Theta = 0.7
VcmaxRef = 9999
8
6
4
2
0
−2
−4
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

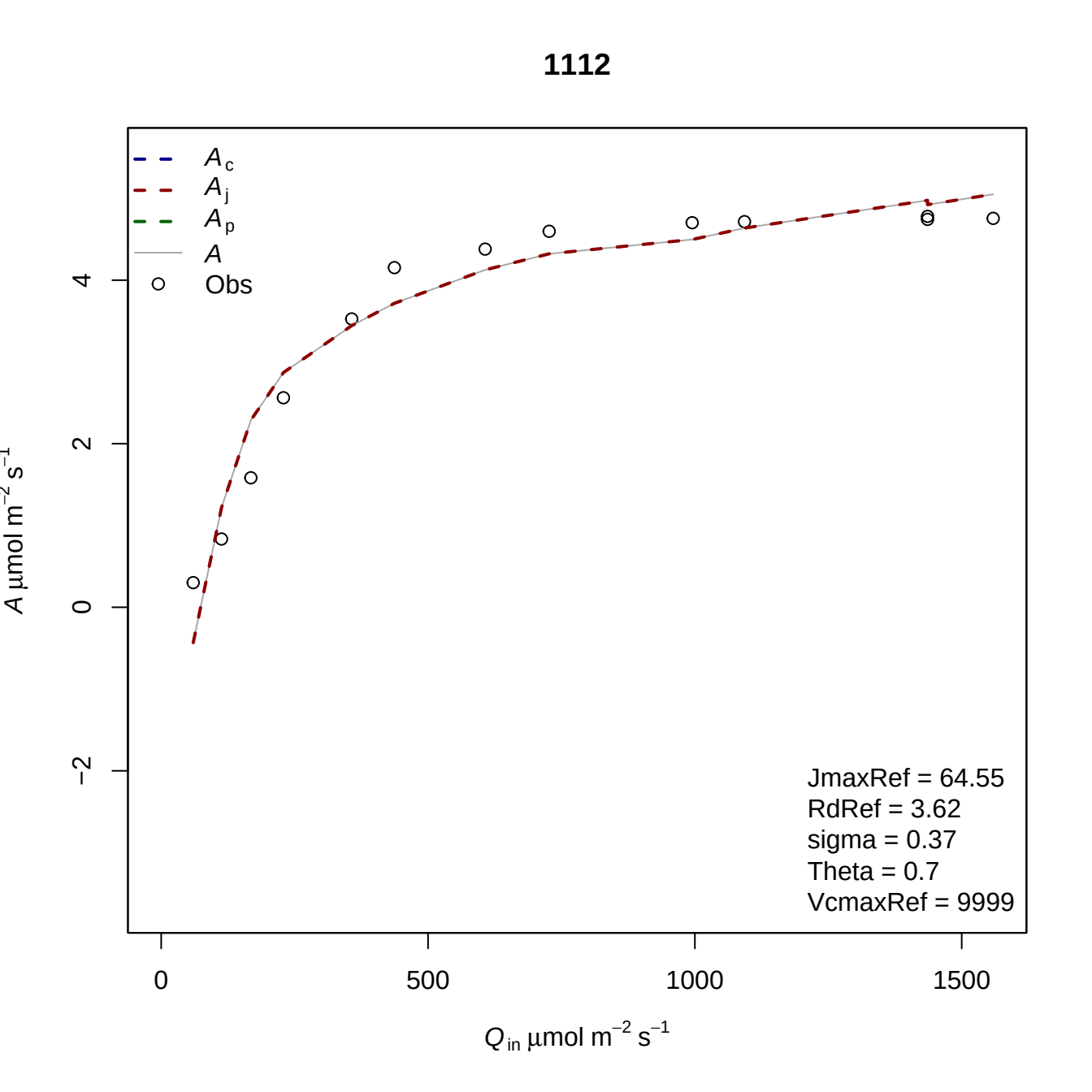

1107
A_c
A_j
A_p
A
Obs
JmaxRef = 20.24
RdRef = 1.35
sigma = 0.84
Theta = 0.7
VcmaxRef = 14.51
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500
2000

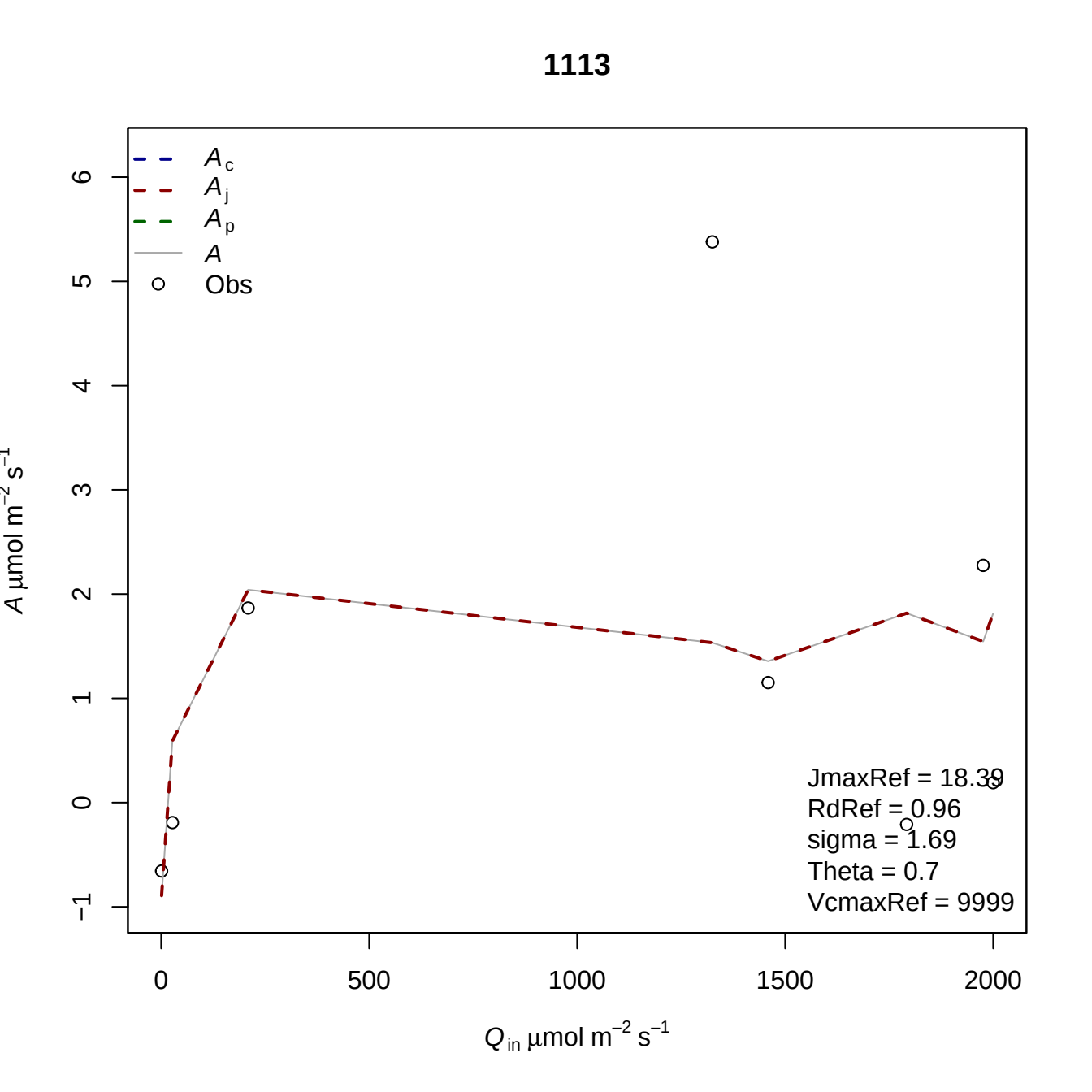

1108
A_c
A_j
A_p
A
Obs
JmaxRef = 95.94
RdRef = 4.3
sigma = 0.96
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500
2000

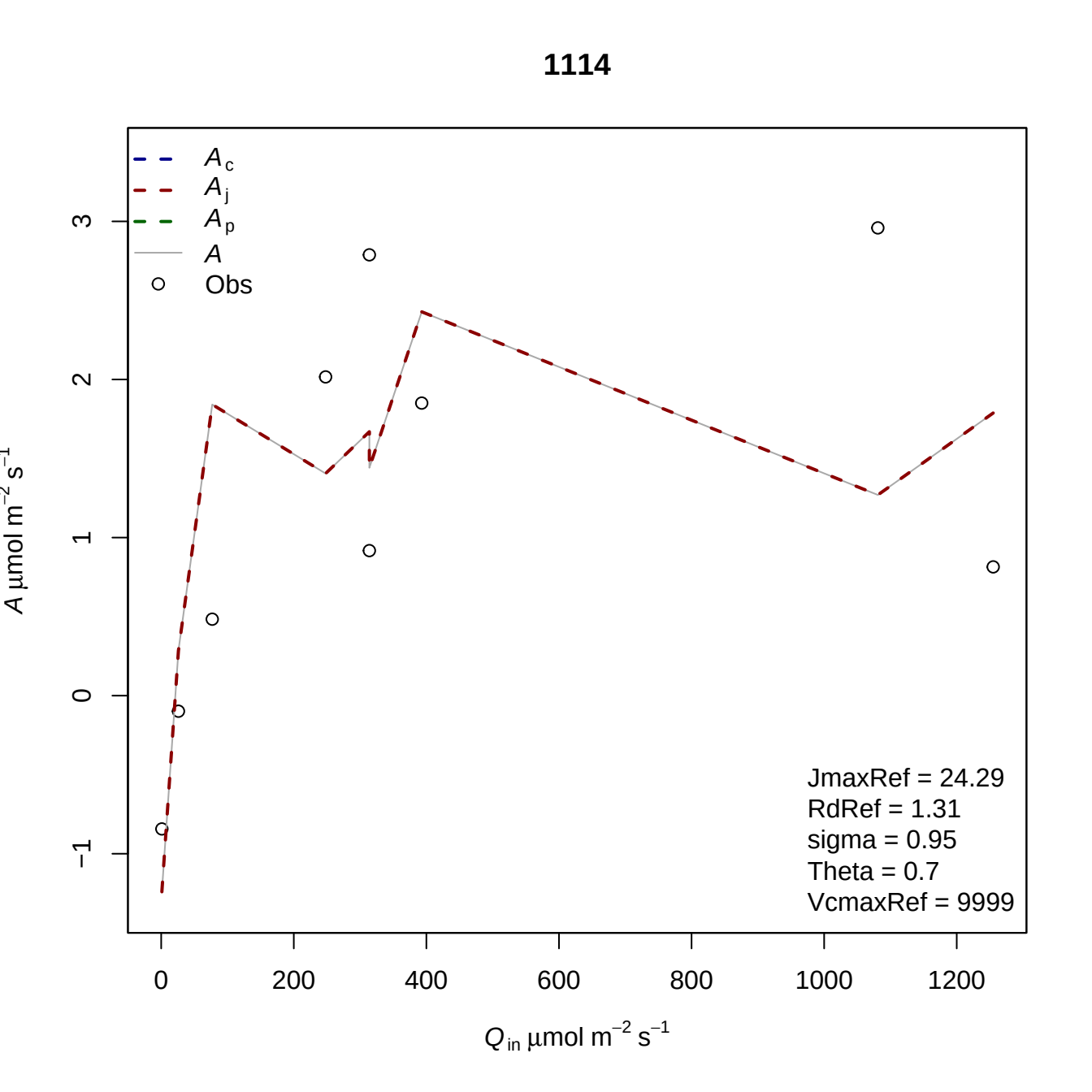

1109
A_c
A_j
A_p
A
Obs
JmaxRef = 36.81
RdRef = 2.52
sigma = 1.56
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
200
400
600
800
1000

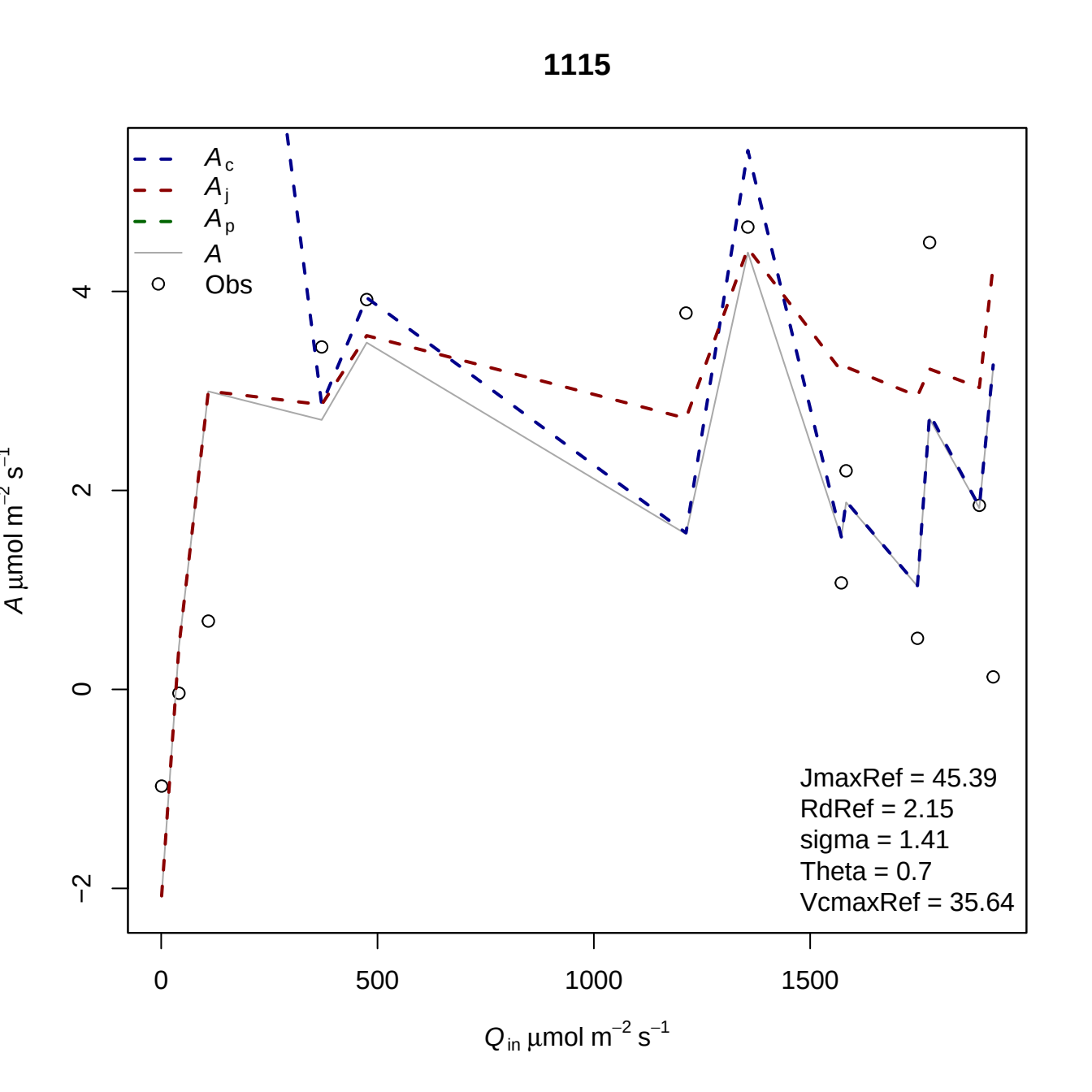

1110
A_c
A_j
A_p
A
Obs
JmaxRef = 41.25
RdRef = 2.4
sigma = 1.78
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800
1000
1200

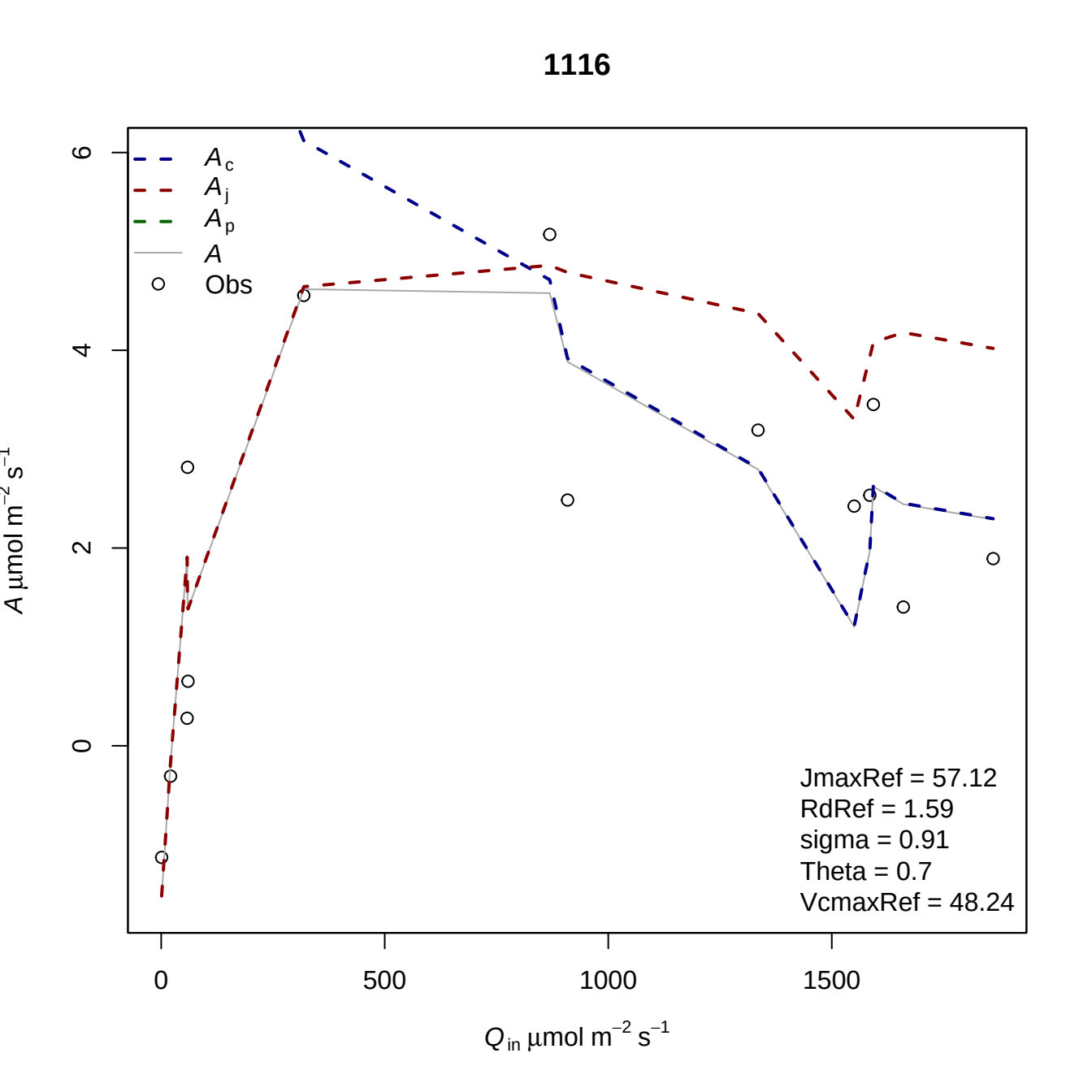

1111
A_c
A_j
A_p
A
Obs
JmaxRef = 65.82
RdRef = 4.92
sigma = 1.81
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
200
400
600
800
1000

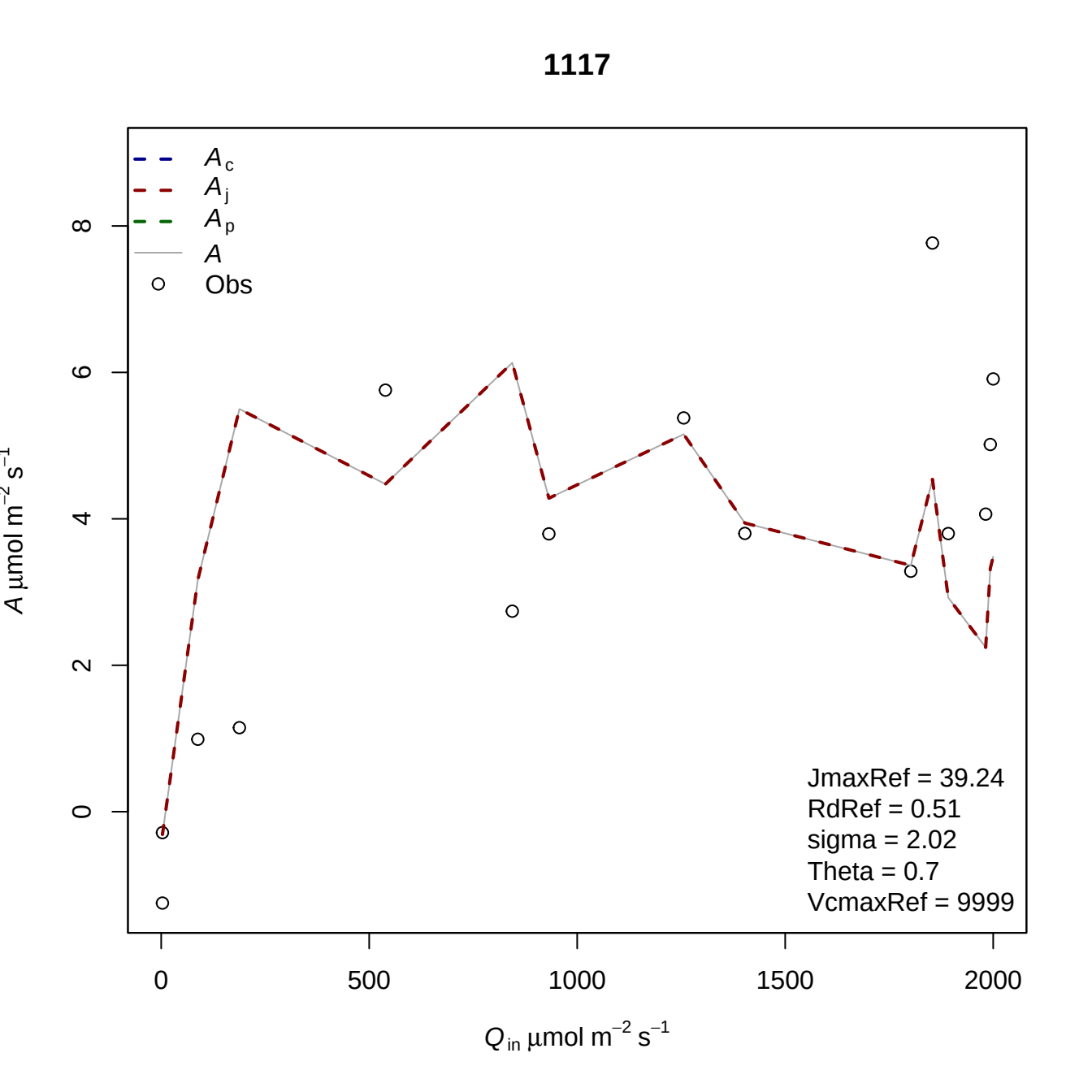

1112
A_c
A_j
A_p
A
Obs
JmaxRef = 64.55
RdRef = 3.62
sigma = 0.37
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

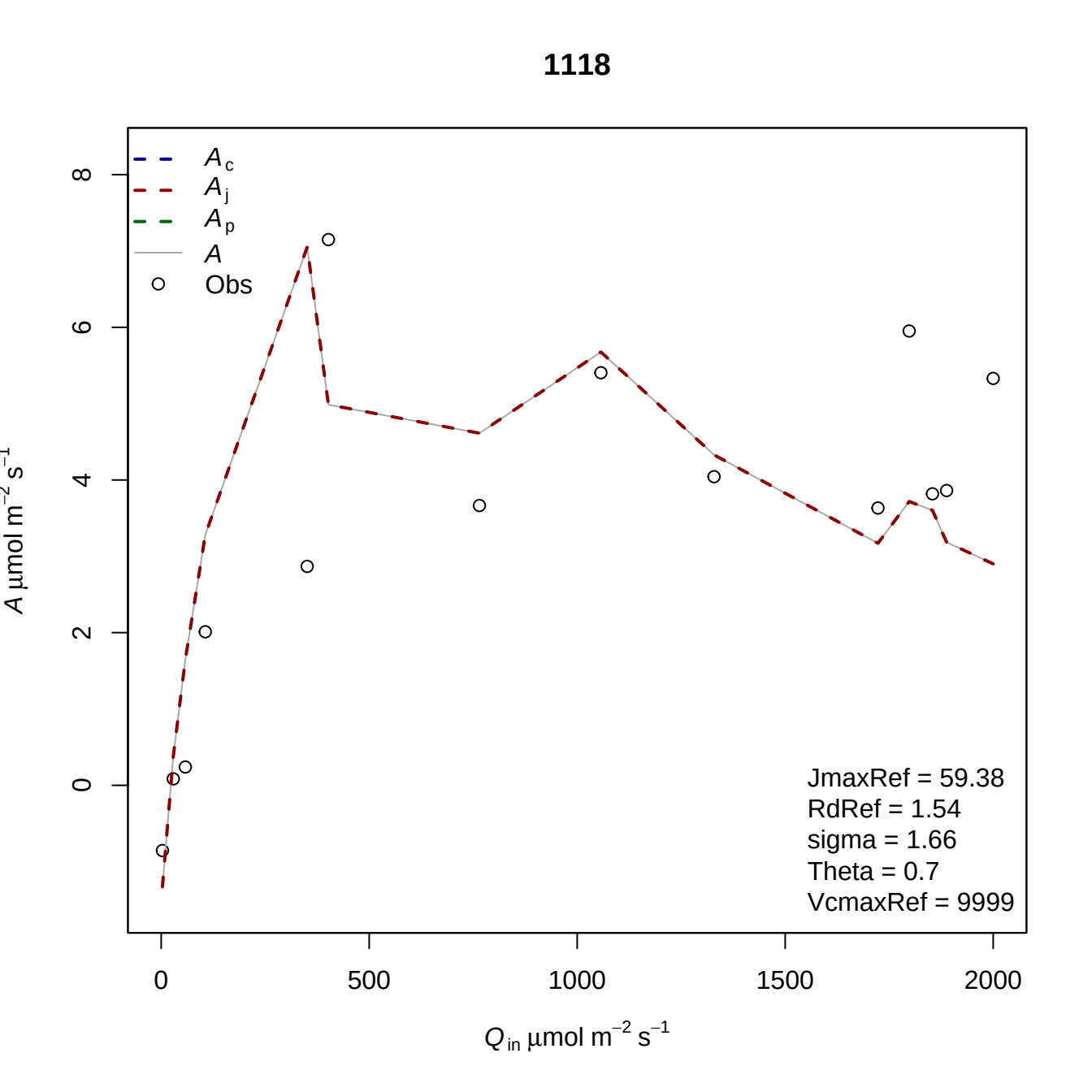

1113
A_c
A_j
A_p
A
Obs
JmaxRef = 18.39
RdRef = 0.96
sigma = 1.69
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500
2000

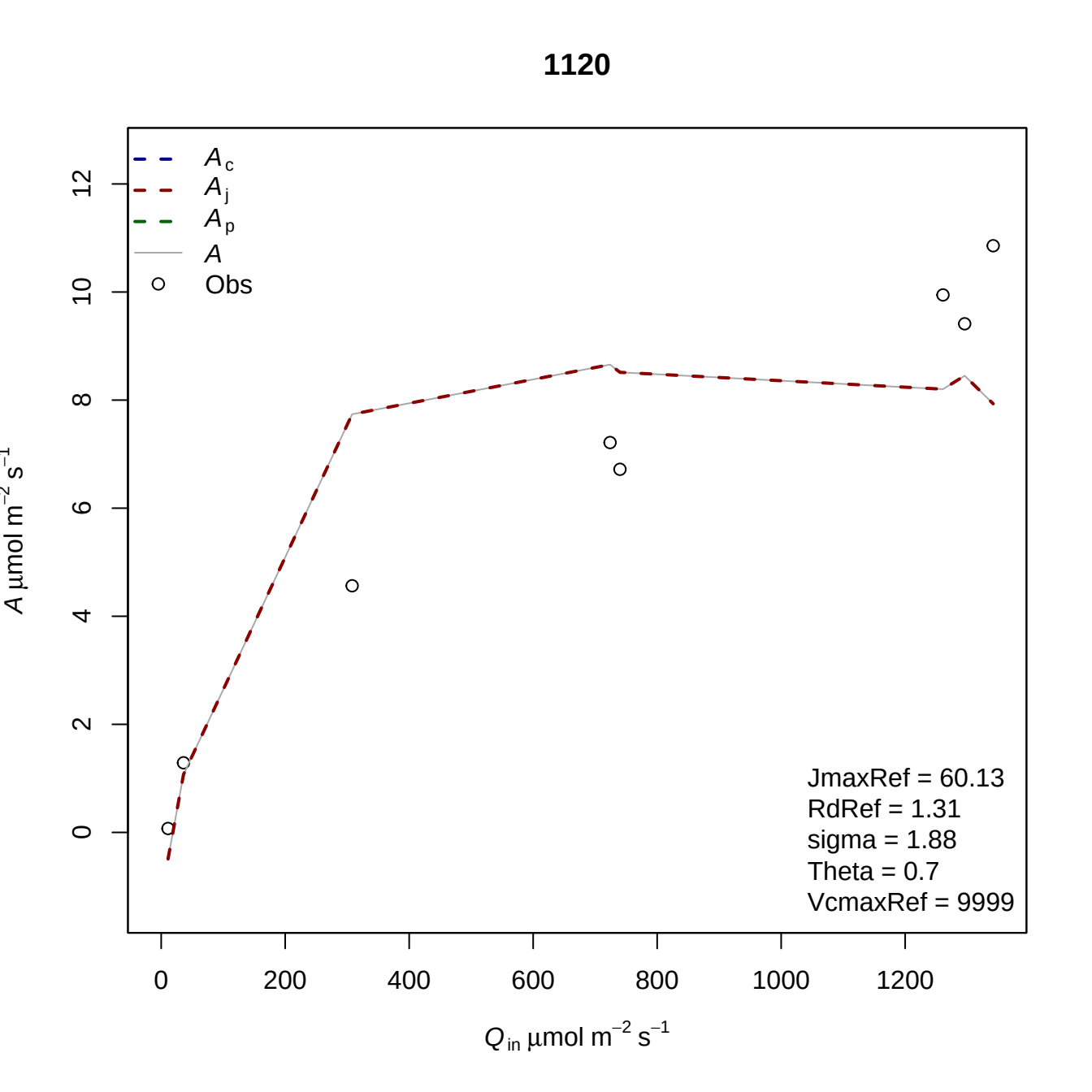

1114
A_c
A_j
A_p
A
Obs
JmaxRef = 24.29
RdRef = 1.31
sigma = 0.95
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800
1000
1200

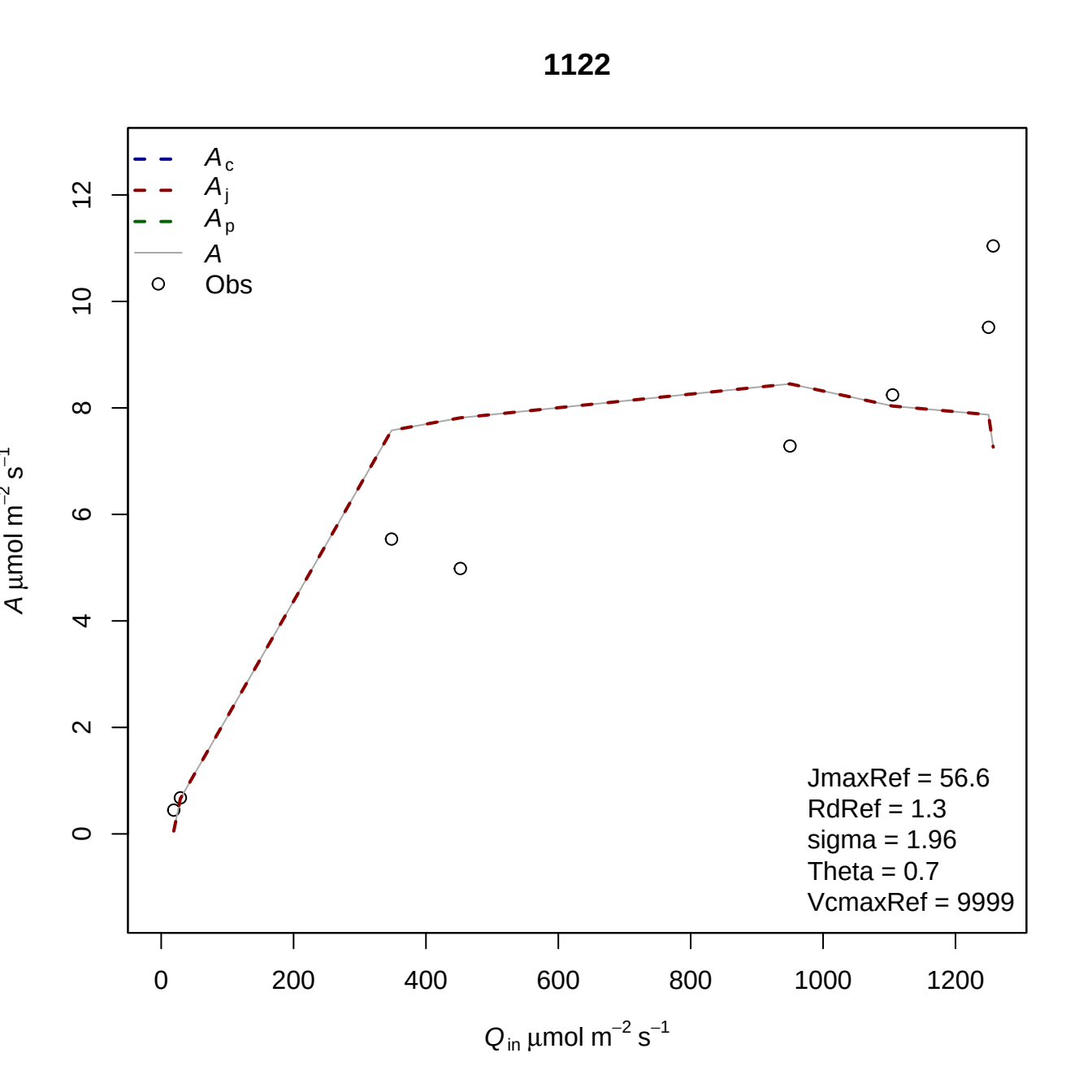

1115
A_c
A_j
A_p
A
Obs
JmaxRef = 45.39
RdRef = 2.15
sigma = 1.41
Theta = 0.7
VcmaxRef = 35.64
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500

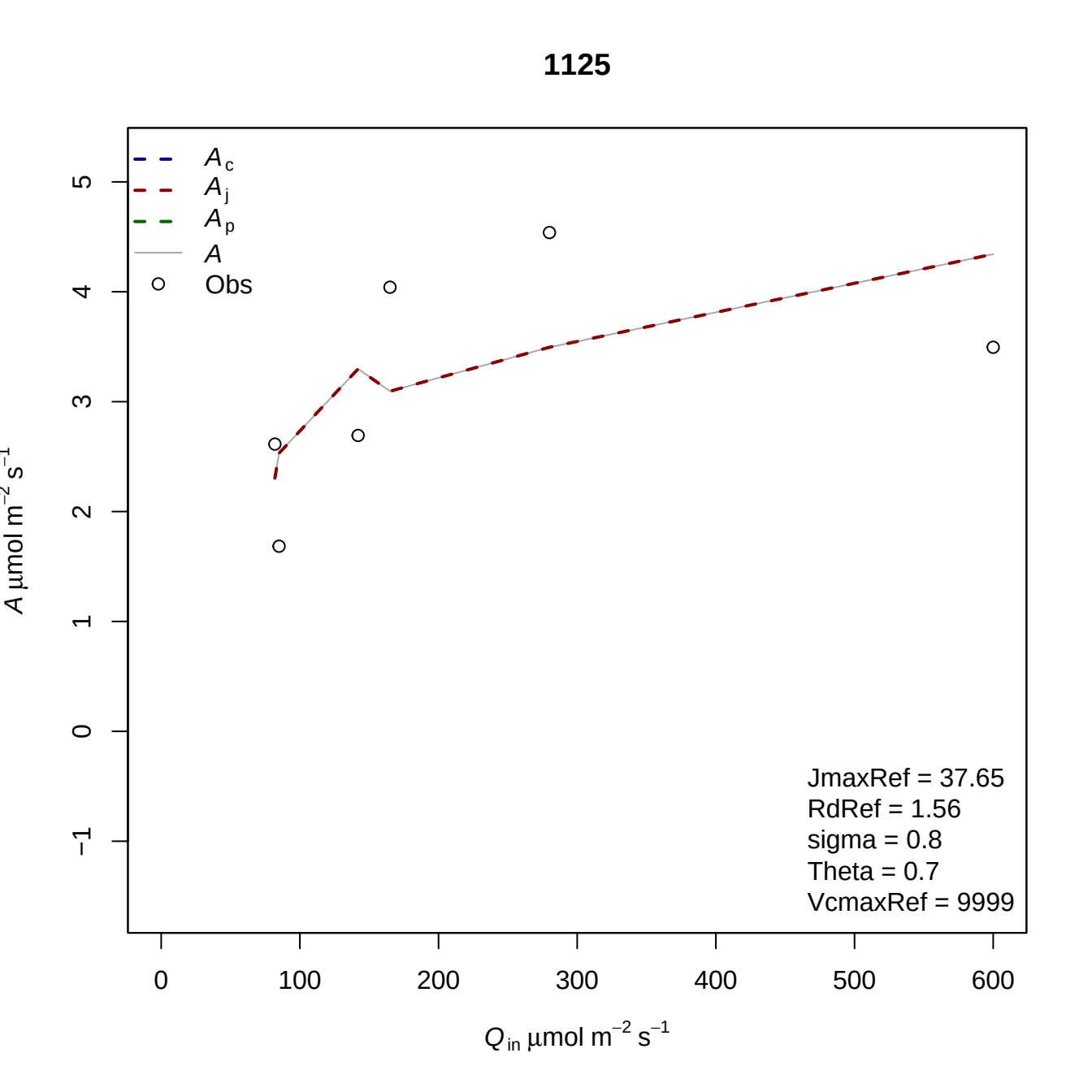

1116
A_c
A_j
A_p
A
Obs
JmaxRef = 57.12
RdRef = 1.59
sigma = 0.91
Theta = 0.7
VcmaxRef = 48.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

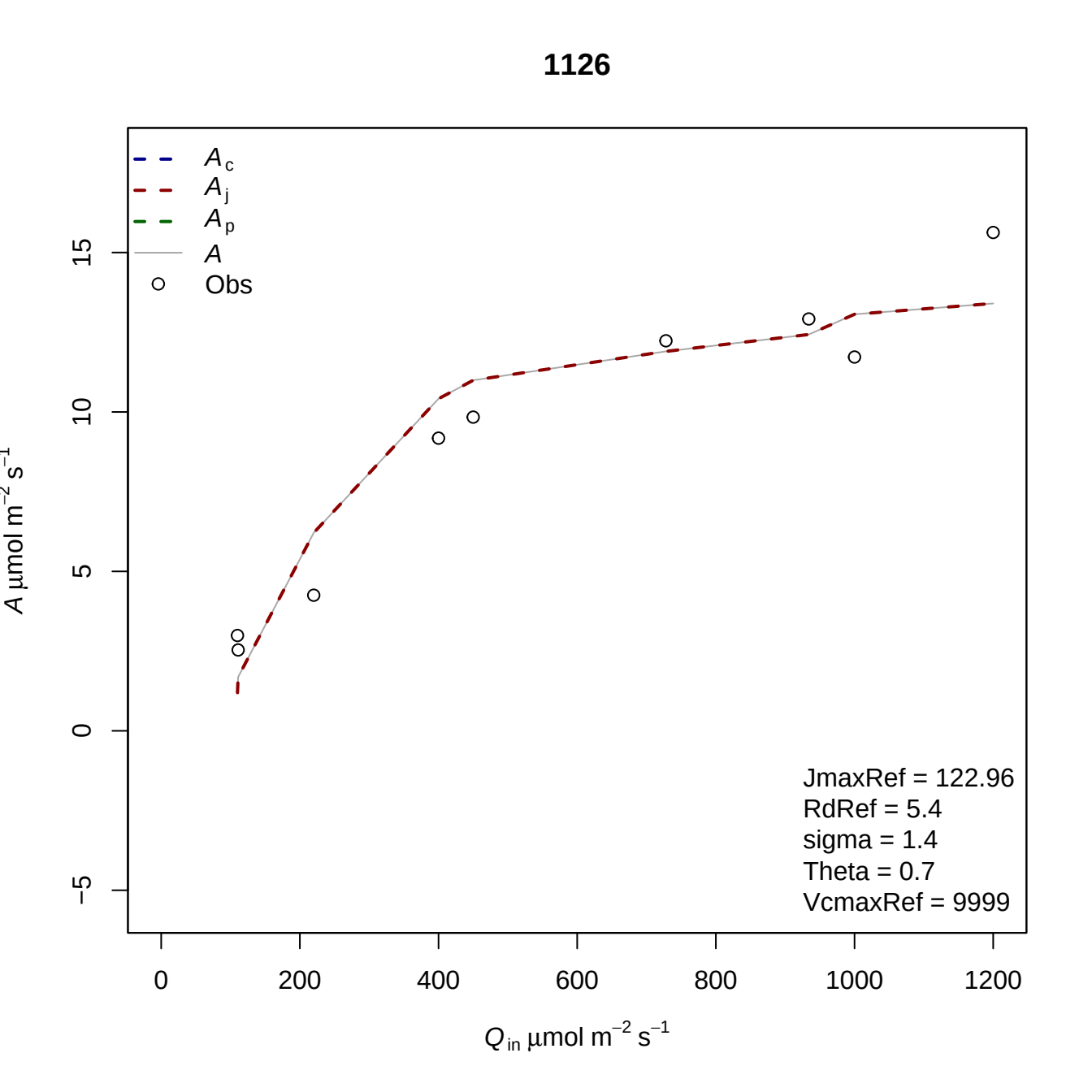

1117
A_c
A_j
A_p
A
Obs
JmaxRef = 39.24
RdRef = 0.51
sigma = 2.02
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

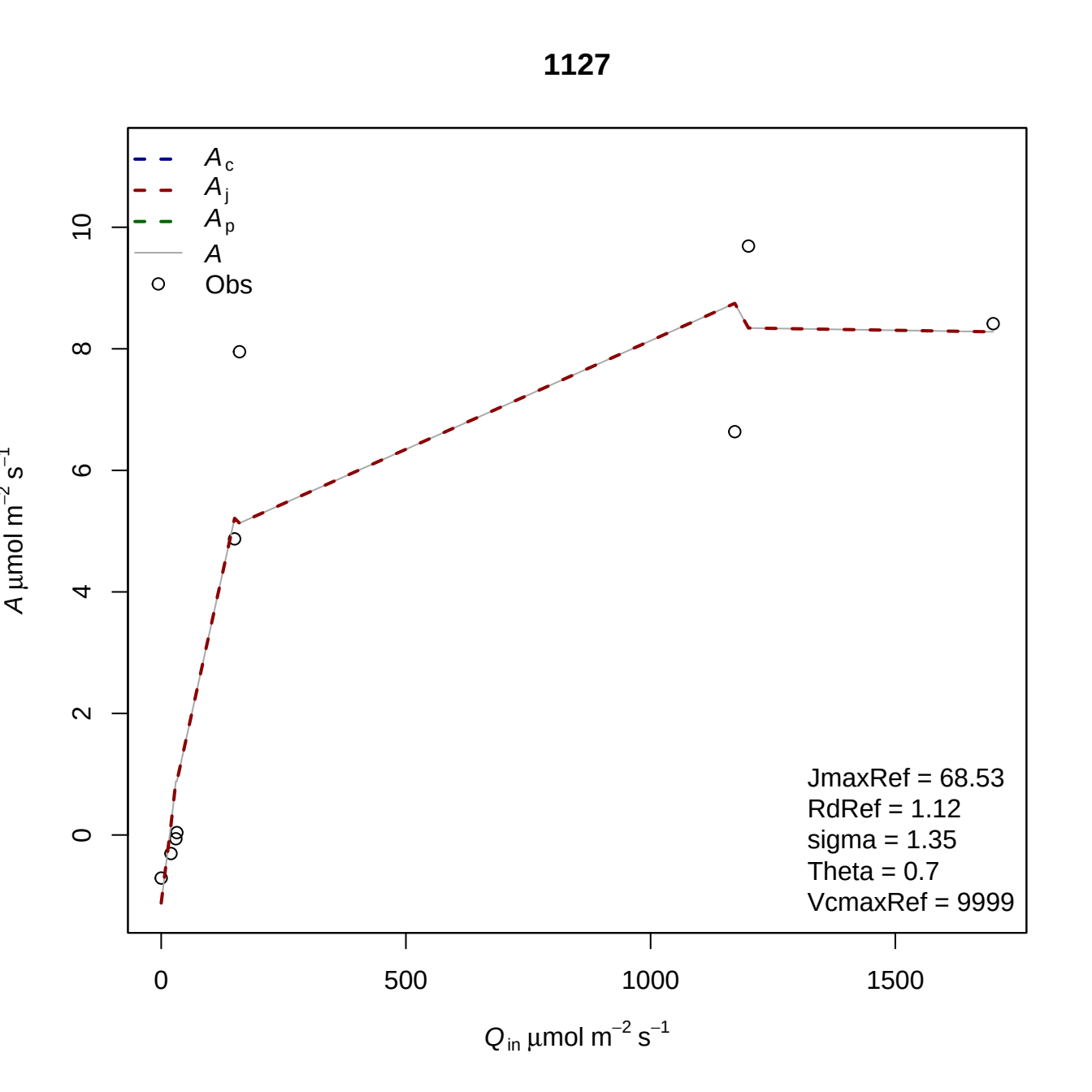

1118
A_c
A_j
A_p
A
Obs
JmaxRef = 59.38
RdRef = 1.54
sigma = 1.66
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

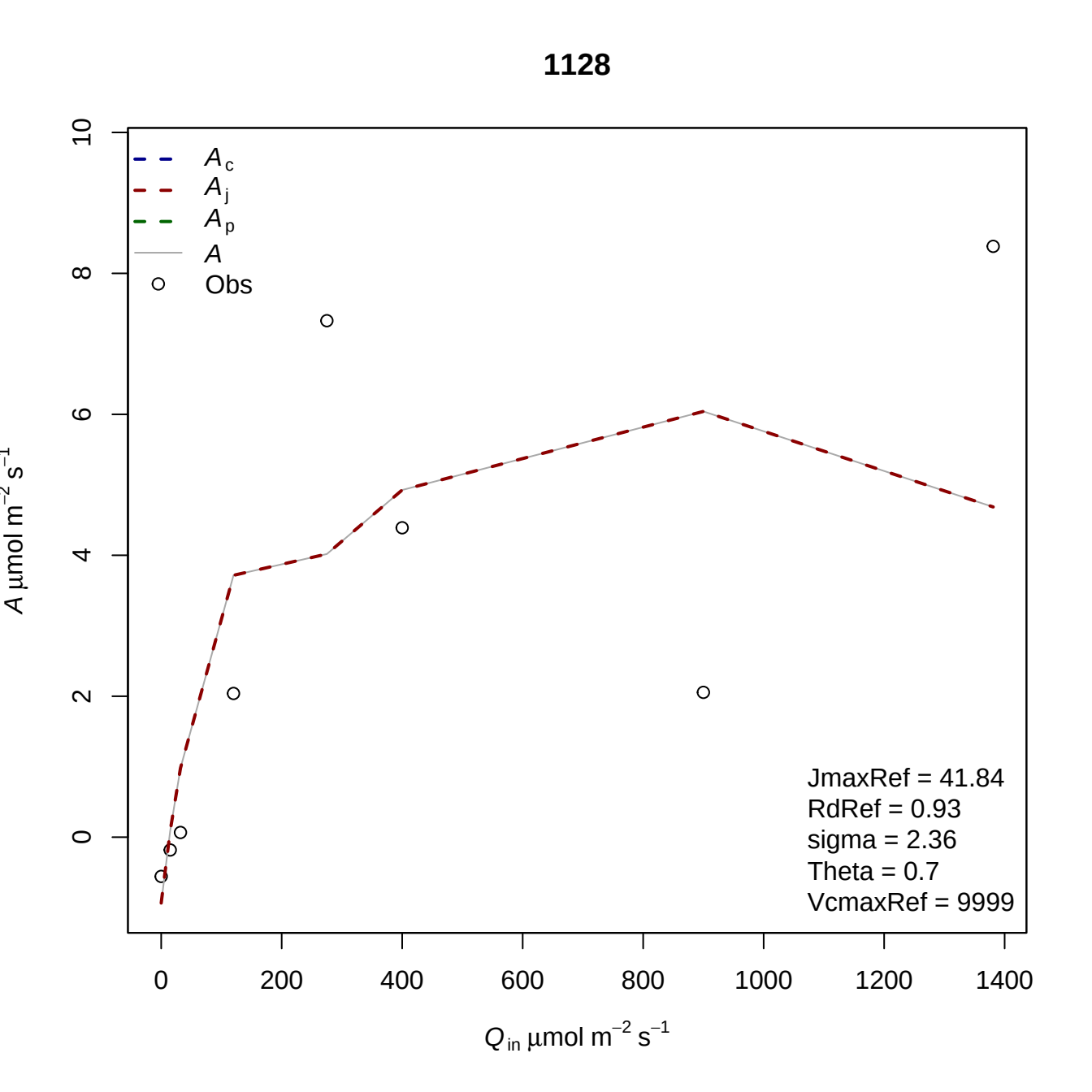

1120
A_c
A_j
A_p
A
Obs
JmaxRef = 60.13
RdRef = 1.31
sigma = 1.88
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
200
400
600
800
1000
1200
Q_{in} µmol m^{-2} s^{-1}

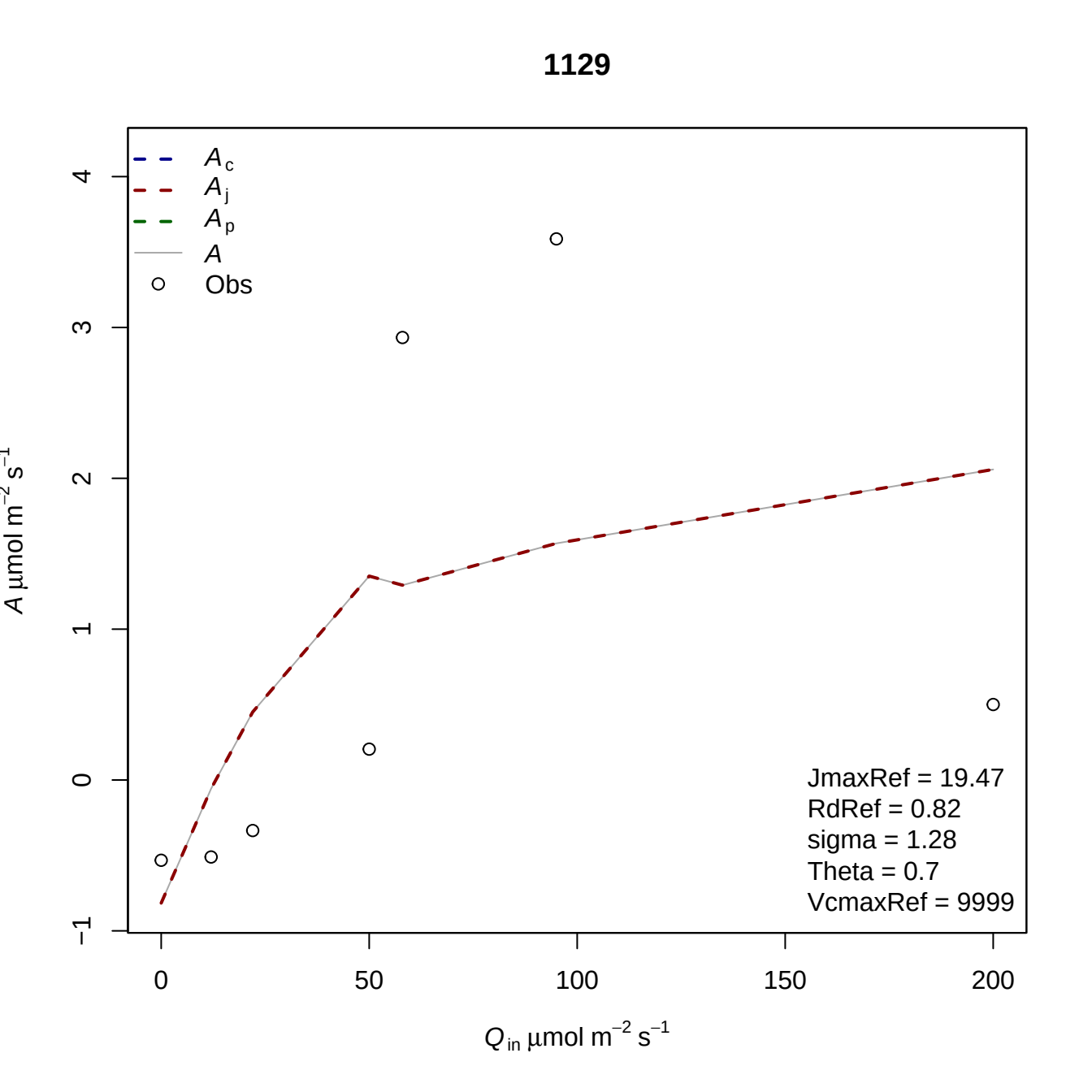

1122
A_c
A_j
A_p
A
Obs
JmaxRef = 56.6
RdRef = 1.3
sigma = 1.96
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200

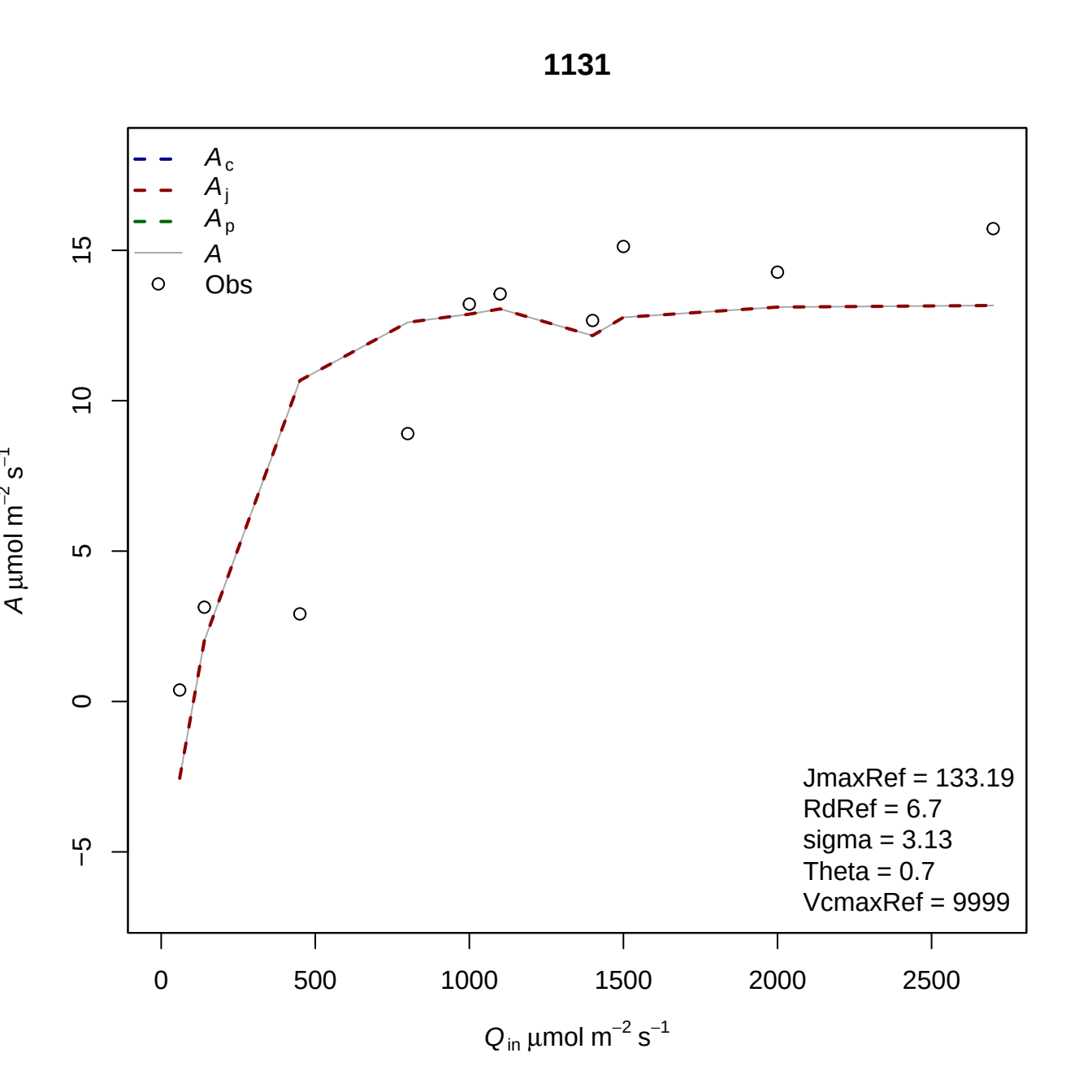

1125
A_c
A_j
A_p
A
Obs
JmaxRef = 37.65
RdRef = 1.56
sigma = 0.8
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
100
200
300
400
500
600

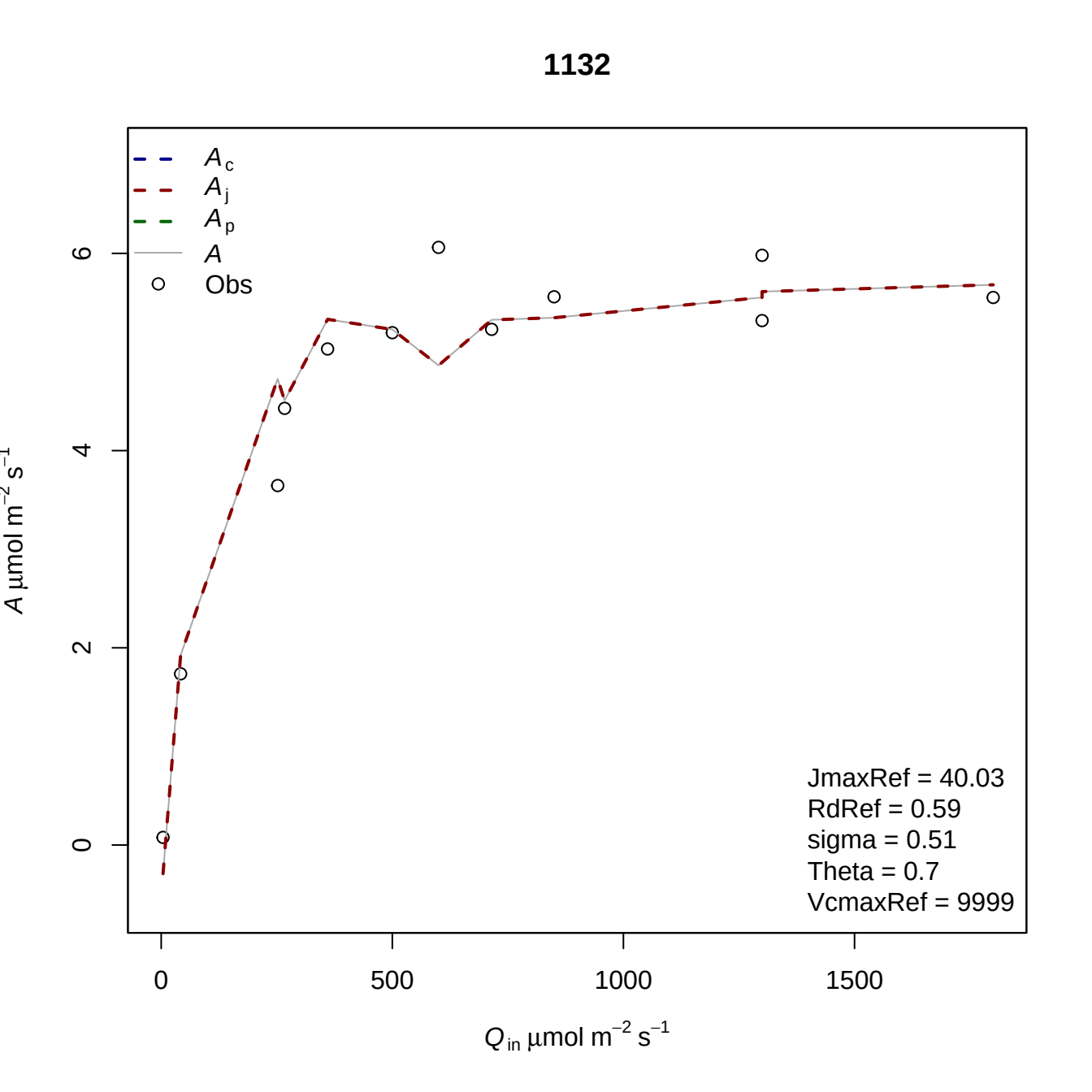

1126
A_c
A_j
A_p
A
Obs
JmaxRef = 122.96
RdRef = 5.4
sigma = 1.4
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
200
400
600
800
1000
1200

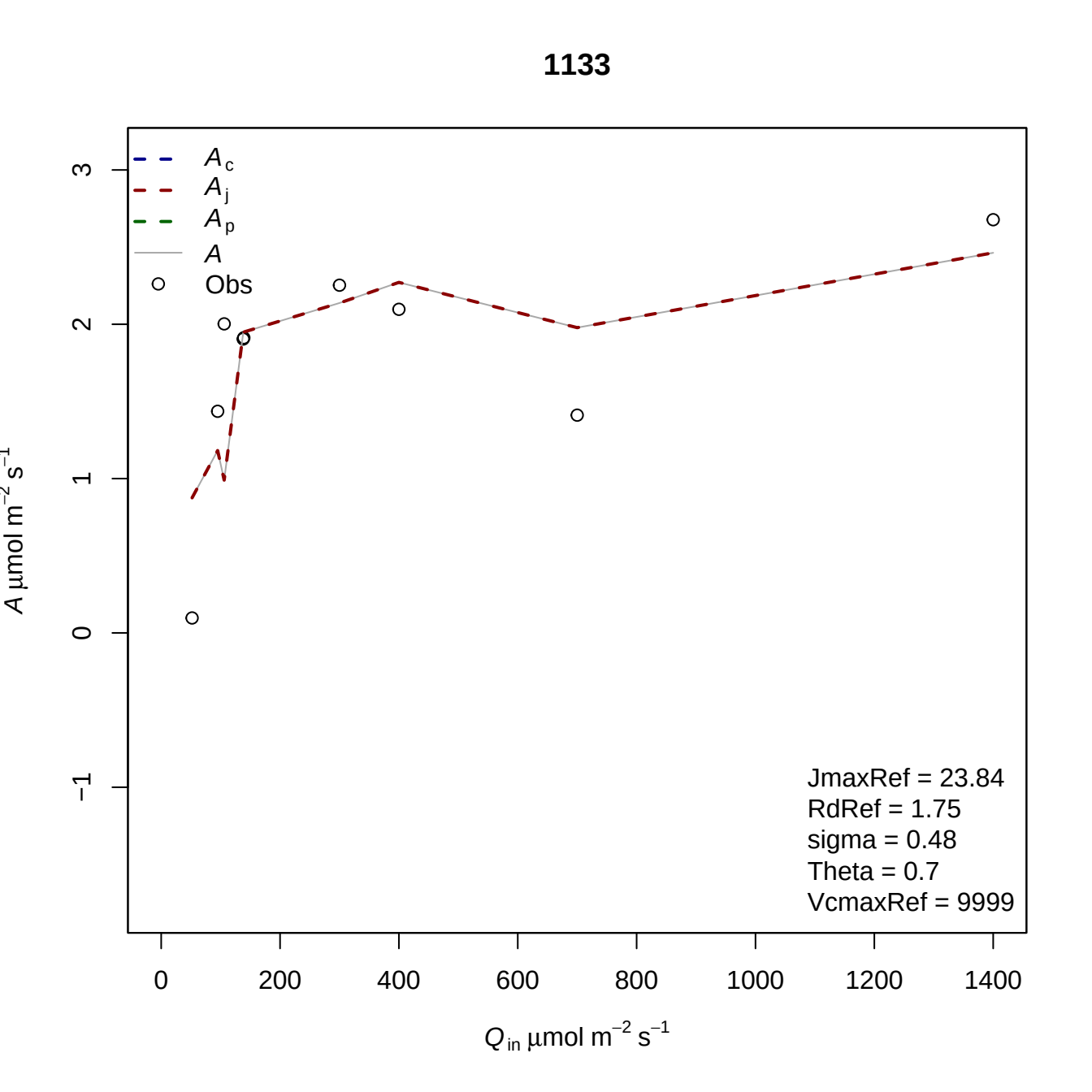

1127
A_c
A_j
A_p
A
Obs
JmaxRef = 68.53
RdRef = 1.12
sigma = 1.35
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

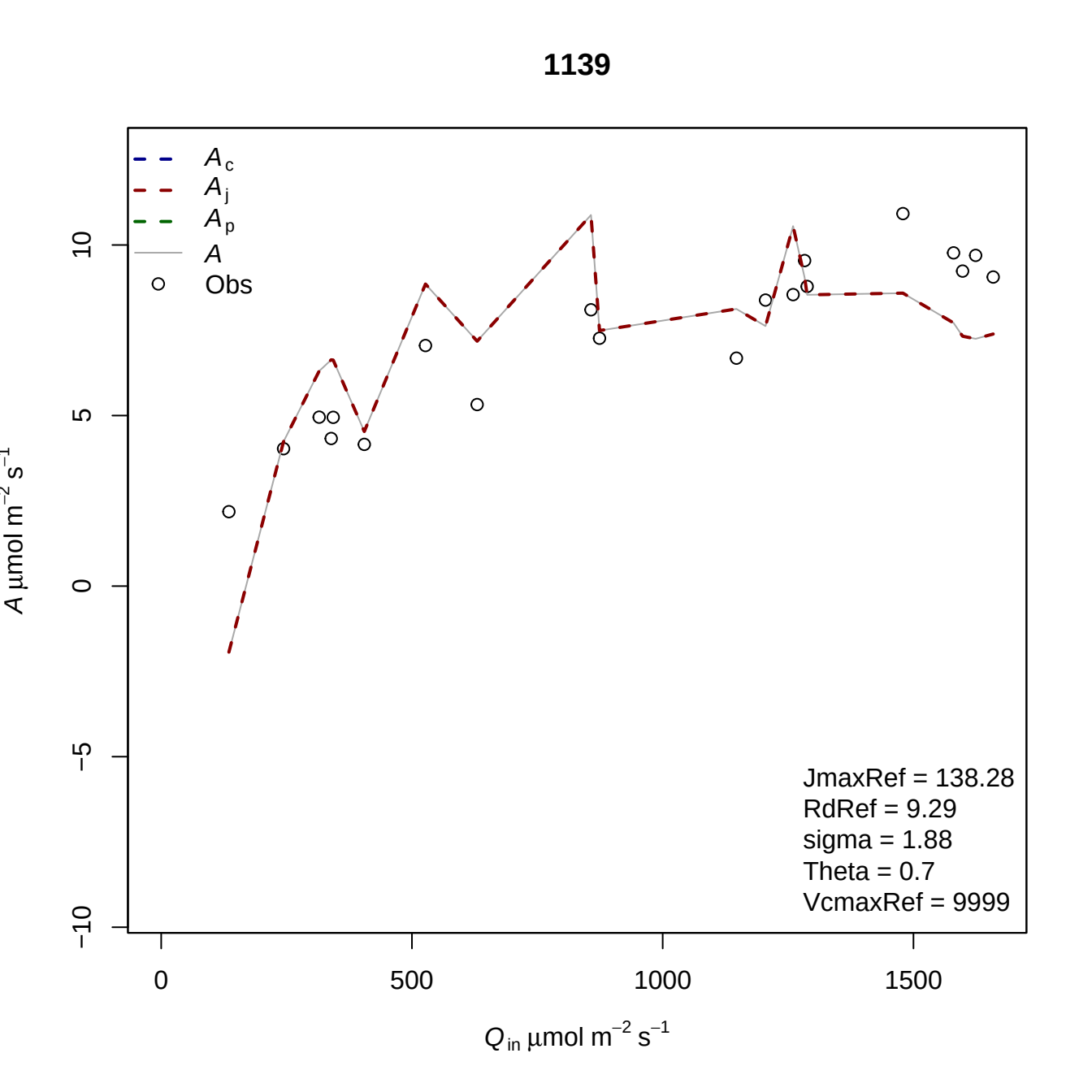

1128
A_c
A_j
A_p
A
Obs
JmaxRef = 41.84
RdRef = 0.93
sigma = 2.36
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
10
A µmol m^{-2} s^{-1}
0
200
400
600
800
1000
1200
1400
Q_{in} µmol m^{-2} s^{-1}

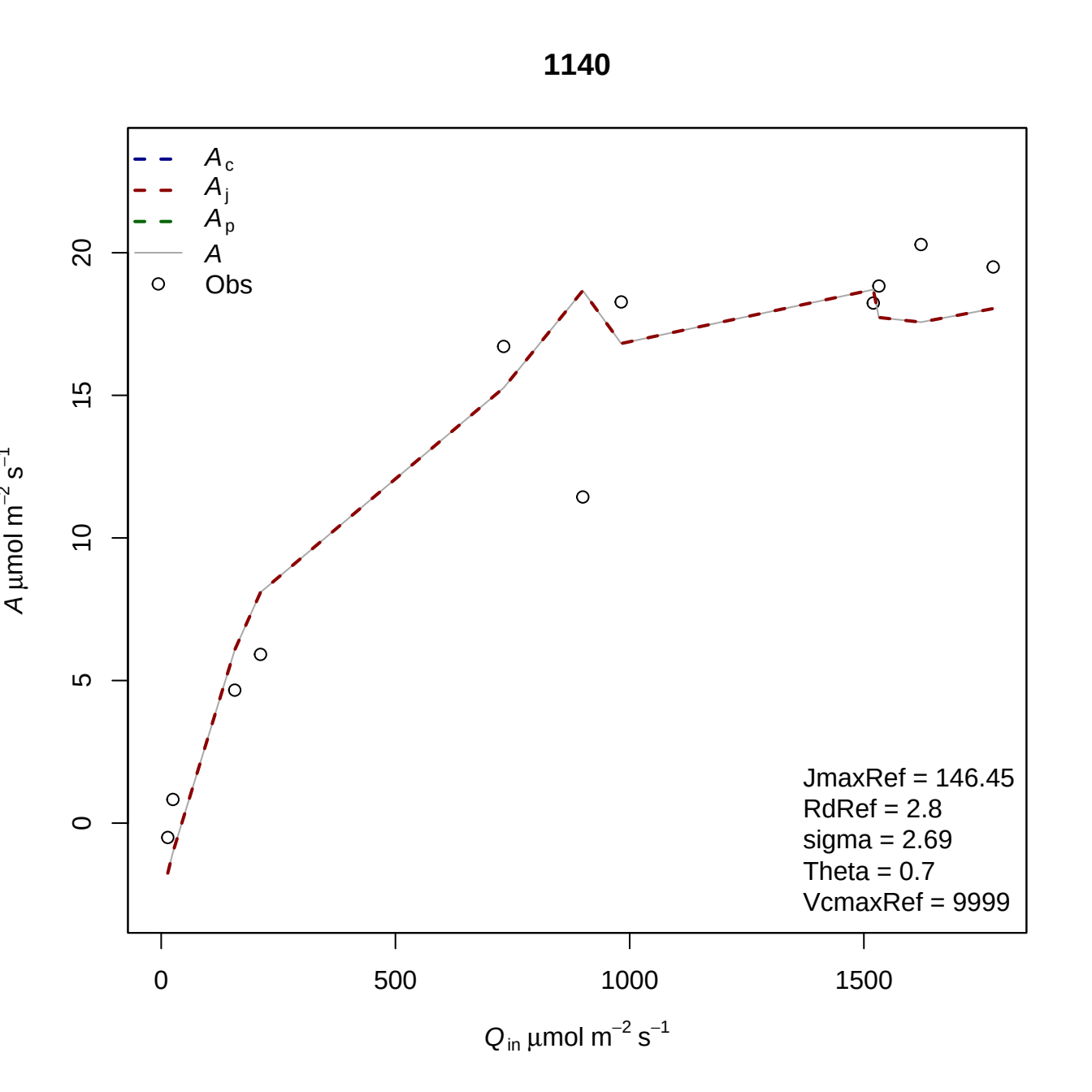

1129
A_c
A_j
A_p
A
Obs
JmaxRef = 19.47
RdRef = 0.82
sigma = 1.28
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
50
100
150
200

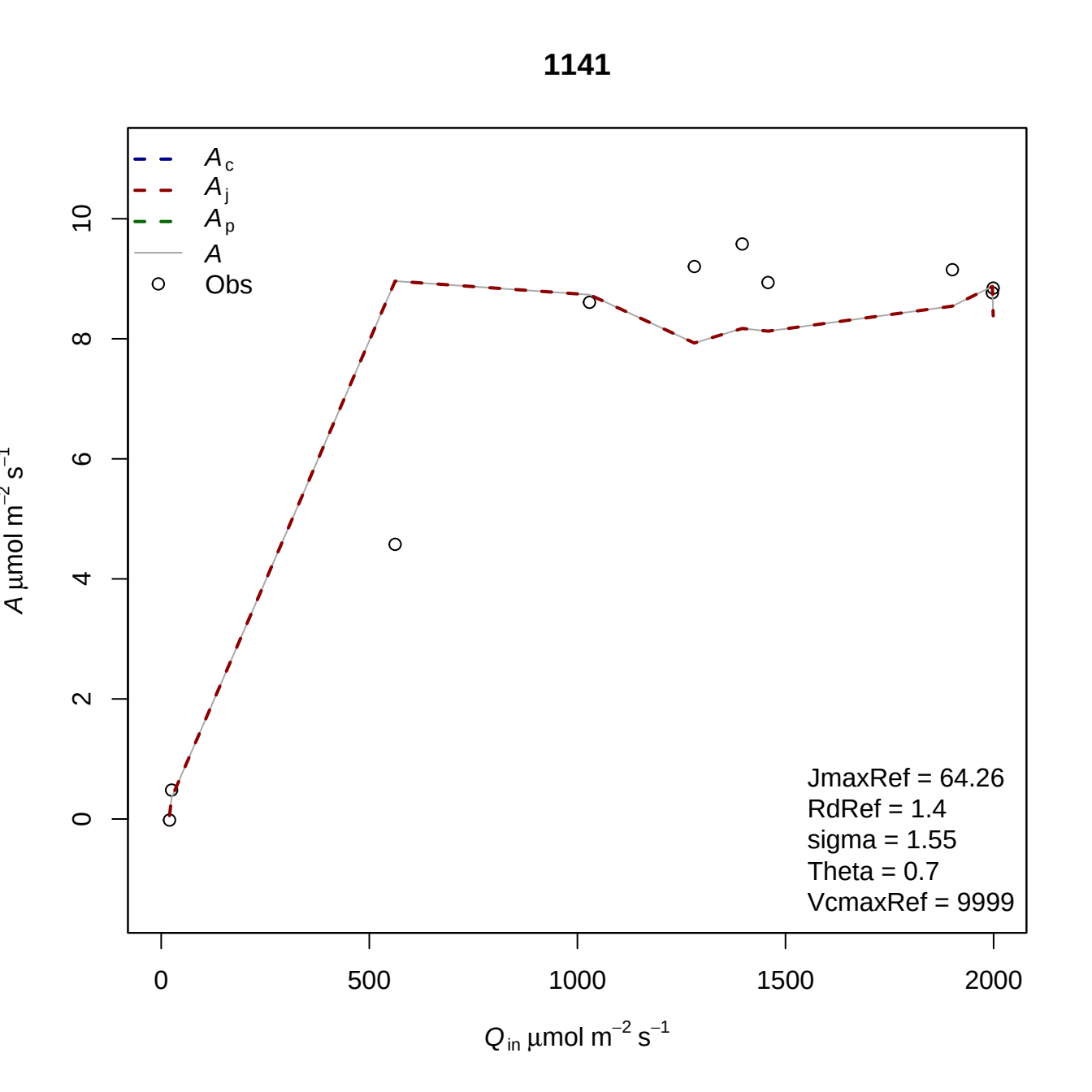

1131
A_c
A_j
A_p
A
Obs
JmaxRef = 133.19
RdRef = 6.7
sigma = 3.13
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
15
10
5
0
−5
0
500
1000
1500
2000
2500

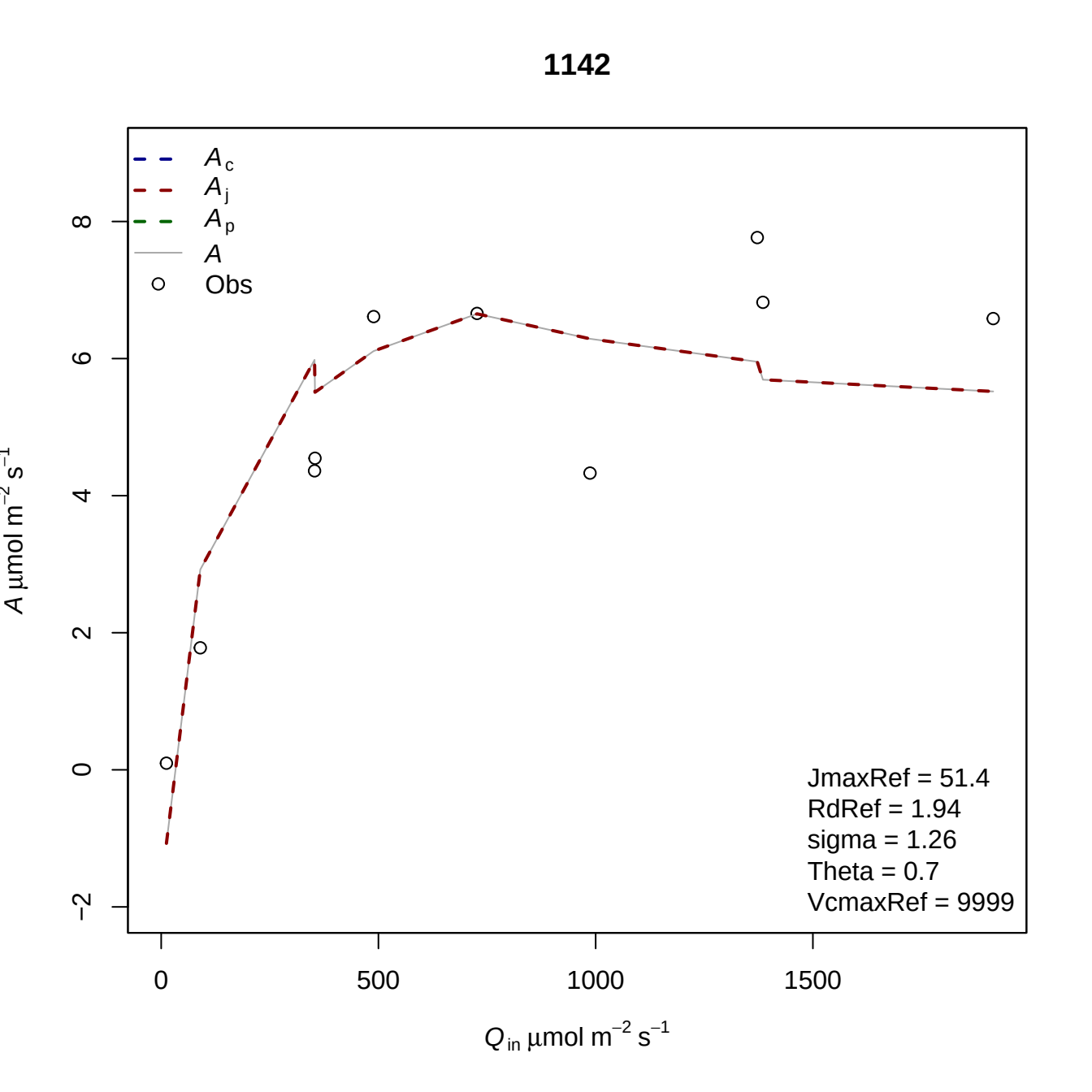

1132
A_c
A_j
A_p
A
Obs
JmaxRef = 40.03
RdRef = 0.59
sigma = 0.51
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

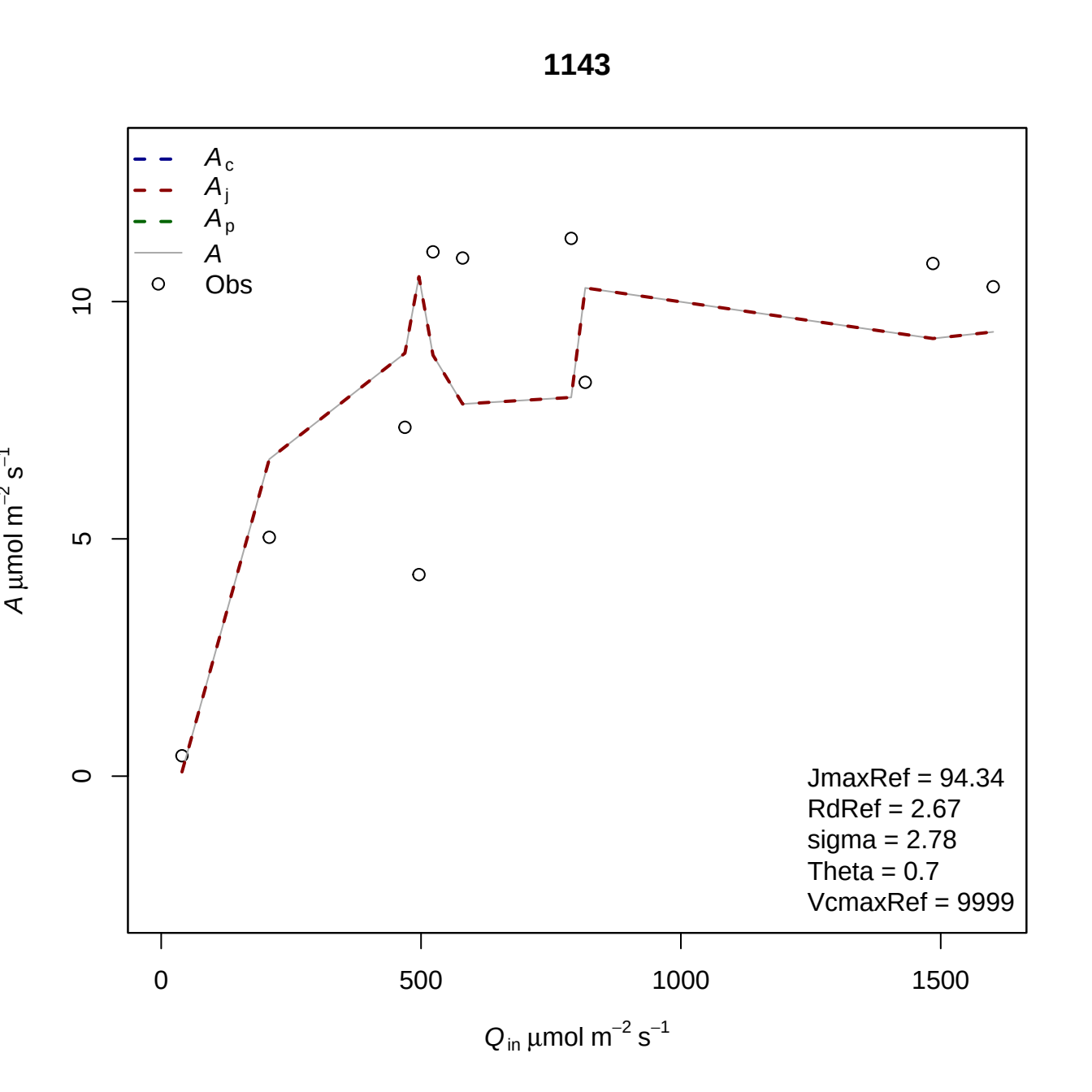

1133
A_c
A_j
A_p
A
Obs
JmaxRef = 23.84
RdRef = 1.75
sigma = 0.48
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
0
200
400
600
800
1000
1200
1400

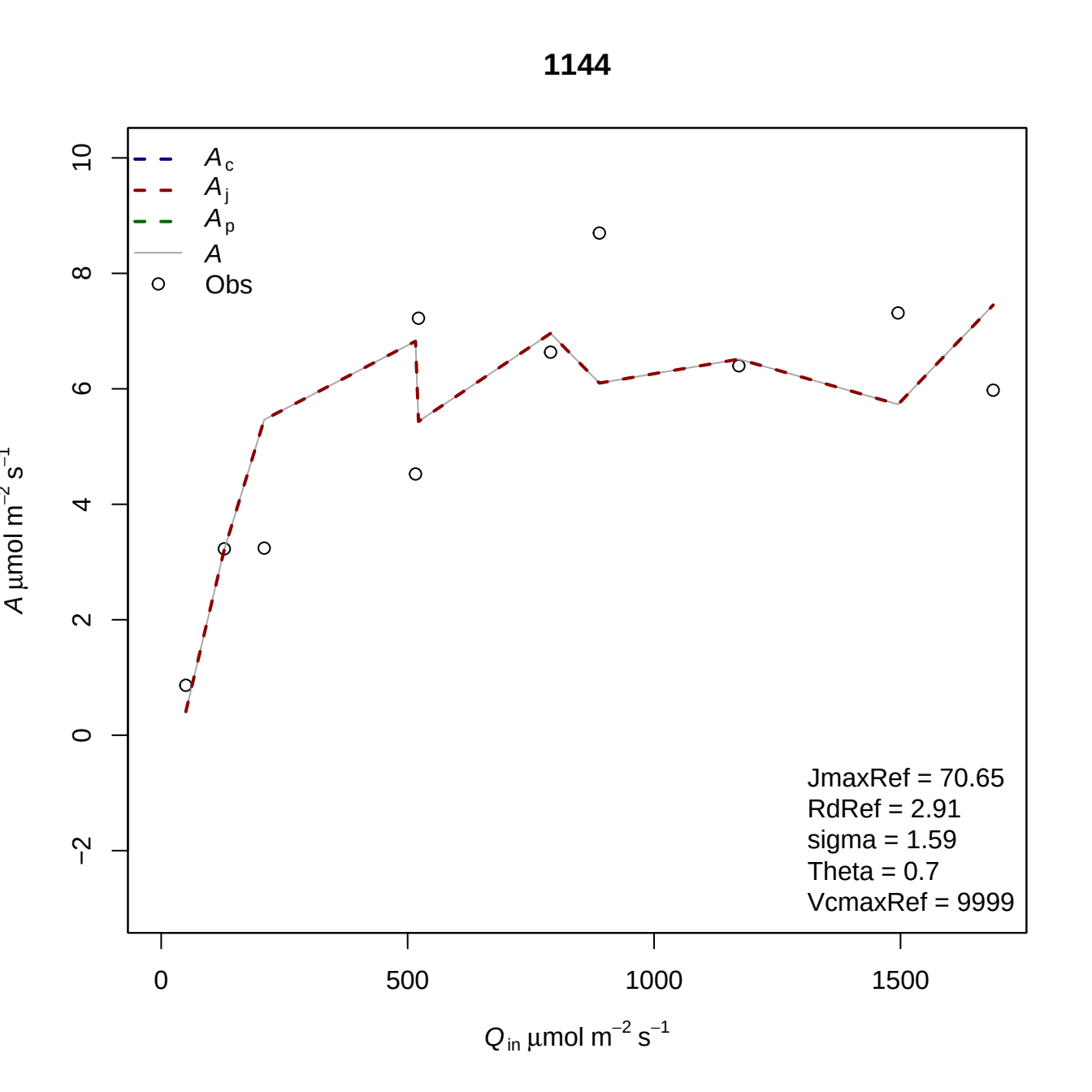

1139
A_c
A_j
A_p
A
Obs
JmaxRef = 138.28
RdRef = 9.29
sigma = 1.88
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
5
0
−5
−10
0
500
1000
1500

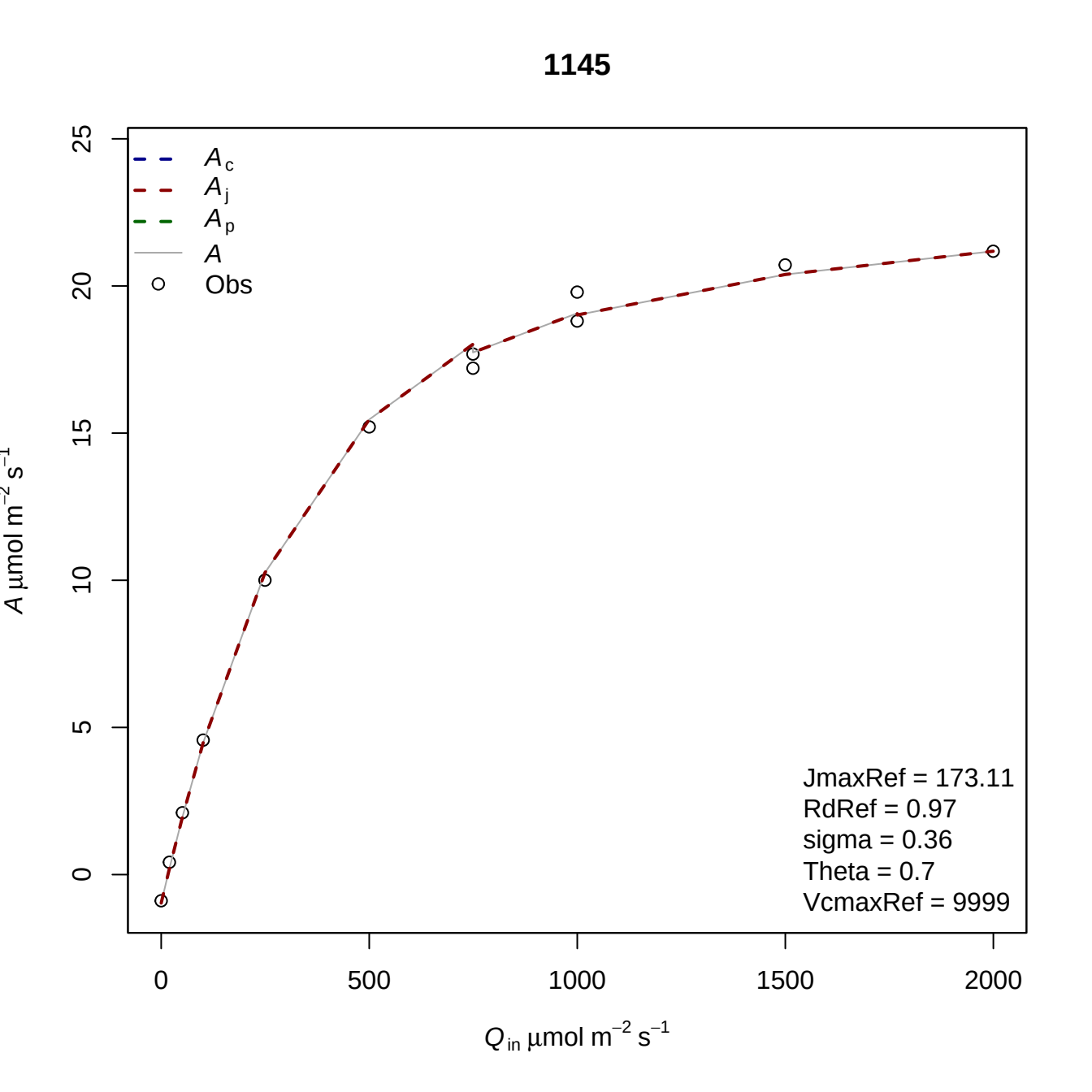

1140
A_c
A_j
A_p
A
Obs
JmaxRef = 146.45
RdRef = 2.8
sigma = 2.69
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

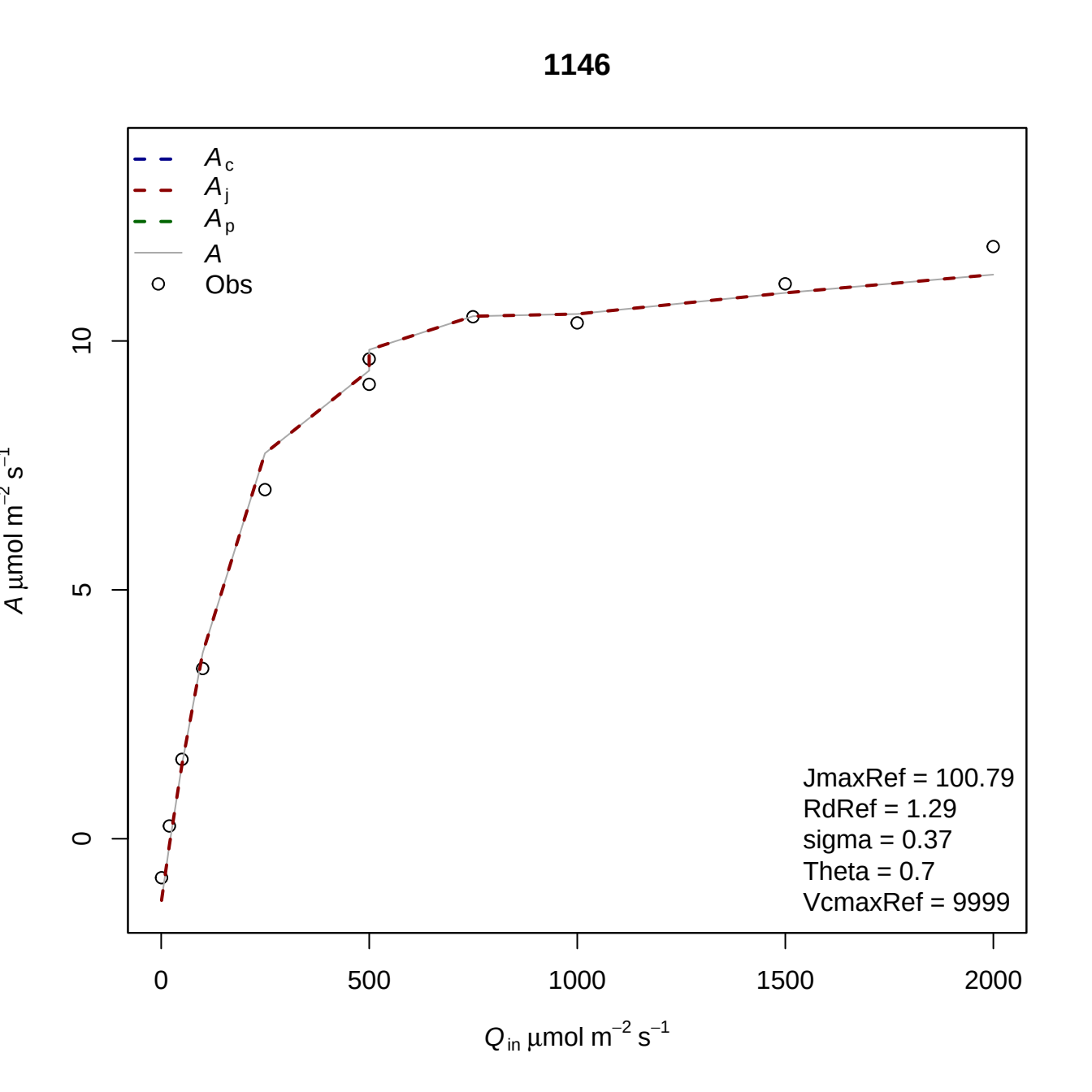

1141
A_c
A_j
A_p
A
Obs
JmaxRef = 64.26
RdRef = 1.4
sigma = 1.55
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

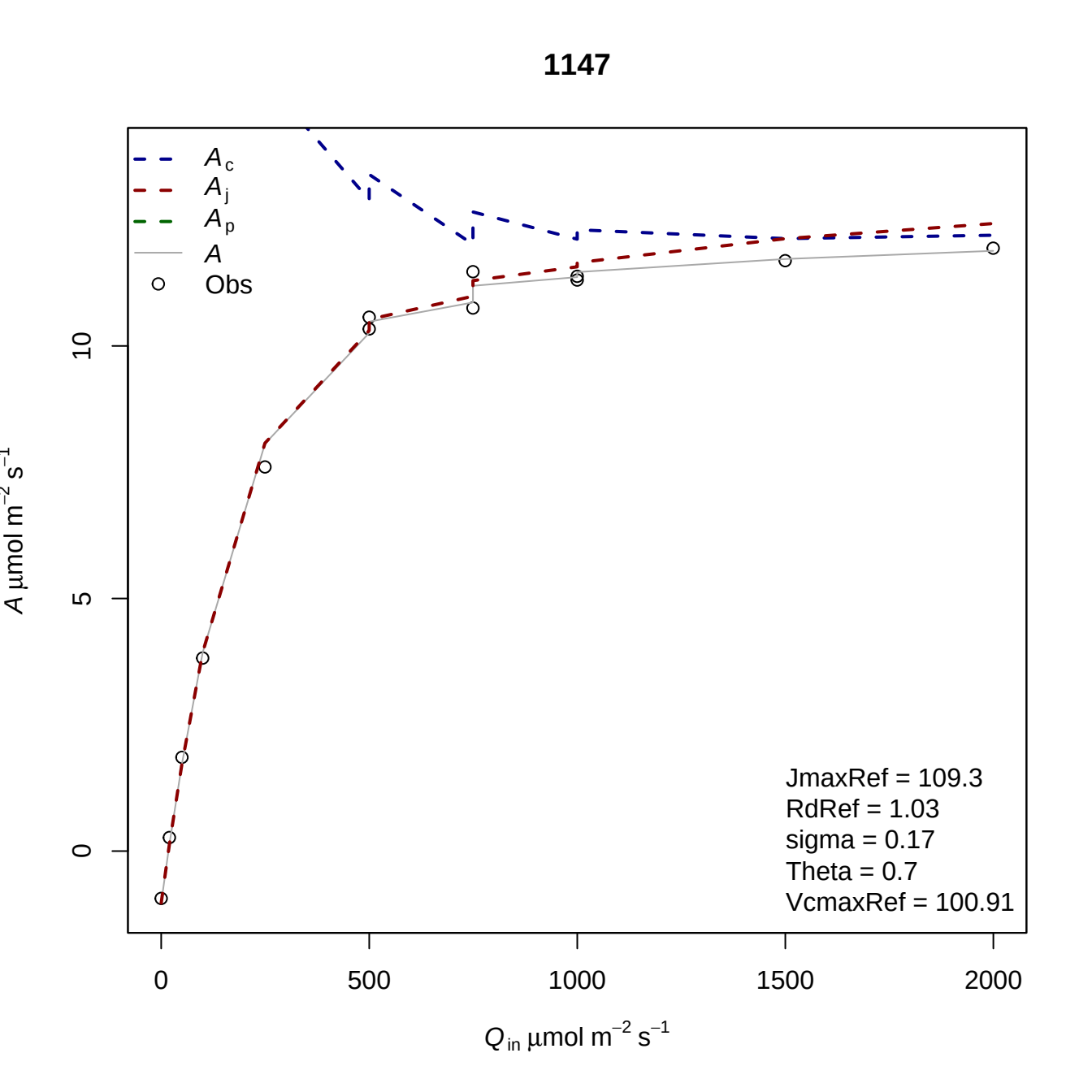

1142
A_c
A_j
A_p
A
Obs
JmaxRef = 51.4
RdRef = 1.94
sigma = 1.26
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

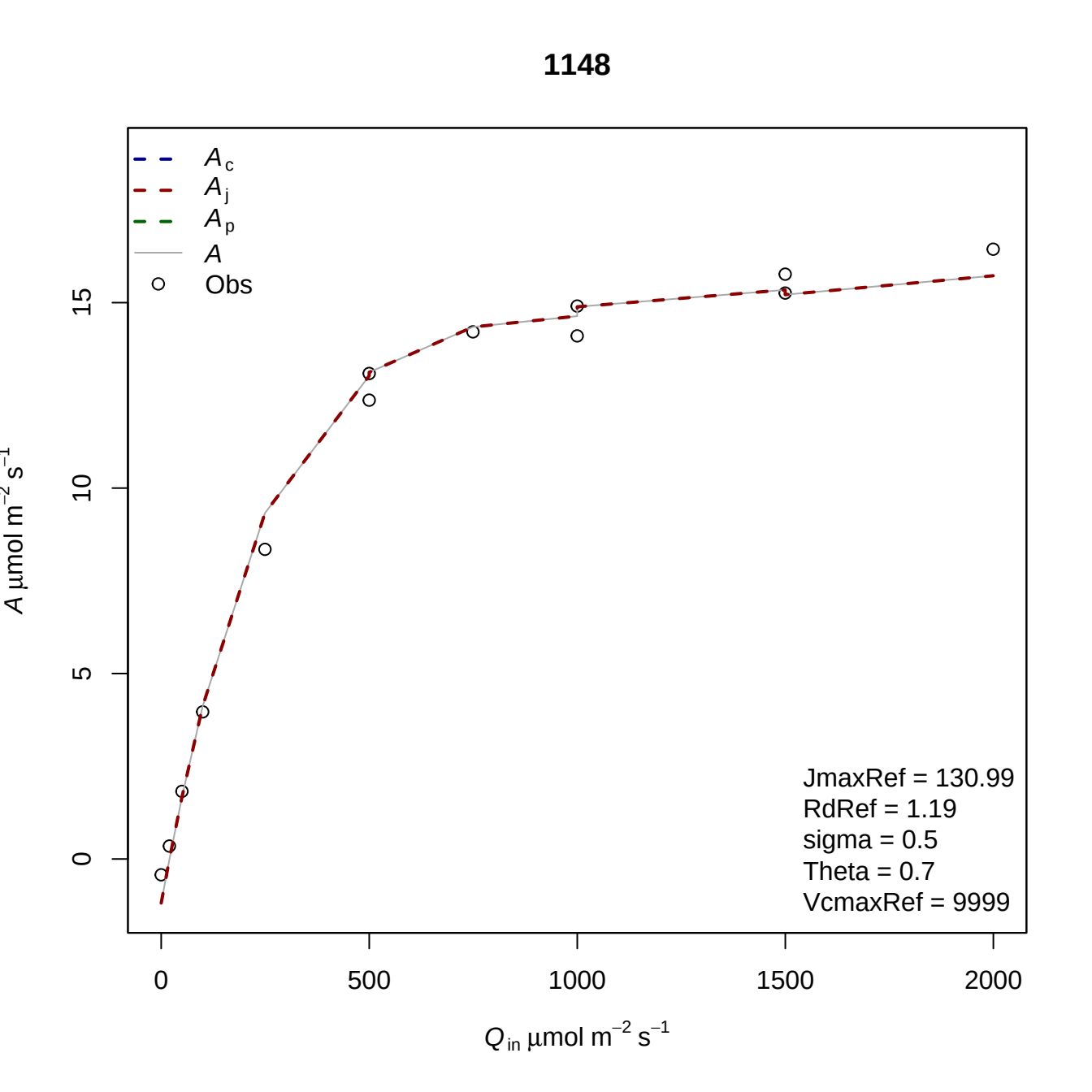

1143
A_c
A_j
A_p
A
Obs
JmaxRef = 94.34
RdRef = 2.67
sigma = 2.78
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

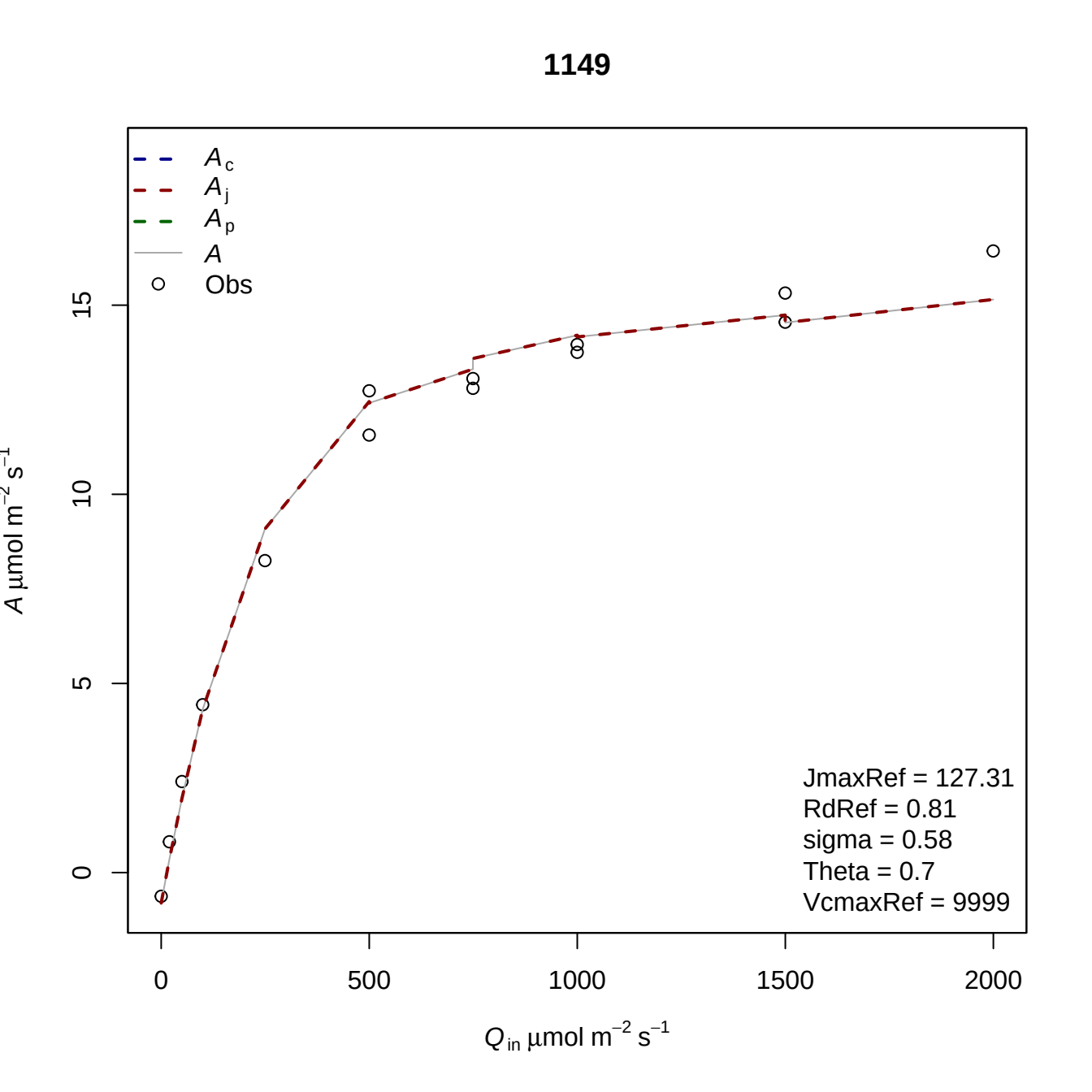

1144
A_c
A_j
A_p
A
Obs
JmaxRef = 70.65
RdRef = 2.91
sigma = 1.59
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

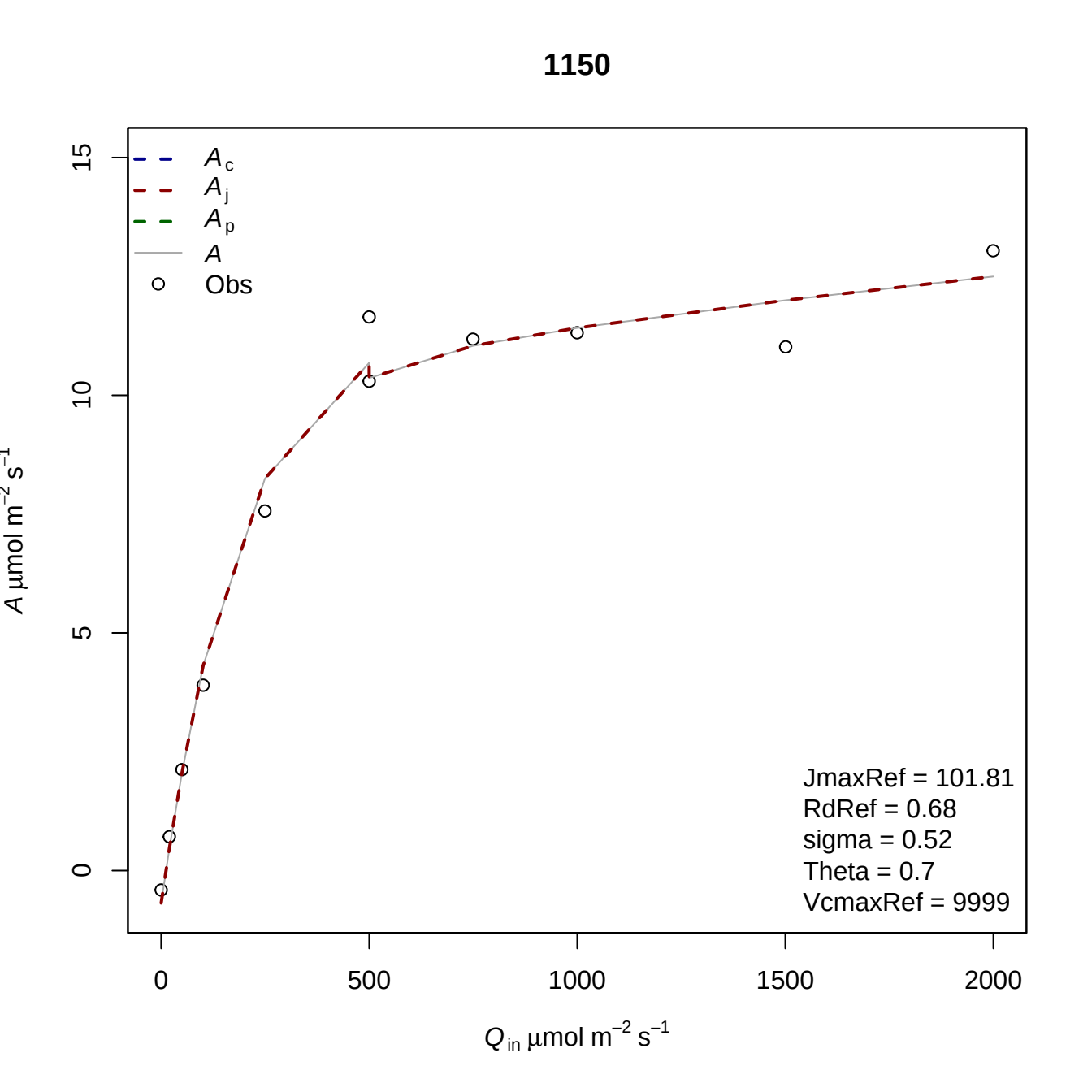

1145
A_c
A_j
A_p
A
Obs
JmaxRef = 173.11
RdRef = 0.97
sigma = 0.36
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

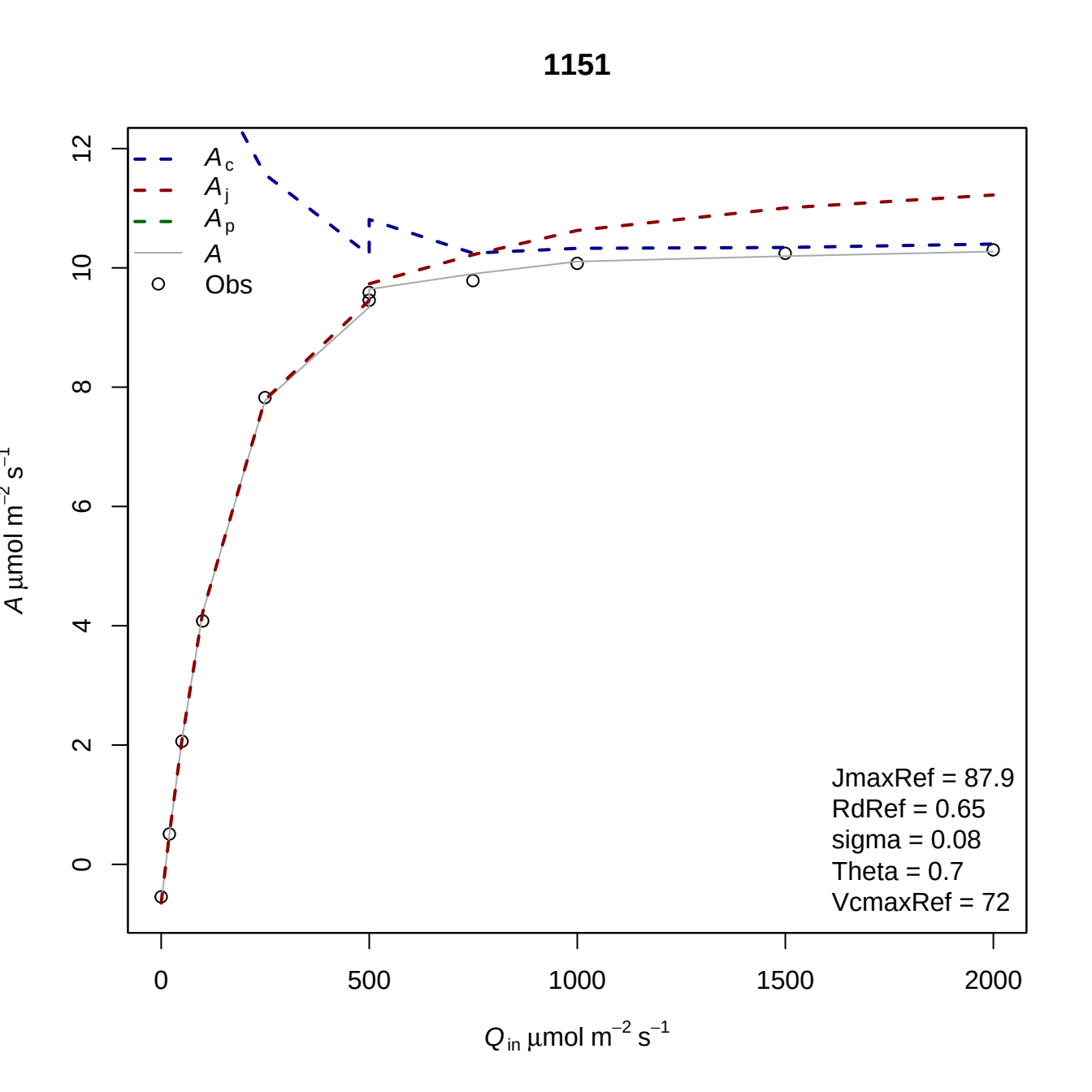

1146
A_c
A_j
A_p
A
Obs
JmaxRef = 100.79
RdRef = 1.29
sigma = 0.37
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

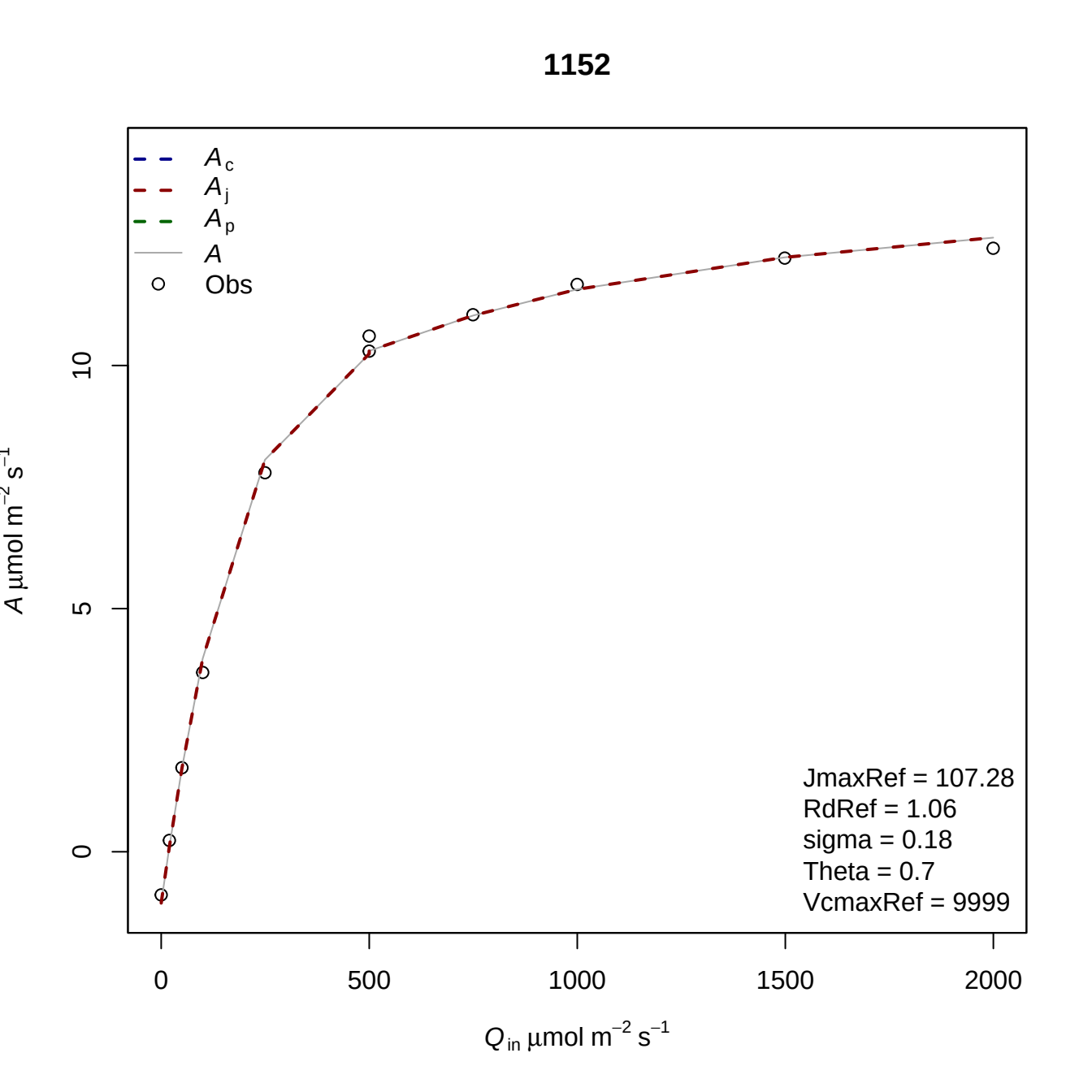

1147
A_c
A_j
A_p
A
Obs
JmaxRef = 109.3
RdRef = 1.03
sigma = 0.17
Theta = 0.7
VcmaxRef = 100.91
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

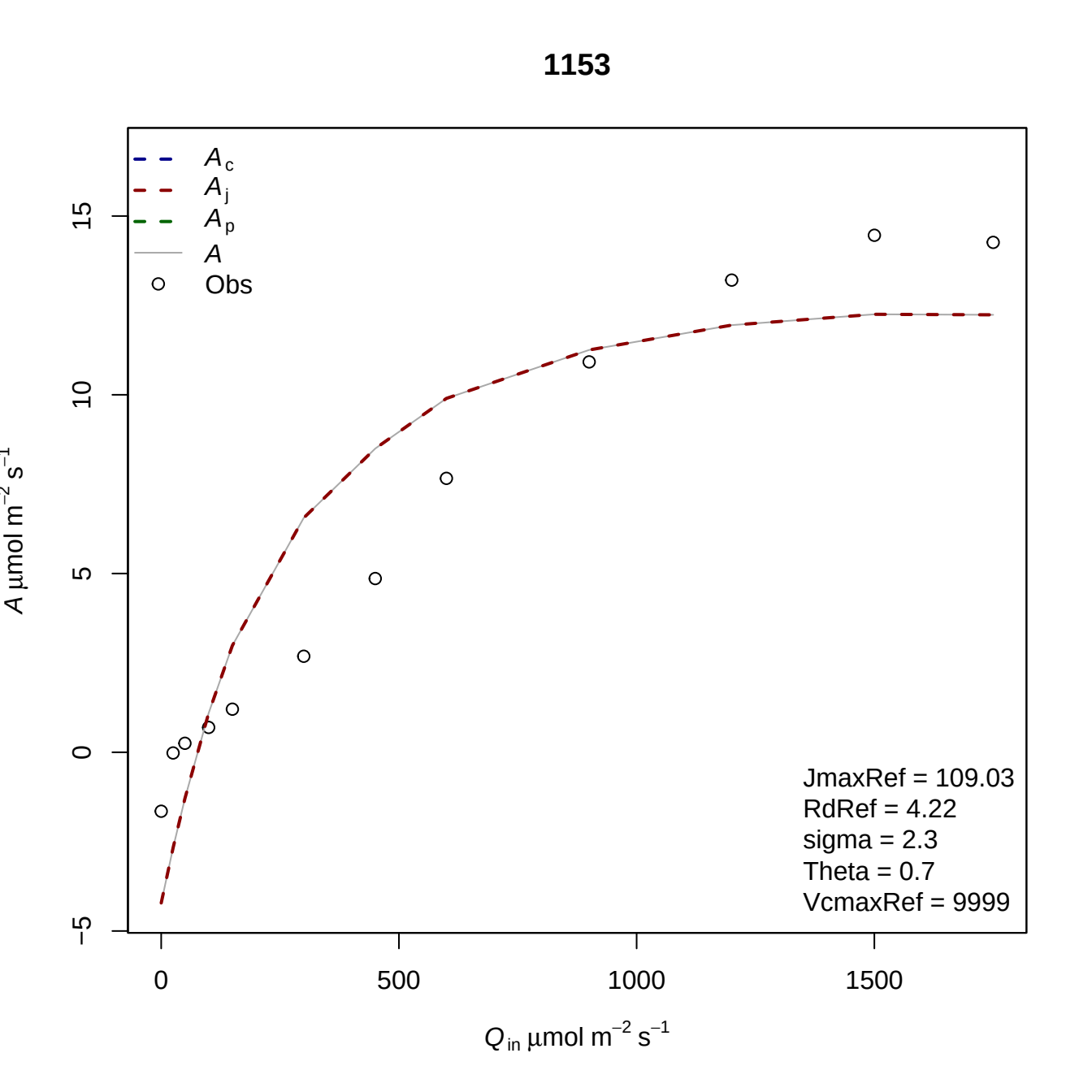

1148
A_c
A_j
A_p
A
Obs
JmaxRef = 130.99
RdRef = 1.19
sigma = 0.5
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

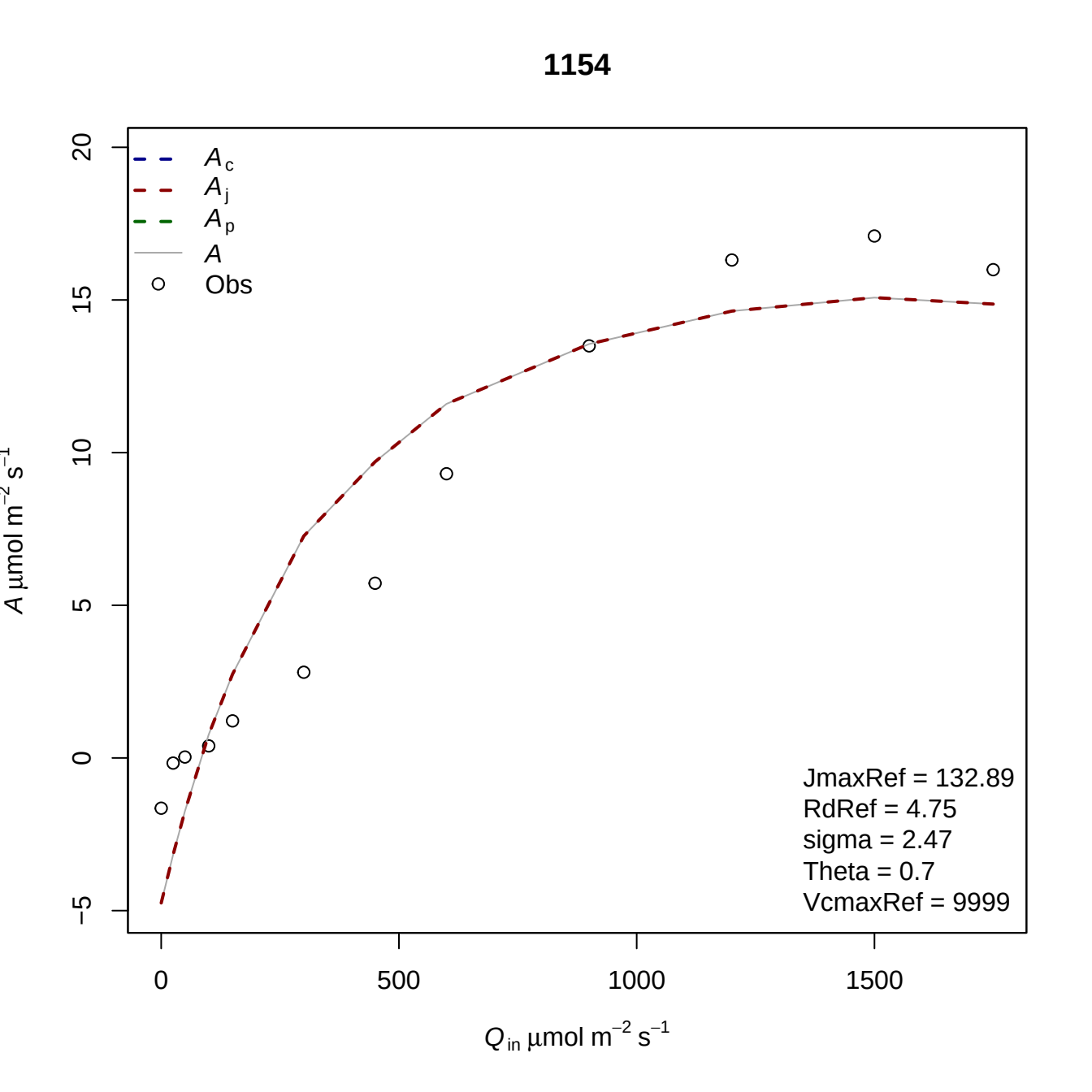

1149
A_c
A_j
A_p
A
Obs
JmaxRef = 127.31
RdRef = 0.81
sigma = 0.58
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

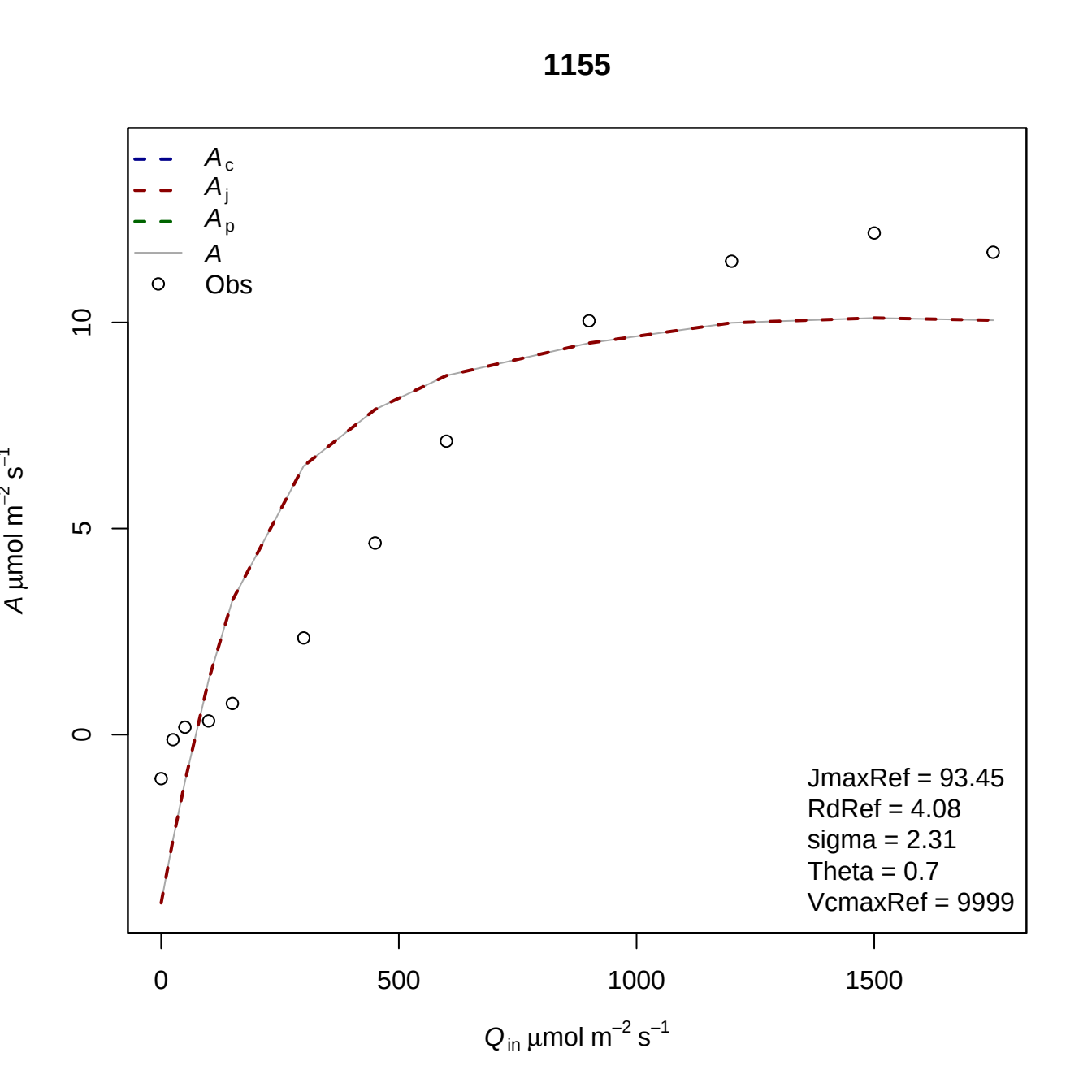

1150
A_c
A_j
A_p
A
Obs
JmaxRef = 101.81
RdRef = 0.68
sigma = 0.52
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

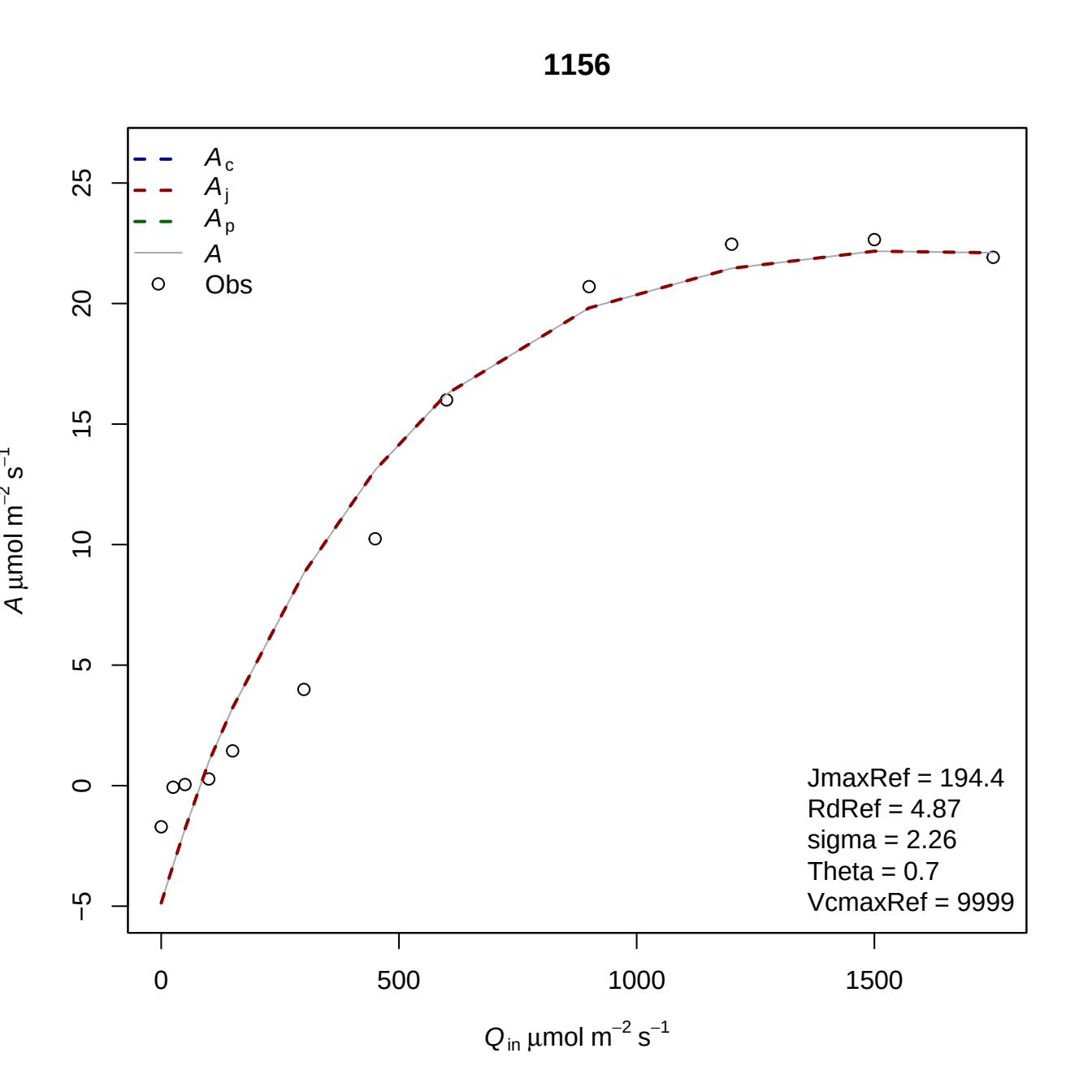

1151
A_c
A_j
A_p
A
Obs
JmaxRef = 87.9
RdRef = 0.65
sigma = 0.08
Theta = 0.7
VcmaxRef = 72
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

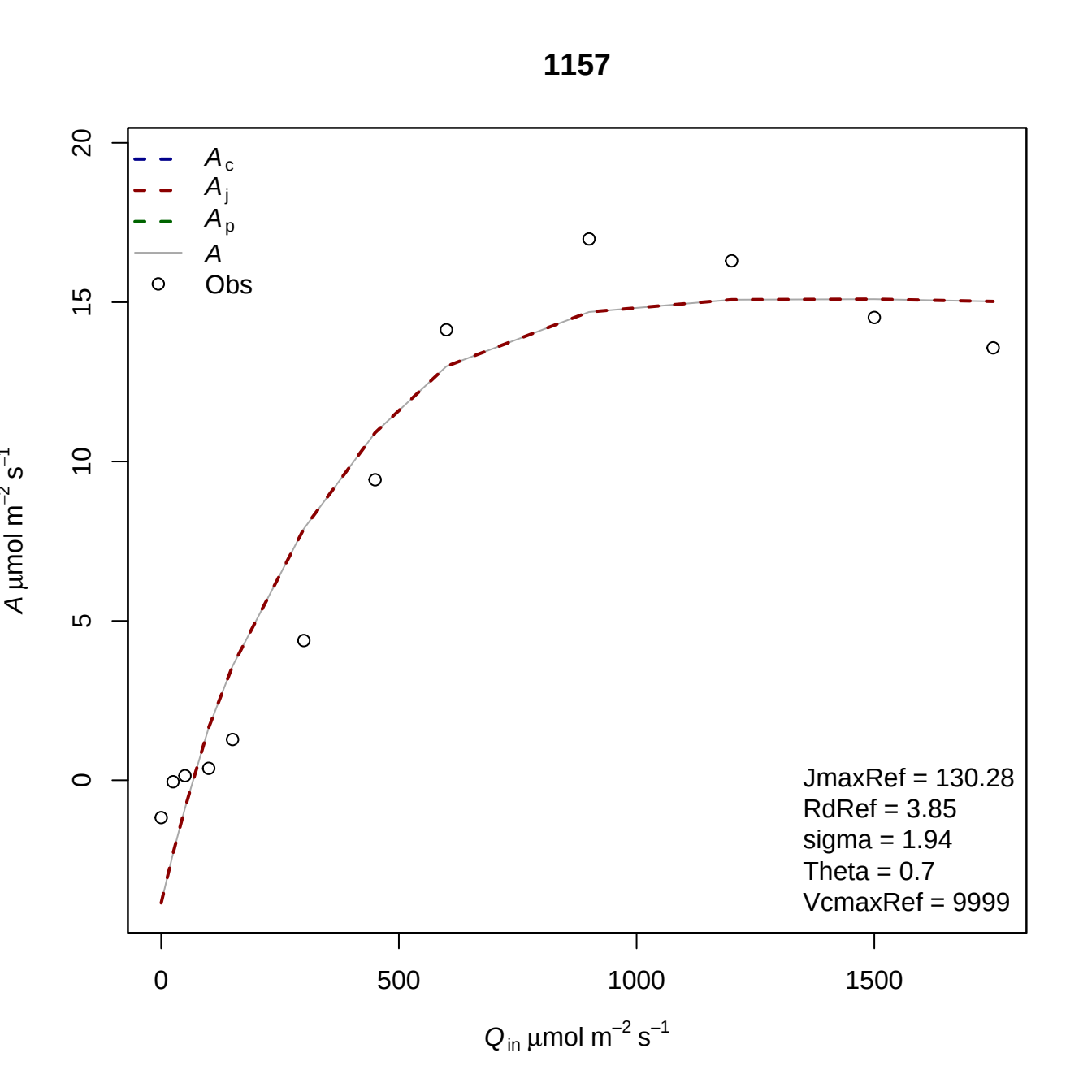

1152
A_c
A_j
A_p
A
Obs
JmaxRef = 107.28
RdRef = 1.06
sigma = 0.18
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

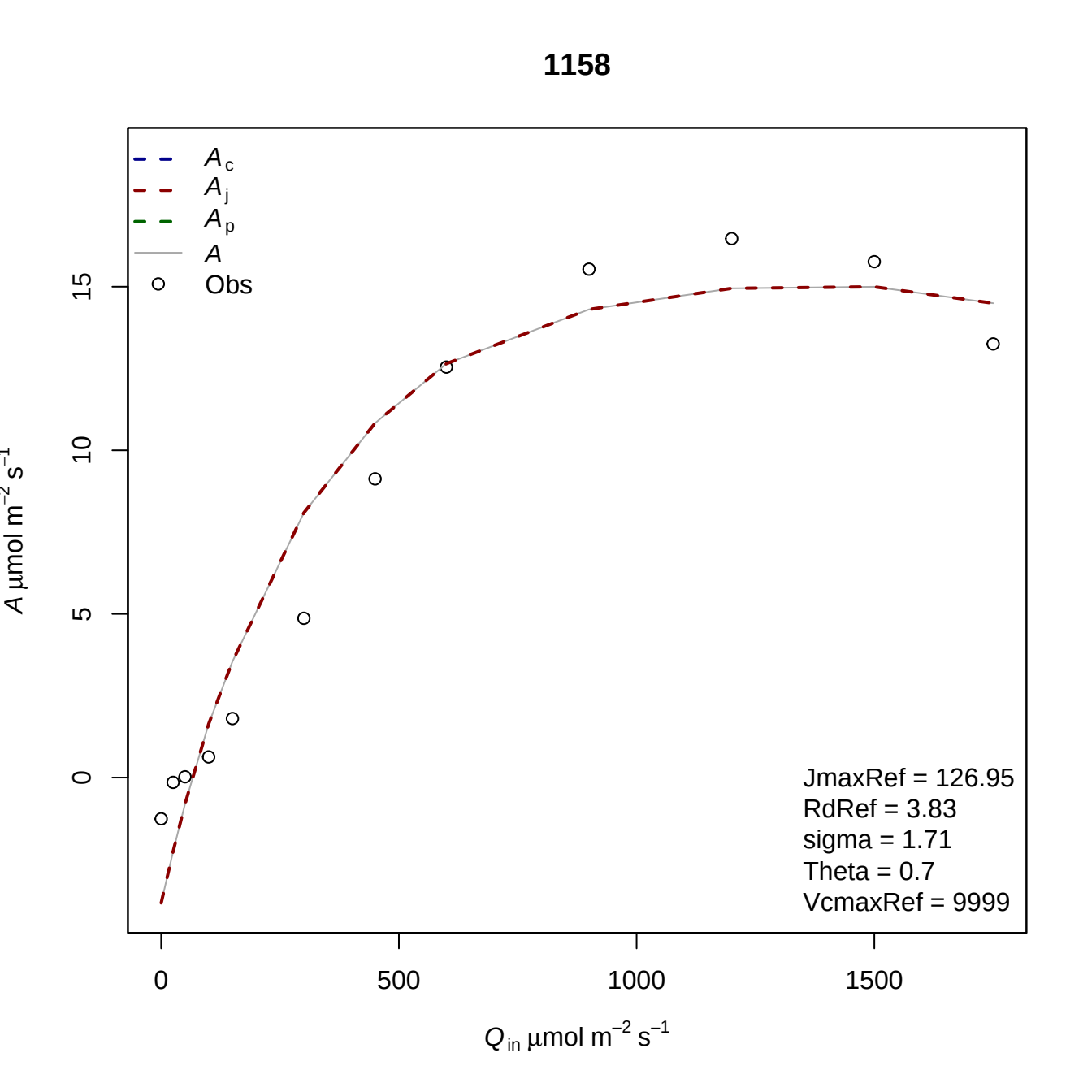

1153
A_c
A_j
A_p
A
Obs
JmaxRef = 109.03
RdRef = 4.22
sigma = 2.3
Theta = 0.7
VcmaxRef = 9999
15
10
5
0
−5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

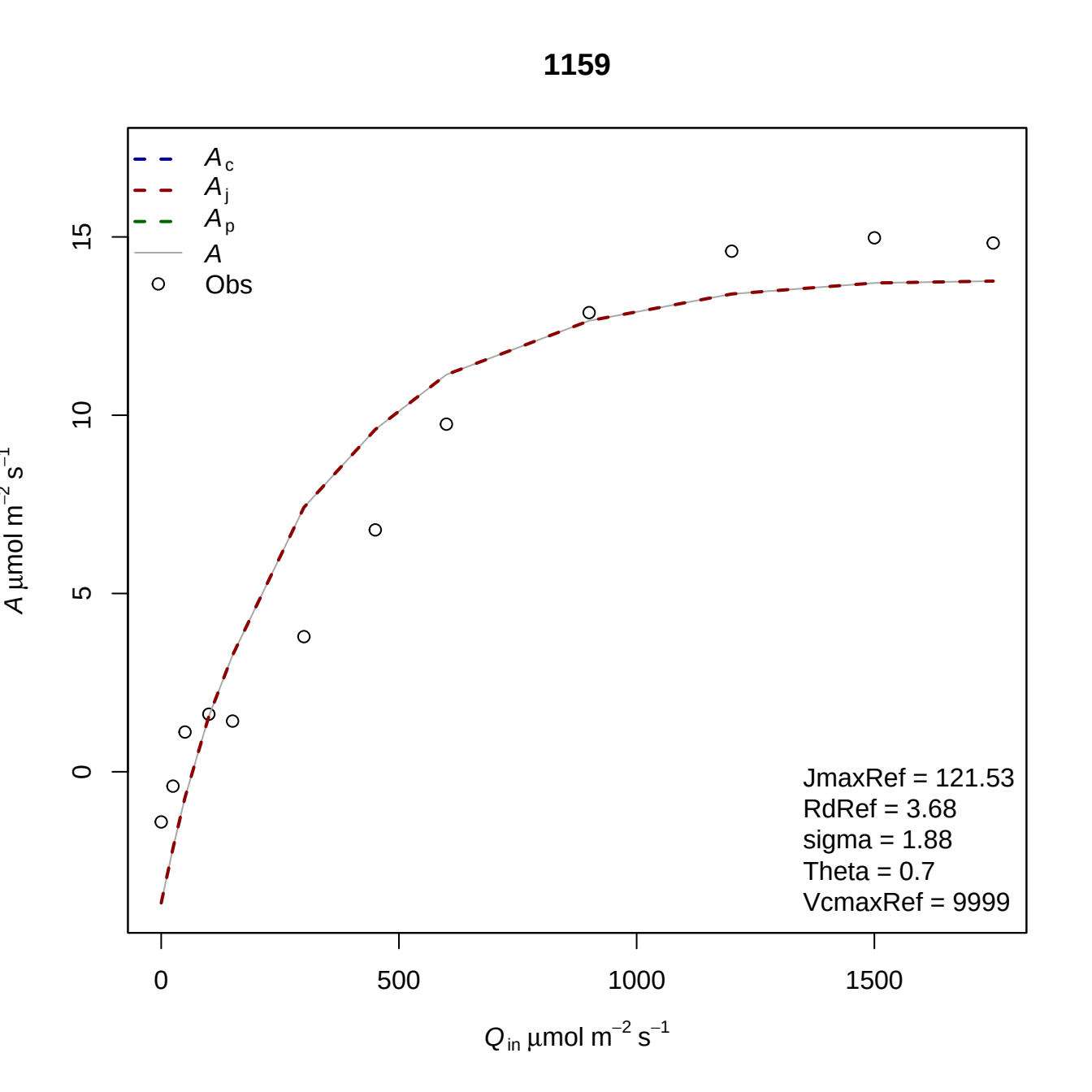

1154
A_c
A_j
A_p
A
Obs
JmaxRef = 132.89
RdRef = 4.75
sigma = 2.47
Theta = 0.7
VcmaxRef = 9999
20
15
10
5
0
−5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

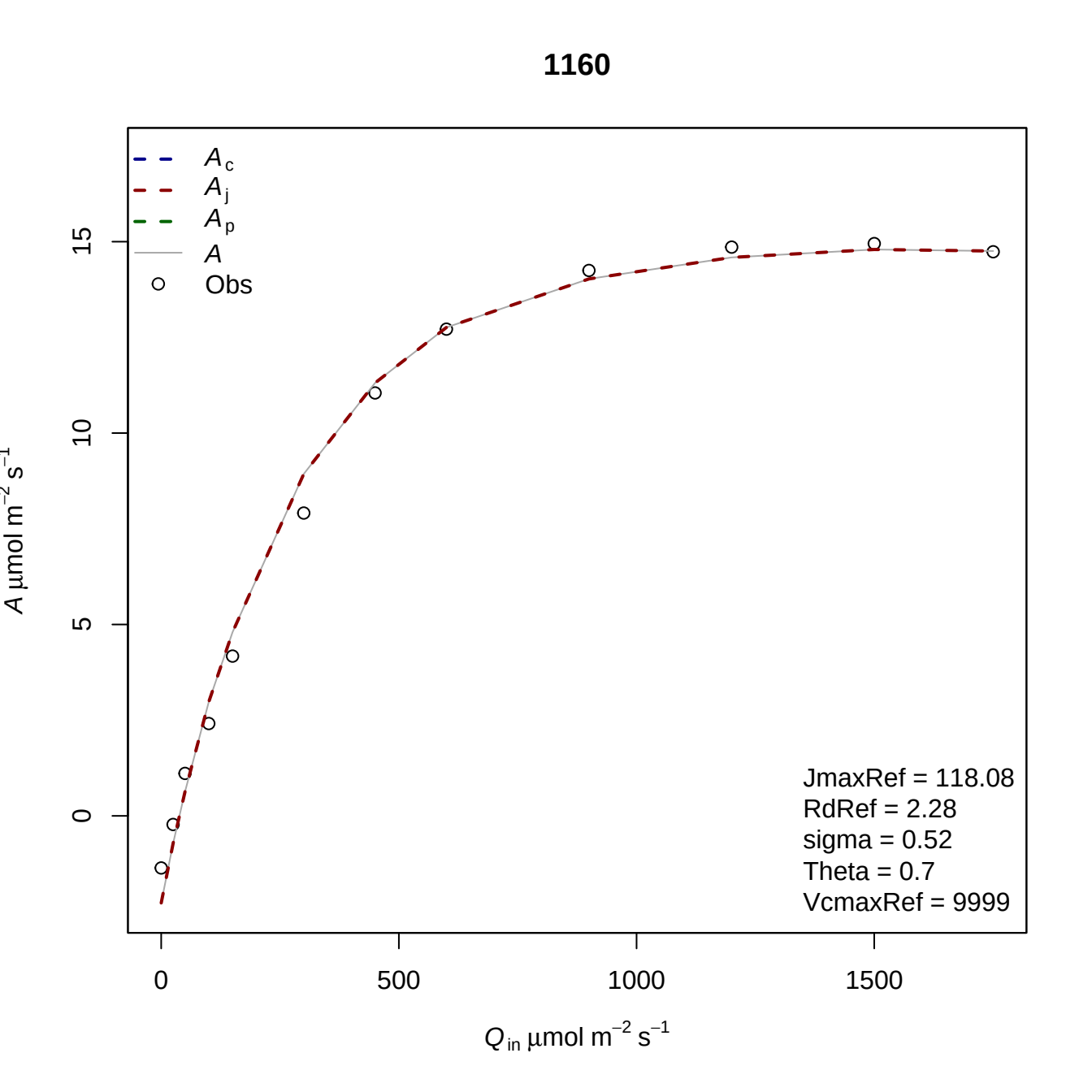

1155
A_c
A_j
A_p
A
Obs
JmaxRef = 93.45
RdRef = 4.08
sigma = 2.31
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

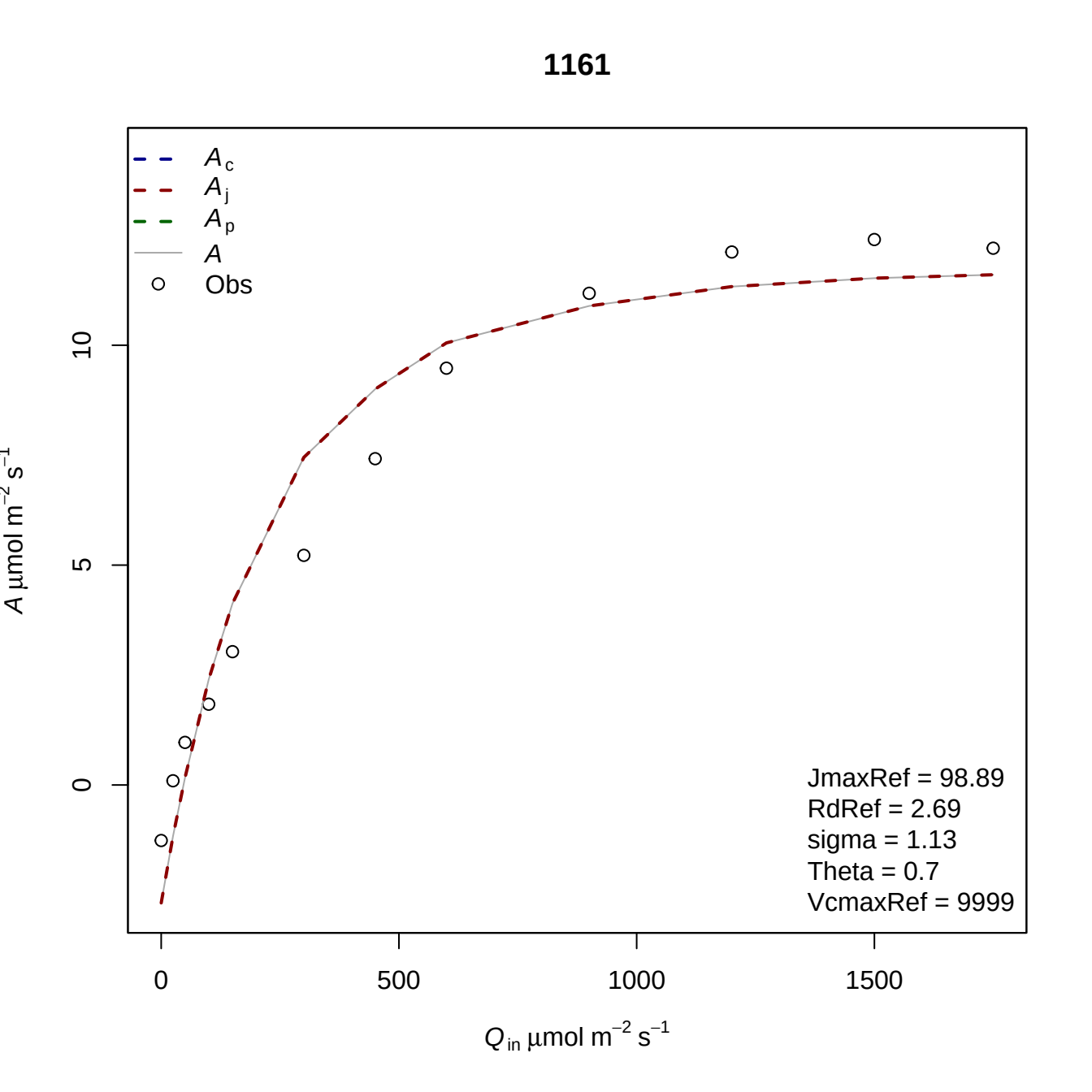

1156
A_c
A_j
A_p
A
Obs
JmaxRef = 194.4
RdRef = 4.87
sigma = 2.26
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
25
20
15
10
5
0
−5
0
500
1000
1500

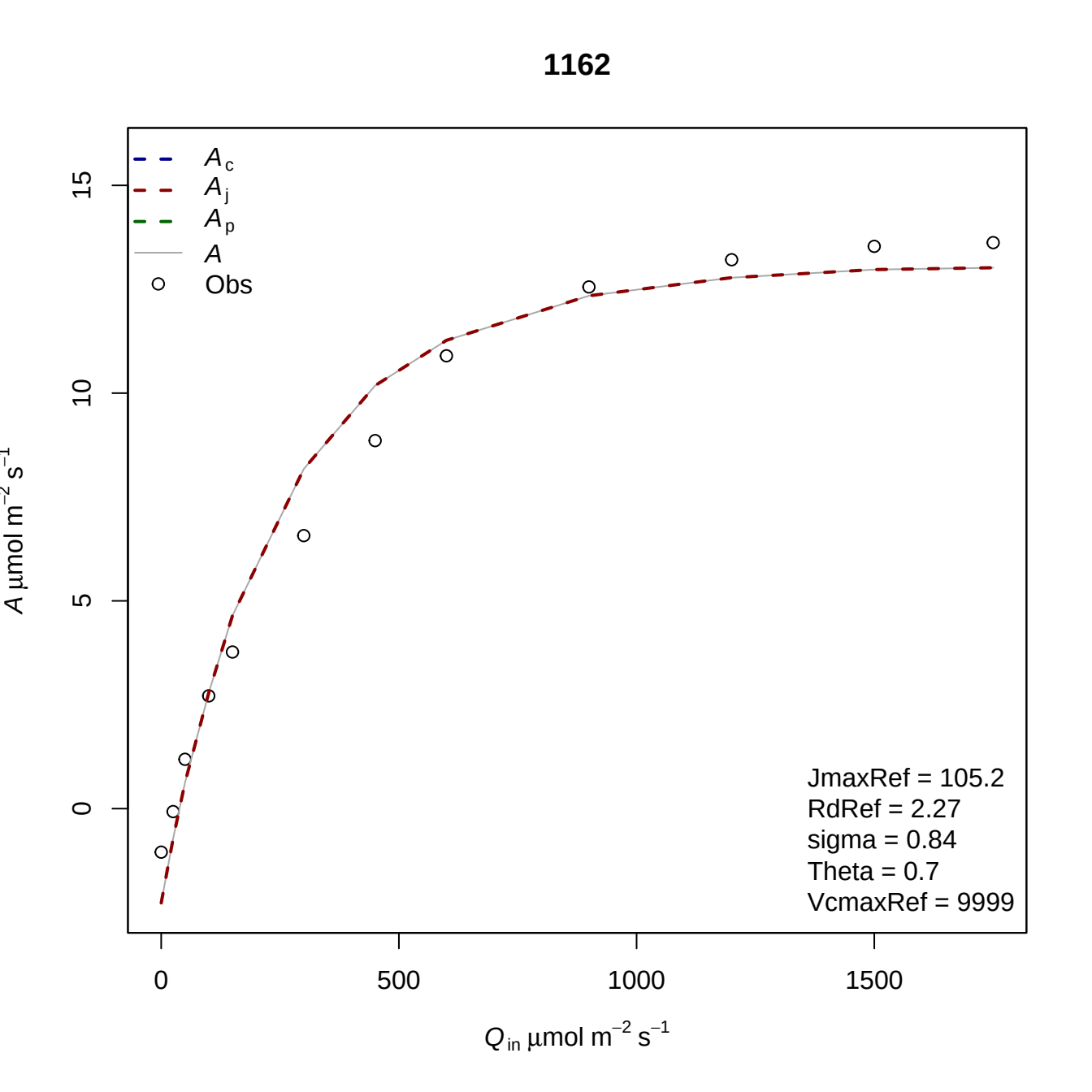

1157
A_c
A_j
A_p
A
Obs
JmaxRef = 130.28
RdRef = 3.85
sigma = 1.94
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

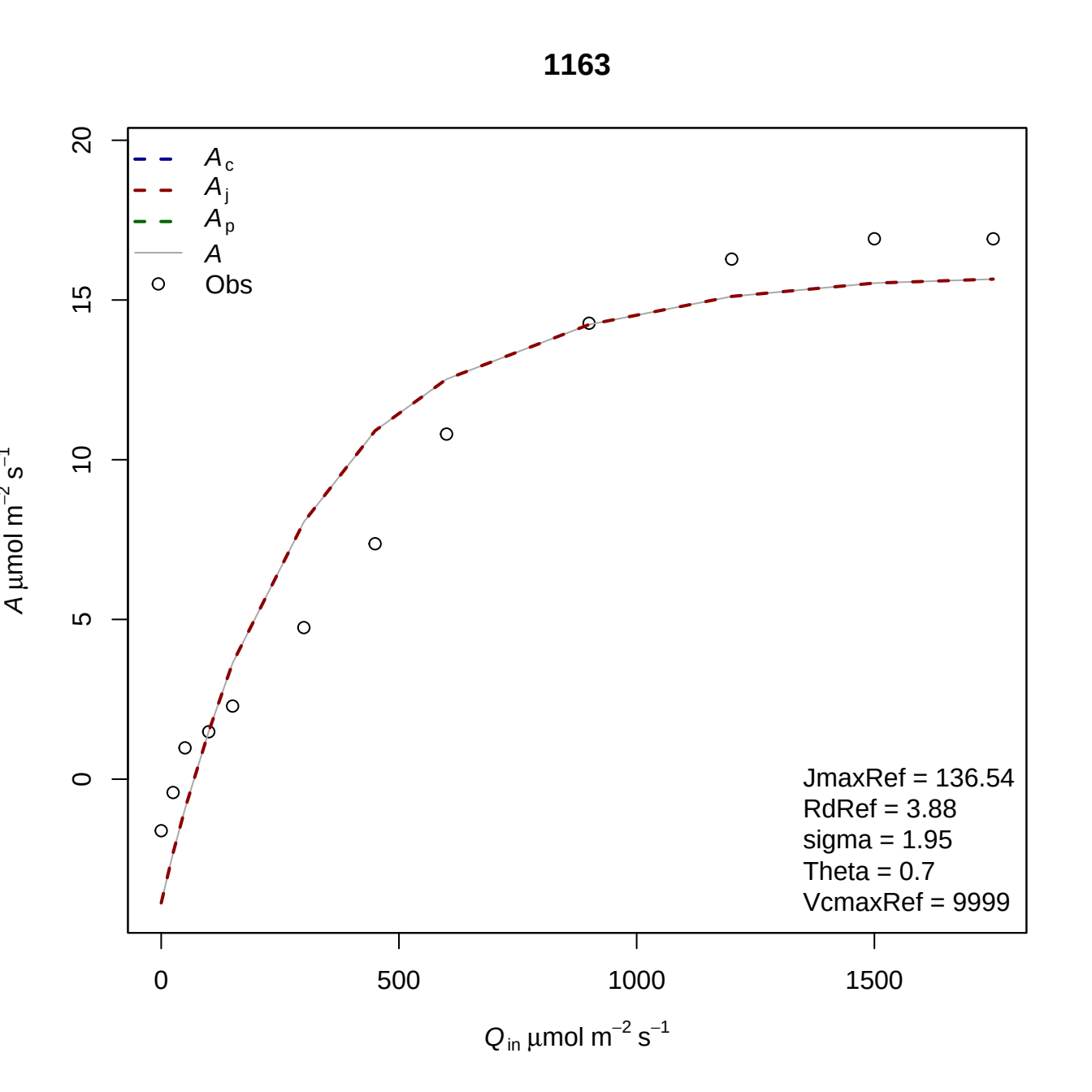

1158
A_c
A_j
A_p
A
Obs
JmaxRef = 126.95
RdRef = 3.83
sigma = 1.71
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

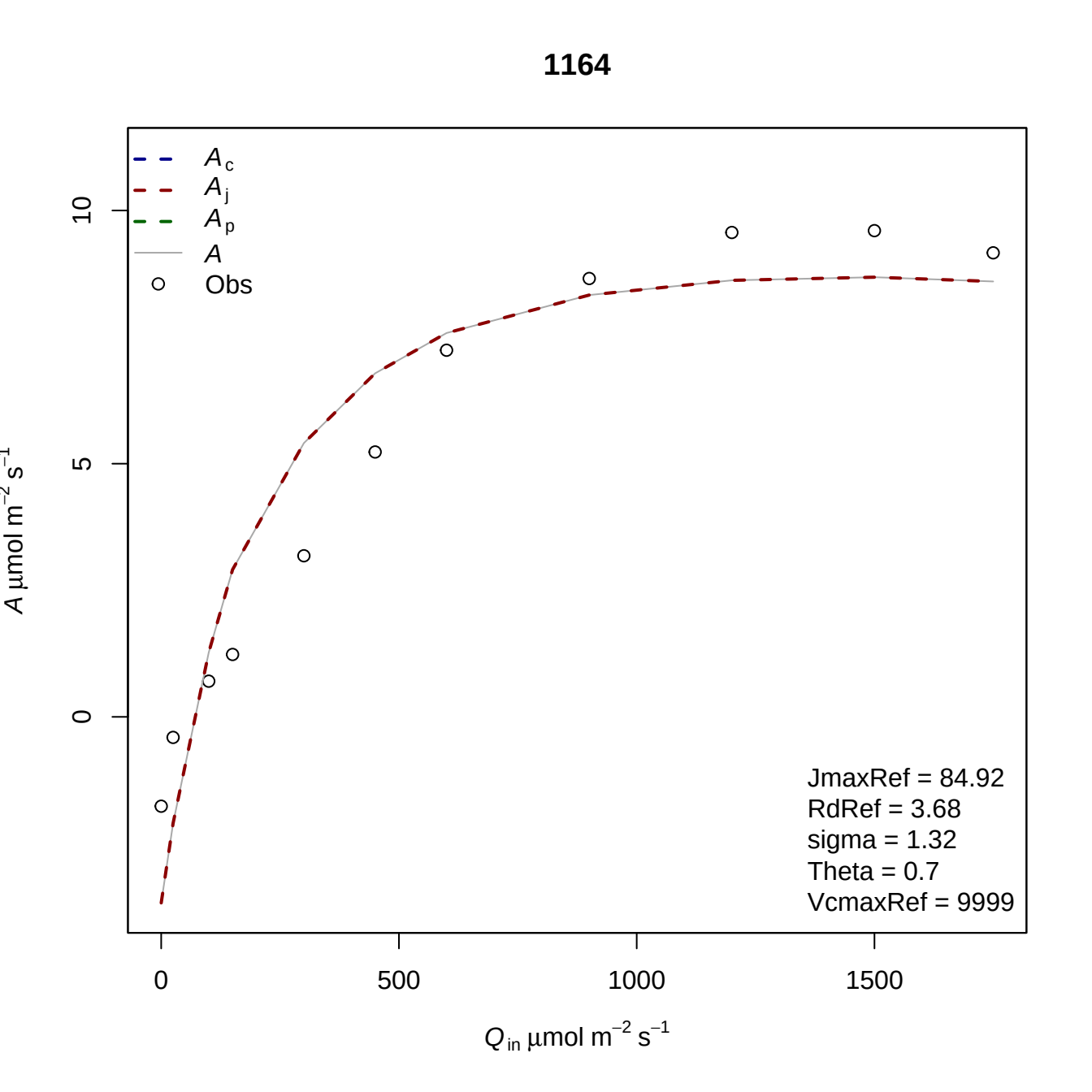

1159
A_c
A_j
A_p
A
Obs
JmaxRef = 121.53
RdRef = 3.68
sigma = 1.88
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

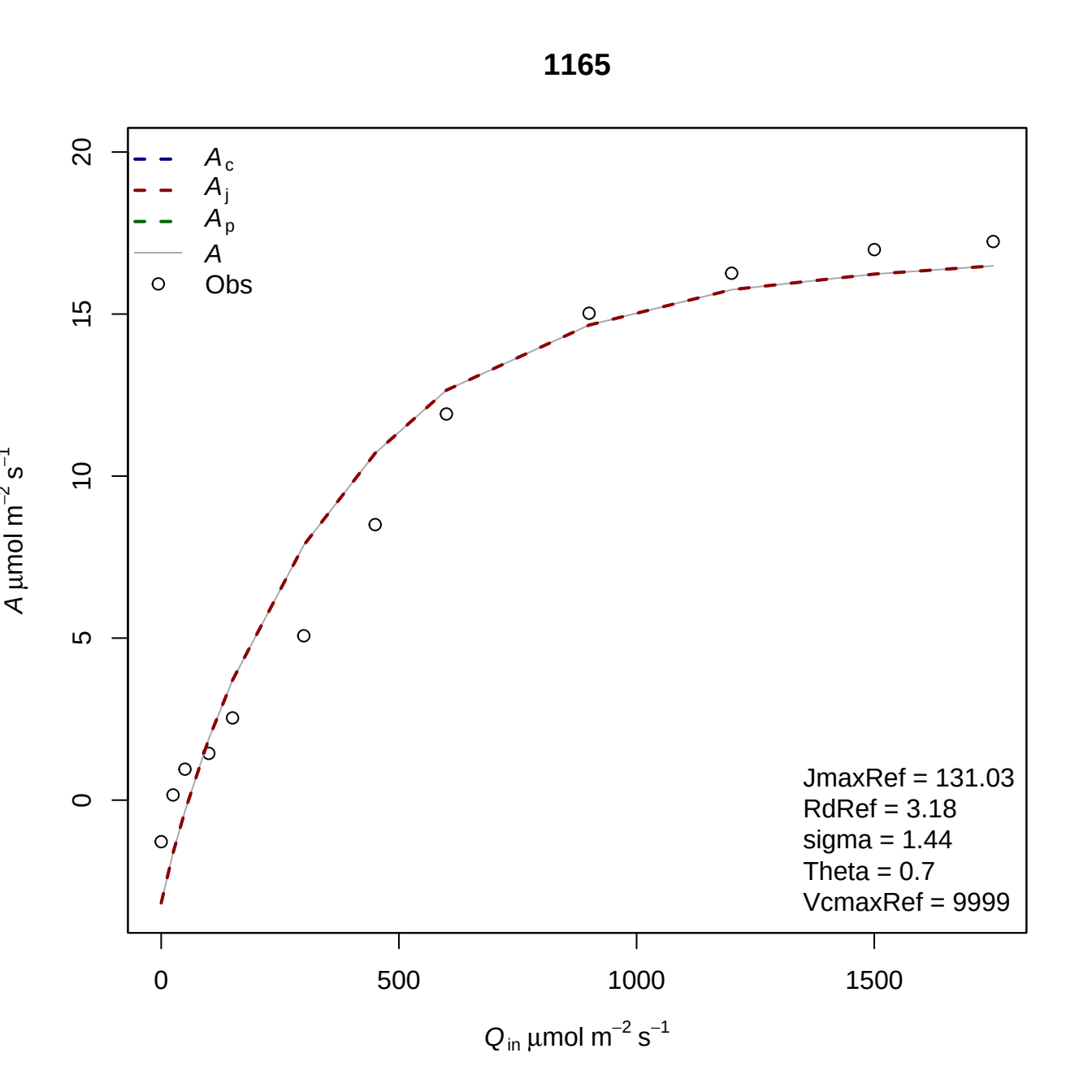

1160
A_c
A_j
A_p
A
Obs
JmaxRef = 118.08
RdRef = 2.28
sigma = 0.52
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

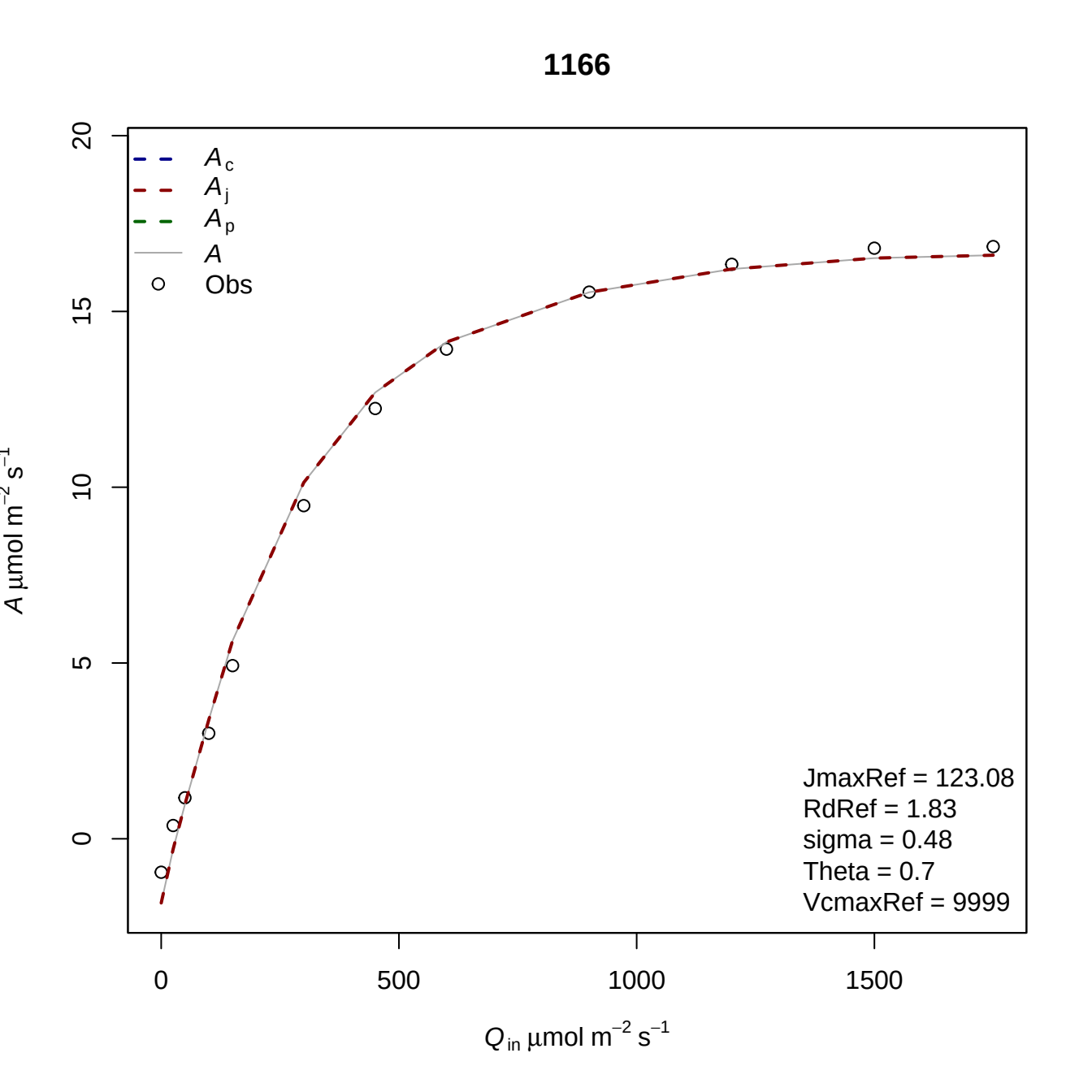

1161
A_c
A_j
A_p
A
Obs
JmaxRef = 98.89
RdRef = 2.69
sigma = 1.13
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

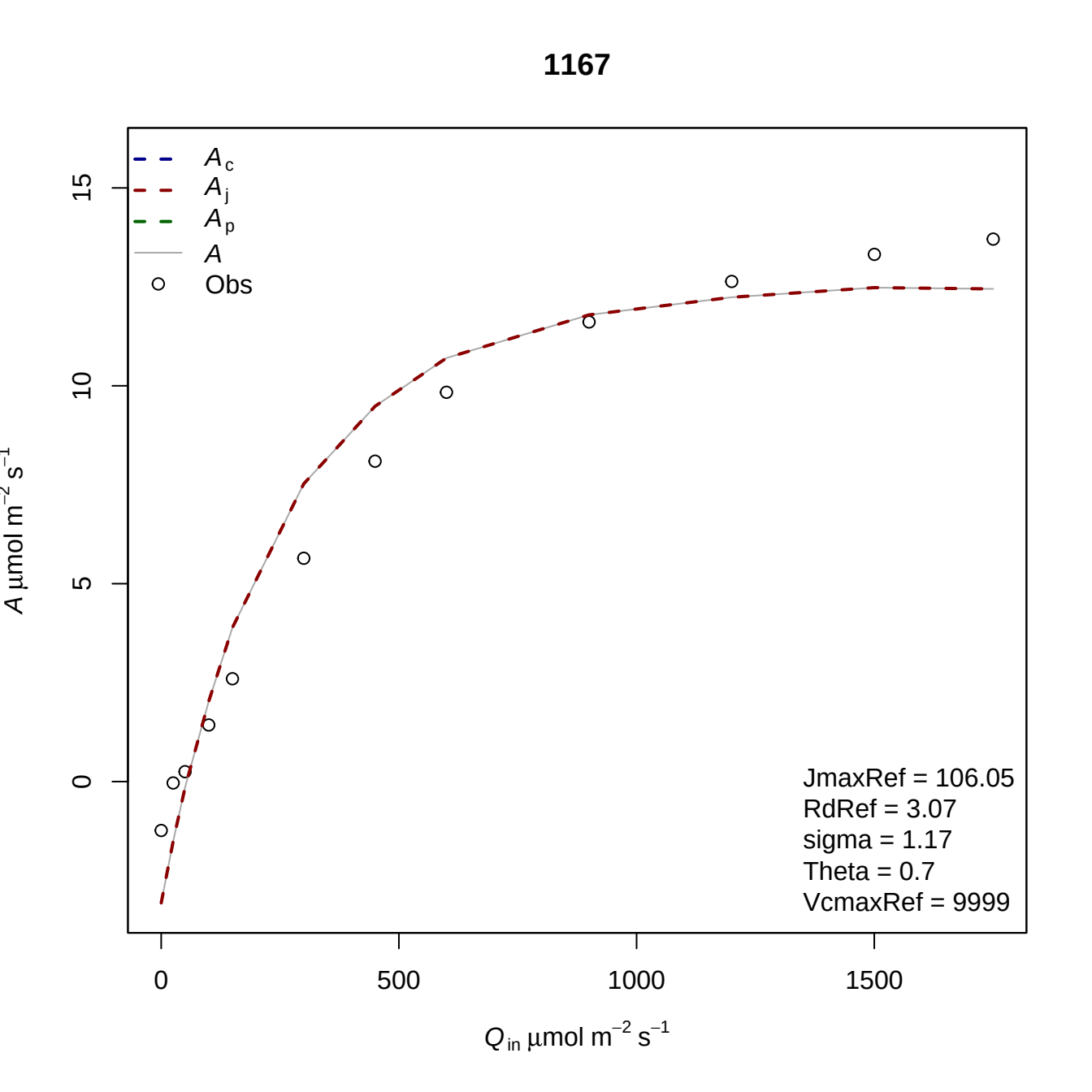

1162
A_c
A_j
A_p
A
Obs
JmaxRef = 105.2
RdRef = 2.27
sigma = 0.84
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

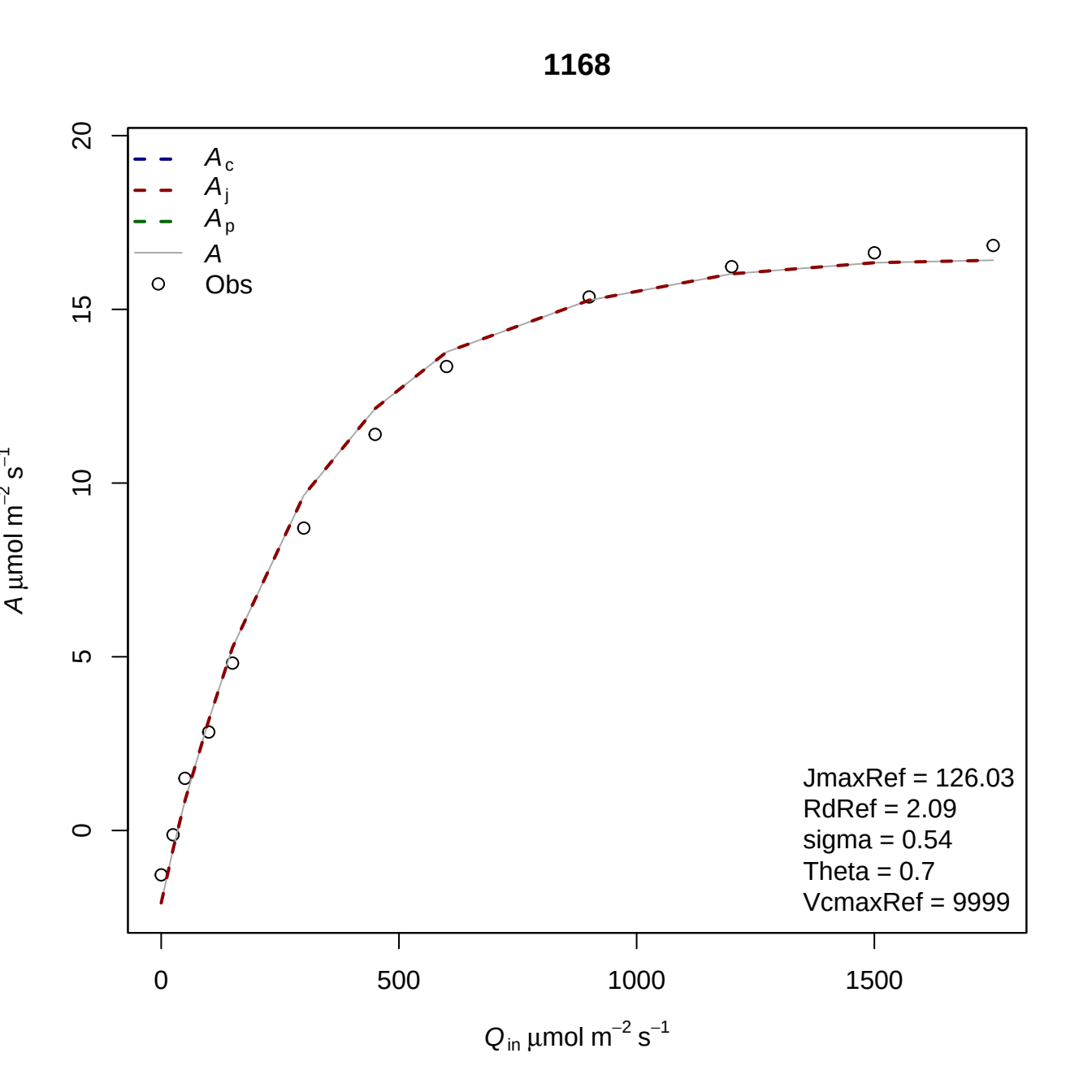

1163
A_c
A_j
A_p
A
Obs
JmaxRef = 136.54
RdRef = 3.88
sigma = 1.95
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

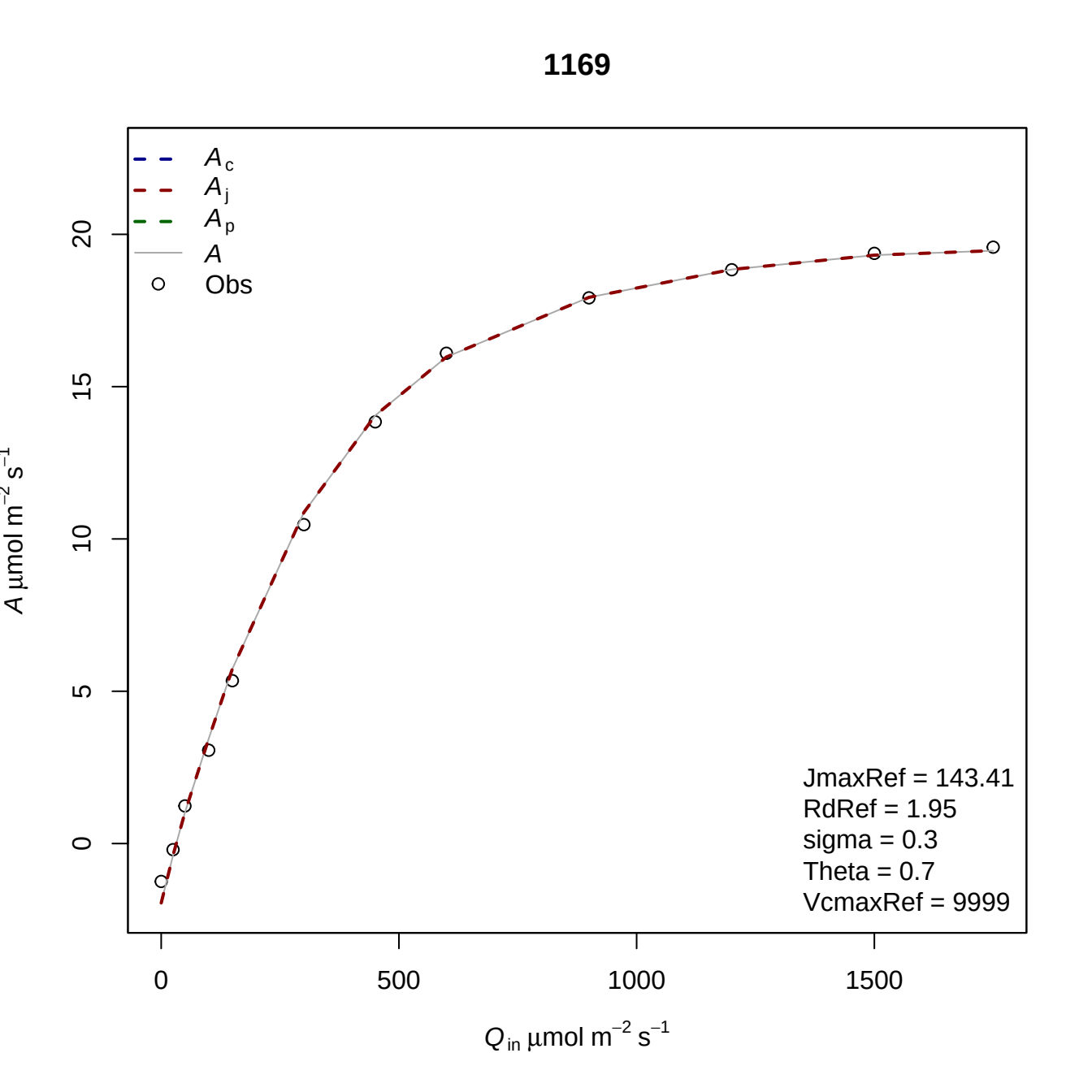

1164
A_c
A_j
A_p
A
Obs
JmaxRef = 84.92
RdRef = 3.68
sigma = 1.32
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

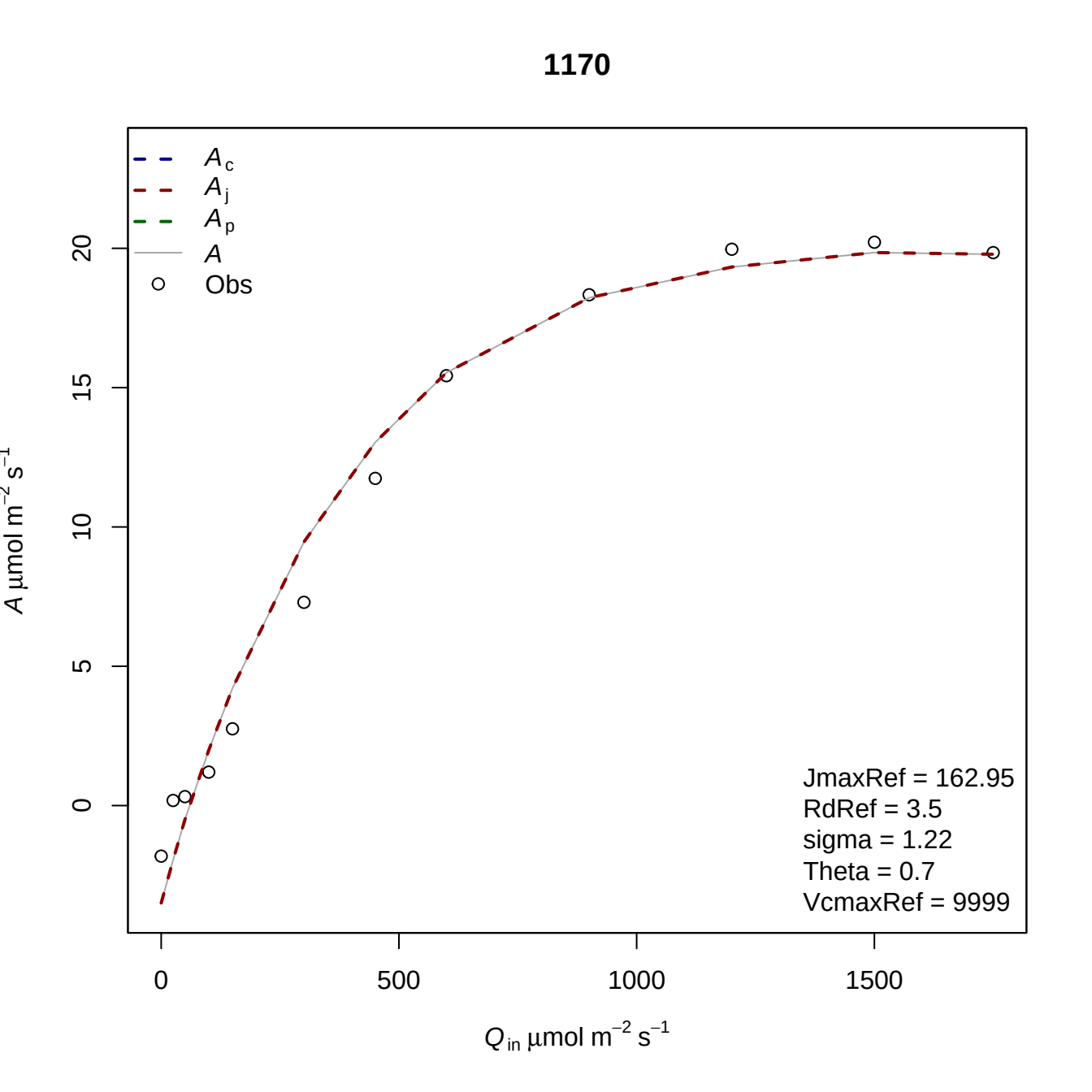

1165
A_c
A_j
A_p
A
Obs
JmaxRef = 131.03
RdRef = 3.18
sigma = 1.44
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

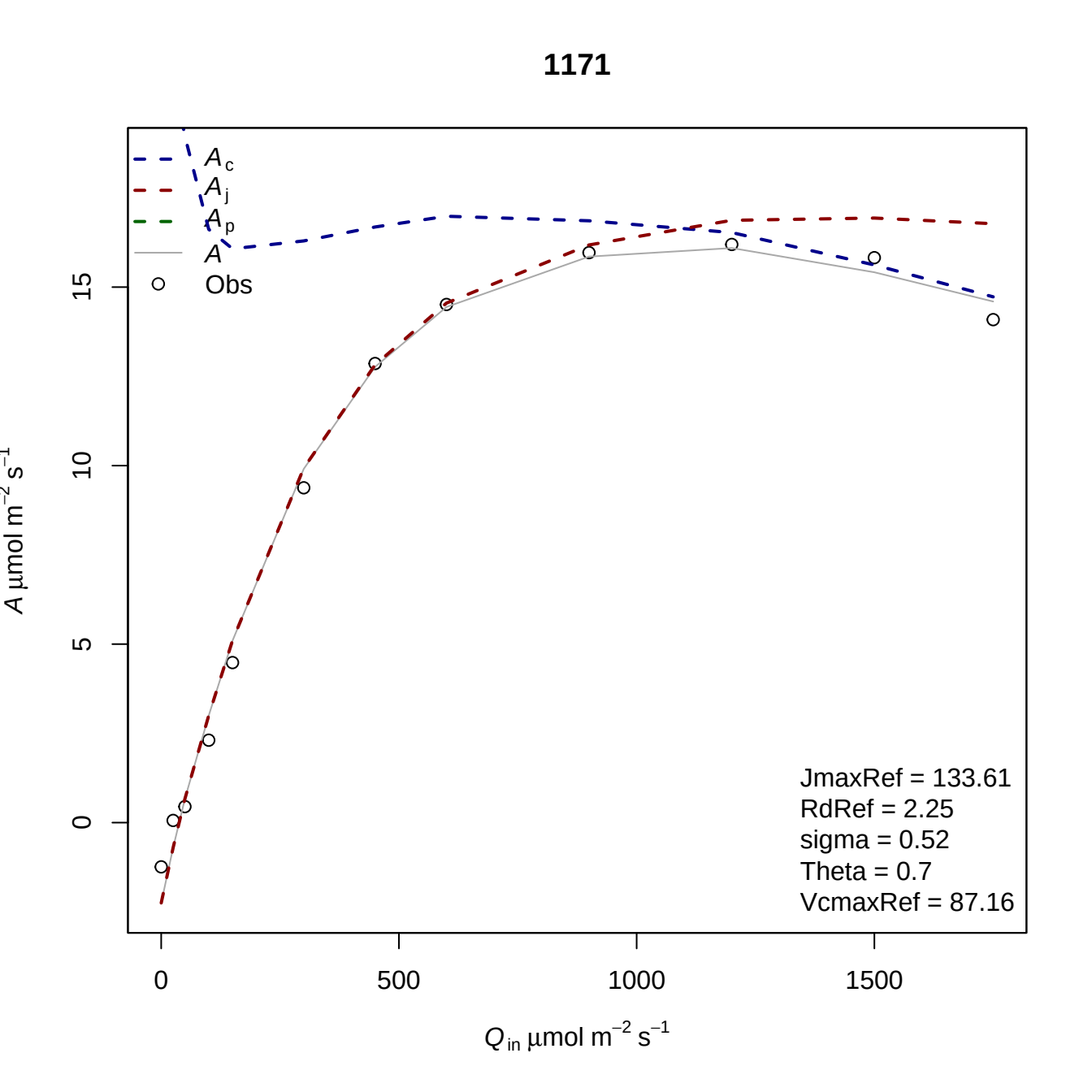

1166
A_c
A_j
A_p
A
Obs
JmaxRef = 123.08
RdRef = 1.83
sigma = 0.48
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

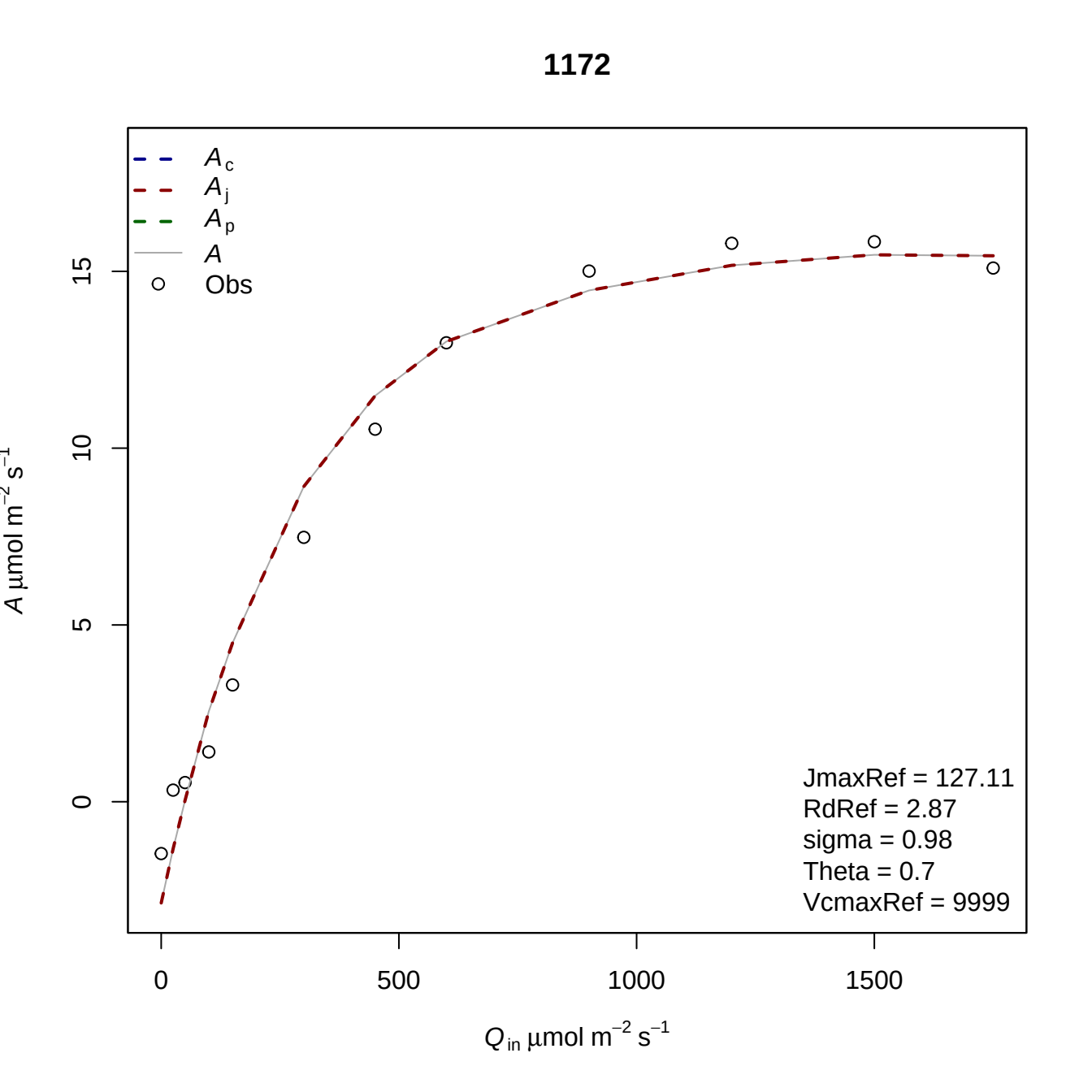

1167
A_c
A_j
A_p
A
Obs
JmaxRef = 106.05
RdRef = 3.07
sigma = 1.17
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

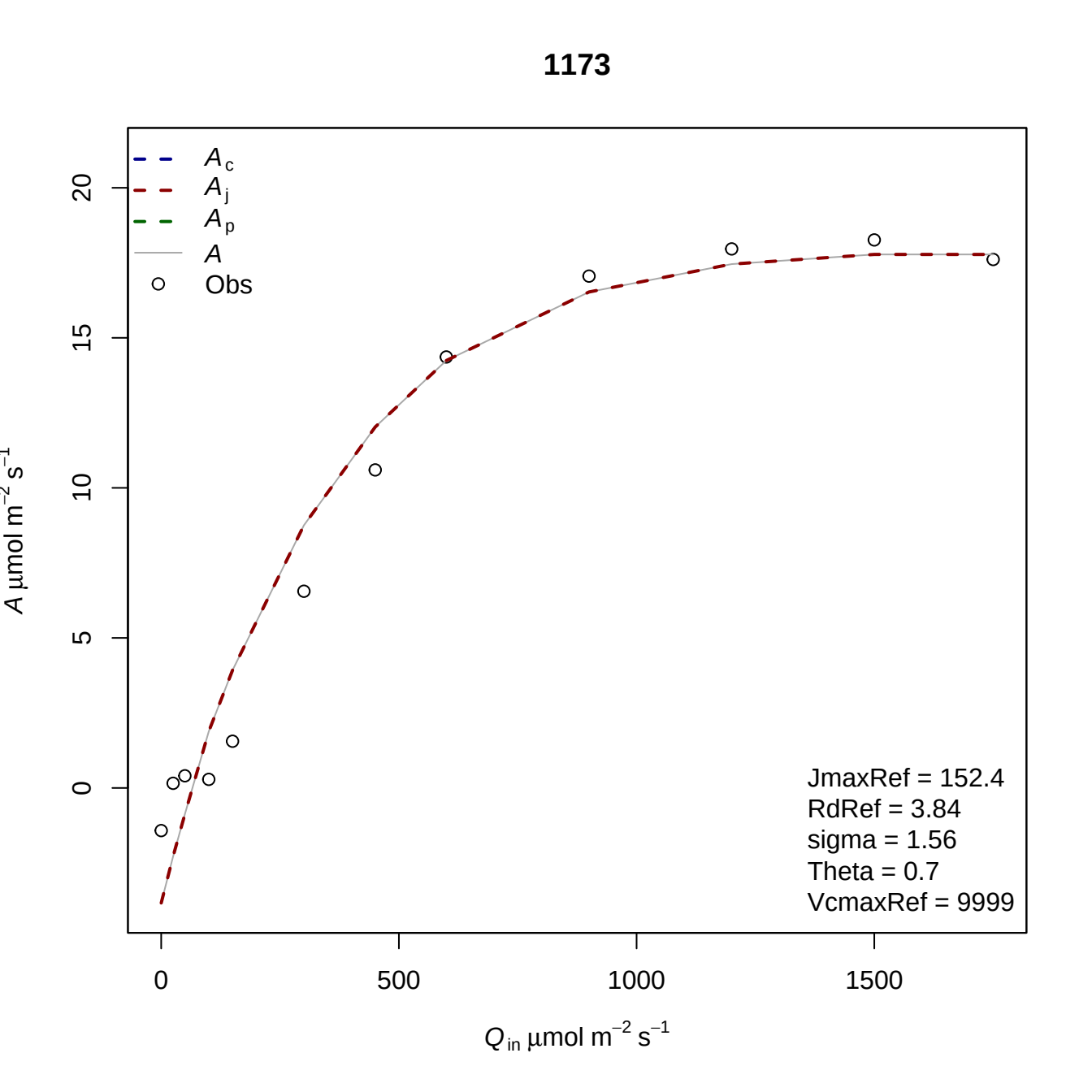

1168
A_c
A_j
A_p
A
Obs
JmaxRef = 126.03
RdRef = 2.09
sigma = 0.54
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

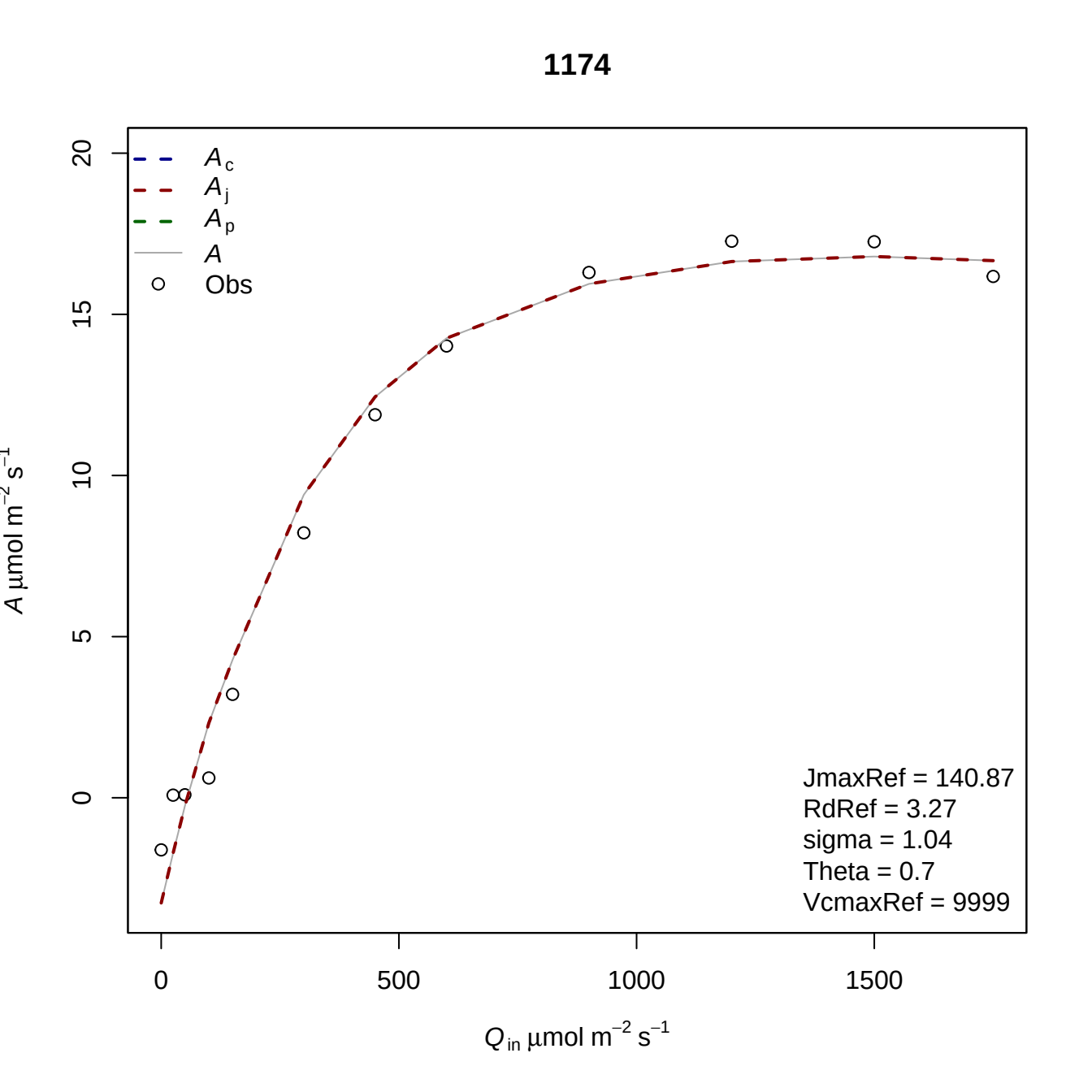

1169
A_c
A_j
A_p
A
Obs
JmaxRef = 143.41
RdRef = 1.95
sigma = 0.3
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

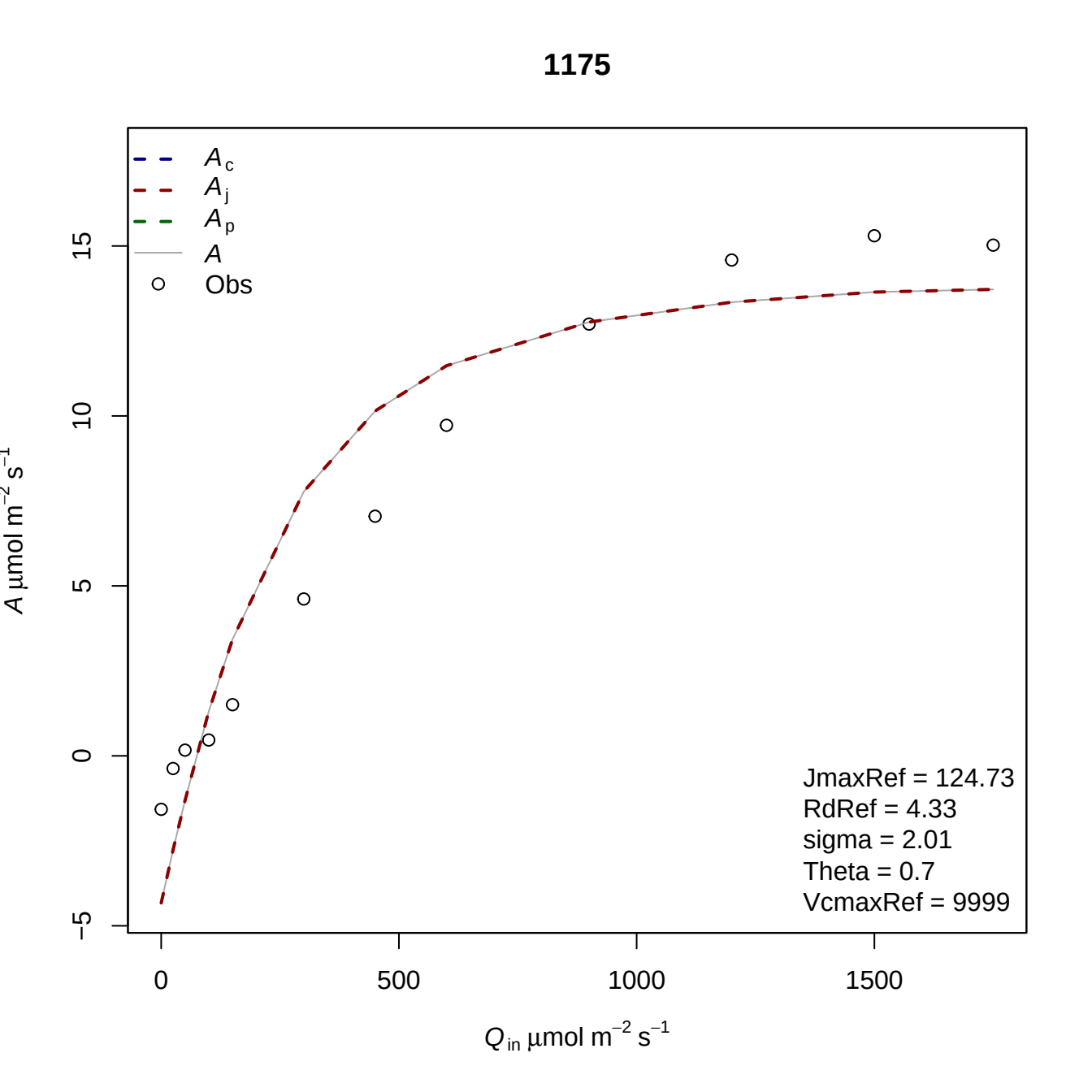

1170
A_c
A_j
A_p
A
Obs
JmaxRef = 162.95
RdRef = 3.5
sigma = 1.22
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

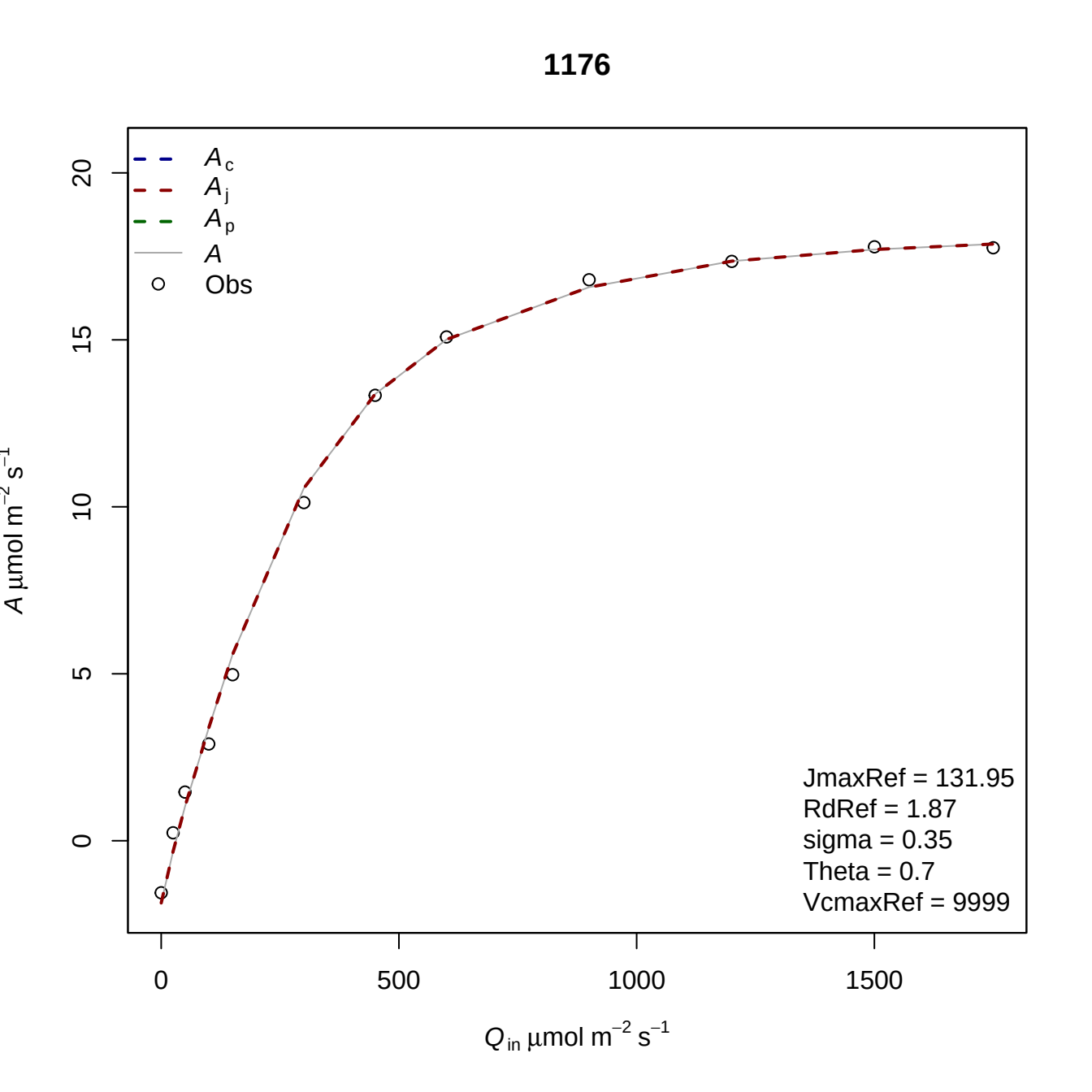

1171
A_c
A_j
A_p
A
Obs
JmaxRef = 133.61
RdRef = 2.25
sigma = 0.52
Theta = 0.7
VcmaxRef = 87.16
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

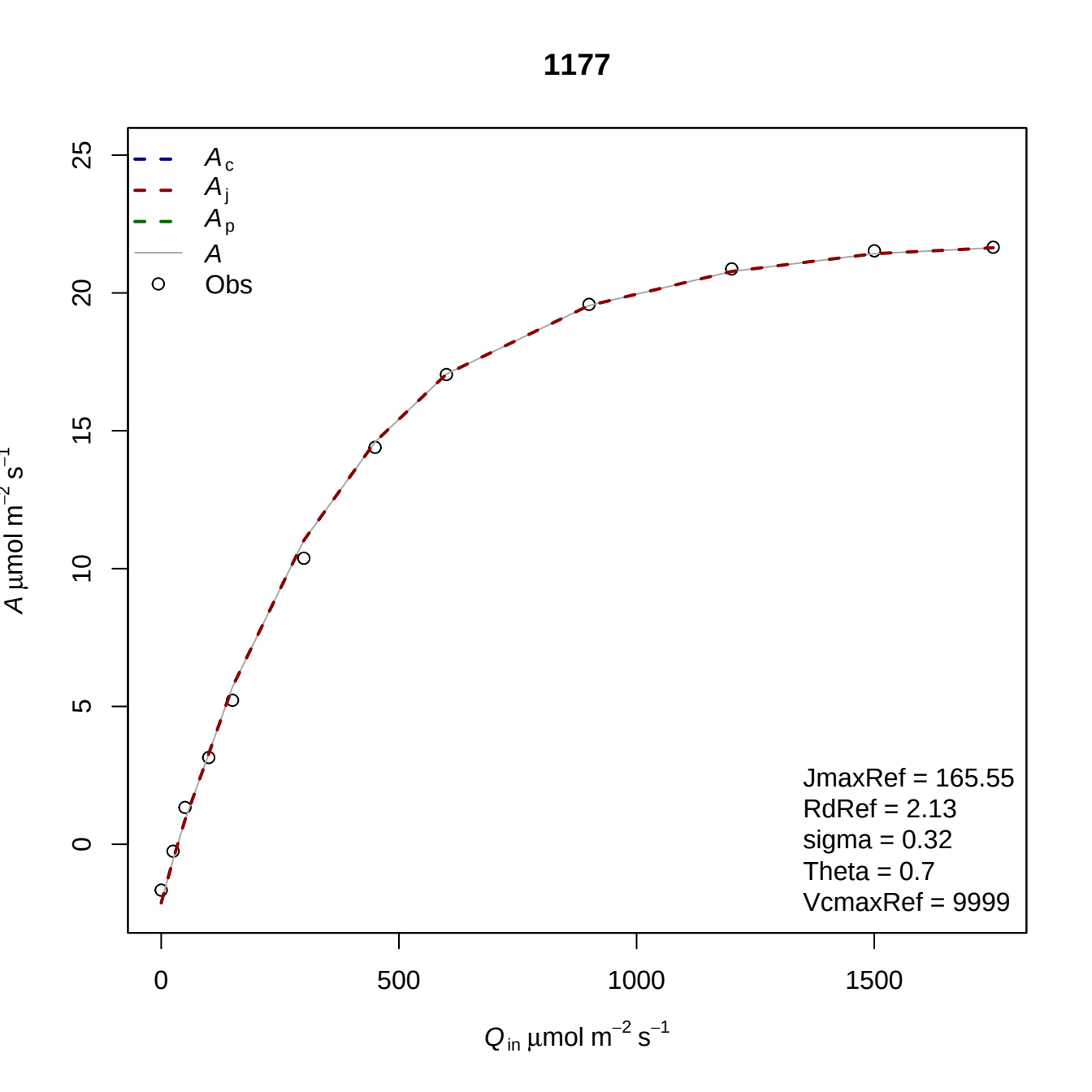

1172
A_c
A_j
A_p
A
Obs
JmaxRef = 127.11
RdRef = 2.87
sigma = 0.98
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

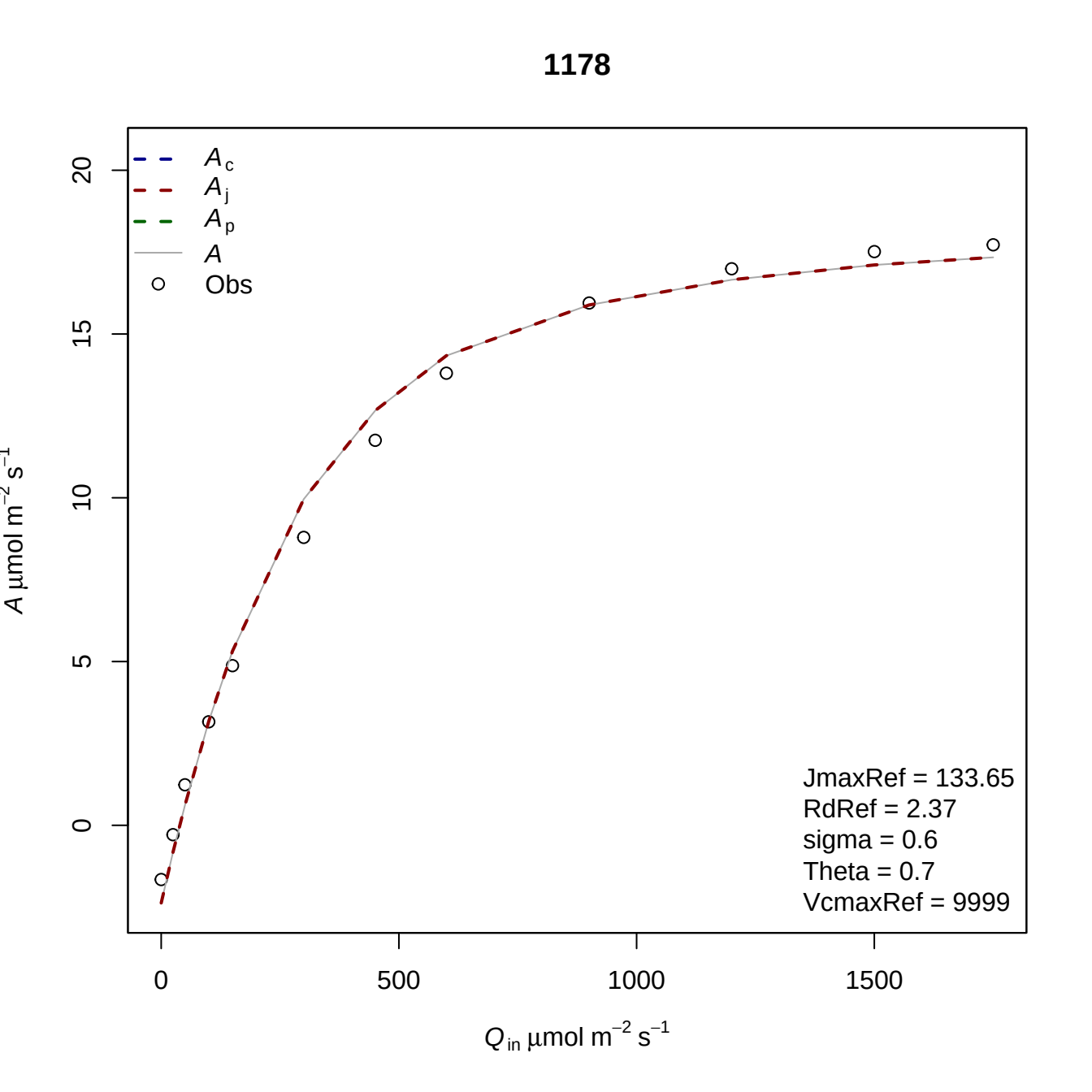

1173
A_c
A_j
A_p
A
Obs
JmaxRef = 152.4
RdRef = 3.84
sigma = 1.56
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

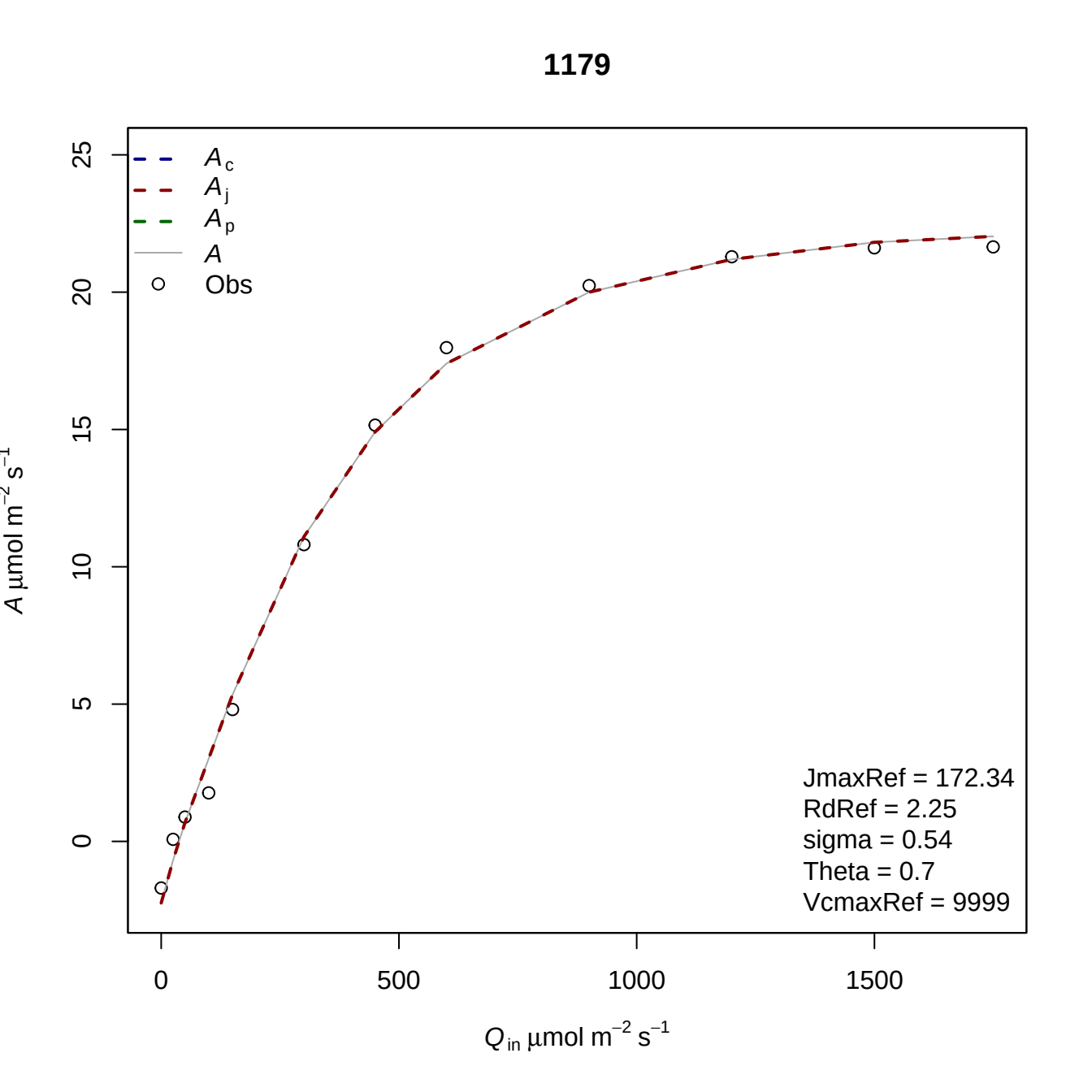

1174
A_c
A_j
A_p
A
Obs
JmaxRef = 140.87
RdRef = 3.27
sigma = 1.04
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

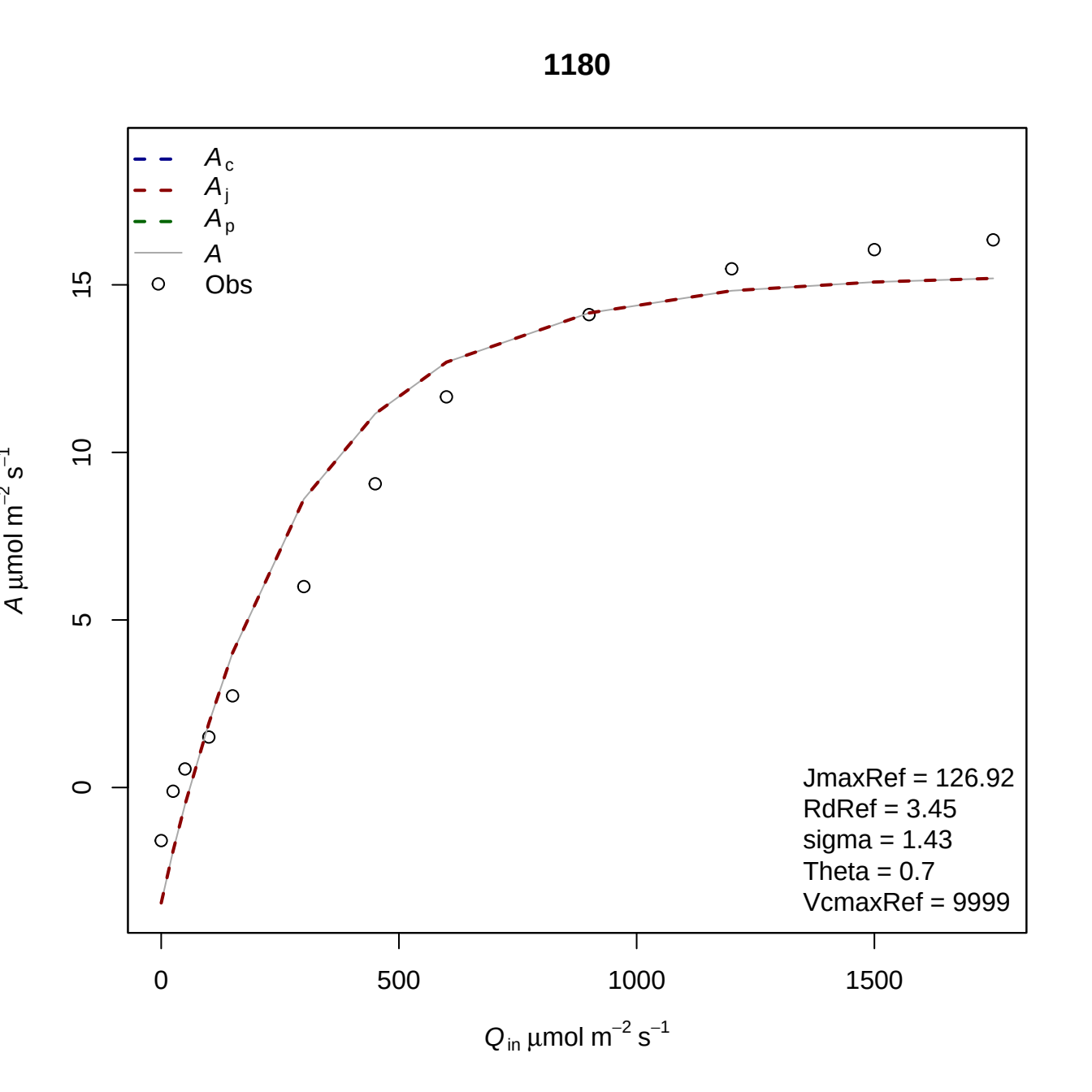

1175
A_c
A_j
A_p
A
Obs
JmaxRef = 124.73
RdRef = 4.33
sigma = 2.01
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500

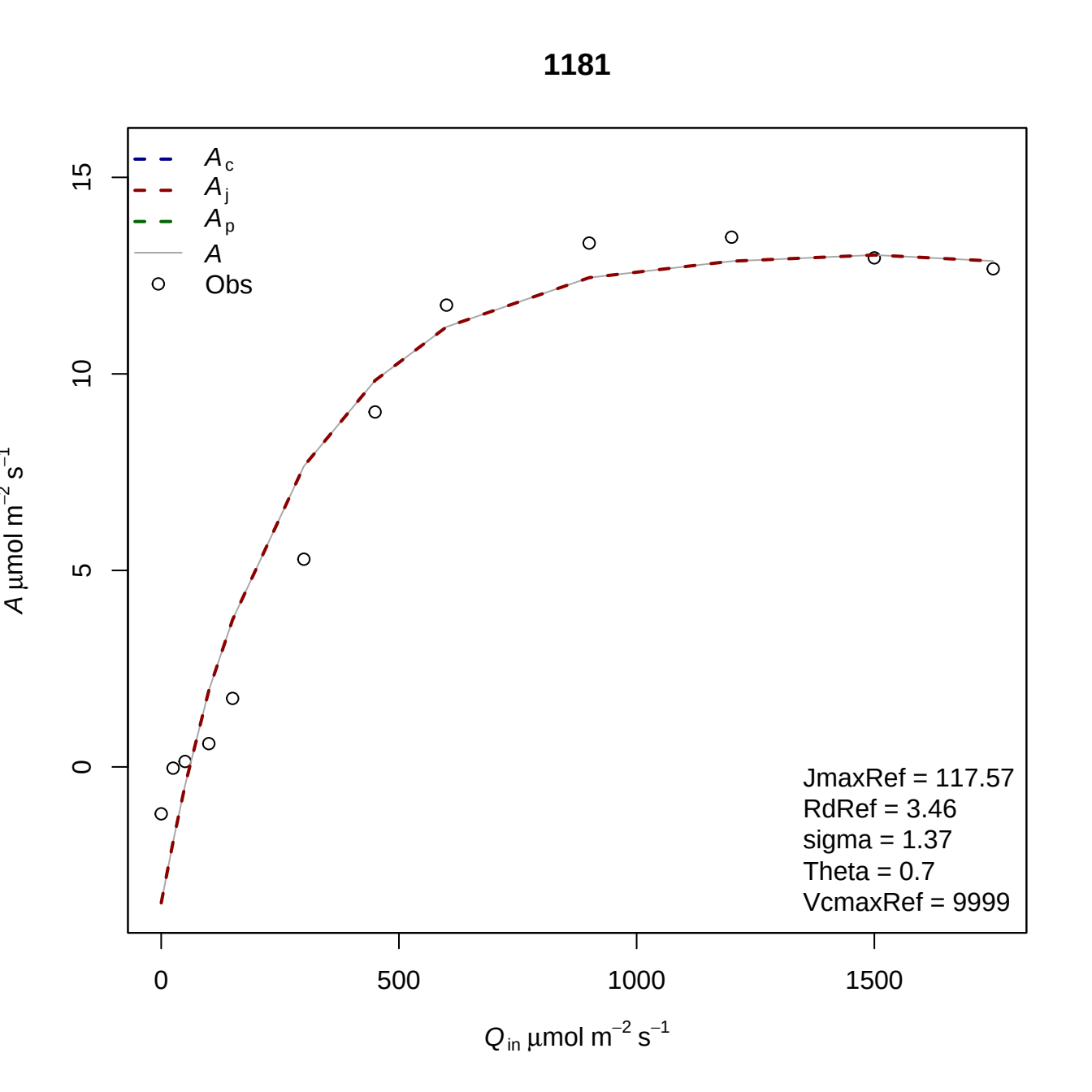

1176
A_c
A_j
A_p
A
Obs
JmaxRef = 131.95
RdRef = 1.87
sigma = 0.35
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

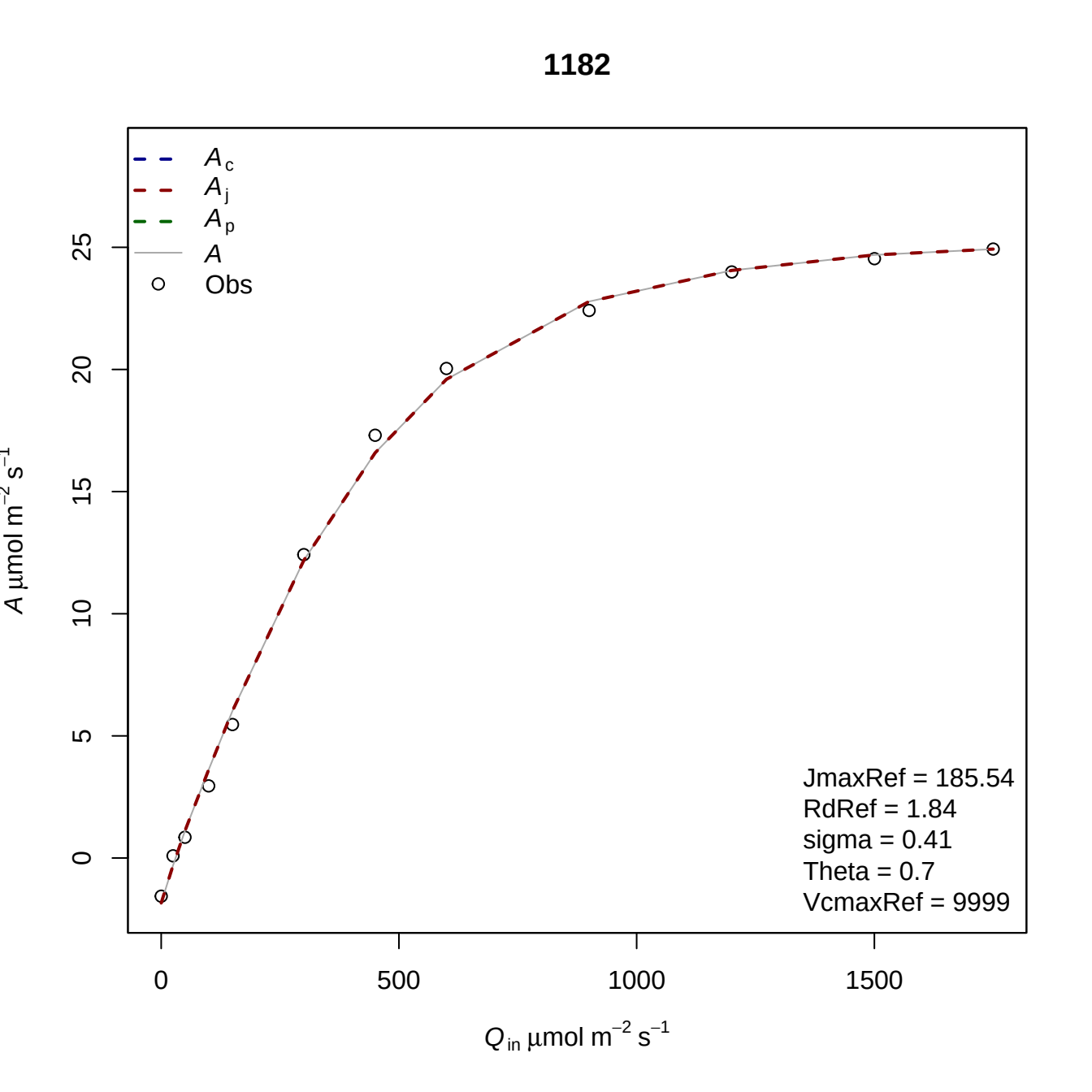

1177
A_c
A_j
A_p
A
Obs
JmaxRef = 165.55
RdRef = 2.13
sigma = 0.32
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

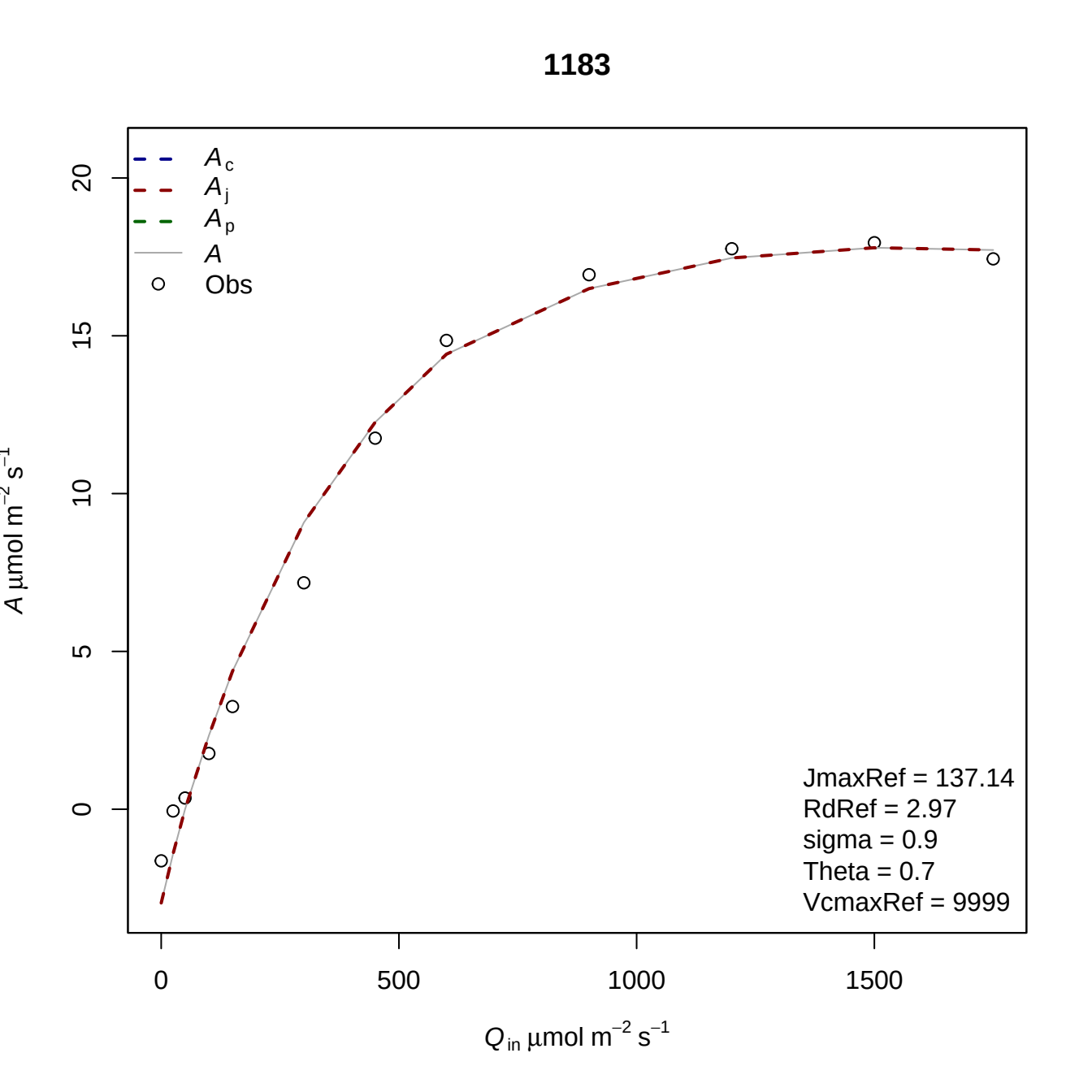

1178
A_c
A_j
A_p
A
Obs
JmaxRef = 133.65
RdRef = 2.37
sigma = 0.6
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

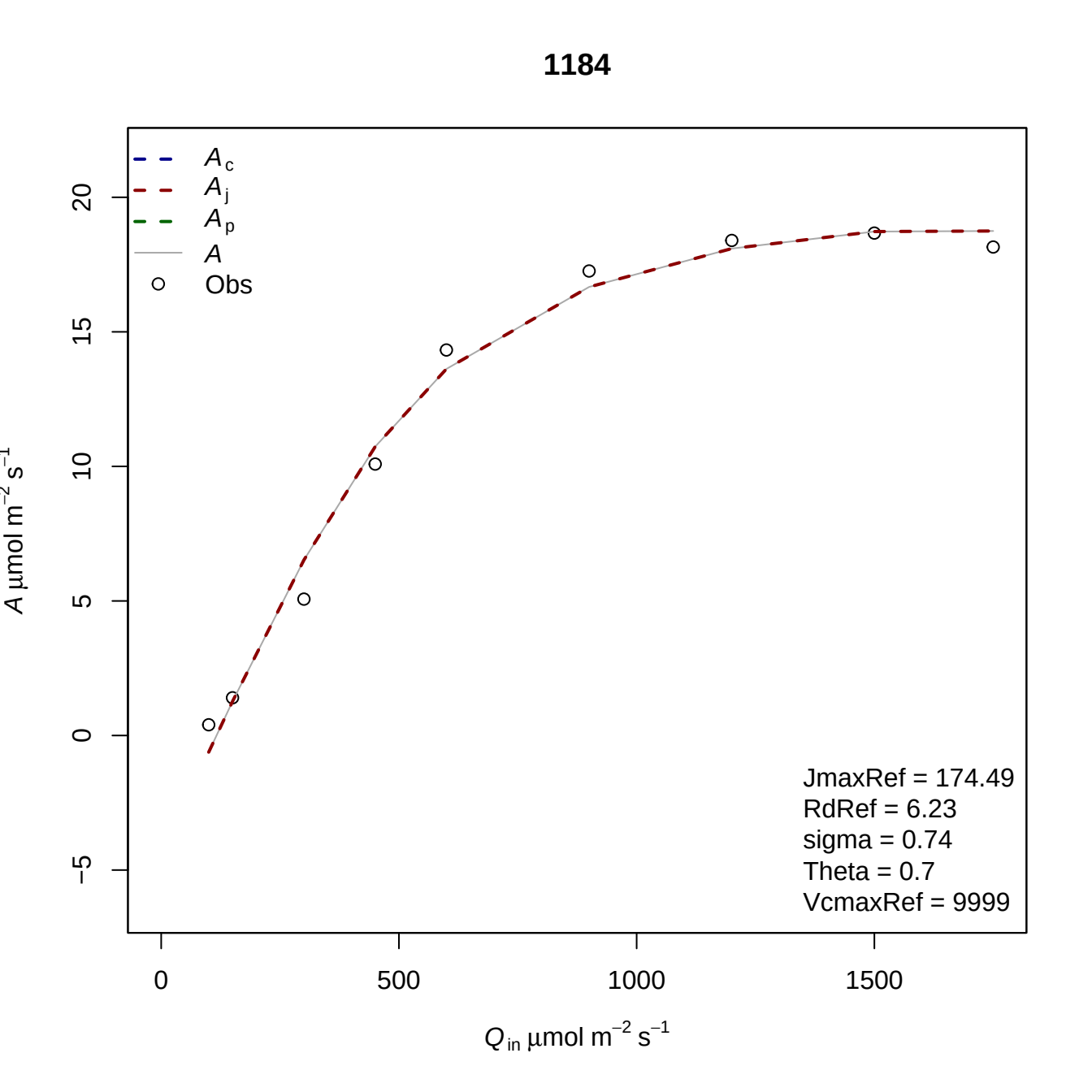

1179
A_c
A_j
A_p
A
Obs
JmaxRef = 172.34
RdRef = 2.25
sigma = 0.54
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

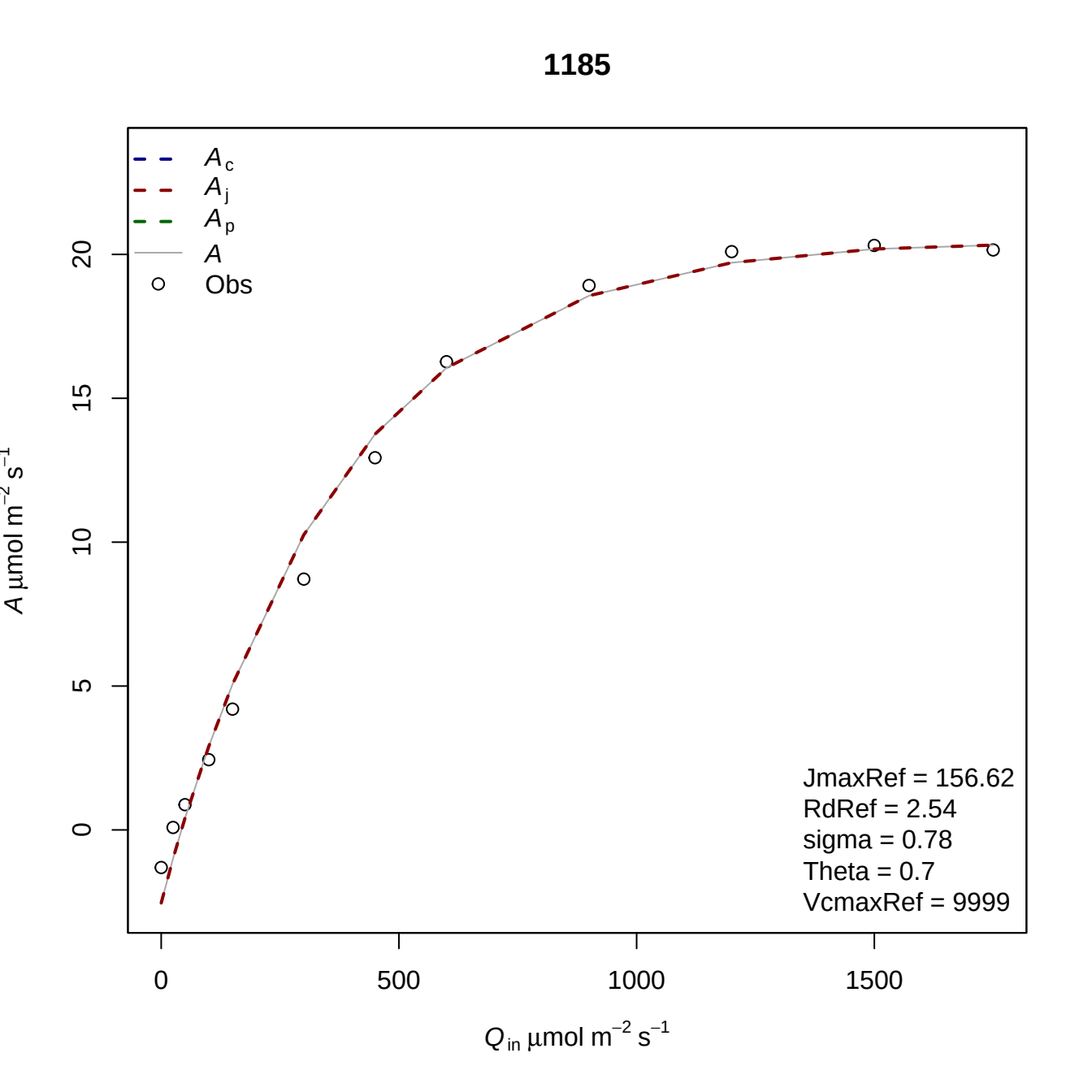

1180
A_c
A_j
A_p
A
Obs
JmaxRef = 126.92
RdRef = 3.45
sigma = 1.43
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

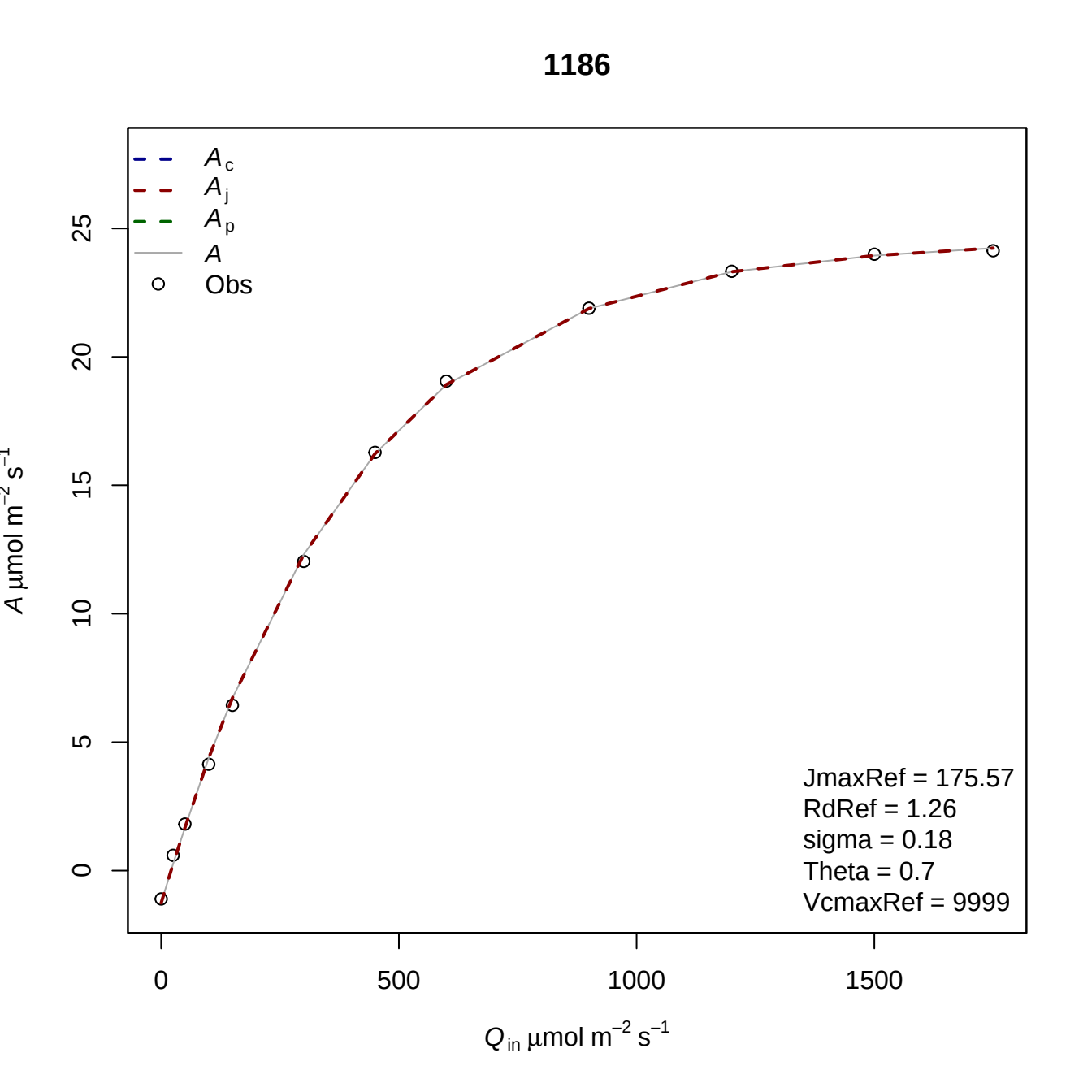

1181
A_c
A_j
A_p
A
Obs
JmaxRef = 117.57
RdRef = 3.46
sigma = 1.37
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

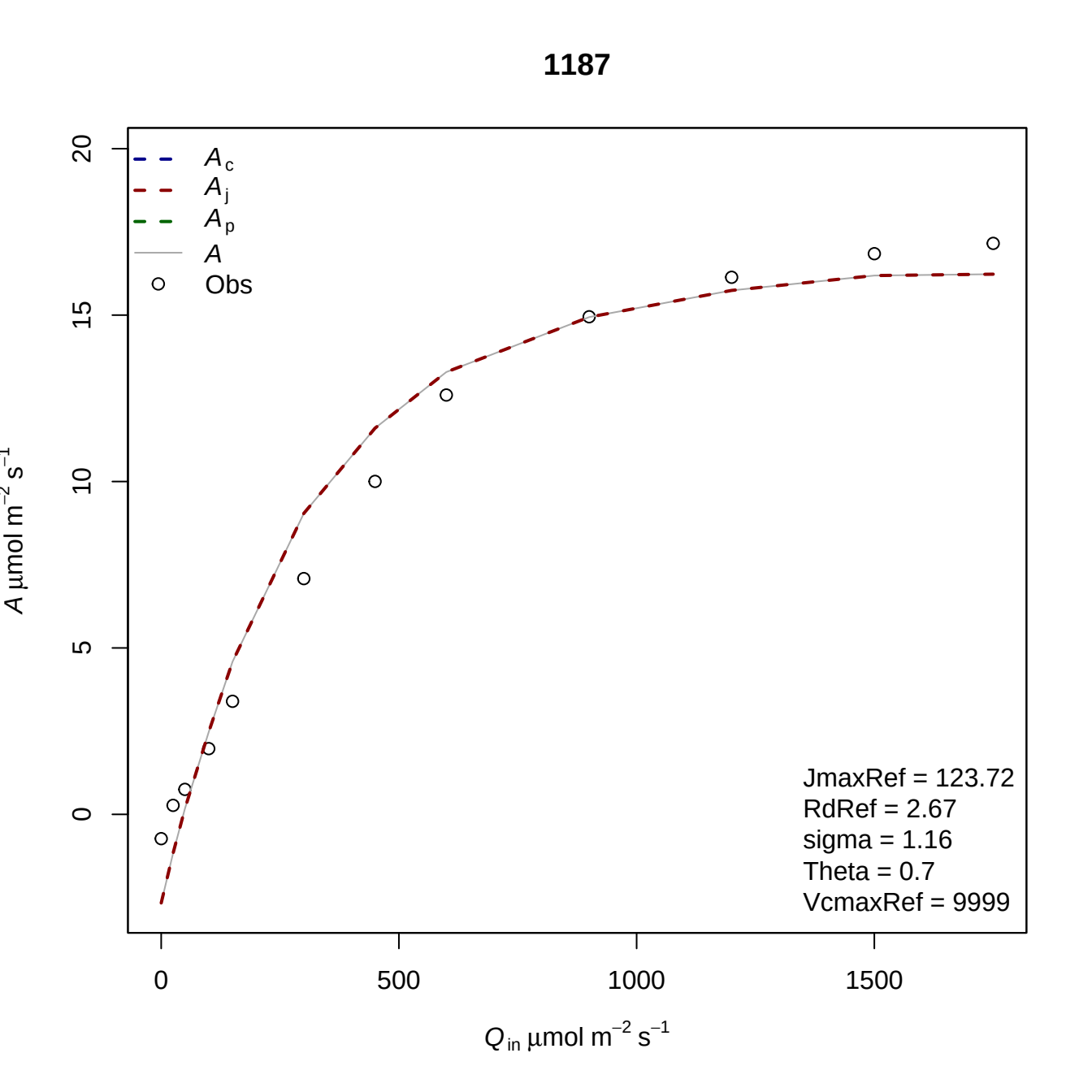

1182
A_c
A_j
A_p
A
Obs
JmaxRef = 185.54
RdRef = 1.84
sigma = 0.41
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

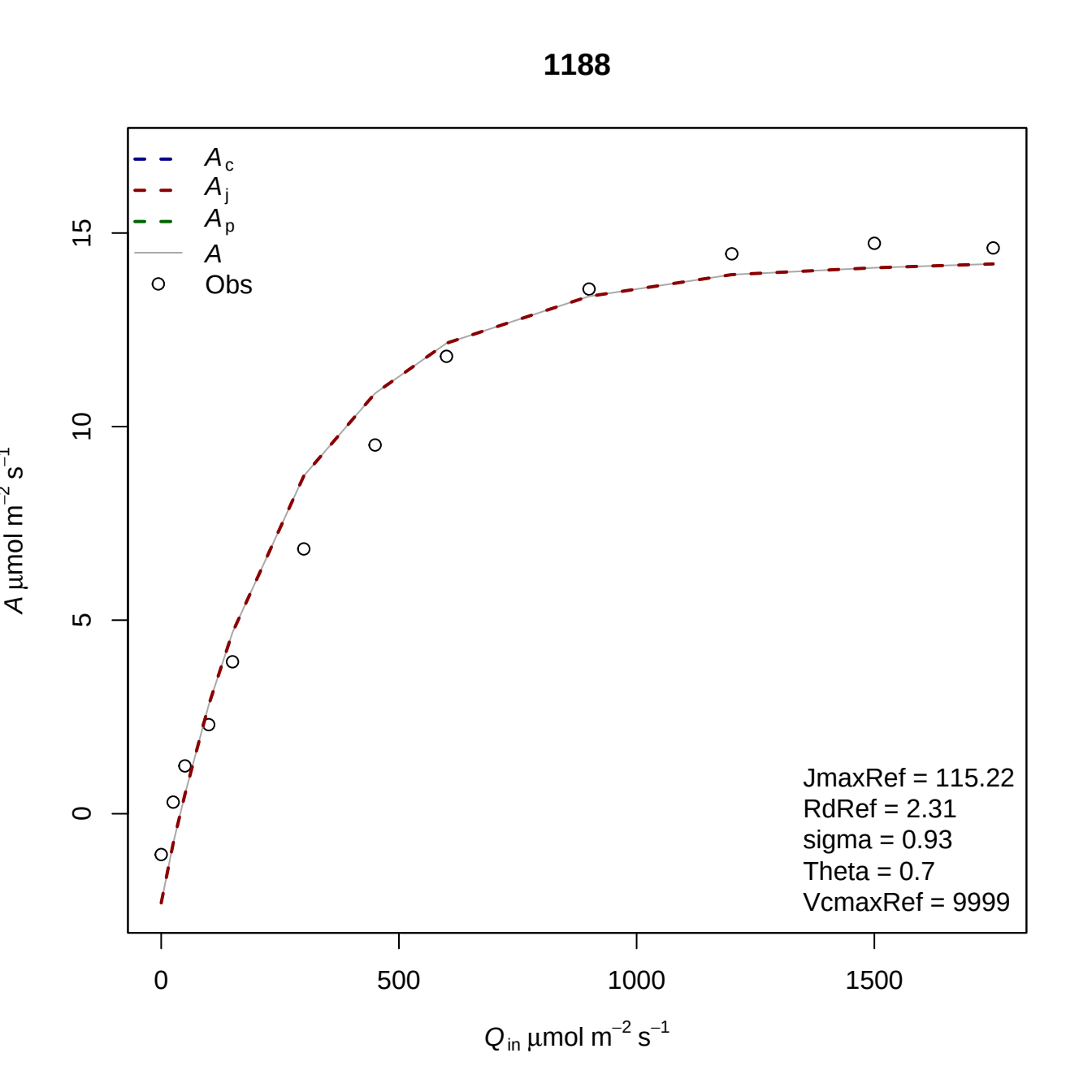

1183
A_c
A_j
A_p
A
Obs
JmaxRef = 137.14
RdRef = 2.97
sigma = 0.9
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

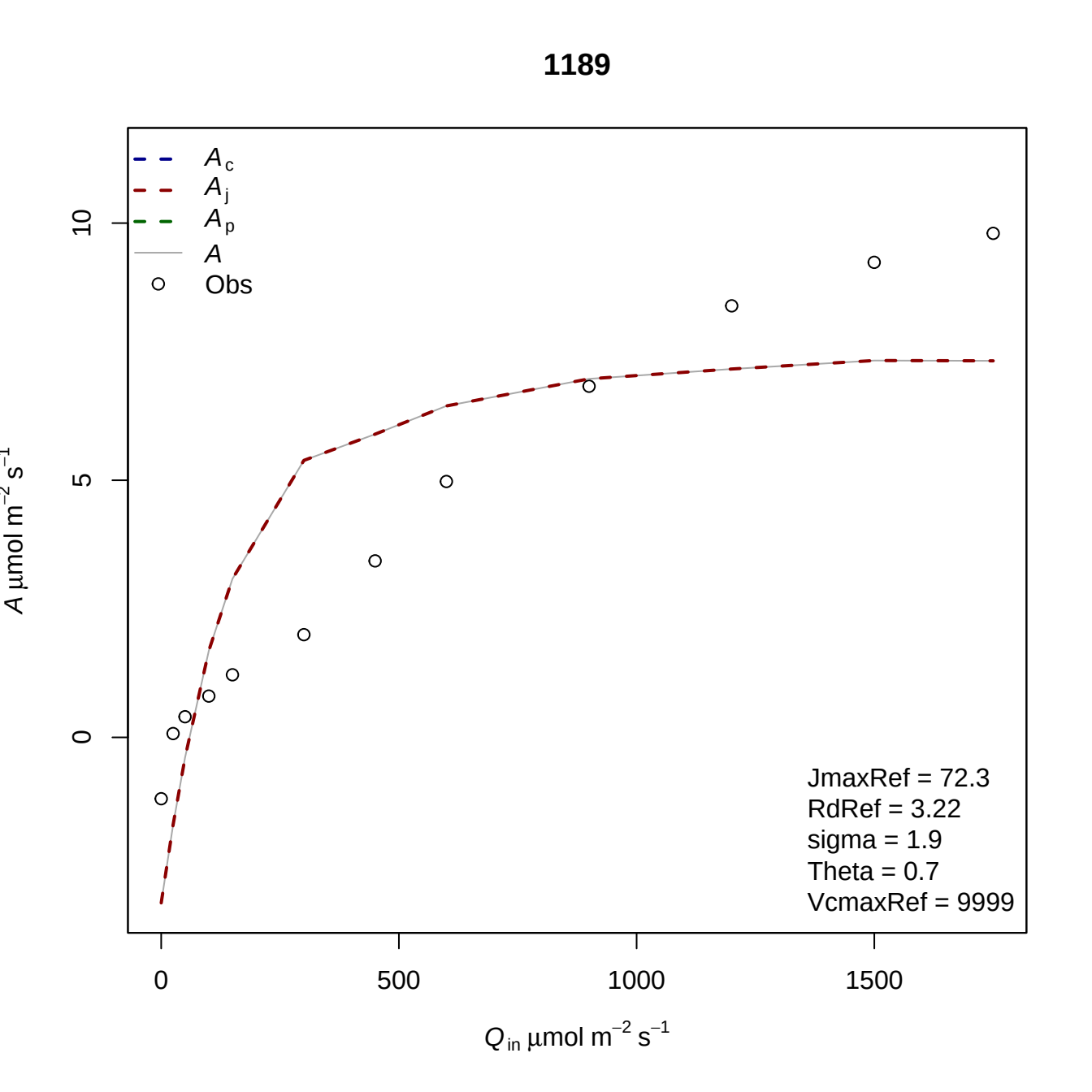

1184
A_c
A_j
A_p
A
Obs
JmaxRef = 174.49
RdRef = 6.23
sigma = 0.74
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
20
15
10
5
0
−5
0
500
1000
1500

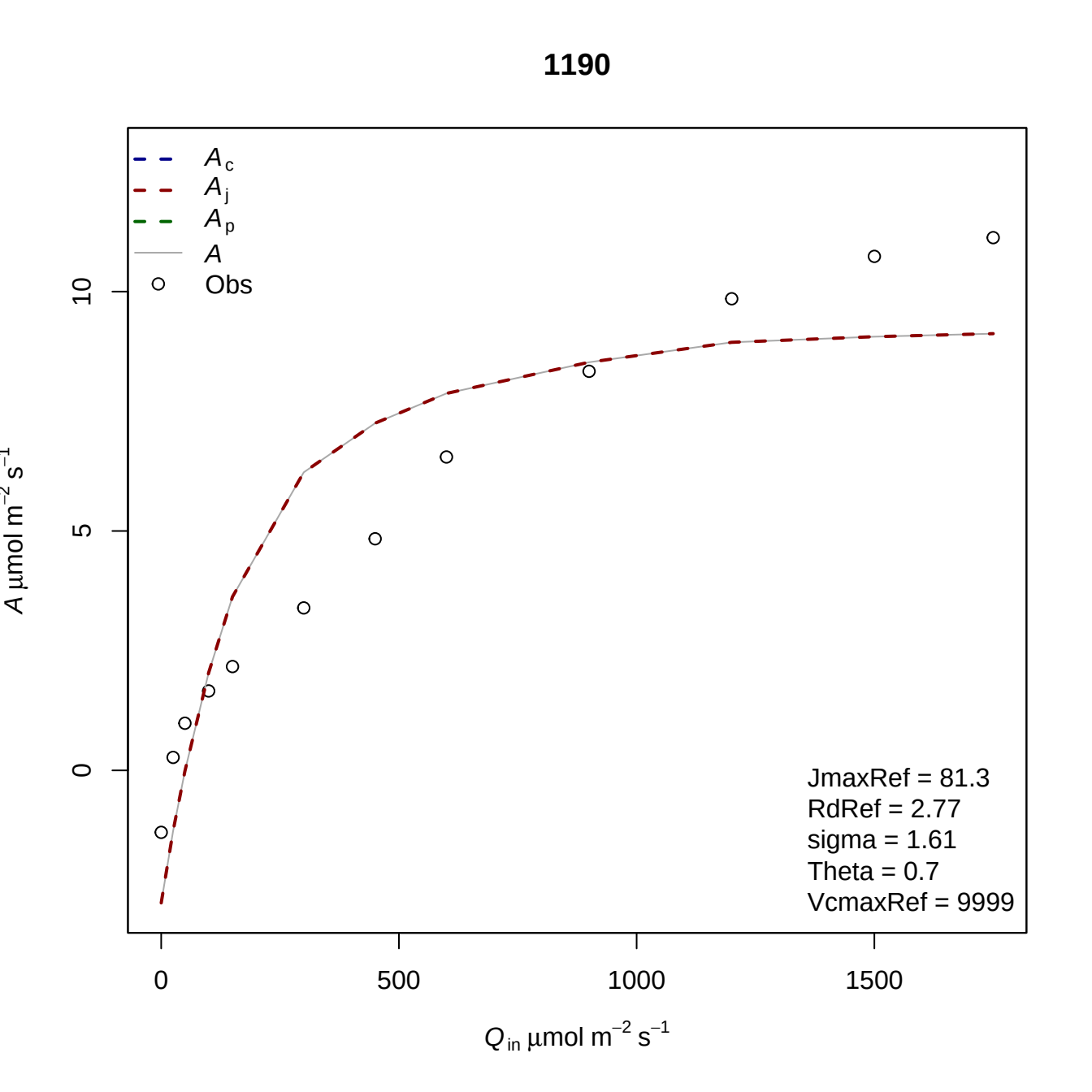

1185
A_c
A_j
A_p
A
Obs
JmaxRef = 156.62
RdRef = 2.54
sigma = 0.78
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

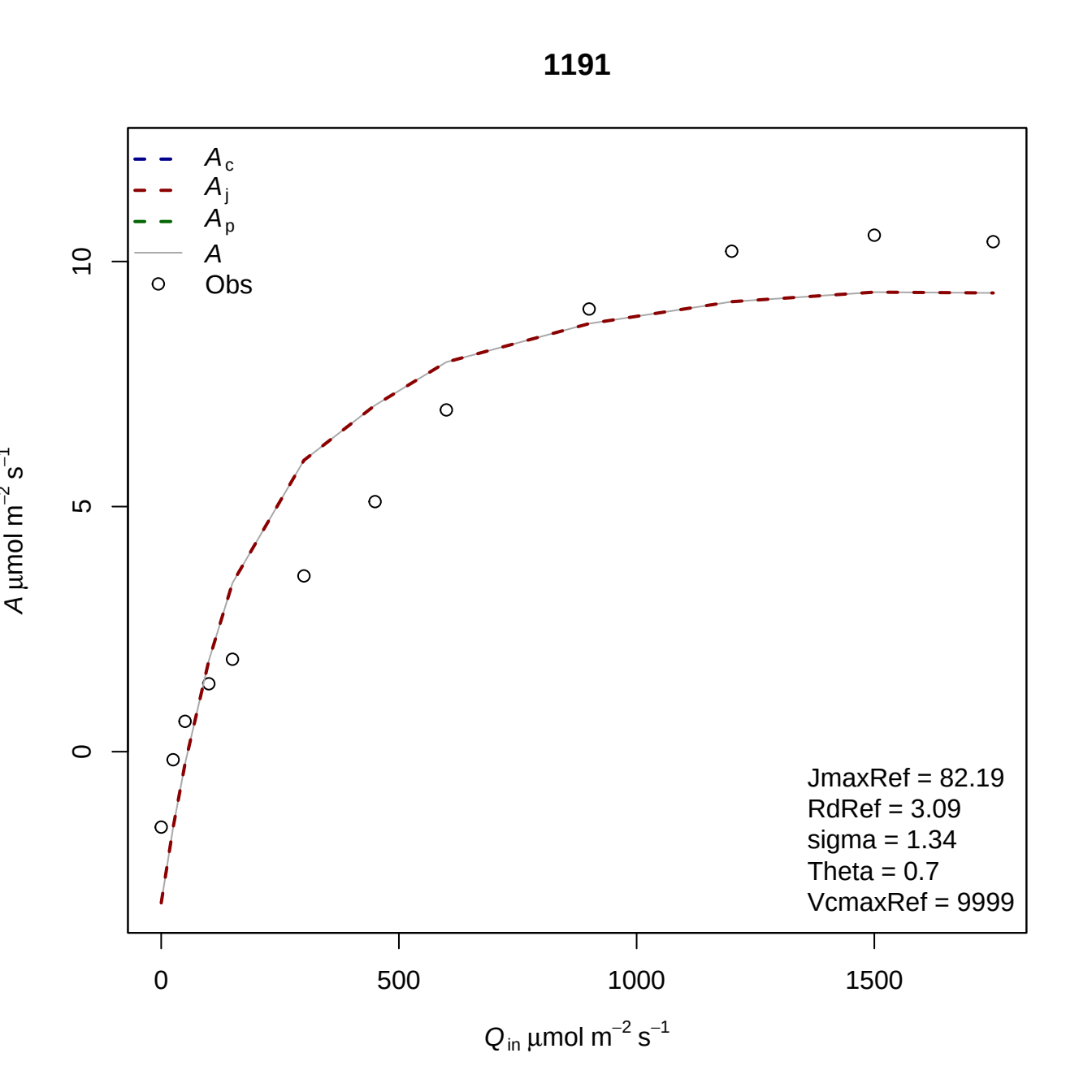

1186
A_c
A_j
A_p
A
Obs
JmaxRef = 175.57
RdRef = 1.26
sigma = 0.18
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

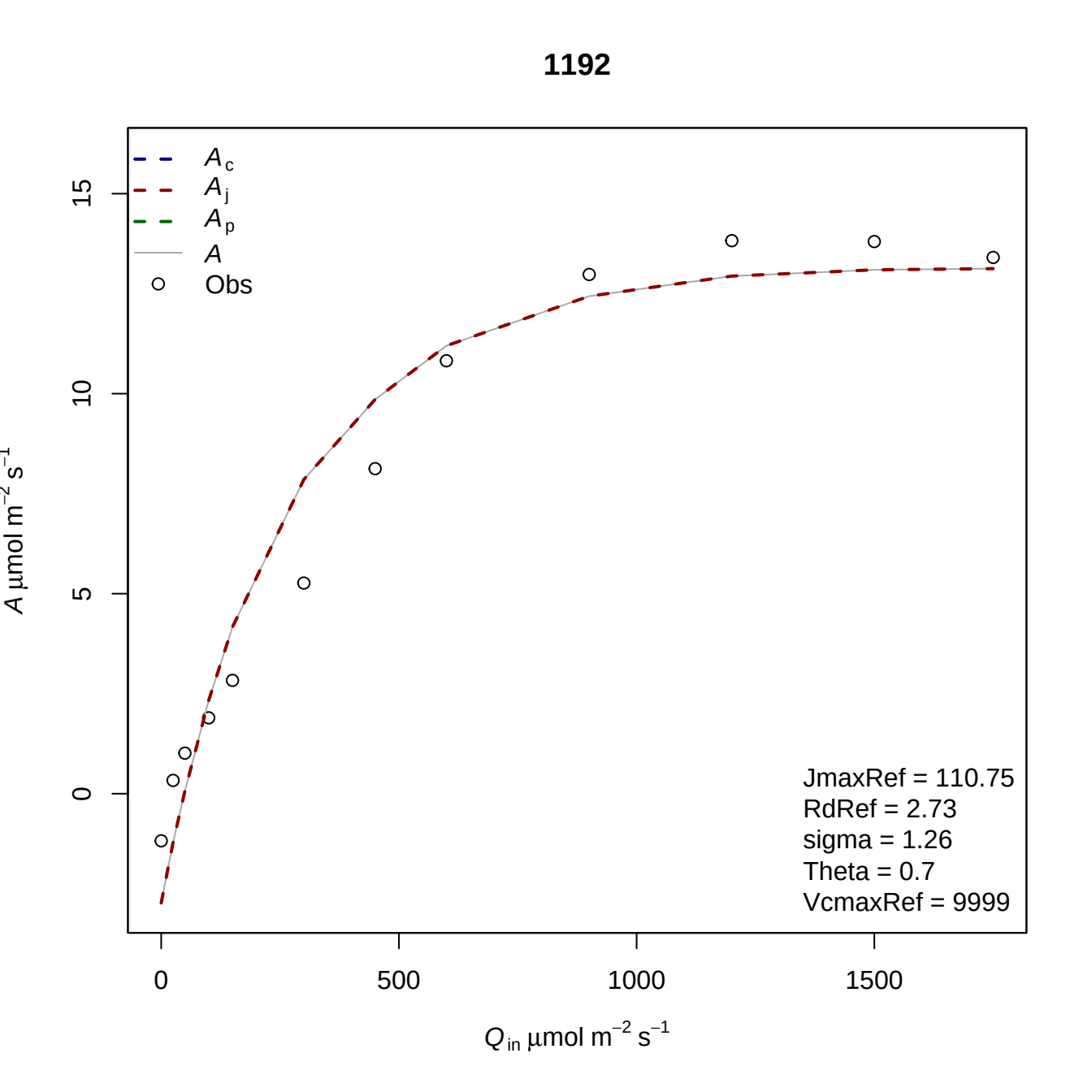

1187
A_c
A_j
A_p
A
Obs
JmaxRef = 123.72
RdRef = 2.67
sigma = 1.16
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

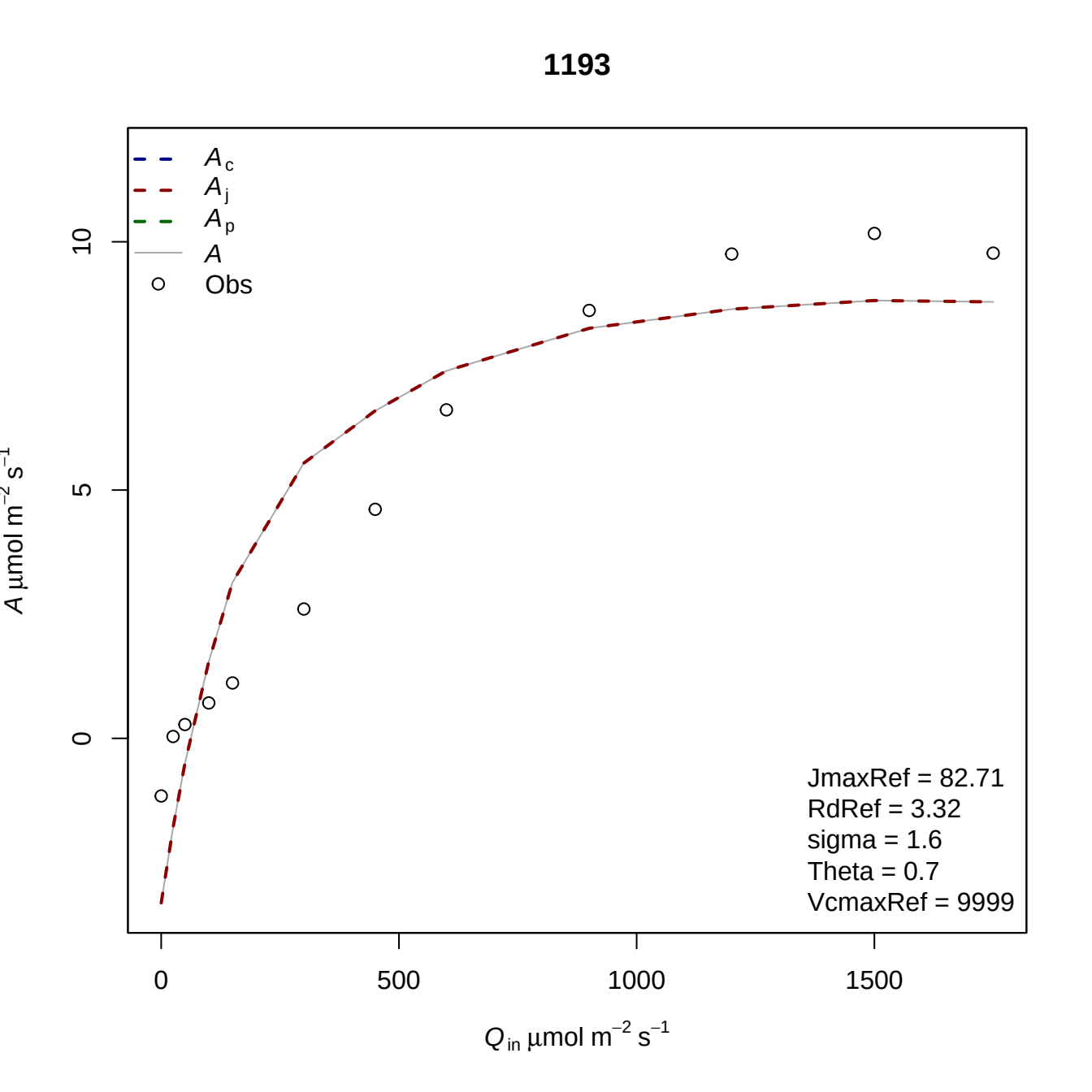

1188
A_c
A_j
A_p
A
Obs
JmaxRef = 115.22
RdRef = 2.31
sigma = 0.93
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

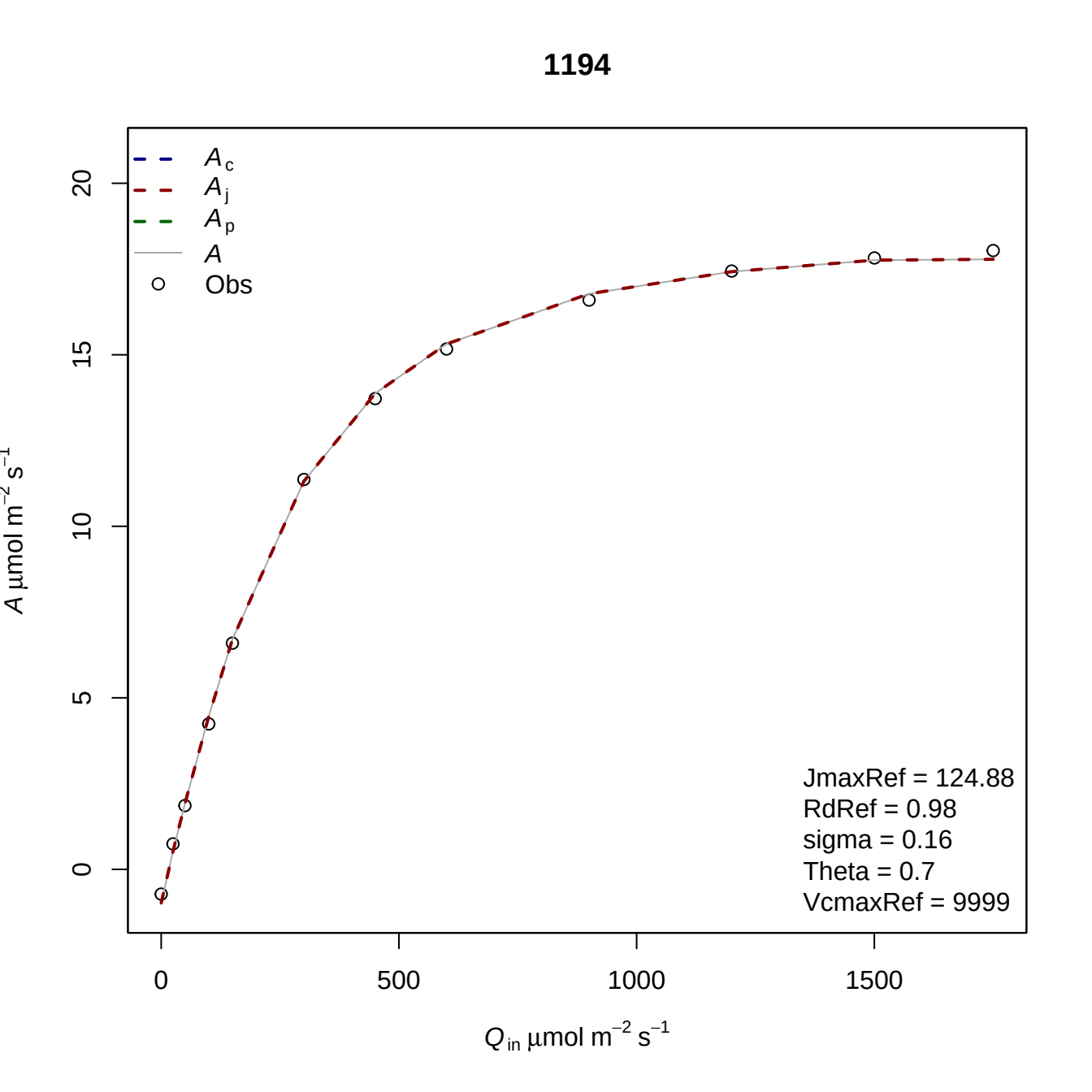

1189
A_c
A_j
A_p
A
Obs
JmaxRef = 72.3
RdRef = 3.22
sigma = 1.9
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

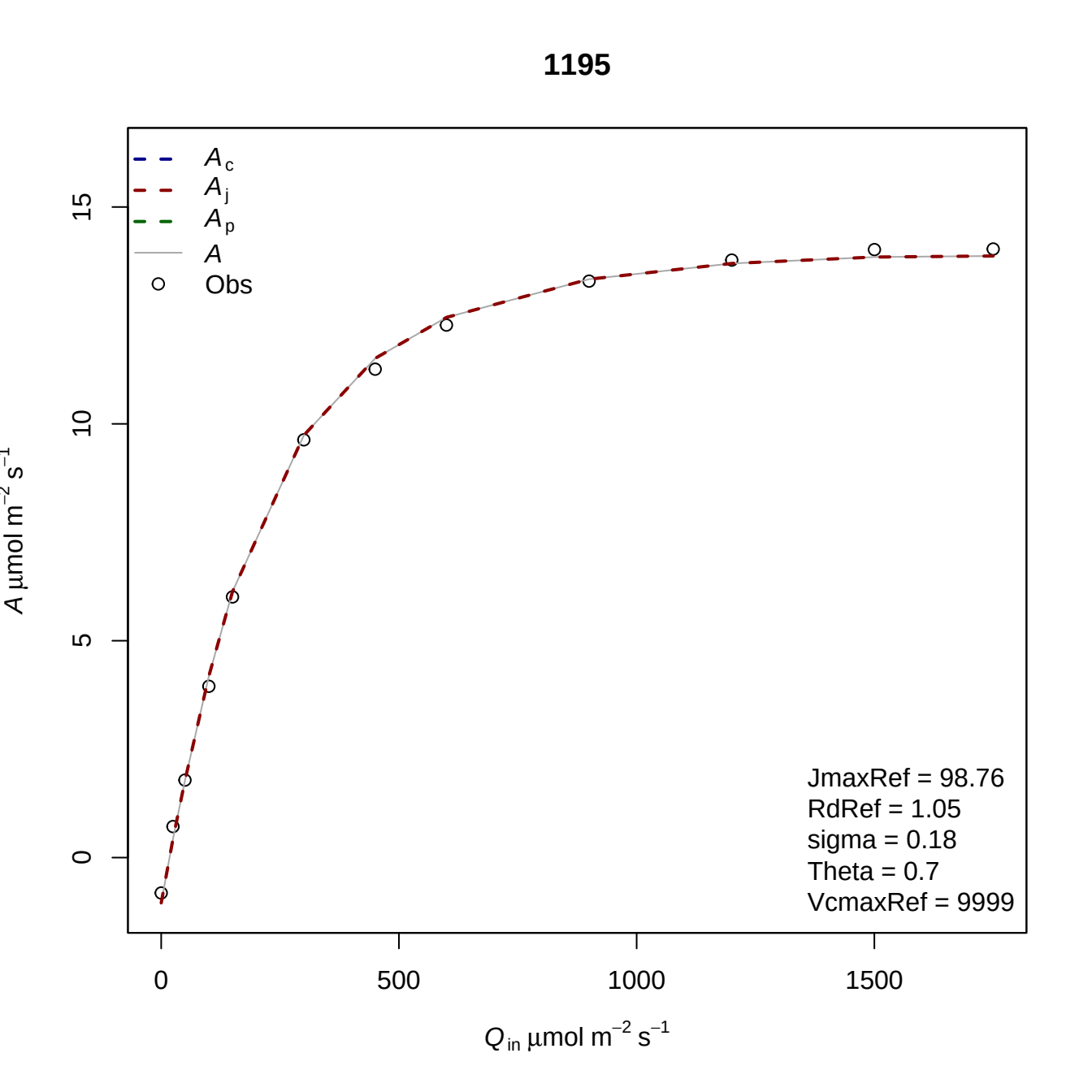

1190
A_c
A_j
A_p
A
Obs
JmaxRef = 81.3
RdRef = 2.77
sigma = 1.61
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

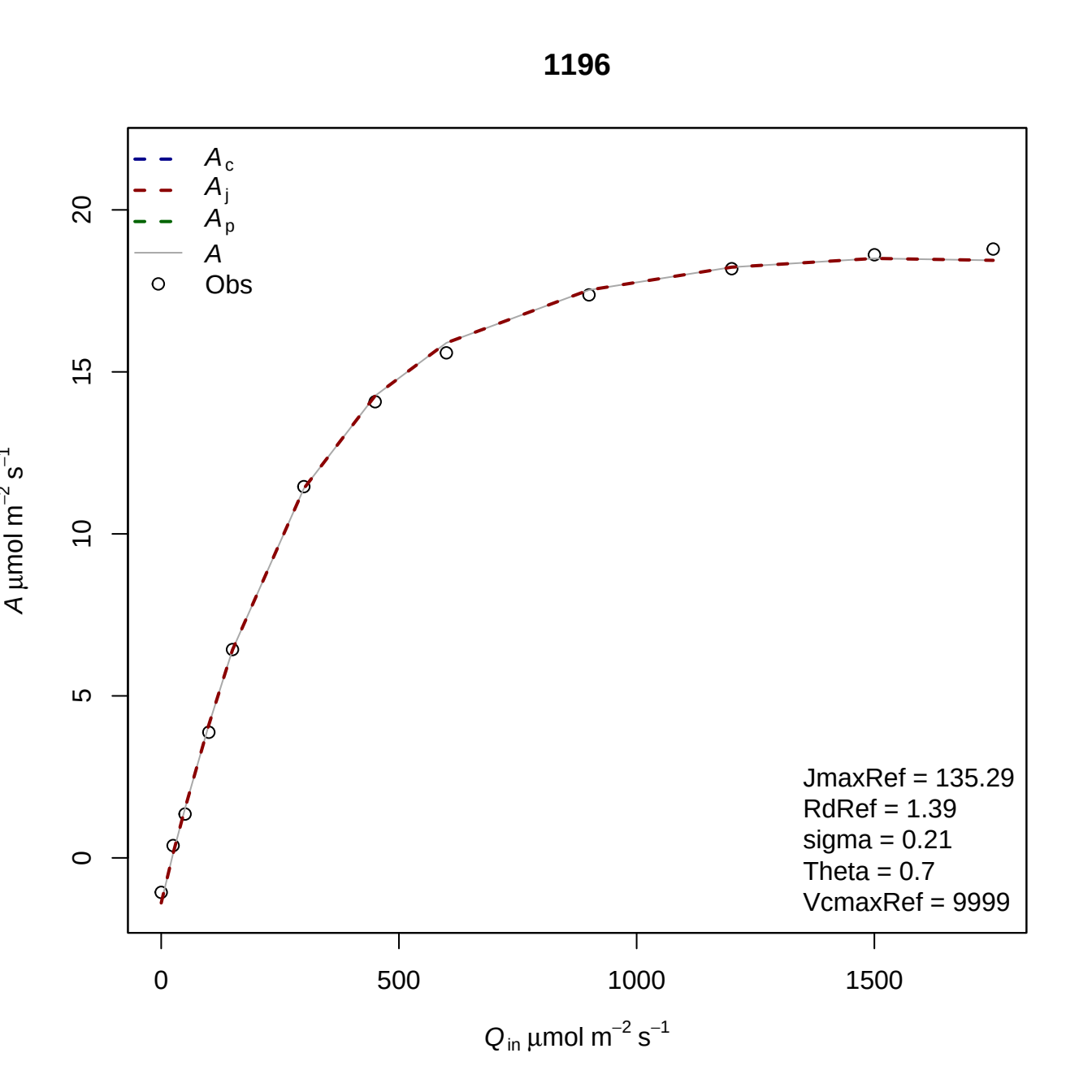

1191
A_c
A_j
A_p
A
Obs
JmaxRef = 82.19
RdRef = 3.09
sigma = 1.34
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

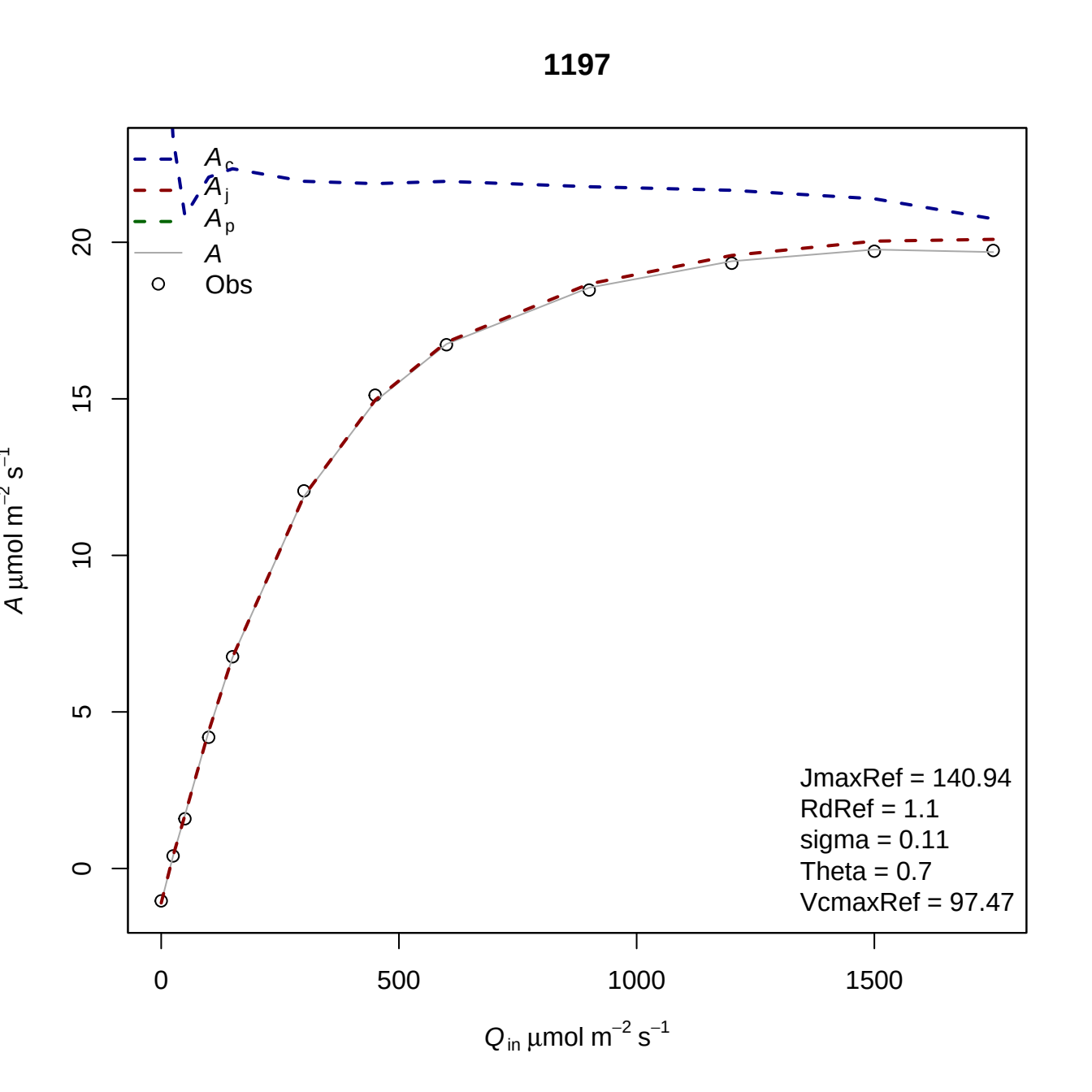

1192
A_c
A_j
A_p
A
Obs
JmaxRef = 110.75
RdRef = 2.73
sigma = 1.26
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

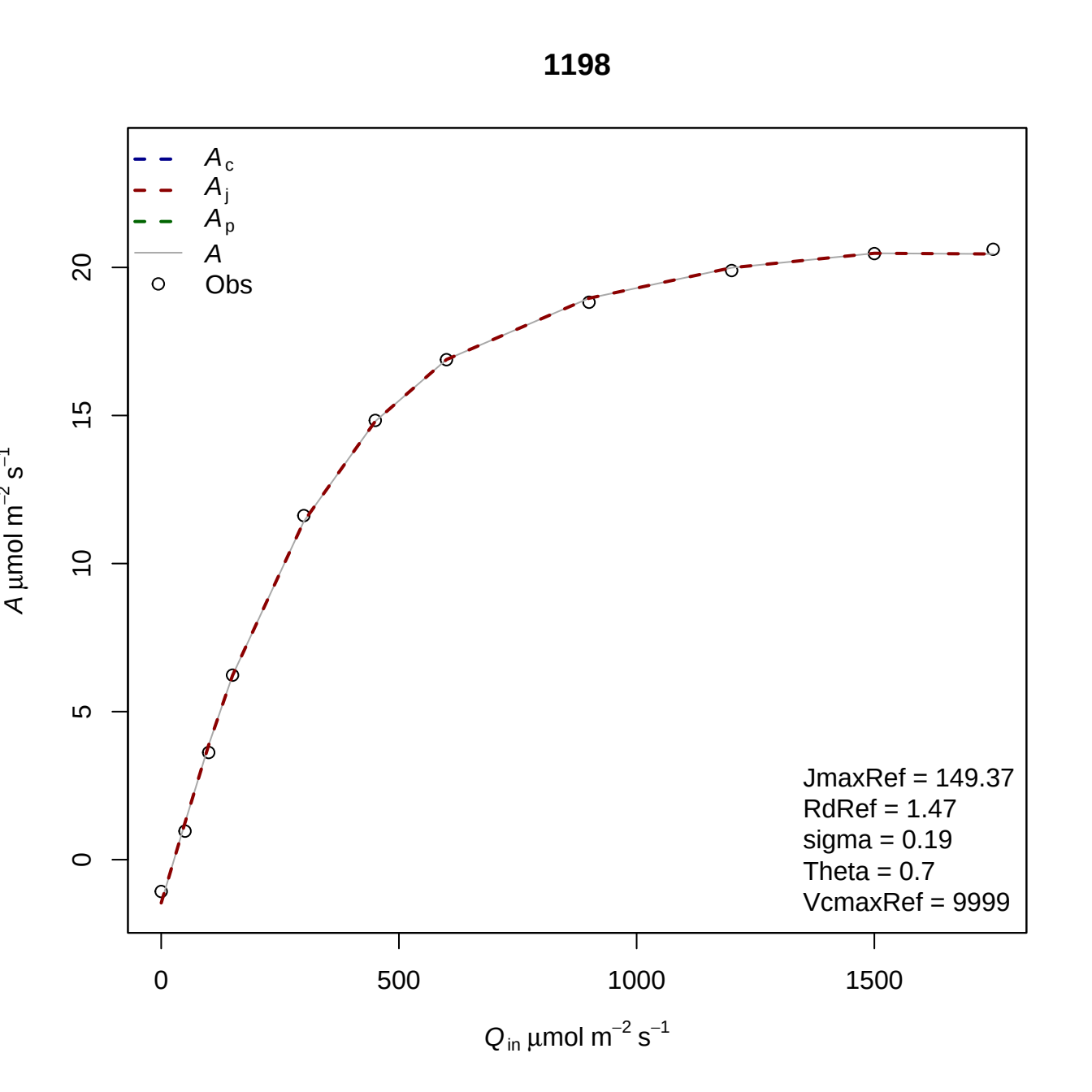

1193
A_c
A_j
A_p
A
Obs
JmaxRef = 82.71
RdRef = 3.32
sigma = 1.6
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

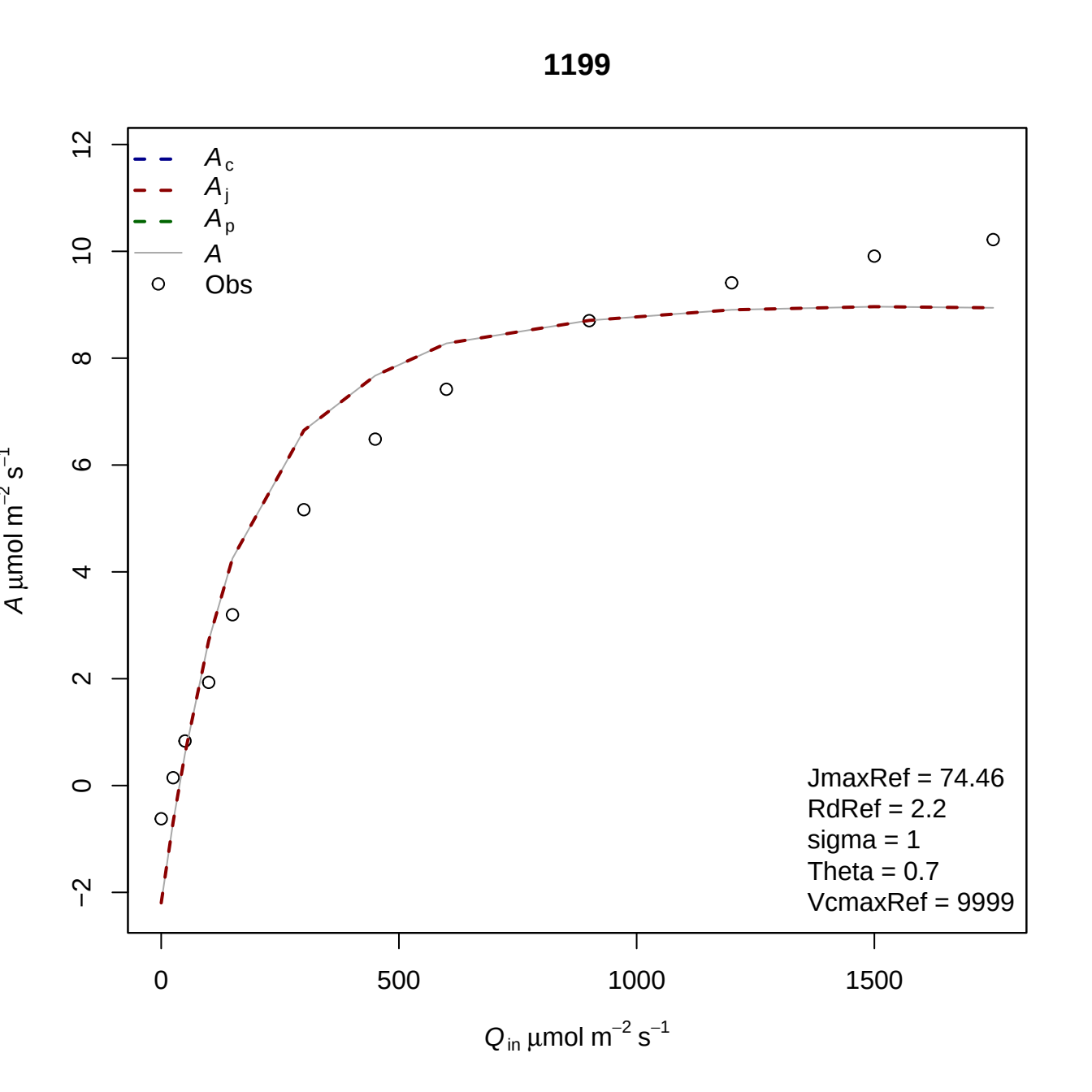

1194
A_c
A_j
A_p
A
Obs
JmaxRef = 124.88
RdRef = 0.98
sigma = 0.16
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

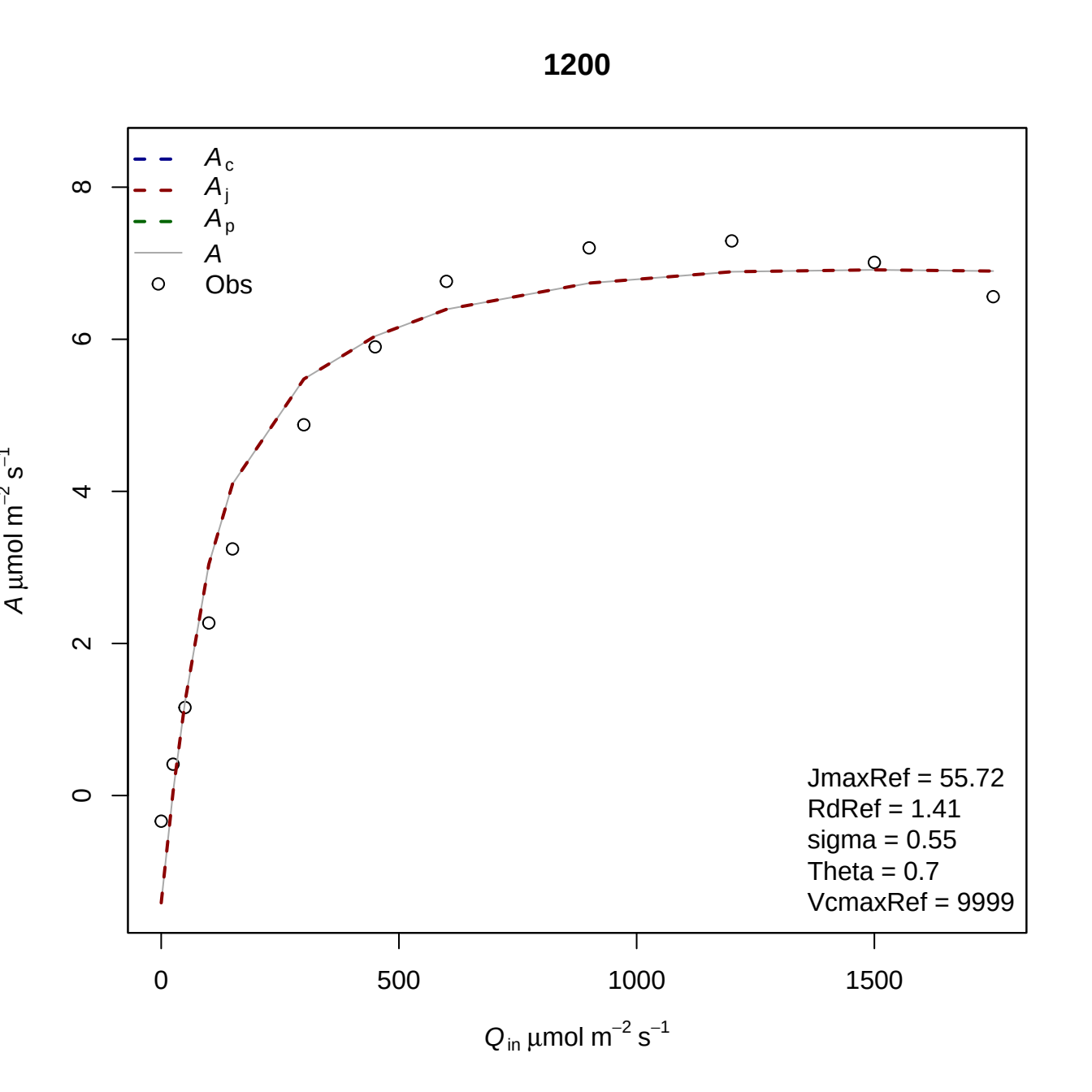

1195
A_c
A_j
A_p
A
Obs
JmaxRef = 98.76
RdRef = 1.05
sigma = 0.18
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

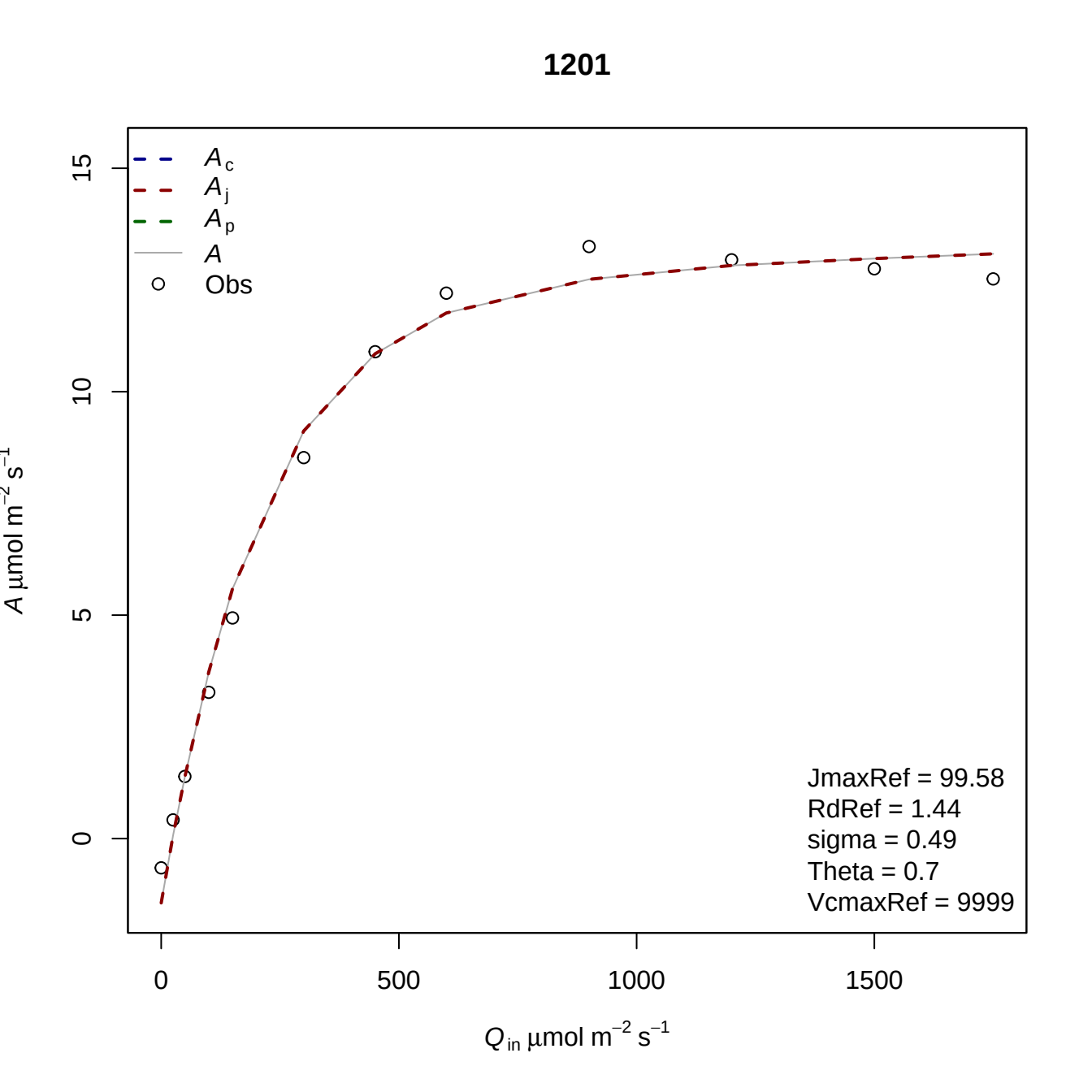

1196
A_c
A_j
A_p
A
Obs
JmaxRef = 135.29
RdRef = 1.39
sigma = 0.21
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

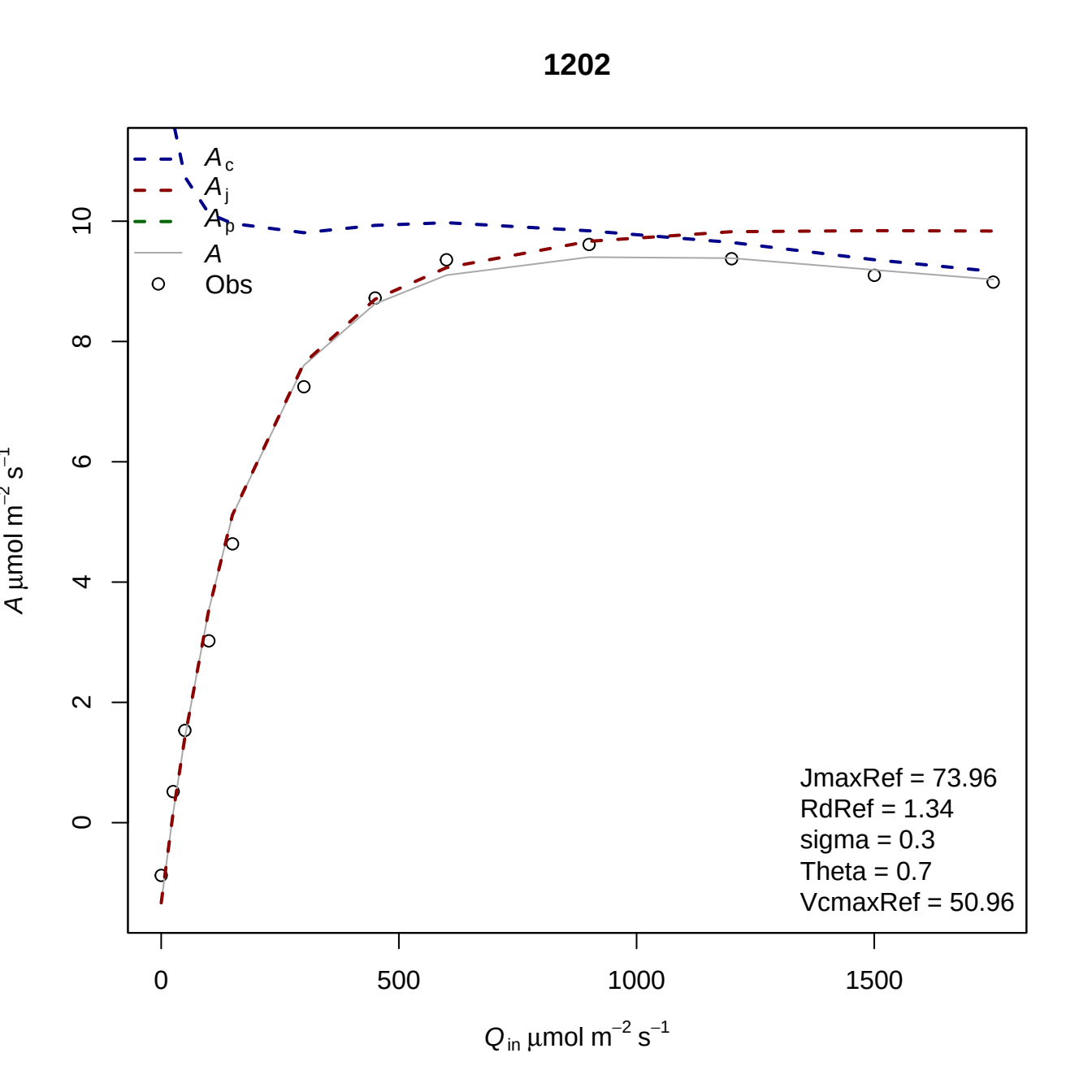

1197
A_c
A_j
A_p
A
Obs
JmaxRef = 140.94
RdRef = 1.1
sigma = 0.11
Theta = 0.7
VcmaxRef = 97.47
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

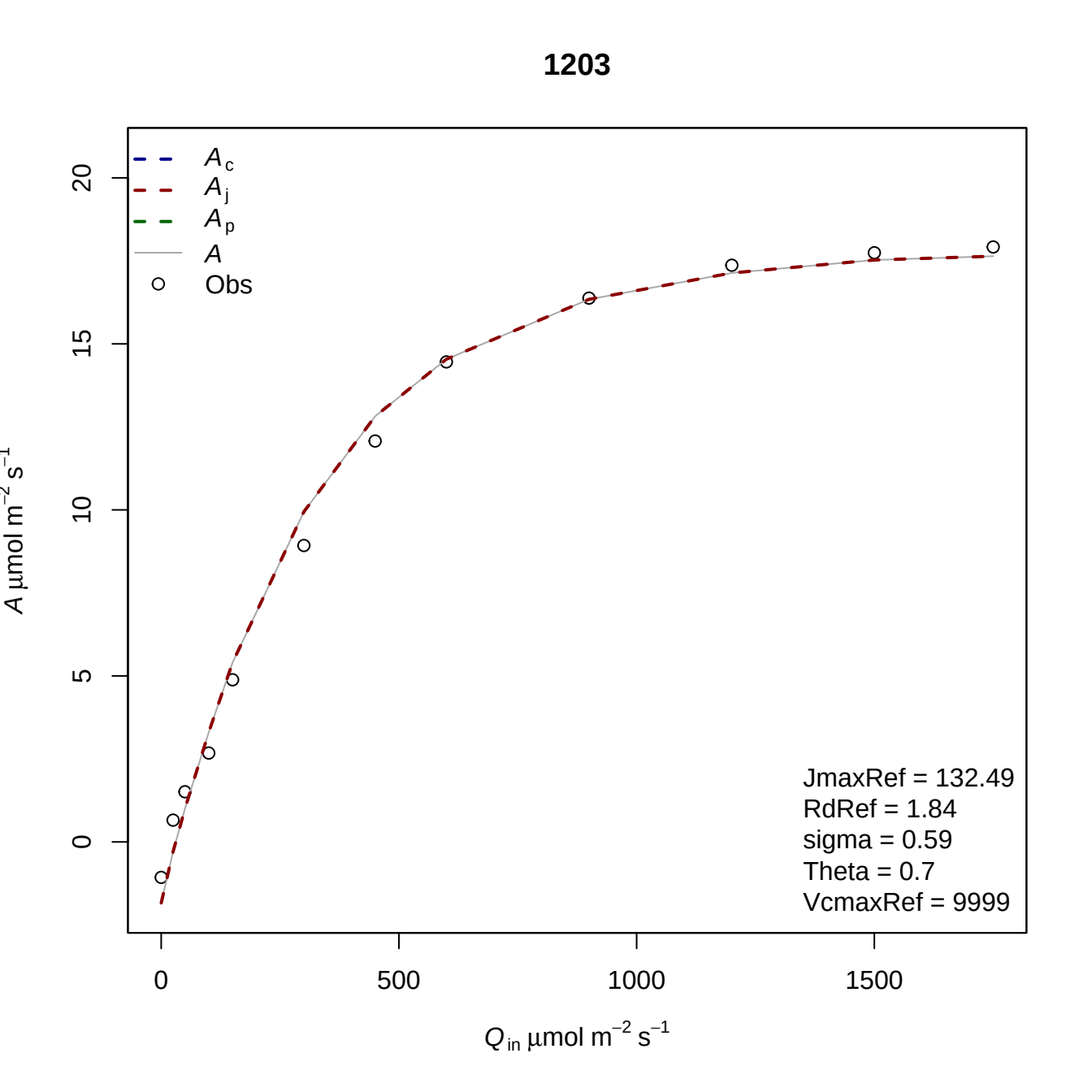

1198
A_c
A_j
A_p
A
Obs
JmaxRef = 149.37
RdRef = 1.47
sigma = 0.19
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

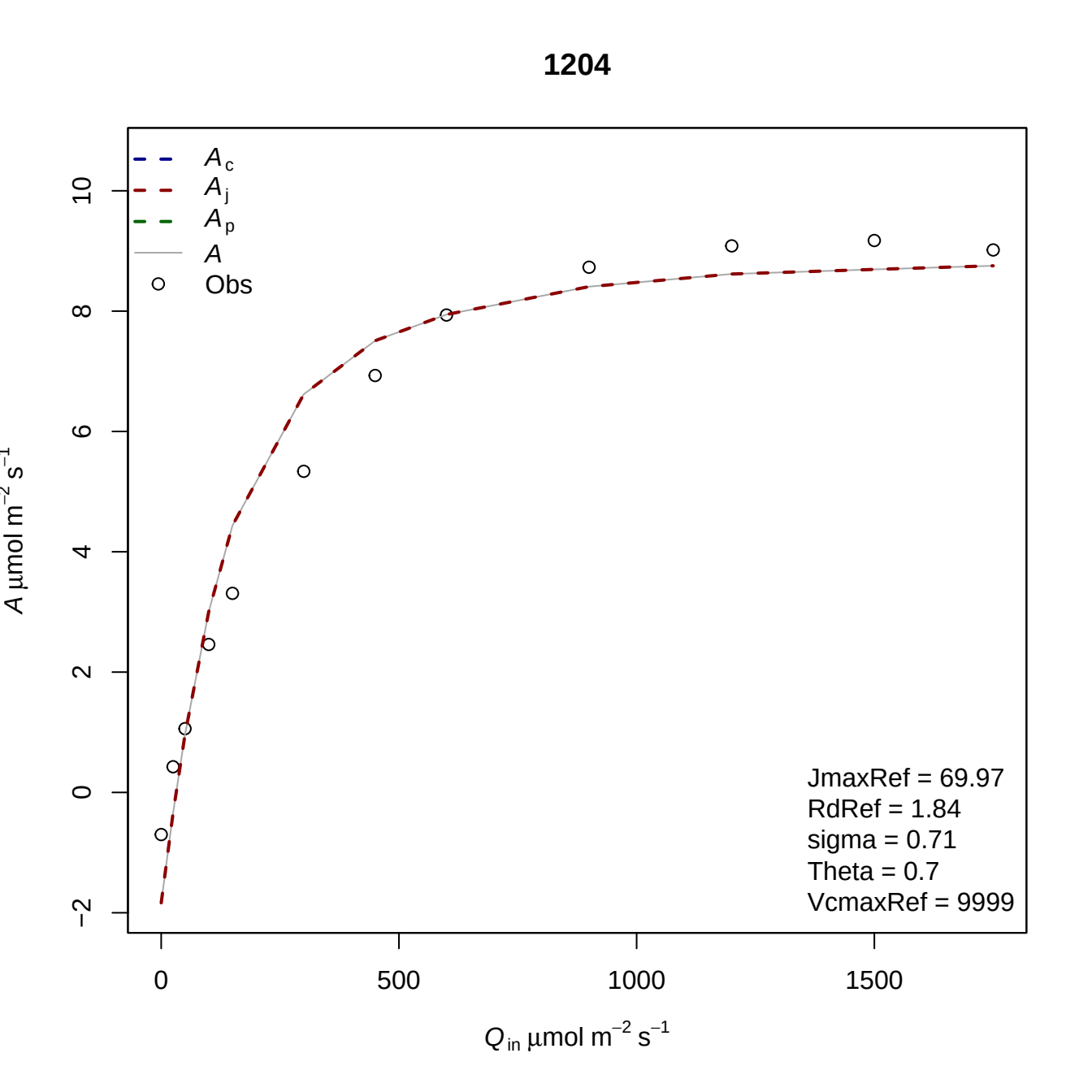

1199
A_c
A_j
A_p
A
Obs
JmaxRef = 74.46
RdRef = 2.2
sigma = 1
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

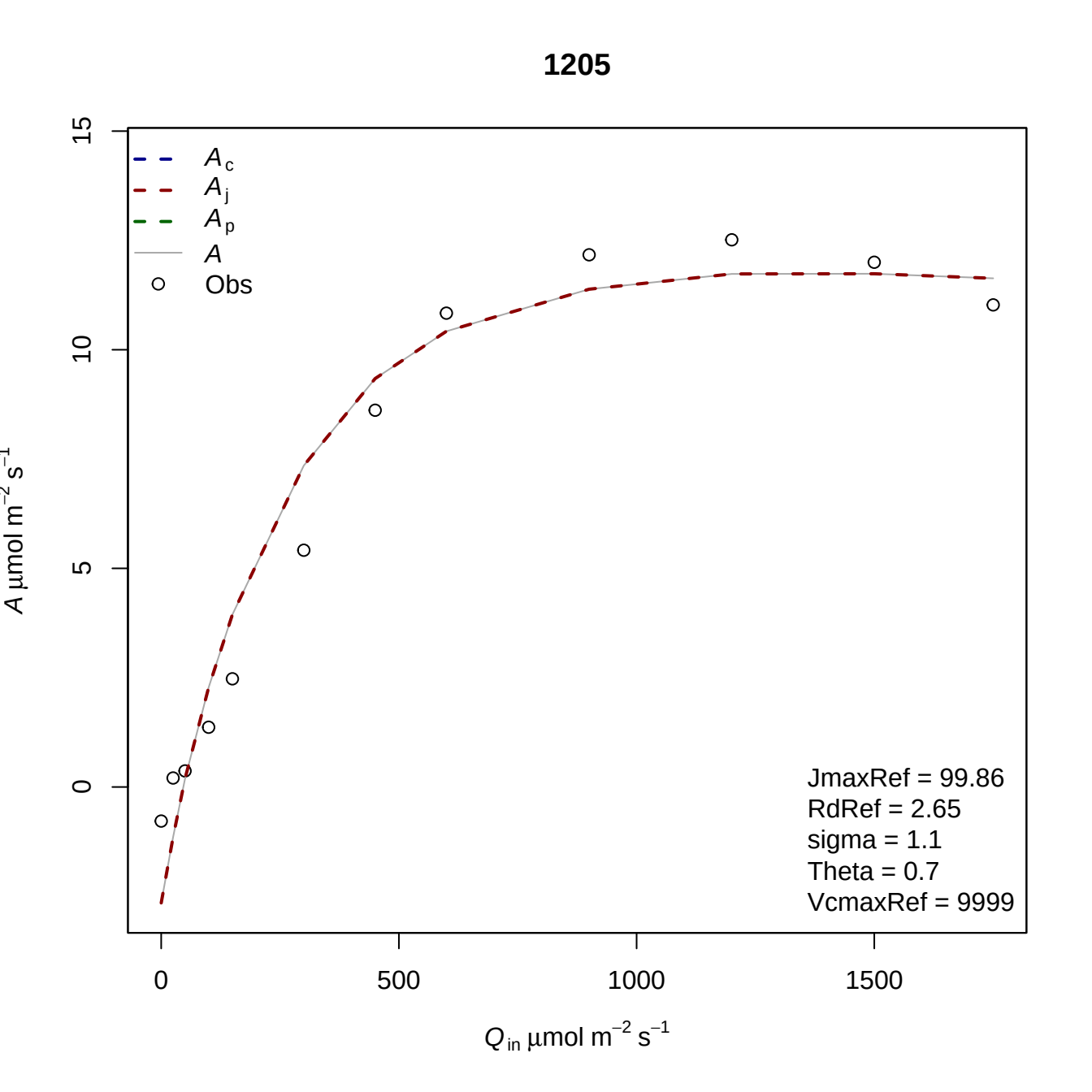

1200
A_c
A_j
A_p
A
Obs
JmaxRef = 55.72
RdRef = 1.41
sigma = 0.55
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

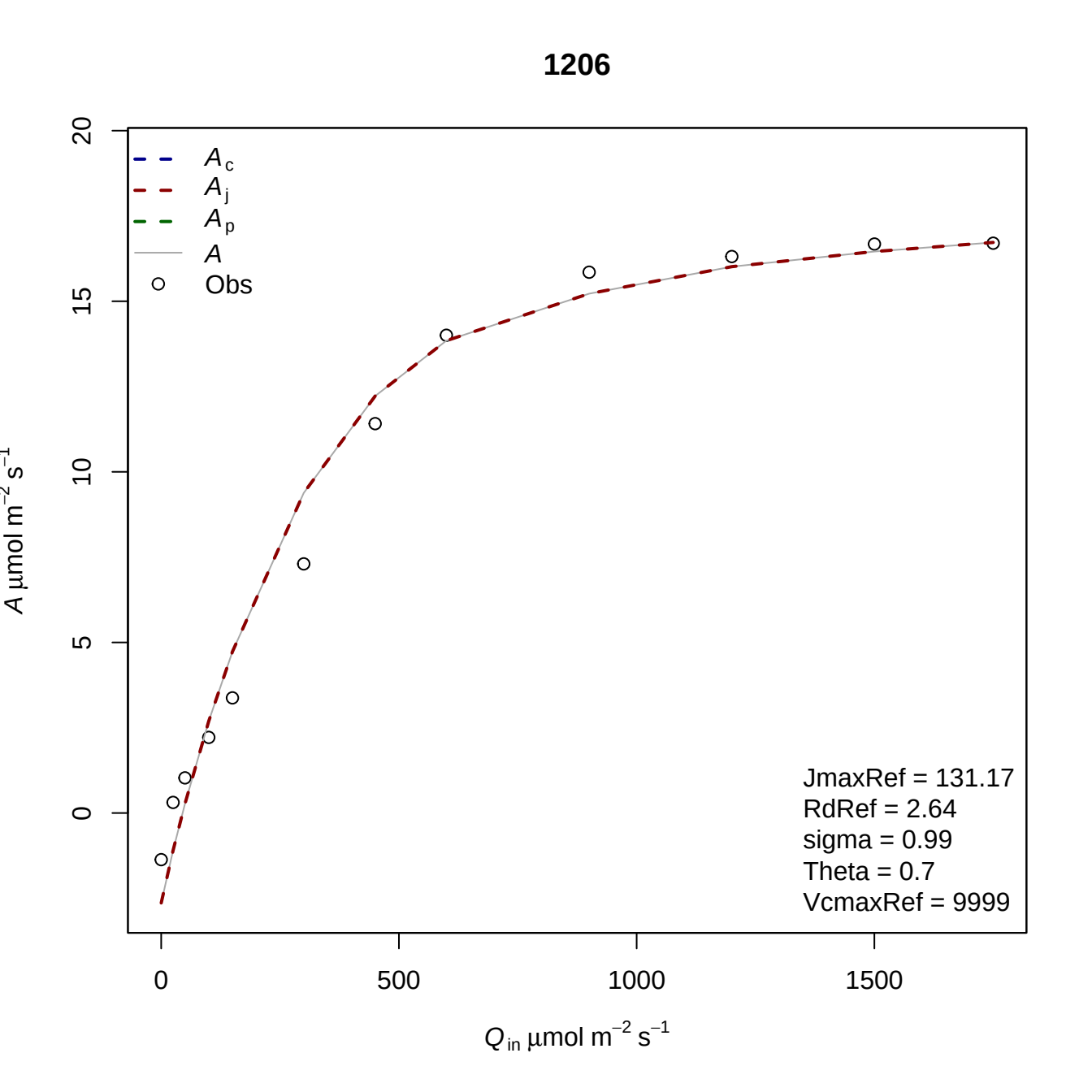

1201
A_c
A_j
A_p
A
Obs
JmaxRef = 99.58
RdRef = 1.44
sigma = 0.49
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

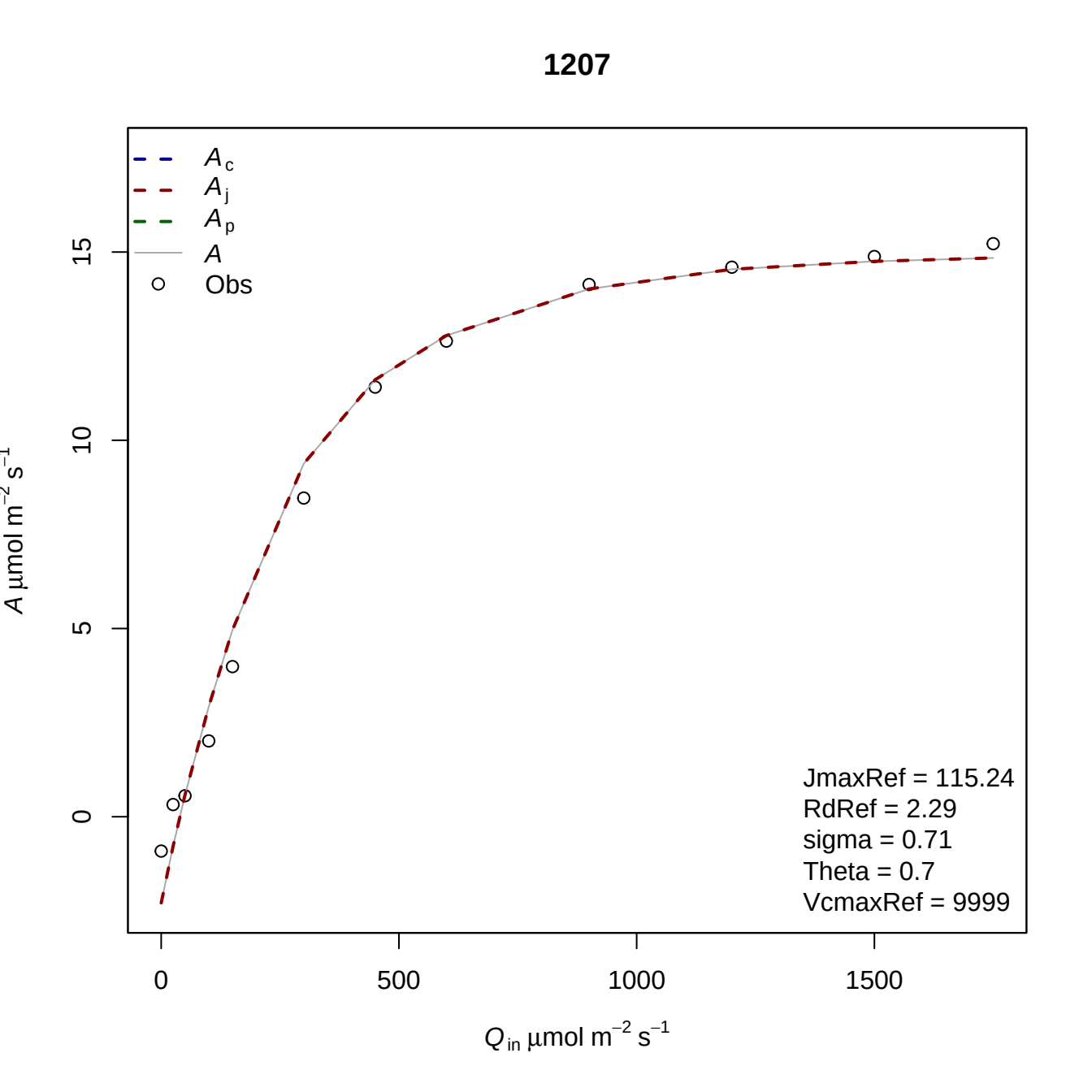

1202
A_c
A_j
A_p
A
Obs
JmaxRef = 73.96
RdRef = 1.34
sigma = 0.3
Theta = 0.7
VcmaxRef = 50.96
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

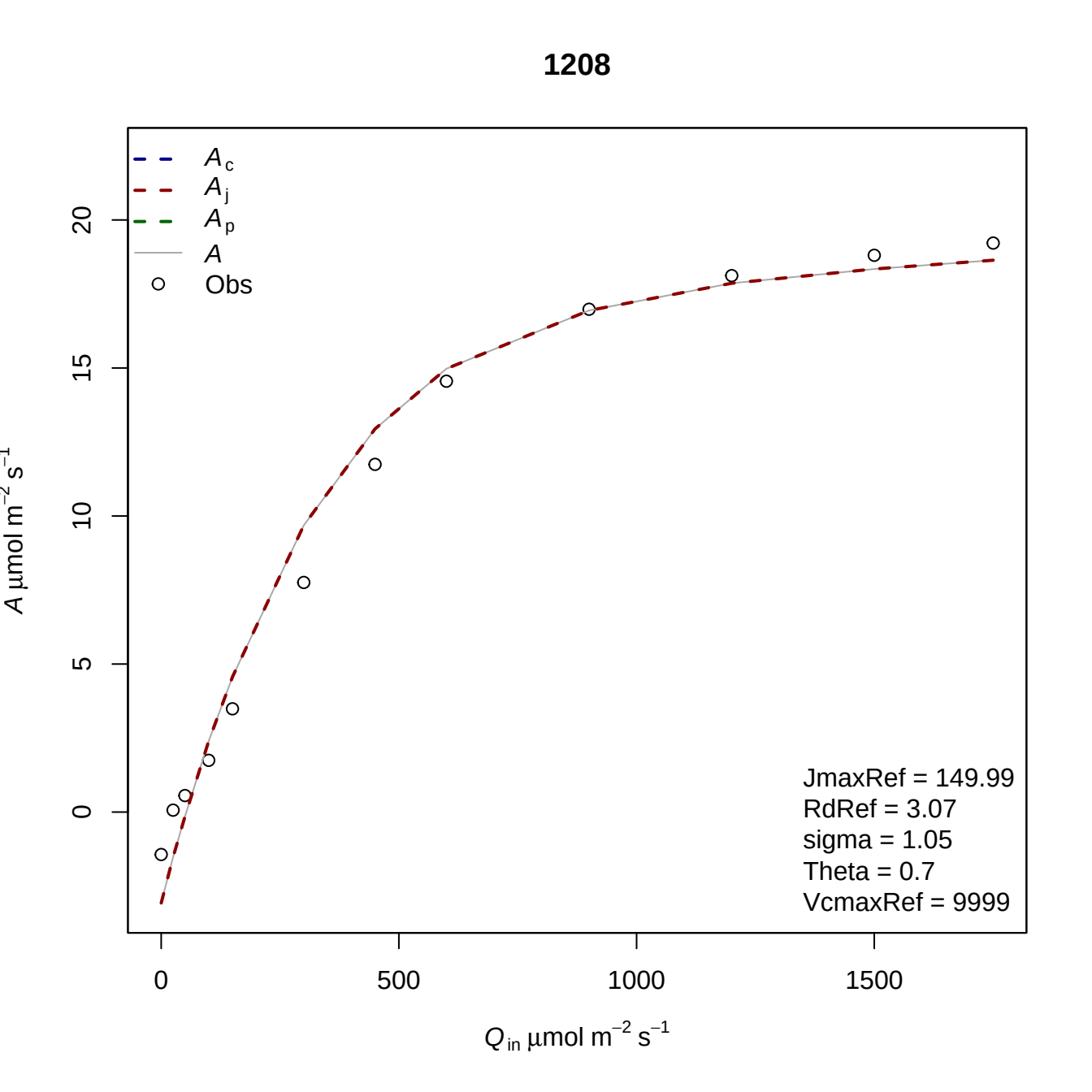

1203
A_c
A_j
A_p
A
Obs
JmaxRef = 132.49
RdRef = 1.84
sigma = 0.59
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

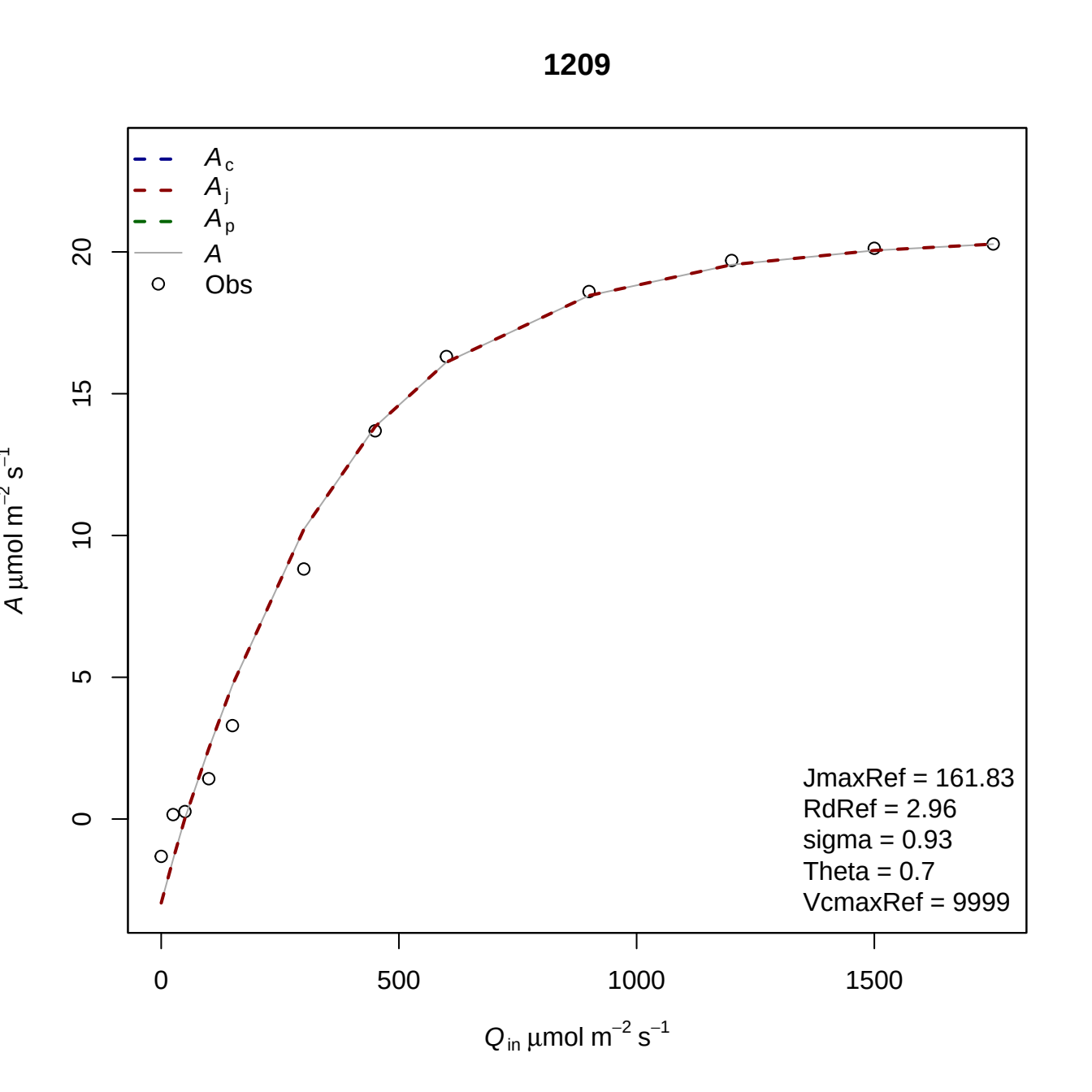

1204
A_c
A_j
A_p
A
Obs
JmaxRef = 69.97
RdRef = 1.84
sigma = 0.71
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

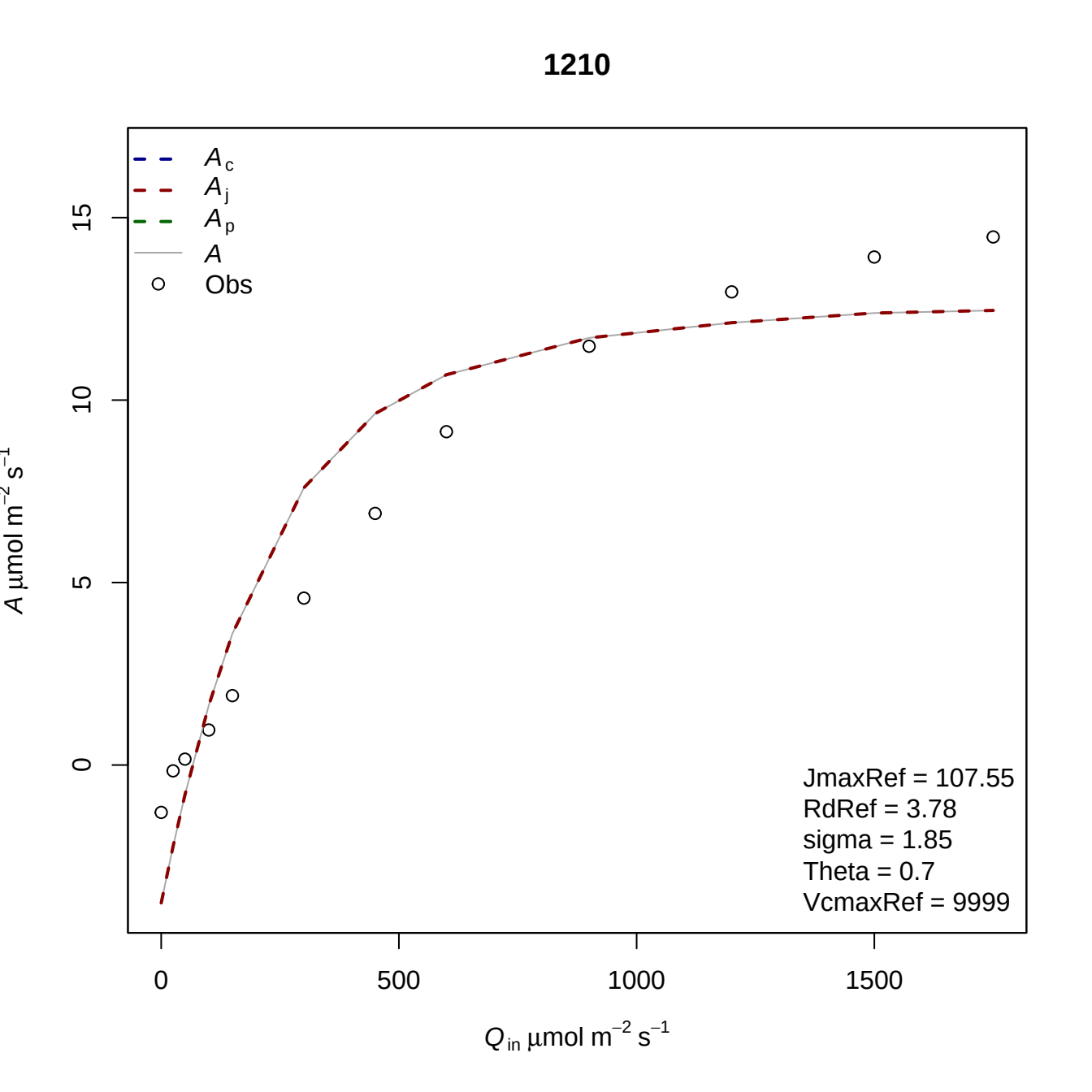

1205
A_c
A_j
A_p
A
Obs
JmaxRef = 99.86
RdRef = 2.65
sigma = 1.1
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

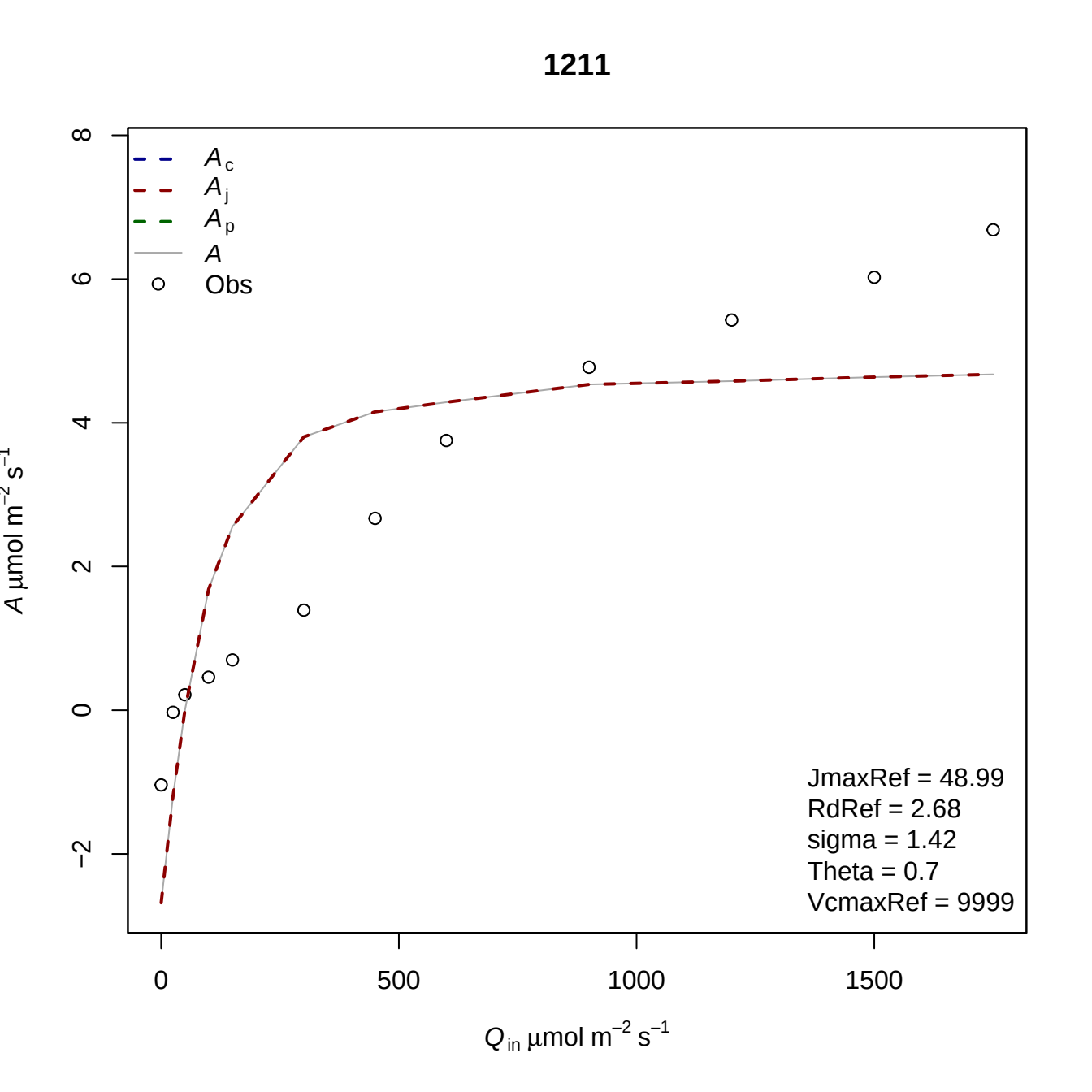

1206
A_c
A_j
A_p
A
Obs
JmaxRef = 131.17
RdRef = 2.64
sigma = 0.99
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

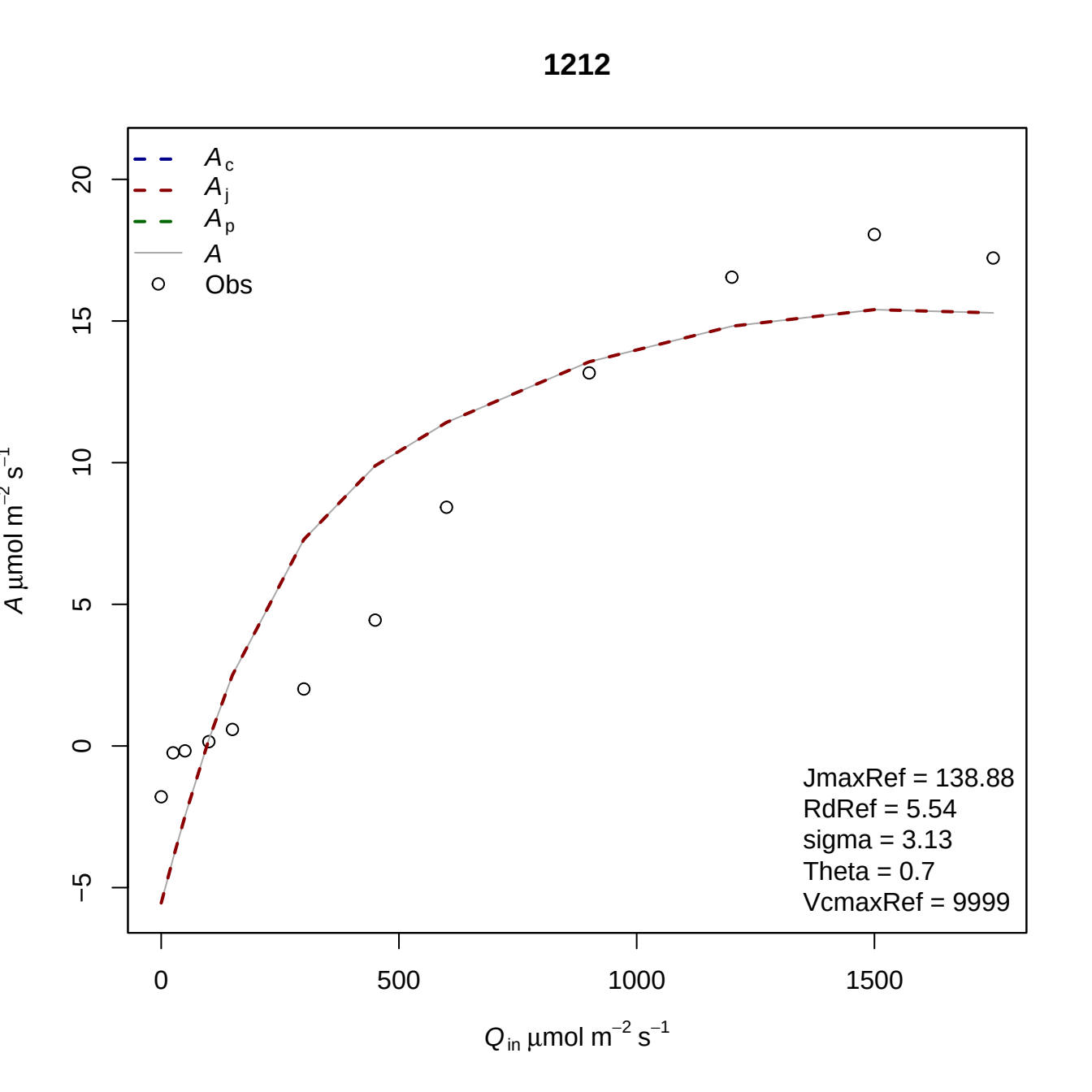

1207
A_c
A_j
A_p
A
Obs
JmaxRef = 115.24
RdRef = 2.29
sigma = 0.71
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

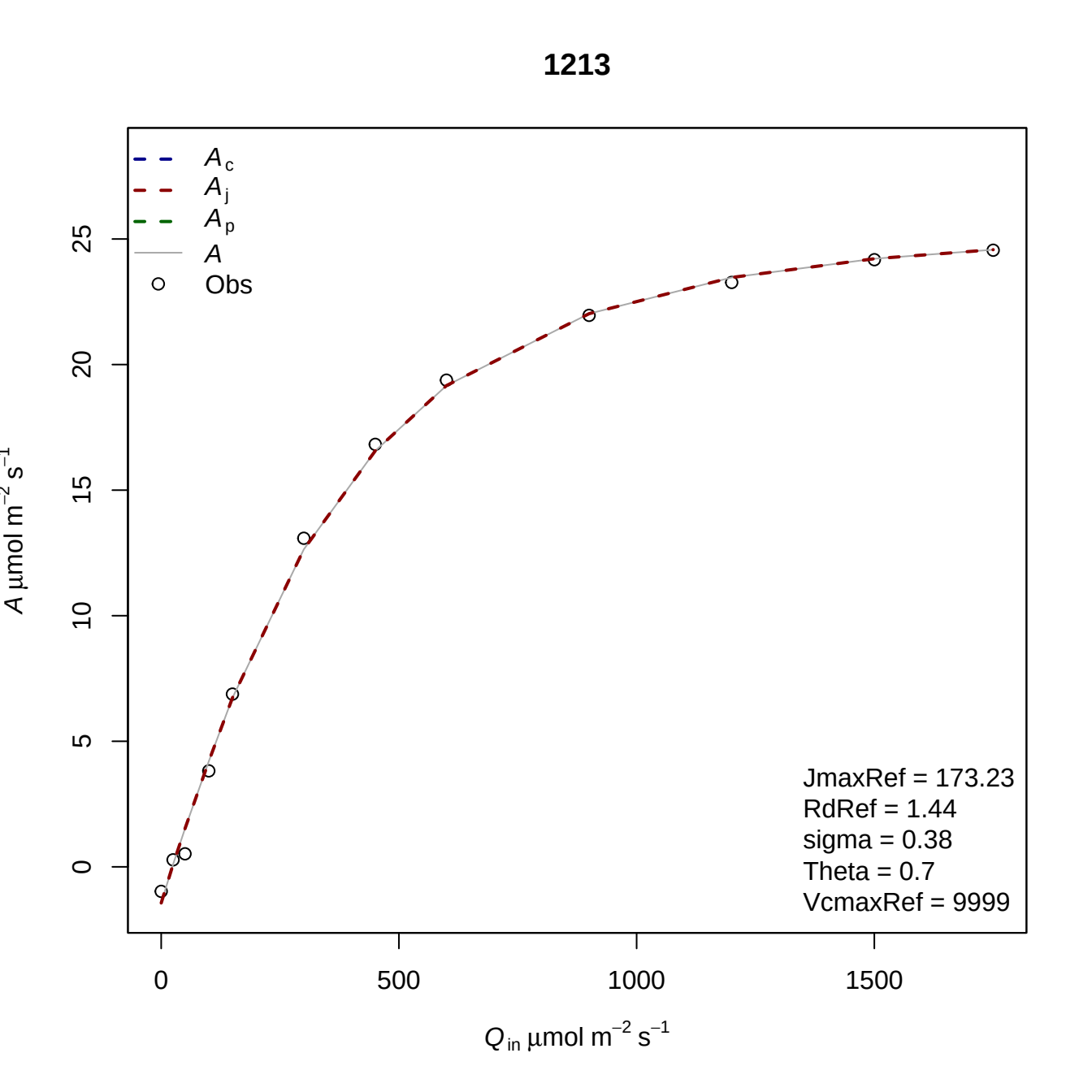

1208
A_c
A_j
A_p
A
Obs
JmaxRef = 149.99
RdRef = 3.07
sigma = 1.05
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

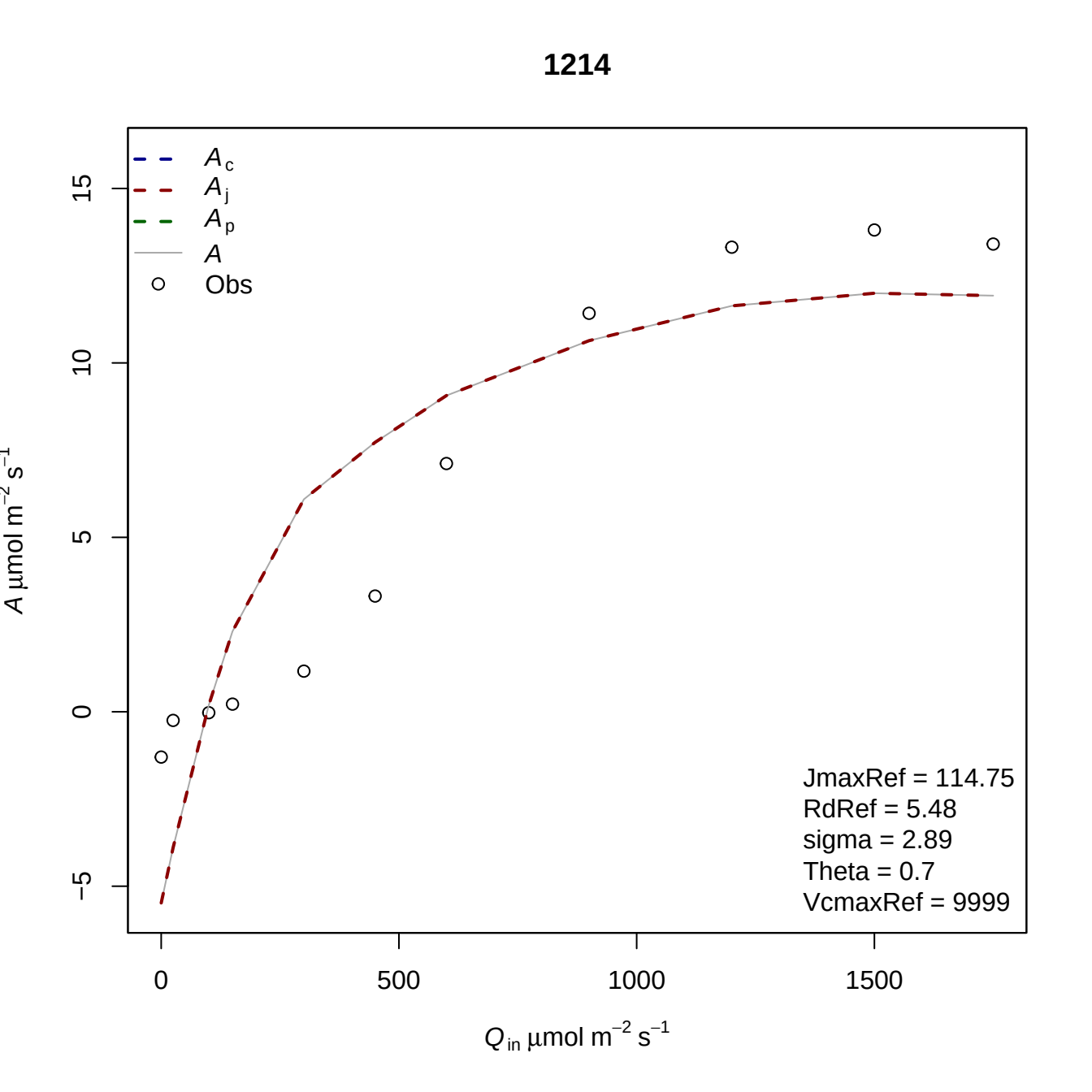

1209
A_c
A_j
A_p
A
Obs
JmaxRef = 161.83
RdRef = 2.96
sigma = 0.93
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

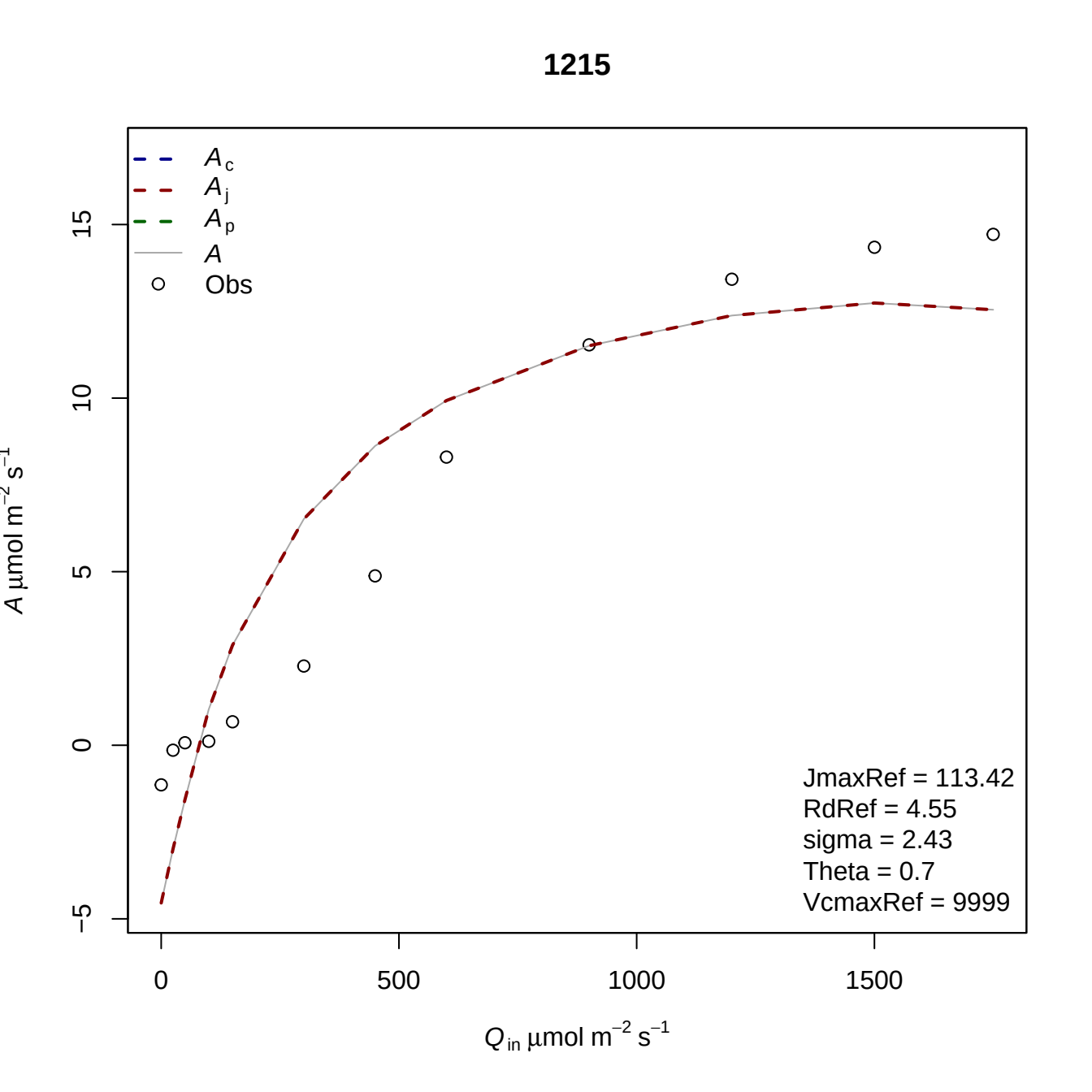

1210
A_c
A_j
A_p
A
Obs
JmaxRef = 107.55
RdRef = 3.78
sigma = 1.85
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

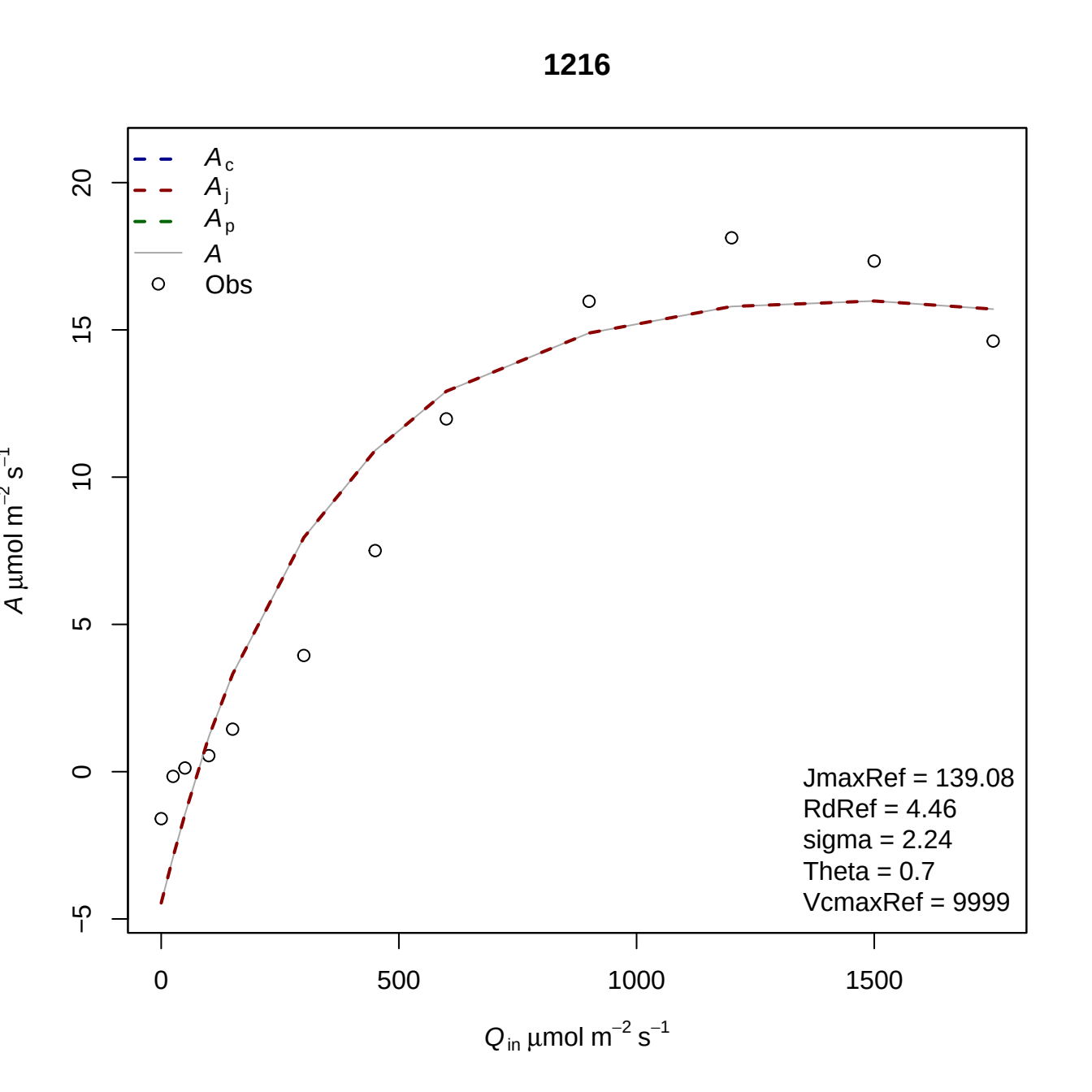

1211
A_c
A_j
A_p
A
Obs
JmaxRef = 48.99
RdRef = 2.68
sigma = 1.42
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500

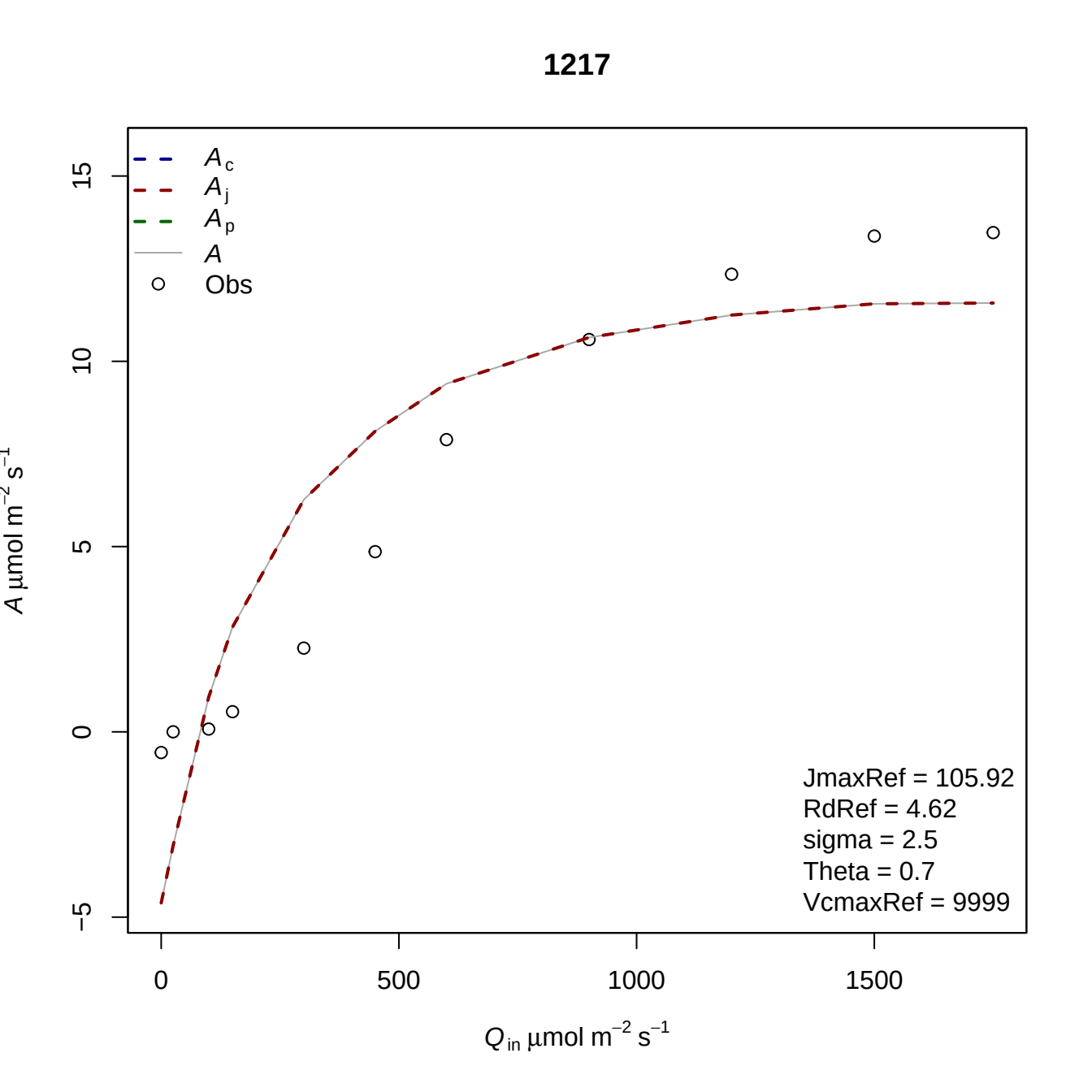

1212
A_c
A_j
A_p
A
Obs
JmaxRef = 138.88
RdRef = 5.54
sigma = 3.13
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
20
15
10
5
0
−5
0
500
1000
1500

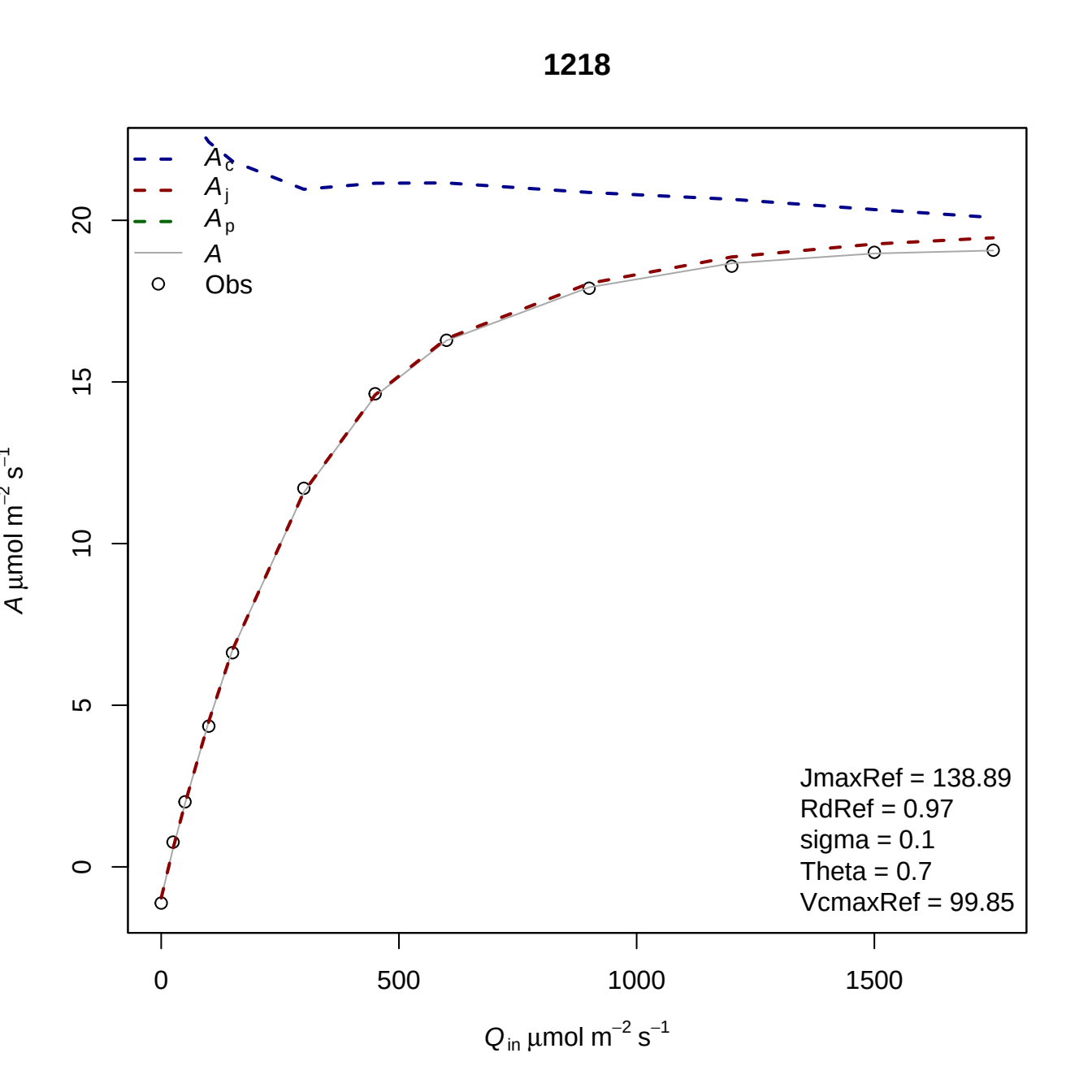

1213
A_c
A_j
A_p
A
Obs
JmaxRef = 173.23
RdRef = 1.44
sigma = 0.38
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
25
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

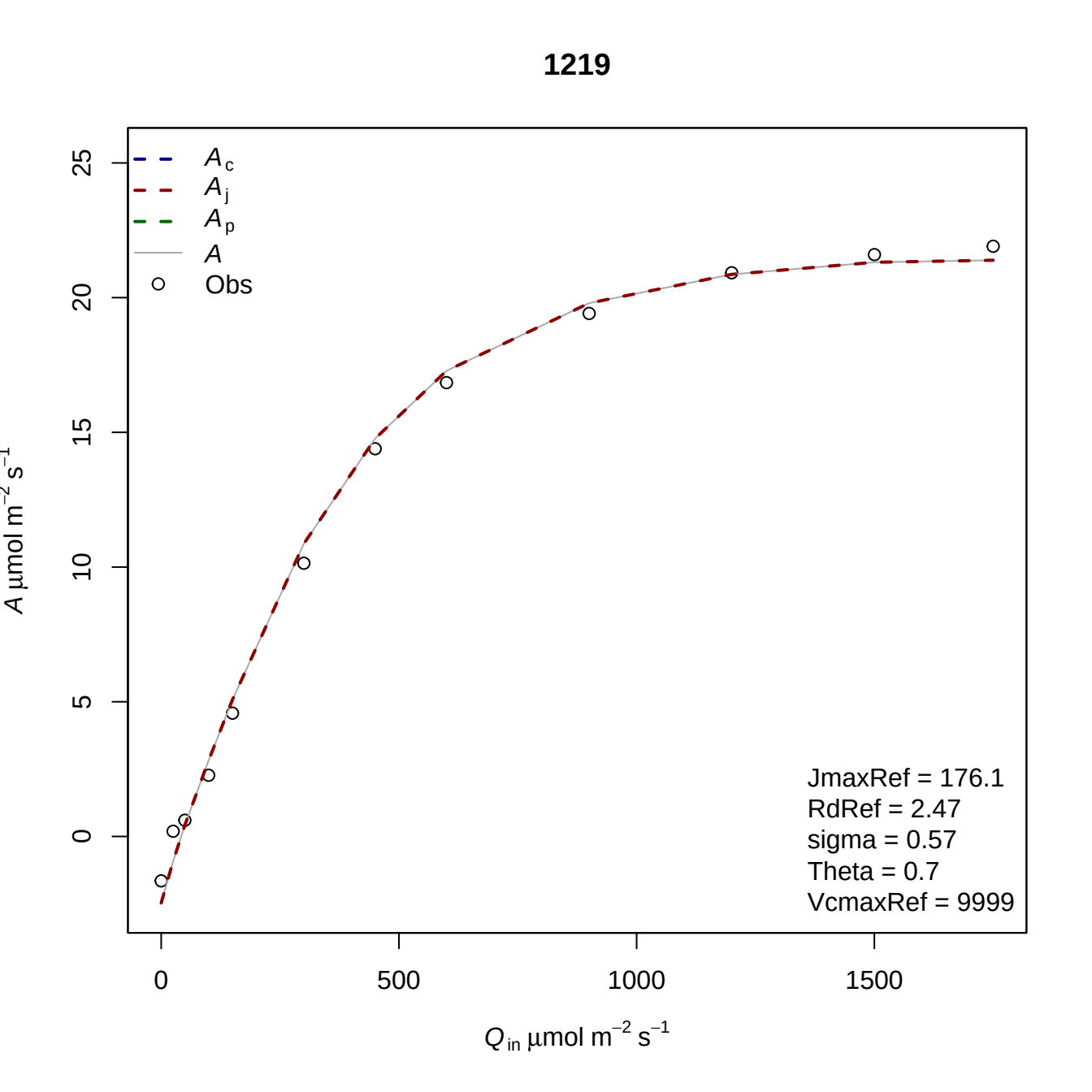

1214
A_c
A_j
A_p
A
Obs
JmaxRef = 114.75
RdRef = 5.48
sigma = 2.89
Theta = 0.7
VcmaxRef = 9999
15
10
5
0
−5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

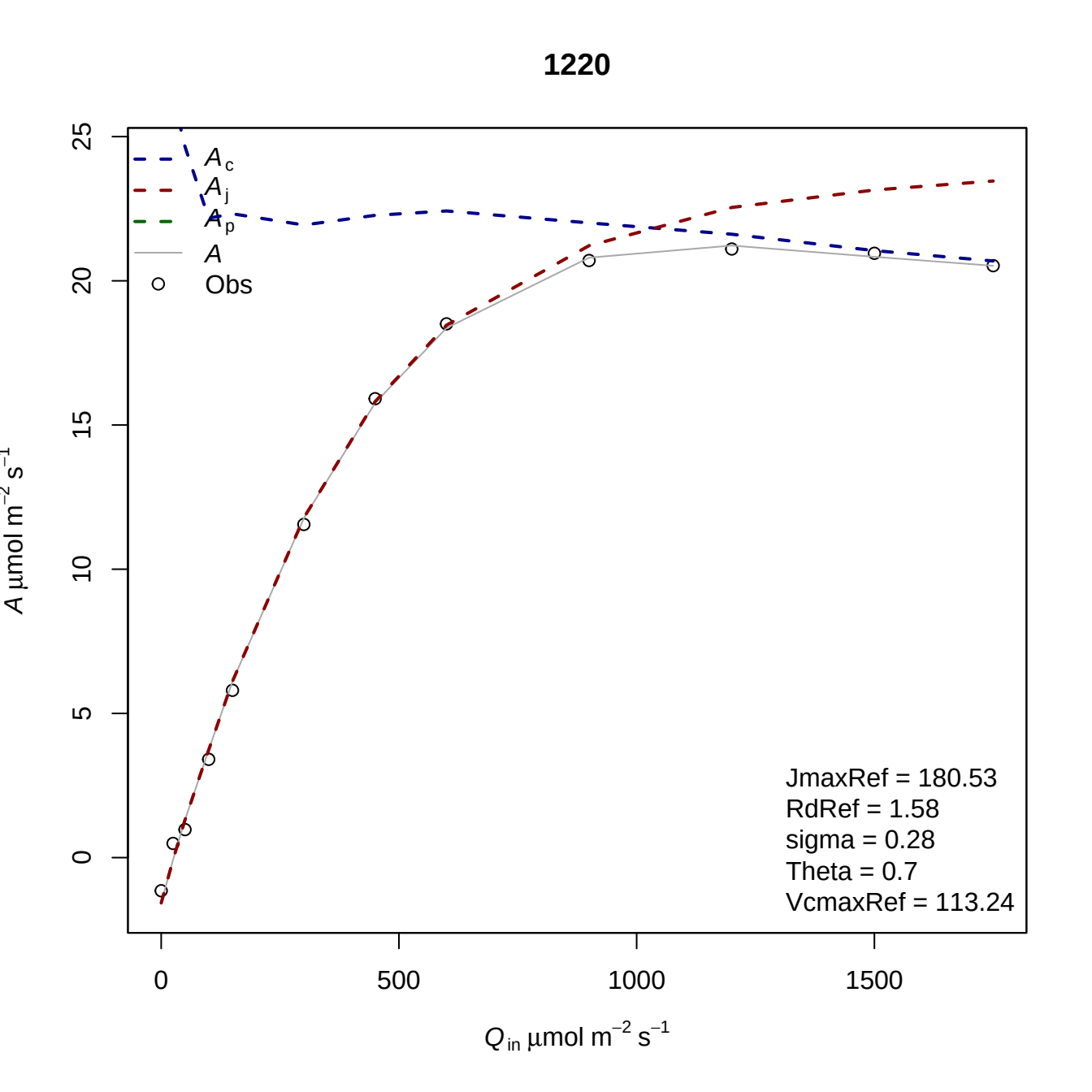

1215
A_c
A_j
A_p
A
Obs
JmaxRef = 113.42
RdRef = 4.55
sigma = 2.43
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
15
10
5
0
−5
0
500
1000
1500

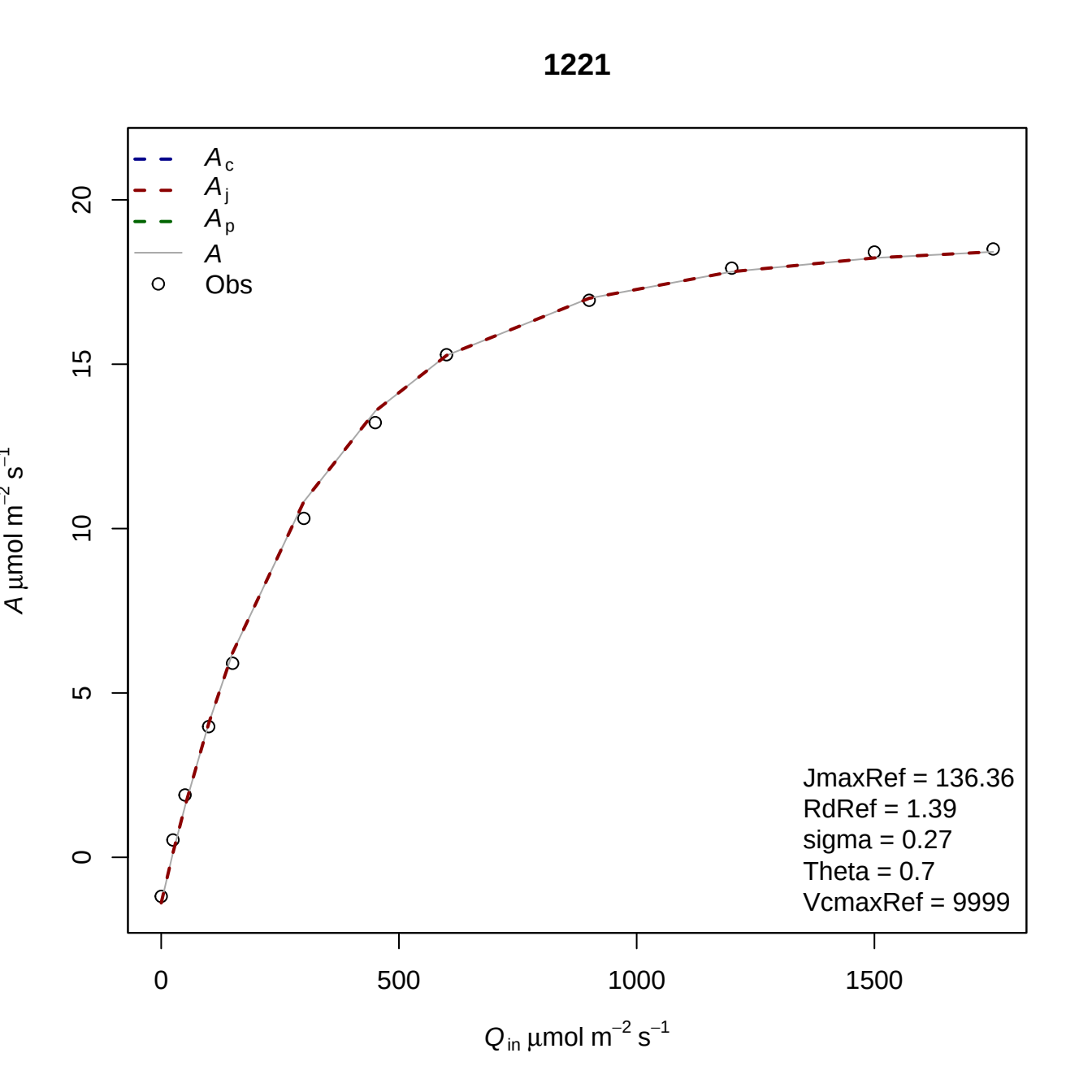

1216
A_c
A_j
A_p
A
Obs
JmaxRef = 139.08
RdRef = 4.46
sigma = 2.24
Theta = 0.7
VcmaxRef = 9999
20
15
10
5
0
−5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

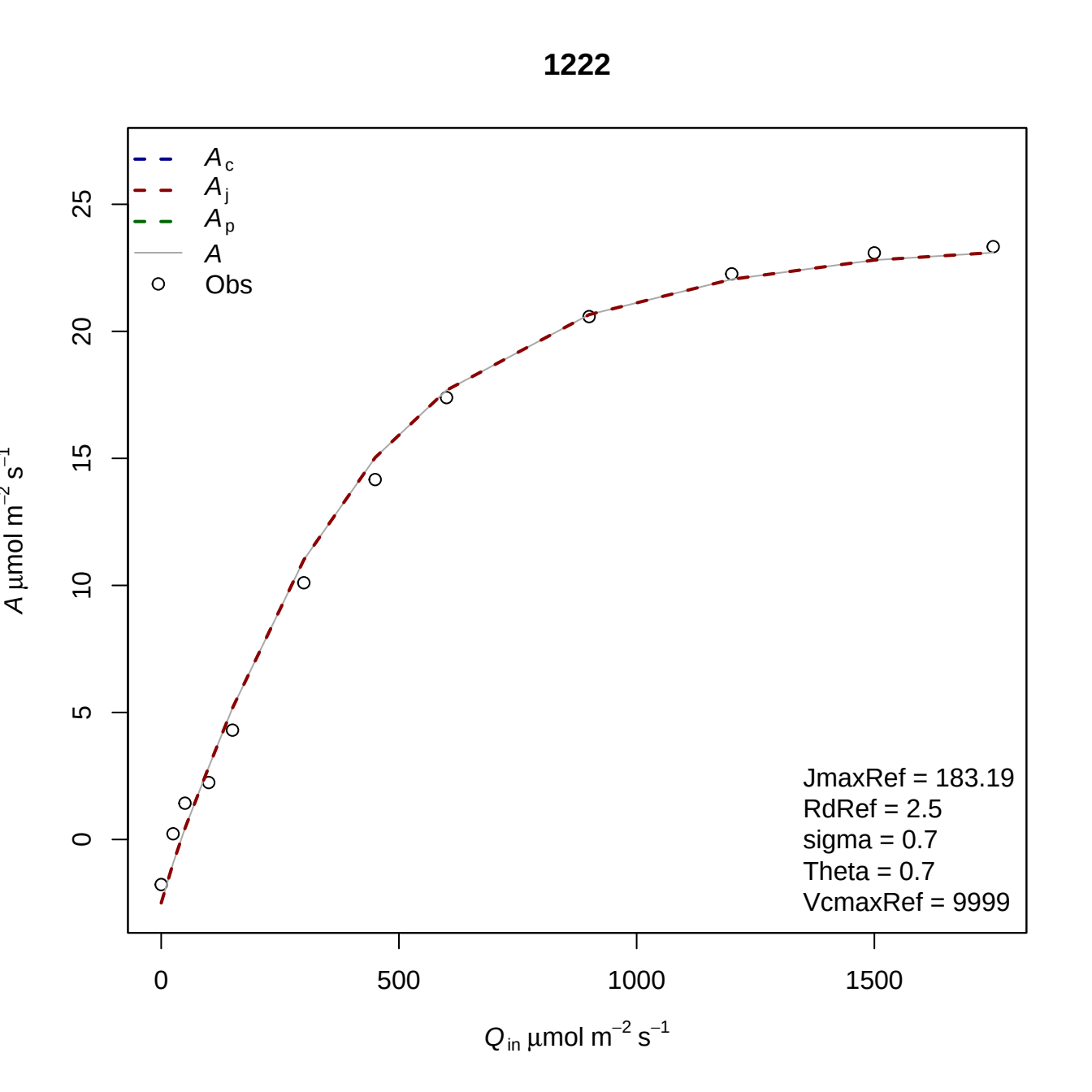

1217
A_c
A_j
A_p
A
Obs
JmaxRef = 105.92
RdRef = 4.62
sigma = 2.5
Theta = 0.7
VcmaxRef = 9999
15
10
5
0
−5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

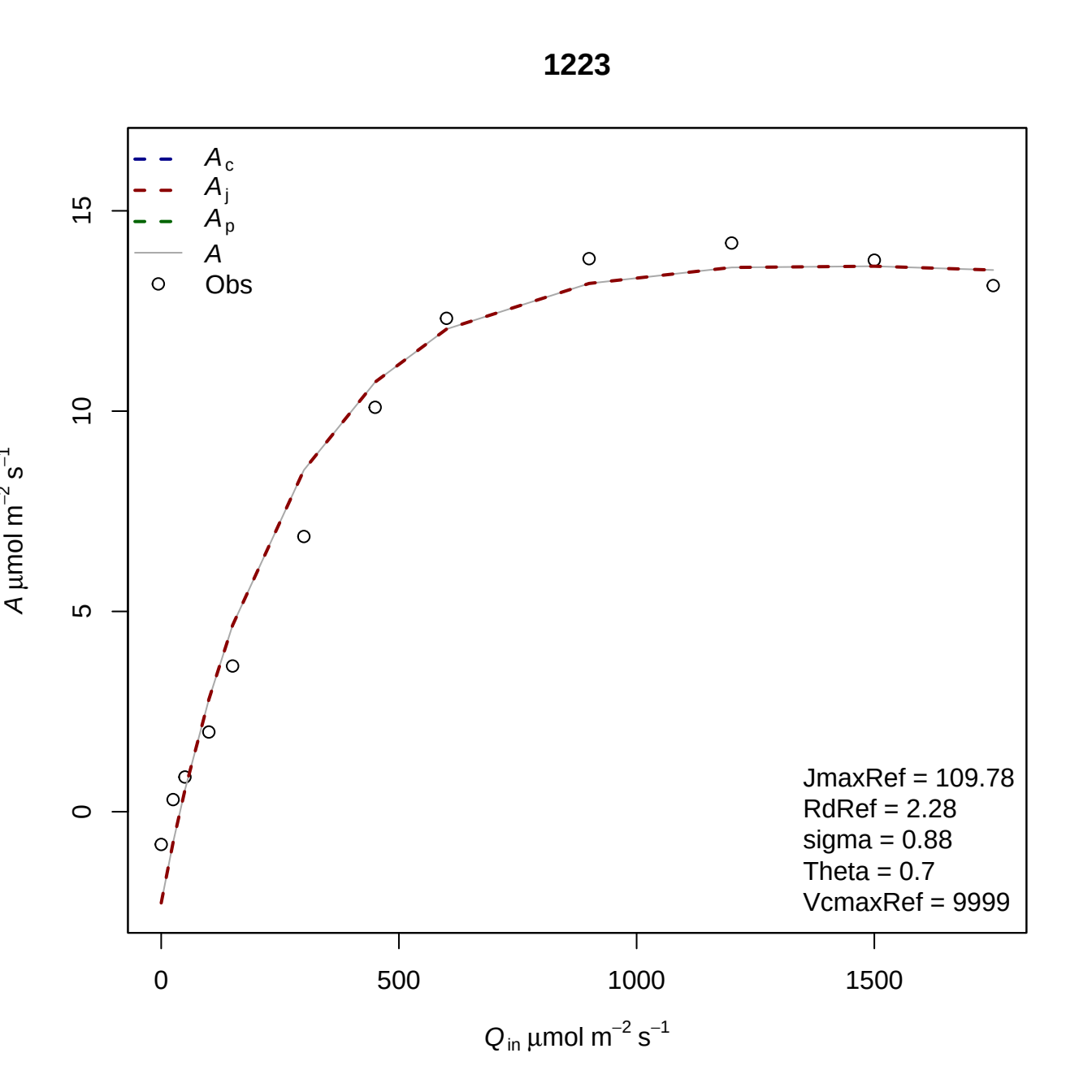

1218
A_c
A_j
A_p
A
Obs
JmaxRef = 138.89
RdRef = 0.97
sigma = 0.1
Theta = 0.7
VcmaxRef = 99.85
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

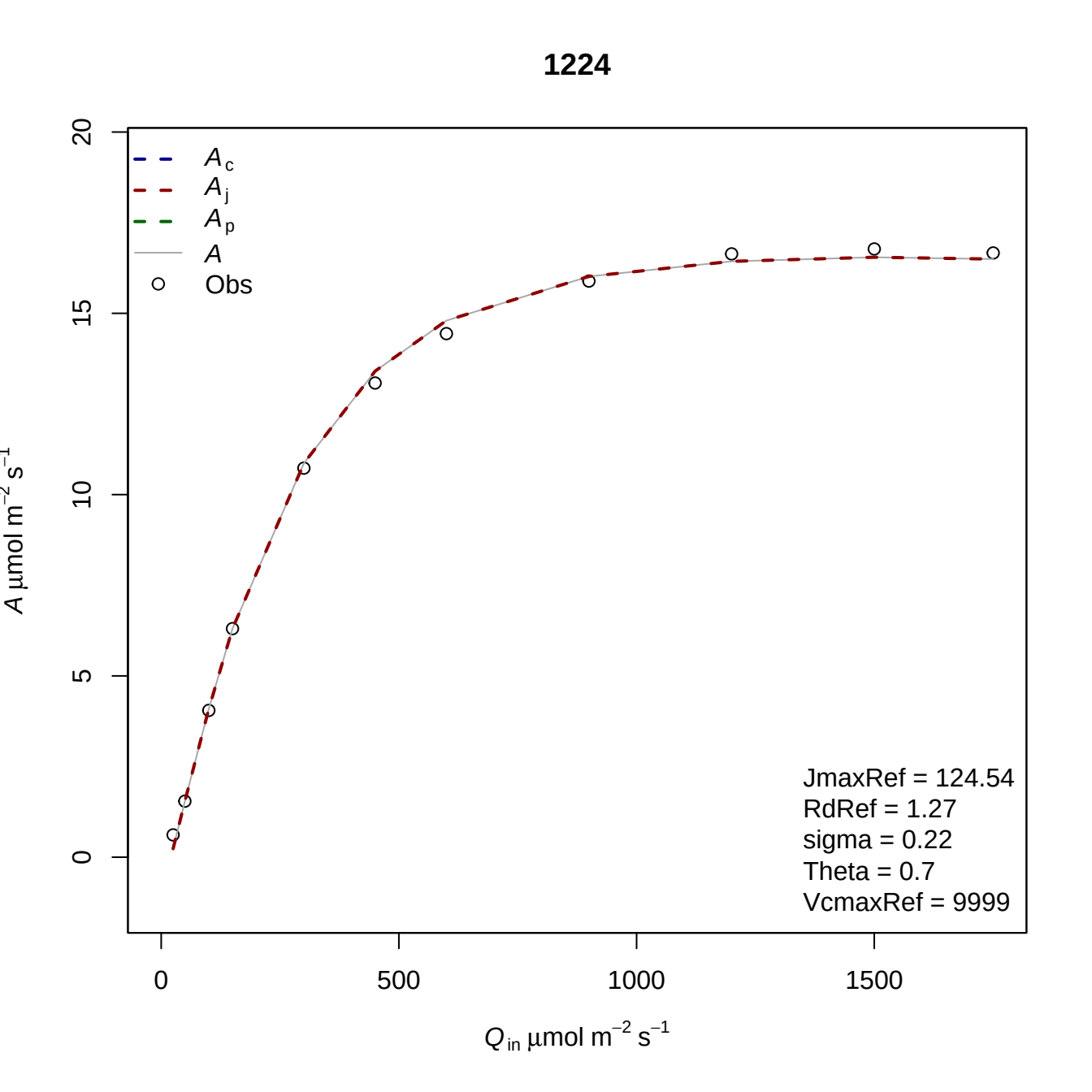

1219
A_c
A_j
A_p
A
Obs
JmaxRef = 176.1
RdRef = 2.47
sigma = 0.57
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

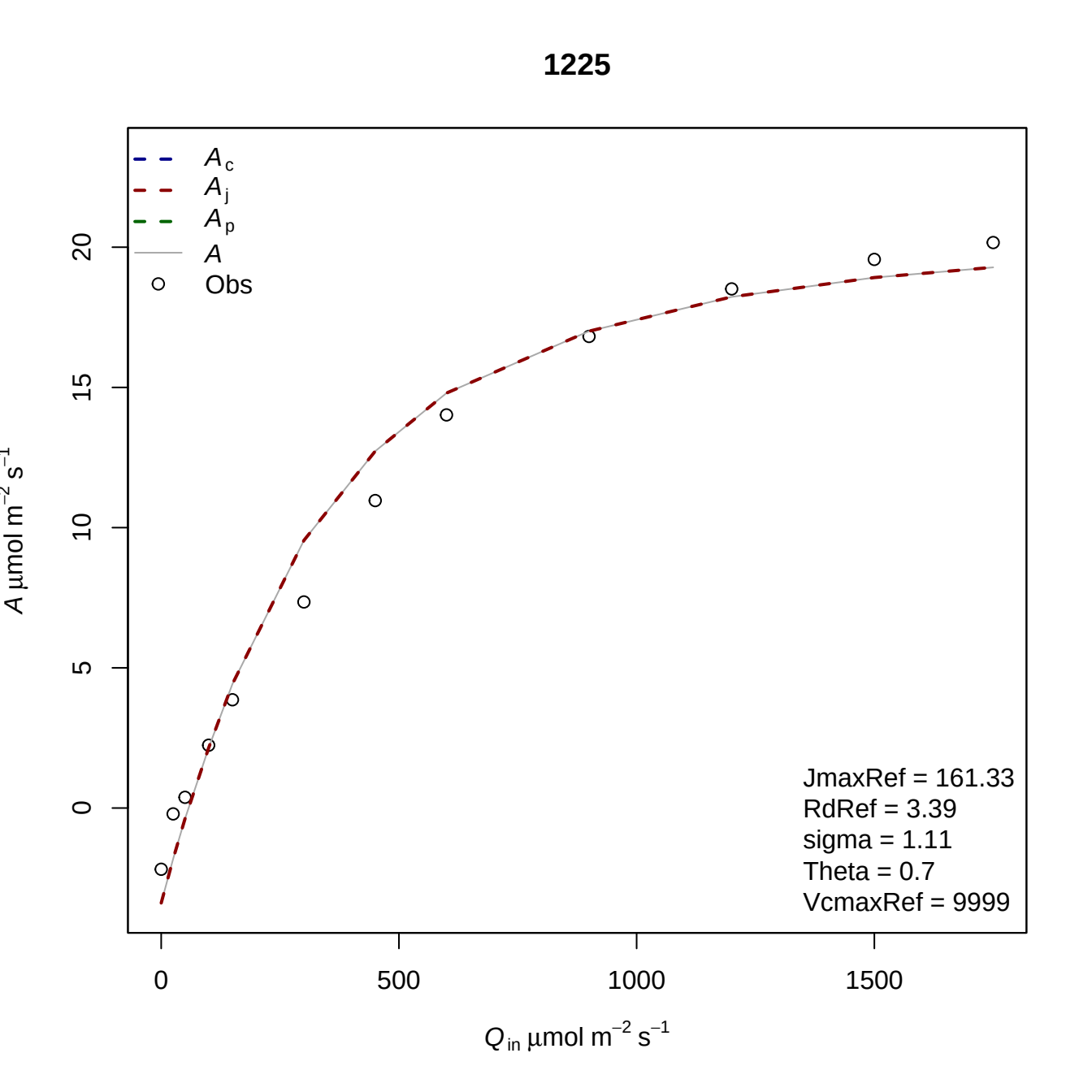

1220
A_c
A_j
A_p
A
Obs
JmaxRef = 180.53
RdRef = 1.58
sigma = 0.28
Theta = 0.7
VcmaxRef = 113.24
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

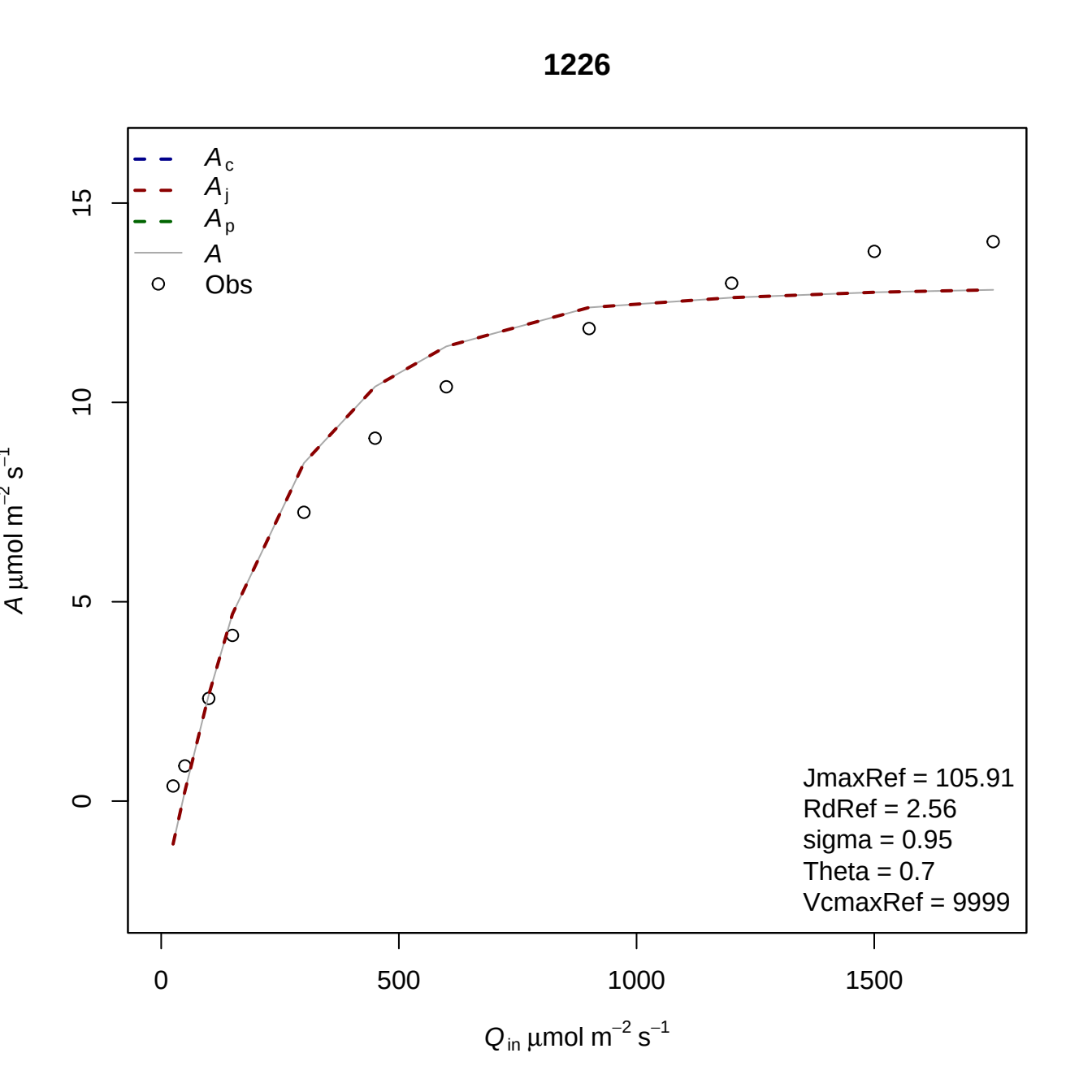

1221
A_c
A_j
A_p
A
Obs
JmaxRef = 136.36
RdRef = 1.39
sigma = 0.27
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

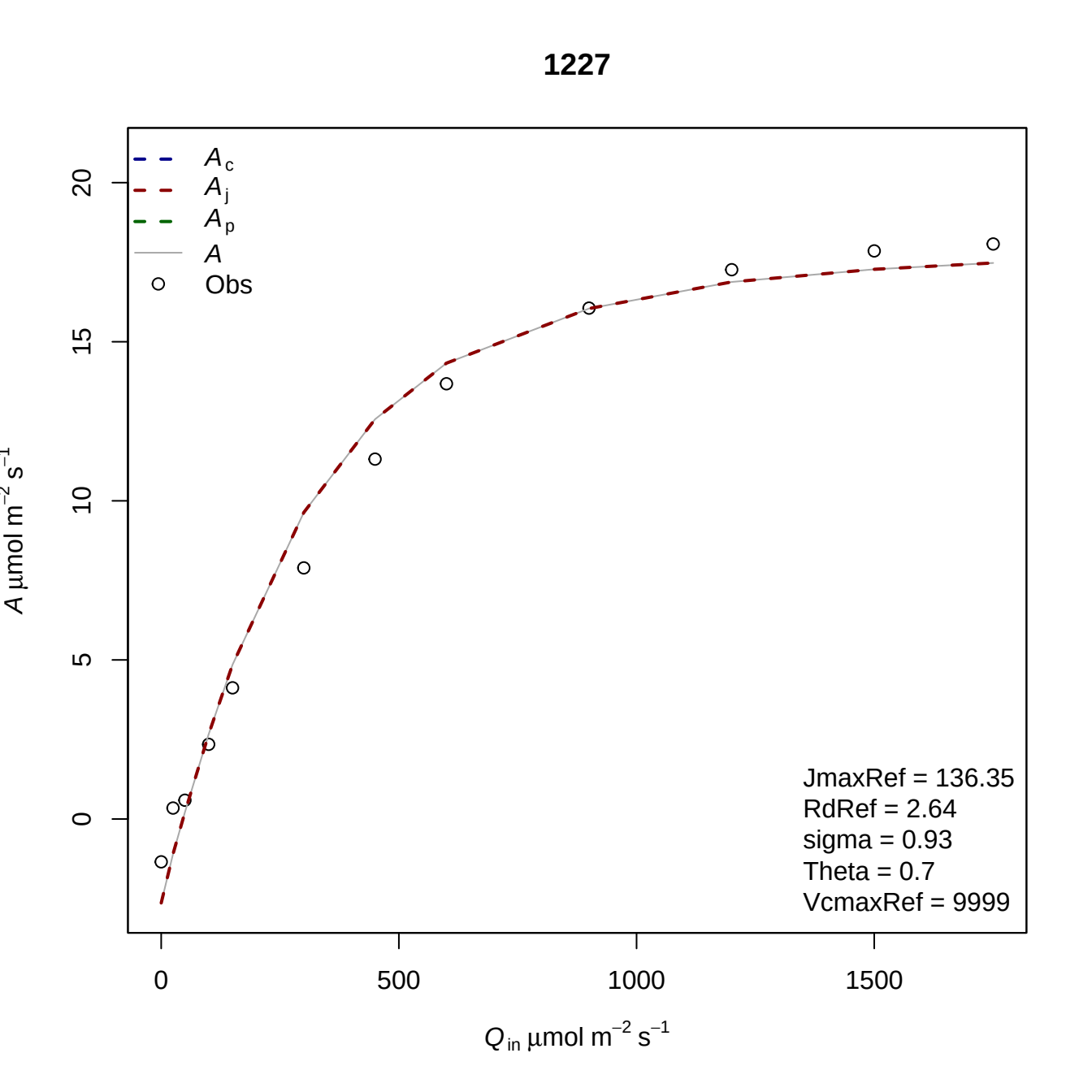

1222
A_c
A_j
A_p
A
Obs
JmaxRef = 183.19
RdRef = 2.5
sigma = 0.7
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
25
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

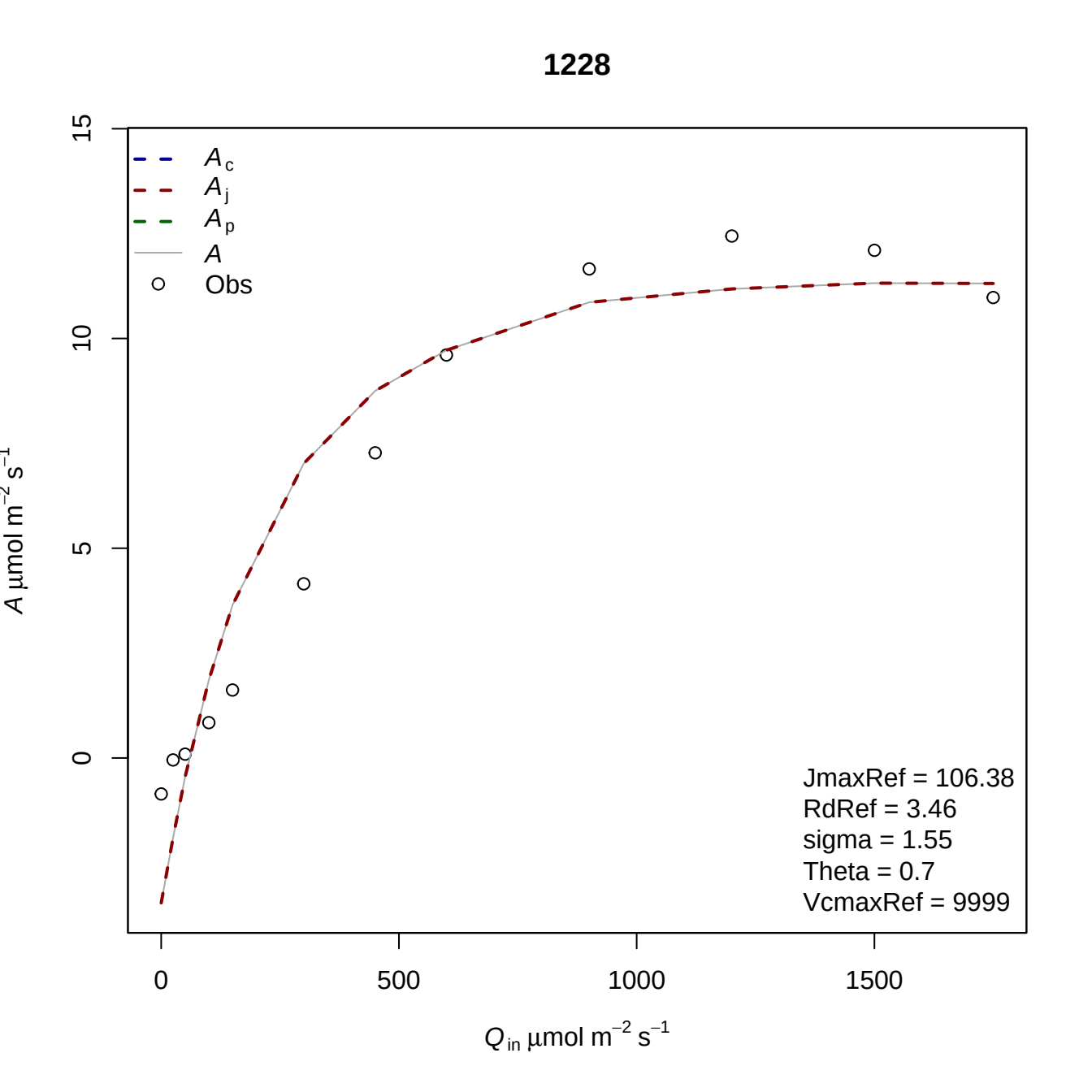

1223
A_c
A_j
A_p
A
Obs
JmaxRef = 109.78
RdRef = 2.28
sigma = 0.88
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

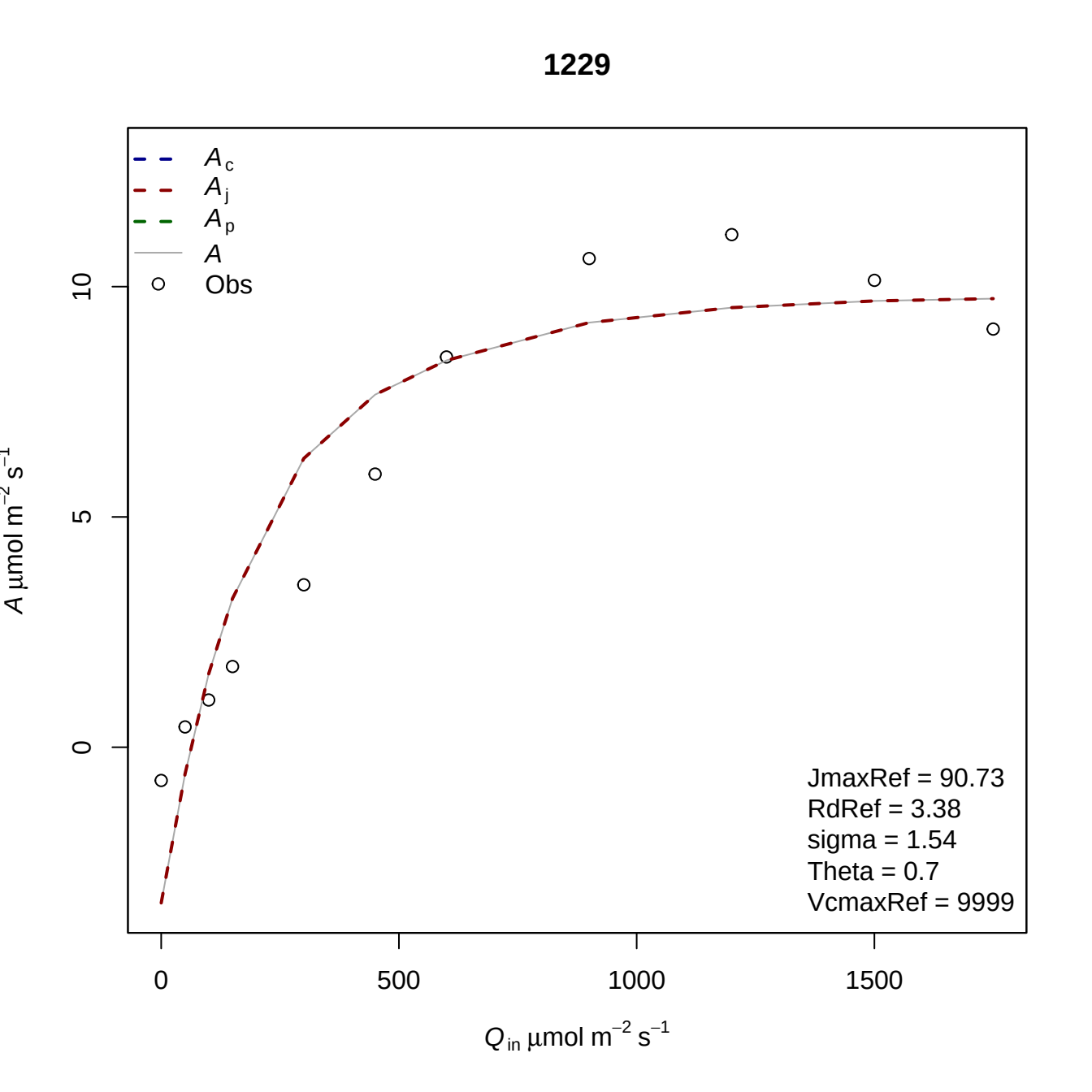

1224
A_c
A_j
A_p
A
Obs
JmaxRef = 124.54
RdRef = 1.27
sigma = 0.22
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

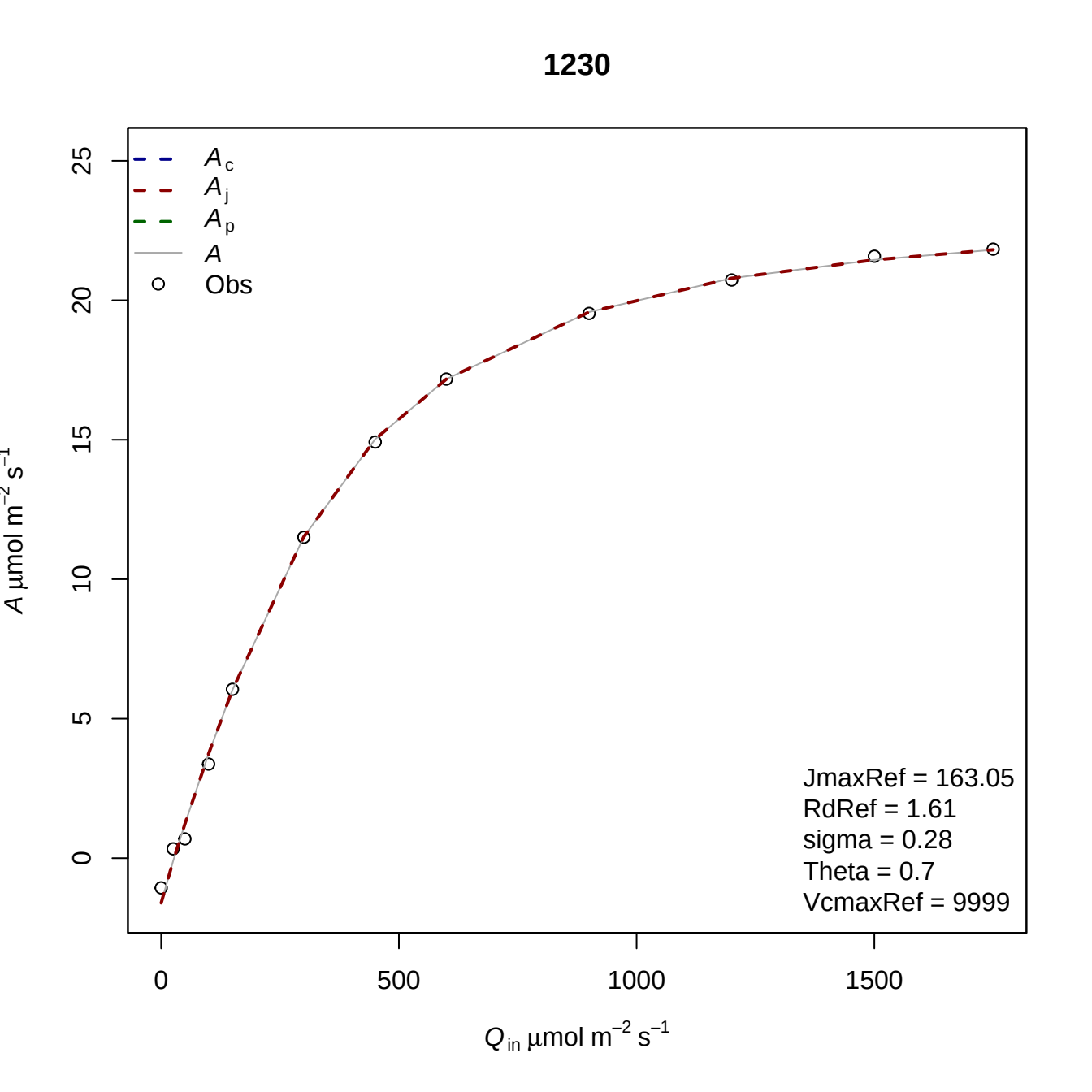

1225
A_c
A_j
A_p
A
Obs
JmaxRef = 161.33
RdRef = 3.39
sigma = 1.11
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

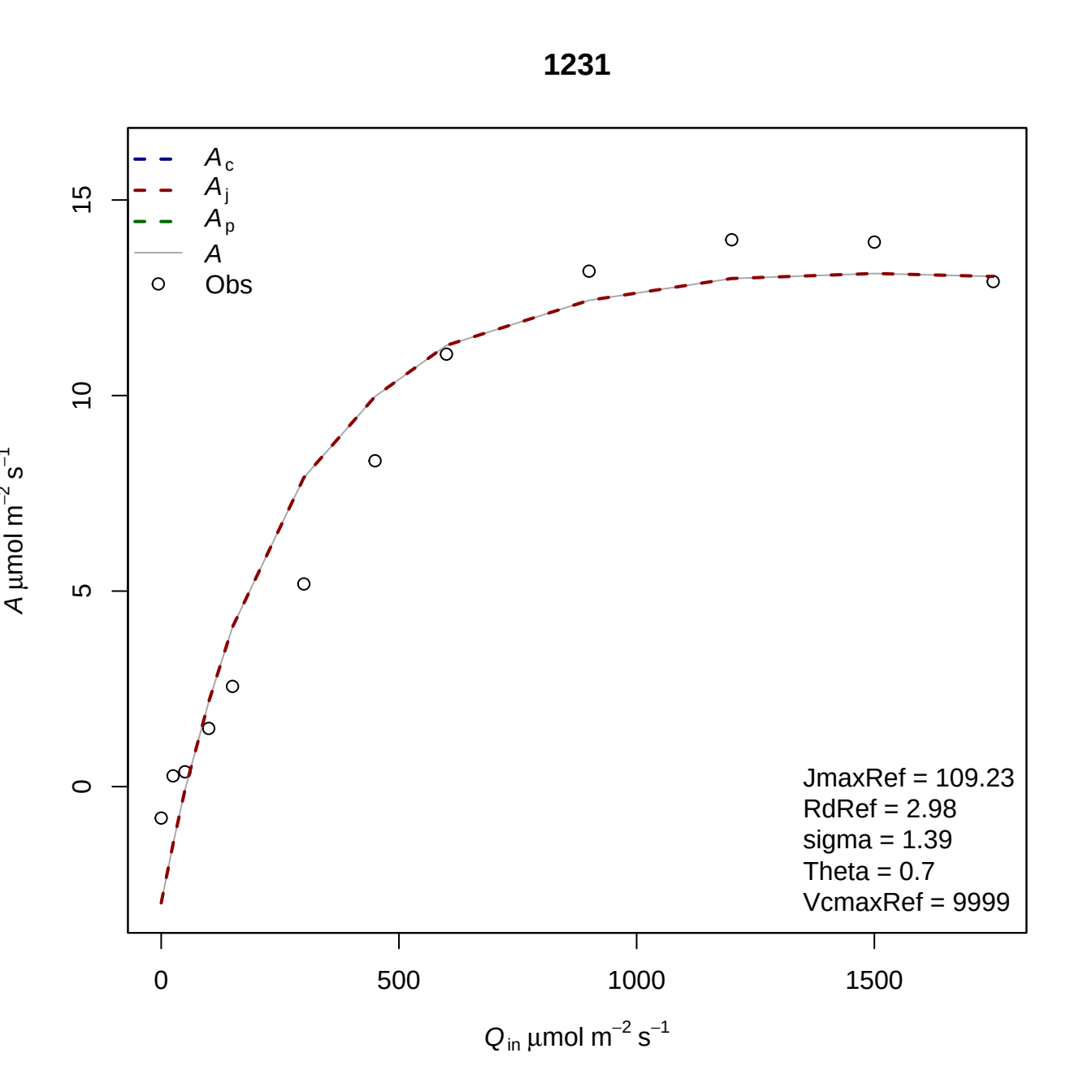

1226
A_c
A_j
A_p
A
Obs
JmaxRef = 105.91
RdRef = 2.56
sigma = 0.95
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

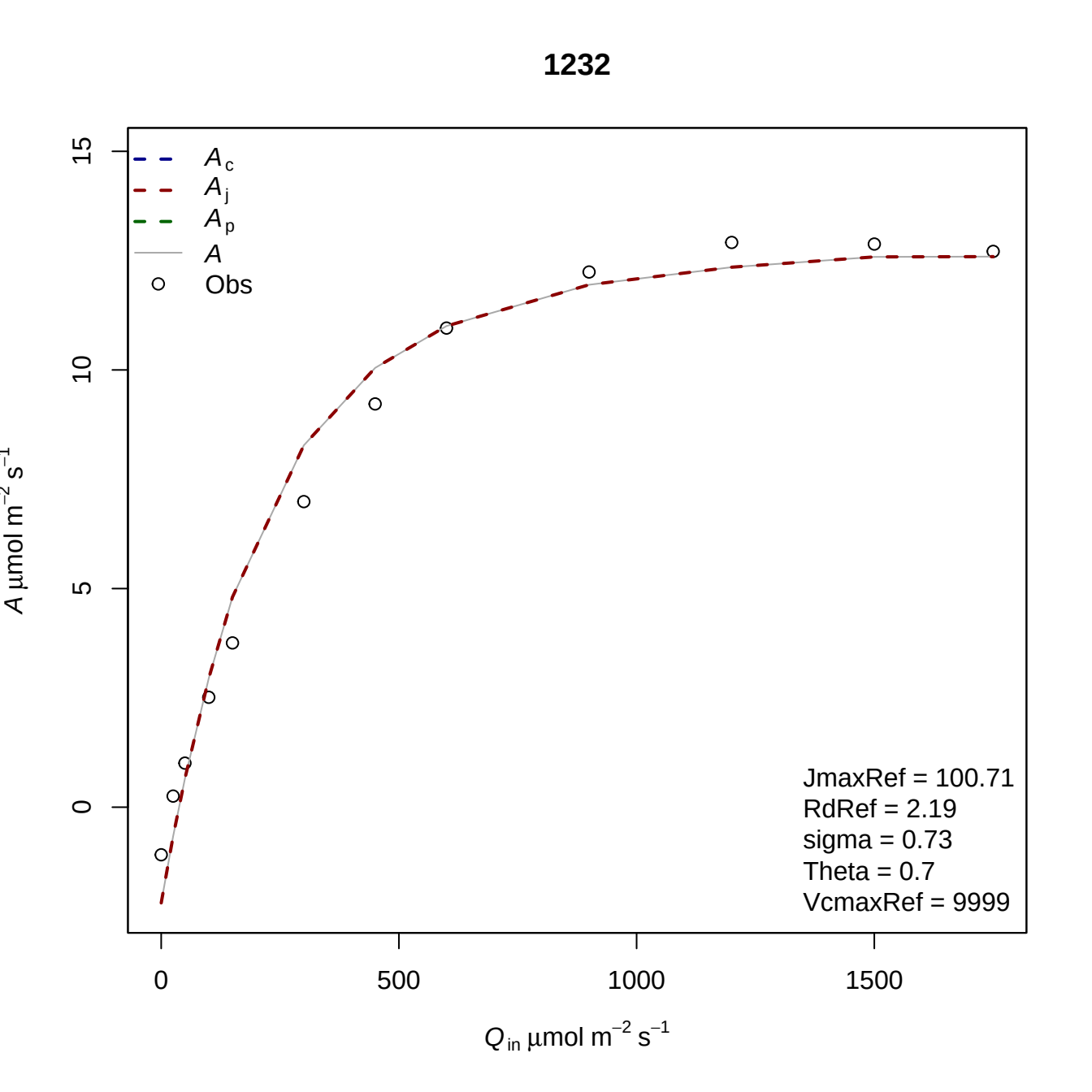

1227
A_c
A_j
A_p
A
Obs
JmaxRef = 136.35
RdRef = 2.64
sigma = 0.93
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

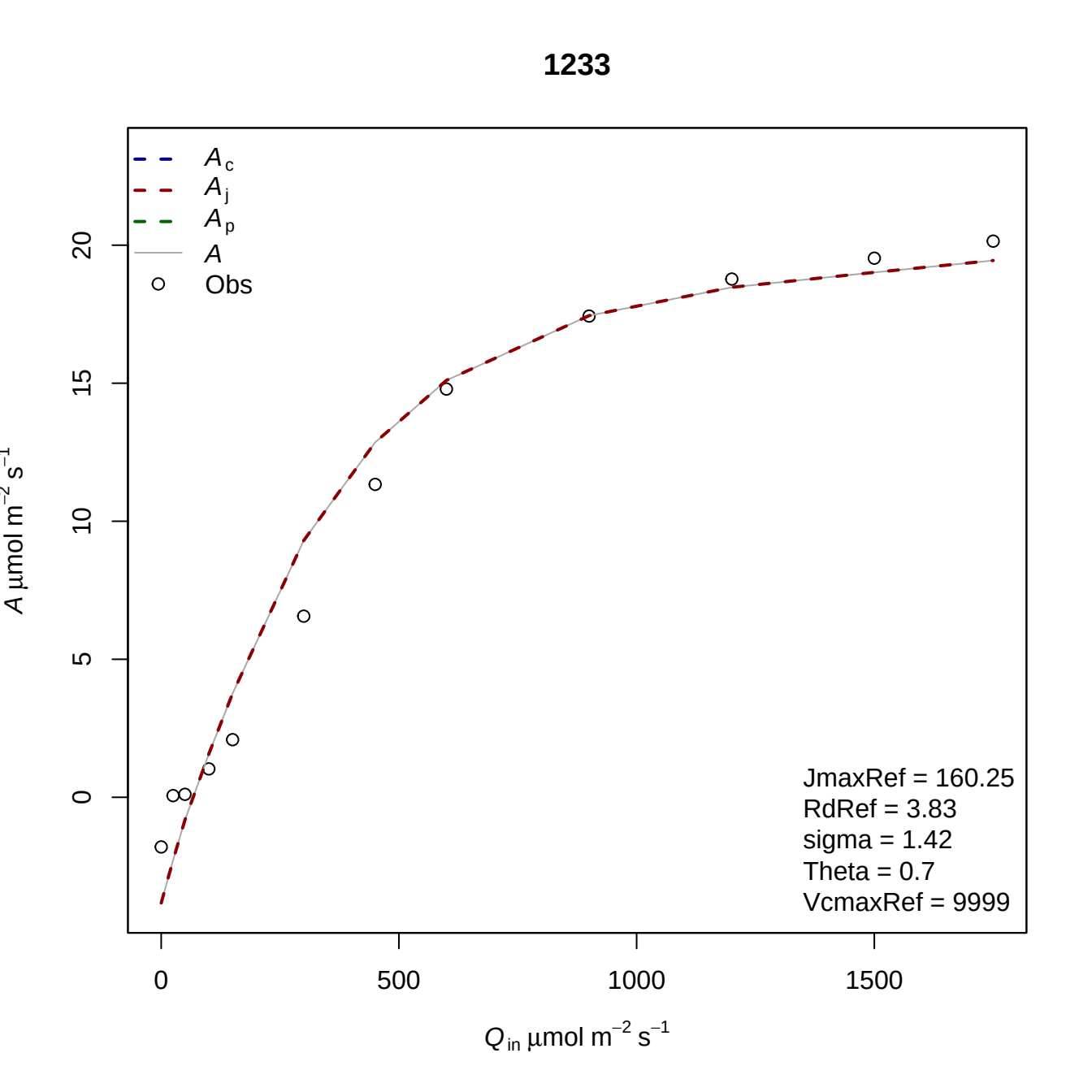

1228
A_c
A_j
A_p
A
Obs
JmaxRef = 106.38
RdRef = 3.46
sigma = 1.55
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

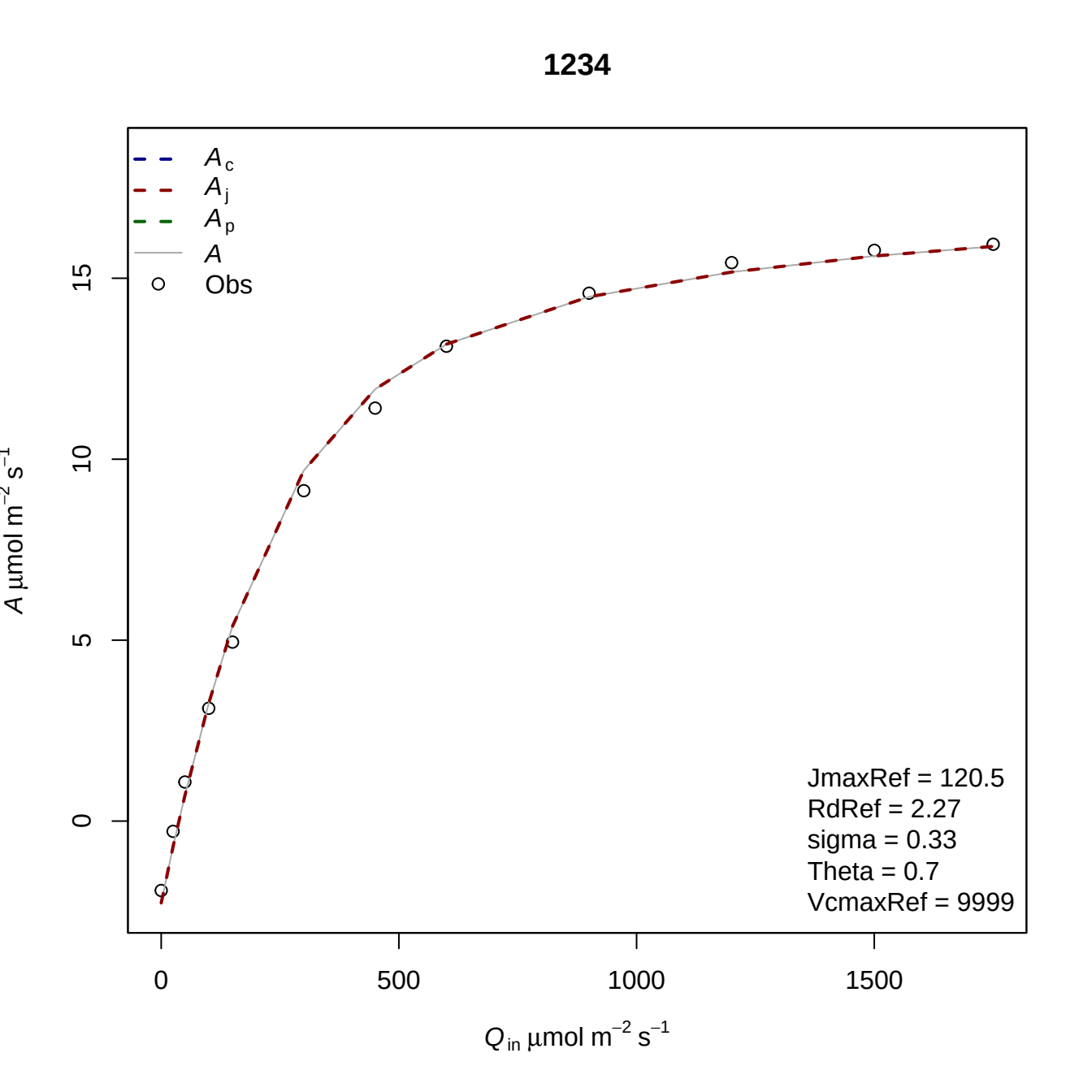

1229
A_c
A_j
A_p
A
Obs
JmaxRef = 90.73
RdRef = 3.38
sigma = 1.54
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

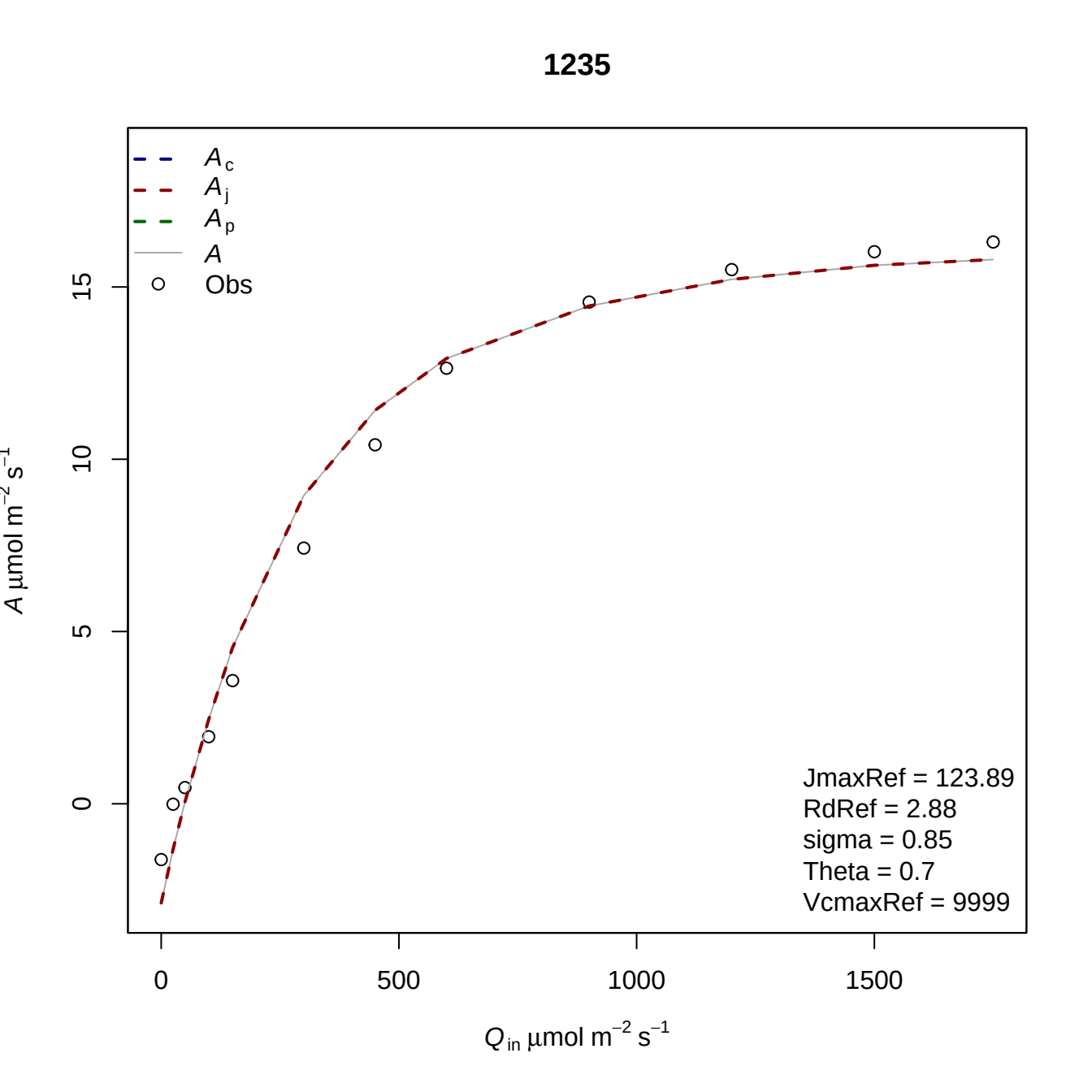

1230
A_c
A_j
A_p
A
Obs
JmaxRef = 163.05
RdRef = 1.61
sigma = 0.28
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

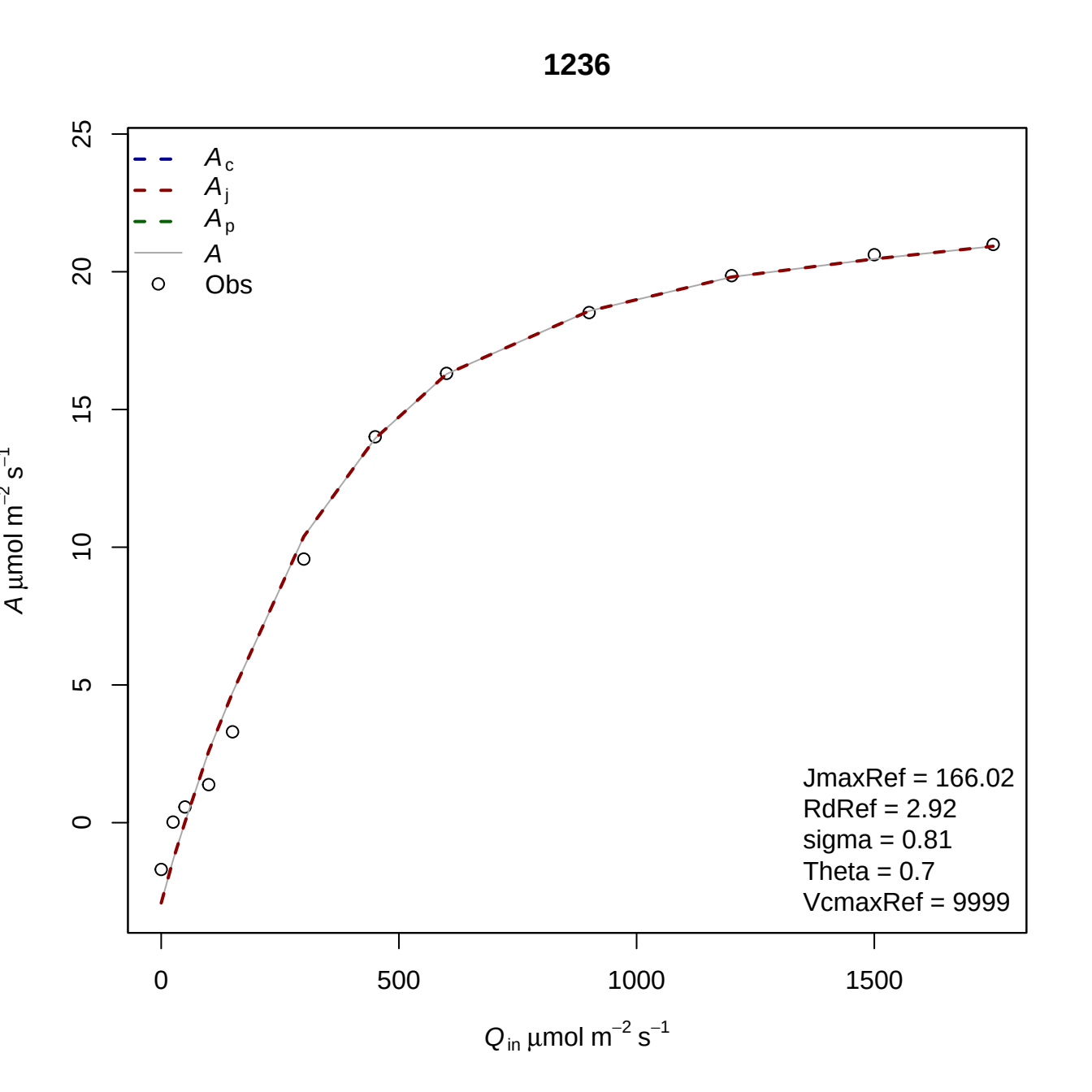

1231
A_c
A_j
A_p
A
Obs
JmaxRef = 109.23
RdRef = 2.98
sigma = 1.39
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

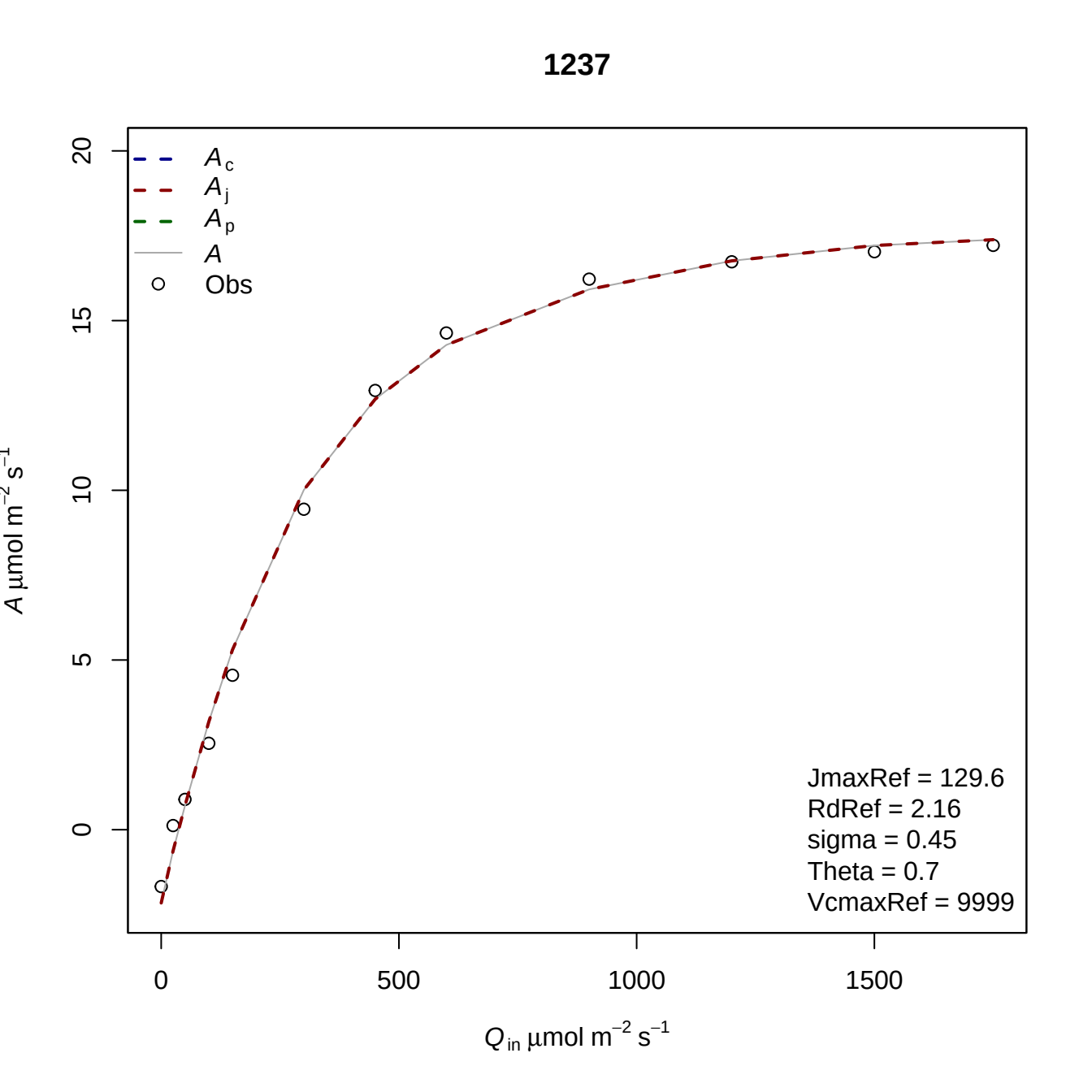

1232
A_c
A_j
A_p
A
Obs
JmaxRef = 100.71
RdRef = 2.19
sigma = 0.73
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

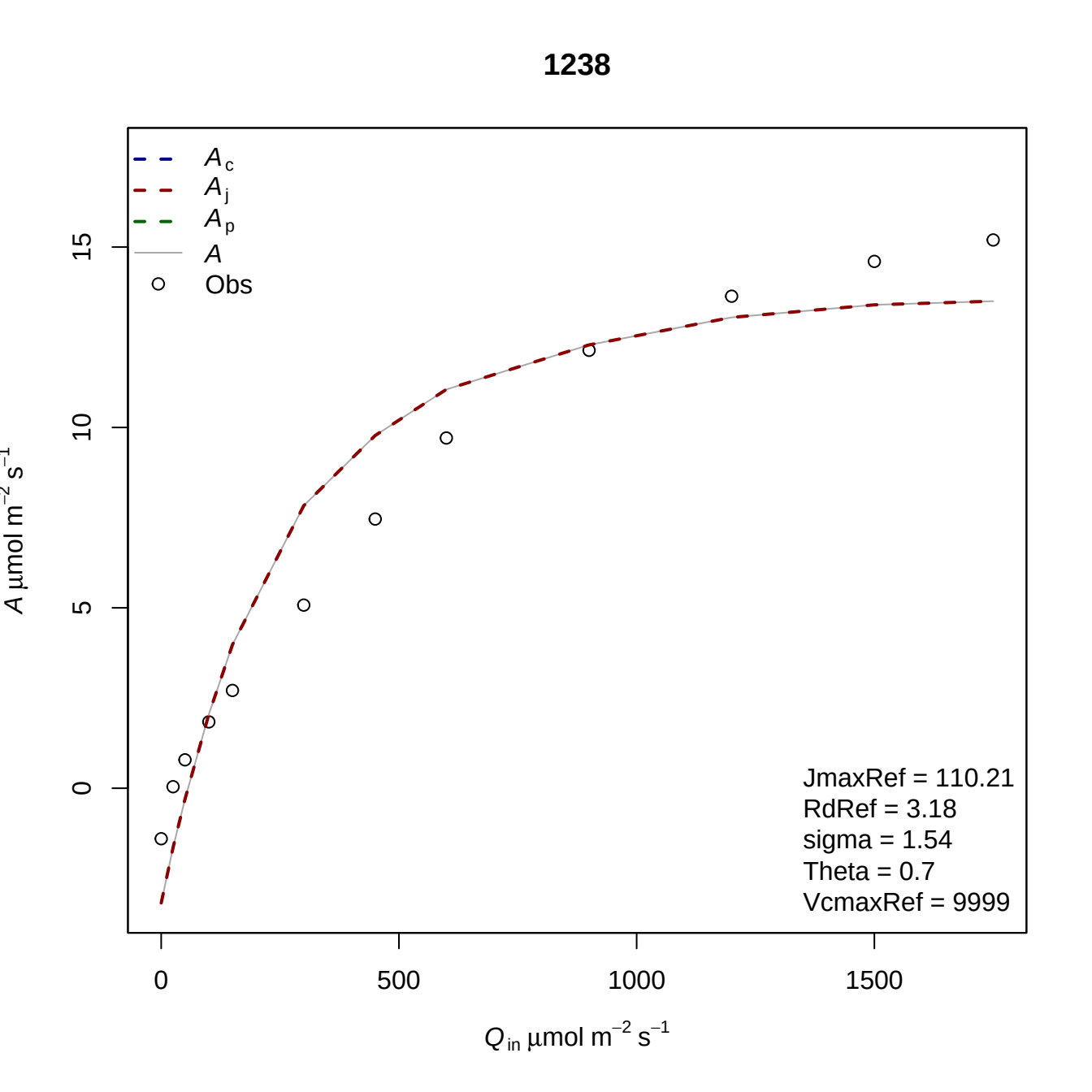

1233
A_c
A_j
A_p
A
Obs
JmaxRef = 160.25
RdRef = 3.83
sigma = 1.42
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

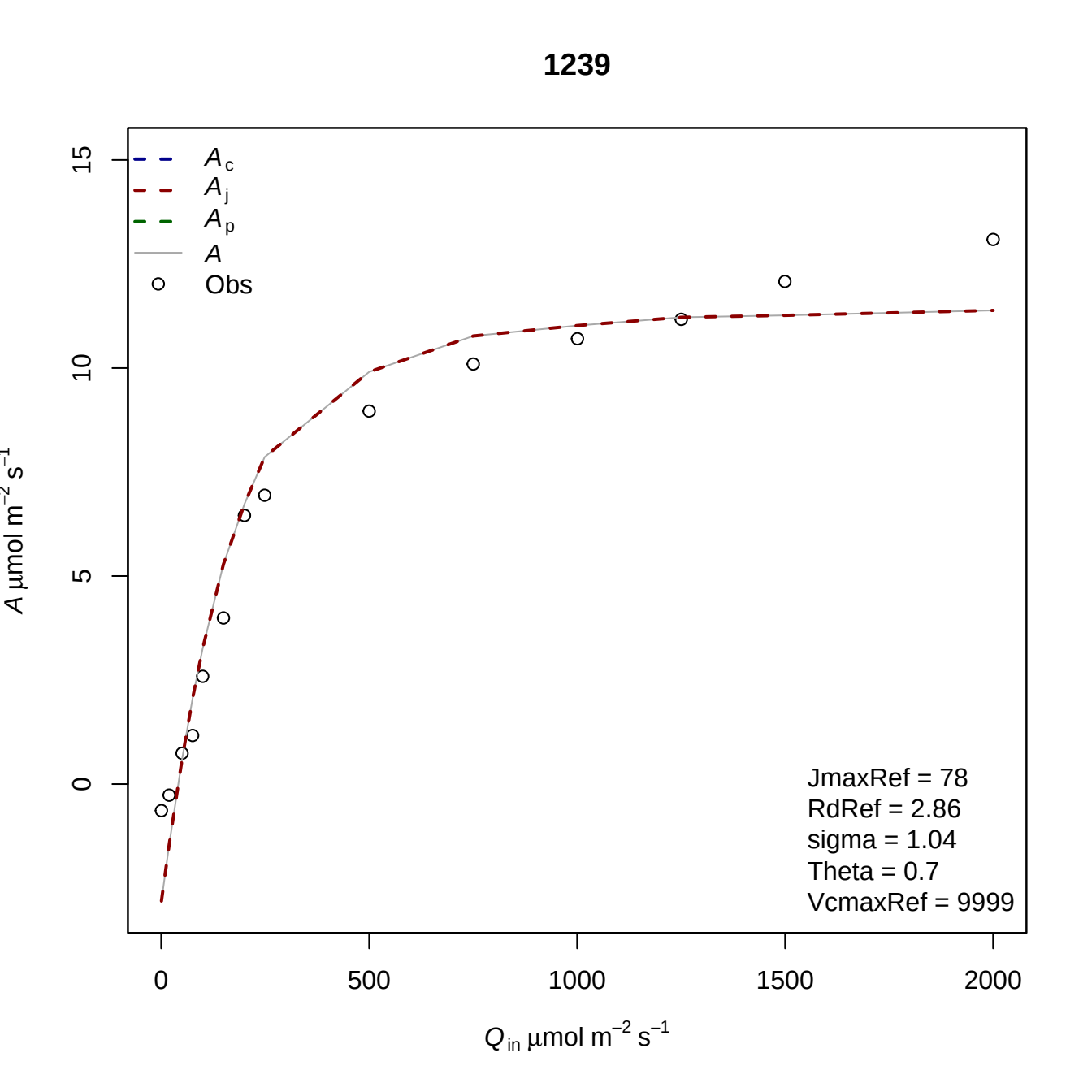

1234
A_c
A_j
A_p
A
Obs
JmaxRef = 120.5
RdRef = 2.27
sigma = 0.33
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

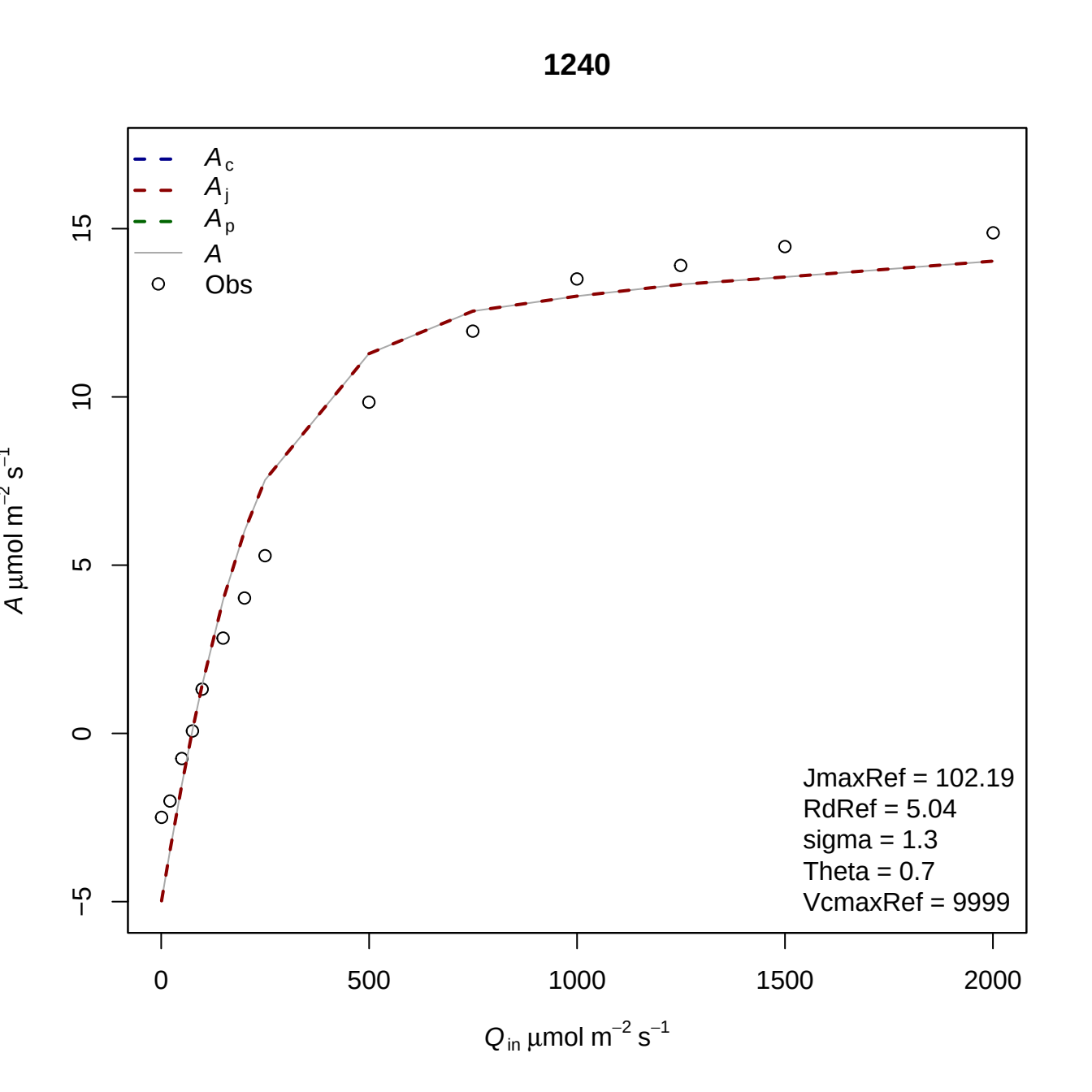

1235
A_c
A_j
A_p
A
Obs
JmaxRef = 123.89
RdRef = 2.88
sigma = 0.85
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

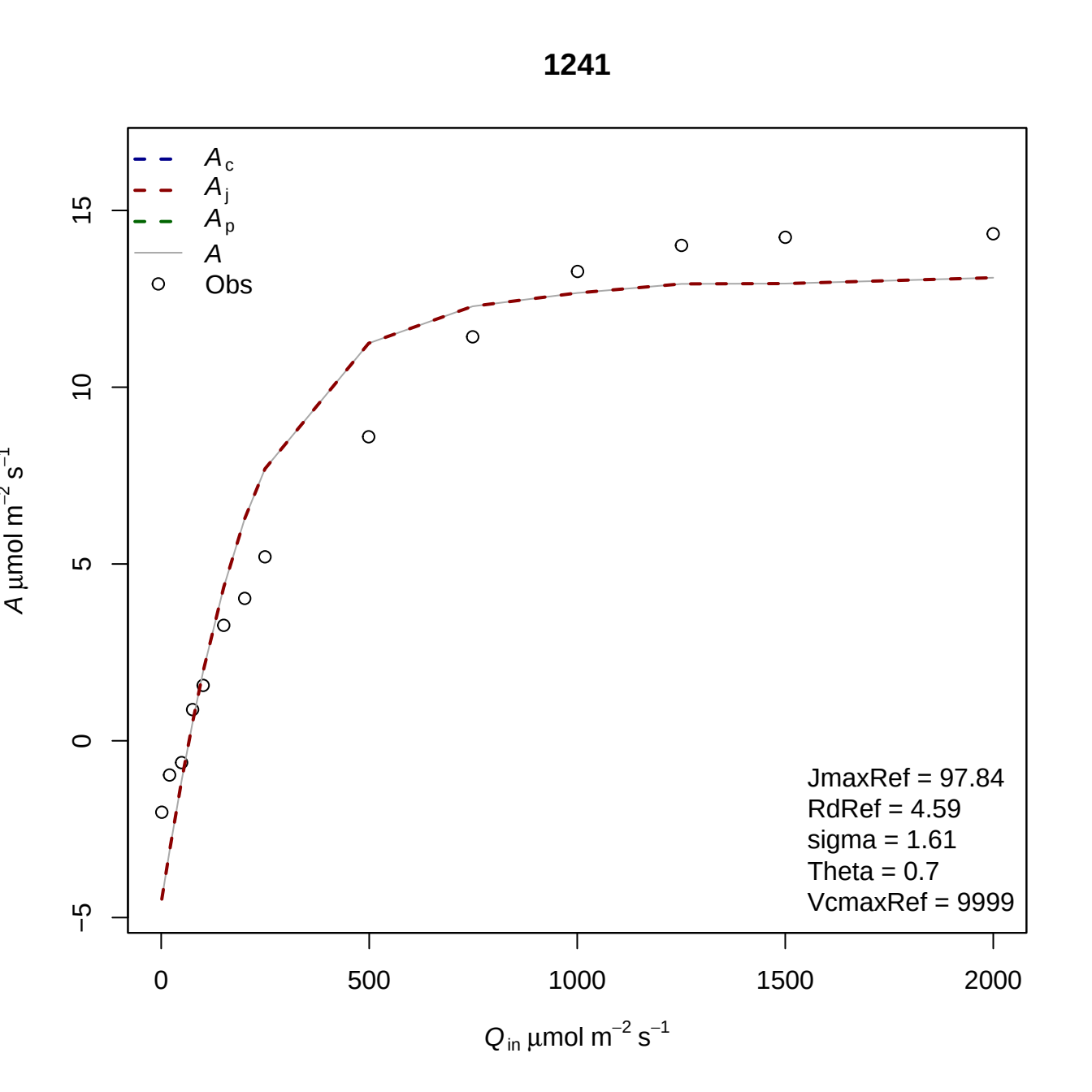

1236
A_c
A_j
A_p
A
Obs
JmaxRef = 166.02
RdRef = 2.92
sigma = 0.81
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

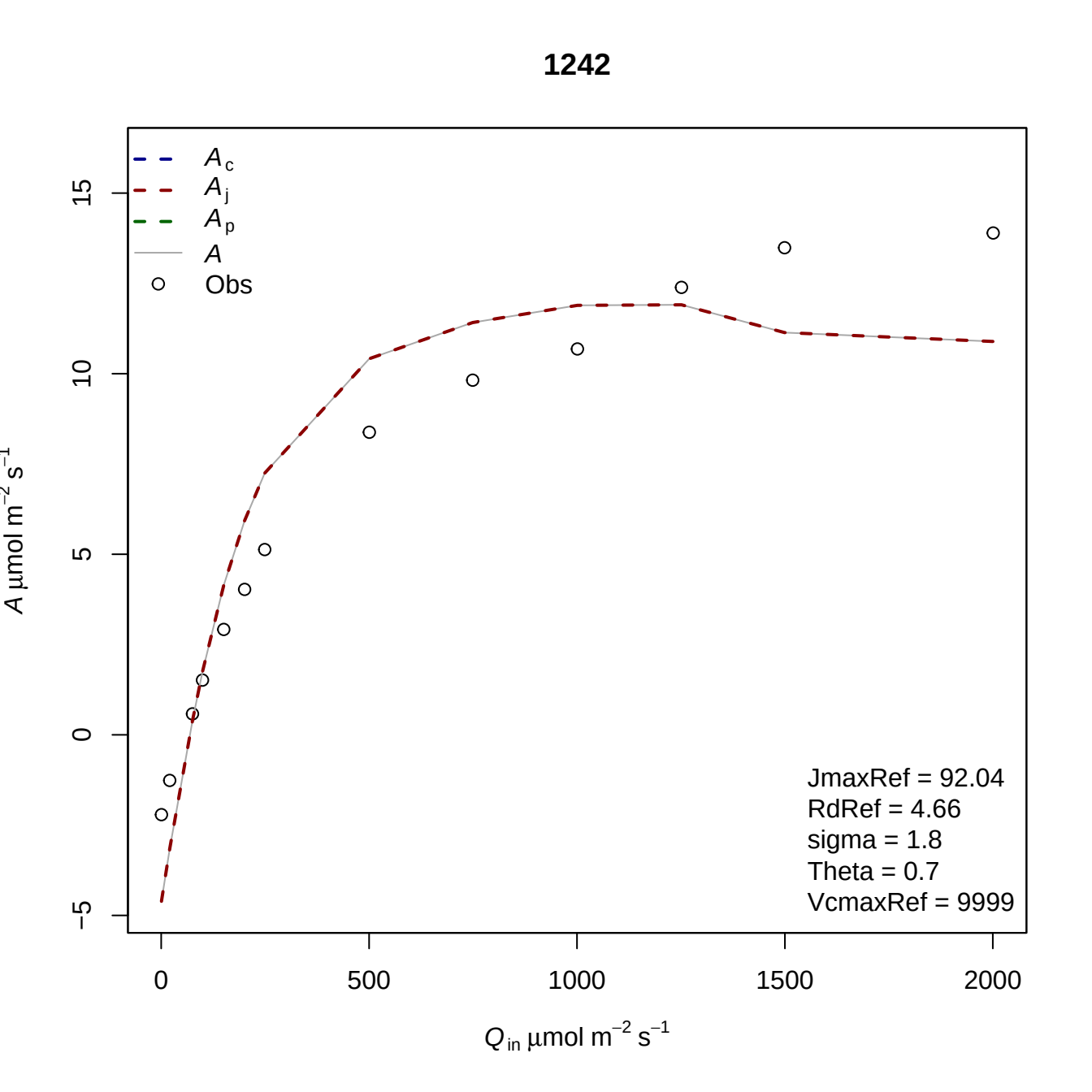

1237
A_c
A_j
A_p
A
Obs
JmaxRef = 129.6
RdRef = 2.16
sigma = 0.45
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500

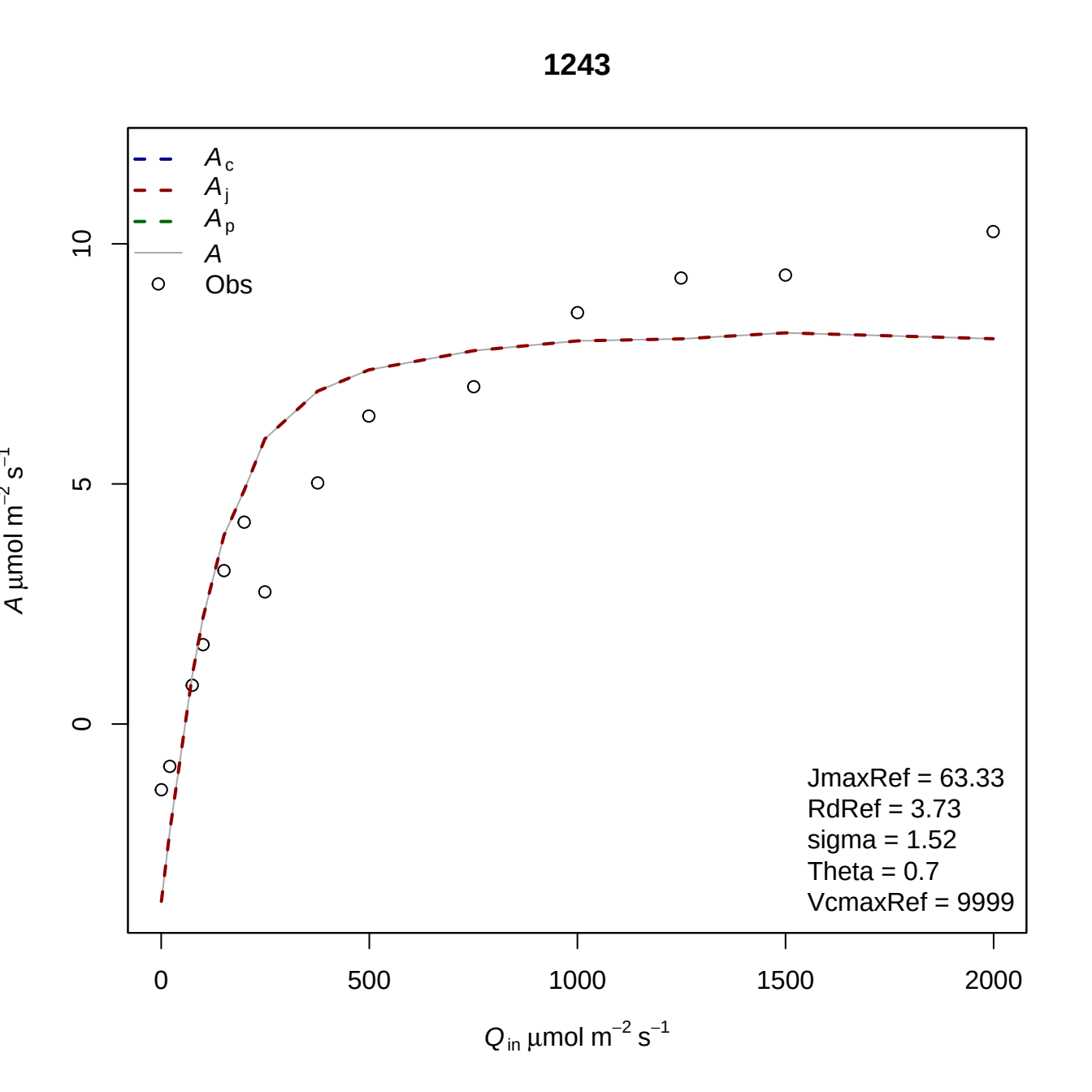

1238
A_c
A_j
A_p
A
Obs
JmaxRef = 110.21
RdRef = 3.18
sigma = 1.54
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

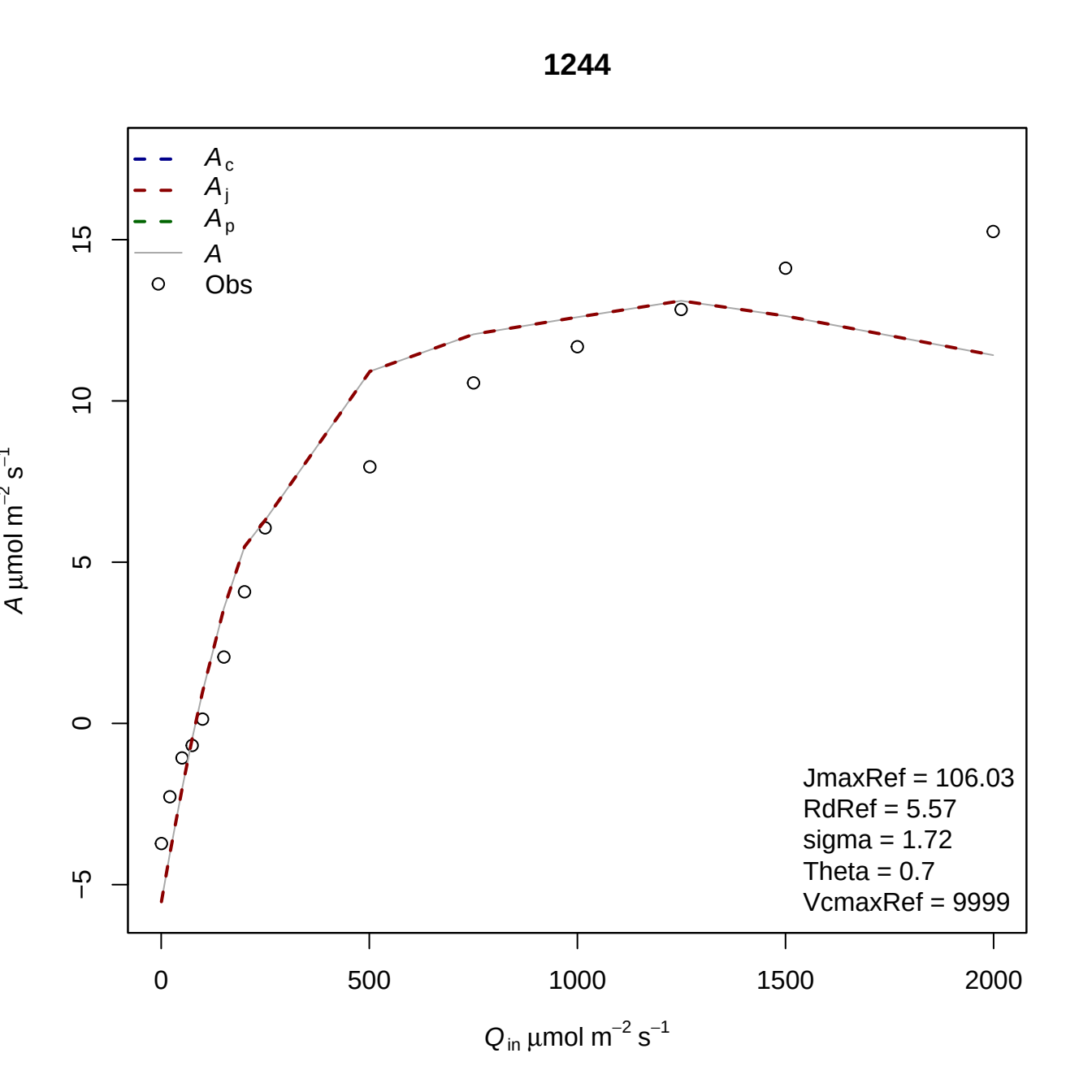

1239
A_c
A_j
A_p
A
Obs
JmaxRef = 78
RdRef = 2.86
sigma = 1.04
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

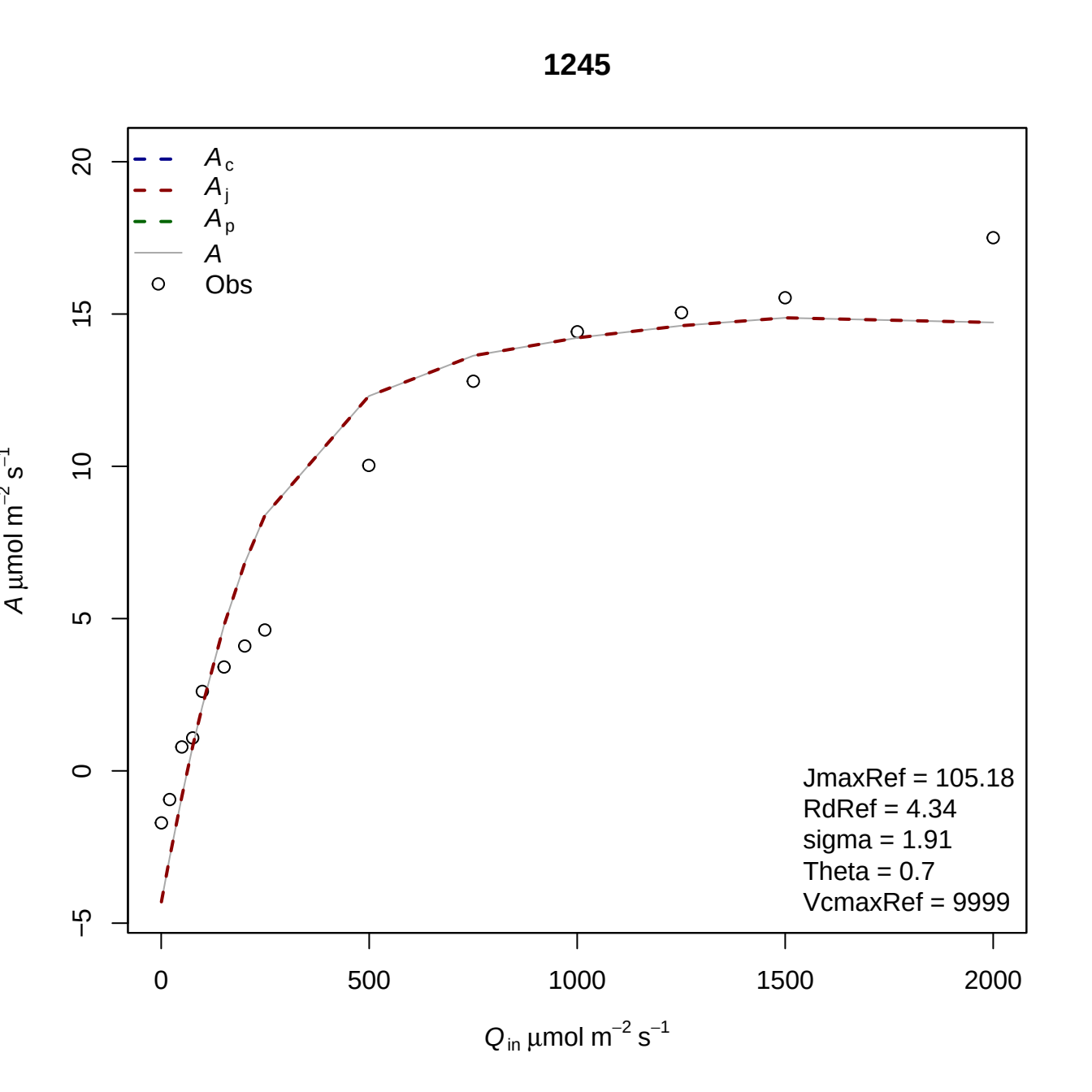

1240
A_c
A_j
A_p
A
Obs
JmaxRef = 102.19
RdRef = 5.04
sigma = 1.3
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

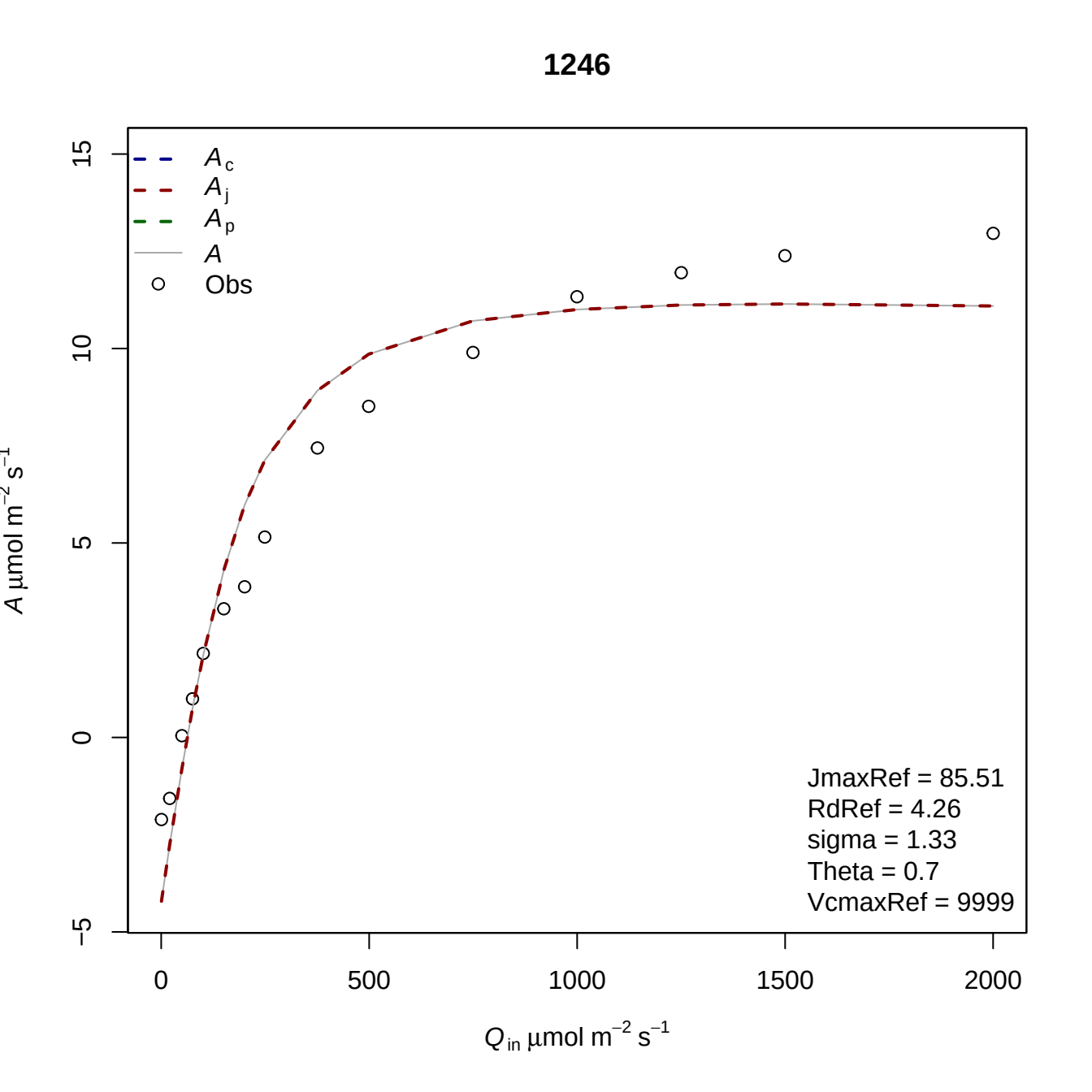

1241
A_c
A_j
A_p
A
Obs
JmaxRef = 97.84
RdRef = 4.59
sigma = 1.61
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

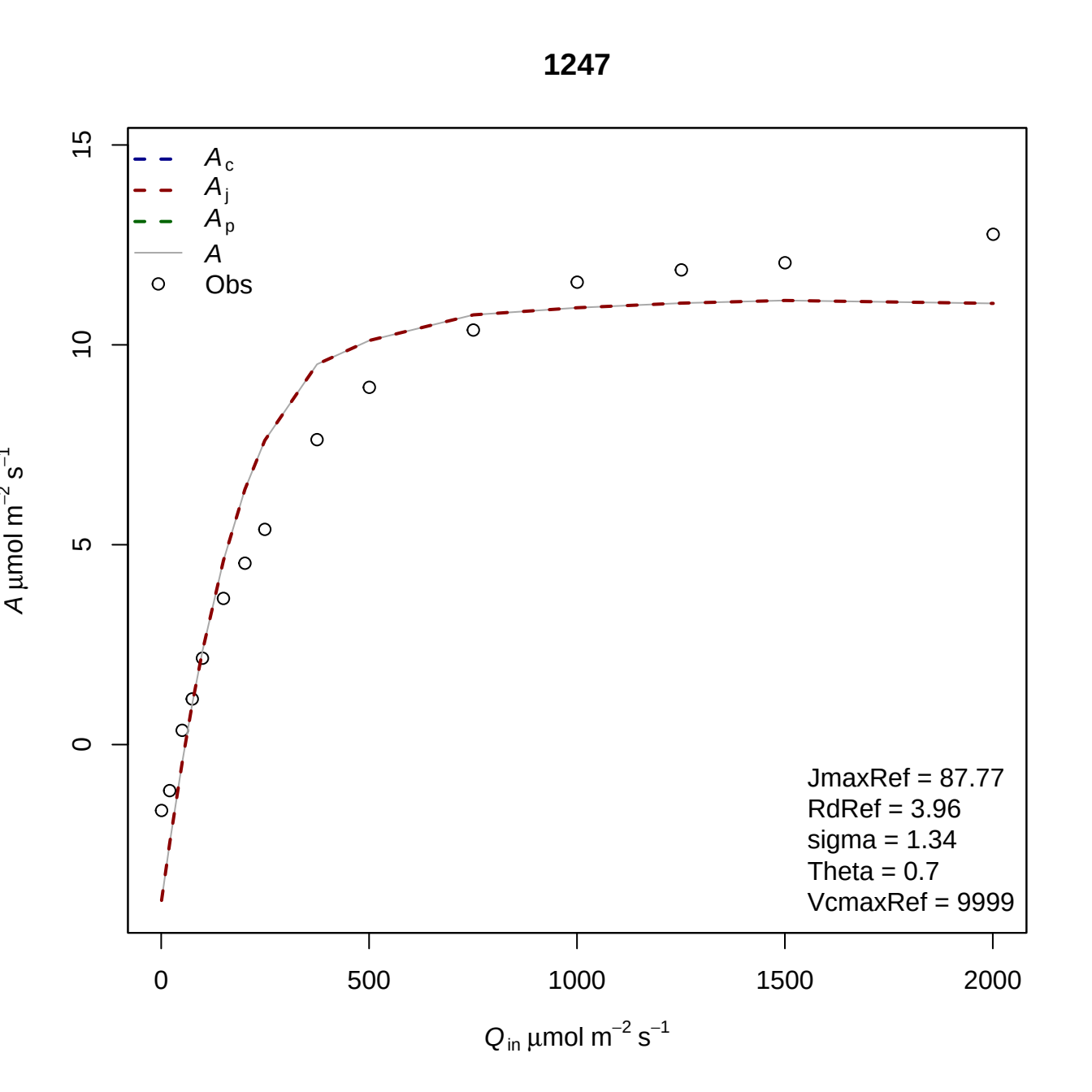

1242
A_c
A_j
A_p
A
Obs
JmaxRef = 92.04
RdRef = 4.66
sigma = 1.8
Theta = 0.7
VcmaxRef = 9999
15
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

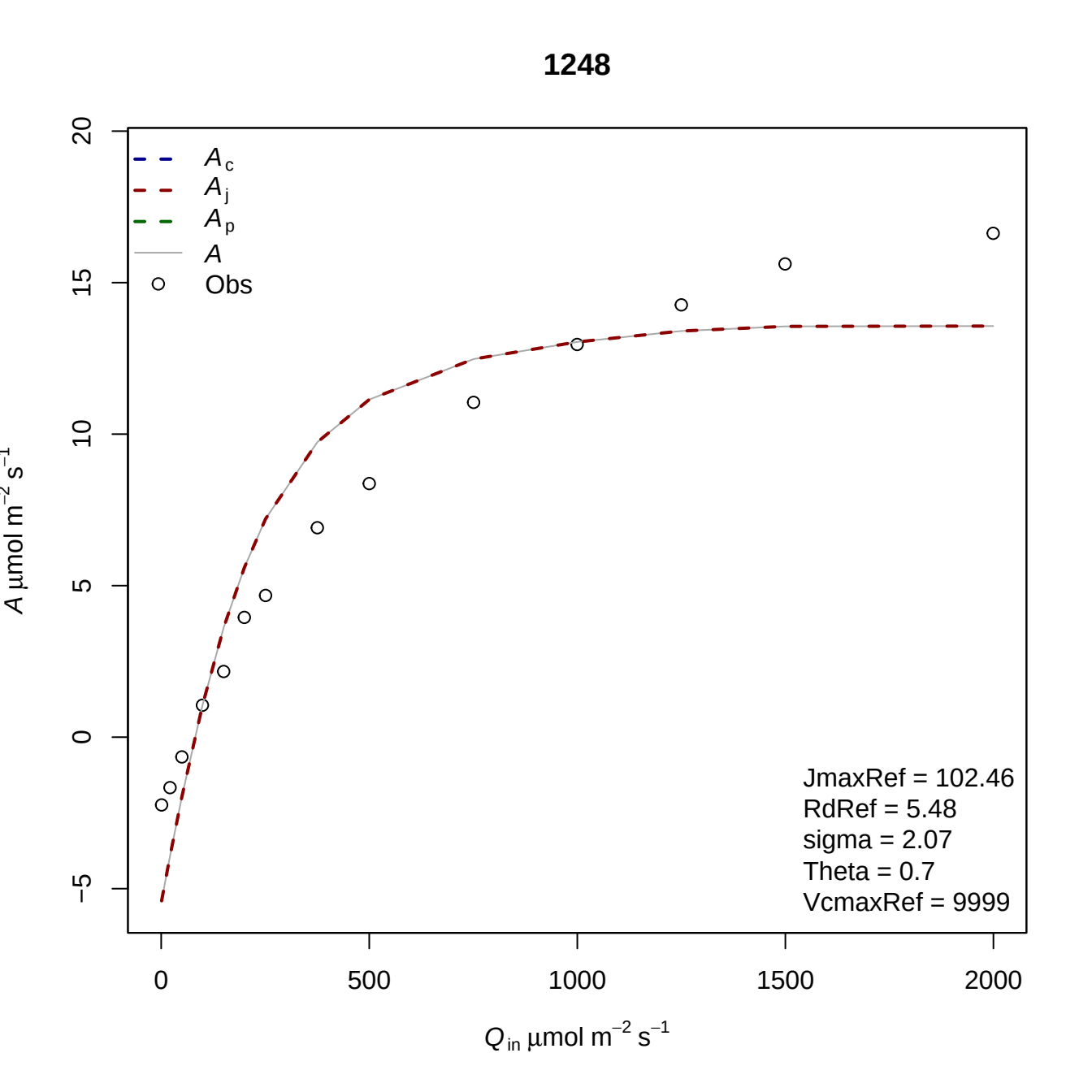

1243
A_c
A_j
A_p
A
Obs
JmaxRef = 63.33
RdRef = 3.73
sigma = 1.52
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

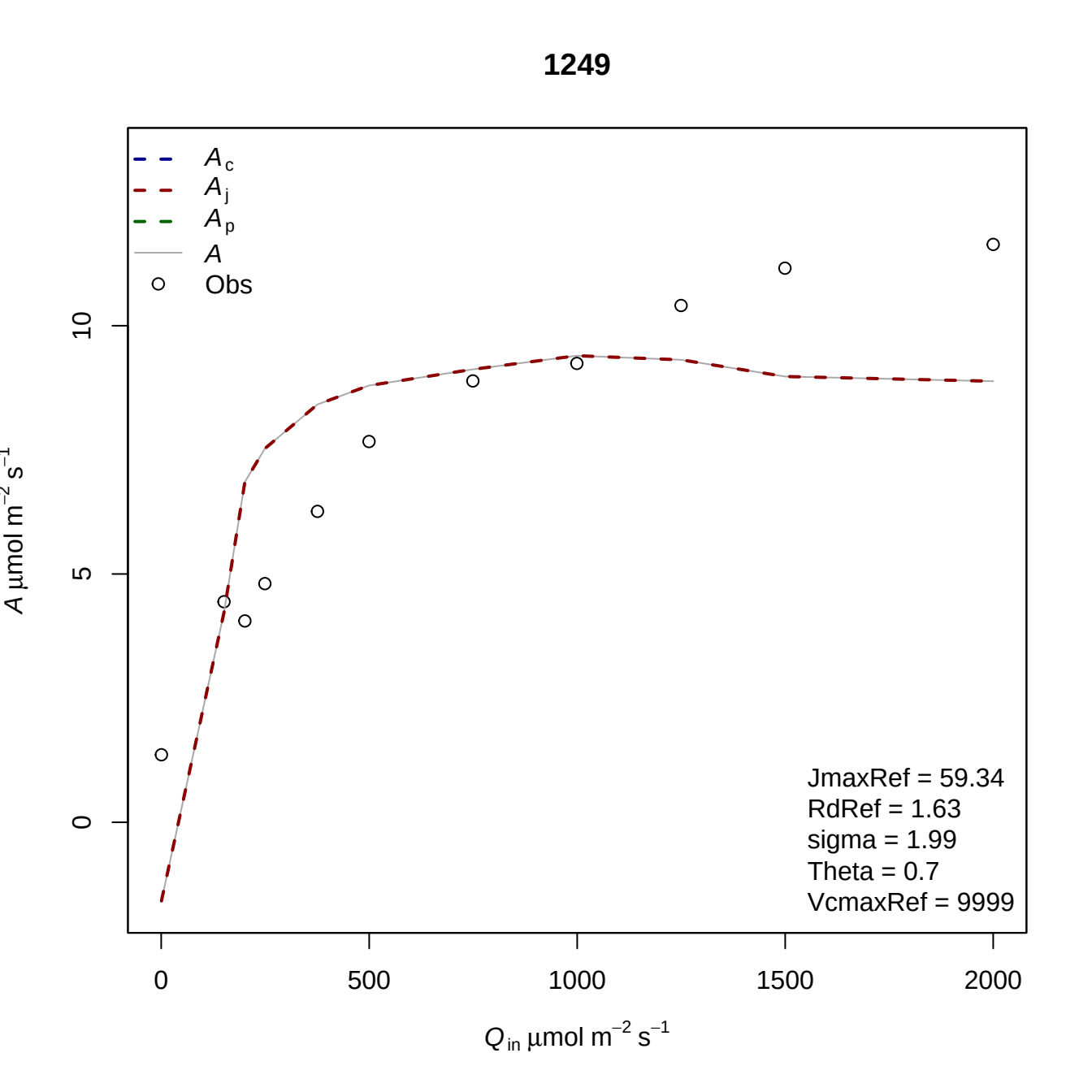

1244
A_c
A_j
A_p
A
Obs
JmaxRef = 106.03
RdRef = 5.57
sigma = 1.72
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

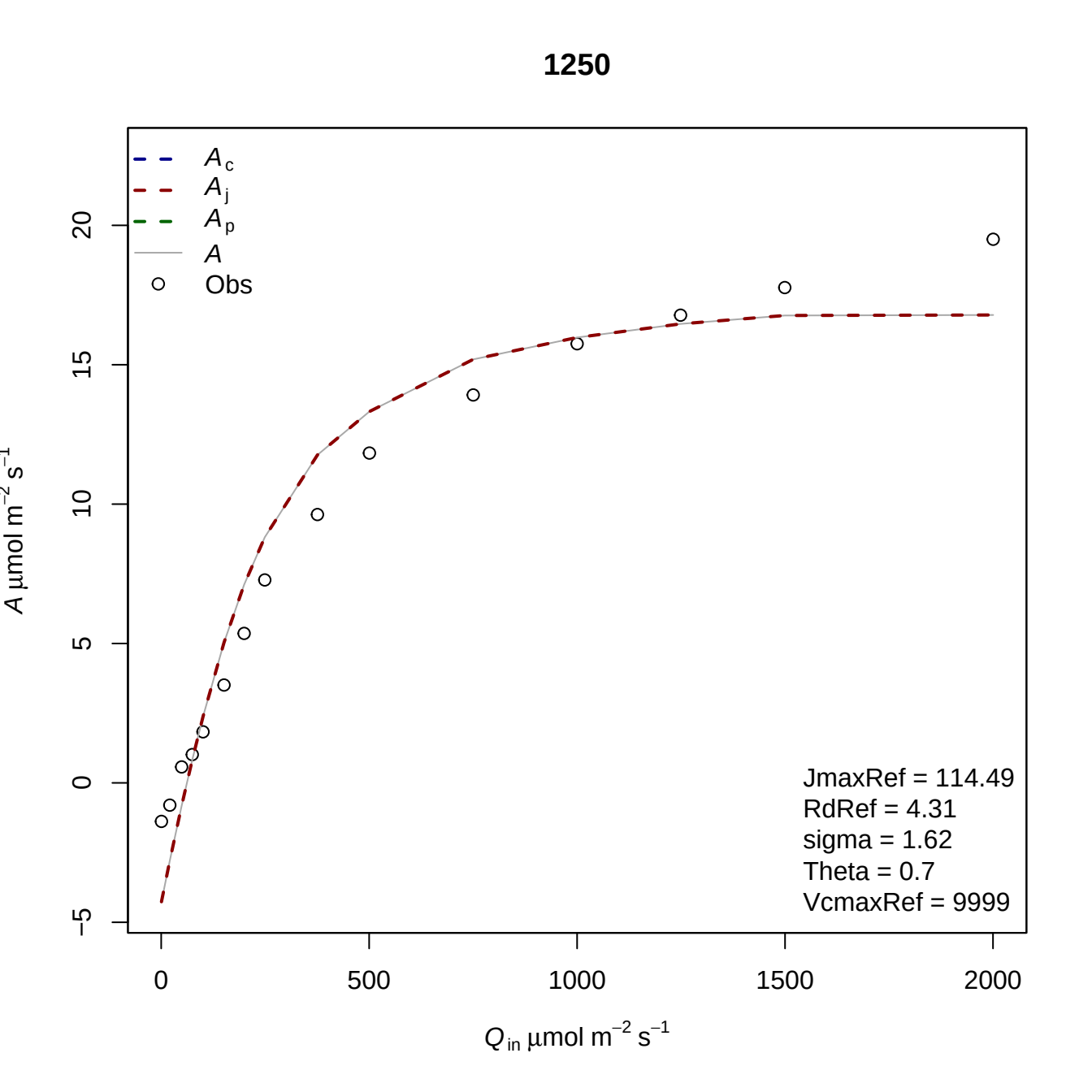

1245
A_c
A_j
A_p
A
Obs
JmaxRef = 105.18
RdRef = 4.34
sigma = 1.91
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
20
15
10
5
0
−5
0
500
1000
1500
2000

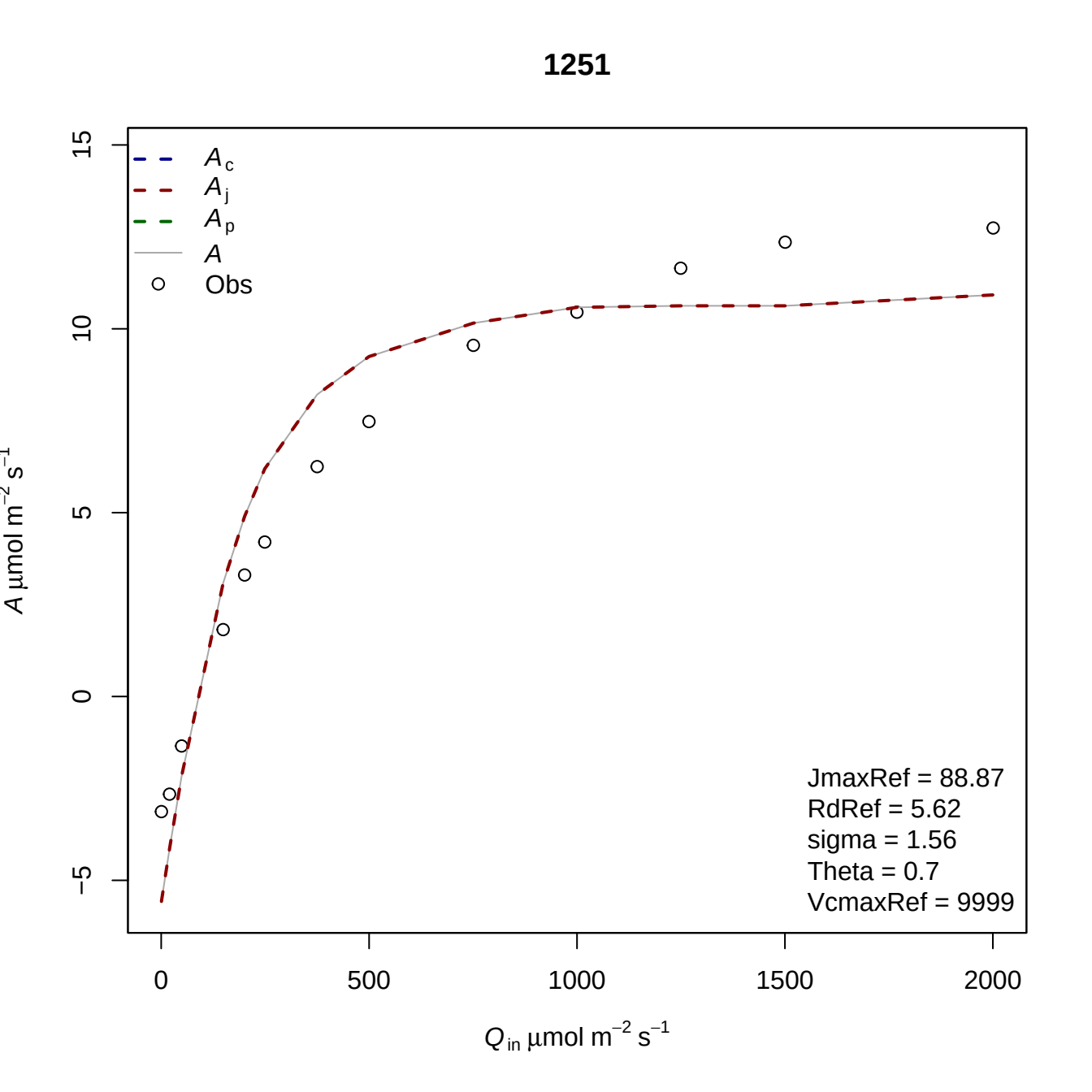

1246
A_c
A_j
A_p
A
Obs
JmaxRef = 85.51
RdRef = 4.26
sigma = 1.33
Theta = 0.7
VcmaxRef = 9999
15
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

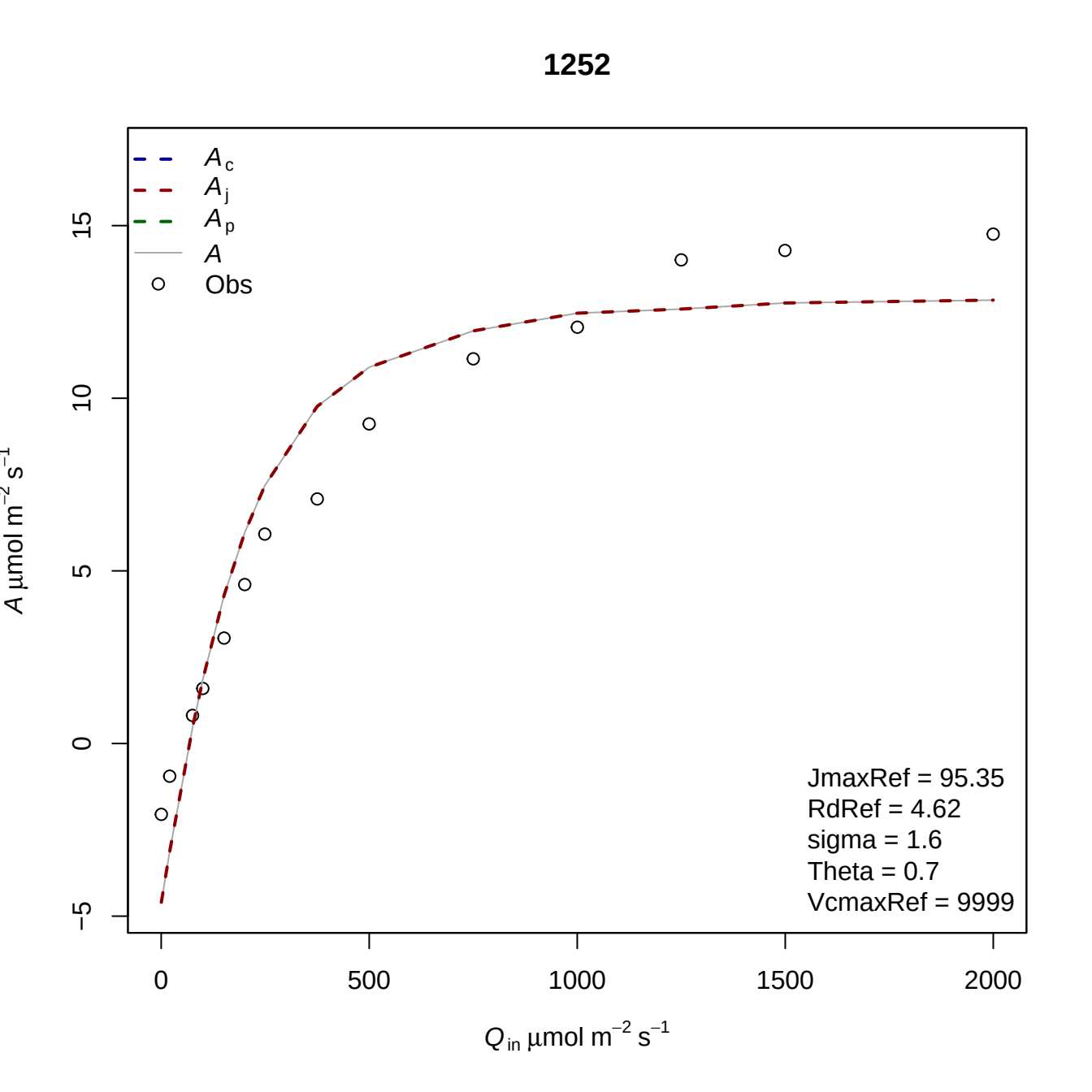

1247
A_c
A_j
A_p
A
Obs
JmaxRef = 87.77
RdRef = 3.96
sigma = 1.34
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

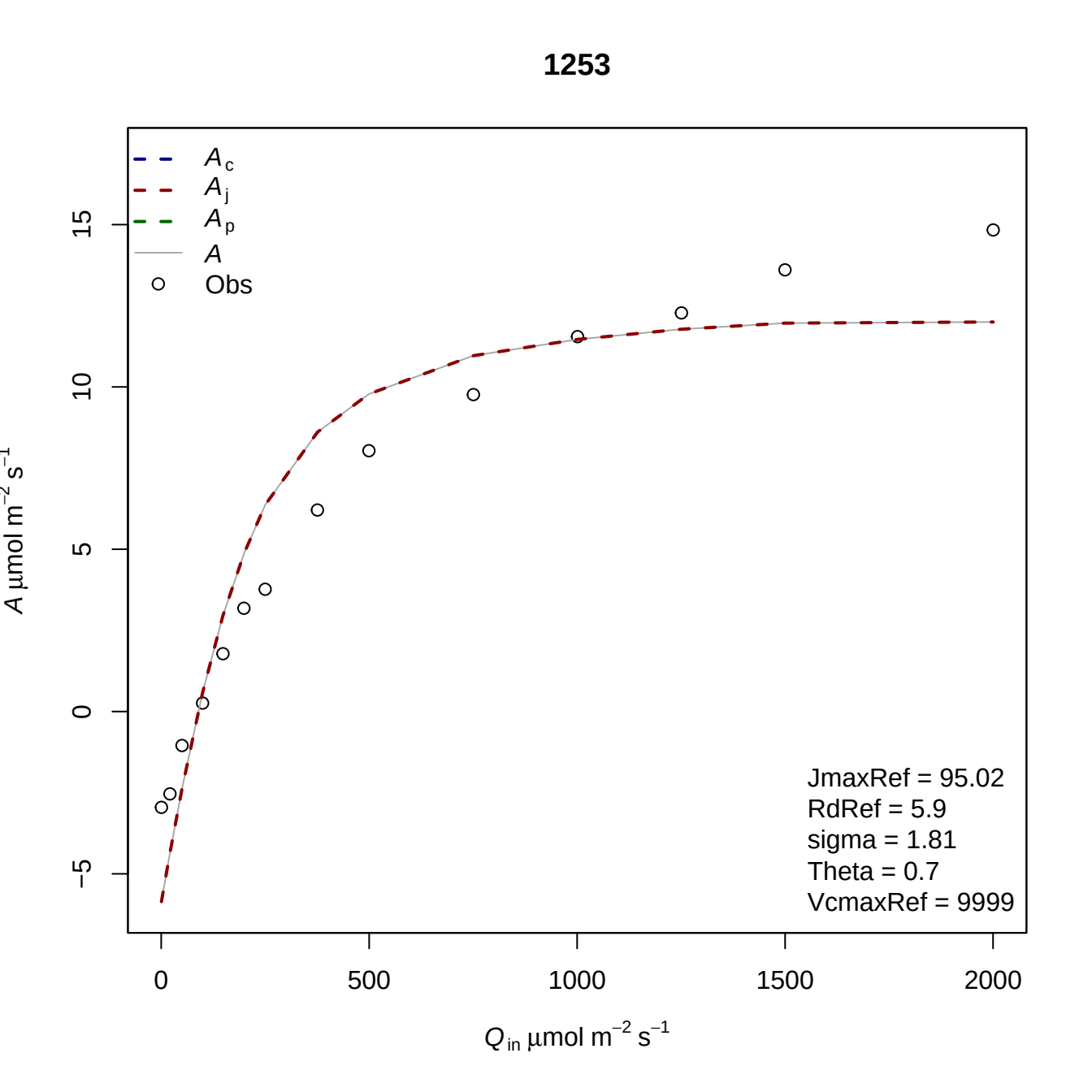

1248
A_c
A_j
A_p
A
Obs
JmaxRef = 102.46
RdRef = 5.48
sigma = 2.07
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
20
15
10
5
0
−5
0
500
1000
1500
2000

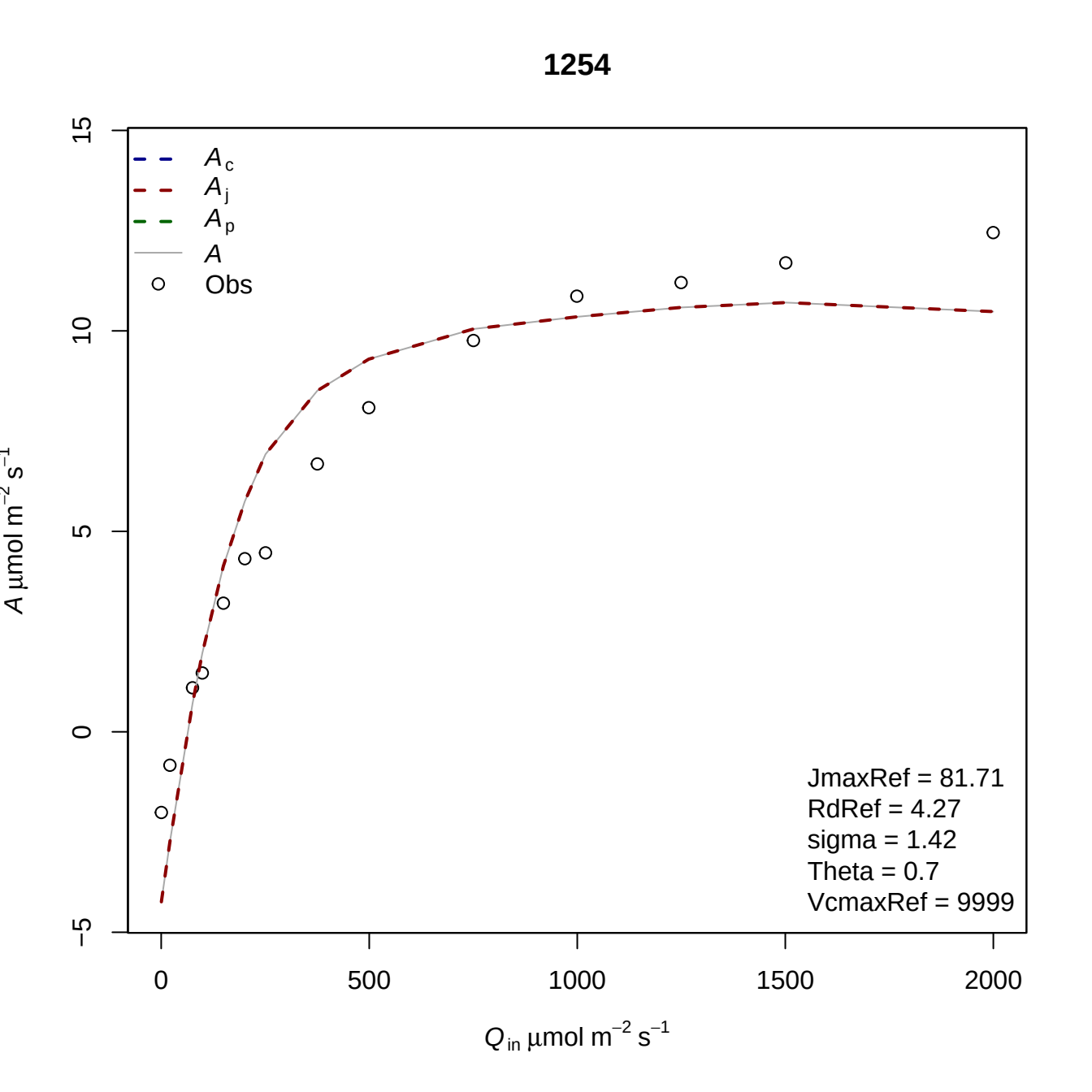

1249
A_c
A_j
A_p
A
Obs
JmaxRef = 59.34
RdRef = 1.63
sigma = 1.99
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

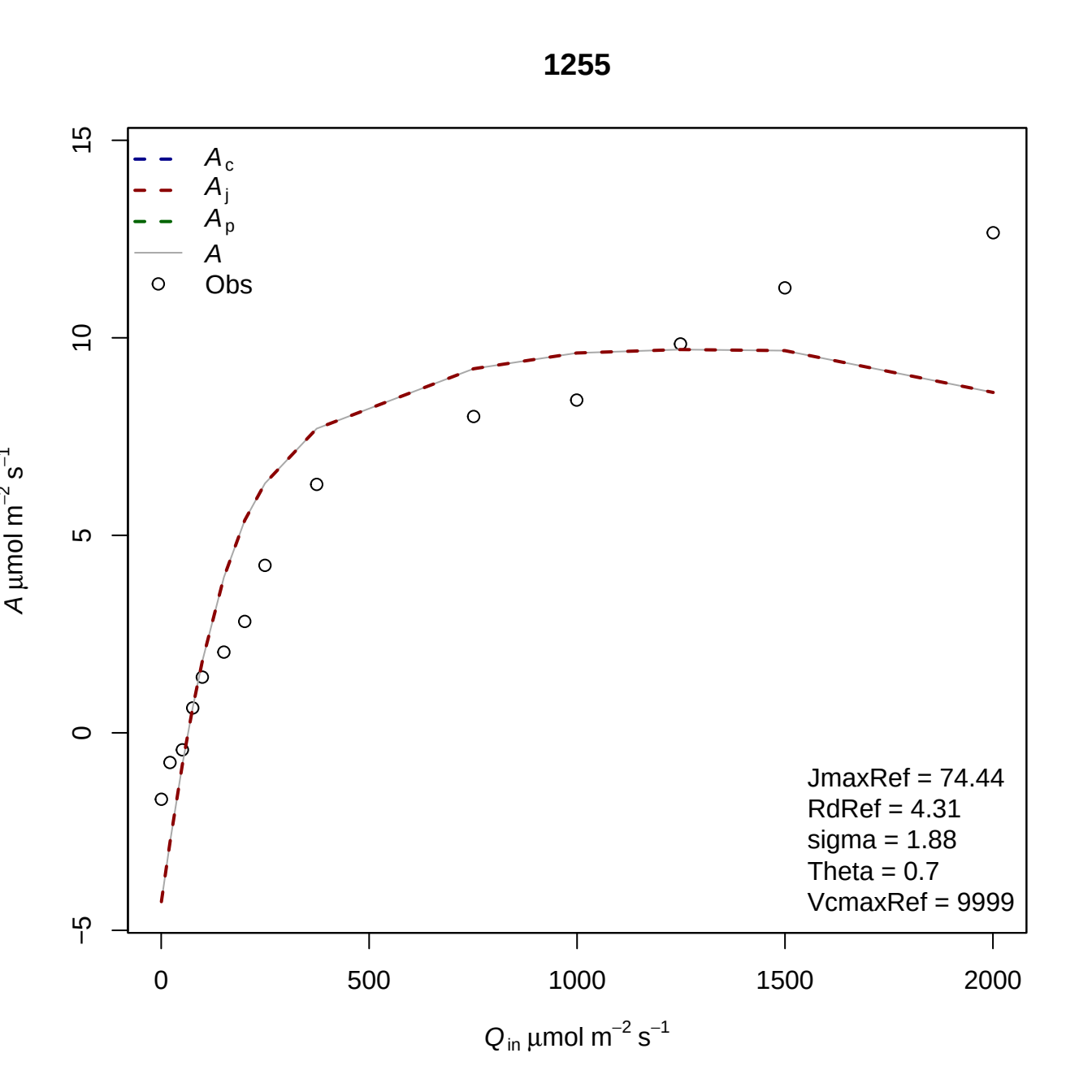

1250
A_c
A_j
A_p
A
Obs
JmaxRef = 114.49
RdRef = 4.31
sigma = 1.62
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

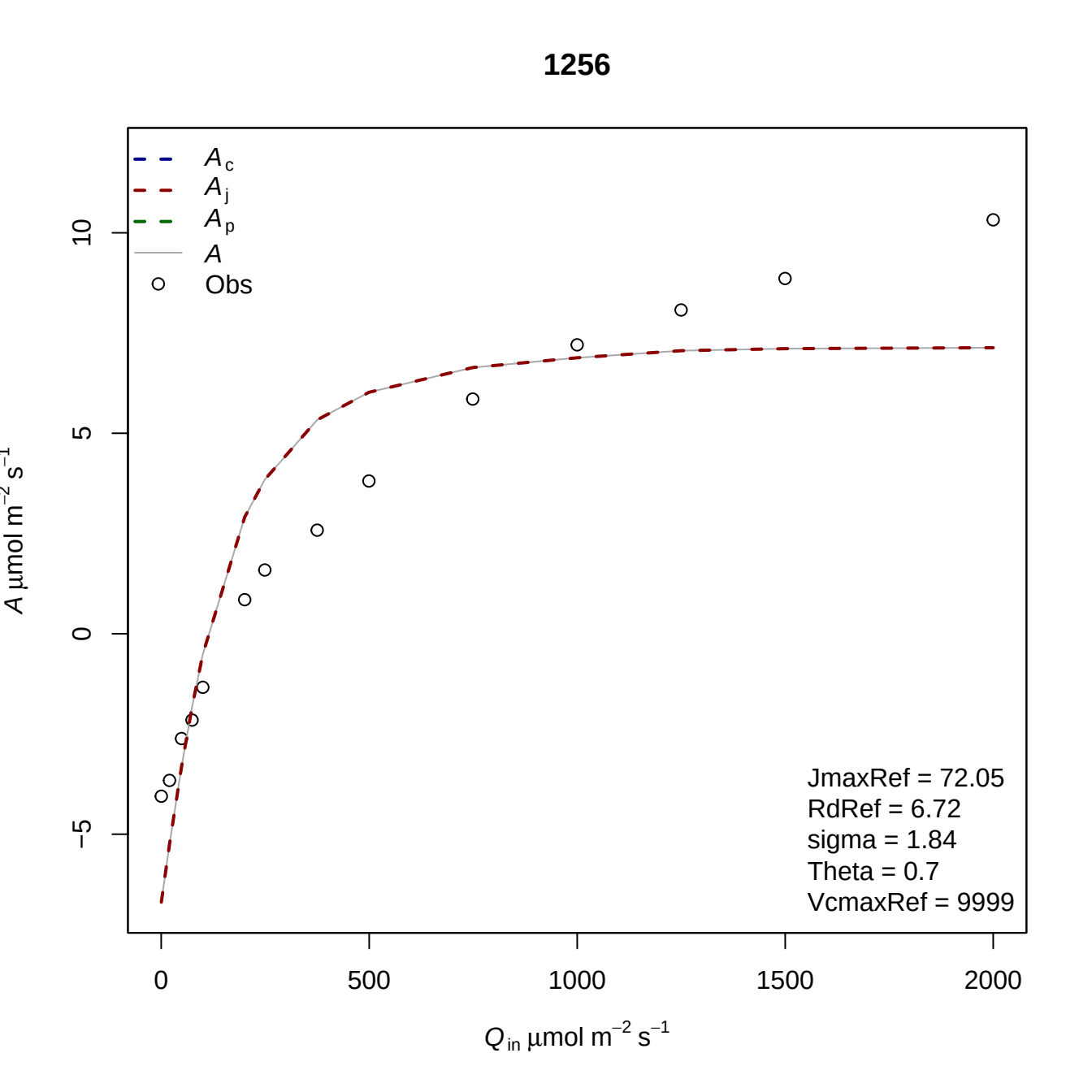

1251
A_c
A_j
A_p
A
Obs
JmaxRef = 88.87
RdRef = 5.62
sigma = 1.56
Theta = 0.7
VcmaxRef = 9999
15
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

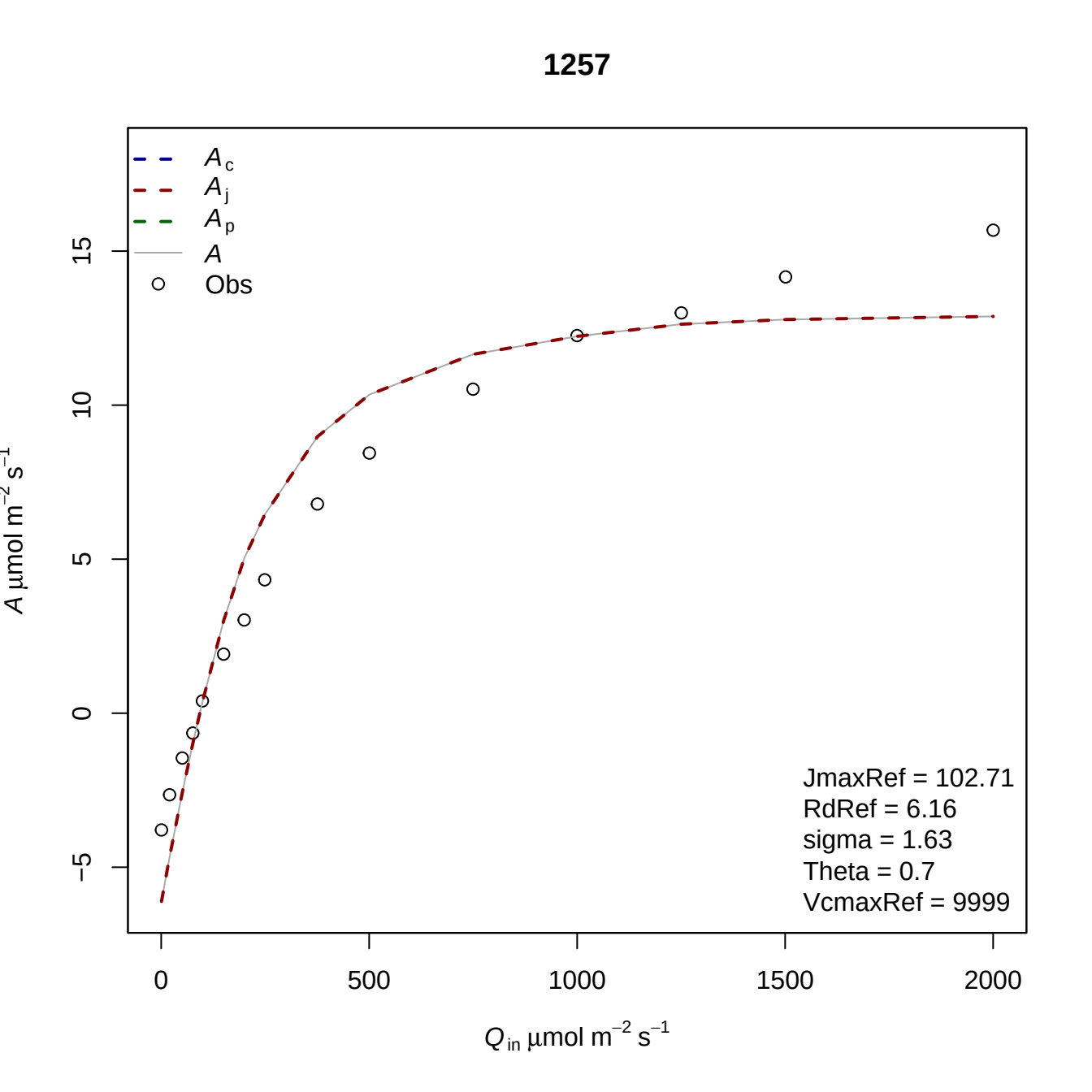

1252
A_c
A_j
A_p
A
Obs
JmaxRef = 95.35
RdRef = 4.62
sigma = 1.6
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

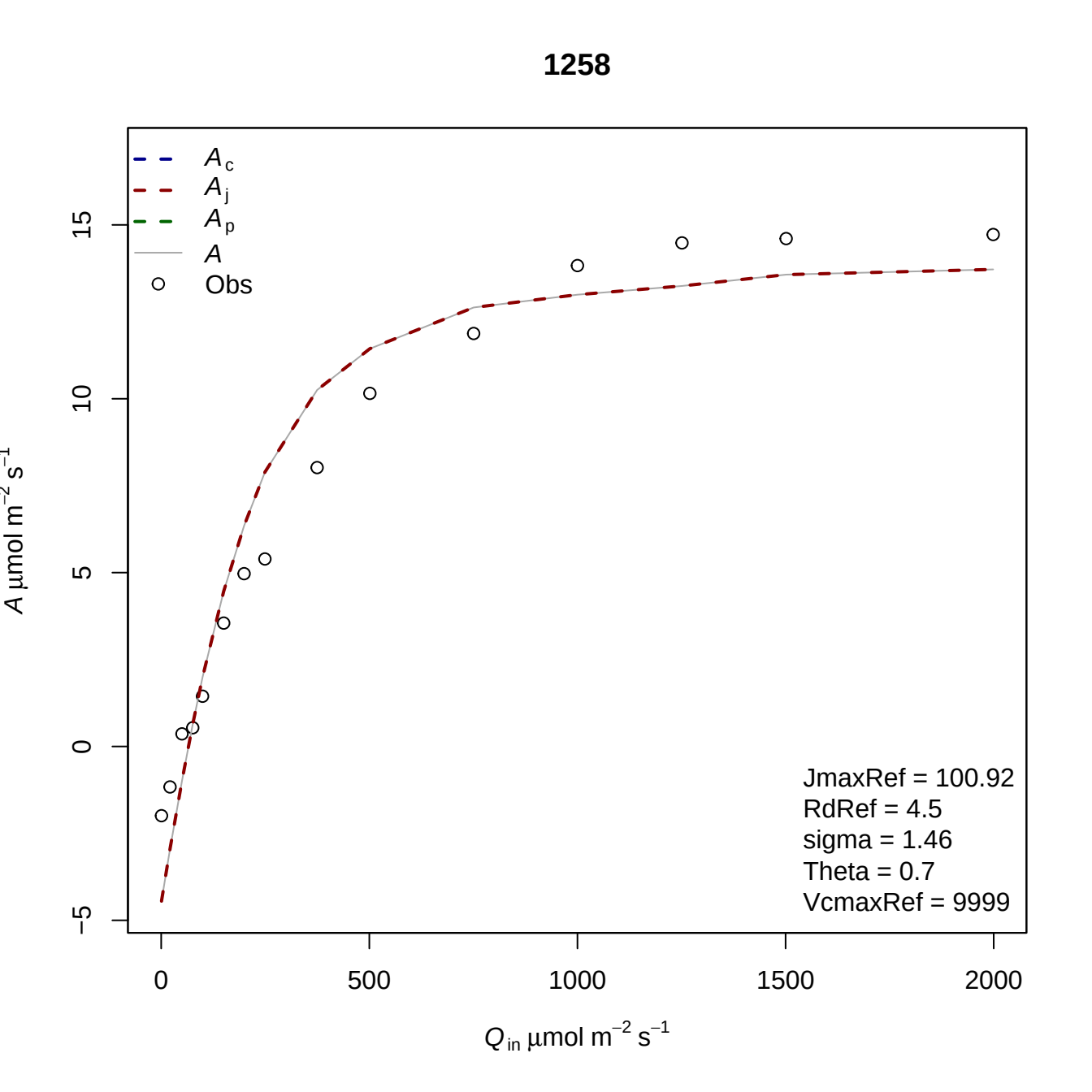

1253
A_c
A_j
A_p
A
Obs
JmaxRef = 95.02
RdRef = 5.9
sigma = 1.81
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
15
10
5
0
−5
0
500
1000
1500
2000

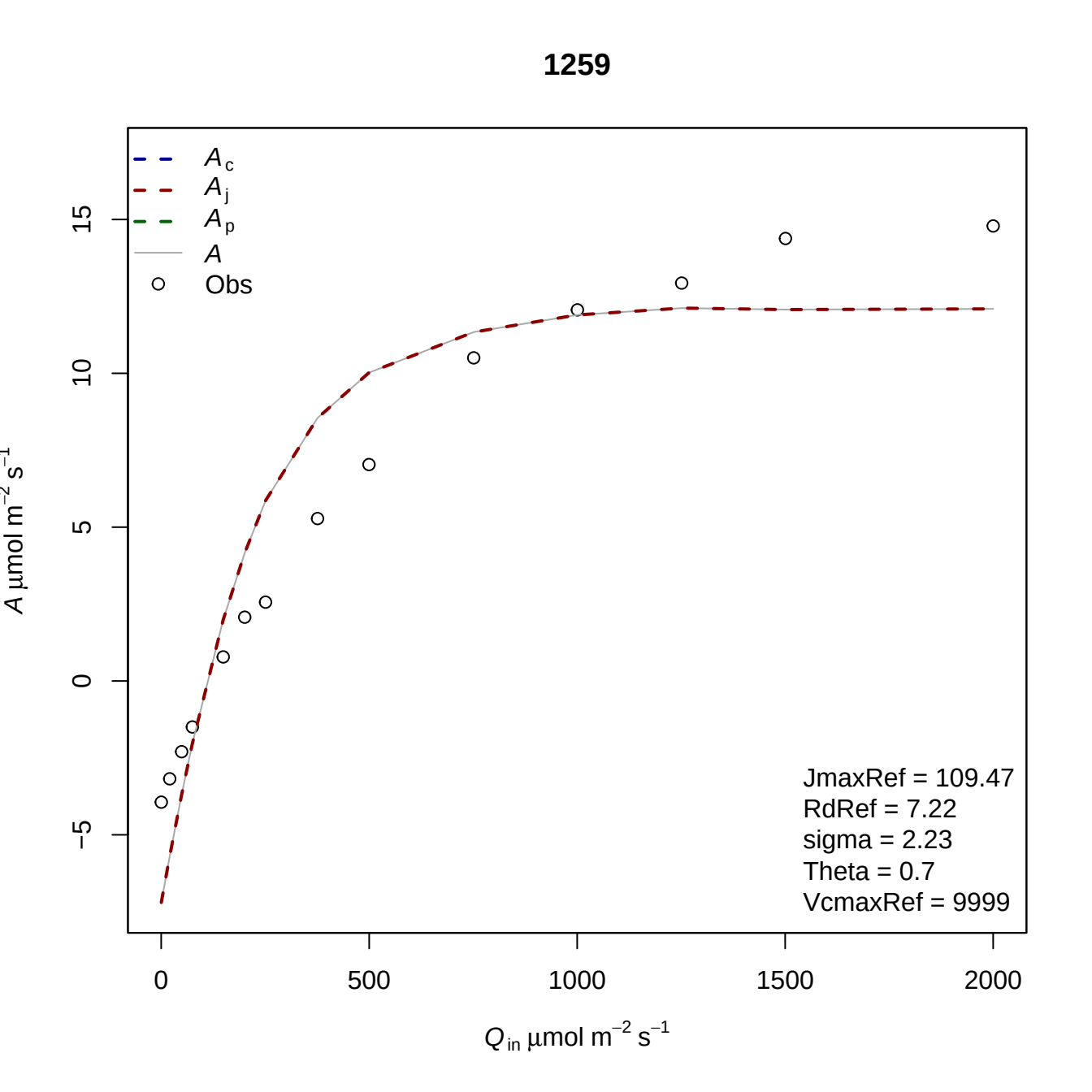

1254
A_c
A_j
A_p
A
Obs
JmaxRef = 81.71
RdRef = 4.27
sigma = 1.42
Theta = 0.7
VcmaxRef = 9999
15
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

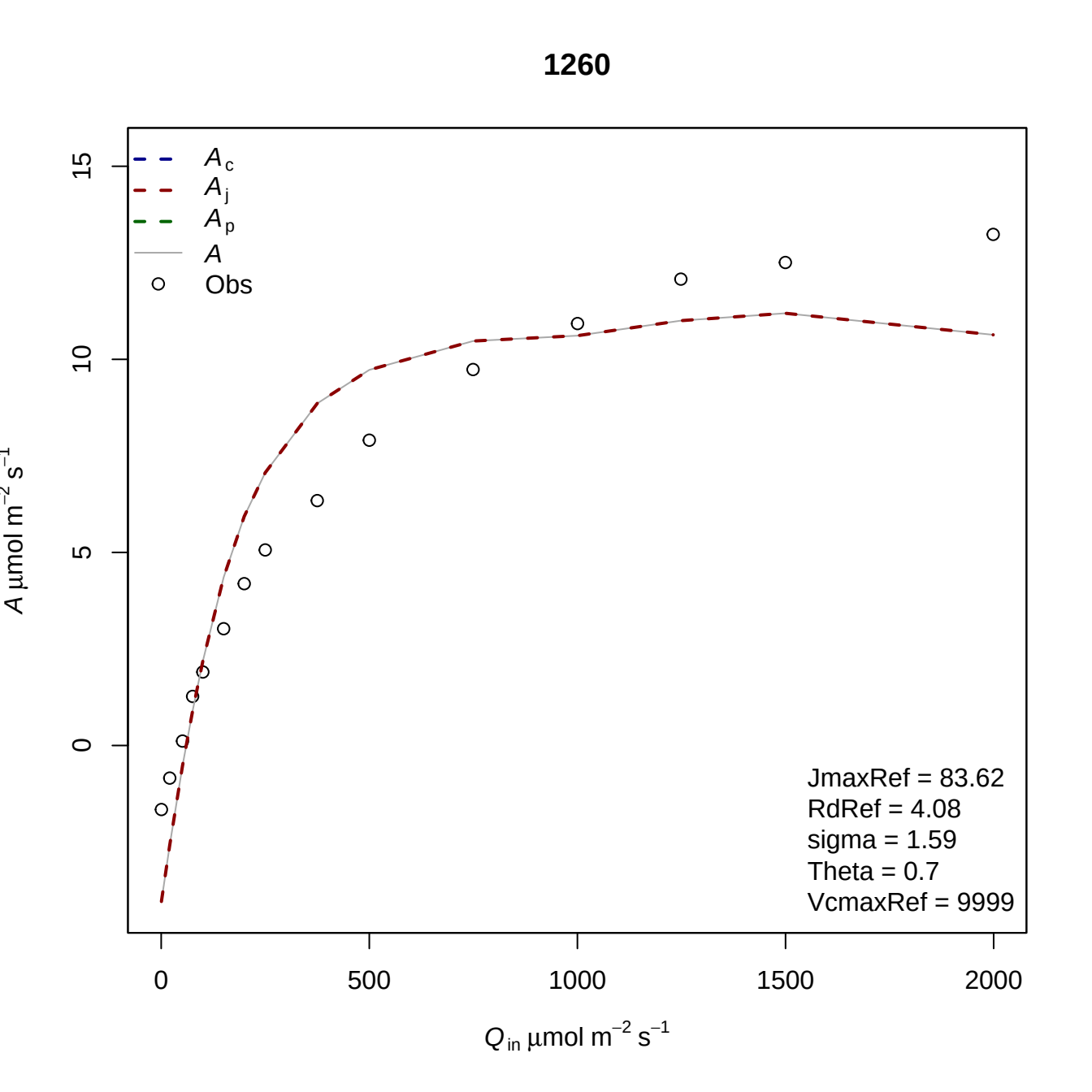

1255
A_c
A_j
A_p
A
Obs
JmaxRef = 74.44
RdRef = 4.31
sigma = 1.88
Theta = 0.7
VcmaxRef = 9999
15
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

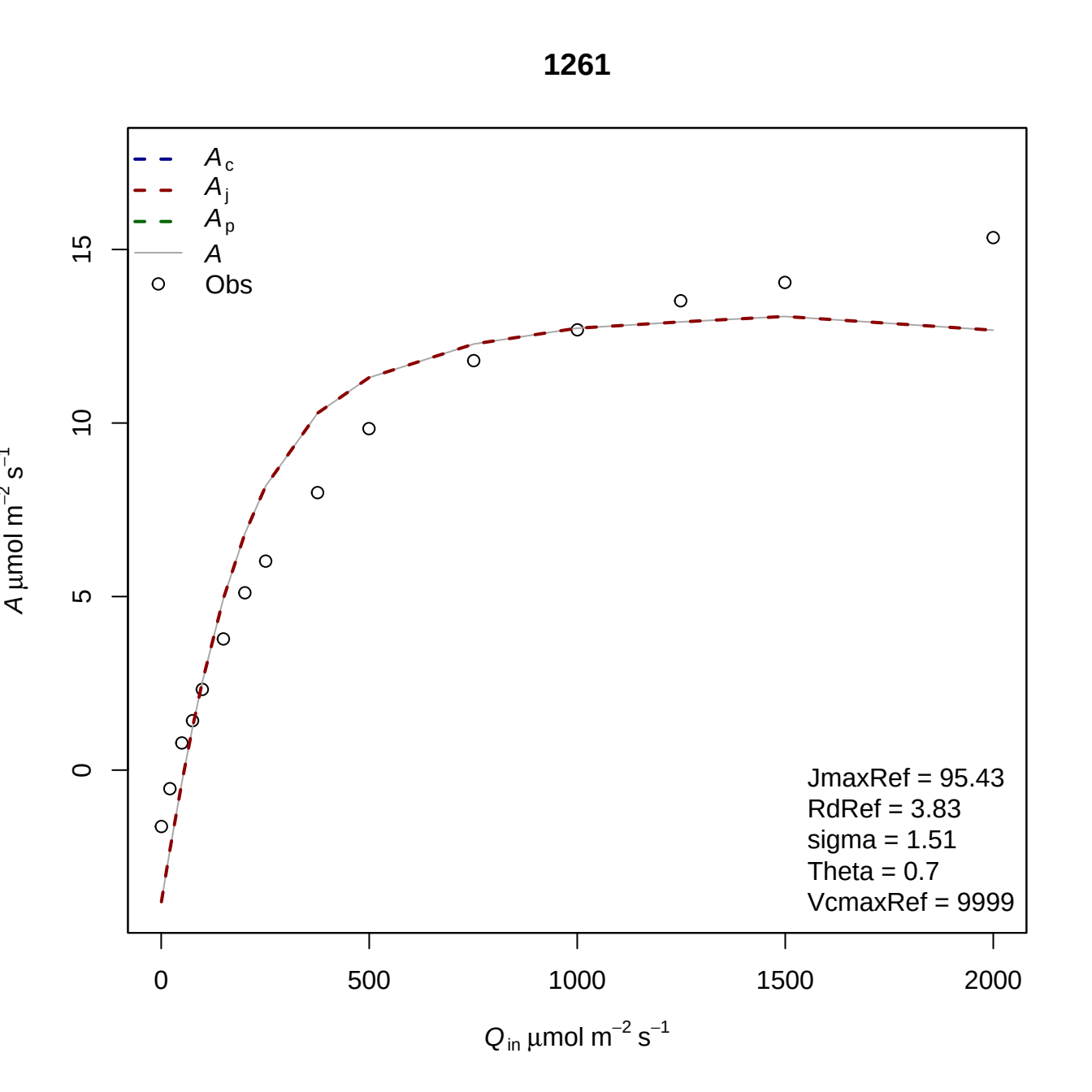

1256
A_c
A_j
A_p
A
Obs
JmaxRef = 72.05
RdRef = 6.72
sigma = 1.84
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
5
0
−5
0
500
1000
1500
2000

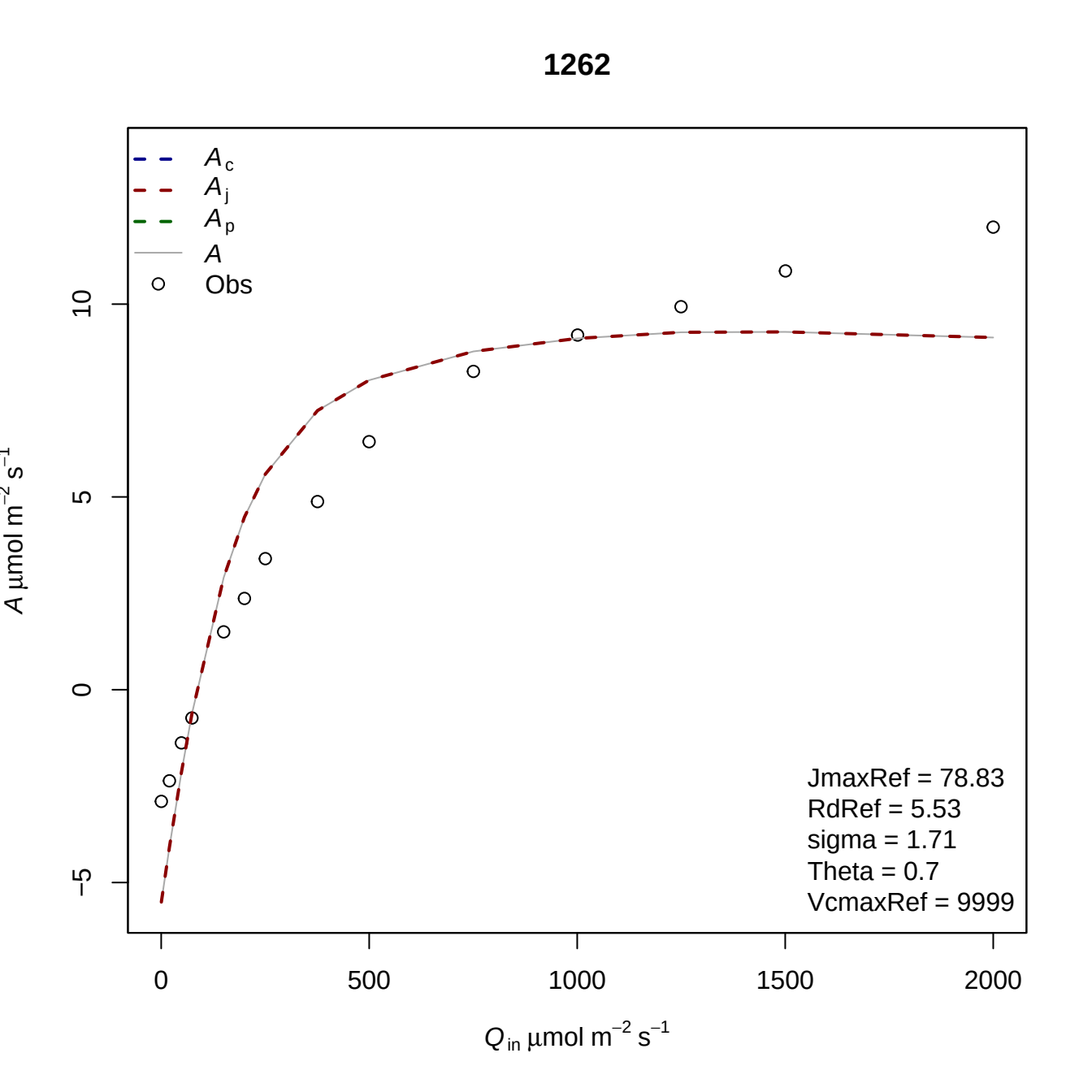

1257
A_c
A_j
A_p
A
Obs
JmaxRef = 102.71
RdRef = 6.16
sigma = 1.63
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

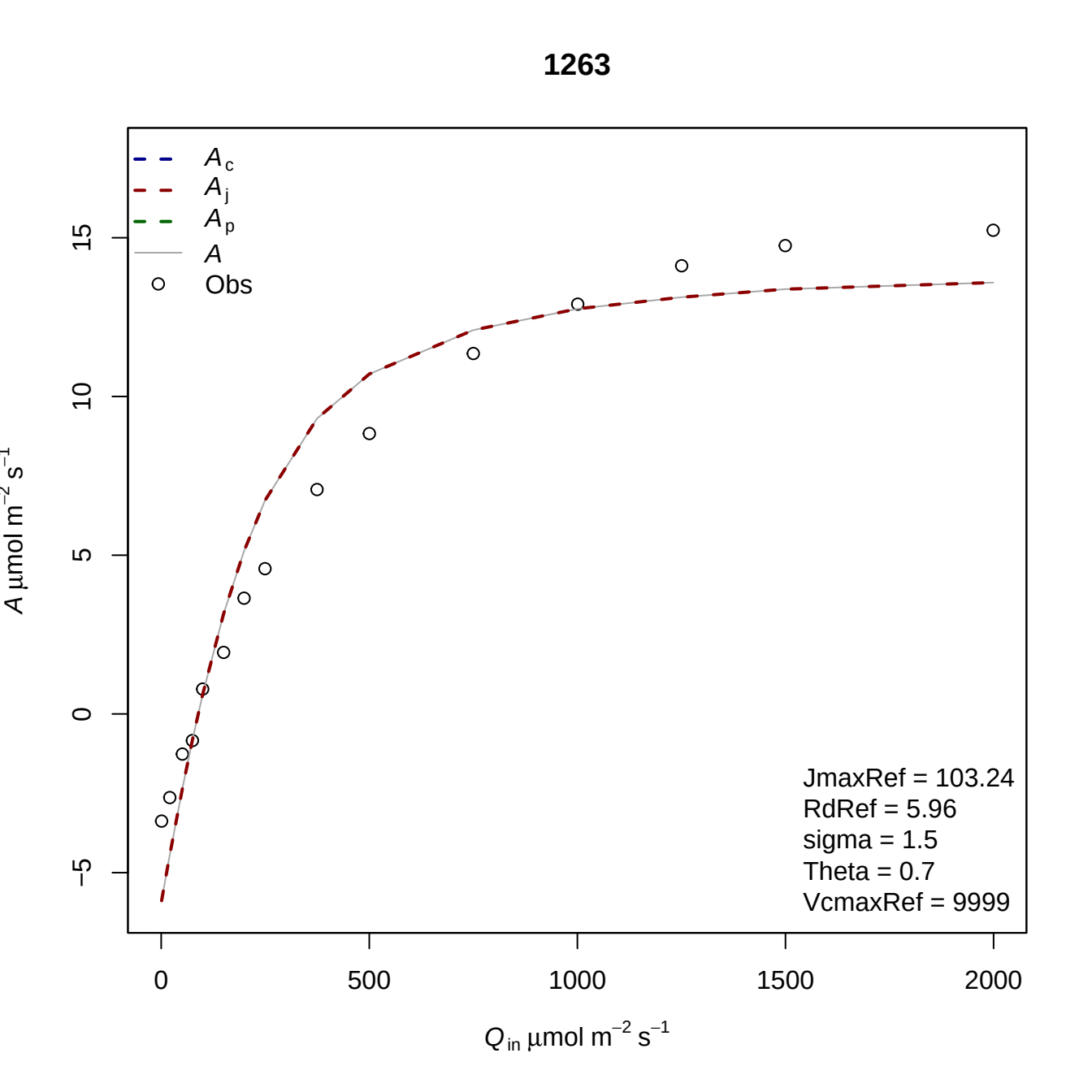

1258
A_c
A_j
A_p
A
Obs
JmaxRef = 100.92
RdRef = 4.5
sigma = 1.46
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

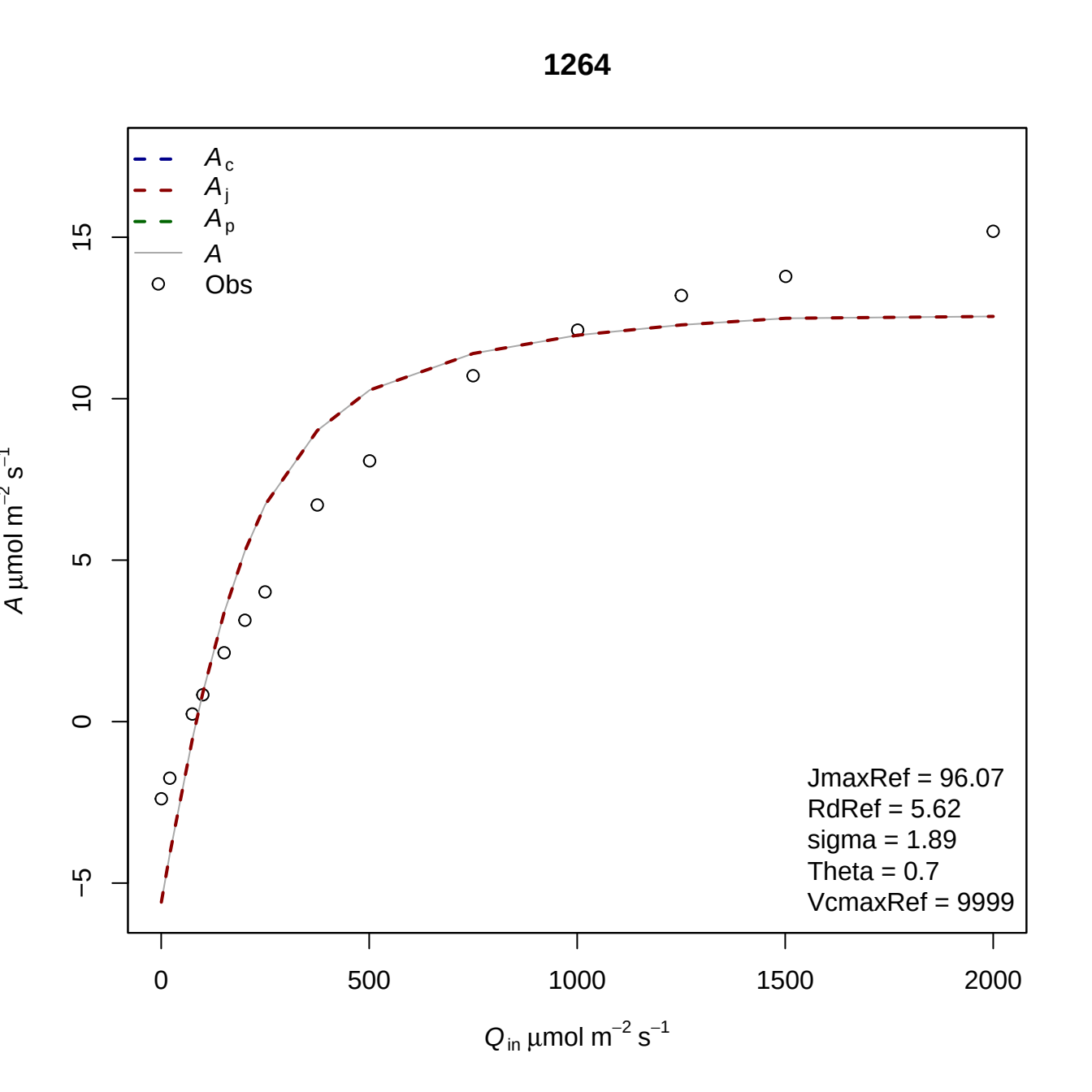

1259
A_c
A_j
A_p
A
Obs
JmaxRef = 109.47
RdRef = 7.22
sigma = 2.23
Theta = 0.7
VcmaxRef = 9999
15
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

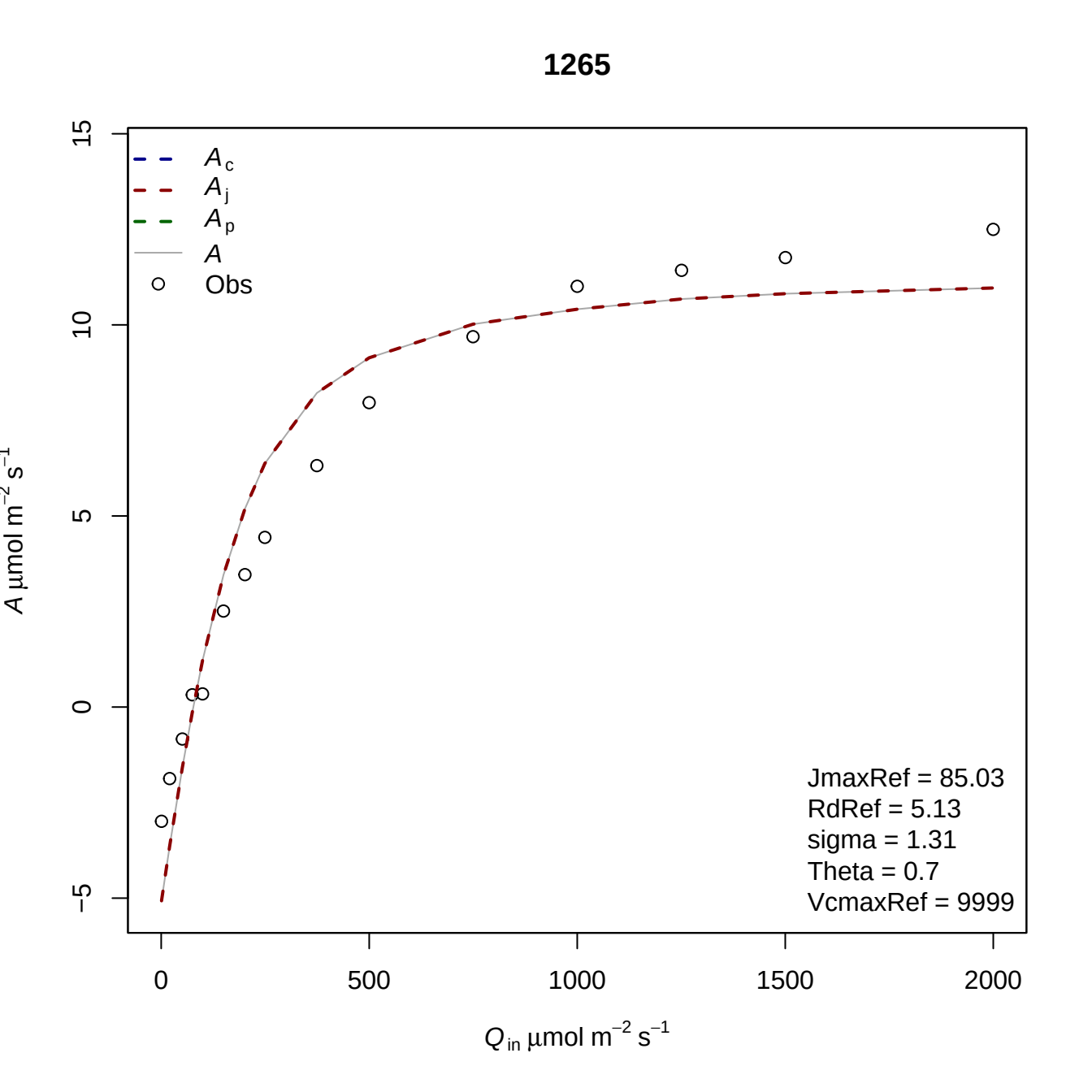

1260
A_c
A_j
A_p
A
Obs
JmaxRef = 83.62
RdRef = 4.08
sigma = 1.59
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

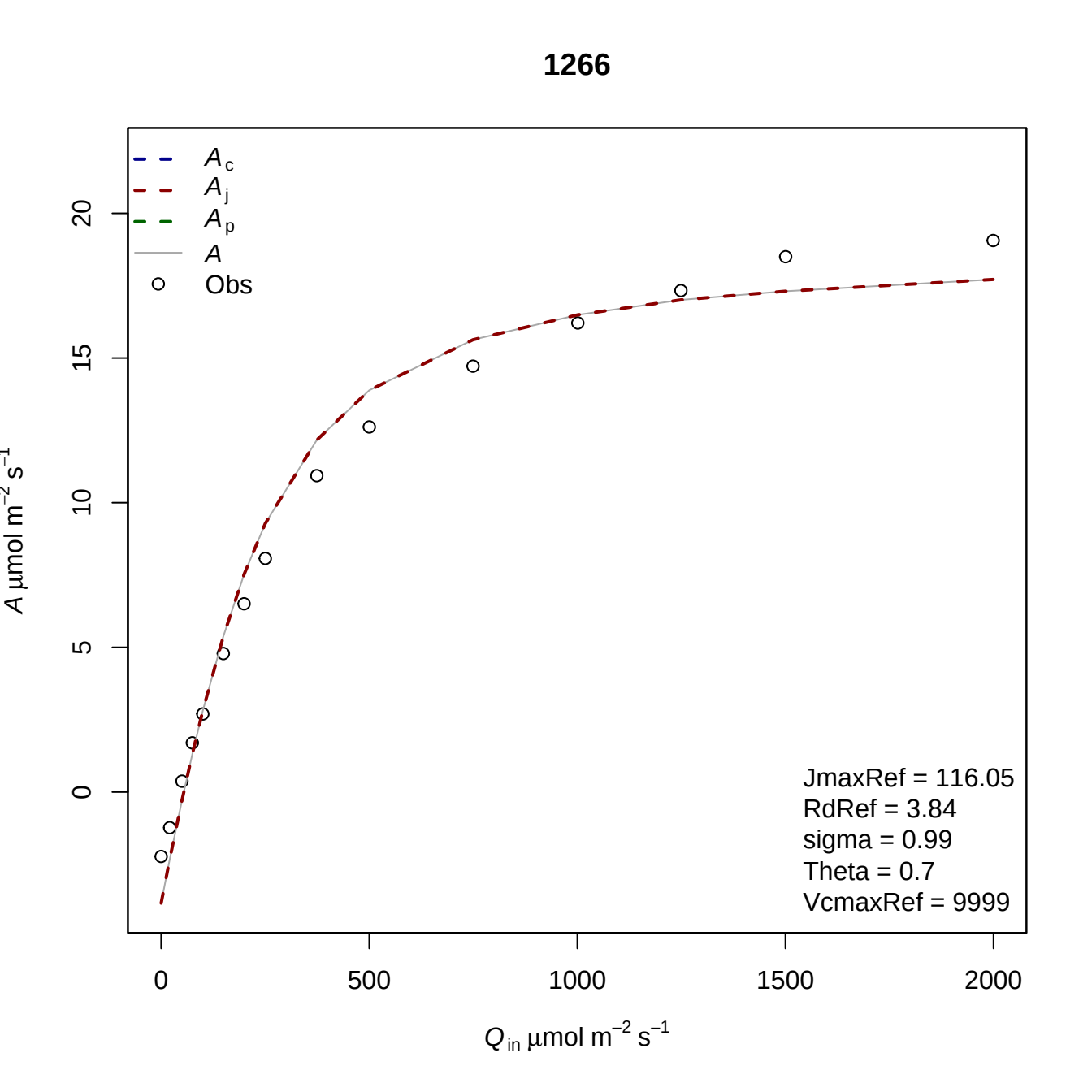

1261
A_c
A_j
A_p
A
Obs
JmaxRef = 95.43
RdRef = 3.83
sigma = 1.51
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

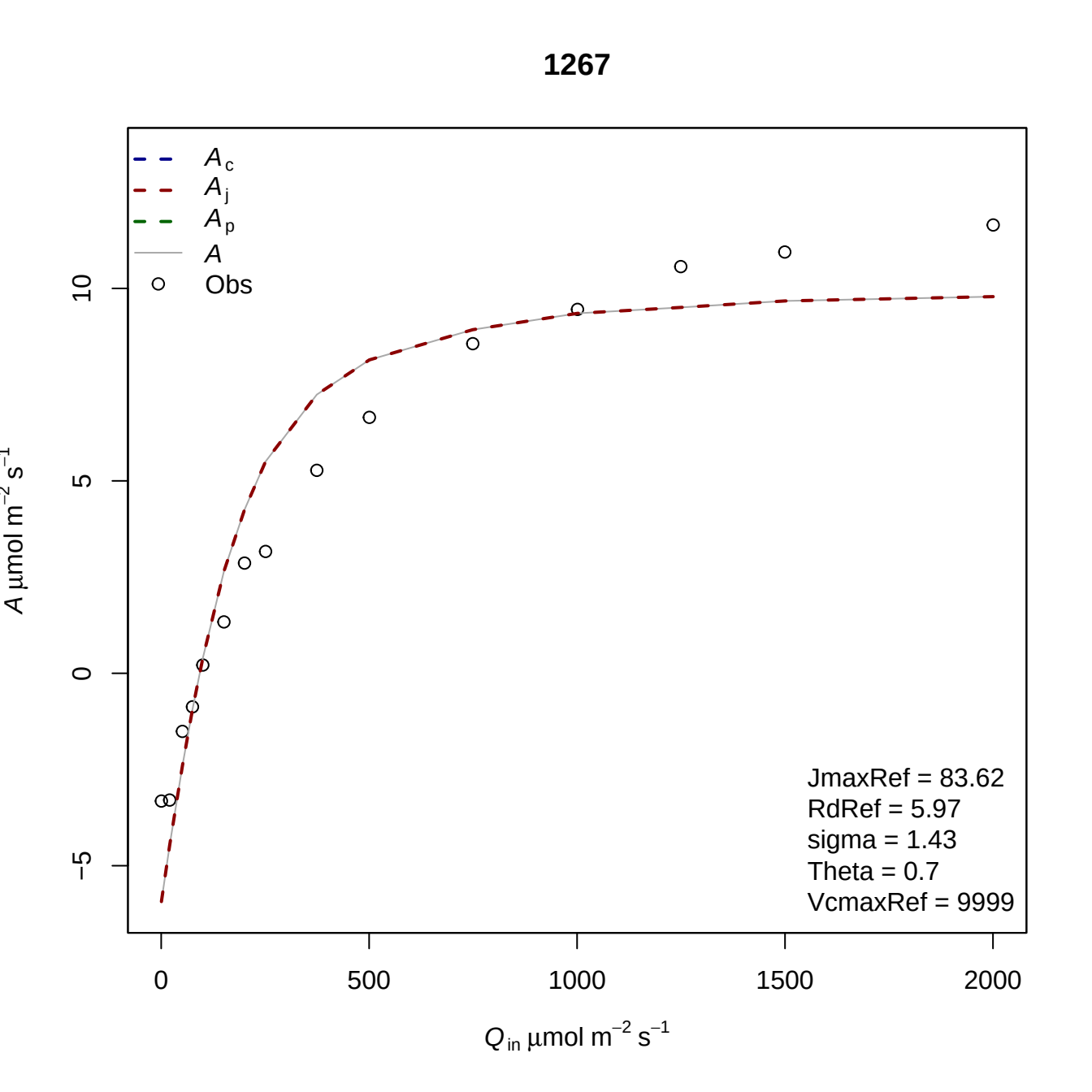

1262
A_c
A_j
A_p
A
Obs
JmaxRef = 78.83
RdRef = 5.53
sigma = 1.71
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

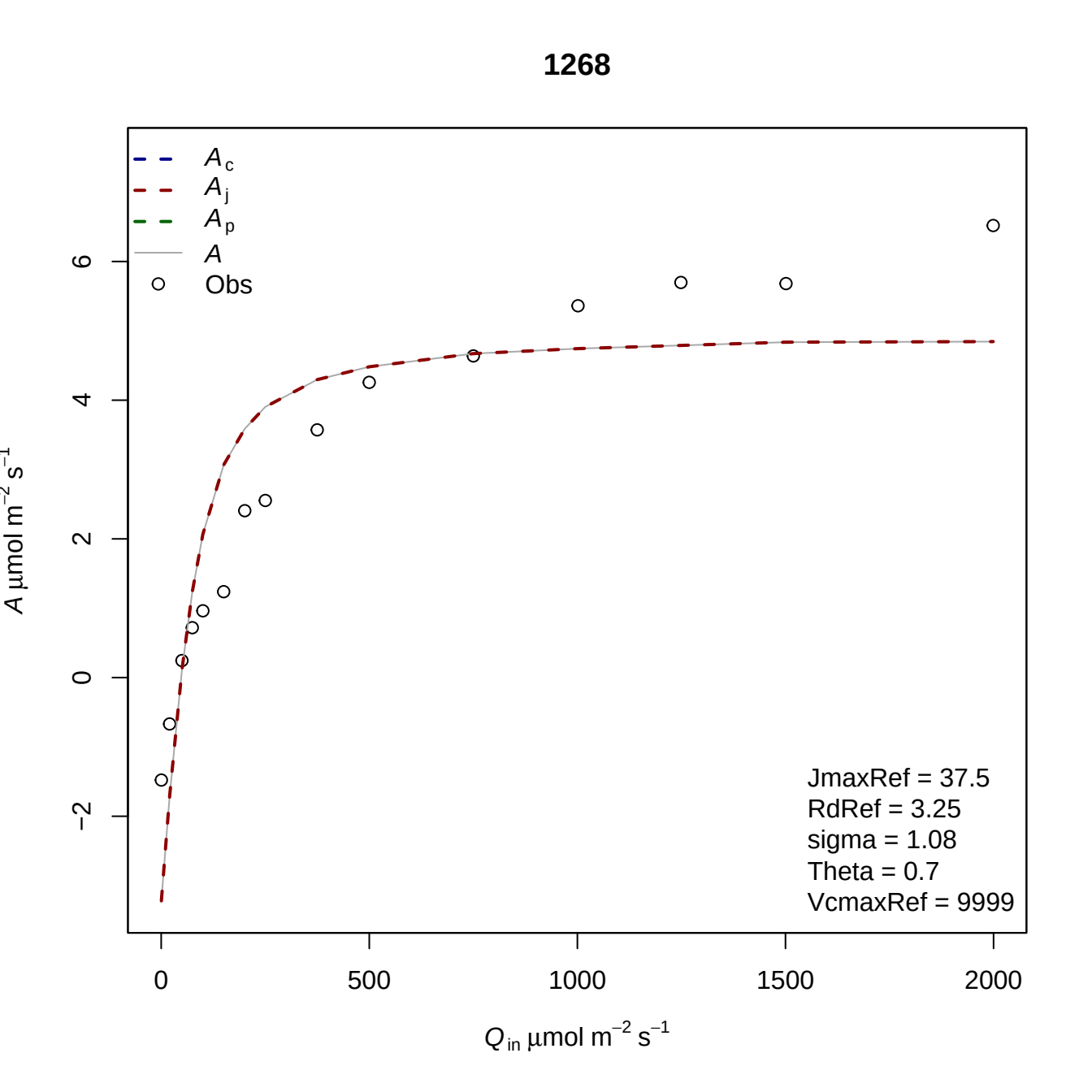

1263
A_c
A_j
A_p
A
Obs
JmaxRef = 103.24
RdRef = 5.96
sigma = 1.5
Theta = 0.7
VcmaxRef = 9999
15
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

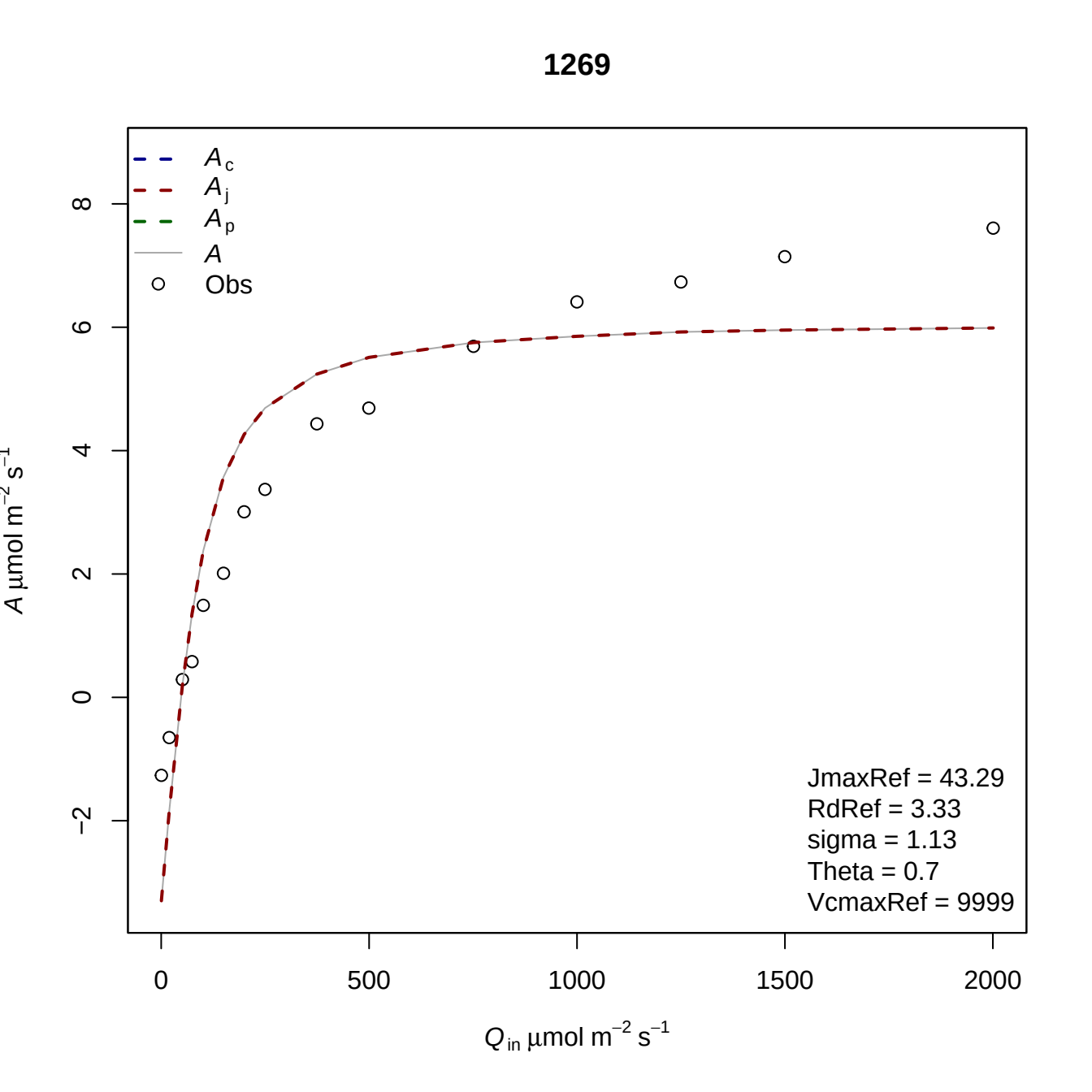

1264
A_c
A_j
A_p
A
Obs
JmaxRef = 96.07
RdRef = 5.62
sigma = 1.89
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

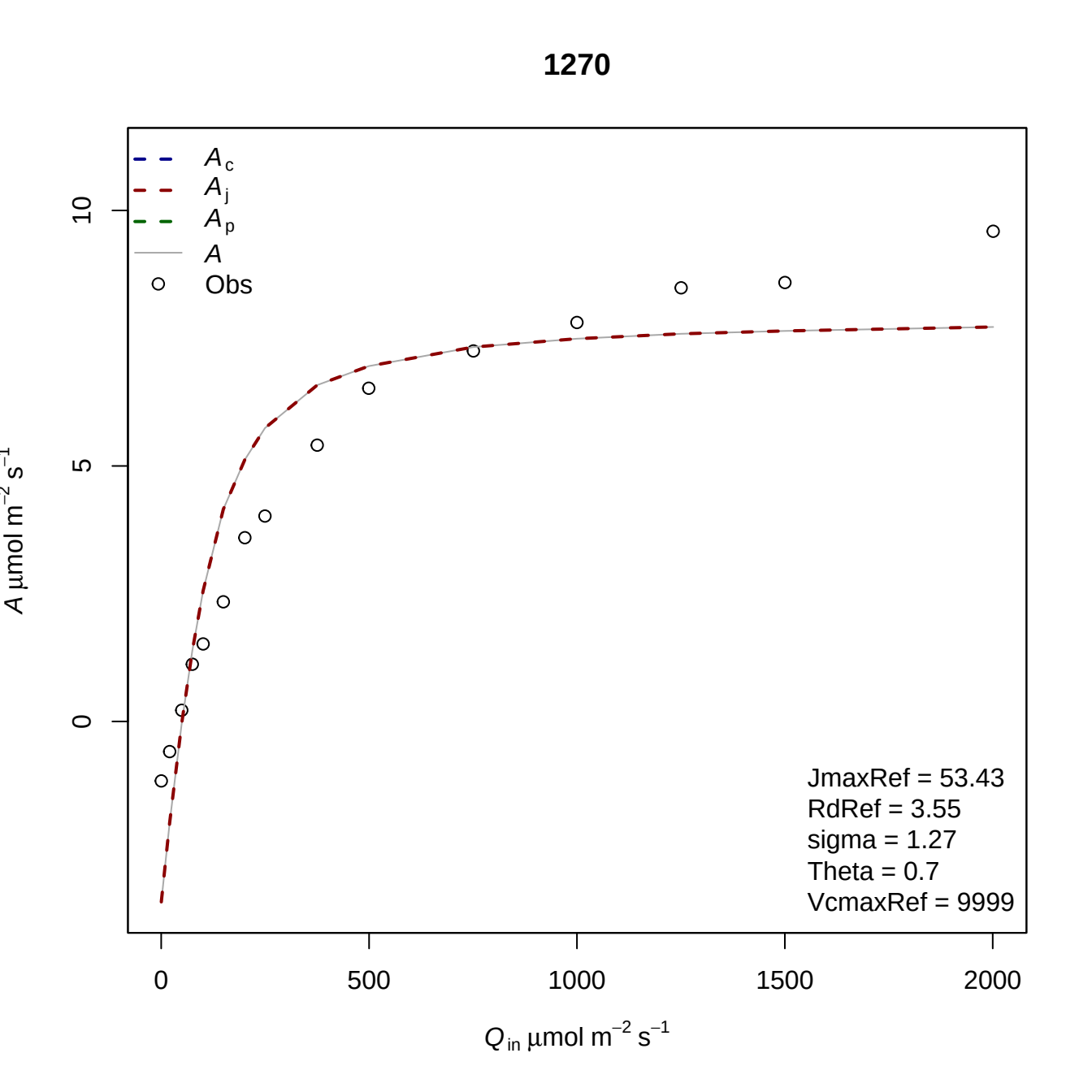

1265
A_c
A_j
A_p
A
Obs
JmaxRef = 85.03
RdRef = 5.13
sigma = 1.31
Theta = 0.7
VcmaxRef = 9999
15
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

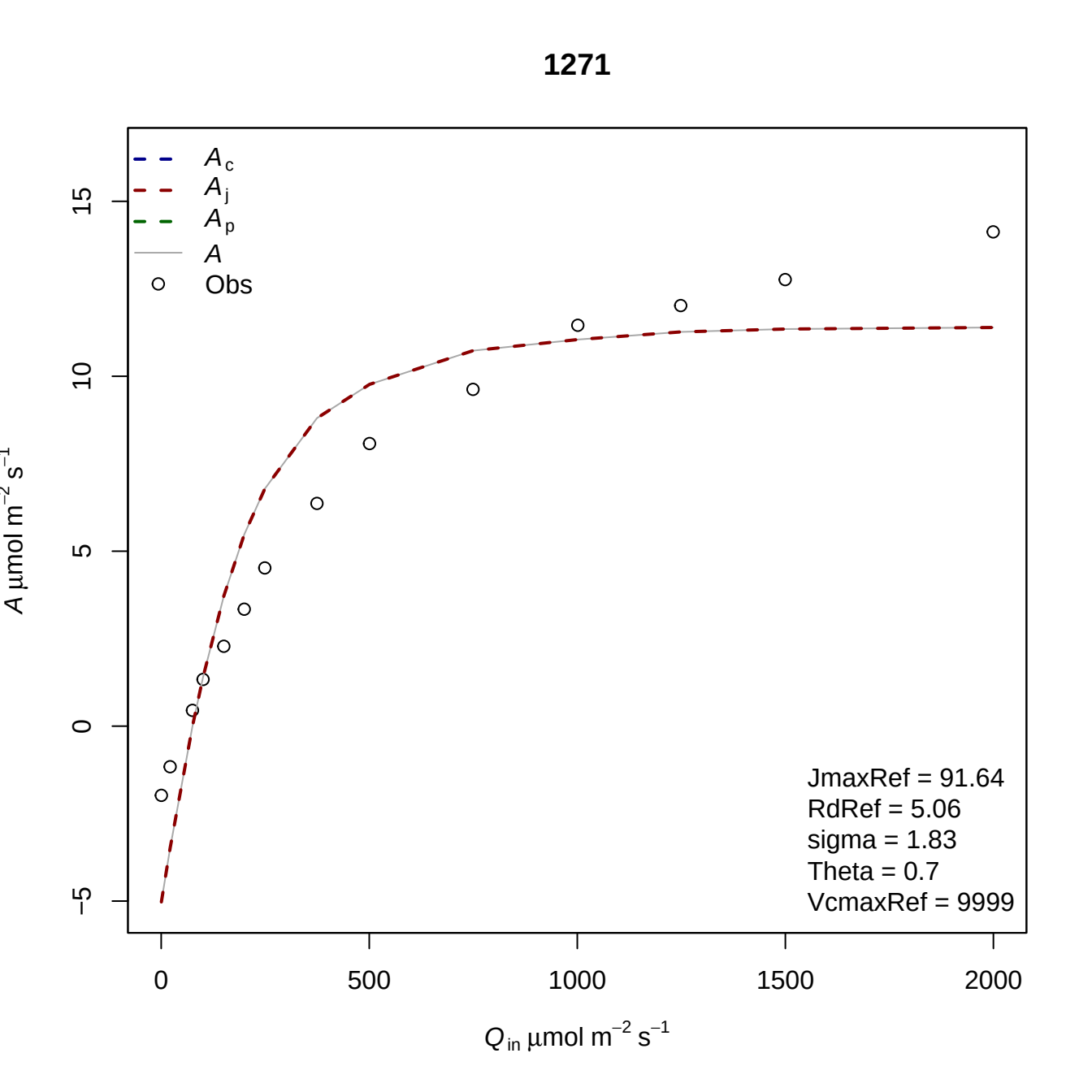

1266
A_c
A_j
A_p
A
Obs
JmaxRef = 116.05
RdRef = 3.84
sigma = 0.99
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

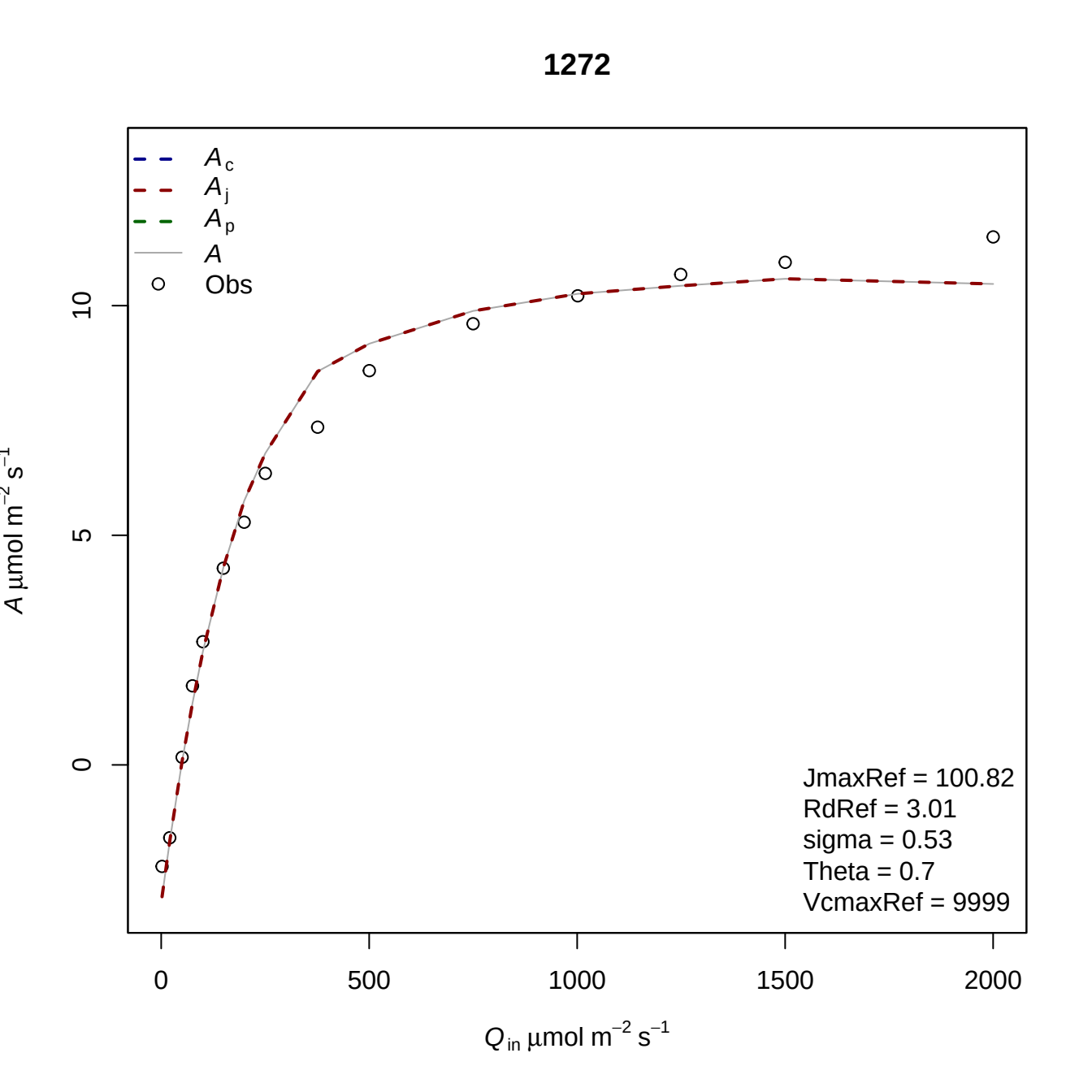

1267
A_c
A_j
A_p
A
Obs
JmaxRef = 83.62
RdRef = 5.97
sigma = 1.43
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

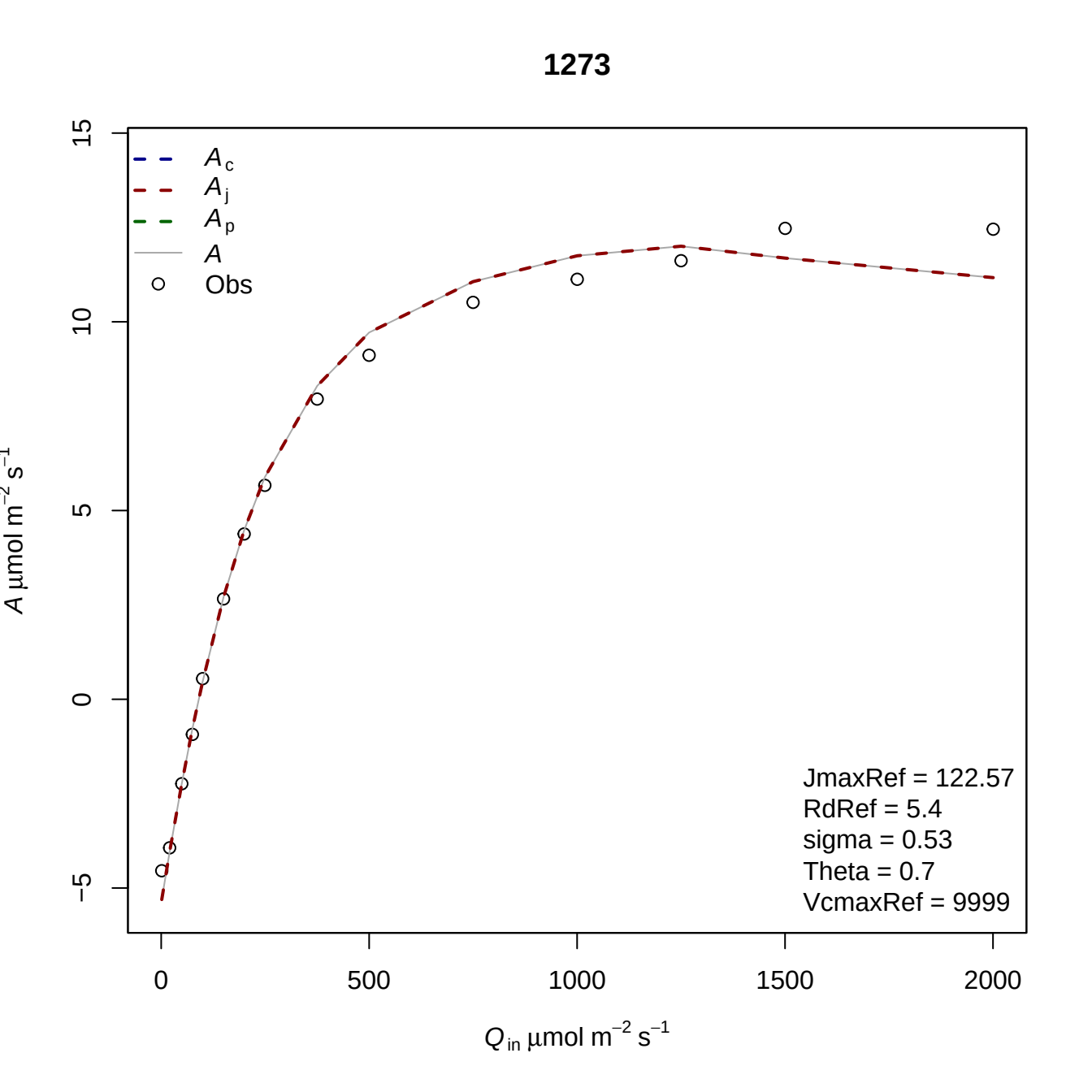

1268
A_c
A_j
A_p
A
Obs
JmaxRef = 37.5
RdRef = 3.25
sigma = 1.08
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

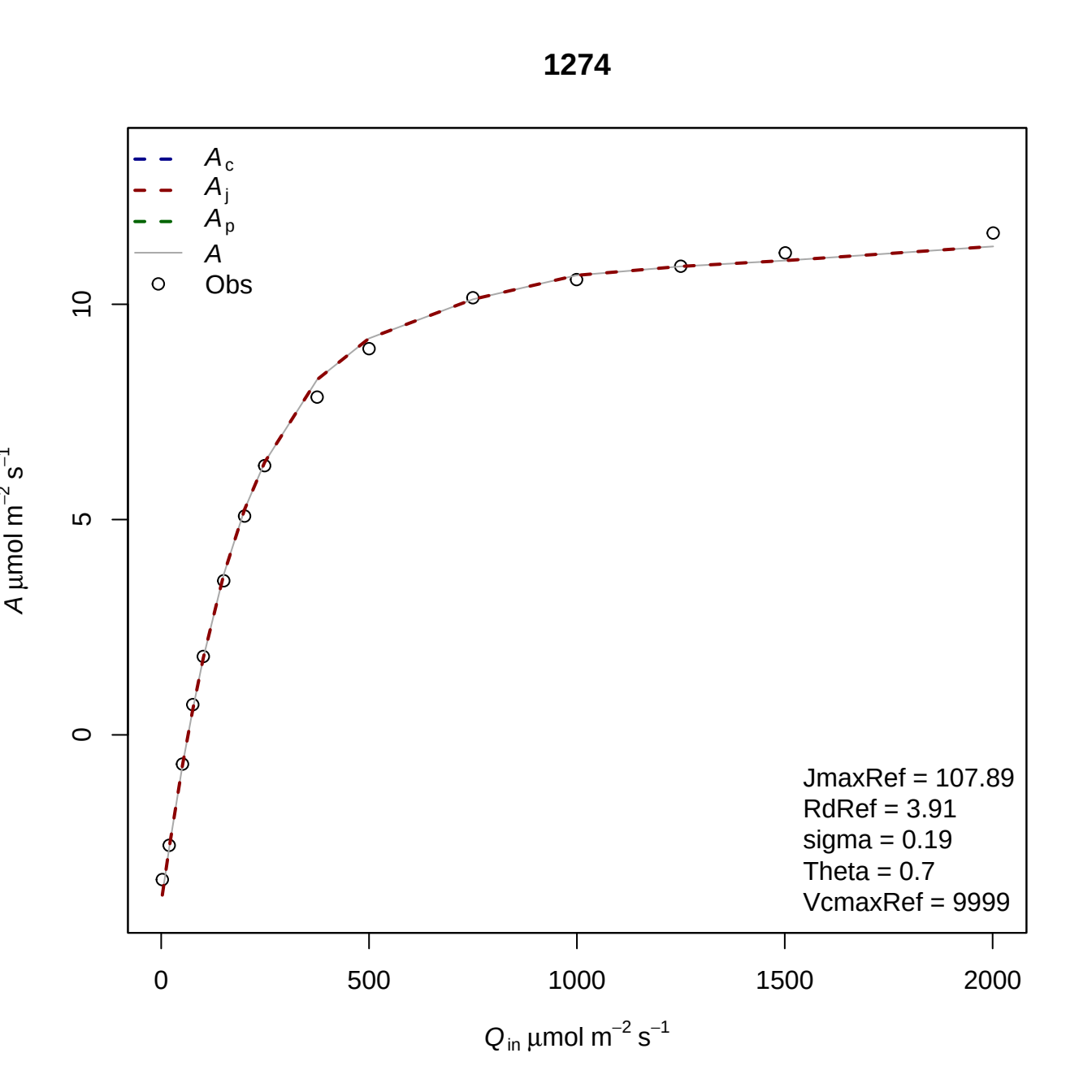

1269
A_c
A_j
A_p
A
Obs
JmaxRef = 43.29
RdRef = 3.33
sigma = 1.13
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

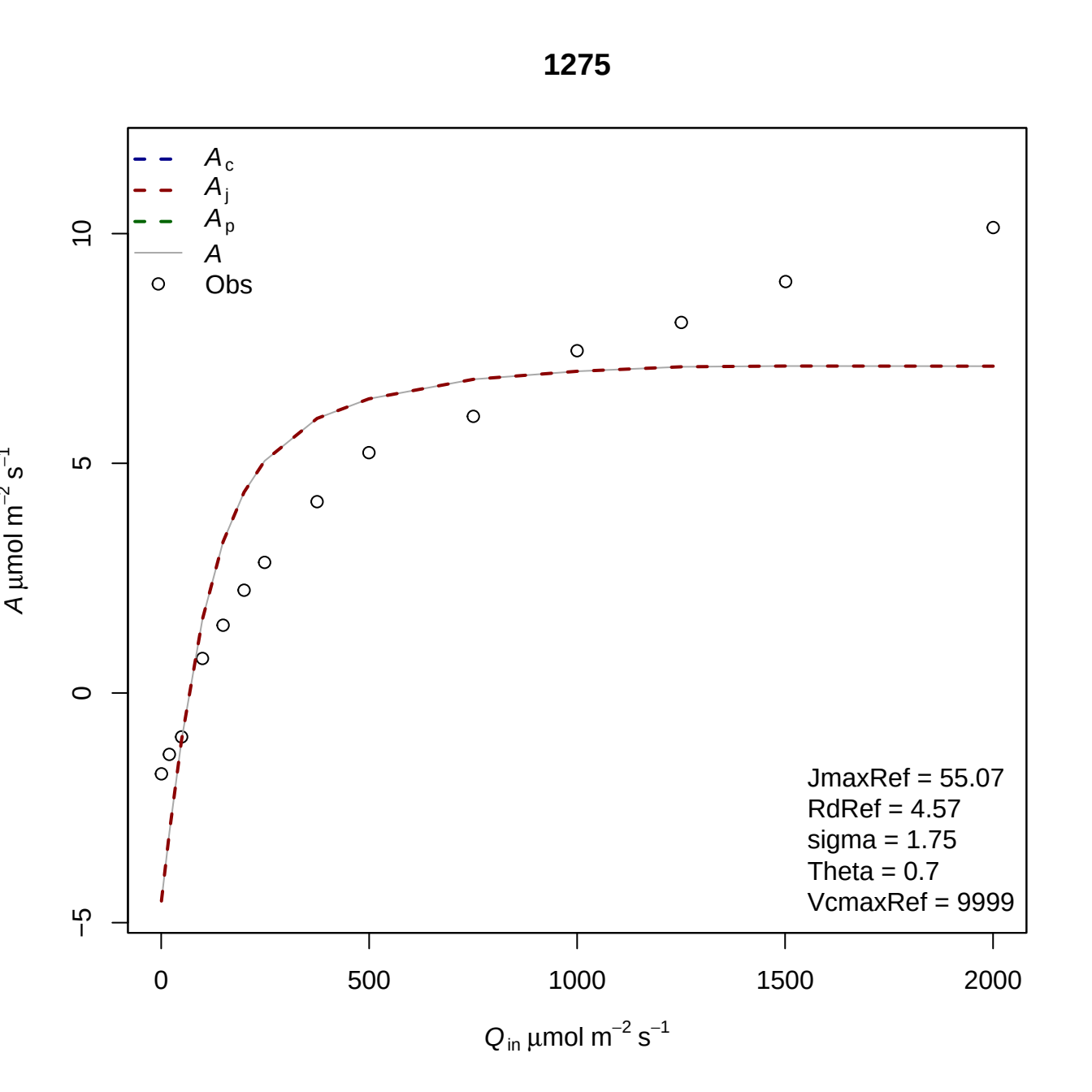

1270
A_c
A_j
A_p
A
Obs
JmaxRef = 53.43
RdRef = 3.55
sigma = 1.27
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

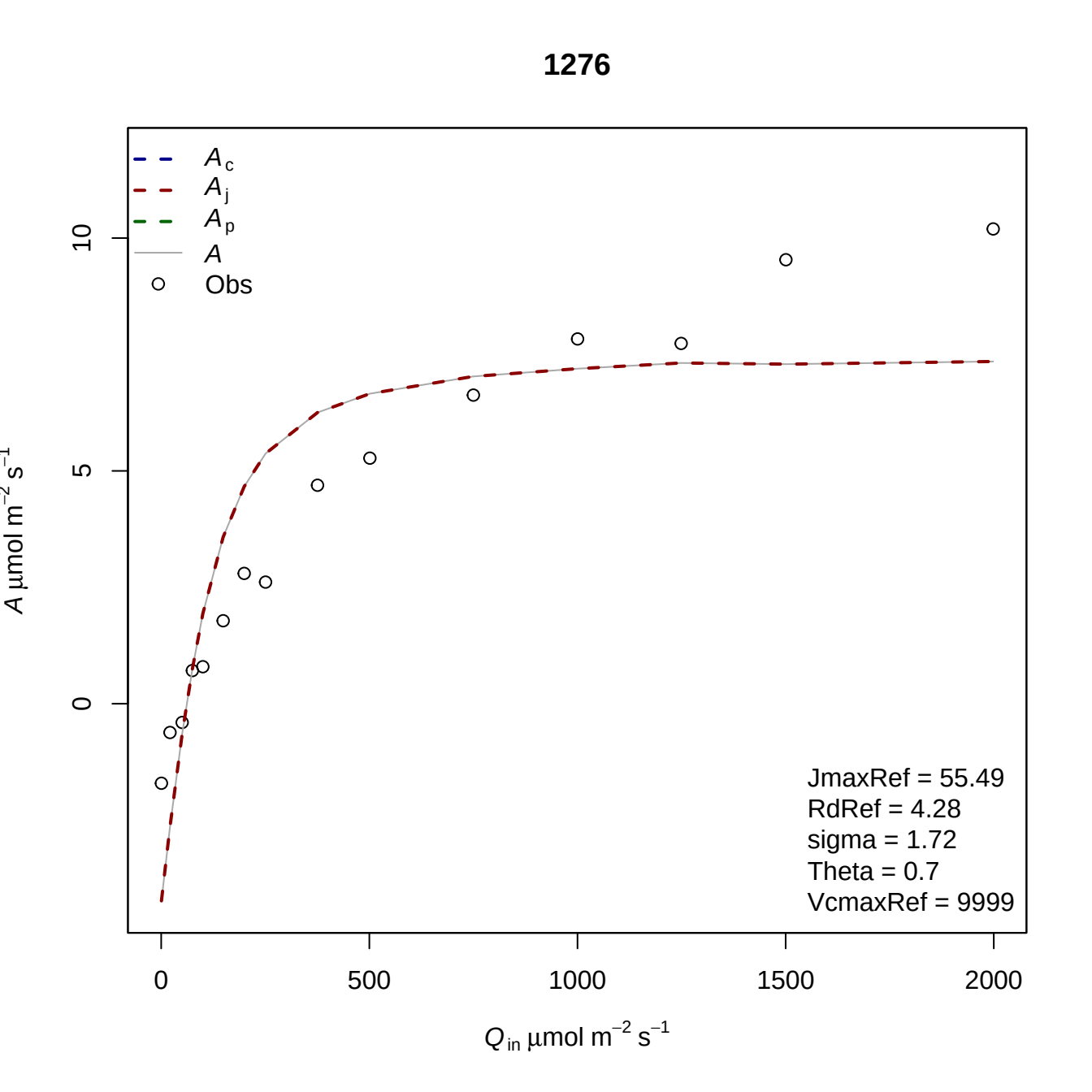

1271
A_c
A_j
A_p
A
Obs
JmaxRef = 91.64
RdRef = 5.06
sigma = 1.83
Theta = 0.7
VcmaxRef = 9999
15
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

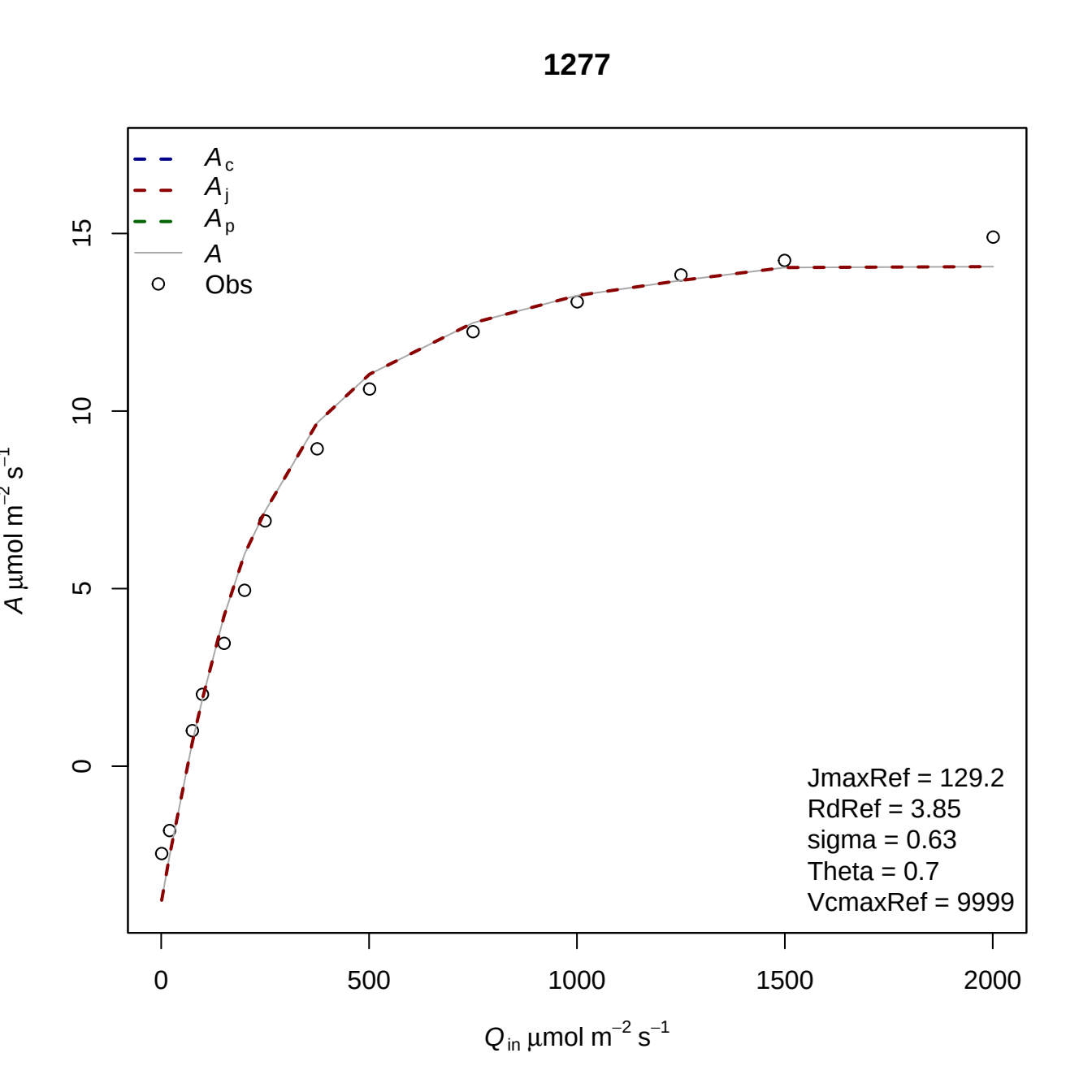

1272
A_c
A_j
A_p
A
Obs
JmaxRef = 100.82
RdRef = 3.01
sigma = 0.53
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

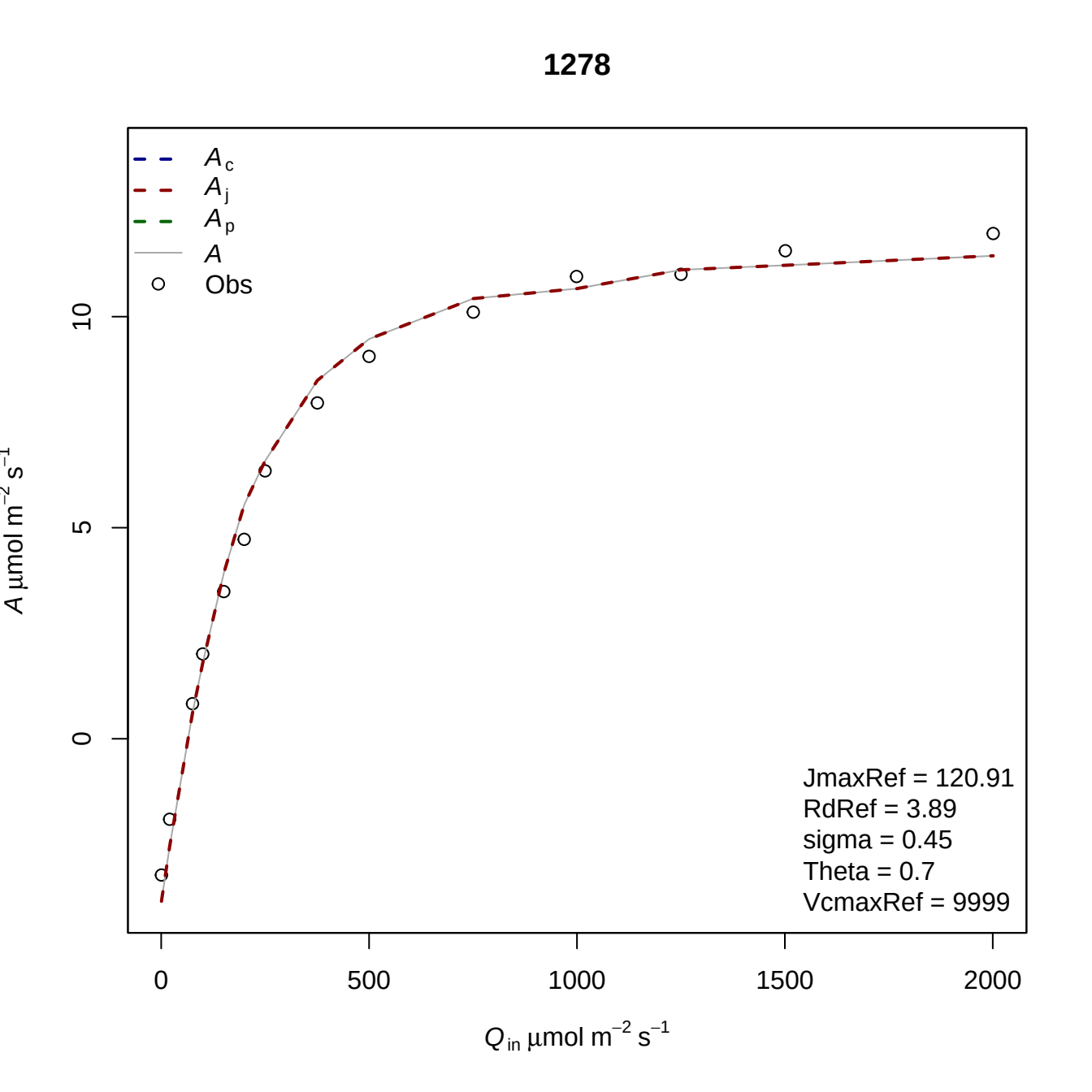

1273
A_c
A_j
A_p
A
Obs
JmaxRef = 122.57
RdRef = 5.4
sigma = 0.53
Theta = 0.7
VcmaxRef = 9999
15
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

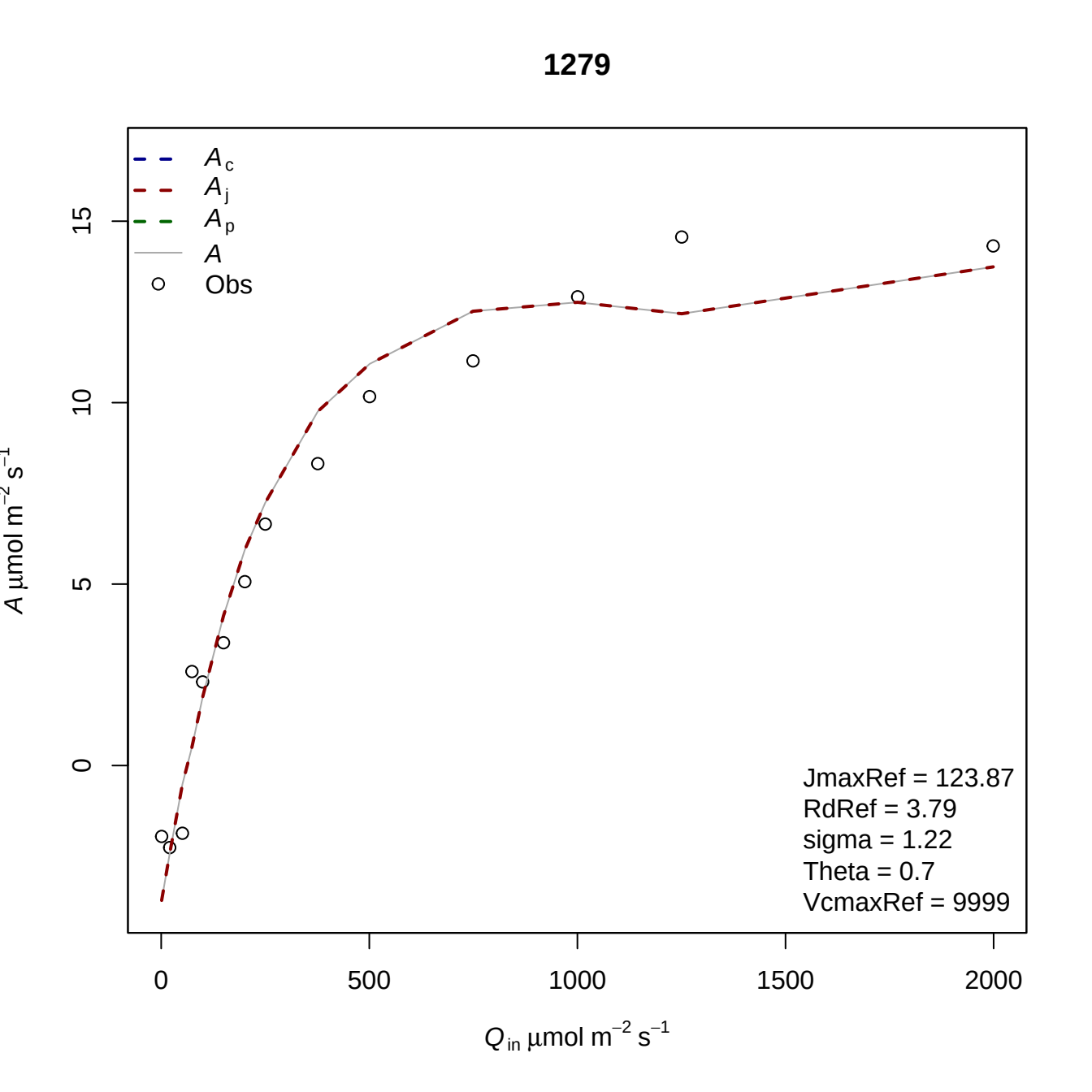

1274
A_c
A_j
A_p
A
Obs
JmaxRef = 107.89
RdRef = 3.91
sigma = 0.19
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

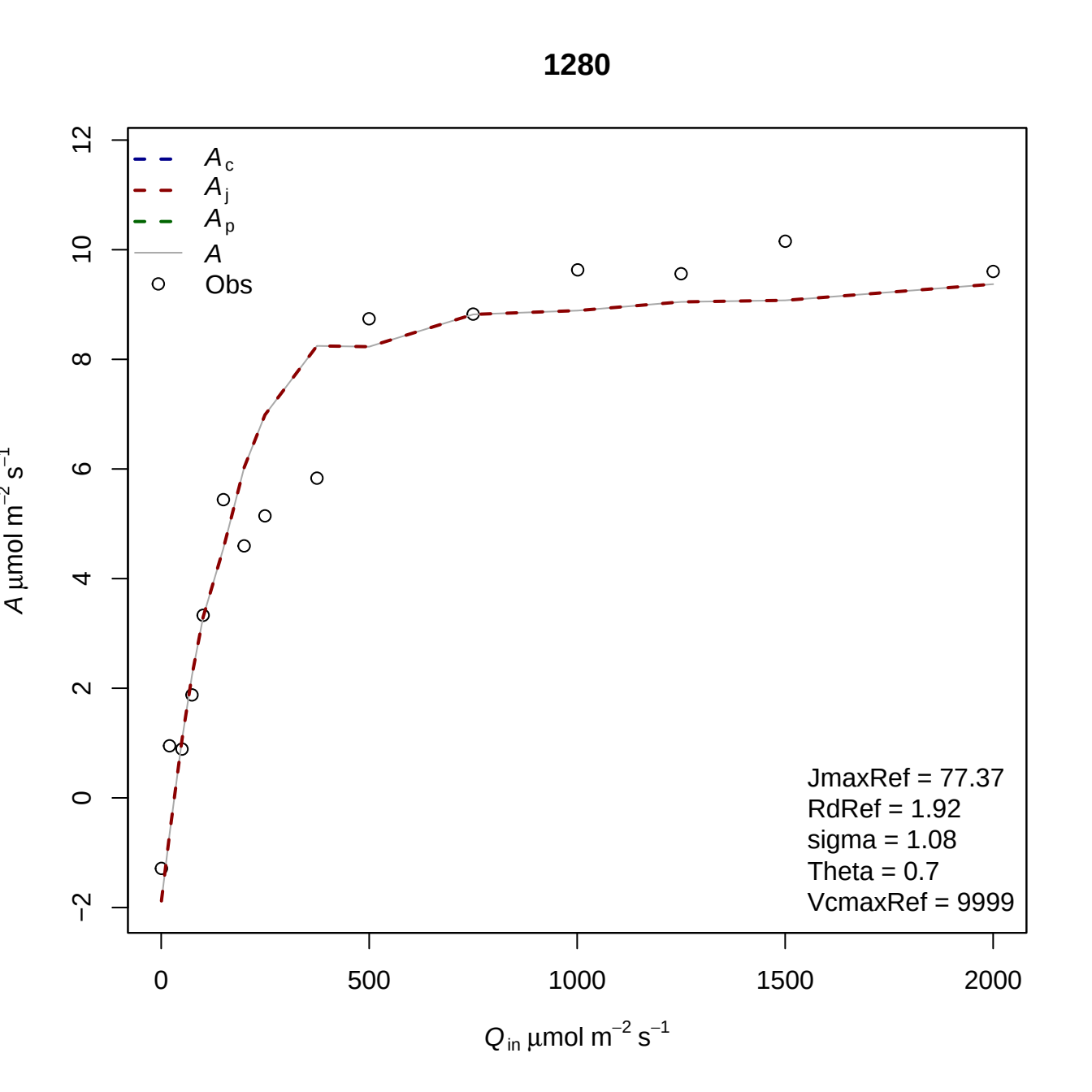

1275
A_c
A_j
A_p
A
Obs
JmaxRef = 55.07
RdRef = 4.57
sigma = 1.75
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

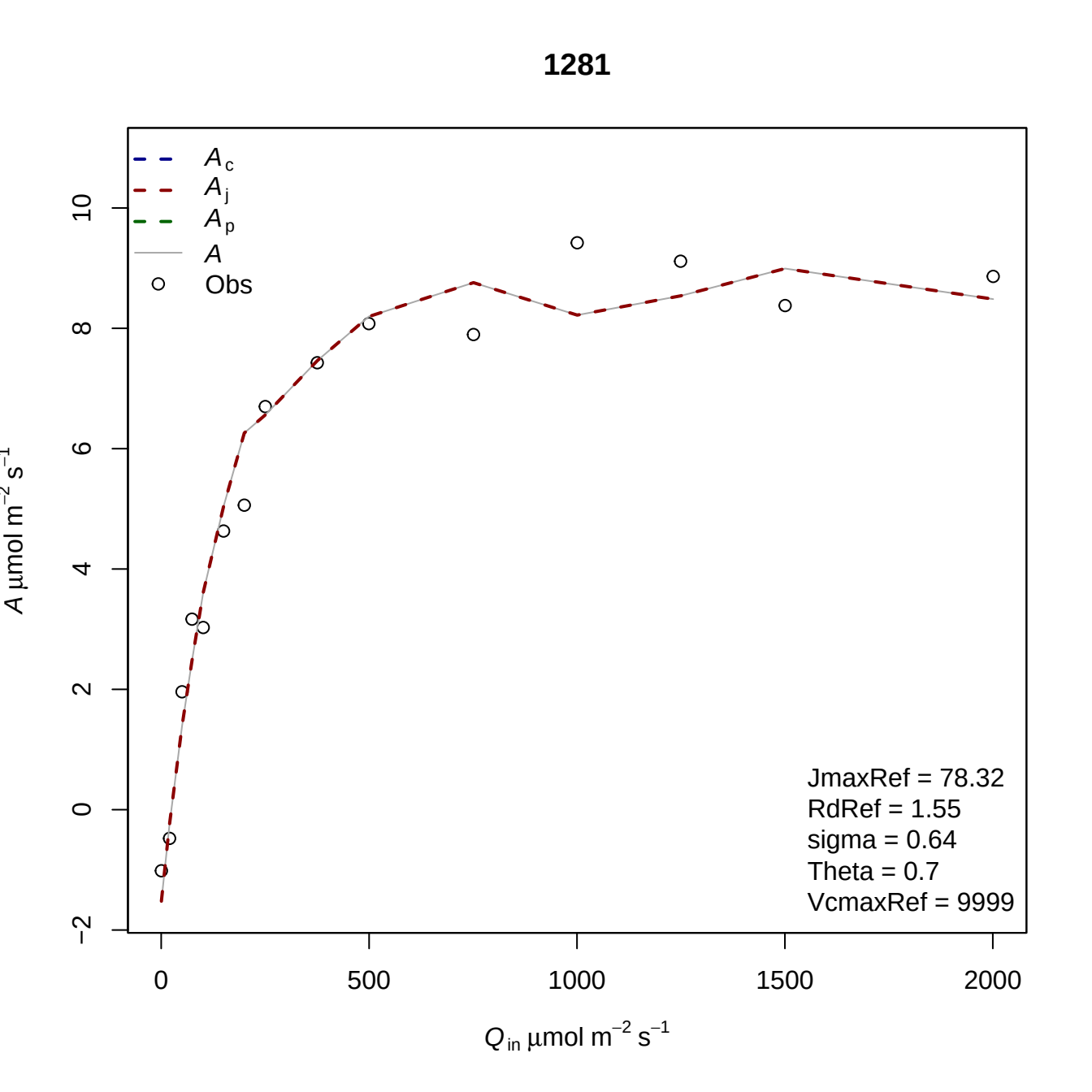

1276
A_c
A_j
A_p
A
Obs
JmaxRef = 55.49
RdRef = 4.28
sigma = 1.72
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

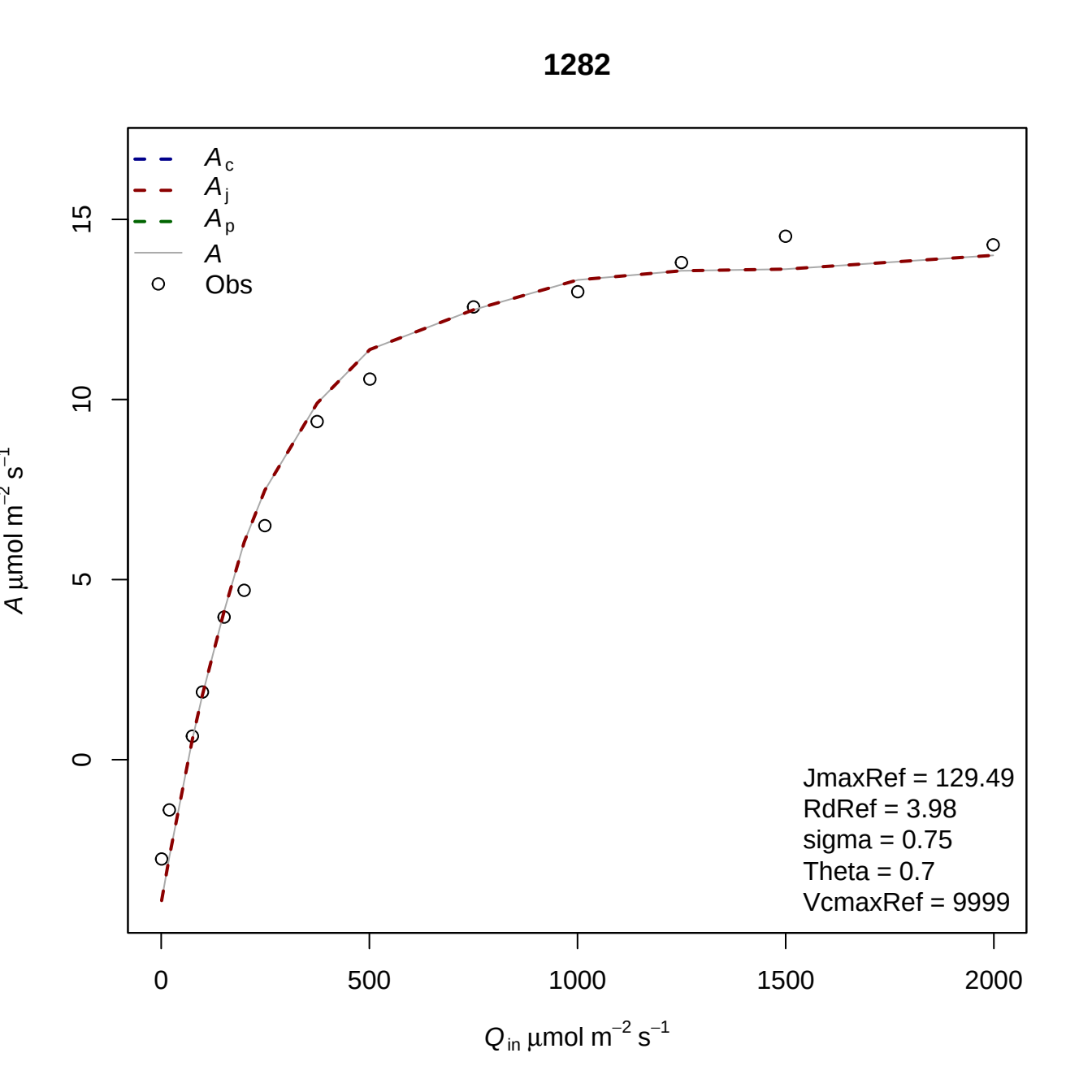

1277
A_c
A_j
A_p
A
Obs
JmaxRef = 129.2
RdRef = 3.85
sigma = 0.63
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

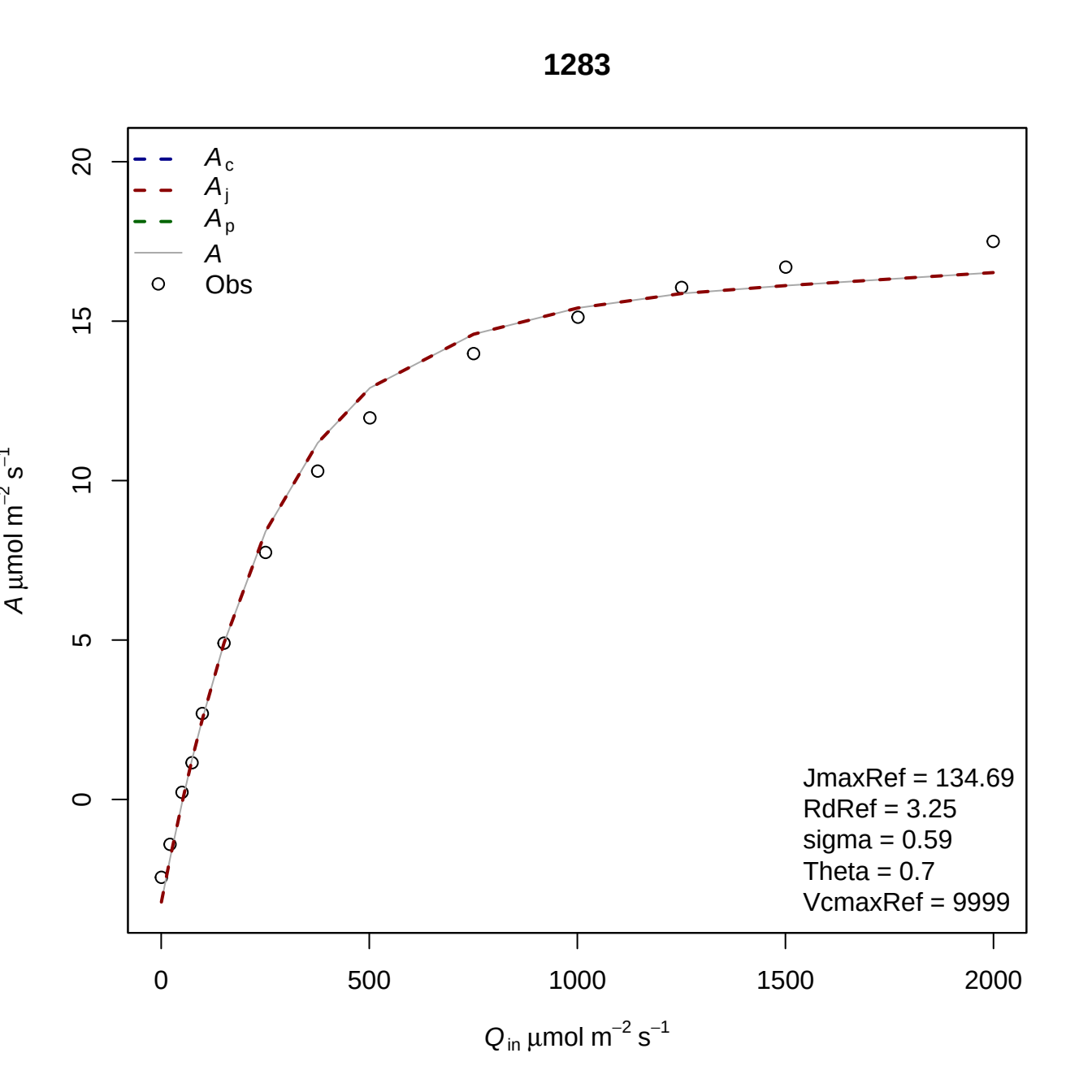

1278
A_c
A_j
A_p
A
Obs
JmaxRef = 120.91
RdRef = 3.89
sigma = 0.45
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

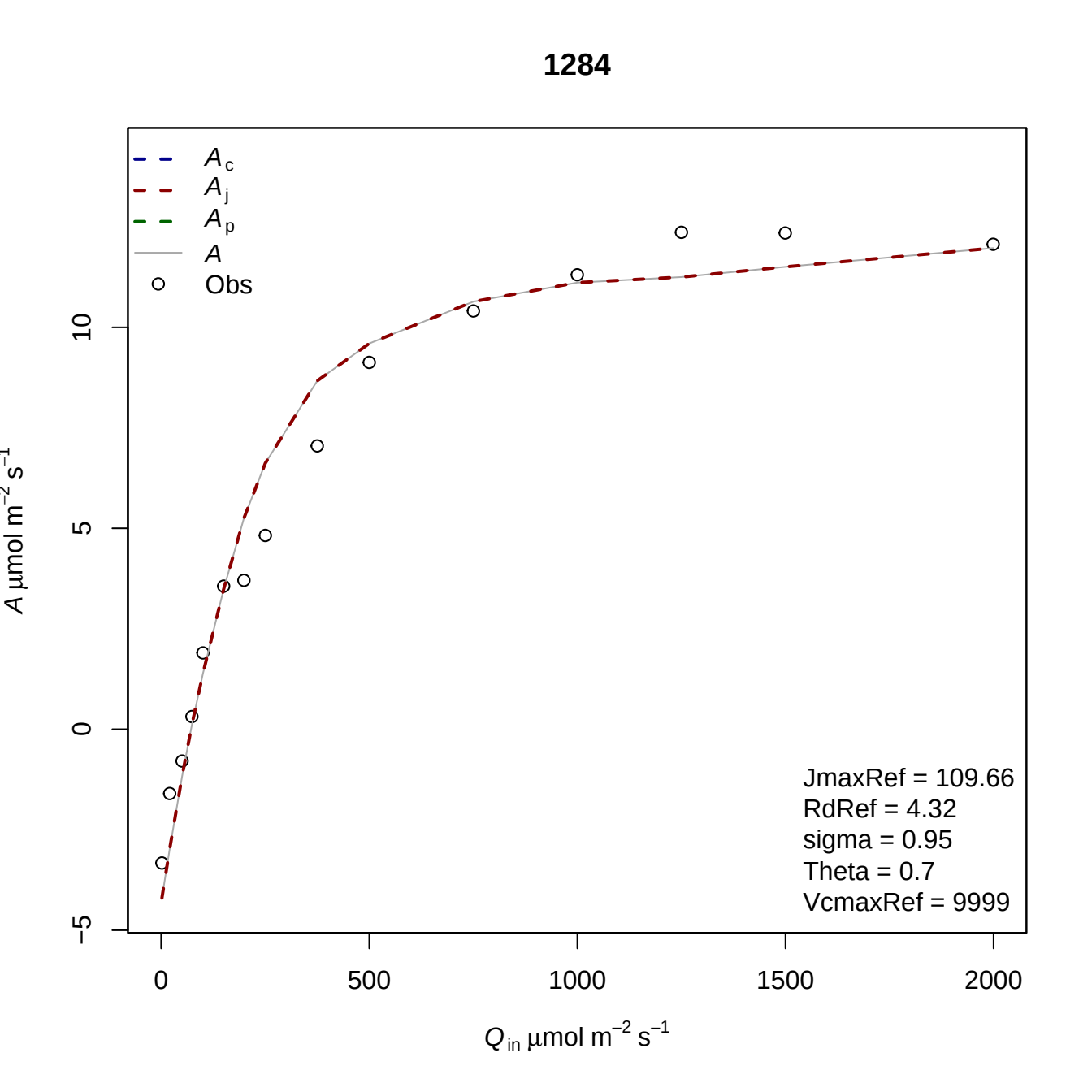

1279
A_c
A_j
A_p
A
Obs
JmaxRef = 123.87
RdRef = 3.79
sigma = 1.22
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

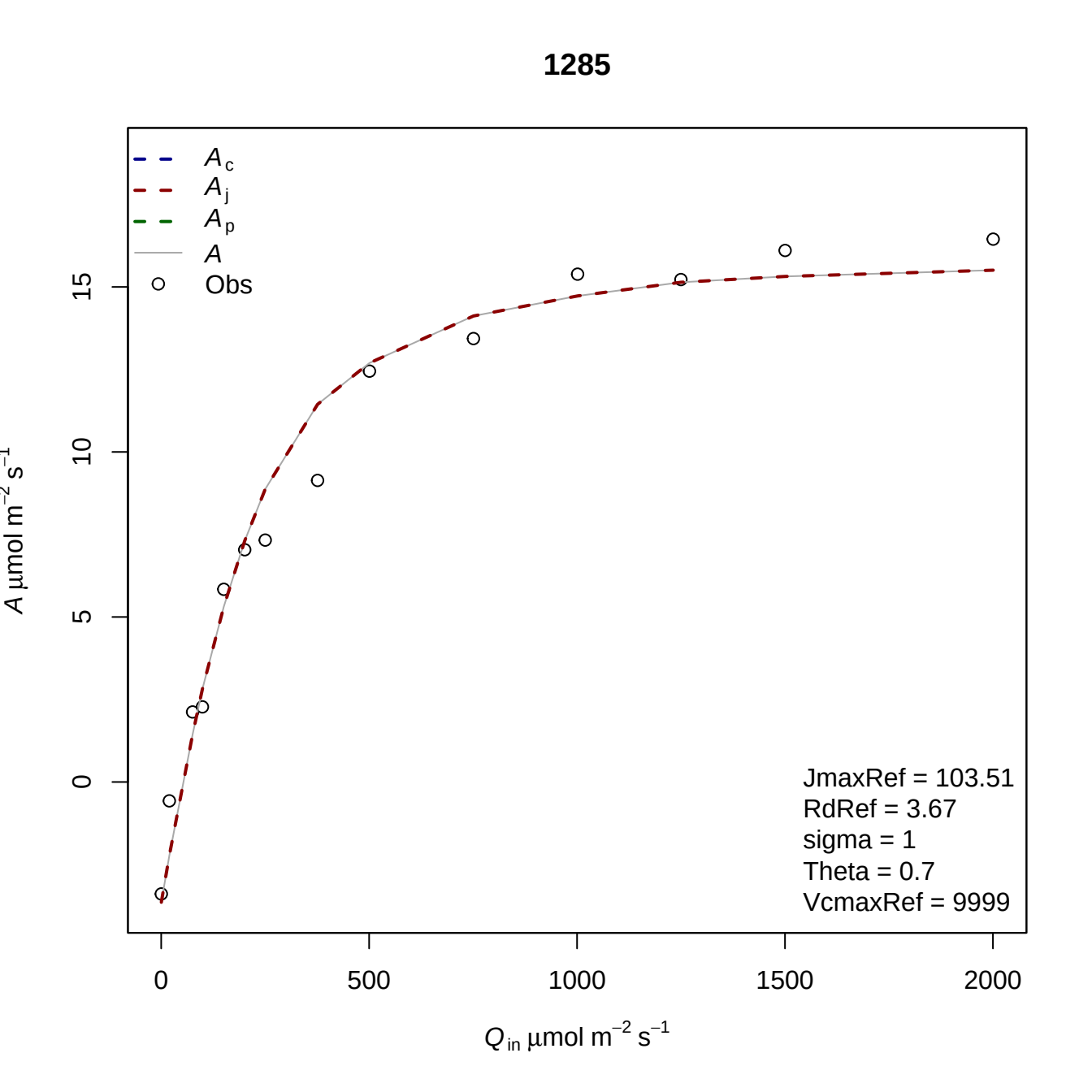

1280
A_c
A_j
A_p
A
Obs
JmaxRef = 77.37
RdRef = 1.92
sigma = 1.08
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

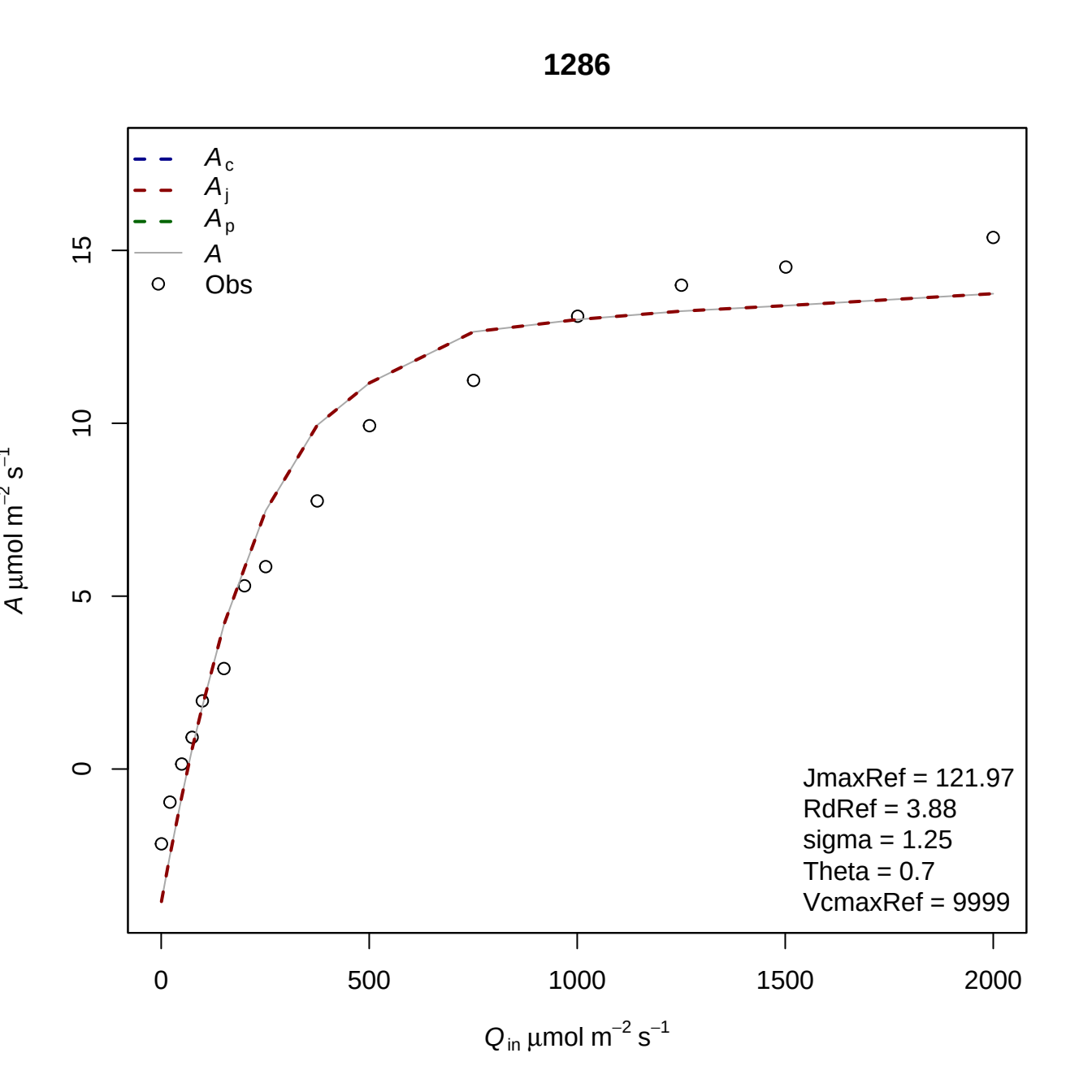

1281
A_c
A_j
A_p
A
Obs
JmaxRef = 78.32
RdRef = 1.55
sigma = 0.64
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

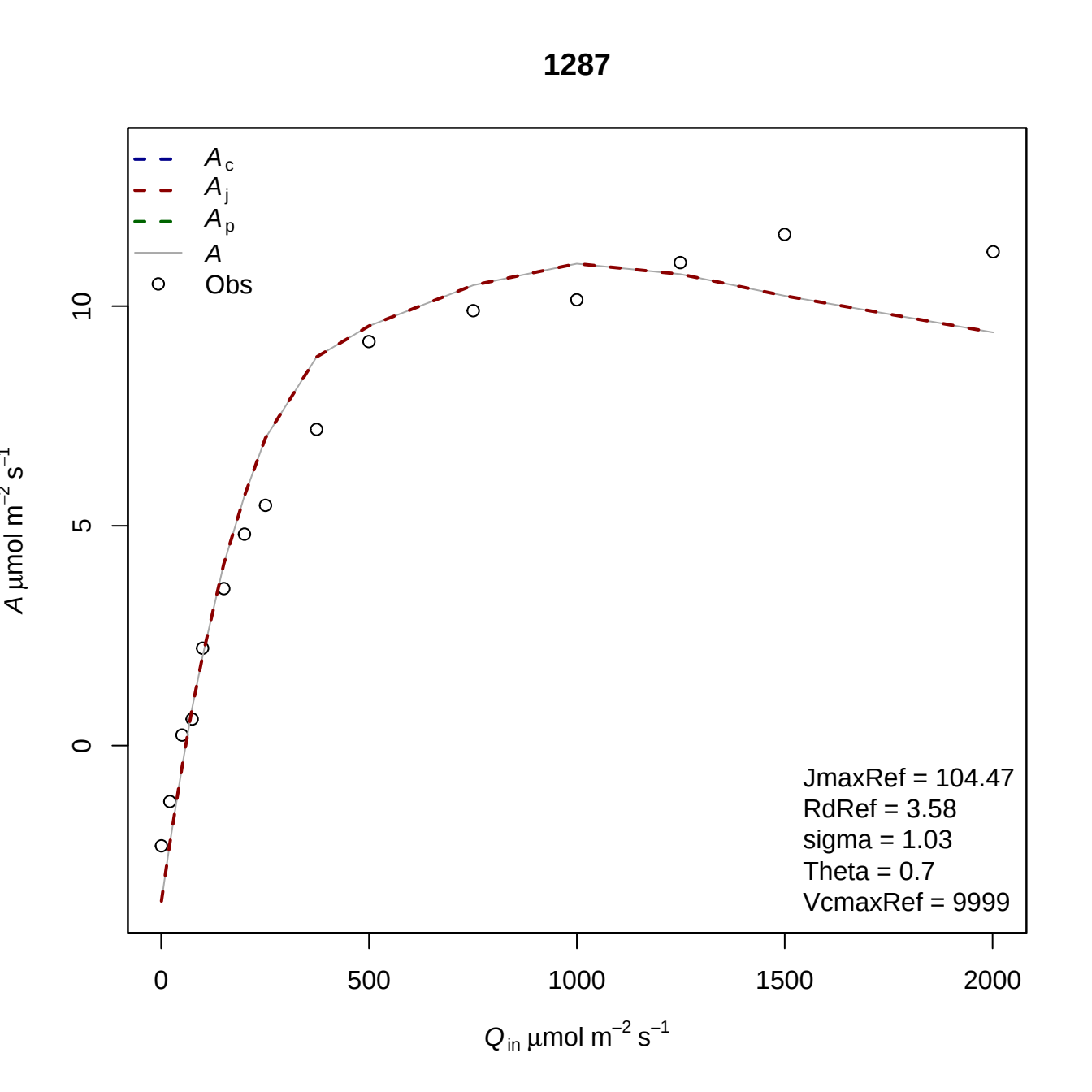

1282
A_c
A_j
A_p
A
Obs
JmaxRef = 129.49
RdRef = 3.98
sigma = 0.75
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

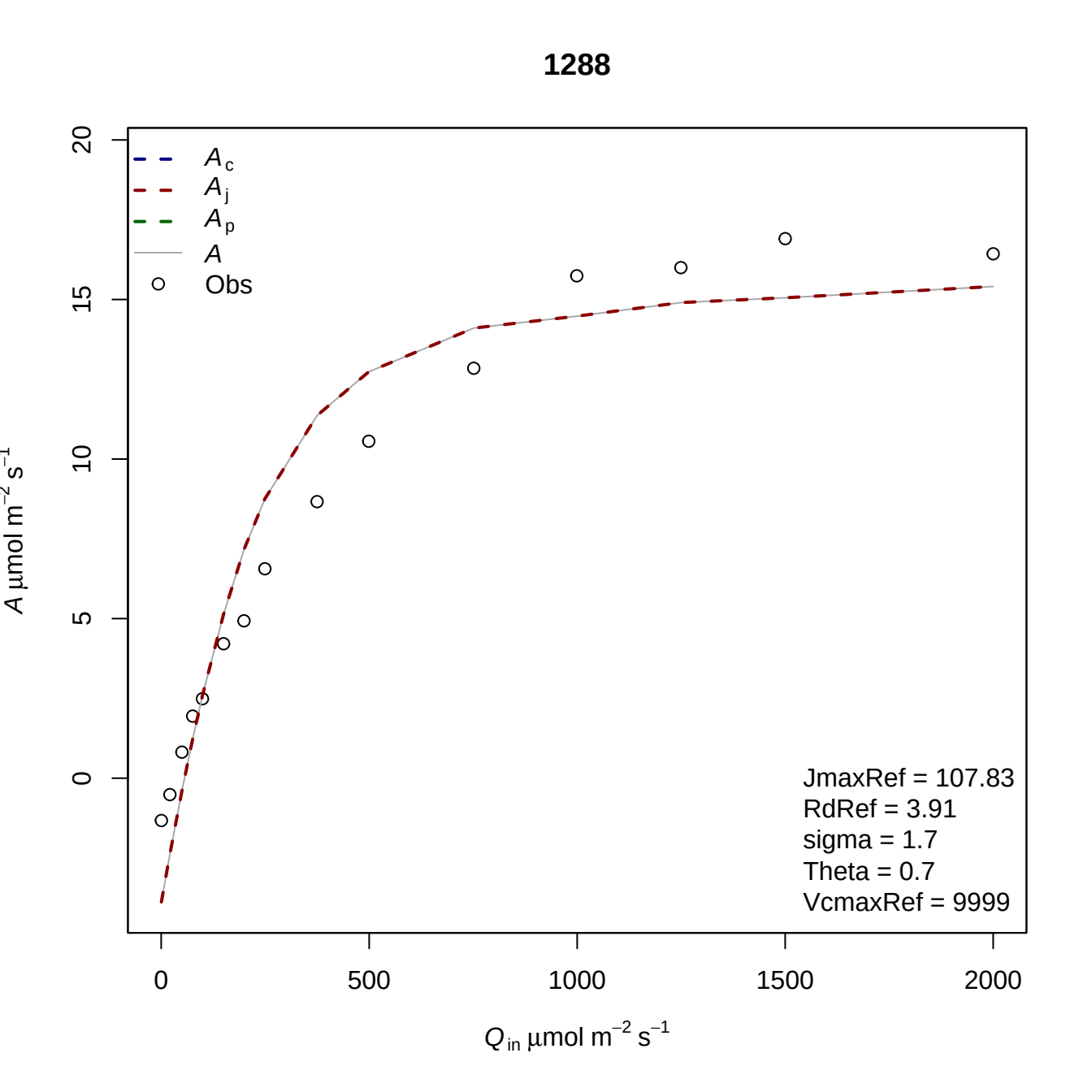

1283
A_c
A_j
A_p
A
Obs
JmaxRef = 134.69
RdRef = 3.25
sigma = 0.59
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

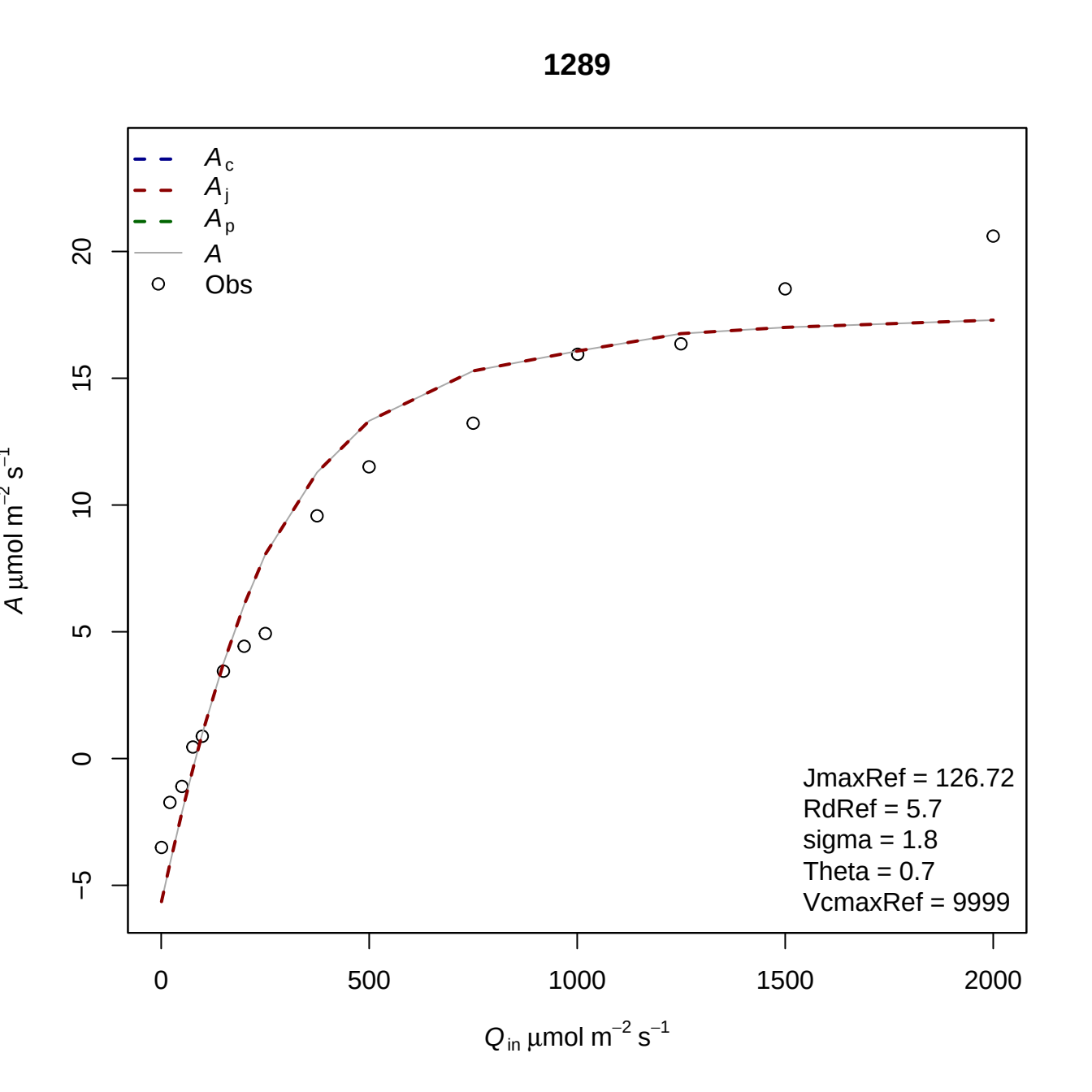

1284
A_c
A_j
A_p
A
Obs
JmaxRef = 109.66
RdRef = 4.32
sigma = 0.95
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

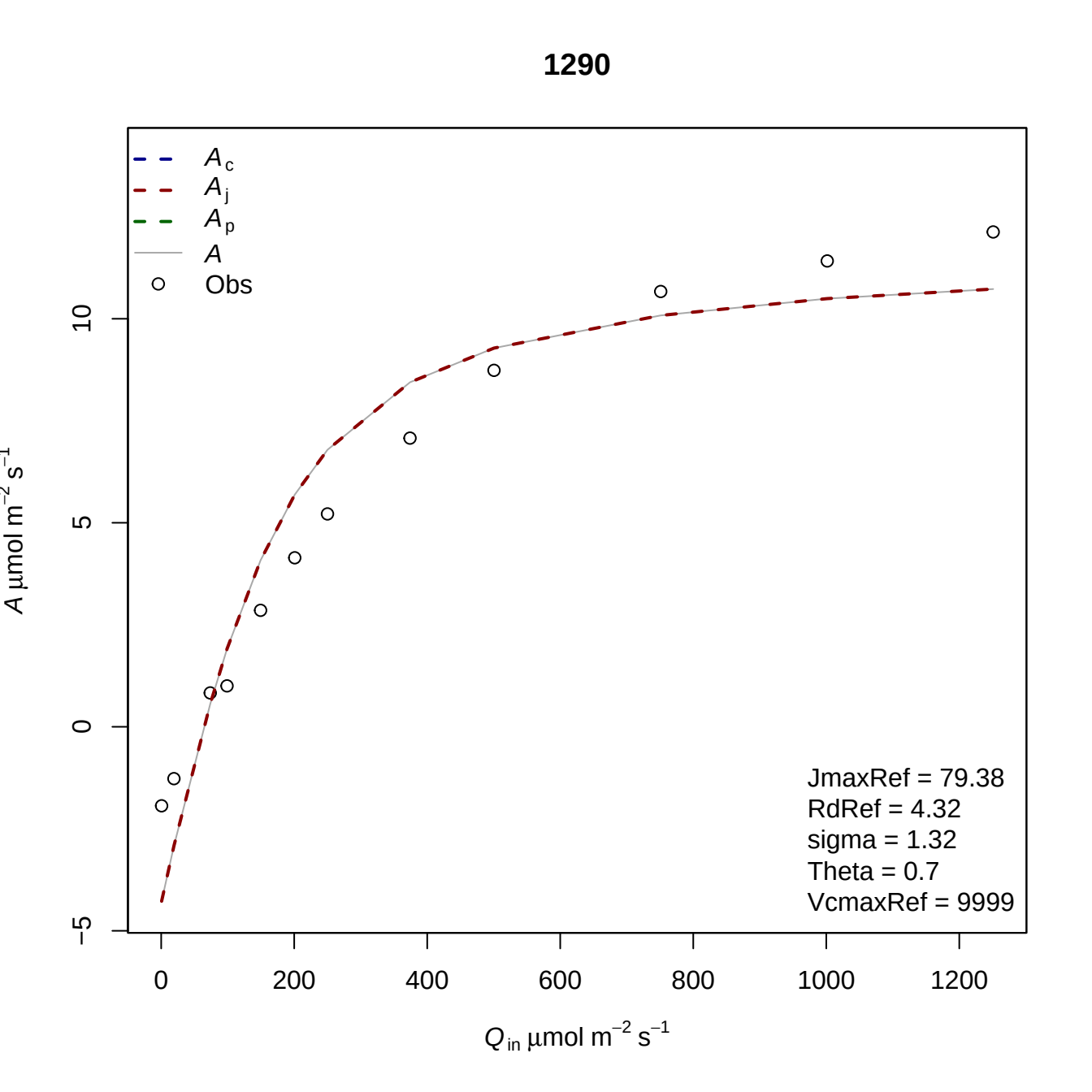

1285
A_c
A_j
A_p
A
Obs
JmaxRef = 103.51
RdRef = 3.67
sigma = 1
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

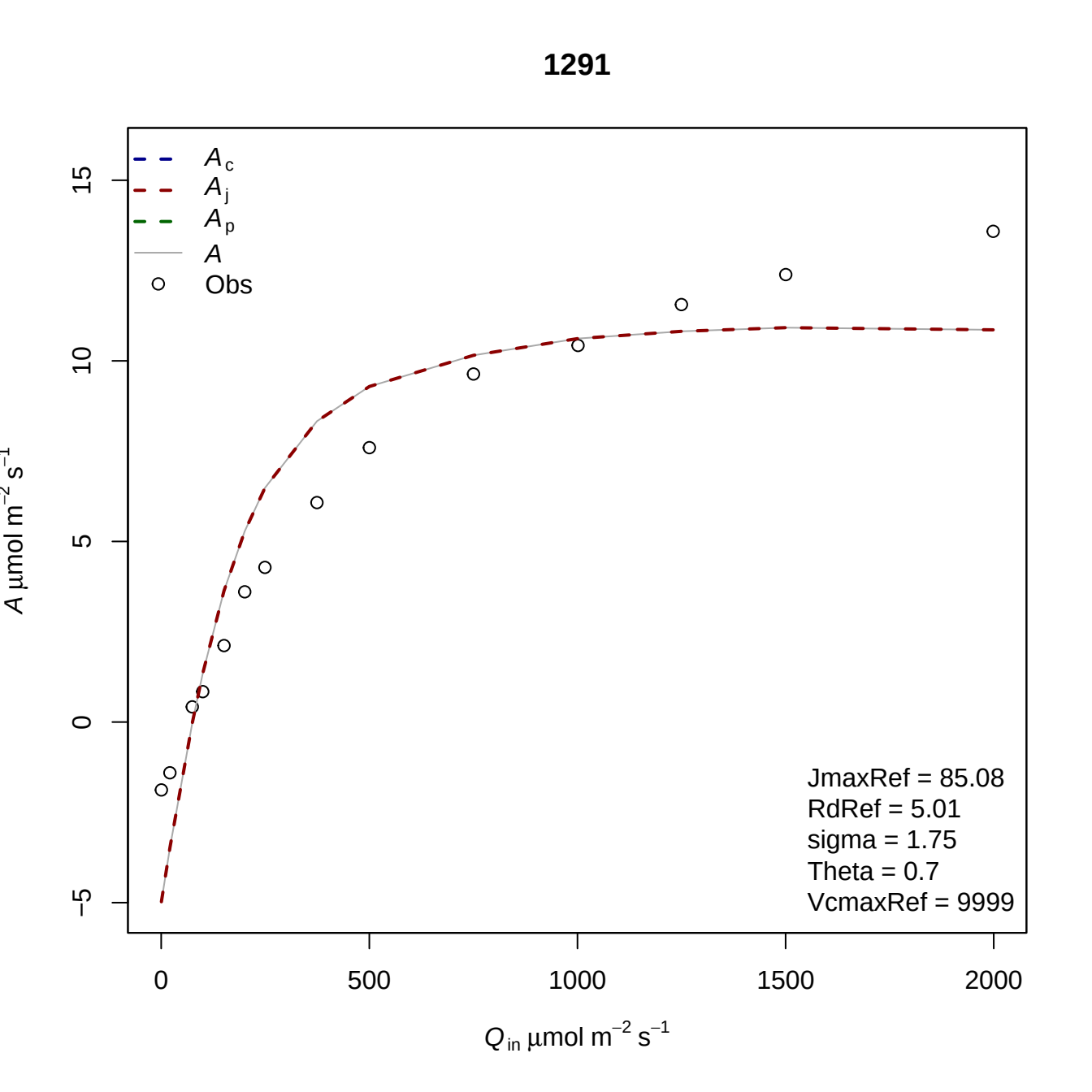

1286
A_c
A_j
A_p
A
Obs
JmaxRef = 121.97
RdRef = 3.88
sigma = 1.25
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

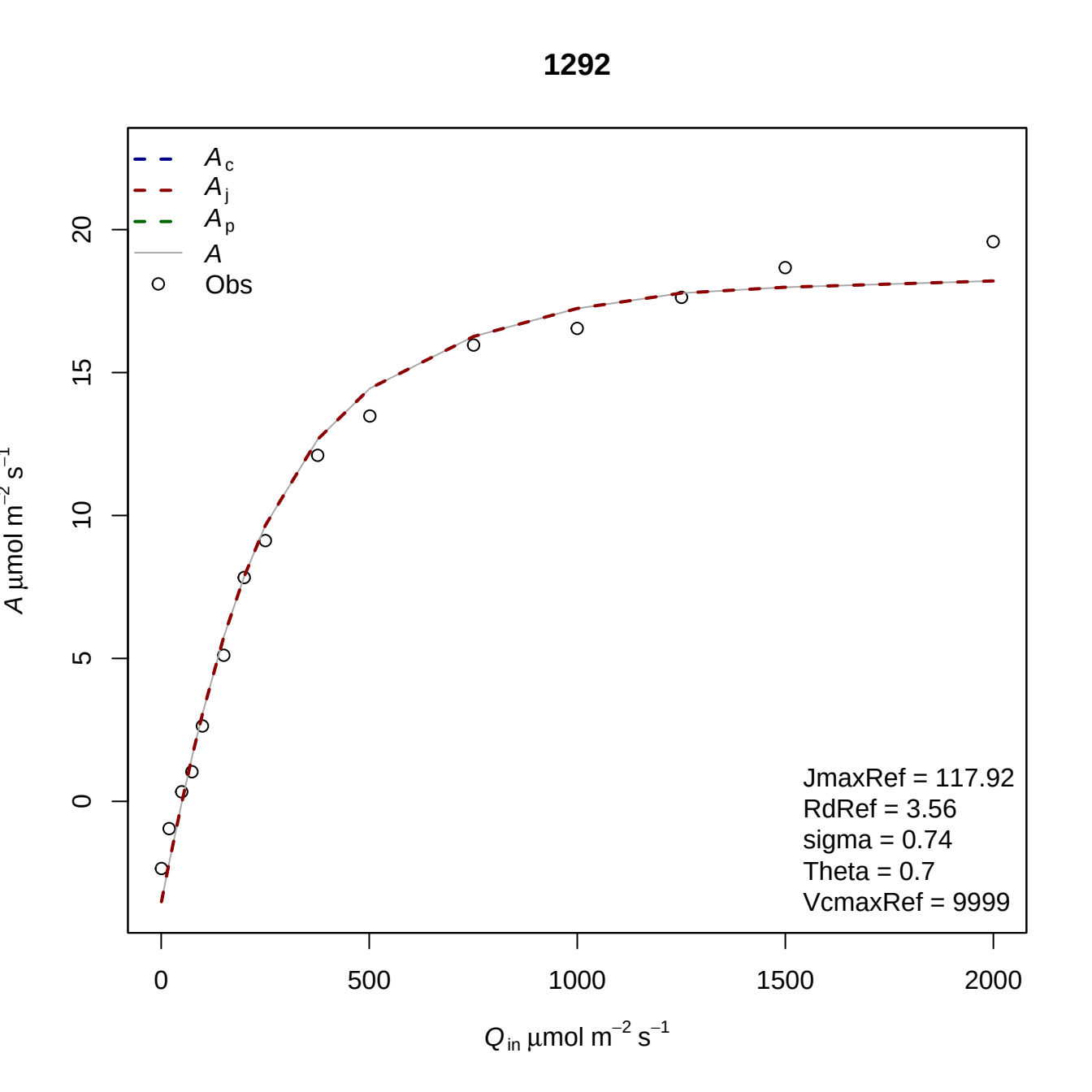

1287
A_c
A_j
A_p
A
Obs
JmaxRef = 104.47
RdRef = 3.58
sigma = 1.03
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

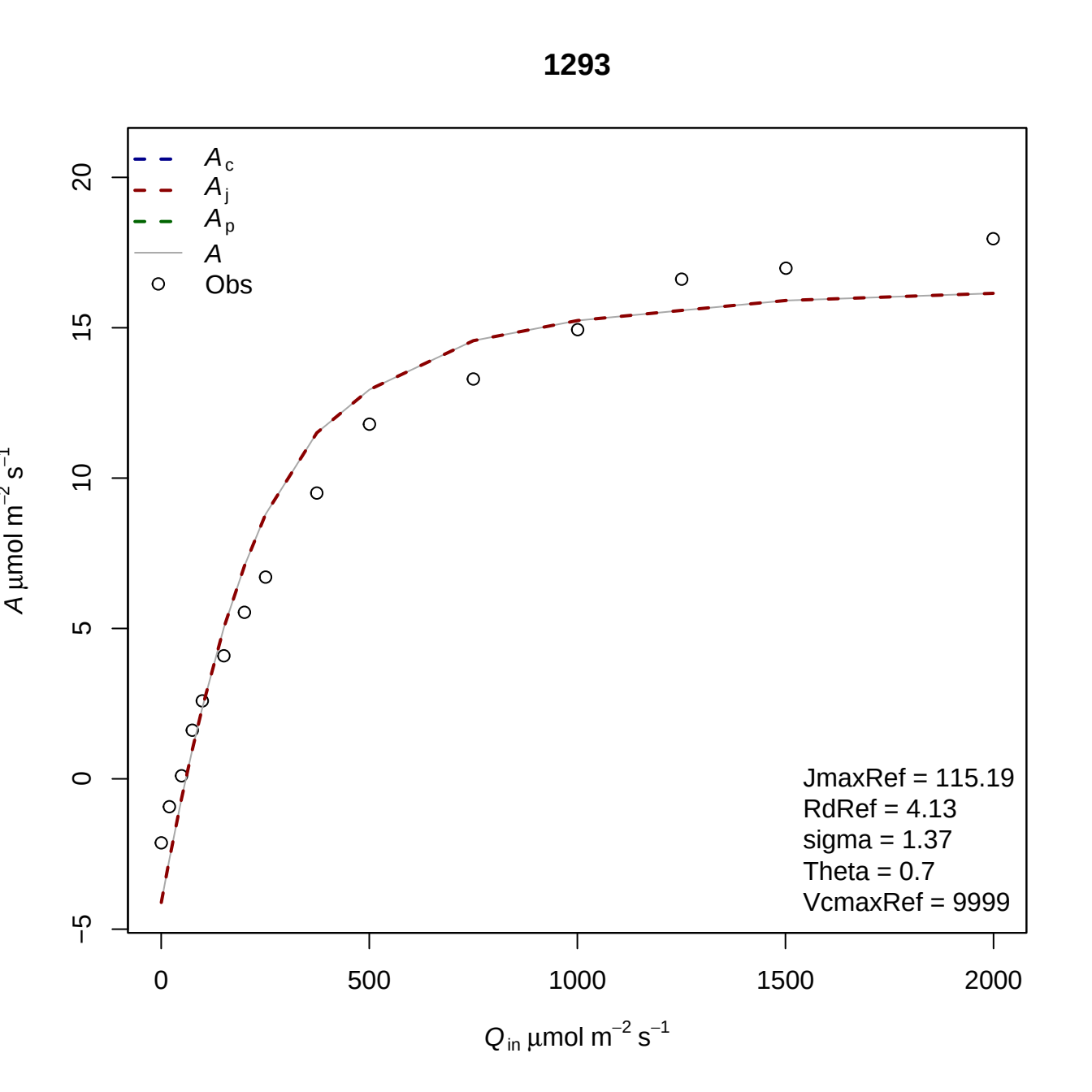

1288
A_c
A_j
A_p
A
Obs
JmaxRef = 107.83
RdRef = 3.91
sigma = 1.7
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

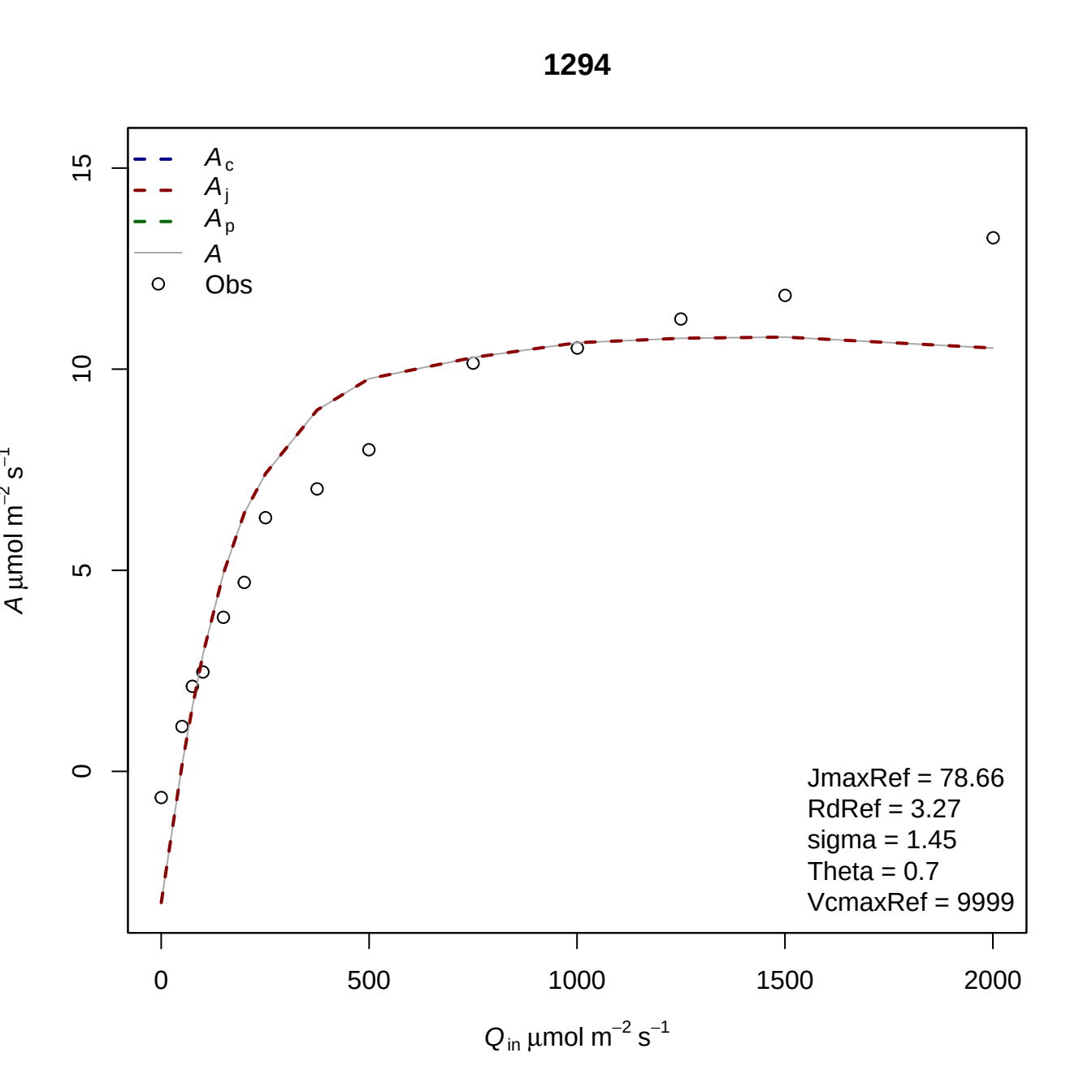

1289
A_c
A_j
A_p
A
Obs
JmaxRef = 126.72
RdRef = 5.7
sigma = 1.8
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

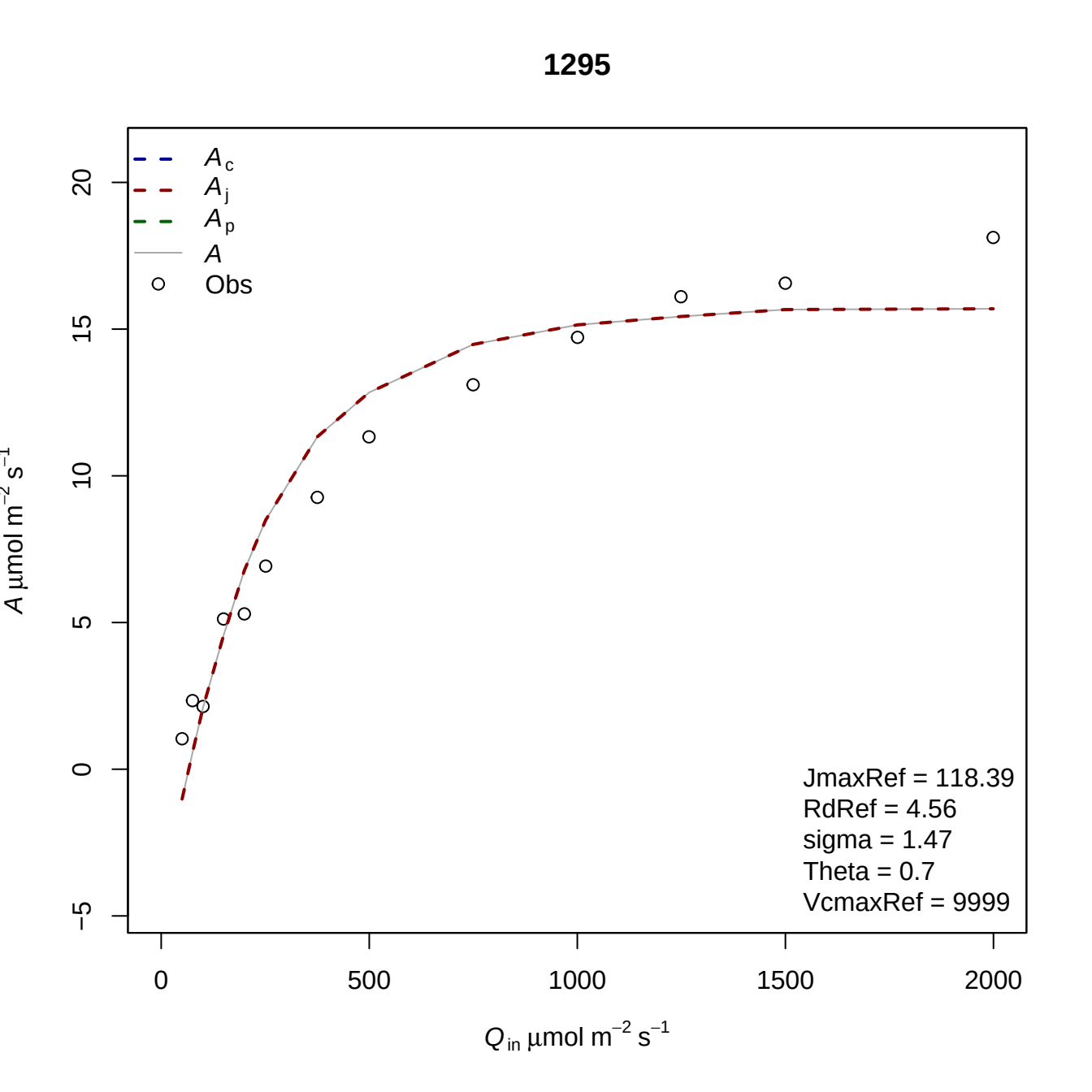

1290
A_c
A_j
A_p
A
Obs
JmaxRef = 79.38
RdRef = 4.32
sigma = 1.32
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
200
400
600
800
1000
1200

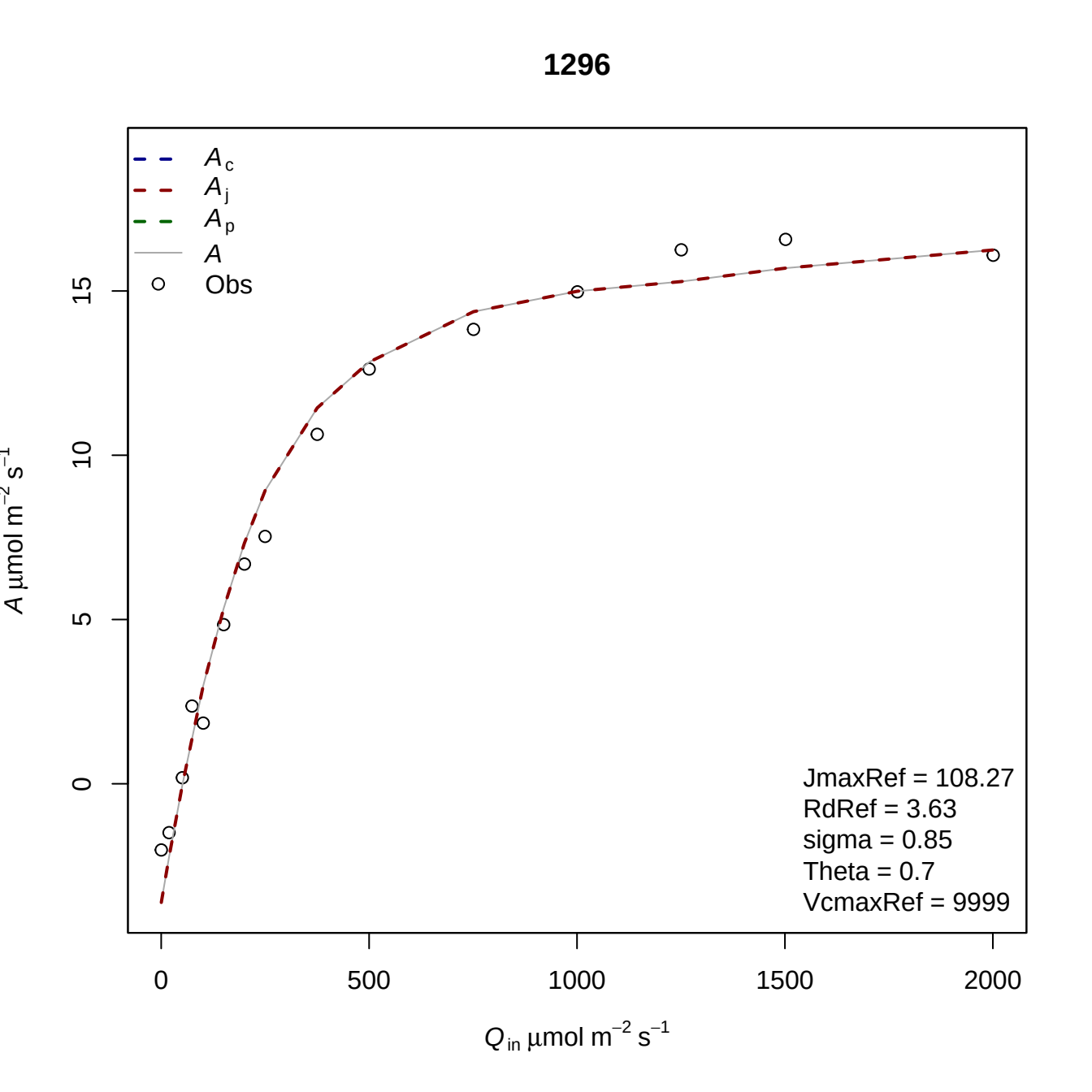

1291
A_c
A_j
A_p
A
Obs
JmaxRef = 85.08
RdRef = 5.01
sigma = 1.75
Theta = 0.7
VcmaxRef = 9999
15
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

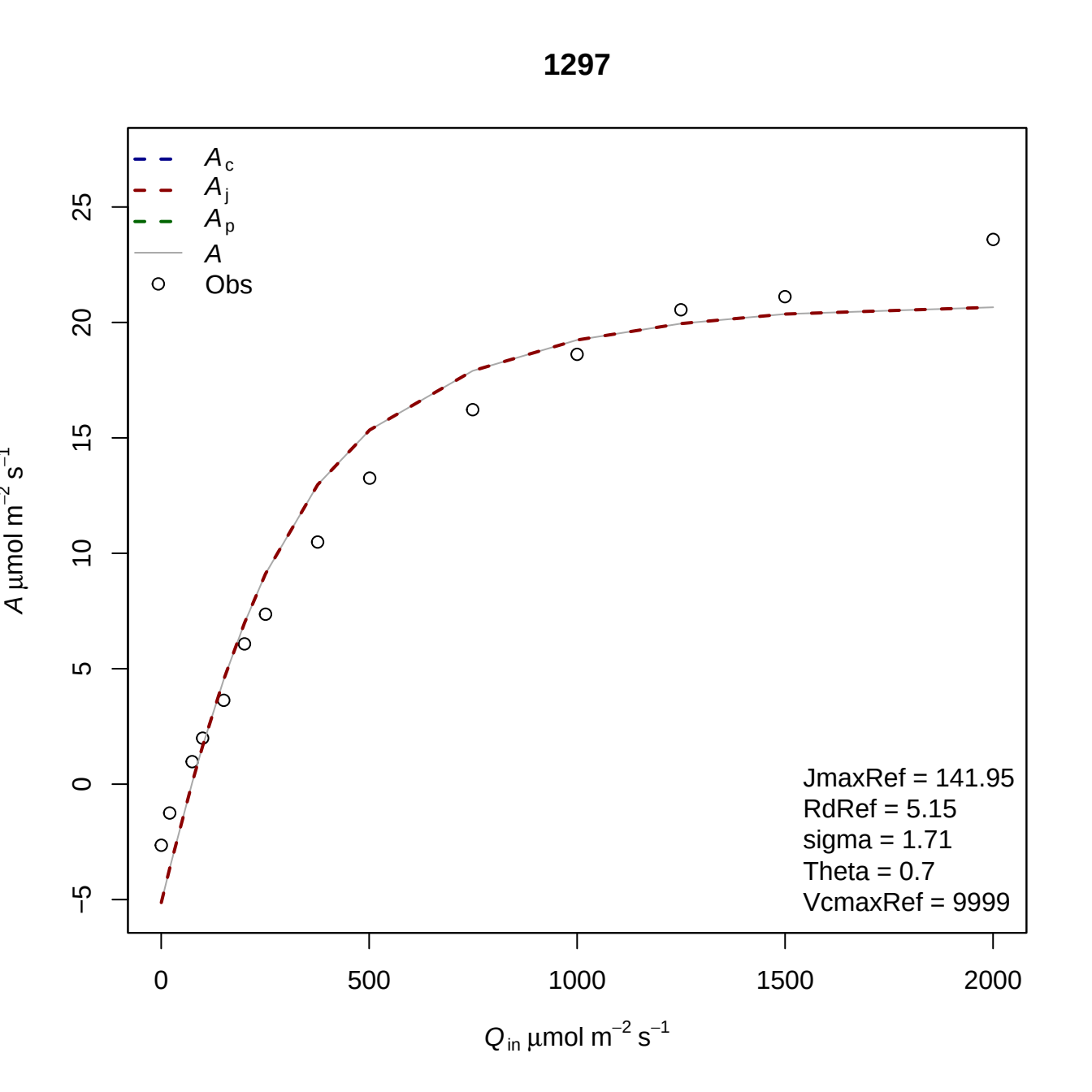

1292
A_c
A_j
A_p
A
Obs
JmaxRef = 117.92
RdRef = 3.56
sigma = 0.74
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

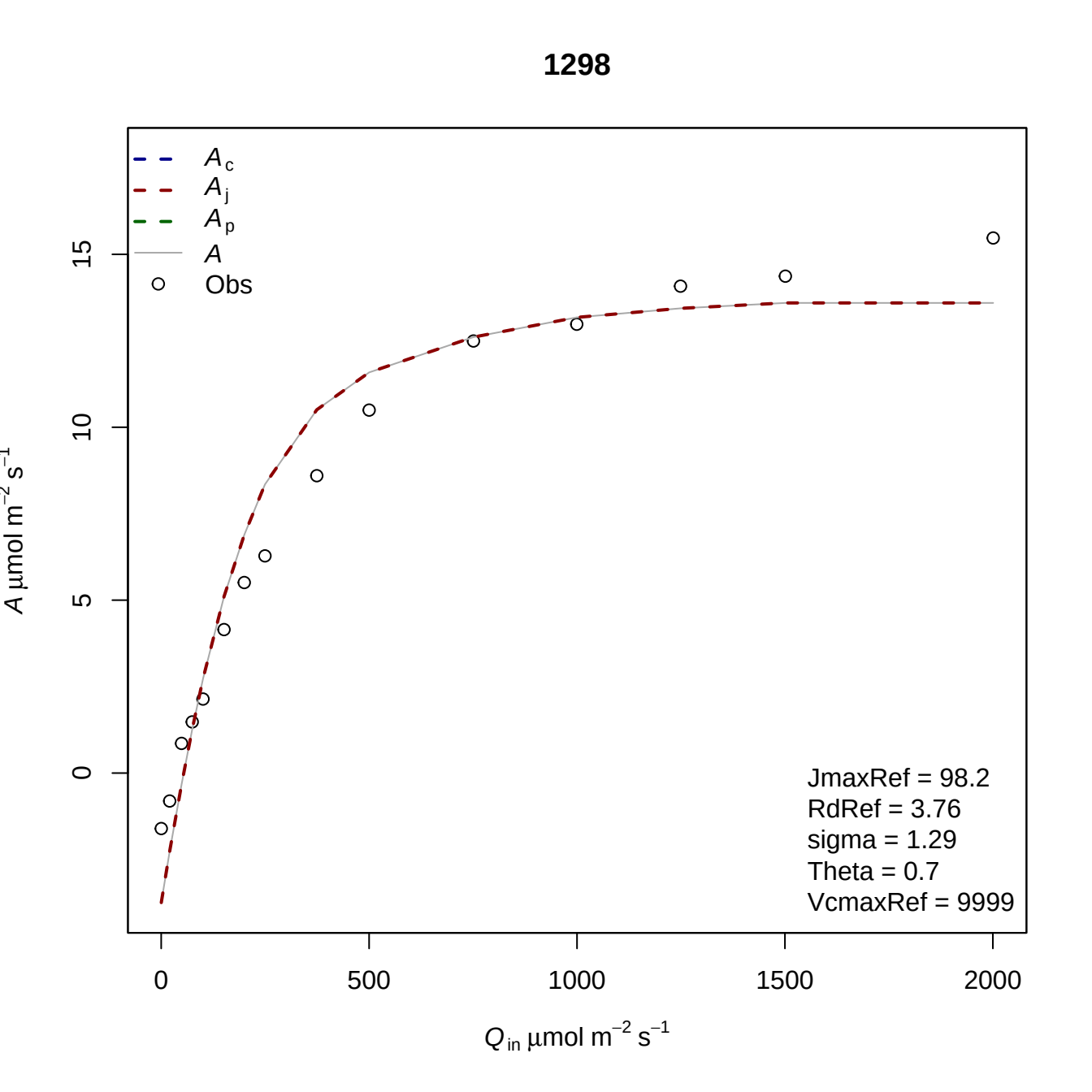

1293
A_c
A_j
A_p
A
Obs
JmaxRef = 115.19
RdRef = 4.13
sigma = 1.37
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

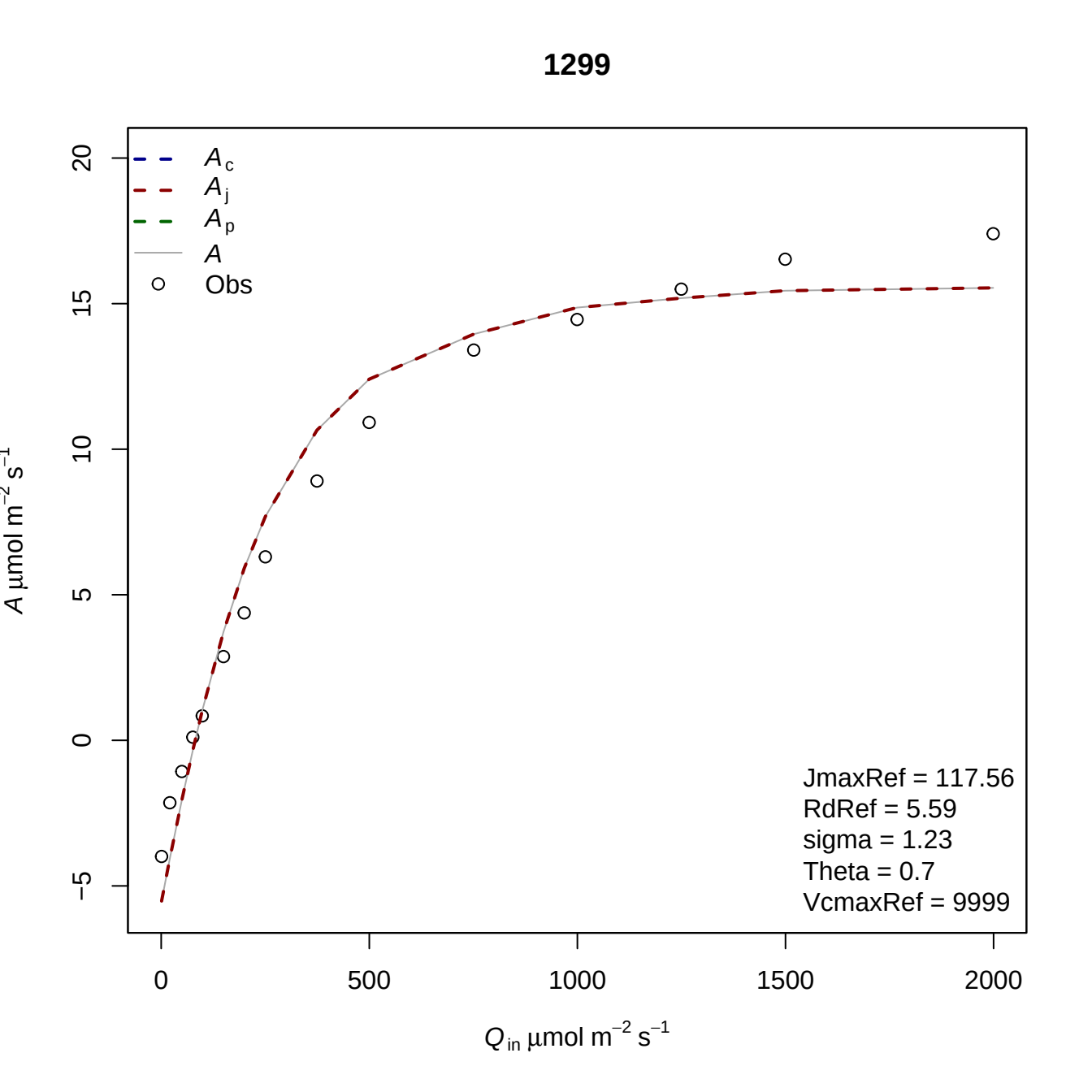

1294
A_c
A_j
A_p
A
Obs
JmaxRef = 78.66
RdRef = 3.27
sigma = 1.45
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

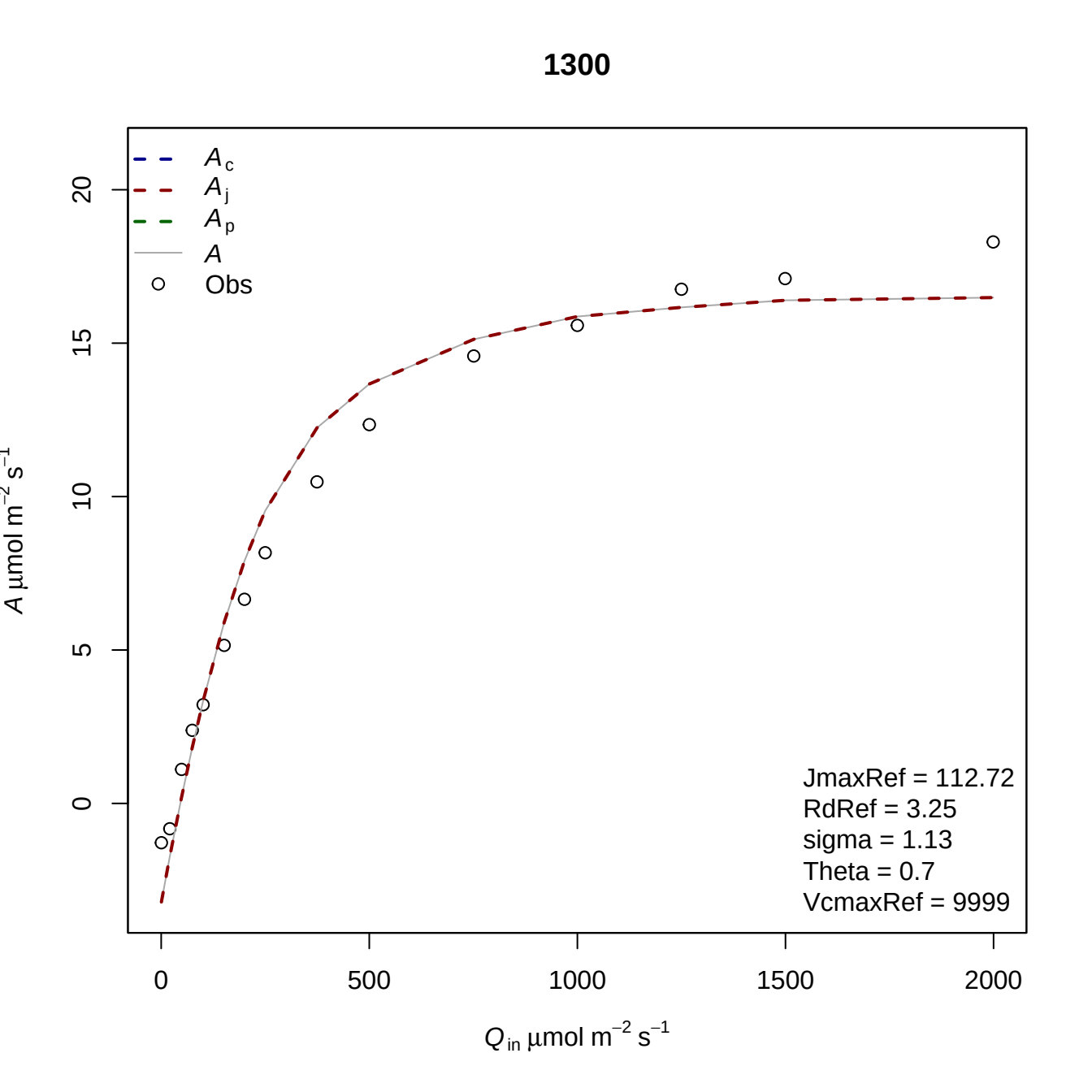

1295
A_c
A_j
A_p
A
Obs
JmaxRef = 118.39
RdRef = 4.56
sigma = 1.47
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

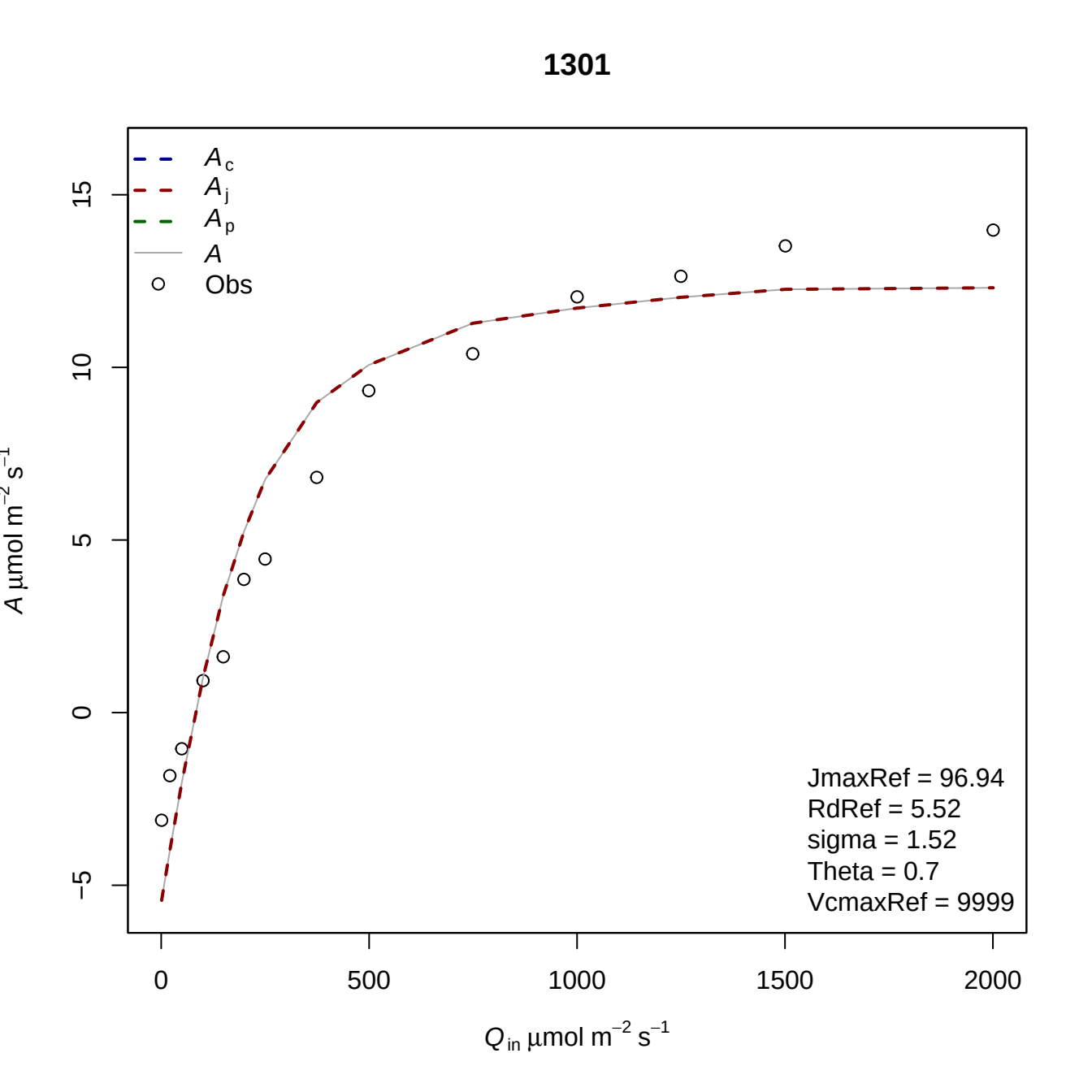

1296
A_c
A_j
A_p
A
Obs
JmaxRef = 108.27
RdRef = 3.63
sigma = 0.85
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

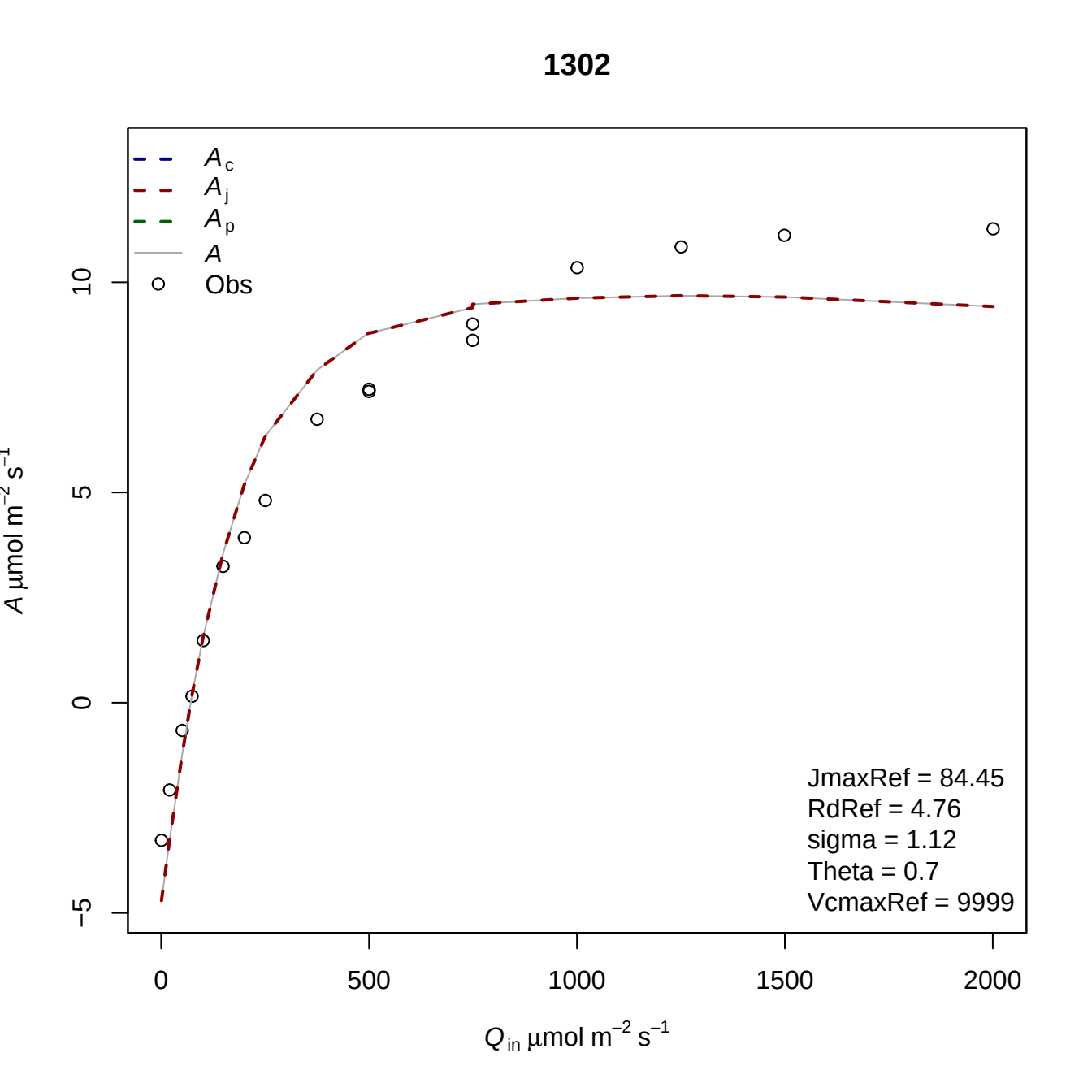

1297
A_c
A_j
A_p
A
Obs
JmaxRef = 141.95
RdRef = 5.15
sigma = 1.71
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
25
20
15
10
5
0
−5
0
500
1000
1500
2000

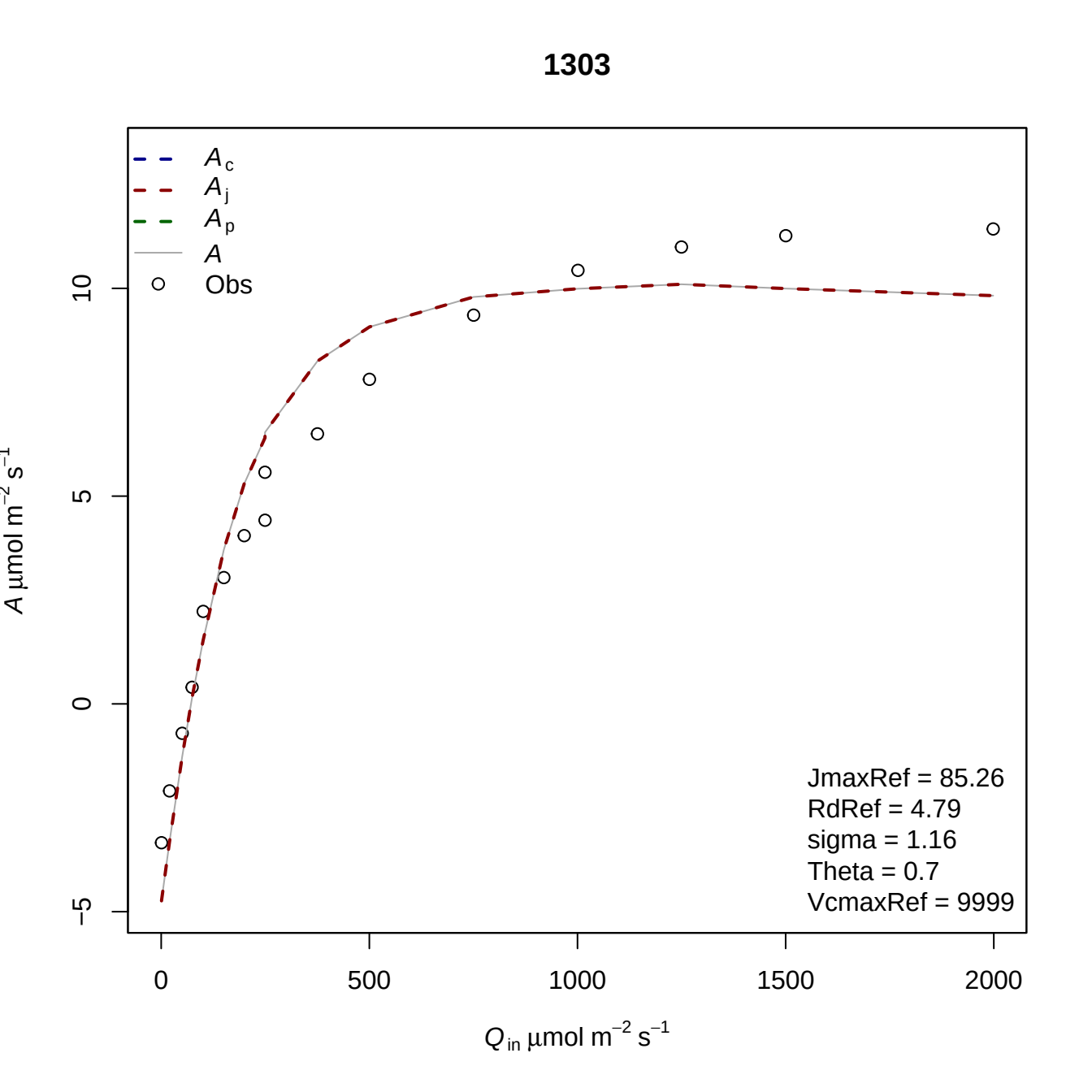

1298
A_c
A_j
A_p
A
Obs
JmaxRef = 98.2
RdRef = 3.76
sigma = 1.29
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

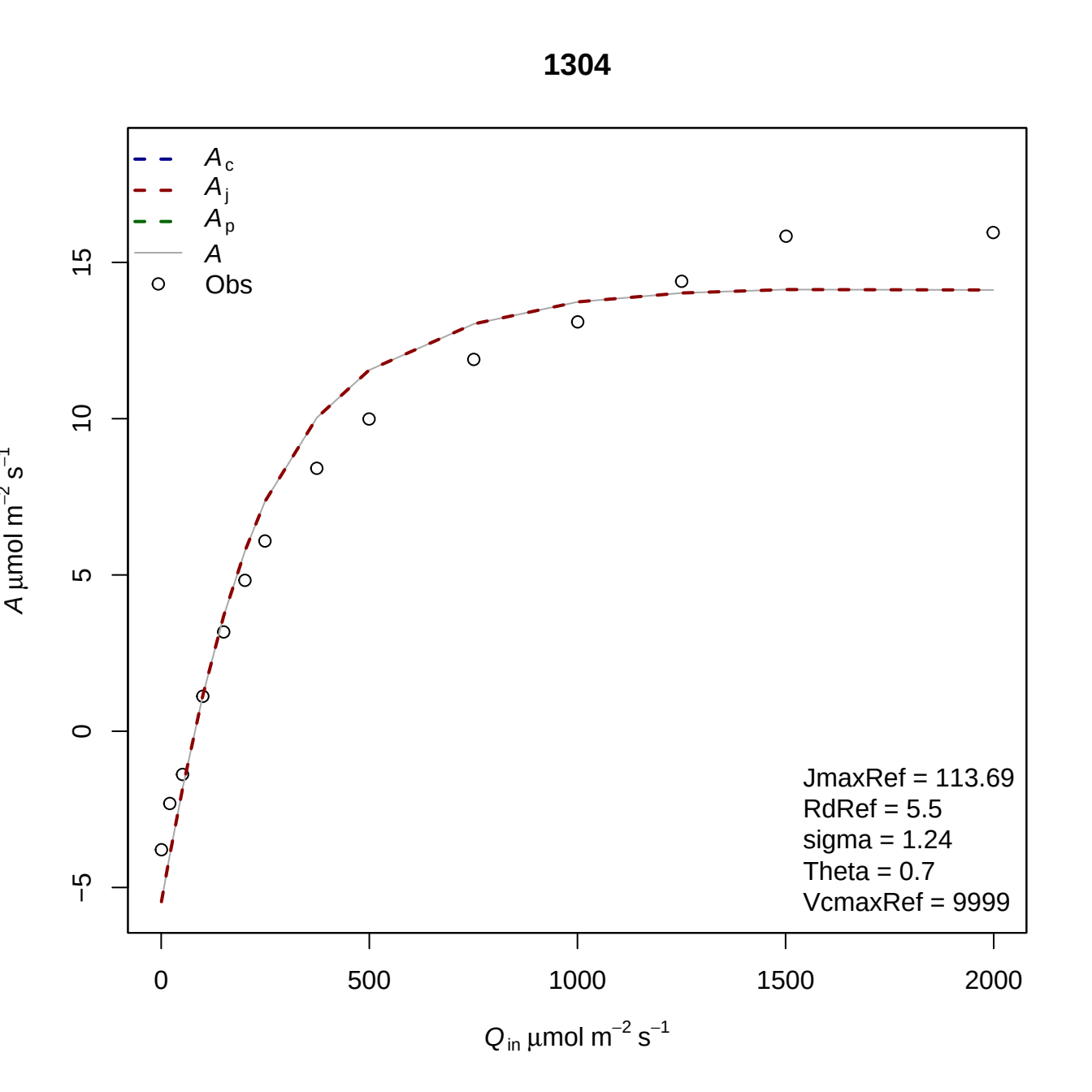

1299
A_c
A_j
A_p
A
Obs
JmaxRef = 117.56
RdRef = 5.59
sigma = 1.23
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
20
15
10
5
0
−5
0
500
1000
1500
2000

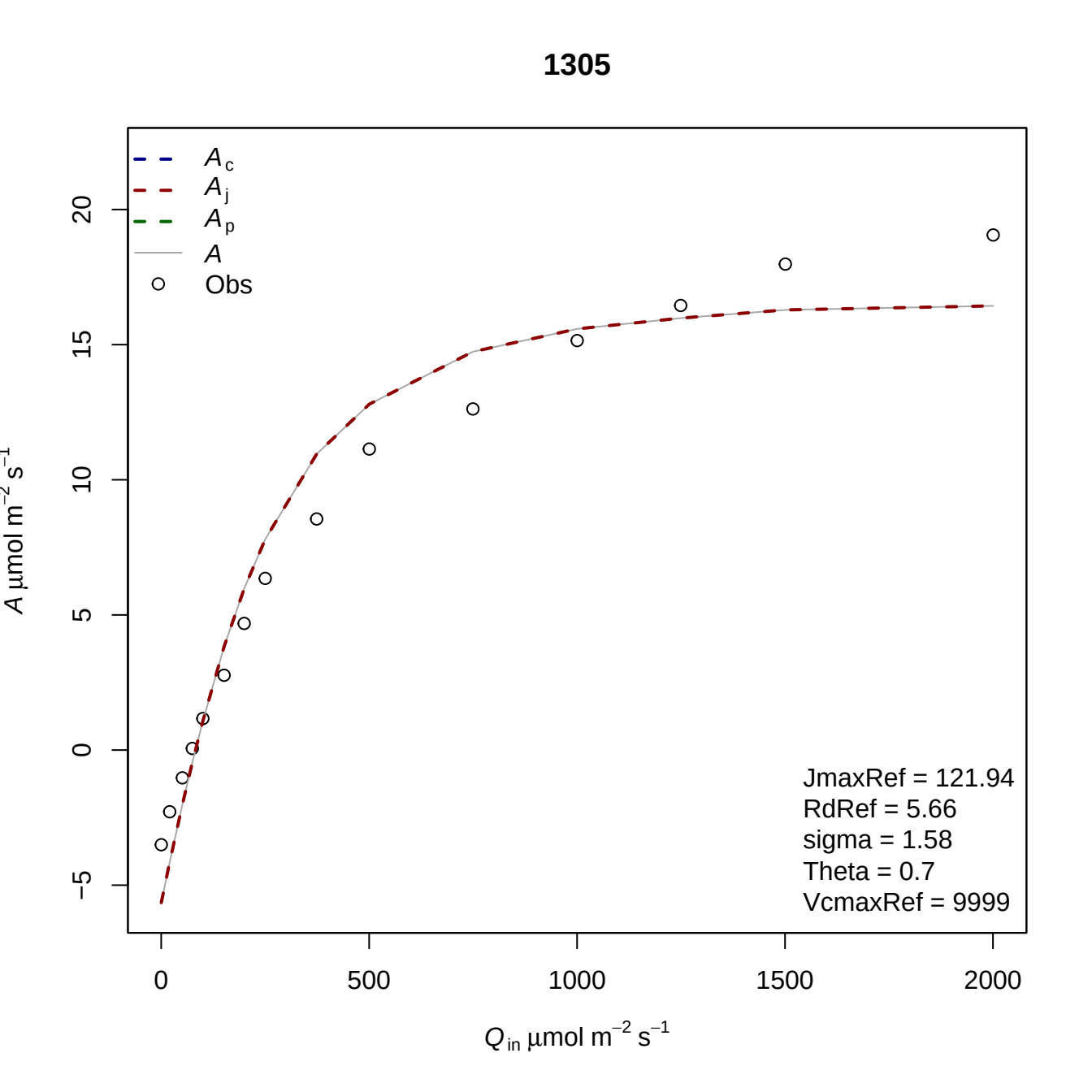

1300
A_c
A_j
A_p
A
Obs
JmaxRef = 112.72
RdRef = 3.25
sigma = 1.13
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

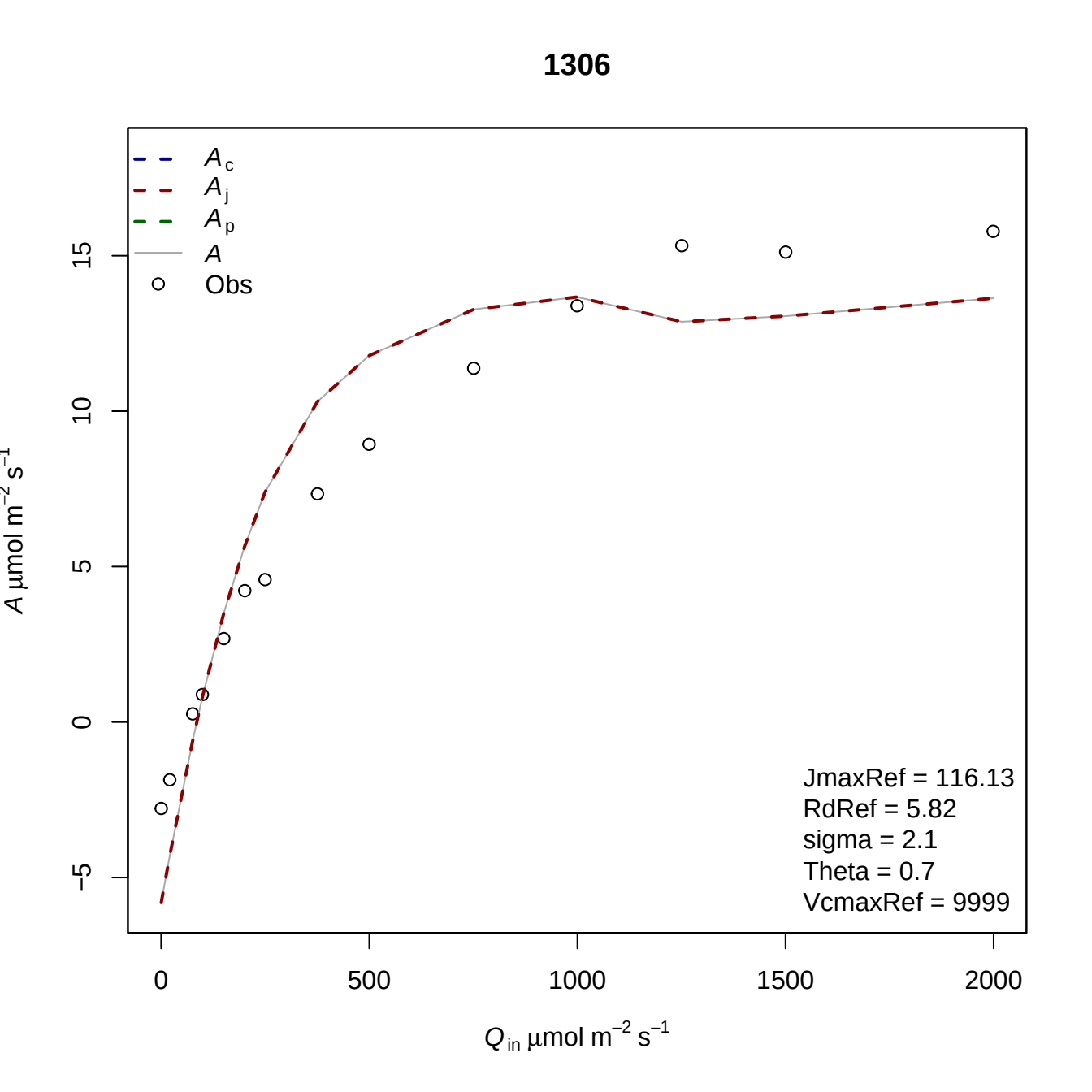

1301
A_c
A_j
A_p
A
Obs
JmaxRef = 96.94
RdRef = 5.52
sigma = 1.52
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
15
10
5
0
−5
0
500
1000
1500
2000

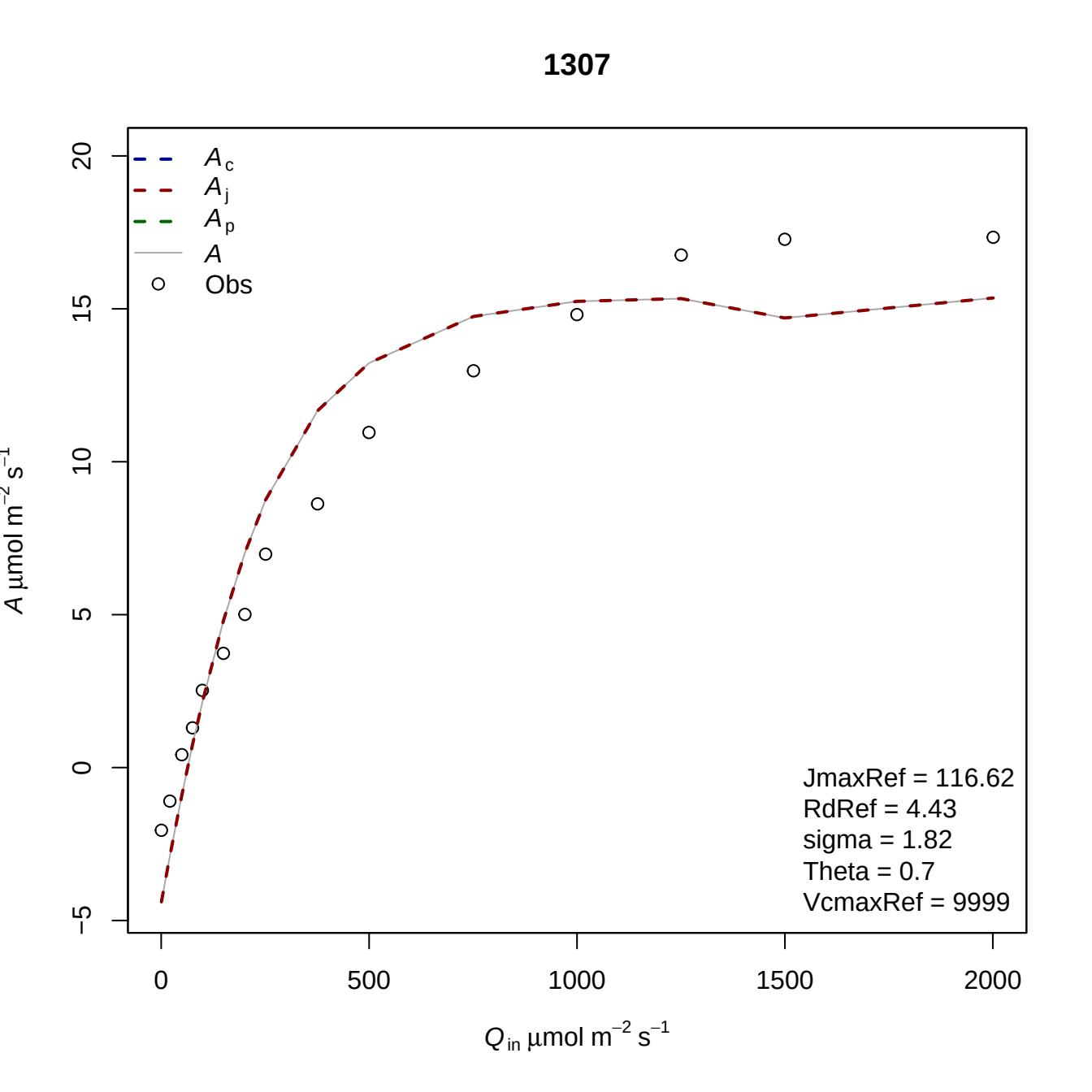

1302
A_c
A_j
A_p
A
Obs
JmaxRef = 84.45
RdRef = 4.76
sigma = 1.12
Theta = 0.7
VcmaxRef = 9999
−5
0
5
10
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

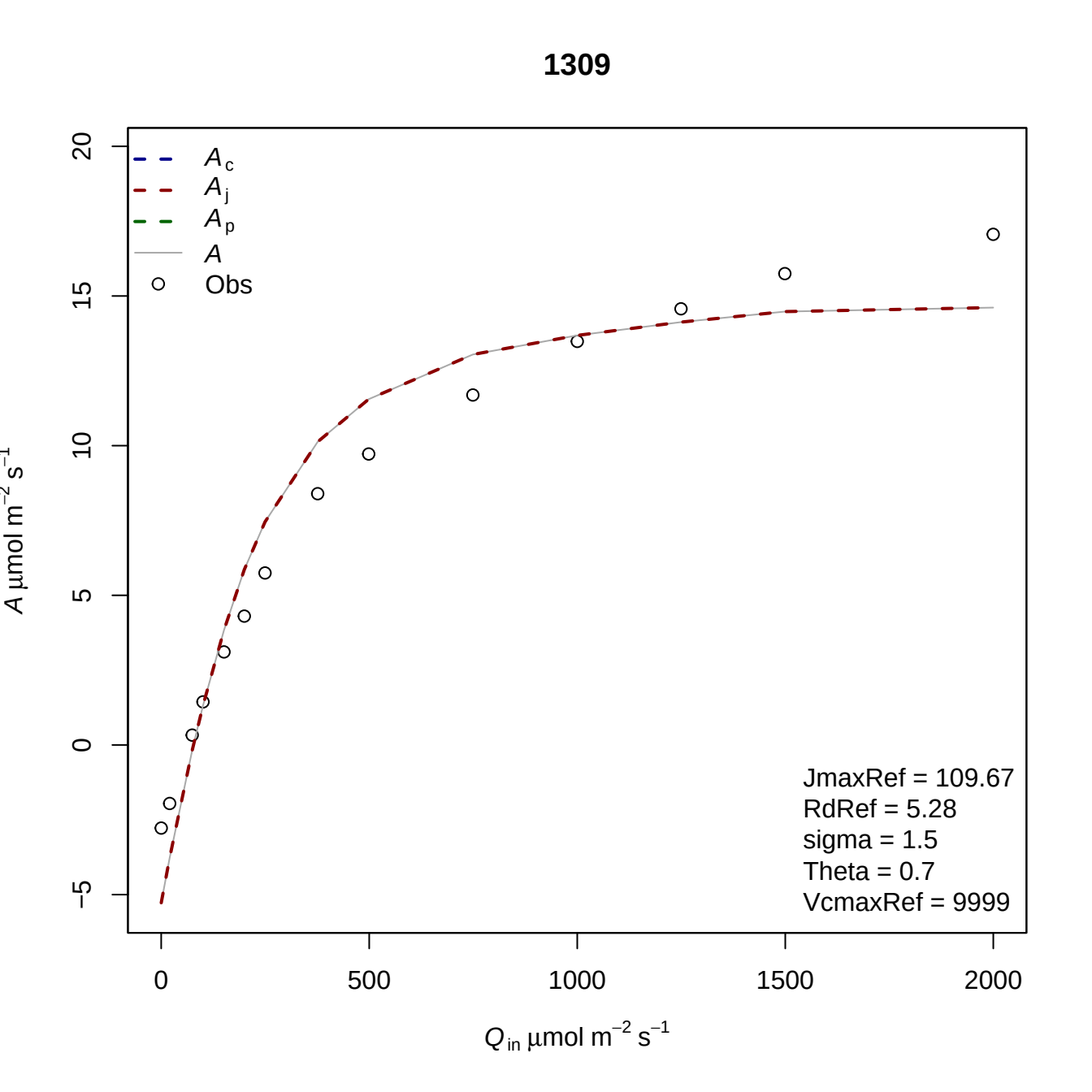

1303
A_c
A_j
A_p
A
Obs
JmaxRef = 85.26
RdRef = 4.79
sigma = 1.16
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

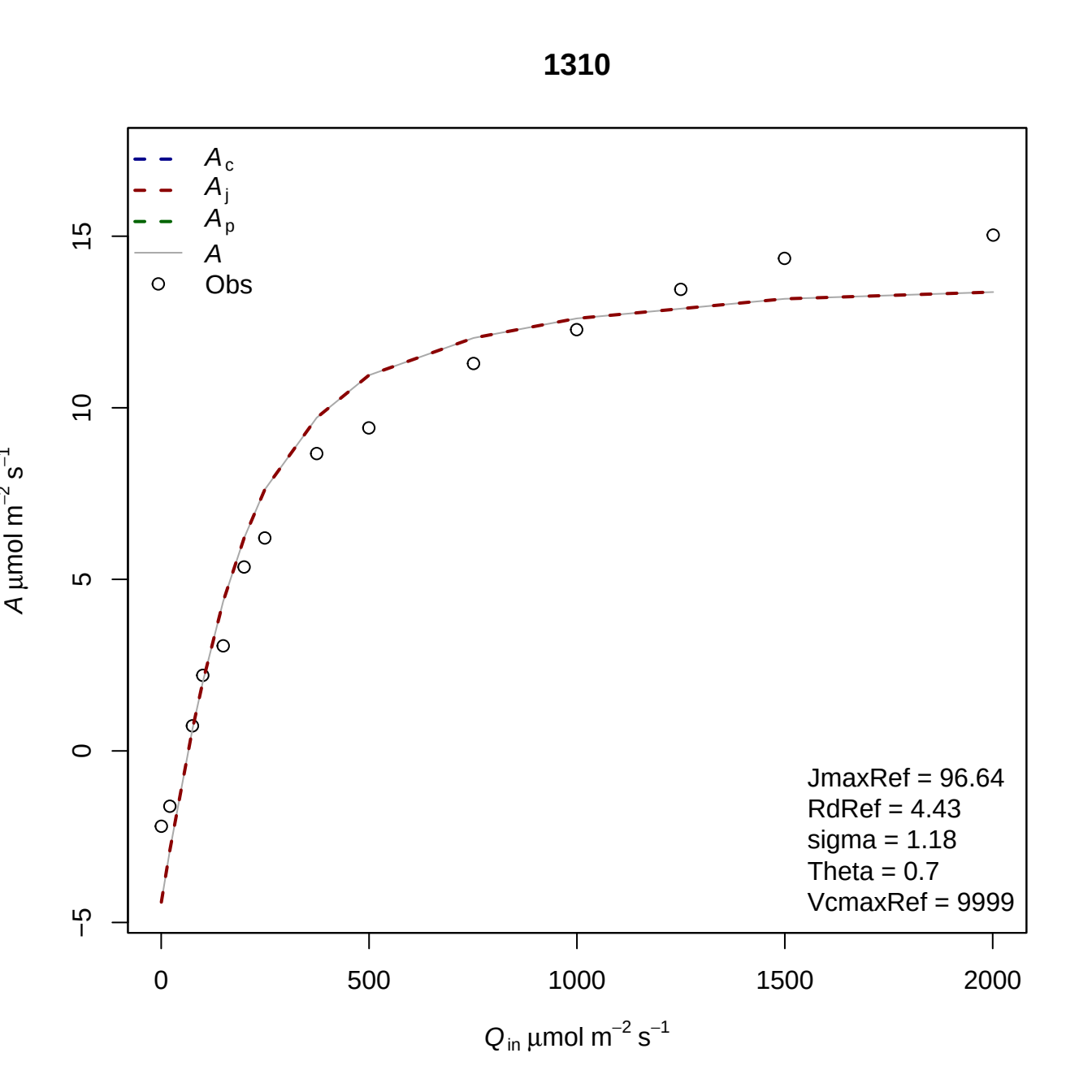

1304
A_c
A_j
A_p
A
Obs
JmaxRef = 113.69
RdRef = 5.5
sigma = 1.24
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

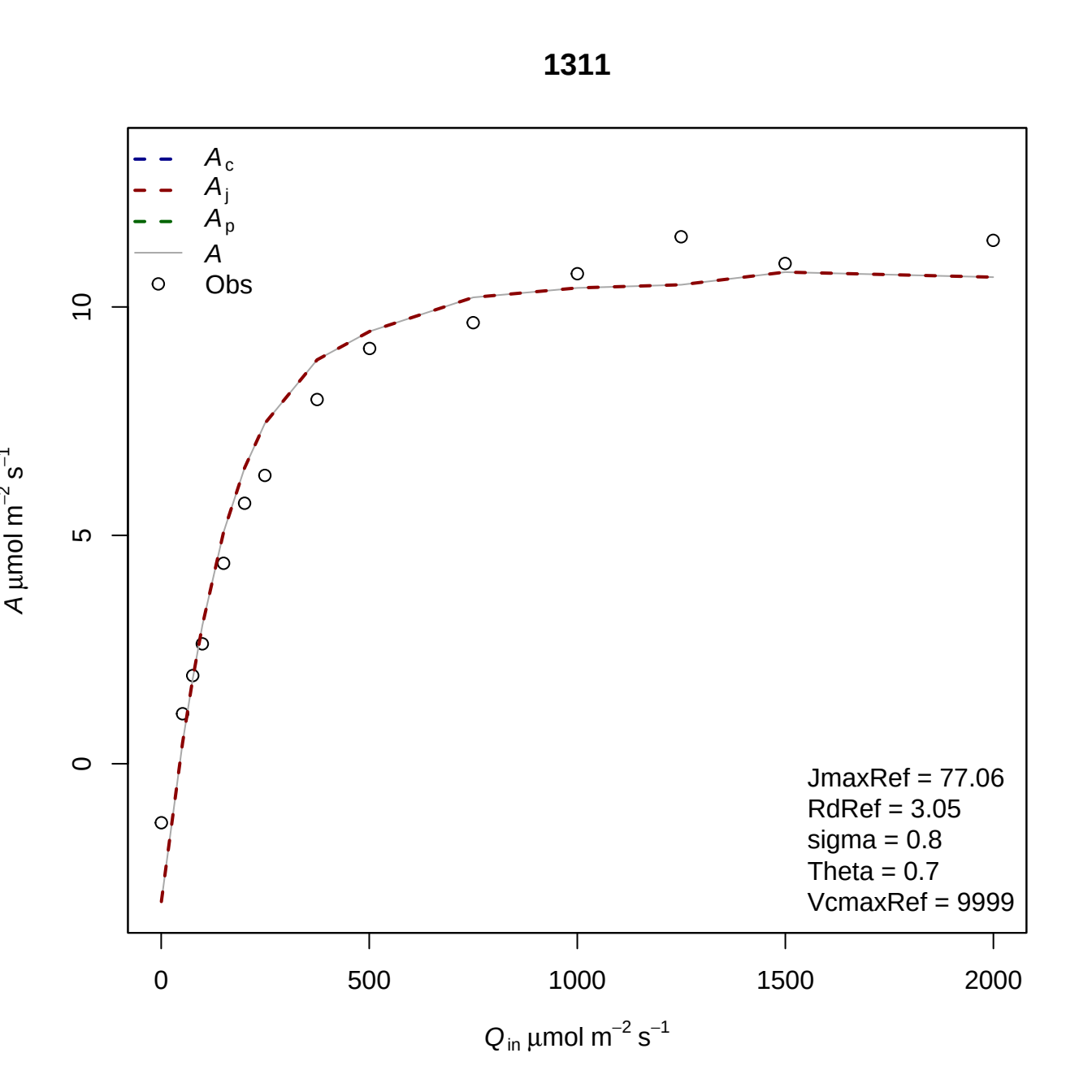

1305
A_c
A_j
A_p
A
Obs
JmaxRef = 121.94
RdRef = 5.66
sigma = 1.58
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
20
0
500
1000
1500
2000

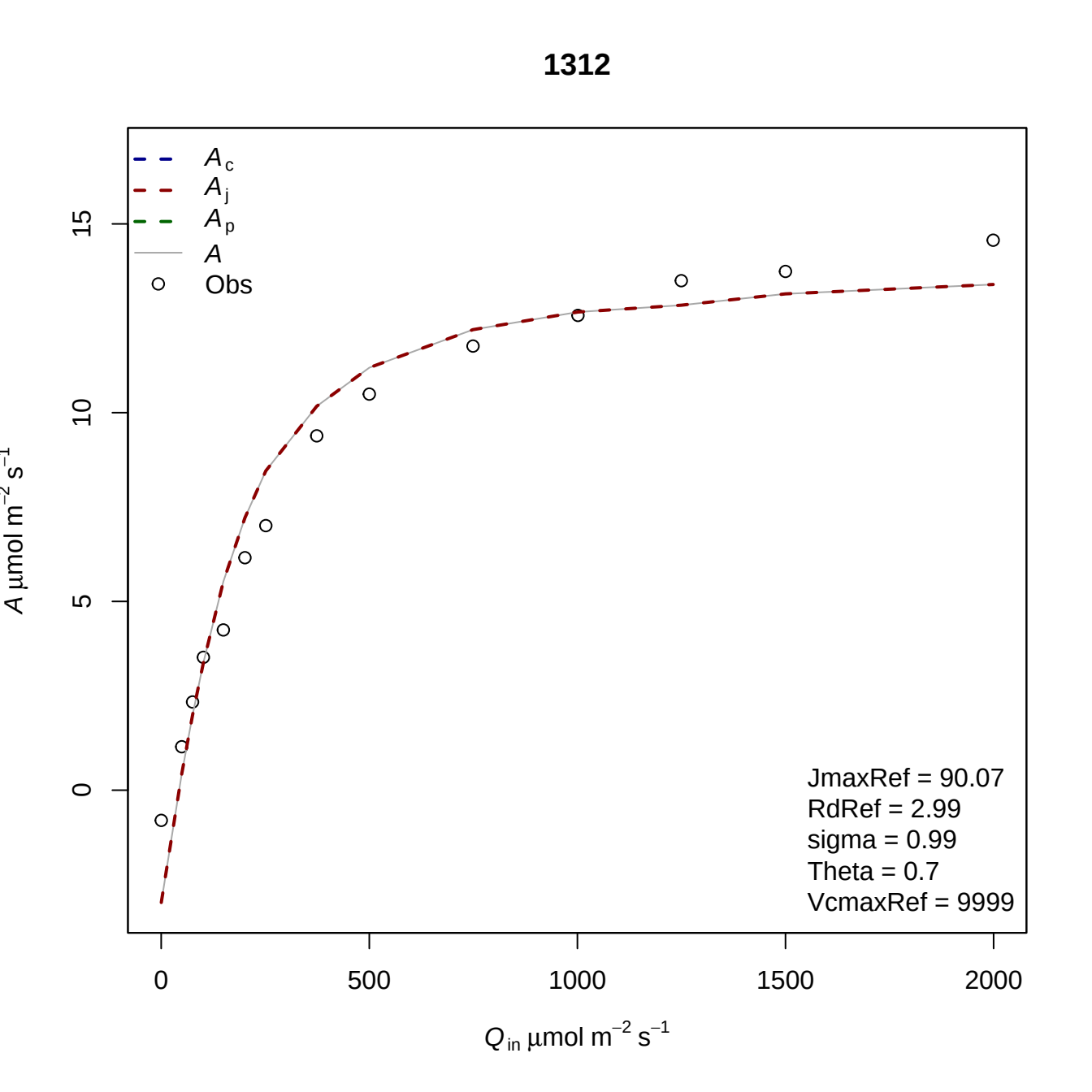

1306
A_c
A_j
A_p
A
Obs
JmaxRef = 116.13
RdRef = 5.82
sigma = 2.1
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

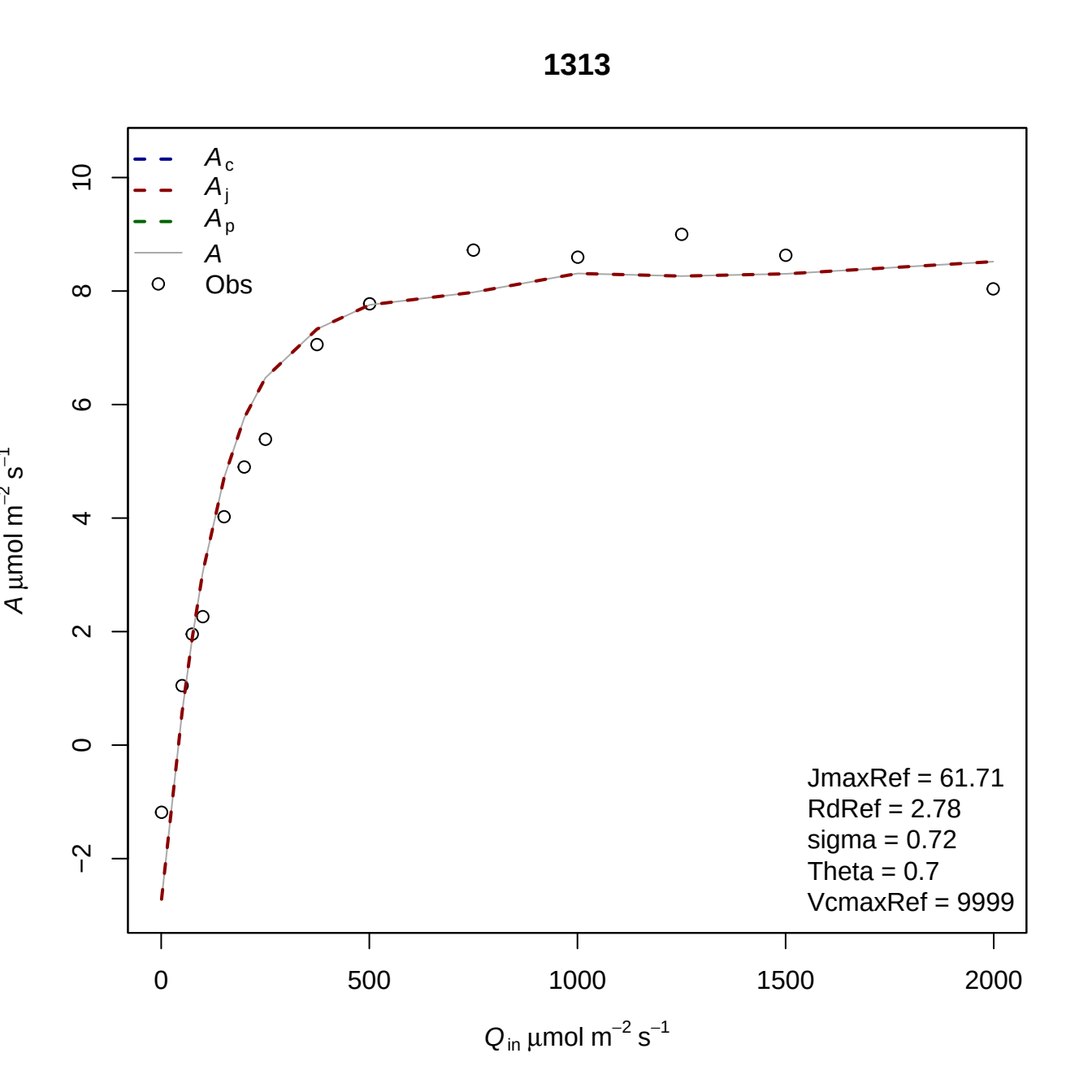

1307
A_c
A_j
A_p
A
Obs
JmaxRef = 116.62
RdRef = 4.43
sigma = 1.82
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
20
15
10
5
0
−5
0
500
1000
1500
2000

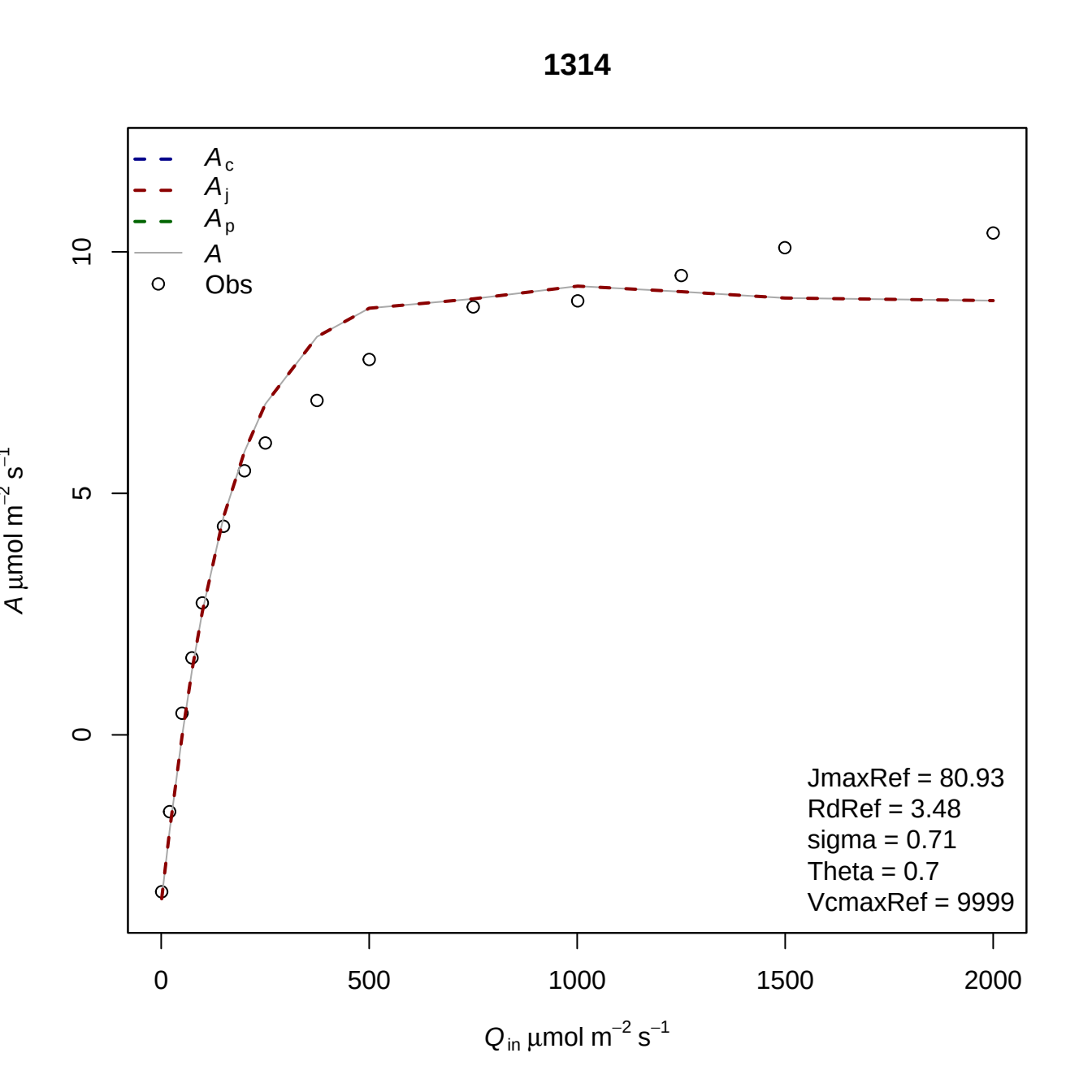

1309
A_c
A_j
A_p
A
Obs
JmaxRef = 109.67
RdRef = 5.28
sigma = 1.5
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
20
15
10
5
0
−5
0
500
1000
1500
2000

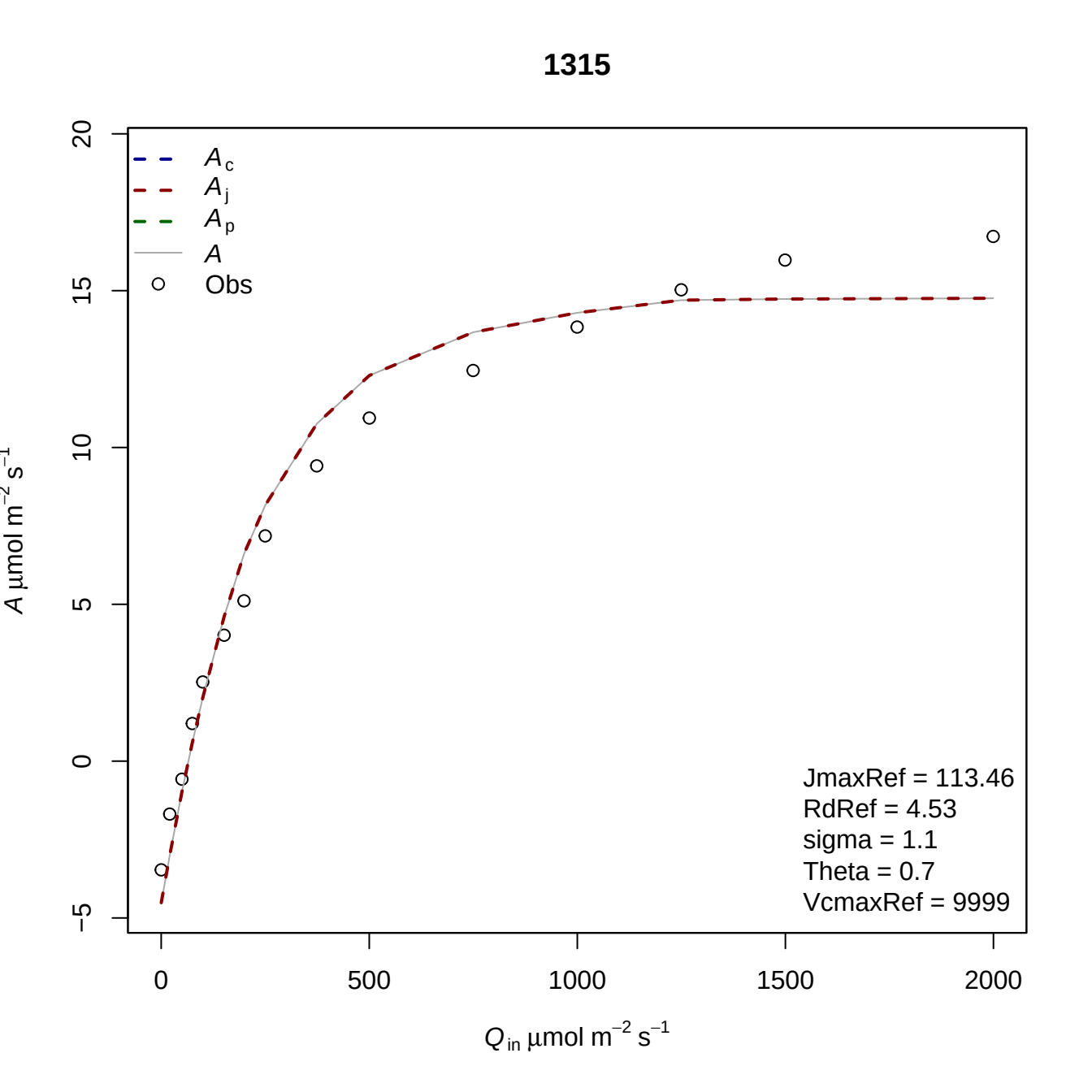

1310
A_c
A_j
A_p
A
Obs
JmaxRef = 96.64
RdRef = 4.43
sigma = 1.18
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

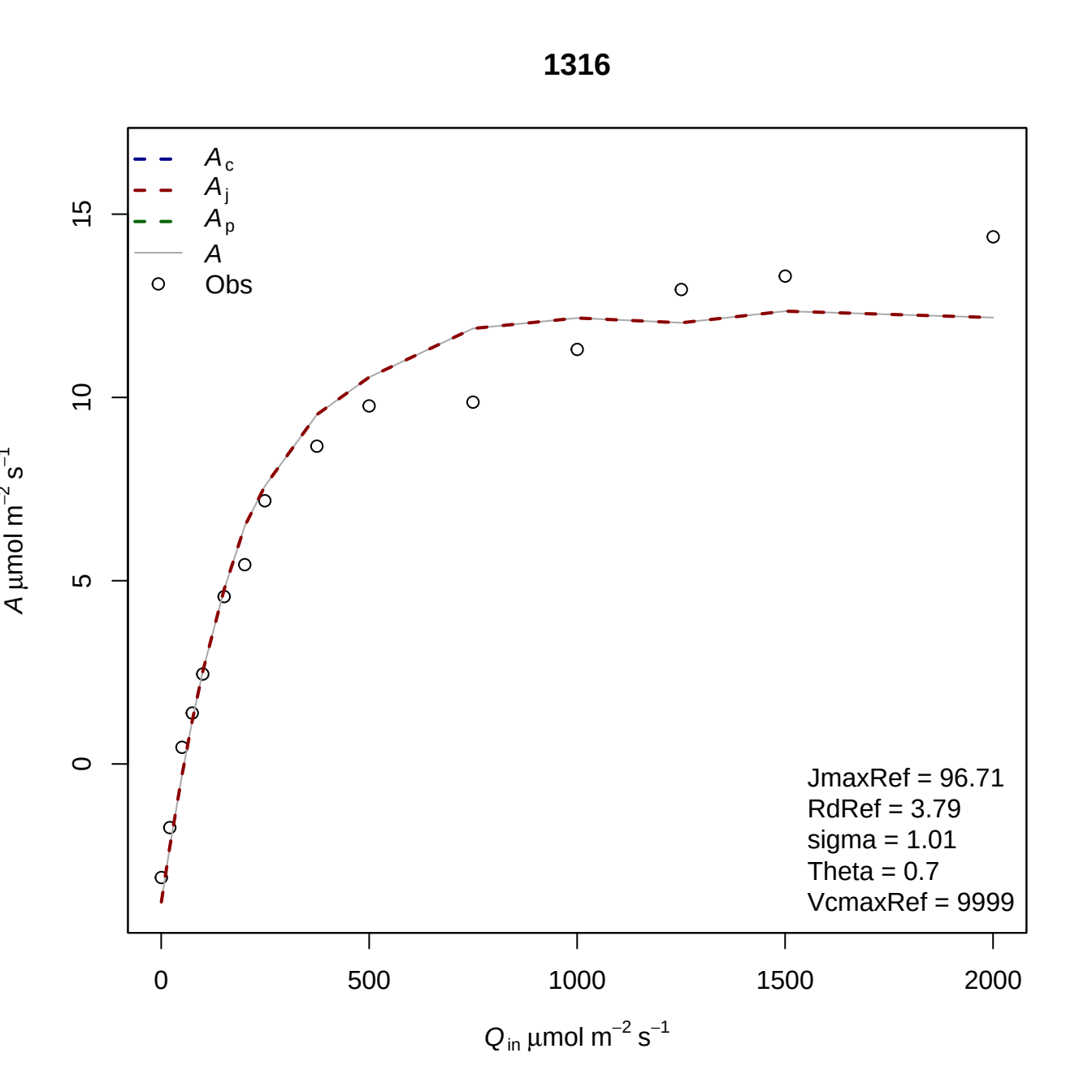

1311
A_c
A_j
A_p
A
Obs
JmaxRef = 77.06
RdRef = 3.05
sigma = 0.8
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

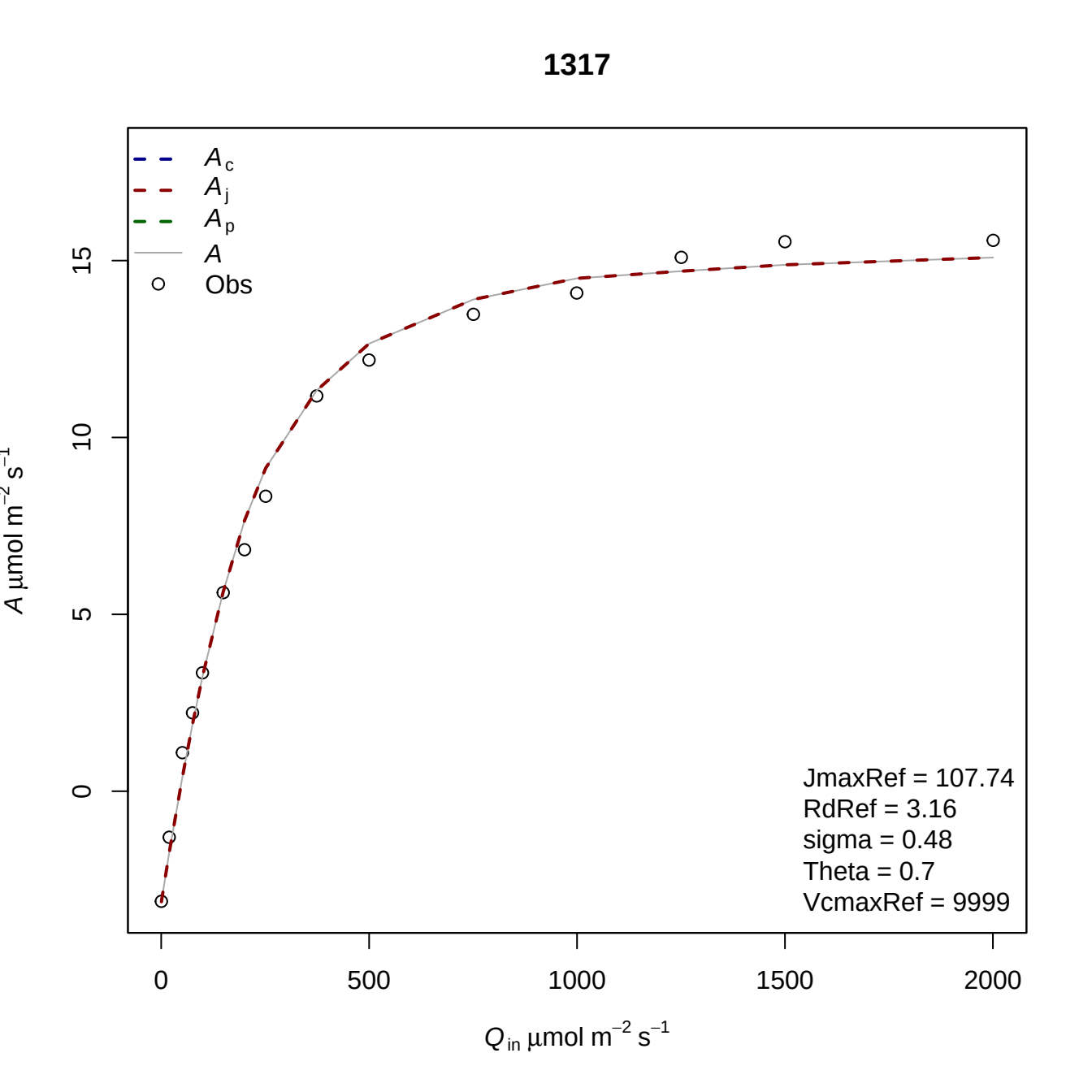

1312
A_c
A_j
A_p
A
Obs
JmaxRef = 90.07
RdRef = 2.99
sigma = 0.99
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

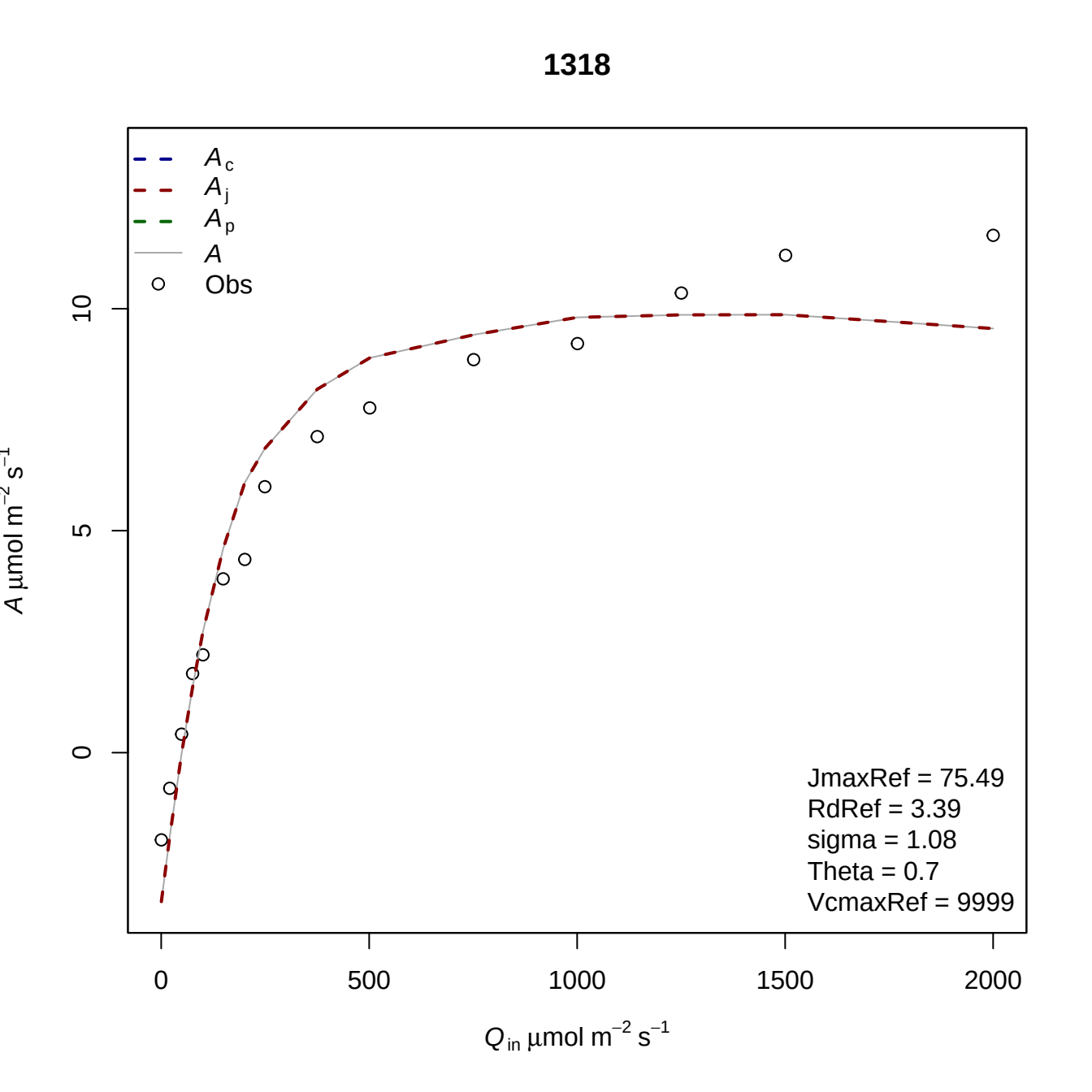

1313
A_c
A_j
A_p
A
Obs
JmaxRef = 61.71
RdRef = 2.78
sigma = 0.72
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

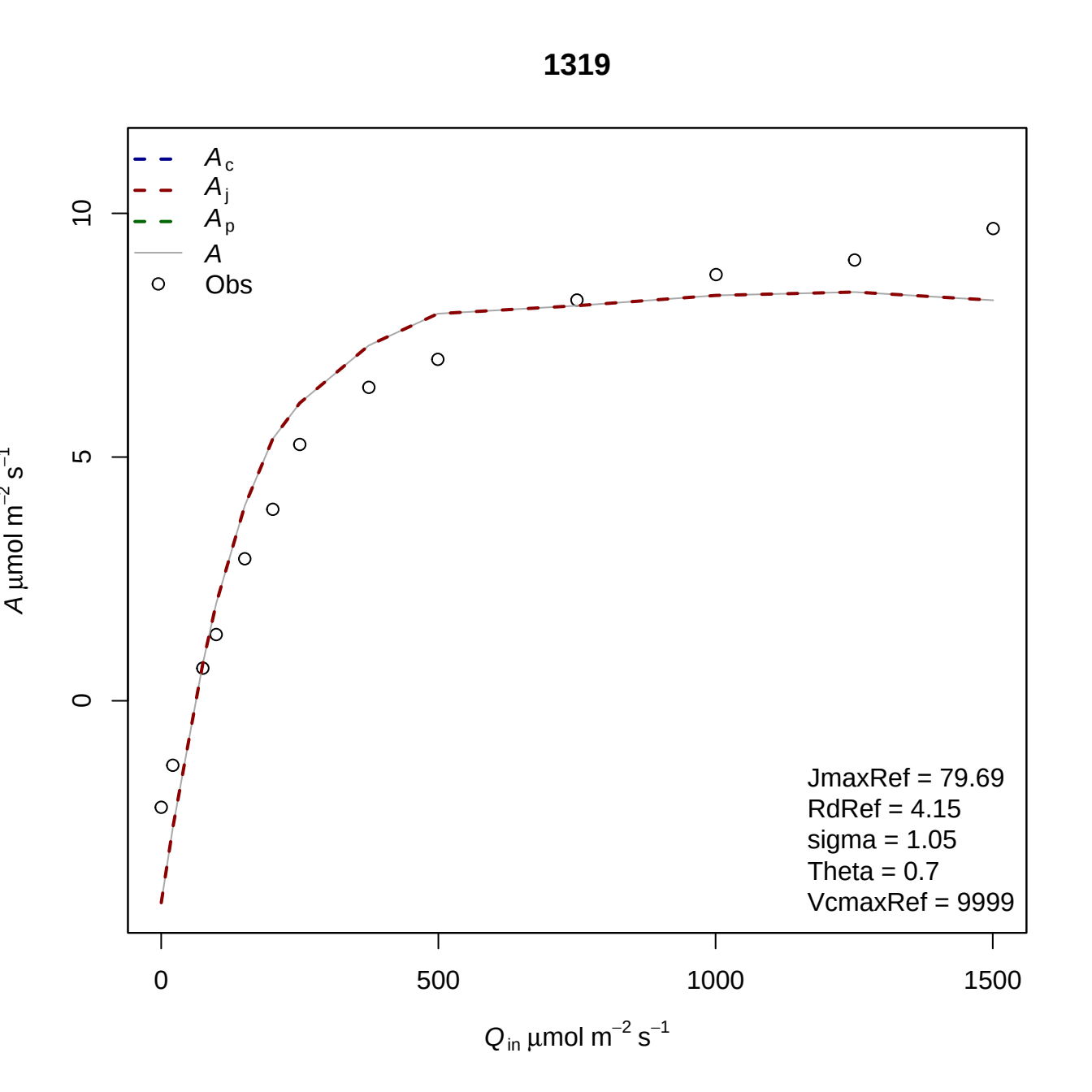

1314
A_c
A_j
A_p
A
Obs
JmaxRef = 80.93
RdRef = 3.48
sigma = 0.71
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

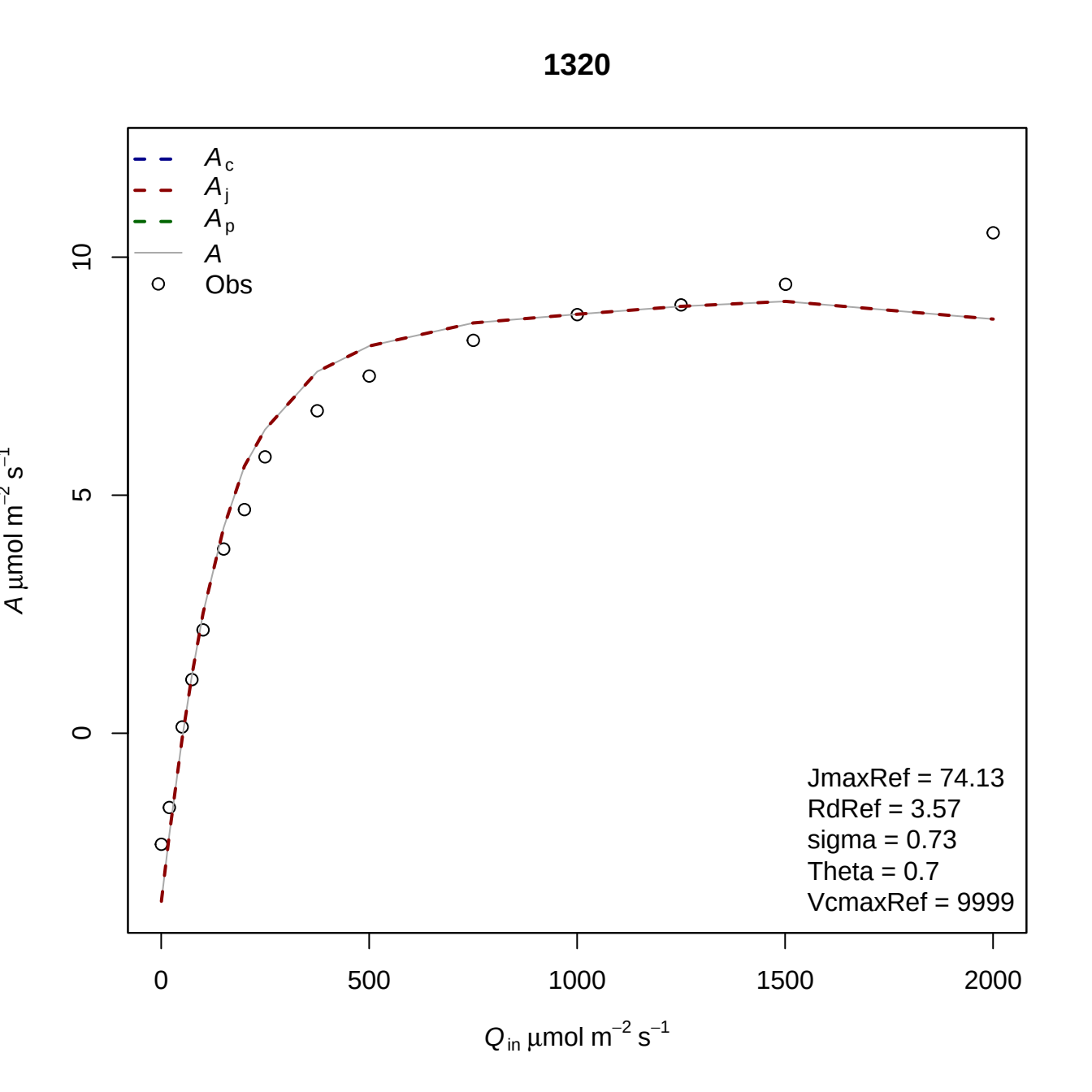

1315
A_c
A_j
A_p
A
Obs
JmaxRef = 113.46
RdRef = 4.53
sigma = 1.1
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
20
15
10
5
0
−5
0
500
1000
1500
2000

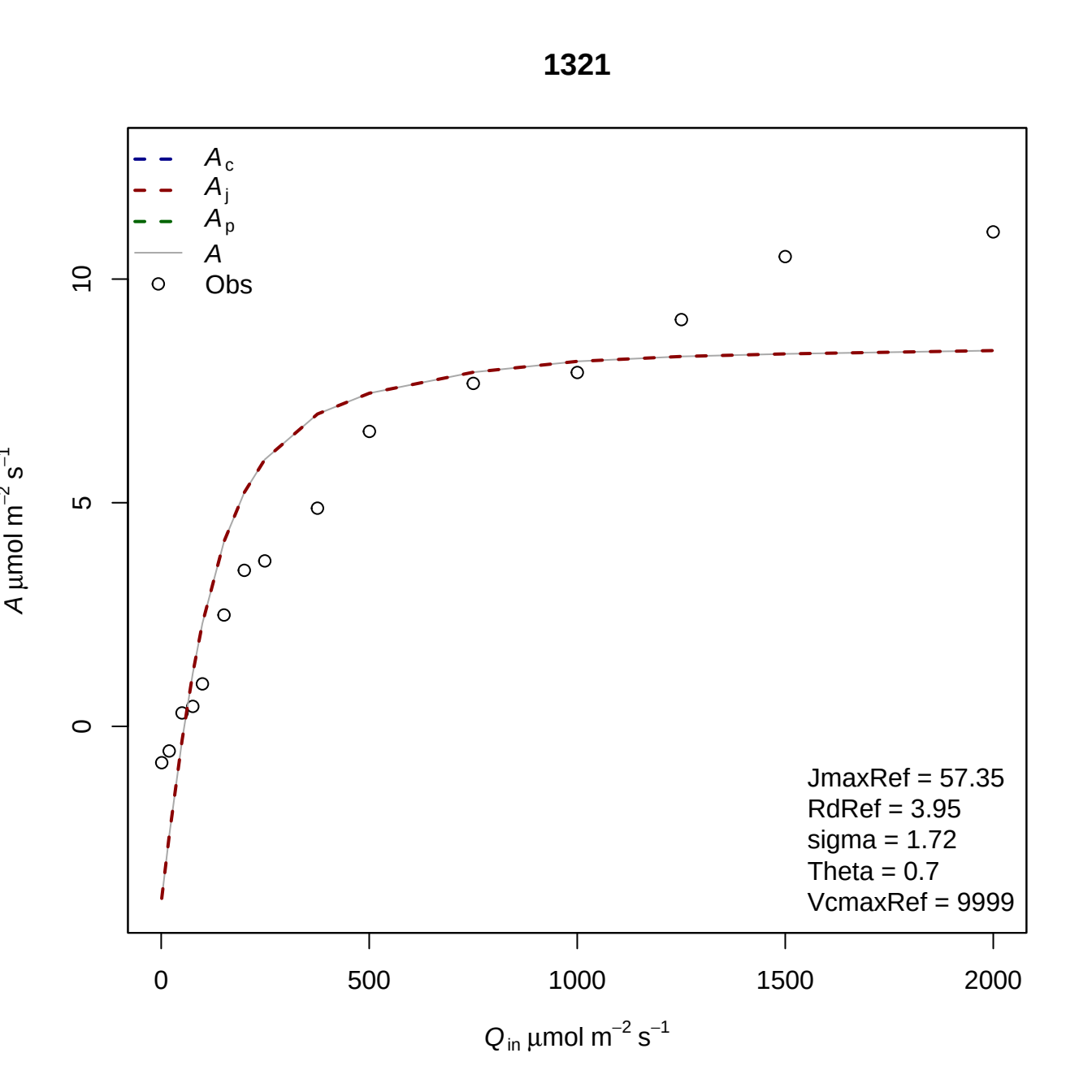

1316
A_c
A_j
A_p
A
Obs
JmaxRef = 96.71
RdRef = 3.79
sigma = 1.01
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

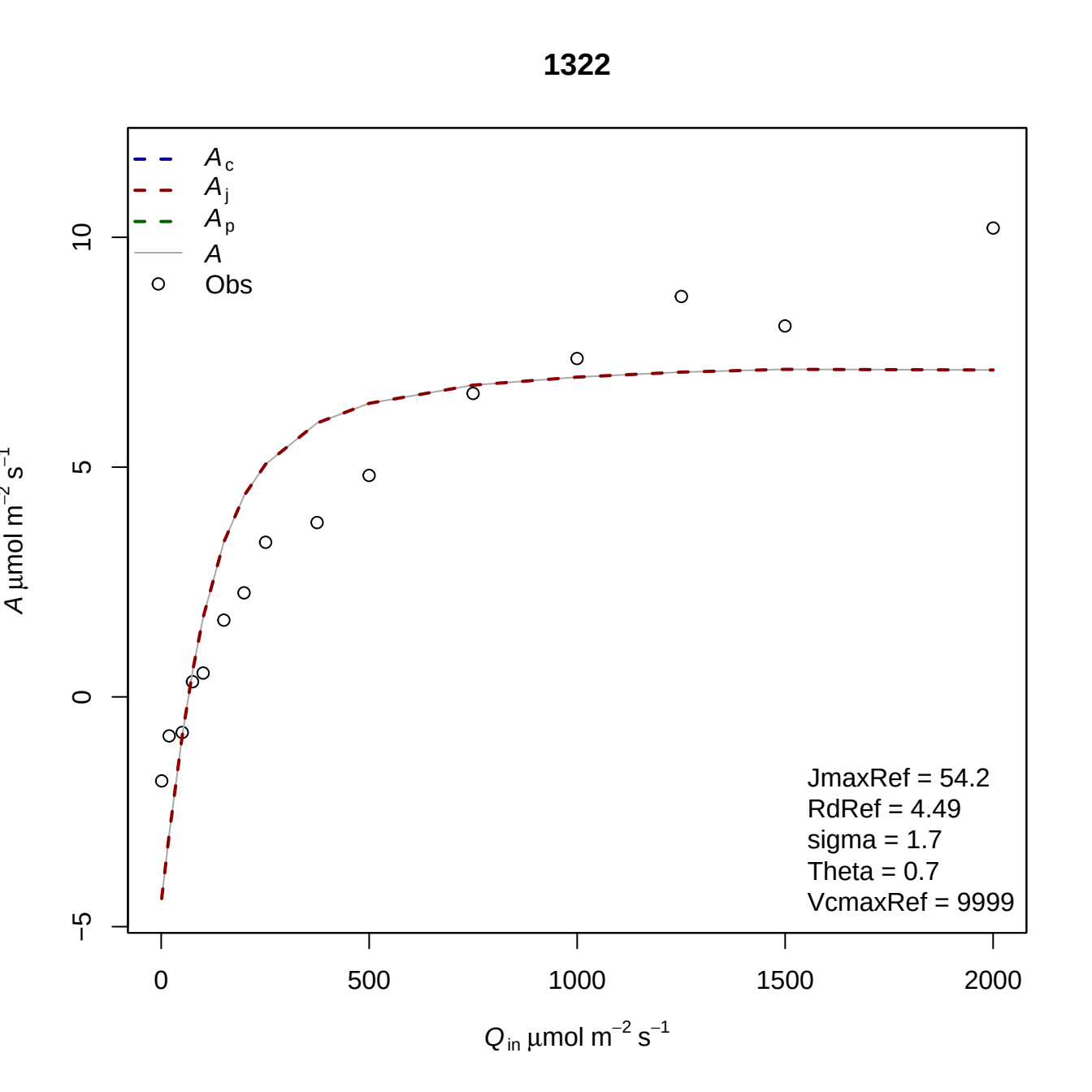

1317
A_c
A_j
A_p
A
Obs
JmaxRef = 107.74
RdRef = 3.16
sigma = 0.48
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

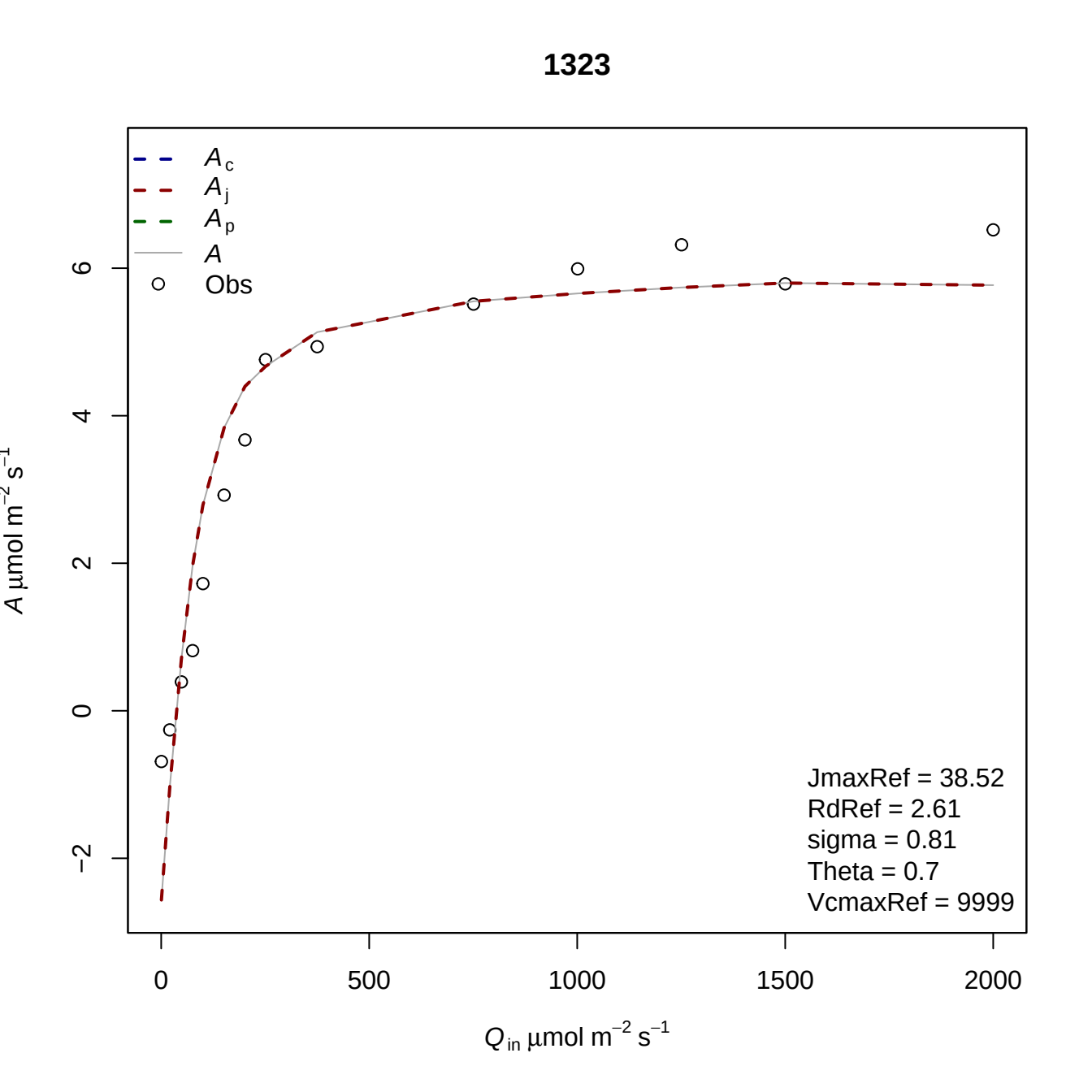

1318
A_c
A_j
A_p
A
Obs
JmaxRef = 75.49
RdRef = 3.39
sigma = 1.08
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

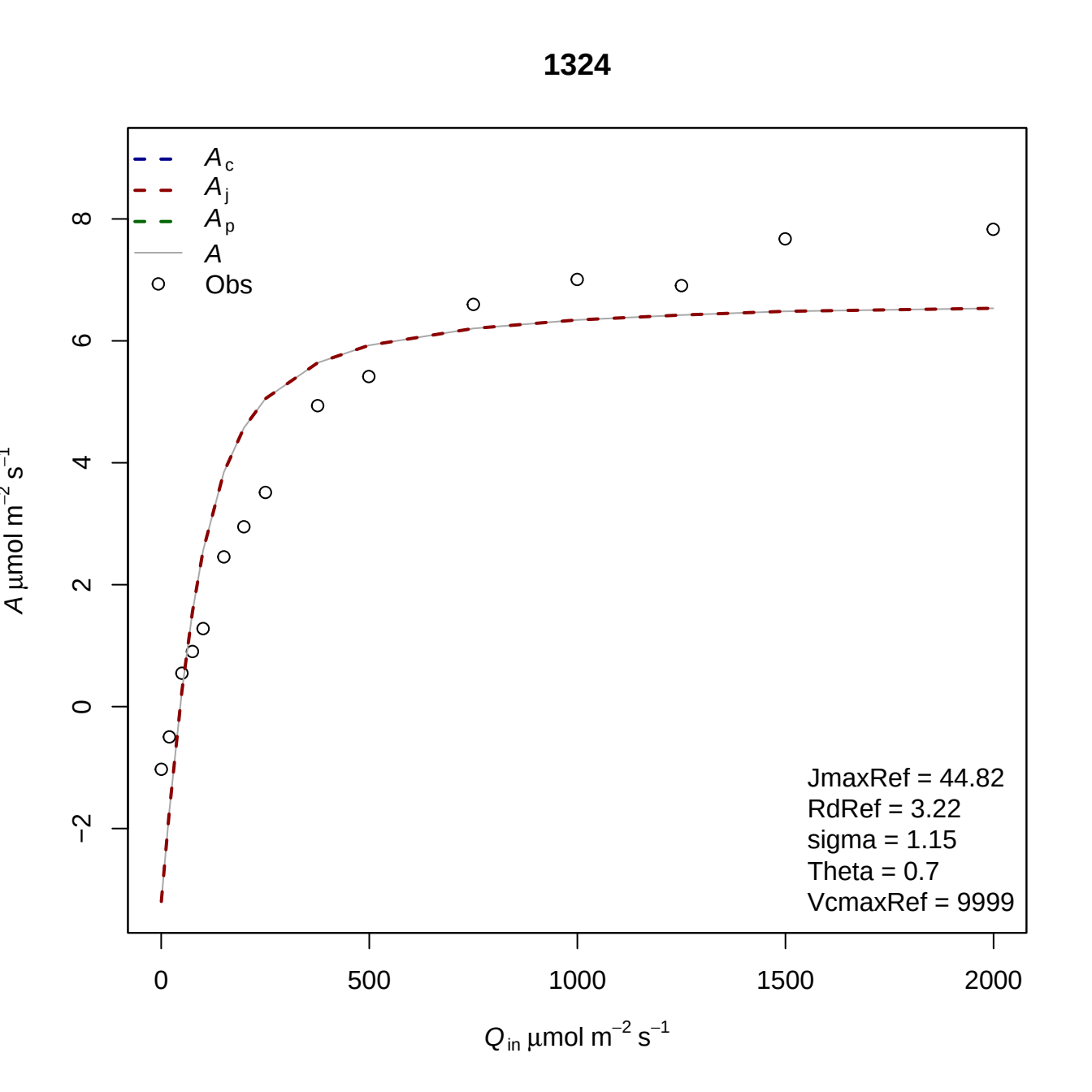

1319
A_c
A_j
A_p
A
Obs
JmaxRef = 79.69
RdRef = 4.15
sigma = 1.05
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

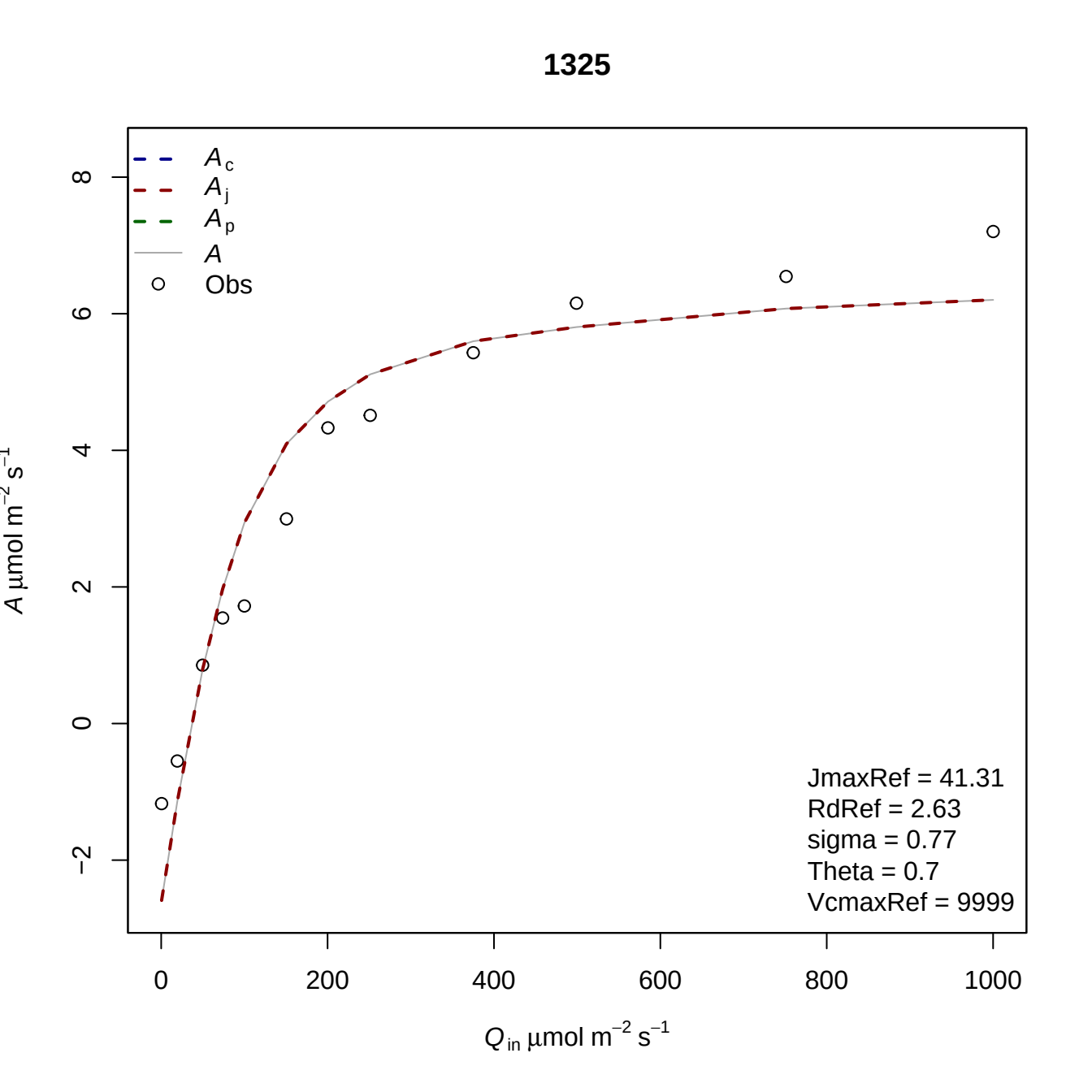

1320
A_c
A_j
A_p
A
Obs
JmaxRef = 74.13
RdRef = 3.57
sigma = 0.73
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

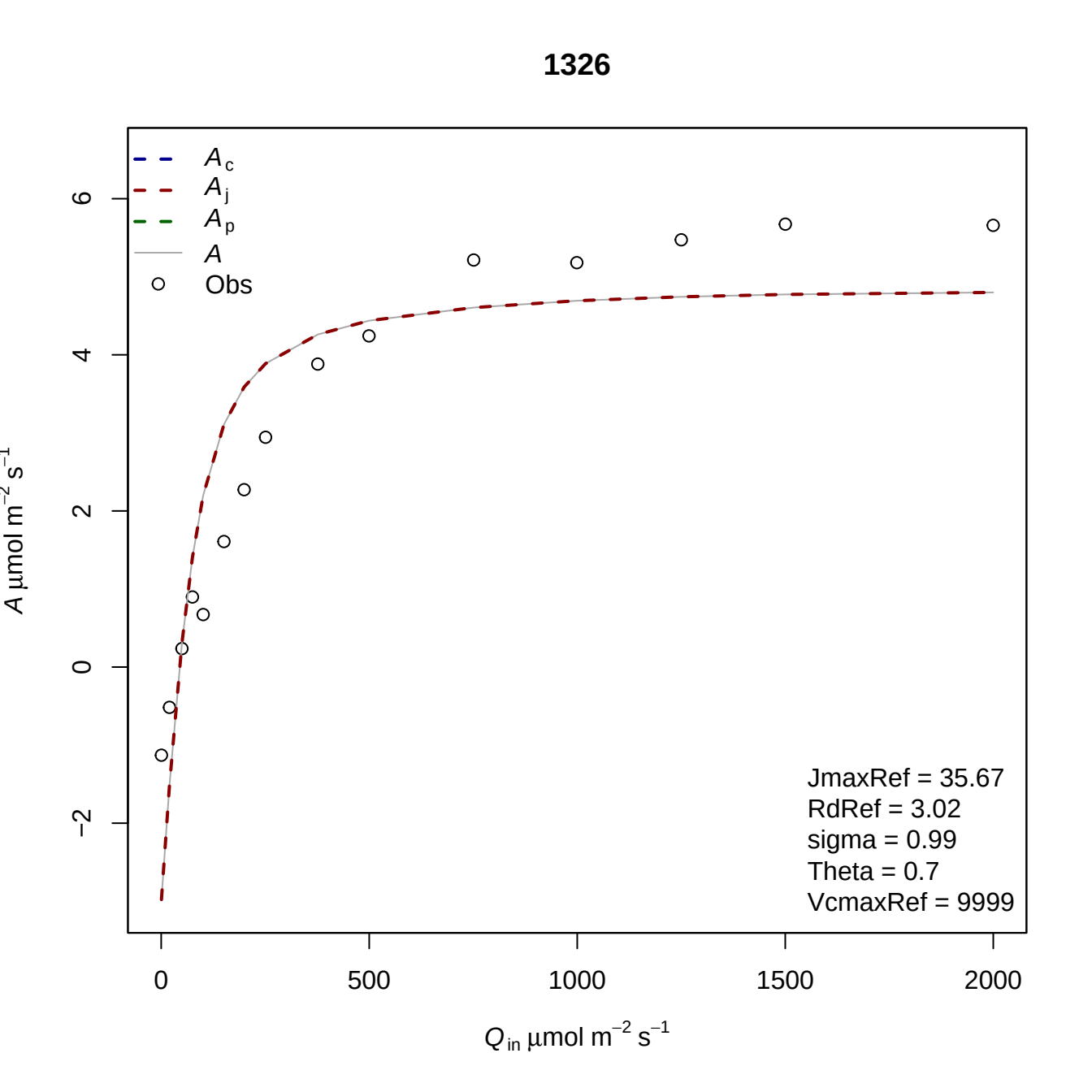

1321
A_c
A_j
A_p
A
Obs
JmaxRef = 57.35
RdRef = 3.95
sigma = 1.72
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

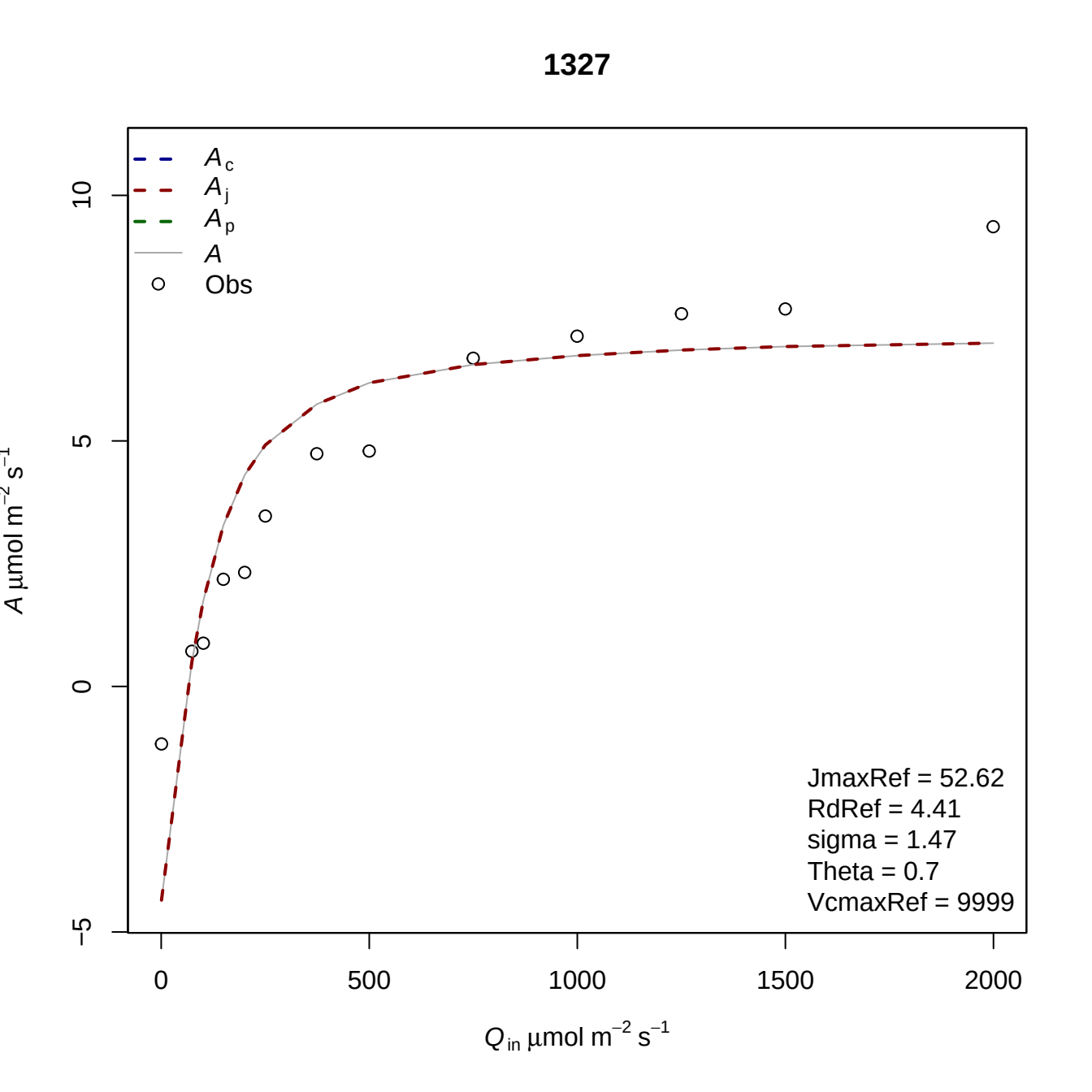

1322
A_c
A_j
A_p
A
Obs
JmaxRef = 54.2
RdRef = 4.49
sigma = 1.7
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
5
0
−5
0
500
1000
1500
2000

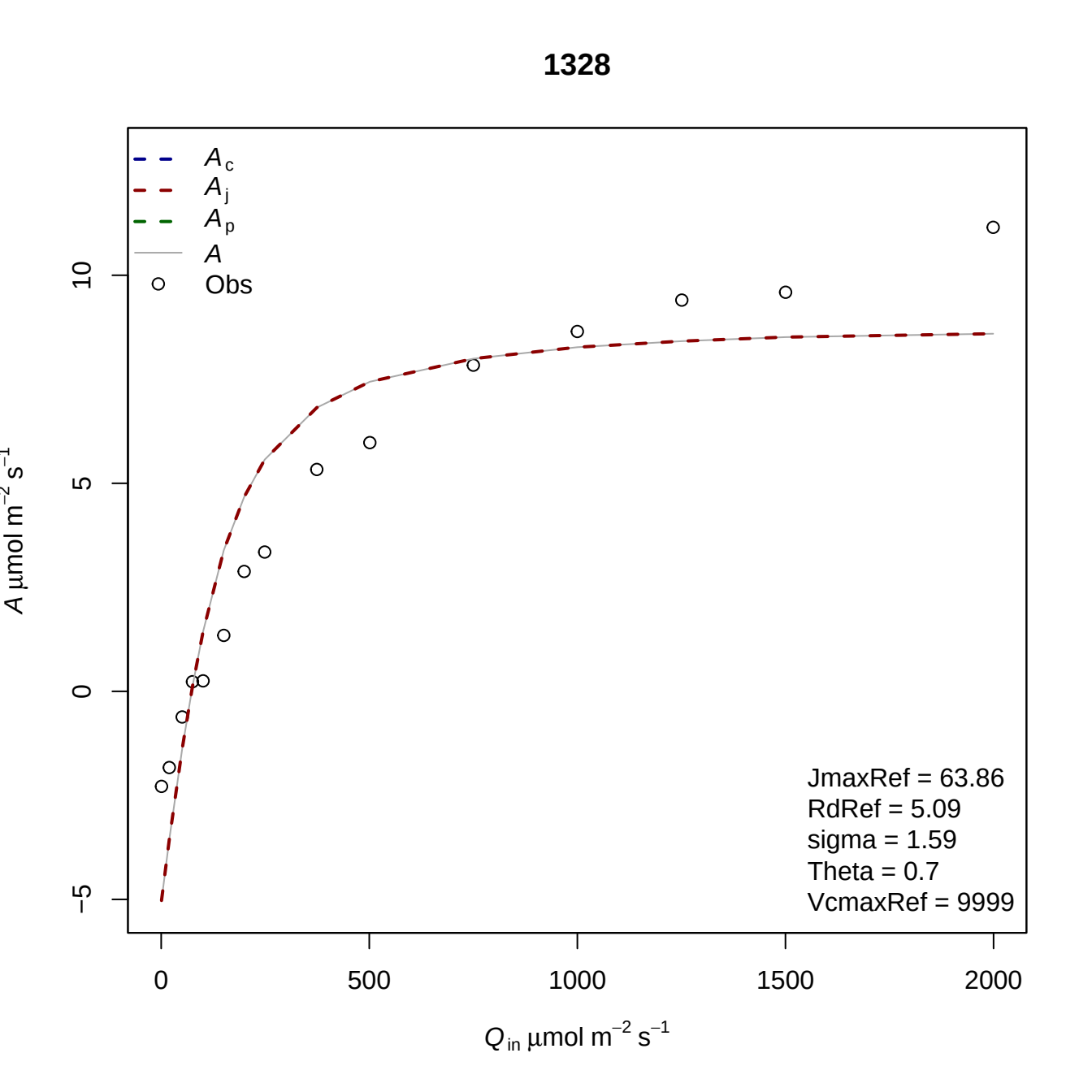

1323
A_c
A_j
A_p
A
Obs
JmaxRef = 38.52
RdRef = 2.61
sigma = 0.81
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

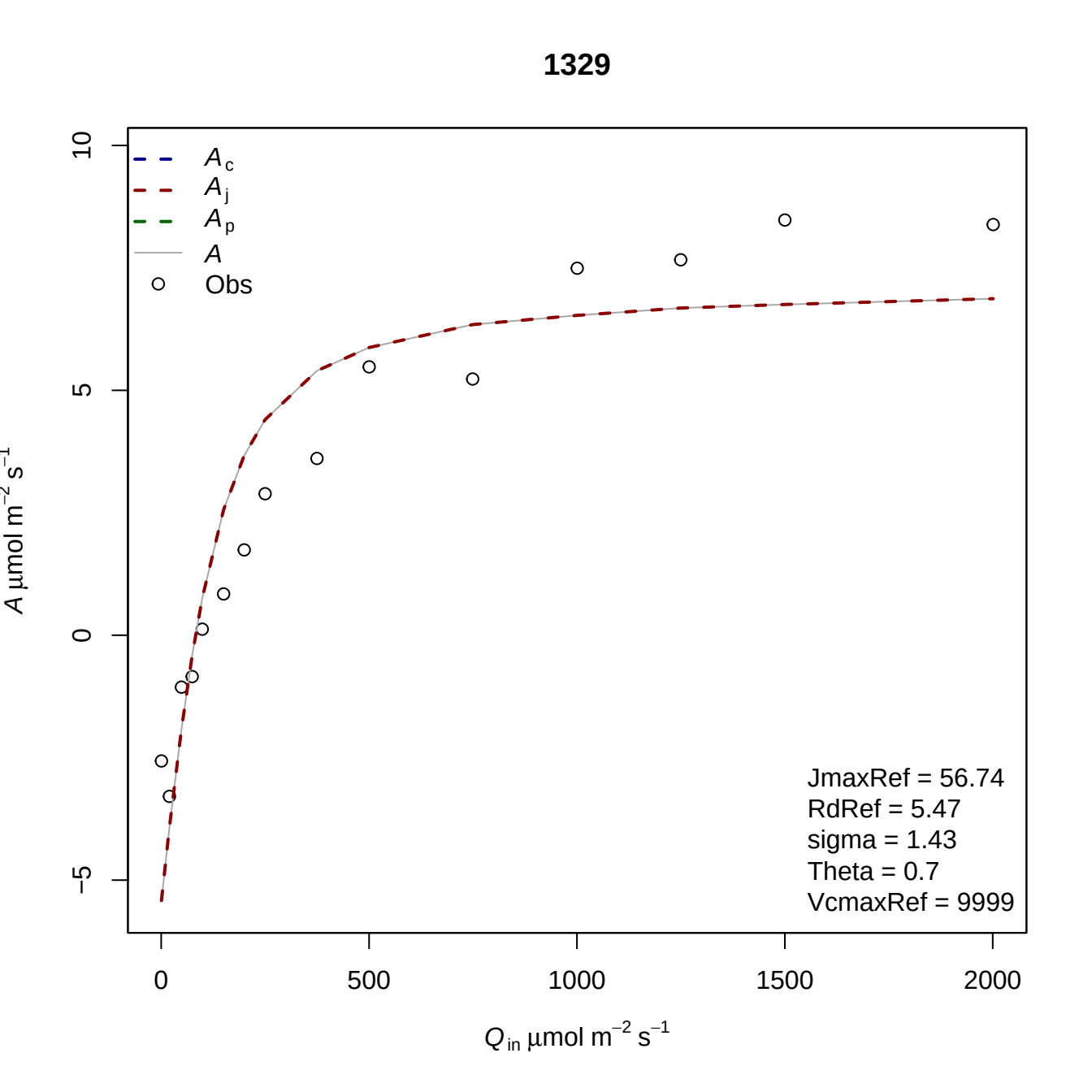

1324
A_c
A_j
A_p
A
Obs
JmaxRef = 44.82
RdRef = 3.22
sigma = 1.15
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

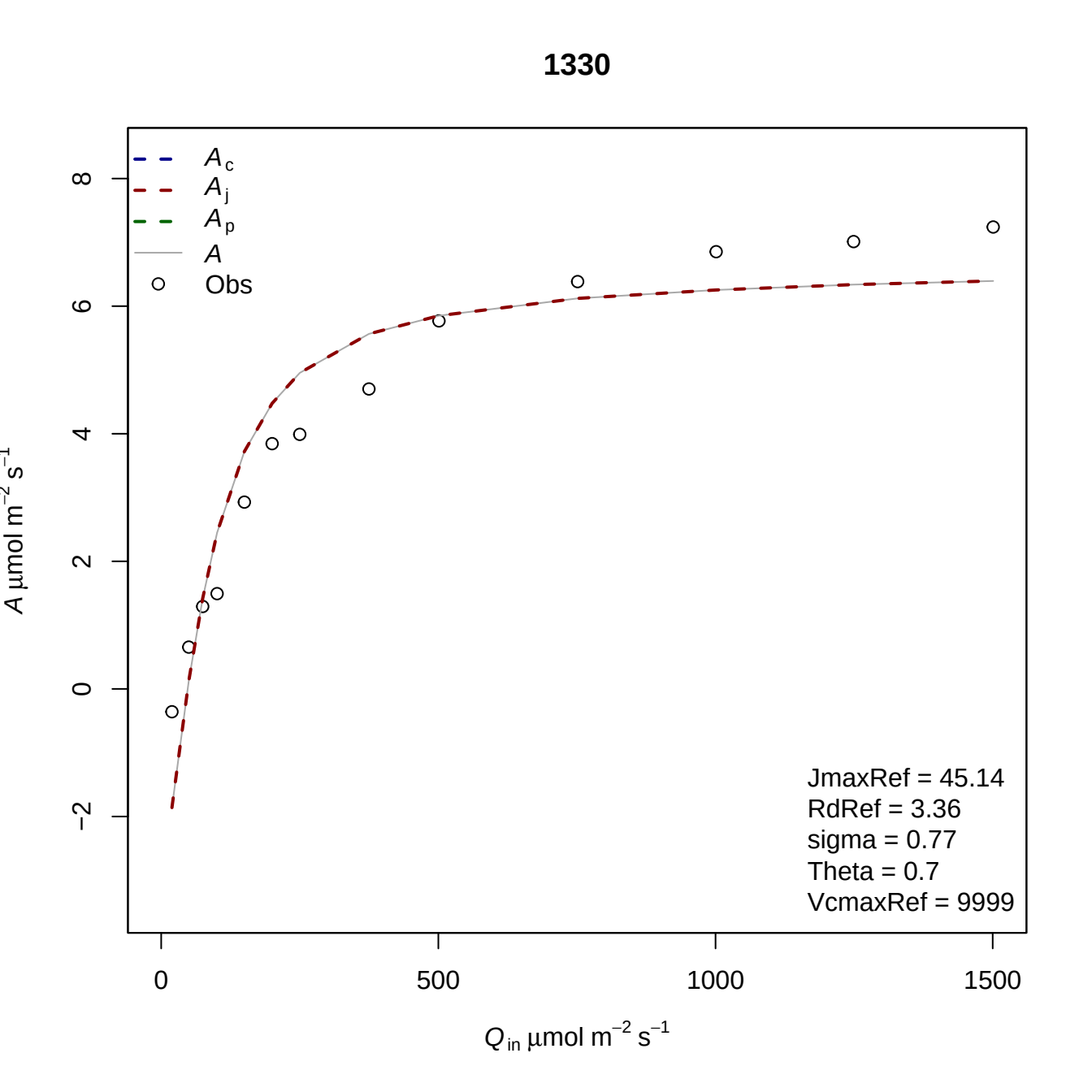

1325
A_c
A_j
A_p
A
Obs
JmaxRef = 41.31
RdRef = 2.63
sigma = 0.77
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
200
400
600
800
1000

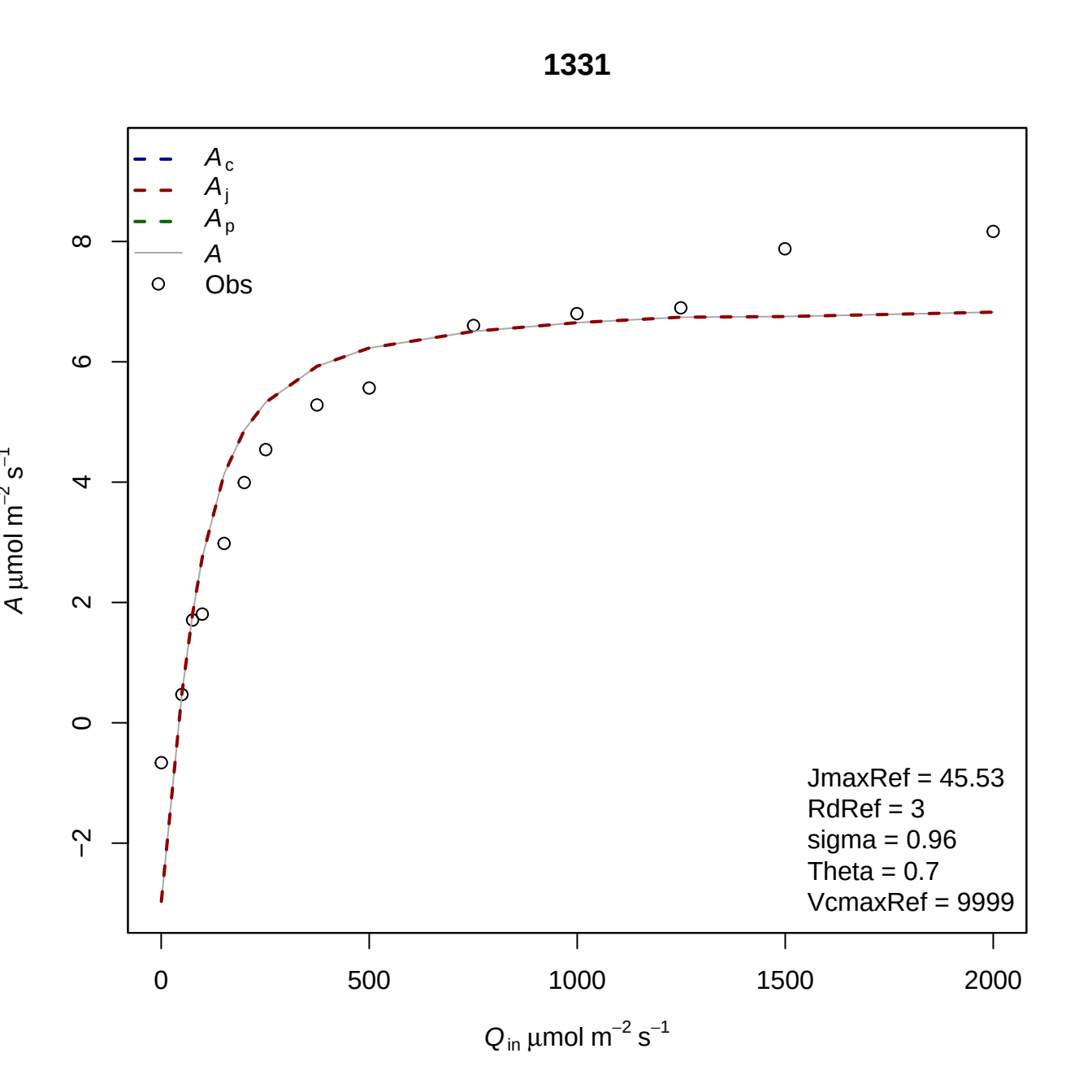

1326
A_c
A_j
A_p
A
Obs
JmaxRef = 35.67
RdRef = 3.02
sigma = 0.99
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

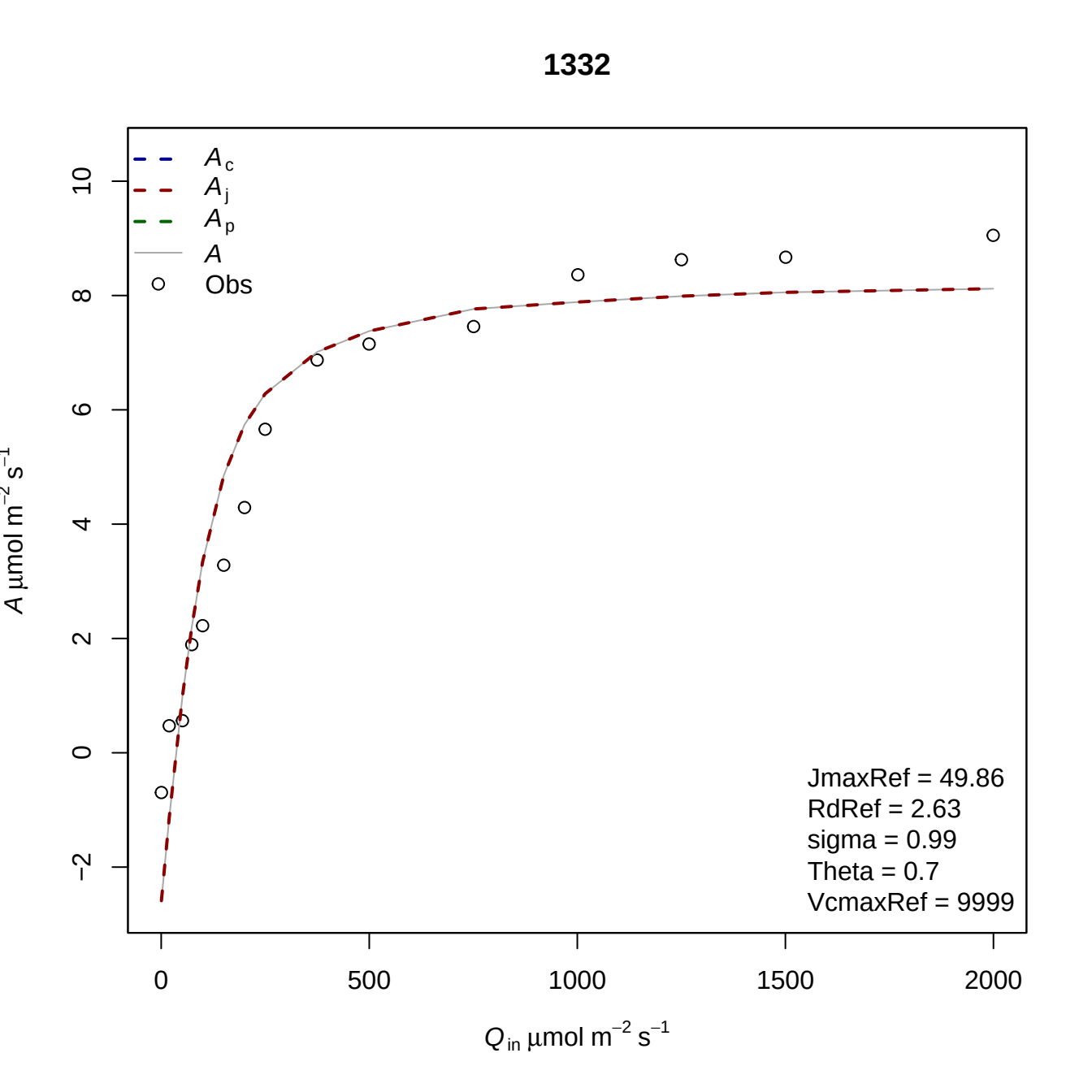

1327
A_c
A_j
A_p
A
Obs
JmaxRef = 52.62
RdRef = 4.41
sigma = 1.47
Theta = 0.7
VcmaxRef = 9999
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

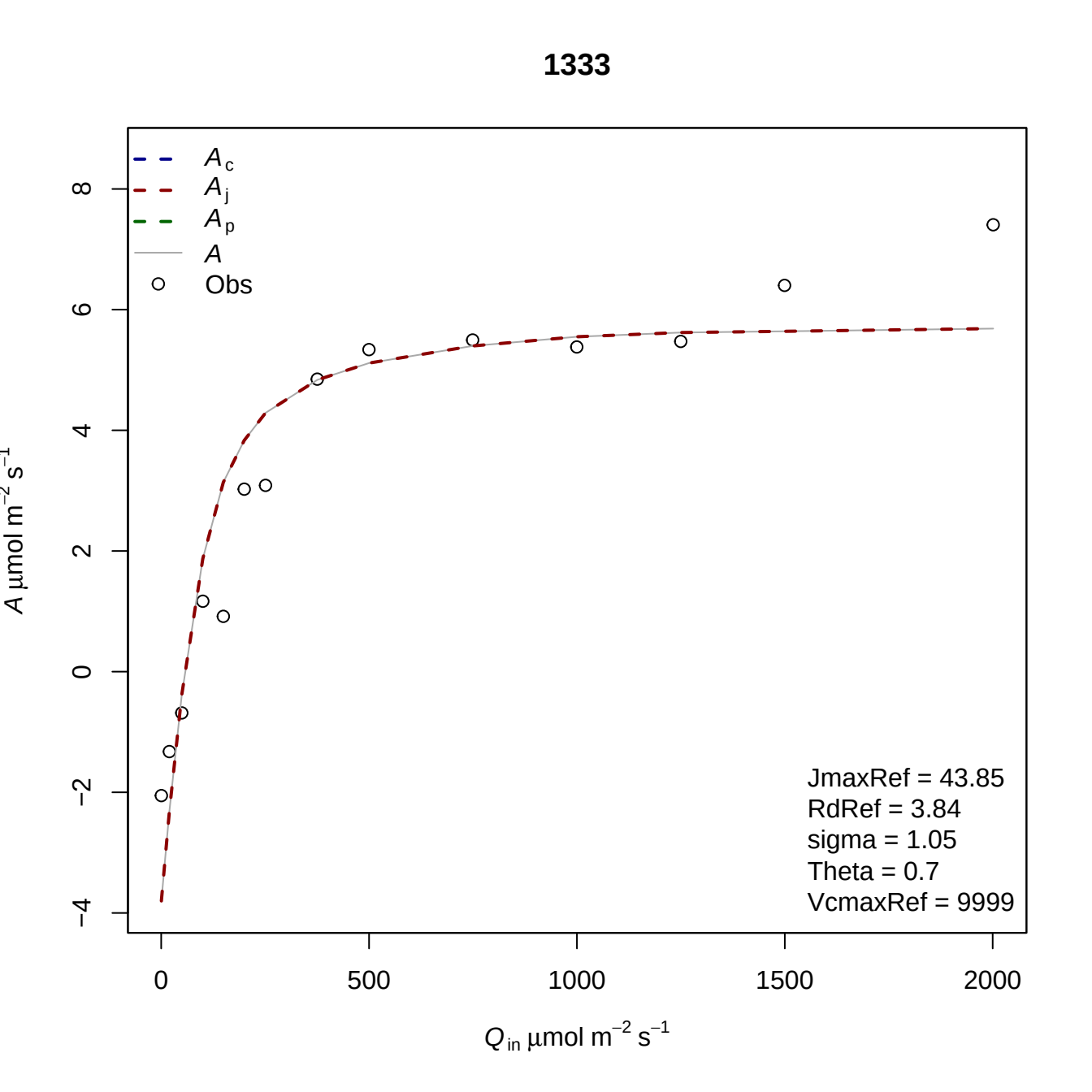

1328
A_c
A_j
A_p
A
Obs
JmaxRef = 63.86
RdRef = 5.09
sigma = 1.59
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

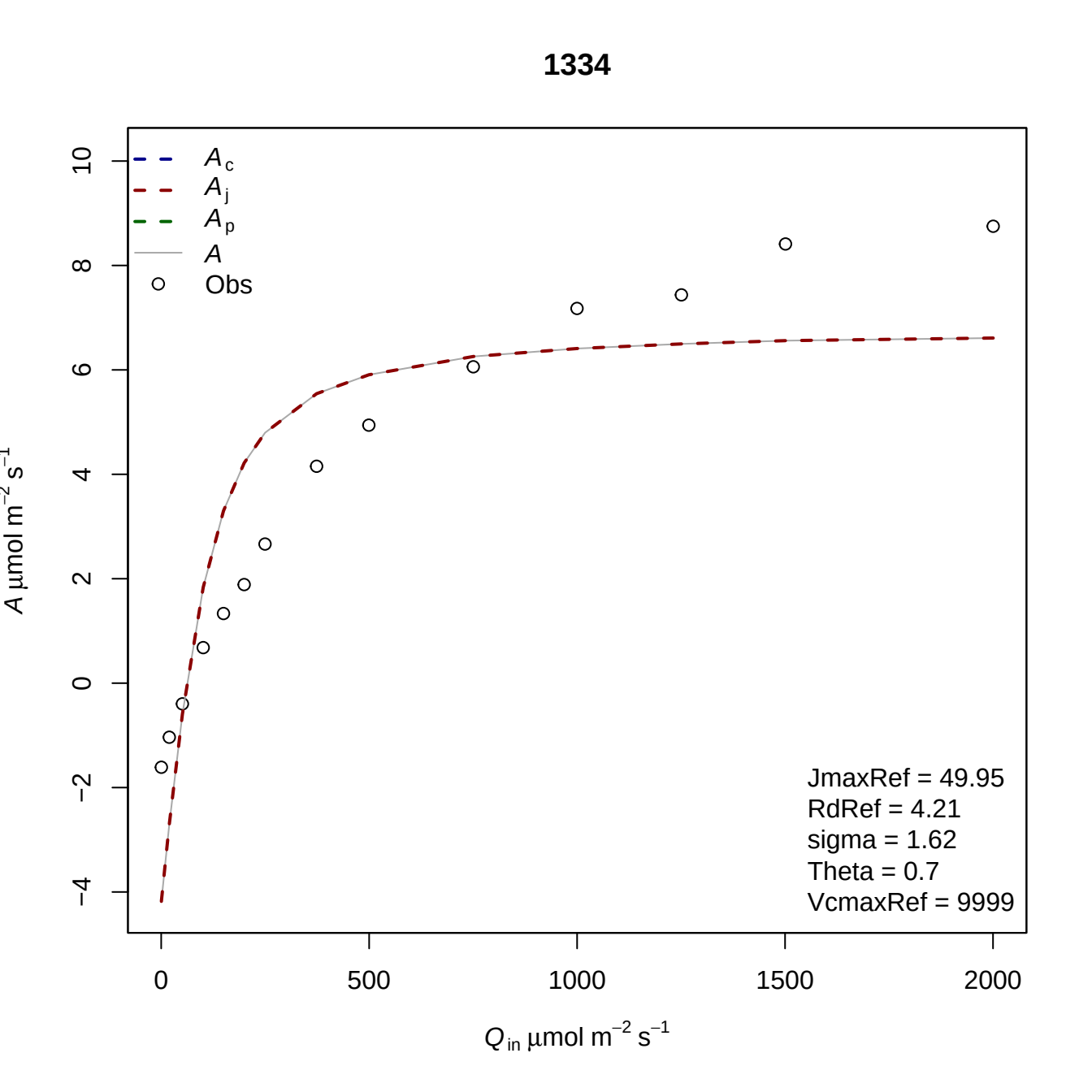

1329
A_c
A_j
A_p
A
Obs
JmaxRef = 56.74
RdRef = 5.47
sigma = 1.43
Theta = 0.7
VcmaxRef = 9999
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

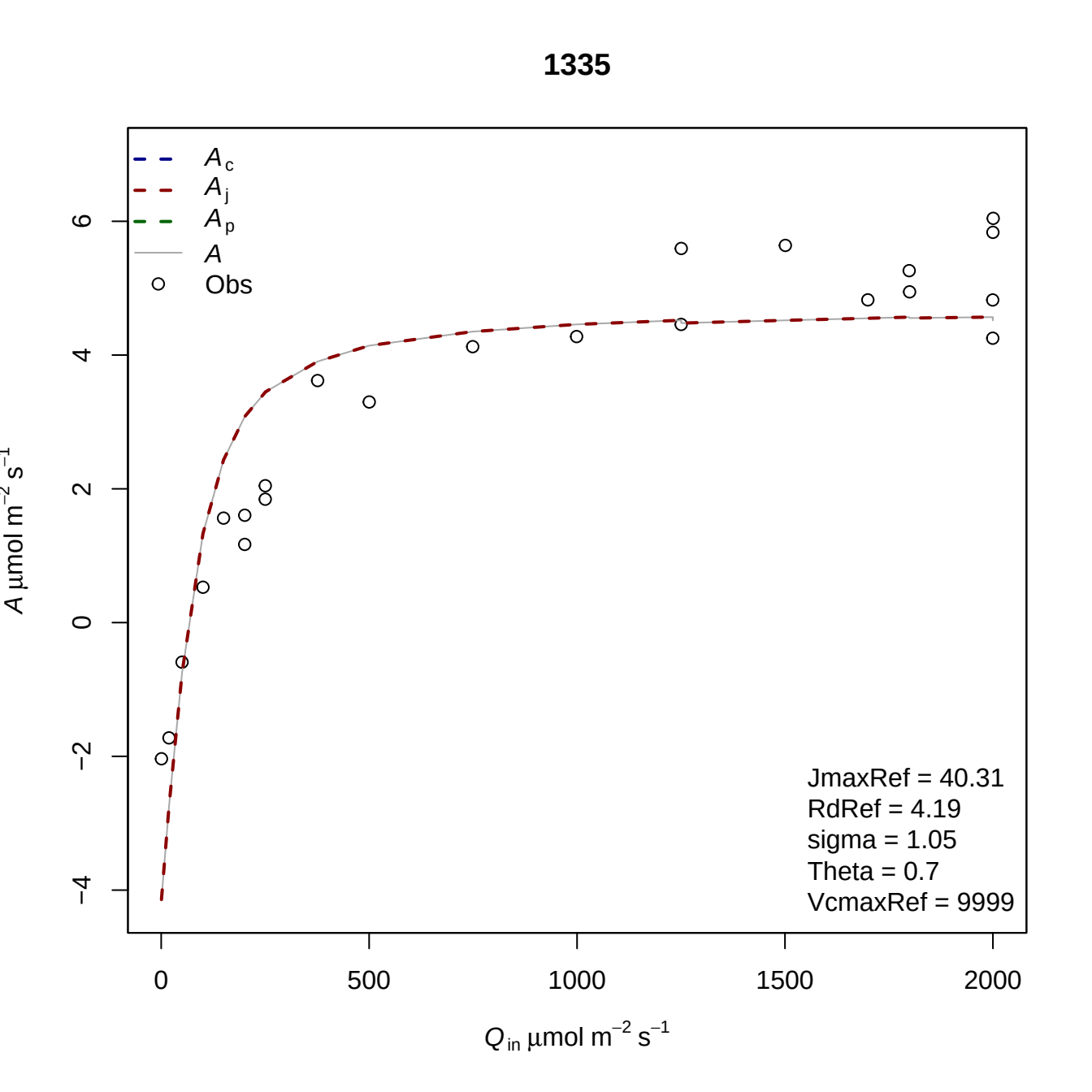

1330
A_c
A_j
A_p
A
Obs
JmaxRef = 45.14
RdRef = 3.36
sigma = 0.77
Theta = 0.7
VcmaxRef = 9999
8
6
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

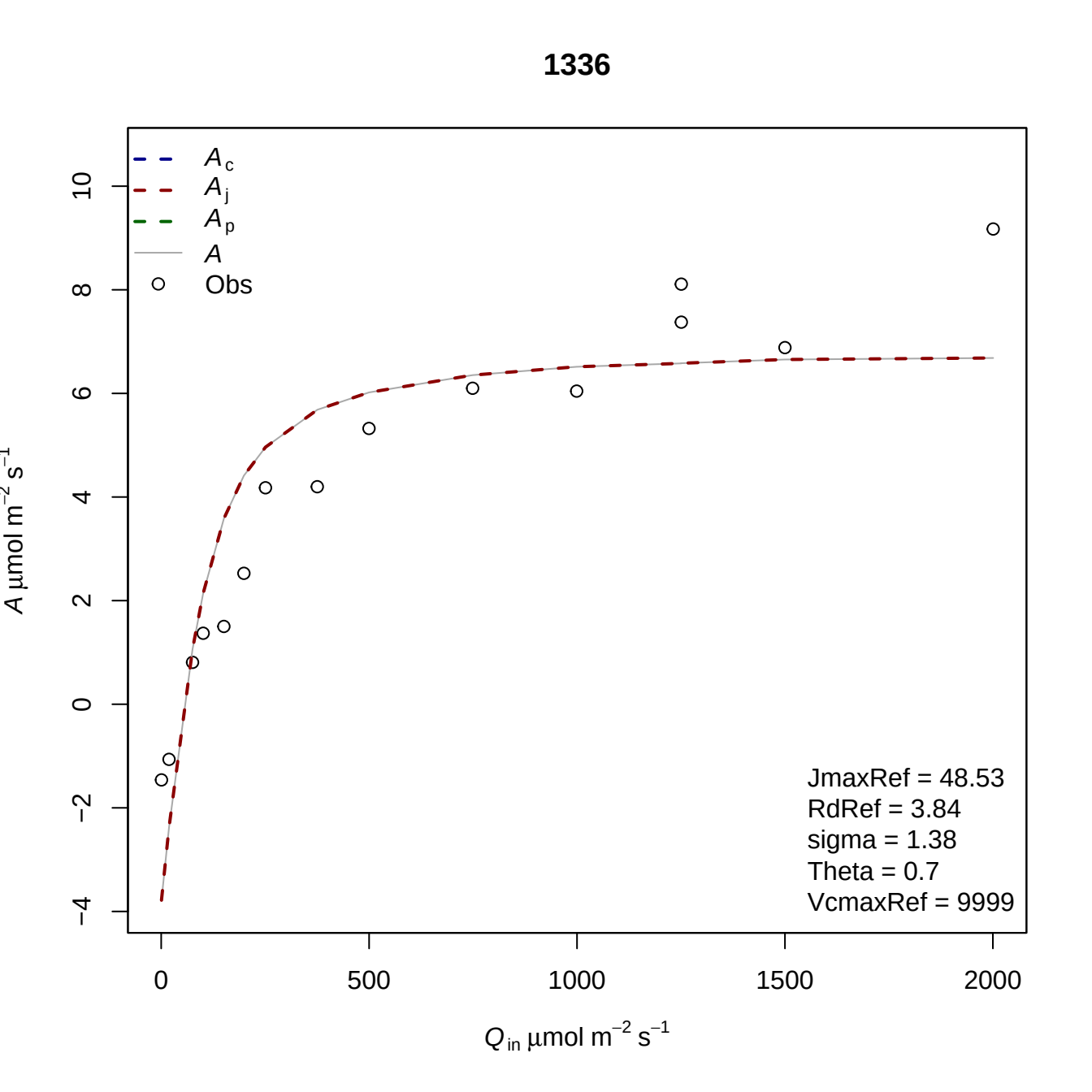

1331
A_c
A_j
A_p
A
Obs
JmaxRef = 45.53
RdRef = 3
sigma = 0.96
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

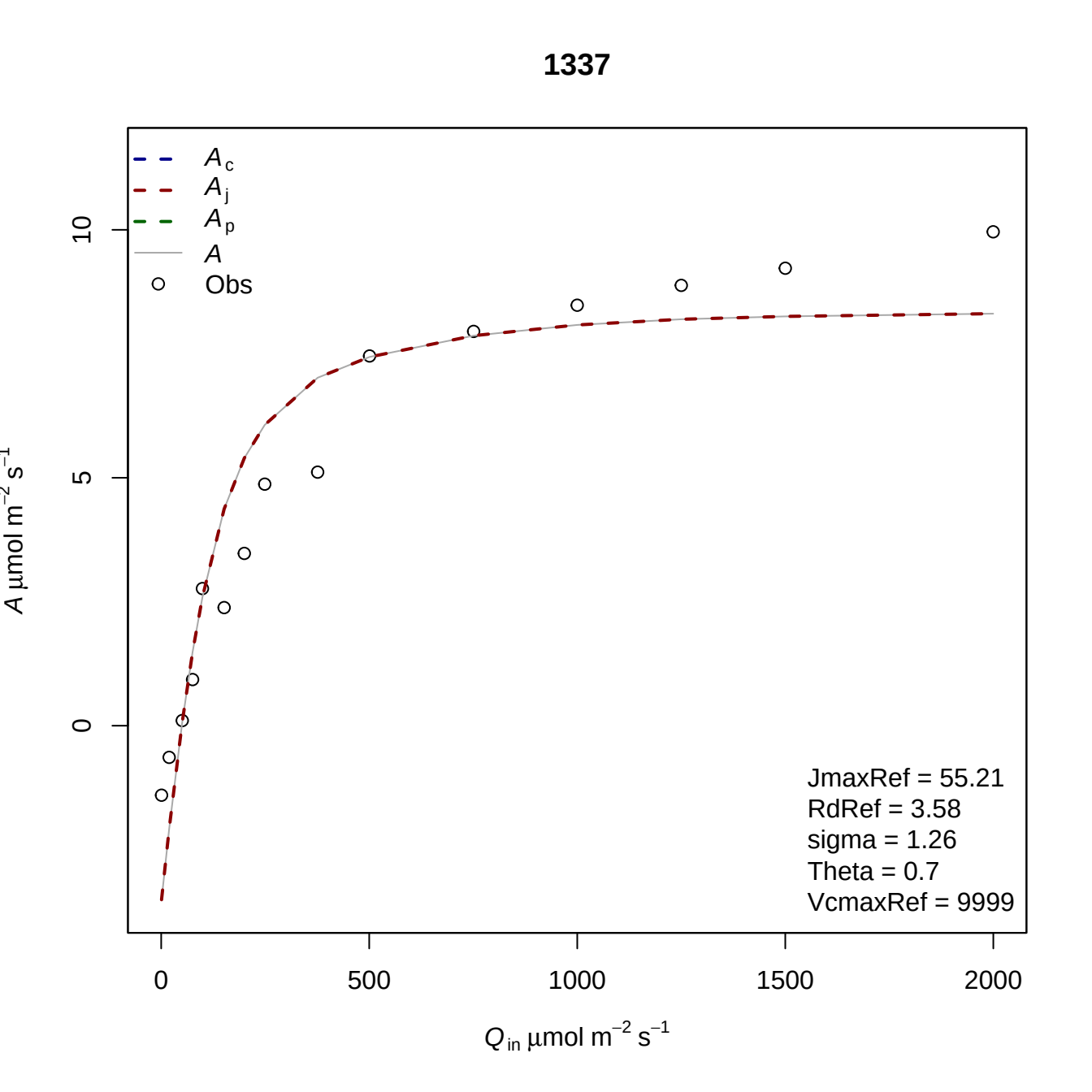

1332
A_c
A_j
A_p
A
Obs
JmaxRef = 49.86
RdRef = 2.63
sigma = 0.99
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

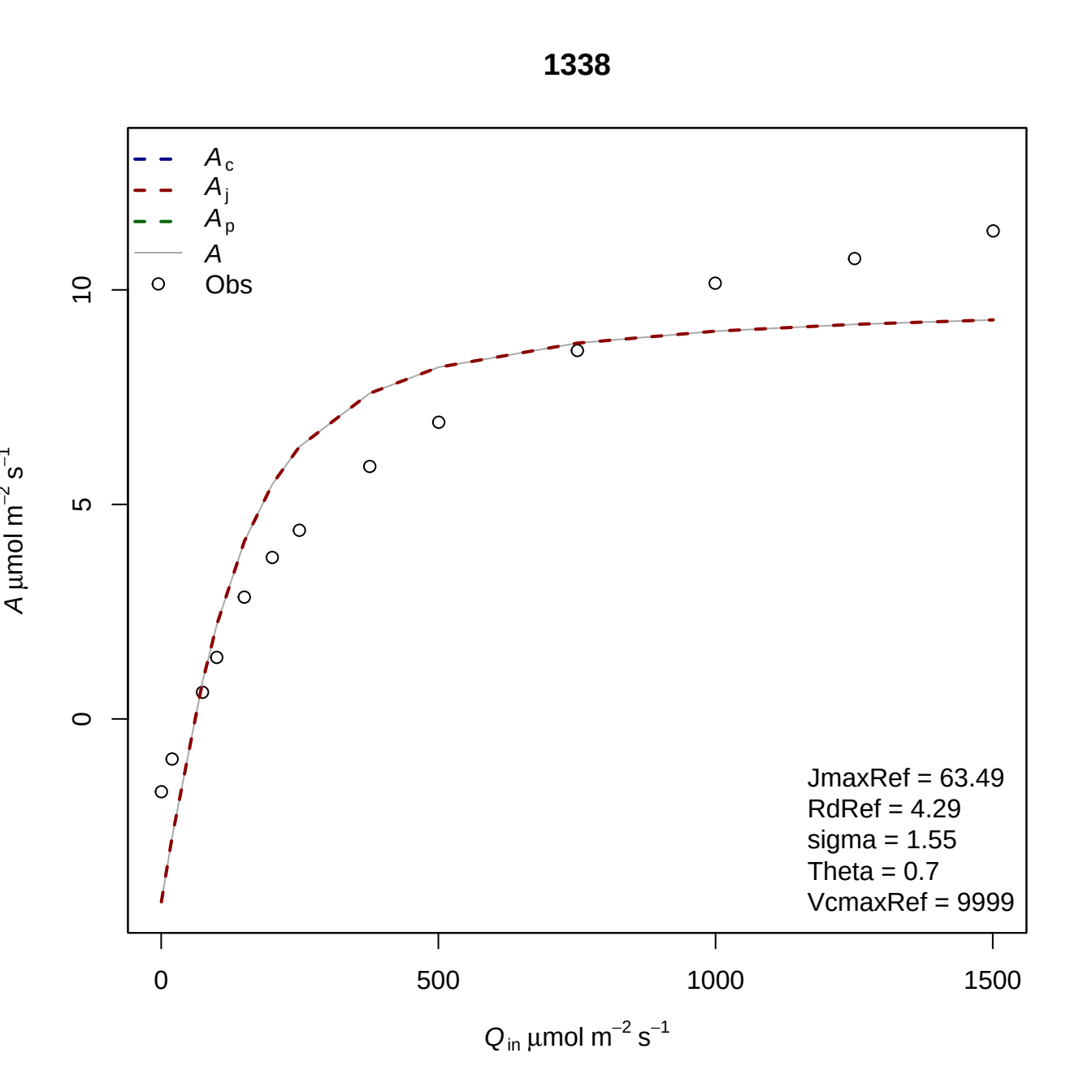

1333
A_c
A_j
A_p
A
Obs
JmaxRef = 43.85
RdRef = 3.84
sigma = 1.05
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

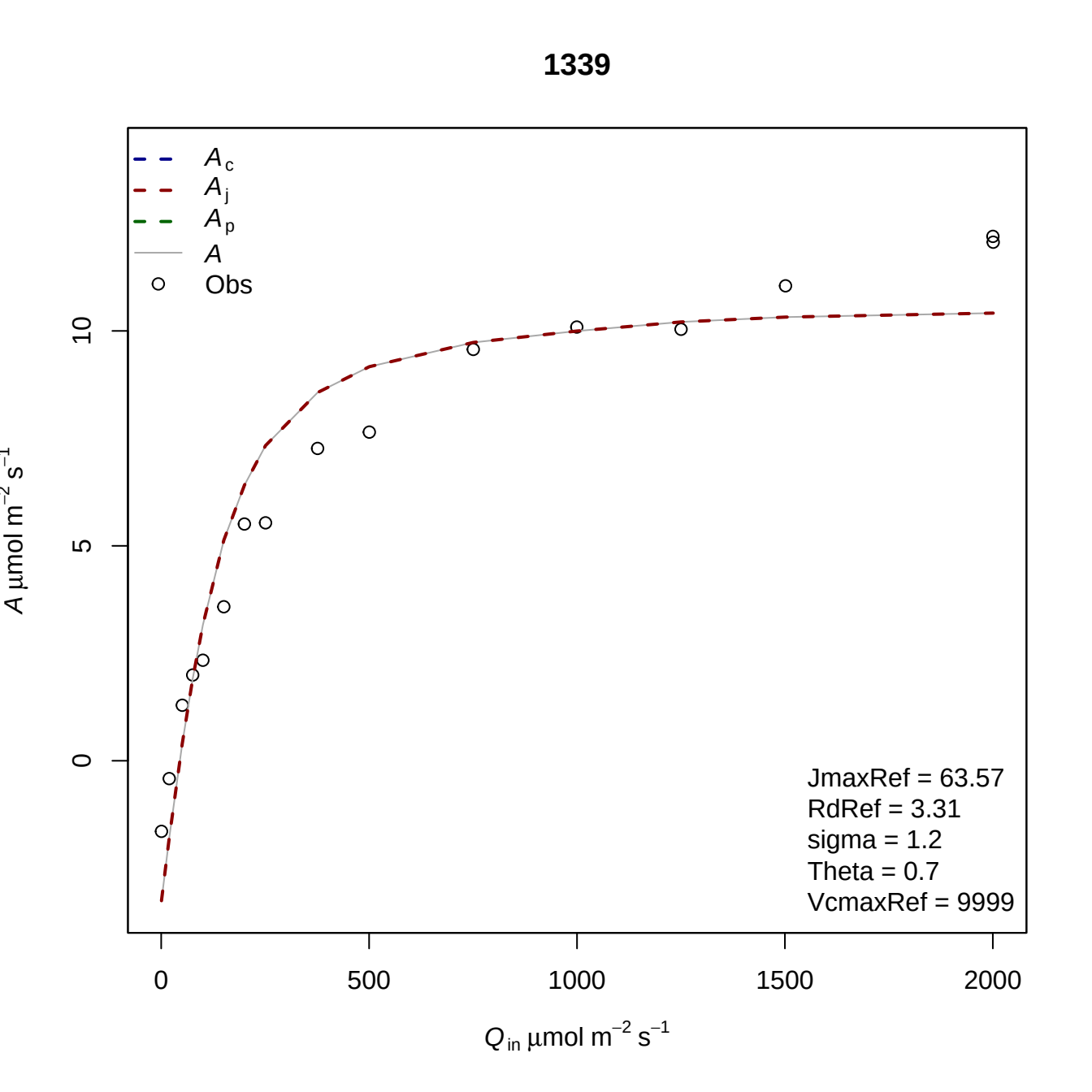

1334
A_c
A_j
A_p
A
Obs
JmaxRef = 49.95
RdRef = 4.21
sigma = 1.62
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

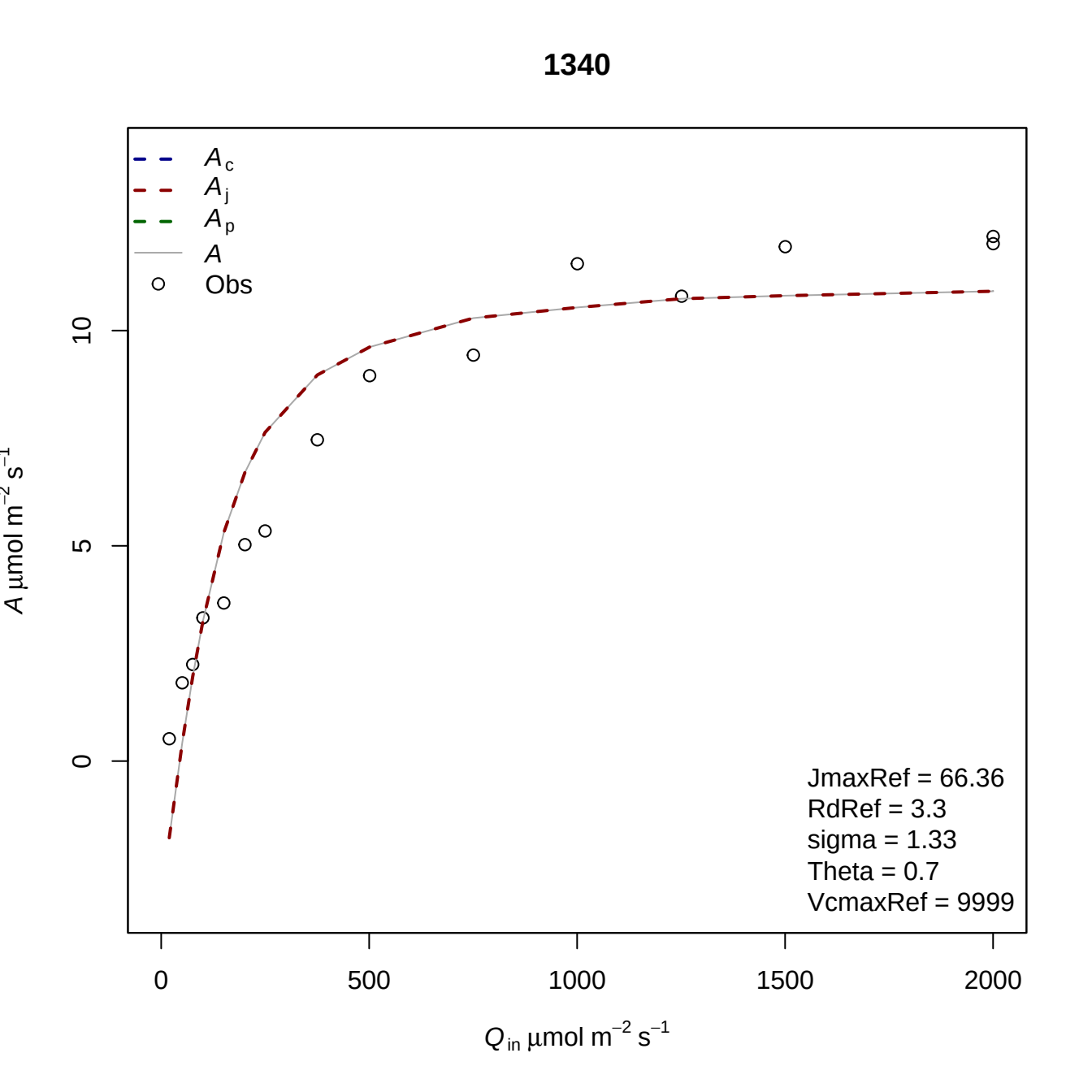

1335
A_c
A_j
A_p
A
Obs
JmaxRef = 40.31
RdRef = 4.19
sigma = 1.05
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500
2000

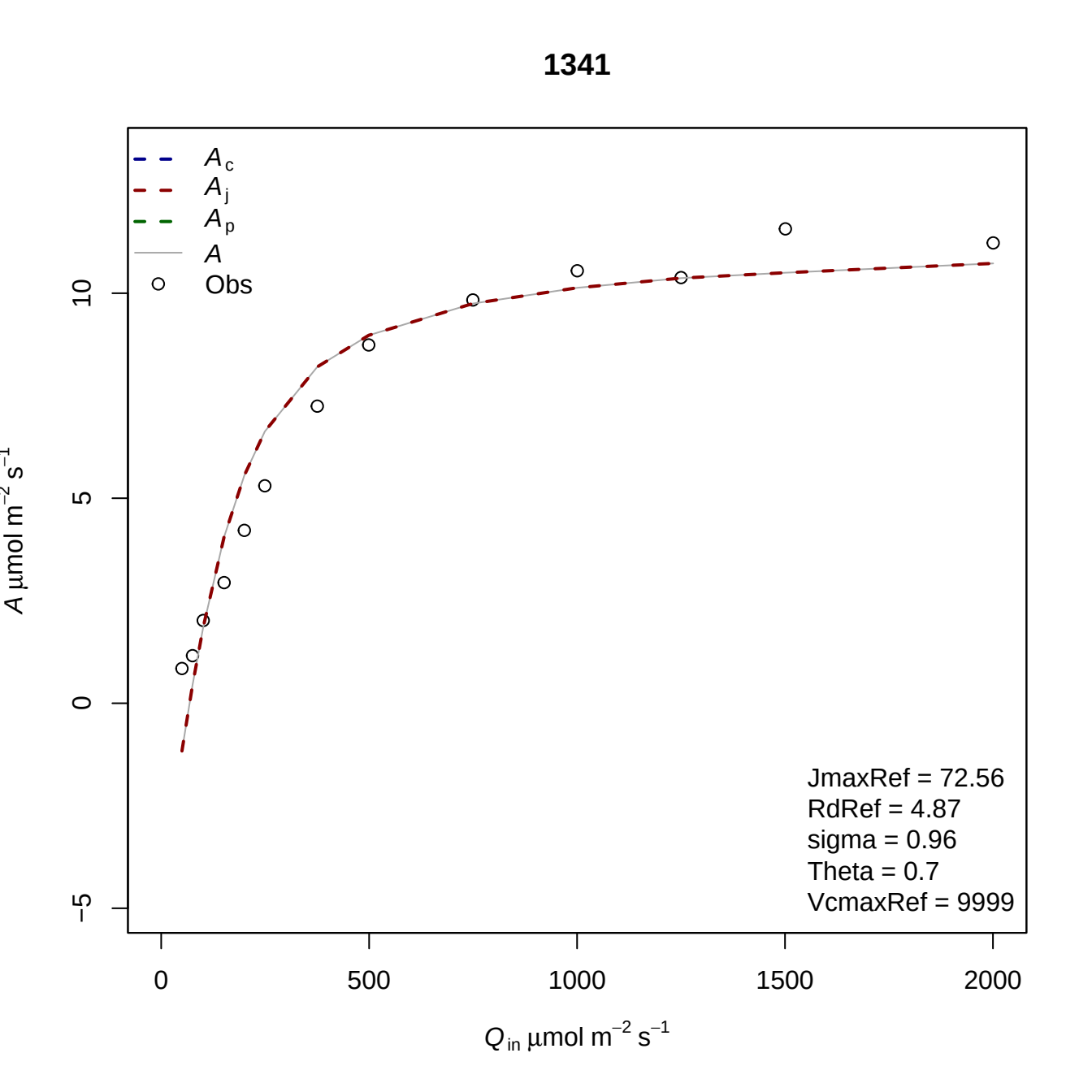

1336
A_c
A_j
A_p
A
Obs
JmaxRef = 48.53
RdRef = 3.84
sigma = 1.38
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

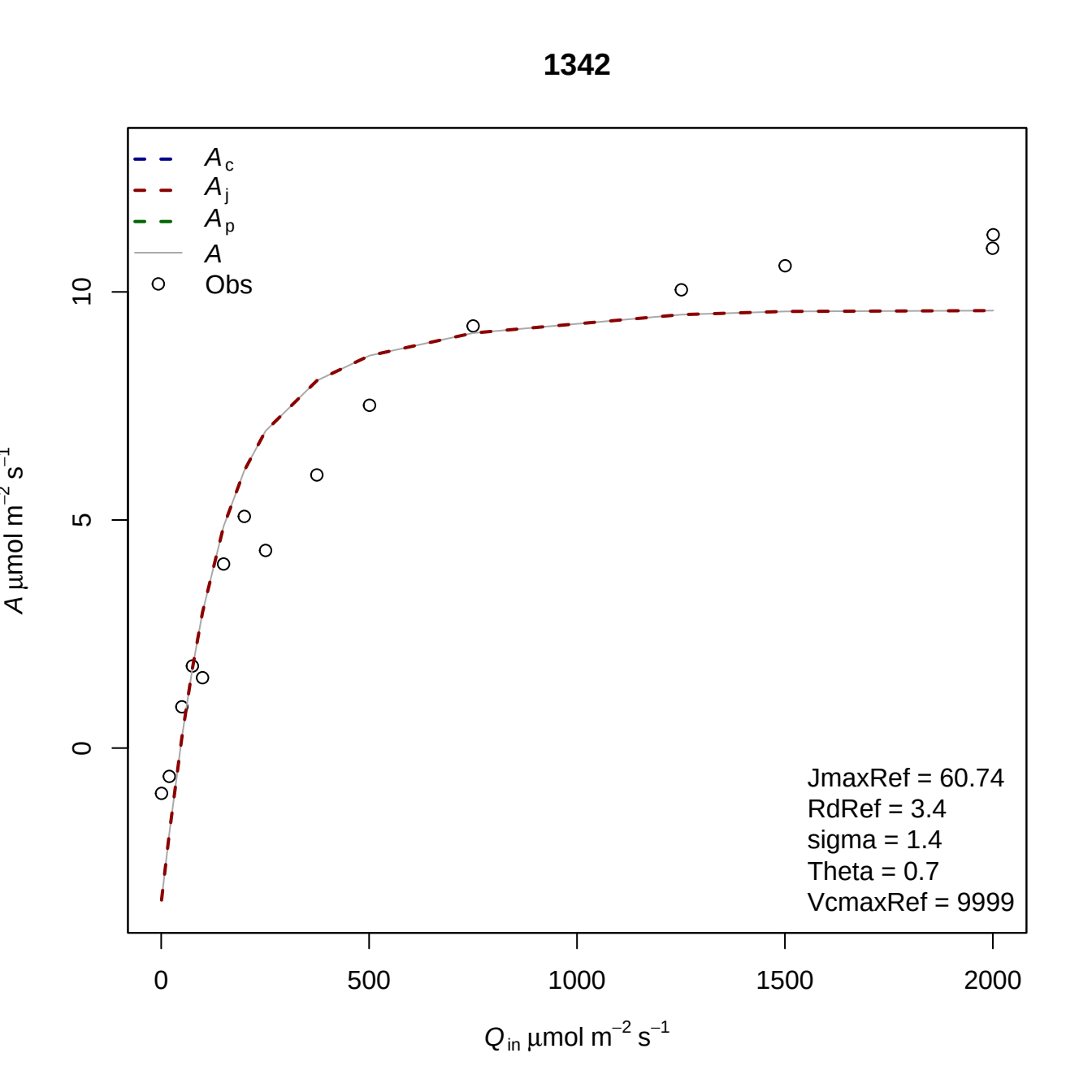

1337
A_c
A_j
A_p
A
Obs
JmaxRef = 55.21
RdRef = 3.58
sigma = 1.26
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

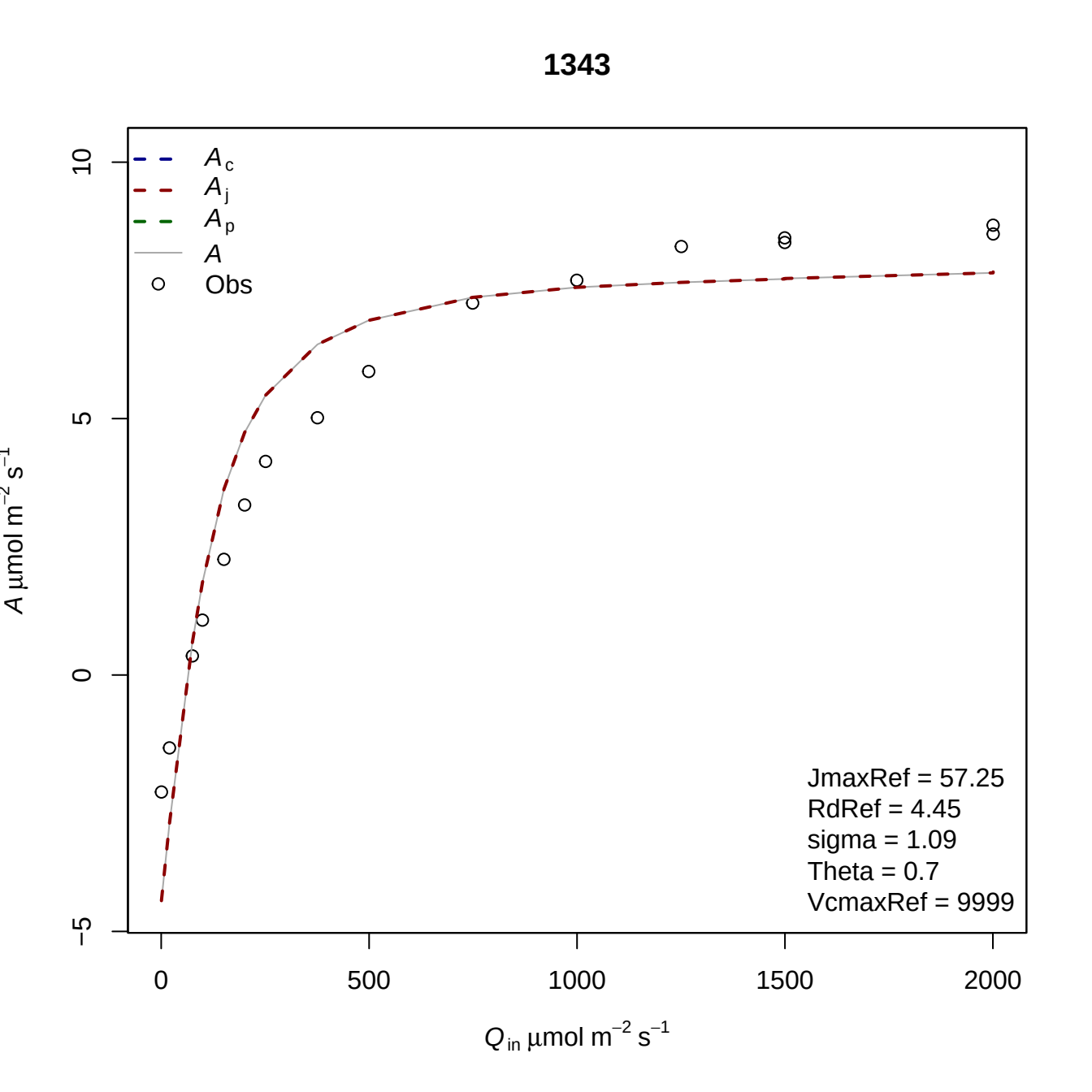

1338
A_c
A_j
A_p
A
Obs
JmaxRef = 63.49
RdRef = 4.29
sigma = 1.55
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

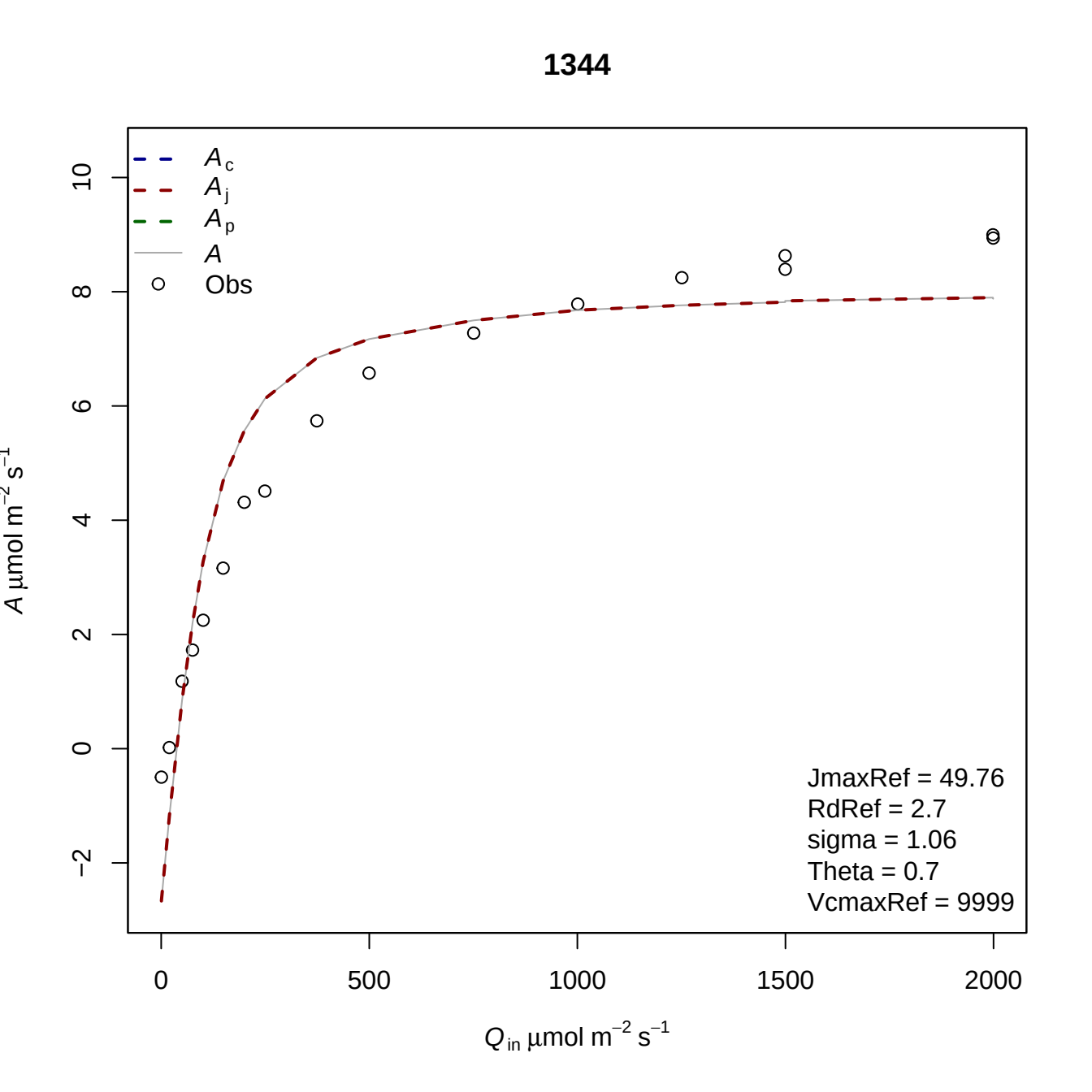

1339
A_c
A_j
A_p
A
Obs
JmaxRef = 63.57
RdRef = 3.31
sigma = 1.2
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

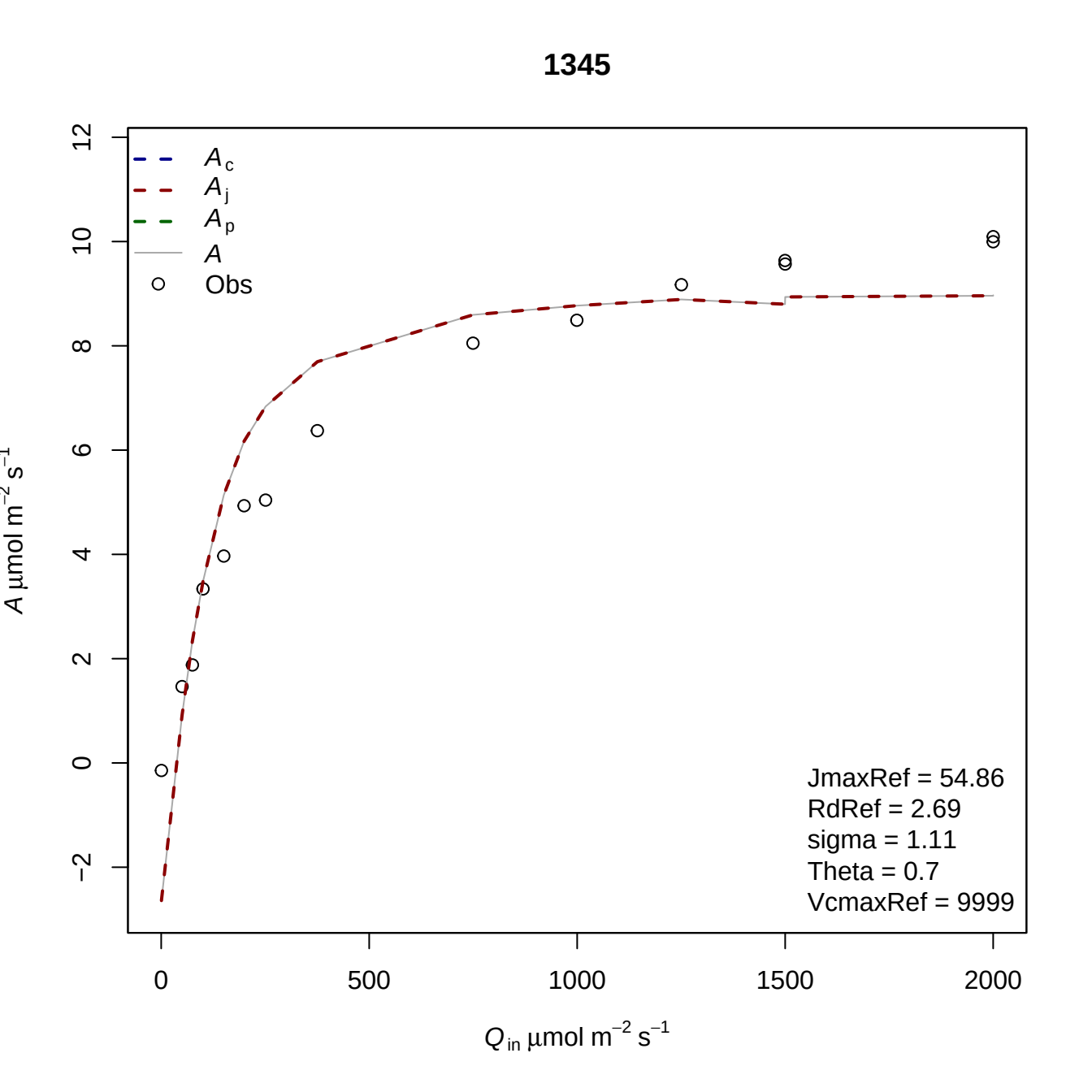

1340
A_c
A_j
A_p
A
Obs
JmaxRef = 66.36
RdRef = 3.3
sigma = 1.33
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

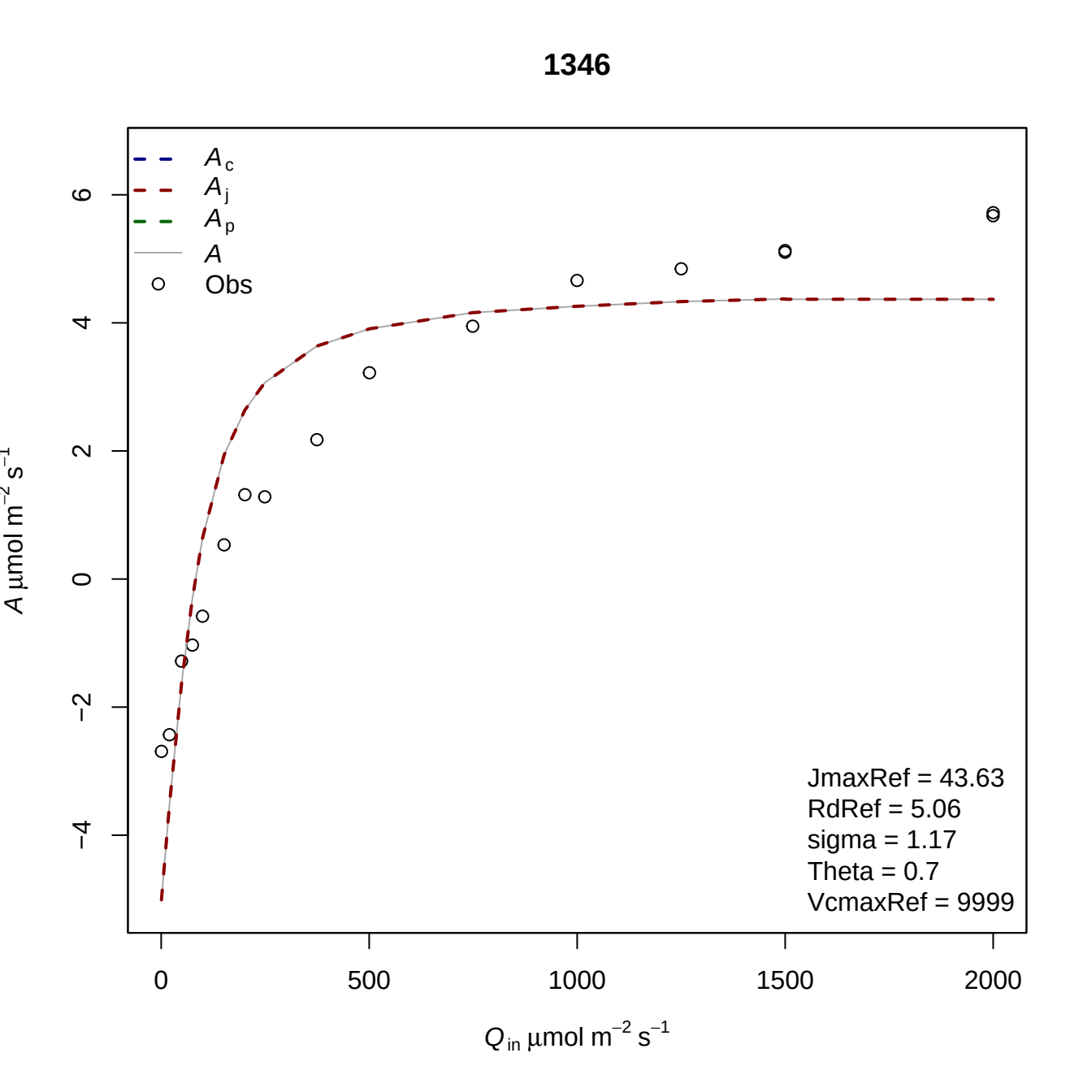

1341
A_c
A_j
A_p
A
Obs
JmaxRef = 72.56
RdRef = 4.87
sigma = 0.96
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

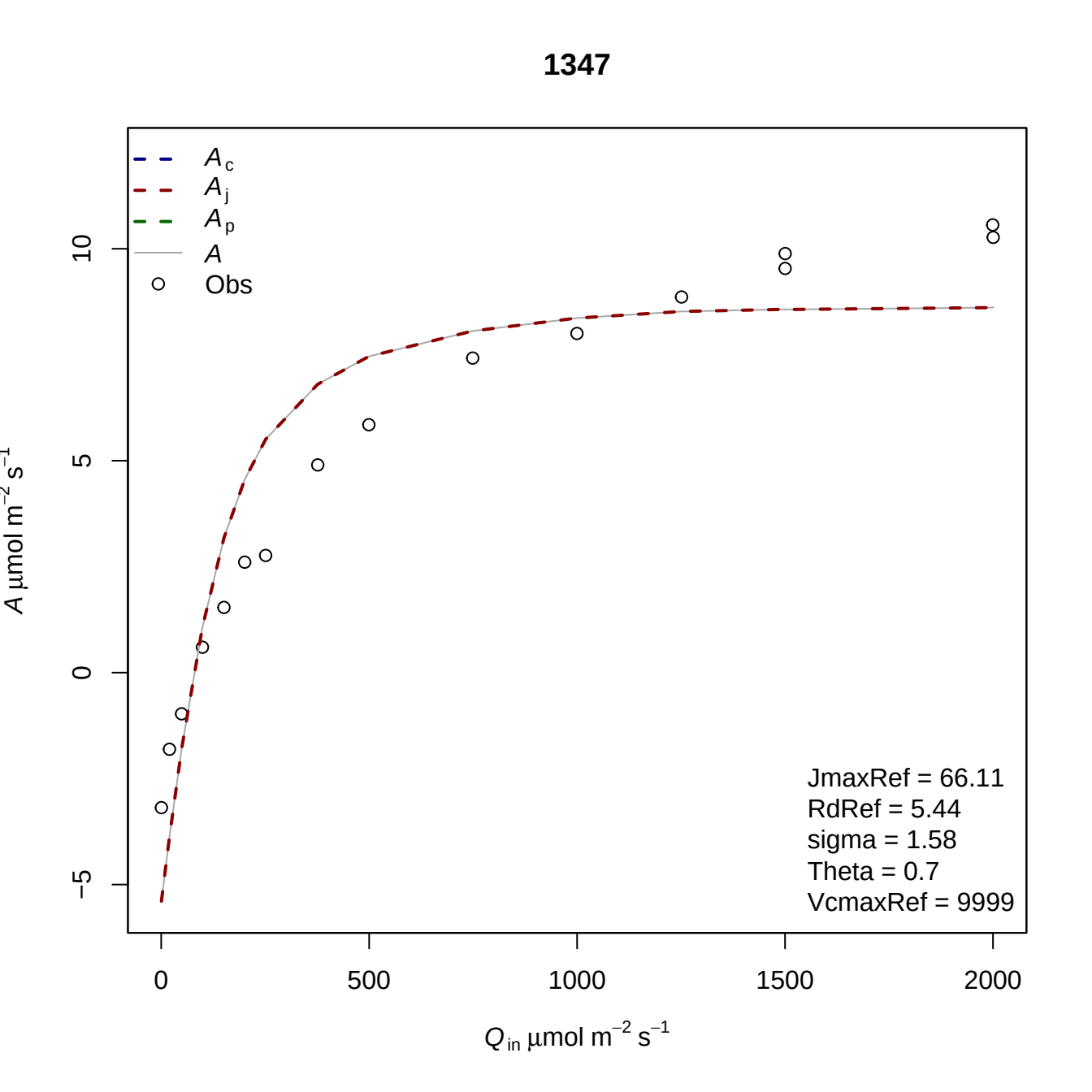

1342
A_c
A_j
A_p
A
Obs
JmaxRef = 60.74
RdRef = 3.4
sigma = 1.4
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

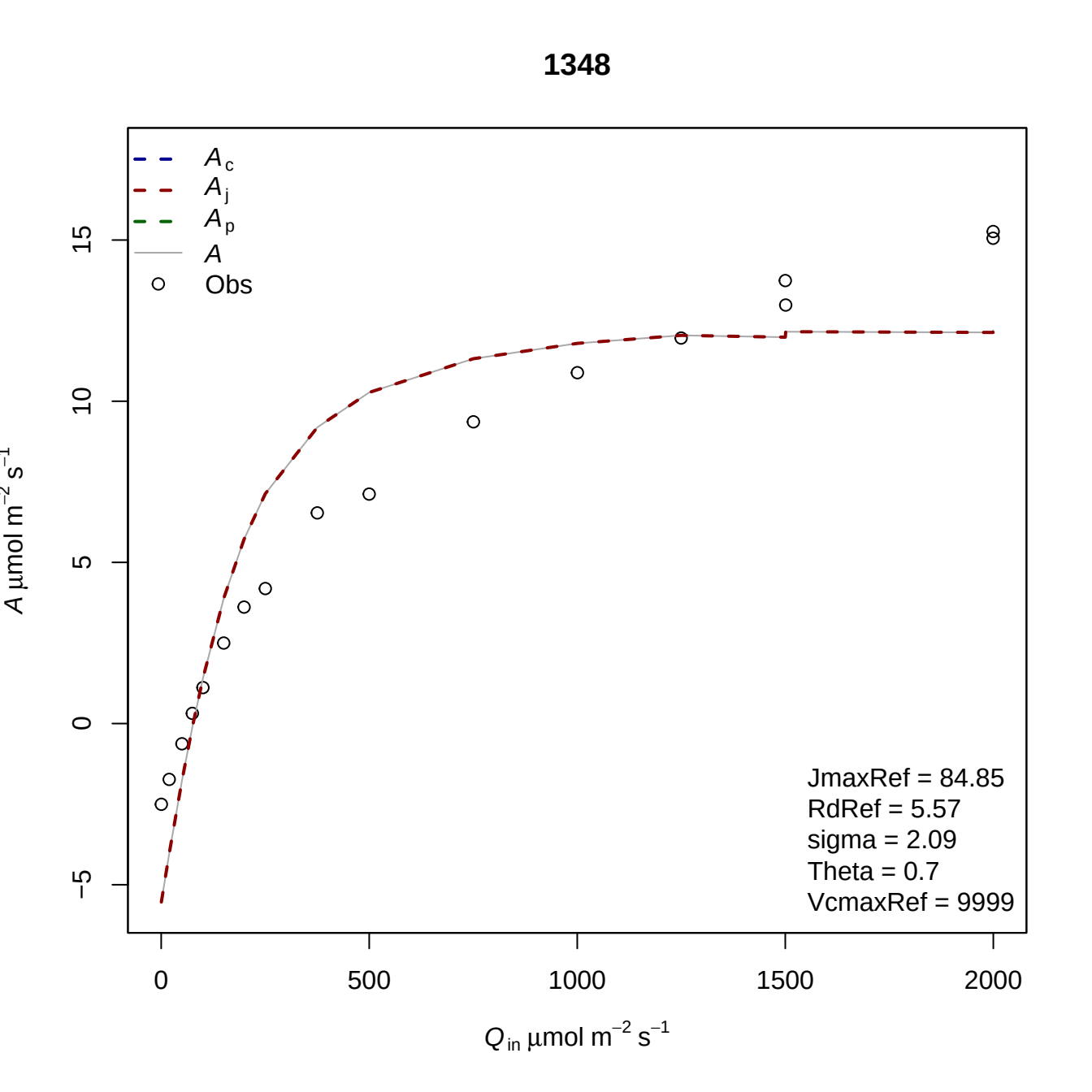

1343
A_c
A_j
A_p
A
Obs
JmaxRef = 57.25
RdRef = 4.45
sigma = 1.09
Theta = 0.7
VcmaxRef = 9999
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

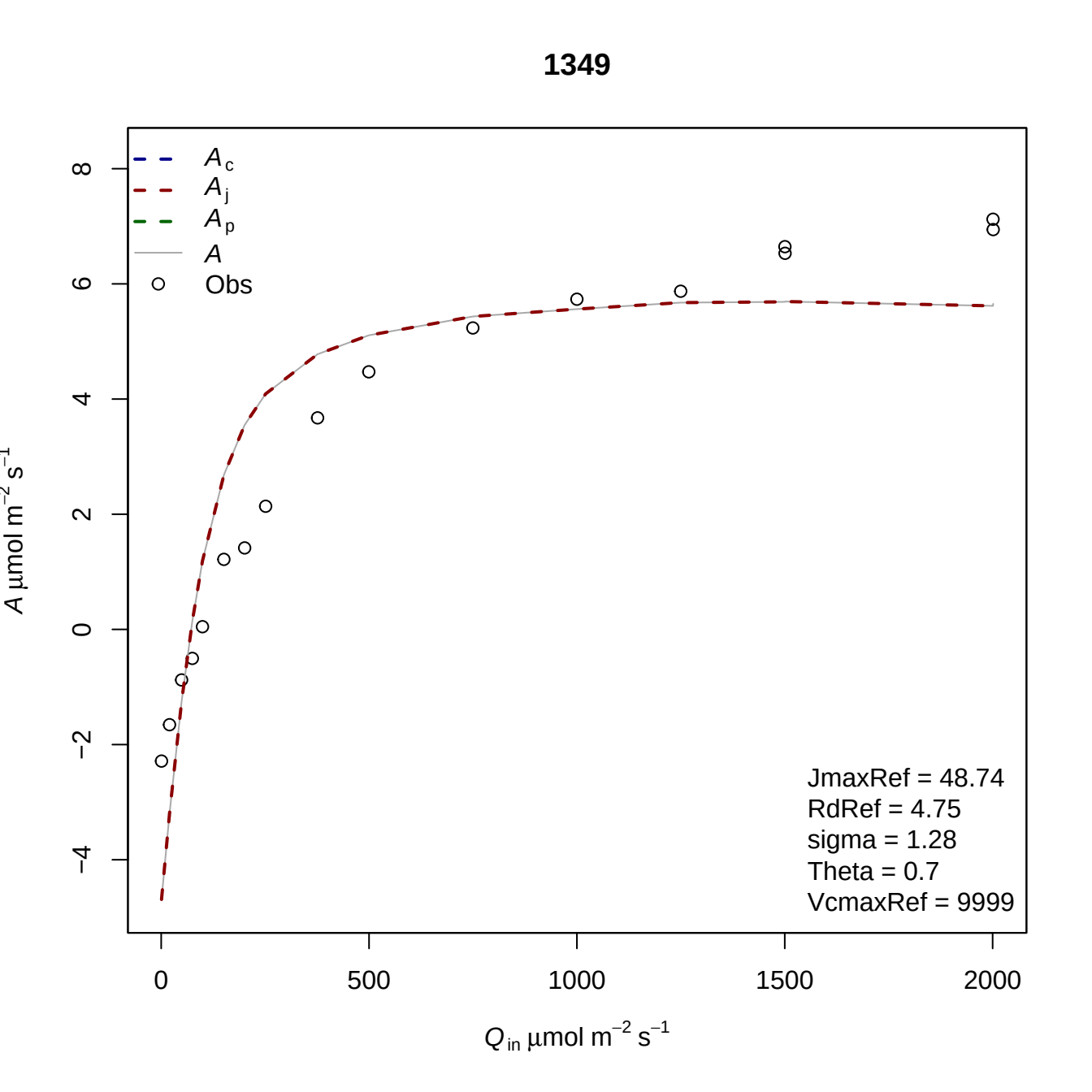

1344
A_c
A_j
A_p
A
Obs
JmaxRef = 49.76
RdRef = 2.7
sigma = 1.06
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

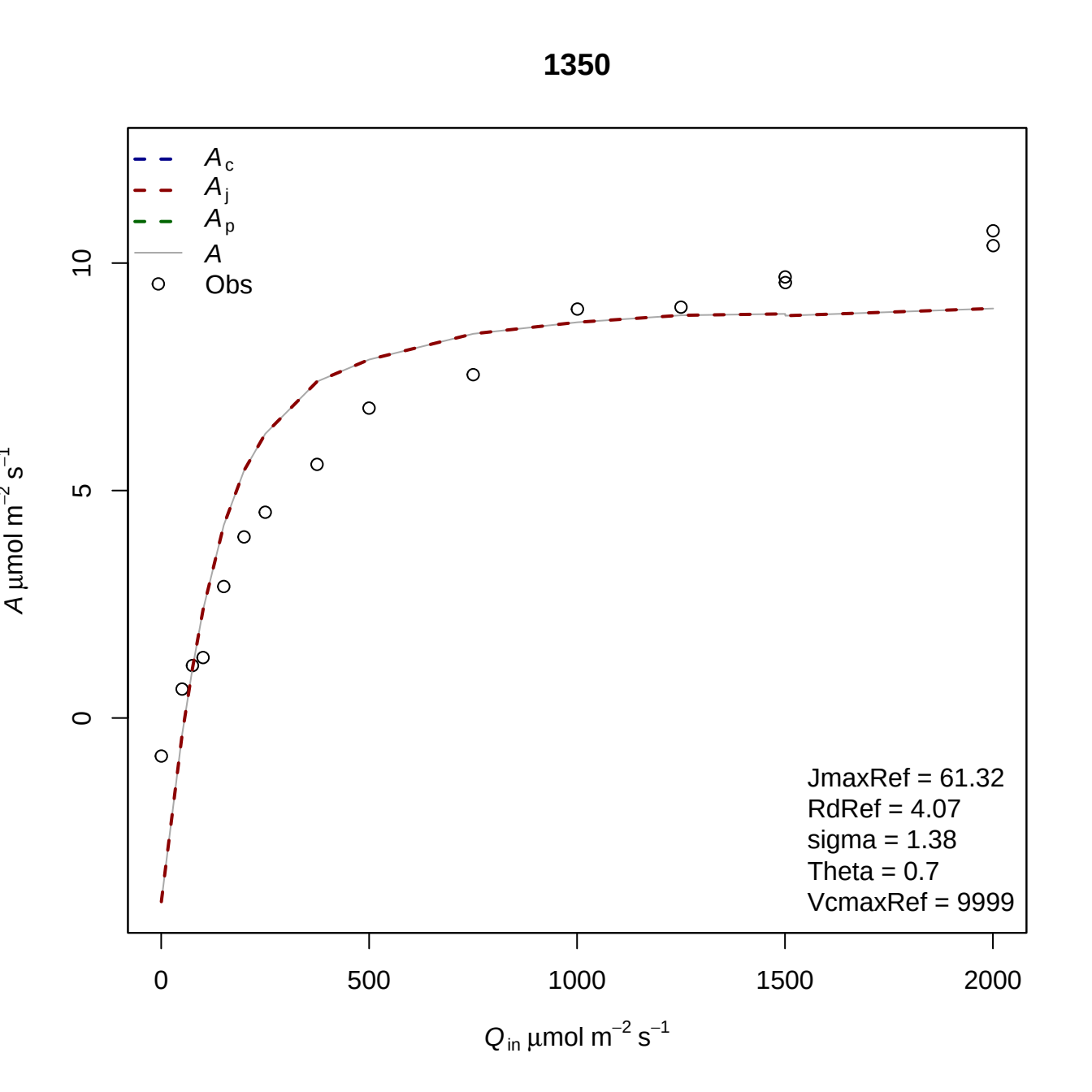

1345
A_c
A_j
A_p
A
Obs
JmaxRef = 54.86
RdRef = 2.69
sigma = 1.11
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

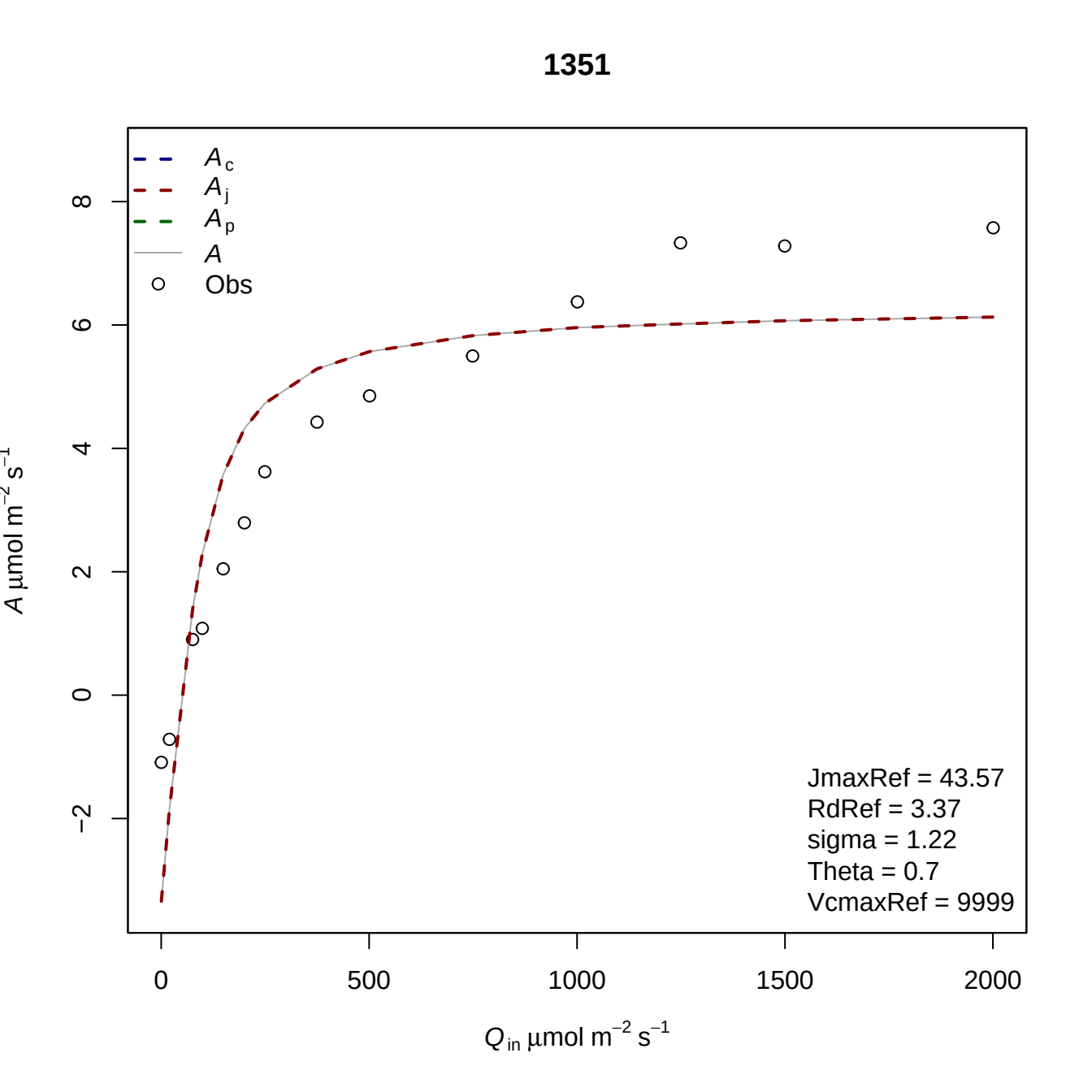

1346
A_c
A_j
A_p
A
Obs
JmaxRef = 43.63
RdRef = 5.06
sigma = 1.17
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500
2000

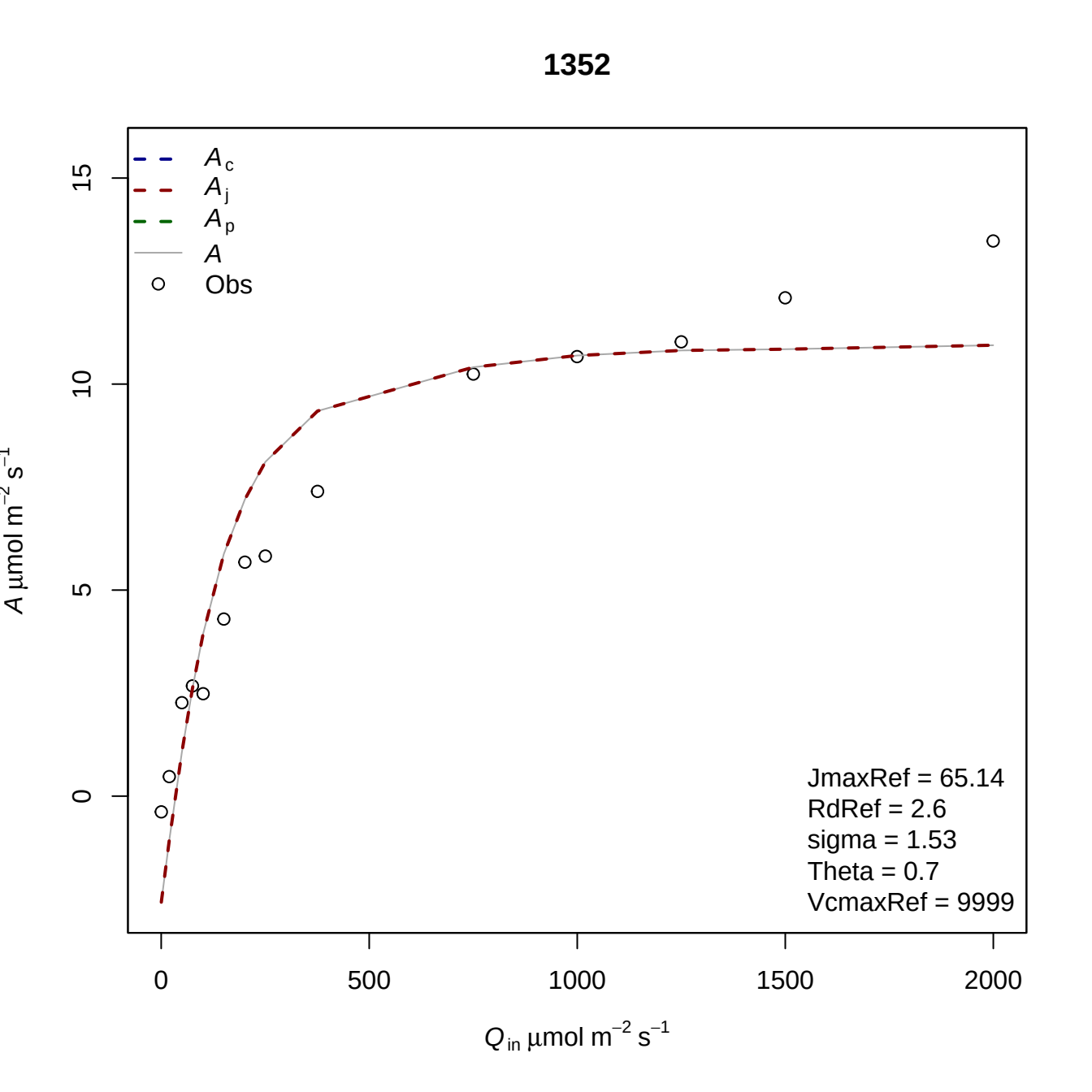

1347
A_c
A_j
A_p
A
Obs
JmaxRef = 66.11
RdRef = 5.44
sigma = 1.58
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
5
0
−5
0
500
1000
1500
2000

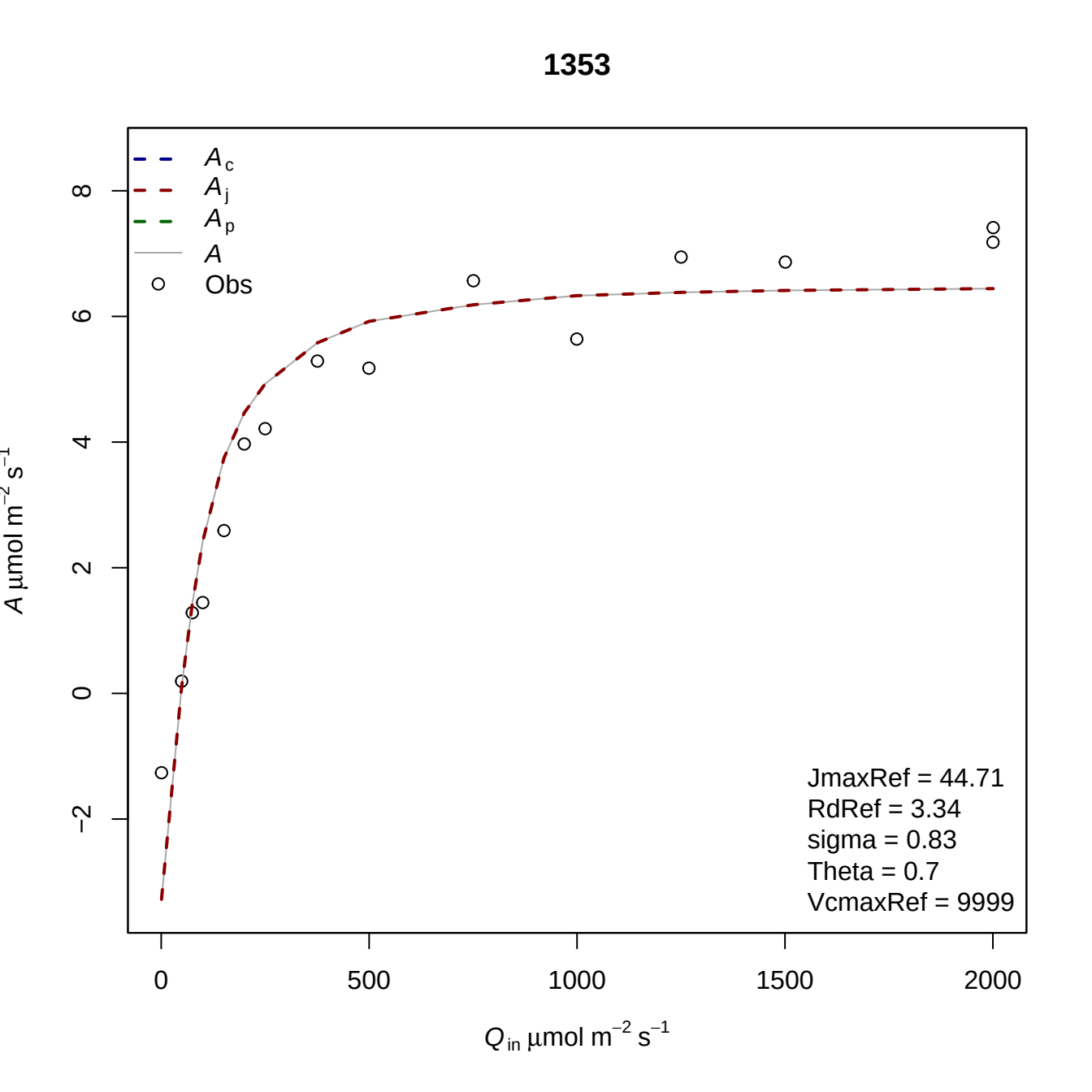

1348
A_c
A_j
A_p
A
Obs
JmaxRef = 84.85
RdRef = 5.57
sigma = 2.09
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
15
0
500
1000
1500
2000

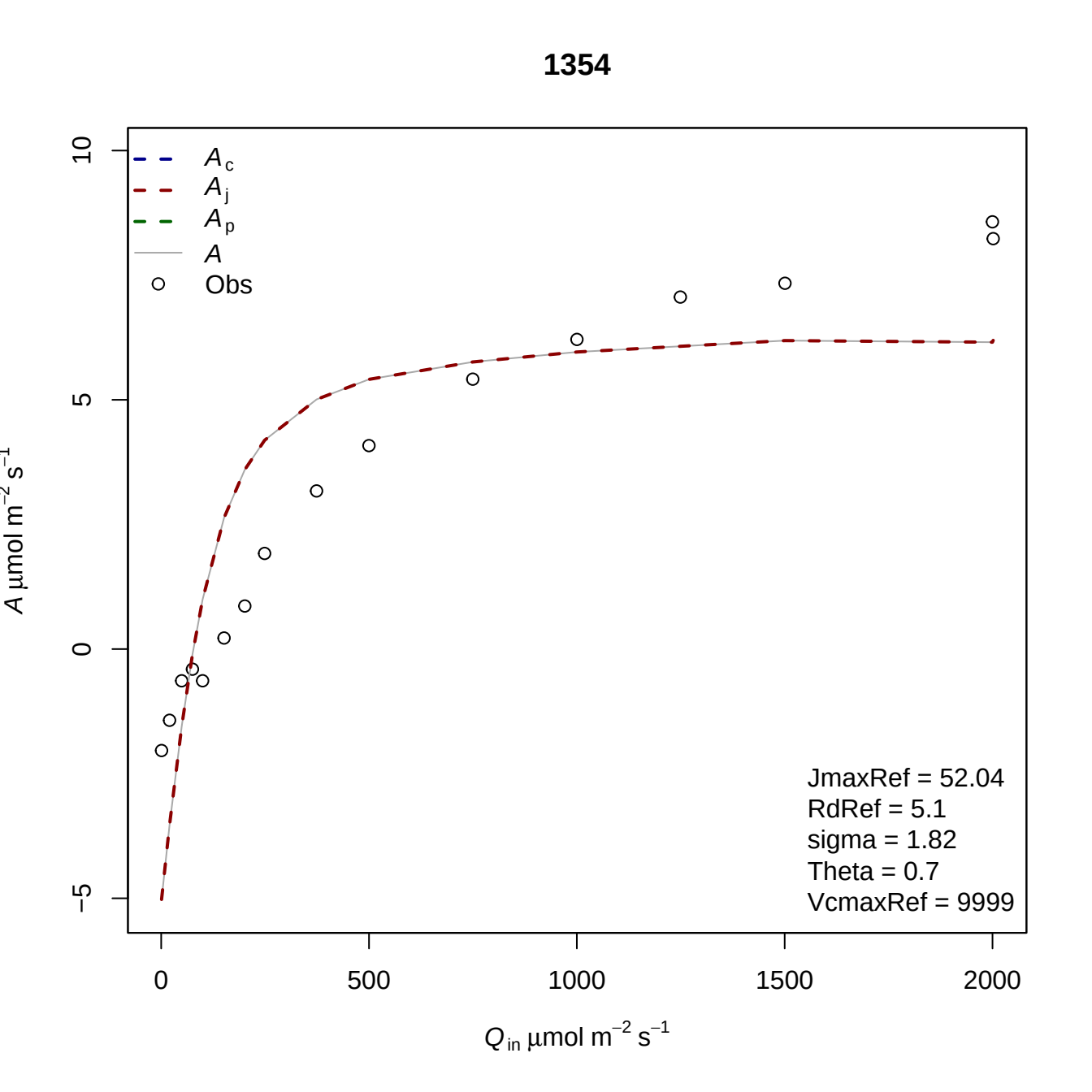

1349
A_c
A_j
A_p
A
Obs
JmaxRef = 48.74
RdRef = 4.75
sigma = 1.28
Theta = 0.7
VcmaxRef = 9999
8
6
4
2
0
−2
−4
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

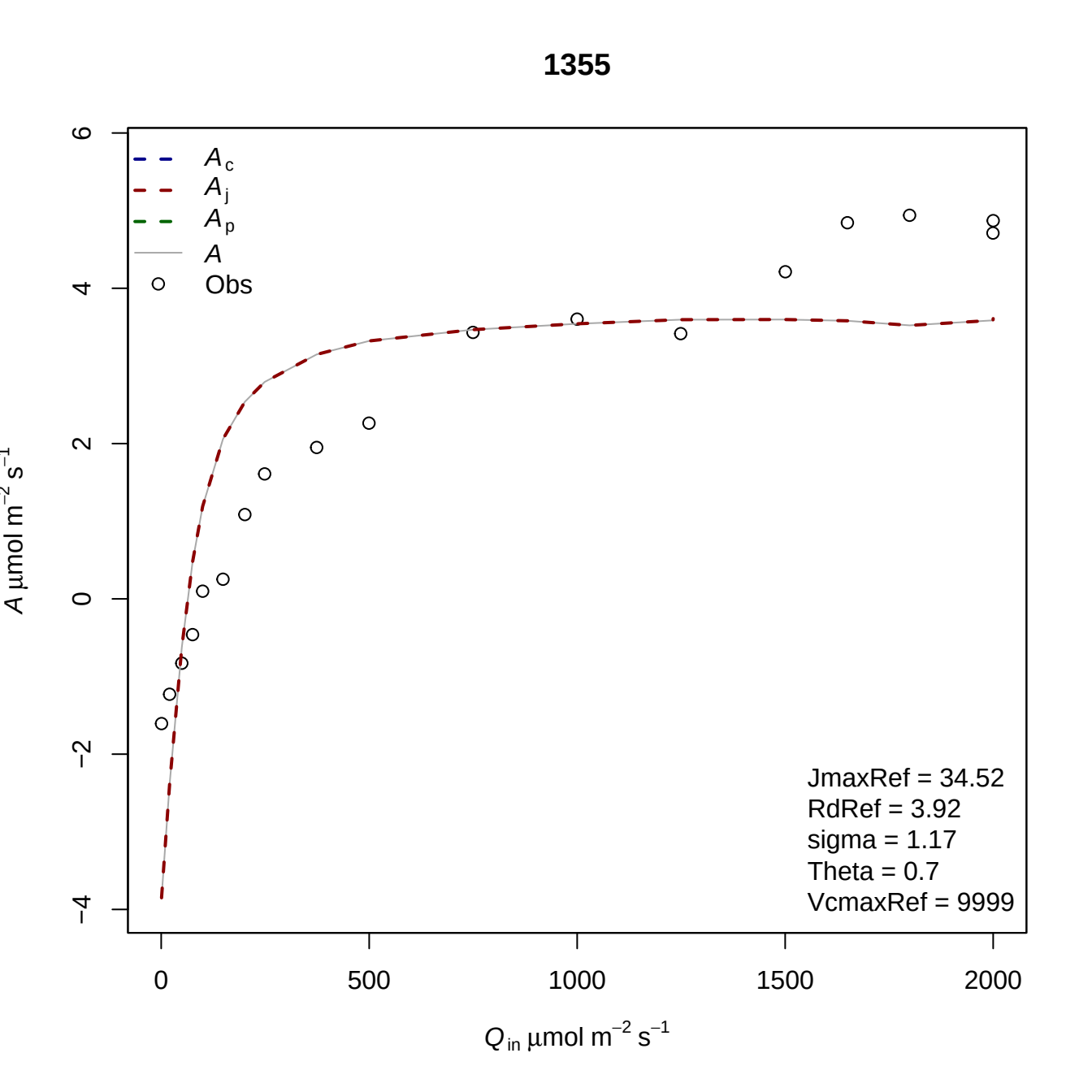

1350
A_c
A_j
A_p
A
Obs
JmaxRef = 61.32
RdRef = 4.07
sigma = 1.38
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

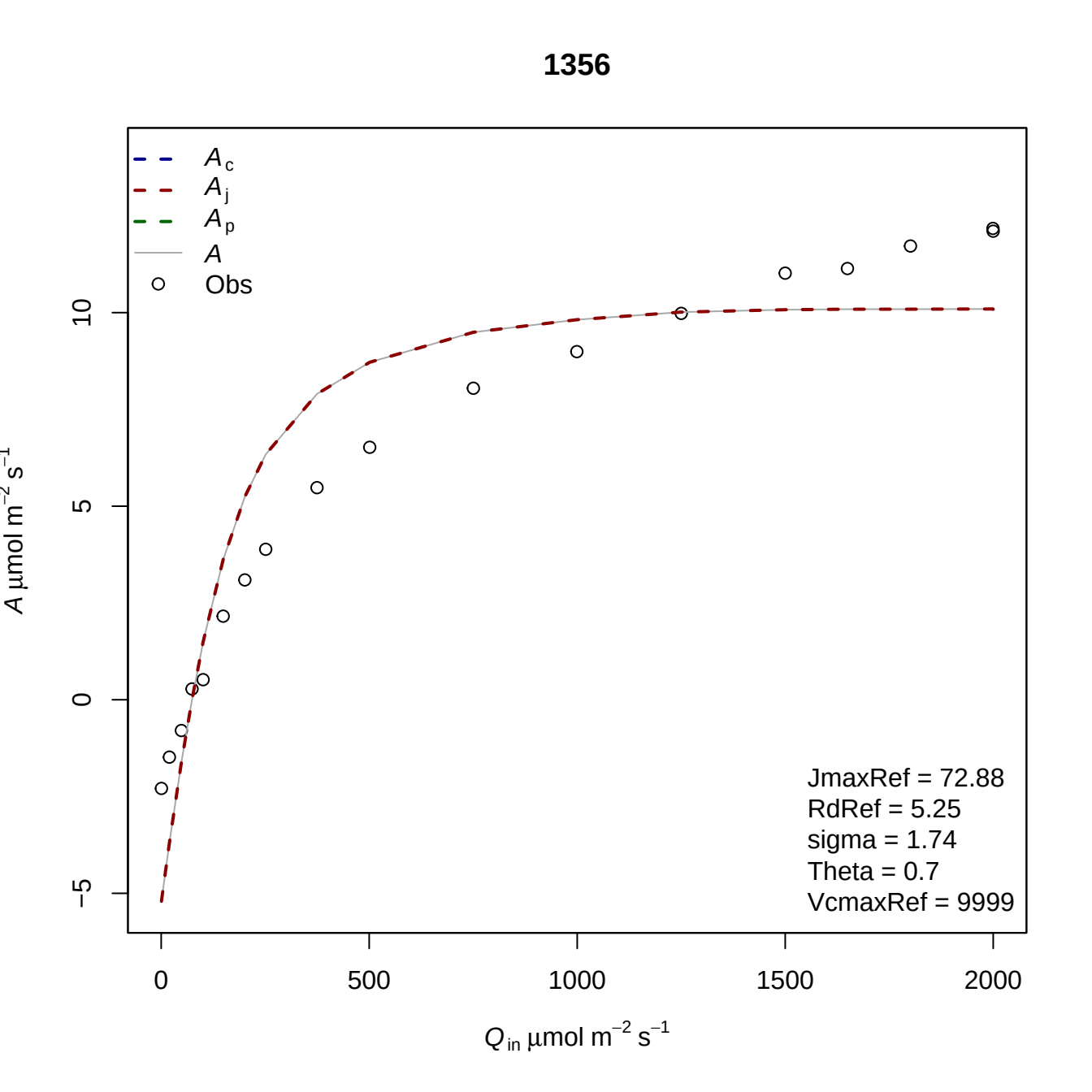

1351
A_c
A_j
A_p
A
Obs
JmaxRef = 43.57
RdRef = 3.37
sigma = 1.22
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

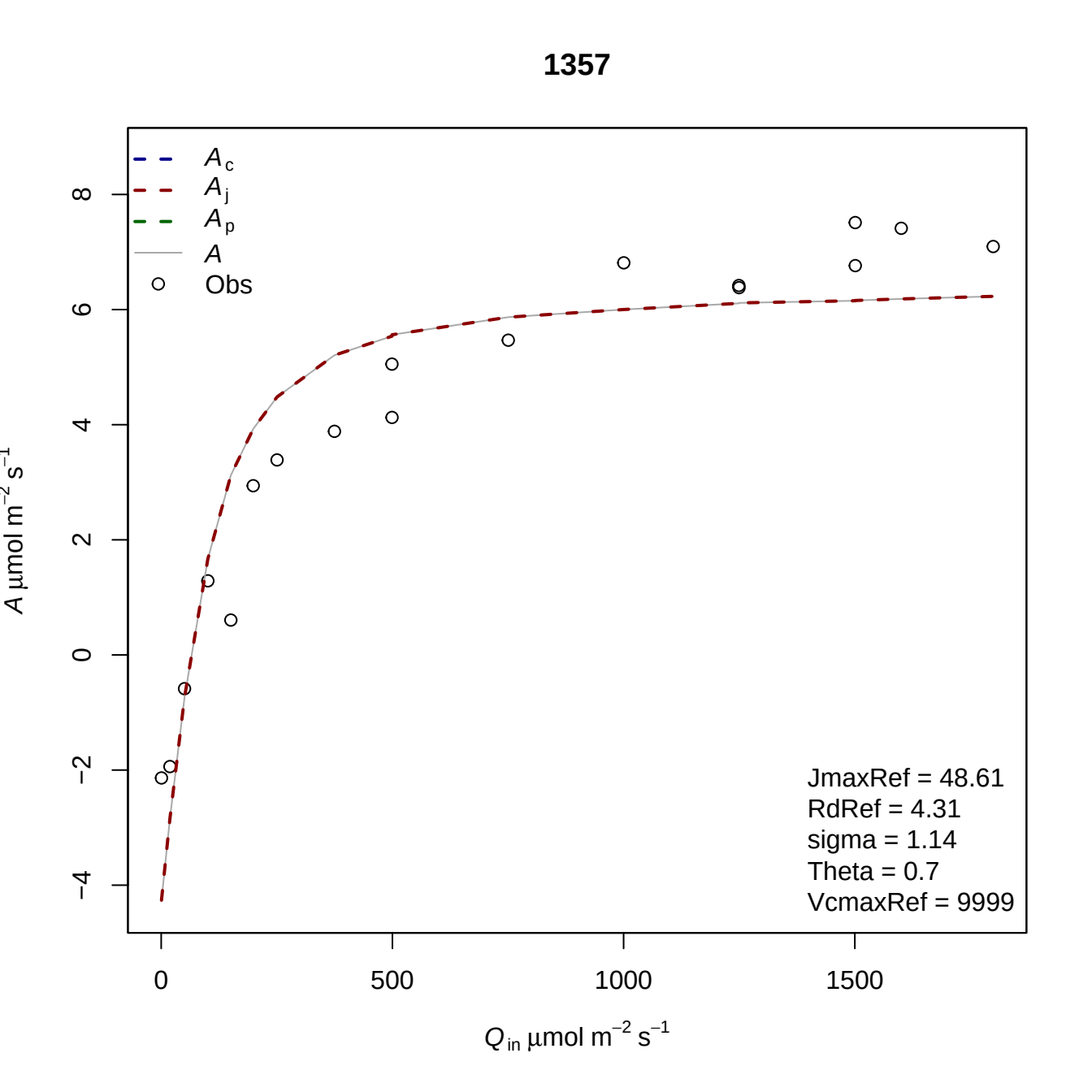

1352
A_c
A_j
A_p
A
Obs
JmaxRef = 65.14
RdRef = 2.6
sigma = 1.53
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

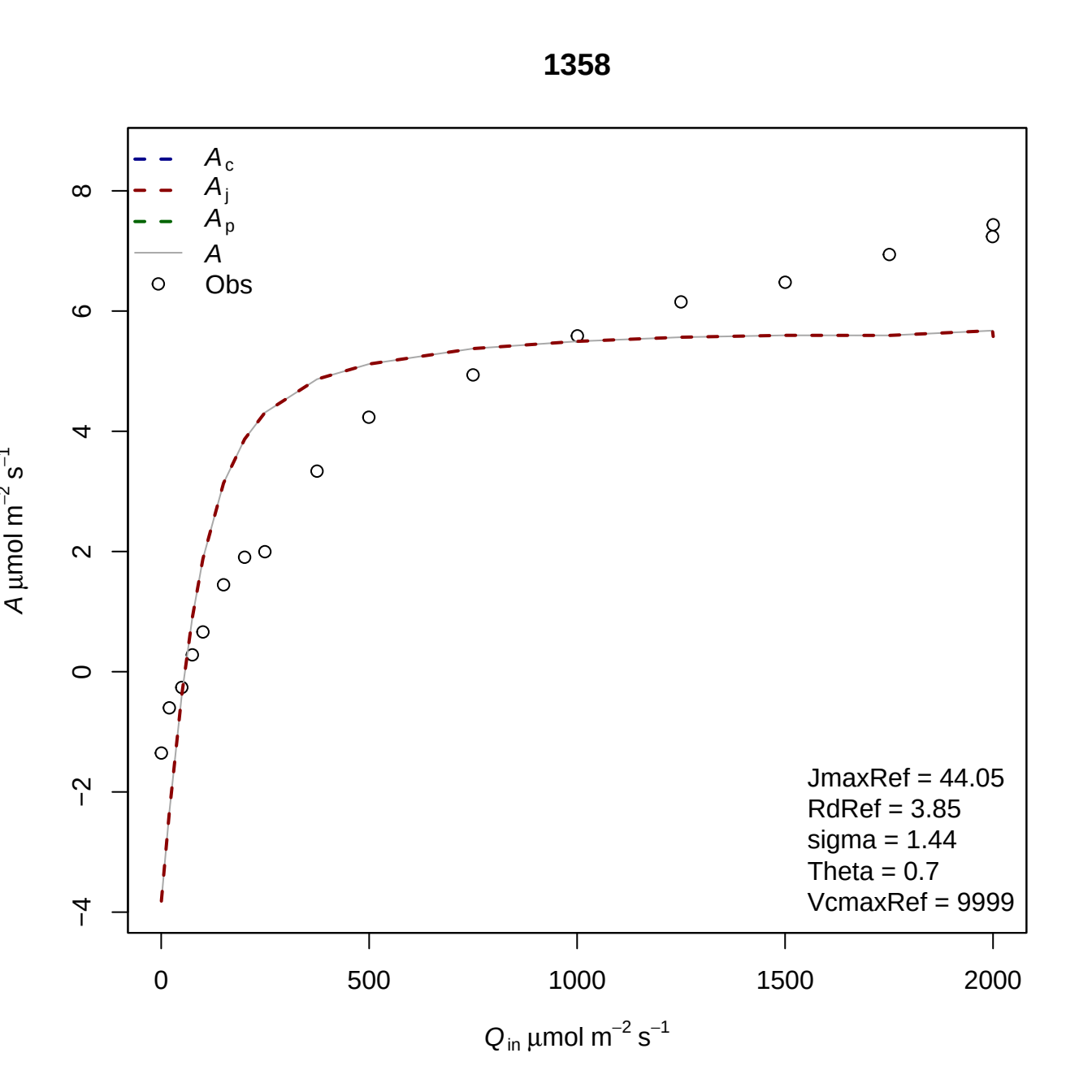

1353
A_c
A_j
A_p
A
Obs
JmaxRef = 44.71
RdRef = 3.34
sigma = 0.83
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

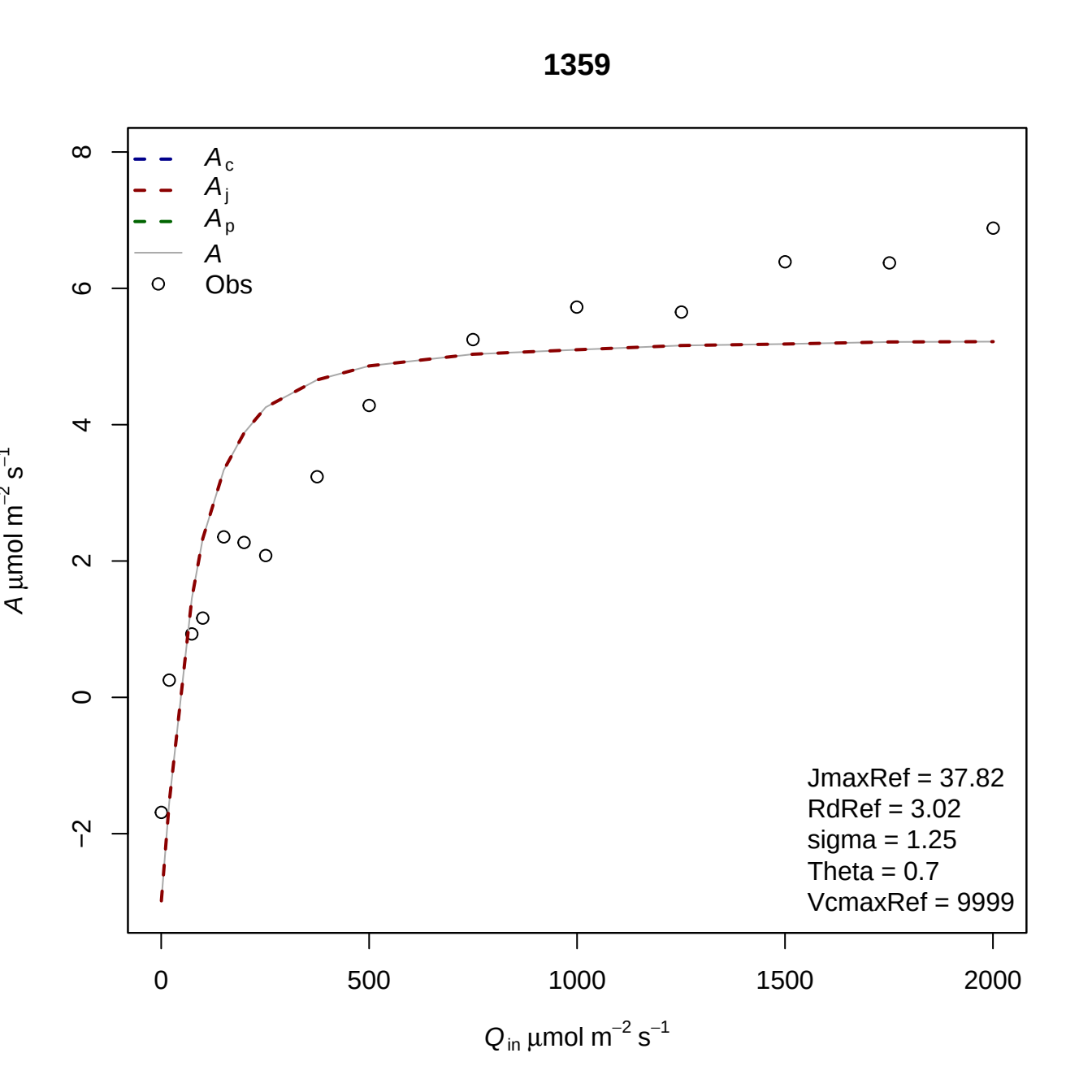

1354
A_c
A_j
A_p
A
Obs
JmaxRef = 52.04
RdRef = 5.1
sigma = 1.82
Theta = 0.7
VcmaxRef = 9999
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

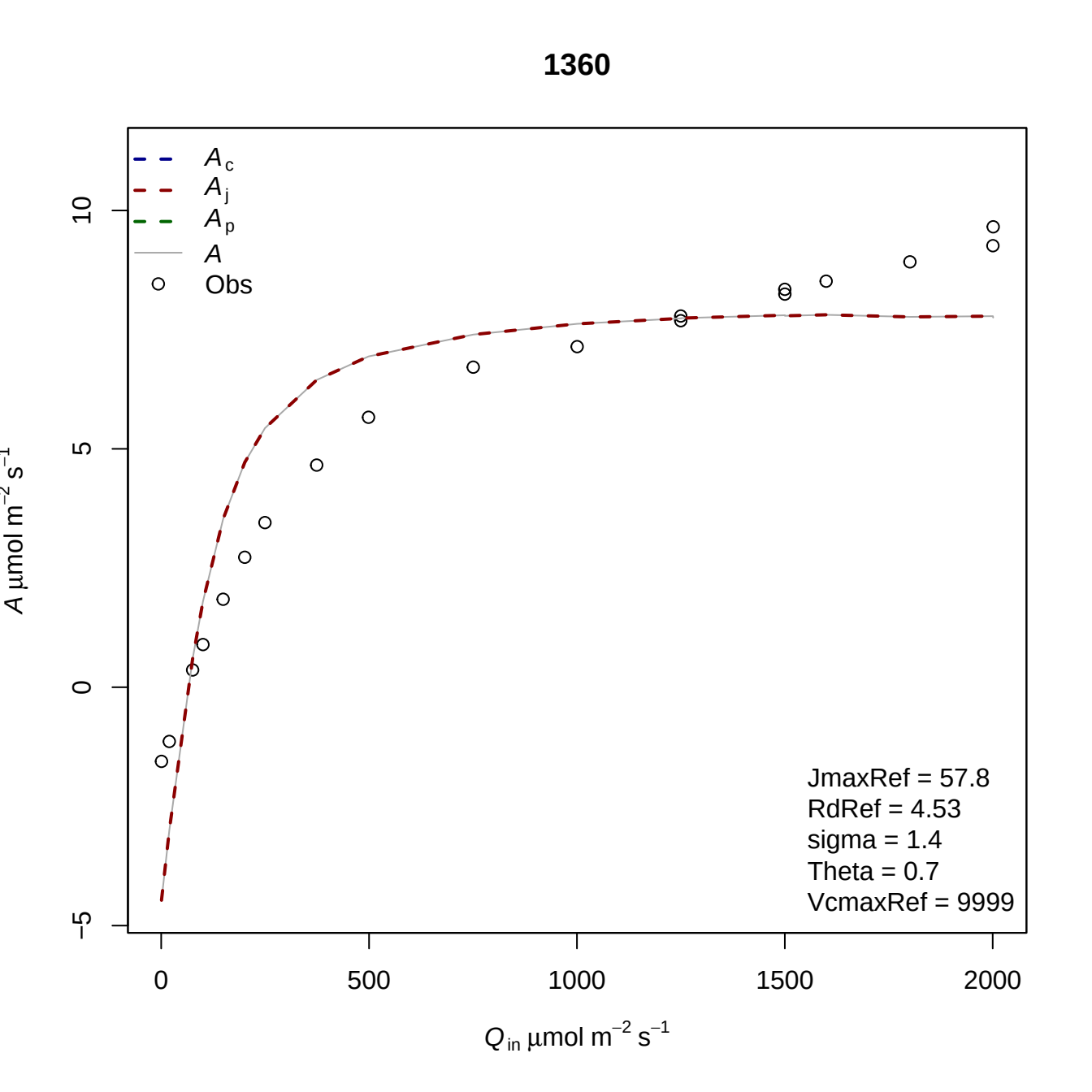

1355
A_c
A_j
A_p
A
Obs
JmaxRef = 34.52
RdRef = 3.92
sigma = 1.17
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500
2000

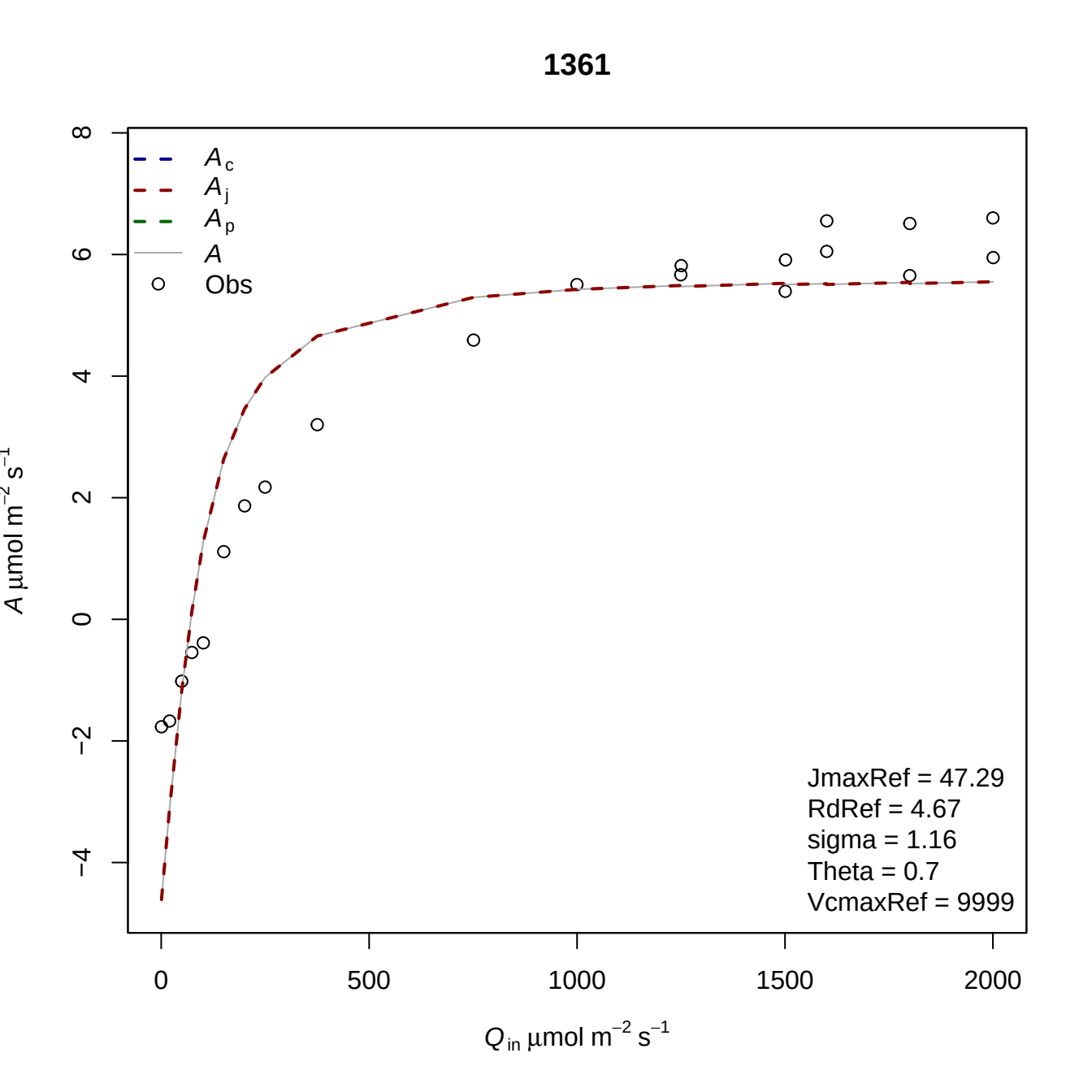

1356
A_c
A_j
A_p
A
Obs
JmaxRef = 72.88
RdRef = 5.25
sigma = 1.74
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500
2000

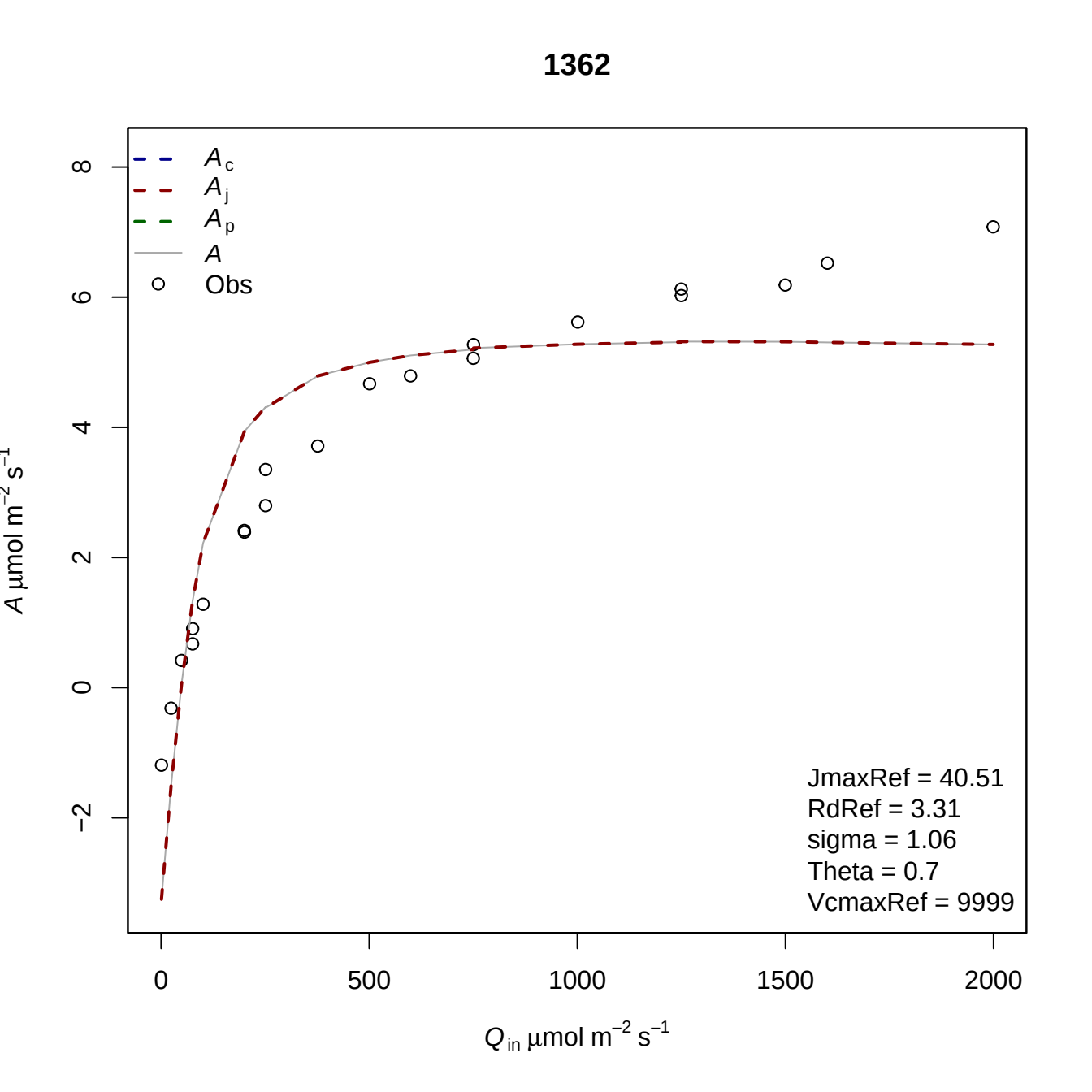

1357
A_c
A_j
A_p
A
Obs
JmaxRef = 48.61
RdRef = 4.31
sigma = 1.14
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

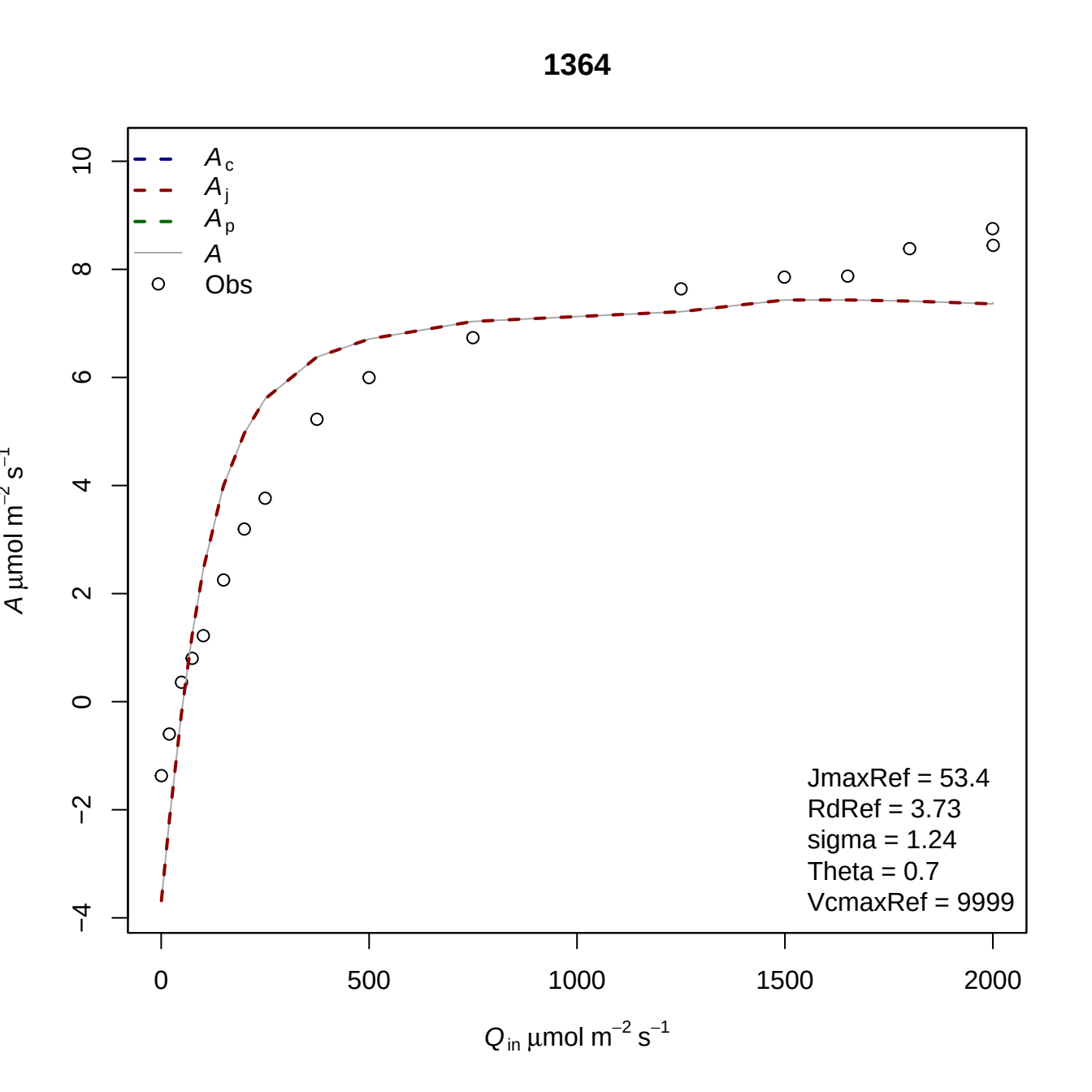

1358
A_c
A_j
A_p
A
Obs
JmaxRef = 44.05
RdRef = 3.85
sigma = 1.44
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

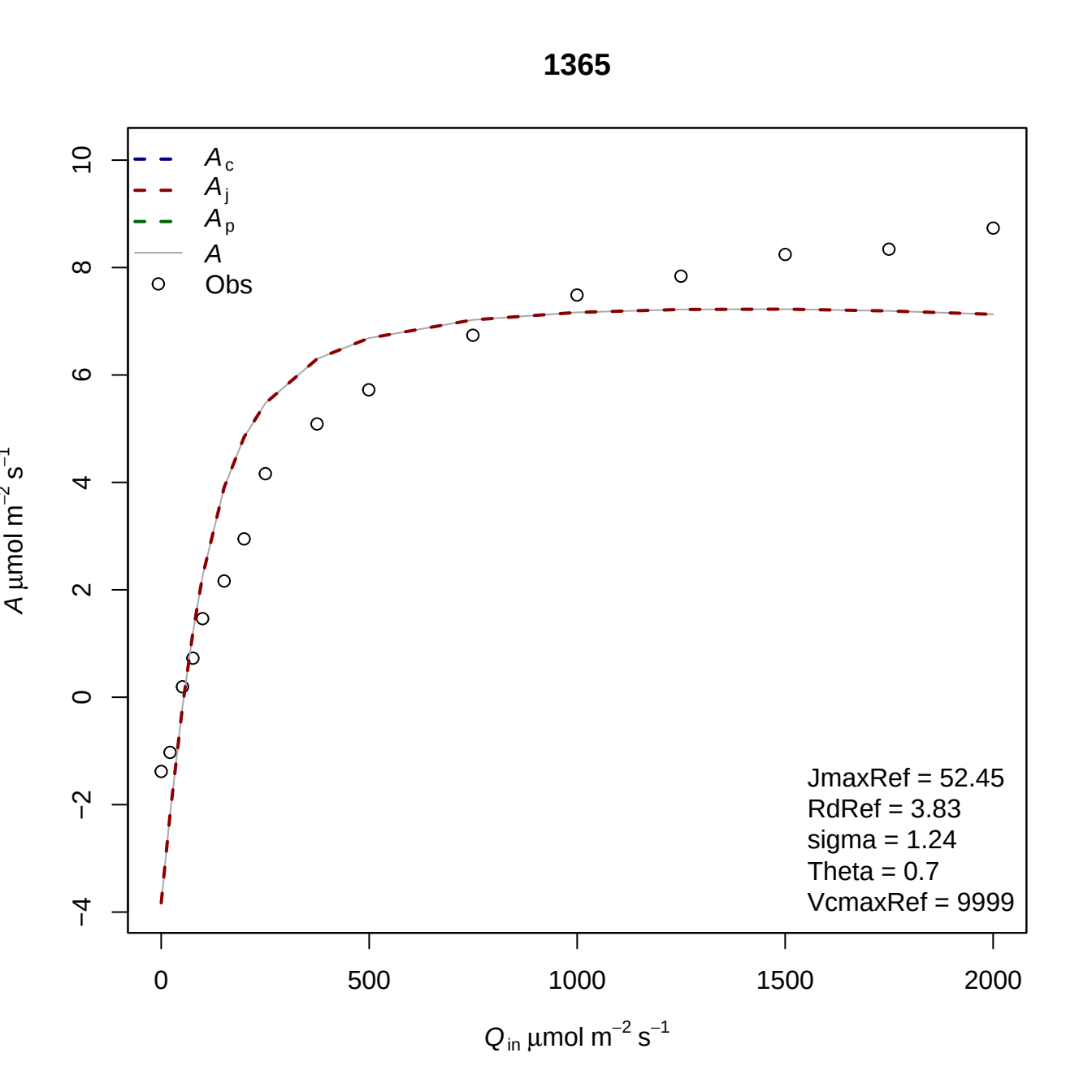

1359
A_c
A_j
A_p
A
Obs
JmaxRef = 37.82
RdRef = 3.02
sigma = 1.25
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

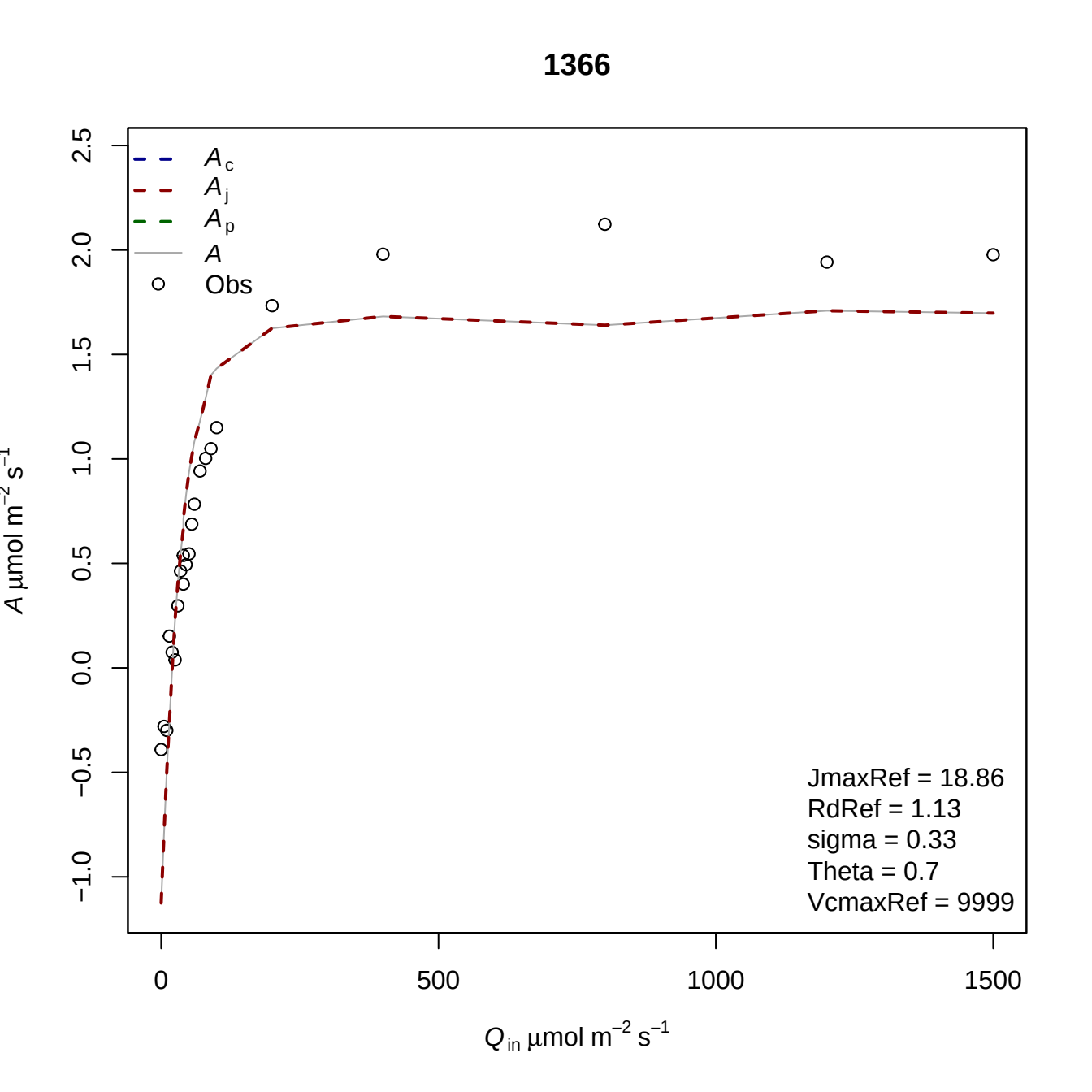

1360
A_c
A_j
A_p
A
Obs
JmaxRef = 57.8
RdRef = 4.53
sigma = 1.4
Theta = 0.7
VcmaxRef = 9999
10
5
0
−5
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

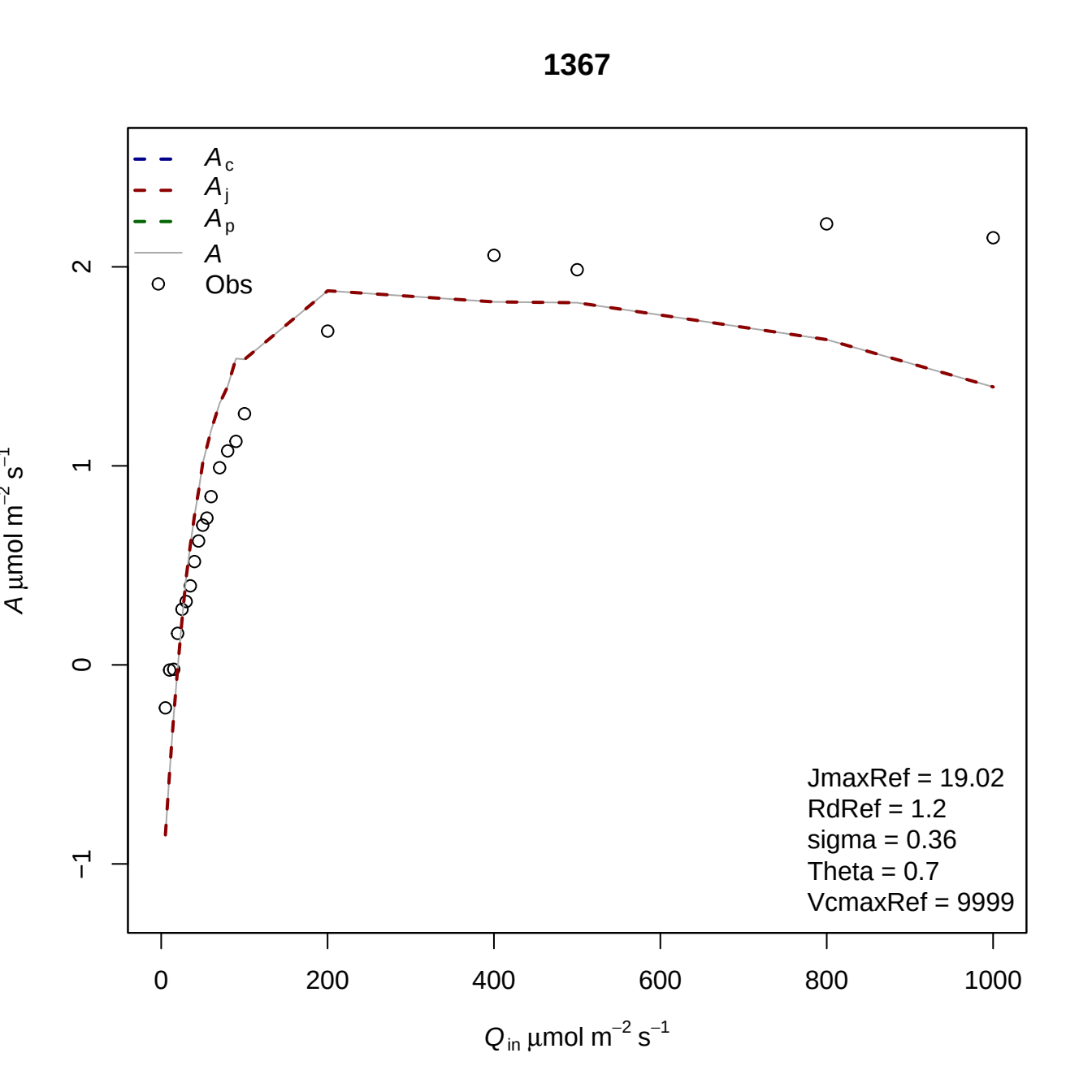

1361
A_c
A_j
A_p
A
Obs
JmaxRef = 47.29
RdRef = 4.67
sigma = 1.16
Theta = 0.7
VcmaxRef = 9999
8
6
4
2
0
−2
−4
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

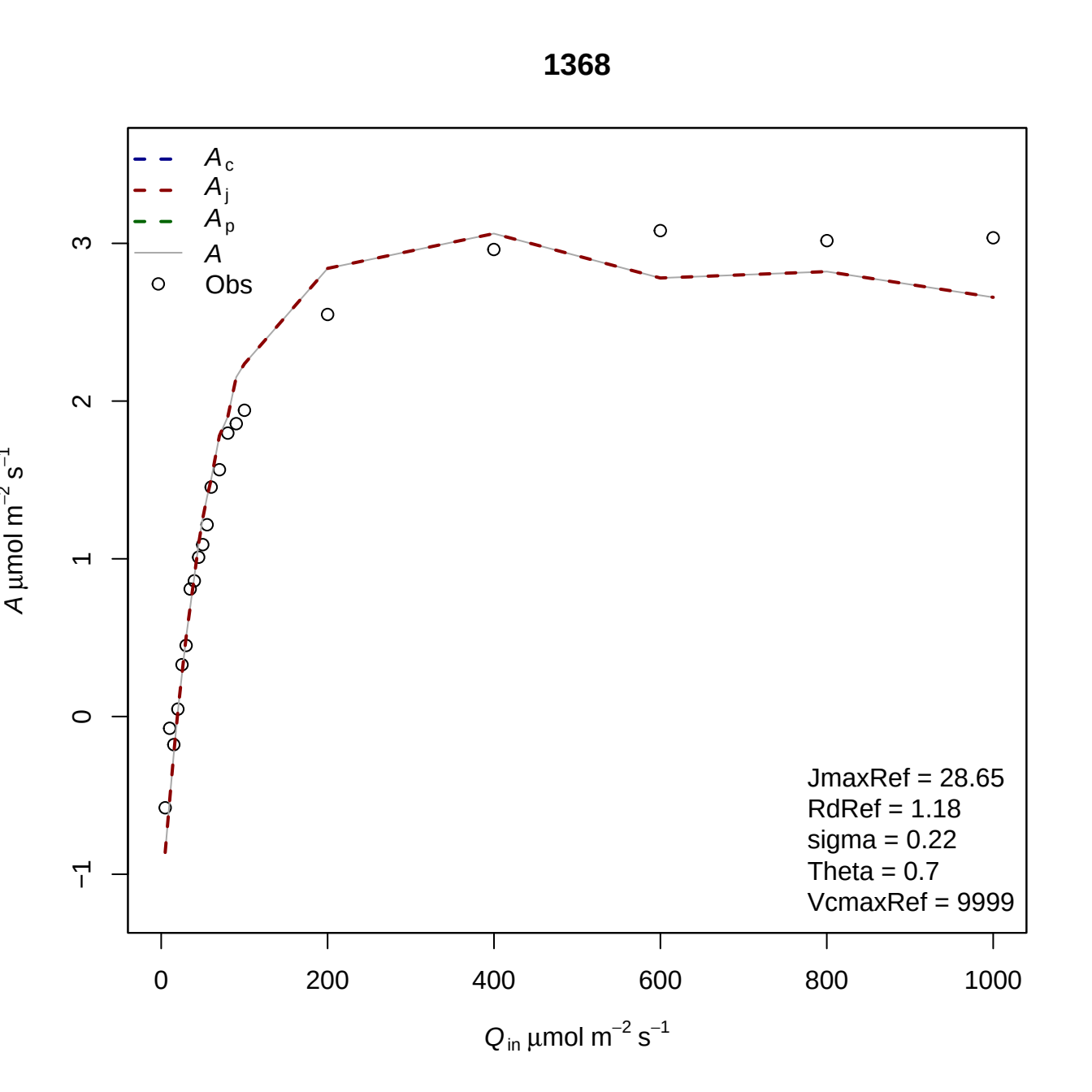

1362
A_c
A_j
A_p
A
Obs
JmaxRef = 40.51
RdRef = 3.31
sigma = 1.06
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
0
500
1000
1500
2000

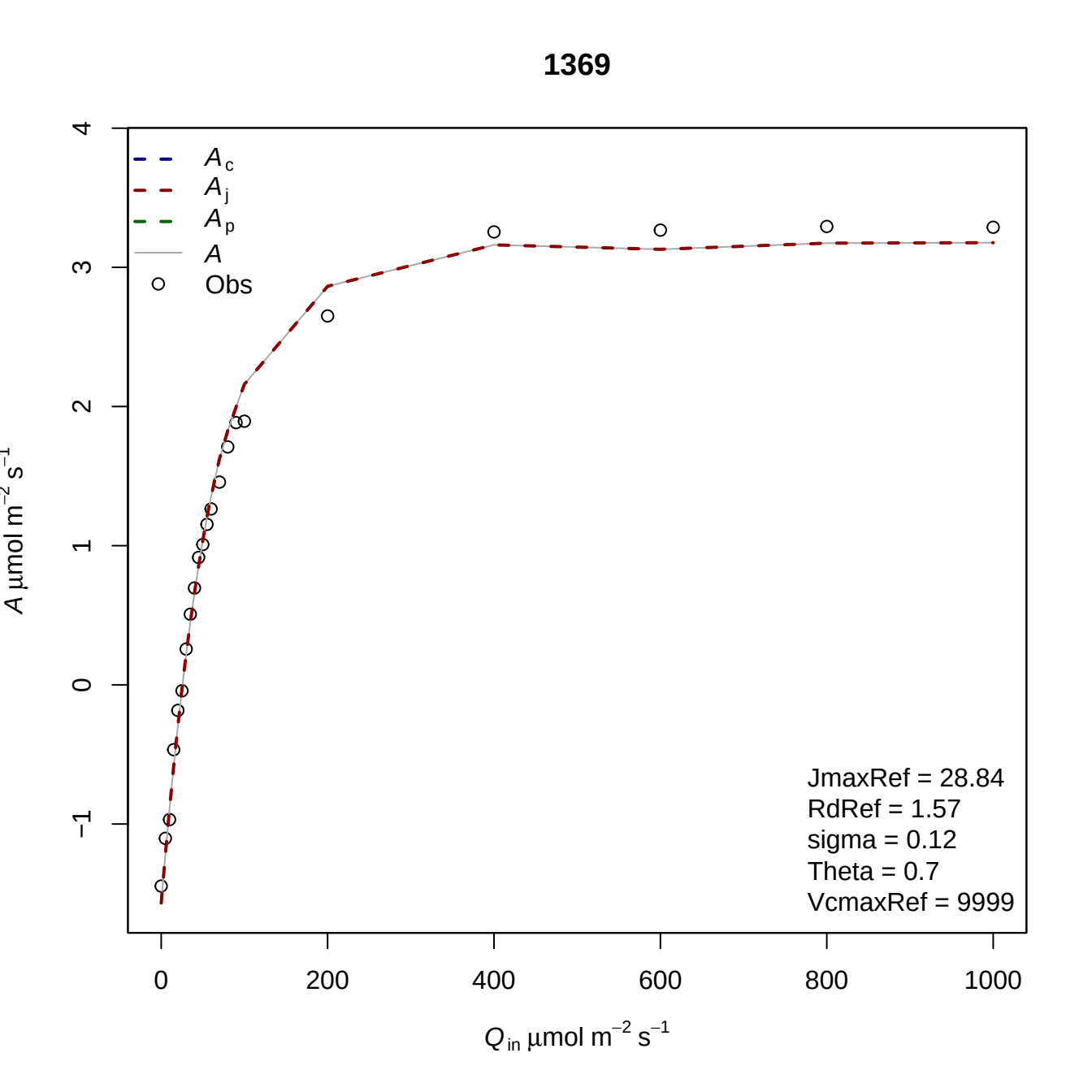

1364
A_c
A_j
A_p
A
Obs
JmaxRef = 53.4
RdRef = 3.73
sigma = 1.24
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

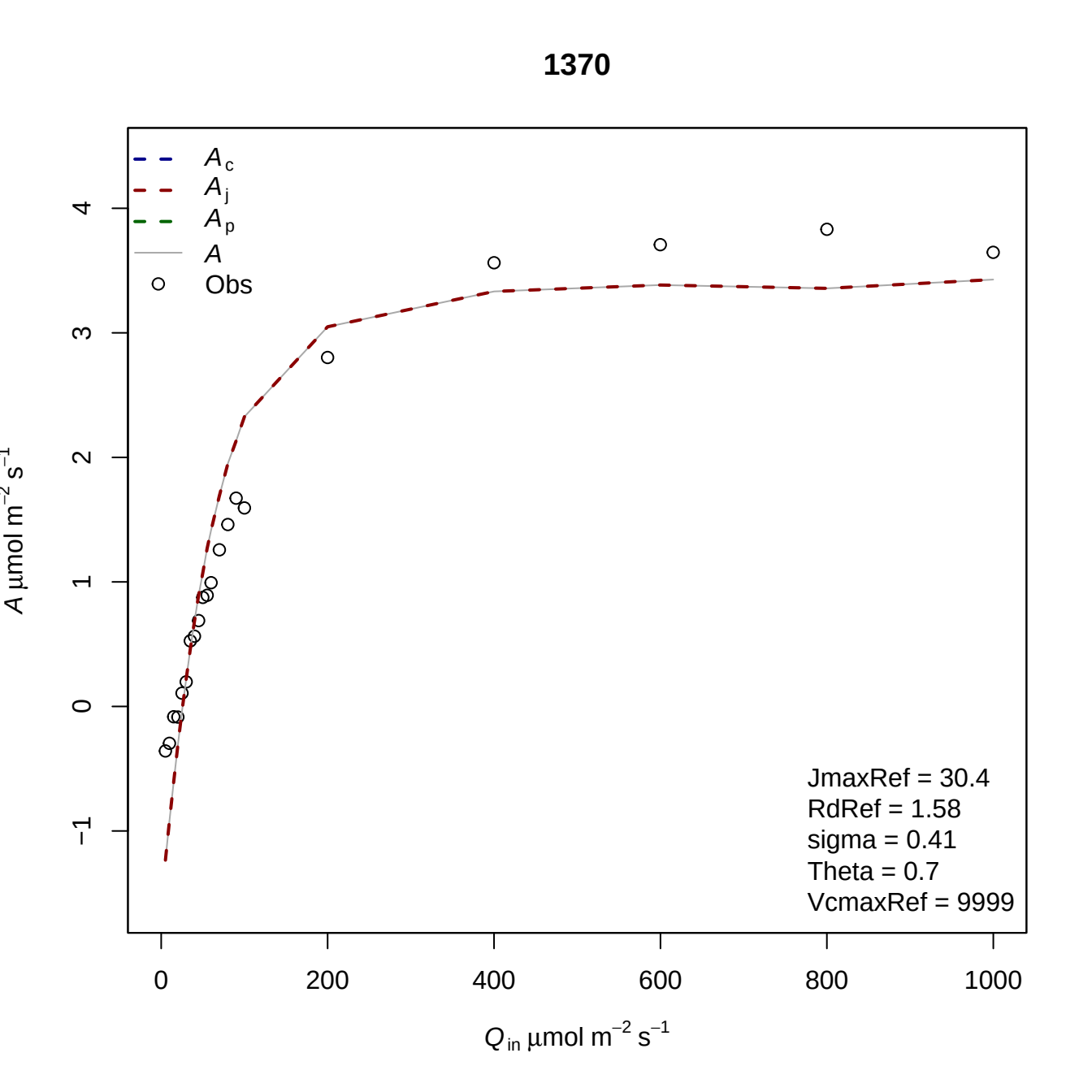

1365
A_c
A_j
A_p
A
Obs
JmaxRef = 52.45
RdRef = 3.83
sigma = 1.24
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500
2000

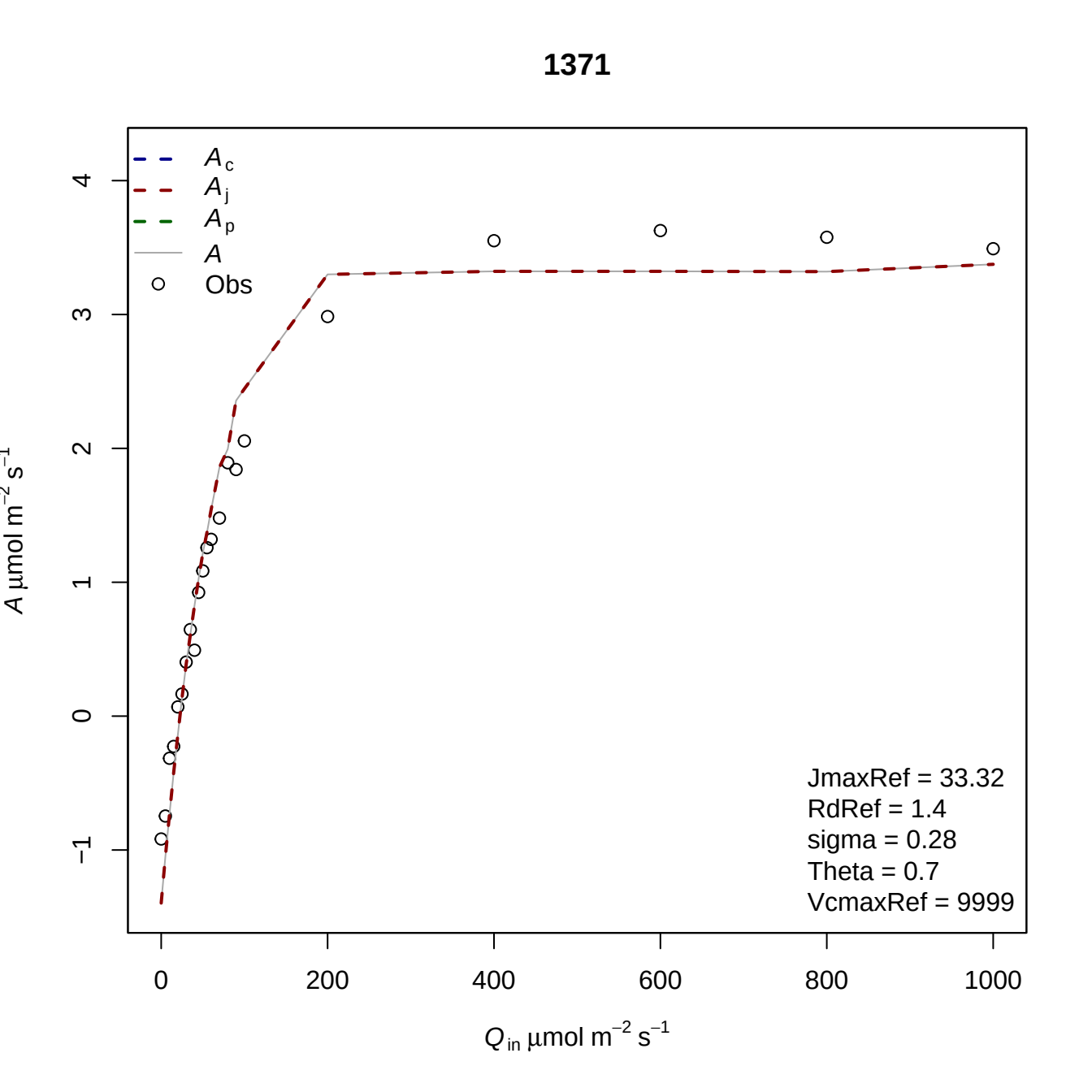

1366
A_c
A_j
A_p
A
Obs
JmaxRef = 18.86
RdRef = 1.13
sigma = 0.33
Theta = 0.7
VcmaxRef = 9999
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

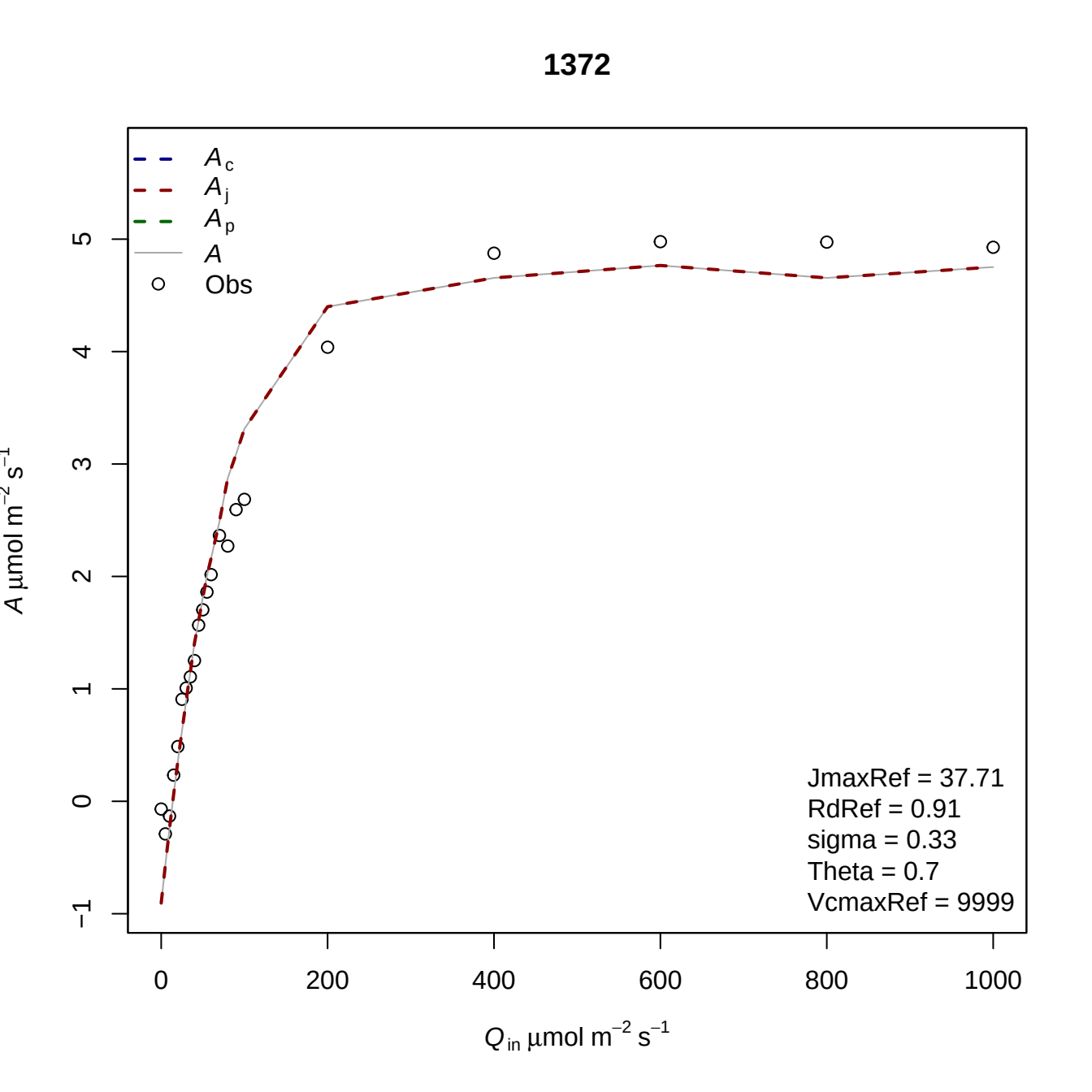

1367
A_c
A_j
A_p
A
Obs
JmaxRef = 19.02
RdRef = 1.2
sigma = 0.36
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
200
400
600
800
1000

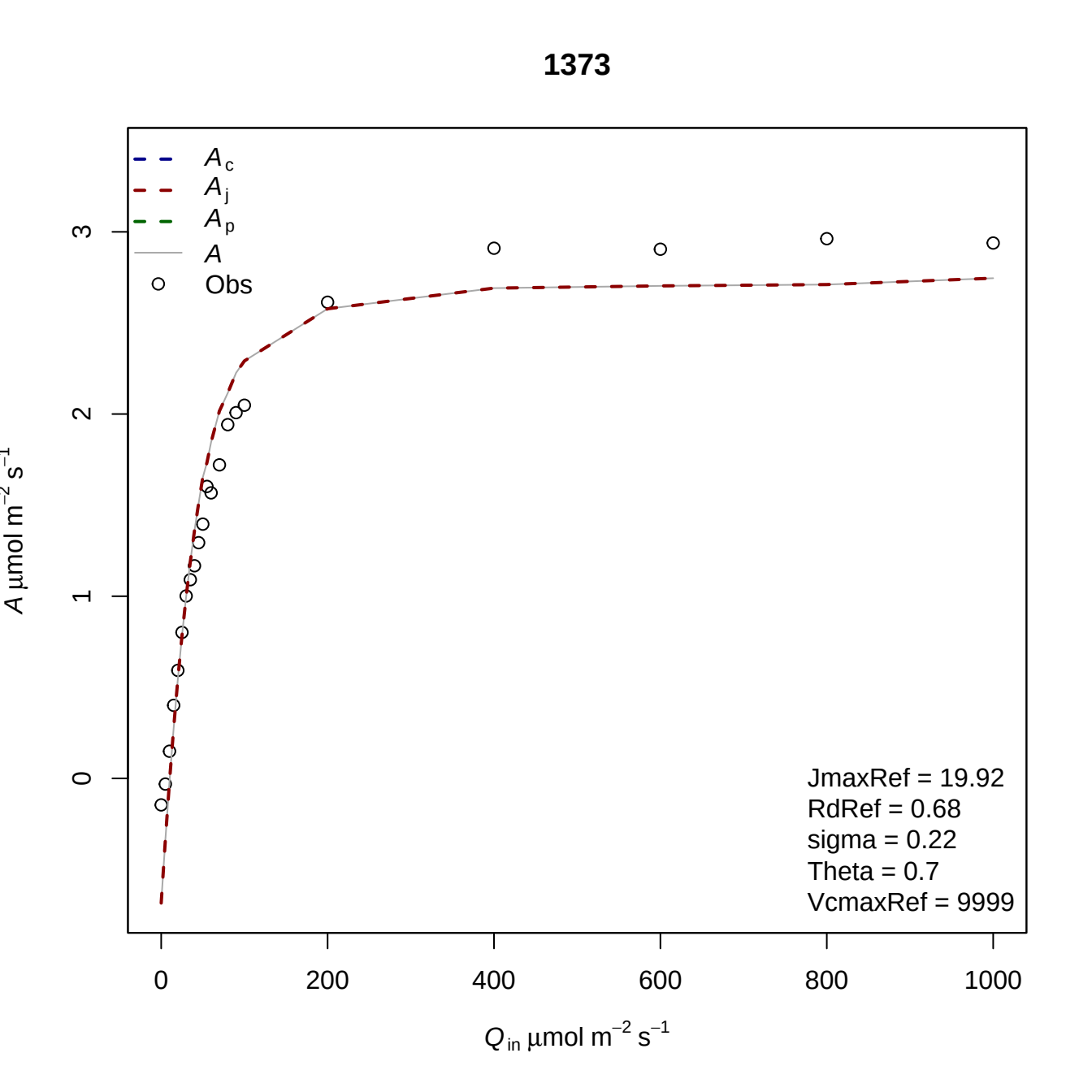

1368
A_c
A_j
A_p
A
Obs
JmaxRef = 28.65
RdRef = 1.18
sigma = 0.22
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800
1000

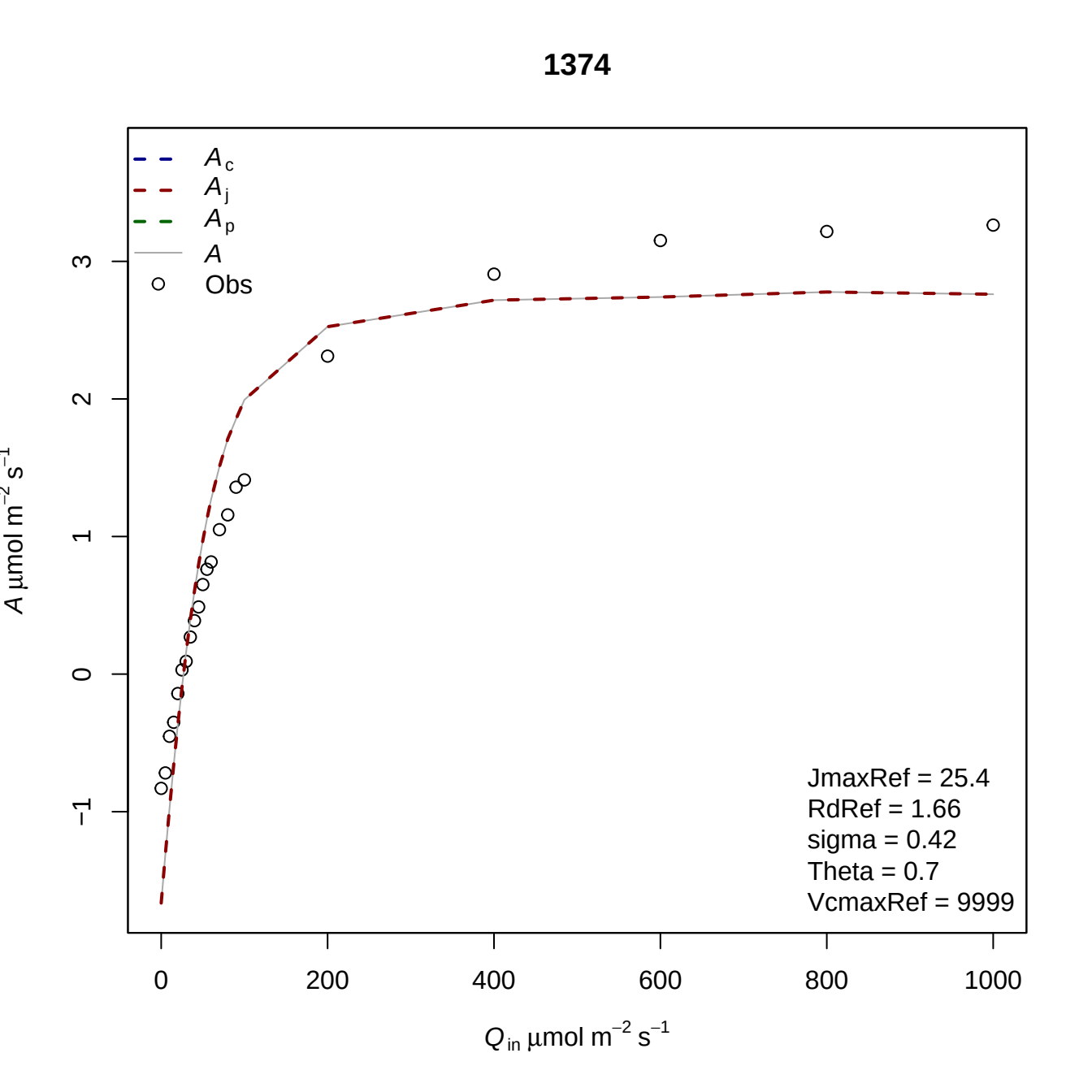

1369
A_c
A_j
A_p
A
Obs
JmaxRef = 28.84
RdRef = 1.57
sigma = 0.12
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
0
200
400
600
800
1000

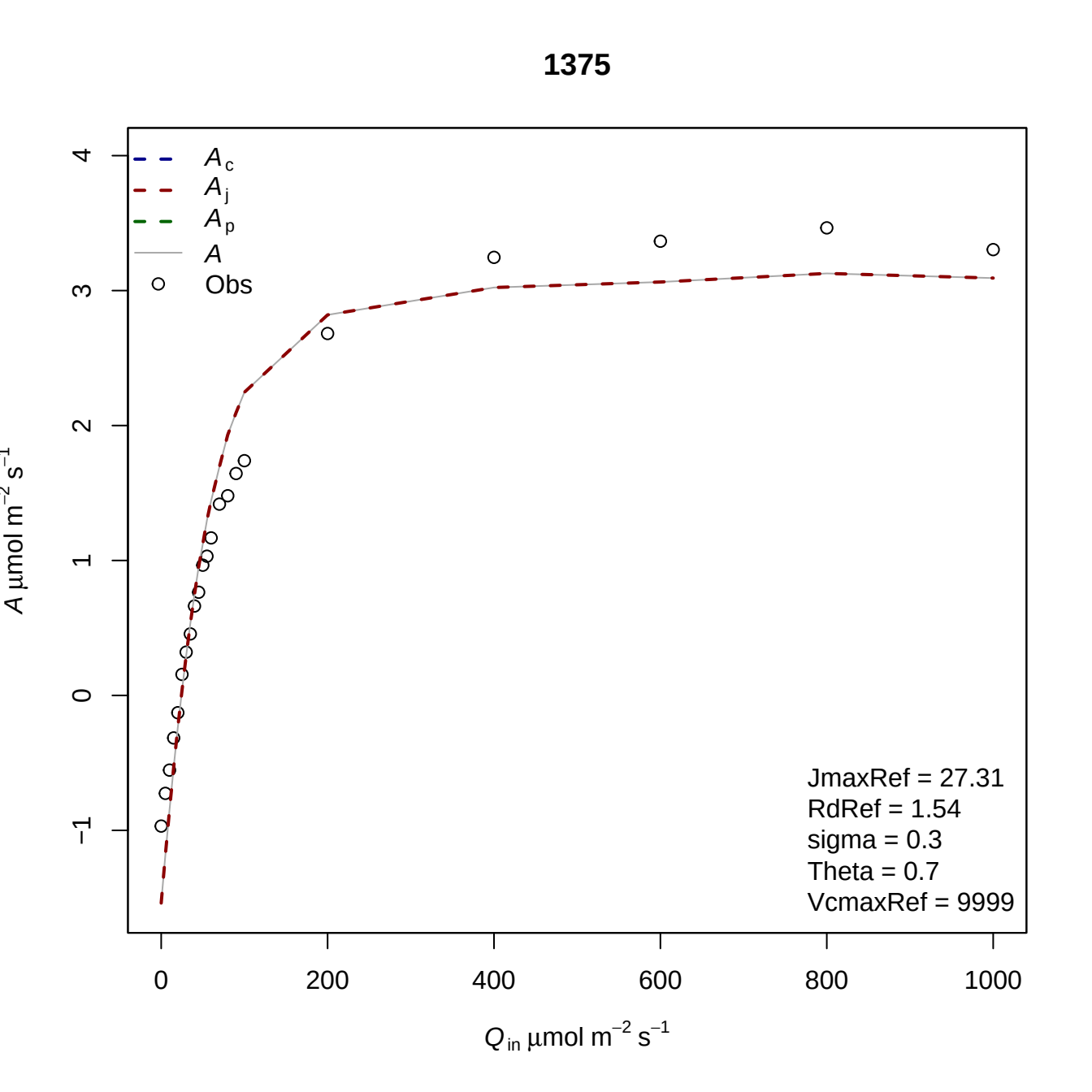

1370
A_c
A_j
A_p
A
Obs
JmaxRef = 30.4
RdRef = 1.58
sigma = 0.41
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000

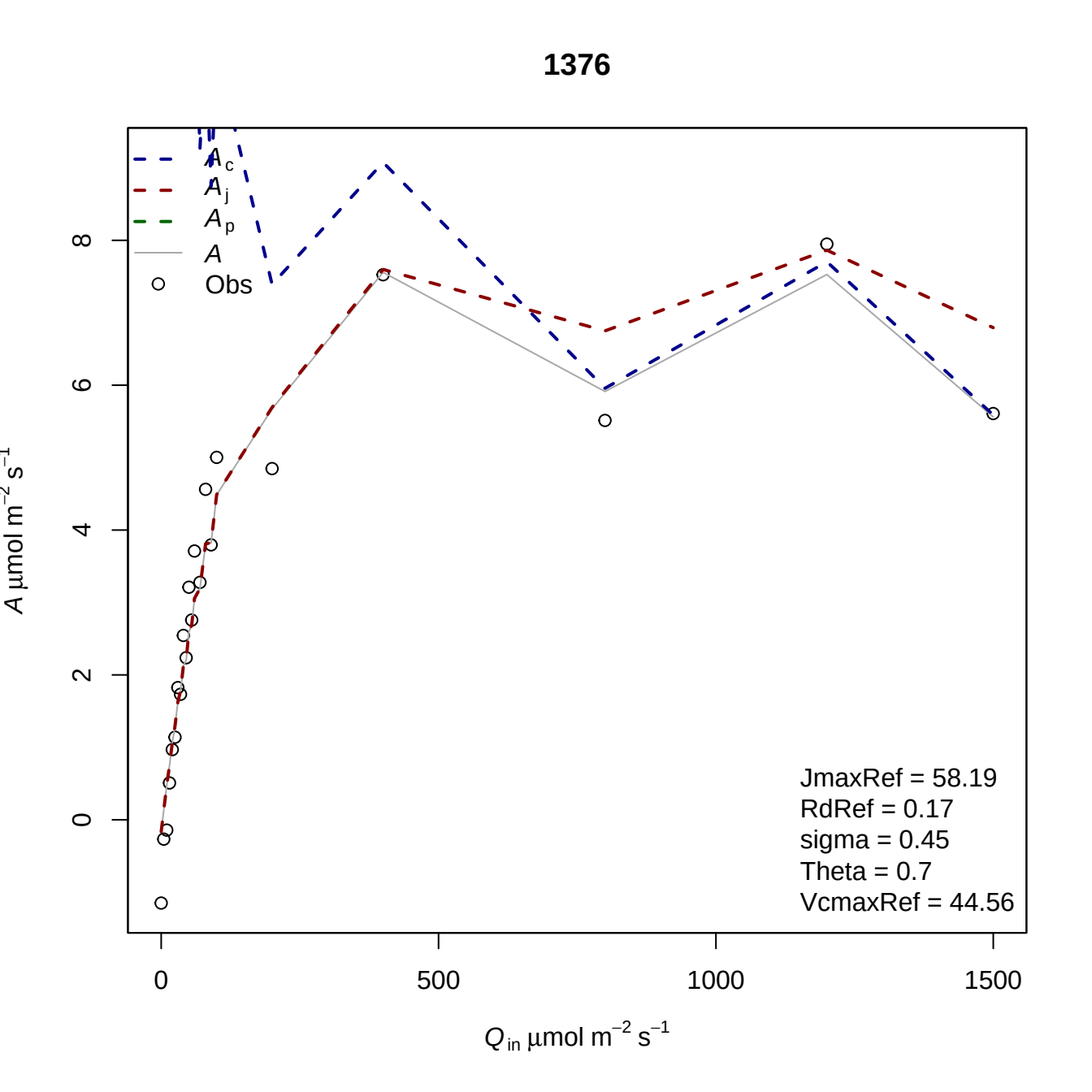

1371
A_c
A_j
A_p
A
Obs
JmaxRef = 33.32
RdRef = 1.4
sigma = 0.28
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000

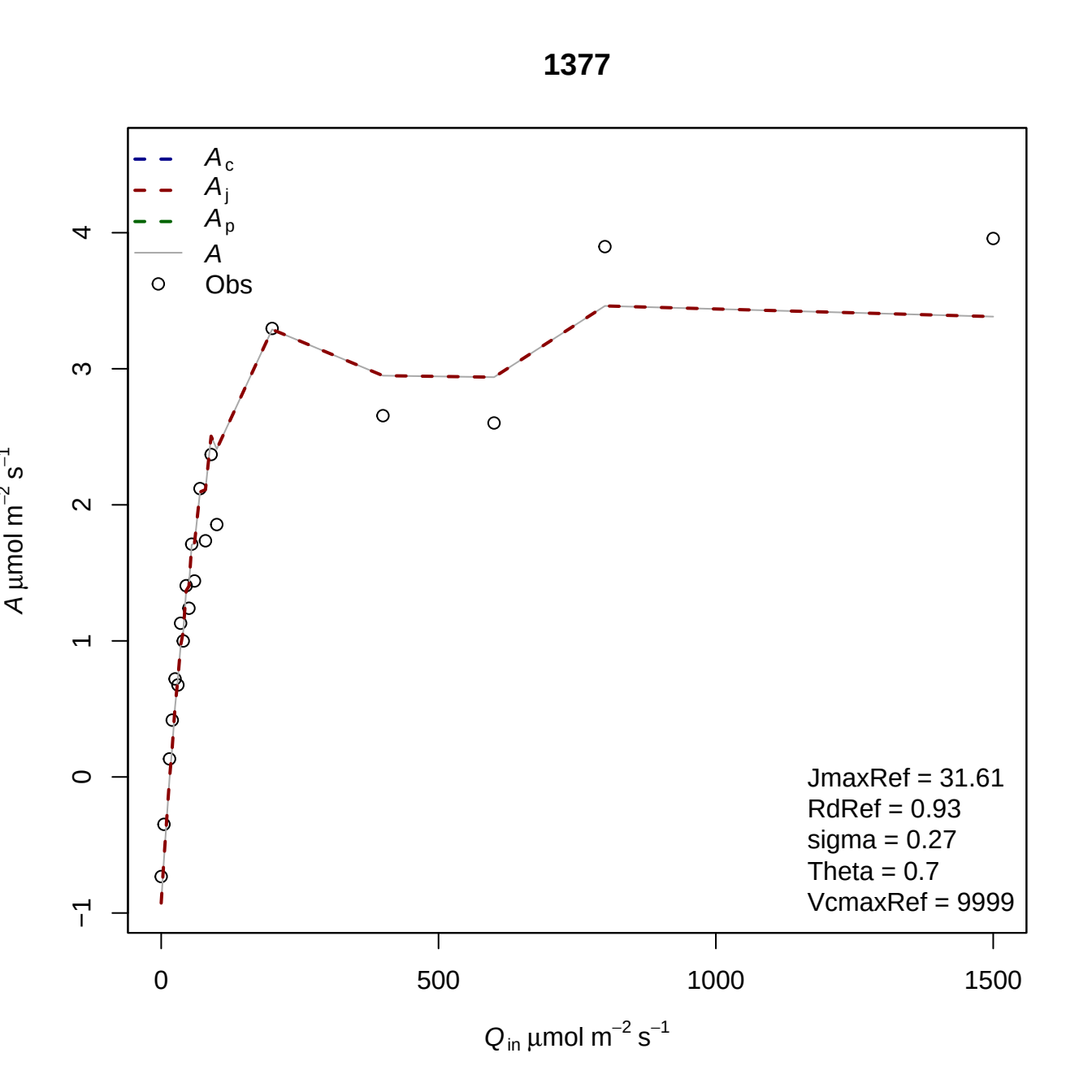

1372
A_c
A_j
A_p
A
Obs
JmaxRef = 37.71
RdRef = 0.91
sigma = 0.33
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
200
400
600
800
1000

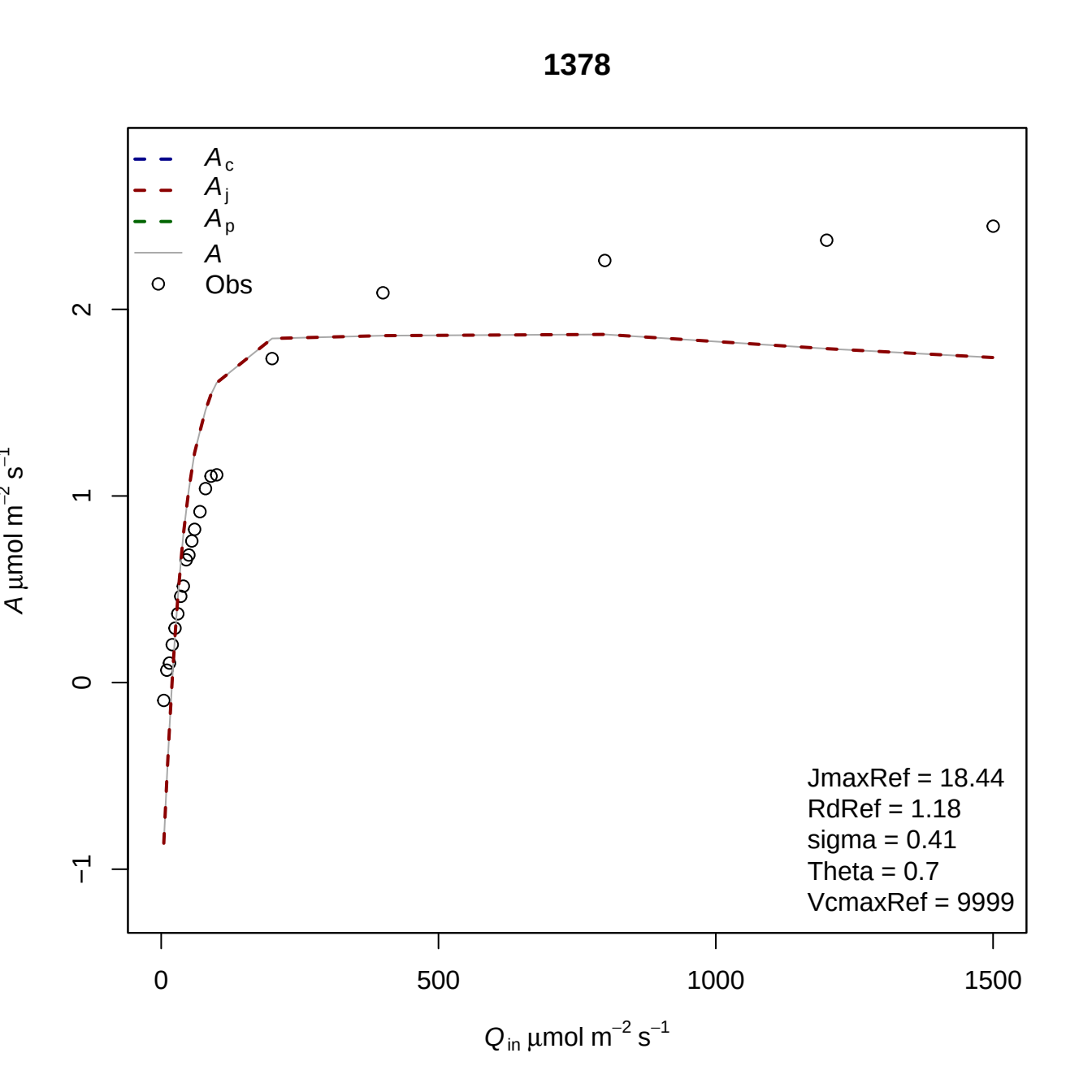

1373
A_c
A_j
A_p
A
Obs
JmaxRef = 19.92
RdRef = 0.68
sigma = 0.22
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
200
400
600
800
1000

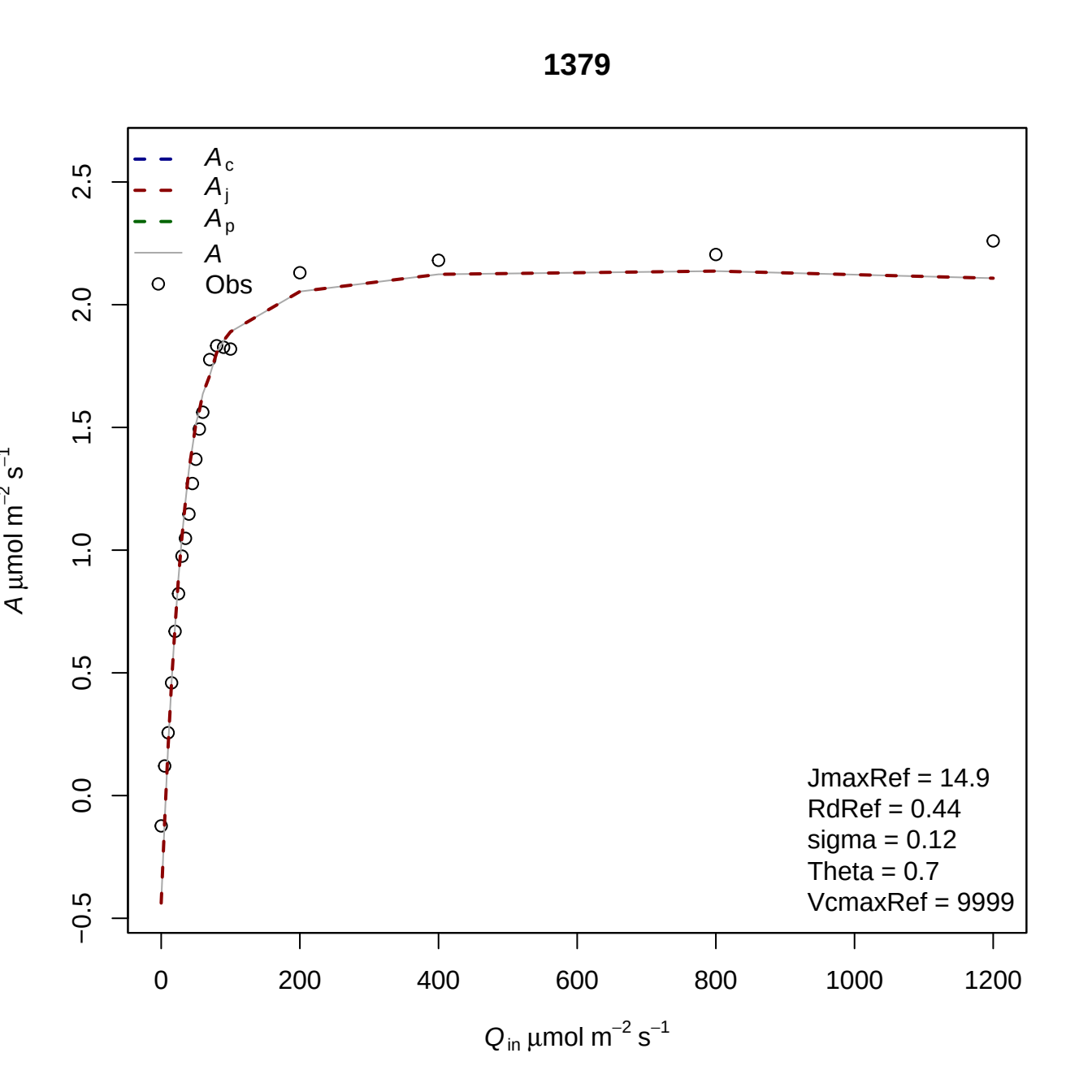

1374
A_c
A_j
A_p
A
Obs
JmaxRef = 25.4
RdRef = 1.66
sigma = 0.42
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800
1000

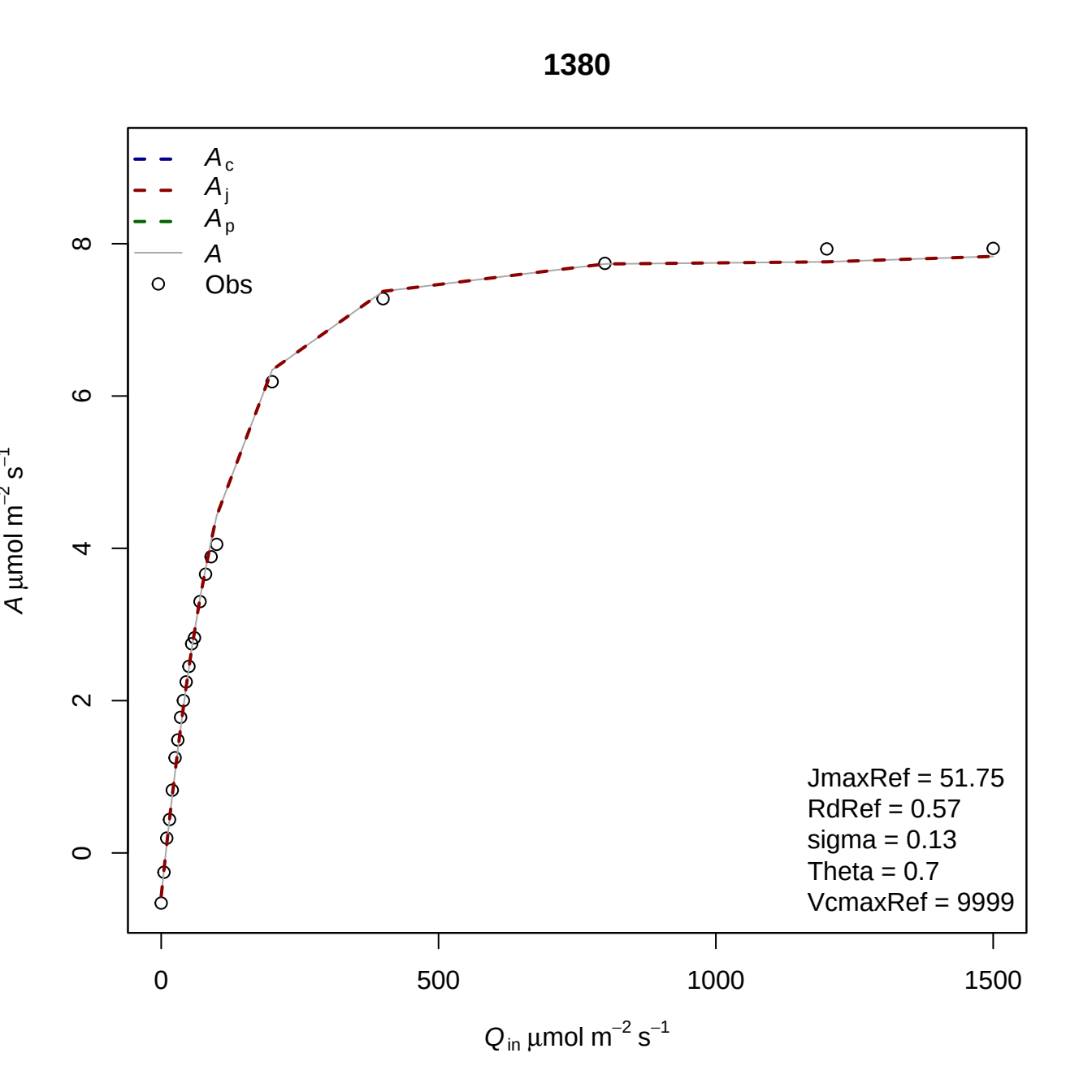

1375
A_c
A_j
A_p
A
Obs
JmaxRef = 27.31
RdRef = 1.54
sigma = 0.3
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
0
200
400
600
800
1000

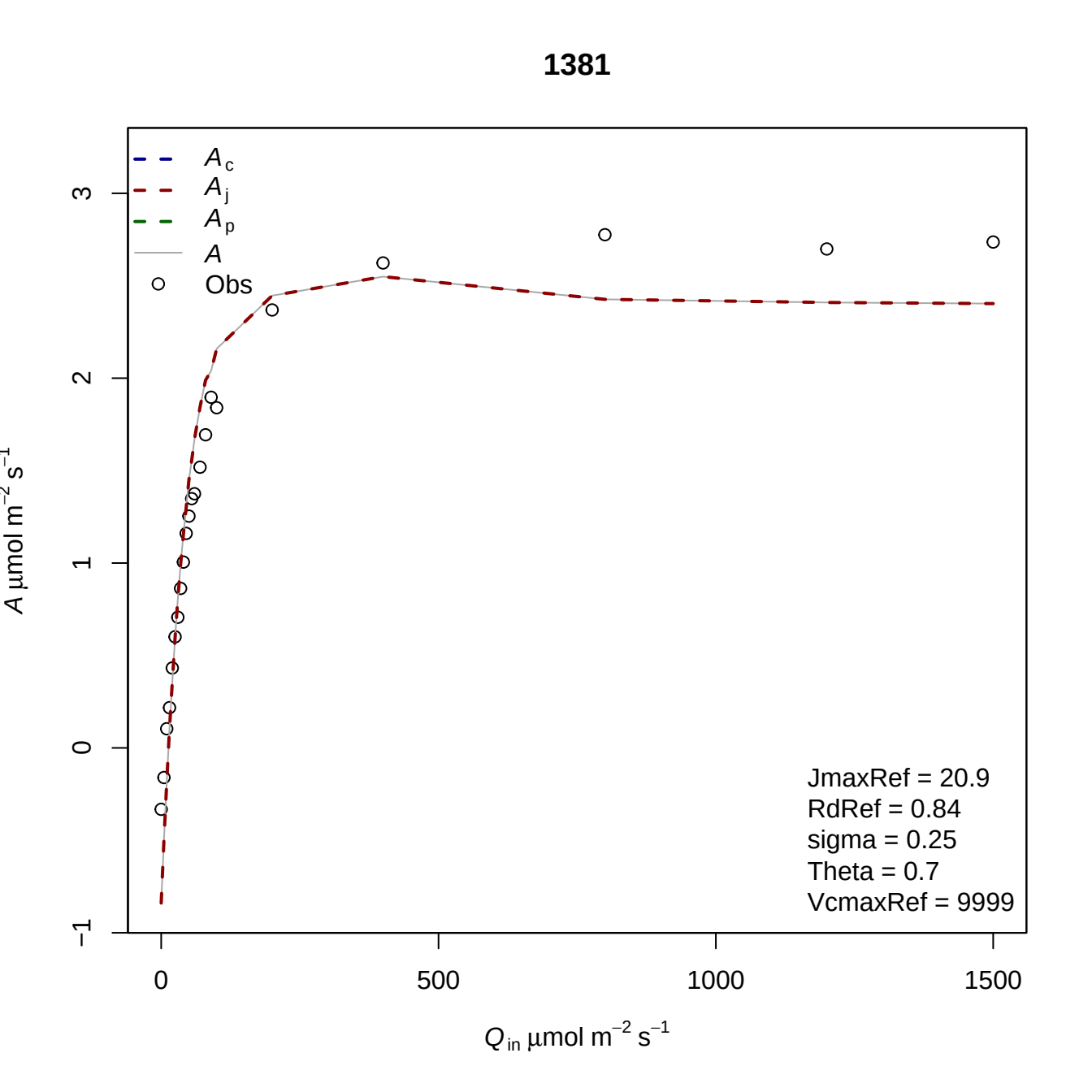

1376
A_c
A_j
A_p
A
Obs
JmaxRef = 58.19
RdRef = 0.17
sigma = 0.45
Theta = 0.7
VcmaxRef = 44.56
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

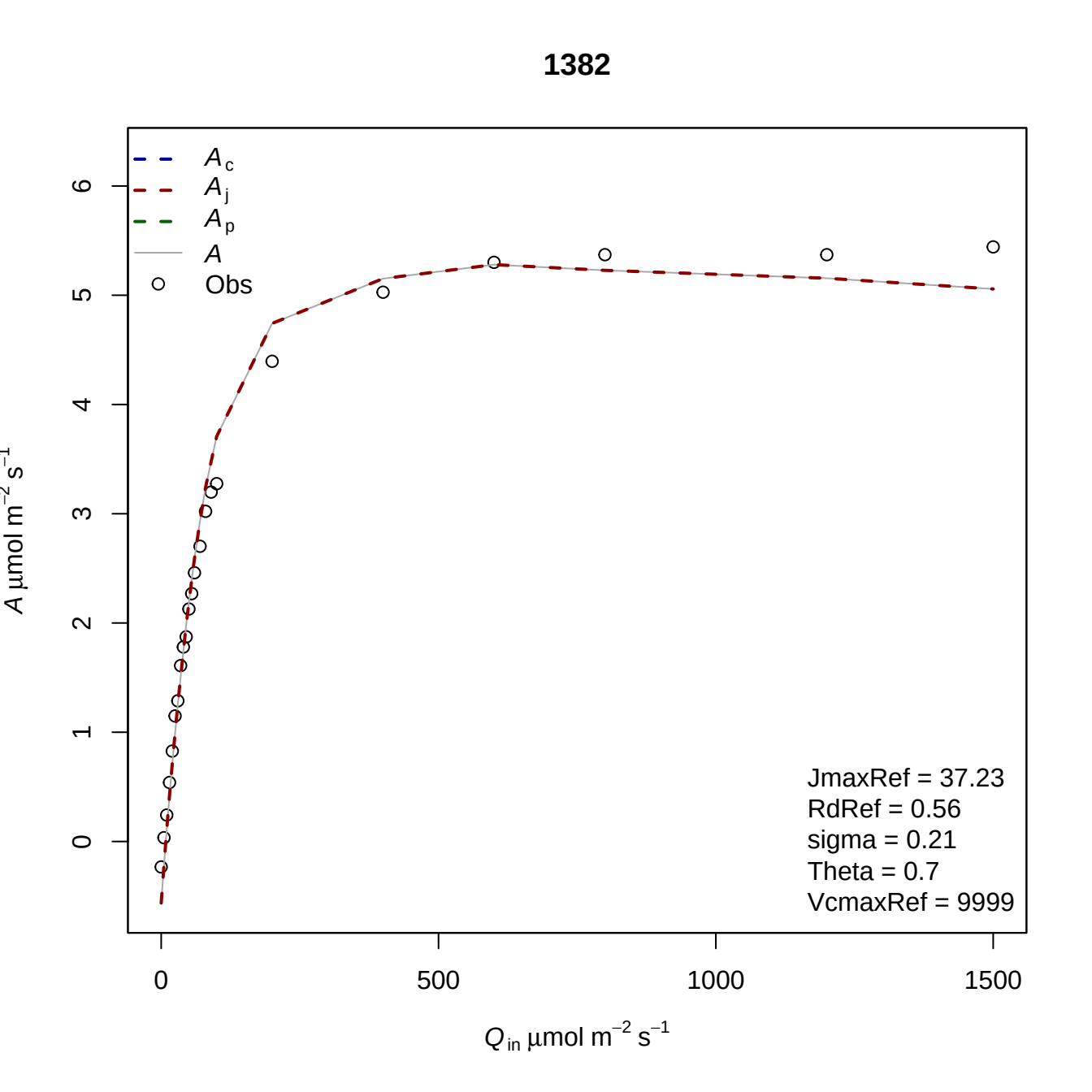

1377
A_c
A_j
A_p
A
Obs
JmaxRef = 31.61
RdRef = 0.93
sigma = 0.27
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

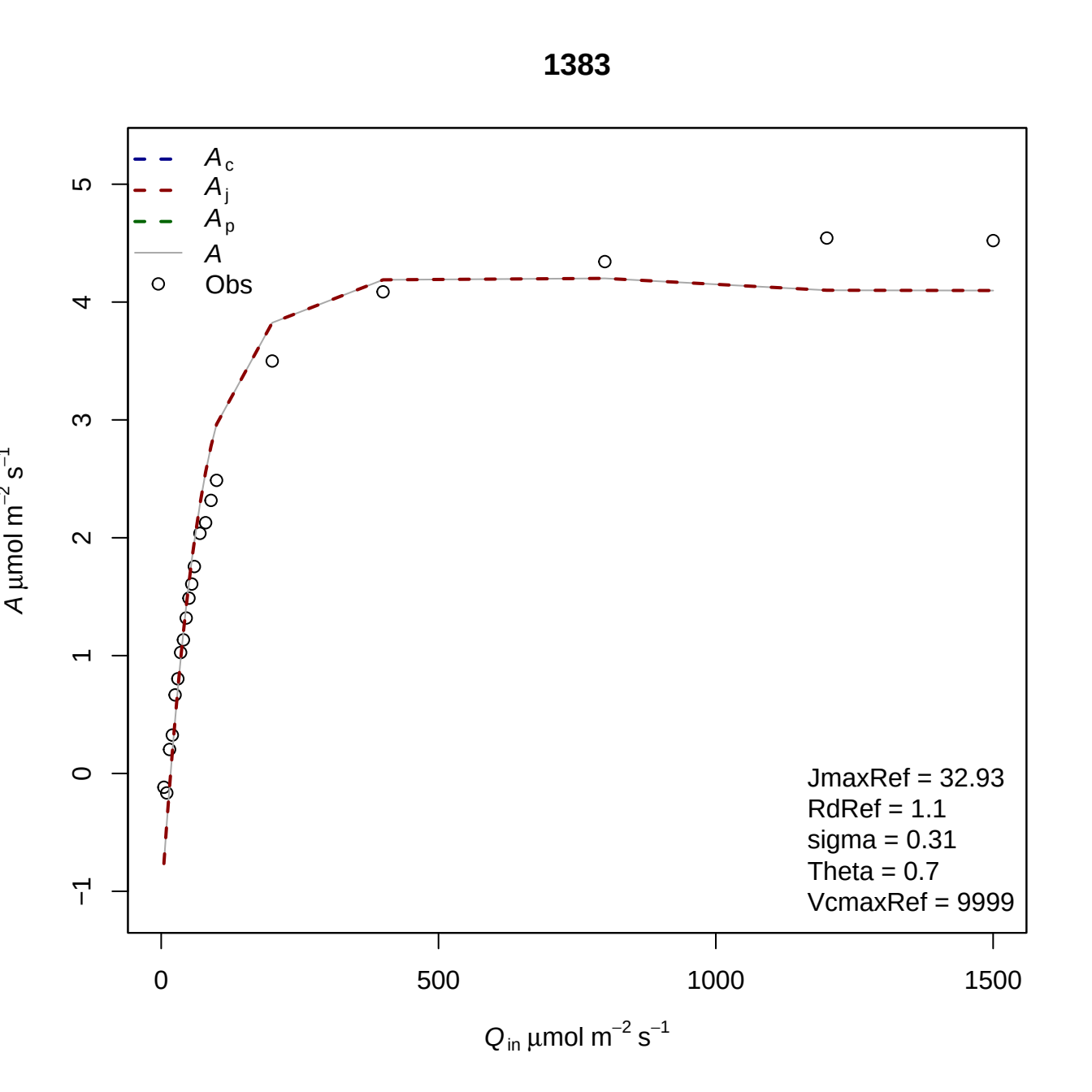

1378
A_c
A_j
A_p
A
Obs
JmaxRef = 18.44
RdRef = 1.18
sigma = 0.41
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
500
1000
1500

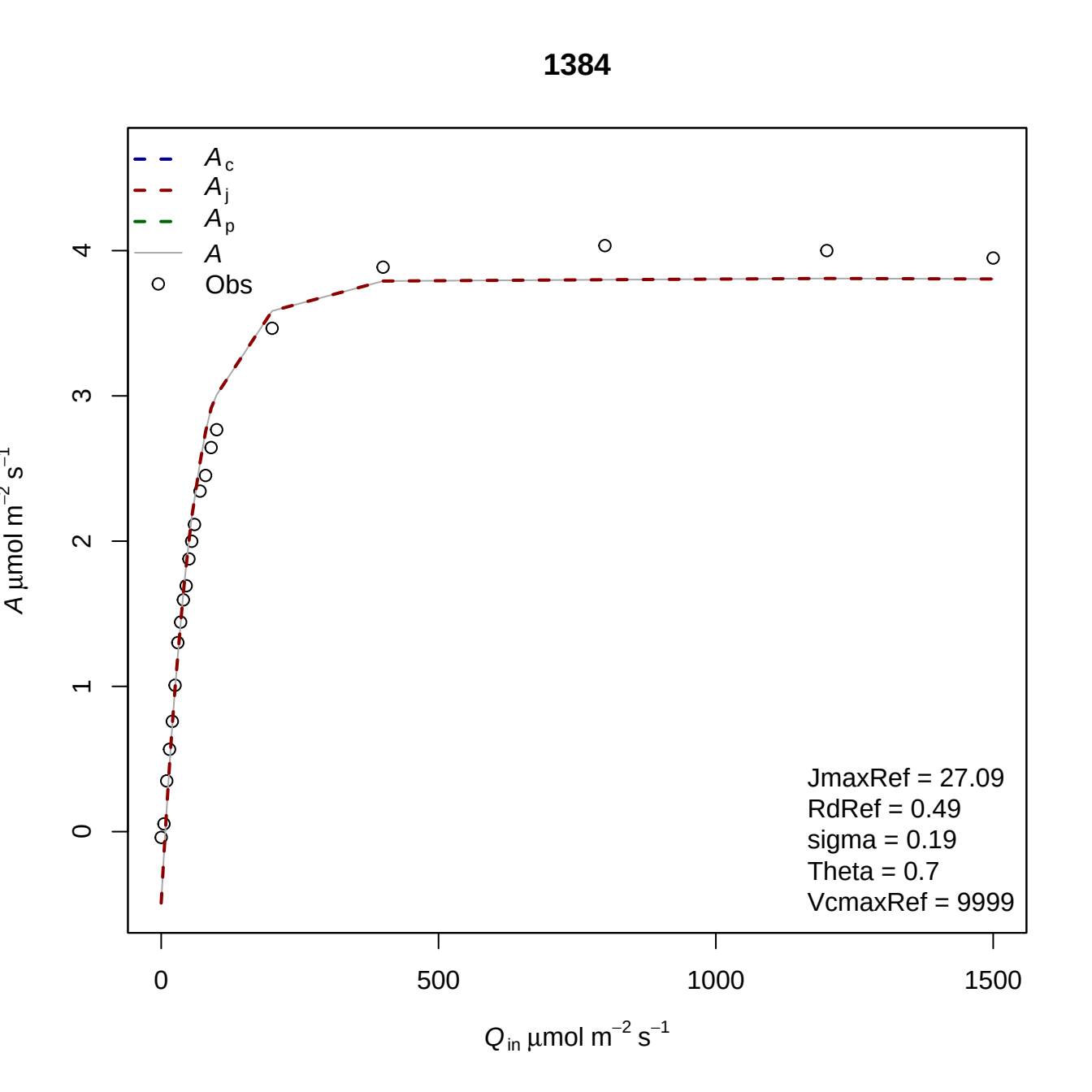

1379
A_c
A_j
A_p
A
Obs
JmaxRef = 14.9
RdRef = 0.44
sigma = 0.12
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
0
200
400
600
800
1000
1200

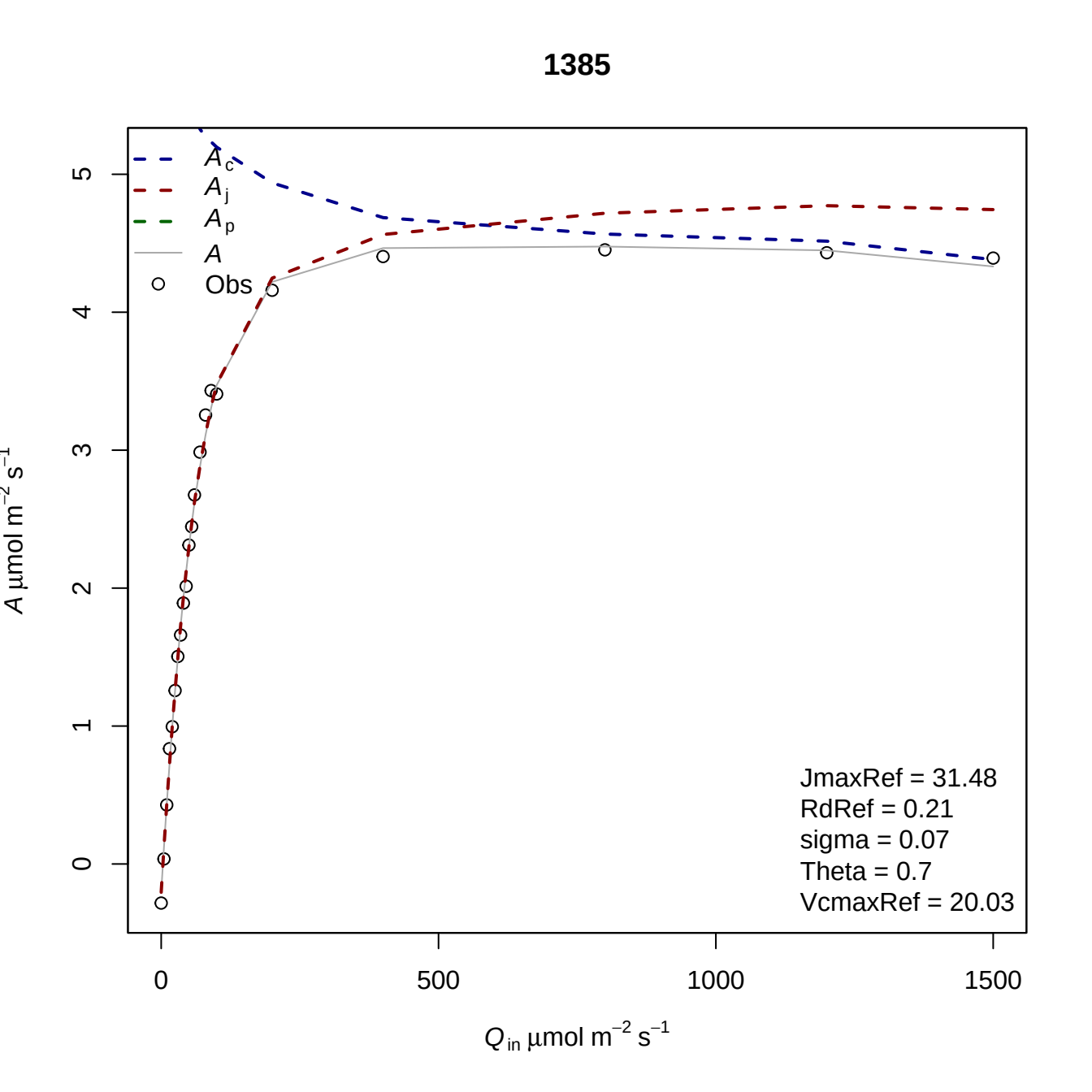

1380
A_c
A_j
A_p
A
Obs
JmaxRef = 51.75
RdRef = 0.57
sigma = 0.13
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

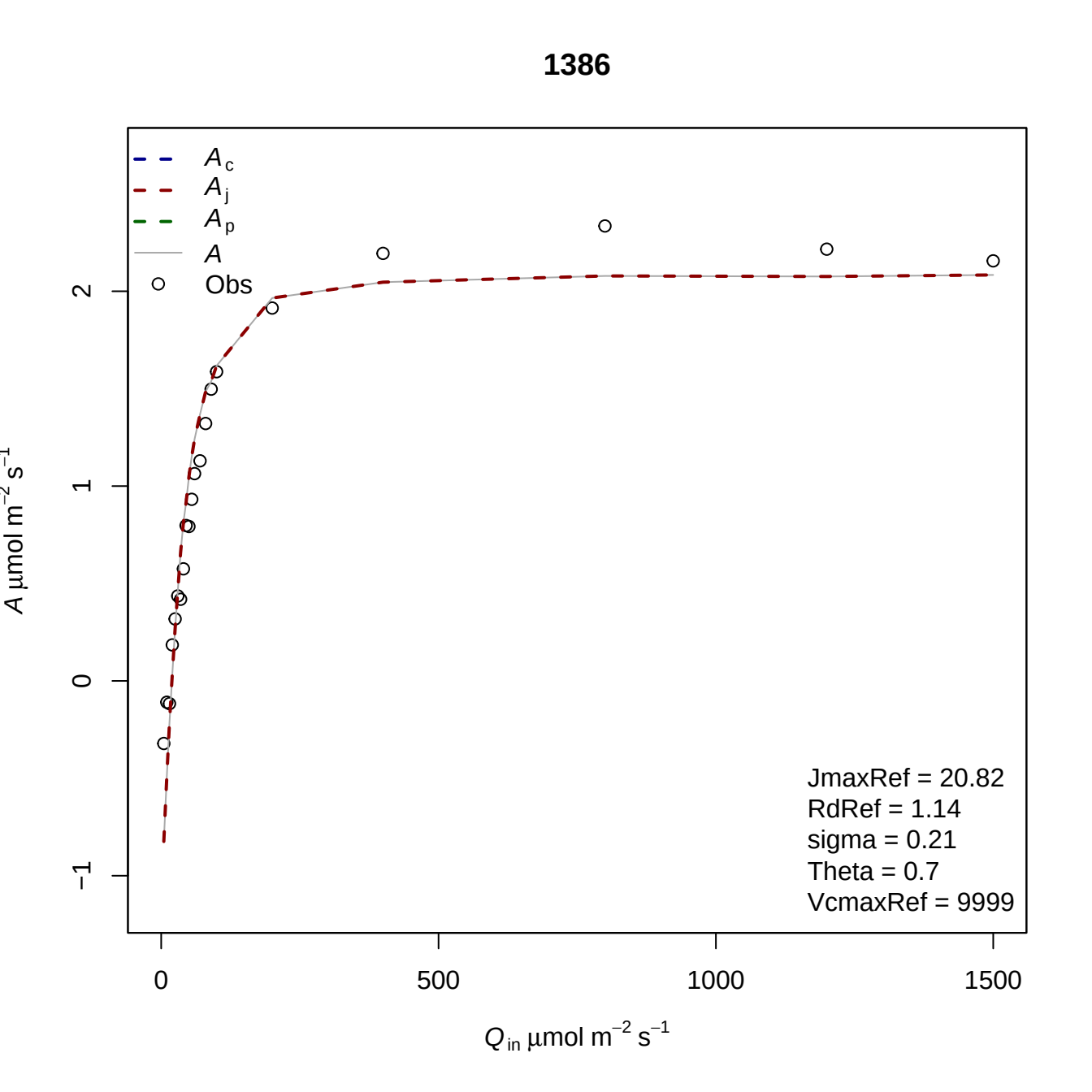

1381
A_c
A_j
A_p
A
Obs
JmaxRef = 20.9
RdRef = 0.84
sigma = 0.25
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

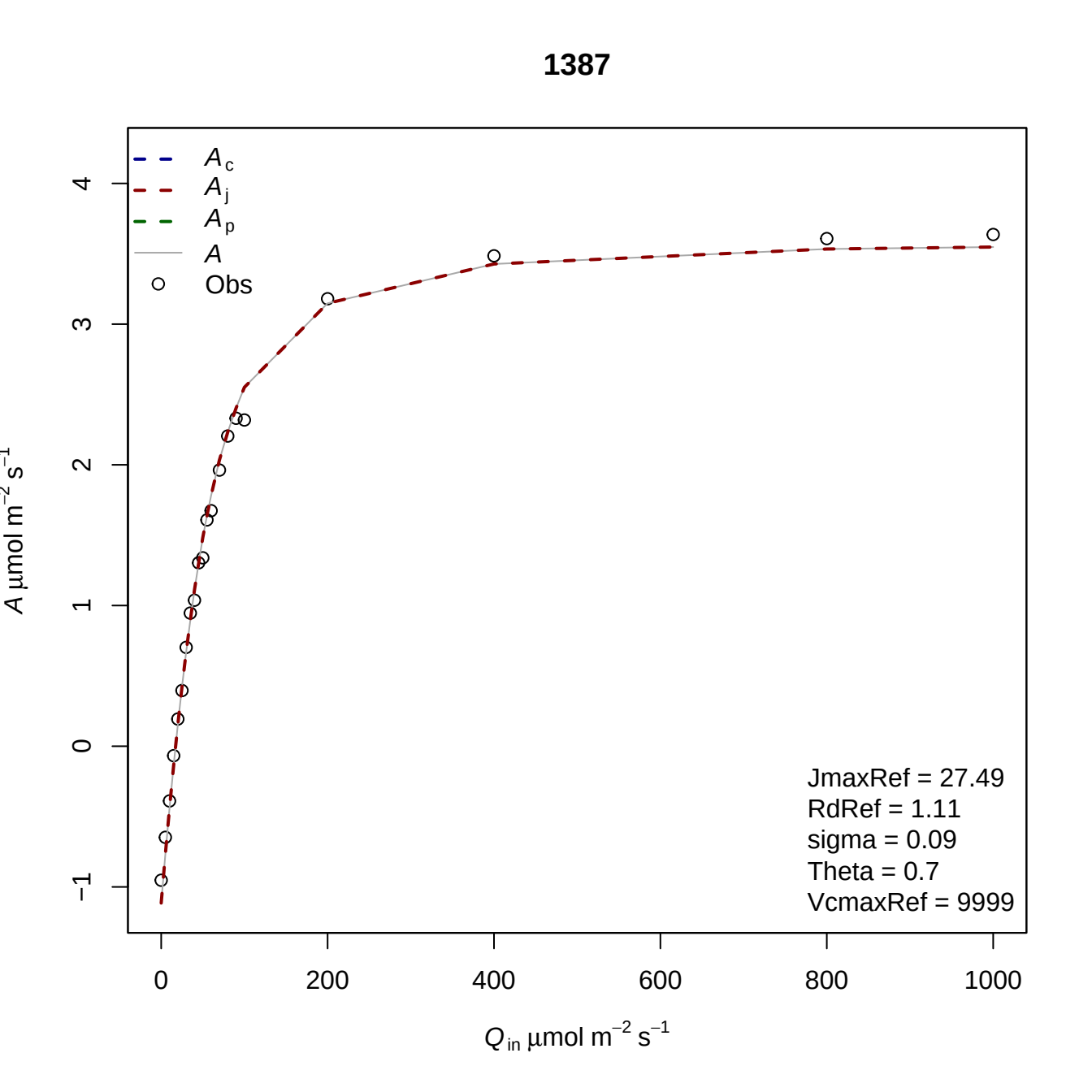

1382
A_c
A_j
A_p
A
Obs
JmaxRef = 37.23
RdRef = 0.56
sigma = 0.21
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

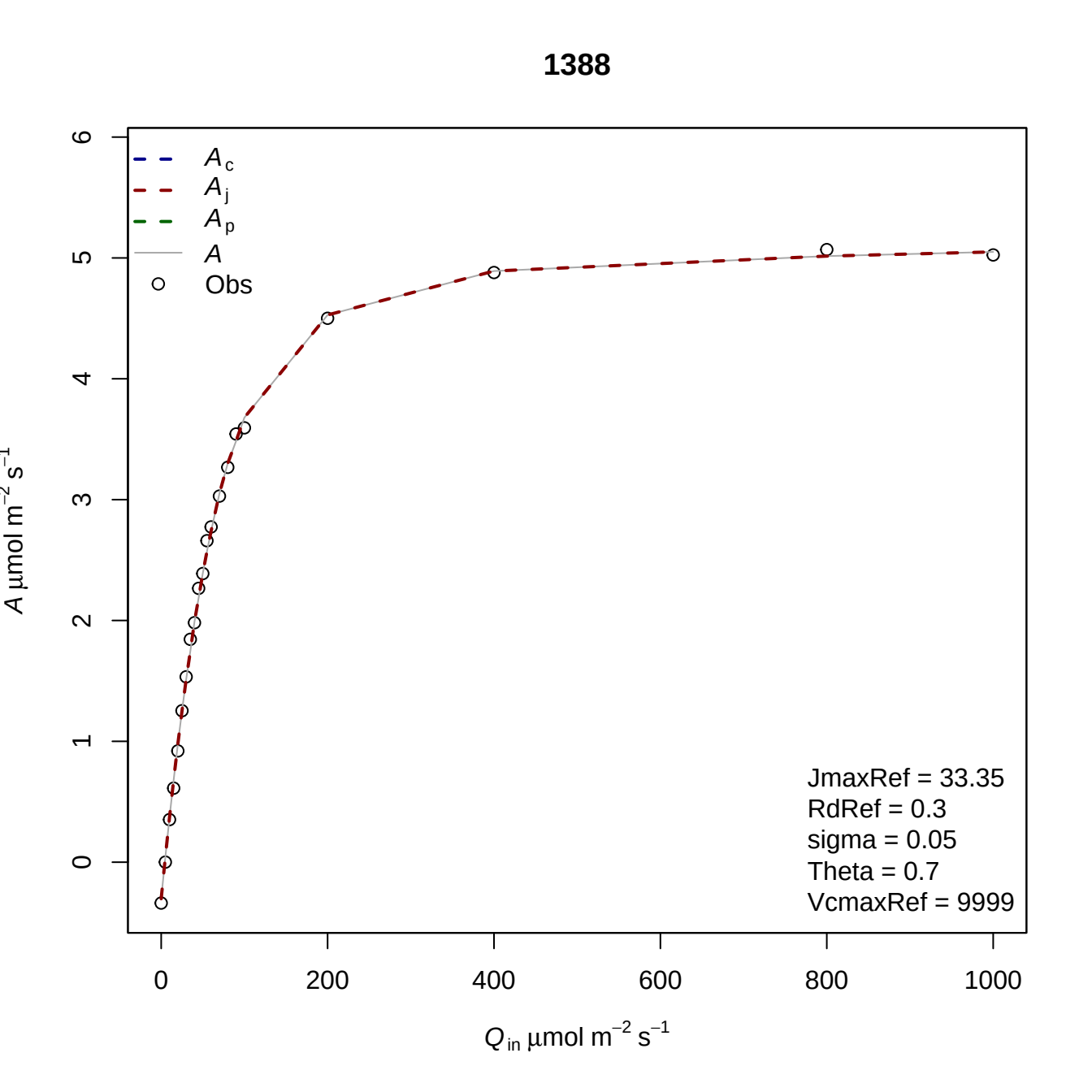

1383
A_c
A_j
A_p
A
Obs
JmaxRef = 32.93
RdRef = 1.1
sigma = 0.31
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

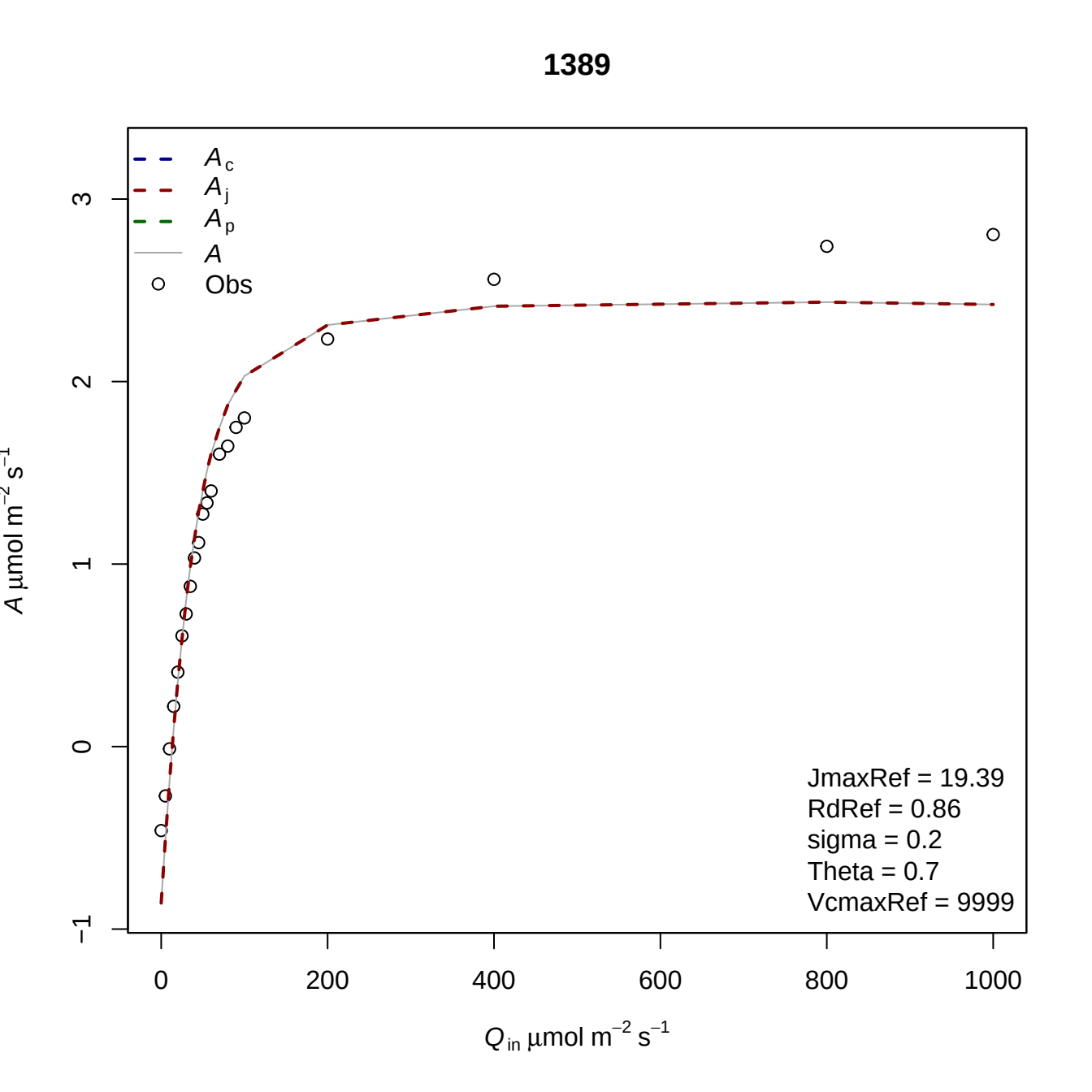

1384
A_c
A_j
A_p
A
Obs
JmaxRef = 27.09
RdRef = 0.49
sigma = 0.19
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500

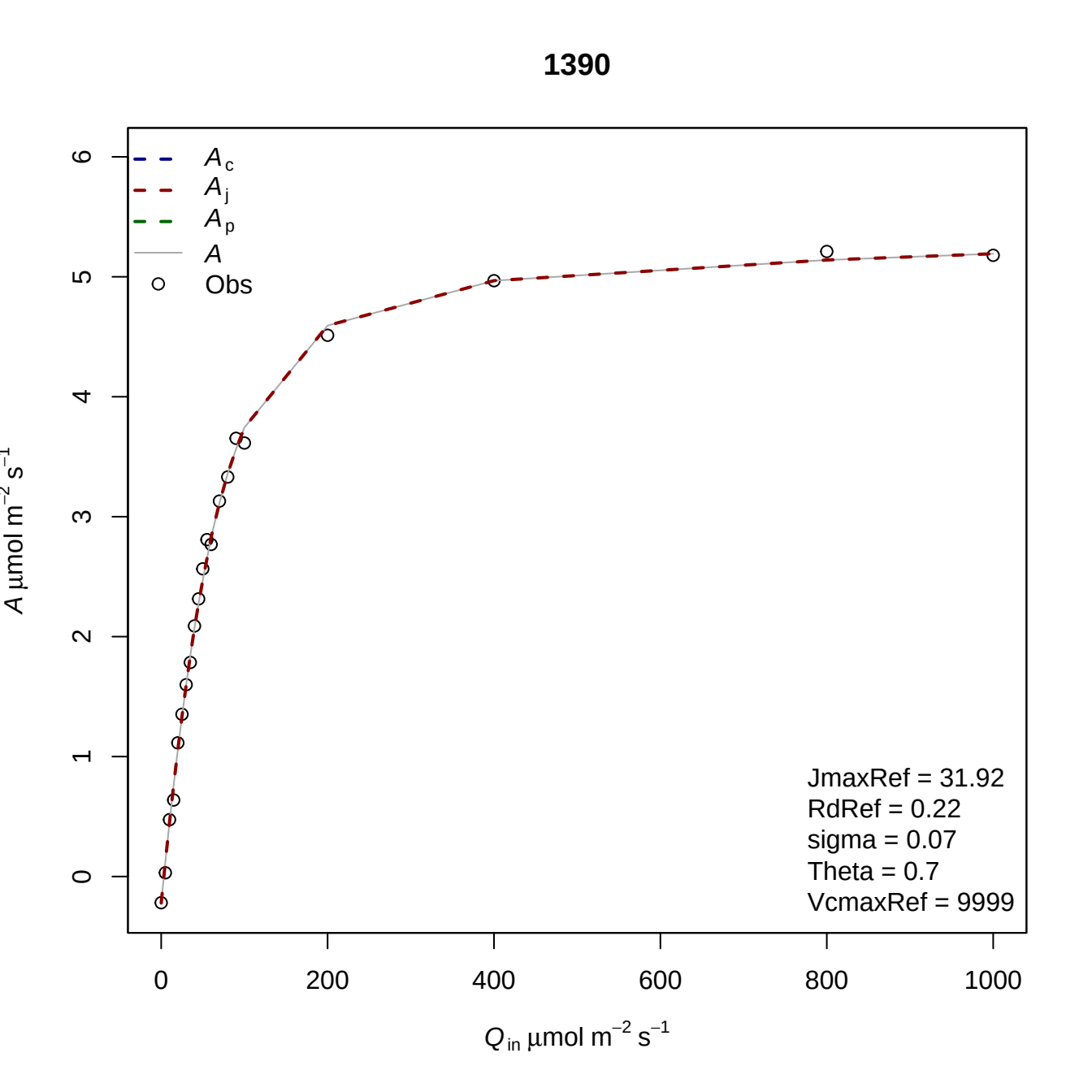

1385
A_c
A_j
A_p
A
Obs
JmaxRef = 31.48
RdRef = 0.21
sigma = 0.07
Theta = 0.7
VcmaxRef = 20.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

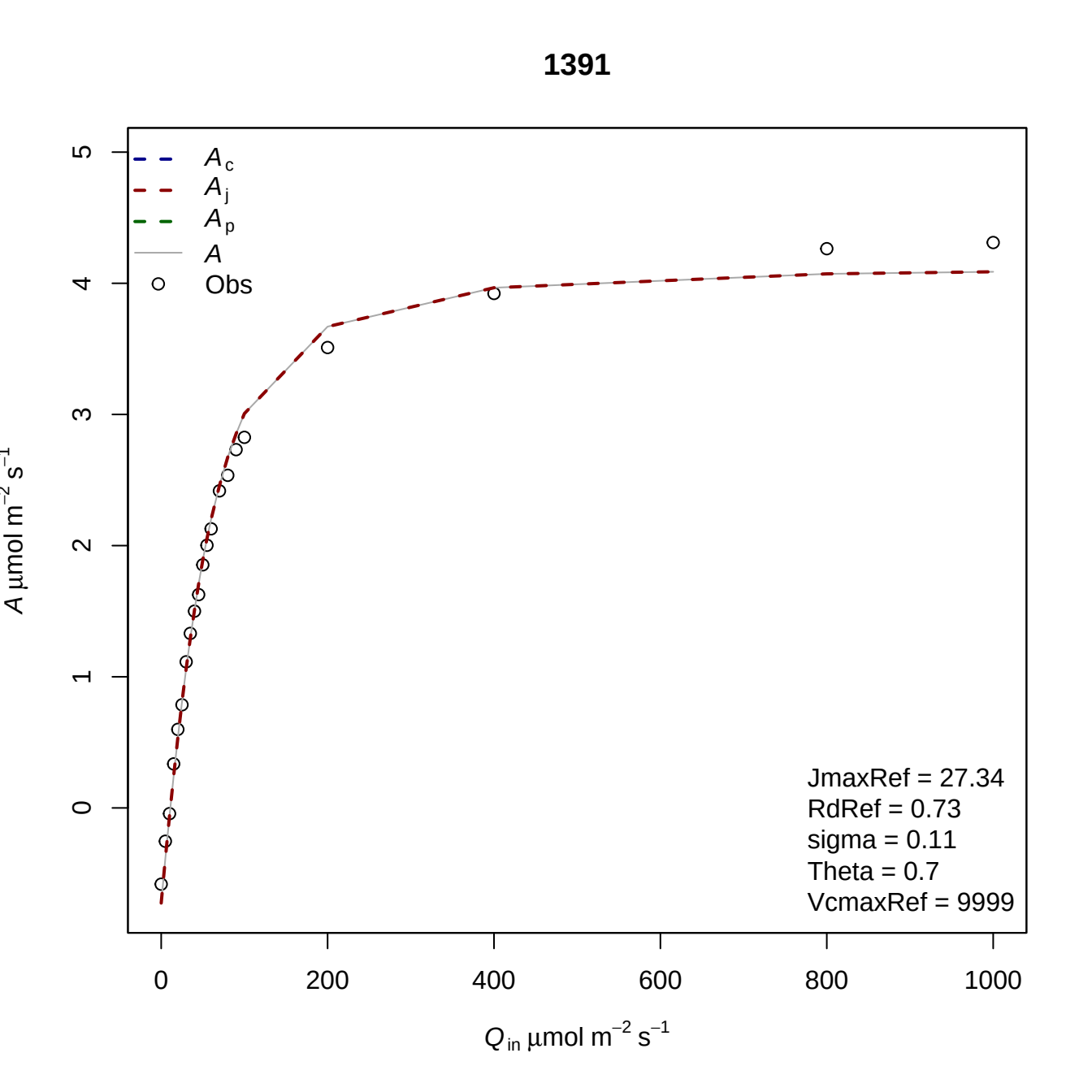

1386
A_c
A_j
A_p
A
Obs
JmaxRef = 20.82
RdRef = 1.14
sigma = 0.21
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
500
1000
1500

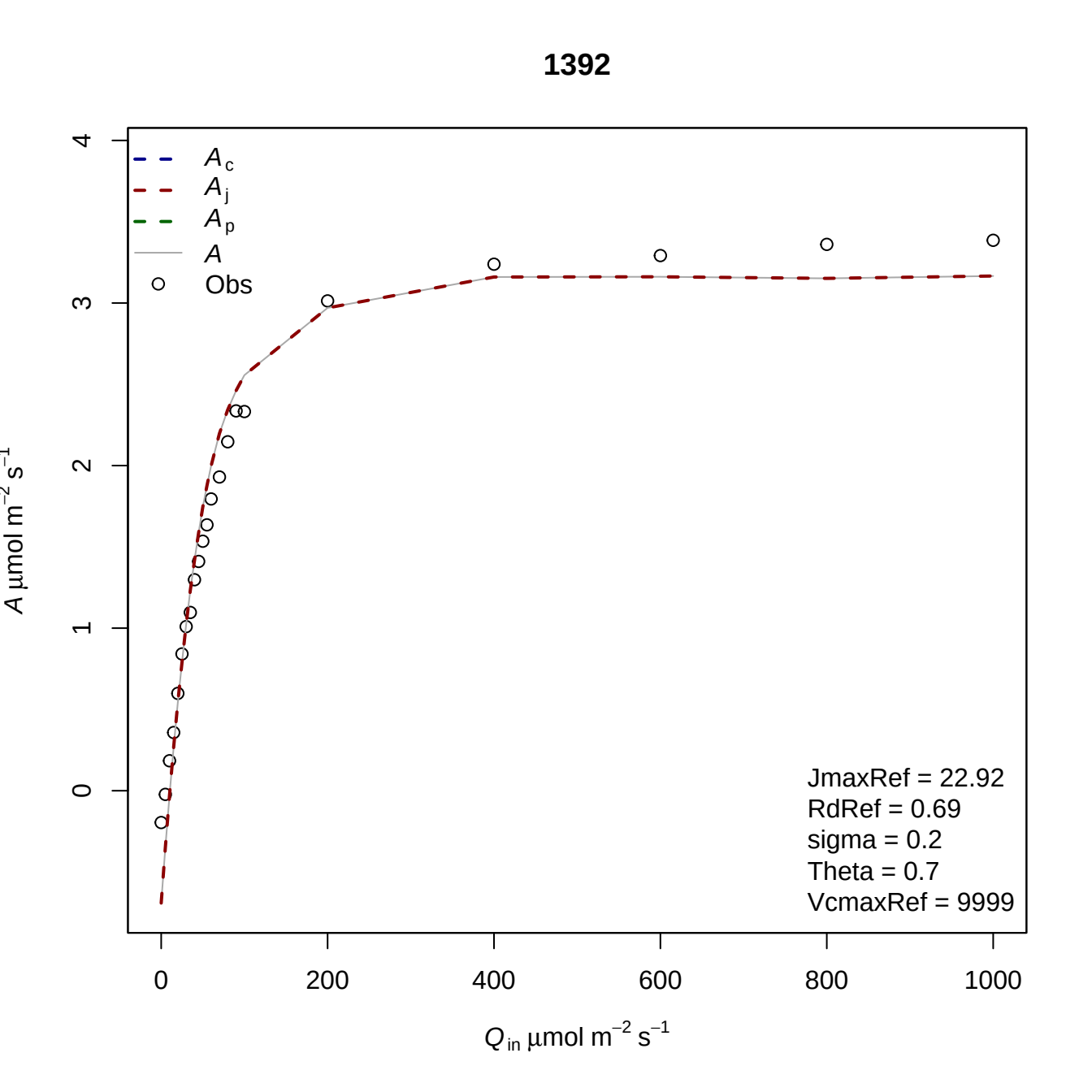

1387
A_c
A_j
A_p
A
Obs
JmaxRef = 27.49
RdRef = 1.11
sigma = 0.09
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
200
400
600
800
1000

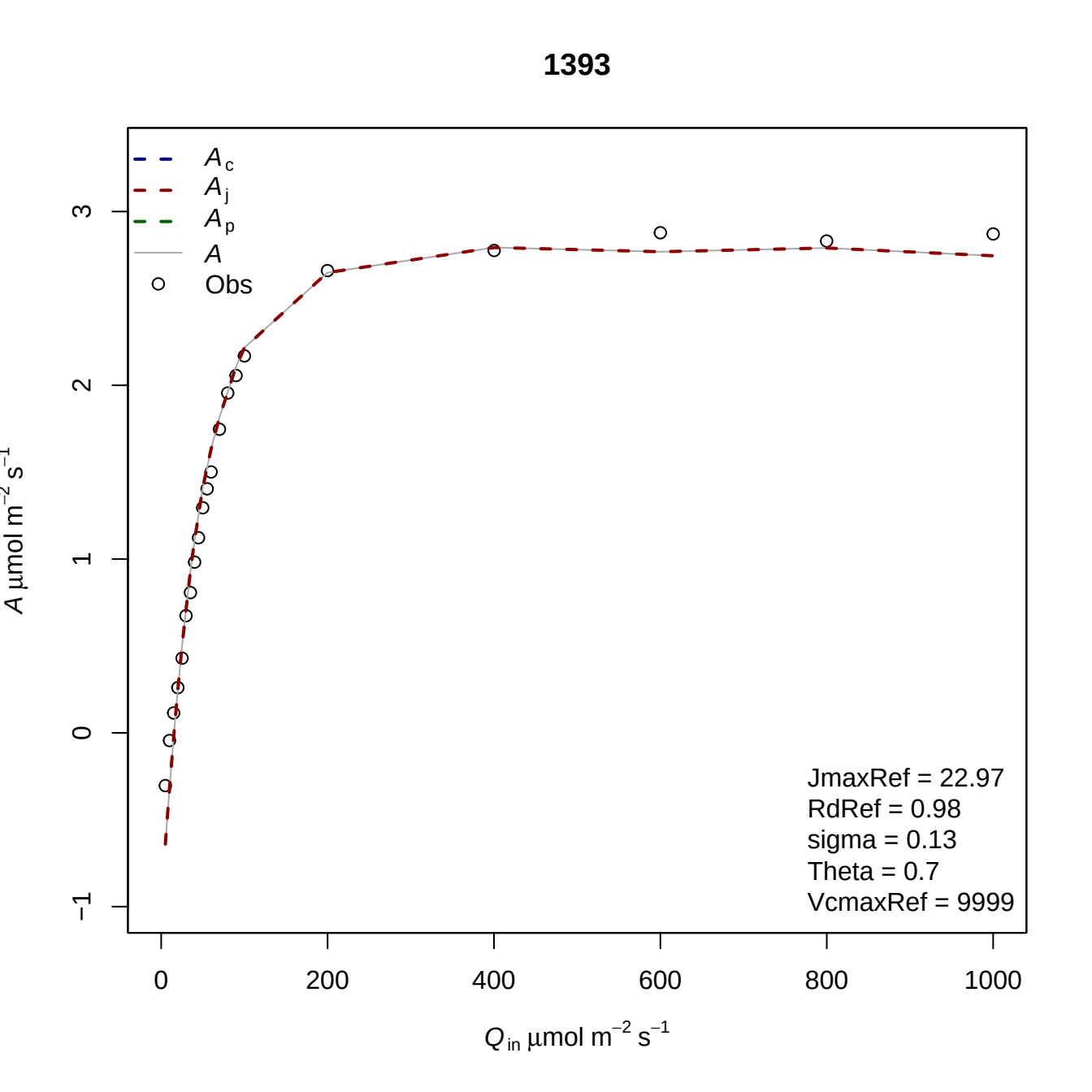

1388
A_c
A_j
A_p
A
Obs
JmaxRef = 33.35
RdRef = 0.3
sigma = 0.05
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
200
400
600
800
1000

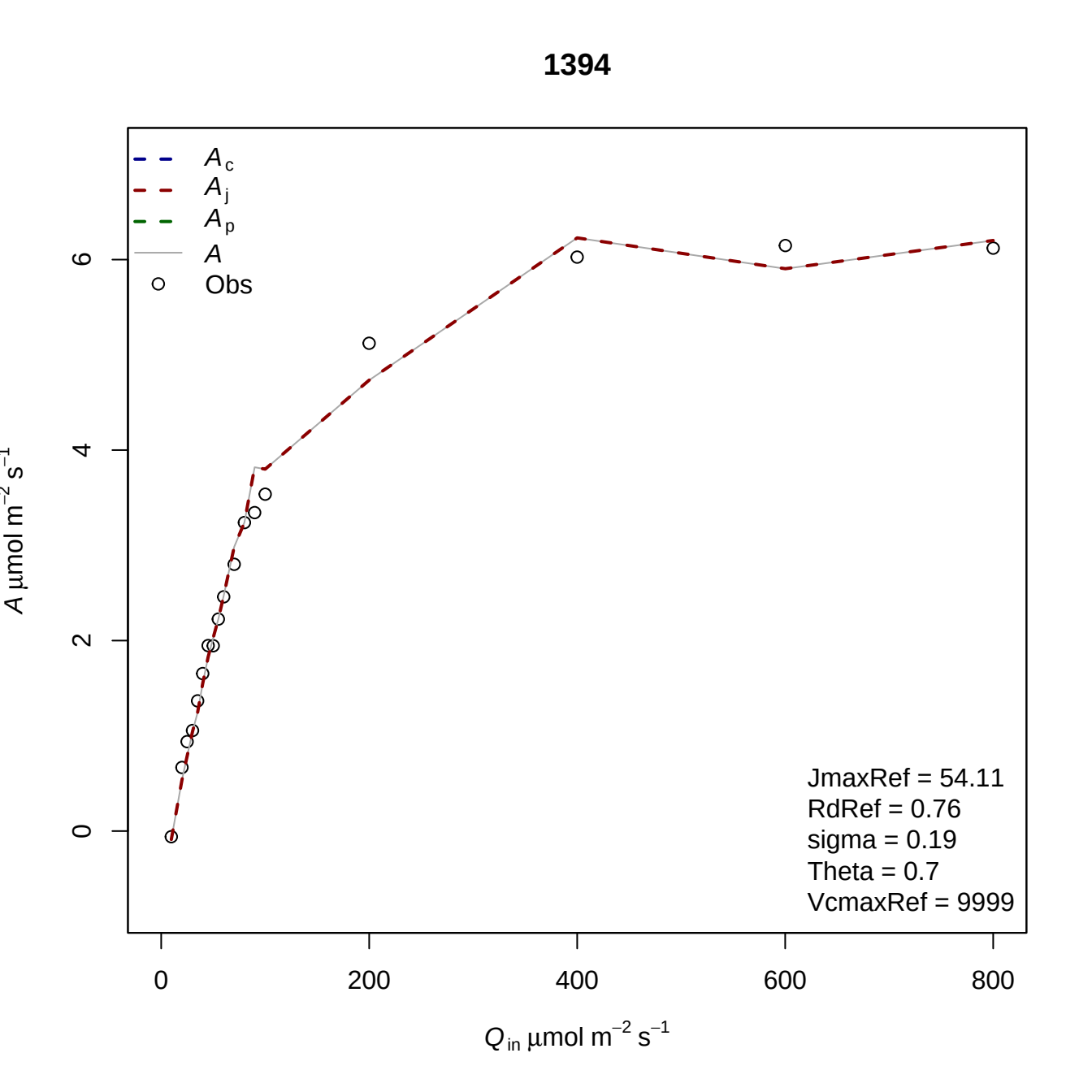

1389
A_c
A_j
A_p
A
Obs
JmaxRef = 19.39
RdRef = 0.86
sigma = 0.2
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800
1000

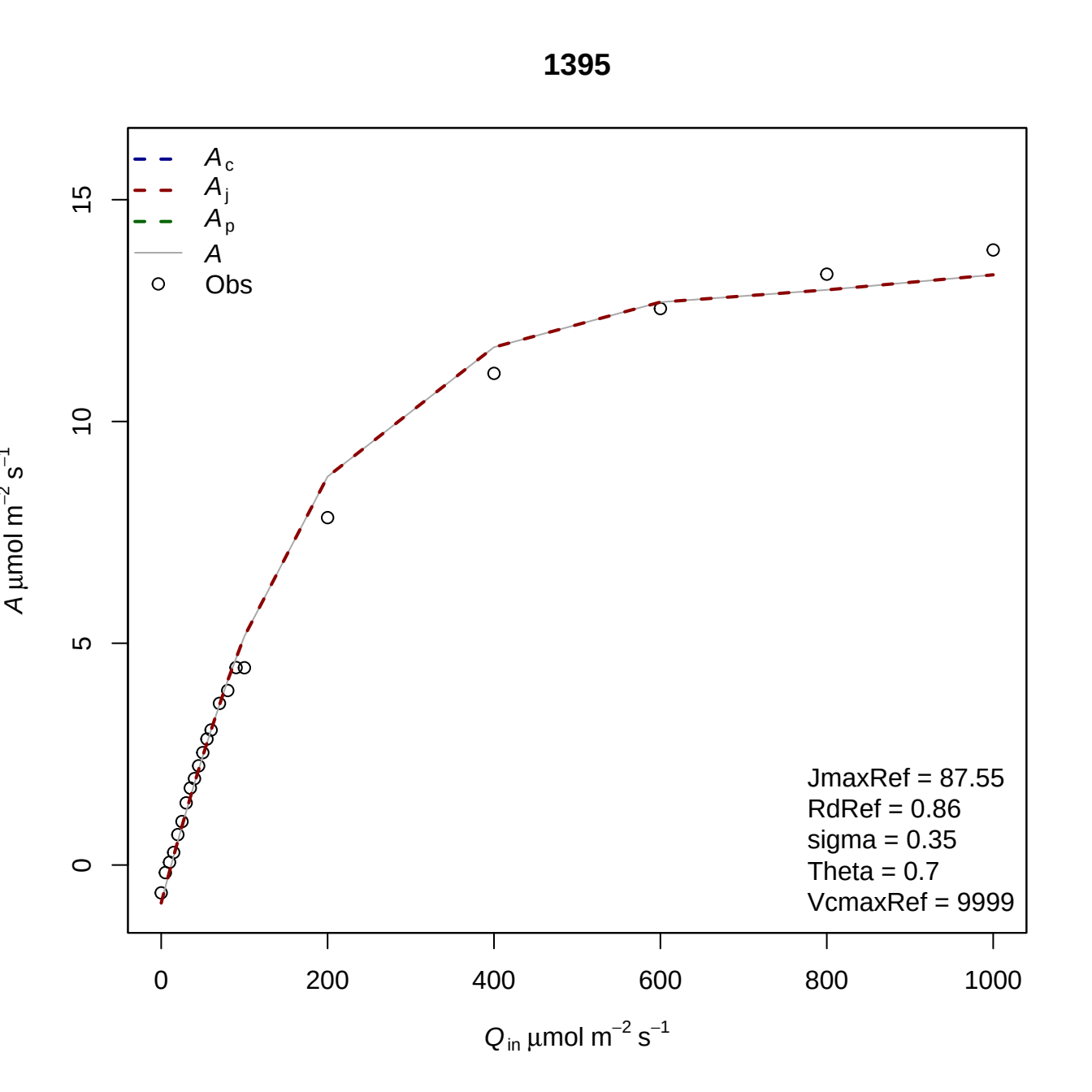

1390
A_c
A_j
A_p
A
Obs
JmaxRef = 31.92
RdRef = 0.22
sigma = 0.07
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
200
400
600
800
1000

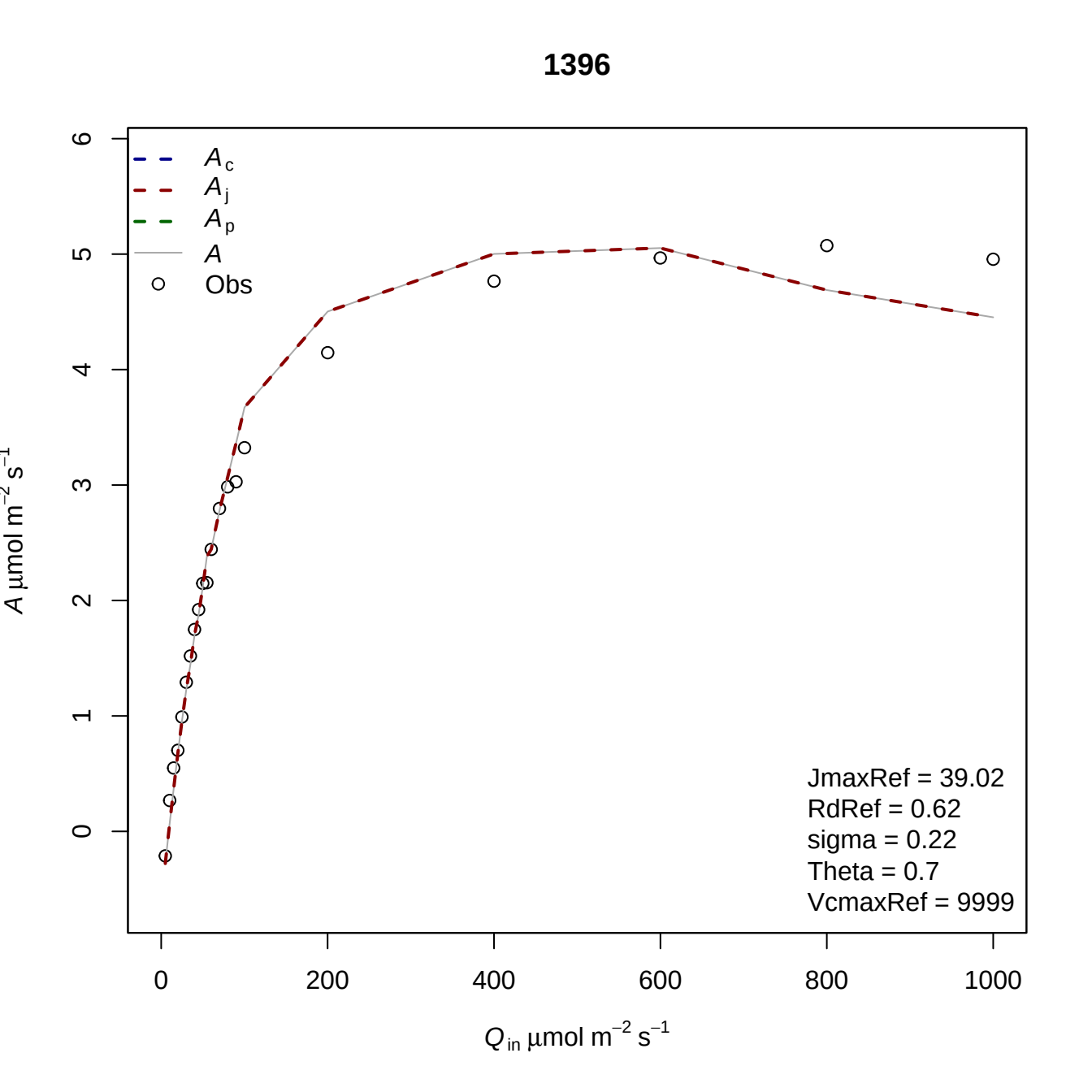

1391
A_c
A_j
A_p
A
Obs
JmaxRef = 27.34
RdRef = 0.73
sigma = 0.11
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
200
400
600
800
1000

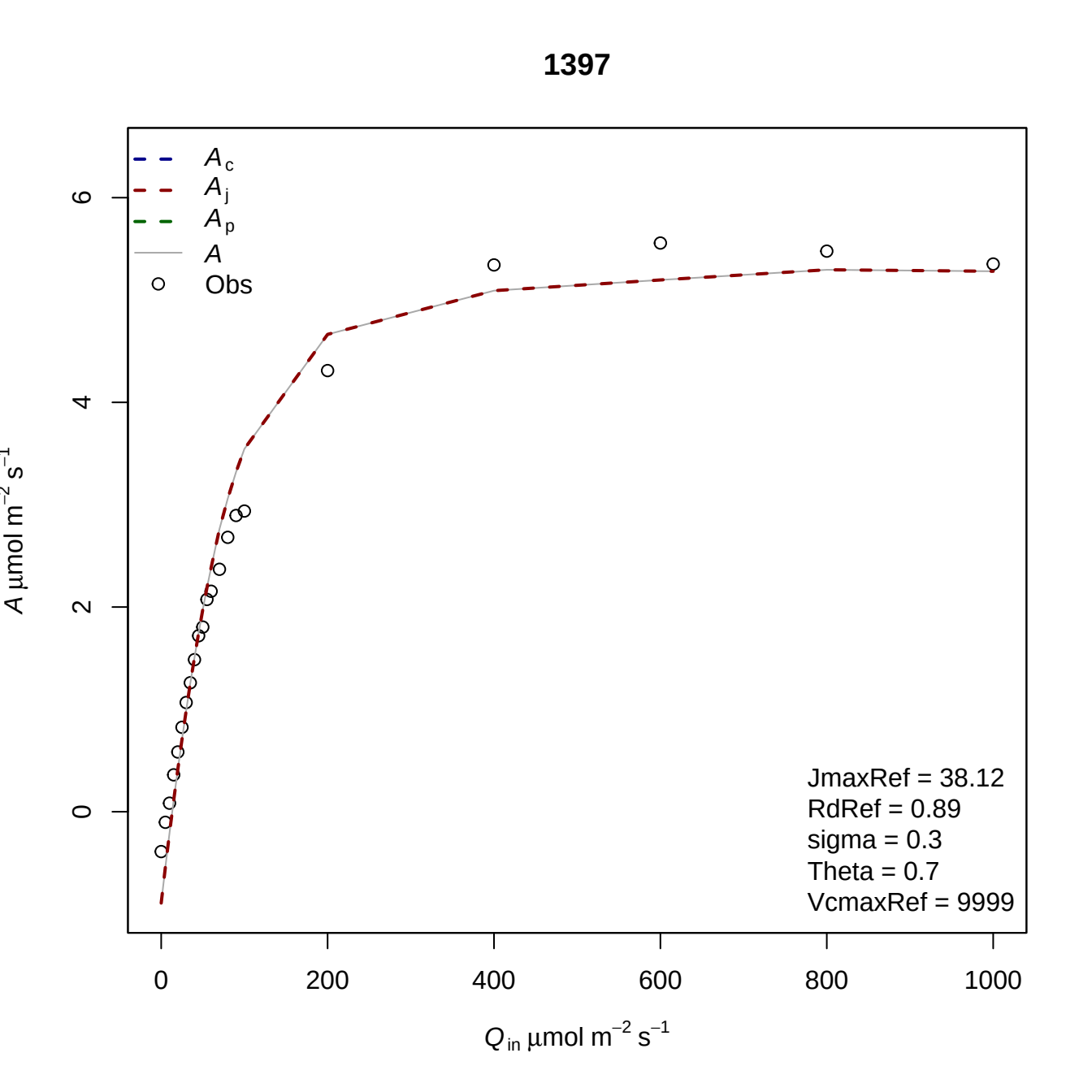

1392
A_c
A_j
A_p
A
Obs
JmaxRef = 22.92
RdRef = 0.69
sigma = 0.2
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
200
400
600
800
1000

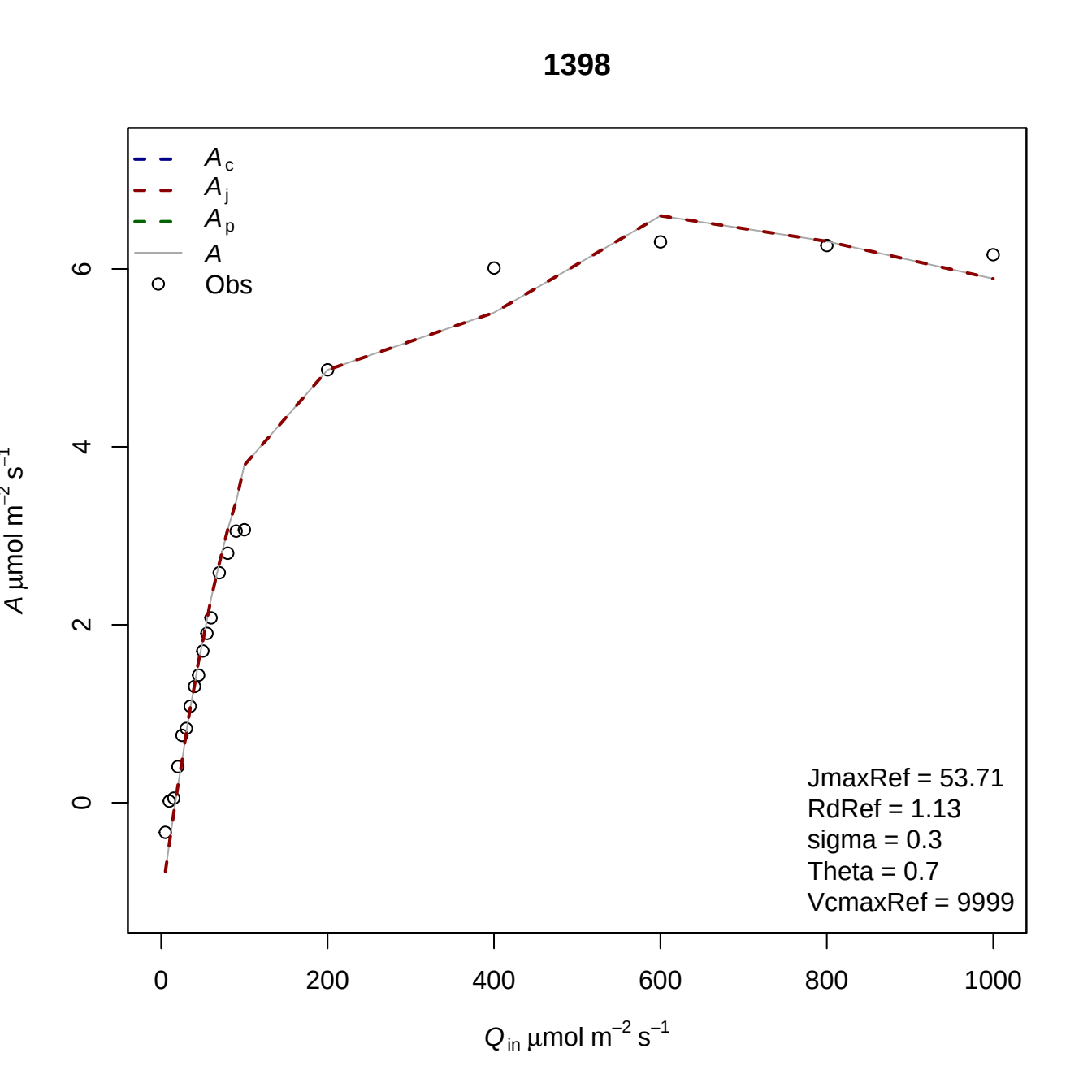

1393
A_c
A_j
A_p
A
Obs
JmaxRef = 22.97
RdRef = 0.98
sigma = 0.13
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800
1000

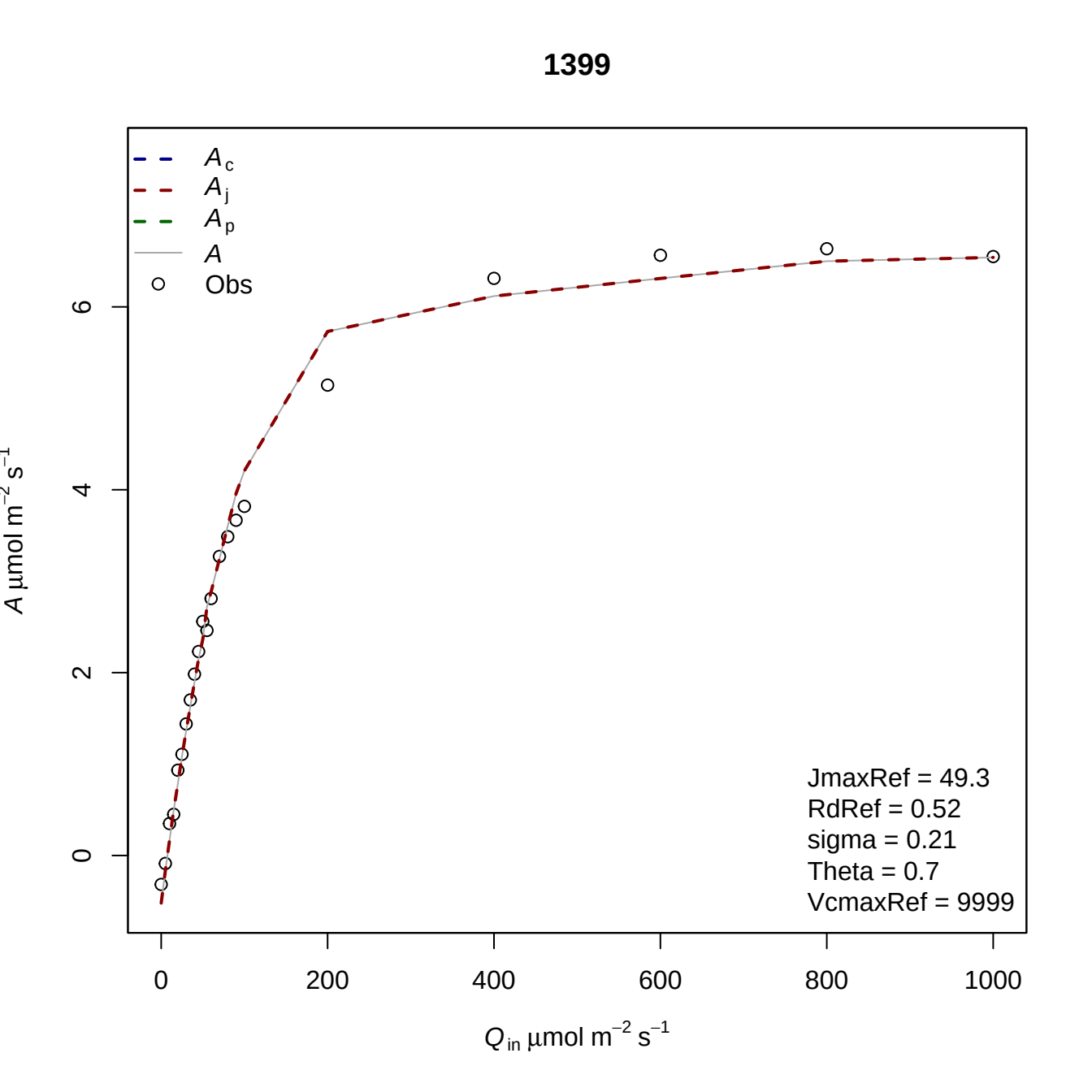

1394
A_c
A_j
A_p
A
Obs
JmaxRef = 54.11
RdRef = 0.76
sigma = 0.19
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
200
400
600
800

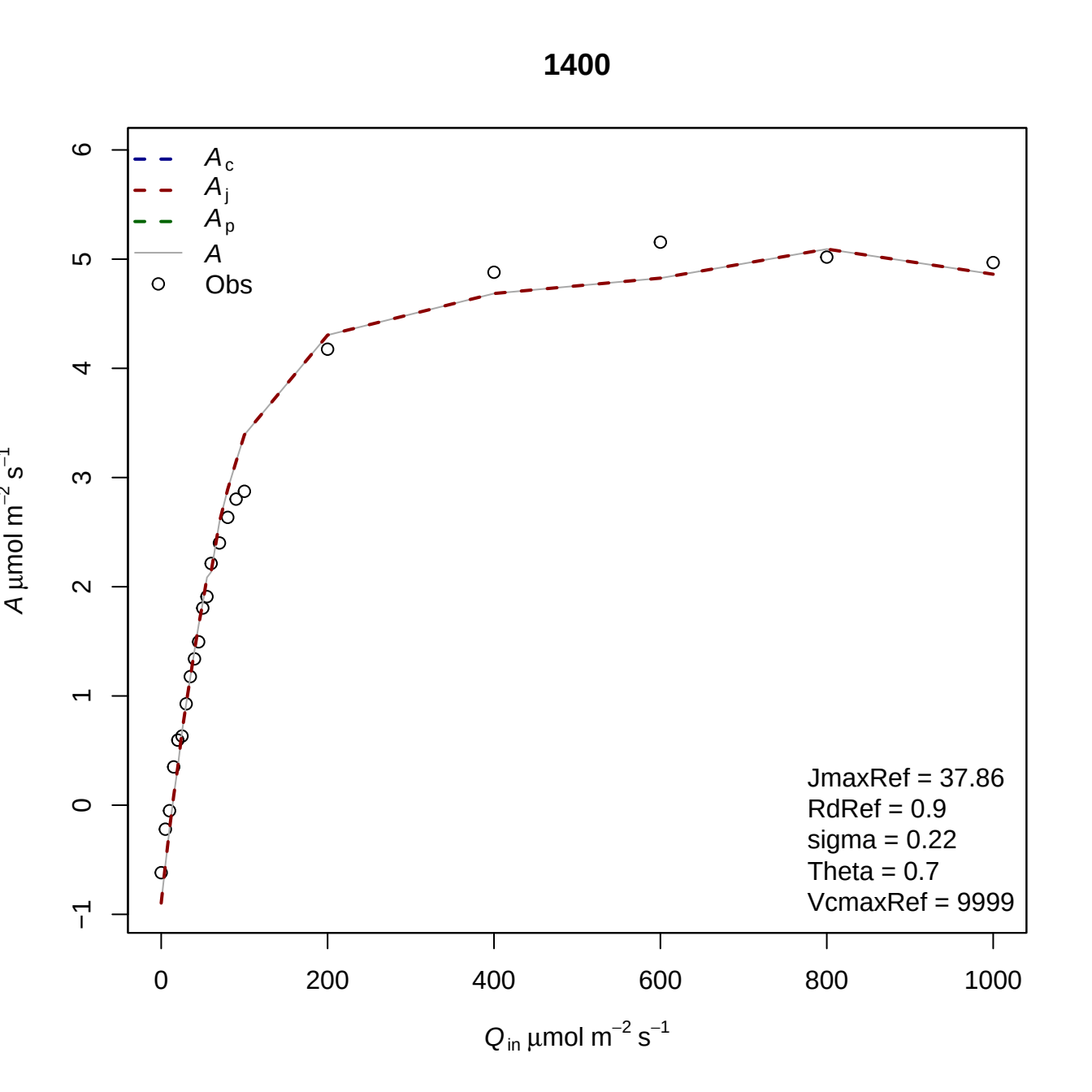

1395
A_c
A_j
A_p
A
Obs
JmaxRef = 87.55
RdRef = 0.86
sigma = 0.35
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
200
400
600
800
1000

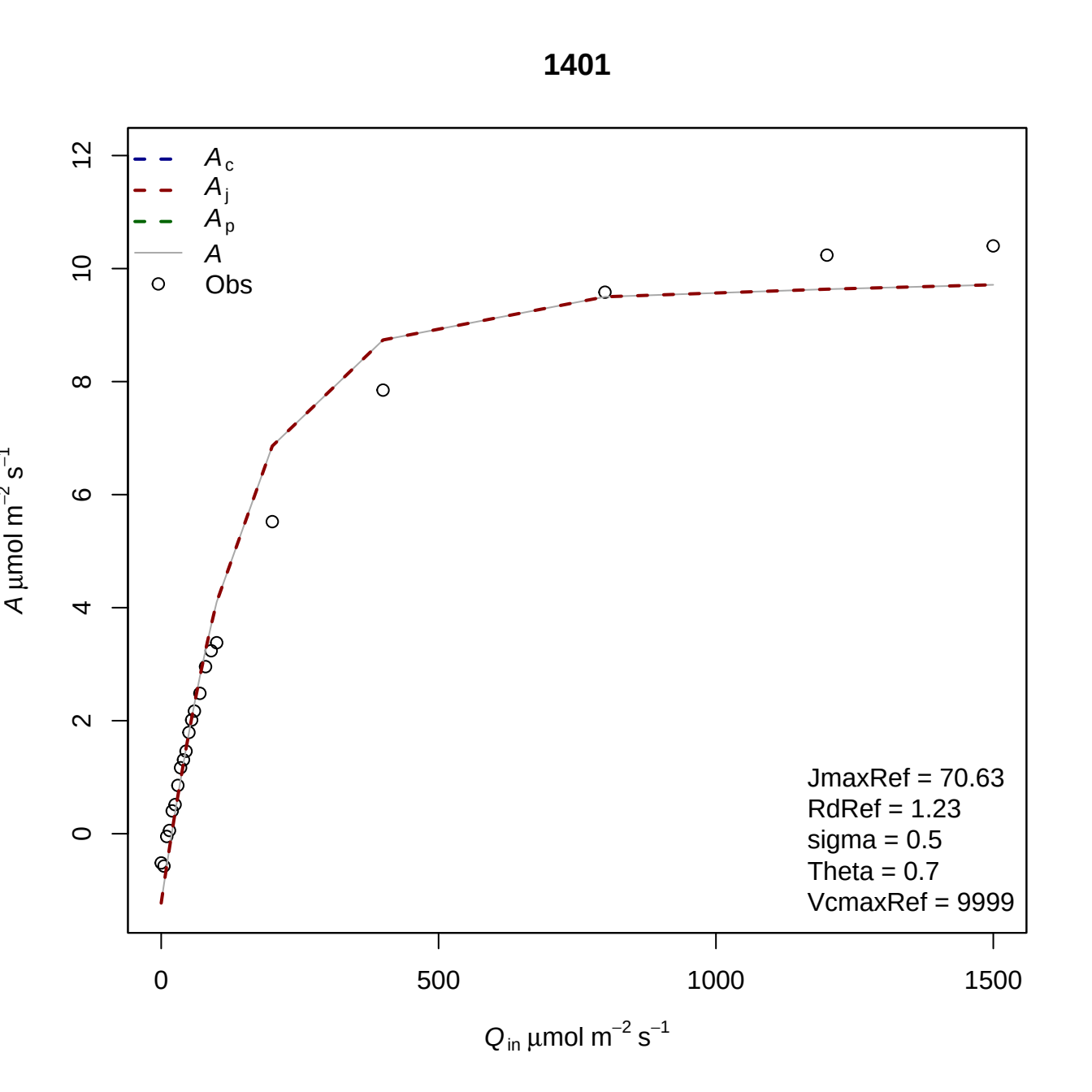

1396
A_c
A_j
A_p
A
Obs
JmaxRef = 39.02
RdRef = 0.62
sigma = 0.22
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
200
400
600
800
1000

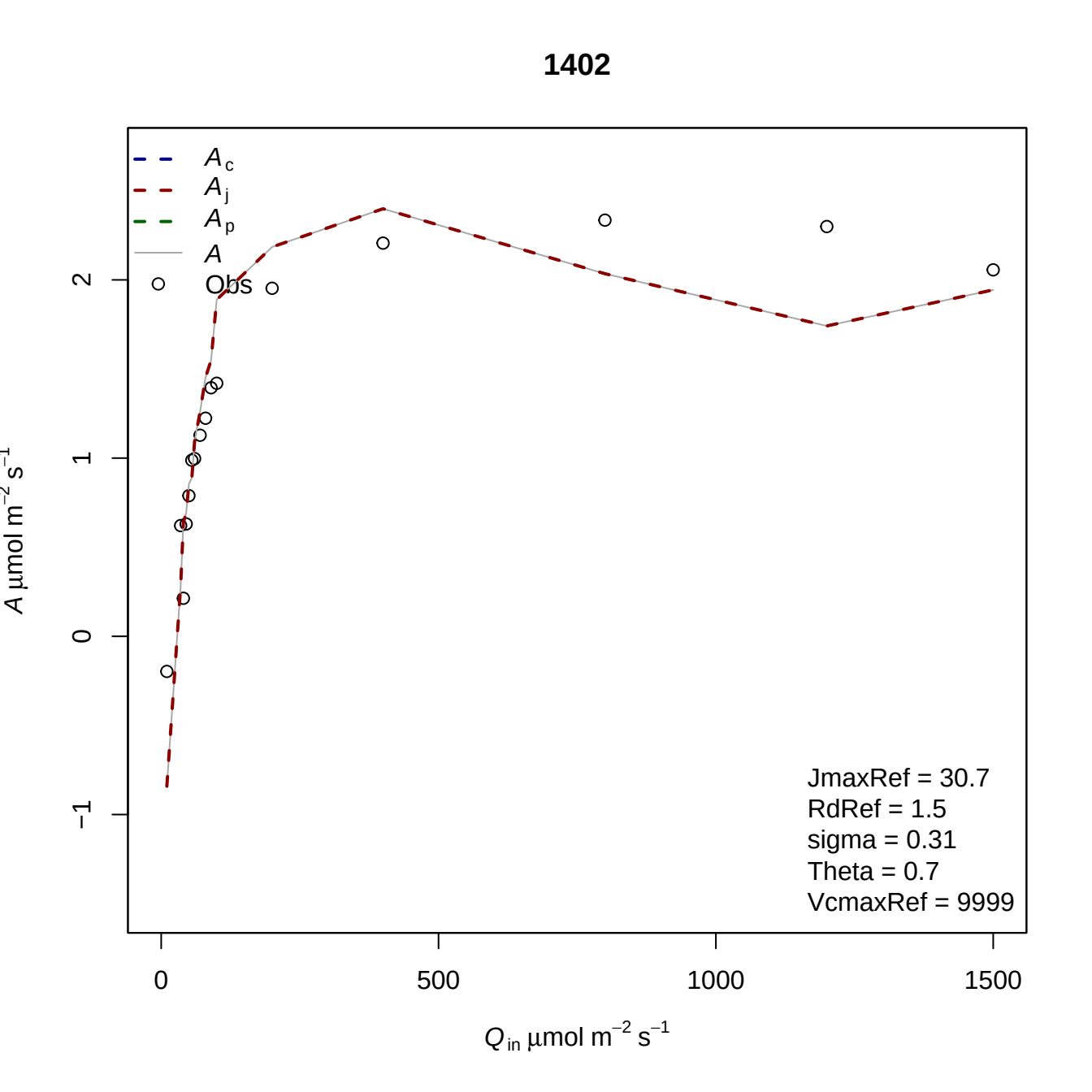

1397
A_c
A_j
A_p
A
Obs
JmaxRef = 38.12
RdRef = 0.89
sigma = 0.3
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
200
400
600
800
1000

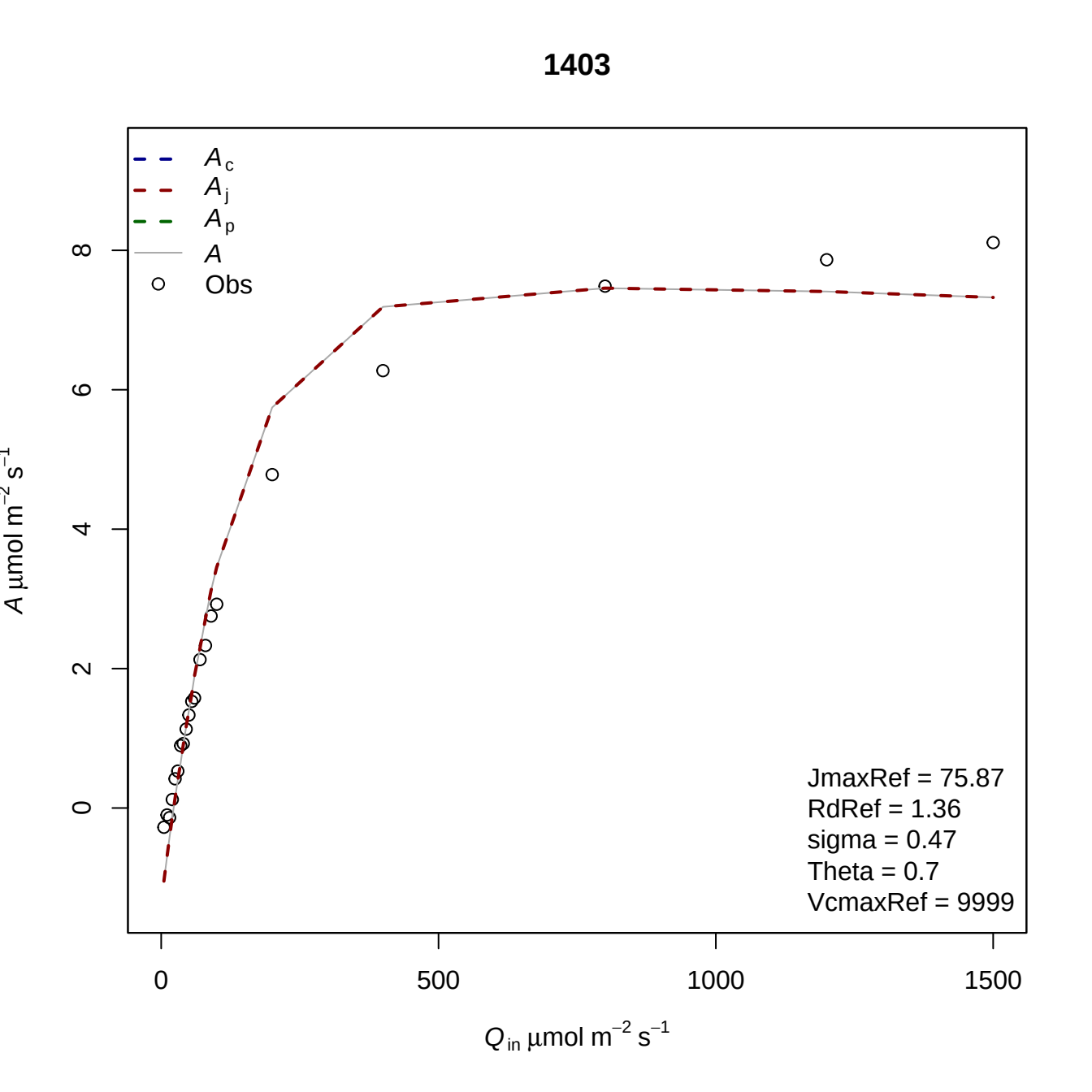

1398
A_c
A_j
A_p
A
Obs
JmaxRef = 53.71
RdRef = 1.13
sigma = 0.3
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
200
400
600
800
1000

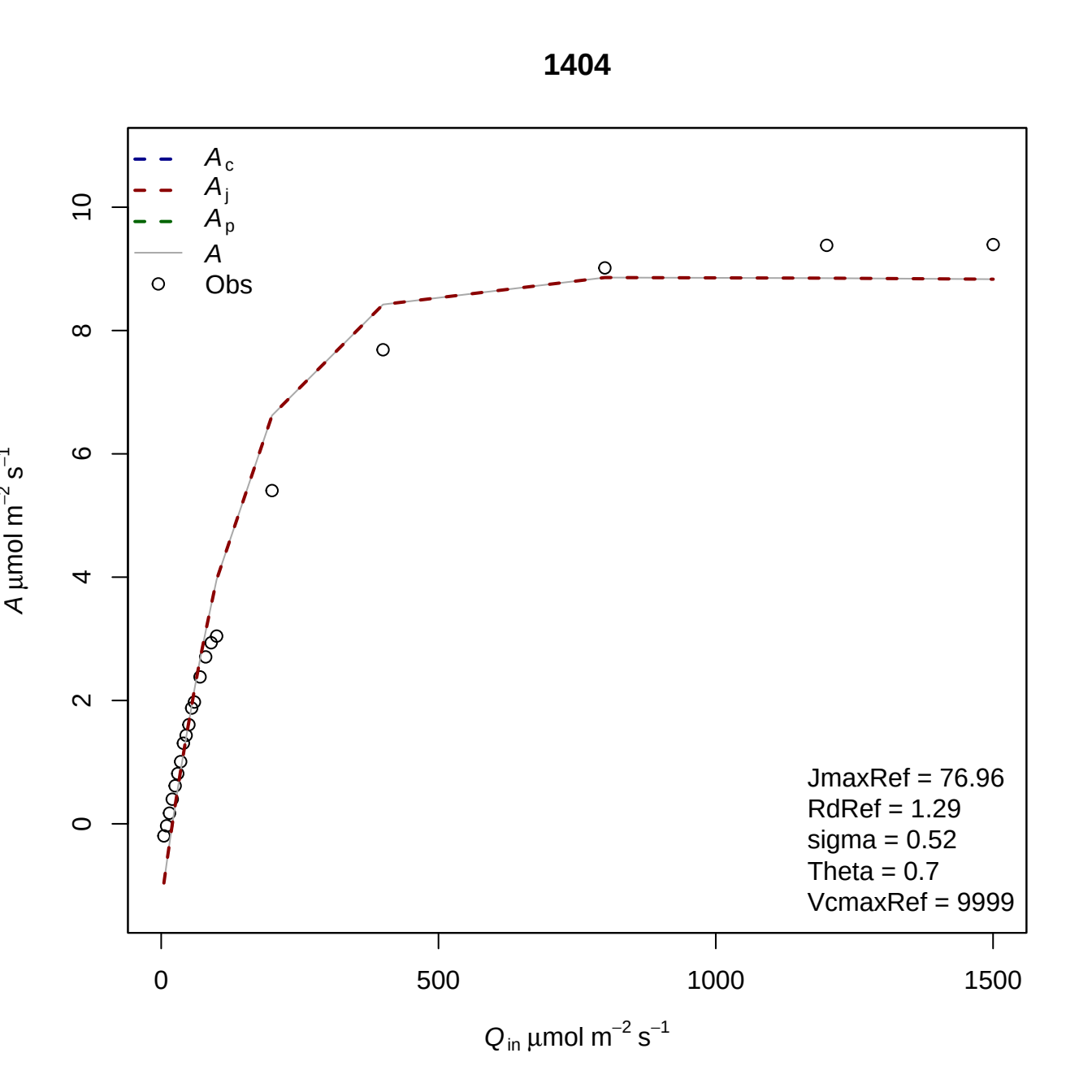

1399
A_c
A_j
A_p
A
Obs
JmaxRef = 49.3
RdRef = 0.52
sigma = 0.21
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
200
400
600
800
1000

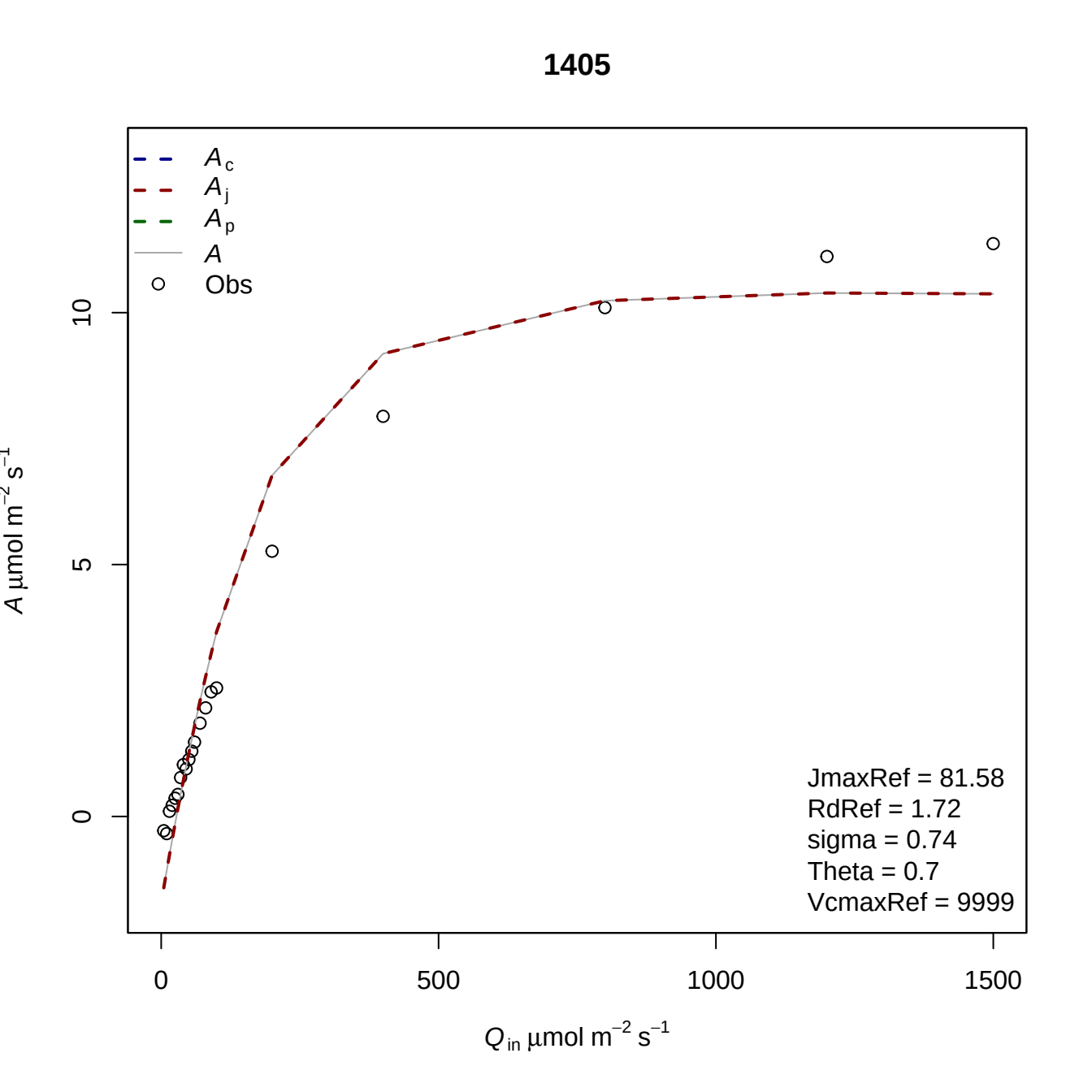

1400
A_c
A_j
A_p
A
Obs
JmaxRef = 37.86
RdRef = 0.9
sigma = 0.22
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
200
400
600
800
1000

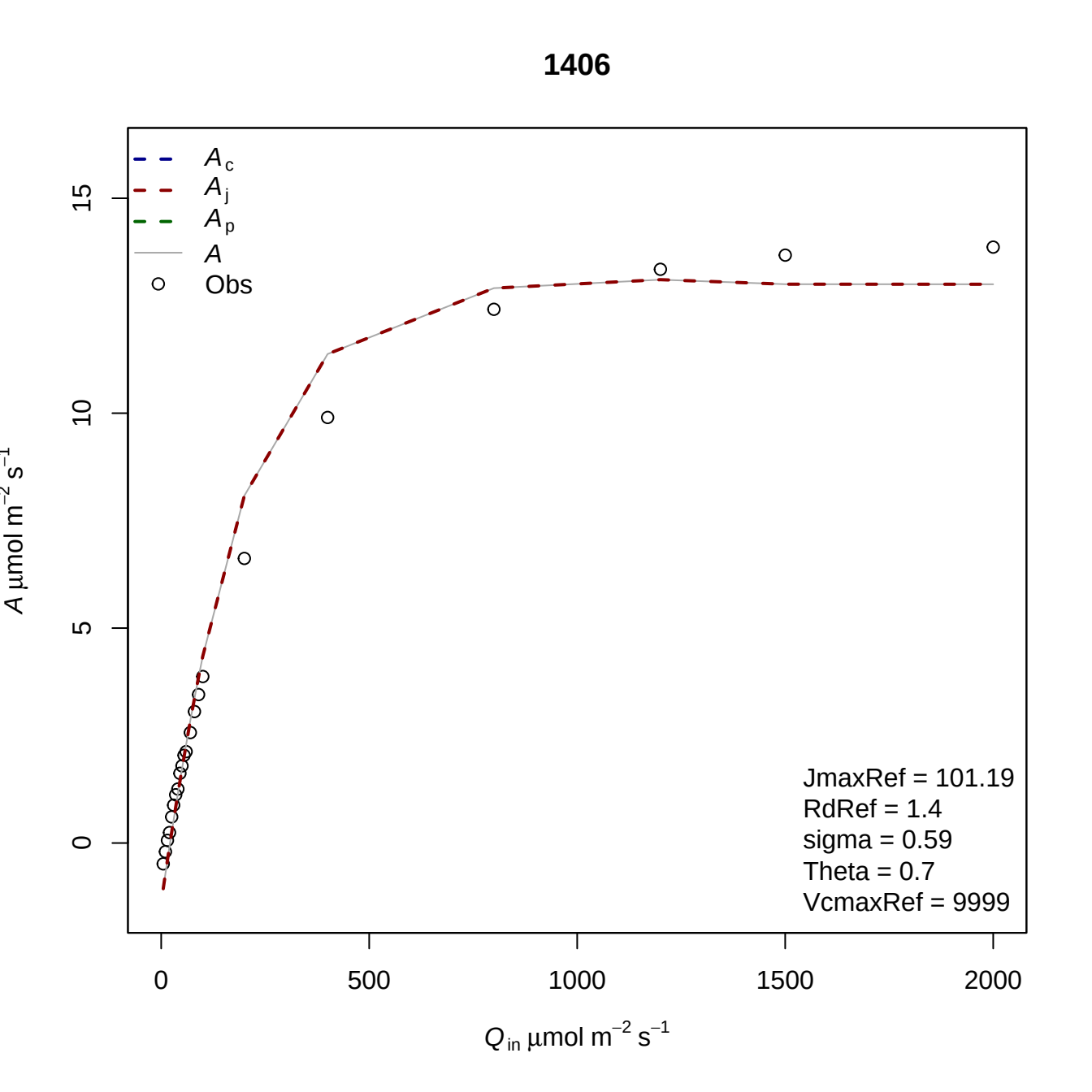

1401
A_c
A_j
A_p
A
Obs
JmaxRef = 70.63
RdRef = 1.23
sigma = 0.5
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

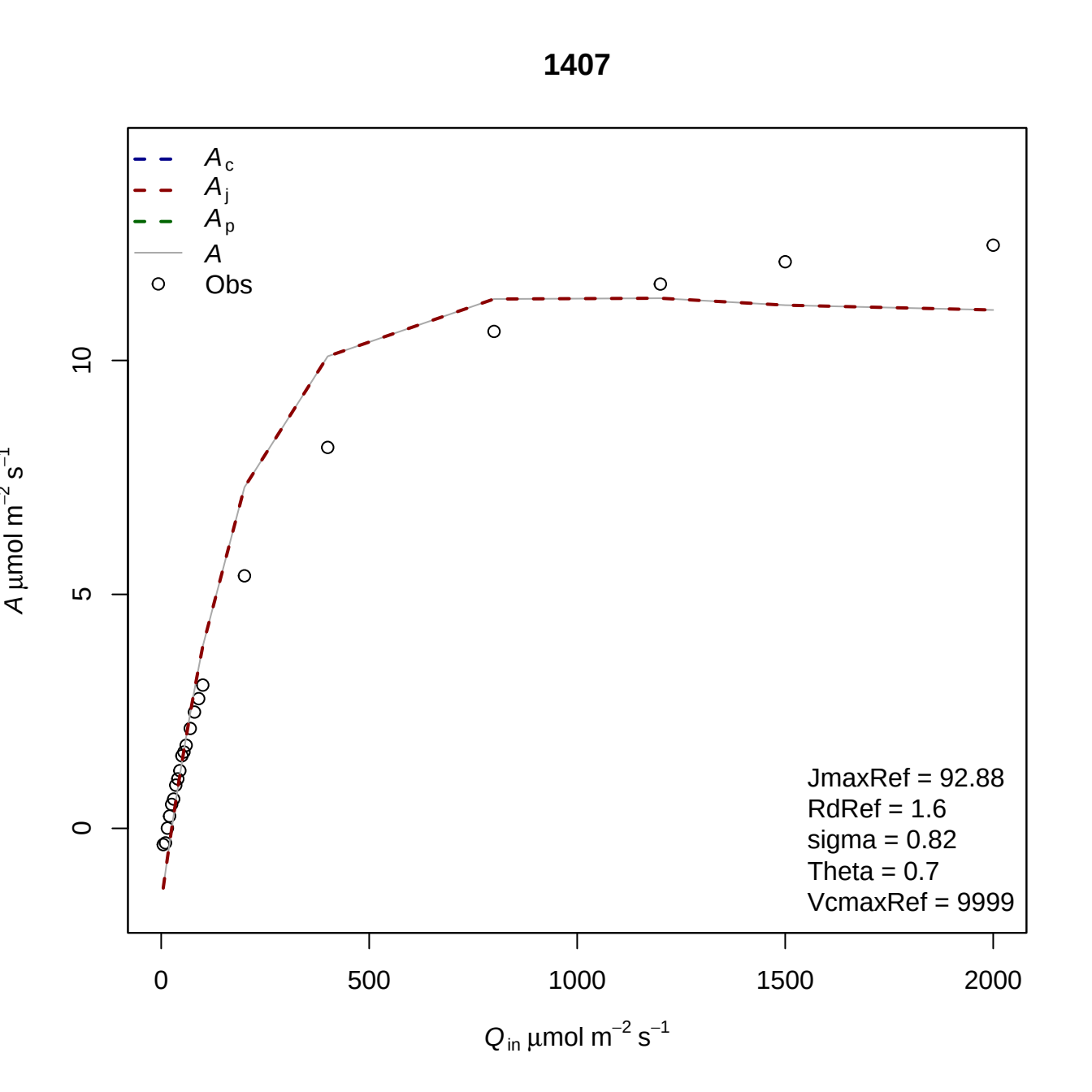

1402
A_c
A_j
A_p
A
Obs
JmaxRef = 30.7
RdRef = 1.5
sigma = 0.31
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
0
500
1000
1500

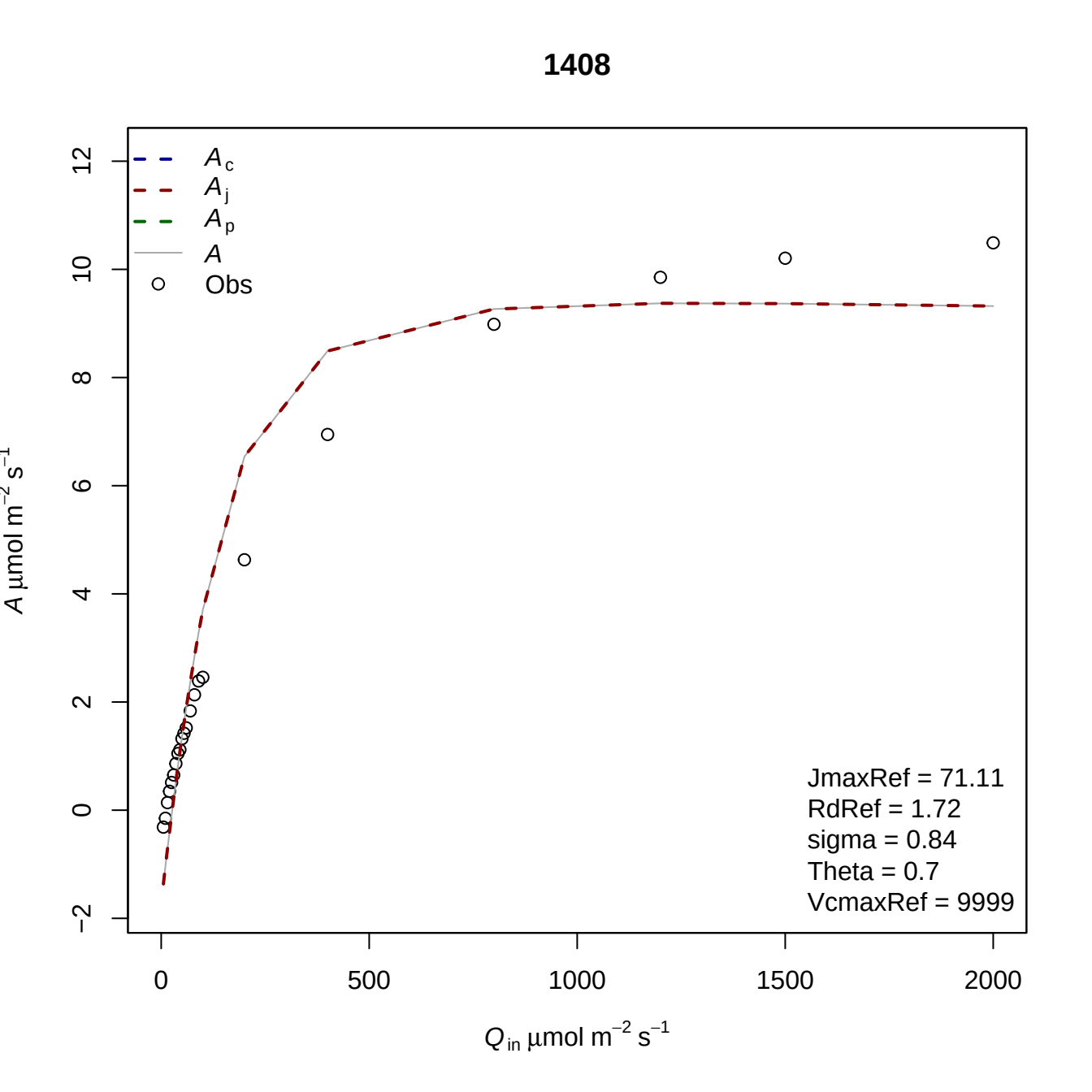

1403
A_c
A_j
A_p
A
Obs
JmaxRef = 75.87
RdRef = 1.36
sigma = 0.47
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

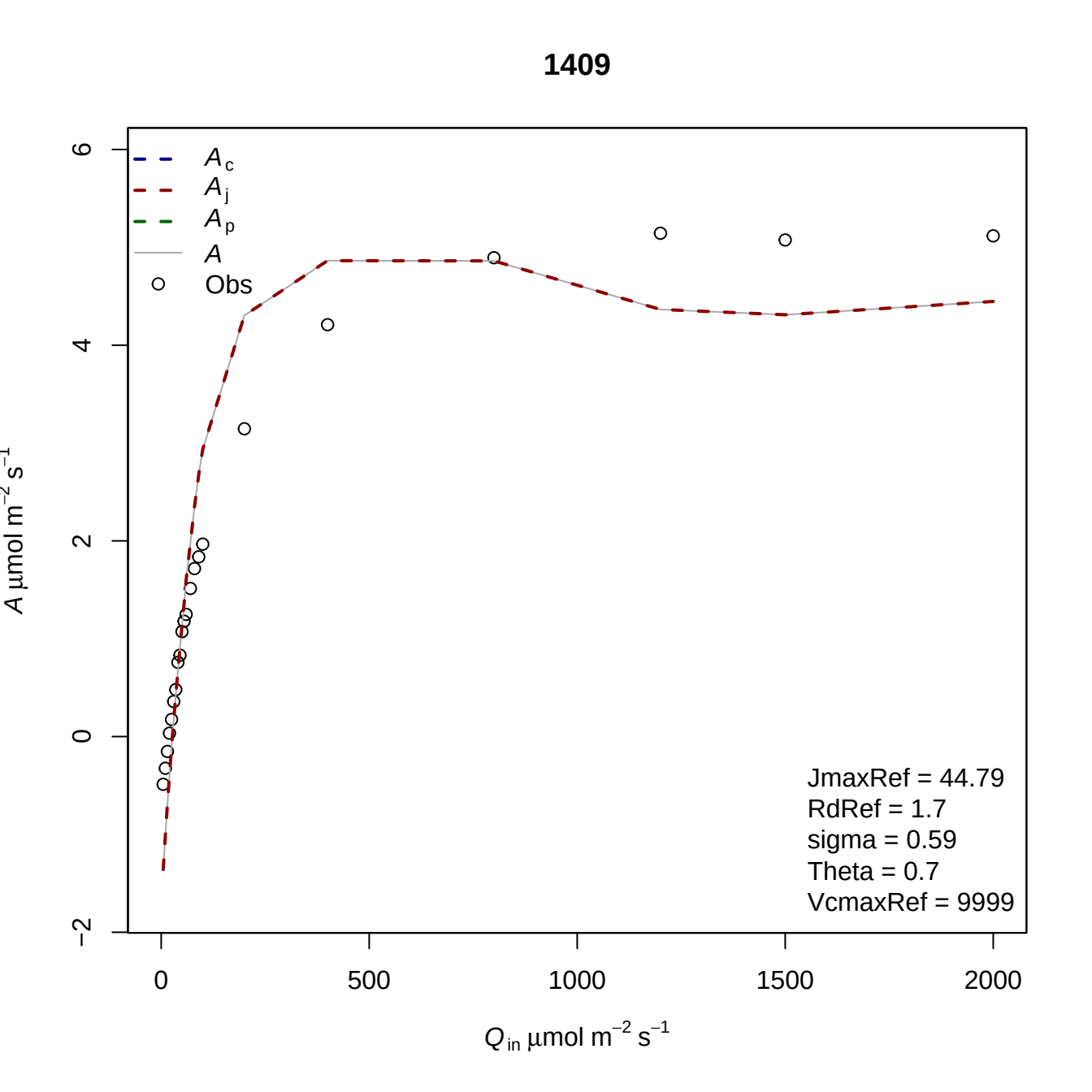

1404
A_c
A_j
A_p
A
Obs
JmaxRef = 76.96
RdRef = 1.29
sigma = 0.52
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

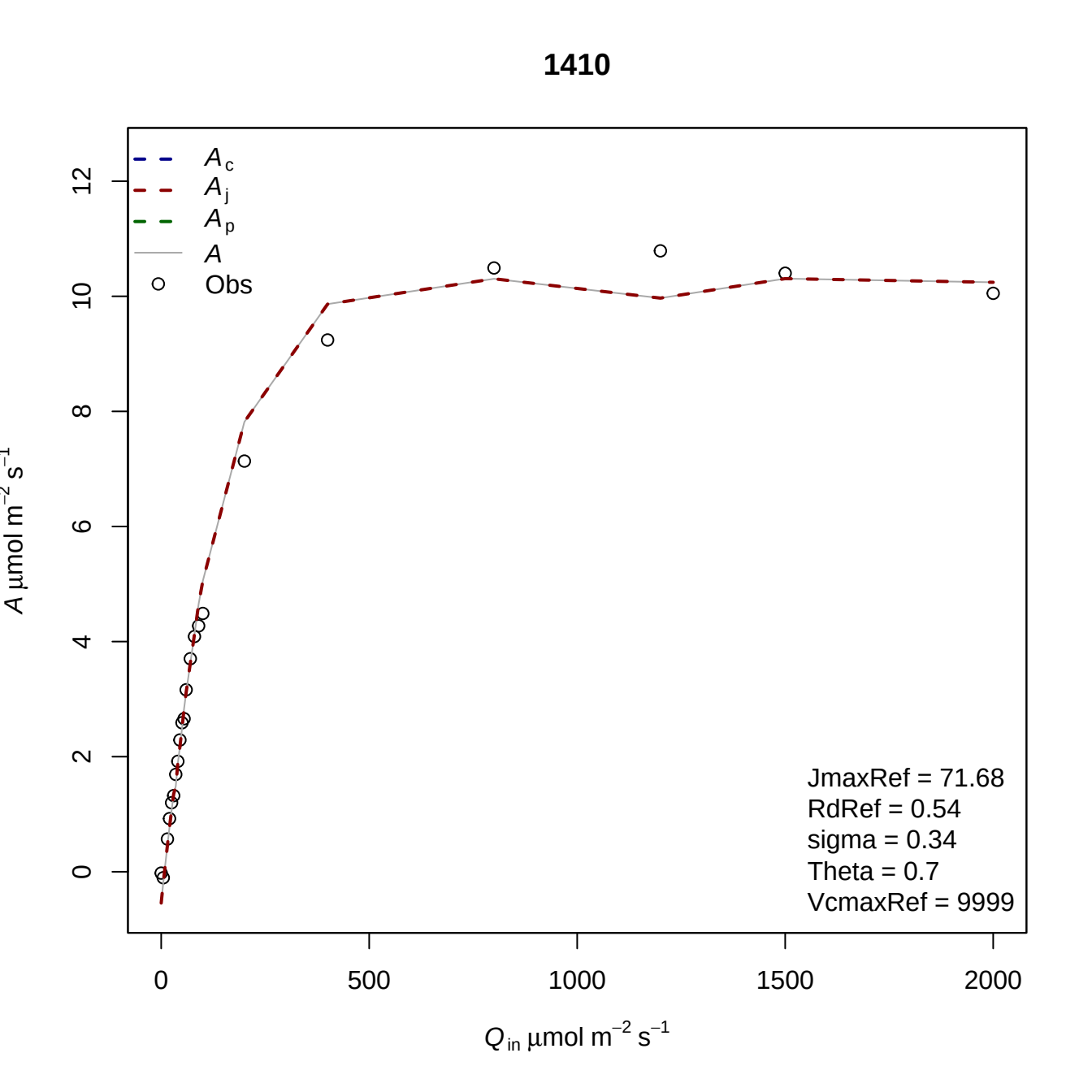

1405
A_c
A_j
A_p
A
Obs
JmaxRef = 81.58
RdRef = 1.72
sigma = 0.74
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

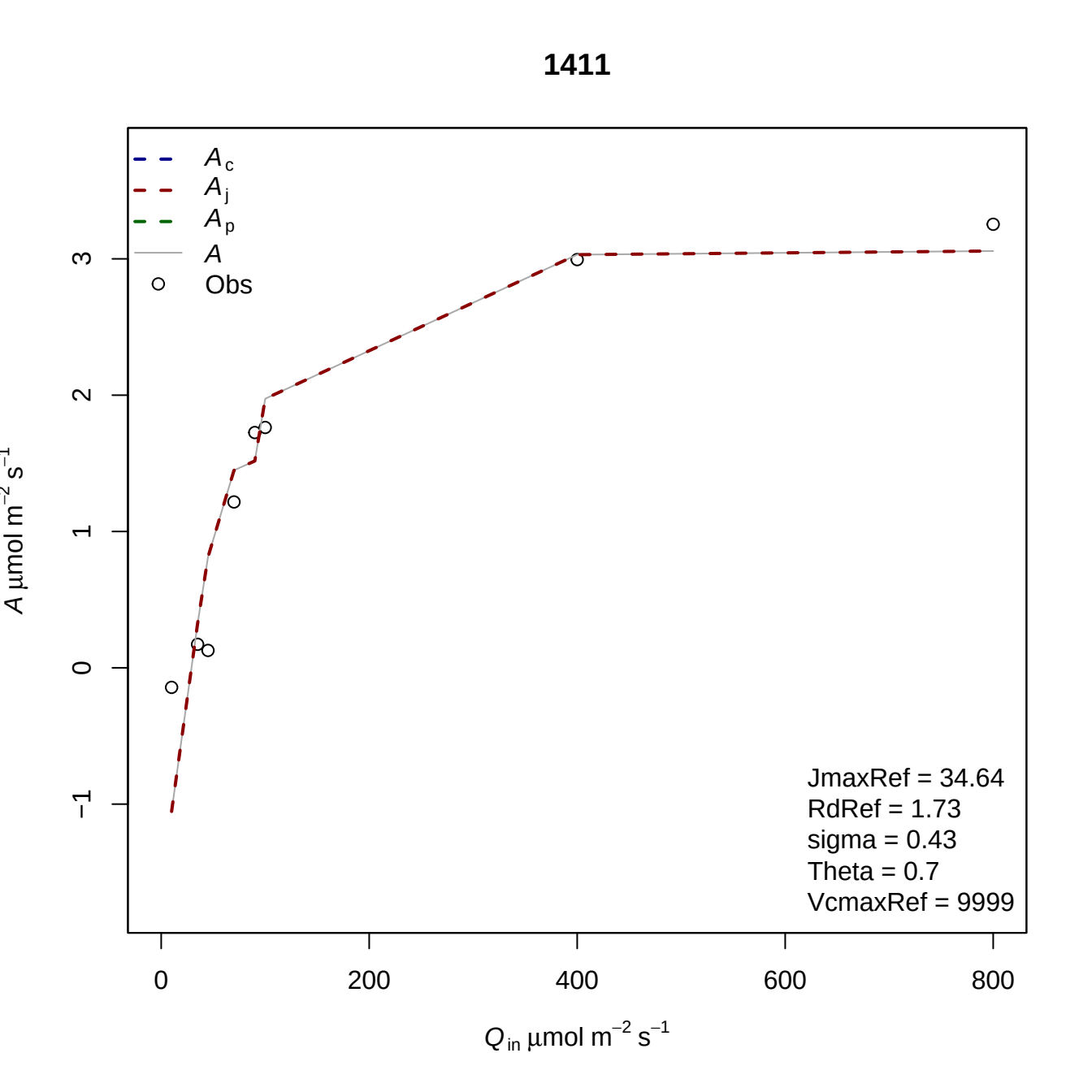

1406
A_c
A_j
A_p
A
Obs
JmaxRef = 101.19
RdRef = 1.4
sigma = 0.59
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

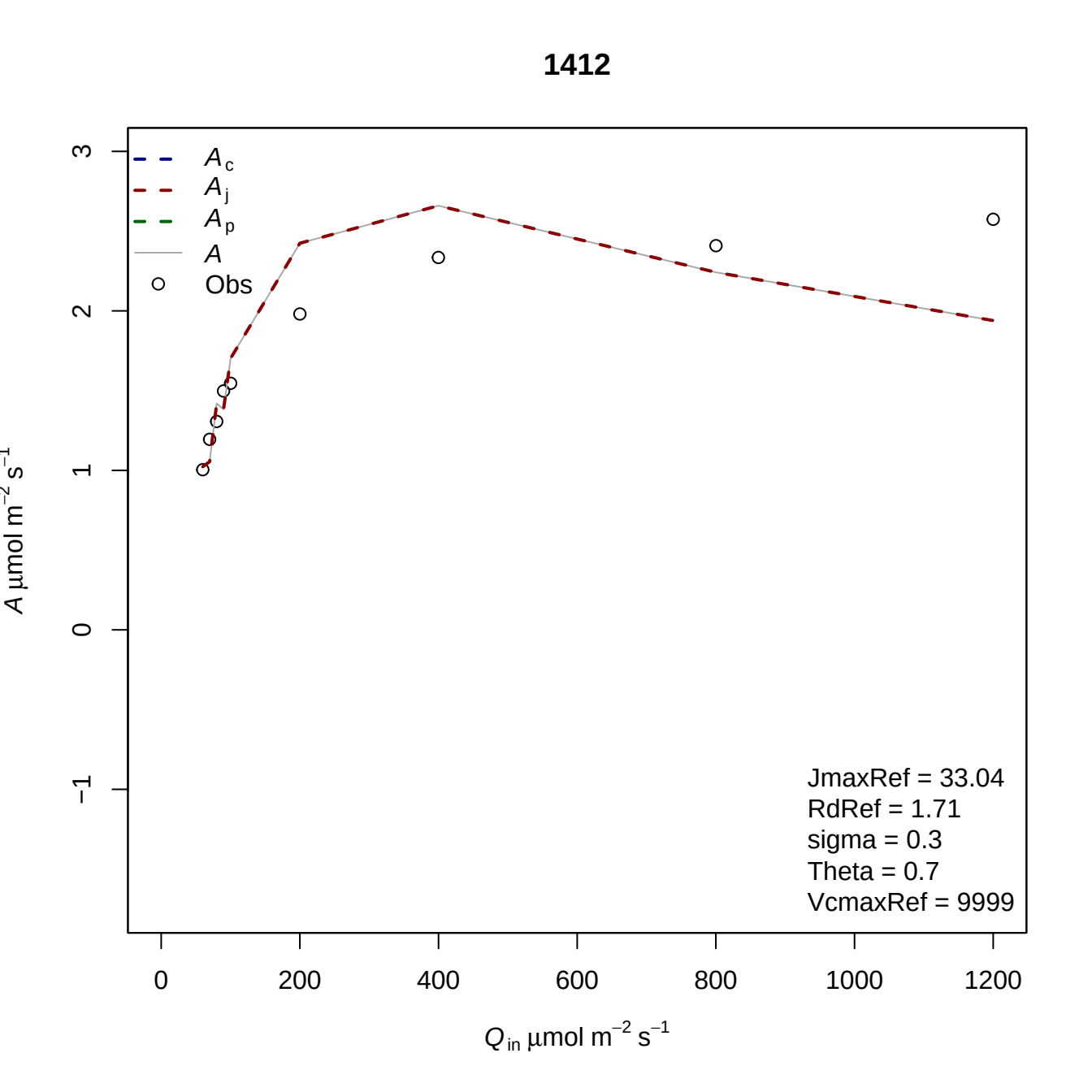

1407
A_c
A_j
A_p
A
Obs
JmaxRef = 92.88
RdRef = 1.6
sigma = 0.82
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
2000

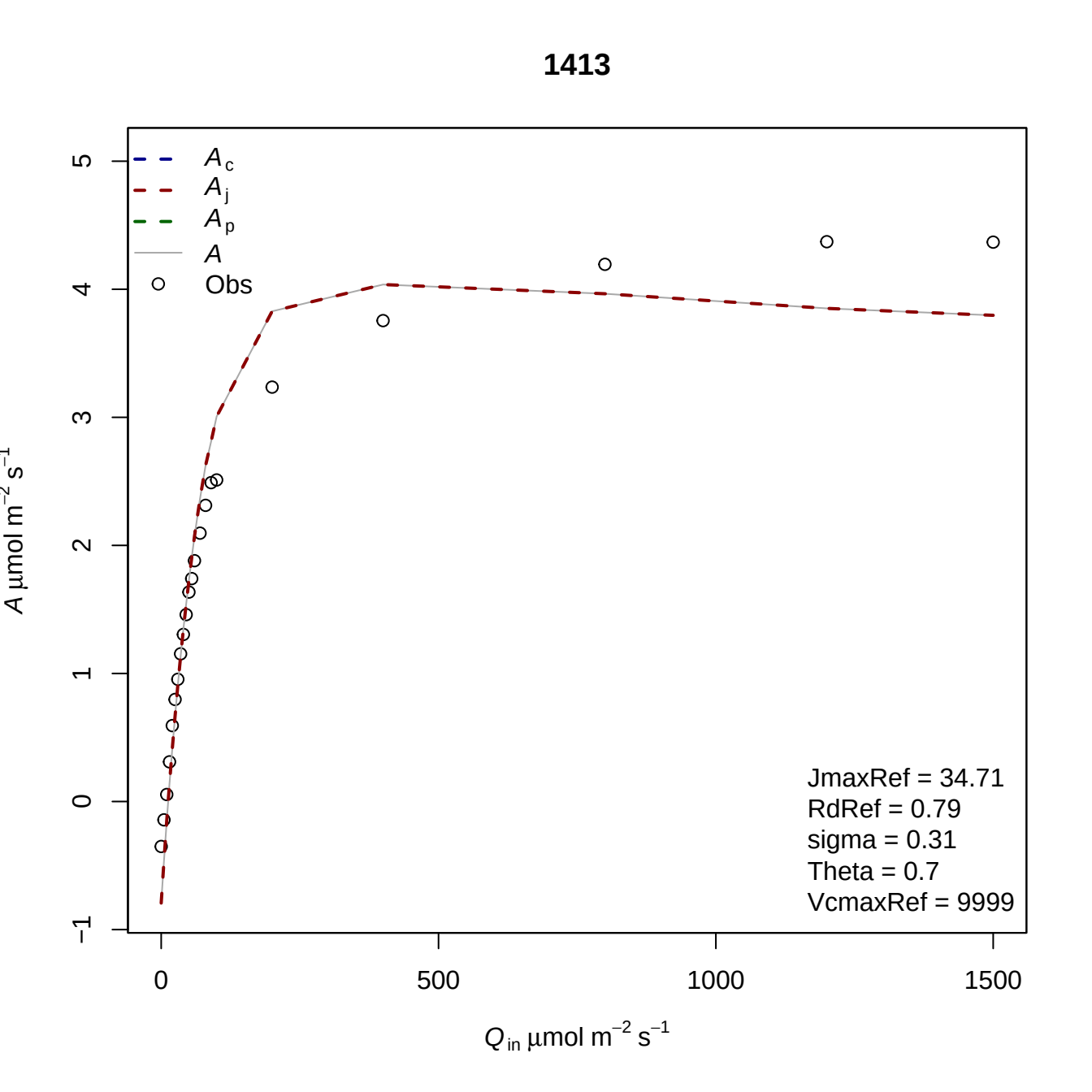

1408
A_c
A_j
A_p
A
Obs
JmaxRef = 71.11
RdRef = 1.72
sigma = 0.84
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

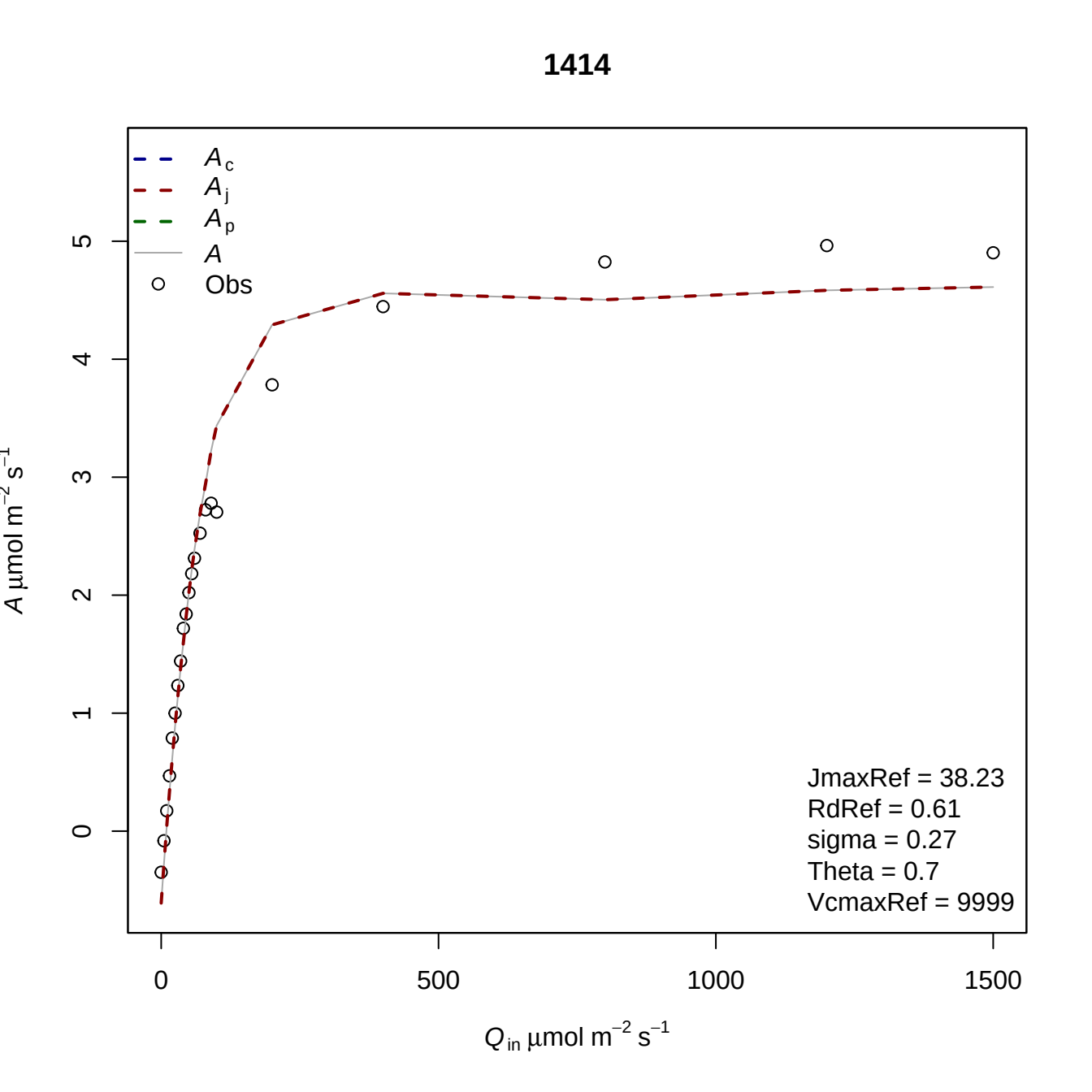

1409
A_c
A_j
A_p
A
Obs
JmaxRef = 44.79
RdRef = 1.7
sigma = 0.59
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
500
1000
1500
2000

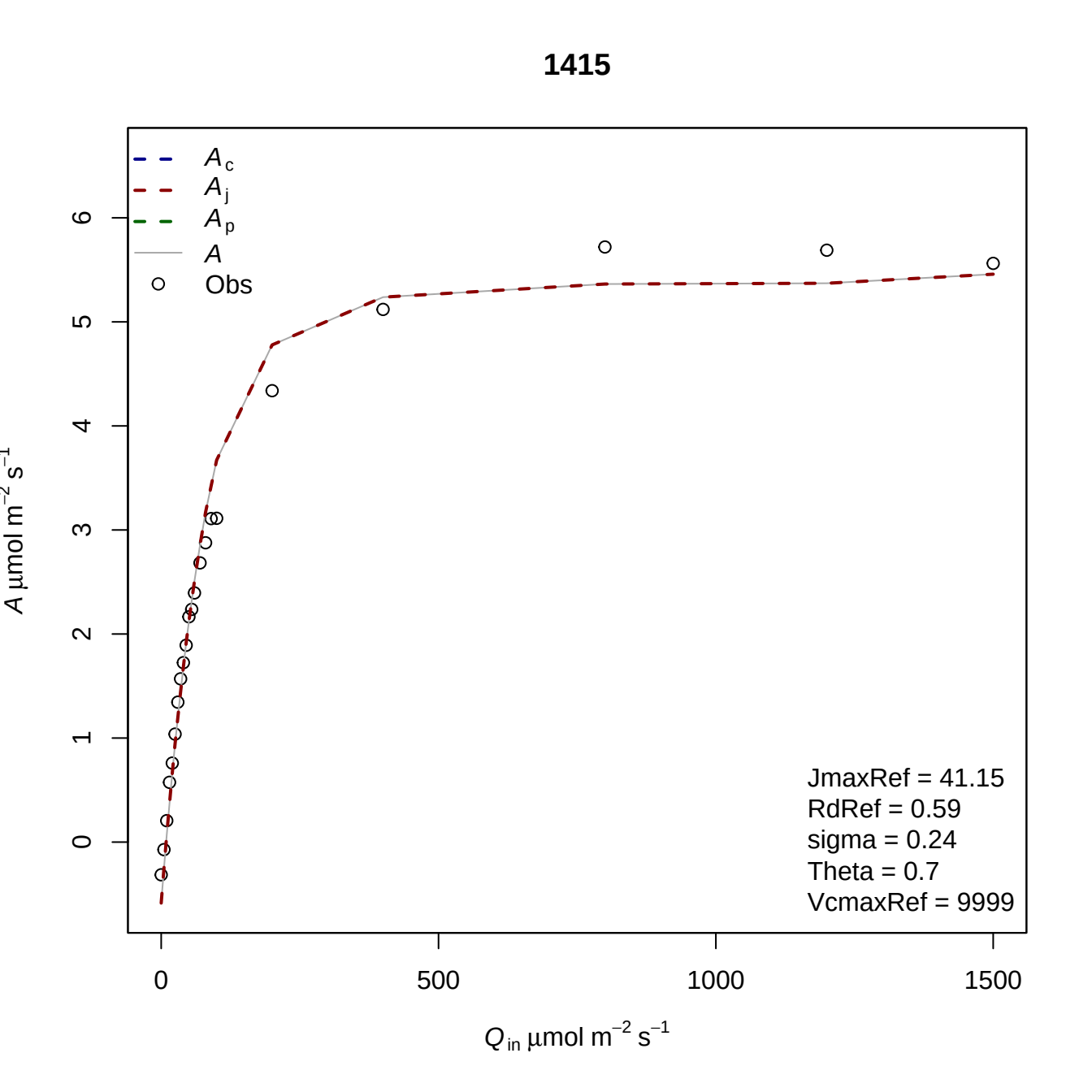

1410
A_c
A_j
A_p
A
Obs
JmaxRef = 71.68
RdRef = 0.54
sigma = 0.34
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

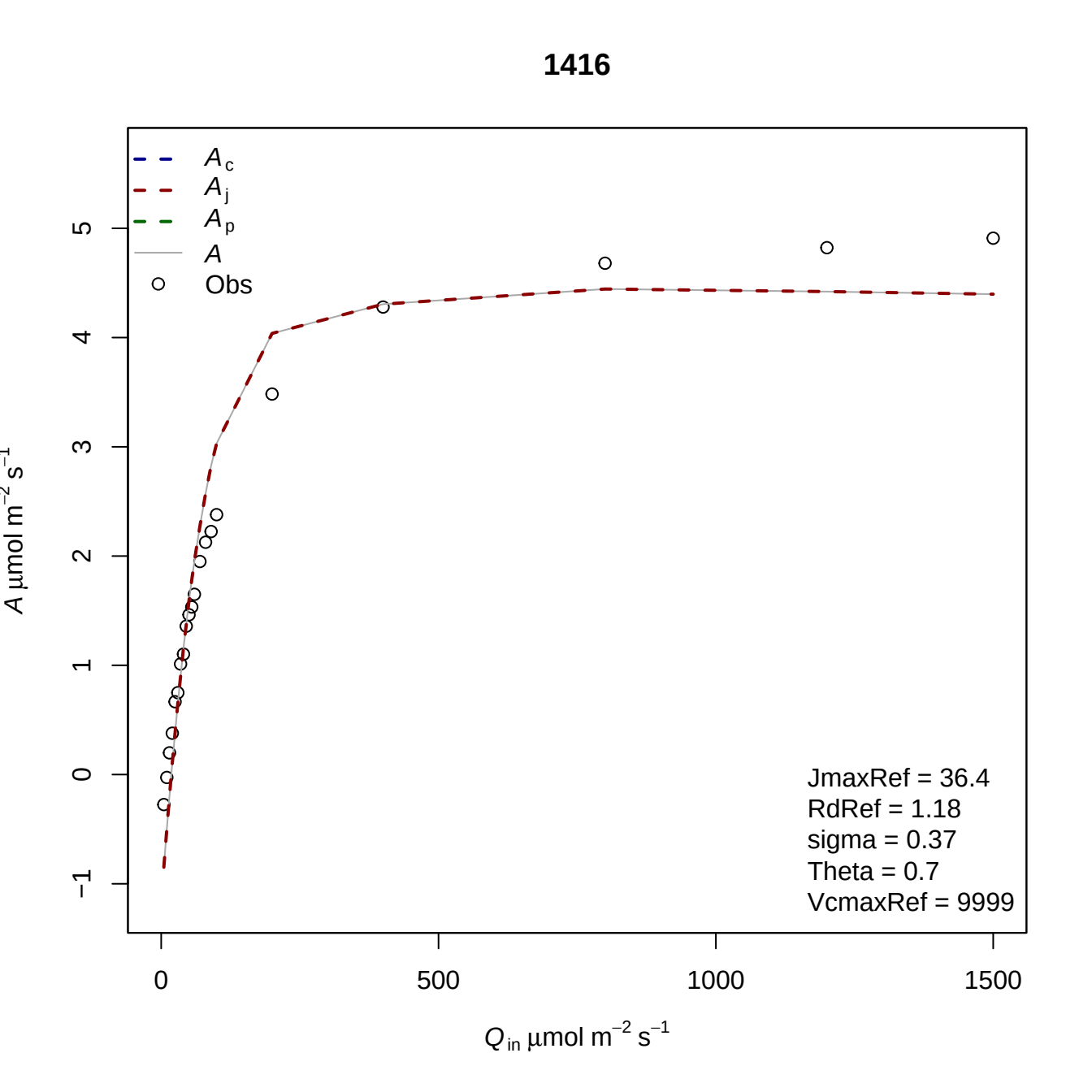

1411
A_c
A_j
A_p
A
Obs
JmaxRef = 34.64
RdRef = 1.73
sigma = 0.43
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
200
400
600
800

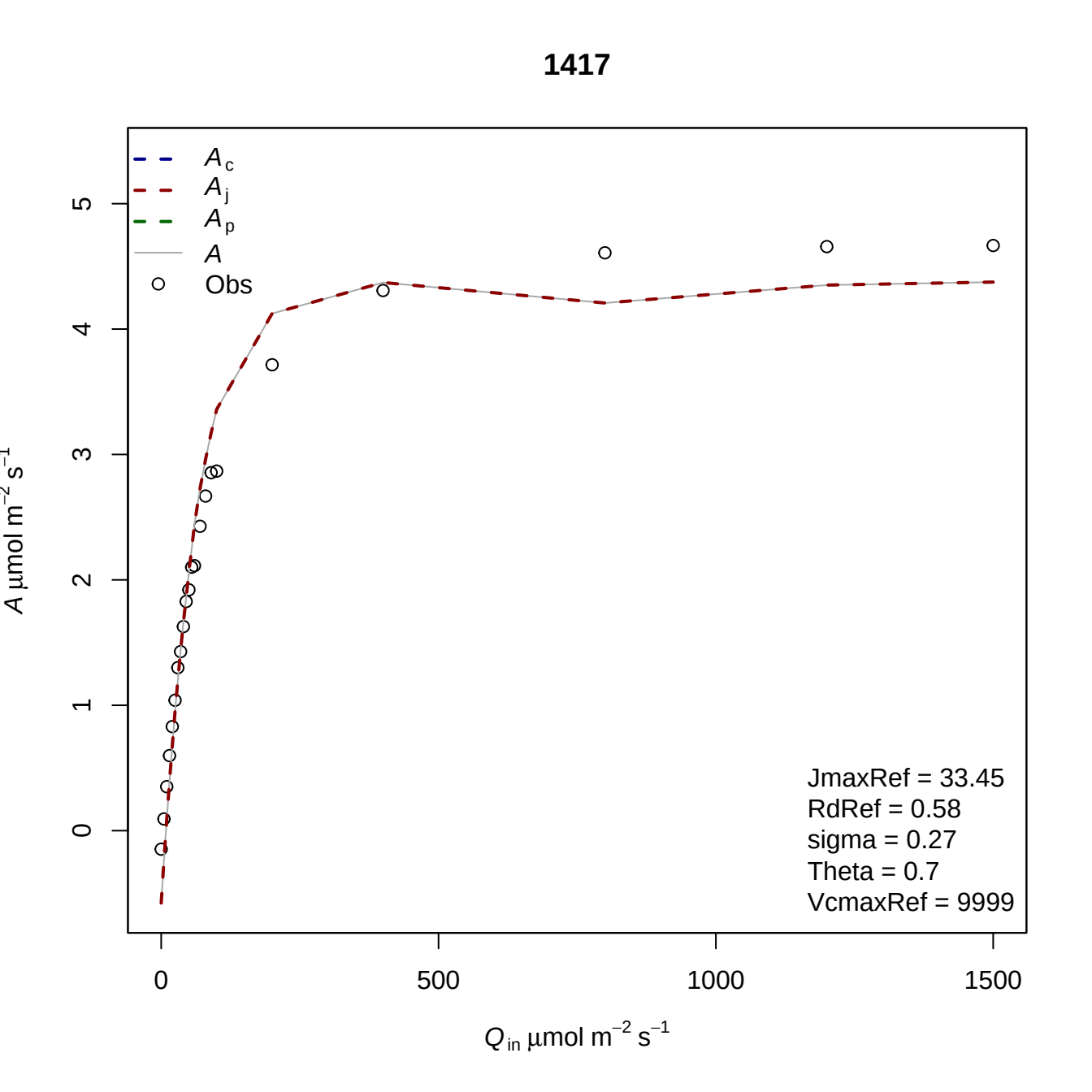

1412
A_c
A_j
A_p
A
Obs
JmaxRef = 33.04
RdRef = 1.71
sigma = 0.3
Theta = 0.7
VcmaxRef = 9999
3
2
1
0
−1
0
200
400
600
800
1000
1200
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

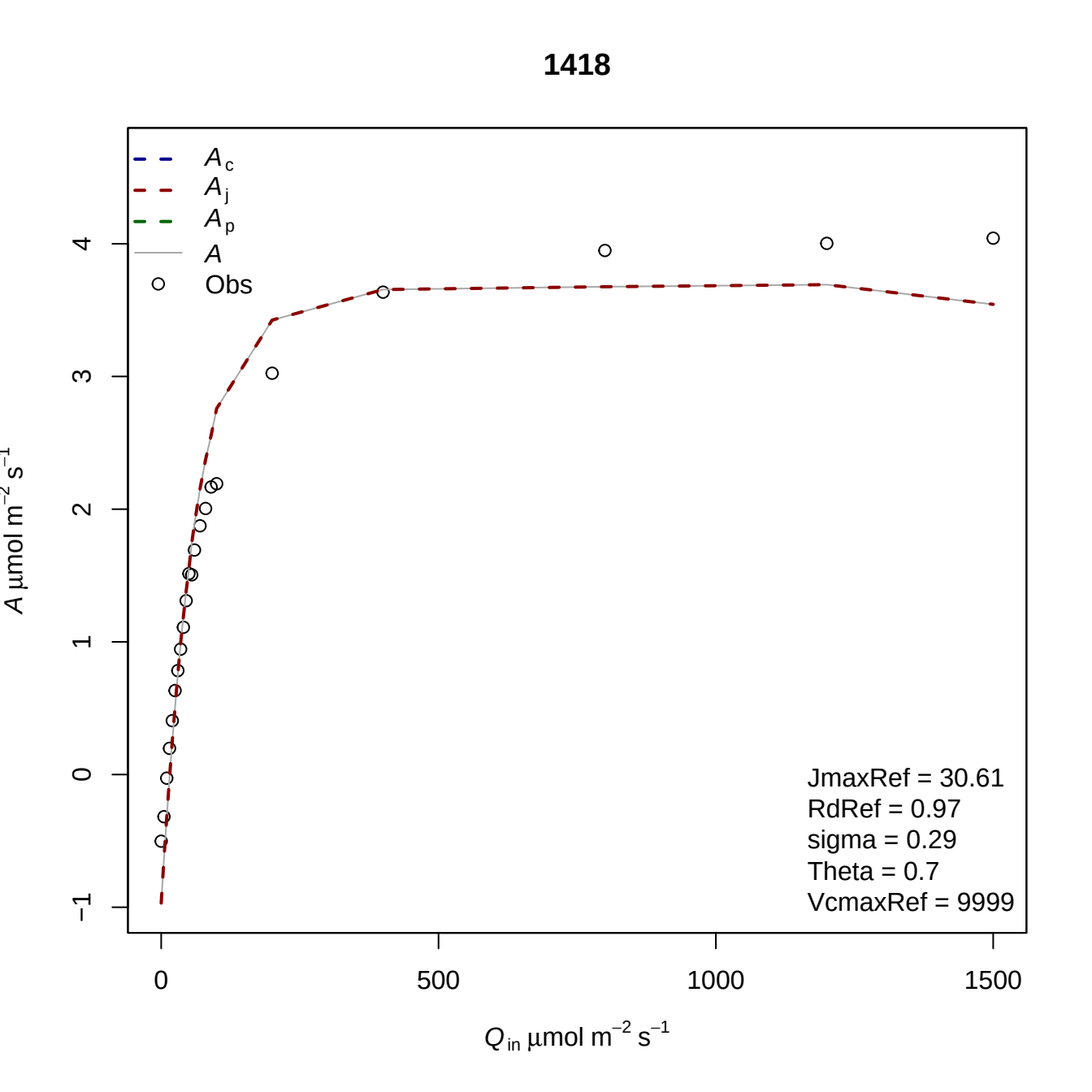

1413
A_c
A_j
A_p
A
Obs
JmaxRef = 34.71
RdRef = 0.79
sigma = 0.31
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

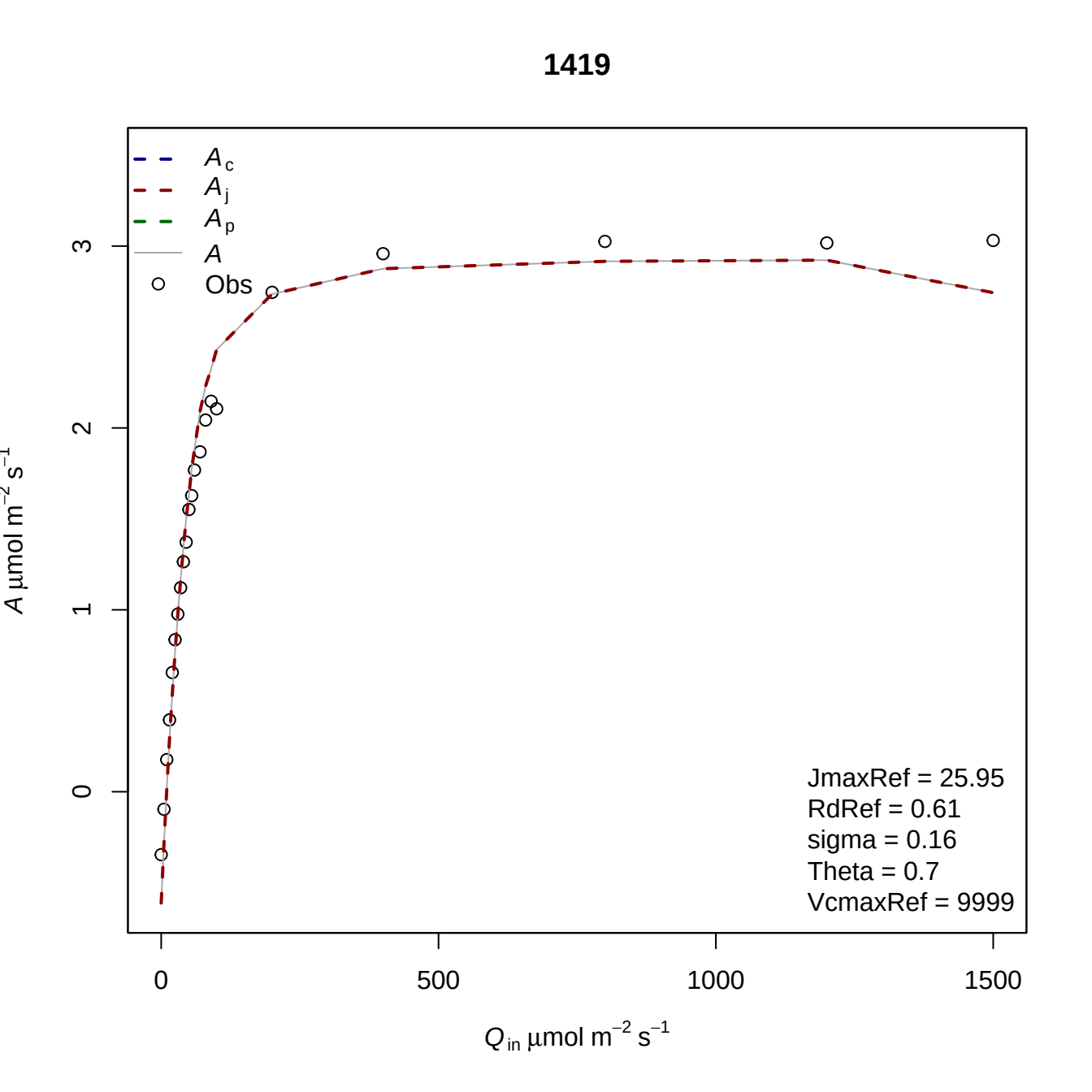

1414
A_c
A_j
A_p
A
Obs
JmaxRef = 38.23
RdRef = 0.61
sigma = 0.27
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

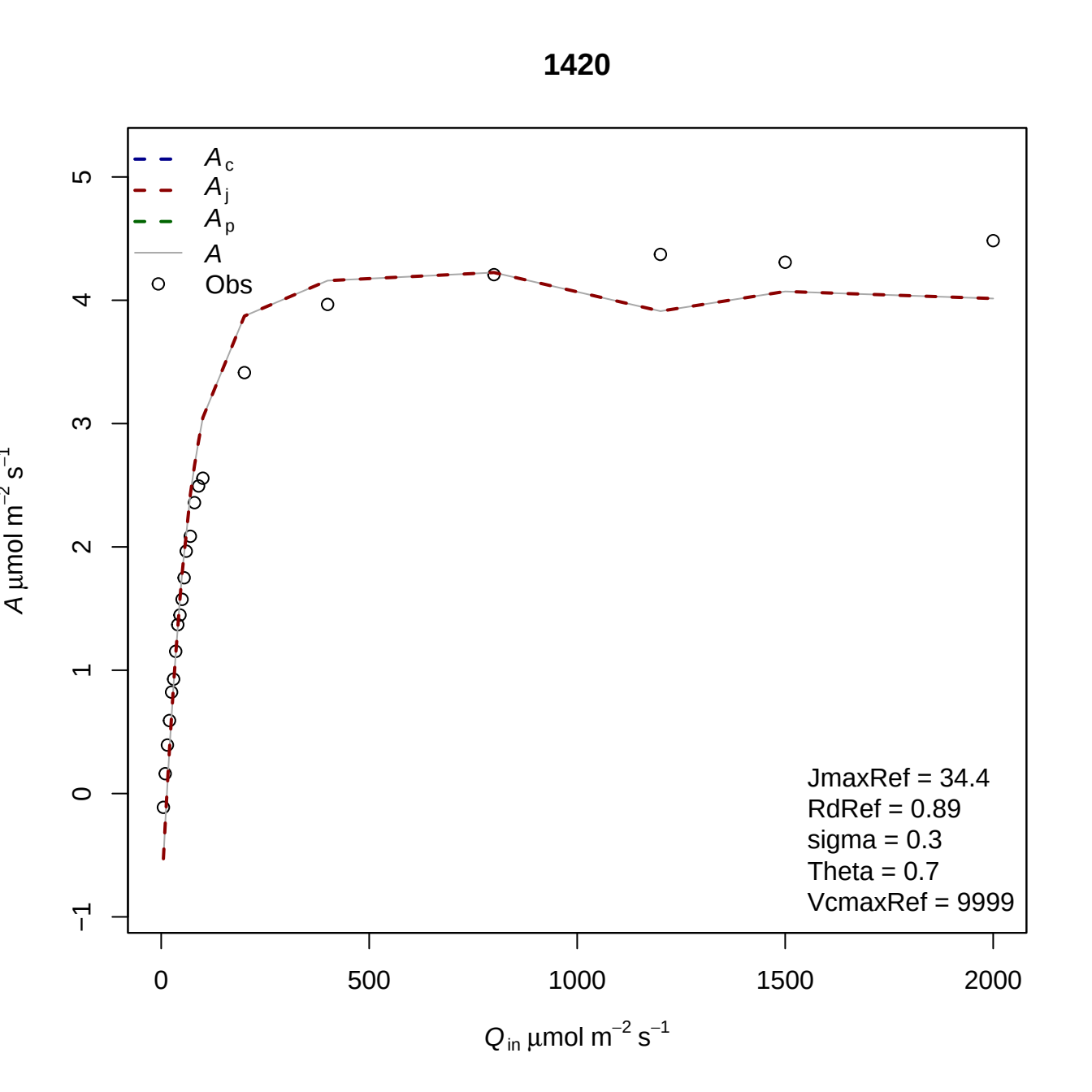

1415
A_c
A_j
A_p
A
Obs
JmaxRef = 41.15
RdRef = 0.59
sigma = 0.24
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

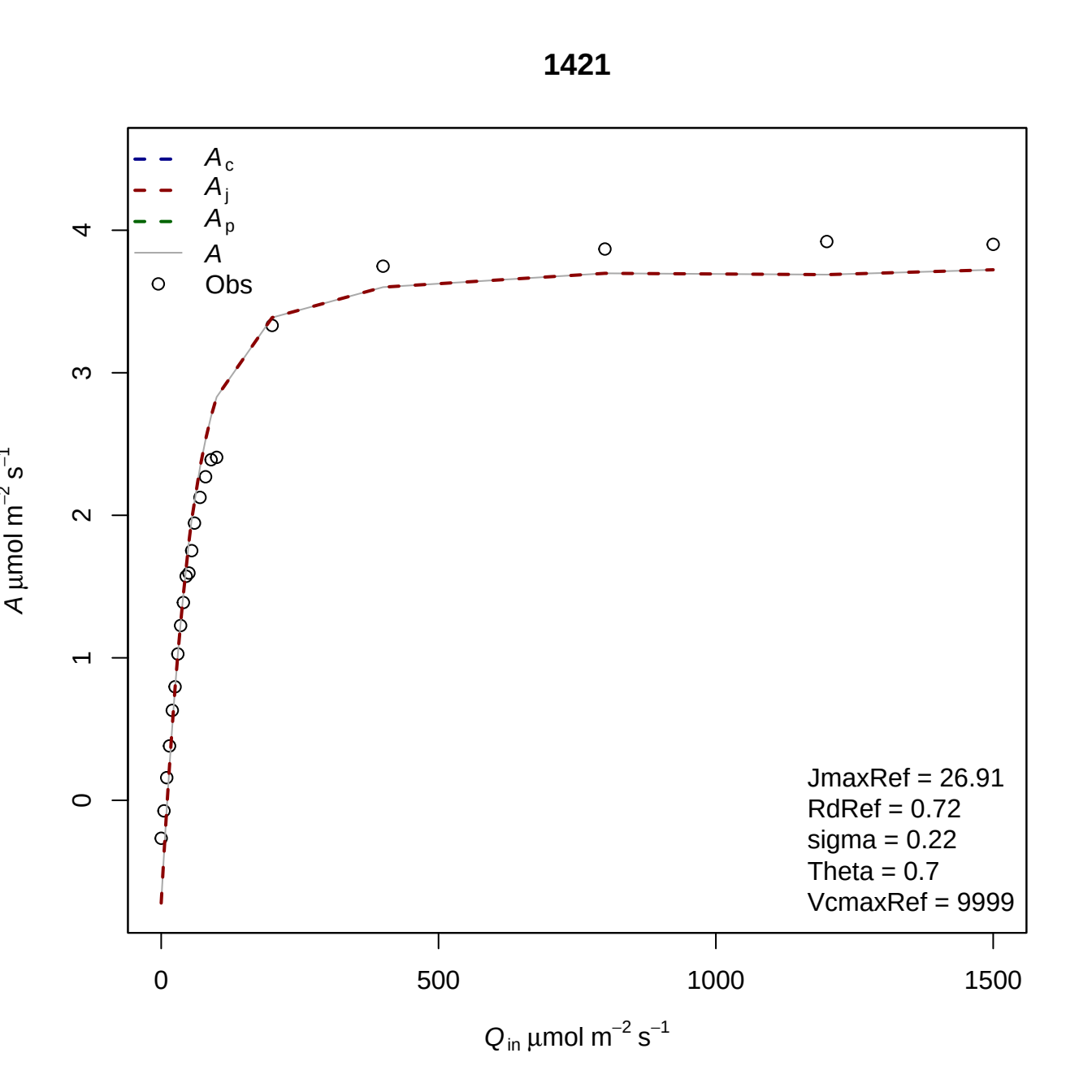

1416
A_c
A_j
A_p
A
Obs
JmaxRef = 36.4
RdRef = 1.18
sigma = 0.37
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

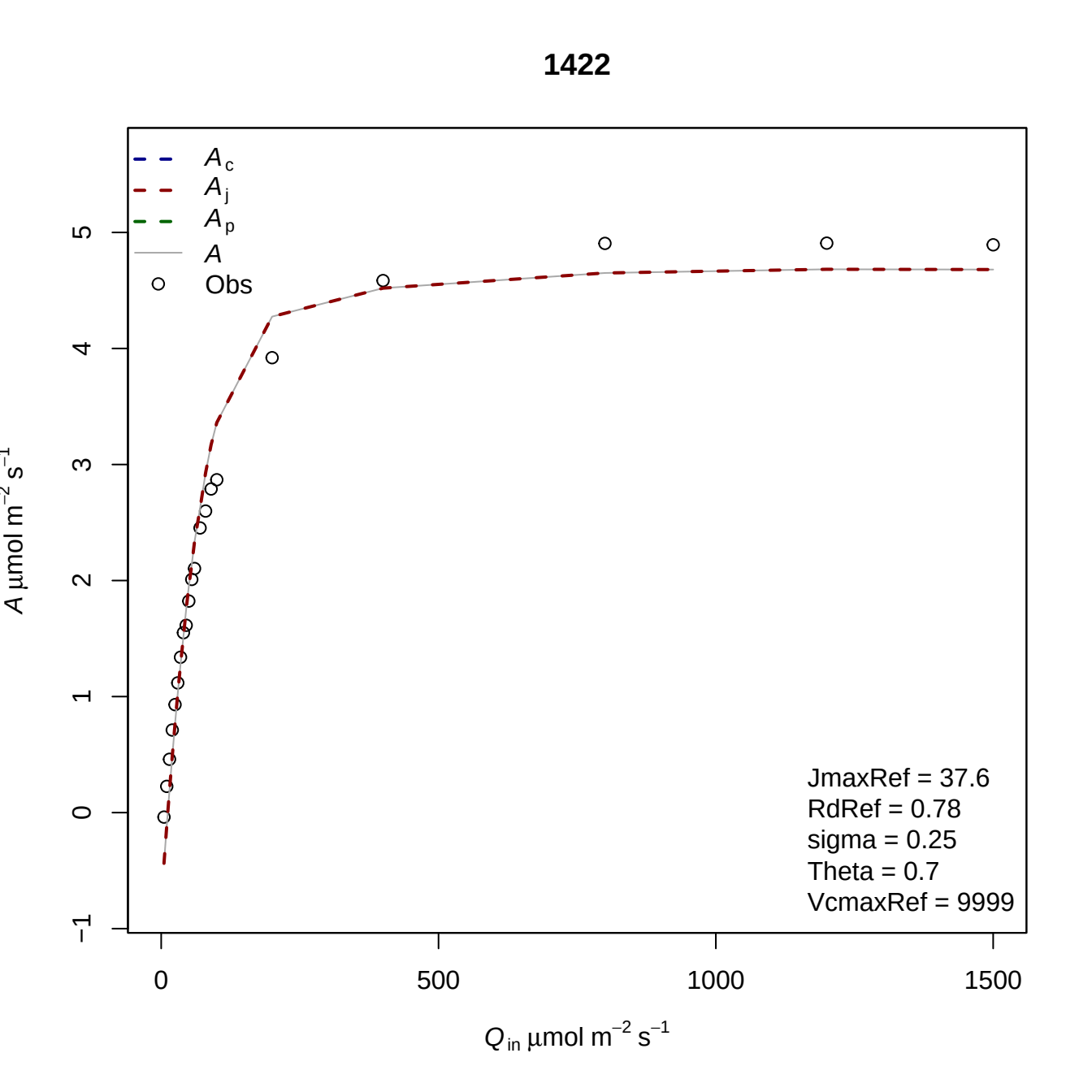

1417
A_c
A_j
A_p
A
Obs
JmaxRef = 33.45
RdRef = 0.58
sigma = 0.27
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

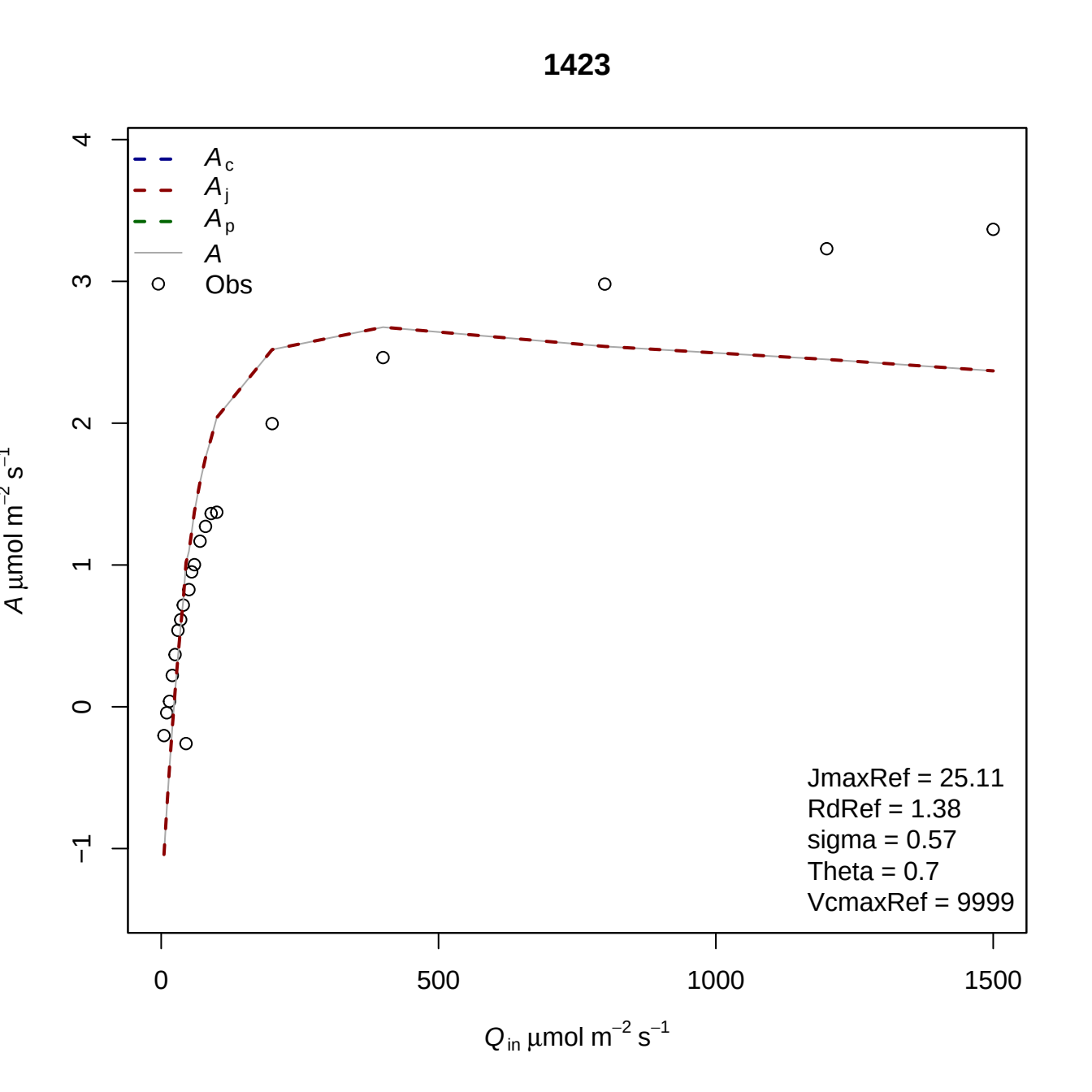

1418
A_c
A_j
A_p
A
Obs
JmaxRef = 30.61
RdRef = 0.97
sigma = 0.29
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

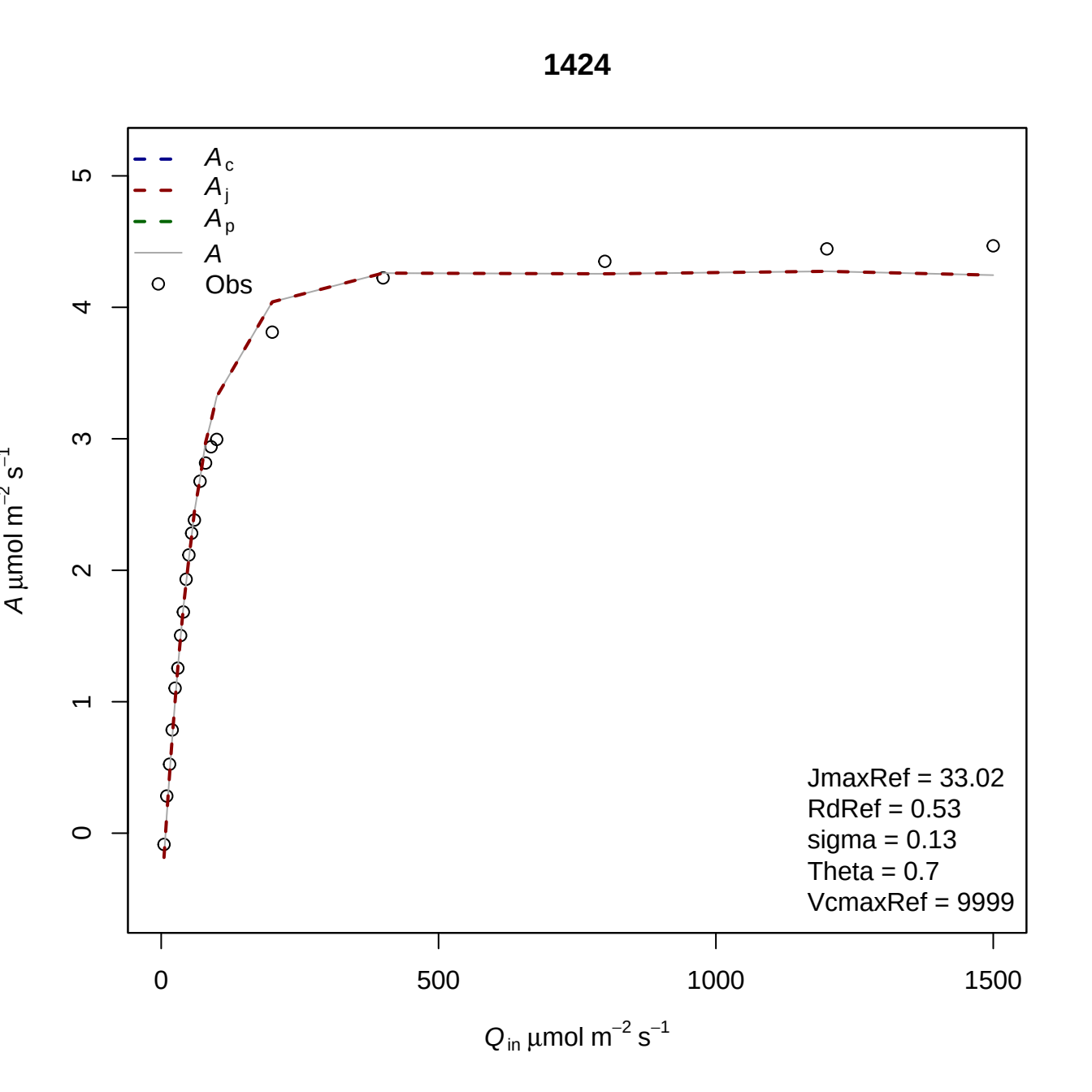

1419
A_c
A_j
A_p
A
Obs
JmaxRef = 25.95
RdRef = 0.61
sigma = 0.16
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
0
500
1000
1500

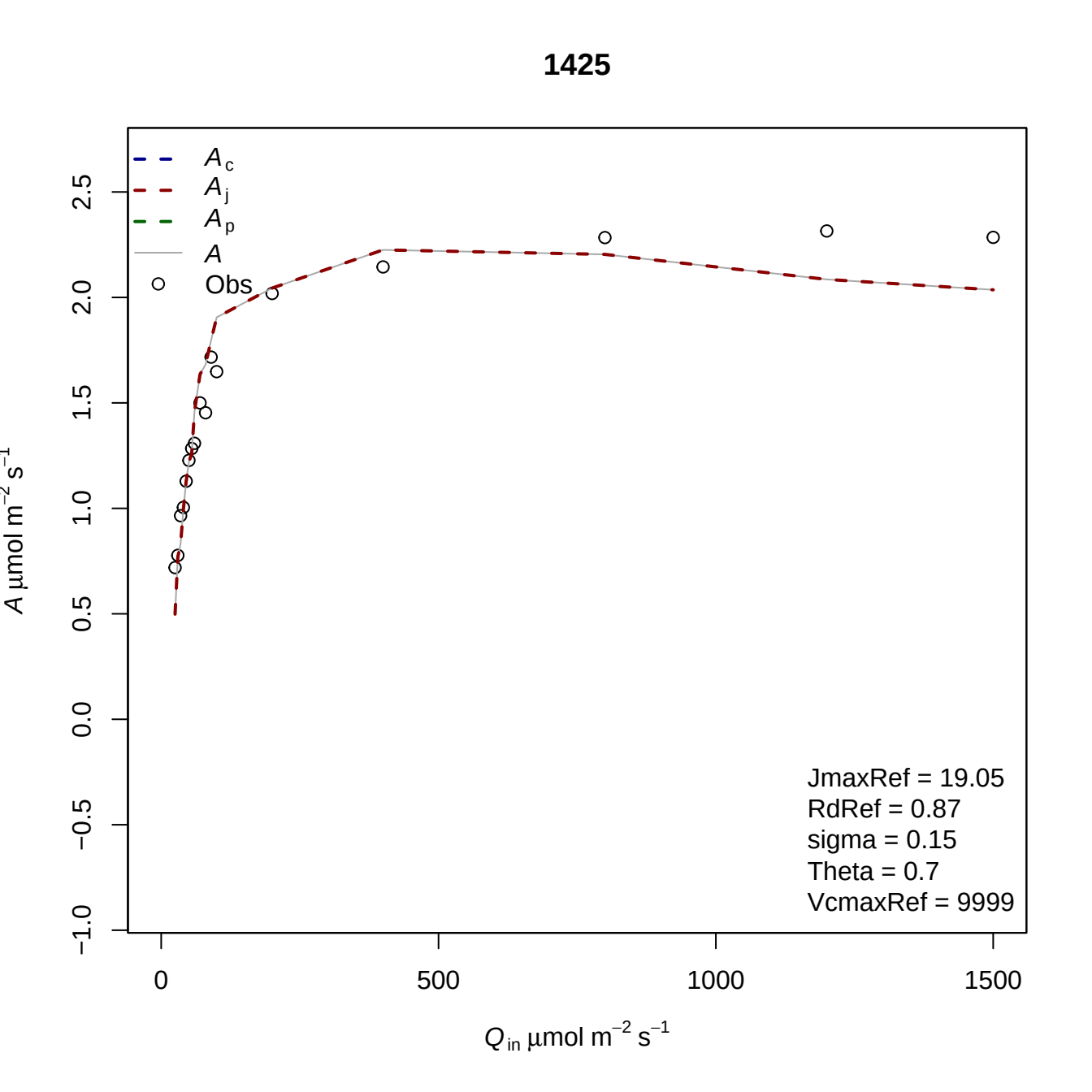

1420
A_c
A_j
A_p
A
Obs
JmaxRef = 34.4
RdRef = 0.89
sigma = 0.3
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500
2000

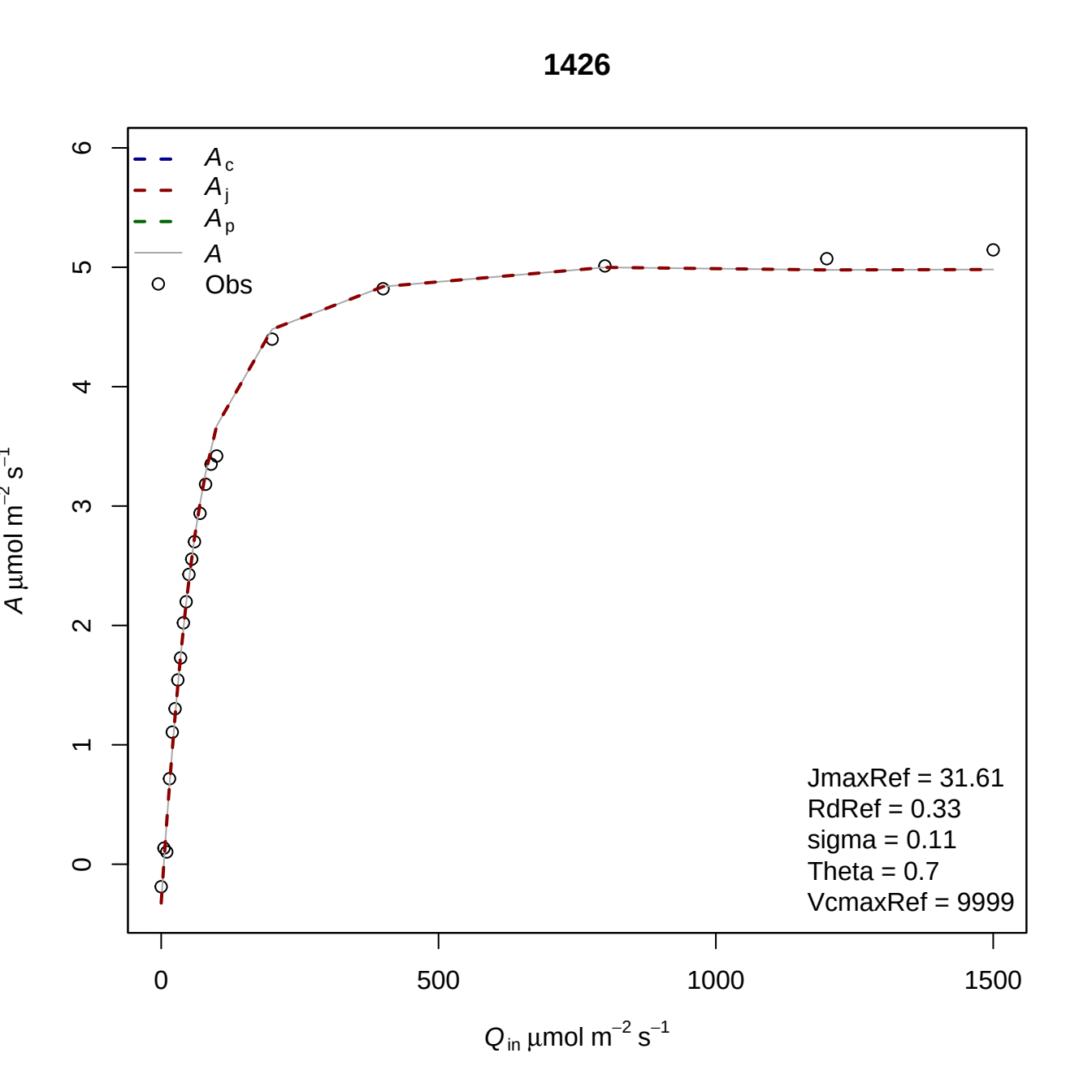

1421
A_c
A_j
A_p
A
Obs
JmaxRef = 26.91
RdRef = 0.72
sigma = 0.22
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
0
500
1000
1500

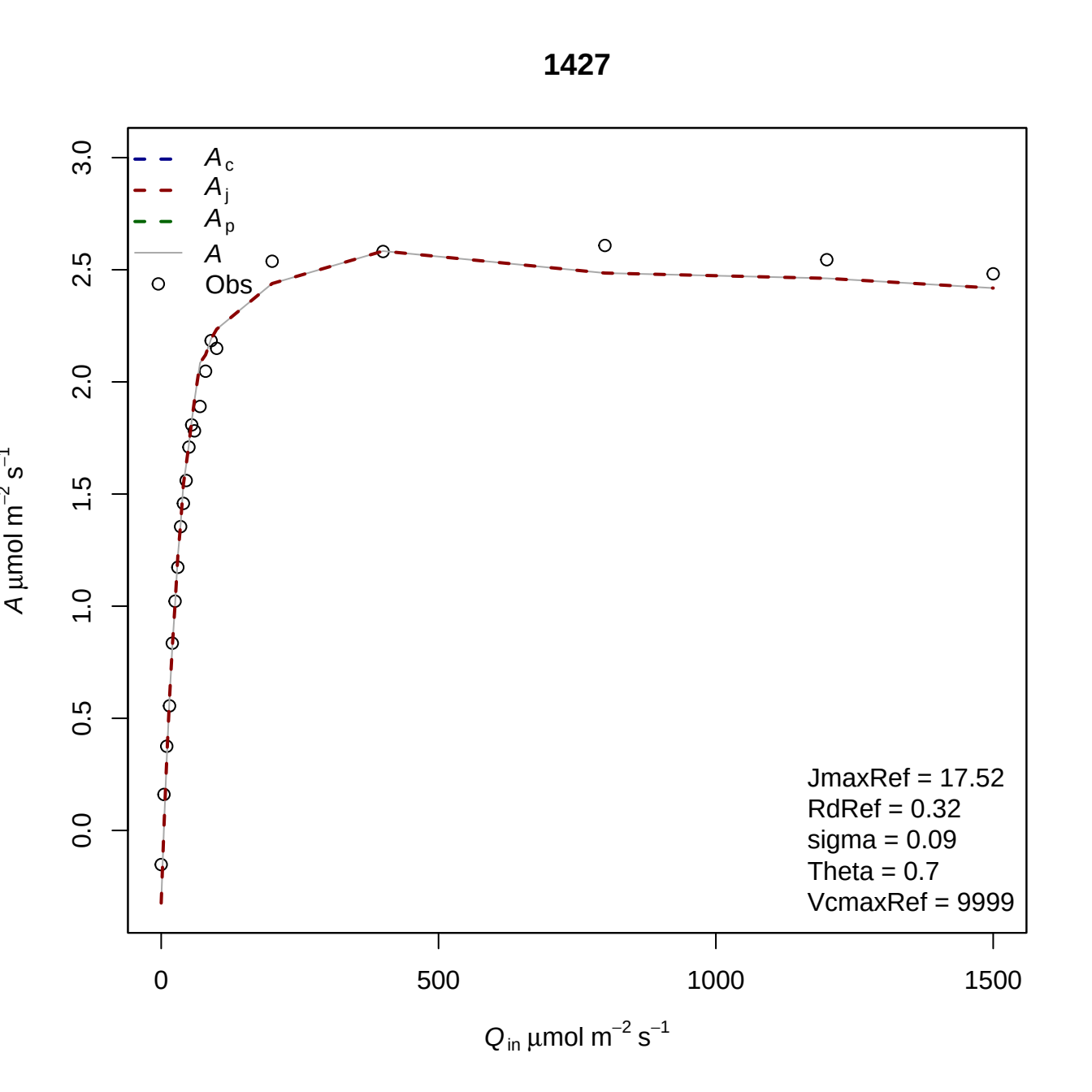

1422
A_c
A_j
A_p
A
Obs
JmaxRef = 37.6
RdRef = 0.78
sigma = 0.25
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

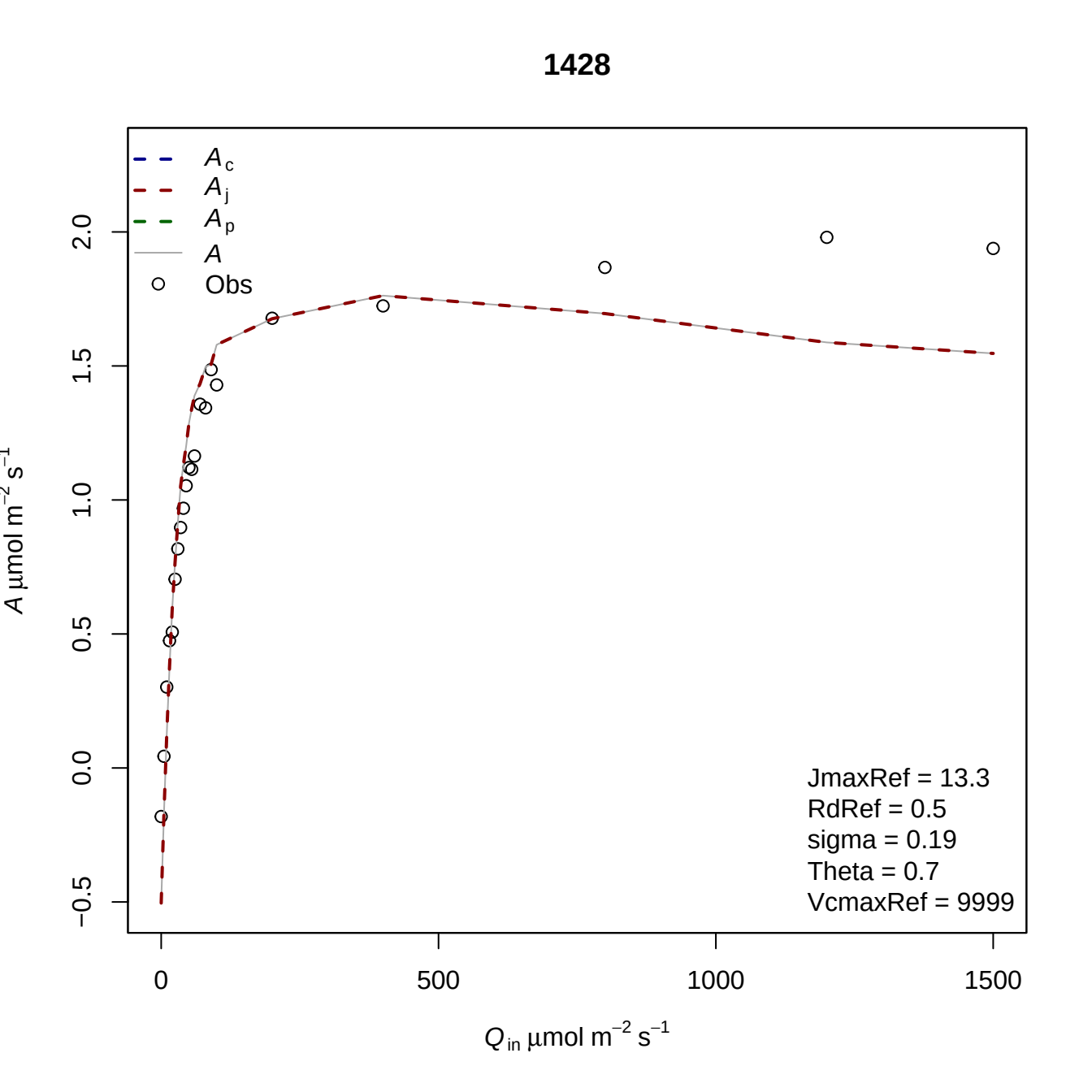

1423
A_c
A_j
A_p
A
Obs
JmaxRef = 25.11
RdRef = 1.38
sigma = 0.57
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
0
500
1000
1500

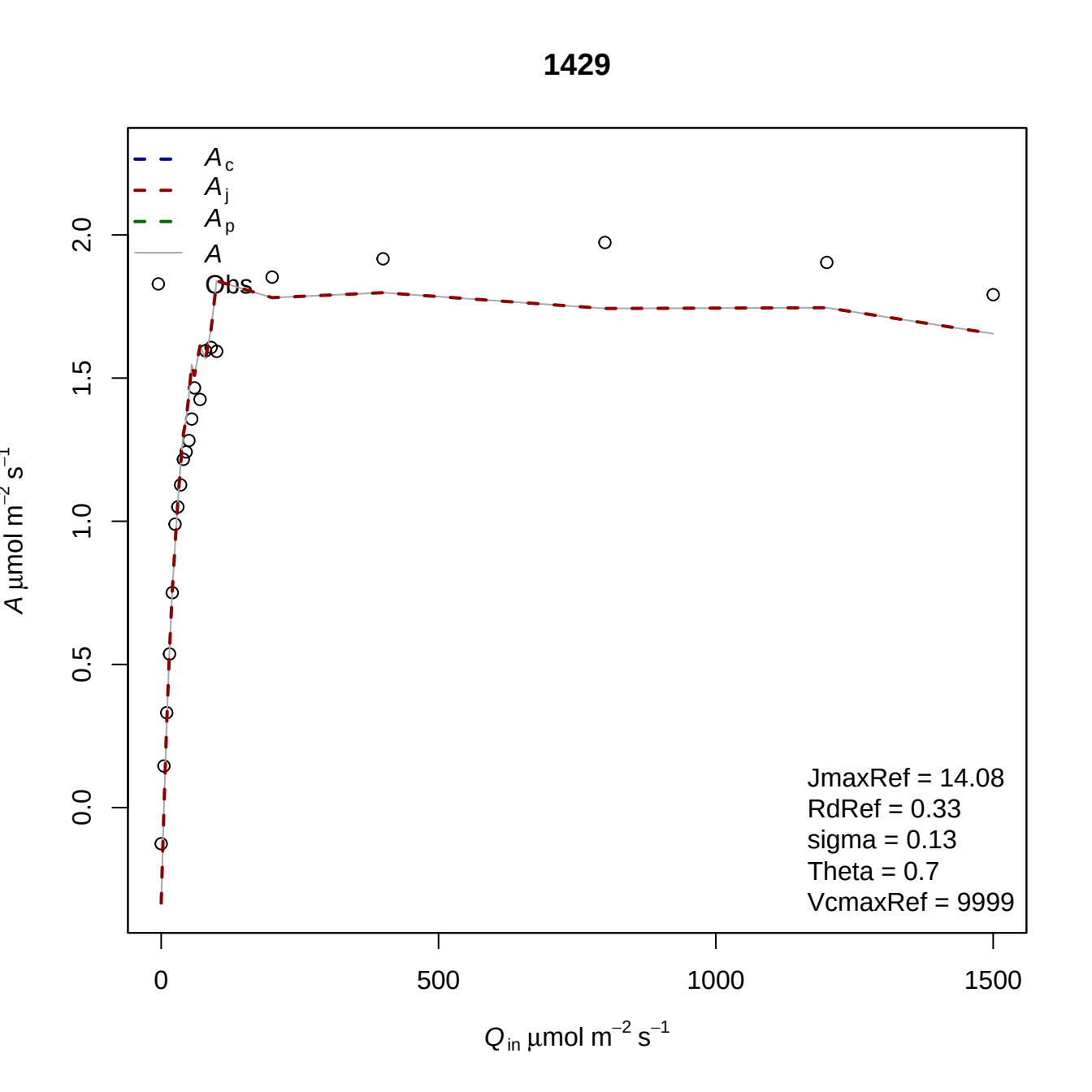

1424
A_c
A_j
A_p
A
Obs
JmaxRef = 33.02
RdRef = 0.53
sigma = 0.13
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

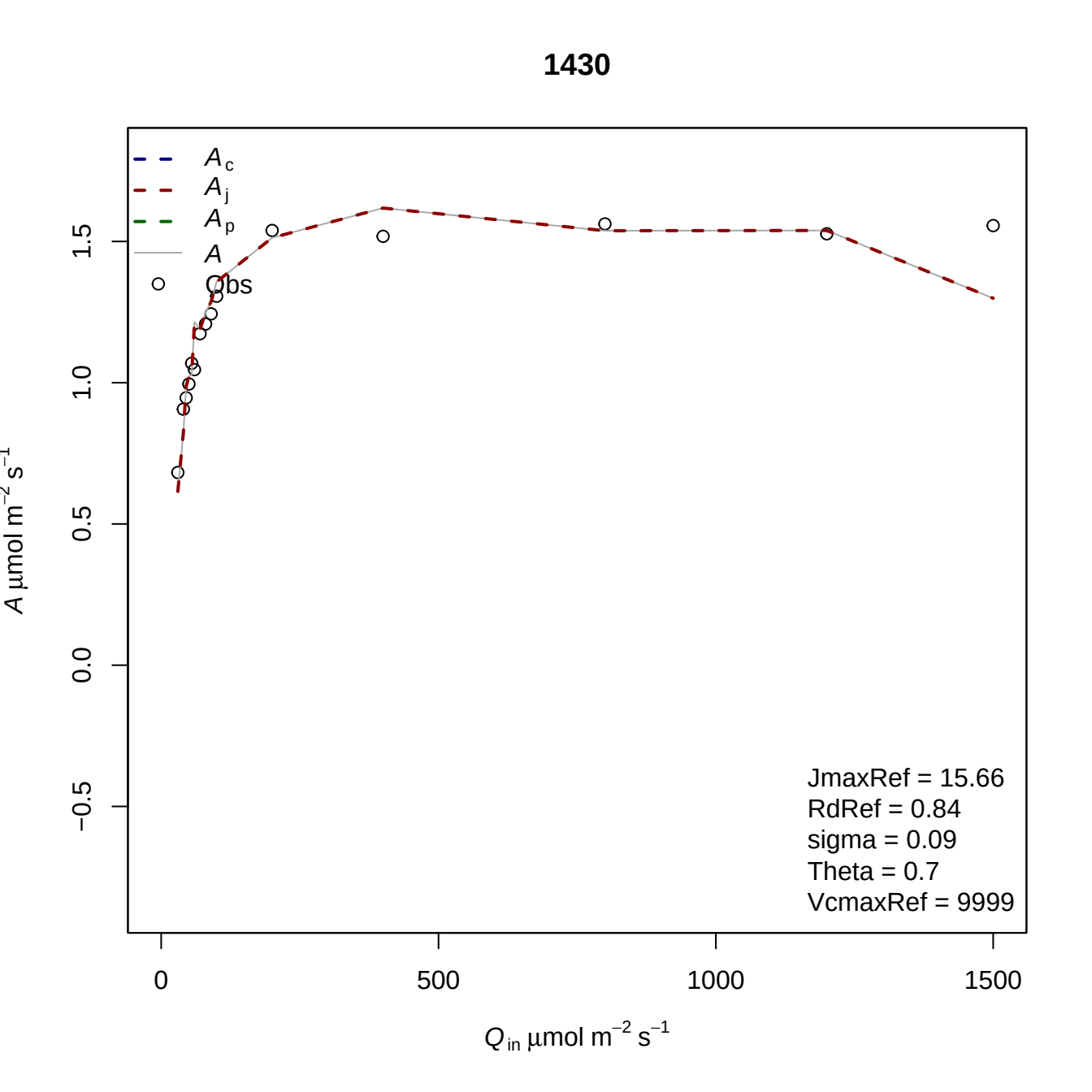

1425
A_c
A_j
A_p
A
Obs
JmaxRef = 19.05
RdRef = 0.87
sigma = 0.15
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
0
500
1000
1500

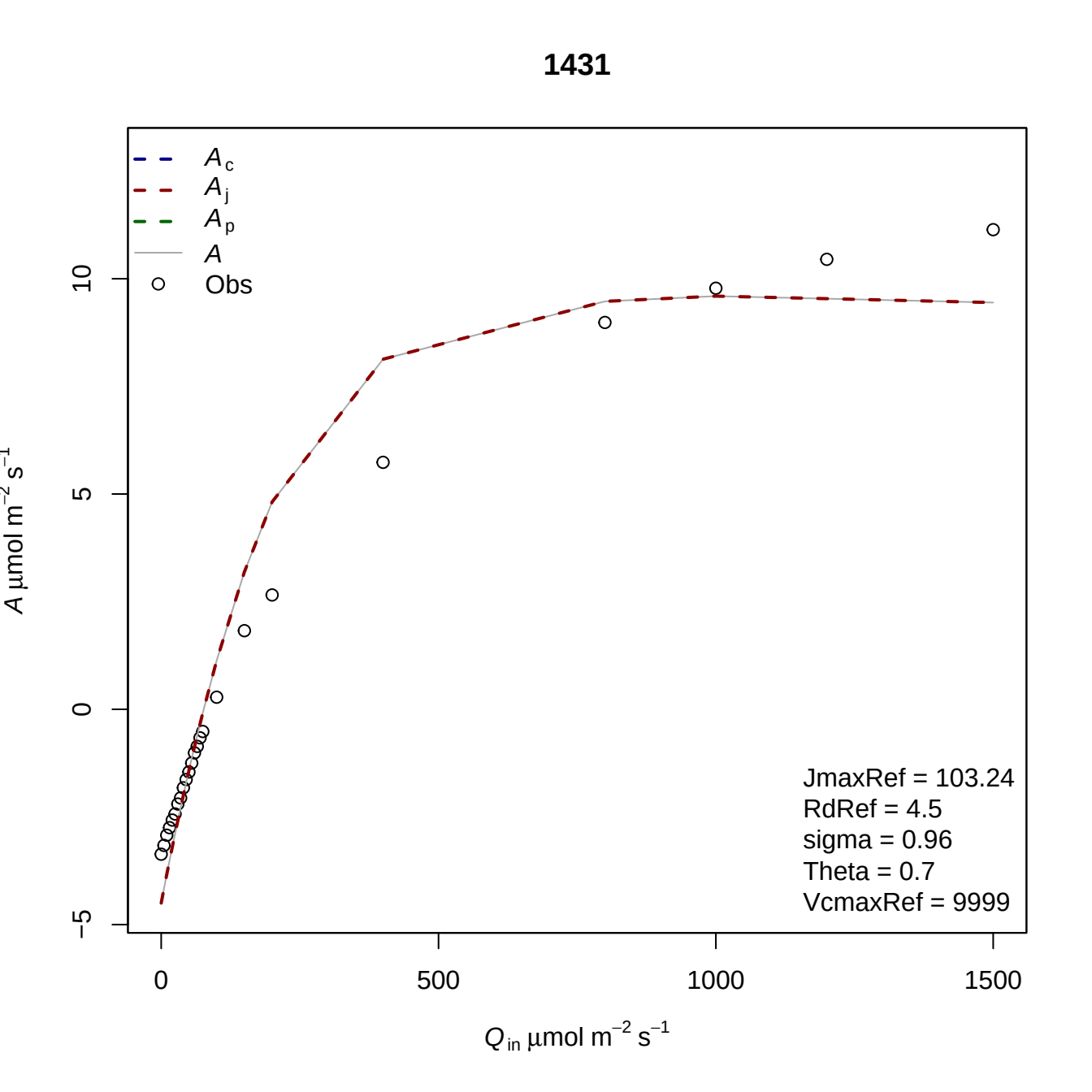

1426
A_c
A_j
A_p
A
Obs
JmaxRef = 31.61
RdRef = 0.33
sigma = 0.11
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
6
0
500
1000
1500

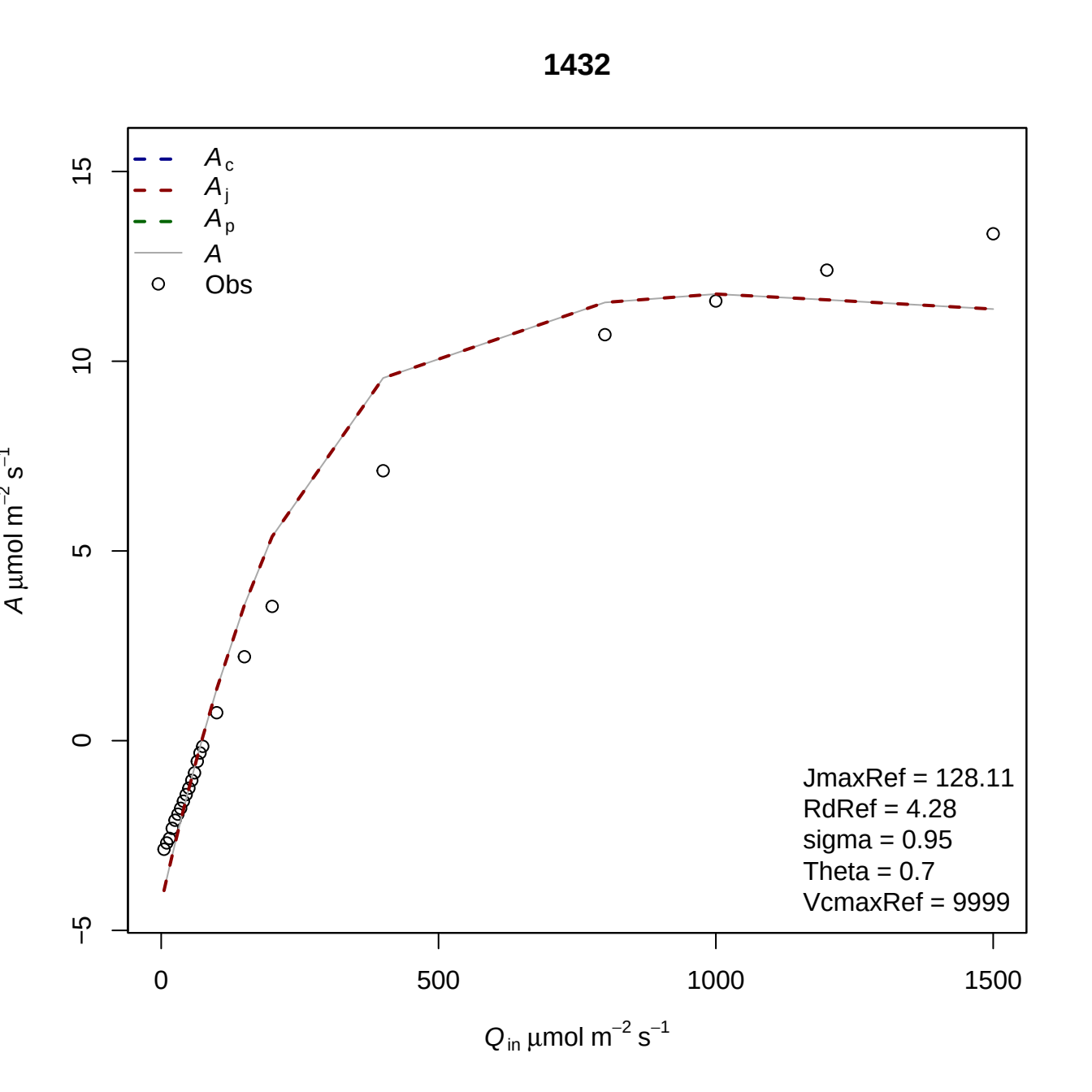

1427
A_c
A_j
A_p
A
Obs
JmaxRef = 17.52
RdRef = 0.32
sigma = 0.09
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
500
1000
1500

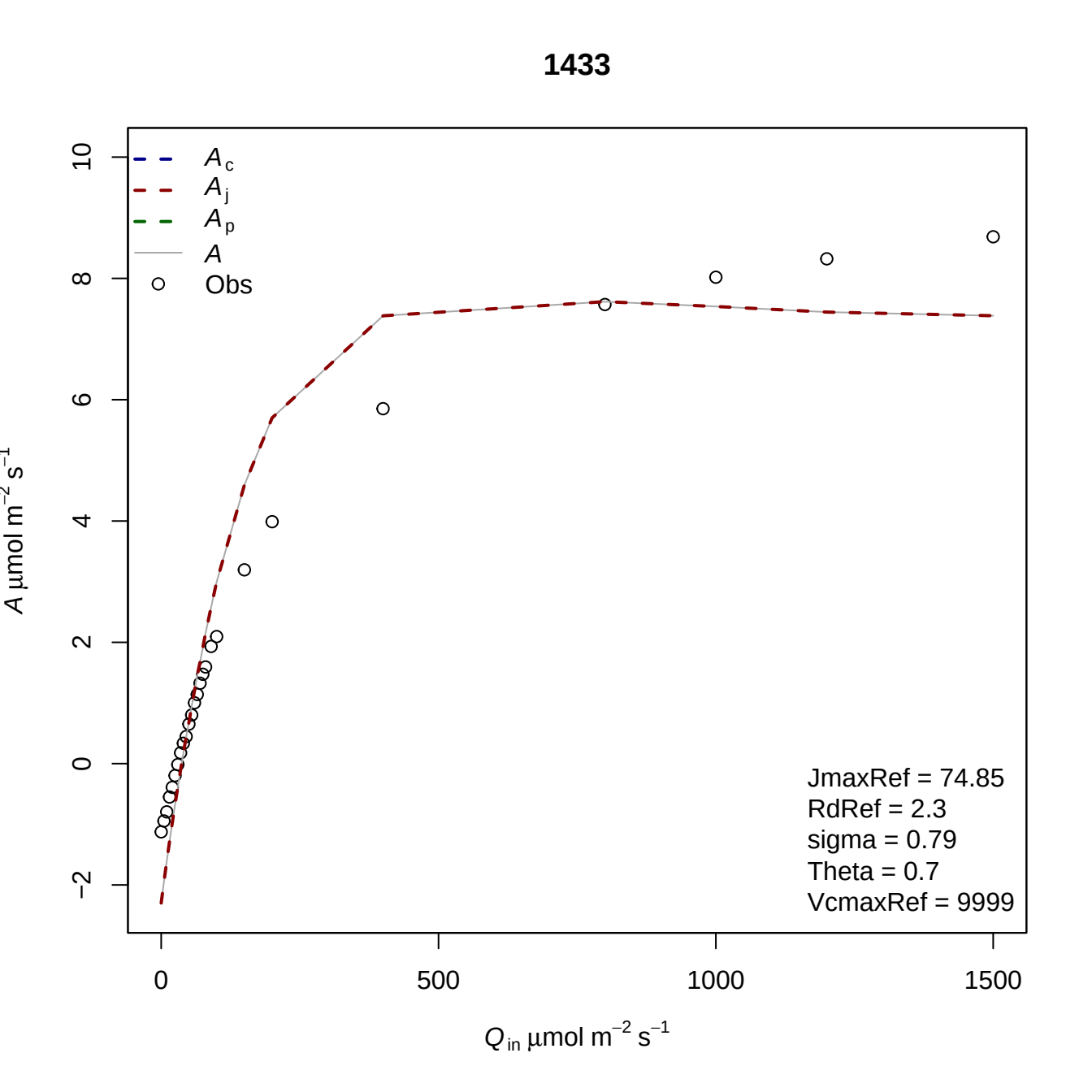

1428
A_c
A_j
A_p
A
Obs
JmaxRef = 13.3
RdRef = 0.5
sigma = 0.19
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
−0.5
0.0
0.5
1.0
1.5
2.0
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

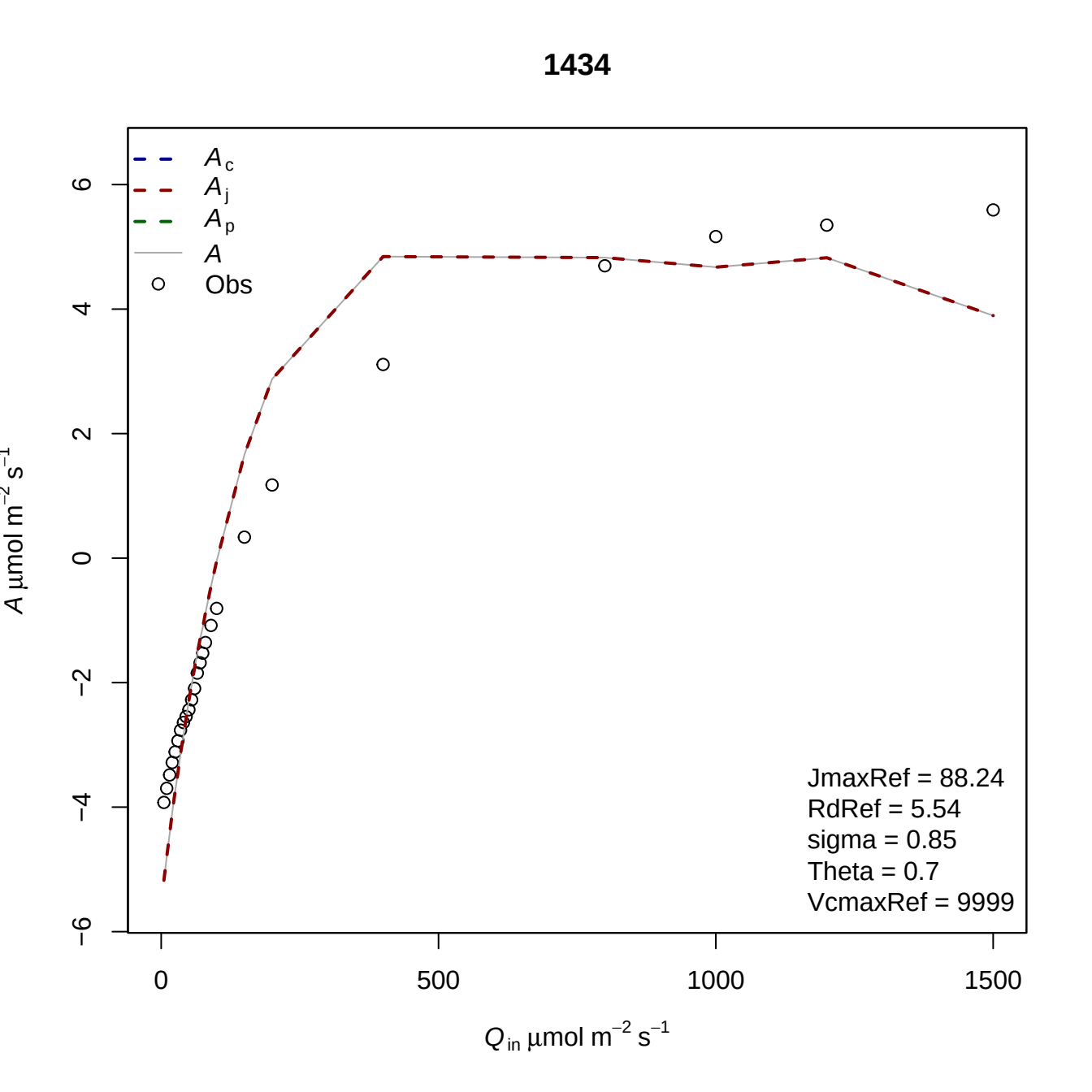

1429
A_c
A_j
A_p
A
Obs
JmaxRef = 14.08
RdRef = 0.33
sigma = 0.13
Theta = 0.7
VcmaxRef = 9999
0.0
0.5
1.0
1.5
2.0
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

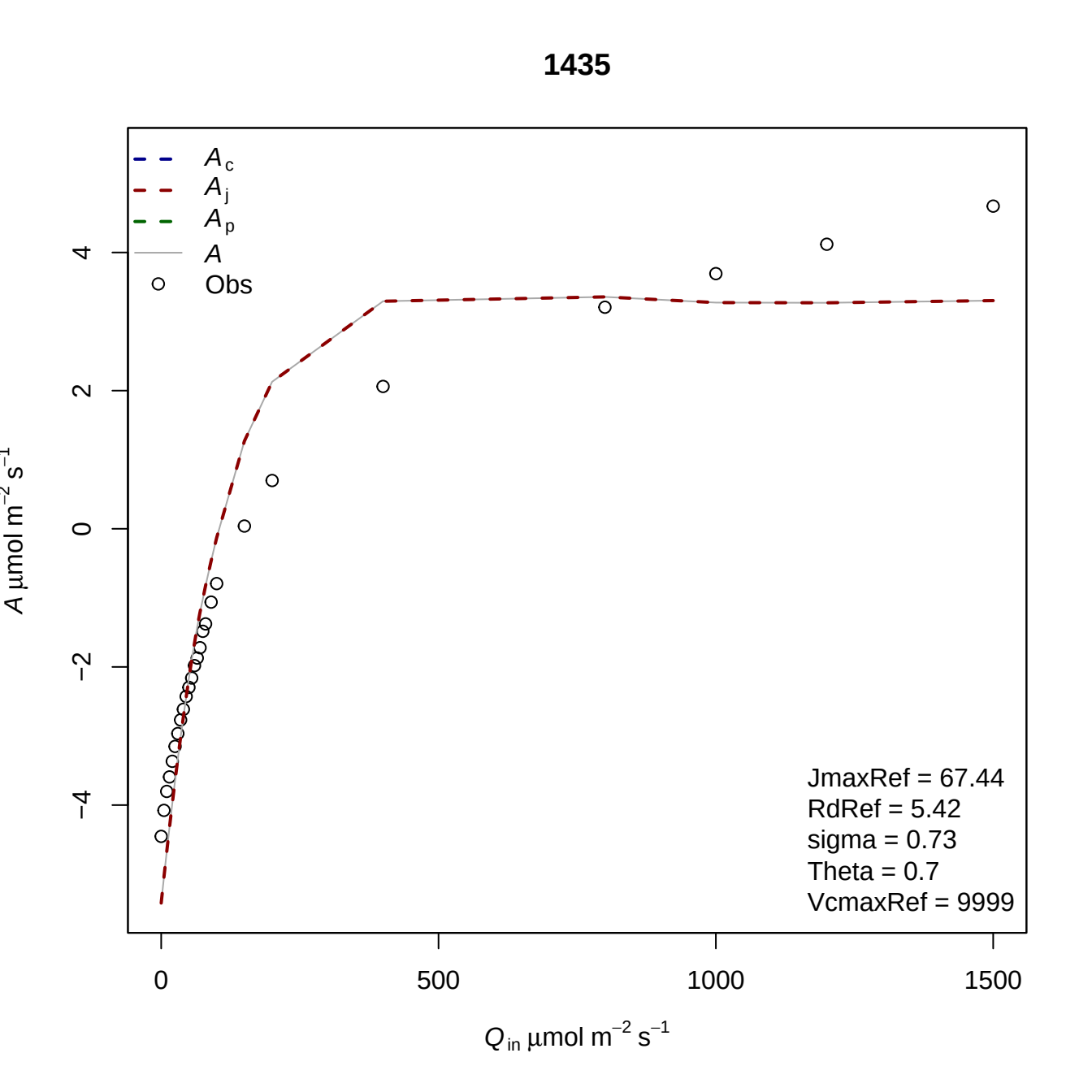

1430
A_c
A_j
A_p
A
Obs
JmaxRef = 15.66
RdRef = 0.84
sigma = 0.09
Theta = 0.7
VcmaxRef = 9999
−0.5
0.0
0.5
1.0
1.5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

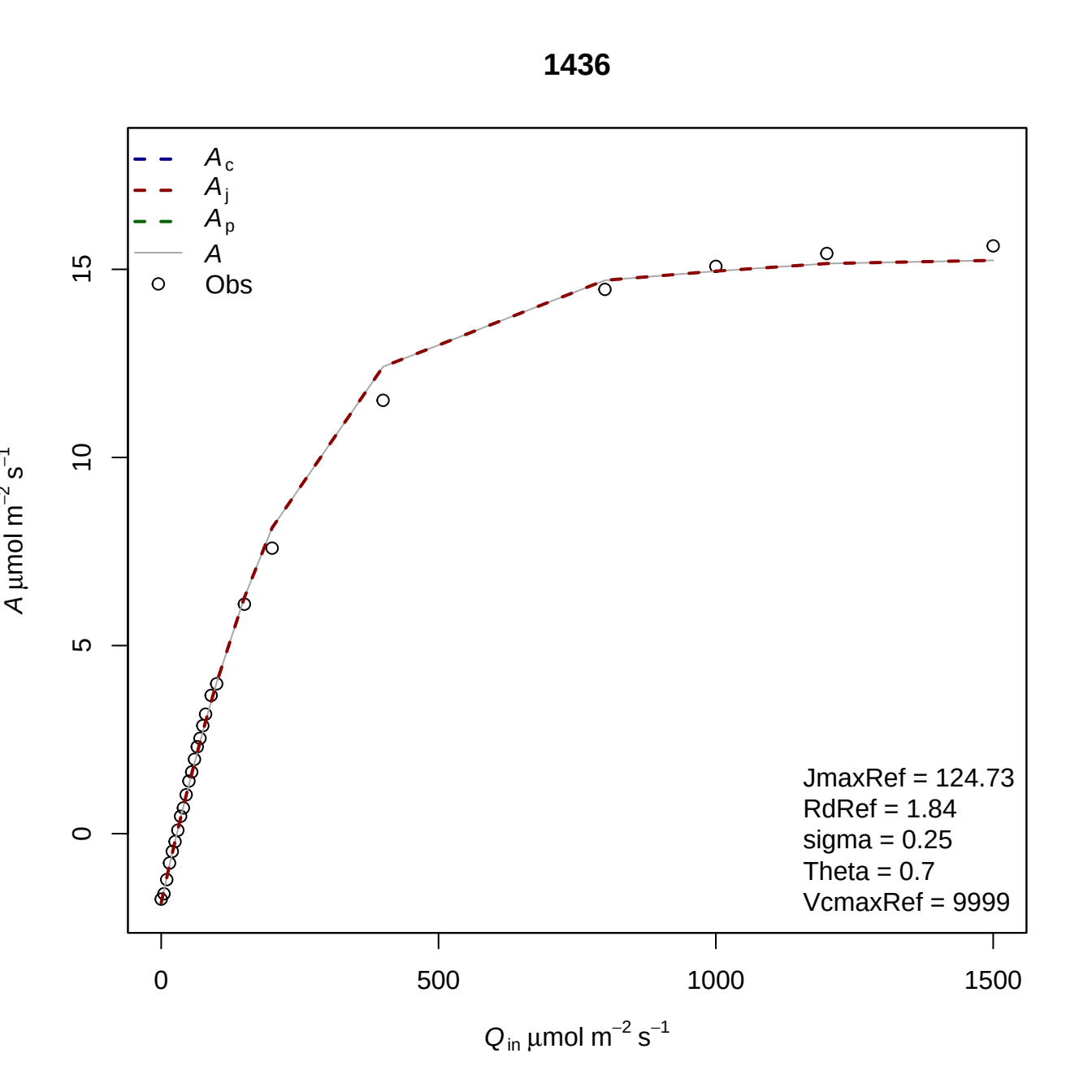

1431
A_c
A_j
A_p
A
Obs
JmaxRef = 103.24
RdRef = 4.5
sigma = 0.96
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500

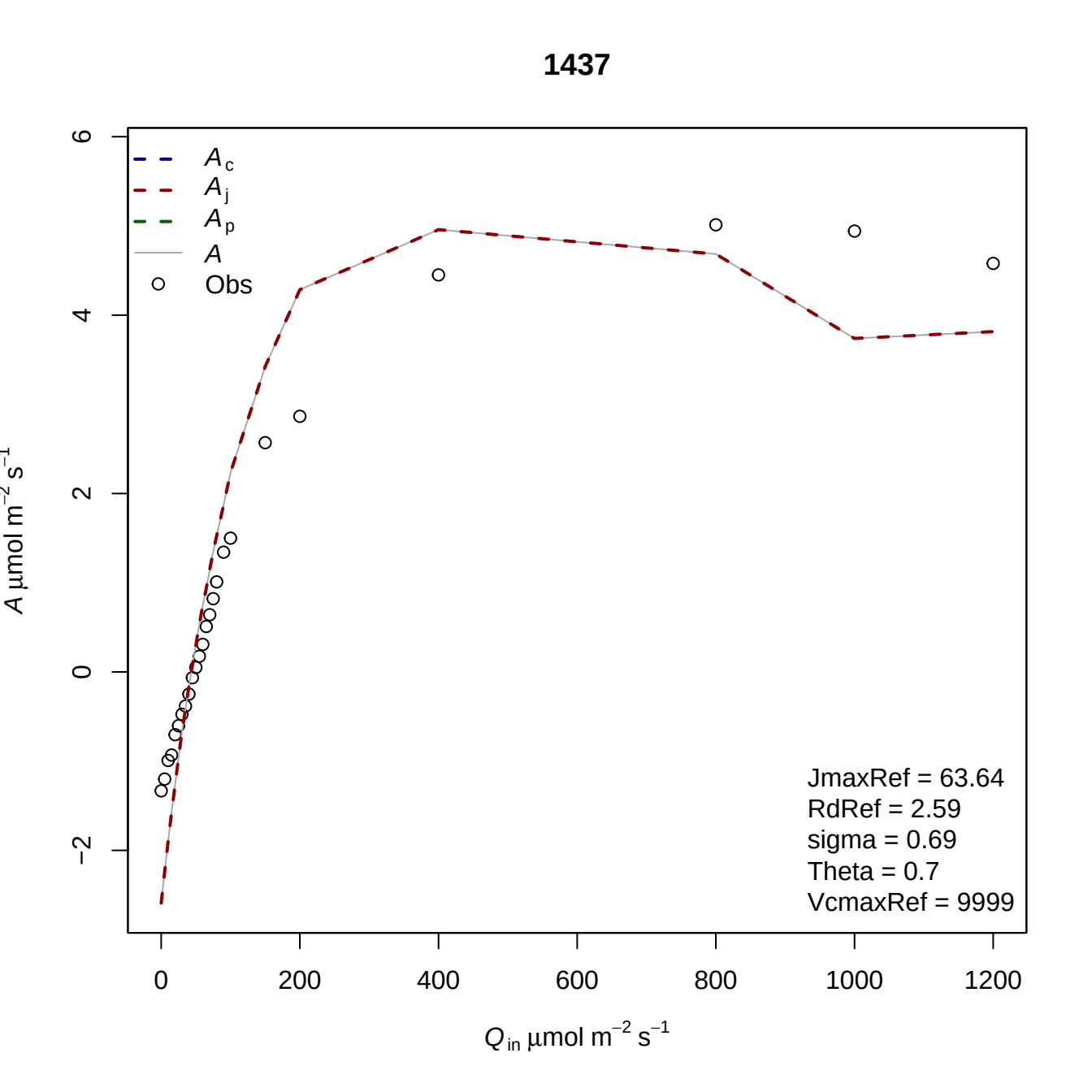

1432
A_c
A_j
A_p
A
Obs
JmaxRef = 128.11
RdRef = 4.28
sigma = 0.95
Theta = 0.7
VcmaxRef = 9999
15
10
5
0
−5
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

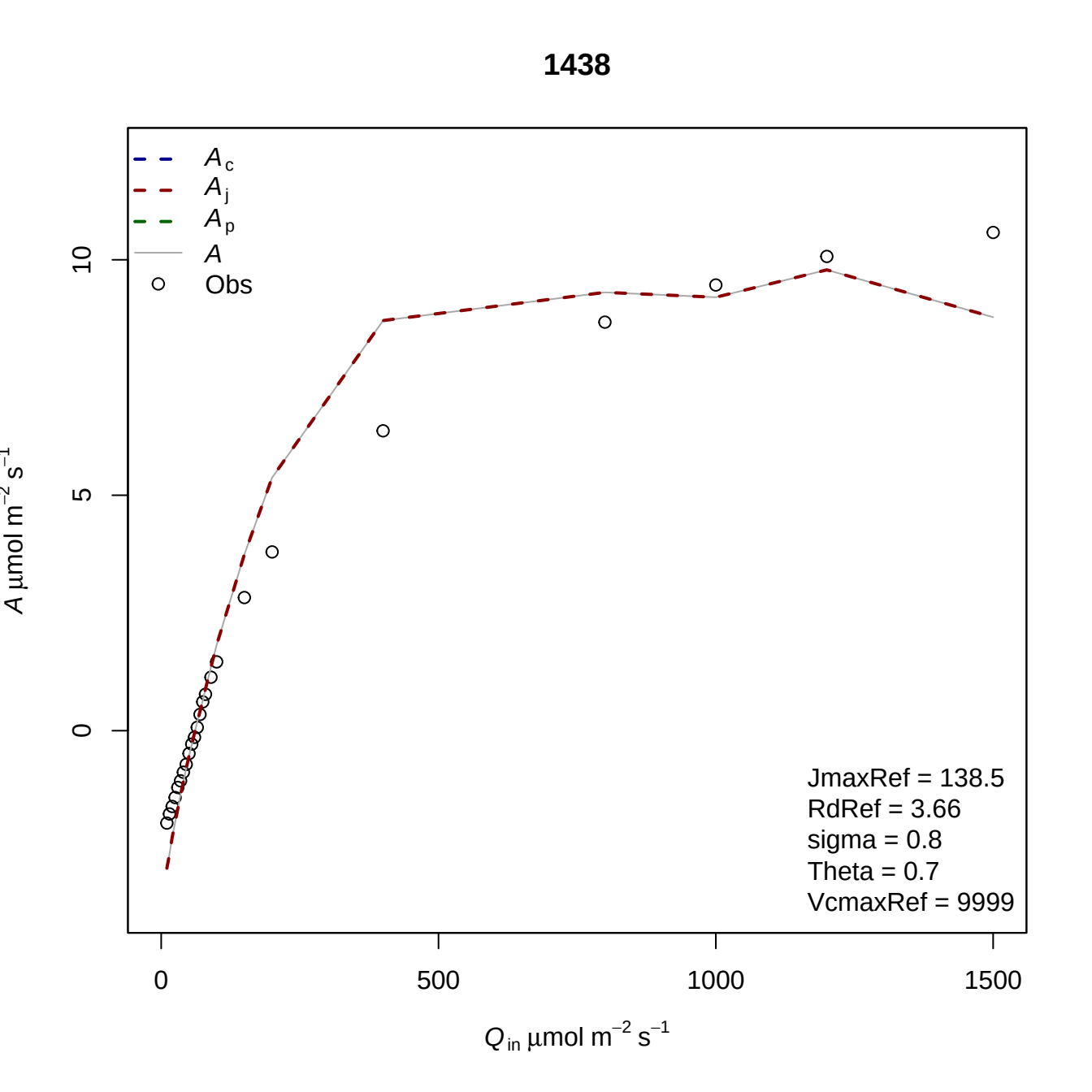

1433
A_c
A_j
A_p
A
Obs
JmaxRef = 74.85
RdRef = 2.3
sigma = 0.79
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

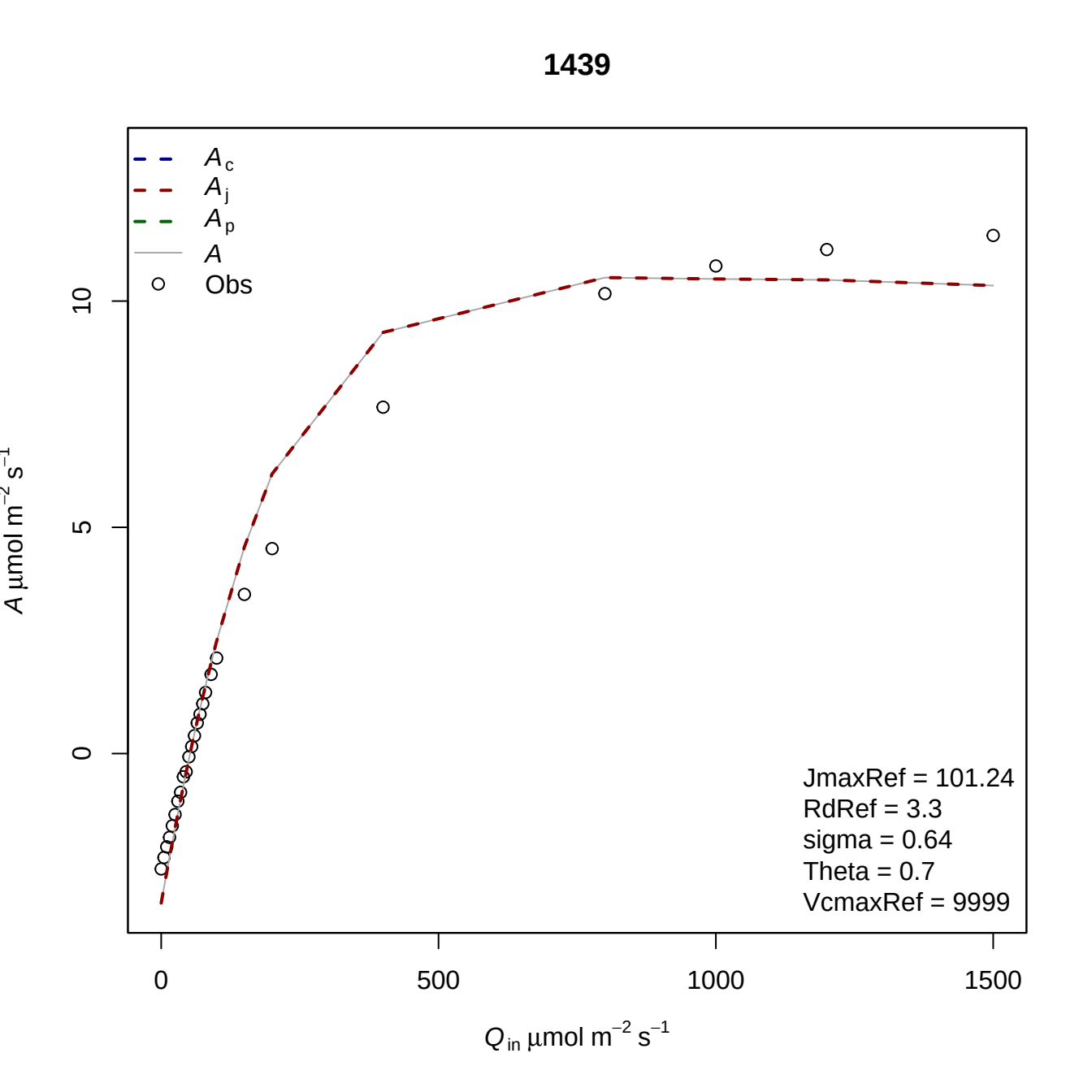

1434
A_c
A_j
A_p
A
Obs
JmaxRef = 88.24
RdRef = 5.54
sigma = 0.85
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
−6
0
500
1000
1500

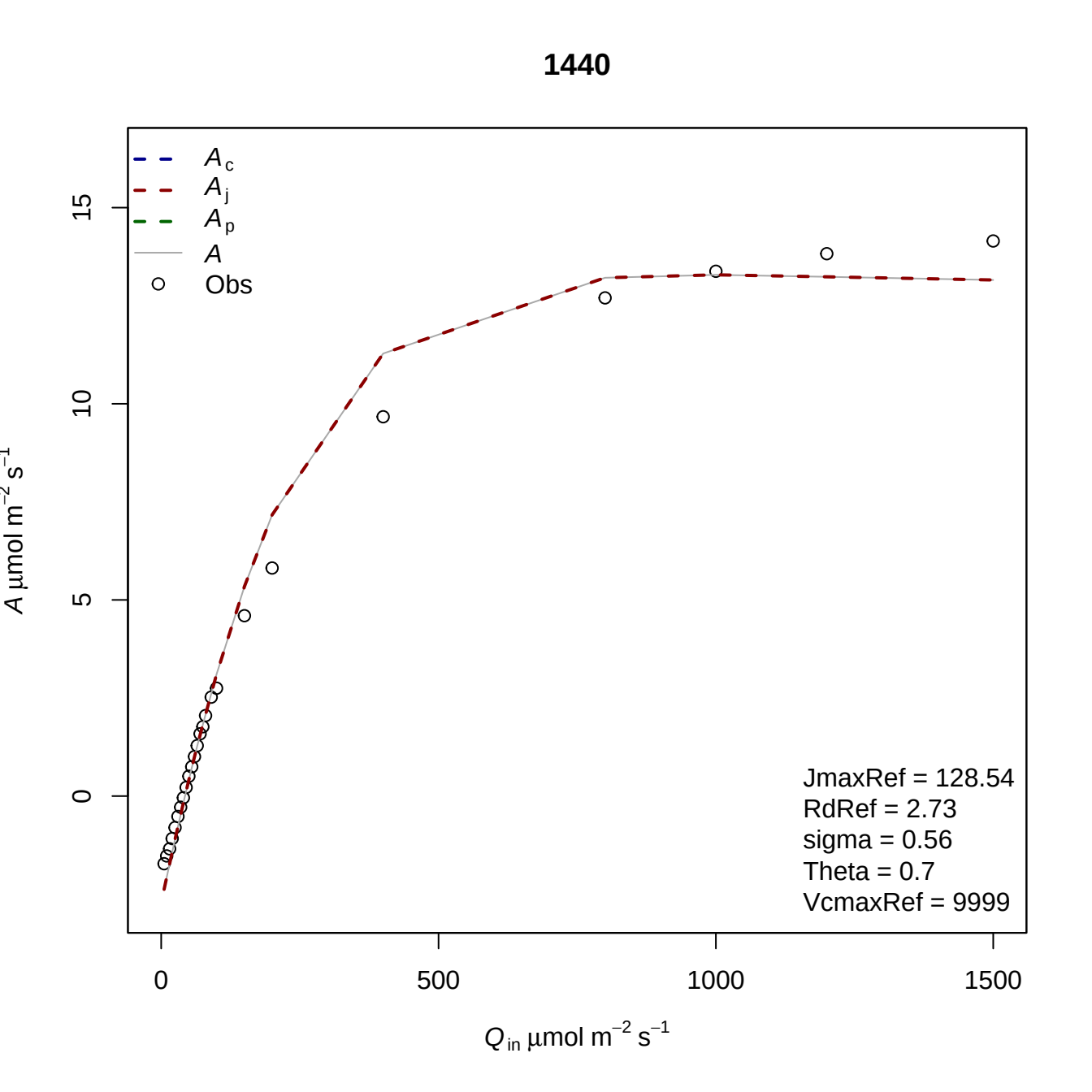

1435
A_c
A_j
A_p
A
Obs
JmaxRef = 67.44
RdRef = 5.42
sigma = 0.73
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
−4
0
500
1000
1500

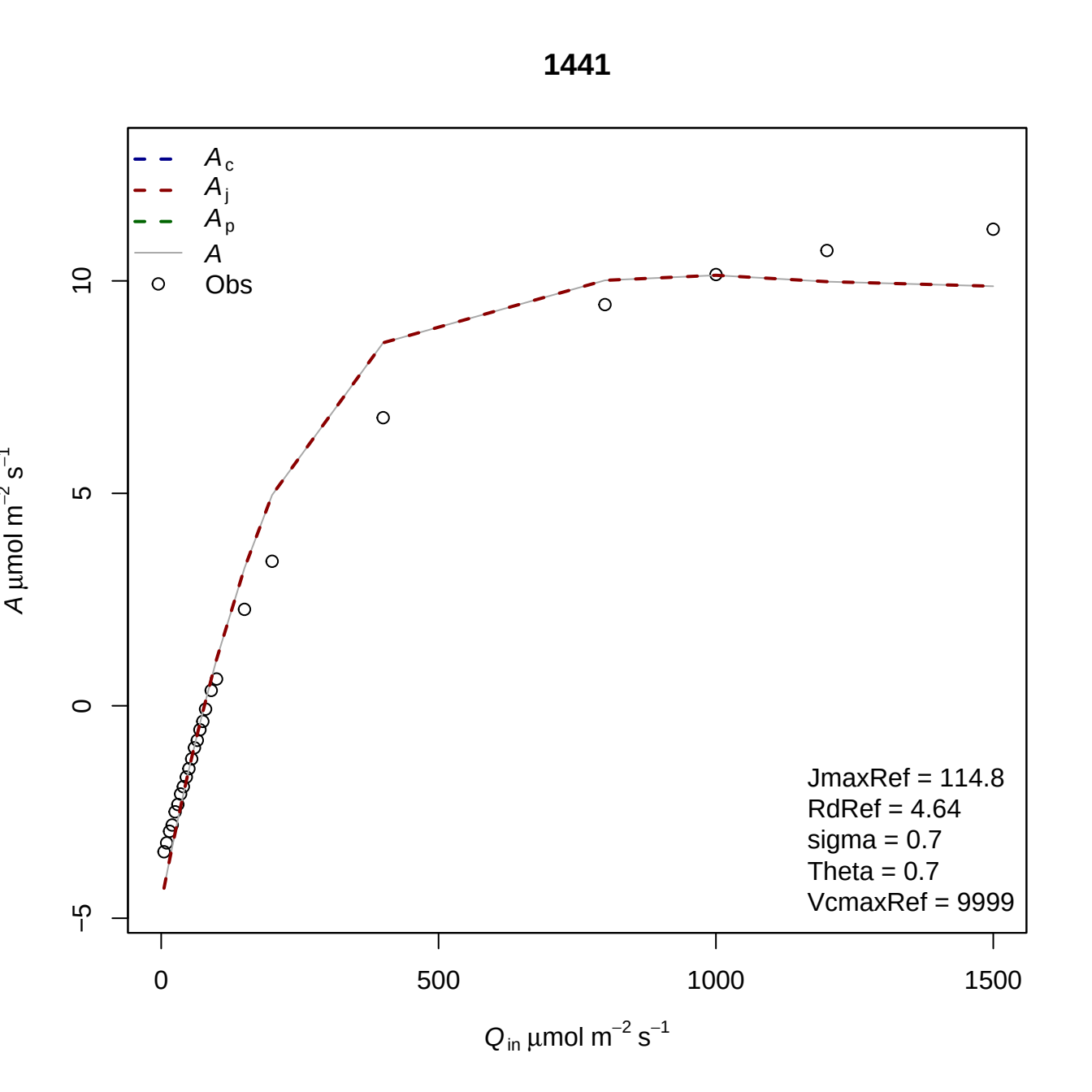

1436
A_c
A_j
A_p
A
Obs
JmaxRef = 124.73
RdRef = 1.84
sigma = 0.25
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

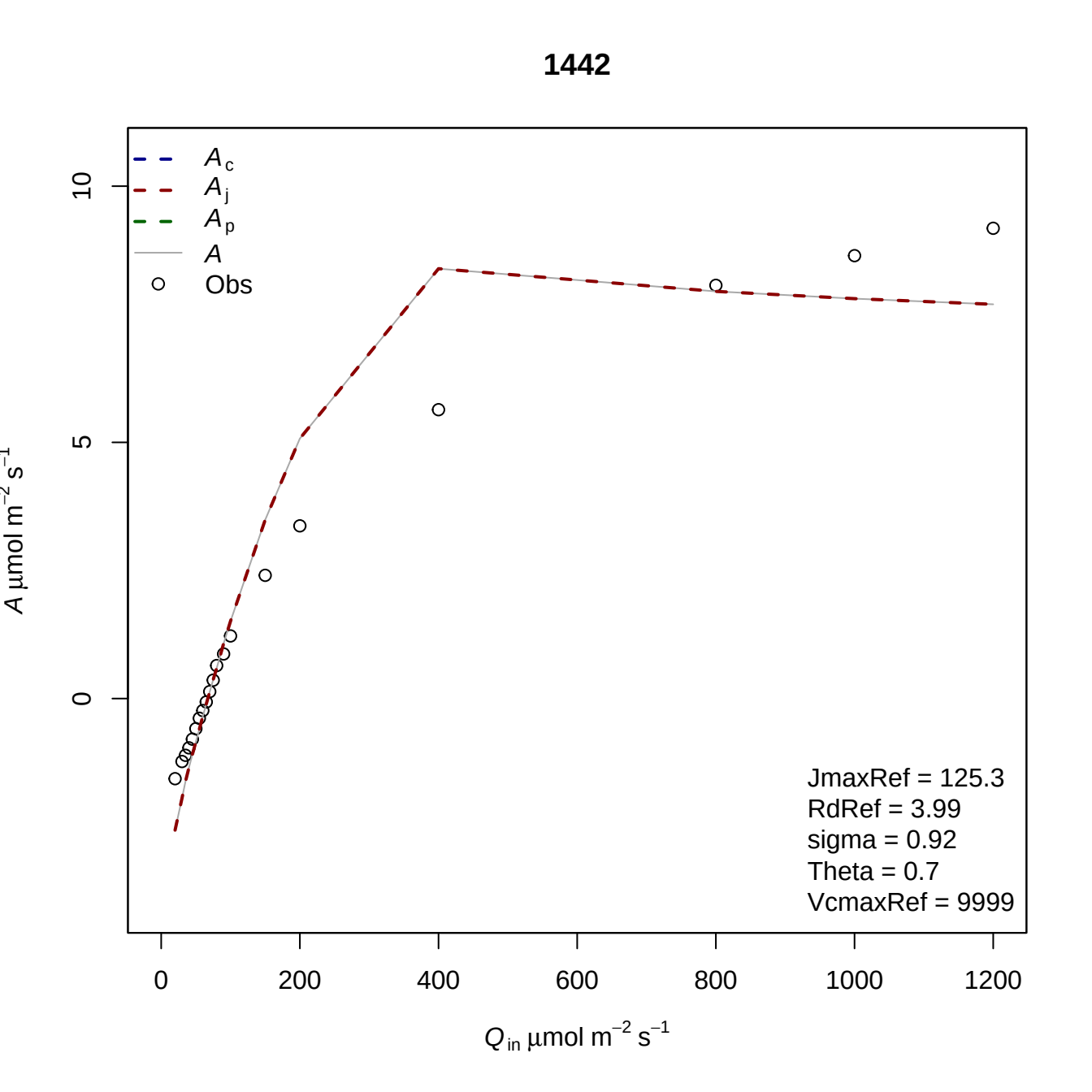

1437
A_c
A_j
A_p
A
Obs
JmaxRef = 63.64
RdRef = 2.59
sigma = 0.69
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
0
200
400
600
800
1000
1200

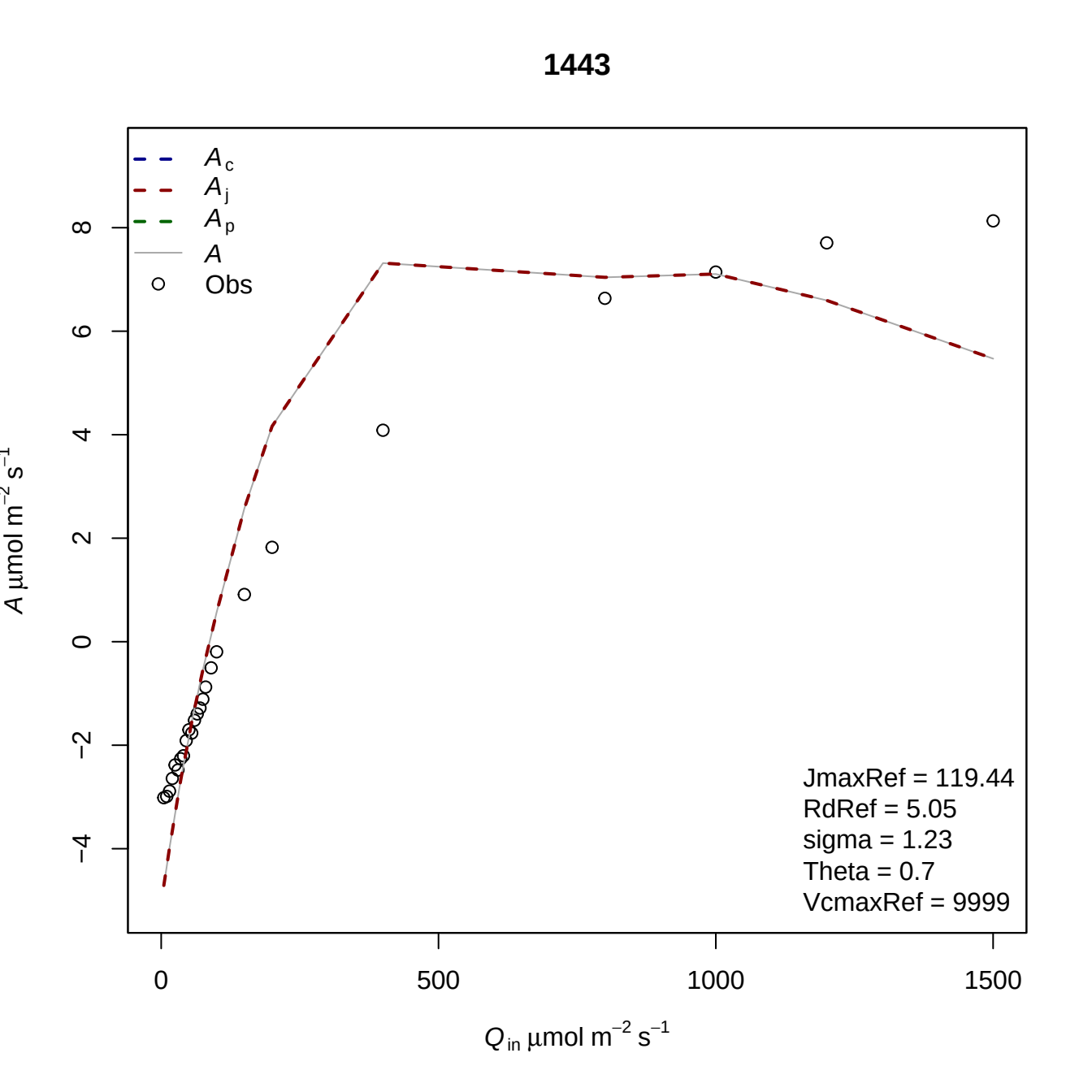

1438
A_c
A_j
A_p
A
Obs
JmaxRef = 138.5
RdRef = 3.66
sigma = 0.8
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

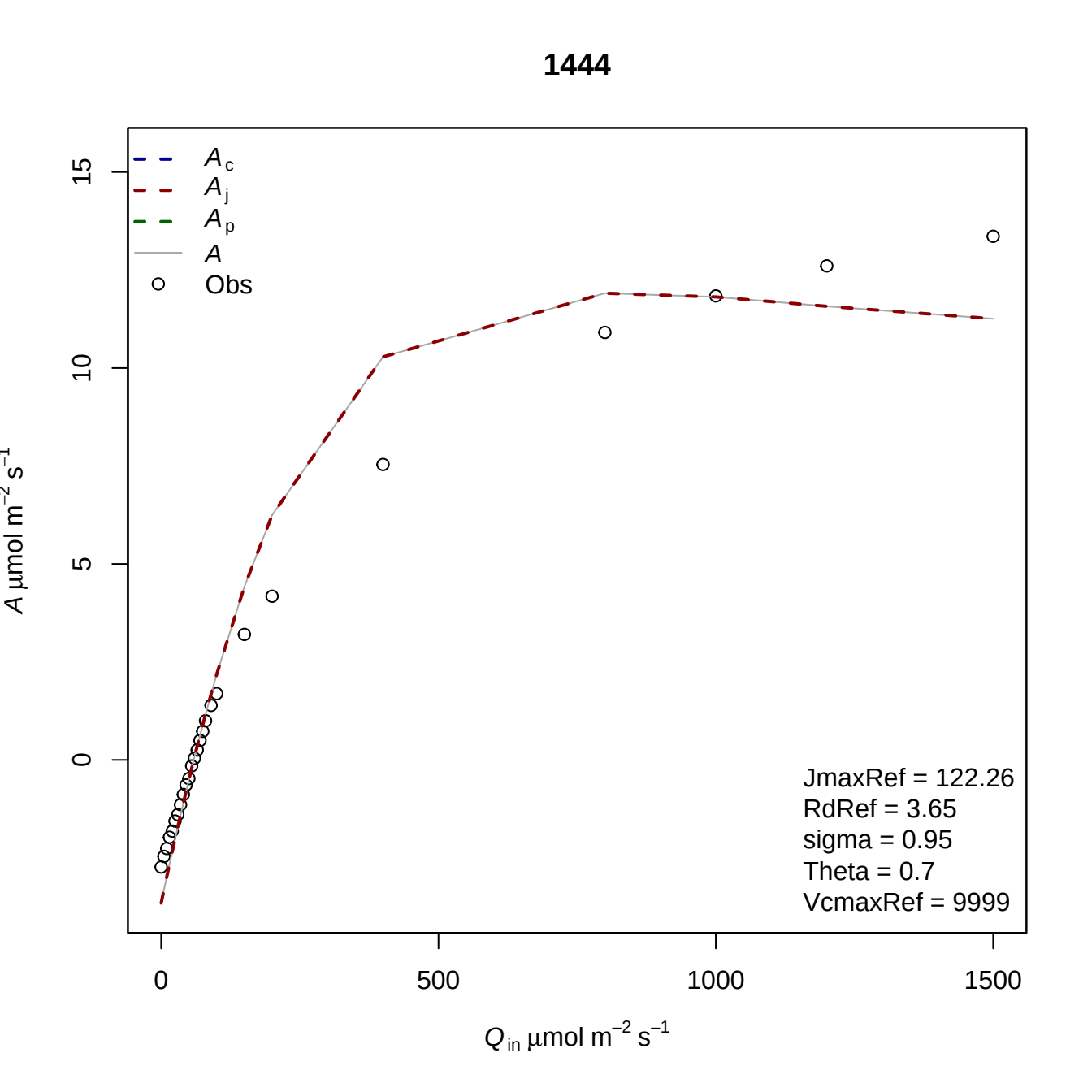

1439
A_c
A_j
A_p
A
Obs
JmaxRef = 101.24
RdRef = 3.3
sigma = 0.64
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

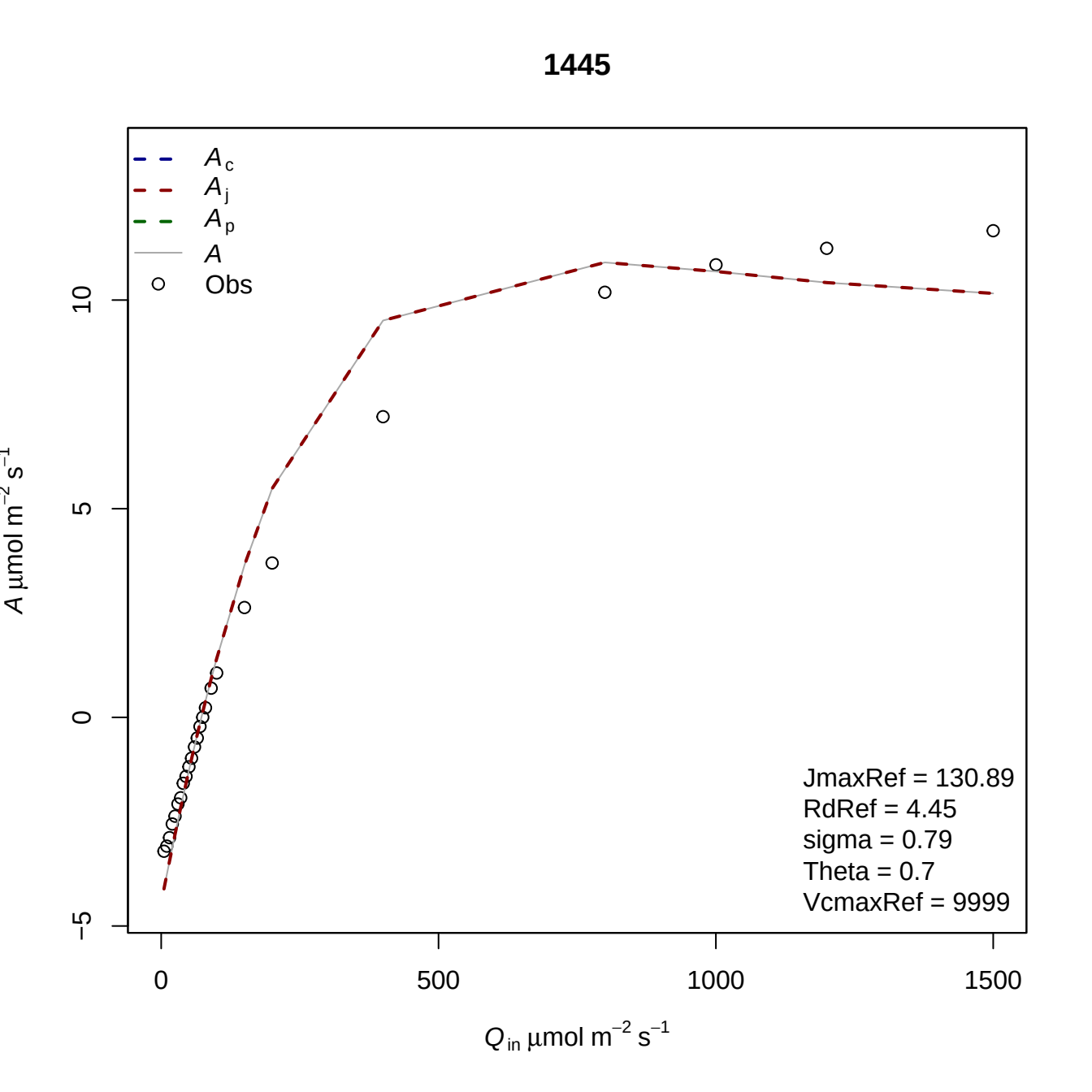

1440
A_c
A_j
A_p
A
Obs
JmaxRef = 128.54
RdRef = 2.73
sigma = 0.56
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

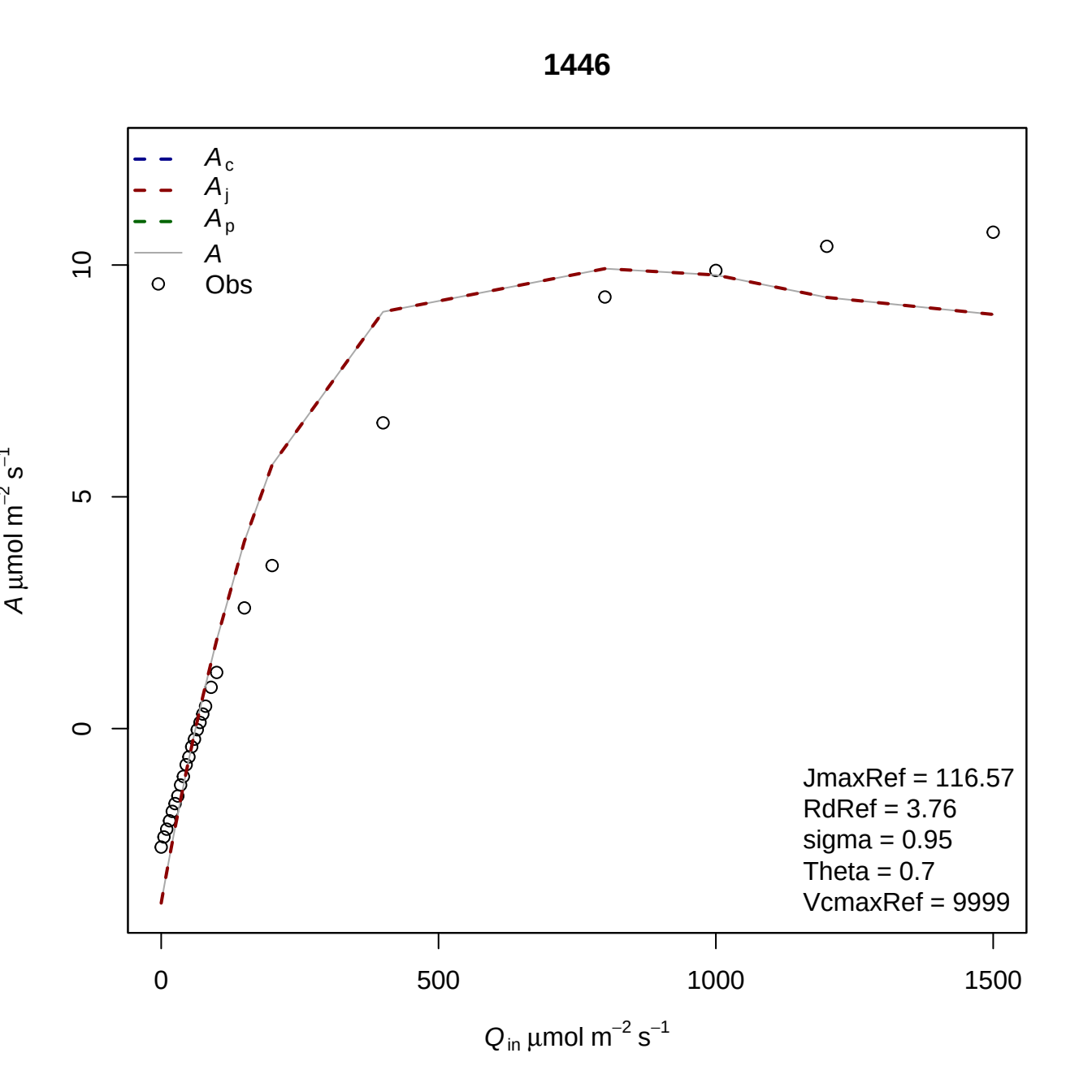

1441
A_c
A_j
A_p
A
Obs
JmaxRef = 114.8
RdRef = 4.64
sigma = 0.7
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500

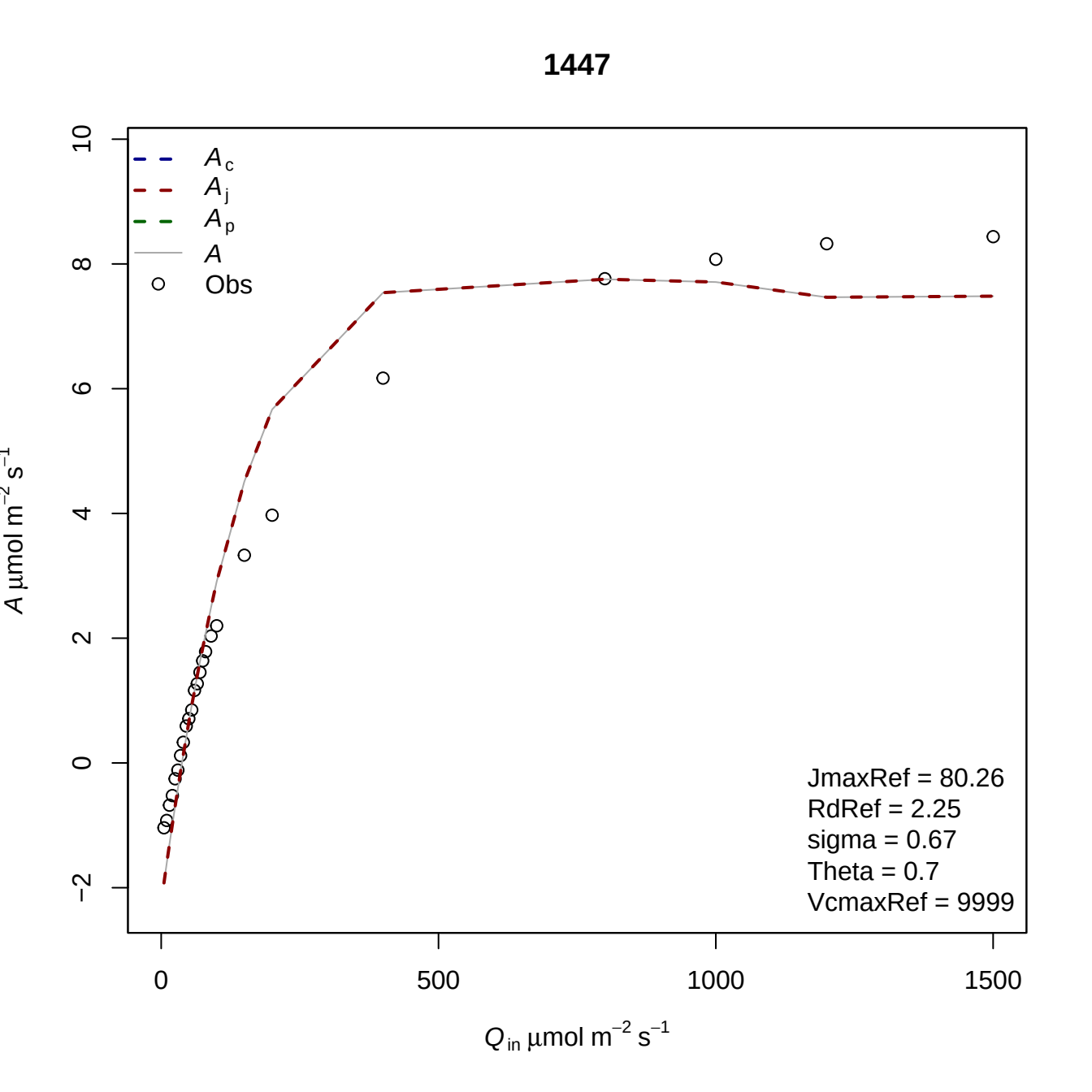

1442
A_c
A_j
A_p
A
Obs
JmaxRef = 125.3
RdRef = 3.99
sigma = 0.92
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
200
400
600
800
1000
1200

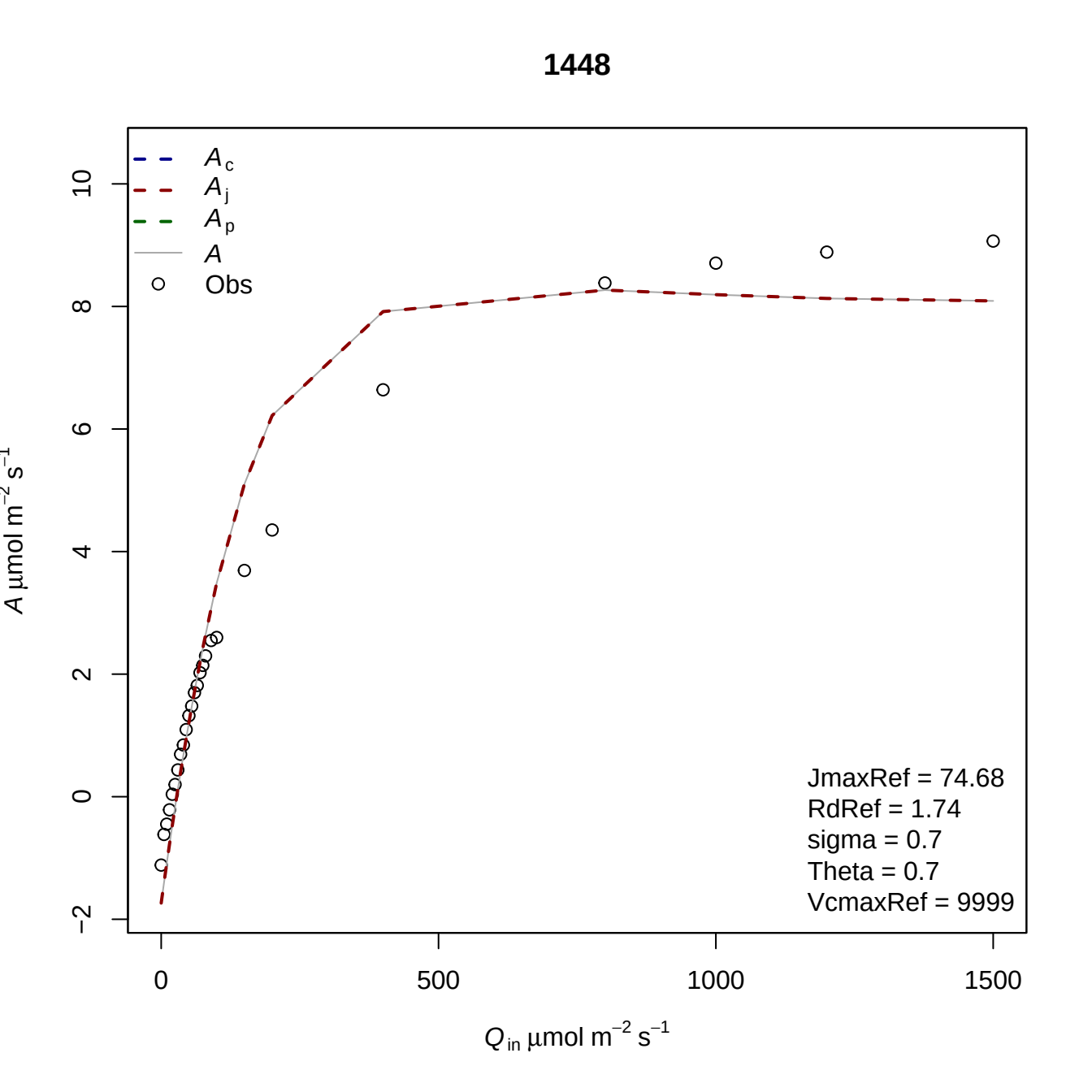

1443
A_c
A_j
A_p
A
Obs
JmaxRef = 119.44
RdRef = 5.05
sigma = 1.23
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

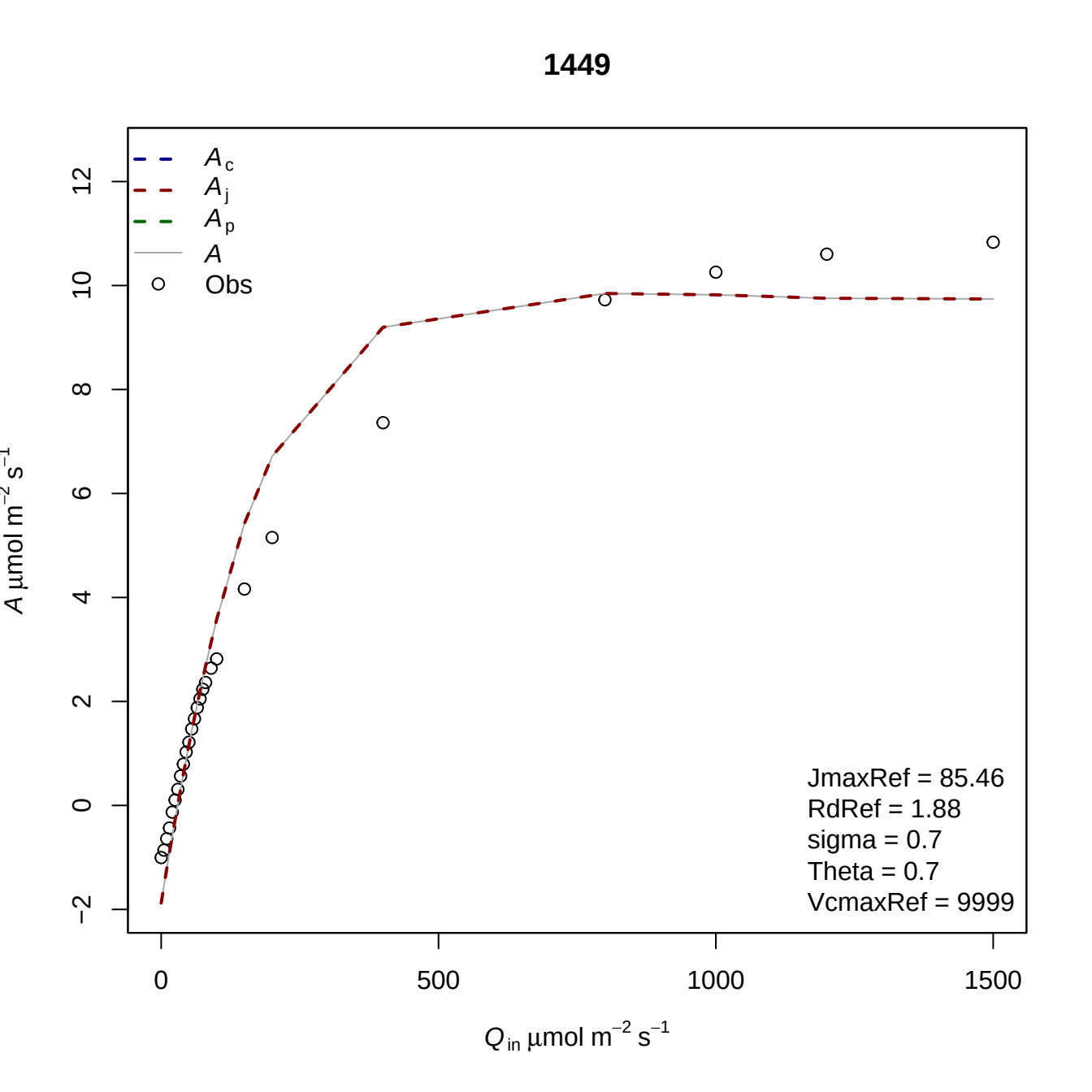

1444
A_c
A_j
A_p
A
Obs
JmaxRef = 122.26
RdRef = 3.65
sigma = 0.95
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

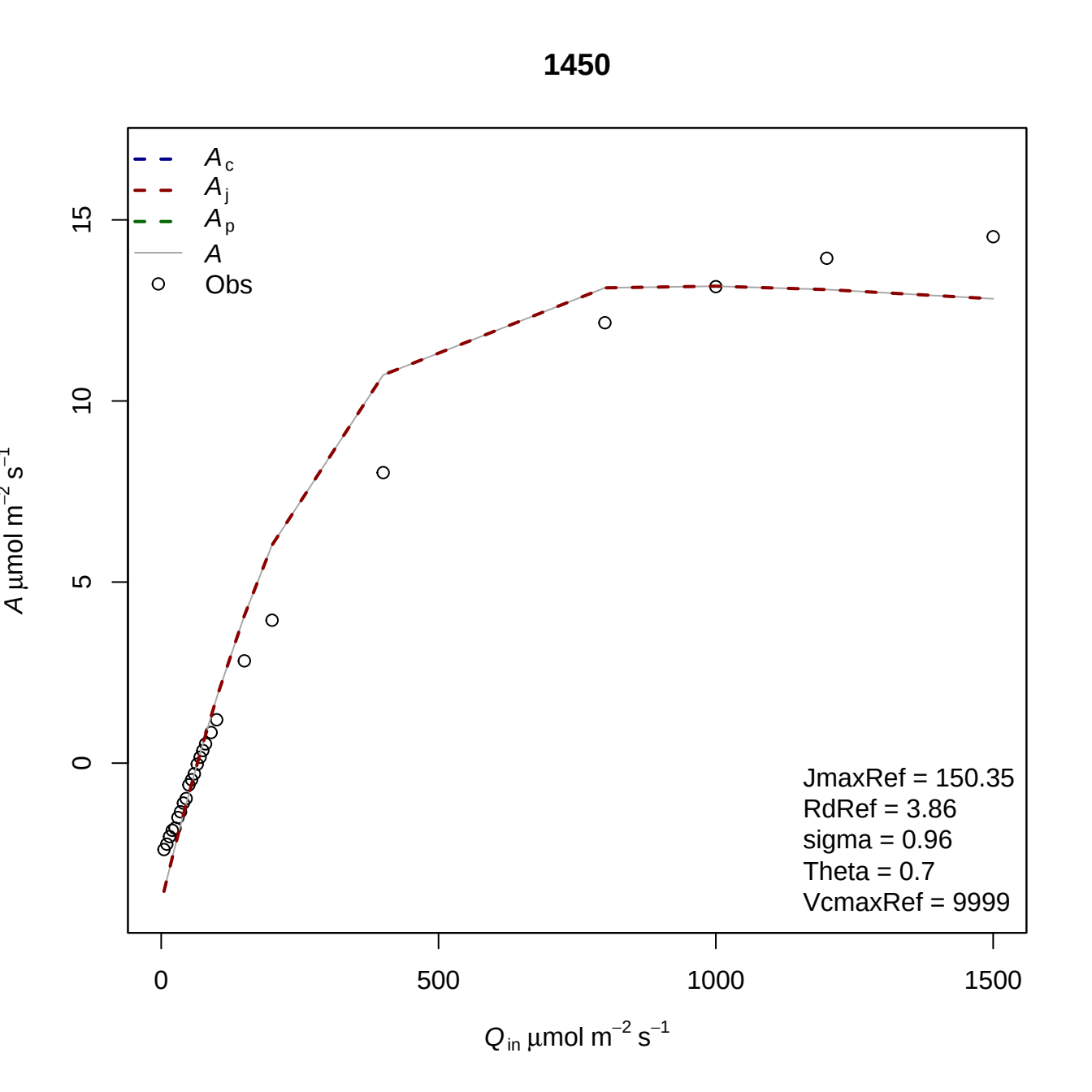

1445
A_c
A_j
A_p
A
Obs
JmaxRef = 130.89
RdRef = 4.45
sigma = 0.79
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500

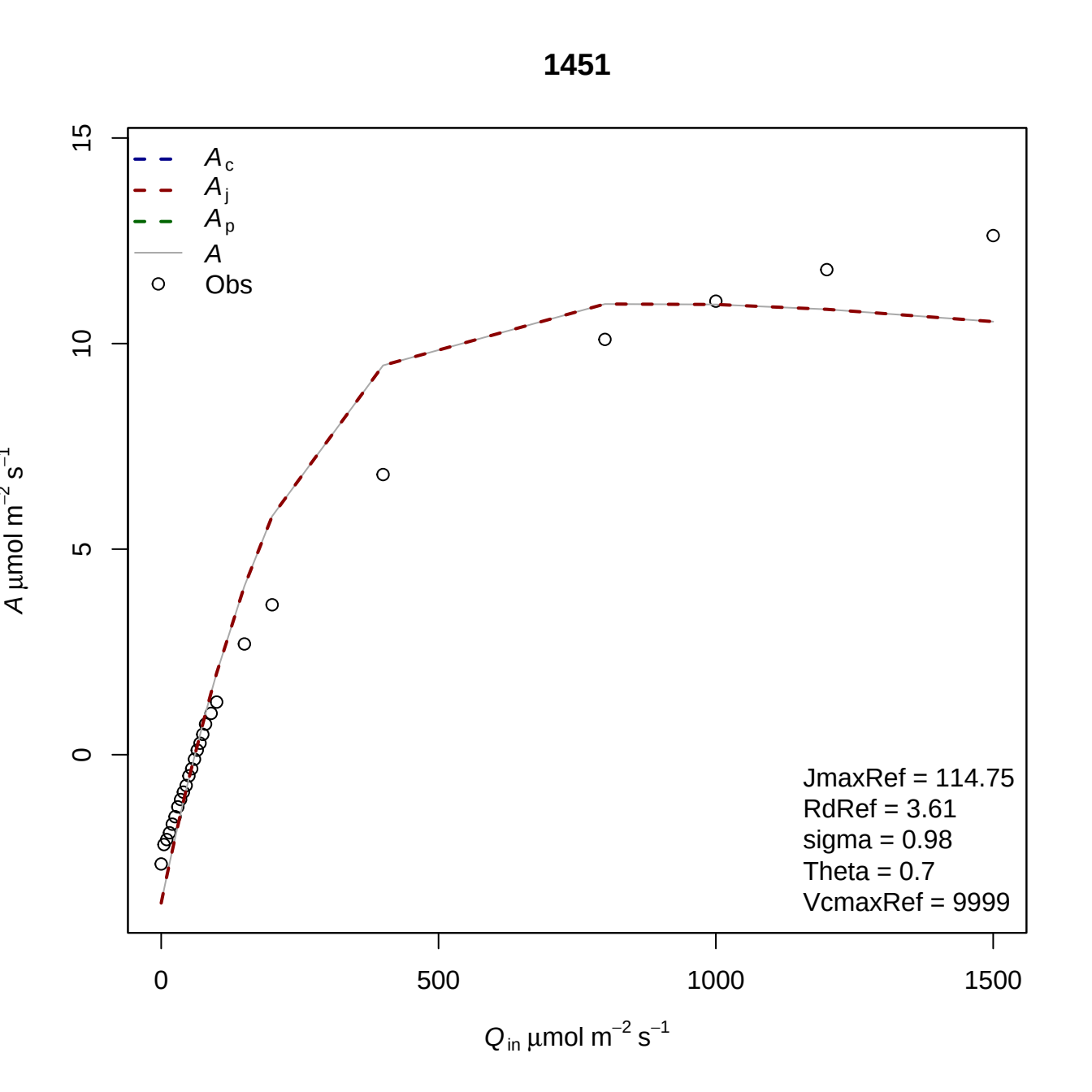

1446
A_c
A_j
A_p
A
Obs
JmaxRef = 116.57
RdRef = 3.76
sigma = 0.95
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

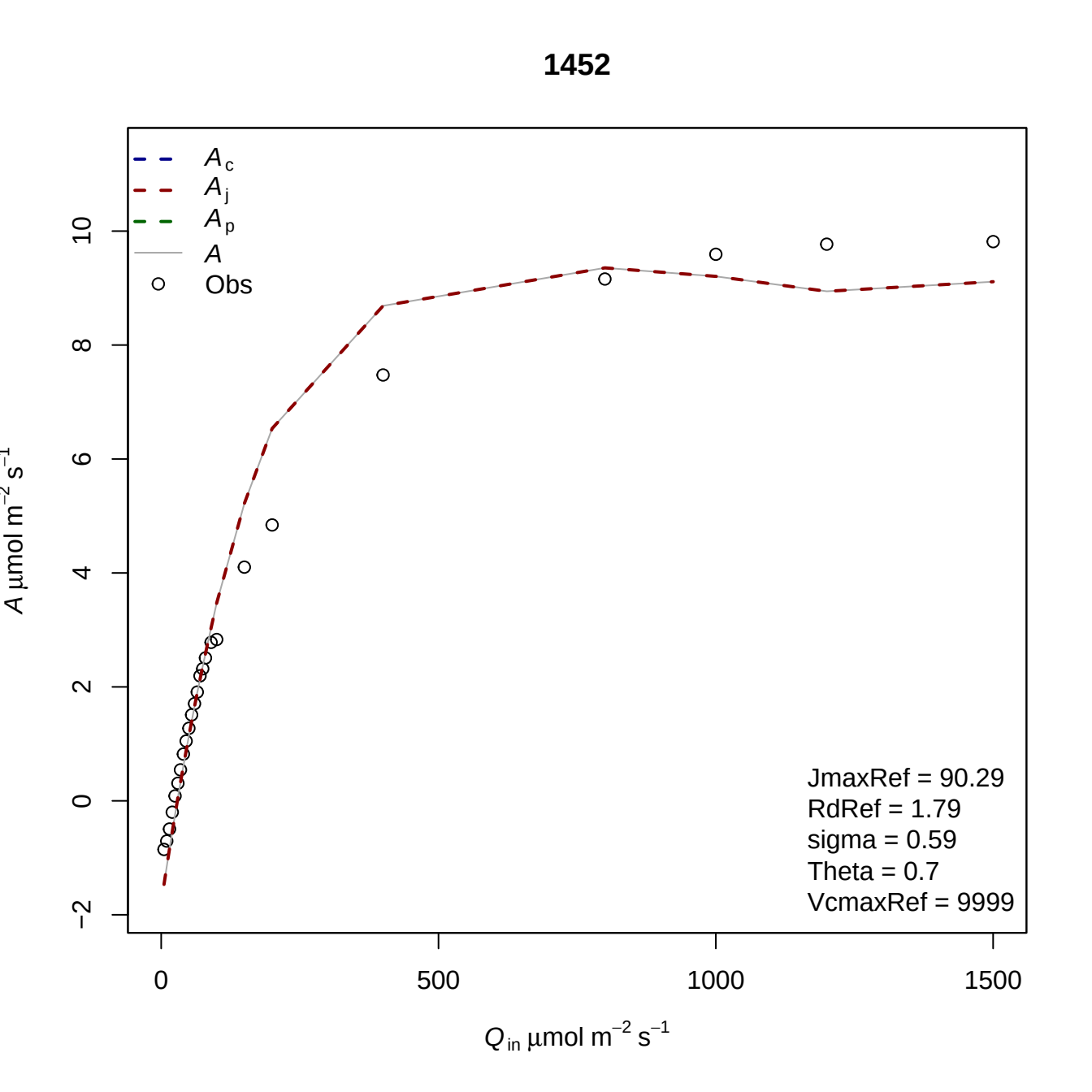

1447
A_c
A_j
A_p
A
Obs
JmaxRef = 80.26
RdRef = 2.25
sigma = 0.67
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

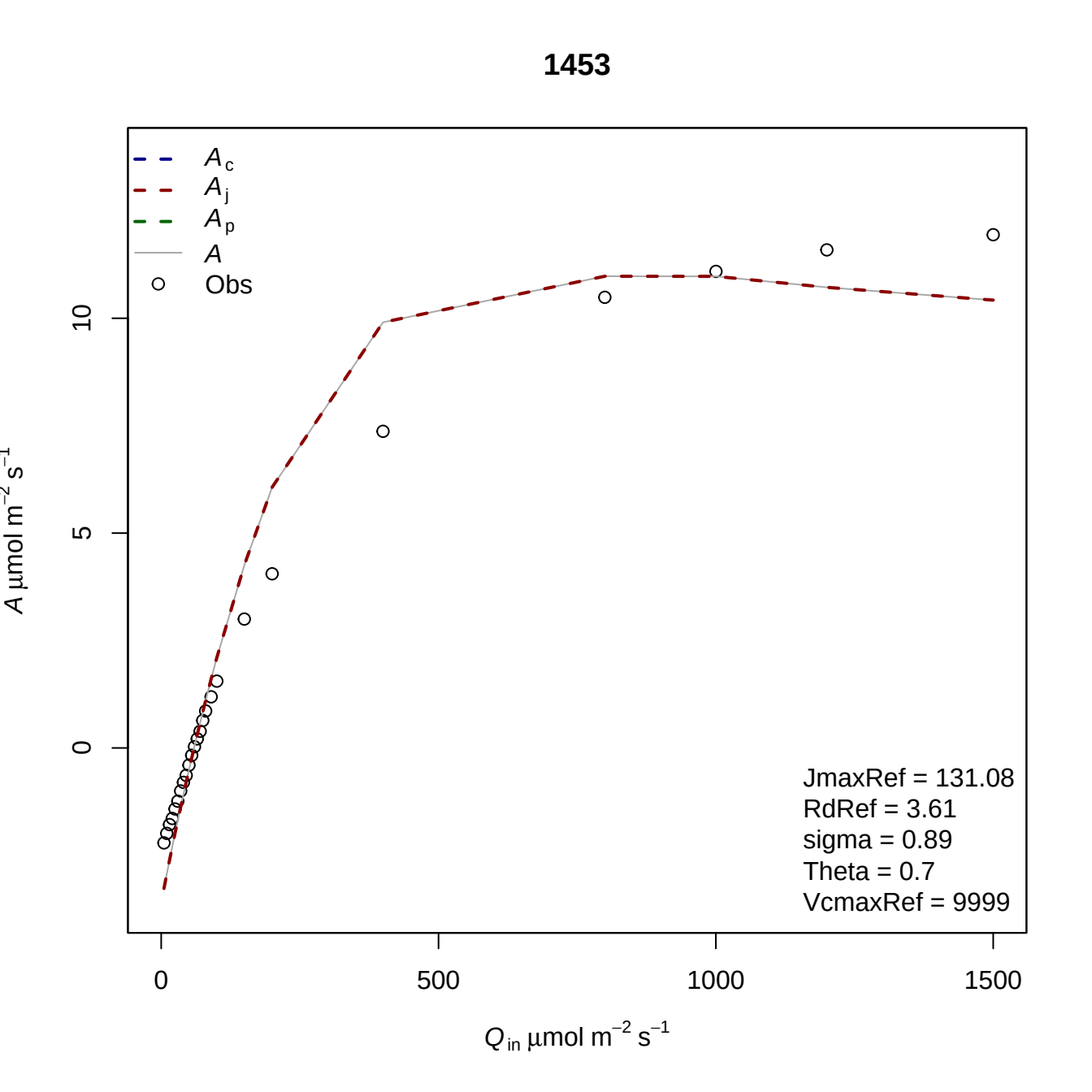

1448
A_c
A_j
A_p
A
Obs
JmaxRef = 74.68
RdRef = 1.74
sigma = 0.7
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

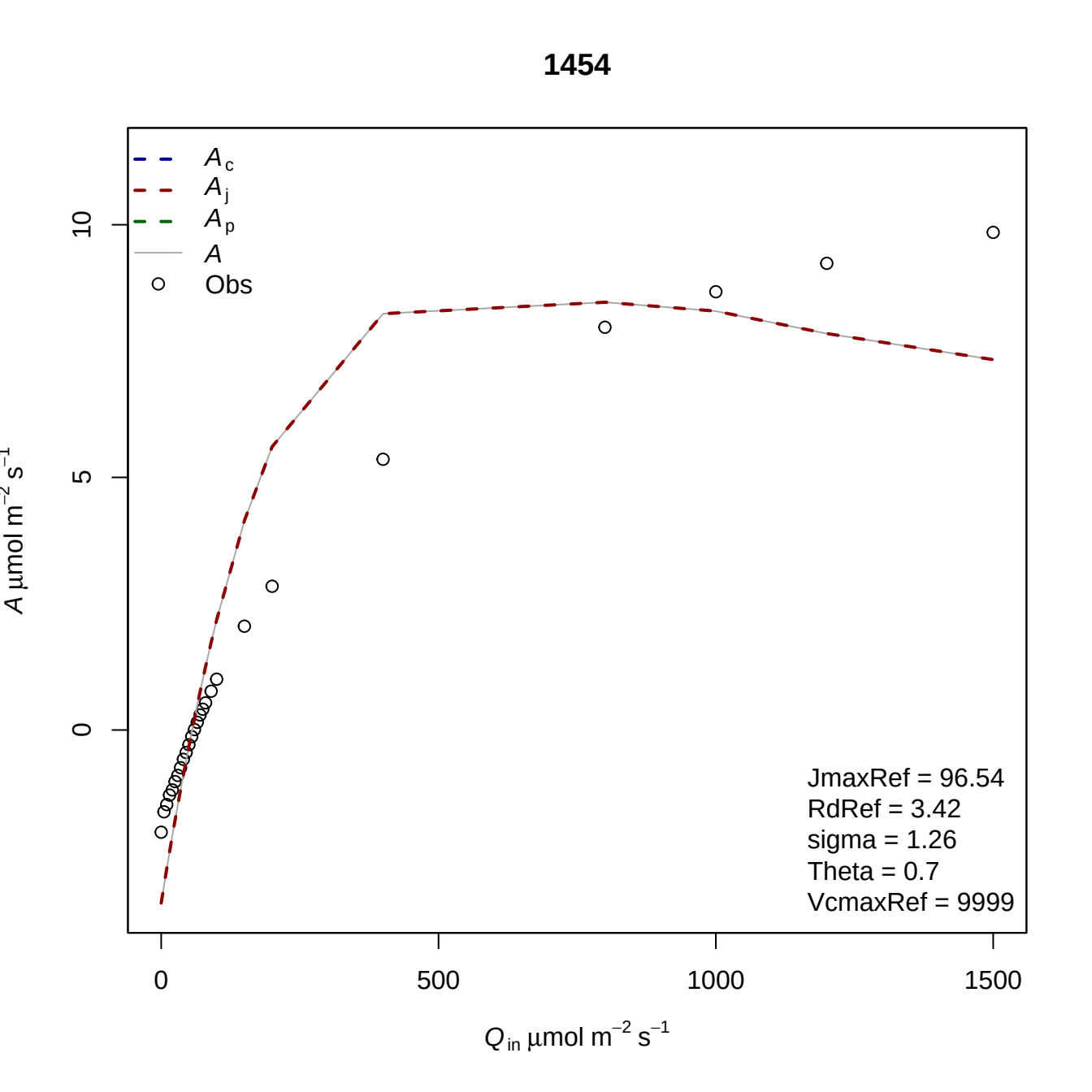

1449
A_c
A_j
A_p
A
Obs
JmaxRef = 85.46
RdRef = 1.88
sigma = 0.7
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500

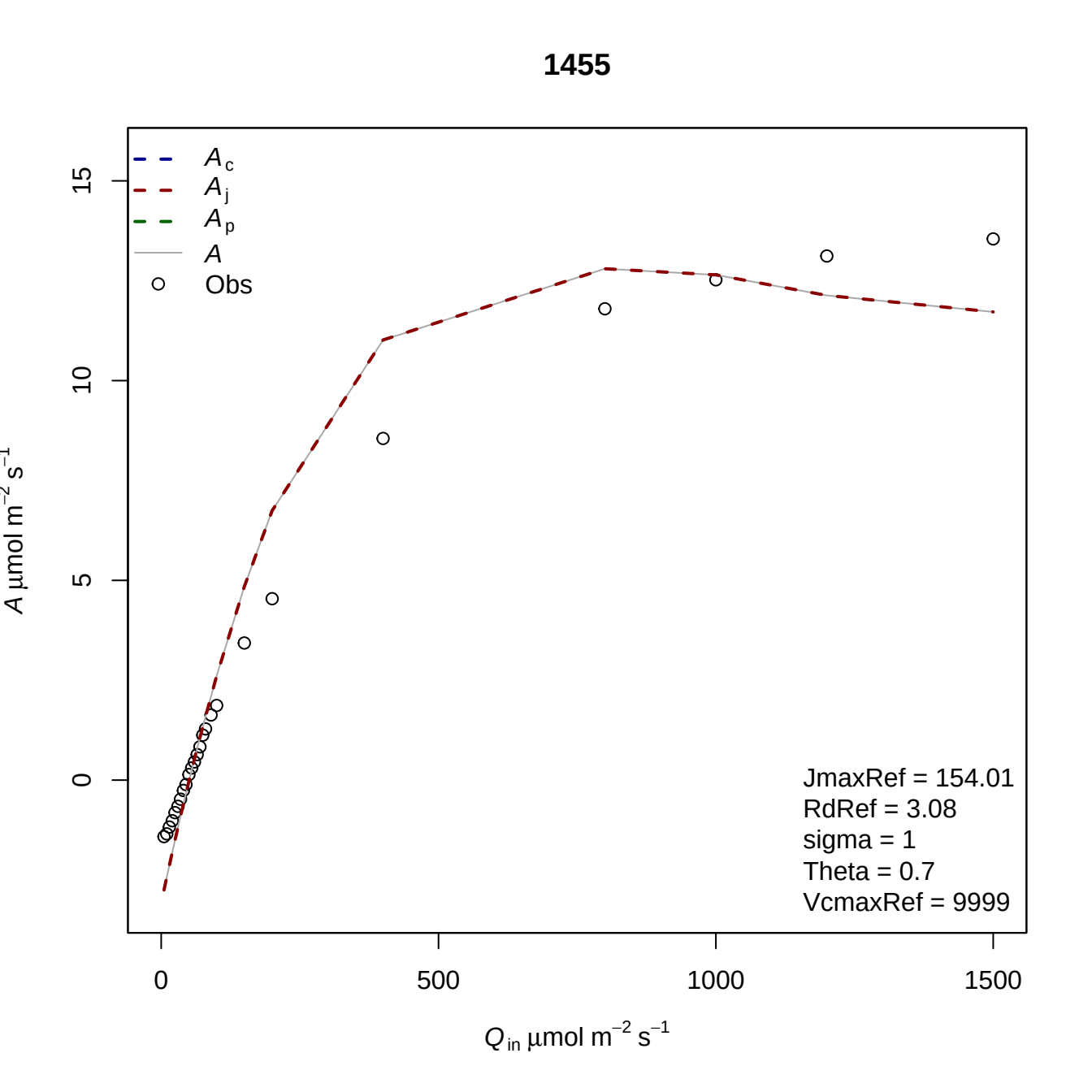

1450
A_c
A_j
A_p
A
Obs
JmaxRef = 150.35
RdRef = 3.86
sigma = 0.96
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

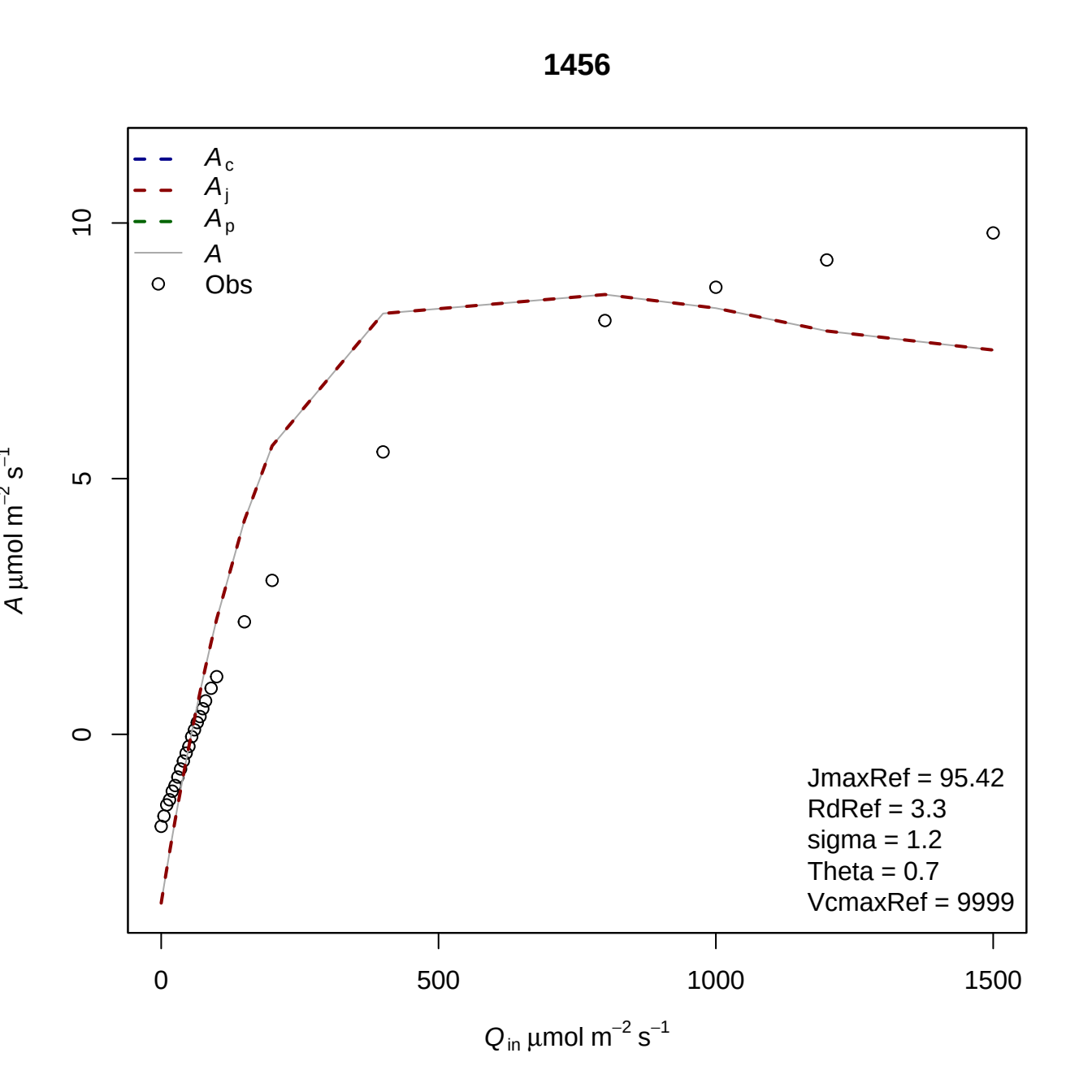

1451
A_c
A_j
A_p
A
Obs
JmaxRef = 114.75
RdRef = 3.61
sigma = 0.98
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

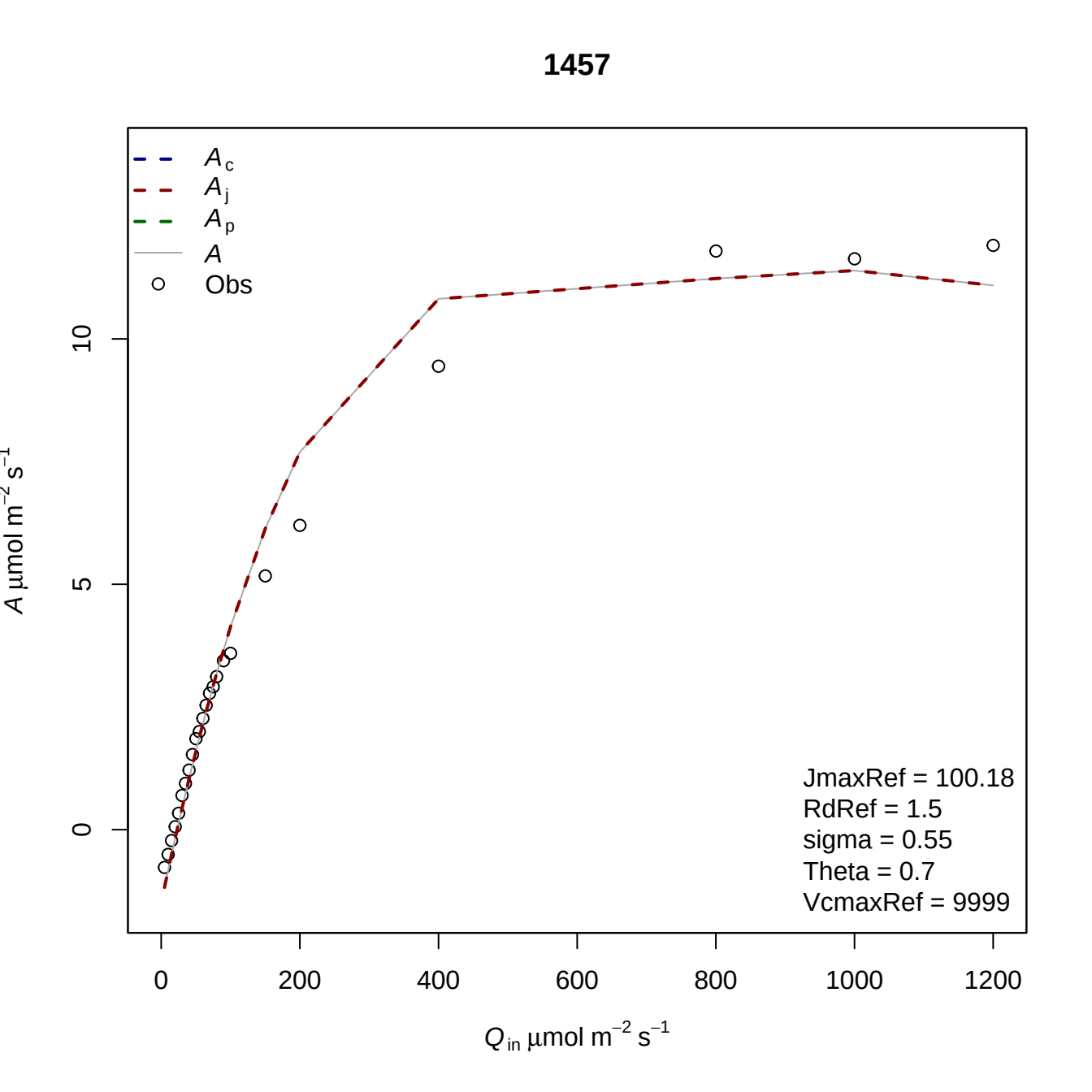

1452
A_c
A_j
A_p
A
Obs
JmaxRef = 90.29
RdRef = 1.79
sigma = 0.59
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

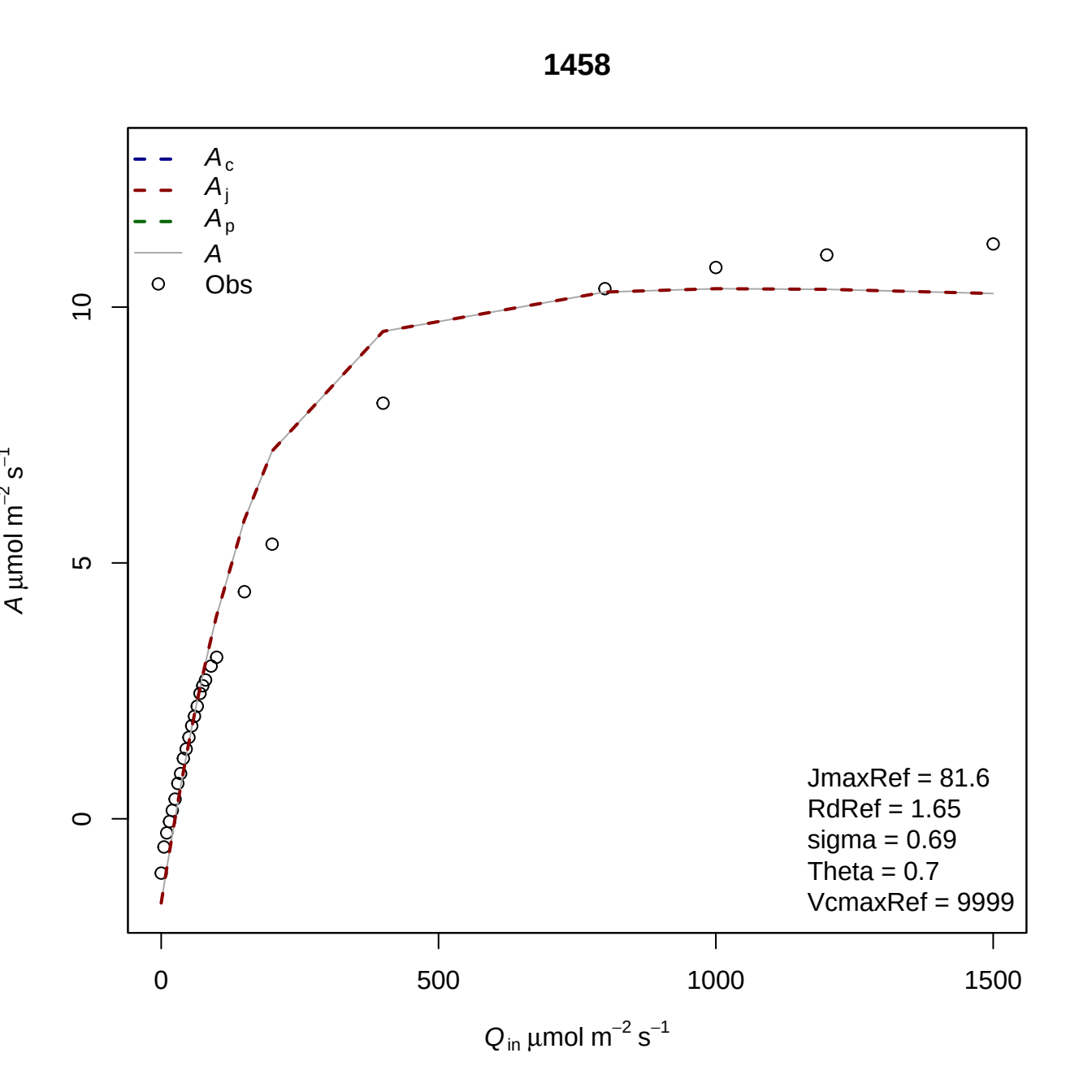

1453
A_c
A_j
A_p
A
Obs
JmaxRef = 131.08
RdRef = 3.61
sigma = 0.89
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

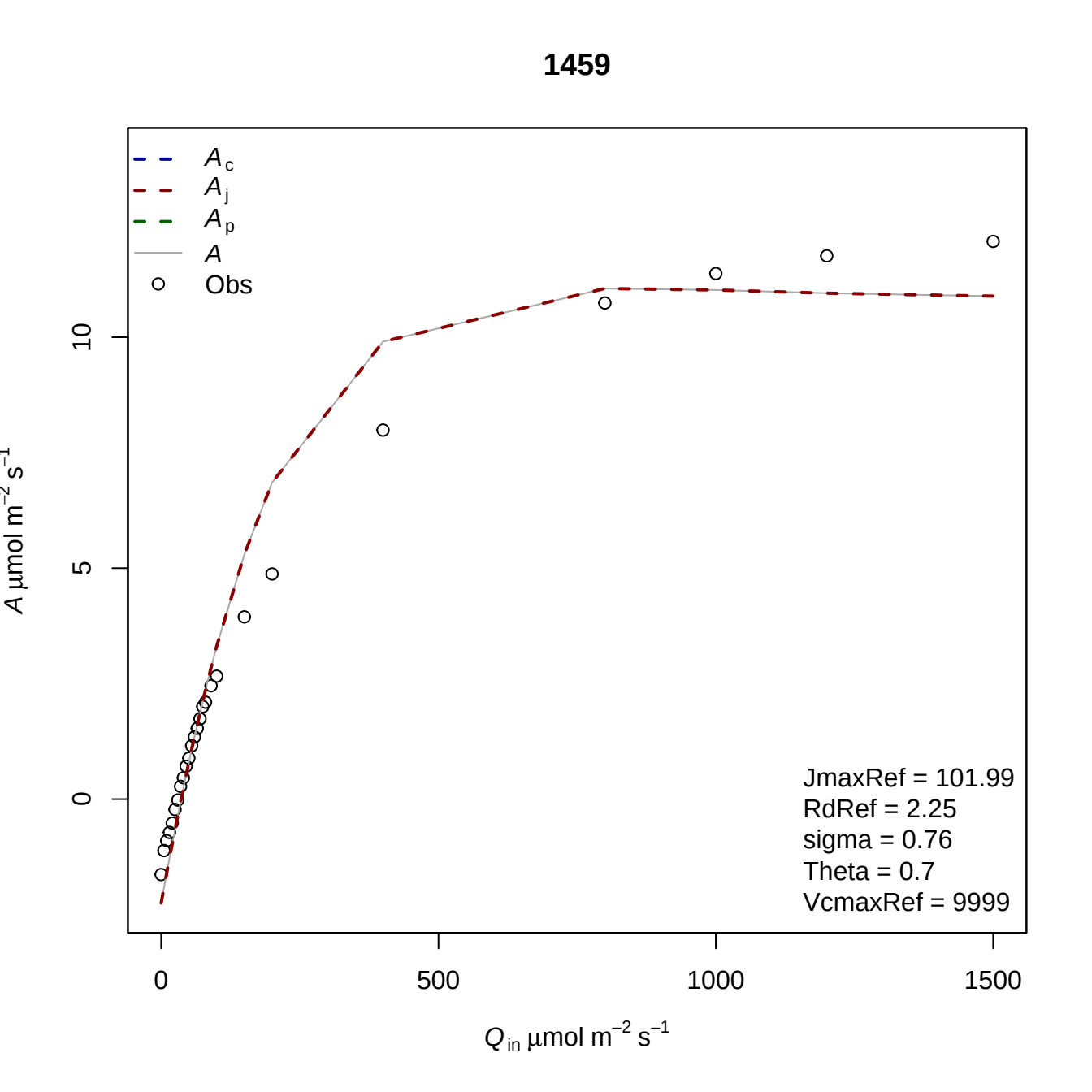

1454
A_c
A_j
A_p
A
Obs
JmaxRef = 96.54
RdRef = 3.42
sigma = 1.26
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

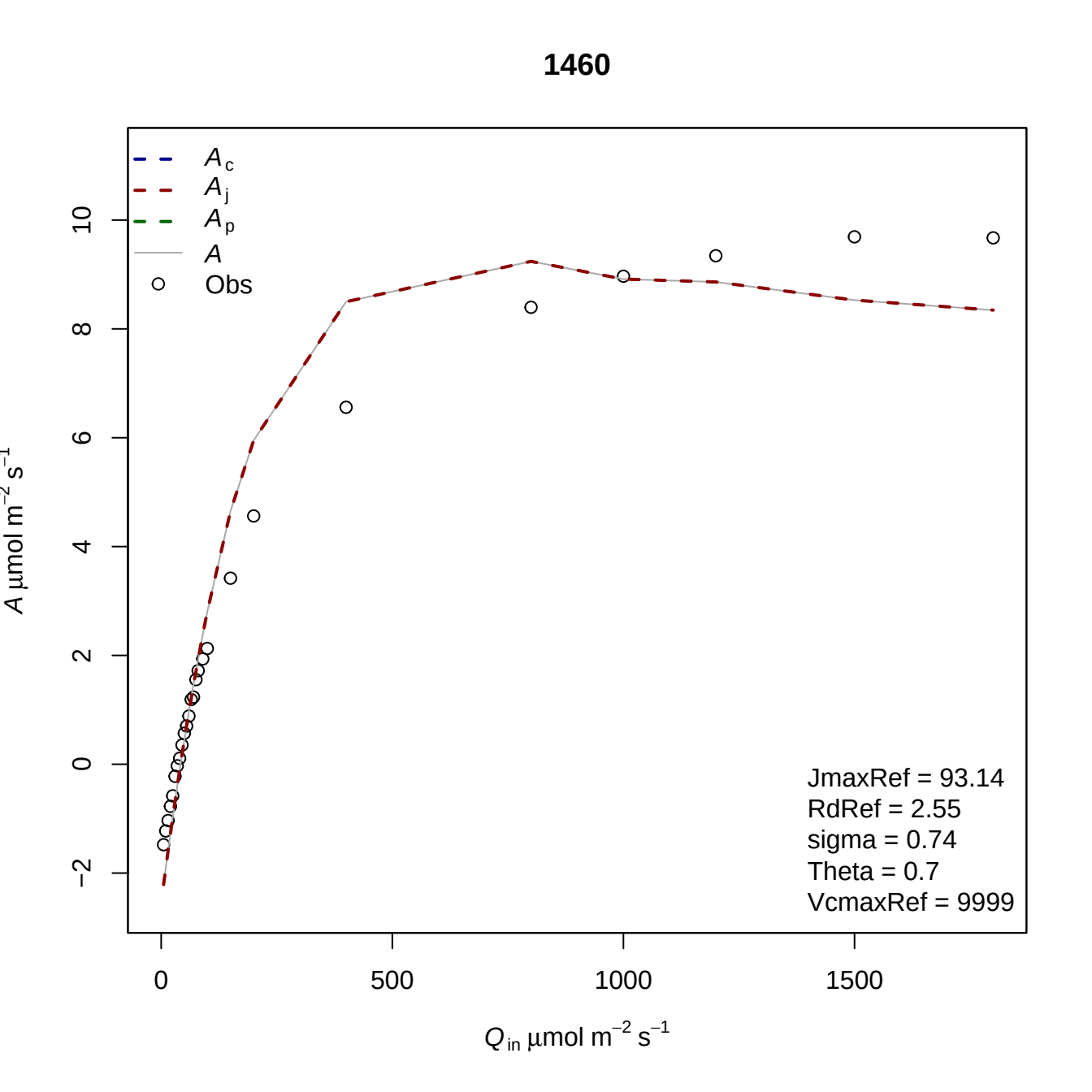

1455
A_c
A_j
A_p
A
Obs
JmaxRef = 154.01
RdRef = 3.08
sigma = 1
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

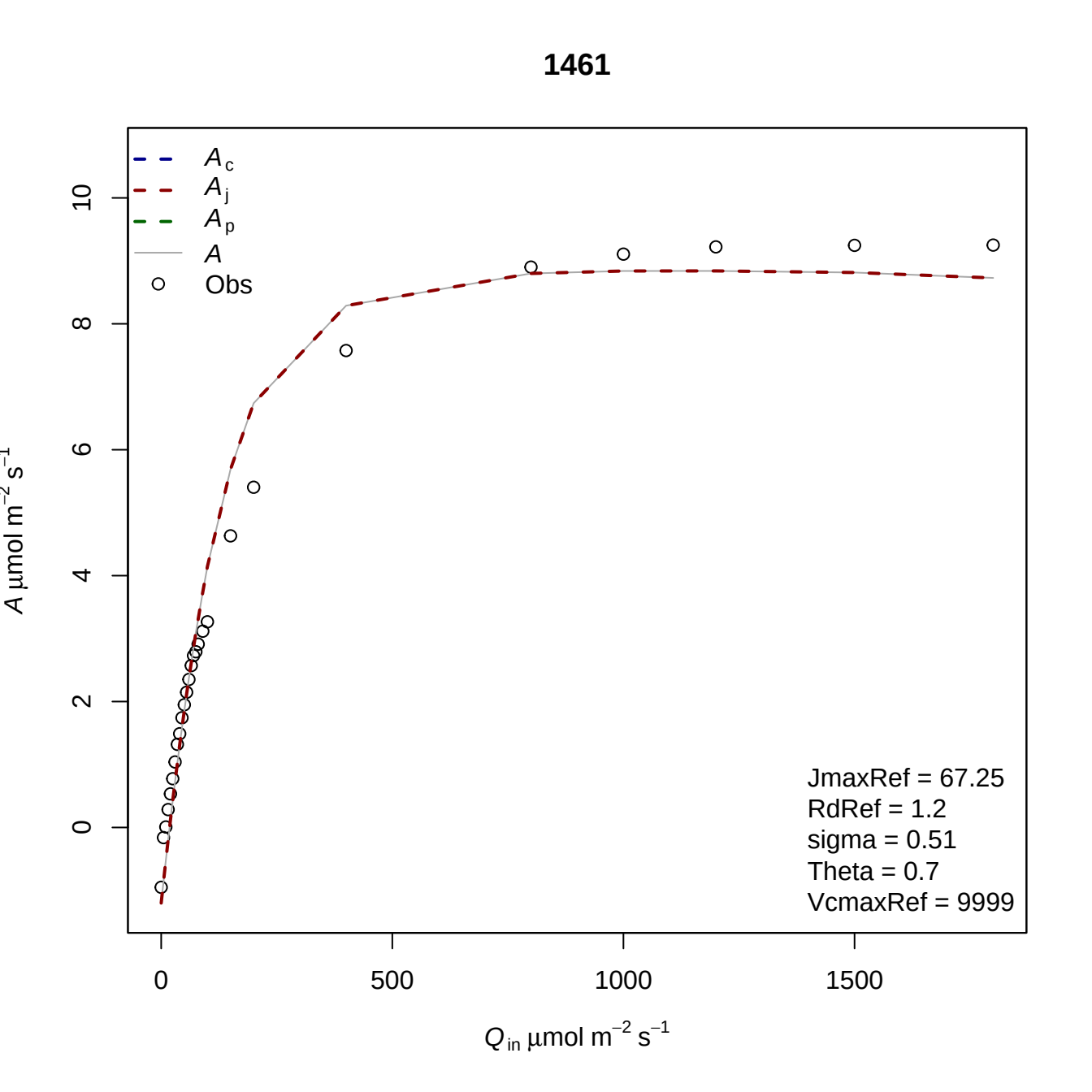

1456
A_c
A_j
A_p
A
Obs
JmaxRef = 95.42
RdRef = 3.3
sigma = 1.2
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

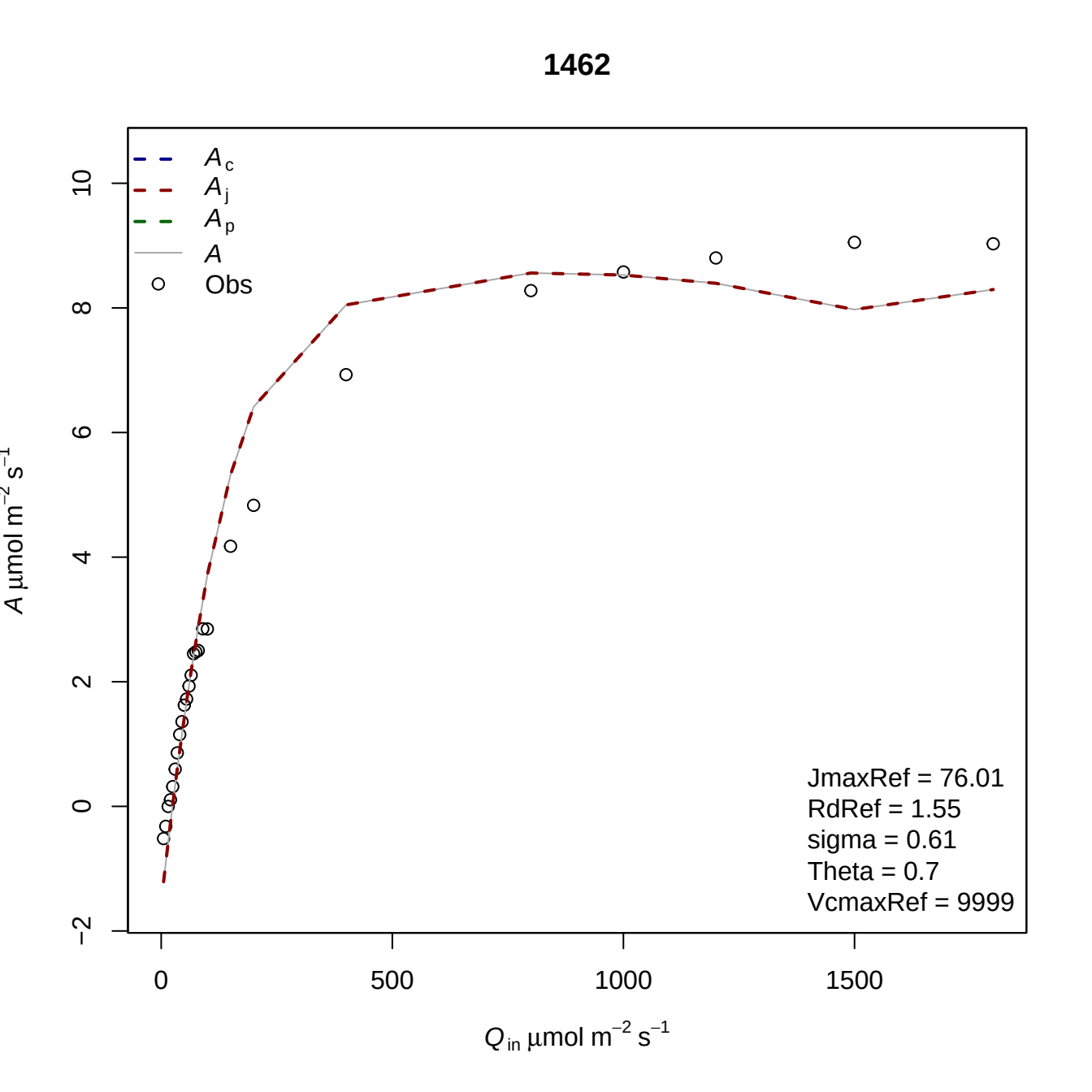

1457
A_c
A_j
A_p
A
Obs
JmaxRef = 100.18
RdRef = 1.5
sigma = 0.55
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
200
400
600
800
1000
1200
Q_{in} µmol m^{-2} s^{-1}

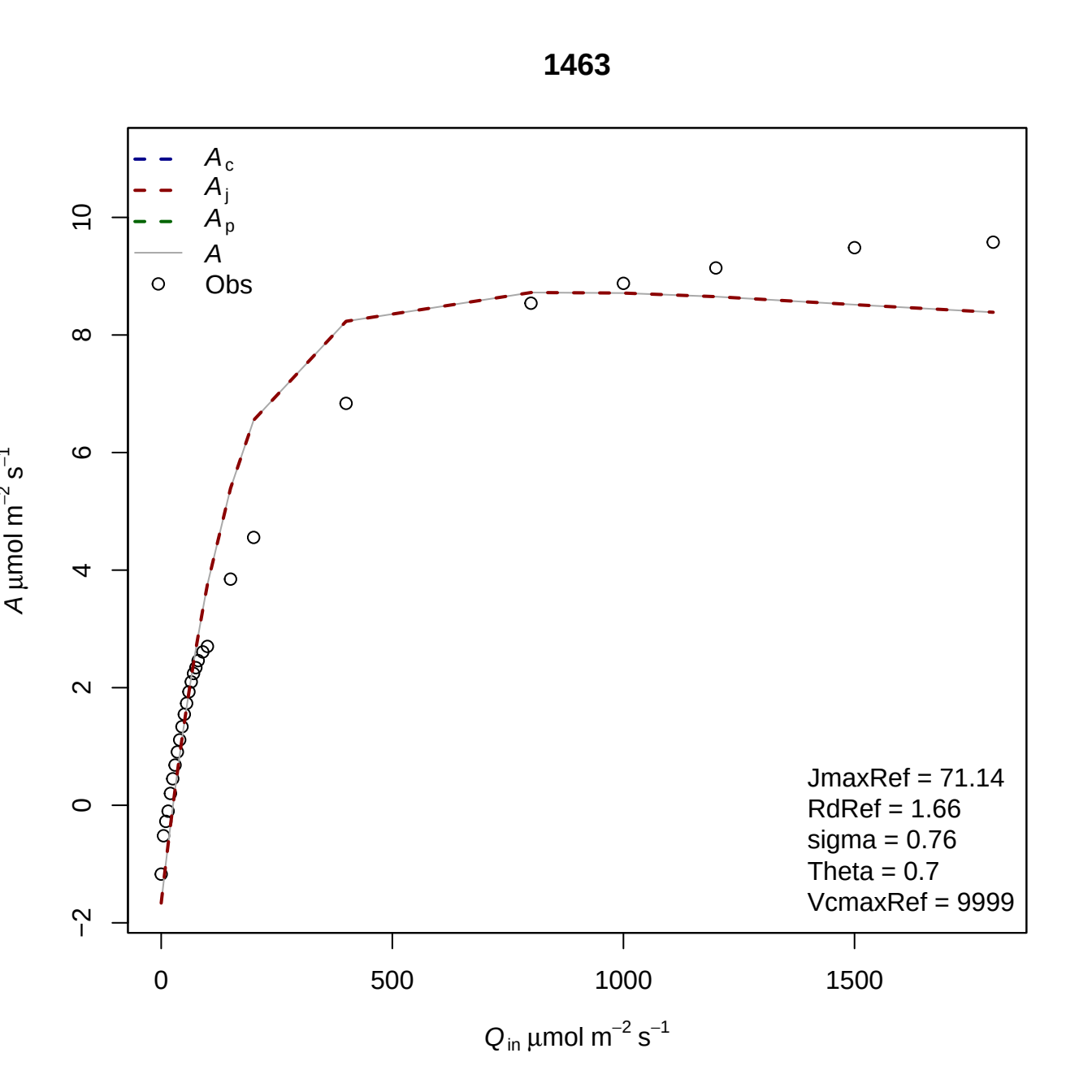

1458
A_c
A_j
A_p
A
Obs
JmaxRef = 81.6
RdRef = 1.65
sigma = 0.69
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

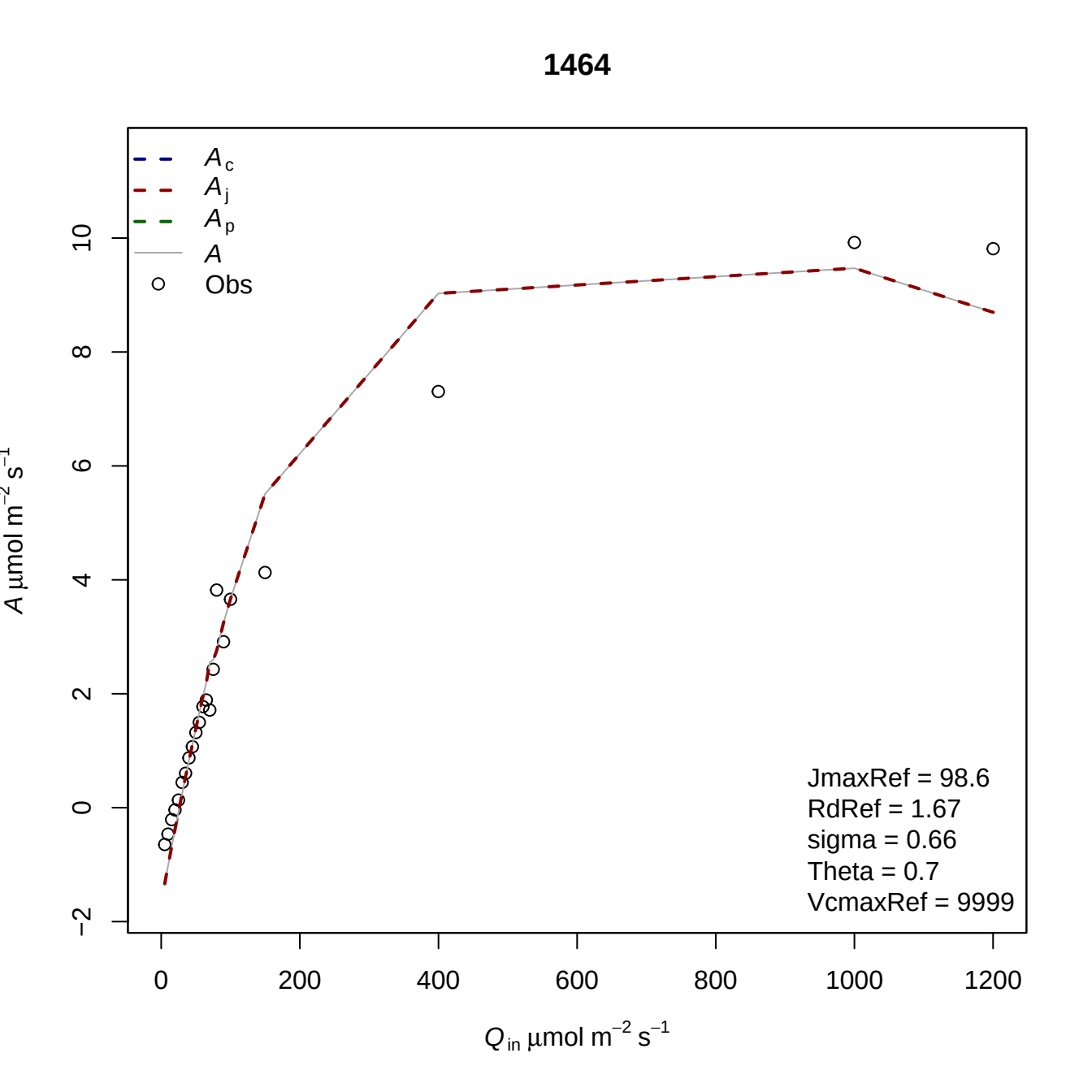

1459
A_c
A_j
A_p
A
Obs
JmaxRef = 101.99
RdRef = 2.25
sigma = 0.76
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

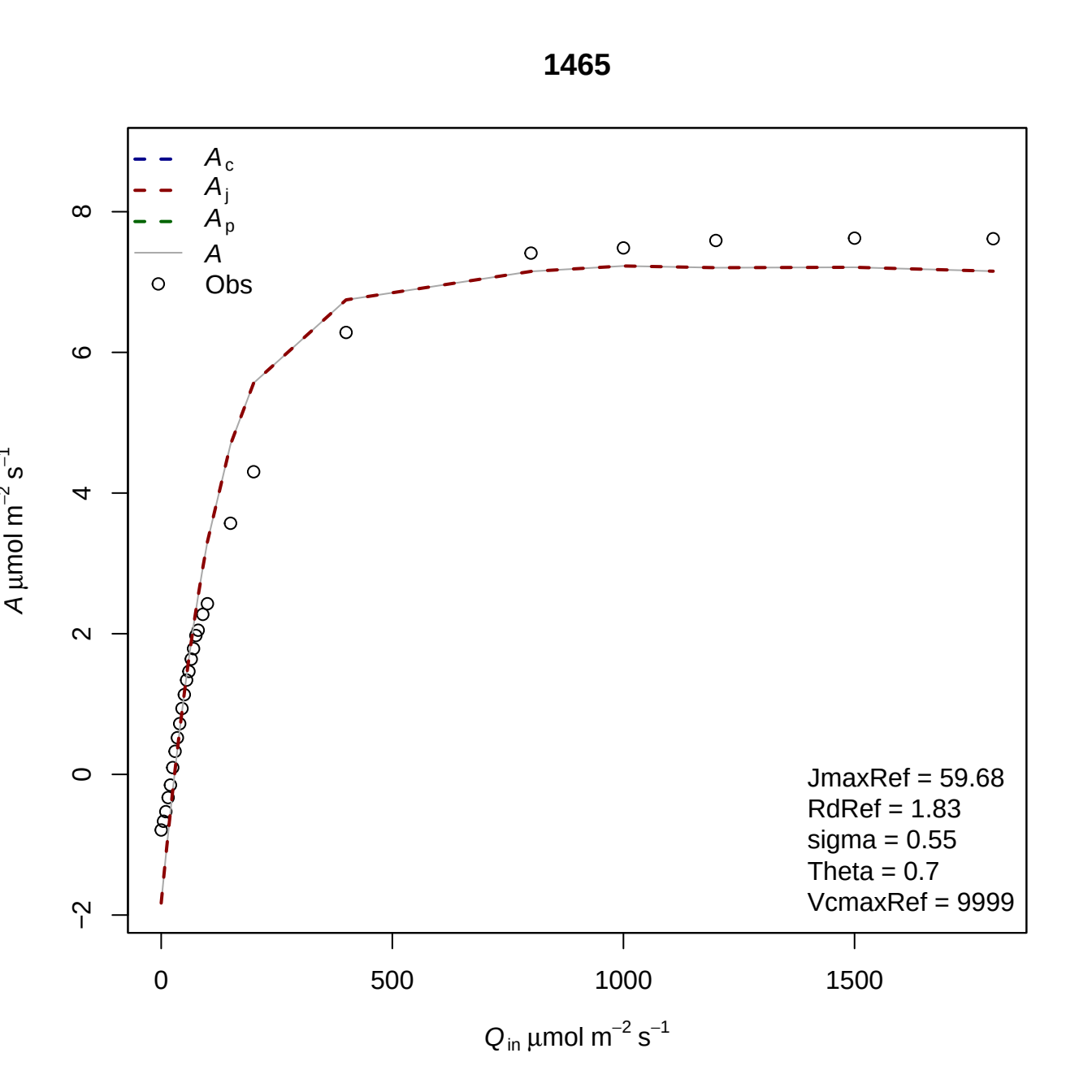

1460
A_c
A_j
A_p
A
Obs
JmaxRef = 93.14
RdRef = 2.55
sigma = 0.74
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

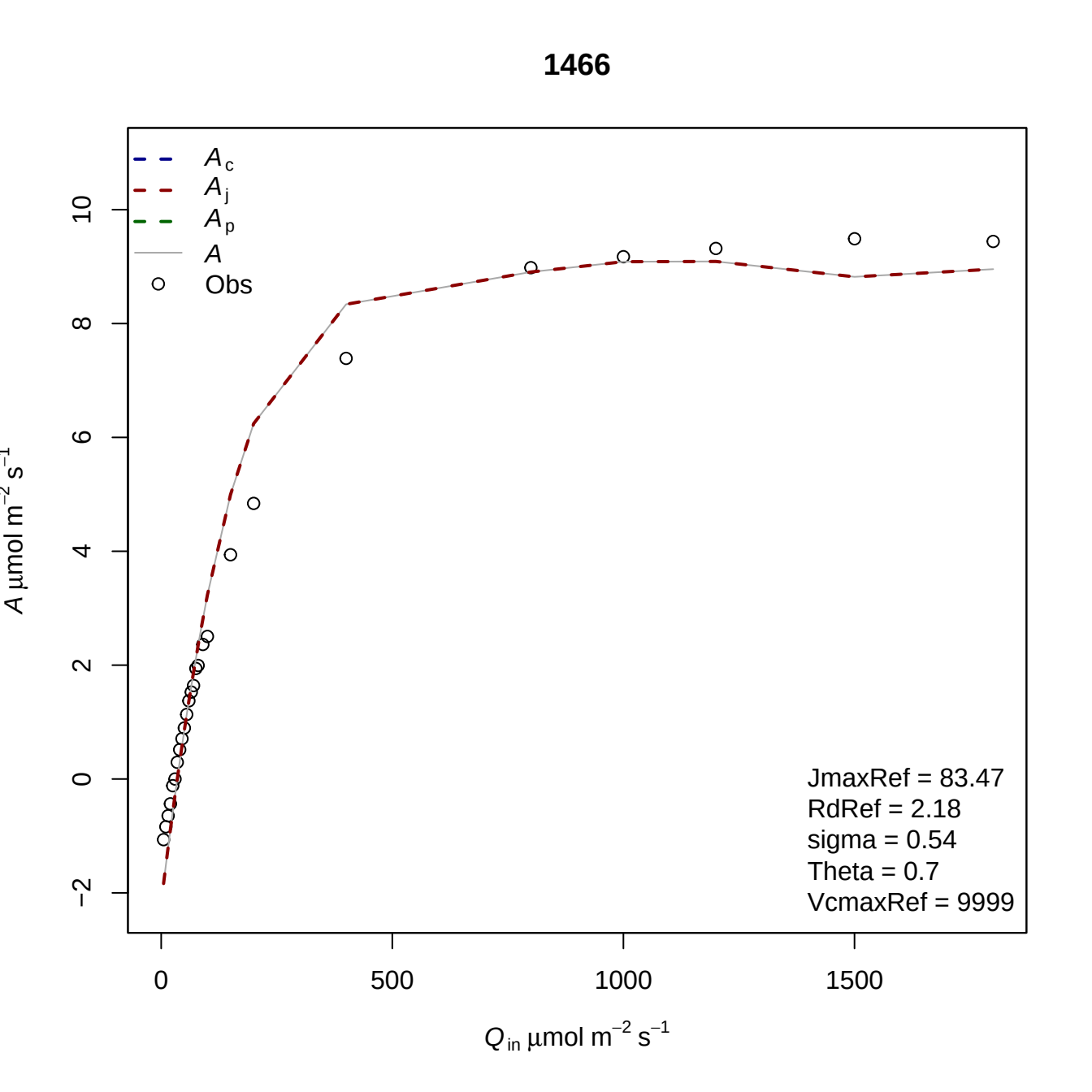

1461
A_c
A_j
A_p
A
Obs
JmaxRef = 67.25
RdRef = 1.2
sigma = 0.51
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

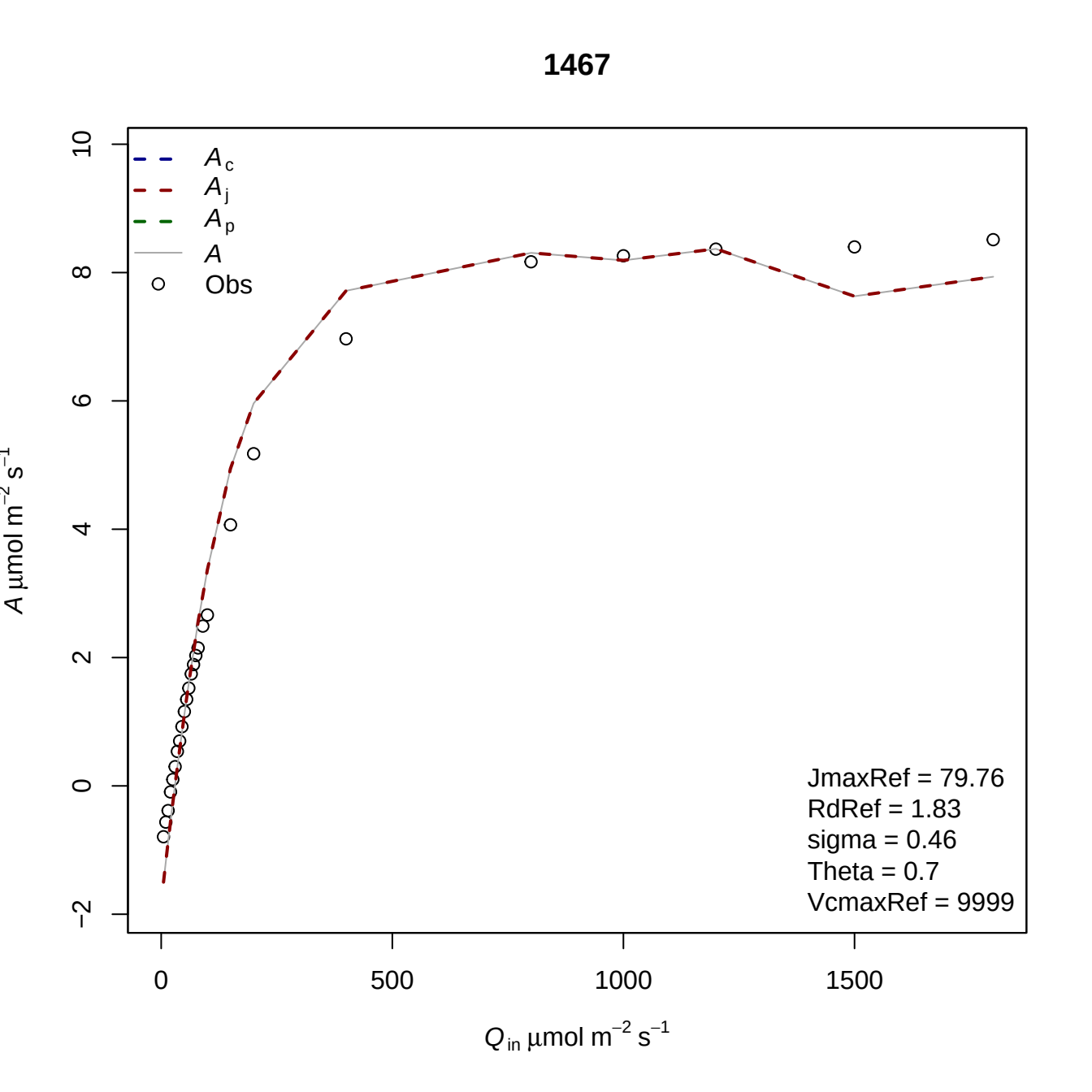

1462
A_c
A_j
A_p
A
Obs
JmaxRef = 76.01
RdRef = 1.55
sigma = 0.61
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

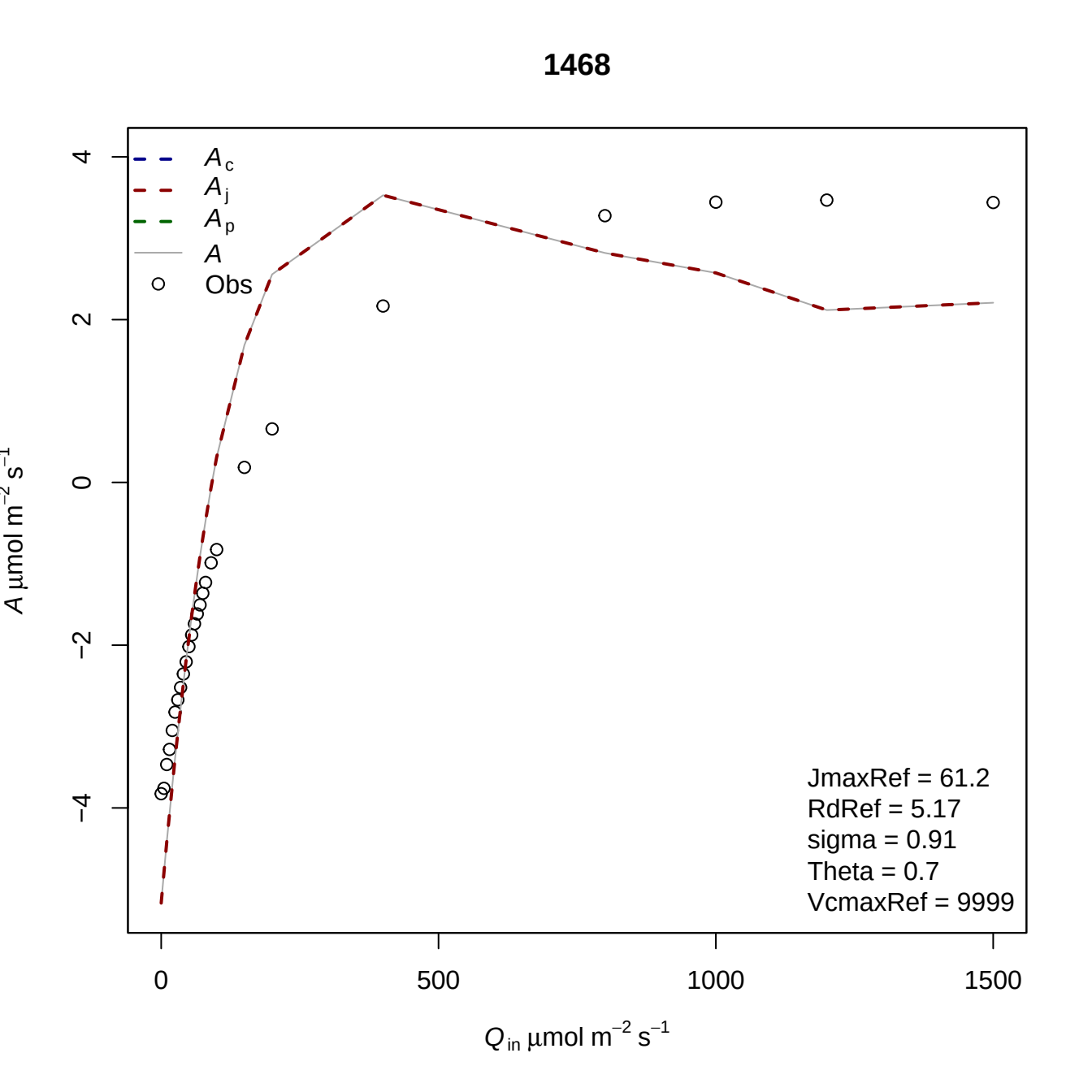

1463
A_c
A_j
A_p
A
Obs
JmaxRef = 71.14
RdRef = 1.66
sigma = 0.76
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

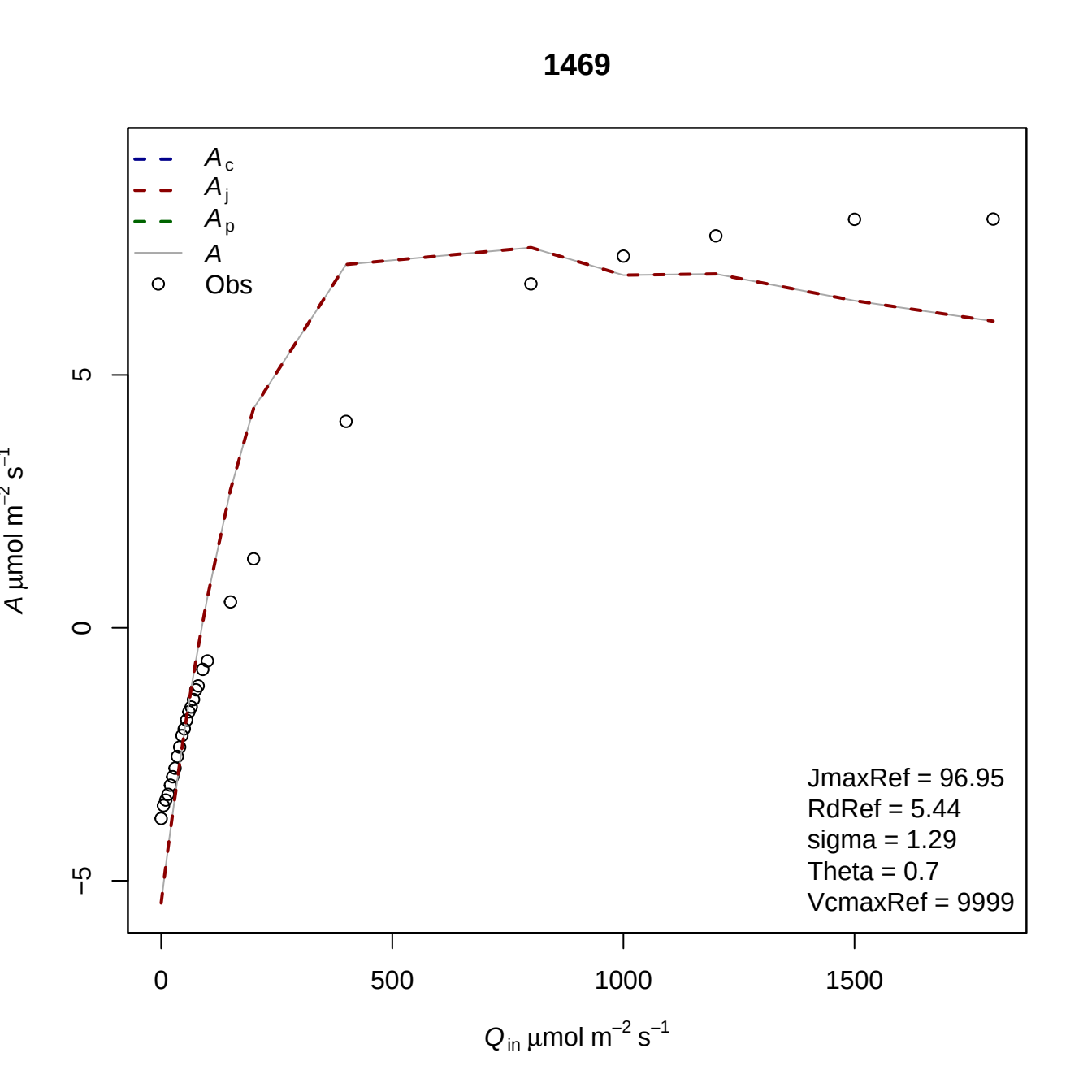

1464
A_c
A_j
A_p
A
Obs
JmaxRef = 98.6
RdRef = 1.67
sigma = 0.66
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
200
400
600
800
1000
1200

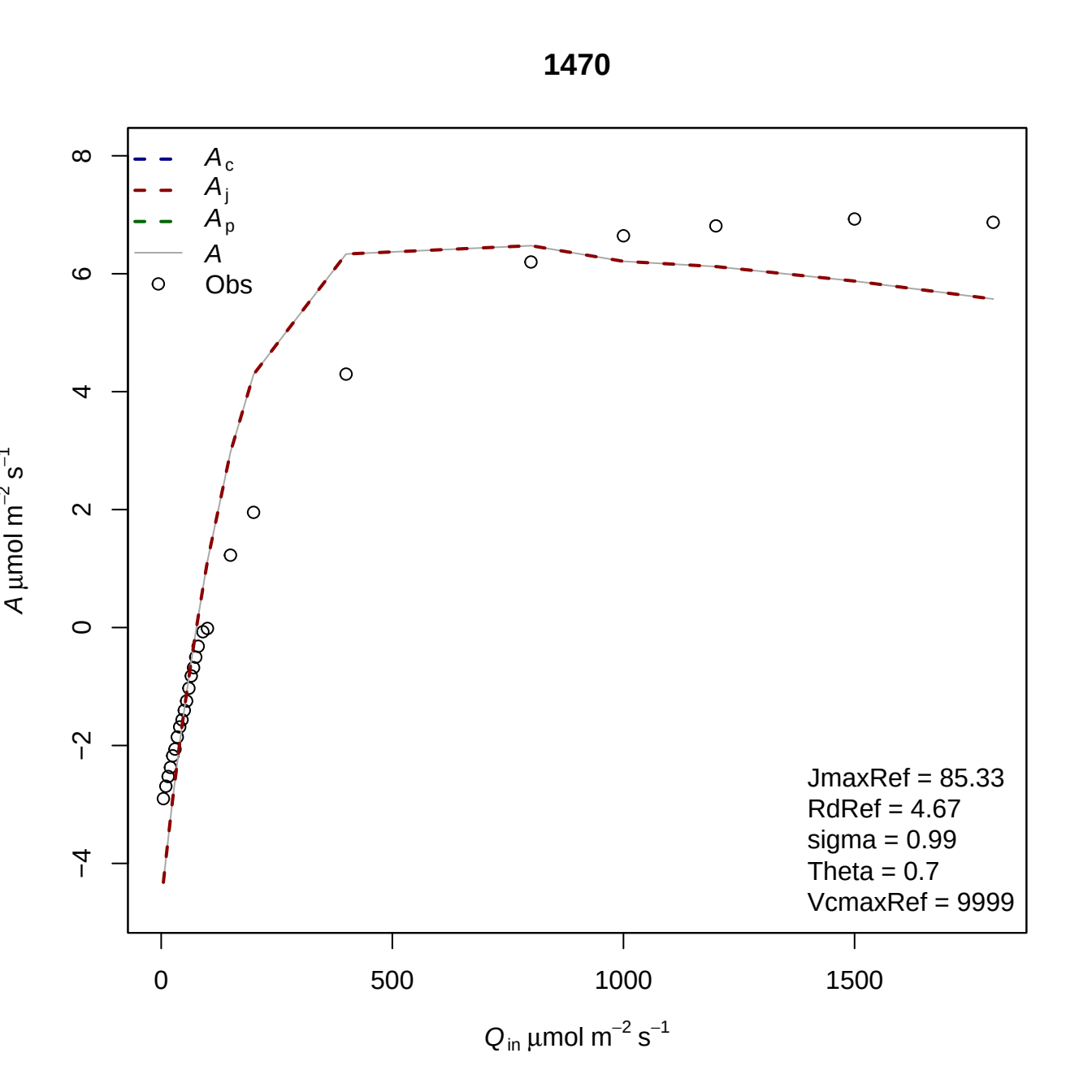

1465
A_c
A_j
A_p
A
Obs
JmaxRef = 59.68
RdRef = 1.83
sigma = 0.55
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

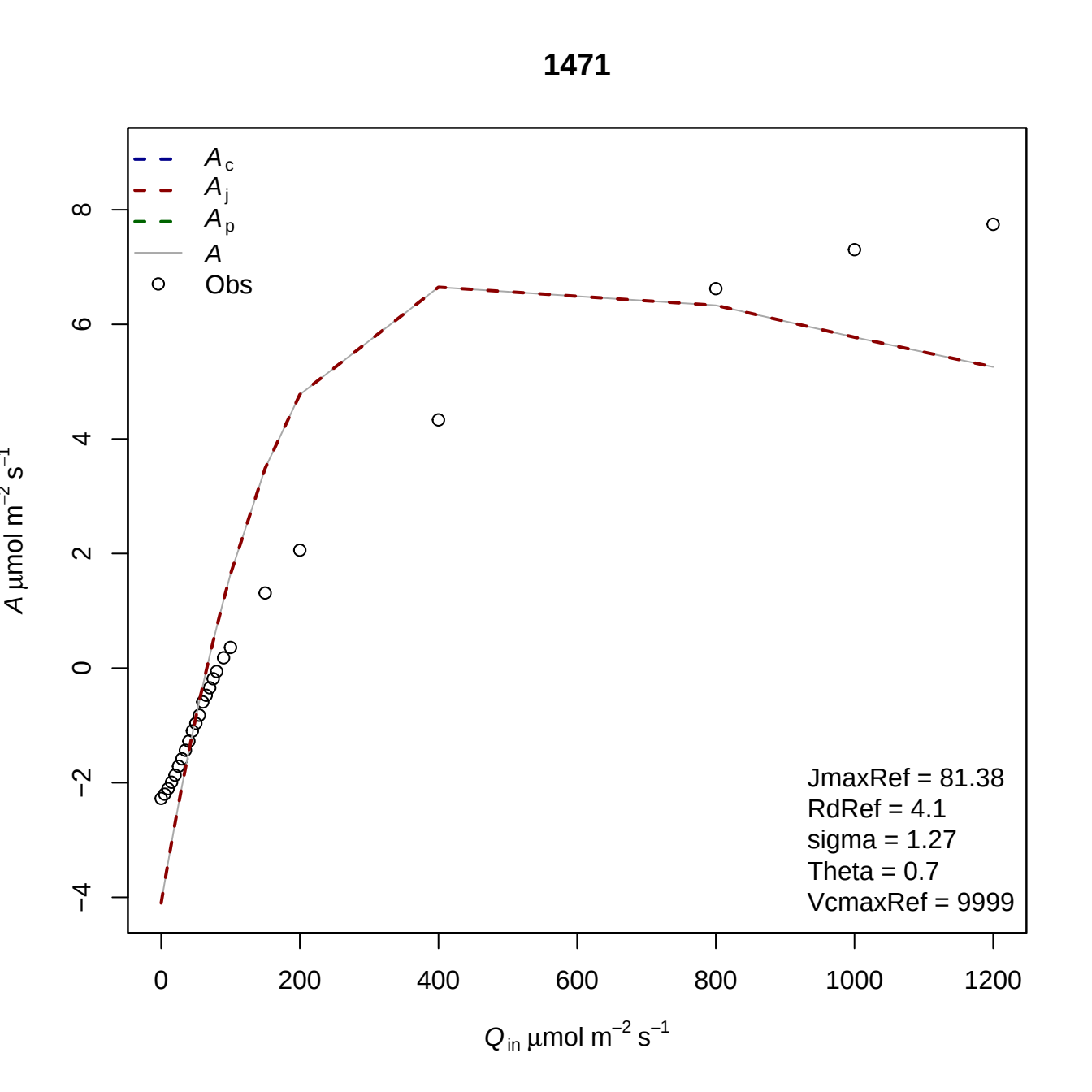

1466
A_c
A_j
A_p
A
Obs
JmaxRef = 83.47
RdRef = 2.18
sigma = 0.54
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

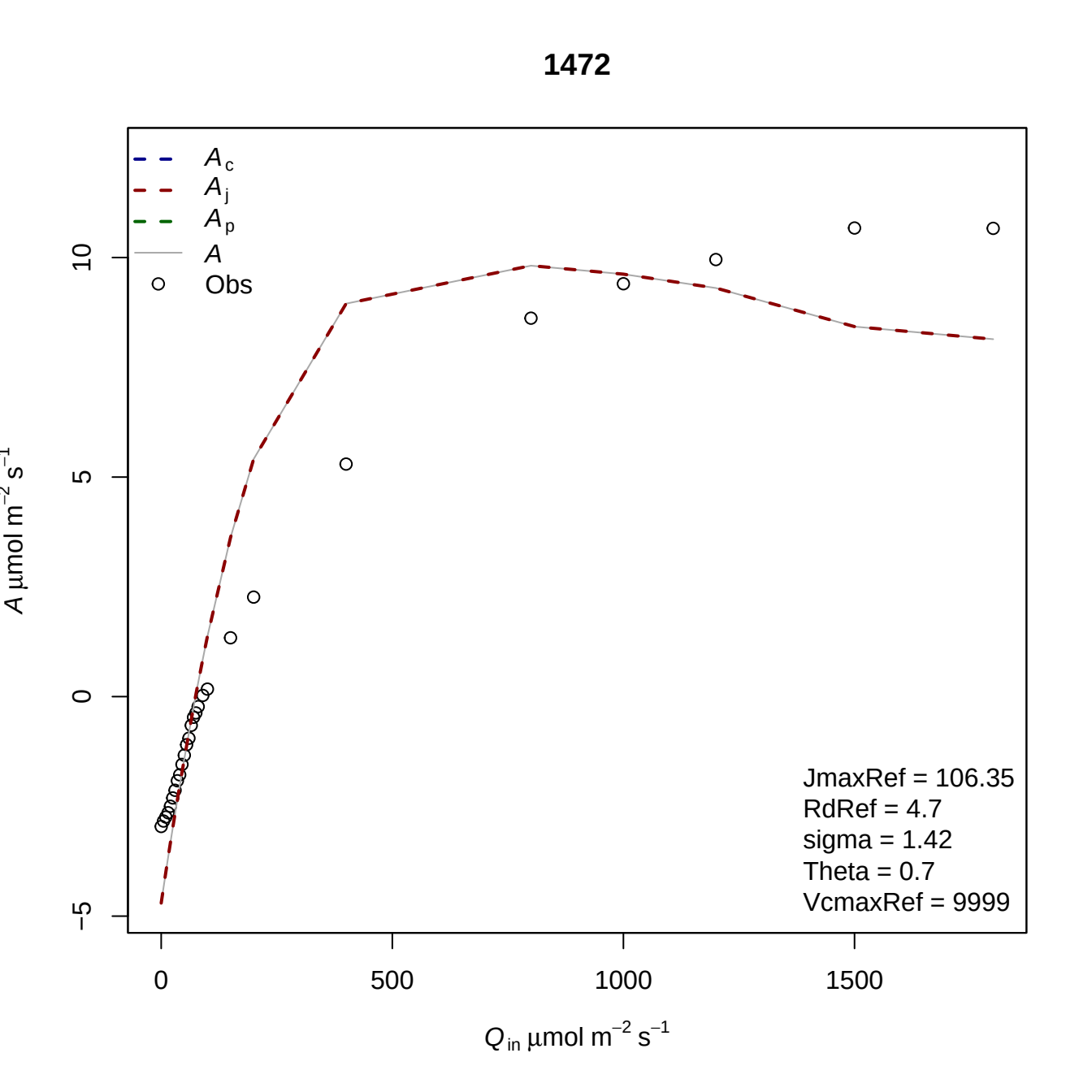

1467
A_c
A_j
A_p
A
Obs
JmaxRef = 79.76
RdRef = 1.83
sigma = 0.46
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

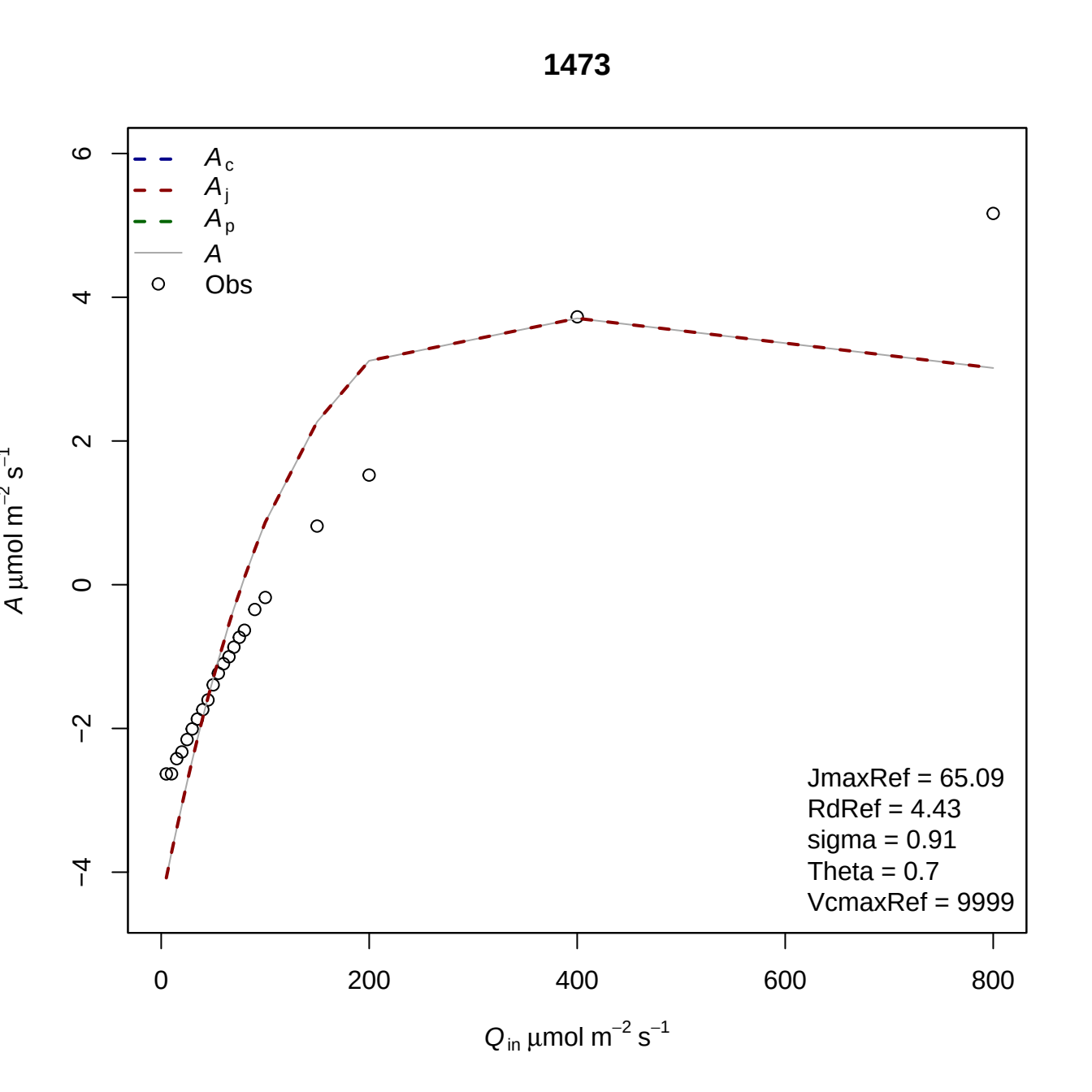

1468
A_c
A_j
A_p
A
Obs
JmaxRef = 61.2
RdRef = 5.17
sigma = 0.91
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
−4
0
500
1000
1500

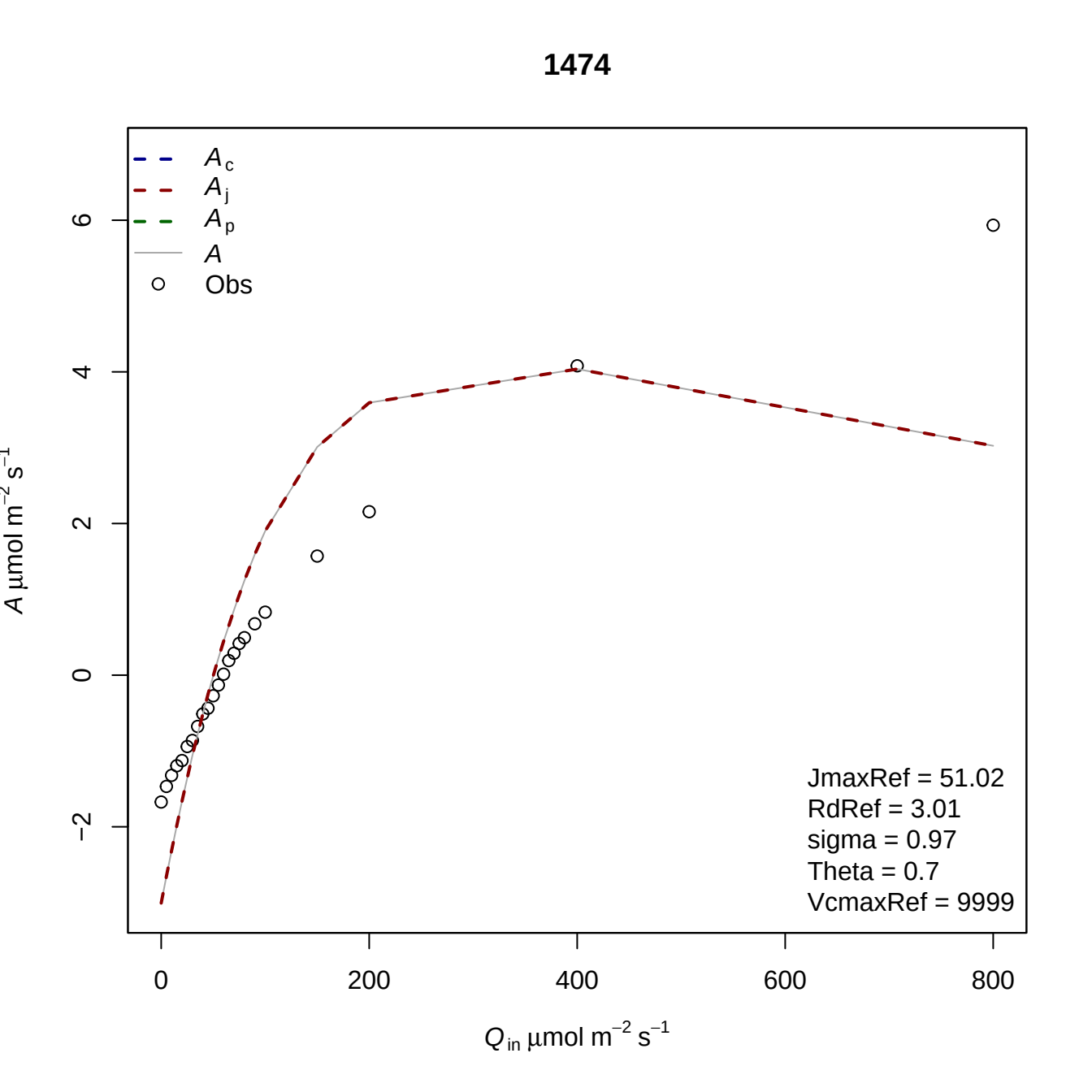

1469
A_c
A_j
A_p
A
Obs
JmaxRef = 96.95
RdRef = 5.44
sigma = 1.29
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
−5
0
5
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

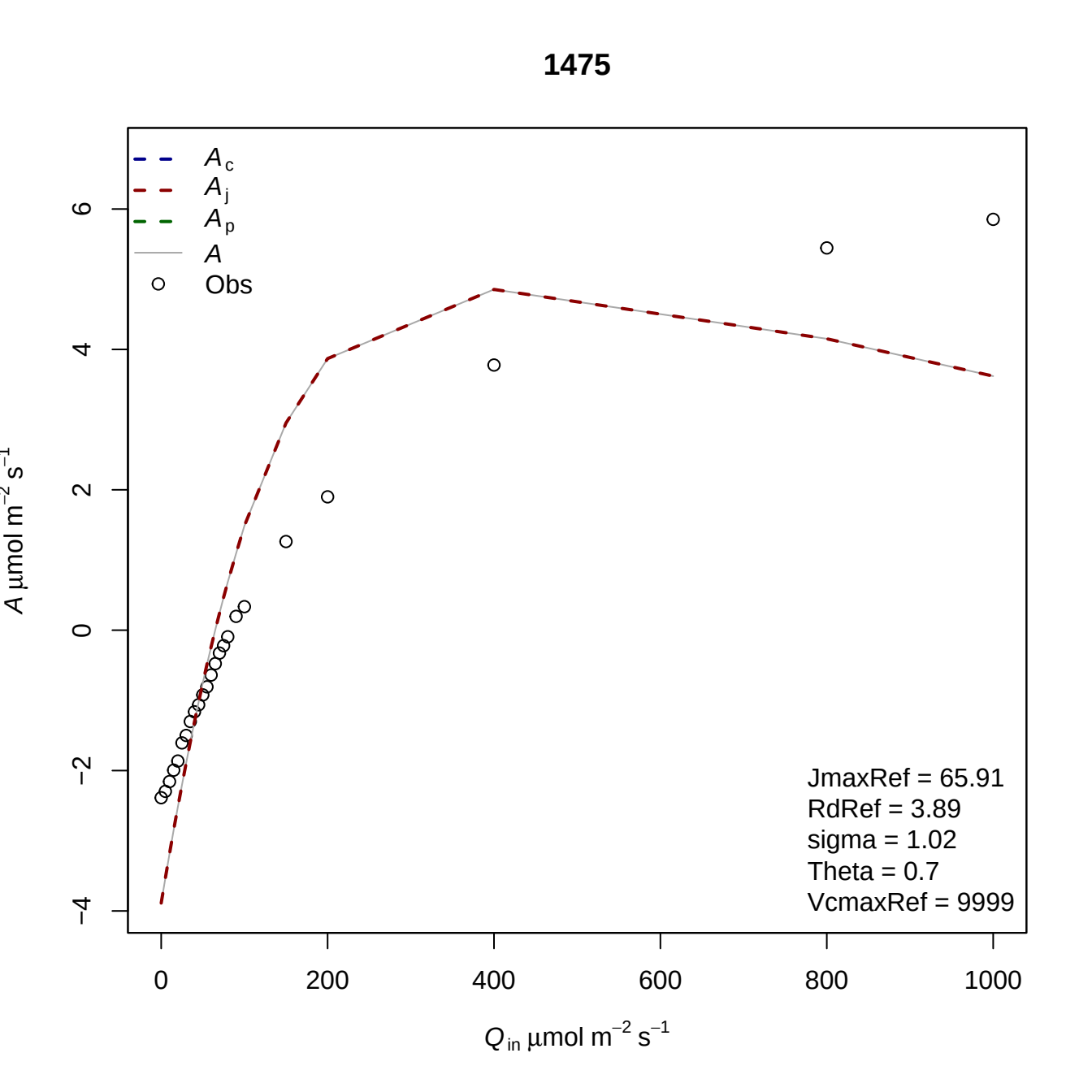

1470
A_c
A_j
A_p
A
Obs
JmaxRef = 85.33
RdRef = 4.67
sigma = 0.99
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

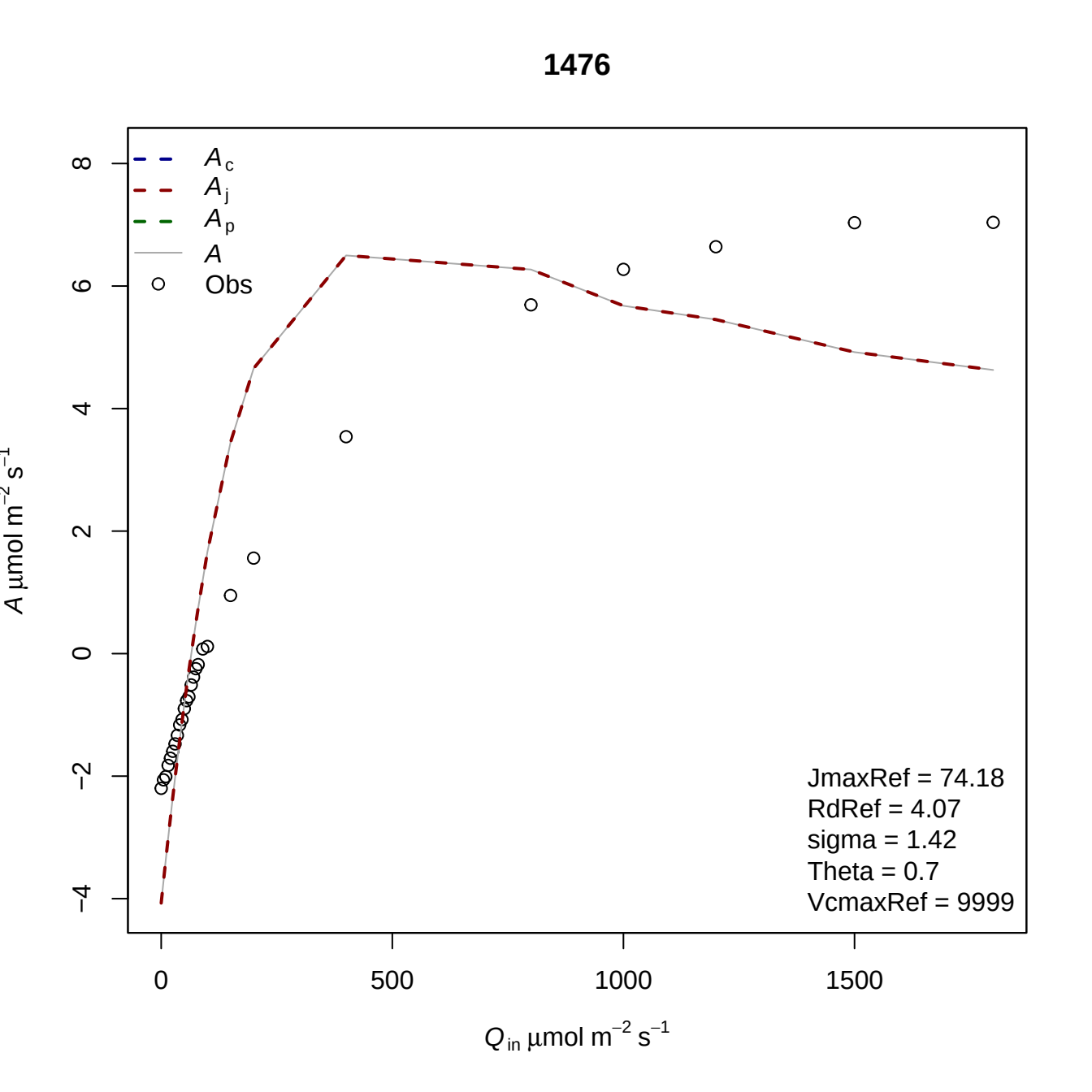

1471
A_c
A_j
A_p
A
Obs
JmaxRef = 81.38
RdRef = 4.1
sigma = 1.27
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
200
400
600
800
1000
1200

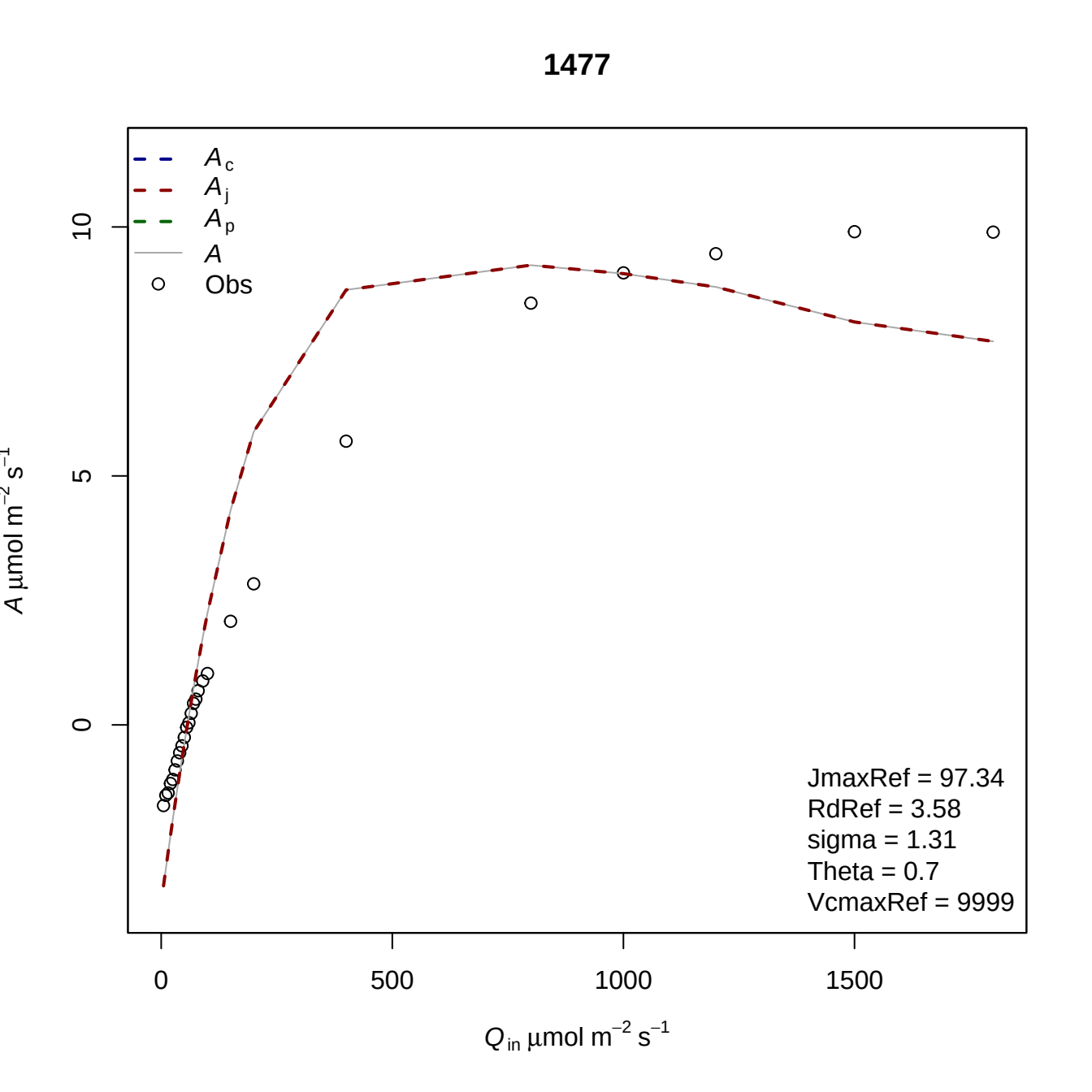

1472
A_c
A_j
A_p
A
Obs
JmaxRef = 106.35
RdRef = 4.7
sigma = 1.42
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
10
0
500
1000
1500

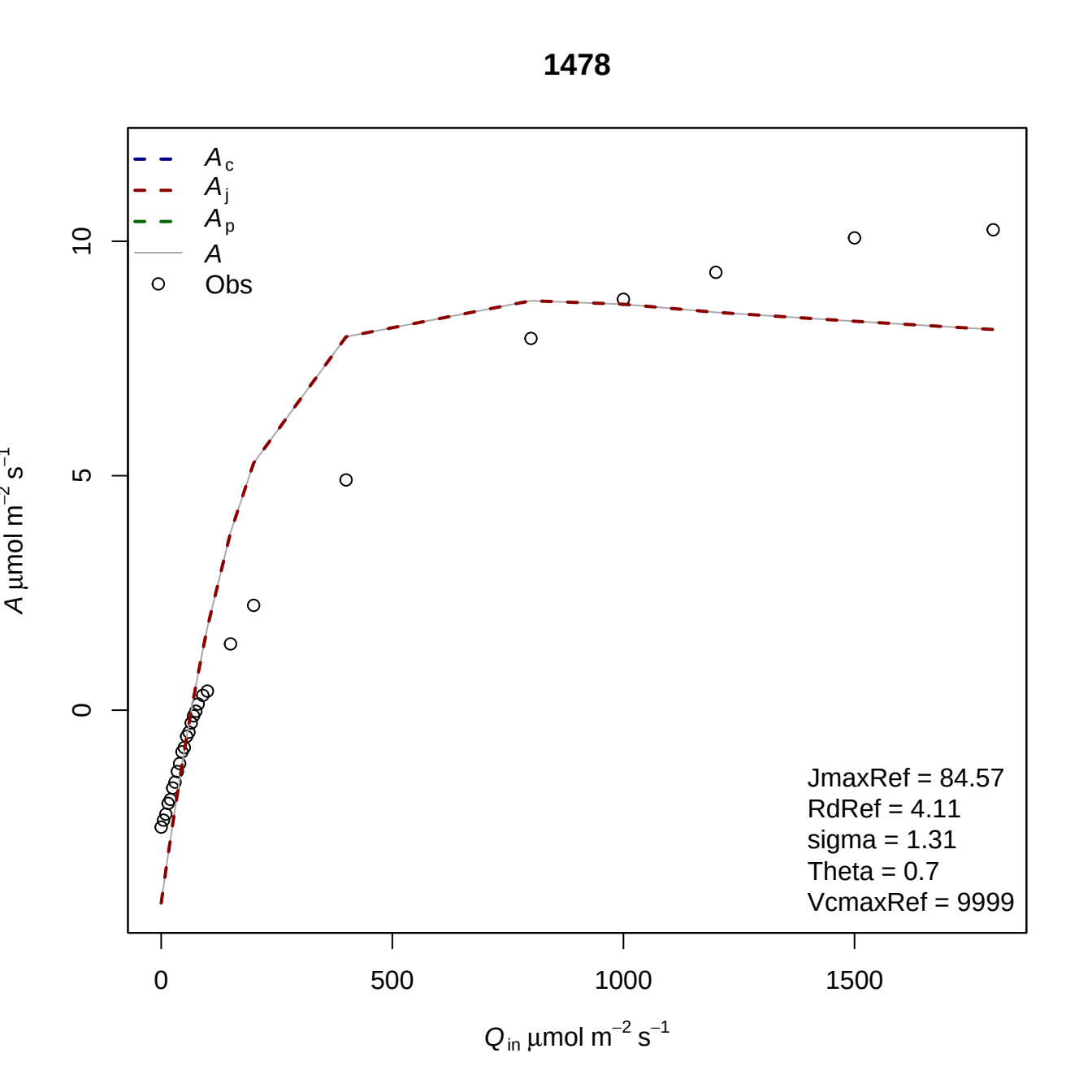

1473
A_c
A_j
A_p
A
Obs
JmaxRef = 65.09
RdRef = 4.43
sigma = 0.91
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
200
400
600
800

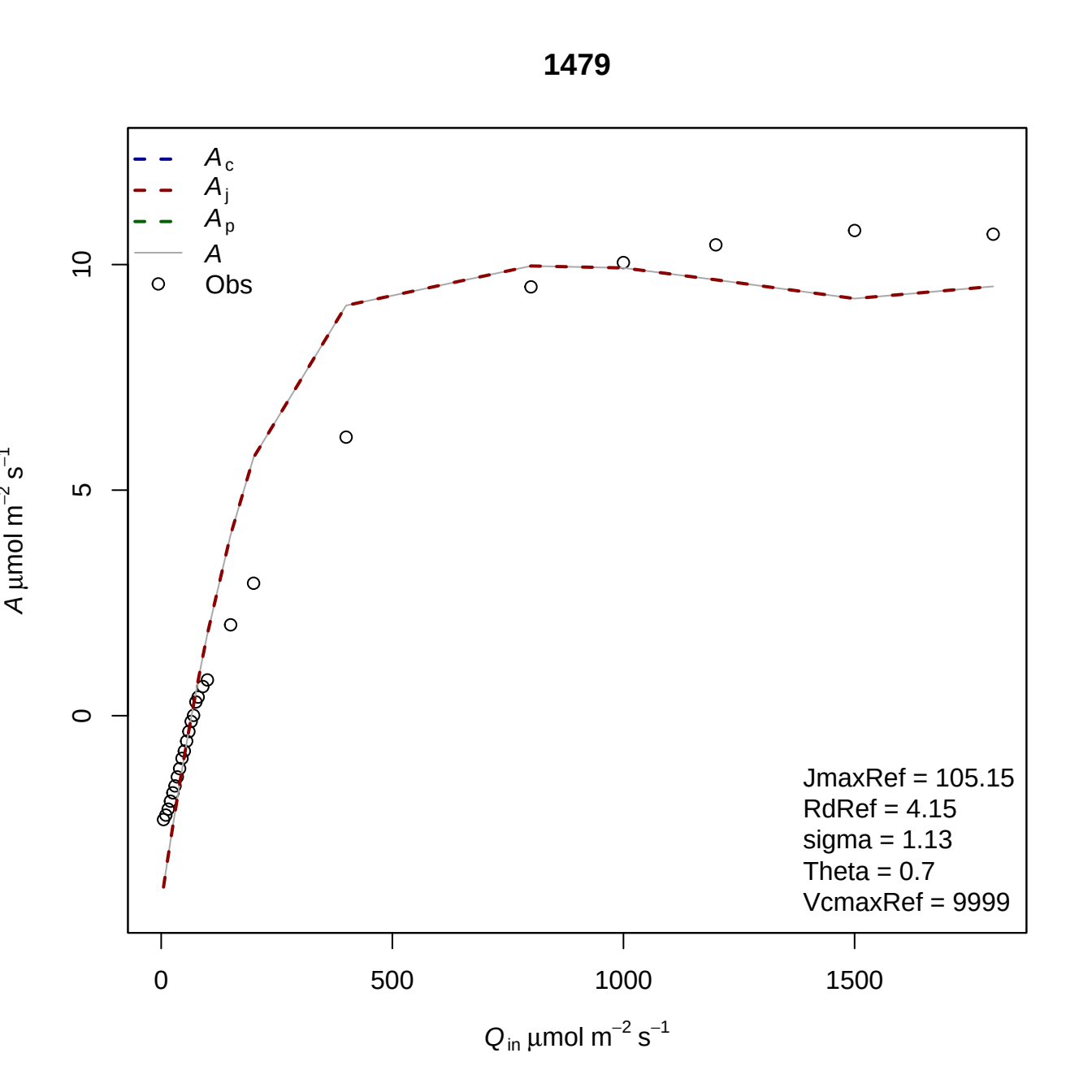

1474
A_c
A_j
A_p
A
Obs
JmaxRef = 51.02
RdRef = 3.01
sigma = 0.97
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
200
400
600
800

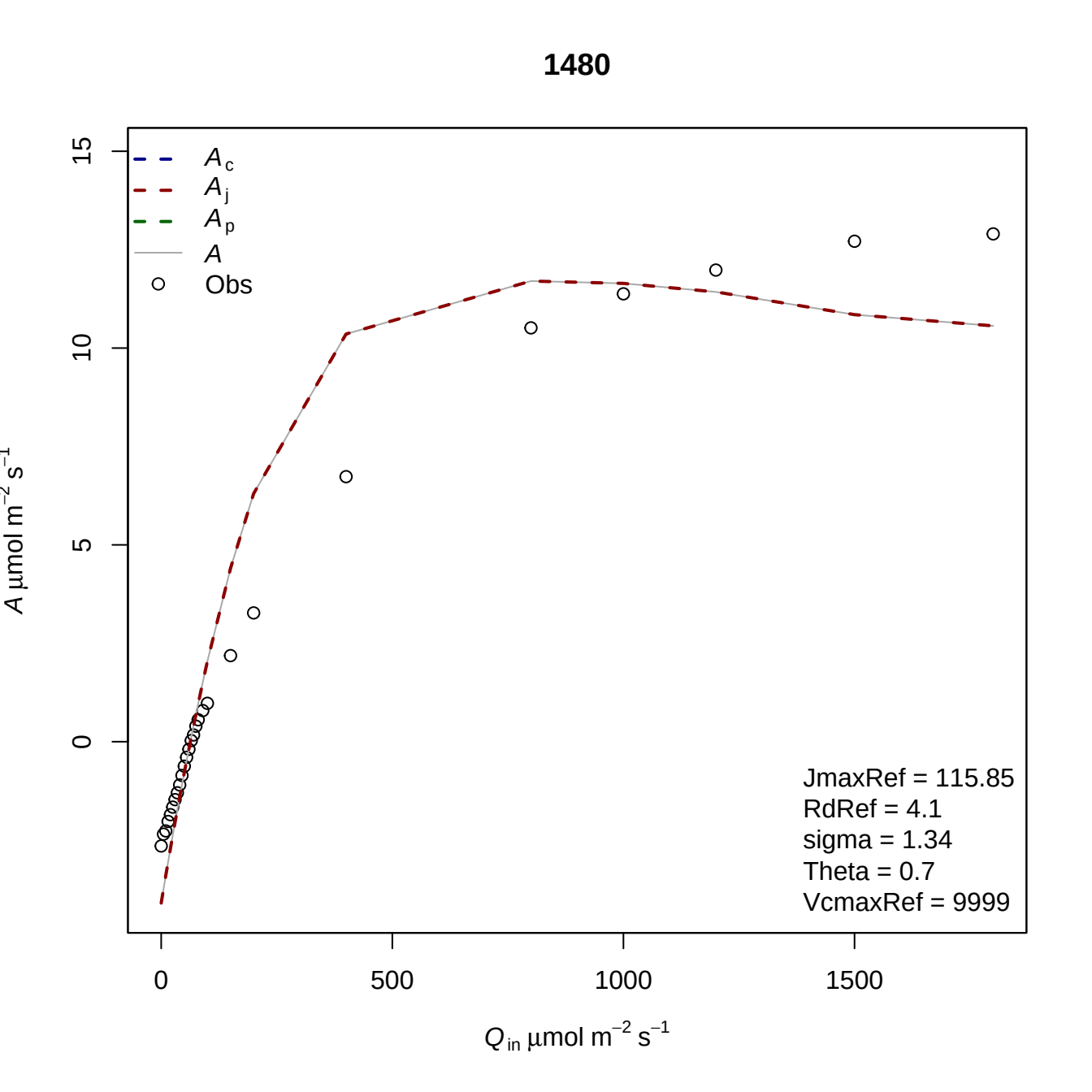

1475
A_c
A_j
A_p
A
Obs
JmaxRef = 65.91
RdRef = 3.89
sigma = 1.02
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
200
400
600
800
1000

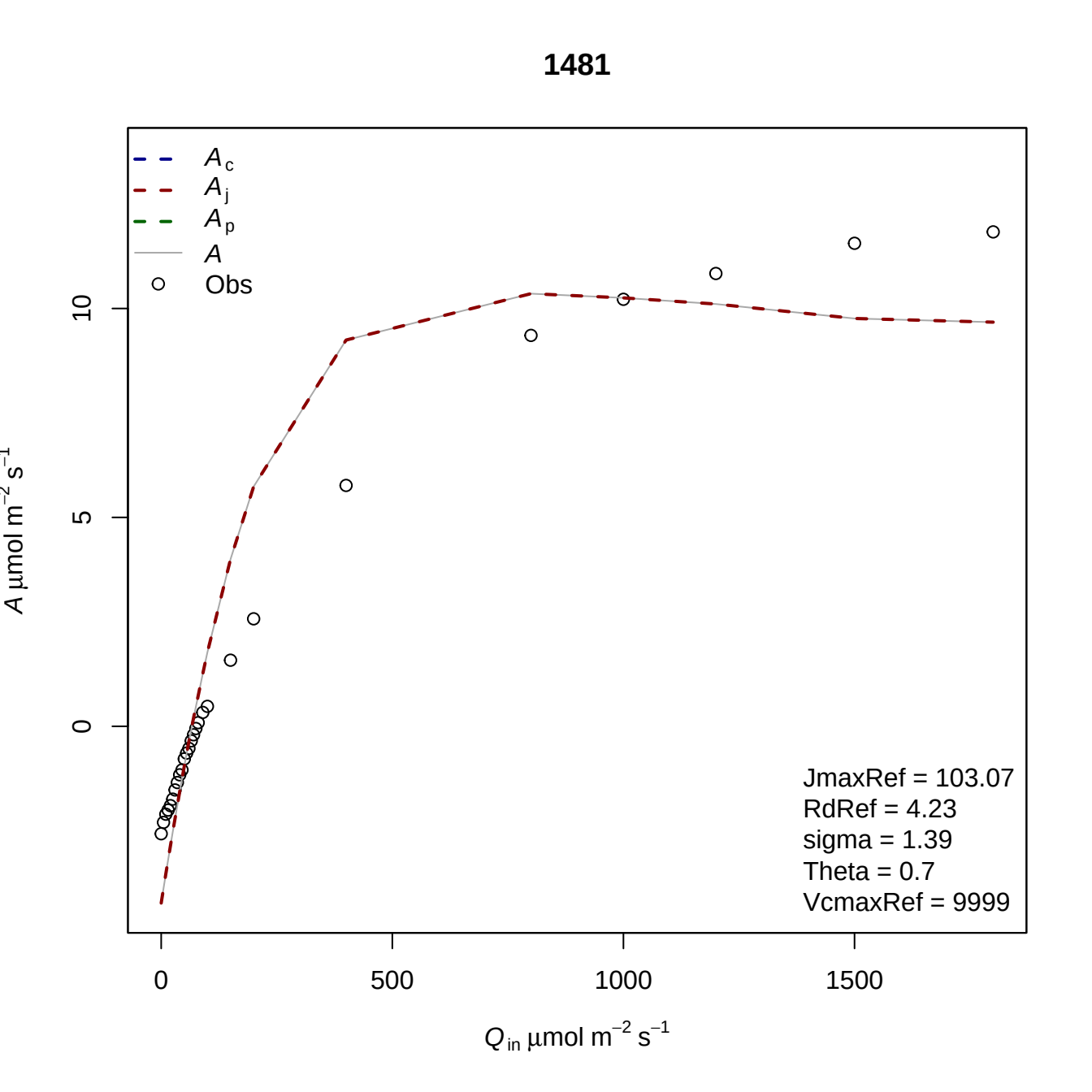

1476
A_c
A_j
A_p
A
Obs
JmaxRef = 74.18
RdRef = 4.07
sigma = 1.42
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

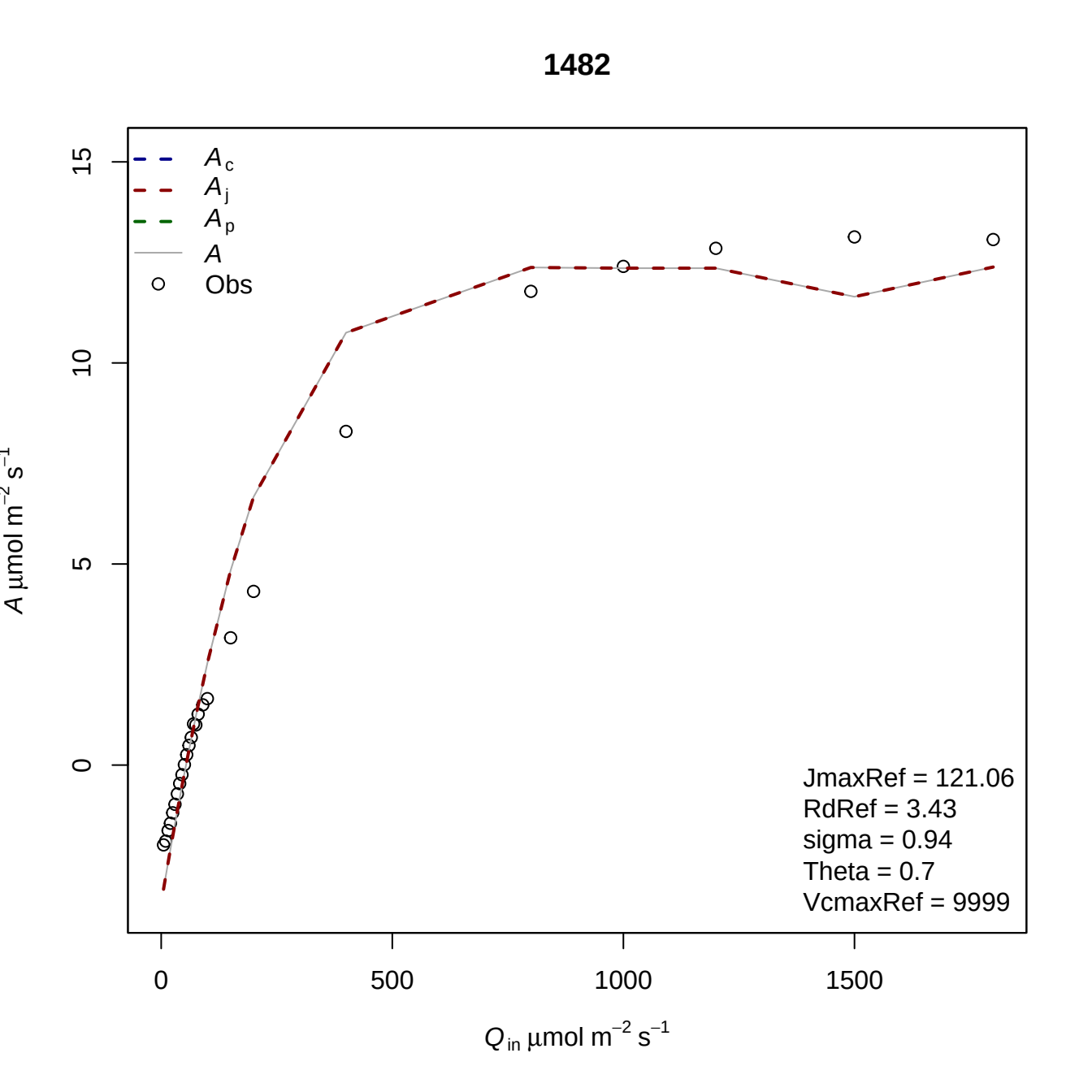

1477
A_c
A_j
A_p
A
Obs
JmaxRef = 97.34
RdRef = 3.58
sigma = 1.31
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

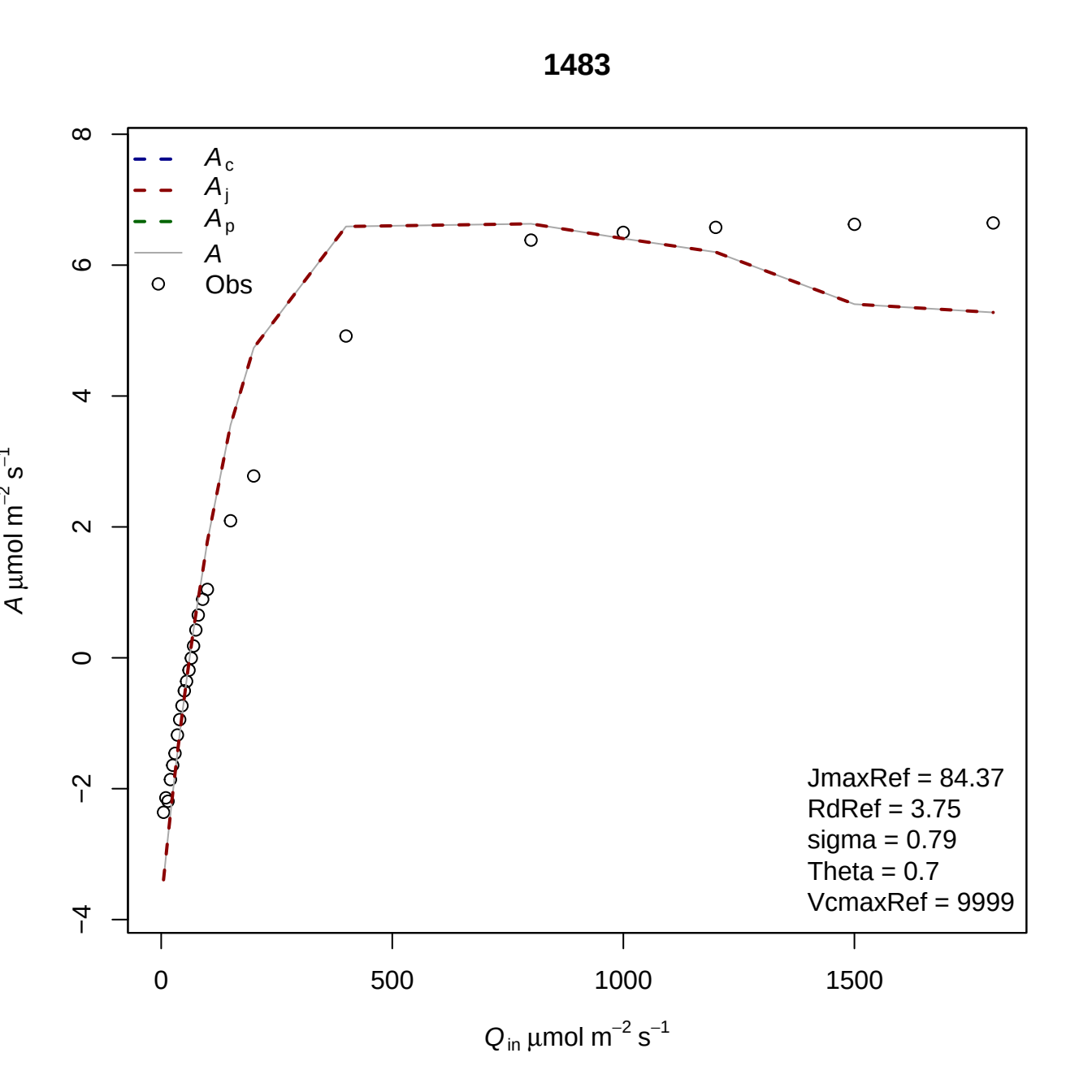

1478
A_c
A_j
A_p
A
Obs
JmaxRef = 84.57
RdRef = 4.11
sigma = 1.31
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

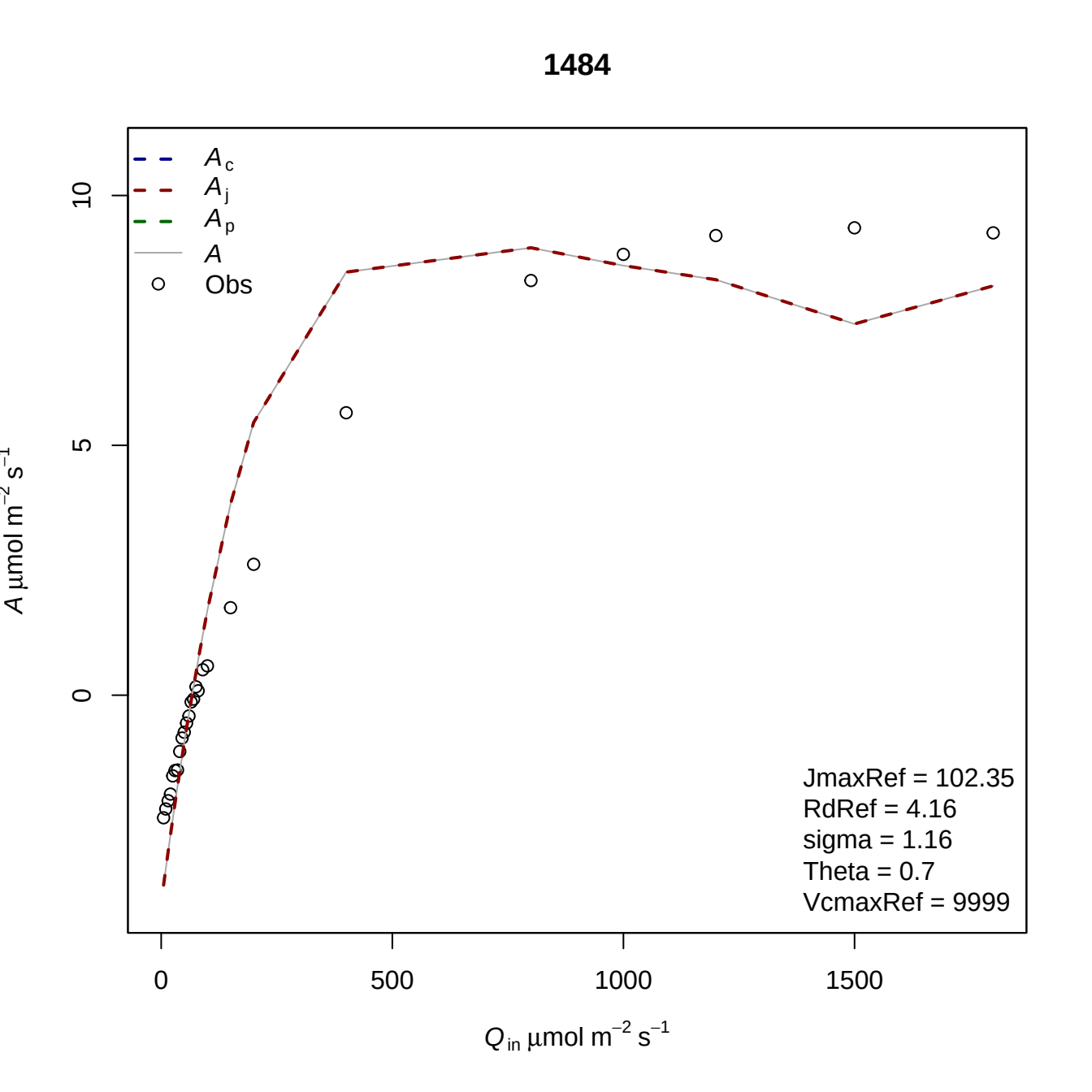

1479
A_c
A_j
A_p
A
Obs
JmaxRef = 105.15
RdRef = 4.15
sigma = 1.13
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

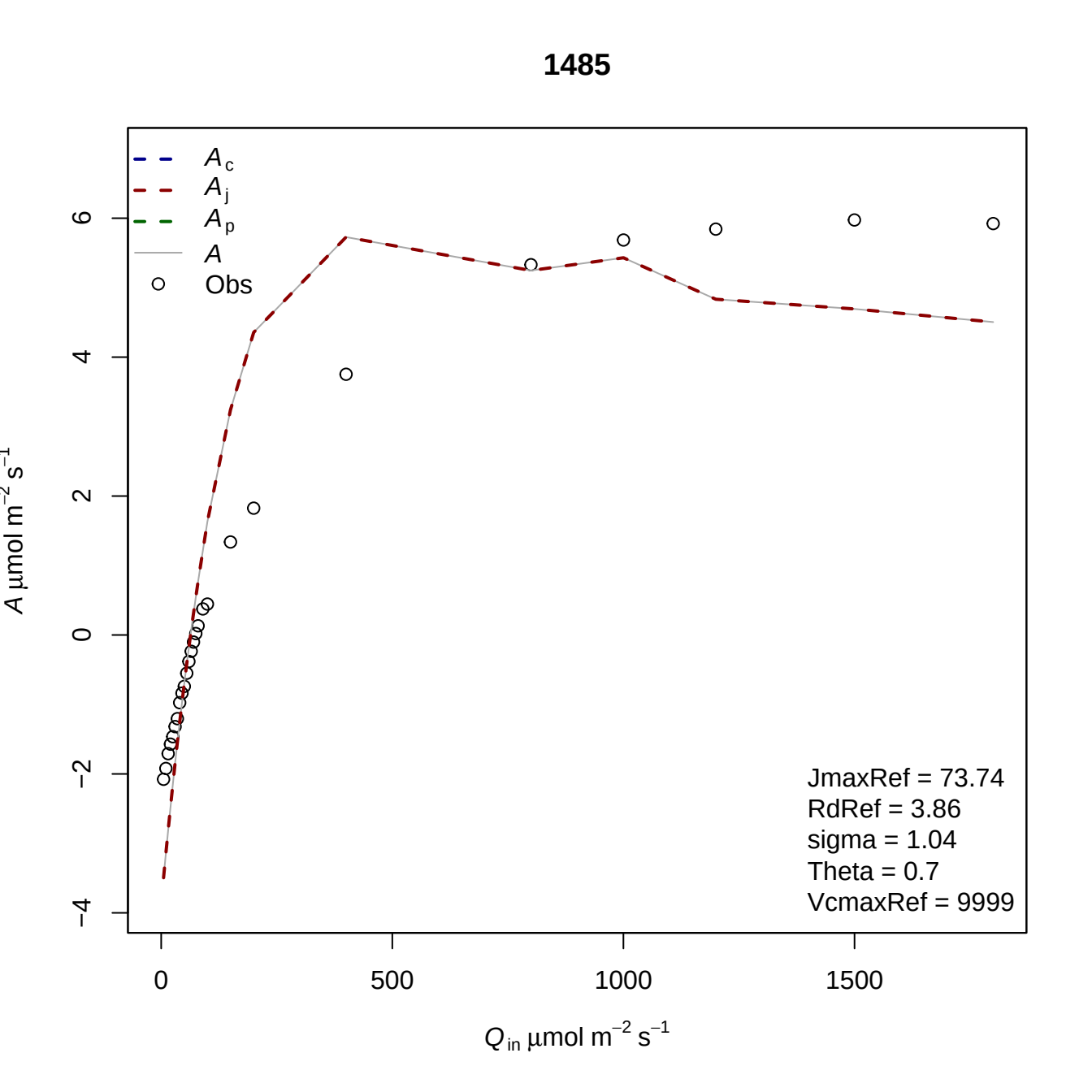

1480
A_c
A_j
A_p
A
Obs
JmaxRef = 115.85
RdRef = 4.1
sigma = 1.34
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

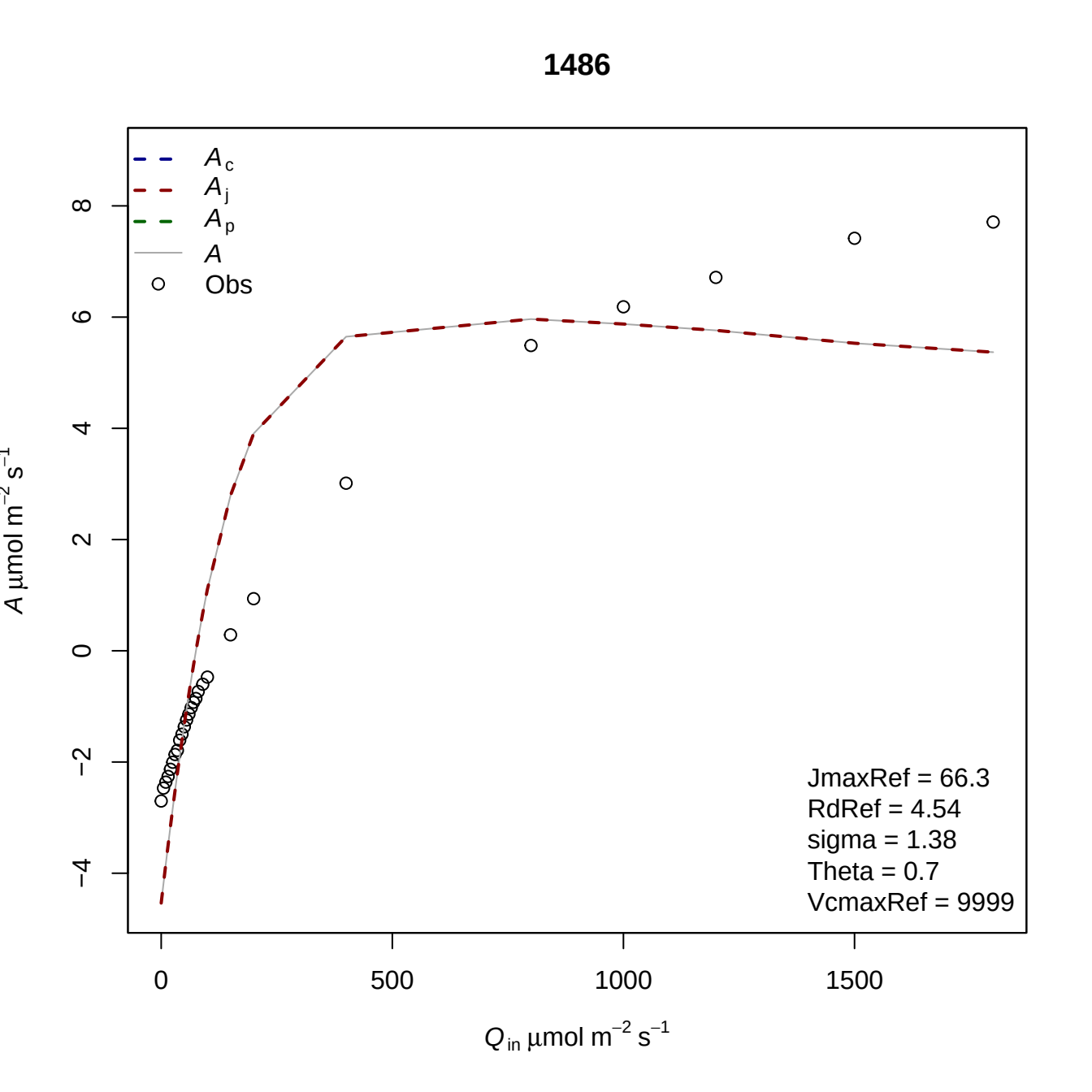

1481
A_c
A_j
A_p
A
Obs
JmaxRef = 103.07
RdRef = 4.23
sigma = 1.39
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

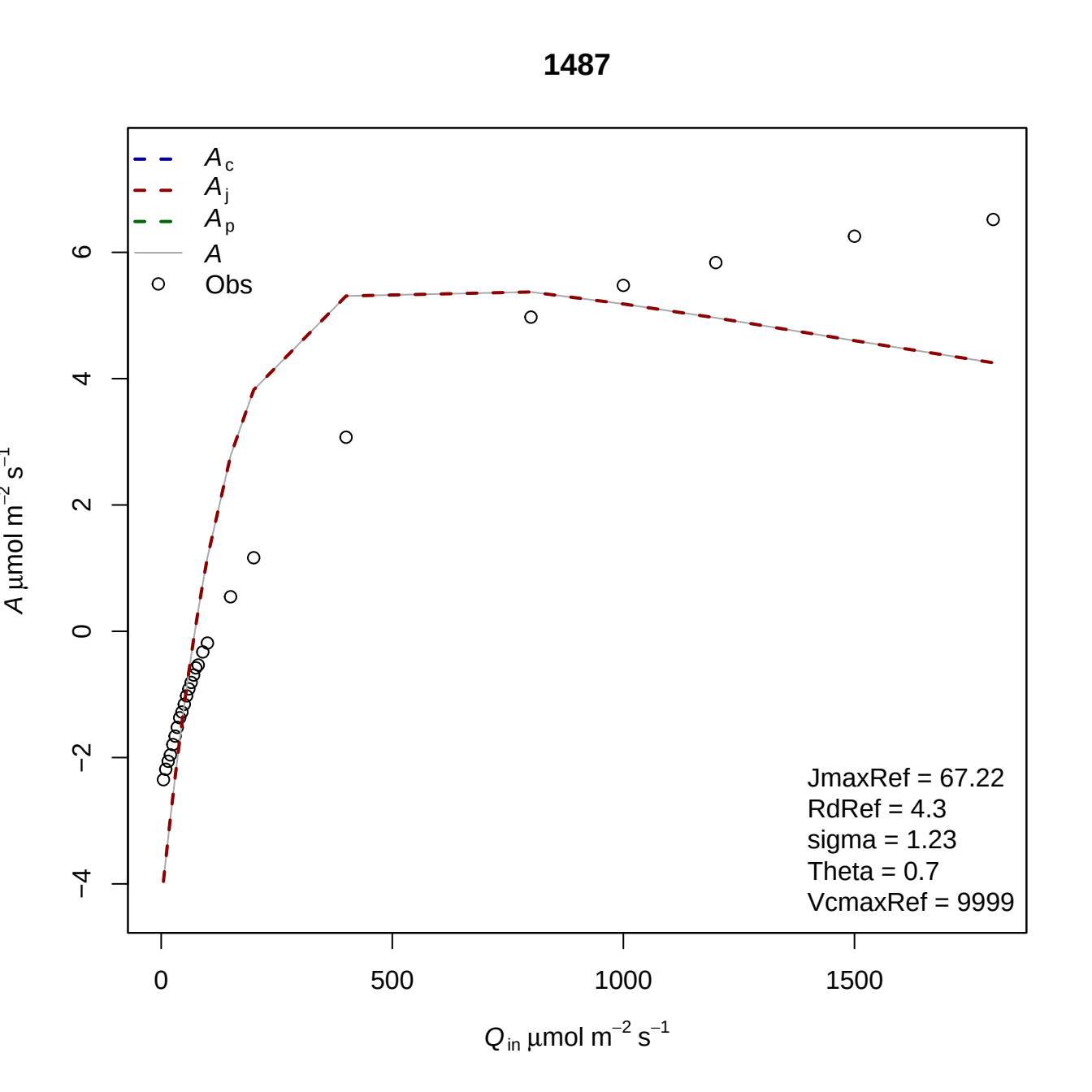

1482
A_c
A_j
A_p
A
Obs
JmaxRef = 121.06
RdRef = 3.43
sigma = 0.94
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

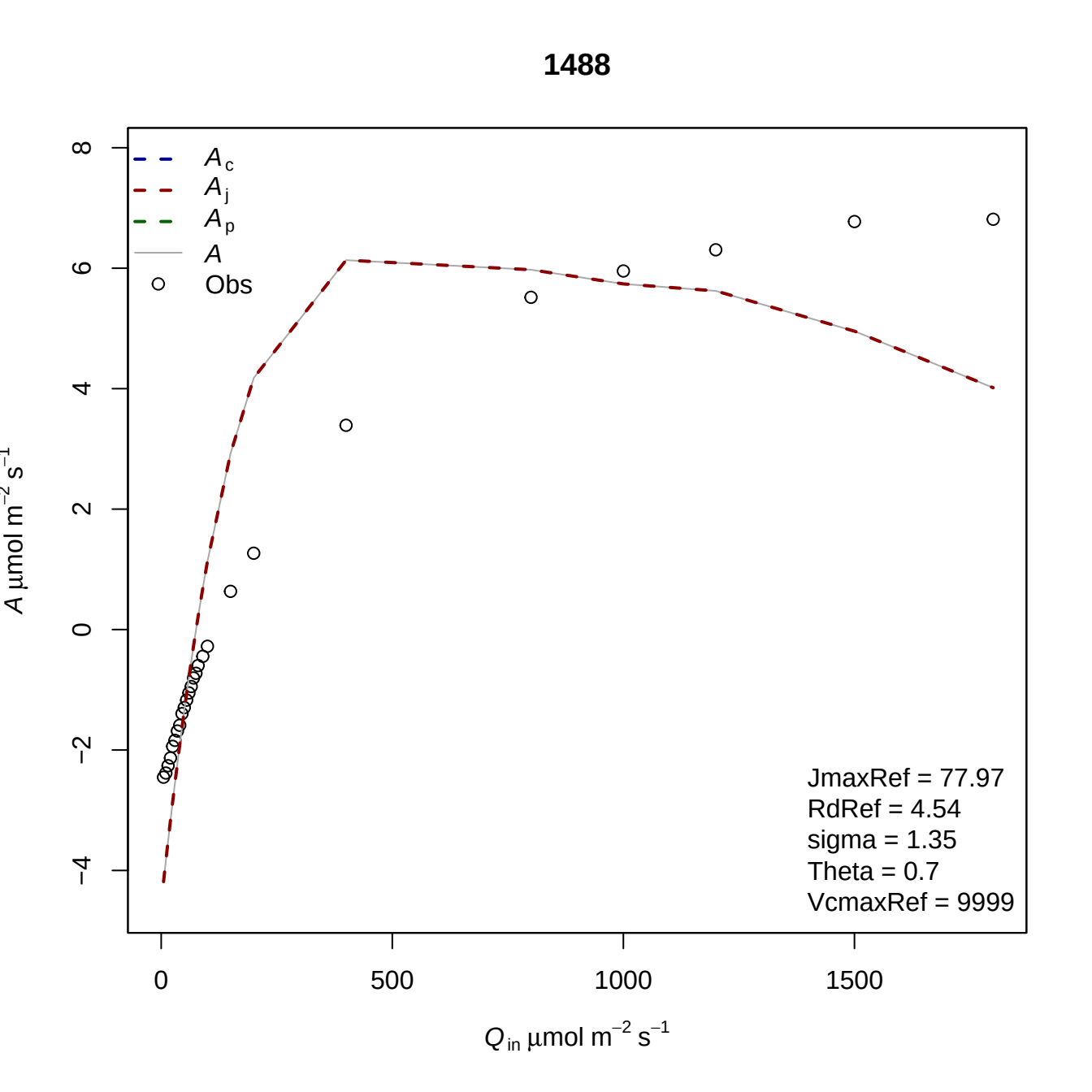

1483
A_c
A_j
A_p
A
Obs
JmaxRef = 84.37
RdRef = 3.75
sigma = 0.79
Theta = 0.7
VcmaxRef = 9999
8
6
4
2
0
−2
−4
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

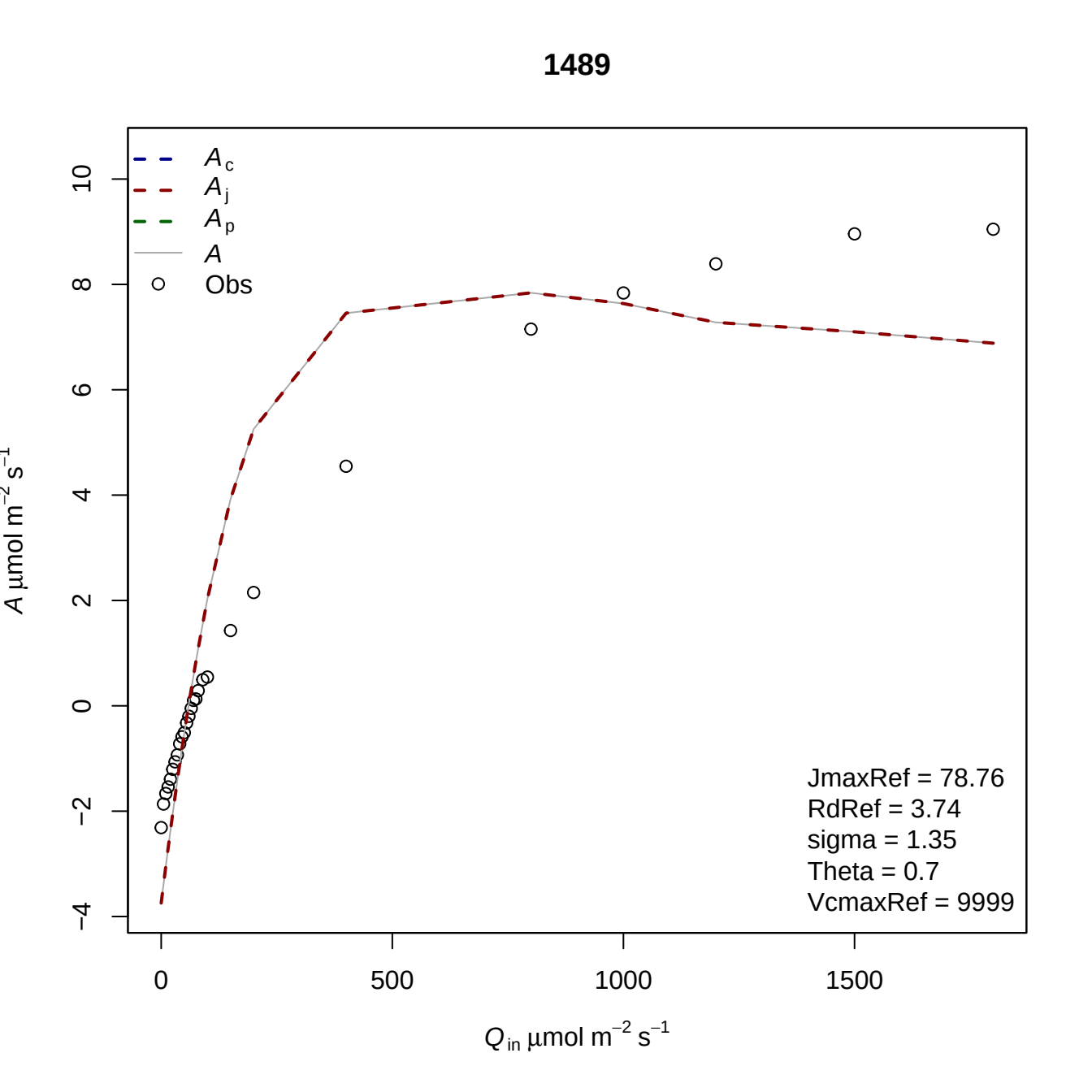

1484
A_c
A_j
A_p
A
Obs
JmaxRef = 102.35
RdRef = 4.16
sigma = 1.16
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

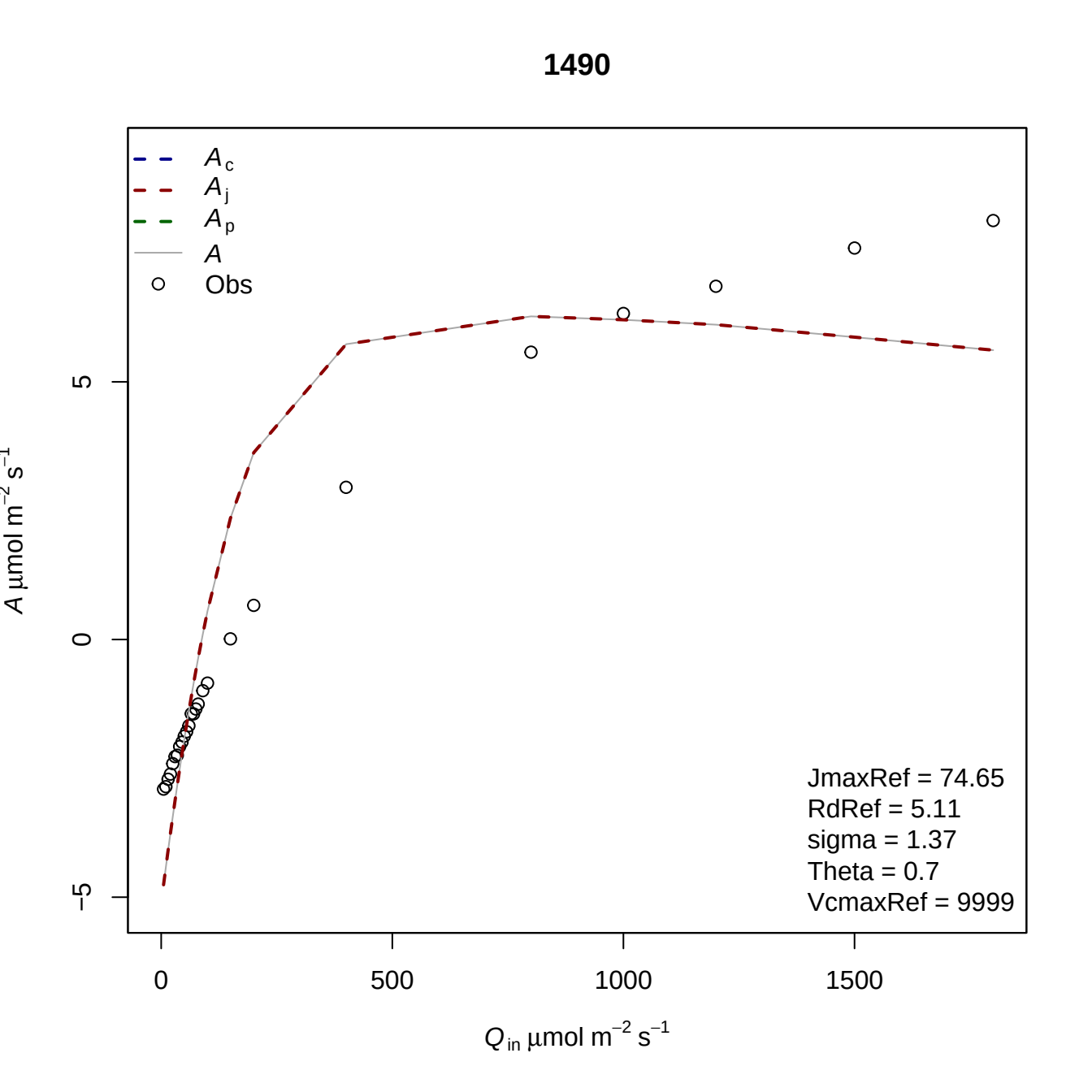

1485
A_c
A_j
A_p
A
Obs
JmaxRef = 73.74
RdRef = 3.86
sigma = 1.04
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500

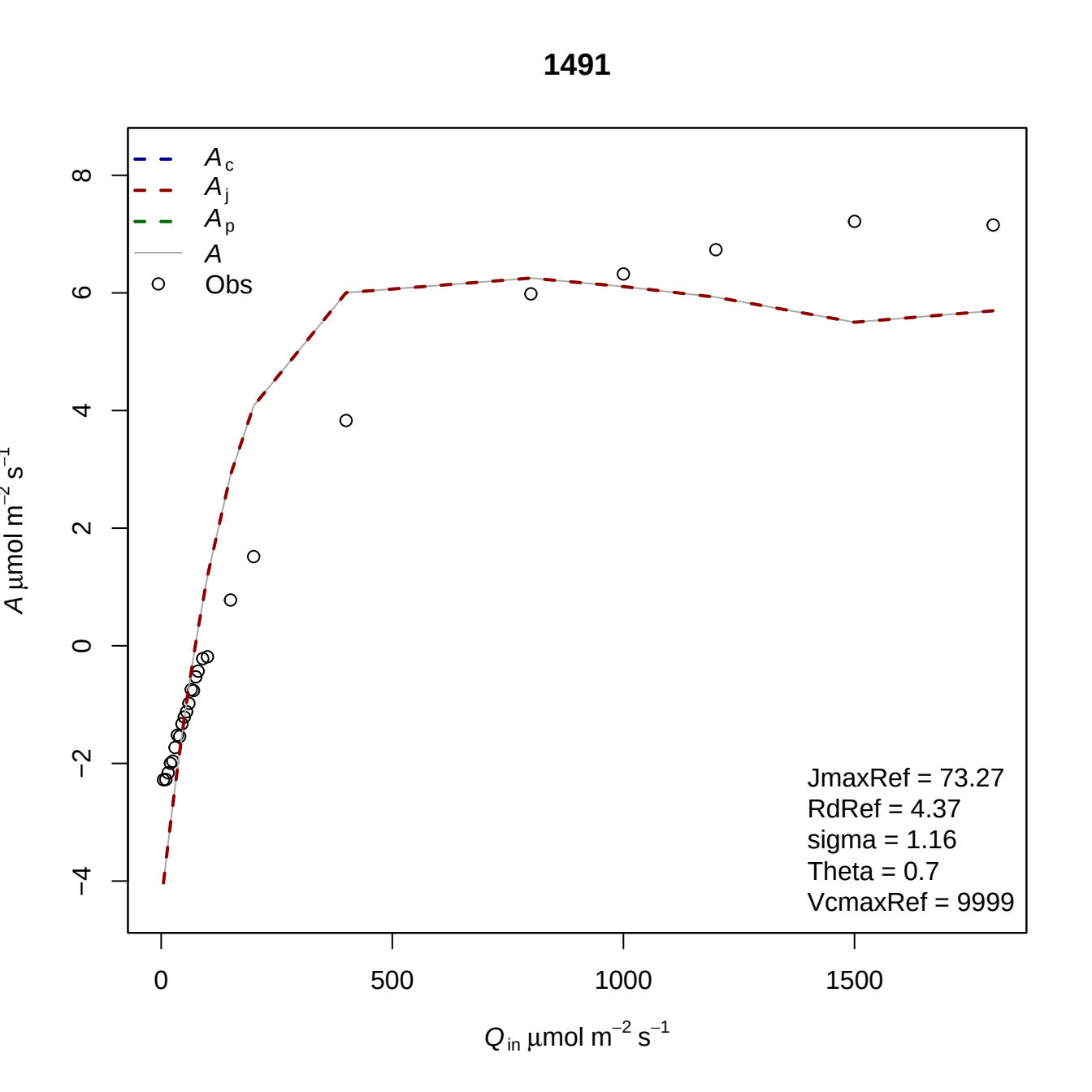

1486
A_c
A_j
A_p
A
Obs
JmaxRef = 66.3
RdRef = 4.54
sigma = 1.38
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

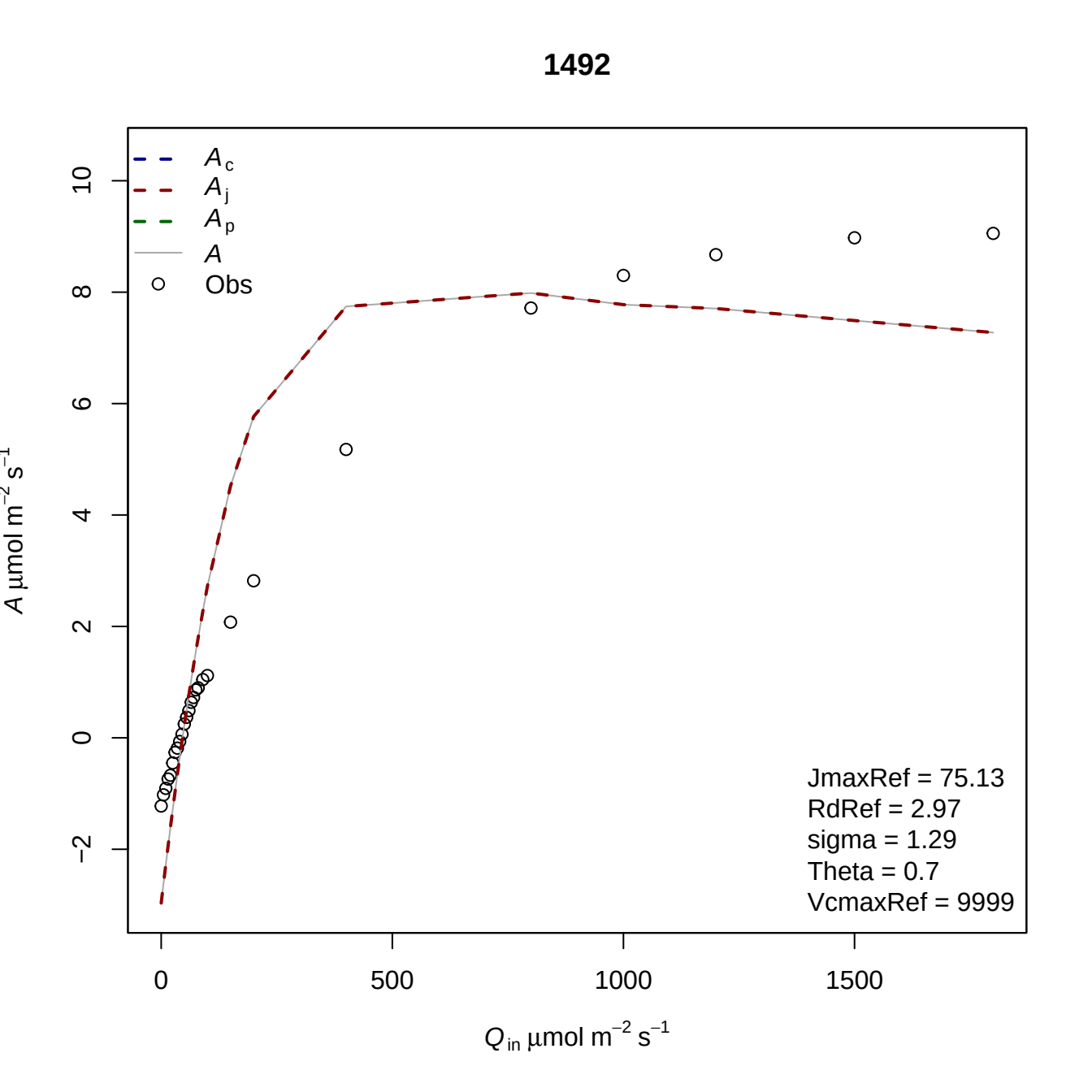

1487
A_c
A_j
A_p
A
Obs
JmaxRef = 67.22
RdRef = 4.3
sigma = 1.23
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500

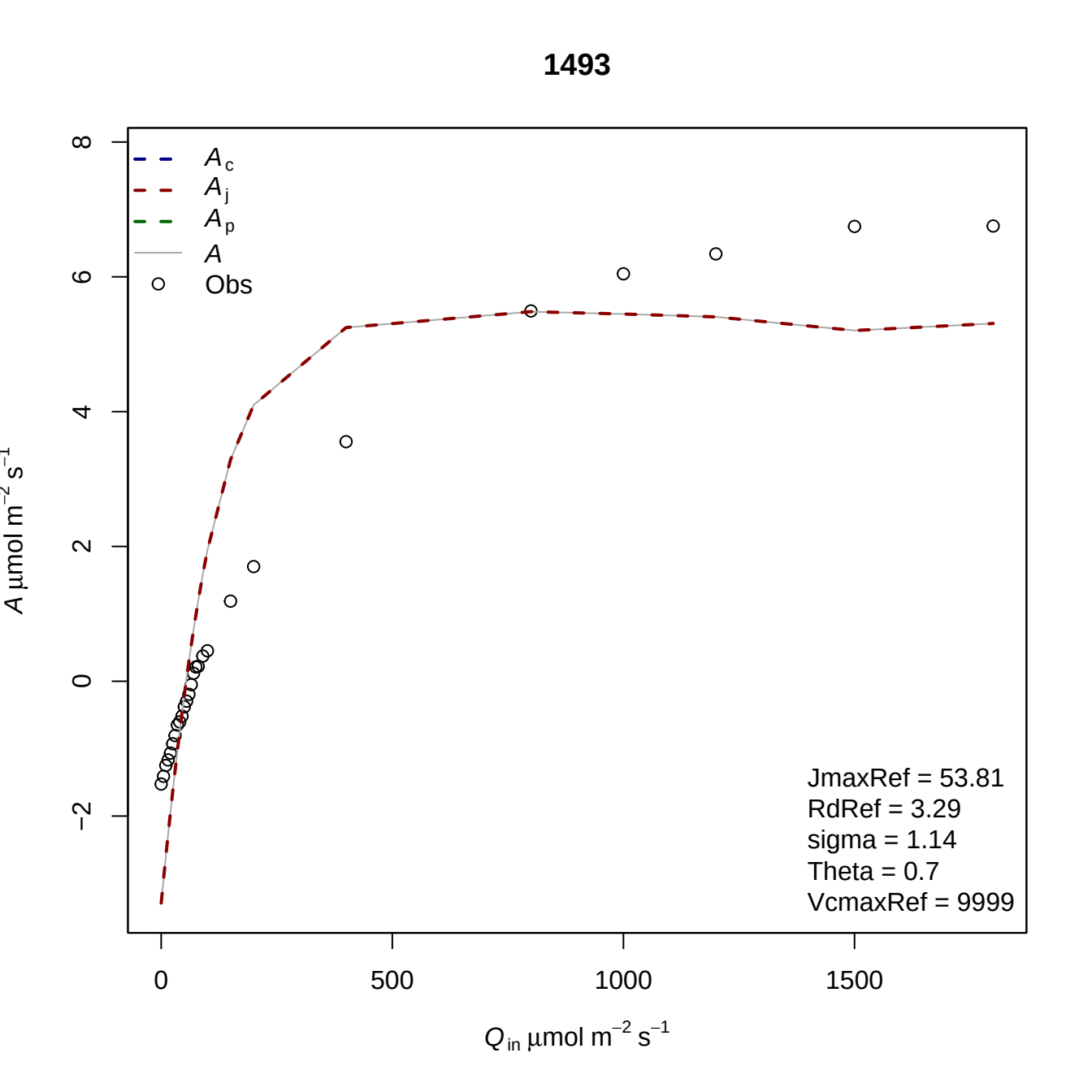

1488
A_c
A_j
A_p
A
Obs
JmaxRef = 77.97
RdRef = 4.54
sigma = 1.35
Theta = 0.7
VcmaxRef = 9999
8
6
4
2
0
−2
−4
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

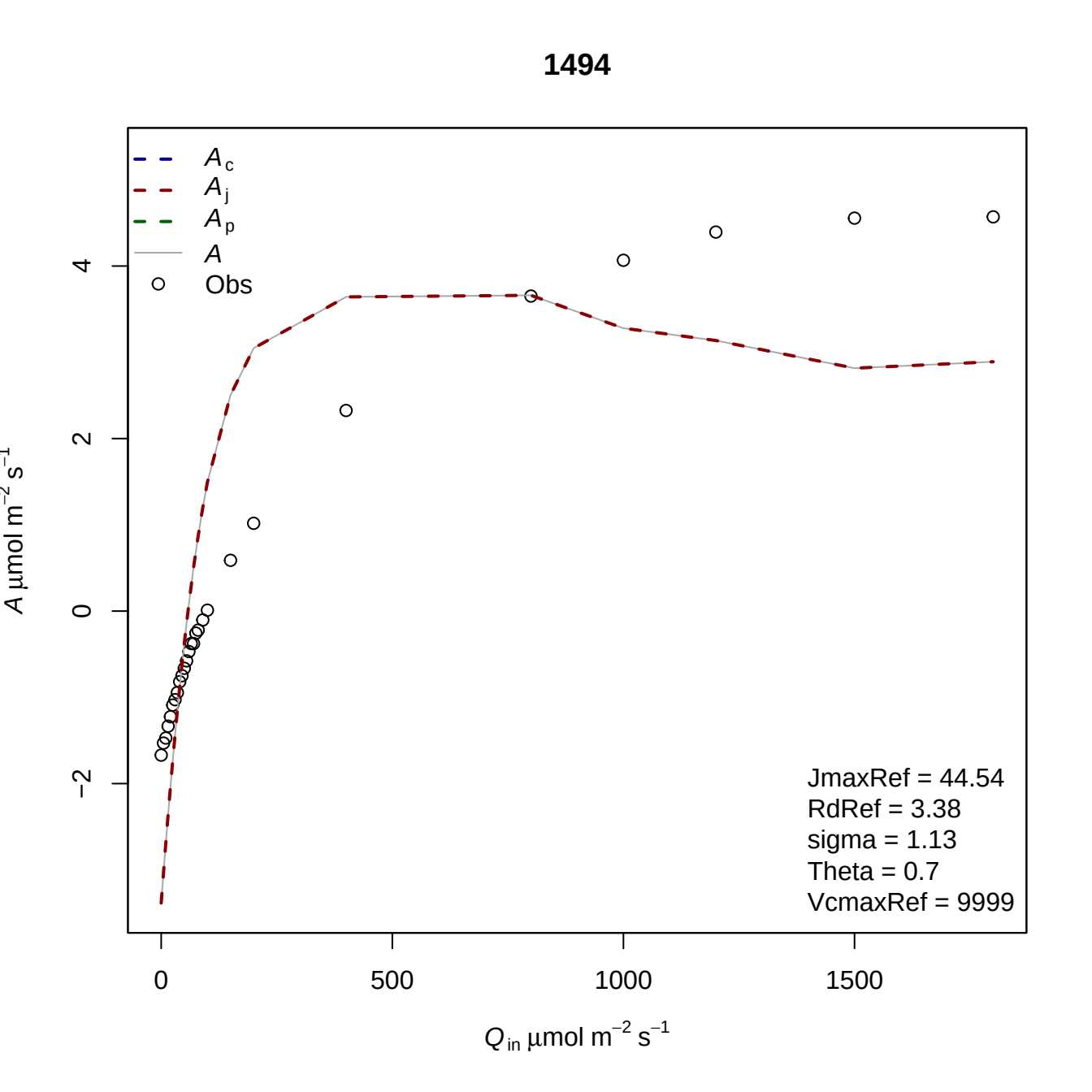

1489
A_c
A_j
A_p
A
Obs
JmaxRef = 78.76
RdRef = 3.74
sigma = 1.35
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500

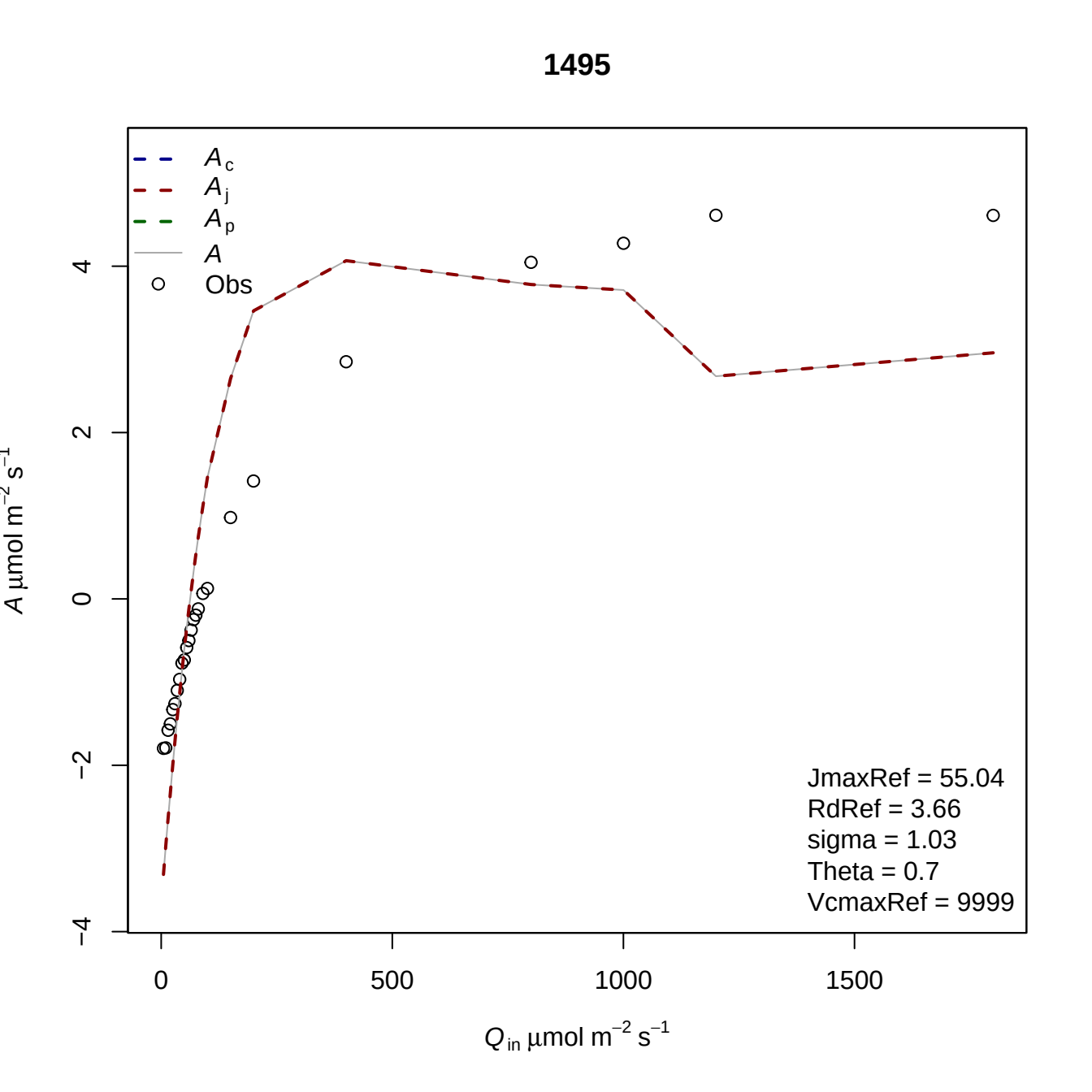

1490
A_c
A_j
A_p
A
Obs
JmaxRef = 74.65
RdRef = 5.11
sigma = 1.37
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−5
0
5
0
500
1000
1500

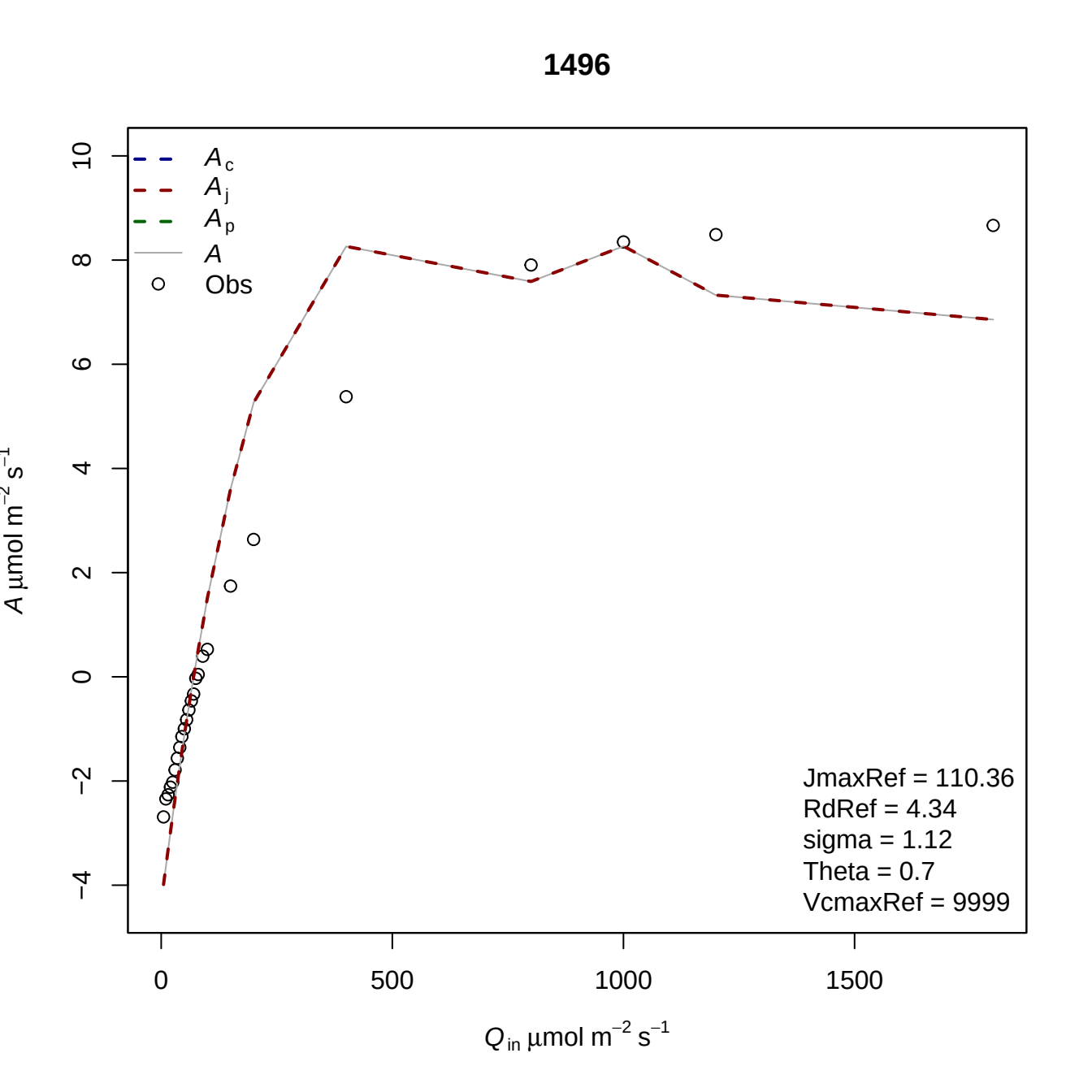

1491
A_c
A_j
A_p
A
Obs
JmaxRef = 73.27
RdRef = 4.37
sigma = 1.16
Theta = 0.7
VcmaxRef = 9999
8
6
4
2
0
−2
−4
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

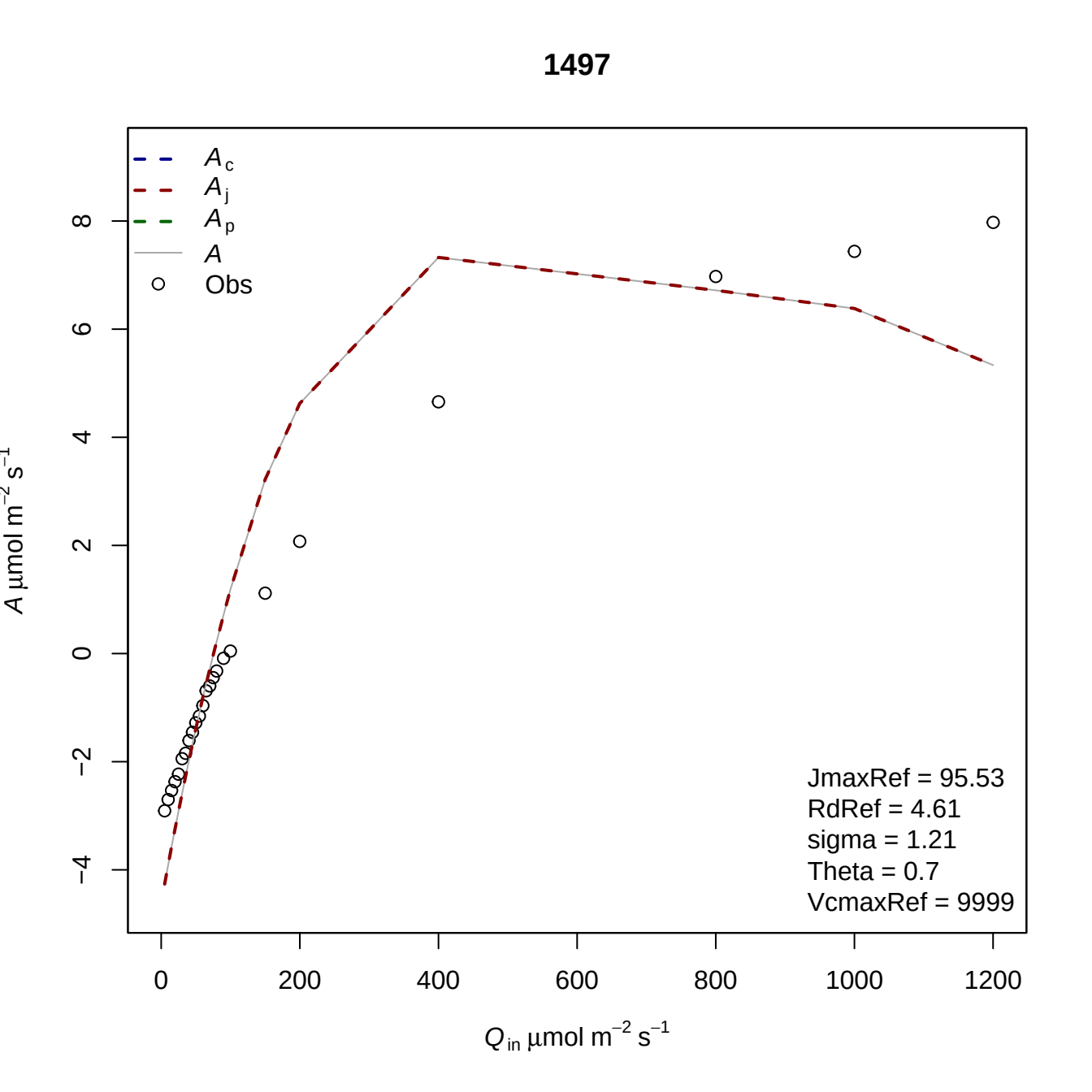

1492
A_c
A_j
A_p
A
Obs
JmaxRef = 75.13
RdRef = 2.97
sigma = 1.29
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

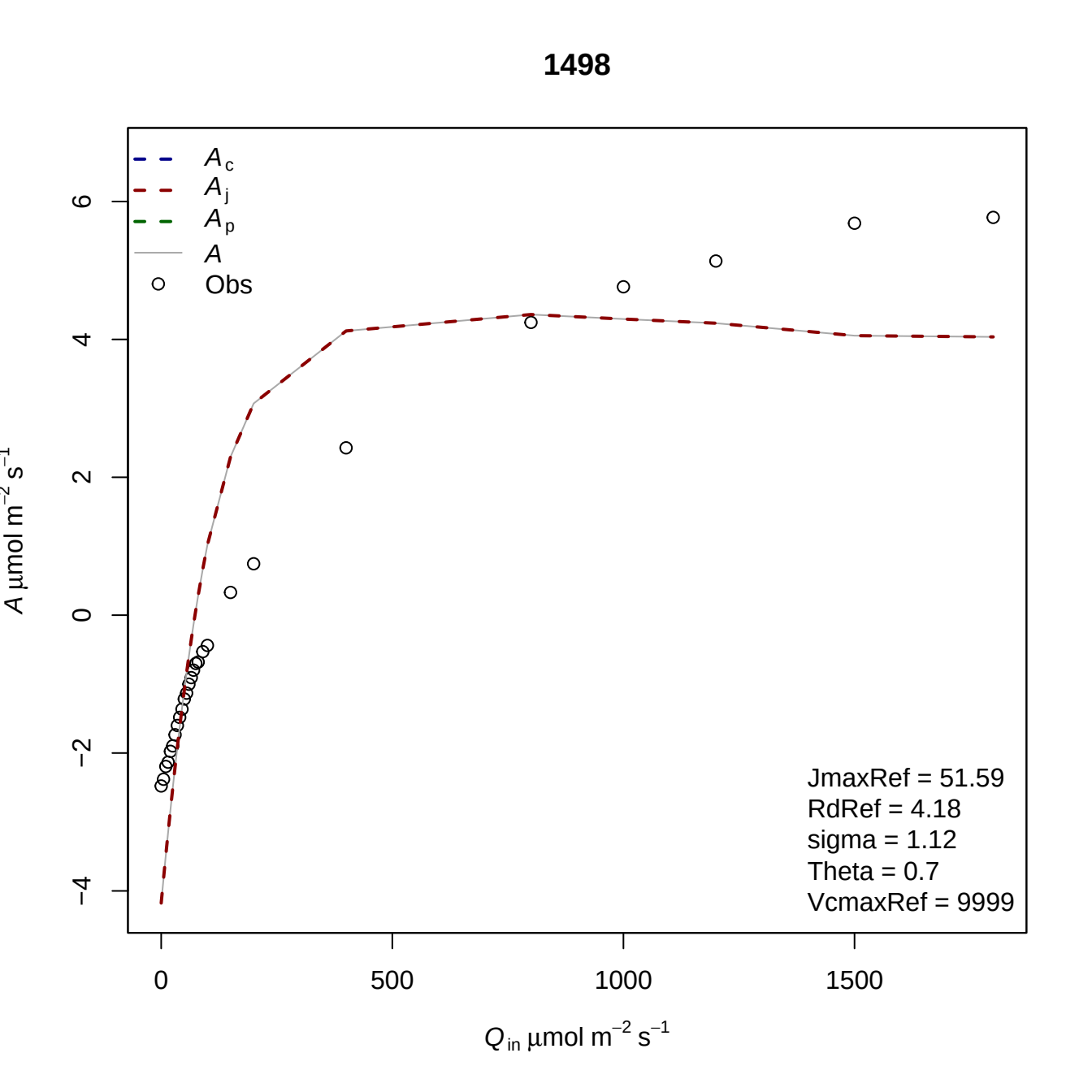

1493
A_c
A_j
A_p
A
Obs
JmaxRef = 53.81
RdRef = 3.29
sigma = 1.14
Theta = 0.7
VcmaxRef = 9999
8
6
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

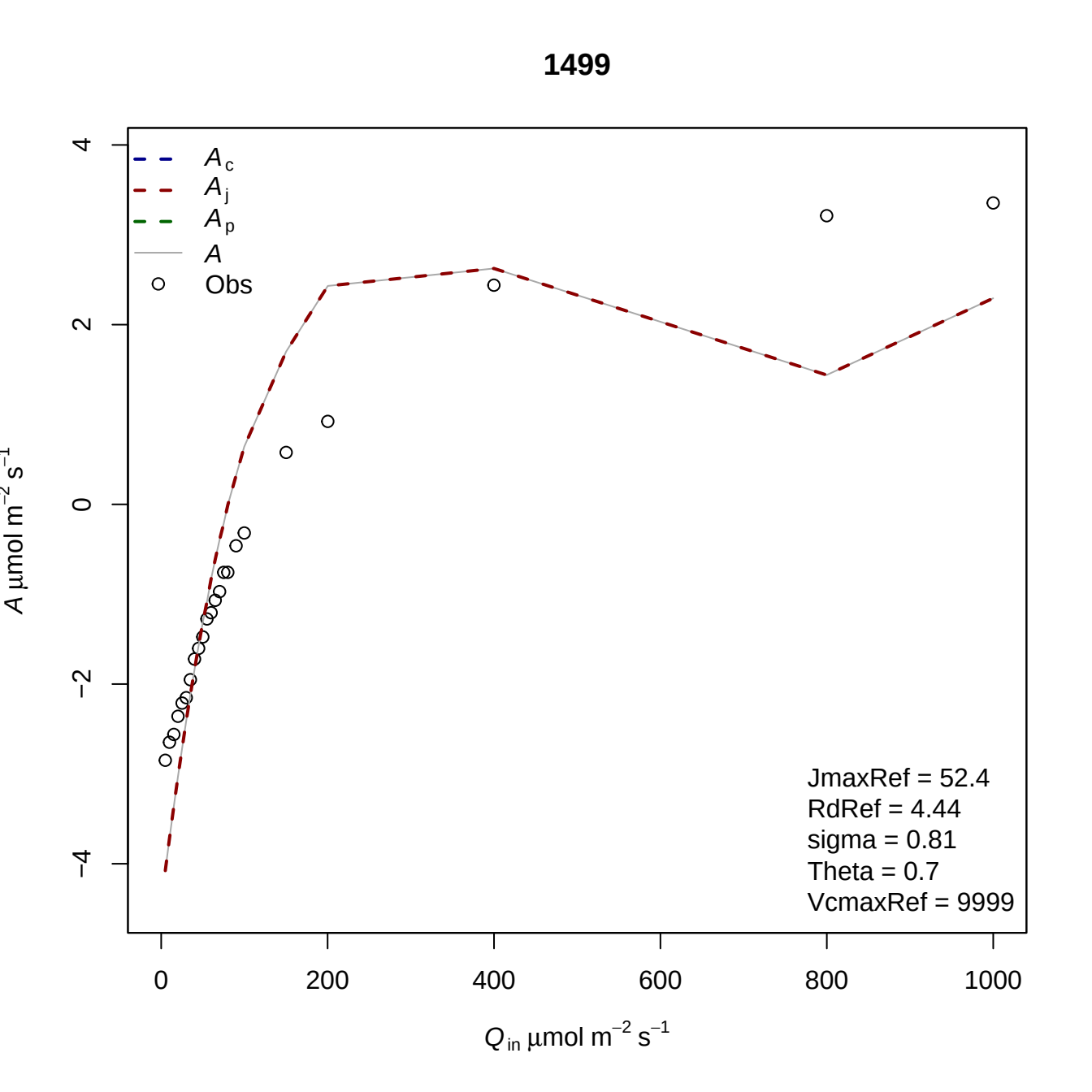

1494
A_c
A_j
A_p
A
Obs
JmaxRef = 44.54
RdRef = 3.38
sigma = 1.13
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
0
500
1000
1500

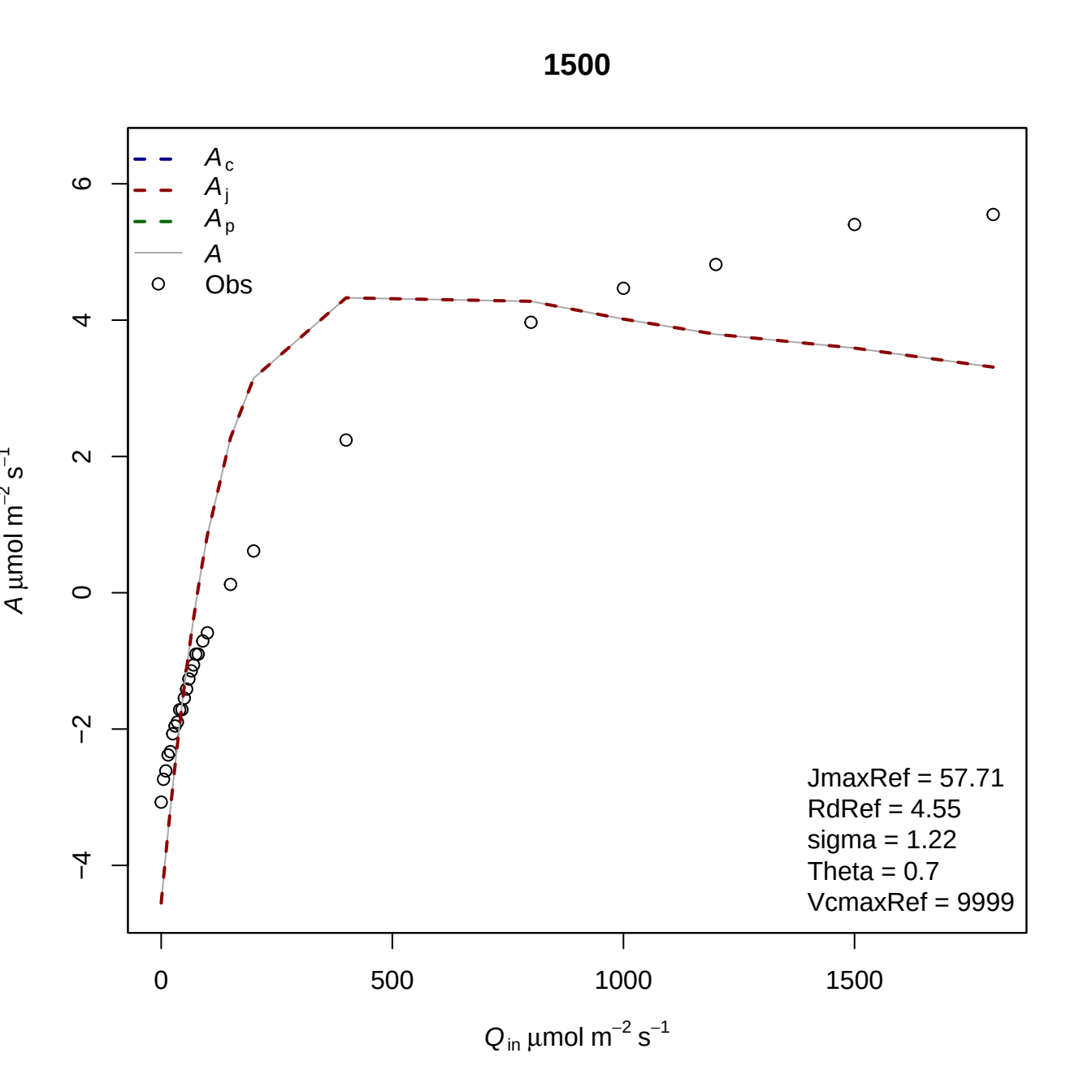

1495
A_c
A_j
A_p
A
Obs
JmaxRef = 55.04
RdRef = 3.66
sigma = 1.03
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
−4
0
500
1000
1500

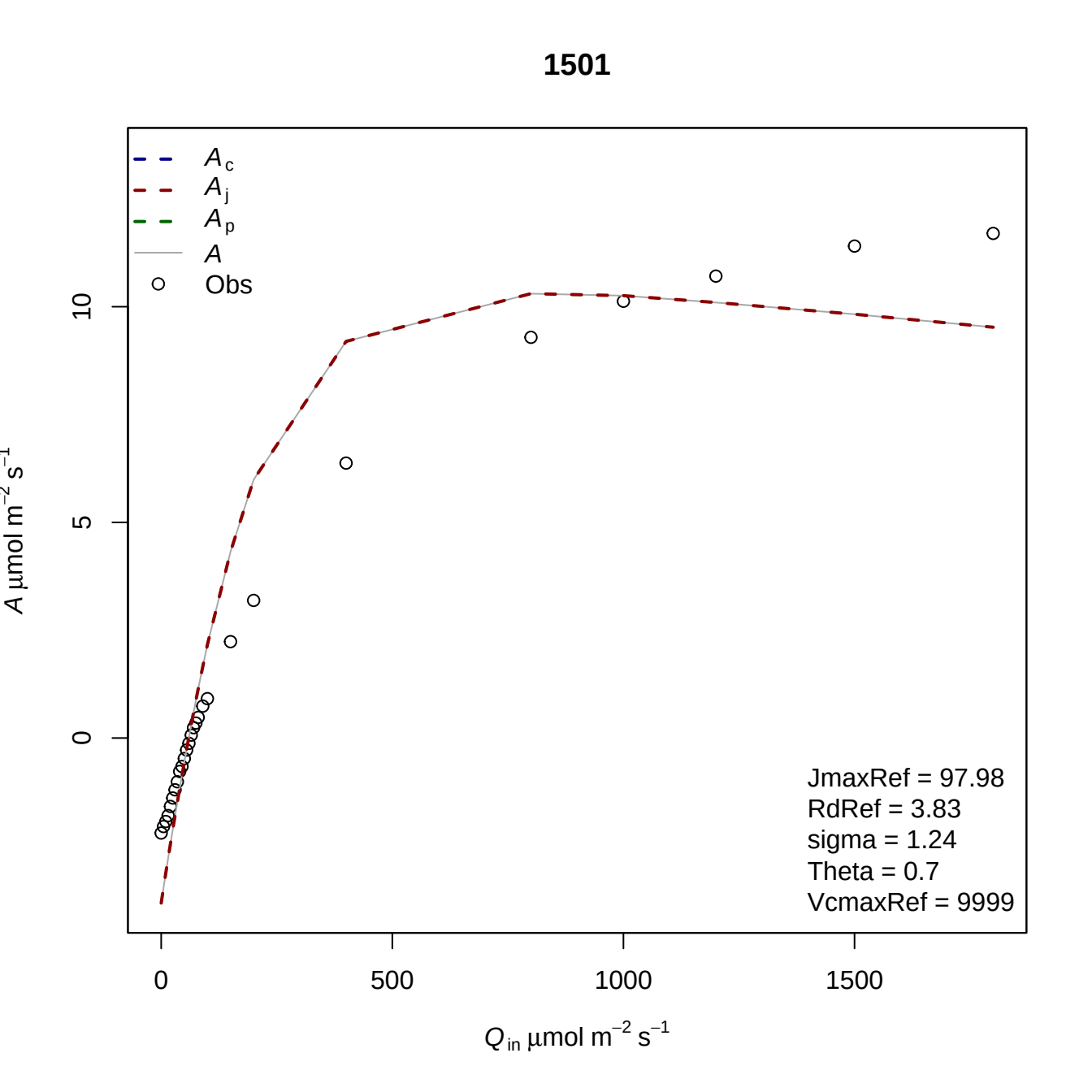

1496
A_c
A_j
A_p
A
Obs
JmaxRef = 110.36
RdRef = 4.34
sigma = 1.12
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
−4
0
500
1000
1500

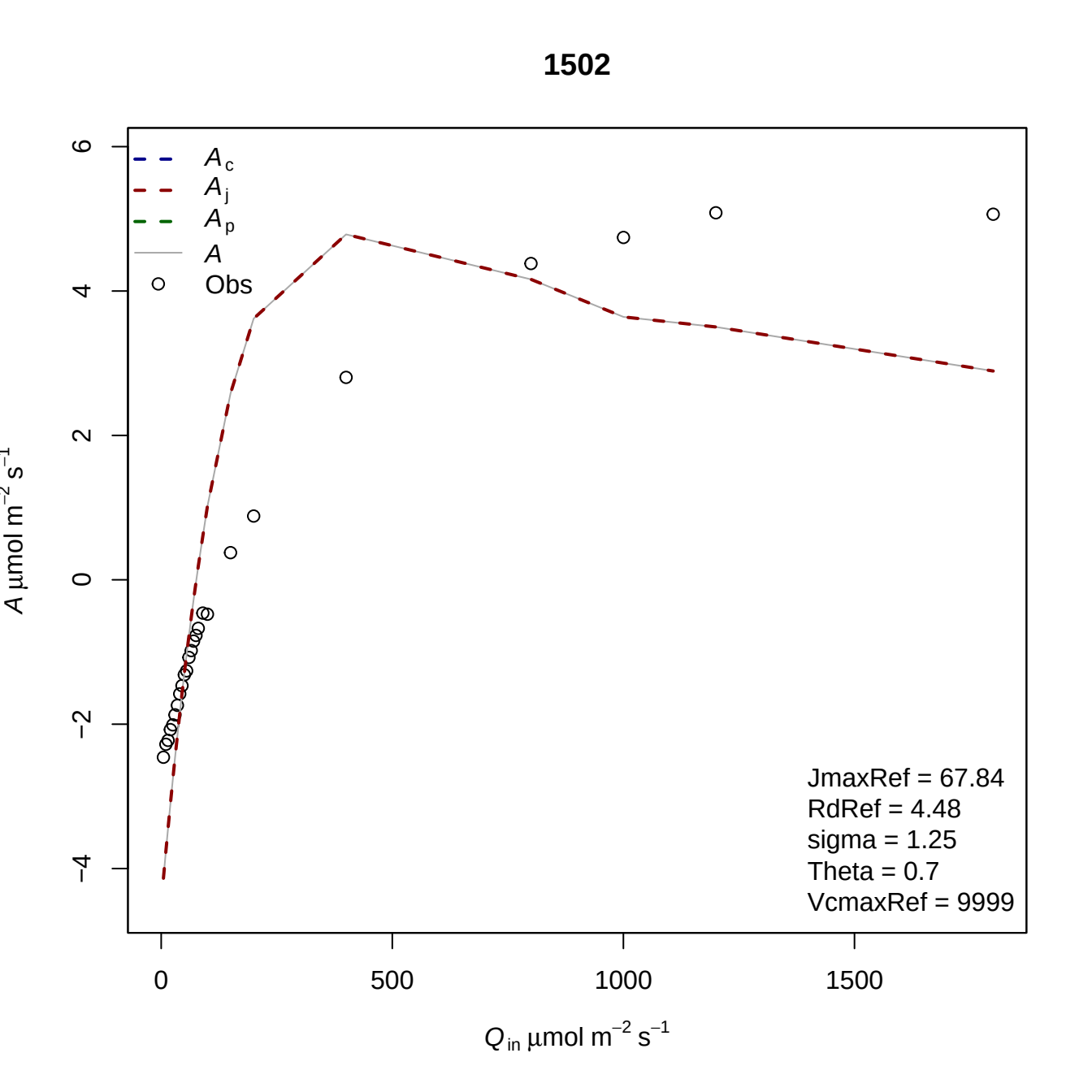

1497
A_c
A_j
A_p
A
Obs
JmaxRef = 95.53
RdRef = 4.61
sigma = 1.21
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
200
400
600
800
1000
1200

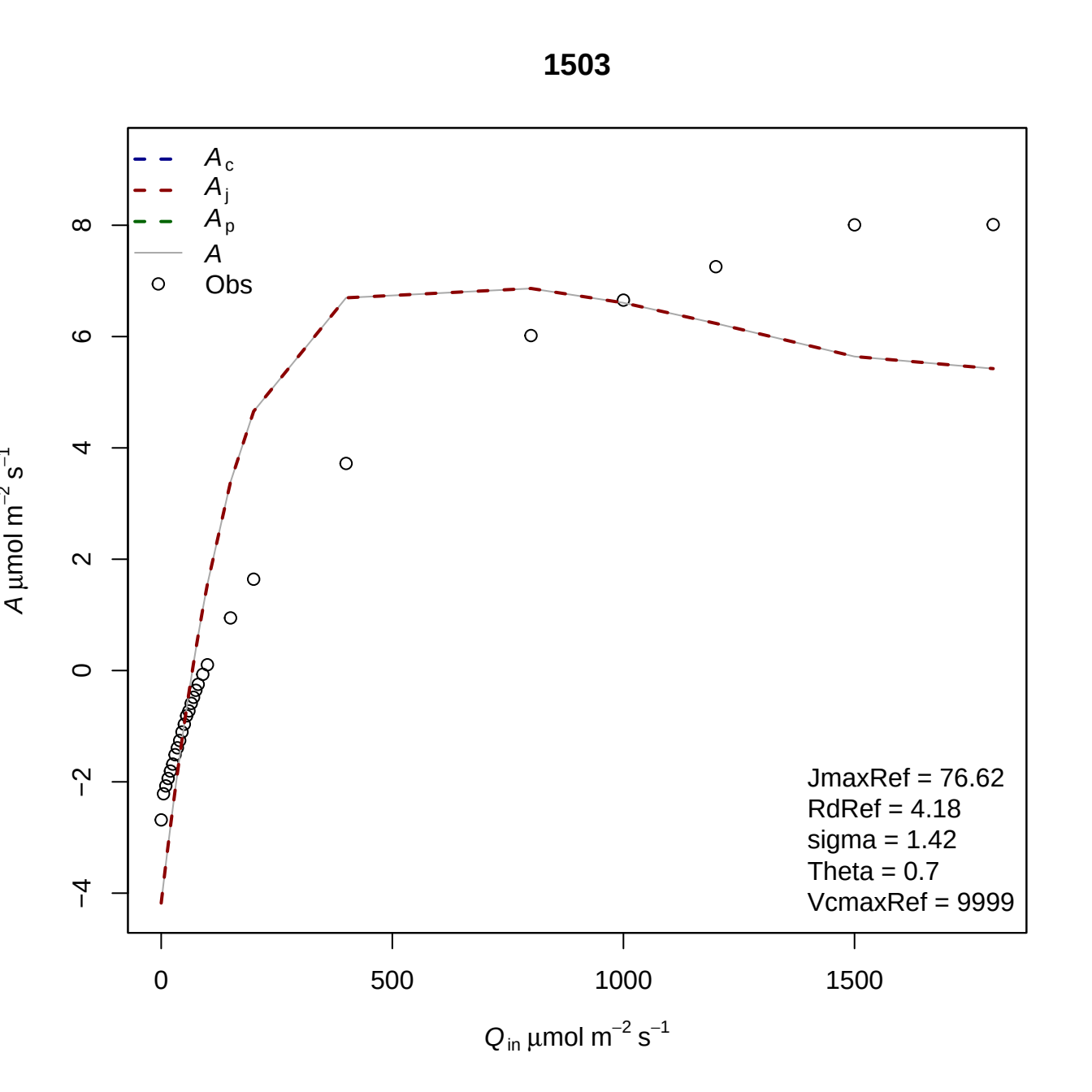

1498
A_c
A_j
A_p
A
Obs
JmaxRef = 51.59
RdRef = 4.18
sigma = 1.12
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500

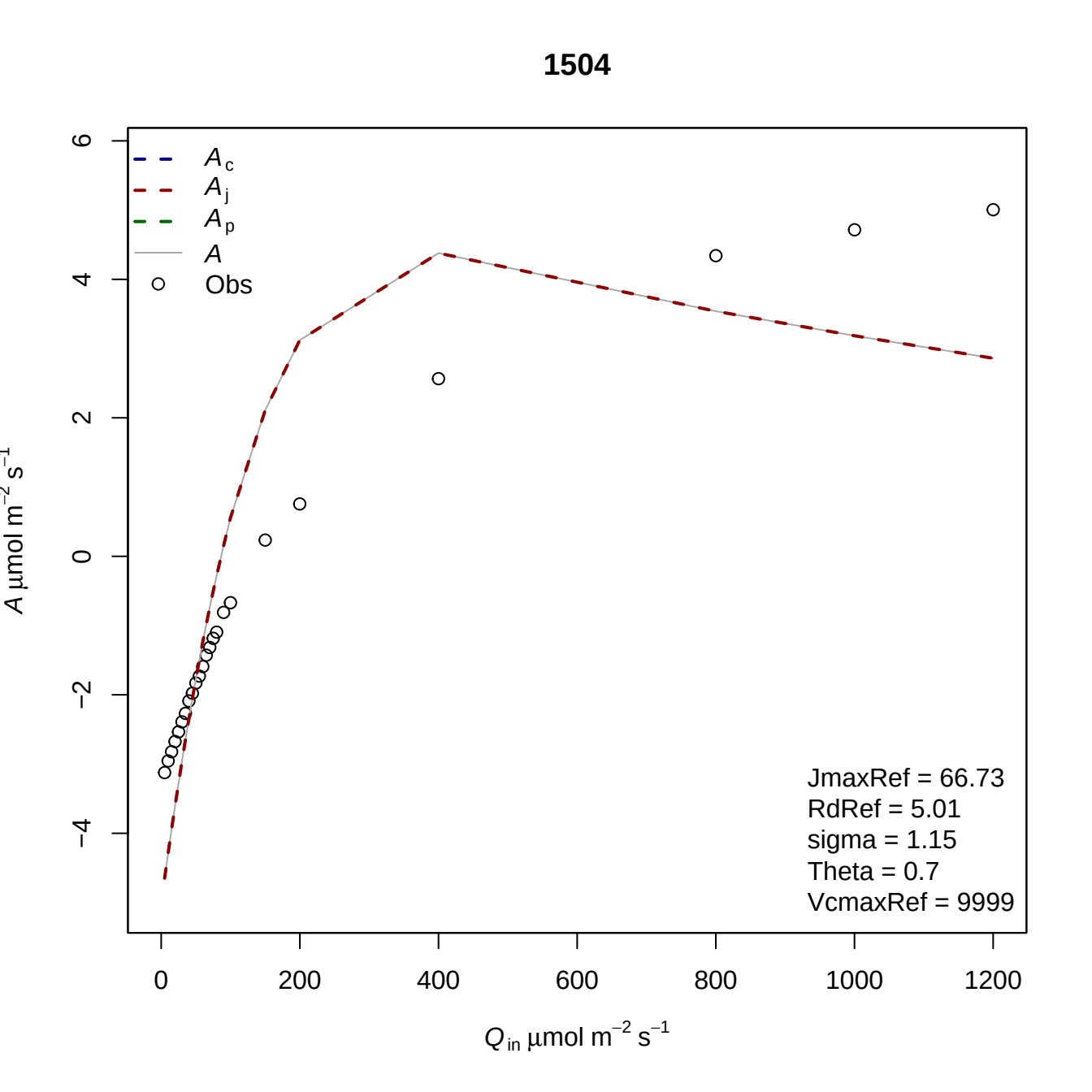

1499
A_c
A_j
A_p
A
Obs
JmaxRef = 52.4
RdRef = 4.44
sigma = 0.81
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
2
0
−2
−4
0
200
400
600
800
1000

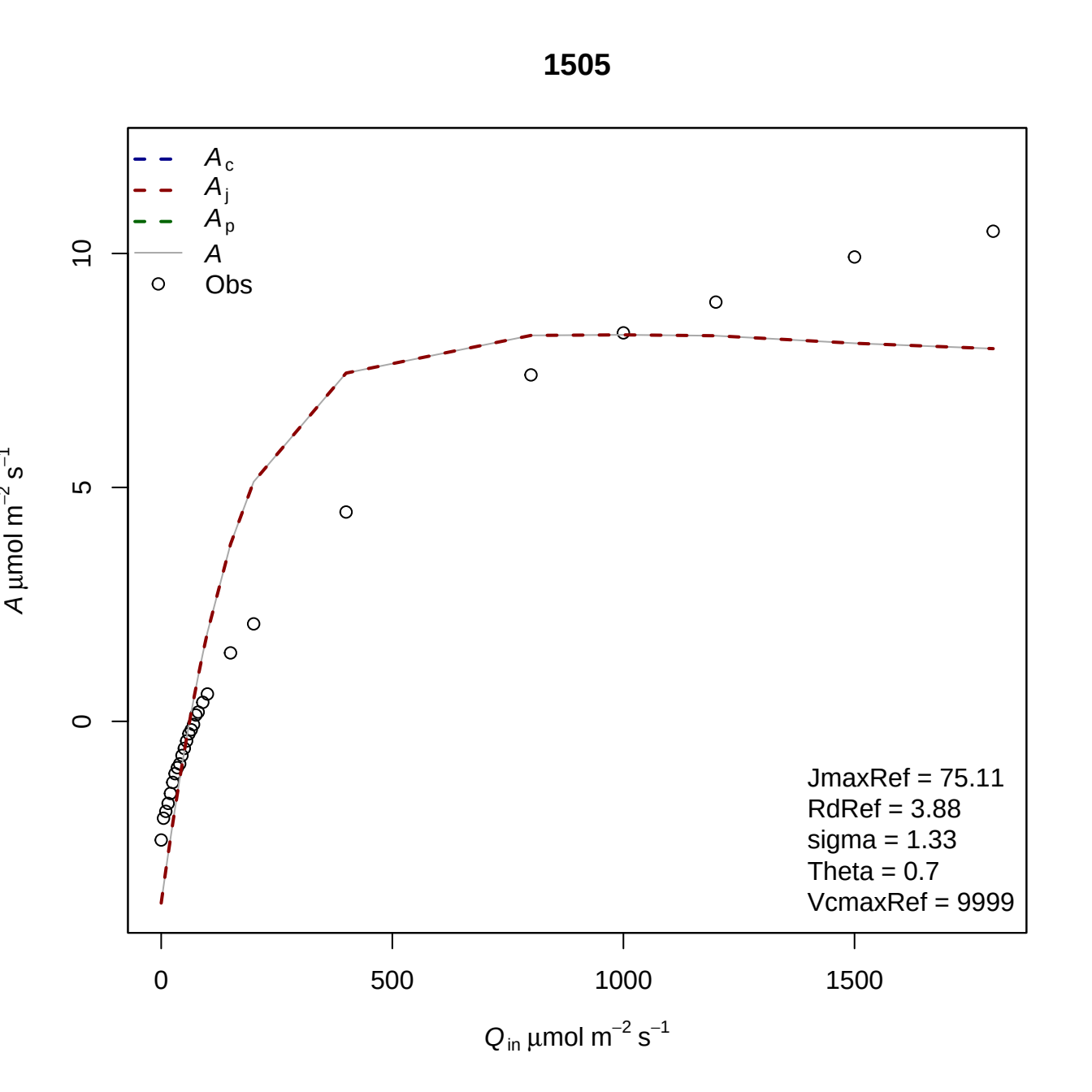

1500
A_c
A_j
A_p
A
Obs
JmaxRef = 57.71
RdRef = 4.55
sigma = 1.22
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500

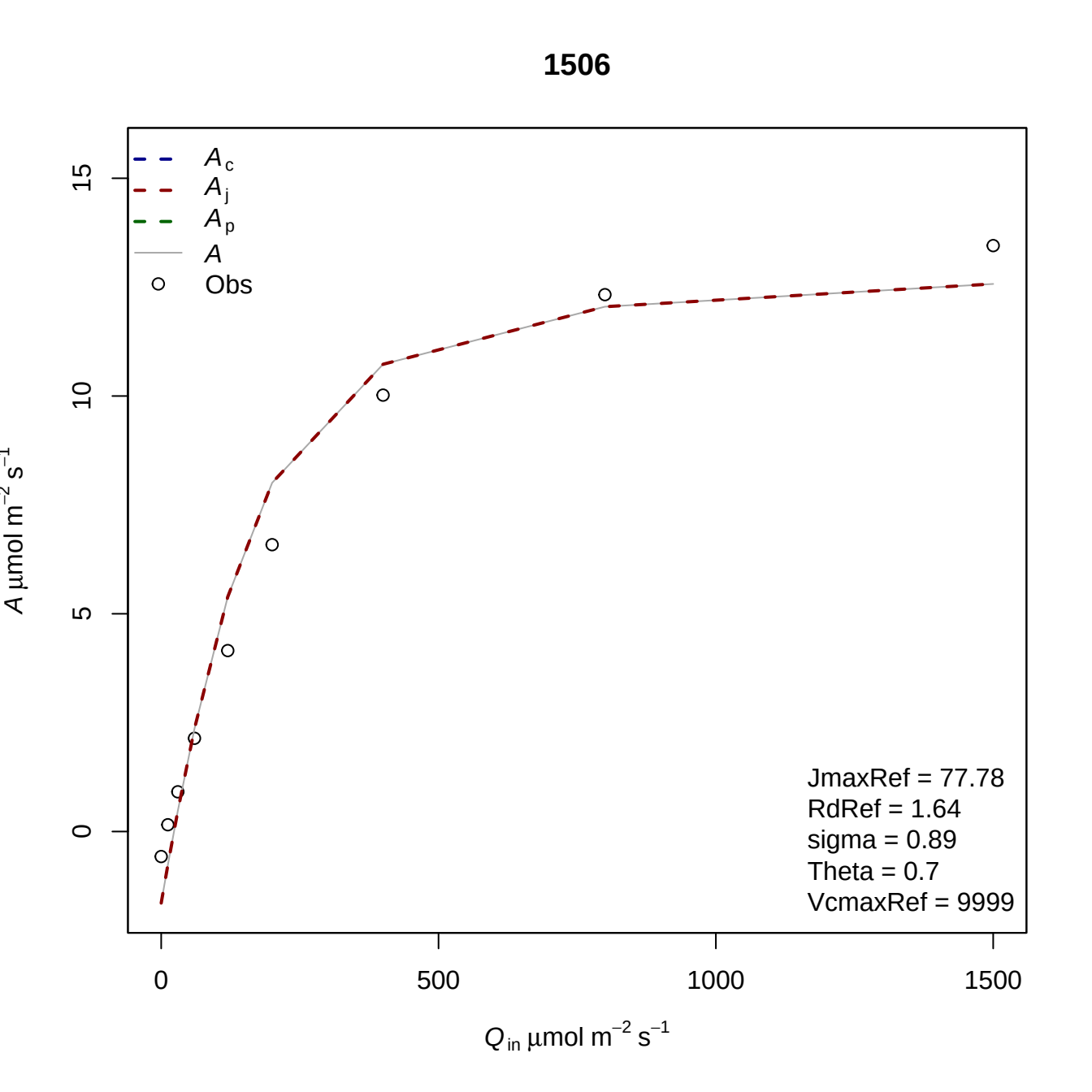

1501
A_c
A_j
A_p
A
Obs
JmaxRef = 97.98
RdRef = 3.83
sigma = 1.24
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

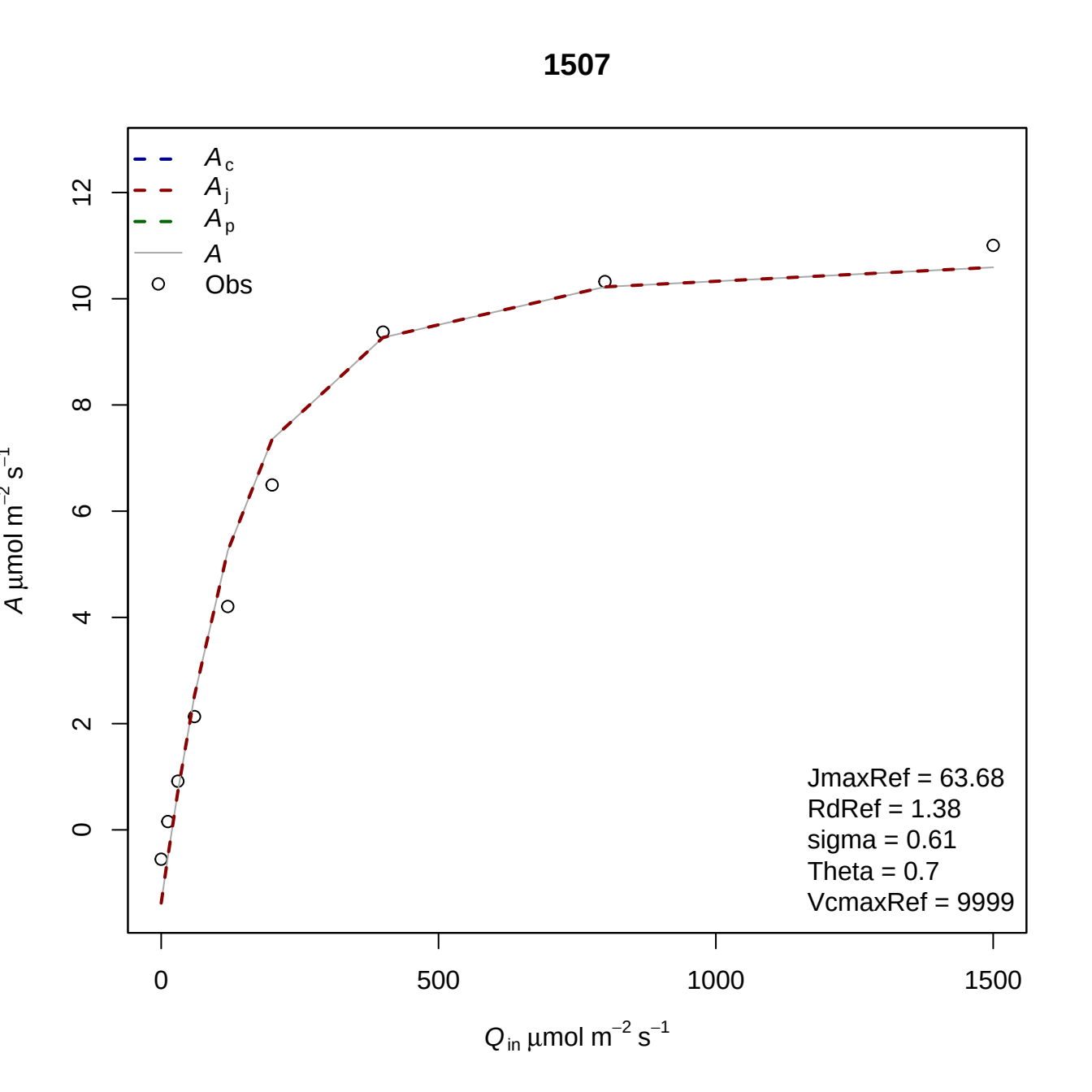

1502
A_c
A_j
A_p
A
Obs
JmaxRef = 67.84
RdRef = 4.48
sigma = 1.25
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
500
1000
1500

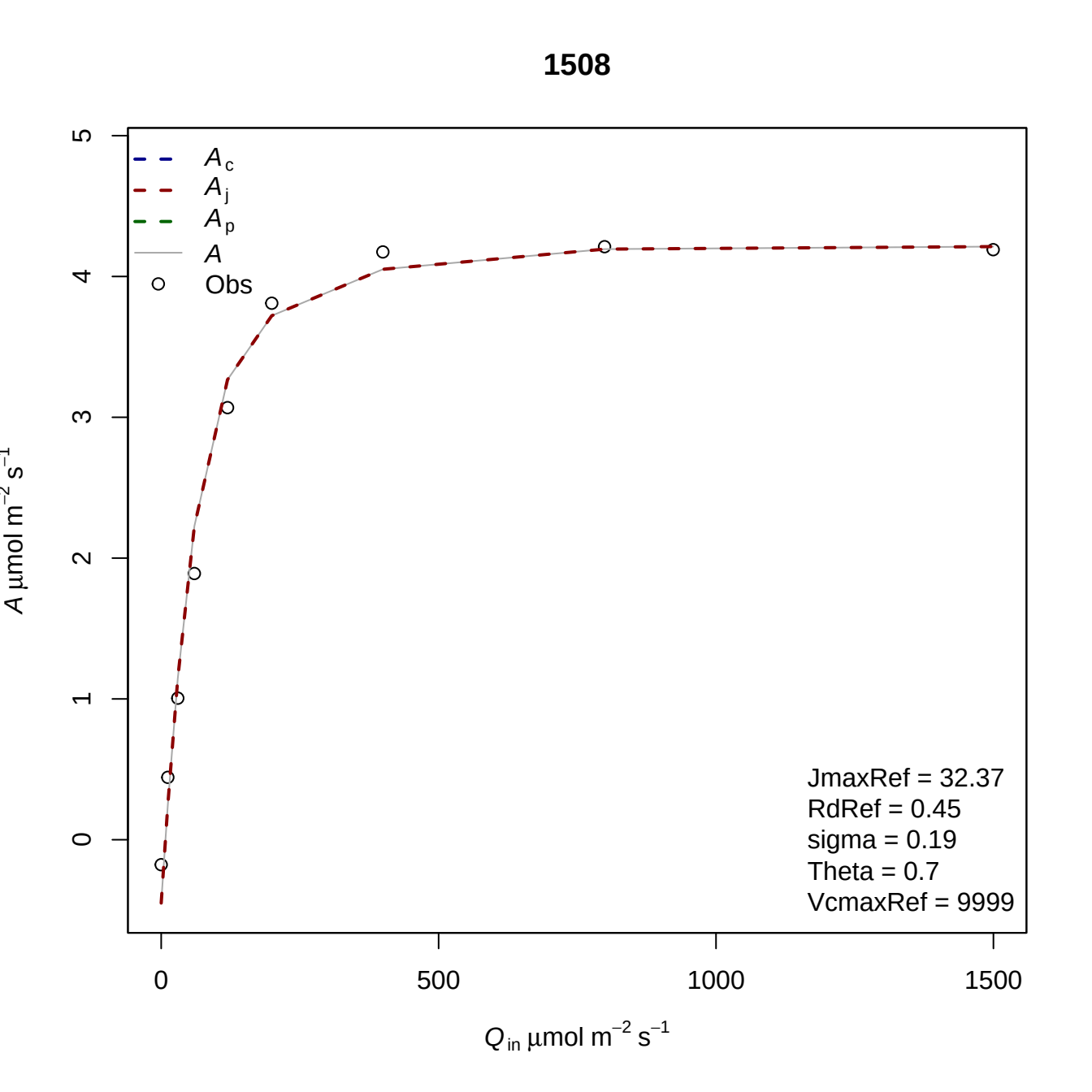

1503
A_c
A_j
A_p
A
Obs
JmaxRef = 76.62
RdRef = 4.18
sigma = 1.42
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
8
6
4
2
0
−2
−4
0
500
1000
1500

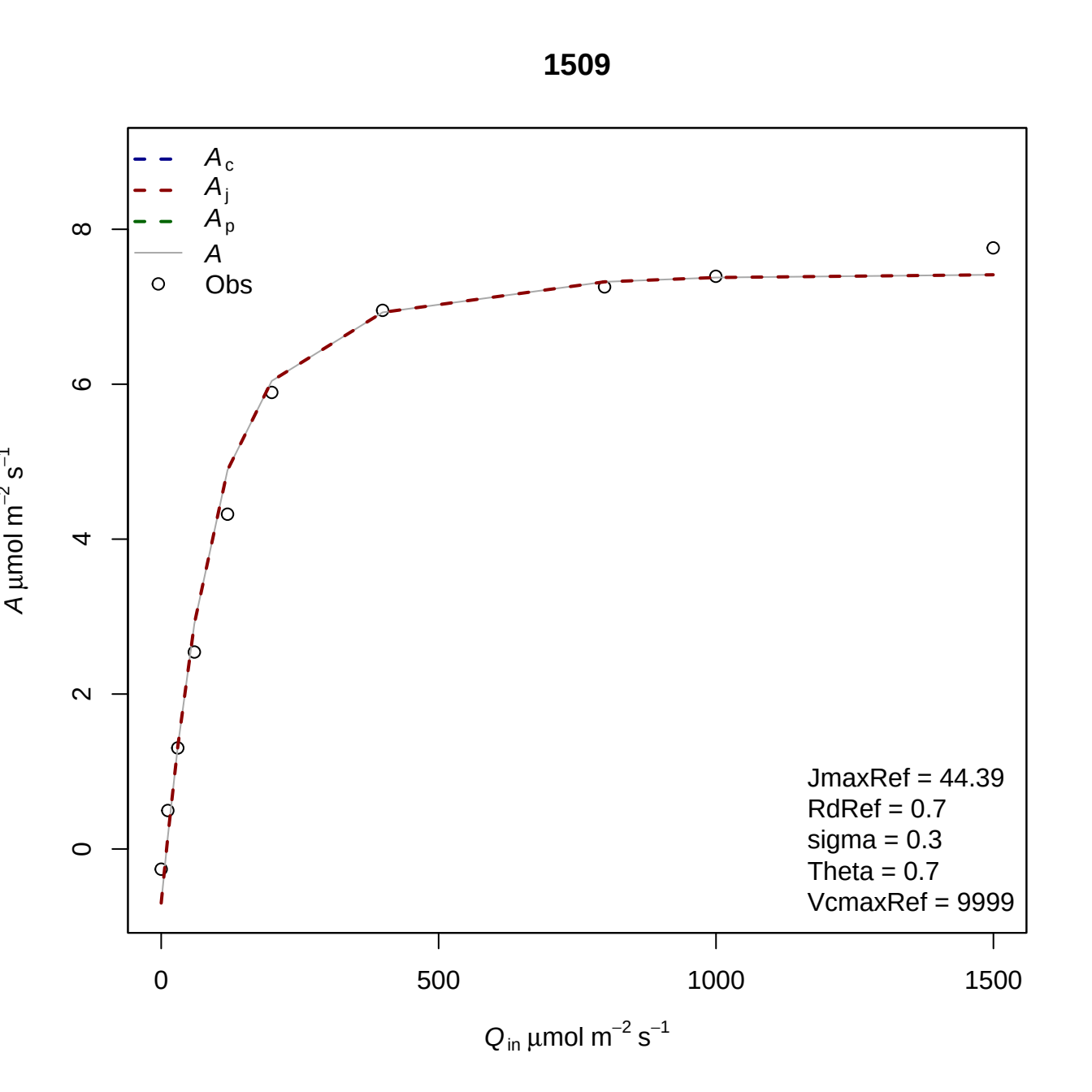

1504
A_c
A_j
A_p
A
Obs
JmaxRef = 66.73
RdRef = 5.01
sigma = 1.15
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
6
4
2
0
−2
−4
0
200
400
600
800
1000
1200

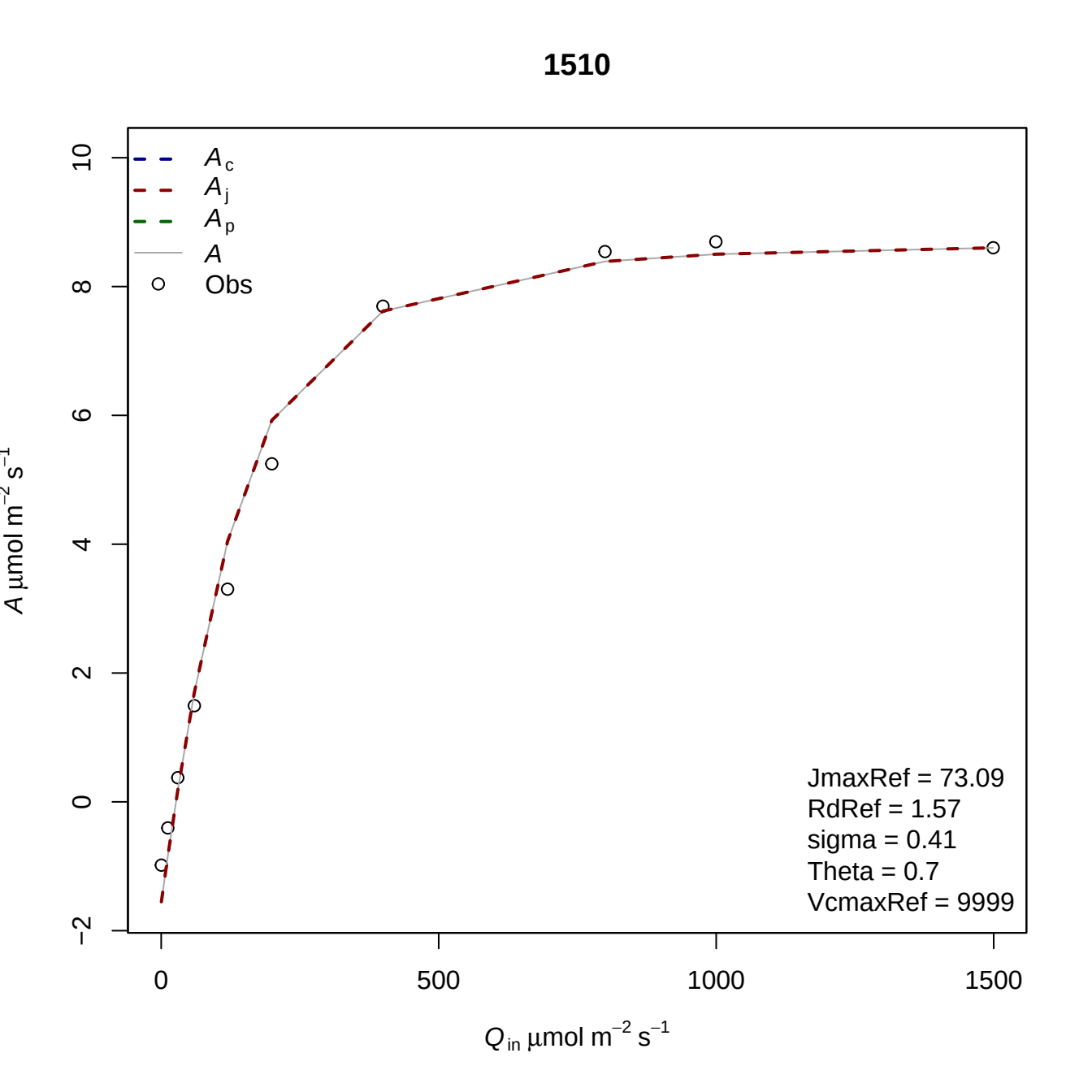

1505
A_c
A_j
A_p
A
Obs
JmaxRef = 75.11
RdRef = 3.88
sigma = 1.33
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

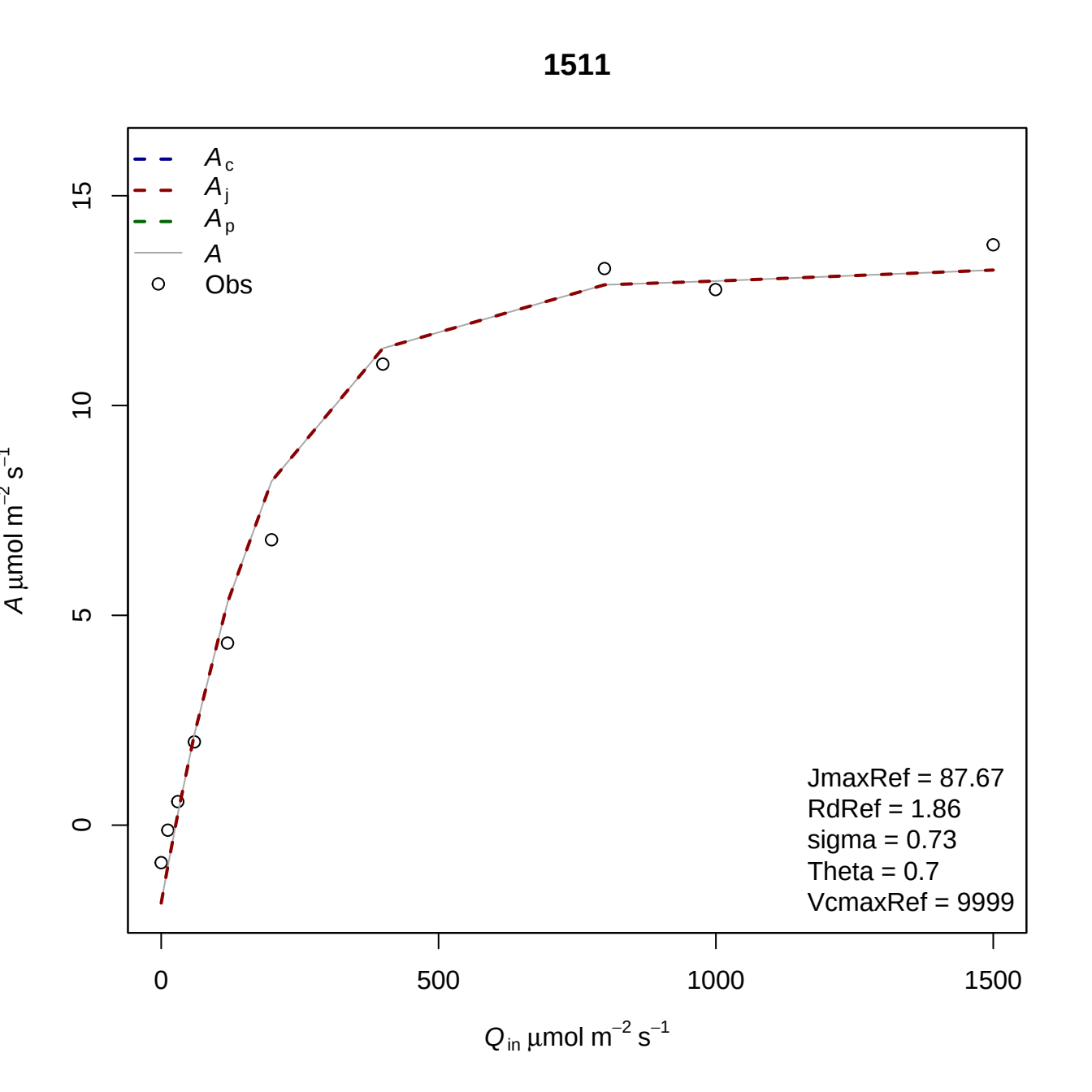

1506
A_c
A_j
A_p
A
Obs
JmaxRef = 77.78
RdRef = 1.64
sigma = 0.89
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

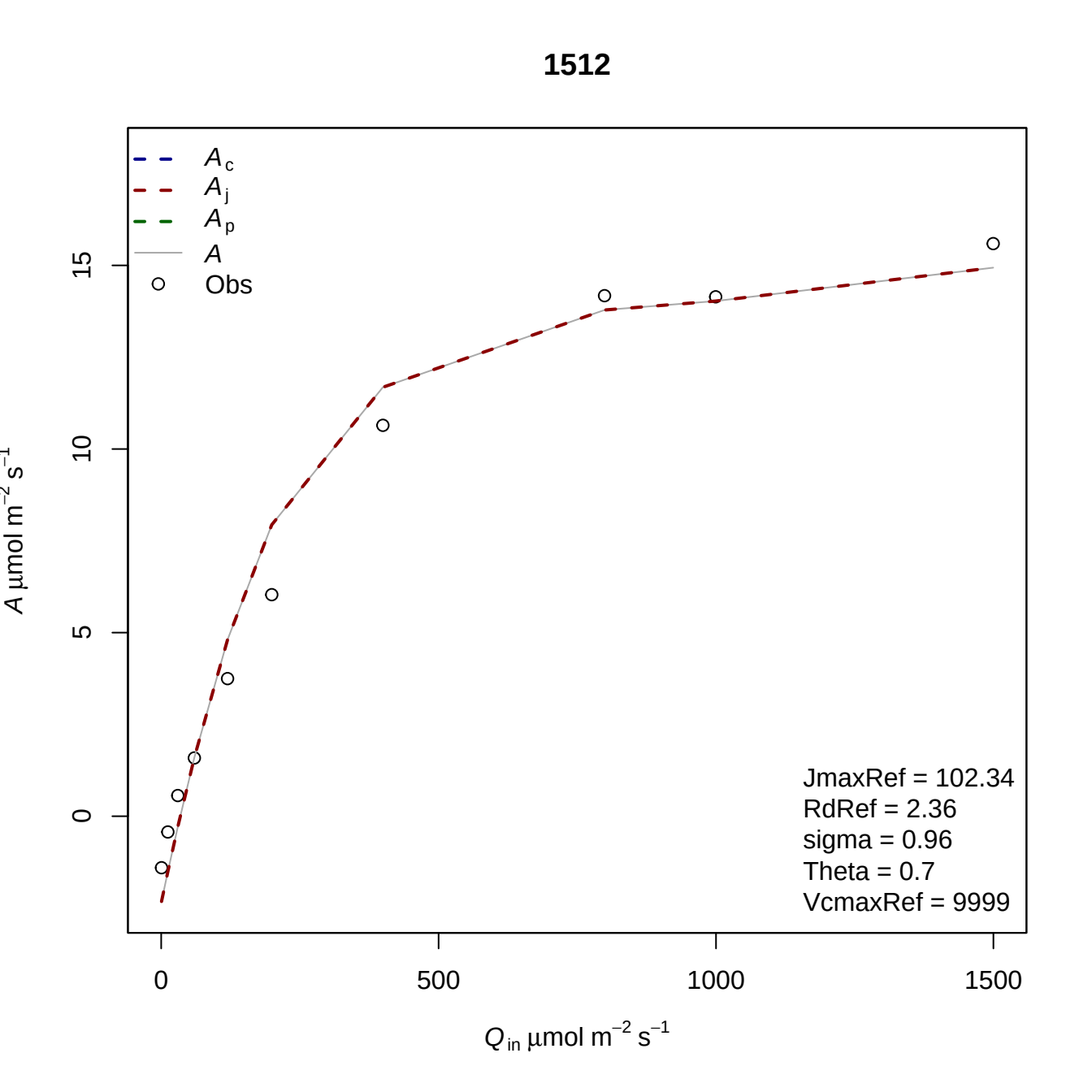

1507
A_c
A_j
A_p
A
Obs
JmaxRef = 63.68
RdRef = 1.38
sigma = 0.61
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

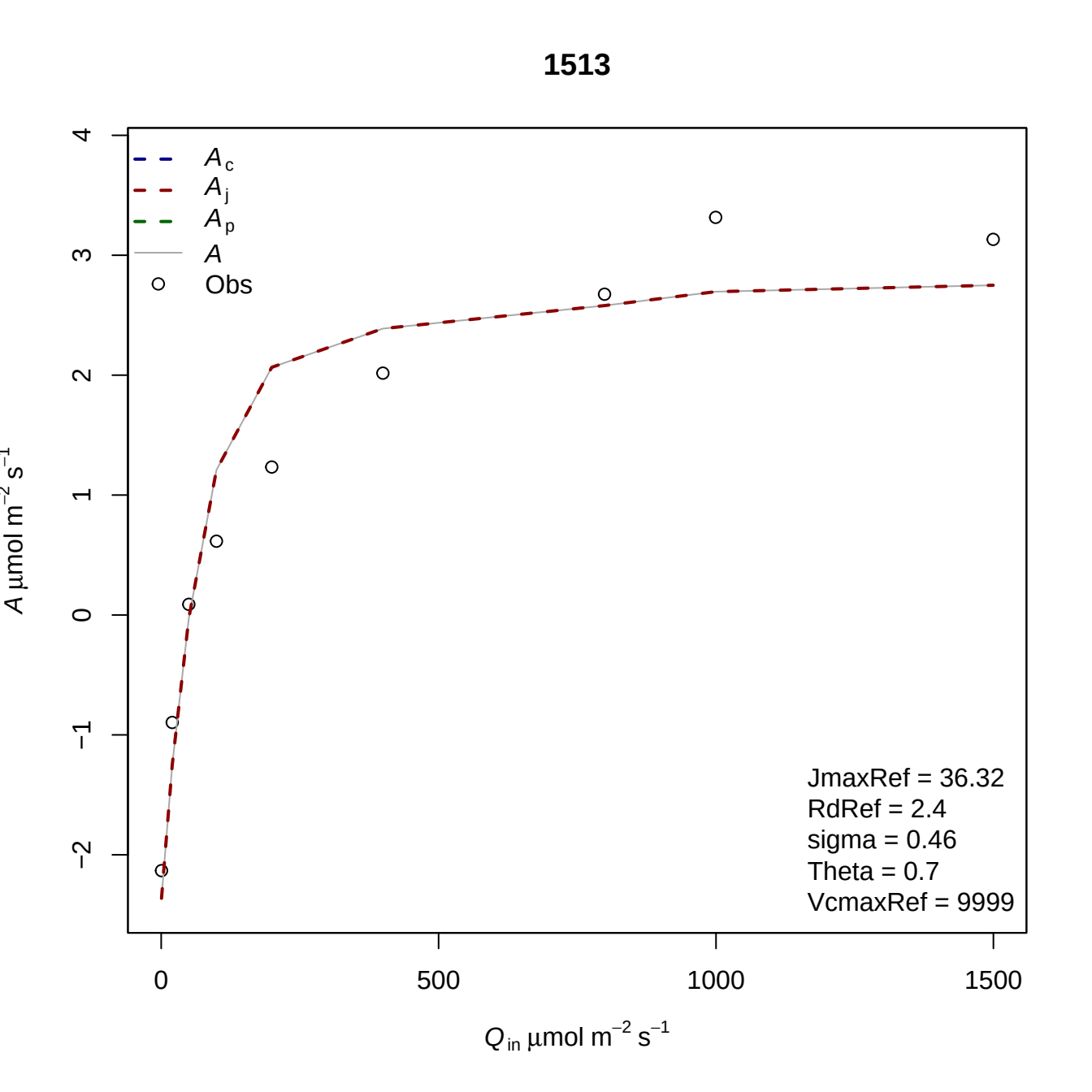

1508
A_c
A_j
A_p
A
Obs
JmaxRef = 32.37
RdRef = 0.45
sigma = 0.19
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

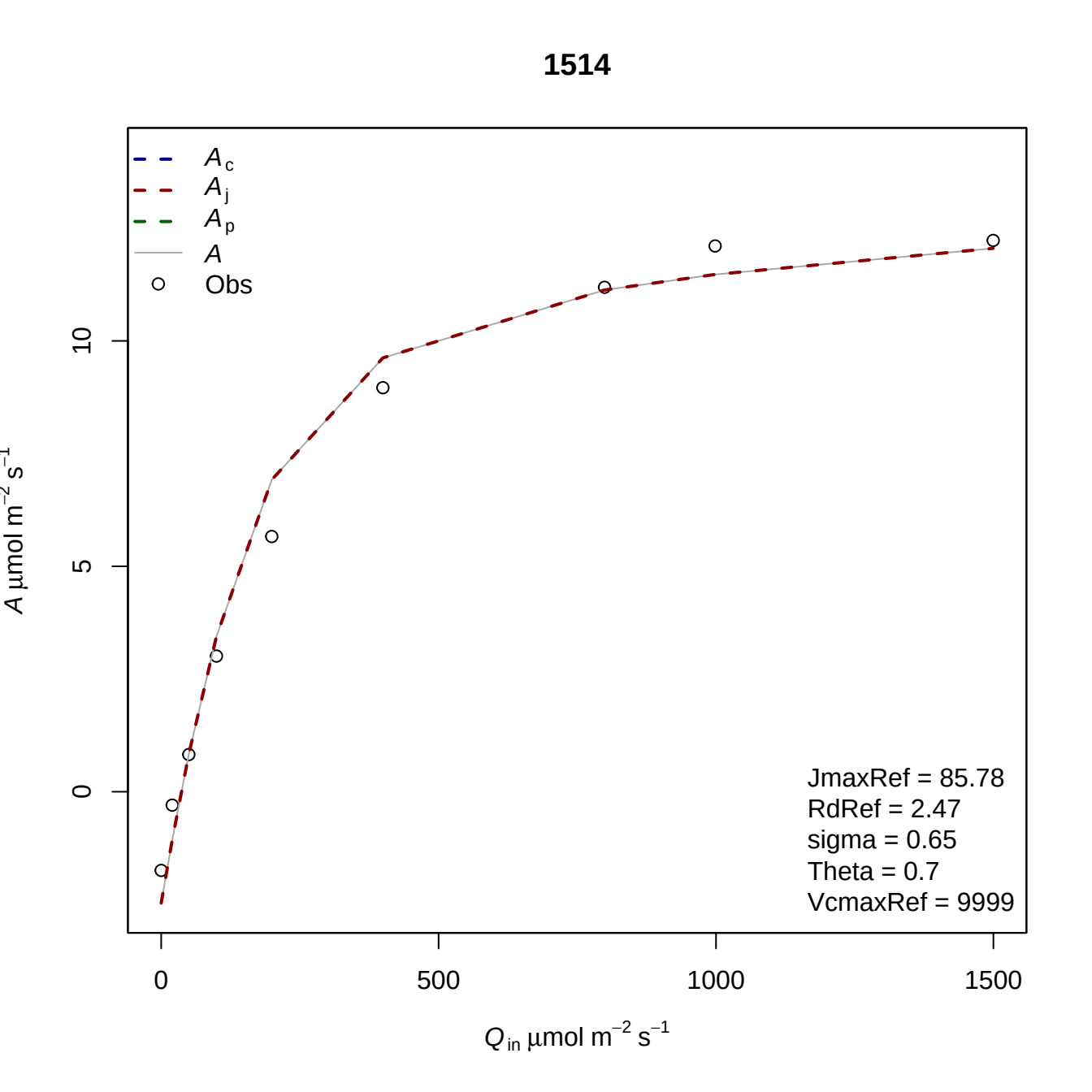

1509
A_c
A_j
A_p
A
Obs
JmaxRef = 44.39
RdRef = 0.7
sigma = 0.3
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

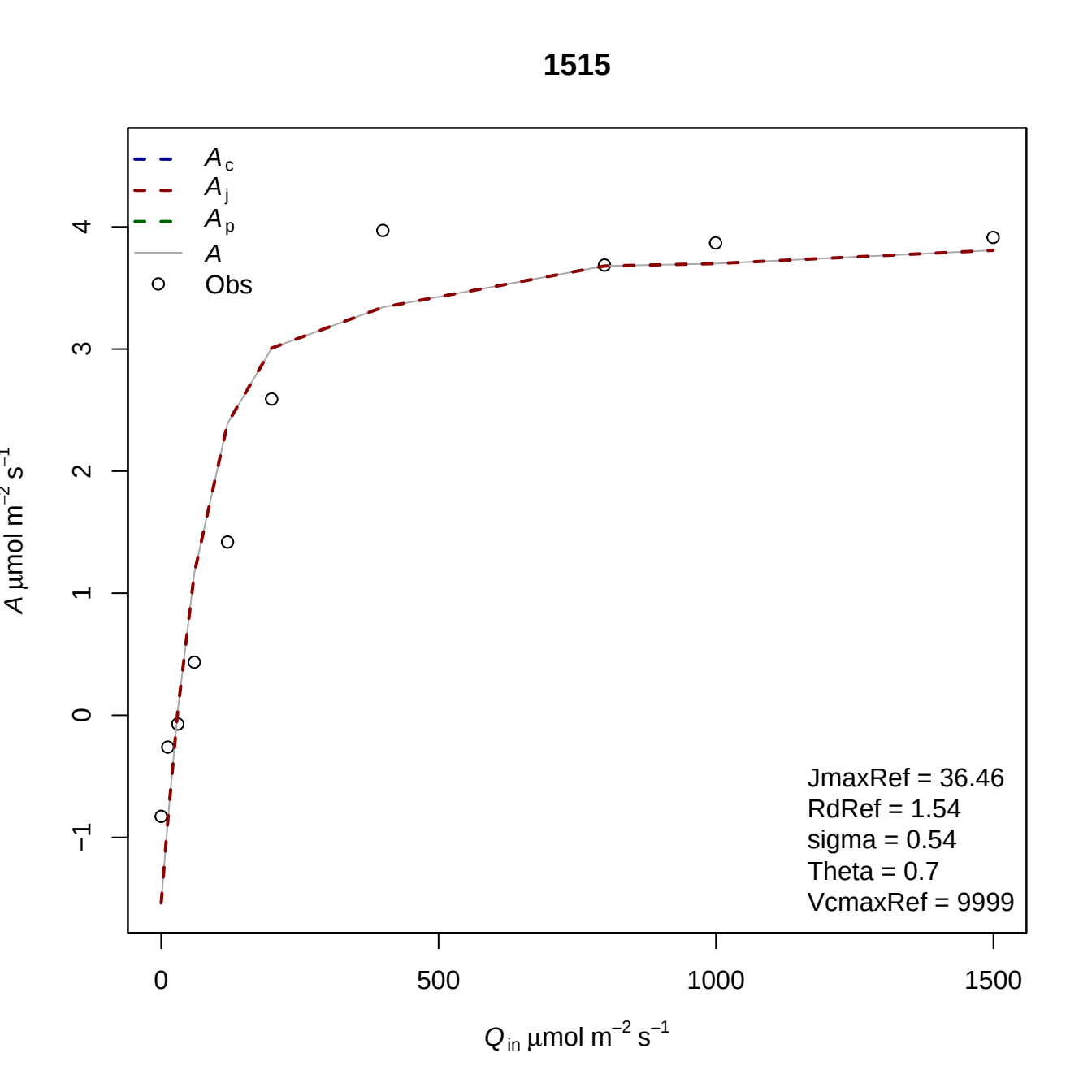

1510
A_c
A_j
A_p
A
Obs
JmaxRef = 73.09
RdRef = 1.57
sigma = 0.41
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500

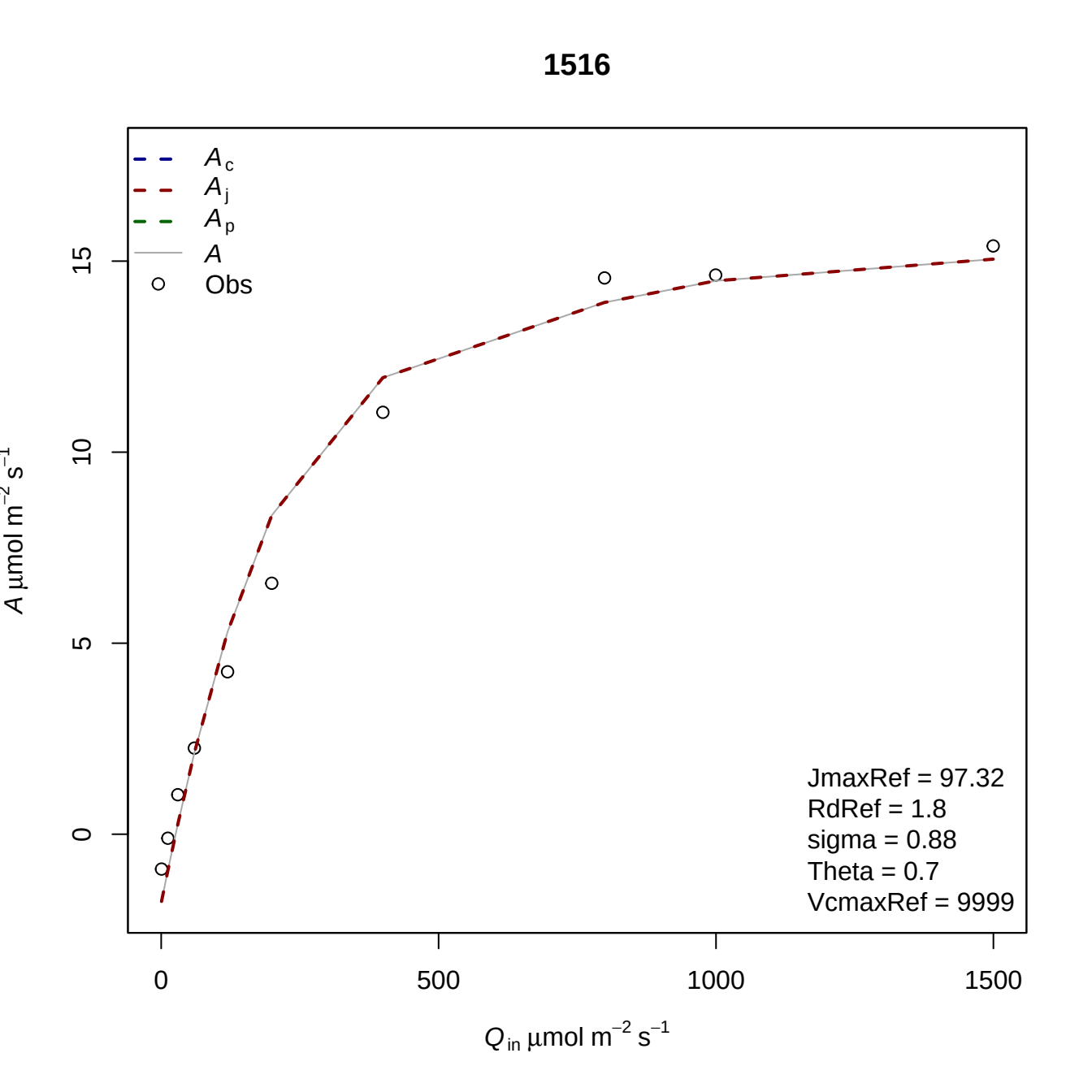

1511
A_c
A_j
A_p
A
Obs
JmaxRef = 87.67
RdRef = 1.86
sigma = 0.73
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

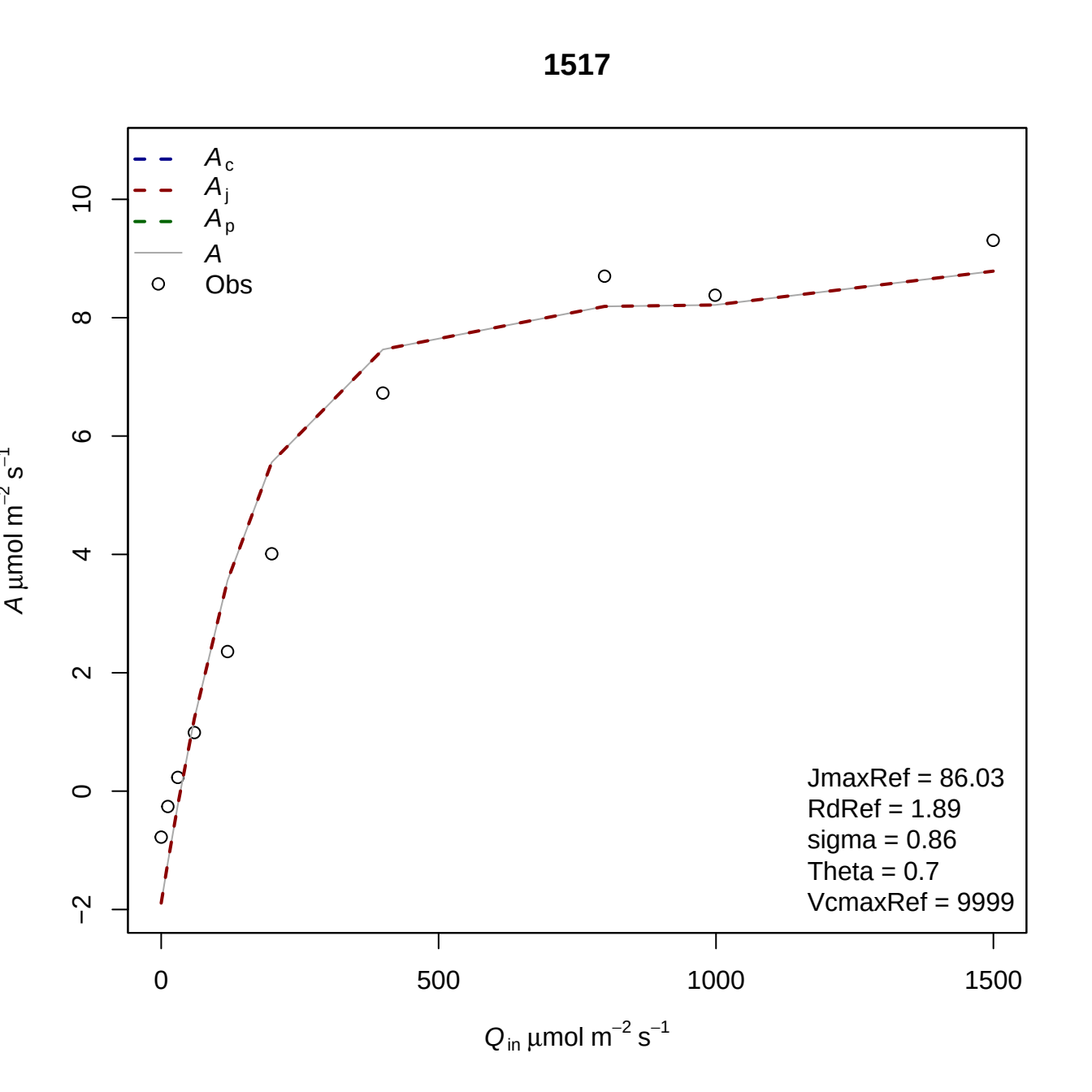

1512
A_c
A_j
A_p
A
Obs
JmaxRef = 102.34
RdRef = 2.36
sigma = 0.96
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

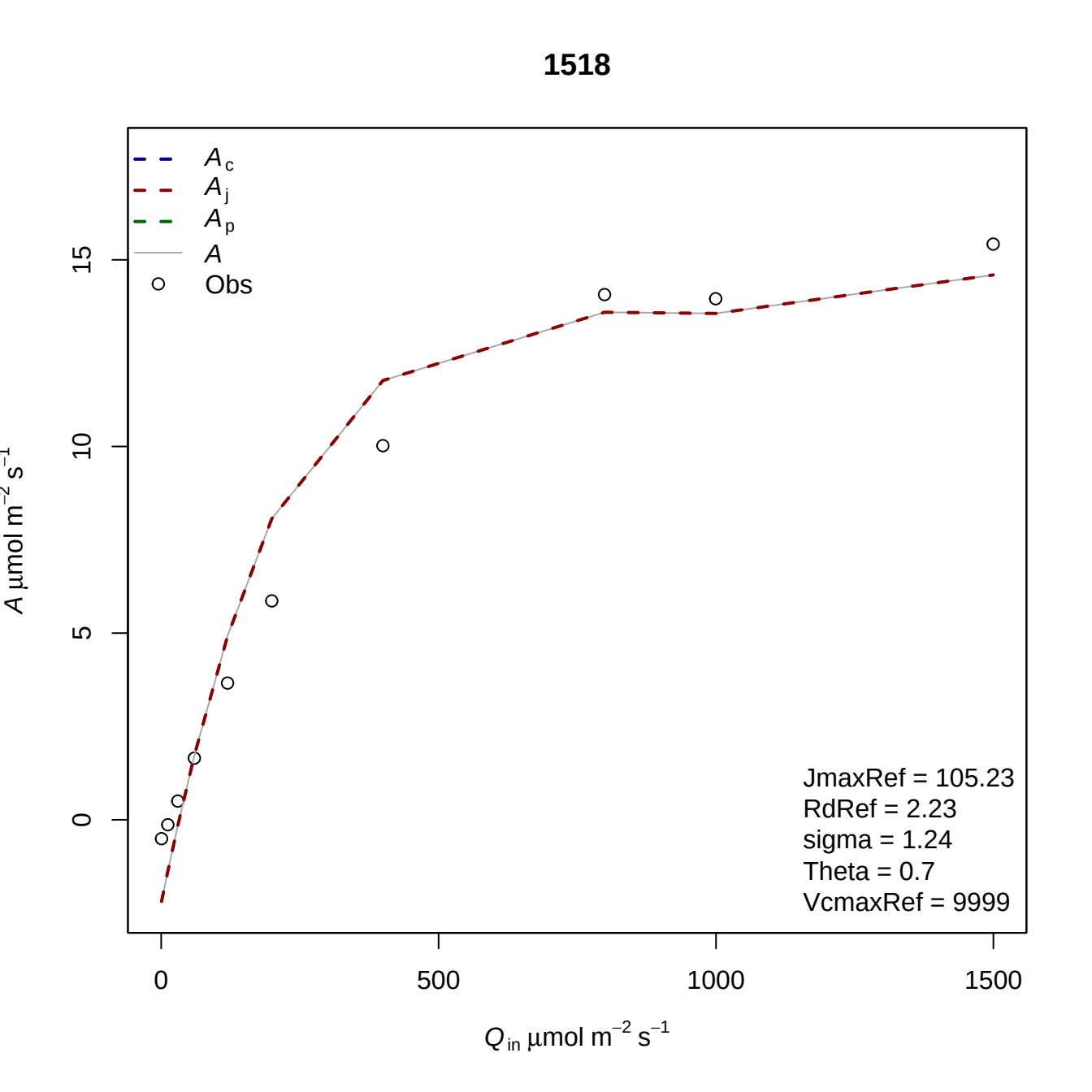

1513
A_c
A_j
A_p
A
Obs
JmaxRef = 36.32
RdRef = 2.4
sigma = 0.46
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
4
3
2
1
0
−1
−2
0
500
1000
1500

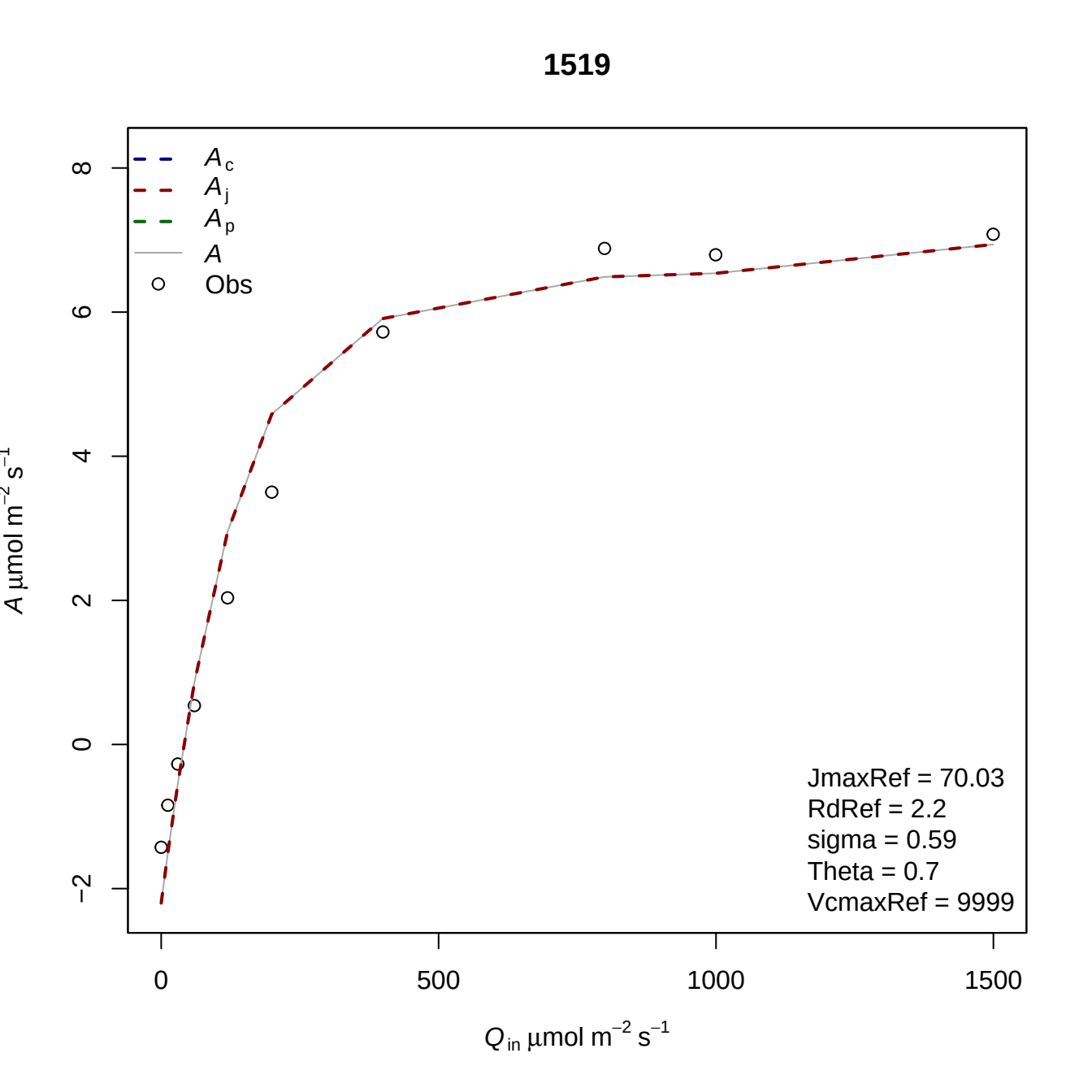

1514
A_c
A_j
A_p
A
Obs
JmaxRef = 85.78
RdRef = 2.47
sigma = 0.65
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

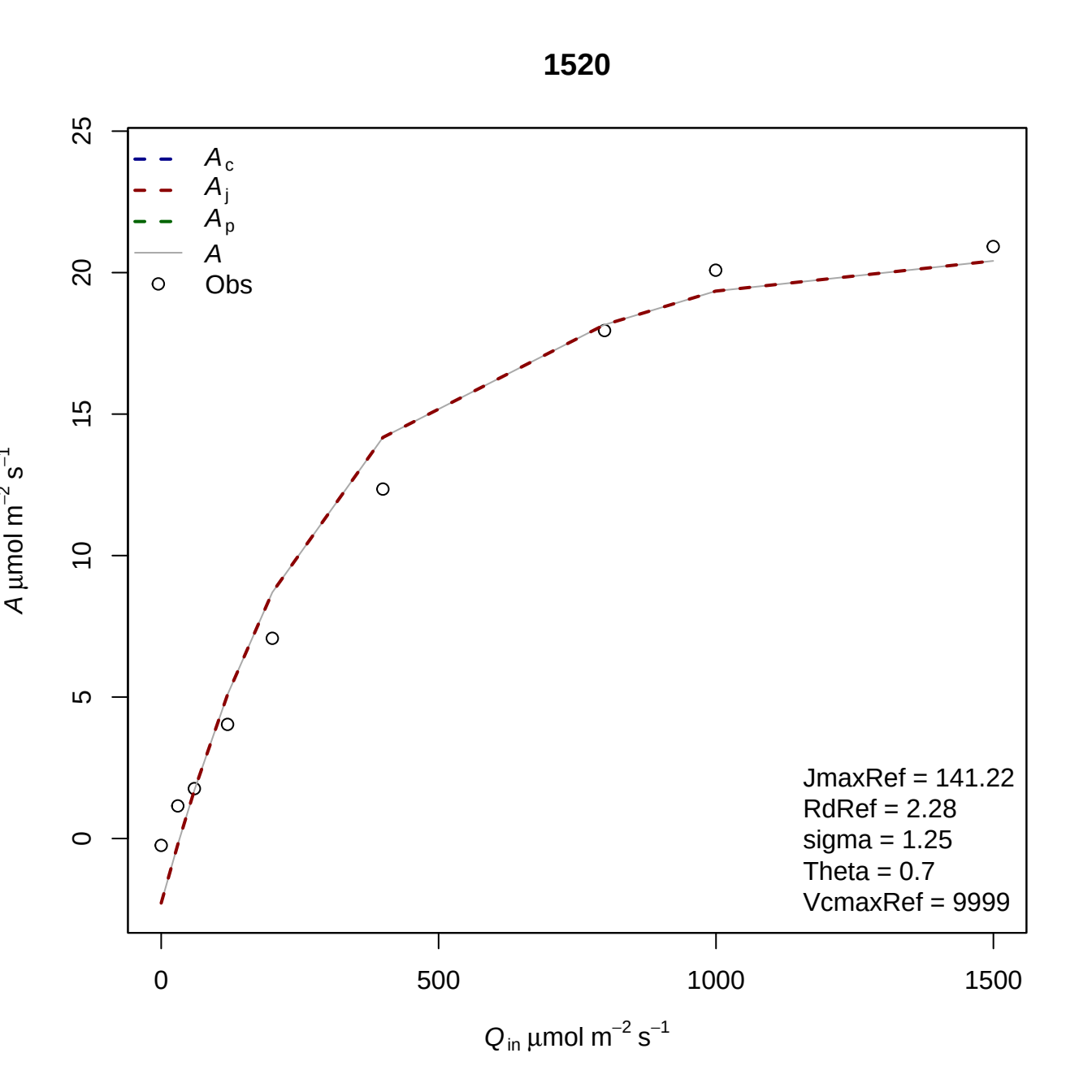

1515
A_c
A_j
A_p
A
Obs
JmaxRef = 36.46
RdRef = 1.54
sigma = 0.54
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

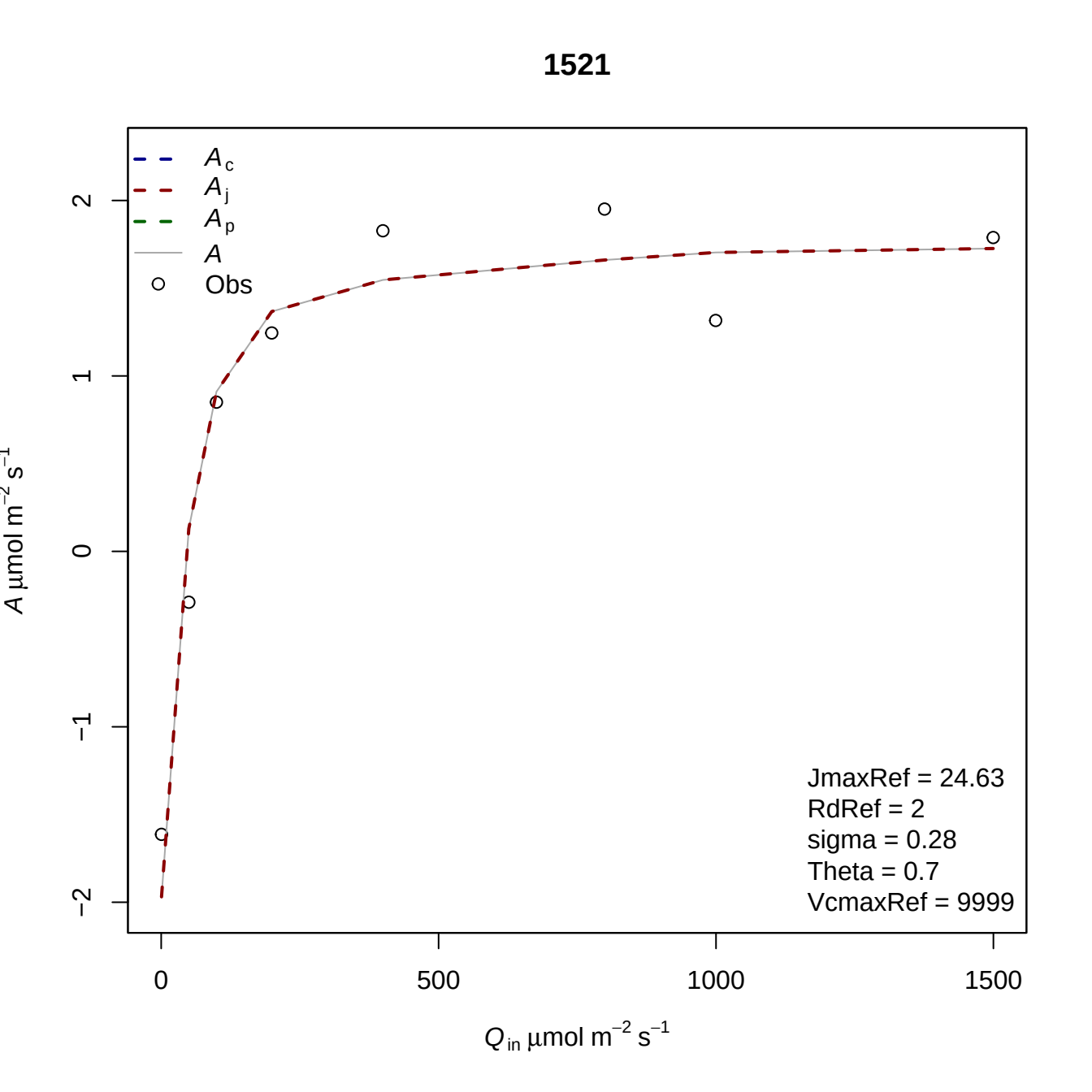

1516
A_c
A_j
A_p
A
Obs
JmaxRef = 97.32
RdRef = 1.8
sigma = 0.88
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

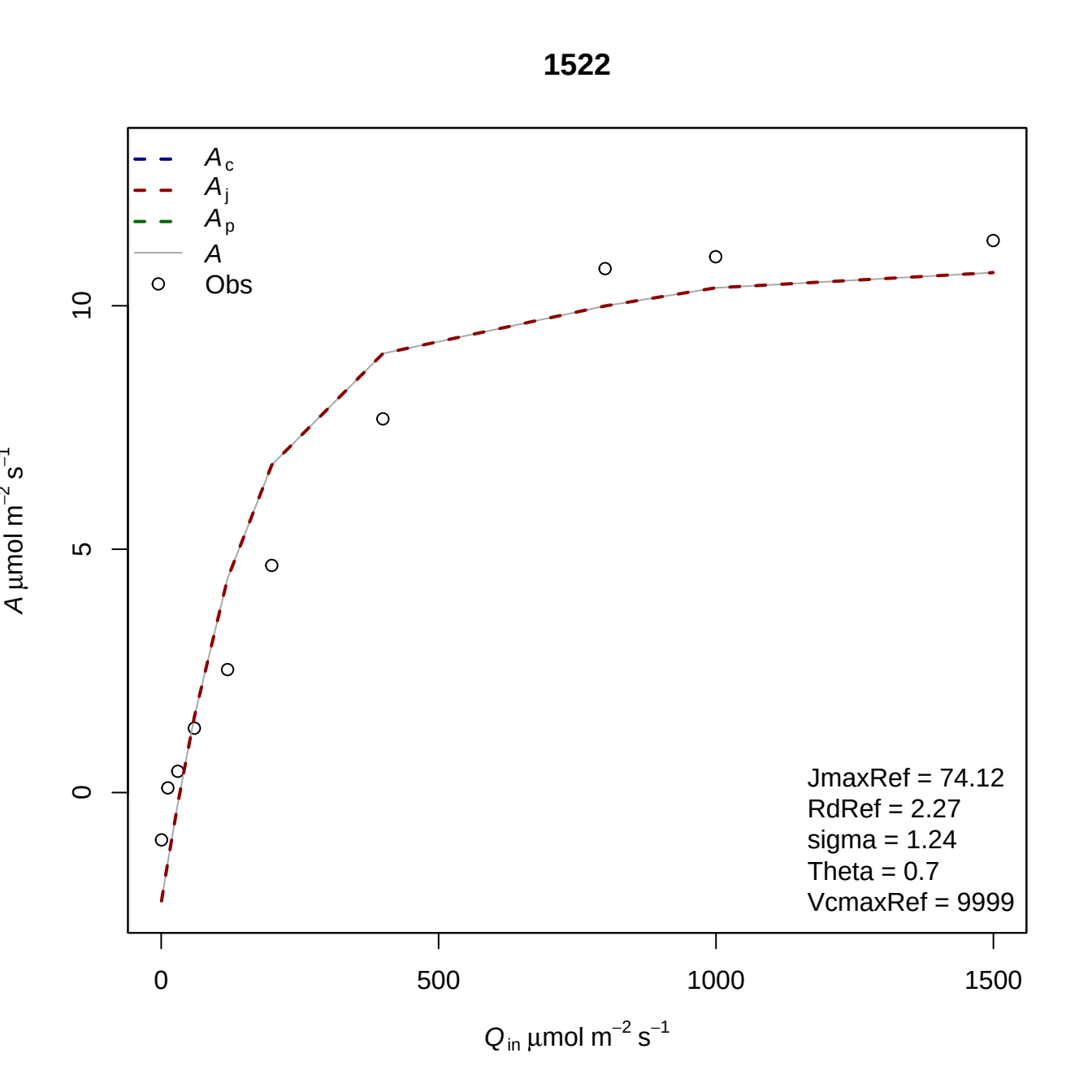

1517
A_c
A_j
A_p
A
Obs
JmaxRef = 86.03
RdRef = 1.89
sigma = 0.86
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500

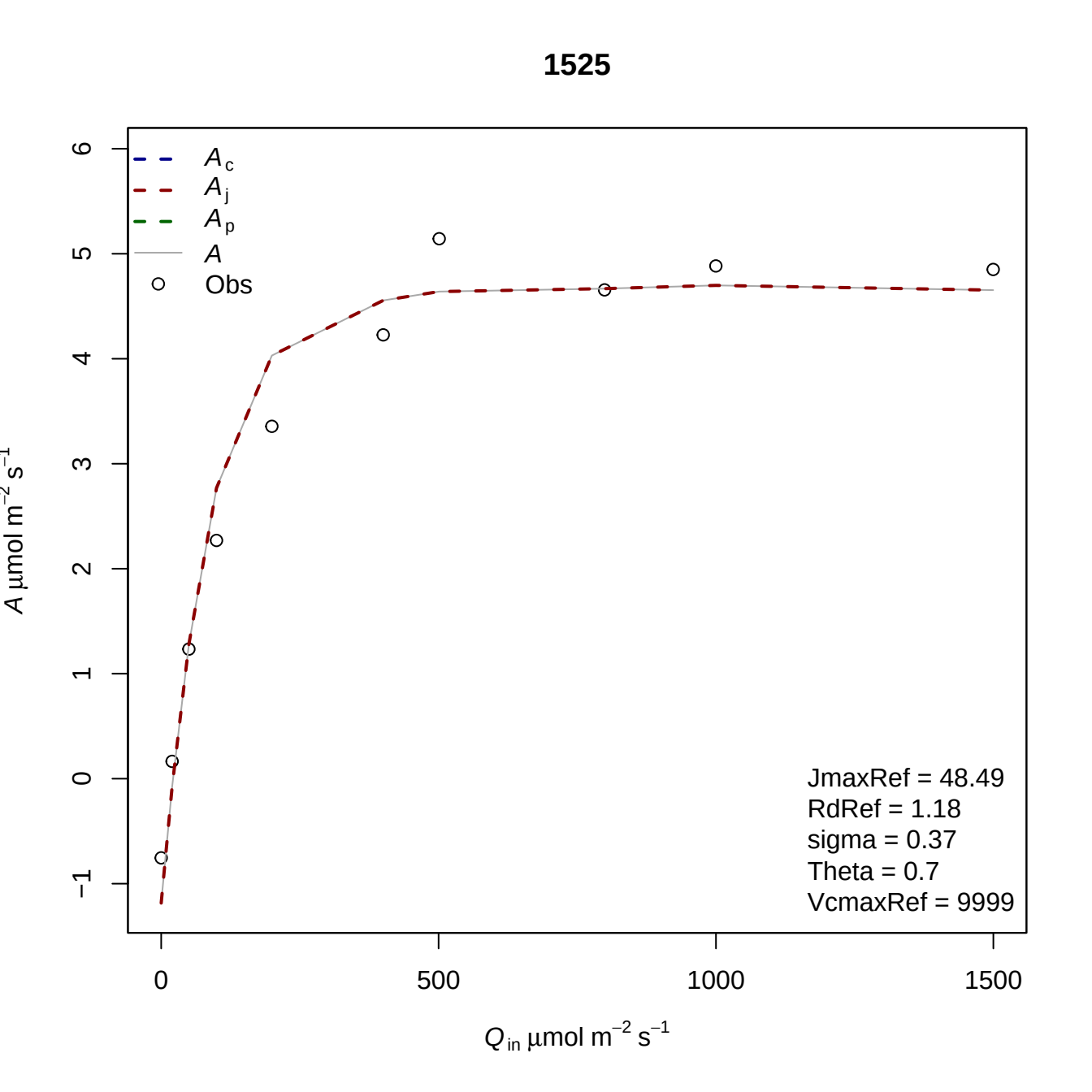

1518
A_c
A_j
A_p
A
Obs
JmaxRef = 105.23
RdRef = 2.23
sigma = 1.24
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

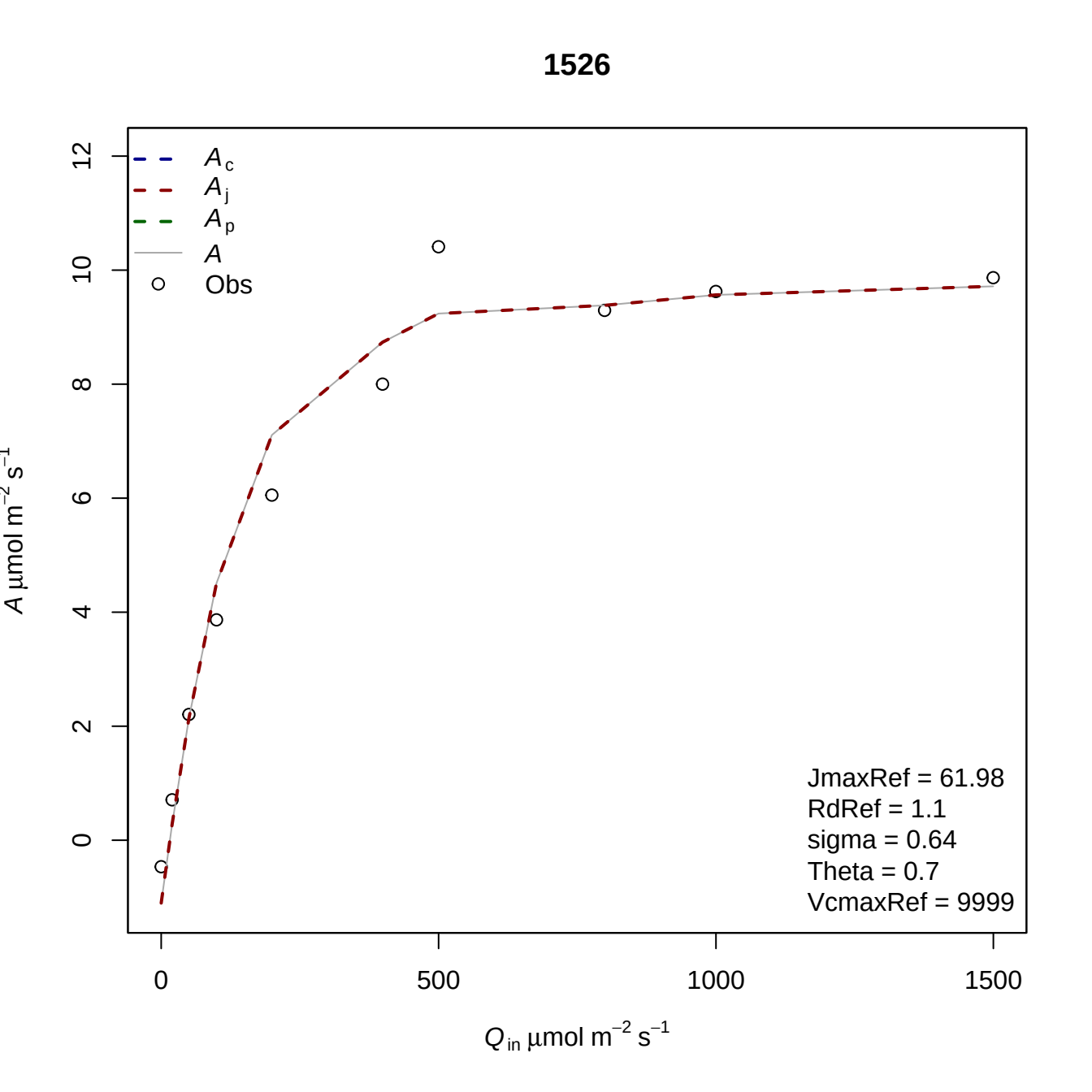

1519
A_c
A_j
A_p
A
Obs
JmaxRef = 70.03
RdRef = 2.2
sigma = 0.59
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500

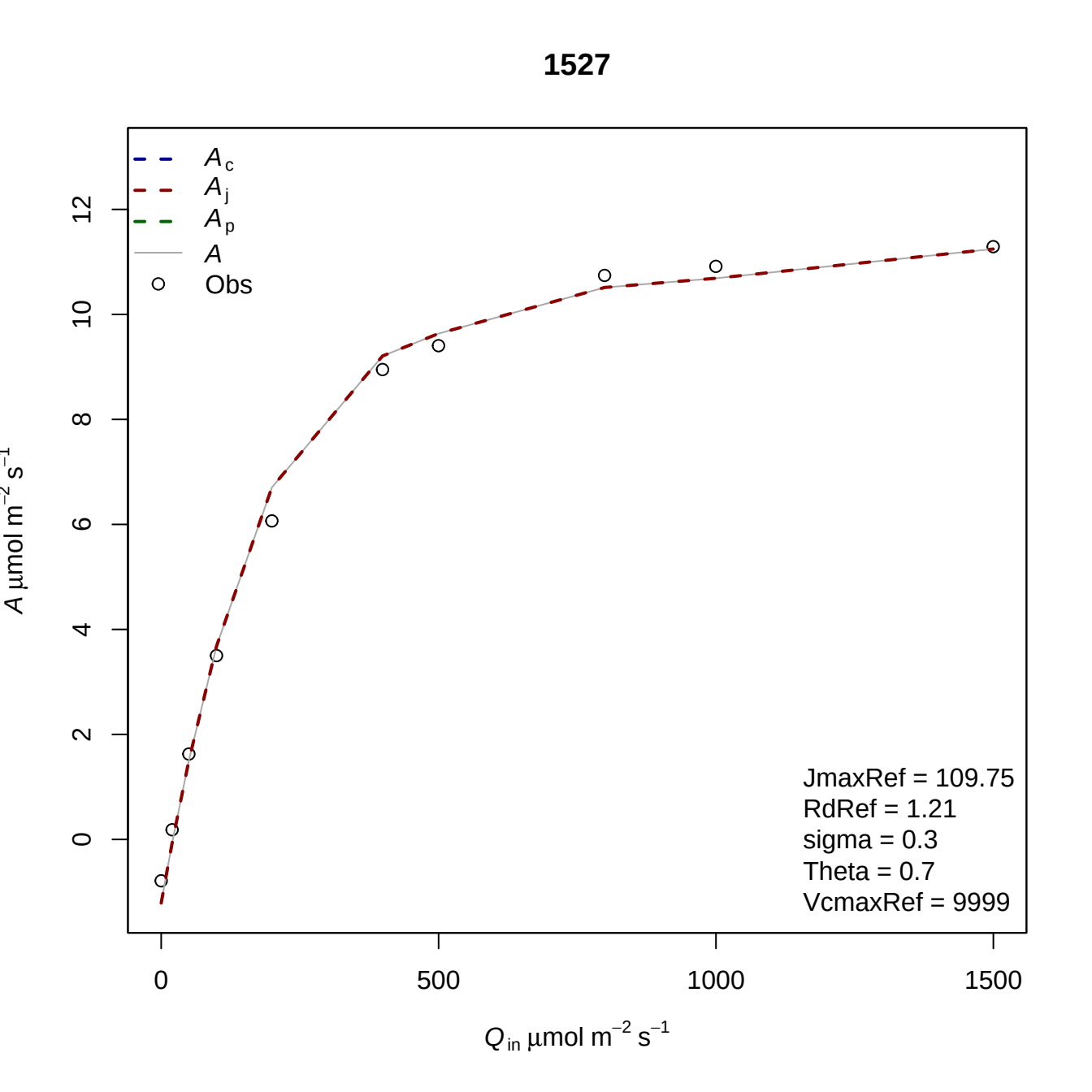

1520
A_c
A_j
A_p
A
Obs
JmaxRef = 141.22
RdRef = 2.28
sigma = 1.25
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500

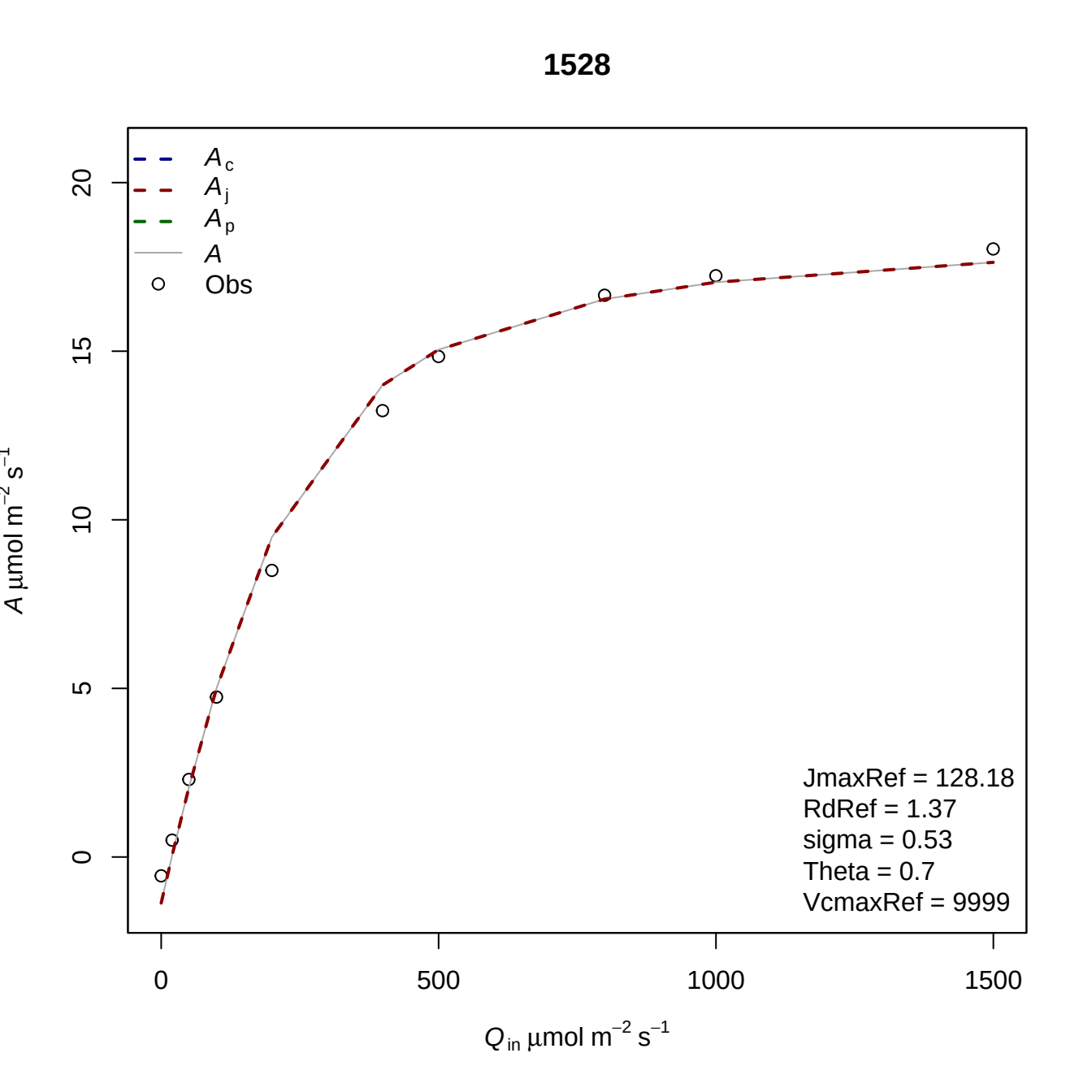

1521
A_c
A_j
A_p
A
Obs
JmaxRef = 24.63
RdRef = 2
sigma = 0.28
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
2
1
0
−1
−2
0
500
1000
1500

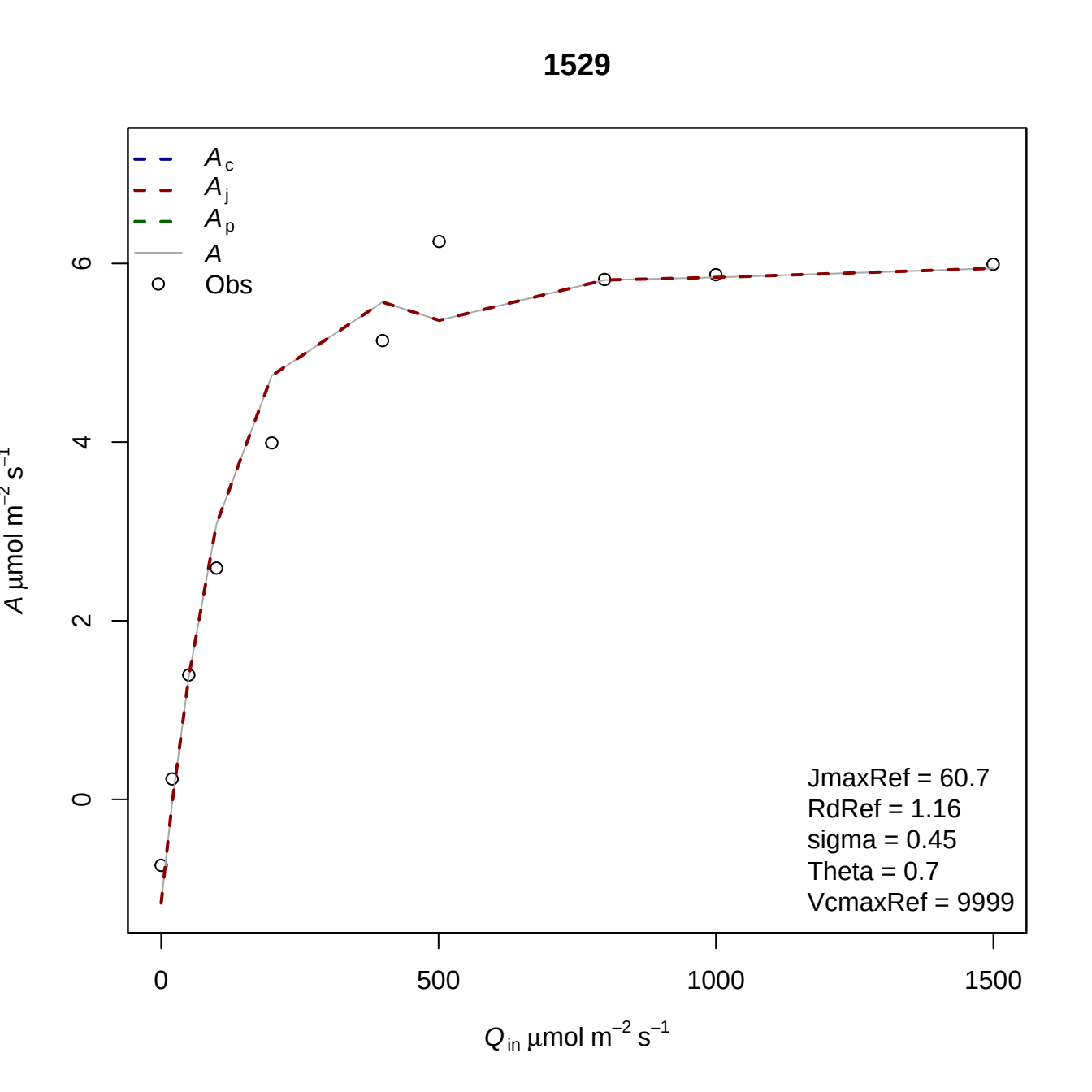

1522
A_c
A_j
A_p
A
Obs
JmaxRef = 74.12
RdRef = 2.27
sigma = 1.24
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500

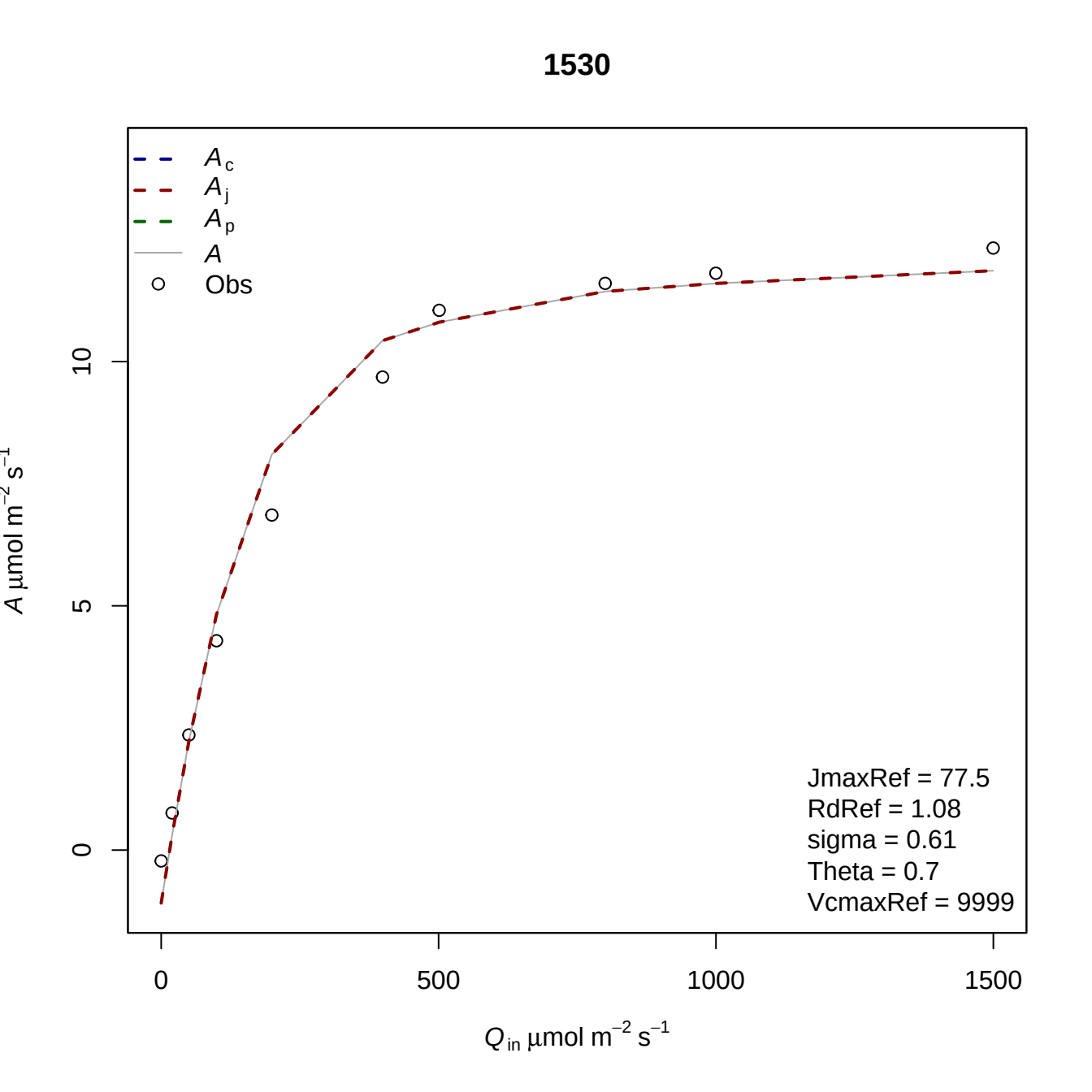

1525
A_c
A_j
A_p
A
Obs
JmaxRef = 48.49
RdRef = 1.18
sigma = 0.37
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

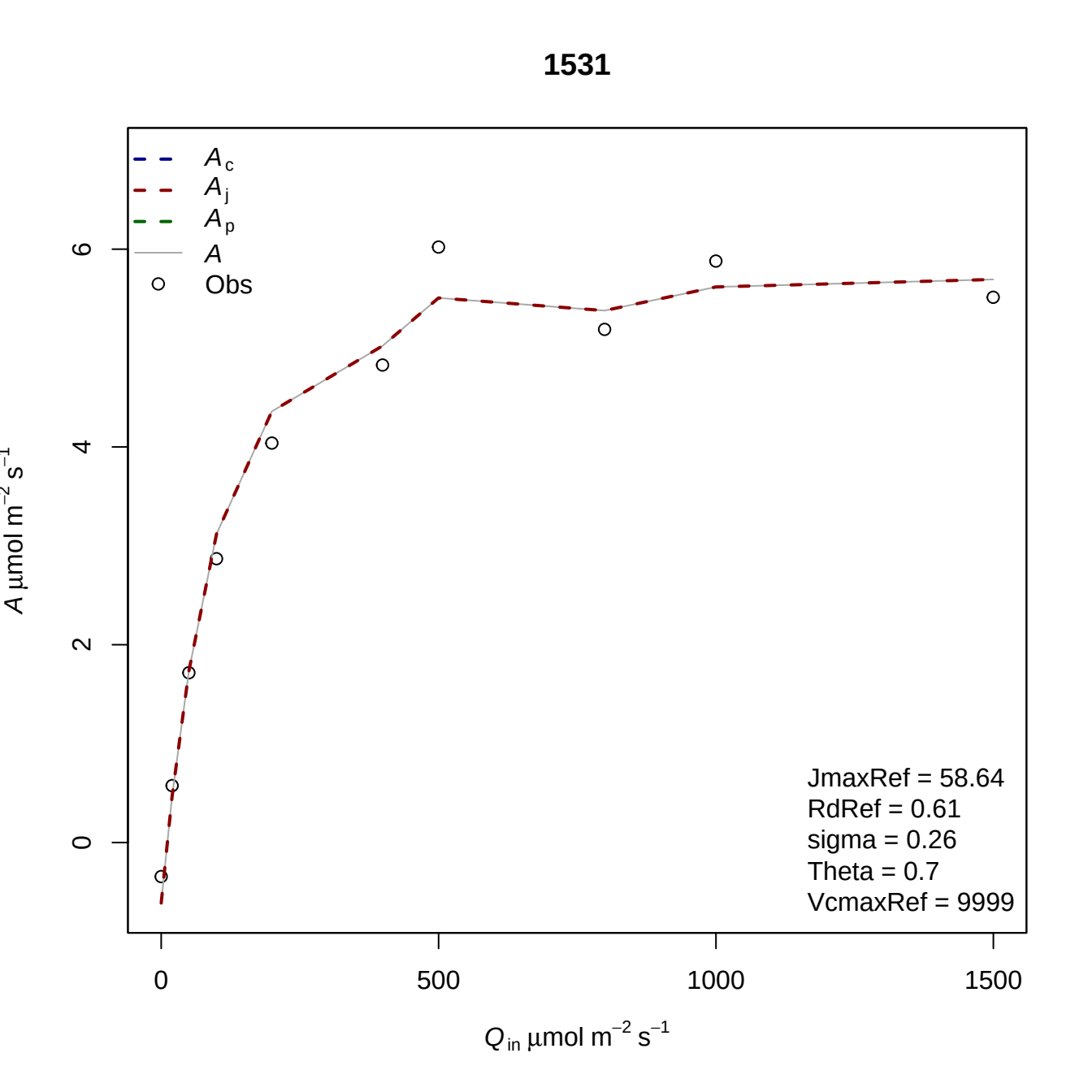

1526
A_c
A_j
A_p
A
Obs
JmaxRef = 61.98
RdRef = 1.1
sigma = 0.64
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

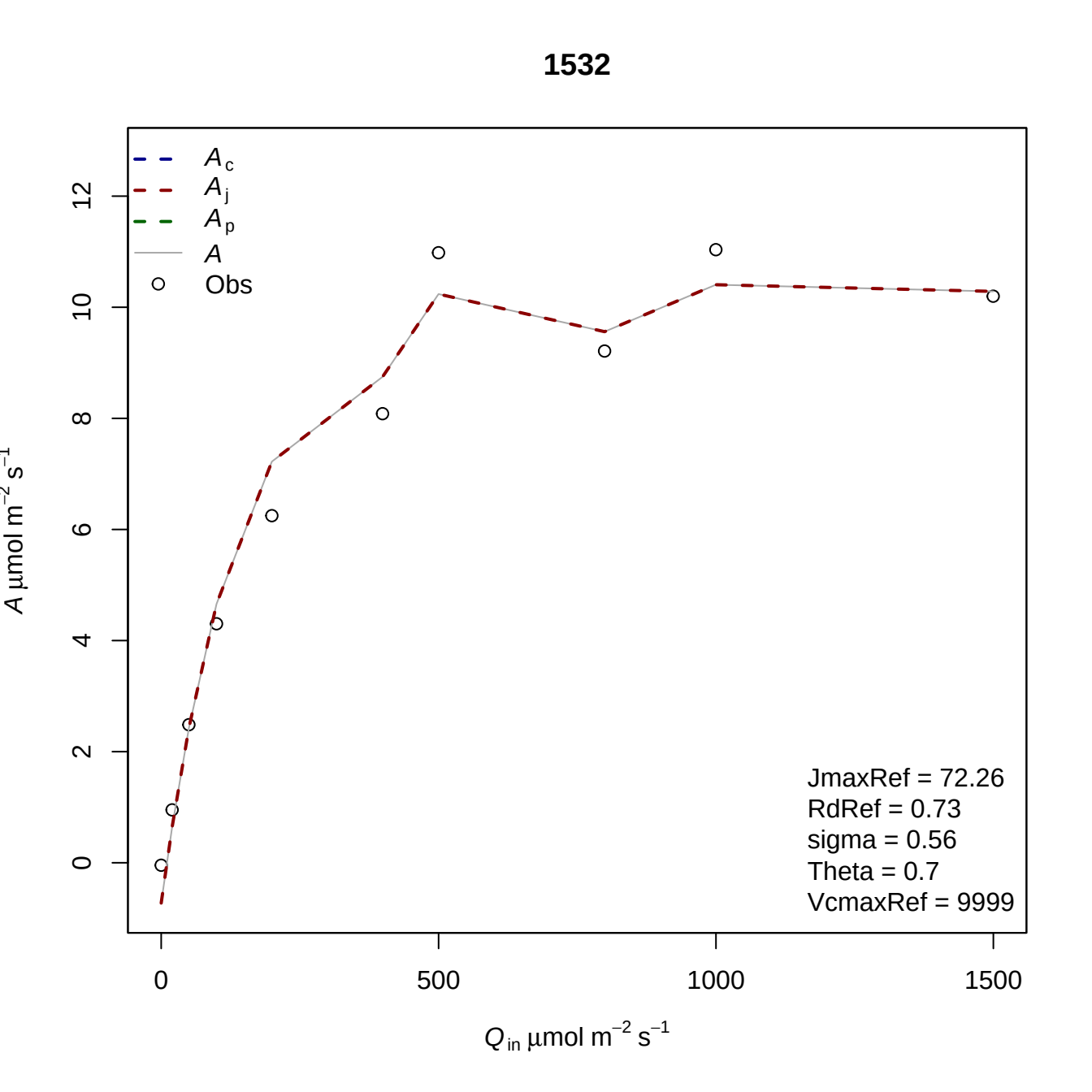

1527
A_c
A_j
A_p
A
Obs
JmaxRef = 109.75
RdRef = 1.21
sigma = 0.3
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

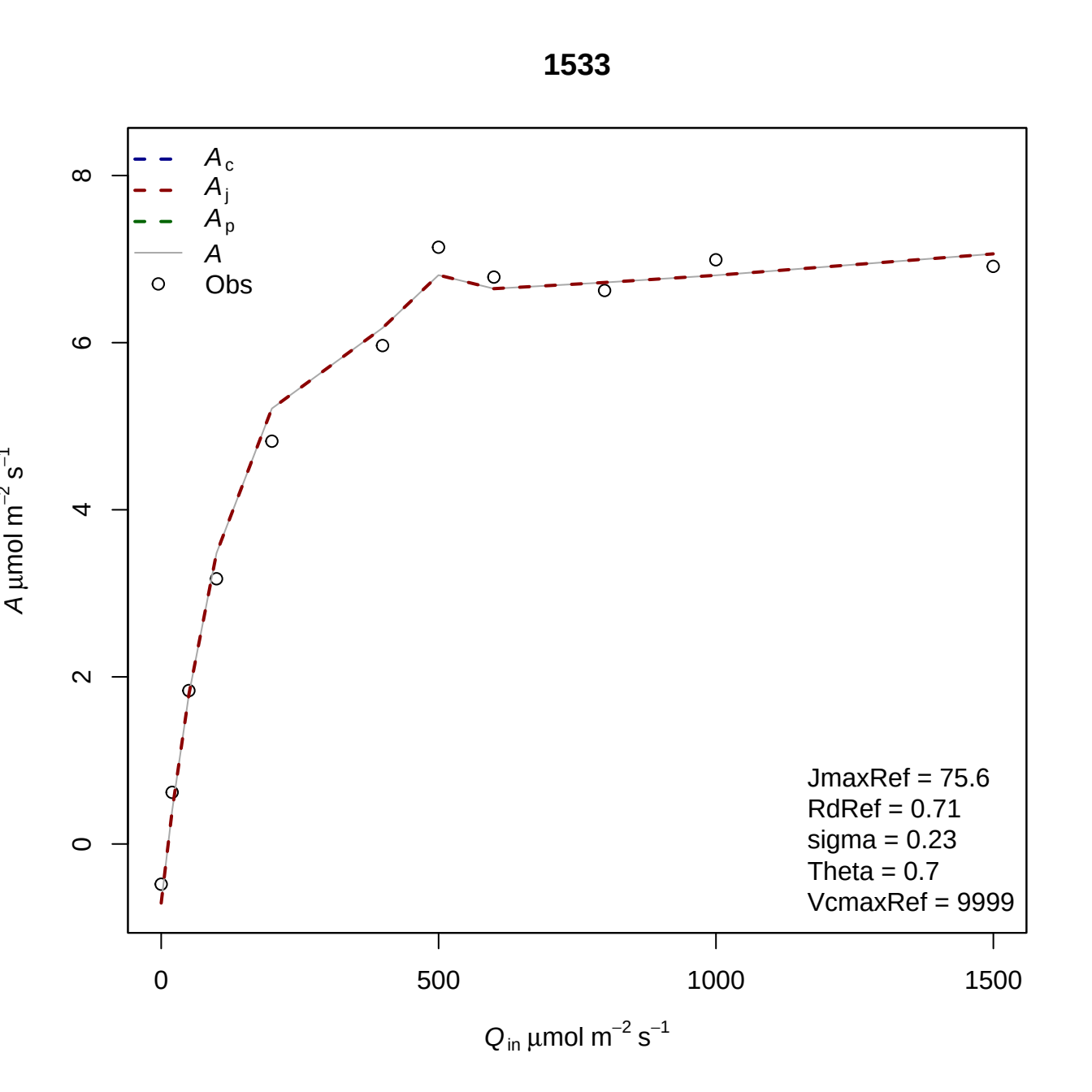

1528
A_c
A_j
A_p
A
Obs
JmaxRef = 128.18
RdRef = 1.37
sigma = 0.53
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

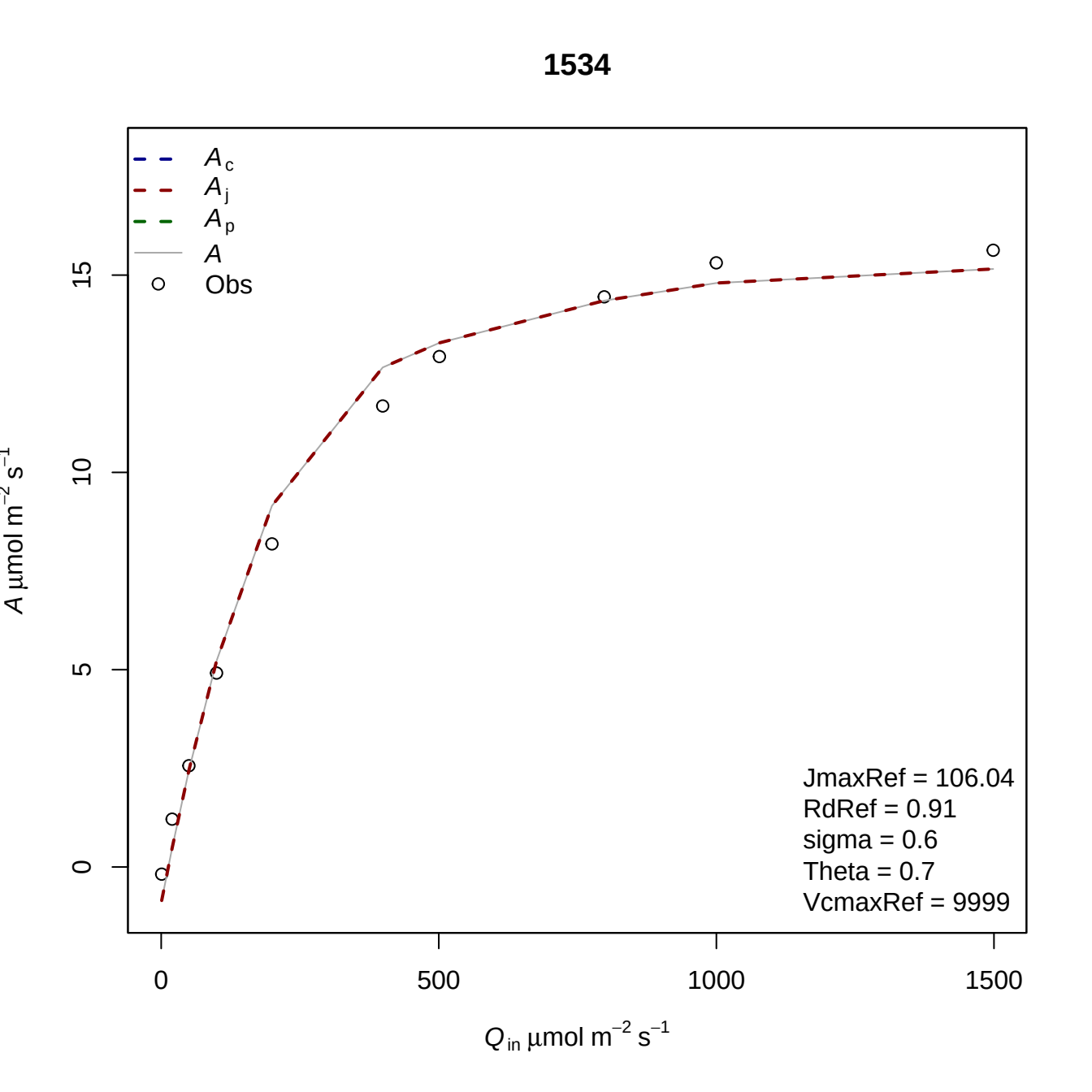

1529
A_c
A_j
A_p
A
Obs
JmaxRef = 60.7
RdRef = 1.16
sigma = 0.45
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

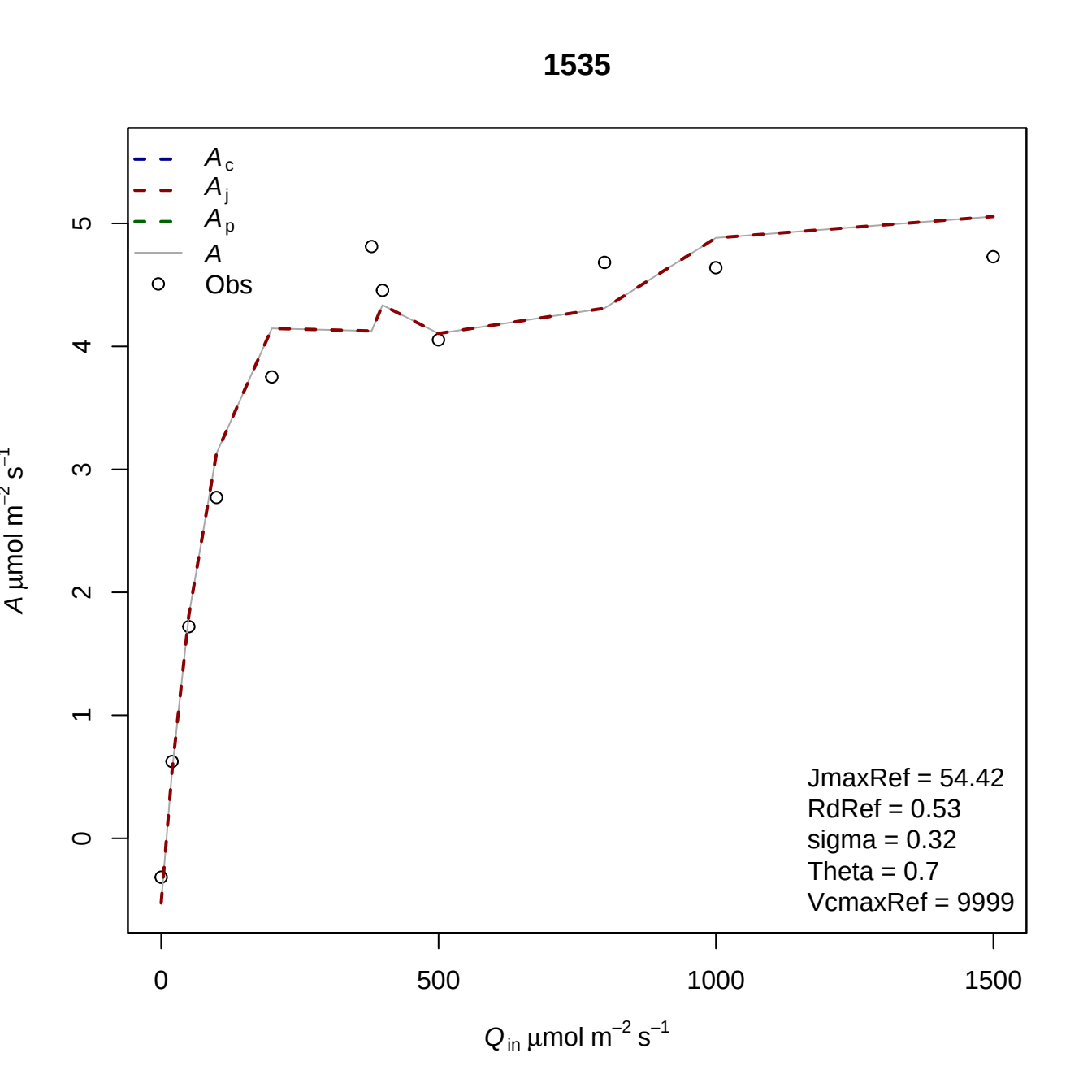

1530
A_c
A_j
A_p
A
Obs
JmaxRef = 77.5
RdRef = 1.08
sigma = 0.61
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

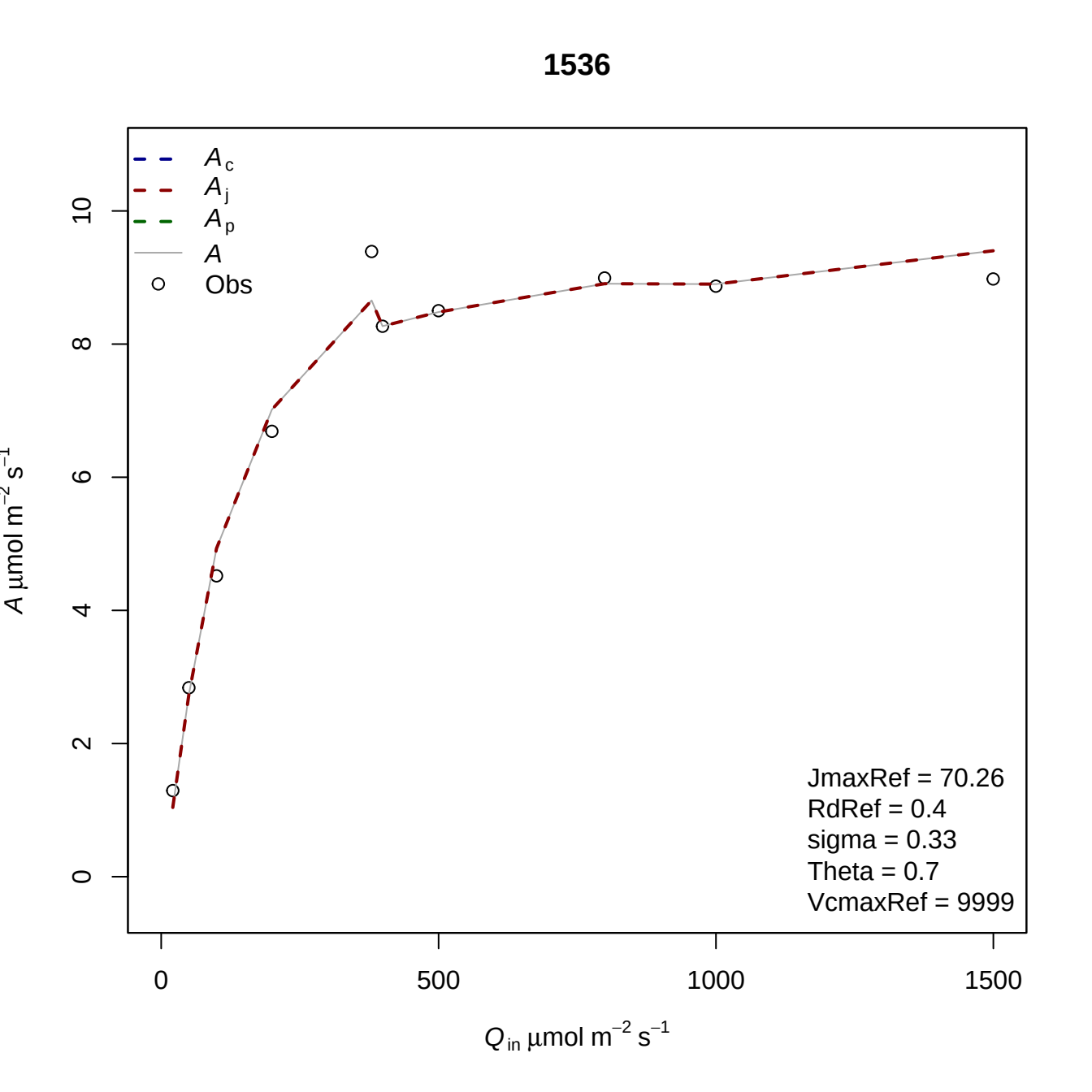

1531
A_c
A_j
A_p
A
Obs
JmaxRef = 58.64
RdRef = 0.61
sigma = 0.26
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

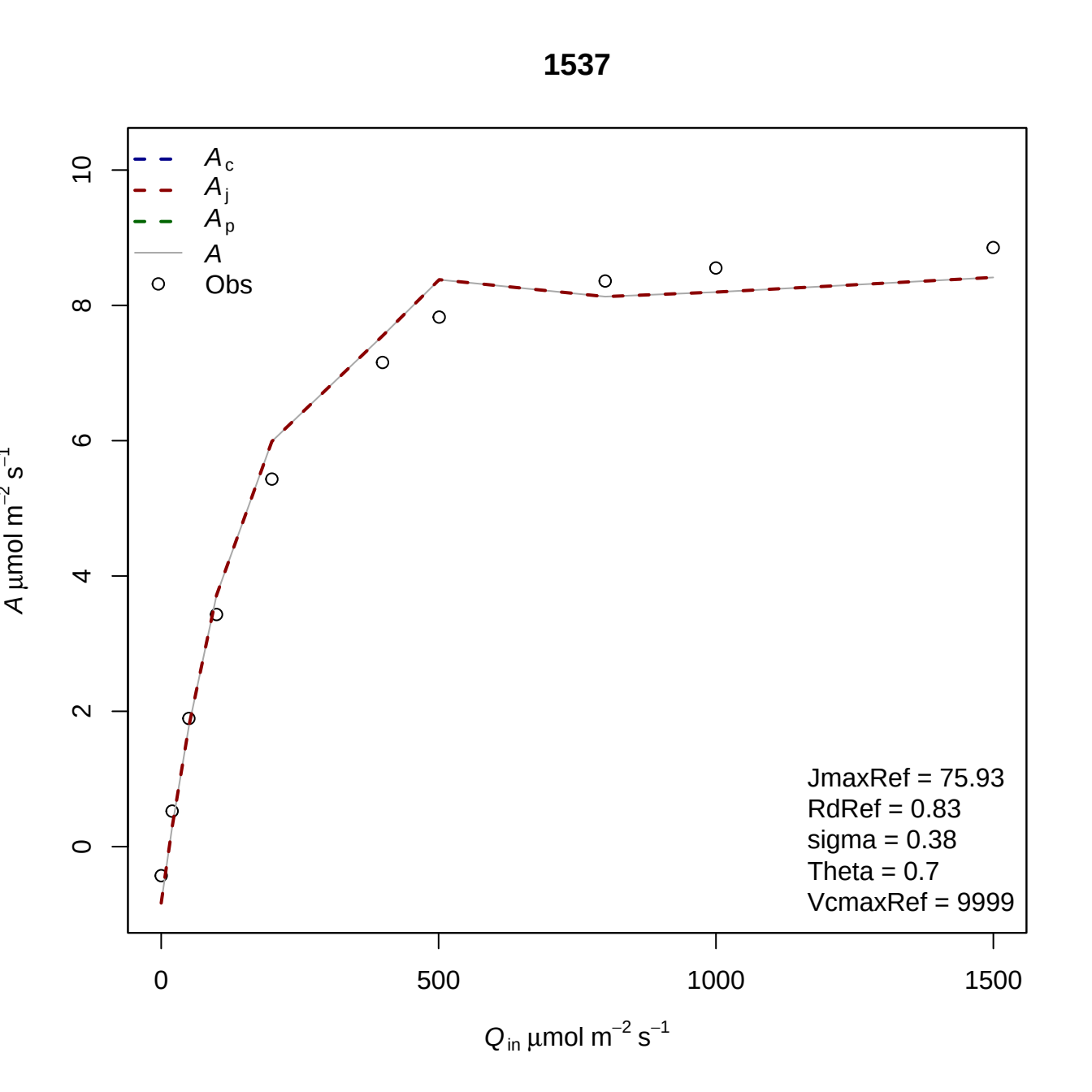

1532
A_c
A_j
A_p
A
Obs
JmaxRef = 72.26
RdRef = 0.73
sigma = 0.56
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

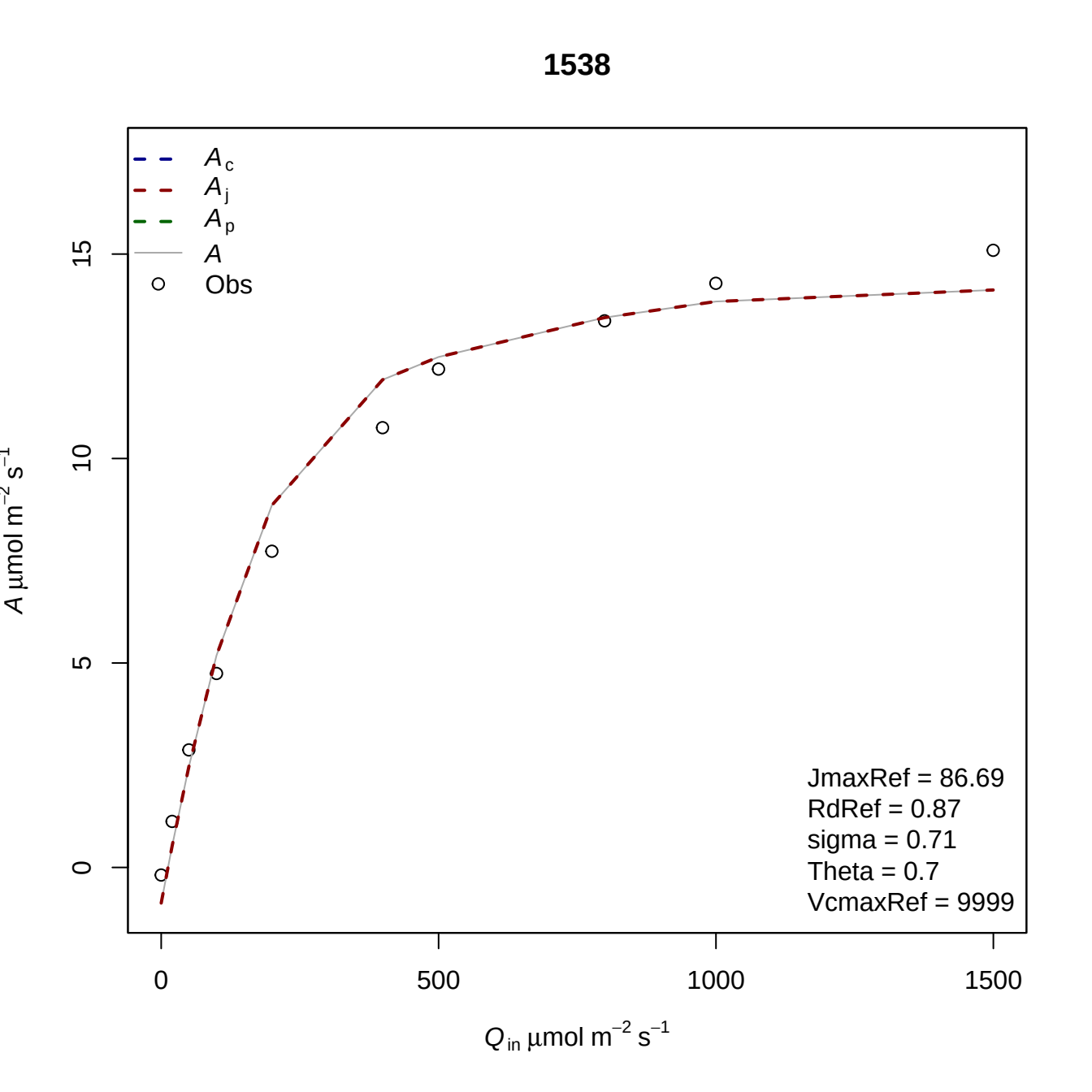

1533
A_c
A_j
A_p
A
Obs
JmaxRef = 75.6
RdRef = 0.71
sigma = 0.23
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

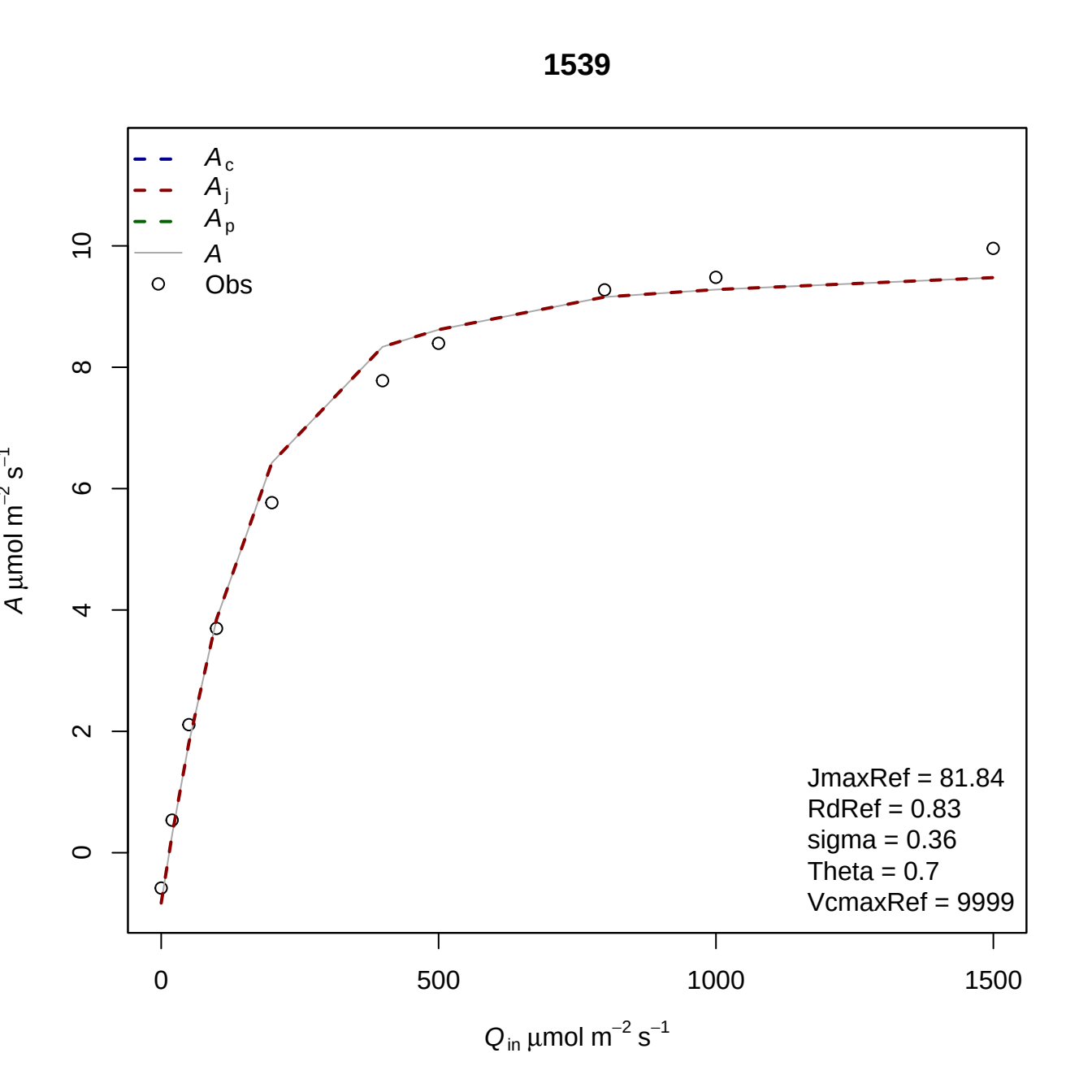

1534
A_c
A_j
A_p
A
Obs
JmaxRef = 106.04
RdRef = 0.91
sigma = 0.6
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

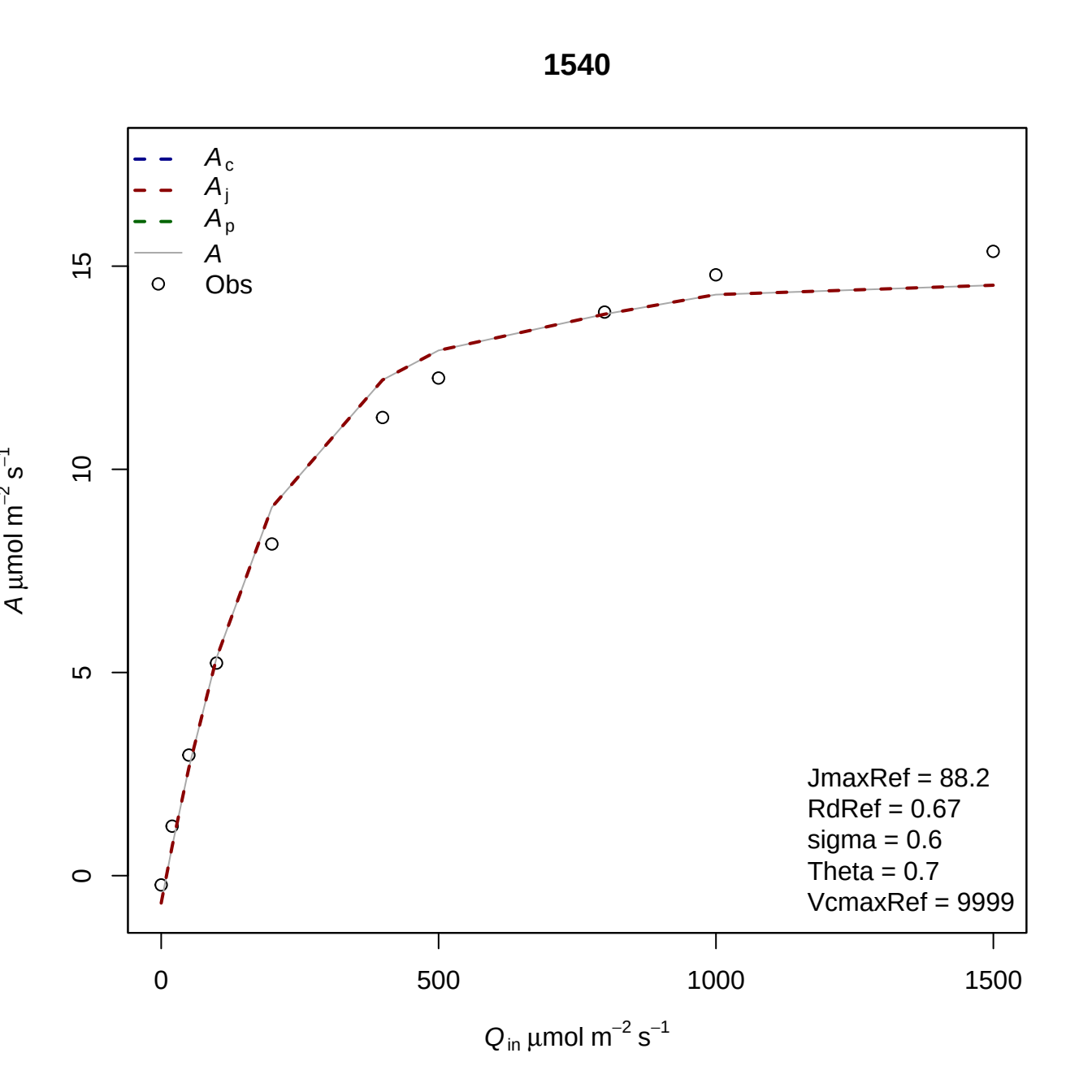

1535
A_c
A_j
A_p
A
Obs
JmaxRef = 54.42
RdRef = 0.53
sigma = 0.32
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
1
2
3
4
5
0
500
1000
1500

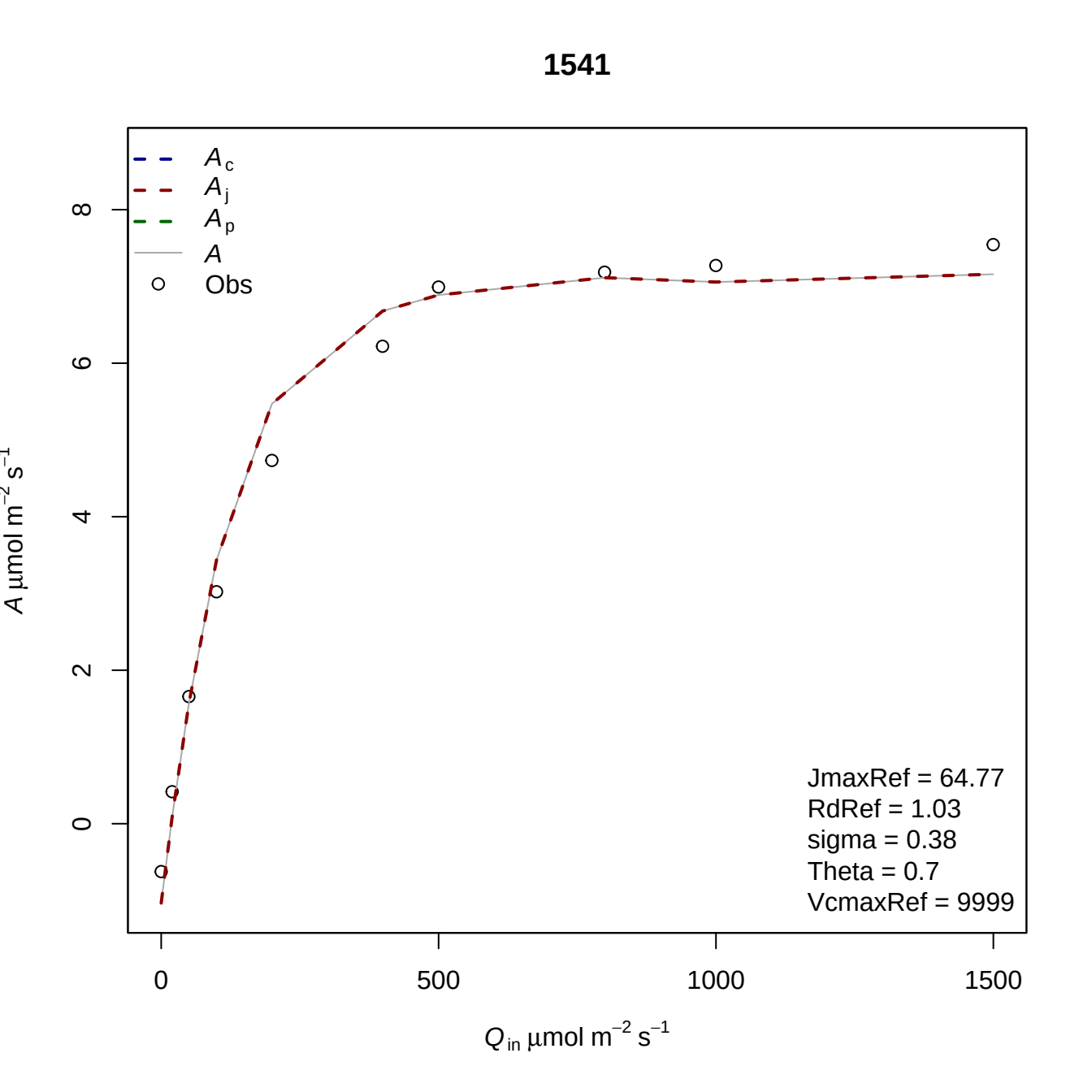

1536
A_c
A_j
A_p
A
Obs
JmaxRef = 70.26
RdRef = 0.4
sigma = 0.33
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

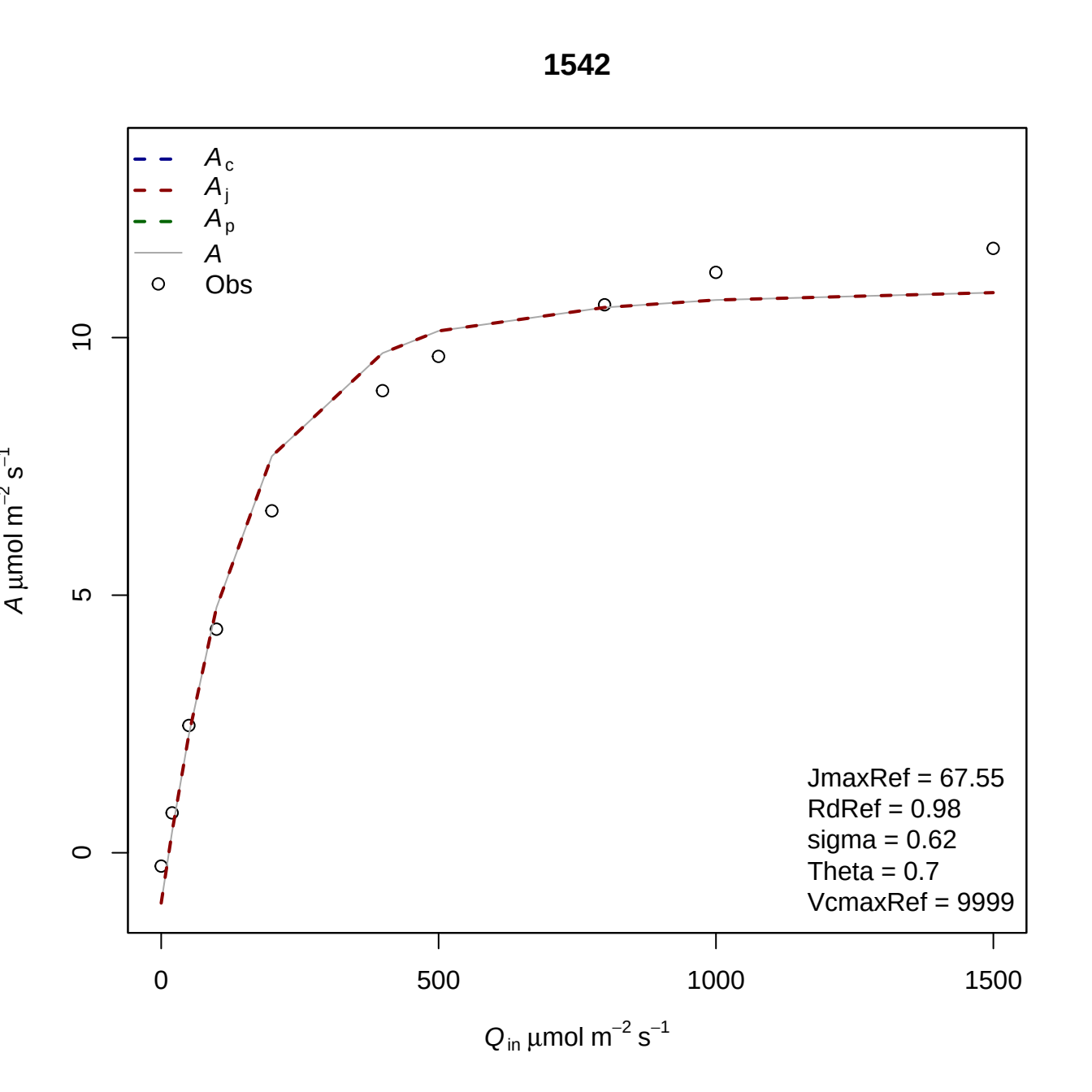

1537
A_c
A_j
A_p
A
Obs
JmaxRef = 75.93
RdRef = 0.83
sigma = 0.38
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

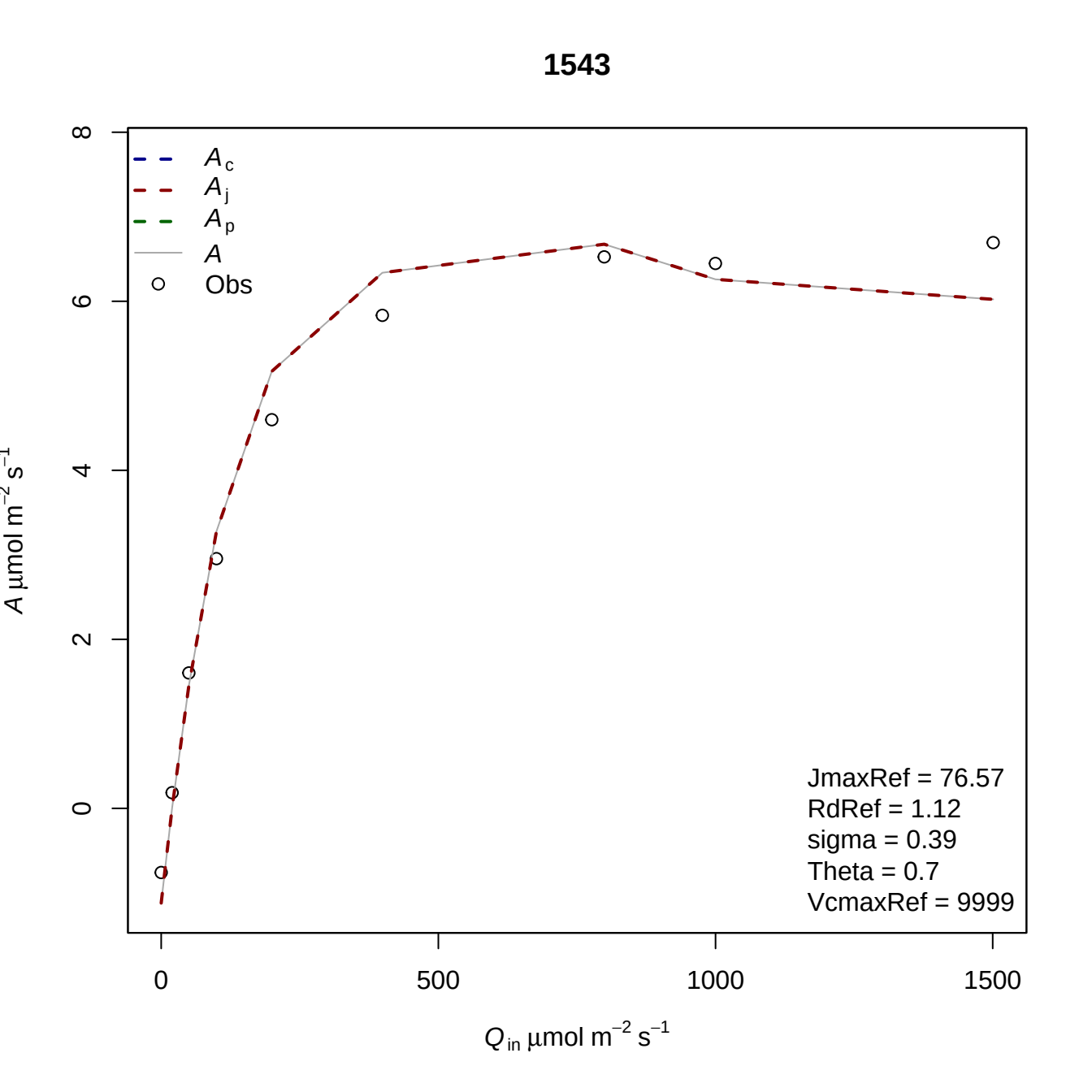

1538
A_c
A_j
A_p
A
Obs
JmaxRef = 86.69
RdRef = 0.87
sigma = 0.71
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

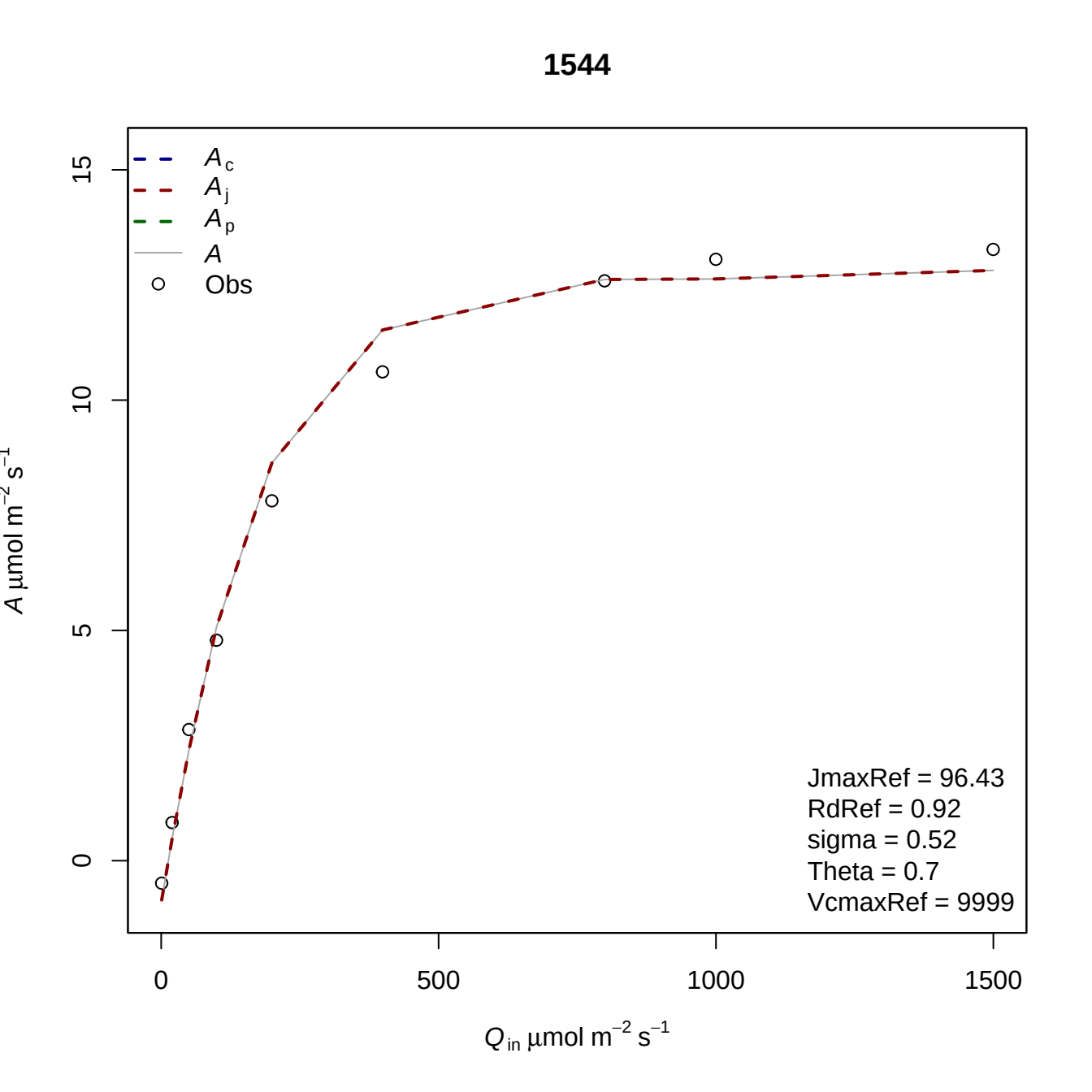

1539
A_c
A_j
A_p
A
Obs
JmaxRef = 81.84
RdRef = 0.83
sigma = 0.36
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

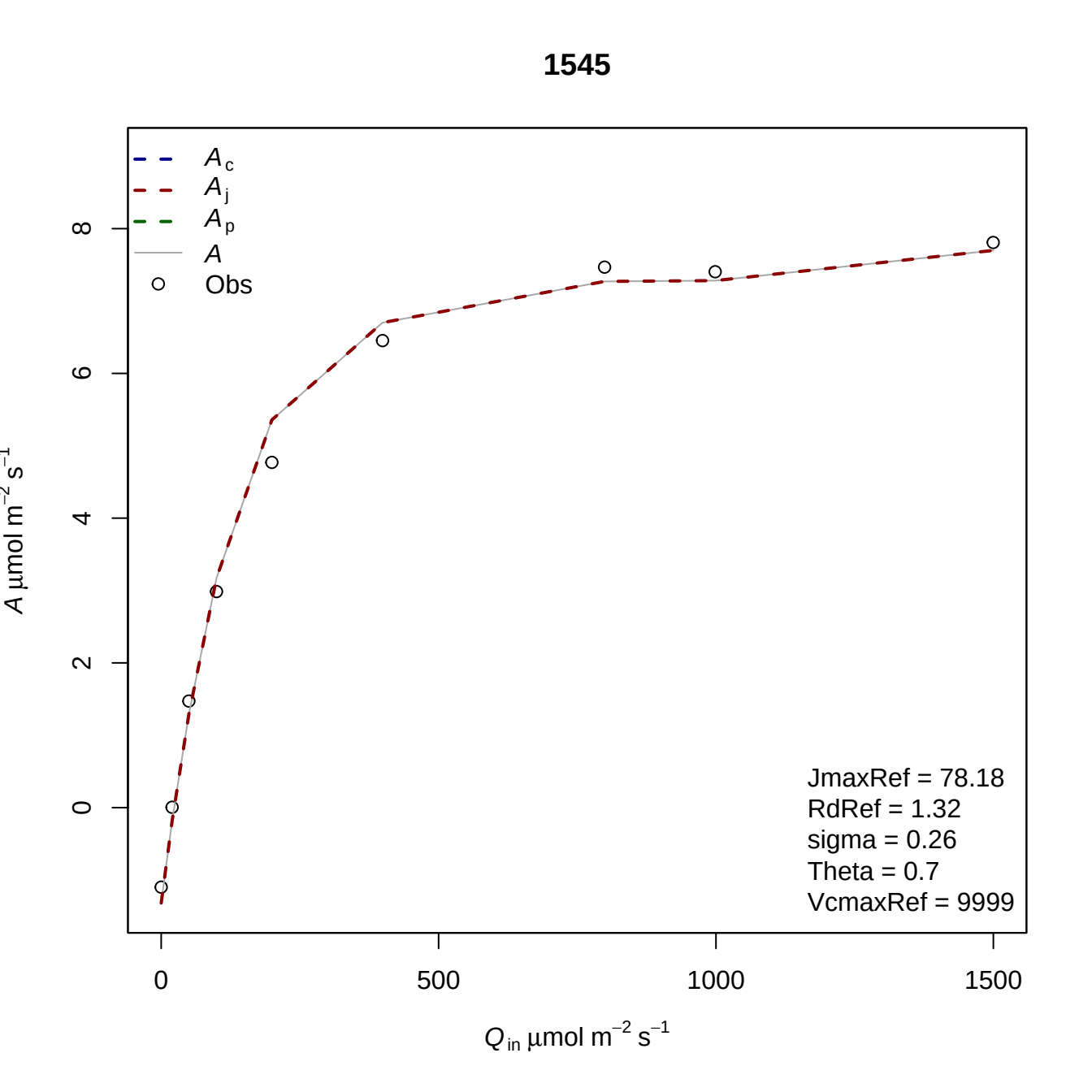

1540
A_c
A_j
A_p
A
Obs
JmaxRef = 88.2
RdRef = 0.67
sigma = 0.6
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

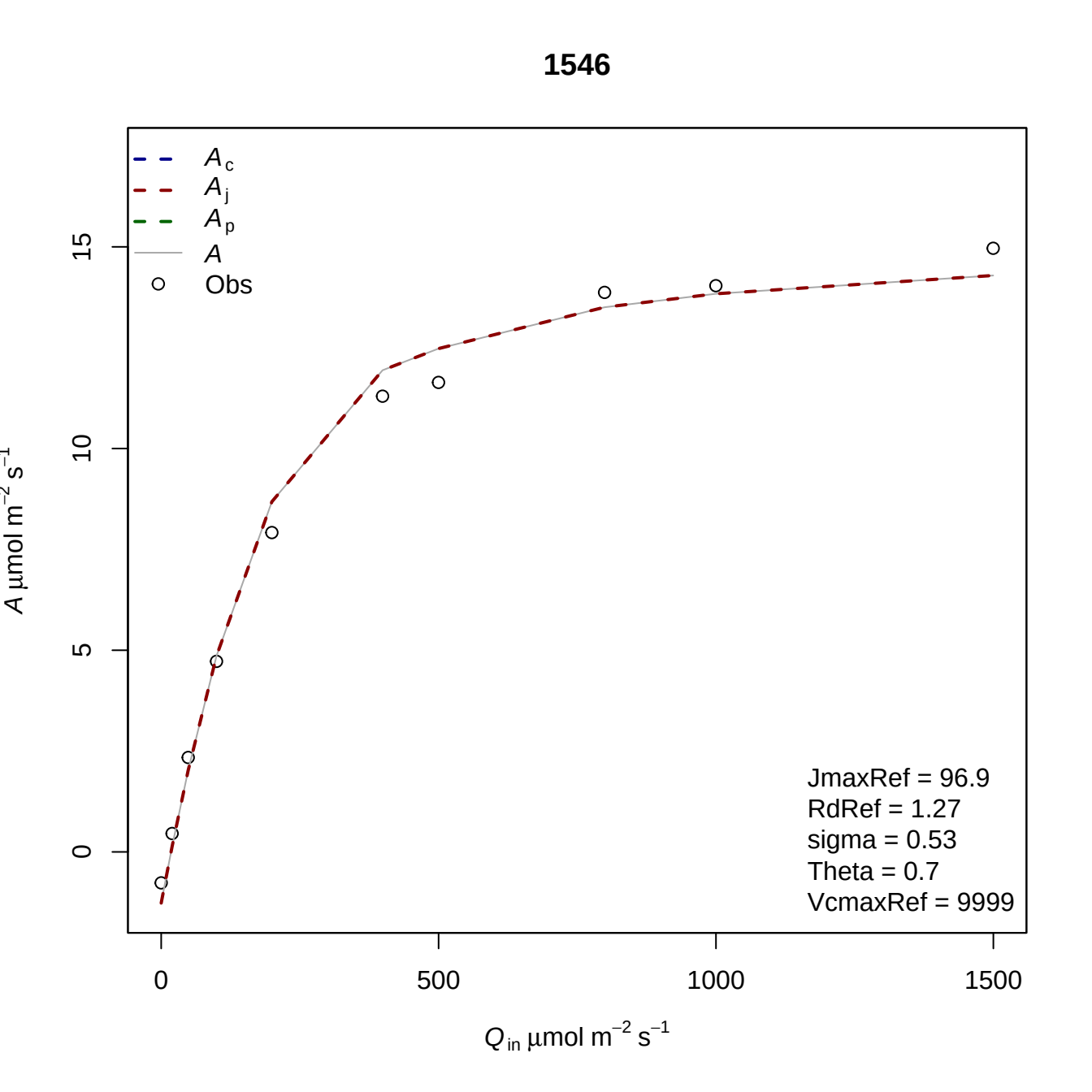

1541
A_c
A_j
A_p
A
Obs
JmaxRef = 64.77
RdRef = 1.03
sigma = 0.38
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

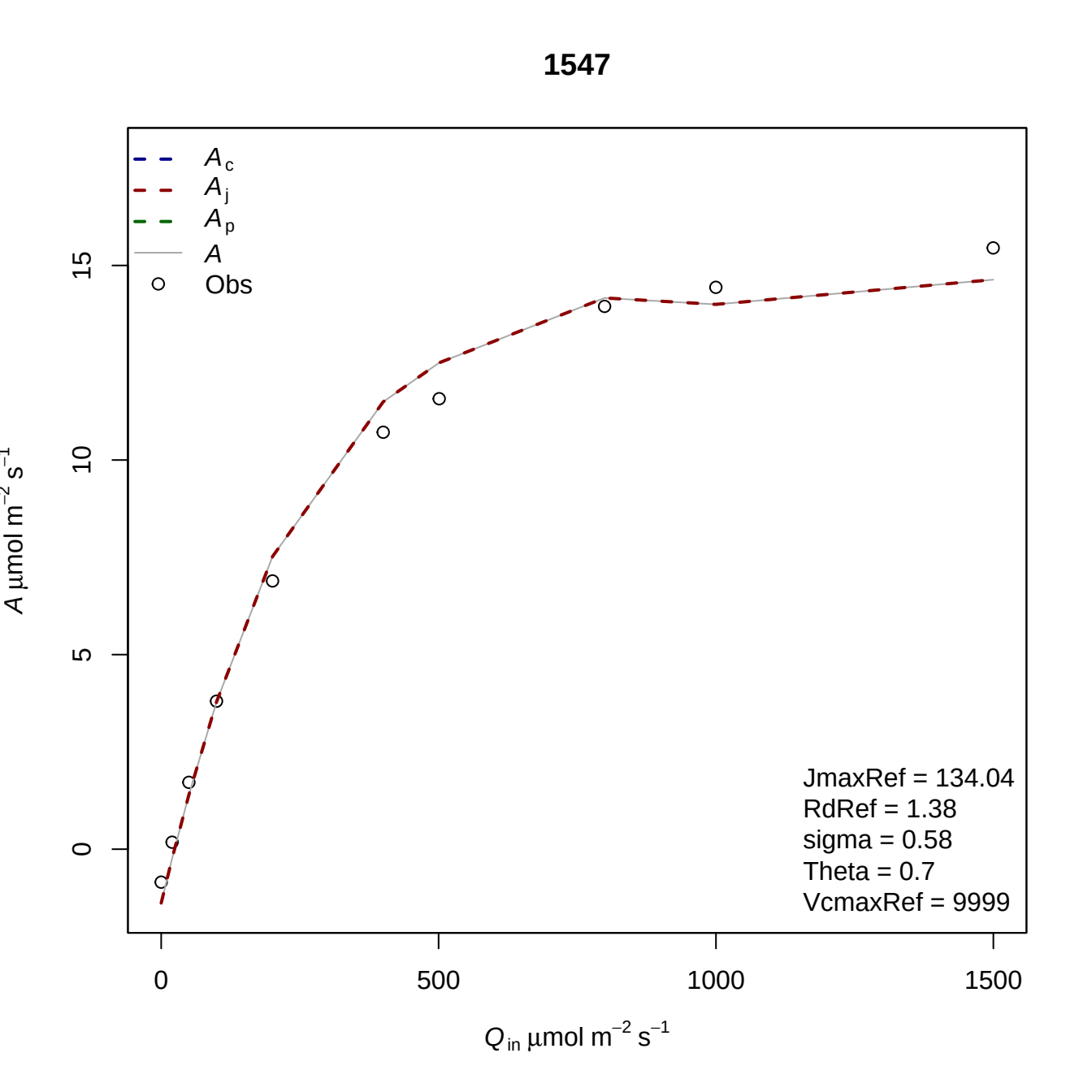

1542
A_c
A_j
A_p
A
Obs
JmaxRef = 67.55
RdRef = 0.98
sigma = 0.62
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

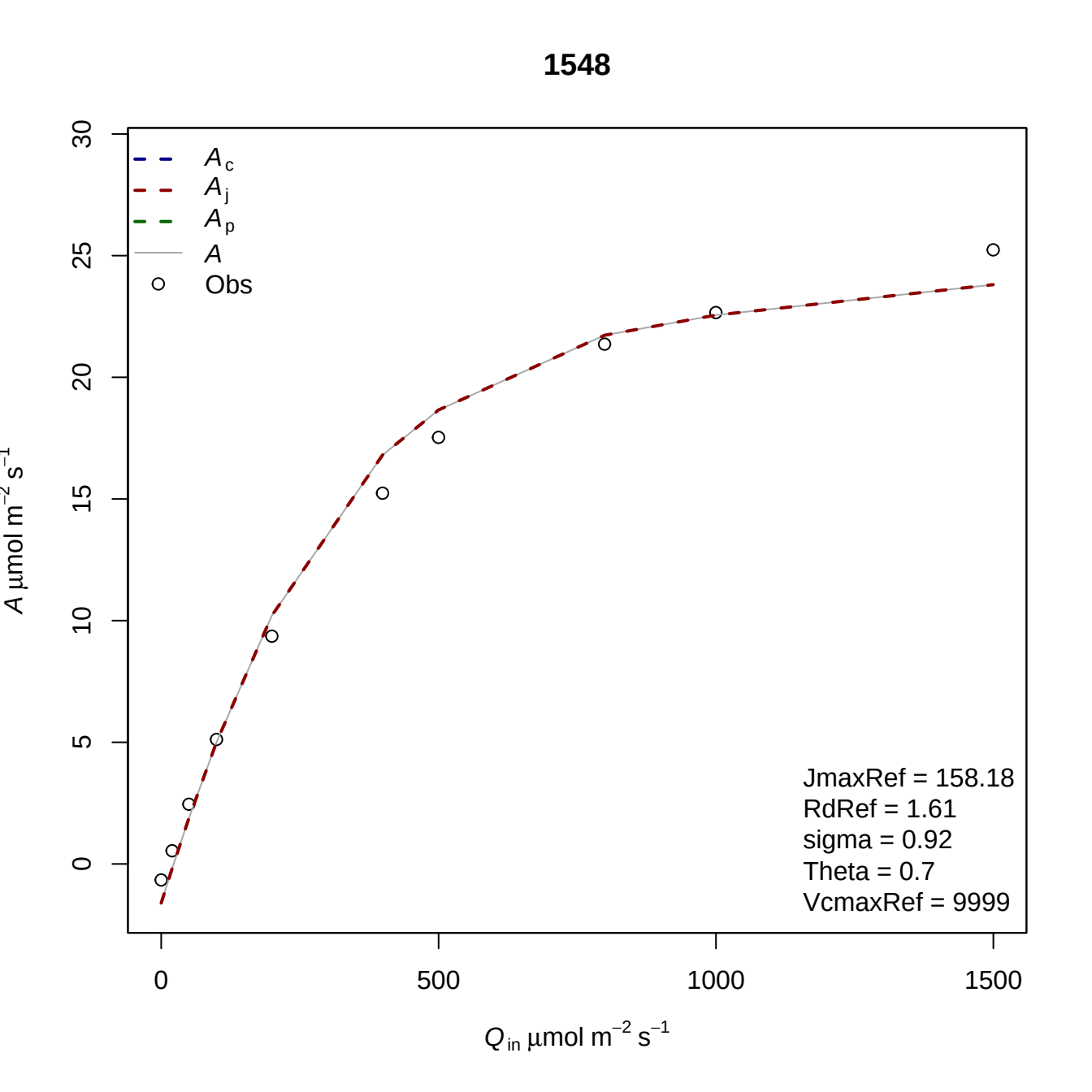

1543
A_c
A_j
A_p
A
Obs
JmaxRef = 76.57
RdRef = 1.12
sigma = 0.39
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

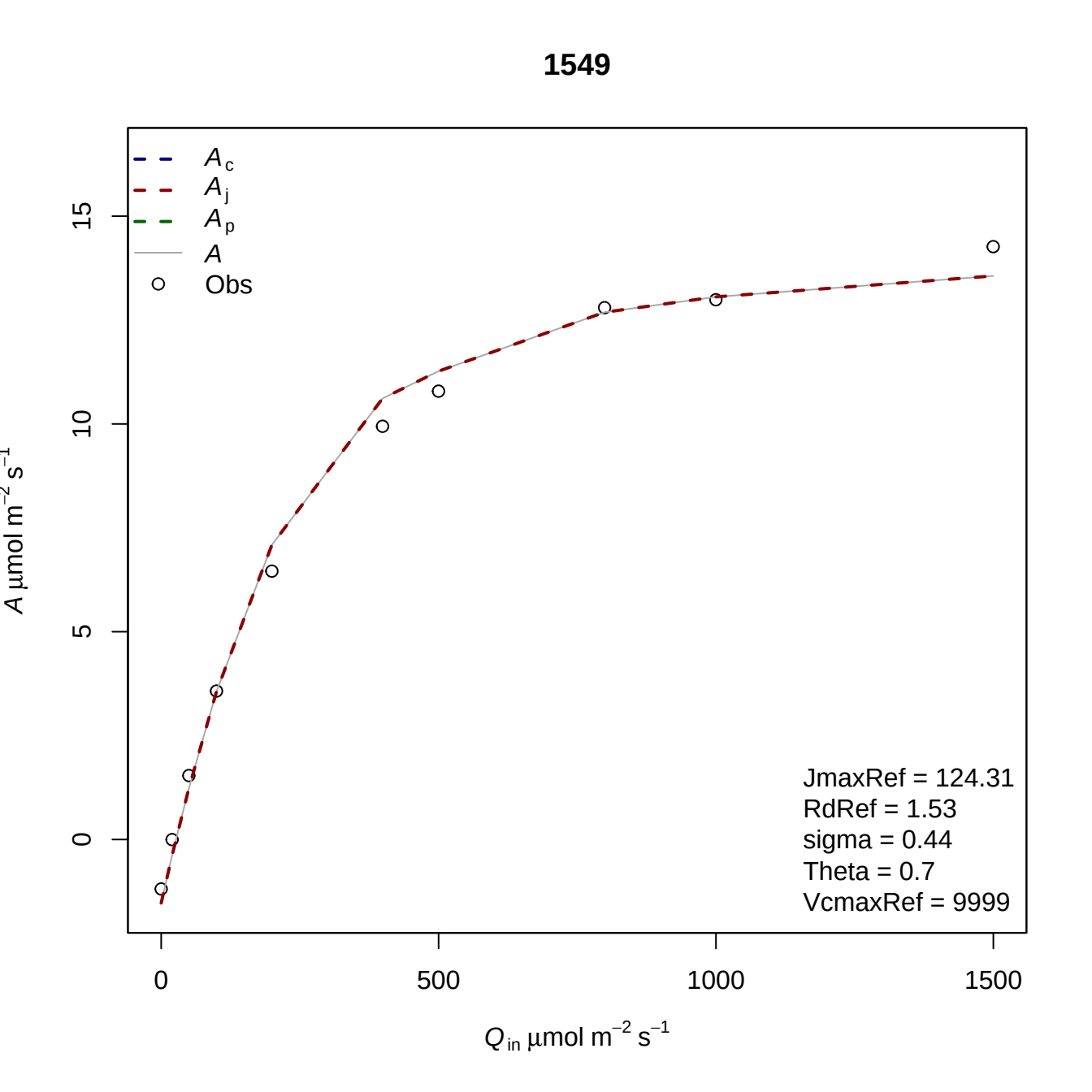

1544
A_c
A_j
A_p
A
Obs
JmaxRef = 96.43
RdRef = 0.92
sigma = 0.52
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500

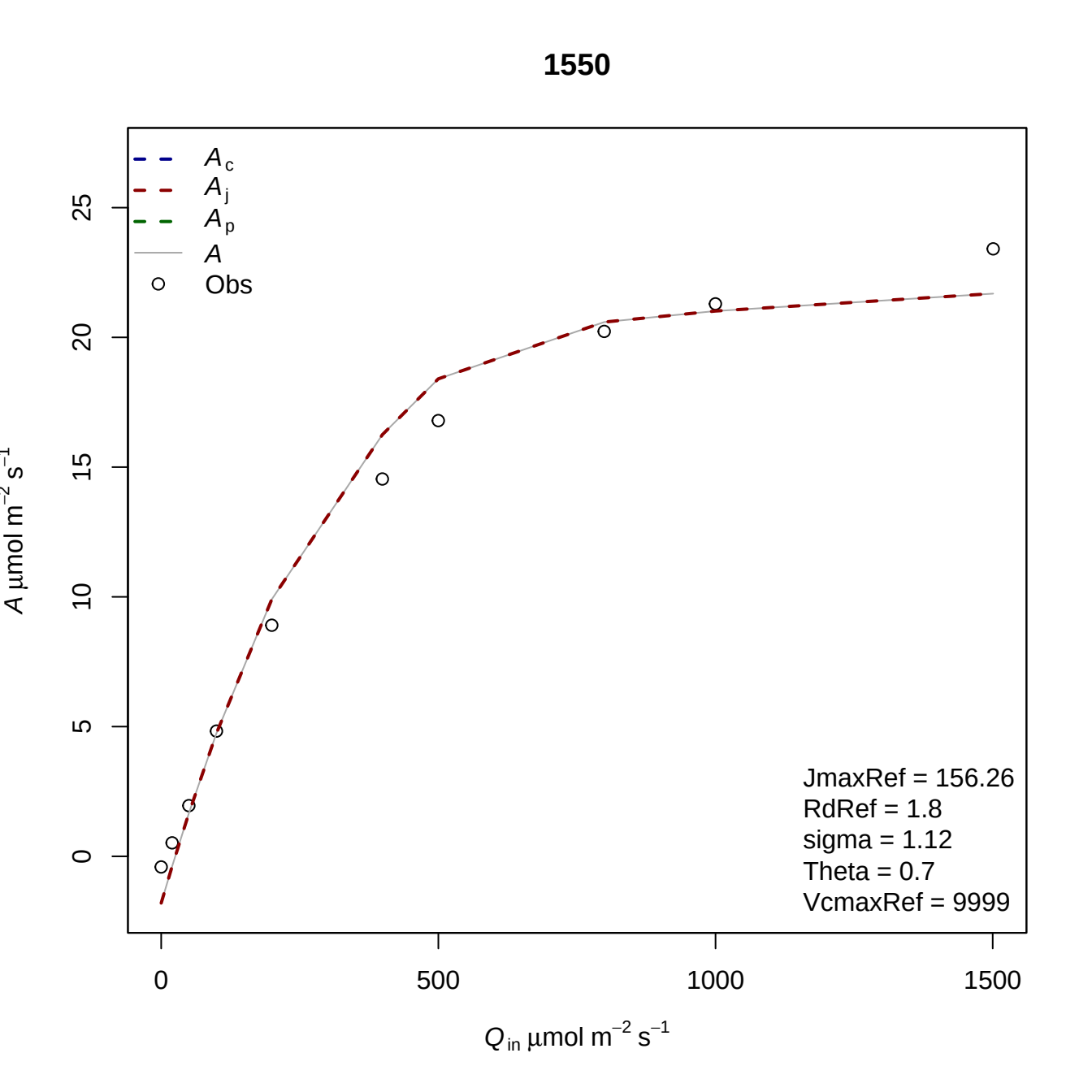

1545
A_c
A_j
A_p
A
Obs
JmaxRef = 78.18
RdRef = 1.32
sigma = 0.26
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

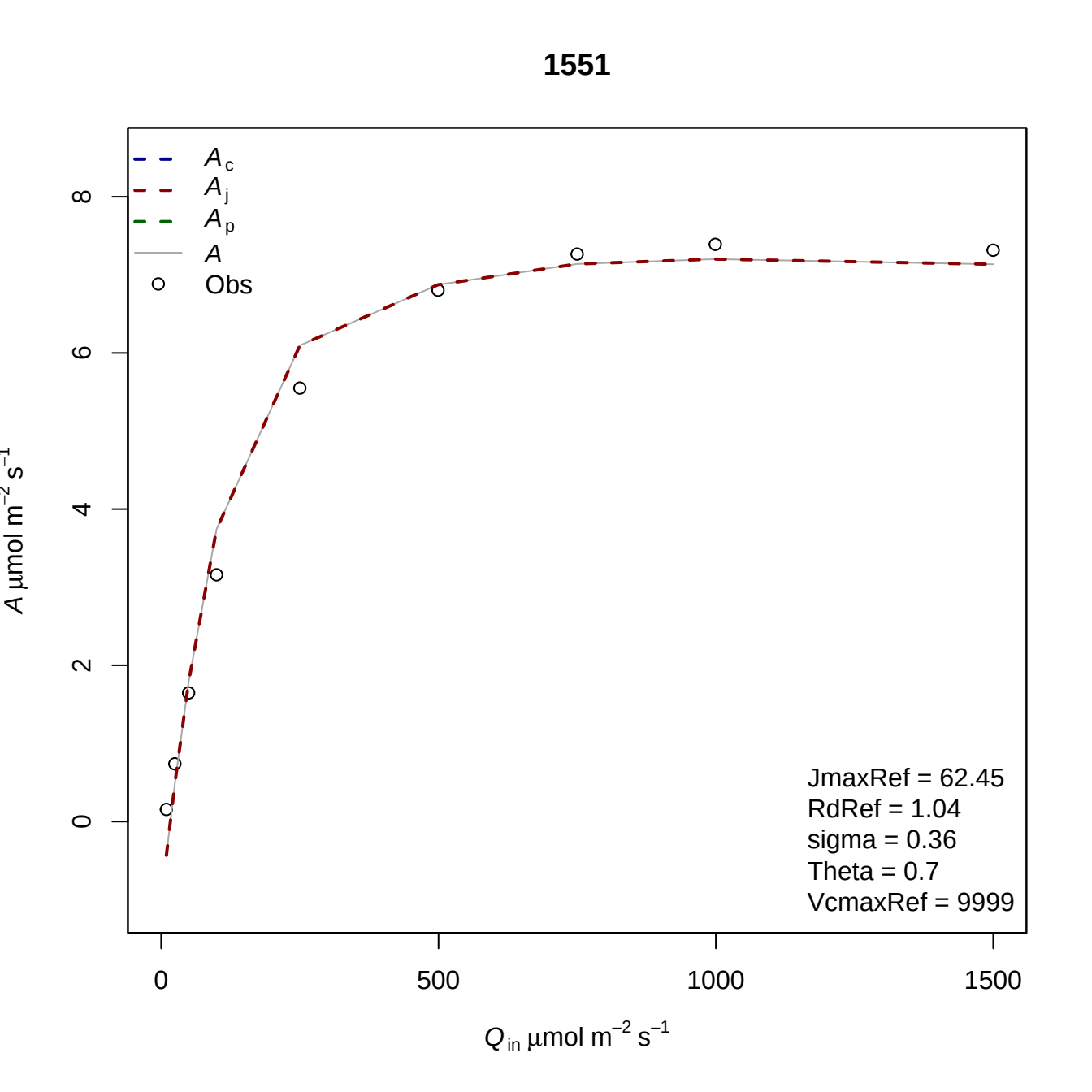

1546
A_c
A_j
A_p
A
Obs
JmaxRef = 96.9
RdRef = 1.27
sigma = 0.53
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

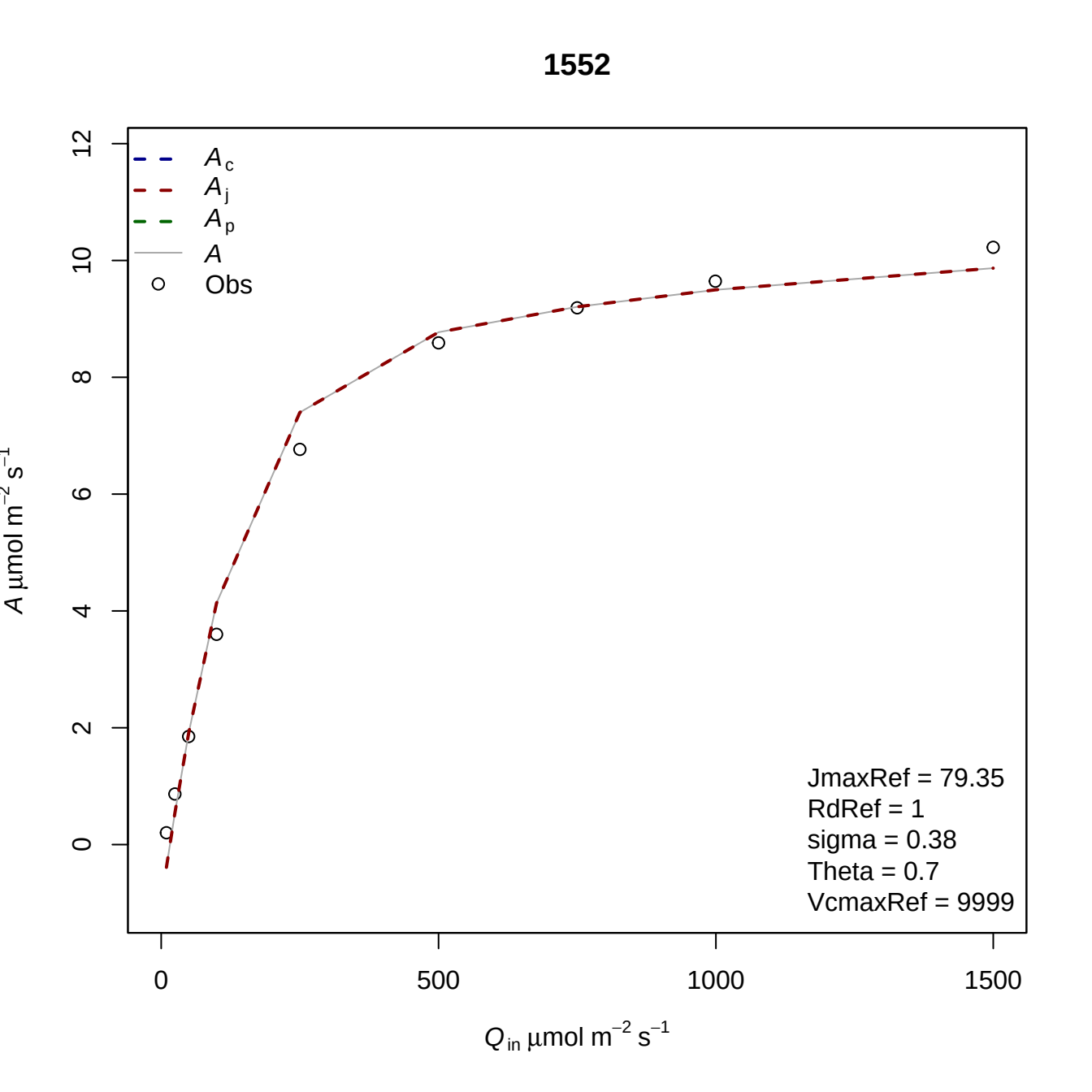

1547
A_c
A_j
A_p
A
Obs
JmaxRef = 134.04
RdRef = 1.38
sigma = 0.58
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

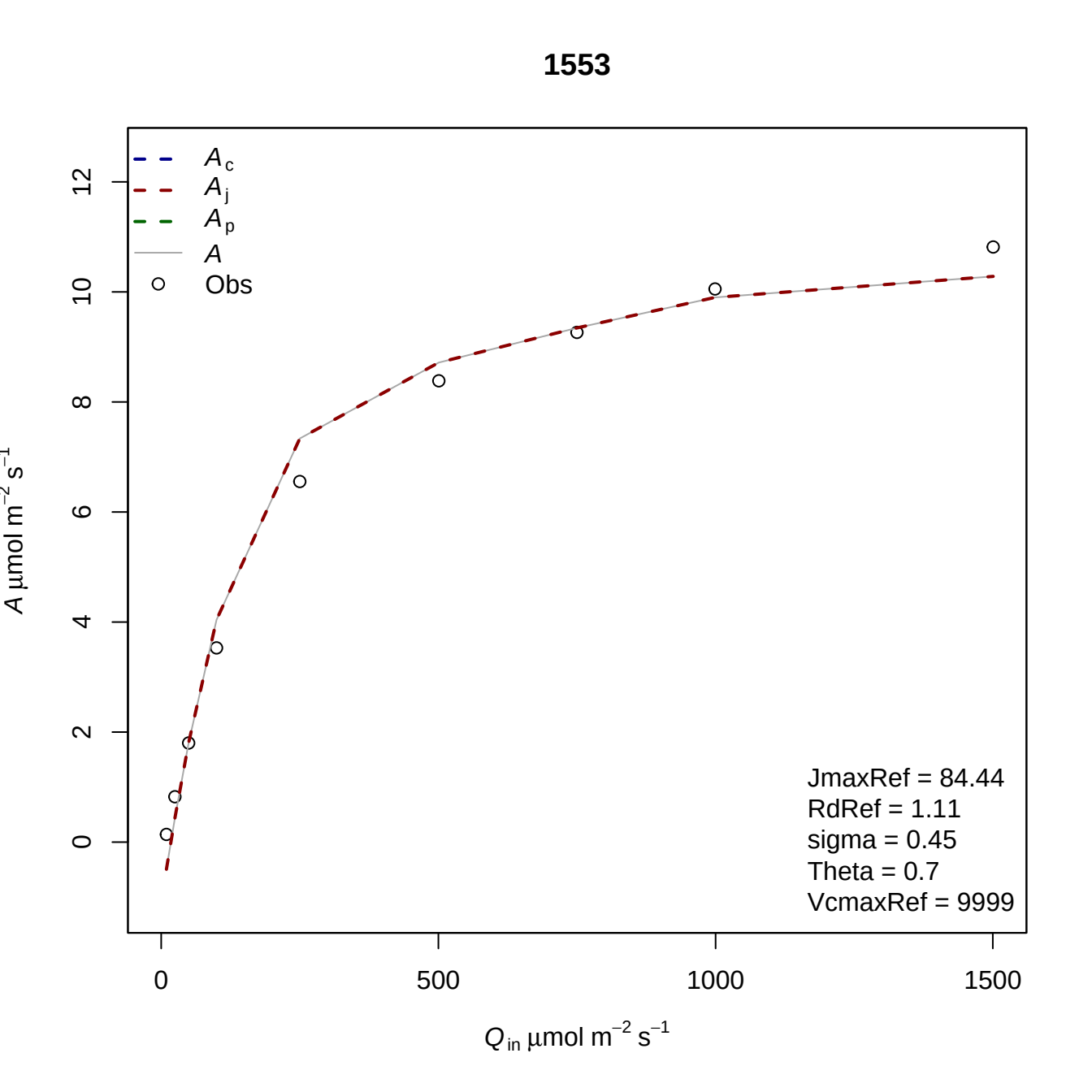

1548
A_c
A_j
A_p
A
Obs
JmaxRef = 158.18
RdRef = 1.61
sigma = 0.92
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500

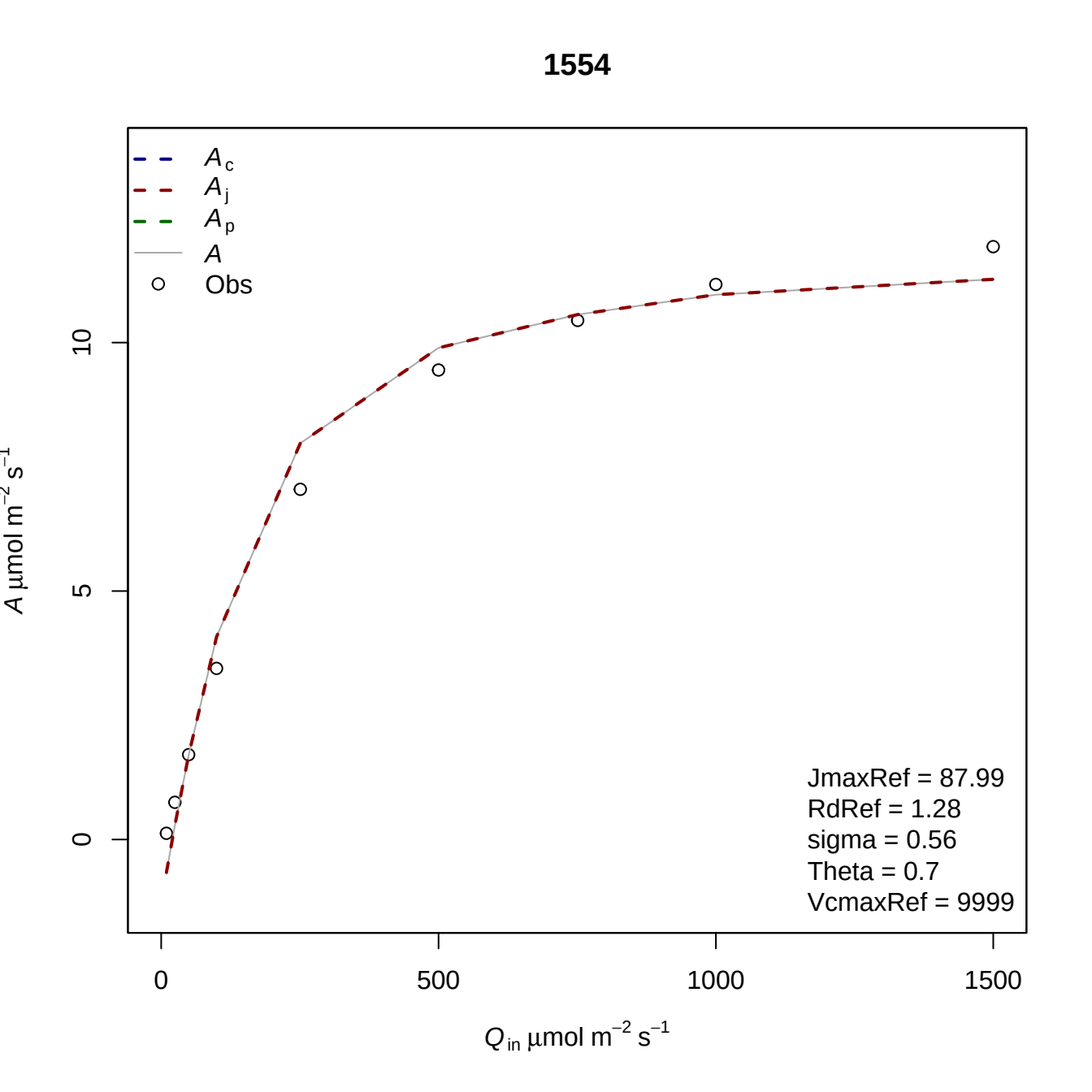

1549
A_c
A_j
A_p
A
Obs
JmaxRef = 124.31
RdRef = 1.53
sigma = 0.44
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

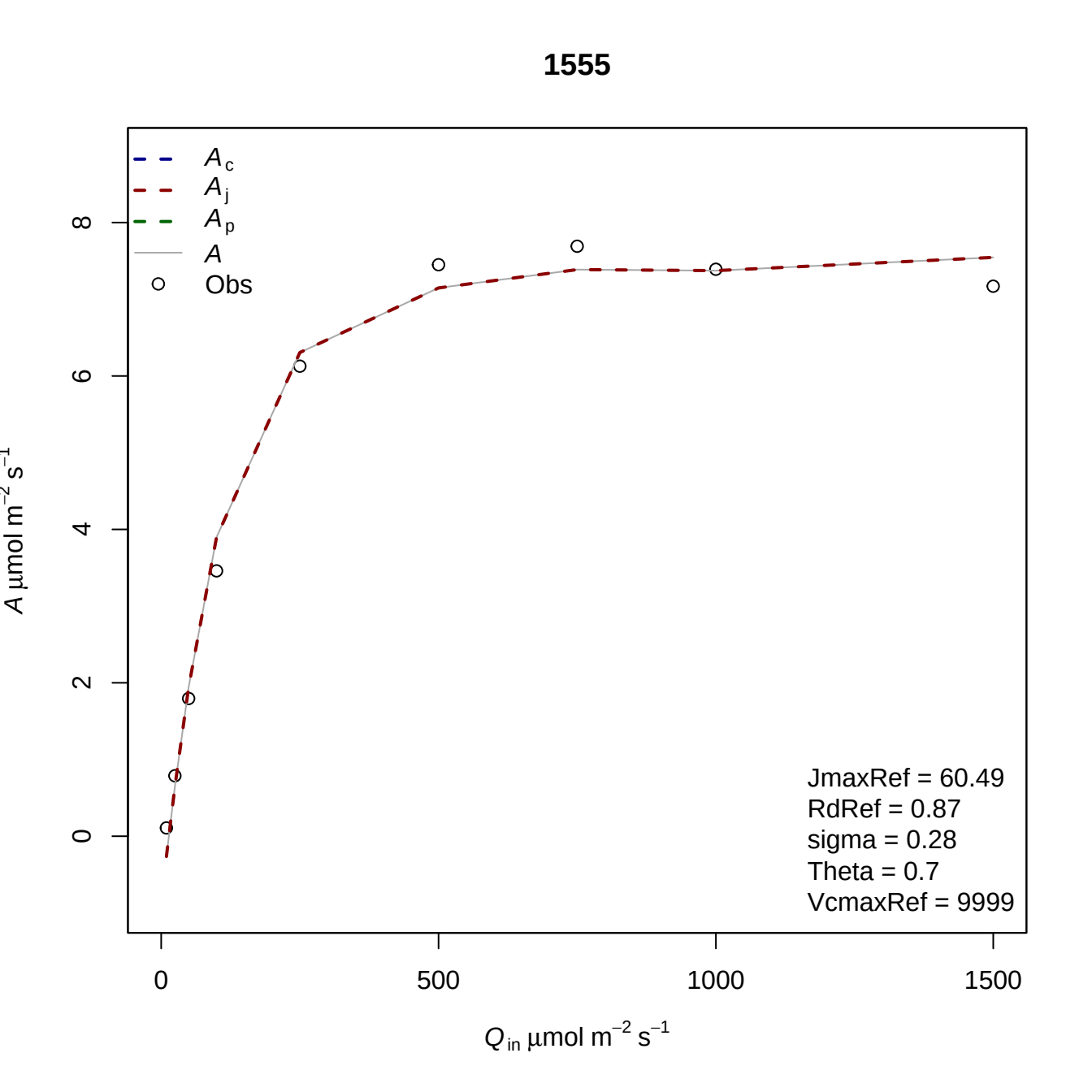

1550
A_c
A_j
A_p
A
Obs
JmaxRef = 156.26
RdRef = 1.8
sigma = 1.12
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
25
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

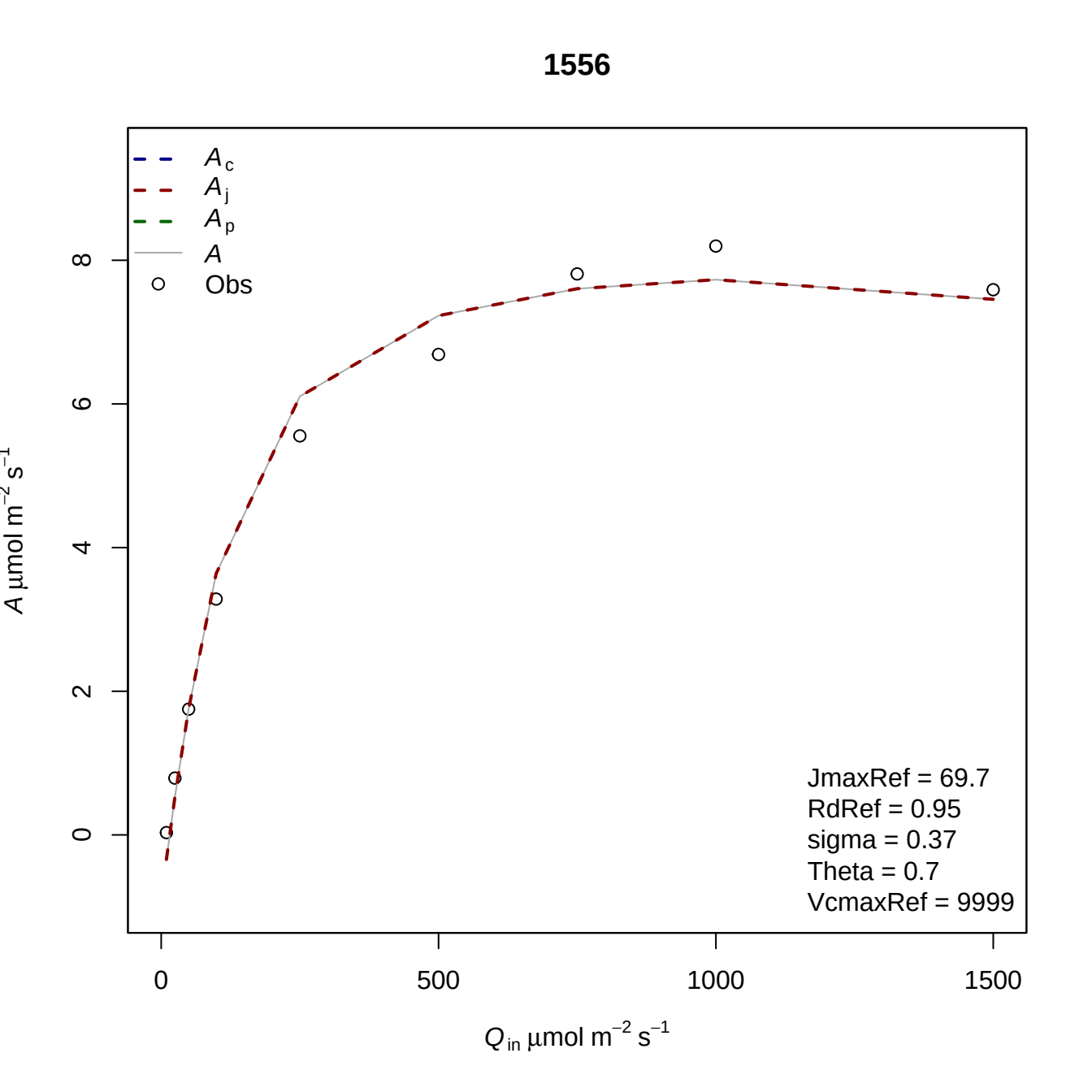

1551
A_c
A_j
A_p
A
Obs
JmaxRef = 62.45
RdRef = 1.04
sigma = 0.36
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

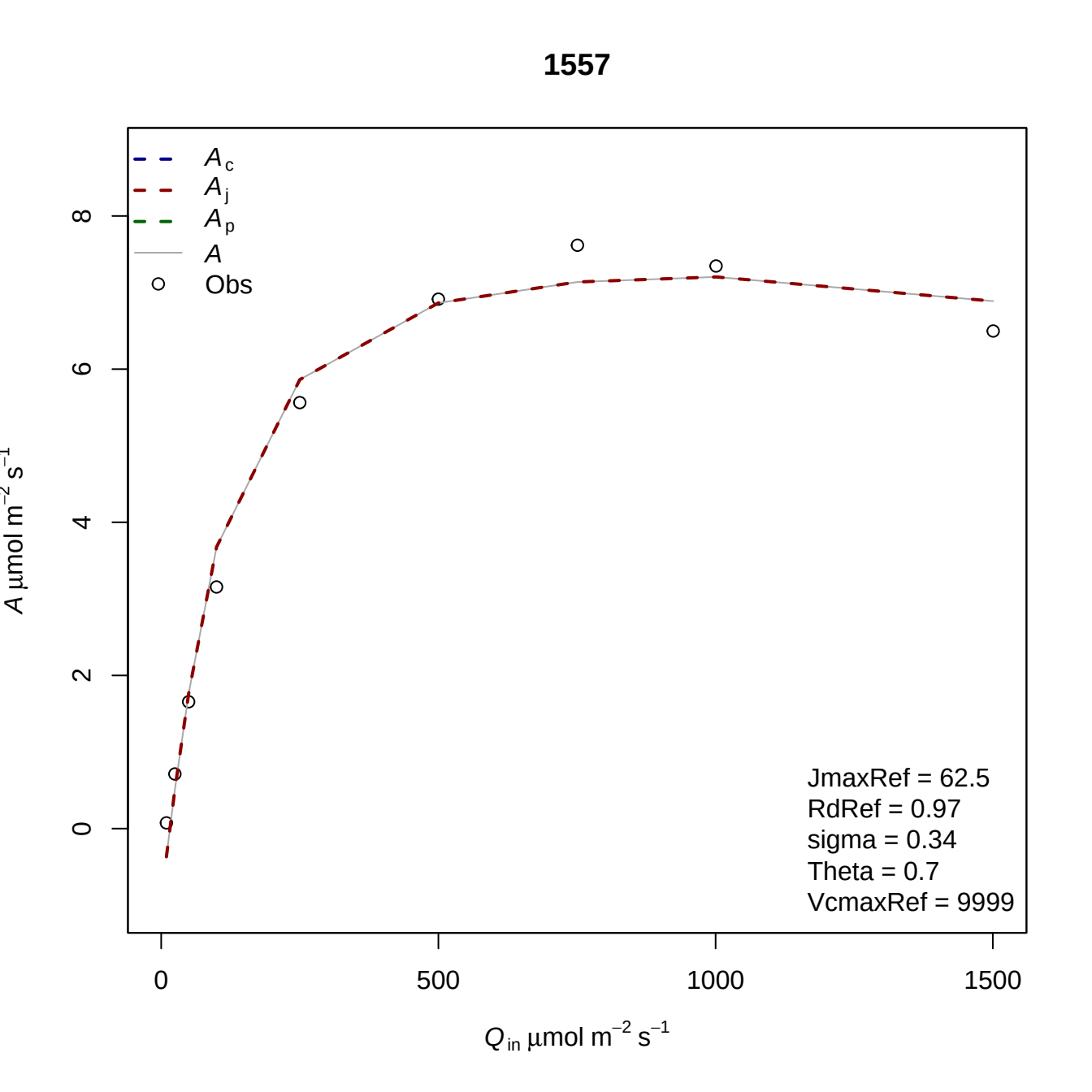

1552
A_c
A_j
A_p
A
Obs
JmaxRef = 79.35
RdRef = 1
sigma = 0.38
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

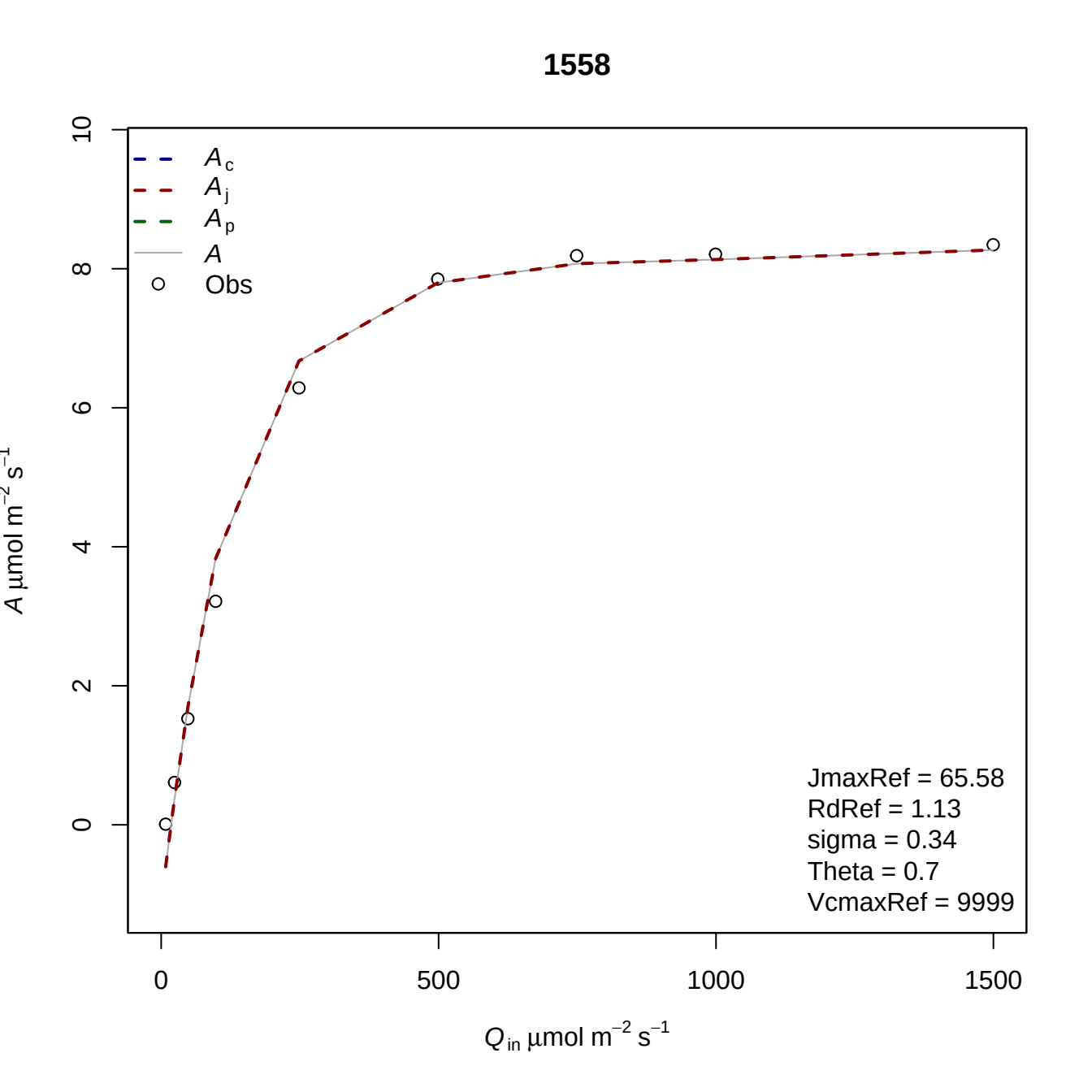

1553
A_c
A_j
A_p
A
Obs
JmaxRef = 84.44
RdRef = 1.11
sigma = 0.45
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

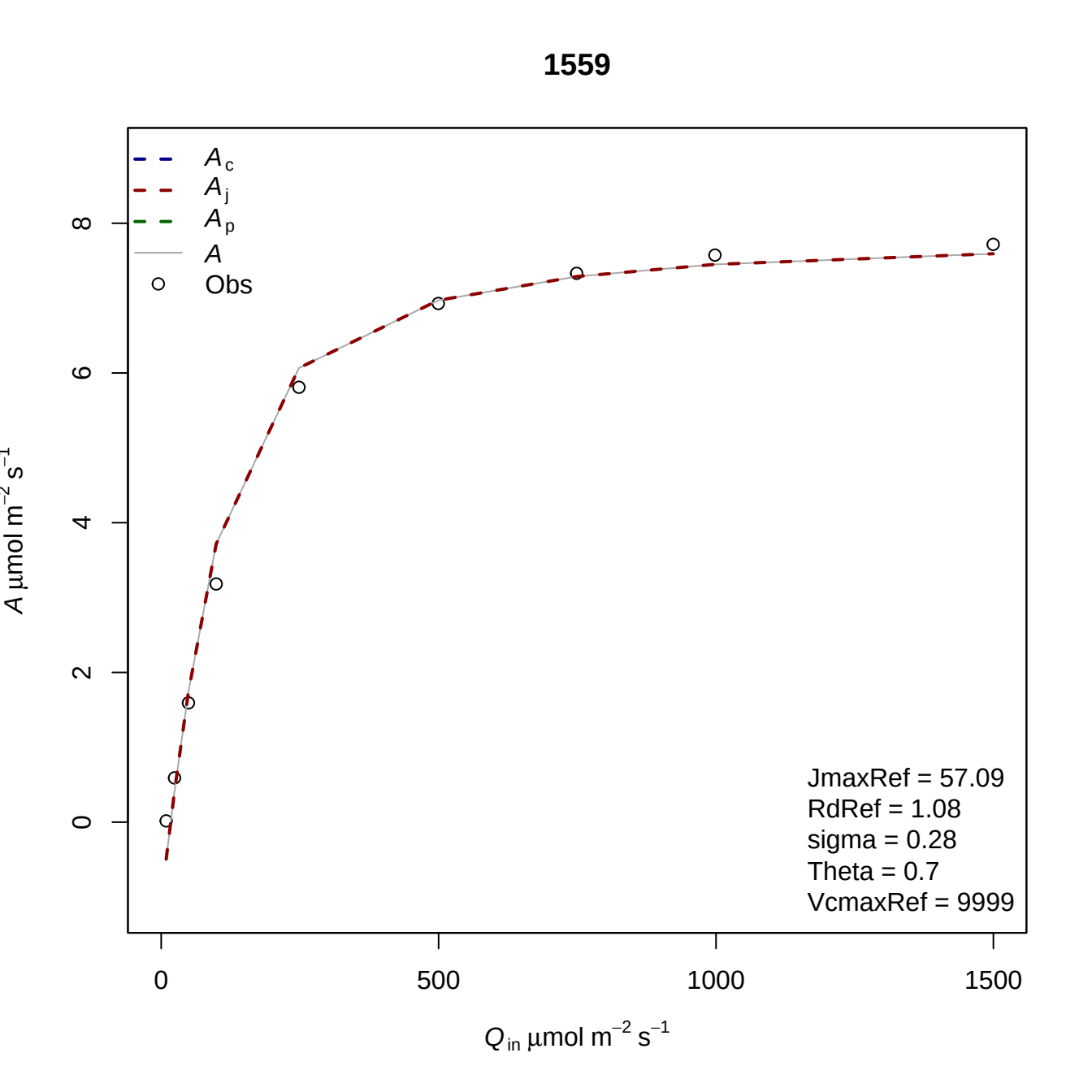

1554
A_c
A_j
A_p
A
Obs
JmaxRef = 87.99
RdRef = 1.28
sigma = 0.56
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

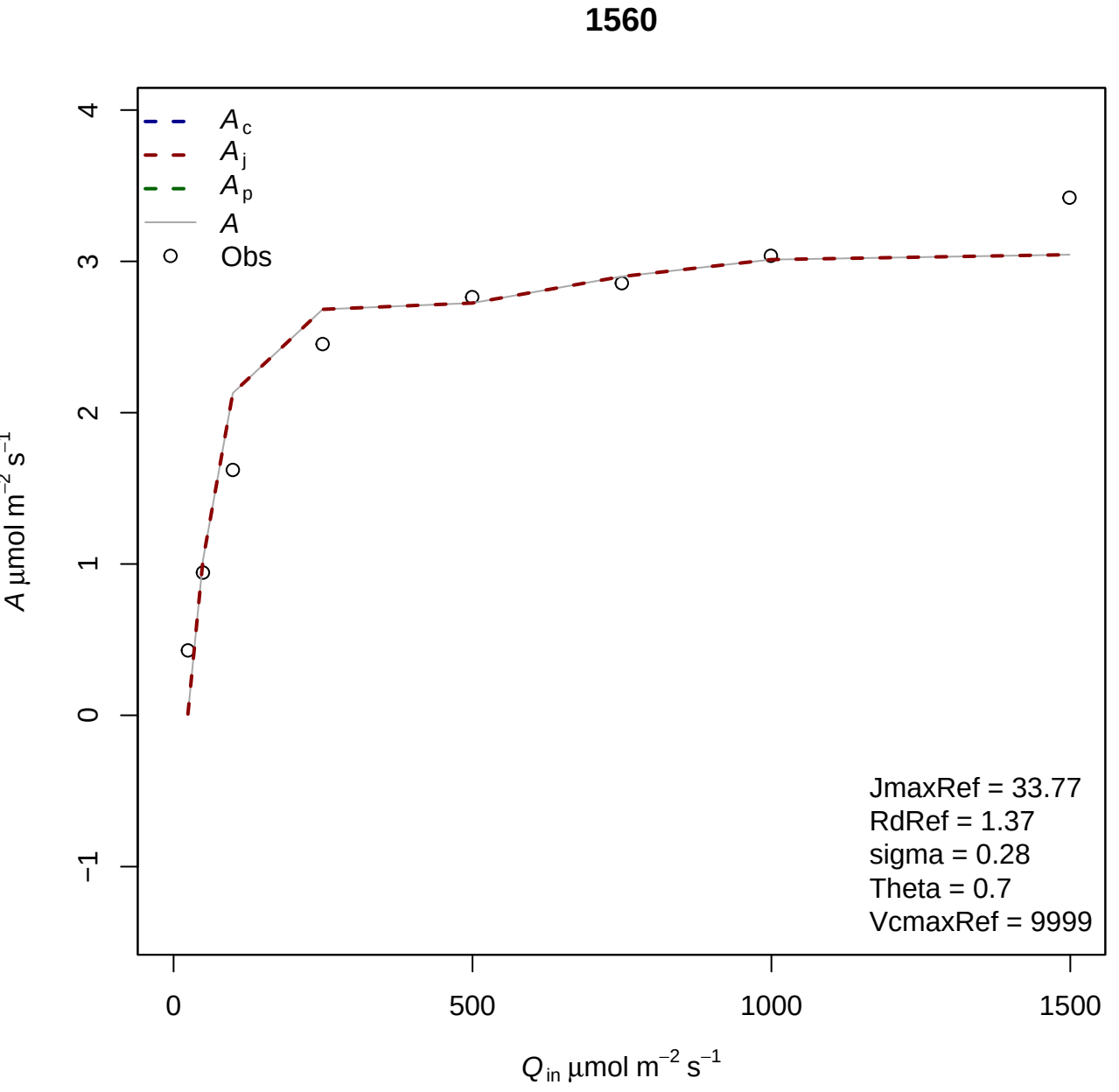

1555
A_c
A_j
A_p
A
Obs
JmaxRef = 60.49
RdRef = 0.87
sigma = 0.28
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

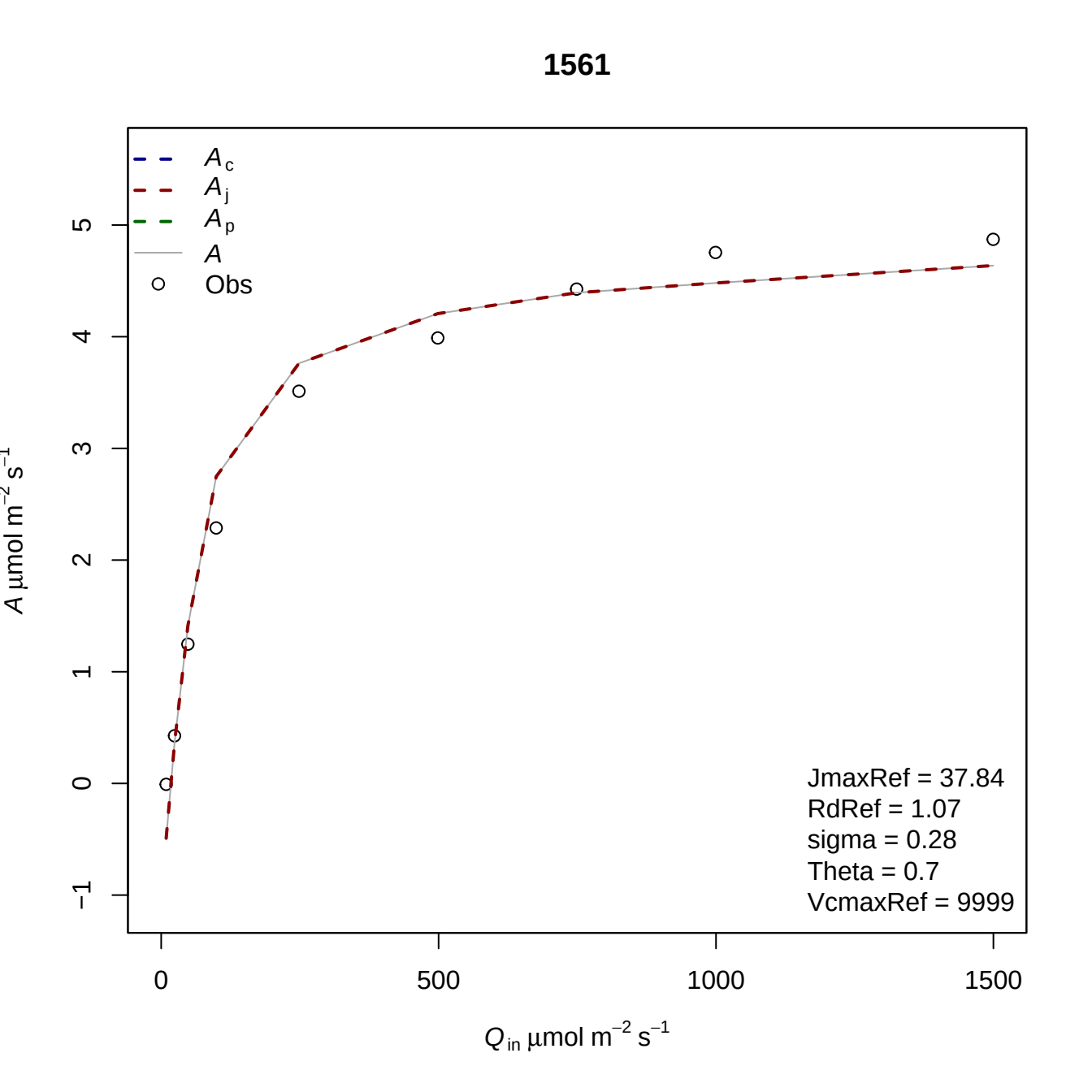

1556
A_c
A_j
A_p
A
Obs
JmaxRef = 69.7
RdRef = 0.95
sigma = 0.37
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

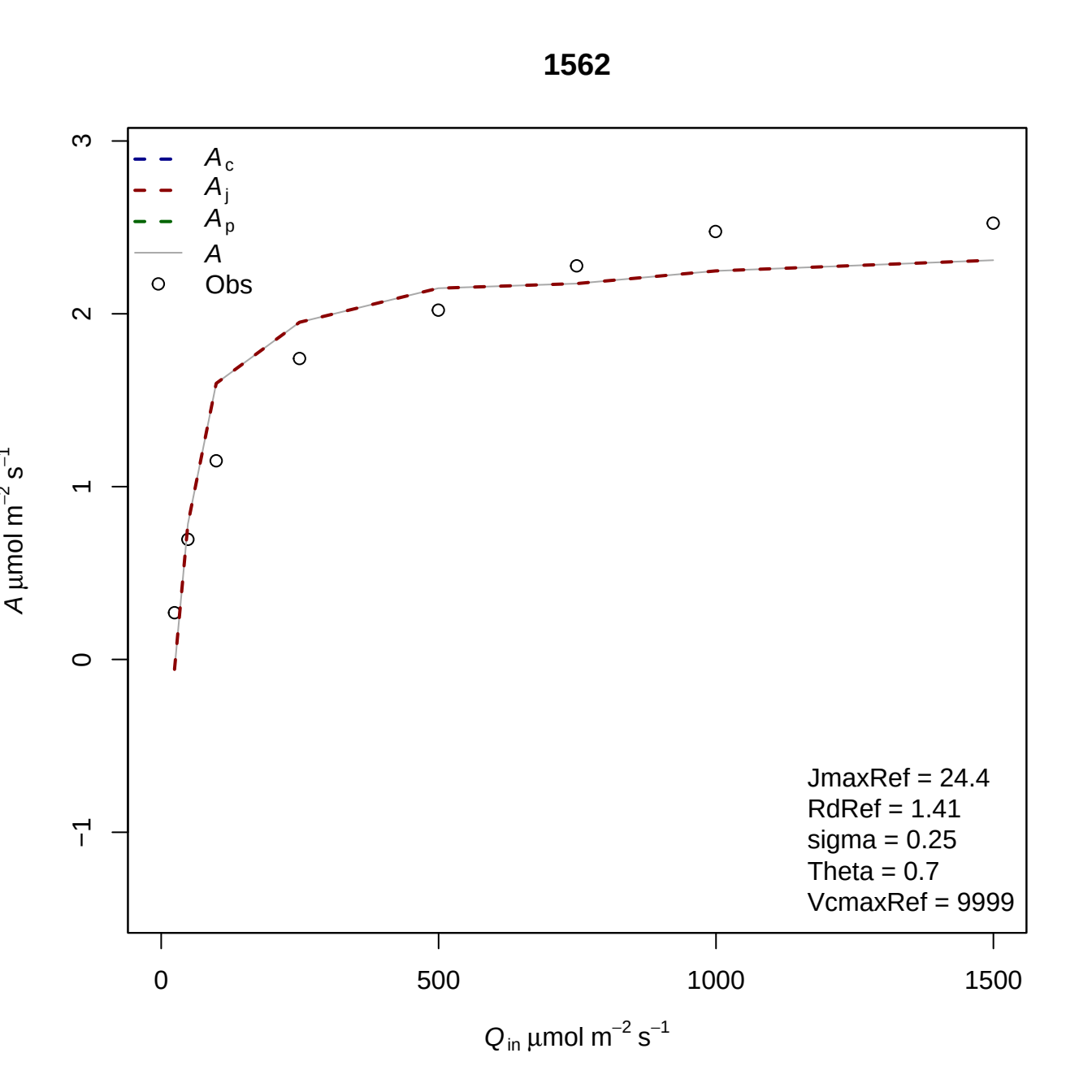

1557
A_c
A_j
A_p
A
Obs
JmaxRef = 62.5
RdRef = 0.97
sigma = 0.34
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

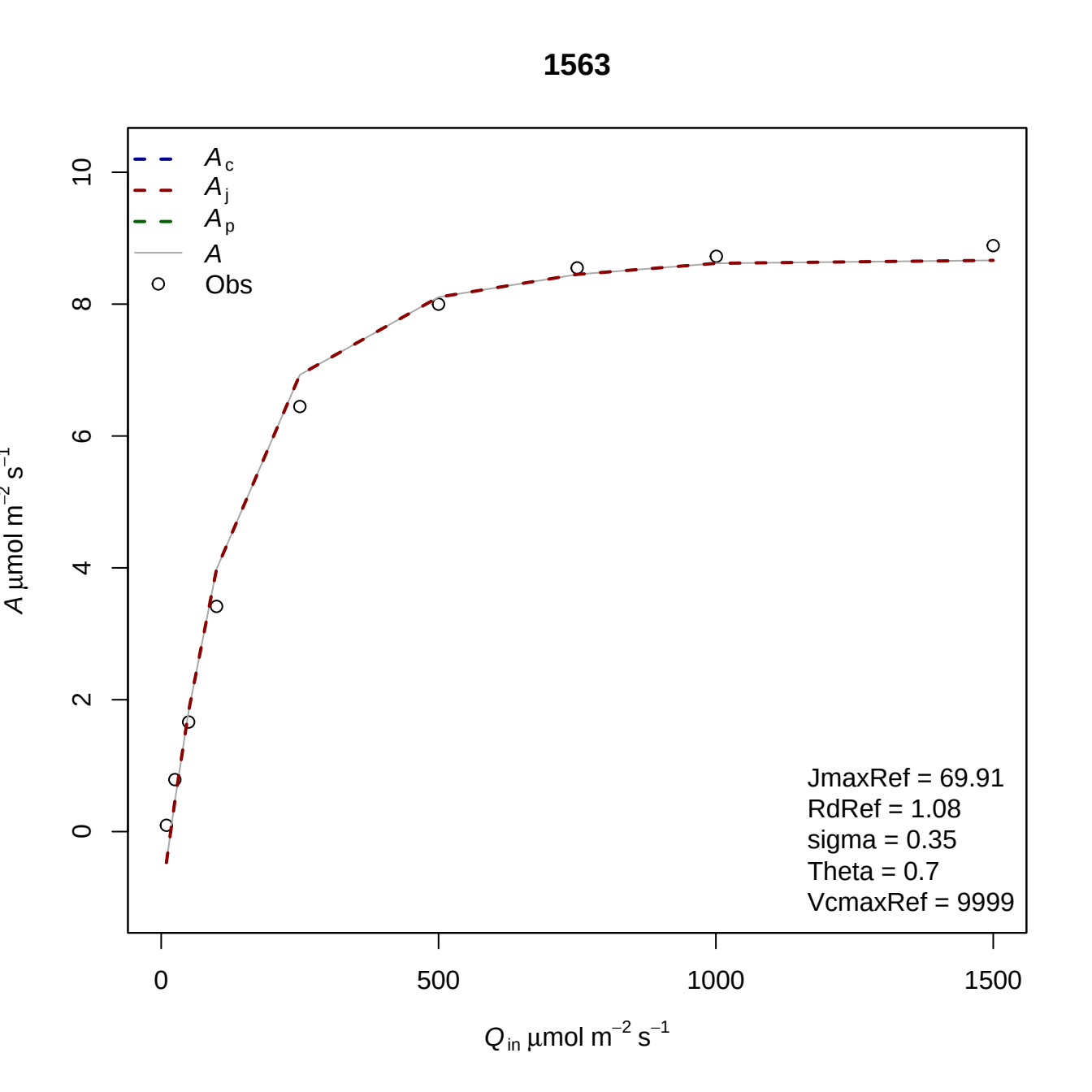

1558
A_c
A_j
A_p
A
Obs
JmaxRef = 65.58
RdRef = 1.13
sigma = 0.34
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

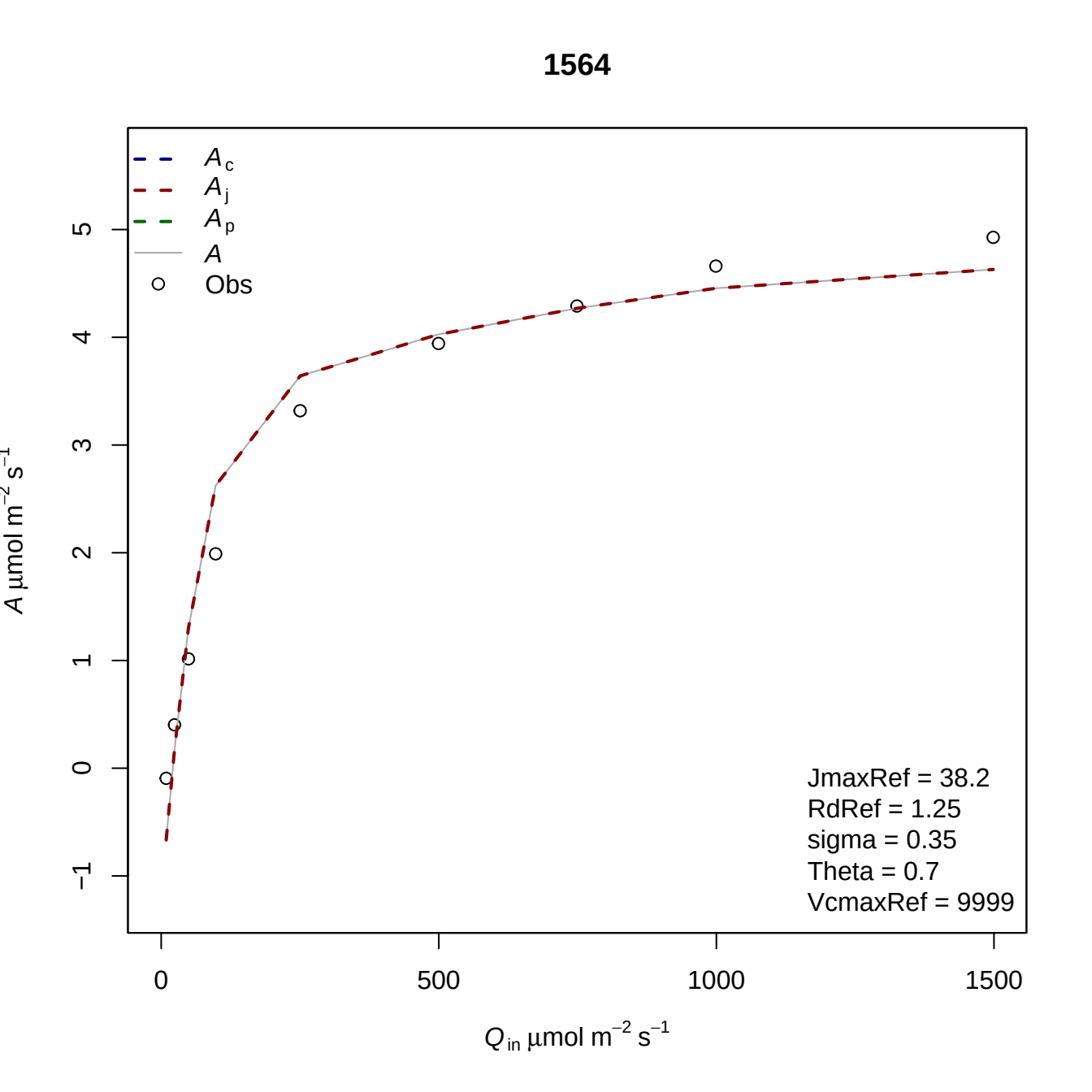

1559
A_c
A_j
A_p
A
Obs
JmaxRef = 57.09
RdRef = 1.08
sigma = 0.28
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

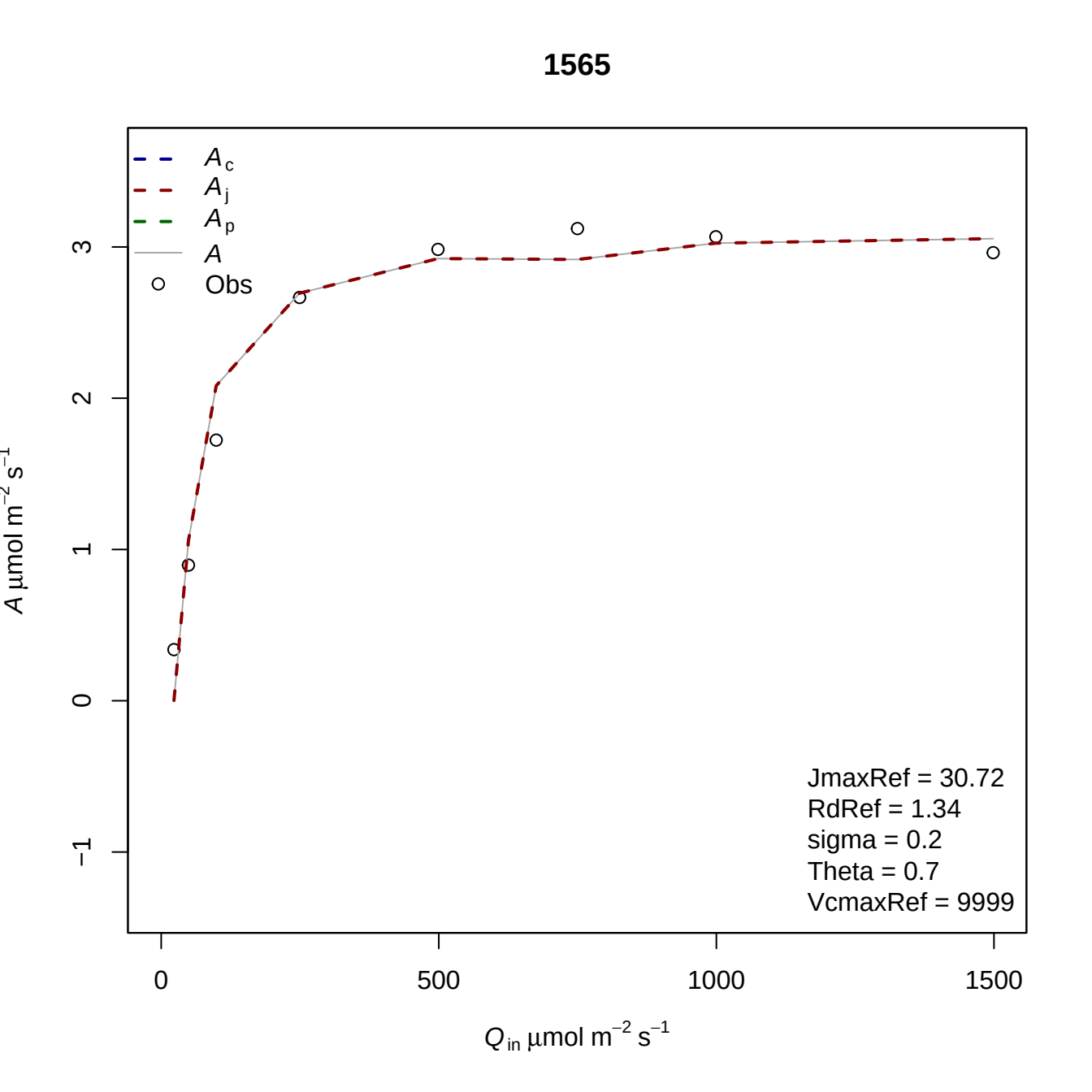

1560
A_c
A_j
A_p
A
Obs
JmaxRef = 33.77
RdRef = 1.37
sigma = 0.28
Theta = 0.7
VcmaxRef = 9999
4
3
2
1
0
−1
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

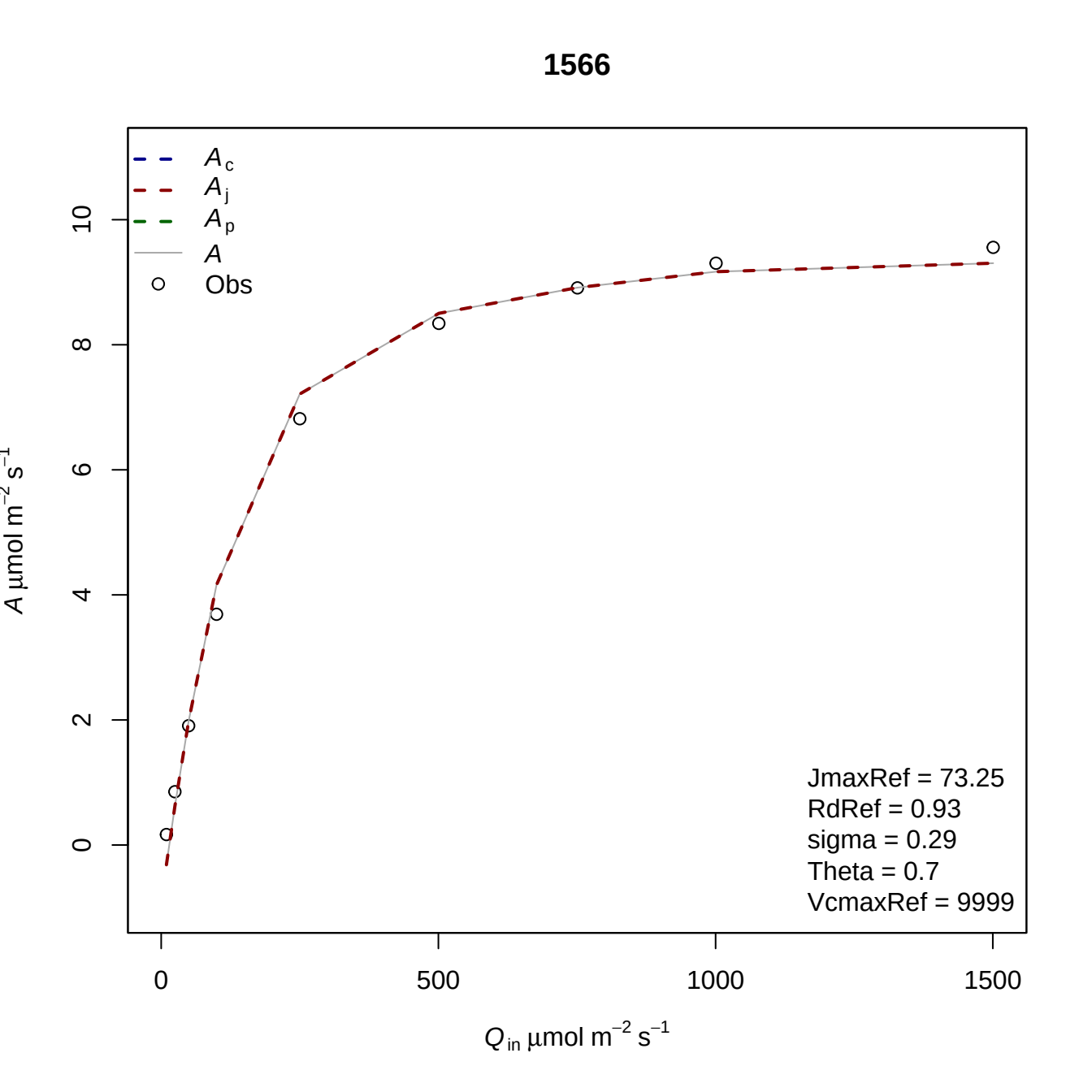

1561
A_c
A_j
A_p
A
Obs
JmaxRef = 37.84
RdRef = 1.07
sigma = 0.28
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

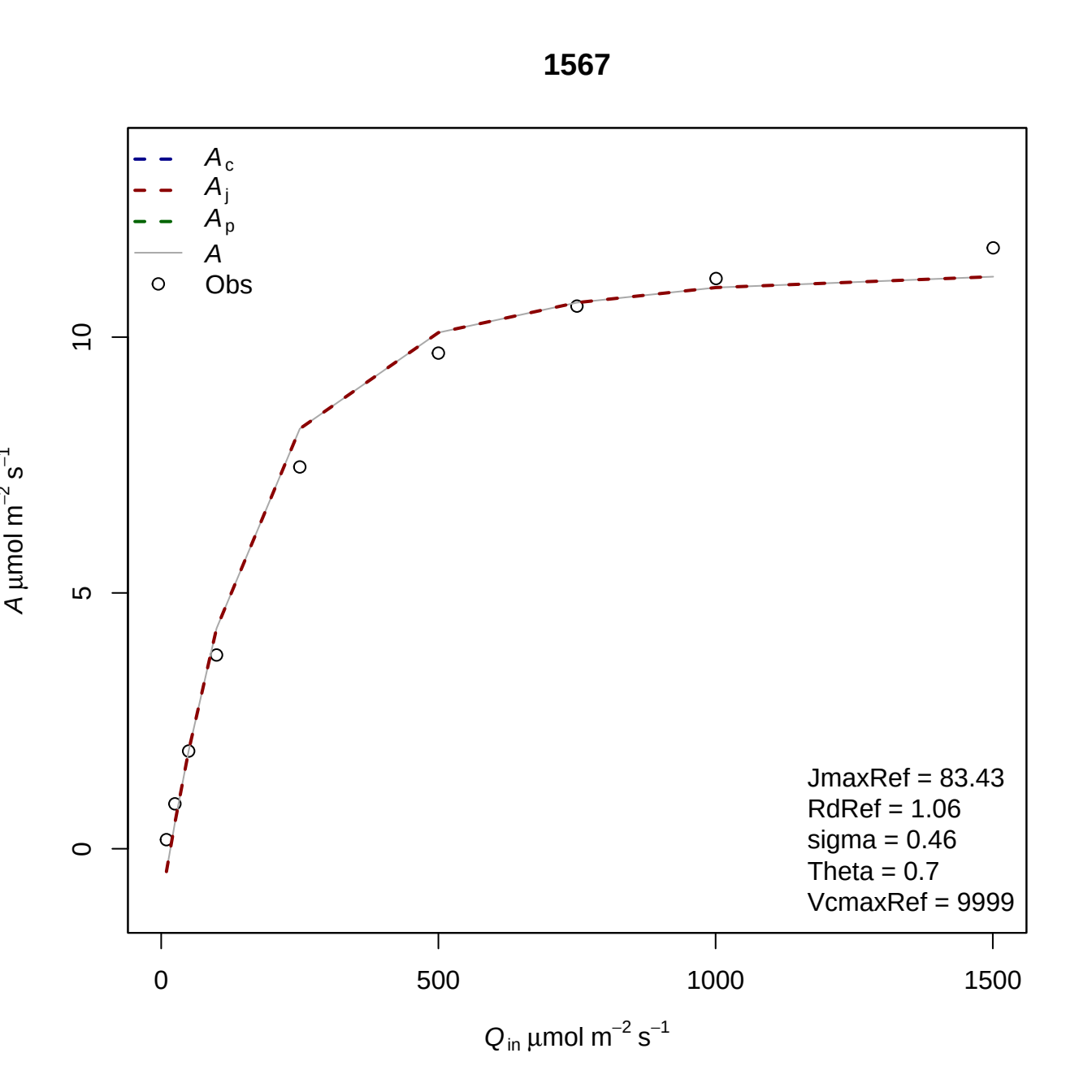

1562
A_c
A_j
A_p
A
Obs
JmaxRef = 24.4
RdRef = 1.41
sigma = 0.25
Theta = 0.7
VcmaxRef = 9999
3
2
1
0
−1
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

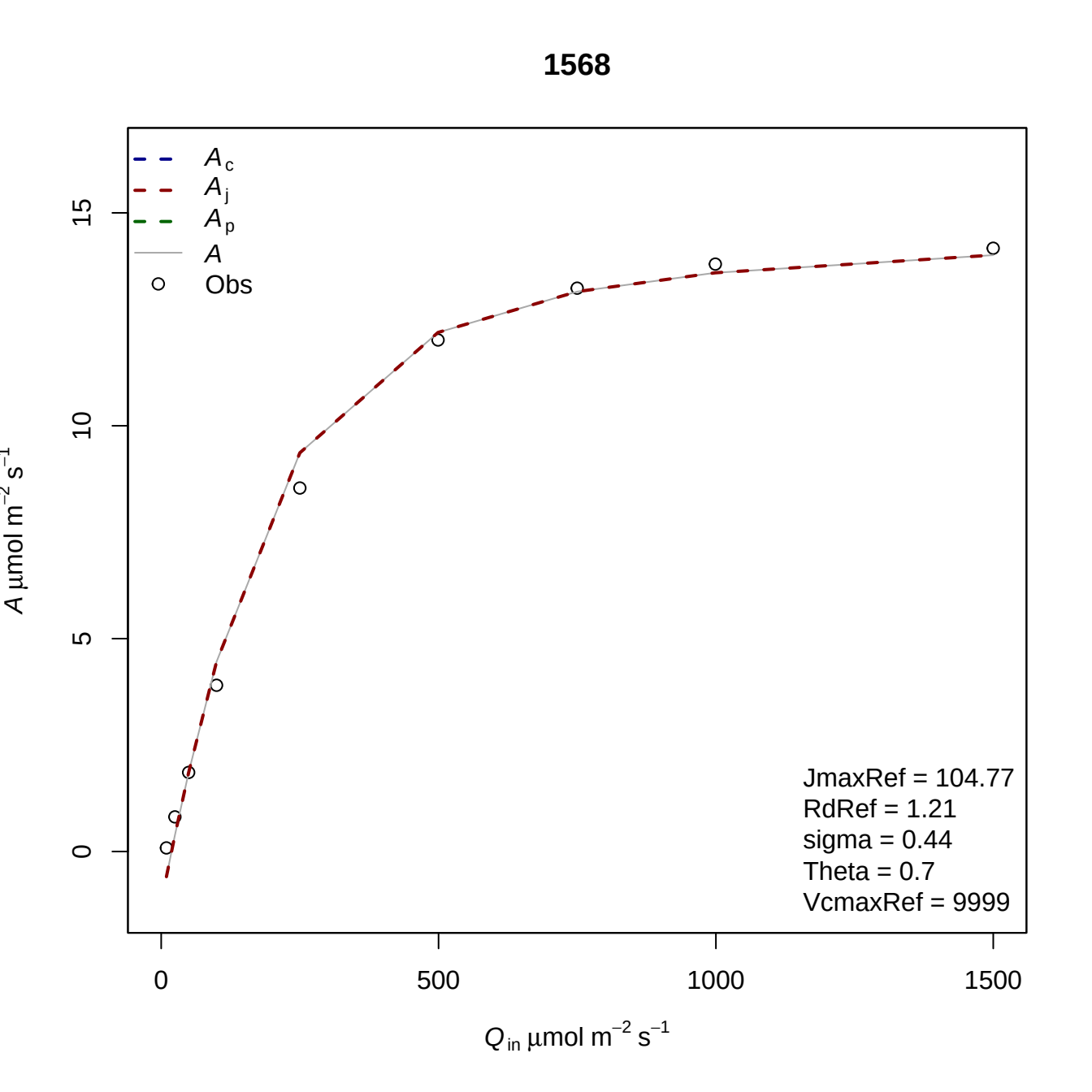

1563
A_c
A_j
A_p
A
Obs
JmaxRef = 69.91
RdRef = 1.08
sigma = 0.35
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

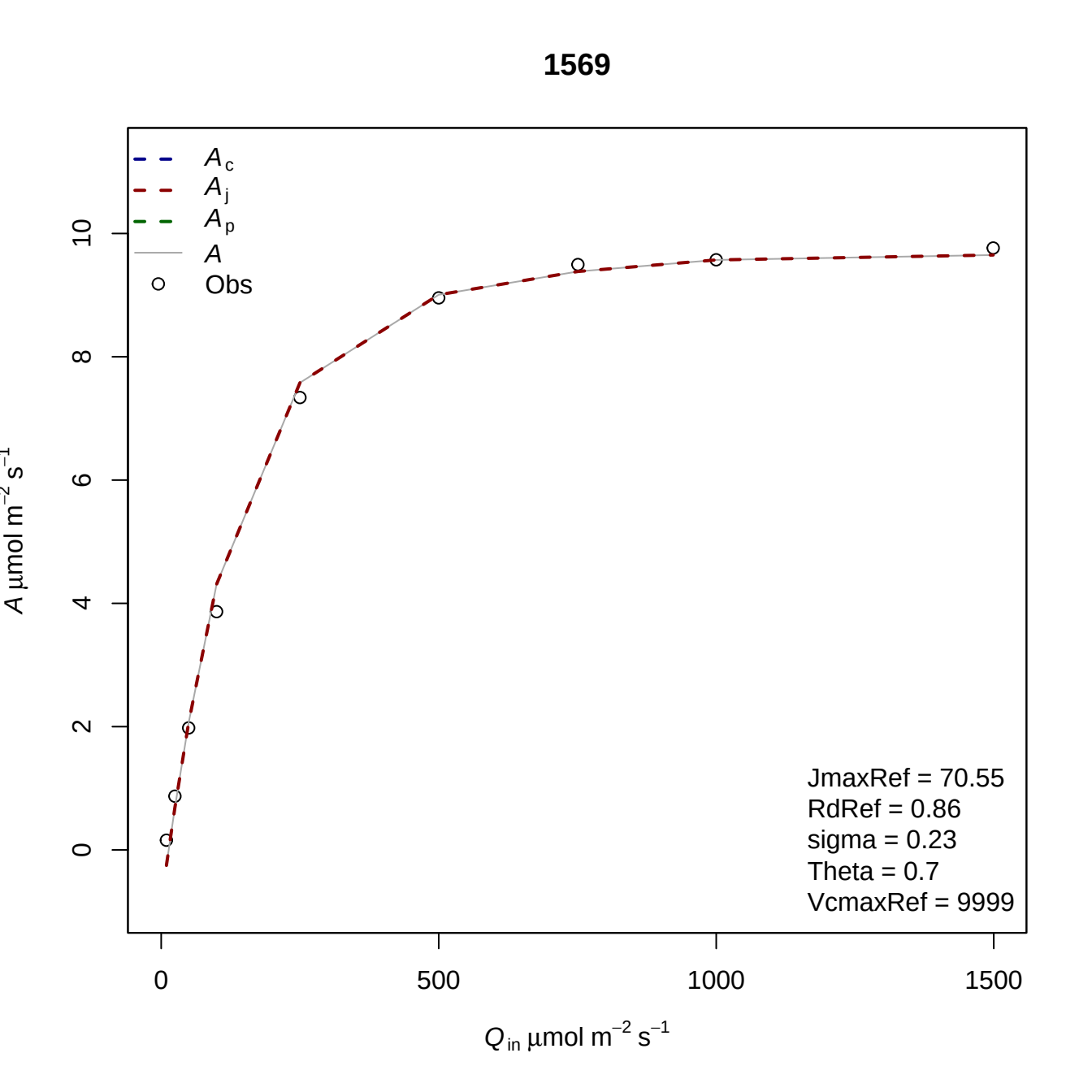

1564
A_c
A_j
A_p
A
Obs
JmaxRef = 38.2
RdRef = 1.25
sigma = 0.35
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

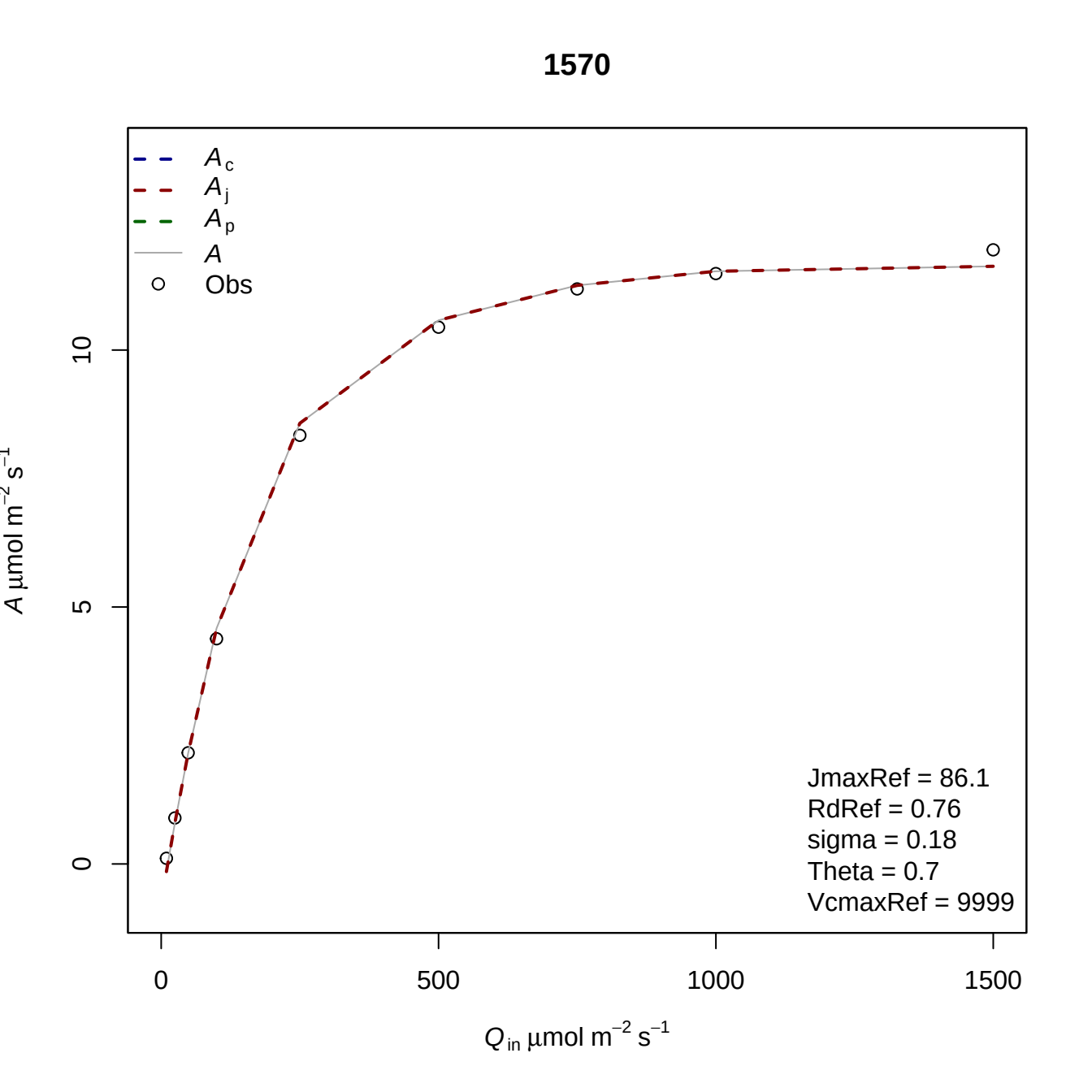

1565
A_c
A_j
A_p
A
Obs
JmaxRef = 30.72
RdRef = 1.34
sigma = 0.2
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
0
500
1000
1500

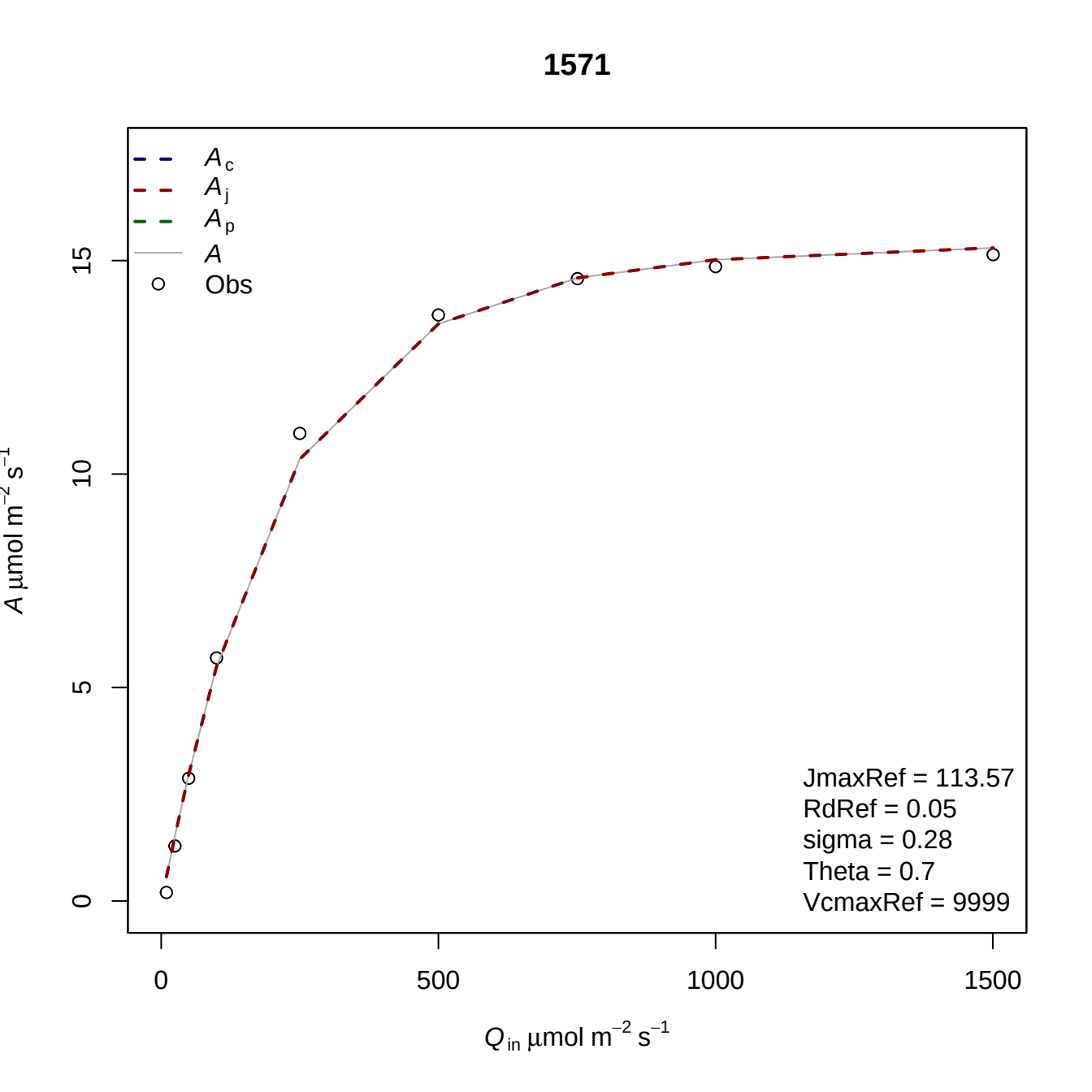

1566
A_c
A_j
A_p
A
Obs
JmaxRef = 73.25
RdRef = 0.93
sigma = 0.29
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

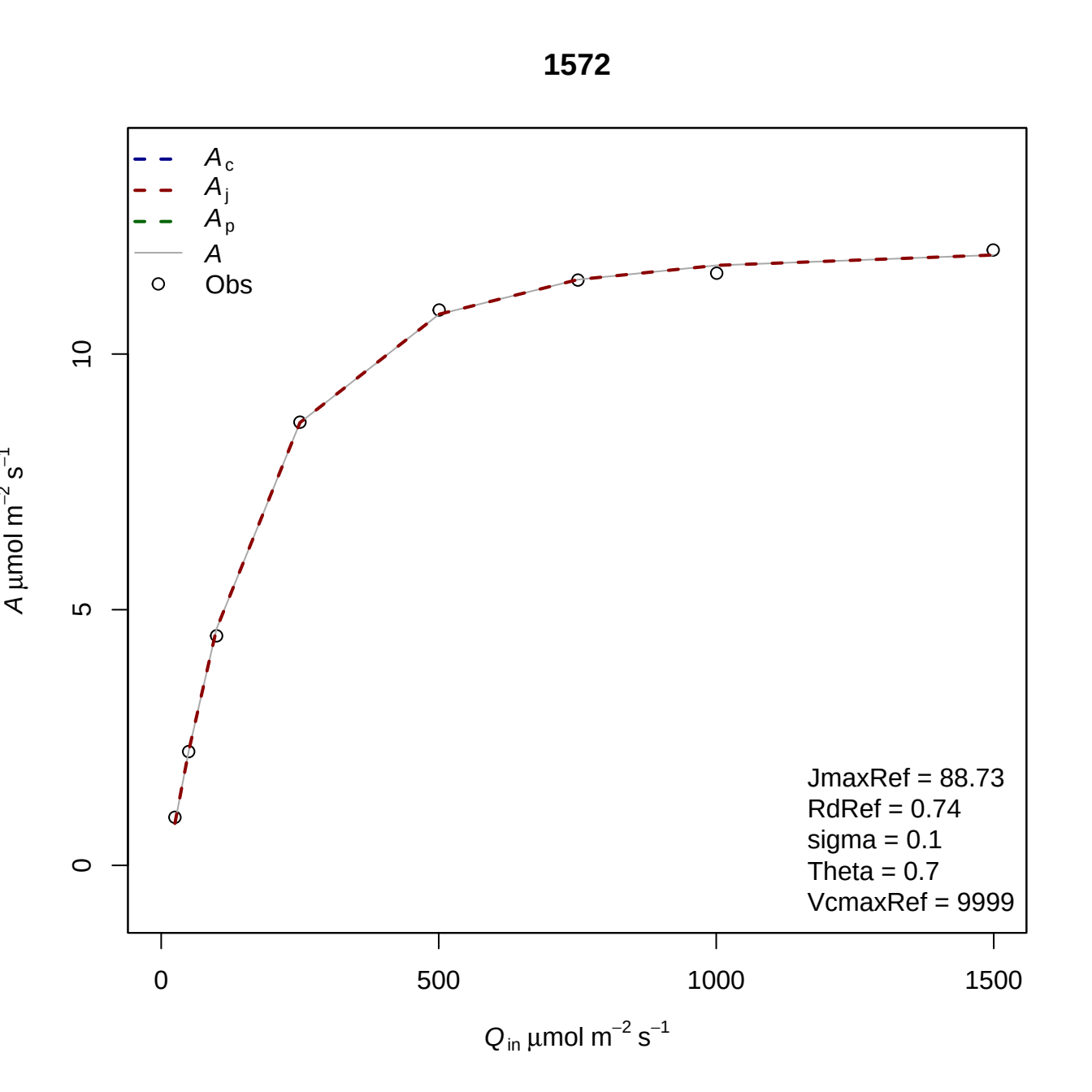

1567
A_c
A_j
A_p
A
Obs
JmaxRef = 83.43
RdRef = 1.06
sigma = 0.46
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

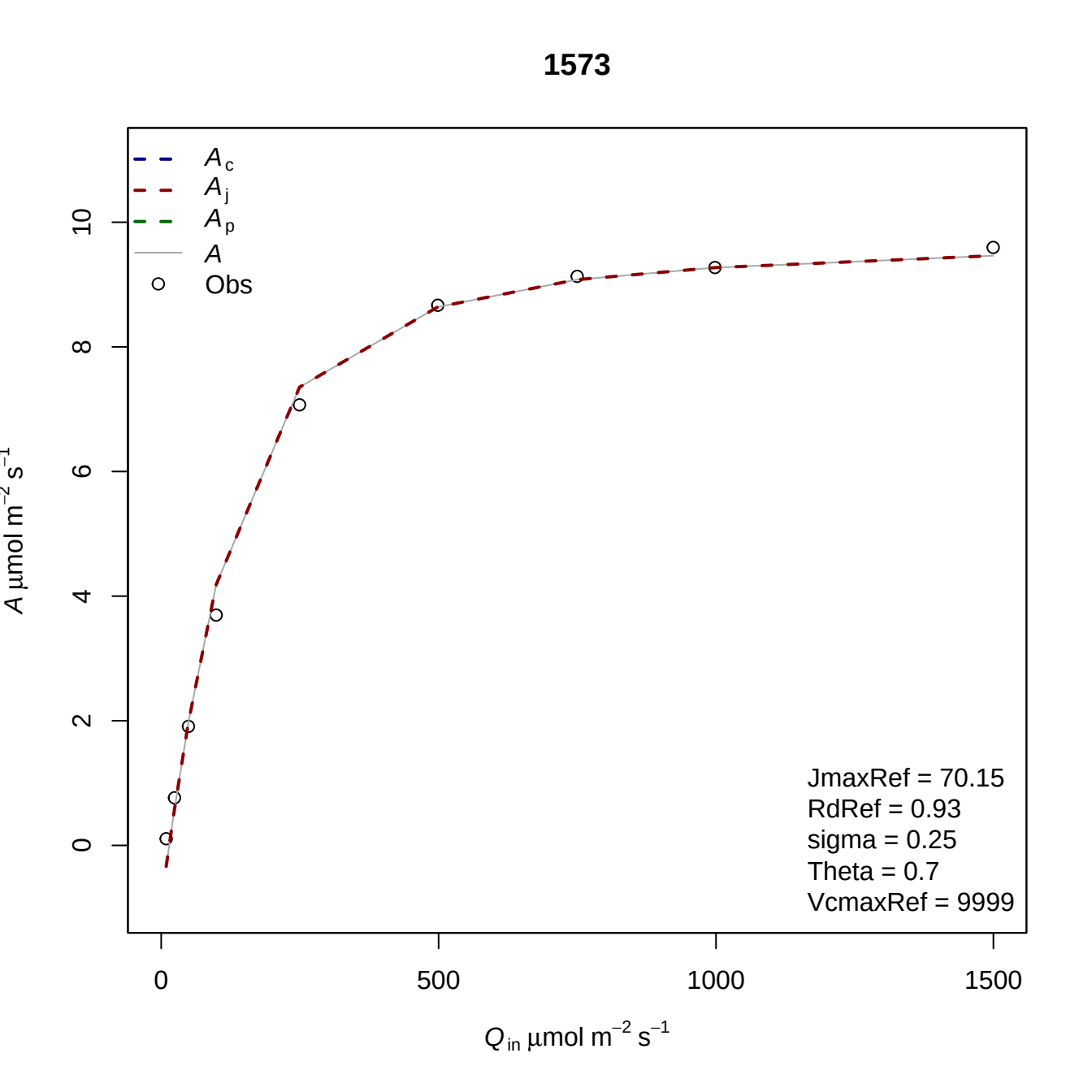

1568
A_c
A_j
A_p
A
Obs
JmaxRef = 104.77
RdRef = 1.21
sigma = 0.44
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

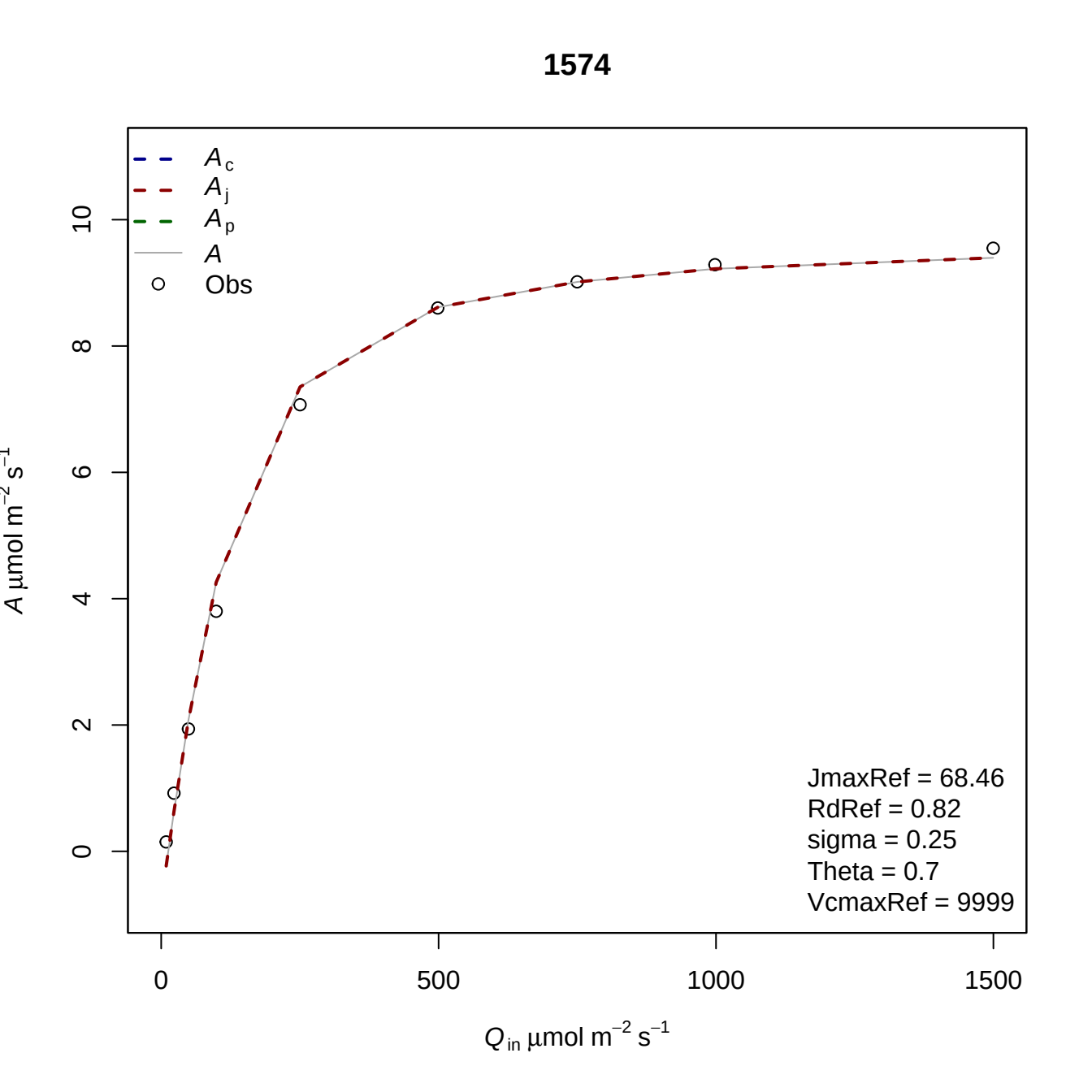

1569
A_c
A_j
A_p
A
Obs
JmaxRef = 70.55
RdRef = 0.86
sigma = 0.23
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

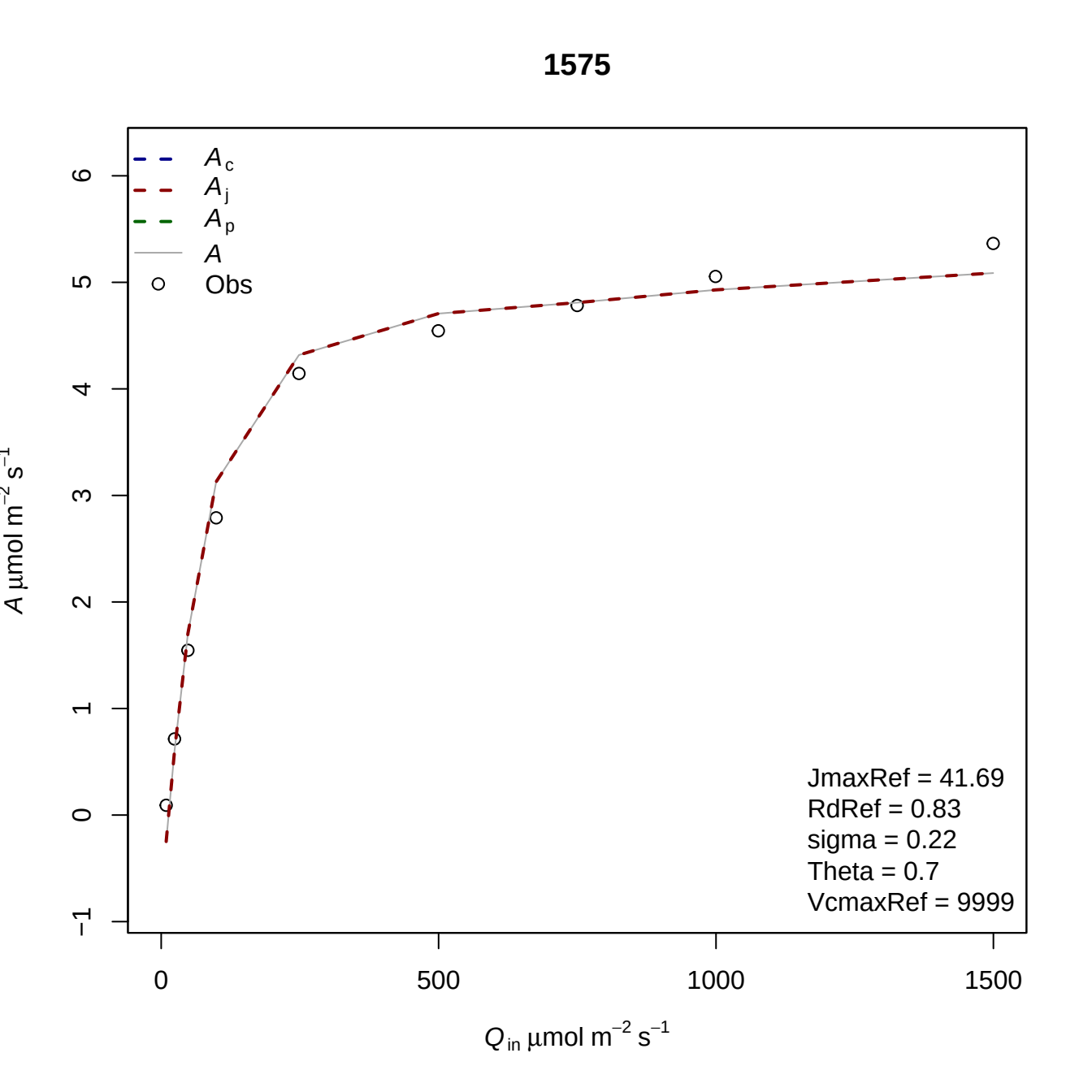

1570
A_c
A_j
A_p
A
Obs
JmaxRef = 86.1
RdRef = 0.76
sigma = 0.18
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

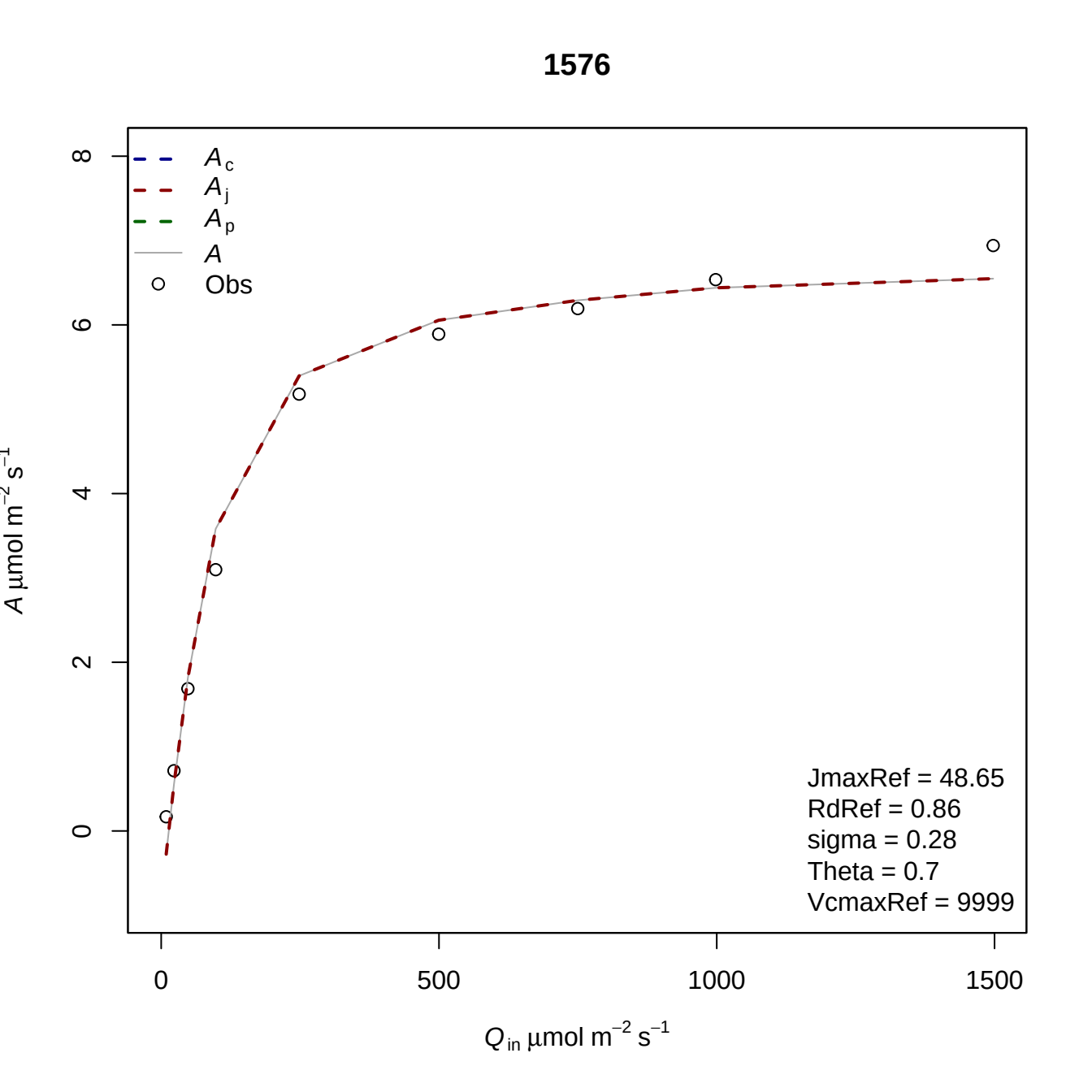

1571
A_c
A_j
A_p
A
Obs
JmaxRef = 113.57
RdRef = 0.05
sigma = 0.28
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

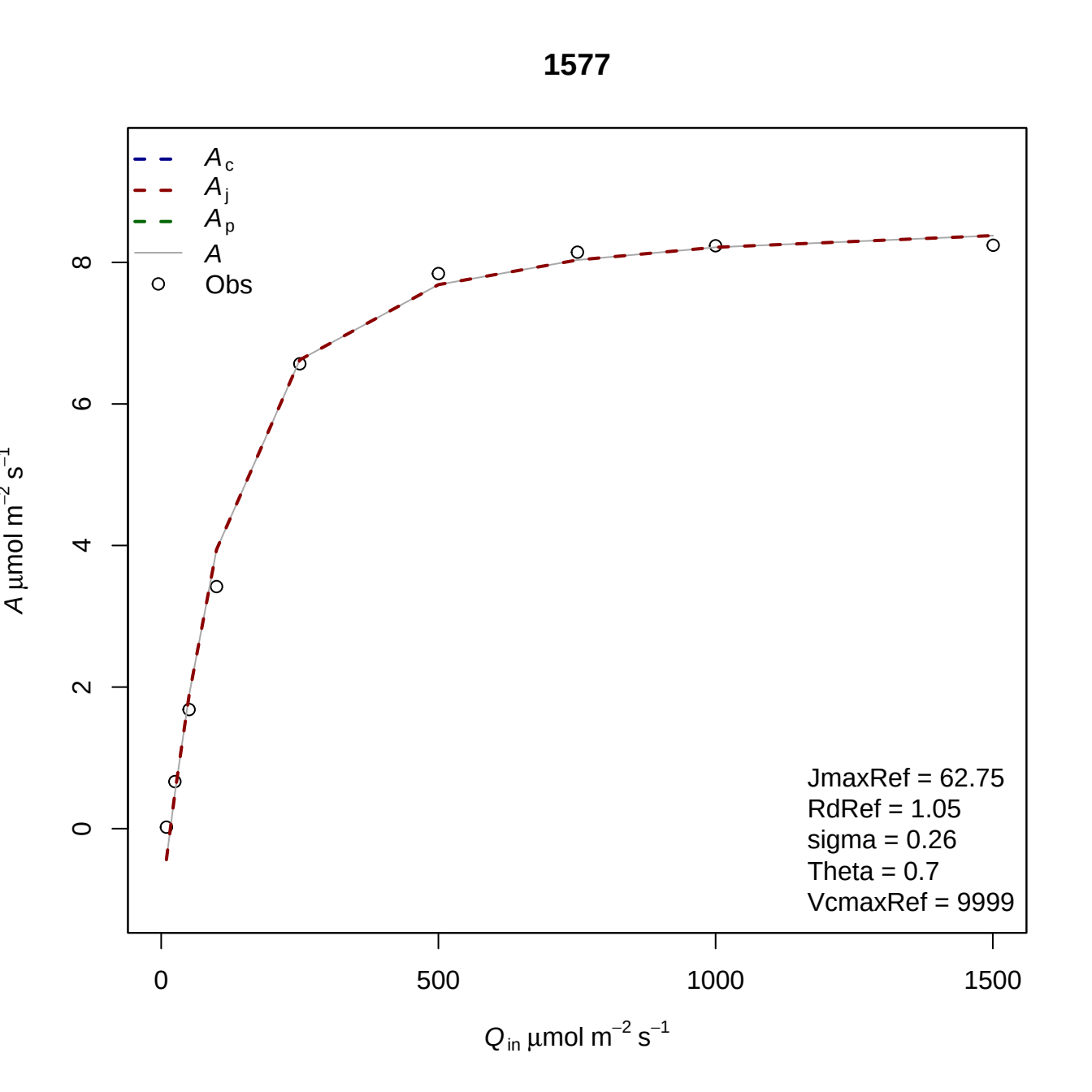

1572
A_c
A_j
A_p
A
Obs
JmaxRef = 88.73
RdRef = 0.74
sigma = 0.1
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

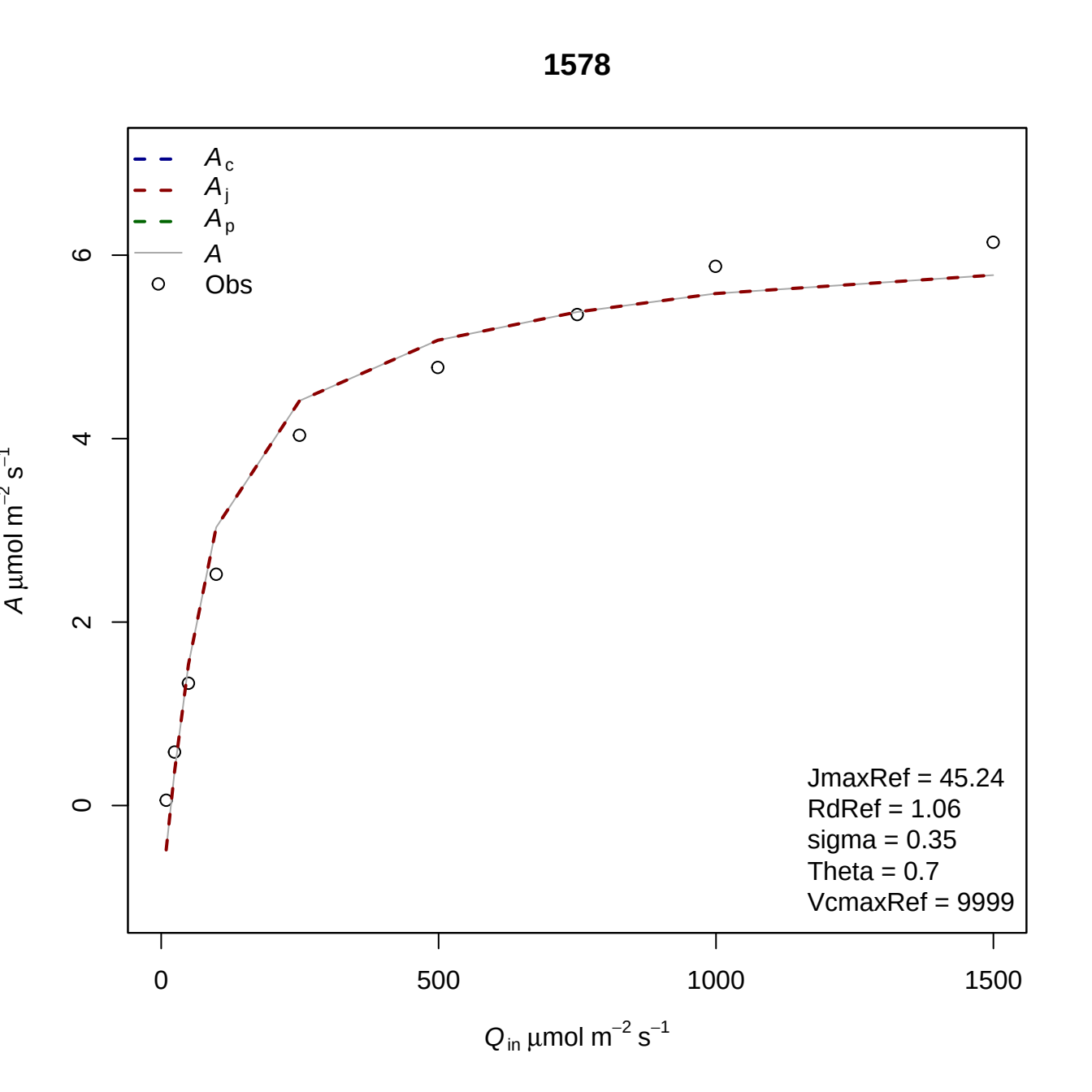

1573
A_c
A_j
A_p
A
Obs
JmaxRef = 70.15
RdRef = 0.93
sigma = 0.25
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

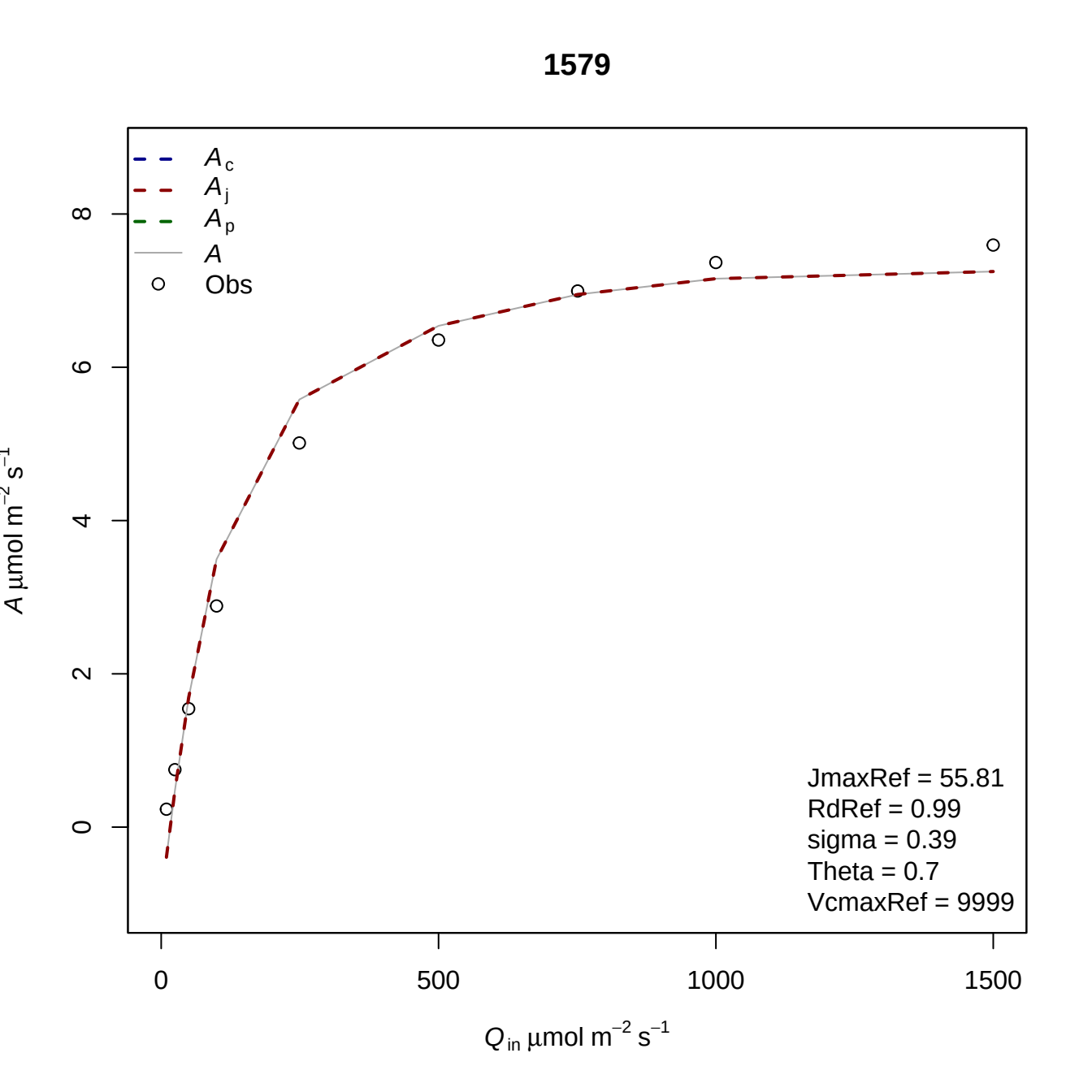

1574
A_c
A_j
A_p
A
Obs
JmaxRef = 68.46
RdRef = 0.82
sigma = 0.25
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

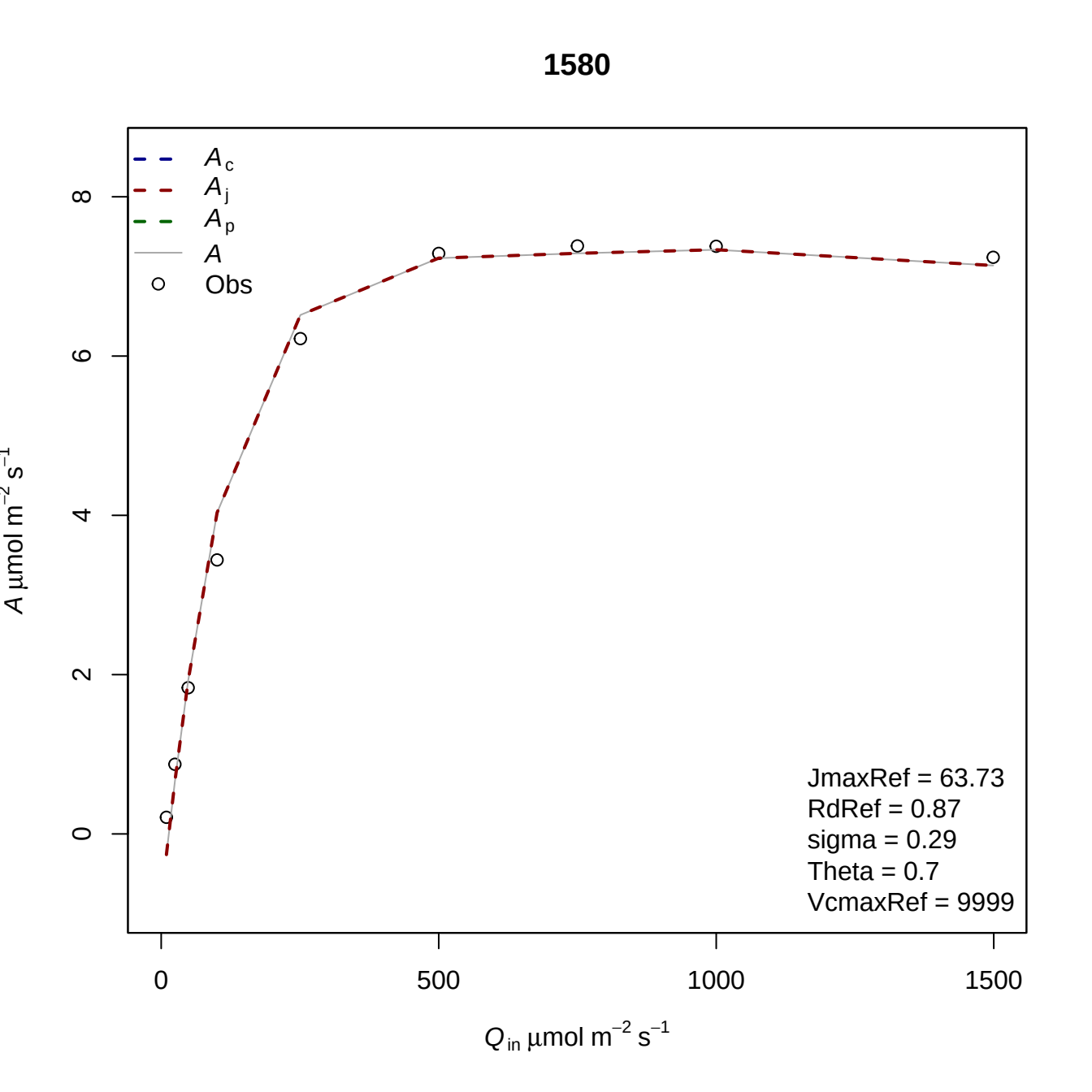

1575
A_c
A_j
A_p
A
Obs
JmaxRef = 41.69
RdRef = 0.83
sigma = 0.22
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
6
0
500
1000
1500

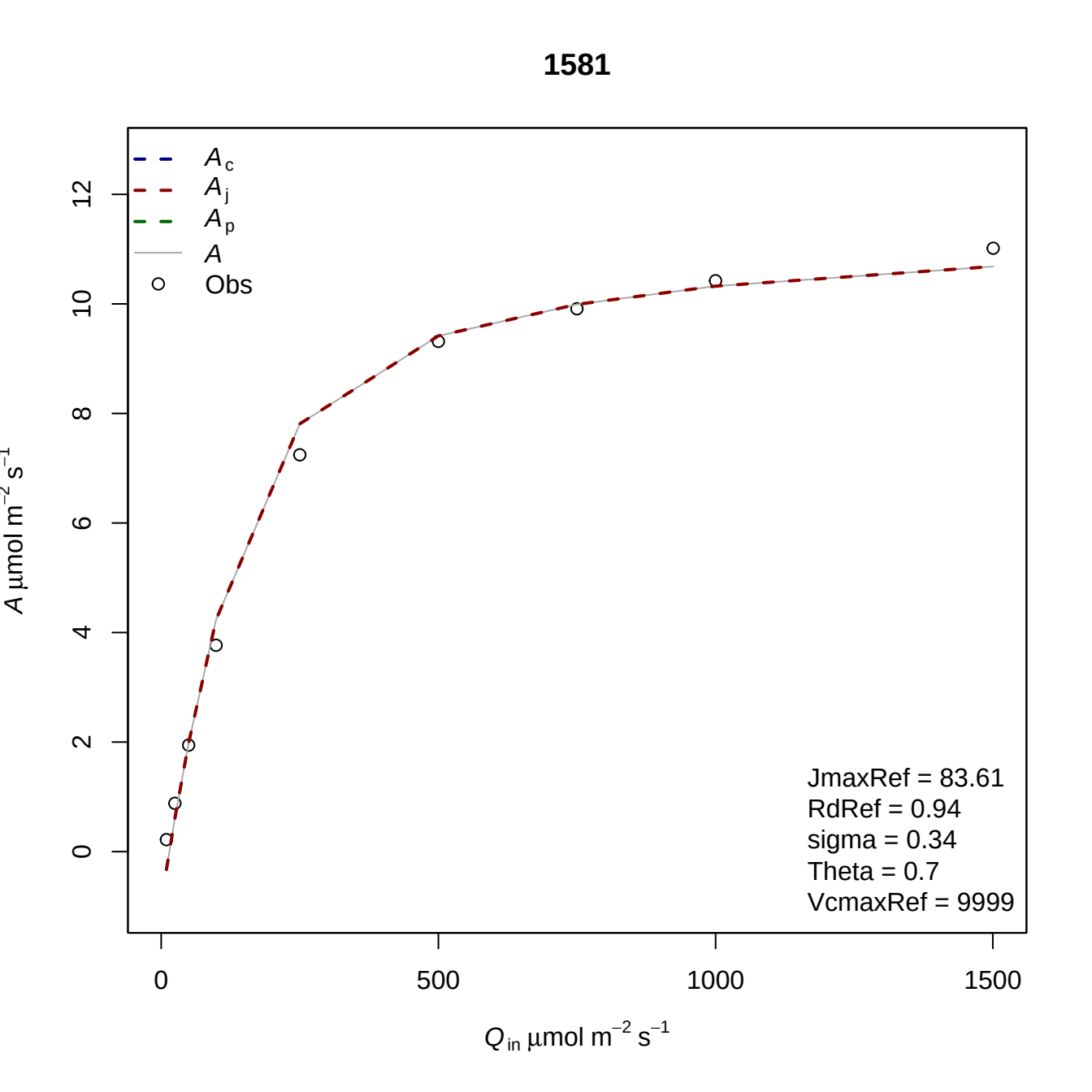

1576
A_c
A_j
A_p
A
Obs
JmaxRef = 48.65
RdRef = 0.86
sigma = 0.28
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

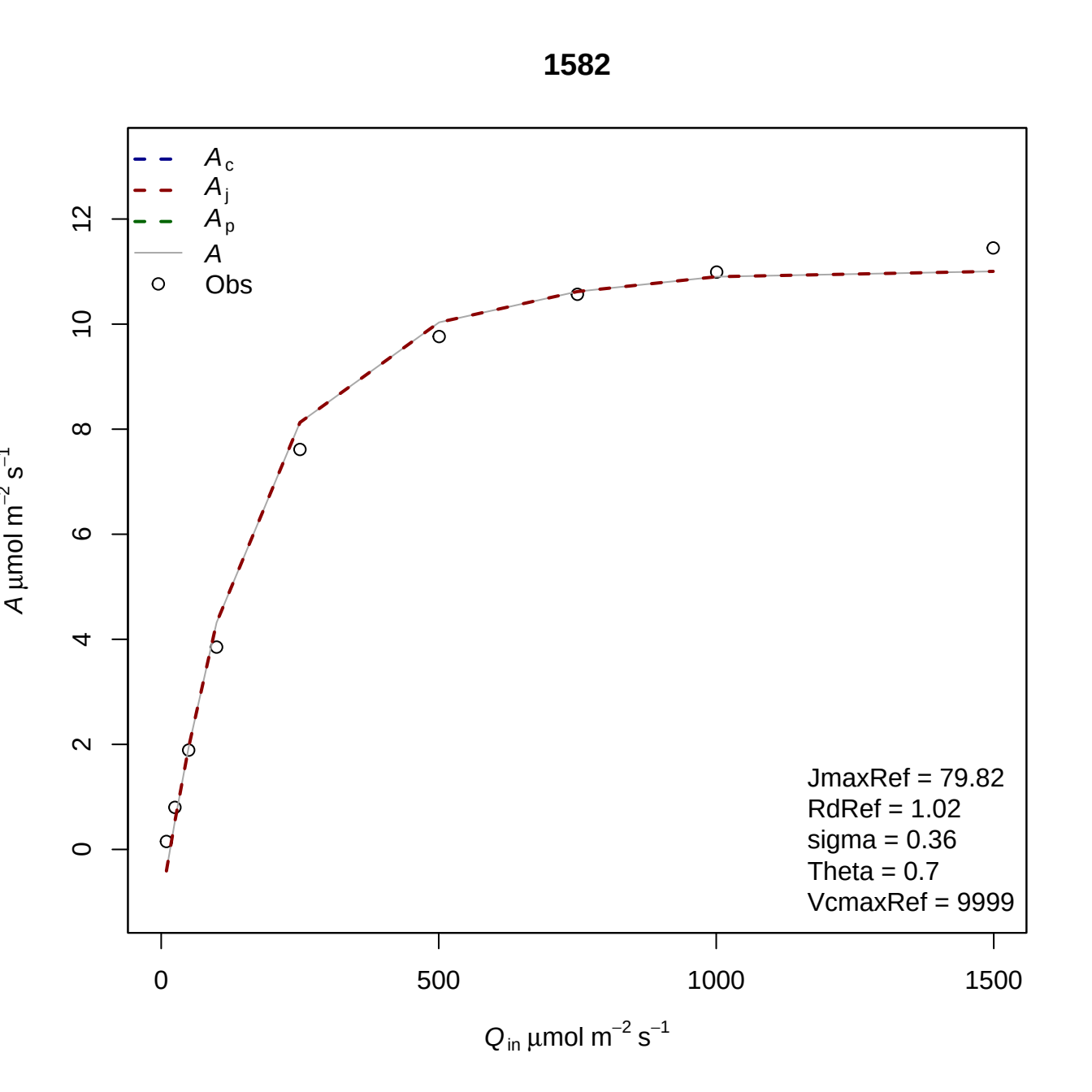

1577
A_c
A_j
A_p
A
Obs
JmaxRef = 62.75
RdRef = 1.05
sigma = 0.26
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

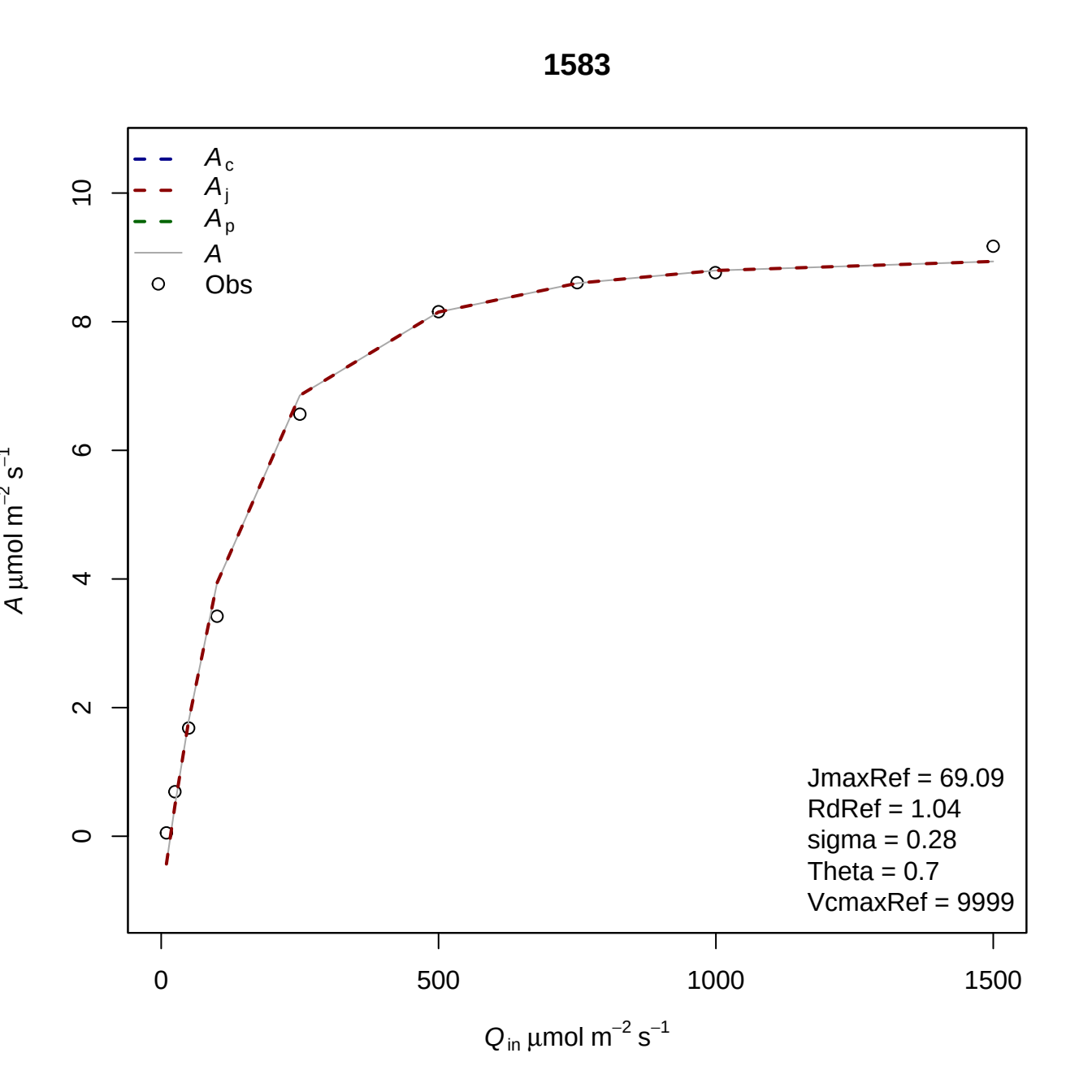

1578
A_c
A_j
A_p
A
Obs
JmaxRef = 45.24
RdRef = 1.06
sigma = 0.35
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

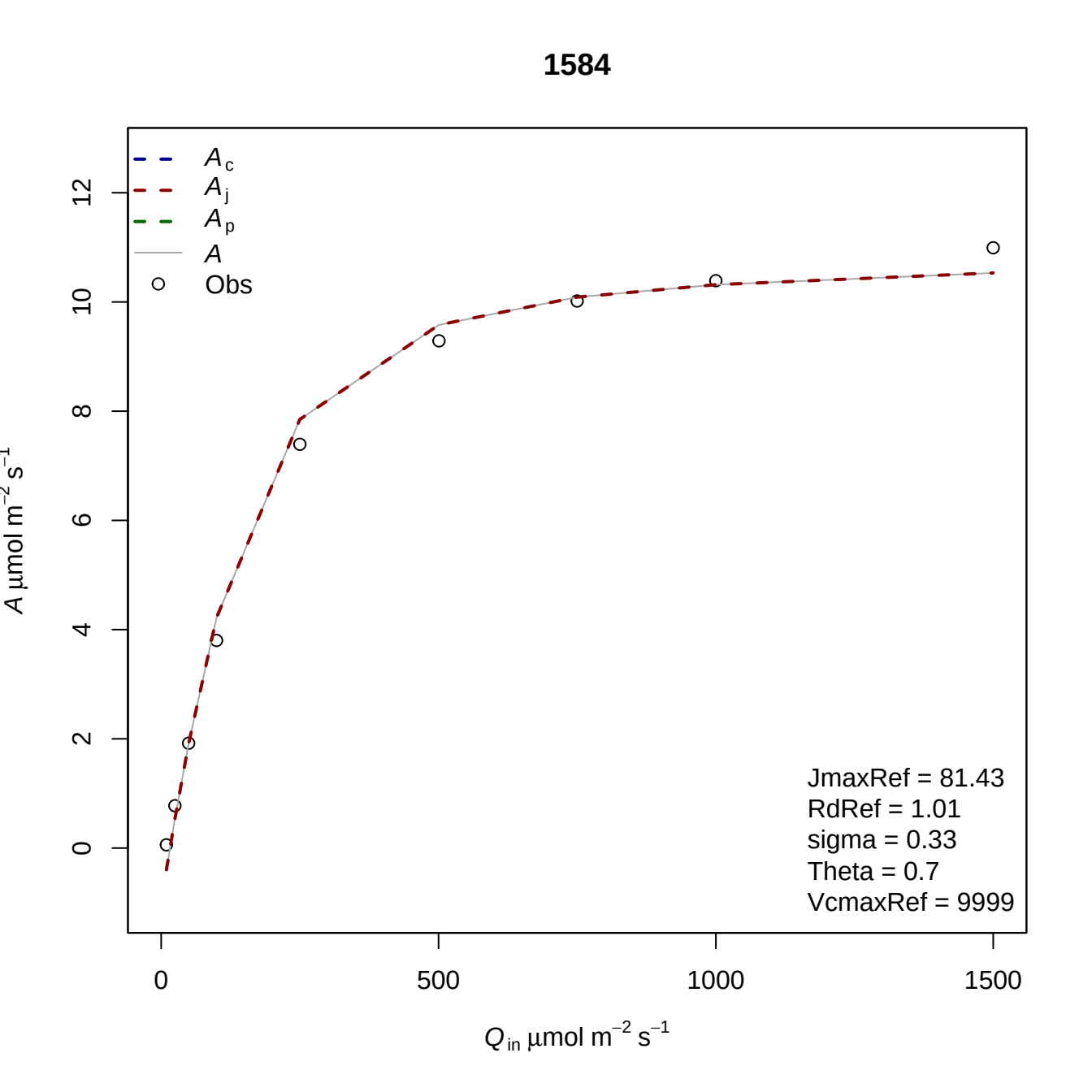

1579
A_c
A_j
A_p
A
Obs
JmaxRef = 55.81
RdRef = 0.99
sigma = 0.39
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

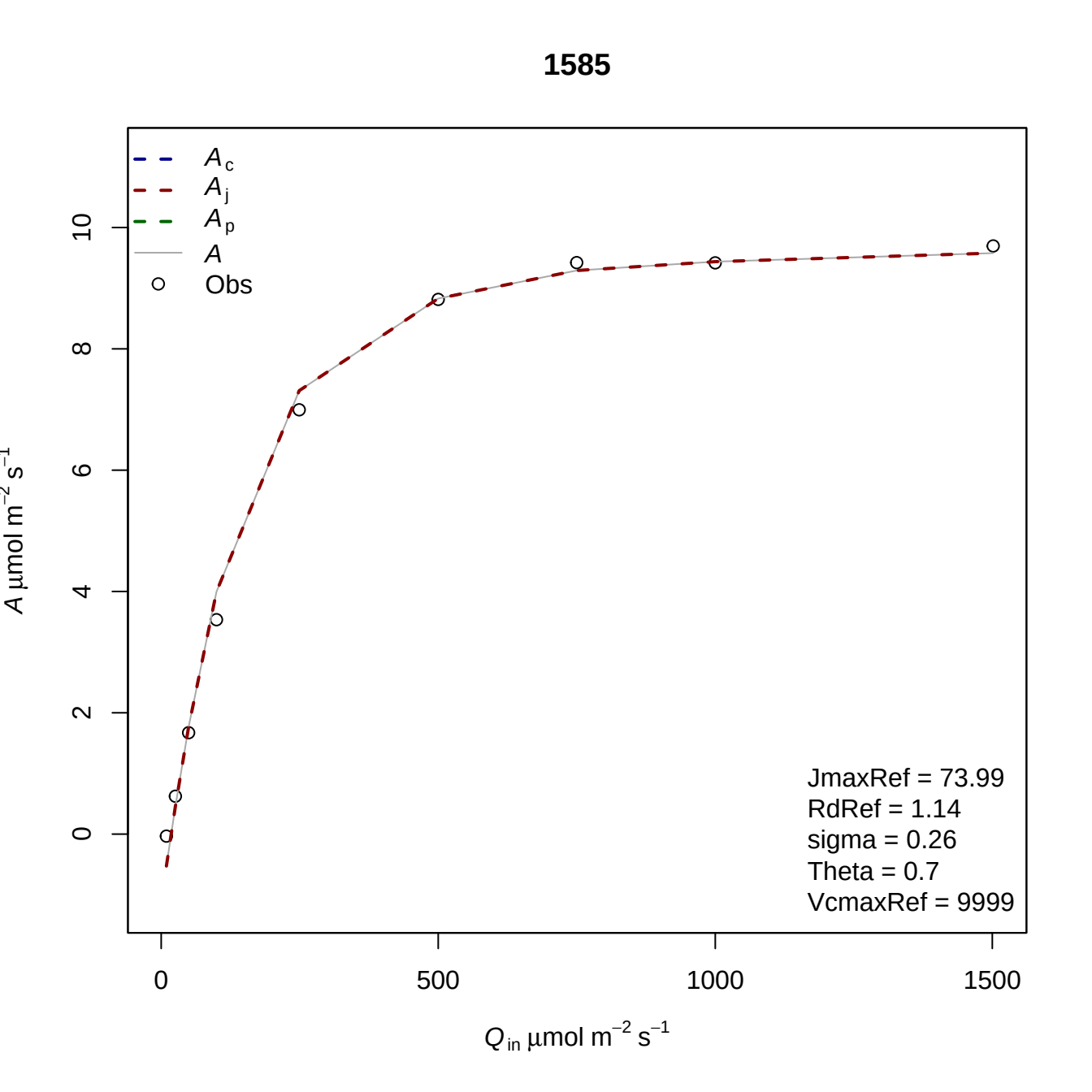

1580
A_c
A_j
A_p
A
Obs
JmaxRef = 63.73
RdRef = 0.87
sigma = 0.29
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

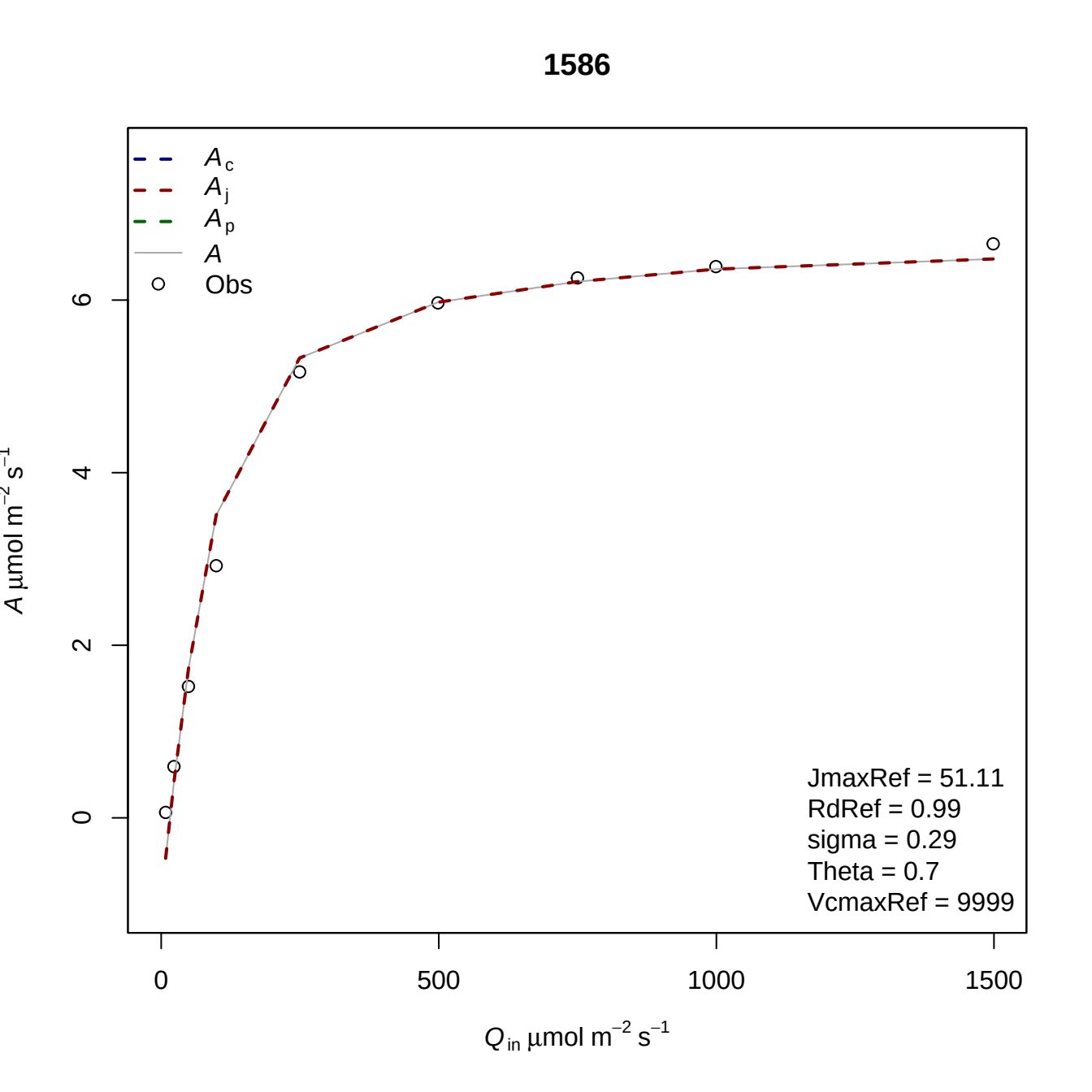

1581
A_c
A_j
A_p
A
Obs
JmaxRef = 83.61
RdRef = 0.94
sigma = 0.34
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

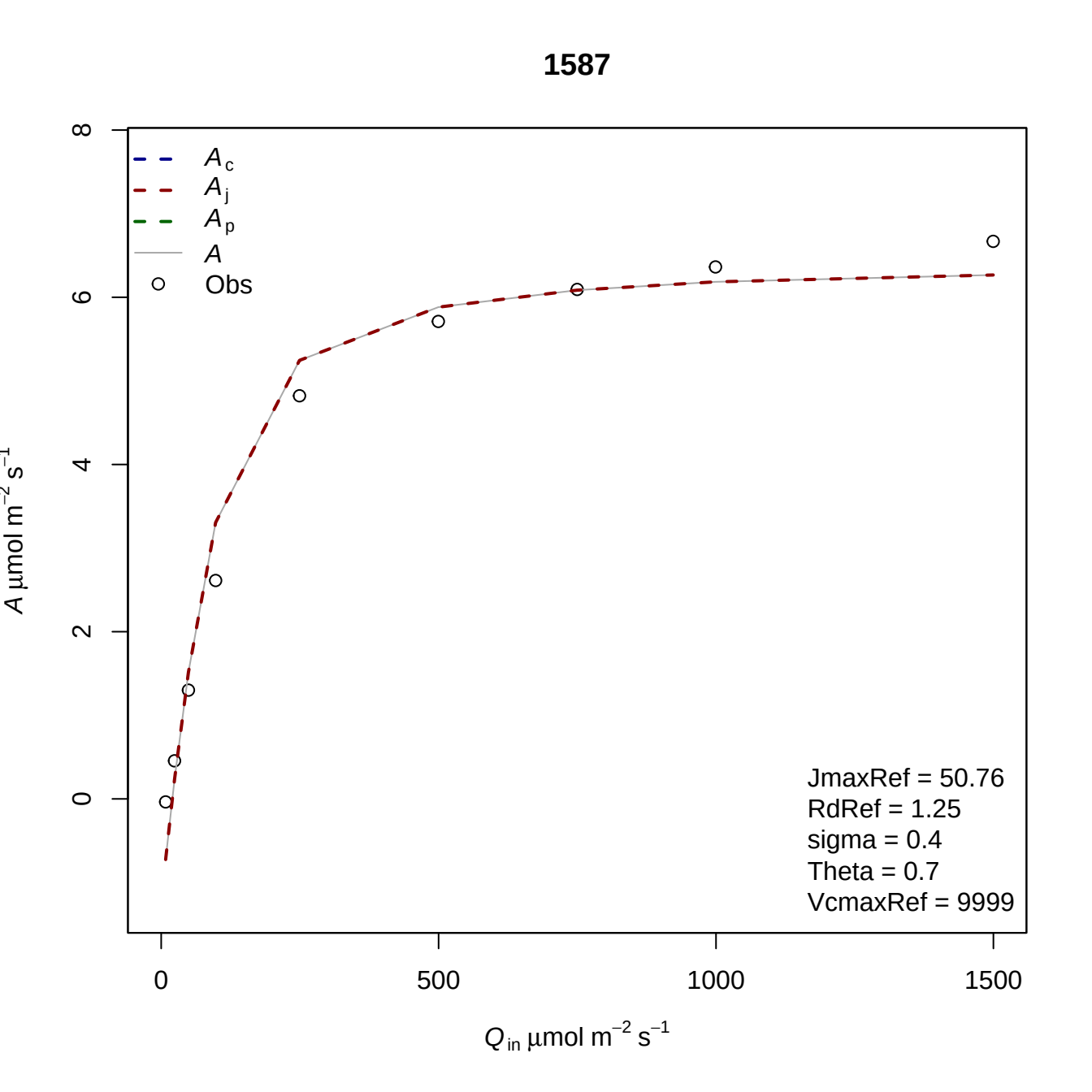

1582
A_c
A_j
A_p
A
Obs
JmaxRef = 79.82
RdRef = 1.02
sigma = 0.36
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

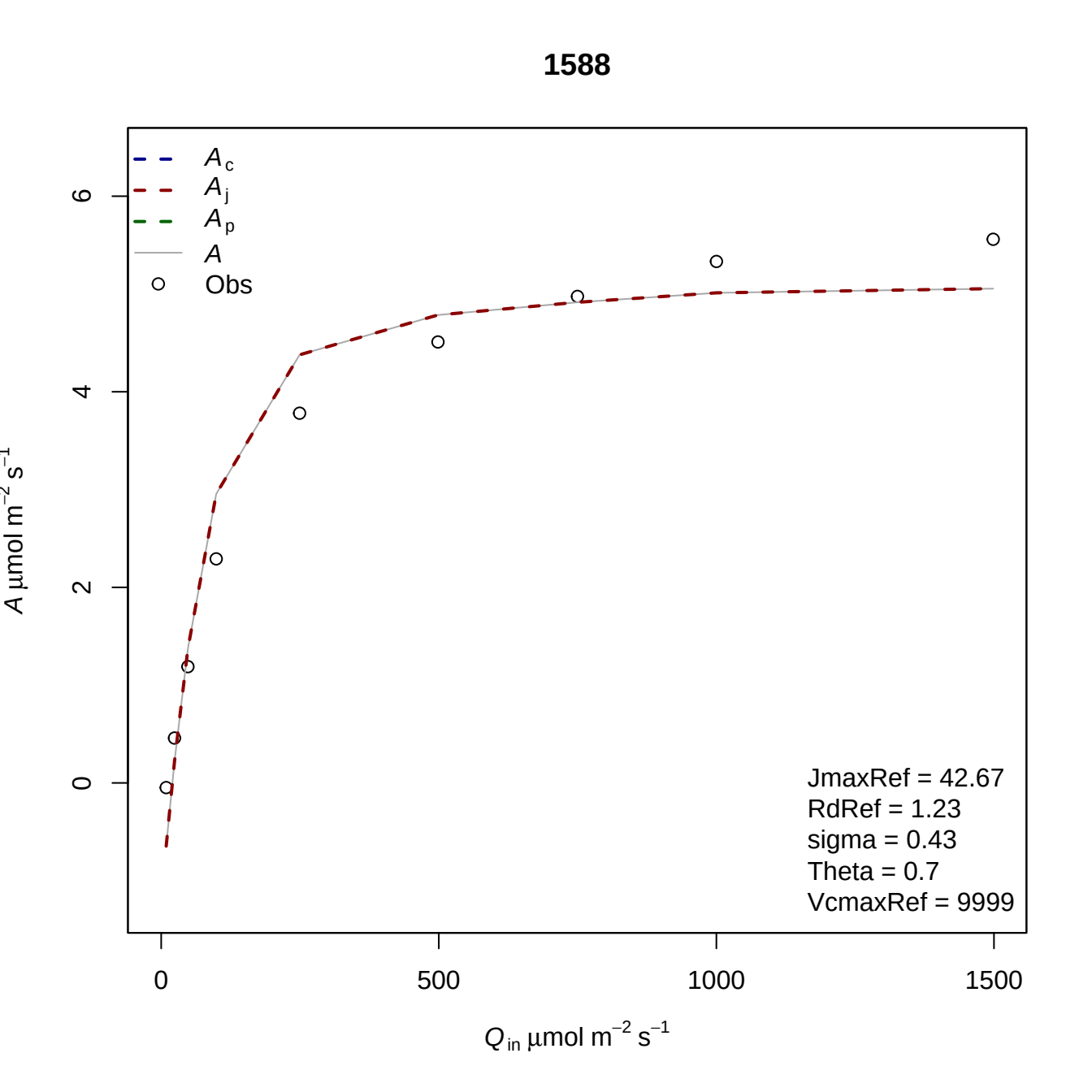

1583
A_c
A_j
A_p
A
Obs
JmaxRef = 69.09
RdRef = 1.04
sigma = 0.28
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

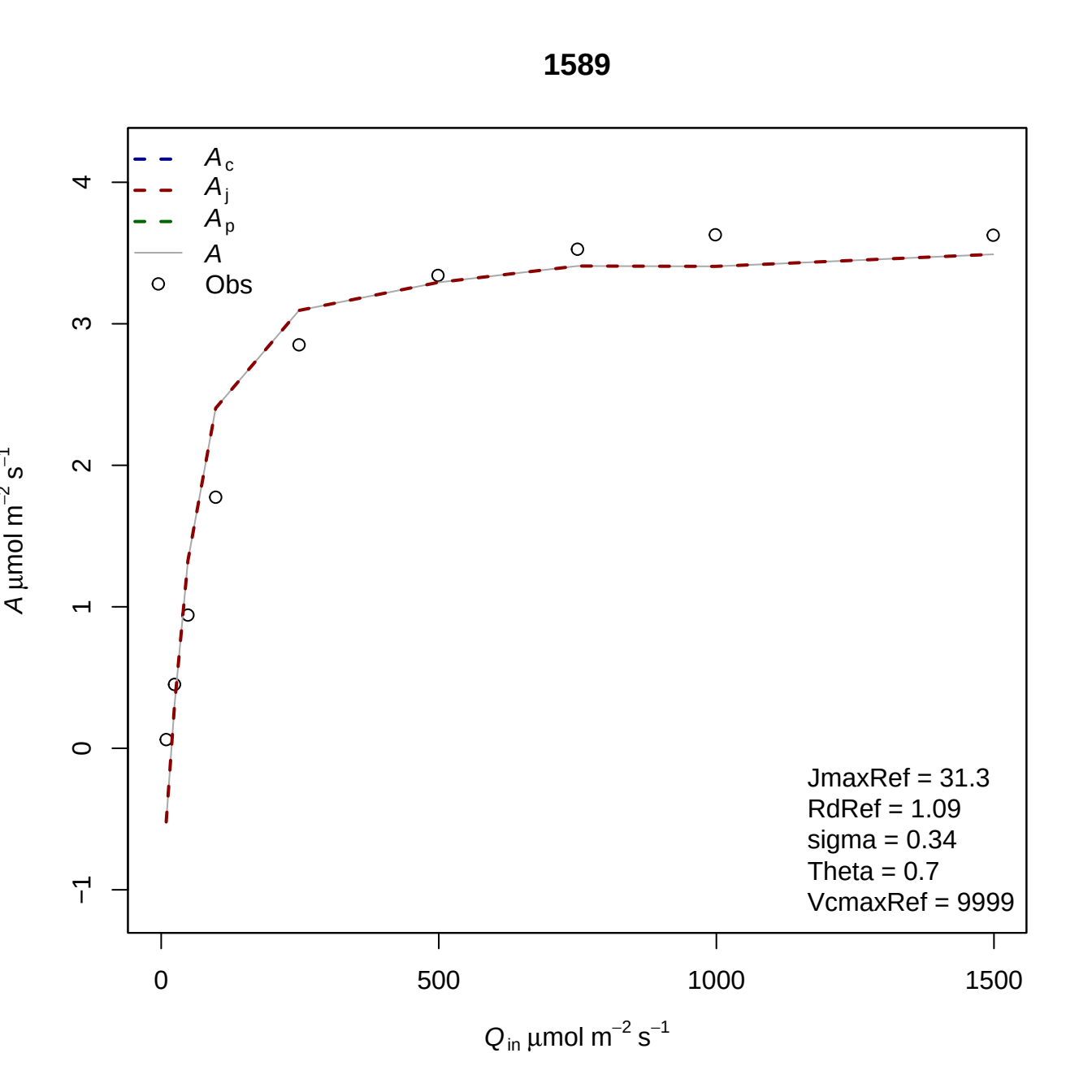

1584
A_c
A_j
A_p
A
Obs
JmaxRef = 81.43
RdRef = 1.01
sigma = 0.33
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500

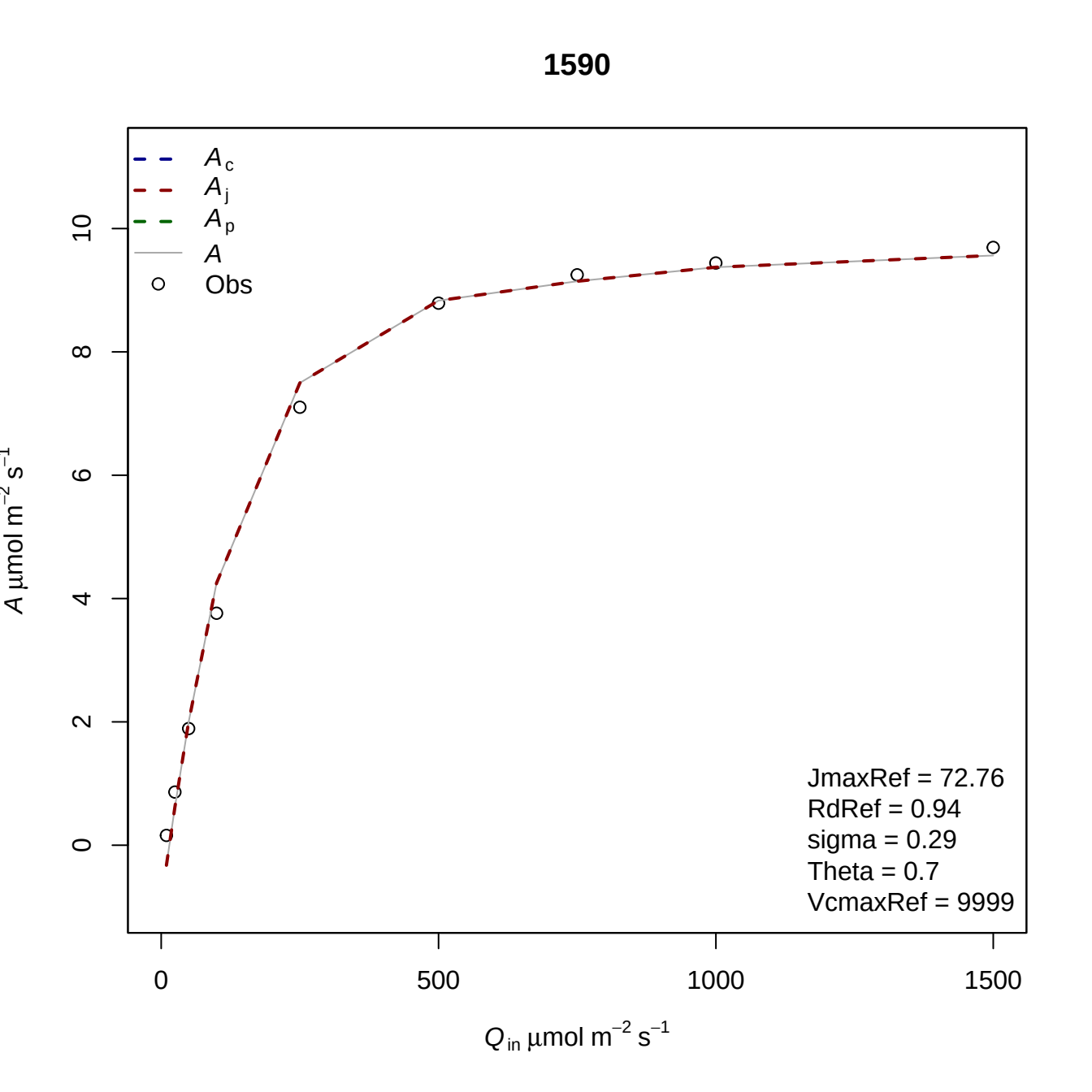

1585
A_c
A_j
A_p
A
Obs
JmaxRef = 73.99
RdRef = 1.14
sigma = 0.26
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

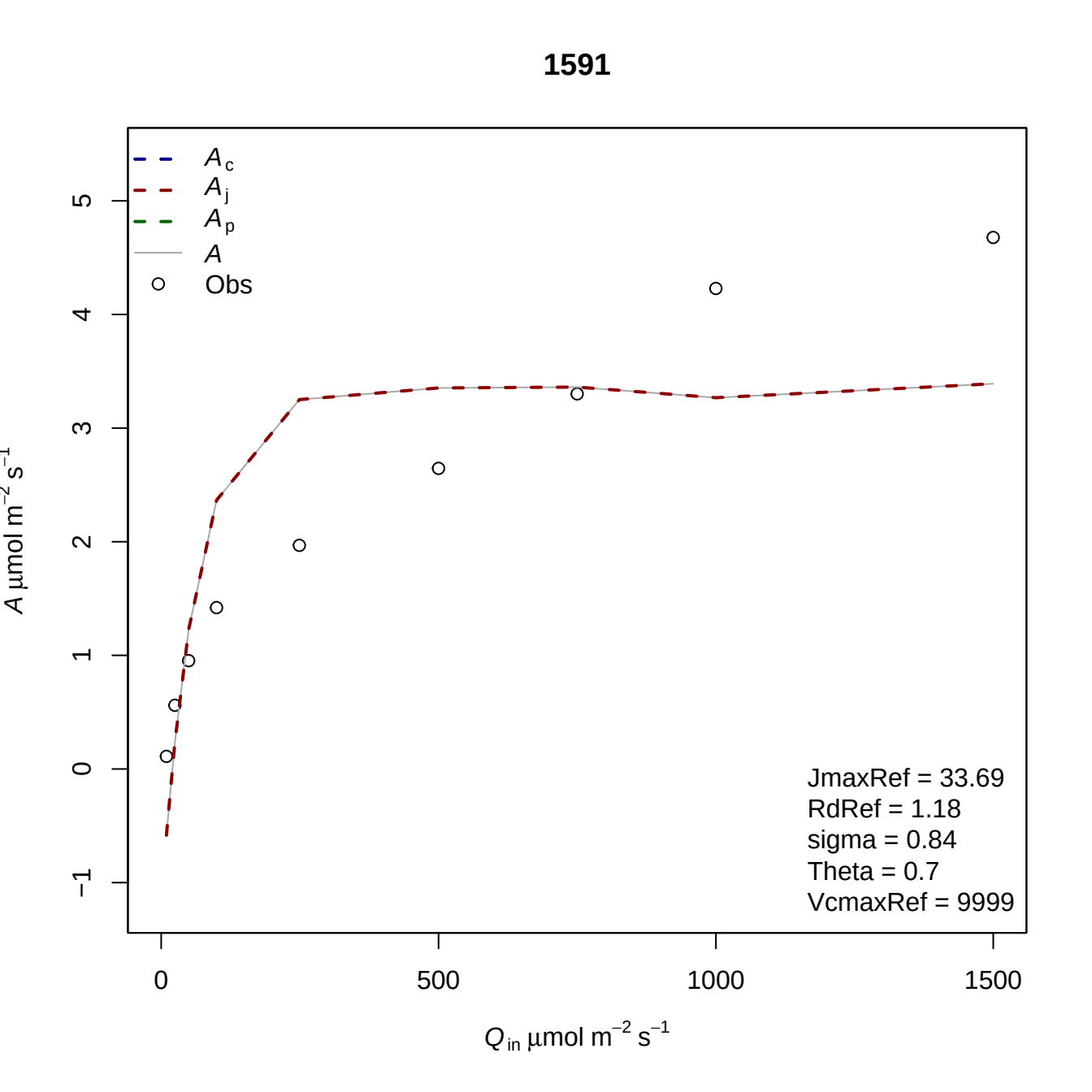

1586
A_c
A_j
A_p
A
Obs
JmaxRef = 51.11
RdRef = 0.99
sigma = 0.29
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

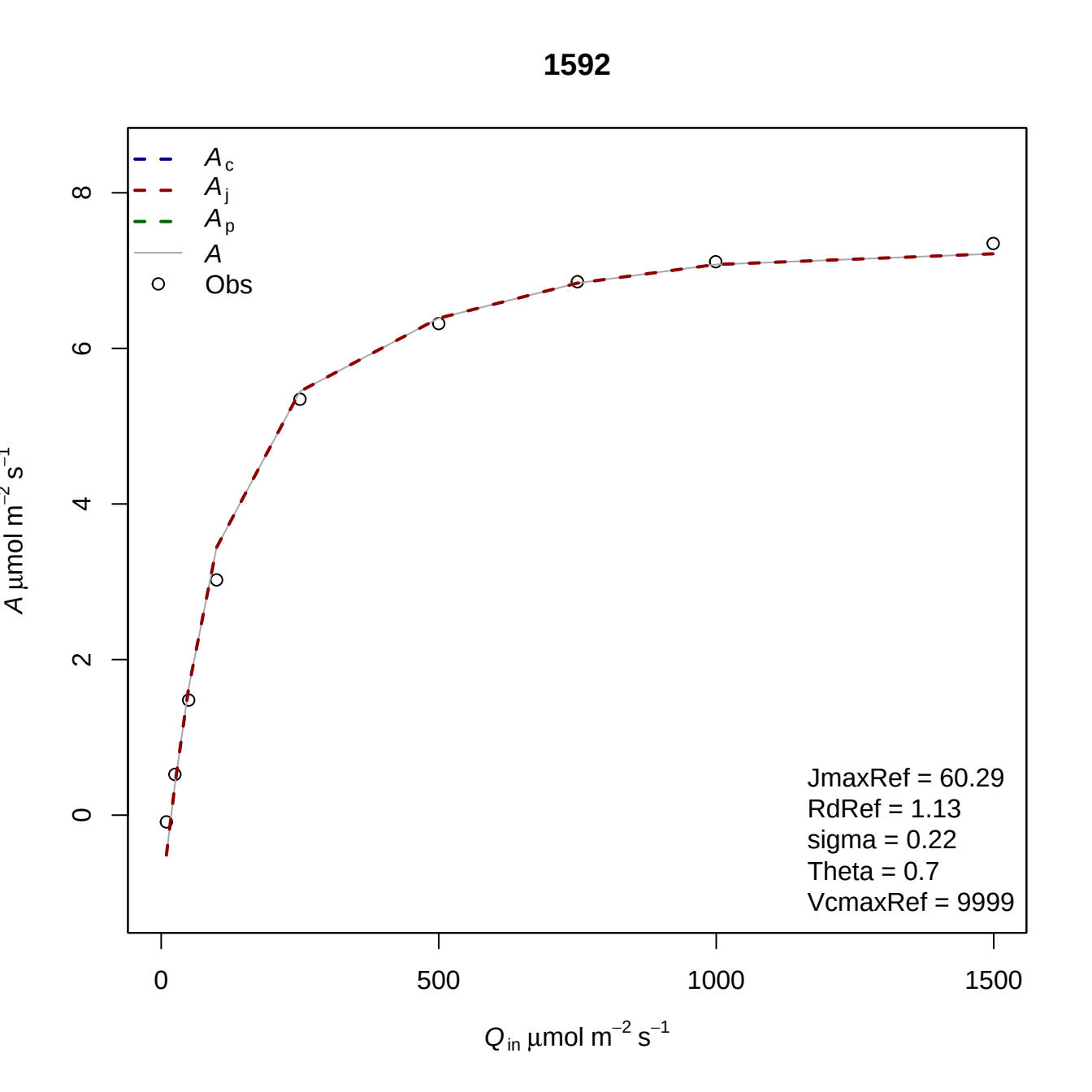

1587
A_c
A_j
A_p
A
Obs
JmaxRef = 50.76
RdRef = 1.25
sigma = 0.4
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500

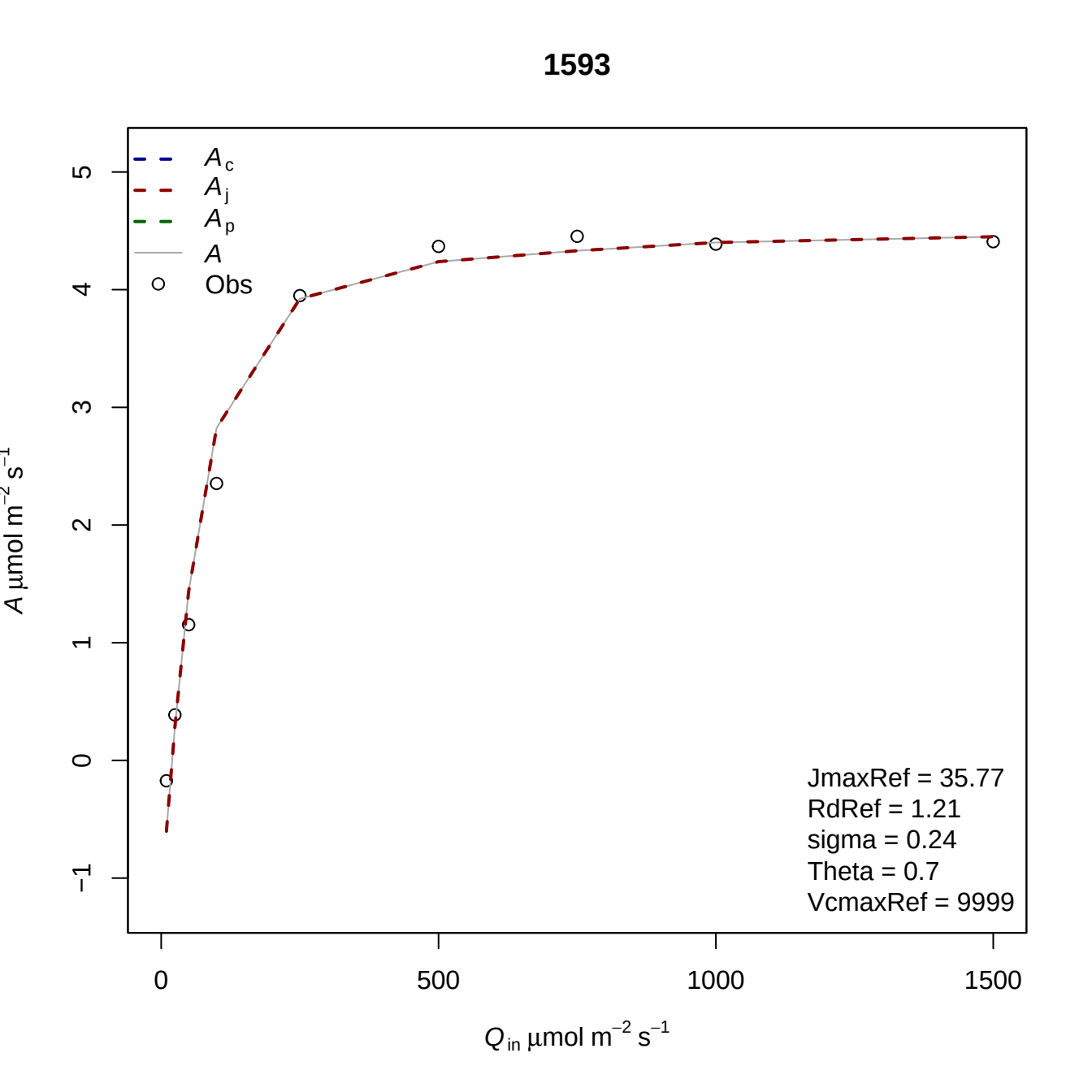

1588
A_c
A_j
A_p
A
Obs
JmaxRef = 42.67
RdRef = 1.23
sigma = 0.43
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

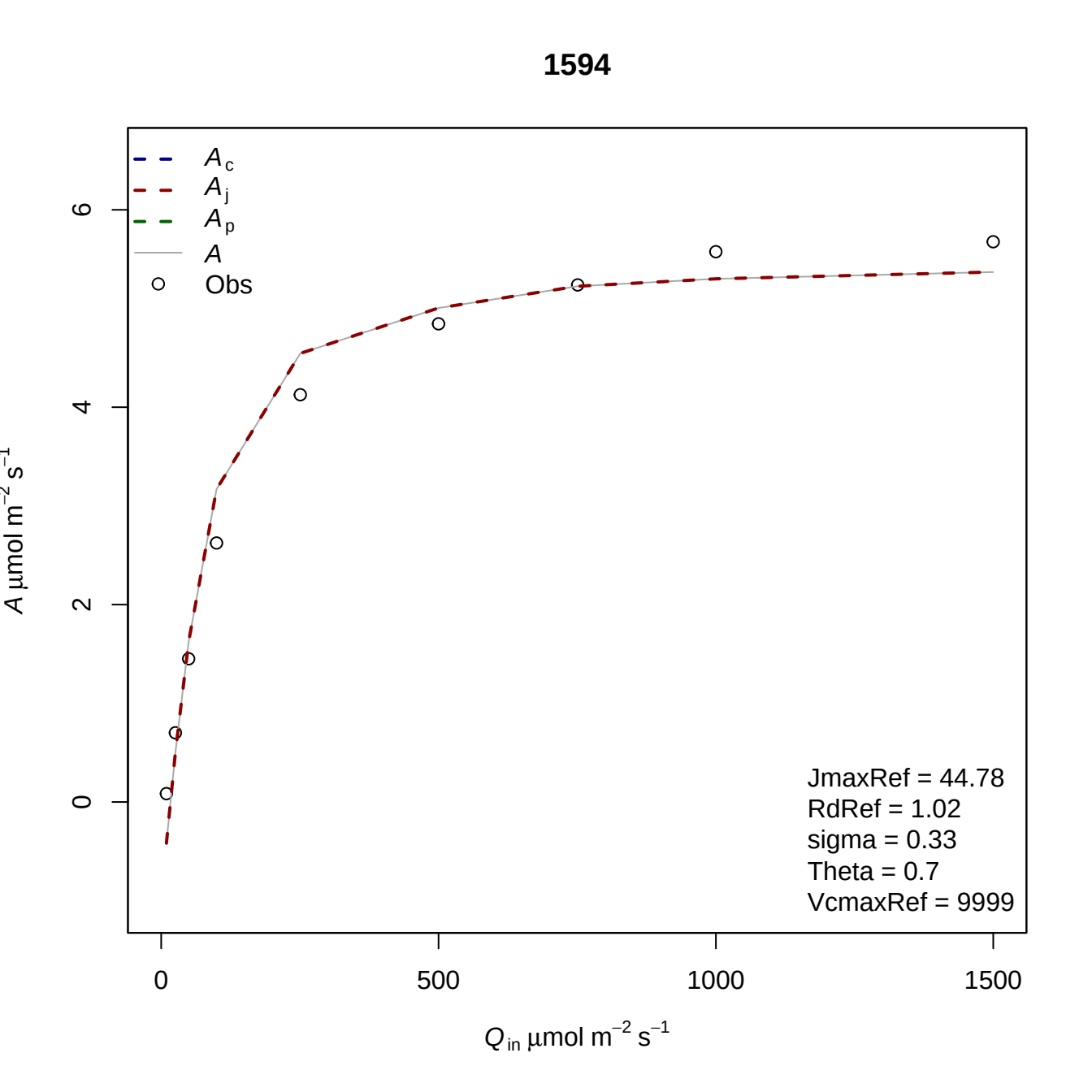

1589
A_c
A_j
A_p
A
Obs
JmaxRef = 31.3
RdRef = 1.09
sigma = 0.34
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
0
500
1000
1500

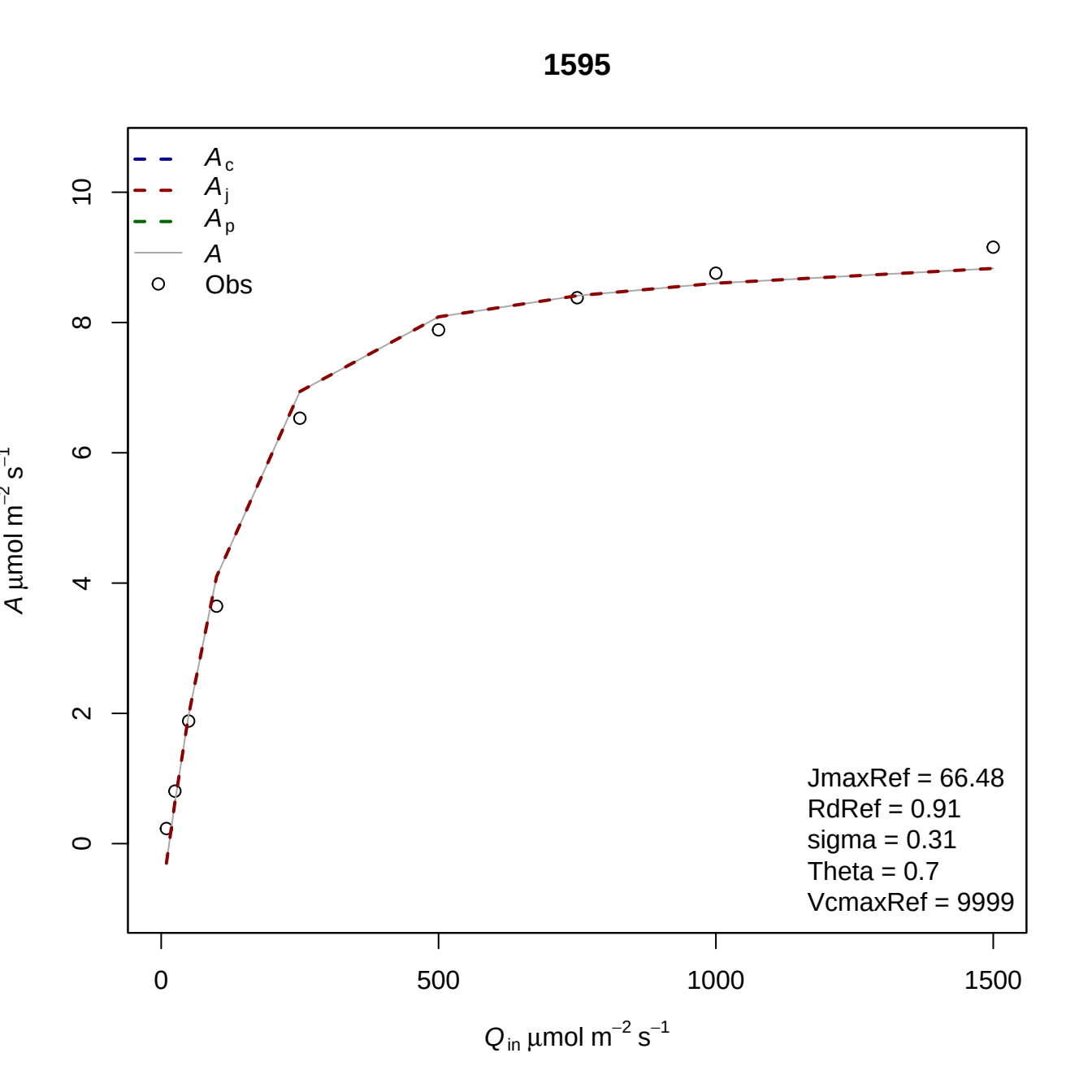

1590
A_c
A_j
A_p
A
Obs
JmaxRef = 72.76
RdRef = 0.94
sigma = 0.29
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500

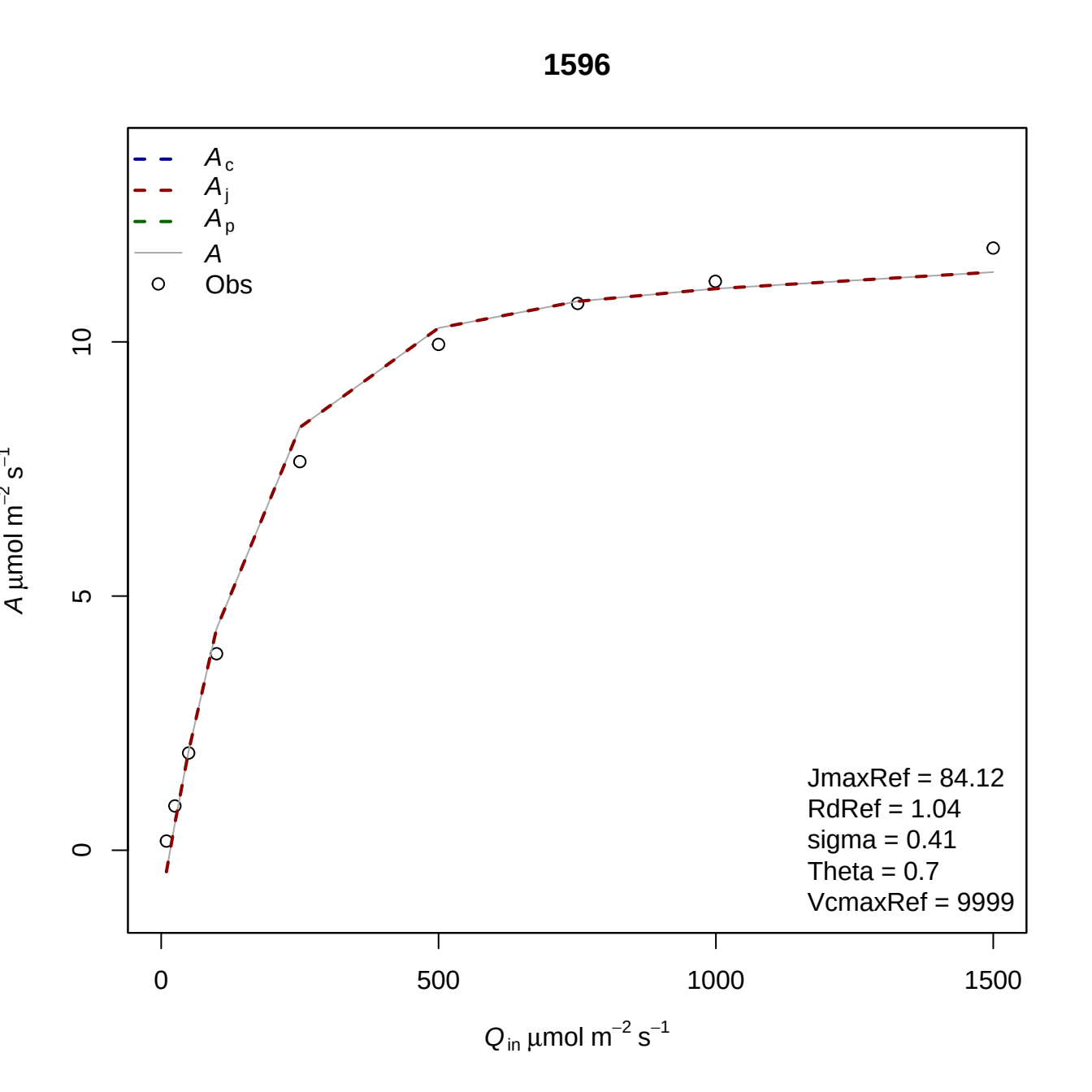

1591
A_c
A_j
A_p
A
Obs
JmaxRef = 33.69
RdRef = 1.18
sigma = 0.84
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

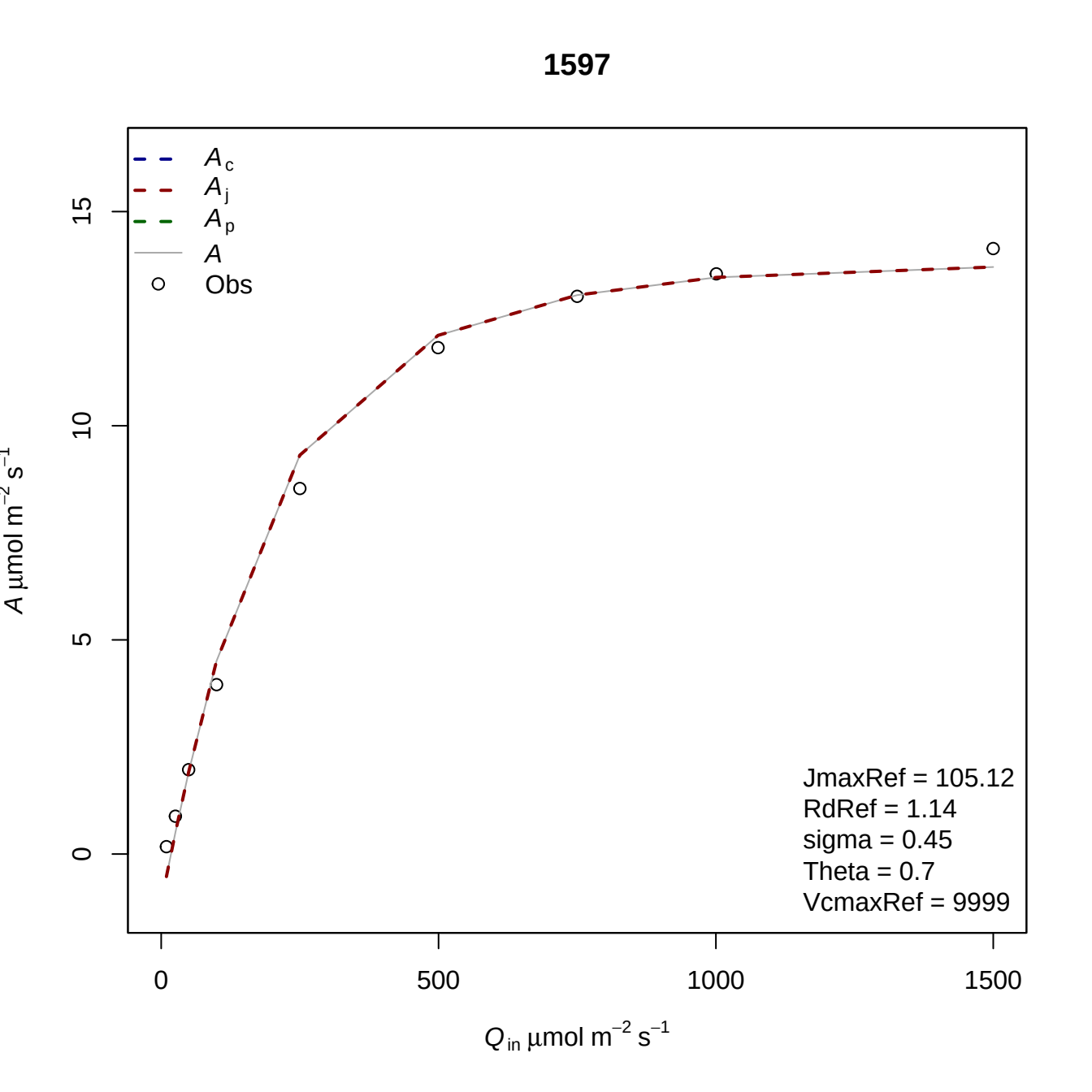

1592
A_c
A_j
A_p
A
Obs
JmaxRef = 60.29
RdRef = 1.13
sigma = 0.22
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

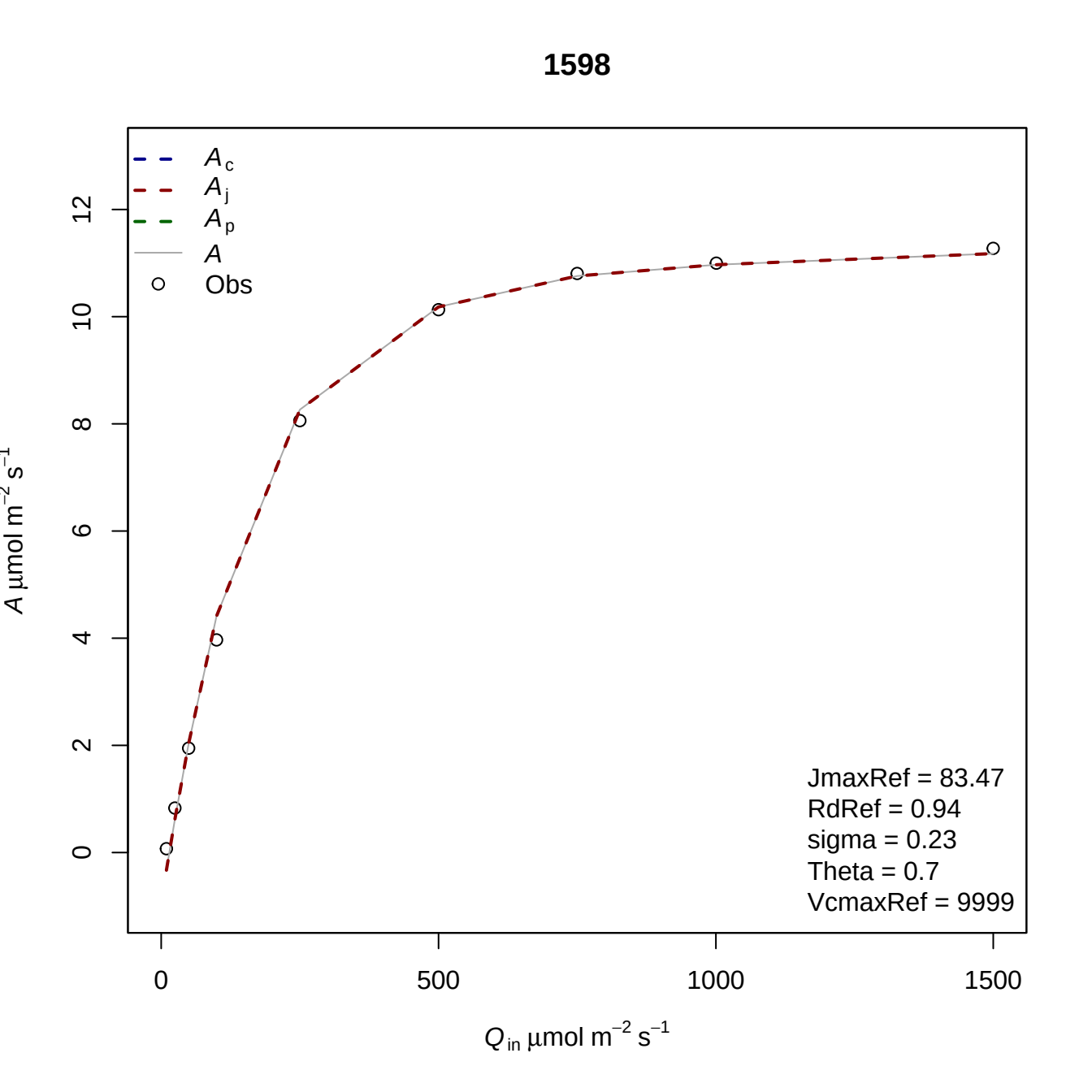

1593
A_c
A_j
A_p
A
Obs
JmaxRef = 35.77
RdRef = 1.21
sigma = 0.24
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−1
0
1
2
3
4
5
0
500
1000
1500

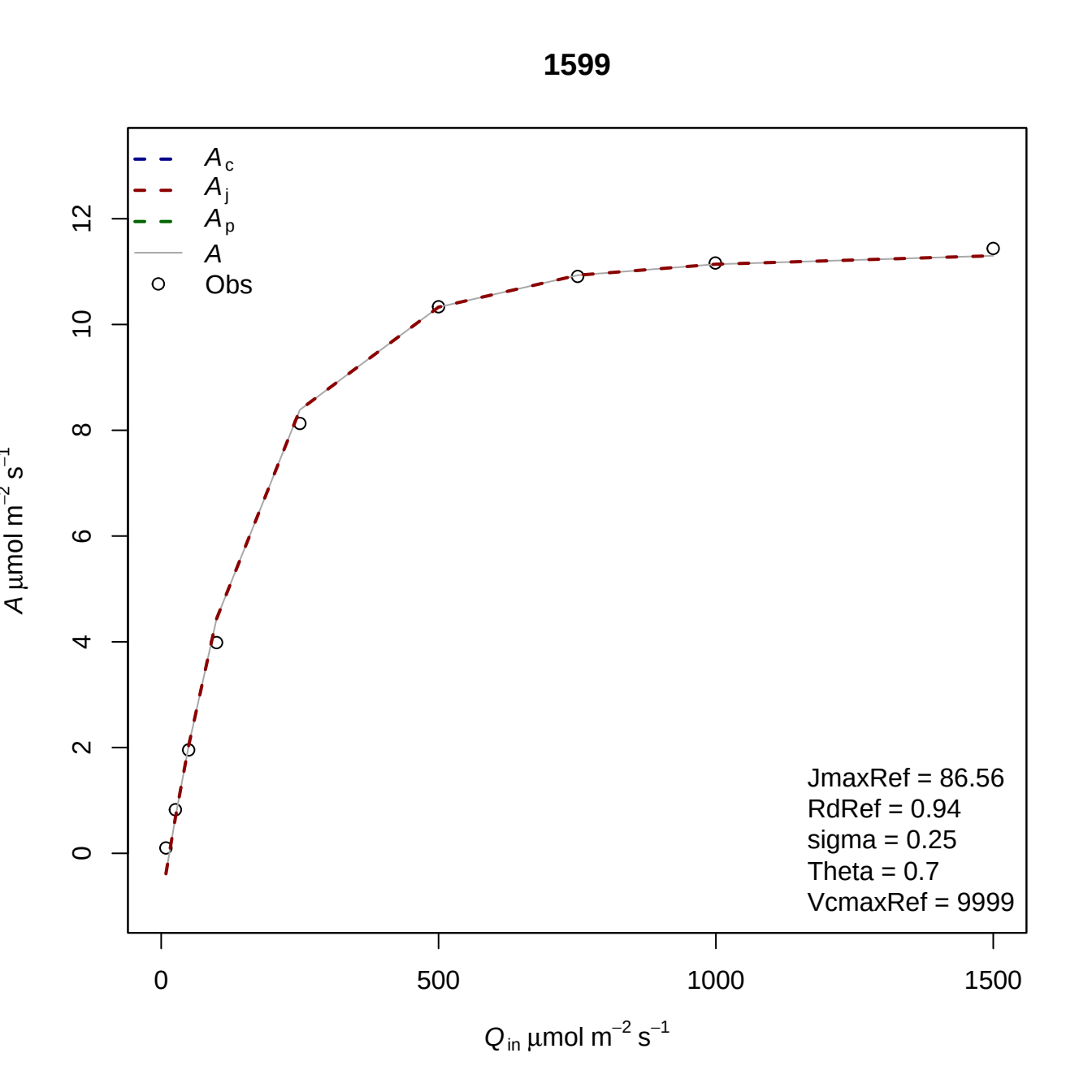

1594
A_c
A_j
A_p
A
Obs
JmaxRef = 44.78
RdRef = 1.02
sigma = 0.33
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
0
500
1000
1500

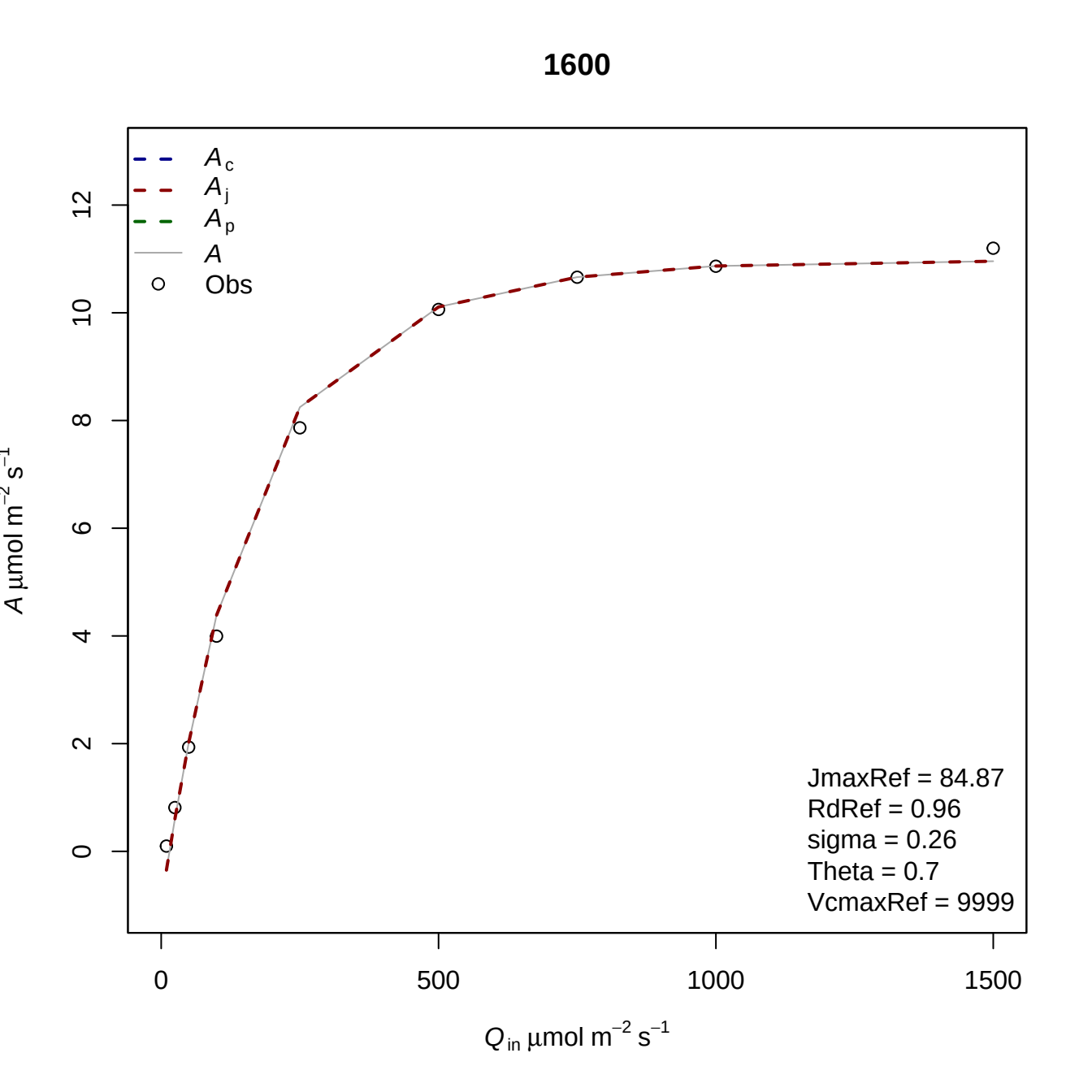

1595
A_c
A_j
A_p
A
Obs
JmaxRef = 66.48
RdRef = 0.91
sigma = 0.31
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

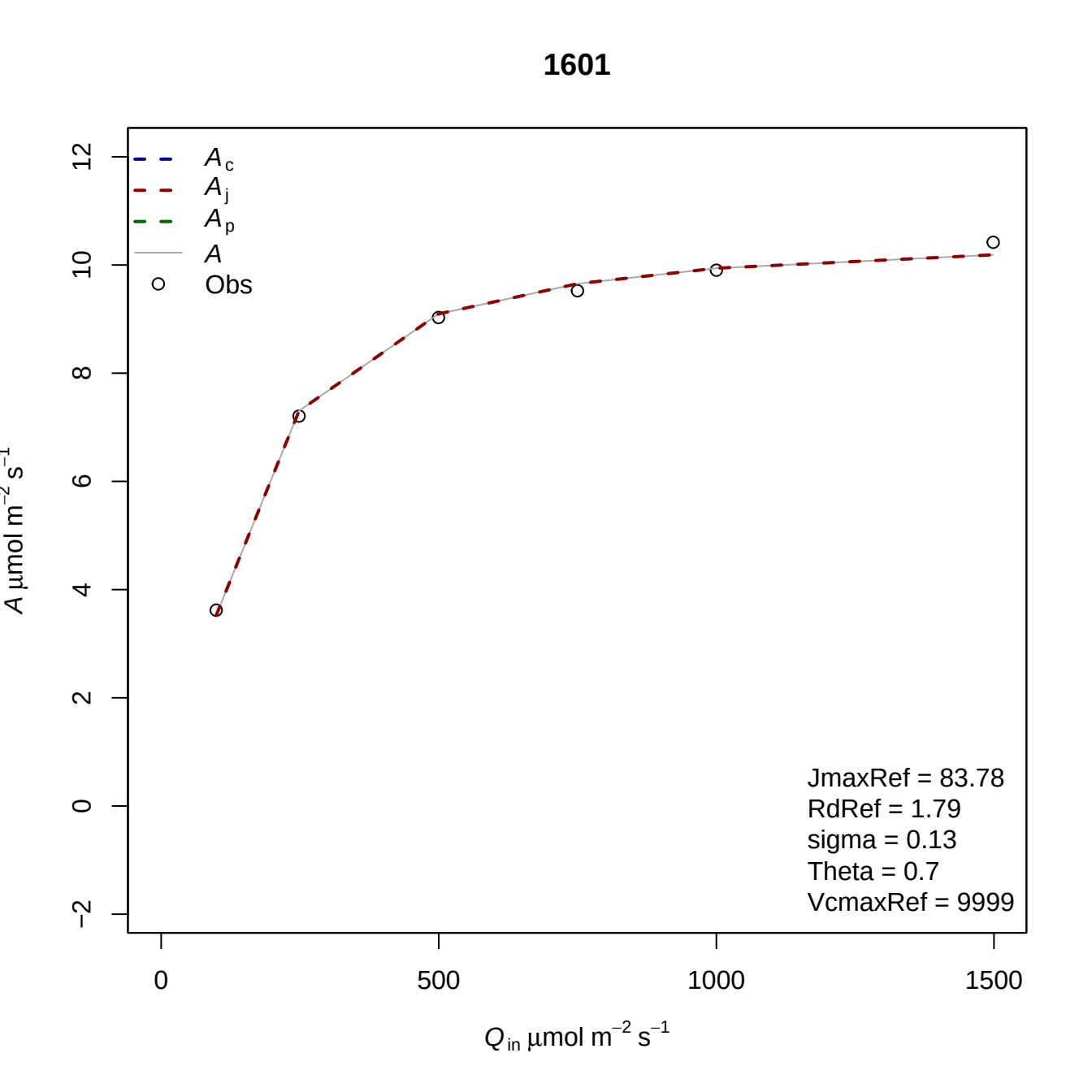

1596
A_c
A_j
A_p
A
Obs
JmaxRef = 84.12
RdRef = 1.04
sigma = 0.41
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
0
5
10
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

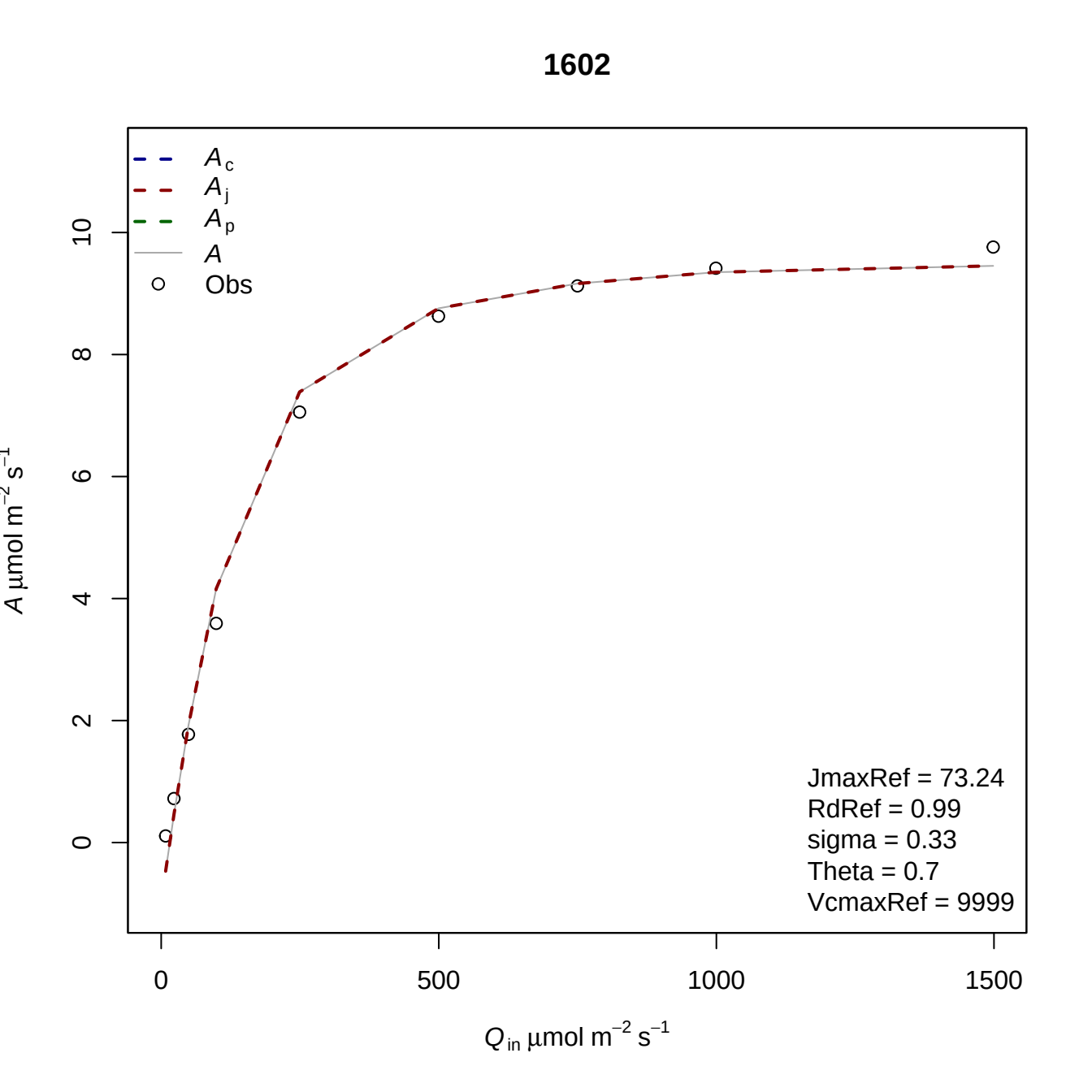

1597
A_c
A_j
A_p
A
Obs
JmaxRef = 105.12
RdRef = 1.14
sigma = 0.45
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

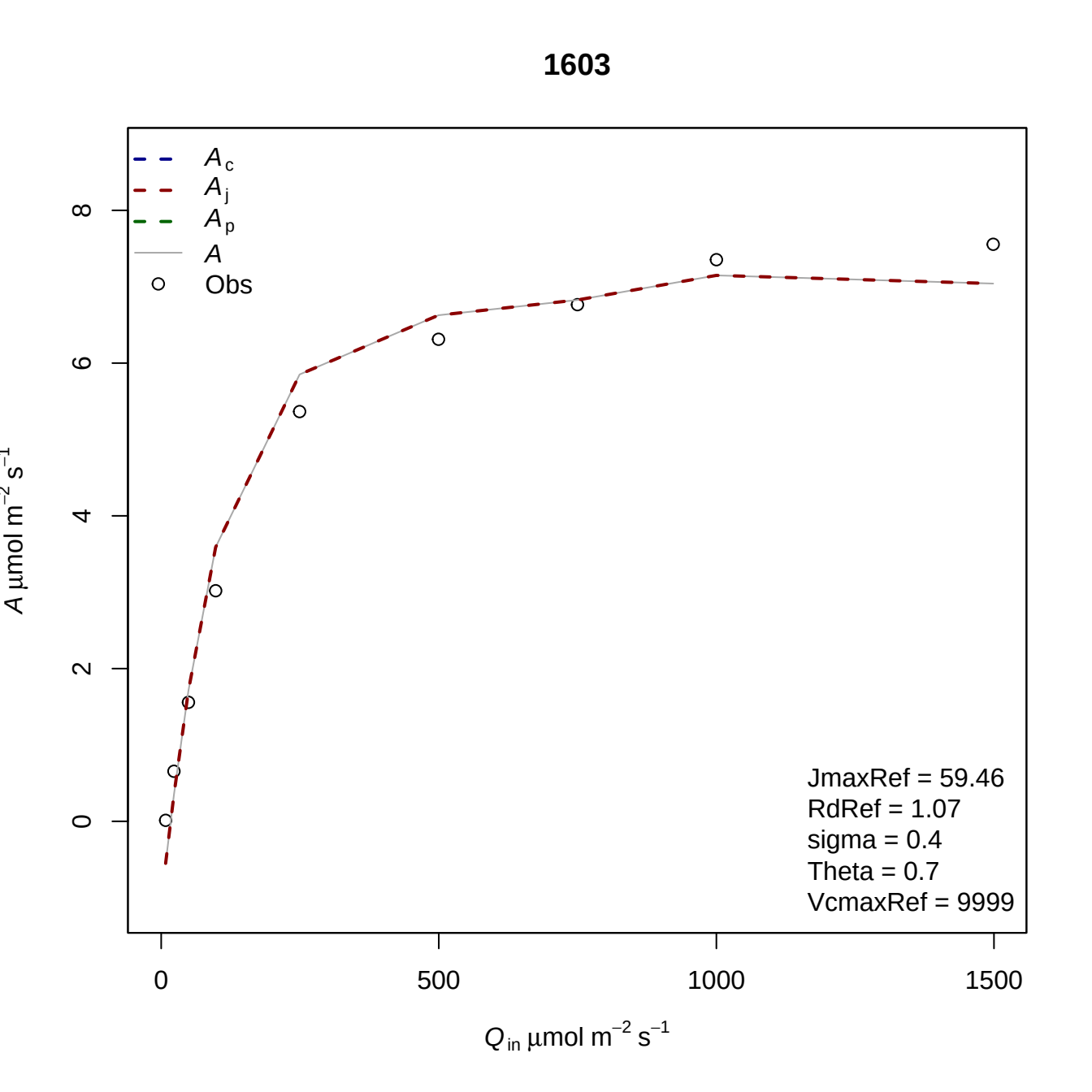

1598
A_c
A_j
A_p
A
Obs
JmaxRef = 83.47
RdRef = 0.94
sigma = 0.23
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

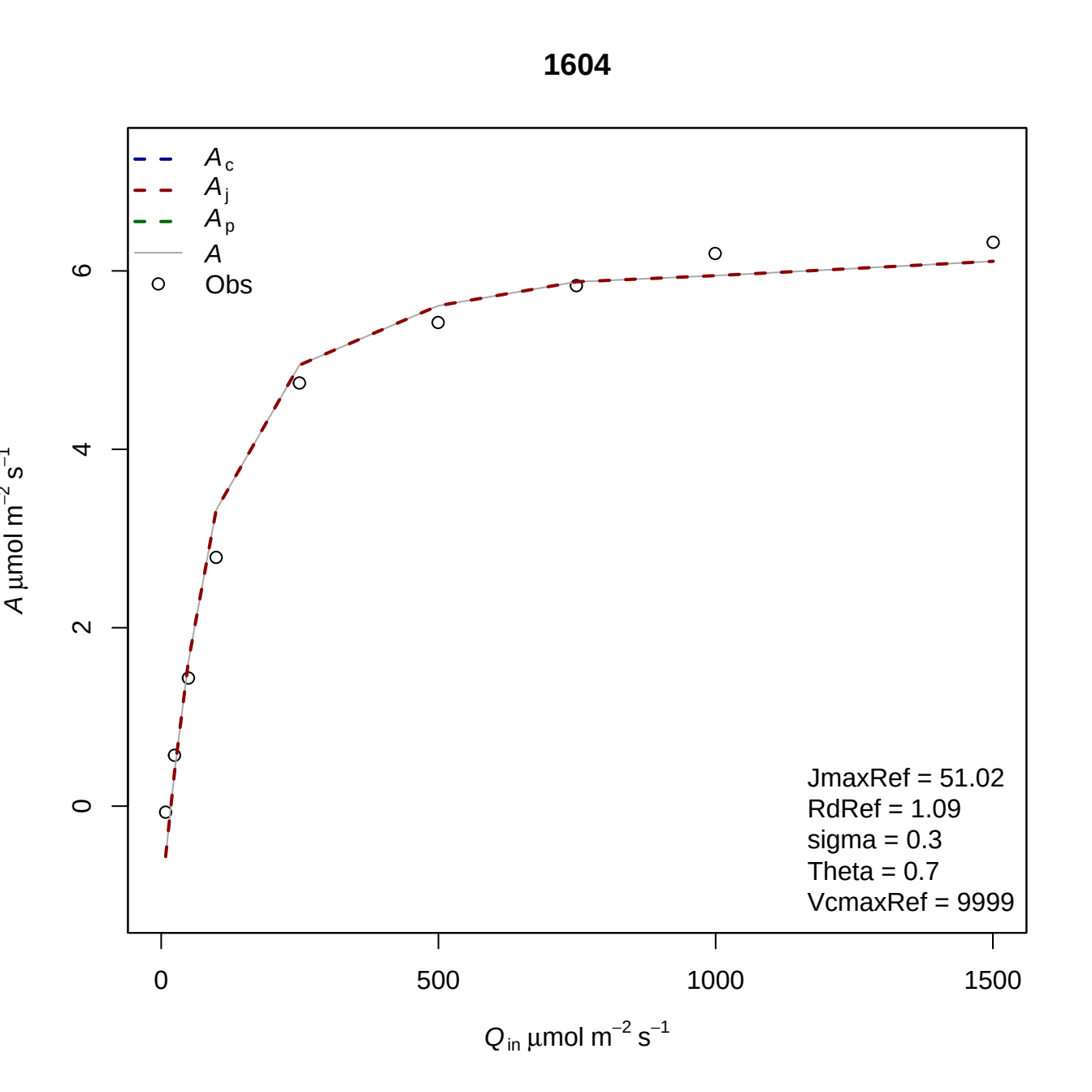

1599
A_c
A_j
A_p
A
Obs
JmaxRef = 86.56
RdRef = 0.94
sigma = 0.25
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

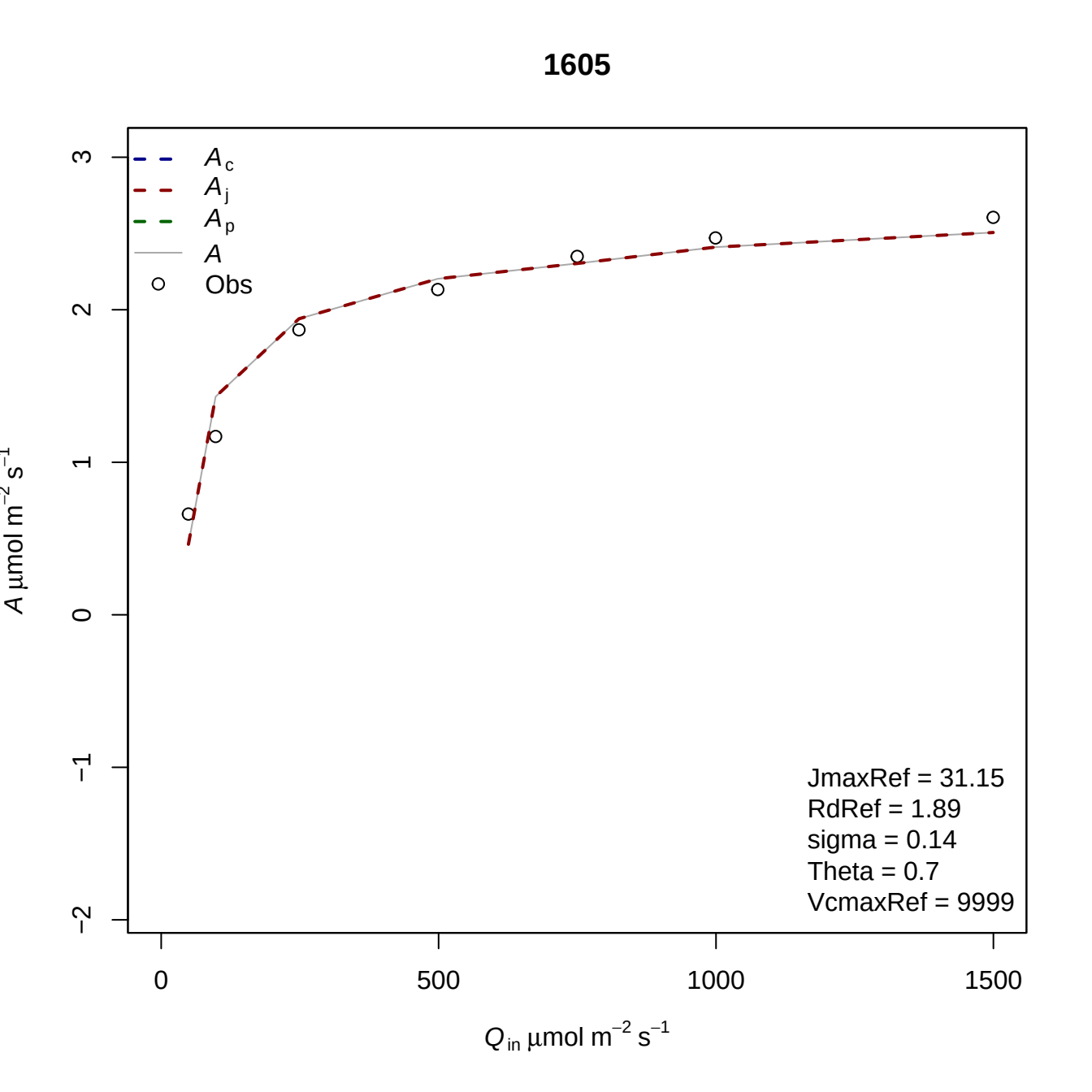

1600
A_c
A_j
A_p
A
Obs
JmaxRef = 84.87
RdRef = 0.96
sigma = 0.26
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

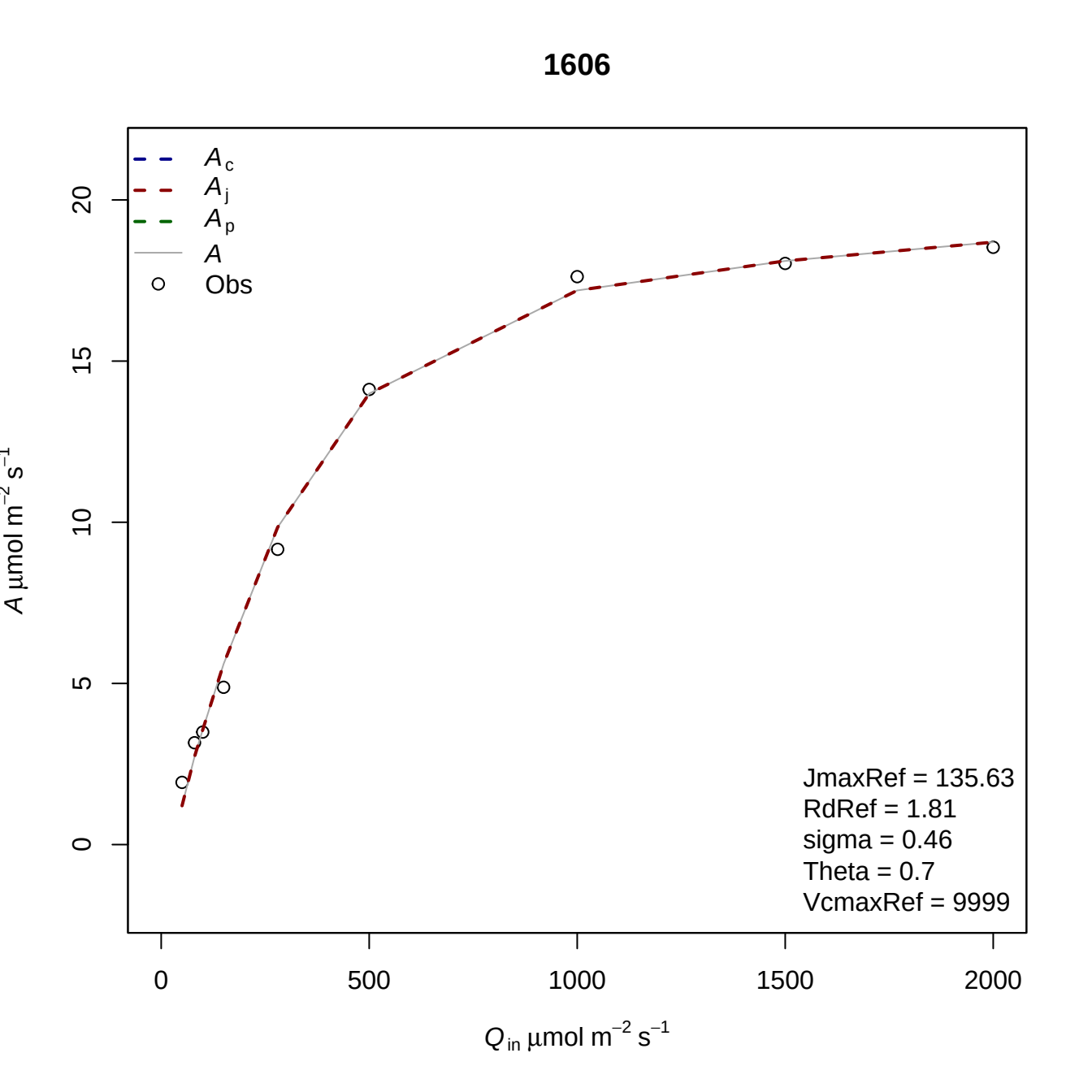

1601
A_c
A_j
A_p
A
Obs
JmaxRef = 83.78
RdRef = 1.79
sigma = 0.13
Theta = 0.7
VcmaxRef = 9999
12
10
8
6
4
2
0
−2
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

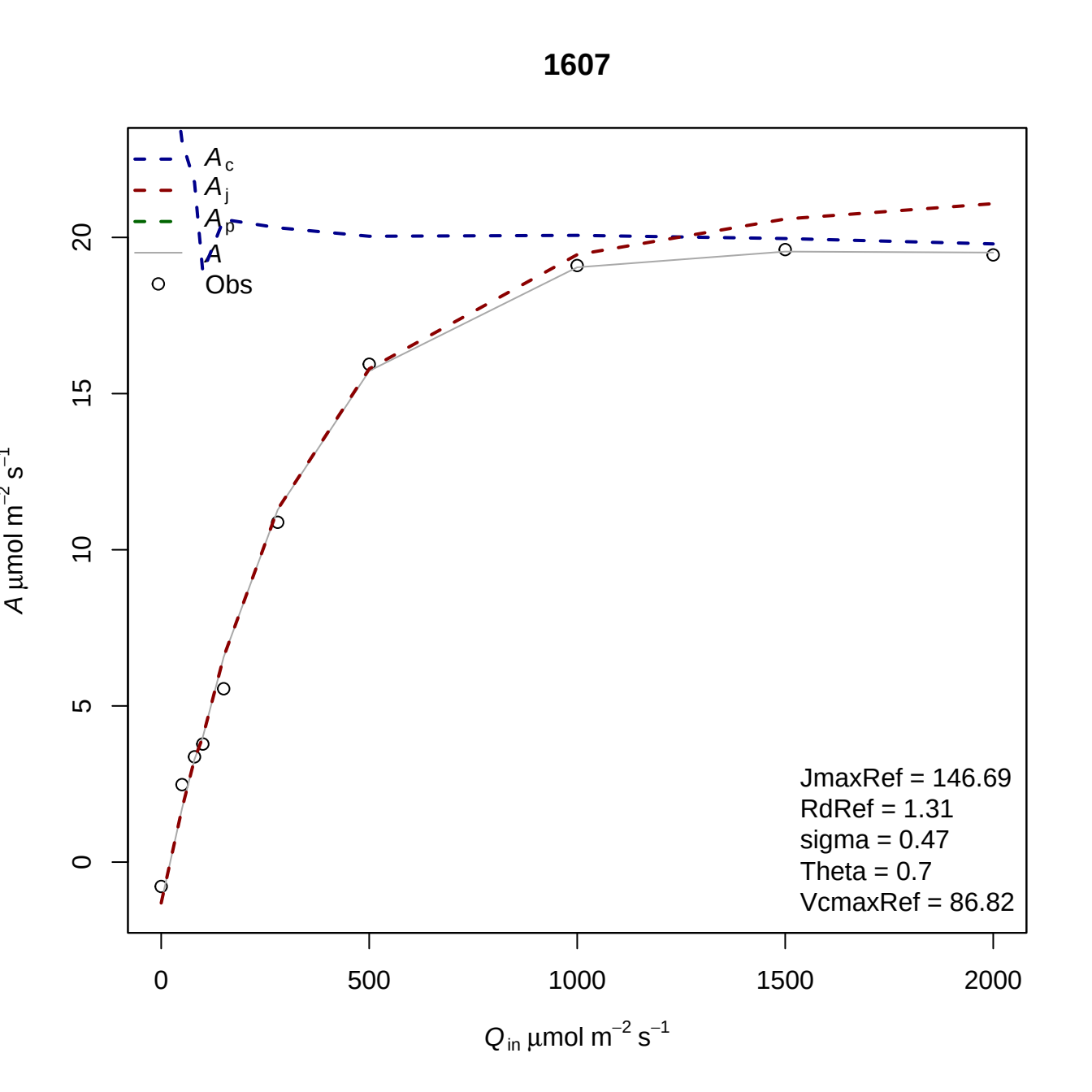

1602
A_c
A_j
A_p
A
Obs
JmaxRef = 73.24
RdRef = 0.99
sigma = 0.33
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
10
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

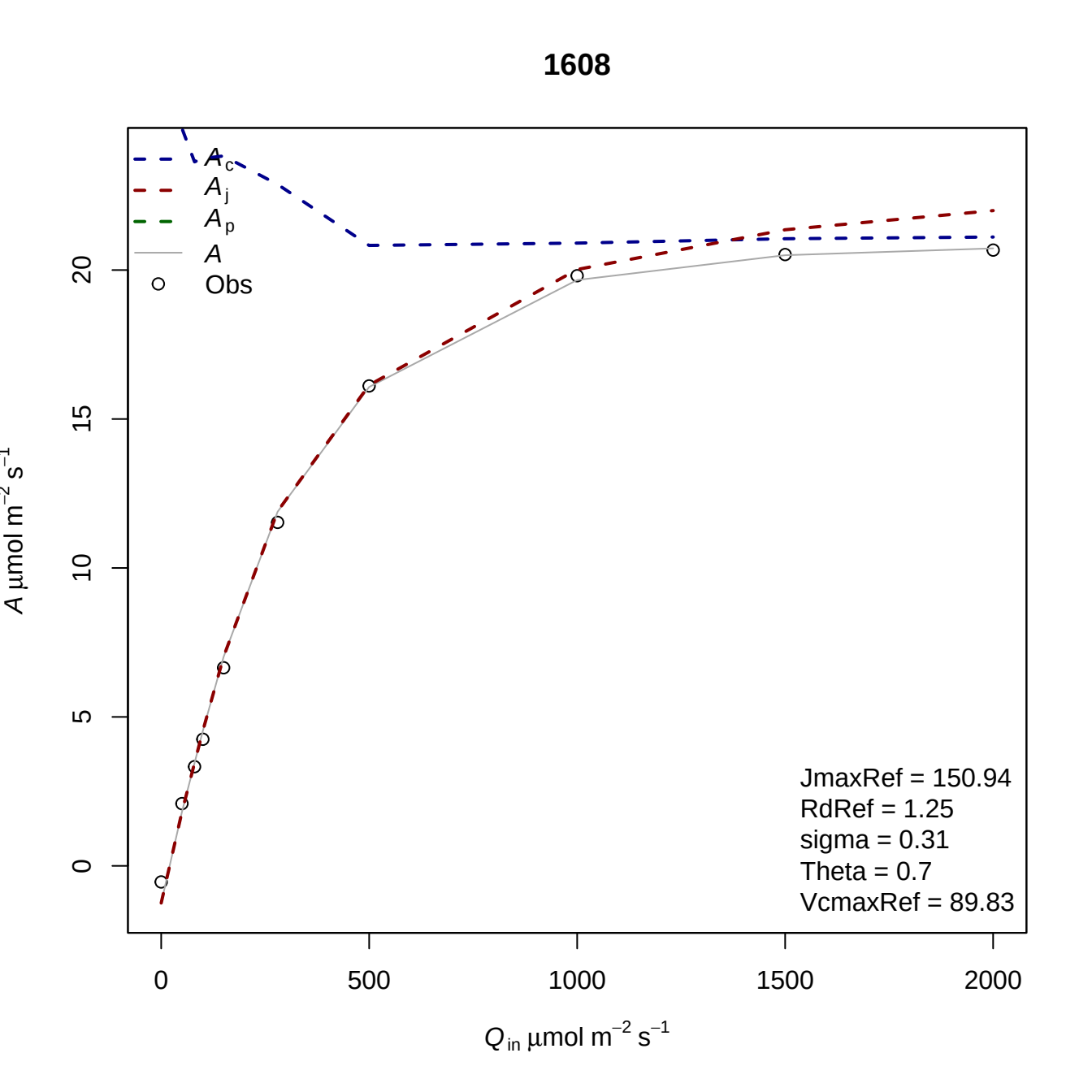

1603
A_c
A_j
A_p
A
Obs
JmaxRef = 59.46
RdRef = 1.07
sigma = 0.4
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
8
A µmol m^{-2} s^{-1}
0
500
1000
1500
Q_{in} µmol m^{-2} s^{-1}

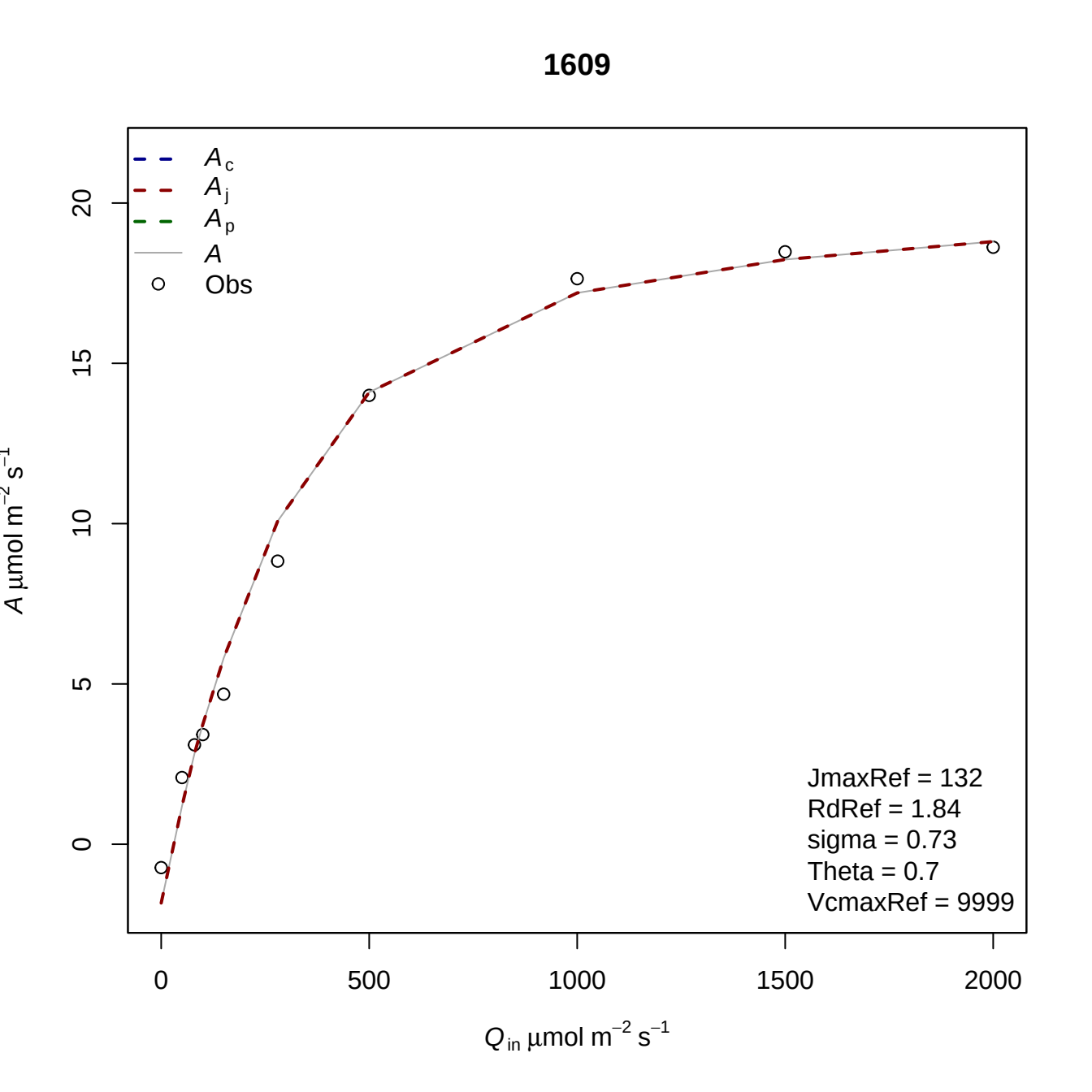

1604
A_c
A_j
A_p
A
Obs
JmaxRef = 51.02
RdRef = 1.09
sigma = 0.3
Theta = 0.7
VcmaxRef = 9999
0
2
4
6
0
500
1000
1500
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

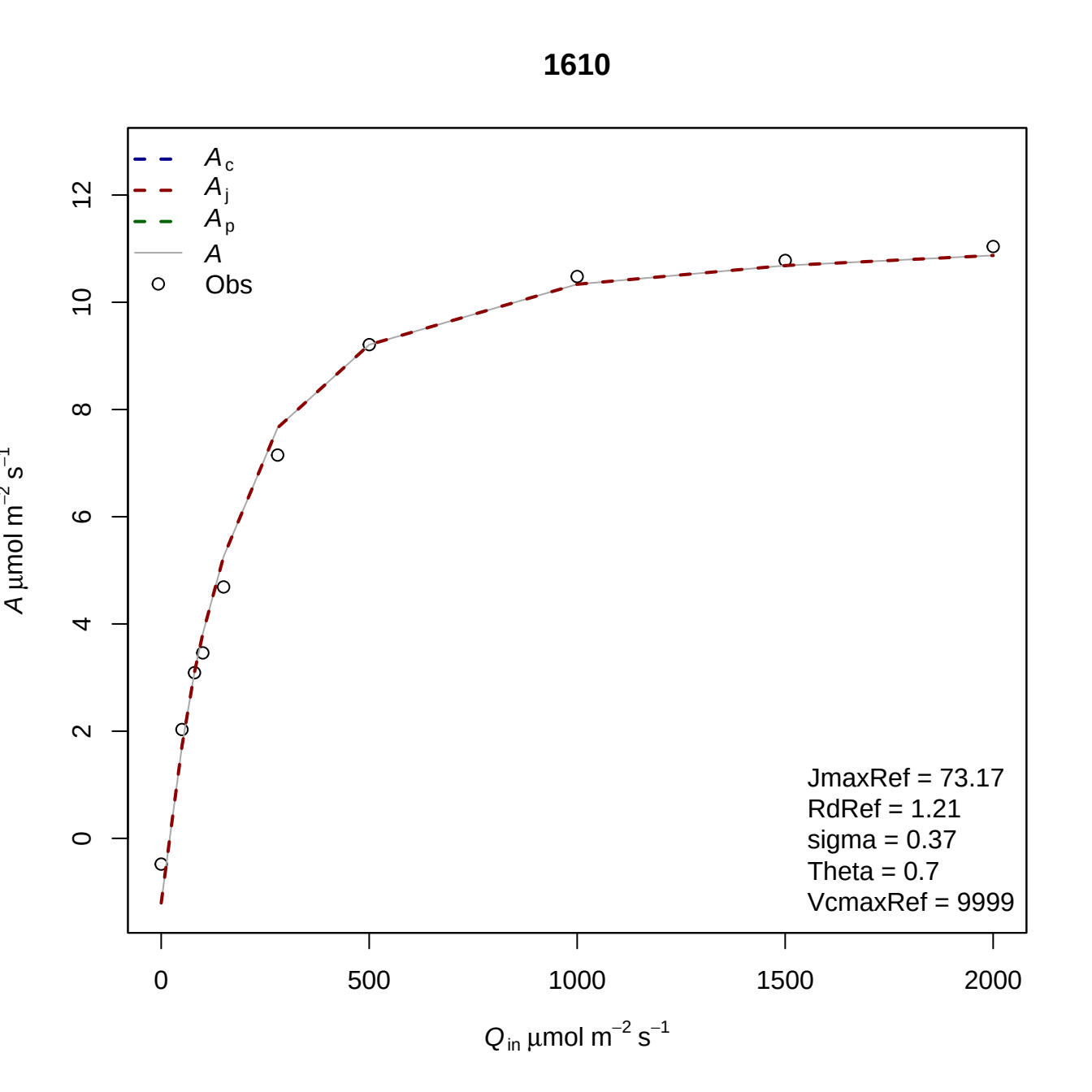

1605
A_c
A_j
A_p
A
Obs
JmaxRef = 31.15
RdRef = 1.89
sigma = 0.14
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
3
2
1
0
−1
−2
0
500
1000
1500

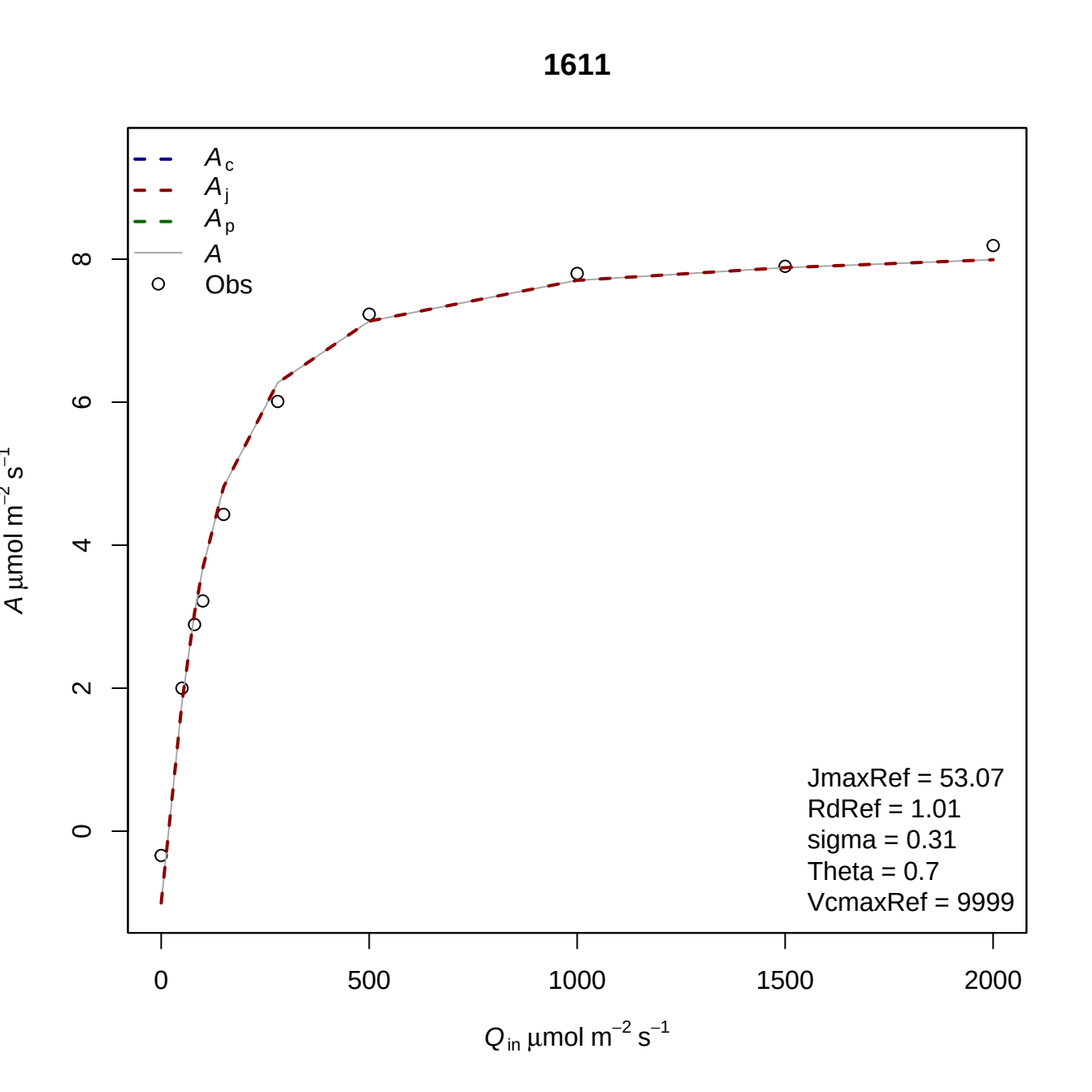

1606
A_c
A_j
A_p
A
Obs
JmaxRef = 135.63
RdRef = 1.81
sigma = 0.46
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

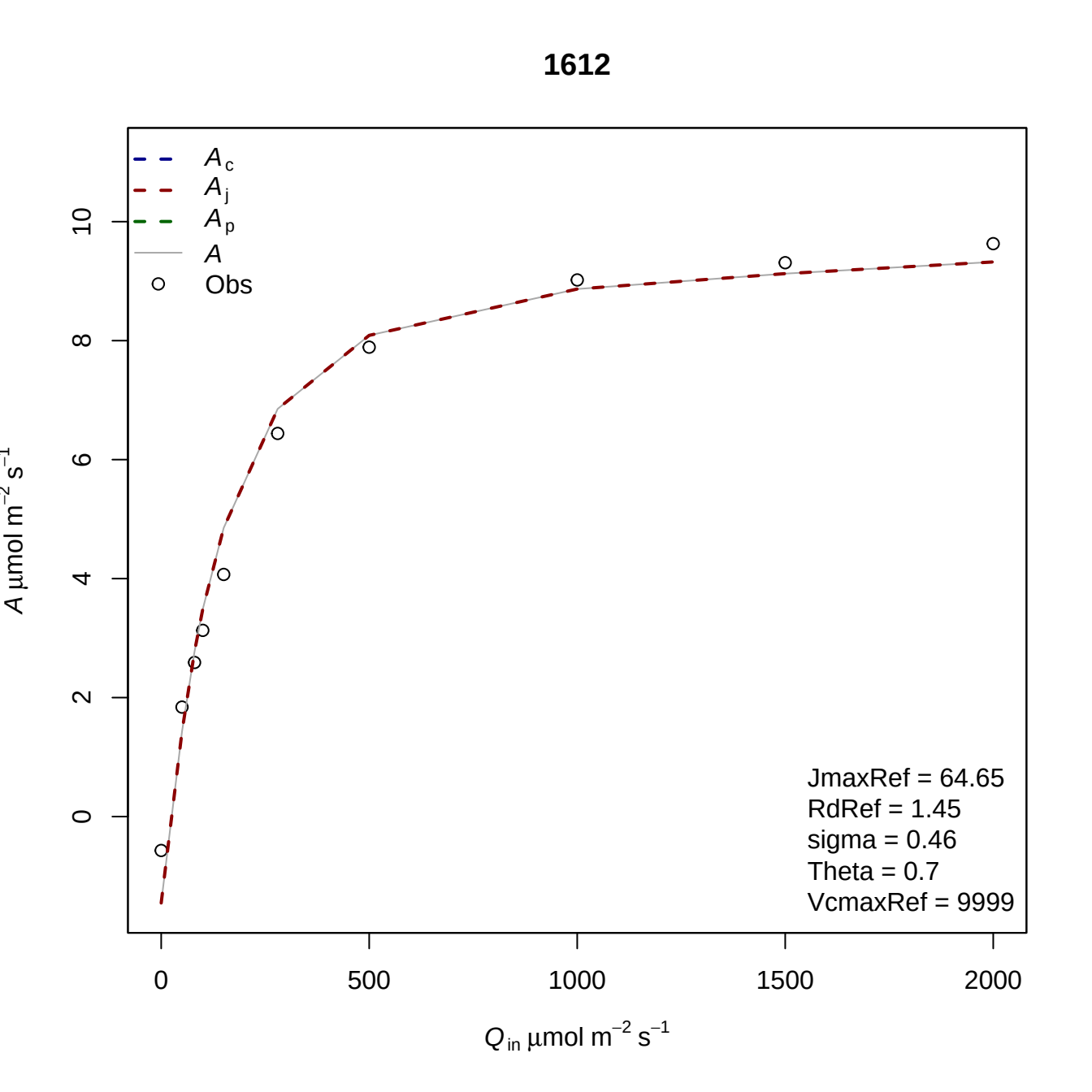

1607
A_c
A_j
A_p
A
Obs
JmaxRef = 146.69
RdRef = 1.31
sigma = 0.47
Theta = 0.7
VcmaxRef = 86.82
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

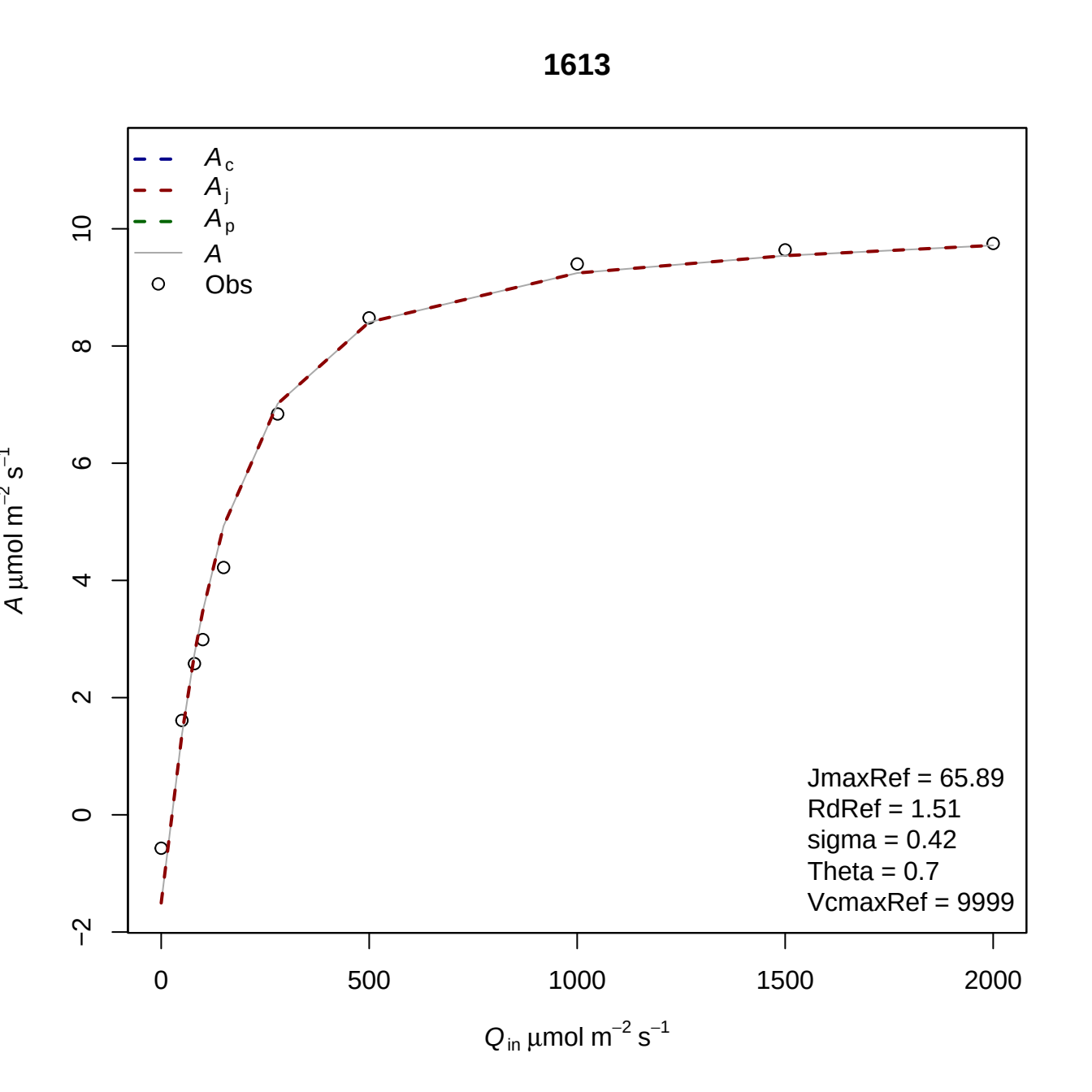

1608
A_c
A_j
A_p
A
Obs
JmaxRef = 150.94
RdRef = 1.25
sigma = 0.31
Theta = 0.7
VcmaxRef = 89.83
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

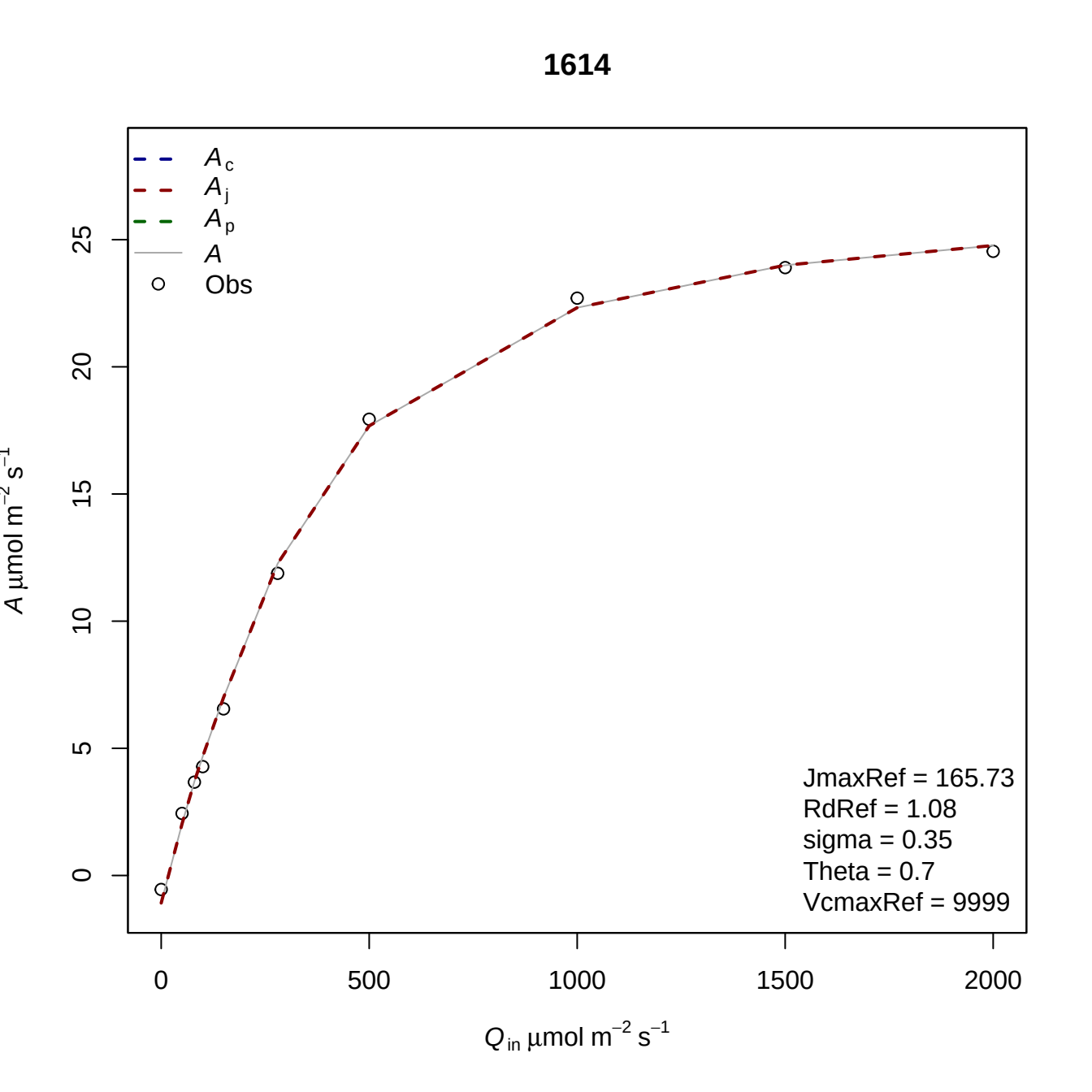

1609
A_c
A_j
A_p
A
Obs
JmaxRef = 132
RdRef = 1.84
sigma = 0.73
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

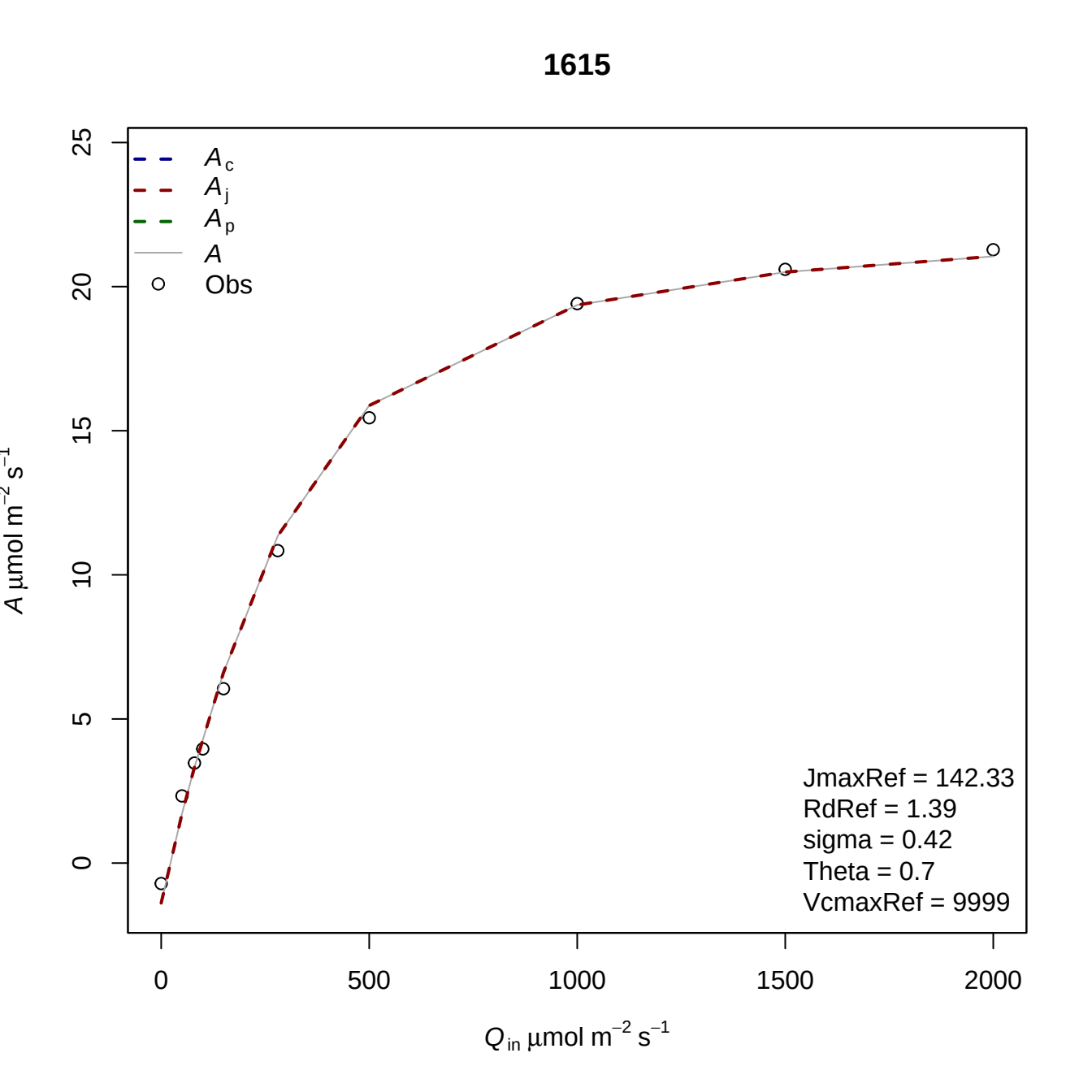

1610
A_c
A_j
A_p
A
Obs
JmaxRef = 73.17
RdRef = 1.21
sigma = 0.37
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
12
0
500
1000
1500
2000

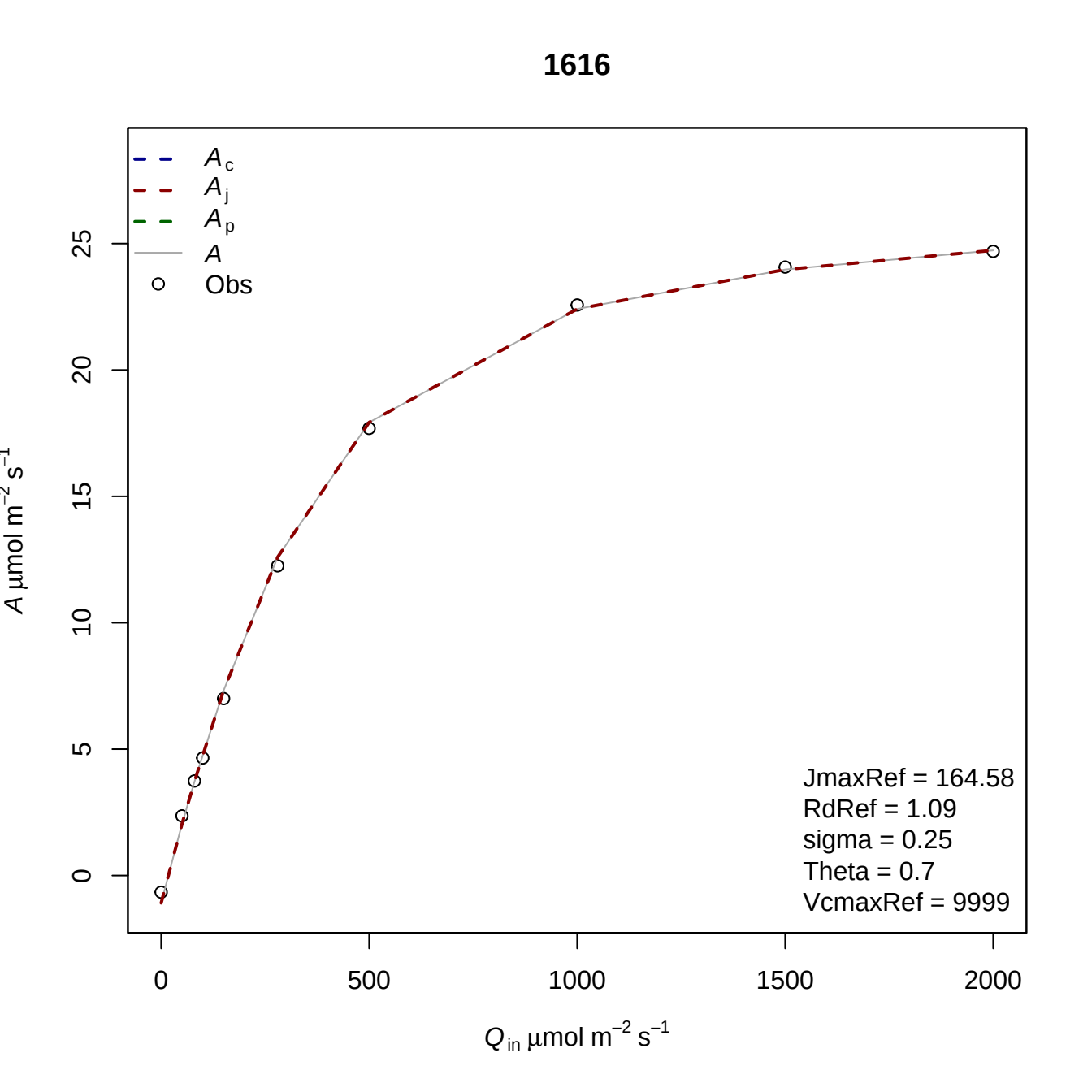

1611
A_c
A_j
A_p
A
Obs
JmaxRef = 53.07
RdRef = 1.01
sigma = 0.31
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

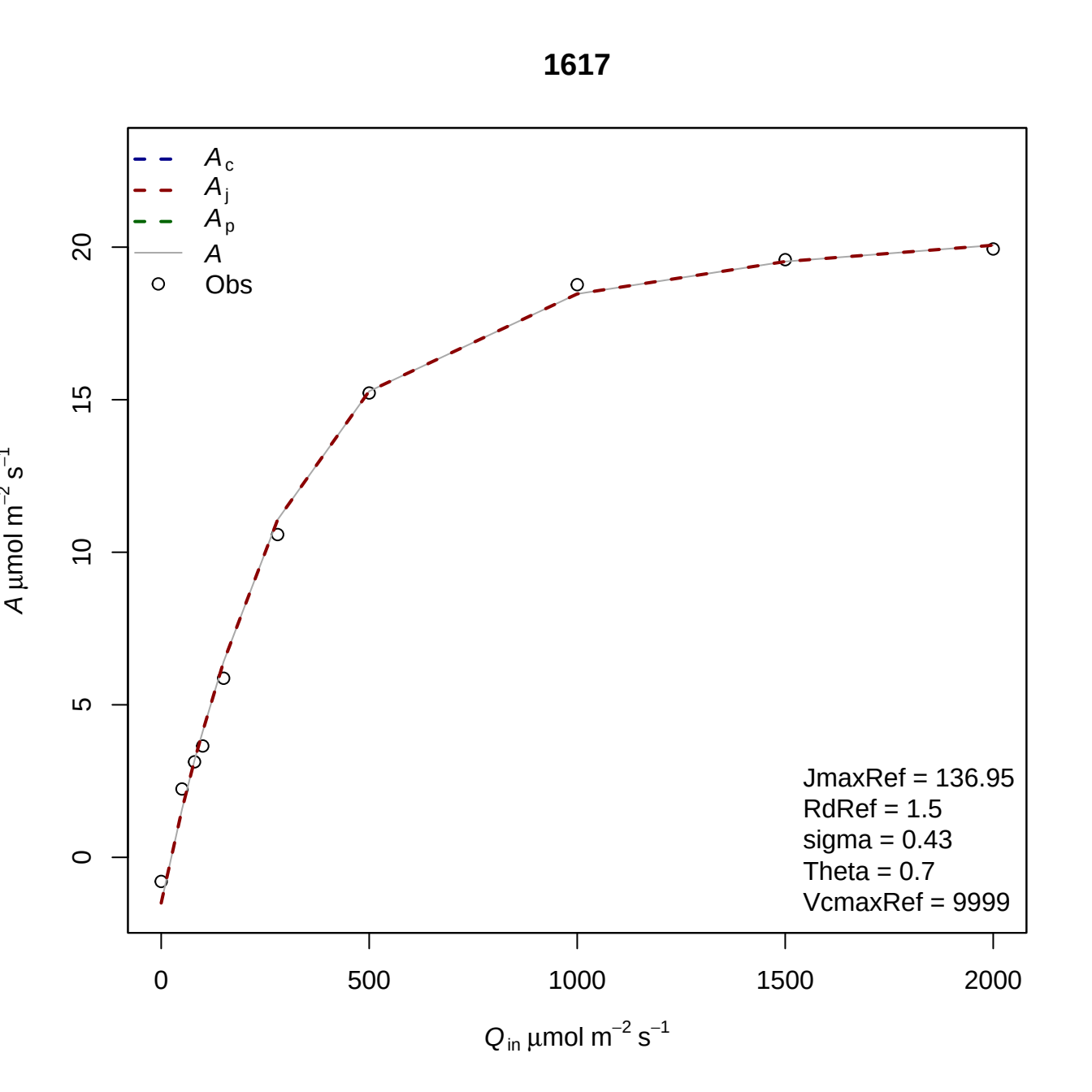

1612
A_c
A_j
A_p
A
Obs
JmaxRef = 64.65
RdRef = 1.45
sigma = 0.46
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

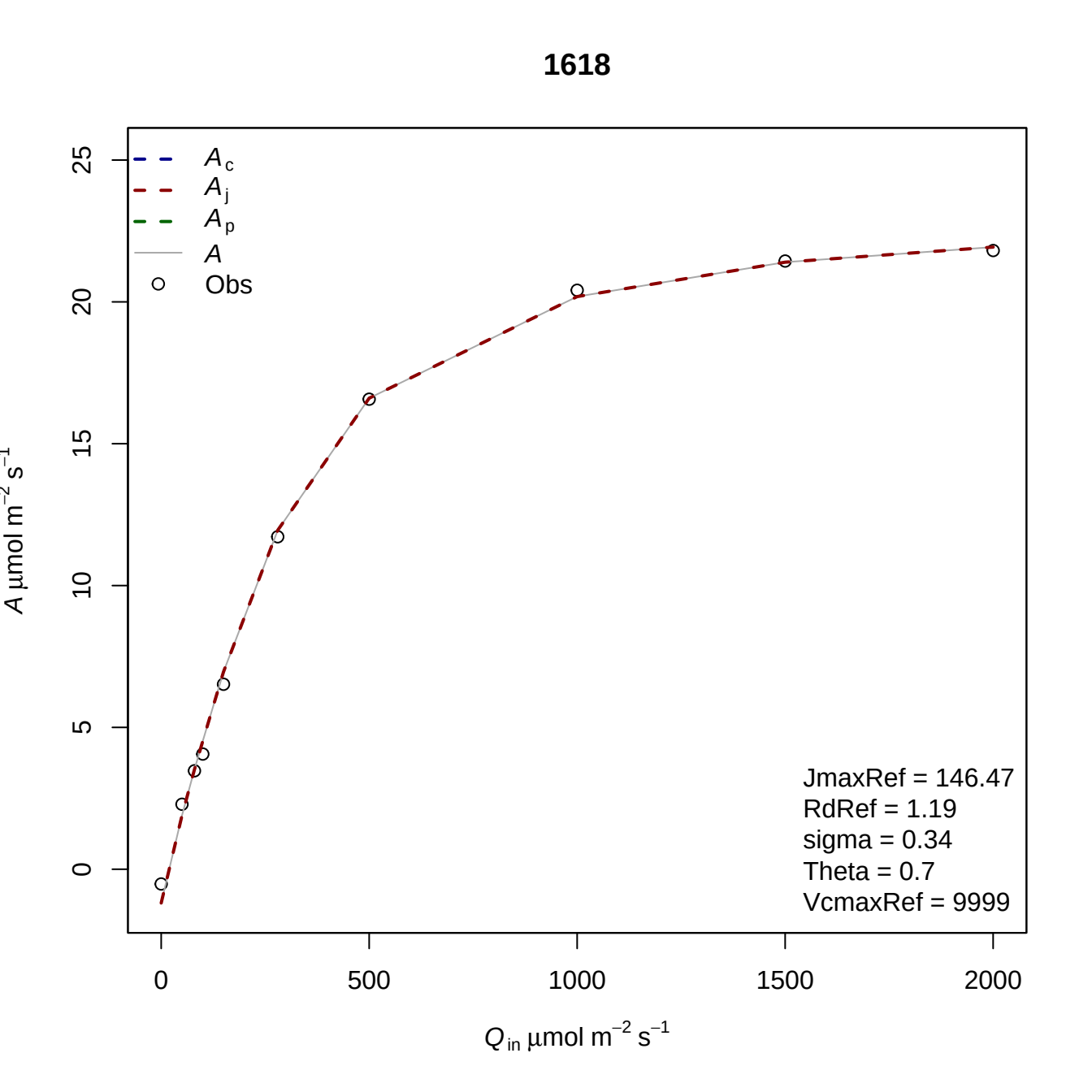

1613
A_c
A_j
A_p
A
Obs
JmaxRef = 65.89
RdRef = 1.51
sigma = 0.42
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
10
0
500
1000
1500
2000

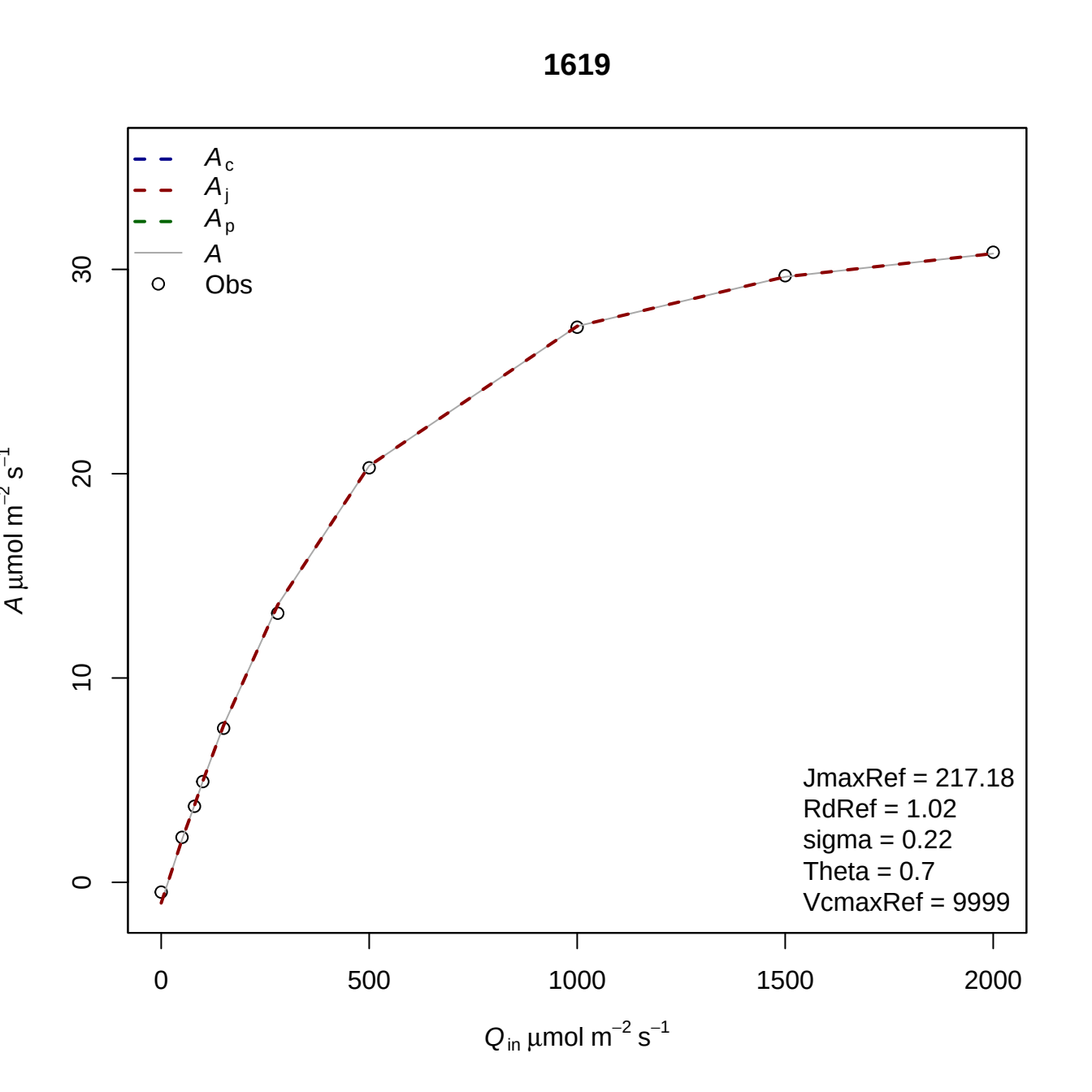

1614
A_c
A_j
A_p
A
Obs
JmaxRef = 165.73
RdRef = 1.08
sigma = 0.35
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

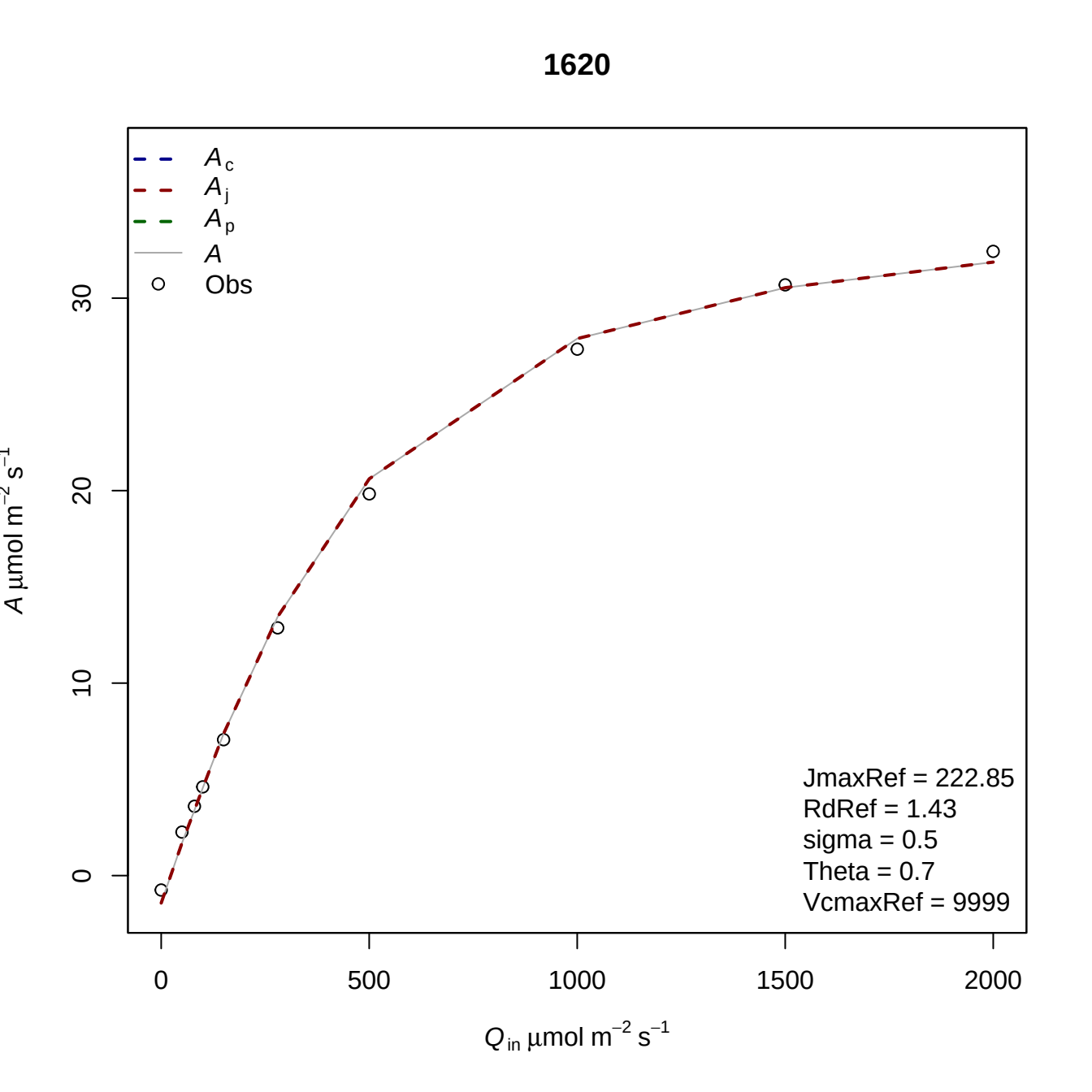

1615
A_c
A_j
A_p
A
Obs
JmaxRef = 142.33
RdRef = 1.39
sigma = 0.42
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

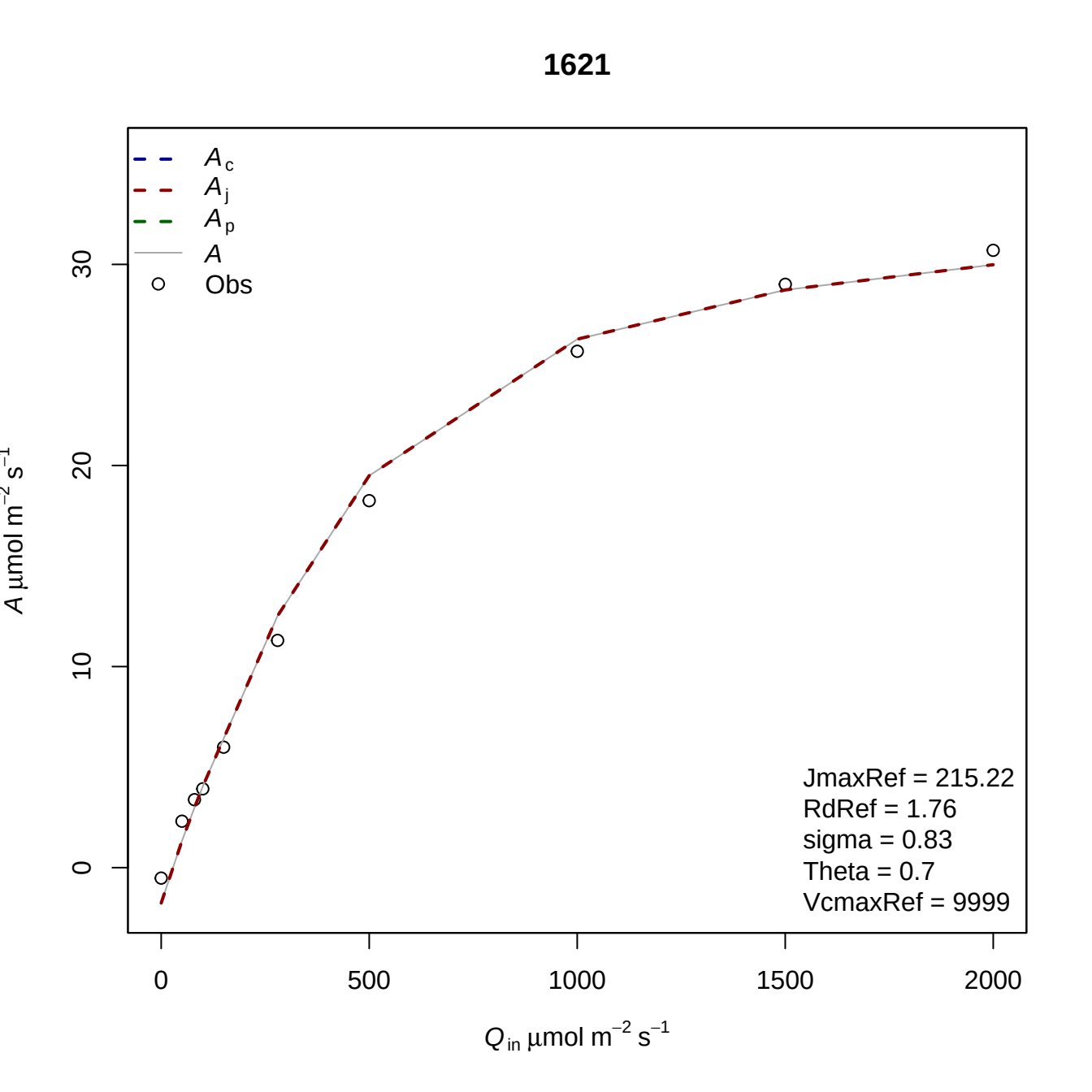

1616
A_c
A_j
A_p
A
Obs
JmaxRef = 164.58
RdRef = 1.09
sigma = 0.25
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

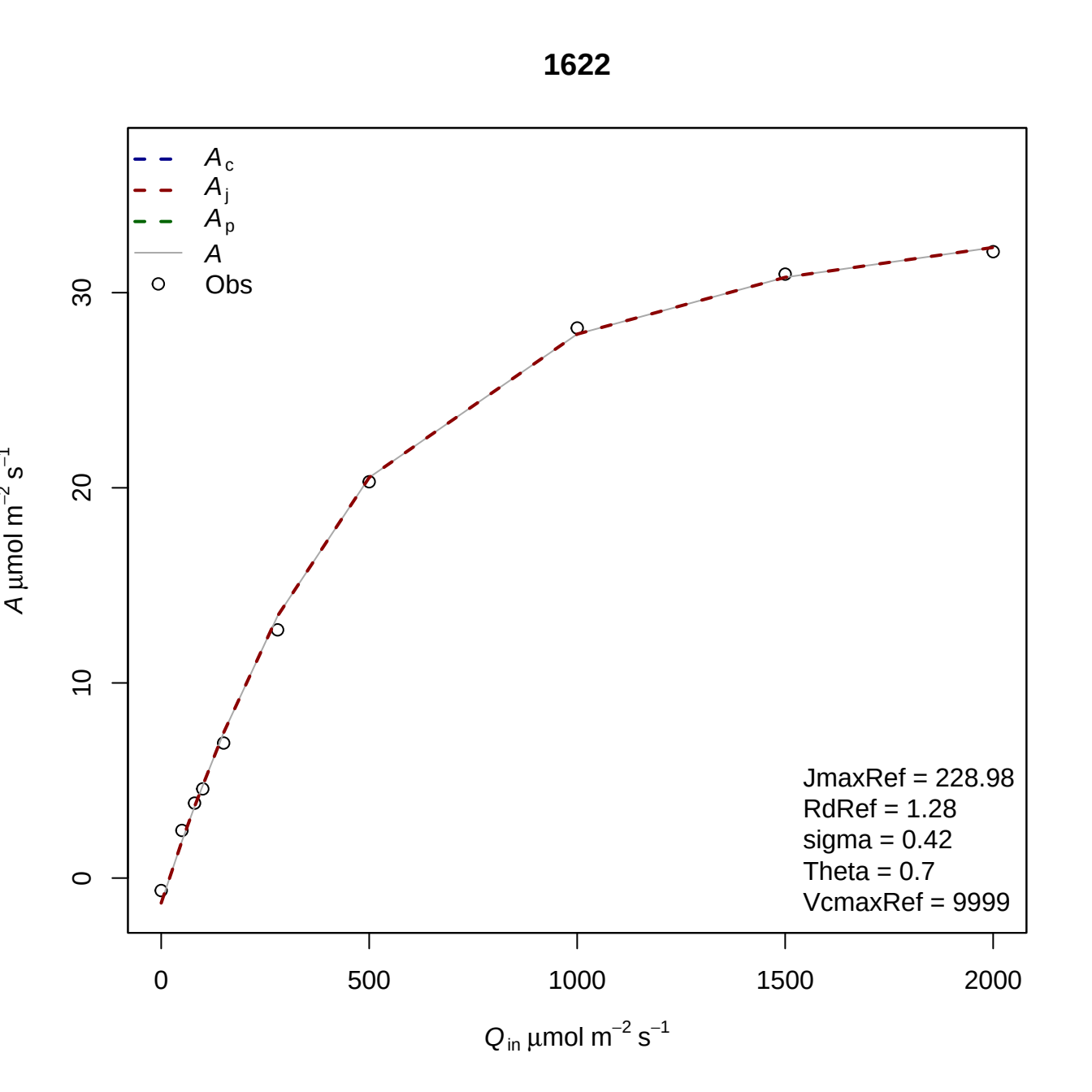

1617
A_c
A_j
A_p
A
Obs
JmaxRef = 136.95
RdRef = 1.5
sigma = 0.43
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000

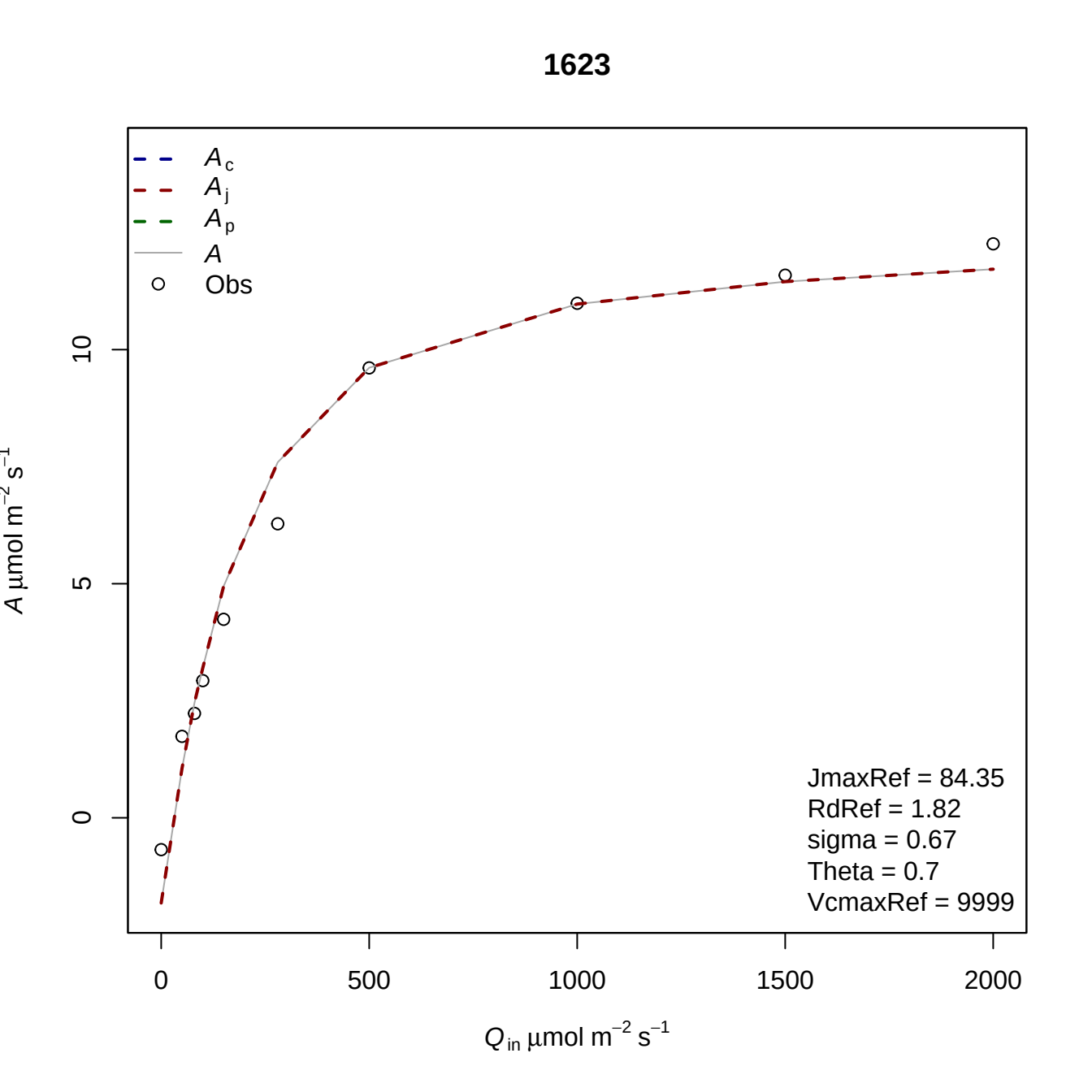

1618
A_c
A_j
A_p
A
Obs
JmaxRef = 146.47
RdRef = 1.19
sigma = 0.34
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

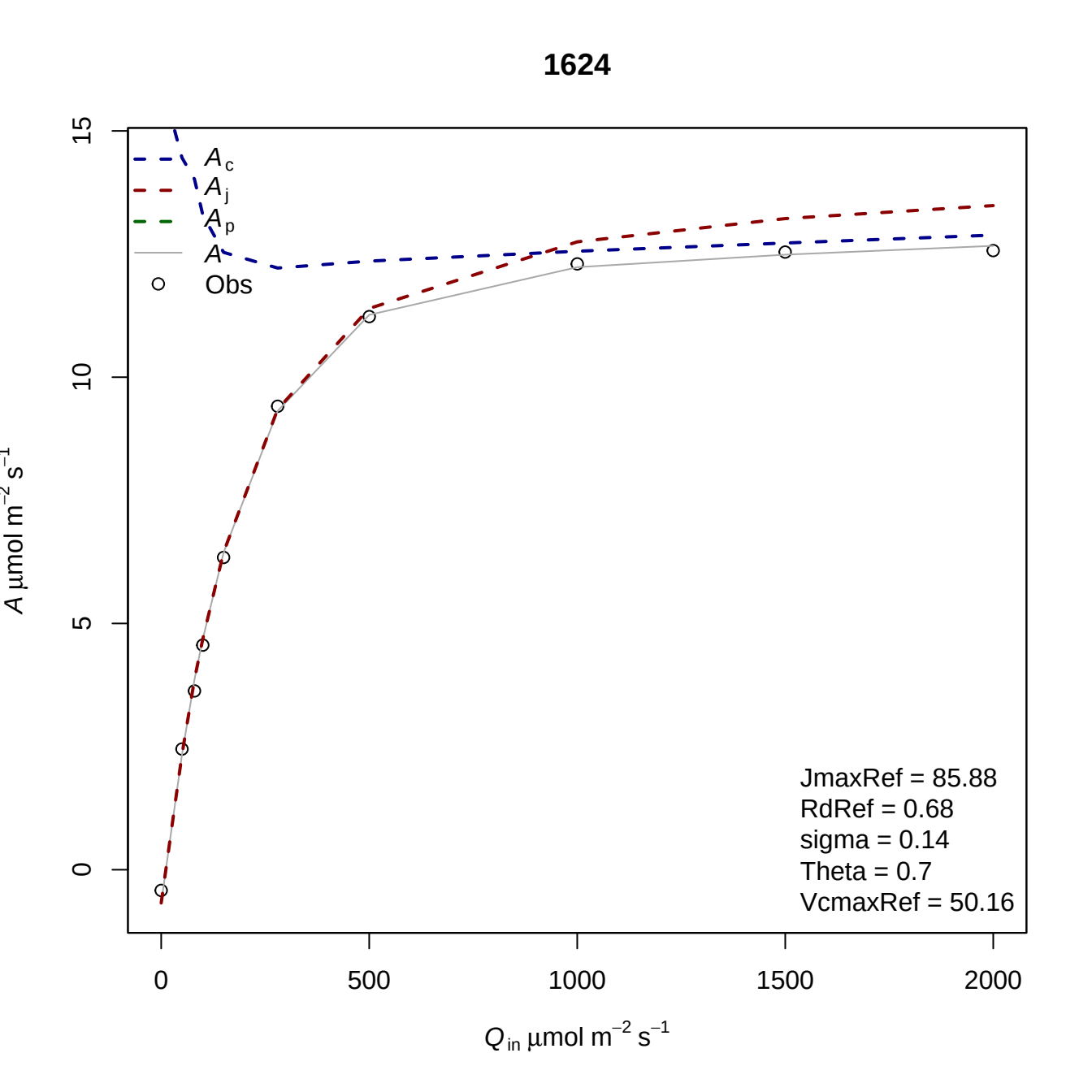

1619
A_c
A_j
A_p
A
Obs
JmaxRef = 217.18
RdRef = 1.02
sigma = 0.22
Theta = 0.7
VcmaxRef = 9999
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

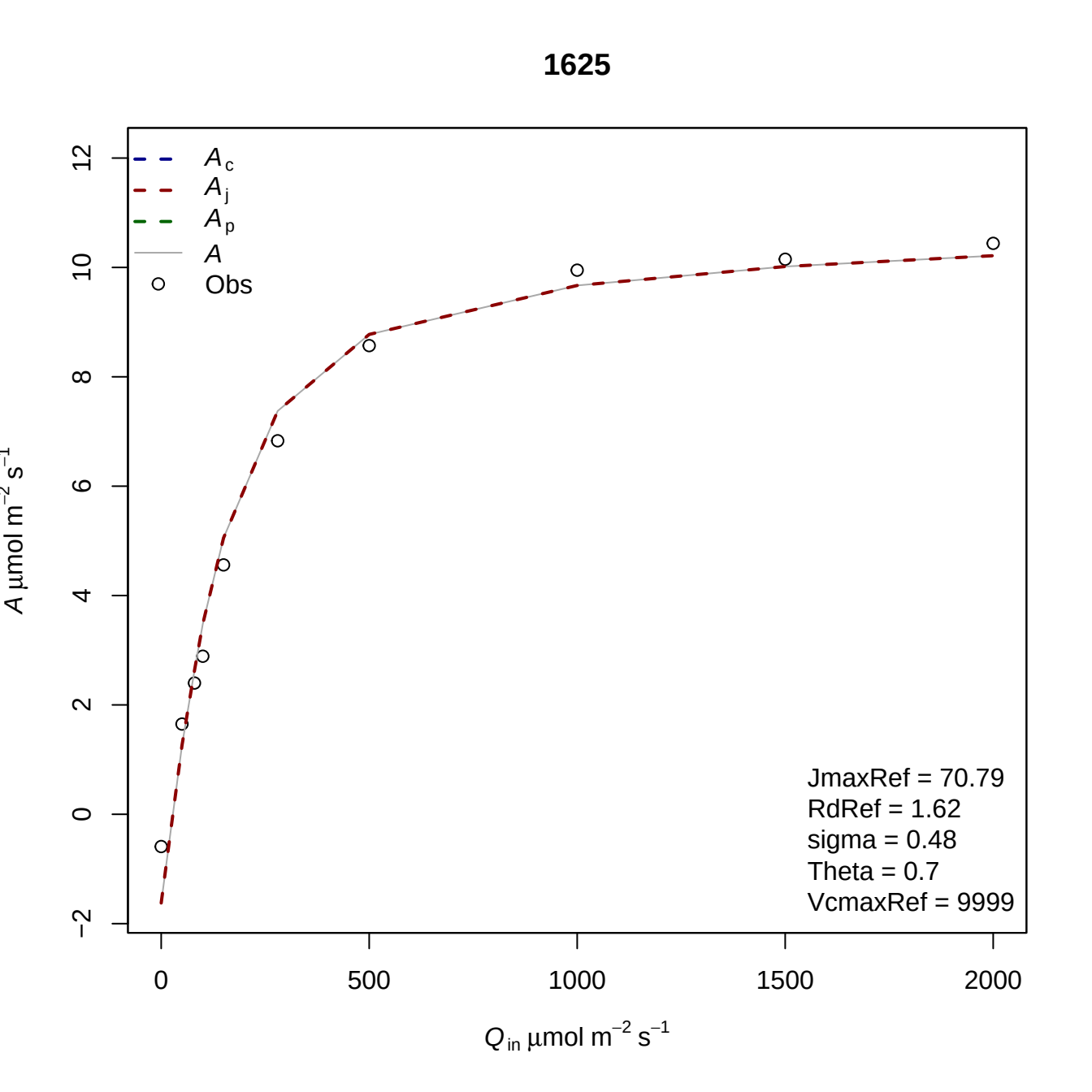

1620
A_c
A_j
A_p
A
Obs
JmaxRef = 222.85
RdRef = 1.43
sigma = 0.5
Theta = 0.7
VcmaxRef = 9999
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

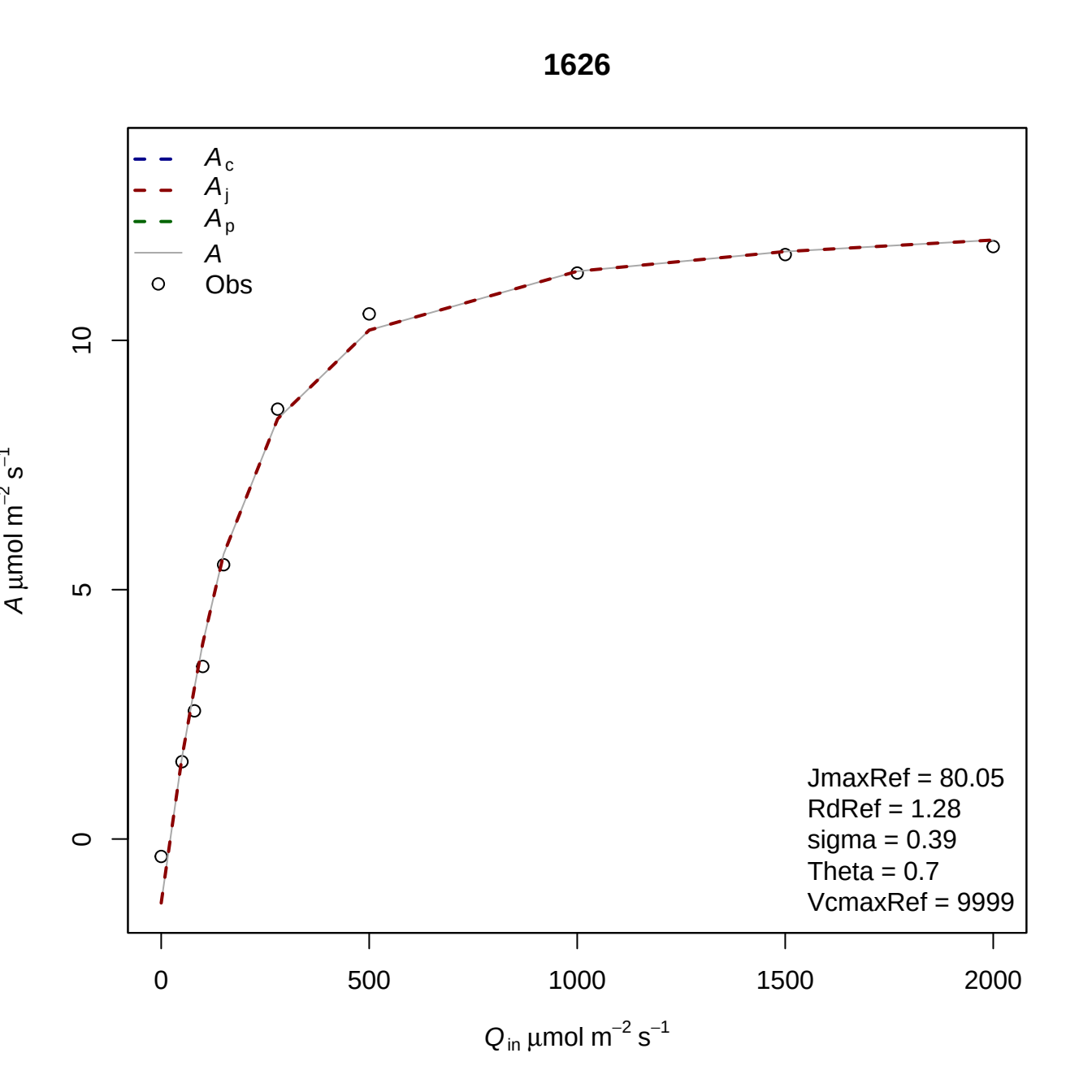

1621
A_c
A_j
A_p
A
Obs
JmaxRef = 215.22
RdRef = 1.76
sigma = 0.83
Theta = 0.7
VcmaxRef = 9999
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

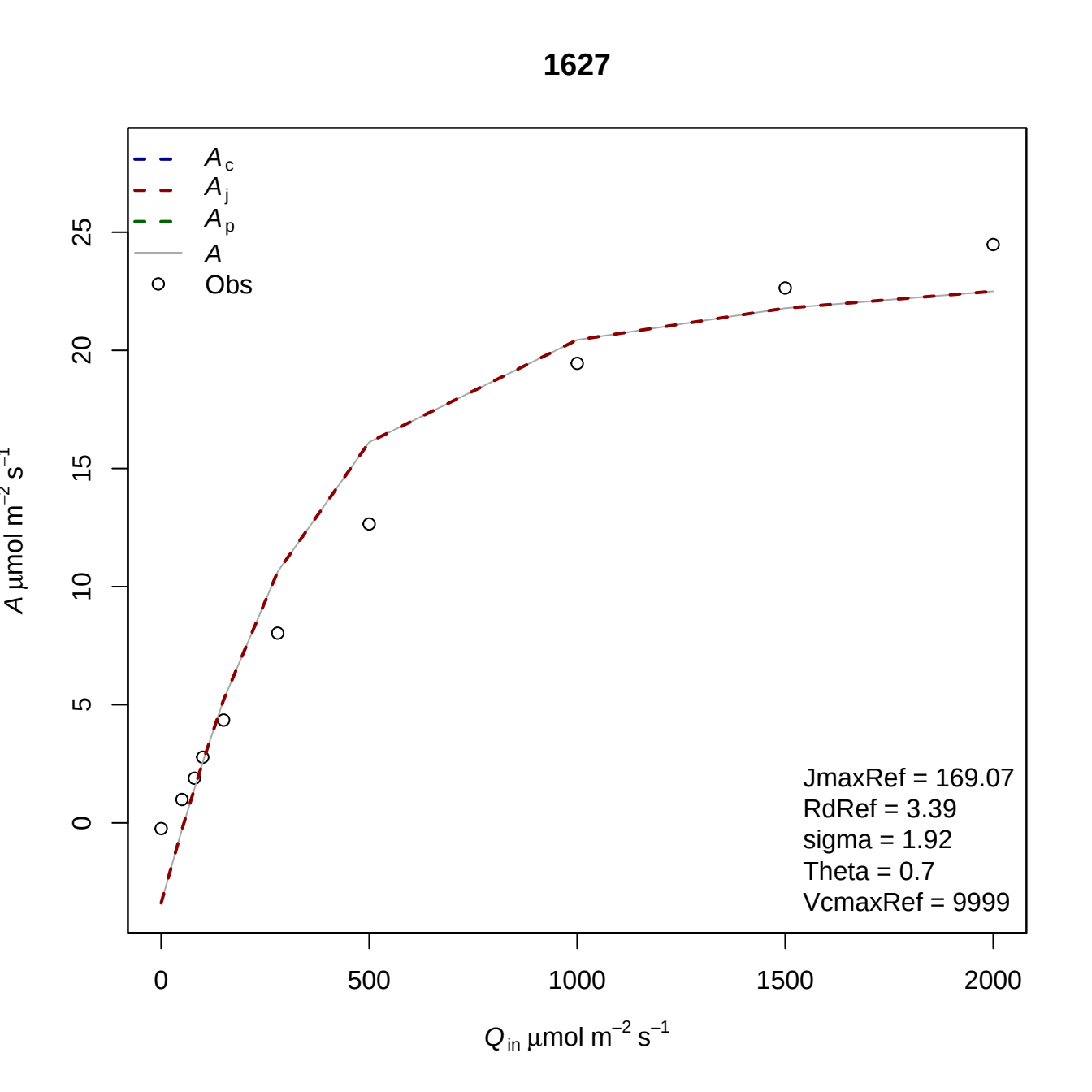

1622
A_c
A_j
A_p
A
Obs
JmaxRef = 228.98
RdRef = 1.28
sigma = 0.42
Theta = 0.7
VcmaxRef = 9999
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

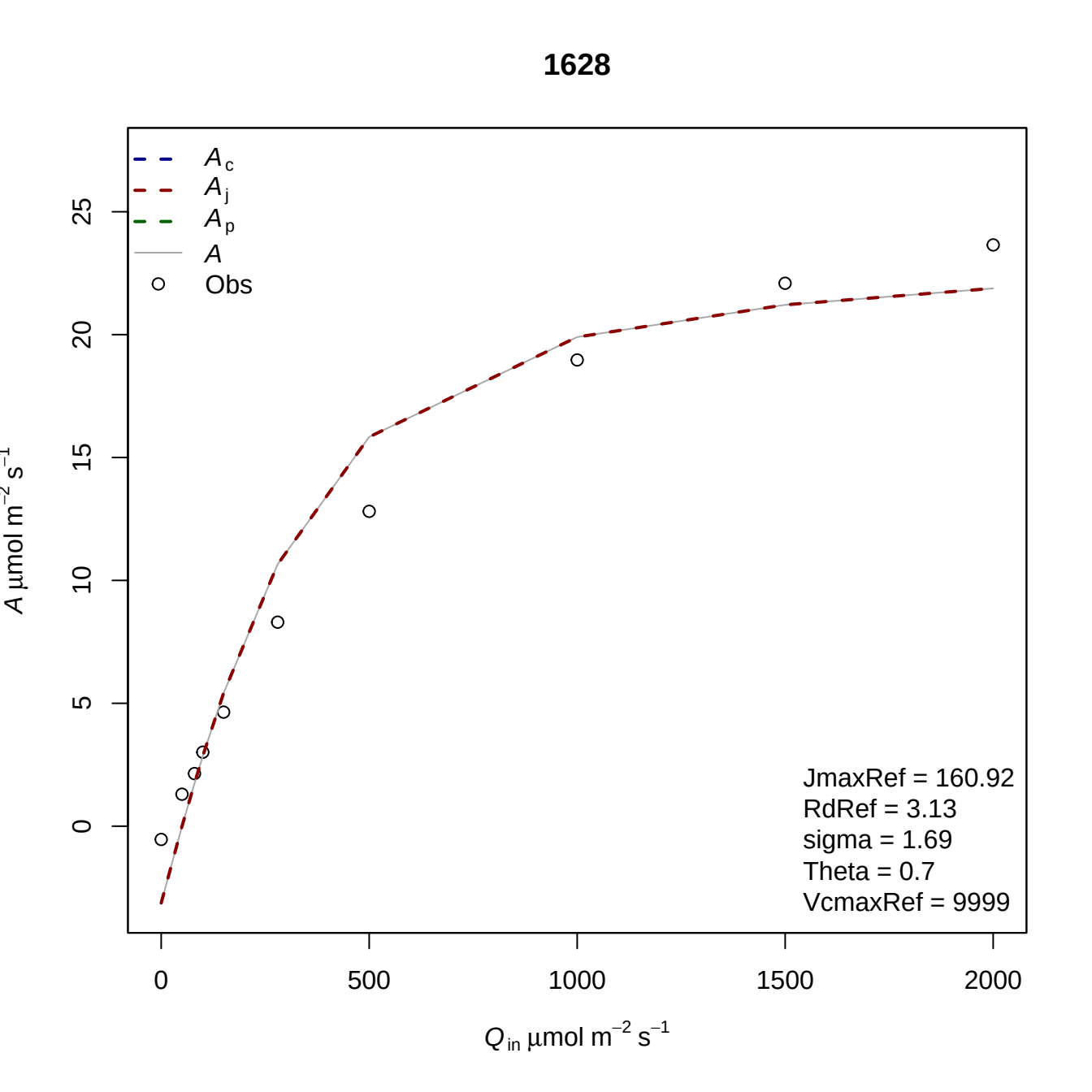

1623
A_c
A_j
A_p
A
Obs
JmaxRef = 84.35
RdRef = 1.82
sigma = 0.67
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

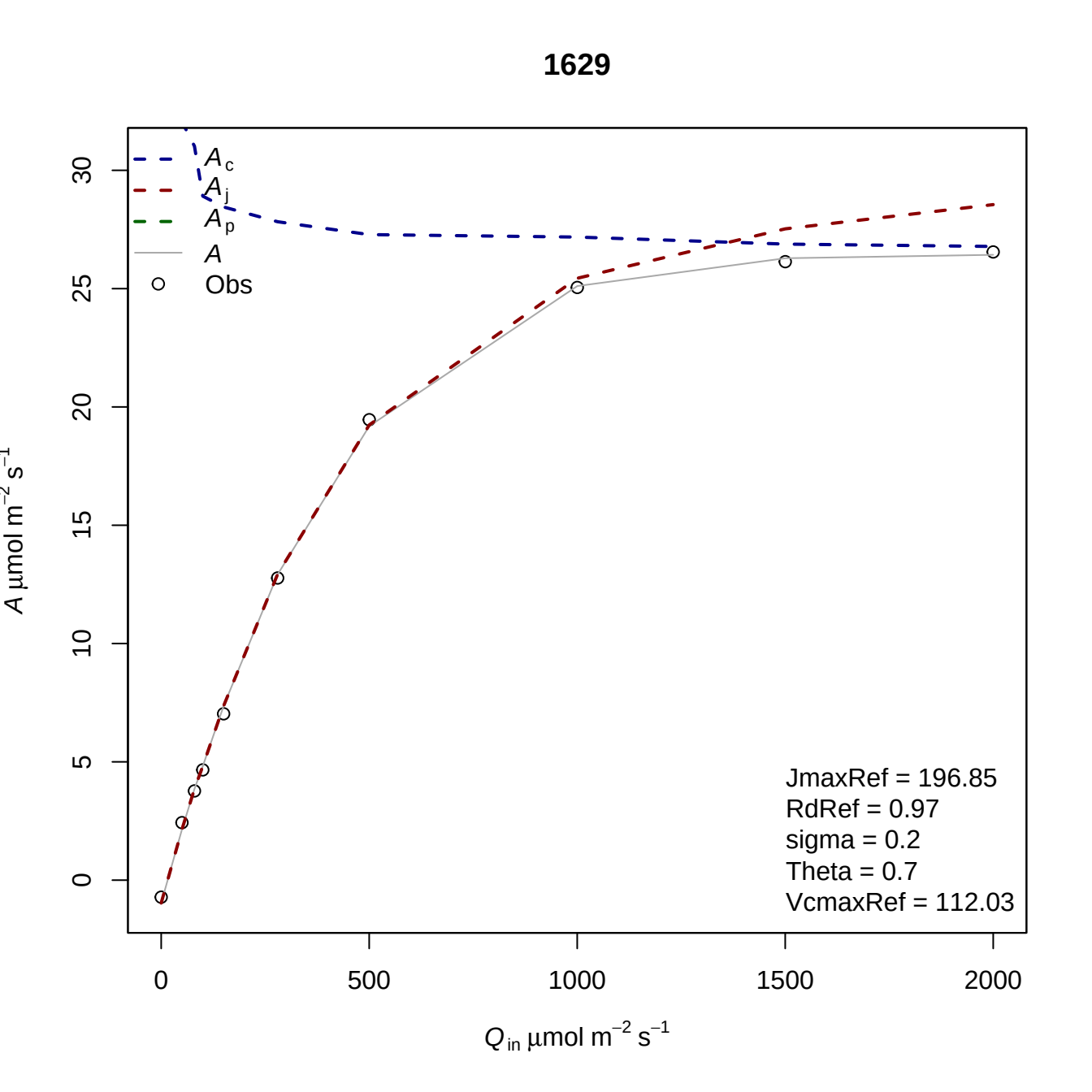

1624
A_c
A_j
A_p
A
Obs
JmaxRef = 85.88
RdRef = 0.68
sigma = 0.14
Theta = 0.7
VcmaxRef = 50.16
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

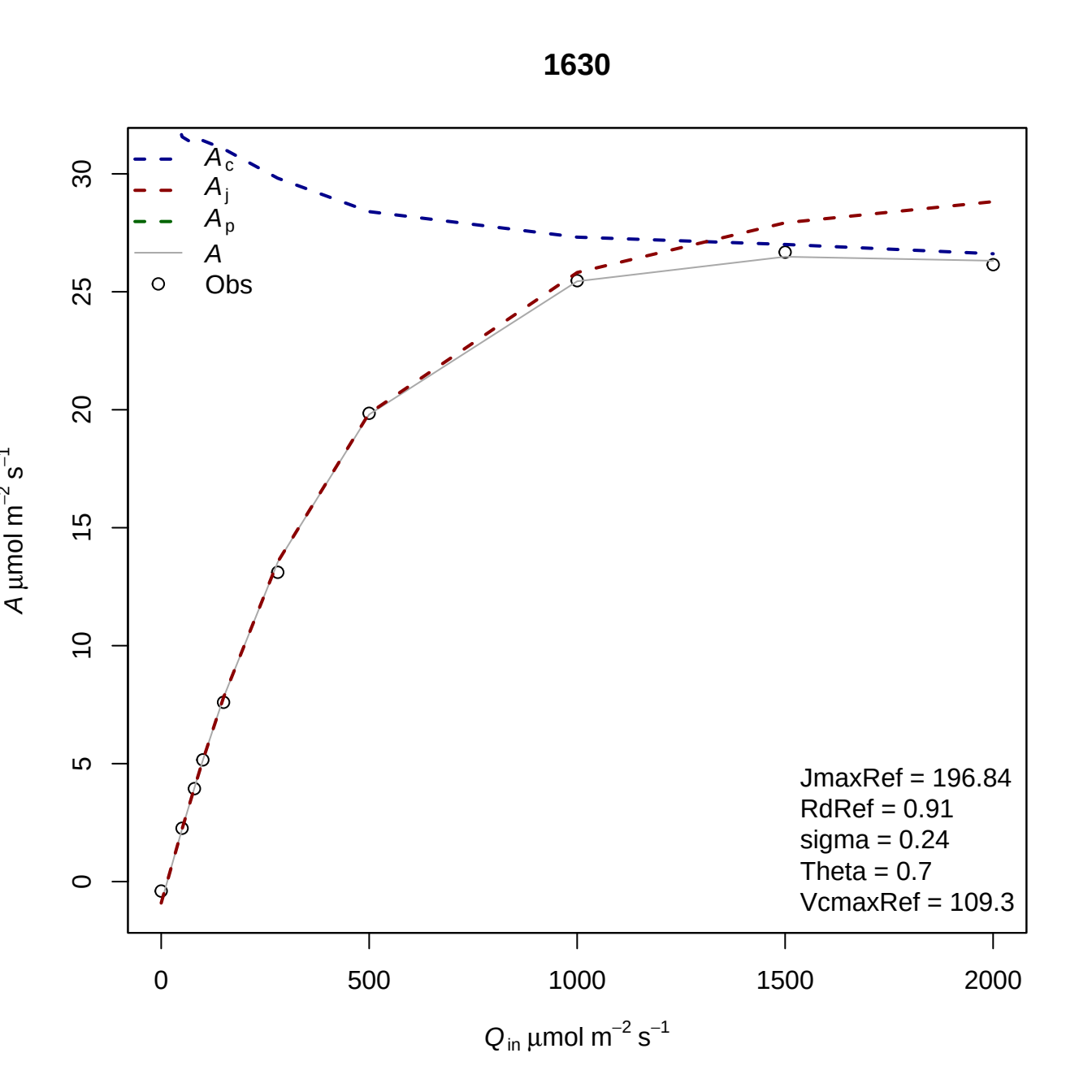

1625
A_c
A_j
A_p
A
Obs
JmaxRef = 70.79
RdRef = 1.62
sigma = 0.48
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
12
10
8
6
4
2
0
−2
0
500
1000
1500
2000

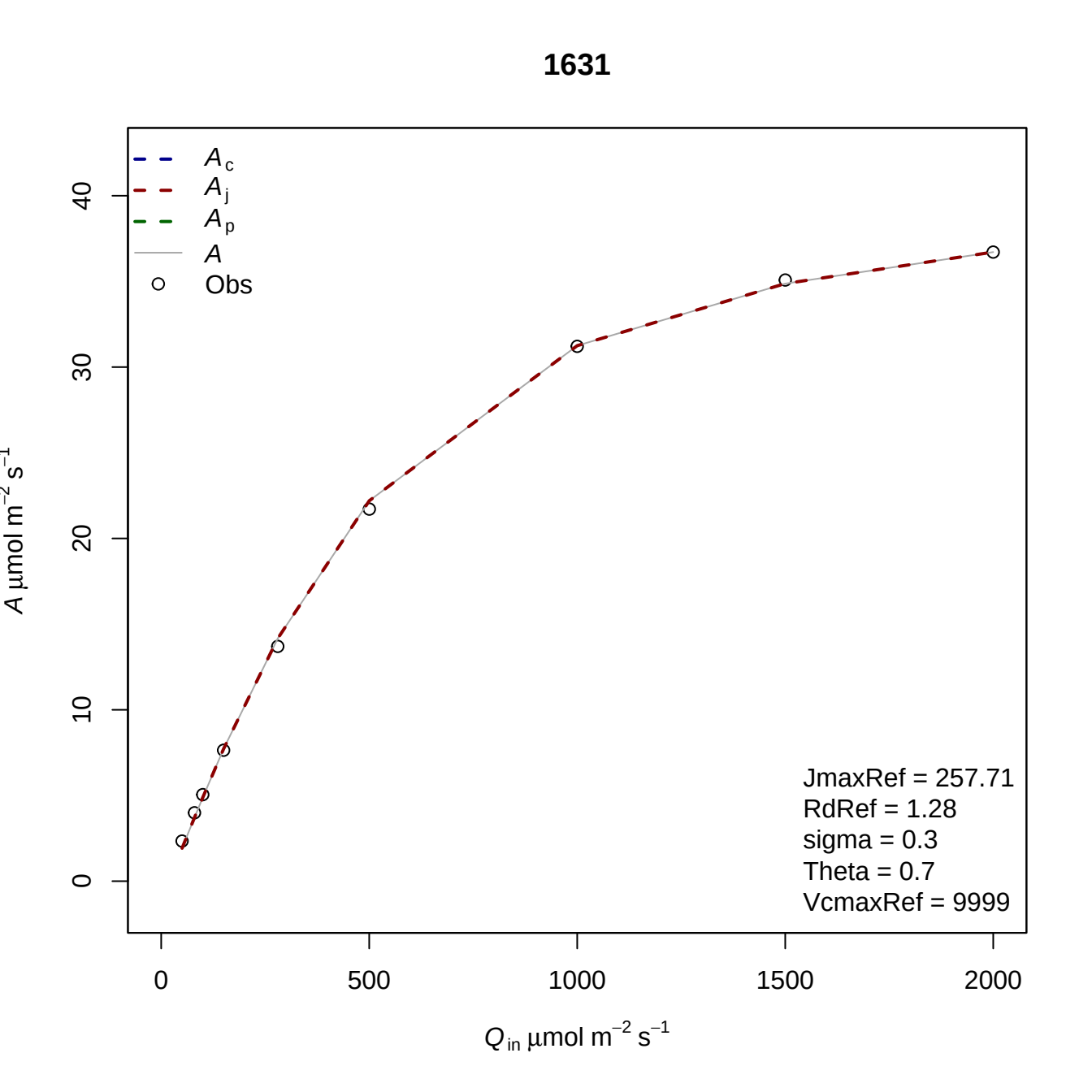

1626
A_c
A_j
A_p
A
Obs
JmaxRef = 80.05
RdRef = 1.28
sigma = 0.39
Theta = 0.7
VcmaxRef = 9999
0
5
10
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

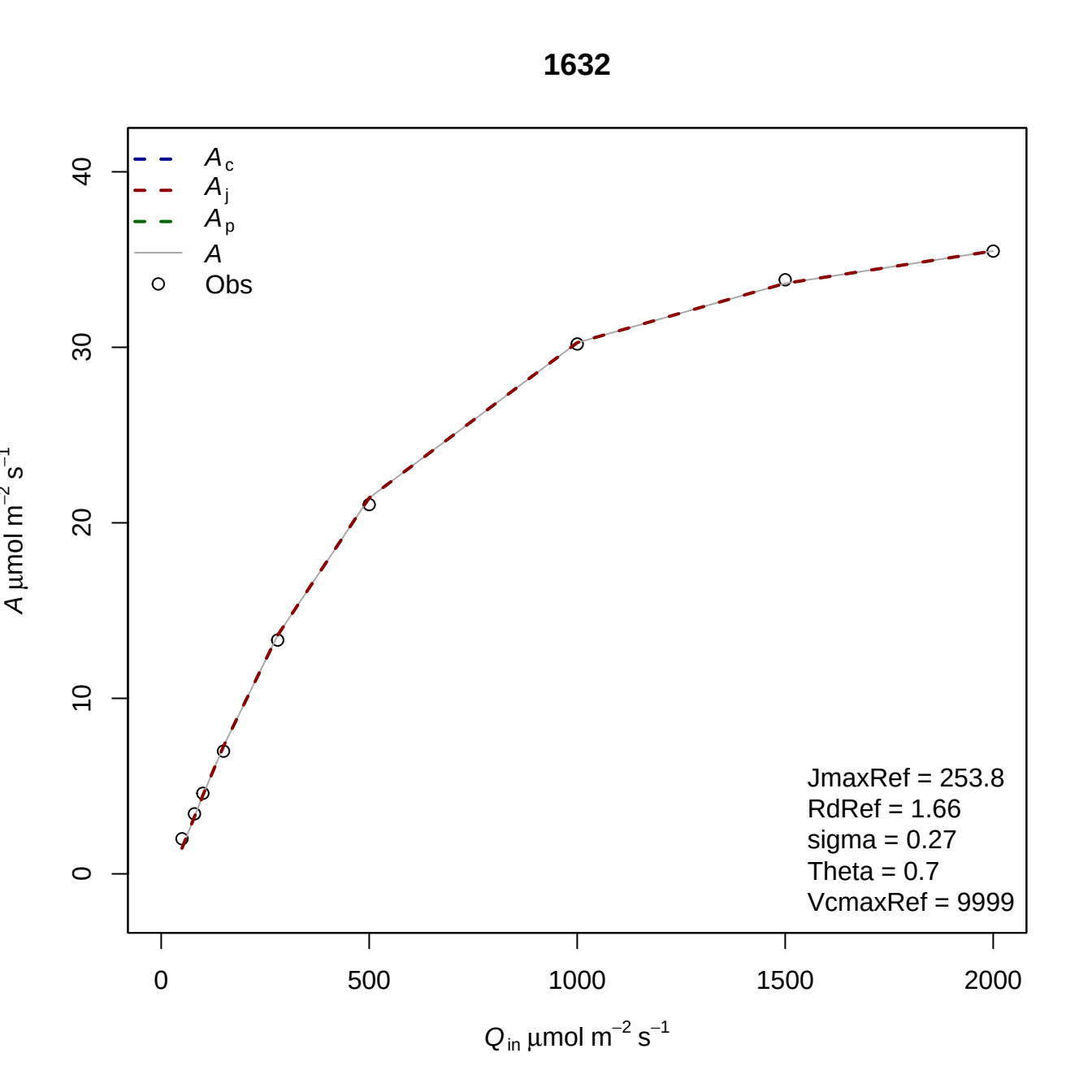

1627
A_c
A_j
A_p
A
Obs
JmaxRef = 169.07
RdRef = 3.39
sigma = 1.92
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

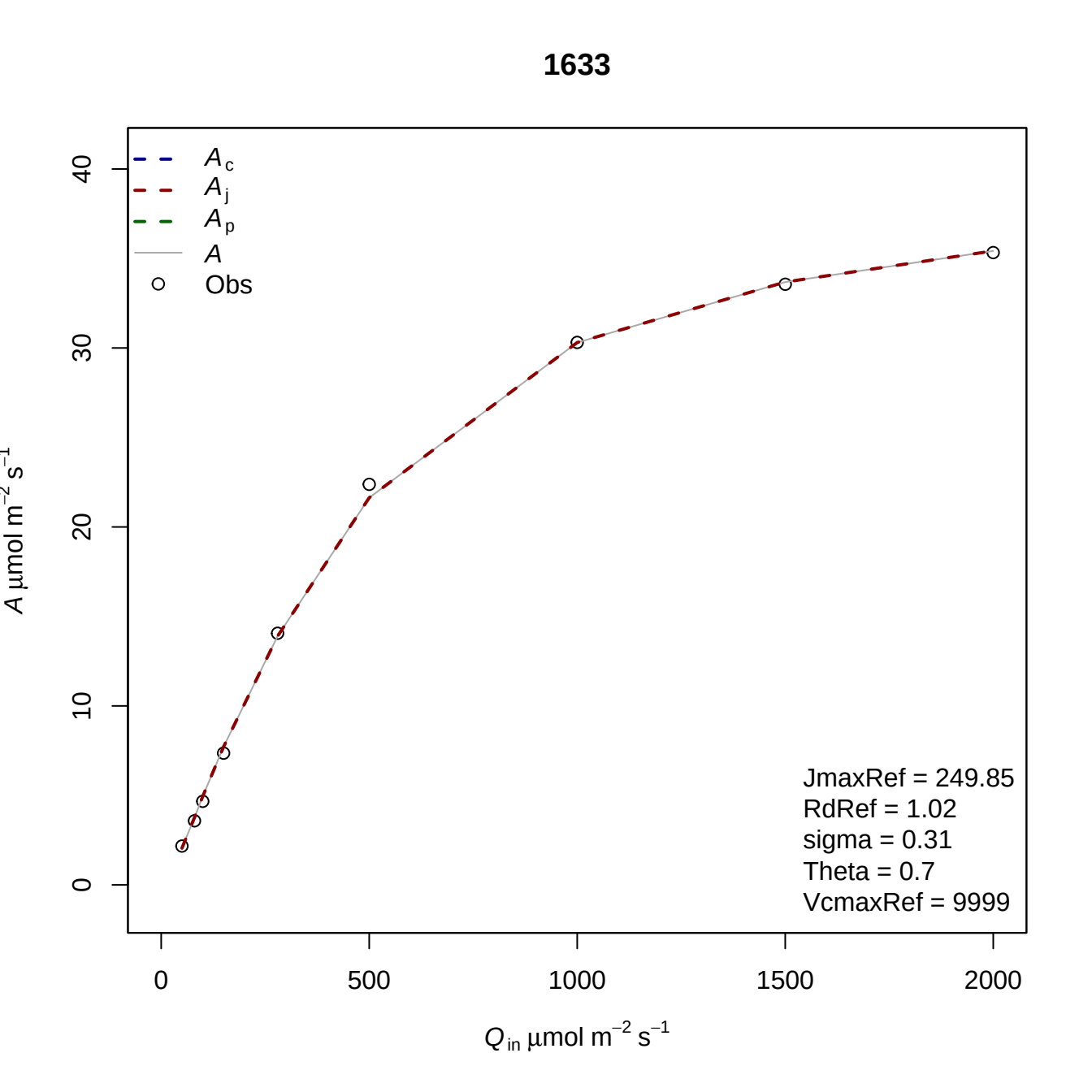

1628
A_c
A_j
A_p
A
Obs
JmaxRef = 160.92
RdRef = 3.13
sigma = 1.69
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

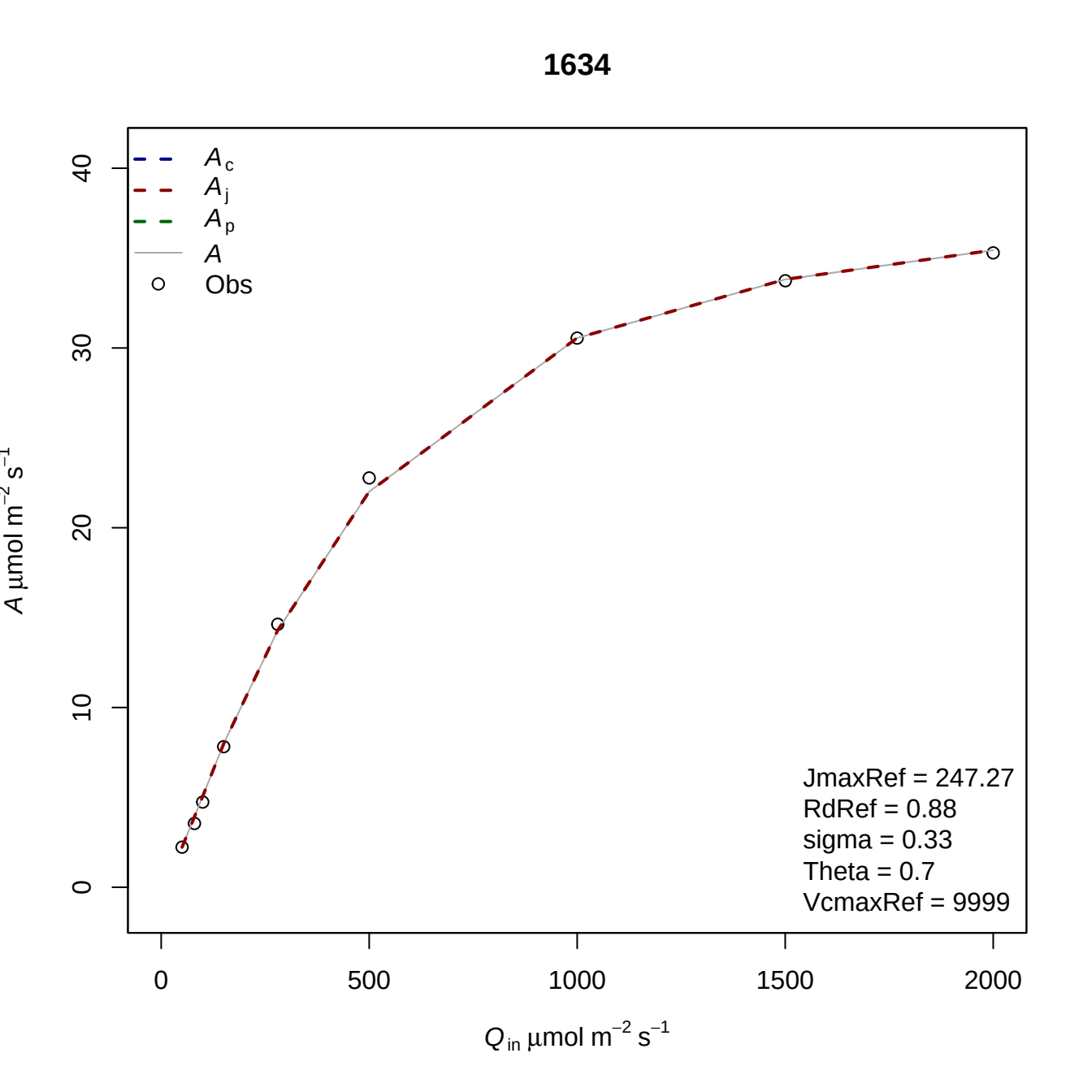

1629
A_c
A_j
A_p
A
Obs
JmaxRef = 196.85
RdRef = 0.97
sigma = 0.2
Theta = 0.7
VcmaxRef = 112.03
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

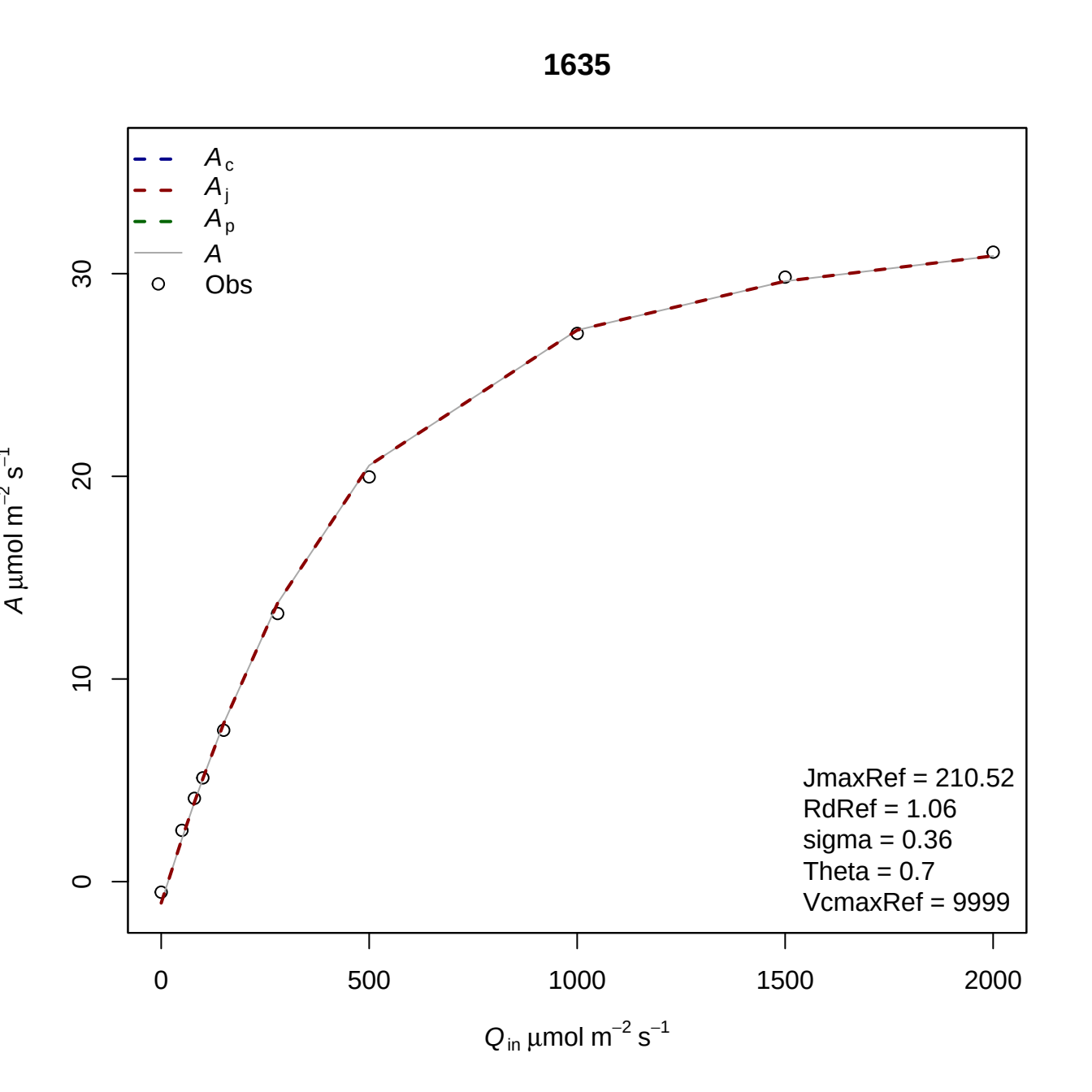

1630
A_c
A_j
A_p
A
Obs
JmaxRef = 196.84
RdRef = 0.91
sigma = 0.24
Theta = 0.7
VcmaxRef = 109.3
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

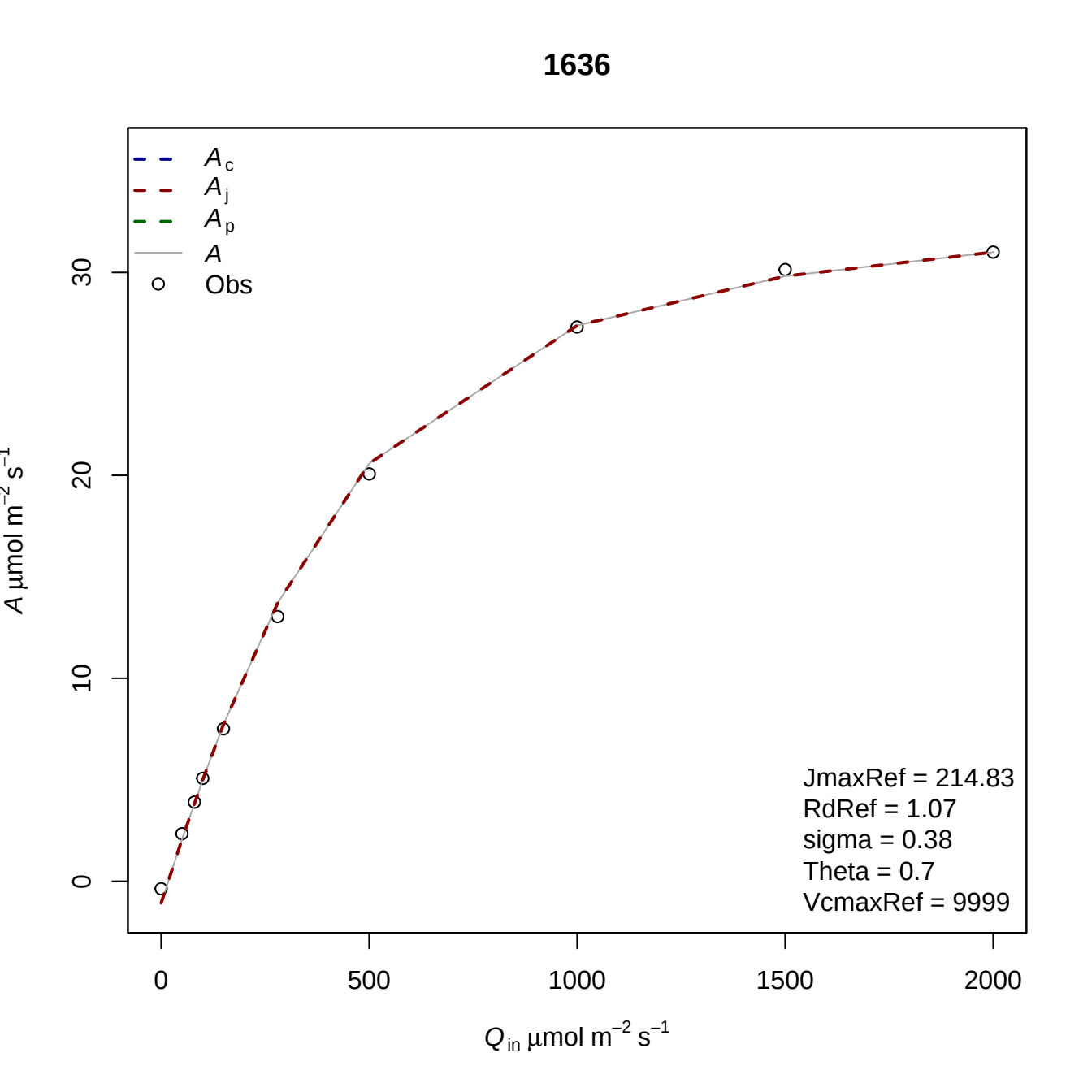

1631
A_c
A_j
A_p
A
Obs
JmaxRef = 257.71
RdRef = 1.28
sigma = 0.3
Theta = 0.7
VcmaxRef = 9999
0
10
20
30
40
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

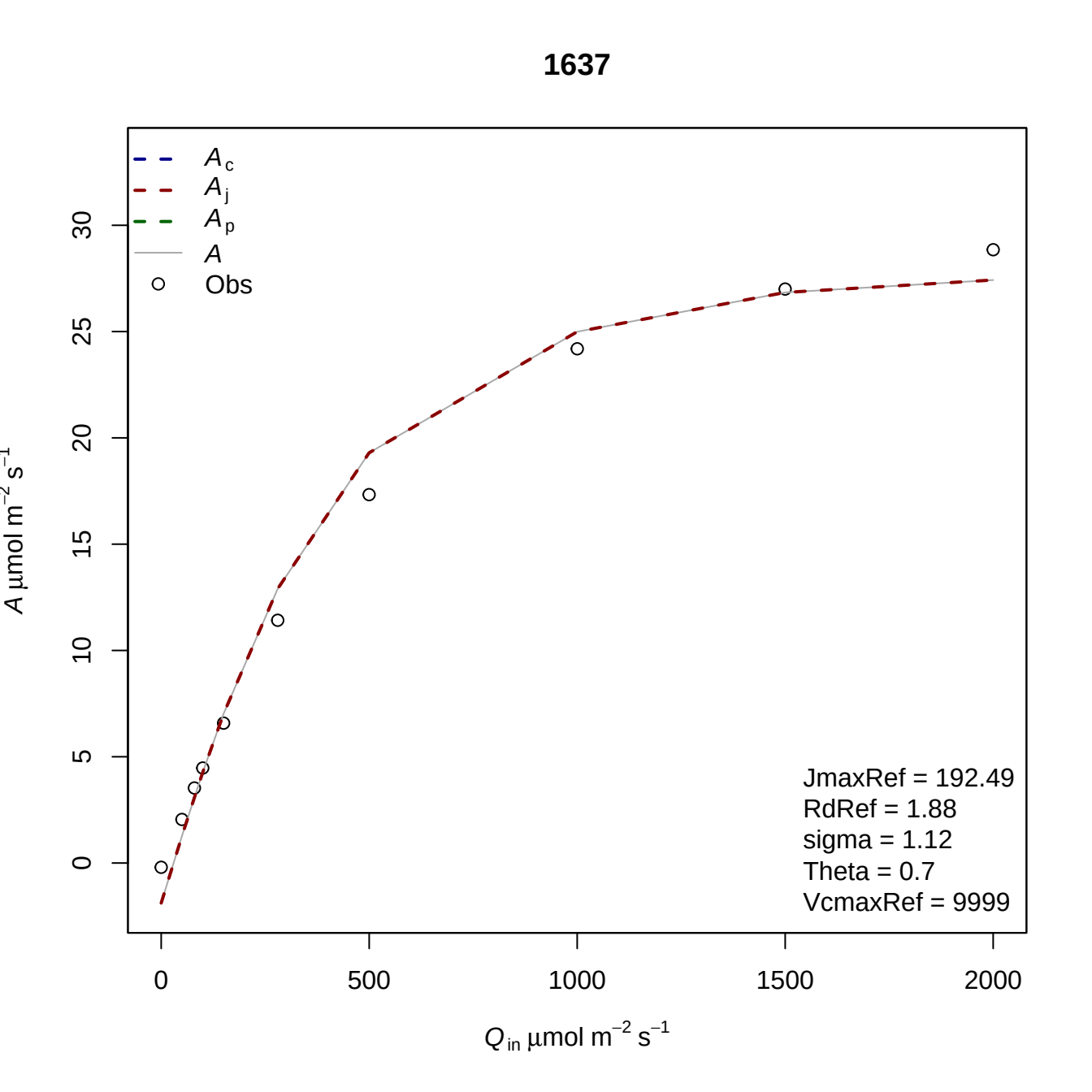

1632
A_c
A_j
A_p
A
Obs
JmaxRef = 253.8
RdRef = 1.66
sigma = 0.27
Theta = 0.7
VcmaxRef = 9999
0
10
20
30
40
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

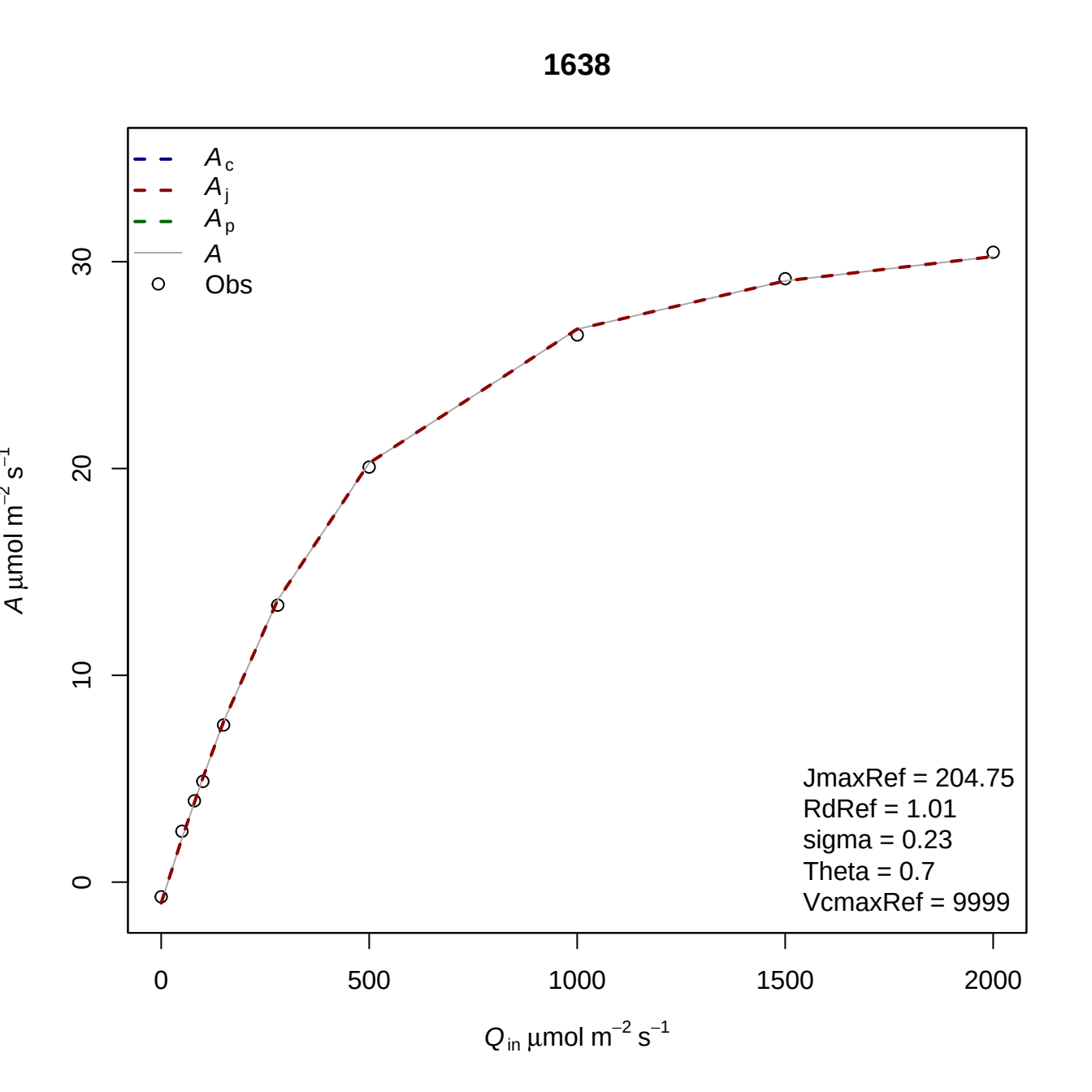

1633
A_c
A_j
A_p
A
Obs
JmaxRef = 249.85
RdRef = 1.02
sigma = 0.31
Theta = 0.7
VcmaxRef = 9999
0
10
20
30
40
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

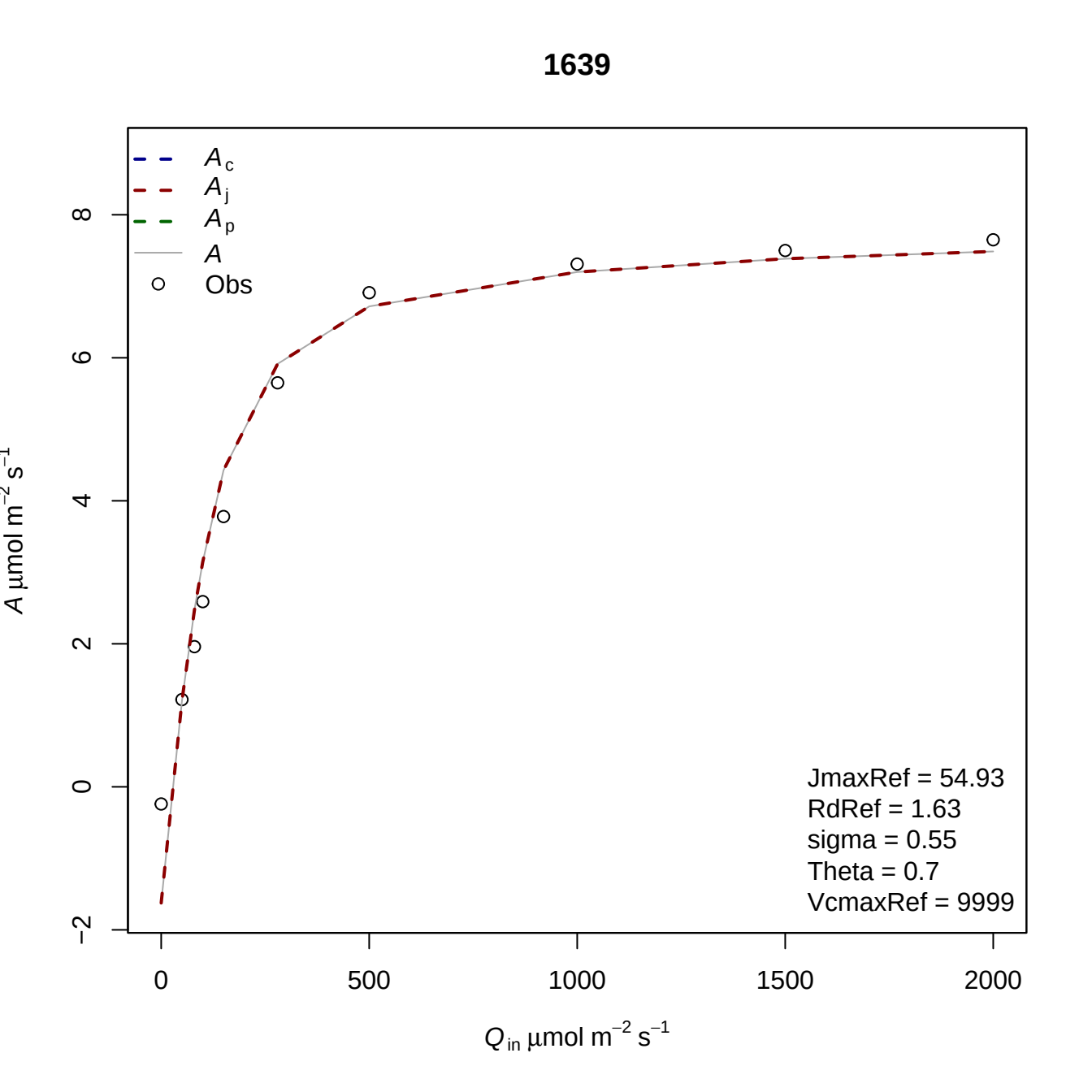

1634
A_c
A_j
A_p
A
Obs
JmaxRef = 247.27
RdRef = 0.88
sigma = 0.33
Theta = 0.7
VcmaxRef = 9999
0
10
20
30
40
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

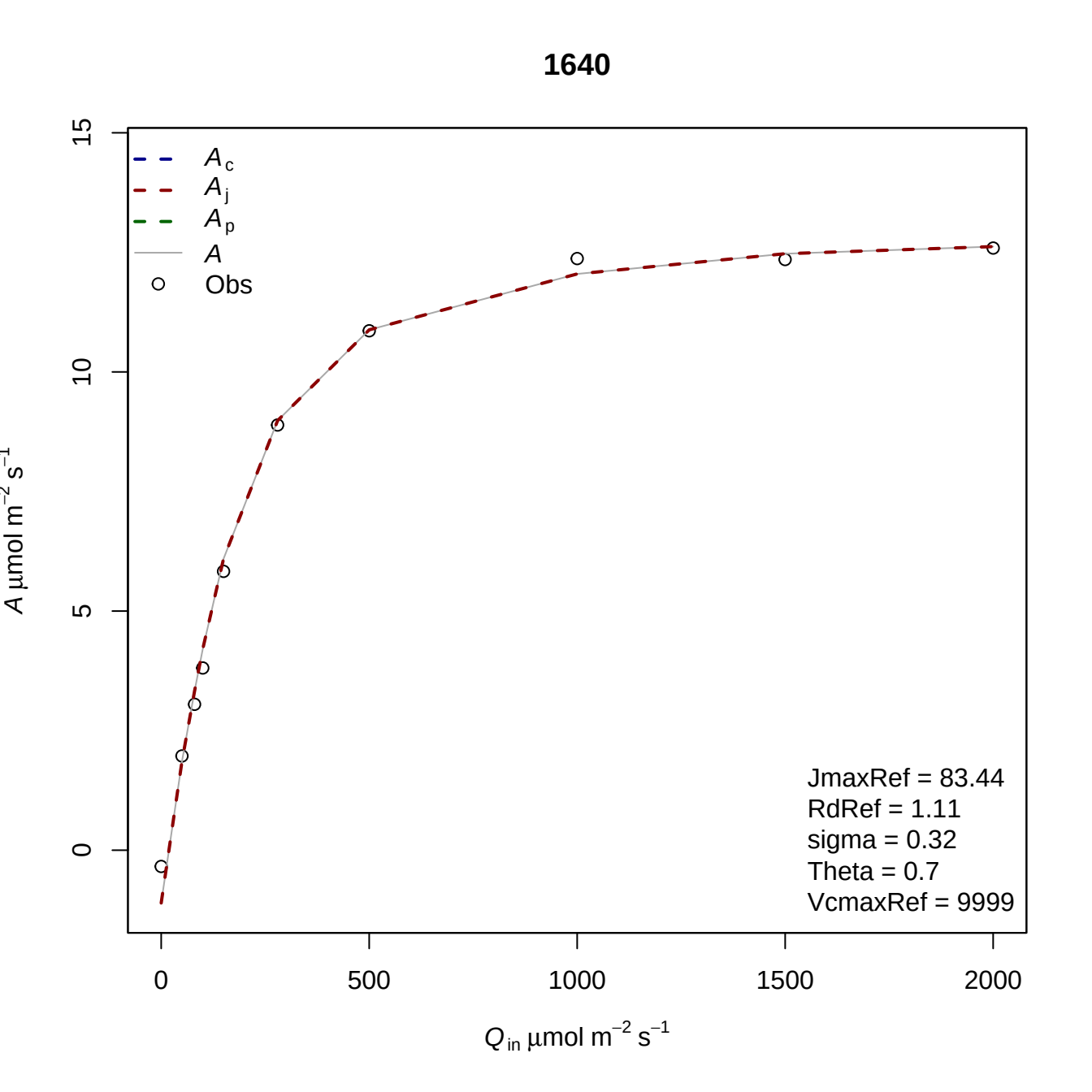

1635
A_c
A_j
A_p
A
Obs
JmaxRef = 210.52
RdRef = 1.06
sigma = 0.36
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
0
500
1000
1500
2000

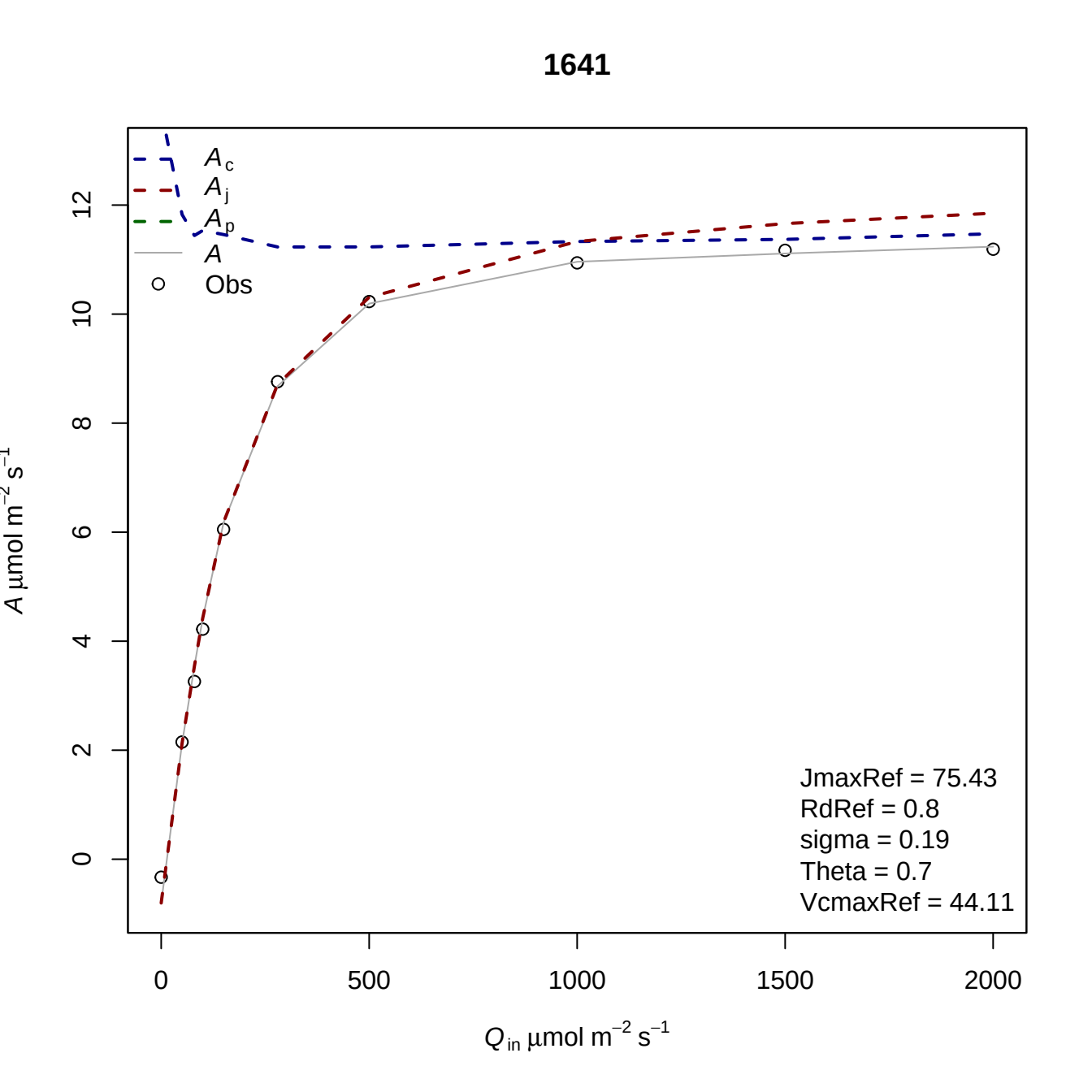

1636
A_c
A_j
A_p
A
Obs
JmaxRef = 214.83
RdRef = 1.07
sigma = 0.38
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
0
500
1000
1500
2000

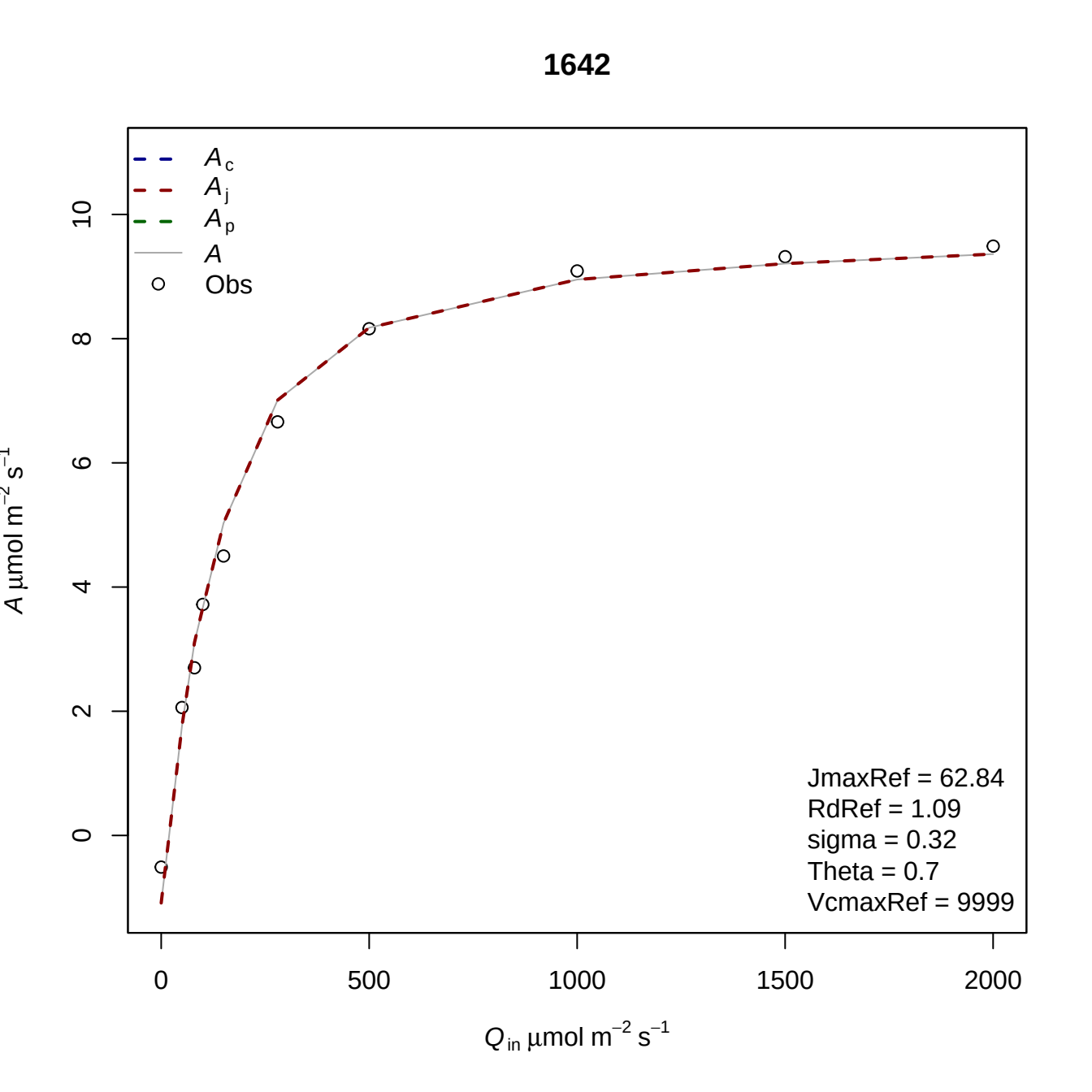

1637
A_c
A_j
A_p
A
Obs
JmaxRef = 192.49
RdRef = 1.88
sigma = 1.12
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

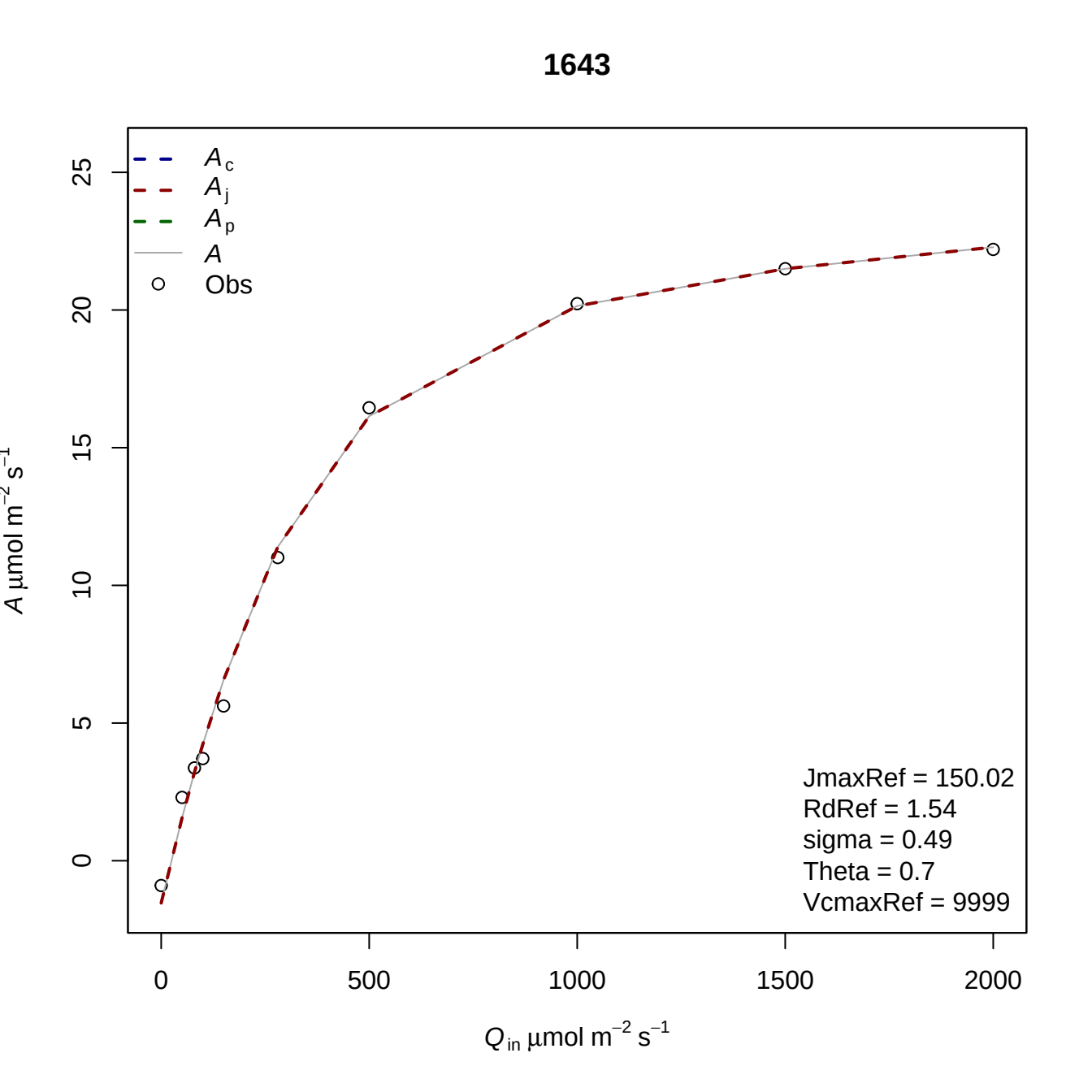

1638
A_c
A_j
A_p
A
Obs
JmaxRef = 204.75
RdRef = 1.01
sigma = 0.23
Theta = 0.7
VcmaxRef = 9999
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

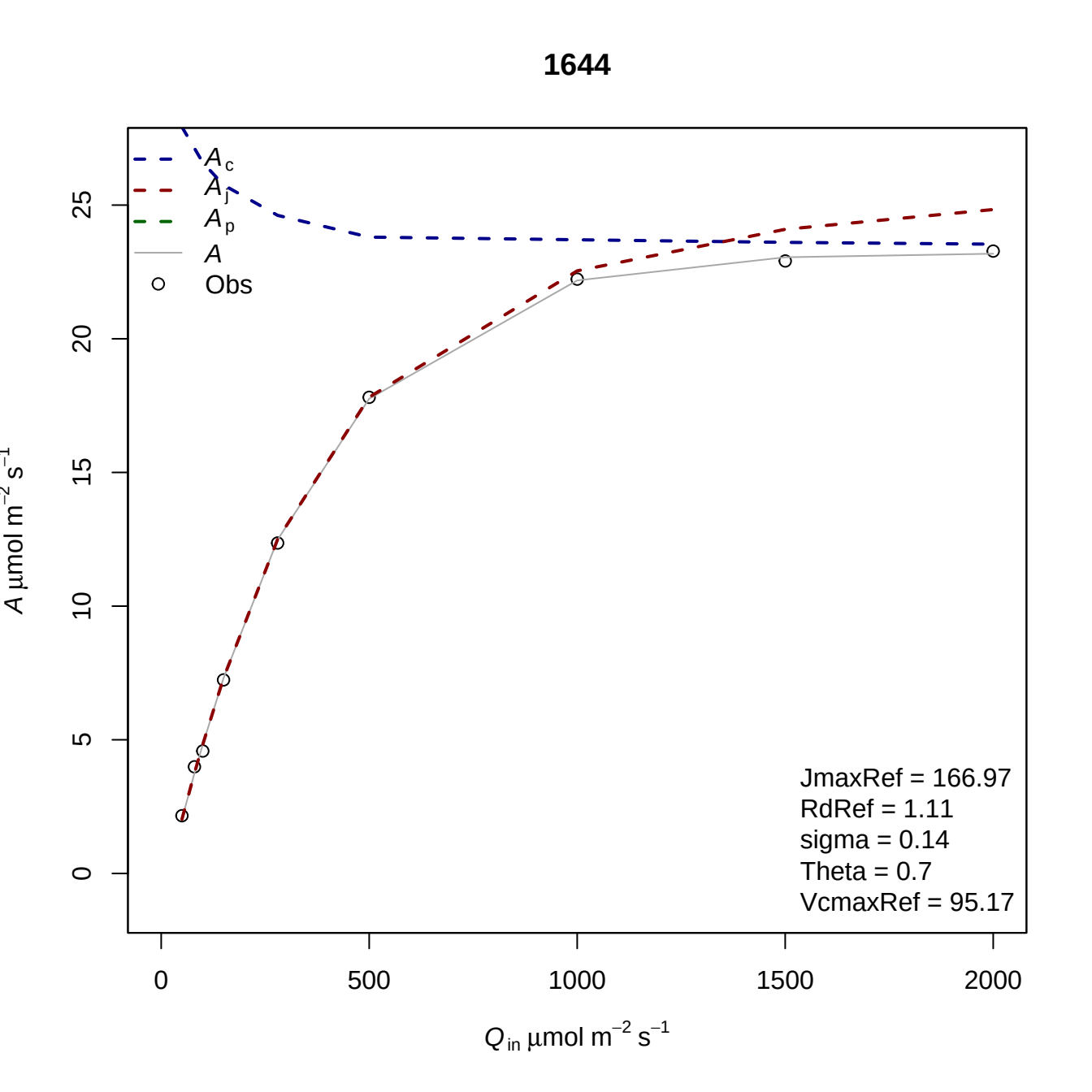

1639
A_c
A_j
A_p
A
Obs
JmaxRef = 54.93
RdRef = 1.63
sigma = 0.55
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
8
0
500
1000
1500
2000

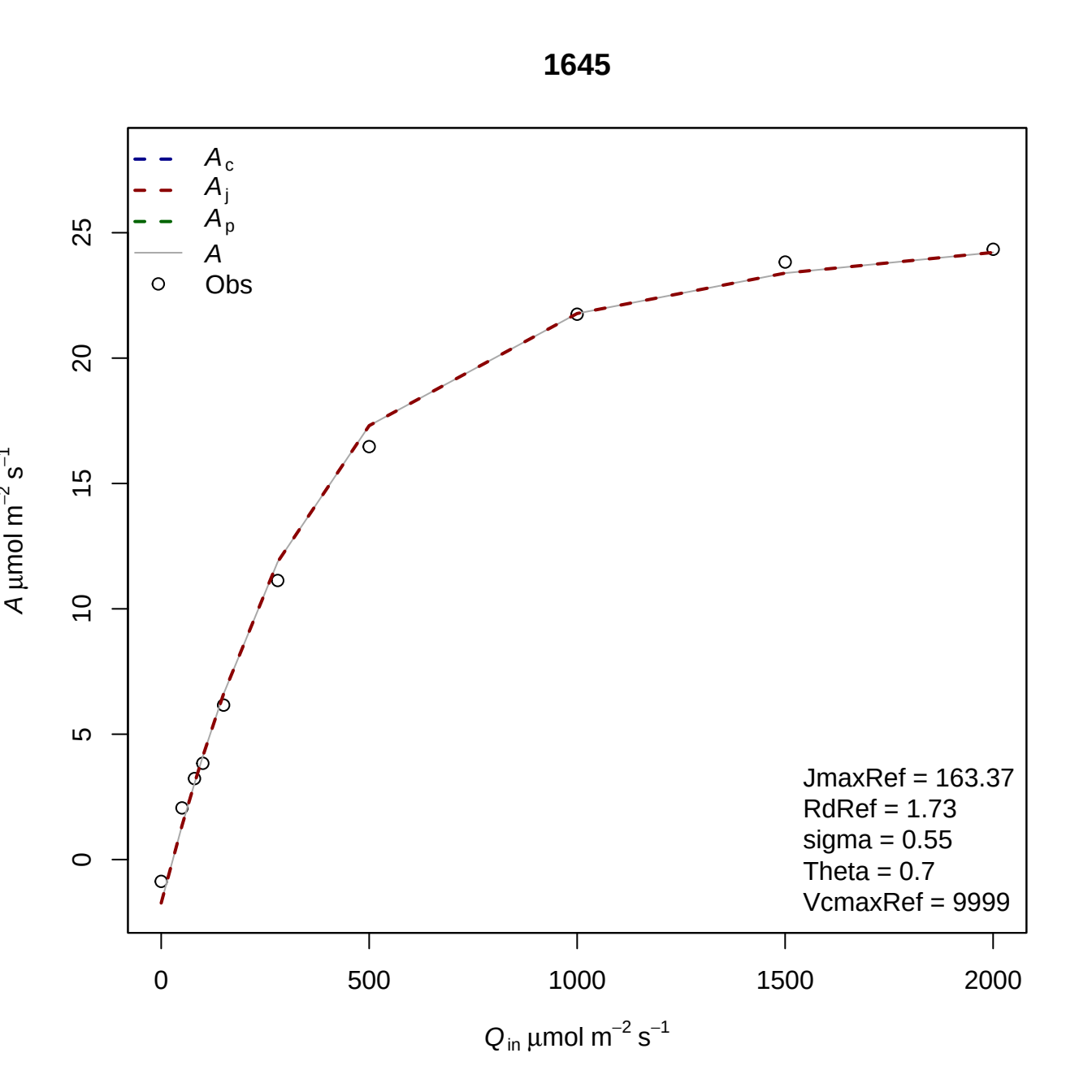

1640
A_c
A_j
A_p
A
Obs
JmaxRef = 83.44
RdRef = 1.11
sigma = 0.32
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

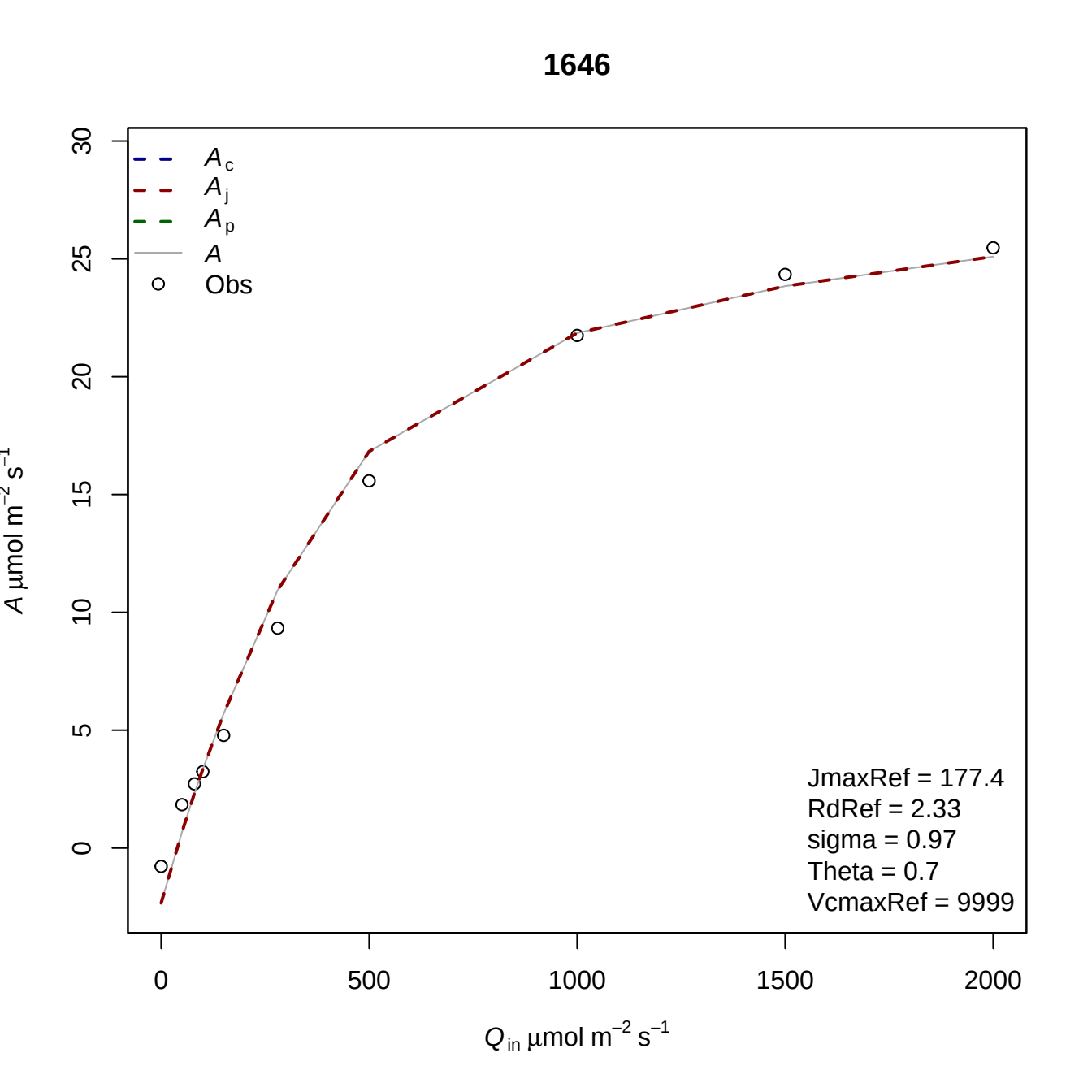

1641
A_c
A_j
A_p
A
Obs
JmaxRef = 75.43
RdRef = 0.8
sigma = 0.19
Theta = 0.7
VcmaxRef = 44.11
0
2
4
6
8
10
12
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

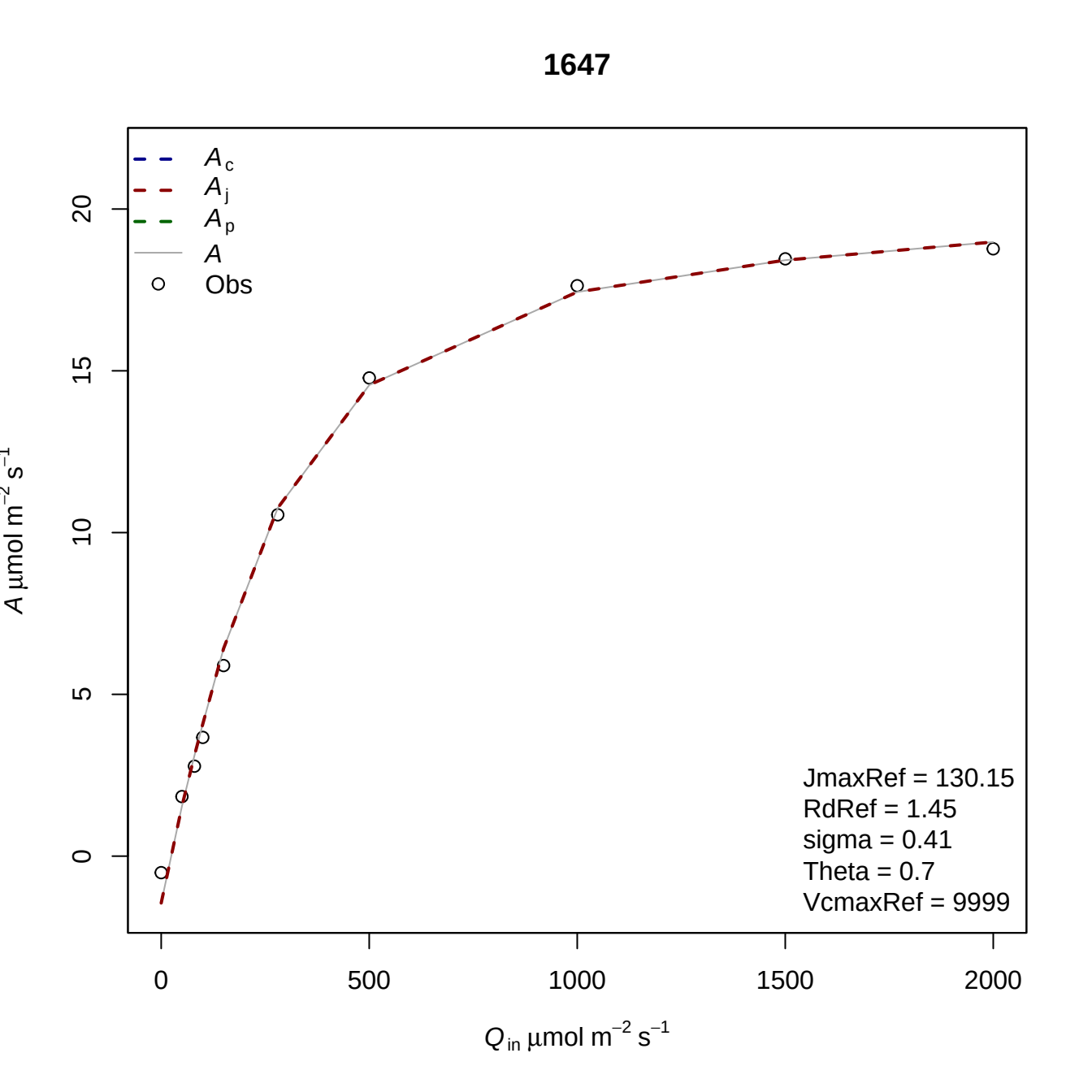

1642
A_c
A_j
A_p
A
Obs
JmaxRef = 62.84
RdRef = 1.09
sigma = 0.32
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

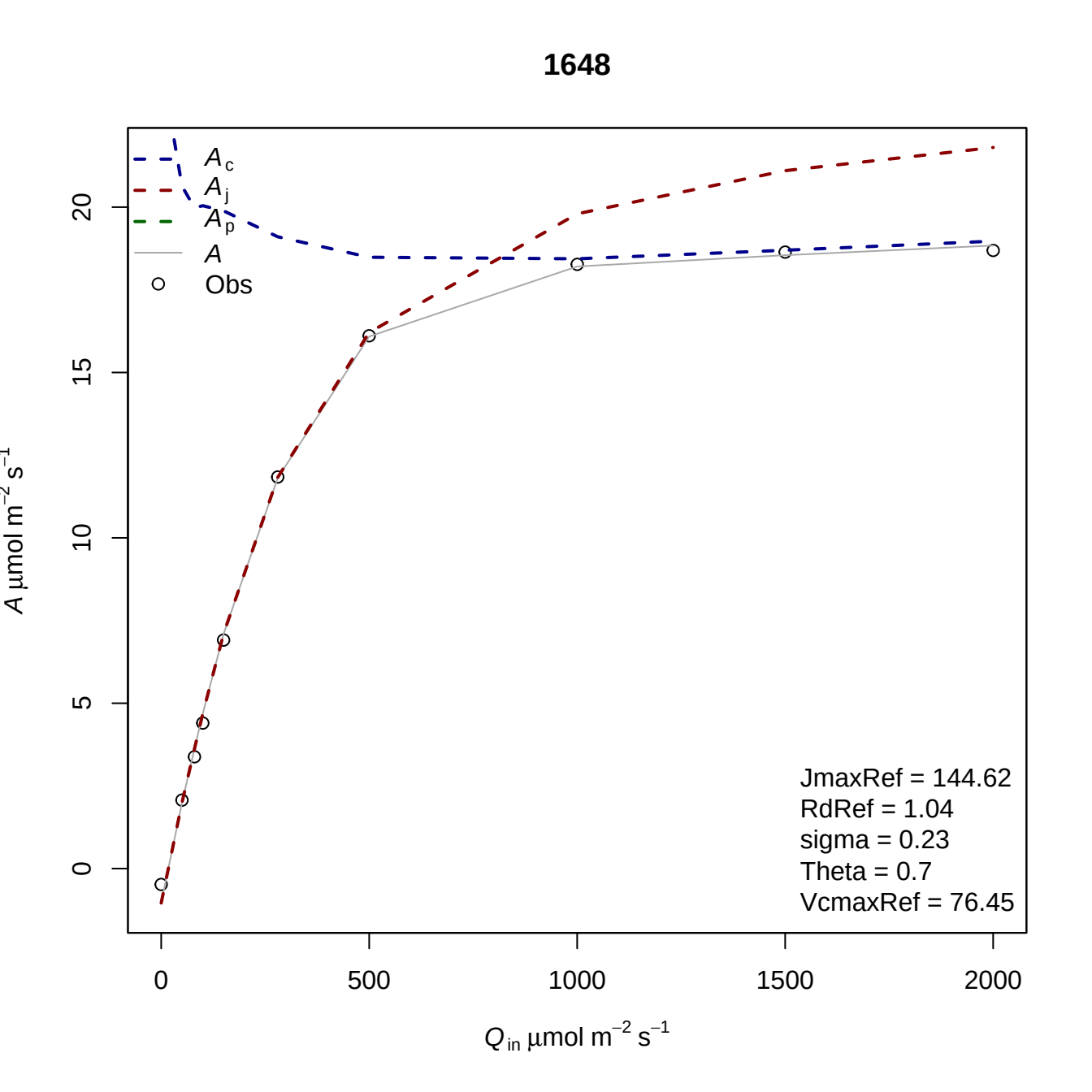

1643
A_c
A_j
A_p
A
Obs
JmaxRef = 150.02
RdRef = 1.54
sigma = 0.49
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

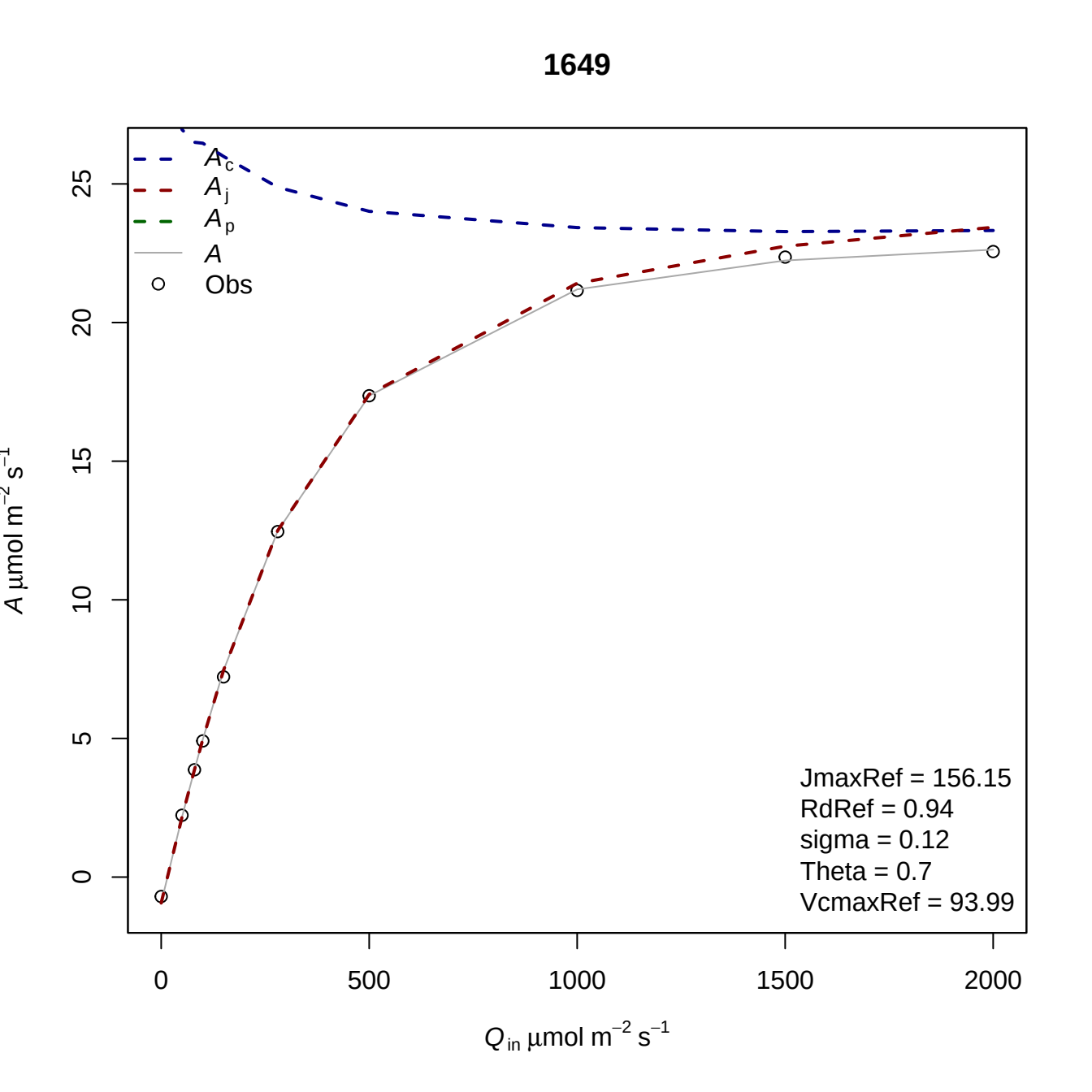

1644
A_c
A_j
A_p
A
Obs
JmaxRef = 166.97
RdRef = 1.11
sigma = 0.14
Theta = 0.7
VcmaxRef = 95.17
0
5
10
15
20
25
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

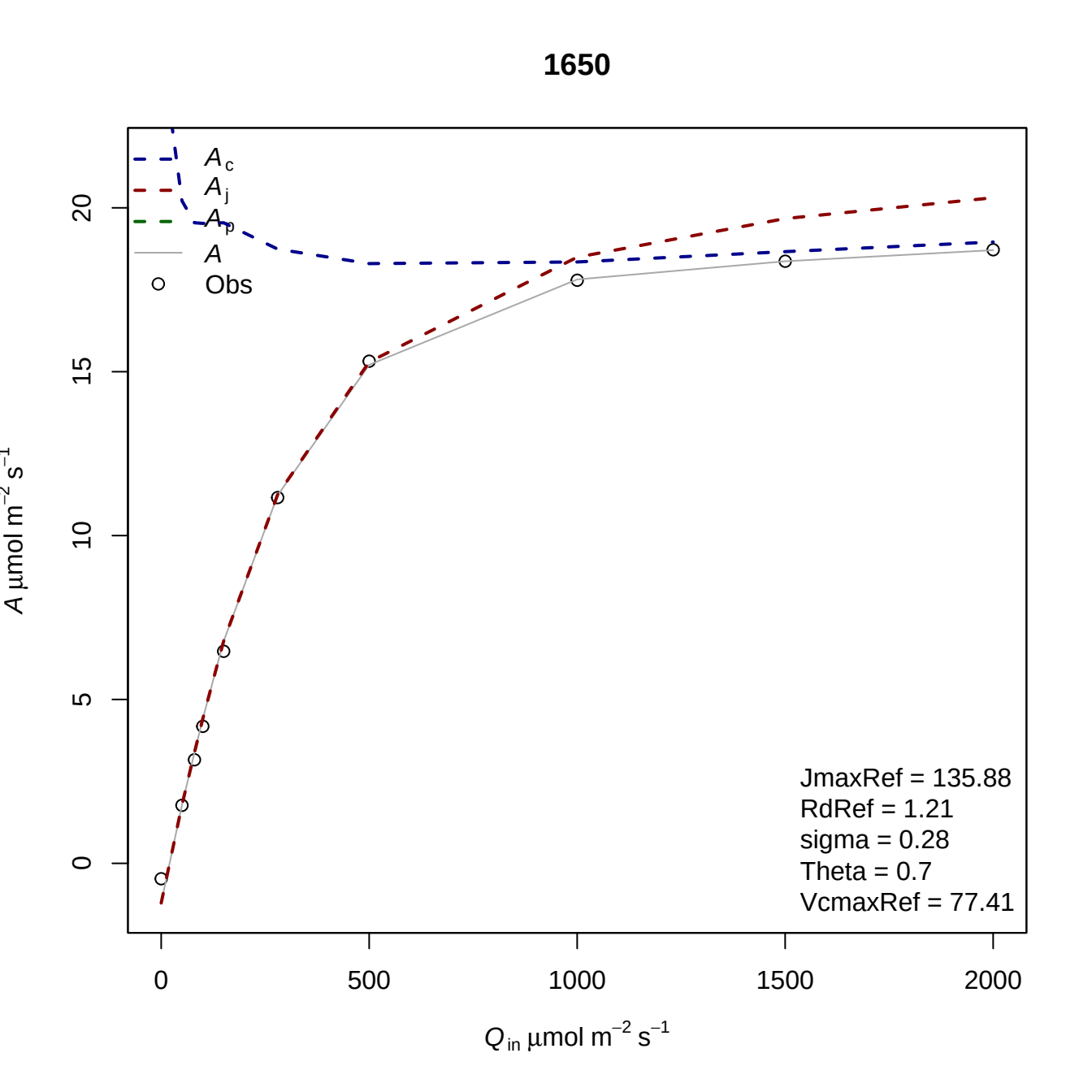

1645
A_c
A_j
A_p
A
Obs
JmaxRef = 163.37
RdRef = 1.73
sigma = 0.55
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

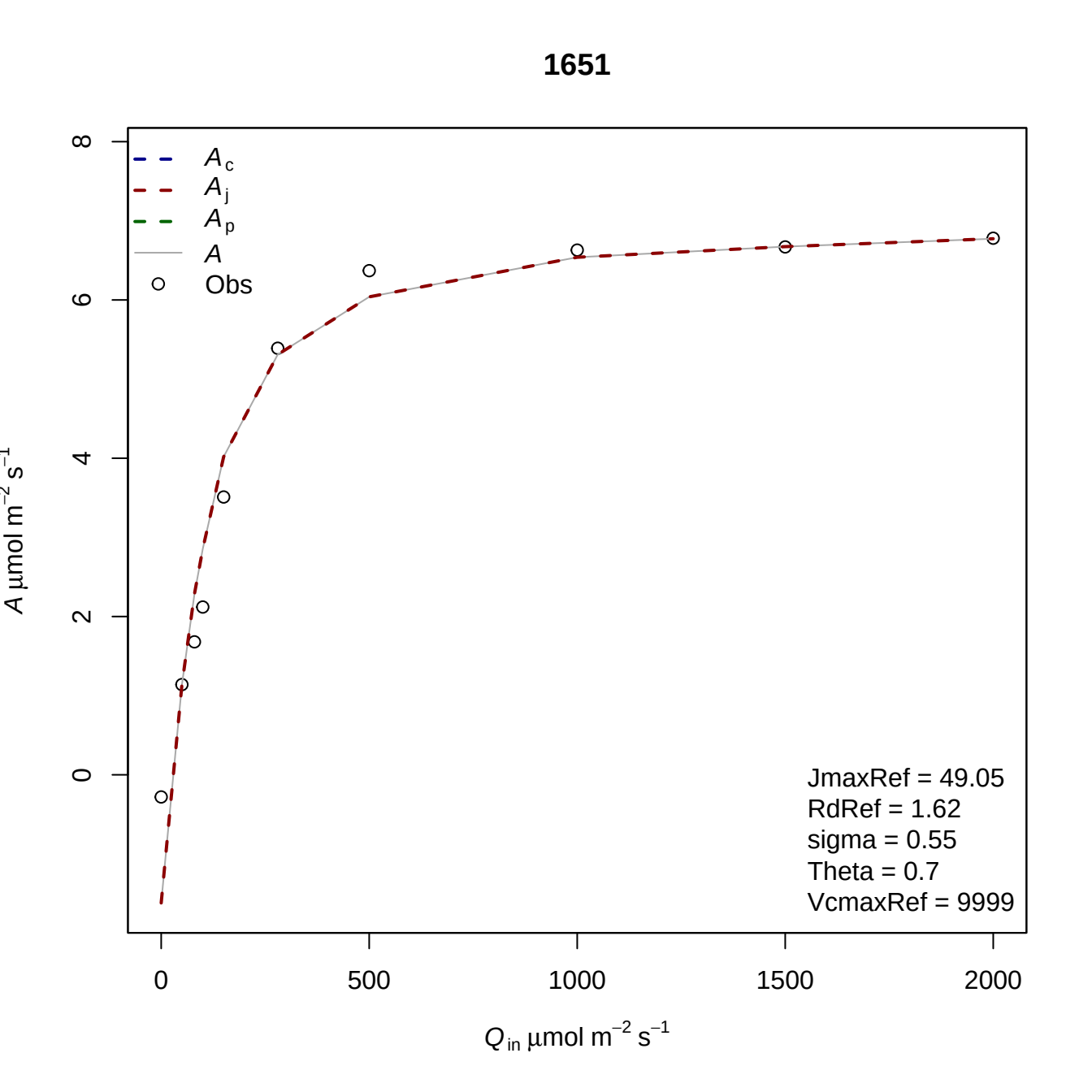

1646
A_c
A_j
A_p
A
Obs
JmaxRef = 177.4
RdRef = 2.33
sigma = 0.97
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
30
0
500
1000
1500
2000

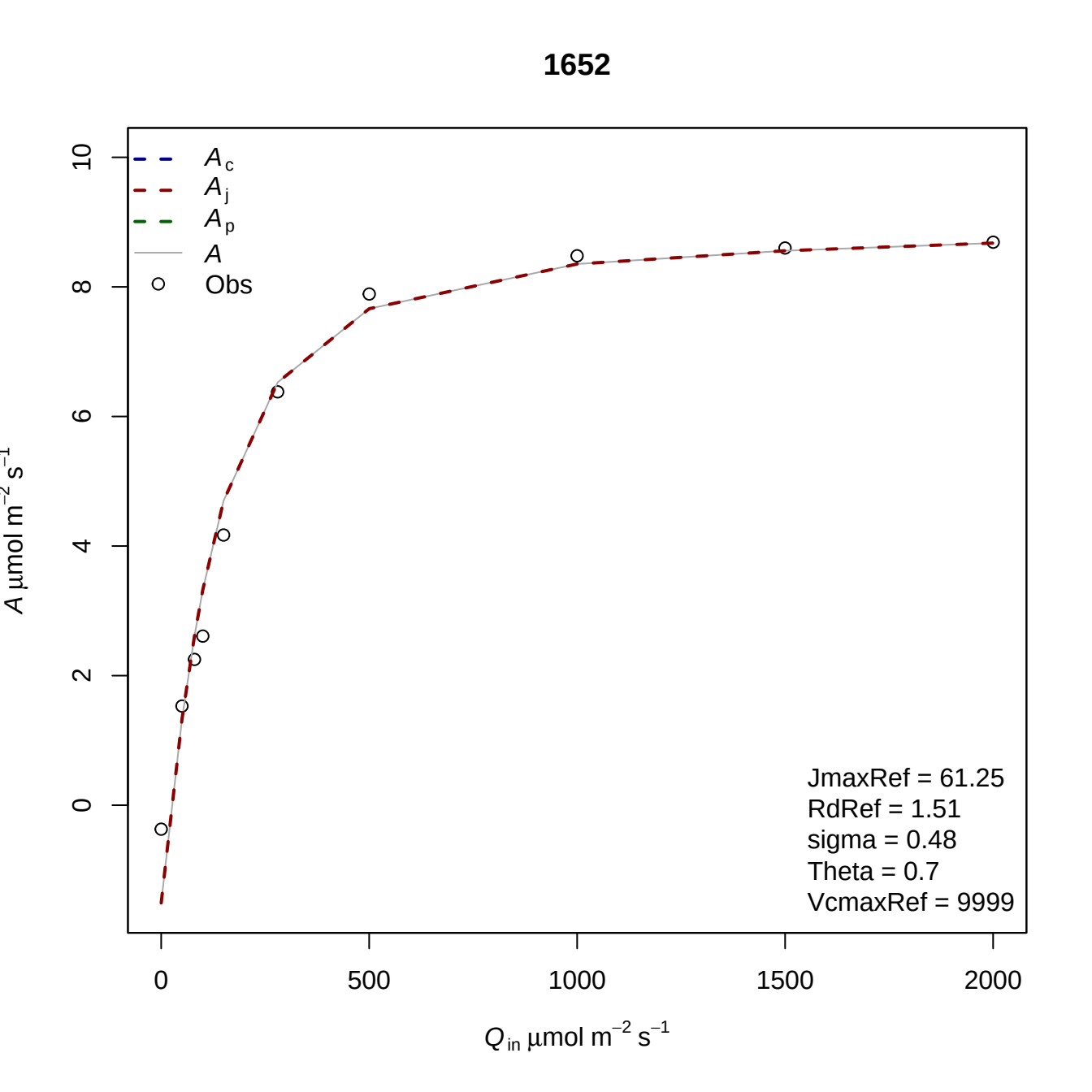

1647
A_c
A_j
A_p
A
Obs
JmaxRef = 130.15
RdRef = 1.45
sigma = 0.41
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

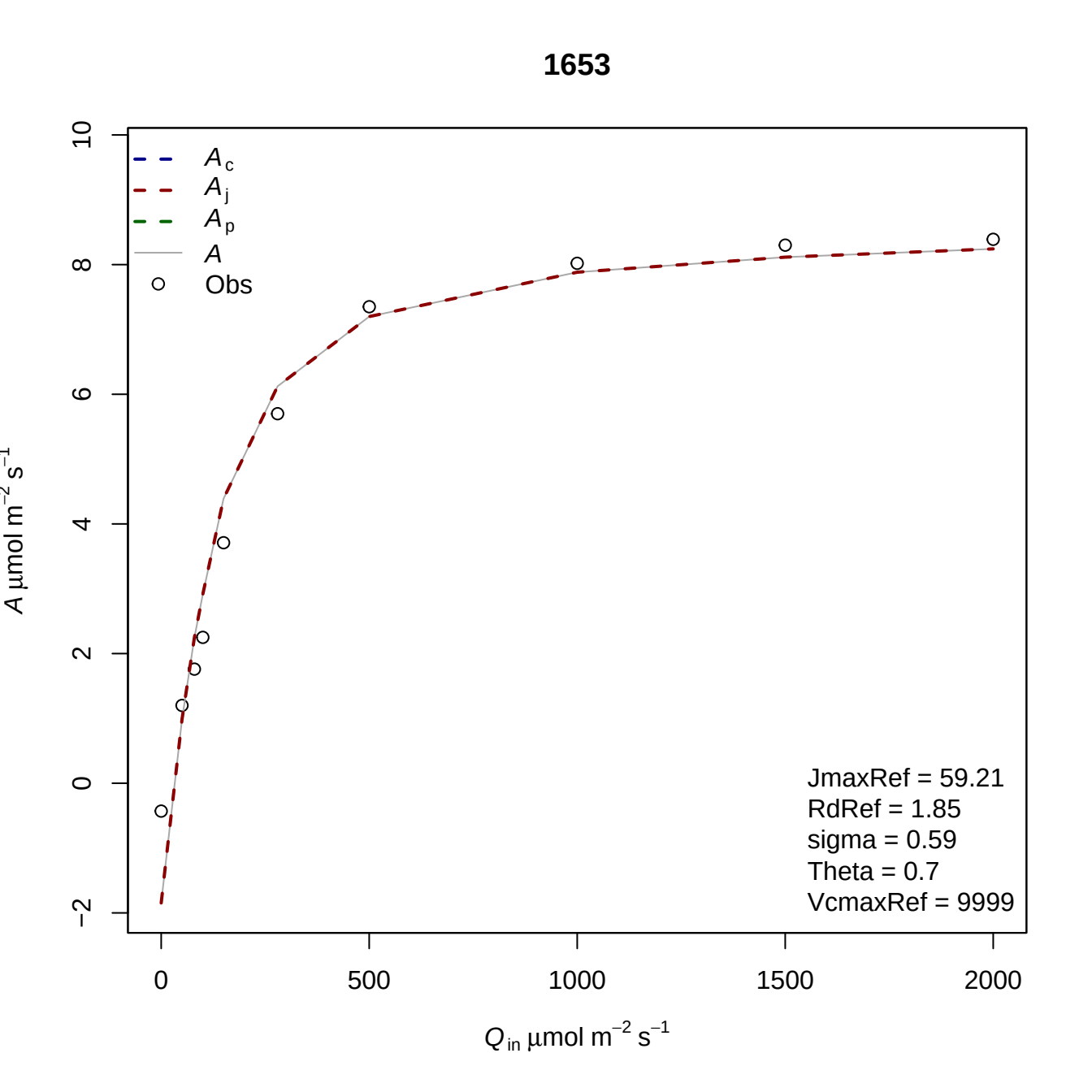

1648
A_c
A_j
A_p
A
Obs
JmaxRef = 144.62
RdRef = 1.04
sigma = 0.23
Theta = 0.7
VcmaxRef = 76.45
0
5
10
15
20
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

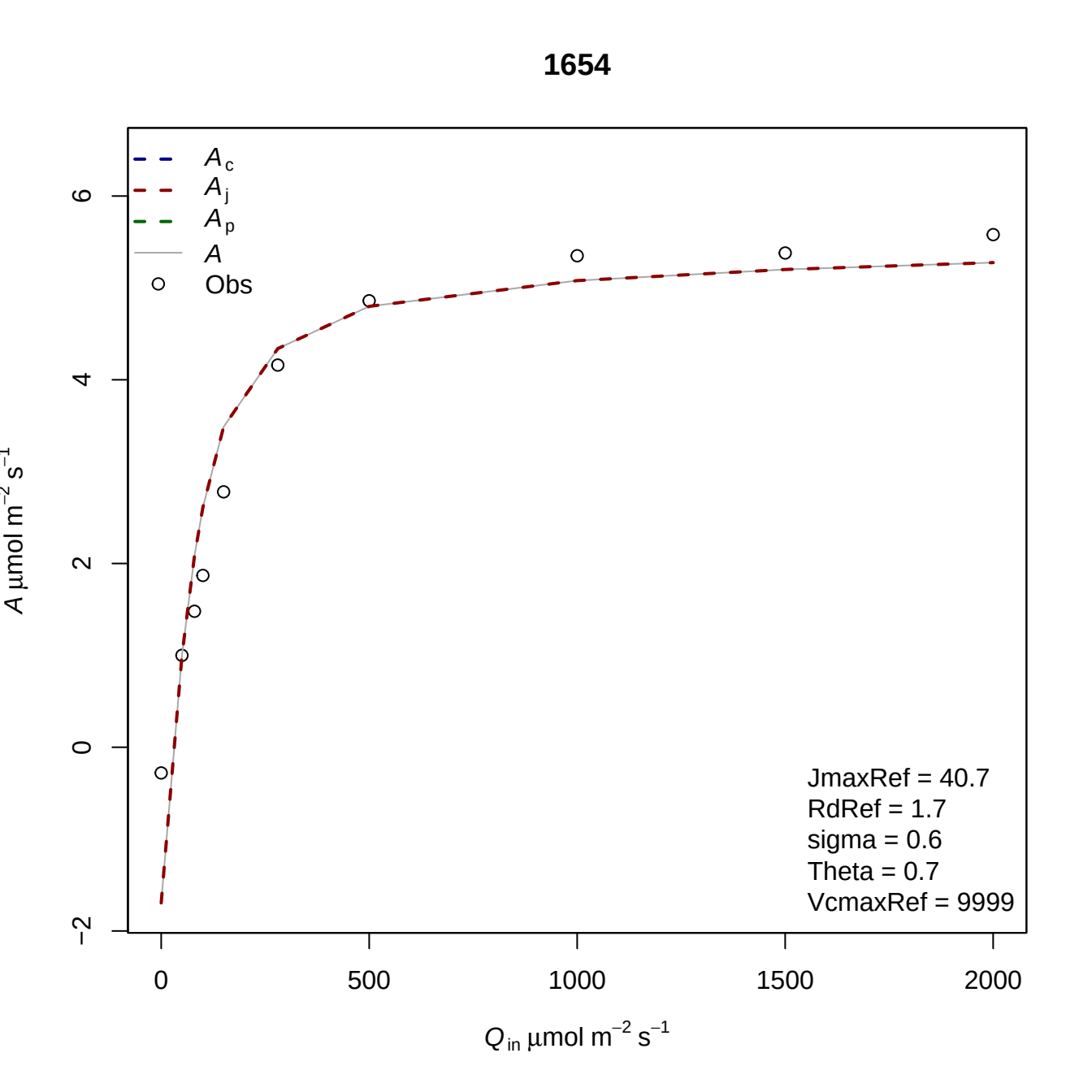

1649
A_c
A_j
A_p
A
Obs
JmaxRef = 156.15
RdRef = 0.94
sigma = 0.12
Theta = 0.7
VcmaxRef = 93.99
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
25
0
500
1000
1500
2000

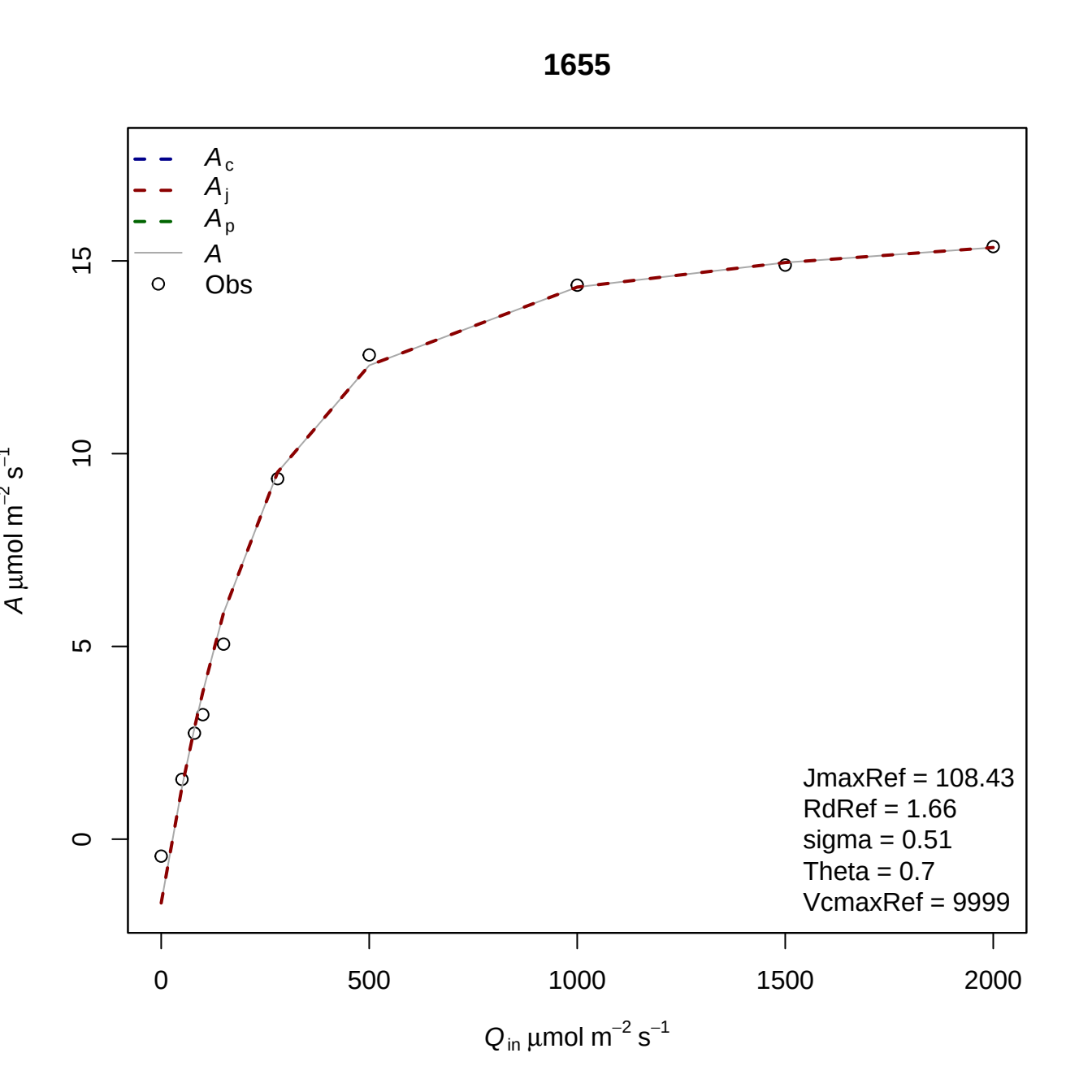

1650
A_c
A_j
A_p
A
Obs
JmaxRef = 135.88
RdRef = 1.21
sigma = 0.28
Theta = 0.7
VcmaxRef = 77.41
0
5
10
15
20
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}

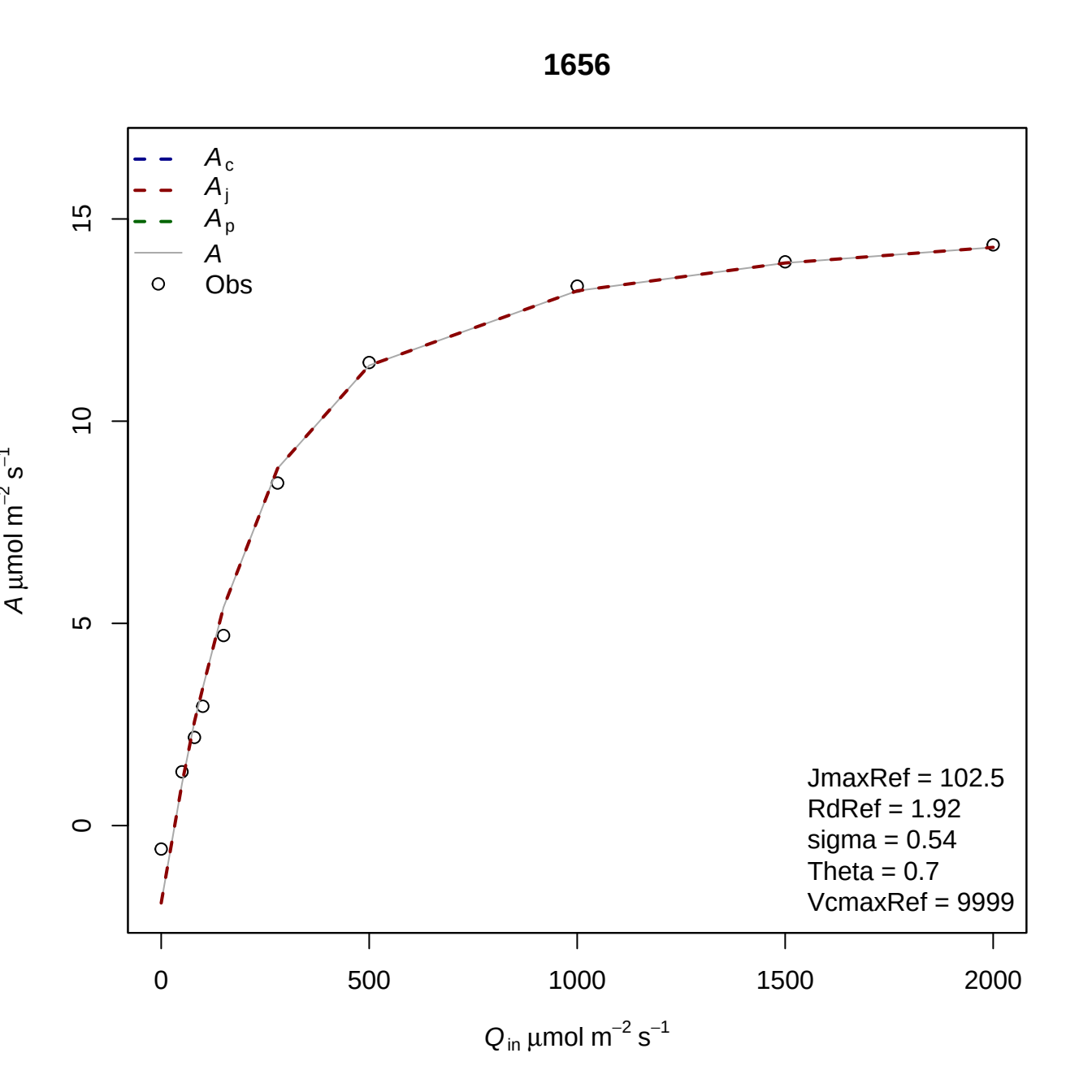

1651
A_c
A_j
A_p
A
Obs
JmaxRef = 49.05
RdRef = 1.62
sigma = 0.55
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
0
500
1000
1500
2000

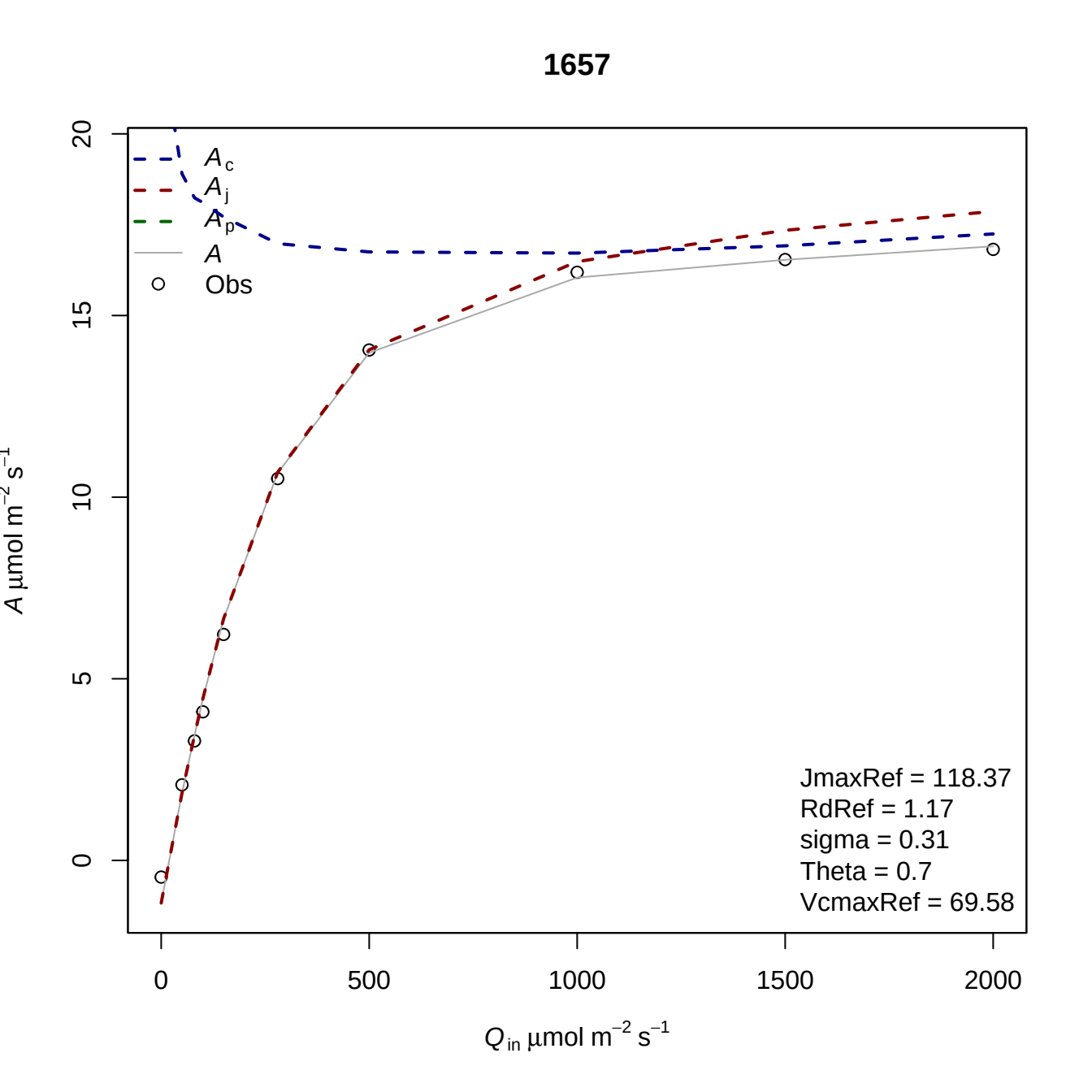

1652
A_c
A_j
A_p
A
Obs
JmaxRef = 61.25
RdRef = 1.51
sigma = 0.48
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
2
4
6
8
10
0
500
1000
1500
2000

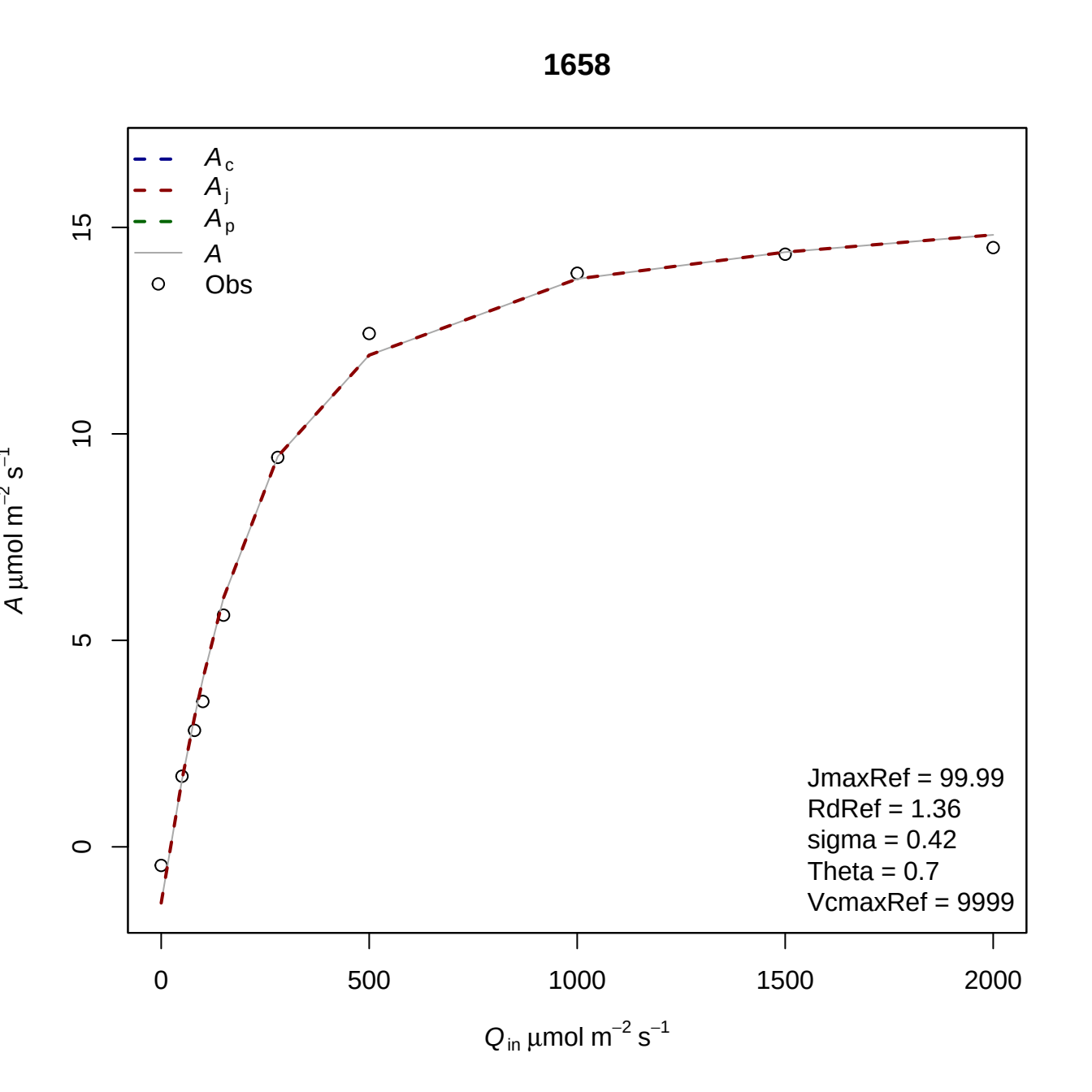

1653
A_c
A_j
A_p
A
Obs
JmaxRef = 59.21
RdRef = 1.85
sigma = 0.59
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
10
8
6
4
2
0
−2
0
500
1000
1500
2000

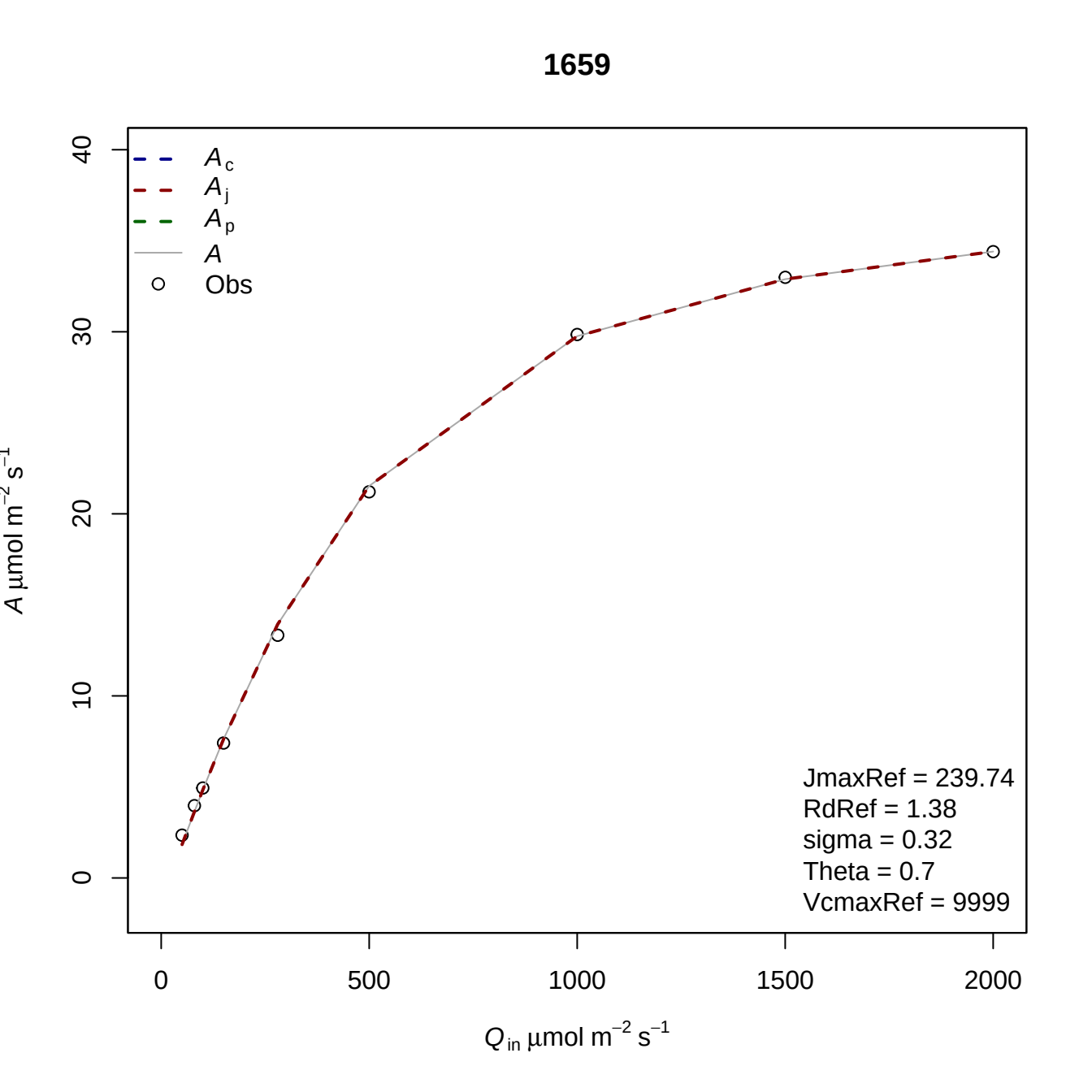

1654
A_c
A_j
A_p
A
Obs
JmaxRef = 40.7
RdRef = 1.7
sigma = 0.6
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
−2
0
2
4
6
0
500
1000
1500
2000

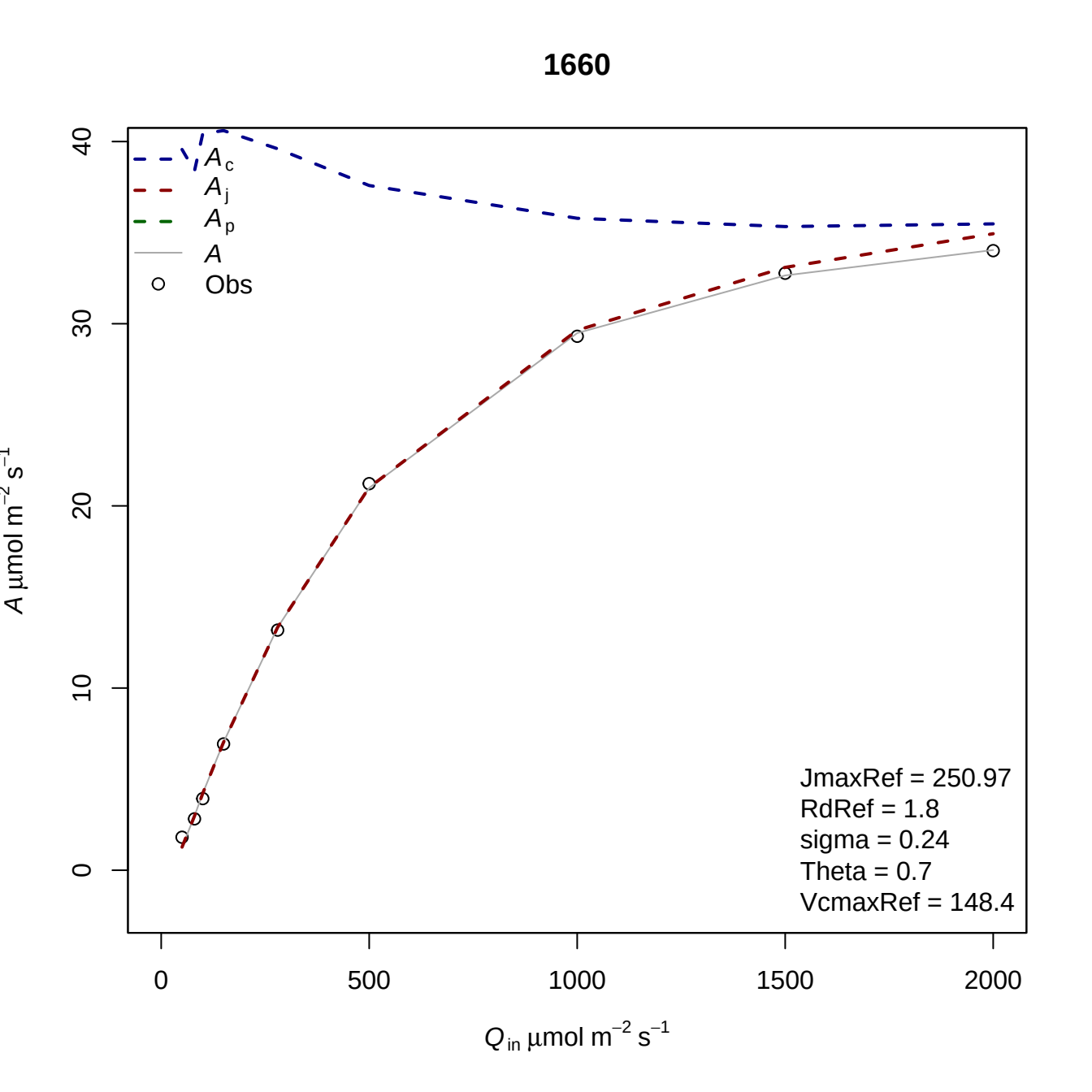

1655
A_c
A_j
A_p
A
Obs
JmaxRef = 108.43
RdRef = 1.66
sigma = 0.51
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
0
500
1000
1500
2000
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}

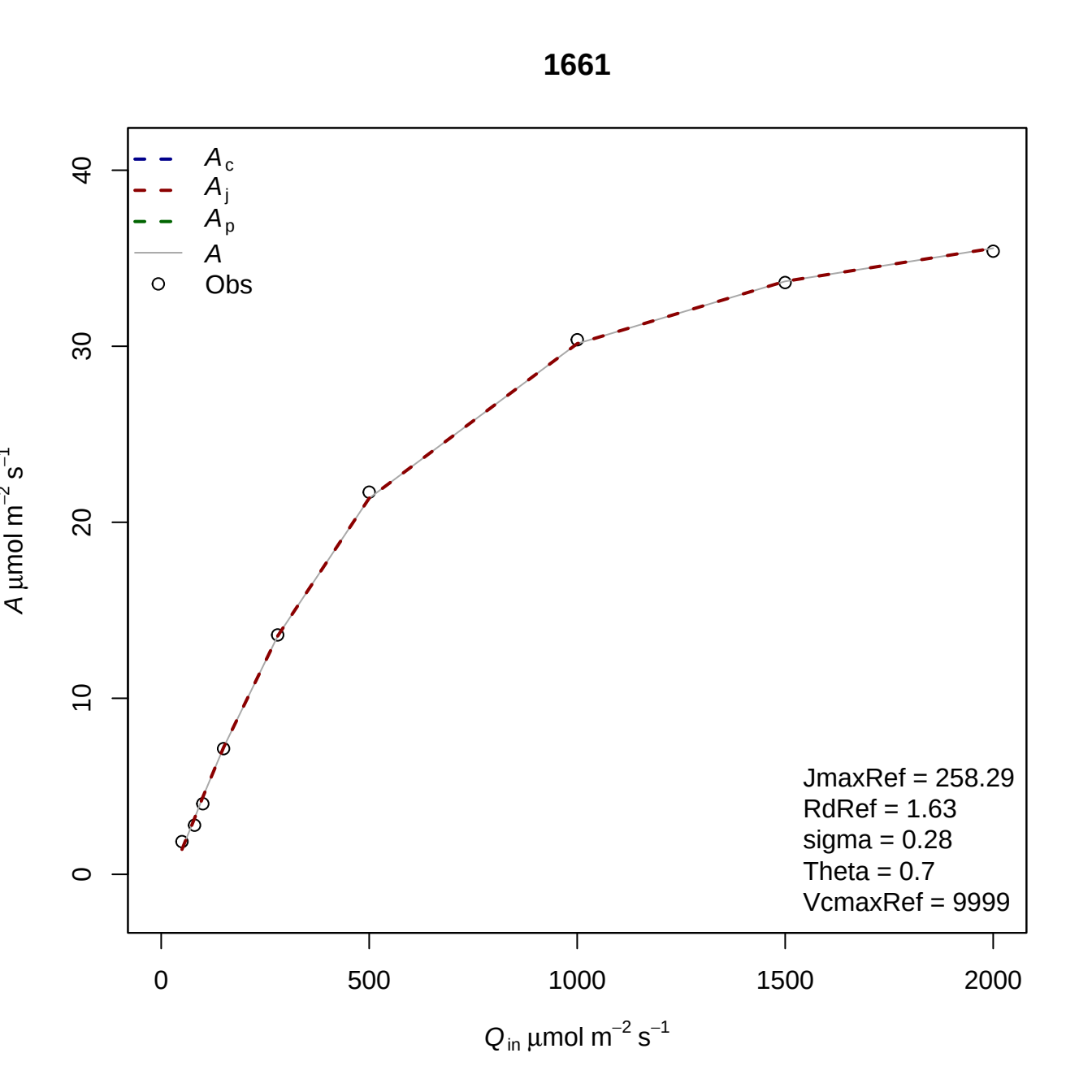

1656
A_c
A_j
A_p
A
Obs
JmaxRef = 102.5
RdRef = 1.92
sigma = 0.54
Theta = 0.7
VcmaxRef = 9999
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
0
500
1000
1500
2000

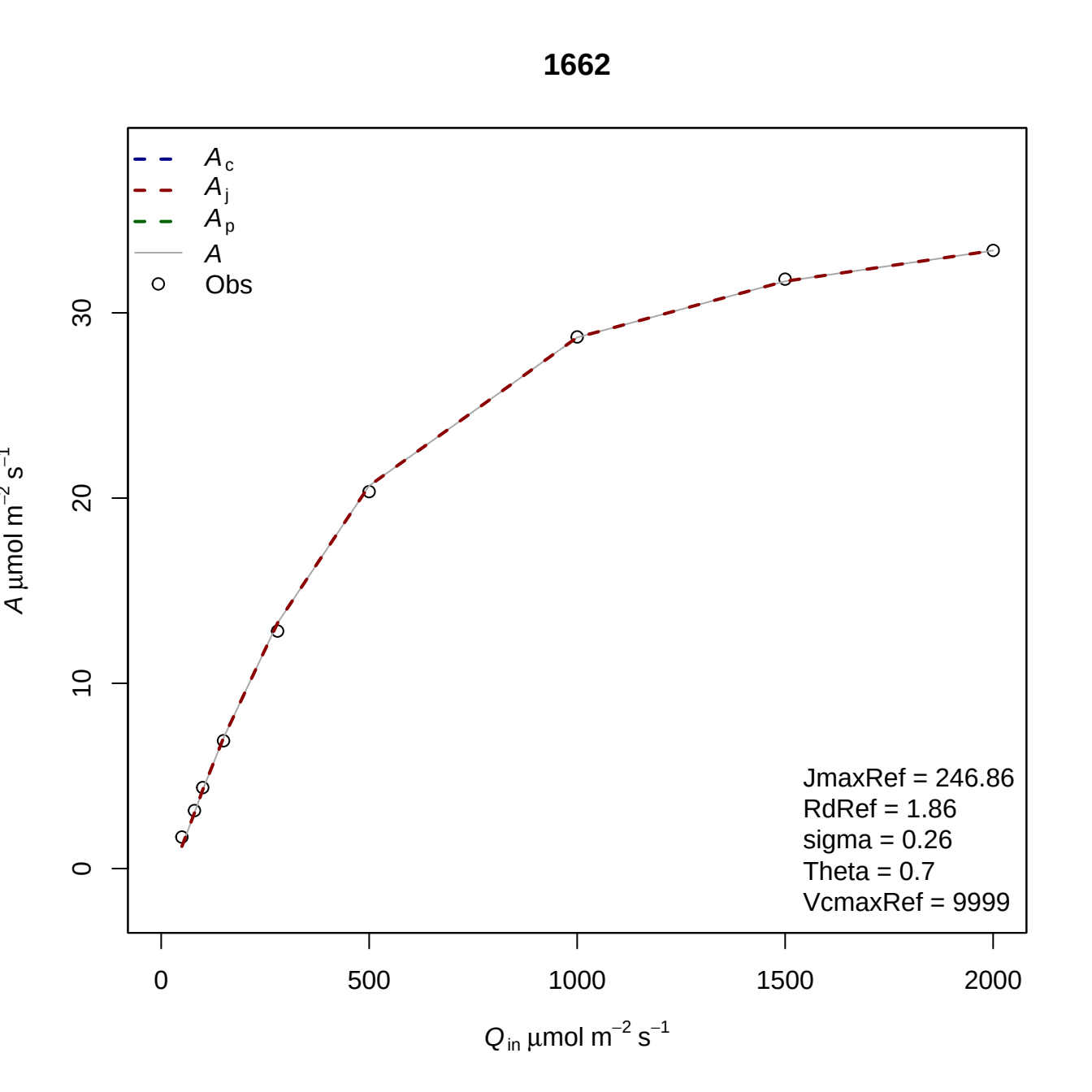

1657
A_c
A_j
A_p
A
Obs
JmaxRef = 118.37
RdRef = 1.17
sigma = 0.31
Theta = 0.7
VcmaxRef = 69.58
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
5
10
15
20
0
500
1000
1500
2000


1658
A_c
A_j
A_p
A
Obs
JmaxRef = 99.99
RdRef = 1.36
sigma = 0.42
Theta = 0.7
VcmaxRef = 9999
0
5
10
15
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}


1659
A_c
A_j
A_p
A
Obs
JmaxRef = 239.74
RdRef = 1.38
sigma = 0.32
Theta = 0.7
VcmaxRef = 9999
0
10
20
30
40
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}


1660
A_c
A_j
A_p
A
Obs
JmaxRef = 250.97
RdRef = 1.8
sigma = 0.24
Theta = 0.7
VcmaxRef = 148.4
A µmol m^{-2} s^{-1}
Q_{in} µmol m^{-2} s^{-1}
0
10
20
30
40
0
500
1000
1500
2000


1661
A_c
A_j
A_p
A
Obs
JmaxRef = 258.29
RdRef = 1.63
sigma = 0.28
Theta = 0.7
VcmaxRef = 9999
0
10
20
30
40
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}


1662
A_c
A_j
A_p
A
Obs
JmaxRef = 246.86
RdRef = 1.86
sigma = 0.26
Theta = 0.7
VcmaxRef = 9999
0
10
20
30
A µmol m^{-2} s^{-1}
0
500
1000
1500
2000
Q_{in} µmol m^{-2} s^{-1}